

Current typed-source status, 24 September 2026.

The current source has ungraded Z_0 absence and Z_1 presence, with no addition supplied at Z_1 . The user retracted the earlier additive equation on presence and the mixed-expression comparison. The former total-addition model is historical auxiliary mathematics, not a programme premise. The former TAC chapter is replaced by a withdrawal notice. The current TRZ theorem proves integer translation representation and the precise identity-only factorization obstruction; no $\mathbb{F}_1$ or prismatic conclusion is asserted.

Full lattice geometry in the Connes–Consani programme

Divisors, arithmetic sites, absolute curves and spectral receivers

Programme derivations from the cited human sources

23 September 2026

The retained receiving lattice object and its two exact coordinates are

$$G_L(R) = \{(0, \lambda) : \lambda \in L\} \cup (R \times \{1_L\}), \quad (q, \sigma)(r, \lambda) = (r, \lambda).$$

The actual characteristic-one comparison retains the entire lattice:

$$K_L = K \otimes_{\mathbb{B}} L, \quad \iota_L(\lambda) = 1_K \otimes \lambda, \quad \Phi_R : G_L(R) \hookrightarrow R \times K_L, \quad (r, \lambda) \mapsto (r, \iota_L(\lambda)).$$

Here K is the tropical semifield with zero. The complete family of lattice characters proves that ι_L is injective; its exact universal property and the constrained image of Φ_R are proved in ASL18–22. This is the coefficient map used below, including its sheaf, operator, trace and homology constructions.

Throughout, L is the entire bounded distributive lattice. Every label, both distinguished zeros, each join and meet, and the original nonzero-amplitude constraint are retained. The finite-lattice prime classification states its finite hypothesis separately; the coefficient and sheaf constructions apply to arbitrary nontrivial L .

The human sources are Alain Connes and Caterina Consani’s

Absolute algebra and Segal’s Γ -sets,

Riemann–Roch for the ring $\mathbb{Z}$,

Geometry of the Arithmetic Site,

On the Absolute Geometry of $\mathrm{Spec} \, Z$ and the Fargues–Fontaine curve, and

The Absolute Twistor Line and the Geometry of $\overline{\mathrm{Spec} \, \mathbb{Z}}$,

Segal’s Gamma rings and universal arithmetic, and

Homological algebra in characteristic one. The subsequent comparisons also use Uwe Jannsen’s

Weights in Arithmetic Geometry

and Brad Rodgers and Terence Tao’s

The De Bruijn–Newman constant is non-negative, with the actual bounded reading coverage recorded at the point of use.

The exact original TeX versions are retained with archive receipts.

Every application below includes its proof locators. The full lattice foundation is The Clankers,

Split Support Geometry, v11, Definition `def:lattice-split`.

The user attributes the earlier scalar supported-zero prime chain to Gemini 2.5 Pro; the present text does not claim its rediscovery.

This cumulative component contains the complete derivations currently proved: coefficient functors and cancellation, bounded sections, divisor multiplication and morphisms, tropical base change and primes, absolute-curve characters and sheaf comparisons, the original

spectral receiver with its tensor and nilpotent sectors, full-support Adams and Witt operations, monodromy compatibility, supported homology, and the exact zero-inversion comparison with the zeta heat family. It is not a claim to have completed the entire corpus synthesis, a Deligne purity comparison, or the Riemann hypothesis.

Original zeta and theta scope. The original arithmetic functions remain $\zeta(s)$ and the specified

shifted family $F_t(s) = \zeta(s, 1 + t)$. A support-preserving lift of the classical theta kernel is not a derivation of that kernel from the entire base. ZTC1–21 reconstruct the direct scalar theta/Mellin map in both directions, every endpoint term, the retained boundary pair, and the original divisor/trace. CFD1–27 proves the exceptional module and jet comparisons and the difference between the heat and Hurwitz trajectories on the actual initial function. No completion multiplier or supported-zero stratum is discarded by these comparisons.

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The whole lattice map and its prime fibres

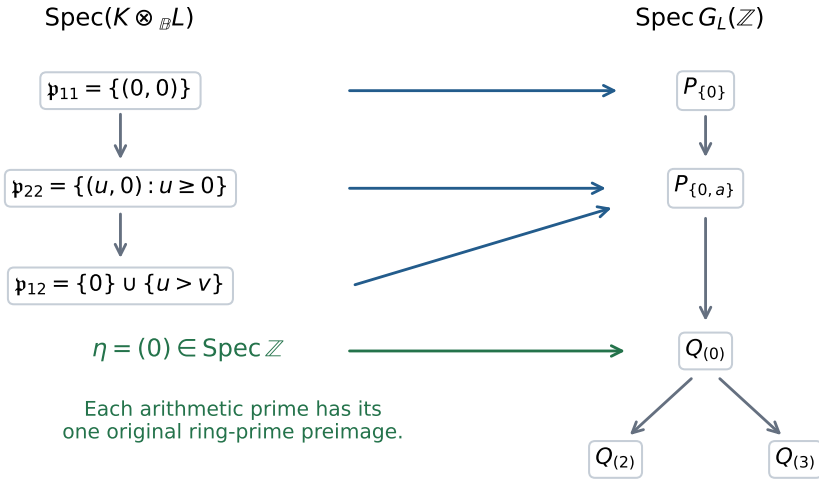
$L = \{0 < a < 1\}$, with join and meet; $K = ([0, \infty), \max, \cdot)$.

$$G_L(\mathbb{Z}) \longrightarrow \mathbb{Z} \times (K \otimes_{\mathbb{B}} L)$$

$$(r, \lambda) \longmapsto (r, \mathbf{1}_{\{j \leq \lambda\}})$$

$\tau = z_0$	$\longrightarrow$	$(0; (0, 0))$
z_a	$\longrightarrow$	$(0; (1, 0))$
$e = z_1$	$\longrightarrow$	$(0; (1, 1))$
$\hat{n} \ (n \neq 0)$	$\longrightarrow$	$(n; (1, 1))$

Prime contraction: the complete fibres



Grey arrows: strict prime inclusion. Blue/green arrows: contraction.
Only primes 2 and 3 are drawn; every rational prime remains.

The additional prime is retained

$$(2, 1) \in \mathfrak{p}_{12}, \quad (2, 1) \notin \mathfrak{p}_{11} \cup \mathfrak{p}_{22}.$$

$$\text{Fr}_0^{-1}(\mathfrak{p}_{12}) = \mathfrak{p}_{22}, \quad \text{Fr}_t^{-1}(\mathfrak{p}_{ij}) = \mathfrak{p}_{ij} \ (t > 0).$$

Proofs: ASL18–22, ASL31; LSP3–27. Tropical coefficients: Connes–Consani,
Geometry of the Arithmetic Site, arXiv:1502.05580v1, Definition defnpt.
Full lattice base: The Clankers, Split Support Geometry v11, def:lattice-split.

Figure 1: The full lattice map in the smallest example with an intermediate label. The three tropical primes map to two support primes; the arithmetic primes are supplied by the ring factor. Both coordinates of the original carrier remain. All displayed maps, fibres and Frobenius actions are proved in ASL18–22 and LSP3–27.

1 The full support lattice and the arithmetic site's coefficient extension

The human source used here is Alain Connes and Caterina Consani, *Geometry of the Arithmetic Site*, original author source `arithmeticsite_Adv_final1.tex`, 2074 lines, SHA-256 `bd9da2b9e3c8846e306af18dbf745d7cf603450d6630a00e239da4235a7320b4`.

The source mathematics actually read for this calculation is lines 331–407, 440–503, 598–704, and 850–1010. These ranges include the definition of the characteristic-one semirings and Frobenius, the rank-one description of points, the arithmetic-site definition, the complete stalk proof, global sections, the definition of valued points, and the stated pullback to the ordinary arithmetic spectrum. This is bounded reading, not a claim to have read the whole paper.

The known programme input is the full, arbitrary bounded distributive lattice L , not a Boolean lattice in its place. Its carrier, complete prime classification, and supported-zero localization are already proved in 031, Theorem `thm:rebuilt-full-support`, and Theorem `thm:e-stalk`, equation `eq:e-element-localization`. The latter states $G_L(R)[e^{-1}] \simeq L$. It is an existing result, not a new assertion of the present calculation. We prove its precise universal consequence for characteristic-one targets, and construct the resulting full-lattice coefficient extension of the arithmetic site, including its points, stalks, and Frobenius.

1.1 The characteristic-one universal property

Let R be a nonzero commutative unital ring and let L be a nontrivial bounded distributive lattice, finite or infinite. Retain

$$\begin{aligned} S &= G_L(R) = \{(0, \lambda) : \lambda \in L\} \cup \{(r, 1_L) : r \in R\}, \\ (r, \lambda) + (s, \mu) &= (r + s, \lambda \vee \mu), \quad (r, \lambda)(s, \mu) = (rs, \lambda \wedge \mu), \\ z_\lambda &= (0, \lambda), \quad \tau = z_{0_L}, \quad e = z_{1_L}, \quad 1_S = (1_R, 1_L), \quad \sigma : S \longrightarrow L, \quad \sigma(r, \lambda) = \lambda. \end{aligned} \tag{ASL1}$$

The operations on the semiring L are join and meet, with zero 0_L and unit 1_L . A semiring homomorphism below preserves both zero and unit.

For every commutative semiring K whose addition is idempotent, the map σ induces a natural bijection

$$\mathrm{Hom}_{\mathrm{sr}}(L, K) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{sr}}(S, K), \quad g \longmapsto g\sigma. \tag{ASL2}$$

To prove this, let $f : S \rightarrow K$. In S , write $v = (-1_R, 1_L)$, so $e = 1_S + v$ and $e + 1_S = 1_S$. Idempotence in K gives

$$f(e) + 1_K = (1_K + f(v)) + 1_K = 1_K + f(v) = f(e),$$

whereas the second source relation makes this equal to 1_K . Thus $f(e) = 1_K$. Since $e(r, 1_L) = e$, every supported amplitude satisfies $f(r, 1_L) = 1_K$. Define $g(\lambda) = f(z_\lambda)$. Addition and multiplication of the z_λ show that g preserves join and meet, and $g(0_L) = 0_K$, $g(1_L) = 1_K$. It follows that $f = g\sigma$. Conversely every such composite is a homomorphism. Surjectivity of σ gives uniqueness. This proves (ASL2) directly, with exactly the same localization map as the established result.

There is a more precise description for an arbitrary idempotent K . Use its natural partial order $x \leq y \iff x + y = y$ and let

$$\mathrm{Id}_{\leq 1}(K) = \{a \in K : a^2 = a, a + 1_K = 1_K\}. \tag{ASL3}$$

This is a bounded distributive lattice under addition and multiplication. For a, b in this set, $ab \leq a, b$, while any other c in the set with $c \leq a, b$ satisfies $c = c^2 \leq ab$. Thus multiplication is their meet. Also $(a + b)^2 = a + b + ab = a + b$, since $ab \leq a + b$, so addition gives the join in the same set. Distributivity is inherited from K . Every unital homomorphism from L has its image here, because $\lambda^2 = \lambda$ and $\lambda \vee 1_L = 1_L$. Consequently (ASL2) is also the exact bijection

$$\mathrm{Hom}_{\mathrm{sr}}(S, K) \simeq \mathrm{Hom}_{\mathrm{BLat}}(L, \mathrm{Id}_{\leq 1}(K)). \tag{ASL4}$$

1.2 All maps to the actual tropical coefficient semirings

Connes–Consani define their structure semiring as $\mathbb{Z}_{\max}$. Its stalk at the ordered group H is

$$H_{\max} = (H \cup \{-\infty\}, \max, +).$$

See their Definition `site`, lines 618–622, and Theorem `structure2`, lines 626–659. Write $\mathbf{0} = -\infty$ and $\mathbf{1} = 0_H$ in this additive presentation, so that the multiplicative unit is never confused with the absorbing zero. For any totally ordered abelian group H , the only multiplicative idempotents of $H_{\max}$ are $\mathbf{0}$ and $\mathbf{1}$: for $h \in H$, $h + h = h$ forces $h = 0_H$. The same assertion holds for their nonnegative min-version $(H_{\geq 0} \cup \{\infty\}, \min, +)$ (source lines 662–665), and for the multiplicative tropical real semifield $\mathbb{R}_+^{\max} = ([0, \infty), \max, \cdot)$.

Let $X = \text{SpecLat } L$ be the set of proper lattice prime ideals. For $P \in X$ set

$$\chi_P(\lambda) = \begin{cases} 0, & \lambda \in P, \\ 1, & \lambda \notin P. \end{cases} \quad \chi_P : L \longrightarrow \mathbb{B} = \{0, 1\}. \quad (\text{ASL5})$$

A lattice ideal is closed under finite joins and downward closed; primality means $\lambda \wedge \mu \in P$ forces one factor into P . These properties prove exactly that (ASL5) preserves zero, unit, join, and meet. Conversely the zero fibre of a bounded lattice homomorphism to $\mathbb{B}$ is such a prime ideal, and determines the homomorphism. Combining this fact with (ASL4) gives, for each of the tropical targets K just specified,

$$\text{Hom}_{\text{sr}}(S, K) \simeq X, \quad f_P = \iota_{\mathbb{B}, K} \chi_P \sigma. \quad (\text{ASL6})$$

Here the canonical inclusion sends the two Boolean elements to $\mathbf{0}_K, \mathbf{1}_K$. The complete fibres are

$$\begin{aligned} f_P^{-1}(\mathbf{0}_K) &= \{z_\lambda : \lambda \in P\}, \\ f_P^{-1}(\mathbf{1}_K) &= \{z_\lambda : \lambda \notin P\} \cup \{(r, 1_L) : r \in R \setminus \{0_R\}\}, \\ f_P^{-1}(a) &= \emptyset \quad (a \notin \{\mathbf{0}_K, \mathbf{1}_K\}). \end{aligned} \quad (\text{ASL7})$$

In particular all ordinary nonzero amplitudes and the supported zero e have value $\mathbf{1}_K$. This is forced by the universal property; it is not an assumption that the added prime has no effect.

The source’s Frobenius maps are $x \mapsto x^n$, $n \geq 1$, and on $\mathbb{R}_+^{\max}$ also $x \mapsto x^t$, $t > 0$ (source lines 395–406). They fix both Boolean elements. Thus

$$\text{Fr}_n^K f_P = f_P, \quad f_P(x^n) = f_P(x), \quad \text{Fr}_t^{\mathbb{R}_+^{\max}} f_P = f_P. \quad (\text{ASL8})$$

The middle identity holds for the actual powers in S , by multiplicativity. It does not assert that $x \mapsto x^n$ is an additive endomorphism of S ; that fails, for example, for $R = \mathbb{Z}$ and $n = 2$. On the reflected lattice, $\lambda^n = \lambda$, so its Frobenius is exactly the identity. Equations (ASL8) are the precise compatibility statements, with no additional action imposed on R .

1.3 The full lattice is recovered by its whole family of characters

Define the basic opens $D(\lambda) = \{P \in X : \lambda \notin P\}$. The map

$$L \longrightarrow \mathbb{B}^X, \quad \lambda \longmapsto (\chi_P(\lambda))_{P \in X} = \mathbf{1}_{D(\lambda)} \quad (\text{ASL9})$$

is an injective lattice homomorphism. Here and below we use the usual prime-ideal theorem for bounded distributive lattices, for which the needed separation proof is as follows. If $\lambda \not\leq \mu$, the ideal $\downarrow \mu$ and filter $\uparrow \lambda$ are disjoint. By Zorn’s lemma choose an ideal maximal among those containing the first and disjoint from the second. If a, b are outside it but $a \wedge b$ lies in it, adjoining a and adjoining b each produces an intersection with the filter. Thus filter elements are bounded above by $i \vee a$ and $j \vee b$ for some ideal elements i, j . Their meet is bounded above by $i \vee j \vee (a \wedge b)$,

by distributivity, and hence is in the ideal, a contradiction. The maximal ideal is therefore prime, contains μ , and avoids λ . This separates the two elements in (ASL9).

The same proof shows $D(\lambda)$ is compact. If it is covered by $D(\mu_i)$ but no finite subfamily covers it, then λ is not in the ideal generated by all the μ_i . The preceding ideal/filter argument gives a prime containing all μ_i and avoiding λ , contradicting the cover. Thus a finite subfamily covers it. Since

$$D(\lambda \vee \mu) = D(\lambda) \cup D(\mu), \quad D(\lambda \wedge \mu) = D(\lambda) \cap D(\mu), \quad (\text{ASL10})$$

every compact open of X equals a unique $D(\lambda)$. Consequently (ASL9) identifies L with the lattice of compact opens, not with the entire power set of X .

For finite L this is especially explicit. Every nonzero join-irreducible j is join-prime: if $j \leq a \vee b$, then $j = (j \wedge a) \vee (j \wedge b)$ forces $j \leq a$ or $j \leq b$. Its character is $\chi_j(\lambda) = 1$ exactly when $j \leq \lambda$. Conversely the complement of any prime ideal is a finite filter; its least element j is nonzero and join-prime, so every character arises this way. Thus

$$X \simeq J(L), \quad L \simeq \{\text{downward-closed subsets of } J(L)\}, \quad \lambda \mapsto \{j : j \leq \lambda\}. \quad (\text{ASL11})$$

For surjectivity in the last map, a downward-closed subset A has $\lambda = \bigvee A$; join-primality puts every irreducible below λ in A . No Boolean support assumption occurs. For the three-element chain, the character vectors are $(0,0), (1,0), (1,1)$; the vector $(0,1)$ is absent. Replacing this lattice by two independent Boolean channels would therefore change the object.

1.4 Prime centres, localization, and the support structure sheaf

The complete prime data of the original semiring, already proved in 031, is

$$\begin{aligned} P^{\text{supp}} &= \{z_\lambda : \lambda \in P\} \quad (P \in X), \\ Q_{\mathfrak{p}} &= \{z_\lambda : \lambda \in L\} \cup \{(r, 1_L) : r \in \mathfrak{p}\} \quad (\mathfrak{p} \in \text{Spec } R), \\ \text{Spec } S &= X \blacktriangleright \text{Spec } R, \quad D_S(e) = X, \quad V_S(e) = \text{Spec } R. \end{aligned} \quad (\text{ASL12})$$

For clarity, exhaustion follows directly from the source operations. An ideal containing a supported element contains e , hence every z_λ ; its supported amplitudes form a ring ideal. An ideal containing no supported element consists of the z_λ from a proper lattice ideal. The two primality tests are respectively ring primality and meet-primality. Every support prime lies strictly below every arithmetic prime, and the basic opens are $D_S(z_\lambda) = D(\lambda)$ and $D_S((r, 1_L)) = X \cup D_R(r)$. This proves the stated topology.

A tropical semifield has only its zero ideal as a proper ideal, because every nonzero element is a unit. The spectral map induced by f_P therefore has the single image P^{supp} , by (ASL7). More generally, every proper prime of any target in (ASL6) contracts to that same prime, since it contains $\mathbf{0}$ and excludes $\mathbf{1}$. In particular these maps do not centre on an arithmetic prime $Q_{\mathfrak{p}}$ containing e .

For a multiplicative subset $T \subset S$, a character map f_P extends to $T^{-1}S$ exactly when

$$\sigma(T) \cap P = \emptyset; \quad f_P(s/t) = f_P(s) \quad (t \in T). \quad (\text{ASL13})$$

Indeed all target values of denominators are either zero or unit one. For the universal property with arbitrary idempotent targets, localization commutes with the reflection:

$$(T^{-1}S)_{\text{idemp}} \simeq \sigma(T)^{-1}L. \quad (\text{ASL14})$$

Both sides represent maps from S to idempotent semirings in which all elements of T become units, using (ASL2), so the displayed isomorphism follows from the two inverse universal maps.

The full support localizations are explicit:

$$L[\lambda^{-1}] \simeq \downarrow \lambda, \quad \mu \mapsto \mu \wedge \lambda, \quad 1_{\downarrow \lambda} = \lambda. \quad (\text{ASL15})$$

The equivalence relation is $\mu \sim \nu \iff \mu \wedge \lambda = \nu \wedge \lambda$; all positive powers of the denominator are λ itself. For a prime P , the stalk has the description

$$L_P = L/(\mu \sim_P \nu), \quad \mu \sim_P \nu \iff \exists t \notin P : \mu \wedge t = \nu \wedge t, \quad S_{P^{\text{supp}}} \simeq L_P. \quad (\text{ASL16})$$

Every denominator in $L \setminus P$ is idempotent, so after being inverted it becomes the unit; this gives the stated equivalence and exhausts fractions. In $S_{P^{\text{supp}}}$, e is inverted first, giving L , and the remaining denominators are precisely those represented by $L \setminus P$. This proves the last isomorphism.

These semirings of sections form a sheaf on the support spectrum. On compact basic opens set

$$\mathcal{L}(D(\lambda)) = \downarrow \lambda, \quad \text{res}_{D(\mu)}(a) = a \wedge \mu \quad (\mu \leq \lambda). \quad (\text{ASL17})$$

For a finite cover with $\lambda = \bigvee_i \lambda_i$, compatible sections $a_i \leq \lambda_i$ obey $a_i \wedge \lambda_j = a_j \wedge \lambda_i$. Their unique gluing is $a = \bigvee_i a_i$: distributivity proves $a \wedge \lambda_i = a_i$, and if a second section has these restrictions then intersecting it with $\bigvee_i \lambda_i$ proves equality. Compactness gives a finite subcover for any cover of a basic open, and the same compatibility checks every remaining member. Thus this is a sheaf on the basis and has stalk (ASL16). It is the restriction of the localization structure sheaf of S to $D_S(e)$.

The stalk L_P generally retains more than a Boolean value. For the chain $0 < a < 1$ at the prime $P = \{0, a\}$, only 1 is inverted, so $L_P = L$ still has three elements. Its local residue map to $\mathbb{B}$ sends $0, a$ to zero and 1 to one. In general the extension of χ_P sends a class to one precisely when that class is the unit: $\mu \notin P$ becomes invertible and hence equals one; if $\mu \in P$ could equal one, then some $t \notin P$ would satisfy $t \leq \mu$, contradicting downward closure. The Bourne quotient by its zero fibre is exactly $\mathbb{B}$, because it collapses all nonunits to zero while leaving the unit distinct. Thus the residue character is related to the full support stalk by an exact local map; it is not a replacement for that stalk.

1.5 The universal tropical coefficient extension retains every support

Let K be a tropical semifield as above and form the commutative semiring tensor product

$$K_L = K \otimes_{\mathbb{B}} L. \quad (\text{ASL18})$$

There is no unital map $\mathbb{B} \rightarrow S$: it would require $1_S + 1_S = 1_S$, which in the nonzero ring amplitude would say $2 \cdot 1_R = 1_R$. The correct direct base change from S uses the initial semiring $\mathbb{N}_0$, and (ASL2) gives the canonical isomorphism

$$S \otimes_{\mathbb{N}_0} K \simeq L \otimes_{\mathbb{B}} K. \quad (\text{ASL19})$$

Indeed any semiring receiving a unital map from K is additively idempotent: $1 + 1 = 1$ implies $x + x = x$ for every element. Its map from S therefore factors uniquely through L . Maps from L and K agree on $\mathbb{B}$, and these are exactly the universal maps on the right side of (ASL19). This proves the isomorphism without pretending that S was already a Boolean algebra or a $\mathbb{B}$ -algebra.

The simultaneous character evaluations give a faithful model

$$K_L \longrightarrow K^X, \quad w = \sum_{i=1}^r a_i \otimes \lambda_i \longmapsto \left(P \longmapsto \max_{i: \lambda_i \notin P} a_i \right), \quad (\text{ASL20})$$

with the empty maximum equal to $\mathbf{0}_K$. Both operations are pointwise on the displayed functions. The map exists by the tensor universal property. Every tensor expression is a finite sum of pure tensors, because products of pure tensors are pure tensors. To prove injectivity, discard zero-coefficient terms and list the finitely many distinct remaining coefficients as $a_1 < \dots < a_r$ in

the natural total order of K . Let μ_i be the join of the supports whose coefficients are at least a_i . The original tensor expression equals

$$\sum_{i=1}^r a_i \otimes \mu_i, \quad \mu_1 \geq \cdots \geq \mu_r. \quad (\text{ASL21})$$

For verification, distributing each μ_i merely adds terms $a_i \otimes \lambda$ already bounded above by a term $a_j \otimes \lambda$ with $j \geq i$; their sum is the latter because $a_i + a_j = a_j$. Thus all original data is related by actual tensor relations. In the image function $w(P)$, the set where $w(P) \geq a_i$ is exactly $D(\mu_i)$. For two expressions, use the union of their finite coefficient lists in (ASL21). Equality of functions gives equality of every $D(\mu_i)$ and therefore of every μ_i by (ASL9). Hence the two tensors are equal, proving faithfulness.

Equivalently, (ASL20) identifies K_L with the finite-valued functions $w : X \rightarrow K$ whose sets $\{P : w(P) \geq a\}$ are compact open for every nonzero attained value a . Conversely, the finitely many nonzero values of such a function and their compact opens supply (ASL21), by (ASL10). In particular $1 \otimes L$ embeds faithfully as its compact-open characteristic functions. The fibre of evaluation at a specified P is completely described by the displayed maximum: it is zero exactly when all nonzero-coefficient supports belong to P ; a nonzero value a occurs exactly when a is the largest active coefficient. In level-set language this is $P \in D(\mu_{\geq a}) \setminus D(\mu_{>a})$.

The coefficient extension can also accompany the original ring amplitudes without losing them. The map

$$S \longrightarrow R \times K_L, \quad (r, \lambda) \longmapsto (r, 1 \otimes \lambda) \quad (\text{ASL22})$$

is an injective semiring homomorphism: its first coordinate retains the amplitude and (ASL20) makes its second coordinate faithful on the entire lattice. Its image consists exactly of pairs with a compact-open characteristic function in the second coordinate and with that function identically one whenever $r \neq 0$. This is an actual correspondence retaining both coordinates, not a claim that the ring amplitude itself embeds into a characteristic-one semiring.

1.6 The arithmetic site with full lattice coefficients

Let $\mathcal{A} = \widehat{\mathbb{N}}^\times$ be the source topos and $\mathcal{O} = \mathbb{Z}_{\max}$ with its Frobenius action, exactly as in Connes–Consani’s Definition **site**. Give the constant semiring object $\underline{L}$ the trivial action and set

$$\mathcal{O}_L = \mathcal{O} \otimes_{\mathbb{B}} \underline{L}, \quad (\mathcal{A}, \mathcal{O}_L) \longrightarrow (\mathcal{A}, \mathcal{O}). \quad (\text{ASL23})$$

The underlying geometric morphism is the identity, and its structure map is $\mathcal{O} \rightarrow \mathcal{O}_L$, $a \mapsto a \otimes 1$. By (ASL19) it is also the universal coefficient object receiving $\mathcal{O}$ and the constant full semiring $\underline{S}$ over $\underline{\mathbb{N}}_0$. The latter is expressly a constant object; no unproved additive Frobenius on the ring R is asserted.

The Frobenius action and joint character maps are

$$\text{Fr}_n \left(\sum_i a_i \otimes \lambda_i \right) = \sum_i a_i^n \otimes \lambda_i, \quad \text{ev}_P : \mathcal{O}_L \rightarrow \mathcal{O}. \quad (\text{ASL24})$$

Evaluation is (ASL20), so these are equivariant semiring maps and their family is jointly injective. In the additive notation of $\mathbb{Z}_{\max}$ the coefficient operation a_i^n means na_i ; the absorbing zero stays absorbing zero. For each P the exact coefficient specialization is

$$\mathcal{O}_L \otimes_{\underline{L}, \chi_P} \mathbb{B} \simeq \mathcal{O}. \quad (\text{ASL25})$$

This follows from the two tensor universal properties, or directly by substituting $\lambda \mapsto \chi_P(\lambda)$ in every pure tensor. All supports are present before this specified specialization.

At the point corresponding to a nonzero ordered subgroup $H \subset \mathbb{Q}$, the stalk is

$$(\mathcal{O}_L)_H \simeq H_{\max} \otimes_{\mathbb{B}} L. \quad (\text{ASL26})$$

To justify this precisely, the inverse-image functor of the point preserves finite products and arbitrary colimits. A semiring tensor product is constructed from finite products, coproducts and the coequalizers imposing its addition and multiplication relations. It is therefore preserved by that inverse-image functor. A constant object with trivial action has the same stalk: in the source's description $(X \times H_{>0})/\sim$ at lines 631–649, any two positive parameters have a common divisor in $H_{>0}$, and the trivial action identifies them while retaining the value in X . Finally the source's Theorem **structure2** identifies the original structure stalk as $H_{\max}$. These facts prove (ASL26). Its elements have exactly the full-lattice function description (ASL20)–(ASL21), now with coefficients in $H_{\max}$.

The global sections of the extended site are exactly

$$\Gamma(\mathcal{A}, \mathcal{O}_L) \simeq L. \quad (\text{ASL27})$$

For a one-object action topos, a global section is a simultaneous fixed element; this is the source's argument at lines 668–673. Under the faithful function representation (ASL20), invariance under Fr_2 forces each value a to satisfy $a^2 = a$. In $\mathbb{Z}_{\max}$ these are precisely **0**, **1**. Such a finite-valued function is a section of $\mathcal{O}_L$ exactly when its nonzero set is $D(\lambda)$, and then it is $1 \otimes \lambda$. Conversely these sections are fixed by every Frobenius. This proves (ASL27); the source's unextended case is $\Gamma(\mathcal{A}, \mathcal{O}) = \mathbb{B}$.

1.7 All tropical-valued points of the extended site

By the definition of a valued point in the source, lines 685–690, one supplies a topos point H and a semiring map from its structure stalk to $\mathbb{R}_+^{\max}$. The tensor universal property and (ASL6) therefore give the exact classification

$$\begin{aligned} \text{Hom}_{\text{sr}}(H_{\max} \otimes_{\mathbb{B}} L, \mathbb{R}_+^{\max}) &\simeq \text{Hom}_{\text{sr}}(H_{\max}, \mathbb{R}_+^{\max}) \times X, \\ (\beta, P) : \sum_i a_i \otimes \lambda_i &\longmapsto \max_{i: \lambda_i \notin P} \beta(a_i). \end{aligned} \quad (\text{ASL28})$$

There is no missing compatibility condition: every unital semiring map agrees on the two distinguished elements of $\mathbb{B}$. Thus the set of isomorphism classes of valued points of the extended site is the product of the original valued-point set with X . An isomorphism of underlying topos points can change H and its map β as in the source's equivalence relation, but it acts identically on the constant object L , hence cannot identify different P 's.

For a fixed embedding $H \subset \mathbb{Q}$, the first factor in (ASL28) can itself be given exactly. Every map has the form

$$\beta_c(-\infty) = 0, \quad \beta_c(h) = e^{ch} \ (h \in H), \quad c \geq 0. \quad (\text{ASL29})$$

Indeed group units must map to positive real units. Taking their real logarithms gives an additive order-preserving map $\ell : H \rightarrow \mathbb{R}$. Fix $h_0 > 0$. Since every h/h_0 is rational, additivity forces $\ell(h) = h\ell(h_0)/h_0$; order preservation makes $c = \ell(h_0)/h_0 \geq 0$. Conversely every formula (ASL29) preserves both multiplication and maximum. The value $c = 0$ is the degenerate map through $\mathbb{B}$; $c > 0$ is injective on the ordered group.

Under the source's real Frobenius the exact point action is

$$\text{Fr}_t : (H, \beta_c, P) \longmapsto (H, \beta_{tc}, P), \quad t > 0. \quad (\text{ASL30})$$

Thus the arithmetic site's original nontrivial coefficient dynamics is retained, and the full lattice contributes its whole space of prime characters as a fixed parameter. A direct map from S to a single tropical stalk sees only its one character, by (ASL7), whereas the extended coefficient sheaf, all its stalks, and their joint evaluations retain every element and every relation of L . Equations (ASL19), (ASL23), (ASL26), and (ASL28) are the exact connecting maps. No purity, cohomology comparison, or Riemann-hypothesis conclusion is inferred from them.

1.8 A precise distinction between prime ideals and valued points

The preceding statements classify all support primes of S , all lattice primes, and all tropical-valued points of the coefficient extension. They do not identify the ordinary semiring prime spectrum of K_L with X . There can be additional primes, with an explicit example already for the three-element chain $L = \{0 < a < 1\}$. For $K = \mathbb{R}_+^{\max}$, (ASL20) gives

$$K_L \simeq \{(u, v) \in [0, \infty)^2 : u \geq v\}, \quad \mathfrak{q} = \{(0, 0)\} \cup \{(u, v) : u > v\}. \quad (\text{ASL31})$$

The operations are coordinatewise maximum and multiplication. The set $\mathfrak{q}$ is a proper prime ideal. It is closed under maximum: whichever of two first coordinates is larger also exceeds both second coordinates. Multiplying (u, v) with $u > v$ by a nonzero (r, s) with $r \geq s$ gives $ur > vs$; multiplication by zero gives zero. Its complement is the set of positive diagonal pairs, which is multiplicatively closed. These observations prove the ideal and prime assertions. But $\mathfrak{q}$ is neither evaluation kernel: the first evaluation has only the zero kernel, the second kernel is $\{(u, 0) : u \geq 0\}$, and $(2, 1)$ belongs to $\mathfrak{q}$ but to neither of those. Thus the full support and valued-point comparison retains its precise stated scope even in this smallest non-Boolean case. It does not erase other prime geometry introduced by the tropical coefficient extension.

1.9 The exact pullback over the ordinary arithmetic spectrum

Connes–Consani’s geometric morphism over $\text{Spec } \mathbb{Z}$ uses the monoid $\mathbb{N}_0^\times$ that includes zero, not just $\mathbb{N}^\times$. This distinction is explicit in their Remark `caseno`, source line 660, and Theorem `thmspz`, lines 951–961. Put $\mathcal{A}_0 = \widehat{\mathbb{N}_0^\times}$ and let $\mathcal{O}^0$ be their extended structure semiring. In additive exponent notation its additional action is

$$\text{Fr}_0(-\infty) = -\infty, \quad \text{Fr}_0(h) = 0_H \quad (h \in H). \quad (\text{ASL32})$$

This preserves both operations: a maximum is absorbing zero exactly when both inputs are absorbing zero, and a product is absorbing zero exactly when one factor is. The extension $\mathcal{O}_L^0 = \mathcal{O}^0 \otimes_{\mathbb{B}} \underline{L}$ has

$$\text{Fr}_0\left(\sum_i a_i \otimes \lambda_i\right) = 1 \otimes \rho\left(\sum_i a_i \otimes \lambda_i\right), \quad \rho\left(\sum_i a_i \otimes \lambda_i\right) = \bigvee_{a_i \neq 0} \lambda_i. \quad (\text{ASL33})$$

The map ρ is well defined by the tensor universal property, using the semiring map $H_{\max} \rightarrow \mathbb{B}$ that sends every nonzero element to one. Equivalently, in (ASL20), $D(\rho(w))$ is exactly the set where w is nonzero. Thus (ASL33) retains a specific element of L , rather than only whether any support exists. It obeys $\text{Fr}_0 \text{Fr}_n = \text{Fr}_n \text{Fr}_0 = \text{Fr}_0$.

The original source constructs $\Theta : \text{Sh}(\text{Spec } \mathbb{Z}) \rightarrow \mathcal{A}_0$. At a rational prime p , its point is $H(p) = \mathbb{Z}[1/p] \subset \mathbb{Q}$, as defined at source lines 910–914; the generic point η goes to the trivial group. Applying its inverse-image functor to the full coefficient extension gives the exact sheaf and stalk formulas

$$\begin{aligned} \mathcal{T}_L &:= \Theta^* \mathcal{O}_L^0 \simeq \Theta^* \mathcal{O}^0 \otimes_{\mathbb{B}} \underline{L}, \\ (\mathcal{T}_L)_p &\simeq \mathbb{Z}[1/p]_{\max} \otimes_{\mathbb{B}} L, \quad (\mathcal{T}_L)_\eta \simeq L. \end{aligned} \quad (\text{ASL34})$$

The preservation argument proving (ASL26) applies to Θ^* as well. Connes–Consani’s Proposition `sheavesonspz0`, source lines 965–975, gives the original stalks $H(p)_{\max}$ and $\mathbb{B}$, so tensoring proves (ASL34). This is a pullback of semiring coefficients on the underlying topos of the arithmetic spectrum. It does not identify these coefficients with the ordinary ring structure sheaf $\mathcal{O}_{\text{Spec } \mathbb{Z}}$.

All its sections can also be described without suppressing the prime or support labels. For a nonempty open U of $\text{Spec } \mathbb{Z}$, a section of $\mathcal{T}_L$ is exactly the data

$$\begin{aligned} \lambda \in L, \quad w_p \in \mathbb{Z}[1/p]_{\max} \otimes_{\mathbb{B}} L \quad (p \in U \text{ a rational prime}), \\ \rho(w_p) = \lambda \quad \text{for every } p, \quad w_p = 1 \otimes \lambda \quad \text{for all but finitely many } p. \end{aligned} \quad (\text{ASL35})$$

The value at η is λ . Addition and multiplication are pointwise, with join and meet at η ; restriction discards the coordinates at the removed primes. The integer Frobenius applies $h \mapsto nh$ to every coefficient at every retained prime and acts identically on λ . Its zero action is (ASL33).

Here is a complete gluing proof. The source's Proposition **sheavesonspz0(ii)** says that an original section is either absorbing zero everywhere or a family of nonzero coefficients equal to the unit outside a finite set of primes. A local finite tensor expression therefore has a single generic support $\lambda = \bigvee_{a_i \neq 0} \lambda_i$; all its prime values have that same support, and outside a finite set every value is $1 \otimes \lambda$. Every sheaf tensor section has such expressions locally, by the construction of the tensor as the sheafification of the presheaf tensor. Every nonempty open U here is quasi-compact: a nonempty member of any open cover misses only finitely many points of U , which finitely many other members can cover. Thus finitely many local expressions suffice. Their generic values agree since every nonempty intersection contains η . Their finite exceptional sets have finite union. This proves that every section yields (ASL35).

Conversely take (ASL35), with finite exceptional set E . For each $p \in E$ write w_p as a finite tensor sum using nonzero coefficients $a_i \in \mathbb{Z}[1/p]_{\max}$. Its support labels have join λ . On the open $U_p = U \setminus (E \setminus \{p\})$ extend a_i to the original coefficient section with value a_i at p and the unit at every other prime; the source's section description makes this an actual section. Tensor these sections with their labels and add. The result has prescribed value w_p at p and value $1 \otimes \lambda$ elsewhere on U_p . The case $\lambda = 0_L$ has only zero tensors and is obtained from the zero section. Add the constant section $1 \otimes \lambda$ on $U \setminus E$. These local sections agree on their intersections, so the sheaf gluing axiom constructs the required global section. Uniqueness follows because sections of a sheaf on a topological space are determined by all their germs. This proves (ASL35) and its operation and restriction formulas.

2 The complete finite-lattice prime spectrum after tropical base change

The coefficient semiring throughout this calculation is exactly $K = \mathbb{R}_+^{\max} = ([0, \infty), \max, \cdot)$, with absorbing zero 0 and unit 1. This is the tropical coefficient semifield of Alain Connes and Caterina Consani, *Geometry of the Arithmetic Site*. The exact locators are lines 395–406 and Definition **defnpt**, lines 685–690 of the original author source:

`arithmeticsite_Adv_final1.tex.`

The complete support base $G_L(R)$ and its original spectrum are the existing programme construction cited and reproduced in (ASL1), (ASL12), and 031, Theorem **thm:rebuilt-full-support**. The ordinary-prime classification proved below is a derivation for the coefficient extension; it is not attributed to Connes–Consani.

Let L be any nontrivial finite bounded distributive lattice. Its original elements are retained through the exact isomorphism

$$L \longrightarrow \text{Down}(J), \quad \lambda \longmapsto \{j \in J : j \leq \lambda\}, \quad J = J(L), \quad (\text{LSP1})$$

whose inverse takes the join of a downset. The proof with all maps is (ASL11): join-irreducibles in a distributive lattice are join-prime, and every element is the join of those below it. In particular this is not a replacement by independent Boolean coordinates. By the faithful tensor calculation (ASL18)–(ASL21), the full coefficient extension is exactly

$$A_J := K \otimes_{\mathbb{B}} L \simeq \{f : J \rightarrow [0, \infty) : x \leq y \implies f(x) \geq f(y)\}, \quad (\text{LSP2})$$

with pointwise maximum and multiplication. For completeness the finite version follows directly: the nonzero superlevel sets of an antitone function are downsets, and its finitely many values express it as a sum of those values tensored with those downsets. Sorting the coefficient values and

using their cumulative downsets proves uniqueness by character evaluation, exactly as (ASL21). For every downset $D \subset J$, write e_D for its characteristic function. These are precisely the multiplicative idempotents of A_J ; each coordinate of an idempotent is zero or one.

An ideal here means an ordinary semiring ideal, closed under addition and multiplication by every element. It is not required to be subtractive. A prime ideal is proper and its multiplicative complement is closed. Retaining that distinction is necessary for the complete answer.

2.1 Exhaustive classification, including the primes with several labels

For $j \in J$, put $J_{\leq j} = \{x : x \leq j\}$. Let U be an upward-closed subset of $J_{\leq j}$ containing j . Define

$$\mathfrak{p}_{j,U} = \{f \in A_J : f(j) = 0 \text{ or } \exists i \in U : f(i) > f(j)\}. \quad (\text{LSP3})$$

Then every ordinary prime ideal of A_J is exactly one of (LSP3), and the pair (j, U) is unique.

First prove existence. If f, g belong to (LSP3), put $h = \max(f, g)$. When $h(j) = 0$, it belongs to the ideal. Otherwise one of f, g , say f , attains $h(j) > 0$ at j . Since f is in the ideal, there is $i \in U$ with $f(i) > f(j) = h(j)$, whence $h(i) > h(j)$. This proves additive closure. For a product fg , a zero at j persists. If $f(i) > f(j) > 0$ and $g(j) > 0$, then $g(i) \geq g(j) > 0$, so $f(i)g(i) > f(j)g(j)$. If $g(j) = 0$, the product has zero at j . This proves absorption. The constant unit is outside the ideal. Finally its complement is exactly

$$f(j) > 0, \quad f(i) = f(j) \quad (i \in U), \quad (\text{LSP4})$$

and is multiplicatively closed. Thus (LSP3) is prime.

For exhaustion, let $\mathfrak{p}$ be any prime of A_J . Its intersection with the e_D is a prime of the lattice of downsets. The complement of that lattice prime is a finite filter. Its least member D_0 is nonempty and join-prime: if $D_0 \subset B \cup C$ then $D_0 \subset B$ or $D_0 \subset C$, by primality of the complementary ideal. Since D_0 is the union of its principal downsets, $D_0 = J_{\leq j}$ for a unique j . It follows that

$$e_D \notin \mathfrak{p} \iff j \in D. \quad (\text{LSP5})$$

In particular $e_{J_{\leq j}} \notin \mathfrak{p}$, and primality gives

$$f \in \mathfrak{p} \iff e_{J_{\leq j}} f \in \mathfrak{p}.$$

We can therefore work on the full subposet $B = J_{\leq j}$, with unit e_B . This is the exact truncation map $f \mapsto f|_B$, followed by extension by zero outside B when viewed in A_J . The latter extension is a homomorphism with unit e_B in its image.

Every function on B with value zero at j is in the prime: it is multiplied identically by $e_{B \setminus \{j\}}$, which belongs to the prime by (LSP5). All remaining functions are positive at every coordinate, because j is the maximum of B . The positive constant functions are units, so multiplying by such a function does not change ideal membership. For a positive function f define the exact coordinates

$$a_x = \log(f(x)/f(j)), \quad C_j = \{a : B \rightarrow [0, \infty) : a_j = 0, a_x \geq a_y \text{ if } x \leq y\}. \quad (\text{LSP6})$$

This retains f through $f(x) = f(j) \exp(a_x)$ and a positive unit factor $f(j)$. Write $h_a(x) = \exp(a_x)$ and set

$$F = \{a \in C_j : h_a \notin \mathfrak{p}\}.$$

The set F contains zero and is closed under addition, by $h_{a+b} = h_a h_b$ and the multiplicative closure of the complement. It also has the stronger property

$$b \in F, \quad a \in C_j, \quad a_x \leq b_x \quad (x \in B) \implies a \in F. \quad (\text{LSP7})$$

Indeed let $g(x) = \exp(b_x)$ for $x \neq j$ and $g(j) = 0$. It is antitone, is supported on $B \setminus \{j\}$, and belongs to the prime. If h_a also belonged to it, then $\max(h_a, g) = h_b$ would belong, a contradiction. This proves (LSP7).

Let D be the union of the sets $\{x : a_x > 0\}$ over $a \in F$. It is a downset of B and excludes j . Since B is finite, there is a single $b \in F$ whose positive-coordinate set is exactly D : choose one member positive at each coordinate of D and add them. When D is empty take $b = 0$. For any $a \in C_j$ vanishing outside D , some positive integer N satisfies $a_x \leq Nb_x$ at every coordinate; take an integer at least every nonzero ratio a_x/b_x . Then $Nb \in F$ and (LSP7) gives $a \in F$. Conversely every member of F vanishes outside D by the definition of D . Hence

$$F = \{a \in C_j : a_x = 0 \text{ for every } x \in U\}, \quad U = B \setminus D. \quad (\text{LSP8})$$

The set U is upward closed and contains j . Returning through (LSP6) and retaining the functions with zero at j proves $\mathfrak{p} = \mathfrak{p}_{j,U}$.

Uniqueness of j follows from (LSP5). If $U \neq V$ have the same j , take $i \in U \setminus V$, interchanging them if necessary. Then $i \neq j$ and the principal downset $J_{\leq i}$ is disjoint from V . The function equal to $\exp(1)$ on $J_{\leq i}$ and to 1 elsewhere belongs to $\mathfrak{p}_{j,U}$ and not to $\mathfrak{p}_{j,V}$. This proves uniqueness completely.

For a three-point poset with incomparable a, b both below j , the upset $U = \{a, b, j\}$ gives the additional prime

$$\{f : f(j) = 0 \text{ or } f(a) > f(j) \text{ or } f(b) > f(j)\}. \quad (\text{LSP9})$$

It is distinct from both comparable-pair primes with $U = \{a, j\}$ or $U = \{b, j\}$, by the uniqueness proof. Thus comparable pairs alone do not exhaust the spectrum for a general finite lattice.

2.2 Specialization and every basic open

The complete specialization criterion is

$$\mathfrak{p}_{j,U} \subseteq \mathfrak{p}_{k,V} \iff j \leq k \text{ and } U \setminus \{j\} \subseteq V. \quad (\text{LSP10})$$

Necessity of $j \leq k$ follows by contracting to the idempotents: the lattice prime of j is contained in that of k exactly when $j \leq k$. Explicitly, if $j \not\leq k$, the downset $J_{\leq k}$ contains k and excludes j , contradicting the required inclusion. If $i \in U \setminus \{j\}$ but $i \notin V$, the function equal to $\exp(1)$ on $J_{\leq i}$ and 1 elsewhere has value 1 at j, k and all of V , but a strict excess at i . It contradicts prime inclusion. This proves the other necessity.

For sufficiency, take $f \notin \mathfrak{p}_{k,V}$. Then $f(k) > 0$ and $f(v) = f(k)$ for all $v \in V$. If $U = \{j\}$, the inequality $f(j) \geq f(k) > 0$ already puts f outside $\mathfrak{p}_{j,U}$. Otherwise every $i \in U \setminus \{j\}$ lies in V , and upward closure with $i \leq j \leq k$ also puts j in V . Thus $f(i) = f(j) = f(k)$ for every $i \in U$, proving the implication. The basic open of any original function is exactly

$$D(f) = \{(j, U) : f(j) > 0, f(i) = f(j) \text{ for all } i \in U\}. \quad (\text{LSP11})$$

Equations (LSP10)–(LSP11) determine the whole finite topological spectrum, with every original support label still present.

For the chain $J = \{1 < \dots < n\}$, every possible upset in $J_{\leq j}$ is $\{i, i+1, \dots, j\}$, with $1 \leq i \leq j \leq n$. The list becomes

$$\mathfrak{p}_{ij} = \{f : f_i = 0 \text{ or } f_i > f_j\}, \quad 1 \leq i \leq j \leq n, \quad |\text{Spec } A_J| = \frac{n(n+1)}{2}. \quad (\text{LSP12})$$

This formula equals (LSP3): when $f_j = 0$, either $f_i = 0$ or $f_i > f_j$, and when $f_j > 0$ the largest value in the indicated upset is f_i . The inclusion criterion is

$$\mathfrak{p}_{ij} \subseteq \mathfrak{p}_{k\ell} \iff j \leq \ell \text{ and } (i = j \text{ or } k \leq i). \quad (\text{LSP13})$$

In particular, for $n = 2$ the entire spectrum is the three-element chain $\{0\} = \mathfrak{p}_{11} \subsetneq \mathfrak{p}_{22} \subsetneq \mathfrak{p}_{12}$, including (ASL31). For general n its Krull dimension is exactly $2n - 2$. The integer $2j - i - 1$ strictly increases under each strict inclusion in (LSP13) and ranges from 0 to $2n - 2$, giving the upper bound. The chain

$$\mathfrak{p}_{11} \subsetneq \mathfrak{p}_{22} \subsetneq \cdots \subsetneq \mathfrak{p}_{nn} \subsetneq \mathfrak{p}_{n-1,n} \subsetneq \cdots \subsetneq \mathfrak{p}_{1n} \quad (\text{LSP14})$$

has exactly $2n - 2$ strict inclusions, proving equality, including the single-point case $n = 1$.

2.3 All localized coordinates and the exact maps between them

Fix (j, U) and put $B = J_{\leq j}$. The localization at its prime has the following concrete realization, with pointwise operations:

$$(A_J)_{\mathfrak{p}_{j,U}} \simeq \left\{ h : B \rightarrow [0, \infty) : \begin{array}{l} \{x : h(x) > 0\} \text{ is a downset of } B, \\ h|_U \text{ is antitone} \end{array} \right\}. \quad (\text{LSP15})$$

The map sends a fraction f/g , $g \notin \mathfrak{p}_{j,U}$, to $h(x) = f(x)/g(x)$ for $x \in B$. By (LSP4), every such denominator is positive throughout B and constant on U . Thus the indicated support and order properties hold. Equality of fractions implies coordinatewise equality there. Conversely coordinatewise equality implies equality after multiplication by e_B , which is outside the prime; hence this map is injective. This also shows explicitly why coordinates outside B disappear at this specified localization.

To prove surjectivity without guessing denominators, take any h on the right side. Set $g(x) = 1$ for $x \in U$, and construct its other positive values in reverse order, processing all strict successors of x first. Choose

$$g(x) = \max(\{1\} \cup \{g(y) : y > x\} \cup \{h(y)g(y)/h(x) : y > x\}) \quad (h(x) > 0), \quad (\text{LSP16})$$

and omit the last set when $h(x) = 0$. Every set here is finite. For $x \in U$, its successors are also in U and the antitone condition on $h|_U$ already verifies the required inequalities with $g(x) = 1$. For every other x , formula (LSP16) makes both g and hg antitone. In the zero case every successor also has $h(y) = 0$ because the nonzero support is a downset, so no inequality was lost by omitting the ratios. Thus g is positive, antitone and constant on U , while $f = hg$ is antitone. Extend both by zero outside B . They are elements of A_J , g is outside the prime, and $f/g = h$. This proves (LSP15) with an explicit inverse construction.

The units and maximal ideal in this local semiring are therefore

$$\begin{aligned} h \text{ is a unit} &\iff h(x) > 0 \ (x \in B), \quad h(i) = h(j) \ (i \in U), \\ \mathfrak{m}_{j,U} &= \{h : h(j) = 0 \text{ or } \exists i \in U : h(i) > h(j)\}. \end{aligned} \quad (\text{LSP17})$$

Indeed an inverse must be the coordinatewise reciprocal, so support must be all of B . Both $h|_U$ and its reciprocal are antitone exactly when values agree along every comparable pair. Every element of U is below j , forcing the displayed constant value. Conversely these conditions make the reciprocal belong to (LSP15). The complement of the units is the ideal displayed in (LSP17), so it is the unique maximal ideal.

If (LSP10) holds, the canonical map between the localizations is

$$(A_J)_{\mathfrak{p}_{k,V}} \longrightarrow (A_J)_{\mathfrak{p}_{j,U}}, \quad h \longmapsto h|_{J_{\leq j}}. \quad (\text{LSP18})$$

It has this value on every original fraction, hence is the required localization map. Its image consists exactly of the functions in (LSP15) whose restriction to $V \cap J_{\leq j}$ is antitone. Necessity follows by restriction. For sufficiency extend such a function by zero on $J_{\leq k} \setminus J_{\leq j}$. Its nonzero support remains a downset. Its restriction to V is antitone: the only cross-boundary comparisons

start in $J_{\leq j}$ and end outside, and then the latter value is zero. It is therefore an element of the source of (LSP18). The fibres of the map are exactly the admissible functions agreeing on $J_{\leq j}$. No additional surjectivity assertion is needed or made.

For completeness, primes of the localized semiring correspond to the pairs (k, V) whose prime is contained in $\mathfrak{p}_{j,U}$. Extension takes such a prime to the fractions with numerator in it. This is well defined: if $f/g = f'/g'$ then some inverted t satisfies $tf g' = tf' g$; all inverted factors are outside the smaller prime, so primality makes f belong exactly when f' does. The complement is multiplicatively closed, and contraction recovers the same prime. Conversely a prime of the localization cannot contain an inverted element; its contraction has this property and the fraction rule recovers it. Together with (LSP10), this proves the entire localized prime geometry, not just the local coordinate formulas.

2.4 Subtractive primes, quotient maps, and actual tropical residue maps

An ideal I is subtractive when $a + b \in I$ and $a \in I$ imply $b \in I$. Precisely the primes

$$\mathfrak{p}_{j,\{j\}} = \{f : f(j) = 0\} \quad (\text{LSP19})$$

are subtractive. They are kernels of evaluation at j , and the subtractive property follows by evaluating the defining implication in K . For any $U \neq \{j\}$ choose $i \in U \setminus \{j\}$. The function b equal to $\exp(1)$ on $J_{\leq i}$ and 1 elsewhere satisfies

$$b \in \mathfrak{p}_{j,U}, \quad \max(1, b) = b, \quad 1 \notin \mathfrak{p}_{j,U}. \quad (\text{LSP20})$$

This violates subtractivity. It also proves that the Bourne quotient by this prime is the zero semiring: the relation generated by $b = 0$ and $1 + b = b$ forces $1 = 0$. This is a property of that precise quotient construction; the original prime and localization in (LSP15) are still proper and nonzero.

For the subtractive prime the complete global Bourne quotient is

$$A_J / \mathfrak{p}_{j,\{j\}} \simeq A_{J_{\geq j}}, \quad f \mapsto f|_{J_{\geq j}}. \quad (\text{LSP21})$$

Here the quotient identifies f, g when $f + a = g + b$ for some a, b in the ideal. Such functions a, b vanish on $J_{\geq j}$, so equivalence forces equality on that whole upset. Conversely, if the restrictions agree, choose a real M at least all the finitely many values of f, g . The function $Me_{J \setminus J_{\geq j}}$ is in the ideal, and adding it to both functions makes them equal everywhere. This proves the kernel-congruence assertion. Every antitone function on $J_{\geq j}$ extends by a sufficiently large constant on its complementary downset, proving surjectivity. Thus (LSP21) is an actual quotient isomorphism; it need not equal K before localization.

After localization at this prime the residue quotient is exactly

$$(A_J)_{\mathfrak{p}_{j,\{j\}}} / \mathfrak{m}_{j,\{j\}} \simeq K, \quad h \mapsto h(j). \quad (\text{LSP22})$$

Evaluation is surjective on constants. If $h(j) = h'(j)$, choose a constant M dominating both functions on $B \setminus \{j\}$. The function equal to M there and zero at j belongs to the local maximal ideal and to (LSP15); adding it makes h, h' equal. Conversely any such equivalence preserves the value at j . This proves (LSP22). For $U \neq \{j\}$ the local Bourne quotient is also zero, by the image of (LSP20).

Any unital homomorphism to a tropical semifield has a subtractive zero kernel: if the image of $a + b$ and of a is zero, the image of b is zero. Consequently none of the additional primes can be the zero kernel of such a map. The exact tropical-valued-point classification (ASL28) and the larger ordinary-prime classification (LSP3) therefore agree through the specified kernel map; neither geometry has been substituted for the other.

2.5 The full original prime spectrum is recovered from the companion map

For a nonzero commutative ring R , retain the original, injective map (ASL22), with all amplitude and support coordinates:

$$G_L(R) \longrightarrow R \times A_J, \quad (r, \lambda) \longmapsto (r, e_{\{j \in J: j \leq \lambda\}}). \quad (\text{LSP23})$$

Injectivity follows because both displayed coordinates are retained and (LSP1) is injective; preservation of operations follows from join being union and meet being intersection of these downsets. Every prime of $R \times A_J$ comes from exactly one factor. Indeed the two idempotents $(1, 0), (0, 1)$ multiply to zero and sum to the unit; a proper prime therefore contains exactly one of them. It then contains the whole corresponding factor and is the inverse image of a prime of the other factor. Hence

$$\text{Spec}(R \times A_J) = \text{Spec } R \sqcup \text{Spec } A_J. \quad (\text{LSP24})$$

The contractions under (LSP23) are exactly

$$\begin{aligned} \mathfrak{r} \in \text{Spec } R &\longmapsto \{(0, \lambda) : \lambda \in L\} \cup \{(r, 1_L) : r \in \mathfrak{r}\}, \\ (j, U) &\longmapsto \{(0, \lambda) : j \not\leq \lambda\}. \end{aligned} \quad (\text{LSP25})$$

To check the second formula, an idempotent e_D belongs to $\mathfrak{p}_{j,U}$ exactly when $j \notin D$, by (LSP5). Every nonzero supported amplitude, and the supported zero at top support, maps to the unit function. The first formula follows directly from the amplitude coordinate. The original prime classification (ASL12) now proves that (LSP24) maps surjectively onto the whole original spectrum of $G_L(R)$. The fibre over an arithmetic prime is its one R -prime; the fibre over a support prime labelled j is exactly all upsets $U \subseteq J_{\leq j}$ containing j . This is the precise simultaneous relation between the ring amplitudes, every support prime, and the additional tropical prime geometry. It makes no cohomological or purity identification.

2.6 The complete Frobenius action on these primes

For every real $t > 0$, coefficient Frobenius is the automorphism $\text{Fr}_t(f)(x) = f(x)^t$. It preserves every zero, equality, and strict inequality in (LSP3), so

$$\text{Fr}_t^{-1}(\mathfrak{p}_{j,U}) = \mathfrak{p}_{j,U} \quad (t > 0). \quad (\text{LSP26})$$

Thus its action on this ordinary prime spectrum is the identity, although its action on coefficient functions and on the local semirings remains the indicated power operation. At the additional zero element of $\mathbb{N}_0^\times$, use the actual convention (ASL32), which gives $\text{Fr}_0(f) = e_{\text{supp } f}$. By (LSP5), the spectral action is precisely

$$\text{Fr}_0^{-1}(\mathfrak{p}_{j,U}) = \mathfrak{p}_{j,\{j\}}, \quad (j, U) \longmapsto (j, \{j\}). \quad (\text{LSP27})$$

More explicitly, the semiring maps $L \xrightarrow{D \mapsto e_D} A_J \xrightarrow{f \mapsto \text{supp } f} L$ compose to the identity on L , while their reverse composite is Fr_0 . The corresponding spectral maps are the projection $(j, U) \mapsto j$ and its section $j \mapsto (j, \{j\})$. This proves an actual retraction onto the evaluation primes. It does not identify the additional primes with them: their distinct specializations and localized functions remain (LSP10)–(LSP18).

2.7 Exact naturality for every finite bounded-lattice morphism

Let $\phi : L \rightarrow L'$ preserve zero, unit, join, and meet, with both lattices finite, nontrivial, and distributive. Put $J = J(L)$ and $J' = J(L')$. The dual monotone map $g : J' \rightarrow J$ is uniquely specified by

$$j' \leq \phi(\lambda) \iff g(j') \leq \lambda \quad (\lambda \in L). \quad (\text{LSP28})$$

To prove existence, evaluate ϕ at the character of j' . The resulting bounded lattice character of L is the character of a unique $j \in J$, by the proof of (LSP5), and this j defines $g(j')$. For monotonicity, if $j' \leq k'$ then $k' \leq \phi(\lambda)$ implies $j' \leq \phi(\lambda)$ for every λ . Taking $\lambda = g(k')$ in (LSP28) gives $g(j') \leq g(k')$. Under (LSP1) the entire map is therefore $\phi(D) = g^{-1}(D)$. Conversely inverse image under any monotone g preserves downsets, empty and full sets, unions, and intersections, and hence gives exactly such a ϕ .

The induced coefficient map has the full coordinate formula

$$\Phi = \text{id}_K \otimes \phi : A_J \longrightarrow A_{J'}, \quad \Phi(f) = f \circ g. \quad (\text{LSP29})$$

On a pure tensor $a \otimes D$ it sends its characteristic support to $g^{-1}(D)$ and keeps the coefficient a , so (LSP29) follows on all finite sums. Pointwise operations also prove directly that this is a unital semiring map. It commutes with every positive real Frobenius and with Fr_0 , since both power and the nonzero-support test commute with precomposition.

Fix an arbitrary target prime $\mathfrak{p}_{j',U'}$. Its exact contraction is

$$\Phi^{-1}(\mathfrak{p}_{j',U'}) = \mathfrak{p}_{j,U}, \quad j = g(j'), \quad U = \{x \leq j : \exists u' \in U', g(u') \leq x\}. \quad (\text{LSP30})$$

The set U is the upward closure of $g(U')$ inside $J_{\leq j}$, is upward closed there, and contains j . Membership of $\Phi(f)$ in the target prime says either $f(g(j')) = 0$ or $f(g(u')) > f(g(j'))$ for some $u' \in U'$. Every $g(u')$ belongs to U , so this implies membership on the right of (LSP30). Conversely, if $f(x) > f(j)$ for $x \in U$, choose $u' \in U'$ with $g(u') \leq x$. Antitonicity gives $f(g(u')) \geq f(x) > f(j)$. The zero alternative is identical on both sides. This proves equality of the two ideals, with no loss of the upward-closure labels.

The associated local map is just as explicit:

$$(A_J)_{\mathfrak{p}_{j,U}} \longrightarrow (A_{J'})_{\mathfrak{p}_{j',U'}}, \quad h \longmapsto h \circ (g|_{J'_{\leq j'}}). \quad (\text{LSP31})$$

Its domain is correct because $g(J'_{\leq j'}) \subseteq J_{\leq j}$. An allowed nonzero-support downset in (LSP15) pulls back to a downset. On U' , the arguments of h lie in $g(U') \subseteq U$, where h is antitone, so the image satisfies the target order condition. Formula (LSP30) makes every denominator outside the source prime map outside the target prime. Thus (LSP31) is the canonical localization map, as can also be checked on its fractions $\Phi(f)/\Phi(q) = (f/q) \circ g$. It is local: a value function h is a unit precisely when $h(j) > 0$ and it is constant on U , by (LSP17). Its image is a unit precisely when $h(g(j')) > 0$ and it is constant on $g(U')$. These conditions are equivalent, since each $x \in U$ lies between some $g(u')$ and j , all in U ; the antitone inequalities force $h(x) = h(j)$ when both endpoint values equal $h(j)$. This proves the inverse image of the target maximal ideal is the source maximal ideal.

All maps compose with their full labels. If a second bounded-lattice map $L' \rightarrow L''$ has dual $r : J'' \rightarrow J'$, the dual of its composite with ϕ is $g \circ r$ by (LSP28), and the coefficient maps compose by the ordinary identity $(f \circ g) \circ r = f \circ (g \circ r)$. The required equality for prime labels is

$$\uparrow_{\leq g(r(j''))} g \left(\uparrow_{\leq r(j'')} r(U'') \right) = \uparrow_{\leq g(r(j''))} (g \circ r)(U''). \quad (\text{LSP32})$$

Indeed an intermediate element y' obeys $r(u'') \leq y' \leq r(j'')$ for some $u'' \in U''$, hence $g(r(u'')) \leq g(y') \leq g(r(j''))$, proving one inclusion. The other follows because each $r(u'')$ is itself in the intermediate upward closure. This proves (LSP32) and the composition of (LSP30). Localized precomposition in (LSP31) then composes identically on every retained coordinate.

The maps also fit the original split-support base, not only its tropical coefficient extension. For any fixed nonzero commutative ring R , define

$$G_\phi : G_L(R) \longrightarrow G_{L'}(R), \quad (r, \lambda) \longmapsto (r, \phi(\lambda)). \quad (\text{LSP33})$$

This is defined on the original carriers: nonzero amplitudes have top support, which ϕ preserves, while zero amplitudes admit every image support. The operations in (ASL1) prove that it preserves addition, multiplication, zero, and unit. It also sends the global zero τ to τ and the supported zero e to e . The companion maps (LSP23) obey the exact equality

$$(\mathrm{id}_R \times \Phi) \circ \iota_L = \iota_{L'} \circ G_\phi. \quad (\text{LSP34})$$

For verification at (r, λ) , both sides have amplitude r and characteristic support $g^{-1}(\{j : j \leq \lambda\})$, which equals $\{j' : j' \leq \phi(\lambda)\}$ by (LSP28).

Write $\mathfrak{s}_j = \{(0, \lambda) : j \not\leq \lambda\}$ for an original support prime and $Q_{\mathfrak{r}}$ for its original arithmetic prime at $\mathfrak{r} \in \mathrm{Spec} R$, as in (ASL12). Direct contraction gives

$$G_\phi^{-1}(\mathfrak{s}_{j'}) = \mathfrak{s}_{g(j')}, \quad G_\phi^{-1}(Q_{\mathfrak{r}}) = Q_{\mathfrak{r}}. \quad (\text{LSP35})$$

The first follows from (LSP28), with no nonzero supported amplitude entering a support prime. The second follows because amplitude is unchanged and all zero-supported elements are in every arithmetic prime. Consequently the spectral contraction square associated to (LSP34) commutes explicitly: on tropical primes it is $(j', U') \mapsto (g(j'), U) \mapsto g(j')$, which also results from $(j', U') \mapsto j' \mapsto g(j')$; on arithmetic primes both paths retain $\mathfrak{r}$. Each entire fibre of upsets in (LSP25) therefore maps by the exact upward-closure formula (LSP30), not merely by the map on its base point.

Finally the full original support stalks participate in the same local square. In finite-lattice coordinates, (ASL15)–(ASL16) give

$$(G_L(R))_{\mathfrak{s}_j} \simeq \mathrm{Down}(J_{\leq j}), \quad (r, \lambda) \mapsto \{x \leq j : x \leq \lambda\}. \quad (\text{LSP36})$$

Indeed among the lattice denominators containing j , the least is $J_{\leq j}$, and localizing that idempotent restricts every downset to it. The map of these stalks induced by G_ϕ is

$$D \mapsto \{x' \leq j' : g(x') \in D\}. \quad (\text{LSP37})$$

Their embeddings into the two local tropical semirings send D to e_D , which satisfies every support and order condition in (LSP15). Applying (LSP31) to such an idempotent yields exactly the characteristic function of (LSP37). This proves commutation of the local coefficient square, retaining each original stalk and each chosen prime label U, U' in its proper place.

3 The absolute arithmetic curve over the full support spectrum

3.1 Sources, conventions, and the objects retained

Alain Connes and Caterina Consani’s original author source `FF.tex`, *On the Absolute Geometry of Spec Z and the Fargues–Fontaine curve*, arXiv:2606.06604, was read in full, lines 1–1764. Its SHA-256 is

14c17a95040be7bfc57bb5f87e58907a05
cacd0bcb1b6967b2892cf3d9bc1229.

The precise inputs are Section `sect9.1`, Proposition `prop12`, and the following source locators:

```
prop:monoid_structure_morphisms
defnloc, propflat
prop:archimedean_points_C
prop:nonarchimedean_points_C.
```

The notation of that source is preserved:

$$\theta(\eta) = [\mathbb{Z}], \quad \Theta(\eta) = [\{0\}], \quad \Theta(p) = [\mathbb{Z}[1/p]], \quad \kappa\Theta = \theta. \quad (\text{ACF1})$$

The first equality concerns an arithmetic divisor class; the second concerns a point of the topos. In particular the source sheaf is

$$\begin{aligned} \mathcal{F} &= \Theta^{-1}\mathcal{O}_{\mathbf{F}_1}, & \mathcal{O}_{\mathbf{F}_1} &= \mathbf{F}_1[T], \\ \mathcal{F}_\eta &= \mathbf{F}_1, & \mathcal{F}_p &= \mathbf{F}_1[T^{H_p^+}], & H_p &= \mathbb{Z}[1/p], & H_p^+ &= H_p \cap [0, \infty). \end{aligned} \quad (\text{ACF2})$$

The original sheaf on the topos and its generic stalk are different objects in this formula.

For the coefficient foundation the input is *Split Support Geometry: Universal Zero Fibres, Arithmetic Curves, and Frobenius-Perfect Quantization*, source byline *The Clankers*, June 2026, pre-publication Version 11. The complete original proof file is pinned at commit a2851f9e. The locators used here are Definition `def:lattice-split`, Theorems `thm:lattice-ideal-classification`, `thm:lattice-prime-classification`, `thm:spectral-ordinal-sum`, Proposition `prop:classical-collapse`, and Definition `def:split-absolute-point-map`. Their relevant source ranges 1221–1311, 1606–1749, and 4591–4788 were read. All extensions proved below retain every lattice label and reproduce the necessary arguments.

Let L be a bounded distributive lattice, including the possible one-element lattice. For a nonzero commutative ring R set

$$\begin{aligned} S_R &= G_L(R) = \{z_\lambda = (0, \lambda) : \lambda \in L\} \cup \{\widehat{a} = (a, 1_L) : a \in R\}, \\ (a, \lambda) + (b, \mu) &= (a + b, \lambda \vee \mu), & (a, \lambda)(b, \mu) &= (ab, \lambda \wedge \mu), \\ \tau &= z_{0_L}, & e &= z_{1_L} = \widehat{0}, & 1 &= \widehat{1}, & q(a, \lambda) &= a. \end{aligned} \quad (\text{ACF3})$$

These operations close on the displayed carrier: an amplitude different from zero forces support 1_L , while a factor with smaller support has zero amplitude. The ring and lattice distributive laws give the semiring laws component by component. Thus q is a unital semiring map. The notation $\widehat{0}$ always denotes e , not τ .

3.2 The full prime spectrum and its arithmetic retraction

Here Spec means prime ideals of the additive semiring: an ideal contains τ , is closed under addition, and absorbs multiplication by every element. It does not mean prime ideals of only the underlying multiplicative monoid. Put $S = S_{\mathbb{Z}}$, $X_L = \text{Spec } S$, and $Y = \text{Spec } \mathbb{Z}$.

Proposition 1 (All primes and all basic opens). *Write $\text{Spec}_{\text{lat}} L$ for the proper prime lattice ideals, where ideals are lower, contain 0_L , and are closed under finite joins. The points of X_L are exactly*

$$\begin{aligned} P_J &= \{z_\lambda : \lambda \in J\}, & J &\in \text{Spec}_{\text{lat}} L, \\ Q_{\mathfrak{p}} &= q^{-1}(\mathfrak{p}), & \mathfrak{p} &\in Y. \end{aligned} \quad (\text{ACF4})$$

Every P_J is strictly contained in $Q_{(0)}$, and $Q_{(0)} \subsetneq Q_{(p)}$ for every rational prime p . Within the support stratum inclusion of primes is exactly inclusion of J . With $A = \text{Spec}_{\text{lat}} L$, the basic opens are

$$\begin{aligned} D_S(z_\lambda) &= \{P_J : \lambda \notin J\}, \\ D_S(\widehat{n}) &= A \cup i(D_{\mathbb{Z}}(n)) \quad (n \in \mathbb{Z}), \\ D_S(e) &= A, & V_S(e) &= i(Y), & i(\mathfrak{p}) &= Q_{\mathfrak{p}}. \end{aligned} \quad (\text{ACF5})$$

In the second line $n = 0$ is allowed, and then the right side is A .

Proof. If an ideal I contains $\widehat{n}$ with $n \neq 0$, then multiplication by $\widehat{-1}$ and addition give $e \in I$. If it contains e initially the same conclusion is already available. Then $ez_\lambda = z_\lambda$ puts the entire

zero fibre in I . The amplitudes whose supported representatives belong to I form a ring ideal, so $I = q^{-1}(\mathfrak{a})$ for a unique ring ideal $\mathfrak{a}$. Otherwise I consists solely of z_λ for a proper lattice ideal J : addition gives join closure and $z_\mu z_\lambda = z_\mu$ when $\mu \leq \lambda$ gives downward closure. Conversely these constructions are ideals. The amplitude ideals are prime exactly when $\mathfrak{a}$ is prime, by surjectivity of q . For a support ideal, a product of two top-supported elements is still top-supported, even if its amplitude is zero. A product of a top-supported element with z_λ has support λ . For two zero elements primality is precisely $\lambda \wedge \mu \in J \Rightarrow \lambda \in J$ or $\mu \in J$. This proves exhaustion and primality. The strict cross-stratum inclusion follows because every $Q_{\mathfrak{p}}$ contains e and no P_J does. Membership in the displayed ideals proves every formula for the basic opens directly. $\square$

Proposition 2 (A retraction which retains the support stratum). *The map $i : Y \rightarrow X_L$ is a homeomorphism onto the closed subspace $V(e)$. There is a continuous retraction*

$$\begin{aligned} r : X_L &\longrightarrow Y, & r(P_J) &= \eta, & r(Q_{(0)}) &= \eta, & r(Q_{(p)}) &= (p), \\ & & ri &= \text{id}_Y. \end{aligned} \tag{ACF6}$$

Its fibres are $r^{-1}(\eta) = A \cup \{Q_{(0)}\}$ and $r^{-1}((p)) = \{Q_{(p)}\}$. For $n \neq 0$,

$$r^{-1}D_{\mathbb{Z}}(n) = D_S(\widehat{n}). \tag{ACF7}$$

For $n = 0$ the left side is empty, whereas $D_S(\widehat{0}) = A$.

Proof. The formulas in ACF5 give the homeomorphism onto $V(e)$ and show that ACF7 holds for every nonzero integer, since every nonempty basic open of Y contains η . The inverse image of the empty open is empty. These opens form a basis, proving continuity. All statements about fibres and the composite follow from the explicit assignments. In particular r is a quotient map: if $r^{-1}U$ is open, then $U = i^{-1}r^{-1}U$ is open. $\square$

For $0_L \neq 1_L$, this continuous retraction is not induced by a unital semiring homomorphism $\mathbb{Z} \rightarrow S$. Indeed any such homomorphism would send 1 to $\widehat{1}$. If u were the image of -1 , then $\widehat{1} + u = \tau$ would be required. Its left support is 1_L , which cannot equal 0_L . The continuous map and its sheaf pullback nevertheless exist exactly as proved.

3.3 The pullback sheaf and its exact comparison

Define a spherical-algebra sheaf on the actual full base by

$$\mathcal{F}_L = r^{-1}\mathcal{F} = (\Theta \circ r)^{-1}\mathcal{O}_{\mathbf{F}_1}. \tag{ACF8}$$

For a continuous map, the inverse image is the sheafification of the presheaf assigning to $W \subseteq X_L$ the colimit of $\mathcal{F}(V)$ over opens $V \subseteq Y$ for which $W \subseteq r^{-1}V$. This specifies the sheaf, not only a list of stalks.

Proposition 3 (Exact restriction, stalks, and direct image). *There are canonical isomorphisms*

$$i^{-1}\mathcal{F}_L \simeq \mathcal{F}, \quad r_*\mathcal{F}_L \simeq \mathcal{F}, \quad (\mathcal{F}_L)_x \simeq \mathcal{F}_{r(x)}. \tag{ACF9}$$

Consequently

$$(\mathcal{F}_L)_{P_J} = \mathbf{F}_1, \quad (\mathcal{F}_L)_{Q_{(0)}} = \mathbf{F}_1, \quad (\mathcal{F}_L)_{Q_{(p)}} = \mathbf{F}_1[T^{H_p^+}]. \tag{ACF10}$$

All generic-direction stalk maps from the last object to either of the first two are

$$\epsilon_p(0) = 0, \quad \epsilon_p(T^h) = 1 \quad (h \in H_p^+). \tag{ACF11}$$

Maps between stalks $\mathbf{F}_1$ in ACF10 are identities.

Proof. The stalk of the inverse-image presheaf at x is the colimit of $\mathcal{F}(V)$ over neighbourhoods of $r(x)$: every such V yields the neighbourhood $r^{-1}V$ of x , and every representative of the presheaf stalk is already represented on some such V . The two identifications respect equality of germs. Sheafification preserves this stalk, proving the third isomorphism. The first follows either by the same construction and $ri = \text{id}$ or by the canonical composition isomorphism for inverse images.

The second isomorphism holds for any sheaf E on Y in place of $\mathcal{F}$. The adjunction unit sends $E(U)$ into $(r^{-1}E)(r^{-1}U)$. Restriction along i gives a left inverse using $i^{-1}r^{-1}E = E$. If two sections over $r^{-1}U$ have the same restriction, their germs agree at every point $Q_{\mathfrak{p}}$ with $\mathfrak{p} \in U$. When U is nonempty it contains η . Each support point is a generization of $Q_{(0)}$, so the germ at that support point is the image of the germ at $Q_{(0)}$. The two sections therefore agree at all stalks and are equal. The case $U = \emptyset$ is automatic. This proves bijectivity, respecting all sheaf operations and restrictions.

The source has all endomorphisms $\text{Fr}_m(T^h) = T^{mh}$, including $m = 0$, with the absorbing zero fixed. Every sheaf generization map commutes with them. On $\mathbf{F}_1$, Fr_0 is the identity. Thus a generization g to $\mathbf{F}_1$ satisfies $g(T^h) = \text{Fr}_0 g(T^h) = g(T^0) = 1$. It sends the absorbing zero to zero. This proves ACF11 without selecting an unsupported alternative augmentation. All other displayed maps are unital maps $\mathbf{F}_1 \rightarrow \mathbf{F}_1$, hence identities. $\square$

Proposition 4 (The actual semiring structure sheaf). *Let $\mathcal{O}_X$ be the sheaf of locally represented fractions of S on X_L . For a lattice prime J define*

$$\begin{aligned} L_J &= L[(L \setminus J)^{-1}], \\ [\lambda] &= [\mu] \iff \text{there is } \nu \notin J \text{ with } \lambda \wedge \nu = \mu \wedge \nu. \end{aligned} \tag{ACF12}$$

Its operations are induced join and meet. The stalks are

$$(\mathcal{O}_X)_{P_J} = L_J, \quad (\mathcal{O}_X)_{Q_{(0)}} = G_L(\mathbb{Q}), \quad (\mathcal{O}_X)_{Q_{(p)}} = G_L(\mathbb{Z}_{(p)}). \tag{ACF13}$$

The generic-direction map from an arithmetic stalk to L_J is $(a, \lambda) \mapsto [\lambda]$. The maps between arithmetic stalks are the usual localization of amplitudes with support unchanged. If $J \subseteq K$, the map $L_K \rightarrow L_J$ is the canonical localization.

Proof. The locally represented fraction definition gives the stalk S_P at a prime P : a fraction near P has denominator outside P , and equality of fraction germs is the localization equality witnessed by another such denominator. At $Q_{\mathfrak{p}}$, every allowed denominator is $\hat{n}$ with $n \notin \mathfrak{p}$ and $n \neq 0$. Thus

$$(a, \lambda) / \hat{n} \longmapsto (a/n, \lambda)$$

is onto $G_L(\mathbb{Z}_{\mathfrak{p}})$. The usual localization cross-multiplication criterion proves injectivity: an additional denominator is top-supported, so equality preserves the supports exactly, and its amplitude is a nonzero integer. Operations match component by component.

At P_J , e is an allowed denominator. The relation $e^2 = e$ makes $e = 1$ after localization. Hence every (a, λ) equals its product with e , namely z_{λ} . This first localization is exactly L : $z_{\lambda} \mapsto \lambda$ is the isomorphism. The remaining allowed zero denominators are the z_{ν} with $\nu \notin J$, so the second localization is L_J . An invertible idempotent is 1; thus every fraction is represented by its numerator, and the usual fraction equality becomes precisely ACF12. This also proves the formula for all maps, because they are induced by the original localization maps. The top and bottom of L_J remain different: equality would require $\nu \notin J$ with $\nu = 0_L$, impossible for a lattice ideal. $\square$

Proposition 5 (Coefficient comparison over the whole base). *On Y define the sheaf B_L by $B_L(U) = G_L(\mathcal{O}_Y(U))$ for nonempty opens U , and the terminal semiring for the empty open. There is a canonical sheaf morphism*

$$\Phi : r^{-1}B_L \longrightarrow \mathcal{O}_X \tag{ACF14}$$

which is an isomorphism at every arithmetic stalk and a surjection

$$G_L(\mathbb{Q}) \longrightarrow L_J, \quad (a, \lambda) \longmapsto [\lambda] \quad (\text{ACF15})$$

at P_J . Its exact equality relation at that stalk is equality of the classes $[\lambda]$ and $[\mu]$ in ACF12, regardless of amplitudes.

Proof. Every two nonempty opens of Y meet, and their rings of sections embed in $\mathbb{Q}$. Compatible local sections therefore have one support label on all members of a nonempty cover and glue their amplitudes by the ring sheaf axiom. A nonzero glued amplitude forces top support. This proves the claimed sheaf description. Its stalks are $G_L(\mathbb{Z}_{(p)})$ and $G_L(\mathbb{Q})$: passage to the stalk keeps every support label and takes the corresponding filtered localization of the amplitude.

For $D_{\mathbb{Z}}(n)$ with $n \neq 0$, a section of B_L is represented by $(a/n^k, \lambda)$. Send it to the local fraction $(a, \lambda)/\widehat{n^k}$ over $D_S(\widehat{n})$, which is its full inverse image by ACF7. Equality and restriction are preserved by the fraction rules. These maps on a basis give $B_L \rightarrow r_*\mathcal{O}_X$ and hence ACF14. At arithmetic stalks the map is the isomorphism just proved. At a support stalk all nonzero integers have been inverted, and the subsequent inversion of e and z_ν gives ACF15. Every $[\lambda]$ is hit by z_λ . Formula ACF12 gives precisely its fibres, proving all assertions. $\square$

The functor used here is Connes–Consani’s H from *Absolute algebra and Segal’s Γ -rings: au dessous de $\text{Spec } \mathbb{Z}$* , original `F1_corr.tex`, formula `functsum` (lines 433–442) and Lemma `ssalg` (lines 519–533). That source was read completely, lines 1–1336, for the preceding coefficient derivation. Applying the semiring functor H degree by degree gives the corresponding spherical-algebra map $r^{-1}HB_L \rightarrow H\mathcal{O}_X$. Here $(HA)(k_+) = A^k$, a pointed map sums coordinates over its fibres, and multiplication sends $(a_i), (b_j)$ to $(a_ib_j)_{i,j}$. These formulas show directly that the comparison commutes with all pointed maps and multiplications. Stalks and inverse images commute with this finite coordinate construction. Thus ACF14 is also the exact connection with the split coefficient sheaf, rather than an identification with $\mathcal{F}_L$.

Proposition 6 (The spherical comparison is forced by the generic stalk). *There is exactly one unital spherical-algebra sheaf map*

$$\mathcal{F}_L \longrightarrow H\mathcal{O}_X. \quad (\text{ACF16})$$

At $Q_{(p)}$ it sends every T^h to 1 and the source absorbing zero to the target additive zero. At all other stalks it is the unit map from $\mathbf{F}_1$.

Proof. The original endomorphism Fr_0 of $\mathcal{O}_{\mathbf{F}_1} = \mathbf{F}_1[T]$ factors through the constant spherical algebra $\mathbf{F}_1$: every monomial goes to 1 and the absorbing zero to 0. This factorization is equivariant for every Fr_m , so it pulls back to a sheaf augmentation $\mathcal{F}_L \rightarrow \mathbf{F}_1$. Compose with the unit map to $H\mathcal{O}_X$ to obtain ACF16. For uniqueness, any such morphism has the unique unit map at $Q_{(0)}$. The generization $G_L(\mathbb{Z}_{(p)}) \rightarrow G_L(\mathbb{Q})$ is injective. Compatibility with ACF11 forces the image of every T^h at $Q_{(p)}$ to be the unique element whose image in $G_L(\mathbb{Q})$ is 1, namely 1. At support points the source is $\mathbf{F}_1$, whose unital map is unique. A spherical monoid-algebra map is determined by its values on monomials, as is also proved explicitly below. Hence the maps agree on every stalk and are equal as sheaf maps. $\square$

3.4 Every point of the finite-prime stalk, with every zero label

Put $S_{\mathbb{C}} = G_L(\mathbb{C})$. A spherical-algebra point of $\mathbf{F}_1[T^{H_p^+}]$ with values in $HS_{\mathbb{C}}$ is exactly a pointed unital multiplicative monoid map to $S_{\mathbb{C}}$. Explicitly, for a pointed monoid M , the source in degree k_+ is $M \wedge k_+$. A monoid map $f : M \rightarrow S_{\mathbb{C}}$ induces the map which sends (m, j) to the tuple with $f(m)$ in coordinate j and τ elsewhere. The wedge basepoint goes to the all- τ tuple. A pointed map between finite pointed sets moves this one occupied coordinate or sends it to the basepoint; the coordinate-sum formula of H does exactly the same. Products agree because f is

multiplicative. Conversely the maps $1_+ \rightarrow k_+$ selecting one coordinate force this formula for every natural map. Its degree-one component must preserve multiplication, the unit and the absorbing zero. This proves the source adjunction in the present case, including its zero convention.

Consequently a point is a character

$$\chi : H_p^+ \longrightarrow S_{\mathbb{C}}, \quad \chi(0) = 1, \quad \chi(h + h') = \chi(h)\chi(h'), \quad (\text{ACF17})$$

with the additional source absorbing zero mapped to τ . The exponent 0 and the absorbing zero are distinct source elements.

Proposition 7 (Complete root-sequence classification). *Define*

$$\mathcal{U}_p = \{(u_n)_{n \geq 0} \in (\mathbb{C}^\times)^\mathbb{N} : u_{n+1}^p = u_n\}. \quad (\text{ACF18})$$

Then the full monoid of points is the disjoint union

$$\begin{aligned} \mathcal{P}_{p,L} &= \mathcal{U}_p \sqcup \{Z_\lambda : \lambda \in L\}, \\ Z_\lambda(0) &= 1, \quad Z_\lambda(h) = z_\lambda \quad (h > 0). \end{aligned} \quad (\text{ACF19})$$

A sequence $u \in \mathcal{U}_p$ gives the character

$$\chi_u(a/p^n) = \widehat{u_n^a} \quad (a \in \mathbb{N}), \quad \chi_u(0) = 1. \quad (\text{ACF20})$$

Products and the amplitude map are exactly

$$\begin{aligned} \chi_u \chi_v &= \chi_{uv}, \quad \chi_u Z_\lambda = Z_\lambda, \quad Z_\lambda Z_\mu = Z_{\lambda \wedge \mu}, \\ q_* : \mathcal{P}_{p,L} &\longrightarrow \varprojlim_{x \mapsto x^p} \mathbb{C} = \mathcal{U}_p \sqcup \{\mathbf{0}\}, \\ q_*^{-1}(\mathbf{0}) &= \{Z_\lambda : \lambda \in L\}, \quad q_*^{-1}(u) = \{\chi_u\} \quad (u \in \mathcal{U}_p). \end{aligned} \quad (\text{ACF21})$$

This inverse limit is asserted as a multiplicative monoid, not as a perfectoid field or as the ordinary field $\mathbb{C}$.

Proof. Given a character put $s_n = \chi(1/p^n)$. Then $s_{n+1}^p = s_n$. Conversely, from any sequence with this property define $\chi(a/p^n) = s_n^a$, with $s_n^0 = 1$. To check independence of the representation, pass to a common denominator p^N : $s_n^a = s_N^{ap^{N-n}}$; equal rational exponents give equal integer powers. The same common-denominator calculation proves multiplicativity.

Write $s_n = (u_n, \lambda_n)$. Idempotence of lattice meet gives $\lambda_{n+1} = \lambda_n$ for every n . If one u_n is zero, the relations and the absence of nonzero nilpotents in $\mathbb{C}$ force all $u_n = 0$. Thus all $s_n = z_\lambda$ for a single arbitrary label λ , producing Z_λ . Otherwise all u_n are nonzero and every support is 1_L , producing exactly ACF18 and ACF20. All products follow by evaluating at the generators $1/p^n$. The unit is the constant character 1 in $\mathcal{U}_p$, and the absorbing point is Z_{0_L} . The amplitude formula follows coordinate by coordinate; a complex coherent sequence is either everywhere zero or everywhere nonzero. This proves exhaustion, injectivity, and every fibre statement. $\square$

The labelled zeros form a multiplicatively absorbing ideal of the point monoid, with multiplication given by meet. They remain distinct before the explicitly stated amplitude quotient. Each support prime P_J has a unique algebraic $HS_{\mathbb{C}}$ -valued stalk point, because its stalk is $\mathbf{F}_1$; these points are still indexed by their different base points P_J . The map on point sets induced by the stalk augmentation ACF11 sends this unique generic point to the constant unit character in ACF19. It does not select a zero branch.

Proposition 8 (Every Frobenius, including exponent zero). *For every integer $m \geq 1$, source Frobenius induces*

$$\text{Fr}_m^*(\chi_u) = \chi_{(u_n^m)_n}, \quad \text{Fr}_m^*(Z_\lambda) = Z_\lambda. \quad (\text{ACF22})$$

For $m = 0$, every point is sent to the constant unit character. The p -th Frobenius is a bijection, with inverse on the group part $(u_n) \mapsto (u_{n+1})$, and with every Z_λ fixed. More generally, write $m = p^a d$ with $(d, p) = 1$. On $\mathcal{U}_p$, Fr_m^* is surjective with kernel canonically isomorphic to $\mu_d(\mathbb{C})$ by the zeroth coordinate after removal of the p -power automorphism. For $d > 1$ it is not injective. All these maps commute with the sheaf generizations ACF11.

Proof. Precomposition sends $\chi(h)$ to $\chi(mh) = \chi(h)^m$, giving ACF22. For $m = 0$ it gives $\chi(0) = 1$ at every exponent, while the extra source absorbing zero is still mapped to τ . The inverse for p follows from $u_{n+1}^p = u_n$ and fixes the constant support sequences. For d prime to p , to lift a sequence u choose $v_0^d = u_0$. Having chosen v_n , choose a p -th root w of v_n . Then w^d/u_{n+1} is a p -th root of unity. Since raising to the d -th power permutes μ_p , multiply w by the unique suitable p -th root of unity to obtain v_{n+1} with both $v_{n+1}^p = v_n$ and $v_{n+1}^d = u_{n+1}$. This constructs a preimage. In the kernel every coordinate belongs to μ_d , and raising to p is an automorphism of that finite group, so v_0 determines the whole sequence uniquely. Combining with the p -power bijection proves the assertion for m . Finally $\epsilon_p(T^{mh}) = 1$ for every h, m , which proves the compatibility even at $m = 0$. $\square$

3.5 The source's actual locality condition

In `defnloc`, Connes–Consani define a local character using a homomorphism from the ordered group H into a complete Hausdorff topological group. It must be continuous for the topology induced by its place and then extends uniquely to the corresponding local field. For H_p the places are p and ∞ ; for the trivial group there are no such places (source Proposition `classpla`). We apply this definition exactly to the unit group of the present target.

Proposition 9 (The exact boundary excluded by group extension). *The units of $S_{\mathbb{C}}$ are exactly $\{\widehat{a} : a \in \mathbb{C}^\times\}$, identified with the usual topological group $\mathbb{C}^\times$. A character ACF17 extends to a group homomorphism $H_p \rightarrow S_{\mathbb{C}}^\times$ if and only if it belongs to $\mathcal{U}_p$. Every Z_λ fails this extension, including Z_{1_L} and Z_{0_L} .*

Proof. If $(a, \lambda)(b, \mu) = 1$, then $ab = 1$, so a, b are nonzero and both supports are 1_L . Conversely $\widehat{a}^{-1} = \widehat{a^{-1}}$. A character extending to a group must take every $h > 0$ to a unit, since its value at $-h$ is an inverse. This excludes every zero branch. A character from $\mathcal{U}_p$ extends by $\rho(h-k) = \chi(h)\chi(k)^{-1}$ for $h, k \in H_p^+$; if $h-k = h'-k'$, then $h+k' = h'+k$, proving independence of the representation. The group law and uniqueness follow from the same calculation. Thus the obstruction object is exactly the retained lattice ideal $\{Z_\lambda\}$ of ACF21. $\square$

This argument does not discard the algebraic boundary: it identifies its exact failure to satisfy the source's group-valued definition. For an explicit topological semiring realization give L the discrete topology and $S_{\mathbb{C}} \subseteq \mathbb{C} \times L$ the subspace topology. Its operations are continuous, since the ring operations are continuous and maps between finite products of discrete sets are continuous. Its unit group has exactly the stated topology. Even without requiring group extension, a zero-branch character is not continuous at exponent 0: at the archimedean place $1/p^n \rightarrow 0$, and at the p -adic place $p^n \rightarrow 0$, whereas its amplitudes are constantly 0 along these sequences and its amplitude at 0 is 1.

Proposition 10 (All archimedean local characters). *The archimedean local points form the family*

$$\chi_z(h) = \widehat{\exp(zh)}, \quad z \in \mathbb{C}, \quad h \in H_p^+. \quad (\text{ACF23})$$

Different z give different characters. The trivial point is $z = 0$. Source Frobenius Fr_m^ acts by $z \mapsto mz$.*

Proof. The group H_p is dense in $\mathbb{R}$: rounding $p^n t$ to the nearest integer approximates any real t with error at most $1/(2p^n)$. A continuous homomorphism $H_p \rightarrow \mathbb{C}^\times$ is uniformly continuous for the group uniformities. Completeness of $\mathbb{C}^\times$ for its group uniformity follows from $\mathbb{C}^\times \simeq \mathbb{R} \times S^1$, by logarithm of the modulus and phase. Therefore the homomorphism extends uniquely to $\mathbb{R}$: apply it to approximating Cauchy sequences, and use uniform continuity to prove independence and continuity of the limit. For a continuous homomorphism $f : \mathbb{R} \rightarrow \mathbb{C}^\times$, choose a small interval around zero on which f takes values in a neighbourhood of 1 with a continuous logarithm g and $g(0) = 0$. For sufficiently small $s, t, s+t$, the difference $g(s+t) - g(s) - g(t)$ belongs to $2\pi i\mathbb{Z}$ and is continuous; on the connected neighbourhood of $(0, 0)$ it is therefore zero. Continuous local additivity gives $g(t) = zt$: rational subdivisions first give this equality on rational multiples of a fixed sufficiently small nonzero real argument, and continuity gives it on the entire small interval. Subdividing any real t into sufficiently small equal pieces gives $f(t) = \exp(zt)$ everywhere. Conversely each such function is continuous and multiplicative. If two parameters give the same character on H_p , density makes them agree on $\mathbb{R}$; their local logarithms have equal slopes. Frobenius follows by substituting mh . $\square$

For the p -adic family fix its parameter convention explicitly. For $x \in \mathbb{Q}_p$, let $\{x\}_p \in \mathbb{Z}[1/p] \cap [0, 1)$ be its finite sum of negative p -adic digits. Its class modulo $\mathbb{Z}$ depends only on $x \bmod \mathbb{Z}_p$. Set

$$\psi_p(x) = \exp(2\pi i \{x\}_p). \quad (\text{ACF24})$$

The map $\mathbb{Z}[1/p]/\mathbb{Z} \rightarrow \mathbb{Q}_p/\mathbb{Z}_p$ is a bijection: every p -adic class has the finite negative-digit representative and its kernel is $\mathbb{Z}[1/p] \cap \mathbb{Z}_p = \mathbb{Z}$. It respects addition, so ACF24 is a continuous additive character with kernel $\mathbb{Z}_p$.

Proposition 11 (All p -adic local characters). *The p -adic local points form the family*

$$\chi_a(h) = \widehat{\psi_p(ah)}, \quad a \in \mathbb{Q}_p, \quad h \in H_p^+. \quad (\text{ACF25})$$

The parameter is unique, the trivial point is $a = 0$, and source Frobenius acts by $a \mapsto ma$.

Proof. The group H_p is dense in $\mathbb{Q}_p$: truncate the p -adic expansion of any number at successively larger nonnegative exponents. The completeness argument just used extends each continuous character uniquely to $\mathbb{Q}_p$. A continuous homomorphism $f : \mathbb{Q}_p \rightarrow \mathbb{C}^\times$ maps a sufficiently small open subgroup $p^N \mathbb{Z}_p$ into a neighbourhood of 1 containing no nontrivial subgroup. Such a neighbourhood exists: use a bounded open interval for $\log |z|$ and an open phase arc of length less than one half-turn; powers of any nonidentity element leave it. Continuity therefore gives $f(p^N \mathbb{Z}_p) = 1$. For each $x \in \mathbb{Q}_p$, some p -power multiple belongs to $p^N \mathbb{Z}_p$, so $f(x)$ is a root of unity of p -power order.

Write $x_n = f(1/p^n)$ and use the fixed isomorphism $\mathbb{Q}_p/\mathbb{Z}_p \rightarrow \mu_{p^\infty}(\mathbb{C})$ induced by ψ_p to write $x_n = \psi_p(b_n)$, where b_n is specified modulo $\mathbb{Z}_p$. The relation $x_{n+1}^p = x_n$ gives $pb_{n+1} \equiv b_n \pmod{\mathbb{Z}_p}$. Thus $a_n = p^n b_n$ is specified modulo $p^n \mathbb{Z}_p$ and satisfies $a_{n+1} \equiv a_n \pmod{p^n \mathbb{Z}_p}$. Choose representatives recursively; their differences have valuations at least n , so they form a Cauchy sequence in $\mathbb{Q}_p$. Its limit a satisfies $a \equiv a_n \pmod{p^n \mathbb{Z}_p}$ for every n . It follows that $x_n = \psi_p(a/p^n)$. Agreement on these generators gives agreement on H_p , and continuity gives agreement on $\mathbb{Q}_p$. If two parameters work, their difference belongs to $\bigcap_{n \geq 0} p^n \mathbb{Z}_p = \{0\}$, proving uniqueness. Conversely ACF24 shows that every a gives a continuous character. Replacing h by mh proves the Frobenius action. $\square$

Proposition 12 (The exact intersection of the two local families). *As subsets of the algebraic point set ACF19, their intersection is*

$$\{\chi_{2\pi i b} : b \in \mathbb{Z}[1/p]\} = \{\chi_b \text{ of ACF25} : b \in \mathbb{Z}[1/p]\}. \quad (\text{ACF26})$$

The place-labelled local space therefore has two copies of these characters when the labels p and ∞ are retained. Forgetting the place label identifies exactly these copies. The zero-labelled ideal is disjoint from both families.

Proof. If an archimedean character $\exp(zh)$ is also p -adic local, then its value at $h = 1$ is a root of unity of p -power order by the preceding proof. Therefore $\exp z = \exp(2\pi ic/p^n)$ for integers c, n , so $z = 2\pi ib$ with $b \in \mathbb{Z}[1/p]$. For such b and $h \in H_p$, the rational number bh belongs to $\mathbb{Z}[1/p]$ and differs from $\{bh\}_p$ by an integer. Hence $\exp(2\pi ibh) = \psi_p(bh)$. This proves both sufficiency and the exact match of parameters; uniqueness follows from the preceding propositions. Exclusion of the zero-labelled ideal was proved by the unit criterion. $\square$

3.6 Comparison with the foundation's resolved point diagram

The foundation's Definition `def:split-absolute-point-map` assigns, in the two-element support case, the ordered groups $\{0\}, \mathbb{Z}, H_p$ to $P_\tau, Q_{(0)}, Q_{(p)}$, respectively. Its Corollary `cor:resolved-stalk-chain` gives the forward maps $\{0\} \hookrightarrow \mathbb{Z} \hookrightarrow H_p$, and hence $\mathbf{F}_1 \rightarrow \mathbf{F}_1[T] \rightarrow \mathbf{F}_1[T^{H_p^+}]$. This is a point-and-stalk diagram explicitly described as such in the source. The sheaf pulled back along r instead has the stalks and the reverse, generic-direction maps in ACF10–ACF11. In particular it restricts exactly to CC's original sheaf on $V(e)$; its value at $Q_{(0)}$ is not $\mathbf{F}_1[T]$.

There are exact objectwise maps relating the two intermediate objects:

$$\mathbf{F}_1 \xrightarrow{\text{unit}} \mathbf{F}_1[T] \xrightarrow{T \mapsto 1} \mathbf{F}_1, \quad \text{whose composite is the identity.} \quad (\text{ACF27})$$

The following calculation records the precise defect when this augmentation is compared with the forward inclusion into the finite stalk. It does not assert a sheaf identification.

Proposition 13 (The coequalizer of the intermediate comparison). *Let $j, c : \mathbf{F}_1[T] \rightrightarrows \mathbf{F}_1[T^{H_p^+}]$ be the maps $j(T) = T^1$ and $c(T) = 1$. At the level of pointed multiplicative monoids, their coequalizer is*

$$\mathbf{F}_1[T^{H_p^+}] \longrightarrow \mathbf{F}_1[H_p/\mathbb{Z}], \quad T^h \longmapsto [h], \quad 0 \longmapsto 0. \quad (\text{ACF28})$$

The target is the group $H_p/\mathbb{Z}$ with an extra absorbing zero; its identity is $[0]$, distinct from that zero. Applying the spherical monoid-algebra functor gives the spherical-algebra coequalizer. Its $HS_{\mathbb{C}}$ -valued points are exactly the characters ACF25 with $a \in \mathbb{Z}_p$. They have no extra zero-support branches.

Proof. The imposed relation is $T^1 = 1$. Every $h = a/p^n \in H_p^+$ then satisfies $(T^h)^{p^n} = T^a = 1$, so all monomials become invertible. The resulting group is the quotient of the group completion H_p by $\mathbb{Z}$. More explicitly, if $h - k \in \mathbb{Z}$, multiply the relation $T^1 = 1$ by the appropriate nonnegative monomial to identify T^h and T^k . Conversely their classes in $H_p/\mathbb{Z}$ can be equal only under that condition. The extra absorbing zero remains separate, witnessed by the displayed map to the pointed group. Any monoid map equalizing j, c sends T^1 to 1, makes every image monomial a unit by the displayed power identity, and therefore factors uniquely through this quotient. This proves the universal property. The monoid-algebra adjunction transports that same universal property to spherical algebras: the coordinate-inclusion argument proving it above applies to every spherical target using its degree-one multiplication and its structure maps.

A point of this quotient is a coherent sequence of roots of unity $x_{n+1}^p = x_n$ with $x_0 = 1$. Since $x_n^{p^n} = 1$, the parameter construction in the proof of ACF25 yields a unique $a \in \mathbb{Z}_p$; conversely such a satisfies $\psi_p(a) = 1$ and gives a point. All monomials in a group must map to units, excluding every Z_λ . Thus the entire point set and the exclusion are exact. $\square$

This coequalizer is an additional explicitly stated quotient, not a replacement of ACF19. It records the relation required to make the two forward maps agree. Its p -Frobenius on points is $a \mapsto pa$ on $\mathbb{Z}_p$, which is injective and not surjective; in contrast ACF22 is bijective on the

unquotiented full point set. The loss of the inverse Frobenius is exact: $T^1 = 1$ does not imply $T^{1/p} = 1$.

The support-local semiring need not be Boolean. For the three-element chain $L = \{0 < a < 1\}$ and the prime ideal $J = \{0, a\}$, the only allowed lattice denominator is 1, so ACF12 gives $L_J = L$ with all three elements distinct. For $J = \{0\}$, inverting a identifies a with 1 and leaves two elements. Thus the actual support stalks retain the different local lattice geometries even though every spherical stalk in ACF10 over them is $\mathbf{F}_1$.

The comparison quotient also gives an exact Frobenius calculation. Under the character parameter its point multiplication is addition in $\mathbb{Z}_p$, because $\psi_p((a+b)h) = \psi_p(ah)\psi_p(bh)$. Writing $m = p^a d > 0$ with $(p, d) = 1$, its Frobenius therefore has

$$\begin{aligned} \ker(m : \mathbb{Z}_p \rightarrow \mathbb{Z}_p) &= 0, & \text{coker}(m : \mathbb{Z}_p \rightarrow \mathbb{Z}_p) &= \mathbb{Z}_p/p^a \mathbb{Z}_p \simeq \mathbb{Z}/p^a \mathbb{Z}, \\ \{\text{quotient characters}\} \cap \{\text{archimedean local characters}\} &= \{\chi_{2\pi i k} : k \in \mathbb{Z}\}. \end{aligned} \quad (\text{ACF32})$$

The first line follows because $\mathbb{Z}_p$ is an integral domain and d is a unit, with the last quotient identified by its first a p -adic digits. The second follows from ACF26 and $\mathbb{Z}_p \cap \mathbb{Z}[1/p] = \mathbb{Z}$. Thus the full algebraic point group has the prime-to- p kernel μ_d in ACF22, whereas this specific comparison quotient has the p -primary cokernel in ACF32. Both statements retain their respective original domains.

3.7 An exact diagram of the retained data

For $L = \{0, a, b, 1\}$ with $a \wedge b = 0$ and $a \vee b = 1$, there are two support primes $P_{\{0,a\}}$ and $P_{\{0,b\}}$, both below $Q_{(0)}$. The following diagram displays the maps and stalk values; every horizontal arrow points in the generization direction for stalk maps:

$$\begin{array}{ccccc} \mathbf{F}_1[T^{H_p^+}] & \xrightarrow{\epsilon_p} & \mathbf{F}_1 & \xrightarrow{\text{id}} & \mathbf{F}_1 \\ T^h \mapsto 1 \downarrow & & \downarrow \text{unit} & & \downarrow \text{unit} \\ HG_L(\mathbb{Z}_{(p)}) & \longrightarrow & HG_L(\mathbb{Q}) & \longrightarrow & HL_J. \end{array} \quad (\text{ACF29})$$

The three columns lie over $Q_{(p)}$, $Q_{(0)}$, P_J , and the last column is present separately for each of the two support primes. Its bottom horizontal map is support localization, not an amplitude map. For this Boolean lattice $L_J \cong \{0, 1\}$: localizing the unique nontrivial element outside J makes it 1, and its complementary element becomes 0. Before this localization the finite-prime point boundary consists of the four distinct characters Z_0, Z_a, Z_b, Z_1 . Amplitude collapses those four to one zero character; locality selects only the group part; neither operation silently changes the other. The source locators for the top row are FF Proposition [prop12](#) and its Frobenius structure; the bottom row is proved in ACF12–ACF16, and the four labelled characters in ACF19–ACF21.

3.8 The assembled point object and its exact local map

The base, all its spherical stalks, and all their algebraic characters can be assembled without any implicit identification. Define the set of base-labelled algebraic points, with its displayed projection to X_L , by

$$\begin{aligned} \mathfrak{P}_L &= \coprod_{J \in \text{Spec}_{\text{lat}} L} \{(P_J, *)\} \sqcup \{(Q_{(0)}, *)\} \\ &\sqcup \coprod_{p \text{ prime}} (\{Q_{(p)}\} \times (\mathcal{U}_p \sqcup L)). \end{aligned} \quad (\text{ACF30})$$

This is a set with explicitly specified fibre monoids; no additional topology on this set is asserted. Its structure is fixed by the sheaf $\mathcal{F}_L$, the maps ACF11, and the products ACF21. The point

map for a generization from $Q_{(p)}$ to $Q_{(0)}$ or P_J goes in the opposite direction and selects the constant unit character in the finite-prime fibre. The multiplicative point at each support prime retains its base index J even when the corresponding spherical stalk map is identical to that at another support prime.

Define the place-labelled local object, retaining the trivial character at every place, by

$$\begin{aligned} \mathfrak{P}_L^{\text{loc}} &= \coprod_{p \text{ prime}} ((\{Q_{(p)}, \infty\} \times \mathbb{C}) \sqcup (\{Q_{(p)}, p\} \times \mathbb{Q}_p)), \\ \ell : \mathfrak{P}_L^{\text{loc}} &\longrightarrow \mathfrak{P}_L, \quad (Q_{(p)}, \infty, z) \longmapsto (Q_{(p)}, \chi_z), \quad (Q_{(p)}, p, a) \longmapsto (Q_{(p)}, \chi_a). \end{aligned} \tag{ACF31}$$

All fibres of ℓ have size at most two. They have size two exactly at $\chi_{2\pi i b} = \chi_b$ with $b \in \mathbb{Z}[1/p]$, by ACF26; their two preimages are the displayed distinct place-labelled points. The map has no image in the zero-labelled boundary, and there are no place-labelled points above P_J or $Q_{(0)}$, because their group is $\{0\}$ and its place set is empty. These statements follow respectively from ACF19, the unit criterion, ACF23–ACF26, and the source place definition. Thus ACF30–ACF31 give the full object and its exact local subfamilies, rather than a collapse to its classical amplitude.

Every positive source Frobenius fixes the base, fixes every boundary label, and acts on the local parameters by $z \mapsto mz$ and $a \mapsto ma$. Exponent-zero Frobenius sends every finite-prime algebraic point to its constant unit character and sends both local parameters to zero while retaining their place labels. Consequently ℓ is equivariant for every $m \geq 0$, directly by ACF22–ACF25.

4 A common sheaf for spherical, tropical, and full support coefficients

4.1 Sources and the exact base

The human sources are Alain Connes and Caterina Consani, *Geometry of the Arithmetic Site*, original `arithmetic-site_Adv_final1.tex`, specifically Definition `site`, Theorem `structure2`, Remark `caseno`, Theorem `thmspz`, and Proposition `sheavesonspz0`; and their *On the Absolute Geometry of Spec Z and the Fargues–Fontaine curve*, original `FF.tex`, Proposition `prop12`, the monoid-algebra adjunction in Section 2.3, and the Frobenius formula at source lines 150–152. The arithmetic-site source was read here at lines 598–704 and 900–1017, without a claim to exhaustive reading; FF was previously read completely, lines 1–1764. The spherical functor H is the explicit coordinate functor of Connes–Consani’s *Absolute algebra and Segal’s Γ -rings*, original `F1_corr.tex`, formula `functsum` and Lemma `sssalg`. That source was previously read completely, lines 1–1336. The coefficient foundation is *The Clankers, Split Support Geometry*, Version 11, Definition `def:lattice-split` and its complete ideal and prime classification, in the complete pinned original source. The constructions below give the common diagram for the companion calculations ACF and ASL, with the required definitions and proofs included here.

Let L be a nontrivial bounded distributive lattice. It need not be finite or Boolean. For a nonzero commutative ring R retain

$$\begin{aligned} G_L(R) &= \{(0, \lambda) : \lambda \in L\} \cup \{(a, 1_L) : a \in R\}, \\ (a, \lambda) + (b, \mu) &= (a + b, \lambda \vee \mu), \quad (a, \lambda)(b, \mu) = (ab, \lambda \wedge \mu), \\ z_\lambda &= (0, \lambda), \quad \tau = z_{0_L}, \quad e = z_{1_L}, \quad 1 = (1_R, 1_L), \quad q(a, \lambda) = a, \quad \sigma(a, \lambda) = \lambda. \end{aligned} \tag{FSC1}$$

Put $S = G_L(\mathbb{Z})$, $X = \text{Spec } S$, $Y = \text{Spec } \mathbb{Z}$, and $A = \text{Spec}_{\text{lat}} L$. The additive semiring prime spectrum has points

$$P_J = \{z_\lambda : \lambda \in J\} \quad (J \in A), \quad Q_{\mathfrak{p}} = q^{-1}(\mathfrak{p}) \quad (\mathfrak{p} \in Y). \tag{FSC2}$$

For completeness, an ideal containing a top-supported element contains e : add its product by $(-1, 1_L)$ to itself if its amplitude is nonzero. Then $ez_\lambda = z_\lambda$ puts the whole zero fibre in it, and

its amplitudes form a ring ideal. All other ideals are exactly the proper lattice ideals in the zero fibre. For an amplitude ideal primality is ring primality; for a support ideal it is meet primality. This proves FSC2 directly. Their basic opens are

$$D_S(z_\lambda) = D_L(\lambda), \quad D_S((n, 1_L)) = A \cup i(D_{\mathbb{Z}}(n)), \quad A = D_S(e), \quad i(Y) = V_S(e). \quad (\text{FSC3})$$

Here $i(\mathfrak{p}) = Q_{\mathfrak{p}}$ is the closed inclusion and $j : A \hookrightarrow X$ the open inclusion. Every open meeting $i(Y)$ is uniquely $W_U = A \cup i(U)$ for a nonempty open $U \subseteq Y$; the other opens are exactly the opens $V \subseteq A$. This follows from FSC3 because every nonempty open of Y contains its generic point and all support points are generizations of every arithmetic point. There is a continuous retraction

$$r(P_J) = r(Q_{(0)}) = \eta, \quad r(Q_{(p)}) = (p), \quad ri = \text{id}. \quad (\text{FSC4})$$

Indeed $r^{-1}D_{\mathbb{Z}}(n) = D_S((n, 1_L))$ for $n \neq 0$; $r^{-1}\emptyset = \emptyset$. These constitute the basis test. In particular $r^{-1}U = W_U$ for every nonempty open U .

4.2 The support sheaf, its direct image, and all their sections

For a lattice prime J define

$$L_J = L[(L \setminus J)^{-1}], \quad [\lambda]_J = [\mu]_J \iff \exists \nu \notin J : \lambda \wedge \nu = \mu \wedge \nu. \quad (\text{FSC5})$$

Every invertible idempotent is the unit; hence this formula follows from the localization equality, and every fraction is represented by its numerator. It makes explicit the possible loss of a label in an individual support stalk.

Lemma 14 (Sections of the support sheaf). *The restriction $\mathcal{O}_A = j^{-1}\mathcal{O}_X$ of the actual localization sheaf of S satisfies*

$$\mathcal{O}_A(D_L(\lambda)) = \downarrow \lambda, \quad a|_{D_L(\mu)} = a \wedge \mu \quad (\mu \leq \lambda), \quad (\mathcal{O}_A)_J = L_J. \quad (\text{FSC6})$$

In particular $\Gamma(A, \mathcal{O}_A) = L$. For every open $V \subseteq A$ its sections are exactly

$$\{(a_\lambda)_{D(\lambda) \subseteq V} : a_\lambda \leq \lambda, \quad a_\lambda \wedge \mu = a_\mu \text{ whenever } \mu \leq \lambda \text{ and } D(\lambda) \subseteq V\}. \quad (\text{FSC7})$$

The two operations are coordinatewise join and meet.

Proof. First $S[e^{-1}] = L$ because $e^2 = e$ makes $e = 1$ and $e(a, \lambda) = z_\lambda$. Then $L[\lambda^{-1}] = \downarrow \lambda$ by $a \mapsto a \wedge \lambda$, with unit λ . Indeed the equality of localized elements is exactly equality after meeting with λ . To verify the sheaf assertion, the prime-ideal separation theorem for bounded distributive lattices gives an injective map $\lambda \mapsto D_L(\lambda)$ preserving finite joins and meets, and these basic opens are compact. Here is the needed argument. If an ideal and filter are disjoint, an ideal maximal subject to avoiding the filter exists by Zorn's lemma and is prime: if it contains $a \wedge b$ but contains neither a nor b , adjoining either meets the filter, and distributivity makes the meet of those two filter elements belong to the original ideal, a contradiction. Apply this with the ideal $\downarrow \mu$ and filter $\uparrow \lambda$ when $\lambda \not\leq \mu$ to obtain separation. For compactness, a cover $D(\lambda) \subseteq \bigcup_i D(\mu_i)$ without a finite subcover would leave the ideal generated by all μ_i disjoint from $\uparrow \lambda$. The same prime would contradict the cover. For a finite cover $\lambda = \bigvee_i \lambda_i$, sections $a_i \leq \lambda_i$ agree on overlaps precisely when $a_i \wedge \lambda_j = a_j \wedge \lambda_i$. Their unique gluing is $\bigvee_i a_i$, by distributivity. Compactness reduces any cover of a basic open to this case. Thus FSC6 defines the localization sheaf on a basis and has the stalk FSC5. Restricting a section to every basis open gives FSC7; conversely those compatible values glue and are unique. This proves the formula for all opens, including $V = A$. $\square$

Set $\mathcal{D} = j_*\mathcal{O}_A$. There is an actual unital restriction map $d : \mathcal{O}_X \rightarrow \mathcal{D}$. Its sections and stalks are

$$\begin{array}{c|cc} & W_U = A \cup i(U) & V \subseteq A \\ \hline \mathcal{D} & L & \mathcal{O}_A(V) \\ \mathcal{O}_X & G_L(\mathcal{O}_Y(U)) & \mathcal{O}_A(V) \end{array} \quad \mathcal{D}_{Q_p} = L, \quad \mathcal{D}_{P_J} = L_J. \quad (\text{FSC8})$$

On W_U , d is σ ; on V , it is the identity. To justify the remaining entry for $\mathcal{O}_X$, localizing at an arithmetic prime gives $G_L(\mathbb{Z}_p)$, including $G_L(\mathbb{Q})$ at $Q_{(0)}$: all denominators there are nonzero integer amplitudes with top support, so labels never change. A section over W_U has one rational amplitude and one label at $Q_{(0)}$. Generization from each $Q_{(p)}$ injects $G_L(\mathbb{Z}_{(p)}) \hookrightarrow G_L(\mathbb{Q})$, forcing its amplitude to lie in $\mathcal{O}_Y(U)$ and its label to be the same. At all support points the section is determined by the germ at $Q_{(0)}$, and equals the localization of that label. Conversely the corresponding element of $G_L(\mathcal{O}_Y(U))$ gives these local fractions. This proves FSC8. Every neighbourhood of an arithmetic point contains all of A , so the direct-image stalk there is L ; at a support point it is the ordinary support stalk. Restriction from W_U to V sends (a, λ) to the section represented by λ , whose component on $D(\mu) \subseteq V$ is $\lambda \wedge \mu$.

Write $\underline{L}_X$ for the constant sheaf: its sections are the locally constant maps into the discrete set L , with pointwise lattice operations. There is a canonical unital sheaf map

$$\ell : \underline{L}_X \longrightarrow \mathcal{D}, \quad \ell_{Q_p} = \text{id}_L, \quad \ell_{P_J}(\lambda) = [\lambda]_J. \quad (\text{FSC9})$$

On W_U a locally constant map is constant: every point of A is a generization of $Q_{(0)}$ and every other arithmetic point specializes from $Q_{(0)}$, whereas a locally constant map into a discrete space is constant along either relation. Thus its sections there are L . On $V \subseteq A$ send a locally constant label function to the section whose germ at J is the localized label there; it is locally represented by a single label and therefore is a section of $\mathcal{O}_A$. This proves FSC9 on sections and under restrictions.

4.3 Tropical coefficients, labels, and Frobenius zero

For each rational prime p , put $H_p = \mathbb{Z}[1/p]$ and $K_p = H_{p,\max} = H_p \cup \{-\infty\}$ with operations maximum and ordinary addition. Its semiring zero is $\mathbf{0} = -\infty$, its semiring unit is $\mathbf{1} = 0 \in H_p$. Define

$$K_{p,L} = K_p \otimes_{\mathbb{B}} L, \quad \kappa(\lambda) = \mathbf{1} \otimes \lambda, \quad \rho \left(\bigoplus_i h_i \otimes \lambda_i \right) = \bigvee_{h_i \neq -\infty} \lambda_i. \quad (\text{FSC10})$$

The map $K_p \rightarrow \mathbb{B}$ sending every finite exponent to 1 and $-\infty$ to 0 is a semiring homomorphism. Tensoring it proves that ρ is well defined and a homomorphism; the tensor relations prove the same for κ , and $\rho\kappa = \text{id}$. Consequently κ is injective without collapsing any label. Positive Frobenius multiplies every finite h_i by its integer index n , while fixing $-\infty$. Exponent-zero Frobenius sends every finite exponent to 0 and fixes $-\infty$, so

$$\text{Fr}_0 = \kappa\rho, \quad \rho \text{Fr}_n = \rho \quad (n \geq 0), \quad \text{Fr}_n \kappa = \kappa. \quad (\text{FSC11})$$

These are semiring maps: scaling preserves maximum and exponent addition, and the zero action preserves which coefficients are absorbing zero. The last assertions also follow directly from the tensor formulas.

The source geometric morphism $\Theta : Y \rightarrow \widehat{\mathbb{N}}_0^\times$ has $\Theta(\eta) = \{0\}$ and $\Theta(p) = H_p$. Let $\mathcal{T}_L = \Theta^*(\mathcal{O}^0 \otimes_{\mathbb{B}} \underline{L})$. The source Proposition `sheavesonspz0`, including the zero Frobenius convention in Remark `caseno`, gives

$$(\mathcal{T}_L)_\eta = L, \quad (\mathcal{T}_L)_p = K_{p,L}. \quad (\text{FSC12})$$

The tensor commutes with inverse images: its finite products, coproducts, and coequalizers imposing the semiring relations are preserved by those functors. More explicitly, for every nonempty open $U \subseteq Y$ the complete section description is

$$\begin{aligned} \mathcal{T}_L(U) &= \{(\lambda, (w_p)_{p \in U}) : \lambda \in L, w_p \in K_{p,L}, \rho(w_p) = \lambda, \\ &w_p = \kappa(\lambda) \text{ for all but finitely many primes } p \in U\}. \end{aligned} \quad (\text{FSC13})$$

Operations are componentwise and restrictions discard removed prime coordinates. To prove this rather than infer a sheaf from stalks, use the source description: an original coefficient section is either absorbing zero everywhere, or finite exponents with value 0 outside a finite set of primes. Each local tensor expression therefore has a single generic label λ and is $\kappa(\lambda)$ outside a finite set. Every open of Y is quasi-compact, since a single nonempty member of a cover misses only finitely many points. A finite cover of local expressions gives precisely FSC13. Conversely, for each prime in its finite exceptional set, express w_p as a finite tensor sum; extend each finite coefficient to the source section with that value at p and exponent 0 elsewhere. On the open excluding the other exceptional primes this realizes w_p and is $\kappa(\lambda)$ elsewhere. These sections agree on overlaps and glue uniquely. Zero coefficients can be omitted, including the all-zero case. This proves both directions and every stated restriction map.

Set $\mathcal{E} = r^{-1}\mathcal{T}_L$. It has maps $\rho : \mathcal{E} \rightarrow \underline{L}_X$ and $\kappa : \underline{L}_X \rightarrow \mathcal{E}$ with $\rho\kappa = \text{id}$. Its sections on the two types of open, and its stalks, are

$$\begin{aligned} \mathcal{E}(W_U) &= \mathcal{T}_L(U), & \mathcal{E}(V) &= \text{LC}(V, L) \quad (V \subseteq A), \\ \mathcal{E}_{Q(p)} &= K_{p,L}, & \mathcal{E}_{Q(0)} &= L, \quad \mathcal{E}_{P_J} = L. \end{aligned} \quad (\text{FSC14})$$

Here LC denotes locally constant maps. For the first identity the inverse-image adjunction gives a map in one direction and restriction along i a left inverse. Sections with equal arithmetic restriction have equal germs at $Q_{(0)}$, hence at every support generalization, so the left inverse is also injective. For the second, $r|_A$ factors through the point η ; inverse image to that point is the stalk L , and pulling it back to A is the constant sheaf. The stalk formula follows either way. Restriction from W_U to V remembers the constant label λ in FSC13. Finally put

$$t = \ell\rho : \mathcal{E} \longrightarrow \mathcal{D}. \quad (\text{FSC15})$$

This is the promised unital tropical-to-support map: at arithmetic points it is ρ , at P_J it is $L \rightarrow L_J$, and on all sections it is fixed by every Frobenius, including zero.

4.4 The actual fibre product and its constrained carrier

Form the following fibre product of sheaves of unital semirings:

$$\mathcal{C} = \mathcal{O}_X \times_{\mathcal{D}} \mathcal{E}, \quad \mathcal{C}(W) = \{(s, w) : d(s) = t(w) \text{ in } \mathcal{D}(W)\}. \quad (\text{FSC16})$$

All operations are componentwise. Compatible pairs glue uniquely in the two factor sheaves, and the equality in $\mathcal{D}$ can be checked locally, so FSC16 is already a sheaf. The projections are unital homomorphisms. For any other sheaf with maps to both factors whose composites to $\mathcal{D}$ agree, their pair is the unique map to FSC16. This proves the fibre-product universal property.

Its sections on every open and its stalks are as follows:

$$\begin{aligned} \mathcal{C}(W_U) &= \{(a, \lambda, (w_p)) : a \in \mathcal{O}_Y(U), (\lambda, (w_p)) \in \mathcal{T}_L(U), a \neq 0 \Rightarrow \lambda = 1_L\}, \\ \mathcal{C}(V) &= \text{LC}(V, L) \quad (V \subseteq A). \end{aligned} \quad (\text{FSC17})$$

In the second line a label function f corresponds to the pair $(\ell f, f)$, not only to its localized value ℓf . Restrictions among the W_U localize the amplitude and remove prime coordinates. Restriction

to $V \subseteq A$ forgets the amplitude and prime coordinates and keeps the constant function λ . On the arithmetic finite stalk, abbreviate $R_p = \mathbb{Z}_{(p)}$. Then

$$\begin{aligned} \mathcal{C}_{Q_{(p)}} &\simeq \{(a, w) \in R_p \times K_{p,L} : a \neq 0 \Rightarrow \rho(w) = 1_L\}, \\ \mathcal{C}_{Q_{(0)}} &\simeq G_L(\mathbb{Q}), \quad \mathcal{C}_{P_J} \simeq L. \end{aligned} \tag{FSC18}$$

The finite-stalk pair (a, w) corresponds to $((a, \rho(w)), w)$ in the original fibre product. Its sum and product are $(a + b, w \oplus v)$ and (ab, wv) , with their exact original amplitudes and tropical coefficients. The support stalk λ corresponds to $([\lambda]_J, \lambda)$. To prove these assertions, substitute FSC8 and FSC14 into FSC16. For stalks, every finite set of section representatives can be put on one common neighbourhood, and an equality of their germs holds on a smaller common neighbourhood. Thus a stalk commutes with this finite fibre product. At a support prime the two maps to L_J are the identity on L_J and the localization $L \rightarrow L_J$, so their fibre product is L , by precisely the pair displayed above. At an arithmetic point the two maps are σ and ρ , which give the first two lines. This proves all sections and stalks rather than identifying the three input sheaves.

Write $\mathcal{R} = i_*\mathcal{O}_Y$. On W_U it is $\mathcal{O}_Y(U)$ and on every $V \subseteq A$ it is the zero ring (the singleton ring with $0 = 1$). The amplitude of an actual local fraction gives a unital map $q : \mathcal{O}_X \rightarrow \mathcal{R}$. There is an injective unital comparison

$$\mathcal{C} \hookrightarrow \mathcal{R} \times \mathcal{E}, \quad (s, w) \longmapsto (q(s), w). \tag{FSC19}$$

At finite arithmetic stalks its image is exactly the constrained subset in FSC18; at the generic arithmetic stalk it is $G_L(\mathbb{Q}) \subset \mathbb{Q} \times L$; at support stalks it is all of $\{0\} \times L$. Injectivity follows directly from these stalk descriptions: on arithmetic stalks the amplitude and $\rho(w)$ recover s , while on support stalks s is the localization of w . A sheaf map injective on stalks is injective on sections, since equality of germs gives local equality and then equality. The additional elements in the unrestricted finite-stalk ambient product are exactly

$$\{(a, w) : a \neq 0, \rho(w) \neq 1_L\}. \tag{FSC20}$$

They are not present in $\mathcal{C}$. At the generic stalk the corresponding extra pairs are $a \neq 0$ with support below 1_L . Thus the correspondence never silently enlarges the original nonzero-amplitude support constraint.

4.5 Original coefficients, the tropical ideal, and their maps

Let B_L be the coefficient sheaf on Y with $B_L(U) = G_L(\mathcal{O}_Y(U))$ for nonempty U . A compatible cover has one support label because all nonempty intersections contain η ; the ring amplitudes glue and retain top support when nonzero. This proves the sheaf assertion. Preserve the actual inverse image under the separate name

$$\mathcal{P} = r^{-1}B_L.$$

Its sections on W_U are $G_L(\mathcal{O}_Y(U))$, while its sections on $V \subseteq A$ are $\text{LC}(V, G_L(\mathbb{Q}))$. Its arithmetic stalks are $G_L(R_p)$ and $G_L(\mathbb{Q})$, and its stalk at *every* support point is also $G_L(\mathbb{Q})$. These follow from the inverse-image stalk formula and the factorization of $r|_A$ through η , whose B_L -stalk is $G_L(\mathbb{Q})$. Thus this original pullback still retains rational amplitudes on the support open. It is not the fixed sheaf constructed next.

Define the separate sheaf

$$\mathcal{B} = \mathcal{O}_X \times_{\mathcal{D}} \underline{L}_X.$$

Its sections on W_U are $G_L(\mathcal{O}_Y(U))$, represented by $((a, \lambda), \lambda)$; its sections on $V \subseteq A$ are $\text{LC}(V, L)$, represented by $(\ell f, f)$. Its stalks are $G_L(R_p)$, $G_L(\mathbb{Q})$, and L at support points. These formulas

are direct substitutions from FSC8–FSC9, with the same sheaf and finite-stalk argument as for FSC16–FSC18. There is an exact sequence of comparison maps (no additive exactness terminology is imposed)

$$\mathcal{P} \xrightarrow{\alpha} \mathcal{B} \xrightarrow{\Phi} \mathcal{O}_X. \quad (\text{FSC33})$$

Here Φ is the first projection. On arithmetic stalks both maps are the identity under the displayed identifications; at P_J they are exactly

$$G_L(\mathbb{Q}) \xrightarrow{\sigma} L \longrightarrow L_J, \quad (a, \lambda) \longmapsto \lambda \longmapsto [\lambda]_J. \quad (\text{FSC34})$$

To construct them on sections, let α be the identity on W_U and the coordinatewise support map on locally constant sections $V \rightarrow G_L(\mathbb{Q})$. Restriction of an arithmetic-open section to V is its constant rational-amplitude germ in $\mathcal{P}$, and its constant support label in $\mathcal{B}$; hence these formulas commute with that restriction as well as restrictions within each kind of open. Operations are preserved by σ . This proves α as an actual unital sheaf map and proves FSC34. Both maps are epimorphisms because all their stalk maps are surjective. The equality relation of α at support points is equality of support labels, with amplitudes arbitrary within the original constrained carrier; at arithmetic points it is equality of the whole element. The equality relation of Φ at support points is FSC5, and at arithmetic points again equality. Their composite is the previous full coefficient-to-localization map, with the support-only loss of amplitude occurring at the first arrow and lattice localization at the second. All three sheaves remain explicit.

There is an injective unital map and a unital retraction

$$\begin{aligned} \nu : \mathcal{B} &\hookrightarrow \mathcal{C}, & (a, \lambda) &\longmapsto (a, \kappa(\lambda)), \\ \beta : \mathcal{C} &\longrightarrow \mathcal{B}, & (a, w) &\longmapsto (a, \rho(w)), & \beta\nu &= \text{id}. \end{aligned} \quad (\text{FSC21})$$

The displayed formulas apply at arithmetic stalks and to the corresponding prime coordinates of sections; at support points both maps are the identity of L . They respect every restriction formula already proved. Their operations follow because κ and ρ are homomorphisms and $\rho\kappa = \text{id}$. The composite $\mathcal{B} \rightarrow \mathcal{C} \rightarrow \mathcal{O}_X$ is the isomorphism at arithmetic stalks and the localization $L \rightarrow L_J$ at support stalks. These are the exact places where the fixed coefficient comparison and the actual localization sheaf differ.

Frobenius on the common object is

$$F_n(s, w) = (s, \text{Fr}_n w) \quad (n \geq 0), \quad F_0 = \nu\beta, \quad \text{Fix}(F_0) = \nu(\mathcal{B}). \quad (\text{FSC22})$$

It is a sheaf endomorphism because $\rho \text{Fr}_n = \rho$. Every amplitude is retained unchanged; no claim is made that ordinary ring powering is an additive endomorphism. The last two assertions follow from FSC11 and FSC21. They also recover the original constrained $G_L(R_p)$ exactly at every arithmetic stalk. As sheaves of semirings, F_0 is an actual idempotent with the indicated image, not an assumed identification of the spherical, tropical and arithmetic structures.

There is a second exact map

$$\delta : \mathcal{E} \longrightarrow \mathcal{C}, \quad \delta(w) = (z_{\rho(w)}, w) \text{ ext at arithmetic stalks}, \quad \delta(\lambda) = \lambda \text{ at support stalks}. \quad (\text{FSC23})$$

On W_U it sets the amplitude equal to zero and retains all tropical coordinates. On V it is the identity on locally constant labels. These descriptions commute with restrictions and give an injective map preserving zero, addition, and multiplication. It is nonunital: its unit image has arithmetic amplitude zero. More precisely let $b = \delta(1_{\mathcal{E}})$, the global section whose pair in FSC16 is $(e, 1_{\mathcal{E}})$. Then

$$b^2 = b, \quad b\mathcal{C} = \delta(\mathcal{E}) = \ker(\mathcal{C} \longrightarrow \mathcal{R}), \quad \delta : \mathcal{E} \xrightarrow{\sim} b\mathcal{C}. \quad (\text{FSC24})$$

The last map is unital when the principal ideal is given its own identity b . Indeed multiplication by b sends (a, w) to $(0, w)$, which proves both equalities and the asserted identity on the ideal; at

support points $b = 1$ and the same formula holds in the zero-amplitude ambient description. All inverse maps are given by the tropical projection. There cannot be a unital sheaf map $\mathcal{E} \rightarrow \mathcal{C}$: addition is idempotent in $\mathcal{E}$, whereas $1_{\mathcal{C}} + 1_{\mathcal{C}} \neq 1_{\mathcal{C}}$ at the arithmetic generic stalk, since its amplitudes are 2 and 1 in $\mathbb{Q}$. Thus FSC23–FSC24 are the exact replacement for the invalid unital inclusion, while retaining the entire ideal.

The amplitude map is the ring reflection of this common sheaf:

$$\mathcal{C}/\sim_{\delta\mathcal{E}} \simeq \mathcal{R}, \quad x \sim_{\delta\mathcal{E}} y \iff \text{locally } x + u = y + v \text{ for } u, v \in \delta\mathcal{E}. \quad (\text{FSC25})$$

It is surjective on sections of W_U , since a has a lift $(a, \kappa(1_L))$, and on sections of support opens, whose target is one element. If two arithmetic-open sections (a, w) and (a, w') have the same amplitude, then

$$(a, w) + \delta(w') = (a, w') + \delta(w)$$

in the full pair presentation: supports join on both sides and the tropical sums agree. At support opens the whole semiring is the ideal. Conversely adding ideal sections does not change amplitude. This proves the exact congruence on all opens, hence its sheaf version. Any unital map to a sheaf of rings annihilates this ideal because its elements are additively idempotent; it therefore factors uniquely through FSC25. This proves the universal property, without any unsupported stable-spectrum consequence.

4.6 The spherical map into the entire common object

Let $\mathcal{F}_L = r^{-1}\Theta^{-1}\mathbf{F}_1[T]$, with $\mathcal{F}_\eta = \mathbf{F}_1$ and $\mathcal{F}_p = \mathbf{F}_1[T^{H_p^+}]$ as in FF Proposition prop12. On the source topos the pointed multiplicative map

$$\mathbf{F}_1[T] \longrightarrow H(\mathbb{Z}_{\max} \otimes_{\mathbb{B}} L), \quad T^n \longmapsto n \otimes 1_L, \quad 0 \longmapsto \mathbf{0} \quad (\text{FSC26})$$

preserves the unit: T^0 goes to exponent 0, the tropical unit, not to the absorbing $-\infty$. It preserves multiplication because $n + m$ is the tropical product, and it commutes with every source Frobenius including zero. To see its extension to spherical algebras explicitly, a monoid-algebra element in degree k_+ has a single monomial and one nonbasepoint coordinate. Put its image in that coordinate and the additive zero elsewhere. Pointed maps move or discard that coordinate, agreeing with the fibre-sum formula of H ; products agree by the proved multiplicative formula. Conversely coordinate inclusions force this construction. Thus FSC26 is an actual spherical-algebra morphism. Inverse image gives

$$u : \mathcal{F}_L \longrightarrow H\mathcal{E}, \quad u_{Q_{(p)}}(T^h) = h \otimes 1_L, \quad u_{P_J} : \mathbf{F}_1 \rightarrow HL \text{ the unit map.} \quad (\text{FSC27})$$

The same unit formula holds at $Q_{(0)}$.

There is also the augmentation $v : \mathcal{F}_L \rightarrow H\mathcal{O}_X$, with $T^h \mapsto 1$ and the absorbing zero mapped to the target additive zero. It exists by pulling back the equivariant source augmentation $T^n \mapsto 1$. It is unique: any such map is the unit map at $Q_{(0)}$, whose source is $\mathbf{F}_1$; the injection $G_L(R_p) \rightarrow G_L(\mathbb{Q})$ then forces every monomial's image at $Q_{(p)}$ to be 1. The source generalization has $T^h \mapsto 1$, because it commutes with Fr_0 , which is the identity on $\mathbf{F}_1$ and sends T^h to T^0 . At support primes the source is also $\mathbf{F}_1$, so the unit map is unique there. Equality on stalks proves equality of sheaf maps.

The composites $Hd \circ v$ and $Ht \circ u$ are equal. On every finite stalk both send each monomial to 1_L and the absorbing zero to 0_L ; at support points both give the unit map to HL_J . The monoid-algebra coordinate argument makes this equality hold in every finite pointed degree. Since H takes the semiring fibre product to the spherical fibre product degree by degree, there is a unique map

$$\begin{aligned} \gamma : \mathcal{F}_L &\longrightarrow HC, \\ \gamma_{Q_{(p)}}(T^h) &= (1, h \otimes 1_L), \quad \gamma_{Q_{(p)}}(0) = (0, \mathbf{0}), \\ \gamma_{Q_{(0)}}(1) &= (1, 1_L), \quad \gamma_{Q_{(0)}}(0) = (0, 0_L), \quad \gamma_{P_J} : \mathbf{F}_1 \rightarrow HL \text{ the unit map.} \end{aligned} \quad (\text{FSC28})$$

Here $(1, h \otimes 1_L)$ denotes exactly the constrained pair of FSC18. This map is injective. At a finite stalk, distinct h stay distinct: a lattice prime exists by the separation proof above, and its Boolean character gives an evaluation $K_{p,L} \rightarrow K_p$ taking $h \otimes 1_L$ to h . The absorbing zero has a different arithmetic amplitude. At the generic and support stalks the two values are also distinct, since L is nontrivial. In each finite pointed degree the coordinate description has at most one nonzero entry, preserving its coordinate and value. Thus it is injective in every degree and every stalk, hence as a sheaf map. This proves a faithful spherical comparison retaining positive exponents, even though its projection to the ordinary split structure sheaf is the forced augmentation. Finally $F_n \gamma = \gamma \text{Fr}_n$ for all $n \geq 0$ by FSC22 and the displayed monomial formulas; no ordinary amplitude Frobenius has been introduced.

The complete diagram is

$$\begin{array}{ccccc}
\mathcal{F}_L & \xrightarrow{\gamma} & HC & \longrightarrow & H\mathcal{E} \\
v \downarrow & & \downarrow & & \downarrow Ht \\
HO_X & \xlongequal{\quad} & HO_X & \xrightarrow{Hd} & H\mathcal{D}.
\end{array} \tag{FSC29}$$

The right square is a fibre product, all arrows are actual unital spherical sheaf maps, and the upper-left map is injective. The ideal map δ of FSC23 is deliberately outside this diagram of unital maps; it becomes a unital isomorphism only onto the ideal with identity b in FSC24. Exact proof locators for the topological base and bottom row are FSC2–FSC9; for the right-hand maps FSC10–FSC16; and for the spherical monomial maps FSC26–FSC28.

4.7 The precise obstruction to a global section and its replacement

The projection $\pi : \mathcal{C} \rightarrow \mathcal{O}_X$ is an epimorphism of sheaves, but for the full lattice its arithmetic section need not extend over the support open. Its exact stalk fibres are

$$\begin{aligned}
\pi_{Q_{(p)}}^{-1}(a, \lambda) &= \{w \in K_{p,L} : \rho(w) = \lambda\}, & \pi_{Q_{(0)}} &= \text{id}_{G_L(\mathbb{Q})}, \\
\pi_{P_J}^{-1}([\mu]_J) &= \{\lambda \in L : \exists \nu \notin J, \lambda \wedge \nu = \mu \wedge \nu\}.
\end{aligned} \tag{FSC30}$$

The finite fibres are nonempty, containing $\kappa(\lambda)$; the support fibres are nonempty because localization is represented by a numerator. Thus it is surjective on stalks. To see directly that this implies the sheaf epimorphism assertion, any section of the target has a preimage germ at each point; choose local representative sections of the domain and shrink so their images equal the given target section. These local lifts prove local surjectivity, the epimorphism condition for sheaves of sets with algebraic operations.

A unital sheaf section of π exists if and only if L is the two-element lattice. First suppose one exists. Compose it with β to obtain a section of $\Phi : \mathcal{B} \rightarrow \mathcal{O}_X$ from FSC21. At $Q_{(0)}$, Φ is the identity, so this section is forced to be the identity there. Compatibility with generization to P_J forces its support-stalk component to satisfy

$$s_J([\lambda]_J) = \lambda \quad \text{for every } \lambda \in L. \tag{FSC31}$$

Indeed apply the compatibility square to z_λ at $Q_{(0)}$: in $\mathcal{O}_X$ it generizes to $[\lambda]_J$, whereas in $\mathcal{B}$ it generizes to the unlocalized label λ . If $0_L < \lambda < 1_L$, prime-ideal separation supplies a prime J avoiding λ . In L_J , $[\lambda]_J = [1_L]_J$, which contradicts FSC31. Thus no intermediate element exists and $L = \{0_L, 1_L\}$. Conversely for this two-element lattice its sole prime is $\{0_L\}$, and localization there is the identity. Therefore Φ is an isomorphism at every stalk, hence a sheaf isomorphism: local inverses exist and glue uniquely by injectivity. The map $\nu\Phi^{-1}$ then is a unital section of π . This proves both directions.

The obstruction therefore defines the exact replacement rather than removing the support geometry: $\mathcal{B}$ is the coefficient sheaf which retains the unlocalized support labels, embeds by ν into

$\mathcal{C}$, and maps onto the actual $\mathcal{O}_X$ by precisely FSC31's localization quotient. On the arithmetic closed stratum,

$$i^{-1}\mathcal{C} = B_L \times_{\underline{L}_Y} \mathcal{T}_L, \quad B_L \hookrightarrow i^{-1}\mathcal{C}, \quad (a, \lambda) \longmapsto (a, \kappa\lambda). \quad (\text{FSC32})$$

To verify this, inverse image commutes with finite products and equalizers, or one may substitute the already proved sections and stalks and use the same finite-representative germ argument as in FSC18. Here $i^{-1}\mathcal{O}_X = B_L$, $i^{-1}\mathcal{D} = \underline{L}_Y$, and $i^{-1}\mathcal{E} = \mathcal{T}_L$. Thus the original constrained coefficient sheaf survives exactly, while its globally extendable replacement and the actual localization map remain visible.

For the three-element chain $0 < a < 1$, the support primes are $J_0 = \{0\}$ and $J_a = \{0, a\}$. At J_0 , localization sends a to 1 and gives $L_{J_0} = \{0, 1\}$; at J_a , only 1 is inverted and $L_{J_a} = L$. Accordingly $\mathcal{C}$ has the three-element stalk L at both support points, whereas $\mathcal{O}_X$ has respectively two and three elements. The fibre over $1 \in L_{J_0}$ in FSC30 is exactly $\{a, 1\}$. This proves the obstruction concretely without replacing the chain lattice by independent Boolean coordinates. At a finite arithmetic stalk, $w = \mathbf{1} \otimes a$ has $\rho(w) = a$. The pair $(1, w)$ therefore exists in the unrestricted ambient product but is excluded by FSC20, whereas $(0, w)$ exists in $\mathcal{C}$. The original three zero labels appear exactly as $(0, \kappa 0)$, $(0, \kappa a)$, $(0, \kappa 1)$ in its fixed coefficient subobject. These examples describe the map, its fibre, and the excluded amplitudes using the same original lattice throughout.

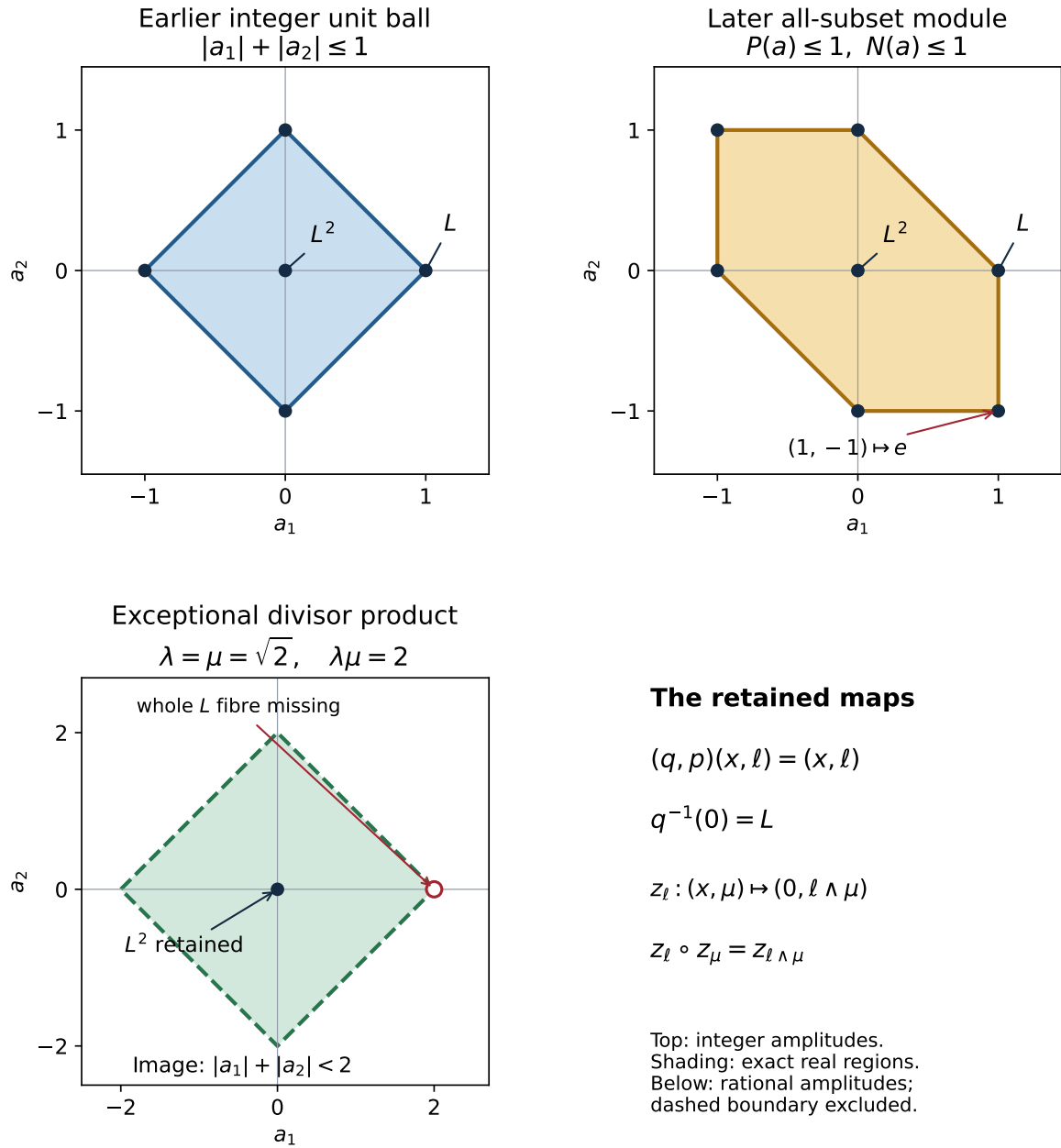


Figure 2: The exact older and later section bounds and the exceptional divisor product. Every zero-amplitude coordinate retains its full L fibre. Complete proofs: HG39–41 and SDM6–8.

5 The full Split-Zero coefficient object as a Connes–Consani Γ -algebra

The construction used here is Alain Connes and Caterina Consani, *Absolute algebra and Segal’s Γ -rings: au dessous de $\overline{\text{Spec } \mathbb{Z}}$* .

The original author source is `F1_corr.tex`, associated with arXiv:1502.05585v2. The whole source, including its bibliography and final empty lines, was read: lines 1–1336. The read file has SHA-256

3b1319ec1df4ed6e2412ace960c2813b683a
36c88509234c9abd0a39f98e8913.

The local source has duplicate labels `sssalg1`; below the line numbers distinguish the full-faithfulness proposition (line 536) from the Boolean-semiring lemma (line 575). The formula `functsum`, lines 433–442, and Lemma `sssalg`, lines 519–533, specify the actual functor and multiplication. Proposition `sssalg2`, lines 905–932, specifies the norm construction. The conclusions below are direct calculations with those formulas, not an identification of every spherical algebra with a semiring.

Let R be a nonzero commutative ring with identity and let L be a bounded distributive lattice. Write its bottom and top as $0_L, 1_L$. Keep the entire coefficient object

$$\begin{aligned} G &= G_L(R) = \{(0, \lambda) : \lambda \in L\} \cup \{(r, 1_L) : r \in R\}, \\ (r, \lambda) + (s, \mu) &= (r + s, \lambda \vee \mu), \quad (r, \lambda)(s, \mu) = (rs, \lambda \wedge \mu), \\ \tau &= (0, 0_L), \quad e = (0, 1_L), \quad \hat{r} = (r, 1_L), \quad z_\lambda = (0, \lambda). \end{aligned} \tag{HG1}$$

Thus $\hat{0} = e$. No occurrence of e in this fragment is silently replaced by τ . The semiring identity is $\hat{1}$. The formulas close on the stated carrier, and associativity and distributivity follow coordinate by coordinate from the ring and lattice laws. Regard L itself as the semiring with addition $\vee$ and multiplication $\wedge$.

5.1 Every finite pointed set, every fold, and multiplication

For a finite pointed set X let $X^\circ = X \setminus \{*\}$. The base point of $HG(X)$ is the tuple constantly equal to τ . For $X = 0_+$ it is the unique empty tuple. The exact formula is

$$HG(X) = \{((r_x, \lambda_x))_{x \in X^\circ} : r_x \neq 0 \implies \lambda_x = 1_L\}. \tag{HG2}$$

For a pointed map $f : X \rightarrow Y$ set

$$(HG(f)(r, \lambda))_y = \left(\sum_{x \in X^\circ : f(x)=y} r_x, \bigvee_{x \in X^\circ : f(x)=y} \lambda_x \right), \quad y \in Y^\circ. \tag{HG3}$$

An empty sum is 0_R and an empty join is 0_L . Coordinates sent to the base point of Y are discarded, rather than added to a retained coordinate.

Theorem 15. *Equations (HG2)–(HG3), together with*

$$\mu_{X,Y}((r, \lambda), (s, \mu))_{(x,y)} = (r_x s_y, \lambda_x \wedge \mu_y), \quad (x, y) \in X^\circ \times Y^\circ, \tag{HG4}$$

define the Connes–Consani commutative $\mathbb{S}$ -algebra HG . Its unit $\mathbb{S}(X) = X \rightarrow HG(X)$ sends $x \neq *$ to the tuple with $\hat{1}$ at x and τ elsewhere, and sends $*$ to the all- τ tuple.

Proof. If a sum in (HG3) is nonzero, at least one summand is nonzero, so its support is 1_L and the output join is 1_L . Thus the output satisfies (HG2), even when other terms cancel. Identity maps act identically. For $g : Y \rightarrow Z$ and $z \neq *$, the fibres $f^{-1}(y)$ for $g(y) = z$ partition $(gf)^{-1}(z)$; no point discarded by f can reappear, since $g(*) = *$. Associativity of finite sums and joins now gives $HG(g)HG(f) = HG(gf)$.

Formula (HG4) lands in (HG2) because a nonzero product requires both amplitudes to be nonzero. An all- τ input produces an all- τ output, so this map factors through the smash product of pointed sets. Naturality in $f : X \rightarrow X'$ and $g : Y \rightarrow Y'$ is the equality, at (x', y') , between the sum over $f^{-1}(x') \times g^{-1}(y')$ of the products and the product of the two fibre sums. The amplitude equality is ring distributivity; the support equality is

$$\bigvee_{x,y} (\lambda_x \wedge \mu_y) = \left(\bigvee_x \lambda_x \right) \wedge \left(\bigvee_y \mu_y \right).$$

This natural bifunctor map is precisely the multiplication map for the Day smash product, by the source's definition preceding `colim`, lines 445–464. The two triple products agree coordinatewise, as do the switched products. The unit rule is natural because each unit tuple has at most one non- τ coordinate; the product with that tuple has its original value on its single selected coordinate and τ elsewhere. This proves all claimed algebra axioms directly. $\square$

5.2 Amplitude and lattice support form one exact object

Define $q : G \rightarrow R$ and $p : G \rightarrow L$ by

$$q(r, \lambda) = r, \quad p(r, \lambda) = \lambda. \quad (\text{HG5})$$

Both maps preserve zero, identity, addition and multiplication. Consequently $Hq : HG \rightarrow HR$ and $Hp : HG \rightarrow HL$ are morphisms of $\mathbb{S}$ -algebras, not merely maps on level 1.

Theorem 16. *The joint map*

$$(Hq, Hp) : HG \hookrightarrow HR \times HL \quad (\text{HG6})$$

is injective at every finite pointed set. Its image is exactly

$$\{(r, \lambda) \in R^{X^\circ} \times L^{X^\circ} : r_x \neq 0 \implies \lambda_x = 1_L\}. \quad (\text{HG7})$$

For fixed amplitude r and fixed support λ , respectively, the fibres are

$$\begin{aligned} (Hq)_X^{-1}(r) &\cong \prod_{x:r_x=0} L, \\ (Hp)_X^{-1}(\lambda) &\cong \prod_{x:\lambda_x=1_L} R. \end{aligned} \quad (\text{HG8})$$

All omitted support coordinates in the first line are 1_L ; all omitted amplitude coordinates in the second line are 0_R . These descriptions also cover $0_L = 1_L$, when $G_L(R) = R$.

Proof. A coefficient is exactly its amplitude and support pair, proving injectivity and the image claim. The naturality of the two projections follows from (HG3): the first takes fibre sums and the second takes fibre joins, which are the defining actions of HR and HL . Equation (HG4) proves multiplication compatibility and the unit tuples give unit compatibility. At each zero amplitude coordinate every support is allowed; at each nonzero amplitude coordinate only top is allowed. Conversely, at a support different from top the amplitude must be zero, and at top every amplitude is allowed. Independence of the coordinates proves (HG8). $\square$

The pointwise inverse image of zero under Hq is the embedded full lattice object

$$\iota : HL \longrightarrow HG, \quad (\lambda_x)_x \longmapsto (z_{\lambda_x})_x, \quad Hp \circ \iota = \text{id}_{HL}. \quad (\text{HG9})$$

This is a natural injective map of Γ -sets and preserves products, but when $R \neq 0$ it is not a unital map of $\mathbb{S}$ -algebras: its unit is sent to e rather than $\hat{1}$. Its image is an ideal for (HG4), with the explicit action

$$(r_x, \lambda_x) z_{\mu_y} = z_{\lambda_x \wedge \mu_y}. \quad (\text{HG10})$$

The map Hp has only the base point in its zero fibre when $0_L \neq 1_L$, yet it is not injective: e and $\hat{1}$ have the same support. No argument appropriate only to abelian-group exact sequences can erase these fibres.

Proposition 17. *Fix $r \in HR(X)$ and $f : X \rightarrow Y$. Write $r' = HR(f)r$ and $I_y = \{x \in X^\circ : f(x) = y\}$. The image of the entire fibre $(Hq)_X^{-1}(r)$ in $(Hq)_Y^{-1}(r')$ consists exactly of the following support choices, independently at each y :*

$$\lambda'_y \in \begin{cases} \{0_L\}, & I_y = \emptyset, \\ \{1_L\}, & \text{some } x \in I_y \text{ has } r_x \neq 0, \\ L, & I_y \neq \emptyset \text{ and all } r_x = 0 \text{ on } I_y. \end{cases} \quad (\text{HG11})$$

In particular nontrivial cancellation in a fold produces e , not τ , while a fold of labelled zeros produces their exact lattice join.

Proof. The first case is the empty-join convention. In the second case at least one input support is top, so the output support is top regardless of whether r'_y vanishes. In the third case all input supports are unrestricted. Every $\lambda \in L$ occurs as the join by assigning it to any one input coordinate and assigning bottom to the others. Distinct I_y are disjoint, so all these choices can be made independently. The last assertions are exactly (HG3). $\square$

5.3 The failed section and its complete correction object

There is always the pointed set section

$$b : R \rightarrow G, \quad b(0) = \tau, \quad b(r) = \hat{r} \ (r \neq 0), \quad q \circ b = \text{id}_R. \quad (\text{HG12})$$

Let $b_X : R^{X^\circ} \rightarrow G^{X^\circ}$ apply this formula coordinatewise. This is a family of pointed maps, not yet a natural transformation.

Theorem 18. *Assume $0_L \neq 1_L$. Every natural transformation of Γ -sets $HR \rightarrow HG$ is the constant base-point map. In particular (HG12) does not lift to a Γ -map, and Hq has no Γ -section. The pointed section b is multiplicative precisely when R has no nonzero zero divisors.*

Proof. Let $\eta : HR \rightarrow HG$ be natural and write $u = \eta_{1_+}$. For each $x \in X^\circ$, the pointed projection $p_x : X \rightarrow 1_+$ takes x to 1 and every other point to 0. Naturality forces $(\eta_X(r))_x = u(r_x)$. Apply this first to $X = 2_+$ and then to its fold $1, 2 \mapsto 1$. One obtains $u(a + b) = u(a) + u(b)$, with $u(0) = \tau$. This is the same recovery of addition used by Connes–Consani in (addition) and the proof at lines 536–558, here proved for a Γ -map without assuming it preserves multiplication.

If $v + w = \tau$ in G , their supports satisfy $p(v) \vee p(w) = 0_L$, so both supports are bottom. Since $0_L \neq 1_L$, the carrier (HG1) then forces $v = w = \tau$. Thus τ is the only additively invertible element of G . For every $r \in R$ the equality $u(r) + u(-r) = u(0) = \tau$ forces $u(r) = \tau$. The coordinate projections now force every η_X to be constant.

For multiplicativity of b , if either input vanishes the identity follows from absorption by τ . If both are nonzero and their product is nonzero, both sides equal the top-supported product. If $r, s \neq 0$ but $rs = 0$, the left side $b(r)b(s)$ is e and the right side $b(rs)$ is τ . This proves exactly the asserted criterion. $\square$

For the obstruction formulas (HG13)–(HG16), take $0_L \neq 1_L$. When L has one element, $G_L(R) = R$, b is the identity, and each correction is the unique zero element of HL . To calculate the obstruction at every arity, use the two-element subset $\{0_L, 1_L\}$ inside L . For $a \in R^{X^\circ}$ put $\nu(a)_x = 1_L$ when $a_x \neq 0$ and 0_L otherwise. For a pointed map $f : X \rightarrow Y$ define

$$C_f(a)_y = 1_L \iff \sum_{x:f(x)=y} a_x = 0 \text{ and } \text{some } x \in X^\circ \text{ with } f(x) = y \text{ has } a_x \neq 0. \quad (\text{HG13})$$

otherwise put $C_f(a)_y = 0_L$. This defines an actual element of $HL(Y)$, including empty fibres and discarded coordinates. The exact failure-of-naturality equation is

$$HG(f)b_X(a) = b_Y(HR(f)a) + \iota_Y(C_f(a)). \quad (\text{HG14})$$

Indeed both amplitudes are the fibre sum. If that sum is nonzero, both supports are top. If it is zero, the left support is top exactly when at least one input amplitude is nonzero, which is precisely (HG13). In particular, for $a \neq 0$ the binary fold of $(b(a), b(-a))$ is e and its correction is z_{1_L} .

$$\begin{aligned} (a, -a) &\xrightarrow{b_{2+}} (\widehat{a}, \widehat{-a}) \xrightarrow{HG(\text{fold})} e = (0, 1_L), \\ (a, -a) &\xrightarrow{HR(\text{fold})} 0 \xrightarrow{b_{1+}} \tau = (0, 0_L). \end{aligned}$$

Figure 3: The two exact paths compared by naturality, for any $a \neq 0$ and $0_L \neq 1_L$. Their endpoints are different. Equation (HG14) supplies their full correction inside the embedded HL , and (HG15) computes its behavior under further folds. The fold operation is Connes–Consani’s **functsum**, source lines 433–442. This is a diagram of the calculated noncommutation, not a commuting square.

The correction itself has an exact composition rule. Put $Z(a)_x = 1_L$ when $a_x = 0$, and bottom otherwise. For $g : Y \rightarrow Z$ one has

$$C_{gf}(a) = C_g(HR(f)a) \vee (Z(HR(gf)a) \wedge HL(g)(C_f(a))), \quad (\text{HG15})$$

where meet and join are coordinatewise. To prove it, a final coordinate with nonzero amplitude has zero on both sides. At a final coordinate with zero amplitude, an original nonzero summand is either in an intermediate fibre whose amplitude is nonzero, or in an intermediate fibre that cancelled to zero. The first event is exactly the first term on the right, and the second is exactly its second term. This includes their possible simultaneous occurrence and proves (HG15).

There is also a complete multiplication correction for general rings with zero divisors:

$$\begin{aligned} D(a, b)_{x,y} = 1_L &\iff a_x \neq 0, b_y \neq 0, a_x b_y = 0, \\ \mu(b_X(a), b_Y(b)) &= b_{X \wedge Y}((a_x b_y)_{x,y}) + \iota_{X \wedge Y}(D(a, b)), \end{aligned} \quad (\text{HG16})$$

with $D = 0_L$ at all other coordinates. At a nonzero product both sides have top support; at a zero product only the case where both factors were nonzero contributes e . This proves the formula without a domain assumption.

5.4 The exact spherical replacement and universal ring image

Let $M(R)$ be the multiplicative monoid of R , pointed by its ring zero. The source’s spherical monoid algebra is $\mathbb{S}M(R)(X) = M(R) \wedge X$; see **mono2sss** and **monomono**, lines 489–511. The pointed family

$$\widetilde{b}_X : M(R) \wedge X \rightarrow HG(X), \quad (r, x) \mapsto [b(r) \text{ at } x, \tau \text{ elsewhere}] \quad (\text{HG17})$$

is a Γ -map for every R : an ordinary pointed map either retains the single selected coordinate, carrying its value with it, or discards it entirely. Its compatibility with the multiplication of spherical algebras is equivalent to $b(rs) = b(r)b(s)$ at level 1, hence holds exactly under the criterion in Theorem 18. It is a unital $\mathbb{S}$ -algebra map for a domain. When $0_L \neq 1_L$ it is not such a map for a ring with nonzero zero divisors; when L has one element it is the source's usual map $\mathbb{S}M(R) \rightarrow HR$ for every ring.

For every ring, including rings with zero divisors, the required multiplicative replacement retains the two zero states. Let

$$M^+(R) = R \sqcup \{\perp\}, \quad \perp m = m \perp = \perp, \quad (\text{HG18})$$

with the original ring product on R , identity 1_R , and distinguished pointed zero $\perp$. The original element $0_R \in R$ is not $\perp$. Define

$$c : M^+(R) \rightarrow G, \quad c(\perp) = \tau, \quad c(r) = \widehat{r}, \quad d : M^+(R) \rightarrow M(R), \quad d(\perp) = 0, \quad d(r) = r. \quad (\text{HG19})$$

Both are pointed unital multiplicative maps, including products of two nonzero zero divisors, because such a product lands at the original 0_R and hence at e under c . The source adjunction gives the actual commuting diagram

$$\begin{array}{ccc} \mathbb{S}M^+(R) & \xrightarrow{\tilde{c}} & HG \\ \text{\textsf{Sd}} \downarrow & & \downarrow Hq \\ \mathbb{S}M(R) & \xrightarrow{\rho_R} & HR. \end{array} \quad (\text{HG20})$$

Here ρ_R inserts one ring coefficient into its selected coordinate, and $\tilde{c}$ does the same with c . Their naturality follows from the single-coordinate argument just given; their multiplicativity and units follow from (HG19), so (HG20) is proved coordinatewise. This is a precise multiplicative replacement, with its correction state visible.

Amplitude also has a universal characterization. For any ring A , every unital semiring homomorphism $u : G \rightarrow A$ factors uniquely through $q : G \rightarrow R$. Indeed every z_λ is additively idempotent, so $u(z_\lambda) = 0$ in the additive group of A . Therefore $v(r) = u(\widehat{r})$ satisfies $v(0) = 0$, $v(1) = 1$, and preserves sums and products by (HG1), giving a ring map $v : R \rightarrow A$ with $u = vq$. More generally this same proof, omitting multiplication, shows that q is the universal map from the additive monoid of G into an abelian group. Thus

$$G_{\text{additive}}^{\text{gp}} \cong (R, +), \quad \text{Hom}_{\text{semiring}}(G, A) \cong \text{Hom}_{\text{ring}}(R, A). \quad (\text{HG21})$$

This explains exactly which coefficient information a ring-valued realization forgets: the full ideal (HG9). It does not provide a Γ -section in the reverse direction.

Proposition 19. *Degreewise additive Grothendieck group completion gives the exact natural isomorphism*

$$(HG(X))^{\text{gp}} \xrightarrow{\cong} HR(X), \quad [(r, \lambda)] \mapsto r. \quad (\text{HG21a})$$

It preserves all pointed-set maps and the bilinear multiplication after group completion. The additive congruence that effects this quotient is exactly

$$(r, \lambda) \sim (s, \mu) \iff r = s \iff (r, \lambda) + (e)_{x \in X^\circ} = (s, \mu) + (e)_{x \in X^\circ}. \quad (\text{HG21b})$$

Thus group completion identifies all support choices over each fixed amplitude tuple and makes no further identifications.

Proof. For a single coordinate, additive idempotence gives $[z_\lambda] = [z_\lambda] + [z_\lambda]$, hence $[z_\lambda] = 0$ in a group. The identity $\widehat{r} + \widehat{-r} = e$ gives $[\widehat{-r}] = -[\widehat{r}]$. The map $r \mapsto [\widehat{r}]$ is additive, because the sum of any two top-supported coefficients is the top-supported coefficient of their ring sum, including cancellation. Its value at zero is $[e] = 0$. It is inverse to the map induced by q . For finitely many coordinates, every tuple is the sum of its one-coordinate insertions; the same argument therefore gives the inverse $r \mapsto \sum_x [\widehat{r}_x \text{ at } x, \tau \text{ elsewhere}]$ and proves (HG21a) with its universal property. More explicitly, any additive map from G^{X° to an abelian group kills every tuple of labelled zeros and is determined uniquely by these coordinate classes, so factors uniquely through q^{X° .

Adding e at a coordinate replaces its support by 1_L without changing its amplitude. This proves (HG21b). Conversely the congruence must identify all labelled zeros because their classes vanish; its remaining quotient is already the group R^{X° by the displayed inverse, so no larger congruence is present.

Applying a pointed-set map to a tuple class replaces the amplitude at y by $\sum_{f(x)=y} r_x$ by (HG3), which is exactly $HR(f)$. The isomorphism is therefore natural on the entire category of finite pointed sets, not only on projections. The product on the additive completions is determined by its values on coefficient classes, where (HG4) becomes $r_x s_y$; its bilinear extension is precisely the usual product of HR . This proves the asserted compatibility. $\square$

The conclusion is the following pair of proved maps, with the operation on the right explicitly stated:

$$HL \xrightarrow{\iota} HG_L(R) \xrightarrow{Hq} HR \cong (HG_L(R))_{\text{degree-wise}}^{\text{gp}}. \quad (\text{HG21c})$$

It is not a claim that an arbitrary stable realization of $HG_L(R)$ is equivalent to the Eilenberg–Mac Lane spectrum of R . It is also not an unmentioned group completion of the bounded objects introduced next: those bounds need not be closed under coordinatewise addition.

5.5 The original seminorm constraint retains the full support ideal

Give R a submultiplicative seminorm with the exact axioms in Connes–Consani (`subverti1`), source lines 895–903: $|0| = 0$, $|1| = 1$, $|r + s| \leq |r| + |s|$, and $|rs| \leq |r||s|$. Define

$$|(r, \lambda)|_G = |r|. \quad (\text{HG22})$$

This is a submultiplicative seminorm on G , as each required inequality reduces to its amplitude counterpart. For $c > 0$ define the closed and strict functors

$$\begin{aligned} B_c^L(X) &= \{(r, \lambda) \in HG(X) : \sum_{x \in X^\circ} |r_x| \leq c\}, \\ B_{< c}^L(X) &= \{(r, \lambda) \in HG(X) : \sum_{x \in X^\circ} |r_x| < c\}. \end{aligned} \quad (\text{HG23})$$

Theorem 20. B_1^L is a unital $\mathbb{S}$ -subalgebra of HG . Both functors in (HG23) are modules over B_1^L and the multiplication maps restrict to

$$B_c^L \wedge B_d^L \rightarrow B_{cd}^L, \quad B_{< c}^L \wedge B_d^L \rightarrow B_{< cd}^L \quad (c, d > 0). \quad (\text{HG24})$$

Under amplitude the fibres are still exactly (HG8), for every amplitude tuple obeying its bound. Under support the fibre at λ is exactly

$$\{(r_x)_{x: \lambda_x = 1_L} : \sum_{x: \lambda_x = 1_L} |r_x| \leq c\}, \quad (\text{HG25})$$

with the corresponding strict inequality for $B_{< c}^L$. The entire HL embeds into every one of these functors, and is precisely the zero-amplitude fibre. In particular every support label has zero seminorm cost.

Proof. The fibre-sum inequality in the source becomes, with all supports retained,

$$\sum_{y \in Y^\circ} \left| \sum_{x: f(x)=y} r_x \right| \leq \sum_{x: f(x) \neq *} |r_x| \leq \sum_{x \in X^\circ} |r_x|. \quad (\text{HG26})$$

Thus (HG3) preserves both bounds. The exact support joins remain those of (HG3), since no inequality has changed any coefficient. Likewise

$$\sum_{x,y} |r_x s_y| \leq \left(\sum_x |r_x| \right) \left(\sum_y |s_y| \right) \quad (\text{HG27})$$

proves (HG24), including the strict case since $d > 0$. Unit tuples have seminorm sum 1, so are in B_1^L . The module identities are the restrictions of the already proved algebra identities. Since (HG23) depends only on amplitude, it removes none of the support choices over any allowed tuple. Substituting the support fibre from (HG8) gives (HG25). Zero tuples have total seminorm zero, so all of HL remains in every positive closed or strict bound. $\square$

Let $N = \{r \in R : |r| = 0\}$. The triangle and product inequalities make N an ideal: closure under additive inverses follows from $|-r| \leq |-1||r| = 0$. At the closed zero bound the exact object is

$$B_0^L(X) = \{(r, \lambda) \in HG(X) : r_x \in N \text{ for all } x\}. \quad (\text{HG28})$$

It equals HL precisely when $N = \{0\}$. This distinction prevents a degenerate seminorm from silently erasing nonzero amplitudes. The strict unit ball is not unital, because the unit tuple has seminorm sum exactly 1; it still retains every element of HL .

Theorem 21. *Assume $0_L \neq 1_L$ and let $B_c(R)$ denote the source's classical amplitude functor with the same bound $c > 0$. A Γ -section of $q : B_c^L \rightarrow B_c(R)$ exists exactly when*

$$|r| + |-r| > c \quad \text{for every nonzero } r \in R. \quad (\text{HG34})$$

When it exists it is unique and is the coordinatewise map b_X . For strict balls $B_{<c}$ the exact criterion is $|r| + |-r| \geq c$ for all nonzero r . At $c = 1$, an existing Γ -section is a unital $\mathbb{S}$ -algebra section exactly when, in addition,

$$r, s \neq 0, \quad |r| \leq 1, \quad |s| \leq 1 \implies rs \neq 0. \quad (\text{HG35})$$

Proof. Projection to any coordinate is a map of the bounded functors, because discarding terms cannot increase the sum of seminorms. Thus any section is determined coordinatewise by its value at 1_+ . A nonzero amplitude has only one preimage in G , and pointedness forces the zero amplitude to go to τ . Therefore b_X is the only possible section.

If $r \neq 0$ and $|r| + |-r| \leq c$, the tuple $(r, -r)$ is admissible; its two paths in the figure following (HG14) have endpoints e and τ . Naturality fails. Conversely, suppose (HG34). A nonzero correction $C_f(a)_y$ for an admissible tuple would give a zero-sum fibre with some nonzero summand a_i . In the ring, $-a_i = \sum_{x \in I_y \setminus \{i\}} a_x$. The triangle inequality gives

$$|a_i| + |-a_i| \leq \sum_{x \in I_y} |a_x| \leq \sum_{x \in X^\circ} |a_x| \leq c,$$

contrary to (HG34). Hence every correction vanishes and (HG14) proves naturality. The strict case has the same proof with the final strict inequality; its failed pair condition is correspondingly $|r| + |-r| < c$.

Unit preservation holds for the coordinatewise b , and (HG16) shows that multiplication is preserved if (HG35) holds, because all individual amplitudes in a unit-ball tuple have seminorm at most 1. If (HG35) fails, place the two nonzero zero divisors at level 1. Each is allowed by the unit bound and their product produces e instead of τ , proving failure of multiplication. This proves both directions of the criterion. $\square$

5.6 The sheaf calculation on the source's compactified base

The indexing space here is the source's $X = \overline{\text{Spec } \mathbb{Z}}$. Its topology is defined in `sectstructsh`, lines 938–946. This section constructs a coefficient sheaf on that specified space; no identification with a prime spectrum of $G_L(\mathbb{Z})$ is used. For a nonempty open U put

$$R_U = \{r \in \mathbb{Q} : v_p(r) \geq 0 \text{ for every finite prime } p \in U\}. \quad (\text{HG29})$$

Define a sheaf candidate, with empty-open value the one-point algebra, by

$$\mathcal{A}_L(U)(K) = \begin{cases} HG_L(R_U)(K), & \infty \notin U, \\ \{(r, \lambda) \in HG_L(R_U)(K) : \sum_{k \in K^\circ} |r_k| \leq 1\}, & \infty \in U. \end{cases} \quad (\text{HG30})$$

The absolute value is the usual one on $\mathbb{Q}$. The restriction map for nonempty $V \subset U$ includes R_U into R_V and retains each support coordinate unchanged.

Theorem 22. *Equation (HG30) is a sheaf of unital $\mathbb{S}$ -algebras, with amplitude morphism to the exact Connes–Consani sheaf $\mathcal{O}_X$ of `(subsalg)` and Proposition `propspecz`, lines 944–977. Its amplitude-zero sheaf is the constant HL on nonempty opens, including opens containing ∞ . Its stalks are*

$$\begin{aligned} (\mathcal{A}_L)_\eta &= HG_L(\mathbb{Q}), \\ (\mathcal{A}_L)_p &= HG_L(\mathbb{Z}_{(p)}) \quad (p \text{ finite}), \\ (\mathcal{A}_L)_\infty &= B_1^L(\mathbb{Q}). \end{aligned} \quad (\text{HG31})$$

All these identities retain the full lattice L .

Proof. Every value in (HG30) is a unital algebra by Theorem 15 or Theorem 20; the restrictions are the stated algebra inclusions. Every pair of nonempty opens intersects in the generic point. Thus compatible local sections over a cover have literally the same rational amplitudes and the same support coordinates in the ambient $HG_L(\mathbb{Q})(K)$. An amplitude integral at all primes in each covering open is integral at every prime of their union. If the union contains ∞ , one covering open does too, and imposes the norm bound on that common tuple. The common tuple therefore belongs to (HG30) on the union, and is its unique gluing. This proves the sheaf condition at every K , with multiplication and units already compatible. Amplitude removes precisely the support coordinates and gives the source sheaf formulas. Its zero fibre is HL on every nonempty open, with identity restriction maps.

At the generic point one can take opens excluding ∞ and any finite set of denominator primes, so every rational tuple occurs. At a finite prime p the same procedure allows all denominator primes other than p and excludes ∞ , giving exactly $\mathbb{Z}_{(p)}$. At ∞ every open retains the norm bound, but any finite collection of rational denominators is allowed after removing its finite prime divisors. Hence the union of these bounded sets is exactly $B_1^L(\mathbb{Q})(K)$ at each K . The restrictions never change supports, so the stalk descriptions include all of L . $\square$

The global sections at a finite pointed set k_+ are especially explicit. Since $R_X = \mathbb{Z}$, the inequality $\sum_{i=1}^k |r_i| \leq 1$ allows either the zero amplitude tuple or a single coordinate with value $+1$ or -1 . Consequently

$$\mathcal{A}_L(X)(k_+) = L^k \sqcup \coprod_{i=1}^k \coprod_{\varepsilon \in \{+1, -1\}} \{r_i = \varepsilon, \lambda_i = 1_L, r_j = 0, \lambda_j \in L \ (j \neq i)\}. \quad (\text{HG32})$$

These are disjoint subsets; the first is the entire zero-amplitude fibre, whose own pointed zero is the all-bottom tuple. If L is finite of cardinality N , then for $k \geq 1$

$$|\mathcal{A}_L(X)(k_+)| = N^k + 2kN^{k-1}. \quad (\text{HG33})$$

For $k = 0$ there is one section. At level 1, the global sections are all z_λ together with $\widehat{1}$ and $\widehat{-1}$, so their number is $N + 2$ when L is finite. This is a direct application of the original unit bound, with no added positivity assumption.

Proposition 23. *For $0_L \neq 1_L$, the global-sections amplitude map*

$$\mathcal{A}_L(X) \longrightarrow \mathcal{O}_X(X) \quad (\text{HG36})$$

has a unique unital $\mathbb{S}$ -algebra section, given by b in every coordinate. Nevertheless the sheaf amplitude map $\mathcal{A}_L \rightarrow \mathcal{O}_X$ has no section even as a sheaf of Γ -sets. Already on the single open $U = X \setminus \{2\}$ there is no Γ -section. There is likewise no Γ -section at the infinity stalk.

Proof. For a nonzero integer r , $|r| + |-r| = 2|r| \geq 2 > 1$, and $\mathbb{Z}$ has no nonzero zero divisors. Theorems 20 and 21 therefore prove the unique algebra section on global sections. Its image has all unused coordinates equal to τ , leaving the other labelled-zero points outside this chosen section.

On $U = X \setminus \{2\}$ the coefficient ring is $\mathbb{Z}[1/2]$ and the archimedean bound is still exactly 1. Thus $(1/2, -1/2)$ is an allowed tuple and

$$HG(\text{fold})(\widehat{1/2}, \widehat{-1/2}) = e, \quad b(HR(\text{fold})(1/2, -1/2)) = \tau. \quad (\text{HG37})$$

Pointedness and the coordinate projections force any putative section to use those two lifts, so there is no alternative choice that could repair this failure. A section of sheaves would evaluate to a section on U , which is impossible. The same allowed tuple in $B_1(\mathbb{Q})$, the source infinity stalk from (HG31), proves its asserted failure directly. $\square$

For an arbitrary bound $c > 0$, the same proof replaces (HG32) by all integer vectors satisfying $\sum_i |r_i| \leq c$, each with fibre $L^{\{i:r_i=0\}}$. For $0 < c < 1$ the entire global bounded object is exactly HL ; at the strict bound $c = 1$ it is again exactly HL . At a rational or complex archimedean coefficient ring, a pair $r, -r$ with $r \neq 0$ and $2|r| \leq 1$ is allowed by the unit ball. Folding its two top-supported coordinates gives e , whereas folding the zero tuple gives τ . Thus cancellation-labelled zero survives the actual archimedean restriction at precisely the original norm bound.

The complete retained object is therefore (HG7) with (HG3), (HG4), and (HG23), and its nontrivial zero-amplitude ideal is (HG9)–(HG10). The original seminorm does not force that ideal to vanish. Ring-valued realization removes it by (HG21), while the Γ -algebra and every positive archimedean bound retain it explicitly.

5.7 Exact comparison with the later all-subset Riemann–Roch module

The following second source uses a different, explicitly stated module: Alain Connes and Caterina Consani, *Riemann–Roch for the ring $\mathbb{Z}$* , arXiv:2306.00456, original `RRZ.tex`, equation `normdefn`, Lemma `xlem`, and equation `xdefn`, source lines 185–208. That source records the older sum-of-absolute-values bound and then defines the new module by all subset sums. These definitions give an exact inclusion without identifying the two modules or changing either original bound.

For an integer $m \geq 0$ set

$$E_{L,m}(k_+) = \{((a_i, \lambda_i)) \in HG_L(\mathbb{Z})(k_+) : -m \leq \sum_{i \in I} a_i \leq m \quad \text{for every } I \subseteq \{1, \dots, k\}\}. \quad (\text{HG38})$$

This is the same complete support lift as (SRR2) in the accompanying programme calculation, here defined in full so that the comparison needs none of its dimension claims. An output subset under a pointed map has as preimage a subset of input coordinates, so (HG3) preserves (HG38). Thus it is a Γ -subfunctor. Its amplitude version is denoted E_m .

Proposition 24. For $P(a) = \sum_{a_i > 0} a_i$ and $N(a) = \sum_{a_i < 0} (-a_i)$, the two exact bounds are

$$\begin{aligned} (a, \lambda) \in B_m^L(\mathbb{Z})(k_+) &\iff P(a) + N(a) \leq m, \\ (a, \lambda) \in E_{L,m}(k_+) &\iff P(a) \leq m \text{ and } N(a) \leq m. \end{aligned} \tag{HG39}$$

There are natural inclusions preserving every amplitude and support coordinate:

$$B_m^L(\mathbb{Z}) \hookrightarrow E_{L,m} \hookrightarrow B_{2m}^L(\mathbb{Z}). \tag{HG40}$$

For $m \geq 1$ both inclusions are strict; for $m = 0$ all three objects are exactly HL . The amplitude diagrams for these inclusions commute, and their zero-amplitude fibres are identically HL , with no quotient of the lattice labels.

Proof. The first equivalence is the identity $\sum_i |a_i| = P(a) + N(a)$. Every subset sum is between $-N(a)$ and $P(a)$, and these endpoints are attained by taking all negative or all positive coordinates respectively. This proves the second equivalence, including empty subsets, and is the calculation in the source's subsequent `xdefnr`. The numerical implications $P + N \leq m \Rightarrow (P \leq m, N \leq m) \Rightarrow P + N \leq 2m$ prove (HG40). All three use exactly the same support constraint (HG2) and the same pushforward (HG3), proving naturality and compatibility with amplitude.

The tuple $(\widehat{m}, \widehat{-m})$ belongs to $E_{L,m}$ but has absolute-value sum $2m > m$, so is outside the first ball. The one-coordinate tuple $\widehat{2m}$ belongs to the second ball but its singleton subset sum exceeds m , so is outside $E_{L,m}$. At $m = 0$, each individual coordinate is a subset sum, hence every amplitude is zero; all support tuples remain, giving HL . The same observation proves the assertion about zero-amplitude fibres for every m . $\square$

Proposition 25. For $0_L \neq 1_L$ and every $m \geq 1$, the map $E_{L,m} \rightarrow E_m$ has no Γ -section. In particular, the unique global Γ -algebra section on the older $B_1(\mathbb{Z})$ from (HG36) does not extend to the later E_1 module, even though their level-1 point sets agree.

Proof. Pointed coordinate projections are available in the all-subset module. Therefore any section must send zero to τ and every nonzero coordinate a to its unique lift $\widehat{a}$, exactly as in the proof of Theorem 21. For every $m \geq 1$, the tuple $(1, -1)$ belongs to $E_m(2_+)$: its subset sums are $0, 1, -1, 0$, all in $[-m, m]$. The required naturality equation would be

$$e = HG(\text{fold})(\widehat{1}, \widehat{-1}) = b(E_m(\text{fold})(1, -1)) = b(0) = \tau, \tag{HG41}$$

which is false for $0_L \neq 1_L$. At $m = 1$ the tuple $(1, -1)$ is exactly an additional tuple supplied by the first strict inclusion in (HG40); it was disallowed by the older absolute-value sum bound. At level 1 both bounds require only $|a| \leq 1$, so their point sets do agree there. The failure therefore occurs in their actual higher-level fold structures, as calculated, and does not change any coordinate or bound. $\square$

6 The full support contribution to the bounded section module

Alain Connes and Caterina Consani, *Riemann–Roch for the ring $\mathbb{Z}$* , arXiv:2306.00456, define $(HA)_X$ by requiring every partial sum to belong to X (Lemma `xlem`, equation `xdefn`). Their Definition `generator` defines dimension over the spherical base $\mathbb{S}$ by the smallest number of distinct generators, with coefficients in $\{0, 1\}$ and with the stated tolerance relation. We apply those exact definitions to the additive monoid of the full Split-Zero semiring. The following calculation concerns that explicitly constructed module. It does not presume a new definition of H^1 or a Riemann–Roch identity for an unspecified cohomology theory.

6.1 All levels, all fold maps, and the amplitude fibre

Let L be a bounded distributive lattice with $0_L \neq 1_L$. Its full carrier and operations are those of *Split Support Geometry*, version 11, by The Clankers, Definition `def:lattice-split`. Keep

$$M = G_L(\mathbb{Z}) = \{z_\lambda = (0, \lambda) : \lambda \in L\} \cup \{\hat{a} = (a, 1_L) : a \in \mathbb{Z}\}, \quad \tau = z_{0_L}, \quad e = z_{1_L} = \hat{0}. \quad (\text{SRR1})$$

The operations remain $(a, \lambda) + (b, \mu) = (a + b, \lambda \vee \mu)$ and $(a, \lambda)(b, \mu) = (ab, \lambda \wedge \mu)$. The additive identity is τ , and the amplitude map is $q(a, \lambda) = a$. Fix an integer $m \geq 0$ and put $X_m = q^{-1}([-m, m] \cap \mathbb{Z})$.

For the finite pointed set $r_+ = \{*, 1, \dots, r\}$ define

$$\mathcal{E}_{L,m}(r_+) = \left\{ (x_1, \dots, x_r) \in M^r : \sum_{j \in Z} x_j \in X_m \text{ for every } Z \subseteq \{1, \dots, r\} \right\}. \quad (\text{SRR2})$$

An empty sum is τ . A pointed map $f : r_+ \rightarrow s_+$ acts by

$$(x_j = (a_j, \lambda_j))_{j=1}^r \mapsto \left(\sum_{f(j)=i} a_j, \bigvee_{f(j)=i} \lambda_j \right)_{i=1}^s. \quad (\text{SRR3})$$

An empty join is 0_L . The pointed value in every level is the tuple whose entries are all τ . For a subset W of the target indices, the sum of the resulting entries is the sum over the disjoint union $\bigcup_{i \in W} f^{-1}(i)$. Thus (SRR3) preserves (SRR2). Associativity of finite sums and joins proves composition, and the identity map acts identically. These are all the structure maps of the resulting Γ -set, or $\mathbb{S}$ -module.

Write $\mathcal{E}_m = (H\mathbb{Z})_{[-m, m]}$, with the same all-subset definition used by Connes–Consani. Its full relation to (SRR2) is

$$\begin{aligned} \mathcal{E}_{L,m}(r_+) &= \{((a_j), (\lambda_j)) \in \mathcal{E}_m(r_+) \times L^r : a_j \neq 0 \implies \lambda_j = 1_L\}, \\ q_r : \mathcal{E}_{L,m}(r_+) &\longrightarrow \mathcal{E}_m(r_+), \quad ((a_j), (\lambda_j)) \longmapsto (a_j). \end{aligned} \quad (\text{SRR4})$$

Indeed the amplitude of a subset sum is its integer sum, independently of its retained join support. This proves the equality. The maps q_r are surjective: assign 1_L to every nonzero coordinate and 0_L to every zero coordinate. Formula (SRR3) proves naturality. This last choice is a set-theoretic section in each level. For $m \geq 1$ no pointed natural section exists: the only possible lifts of 1 and -1 are $\hat{1}$ and $\widehat{-1}$. Naturality for the two coordinate projections forces their two-coordinate lift to be that pair. It is admissible, but its fold is e , while pointedness forces the lift of the integer zero to be τ . Since $e \neq \tau$, fold naturality fails. This proves the exact obstruction to a section of (SRR4).

For a fixed integer tuple $a \in \mathcal{E}_m(r_+)$, its fibre is exactly $L^{\{j:a_j=0\}}$: all the other supports are forced to be 1_L . In particular

$$q_r^{-1}(0) = \{(z_{\lambda_1}, \dots, z_{\lambda_r}) : (\lambda_j) \in L^r\}. \quad (\text{SRR5})$$

The maps induced on this fibre are finite joins along f -fibres. It is therefore exactly the Γ -set $H(L, \vee, 0_L)$, with its explicit inclusion into $\mathcal{E}_{L,m}$. No support label is identified at this stage.

6.2 The original dimension definition on this module

Give $\mathcal{E}_{L,m}(1_+) = X_m$ the equality tolerance relation. For a finite set $F \subset X_m$ and a subset $Z \subseteq F$, the Connes–Consani sum $\sum_Z x$ exists precisely when every partial sum of Z has amplitude in $[-m, m]$. This follows because the projection maps $\delta(j, r)$ in their equation `defnsum` force the tuple in (SRR2) to be exactly the listed summands, while the fold map is (SRR3). With equality

tolerance, F generates if and only if every $x \in X_m$ is one of these admissible sums. The empty set generates only τ .

Let $J(L)$ denote the set of nonzero join-irreducible elements: $j \neq 0_L$ is in $J(L)$ if $j = u \vee v$ implies $j = u$ or $j = v$. Put $J_{<}(L) = J(L) \setminus \{1_L\}$. Every element of a finite lattice is a join of join-irreducible elements below it. To prove this, induct on the number of elements below x . The bottom is the empty join. If x is join-irreducible there is nothing further to prove; otherwise $x = u \vee v$ with both $u, v < x$, and the induction hypothesis on those two smaller lower sets gives the required finite join. If a finite join equals a join-irreducible j , at least one of its terms equals j , by repeated application of its defining binary property.

Theorem 26. *For finite L , the dimension defined above is exactly*

$$\dim_{\mathbb{S}} \mathcal{E}_{L,m} = \begin{cases} |J(L)|, & m = 0, \\ n + 1 + |J_{<}(L)|, & 2^{n-1} \leq m < 2^n, \quad m \geq 1. \end{cases} \quad (\text{SRR6})$$

In the second line $n \geq 1$ is the unique integer with the displayed inequalities. In particular the number $n + 1$ is the unchanged Connes–Consani value for $\mathcal{E}_m$; the additional term is the exact contribution of the proper nonzero join-irreducible supports.

Proof. When $m = 0$, every point is some z_λ and every tuple is admissible. Sums are exactly joins. The preceding finite-lattice argument proves that the points z_j for $j \in J(L)$ generate, and that every generating set must contain each one. This proves the first line.

Now let $m \geq 1$. Partition any generating set into its nonzero amplitude elements and its zero amplitude elements. A sum containing a nonzero amplitude summand has support 1_L , even when its amplitudes cancel. Thus any representation of z_j , for $j \in J_{<}(L)$, uses only zero amplitude summands. Their labels have join j , so some summand must be z_j itself. Every such generator is therefore compulsory.

Project the nonzero amplitude generators to a subset $A \subset [-m, m] \setminus \{0\}$. This is a bijection on those generators, because nonzero amplitudes have unique lifts. For each integer $a \in [-m, m]$, project an admissible representation of $\hat{a}$ and delete its zero amplitude terms. Every partial sum of the remaining terms is still in $[-m, m]$, so A generates $\mathcal{E}_m$. The empty subset represents integer zero. Connes–Consani, Lemma [sequencebleft\(ii\)](#), proves that the least number of such integer generators is $n + 1$. Their counting lower bound can also be seen directly here: $2^{|A|}$ subsets must represent $2m + 1 > 2^n$ different integers, hence $|A| \geq n + 1$. Together with the compulsory distinct zero generators this proves the lower bound in (SRR6).

For the upper bound take an optimal integer generating set A_m from that same lemma. It can be chosen to admit a nonempty admissible zero sum. We verify this extra property, which is needed to generate the supported zero e rather than just integer zero. For $n > 4$ the source construction is $A_m = \{-m\} \cup \{\alpha_1, \dots, \alpha_n\}$, where the distinct α_i are positive, sum to m , and have subset sums covering $[0, m]$. The full set has sum zero, and every partial sum lies in $[-m, m]$, since the positive total is m and its only negative summand is $-m$. This proves the required property in that range, using the exact source construction.

For the remaining values the following are the source’s sets, with their values of m stated explicitly:

m	A_m	m	A_m
1	$\{-1, 1\}$	2	$\{-2, 1, 2\}$
3	$\{-3, 1, 2\}$	4	$\{-3, -1, 1, 3\}$
5	$\{-4, -1, 2, 3\}$	6	$\{-6, 1, 2, 3\}$
7	$\{-7, 1, 2, 4\}$	8	$\{-7, -1, 1, 2, 5\}$
9	$\{-8, -1, 1, 3, 5\}$	10	$\{-10, 1, 2, 3, 4\}$
11	$\{-11, 1, 2, 3, 5\}$	12	$\{-12, 1, 2, 3, 6\}$
13	$\{-13, 1, 2, 3, 7\}$	14	$\{-14, 1, 2, 4, 7\}$
15	$\{-15, 1, 2, 4, 8\}$		

(SRR7)

Except at $m = 2$, each whole set has sum zero, positive total m and negative total $-m$; hence its full zero sum is admissible. At $m = 2$, the subset $\{-2, 2\}$ is the required admissible zero sum. The source proves these sets generate their respective integer intervals with the stated admissibility condition.

Lift A_m to $\hat{A}_m \subset M$ and adjoin $\{z_j : j \in J_<(L)\}$. All nonzero amplitude targets are generated by the lifted admissible integer sums. Every proper support zero is generated by its join-irreducible decomposition, and τ is the empty sum. Finally the nonempty admissible zero sum just verified generates e , since all its summands have top support. Thus this set generates every point of X_m and has exactly $n + 1 + |J_<(L)|$ elements. This attains the lower bound. $\square$

For infinite L , the construction and exact maps (SRR1)–(SRR5) remain unchanged, and no finite set generates $\mathcal{E}_{L,m}$ for any $m \geq 0$. Indeed a generating set of size t has at most 2^t subset sums, whereas the target contains the distinct points z_λ for every $\lambda \in L$. If one extends the minimum-cardinality definition to arbitrary generating sets with finitely supported sums, its value is exactly $|L|$: an infinite set of size κ has κ finite subsets, forcing $\kappa \geq |L|$, and the full point set X_m has cardinality $|L|$ and generates by its singletons. This states the cardinal convention explicitly; the failure of finite generation needs no such extension of the convention.

6.3 Exact consequences for the stated archimedean bound

Retain a real parameter $\lambda > 0$ and the original bound $[-\lambda, \lambda] \cap \mathbb{Z}$. Put $m = \lfloor \lambda \rfloor$; this equality identifies precisely the same subset of integers and does not rescale its metric or change its endpoints. Define $\mathcal{E}_{L,\lambda}$ by (SRR2) using that interval. Then

$$\dim_{\mathbb{S}} \mathcal{E}_{L,\lambda} = \begin{cases} |J(L)|, & 0 < \lambda < 1, \\ n + 1 + |J_<(L)|, & 2^{n-1} \leq \lambda < 2^n, \quad \lambda \geq 1. \end{cases} \quad (\text{SRR8})$$

For the second line, since each 2^{n-1} and 2^n is an integer, the displayed inequalities on λ are equivalent to the same inequalities on m . Formula (SRR6) proves (SRR8). For the particular divisor $D = \delta\{\infty\}$ used in the source’s calculation, keep $\lambda = e^\delta$ exactly. Thus the full-support module over its integer sections has dimension $|J(L)|$ for $\delta < 0$, and $n + 1 + |J_<(L)|$ for $(n - 1) \log 2 \leq \delta < n \log 2$. This is a calculation of the specified module and its surjection (SRR4), with the entire support fibre (SRR5) retained.

For the two-element lattice, $J(L) = \{1_L\}$: the dimension is one when $0 < \lambda < 1$ and equals the original value $n + 1$ for $\lambda \geq 1$. For a chain with four elements $0_L < u < v < 1_L$, $J(L) = \{u, v, 1_L\}$: the two respective values are 3 and $n + 3$. For the Boolean lattice of subsets of $\{u, v\}$, $J(L) = \{\{u\}, \{v\}\}$ and its top is not join-irreducible: the two values are 2 and $n + 3$. These follow from the definition of join-irreducibility; they show why retaining only the two-element support lattice would miss the exact dimension contribution.

7 Divisor multiplication and degree with the full lattice fibre

The source construction is Alain Connes and Caterina Consani, *Absolute algebra and Segal’s Γ -sets*, original `F1_corr.tex`, §5.2: Proposition `propspecz`, equations `propsmash1`, `propsmash2`, and Proposition `proppic`, source lines 938–1053. Their divisor modules and exceptional multiplication rule are the source results. Here we calculate the rule for the full lattice carrier and replace the global-section test by an intrinsic test that retains every labelled zero. The split carrier is from The Clankers, *Split Support Geometry*, v11, Definition `def:lattice-split`.

7.1 The original divisor and its entire support lift

Fix a nontrivial bounded distributive lattice L and write $z_\ell = (0, \ell)$, $\tau = z_{0_L}$, $e = z_{1_L}$, and $\hat{x} = (x, 1_L)$ for $x \neq 0$. Retain the original divisor

$$D = \sum_p n_p[p] + a[\infty], \quad n_p \in \mathbb{Z}, \quad n_p = 0 \text{ for all but finitely many } p, \quad a \in \mathbb{R}. \quad (\text{SDM1})$$

The underlying space is the source's $X = \overline{\text{Spec } \mathbb{Z}}$, whose nonempty opens are complements of finite sets of closed primes, possibly including ∞ . This section constructs sheaves on that specified space; it does not identify it with $\text{Spec } G_L(\mathbb{Z})$. For nonempty U define

$$\begin{aligned} R_U &= \{x \in \mathbb{Q} : v_p(x) \geq 0 \text{ for finite } p \in U\}, \\ I_D(U) &= \{x \in \mathbb{Q} : v_p(x) + n_p \geq 0 \text{ for finite } p \in U\}, \\ G_L(I_D(U)) &= \{(0, \ell) : \ell \in L\} \cup \{(x, 1_L) : x \in I_D(U)\}. \end{aligned} \quad (\text{SDM2})$$

Here $v_p(0) = +\infty$. Addition joins the supports, and the scalar action of $G_L(R_U)$ multiplies amplitudes and meets supports. For a finite pointed set K , with $K^\circ = K \setminus \{*\}$, put

$$\mathcal{M}_L(D)(U)(K) = \left\{ (x_j, \ell_j)_{j \in K^\circ} \in G_L(I_D(U))^{K^\circ} : \infty \in U \implies \sum_j |x_j| \leq e^a \right\}. \quad (\text{SDM3})$$

The empty-open value is the singleton. A pointed map sums the amplitudes and joins the supports over each nonbasepoint fibre; an empty fibre has value τ . Restriction includes the rational amplitudes and leaves every support unchanged. Write $\mathcal{A}_L = \mathcal{M}_L(0)$. Also define $\mathcal{M}_L^<(D)$ by replacing $\leq e^a$ by $< e^a$ on opens containing ∞ , with no change elsewhere.

Theorem 27. *These are sheaves of $\mathcal{A}_L$ -modules. The amplitude-zero subsheaf is HL on every nonempty open, for both inequalities. Their stalks are*

$$\begin{aligned} \mathcal{M}_L(D)_\eta &= HG_L(\mathbb{Q}), \\ \mathcal{M}_L(D)_p &= HG_L(p^{-n_p} \mathbb{Z}_{(p)}), \\ \mathcal{M}_L(D)_\infty &= B_{e^a}^L(\mathbb{Q}). \end{aligned} \quad (\text{SDM4})$$

For the strict sheaf the last stalk is $B_{<e^a}^L(\mathbb{Q})$. The notation H on an additive module denotes its finite-tuple Γ -set; it does not assert that $I_D(U)$ is a ring.

Proof. The valuation inequalities make $I_D(U)$ an R_U -module. Addition and scalar action on the displayed carrier satisfy their axioms by distributivity in L and in $\mathbb{Q}$. A nonzero product has both input supports equal to 1_L , as required. The triangle inequality proves closure under pointed maps, including maps that discard input coordinates. For the external module action the coordinates are $(r_i x_j, \lambda_i \wedge \ell_j)$; its bound is $\sum_{i,j} |r_i x_j| = (\sum_i |r_i|)(\sum_j |x_j|) \leq e^a$, and is strict when the second bound is strict. The valuation inequalities add. This proves the module identities coordinatewise.

Every two nonempty opens intersect. Compatible local sections therefore have exactly the same rational amplitudes and lattice labels. All valuation conditions required on the union hold on a covering open; if the union contains infinity, one such open supplies the required bound. This proves existence and uniqueness of gluing. Every zero-amplitude tuple satisfies either positive bound, proving the subsheaf assertion. At a finite prime p , all other primes q where one of the finitely many inequalities $v_q(x_j) + n_q \geq 0$ fails can be excluded, together with infinity. This is a finite set: it includes denominator primes and may also include primes with negative divisor coefficient. At infinity every such finite prime can be excluded, while the original bound remains. At the generic point every such finite prime and infinity can be excluded. Taking these unions gives (SDM4) without changing any label. $\square$

7.2 The exact tensor law, including its boundary

Let $D' = \sum_p n'_p[p] + a'[\infty]$, and set $\lambda = e^a$, $\mu = e^{a'}$. Multiplication defines

$$m_{D,D'} : \mathcal{M}_L(D) \wedge_{\mathcal{A}_L} \mathcal{M}_L(D') \longrightarrow \mathcal{M}_L(D + D'), \quad (x_i, \ell_i), (y_j, t_j) \longmapsto (x_i y_j, \ell_i \wedge t_j). \quad (\text{SDM5})$$

The relative smash product is the sheafification of the module coequalizer for the Day smash product. The displayed product is balanced and respects all pointed maps by the two distributive laws.

Theorem 28. *The map (SDM5) is injective as a map of sheaves and identifies its source with*

$$\begin{cases} \mathcal{M}_L^<(D + D'), & \lambda, \mu \notin \mathbb{Q} \text{ and } \lambda\mu \in \mathbb{Q}, \\ \mathcal{M}_L(D + D'), & \text{otherwise.} \end{cases} \quad (\text{SDM6})$$

In particular its image contains the whole HL in every case. The exceptional missing stalk tuples are exactly those at infinity whose rational amplitudes satisfy $\sum_j |x_j| = \lambda\mu$; every permissible choice of support on their zero coordinates is missing with them. The finite stalks have no such defect.

Proof. We give the module calculation, rather than infer it from level one. At a finite prime, multiplication by the rational p^{-n_p} gives an isomorphism from $HG_L(\mathbb{Z}_{(p)})$ to the second object in (SDM4). The map multiplies every amplitude and fixes every label; its inverse multiplies by p^{n_p} . The same holds for D' . The unit identity $A \wedge_A A \simeq A$ now proves the claimed isomorphism on that stalk. At the generic point both modules are the coefficient module itself.

At infinity put $A = B_1^L(\mathbb{Q})$. For each rational $r > 0$,

$$s_r : A \xrightarrow{\sim} B_r^L(\mathbb{Q}), \quad (x_j, \ell_j)_j \longmapsto (rx_j, \ell_j)_j \quad (\text{SDM7})$$

is an A -module isomorphism, with inverse $s_{1/r}$ on the indicated codomain. For $0 < r \leq t$ the inclusion $B_r^L \subset B_t^L$ corresponds under these isomorphisms to multiplication by $\widehat{r/t}$ in $A(1_+)$. Thus the multiplication map induces $B_r^L \wedge_A B_s^L \simeq B_{rs}^L$ and the transition maps are exactly the coordinatewise inclusions, including all zero labels. For completeness, the inverse of the balanced-unit multiplication is $A \simeq A \wedge \mathbb{S} \rightarrow A \wedge A \rightarrow A \wedge_A A$, using the unit $\mathbb{S} \rightarrow A$ in the middle map. One composite is the identity by the unit law; the other is the identity because the balanced relation transfers the first A -factor to the second. A tuple $(z_{\ell_j})_j$ is represented by that tuple paired with $\widehat{1}$ and is recovered with supports $\ell_j \wedge 1_L = \ell_j$.

For any real $c > 0$ the full bounded module is the filtered union

$$B_c^L(\mathbb{Q}) = \operatorname{colim}_{r \in \mathbb{Q}, 0 < r \leq c} B_r^L(\mathbb{Q}). \quad (\text{SDM8})$$

Indeed each tuple has rational total absolute amplitude. If that total is positive it may be used as r ; if it is zero any rational $0 < r \leq c$ works, retaining its arbitrary lattice labels. The relative smash product preserves colimits in each variable: the Day smash is a left adjoint in each variable, and the defining balanced coequalizer commutes with colimits. Hence the product at infinity is the union of B_{rs}^L over positive rational $r \leq \lambda$, $s \leq \mu$. A tuple of rational total amplitude $t < \lambda\mu$ belongs to this union by choosing r, s sufficiently close to their upper endpoints. The same is true for $t = 0$. The endpoint $t = \lambda\mu$ is attained precisely when both endpoints λ, μ are rational: since $r/\lambda, s/\mu \leq 1$, their product is one only when both equal one. If exactly one endpoint is irrational the product is irrational, so no rational tuple total equals it. If both are irrational and their product is irrational, the same observation applies. This proves (SDM6) at infinity. All these are injections into the same tuple module.

Stalks commute with the tensor construction used here. To verify this directly, the stalk is a filtered colimit over neighborhoods; each Day representative uses finite pointed sets and finitely

many sections, so all its entries occur on a common neighborhood. Every defining relation between representatives also occurs on a common neighborhood. The same finite-representative argument applies to the balanced coequalizer. Sheafification leaves stalks unchanged. Consequently the stalk computations prove the sheaf statement. $\square$

7.3 Recovering degree without deleting labelled zeros

For a pointed Γ -set M , call $x \in M(1_+)$ additively idempotent when there exists $w \in M(2_+)$ with

$$M(\pi_1)w = x = M(\pi_2)w, \quad M(\nabla)w = x, \quad (\text{SDM9})$$

where π_1, π_2 are the two pointed coordinate projections and ∇ is the fold. This property uses only the given Γ -structure and is preserved by isomorphisms.

Lemma 29. *For every nonempty open in (SDM3), its additively idempotent level-one elements are exactly $\{z_\ell : \ell \in L\}$, for either inequality. The same holds for the image sheaf in (SDM6).*

Proof. The two projections force $w = (x, x)$. If $x = (u, \ell)$, the fold equation says $2u = u$ and $\ell \vee \ell = \ell$. In $\mathbb{Q}$ the first equality is equivalent to $u = 0$. Conversely (z_ℓ, z_ℓ) always satisfies the valuation conditions and either positive bound, and its fold is z_ℓ . The image description (SDM6) supplies the last assertion. $\square$

Keep all the coefficients of (SDM1), and define

$$c_D = \prod_p p^{-n_p} > 0, \quad \deg D = a + \sum_p n_p \log p = a - \log c_D. \quad (\text{SDM10})$$

These formulas do not replace D by another divisor. The actual global amplitude module $I_D(X)$ is $c_D \mathbb{Z}$: for a rational x , all $v_p(x/c_D) \geq 0$ hold exactly when $x/c_D \in \mathbb{Z}$. For $D_b = b[\infty]$, set $\mathcal{T}_b = \mathcal{M}_L(D) \wedge_{\mathcal{A}_L} \mathcal{M}_L(D_b)$. Define the following invariant of an abstract $\mathcal{A}_L$ -module sheaf of the displayed form:

$$\alpha_L(\mathcal{M}_L(D)) = \inf \{b \in \mathbb{R} : \mathcal{T}_b(X)(1_+) \text{ contains a non-idempotent element}\}. \quad (\text{SDM11})$$

Theorem 30. *One has the exact formula and classification*

$$\alpha_L(\mathcal{M}_L(D)) = \log c_D - a = -\deg D, \quad \mathcal{M}_L(D) \simeq \mathcal{M}_L(D') \iff D' - D \text{ is principal.} \quad (\text{SDM12})$$

The invariant in (SDM11) uses the complete support sheaf and no quotient of its zero fibre.

Proof. By (SDM6), a non-idempotent global section is a nonzero element of $c_D \mathbb{Z}$ with $|x| \leq e^{a+b}$, with strict inequality in the exceptional case. Its least possible absolute value is exactly c_D . Thus it does not exist for $b < \log c_D - a$ and does exist for every $b > \log c_D - a$. At equality it exists precisely when the tensor law is not exceptional, a distinction that leaves the infimum unchanged. More precisely, the set in (SDM11) is $[-\deg D, \infty)$ when $e^a \in \mathbb{Q}$ and $(-\deg D, \infty)$ otherwise: at equality the product of the two radius bounds is $c_D \in \mathbb{Q}$, and $e^b = c_D/e^a$ is rational exactly when e^a is rational. This proves the first formula with the endpoint fully accounted for. An isomorphism of module sheaves induces an isomorphism after tensoring with $\mathcal{M}_L(D_b)$; it preserves (SDM9), and therefore preserves this infimum. It follows that isomorphic sheaves have $\deg D = \deg D'$.

Put $q = \prod_p p^{n'_p - n_p} \in \mathbb{Q}_{>0}$. Equality of degrees gives $a' - a = -\log q$, while $v_p(q) = n'_p - n_p$ for every p . Thus $D' = D + (q)$ with the original principal-divisor convention $(q) = \sum_p v_p(q)[p] - \log |q|[\infty]$. Conversely, for any $q \in \mathbb{Q}^\times$ and $D' = D + (q)$, the explicit map

$$\mathcal{M}_L(D)(U)(K) \longrightarrow \mathcal{M}_L(D')(U)(K), \quad (x_j, \ell_j)_j \longmapsto (q^{-1}x_j, \ell_j)_j \quad (\text{SDM13})$$

is an isomorphism. Indeed $v_p(q^{-1}x_j) + n_p + v_p(q) = v_p(x_j) + n_p$, and the new archimedean bound is $e^a/|q|$. Its inverse multiplies by q . It commutes with all restrictions, pointed maps and module actions, and fixes every z_ℓ . This proves both implications. $\square$

For nontrivial L , replacing “non-idempotent” in (SDM11) by “different from the pointed zero” would give $-\infty$, since $e \neq \tau$ is always present. This is the precise defect of transporting the source’s nonzero-section test without its lattice fibre. The intrinsic test proves (SDM12) while preserving that fibre. The tensor calculation concerns the original sum-of-absolute-values divisor sheaves. It does not identify them with the later all-subset modules of *Riemann–Roch for the ring $\mathbb{Z}$* ; their exact comparison is (HG38)–(HG41). No weight or cohomological purity statement is inferred from these module calculations.

7.4 All abstract divisor-module morphisms and their full zero fibre

Retain the original divisors, lattice, space, coefficient sheaf and modules of SDM1–4. In particular, no coefficient of either divisor is changed. The divisors are

$$D = \sum_p n_p[p] + a[\infty], \quad D' = \sum_p n'_p[p] + a'[\infty].$$

The construction of these modules is that of Alain Connes and Caterina Consani, *Absolute algebra and Segal’s Γ -sets*, arXiv:1502.05585v2, Propositions **propspecz** and **proppic**, with the full lattice lift calculated in SDM1–13. The following proof determines every morphism in this lifted construction, including the maps with zero amplitude.

Define the subset of the original generic scalar semiring $G_L(\mathbb{Q})$ by

$$C_L(D, D') = \{z_\ell : \ell \in L\} \cup \left\{ \widehat{r} : r \in \mathbb{Q}^\times, \quad v_p(r) + n'_p - n_p \geq 0 \text{ for every finite } p, \quad |r| \leq e^{a'-a} \right\}. \quad (\text{SDM14})$$

It is a set of permitted scalars; no closure under addition at the archimedean bound is asserted. For every such scalar, define the coordinatewise map f_u on every nonempty open and every finite pointed set by multiplication by u . Its exact coordinate formulas are

$$\begin{aligned} f_{\widehat{r}}(x, \mu) &= (rx, \mu), \\ f_{z_\ell}(x, \mu) &= (0, \ell \wedge \mu). \end{aligned} \quad (\text{SDM15})$$

In particular the supported-zero scalar e gives $(x, \mu) \mapsto (0, \mu)$, whereas τ gives the constant map to the actual pointed zero. Both occur and are distinct.

Theorem 31. *Multiplication gives the canonical bijection of pointed sets*

$$\begin{aligned} C_L(D, D') &\xrightarrow{\sim} \text{Hom}_{\mathcal{A}_L}(\mathcal{M}_L(D), \mathcal{M}_L(D')), \\ u &\longmapsto f_u. \end{aligned} \quad (\text{SDM16})$$

$$C_L(D, D') = \mathcal{M}_L(D' - D)(X)(1_+).$$

The equality on the right uses the full labelled-zero fibre. For a third original divisor D'' and $v \in C_L(D', D'')$, the exact composition law is

$$f_v \circ f_u = f_{vu}. \quad (\text{SDM17})$$

Thus multiplication of zero scalars meets their lattice labels; it does not collapse them to the actual zero.

Proof. First prove that every permitted scalar defines the displayed module morphism. For $u = \widehat{r}$ and a local amplitude x satisfying the source valuation inequalities, one has, for every finite p in the open,

$$v_p(rx) + n'_p = (v_p(x) + n_p) + (v_p(r) + n'_p - n_p) \geq 0.$$

On an open containing infinity, its finite tuple satisfies

$$\sum_j |rx_j| = |r| \sum_j |x_j| \leq |r|e^a \leq e^{a'}.$$

These are the original inequalities, with their constants retained. For $u = z_\ell$, every output amplitude is zero, so every valuation and positive archimedean bound is satisfied. Formula SDM15 gives its support exactly.

Multiplication by a fixed scalar commutes with all pointed pushforwards. For the zero scalar this uses the explicit identity

$$\ell \wedge \bigvee_{j \in F} \mu_j = \bigvee_{j \in F} (\ell \wedge \mu_j)$$

for every finite fibre F , including the empty fibre; its empty join is 0_L . For the nonzero scalar it uses distributivity of rational multiplication over each amplitude sum, with unchanged support joins. The maps commute with restrictions because those are inclusions of the same amplitudes and labels. They commute with the external $\mathcal{A}_L$ -module action by associativity and commutativity of scalar multiplication and lattice meet. They therefore define morphisms of the required sheaves of modules.

We now prove that these are every abstract morphism; no preservation of an amplitude projection or of an individual zero label is assumed. Let

$$f : \mathcal{M}_L(D) \longrightarrow \mathcal{M}_L(D')$$

be any morphism of $\mathcal{A}_L$ -module sheaves. The canonical generic-stalk identifications in SDM4 identify its source, target and coefficient algebra with $A_\eta = HG_L(\mathbb{Q})$. Write $u = f_\eta(1_+)(\widehat{1})$, where the input is the actual multiplicative unit $\widehat{1} = (1, 1_L)$ of that generic module. For every finite pointed set K and every $x \in A_\eta(K)$, the external action of x on the module unit at 1_+ returns the same tuple x , using $K \wedge 1_+ \simeq K$. Module compatibility therefore gives

$$f_\eta(K)(x) = x u.$$

This proves the scalar formula at every finite arity, not only at level one.

For every nonempty open U , the map from $\mathcal{M}_L(D')(U)(K)$ to its generic stalk is injective: all restriction maps are coordinatewise inclusions in the common tuple carrier $G_L(\mathbb{Q})^{K^\circ}$, and the stalk is their filtered union. Naturality with respect to these maps now forces $f_U(K)$ to be the restriction of multiplication by that same generic scalar u . This proves uniqueness once its permitted values have been determined.

Every element of $G_L(\mathbb{Q})$ is either z_ℓ for a unique $\ell \in L$, or $\widehat{r}$ for a unique nonzero rational r . The zero-amplitude elements have already been shown to work without any further inequality. Suppose $u = \widehat{r}$. At the finite stalk p , apply the map to the literal input $\widehat{p^{-n_p}}$. It belongs to the source module $HG_L(p^{-n_p}\mathbb{Z}_{(p)})$. Its output must belong to $HG_L(p^{-n'_p}\mathbb{Z}_{(p)})$, so

$$v_p(r) - n_p \geq -n'_p.$$

This is exactly the finite inequality in SDM14.

At infinity, every positive rational $t \leq e^a$ is a permitted level-one amplitude. If $|r|e^a > e^{a'}$, density of the rationals provides

$$\frac{e^{a'}}{|r|} < t < e^a.$$

Its image has absolute amplitude greater than $e^{a'}$, a contradiction. Hence $|r| \leq e^{a'-a}$ is necessary, including the original endpoint. The sufficiency of precisely these inequalities was proved at the start. This proves the bijection. Its basepoint is preserved because τ acts as the actual constant-zero morphism.

The equality with global sections in SDM16 is now obtained by substituting the coefficients $n'_p - n_p$ and $a' - a$ of the actual difference divisor into SDM2–3. The same full zero lattice appears on both sides. Finally, composition of the constructed maps is scalar multiplication by vu . If both amplitudes are nonzero, their finite inequalities add and their archimedean bounds multiply. If either amplitude is zero, the product is its exact lattice-zero scalar. In either case $vu \in C_L(D, D'')$, proving SDM17. $\square$

Corollary 32. *Every abstract isomorphism of the displayed $\mathcal{A}_L$ -module sheaves is the scaling in SDM13 and fixes each labelled zero. More precisely the isomorphisms are exactly*

$$\begin{aligned} f_{\widehat{r}} \quad (r \in \mathbb{Q}^\times), \quad n'_p - n_p = -v_p(r) \quad \text{for every } p, \quad a' - a = \log |r|, \\ D' = D + (r^{-1}), \quad f_{\widehat{r}}(z_\ell) = z_\ell \quad (\ell \in L). \end{aligned} \quad (\text{SDM18})$$

For every original D , its complete endomorphism monoid and automorphism group are

$$\begin{aligned} \text{End}_{\mathcal{A}_L}(\mathcal{M}_L(D)) &\simeq \{z_\ell : \ell \in L\} \cup \{\widehat{1}, \widehat{-1}\}, \\ \text{Aut}_{\mathcal{A}_L}(\mathcal{M}_L(D)) &= \{f_{\widehat{1}}, f_{\widehat{-1}}\}. \end{aligned} \quad (\text{SDM19})$$

The first isomorphism retains multiplication of the original split scalars; it does not assert an additive ring structure on this bounded set.

Proof. An inverse pair induces inverse generic scalar multiplications, by SDM16–17. A zero-amplitude scalar cannot be invertible, since its product with any scalar has zero amplitude. A nonzero generic scalar $\widehat{r}$ has the unique inverse $\widehat{r^{-1}}$ in $G_L(\mathbb{Q})$. Applying the finite inequalities of SDM14 to the map and its inverse yields

$$v_p(r) + n'_p - n_p \geq 0, \quad -v_p(r) + n_p - n'_p \geq 0.$$

Their left sides sum to zero, so both equal zero. Applying the two archimedean inequalities gives

$$|r| \leq e^{a'-a}, \quad |r|^{-1} \leq e^{a-a'}.$$

The second is $|r| \geq e^{a'-a}$; hence equality holds. These are exactly the conditions in SDM18. Conversely those equalities put both scalars in the required sets SDM14, so SDM17 proves that they are inverse sheaf maps. The principal-divisor convention in SDM12 gives

$$(r^{-1}) = \sum_p (-v_p(r)) [p] + \log |r| [\infty],$$

which proves the claimed identity of divisors with the signs unchanged. Formula SDM15 proves fixation of every z_ℓ .

For $D' = D$, the nonzero scalars in SDM14 are rational numbers with every finite valuation nonnegative and with $|r| \leq 1$. Such a rational number is an integer: a reduced denominator greater than one would have a prime divisor giving a negative valuation. The only nonzero integers of absolute value at most one are 1 and -1 . All zero scalars remain allowed. This proves the endomorphism assertion and, by the preceding inverse calculation, its exact automorphism subgroup. $\square$

7.5 The complete category of the original divisors

Define a category Div_L retaining each original divisor D of SDM1 as an object. Its morphisms, composition and identities are the specified data

$$\begin{aligned} \text{Hom}_{\text{Div}_L}(D, D') &= \mathcal{M}_L(D' - D)(X)(1_+) = C_L(D, D'), \\ (v, u) &\longmapsto v \circ u := vu, \quad \text{id}_D = \widehat{1}. \end{aligned} \quad (\text{SDM20})$$

The product on the second line is the actual split multiplication: $\widehat{s}\widehat{r} = \widehat{s}r$, $\widehat{r}z_\ell = z_\ell\widehat{r} = z_\ell$ for $r \neq 0$, and $z_\ell z_\mu = z_{\ell \wedge \mu}$. Every z_ℓ is a morphism between every ordered pair of divisors; no such label has been suppressed.

These data satisfy the category axioms. Closure of composition is the scalar inequality calculation in the proof of SDM17. The identity scalar belongs to $C_L(D, D)$ by SDM14, since its finite valuations are zero and its absolute value is one. Associativity and the two identity laws follow directly from associativity and the unit of $G_L(\mathbb{Q})$. The distinguished zero morphism is τ ; its products on both sides are τ . Other z_ℓ remain separate morphisms and have the meet-composition law just displayed. No additive closure or ring enrichment is asserted: already in $\text{End}_{\text{Div}_L}(D)$, the amplitude sum $1 + 1 = 2$ fails the archimedean bound in SDM19.

Let ModDiv_L be the full subcategory of $\mathcal{A}_L$ -module sheaves with objects the displayed $\mathcal{M}_L(D)$ for original divisors D . Then

$$\begin{aligned} \mathcal{F} : \text{Div}_L &\longrightarrow \text{ModDiv}_L, \\ D &\longmapsto \mathcal{M}_L(D), \\ u &\longmapsto f_u \end{aligned} \tag{SDM21}$$

is fully faithful and essentially surjective. Indeed SDM17 proves preservation of composition and SDM15 proves preservation of identities. The bijection SDM16 is exactly the bijection on every Hom set required for full faithfulness. Every object of ModDiv_L is the image of its displayed original divisor, proving essential surjectivity. If the full subcategory is enlarged to contain all module sheaves isomorphic to those displayed objects, the same argument proves essential surjectivity onto that enlargement as well: its defining isomorphism is an isomorphism to an image of $\mathcal{F}$.

Consequently the original divisor category, with its complete lattice of zero-amplitude morphisms, is equivalent to the specified full divisor-module subcategory. The functor retains the original divisor coefficients and takes each z_ℓ to its exact action $(x, \mu) \mapsto (0, \ell \wedge \mu)$. Its isomorphisms and automorphisms are precisely SDM18–19; passing to this category does not identify the labelled zero morphisms or assume an additional purity property.

8 Full-lattice unit orbits and the Cartier valuation sheaf

The human source for the Cartier quotient and its arithmetic-site comparison is Alain Connes and Caterina Consani, *Geometry of the arithmetic site*, arXiv:1502.05580v1, original author source `arithmetic-site_Adv_final1.tex`, Lemma `hyper`, source lines 987–1001, and Proposition `sheavesonspz`, source lines 1003–1024, in the subsection identifying the structure sheaf with Cartier divisors on the cyclotomic spectrum. Their quotient $\mathcal{K}/\mathcal{O}^\times$, normalization $v(p) = p - 1$ and cyclotomic value group are retained. The local equal-value sum requires a residue-field correction, proved below. The full split carrier and its two kinds of prime stalks are those of The Clankers, *Split Support Geometry*, version 11, Definition `def:lattice-split` and its prime-localization theorems, calculated explicitly in FSC2–9 of this reader. No purity hypothesis is made in this construction.

8.1 The genuine unit quotient

Let K be a field and $v : K^\times \rightarrow \Gamma$ a surjective valuation to an ordered abelian group, written additively, with $v(0) = +\infty$. Write

$$\begin{aligned} \mathcal{O}_v &= \{x : v(x) \geq 0\}, & \mathfrak{m}_v &= \{x : v(x) > 0\}, \\ \kappa_v &= \mathcal{O}_v / \mathfrak{m}_v, & U_v &= \mathcal{O}_v^\times = \{u : v(u) = 0\}. \end{aligned} \tag{FCQ1}$$

The trivial valuation is permitted. Fix the original nontrivial bounded distributive lattice L and retain

$$\begin{aligned} S = G_L(K) &= \{z_\ell = (0, \ell) : \ell \in L\} \cup \{\widehat{x} = (x, 1_L) : x \in K^\times\}, \\ \tau &= z_{0_L}, \quad e = z_{1_L}, \\ (x, \ell) + (y, m) &= (x + y, \ell \vee m), \quad (x, \ell)(y, m) = (xy, \ell \wedge m). \end{aligned} \tag{FCQ2}$$

The notation for a zero-amplitude pair in the last line always means its displayed z -label. The actual multiplicative units of $G_L(\mathcal{O}_v)$ are precisely $\widehat{u}$ with $u \in U_v$: an invertible element must have nonzero amplitude, and its amplitude and inverse must both belong to $\mathcal{O}_v$. Conversely $\widehat{u}^{-1} = \widehat{u^{-1}}$ verifies the assertion.

These units act by multiplication on S . Their action is

$$\widehat{u}\widehat{x} = \widehat{ux}, \quad \widehat{u}z_\ell = z_\ell. \tag{FCQ3}$$

It fixes every labelled zero. Multiplication by a unit is an additive automorphism; it is not being called a unital semiring automorphism, since it need not fix the multiplicative unit. Let $\mathcal{Q}_L(K, v)$ be its orbit set. Nonzero amplitudes have the same orbit exactly when their valuations agree: $v(x) = v(y)$ is equivalent to $x/y \in U_v$. Thus the exact orbit set is

$$\mathcal{Q}_L(K, v) = \{z_\ell : \ell \in L\} \sqcup \{\mathbf{a} : a \in \Gamma\}, \quad \mathbf{a} = [\widehat{x}] \iff a = -v(x). \tag{FCQ4}$$

In particular the symbol $\mathbf{0}$ is the nonzero-amplitude unit orbit. It is neither τ nor e .

Define orbit multiplication and multivalued addition by

$$[x][y] = [xy], \quad [x] \boxplus [y] = \{\widehat{u}x + \widehat{w}y : u, w \in U_v\}. \tag{FCQ5}$$

Changing either representative changes neither set: absorb its unit into u or w . Multiplication is well defined because the product of two such unit factors is a unit. The operations on arbitrary subsets below mean union over their elements.

8.2 The complete addition table and the residue-field correction

The operations retaining every lattice element are

$$\begin{aligned} z_\ell z_m &= z_{\ell \wedge m}, & z_\ell \mathbf{a} &= z_\ell, & \mathbf{a}\mathbf{b} &= \mathbf{a} + \mathbf{b}, \\ z_\ell \boxplus z_m &= \{z_{\ell \vee m}\}, & z_\ell \boxplus \mathbf{a} &= \{\mathbf{a}\}, \\ \mathbf{a} \boxplus \mathbf{b} &= \{\max(\mathbf{a}, \mathbf{b})\} \quad (a \neq b). \end{aligned} \tag{FCQ6}$$

For the equal-value case, the exact formula is

$$\mathbf{a} \boxplus \mathbf{a} = \begin{cases} \{e\} \cup \{\mathbf{b} : b < a\}, & |\kappa_v| = 2, \\ \{e\} \cup \{\mathbf{b} : b \leq a\}, & |\kappa_v| > 2. \end{cases} \tag{FCQ7}$$

The set $b < a$ may be empty for the trivial valuation.

Proof. The first two lines of FCQ6 follow by the original split laws. For different valuations the nonarchimedean equality $v(x + y) = \min(v(x), v(y))$ gives its last line for every choice of unit representatives.

For equal values, divide both representatives by an element of valuation $-a$. The possible sums reduce to sums of two units. Their valuations are nonnegative, or $+\infty$. Every strictly positive value $\gamma \in \Gamma$ occurs: choose $t \in K^\times$ with $v(t) = \gamma$, and write $t = 1 + (t - 1)$. Both 1 and $t - 1$ are units. Exact cancellation is $1 + (-1) = 0$. In the split carrier, this cancellation has support $1_L \vee 1_L = 1_L$, and hence gives e , never τ .

A sum of two units has valuation zero exactly when the sum of their two nonzero residues is nonzero. Over $\mathbf{F}_2$ the only two nonzero residues are both 1, whose sum is zero, so valuation zero cannot occur. If $|\kappa_v| > 2$, choose a nonzero residue different from -1 and lift it to a unit u . Then $1 + u$ has nonzero residue and valuation zero. Multiplying all these constructions by the chosen element of valuation $-a$ proves both directions of FCQ7. $\square$

This corrects the generality of Lemma **hyper** in the source. For $K = \mathbb{Q}$ at the prime 2, with its required scale $v(2) = 2 - 1 = 1$, the units have odd numerator and odd denominator in reduced form. The sum of two units has even numerator and odd denominator and cannot be a unit. Consequently

$$\mathbf{0} \boxplus \mathbf{0} = \{e, -1, -2, \dots\}, \quad \mathbf{0} \notin \mathbf{0} \boxplus \mathbf{0}. \quad (\text{FCQ8})$$

Every displayed negative value occurs by $1 + (2^j - 1) = 2^j$, $j \geq 1$, and e occurs by cancellation. Replacing e by the ordinary zero gives the same counterexample to the source's claimed equal-value interval including its upper endpoint. The correction concerns that finite-stage lemma, not an assertion that the cyclotomic endpoint fails.

8.3 The exact multivalued algebra and its projections

The orbit object is a commutative associative multivalued additive monoid with identity τ . Its multiplication is a commutative monoid with identity $\mathbf{0}$ and absorber τ , and it distributes over the multivalued addition with equality of sets:

$$c(a \boxplus b) = ca \boxplus cb. \quad (\text{FCQ9})$$

Here these axioms specify the resulting multivalued algebra; no additive-inverse axiom is tacitly included.

Proof. Commutativity follows from FCQ5. An iterated three-term orbit sum on either parenthesization is exactly

$$\{\widehat{u}x + \widehat{w}y + \widehat{t}z : u, w, t \in U_v\}.$$

To verify this assertion, a representative of an intermediate orbit differs from its defining two-term sum by a unit, which multiplies both corresponding coefficients. Conversely one may take that extra unit to be 1. This proves associativity including every possible zero-amplitude intermediate sum. The identity equation follows because a unit multiple of τ is τ and addition by τ is the original identity. The multiplicative monoid laws descend from FCQ2. For distributivity, fixing a representative c , both sides of FCQ9 are exactly

$$\{[c(\widehat{u}x + \widehat{w}y)] : u, w \in U_v\} = \{[\widehat{u}cx + \widehat{w}cy] : u, w \in U_v\}.$$

This uses original distributivity and proves equality, not merely one inclusion. $\square$

It is not an ordinary hyperring with zero τ . Indeed $\tau \in a \boxplus b$ forces both amplitudes to vanish and their support join to be 0_L , so $a = b = \tau$. Thus no other element has an additive inverse to τ . The zero-amplitude part $Z_L = \{z_\ell : \ell \in L\}$ is closed under addition and absorbs multiplication into itself. Its addition and multiplication are exactly join and meet.

There is nevertheless an exact ordinary hyperfield inside this object, with its different additive zero:

$$\mathcal{H}(K, v) = \{e\} \sqcup \{\mathbf{a} : a \in \Gamma\} \cong K/U_v. \quad (\text{FCQ10})$$

On this subset, e is the additive identity and the multiplicative absorber. Every $\mathbf{a}$ has multiplicative inverse $-\mathbf{a}$. Since $-1 \in U_v$, each is its own additive opposite, with e in its self-sum; FCQ6 shows that no other value can be its opposite. For completeness, reversibility of this hyperaddition follows directly from a field equation. The assertion $[c] \in [a] \boxplus [b]$ in K/U_v means $wc = ua + tb$ for some $u, t, w \in U_v$. Solving for a gives $a = u^{-1}wc - u^{-1}tb$, which is precisely $[a] \in [c] \boxplus [-b]$. Solving back gives the converse. Associativity and distributivity were already proved. This verifies all the hyperfield axioms of FCQ10 without assuming the incorrect residue-independent addition formula.

Write $0_{\mathcal{H}}$ for e when it is regarded as the zero of FCQ10. There are exact projections

$$\begin{aligned} \pi : \mathcal{Q}_L(K, v) &\longrightarrow \mathcal{H}(K, v), & \pi(z_\ell) &= 0_{\mathcal{H}}, & \pi(\mathbf{a}) &= \mathbf{a}, \\ \sigma : \mathcal{Q}_L(K, v) &\longrightarrow L, & \sigma(z_\ell) &= \ell, & \sigma(\mathbf{a}) &= 1_L. \end{aligned} \quad (\text{FCQ11})$$

Both preserve multiplication and their respective units and zeros. For addition they preserve the complete image sets: $\pi(a \boxplus b) = \pi(a) \boxplus \pi(b)$ and $\sigma(a \boxplus b) = \{\sigma(a) \vee \sigma(b)\}$. These statements follow in each case from FCQ6–7. The second formula retains the support of cancellation.

The two maps together are injective and have precisely the following image:

$$\mathcal{Q}_L(K, v) \xrightarrow{\sim} \{(h, \ell) \in \mathcal{H}(K, v) \times L : h \neq 0_{\mathcal{H}} \Rightarrow \ell = 1_L\}. \quad (\text{FCQ12})$$

The inverse sends $(0_{\mathcal{H}}, \ell)$ to z_ℓ and $(\mathbf{a}, 1_L)$ to $\mathbf{a}$. Under this bijection the sum is

$$(h, \ell) \boxplus (h', m) = \{(t, \ell \vee m) : t \in h \boxplus h'\},$$

and the product is $(hh', \ell \wedge m)$. These sets obey the displayed admissibility condition: if a sum has nonzero amplitude at least one input has nonzero amplitude and hence top support. A product of nonzero amplitudes has both supports top. Thus FCQ12 is an exact construction of the full object from its ordinary Cartier hyperfield and its complete support lattice, not an identification of their distinct zero elements.

The lattice inclusion $\ell \mapsto z_\ell$ preserves join, meet and the actual zero, but sends the lattice unit to e , not to $\mathbf{0}$. The inclusion of FCQ10 preserves its full operations and its multiplicative unit, but sends its additive zero to e , not to τ . Their composites with the corresponding projections are identities. The original orbit map $q : G_L(K) \rightarrow \mathcal{Q}_L(K, v)$ preserves products exactly and satisfies $q(x + y) \in q(x) \boxplus q(y)$; its unit-saturated image of sums is exactly FCQ5. The commutative square

$$\begin{array}{ccc} G_L(K) & \longrightarrow & \mathcal{Q}_L(K, v) \\ \downarrow \text{amplitude} & & \downarrow \pi \\ K & \longrightarrow & K/U_v \end{array} \quad (\text{FCQ13})$$

uses the ordinary field quotient on the bottom row.

Every bounded lattice homomorphism $\epsilon : L \rightarrow L'$ induces the exact operation-preserving map $z_\ell \mapsto z_{\epsilon(\ell)}$, $\mathbf{a} \mapsto \mathbf{a}$. It commutes with both projections in FCQ11 and with FCQ13. In particular this gives all Boolean-branch maps, without replacing the original L by a chosen branch.

8.4 The actual valuation scale and the divisor modules

At a prime $\wp$ of a number field K above the rational prime p , retain the source's scale $v_\wp(p) = p - 1$. If $e_\wp$ is its ramification index and $\text{ord}_\wp$ takes value one at a uniformizer, then

$$v_\wp = \alpha_\wp \text{ord}_\wp, \quad \alpha_\wp = \frac{p-1}{e_\wp}, \quad \Gamma_\wp = \alpha_\wp \mathbb{Z}. \quad (\text{FCQ14})$$

This follows because $\text{ord}_\wp(p) = e_\wp$ and the required scale fixes the multiplier. If the residue degree is $f_\wp$, the residue field has $p^{f_\wp}$ elements. The second line of FCQ7 applies except exactly when $p = 2$ and $f_\wp = 1$; in that case its first line applies. All intervals in the source's negative-valuation coordinate are therefore intervals in the actual group $\alpha_\wp \mathbb{Z}$, with the specified endpoint rule.

For a discrete valuation, let ϖ be a uniformizer and $\alpha = v(\varpi) > 0$. The original fractional ideal $\varpi^{-n} \mathcal{O}_v$ and its full split lift have the following exact orbit set under U_v :

$$G_L(\varpi^{-n} \mathcal{O}_v)/U_v = \{z_\ell : \ell \in L\} \sqcup \{\mathbf{a} : a \in \Gamma, a \leq n\alpha\}. \quad (\text{FCQ15})$$

Indeed membership of a nonzero amplitude is $v(x) \geq -n\alpha$, which is exactly the displayed inequality for $a = -v(x)$. Every such value occurs by the definition of the value group. The

whole zero lattice occurs and is fixed by the unit action. The addition table FCQ6–7 preserves this bound, and multiplication by the coefficient quotient, whose nonzero values satisfy $a \leq 0$, preserves it as well.

For the finite stalk at a rational prime in SDM4, $I_D = p^{-n_p} \mathbb{Z}_{(p)}$, the formula is

$$G_L(I_D)/\mathbb{Z}_{(p)}^\times = \{z_\ell : \ell \in L\} \sqcup \{\mathbf{a} : a \in (p-1)\mathbb{Z}, a \leq (p-1)n_p\}. \quad (\text{FCQ16})$$

The coefficient n_p has not been replaced: the multiplier $p-1$ records the source's valuation scale. For the principal change $D' = D + (q)$, the actual map $x \mapsto q^{-1}x$ in SDM13 sends

$$\mathbf{a} \mapsto \mathbf{a} + \mathbf{v}_p(\mathbf{q}), \quad z_\ell \mapsto z_\ell, \quad v_p(q) = (p-1) \text{ord}_p(q).$$

Its target bound is $(p-1)n'_p = (p-1)n_p + v_p(q)$, exactly the translated source bound. This proves the local Cartier receiver of the original divisor-module isomorphism, with all signs and all support labels retained.

8.5 Why the cyclotomic stalk has the full closed interval

Let $K^{\text{cyc}} = \mathbb{Q}^{\text{cyc}}$ be the union of the cyclotomic number fields, and let v extend the p -adic valuation with $v(p) = p-1$. Let $\mathcal{O}_v$ now be its valuation ring in this full field. Its residue field is

$$\kappa_v = \overline{\mathbb{F}}_p. \quad (\text{FCQ17})$$

Here the equality means an isomorphism inside a fixed algebraic residue-field closure.

Proof. Every residue of an algebraic element of K^{cyc} is algebraic over $\mathbb{F}_p$, so the residue field is an algebraic extension of $\mathbb{F}_p$. For every $f \geq 1$ the field contains all roots of unity of order $n = p^f - 1$. Their reductions are distinct. To prove this, if a nontrivial n -th root ζ reduced to 1, then the equation $1 + \zeta + \dots + \zeta^{n-1} = 0$ would reduce to $n = 0$ in characteristic p , contrary to $p \nmid n$. Applying this to the ratio of any two roots proves distinctness. Their n distinct nonzero reductions are all the roots of $X^n - 1$; together with zero they are the roots of $X^{p^f} - X$. In characteristic p this root set is closed under addition, subtraction, multiplication and inversion of nonzero elements, because the p^f -power map is a field homomorphism. It is therefore a copy of $\mathbb{F}_{p^f}$ inside the residue field. Taking every f proves FCQ17. $\square$

In particular this residue field always has more than two elements. At $p = 2$, the correction is already visible after adjoining a primitive cube root of unity: its residue lies in $\mathbb{F}_4 \setminus \mathbb{F}_2$, and its sum with 1 is nonzero. This supplies the equal-value endpoint missing over the finite field $\mathbb{F}_2$.

The source's cyclotomic value-group calculation, with its scale unchanged, is

$$v((K^{\text{cyc}})^\times) = H(p) = \mathbb{Z}[1/p]. \quad (\text{FCQ18})$$

Its local-field input, explicitly used in the proof of Proposition **sheavesonspz**, is that the local cyclotomic extension is obtained by adjoining the unramified roots of unity of order prime to p and the totally ramified p -power roots of unity. The source cites Jean-Pierre Serre, *Corps locaux*, second edition (1968), for this local-field statement. We retain that cited input rather than assert an unreferenced new ramification theorem. Its consequences for the value group can be checked with the exact constants as follows.

For a primitive p^m -th root ζ , $\Phi_{p^m}(1+T)$ is Eisenstein at p : its constant term is p , and its reduction modulo p is $T^{(p-1)p^{m-1}}$. The latter identity follows from $\Phi_{p^m}(X) = \sum_{j=0}^{p-1} X^{jp^{m-1}}$ by reducing in characteristic p and substituting $X = 1+T$. The valuation of $\pi = \zeta - 1$ is positive, since a p -power root of unity reduces to 1 in residue characteristic p . In its Eisenstein equation, all intermediate terms then have value strictly greater than $v(p)$ if the leading term has value $v(p)$; more directly, equality of the minimal values in that equation forces

$$((p-1)p^{m-1})v(\zeta - 1) = v(p) = p-1.$$

Indeed, if the leading value were smaller than $v(p)$, it would be the unique minimum, since an intermediate coefficient has value at least $v(p)$; if it were larger, the constant term would be the unique minimum. Thus $v(\zeta - 1) = p^{-(m-1)}$. The unramified extensions do not enlarge the value group; the finite totally ramified cyclotomic stage has index $(p-1)p^{m-1}$, so its group is exactly $p^{-(m-1)}\mathbb{Z}$. Taking their union proves FCQ18. This is also why replacing $v(p)$ by 1 without recording the multiplier would give the wrong displayed group.

Combining FCQ7, FCQ17 and FCQ18 proves the full cyclotomic stalk, rather than importing an assumed tropical addition:

$$\begin{aligned}\mathcal{Q}_L(K^{\text{cyc}}, v) &= \{z_\ell : \ell \in L\} \sqcup \{\mathbf{a} : a \in \mathbb{Z}[1/p]\}, \\ \mathbf{a} \boxplus \mathbf{a} &= \{e\} \cup \{\mathbf{b} : b \in \mathbb{Z}[1/p], b \leq a\}.\end{aligned}\tag{FCQ19}$$

Collapsing the full zero lattice by π gives precisely the source's $H(p)_{\text{hmax}}$, with its ordinary zero written $-\infty$. The same closed interval is not valid at every finite residue- $\mathbf{F}_2$ stage, as FCQ8 shows.

For an extension of valued fields with the same stated scale, the induced map of quotient orbit sets is injective: the nonzero orbits are determined by the unchanged valuation and the zero labels are fixed. Its additive image is contained in the target hyperaddition, but equality need not hold. New value-group elements or a larger residue field can add new target sums. At each cyclotomic stalk, every element, unit witness and finite sum relation occurs in a finite cyclotomic subfield, after adjoining the finitely many elements involved. Thus the filtered union of these exact finite-stage quotient relations is the relation FCQ19. This proves how the corrected finite-stage rule yields the source's final cyclotomic hyperfield.

8.6 The quotient sheaf over the original arithmetic scheme

Let A be an integral domain with fraction field K , and $X = \text{Spec } A$. On this irreducible space define the split meromorphic sheaf $G_L(\underline{K})$ by the constant full carrier on each nonempty open, with its coordinatewise operations. The ordinary unit sheaf $\mathcal{O}_X^\times$ acts by FCQ3. Let

$$\mathcal{C}_L(X) = (G_L(\underline{K})/\mathcal{O}_X^\times)^\#, \quad \mathcal{C}(X) = (\underline{K}/\mathcal{O}_X^\times)^\#.\tag{FCQ20}$$

The superscript means sheafification of the orbit presheaf. The second sheaf is the source's Cartier hyperring sheaf. In the first, the relation of multivalued addition is the sheafification of the orbit-sum relation FCQ5; equivalently, membership is locally witnessed by unit representatives. The multiplication and the support and amplitude maps are compatible with restriction, so they give sheaf maps.

At every point $x \in X$, the exact stalk is

$$\mathcal{C}_L(X)_x = G_L(K)/\mathcal{O}_{X,x}^\times.\tag{FCQ21}$$

To prove the orbit assertion, an equality of two stalk orbits is witnessed by one unit germ, and that unit and its inverse are represented on some common neighborhood. After shrinking, their product is the identity, so the germ is represented by an actual section of the unit sheaf there. The converse follows by taking germs. The same argument with the finitely many representatives and unit witnesses in FCQ5 proves the stalk formula for hyperaddition. Sheafification leaves these stalks unchanged. Thus no relation is introduced merely by asserting a stalk calculation.

When $\mathcal{O}_{X,x}$ is a valuation ring, FCQ21 has the full table FCQ6–7. At the generic point its unit group is $K^\times$, and the quotient has just one nonzero-amplitude orbit, together with the full L :

$$\mathcal{C}_L(X)_\eta = \{z_\ell : \ell \in L\} \sqcup \{\mathbf{0}\}, \quad \mathbf{0} \boxplus \mathbf{0} = \begin{cases} \{e\}, & K = \mathbf{F}_2, \\ \{e, \mathbf{0}\}, & |K| > 2. \end{cases}\tag{FCQ22}$$

This follows either from the same residue argument for the trivial valuation or directly by summing two nonzero field elements. For the number fields and cyclotomic field here, the second case applies and the collapsed generic stalk is the Krasner hyperfield. The map from a valuation stalk to the generic stalk sends every $\mathbf{a}$ to $\mathbf{0}$ and fixes every z_ℓ . If the valuation is nontrivial, its group has an element less than any given a , which supplies the nonzero generic self-sum even in the residue- $\mathbf{F}_2$ case. If it is trivial, FCQ22 already gives the exact rule. This arithmetic generalization preserves hyperaddition in the inclusion sense, and is not in general strong. For distinct $a, b \in \Gamma$, the image of $\mathbf{a} \boxplus \mathbf{b}$ is the singleton $\{\mathbf{0}\}$, whereas the generic sum $\mathbf{0} \boxplus \mathbf{0}$ is $\{e, \mathbf{0}\}$ for the fields in this application. The larger generic unit group has added a cancellation. Thus the sheaf relations and their restriction maps retain this distinction; no equality of these two sets is assumed.

There is an exact sheaf version of FCQ12:

$$\mathcal{C}_L(X) \cong \{(h, \ell) \in \mathcal{C}(X) \times \underline{L} : h \neq 0 \implies \ell = 1_L\}. \quad (\text{FCQ23})$$

The condition is understood locally, or equivalently on germs, and the operations are the componentwise formulas following FCQ12. The natural map sends an orbit to its ordinary Cartier orbit and its retained support. It is bijective on every stalk by FCQ21 and FCQ12, so it is an isomorphism of sheaves; the stalk calculation also proves exact preservation of their multivalued relations. This is the explicit sheaf map to the original Cartier construction, retaining all of its zero-fibre data.

For $A = \mathcal{O}_{\mathbb{Q}^{\text{cyc}}}$, a prime above p has local ring equal to the valuation ring used in FCQ19. Indeed both are the union of the compatible finite localizations at that prime: every element and its denominator occur in a finite cyclotomic subfield. The ordinary collapsed stalks in FCQ23 are consequently $H(p)_{\text{hmax}}$ at closed primes and the Krasner hyperfield at the generic point, exactly as in the source's Proposition **sheavesonspz**, now with the corrected finite-stage addition. Applying FCQ23 to that source comparison gives the full supported sheaf over its arithmetic-site pullback. The finite-support valuation sections and the normalization remain those constructed in that proposition; no numerical value has been assigned to a lattice label.

8.7 Extension to the actual spectrum of the split carrier

The arithmetic scheme X in FCQ20 is not the whole prime spectrum of $S_A = G_L(A)$. To retain that additional geometry, put $Y = \text{Spec } S_A$, with its local-fraction structure sheaf $\mathcal{O}_Y$. The original prime classification and localization give

$$\begin{aligned} Q_{\mathfrak{q}} &= \{z_\ell : \ell \in L\} \cup \{\widehat{a} : a \in \mathfrak{q}, a \neq 0\}, & P_{\mathfrak{p}} &= \{z_\ell : \ell \in \mathfrak{p}\}, \\ (\mathcal{O}_Y)_{Q_{\mathfrak{q}}} &= G_L(A_{\mathfrak{q}}), & (\mathcal{O}_Y)_{P_{\mathfrak{p}}} &= L[(L \setminus \mathfrak{p})^{-1}], \\ V_Y(e) &\cong X, & D_Y(e) &\cong \text{SpecLat } L. \end{aligned} \quad (\text{FCQ24})$$

Here $\mathfrak{q}$ is an ordinary prime of A and $\mathfrak{p}$ is a lattice prime. All of these primes are present.

For explicitness, at an arithmetic prime every denominator has nonzero amplitude and top support, so fraction equality is exactly the amplitude fraction equality with each z_ℓ retained; this proves the first localization. At a support prime, e is invertible. As a multiplicative idempotent it becomes the unit; since $e\widehat{a} = e$ for every supported amplitude, all such amplitudes become that unit. The remaining lattice localization is the second displayed stalk. Writing $F = L \setminus \mathfrak{p}$, its elements have the exact equality relation

$$\ell \sim_F m \iff \exists \nu \in F : \nu \wedge \ell = \nu \wedge m. \quad (\text{FCQ25})$$

The relation is just localization equality after clearing one denominator. Finite distributivity proves that it is a lattice congruence. Elements of F become top, and bottom does not become top: a witness to that equality would force a member of F to be bottom. This proves the relevant support stalks without replacing them by Boolean residues.

Let $\mathcal{M}_Y$ be the meromorphic S_A -algebra sheaf obtained by inverting all supported nonzero amplitudes $T = \{\widehat{a} : 0 \neq a \in A\}$. Concretely it is the sheaf of local fractions of $T^{-1}S_A = G_L(K)$ as an S_A -algebra: on each principal-open local chart one also inverts that chart's denominators. These local fractions and their agreement relations define the sheaf, and give

$$(\mathcal{M}_Y)_{Q_q} = G_L(K), \quad (\mathcal{M}_Y)_{P_p} = L[(L \setminus \mathfrak{p})^{-1}]. \quad (\text{FCQ26})$$

The second equality holds because those additional supported denominators are already the unit in the support stalk, by FCQ24.

Define the full quotient on the actual split spectrum by

$$\widetilde{\mathcal{C}}_L(Y) = (\mathcal{M}_Y / \mathcal{O}_Y^\times)^\# . \quad (\text{FCQ27})$$

The orbit and addition relations are sheafified as in FCQ20. At an arithmetic stalk its unit group is $\widehat{A}_q^\times$, and at a support stalk the only multiplicative unit of the lattice semiring is its top: an invertible idempotent equals the unit. Hence its exact two kinds of stalks are

$$\begin{aligned} \widetilde{\mathcal{C}}_L(Y)_{Q_q} &= G_L(K) / A_q^\times, \\ \widetilde{\mathcal{C}}_L(Y)_{P_p} &= L[(L \setminus \mathfrak{p})^{-1}]. \end{aligned} \quad (\text{FCQ28})$$

The quotient–stalk commutation is proved by the same finite-unit-witness argument as FCQ21. In the second line the orbit action is trivial, and addition and multiplication are the single-valued join and meet. Thus both the arithmetic Cartier data and the support open are calculated.

On the closed arithmetic subspace $i : X \cong V_Y(e) \hookrightarrow Y$, restriction of FCQ27 is canonically FCQ20. On a basic open $D_Y(\widehat{a})$ its arithmetic part is $D_X(a)$; the fraction maps restrict to the identical split amplitudes and unit sections there. These charts give the map, and FCQ21 and FCQ28 prove it an isomorphism on every arithmetic stalk. On the open $D_Y(e)$, the restriction of FCQ27 is the ordinary local-fraction sheaf of the lattice semiring L , because inverting e identifies both $\mathcal{O}_Y$ and $\mathcal{M}_Y$ with that sheaf and its unit quotient is trivial.

The generization map from an arithmetic stalk to a support stalk has the exact formula

$$G_L(K) / A_q^\times \longrightarrow L[(L \setminus \mathfrak{p})^{-1}], \quad z_\ell \longmapsto [\ell]_F, \quad [\widehat{x}] \longmapsto [1_L]_F. \quad (\text{FCQ29})$$

It is well defined because every ordinary unit and every nonzero amplitude becomes top there. It sends each multivalued sum to the singleton join of its images, including cancellation, since cancellation has support 1_L . Multiplication also commutes. This is the localized support projection, and is the actual connecting map between the two strata, not an assertion that the support stratum is absent.

Equations FCQ23 and FCQ27–29 supply both requested Cartier bridges. Over the ordinary arithmetic subspace they retain the original unit-orbit Cartier hyperfield together with every z_ℓ . Over the full split spectrum they additionally retain the localized support sheaf and its exact generization maps. The independent additive zero τ and cancellation zero e remain distinct at arithmetic stalks; their behavior on the support open is the explicitly calculated localization $e = 1$. None of these constructions assumes a Frobenius weight, positivity theorem or purity.

9 Exact checks and repairs in the classical circle calculation

This section concerns the original author source `RRZ.tex` of Alain Connes and Caterina Consani, *Riemann–Roch for the ring $\mathbb{Z}$* . The source SHA–256 is

1975ade5af5d3facf0f954da414a85c07c8adb7a83d1cc08e38f3112c5d072f9.

The exact locations are the sequence definition at lines 309–311, Lemma `sequenceb` at lines 316–322, and Proposition `caseh1` and its proof at lines 324–338. The sequence formula is verified below. The minimum-distance assertion and the choice of a nearest integer in a fixed interval require the indicated repairs. The resulting classical dimension statement is preserved. Nothing here identifies a first cohomology for the support-sensitive carrier.

9.1 The interval formula, with its actual indices

The author's formula is

$$j(n) = \frac{(-2)^n}{3} - \frac{(-1)^n}{2} + \frac{1}{6}, \quad k(n) = 2\lfloor n/2 \rfloor + 1. \quad (\text{SFC1})$$

In particular $j(1) = 0$, $j(2) = 1$, and $j(3) = -2$. The displayed value list in the source starts at $n = 1$. For $n = 2$, its interval formula is therefore

$$[j(k(2)), j(k(2)) + 2^2 - 1] = [j(3), j(3) + 3] = [-2, 1], \quad (\text{SFC2})$$

exactly the subset sums of $\{1, -2\}$. No change to $k(n)$ is needed.

For all $n \geq 0$ define $G(n) = \{(-2)^r : 0 \leq r < n\}$. The least and greatest subset sums are

$$L_n = -\frac{2(4^{\lfloor n/2 \rfloor} - 1)}{3}, \quad U_n = \frac{4^{\lfloor n/2 \rfloor} - 1}{3}, \quad L_n = j(k(n)), \quad U_n - L_n = 2^n - 1. \quad (\text{SFC3})$$

The first two formulas are the sums of all negative and all positive powers, respectively. Substitution of the odd number $k(n)$ into (SFC1) proves $L_n = j(k(n))$. The two parity cases of n give the last equality. The subset sums are distinct: in the difference of two unequal sums, take the smallest index r with nonzero coefficient. The coefficient is 1 or -1 , so the difference is nonzero modulo 2^{r+1} . There are consequently 2^n distinct integer sums in an interval of exactly that cardinality. They fill $[L_n, U_n]$. The repeated label $\Delta(4)$ on the last listed interval $[-10, 21]$ at source line 320 should be $\Delta(5)$; (SFC3) gives $L_5 = -10$ and $U_5 = 21$ directly.

9.2 The exact circular minimum distance

Use the circle $\mathbb{R}/\mathbb{Z}$ with $d(x, y) = \min_{r \in \mathbb{Z}} |x - y - r|$, and the actual set

$$F(m) = \{(-2)^{-j} \bmod \mathbb{Z} : 1 \leq j \leq m\}. \quad (\text{SFC4})$$

For $m = 0$ it is empty, and for $m = 1$ it is a singleton, so the pairwise-separation requirement is vacuous. For every $m \geq 2$,

$$\min_{x \neq y \in F(m)} d(x, y) = \min \left\{ \frac{1}{4}, 3 \cdot 2^{-m} \right\}. \quad (\text{SFC5})$$

For proof, write $x_j = (-1)^j 2^{-j}$. The distance between $x_1 = -1/2$ and x_j is $1/2 - 2^{-j}$ for every $j \geq 2$: if j is even one uses the complementary circular arc. Its minimum is $1/4$, attained at $j = 2$. For $2 \leq i < j \leq m$, the absolute difference is at most $3/8 < 1/2$, so it already equals the circular distance. It is

$$2^{-j}(2^{j-i} - 1) \quad \text{when } i, j \text{ have equal parity,} \quad 2^{-j}(2^{j-i} + 1) \quad \text{otherwise.}$$

Equal parity implies $j - i \geq 2$; different parity implies $j - i \geq 1$. Both expressions are at least $3 \cdot 2^{-j} \geq 3 \cdot 2^{-m}$. For $m \geq 3$, the adjacent pair $i = m - 1, j = m$ attains this value. For $m = 2$, only the first pair exists. This proves (SFC5).

In particular, at $m = 2$ the two points are $-1/2$ and $1/4$, whose distance is $1/4$, not $3/4$. At $m = 3$ the minimum is still $1/4$, not $3/8$. These are the small cases missed by the minimum-distance expression in source line 331. For $m \geq 4$, its expression $3 \cdot 2^{-m}$ does equal (SFC5).

9.3 The periodic grid proves the covering step

The subset sums of (SFC4) are exactly the dyadic grid

$$\left\{ \sum_{j=1}^m \alpha_j (-2)^{-j} \bmod \mathbb{Z} : \alpha_j \in \{0, 1\} \right\} = \left\{ \frac{r}{2^m} \bmod \mathbb{Z} : 0 \leq r < 2^m \right\}. \quad (\text{SFC6})$$

To verify this without an endpoint assumption, multiply two proposed equal subset sums by $(-2)^m$. Their difference is an integer combination of $1, -2, \dots, (-2)^{m-1}$ with coefficients in $\{-1, 0, 1\}$, divisible by 2^m . A smallest nonzero coefficient would give valuation less than m , by the argument after (SFC3), which is impossible. The 2^m subset sums therefore give distinct elements of the group of 2^m grid points, proving equality.

For any $x \in \mathbb{R}$, choose an integer r nearest to $2^m x$, so $|2^m x - r| \leq 1/2$. Reducing r modulo 2^m does not alter the circle class of $r/2^m$. Hence

$$\max_{x \in \mathbb{R}/\mathbb{Z}} \min_{y \text{ a subset sum of } F(m)} d(x, y) = 2^{-m-1}. \quad (\text{SFC7})$$

The upper bound follows from the nearest-integer construction; the midpoints of adjacent grid points attain it.

The fixed nonperiodic nearest-integer step at source lines 336–337 cannot replace this reduction. For $m = 2$, its integer interval is $\Delta(2) = [-2, 1]$. The point $x = 7/16$ belongs to the corresponding half-open real interval $[-1/2, 1/2)$, but

$$\min_{q \in \{-2, -1, 0, 1\}} |4x - q| = \frac{3}{4} > \frac{1}{2}. \quad (\text{SFC8})$$

The nearest integer is 2, outside that chosen interval. Modulo four it is represented by -2 , and the associated circle point is $-1/2 \bmod \mathbb{Z}$, at distance $1/16$ from x . Thus the periodic-grid argument gives the required covering even at this endpoint case.

9.4 The classical dimension conclusion is retained

For the original tolerant circle object in Connes–Consani’s Proposition **caseh1**, all group subset sums are admissible, and the degree-one tolerance is $d(x, y) \leq \lambda$. When $\lambda \geq 1/2$, the empty set generates because every circle point is within tolerance of zero. When

$$2^{-m-1} \leq \lambda < 2^{-m}, \quad m \geq 1, \quad (\text{SFC9})$$

the m -element set $F(m)$ generates by (SFC7). Its separation condition is also valid: it is vacuous for $m = 1$; for $m = 2$, (SFC5) gives $1/4 > \lambda$; and for $m \geq 3$ both entries in the minimum in (SFC5) exceed $\lambda < 2^{-m}$.

Conversely, a set of r generators has at most 2^r subset sums. For $\lambda < 1/2$, each closed circular tolerance ball has measure 2λ , in the length-one Haar measure. Covering the circle therefore requires $2\lambda 2^r \geq 1$. If $r \leq m - 1$ in (SFC9), this left side is strictly less than 1. Thus $r \geq m$, and the original formula is proved with the repaired steps:

$$\dim_{\mathbb{S}} U(1)_{\lambda} = \begin{cases} 0, & \lambda \geq 1/2, \\ m, & 2^{-m-1} \leq \lambda < 2^{-m}, \quad m \geq 1. \end{cases} \quad (\text{SFC10})$$

This calculation supplies corrected finite-circle distances and a periodic covering proof for the existing classical object. It makes no claim that the support fibres calculated in (SRR4)–(SRR5) have disappeared in a new first cohomology construction.

The source is Alain Connes and Caterina Consani, *The Absolute Twistor Line and the Geometry of $\overline{\text{Spec } \mathbb{Z}}$* , arXiv:2609.00299v1, specifically the original author file `CC.tex`. Its complete 1,228-line source was read for this calculation. The source labels used below are

```
opens; eq:restriction_maps;
def:alpha_symmetry; thm:twistor_space_scheme;
prop:twistor_frobenius.
```

Their local signs and generators are retained. The source file's SHA-256 is

```
57a10dcef5cd758e6b2f1c54b1c0a11b0fb0
93638f74fb1be515ec3bc7080ba4.
```

This note evaluates the source's *pointed multiplicative chart monoids*, glues the resulting chart-point sets, and takes their explicit α -orbits. These constructions are defined below. No assertion that this calculation classifies all geometric morphisms into a quotient topos, or all torsors over a general target, is needed or made. This specification is essential because the target is not a ring. No Riemann-hypothesis conclusion is asserted.

10 The target and its exact morphism to the complex numbers

Let L be a bounded distributive lattice with bottom 0_L and top 1_L . The target is the given Split-Zero semiring

$$\begin{aligned} S = G_L(\mathbb{C}) &= \{(0, \lambda) : \lambda \in L\} \cup \{(a, 1_L) : a \in \mathbb{C}\}, \\ (a, \lambda) + (b, \mu) &= (a + b, \lambda \vee \mu), \quad (a, \lambda)(b, \mu) = (ab, \lambda \wedge \mu). \end{aligned} \tag{CSP1}$$

Pairs in the displayed union have their literal equality; thus the two occurrences of $(0, 1_L)$ are one element. Put

$$z_\lambda = (0, \lambda), \quad \tau = z_{0_L}, \quad e = z_{1_L}, \quad \widehat{a} = (a, 1_L), \quad 1 = \widehat{1}, \quad \epsilon_S = -\widehat{1}. \tag{CSP2}$$

The multiplicative monoid is pointed by its absorbing element τ . In particular e is not its distinguished absorbing element when $0_L \neq 1_L$. Addition and multiplication in (CSP1) are closed: a nonzero output amplitude forces every factor with a nonzero amplitude to have top support. The lattice laws and complex arithmetic prove the semiring laws, with distributivity using $\lambda \wedge (\mu \vee \nu) = (\lambda \wedge \mu) \vee (\lambda \wedge \nu)$.

Lemma 33. *The units of S and the solutions of the imaginary-generator equation are*

$$S^\times = \{\widehat{a} : a \in \mathbb{C}^\times\}, \quad \widehat{a}^{-1} = \widehat{a^{-1}}, \quad \{j \in S : j^2 = \epsilon_S\} = \{\widehat{i}, \widehat{-i}\}. \tag{CSP3}$$

For every $\lambda \in L$ and positive integer n ,

$$z_\lambda^n = z_\lambda, \quad \epsilon_S z_\lambda = z_\lambda, \quad z_\lambda z_\mu = z_{\lambda \wedge \mu}. \tag{CSP4}$$

Proof. An inverse equation $xy = 1$ forces the product of the amplitudes to be 1. Both amplitudes are nonzero, so both supports equal 1_L and the inverse has the displayed value. Conversely these values multiply to 1. If $j^2 = -\widehat{1}$, its amplitude is nonzero and squares to -1 , so it equals i or $-i$ and its support is 1_L . Formula (CSP4) follows directly from lattice idempotence and (CSP1). $\square$

Proposition 34. *The amplitude map and a pointed multiplicative section are*

$$q : S \longrightarrow \mathbb{C}, \quad q(a, \lambda) = a, \quad b : \mathbb{C} \longrightarrow S, \quad b(0) = \tau, \quad b(a) = \widehat{a} \ (a \neq 0). \tag{CSP5}$$

Here q is a unital semiring homomorphism; b is the unique pointed unital multiplicative section of q and preserves -1 . For $0_L \neq 1_L$, b is not additive, and no unital semiring homomorphism from a nonzero ring to S exists.

Proof. The formula for q preserves each operation in (CSP1), together with zero and one. Multiplicativity of b holds when both factors are nonzero because $\mathbb{C}$ is a field; when at least one is zero it follows from absorption by τ . Any section must take each nonzero a to the only point with amplitude a , namely $\widehat{a}$, and pointedness forces its value at zero. However $b(1) + b(-1) = e \neq \tau = b(0)$. More generally, a unital semiring homomorphism $f : R \rightarrow S$ from a ring would give $1 + f(-1) = f(0) = \tau$. The support of the left side is $1_L \vee \text{supp}(f(-1)) = 1_L$, contradicting the bottom support of τ when $0_L \neq 1_L$. $\square$

Complex conjugation is the semiring automorphism $c_S(a, \lambda) = (\overline{a}, \lambda)$. It fixes every z_λ , preserves ϵ_S , and exchanges $\widehat{i}$ with $\widehat{-i}$.

11 All chart points and all overlaps

Let $B = \{0, 1, \epsilon\}$ with $\epsilon^2 = 1$ be the source's signed pointed base monoid. Its target structure sends 0 to τ , 1 to 1, and ϵ to ϵ_S . The chart monoids, with the source's generators unchanged, are

$$M_+ = B[T, J_+]/(J_+^2 = \epsilon), \quad M_- = B[S, J_-]/(J_-^2 = \epsilon), \quad M_0 = B[T, T^{-1}, J]/(J^2 = \epsilon). \quad (\text{CSP6})$$

Each nonzero element of M_+ is uniquely $J_+^a T^m$ with $a \in \mathbb{Z}/4\mathbb{Z}$ and $m \in \mathbb{N}$; M_- has the corresponding form in J_- , S , and M_0 has $m \in \mathbb{Z}$. Thus these presentations have no additive relation identifying $1 + \epsilon$ with the distinguished zero. All homomorphisms in this section are pointed unital multiplicative B -homomorphisms.

The restrictions are exactly the source's restrictions:

$$\rho_+(T) = T, \quad \rho_+(J_+) = J, \quad \rho_-(S) = T^{-1}, \quad \rho_-(J_-) = \epsilon J. \quad (\text{CSP7})$$

Theorem 35. *The complete point sets and overlap maps are*

$$U_\pm(S) = \text{Hom}_B(M_\pm, S) = S \times \{\widehat{i}, \widehat{-i}\}, \quad U_0(S) = S^\times \times \{\widehat{i}, \widehat{-i}\}, \quad (\text{CSP8})$$

$$\rho_+^*(t, j) = (t, j)_+, \quad \rho_-^*(t, j) = (t^{-1}, \epsilon_S j)_-. \quad (\text{CSP9})$$

No z_λ belongs to the overlap, including the supported zero e . The glued chart-point set

$$X_L = U_+(S) \coprod_{U_0(S)} U_-(S) \quad (\text{CSP10})$$

is the disjoint union of two copies of

$$P_L = \mathbb{C}^\times \sqcup \{0_\lambda : \lambda \in L\} \sqcup \{\infty_\lambda : \lambda \in L\}. \quad (\text{CSP11})$$

The symbols 0_λ and ∞_μ are distinct for all λ, μ ; equality within either family is precisely equality of the lattice labels.

Proof. A map on M_+ is determined by $t = \phi(T)$ and $j = \phi(J_+)$. The relation forces $j^2 = \epsilon_S$, whose solutions are (CSP3), and imposes no condition on t . Conversely, for any such pair, the formula $\phi(J_+^a T^m) = j^a t^m$ and $\phi(0) = \tau$ defines a unique map. The same proof applies to M_- . On M_0 , the equation $TT^{-1} = 1$ forces t to be a unit, and (CSP3) gives all possibilities. Precomposition with (CSP7) yields (CSP9).

Both overlap maps are injective. Their gluing relation pairs exactly $(t, j)_+$ with $(t^{-1}, -j)_-$ for $t \in S^\times$. The first component consists of U_+ with $j = \widehat{i}$ and U_- with $j = \widehat{-i}$. In it, a nonzero complex a is represented by $(\widehat{a}, \widehat{i})_+$, equivalently $(\widehat{a^{-1}}, \widehat{-i})_-$. Its two families of additional points are represented by $(z_\lambda, \widehat{i})_+$ and $(z_\lambda, \widehat{-i})_-$, which define respectively 0_λ and ∞_λ . None is in an overlap, so none can be identified by a chain of gluing relations. Reversing the two imaginary signs gives the second component and proves the assertion. $\square$

This proof constructs P_L as the two-chart gluing $S \coprod_{S^\times} S$ with reciprocal transition. It neither declares a labelled zero invertible nor replaces the pointed zero by e .

12 The geometric quotient and its amplitude fibres

The source's symmetry is given on chart points by

$$\alpha(t, j)_+ = (\epsilon_S t, j)_-, \quad \alpha(s, j)_- = (\epsilon_S s, j)_+, \quad (\text{CSP12})$$

and on the overlap by

$$\alpha(t, j) = (\epsilon_S t^{-1}, \epsilon_S j). \quad (\text{CSP13})$$

These formulas are evaluations of $\alpha_+^*(S) = \epsilon T$, $\alpha_+^*(J_-) = J_+$ and their minus-chart counterparts, exactly as in the source.

Theorem 36. *Formulas (CSP12)–(CSP13) define an involution of X_L exchanging its two components. Consequently the orbit set $C_L = X_L / \langle \alpha \rangle$ is in bijection with P_L , by the unique representative in the first component. Amplitude gives the surjection*

$$\pi_L : C_L \cong P_L \longrightarrow \mathbb{P}^1(\mathbb{C}), \quad a \longmapsto a \ (a \neq 0), \quad 0_\lambda \longmapsto 0, \quad \infty_\lambda \longmapsto \infty. \quad (\text{CSP14})$$

Its fibre is a singleton over every $a \in \mathbb{C}^\times$ and an exact copy of L over each of 0 and ∞ .

Proof. To check the only gluing issue, start with the equivalent points $(t, j)_+$ and $(t^{-1}, -j)_-$. Their images are $(-t, j)_-$ and $(-t^{-1}, -j)_+$. These are equivalent because the inverse of $-t^{-1}$ is $-t$ and the imaginary sign changes in the transition. Applying (CSP12) twice multiplies the spatial coordinate by $\epsilon_S^2 = 1$ and leaves j unchanged. A first-component plus point has $j = i$ and maps to a minus point with $j = i$, which lies in the second component; the other three chart/sign combinations give the same exchange. Every orbit therefore meets the first component exactly once. The map q in (CSP5) preserves all chart relations, the reciprocal transition, and the symmetry. Its resulting quotient map is exactly (CSP14); Theorem 35 supplies all its fibres. $\square$

$$\begin{array}{ccc} \{0_\lambda : \lambda \in L\} & \mathbb{C}^\times & \{\infty_\lambda : \lambda \in L\} \\ \pi_L \downarrow & \parallel & \downarrow \pi_L \\ \{0\} & \mathbb{C}^\times & \{\infty\} \end{array}$$

Figure 4: Exact disjoint decomposition of the amplitude map, proved in Theorems 35 and 36. Every support label survives in the upper row; only the map to the lower row identifies them. The real structure exchanges 0_λ and ∞_λ with the same label (Theorem 37). The chart and sign conventions come from Connes–Consani [1], formulas `opens` and `eq:restriction_maps`.

For every $\lambda \in L$, the formula

$$s_\lambda : \mathbb{P}^1(\mathbb{C}) \rightarrow P_L, \quad s_\lambda(a) = a \ (a \neq 0), \quad s_\lambda(0) = 0_\lambda, \quad s_\lambda(\infty) = \infty_\lambda \quad (\text{CSP15})$$

is an injective set-theoretic section of π_L . Only s_{0_L} is induced by a pointed multiplicative section of q : it is induced by b in (CSP5). Indeed a pointed map must send zero to τ . Thus using s_{1_L} to retain supported zeros is a well-defined section of the point map but is not an unmentioned pointed-monoid section. All of these distinctions disappear when L has one element, in which case $S = \mathbb{C}$ and $P_L = \mathbb{P}^1(\mathbb{C})$.

If $f : L \rightarrow L'$ is a bounded lattice homomorphism, the map

$$G_f : S \rightarrow G_{L'}(\mathbb{C}), \quad z_\lambda \mapsto z_{f(\lambda)}, \quad \hat{a} \mapsto \hat{a} \ (a \neq 0) \quad (\text{CSP16})$$

preserves addition, multiplication, zero, one, and ϵ . For addition of two nonzero amplitudes with a zero sum, the image is $z_{f(1_L)} = z_{1_{L'}}$, so cancellation is also checked. All other cases follow directly from meet and join preservation. It therefore induces $P_L \rightarrow P_{L'}$ taking each pole label λ to $f(\lambda)$ and fixing $\mathbb{C}^\times$. This is the exact change-of- support map, including every zero.

13 Real structure and every odd Frobenius

Define two bijections of P_L :

$$\begin{aligned} \beta(a) &= -a^{-1}, & \beta(0_\lambda) &= \infty_\lambda, & \beta(\infty_\lambda) &= 0_\lambda, \\ \sigma(a) &= -\bar{a}^{-1}, & \sigma(0_\lambda) &= \infty_\lambda, & \sigma(\infty_\lambda) &= 0_\lambda. \end{aligned} \tag{CSP17}$$

These are formulas on disjoint subsets; no inverse of z_λ is present.

Theorem 37. *Postcomposition of chart points with c_S commutes with α , and the induced real structure on $C_L = P_L$ is σ of (CSP17). It has no fixed points, retains every support label, and obeys*

$$\pi_L \sigma(a) = -1/\overline{\pi_L(a)}, \quad \sigma s_\lambda = s_\lambda \sigma_{\mathbb{C}}. \tag{CSP18}$$

The first formula uses the usual projective exchange of 0 and ∞ on its right side.

Proof. Complex conjugation preserves multiplication, inverse of units, and ϵ_S , hence commutes with (CSP12)–(CSP13) and respects (CSP9). It exchanges the two components. On a first-component plus point $(t, i)_+$, apply conjugation and then α to return to the first component: $(t, i)_+ \mapsto (\bar{t}, -i)_+ \mapsto (-\bar{t}, -i)_-$. If $t = \hat{a}$ is a unit, the first-component global coordinate is $-1/\bar{a}$; if $t = z_\lambda$, the point is ∞_λ . The minus-chart calculation similarly takes ∞_λ to 0_λ . For a unit, a fixed point would satisfy $a\bar{a} = -1$, which is impossible in $\mathbb{C}$. The two labelled pole families are disjoint, so there are no fixed points there either. Equation (CSP18) follows on all three disjoint subsets. $\square$

For a positive integer n let P_n on P_L raise $a \in \mathbb{C}^\times$ to a^n and fix every labelled pole. Let $d(n) \in \{0, 1\}$ be the residue of $(n-1)/2$ modulo 2 for odd n .

Theorem 38. *Raising every element of the source chart monoids to its n th power is a pointed B -endomorphism precisely when n is odd. These maps commute with restriction and α . Their induced action on P_L is*

$$F_n = \beta^{d(n)} P_n, \quad F_n(a) = \begin{cases} a^n & n \equiv 1 \pmod{4}, \\ -a^{-n} & n \equiv 3 \pmod{4}, \end{cases} \tag{CSP19}$$

	$F_n(0_\lambda)$	$F_n(\infty_\lambda)$	
$n \equiv 1 \pmod{4}$	0_λ	∞_λ	.
$n \equiv 3 \pmod{4}$	∞_λ	0_λ	

(CSP20)

They satisfy, without any removal of labelled poles,

$$F_m F_n = F_{mn}, \quad F_n \sigma = \sigma F_n, \quad \pi_L F_n = \text{Fr}_n \pi_L, \quad F_n s_\lambda = s_\lambda \text{Fr}_n. \tag{CSP21}$$

Here Fr_n is the source's complex twistor action, from the proposition labelled `prop:twistor_frobenius`.

Proof. Power maps are monoid endomorphisms because the chart monoids are commutative. Their restriction to B fixes ϵ precisely when $\epsilon^n = \epsilon$, namely for odd n . For odd n the relation $(J^n)^2 = \epsilon^n = \epsilon$ is preserved. On the overlap, the two orders of restriction send J_- to $(\epsilon J)^n = \epsilon J^n$ and send S to T^{-n} , so they agree. Power maps commute with every monoid homomorphism, in particular the maps defining α .

On chart points they take (t, j) to (t^n, j^n) . Since $j^n = (-1)^{(n-1)/2} j$, for $n \equiv 1 \pmod{4}$ the first component is preserved, while for $n \equiv 3 \pmod{4}$ it is exchanged with the second. In the latter case applying α returns the image to the first component and changes a unit coordinate to $-a^{-n}$. For $t = z_\lambda$, (CSP4) gives $t^n = z_\lambda$ and multiplication by ϵ_S does not change it. The chart exchange gives exactly (CSP20), including $\lambda = 0_L, 1_L$.

For odd n , $\beta P_n = P_n \beta$: on units this is the identity $(-a^{-1})^n = -a^{-n}$, and both sides have the same pole exchange. Since $P_m P_n = P_{mn}$ and $d(mn) = d(m) + d(n)$ modulo 2, (CSP19) proves the first identity in (CSP21). The map σ is β followed by amplitude conjugation; the latter commutes with every P_n and β . This proves the second identity. The other two follow from (CSP14), (CSP15), (CSP19), and (CSP20) on every part of P_L . $\square$

Proposition 39. *For odd n , every unit point of P_L has exactly n distinct F_n -preimages, and each labelled pole has exactly one preimage. For $n > 1$ the fixed-point sets are*

$$\text{Fix}(F_n) = \begin{cases} \{a \in \mathbb{C}^\times : a^{n-1} = 1\} \sqcup \{0_\lambda, \infty_\lambda : \lambda \in L\}, & n \equiv 1 \pmod{4}, \\ \{a \in \mathbb{C}^\times : a^{n+1} = -1\}, & n \equiv 3 \pmod{4}. \end{cases} \quad (\text{CSP22})$$

In particular

$$\bigcap_{n \text{ odd}} \text{Fix}(F_n) = \emptyset. \quad (\text{CSP23})$$

Proof. For a nonzero target w , the preimage equation is $a^n = w$ or $a^n = -1/w$. Its right side is nonzero and its n roots in $\mathbb{C}$ are distinct, since a repeated root would also annihilate na^{n-1} . No labelled pole maps to a unit. Formula (CSP20) gives the single preimage of each pole, with its unchanged support label. Setting the expressions in (CSP19) equal to a gives (CSP22); (CSP20) settles the poles. A simultaneous fixed point cannot be a pole because F_3 exchanges the two pole families. A unit fixed by F_3 satisfies $a^4 = -1$, whereas one fixed by F_5 satisfies $a^4 = 1$. Their intersection is empty, proving (CSP23). $\square$

Statement (CSP23) concerns *points*. It does not contradict the source's Frobenius-invariant *sections* calculation: those are different explicitly defined functors. On the source monoid M_+ the simultaneous fixed elements are exactly B : a nonzero element $J^a T^m$ fixed by its third power has $3m = m$, whence $m = 0$, and $3a = a$ modulo 4, whence $a = 0$ or 2. Zero is also fixed. On evaluated points, however, the image of J must be one of the two square roots of -1 ; neither is fixed by its third power. This calculates the difference directly.

14 The precise Hopf limitation and the surviving torsor law

The spatial core $B[T, T^{-1}]$ has the source's coproduct $T \mapsto T \otimes T$, with the base sign fixed. Its S -points form exactly $S^\times = \mathbb{C}^\times$. The action on the enriched overlap obtained by multiplying its spatial coordinate is

$$D_a(t, j) = (\widehat{at}, j), \quad a \in \mathbb{C}^\times. \quad (\text{CSP24})$$

It extends to X_L by multiplying the plus coordinate by a and the minus coordinate by a^{-1} , preserving each component. It obeys $\alpha D_a = D_{a^{-1}} \alpha$. If $a^2 \neq 1$, it does not induce an action on C_L : for a generic point $x = (t, i)$ the two points $D_a x$ and $D_a \alpha x$ would have to be α -related, but their second coordinates force that relation and their first coordinates then force $-a/t = -1/(at)$, or $a^2 = 1$.

There is nevertheless an exact lift of the usual quotient scaling. Write $\chi(\widehat{i}) = 1$, $\chi(\widehat{-i}) = -1$. Define on each affine chart, using its own local imaginary generator,

$$A_a(t, j)_+ = (\widehat{a^{\chi(j)} t}, j)_+, \quad A_a(s, j)_- = (\widehat{a^{\chi(j)} s}, j)_-. \quad (\text{CSP24a})$$

On the overlap the sign of j changes in the transition and t is inverted, so the two formulas agree. Formula (CSP12) preserves the local j while exchanging charts and multiplying by -1 ; therefore A_a commutes with α . Also $A_a A_b = A_{ab}$. On the first component, (CSP24a) scales the global unit coordinate by a and fixes every labelled pole, hence descends to that precise action $a \cdot x$ on P_L . This is a lift that uses the imaginary branch; it is not the unmodified action (CSP24). It obeys

$$\sigma(a \cdot x) = \bar{a}^{-1} \cdot \sigma(x) \quad (a \in \mathbb{C}^\times, x \in P_L). \quad (\text{CSP25})$$

For unit x this is complex arithmetic, and for a pole it follows because both scaling maps fix the poles and σ exchanges their two families.

Proposition 40. *The prescription $J \mapsto J \otimes J$ does not extend the spatial diagonal to a B -coproduct on M_0 . On its full enriched point set $U_0(S) = \mathbb{C}^\times \times \{\widehat{i}, \widehat{-i}\}$ the choice-free operation that survives is the ternary law*

$$[(t_1, j_1), (t_2, j_2), (t_3, j_3)] = (t_1 t_2^{-1} t_3, j_1 j_2^{-1} j_3). \quad (\text{CSP26})$$

It satisfies

$$[x, y, y] = x = [y, y, x], \quad [[x, y, z], u, v] = [x, y, [z, u, v]]. \quad (\text{CSP27})$$

*Both α and complex conjugation preserve it, as does every odd power map. Selecting any one point o gives the group law $x *_o y = [x, o, y]$; selecting $o = (1, \widehat{i})$ gives the explicit law*

$$(t_1, j_1) *_o (t_2, j_2) = (t_1 t_2, j_1 \widehat{-i} j_2). \quad (\text{CSP28})$$

Proof. In the balanced monoid tensor product over B , $\epsilon \otimes 1 = 1 \otimes \epsilon$. Therefore $(J \otimes J)^2 = \epsilon \otimes \epsilon = 1$, while a B -map must send $J^2 = \epsilon$ to $\epsilon \otimes 1$. These two elements are different: evaluate both tensor factors at $T = 1, J = \widehat{i}$ in S . The balanced tensor evaluation sends 1 to 1 and $\epsilon \otimes 1$ to $\widehat{-1} \neq 1$. This is an obstruction to this exact diagonal on the *enriched* monoid; it is no obstruction to the coproduct on the spatial core used in (CSP24).

In (CSP26), every displayed inverse exists, and $(j_1 j_2^{-1} j_3)^2 = (-1)(-1)^{-1}(-1) = -1$, so the law lands in the enriched point set. The cancellation and association identities (CSP27) follow by multiplying the coordinates; for example both association expressions have first coordinate $t_x t_y^{-1} t_z t_u^{-1} t_v$ and the corresponding second coordinate. Equivalently its source ternary co-operation is $T \mapsto T \otimes T^{-1} \otimes T$ and $J \mapsto J \otimes J^{-1} \otimes J$: the square of the latter is $\epsilon \otimes \epsilon^{-1} \otimes \epsilon = \epsilon$, so the source relation is respected. The image of ϵ is the base element, as required.

On the overlap $\alpha(t, j) = (-t^{-1}, j^{-1})$. Substitution into (CSP26) shows the first coordinate becomes $-t_1^{-1} t_2 t_3^{-1}$ in either order, and the second becomes $j_1^{-1} j_2 j_3^{-1}$ in either order. Conjugation preserves products and inverses, and power maps do so in this commutative group; for odd powers they also preserve the equation $j^2 = -1$. Hence they preserve (CSP26). Formula (CSP27) gives identity o , associativity, and inverse $[o, x, o]$ for $*_o$, proving it is a group. Formula (CSP28) uses $(\widehat{i})^{-1} = \widehat{-i}$. $\square$

The last proposition supplies the exact enriched replacement when the group-like imaginary-generator prescription fails. A binary law requires the stated choice of o ; the ternary law and all chart, support, real-structure, and Frobenius maps above require no such choice. The calculation retains the source's distinguished choice of i only when identifying the quotient with its first component.

References

- [1] Alain Connes and Caterina Consani, *The Absolute Twistor Line and the Geometry of $\overline{\text{Spec } \mathbb{Z}}$* , arXiv:2609.00299v1 (2026). Original author source `CC.tex`; source formulas and labels are specified at the beginning and at their mathematical uses.

Source and scope. The source read is Alain Connes and Caterina Consani, *The Absolute Twistor Line and the Geometry of $\overline{\text{Spec } \mathbb{Z}}$* , arXiv:2609.00299v1, original author source `CC.tex`, 1228 lines, SHA-256 57a10dcef5cd758e6b2f1c54b1c0a11b0fb093638f74fb1be515ec3bc7080ba4. Reading coverage is the complete file, including its bibliography. The signed base, imaginary chart, geometric involution, and odd maps are Connes–Consani's constructions, not new programme constructions. All assertions below are proved in the specified category of pointed commutative monoids, or in ordinary algebraic $K_0(\mathbb{P}_{\mathbb{C}}^1)$. No claim about a more general category of spherical algebras is inferred merely from this monoid calculation.

15 The actual base and tensor product

Let

$$B = \{0, 1, \epsilon\}, \quad \epsilon^2 = 1, \quad \epsilon \neq 1, \quad D = B[J]/(J^2 = \epsilon). \quad (\text{HA1})$$

Here 0 is absorbing and each displayed presentation is a presentation of a pointed commutative monoid. In particular there is no additive relation $1 + \epsilon = 0$. A B -monoid is a pointed commutative monoid with a specified map from B , and a B -morphism fixes that map. Put

$$C_+ = B[T], \quad C_\times = B[T, T^{-1}], \quad A_+ = D[T], \quad A_\times = D[T, T^{-1}]. \quad (\text{HA2})$$

These are the spatial chart and overlap, and their imaginary enrichments. The relation in (HA1) is the one in [1, original `CC.tex`, lines 576–600].

The tensor product $P \otimes_B Q$ means the smash product of pointed monoids modulo $(bp) \otimes q = p \otimes (bq)$ for $b \in B$. Its multiplication is $(p \otimes q)(p' \otimes q') = pp' \otimes qq'$. In particular

$$\epsilon \otimes 1 = 1 \otimes \epsilon = \epsilon, \quad \epsilon \otimes \epsilon = 1. \quad (\text{HA3})$$

These are exactly the balancing relations specified in [1, original `CC.tex`, lines 437–465].

Lemma 41. *Every nonzero element of A_+ has a unique expression $J^a T^b$ with $a \in \mathbb{Z}/4\mathbb{Z}$ and $b \in \mathbb{N}$. For $A_\times$, replace $\mathbb{N}$ by $\mathbb{Z}$. The units of $A_+ \otimes_B A_+$ have the presentation*

$$\langle j_1, j_2, \epsilon \mid j_1^2 = j_2^2 = \epsilon, \quad \epsilon^2 = 1, \quad [j_1, j_2] = 1 \rangle \cong (\mathbb{Z}/4\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z}).$$

In this tensor product $\epsilon \neq 1$.

Proof. The pointed monoid obtained by adjoining an absorbing zero to $(\mathbb{Z}/4\mathbb{Z}) \times \mathbb{N}$, with $J = (1, 0)$, $T = (0, 1)$ and $\epsilon = (2, 0)$, satisfies (HA1)–(HA2). Conversely those presentations reduce uniquely to these coordinates. The same construction works with $\mathbb{Z}$ in the second coordinate. In the tensor product use $j_1 = J \otimes 1$, $j_2 = 1 \otimes J$. The only additional relation between the two factors is $j_1^2 = j_2^2$. Setting $h = j_2 j_1^{-1}$ gives $h^2 = 1$; the inverse change of generators is $j_2 = j_1 h$. Thus the unit group is the asserted direct product. Its element $\epsilon = j_1^2$ has order two. Alternatively, the compatible pair of maps to the multiplicative monoid of $\mathbb{C}$, sending both J 's to i and ϵ to -1 , induces a map on the balanced tensor product which separates ϵ and 1. $\square$

Proposition 42 (The imaginary diagonal does not descend). *The assignments*

$$\Delta(T) = T \otimes T, \quad \Delta(J) = J \otimes J, \quad \Delta(\epsilon) = \epsilon \quad (\text{HA4})$$

do not define a B -morphism on A_+ or $A_\times$. Moreover, neither of those two B -monoids admits a counit to B .

Proof. A B -morphism must preserve $J^2 = \epsilon$. But (HA3) gives

$$(J \otimes J)^2 = \epsilon \otimes \epsilon = 1 \neq \epsilon = \Delta(J^2). \quad (\text{HA5})$$

For the second assertion, the image of the unit J under a unital monoid map is a unit. The only units of B are 1 and ϵ , both with square 1. Neither squares to ϵ . Consequently there is no B -morphism from either enriched chart to B at all. A counital comonoid structure, and hence a Hopf structure over B , would include such a morphism. $\square$

The scope matters: the coproduct in the author's definition is on the *spatial* monoid M of $B[M]$ ([1, original `CC.tex`, lines 462–465]), not an assertion that a diagonal on an additional generator survives every signed quotient relation. Equation (HA5) identifies the exact relation which prevents that extension; it does not erase the spatial coproduct.

16 The surviving coaction and the stronger torsor correspondence

Proposition 43 (Spatial coaction). *There is a counital and coassociative B -morphism*

$$\rho : A_+ \longrightarrow C_+ \otimes_B A_+, \quad T \longmapsto T \otimes T, \quad J \longmapsto 1 \otimes J. \quad (\text{HA6})$$

It extends to $A_\times \rightarrow C_\times \otimes_B A_\times$. The spatial bialgebra has $\Delta_C(T) = T \otimes T$ and counit $\eta_C(T) = 1$. On $C_\times$ the antipode is $T \mapsto T^{-1}$.

Proof. The image of J in (HA6) has square $1 \otimes \epsilon = \epsilon$, as required. There is no other relation to check beyond invertibility of T on the overlap, where its image is invertible. Both coassociativity composites send T to $T \otimes T \otimes T$ and J to $1 \otimes 1 \otimes J$. Applying the spatial counit sends them back to T and J . The assertions about C follow by the same checks on T and the base. $\square$

For a commutative B -monoid R let

$$X(R) = \text{Hom}_B(A_\times, R) = \{(t, j) \in R^\times \times R^\times : j^2 = \epsilon_R\}.$$

The spatial coaction is the actual map

$$R^\times \times X(R) \longrightarrow X(R), \quad (u, (t, j)) \longmapsto (ut, j). \quad (\text{HA7})$$

This includes, on complex points, $\mathbb{C}^\times$ acting without changing the chosen root of -1 . It proves the precise action on the unfolded chart.

There is a stronger intrinsic relation. Define

$$H = B[u, u^{-1}, h]/(h^2 = 1), \quad G(R) = R^\times \times \{h \in R^\times : h^2 = 1\}. \quad (\text{HA8})$$

The letter h in (HA8) is an independent generator; it is not the base element ϵ . The Hopf structure on H sends u and h to their tensor diagonals, with counit $u, h \mapsto 1$, and antipode $u \mapsto u^{-1}$, $h \mapsto h$.

Theorem 44 (Exact torsor identities and ternary product). *There is a coaction and a canonical isomorphism*

$$\rho_H : A_\times \longrightarrow H \otimes_B A_\times, \quad T \longmapsto u \otimes T, \quad J \longmapsto h \otimes J, \quad (\text{HA9})$$

$$\text{can} : A_\times \otimes_B A_\times \xrightarrow{\sim} H \otimes_B A_\times, \quad (\text{HA10})$$

where can sends

$$T_1 \mapsto uT, \quad T_2 \mapsto T, \quad J_1 \mapsto hJ, \quad J_2 \mapsto J.$$

When $X(R)$ is nonempty, the action $(u, h)(t, j) = (ut, hj)$ makes it a principal homogeneous set for $G(R)$. The ternary operation below requires no choice of a root:

$$[x, y, z] = (t_x t_y^{-1} t_z, j_x j_y^{-1} j_z), \quad X(R)^3 \longrightarrow X(R). \quad (\text{HA11})$$

This operation satisfies $[x, y, y] = x = [y, y, x]$ and $[[x, y, z], u, v] = [x, y, [z, u, v]]$.

Proof. For (HA9), $(h \otimes J)^2 = 1 \otimes \epsilon = \epsilon$; coassociativity and the counit follow on each generator. The inverse to (HA10) is the explicitly defined B -morphism

$$u \longmapsto T_1 T_2^{-1}, \quad h \longmapsto J_1 J_2^{-1}, \quad T \longmapsto T_2, \quad J \longmapsto J_2. \quad (\text{HA12})$$

Indeed $(J_1 J_2^{-1})^2 = \epsilon \epsilon^{-1} = 1$, so (HA12) respects the defining equation of h ; each composite fixes the displayed generators. Given $(t, j), (t', j') \in X(R)$, the unique group element carrying the second to the first is (tt'^{-1}, jj'^{-1}) , whose second coordinate has square 1. The square of the second coordinate in (HA11) is $\epsilon_R \epsilon_R^{-1} \epsilon_R = \epsilon_R$. The stated identities follow by cancellation in the group of units of R . In particular they are identities of the actual functor X , not only of its complex points. The coordinate morphism for (HA11) is $T \mapsto T_1 T_2^{-1} T_3$, $J \mapsto J_1 J_2^{-1} J_3$. $\square$

Here “torsor identities” means the proved coaction and canonical-map isomorphism (HA9)–(HA12); no unspecified Grothendieck topology or descent theorem is being used.

Proposition 45 (The precise effect of choosing an imaginary root). *After the explicit base extension $B \rightarrow D$, let j_0 denote the root in the base copy of D . There is an isomorphism of D -monoids*

$$D \otimes_B A_\times \cong D[T, T^{-1}, k]/(k^2 = 1), \quad k = Jj_0^{-1}. \quad (\text{HA13})$$

Transporting the usual Hopf structure on the right gives

$$\Delta_D(J) = j_0^{-1}(J \otimes J), \quad \eta_D(J) = j_0, \quad \mathcal{S}_D(J) = J, \quad \Delta_D(T) = T \otimes T. \quad (\text{HA14})$$

Replacing j_0 by ϵj_0 changes the displayed coproduct on J by the factor ϵ and changes its counit by the same factor.

Proof. One has $k^2 = J^2 j_0^{-2} = \epsilon \epsilon^{-1} = 1$. The inverse to (HA13) sends J to $j_0 k$. The Hopf formula $\Delta(k) = k \otimes k$ gives $\Delta(J) = j_0 k \otimes k = j_0^{-1} J \otimes J$; the counit gives $\eta(J) = j_0$. Since $k^{-1} = k$, the antipode fixes $J = j_0 k$. The last assertion follows from $\epsilon^{-1} = \epsilon$. This calculation also shows explicitly what root choice the missing B -counit would have had to make. $\square$

17 The quotient symmetry and the spatial action

On the unfolded complex overlap the source defines

$$\alpha(z, j) = (-1/z, -j), \quad j \in \{i, -i\}, \quad (\text{HA15})$$

by $\alpha^*(T) = \epsilon T^{-1}$, $\alpha^*(J) = \epsilon J$ ([1, original `CC.tex`, lines 773–806]). If $r_\lambda(z, j) = (\lambda z, j)$, then

$$\alpha r_\lambda = r_{\lambda^{-1}} \alpha. \quad (\text{HA16})$$

This is verified by evaluating both sides at (z, j) : both give $(-\lambda^{-1}/z, -j)$. Thus the spatial action and involution have an exact semidirect-product relation.

The same uniform map r_λ does not descend to the orbit quotient for arbitrary λ . Indeed two representatives (z, j) and $(-1/z, -j)$ are sent to $(\lambda z, j)$ and $(-\lambda/z, -j)$. The latter is in the orbit of the former, for all $z \neq 0, \infty$, exactly when $-\lambda/z = -1/(\lambda z)$, that is, $\lambda^2 = 1$. After choosing the representative component $j = i$, the action which *does* lift the usual quotient scaling is

$$\tilde{r}_\lambda(z, i) = (\lambda z, i), \quad \tilde{r}_\lambda(z, -i) = (\lambda^{-1} z, -i). \quad (\text{HA17})$$

Its commutation with α follows from (HA16), and its multiplication law follows component by component. This supplies the exact additional branch rule needed to transport scaling through the geometric fold. It does not replace the underlying spatial-core coaction by a diagonal on J .

18 Odd Frobenius maps on the projective line

For a positive odd integer n , define

$$\chi(n) = (-1)^{(n-1)/2}, \quad e(n) = \frac{1 - \chi(n)}{2}, \quad a(z) = -\frac{1}{z}, \quad p_n(z) = z^n, \quad F_n = a^{e(n)} p_n. \quad (\text{HA18})$$

Thus $F_n(z) = z^n$ for $n \equiv 1 \pmod{4}$ and $F_n(z) = -1/z^n$ for $n \equiv 3 \pmod{4}$, as in Connes–Consani’s Proposition `prop:twistor_frobenius` ([1, original `CC.tex`, lines 1159–1191]). In homogeneous coordinates $z = X/Y$, these are everywhere-defined maps

$$F_n[X : Y] = \begin{cases} [X^n : Y^n], & \chi(n) = 1, \\ [-Y^n : X^n], & \chi(n) = -1. \end{cases} \quad (\text{HA19})$$

The two coordinate polynomials never vanish simultaneously on $\mathbb{P}^1$, so each map has degree n .

Theorem 46 (Composition, real structure, and the exact equivariance). *For all positive odd m, n ,*

$$F_m F_n = F_{mn}, \quad F_1 = \text{id}. \quad (\text{HA20})$$

For $\sigma(z) = -1/\bar{z}$ one has

$$F_n \sigma = \sigma F_n. \quad (\text{HA21})$$

For the usual scaling action of $\mathbb{C}^\times$ the exact formula is

$$F_n(\lambda z) = \theta_n(\lambda) F_n(z), \quad \theta_n(\lambda) = \lambda^{n\chi(n)}. \quad (\text{HA22})$$

Proof. For odd n , $p_n a(z) = (-1/z)^n = -1/z^n = a p_n(z)$ on the nonzero affine coordinate, hence on $\mathbb{P}^1$ by (HA19). Also $a^2 = \text{id}$ and $p_m p_n = p_{mn}$. The identity $\chi(mn) = \chi(m)\chi(n)$ follows by checking the two invertible residue classes modulo four. Consequently $e(mn) \equiv e(m) + e(n) \pmod{2}$, proving (HA20). Complex conjugation c commutes with a and each p_n , which have real coefficients; $\sigma = ac$ gives (HA21). Finally (HA22) follows by substitution in the two cases of (HA18). The homogeneous formulas prove all these equalities at 0 and ∞ as well. $\square$

19 A complete computation of the ordinary K_0 action

Here $K_0(\mathbb{P}_{\mathbb{C}}^1)$ is the Grothendieck ring of algebraic vector bundles, with addition from exact sequences and multiplication from tensor product. Let $L = [\mathcal{O}(1)]$ and $\eta = L - 1$.

Lemma 47.

$$K_0(\mathbb{P}_{\mathbb{C}}^1) = \mathbb{Z}[\eta]/(\eta^2), \quad [\mathcal{O}(d)] = 1 + d\eta \quad (d \in \mathbb{Z}). \quad (\text{HA23})$$

Every vector bundle E has

$$[E] = r + q\eta, \quad r = \text{rank } E, \quad q = \chi(E) - r. \quad (\text{HA24})$$

Proof. The Euler sequence on $\mathbb{P}^1$ is $0 \rightarrow \mathcal{O}(-1) \rightarrow \mathcal{O}^{\oplus 2} \rightarrow \mathcal{O}(1) \rightarrow 0$; its maps in homogeneous coordinates are $(-Y, X)$ and (X, Y) . The composition is zero, and checking the charts $X \neq 0$ and $Y \neq 0$ shows that its kernel and cokernel have the asserted form. Hence $L + L^{-1} = 2$ and $(L - 1)^2 = 0$.

For completeness, generation of K_0 by $1, L^{-1}$ follows without assuming that every bundle has already been split. On $\mathbb{P}^1 \times \mathbb{P}^1$, with projections q_1, q_2 , the diagonal is cut out by $X_1 Y_2 - Y_1 X_2$, giving the exact sequence

$$0 \longrightarrow \mathcal{O}(-1, -1) \longrightarrow \mathcal{O} \longrightarrow \mathcal{O}_{\Delta} \longrightarrow 0.$$

Tensor it by $q_1^* E$ and apply the alternating direct image under q_2 . The diagonal term is E . The other two terms have direct images $H^j(E) \otimes \mathcal{O}$ and $H^j(E(-1)) \otimes \mathcal{O}(-1)$ respectively. These identities can be checked by the two-chart Čech complex on the first factor, tensored with either indicated line bundle on the second; there are no terms in degrees greater than one. The alternating identity of the resulting long exact sequence gives

$$[E] = \chi(E)[\mathcal{O}] - \chi(E(-1))[\mathcal{O}(-1)]. \quad (\text{HA25})$$

Thus $1, L^{-1}$ generate K_0 , and so do $1, \eta$. The same two-chart Čech calculation gives $H^0(\mathcal{O}) = \mathbb{C}$, $H^1(\mathcal{O}) = 0$, and $H^0(\mathcal{O}(-1)) = H^1(\mathcal{O}(-1)) = 0$. Rank and Euler characteristic therefore map $[\mathcal{O}], [\mathcal{O}(-1)]$ to $(1, 1), (1, 0)$ in $\mathbb{Z}^2$. Their determinant is -1 , so there are no additional additive relations. This proves the ring presentation. For $d \geq 0$, $(1 + \eta)^d = 1 + d\eta$; for negative d use $(1 + \eta)^{-1} = 1 - \eta$. Finally rank sends $1, \eta$ to $1, 0$, while Euler characteristic sends them to $1, 1$ (the Euler sequence gives $\chi(\mathcal{O}(1)) = 2$). Formula (HA24) follows. $\square$

Theorem 48 (Exact ordinary Adams equality). *For every positive odd n ,*

$$F_n^*(1 + q\eta) = 1 + nq\eta, \quad F_n^*(r + q\eta) = r + nq\eta, \quad F_n^* = \psi^n \quad \text{on } K_0(\mathbb{P}_{\mathbb{C}}^1). \quad (\text{HA26})$$

Proof. The two homogeneous coordinates of (HA19) are sections of $\mathcal{O}(n)$ with no common zero. The universal tautological quotient defining the map therefore gives $F_n^*\mathcal{O}(1) \simeq \mathcal{O}(n)$. This can equally be checked by pulling back the transition ratio X/Y on the inverse images of the two standard charts, with their interchange in the second case of (HA19). Thus $F_n^*\eta = L^n - 1 = n\eta$, proving the two explicit formulas.

Define the Adams operations from the usual exterior-power operations by the formal identity

$$\frac{d}{dt} \log \lambda_t(x) = \sum_{n \geq 1} (-1)^{n-1} \psi^n(x) t^{n-1}, \quad \lambda_t(x+y) = \lambda_t(x) \lambda_t(y).$$

It shows additivity of each ψ^n . For a line-bundle class M , $\lambda_t(M) = 1 + Mt$, so $\psi^n(M) = M^n$. The additive generators $1, L^{-1}$ from (HA25) therefore have identical images under F_n^* and ψ^n , proving the equality on every class. In particular the holomorphic involution a acts trivially on this K_0 , because $a^*\mathcal{O}(1) \simeq \mathcal{O}(1)$. $\square$

This is the exact ordinary- K_0 statement justified by the maps. It proves a comparison on the receiving group in (HA23), rather than an identity of all additional structures which a bundle may carry.

20 What happens to equivariant weights and real structures

Let E be a $\mathbb{C}^\times$ -equivariant bundle for the usual scaling action on $\mathbb{P}^1$. Pullback by F_n acquires a specified equivariant structure using the group homomorphism θ_n in (HA22):

$$\lambda \cdot (z, v) = (\lambda z, \theta_n(\lambda) \cdot v), \quad v \in E_{F_n(z)}. \quad (\text{HA27})$$

Indeed the right-hand side belongs to $E_{F_n(\lambda z)}$ by (HA22), and associativity follows from multiplicativity of θ_n . If the character weights at 0 and ∞ are the multisets W_0, W_∞ , the exact transformed multisets are

$$(W'_0, W'_\infty) = \begin{cases} (nW_0, nW_\infty), & n \equiv 1 \pmod{4}, \\ (-nW_\infty, -nW_0), & n \equiv 3 \pmod{4}. \end{cases} \quad (\text{HA28})$$

The fibre identification is $(F_n^*E)_z = E_{F_n(z)}$; F_n fixes the poles in the first case and interchanges them in the second. A target character λ^m becomes $(\lambda^{n\chi(n)})^m$, which proves (HA28).

There is an explicit test showing that (HA26) cannot simply be promoted to the same assertion about this equivariant pullback. Give the trivial line bundle the fibre action of character λ . Under (HA27) its character becomes $\lambda^{n\chi(n)}$. Its equivariant Adams image has character λ^n . For $n \equiv 3 \pmod{4}$ these are different: restricting to the fixed point 0 detects the distinct Laurent monomials t^{-n} and t^n in the representation ring $R(\mathbb{C}^\times) = \mathbb{Z}[t, t^{-1}]$. Both forget to the identical class 1 in the ordinary K_0 used in (HA26).

Suppose now that a bundle is supplied with an actual antilinear lift $\tau : E_z \rightarrow E_{\sigma(z)}$ of the antipodal real structure. The pullback has the exact antilinear lift

$$\tau_n(z, v) = (\sigma(z), \tau v). \quad (\text{HA29})$$

Its target fibre is correct because $\sigma F_n(z) = F_n \sigma(z)$ by (HA21). If the supplied lift squares to $+\text{id}$ or $-\text{id}$, its pullback squares to the same operator. This is a construction from an existing lift; it does not conjure one from the class of the underlying vector bundle. For example an actual isomorphism $E \simeq \mathcal{O}(w)^{\oplus r}$ pulls back to $\mathcal{O}(nw)^{\oplus r}$, while the rank and degree calculation

alone does not specify a filtration, a polarization, a connection, or the equivariant fibre weights. Equations (HA27)–(HA29) specify precisely the further structures transported here.

The source’s discussion of real Hodge structures and Adams operations appears in [1, original CC.tex, lines 416–417] and in its section *Frobenius and Twistor Equivariance*. The calculation here supplies the exact ordinary- K_0 map, the character correction for the second residue class, and the transport of any supplied real lift. No equality with a programme cohomology group, no positivity theorem for a trace pairing, and no Riemann-hypothesis conclusion is asserted by these identities.

References

- [1] Alain Connes and Caterina Consani, *The Absolute Twistor Line and the Geometry of $\overline{\text{Spec } \mathbb{Z}}$* , arXiv:2609.00299v1 (2026), original author LaTeX source CC.tex. Versioned paper identity; original source archive. The exact author-source line locations used above refer to the SHA-256 edition identified at the beginning of this derivation.

21 The odd twistor dynamics on the original annulus object

We apply the concrete dynamics in Alain Connes and Caterina Consani, *The Absolute Twistor Line and the Geometry of $\overline{\text{Spec } \mathbb{Z}}$* , Proposition prop:twistor_frobenius, to the exact companion and annulus space of 033, SBT1–23. This section proves the receiving maps and their complete metric defects. Neither a spherical-algebra point functor nor an arithmetic cohomology comparison is silently substituted for these maps.

21.1 The actual circle map and its linear pullback

For every positive odd integer d put

$$\chi_d = (-1)^{(d-1)/2}, \quad m_d = \chi_d d, \quad f_d(z) = \chi_d z^{m_d} \quad (z \in \mathbb{C}^\times). \quad (\text{TOB1})$$

At 0 and ∞ use the unique continuous extension to $\mathbb{P}^1(\mathbb{C})$. Thus $f_d = z^d$ when $d \equiv 1 \pmod{4}$, and $f_d = -z^{-d}$ when $d \equiv 3 \pmod{4}$, exactly as in the source. For odd c, d , the equality $\chi_{cd} = \chi_c \chi_d$ and the oddness of m_c give

$$f_c(f_d(z)) = \chi_c \chi_d^{m_c} z^{m_c m_d} = \chi_{cd} z^{m_{cd}} = f_{cd}(z).$$

The formula extends at both poles. The real structure $\sigma(z) = -1/\bar{z}$ satisfies $f_d \sigma = \sigma f_d$: substituting either sign of χ_d proves this equality on $\mathbb{C}^\times$, and continuity proves it at the poles.

Give S^1 its probability Haar measure $dt/(2\pi)$, and write $e_j(z) = z^j$, $j \in \mathbb{Z}$. Define the pullback

$$V_d : L^2(S^1) \longrightarrow L^2(S^1), \quad (V_d \varphi)(z) = \varphi(f_d(z)), \quad V_d e_j = \chi_d^j e_{m_d j}. \quad (\text{TOB2})$$

It is an isometry: substitution in the integral of a trigonometric polynomial, or orthogonality of the displayed distinct basis vectors, proves the norm equality on their span; that span is dense, so the operator extends uniquely to all of L^2 . Pullback by the measurable degree- d circle covering is the same extension, since that covering preserves Haar measure. The adjoint and range projection are exactly

$$V_d^* e_j = \begin{cases} \chi_d^{j/m_d} e_{j/m_d}, & m_d \mid j, \\ 0, & m_d \nmid j, \end{cases} \quad V_d V_d^* e_j = \mathbf{1}_{d \mid j} e_j. \quad (\text{TOB3})$$

For $S e_j = e_{j+1}$, direct substitution yields

$$V_d S = \chi_d S^{m_d} V_d, \quad V_c V_d = V_{cd}. \quad (\text{TOB4})$$

These identities include all signs for the 3 (mod 4) case.

We use the same coefficient operator on E -valued distributions $E^{\mathbb{Z}}$, paired with finitely supported covectors:

$$(V_d x)_j = \begin{cases} \chi_d^{j/m_d} x_{j/m_d}, & m_d \mid j, \\ 0, & m_d \nmid j. \end{cases} \quad (\text{TOB5})$$

Every pairing is a finite sum; no convergence hypothesis is needed. Equation (TOB4) continues to hold by its coordinate calculation. On $\ell^2(\mathbb{Z}; E, G_N)$ the same map is an isometry.

21.2 An exact finite receiver with the original metric

Keep the original parameters and operators:

$$\begin{aligned} k \geq 17, \quad k \equiv 1 \pmod{4}, \quad q = (k+1)^2, \quad \delta, \gamma > 0, \quad q-1 \leq N \leq 2q, \\ \lambda_{ab} = (2b-k)\gamma - i(2a-k)\delta, \quad Q_k(y) = \prod_{a,b=0}^k (y - \lambda_{ab}), \end{aligned} \quad (\text{TOB6})$$

$$E = \mathbb{C}[y]/(Q_k), \quad M[v] = [yv], \quad F_p = e^{i(\log p)M}, \quad \tilde{F}_p = p^{k/2} F_p, \quad \langle v, w \rangle_N = v^* G_N w.$$

The positive matrix G_N is the full original metric at the stated source cutoff. The four cutoffs $q-1, q, 2q-1, 2q$ are included separately. The new integer d is neither a source cutoff nor the amplification parameter k . Let P_{ab} and ℓ_{ab} be the original Lagrange projectors and vectors. Thus

$$F_p \ell_{ab} = \omega_{ab} \ell_{ab}, \quad \omega_{ab} = p^{(2a-k)\delta} e^{i(2b-k)\gamma \log p}, \quad (\mathcal{J}_p v)_j = F_p^{-j} v. \quad (\text{TOB7})$$

The original embedding satisfies $S\mathcal{J}_p = \mathcal{J}_p F_p$ and $\text{ev}_0 \mathcal{J}_p = I$ by substitution.

Define

$$A_d = \chi_d F_p^{\chi_d}, \quad E_d = E^{\oplus d}, \quad G_{N,d} = \text{diag}(G_N, \dots, G_N), \quad \mathcal{W}_d = V_d \mathcal{J}_p. \quad (\text{TOB8})$$

For every integer ℓ ,

$$(\mathcal{W}_d v)_{d\ell} = A_d^{-\ell} v, \quad (\mathcal{W}_d v)_j = 0 \quad (d \nmid j), \quad S^d \mathcal{W}_d = \mathcal{W}_d A_d. \quad (\text{TOB9})$$

For $\chi_d = 1$ this is (TOB5) directly. For $\chi_d = -1$, $j/m_d = -\ell$, so the coefficient is $(-1)^{-\ell} F_p^{\ell} v = (-F_p^{-1})^{-\ell} v$. This proves both cases without changing any original coordinate.

Let the receiving companion and embedding be

$$\begin{aligned} C_d(v_0, \dots, v_{d-1}) &= (A_d v_{d-1}, v_0, \dots, v_{d-2}), \\ \mathcal{I}_d(v) &= \sum_{r=0}^{d-1} S^r \mathcal{W}_d v_r, \quad (\mathcal{I}_d v)_{d\ell+r} = A_d^{-\ell} v_r \quad (0 \leq r < d). \end{aligned} \quad (\text{TOB10})$$

Then

$$\begin{aligned} \text{ev}_{[0,d-1]} \mathcal{I}_d &= I_{E_d}, \quad S\mathcal{I}_d = \mathcal{I}_d C_d, \quad C_d^d = \text{diag}(A_d, \dots, A_d), \\ \text{im } \mathcal{I}_d &= \{x \in E^{\mathbb{Z}} : x_{j-d} = A_d x_j \text{ for all } j\}. \end{aligned} \quad (\text{TOB11})$$

The first identity follows at the d indicated indices. Shifting the sum in (TOB10), and using its last term (TOB9), proves the second. Following each block through d steps proves the third. In the displayed recurrence, division with remainder $j = d\ell + r$ forces $x_j = A_d^{-\ell} x_r$ for positive and negative ℓ , proving the last equality and its inverse. The image is also the smallest S - and S^{-1} -stable linear subspace containing $\mathcal{W}_d(E)$: its d shifts generate the image, and (TOB9), with invertible A_d , proves stability in both directions. In particular its dimension is exactly dq , with all original q labels retained.

The original metric is transported by the exact first-block evaluation:

$$\langle \mathcal{I}_d v, \mathcal{I}_d w \rangle_{d,N} = \sum_{r=0}^{d-1} v_r^* G_N w_r. \quad (\text{TOB12})$$

This is not the infinite sum of annulus norms. Each original copy has precisely its original norm; no averaging by d is performed. Direct subtraction gives the full defect

$$\begin{aligned} \|C_d v\|_{N,d}^2 - \|v\|_{N,d}^2 &= v_{d-1}^* (A_d^* G_N A_d - G_N) v_{d-1}, \\ \|p^{k/2} C_d v\|_{N,d}^2 - p^k \|v\|_{N,d}^2 &= p^k v_{d-1}^* (A_d^* G_N A_d - G_N) v_{d-1}. \end{aligned} \quad (\text{TOB13})$$

The cancellation concerns the unchanged first $d-1$ summands only; the last block is the complete remaining quadratic form. Equivalently, the singular values of C_d in $G_{N,d}$ consist of $(d-1)q$ copies of 1 and the q singular values of A_d in G_N . Indeed C_d is a unitary block permutation after the block diagonal operator $\text{diag}(I, \dots, I, A_d)$. This proves

$$\|C_d\|_{N,d} = \max\{1, \|A_d\|_N\}, \quad \|C_d^{-1}\|_{N,d} = \max\{1, \|A_d^{-1}\|_N\} \quad (d > 1). \quad (\text{TOB14})$$

For $d=1$, the corresponding norms are just $\|F_p\|_N$ and $\|F_p^{-1}\|_N$. Since $\delta > 0$, these latter norms already exceed 1, by the original eigenvectors of largest and smallest modulus, respectively.

For the last-block vector $v = (0, \dots, 0, \ell_{ab})$, the exact right side in (TOB13) is

$$p^k (p^{2\chi_d(2a-k)\delta} - 1) \|\ell_{ab}\|_N^2. \quad (\text{TOB15})$$

At every odd d , both signs occur as a varies. For either fixed residue class modulo four this expression is independent of d . Thus the original metric defect has not vanished in a growing covering degree.

21.3 All roots, labels, phases, and finite boundaries

Put $b_d = (1 - \chi_d)/2$. The eigenvectors and eigenvalues of C_d are explicitly

$$\begin{aligned} \zeta_{ab,t} &= \exp\left(\frac{\chi_d i(\log p) \lambda_{ab} + i\pi b_d + 2\pi i t}{d}\right), \quad 0 \leq t < d, \\ v_{ab,t} &= (\ell_{ab}, \zeta_{ab,t}^{-1} \ell_{ab}, \dots, \zeta_{ab,t}^{-(d-1)} \ell_{ab}), \\ C_d v_{ab,t} &= \zeta_{ab,t} v_{ab,t}, \quad |\zeta_{ab,t}| = p^{\chi_d(2a-k)\delta/d}. \end{aligned} \quad (\text{TOB16})$$

The first formula specifies the phase explicitly, without a choice of a new logarithm. Its d th power is the eigenvalue $\chi_d \omega_{ab}^{\chi_d}$ of A_d . This proves the second line by substitution in (TOB10). For each fixed (a, b) the d roots are distinct and the corresponding Vandermonde matrix is invertible. The original P_{ab} split E_d into their d -block spaces, so these dq vectors form a basis even if roots from different original labels have equal phases or eigenvalues. Their full Gram entries are

$$\langle v_{ab,t}, v_{ce,u} \rangle_{N,d} = \langle \ell_{ab}, \ell_{ce} \rangle_N \sum_{r=0}^{d-1} (\overline{\zeta_{ab,t}} \zeta_{ce,u})^{-r}. \quad (\text{TOB17})$$

All original off-diagonal metric entries remain in this formula.

The original annulus functions and operator are

$$h_j(\mathbf{x}) = p^{kj/2} (1 - p^{-1})^{-k/2} \mathbf{1}_{(p^j \mathbb{Z}_p^\times)^k}(\mathbf{x}), \quad \mathcal{T}_{p,k} = p^{k/2} \mathcal{U}_p^{\otimes k}, \quad \mathcal{U}_p F(x) = \sqrt{p} F(x/p). \quad (\text{TOB18})$$

Here Haar measure gives $\mu(\mathbb{Z}_p) = 1$. The support in h_j has measure $p^{-kj} (1 - p^{-1})^k$; these functions are orthonormal, and substitution proves $\mathcal{T}_{p,k} h_j = p^{k/2} h_{j+1}$. For an integer $H \geq 0$ define

$$I_{d,H} v = \sum_{j=-H}^H h_j \otimes (\mathcal{I}_d v)_j. \quad (\text{TOB19})$$

Its squared norm is exactly $\sum_{j=-H}^H \|A_d^{-[j/d]} v_{j \bmod d}\|_N^2$, where the remainder lies in $\{0, \dots, d-1\}$ also for negative j . The complete boundary is

$$\begin{aligned} B_{d,H}v &= (\mathcal{I}_{p,k} \otimes I)I_{d,H}v - I_{d,H}(p^{k/2}C_d v) \\ &= p^{k/2}(h_{H+1} \otimes (\mathcal{I}_d v)_H - h_{-H} \otimes (\mathcal{I}_d v)_{-H-1}), \\ \|B_{d,H}v\|^2 &= p^k(\|(\mathcal{I}_d v)_H\|_N^2 + \|(\mathcal{I}_d v)_{-H-1}\|_N^2). \end{aligned} \quad (\text{TOB20})$$

The shift relation (TOB11) cancels every common summation index; the two remaining annuli are disjoint, proving the norm formula. For $d > 2$, this boundary map need not be injective: it observes only the two indicated residue blocks. Its kernel is exactly the subspace whose entries at residues $H \bmod d$ and $(-H-1) \bmod d$ vanish. This retains a new boundary kernel that cannot be removed by citing the injective $d = 1$ boundary.

For the eigenvector $v_{ab,t}$, (TOB10) gives the full sequence $(\mathcal{I}_d v_{ab,t})_j = \zeta_{ab,t}^{-j} \ell_{ab}$. With $r = |\zeta_{ab,t}|$, equation (TOB20) becomes

$$\frac{\|B_{d,H}v_{ab,t}\|^2}{\|I_{d,H}v_{ab,t}\|^2} = p^k \frac{r^{-2H} + r^{2H+2}}{\sum_{j=-H}^H r^{-2j}} \rightarrow p^k |r^2 - 1| = p^k \left| p^{2\chi_d(2a-k)\delta/d} - 1 \right|. \quad (\text{TOB21})$$

For $r > 1$ divide numerator and denominator by r^{2H} ; for $r < 1$ divide by r^{-2H} . The resulting geometric series prove the displayed limits. They are positive for every finite d , since k is odd. As d tends to infinity they tend to zero, but the exact d -step return (TOB11) and original defect (TOB15) remain. These statements concern different, explicitly related operators; the root estimate has not been substituted for its unchanged return.

For a nonzero $v \in E_d$ define

$$R_{p,d}(v) = \frac{\delta \log p}{d} \max\{|2a - k| : P_{ab}v_r \neq 0 \text{ for some } r\}. \quad (\text{TOB22})$$

Then

$$\lim_{H \rightarrow \infty} \frac{1}{H} \log \max_{|j| \leq H} \|(\mathcal{I}_d v)_j\|_N = R_{p,d}(v) > 0. \quad (\text{TOB23})$$

For the upper bound expand each of the d blocks with the original P_{ab} and use $\lfloor |j/d| \rfloor \leq H/d + 1$. The finite sum is bounded by a constant times $\exp(HR_{p,d})$. For the lower bound choose a nonzero projected block attaining the maximum. Choose $j = d\ell + r$ within $[-H, H]$ with the sign of ℓ that makes $A_d^{-\ell}$ grow on that block and with $|\ell| \geq H/d - 2$. Projection bounds its norm from below by the growing projected norm divided by $\|P_{ab}\|_N$. The constants are independent of H and their logarithms disappear after division by H . This proves (TOB23), retaining the original metric. Consequently the image contains no nonzero polynomial-growth sequence for any fixed finite d .

21.4 Exact observation-kernel and support transport

Retain the original isometric section $J : B \rightarrow E$, $P_B = JJ^\dagger$, $Q = I - P_B$, and $K = \ker \Lambda = QE$. Let $J_d = \text{diag}(J, \dots, J)$, and define $\widehat{C}_d = J_d^\dagger C_d J_d$. It is (TOB10) with the last corner A_d replaced by $J^\dagger A_d J$. Therefore

$$C_d J_d - J_d \widehat{C}_d : (b_0, \dots, b_{d-1}) \mapsto (Q A_d J b_{d-1}, 0, \dots, 0). \quad (\text{TOB24})$$

This follows by multiplying each block, using $J^\dagger J = I$. Applying the injective $\mathcal{I}_d$ gives the full corresponding defect for the distribution comparison. More explicitly,

$$p^{k/2} S \mathcal{I}_d J_d - \mathcal{I}_d J_d (p^{k/2} \widehat{C}_d) = p^{k/2} \mathcal{I}_d (Q A_d J \text{pr}_{d-1}, 0, \dots, 0). \quad (\text{TOB25})$$

The finite-cutoff version is the sum of this compressed-kernel term and the two annulus terms (TOB20); thus neither contribution is lost by compression.

For the full bounded distributive lattice L , every vector space V here has its synchronized carrier $V^\uparrow = \{(0, \lambda) : \lambda \in L\} \cup \{(v, 1_L) : v \in V\}$. A linear map T lifts as $(v, \lambda) \mapsto (Tv, \lambda)$. Substitution in the addition and scalar multiplication laws proves linearity over $G_L(\mathbb{C})$. Applying this to (TOB10)–(TOB11) proves that the first-block evaluation recovers every $(0, \lambda)$ and every nonzero original vector. In (TOB24)–(TOB25), zero amplitude in the exact defect is still a labelled zero. All these identities therefore have exact full- L lifts, including the observation-kernel inclusion and the growing finite receiver; no free-semimodule basis change is assumed.

21.5 The complete boundary window restores the original exponent

A single boundary in (TOB20) reads at most two residue blocks. The exact joint map through a complete window reads every block. For an integer $h \geq 0$ put

$$\mathcal{B}_{d,h}v = (B_{d,dh}v, B_{d,dh+1}v, \dots, B_{d,dh+d-1}v). \quad (\text{TOB26})$$

Give its target the direct-sum Hilbert norm. For $H = dh + r$, equation (TOB10) gives $x_H = A_d^{-h}v_r$ and $x_{-H-1} = A_d^{h+1}v_{d-1-r}$. Summing the exact two-annulus identity proves

$$\|\mathcal{B}_{d,h}v\|^2 = p^k \sum_{r=0}^{d-1} (\|A_d^{-h}v_r\|_N^2 + \|A_d^{h+1}v_r\|_N^2). \quad (\text{TOB27})$$

In particular this joint map is injective for every h : every summand is nonnegative, and invertibility of A_d implies that a zero sum forces each $v_r = 0$.

For comparison with exactly the same complete blocks, define

$$L_{d,h}v = \sum_{\ell=-h}^h \sum_{r=0}^{d-1} h_{d\ell+r} \otimes A_d^{-\ell}v_r, \quad \|L_{d,h}v\|^2 = \sum_{\ell=-h}^h \sum_{r=0}^{d-1} \|A_d^{-\ell}v_r\|_N^2. \quad (\text{TOB28})$$

This is the explicitly specified annulus interval $[-dh, dh + d - 1]$, not a redefinition of the symmetric truncation (TOB19). Its norm formula follows from orthonormality of all the different annuli in (TOB18). For any nonzero v with each block in the same original one-dimensional subspace $\mathbb{C}\ell_{ab}$, put $R = |\omega_{ab}|^{\chi_d} = p^{\chi_d(2a-k)\delta}$. Equations (TOB27)–(TOB28) yield

$$\frac{\|\mathcal{B}_{d,h}v\|^2}{\|L_{d,h}v\|^2} = p^k \frac{R^{-2h} + R^{2h+2}}{\sum_{\ell=-h}^h R^{-2\ell}} \longrightarrow p^k |R^2 - 1| = p^k \left| p^{2\chi_d(2a-k)\delta} - 1 \right| > 0. \quad (\text{TOB29})$$

The common factor $\sum_r \|v_r\|_N^2$ cancels from this ratio; no metric was changed. The geometric-series proof after (TOB21) applies to R . The formula includes every $v_{ab,t}$ of (TOB16). It is independent of the covering degree within each residue class modulo four. Therefore the full boundary window, with all residue classes restored, retains the original nonzero radial defect exactly, while the one-step eigenvalue radius in (TOB16) has its explicitly proved $1/d$ exponent. Equations (TOB10), (TOB26) and (TOB28) are the connecting maps between these two quantities.

22 Exact prime traces and determinants of the odd twistor receiver

The input dynamics is due to Alain Connes and Caterina Consani, *The Absolute Twistor Line and the Geometry of $\overline{\text{Spec } \mathbb{Z}}$* , Proposition `prop:twistor_frobenius`; its original author file `CC.tex`, lines 1159–1191, gives the two odd residue classes and their compatibility with the real structure. The source’s spatial coproduct restriction is at lines 462–465. The maps used here are the explicit receiving maps (TOB1)–(TOB11), constructed from the original companion in 033, SBT1–SBT23. The calculation concerns one fixed prime p . No product over primes is introduced.

22.1 The complete cyclic receiver and its labelled traces

Retain $k, \delta, \gamma, q, N, E, G_N$ and all original Lagrange labels (a, b) from (TOB6)–(TOB7). In particular

$$\omega_{ab} = p^{(2a-k)\delta} e^{i(2b-k)\gamma \log p}, \quad 0 \leq a, b \leq k, \quad q = (k+1)^2. \quad (\text{TPT1})$$

For a positive odd integer d , put $\chi = \chi_d = (-1)^{(d-1)/2}$, $A = \chi F_p^\chi$, and

$$C = C_d : E^{\oplus d} \longrightarrow E^{\oplus d}, \quad C(v_0, \dots, v_{d-1}) = (Av_{d-1}, v_0, \dots, v_{d-2}), \quad \tilde{C} = p^{k/2} C. \quad (\text{TPT2})$$

The metric is the unchanged direct sum of d copies of G_N . Trace and determinant below are the ordinary trace and determinant of these actual finite-dimensional complex-linear operators; their definitions do not replace this metric or its defects. Let $P_{ab}^{(d)} = \text{diag}(P_{ab}, \dots, P_{ab})$.

For every integer n , including negative integers, the exact formula is

$$\text{tr}(C^n) = \begin{cases} 0, & d \nmid n, \\ d\chi^m \sum_{a,b=0}^k \omega_{ab}^{\chi^m}, & n = dm. \end{cases} \quad (\text{TPT3})$$

For $n = 0$ the second line means dq . The stronger labelled identity is

$$\text{tr}(P_{ab}^{(d)} C^n) = \begin{cases} 0, & d \nmid n, \\ d\chi^m \omega_{ab}^{\chi^m}, & n = dm. \end{cases} \quad \text{tr}(P_{ab}^{(d)} \tilde{C}^n) = p^{kn/2} \text{tr}(P_{ab}^{(d)} C^n). \quad (\text{TPT4})$$

To prove these statements, let $\iota_r : E \rightarrow E^{\oplus d}$ be the inclusion into block r and let π_r be its projection. Iteration of (TPT2) sends every block to the block whose index is increased by n modulo d . If $d \nmid n$, no diagonal block is present, so the trace is zero. If $n = dm$, the exact identity $C^d = \text{diag}(A, \dots, A)$ gives A^m in every diagonal block. This also holds for negative m , because A and C are invertible. Finally $P_{ab} A^m = \chi^m \omega_{ab}^{\chi^m} P_{ab}$ and $\text{tr} P_{ab} = 1$. This proves (TPT3)–(TPT4) without an orthogonality assumption on the Lagrange basis.

For later comparison define the original, fully labelled power sum

$$t_p(m) = \text{tr}(F_p^m) = \sum_{a=0}^k \sum_{b=0}^k p^{m(2a-k)\delta} e^{im(2b-k)\gamma \log p} \quad (m \in \mathbb{Z}). \quad (\text{TPT5})$$

Thus the surviving trace at time $n = dm$ samples original exponent $\chi m = \chi n/d$, not exponent n . Its multiplicity is exactly d . For the weighted receiver the same trace is

$$\text{tr}(\tilde{C}^{dm}) = d\chi^m p^{kdm/2} t_p(\chi m) = d\chi^m p^{k(d-\chi)m/2} \text{tr}(\tilde{F}_p^{\chi m}), \quad \tilde{F}_p = p^{k/2} F_p. \quad (\text{TPT6})$$

All signs, original phases, and both powers of p are retained.

22.2 Characteristic polynomial and the two local determinants

The characteristic polynomials and local determinant polynomials are

$$\det(zI - C) = \det(z^d I - A) = \prod_{a,b=0}^k (z^d - \chi \omega_{ab}^\chi), \quad (\text{TPT7})$$

$$\det(zI - \tilde{C}) = \prod_{a,b=0}^k (z^d - \chi p^{kd/2} \omega_{ab}^\chi), \quad (\text{TPT8})$$

$$D_{p,d}(u) := \det(I - uC) = \prod_{a,b=0}^k (1 - \chi \omega_{ab}^\chi u^d), \quad (\text{TPT9})$$

$$\tilde{D}_{p,d}(u) := \det(I - u\tilde{C}) = \prod_{a,b=0}^k (1 - \chi p^{kd/2} \omega_{ab}^\chi u^d). \quad (\text{TPT10})$$

Indeed $P_{ab}^{(d)} E^{\oplus d}$ is an invariant d -dimensional space, and C on it is the cyclic companion with scalar corner $\chi \omega_{ab}^\chi$. Its eigenvalues are all d roots of that nonzero scalar, with the labelled eigenvectors of (TOB16). The roots within this block are distinct. The original projectors decompose the full space as the direct sum of these invariant blocks, so the products above include the correct multiplicities even when roots from different original labels coincide. Multiplication by $p^{k/2}$ gives (TPT8); putting the eigenvalues in $\det(I - uC)$ gives (TPT9)–(TPT10).

Write, with the indicated original operator retained,

$$D_p^\pm(v) = \det(I - vF_p^{\pm 1}), \quad \tilde{D}_p^\pm(v) = \det(I - v\tilde{F}_p^{\pm 1}). \quad (\text{TPT11})$$

Here D_p^+ is the original $\det(I - vF_p)$, and D_p^- is its inverse-operator comparison. Equations (TPT9)–(TPT10) give the exact substitutions

$$\begin{aligned} D_{p,a}(u) &= D_p^\chi(\chi u^d), \\ \tilde{D}_{p,a}(u) &= D_p^\chi(\chi p^{kd/2} u^d) = \tilde{D}_p^\chi(\chi p^{k(d-\chi)/2} u^d). \end{aligned} \quad (\text{TPT12})$$

For $d \equiv 1 \pmod{4}$, the argument of $\tilde{D}_p^+$ is $p^{k(d-1)/2} u^d$. For $d \equiv 3 \pmod{4}$, the argument of $\tilde{D}_p^-$ is $-p^{k(d+1)/2} u^d$.

There is also an exact relation between the two original labelled spectra. Define the involution $R : E \rightarrow E$ on the original basis by

$$R\ell_{ab} = \ell_{k-a, k-b}, \quad R^2 = I, \quad \omega_{k-a, k-b} = \omega_{ab}^{-1}, \quad RF_p R = F_p^{-1}. \quad (\text{TPT13})$$

The scalar identity follows directly from (TPT1). Applying both operators in the last identity to each ℓ_{ab} proves it. Consequently $D_p^- = D_p^+$ and $t_p(-m) = t_p(m)$ as scalar invariants. These equalities are mediated by the explicit label permutation R ; they do not identify an original labelled eigenline with its reflected one, and they do not assert that R is unitary for G_N . The weighted counterpart is

$$R\tilde{F}_p R = p^k \tilde{F}_p^{-1}, \quad \tilde{D}_p^-(v) = \tilde{D}_p^+(p^{-k}v). \quad (\text{TPT14})$$

This follows by multiplying (TPT13) by $p^{k/2}$; the displayed p^k must not be dropped.

For completeness, the product of all original eigenvalues is 1: the sums of $2a - k$ and $2b - k$ over their full ranges are zero in (TPT1). Since $q = (k+1)^2$ is even and d is odd,

$$\det F_p = 1, \quad \det C = \det A = 1, \quad \det \tilde{C} = p^{kdq/2}, \quad D_p^+(v) = v^q D_p^+(v^{-1}) \quad (v \neq 0). \quad (\text{TPT15})$$

The determinant of a d -block cyclic companion is $(-1)^{(d-1)q} \det A$, which here equals $\det A$. For the final equality, multiply $\prod_{a,b} (1 - v\omega_{ab})$ by writing each factor as $-v\omega_{ab}(1 - v^{-1}\omega_{ab}^{-1})$, and use (TPT13).

22.3 Formal logarithmic series and their precise convergence

In the formal power-series ring $\mathbb{C}[[u]]$, the logarithm of a series with constant coefficient 1 is defined by its zero constant coefficient and the usual formal derivative. Thus no analytic branch

is needed for the following identities:

$$-\log D_{p,d}(u) = \sum_{m \geq 1} \frac{\chi^m t_p(\chi m)}{m} u^{dm}, \quad (\text{TPT16})$$

$$-\log \tilde{D}_{p,d}(u) = \sum_{m \geq 1} \frac{\chi^m p^{kdm/2} t_p(\chi m)}{m} u^{dm}, \quad (\text{TPT17})$$

$$-u \frac{D'_{p,d}(u)}{D_{p,d}(u)} = d \sum_{m \geq 1} \chi^m t_p(\chi m) u^{dm} = \sum_{a,b=0}^k \frac{d \chi \omega_{ab}^\chi u^d}{1 - \chi \omega_{ab}^\chi u^d}, \quad (\text{TPT18})$$

$$-u \frac{\tilde{D}'_{p,d}(u)}{\tilde{D}_{p,d}(u)} = d \sum_{m \geq 1} \chi^m p^{kdm/2} t_p(\chi m) u^{dm} = \sum_{a,b=0}^k \frac{d \chi p^{kd/2} \omega_{ab}^\chi u^d}{1 - \chi p^{kd/2} \omega_{ab}^\chi u^d}. \quad (\text{TPT19})$$

To prove them, expand each factor in (TPT9)–(TPT10) with $-\log(1-x) = \sum_{m \geq 1} x^m/m$, a valid formal identity, and then differentiate term by term. Equivalently, for either receiver $B = C$ or $B = \tilde{C}$,

$$\begin{aligned} -\log \det(I - uB) &= \sum_{n \geq 1} \frac{\text{tr}(B^n)}{n} u^n, \\ -\frac{d}{du} \log \det(I - uB) &= \text{tr}(B(I - uB)^{-1}) = \sum_{n \geq 1} \text{tr}(B^n) u^{n-1}. \end{aligned} \quad (\text{TPT20})$$

Substitution of (TPT3) or (TPT6) into (TPT20) retains both the d -fold trace multiplicity and the factor $1/(dm)$; their product is the exact coefficient $1/m$ in (TPT16)–(TPT17).

The exact radii of these Taylor series about zero are

$$\mathcal{R}_C = p^{-k\delta/d}, \quad \mathcal{R}_{\tilde{C}} = p^{-k/2-k\delta/d}. \quad (\text{TPT21})$$

Both logarithmic series and logarithmic-derivative series converge absolutely on the corresponding open disk. On its boundary the logarithmic series converges conditionally precisely at the points which are not zeros of the determinant; at its zeros it diverges. The logarithmic-derivative Taylor series diverges at every point of that boundary. Neither series converges outside the closed disk.

Here is a proof including boundary behaviour. The eigenvalues of C are those of (TOB16), so their largest modulus is $p^{k\delta/d}$; the labels $a = 0, k$ ensure that this maximum is attained. Multiplication by $p^{k/2}$ gives the second radius. For a fixed receiver B , list its eigenvalues with algebraic multiplicity as β_j . For $|u| < \rho(B)^{-1}$ every $|u\beta_j| < 1$, giving absolute convergence by a finite sum of scalar geometric or logarithmic series. On $|u| = \rho(B)^{-1}$, the eigenvalues of smaller modulus still give absolutely convergent series. Each eigenvalue of maximal modulus gives $\sum_{n \geq 1} (u\beta_j)^n/n$. If $u\beta_j \neq 1$, its partial sums are bounded geometric sums and summation by parts proves convergence. If $u\beta_j = 1$, it contributes the harmonic series with positive multiplicity; all other terms converge, so it cannot be cancelled. This proves the claimed convergence and divergence for the logarithm.

To show that the convergence at the other boundary points is not absolute, group equal maximal phases into distinct unit complex numbers $\alpha_1, \dots, \alpha_s$ with positive integer multiplicities $c_1, \dots, c_s$. Put $b_n = \sum_j c_j \alpha_j^n$. Finite geometric summation gives

$$\lim_{M \rightarrow \infty} \frac{1}{M} \sum_{n=1}^M |b_n|^2 = \sum_{j=1}^s c_j^2 > 0.$$

Since $|b_n| \leq \sum_j c_j$, the average of $|b_n|$ has a positive lower bound. The smaller-modulus contributions tend to zero exponentially and do not change that bound. Summation by parts then shows that the series of absolute values divided by n diverges. The same positive mean square

shows that the terms of the logarithmic-derivative series fail to tend to zero on the boundary. If $|u| > \rho(B)^{-1}$, factor the dominant growth $(|u|\rho(B))^n$ out of its coefficients. The same finite-phase mean-square argument provides a subsequence bounded away from zero; exponentially smaller spectral terms cannot cancel it. Along this subsequence even division by n does not make the terms tend to zero. This proves the asserted exterior divergence. Finally every nearest determinant zero gives a genuine pole of $-D'/D$ whose residue is minus its positive multiplicity. Thus (TPT21) is also the exact analytic Taylor radius, without a possible pole cancellation.

22.4 The exact prime variable and reverse recovery

Only for this fixed prime define

$$Z_{p,d}(s) = \tilde{D}_{p,d}(p^{-s})^{-1}, \quad p^{-s} = e^{-s \log p}. \quad (\text{TPT22})$$

It is a meromorphic function of s . In the precise open half-plane $\text{Re } s > k/2 + k\delta/d$, its branch with the indicated convergent series satisfies

$$\begin{aligned} \log Z_{p,d}(s) &= \sum_{m \geq 1} \frac{\chi^m t_p(\chi m)}{m} p^{-dm(s-k/2)}, \\ -\frac{Z'_{p,d}(s)}{Z_{p,d}(s)} &= d \log p \sum_{m \geq 1} \chi^m t_p(\chi m) p^{-dm(s-k/2)}. \end{aligned} \quad (\text{TPT23})$$

The second identity follows by the chain rule from (TPT19), including the sign $d(p^{-s})/ds = -(\log p)p^{-s}$. The original prime powers which appear are indexed by dm , the retained mass is $p^{kdm/2}$, and the derivative contributes $d \log p$. The support of these prime-power indices is not replaced by all positive integers.

Put $b_d = (1 - \chi)/2$. Every zero of the denominator in (TPT22), with its original label retained, is

$$s_{ab,\ell} = \frac{k}{2} + \frac{\chi(2a-k)\delta}{d} + i \left(\frac{\chi(2b-k)\gamma}{d} + \frac{\pi b_d + 2\pi\ell}{d \log p} \right), \quad \ell \in \mathbb{Z}. \quad (\text{TPT24})$$

Indeed substituting this expression into $\chi \omega_{ab}^x p^{-d(s-k/2)} = 1$ proves the equality; conversely every solution differs by one indicated logarithmic period. If labels produce the same point, their positive multiplicities add. These are poles of the specified finite local function $Z_{p,d}$, not a statement about a global zeta function.

The trace information has an exact reverse map. For every integer m ,

$$t_p(\chi m) = \frac{\chi^m}{d} \text{tr}(C^{dm}) = \frac{\chi^m p^{-kdm/2}}{d} \text{tr}(\tilde{C}^{dm}). \quad (\text{TPT25})$$

For a specified original projector, even its eigenvalue is recovered by

$$\omega_{ab} = \left(\frac{\chi}{d} \text{tr}(P_{ab}^{(d)} C^d) \right)^x = \left(\frac{\chi p^{-kd/2}}{d} \text{tr}(P_{ab}^{(d)} \tilde{C}^d) \right)^x. \quad (\text{TPT26})$$

These follow directly from (TPT3)–(TPT4), using $\chi^2 = 1$ and the nonzero original eigenvalues.

Even without the labelled projectors, the first q nonzero-time trace slots determine the characteristic polynomial of A . Set $a_m = d^{-1} \text{tr}(C^{dm})$ and recursively define

$$e_0 = 1, \quad j e_j = \sum_{m=1}^j (-1)^{m-1} e_{j-m} a_m \quad (1 \leq j \leq q). \quad (\text{TPT27})$$

Then $\det(I - vA) = \sum_{j=0}^q (-1)^j e_j v^j$. For proof, the identity $-vD'_A(v)/D_A(v) = \sum_{m \geq 1} a_m v^m$ follows from its finite eigenvalue product. Multiplying by D_A and comparing coefficients gives

exactly (TPT27), which recursively determines all coefficients. The eigenvalues of $A = \chi F_p^\chi$ then recover the original eigenvalue multiset by $\omega = (\chi\alpha)^\chi$. Trace data alone do not recover an ordering of its labels or the original metric G_N ; (TPT26) uses the actual retained projectors when such labelled recovery is intended.

Finally the exact operator recovery, stronger than spectral recovery, uses the original block inclusions and projections:

$$\begin{aligned} \pi_r C^d_{\iota_r} &= A, & \pi_r \tilde{C}^d_{\iota_r} &= p^{kd/2} A, \\ F_p &= \chi(\pi_r C^d_{\iota_r})^\chi, & \tilde{F}_p &= p^{k/2} \chi(p^{-kd/2} \pi_r \tilde{C}^d_{\iota_r})^\chi \quad (0 \leq r < d). \end{aligned} \tag{TPT28}$$

Substituting $C^d = \text{diag}(A, \dots, A)$ proves every equality. The inclusions remain isometries for the original block metric, and the projections remain their adjoints. Thus the $1/d$ appearing in the labelled radii (TPT24) has exact return maps: it retains the original operator, its inverse twist, all mass factors, all labelled traces, and the metric defects of (TOB13). The finite local determinant calculation therefore does not discard the complete return while retaining only its contracted root radii.

Tensor sectors retained by the original operators

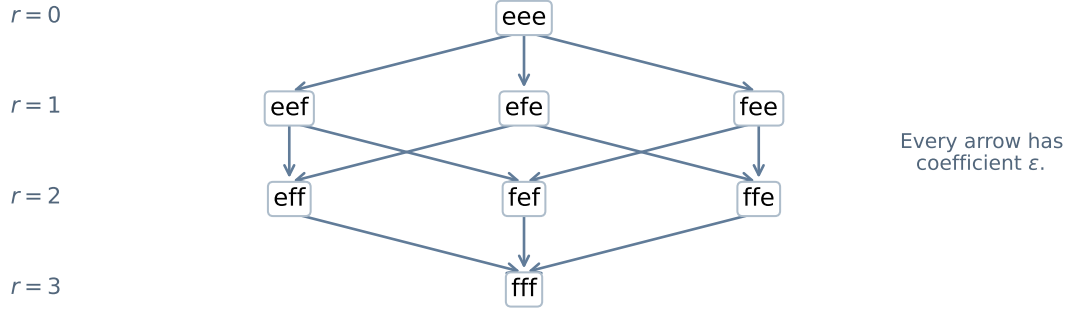
$$A_p \ell_{ab} = \alpha_{ab} \ell_{ab}, \quad \alpha_{ab} = (2a - k)\delta \log p + i(2b - k)\gamma \log p$$

$$\begin{array}{ccc} \ell_{ab} & & \ell_{k-a, k-b} \\ \alpha_{ab} & & -\alpha_{ab} \\ & \searrow \quad \swarrow & \\ & A_p^{[2]}(\ell_{ab} \otimes \ell_{k-a, k-b}) = 0 & \end{array}$$

$q = (k+1)^2$ independent vectors; uncentred exponent $k \log p$ is retained.

Nilpotent tensor block: $Re = \varepsilon f, Rf = 0, \varepsilon > 0$

$$X_3 = R \otimes I \otimes I + I \otimes R \otimes I + I \otimes I \otimes R$$



$$\ker X_3 = \text{span}\{ffe - fef, fef - eff, fff\}$$

Words denote the displayed tensor factors. The original orthonormal e, f and their tensor metric are retained; the three kernel vectors need not be orthogonal.

$$\dim \ker R^{[t]} = \sum_{j=0}^t \binom{t}{j} (q-2)^{t-j} \binom{j}{\lfloor j/2 \rfloor}$$

Complete receiver proofs: OLR23–31. Lattice lifts: OLR16–17.
Source locality: Connes–Consani, arXiv:2606.06604v1, Definition defnloc.
These local tensor sectors do not impose a purity theorem on the original object.

Figure 5: Two tensor mechanisms in the unchanged spectral receiver. Opposite original labels give zero centred exponent; the uncentred mass is still $k \log p$. In the three active nilpotent slots every arrow carries the actual coefficient ε . Its full kernel and the general tensor formula are proved in OLR23–31.

23 Locality of the original finite spectral receiver

Sources and scope of the calculation. Alain Connes and Caterina Consani, *On the Absolute Geometry of Spec $\mathbb{Z}$ and the Fargues–Fontaine curve*, arXiv:2606.06604, original author file `FF.tex`, Definition `defnloc` and the following extension proposition (lines 693–774), define locality by continuity on $H_p = \mathbb{Z}[1/p]$ for its specified place topology. Their archimedean complex points are in `prop:archimedean_points_C` (lines 1465–1500), and their p -adic complex points in `prop:nonarchimedean_points_C` (lines 1643–1659). Those original TeX passages were read in this derivation. ACF23–ACF26 retain their scalar local families on the full support base. The matrix analogues needed below are proved here, including their Jordan terms, rather than inferred from eigenvalue moduli alone.

The original receiver, metric, odd twistor return and complete boundary are TOB6–TOB29 in `TWISTOR_TO_ORIGINAL_BOUNDARY.tex`; its labelled prime traces and reverse maps are TPT1–TPT28 in `TWISTOR_PRIME_TRACE.tex`. Their original programme input is 033, SBT1–SBT23. The rank-one nilpotent and original observation maps below are TDP4, TDP11–TDP13 and TDP23 in the retained `TAU_DELiGNE_PRIME_MEMORY.tex`. These local proof files are programme derivations, not author papers by Connes or Consani.

23.1 The exact matrix extension criterion

Let V be a finite-dimensional complex vector space and $A \in \text{End}_{\mathbb{C}}(V)$. Give $\text{GL}(V)$ its usual complex topology. Fix a rational prime p and put

$$\rho_A(h) = \exp(hA), \quad h \in H_p = \mathbb{Z}[1/p]. \quad (\text{OLR1})$$

This is an additive-group homomorphism into $\text{GL}(V)$: the absolutely convergent exponential series for the commuting matrices hA, tA gives $\rho_A(h+t) = \rho_A(h)\rho_A(t)$, and $\rho_A(-h)$ is its inverse.

The following three statements are equivalent:

$$\begin{aligned} \rho_A \text{ extends continuously to } \mathbb{Q}_p; \quad \exp(p^r A) = I_V \text{ for some integer } r \geq 0; \\ A \text{ is semisimple and every eigenvalue belongs to } 2\pi i\mathbb{Z}[1/p]. \end{aligned} \quad (\text{OLR2})$$

Here semisimple means diagonalizable over $\mathbb{C}$; it is not a claim that the original metric makes the eigenbasis orthogonal. The same statements are equivalent to continuity of OLR1 on H_p for its p -adic topology. The extension is unique.

Here is an elementary proof of necessity that retains nilpotents. For any induced matrix norm, the neighbourhood $\mathcal{U} = \{T : \|T - I_V\| < 1/2\}$ contains no nontrivial subgroup. Indeed, if a subgroup lies in $\mathcal{U}$, all positive and negative integer powers of any one of its elements T lie there. For every eigenvalue λ of T this implies $|\lambda^m - 1| < 1/2$ for all integers m . Positive or negative powers exclude $|\lambda| \neq 1$. For $|\lambda| = 1$, the same bound implies $\Re \lambda^m > 7/8$. If $\lambda \neq 1$, the geometric sum gives $n^{-1} \sum_{m=1}^n \lambda^m \rightarrow 0$, contradicting that inequality. Thus every eigenvalue is one, and $T = I_V + D$ with D nilpotent. If $D \neq 0$, choose the largest j with $D^j \neq 0$. Some entry of $T^m = \sum_{a=0}^j \binom{m}{a} D^a$ is a polynomial of degree j with nonzero leading coefficient, so its powers are unbounded. This again contradicts membership in $\mathcal{U}$. Hence $T = I_V$.

Continuity at zero of an extension provides an integer $r \geq 0$ with $\rho_A(p^r \mathbb{Z}_p) \subset \mathcal{U}$; this image is a subgroup, so it is trivial. In particular $\exp(p^r A) = I_V$. The identical argument applied to the subgroup $p^r \mathbb{Z}$ inside H_p proves this conclusion from continuity on H_p alone.

On a Jordan primary space write $A = \alpha I + D$, $D^s = 0$. The equality $\exp(p^r A) = I$ first implies $\exp(p^r \alpha) = 1$, so $\alpha = 2\pi i a / p^r$ for an integer a . It then requires $\exp(p^r D) = I$. But

$$\exp(p^r D) - I = p^r D \left(\sum_{j=0}^{s-1} \frac{(p^r D)^j}{(j+1)!} \right). \quad (\text{OLR3})$$

The factor in parentheses is I plus a nilpotent and is invertible, with inverse given by a finite geometric sum. Thus $D = 0$. Conversely finitely many eigenvalues in $2\pi i\mathbb{Z}[1/p]$ have a common denominator p^r , and diagonalization proves $\exp(p^r A) = I$.

Sufficiency and uniqueness can be proved directly without assuming completeness of a particular matrix norm on the open set $\mathrm{GL}(V)$. The inclusion induces a group isomorphism

$$H_p/p^r\mathbb{Z} \xrightarrow{\simeq} \mathbb{Q}_p/p^r\mathbb{Z}_p. \quad (\text{OLR4})$$

Its kernel is $H_p \cap p^r\mathbb{Z}_p = p^r\mathbb{Z}$. It is onto because truncating the p -adic digits of any x just before exponent r gives a finite element $h_x \in H_p$ with $x - h_x \in p^r\mathbb{Z}_p$. Define $\tilde{\rho}_A(x) = \exp(h_x A)$. Changing h_x changes it by an integer multiple of p^r , so this definition is independent of the choice. OLR4 proves the group law. It is constant on the open cosets of $p^r\mathbb{Z}_p$, hence continuous. Truncated digit expansions also prove density of H_p in $\mathbb{Q}_p$, and agreement on a dense subset of a Hausdorff target proves uniqueness. This completes OLR2.

Using the original ACF24 convention $\psi_p(x) = \exp(2\pi i\{x\}_p)$, the explicit spectral formula is

$$A = \sum_j 2\pi i b_j P_j \implies \tilde{\rho}_A(x) = \sum_j \psi_p(b_j x) P_j, \quad b_j \in \mathbb{Z}[1/p]. \quad (\text{OLR5})$$

For $h \in H_p$, $b_j h - \{b_j h\}_p$ is an integer, proving agreement with the original exponential. No phases or eigenspace projectors have been changed. For comparison, every A , including every Jordan nilpotent, gives the continuous archimedean representation $t \mapsto \exp(tA)$ on $\mathbb{R}$. Thus OLR2 describes the intersection of that matrix family with the p -adic local family; it does not exclude the archimedean local points admitted in the Connes–Consani source.

23.2 The maximal admissible subspace and its exact defect

For arbitrary A define $D_r(A) = \exp(p^r A) - I$. Then

$$\begin{aligned} V_{p\text{-loc}}(A) &= \bigcup_{r \geq 0} \ker D_r(A) \\ &= \bigoplus_{\alpha \in \mathrm{Spec}(A) \cap 2\pi i H_p} \ker(A - \alpha I). \end{aligned} \quad (\text{OLR6})$$

The kernels increase because $T^p - I = (T - I) \sum_{j=0}^{p-1} T^j$. On an α -primary space, if $e^{p^r \alpha} \neq 1$, then $D_r(A)$ is a nonzero scalar times I plus a nilpotent, so is invertible. If $e^{p^r \alpha} = 1$, OLR3 proves that its kernel is exactly the kernel of the primary nilpotent, the actual eigenspace. This proves OLR6. Finitely many allowable α have a common denominator, so the union is already $\ker D_{r_0}(A)$ for one $r_0 \geq 0$ and remains that kernel for all $r \geq r_0$. The restriction to this subspace satisfies OLR2. Conversely every A -invariant subspace on which the restriction is p -adic local is contained in OLR6 by OLR2. Moreover every linear subspace invariant under $\rho_A(H_p)$ is A -invariant: for its vector v the vectors $p^n(\rho_A(p^{-n}) - I)v$ belong to that subspace and converge to Av by the exponential series; finite-dimensional subspaces are closed. Thus maximality applies to the original group action itself.

The defect is the original quotient, with its exact connecting map:

$$\begin{aligned} 0 \longrightarrow V_{p\text{-loc}}(A) \longrightarrow V \xrightarrow{\pi} \mathfrak{D}_p(A) := V/V_{p\text{-loc}}(A) \longrightarrow 0, \\ \overline{D}_r : \mathfrak{D}_p(A) \xrightarrow{\simeq} \mathrm{im} D_r(A), \quad \overline{D}_r([v]) = D_r(A)v \quad (r \geq r_0). \end{aligned} \quad (\text{OLR7})$$

It is well-defined and injective because its kernel before taking the quotient is exactly OLR6; it is onto its displayed image. All maps intertwine A and $\exp(hA)$ because $D_r(A)$ commutes with them. A good-eigenvalue Jordan block can have a local quotient even though the entire block is not local. OLR7 retains that extension through the original nilpotent; it does not claim a decomposition into a local summand and a complementary invariant summand.

23.3 Application to every label and cutoff of the original receiver

Retain exactly

$$k \geq 17, \quad k \equiv 1 \pmod{4}, \quad q = (k+1)^2, \quad \delta, \gamma > 0, \quad q-1 \leq N \leq 2q,$$

$$\lambda_{ab} = (2b-k)\gamma - i(2a-k)\delta, \quad Q_k(y) = \prod_{a,b=0}^k (y - \lambda_{ab}), \quad (\text{OLR8})$$

$$E = \mathbb{C}[y]/(Q_k), \quad M[v] = [yv], \quad A_p = i(\log p)M, \quad F_p = e^{A_p}, \quad \langle v, w \rangle_N = v^* G_N w.$$

The four cutoffs $q-1, q, 2q-1, 2q$ remain individual members of this range. No replacement of G_N is used. The roots are distinct, since equality forces equality of real and imaginary parts and then of both integer labels. Let ℓ_{ab} be the original Lagrange polynomial with value one at λ_{ab} and zero at the other roots, and let P_{ab} be the projector onto its coefficient. Evaluation proves that these vectors form a basis and $M\ell_{ab} = \lambda_{ab}\ell_{ab}$. Thus the complete exponent and its locality are

$$A_p \ell_{ab} = ((2a-k)\delta \log p + i(2b-k)\gamma \log p) \ell_{ab},$$

$$E_{p\text{-loc}}(A_p) = 0, \quad \mathfrak{D}_p(A_p) = E,$$

$$D_r(A_p) = \sum_{a,b} (\omega_{ab}^{p^r} - 1) P_{ab}, \quad D_r(A_p)^{-1} = \sum_{a,b} (\omega_{ab}^{p^r} - 1)^{-1} P_{ab}, \quad (\text{OLR9})$$

$$\omega_{ab} = p^{(2a-k)\delta} e^{i(2b-k)\gamma \log p}.$$

Indeed k is odd, so $2a-k \neq 0$. Every exponent in OLR9 has nonzero real part, excluding every label from OLR6. Every $\omega_{ab}^{p^r} - 1$ is nonzero for the same reason, proving the displayed inverse. This is the entire original defect, not only an observation of a selected eigenvalue.

For $v = \sum_i c_i \ell_i$ and $w = \sum_j d_j \ell_j$, the complete original Gram form of the defect is

$$\langle D_r v, D_r w \rangle_N = \sum_{i,j} \bar{c}_i d_j \overline{(\omega_i^{p^r} - 1)} (\omega_j^{p^r} - 1) \langle \ell_i, \ell_j \rangle_N. \quad (\text{OLR10})$$

All off-diagonal Gram entries survive. On one original vector its literal squared factor is

$$\|D_r \ell_{ab}\|_N^2 = [p^{2p^r(2a-k)\delta} - 2p^{p^r(2a-k)\delta} \cos(p^r(2b-k)\gamma \log p) + 1] \|\ell_{ab}\|_N^2. \quad (\text{OLR11})$$

The topology failure therefore keeps the full phase as well as the radial exponent. More precisely $D_r v \not\rightarrow 0$ for any nonzero v . If a coefficient with $2a-k > 0$ is nonzero, applying its fixed bounded projector to $D_r v$ proves divergence of that component. Otherwise every occupied label has $2a-k < 0$, and the finite spectral sum gives $D_r v \rightarrow -v \neq 0$.

The uncentred action is also retained:

$$\tilde{A}_p = A_p + \frac{k \log p}{2} I, \quad \tilde{F}_p = e^{\tilde{A}_p} = p^{k/2} F_p. \quad (\text{OLR12})$$

Its admissible subspace is exactly the sum of those original $\mathbb{C}\ell_{ab}$ for which $k/2 + (2a-k)\delta = 0$ and $(2b-k)\gamma \log p / (2\pi) \in H_p$. This follows directly from OLR6 and keeps the mass factor. For the stated general hypothesis $\delta > 0$ one must retain both conditions; discarding $k/2$ would be a different calculation.

23.4 The original nilpotent and observation-kernel return

Retain the programme vectors e, f and the literal coefficient $\epsilon > 0$ with $e^\dagger f = 0$ and $\|e\|_N = \|f\|_N = 1$. For $R = \epsilon f e^\dagger$, $R^2 = 0$, and the complete locality calculation is

$$E_{p\text{-loc}}(R) = \ker R = e^\perp, \quad D_r(R) = p^r \epsilon f e^\dagger, \quad \|D_r(R)v\|_N^2 = p^{2r} \epsilon^2 |e^\dagger v|^2. \quad (\text{OLR13})$$

The exponential is exactly $I + hR$, proving the first two statements by OLR6, and the original unit norm of f proves the third. Thus the nonzero nilpotent is not removed by the fact that its sole eigenvalue is zero. This calculation concerns this specified R ; it does not replace $M + R$ by M when the two fail to commute. For any such sum, OLR3–OLR7 apply to its actual primary spaces.

For the original isometric section $J : B \rightarrow E$, retain $P_B = JJ^\dagger$, $Q = I - P_B$, and $K = QE = \ker \Lambda$. Put $U_r = J^\dagger F_p^{p^r} J$. The complete visible and kernel defect is

$$\begin{aligned} D_r(A_p)Jb &= J(U_r - I)b + QF_p^{p^r}Jb, \\ \|D_r(A_p)Jb\|_N^2 &= \|(U_r - I)b\|_B^2 + \|QF_p^{p^r}Jb\|_N^2. \end{aligned} \quad (\text{OLR14})$$

Insert $I = P_B + Q$ and use $QJ = 0$ for the first identity. The two terms are orthogonal in the original metric for the second. The combined map on the right is injective: its vanishing gives $D_r(A_p)Jb = 0$, then OLR9 gives $Jb = 0$, hence $b = 0$. No such conclusion about the visible term alone is asserted. For arbitrary h, t the original compression has the exact product defect

$$J^\dagger \rho_{A_p}(h + t)J - J^\dagger \rho_{A_p}(h)JJ^\dagger \rho_{A_p}(t)J = J^\dagger \rho_{A_p}(h)Q\rho_{A_p}(t)J. \quad (\text{OLR15})$$

Insertion of $I = P_B + Q$ proves it. Thus treating compression as a group representation would remove precisely the displayed return.

23.5 Full lattice support and the exact lifted correspondence

For the full bounded distributive lattice L , keep $V_L^\uparrow = \{(0, \lambda) : \lambda \in L\} \cup \{(v, 1_L) : v \in V\}$ over $G_L(\mathbb{C})$, and $T^\uparrow(v, \lambda) = (Tv, \lambda)$ for every linear T . The component laws give $(TU)^\uparrow = T^\uparrow U^\uparrow$ and $(T + U)^\uparrow = T^\uparrow + U^\uparrow$ directly. In particular the zero linear map lifts to $\varepsilon_V(v, \lambda) = (0, \lambda)$, not to the map constant at $\tau = (0, 0_L)$. The exact support-retaining version of OLR7 is

$$\begin{aligned} D_r(A)^\uparrow &= \overline{D_r}^\uparrow \pi^\uparrow, & (D_r(A)^\uparrow)^{-1}(Z_L) &= V_{p\text{-loc}}(A)_L^\uparrow \quad (r \geq r_0), \\ Z_L &= \{(0, \lambda) : \lambda \in L\}, & \pi^\uparrow(v, \lambda) &= ([v], \lambda). \end{aligned} \quad (\text{OLR16})$$

Equality is checked on each (v, λ) ; membership in the inverse image is exactly $D_r(A)v = 0$ with its support unchanged. The fibre of $\pi^\uparrow$ over $(0, \lambda)$ for $\lambda < 1_L$ is the singleton $(0, \lambda)$; over $(0, 1_L)$ it is $V_{p\text{-loc}}(A) \times \{1_L\}$. For the original A_p , OLR9 says that the inverse image in OLR16 is precisely Z_L , and the defect lift is an automorphism on the whole synchronized carrier. In particular no support label is erased.

With L discrete and the carrier a subspace of $V \times L$, the lifted representation is continuous whenever its matrix representation is continuous: its action is $(h, (v, \lambda)) \mapsto (\rho_A(h)v, \lambda)$. Every zero state is fixed. This does not turn a zero-valued monoid character $T^h \mapsto z_\lambda$ into a group-valued local point: the fixed vector of a representation and the value of a character are different specified maps, and z_λ is not a unit.

The original free-coordinate support information has its exact correspondence as well for a retained basis of positive dimension. In that basis write a free vector as $(v, \boldsymbol{\mu})$, where $v_i \neq 0$ forces $\mu_i = 1_L$. The idempotent synchronization is

$$P_{\text{syn}}(v, \boldsymbol{\mu}) = (v, \bigvee_i \mu_i), \quad P_{\text{syn}}^{-1}(v, \lambda) = \{(v, \boldsymbol{\mu}) : \bigvee_i \mu_i = \lambda, v_i \neq 0 \Rightarrow \mu_i = 1_L\}. \quad (\text{OLR17})$$

This is the matrix pair $(I, \mathbf{1})$ on free coordinates: its amplitude is I and its support sends every output coordinate to the full finite join. Its image is exactly the synchronized carrier, and its square is itself, proving OLR17. Taking the inverse image of OLR16 retains all masks with $v \in V_{p\text{-loc}}(A)$. In the original case these are all zero-amplitude masks, with arbitrary $\boldsymbol{\mu}$. No group action with the free-module identity has been inferred from an action on this retract. For

dimension zero the free zero-coordinate semimodule has one element, with empty support join 0_L , whereas the synchronized carrier of the zero complex vector space still contains every $(0, \lambda)$, $\lambda \in L$. Thus OLR17 is not a surjection onto that latter carrier in dimension zero unless L has one element. The original receiver and its positive tensor powers have dimensions $q > 0$ and $q^t > 0$, so the stated correspondence applies to them, including their zero-amplitude fibres.

23.6 Frobenius, the complete twistor return, and the boundary

Source Frobenius acts by precomposition $h \mapsto mh$, so its operator parameter is exactly $A \mapsto mA$. For $m = p^j$, $j \in \mathbb{Z}$, OLR6 gives $V_{p\text{-loc}}(p^j A) = V_{p\text{-loc}}(A)$, because multiplication by p^j preserves H_p and the eigenspace kernels. OLR5 gives the corresponding p -adic parameters $b_j \mapsto p^j b_j$. For a general positive integer m the exact condition is $m\alpha \in 2\pi i H_p$; primes dividing m other than p can enlarge the admissible subspace. For $m = 0$ the action is identity on the whole space. These statements retain every source Frobenius.

Keep every positive odd d , $\chi_d = (-1)^{(d-1)/2}$, and the original receiver

$$C_d(v_0, \dots, v_{d-1}) = (\chi_d F_p^{\chi_d} v_{d-1}, v_0, \dots, v_{d-2}), \quad G_{N,d} = \text{diag}(G_N, \dots, G_N). \quad (\text{OLR18})$$

Let ι_s, π_s be the literal block inclusion and projection. The identity $C_d^d = \text{diag}(\chi_d F_p^{\chi_d}, \dots)$ proves the exact recovery of every locality defect:

$$\begin{aligned} D_r(A_p) &= \chi_d^{p^r} \pi_s C_d^{\chi_d p^r} \iota_s - I_E \\ &= \chi_d^{p^r} p^{-k\chi_d p^r/2} \pi_s (p^{k/2} C_d)^{\chi_d p^r} \iota_s - I_E. \end{aligned} \quad (\text{OLR19})$$

Negative powers in the $\chi_d = -1$ case are legitimate because the receiver is invertible. The powers of the sign multiply to one, and the scalar mass in the second equality is explicitly cancelled by its displayed inverse factor; it has not been omitted. Thus the original nonzero defect has an exact return through every odd receiver and every original block.

The explicit logarithm on the full TOB16 eigenbasis has eigenvalues

$$\nu_{ab,t} = \frac{\chi_d^i (\log p) \lambda_{ab} + i\pi(1 - \chi_d)/2 + 2\pi i t}{d}, \quad 0 \leq t < d. \quad (\text{OLR20})$$

Its exponential is C_d . The eigenvectors $(\ell_{ab}, e^{-\nu_{ab,t}} \ell_{ab}, \dots, e^{-(d-1)\nu_{ab,t}} \ell_{ab})$ form a basis: within each label the d distinct roots give an invertible Vandermonde matrix. Every real part is $\chi_d(2a - k)\delta \log p/d \neq 0$. OLR6 therefore gives zero admissible subspace for this entire dq -dimensional exponential family, despite its contracted real parts. The complete Gram entries remain TOB17, not a diagonal substitute for $G_{N,d}$.

The same return is visible in the complete original boundary. Put $T_d = \chi_d F_p^{\chi_d}$ and keep the TOB26–TOB28 maps $\mathcal{B}_{d,H}$ and $L_{d,H}$, for $H \geq 0$. Their exact norms are

$$\begin{aligned} \|\mathcal{B}_{d,H} v\|^2 &= p^k \sum_{s=0}^{d-1} (\|T_d^{-H} v_s\|_N^2 + \|T_d^{H+1} v_s\|_N^2), \\ \|L_{d,H} v\|^2 &= \sum_{j=-H}^H \sum_{s=0}^{d-1} \|T_d^{-j} v_s\|_N^2. \end{aligned} \quad (\text{OLR21})$$

For completeness these maps use the original orthonormal annuli

$$h_j(\mathbf{x}) = p^{kj/2} (1 - p^{-1})^{-k/2} \mathbf{1}_{(p^j \mathbb{Z}_p^\times)^k}(\mathbf{x}).$$

The complete finite receiver has coordinates $x_{d\ell+s} = T_d^{-\ell} v_s$. The boundary at $dH + s$ consists exactly of $p^{k/2} (h_{dH+s+1} \otimes x_{dH+s} - h_{-dH-s} \otimes x_{-dH-s-1})$. Its two annuli are disjoint; summing

over s gives the first line. The interval $[-dH, dH + d - 1]$ gives the second line by the same orthogonality. Invertibility of T_d makes the joint boundary injective. On blocks in $\mathbb{C}\ell_{ab}$, finite geometric summation gives

$$\frac{\|\mathcal{B}_{d,H}v\|^2}{\|L_{d,H}v\|^2} \longrightarrow p^k \left| p^{2\chi_d(2a-k)\delta} - 1 \right| > 0. \quad (\text{OLR22})$$

Indeed the ratio is $p^k(R^{-2H} + R^{2H+2})/\sum_{j=-H}^H R^{-2j}$, $R = p^{\chi_d(2a-k)\delta} \neq 1$; divide by the largest geometric term to obtain the limit. OLR19 and OLR21 are the explicit maps linking this complete boundary with locality's unchanged original prime return.

23.7 All tensor labels and the nonzero local tensor sector

For $t \geq 1$, retain the entire original tensor object $E^{\otimes t}$ with metric $G_N^{\otimes t}$ and generator $A_p^{[t]} = \sum_{j=1}^t I^{\otimes(j-1)} \otimes A_p \otimes I^{\otimes(t-j)}$. The commuting summands give $e^{hA_p^{[t]}} = \rho_{A_p}(h)^{\otimes t}$. On the original vector $\ell_{a_1b_1} \otimes \cdots \otimes \ell_{a_tb_t}$ its exponent is

$$(2\sum_j a_j - tk)\delta \log p + i(2\sum_j b_j - tk)\gamma \log p. \quad (\text{OLR23})$$

The tensor of the original Lagrange bases is a basis, so OLR6 proves that the exact admissible projector is

$$\Pi_{p,t} = \sum_{\substack{\sum a_j = tk/2 \\ (2\sum b_j - tk)\gamma \log p / (2\pi) \in H_p}} P_{a_1b_1} \otimes \cdots \otimes P_{a_tb_t}. \quad (\text{OLR24})$$

An empty index set means the zero projector. These spectral projectors need not be orthogonal for $G_N^{\otimes t}$; that metric remains unchanged, with entries the products of the original Gram entries. The entire complementary defect also has an explicit receiver: $[v] \mapsto (I - \Pi_{p,t})v$ is an isomorphism from $E^{\otimes t}/\text{im } \Pi_{p,t}$ to $\ker \Pi_{p,t}$, with inverse sending a vector to its quotient class. Its kernel before passing to the quotient is exactly the admissible range; $\Pi_{p,t}$ commutes with the tensor generator, so both maps intertwine the complete exponential action. This is a spectral direct sum, not an orthogonal replacement of the metric. OLR7 and OLR16 apply to all its retained components. Since k is odd, the admissible space is zero for odd t . For even t , define $n_{t,j} = [x^j](1+x+\cdots+x^k)^t$. Counting the independent a - and b -tuples gives its exact dimension

$$\dim \text{im } \Pi_{p,t} = n_{t,tk/2} \sum_{j=0}^{tk} n_{t,j} \mathbf{1}_{(2j-tk)\gamma \log p / (2\pi) \in H_p}. \quad (\text{OLR25})$$

Each coefficient counts precisely the tuples with the indicated sum by expansion of the polynomial product, proving the count.

In particular at $t = 2$ the original q independent vectors

$$\ell_{ab} \otimes \ell_{k-a,k-b}, \quad 0 \leq a, b \leq k, \quad (\text{OLR26})$$

have exponent zero and hence the trivial local character. There can be additional labels in OLR24 with nonzero allowed phase, and OLR25 retains them exactly. Thus the local tensor sector is nonzero although the individual receiver has no nonzero local amplitude. This calculation cannot be replaced by tensoring the zero admissible subspace of the individual factor. The involution $R_0\ell_{ab} = \ell_{k-a,k-b}$ satisfies $R_0MR_0 = -M$ by evaluation on each basis vector, so $R_0F_pR_0 = F_p^{-1}$. It identifies explicitly the reverse label in every vector OLR26. Their Gram entry at labels i, j is $\langle \ell_i, \ell_j \rangle_N \langle R_0\ell_i, R_0\ell_j \rangle_N$; no unproved isometry of R_0 in the original metric is assumed.

All mass factors also tensor. For the uncentred generator

$$\tilde{A}_p^{[t]} = A_p^{[t]} + \frac{tk \log p}{2} I, \quad (\text{OLR27})$$

the first condition in OLR24 becomes $tk/2 + (2\sum a_j - tk)\delta = 0$; its phase condition is unchanged. In particular the vectors OLR26 have uncentred exponent $k \log p$, so their centred trivial characters do not make the uncentred characters p -adic local.

For a general primary tensor block, the same calculation retains all nilpotents. If its factor generators are $\alpha_j I + D_j$, the tensor generator is $(\sum_j \alpha_j)I + D_{\text{tot}}$ with $D_{\text{tot}} = \sum_j I \otimes D_j \otimes I$ nilpotent. OLR3 proves that its admissible space is exactly $\ker D_{\text{tot}}$ when $\sum_j \alpha_j \in 2\pi i H_p$, and zero otherwise. For the original rank-one R , its tensor generator $R^{[t]} = \sum_j I \otimes R \otimes I$ satisfies

$$e^{hR^{[t]}} = (I + hR)^{\otimes t}, \quad (R^{[t]})^{t+1} = 0, \quad (R^{[t]})^t = t! R^{\otimes t} \neq 0, \quad E_{p\text{-loc}}(R^{[t]}) = \ker R^{[t]}. \quad (\text{OLR28})$$

Commutativity of the factor operators gives the exponential; every repeated slot in a product vanishes since $R^2 = 0$, and the $t!$ orders using every slot give the last nonzero power. OLR3 gives the final kernel. All these tensor maps admit the support-preserving lifts OLR16–OLR17 on their actual full carriers.

Its kernel dimension can be calculated exactly, with the original rank-two support projector $\Pi = ee^\dagger + ff^\dagger$:

$$\dim \ker R^{[t]} = \sum_{j=0}^t \binom{t}{j} (q-2)^{t-j} \binom{j}{\lfloor j/2 \rfloor}. \quad (\text{OLR29})$$

Here R and ϵ are precisely those in OLR13. To prove the formula, let $W = \text{im } \Pi = \mathbb{C}e \oplus \mathbb{C}f$ and $U = \ker \Pi$, an orthogonal decomposition for the unchanged G_N . The original vectors e, f are orthonormal; $Re = \epsilon f$, $Rf = 0$, and $R|_U = 0$. Tensoring the exact decomposition $E = W \oplus U$ gives a direct orthogonal sum indexed by the set of slots occupied by W . A sector with j such slots has operator X_j on $W^{\otimes j}$, the sum of the original R over these slots, tensored with the identity on the remaining U factors.

Grade $W^{\otimes j}$ by the number r of factors equal to f . Its grade- r subspace V_r has dimension $\binom{j}{r}$. The operator X_j maps V_r to V_{r+1} , with coefficient ϵ on every replacement of one e by f . Its original-metric adjoint $Y_j = X_j^\dagger$ reverses those replacements with the same real coefficient ϵ . Terms in different slots commute. In one slot $(R^\dagger R - RR^\dagger)e = \epsilon^2 e$ and $(R^\dagger R - RR^\dagger)f = -\epsilon^2 f$. Consequently

$$(Y_j X_j - X_j Y_j)|_{V_r} = \epsilon^2 (j - 2r) I_{V_r}, \quad \|X_j v\|^2 - \|Y_j v\|^2 = \epsilon^2 (j - 2r) \|v\|^2. \quad (\text{OLR30})$$

Thus $X_j : V_r \rightarrow V_{r+1}$ is injective when $r < j/2$. The isometry interchanging e and f in each of the j slots interchanges X_j and Y_j and sends grade $r + 1$ to grade $j - r - 1$. For $r \geq j/2$ that latter grade is less than $j/2$, so $Y_j : V_{r+1} \rightarrow V_r$ is injective by the result just proved. Its adjoint $X_j : V_r \rightarrow V_{r+1}$ is then surjective. The total kernel dimension therefore telescopes:

$$\dim \ker X_j = \sum_{r=\lfloor j/2 \rfloor}^j \left(\binom{j}{r} - \binom{j}{r+1} \right) = \binom{j}{\lfloor j/2 \rfloor}. \quad (\text{OLR31})$$

For odd j , the two middle grades have equal dimension and the injective intervening map is an isomorphism, consistent with this sum. For $j = 0$, $W^{\otimes 0} = \mathbb{C}$ and $X_0 = 0$, so the same formula gives one. Each j -slot sector has the further multiplicity $(q-2)^{t-j}$ and there are $\binom{t}{j}$ such sectors. Their direct sum proves OLR29. No coefficient ϵ , tensor slot, or original metric term was removed.

The original single receiver, its nilpotent channels, its complete observation return, its odd twistor receivers, and its tensor local sector have thus been placed in the specified local-point framework by exact maps. The calculation selects no locality condition that Connes–Consani impose on all points: their archimedean points still include every original exponential above. It therefore neither proves RH by imposing p -locality nor produces an RH counterexample by exhibiting a nonlocal receiver.

24 Complete mixed-tensor nilpotent ladders in the original metric

Programme input and scope. This calculation extends OLR13 and OLR28–OLR31 of OPERATOR_LOCALITY_PRIME_RECEIVER.tex, and TDP11, TDP12 and TDP23 of the retained programme proof TAU_DELiGNE_PRIME_MEMORY.tex. Those are the actual input proofs; no theorem from SGA or from Deligne is invoked here. The original source has $k \geq 17$, $k \equiv 1 \pmod{4}$, $q = (k+1)^2$ and each literal cutoff $q-1 \leq N \leq 2q$, with its positive metric G_N on the q -dimensional complex space E . Keep the actual vectors $e, f \in E$ and coefficient $\epsilon > 0$:

$$\begin{aligned} \langle v, w \rangle_N &= v^* G_N w, & \|e\|_N &= \|f\|_N = 1, & e^\dagger f &= 0, \\ R &= \epsilon f e^\dagger, & \Pi &= \epsilon e^\dagger + f f^\dagger, & W &= \text{im } \Pi = \mathbb{C}e \oplus \mathbb{C}f, & U &= \ker \Pi, & E &= W \perp U. \end{aligned} \quad (\text{OLM1})$$

Thus $Re = \epsilon f$, $Rf = 0$, $R|_U = 0$, and $\dim U = q - 2$. Every adjoint below uses this metric or its original tensor product. The grade introduced below counts slots occupied by f . It is not identified with an arithmetic weight or a purity filtration.

24.1 Every active sector and its original adjoint

For a positive integer t let

$$X = R^{[t]} = \sum_{a=1}^t I^{\otimes(a-1)} \otimes R \otimes I^{\otimes(t-a)}, \quad E_t = E^{\otimes t}, \quad G_t = G_N^{\otimes t}. \quad (\text{OLM2})$$

For each $J \subseteq \{1, \dots, t\}$ let E_J be the tensor subspace with factor W at slots in J and factor U elsewhere. Then

$$E_t = \bigoplus_{J \subseteq \{1, \dots, t\}}^\perp E_J, \quad E_J \simeq W^{\otimes j} \otimes U^{\otimes(t-j)}, \quad j = |J|. \quad (\text{OLM3})$$

The isomorphism moves the W factors into increasing order before the U factors, without changing the order within either family. It is an isometry, since the tensor inner product is the product of the factor inner products. Distinct sectors are orthogonal because some slot pairs W with U . Expanding the identity $I_E = \Pi + (I_E - \Pi)$ in every slot proves exhaustiveness. Every sector reduces both X and $X^\dagger$, and the restriction of X is $X_j \otimes I$, where X_j is the sum of the same original R over the j active slots.

For $0 \leq s \leq j$, let $V_{j,s} \subset W^{\otimes j}$ be the span of the original tensor words containing s copies of f and $j-s$ copies of e . These words are orthonormal already by OLM1. Therefore the $V_{j,s}$ are mutually orthogonal and have dimension $\binom{j}{s}$. Put $Y_j = X_j^\dagger$. The operator X_j replaces each occurrence of e in turn by ϵf , and Y_j replaces each f in turn by ϵe . Terms from different slots commute. In one slot $R^\dagger R - R R^\dagger = \epsilon^2(e e^\dagger - f f^\dagger)$, so the complete commutator is

$$[Y_j, X_j]|_{V_{j,s}} = \epsilon^2(j-2s)I_{V_{j,s}}, \quad \|X_j v\|^2 - \|Y_j v\|^2 = \epsilon^2(j-2s)\|v\|^2. \quad (\text{OLM4})$$

The second equality follows by pairing the first with v . In particular $X_j : V_{j,s} \rightarrow V_{j,s+1}$ is injective for $s < j/2$. The isometry interchanging e and f in every slot interchanges X_j and Y_j . It sends $V_{j,s}$ to $V_{j,j-s}$. Thus $Y_j : V_{j,s+1} \rightarrow V_{j,s}$ is injective for $s \geq j/2$, and its adjoint X_j is surjective there. These facts use the original ϵ ; it has not been replaced by one.

24.2 Actual primitive spaces and their orthogonal projections

For $0 \leq r \leq \lfloor j/2 \rfloor$ define

$$\mathcal{P}_{j,r} = \ker(Y_j : V_{j,r} \rightarrow V_{j,r-1}), \quad m_{j,r} = \dim \mathcal{P}_{j,r} = \binom{j}{r} - \binom{j}{r-1}, \quad \binom{j}{-1} = 0. \quad (\text{OLM5})$$

For $r = 0$, interpret $V_{j,-1} = 0$. For $r \geq 1$, the map $X_j : V_{j,r-1} \rightarrow V_{j,r}$ is injective by OLM4. Its adjoint Y_j is consequently onto $V_{j,r-1}$: the orthogonal complement of its image is the kernel of X_j , which is zero. Rank and nullity prove the dimension in OLM5. These are concrete kernels of the actual adjoint, not postulated primitive classes.

The corresponding orthogonal projector is explicit. With all restrictions to the displayed grades understood, put

$$P_{j,0} = I_{V_{j,0}}, \quad P_{j,r} = I_{V_{j,r}} - X_j(Y_j X_j|_{V_{j,r-1}})^{-1} Y_j \quad (r \geq 1). \quad (\text{OLM6})$$

The inverse exists because $\langle v, Y_j X_j v \rangle = \|X_j v\|^2 > 0$ for nonzero $v \in V_{j,r-1}$. The subtracted operator is self-adjoint and idempotent, by multiplication and $Y_j = X_j^\dagger$. It fixes $\text{im}(X_j|_{V_{j,r-1}})$ and vanishes on $\ker Y_j$. These two subspaces are orthogonal complements, so OLM6 is exactly the claimed projector.

24.3 Exact ladder lengths, lowering coefficients and norms

Fix j, r as above and write $d = j - 2r$. For every $u \in \mathcal{P}_{j,r}$ one has

$$Y_j X_j^a u = \epsilon^2 a(d - a + 1) X_j^{a-1} u \quad (a \geq 1). \quad (\text{OLM7})$$

To prove it, expand $[Y_j, X_j^a] = \sum_{b=0}^{a-1} X_j^b [Y_j, X_j] X_j^{a-1-b}$. The vector to which the commutator in summand b is applied has grade $r + a - 1 - b$. OLM4 makes its coefficient $\epsilon^2(j - 2r - 2a + 2 + 2b)$. Summing those a coefficients gives $\epsilon^2 a(d - a + 1)$, and the term $X_j^a Y_j u$ is zero. This proves OLM7 without deleting any power of ϵ .

Define the literal factors

$$s_{d,a} = \epsilon^{2a} a! \frac{d!}{(d-a)!} \quad (0 \leq a \leq d), \quad s_{d,0} = 1.$$

Then the original inner products are

$$\boxed{\langle X_j^a u, X_j^b v \rangle = \mathbf{1}_{a=b} s_{d,a} \langle u, v \rangle, \quad u, v \in \mathcal{P}_{j,r}, \quad 0 \leq a, b \leq d.} \quad (\text{OLM8})$$

Different powers have different grades, hence are orthogonal. At equal powers, move one X_j from the first entry to the second by its actual adjoint and use OLM7. Induction gives $s_{d,a} = \epsilon^2 a(d - a + 1) s_{d,a-1}$, the displayed factorial product. It is strictly positive for $a \leq d$. At the next power OLM7 gives $Y_j X_j^{d+1} u = 0$, and therefore $\|X_j^{d+1} u\|^2 = \langle X_j^d u, Y_j X_j^{d+1} u \rangle = 0$. Thus a nonzero primitive u has the exact ladder

$$u, X_j u, \dots, X_j^d u \neq 0, \quad X_j^{d+1} u = 0, \quad \text{length } d + 1 = j - 2r + 1. \quad (\text{OLM9})$$

For its unaltered power vectors the raising coefficient is one, while its lowering coefficient at level a is exactly $\epsilon^2 a(d - a + 1)$ from OLM7. The metric entries OLM8 record why these are adjoints. No power vector has been divided by its norm, and no source metric has been changed.

Ladders arising from different primitive grades are orthogonal. It suffices to examine equal total grades: let $r < r'$ and $r + a = r' + b$, so $a > b$. Moving X_j^b from the second vector to the first gives $Y_j^b X_j^a u$, a scalar multiple of $X_j^{a-b} u$ by OLM7. Its pairing with $v \in \mathcal{P}_{j,r'}$ is zero because the adjoint expression contains $Y_j^{a-b} v = 0$. Ladders from orthogonal primitive vectors in the same primitive space are orthogonal by OLM8.

They fill every original grade. At grade s the nonzero summands are exactly $X_j^{s-r} \mathcal{P}_{j,r}$ for $0 \leq r \leq \min(s, j - s)$, and their dimensions sum to

$$\sum_{r=0}^{\min(s, j-s)} m_{j,r} = \binom{j}{\min(s, j-s)} = \binom{j}{s} = \dim V_{j,s}. \quad (\text{OLM10})$$

Their orthogonality and the nonzero factors in OLM8 prove independence, so this dimension identity proves equality. Equivalently the map

$$\mathcal{L}_j : \bigoplus_{r=0}^{\lfloor j/2 \rfloor} \bigoplus_{a=0}^{j-2r} \mathcal{P}_{j,r} \longrightarrow W^{\otimes j}, \quad (u_{r,a}) \longmapsto \sum_{r,a} X_j^a u_{r,a} \quad (\text{OLM11})$$

is bijective and satisfies $\|\mathcal{L}_j(u)\|^2 = \sum_{r,a} s_{j-2r,a} \|u_{r,a}\|^2$. This identity specifies the domain metric transported from the original one. It is not an assertion that the unweighted domain direct sum has that metric.

The inverse has an exact formula in the original operators. Let v_s be the orthogonal component in grade s of v . For $a = s - r$ in the indicated range, its primitive coordinate is

$$u_{r,a} = s_{j-2r,a}^{-1} P_{j,r} Y_j^a v_{r+a}. \quad (\text{OLM12})$$

For its own summand, OLM7 iterated gives $Y_j^a X_j^a u = s_{j-2r,a} u$. For a summand of a lower primitive grade, the same iteration leaves a positive power of X_j in grade r , which OLM6 annihilates. For a higher primitive grade it applies more lowerings than available powers and the result vanishes. This proves OLM12 and supplies the connecting inverse, including all factorials and powers of ϵ .

24.4 Full Jordan multiplicities over all original slots

The decomposition in OLM11, followed by the sector maps OLM3, proves that for each pair (j, r) the original $X = R^{[t]}$ has Jordan blocks of length $j - 2r + 1$ with total multiplicity

$$M_{t;j,r} = \binom{t}{j} (q-2)^{t-j} \left(\binom{j}{r} - \binom{j}{r-1} \right), \quad 0 \leq j \leq t, \quad 0 \leq r \leq \lfloor j/2 \rfloor. \quad (\text{OLM13})$$

There are $\binom{t}{j}$ literal choices of active slots. For each, the inactive factor has dimension $(q-2)^{t-j}$, and each primitive vector supplies exactly the one ladder OLM9. The sector decompositions exhaust E_t , so no blocks are missing. For a fixed length $\ell \geq 1$, the multiplicity, with equal lengths combined, is

$$M_{t,\ell} = \sum_{\substack{0 \leq j \leq t \\ j \geq \ell-1, j-\ell+1 \text{ even}}} \binom{t}{j} (q-2)^{t-j} \left[\binom{j}{(j-\ell+1)/2} - \binom{j}{(j-\ell-1)/2} \right]. \quad (\text{OLM14})$$

Binomial coefficients with negative lower index mean zero. The longest block has length $t+1$ and multiplicity one, arising from $j = t, r = 0$. The exact dimensions of every power kernel and image follow:

$$\dim \ker X^a = \sum_{\ell=1}^{t+1} M_{t,\ell} \min(a, \ell), \quad \text{rank } X^a = \sum_{\ell=1}^{t+1} M_{t,\ell} \max(\ell - a, 0) \quad (a \geq 0). \quad (\text{OLM15})$$

Within a length- ℓ ladder, applying X^a shifts each unchanged power vector by a and kills exactly its last $\min(a, \ell)$ positions; independence proves these identities. In particular $\sum_{\ell} \ell M_{t,\ell} = q^t$. At $a = 1$, summing $m_{j,r}$ over r telescopes to $\binom{j}{\lfloor j/2 \rfloor}$ and recovers OLR29 exactly.

24.5 Every exponential entry and its complete Gram energy

Let $h \in \mathbb{C}$. On a ladder with primitive grade r and $d = j - 2r$, the original exponential has the finite exact action

$$e^{hX_j} X_j^a u = \sum_{b=a}^d \frac{h^{b-a}}{(b-a)!} X_j^b u, \quad \langle X_j^b u, e^{hX_j} X_j^a v \rangle = \begin{cases} s_{d,b} \frac{h^{b-a}}{(b-a)!} \langle u, v \rangle, & b \geq a, \\ 0, & b < a. \end{cases} \quad (\text{OLM16})$$

The first identity is the exponential series terminated by OLM9; OLM8 gives the second with the stated complex phase. No additional eigenvalue or unipotent coefficient has been introduced: all powers of ϵ remain in the literal vectors and the factors $s_{d,b}$.

The formula works directly with primitive-space-valued coordinates. For one fixed sector and primitive grade, let $v = \sum_{a=0}^d X_j^a u_a$, where $u_a \in \mathcal{P}_{j,r} \otimes U^{\otimes(t-j)}$ and X_j acts on its active factor. Then

$$\|e^{hX} v\|_{G_t}^2 = \sum_{b=0}^d s_{d,b} \left\| \sum_{a=0}^b \frac{h^{b-a}}{(b-a)!} u_a \right\|^2. \quad (\text{OLM17})$$

This follows by collecting the coefficient of X_j^b in OLM16 and applying OLM8. The norm inside uses the original primitive and inactive-factor metrics. The expression retains every interference term between different positions in the same ladder; such terms must not be discarded merely because the original ladder positions were orthogonal.

More explicitly, define

$$\mathbf{G}_{d;a,c}(h, z) = \sum_{b=\max(a,c)}^d s_{d,b} \frac{\bar{h}^{b-a} z^{b-c}}{(b-a)!(b-c)!}. \quad (\text{OLM18})$$

For $v = \sum_a X_j^a u_a$ and $w = \sum_c X_j^c v_c$ in the same primitive sector, finite expansion gives

$$\langle e^{hX} v, e^{zX} w \rangle_{G_t} = \sum_{a,c=0}^d \mathbf{G}_{d;a,c}(h, z) \langle u_a, v_c \rangle. \quad (\text{OLM19})$$

Indeed the coefficient in the first argument is conjugated and the second is not, which fixes the order of the powers in OLM18. At $h = z = 0$ the matrix is the original diagonal $s_{d,a} \mathbf{1}_{a=c}$. For every h , its diagonal specialization $\mathbf{G}_d(h, h)$ is positive definite: the triangular coefficient map in OLM17 has diagonal entries one and inverse obtained with $-h$, and every $s_{d,b}$ is positive. Its determinant is $\prod_{b=0}^d s_{d,b}$, since the triangular map and its adjoint have determinant one. These statements retain the original Gram matrix and do not impose an arithmetic purity condition.

All sectors and primitive grades are orthogonal and reduce both X and $X^\dagger$. Consequently the norm and pairing on the entire E_t are exactly the sums of OLM17 and OLM19 over the actual $J \subseteq \{1, \dots, t\}$ and all its primitive grades. These formulas give the full energy, including every inactive slot, rather than only its largest Jordan block.

The exact exponential displacement is particularly explicit:

$$\|(e^{hX} - I)v\|_{G_t}^2 = \sum_{b=1}^d s_{d,b} \left\| \sum_{a=0}^{b-1} \frac{h^{b-a}}{(b-a)!} u_a \right\|^2 \quad (\text{OLM20})$$

on each primitive sector, with the same sums over sectors for a general vector. It follows by removing the $a = b$ term in OLM17. For $h \neq 0$, the kernel is precisely the top position $X_j^d \mathcal{P}_{j,r}$ in each ladder: the coefficient at $b = 1$ first forces $u_0 = 0$, then $b = 2$ forces $u_1 = 0$, and induction forces $u_0 = \dots = u_{d-1} = 0$. The coefficient u_d is unrestricted. Thus the displacement has the same kernel as X , exactly as in the locality calculation OLR28.

For a nonzero vector in one ladder sector let a_0 be its smallest occupied position. Along positive real $h \rightarrow \infty$, the exact polynomial OLM17 has leading term

$$\|e^{hX} v\|_{G_t}^2 = \frac{s_{d,d}}{(d-a_0)!^2} \|u_{a_0}\|^2 h^{2(d-a_0)} + O(h^{2(d-a_0)-1}) \quad (a_0 < d). \quad (\text{OLM21})$$

Only $b = d$ and its smallest occupied $a = a_0$ contribute this highest degree; every other term has smaller degree, proving the formula without a possible cancellation of its positive leading coefficient. If $a_0 = d$, the vector is fixed and its energy is exactly $s_{d,d} \|u_d\|^2$. The same leading term holds for OLM20 when $a_0 < d$. Taking $h = p^n$ retains the exact p -adic-to-complex locality failure in every nonfixed ladder position.

24.6 Exact global singular energy as an independent consequence

The tensor exponential identity in TDP23 can be proved directly: the slot operators commute and each has square zero, so

$$e^{hX} = (I_E + hR)^{\otimes t}. \quad (\text{OLM22})$$

On the original orthonormal pair (e, f) , the one-factor matrix of $I + hR$ is $\begin{pmatrix} 1 & 0 \\ h\epsilon & 1 \end{pmatrix}$; on U it is I_U . Writing $a = \epsilon^2|h|^2$, the two eigenvalues of its actual positive Gram operator on W are

$$\lambda_{\pm}(h) = \frac{2+a \pm \sqrt{a^2+4a}}{2}, \quad \lambda_+ \lambda_- = 1. \quad (\text{OLM23})$$

Indeed that Gram matrix is $\begin{pmatrix} 1+a & \epsilon\bar{h} \\ \epsilon h & 1 \end{pmatrix}$, whose trace is $2+a$ and determinant is one. Its characteristic polynomial proves OLM23. The other $q-2$ eigenvalues are one. The squared singular values of the entire tensor exponential therefore consist of

$$\lambda_+^b \lambda_-^c, \quad \text{multiplicity } \frac{t!}{b!c!(t-b-c)!} (q-2)^{t-b-c}, \quad b, c \geq 0, \quad b+c \leq t. \quad (\text{OLM24})$$

To prove this, take orthogonal eigenspaces of the positive one-factor Gram operator, tensor them in the original slots, and count choices of the two one-dimensional factors and the $(q-2)$ -dimensional factor. The tensor Gram operator is the tensor of the original Gram operators by OLM22. If eigenvalues coincide, the displayed positive multiplicities add; the labelling remains valid, also at $h=0$. Thus the complete operator norm and squared Hilbert–Schmidt energy are exactly

$$\boxed{\|e^{hX}\|_{G_t} = \lambda_+(h)^{t/2}, \quad \|e^{hX}\|_{\text{HS}, G_t}^2 = (q + \epsilon^2|h|^2)^t.} \quad (\text{OLM25})$$

The first identity follows from the largest product eigenvalue, and the second follows by summing OLM24, or multiplying the one-factor traces. These are consequences in the same metric as the full ladder Gram formulas; no norm estimate has replaced the exact ladder entries.

24.7 The full support lifts and free-mask correspondence

For the full bounded distributive lattice L keep $V_L^\uparrow = \{(0, \lambda) : \lambda \in L\} \cup \{(v, 1_L) : v \in V\}$ and the original lift $T^\uparrow(v, \lambda) = (Tv, \lambda)$ over $G_L(\mathbb{C})$. The componentwise ring and lattice operations prove directly that $T^\uparrow$ is linear and that composition is preserved. Apply this to the complete isomorphism formed by OLM3 and OLM11 and to its explicit inverse OLM12. They become mutually inverse maps on the full synchronized carriers, recovering every support label as well as every primitive coordinate. The domain of this lift is the synchronized carrier of the entire direct sum in OLM11 and OLM3. It is not a product assigning unrelated support labels to different summands.

All metric identities lift with the same support meet:

$$\langle (v, \lambda), (w, \mu) \rangle_L = \langle (v, w)_{G_t}, \lambda \wedge \mu \rangle, \quad (X^a)^\uparrow (X^b)^\uparrow = (X^{a+b})^\uparrow. \quad (\text{OLM26})$$

The pair on the right of the first formula lies in $G_L(\mathbb{C})$: a nonzero amplitude requires both vectors nonzero and hence both supports top. At a zero amplitude the literal meet is retained. For $a+b \geq t+1$, the second formula is the supported-zero operator $(v, \lambda) \mapsto (0, \lambda)$, not the map constant at τ . All factorial coefficients in OLM7–OLM21 are unchanged on amplitudes.

For $h \neq 0$, the exact locality-defect correspondence is

$$\begin{aligned} ((e^{hX} - I)^\uparrow)^{-1}(Z_L) &= (\ker X)_L^\uparrow, \\ \ker X &= \bigoplus_{J,r}^\perp X_J^{j-2r} \mathcal{P}_{j,r} \otimes U^{\otimes(t-j)}, \quad Z_L = \{(0, \lambda) : \lambda \in L\}. \end{aligned} \quad (\text{OLM27})$$

OLM20 proves the amplitude kernel and its complete ladder description. The lift retains the support of every preimage, which proves the first equality. The quotient map by this kernel, followed by its induced injection into $\text{im}(e^{hX} - I)$, is exactly the lift of OLR7; its formula on a class is $[v] \mapsto (e^{hX} - I)v$. It is well-defined, injective and onto its displayed image because its amplitude kernel is precisely the stated one.

In the original free coordinates of the positive-dimensional space E_t of dimension q^t , write a vector as $(v, \boldsymbol{\mu})$, with $v_i \neq 0 \Rightarrow \mu_i = 1_L$. Keep the original synchronization and its complete fibre:

$$\begin{aligned} P_{\text{syn}}(v, \boldsymbol{\mu}) &= (v, \bigvee_i \mu_i), \\ P_{\text{syn}}^{-1}(v, \lambda) &= \{(v, \boldsymbol{\mu}) : \bigvee_i \mu_i = \lambda, v_i \neq 0 \Rightarrow \mu_i = 1_L\}. \end{aligned} \tag{OLM28}$$

The free operator has amplitude I and all-top support matrix, so it sends every output support coordinate to the displayed join. Repeating that join leaves it unchanged, proving it is an idempotent with precisely the displayed fibres. The free masks over OLM27 are exactly those whose original amplitude belongs to the explicitly computed $\ker X$; all zero-amplitude masks are retained. There is no asserted change of free-semimodule basis to a ladder basis. The exact relation is the correspondence OLM28 followed by the synchronized isomorphism OLM11 and inverse OLM12. This keeps every original mask while using the proved metric ladder coordinates for its amplitude. If an individual zero-dimensional summand is considered on its own, its free zero-coordinate semimodule has one element whereas its synchronized zero-vector carrier still has all of L ; the positive-dimensional assertion just made concerns the full original E_t and does not suppress that distinction.

These constructions give the complete original-metric nilpotent decomposition, every Jordan multiplicity, every exponential coefficient and its joint energy, and the exact full-support maps. Their slot-counting grades and finite ladder commutator have not been identified with an arithmetic weight, monodromy filtration or a Frobenius purity statement.

25 Exterior powers and Adams operations with every split-support label retained

The human source for the Adams-to-arithmetic-topos construction is Alain Connes and Caterina Consani, *Segal's Gamma rings and universal arithmetic*, original author source `Froblifts.tex`. The source ranges read for the present calculation are 600–704, with the Adams formulas and their full topos argument at 632–681, and the Atiyah–Tall bibliography entry at 1242–1243. The source SHA–256 is

226eecb4cfde540362b1d5073fb23bad2c3de6e0b8c3cf267b47447e3166298d.

Their mathematical reference is M. F. Atiyah and D. O. Tall, *Group representations, λ -rings and the J -homomorphism*, *Topology* **8** (1969), 253–297, especially their Section 5 and Proposition 5.2 as identified in Connes–Consani’s text. No claim to have read that entire reference is made here. The twistor maps and real structure are Connes–Consani’s *The Absolute Twistor Line and the Geometry of $\overline{\text{Spec } \mathbb{Z}}$* ; the original `CC.tex` was read in full for the preceding HA1–29 derivation. Its ordinary K_0 calculation, precise equivariant correction, and real-lift maps are HA23–29 of the retained proof `SIGNED_HOPF_AND_ADAMS.tex`. The operations below are a full-support extension derived here, not an assertion attributed to either human source.

25.1 The original ring and its integral exterior-power formula

Retain the original generator, without changing its sign:

$$B = K_0(\mathbb{P}_{\mathbb{C}}^1) = \mathbb{Z}[\eta]/(\eta^2), \quad \eta = [\mathcal{O}(1)] - 1, \quad b = r + a\eta \quad (r, a \in \mathbb{Z}). \quad (\text{FLA1})$$

The presentation and uniqueness of these coordinates are proved in HA23–25 by the Euler sequence and the diagonal resolution. In particular (r, a) here is rank and degree, including negative virtual rank and degree. Let $\ell = [\mathcal{O}(1)] = 1 + \eta$. The ordinary exterior-power operations give $\lambda_t(1) = 1 + t$ and $\lambda_t(\ell) = 1 + \ell t$. For an exact sequence of vector bundles, the exterior powers have the filtration with associated graded terms $\bigwedge^i E' \otimes \bigwedge^{n-i} E''$; it is obtained on locally split charts and is independent of the splitting. Taking classes proves multiplicativity of total exterior power under addition in K_0 . Extension to virtual differences uses the formal inverse of a series with constant coefficient one. Since $b = (r - a)1 + a\ell$, this proves the exact integral formula

$$\lambda_t^B(r + a\eta) = (1 + t)^{r-a}(1 + (1 + \eta)t)^a = (1 + t)^r \left(1 + \frac{a\eta t}{1 + t}\right). \quad (\text{FLA2})$$

Every inverse here is a formal power-series inverse over B , not division by an integer. In particular $1/(1 + t) = \sum_{q \geq 0} (-t)^q$. For every integer a , the identity $(1 + v)^a = 1 + av$ holds when $v^2 = 0$: positive exponents follow by multiplication, and negative ones follow from $(1 + v)^{-1} = 1 - v$. This proves the second equality for all the stated coordinates.

For $r \in \mathbb{Z}$ and $n \geq 0$, define the integer $\binom{r}{n}$ by the coefficients of $(1 + t)^r$. Equivalently it is the usual falling-product formula, with integrality for negative r given by $\binom{r}{n} = (-1)^n \binom{n-r-1}{n}$. Thus (FLA2) gives

$$\lambda_0^B(b) = 1, \quad \lambda_n^B(r + a\eta) = \binom{r}{n} + a \binom{r-1}{n-1} \eta \quad (n \geq 1). \quad (\text{FLA3})$$

The expressions remain valid for $r = 0$, $a = 0$, or either coordinate negative. Direct multiplication in (FLA2), with $\eta^2 = 0$, proves

$$\lambda_t^B(b + c) = \lambda_t^B(b) \lambda_t^B(c) \quad (b, c \in B). \quad (\text{FLA4})$$

This verifies the required identity without omitting the virtual classes.

25.2 The support-preserving operations and every zero fibre

Let L be any nontrivial bounded distributive lattice, finite or infinite. Keep the original semiring

$$\begin{aligned} S &= G_L(B) = \{(0, \lambda) : \lambda \in L\} \cup \{(b, 1_L) : b \in B\}, \\ (b, \lambda) + (c, \mu) &= (b + c, \lambda \vee \mu), \quad (b, \lambda)(c, \mu) = (bc, \lambda \wedge \mu), \\ z_\lambda &= (0, \lambda), \quad \tau = z_{0_L}, \quad e = z_{1_L}, \quad 1_S = (1, 1_L), \quad \overline{(b, \lambda)} = (-b, \lambda). \end{aligned} \quad (\text{FLA5})$$

The bar is the actual support-preserving signed operation; it is not a global additive inverse. In particular $x + \bar{x} = z_\lambda$ when $x = (b, \lambda)$. Define

$$\lambda_0^\uparrow(x) = 1_S, \quad \lambda_n^\uparrow(b, \lambda) = (\lambda_n^B(b), \lambda) \quad (n \geq 1), \quad \Lambda_t(x) = \sum_{n \geq 0} \lambda_n^\uparrow(x) t^n. \quad (\text{FLA6})$$

These are defined on the original carrier: if $\lambda \neq 1_L$ then $b = 0$ and $\lambda_n^B(0) = 0$ for $n \geq 1$. The coefficient at every positive degree therefore has exactly the stated support, even when its amplitude vanishes. In particular

$$\begin{aligned} \lambda_1^\uparrow(x) &= x, \quad \lambda_1^\uparrow(\tau) = \tau, \quad \lambda_n^\uparrow(z_\lambda) = z_\lambda \quad (n \geq 1), \\ \lambda_1^\uparrow(1_S) &= 1_S, \quad \lambda_n^\uparrow(1_S) = e \quad (n \geq 2). \end{aligned} \quad (\text{FLA7})$$

In the formal semiring $S[[t]]$, the constant series 1 has zero coefficient τ at positive degrees. Accordingly (FLA7) does not identify $\Lambda_t(1_S)$ with the polynomial $1 + t$.

The full additive exterior-power law is nevertheless exact:

$$\Lambda_t(x + y) = \Lambda_t(x)\Lambda_t(y), \quad \Lambda_t(\tau) = 1. \quad (\text{FLA8})$$

To check all labels, write $x = (b, \lambda)$ and $y = (c, \mu)$. For degree $n \geq 1$, the terms at indices $(0, n)$ and $(n, 0)$ on the right have supports μ and λ . Every other term has support $\lambda \wedge \mu$. Their join is exactly $\lambda \vee \mu$, the support on the left. Their amplitudes agree by (FLA4). At degree zero both sides are 1_S . This proves the identity in $S[[t]]$, not only after applying an amplitude map. Thus (FLA6)–(FLA8) specify the support-preserving exterior-power laws used here. They do not assert that S is a ring or that the ordinary special- λ -ring axioms hold on it: already the ordinary unit axiom $\lambda_n(1) = 0$ for $n > 1$ would replace e in (FLA7) by the different element τ .

25.3 The exact relative inverses and their support domains

Define the idempotent series

$$E_\lambda(t) = 1 + \sum_{n \geq 1} z_\lambda t^n = \Lambda_t(z_\lambda), \quad E_\lambda E_\mu = E_{\lambda \vee \mu}, \quad E_{0_L} = 1. \quad (\text{FLA9})$$

The middle identity follows by the same support computation as (FLA8), with all positive-degree amplitudes zero. In particular $E_\lambda^2 = E_\lambda$, and $E_\lambda = 1$ holds exactly when $\lambda = 0_L$, as seen in its coefficient at degree one.

The precise multiplicative domains are

$$\mathcal{H}_\lambda = \left\{ 1 + \sum_{n \geq 1} (b_n, \lambda) t^n : (b_n, \lambda) \in S \text{ for every } n \right\}. \quad (\text{FLA10})$$

For $\lambda \neq 1_L$ this is the one-element group $\{E_\lambda\}$. For $\lambda = 1_L$, the amplitude map identifies $\mathcal{H}_{1_L}$, with its own identity E_{1_L} , with the ordinary group $1 + tB[[t]]$. To verify the latter without any assumed inverse, for $A(t) = 1 + \sum b_n t^n$ set $c_0 = 1$ and recursively $c_n = -\sum_{i=1}^n b_i c_{n-i}$. The coefficients are in B and multiplication gives $A(t) \sum c_n t^n = 1$. Lifting every positive coefficient

to support 1_L makes the product E_{1_L} . The amplitude coordinates also prove uniqueness of this inverse. This proves the group assertions in every case. Multiplication of two arbitrary domains satisfies

$$\mathcal{H}_\lambda \mathcal{H}_\mu \subseteq \mathcal{H}_{\lambda \vee \mu}. \quad (\text{FLA11})$$

When $\lambda \leq \nu$, the map $A \mapsto E_\nu A$ is a group homomorphism $\mathcal{H}_\lambda \rightarrow \mathcal{H}_\nu$, sends the identity to E_ν , and simply raises every positive-degree support to ν . These maps compose for nested supports, by (FLA9). Thus all multiplication domains and their connecting maps are specified.

For $x = (b, \lambda)$ the actual signed inverse obeys

$$\Lambda_t(x) \Lambda_t(\bar{x}) = E_\lambda, \quad E_\lambda \Lambda_t(x) = \Lambda_t(x). \quad (\text{FLA12})$$

The first follows from (FLA8) and $x + \bar{x} = z_\lambda$; the second follows directly by multiplying amplitudes and labels. Consequently $\Lambda_t(\bar{x})$ is exactly the group inverse in $\mathcal{H}_\lambda$, and both reflexive inverse laws $A\bar{A}A = A$, $\bar{A}A\bar{A} = \bar{A}$ hold in the full series semiring. Except at τ , it cannot be an ordinary inverse with product 1: the coefficient of degree one in a product of $\Lambda_t(x)$ with any series of constant coefficient 1_S has support at least λ . A global inverse would have that constant coefficient, since the only possible constant inverse of 1_S is 1_S itself. For $\lambda \neq 0_L$ this precludes a zero coefficient τ at degree one. Finally Λ_t is injective because its first coefficient is x . Equations (FLA8)–(FLA12) therefore describe the whole additive split-support object by an exact series embedding, with the correct relative, rather than global, inverses.

25.4 Adams endomorphisms and the division-free Newton calculation

For every integer $n \geq 1$ define

$$\psi_n^B(r + a\eta) = r + na\eta, \quad \Psi_n(r + a\eta, \lambda) = (r + na\eta, \lambda). \quad (\text{FLA13})$$

The lifted map is defined on all of S for the same carrier reason as (FLA6). It is a unital semiring endomorphism. Additivity follows from the displayed formula and the unchanged join of the two supports. For multiplication, in the original coordinates,

$$(r + a\eta)(s + b\eta) = rs + (rb + sa)\eta,$$

and both ψ_n^B of this product and the product of the two images equal $rs + n(rb + sa)\eta$. The support is the unchanged meet. Zero and unit are preserved. The same calculation proves

$$\Psi_1 = \text{id}, \quad \Psi_m \Psi_n = \Psi_{mn}, \quad \Psi_m \lambda_n^\uparrow = \lambda_n^\uparrow \Psi_m \quad (n \geq 0), \quad \Psi_m(z_\lambda) = z_\lambda. \quad (\text{FLA14})$$

For the commutation with λ_n , use (FLA3): both maps keep the binomial rank term and replace its η coefficient by m times that coefficient, with the same support. The case $n = 0$ is the preserved unit. Thus these operations define an actual $\mathbb{N}^\times$ -action by semiring endomorphisms, and hence the corresponding semiring object in the presheaf topos underlying the arithmetic site. This is the semiring version of the exact action mechanism in Connes–Consani’s Lemma `lambda1`, source lines 649–655, proved here on the full original carrier.

No division by a positive integer is needed to identify these as the Newton operations of (FLA6). Formal differentiation of (FLA2) in the ring $B[[t]]$ gives the integral identity

$$t \frac{d}{dt} \lambda_t^B(b) \lambda_t^B(-b) = \frac{rt}{1+t} + \frac{a\eta t}{(1+t)^2} = \sum_{n \geq 1} (-1)^{n-1} (r + na\eta) t^n. \quad (\text{FLA15})$$

For the first equality, differentiate the integer powers in (FLA2), including their formal inverses, and use $\eta^2 = 0$. For the second, differentiate the integral geometric series for $(1+t)^{-1}$ or multiply it by itself. Its squared inverse has coefficient $(-1)^q(q+1)$ at degree q . All the operations take place over $\mathbb{Z}$ and all denominators displayed are units of $\mathbb{Z}[[t]]$, so no rationalization is involved.

For an element of support λ , define a signed multiple in its additive support fibre by

$$[c]_\lambda(b, \lambda) = (cb, \lambda) \quad (c \in \mathbb{Z}).$$

For $c > 0$ this is repeated addition; $[-1]_\lambda x = \bar{x}$. For $c = 0$ it is the identity z_λ of that fibre, not an instruction to change it to the global zero τ . This convention records exactly the support of signed polynomial terms. Let

$$W_x(t) = \sum_{n \geq 1} [-1]_\lambda^{n-1} \Psi_n(x) t^n \quad (x = (b, \lambda)), \quad (\text{FLA16})$$

where the superscript denotes the iterated signed operation and the constant coefficient is τ . Every positive coefficient has support λ . If D denotes the formal derivative in the semiring, with repeated-addition coefficient n , then the exact lifted identities are

$$tD\Lambda_t(x) = \Lambda_t(x)W_x(t), \quad W_x(t) = \Lambda_t(\bar{x})tD\Lambda_t(x). \quad (\text{FLA17})$$

For the first identity, every positive-degree coefficient on each side has support λ , and the amplitudes agree by (FLA15) after multiplying there by $\lambda_t^B(b)$. Both constant coefficients are τ . For the second, multiply by the exact relative inverse in (FLA12). Its extra factor E_λ acts identically on W_x : its constant coefficient is 1_S , its positive amplitudes vanish, and all resulting supports remain λ . This proves the second identity without treating E_λ as the constant series 1.

Taking degree $n \geq 1$ in the first identity gives the complete division-free Newton formula in S :

$$n\lambda_n^\uparrow(x) = \sum_{i=1}^n [-1]_\lambda^{i-1} (\lambda_{n-i}^\uparrow(x) \Psi_i(x)). \quad (\text{FLA18})$$

Every term has support λ , including $i = n$, since $\lambda_0^\uparrow(x) = 1_S$ has top support. The sum is therefore in that same additive fibre. Each such fibre is a group under addition with identity z_λ : it is B at top support and a singleton at every other support. Thus solving for the last term uses only a coefficient $+1$ or -1 and gives

$$\Psi_n(x) = [-1]_\lambda^{n-1} (n\lambda_n^\uparrow(x)) + \sum_{i=1}^{n-1} [-1]_\lambda^{n+i-1} (\lambda_{n-i}^\uparrow(x) \Psi_i(x)). \quad (\text{FLA19})$$

For $n = 1$ the sum is absent and the identity is $\Psi_1 = x$. This recursive formula defines a unique sequence in the stated support fibres. It proves that (FLA13) is the actual Newton sequence of the lifted exterior operations. It does not attempt to reconstruct exterior operations by dividing n in (FLA18).

There is a local sign correction to the human source's displayed equation `lognewton`, line 645. It prints $(-1)^n$ on the right, while lines 637–640 state $\psi_1 = \lambda_1$ and the usual Newton polynomials. With the source's displayed derivative $t\partial_t \lambda(t)/\lambda(t)$, its coefficient of t is λ_1 and therefore requires $(-1)^{n-1}$. For the ordinary unit class $b = 1$, (FLA15) gives $t/(1+t) = t - t^2 + t^3 - \dots$; the printed sign would give its negative. Equations (FLA15)–(FLA19) prove the corrected formula directly. This correction is to that display and does not negate the source's topos-action argument.

25.5 Actual characteristic-prime Frobenius lifts with support retained

Let p be a rational prime and use the exact quotient map

$$q_p : S \longrightarrow G_L(B/pB), \quad (b, \lambda) \longmapsto (b \bmod pB, \lambda). \quad (\text{FLA20})$$

It preserves both operations and is surjective: every nonzero amplitude class has a representative at top support, while every zero label already has its designated representative. Its precise kernel

congruence is equality of supports together with amplitude difference in pB . Thus it does not identify distinct zero labels. In B , for $b = r + a\eta$,

$$b^p = r^p + pr^{p-1}a\eta, \quad \psi_p^B(b) - b^p = (r - r^p) + pa(1 - r^{p-1})\eta \in pB. \quad (\text{FLA21})$$

The inclusion uses $r^p \equiv r \pmod{p}$ for every integer r . For a direct proof, the intermediate coefficients of $(x+1)^p$ are divisible by p , since p divides its numerator factorial and neither denominator factorial. Induction from 0 gives the congruence for nonnegative integers; applying the same polynomial identity backward gives it for negative integers as well. Since the support of the p th power is $\lambda \wedge \cdots \wedge \lambda = \lambda$, (FLA21) proves the full equality

$$q_p \Psi_p = \text{Fr}_p q_p, \quad \text{Fr}_p(c, \lambda) = (c^p, \lambda) \text{ on } G_L(B/pB). \quad (\text{FLA22})$$

The target map is a semiring endomorphism: its amplitude is the characteristic- p Frobenius, and its labels preserve joins and meets. Thus (FLA22) is an actual Frobenius-lift statement on the full split-support base, not only a congruence in its ring shadow.

25.6 Odd twistor pullbacks, with their real and equivariant data

For positive odd n , put $\chi(n) = 1$ for $n \equiv 1 \pmod{4}$ and $\chi(n) = -1$ for $n \equiv 3 \pmod{4}$. The original twistor maps are exactly

$$F_n(z) = \begin{cases} z^n, & n \equiv 1 \pmod{4}, \\ -1/z^n, & n \equiv 3 \pmod{4}. \end{cases} \quad (\text{FLA23})$$

In homogeneous coordinates they are $[X^n : Y^n]$ and $[-Y^n : X^n]$, respectively. Both pairs are sections of $\mathcal{O}(n)$ with no common zero, proving $F_n^* \mathcal{O}(1) \simeq \mathcal{O}(n)$ and hence $F_n^* \eta = n\eta$. These two formulas therefore prove on every original class, and then on its entire support fibre, that

$$G_L(F_n^*) = \Psi_n, \quad F_n^*(r + a\eta) = r + na\eta. \quad (\text{FLA24})$$

The maps compose as $F_m F_n = F_{mn}$ for odd indices: the involution $a(z) = -1/z$ commutes with odd powering, $a^2 = \text{id}$, and the parity selecting a adds because $\chi(mn) = \chi(m)\chi(n)$. This proves the geometric counterpart of (FLA14) for these maps.

For the scaling group, however, the exact equivariance is $F_n(uz) = u^{n\chi(n)} F_n(z)$. Thus a bundle with pole weight multisets (W_0, W_∞) pulls back with the specific weights

$$(W'_0, W'_\infty) = \begin{cases} (nW_0, nW_\infty), & n \equiv 1 \pmod{4}, \\ (-nW_\infty, -nW_0), & n \equiv 3 \pmod{4}. \end{cases} \quad (\text{FLA25})$$

Indeed the action on (z, v) in the pulled-back fibre is $u \cdot (z, v) = (uz, u^{n\chi(n)} \cdot v)$; its target fibre is correct by the displayed equivariance. The second case exchanges the two poles, proving all labels in (FLA25). The trivial line with fibre character u then has pullback character u^{-n} in the second case, while its equivariant Adams operation has character u^n . Restriction to the fixed point detects distinct Laurent monomials, so (FLA24) does not assert their equality equivariantly. The exact connecting map is the forgetful map $\text{For} : K_0^{\text{C}\times}(\mathbb{P}^1) \rightarrow B$: forgetting the specified pulled-back action gives ordinary pullback, hence

$$G_L(\text{For}) G_L(P_n) = \Psi_n G_L(\text{For}), \quad (\text{FLA26})$$

where P_n is the equivariant pullback just constructed. The equality follows on actual bundles and is preserved under exact-sequence relations, tensor product, and every support label.

For the antipodal real map $\sigma(z) = -1/\bar{z}$ one has $F_n\sigma = \sigma F_n$, by the same odd-power calculation and real coefficients in (FLA23). If an actual bundle has an antilinear lift $\tau : E_z \rightarrow E_{\sigma(z)}$, its transported lift is

$$\tau_n(z, v) = (\sigma(z), \tau v), \quad \tau_n^2(z, v) = (z, \tau^2 v). \quad (\text{FLA27})$$

Commutation of the base maps proves its target fibre, and applying it twice proves the square. It therefore preserves a supplied real or quaternionic sign exactly. It does not construct a lift, filtration, polarization, or connection from an ordinary K_0 class. Equations (FLA25)–(FLA27) give the actual maps on the additional structures that are present.

25.7 Naturality in the entire support lattice

Let $\phi : L \rightarrow L'$ preserve zero, unit, join, and meet, with no finiteness requirement. The original morphism is $G_\phi(b, \lambda) = (b, \phi(\lambda))$. Its well-definedness and operation laws follow directly from (FLA5); in particular nonzero amplitudes remain at top support. The explicit formulas prove

$$\begin{aligned} G_\phi \lambda_n^{\uparrow, L} &= \lambda_n^{\uparrow, L'} G_\phi \quad (n \geq 0), & G_\phi \Psi_n^L &= \Psi_n^{L'} G_\phi, \\ G_\phi(\bar{x}) &= \overline{G_\phi(x)}, & G_\phi(E_\lambda) &= E_{\phi(\lambda)}, & G_\phi(\mathcal{H}_\lambda) &\subseteq \mathcal{H}_{\phi(\lambda)}. \end{aligned} \quad (\text{FLA28})$$

The series maps are coefficientwise; their constant unit is preserved. Thus they preserve the relative inverses with their actual changed identity and carry both sides of (FLA17)–(FLA19) to the corresponding full identities over L' . Reduction modulo p , the ordinary twistor pullback, and its forgetful square commute with these same maps because they change only the amplitude. Composition of lattice maps is exact on each coordinate, proving composition of all these squares. Finally the amplitude and support maps themselves satisfy

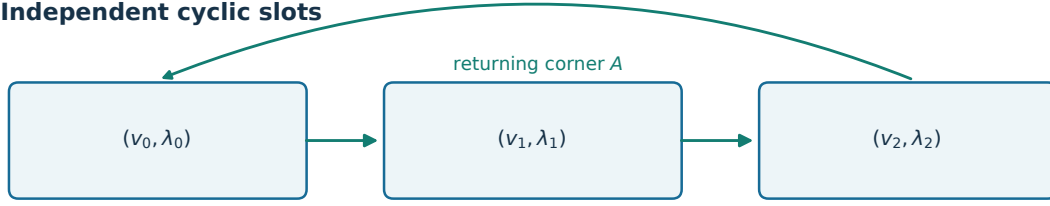
$$q\Psi_n = \psi_n^B q, \quad q\lambda_n^\uparrow = \lambda_n^B q, \quad \sigma\Psi_n = \sigma, \quad \sigma\lambda_n^\uparrow = \sigma \quad (n \geq 1). \quad (\text{FLA29})$$

Here $q(b, \lambda) = b$ and $\sigma(b, \lambda) = \lambda$; the constant operation has separately specified top support. This records the exact relationship of both shadows to the retained full operations. No cohomological or purity identification is assumed or inferred.

A returning zero is not an absent trace

Actual odd receiver at $d = 3$: $A = -F_\rho^{-1}$, original blocks of dimension $q = (k + 1)^2$.

Independent cyclic slots



$$\text{Tr}(C_3^n) = \tau \quad (3 \nmid n)$$

$$\text{Tr}(C_3^{3m}) = (3\text{tr}(A^m), 1_L)$$

If the returning amplitude is zero, this is e , not τ .

The exact synchronization retract



$$rj = I, \quad jr: (\lambda_j)_j \mapsto (\bigvee_j \lambda_j)_j$$

Global synchronization gives trace e also when $3 \nmid n$.

The cycle comparison retains $\Theta_3 - \Theta_1$, with ghost $3\mathbf{1}_{3 \mid n} - 1$.

Proof: FLW7–17, FLW23–31. Every original metric entry and mass factor is retained.
Human source: Alain Connes and Caterina Consani, arXiv:2004.08879v1, lines 454–600.

Figure 6: The actual three-slot projective receiver and its full support trace. The synchronization comparison is a proved retract, with every fibre retained. Proofs and human-source locators: FLW7–17 and FLW23–31.

26 The full-lattice projective Verschiebung and its retained cycle object

Alain Connes and Caterina Consani define the Verschiebung of an endomorphism by cyclically moving its summands, with the original endomorphism at the returning corner, in *Segal's Gamma rings and universal arithmetic*, original author source `Froblifts.tex`, lines 454–681, especially `defninv`, `w0s`, `rabi9`, `rabi10`, `rhoinf`, and `lognewton`. Their construction uses finite pointed sets and their associated $\mathbb{S}$ -modules. The construction below is instead carried out on explicit finitely generated projective semimodules over the programme's full $G_L(\mathbb{C})$. We prove its coefficient maps, the intrinsic trace, and the actual Frobenius/Verschiebung relations; we then construct an exact receiving map to the authors' pointed cycle object. Neither scalar group completion nor a single lattice trace is assumed to retain all projective support. The cited source expressly credits Gert Almkvist for the original endomorphism invariant and Michael Atiyah for the λ -ring framework used with Adams operations. Those earlier papers are human precedents cited through this source; no fresh reading of their complete proofs is claimed here.

The original complex companion, its full metric, and its signed odd receiver are TPT1–TPT28 and TOB6–TOB29 of the present reader. The full carrier and its coordinate-mask correspondence are OLR16–OLR17. Their predecessor source is 033, SBT1–SBT23. The earlier supported-zero prime construction retains its original provenance; it is not a discovery of this section.

26.1 The full coefficient semiring and projective carriers

Let L be a bounded distributive lattice, with $0_L \neq 1_L$, and put

$$\begin{aligned} S = G_L(\mathbb{C}) &= \{(c, 1_L) : c \in \mathbb{C}\} \cup \{z_\lambda = (0, \lambda) : \lambda \in L\}, \\ (a, \lambda) + (b, \mu) &= (a + b, \lambda \vee \mu), \\ (a, \lambda)(b, \mu) &= (ab, \lambda \wedge \mu), \quad \tau = z_{0_L}, \quad e = z_{1_L}, \quad 1_S = (1, 1_L). \end{aligned} \tag{FLW1}$$

The two coefficient maps

$$\rho : S \longrightarrow \mathbb{C}, \quad (c, \lambda) \longmapsto c, \quad s : S \longrightarrow L, \quad (c, \lambda) \longmapsto \lambda \tag{FLW2}$$

are unital semiring homomorphisms, by their displayed component laws. They are jointly injective. Their joint image consists exactly of the pairs with $c \neq 0 \Rightarrow \lambda = 1_L$. In particular $\rho(\tau) = \rho(e) = 0$, whereas $s(\tau) = 0_L$ and $s(e) = 1_L$.

For a positive integer r , let J_r be the all- 1_L support matrix and let

$$\mathbf{P}_r = (I_r, J_r) \in M_r(S), \quad P_r = \text{im } \mathbf{P}_r = \{(v, \lambda) : v \in \mathbb{C}^r, v \neq 0 \Rightarrow \lambda = 1_L\}. \tag{FLW3}$$

Here a matrix pair records its amplitude and support, with every nonzero amplitude entry forcing support 1_L ; matrix multiplication uses ordinary complex multiplication in amplitude and join-of-meets in support. On a free vector $(v, \mu_1, \dots, \mu_r)$, $\mathbf{P}_r$ leaves v unchanged and sets every support coordinate to $\bigvee_j \mu_j$. Thus $\mathbf{P}_r^2 = \mathbf{P}_r$ and its image is exactly FLW3. This proves that P_r is projective, with the indicated splitting of the free module S^r . For any complex b -by- a matrix T , the matrix $(T, J_{b,a})$ on the retracts gives

$$T^\dagger : P_a \longrightarrow P_b, \quad (v, \lambda) \longmapsto (Tv, \lambda). \tag{FLW4}$$

It is S -linear by FLW1, including cancellation of amplitudes. Composition and addition preserve this lift, and an invertible square T gives an automorphism with inverse $(T^{-1})^\dagger$. The lift of the complex zero map is $(v, \lambda) \mapsto (0, \lambda)$, which is not the constant- τ map.

The source dimension throughout is $q = (k+1)^2 > 0$. The actual cyclic carrier with d slots is the direct sum of d such projectives:

$$P_{q,d} = P_q^{\oplus d} = \text{im } \text{diag}(\mathbf{P}_q, \dots, \mathbf{P}_q). \tag{FLW5}$$

An element has independent slot supports $((v_0, \lambda_0), \dots, (v_{d-1}, \lambda_{d-1}))$, with $v_j \neq 0 \Rightarrow \lambda_j = 1_L$. For the original free coordinate masks the complete fibre is

$$\{(\mu_{j,i}) : \bigvee_{i=1}^q \mu_{j,i} = \lambda_j \text{ for each } j, (v_j)_i \neq 0 \Rightarrow \mu_{j,i} = 1_L\}. \quad (\text{FLW6})$$

This follows by applying FLW3 separately to every slot. The tensor of two such synchronized projectives is also explicit: $P_a \otimes_S P_b = \text{im}(P_a \otimes P_b)$ under the standard free-coordinate identification, and the matrix there is (I_{ab}, J_{ab}) . Hence it is P_{ab} . This assertion follows directly by tensoring the two splitting maps: their retraction composite is the identity and their opposite composite is the displayed Kronecker idempotent.

26.2 Intrinsic trace, including both zeros

For any finitely generated projective S -semimodule P , choose a splitting $i : P \rightarrow S^a$, $r : S^a \rightarrow P$, $ri = I_P$. For $T : P \rightarrow P$ define

$$\text{Tr}_P(T) = \text{tr}_{S^a}(iTr) = \sum_{j=1}^a (iTr)_{jj} \in S. \quad (\text{FLW7})$$

This is independent of the chosen splitting. Indeed for a second splitting i', r' through S^b , put $U = iTr'$ and $V = i'r$. Then $UV = iTr$ and $VU = i'Tr'$. For rectangular matrices over a commutative semiring, $\text{tr}(UV) = \sum_{j,l} U_{jl}V_{lj} = \sum_{l,j} V_{lj}U_{jl} = \text{tr}(VU)$. This proves independence without subtraction or a chosen basis of the projective. The same proof gives cyclicity for maps between two projectives, invariance under isomorphism, additivity on direct sums, and multiplicativity on tensor products, the last by the diagonal entries of a Kronecker product.

For the chosen projective lift FLW4, every diagonal support is top, so its intrinsic trace is exactly

$$\text{Tr}_{P_q}(T^\uparrow) = (\text{tr}_{\mathbb{C}} T, 1_L). \quad (\text{FLW8})$$

In particular the supported zero map has trace e and the actual zero endomorphism has trace τ . These are distinct even when the complex trace is zero. More generally, applying ρ or s to the splitting matrices gives the corresponding split projective over $\mathbb{C}$ or L , and

$$\rho(\text{Tr}_P T) = \text{Tr}_{P \otimes_S \mathbb{C}} T_\rho, \quad s(\text{Tr}_P T) = \text{Tr}_{P \otimes_S L} T_s. \quad (\text{FLW9})$$

One may define each base change here as the image of the coefficient image of its splitting idempotent; the split maps prove the stated tensor descriptions. FLW9 then follows by applying the coefficient homomorphism to each finite diagonal sum. In particular $P_q \otimes_S \mathbb{C} = \mathbb{C}^q$ and $P_q \otimes_S L = \text{im } J_q \cong L$, the latter by evaluation at any coordinate of the constant support vector.

26.3 The actual signed cyclic operator and every integer power

Keep every original parameter and metric:

$$\begin{aligned} k \geq 17, \quad k \equiv 1 \pmod{4}, \quad \delta, \gamma > 0, \quad q = (k+1)^2, \quad q-1 \leq N \leq 2q, \\ E = \mathbb{C}[y] / \prod_{a,b=0}^k (y - (2b-k)\gamma + i(2a-k)\delta), \quad M[v] = [yv], \\ F_p = \exp(i(\log p)M), \quad \tilde{F}_p = p^{k/2} F_p, \quad \langle v, w \rangle_N = v^* G_N w, \\ \omega_{ab} = p^{(2a-k)\delta} \exp(i(2b-k)\gamma \log p). \end{aligned} \quad (\text{FLW10})$$

For every positive odd d retain $\chi = \chi_d = (-1)^{(d-1)/2}$, $A = \chi F_p^\chi$. On the carrier FLW5 define the literal Verschiebung

$$\mathcal{C}_d(A)((v_j, \lambda_j)_{j=0}^{d-1}) = ((Av_{d-1}, \lambda_{d-1}), (v_0, \lambda_0), \dots, (v_{d-2}, \lambda_{d-2})). \quad (\text{FLW11})$$

The ambient block matrix has $A^\uparrow$ in its corner, $I^\uparrow$ on its cyclic forward edges, and the actual all- τ zero matrix in all other blocks. Thus its support is $P_d \otimes J_q$, where P_d is the cyclic permutation support matrix; it is not the all-top dq -by- dq matrix. Its inverse reverses the cyclic steps and uses $(A^{-1})^\uparrow$ at the reversed corner. Consequently it is an automorphism of $P_{q,d}$, whose identity is the block idempotent in FLW5.

For every integer n and every input slot $j \in \{0, \dots, d-1\}$, write $j+n = dw + r$, $0 \leq r < d$, using Euclidean division even for negative n . The power map has exactly the following one block in that input column:

$$\mathcal{C}_d(A)^n \iota_j = \iota_r(A^w)^\uparrow, \quad r = j + n \bmod d, \quad w = \left\lfloor \frac{j+n}{d} \right\rfloor. \quad (\text{FLW12})$$

Following one forward step increments the slot and adds one to the exponent only at a wrap. This proves FLW12 for $n \geq 0$ by induction. The stated inverse proves the backward step and hence all $n < 0$. Its support matrix is $P_d^n \otimes J_q$ at every integer n , including $n = 0$, which gives the projective identity in FLW5. The exact intrinsic power traces are therefore

$$\text{Tr}_{P_{q,d}} \mathcal{C}_d(A)^n = \begin{cases} \tau, & d \nmid n, \\ \left(d\chi^m \sum_{a,b=0}^k \omega_{ab}^{\chi^m}, 1_L \right), & n = dm. \end{cases} \quad (\text{FLW13})$$

At a nonreturn time, every diagonal block is the actual zero block, so the result is τ . At a return time $n = dm$, every diagonal block is $(A^m)^\uparrow$; FLW8 and the ordinary eigenvalue trace give the second line. For $n = 0$ it is $(dq, 1_L)$. For cancellation of its complex power sum it is e , not τ . No metric is involved in this intrinsic trace; in particular none of the nonorthogonal Lagrange vectors is silently made orthogonal.

Let P_{ab} be the actual original Lagrange projector and lift its block diagonal as $\mathcal{P}_{ab}$ on $P_{q,d}$. The labelled and mass-weighted identities are

$$\begin{aligned} \text{Tr}(\mathcal{P}_{ab} \mathcal{C}_d(A)^n) &= \begin{cases} \tau, & d \nmid n, \\ (d\chi^m \omega_{ab}^{\chi^m}, 1_L), & n = dm, \end{cases} \\ \text{Tr}(\mathcal{P}_{ab}(p^{k/2}, 1_L)^n \mathcal{C}_d(A)^n) &= (p^{kn/2}, 1_L) \text{Tr}(\mathcal{P}_{ab} \mathcal{C}_d(A)^n). \end{aligned} \quad (\text{FLW14})$$

These follow from $P_{ab} A^m = \chi^m \omega_{ab}^{\chi^m} P_{ab}$ and $\text{tr } P_{ab} = 1$. The projectors are retained as specific maps: for distinct labels their supported product is the lift of the complex zero map on each slot, not the actual zero endomorphism. Thus their complex direct-sum decomposition has not been incorrectly declared an orthogonal projective decomposition over S .

The two full coefficient images of FLW13, as formal ghost series at positive times, are

$$\begin{aligned} \mathfrak{g}_d(t) &= \sum_{n \geq 1} \text{Tr}(\mathcal{C}_d(A)^n) t^n \in S[[t]], \\ \rho(\mathfrak{g}_d(t)) &= d \sum_{m \geq 1} \chi^m \left(\sum_{a,b} \omega_{ab}^{\chi^m} \right) t^{dm}, \\ s(\mathfrak{g}_d(t)) &= \sum_{m \geq 1} 1_L t^{dm} \in L[[t]]. \end{aligned} \quad (\text{FLW15})$$

All absent coefficients of the first and last series are their respective additive zeros. The mass-weighted series inserts the literal $p^{kdm/2}$ into the second line and leaves the third line unchanged. These are formal series, so no analytic convergence is being substituted for a finite coefficient calculation. The support trace records divisibility of the return time but not its integer multiplicity d ; addition in L is idempotent. The stronger cyclic-support object is retained below.

26.4 The unchanged metric, including support left by cancellation

The full block pairing of two vectors in FLW5 is

$$\mathfrak{h}(v, w) = \left(\sum_{j=0}^{d-1} v_j^* G_N w_j, \bigvee_{j=0}^{d-1} (\lambda_j \wedge \mu_j) \right). \quad (\text{FLW16})$$

It takes values in S : a nonzero summand in its complex coordinate requires both involved supports to be top. It is the sum of the original supported Hermitian pairings, with the involution conjugating amplitude and fixing support. Under FLW11 the supports are merely permuted, so the exact signed difference formed using $(-1, 1_L)$ is

$$\begin{aligned} & \mathfrak{h}(\mathcal{C}_d(A)v, \mathcal{C}_d(A)w) + (-1, 1_L)\mathfrak{h}(v, w) \\ &= \left(v_{d-1}^* (A^* G_N A - G_N) w_{d-1}, \bigvee_j (\lambda_j \wedge \mu_j) \right). \end{aligned} \quad (\text{FLW17})$$

The complex cancellation is the original TOB13 identity. Its support does not cancel: the zero-amplitude contributions from the other slots still supply the displayed join. Replacing that join by only $\lambda_{d-1} \wedge \mu_{d-1}$ would be a false semiring equality. For the mass-weighted operator the complex defect against p^k times the original pairing is p^k times the same displayed amplitude, while its support remains this same join. All phases and all entries of G_N in $A^* G_N A$ are unchanged.

26.5 Exact Frobenius and Verschiebung maps on the projectives

For an automorphism T of a projective P define $F_n(P, T) = (P, T^n)$ for $n \geq 1$, and let $V_d(P, T)$ be its cyclic construction on $P^{\oplus d}$ with corner T . We use actual zero off-block maps in this definition. These definitions apply in particular to FLW11 with $P = P_q$ and $T = A^\uparrow$. The equalities $F_m F_n = F_{mn}$ follow by powers. The isomorphism $V_m V_n(P, T) \cong V_{mn}(P, T)$ sends the outer slot i and inner slot j to $i + mj$. One outer wrap advances the inner slot and the final wrap uses T , so the permutation proves the identity on every coordinate. It preserves the unchanged direct sum metric and all independent slot supports.

For the full mixed relation let $g = \gcd(d, n)$, $l = d/g$, and $a = n/g$. There is an explicit isomorphism

$$F_n V_d(P, T) \cong \bigoplus_{t=0}^{g-1} V_l(P, T^a). \quad (\text{FLW18})$$

For $P = P_q$ and $T = A^\uparrow$, its map from the right to the left sends the slot j of the t th summand to the original slot $r_{t,j} = t + jn \bmod d$, with block map $(A^{w_{t,j}})^\uparrow$, where $w_{t,j} = \lfloor (t + jn)/d \rfloor$ and $0 \leq j < l$. The pairs (t, j) enumerate every slot exactly once: the congruence classes modulo g are the disjoint orbits, each of length l . The coordinate action FLW12 adds exponent $w_{t,j+1} - w_{t,j}$ at each step. All intermediate steps therefore intertwine, and $w_{t,l} = a$, $r_{t,l} = t$ proves the returning corner A^a . Invertibility of A proves invertibility of this map. For a general invertible T use its powers in the same formula. In particular $F_d V_d(P, T) = (P, T)^{\oplus d}$ and, when $(d, n) = 1$, $F_n V_d(P, T) \cong V_d F_n(P, T)$.

The amplitude comparison is not asserted to be an isometry for G_N when powers of A occur. Its exact pulled-back form is

$$\sum_{t=0}^{g-1} \sum_{j=0}^{l-1} x_{t,j}^* (A^{w_{t,j}})^* G_N A^{w_{t,j}} y_{t,j}. \quad (\text{FLW19})$$

Its support pairing is the unchanged corresponding join, because each block map preserves its support. Thus the algebraic relation has its complete metric comparison and does not remove a weight defect by changing metrics.

For automorphisms X of P and Y of Q , with product defined by tensoring, the projection formula is the concrete isomorphism

$$V_n(P \otimes Q, X^n \otimes Y) \cong (P, X) \otimes V_n(Q, Y). \quad (\text{FLW20})$$

From left to right use $X^j \otimes I$ in slot j . On the right each nonreturning step uses $X \otimes I$ and the return uses $X \otimes Y$; at the last step their product is $X^n \otimes Y$. This checks the intertwining and gives its inverse $X^{-j} \otimes I$. Likewise

$$V_n(P, X) \otimes V_n(Q, Y) \cong \bigoplus_{t=0}^{n-1} V_n(P \otimes Q, X \otimes Y). \quad (\text{FLW21})$$

On the t th summand send slot j to the pair $(j, t + j \bmod n)$, using block map $I \otimes Y^{\lfloor (t+j)/n \rfloor}$. The simultaneous cycle steps preserve t ; when either coordinate wraps, its X or Y is exactly supplied by this formula, and the full return uses $X \otimes Y$. The orbits partition all n^2 slots and all block maps are invertible, proving FLW21. For $P = P_a$, $Q = P_b$ with their retained complex metrics G_P, G_Q , the exact metric comparisons are respectively $(X^j)^* G_P X^j \otimes G_Q$ and $G_P \otimes (Y^{\lfloor (t+j)/n \rfloor})^* G_Q Y^{\lfloor (t+j)/n \rfloor}$. The support maps are permutations of the actual slot labels, with the indicated tensor meet within each slot. No cancellation of support or unproved unitary conjugacy is used.

The trace ghosts obey these isomorphisms in S , by FLW7. For example FLW18 says that the r th trace of its left side is the sum of g copies of the r th trace of $V_l(P, T^a)$. For FLW11 the support condition on both sides is $d \mid nr$, equivalently $l \mid r$; at return the complex multiplicity is $gl = d$. Thus the integer multiplier survives in amplitude while the support join still has only the values $0_L, 1_L$.

The same formulas can be used without discarding the original mass. For any supported nonzero scalar η , the diagonal gauge $\text{diag}(I, \eta I, \dots, \eta^{d-1} I)$ gives

$$\eta \mathcal{C}_d(A) \cong \mathcal{C}_d(\eta^d A). \quad (\text{FLW22})$$

Indeed a forward step introduces one η and the full return introduces η^d . For $\eta = p^{k/2}$ this is precisely the corner $p^{kd/2} A$, not $p^{k/2} A$. The gauge is not declared unitary: its pulled-back block metric has the factors $p^{kj} G_N$. Its power trace is exactly $p^{kn/2}$ times FLW13.

The signed odd family has an additional exact composition statement; it must not be confused with Verschiebung at one fixed corner. For an invertible original F put $\mathcal{W}_d(F) = V_d(\chi_d F^{\chi_d})$ for odd d . For odd c, d there is an explicit full-support isomorphism

$$\mathcal{W}_c(\mathcal{W}_d(F)) \cong \mathcal{W}_{cd}(F). \quad (\text{FLW36})$$

Here the formula means an operator construction on the corresponding direct sums of projectives; each newly introduced slot remains independent. If $\chi_c = 1$, flatten the two cyclic sums as before, using $\chi_{cd} = \chi_d$. If $\chi_c = -1$, write $A = \chi_d F^{\chi_d}$ and first compare $-\mathcal{C}_d(A)^{-1}$ with $\mathcal{C}_d(-A^{-1})$. The intertwiner from the latter to the former sends slot j to $-j \bmod d$ with block $(-1)^j (A^{\lfloor -j/d \rfloor})^\uparrow$. FLW12 with step -1 proves the inverse-cycle part. The successive signs supply a factor -1 at each step, including the return since d is odd. Thus the map and its inverse are proved directly. Its returning corner is $-A^{-1} = -\chi_d F^{-\chi_d} = \chi_{cd} F^{\chi_{cd}}$. Apply this same intertwiner in each of the c outer slots and then flatten the cycles; the result is FLW36. In this inverse case its metric contains G_N in slot zero and $(A^{-1})^* G_N A^{-1}$ in each other slot, followed by the stated permutation; the signs have absolute value one. It is not silently made an isometry.

For general positive n , the exact Frobenius result for the signed family is instead

$$F_n \mathcal{W}_d(F) \cong \bigoplus_{t=0}^{g-1} V_{d/g}(\chi_d^{n/g} F^{\chi_d n/g}), \quad g = \gcd(d, n). \quad (\text{FLW37})$$

This is FLW18 with its literal corner substituted. For $(d, n) = 1$ and odd n it agrees with $\mathcal{W}_d(F^n)$. For coprime even n and $\chi_d = -1$, its corner is F^{-n} whereas the latter proposed replacement has corner $-F^{-n}$. Their exact amplitude difference at that corner is $2F^{-n}$, an invertible map, and both have the same cyclic support. On the identified d slots the signed difference of the two endomorphisms has this last-to-first amplitude, while every cyclic edge still retains its supported-zero support when its other amplitudes cancel. Thus FLW37, rather than an unqualified commutation assertion for the signed family, records the full parity defect.

26.6 Global synchronization and its exact defect object

The single synchronized carrier of amplitude dimension dq is P_{dq} , whose free idempotent has all-top support in all dq coordinates. It is not FLW5. The exact comparison consists of

$$\begin{aligned} j : P_{dq} &\longrightarrow P_{q,d}, & (v, \lambda) &\longmapsto ((v_0, \lambda), \dots, (v_{d-1}, \lambda)), \\ r : P_{q,d} &\longrightarrow P_{dq}, & ((v_j, \lambda_j)_j) &\longmapsto ((v_j)_j, \bigvee_j \lambda_j), & rj = I. \end{aligned} \quad (\text{FLW23})$$

Substitution in FLW1 proves S -linearity. Their other composite jr synchronizes all slot labels to their join; it generally is not the identity. The complete fibre over (v, λ) is

$$\{(\lambda_j) : \bigvee_j \lambda_j = \lambda, v_j \neq 0 \Rightarrow \lambda_j = 1_L\}. \quad (\text{FLW24})$$

Every vector amplitude is retained, and the exact identifications among labels are displayed rather than removed silently. The complex $C_d(A)$ has a lift $C_d(A)^\uparrow$ on P_{dq} , with all-top support in every block. Because FLW11 permutes the independent labels, the maps j, r intertwine these two operators. Its intrinsic trace is, at every integer n ,

$$\text{Tr}_{P_{dq}}((C_d(A)^n)^\uparrow) = (\text{tr}_{\mathbb{C}} C_d(A)^n, 1_L). \quad (\text{FLW25})$$

At $d \nmid n$ FLW25 is e whereas FLW13 is τ . For $d \mid n$ they agree exactly. Thus replacing independent cyclic slots by the globally synchronized lift loses a proved part of the semiring trace even though the entire original complex operator and all its metric entries are the same.

This difference defines an actual comparison object: the split surjection r , its section j , its idempotent jr , the full fibres FLW24, and the two intertwined automorphisms. For $d > 1$ its label relation is nontrivial. For example choose any slot and place support 1_L only there, with every amplitude zero; all these distinct vectors map to $(0, 1_L)$. There is no assertion that an additive difference of semimodules or a direct-sum complement of j exists.

26.7 What group completion kills, and what it does not kill

The additive group completion of the coefficient semiring S is exactly $\mathbb{C}$. Indeed $z_\lambda + z_\lambda = z_\lambda$ forces every $[z_\lambda]$ to be zero in the group. Every other element has the form $(c, 1_L)$, and $(a, 1_L) + (b, 1_L) = (a + b, 1_L)$. The class of $(0, 1_L)$ is zero, so these classes define the inverse of the homomorphism induced by ρ to the additive group $\mathbb{C}$. Multiplication is compatible by FLW1. In particular

$$S^{\text{gp}} \cong \mathbb{C}, \quad [e] = [\tau] = 0. \quad (\text{FLW26})$$

This proves a statement about scalar coefficients, not a vanishing theorem for projective objects or their endomorphisms.

Here is a direct invariant detecting the distinction. Choose a bounded lattice homomorphism $\beta : L \rightarrow \mathbb{B} = \{0, 1\}$. Such a homomorphism exists for a nontrivial bounded distributive lattice: choose a maximal proper ideal by Zorn's lemma. It is prime, because if x, y are outside it, maximality gives $i \vee x = 1$ and $j \vee y = 1$ for ideal elements i, j ; if $x \wedge y$ were in the ideal,

distributing these equalities would put 1 in it. The complementary indicator is the required bounded homomorphism. This chosen probe does not replace L in FLW1–FLW25.

Base change along $\beta_S : S \rightarrow \mathbb{B}$ gives

$$(P_{q,d}, \mathcal{C}_d(A))_{\mathbb{B}} = (\mathbb{B}^d, \text{cyclic permutation of the } d \text{ coordinates}), \quad (P_{dq}, C_d(A)^\uparrow)_{\mathbb{B}} = (\mathbb{B}, I). \quad (\text{FLW27})$$

Every returning block of $A^\uparrow$ has support all top, so it acts as the identity on the single synchronized Boolean coordinate; its nonzero amplitude, sign, and phase do not alter this map. For the actual source, both objects are automorphisms, so their stable images are their full underlying finite sets.

For any endomorphism T of a finitely generated projective S -semimodule P , its Boolean base change $P_{\mathbb{B}}$ is finite: it is an image of $\mathbb{B}^a$ for a finite a . Let $T_{\mathbb{B}}^\infty(P_{\mathbb{B}})$ be the eventual image of its descending finite image sequence and define

$$b(P, T) = \log_2 |T_{\mathbb{B}}^\infty(P_{\mathbb{B}})|. \quad (\text{FLW28})$$

It is invariant under isomorphism and additive on direct sums, because direct sums of finite semimodules are their Cartesian products and iteration acts componentwise. Null endomorphisms have singleton eventual image and value zero. Thus FLW28 defines a group homomorphism from the group generated by projective endomorphism isomorphism classes with the direct-sum relations, even after quotienting by all null-endomorphism classes. It also respects the image-stabilization relation whenever that image is a projective in this category. Write its factorization as $T = ba$, with $a : P \rightarrow T(P)$ and $b : T(P) \rightarrow P$; the restricted endomorphism on $T(P)$ is ab . After base change the same two factorizations hold, even without assuming that the changed b is injective. For two maps a, b between finite sets, a takes the eventual image of ba to that of ab and intertwines their permutations there. It is injective on that image because $ax = ay$ implies $(ba)x = (ba)y$ and ba is bijective there; the same reasoning makes b injective on the other eventual image. The two finite eventual images therefore have equal cardinality, proving the claimed relation. For the two objects in FLW27 the values are respectively d and 1. Their difference therefore has nonzero invariant $d - 1$ for $d > 1$. No relation from a different, unspecified exact category is assumed.

Even a projective having zero complex amplitude can survive this group. Namely $eS = \text{im}(e : S \rightarrow S)$ is projective because $e^2 = e$. Its amplitude base change is zero, its full support module is L , and its Boolean base change is $\mathbb{B}$. Consequently

$$\text{Tr}_{eS}(I) = e, \quad [e] = 0 \text{ in } S^{\text{gp}}, \quad b(eS, I) = 1. \quad (\text{FLW29})$$

These explicit maps prove that scalar group completion kills its trace coefficient while the projective endomorphism class remains nonzero in the stated additive quotient.

26.8 The exact pointed-cycle receiver and corrected multiplicities

The support module before choosing β is L^d with the literal cyclic permutation of its slots. After the probe it is $\mathbb{B}^d$, whose nonzero join-irreducible elements are precisely the d unit coordinate vectors: a subset with at least two elements is a join of two proper subsets, whereas a singleton has no such expression. An automorphism preserves this property and therefore permutes the same d elements. Adding a fixed base point gives the finite pointed cycle $C(d)$ in the notation of Connes–Consani. For free Boolean modules, this operation respects direct sums and tensor products: singleton coordinates of $\mathbb{B}^{a+b}$ are the disjoint union of those of its factors, and the free tensor $\mathbb{B}^a \otimes \mathbb{B}^b = \mathbb{B}^{ab}$ has coordinate pairs. These identifications intertwine the permutations. Thus it gives an exact additive and multiplicative comparison on the subcategory of the support cycles constructed here.

The original pointed-cycle ghost and the scalar lattice trace now have their precise relation:

$$\text{gh}_n C(d) = d \mathbf{1}_{d|n}, \quad \text{tr}_L(P_d^n) = \begin{cases} 1_L, & d \mid n, \\ 0_L, & d \nmid n. \end{cases} \quad (\text{FLW30})$$

The first counts the d fixed nonbasepoint slots; the second is the join of their diagonal indicators. It is obtained from the nonnegative integer ghost by the homomorphism $\mathbb{N} \rightarrow L$ sending zero to 0_L and every positive integer to 1_L . The actual cycle object retains the multiplicity that this homomorphism loses. For comparison the full fixed set of P_d^n on $\mathbb{B}^d$ has cardinality $2^{\gcd(d,n)}$: a fixed subset is a union of the $\gcd(d,n)$ permutation orbits. Thus counting all subset states would give a different invariant from the pointed atom trace; the explicit atom map specifies the source invariant being used.

The root-of-unity receiver of the authors' theorem is exactly

$$\Theta_d = \sum_{d\alpha=0} [\alpha] \in \mathbb{Z}[\mathbb{Q}/\mathbb{Z}], \quad \widehat{\Theta}_d(n) = \sum_{j=0}^{d-1} e^{2\pi i n j/d} = d \mathbf{1}_{d|n}. \quad (\text{FLW31})$$

The finite geometric sum proves the equality. Hence the full-support comparison defect FLW23 has pointed-cycle image $\Theta_d - \Theta_1$, with ghost $d \mathbf{1}_{d|n} - 1$. It is nonzero for $d > 1$, for example at $n = 1$. This proves its connection with the authors' universal invariant without asserting that the original semiring scalar trace itself is that group-ring invariant.

There are two source formulas that require explicit correction in the cited original version. If $m(k)$ denotes the number of cycles of length k , direct counting gives

$$g_n = \sum_{k|n} k m(k), \quad m(k) = \frac{1}{k} \sum_{d|k} \mu(k/d) g_d. \quad (\text{FLW32})$$

For the inverse formula, substitute the first sum. The coefficient of $j m(j)$ is $\sum_{j|d|k} \mu(k/d)$, which equals one when $j = k$ and zero otherwise. This divisor identity follows by expanding $\prod_{p|(k/j)} (1 - 1)$, with value one for $k/j = 1$. The source at lines 510–512 omits both the cycle length in its preceding count and the factor $1/k$ in its inverse. A single two-cycle has $g_1 = 0, g_2 = 2$ and $m(2) = 1$, which checks the factor directly. The primitive-root formula at line 516 has a mismatched dummy index; its correctly indexed form is $\sum_{d|n} \mu(n/d) \Theta_d$, following from partitioning roots by exact order. These corrections do not alter the cycle invariant FLW31.

Even the corrected signed Möbius coefficient over the full semiring has a retained supported-zero tail. For the operator $\mathcal{C}_d(I)$ on $P_{q,d}$, let t_n denote its FLW13 trace, and form, for $k \geq 1$, the literal finite sum

$$M_k^S = (1/k, 1_L) \sum_{\substack{l|k \\ \mu(k/l) \neq 0}} (\mu(k/l), 1_L) t_l = \begin{cases} \tau, & d \nmid k, \\ (q, 1_L), & k = d, \\ e, & d | k, k \neq d. \end{cases} \quad (\text{FLW33})$$

Its amplitude is FLW32 for q complex copies of the d -cycle. If $d | k$, the summand $l = k$ with Möbius value one has top support; the support join therefore remains top even when all complex terms cancel. If $d \nmid k$, none of its divisors returns and every summand is τ . This proves all cases of FLW33 and exhibits precisely why subtraction after scalar group completion loses more information than the original semiring calculation.

26.9 Adams operations and the actual domain of Newton recovery

For the original amplitude corner $A = \chi F_p^\chi$, retain its labelled eigenvalues $\alpha_{ab} = \chi \omega_{ab}^\chi$ and set $\lambda_A(t) = \det(I + tA) = \prod_{a,b} (1 + \alpha_{ab} t)$. Direct formal differentiation of this finite product proves

$$t \lambda'_A(t) / \lambda_A(t) = \sum_{n \geq 1} (-1)^{n-1} \text{tr}(A^n) t^n, \quad j e_j = \sum_{n=1}^j (-1)^{n-1} e_{j-n} \text{tr}(A^n), \quad (\text{FLW34})$$

where $\lambda_A(t) = \sum_{j=0}^q e_j t^j$ and $e_0 = 1$. For the first identity expand

$$\frac{\alpha t}{1 + \alpha t} = \sum_{n \geq 1} (-1)^{n-1} \alpha^n t^n;$$

for the second multiply by $\lambda_A(t)$ and compare coefficients. The sign at `lognewton`, source lines 644–647, is printed as $(-1)^n$; the coefficient of t already requires the displayed $(-1)^{n-1}$. This agrees with the source’s preceding explicit $\psi_1 = \lambda_1$ and $\psi_2 = \lambda_1^2 - 2\lambda_2$.

For the cyclic amplitude receiver,

$$\det(I - tC_d(A)) = \det(I - t^d A) = \prod_{a,b} (1 - \chi \omega_{ab}^\chi t^d), \quad \text{tr}(A^m) = d^{-1} \rho(\text{Tr } C_d(A)^{dm}). \quad (\text{FLW35})$$

The determinant follows by restricting to each original Lagrange label, whose cyclic matrix has polynomial $z^d - \alpha_{ab}$. Those labelled invariant complex subspaces sum to the full amplitude space, so all multiplicities are retained. The second equality is FLW13. FLW34 then recovers every coefficient of the original $\det(I - tA)$ from its first q returning traces. For the weighted receiver the factor is $\chi p^{kd/2} \omega_{ab}^\chi$ and the returning traces include $p^{kdm/2}$ exactly as in FLW14–FLW15.

FLW34–FLW35 are statements in the complex amplitude ring or its formal series ring. They are not assumed determinant or exterior power identities for the full semiring projectives. Over S , $(1, 1_L) + (-1, 1_L) = e \neq \tau$; an alternating complex cancellation does not give the semiring additive zero. The full coefficient object retained in parallel is FLW15 together with its projective slot action FLW11 and the comparison FLW23–FLW24. Its exact images under both coefficient maps and the surviving cycle invariant have all been proved above. In particular neither the Newton amplitude recovery nor scalar group completion deletes the constructed projective support defect.

27 Exact monodromy compatibility of the original prime receiver

Source convention and original input. Uwe Jannsen, *Weights in Arithmetic Geometry*, arXiv:1003.0927v1, original author file `main.tex`, lines 325–365, was read for this calculation. Lines 335–340 state $\mathcal{N}F = q_{\text{res}}F\mathcal{N}$ for geometric Frobenius and nilpotent monodromy, with q_{res} the residue-field cardinality. Lines 341–352 specify the associated monodromy filtration and twists. The retained source has SHA-256

deadee9fddd84a96a5f8a5ad282f929ad837b1f0a67bbb04907980681b12ab0a.

We test the stated geometric convention with residue field $\mathbb{F}_p$, so its equation becomes $F_p\mathcal{N}F_p^{-1} = p^{-1}\mathcal{N}$. The programme’s symbol $q = (k+1)^2$ remains its vector-space dimension and is not the q_{res} in Jannsen’s equation. This is a finite compatibility calculation, not an assertion that the programme operators already have an arithmetic realization.

The precise programme input is ST1–ST3 of the retained `SUPPORT_TRANSPORT_PUBLIC.md` and TDP1–TDP4, TDP11 of `TAU_DELiGNE_PRIME_MEMORY.tex`. ST3 states the exact source-metric decomposition below and locates its proof in CE2–CE8 of the pinned original effective-cutoff proof. The present derivation read ST1–ST3 and their displayed proof locator; it does not claim a new derivation of those source identities from an unspecified metric. Keep every original parameter and cutoff:

$$\begin{aligned} k \geq 17, \quad k \equiv 1 \pmod{4}, \quad q = (k+1)^2, \quad \delta, \gamma > 0, \quad q-1 \leq N \leq 2q, \\ \lambda_{ab} = (2b-k)\gamma - i(2a-k)\delta, \quad Q_k(y) = \prod_{a,b=0}^k (y - \lambda_{ab}), \quad E = \mathbb{C}[y]/(Q_k), \quad M[v] = [yv], \\ F_p = e^{i(\log p)M}, \quad \tilde{F}_p = p^{k/2}F_p, \quad \langle v, w \rangle_N = v^*G_Nw, \\ M = C + R, \quad C = C^\dagger, \quad R = \epsilon f e^\dagger, \quad e^\dagger f = 0, \quad \|e\|_N = \|f\|_N = 1, \quad \epsilon > 0. \end{aligned} \tag{MRR1}$$

Here the actual ST3 vectors are $e = b_N/\sqrt{E_N}$, $f = b_{N+1}/\sqrt{F_N}$ and $\epsilon = \sqrt{E_N F_N}/\omega_N$. These are the retained source definitions, not substitutions made in this derivation. The four cutoffs $q-1, q, 2q-1, 2q$ remain distinct.

27.1 Two primes determine the complete all-prime space

Let ℓ_{ab} be the original Lagrange cardinal polynomials. Their distinct roots give a basis of E by evaluation, and $M\ell_{ab} = \lambda_{ab}\ell_{ab}$. Let ℓ^{ab} denote the linear evaluation covector at λ_{ab} , so $\ell^{ab}(\ell_{cd}) = \mathbf{1}_{(a,b)=(c,d)}$. It is not silently replaced by a Hermitian adjoint. Put $U_{ab,cd} = \ell_{ab}\ell^{cd}$ and $P_{ab} = U_{ab,ab}$. These are the exact matrix units and spectral projectors in the original coordinates. In particular

$$F_p U_{ab,cd} F_p^{-1} = p^{2(a-c)\delta} e^{2i(b-d)\gamma \log p} U_{ab,cd}, \quad [M, U_{ab,cd}] = (\lambda_{ab} - \lambda_{cd})U_{ab,cd}. \tag{MRR2}$$

Applying either side to every original cardinal vector proves these identities, including their phases.

The complete compatibility space is

$$\begin{aligned} \mathcal{N} &= \{T : F_2 T F_2^{-1} = 2^{-1}T, F_3 T F_3^{-1} = 3^{-1}T\} \\ &= \{T : F_p T F_p^{-1} = p^{-1}T \text{ for every prime } p\} \\ &= \ker(\text{ad}_M - iI). \end{aligned} \tag{MRR3}$$

To prove equality, a nonzero matrix-unit coefficient must satisfy both scalar equations in MRR2. Their moduli force $2(a-c)\delta = -1$. Their phases force $2(b-d)\gamma \log 2 = 2\pi n_2$ and $2(b-d)\gamma \log 3 = 2\pi n_3$, with integers n_2, n_3 . If $(b-d)\gamma \neq 0$, both integers are nonzero, so $\log 2/\log 3 = n_2/n_3$ would be rational. This is impossible: a rational equality gives, after choosing positive numerator

and denominator, $2^v = 3^u$, contradicted by the prime divisors of these integers. Thus $b = d$ because $\gamma > 0$, and the modulus equality gives $\lambda_{ab} - \lambda_{cd} = i$. Conversely this eigenvalue difference makes the scalar in MRR2 equal $e^{-\log p} = p^{-1}$ for every prime, and also gives the commutator equation. Independence of all matrix units proves the equalities for every operator.

Writing $m = c - a$, the space is therefore exactly

$$\left\{ \begin{array}{ll} \mathcal{N} = 0, & \delta \neq 1/(2m) \text{ for all } 1 \leq m \leq k, \\ \mathcal{N} = \bigoplus_{a=0}^{k-m} \bigoplus_{b=0}^k \mathbb{C} U_{ab, a+m, b}, & \delta = 1/(2m), \quad 1 \leq m \leq k. \end{array} \right. \quad (\text{MRR4})$$

Indeed a, c are integers in $[0, k]$ and $\delta > 0$, so the equation $2(a - c)\delta = -1$ has precisely that range of positive m . For a fixed δ at most one m occurs. At resonance its dimension is $(k + 1 - m)(k + 1)$, and otherwise it is zero. Multiplication of F_p by its retained scalar $p^{k/2}$ does not change conjugation, so exactly the same space results from the original uncentred $\tilde{F}_p$.

Every operator in this space is nilpotent, with its whole chain law specified. At resonance write $T = \sum_{a=0}^{k-m} \sum_b n_{ab} U_{ab, a+m, b}$. Then

$$T^s \ell_{ab} = \begin{cases} \left(\prod_{j=1}^s n_{a-jm, b} \right) \ell_{a-sm, b}, & a \geq sm, \\ 0, & a < sm, \end{cases} \quad T^{\lfloor k/m \rfloor + 1} = 0. \quad (\text{MRR5})$$

The first step is the definition, and induction proves the product for every s . It lowers a by m and keeps b fixed, so the last equality follows from $a \leq k$. The nilpotence bound is sharp: taking all $n_{ab} = 1$ keeps the longest chain from $a = k$ nonzero through $\lfloor k/m \rfloor$ steps.

27.2 The exact spectral and metric projections

At resonance the coordinate projection is

$$S(T) = \sum_{a=0}^{k-m} \sum_{b=0}^k P_{ab} T P_{a+m, b}; \quad S = 0 \text{ off resonance.} \quad (\text{MRR6})$$

The matrix-unit relations prove $S^2 = S$ and $\text{im } S = \mathcal{N}$. It commutes with ad_M and all original prime conjugations because it selects their simultaneous coordinate eigenvectors. It is not asserted to be orthogonal in G_N .

The actual metric projection is also explicit. Let V be the invertible matrix whose columns are all original ℓ_{ab} in their retained ordering, and put $H = V^* G_N V$. On $\text{End}(E)$ use the exact Hilbert–Schmidt inner product $\langle A, B \rangle_{\text{HS}, N} = \text{tr}(A^\dagger B)$, $A^\dagger = G_N^{-1} A^* G_N$. In the cardinal coordinates this is $\text{tr}(H^{-1} A^* H B)$. For matrix units labelled by pairs (i, j) , (u, v) , the complete Gram entries are

$$\langle U_{ij}, U_{uv} \rangle_{\text{HS}, N} = H_{iu} (H^{-1})_{vj}. \quad (\text{MRR7})$$

Indeed $U_{ij}^\dagger = H^{-1} U_{ji} H$ in those coordinates; the product $U_{ji} H U_{uv}$ is $H_{iu} U_{jv}$, whose trace after multiplication by H^{-1} is the displayed entry. Thus every off-diagonal metric contribution is retained.

Let $\{K_\alpha\}$ be exactly the compatible matrix units in MRR4 and define $\Gamma_{\alpha\beta} = \langle K_\alpha, K_\beta \rangle_{\text{HS}, N}$, $b_\alpha(T) = \langle K_\alpha, T \rangle_{\text{HS}, N}$. Their independence and the positive metric make Γ positive definite. The exact orthogonal projection and distance are

$$\begin{array}{l} P_N T = \sum_{\alpha, \beta} K_\alpha (\Gamma^{-1})_{\alpha\beta} b_\beta(T), \\ \text{dist}_{\text{HS}, N}(T, \mathcal{N})^2 = \|T\|_{\text{HS}, N}^2 - b(T)^* \Gamma^{-1} b(T). \end{array} \quad (\text{MRR8})$$

The coefficient vector solves precisely $\langle K_\alpha, T - P_N T \rangle = 0$ for every α . This proves orthogonality and the norm formula by the Pythagorean identity. Off resonance both projections vanish, and the distance is $\|T\|_{\text{HS},N}$.

27.3 The complete two-prime defect and its inverse receiver

For $\mathcal{E} = \text{End}(E)$ define

$$\begin{aligned} \mathcal{D}_0(T) &= [M, T] - iT, & \mathcal{D}_p(T) &= F_p T F_p^{-1} - p^{-1}T, & \mathcal{D}_{2,3}(T) &= (\mathcal{D}_2(T), \mathcal{D}_3(T)), \\ \mathcal{E}/\mathcal{N} &\xrightarrow{\cong} \text{im } \mathcal{D}_{2,3}, & [T] &\mapsto \mathcal{D}_{2,3}(T). \end{aligned} \quad (\text{MRR9})$$

MRR3 proves that the last map is well-defined and injective; surjectivity onto its specified image is its definition. The analogous map $[T] \mapsto \mathcal{D}_0(T)$ is likewise an isomorphism onto its image. These are the exact defect objects even when a nonzero compatible space survives.

Write $\mu_{ij} = \lambda_i - \lambda_j - i$ and $\xi_{p,ij} = \omega_{p,i}/\omega_{p,j} - p^{-1}$, where $\omega_{p,ab} = p^{(2a-k)\delta} e^{i(2b-k)\gamma \log p}$. Both maps are diagonal on the same matrix units, with coefficients μ_{ij} and $\xi_{p,ij}$ respectively. Their full original-metric energy is therefore

$$\|\mathcal{D}_{2,3}(T)\|_{\text{HS},N \oplus \text{HS},N}^2 = \sum_{ij,uv} \overline{t_{ij}} t_{uv} H_{iu} (H^{-1})_{vj} (\overline{\xi_{2,ij}} \xi_{2,uv} + \overline{\xi_{3,ij}} \xi_{3,uv}), \quad T = \sum_{ij} t_{ij} U_{ij} \quad (\text{MRR10})$$

This is expansion of each squared norm using MRR7, with no assumption that the matrix units are orthogonal.

There is an entire forward relation and an exact finite reverse relation:

$$\begin{aligned} \mathcal{D}_p &= g_p(\mathcal{D}_0) \mathcal{D}_0, & g_p(z) &= p^{-1} \frac{e^{i(\log p)z} - 1}{z}, & g_p(0) &= p^{-1} i \log p, \\ \mathcal{D}_0(T) &= \sum_{\mu_{ij} \neq 0} \sum_{p \in \{2,3\}} \frac{\mu_{ij} \overline{\xi_{p,ij}}}{|\xi_{2,ij}|^2 + |\xi_{3,ij}|^2} P_i \mathcal{D}_p(T) P_j. \end{aligned} \quad (\text{MRR11})$$

For the first equality, conjugation by e^{zM} equals $e^{z \text{ad}_M}$: differentiating the conjugated matrix yields the linear differential equation with that generator, and its exponential series is its solution. Since $\text{ad}_M = \mathcal{D}_0 + iI$, substituting $z = i \log p$ proves the entire relation, including its removable value at zero. For the second equality, MRR3 shows that every denominator in the stated range is strictly positive. On a given matrix unit the two summands add to μ_{ij} , whereas the common kernel contributes zero. This proves the identity and makes the relation of the finite-prime and infinitesimal defects explicit in both directions.

27.4 The actual nilpotent never belongs to this compatibility space

The exact ST3 decomposition gives $[M, R] = [C, R]$, because $[R, R] = 0$. The operator ad_C is self-adjoint on the original Hilbert–Schmidt space. In fact cyclicity of the finite trace gives

$$\langle X, [C, Y] \rangle_{\text{HS},N} = \text{tr}(X^\dagger C Y - X^\dagger Y C) = \langle [C, X], Y \rangle_{\text{HS},N}. \quad (\text{MRR12})$$

The last equality uses $C = C^\dagger$ from ST3; it would not follow by replacing C with an unspecified amplitude shadow. Consequently $\langle [C, R], R \rangle_{\text{HS},N}$ is real. Expanding the squared norm, its cross term with $-iR$ has zero real part. Since the rank-one norm is $\|R\|_{\text{HS},N}^2 = \epsilon^2 \|f\|_N^2 \|e\|_N^2 = \epsilon^2$, the exact obstruction is

$$\|\mathcal{D}_0(R)\|_{\text{HS},N}^2 = \|[C, R]\|_{\text{HS},N}^2 + \epsilon^2 \geq \epsilon^2 > 0. \quad (\text{MRR13})$$

In particular $R \notin \mathcal{N}$, including every resonance in MRR4. The support nilpotent has not been discarded; its nonzero image in the exact defect quotient MRR9 is exhibited. By MRR11, its two-prime defect cannot vanish either.

There is a more explicit receiver in the unchanged original rank-two plane and its orthogonal complement. Put

$$\begin{aligned} \Pi &= ee^\dagger + ff^\dagger, \quad \alpha = e^\dagger Ce \in \mathbb{R}, \quad \beta = f^\dagger Cf \in \mathbb{R}, \quad c = e^\dagger Cf, \\ u &= (I - \Pi)Ce, \quad v = (I - \Pi)Cf, \quad Ce = \alpha e + \bar{c}f + u, \quad Cf = ce + \beta f + v. \end{aligned} \quad (\text{MRR14})$$

Self-adjointness proves the conjugate coefficient in Ce . Both u and v are orthogonal to e and f . Direct multiplication gives the entire original defect

$$\mathcal{D}_0(R) = \epsilon[c(ee^\dagger - ff^\dagger) + (\beta - \alpha - i)fe^\dagger + ve^\dagger - fu^\dagger]. \quad (\text{MRR15})$$

For example $CR = \epsilon(Cf)e^\dagger$ and $RC = \epsilon f(Ce)^\dagger$; substitution of MRR14 proves every coefficient and sign. The displayed four operator terms are mutually orthogonal in the Hilbert–Schmidt metric. This follows from the rank-one formula $\langle xy^\dagger, zw^\dagger \rangle_{\text{HS},N} = \langle x, z \rangle_N \langle w, y \rangle_N$ and the orthogonalities just stated; the diagonal difference has squared norm two. Hence

$$\|\mathcal{D}_0(R)\|_{\text{HS},N}^2 = \epsilon^2[1 + (\beta - \alpha)^2 + 2|c|^2 + \|u\|_N^2 + \|v\|_N^2]. \quad (\text{MRR16})$$

It is also $\epsilon^2[1 + (\beta - \alpha)^2 + (\|Ce\|_N^2 - \alpha^2) + (\|Cf\|_N^2 - \beta^2)]$, by MRR14. No source variance or cutoff term has been omitted. Applying the actual defect to the original unit vector e gives a stronger operator-norm bound:

$$\begin{aligned} \mathcal{D}_0(R)e &= \epsilon(ce + (\beta - \alpha - i)f + v), \\ \|\mathcal{D}_0(R)\|_N &\geq \epsilon\sqrt{|c|^2 + (\beta - \alpha)^2 + 1 + \|v\|_N^2} \geq \epsilon. \end{aligned} \quad (\text{MRR17})$$

The three vectors in the first line are orthogonal, proving the exact norm on that vector before taking the bound.

For completeness the metric distance of this actual R from every compatible replacement is explicit and positive:

$$\begin{aligned} \text{dist}_{\text{HS},N}(R, \mathcal{N})^2 &= \epsilon^2 - b(R)^* \Gamma^{-1} b(R), \\ \text{dist}_{\text{HS},N}(R, \mathcal{N}) &\geq \frac{\|\mathcal{D}_0(R)\|_{\text{HS},N}}{2\|M\|_N + 1} \geq \frac{\epsilon}{2\|M\|_N + 1} > 0. \end{aligned} \quad (\text{MRR18})$$

Use the off-resonance convention of MRR8 when the compatible basis is empty. For any $T \in \mathcal{N}$, $\mathcal{D}_0(R - T) = \mathcal{D}_0(R)$. The original induced norms satisfy $\|MX\|_{\text{HS},N}, \|XM\|_{\text{HS},N} \leq \|M\|_N \|X\|_{\text{HS},N}$, proved by summing norms on an original-metric orthonormal basis or by adjoints. Thus $\|\mathcal{D}_0(X)\|_{\text{HS},N} \leq (2\|M\|_N + 1)\|X\|_{\text{HS},N}$. Taking the infimum proves the bound, with MRR13 giving strict positivity.

27.5 Complete observation blocks and all support labels

Retain the original isometric section $J : B \rightarrow E$ from ST2, $K = \ker \Lambda$, its isometric inclusion ι_K , and $W_0 : B \oplus K \rightarrow E$, $W_0(b, z) = Jb + \iota_K z$. The original orthogonal decomposition proves that W_0 is unitary for these unchanged metrics. For $D = \mathcal{D}_0(R)$ the exact complete observation is

$$W_0^\dagger D W_0 = \begin{pmatrix} J^\dagger D J & J^\dagger D \iota_K \\ \iota_K^\dagger D J & \iota_K^\dagger D \iota_K \end{pmatrix}, \quad \|D\|_{\text{HS},N}^2 = \sum_{a,b \in \{B,K\}} \|D_{ab}\|_{\text{HS}}^2. \quad (\text{MRR19})$$

This follows by summing squared matrix entries in the orthogonal source and target decomposition, and the inverse map is $D = W_0(D_{ab})W_0^\dagger$. Thus the actual nonzero obstruction MRR16 is retained by these four blocks even if its visible B, B compression alone vanishes. The same statements hold for each component of $\mathcal{D}_{2,3}(R)$ using MRR11.

For the full bounded distributive lattice L , give the complex vector spaces $\mathcal{E}, \mathcal{N}$, their quotient and each defect image their synchronized carriers $V_L^\uparrow = \{(0, \lambda) : \lambda \in L\} \cup \{(v, 1_L) : v \in V\}$. Every linear map above lifts by $T^\uparrow(v, \lambda) = (Tv, \lambda)$. Componentwise addition and support meet prove linearity over $G_L(\mathbb{C})$ and preservation of compositions. In particular

$$\begin{aligned} (\mathcal{D}_0^\uparrow)^{-1}(Z_L) &= (\mathcal{D}_{2,3}^\uparrow)^{-1}(Z_L) = \mathcal{N}_L^\uparrow, \\ (\mathcal{E}/\mathcal{N})_L^\uparrow &\xrightarrow{\cong} (\text{im } \mathcal{D}_{2,3})_L^\uparrow, \quad ([T], \lambda) \mapsto (\mathcal{D}_{2,3}T, \lambda). \end{aligned} \quad (\text{MRR20})$$

In the first line Z_L means the zero-amplitude fibre of the appropriate synchronized target, including the direct sum target for the pair. MRR3 and MRR9 prove the amplitude statements; keeping λ proves the displayed lifts. For every $\lambda < 1_L$, the quotient fibre over $(0, \lambda)$ is its singleton zero state; over $(0, 1_L)$ it is $\mathcal{N} \times \{1_L\}$. The actual $R \neq 0$ starts at top support and has nonzero defect by MRR13, so its defect remains at that same top support. All lower zero labels remain distinct and fixed by the zero-amplitude action.

The original free-coordinate correspondence is the full fibre from OLR17: in any retained basis of the positive dimension $\dim \mathcal{E} = q^2$, synchronization sends (T, μ) to $(T, \bigvee_i \mu_i)$ with $T_i \neq 0 \Rightarrow \mu_i = 1_L$. Its inverse fibre retains exactly those masks with that join and those forced top coordinates. Over the compatibility inverse image in MRR20 the amplitude is exactly a member of MRR4 and all its permitted original masks survive. No change of free-semimodule basis to the spectral units has been asserted. The metric pairing on every synchronized space is $\langle x, y \rangle, \lambda \wedge \mu$, so MRR7–MRR19 retain their original amplitudes and every support meet.

The calculation therefore identifies the full finite space compatible with Jannsen’s stated geometric-Frobenius equation, proves its exact relation to the two original prime operators, and exhibits the complete defect of the actual programme nilpotent. It does not replace that nilpotent by a resonant matrix unit, identify its slot grades with arithmetic weights, or infer a purity statement or RH from finite compatibility.

27.6 Every resonance chain, break, and Jordan block

Fix a resonance $\delta = 1/(2m)$ with $1 \leq m \leq k$, and retain an arbitrary member of MRR4, with its full complex coefficients:

$$\mathcal{N} = \sum_{a=0}^{k-m} \sum_{b=0}^k n_{ab} U_{ab, a+m, b}.$$

For each $b \in \{0, \dots, k\}$ and residue $r \in \{0, \dots, m-1\}$ define

$$\begin{aligned} L_r &= \left\lfloor \frac{k-r}{m} \right\rfloor, \quad v_{r,b,j} = \ell_{r+jm,b} \quad (0 \leq j \leq L_r), \\ c_{r,b,j} &= n_{r+(j-1)m,b} \quad (1 \leq j \leq L_r), \quad \mathcal{N}v_{r,b,j} = c_{r,b,j}v_{r,b,j-1} \quad (j \geq 1), \quad \mathcal{N}v_{r,b,0} = 0. \end{aligned} \quad (\text{MRR21})$$

Every original label occurs in exactly one such chain by division with remainder of a by m . Thus this is the entire original object, with b unchanged; no two spectral labels are identified. For the boundary tests below put $c_{r,b,0} = c_{r,b,L_r+1} = 0$. These two values mark the absence of an edge outside the original chain; they are not additional resonance coefficients.

A segment $[s, u] \subseteq [0, L_r]$ is one Jordan block precisely when

$$c_{r,b,s} = 0, \quad c_{r,b,u+1} = 0, \quad c_{r,b,j} \neq 0 \quad (s < j \leq u). \quad (\text{MRR22})$$

Indeed a zero coefficient deletes exactly its one edge. The remaining connected components are exactly these segments, since the underlying finite graph is a path. Their vector spans are $\mathcal{N}$ -invariant and form a direct sum, because they partition the original basis. In one segment put

$$w_{r,b,[s,u],h} = \mathcal{N}^h v_{r,b,u} = \left(\prod_{j=u-h+1}^u c_{r,b,j} \right) v_{r,b,u-h} \quad (0 \leq h \leq u-s). \quad (\text{MRR23})$$

The empty product is one. Every displayed product is nonzero by MRR22, so these vectors are a basis of the segment. They satisfy $\mathcal{N}w_h = w_{h+1}$ up to its last vector, which is sent to zero. Therefore the segment contributes exactly one Jordan block of length $u - s + 1$. This proves the classification for every coefficient choice, including all zero coefficients and all nonzero phases. The vectors in MRR23 are actual powers of the original operator; they have not been divided by their lengths or norms.

In particular the multiplicity of a Jordan block of length $\ell \geq 1$ is the completely specified integer

$$J_\ell(\mathcal{N}) = \sum_{b=0}^k \sum_{r=0}^{m-1} \sum_{s=0}^{L_r+1-\ell} \mathbf{1}_{c_{r,b,s}=0} \mathbf{1}_{c_{r,b,s+\ell}=0} \cdot \prod_{j=s+1}^{s+\ell-1} \mathbf{1}_{c_{r,b,j} \neq 0}. \quad (\text{MRR24})$$

A sum with negative upper bound is empty. Its term is one exactly for the segment in MRR22 whose upper endpoint is $s + \ell - 1$, and zero otherwise, proving the formula. Every admissible Jordan type is thus exactly the multiset obtained by partitioning each of the retained chain lengths $L_r + 1$ into ordered positive segment lengths, independently for every b . Conversely any such partitions are obtained by setting the separating coefficients to zero and the remaining coefficients to any chosen nonzero complex numbers, so this description is also sufficient.

For every integer $a \geq 0$ the exact ranks and kernels are

$$\begin{aligned} \text{rank } \mathcal{N}^a &= \sum_{\ell \geq 1} J_\ell(\mathcal{N}) \max(\ell - a, 0), & \dim \ker \mathcal{N}^a &= \sum_{\ell \geq 1} J_\ell(\mathcal{N}) \min(a, \ell), \\ \dim \ker \mathcal{N} &= q - \#\{(a, b) : n_{ab} \neq 0\}. \end{aligned} \quad (\text{MRR25})$$

The first two identities follow by shifting each basis in MRR23. The last follows because every nonzero coefficient is one retained edge in a disjoint union of paths, whose number of components is the number of vertices minus edges. Equivalently the nonzero columns of $\mathcal{N}$ have different target basis vectors and are independent. The nilpotence index is the largest segment length, at most $\lfloor k/m \rfloor + 1$; the zero operator has index one.

The original spectral metric remains exact throughout. Write $H_{ij} = \langle \ell_i, \ell_j \rangle_N$ as in MRR7. For segments A, B , let $d_{A,h}$ be the literal product in MRR23 and $i(A, h)$ its original remaining spectral label. Then their entire Gram matrix, including different segments, is

$$\langle w_{A,h}, w_{B,l} \rangle_N = \overline{d_{A,h}} d_{B,l} H_{i(A,h), i(B,l)}. \quad (\text{MRR26})$$

This is substitution of MRR23 into the original sesquilinear pairing. In particular distinct Jordan blocks are not assumed orthogonal. The map from all these power vectors to the original cardinal basis is a permutation followed by the displayed nonzero diagonal coefficients; its inverse divides each coefficient by that same product. This proves an exact coordinate isomorphism and its inverse, with the metric given by MRR26 rather than replaced by an identity matrix.

27.7 The exact semidirect group carried by these chains

For every positive real number x retain the original one-parameter extension

$$F_x = \exp(i(\log x)M), \quad F_x \ell_{ab} = x^{(2a-k)\delta} e^{i(2b-k)\gamma \log x} \ell_{ab}, \quad F_x F_y = F_{xy}. \quad (\text{MRR27})$$

The logarithm here is the unique real logarithm. Its additive law and the commuting exponential series prove the last equality. The original prime operators are exactly the values at $x = p$. For $\mathcal{N}$ in MRR4, the identical matrix-unit calculation MRR2 gives $F_x \mathcal{N} F_x^{-1} = x^{-1} \mathcal{N}$ for every $x > 0$.

Define the set $\mathcal{G} = \mathbb{R}_{>0} \times \mathbb{C}$ with the following full operations:

$$(x, t)(y, s) = (xy, t + x^{-1}s), \quad 1_{\mathcal{G}} = (1, 0), \quad (x, t)^{-1} = (x^{-1}, -xt). \quad (\text{MRR28})$$

Both associations of a triple have second coordinate $t + x^{-1}s + (xy)^{-1}u$, proving associativity. Substitution proves both the identity and the two inverse identities. This is a topological group for the product of the usual real and complex topologies, since these displayed operations are continuous. The exact representation is

$$\boxed{\rho_{\mathcal{N}}(x, t) = \exp(t\mathcal{N})F_x, \quad \rho_{\mathcal{N}}((x, t)(y, s)) = \rho_{\mathcal{N}}(x, t)\rho_{\mathcal{N}}(y, s).} \quad (\text{MRR29})$$

Indeed conjugating the finite exponential by F_x gives $F_x e^{s\mathcal{N}} F_x^{-1} = e^{x^{-1}s\mathcal{N}}$, term by term. Its product with $e^{t\mathcal{N}}$ is $e^{(t+x^{-1}s)\mathcal{N}}$, since both exponents are scalar multiples of the same operator. MRR27 proves the group law. The identity maps to I_E , and the inverse maps to $(e^{t\mathcal{N}} F_x)^{-1}$ by that law, with no change in the order of the two original factors. Continuity follows from the finite exponential polynomial and MRR27.

There is also a precise converse: for any endomorphism T , the expression $e^{tT} F_x$ is a representation of MRR28 if and only if $F_x T F_x^{-1} = x^{-1}T$ for all $x > 0$. Necessity follows by evaluating the representation identity on $(x, 0)(1, s) = (x, x^{-1}s)$ and differentiating at $s = 0$. Sufficiency is the same exponential multiplication just proved, valid for the convergent exponential even without an a priori nilpotence assumption. Differentiating $F_{eu} T F_{eu}^{-1} = e^{-u}T$ at $u = 0$ gives $i[M, T] = -T$, equivalently $[M, T] = iT$. Conversely that commutator equation gives the conjugation identity by exponentiation of ad_M . Together with MRR3–MRR4, this proves the full equivalence

$$\boxed{\rho_T \text{ is a representation of } \mathcal{G} \iff F_x T F_x^{-1} = x^{-1}T \ (x > 0) \iff T \in \mathcal{N}.} \quad (\text{MRR30})$$

For this original M , its last condition implies exactly the nilpotent chain classification already proved. Thus the representation has been constructed on the entire object defined by the obstruction, with a converse characterization rather than a merely formal resemblance.

The representation is faithful exactly when $\mathcal{N} \neq 0$. In the original cardinal ordering, $e^{t\mathcal{N}}$ has all diagonal entries one and only strictly lowering off-diagonal entries. Thus the diagonal of $\rho_{\mathcal{N}}(x, t)$ is the original spectrum of F_x . If it is the identity, any label with $2a - k \neq 0$ gives $x^{(2a-k)\delta} = 1$, hence $x = 1$ since $\delta > 0$. If $\mathcal{N} \neq 0$ and $t \neq 0$, the factorization $e^{t\mathcal{N}} - I = t\mathcal{N} \sum_{j \geq 0} (t\mathcal{N})^j / (j+1)!$ has invertible second factor, since it is I plus a nilpotent. Its value cannot be zero. Therefore $t = 0$. When $\mathcal{N} = 0$ its exact kernel is $\{(1, t) : t \in \mathbb{C}\}$, by the same diagonal argument.

27.8 Every phase, chain entry, and metric under the action

For an original chain label $v_{r,b,j}$ put $\omega_{r,b,j}(x) = x^{(2(r+jm)-k)\delta} e^{i(2b-k)\gamma \log x}$. The full action in the unchanged cardinal basis is

$$\rho_{\mathcal{N}}(x, t)v_{r,b,j} = \omega_{r,b,j}(x) \sum_{h=0}^j \frac{t^h}{h!} \left(\prod_{l=j-h+1}^j c_{r,b,l} \right) v_{r,b,j-h}. \quad (\text{MRR31})$$

This is the finite exponential applied after the original F_x . Zero coefficients automatically terminate every path at its actual break. The formula therefore works without discarding any broken or length-one chain.

On one segment with top label $v_{r,b,u}$, let $\omega_A(x) = \omega_{r,b,u}(x)$ and $d_A = u - s$. The power basis in MRR23 has the exact diagonal action $F_x w_{A,h} = \omega_A(x) x^{-h} w_{A,h}$, either by MRR27 or its conjugation law. Consequently

$$\rho_{\mathcal{N}}(x, t)w_{A,h} = \omega_A(x) x^{-h} \sum_{l=h}^{d_A} \frac{t^{l-h}}{(l-h)!} w_{A,l}. \quad (\text{MRR32})$$

This retains the segment's top spectral phase and every power of x ; its coefficients are not arithmetic weight assignments. Combining MRR32 with the complete Gram matrix $G_{A,l;B,n}$ from MRR26 gives the exact pairing

$$\begin{aligned} \langle \rho_{\mathcal{N}}(x, t) w_{A,h}, \rho_{\mathcal{N}}(y, s) w_{B,j} \rangle_N &= \overline{\omega_A(x)} \omega_B(y) x^{-h} y^{-j} \\ &\cdot \sum_{l=h}^{d_A} \sum_{n=j}^{d_B} \frac{\bar{t}^{l-h} s^{n-j}}{(l-h)!(n-j)!} G_{A,l;B,n}. \end{aligned} \quad (\text{MRR33})$$

This is finite sesquilinear expansion; it preserves cross terms between distinct Jordan segments and all original cutoff-dependent entries of G_N .

The uncentred mass has a full representation as well:

$$\tilde{\rho}_{\mathcal{N}}(x, t) = x^{k/2} \rho_{\mathcal{N}}(x, t), \quad \det \rho_{\mathcal{N}}(x, t) = 1, \quad \det \tilde{\rho}_{\mathcal{N}}(x, t) = x^{kq/2}. \quad (\text{MRR34})$$

The scalar $x^{k/2}$ is a group character of MRR28, so its product with MRR29 obeys the same representation law. Its values at $(p, 0)$ are the actual $\tilde{F}_p$. The nilpotent exponential has determinant one by its triangular diagonal. Also $\sum_{a,b} \lambda_{ab} = 0$ by separately summing $2a - k$ and $2b - k$ from zero to k , so the product of all eigenvalues of F_x is one. This proves both determinants, with the full original dimension q .

27.9 Units, labelled zeros, and the full support correspondence

Every operator in MRR29 is invertible, hence has the exact full-lattice automorphism

$$\rho_{\mathcal{N},L}(x, t)(v, \lambda) = (\rho_{\mathcal{N}}(x, t)v, \lambda), \quad \rho_{\mathcal{N},L}(1, 0) = I_{E_L^\uparrow}. \quad (\text{MRR35})$$

Composition of lifts is composition of their amplitudes with support unchanged. Thus MRR29 proves the group law, and the lift at $(x^{-1}, -xt)$ is its inverse, on every original nonzero vector and on every $(0, \lambda)$. Each labelled zero is fixed, individually. In particular the identity fixes supported zero and actual absence as their distinct states; it does not send them both to one chosen zero. Likewise $(\mathcal{N}_L^\uparrow)^a = (\mathcal{N}^a)_L^\uparrow$. After the largest segment length, this is the supported-zero operator $(v, \lambda) \mapsto (0, \lambda)$ rather than the constant map at τ . The exponential lift in MRR35 retains its identity term and remains an automorphism; nilpotence of the amplitude does not make its exponential a zero-valued character. The pairing of any two lifted vectors is $(\langle v, w \rangle_N, \lambda \wedge \mu)$, so the full metric identities MRR26 and MRR33 carry exactly the same support meet. Isometry is not assumed when the displayed metric amplitudes change.

The correspondence with all original free masks can also be stated with its exact identity. Let $\mathcal{F}_L$ be the free-coordinate carrier in the original positive dimension q , and let $P_{\text{syn}} : \mathcal{F}_L \rightarrow E_L^\uparrow$ be the synchronization map of OLR17. For $g \in \mathcal{G}$ define a relation

$$\mathcal{R}_g = \{(u, v) \in \mathcal{F}_L \times \mathcal{F}_L : P_{\text{syn}}v = \rho_{\mathcal{N},L}(g)P_{\text{syn}}u\}. \quad (\text{MRR36})$$

In original coordinates this means that the output amplitude is exactly $\rho_{\mathcal{N}}(g)$ times the input amplitude, their support joins agree, and every nonzero coordinate forces its own support to 1_L . Every intermediate synchronized state has a free lift: use all support coordinates equal to its label when its amplitude is zero, and all equal to 1_L otherwise. It follows, with relational composition in the indicated order, that

$$\mathcal{R}_g \circ \mathcal{R}_h = \mathcal{R}_{gh}, \quad \mathcal{R}_g^{-1} = \mathcal{R}_{g^{-1}}, \quad \mathcal{R}_{1_{\mathcal{G}}} = \{(u, v) : P_{\text{syn}}u = P_{\text{syn}}v\}. \quad (\text{MRR37})$$

For the forward containment in the first equality, apply the representation law to any intermediate state. For the reverse containment, choose a free lift of $\rho_{\mathcal{N},L}(h)P_{\text{syn}}u$ as the intermediate state; the group law puts the prescribed final state in the second relation. Inverting the defining equality

proves the second identity, and MRR35 proves the third. Thus the identity relation on free masks is precisely equality after synchronization, not the diagonal relation on individual masks. The quotient by that equivalence is exactly the synchronized carrier, on which MRR35 is the proved group action with its genuine identity. This gives the full free-mask correspondence without claiming a new free-semimodule basis change or deleting any original mask.

MRR21–MRR37 construct the chains, group, representation, metric and full-support correspondences determined by the resonance space. The actual ST3 boundary nilpotent R remains outside that space by MRR13–MRR17; none of these constructions silently replaces it by a chosen compatible coefficient pattern or supplies an arithmetic purity identification.

28 Full-lattice null homology and the actual nilpotent receiver

The author source and the exact use made of it. Alain Connes and Caterina Consani, *Homological algebra in characteristic one*, arXiv:1703.02325v1, is used through its original author file `homolalpaper.tex`. The relevant source locators are Proposition `Bmod1`, lines 610–640; Definition `catb2`, lines 1607–1609; Definition `ker&coker`, lines 1757–1765; Proposition `grandis` and its proof, lines 1862–1886; and the definition of a null chain complex, lines 2680–2685. The finite-subset quotient below is proved directly from their kernel and cokernel universal properties. It does not invoke an unread resolution theorem or identify the two coefficient categories. Their involution is written ι here because σ already denotes the original support projection.

The programme inputs are the full coefficient laws HG1, the synchronized maps OLR16–17, the actual rank-one operator OLR13, and the complete tensor-ladder proofs OLM1–15 and OLM26–28. All those objects and formulas used below are restated. The new statements are the explicit localization, the two different null-homology calculations, their exact reconstruction maps, and their application to those original receivers.

28.1 The original coefficient object and its actual null maps

Let L be the full bounded distributive lattice with bottom 0_L and top 1_L . Do not replace it by a chain or by its two endpoints. Put

$$\begin{aligned} S &= G_L(\mathbb{C}) = \{(0, \lambda) : \lambda \in L\} \cup \{(a, 1_L) : a \in \mathbb{C}\}, \\ (a, \lambda) + (b, \mu) &= (a + b, \lambda \vee \mu), \quad (a, \lambda)(b, \mu) = (ab, \lambda \wedge \mu), \\ \tau &= (0, 0_L), \quad z_\lambda = (0, \lambda), \quad e_g = z_{1_L}, \quad \hat{a} = (a, 1_L), \quad 1_S = \hat{1}, \quad \hat{0} = e_g. \end{aligned} \tag{FLH1}$$

These are the original component laws. They make S a commutative semiring; closure follows because a nonzero amplitude requires top support, and all distributive laws follow from the ring and lattice laws. The symbol e_g denotes the original top-labelled zero, not the receiver vector e .

For a complex vector space V retain the synchronized S -semimodule

$$\begin{aligned} V_L^\uparrow &= \{(0, \lambda) : \lambda \in L\} \cup \{(v, 1_L) : v \in V\}, \\ (v, \lambda) + (w, \mu) &= (v + w, \lambda \vee \mu), \quad (a, \eta)(v, \lambda) = (av, \eta \wedge \lambda), \\ \iota_V(v, \lambda) &= (-v, \lambda), \quad \varepsilon_V(v, \lambda) = (0, \lambda) = e_g(v, \lambda). \end{aligned} \tag{FLH2}$$

The semimodule laws follow from the same component computation. Since the scalar field has characteristic zero, $(V_L^\uparrow)^{\iota_V} = Z_L := \{(0, \lambda) : \lambda \in L\}$. For every linear $T : V \rightarrow W$ the original map

$$T^\uparrow(v, \lambda) = (Tv, \lambda) \tag{FLH3}$$

is S -linear and involution-equivariant. Its composition law is $(TU)^\uparrow = T^\uparrow U^\uparrow$, and its pointwise sum law is $(T + U)^\uparrow = T^\uparrow + U^\uparrow$. In particular $0^\uparrow = \varepsilon_V$ with the appropriate target. It is not the map constant at τ when L has a label different from 0_L .

There is a useful stronger statement for arbitrary S -semimodules M . The scalar $u = \widehat{-1}$ defines an involution, and

$$M^\iota = \{m : e_g m = m\}, \quad f \text{ is null} \iff e_g f = f \tag{FLH4}$$

for S -linear f , where null means that its image is fixed by the involution. To prove the first equality, $ue_g = e_g$ gives one implication. Conversely $um = m$ gives $e_g m = (1_S + u)m = m + m = 2m$. Multiplication by the actual unit $\widehat{1/2}$, and $\widehat{1/2}e_g = e_g$, give $e_g m = m$. The second equality now follows pointwise. Composition with linear maps preserves nullity, and every null map factors through its target's fixed subsemimodule. This proves the null ideal used here without assuming that S is Boolean.

28.2 Support is both an idempotent reflection and a localization

Give L its semiring operations $\vee, \wedge$. The support map and its section on labelled zeros are

$$\sigma : S \longrightarrow L, \quad (a, \lambda) \longmapsto \lambda, \quad j_L : L \longrightarrow S, \quad \lambda \longmapsto z_\lambda. \quad (\text{FLH5})$$

The first map is a unital semiring homomorphism. The second preserves addition and multiplication, but takes the unit to e_g , so it is not a unital semiring section unless $e_g = 1_S$, which does not occur over $\mathbb{C}$. The actual identities are $\sigma j_L = I_L$ and $j_L \sigma(s) = e_g s$.

There is an exact unital localization

$$S[e_g^{-1}] \simeq L, \quad V_L^\uparrow[e_g^{-1}] \simeq L. \quad (\text{FLH6})$$

For the semiring assertion, let $f : S \rightarrow A$ be unital and suppose $f(e_g)$ is invertible. It is idempotent, hence equals 1_A . Thus $f(s) = f(e_g s) = f(z_{\sigma(s)})$. The unique factor is $h(\lambda) = f(z_\lambda)$; it is a unital semiring map because $h(1_L) = 1_A$, and the original laws prove preservation of joins and meets. Conversely every unital $h : L \rightarrow A$ makes $h\sigma(e_g) = 1_A$. This is the localization universal property, including the case where inversion forces the zero semiring.

For the semimodule assertion, multiplication by e_g is the idempotent projection ε_V onto Z_L . Let A be an S -semimodule where e_g acts invertibly. It then acts as the identity. Every S -linear $f : V_L^\uparrow \rightarrow A$ satisfies $f(v, \lambda) = f(e_g(v, \lambda)) = f(0, \lambda)$. It factors uniquely through $\sigma_V(v, \lambda) = \lambda$, with the L -action on the target induced by σ . Conversely an L -linear factor gives such an S -linear map. Under $Z_L \simeq L$ its scalar action is exactly meet, which proves the second universal property in FLH6.

The same map is the full additive idempotent reflection: for every commutative idempotent monoid Q , composition gives

$$\text{Hom}_{\text{join},0}(L, Q) \simeq \text{Hom}_{\text{add},0}(V_L^\uparrow, Q). \quad (\text{FLH7})$$

Indeed, write $a_v = f(v, 1_L)$ and $c = f(0, 1_L)$. The identities $(v, 1_L) + (0, 1_L) = (v, 1_L)$ and $(v, 1_L) + (-v, 1_L) = (0, 1_L)$ imply $a_v \vee c = a_v$ and $a_v \vee a_{-v} = c$. Thus $c \leq a_v \leq c$, so $a_v = c$. The restriction $h(\lambda) = f(0, \lambda)$ preserves joins and bottom, and gives the unique factor in FLH7. In particular an additive map into a Boolean module cannot retain two different amplitudes at top support.

Every fibre and the full reconstruction are

$$\begin{aligned} \sigma_V^{-1}(\lambda) &= \{(0, \lambda)\} \quad (\lambda \neq 1_L), & \sigma_V^{-1}(1_L) &= V \times \{1_L\}, \\ \pi_V^{-1}(v) &= \{(v, 1_L)\} \quad (v \neq 0), & \pi_V^{-1}(0) &= \{(0, \lambda) : \lambda \in L\}, \\ (\pi_V, \sigma_V) : V_L^\uparrow &\xrightarrow{\sim} \{(v, \lambda) \in V \times L : v = 0 \text{ or } \lambda = 1_L\}. \end{aligned} \quad (\text{FLH8})$$

The map in the last line and its inverse are literally the ordered pair. Addition, scalar multiplication and involution on its image are FLH2. Thus the amplitude/support joint object recovers every state and operation, not merely the cardinalities of the fibres. The same formulas apply to S with $V = \mathbb{C}$.

28.3 A literal finite complex and its full supported quotient

Let

$$0 \longrightarrow C_m \xrightarrow{d_m} C_{m-1} \longrightarrow \cdots \longrightarrow C_0 \longrightarrow 0, \quad d_n d_{n+1} = 0 \quad (\text{FLH9})$$

be a finite complex of finite-dimensional complex vector spaces. Finite here refers to its length and vector-space dimensions; the underlying sets over $\mathbb{C}$ are not finite objects of the Boolean category. Outside the displayed range put $C_n = 0$ and $d_n = 0$; its synchronized completion has

the full null object $(0)_L^\uparrow = Z_L$ in those degrees. On every original state, without removing its label,

$$d_n^\uparrow d_{n+1}^\uparrow(v, \lambda) = (0, \lambda), \quad \mathcal{Z}_{n,L} = (d_n^\uparrow)^{-1}(Z_L) = (\ker d_n)_L^\uparrow. \quad (\text{FLH10})$$

The composition is null by FLH4. The second formula is also the null-kernel universal property in the entire category of S -semimodules with the scalar involution: a map g has $d_n^\uparrow g$ null exactly when its image is contained in this inverse image, so it factors uniquely through its inclusion.

Write $Z_n = \ker d_n$, $B_n = \text{im } d_{n+1} \subset Z_n$. The full supported homology and quotient are

$$\begin{aligned} \mathcal{H}_{n,L} &= (Z_n/B_n)_L^\uparrow, & q_n : \mathcal{Z}_{n,L} &\longrightarrow \mathcal{H}_{n,L}, & (z, \lambda) &\longmapsto ([z], \lambda), \\ q_n^{-1}([z], 1_L) &= (z + B_n) \times \{1_L\}, & q_n^{-1}(0, \lambda) &= \{(0, \lambda)\} & (\lambda \neq 1_L). \end{aligned} \quad (\text{FLH11})$$

This is the null cokernel of the incoming differential viewed as a map into $\mathcal{Z}_{n,L}$. Here is its full universal property, including targets not of synchronized vector form. Let M be any S -semimodule and $g : \mathcal{Z}_{n,L} \rightarrow M$ be S -linear. Then

$$g d_{n+1}^\uparrow \text{ null} \iff \exists! \bar{g} : \mathcal{H}_{n,L} \rightarrow M \text{ } S\text{-linear, } g = \bar{g} q_n. \quad (\text{FLH12})$$

To prove the forward implication, every $b \in B_n$ has an incoming preimage with top support. FLH4 gives $g(b, 1_L) = e_g g(b, 1_L) = g(0, 1_L)$. Consequently

$$g(z + b, 1_L) = g(z, 1_L) + g(b, 1_L) = g(z, 1_L) + g(0, 1_L) = g(z, 1_L).$$

Thus g is constant on every fibre in FLH11, including the singleton lower fibres, and defines the unique $\bar{g}$. Addition and scalar multiplication descend because q_n is onto and preserves them. Conversely the image of $q_n d_{n+1}^\uparrow$ consists of labelled zero states, which are fixed, and every equivariant map preserves fixed states. This proves FLH12. Its kernel is $B_{n,L}^\uparrow$, so it also proves the exact normal boundary object here.

Every amplitude chain map f_n induces the exact supported homology map

$$([z], \lambda) \longmapsto ([f_n z], \lambda). \quad (\text{FLH13})$$

It is well-defined because it sends cycles to cycles and boundaries to boundaries; FLH12 supplies the unique map. These formulas preserve compositions and identities. If C_n has its original positive Hermitian metric, the amplitude quotient metric is, without altering that metric,

$$\|[z]\|_{H_n}^2 = \min_{b \in B_n} \|z + b\|_{C_n}^2 = \|P_{Z_n \cap B_n^\perp} z\|_{C_n}^2. \quad (\text{FLH14})$$

Finite-dimensional orthogonal decomposition of Z_n proves existence, uniqueness of the minimizing representative, and the equality. The supported pairing uses its original amplitude pairing and the support meet; all labelled zeros have amplitude norm zero but remain distinct.

28.4 The precise Connes–Consani reflection and its homology

Denote by Blnv the Connes–Consani category of join-semilattices with bottom and a join-preserving involution. Its null maps have fixed image. The object L , with its original joins and the identity involution, is a null object. FLH6–7 give a functor from the actual synchronized category to Blnv :

$$\mathcal{R}(V_L^\uparrow) = (L, I_L), \quad \mathcal{R}(T^\uparrow) = I_L, \quad \sigma_W T^\uparrow = \sigma_V. \quad (\text{FLH15})$$

Equivalently it is localization followed by forgetting the meet action. This is a genuine functor, since all identities and compositions become the identity on the same L . In the reflected complex each differential is I_L , which is null because the involution is the identity. Its null kernel is L ; the

null cokernel of the incoming identity is also L , with quotient map the identity: every equivariant map from this null object has fixed image and hence already satisfies the cokernel condition. Thus

$$H_n^{\text{CC}}(\mathcal{R}(C^\uparrow)) = (L, I_L) = \mathcal{R}(\mathcal{H}_{n,L}) \quad \text{in every degree.} \quad (\text{FLH16})$$

This equality, with its explicit maps, is the homology comparison for the additive Boolean reflection. It is not an equality with the amplitude Z_n/B_n . Even if $d_n = 0$ on a nonzero vector space, all its amplitudes are lost under FLH15, exactly as the fibres in FLH8 show. One cannot instead apply the exchange-involution square $L \times L$ and keep the identity differential: its square would be the nonnull identity whenever L has two distinct elements. The null condition must be checked for the actual involution and maps.

The faithful joint object retains the amplitude complex and the reflected support complex together with the incidence condition $v = 0$ or $\lambda = 1_L$. Its homology is exactly

$$\{(h, \lambda) \in (Z_n/B_n) \times L : h = 0 \text{ or } \lambda = 1_L\}, \quad (\text{FLH17})$$

with all operations FLH2. This is proved by the quotient map and every fibre in FLH11, not by interchanging an unspecified limit with homology. The amplitude map on that quotient is $([z], \lambda) \mapsto [z]$, and its support map is the identity on λ . Therefore both the lost data and the reconstruction are completely specified.

28.5 An amplitude-retaining Boolean functor and its exact defect

There is also a different, nonadditive-on-states, functor to $\mathbf{Blnv}$ that retains each original state. Define

$$\begin{aligned} \mathcal{P}_L(V) &= \{(A, \lambda) : A \subset V \text{ finite}, 0 \in A, A \neq \{0\} \Rightarrow \lambda = 1_L\}, \\ (A, \lambda) \vee (A', \mu) &= (A \cup A', \lambda \vee \mu), \quad 0_{\mathcal{P}} = (\{0\}, 0_L), \quad \iota(A, \lambda) = (-A, \lambda), \\ \mathcal{P}_L(T)(A, \lambda) &= (T(A), \lambda), \quad j_V(v, \lambda) = (\{0, v\}, \lambda). \end{aligned} \quad (\text{FLH18})$$

Union and join prove every Boolean-module law; negation is an involution preserving those operations. A linear T sends 0 to 0, preserves unions under direct image, and commutes with negation. It therefore gives a morphism in $\mathbf{Blnv}$, with compositions and identities preserved. The map j_V is injective, pointed, equivariant and natural: $\mathcal{P}_L(T)j_V = j_W T^\uparrow$. It does not preserve vector addition: for a nonzero v , $\{0, v\} \cup \{0, -v\} \neq \{0\}$. Thus FLH18 asserts no additive identification forbidden by FLH7. No scalar-distributive S -module structure on finite sets is being assumed.

The null complex and its cycles are now calculated exactly:

$$\begin{aligned} \mathcal{P}_L(d_n)\mathcal{P}_L(d_{n+1})(A, \lambda) &= (\{0\}, \lambda), \\ \mathcal{K}_n &= \{(A, \lambda) \in \mathcal{P}_L(C_n) : d_n(A) = -d_n(A)\}. \end{aligned} \quad (\text{FLH19})$$

The first image is fixed. The second line is the literal inverse image of fixed points, hence the Connes–Consani null kernel and has their stated universal property. Unions of such sets still satisfy the displayed condition, so it is a subobject.

Here is an evaluated normal form for its entire null homology. Let $\mathcal{O}(B_n) = (B_n \setminus \{0\})/\{b \sim -b\}$; its elements are exact two-element sign orbits, not projective lines and not scalar orbits. Put

$$\begin{aligned} \mathcal{Q}_n &= \{(A_{\text{out}}, \Omega, \lambda) : A_{\text{out}} \subset C_n \setminus B_n \text{ finite}, \Omega \subset \mathcal{O}(B_n) \text{ finite}, \\ \{0\} \cup d_n(A_{\text{out}}) &= -(\{0\} \cup d_n(A_{\text{out}})), (A_{\text{out}} \neq \emptyset \text{ or } \Omega \neq \emptyset) \Rightarrow \lambda = 1_L\}, \\ q_n^{\text{CC}}(A, \lambda) &= (A \setminus B_n, \{\{b, -b\} : b \in A \cap (B_n \setminus \{0\})\}, \lambda). \end{aligned} \quad (\text{FLH20})$$

The joins are union in the first two coordinates and join in the last. Its involution sends $(A_{\text{out}}, \Omega, \lambda)$ to $(-A_{\text{out}}, \Omega, \lambda)$. These formulas define a Boolean module with involution. The map q_n^{CC} preserves

its joins, bottom and involution. It is onto: choose one representative of each of the finitely many orbits in Ω , append A_{out} and 0, and use that $d_n(B_n) = 0$.

Its full fibre equivalence and universal property are

$$(A, \lambda) \sim_{\text{CC}} (A', \mu) \iff \lambda = \mu, \quad A \setminus B_n = A' \setminus B_n, \quad \mathcal{O}(A \cap B_n) = \mathcal{O}(A' \cap B_n), \quad (FLH21)$$

$$H_n^{\text{CC}}(\mathcal{P}_L(C)) \simeq \mathcal{Q}_n.$$

Here the orbit expression omits zero as in FLH20. We prove the cokernel property rather than infer it from ranks. The incoming image in $\mathcal{K}_n$ is precisely $\mathcal{P}_L(B_n)$: lift its finitely many vectors individually through d_{n+1} , preserving the label, and retain 0. Its image under q_n^{CC} has empty outside set and therefore is fixed.

Let $f : \mathcal{K}_n \rightarrow M$ be any morphism in $\mathbf{Blv}$ for which the incoming image is null. For every nonzero $b \in B_n$, equivariance and nullity give

$$f(\{0, b\}, 1_L) = f(\{0, -b\}, 1_L). \quad (FLH22)$$

Joining an arbitrary element of $\mathcal{K}_n$ to both sides permits replacing b by $-b$, or adding the other sign and then removing a redundant sign. Removing boundary elements from a cycle set leaves a cycle set because their differential is zero. Consequently any two finite sets in one fibre of FLH20 are connected by these finitely many replacements, keeping their outside sets and support fixed. The all-zero set with a lower label is a singleton fibre. Thus f is constant on every fibre and factors uniquely through the surjection q_n^{CC} . The resulting map preserves joins and the involution by surjectivity. Conversely every factor makes the incoming image fixed. This proves the cokernel universal property for all targets of the Connes–Consani category, and proves FLH21. In particular a nonzero boundary sign orbit becomes fixed; it is not deleted or identified with the labelled empty orbit. Every fibre is more explicitly

$$\left\{ \left(\{0\} \cup A_{\text{out}} \cup \bigcup_{\omega \in \Omega} F_\omega, \lambda \right) : \emptyset \neq F_\omega \subseteq \omega \text{ for each } \omega \in \Omega \right\}.$$

The sign orbits are disjoint two-element sets, so the fibre has exactly $3^{|\Omega|}$ elements. In particular no amplitude translation by a boundary has entered this quotient. There is a canonical section in the Boolean category:

$$c_n(A_{\text{out}}, \Omega, \lambda) = (\{0\} \cup A_{\text{out}} \cup \bigcup_{\omega \in \Omega} \omega, \lambda).$$

It preserves unions, labels and involution, and $q_n^{\text{CC}} c_n = I$. Its other composition is the exact idempotent $(A, \lambda) \mapsto (A \cup [-(A \cap B_n)], \lambda)$. It adds the missing opposite boundary signs and changes no other vector. Consequently the fixed subobject of $\mathcal{Q}_n$ consists precisely of the triples with $A_{\text{out}} = -A_{\text{out}}$. The normal image of the incoming map, namely the null kernel of its null cokernel, is precisely the subset of $\mathcal{K}_n$ with symmetric outside part. It equals $\mathcal{P}_L(B_n) \vee \mathcal{K}_n^\iota$: one inclusion follows by taking outside parts of a union; for the other decompose a cycle set with symmetric outside part into $A \cap B_n$ and $\{0\} \cup (A \setminus B_n)$, the latter being fixed. A nontrivial such union has top label, and the zero sets already lie in the incoming image at every label. This proves the asserted equality with no label omitted. The construction is natural under every original chain map. For $f_n : C_n \rightarrow C'_n$, take the canonical representative $\{0\} \cup A_{\text{out}} \cup \bigcup \Omega$, apply f_n to every vector, and take its FLH20 normal form for B'_n . The chain identity preserves the balanced-image condition and sends B_n into B'_n ; opposite boundary representatives therefore still give the same orbit, or zero, in the target. This proves well-definition and gives the actual induced map on the complete normal-form homology.

This object contains a precisely describable excess over additive cycles. The original cycles are recovered by the exact marked-state identity

$$j_{C_n}((Z_n)_L^\uparrow) = j_{C_n}((C_n)_L^\uparrow) \cap \mathcal{K}_n. \quad (FLH23)$$

Indeed $\{0, d_n v\}$ is sign-invariant over $\mathbb{C}$ if and only if $d_n v = 0$. The original boundary marked states are $\{j_{C_n}(b, 1_L) : b \in B_n\}$ together with all labels. Their additive quotient must use the original addition graph: (z, λ) and (z', μ) represent the same supported class exactly when $\lambda = \mu$, and either that label is not top, or $z' - z \in B_n$. This is precisely FLH11–12. The singleton marking, its original addition and scalar graphs, and its support therefore reconstruct the additive homology from the actual complex. They are additional specified structure; they are not recovered by declaring the Boolean join to be vector addition.

For example, in a two-term complex $C_1 \xrightarrow{T} C_0$, choose $Tv \neq 0$ and $0 \neq w \in \ker T$. Then

$$A = \{0, v, -v + w\}, \quad T(A) = \{0, Tv, -Tv\}. \quad (\text{FLH24})$$

The set A is a Boolean null cycle. It is not fixed by negation: $-v$ is none of $0, v, -v + w$, since characteristic zero, $Tv \neq 0$, and $w \neq 0$ exclude the three equalities. There is no incoming amplitude boundary in degree one, so FLH21 keeps this nonfixed class. Neither nonzero vector in A is an additive cycle. Thus this functor supplies a genuine characteristic-one null complex but does not silently identify its homology with FLH11. Its extra objects and the maps recovering the actual cycles are now explicit.

28.6 The rank-one receiver, with its original metric

Keep the original receiver E , positive metric G_N , unit vectors e, f , and coefficient $\epsilon > 0$:

$$\begin{aligned} \langle v, w \rangle_N &= v^* G_N w, \quad \|e\|_N = \|f\|_N = 1, \quad e^\dagger f = 0, \quad R = \epsilon f e^\dagger, \quad R^2 = 0, \\ W &= \mathbb{C}e \perp \mathbb{C}f, \quad U = W^\perp, \quad E = W \perp U, \quad \dim E = q. \end{aligned} \quad (\text{FLH25})$$

In the literal two-term complex $C_1 = E \xrightarrow{R} C_0 = E$, completed outside these degrees by the supported null object, the complete supported homology is

$$\mathcal{H}_{1,L} = (\mathbb{C}f \perp U)_L^\dagger, \quad \mathcal{H}_{0,L} = (E/\mathbb{C}f)_L^\dagger \simeq (\mathbb{C}e \perp U)_L^\dagger. \quad (\text{FLH26})$$

The last isomorphism sends a coset to its original orthogonal representative $v - f f^\dagger v$, keeping its label. Indeed $Rv = \epsilon f(e^\dagger v)$ has kernel $e^\perp$ and image $\mathbb{C}f$; FLH11 proves the supported quotient and FLH14 proves its metric. On $ae + bf + u$ the quotient norm is $|a|^2 + \|u\|_N^2$, whereas the kernel norm on $bf + u$ is $|b|^2 + \|u\|_N^2$. The actual inverse from the image to the orthogonal source complement is $bf \mapsto (b/\epsilon)e$, with squared norm $|b|^2/\epsilon^2$.

Here a three-term complex is also genuine: $E \xrightarrow{R} E \xrightarrow{R} E$. Its middle homology is $U_L^\dagger$, its top homology is $(\mathbb{C}f \perp U)_L^\dagger$, and its bottom homology is $(\mathbb{C}e \perp U)_L^\dagger$, with the same quotient maps. This follows from $R^2 = 0$, the displayed kernel and image, and their original orthogonal decomposition. The scalar ϵ remains in its differential and inverse, although it does not change those subspaces.

The Boolean homology of the two-term complex is given without an unidentified quotient by

$$\begin{aligned} H_1^{\text{CC}}(\mathcal{P}_L(C)) &= \{(A, \lambda) : A \subset E \text{ finite}, 0 \in A, \epsilon f e^\dagger A = -\epsilon f e^\dagger A\}, \\ H_0^{\text{CC}}(\mathcal{P}_L(C)) &= \{(A_{\text{out}}, \Omega, \lambda) : A_{\text{out}} \subset E \setminus \mathbb{C}f \text{ finite}, \Omega \subset (\mathbb{C}f \setminus \{0\})/\{\pm 1\} \text{ finite}\}. \end{aligned} \quad (\text{FLH27})$$

In both lines retain exactly the support rule of FLH18 or FLH20, respectively. Their operations and involutions are the displayed normal-form operations, not vector operations. Taking $v = e$ and $0 \neq w \in U$ gives the explicit nonfixed extra cycle FLH24 in the actual receiver whenever $q > 2$, as it is in the programme. The additive reflection of both homologies in FLH26 remains the same null L , by FLH16.

28.7 The mixed tensor receiver as a genuine two-term complex

Retain $k \geq 17$, $k \equiv 1 \pmod{4}$, $q = (k+1)^2$, the original cutoff $q-1 \leq N \leq 2q$, and G_N of OLM1. For $t \geq 1$ the unchanged operator and metric are

$$E_t = E^{\otimes t}, \quad G_t = G_N^{\otimes t}, \quad X = R^{[t]} = \sum_{a=1}^t I^{\otimes(a-1)} \otimes R \otimes I^{\otimes(t-a)}. \quad (\text{FLH28})$$

For $t \geq 2$, this operator is not a differential repeated in adjacent degrees:

$$X^2(e^{\otimes t}) = 2\epsilon^2 \sum_{1 \leq a < b \leq t} e \otimes \cdots \otimes f_a \otimes \cdots \otimes f_b \otimes \cdots \otimes e \neq 0. \quad (\text{FLH29})$$

The distinct tensor words in the sum are orthogonal in the original metric, each coefficient is $2\epsilon^2 \neq 0$, and hence the claimed nonvanishing follows. We use the literal two-term complex $C_1 = E_t \xrightarrow{X} C_0 = E_t$. The next differential is the zero amplitude map with its full support lift, so this is a valid null complex for every t . No sign change is made to the slot-sum operator.

For every actual active-slot set $J \subseteq \{1, \dots, t\}$, put $j = |J|$. The original orthogonal sector is $E_J = W^{\otimes j} \otimes U^{\otimes(t-j)}$, embedded in its actual slots in their increasing order. Let $V_{j,r}$ be the span of words with r copies of f , and X_j the sum of R over these j slots. With $Y_j = X_j^\dagger$, the complete primitive spaces and factors are

$$\begin{aligned} \mathcal{P}_{j,r} &= \ker(Y_j : V_{j,r} \rightarrow V_{j,r-1}), \quad 0 \leq r \leq \lfloor j/2 \rfloor, \quad d = j - 2r, \\ \dim \mathcal{P}_{j,r} &= \binom{j}{r} - \binom{j}{r-1}, \quad s_{d,a} = \epsilon^{2a} a! \frac{d!}{(d-a)!}, \\ E_t &= \bigoplus_{J,r} \bigoplus_{a=0}^d \perp X_j^a \mathcal{P}_{j,r} \otimes U^{\otimes(t-j)}, \quad \langle X_j^a u, X_j^b v \rangle = \mathbf{1}_{a=b} s_{d,a} \langle u, v \rangle. \end{aligned} \quad (\text{FLH30})$$

The sector embeddings keep the original order of all slots. For completeness, the one-slot commutator gives $[Y_j, X_j]|_{V_{j,s}} = \epsilon^2(j-2s)I$. It implies injectivity below the middle grade and, by the slotwise e, f exchange, surjectivity above it. Rank-nullity gives the primitive dimension. Commuting Y_j through X_j^a on a primitive vector yields $Y_j X_j^a u = \epsilon^2 a(d-a+1) X_j^{a-1} u$. Induction gives the full factor $s_{d,a}$, and the next power has zero norm, hence vanishes. Different grades and different primitive ladders are orthogonal: moving the smaller number of powers to the other entry leaves either a positive power paired with a primitive vector or different grades. At total grade s , their dimensions telescope as $\sum_{r=0}^{\min(s, j-s)} [\binom{j}{r} - \binom{j}{r-1}] = \binom{j}{s}$. Thus they exhaust every grade and every slot sector. This proves the decomposition and metric in FLH30 with the unchanged vectors and coefficients.

The full homology is now evaluated as actual subspaces and quotient metrics:

$$\begin{aligned} H_1(C) &= \ker X = \bigoplus_{J,r} \perp X_j^d \mathcal{P}_{j,r} \otimes U^{\otimes(t-j)}, \\ H_0(C) &= E_t / \text{im } X \xrightarrow{\sim} \bigoplus_{J,r} \perp \mathcal{P}_{j,r} \otimes U^{\otimes(t-j)}, \\ \mathcal{H}_{n,L} &= H_n(C)_L^\uparrow \quad (n = 0, 1), \end{aligned} \quad (\text{FLH31})$$

$$\left\| \sum_{J,r} X_j^d u_{J,r} \right\|_{H_1}^2 = \sum_{J,r} \epsilon^{2d} (d!)^2 \|u_{J,r}\|^2, \quad \|[v]\|_{H_0}^2 = \sum_{J,r} \|u_{J,r,0}\|^2.$$

Here $v = \sum_{J,r,a} X_j^a u_{J,r,a}$ is its actual full decomposition, and the second line is the original orthogonal representative of the coset. On each ladder X sends level a to $a+1$ and kills exactly

its top level d . Its image is exactly the sum of levels $1, \dots, d$. Independence and the metric in FLH30 prove every equality and the quotient-minimum assertion in FLH31.

All its ranks, including every inactive-slot multiplicity, are therefore

$$b_t := \dim H_1(C) = \dim H_0(C) = \sum_{j=0}^t \binom{t}{j} (q-2)^{t-j} \binom{j}{\lfloor j/2 \rfloor}, \quad (\text{FLH32})$$

$$\text{rank } X = q^t - b_t, \quad M_{t;j,r} = \binom{t}{j} (q-2)^{t-j} \left(\binom{j}{r} - \binom{j}{r-1} \right).$$

Each primitive vector contributes one top kernel vector and one bottom quotient vector. Summing its exact dimension over r telescopes to the middle binomial coefficient. There are $\binom{t}{j}$ actual choices of J and $(q-2)^{t-j}$ inactive-factor dimensions. This proves FLH32, including $j=0, d=0$. The support fibre over every non-top label in FLH31 remains one labelled zero; its top fibre is the entire displayed homology space, even if a particular summand has dimension zero.

The actual inverse on the image and all its metric factors are also explicit:

$$X^+ \left(\sum_{J,r} \sum_{b=1}^d X_j^b u_{J,r,b} \right) = \sum_{J,r} \sum_{b=1}^d X_j^{b-1} u_{J,r,b}, \quad (\text{FLH33})$$

$$\|X^+ v\|^2 = \sum_{J,r} \sum_{b=1}^d s_{d,b-1} \|u_{J,r,b}\|^2, \quad \frac{s_{d,b}}{s_{d,b-1}} = \epsilon^2 b(d-b+1).$$

This inverse lands in the source orthogonal complement of $\ker X$, so it is the attained minimum lift of v . Orthogonality in FLH30 proves uniqueness and its norm. The superscript $+$ denotes this original-metric Moore–Penrose inverse; it does not divide away ϵ from the original differential. All maps and inverses in FLH31–33 lift by FLH3 and retain their label.

Finally the complete Boolean null homology of this same two-term tensor complex, with no tensor-square-zero error, is

$$H_1^{\text{CC}}(\mathcal{P}_L(C)) = \{(A, \lambda) \in \mathcal{P}_L(E_t) : X(A) = -X(A)\},$$

$$H_0^{\text{CC}}(\mathcal{P}_L(C)) = \{(A_{\text{out}}, \Omega, \lambda) : A_{\text{out}} \subset E_t \setminus \text{im } X \text{ finite}, \Omega \subset (\text{im } X \setminus \{0\})/\{\pm 1\} \text{ finite}\}. \quad (\text{FLH34})$$

Retain the support rules and operations of FLH20. The image appearing here is exactly the sum of levels $1, \dots, d$ in every sector of FLH30. Thus every set, sign orbit, degree, kernel and quotient in FLH34 is determined by the original operator. FLH23 selects its original additive cycles, and FLH11 then gives the supported additive quotient with all metrics FLH31. Localization applies to the synchronized additive complex and its supported homology, and sends those objects to the null L in FLH16. The Boolean normal-form object instead has the explicitly defined support morphism $\rho_{\text{CC}}(A_{\text{out}}, \Omega, \lambda) = \lambda$ to (L, I_L) . It preserves joins and bottom and commutes with involution by FLH20. Its fibre at a non-top label is the single $(\emptyset, \emptyset, \lambda)$, and its top fibre is the entire displayed set of allowed first two coordinates at that label. No S -semimodule localization of this Boolean object is asserted.

The first line of FLH34 has a complete description by actual affine fibres, beyond its kernel presentation. Put $K = \ker X$ as explicitly decomposed in FLH31, and let X^+ be FLH33. Every one of its elements is uniquely of the form

$$A = A_0 \cup \bigcup_{w \in T} (X^+ w + U_w), \quad T \subset \text{im } X \setminus \{0\} \text{ finite}, \quad T = -T,$$

$$A_0 \subset K \text{ finite}, \quad 0 \in A_0, \quad \emptyset \neq U_w \subset K \text{ finite} \quad (w \in T), \quad (\text{FLH35})$$

$$(A_0 \neq \{0\} \text{ or } T \neq \emptyset) \Rightarrow \lambda = 1_L,$$

$$\iota : (A_0, T, (U_w), \lambda) \mapsto (-A_0, T, (-U_{-w})_{w \in T}, \lambda).$$

Indeed the minimum inverse satisfies $XX^+w = w$, so the complete vector fibre is $X^{-1}(w) = X^+w + K$. The displayed union has image $\{0\} \cup T$, and hence is a null cycle. Conversely, for a cycle A , take $T = X(A) \setminus \{0\}$, $A_0 = A \cap K$, and $U_w = (A \cap X^{-1}(w)) - X^+w$. These sets have the displayed properties and recover A . The fibres of different w are disjoint, which proves uniqueness. Linearity of X^+ proves the involution formula. In the rank-one case the literal fibre section is $X^+(bf) = (b/\epsilon)e$; in every tensor sector it is the unaltered level-lowering map FLH33. This retains the numerical coefficients in all the Boolean cycle fibres. In the second line of FLH34, the outside vectors are precisely those with at least one nonzero bottom coordinate $u_{J,r,0}$, whereas the boundary sign orbits are the nonzero vectors whose every bottom coordinate is zero. These statements follow from the evaluated image in FLH31. The Boolean category alone assigns no Hermitian norm to finite sets; all original vector norms remain the exact ones in FLH30–33 and in the joint marked-state data.

28.8 The joint diagram and the original free masks

$$\begin{array}{ccccc}
(Z_n)_L^\uparrow & \xrightarrow{q_n} & (Z_n/B_n)_L^\uparrow & \xrightarrow{\sigma_H} & (L, I_L) \\
\pi_Z \downarrow & & \pi_H \downarrow & & \\
Z_n & \xrightarrow{z \mapsto [z]} & Z_n/B_n & & \\
\\
(Z_n)_L^\uparrow & \xrightarrow{j_Z} & \mathcal{K}_n & \xrightarrow{q_n^{\text{CC}}} & \mathcal{Q}_n \\
(z, \lambda) & \mapsto & (\{0, z\}, \lambda) & & (A_{\text{out}}, \Omega, \lambda)
\end{array}$$

Figure 7: The two proved quotients of the original finite complex. The upper row retains the amplitude coset and every support label (FLH11–17). The bottom row is the Connes–Consani Boolean null quotient, retaining outside-boundary sets and boundary sign orbits (FLH18–23); it is not an arrow identifying those sign orbits with amplitude cosets. The map j_Z marks the original individual states; it is pointed and equivariant, and is not additive. Connes–Consani’s kernel and cokernel definitions are at the author-source locators given at the start of this section.

The square of the first two columns commutes by FLH11. The support on the upper quotient is reconstructed jointly with the amplitude, as in FLH17; the Boolean map in the last row is the different universal quotient proved in FLH21–22.

In the original positive-dimensional free coordinate module, the map to a synchronized state and its whole fibre are

$$\begin{aligned}
P_{\text{syn}}(v, \boldsymbol{\mu}) &= (v, \bigvee_i \mu_i), \\
P_{\text{syn}}^{-1}(v, \lambda) &= \{(v, \boldsymbol{\mu}) : \bigvee_i \mu_i = \lambda, \quad v_i \neq 0 \Rightarrow \mu_i = 1_L\}.
\end{aligned} \tag{FLH36}$$

The join is unchanged when synchronization is repeated, so this is the original idempotent correspondence OLR17. Combining it with FLH10–11 gives every free mask over every cycle and supported homology fibre: the amplitude belongs to the stated cycle or affine boundary coset, while the masks satisfy exactly FLH36. No change of free-semimodule basis to a ladder basis is claimed. The ladder maps are applied after synchronization and have their proved amplitude inverse. If the entire vector space has dimension zero, the zero-coordinate free module has one state, whereas $(0)_L^\uparrow$ has all labels; FLH36 is not declared onto in that case. The actual receiver and its positive tensor powers have dimensions q and $q^t > 0$, so their full masks are covered.

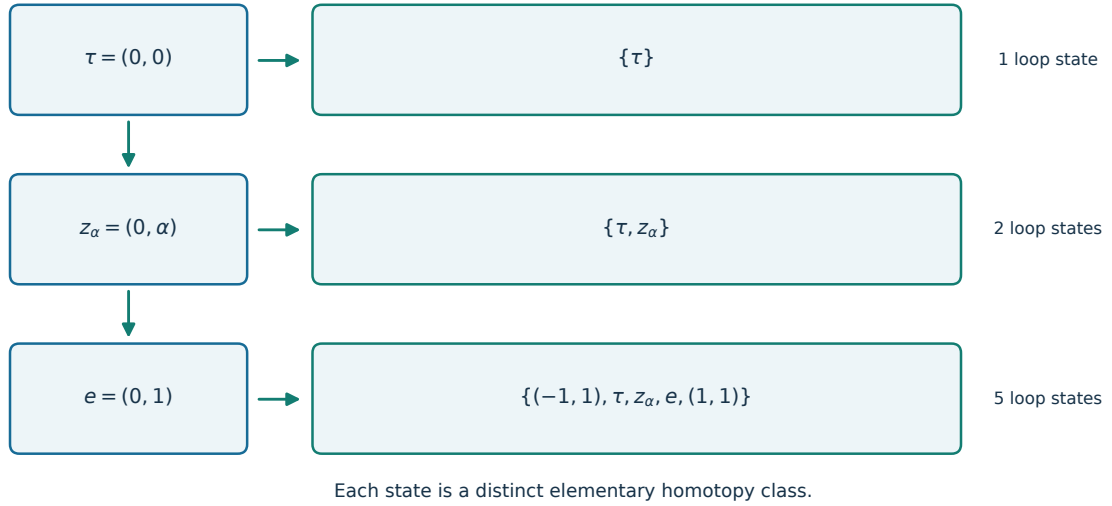
The source localization, the Connes–Consani null quotient, the additive supported quotient, and the actual nilpotent receiver have thus been related by explicit maps and complete fibres.

None of FLH1–36 supplies a Frobenius purity assertion, an RH contradiction, or a counterexample to RH. The calculated amplitude loss under localization and the additional Boolean cycles are part of the resulting objects, with the reconstruction and receiver calculations above retaining their full data.

Loops retain their support basepoint

Exact example: $L = \{0 < \alpha < 1\}$, $D = 0$, $\Lambda_D = \mathbb{Z}$, $e^a = 1$, one coordinate.

All one-spheres, before any amplitude or support is discarded



The complete zero-stage relation

$$(b, \nu) \mathcal{R} (b', \mu) \iff \nu \leq \mu \text{ and } d_{\mathbb{R}/\mathbb{Z}}(b, b') \leq 1$$

Support may increase along an edge. The original distance bound remains.

Proof: SUPPORTED_DOLD_KAN_REVIEW.tex, sphere and directed-relation calculations.
Human source: Alain Connes and Caterina Consani, arXiv:2004.08879v1, lines 875–1231.

Figure 8: The complete one-sphere sets in the indicated finite-lattice example, and the exact directed zero-stage relation. This is the source’s relational homotopy. Complete construction and controlling lattice map: SDKR1–35.

29 The supported relative simplicial construction and relational homotopy

The original relative construction is due to Alain Connes and Caterina Consani, *Segal's Gamma rings and universal arithmetic*, arXiv:2004.08879v1, original `Froblifts.tex`, lines 875–1231. We use Proposition `relativeH` and the interval construction `DKsimple`. The other inputs are the bounded functor `simpconstruct1`, the elementary relation `Maydefn`, and Proposition `prophigher`. Their original amplitude construction and its relational interpretation are antecedents. The full additive carrier is the coefficient construction of The Clankers, *Split Support Geometry*, version 11, Definition `def:lattice-split`.

The source displays `pidegn`, lines 1133–1135, and `pidegnk`, lines 1219–1221, omit one amplitude slot. Degree n has n A slots and one B slot, as its earlier definition and degree-three example require. We derive the faces from the interval construction below. Homotopy means the source's sphere and elementary relation, not homotopy of a geometric realization.

29.1 The original divisor and full additive carriers

Fix

$$\begin{aligned} D = \sum_p n_p[p] + a[\infty], \quad \Lambda_D = H^0(\text{Spec } \mathbb{Z}, \mathcal{O}(D_{\text{fin}})) = u_D \mathbb{Z}, \quad u_D = \prod_p p^{-n_p} > 0, \\ c = e^a > 0, \quad A = \mathbb{R}, \quad B = \mathbb{R}/\Lambda_D, \quad \phi : A \rightarrow B, \quad \phi(r) = r + \Lambda_D. \end{aligned} \tag{SDKR1}$$

The equality for Λ_D follows from all the original conditions $v_p(x) + n_p \geq 0$: x/u_D has every finite valuation nonnegative and is therefore an integer. No divisor coefficient, archimedean bound or period is changed.

Let L be the original nontrivial bounded distributive lattice. For an abelian group C put

$$\begin{aligned} C_L = \{(0, \ell) : \ell \in L\} \cup \{(x, 1_L) : x \in C\}, \\ (x, \ell) + (y, m) = (x + y, \ell \vee m), \quad \tau_C = (0, 0_L), \quad e_C = (0, 1_L). \end{aligned} \tag{SDKR2}$$

The pair $(0, 1_L)$ occurs once. If a sum has nonzero amplitude, a nonzero input forces its support to be top, so the operation is closed. It is an associative commutative monoid law, with identity τ_C ; empty sums mean τ_C . The quotient map lifts additively as

$$\phi_L : A_L \rightarrow B_L, \quad (r, \ell) \mapsto (\phi(r), \ell). \tag{SDKR3}$$

A nonzero $r \in \Lambda_D$ maps to e_B , not τ_B . The construction uses the additive groups; no $\mathbb{R}$ -module structure on B is assumed.

29.2 The relative functor and its exact simplicial coordinates

For a finite pointed pair (X, Y) , $* \in Y \subseteq X$, define

$$\begin{aligned} \mathcal{K}_D^L(X, Y) = \left\{ \psi : Y \setminus \{*\} \rightarrow B_L, \quad X \setminus Y \rightarrow A_L : \right. \\ \left. \sum_{x \in X \setminus Y} |\text{amp } \psi(x)| \leq c \right\}. \end{aligned} \tag{SDKR4}$$

Its basepoint is the all- τ function. For a map of pointed pairs $f : (X, Y) \rightarrow (X', Y')$ its nonbasepoint coordinates are

$$(\mathcal{K}_D^L(f)\psi)(z) = \begin{cases} \sum_{x \in X \setminus Y, f(x)=z} \psi(x), & z \in X' \setminus Y', \\ \sum_{y \in Y \setminus \{*\}, f(y)=z} \psi(y) + \sum_{x \in X \setminus Y, f(x)=z} \phi_L(\psi(x)), & z \in Y' \setminus \{*\}. \end{cases} \tag{SDKR5}$$

In each fibre amplitudes add and supports join. A point outside Y' receives only points outside Y . The triangle inequality therefore bounds the sum of its output absolute amplitudes by the original outside sum; coordinates entering Y' or the basepoint are omitted from this bound. Thus the output lies in SDKR4. For a composite, partition each output fibre into the intermediate fibres. Associativity of sums, additivity of ϕ_L , and associativity of finite joins give exactly the same amplitudes and supports in the two calculations. This argument includes empty fibres. Identity maps have singleton fibres. Hence SDKR5 defines a pointed covariant functor, with no use of additive inverses. Apply it to the source's interval pair $\partial[n]^* = (\{0, \dots, n+1\}, \{0, n+1\})$, pointed at 0, smashed with k_+ . In degree n it gives

$$\mathcal{X}_n(D; k) = \left\{ (a_1, \dots, a_n; b) : a_i \in A_L^k, \quad b \in B_L^k, \right. \\ \left. \sum_{i=1}^n \sum_{j=1}^k |\text{amp}(a_{i,j})| \leq c \right\}. \quad (\text{SDKR6})$$

Thus $\mathcal{X}_0(D; k) = B_L^k$. For $k = 0$ every degree is a singleton. The final endpoint is the B slot and the initial endpoint is discarded. SDKR5 gives

$$\begin{aligned} d_0(a_1, \dots, a_n; b) &= (a_2, \dots, a_n; b), \\ d_i(a_1, \dots, a_n; b) &= (a_1, \dots, a_{i-1}, a_i + a_{i+1}, a_{i+2}, \dots, a_n; b), \quad 1 \leq i < n, \\ d_n(a_1, \dots, a_n; b) &= (a_1, \dots, a_{n-1}; b + \phi_L(a_n)). \end{aligned} \quad (\text{SDKR7})$$

In particular $d_0(a; b) = b$ and $d_1(a; b) = b + \phi_L(a)$. The degeneracies are

$$s_i(a_1, \dots, a_n; b) = (a_1, \dots, a_i, \tau_A^k, a_{i+1}, \dots, a_n; b), \quad 0 \leq i \leq n. \quad (\text{SDKR8})$$

Indeed a collapsed interior fibre adds adjacent A entries; a fibre entering the last endpoint applies ϕ_L and adds to B ; and a degeneracy inserts an empty fibre, hence τ_A . The simplicial identities follow from the interval maps and the proved functoriality of SDKR5. This gives n A slots and one B slot in every degree, including all the support data. For a pointed map $f : k_+ \rightarrow l_+$ put

$$(f_*x)_h = \sum_{\substack{1 \leq j \leq k \\ f(j)=h}} x_j, \quad f_* : \mathcal{X}_n(D; k) \rightarrow \mathcal{X}_n(D; l), \quad (a_i; b) \mapsto (f_*a_i; f_*b). \quad (\text{SDKR9})$$

The triangle inequality proves the bound after merging or discarding coordinates. Additivity of ϕ_L proves commutation with SDKR7–8. Fibre partition proves composition. These are the raw pointed Γ maps with the all- τ basepoint.

29.3 Spheres and elementary homotopy at every supported basepoint

For $s = (s_j)_j \in L^k$ let $z_s = ((0, s_j))_j \in B_L^k$. Its constant simplices are

$$\kappa_s^{(n)} = (\underbrace{\tau_A^k, \dots, \tau_A^k}_{n \text{ slots}}; z_s). \quad (\text{SDKR10})$$

SDKR7–8 prove their basepoint identities. An n -sphere for $n \geq 1$ has every face equal to $\kappa_s^{(n-1)}$. The complete set of one-spheres is

$$\mathcal{S}_1(D; k, s) = \left\{ (a; z_s) : a_j = (r_j, \ell_j) \in (\Lambda_D)_L, \quad \ell_j \leq s_j, \right. \\ \left. \sum_j |r_j| \leq c \right\}. \quad (\text{SDKR11})$$

In fact $d_0(a; b) = z_s$ forces $b = z_s$. The other face equation is exactly $\phi(r_j) = 0$ and $\ell_j \vee s_j = s_j$ in every coordinate. These are precisely the displayed conditions, and conversely they imply both faces. The carrier condition remains: $r_j \neq 0$ forces $\ell_j = s_j = 1_L$. The source elementary relation on these spheres is

$$x\mathcal{R}_{1,s}y \iff \exists w \in \mathcal{X}_2(D; k) : d_0w = \kappa_s^{(1)}, \quad d_1w = x, \quad d_2w = y. \quad (\text{SDKR12})$$

Write $w = (u, v; b)$. The first equation forces $v = \tau_A^k$, $b = z_s$, including their actual support labels. Its other two faces are both $(u; z_s)$. Thus the elementary relation is equality. Conversely, for $x = y = (a; z_s)$ the witness $(a, \tau_A^k; z_s)$ has the required faces and bound. In particular distinct zero-amplitude support labels are distinct relational one-sphere classes. For all higher degrees,

$$\mathcal{S}_n(D; k, s) = \{\kappa_s^{(n)}\} \quad (n \geq 2). \quad (\text{SDKR13})$$

Indeed $d_0x = \kappa_s^{(n-1)}$ forces $a_2 = \dots = a_n = \tau_A^k$ and $b = z_s$. The last face then forces $a_1 = \tau_A^k$. The constant simplex satisfies every face and has its reflexive elementary witness in the next constant simplex. This proves triviality of the source's higher sphere relations directly. The extremes are therefore

$$\begin{aligned} \mathcal{S}_1(D; k, 0_L^k) &= \{(\tau_A^k, \tau_B^k)\}, \\ \mathcal{S}_1(D; k, 1_L^k) &= \{(a; e_B^k) : a \in (\Lambda_D)_L^k, \sum_j |\text{amp } a_j| \leq c\}. \end{aligned} \quad (\text{SDKR14})$$

The first is the loop object at the actual pointed zero. The second, at the supported basepoint, retains the complete bounded Γ -set of global sections in SDM3. The two basepoints give different results and have not been identified.

29.4 The exact finite-lattice count

Suppose L is finite, put $N = \lfloor c/u_D \rfloor$, and let $I_s = \{j : s_j = 1_L\}$. Then

$$\#\mathcal{S}_1(D; k, s) = \sum_{\substack{T \subseteq I_s \\ |T| \leq N}} 2^{|T|} \binom{N}{|T|} \prod_{j \notin T} |\downarrow s_j|, \quad \downarrow s_j = \{\ell \in L : \ell \leq s_j\}. \quad (\text{SDKR15})$$

To prove this, choose the exact set T of nonzero amplitudes. It must lie in I_s , and those supports are forced to be top. Every other coordinate has zero amplitude and independently chooses a label in $\downarrow s_j$. The nonzero amplitudes are $u_D m_j$ for nonzero integers m_j . For $m = |T|$ there are 2^m sign choices. The positive magnitudes sum to at most N . Their successive partial sums give an increasing m -element subset of $\{1, \dots, N\}$, and taking differences is the inverse. Hence there are exactly $\binom{N}{m}$ magnitude choices. The empty set has one empty sign and magnitude choice. Multiplying and summing proves the formula. At full support it becomes

$$\#\mathcal{S}_1(D; k, 1_L^k) = \sum_{m=0}^{\min(k, N)} |L|^{k-m} 2^m \binom{k}{m} \binom{N}{m}, \quad N = \left\lfloor \frac{e^a}{u_D} \right\rfloor. \quad (\text{SDKR16})$$

The original a and u_D remain in the expression. If $e^a < u_D$, every amplitude is zero but there are $|L|^k$ distinct classes. At support 0_L^k there is one class. The source's amplitude count omits these independent label choices; its old symmetry is not asserted for SDKR16.

29.5 Support transport and a genuine pointed Gamma construction

For a pointed map $f : k_+ \rightarrow l_+$, define

$$(f_*s)_h = \bigvee_{\substack{1 \leq j \leq k \\ f(j)=h}} s_j, \quad (\text{SDKR17})$$

with bottom for an empty fibre. The raw map SDKR9 sends κ_s to κ_{f_*s} . In particular full support maps to top in nonempty fibres and bottom in empty fibres. For the insertion $1_+ \rightarrow 2_+$, its B basepoint becomes (e_B, τ_B) , not (e_B, e_B) . All-top basepoints are therefore not natural for the raw maps. The exact support-indexed objects that repair the identity law are

$$\mathcal{X}_n^{\geq s}(D; k) = \{(a_i; b) \in \mathcal{X}_n(D; k) : \text{supp}(b_j) \geq s_j \text{ for all } j\}. \quad (\text{SDKR18})$$

They are simplicial subobjects: only the last face changes the B slot, by joining more labels, and degeneracies leave it unchanged. Their basepoints are κ_s . All the spheres and elementary witnesses above have $B = z_s$, so their sphere objects and relations are unchanged. For $s \leq t$ there is a pointed simplicial transition

$$T_{s,t} : \mathcal{X}^{\geq s}(D; k) \rightarrow \mathcal{X}^{\geq t}(D; k), \quad (a_i; b) \mapsto (a_i; b + z_t). \quad (\text{SDKR19})$$

No A amplitude changes. Commutation with the last face is $b + z_t + \phi_L(a_n) = b + \phi_L(a_n) + z_t$; the other faces and degeneracies commute immediately. It sends z_s to z_t . On its stated domain $T_{s,s} = I$, since each B label is already at least s . For $s \leq t \leq u$ one has $T_{t,u}T_{s,t} = T_{s,u}$. Adding z_s would not be the identity on the unrestricted object; SDKR18 is what proves this required identity. Form the category whose objects are (k_+, s) , $s \in L^k$, and whose morphisms $(k_+, s) \rightarrow (l_+, t)$ are pointed maps f with $f_*s \leq t$. Monotonicity and $(gf)_* = g_*f_*$ on support vectors prove that this is a category. Its action is

$$\Theta_{f;s,t}(a_i; b) = (f_*a_i; f_*b + z_t). \quad (\text{SDKR20})$$

For a subsequent $(g; t, u)$, the composite B slot is

$$g_*(f_*b + z_t) + z_u = (gf)_*b + z_{g_*t \vee u} = (gf)_*b + z_u,$$

and its A slots are $(gf)_*a_i$. Thus

$$\Theta_{g;t,u}\Theta_{f;s,t} = \Theta_{gf;s,u}, \quad \Theta_{\text{id};s,s} = I. \quad (\text{SDKR21})$$

The maps are pointed and simplicial by their construction. On one-spheres their exact action is

$$(a; z_s) \mapsto (f_*a; z_t). \quad (\text{SDKR22})$$

Its supports are at most $f_*s \leq t$ and its absolute-amplitude sum does not increase. For fixed k , upward transition is the inclusion of the same A tuple into the larger permitted support set.

At full support every pointed $f : k_+ \rightarrow l_+$ is permitted with target 1_L^l . Thus SDKR20–21 yield a genuine pointed simplicial Γ -object whose loop object is naturally

$$\Pi_1^{\text{rel}}(\mathcal{X}^{\geq 1_L^\bullet}(D))(k_+) \cong \{(r_j, \ell_j)_j \in (\Lambda_D)_L^k : \sum_j |r_j| \leq e^a\}. \quad (\text{SDKR23})$$

The Γ operation sums amplitudes and joins labels over each fibre, putting τ_A in an empty fibre, by SDKR22. Its pointed zero is the all- τ_A tuple. This proves naturality of the full bounded module rather than assuming naturality of the all-top basepoints in the unrestricted object.

29.6 The complete degree-zero directed relation

A vertex is exactly

$$(b, \nu) \in B^k \times L^k, \quad b_j \neq 0 \implies \nu_j = 1_L. \quad (\text{SDKR24})$$

The source relation is the existence of one simplex with d_0 the first vertex and d_1 the second. It is

$$(b, \nu)\mathcal{R}_0(b', \mu) \iff \nu \leq \mu, \quad \sum_j d_{\Lambda_D}(b_j, b'_j) \leq c, \quad (\text{SDKR25})$$

where the exact original quotient distance is

$$d_{\Lambda_D}(r + \Lambda_D, r' + \Lambda_D) = \min_{m \in \mathbb{Z}} |r - r' + m u_D|. \quad (\text{SDKR26})$$

The minimum exists since the absolute values tend to infinity as $|m| \rightarrow \infty$; it is independent of representatives by reindexing.

For necessity, an edge $(a; (b, \nu))$, $a_j = (r_j, \ell_j)$, ends at $(b + \phi(r), \nu \vee \ell)$. The support order follows. Each r_j represents the amplitude difference, so the distance sum is at most $\sum_j |r_j| \leq c$. Conversely assume SDKR25 and choose minimum-absolute-value representatives r_j of $b'_j - b_j$. If $\mu_j < 1_L$, admissibility gives $b'_j = 0$ and $\nu_j \leq \mu_j < 1_L$ gives $b_j = 0$; choose $r_j = 0$. All other target labels are top. Hence every (r_j, μ_j) is a valid A_L element, its total absolute amplitude is at most c , and the edge with these A coordinates ends at $(b', \nu \vee \mu) = (b', \mu)$. This proves both directions.

In particular the singular relational object retaining both data is

$$\begin{aligned} \mathcal{V}_k &= \{(b, \nu) \in B^k \times L^k : b_j \neq 0 \Rightarrow \nu_j = 1_L\}, \\ \mathcal{R}_0 &= \{((b, \nu), (b', \mu)) \in \mathcal{V}_k^2 : \nu \leq \mu, d_{\Lambda_D}^{(1)}(b, b') \leq c\}, \\ d_{\Lambda_D}^{(1)}(b, b') &= \sum_j d_{\Lambda_D}(b_j, b'_j). \end{aligned} \quad (\text{SDKR27})$$

The amplitude projection maps onto the source distance relation: every amplitude edge lifts with both supports top. Its full-support section preserves the relation in both directions. The support section $\nu \mapsto (0, \nu)$ embeds the preorder L^k with its exact relation, since its amplitude distance is zero. The joint projection is injective by SDKR24.

This relation is reflexive but need not be symmetric: $(0, \nu) \rightarrow (0, \mu)$ exists precisely when $\nu \leq \mu$. For $k \geq 1$ and nontrivial L it is never the complete relation, regardless of c , because an e_B coordinate cannot move to τ_B . It can fail transitivity even on full support: if $c < u_D/4$, the successive first-coordinate amplitudes $0, c, 2c$ modulo Λ_D , all other amplitudes zero, give two permitted edges but first-to-last distance $2c > c$. The one-step relation is retained rather than replaced by its transitive closure.

29.7 Naturality, the amplitude diameter and directed reachability

Every pointed map satisfies

$$d_{\Lambda_D}^{(1)}(f_*b, f_*b') \leq d_{\Lambda_D}^{(1)}(b, b'). \quad (\text{SDKR28})$$

Indeed minimum difference representatives r_j sum to a representative in each output fibre. Its distance is at most that absolute sum, which is at most the sum of the $|r_j|$ in that fibre. Summing over output nonbasepoint fibres proves the inequality. Thus raw Γ maps preserve SDKR25, using monotonicity of support pushforward. For SDKR20 the vertex map is $(b, \nu) \mapsto (f_*b, f_*\nu \vee t)$; joining the same t also preserves support order, proving its relational naturality.

In the full-top subobject the support order is automatic. Its amplitude relation is complete exactly when

$$c \geq k u_D/2 \iff k \leq 2e^{\deg D}, \quad \deg D = a - \log u_D. \quad (\text{SDKR29})$$

The circle $\mathbb{R}/u_D\mathbb{Z}$ has diameter exactly $u_D/2$: representatives in $[-u_D/2, u_D/2]$ give the upper bound and the class of $u_D/2$ attains it. Taking that class in all k coordinates gives exact sum diameter $k u_D/2$. This proves the inclusive endpoint in SDKR29. The second equivalence uses $e^{\deg D} = e^a/u_D$, without changing D .

The separate directed reachability relation, allowing finitely many source edges, is

$$(b, \nu) \mathcal{R}_0^*(b', \mu) \iff \nu \leq \mu. \quad (\text{SDKR30})$$

Necessity follows from support order along a path. For sufficiency first use the zero-amplitude edge $(b, \nu) \rightarrow (b, \mu)$. At fixed support μ all coordinates below top have zero amplitudes at both ends. Choose real minimum difference representatives r_j , zero in those coordinates, and put $R = \sum_j |r_j|$. Since $c > 0$, choose $N \geq 1$ with $R/N \leq c$. The vertices $(b + \phi(ir/N), \mu)$ for $i = 0, \dots, N$ are valid. Each successive edge has A tuple $(r_j/N, \mu_j)_j$ of total absolute amplitude R/N . Its endpoint after N steps is exactly (b', μ) . This proves the finite path. Mutual reachability is therefore equality of supports, with classes indexed by all of L^k and induced order its coordinatewise order. This auxiliary calculation does not change SDKR25 or assert a realization-homotopy interpretation.

29.8 Principal-divisor transport in the original coordinates

For $q \in \mathbb{Q}^\times$, let $D' = D + (q)$ with $(q) = \sum_p v_p(q)[p] - \log |q|[\infty]$. Then

$$\begin{aligned} \Lambda_{D'} &= q^{-1} \Lambda_D, & u_{D'} &= u_D / |q|, & c' &= c / |q|, \\ (r, \ell) &\mapsto (q^{-1}r, \ell), & (b, \ell) &\mapsto (q^{-1}b, \ell). \end{aligned} \tag{SDKR31}$$

The quotient map on B is well defined because its lattice is carried exactly to $q^{-1} \Lambda_D$. Both maps commute with ϕ_L , preserve every join, and scale every absolute amplitude by $|q|^{-1}$. They therefore give inverse isomorphisms of all the relative, simplicial and support-indexed constructions, with inverse multiplication by q . SDKR26 shows that distance scales by the same factor, proving preservation of SDKR25 in both directions. Finally $c'/u_{D'} = c/u_D$, so the counts are unchanged. This proves invariance without first replacing D by another representative.

29.9 The controlling arithmetic-site lattice map

Retain the original tropical coefficient semifield K with zero and the extension of ASL18–22,

$$K_L = K \otimes_{\mathbb{B}} L, \quad \iota_L : L \rightarrow K_L, \quad \ell \mapsto 1_K \otimes \ell, \quad J_L = \iota_L(L). \tag{SDKR32}$$

The target support carrier here is exactly J_L , not unrestricted K_L . In J_L , addition and product are the joins and meets of the original labels: the tensor relations give $\iota_L(\ell \vee m) = \iota_L(\ell) + \iota_L(m)$ and $\iota_L(\ell \wedge m) = \iota_L(\ell) \iota_L(m)$. These elements are multiplicatively idempotent.

For completeness the labels are not identified by this map. The nonzero-indicator map $K \rightarrow \mathbb{B}$, taking zero to zero and each nonzero coefficient to one, is a semiring map: in the positive tropical semifield, a sum is zero exactly when both inputs are zero, and a product is zero exactly when one is zero. Its tensor extension

$$\rho_L : K \otimes_{\mathbb{B}} L \rightarrow L, \quad \rho_L \left(\sum_i a_i \otimes \ell_i \right) = \bigvee_{a_i \neq 0_K} \ell_i$$

is well defined by the tensor universal property and satisfies $\rho_L \iota_L = I_L$. Thus ι_L is injective and identifies L with the bounded distributive sublattice J_L . This also agrees with the joint-character reconstruction in ASL20–22: a lattice character $\chi : L \rightarrow \mathbb{B}$ sends $\iota_L(\ell)$ to $\chi(\ell)1_K$, and those characters separate the original labels.

For either original additive group $C = A$ or $C = B$, form its carrier with support restricted to J_L . The controlling coefficient map and its inverse are

$$\begin{aligned} j_C : C_L &\xrightarrow{\sim} C_{J_L}, & (x, \ell) &\longmapsto (x, 1_K \otimes \ell), \\ j_C^{-1}(x, \iota_L(\ell)) &= (x, \ell). \end{aligned} \tag{SDKR33}$$

This is an isomorphism of additive monoids: amplitude sums are unchanged and ι_L preserves joins and bottom. It retains all coefficients, signs and constants. Its square with the original quotient map commutes exactly:

$$j_B \phi_L(r, \ell) = (\phi(r), \iota_L(\ell)) = \phi_{J_L} j_A(r, \ell).$$

Applying it coordinatewise to SDKR4 therefore gives a bijection $\mathcal{K}_D^L(X, Y) \rightarrow \mathcal{K}_D^{JL}(X, Y)$ with precisely the same amplitude bound. In SDKR5, the B terms commute by that square and the finite support joins commute by the lattice homomorphism. This proves naturality for every map of relative pointed pairs, including every empty fibre. It therefore commutes with every face, degeneracy and raw Γ map in SDKR6–9. Its basepoint is transported by $s \mapsto \iota_L(s)$. Because ι_L preserves and reflects order, it maps SDKR18 bijectively to its corresponding support-indexed subobject. Finally $\iota_L(\ell \vee t) = \iota_L(\ell) \vee \iota_L(t)$ proves exact commutation with SDKR19–22.

The pair of amplitude and encoded support reconstructs the complete relative simplex: in each A or B coordinate it belongs to exactly

$$\{(x, h) \in C \times J_L : x \neq 0 \Rightarrow h = 1_K \otimes 1_L\}, \quad (\text{SDKR34})$$

and the inverse is $(x, h) \mapsto (x, \rho_L(h))$. On the relative object the only additional condition is the unchanged sum of the outside absolute amplitudes. Thus SDKR34 proves both injectivity and the exact joint image, rather than asserting that amplitude and support are unrelated presentations. It carries SDKR11, SDKR25 and their maps to identical reconstructed relations.

More generally, for any bounded lattice homomorphism $\theta : L \rightarrow L'$ the coordinate map $(x, \ell) \mapsto (x, \theta(\ell))$ gives a natural map of the relative functors. It is well defined, since $\theta(1_L) = 1_{L'}$, and additive since θ preserves joins and bottom. It commutes with ϕ , and therefore with SDKR5. Its support basepoint is $\theta(s)$, and $\theta(f_*s) = f_*\theta(s)$, including empty fibres. It preserves the order condition for SDKR20 and obeys

$$\theta_*\Theta_{f;s,t} = \Theta_{f;\theta(s),\theta(t)}\theta_*, \quad (\text{id}_K \otimes \theta)\iota_L = \iota_{L'}\theta. \quad (\text{SDKR35})$$

The first equality follows by evaluating its A coordinates and its B coordinate: both give the pushed amplitude and support $\theta(f_*\nu) \vee \theta(t)$. The second follows on each pure tensor $1_K \otimes \ell$. These formulas prove the full coefficient and support-transport squares for the controlling lattice map. No choice of one Boolean branch or loss of a zero label is needed for SDKR33–34; a noninjective specialization θ has exactly its stated image and does not inherit that bijectivity claim.

30 Supported-zero inversion, the full lattice, and retained amplitudes

Sources and the claim being tested. The user-supplied source `globalization_note.tex`, *A Note on the Globalization Semiring $G(\mathbb{Z})$* , has an empty author field and the date March 2026. Its entire source was read for this calculation. The exact source has SHA-256

ce06ac01c32e0452fe36121ac0f90e4eff8fad469b1cda7a6cf2d73116077a02.

The relevant source locators are:

- Proposition `prop:units-irr`, lines 282–300, and Corollary `cor:zeta`, lines 406–430;
- Theorem `thm:localization-away-from-zero`, lines 436–480;
- Theorem `thm:frac-collapse`, lines 482–534;
- the definition at lines 817–825, followed by Theorem `thm:class-semigroup` and its proof at lines 827–871.

The source’s bibliography names Golan, Deitmar, Lorscheid and Neukirch; those bibliography entries are not represented here as author sources independently read during this calculation. The note’s empty author field does not authorize assigning it a human author.

The full-lattice carrier and its supported-zero localization are existing programme results. Their public proof is 031, Theorems `thm:rebuilt-full-support` and `thm:e-stalk`, equation `eq:e-element-localization`. The earlier foundation is retained under its byline The Clankers in *Split Support Geometry: Universal Zero Fibres, Arithmetic Curves, Frobenius and Perfect Quantization*, v11, Definition `def:lattice-split`. The user’s attribution of the earlier scalar supported-zero prime observation to Gemini Pro 2.5 is retained as user-reported provenance; this calculation does not claim that observation as new.

The precise characteristic-one comparison being used is ASL1–22 of the complete included arithmetic-site lattice-map proof, read in full through ASL22. Its source is

ARITHMETIC_SITE_LATTICE_MAP.tex,
SHA-256 a03b6c40ba06653fe3a481592170d1c4c43ff16d1b4af53959b3e0ab1fe0b492.

ASL identifies its human framework source as Alain Connes and Caterina Consani, *Geometry of the Arithmetic Site*; the following proofs use the displayed ASL constructions and give the required algebra independently. They do not claim a new reading of that entire human paper.

The statement to be tested is the user-supplied wording “zero is invertable this means rh is wrong”. We calculate exactly which element is inverted, what its inverse is, which identities and prime points result, and what information survives in the actual joint coefficient map. No analytic zero is silently identified with the semiring’s absorbing zero.

30.1 The original two zeros and a general idempotent localization

Let R be a nonzero commutative unital ring and L a nontrivial bounded distributive lattice. Its semiring operations are join and meet. Retain the complete carrier

$$\begin{aligned}
 S &= G_L(R) = \{(0_R, \lambda) : \lambda \in L\} \cup \{(r, 1_L) : r \in R\}, \\
 (r, \lambda) + (s, \nu) &= (r + s, \lambda \vee \nu), & (r, \lambda)(s, \nu) &= (rs, \lambda \wedge \nu), \\
 z_\lambda &= (0_R, \lambda), & \tau &= z_{0_L} = 0_S, & e &= z_{1_L}, & 1_S &= (1_R, 1_L), \\
 q : S &\longrightarrow R, & q(r, \lambda) &= r, & \sigma : S &\longrightarrow L, & \sigma(r, \lambda) &= \lambda.
 \end{aligned} \tag{ZIL1}$$

The source operations preserve the displayed carrier. For addition, if $r + s \neq 0$ then at least one of r, s is nonzero, so the joined label is 1_L . For multiplication, $rs \neq 0$ forces both labels to be 1_L . The semiring laws and the two homomorphisms follow from the ring laws and distributivity in L .

The units in this original semiring are exactly

$$S^\times = \{(u, 1_L) : u \in R^\times\}, \quad e^2 = e, \quad e \neq 1_S, \quad e \notin S^\times. \quad (\text{ZIL2})$$

Indeed a product equal to $(1_R, 1_L)$ has ring product 1_R and label meet 1_L , which forces both labels to equal 1_L . Conversely the ordinary ring inverse gives an inverse in S . The inequalities in (ZIL2) use $0_R \neq 1_R$. In particular the note's integer 0 is e , whereas its new symbol τ is 0_S . The Boolean subset $\{\tau, e\}$ has internal multiplicative identity e ; its inclusion into S is not a unit-preserving semiring map.

Here is the localization calculation without any cancellativity assumption. Let A be a commutative unital semiring and let $c \in A$ satisfy $c^2 = c$. The set cA is a semiring with inherited operations, zero 0_A and *internal identity* c . Then

$$A[c^{-1}] \simeq cA, \quad a \mapsto ca, \quad [a, c^n] \mapsto ca \quad (n \geq 0). \quad (\text{ZIL3})$$

To prove this, multiplication by c preserves addition and multiplication because $c^2 = c$, and sends 1_A to the internal unit c . Its image of the denominator c is that unit. Conversely, if $f : A \rightarrow B$ is a zero- and unit-preserving homomorphism and $f(c)$ is invertible, the equality $f(c)^2 = f(c)$ gives $f(c) = 1_B$. The formula $g(ca) = f(a)$ is well defined: $ca = cb$ implies $f(a) = f(ca) = f(cb) = f(b)$. It preserves both operations, zero and the internal unit, and is forced by f . This proves the universal property and the fraction formula. In particular the complete equality relation of the localization is $a \sim b \iff ca = cb$.

Apply (ZIL3) to every labelled zero, without discarding any label:

$$\begin{aligned} S[z_\mu^{-1}] &\simeq \downarrow \mu := \{\nu \in L : \nu \leq \mu\}, & 1_{\downarrow \mu} &= \mu, & 0_{\downarrow \mu} &= 0_L, \\ \ell_\mu : S &\longrightarrow \downarrow \mu, & \ell_\mu(r, \lambda) &= \lambda \wedge \mu, \\ (r, \lambda) &\sim_\mu (s, \nu) & \iff & \lambda \wedge \mu = \nu \wedge \mu. \end{aligned} \quad (\text{ZIL4})$$

Indeed $z_\mu(r, \lambda) = z_{\mu \wedge \lambda}$, and every z_ν with $\nu \leq \mu$ occurs by choosing $(r, \lambda) = z_\nu$. This identifies the entire corner semiring $z_\mu S$, its operations and its internal identity. In particular

$$S[e^{-1}] \simeq L, \quad [e, 1]^{-1} = [e, 1] = [1_S, 1] = 1_L, \quad S[\tau^{-1}] \simeq \{0 = 1\}. \quad (\text{ZIL5})$$

Thus e really is invertible in this localization, and its inverse is exactly its own image, equal to the localized identity. In the first localization τ is still $0_L \neq 1_L$; it has not become invertible. In the second localization the actual absorbing zero is inverted and the entire semiring is trivial. These are different operations.

The complete fibres, including amplitude loss, are

$$\begin{aligned} \ell_\mu^{-1}(\eta) &= \{z_\lambda : \lambda \wedge \mu = \eta\} \\ &\cup \begin{cases} \{(r, 1_L) : r \in R \setminus \{0_R\}\}, & \eta = \mu, \\ \emptyset, & \eta \neq \mu, \end{cases} \quad (\eta \leq \mu), \\ \sigma^{-1}(\lambda) &= \begin{cases} \{z_\lambda\}, & \lambda \neq 1_L, \\ R \times \{1_L\}, & \lambda = 1_L. \end{cases} \end{aligned} \quad (\text{ZIL6})$$

These follow by enumerating the two parts of the carrier in (ZIL1). Thus supported-zero inversion loses every amplitude distinction in the top fibre and preserves all original lattice labels. Inverting a smaller label further identifies exactly the labels having the same meet with it. Formula (ZIL6) also includes $\mu = 0_L$.

For the full multiplicative set $D_R = \{(r, 1_L) : r \in R\}$, including e but excluding τ , one has

$$D_R^{-1}S \simeq L, \quad [(r, \lambda), (a, 1_L)] \mapsto \lambda. \quad (\text{ZIL7})$$

After e is inverted, every member of D_R already equals the unit, by (ZIL4) with $\mu = 1_L$. Conversely any map inverting D_R inverts e . The two universal properties give (ZIL7). For $R = \mathbb{Z}$ and $L = \mathbb{B}$, this is exactly the note's Theorem `thm:frac-collapse`; that theorem is correct. Its symbol $\text{Frac}(S)$ denotes this specific semiring localization, not a field of fractions of the ring $\mathbb{Z}$.

In contrast, if $M \subset \mathbb{Z} \setminus \{0\}$ is a multiplicative set containing 1, then

$$\{(m, 1_L) : m \in M\}^{-1} G_L(\mathbb{Z}) \simeq G_L(\mathbb{Z}_M), \quad [(r, \lambda), (m, 1_L)] \mapsto (r/m, \lambda). \quad (\text{ZIL8})$$

Fraction equality on the left requires a $t \in M$ such that $tm'r = tmr'$ and $\lambda = \lambda'$; multiplication by top-labelled denominators never changes a label. These conditions are exactly equality of the ring fractions and their labels. This proves injectivity. Every top-labelled fraction and every zero label in the right side is attained, proving surjectivity. The formulas for fraction addition and multiplication agree with (ZIL1), so the displayed bijection is an isomorphism. This proves the full-lattice extension of the note's other localization theorem.

30.2 The prime spectrum before and after this localization

All primes of S are the following two kinds:

$$\begin{aligned} P_J &= \{z_\lambda : \lambda \in J\} \quad (J \in \text{SpecLat } L), \\ Q_{\mathfrak{p}} &= \{z_\lambda : \lambda \in L\} \cup \{(r, 1_L) : r \in \mathfrak{p}\} \quad (\mathfrak{p} \in \text{Spec } R), \\ P_J \subsetneq Q_{\mathfrak{p}}, \quad D_S(z_\mu) &= \{P_J : \mu \notin J\}, \quad V_S(e) = \{Q_{\mathfrak{p}} : \mathfrak{p} \in \text{Spec } R\}. \end{aligned} \quad (\text{ZIL9})$$

Here a lattice prime ideal is downward closed, closed under joins, contains 0_L , excludes 1_L , and is prime for meets. For completeness, a prime ideal of S excluding e contains no top-labelled element: multiplication of any such element by e is e . Its remaining labels form a proper lattice ideal, and the primality test is exactly meet-primality. A prime containing e contains every z_λ , since $ez_\lambda = z_\lambda$. Its top-labelled amplitudes form a ring ideal (additive negatives are obtained by multiplying by $(-1_R, 1_L)$); the prime condition is ring primality. These arguments prove exhaustion and both converses. Every P_J is contained in every $Q_{\mathfrak{p}}$, strictly because e is in the latter and not the former. Membership of z_μ now proves the last two formulas.

The spectrum of (ZIL4) identifies exactly with this supported open:

$$\text{Spec}(\downarrow \mu) \simeq \{J \in \text{SpecLat } L : \mu \notin J\} \simeq D_S(z_\mu). \quad (\text{ZIL10})$$

Explicitly J maps to $J \cap \downarrow \mu$. Conversely a prime H of $\downarrow \mu$ maps to $\{\lambda : \lambda \wedge \mu \in H\}$. These are lattice ideals and prime by meet and join preservation. If $\mu \notin J$, primality gives $\lambda \wedge \mu \in J \iff \lambda \in J$, so the maps are inverse. Basic opens correspond under the same formula, proving the topological assertion. Every arithmetic prime disappears under this localization because every $Q_{\mathfrak{p}}$ contains z_μ . This includes $\mu = 1_L$; it does not assert that these primes were absent from the original geometry. For $L = \mathbb{B}$, $R = \mathbb{Z}$, the original strict chain remains $\{\tau\} \subsetneq \{\tau, e\} \subsetneq Q_{(p)}$.

30.3 The actual ASL coefficient map and its localization square

Let K be the tropical semifield with zero used in ASL18–22, and set $K_L = K \otimes_{\mathbb{B}} L$. The zero and unit of K are written $\mathbf{0}_K, \mathbf{1}_K$. The nonzero indicator $\beta_K : K \rightarrow \mathbb{B}$ is a semiring homomorphism: a maximum is zero exactly when both inputs are zero, and a product is zero exactly when a factor is zero. Consequently

$$\begin{aligned} \iota_L : L &\longrightarrow K_L, \quad \iota_L(\lambda) = \mathbf{1}_K \otimes \lambda, \\ \rho_L : K_L &\longrightarrow L, \quad \rho_L \left(\sum_i a_i \otimes \lambda_i \right) = \bigvee_{i: a_i \neq \mathbf{0}_K} \lambda_i, \quad \rho_L \iota_L = \text{id}_L. \end{aligned} \quad (\text{ZIL11})$$

The second map is $\beta_K \otimes \text{id}_L$ followed by the canonical isomorphism $\mathbb{B} \otimes_{\mathbb{B}} L = L$; this proves independence from every tensor presentation. It also proves the displayed formula and the retraction identity. In particular ι_L is injective for arbitrary L , without selecting a single Boolean character or replacing L by one.

Retain the actual original-amplitude map

$$\Phi_R : S \longrightarrow P_R := R \times K_L, \quad \Phi_R(r, \lambda) = (r, \iota_L(\lambda)). \quad (\text{ZIL12})$$

Both coordinates preserve the semiring operations, zero and unit. If two images coincide their ring coordinates agree, and (ZIL11) gives equality of their lattice coordinates. Thus Φ_R is injective. Its image consists exactly of $(r, \iota_L(\lambda))$ with $r \neq 0_R$ implying $\lambda = 1_L$. The source's complete information is retained by this map.

For each $\mu \in L$, write $d_\mu = \iota_L(\mu)$ and $u_\mu = (0_R, d_\mu) = \Phi_R(z_\mu)$. The corresponding localization of the *full product* is

$$\begin{aligned} P_R[u_\mu^{-1}] &\simeq d_\mu K_L, & (r, w) &\longmapsto d_\mu w, \\ 1_{d_\mu K_L} &= d_\mu, & P_R[\Phi_R(e)^{-1}] &\simeq K_L, & (r, w) &\longmapsto w. \end{aligned} \quad (\text{ZIL13})$$

Indeed $u_\mu^2 = u_\mu$, and its corner in P_R is $\{(0_R, d_\mu w) : w \in K_L\}$. Apply (ZIL3) and identify this corner with its second coordinate. In particular the localization of the product is not the zero ring when $\mu \neq 0_L$; it is a semiring whose entire ring factor has been erased.

Here is the exact square, including every label and both new units:

$$\begin{array}{ccc} G_L(R) & \xrightarrow{\Phi_R} & R \times K_L \\ (r, \lambda) \mapsto \lambda \wedge \mu \downarrow & & \downarrow (r, w) \mapsto d_\mu w \quad 1_{\downarrow \mu} = \mu, quad 1_{d_\mu K_L} = d_\mu. \\ \downarrow \mu & \xrightarrow{\eta \mapsto \iota_L(\eta)} & d_\mu K_L. \end{array} \quad (\text{ZIL14})$$

Both composites send (r, λ) to $\iota_L(\lambda \wedge \mu)$, because ι_L preserves meet. The lower arrow takes values in the displayed corner since $\eta \leq \mu$ implies $d_\mu \iota_L(\eta) = \iota_L(\eta)$. It preserves the new unit as well as zero and both operations. Its retraction is the restriction of ρ_L , whose values lie in $\downarrow \mu$ because $\rho_L(d_\mu w) = \mu \wedge \rho_L(w)$.

The square is a pushout of unital semirings:

$$(R \times K_L) \otimes_S S[z_\mu^{-1}] \simeq (R \times K_L)[u_\mu^{-1}] \simeq d_\mu K_L. \quad (\text{ZIL15})$$

To prove the asserted universal property directly, a map $f : R \times K_L \rightarrow A$ agreeing on S with a map from $S[z_\mu^{-1}]$ sends u_μ to an invertible element. By (ZIL3) it factors uniquely through $d_\mu K_L$. Conversely any such factor gives the two agreeing maps in the square. These constructions are inverse, proving (ZIL15). This statement remains valid for $\mu = 0_L$, when every receiving semiring in this universal property is trivial.

At $\mu = 1_L$, the complete relations erased by the two vertical maps are respectively

$$\begin{aligned} (r, \lambda) \sim_e (s, \nu) &\iff \lambda = \nu, \\ (r, w) \sim_{\Phi_R(e)} (s, v) &\iff w = v. \end{aligned} \quad (\text{ZIL16})$$

The first relation is exactly the congruence generated by $e \sim 1_S$: in that congruence $x = 1_S x \sim ex = z_{\sigma(x)}$, whereas σ itself respects $e \sim 1_S$ and distinguishes every label. The second follows in the same way from (ZIL3). Thus no amplitude dependence can be recovered from either localized value alone.

Under the standing assumptions, the square at $\mu = 1_L$ is not a pullback. Its actual pullback has the explicit form

$$(R \times K_L) \times_{K_L} L \simeq R \times L, \quad ((r, \iota_L(\lambda)), \lambda) \longleftrightarrow (r, \lambda). \quad (\text{ZIL17})$$

The original S is precisely the subsemiring of this pullback cut out by $r \neq 0_R \Rightarrow \lambda = 1_L$. For example $(1_R, 0_L)$ belongs to the pullback and not to S . The formula for the pullback follows from equality of the second coordinate with $\iota_L(\lambda)$, and its operations are coordinatewise, proving the asserted isomorphism.

There is, however, a larger explicitly constructed subsemiring on which the faithful map has a retraction:

$$\begin{aligned} T_R &= \{(r, w) \in R \times K_L : r \neq 0_R \Rightarrow \rho_L(w) = 1_L\}, \\ \Psi_R : T_R &\longrightarrow S, \quad \Psi_R(r, w) = (r, \rho_L(w)), \quad \Psi_R \Phi_R = \text{id}_S. \end{aligned} \tag{ZIL18}$$

For closure under addition, a nonzero sum $r + s$ has at least one nonzero summand, whose second-coordinate support is 1_L ; the joined support is then 1_L . A nonzero product rs has both factors nonzero, whose supports have meet 1_L . Zero and unit belong to T_R . Thus it is a subsemiring. The map $\text{id}_R \times \rho_L$ is a semiring homomorphism on the entire product; its values lie in S exactly on T_R , proving both the restriction and its maximal domain for this formula. Formula (ZIL11) proves the retraction. The localization of T_R at u_μ is again $d_\mu K_L$, because T_R contains every $(0_R, w)$ and its u_μ -corner is exactly $\{0_R\} \times d_\mu K_L$. After localization, Ψ_R is the restricted ρ_L in (ZIL14). This gives exact compatible retractions before and after localization on the stated domains.

No retraction $R \times K_L \rightarrow S$ of Φ_R exists on the entire product. Suppose F were one. Put $a = (1_R, \mathbf{0}_{K_L})$ and $b = (0_R, \mathbf{1}_{K_L}) = \Phi_R(e)$. Then $ab = 0$, $a + b = 1$, and $F(b) = e$. In S , the equality $ex = \tau$ forces $x = \tau$ by (ZIL1). Hence $F(a)e = \tau$ forces $F(a) = \tau$, and $1_S = F(a + b) = \tau + e = e$, contradicting (ZIL2). This proves the exact obstruction and identifies the concrete domain (ZIL18) on which the retraction does exist.

The ordinary ring-coordinate square has a different lower target:

$$\begin{array}{ccc} S & \xrightarrow{q} & R \\ \downarrow & & \downarrow \\ S[e^{-1}] \simeq L & \longrightarrow & R[0_R^{-1}] = \{0 = 1\}. \end{array} \tag{ZIL19}$$

Indeed $q(e) = 0_R$, whose inverse forces $1_R = 0_R$. No map from L to a nonzero ring can be a unital semiring homomorphism: the equality $1_L + 1_L = 1_L$ would give $1 + 1 = 1$, hence $1 = 0$, in the target ring. On the original S , however, the combined map $(q, \sigma) : S \hookrightarrow R \times L$ is injective. Equivalently the intersection of their equality congruences is the diagonal. Thus keeping the unlocalized ring coordinate alongside the full support coordinate recovers the original element uniquely, whereas factoring that coordinate through supported-zero inversion is impossible in a nonzero ring.

30.4 Analytic zero locations and an exact non-detection test

Let Ω be any set and $f : \Omega \rightarrow \mathbb{C}$ any function. Form its *top-labelled family* in $G_L(\mathbb{C})$, without asserting that the map $\mathbb{C} \rightarrow G_L(\mathbb{C})$ preserves additive zero. Direct substitution gives

$$\begin{aligned} \widehat{f}(z) &= (f(z), 1_L), \quad \{z : f(z) = 0\} = \widehat{f}^{-1}(\{e\}), \\ \ell_\mu(\widehat{f}(z)) &= \mu \quad \text{for every } z, \quad \Phi_{\mathbb{C}}(\widehat{f}(z)) = (f(z), \mathbf{1}_{K_L}), \\ d_\mu \text{ pr}_2 \Phi_{\mathbb{C}}(\widehat{f}(z)) &= d_\mu \quad \text{for every } z. \end{aligned} \tag{ZIL20}$$

Thus the right vertical map in (ZIL14) sends $(f(z), \mathbf{1}_{K_L})$ to d_μ for every z ; at $\mu = 1_L$ this is the constant unit $\mathbf{1}_{K_L}$. The original family never takes the value τ . Its zero-amplitude locus is the inverse image of e , which localization identifies with the unit. The ring projection of the unlocalized joint family still recovers f exactly. These formulas apply to ζ on any domain avoiding its pole, to ξ , or to a specified H_t wherever they are defined. No analytic theorem is needed for the calculation: it is valid for every function and therefore does not distinguish their different possible zero configurations.

The following finite calculation further determines exactly what inversion alone can detect even when the backward heat equation and a nonnegative real-root threshold are also retained. It is a test of those stated pieces of data, not a replacement for the zeta heat kernel. For $c \in \{0, 1\}$ set

$$P_t^{(c)}(z) = z^2 + c - 2t, \quad \partial_t P_t^{(c)} = -2 = -\partial_z^2 P_t^{(c)}, \quad \{\text{all roots real}\} \iff t \geq c/2. \quad (\text{ZIL21})$$

The differential identity is obtained by differentiating the displayed polynomial. Its roots solve $z^2 = 2t - c$; for real t they are both real, allowing the double root 0, exactly when $2t - c \geq 0$. Thus the two thresholds are 0 and $1/2$, both nonnegative. At $t = 0$ the first polynomial has all roots real and the second has the nonreal roots $\pm i$. Nevertheless every top-labelled family in (ZIL21) has the same localized value in (ZIL20), for every t, z, c . Consequently that localized value, the displayed heat equation, and nonnegativity of the threshold do not together determine whether the threshold is 0 or strictly positive. This is a counterexample to detection by those data only. It says nothing contrary to any additional theorem about the actual H_t .

30.5 Arithmetic ideal norms and the note's scaling error

For $R = \mathbb{Z}$, retain the arithmetic ideals $Q_{(n)} = \{z_\lambda : \lambda \in L\} \cup \{(r, 1_L) : r \in n\mathbb{Z}\}$, for $n \geq 1$. The exact quotient homomorphism and its chosen norm are

$$\pi_n : S \longrightarrow \mathbb{Z}/n\mathbb{Z}, \quad \pi_n(r, \lambda) = r \bmod n, \quad \pi_n^{-1}(0) = Q_{(n)}, \quad N(Q_{(n)}) = n. \quad (\text{ZIL22})$$

The map is the ring projection followed by reduction modulo n , so it is surjective and the displayed kernel is exact. The norm here refers to this specific quotient congruence; an ideal alone must not silently be substituted for a congruence. For $\text{Re } s > 1$, this arithmetic family therefore gives $\sum_{n \geq 1} N(Q_{(n)})^{-s} = \sum_{n \geq 1} n^{-s}$. This proves the retained equality with the Dirichlet-series definition of $\zeta(s)$ on that domain; no continuation or zero-location conclusion follows from the equality alone.

Every ideal in (ZIL22) contains e and hence z_μ . Its extension under ℓ_μ is the entire $\downarrow \mu$, because the image contains the new unit μ . Already $Q_{(2)}$ and $Q_{(3)}$ have different norms and identical extended ideals. Thus there is no function of the extended ideal alone recovering the original norm on this family. This is an exact loss of arithmetic information in the localization, not an assertion that the original prime or zeta data was unaffected.

The note contains a directly related false statement in its class-semigroup definition. Write $A = \mathcal{O}_F$ for a number field F , and $S_A = A \sqcup \{\tau\} = G_{\mathbb{B}}(A)$. This notation avoids confusing the number field with the tropical coefficient semifield K . With the source's own convention, a nonzero semiring ideal means an ideal $I \neq \{\tau\}$. Every such ideal is $I_J = J \cup \{\tau\}$ for an ideal $J \subset A$, including $J = (0)$: multiplying any integer member by the supported zero produces e , and the remaining ideal rules give precisely the ordinary ring ideal. The note defines

$$I \sim J \iff \exists a, b \in S_A \setminus \{\tau\} : aI = bJ. \quad (\text{ZIL23})$$

This allowed scalar set is A , which includes the supported zero $e = 0_A$. For every $I \neq \{\tau\}$,

$$eI = \{\tau, e\}, \quad \{\text{nonzero ideals of } S_A\} / \sim = \{\text{one class}\}. \quad (\text{ZIL24})$$

Indeed $e\tau = \tau$, every ordinary member of I is sent to e , and a nonzero ideal has such a member. Taking $a = b = e$ in (ZIL23) then relates every pair. In particular the sentence claiming these scalars are “exactly the nonzero algebraic integers” is false. The note's subsequent injectivity proof silently changes the scalar domain to $A \setminus \{0_A\}$, and its surjectivity proof silently discards the allowed nonzero semiring ideal $\{\tau, e\} = I_{(0)}$.

There is a precise corrected correspondence, with both restrictions stated explicitly. Use only I_J with $J \neq (0)$ and use only scalars $a, b \in A \setminus \{0_A\}$. Then

$$\begin{aligned} J &\longleftrightarrow I_J = J \cup \{\tau\}, & I_{J_1} I_{J_2} &= I_{J_1 J_2}, \\ a I_{J_1} &= b I_{J_2} &\iff a J_1 &= b J_2 \quad (a, b \neq 0_A). \end{aligned} \tag{ZIL25}$$

For the product identity, the ideal product consists of finite sums of products. Terms involving τ contribute τ ; the other products and sums give exactly the ordinary ideal product, including its integer zero e . The scaling equivalence is proved by intersecting with A in one direction, and adjoining τ in the other. These calculations give an exact isomorphism with the usual nonzero integral-ideal class monoid defined by nonzero scalar equivalence. Equivalence to fractional scaling is also exact: $a J_1 = b J_2$ means $J_2 = (a/b) J_1$, and any element of $F^\times$ has such a representation because F is the fraction field of A . Every nonzero fractional ideal has an integral representative after multiplying by a common nonzero denominator. Thus this restricted quotient is the usual ideal-class quotient of fractional ideals by $F^\times$; in the Dedekind-domain setting of the note it is the ordinary ideal class group.

If only the scalar restriction is repaired but $I_{(0)}$ is retained, its class is an extra absorbing class. Nonzero scalars cannot take a nonzero ideal of the domain A to (0) , so this class is distinct from all classes in (ZIL25); its product with each of them is itself. Thus both source-domain corrections are necessary for the claimed class-group statement, and the actual definition as printed instead has the collapse (ZIL24).

Exact consequence for the proposed RH inference. The checked zero-inversion theorem says that the supported-zero idempotent becomes the *unit of a localized semiring*; it does not give an inverse of $0_{\mathbb{C}}$ in the analytic amplitude field. The complete map to the localized object is (ZIL4), and its entire lost relation is (ZIL6). The coefficient comparison retains the amplitudes before localization by (ZIL12) and has the explicit connecting maps and retraction domain (ZIL14)–(ZIL19). For an analytic family the exact zero locus is retained on that unlocalized amplitude coordinate and is erased by the localization, as (ZIL20) proves. The prime and ideal-norm calculations (ZIL9)–(ZIL10), (ZIL22) identify precisely which arithmetic information has been removed. These established facts supply neither an off-critical zero nor a zero-location contradiction for the actual zeta function. Formula (ZIL21) refutes the narrower attempted inference from inversion, the backward heat equation and threshold nonnegativity alone. The calculation therefore constructs the exact resulting object and its connection to the original one, without treating the localization theorem or the note’s erroneous scaling relation as a disproof of RH.

Supported time-zero is a projection

$$e = (0, 1_L), \quad \tau = (0, 0_L), \quad \Phi(c, \lambda) = (c, 1_K \otimes \lambda)$$

Exact finite example: the polynomial z^4

$$q(U_{(t, 1_L)} z^4) = z^4 - 12tz^2 + 12t^2$$



The full map retains the changed support matrix

$$U_{(1, 1_L)} U_{(-1, 1_L)} = U_e \neq U_\tau = I_5$$

$$\Phi(U_e) = (I_5, B_{1_L})$$

Fixed-image labels keep the two parity chains:

$$\nu_0 \geq \nu_2 \geq \nu_4, \quad \nu_1 \geq \nu_3$$

$$B_{1_L} = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$1 = 1_{K_L}, \quad 0 = 0_{K_L}$

The actual theta family already lies in this fixed image

$$\mathbf{a}(t) = ((h_0(t), 1_L), \tau, (h_1(t), 1_L), \tau, (h_2(t), 1_L), \dots)$$

$$(-1)^j h_j(t) > 0 \quad (t \in \mathbb{R}), \quad U_{(0, \lambda)} \mathbf{a}(t) = \mathbf{a}(t) \quad (\lambda \in L)$$

Every label is retained. The kernel positivity and convergent heat action prove this identity.

Here q is the amplitude map. Proofs: THS3, THS9, THS16–25; full independent-label operator: HSL1–14.

Original heat kernel: Brad Rodgers & Terence Tao, arXiv:1801.05914v5. Full carrier: The Clankers, v11, def:lattice-split.

Figure 9: Supported time-zero changes the full coefficient operator. The exact degree-four example retains every cancellation label. The original theta vector is proved to lie in the resulting fixed image. Complete proofs: HSL1–14 and THS3, THS9, THS19–25.

31 Heat flow on independently labelled coefficients over the full lattice

Sources, carrier and scope. The full carrier is the existing $G_L(R)$ of the programme foundation, under the retained byline The Clankers: *Split Support Geometry: Universal Zero Fibres, Arithmetic Curves, Frobenius and Perfect Quantization*, v11, Definition `def:lattice-split`. Its full-lattice localization is also proved in 031, Theorems `thm:rebuilt-full-support` and `thm:e-stalk`. ASL18–22 of the present reader construct the exact coefficient map $\Phi_R(r, \lambda) = (r, 1_K \otimes \lambda)$; ZIL3–19 prove its localization square and exact amplitude loss. The definitions and the maps needed here are reproduced and proved below. The work in this section is the explicit heat action on that carrier, not an inference obtained by renaming an ordinary scalar heat theorem. The degree cutoff d below is a new polynomial-degree index; it does not replace any retained spectral cutoff or Gaussian metric.

Throughout the finite calculation R is a nonzero commutative unital $\mathbb{Q}$ -algebra, and L is the entire nontrivial bounded distributive lattice. The analytic calculation specifies $R = \mathbb{C}$. No completeness of L is assumed. Write

$$\begin{aligned} S &= G_L(R) = \{(0_R, \lambda) : \lambda \in L\} \cup \{(r, 1_L) : r \in R\}, \\ (r, \lambda) + (s, \mu) &= (r + s, \lambda \vee \mu), \quad (r, \lambda)(s, \mu) = (rs, \lambda \wedge \mu), \\ \tau &= (0_R, 0_L) = 0_S, \quad e = (0_R, 1_L), \quad 1_S = (1_R, 1_L), \quad z_\lambda = (0_R, \lambda), \\ M_d &= S^{d+1}, \quad v = ((a_i, \nu_i))_{i=0}^d, \quad a_i \neq 0_R \implies \nu_i = 1_L. \end{aligned} \tag{HSL1}$$

The labels ν_i are independent at the start. In particular, a zero coefficient may have any label in L , not only 0_L or 1_L . All matrix sums and products in the finite part use these semiring operations.

31.1 The heat matrix, its exact composition and its independent labels

For $x = (t, \lambda) \in S$ define $x^\dagger = (-t, \lambda)$ and

$$U_x(i, j) = \begin{cases} 1_S, & j = i, \\ \left(\frac{(-t)^m j!}{m! i!}, \lambda \right), & j - i = 2m > 0, \\ \tau, & \text{otherwise.} \end{cases} \quad (0 \leq i, j \leq d). \tag{HSL2}$$

These are elements of the original S : if $\lambda \neq 1_L$ then $t = 0_R$, so every displayed off-diagonal amplitude is zero. Let $D_d(i, j) = (j, 1_L)$ when $j = i + 1$, and τ otherwise. Multiplying along the unique possible index path proves that $D_d^n(i, j) = (j! / i!, 1_L)$ for $j - i = n$, and is τ otherwise. Also $D_d^{d+1} = 0$. Thus (HSL2) is the actual finite semiring exponential

$$U_x = I + \sum_{m=1}^{\lfloor d/2 \rfloor} (1/m!, 1_L)(x^\dagger)^m D_d^{2m}.$$

The integer factorials, the factor $1/m!$, the sign and the label are all retained. In this expression the empty higher terms are absent; supported zero is not substituted for the additive zero.

The exact time law is

$$U_x U_y = U_{x+y}, \quad U_\tau = I, \quad x = (t, \lambda), \quad y = (s, \mu). \tag{HSL3}$$

Here is a direct proof, including entries whose amplitudes cancel. For $j - i = 2n > 0$, the amplitude in the product entry is

$$\sum_{m=0}^n \frac{(-t)^m (i + 2m)!}{m! i!} \frac{(-s)^{n-m} j!}{(n - m)! (i + 2m)!} = \frac{j!}{i!} \sum_{m=0}^n \frac{(-t)^m (-s)^{n-m}}{m! (n - m)!} = \frac{(-(t + s))^n j!}{n! i!}.$$

The endpoint summands have labels μ and λ , respectively. Every intermediate summand has label $\lambda \wedge \mu$. Their joined label is therefore exactly $\lambda \vee \mu$. This conclusion does not depend on whether any amplitude vanishes. The diagonal has only the identity summand. An entry below the diagonal or of different parity has no nonzero-support path and is τ . This proves every entry of (HSL3).

Set

$$H_i = \{j : i < j \leq d, j - i \text{ is even}\}, \quad T_i(\nu) = \bigvee_{j \in H_i} \nu_j,$$

with an empty join equal to 0_L . Let E_t be the ordinary ring heat matrix, with the same factorial amplitude as (HSL2) also for $m = 0$, and ring zero elsewhere. Direct matrix action gives

$$\begin{aligned} U_x(a, \nu) &= (E_t a, F_\lambda \nu), \\ (E_t a)_i &= \sum_{\substack{j \geq i \\ j-i=2m}} \frac{(-t)^m j!}{m! i!} a_j, \quad (F_\lambda \nu)_i = \nu_i \vee (\lambda \wedge T_i(\nu)). \end{aligned} \quad (\text{HSL4})$$

The diagonal contributes precisely ν_i ; each higher same-parity coefficient contributes $\lambda \wedge \nu_j$. Distributivity gives the second formula. This proves the action on independent labels, including labelled zeros of the input.

The exact invariant used throughout the support calculation is

$$T_i(F_\lambda \nu) = T_i(\nu), \quad F_\lambda F_\mu = F_{\lambda \vee \mu}, \quad F_\lambda^2 = F_\lambda. \quad (\text{HSL5})$$

For the first equality, $F_\lambda \nu \geq \nu$ gives one inequality. If $j \in H_i$, both ν_j and $\lambda \wedge T_j(\nu)$ are at most $T_i(\nu)$, since $H_j \subset H_i$. Taking their joins proves the other inequality. Consequently $F_\lambda(F_\mu \nu)_i = \nu_i \vee (\mu \wedge T_i(\nu)) \vee (\lambda \wedge T_i(\nu))$; distributivity proves the remaining assertions.

Every supported-zero time is therefore an idempotent:

$$P_\lambda := U_{z_\lambda}, \quad P_\lambda(a, \nu) = (a, F_\lambda \nu), \quad P_\lambda^2 = P_\lambda, \quad U_x U_{x^\dagger} = U_{x^\dagger} U_x = P_\lambda, \quad P_\lambda U_x = U_x P_\lambda = U_x. \quad (\text{HSL6})$$

The matrix identities follow either from (HSL3) or from the explicit formulas. The associated image is

$$\begin{aligned} \text{im } U_x &= \text{im } P_\lambda =: \mathcal{H}_{\lambda, d} \\ &= \{(b, \eta) \in M_d : \lambda \wedge T_i(\eta) \leq \eta_i \text{ for every } i\} \\ &= \{(b, \eta) \in M_d : \lambda \wedge \eta_{i+2} \leq \eta_i \text{ whenever } i + 2 \leq d\}. \end{aligned} \quad (\text{HSL7})$$

An idempotent has image equal to its fixed set; (HSL4) gives the first description. The full-tail inequalities imply the adjacent ones. Conversely the adjacent inequalities, iterated along the parity chain after meeting with λ , imply $\lambda \wedge \eta_j \leq \eta_i$ for every $j \in H_i$. Finite distributivity then gives the full-tail inequalities. Finally $U_x = P_\lambda U_x$ and $P_\lambda = U_x U_{x^\dagger}$ prove equality of the two images.

The complete finite fibres are explicit and require no choice of a least label. For a proposed target (b, η) set

$$\delta_i(\eta) = \lambda \wedge T_i(\eta), \quad a_i = (E_{-t} b)_i = \sum_{\substack{j \geq i \\ j-i=2m}} \frac{t^m j!}{m! i!} b_j. \quad (\text{HSL8})$$

The fibre is empty if the target is outside $\mathcal{H}_{\lambda, d}$. For a target in that image it is exactly

$$U_x^{-1}(b, \eta) = \left\{ ((a_i, \nu_i))_{i=0}^d : \nu_i \vee \delta_i(\eta) = \eta_i \text{ for every } i, \quad a_i \neq 0_R \implies \nu_i = 1_L \right\}. \quad (\text{HSL9})$$

In particular the amplitude vector is unique and the permitted label choices in this formula are independent between coordinates. For necessity, $E_{-t} E_t = I$ follows from the amplitude computation in (HSL3), and tail invariance (HSL5) gives the label equations. For sufficiency,

suppose those label equations hold. Descending induction along either parity gives $T_i(\nu) = T_i(\eta)$. At a terminal index both tails are empty. If $k = i + 2 \leq d$ and the equality is known at k , then

$$\begin{aligned} T_i(\nu) &= \nu_k \vee T_k(\nu) = \nu_k \vee T_k(\eta) \\ &= (\nu_k \vee \delta_k(\eta)) \vee T_k(\eta) = \eta_k \vee T_k(\eta) = T_i(\eta), \end{aligned}$$

because $\delta_k(\eta) \leq T_k(\eta)$. Thus $F_\lambda \nu = \eta$, proving the converse. The carrier condition in (HSL9) is exactly the remaining requirement. The fibre is nonempty: $U_{x^\dagger}(b, \eta) = (a, \eta)$ is a preimage by (HSL6). This also proves that $\eta_i = 1_L$ whenever the recovered a_i is nonzero.

31.2 The flag retract and genuine heat automorphisms on it

Write $U_t = U_{(t, 1_L)}$ and $P = U_e$. Then

$$\begin{aligned} U_t U_{-t} &= P, & \mathcal{H}_{1,d} &= \text{im } P = \{(a, \nu) : \nu_i \geq \nu_{i+2} \text{ when } i + 2 \leq d\}, \\ U_t(a, \nu) &= (E_t a, \nu) \quad \text{on } \mathcal{H}_{1,d}, & U_t|_{\mathcal{H}_{1,d}} &\in \text{Aut}_S(\mathcal{H}_{1,d}), \\ (U_t|_{\mathcal{H}_{1,d}})^{-1} &= U_{-t}|_{\mathcal{H}_{1,d}}. \end{aligned} \tag{HSL10}$$

This is the pair of exact parity flags

$$\nu_0 \geq \nu_2 \geq \nu_4 \geq \cdots, \quad \nu_1 \geq \nu_3 \geq \nu_5 \geq \cdots.$$

The identity of the restricted group is P restricted to its image. The identity on the entire coefficient module is instead $U_\tau = I$. Both statements follow from (HSL6) and the defining fixed-set condition.

The image of every P_λ is a finitely generated projective S -semimodule in the lifting sense for surjective module maps. Indeed it is a retract of the free module M_d , with inclusion and retraction P_λ . Explicitly, if $A \rightarrow B$ is surjective and $f : \mathcal{H}_{\lambda,d} \rightarrow B$ is linear, choose a preimage in A of each $f(P_\lambda \epsilon_i)$, where the ϵ_i are the free coordinate vectors. Extend those choices linearly to M_d and restrict to the retract. The composite to B is f , since P_λ acts identically on the retract. The vectors $P_\lambda \epsilon_i$ generate it, proving finite generation as well as the asserted lifting property.

For $d \geq 2$, the representation $x \mapsto U_x$ is injective: the $(0, 2)$ entry is $(-2t, \lambda)$, so its amplitude divided by -2 recovers t and its label recovers λ . Moreover the only globally invertible operator in this family on all of M_d is U_τ . If $\lambda \neq 0_L$, take two zero-amplitude vectors whose index 2 label is λ and whose other labels are 0_L , except that their index 0 labels are respectively 0_L and λ . They are distinct and have the same image under U_x , by (HSL4). Thus U_x is not injective. If $\lambda = 0_L$, the time carrier forces $t = 0_R$ and $U_x = I$. For $d = 0, 1$ every matrix in (HSL2) is I , and those two qualifications about injectivity and units are not asserted.

For $L = \mathbb{B}$, the zero-amplitude part of the flag retract has exactly

$$(n_{\text{even}} + 1)(n_{\text{odd}} + 1) \quad \text{states}, \quad n_{\text{even}} = \lfloor d/2 \rfloor + 1, \quad n_{\text{odd}} = \lceil d/2 \rceil. \tag{HSL11}$$

A decreasing Boolean chain of length n consists of an initial string of ones and a remaining string of zeros, giving $n + 1$ choices; the two parity chains are independent. At $d = 2$ this gives six states. This Boolean-semimodule retract is not free: a finite free Boolean semimodule has 2^r states for an integer $r \geq 0$, and an infinite free one has infinitely many states. The flag module has a concrete nonfree geometry even though it is a retract of a free coefficient module.

The automorphisms proved here are linear heat automorphisms of a coefficient semimodule. They are not multiplicative automorphisms of polynomial algebras. Already on amplitudes $E_t(z) = z$ but $E_t(z^2) = z^2 - 2t$. Moreover for $d \geq 2$ the ordinarily labelled monomial z lies in $\mathcal{H}_{1,d}$, whereas z^2 does not: its index 0 label is 0_L and its index 2 label is 1_L . Thus closure of this retract under polynomial multiplication has not been assumed.

31.3 Exact amplitude, joint-coefficient and localization morphisms

For the tropical semifield K with zero, put $K_L = K \otimes_{\mathbb{B}} L$ and $\iota_L(\lambda) = 1_K \otimes \lambda$. The map ι_L is injective: the nonzero indicator $\beta : K \rightarrow \mathbb{B}$ preserves maximum and product, and $\beta \otimes \text{id}_L$ is a retraction to L . Thus the original joint map $\Phi_R(r, \lambda) = (r, \iota_L(\lambda))$ is injective. Apply it entrywise to (HSL2). It gives exactly the pair

$$\begin{aligned} \Phi_R(U_x) &= (E_t, \tilde{B}_\lambda), \\ \tilde{B}_\lambda(i, j) &= \begin{cases} 1_{K_L}, & i = j, \\ \iota_L(\lambda), & j > i, j - i \text{ even}, \\ 0_{K_L}, & \text{otherwise.} \end{cases} \quad (\text{HSL12}) \\ q^{d+1}U_x &= E_t q^{d+1}, \quad \Phi_R^{d+1}U_x = (E_t, \tilde{B}_\lambda) \Phi_R^{d+1}. \end{aligned}$$

The last identities follow by entrywise application of semiring homomorphisms to finite matrix multiplication; they also follow directly from (HSL4). They retain the full original amplitude vector and all its independent support labels.

Supported-zero coefficient localization is $S[e^{-1}] \simeq L$, $(r, \nu) \mapsto \nu$: inverting the idempotent e sets it equal to the new unit, and $e(r, \nu) = z_\nu$ gives precisely that quotient. Consequently the matrix specializes to

$$\begin{aligned} B_\lambda(i, j) &= \begin{cases} 1_L, & i = j, \\ \lambda, & j > i, j - i \text{ even}, \\ 0_L, & \text{otherwise,} \end{cases} \quad (\text{HSL13}) \\ B_\lambda B_\mu &= B_{\lambda \vee \mu}, \quad B_\lambda^2 = B_\lambda, \quad \sigma^{d+1}U_x = F_\lambda \sigma^{d+1}. \end{aligned}$$

The composition proof is the support part of (HSL3), or (HSL5). For top-labelled time every U_t has the same specialized matrix B_{1_L} , which is not the global identity for $d \geq 2$. For example at $d = 2$ it is

$$B_{1_L} = \begin{pmatrix} 1_L & 0_L & 1_L \\ 0_L & 1_L & 0_L \\ 0_L & 0_L & 1_L \end{pmatrix}, \quad B_{1_L}(0_L, 0_L, 1_L) = B_{1_L}(1_L, 0_L, 1_L).$$

Thus making the coefficient e invertible does not make the heat matrix U_e invertible on the whole coefficient module. It makes that matrix the displayed support projection.

For any S -semimodule M , localization at e is the corner eM , with scalar action through $eS \simeq L$: the map is $v \mapsto ev$. To check the universal property, any linear map into a module in which e acts invertibly has e acting as the identity, since $e^2 = e$. It therefore factors uniquely through eM . Applied to the free module and the flag retract this proves

$$\begin{aligned} M_d[e^{-1}] &\simeq L^{d+1}, \\ \mathcal{H}_{1,d}[e^{-1}] &\simeq \{(\nu_i) \in L^{d+1} : \nu_i \geq \nu_{i+2}\}, \quad (\text{HSL14}) \\ U_t &\mapsto B_{1_L}, \quad U_t|_{\mathcal{H}_{1,d}} \mapsto \text{id}_{\mathcal{H}_{1,d}[e^{-1}]}. \end{aligned}$$

All these statements retain the original labels; zero-amplitude vectors are the actual corners, rather than an imposed Boolean replacement.

31.4 An analytic completion on which every heat sum converges

Finite polynomial truncation is not a substitute for an infinite heat action. We first construct its amplitude completion exactly. For a coefficient sequence $a = (a_n)_{n \geq 0}$ set

$$\|a\|_h = \sum_{n=0}^{\infty} |a_n| \sqrt{n!} h^n, \quad \mathcal{E} = \{a : \|a\|_h < \infty \text{ for every } h > 0\}. \quad (\text{HSL15})$$

The positive integer h already give a defining countable family of seminorms. Each is a weighted ℓ^1 norm. A Cauchy sequence for all these norms has coordinate limits; completeness of every weighted ℓ^1 space gives the same limit in each of them. Thus $\mathcal{E}$ is complete. Polynomial truncations converge in every norm, because the corresponding positive summable tail tends to zero. This constructs $\mathcal{E}$ as a Fréchet completion of the polynomial amplitudes with this specified topology.

The corresponding entire-function description is exact:

$$f(z) = \sum_{n \geq 0} a_n z^n, \quad \mathcal{E} = \left\{ f \text{ entire} : p_\alpha(f) := \sup_{z \in \mathbb{C}} |f(z)| e^{-\alpha|z|^2} < \infty \text{ for every } \alpha > 0 \right\},$$

$$p_{1/(2h^2)}(f) \leq \|a\|_h, \quad \|a\|_h \leq \frac{p_\alpha(f)}{1 - h\sqrt{2e\alpha}} \quad (h\sqrt{2e\alpha} < 1). \quad (\text{HSL16})$$

The symbol e here is the ordinary real exponential constant, not the supported-zero element e . For the first estimate, each $|z|^n/(h^n \sqrt{n!})$ is at most $\exp(|z|^2/(2h^2))$, since its square is a term in that exponential series. Summing against the weighted coefficients proves uniform convergence on compact sets and the first bound. Conversely Cauchy's coefficient estimate on a circle of radius r gives $|a_n| \leq p_\alpha(f) e^{\alpha r^2} r^{-n}$. For $n \geq 1$ put $r = \sqrt{n/(2\alpha)}$ and use $n! \leq n^n$ to obtain $|a_n| \sqrt{n!} h^n \leq p_\alpha(f) (h\sqrt{2e\alpha})^n$. The $n = 0$ estimate is $|a_0| \leq p_\alpha(f)$. The geometric sum proves the second bound and the equivalence.

For every complex time t the following heat sum is defined on all of $\mathcal{E}$:

$$(T_t a)_i = \sum_{m=0}^{\infty} \frac{(-t)^m (i+2m)!}{m! i!} a_{i+2m}, \quad \|T_t a\|_h \leq \|a\|_{h+\sqrt{2|t|}}. \quad (\text{HSL17})$$

In particular no formal interchange of divergent power series is being used. To prove the bound, take absolute values before any interchange and use nonnegative summation:

$$\begin{aligned} & \sum_{i,m \geq 0} |a_{i+2m}| \frac{|t|^m (i+2m)!}{m! i!} \sqrt{i!} h^i \\ &= \sum_{n \geq 0} |a_n| \sqrt{n!} \sum_{m=0}^{\lfloor n/2 \rfloor} \frac{\sqrt{n!}}{m! \sqrt{(n-2m)!}} |t|^m h^{n-2m}. \end{aligned}$$

The inner coefficient is $\sqrt{\binom{n}{2m}} \sqrt{\binom{2m}{m}}$, at most $\binom{n}{2m} 2^m$: the first binomial coefficient is a positive integer, and $\binom{2m}{m} \leq 2^{2m}$. Thus the inner sum is bounded by the even part of $(h + \sqrt{2|t|})^n$, and hence by the full expression. This proves absolute convergence and the literal estimate in (HSL17).

The resulting operators satisfy

$$T_t T_s = T_{t+s}, \quad T_0 = I, \quad T_t^{-1} = T_{-t}, \quad \frac{\partial}{\partial t} T_t f = -\partial_z^2 T_t f. \quad (\text{HSL18})$$

For composition, the absolute double heat sum is bounded in $\|\cdot\|_h$ by $\|a\|_{h+\sqrt{2|t|}+\sqrt{2|s|}}$. It can therefore be regrouped; for each final total derivative order the finite binomial calculation in (HSL3) gives $t+s$. For time differentiability, (HSL17) applied with $|t| \leq A$ bounds the sum of norms of the power-series terms uniformly on every compact time disc. Applying the same bound on a larger disc justifies termwise differentiation there. The power series is $\sum_{m \geq 0} (-t)^m \partial_z^{2m} f / m!$. Differentiating it gives the displayed heat equation, with the same locally uniform convergence in z by (HSL16). Thus the analytic amplitude flow is an everywhere-defined complex-time group on this completion.

31.5 The exact support completion and the original-lattice domain

An arbitrary L need not admit the countable joins occurring in an infinite heat row. We therefore specify, rather than silently assume, the support completion. Let

$$\widehat{L} = \text{Idl}(L), \quad j(\lambda) = \downarrow \lambda, \quad I \wedge J = I \cap J, \quad \bigvee_{\alpha} I_{\alpha} = \text{the ideal generated by } \bigcup_{\alpha} I_{\alpha}. \quad (\text{HSL19})$$

The ideals are nonempty downward-closed subsets closed under finite joins. The bottom is $\{0_L\}$ and the top is L . The map j is an injective bounded lattice homomorphism: finite joins of principal ideals give the principal ideal of their join, and intersections give the principal ideal of their meet. The displayed arbitrary join exists explicitly: an element belongs to it exactly when it is below a finite join of elements drawn from the indicated ideals.

Finite meets distribute over arbitrary joins in $\widehat{L}$. For example if $a \in I \cap \bigvee_{\alpha} J_{\alpha}$, choose $b_1, \dots, b_r$ from the indicated J_{α} with $a \leq \bigvee b_k$. Then $a = \bigvee_k (a \wedge b_k)$, and each $a \wedge b_k$ lies in $I \cap J_{\alpha_k}$. This proves one inclusion of $I \cap \bigvee J_{\alpha} = \bigvee (I \cap J_{\alpha})$; the opposite inclusion follows from containment. This proves the exact infinite distributivity needed below. It does not identify an infinite join in the original L with one in $\widehat{L}$ without checking that it is principal.

Define the completed, independently labelled coefficient module

$$\widehat{\mathcal{M}} = \{(a, (I_i)_{i \geq 0}) : a \in \mathcal{E}, I_i \in \widehat{L}, a_i \neq 0 \implies I_i = L\}. \quad (\text{HSL20})$$

Addition has amplitude addition and coordinate ideal joins; the original scalar $(r, \lambda) \in S = G_L(\mathbb{C})$ acts by amplitudes ra_i and ideals $j(\lambda) \cap I_i$. These formulas preserve the carrier: a nonzero amplitude after addition has a nonzero contributing summand, and a nonzero amplitude after scalar multiplication has both factors nonzero and hence both relevant labels top. The module laws follow from (HSL19). Every original finite vector embeds by principal coefficient ideals and by $\{0_L\}$ above its degree.

The completed heat action of the original time parameter $x = (t, \lambda) \in S$ is

$$\widehat{U}_x(a, I) = (T_t a, \widehat{F}_{\lambda} I), \quad (\widehat{F}_{\lambda} I)_i = I_i \vee \left(j(\lambda) \cap \bigvee_{\substack{j > i \\ j-i \text{ even}}} I_j \right). \quad (\text{HSL21})$$

The amplitude sum converges by (HSL17), and the support join exists by (HSL19). Admissibility also holds. If $t = 0$, the amplitude at i remains a_i , whose nonvanishing already forces $I_i = L$. If $t \neq 0$, then $\lambda = 1_L$; a nonzero output sum has at least one nonzero summand, whose input amplitude has top support. Its contribution therefore makes the output support top. This proves that the formula defines an actual map of the stated carrier.

With $\widehat{T}_i(I) = \bigvee_{j > i, j-i \text{ even}} I_j$, the proof of (HSL5) uses only nested tails and distributivity. It therefore gives exactly

$$\begin{aligned} \widehat{T}_i(\widehat{F}_{\lambda} I) &= \widehat{T}_i(I), & \widehat{F}_{\lambda} \widehat{F}_{\mu} &= \widehat{F}_{\lambda \vee \mu}, \\ \widehat{U}_x \widehat{U}_y &= \widehat{U}_{x+y}, & \widehat{U}_{\tau} &= I, & \widehat{U}_x \widehat{U}_{x^{\dagger}} &= \widehat{U}_{z_{\lambda}} =: \widehat{P}_{\lambda}. \end{aligned} \quad (\text{HSL22})$$

The amplitude assertion follows from (HSL18). Addition and scalar linearity follow from join preservation and the distributivity proved in (HSL19), together with linearity of T_t . In particular the completed supported-zero operator is again an idempotent with unchanged amplitudes, and every $\widehat{U}_x$ is an automorphism on its exact image $\text{im } \widehat{P}_{\lambda}$.

The complete infinite image and fibres can also be specified without inventing a terminal coordinate at infinity:

$$\text{im } \widehat{U}_x = \{(b, J) \in \widehat{\mathcal{M}} : j(\lambda) \cap J_j \subset J_i \text{ for all } j > i \text{ of the same parity}\}. \quad (\text{HSL23})$$

For a target in this set, put $a = T_{-t}b$. Its fibre consists exactly of ideals I_i satisfying $I_i \subset J_i$, $a_i \neq 0 \Rightarrow I_i = L$, and the following finite-witness generation condition for every i :

for every $c \in J_i$ there are $v_i \in I_i$, $j_1, \dots, j_r > i$ of the same parity, $c_k \in I_{j_k}$ such that

$$c \leq v_i \vee \bigvee_{k=1}^r (\lambda \wedge c_k). \quad (\text{HSL24})$$

An empty list is permitted. These conditions are necessary and sufficient for (HSL21) to have target J : by (HSL19) its right side is exactly the ideal generated by I_i and the elements $\lambda \wedge c$ with c from a higher same-parity I_j . All those generators lie in J_i by (HSL23), and (HSL24) is the reverse containment. The amplitude is unique by (HSL18); existence follows also from the preimage $\widehat{U}_{x^\dagger}(b, J)$. This proves the full infinite fibre, including its boundary condition on arbitrarily high degrees.

The finite independent-coordinate formula (HSL9) must not be extended without this condition. For example let $\lambda = 1_L$, put every $I_i = \{0_L\}$, and let $J_i = L$ at every even index and $\{0_L\}$ at every odd index. The equations $I_i \vee (j(\lambda) \cap J_{i+2}) = J_i$ all hold, but $\widehat{F}_\lambda I$ is zero in every coordinate and is not J . This exhibits the missing information at infinity if the terminating induction used for finite fibres is omitted.

There is an invariant domain entirely within the original lattice. Define

$$\mathcal{M}_L^{\text{tail}} = \{(a, \nu) : a \in \mathcal{E}, \nu_i \in L, a_i \neq 0 \Rightarrow \nu_i = 1_L, \mathcal{T}_i(\nu) \text{ is principal for every } i\},$$

$$\mathcal{T}_i(\nu) = \bigvee_{\substack{j \geq i \\ j-i \text{ even}}} \downarrow \nu_j = \downarrow s_i \quad (s_i \in L). \quad (\text{HSL25})$$

The generator s_i is unique. Being principal here is the precise finite-generation assertion: s_i is a finite join of labels from that tail. Indeed s_i belongs to the generated ideal and hence is below such a finite join; all those labels are at most s_i , giving equality. Every finite polynomial belongs to this domain. For finite L , every label sequence satisfies this condition; no finiteness of L is required for the definition.

On this domain the completed formula has values in the original L and reads

$$(U_x(a, \nu))_i = ((T_t a)_i, \nu_i \vee (\lambda \wedge s_{i+2})), \quad \mathcal{T}_i(F_\lambda \nu) = \mathcal{T}_i(\nu). \quad (\text{HSL26})$$

The ideal $\mathcal{T}_{i+2}$ is the strict tail in (HSL21). Distributivity and principality identify its meet with $\downarrow \lambda$ and its join with $\downarrow \nu_i$ as the principal ideal of the displayed label. The invariant equality follows as in (HSL22), now with the inclusive tail: the output contains each input label, and it imports labels only from that same inclusive tail. Hence the new tails have the same principal generators, proving invariance of the entire domain. It is an S -subsemimodule: tails of a sum are joins of two principal tail ideals, and tails after scalar support multiplication are their meets with a principal ideal. This establishes a heat action preserving the full original lattice on an explicit infinite domain containing all polynomials and the theta family considered below.

31.6 Polynomial approximation, exact truncation defect and the theta lift

Let $\iota_d : M_d \rightarrow \widehat{\mathcal{M}}$ extend a polynomial by amplitude zero and label 0_L above degree d . Then

$$\widehat{U}_x \iota_d = \iota_d U_x. \quad (\text{HSL27})$$

The sums in (HSL21) terminate on this image, proving the identity directly. In contrast let π_d retain only coordinates $0, \dots, d$ of an infinite object. For $i \leq d$, its exact amplitude defect is

$$\begin{aligned} & \left(q\pi_d \widehat{U}_x(a, I) - qU_x \pi_d(a, I) \right)_i \\ &= \sum_{\substack{j > d \\ j-i=2m>0}} \frac{(-t)^m j!}{m! i!} a_j. \end{aligned} \quad (\text{HSL28})$$

Here U_x on the truncated completed module means the same finite formula with ideals as labels. The series is absolutely convergent by (HSL17). The support in the first expression is the support in the second joined with exactly

$$j(\lambda) \cap \bigvee_{\substack{j > d \\ j-i \text{ even}}} I_j. \quad (\text{HSL29})$$

Both identities follow by splitting the row at $j = d$. They give a connecting map for the entire discarded tail, rather than treating truncation as an intertwiner. For the amplitude polynomial $f(z) = z^{d+2}$ the truncated input is zero, whereas the degree- d coefficient after heat is $-t(d+2)(d+1)$. Thus (HSL28) can be nonzero even on polynomial inputs when the chosen truncation is too low.

For a general element of $\widehat{\mathcal{M}}$, amplitude polynomial truncations converge in every norm (HSL15). Their heat images converge in every such norm by (HSL17). For each fixed coordinate their heat supports form an increasing family of ideals whose join is exactly (HSL21), by the explicit definition of joins in (HSL19). Thus the analytic limit and the algebraic support limit are both specified, with no unsupported infinite join in L . Any ideal label can itself be obtained as the directed join of the principal ideals of its elements, so the ideal extension preserves the original labels by an explicit injective map and a defined directed completion.

The actual theta family uses the kernel of Brad Rodgers and Terence Tao, *The De Bruijn–Newman constant is non-negative*, arXiv:1801.05914v5. It provides a particularly exact original-lattice case. Its independently proved kernel calculation has coefficients

$$\begin{aligned} H_t(z) &= \sum_{n \geq 0} a_n(t) z^n, & a_{2j+1}(t) &= 0, \\ a_{2j}(t) &= \frac{(-1)^j}{(2j)!} \int_0^\infty e^{tu^2} \Phi_{\text{theta}}(u) u^{2j} du, \\ (-1)^j a_{2j}(t) &> 0 \quad (t \in \mathbb{R}), & H_t &\in \mathcal{E}. \end{aligned} \quad (\text{HSL30})$$

The symbol Φ_{theta} denotes the analytic theta kernel, not the coefficient map Φ_R . The full kernel proof and its human-source locators are supplied in `THETA_HEAT_FULL_SUPPORT.tex` of this reader: THS9 proves strict real-time nonvanishing, THS15 proves the entire-function bounds, and THS18–23 calculate the original support vector and its heat law. The source read for these locators has SHA-256 `98f9e873b23b2c2b78b856a448e65d65fb9469e26bc36659e49ebef50346f604`. Equation (HSL30) records the exact input used here. The following support and map calculation is independent of any assertion about the zero locations of H_t .

Retain the coefficient lift obtained from these real-time nonzero coefficients and its complex-time continuation:

$$\mathbf{H}_t = ((a_i(t), \nu_i))_{i \geq 0}, \quad \nu_{2j} = 1_L, \quad \nu_{2j+1} = 0_L, \quad s_{2j} = 1_L, \quad s_{2j+1} = 0_L. \quad (\text{HSL31})$$

At real time the even labels are forced by nonvanishing in (HSL30); the odd coefficients are the structural zero τ . At complex time the same even labels are retained even if an even amplitude vanishes. This specifies the lift exactly. Its tail ideals are principal with the generators displayed in (HSL31), so it belongs to the original-lattice domain (HSL25).

The exact heat and coefficient-map laws on this object are

$$\begin{aligned} U_{(r,\lambda)} \mathbf{H}_t &= \mathbf{H}_{t+r} \quad ((r, \lambda) \in G_L(\mathbb{C})), \\ U_{z\lambda} \mathbf{H}_t &= \mathbf{H}_t \quad (\lambda \in L), \quad \Phi_{\mathbb{C}}(a_i(t), \nu_i) = (a_i(t), 1_K \otimes \nu_i). \end{aligned} \quad (\text{HSL32})$$

For amplitudes the actual kernel heat law, or its absolutely convergent coefficient form, gives $T_r H_t = H_{t+r}$. For supports, an even coordinate already has top label from its diagonal term, and an odd coordinate receives only odd structural zeros. Equivalently $\nu_i \vee (\lambda \wedge s_{i+2}) = \nu_i$ for every i . This proves (HSL32) with every label retained. It proves in particular that the actual

theta coefficient family lies in the flag image on which the top-labelled heat operators are genuine inverse automorphisms.

The coefficient map in (HSL32) also commutes with the infinite sums used for this family: every amplitude sum is absolute, while each even support partial sum is top from its diagonal term and every odd support partial sum is zero. Thus no infinite tensor-product join or unproved continuity of $\Phi_{\mathbb{C}}$ is needed. After supported-zero coefficient localization, all $\mathbf{H}_t$ have the same even-top, odd-zero coefficient labels. Before localization their first coordinates retain every actual $a_i(t)$ and therefore the entire function H_t exactly. These proved morphisms and the heat automorphism structure do not assign a value to any real-root threshold or assert an RH conclusion.

31.7 The completed joint coefficient morphism for all support sequences

The exact joint morphism on the general completed module also exists, once its target is specified to admit the required joins. It is not necessary to assume that tensoring with K preserves every infinite join of L . For any commutative additively idempotent semiring A , let $\text{Idl}_{\text{ord}}(A)$ be the nonempty lower subsets for the order $a \leq b \iff a + b = b$ which are closed under finite sums. Define addition and arbitrary joins by the lower ideal generated by the union, and multiplication by the lower ideal generated by the products of elements. The zero is $\downarrow 0_A$ and the unit is $\downarrow 1_A$; the latter need not be the top ideal. These operations form a complete idempotent semiring in which multiplication distributes over arbitrary joins. Indeed both sides of that distributivity statement are the lower ideal generated by the same products; finite distributivity in A supplies equality of their finite generating sums. The same generator argument proves associativity and commutativity of multiplication. The principal-ideal map $a \mapsto \downarrow a$ is an injective unital semiring homomorphism.

Apply this construction to K_L , and put

$$\begin{aligned} \mathfrak{K}_L &= \text{Idl}_{\text{ord}}(K_L), \\ \iota^{\#} : \widehat{L} &\longrightarrow \mathfrak{K}_L, \quad \iota^{\#}(I) = \{w \in K_L : \exists a \in I, w \leq \iota_L(a)\}, \\ \rho^{\#} : \mathfrak{K}_L &\longrightarrow \widehat{L}, \quad \rho^{\#}(J) = \{\lambda \in L : \exists w \in J, \lambda \leq \rho_L(w)\}, \quad \rho^{\#}\iota^{\#} = \text{id}_{\widehat{L}}. \end{aligned} \tag{HSL33}$$

Here $\rho_L = \beta \otimes \text{id}_L$ is the retraction used before (HSL12). The sets displayed in (HSL33) are already ideals: finite sums in the target are bounded by the image of the finite join or sum of their witnesses. A unital semiring homomorphism between idempotent semirings induces this lower-ideal map, and the generator description proves that it preserves arbitrary joins, products, zero and the stated units. For L viewed as a semiring, ideal multiplication is intersection: the products are meets, and any element of an intersection is its own meet. Thus the domain really is precisely (HSL19). The two induced maps compose to the identity because $\rho_L \iota_L = \text{id}_L$; explicitly, $w \leq \iota_L(a)$ implies $\rho_L(w) \leq a$, and every $a \in I$ is obtained by $w = \iota_L(a)$. This proves injectivity of $\iota^{\#}$ as well as its arbitrary-join compatibility.

The completed coefficient map and its restriction to every original coefficient are therefore

$$\begin{aligned} \widehat{\Phi} : G_{\widehat{L}}(\mathbb{C}) &\longrightarrow \mathbb{C} \times \mathfrak{K}_L, \quad (a, I) \longmapsto (a, \iota^{\#}(I)), \\ \widehat{\Phi}(a, \downarrow \nu) &= (a, \downarrow \iota_L(\nu)). \end{aligned} \tag{HSL34}$$

Both coordinates preserve zero, unit and the two operations; the map is injective by the retraction in (HSL33). The second formula proves the exact commuting square with the original $\Phi_{\mathbb{C}}$ and the principal-ideal embedding $K_L \rightarrow \mathfrak{K}_L$. The top ideal L of $\widehat{L}$ maps to $\downarrow 1_{K_L}$, the multiplicative identity of $\mathfrak{K}_L$, as required by the carrier.

Define the infinite support matrix $\mathfrak{B}_{\lambda}$ over $\mathfrak{K}_L$ to have diagonal $\downarrow 1_{K_L}$, strict same-parity upper entries $\downarrow \iota_L(\lambda)$, and zero entries elsewhere. Its action uses the existing arbitrary joins in $\mathfrak{K}_L$. Let $\widehat{\Phi}^{\mathbb{N}}$ apply (HSL34) to every coefficient. Then the exact completed joint intertwiner is

$$\widehat{\Phi}^{\mathbb{N}} \widehat{U}_{(t, \lambda)} = (T_t, \mathfrak{B}_{\lambda}) \widehat{\Phi}^{\mathbb{N}}. \tag{HSL35}$$

The amplitude equality is (HSL17). For the support equality, apply the arbitrary-join and finite-product preserving map $\iota^\#$ to (HSL21); its value is precisely the row action of the stated support matrix. This proves (HSL35) for every completed independent support sequence, not only for the actual theta vector. The theta calculation (HSL31)–(HSL32) descends to the original coefficient target K_L because its relevant joins stabilize, as already proved there. Thus the general completion and the original-family map are connected by an explicit injective morphism, with exact convergence on both coordinates.

32 The theta heat flow through the complete supported coefficient map

Scope correction, 23 September 2026. This chapter lifts the specified classical analytic heat kernel. It does not derive that kernel from the entire absolute arithmetic base. THS22 and THS25 explicitly show that its orbit and evaluated scalar observable do not distinguish all support labels. The exact original zeta and shifted-Hurwitz receiving formulas, retained boundary component, exceptional modules, and inverse Mellin map are reconstructed in ZTC1–21 and CFD1–27. Those comparisons, rather than the scalar theta orbit alone, specify which original arithmetic object is retained.

Original source and actual reading. The original human source is Brad Rodgers and Terence Tao, *The De Bruijn–Newman constant is non-negative*, arXiv:1801.05914v5. The author file `fmp-template.tex`, retained in `zero_inversion_intake/original_tex`, has SHA-256

25b015c0fb151f1613247d82cf53a6561a320d4dab2bb4f809482e7c477a5817.

Lines 1–155 were read, including the exact definitions `phidef`, `htdef`, `hoz` and `sas`. This is a bounded reading, not a reading of the complete paper. The arguments here use the displayed kernel and integral, and prove all convergence, differentiation and coefficient statements below directly. No threshold theorem about the zeros is invoked. The complete coefficient map is ZIL11–12 of the included `ZERO_INVERSION_FULL_LATTICE_REVIEW.tex`; its source at the time of reading has SHA-256

eb87afa8b3ffac41ca0845bb1f2ba49d9dba14e6c6ac0db86d72d0e5963a50b9.

Its lines 1–135 and 231–290 were read, including the complete coefficient laws and the map with its retraction. The needed algebra is proved again here. The existing included ZHR source was read in full for its original coordinates; none of its threshold implications is a premise of this calculation.

32.1 The original kernel and a uniform bound for every moment

Keep the original variables and all constants:

$$\begin{aligned} \Phi(u) &= \sum_{n=1}^{\infty} \Phi_n(u), \quad \Phi_n(u) = (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, \quad u \geq 0, \\ H_t(z) &= \int_0^{\infty} e^{tu^2} \Phi(u) \cos(zu) du, \quad (t, z) \in \mathbb{C}^2. \end{aligned} \tag{THS1}$$

Complex time extends the same integral, without changing its kernel. Set

$$a = e^{-3\pi}, \quad C_4 = \frac{1 + 11a + 11a^2 + a^3}{(1 - a)^5}, \quad C_{\Phi} = 2\pi^2 C_4. \tag{THS2}$$

Then, for every $u \geq 0$,

$$0 < \pi(2\pi - 3)e^{9u - \pi e^{4u}} \leq \Phi(u) \leq C_{\Phi} e^{9u - \pi e^{4u}}. \tag{THS3}$$

Indeed $\Phi_n(u) = \pi n^2 e^{5u} (2\pi n^2 e^{4u} - 3) e^{-\pi n^2 e^{4u}}$ is positive, since $2\pi - 3 > 0$. Its $n = 1$ term is at least the lower bound: $2\pi e^{4u} - 3 \geq (2\pi - 3)e^{4u}$. For the upper bound discard the negative term and use $n^2 - 1 \geq 3(n - 1)$ for $n \geq 1$ and $e^{4u} \geq 1$. Thus

$$\sum_{n \geq 1} n^4 e^{-\pi(n^2 - 1)e^{4u}} \leq \sum_{n \geq 1} n^4 a^{n-1} = C_4.$$

The last identity follows by applying $(a d/da)^4$ to $\sum_{n \geq 1} a^n = a/(1-a)$ and dividing by a ; differentiation is valid on $|a| < 1$ by the geometric series. In particular the kernel series converges absolutely, uniformly on every compact u interval, with the stated global bound.

For $A, B \geq 0$ and every integer $p \geq 0$, define the finite explicit number

$$\mathcal{M}_p(A, B) = \frac{C_\Phi}{2\pi} \exp \left(\frac{3A^2}{32\pi} + \frac{(B+9+p)^2}{8\pi} - \frac{\pi}{2} \right). \quad (\text{THS4})$$

The following pointwise and integral bounds hold:

$$\begin{aligned} u^p e^{Au^2+Bu} \Phi(u) &\leq C_\Phi \exp \left(\frac{3A^2}{32\pi} + \frac{(B+9+p)^2}{8\pi} \right) e^{-\pi e^{4u}/2}, \\ \int_0^\infty u^p e^{Au^2+Bu} \Phi(u) du &\leq \mathcal{M}_p(A, B). \end{aligned} \quad (\text{THS5})$$

To prove this, use $u^p \leq e^{pu}$, with $0^0 = 1$ in this inequality, and the two Taylor bounds $e^{4u} \geq (32/3)u^4$ and $e^{4u} \geq 8u^2$. Completing the square in u^2 and then in u gives

$$Au^2 - \frac{\pi}{4}e^{4u} \leq Au^2 - \frac{8\pi}{3}u^4 \leq \frac{3A^2}{32\pi}, \quad (B+9+p)u - \frac{\pi}{4}e^{4u} \leq (B+9+p)u - 2\pi u^2 \leq \frac{(B+9+p)^2}{8\pi}.$$

The remaining half of the negative exponential proves the first line. Finally $e^{4u} \geq 1 + 4u$ gives $\int_0^\infty e^{-\pi e^{4u}/2} du \leq e^{-\pi/2}/(2\pi)$, proving the second line. These estimates include every moment order, every positive exponential-square weight, and any fixed linear exponential weight.

32.2 All differentiations and infinite heat sums

For $\alpha, \beta \geq 0$ integers and compact bounds $\operatorname{Re} t \leq A$ and $|\operatorname{Im} z| \leq B$, where $A, B \geq 0$, one has

$$\partial_t^\alpha \partial_z^\beta H_t(z) = \int_0^\infty u^{2\alpha+\beta} e^{tu^2} \Phi(u) \cos(zu + \beta\pi/2) du, \quad |\partial_t^\alpha \partial_z^\beta H_t(z)| \leq \mathcal{M}_{2\alpha+\beta}(A, B). \quad (\text{THS6})$$

Indeed $|\cos w|, |\sin w| \leq e^{|\operatorname{Im} w|}$. Thus THS5 dominates every displayed integrand on those compact sets. Difference quotients may be written as integrals of the next derivative along a segment in a slightly larger compact set; the same majorant applies. Dominated convergence therefore proves the first complex derivative and its integral formula. Iteration proves every mixed derivative and commutation of their orders. Alternatively the exponential and cosine series about any (t_0, z_0) converge under the integrable majorant $e^{(\operatorname{Re} t_0 + R)u^2 + (|\operatorname{Im} z_0| + S)u} \Phi(u)$ on a bidisc of radii R, S . This also proves joint entireness. All these arguments also allow the sum over n in THS1 to pass through the integral and derivatives: the sum of the absolute integrands is bounded by the same expression because every $\Phi_n(u)$ is positive. In particular

$$\partial_t H_t(z) = -\partial_z^2 H_t(z), \quad H_t(-z) = H_t(z), \quad H_t(\bar{z}) = \overline{H_t(z)}. \quad (\text{THS7})$$

The last two identities follow directly from the integral and the real positive kernel.

The exact Taylor expansion is

$$H_t(z) = \sum_{j=0}^\infty h_j(t) z^{2j}, \quad h_j(t) = \frac{(-1)^j}{(2j)!} \int_0^\infty e^{tu^2} \Phi(u) u^{2j} du. \quad (\text{THS8})$$

For $|z| \leq R$, the sum of the absolute integrands is $e^{\operatorname{Re} t u^2} \Phi(u) \cosh(Ru)$, bounded by THS5. This proves the expansion, absolute locally uniform convergence, and the absence of every odd coefficient. Every h_j is entire in t by THS6. For real t there is the explicit strict lower bound

$$(-1)^j h_j(t) \geq \frac{\pi(2\pi-3)}{(2j)!(2j+1)} \exp(\min(t, 0) - \pi e^4) > 0. \quad (\text{THS9})$$

Restrict the integral to $0 \leq u \leq 1$. On that interval THS3 is at least $\pi(2\pi - 3)e^{-\pi e^4}$ and $e^{tu^2} \geq e^{\min(t,0)}$; integrate u^{2j} . Thus every even Taylor coefficient is nonzero at every real time. No such nonvanishing statement is assumed at complex time.

For arbitrary complex t, r , the coefficient heat identity is

$$h_j(t+r) = \sum_{m=0}^{\infty} \frac{(-r)^m (2j+2m)!}{m!(2j)!} h_{j+m}(t). \quad (\text{THS10})$$

To verify absolute convergence before summing, THS8 and the triangle inequality give

$$\sum_{m \geq 0} \frac{|r|^m (2j+2m)!}{m!(2j)!} |h_{j+m}(t)| \leq \frac{1}{(2j)!} \int_0^{\infty} u^{2j} e^{(\operatorname{Re} t + |r|)u^2} \Phi(u) du < \infty. \quad (\text{THS11})$$

For a displayed bound replace the exponent coefficient by $\max(\operatorname{Re} t + |r|, 0)$ and apply THS5. After interchanging sum and integral, the signs satisfy $(-1)^m (-1)^{j+m} = (-1)^j$ and the remaining sum is $\sum_m (ru^2)^m / m! = e^{ru^2}$. This proves THS10. The same argument gives, for any finite number of successive complex time steps $r_1, \dots, r_d$, an absolutely convergent multiple series with majorant

$$\frac{1}{(2j)!} \int_0^{\infty} u^{2j} e^{(\operatorname{Re} t + \sum_{a=1}^d |r_a|)u^2} \Phi(u) du. \quad (\text{THS12})$$

Indeed the product of the absolute exponential series is $\prod_a e^{|r_a|u^2}$. Finite partial products increase to that positive bound; their integrals converge by THS5. Tonelli's theorem followed by absolute convergence permits all orders of summation. The binomial or multinomial identity then combines the steps into $\sum_a r_a$, with their original signs.

There is also the function identity, with absolute locally uniform convergence in every complex variable,

$$H_{t+r}(z) = \sum_{m=0}^{\infty} \frac{(-r)^m}{m!} \partial_z^{2m} H_t(z). \quad (\text{THS13})$$

Insert THS6:

$$\partial_z^{2m} \cos(zu) = (-1)^m u^{2m} \cos(zu).$$

The absolute integral majorant is

$$e^{(\operatorname{Re} t + |r|)u^2 + |\operatorname{Im} z|u} \Phi(u);$$

THS5 applies uniformly on compact sets. This proves THS13, and applies after every further t, r, z derivative by increasing the moment order. The group law on this actual family follows either from THS12 or by evaluating its final integral. In particular coefficient differentiation gives

$$h'_j(t) = -(2j+2)(2j+1)h_{j+1}(t). \quad (\text{THS14})$$

For connection with the entire-function domain, every $a > 0$ satisfies

$$\sup_{z \in \mathbb{C}} |H_t(z)| e^{-a|z|^2} \leq \int_0^{\infty} e^{(\operatorname{Re} t + 1/(4a))u^2} \Phi(u) du < \infty. \quad (\text{THS15})$$

This follows from $|z|u \leq a|z|^2 + u^2/(4a)$ and $|\cos(zu)| \leq e^{|z|u}$. Thus the actual family belongs to the space of entire functions satisfying every positive Gaussian bound. This is a proved inclusion, without changing its kernel.

32.3 Every original lattice label and the actual coefficient map

Let L be any bounded distributive lattice, with no completeness assumption. Its bottom and top are $0_L, 1_L$. Retain

$$\begin{aligned} S &= G_L(\mathbb{C}) = \{(0, \lambda) : \lambda \in L\} \cup \{(c, 1_L) : c \in \mathbb{C}\}, \\ (c, \lambda) + (d, \nu) &= (c + d, \lambda \vee \nu), \quad (c, \lambda)(d, \nu) = (cd, \lambda \wedge \nu), \\ \tau &= (0, 0_L), \quad e = (0, 1_L), \quad 1_S = (1, 1_L), \quad \widehat{c} = (c, 1_L). \end{aligned} \quad (\text{THS16})$$

If an amplitude sum is nonzero at least one input label is top; if a product is nonzero both input labels are top. Thus these operations preserve the carrier. The ring laws and lattice distributivity prove all semiring laws. No lower label is identified with another. The construction also covers the one-element lattice, when $\tau = e$ and S is the original ring.

Keep the programme's tropical semifield with zero K and put $K_L = K \otimes_{\mathbb{B}} L$. Its zero and unit are $\mathbf{0}_K, \mathbf{1}_K$. The map that sends nonzero elements of K to $1 \in \mathbb{B}$ and zero to zero preserves addition and multiplication: a tropical sum is zero precisely when all its terms are zero and a product is zero precisely when a factor is zero. Tensoring this map with L proves the well-defined retraction

$$\begin{aligned} \iota_L(\lambda) &= \mathbf{1}_K \otimes \lambda, \quad \rho_L \left(\sum_i b_i \otimes \lambda_i \right) = \bigvee_{b_i \neq \mathbf{0}_K} \lambda_i, \quad \rho_L \iota_L = \text{id}_L, \\ \Phi_{\mathbb{C}} : S &\longrightarrow \mathbb{C} \times K_L, \quad \Phi_{\mathbb{C}}(c, \lambda) = (c, \iota_L(\lambda)). \end{aligned} \quad (\text{THS17})$$

The retraction shows that ι_L is injective. Both coordinates of $\Phi_{\mathbb{C}}$ preserve both operations, zero and unit, and their equality determines both original coordinates. Thus this is the injective original full-lattice map, not a choice of one Boolean character.

For clarity about infinite sums, the following restricted convergence is sufficient. A sequence (c_m, λ_m) in S has a sum here if $\sum_m |c_m| < \infty$ and the finite joins $\lambda_0 \vee \cdots \vee \lambda_M$ are constant for all sufficiently large M . Its sum is the amplitude sum with that eventual label. It belongs to S : if the amplitude sum is nonzero at least one summand has nonzero amplitude and hence top label. On these sums THS17 commutes with summation, because the finite support sums in its target are exactly $\iota_L(\lambda_0 \vee \cdots \vee \lambda_M)$ and are likewise eventually constant. This defines no arbitrary infinite join in L or K_L . Every infinite sum below satisfies this restricted definition from its first summand.

The actual supported coefficient vector is

$$\mathbf{a}(t) = (a_n(t))_{n \geq 0}, \quad a_{2j}(t) = (h_j(t), 1_L), \quad a_{2j+1}(t) = \tau. \quad (\text{THS18})$$

At real times its even labels are forced by THS9 and the carrier condition in THS16. The odd coefficients are structurally absent from the cosine series and have the additive zero τ . For complex time the same labels are retained: if an even coefficient happens to vanish it has value e , not τ . This gives a specified entire amplitude family with its unchanged support vector, not an assertion of nonvanishing at complex times.

For every original supported time $x = (r, \lambda) \in S$, define the upper triangular heat matrix on all degrees by

$$U_x(i, j) = \begin{cases} 1_S, & j = i, \\ \left(\frac{(-r)^m j!}{m! i!}, \lambda \right), & j = i + 2m, \ m \geq 1, \\ \tau, & \text{otherwise.} \end{cases} \quad (\text{THS19})$$

Every entry lies in S . Indeed a nonzero off-diagonal amplitude forces $r \neq 0$, which forces $\lambda = 1_L$. Equivalently that entry is $(-1)^{\widehat{m}j!/(m!i!)} x^m$. Thus if $r = 0$, all off-diagonal amplitudes vanish

but their original label remains λ ; they are not deleted. For $x = (r, \lambda)$ and $y = (s, \mu)$, every matrix-product entry is a finite sum and

$$U_x U_y = U_{x+y}, \quad x + y = (r + s, \lambda \vee \mu), \quad U_\tau = I. \quad (\text{THS20})$$

For an even gap $j - i = 2m > 0$, the amplitude of the product is $\frac{j!}{i!} \sum_{b=0}^m (-r)^b (-s)^{m-b} / (b!(m-b)!)$ which is $(-r - s)^m j! / (m! i!)$. Its two endpoint terms have labels λ and μ ; each interior term has label $\lambda \wedge \mu$. Their join is exactly $\lambda \vee \mu$, including when some or every amplitude is zero. The diagonal has its one unit term; an odd or negative gap has only absorbing zero terms. This proves every entry and the identity statement.

For the actual vector, all coefficient sums converge in the specified sense and satisfy the full supported heat law

$$U_{(r,\lambda)} \mathbf{a}(t) = \mathbf{a}(t + r), \quad t \in \mathbb{C}, \quad (r, \lambda) \in S. \quad (\text{THS21})$$

At row $2j$, its amplitude sum is THS10, absolutely convergent by THS11. The $m = 0$ summand already has label 1_L , so every finite partial sum has that same label. At every odd row every source coefficient is τ , and every product and partial sum is τ . This proves all amplitudes and every label. For successive actions the absolute double or multiple sums are THS12. Their finite support joins again have the displayed labels because they include the diagonal contribution. Thus THS20 acts on this vector with no change of summation order or support completion left unproved.

In particular every zero-amplitude time, with its full original label, acts by

$$U_{(0,\lambda)} \mathbf{a}(t) = \mathbf{a}(t) \quad (\lambda \in L), \quad U_e \mathbf{a}(t) = \mathbf{a}(t), \quad U_\tau \mathbf{a}(t) = \mathbf{a}(t). \quad (\text{THS22})$$

These equalities do not equate the different operators on arbitrary supported vectors. For example a vector whose only nonzero entry is $a_2 = 1_S$ acquires coefficient $(0, \lambda)$ in degree zero under $U_{(0,\lambda)}$. On THS18 the even degree-zero label, and every even tail label, is already top; the same action adds no support. No label has been discarded.

Apply $\Phi_{\mathbb{C}}$ to every coefficient and to every matrix entry; denote the resulting objects by $\mathcal{A}(t)$ and $\mathcal{U}_x$. Then

$$\Phi_{\mathbb{C}}^{\mathbb{N}}(U_x \mathbf{a}(t)) = \mathcal{U}_x \Phi_{\mathbb{C}}^{\mathbb{N}}(\mathbf{a}(t)) = \mathcal{A}(t + r), \quad \mathcal{A}_{2j}(t) = (h_j(t), 1_{K_L}), \quad \mathcal{A}_{2j+1}(t) = (0, 0_{K_L}). \quad (\text{THS23})$$

Finite partial sums commute because THS17 is a semiring map; their absolute amplitude limits and eventually constant support limits commute by the explicit sum definition. Injectivity retains every coefficient, including all supported zeros.

The coefficient heat differential also holds before and after this map. On the fixed support vector in THS18, define the time derivative by its proved entire amplitudes, retaining its even top and odd bottom labels. The formal second derivative in z has row i equal to $(i + 2)(i + 1) a_{i+2}$. Multiplication by $\widehat{-1}$ keeps the labels. THS14 proves

$$\partial_t \mathbf{a}(t) = -D_z^2 \mathbf{a}(t), \quad \Phi_{\mathbb{C}}^{\mathbb{N}}(\partial_t \mathbf{a}(t)) = -D_z^2 \mathcal{A}(t). \quad (\text{THS24})$$

This derivative is not an undeclared subtraction operation in an arbitrary semiring. Each even coordinate lies in the exact additive group $\mathbb{C} \times \{1_L\}$, whose identity is e ; each odd coordinate is the fixed singleton $\{\tau\}$. Their product therefore has its coefficientwise complex vector space structure with zero vector $(e, \tau, e, \tau, \dots)$. Differentiation there is ordinary coordinate differentiation; the preceding integral bounds justify it for this family. The inclusion into the full supported coefficient space retains these identities and is not asserted to preserve its global zero vector $(\tau, \tau, \dots)$. This proves the precise meaning of the differential in THS24 without changing any support.

Evaluation at any original supported point $y = (z, \nu) \in S$ also converges and gives

$$\sum_{n \geq 0} a_n(t) y^n = (H_t(z), 1_L), \quad \sum_{n \geq 0} \Phi_{\mathbb{C}}(a_n(t)) \Phi_{\mathbb{C}}(y)^n = (H_t(z), 1_{K_L}). \quad (\text{THS25})$$

Absolute convergence is THS8; the constant term is already top-supported, so every partial support sum is top, including $z = 0$ and every allowed label ν of that zero. The two equalities follow by THS17 and the ordinary Taylor expansion. In particular an analytic zero has the original value e , and its exact image $(0, 1_{K_L})$, throughout this heat flow.

$$\begin{array}{ccc} \mathbf{a}(t) & \xrightarrow{U_{(r,\lambda)}} & \mathbf{a}(t+r) \\ \Phi_{\mathbb{C}}^{\mathbb{N}} \downarrow & & \downarrow \Phi_{\mathbb{C}}^{\mathbb{N}} \\ \mathcal{A}(t) & \xrightarrow{\mathcal{U}_{(r,\lambda)}} & \mathcal{A}(t+r) \end{array}$$

Figure 10: The actual coefficient heat map, THS19–23. An even coordinate retains label 1_L and its exact moment amplitude; an odd coordinate is τ . Every zero-amplitude time $(0, \lambda)$ fixes this particular vector by THS22. All lattice labels remain in the time object and the full coefficient map; the diagram is not a replacement by a Boolean character. The kernel and coordinates are Rodgers–Tao’s THS1.

32.4 The exact defect of every finite polynomial truncation

Let $P_N H_t(z) = \sum_{j=0}^N h_j(t) z^{2j}$ and let P_N on supported vectors replace all coefficients above degree $2N$ by τ . At $0 \leq j \leq N$ the amplitude of the discrepancy between $P_N H_{t+r}$ and the polynomial heat evolution $e^{-rD_z^2} P_N H_t$ is exactly

$$\begin{aligned} \Delta_{N,j}(t, r) &= \sum_{m=N-j+1}^{\infty} \frac{(-r)^m (2j+2m)!}{m! (2j)!} h_{j+m}(t) \\ &= \frac{(-1)^j}{(2j)!} \int_0^{\infty} e^{tu^2} \Phi(u) u^{2j} \left(e^{ru^2} - \sum_{m=0}^{N-j} \frac{(ru^2)^m}{m!} \right) du. \end{aligned} \quad (\text{THS26})$$

This follows by subtracting the finite beginning of the absolutely convergent series THS10; the polynomial derivative formula gives exactly those retained terms. For complex t, r , writing $M = N - j + 1 \geq 1$, the tail has the explicit estimate

$$|\Delta_{N,j}(t, r)| \leq \frac{|r|^M}{(2j)! M!} \int_0^{\infty} u^{2N+2} e^{(\operatorname{Re} t + |r|)u^2} \Phi(u) du. \quad (\text{THS27})$$

Use $\sum_{m=M}^{\infty} v^m/m! \leq v^M e^v/M!$ for $v \geq 0$: the inequality $(M+\ell)! \geq M!\ell!$ proves it term by term. The right side is finite and explicitly bounded by THS5. If t is real and $r > 0$, then

$$(-1)^j \Delta_{N,j}(t, r) > 0 \quad (0 \leq j \leq N). \quad (\text{THS28})$$

The exponential tail in THS26 is strictly positive for every $u > 0$, and every other factor after removing $(-1)^j$ is positive by THS3. Its finite integral is therefore positive. This proves that finite polynomial truncation does not intertwine the actual heat flow, even at the constant coefficient. Injectivity of THS17 shows that the full coefficient map does not remove this discrepancy. Both retained low coefficients still have their top labels; their amplitudes differ by the explicit nonzero quantity above. Nevertheless the actual finite evolutions do converge to the full flow, with the original coefficients. For $|z| \leq R$ the absolute double-series majorant for their terms is

$$\int_0^{\infty} e^{(\operatorname{Re} t + |r|)u^2} \Phi(u) \cosh(Ru) du < \infty. \quad (\text{THS29})$$

Indeed the indices are $j, m \geq 0$, and the absolute coefficient integrands are bounded by $u^{2j+2m} |r|^m R^{2j} / (m! (2j)!)$ times $e^{\operatorname{Re} t u^2} \Phi(u)$. Summing gives exactly the displayed majorant.

The polynomial evolution retains the triangle $j + m \leq N$. These increasing triangles exhaust the double series, so its absolute tails tend to zero, uniformly on compact sets of (t, r, z) by using their common THS5 bound. Thus $e^{-rD_z^2} P_N H_t \rightarrow H_{t+r}$ locally uniformly, while THS28 excludes exact intertwining at each finite N . At each fixed supported coefficient its label is also eventually the exact label in THS18. This states both the genuine limiting map and the finite defect, with no assumed truncation identity.

32.5 An explicit geometric bound for the entire discarded tail

The convergence in THS29 has an explicit rate that retains the original cutoff. Define

$$E_N(t, r, z) = H_{t+r}(z) - e^{-rD_z^2} P_N H_t(z).$$

For every $\sigma > 1$, $R \geq 0$ and complex t, r , one has

$$\sup_{|z| \leq R} |E_N(t, r, z)| \leq \sigma^{-(N+1)} \mathcal{M}_0(\max(\operatorname{Re} t + \sigma|r|, 0), \sqrt{\sigma}R). \quad (\text{THS30})$$

To prove this, the exact error is the absolutely convergent double series of THS29 over $j + m > N$. For each such term, $1 \leq \sigma^{j+m-(N+1)}$. Multiply its absolute integrand by this bound, then sum over every $j, m \geq 0$. The resulting upper bound is precisely

$$\sigma^{-(N+1)} \int_0^\infty e^{(\operatorname{Re} t + \sigma|r|)u^2} \Phi(u) \cosh(\sqrt{\sigma}Ru) du.$$

Since $\cosh v \leq e^v$ for $v \geq 0$, THS5 gives (THS30). No coefficient, time parameter or cutoff has been rescaled in the actual heat family; σ is only an explicitly stated parameter in the proved upper bound.

Every mixed derivative of this error has a corresponding exact rate. For $\alpha, \beta, \gamma \geq 0$, positive radii $\delta_t, \delta_r, \delta_z$, complex t_0, r_0 , and $R \geq 0$, put

$$A_* = \max\{\operatorname{Re} t_0 + \delta_t + \sigma(|r_0| + \delta_r), 0\}, \quad B_* = \sqrt{\sigma}(R + \delta_z).$$

Then

$$\sup_{|z| \leq R} |\partial_t^\alpha \partial_r^\beta \partial_z^\gamma E_N(t_0, r_0, z)| \leq \frac{\alpha! \beta! \gamma!}{\delta_t^\alpha \delta_r^\beta \delta_z^\gamma} \sigma^{-(N+1)} \mathcal{M}_0(A_*, B_*). \quad (\text{THS31})$$

The error is entire in all three variables by THS6, THS13 and its finite polynomial term. On the product of circles of the three displayed radii about (t_0, r_0, z) , one has $\operatorname{Re} t \leq \operatorname{Re} t_0 + \delta_t$, $|r| \leq |r_0| + \delta_r$ and $|z'| \leq R + \delta_z$. Apply (THS30) there, then apply the one-variable Cauchy integral formula successively in the three variables. Each formula contributes its exact factorial and inverse radius power, proving (THS31). Its constants are finite by THS4 and independent of N , so the bound tends geometrically to zero for the original growing polynomial cutoff. The coefficient supports at every retained even index are already top and the odd indices are already τ , as proved in THS18–29; this amplitude error bound introduces no missing support limit.

The full supported heat law proved here retains the entire amplitude family, all original lattice labels and their joint map. It does not decide whether H_0 has nonreal zeros and asserts no conclusion about RH.

Heat transport: multiplication, flags and actual evaluation

$$\mathcal{E} = \{\sum a_n z^n : \sum |a_n| \sqrt{n!} h^n < \infty \text{ for every } h > 0\}$$

$$T_t = e^{-t\partial_z^2}, \quad f \star_t g = T_t((T_{-t}f)(T_{-t}g))$$

$$f \star_t g = \sum_{n \geq 0} \frac{(-2t)^n}{n!} f^{(n)} g^{(n)}, \quad z \star_t z = z^2 - 2t$$

The complete coefficient algebra keeps two descending support chains:

$$\nu_0 \geq \nu_2 \geq \dots, \quad \nu_1 \geq \nu_3 \geq \dots, \quad a_n \neq 0 \Rightarrow \nu_n = 1_L$$

Multiplication includes the mixed odd-odd contribution

$$(\nu \oplus \mu)_0 = (\nu_0 \wedge \mu_0) \vee (\nu_1 \wedge \mu_1)$$

$$(\nu \oplus \mu)_n = \bigvee_{r=0}^n \nu_r \wedge \mu_{n-r} \quad (n \geq 1)$$

$$\Phi(f, \nu) = (f, (1_K \otimes \nu_n)_n) \in \mathcal{E} \times \mathcal{F}(K_L)$$

The exact character identifies which divisor is transported

$$\chi_{t,z}(f) = (T_{-t}f)(z), \quad \chi_{t,z}(H_a) = H_{a-t}(z)$$

$$\chi_{t,z}(H_t) = H_0(z), \quad H_t(z) = \chi_{t,z}(H_{2t})$$

Every entire function, time parameter and support label remains in these maps.

Proofs: HTP1–24, SHP1–28, HDT1–6. All series converge in the displayed coefficient space.

Kernel: Brad Rodgers & Terence Tao, arXiv:1801.05914v5. Full carrier: The Clankers, v11, defilattice-split.

Figure 11: The convergent transported product and its actual characters, with the entire coefficient flag and the mixed odd-odd term retained. All arrows and identities are proved in HTP1–24, SHP1–28 and HDT1–6.

33 The exact product transported by the original heat group

Sources and scope. The analytic domain and heat operator are the original HSL15–18 in the complete included `FULL_LATTICE_HEAT_OPERATOR.tex`. Lines 366–469 were read directly at source SHA–256

42c1de0bb8991f460f51d8f84c31d271b01fa5f5b9e6a7f2e803a0ea6deb6841.

The norm estimates needed below are proved here as well. The actual theta source is Brad Rodgers and Terence Tao, *The De Bruijn–Newman constant is non-negative*, author file `fmp-template.tex`, SHA–256

25b015c0fb151f1613247d82cf53a6561a320d4dab2bb4f809482e7c477a5817.

The prior actual reading is lines 1–155, including `phidef` and `htdef`; no whole-paper reading is claimed. The positive-kernel proof and all heat integrals are THS1–29, proved and read completely in the included `THETA_HEAT_FULL_SUPPORT.tex`. Its current source SHA–256, including the later tail bounds THS30–31, is

69507dcc5e192fad2023fff6e7fac1f435dbb2dd1809e6c68d1b5b1e712a9957.

No threshold theorem or claim about the location of the theta family’s zeros is a premise. This is the amplitude-ring calculation; the independent coefficient supports and their full lattice map remain the subject of the included supported heat construction, rather than being identified with one label.

33.1 The complete analytic domain and its exact norm bounds

For $h > 0$, retain precisely the coefficient norms

$$\|f\|_h = \sum_{n \geq 0} |a_n| \sqrt{n!} h^n, \quad f(z) = \sum_{n \geq 0} a_n z^n, \quad \mathcal{E} = \{f : \|f\|_h < \infty \text{ for every } h > 0\}. \quad (\text{HTP1})$$

The positive integer h give a countable defining family. A Cauchy sequence in every such norm has coordinate limits; completeness of weighted ℓ^1 gives the same limit in each norm. Thus $\mathcal{E}$ is a complete Fréchet space. Every polynomial truncation converges in all the norms, since each omitted weighted tail is summable. The coefficient domain equals the entire-function domain

$$p_\alpha(f) = \sup_{z \in \mathbb{C}} |f(z)| e^{-\alpha|z|^2}, \quad \mathcal{E} = \{f \text{ entire} : p_\alpha(f) < \infty \ (\alpha > 0)\}, \quad (\text{HTP2})$$

$$p_{1/(2h^2)}(f) \leq \|f\|_h, \quad \|f\|_h \leq \frac{p_\alpha(f)}{1 - h\sqrt{2e\alpha}} \quad (h\sqrt{2e\alpha} < 1).$$

Here e denotes the ordinary exponential constant. For the first inequality, the square of $|z|^n/(h^n \sqrt{n!})$ is a term of $e^{|z|^2/h^2}$. Multiply the resulting bound by the weighted coefficients and sum. This proves entire convergence too. Conversely, Cauchy’s coefficient inequality on radius $r = \sqrt{n/(2\alpha)}$, for $n \geq 1$, and $n! \leq n^n$ give $|a_n| \sqrt{n!} h^n \leq p_\alpha(f) (h\sqrt{2e\alpha})^n$. The degree-zero term is at most $p_\alpha(f)$; the geometric sum proves the second inequality and the stated equivalence.

Ordinary multiplication is a continuous bilinear operation:

$$\|fg\|_h \leq \|f\|_{\sqrt{2}h} \|g\|_{\sqrt{2}h}. \quad (\text{HTP3})$$

Indeed, for coefficient indices $j, k \geq 0$, $\sqrt{(j+k)!} \leq \sqrt{j!k!} 2^{(j+k)/2}$, since $\binom{j+k}{j} \leq 2^{j+k}$. Taking absolute values before the Cauchy-product sum yields HTP3 and its absolute convergence. For every integer $n \geq 0$ and every $b > 0$ one also has

$$\|D^n f\|_h \leq \sqrt{n!} b^{-n} \|f\|_{\sqrt{h^2+b^2}}, \quad D = \frac{d}{dz}. \quad (\text{HTP4})$$

For $k \geq n$, the contribution of a_k to the left side is $|a_k|\sqrt{k!}\sqrt{n!}\sqrt{\binom{k}{n}}h^{k-n}$. The single nonnegative binomial term gives $\binom{k}{n}h^{2(k-n)}b^{2n} \leq (h^2 + b^2)^k$. Its square root proves the bound after summation. Thus every derivative is a continuous operator on this exact domain, with its original factorials.

The heat group is

$$\begin{aligned} T_t f &= \sum_{m \geq 0} \frac{(-t)^m}{m!} D^{2m} f, & (T_t f)_i &= \sum_{m \geq 0} \frac{(-t)^m (i + 2m)!}{m! i!} a_{i+2m}, \\ & \sum_{m \geq 0} \frac{|t|^m}{m!} \|D^{2m} f\|_h \leq \|f\|_{h + \sqrt{2|t|}}, & t &\in \mathbb{C}. \end{aligned} \quad (\text{HTP5})$$

To prove the last inequality, put $k = i + 2m$ in the nonnegative sum of coefficient norms. Its inner factor is

$$\sum_{m=0}^{\lfloor k/2 \rfloor} \frac{\sqrt{k!}}{m! \sqrt{(k-2m)!}} |t|^m h^{k-2m}.$$

The numerical coefficient equals $\sqrt{\binom{k}{2m}} \sqrt{\binom{2m}{m}}$, at most $\binom{k}{2m} 2^m$. The sum is bounded by the even part of $(h + \sqrt{2|t|})^k$, and therefore by that whole expression. Summing against $|a_k|\sqrt{k!}$ proves HTP5. Every heat sum is thus absolutely convergent in every defining norm, locally uniformly in t by enlarging $|t|$. Moreover

$$T_t T_s = T_{t+s}, \quad T_t^{-1} = T_{-t}, \quad T_0 = I, \quad \partial_t T_t f = -D^2 T_t f. \quad (\text{HTP6})$$

The absolute double sum is at most $\|f\|_{h + \sqrt{2|t|} + \sqrt{2|s|}}$ by two applications of HTP5. Regrouping by total derivative order is therefore permitted. Its finite binomial coefficients give $t + s$. The identity at zero and inverse follow. For differentiation, bound the series on a larger time disc by HTP5 and use the convergent power series in the complete normed coordinate spaces. This gives the displayed derivative in every norm.

33.2 A convergent product, with the exact derivative formula

For $a, b > 0$ satisfying $ab > 2|t|$, HTP3–4 prove

$$\sum_{n \geq 0} \frac{(2|t|)^n}{n!} \|f^{(n)} g^{(n)}\|_h \leq \frac{\|f\|_{\sqrt{2h^2+a^2}} \|g\|_{\sqrt{2h^2+b^2}}}{1 - 2|t|/(ab)}. \quad (\text{HTP7})$$

Apply HTP4 with parameters a, b at the norm $\sqrt{2}h$ to the two derivative factors in HTP3. The product contributes $n!(ab)^{-n}$, which cancels the $n!$ in the denominator. The remaining geometric series has the stated sum. Every factor and radius in HTP7 is therefore explicit. The bound is uniform on a compact time set after choosing ab greater than twice its largest $|t|$. It proves absolute convergence in $\mathcal{E}$ of the derivative-product series.

Define the transported product by the actual continuous heat automorphism of the vector space:

$$\begin{aligned} f \star_t g &= T_t((T_{-t}f)(T_{-t}g)), & \mathcal{E}_t &= (\mathcal{E}, \star_t), \\ \|f \star_t g\|_h &\leq \|f\|_{R_t(h)} \|g\|_{R_t(h)}, & R_t(h) &= \sqrt{2}(h + \sqrt{2|t|}) + \sqrt{2|t|}. \end{aligned} \quad (\text{HTP8})$$

First apply HTP5 to the outer heat operator, then HTP3 at $h + \sqrt{2|t|}$, and then HTP5 to each inner operator. This proves the norm bound and bilinear continuity directly.

For polynomials the product has a finite operator calculation. On a polynomial $f(z_1)g(z_2)$ put $D_i = \partial_{z_i}$ and let μ set $z_1 = z_2 = z$. Leibniz's rule gives $D\mu = \mu(D_1 + D_2)$. Every

differential exponential below terminates on this polynomial; the differential operators commute. Consequently

$$\begin{aligned} T_t((T_{-t}f)(T_{-t}g)) &= \mu e^{-t(D_1+D_2)^2} e^{t(D_1^2+D_2^2)}(f \otimes g) \\ &= \mu e^{-2tD_1D_2}(f \otimes g) = \sum_{n \geq 0} \frac{(-2t)^n}{n!} f^{(n)} g^{(n)}. \end{aligned} \quad (\text{HTP9})$$

Combining commuting finite exponentials follows by the finite binomial formula at each order, not by cancelling divergent series. Now approximate $f, g \in \mathcal{E}$ by their polynomial truncations in every norm. HTP8 makes the left side continuous in both arguments. HTP7 makes the sum on the right a continuous bilinear map: apply that bound to the difference in each argument, and note that a convergent sequence is bounded in each input norm. Thus the polynomial identity extends to

$$\boxed{f \star_t g = \sum_{n=0}^{\infty} \frac{(-2t)^n}{n!} f^{(n)} g^{(n)}}. \quad (\text{HTP10})$$

This proof establishes the identity on the exact space, with absolute convergence, without a formal infinite cancellation.

Even the expansion of all three heat operators and every Leibniz term is absolutely bounded. For $x = |t|$ and $R = \sqrt{2}(h + \sqrt{2x}) + \sqrt{2x}$ one has

$$\sum_{a,b,c \geq 0} \sum_{v=0}^{2a} \frac{x^{a+b+c}}{a!b!c!} \binom{2a}{v} \|f^{(2b+v)} g^{(2c+2a-v)}\|_h \leq \|f\|_R \|g\|_R. \quad (\text{HTP11})$$

Let $f^\#(z) = \sum |a_n| z^n$ and define $g^\#$ similarly. They belong to $\mathcal{E}$ with the same norms. Derivatives and products of these series have nonnegative coefficients; they dominate coefficientwise the absolute coefficients of the corresponding derivatives and products of f, g . With $S_x = T_{-x}$, the nonnegative coefficient expansion of $S_x((S_x f^\#)(S_x g^\#))$ has exactly the weights in HTP11, by Leibniz's rule. For any finite set of terms its coefficient sum is bounded by this entire nonnegative series. The norm of the latter is at most $\|f\|_R \|g\|_R$ by HTP8's proof using three positive heat sums. Monotone convergence in the nonnegative coefficient sums proves HTP11. Thus expansions or regroupings of this product, when desired, have a separate finite absolute majorant.

The product is commutative and associative and has unit 1:

$$\begin{aligned} T_{-t}(f \star_t g) &= (T_{-t}f)(T_{-t}g), & (f \star_t g) \star_t k &= f \star_t (g \star_t k), & f \star_t 1 &= f, \\ T_t : (\mathcal{E}, \cdot) &\xrightarrow{\sim} (\mathcal{E}, \star_t) && \text{is a continuous unital algebra isomorphism.} \end{aligned} \quad (\text{HTP12})$$

The first identity is HTP6 applied to the definition. Applying the injective map T_{-t} to either parenthesization gives the same ordinary product of three functions; this proves associativity with no triple derivative-series rearrangement. Commutativity follows the same way. Since $T_t 1 = 1$, the unit claim follows too. Bilinearity and both directions of continuity were already proved. The rings have the same vector-space zero and unit, while their products are the specified ones.

33.3 Characters, principal ideals and the exact finite jets

For $t \neq 0$, ordinary evaluation at any point fails to be a character of this product. With w denoting the coordinate function,

$$w \star_t w = w^2 - 2t, \quad \text{ev}_z(w \star_t w) = z^2 - 2t \neq z^2 = \text{ev}_z(w)^2. \quad (\text{HTP13})$$

Only $n = 0, 1$ contribute in HTP10; the factor $2t$ and its minus sign are therefore fixed.

We first establish the division needed to characterize all complex-valued characters, without assuming their continuity. For every $f \in \mathcal{E}$ and $z \in \mathbb{C}$, the removable quotient belongs to the same space:

$$Q_z f(w) = \begin{cases} (f(w) - f(z))/(w - z), & w \neq z, \\ f'(z), & w = z, \end{cases} \quad Q_z f \in \mathcal{E}, \quad f - f(z) = (w - z)Q_z f. \quad (\text{HTP14})$$

For $|w - z| \geq 1$ its p_α -weighted absolute value is at most $p_\alpha(f) + |f(z)| \leq p_\alpha(f)(1 + e^{\alpha|z|^2})$. For $|w - z| \leq 1$, use $Q_z f(w) = \int_0^1 f'(z + s(w - z)) ds$. Cauchy's derivative bound with radius 1 at each point of this segment is at most $p_\alpha(f)e^{\alpha(|z|+2)^2}$. Multiplication by $e^{-\alpha|w|^2} \leq 1$ preserves that bound. This proves finite $p_\alpha(Q_z f)$ for every $\alpha > 0$, hence HTP14 by HTP2. Repeated division proves the corresponding finite-order Taylor remainder statement.

Every unital $\mathbb{C}$ -algebra character on the ordinary ring $\mathcal{E}$ is evaluation. Indeed, for a character χ put $z = \chi(w)$. Applying χ to HTP14 gives $\chi(f) = f(z)$, because the factor $\chi(w - z)$ is zero. Conversely evaluation is visibly a unital character and is continuous by HTP2. HTP12 now gives the complete transported character formula

$$\chi_{t,z}(f) = (T_{-t}f)(z), \quad z \in \mathbb{C}, \quad \text{Hom}_{\mathbb{C}\text{-alg},1}(\mathcal{E}_t, \mathbb{C}) = \{\chi_{t,z} : z \in \mathbb{C}\}. \quad (\text{HTP15})$$

For a character on $\mathcal{E}_t$, compose with T_t and apply the ordinary classification just proved. Distinct z give distinct characters because $T_{-t}w = w$. The formulas and HTP5/HTP2 show that every one is continuous. No statement about maximal ideals with other residue fields is needed.

For any $F \in \mathcal{E}$, exact principal-ideal transport is

$$\begin{aligned} T_t(F\mathcal{E}) &= (T_t F) \star_t \mathcal{E}, \\ \mathcal{E}/(F\mathcal{E}) &\xrightarrow{\sim} \mathcal{E}_t/((T_t F) \star_t \mathcal{E}), \quad [g] \mapsto [T_t g]. \end{aligned} \quad (\text{HTP16})$$

If $g = Fq$ then $T_t g = (T_t F) \star_t (T_t q)$ by HTP12. Conversely every element on the right has this form because T_t is onto. Its inverse T_{-t} proves the equality of ideals and the well-defined, injective and surjective quotient map, preserving both operations and both identities. This is an algebraic ideal equality; no closure of the ideal is substituted for it.

For a point z and integer $m \geq 1$, write $\mathfrak{m}_z = (w - z)\mathcal{E}$ and $\mathfrak{m}_{t,z} = \ker \chi_{t,z}$. Then

$$\begin{aligned} \mathfrak{m}_{t,z} &= T_t \mathfrak{m}_z = (w - z) \star_t \mathcal{E}, \quad (w - z) \star_t f = (w - z)f - 2tf', \\ T_t((w - z)^m) &= \sum_{b=0}^{\lfloor m/2 \rfloor} \frac{(-t)^b m!}{b!(m - 2b)!} (w - z)^{m-2b} = (w - z)^{\star_t m}, \\ \mathfrak{m}_{t,z}^{\star_t m} &= T_t((w - z)^m \mathcal{E}). \end{aligned} \quad (\text{HTP17})$$

The first identity follows from HTP14–16 and $T_t(w - z) = w - z$. The derivative formula has only two terms in HTP10. The polynomial formula is the finite heat series HTP5. The algebra isomorphism transports powers and products of ideals, proving the last two assertions with no formal interpretation of the displayed polynomial required.

The exact jet map, including its inverse on representatives, is

$$\begin{aligned} J_{t,z,m}(f) &= \sum_{k=0}^{m-1} \frac{(D^k T_{-t} f)(z)}{k!} \epsilon^k \in \mathbb{C}[\epsilon]/(\epsilon^m), \\ \ker J_{t,z,m} &= \mathfrak{m}_{t,z}^{\star_t m}, \quad \mathcal{E}_t/\mathfrak{m}_{t,z}^{\star_t m} \xrightarrow{\sim} \mathbb{C}[\epsilon]/(\epsilon^m), \\ \sum_{k < m} c_k \epsilon^k &\mapsto \left[T_t \left(\sum_{k < m} c_k (w - z)^k \right) \right]. \end{aligned} \quad (\text{HTP18})$$

The ordinary Taylor map is multiplicative modulo ϵ^m by the finite Leibniz formula and preserves zero and unit. Its kernel consists precisely of functions divisible by $(w - z)^m$ in $\mathcal{E}$: apply HTP14 successively to the zero Taylor remainders. Its image contains every polynomial of degree less than m in ϵ . Compose with the isomorphism T_{-t} to prove all lines of HTP18. In particular the transported multiplicity of $T_t F$ at $\chi_{t,z}$ is the original order of F at z . This is an equality of the specified finite jets; it is not an equality with the ordinary Taylor jets of $T_t F$.

33.4 The actual theta family and its complete double integral

Keep the original source kernel, with no change in variables:

$$\begin{aligned}\Phi(u) &= \sum_{n \geq 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, \quad u \geq 0, \\ H_a(z) &= \int_0^\infty e^{au^2} \Phi(u) \cos(zu) du, \quad a \in \mathbb{C}, \\ 0 &< \pi(2\pi - 3)e^{9u - \pi e^{4u}} \leq \Phi(u) \leq C_\Phi e^{9u - \pi e^{4u}}, \\ C_\Phi &= 2\pi^2 \frac{1 + 11c + 11c^2 + c^3}{(1 - c)^5}, \quad c = e^{-3\pi}.\end{aligned}\tag{HTP19}$$

For completeness, every original summand is positive because its prefactor is $\pi n^2 e^{5u} (2\pi n^2 e^{4u} - 3)$. The $n = 1$ term gives the lower bound using $2\pi e^{4u} - 3 \geq (2\pi - 3)e^{4u}$. For the upper bound discard the negative term in each prefactor and use $n^2 - 1 \geq 3(n - 1)$ and $e^{4u} \geq 1$. The remaining sum is $\sum_{n \geq 1} n^4 c^{n-1} = (1 + 11c + 11c^2 + c^3)/(1 - c)^5$; four differentiations of the geometric series give this identity. This proves the entire stated bound directly.

The full moment control, retaining all constants, is

$$\begin{aligned}I_p(A, B) &:= \int_0^\infty u^p e^{Au^2 + Bu} \Phi(u) du \leq \mathcal{M}_p(A, B), \\ \mathcal{M}_p(A, B) &:= \frac{C_\Phi}{2\pi} \exp\left(\frac{3A^2}{32\pi} + \frac{(B + 9 + p)^2}{8\pi} - \frac{\pi}{2}\right), \quad A, B \geq 0, \quad p \geq 0, \\ p_\alpha(H_a) &\leq I_0(\max(\operatorname{Re} a + 1/(4\alpha), 0), 0), \quad H_a \in \mathcal{E}, \quad T_r H_a = H_{a+r}.\end{aligned}\tag{HTP20}$$

The moment proof uses $u^p \leq e^{pu}$ and splits the negative kernel exponent into quarters and a half. Namely,

$$Au^2 - \frac{\pi}{4} e^{4u} \leq \frac{3A^2}{32\pi}, \quad (B + 9 + p)u - \frac{\pi}{4} e^{4u} \leq \frac{(B + 9 + p)^2}{8\pi}.$$

The first follows from $e^{4u} \geq (32/3)u^4$ and completing the square in u^2 ; the second follows from $e^{4u} \geq 8u^2$ and completing the square in u . The remaining integral is at most $e^{-\pi/2}/(2\pi)$, because $e^{4u} \geq 1 + 4u$. This proves the first line. For the second, $|\cos(zu)| \leq e^{|z|u}$ and $|z|u \leq \alpha|z|^2 + u^2/(4\alpha)$ prove the growth bound, so HTP2 gives $H_a \in \mathcal{E}$. Differentiating the cosine under the integral is justified at every order by I_p on compact parameter sets. The sum of the absolute heat-integrands is bounded by $e^{(\operatorname{Re} a + |r|)u^2 + |\operatorname{Im} z|u} \Phi(u)$, with its quadratic coefficient enlarged to zero if negative. It is integrable by the first line. Expanding e^{ru^2} therefore proves $T_r H_a = H_{a+r}$ with the original heat sign; equality of entire functions identifies the same elements of $\mathcal{E}$. No assertion concerning their zero locations occurs in this argument.

For arbitrary complex a, b, t, z , the actual transported product is the following exact double integral:

$$(H_a \star_t H_b)(z) = \frac{1}{2} \int_0^\infty \int_0^\infty \Phi(u) \Phi(v) e^{au^2 + bv^2} \cdot [e^{2tuv} \cos(z(u+v)) + e^{-2tuv} \cos(z(u-v))] du dv.\tag{HTP21}$$

All integrals, derivative sums and their exchanges converge in every norm of $\mathcal{E}$, as we now verify explicitly. For $x \geq 0$, $\sum_{k \geq 0} x^k / \sqrt{k!} \leq \sqrt{2}e^{x^2}$: write each summand as $2^{-k/2}(\sqrt{2}x)^k / \sqrt{k!}$ and apply Cauchy–Schwarz. Thus both the sine and cosine of uz , for real u , have h -norm at most $\sqrt{2}e^{h^2 u^2}$. Since a derivative of order n changes only sine to cosine and supplies a factor u^n , HTP3 gives the majorant

$$\begin{aligned} A_h &= \max(\operatorname{Re} a + 2h^2 + |t|, 0), \quad B_h = \max(\operatorname{Re} b + 2h^2 + |t|, 0), \\ \int_0^\infty \int_0^\infty \Phi(u)\Phi(v)e^{(\operatorname{Re} a)u^2 + (\operatorname{Re} b)v^2} \sum_{n \geq 0} \frac{(2|t|)^n}{n!} \|D^n \cos(uz) D^n \cos(vz)\|_h du dv & \quad (\text{HTP22}) \\ &\leq 2I_0(A_h, 0)I_0(B_h, 0) \leq 2\mathcal{M}_0(A_h, 0)\mathcal{M}_0(B_h, 0) < \infty. \end{aligned}$$

Indeed the norm of the derivative product is at most $2(uv)^n e^{2h^2(u^2+v^2)}$. Its weighted sum is $2e^{2|t|uv+2h^2(u^2+v^2)}$, and $2|t|uv \leq |t|(u^2+v^2)$ separates the variables. For the integrals of the individual derivatives use the same bounds with the finite extra moments in HTP20. Nonnegative summation of the absolute coefficients now justifies integrating every derivative, multiplying the two integrals and interchanging the derivative series with the double integral in each h -norm. The resulting coefficient integrals give a member of every weighted ℓ^1 space, hence the same member of $\mathcal{E}$.

For the value of that sum, the even derivative orders give $\cos(zu)\cos(zv)\cosh(2tuv)$, and the odd orders give $-\sin(zu)\sin(zv)\sinh(2tuv)$. The elementary cosine sum and difference identities transform their sum into the bracket in HTP21 divided by 2. This proves that formula with its signs. Alternatively its two cosine frequencies are eigenvectors of T_t with respective multipliers $e^{t(u+v)^2}$ and $e^{t(u-v)^2}$; the already proved absolute bounds justify the same calculation. On compact sets of the complex parameters, HTP22 supplies one common bound by enlarging their real parts and moduli. Additional parameter derivatives contribute finite powers of u, v and are bounded by the corresponding I_p products.

The requested actual-time specialization and its transport relation are therefore

$$\begin{aligned} H_t \star_t H_s &= T_t(H_0 H_{s-t}), \quad H_t \star_t H_t = T_t(H_0^2), \\ (H_t \star_t H_s)(z) &= \frac{1}{2} \int_0^\infty \int_0^\infty \Phi(u)\Phi(v)e^{tu^2+sv^2} [e^{2tuv} \cos(z(u+v)) + e^{-2tuv} \cos(z(u-v))] du dv. \end{aligned} \quad (\text{HTP23})$$

Use HTP20 in the definition HTP8, or set $a = t, b = s$ in HTP21. For real a, b, t , the value of HTP21 at $z = 0$ is strictly positive: its bracket is $2\cosh(2tuv) > 0$ and the kernel factors are positive. That statement is about this value of the actual integral; it supplies no assertion that the entire function has no zeros at other points.

Finally the character and divisor comparison for the actual family is exact:

$$\begin{aligned} \chi_{t,z}(H_a) &= H_{a-t}(z), \quad \chi_{t,z}(H_t) = H_0(z), \quad H_t(z) = \chi_{t,z}(H_{2t}), \\ T_t(H_0\mathcal{E}) &= H_t \star_t \mathcal{E}, \quad \mathcal{E}/(H_0\mathcal{E}) \simeq \mathcal{E}_t/(H_t \star_t \mathcal{E}), \\ J_{t,z,m}(H_t) &= \sum_{k=0}^{m-1} \frac{H_0^{(k)}(z)}{k!} \epsilon^k, \quad J_{t,z,m}(H_{2t}) = \sum_{k=0}^{m-1} \frac{H_t^{(k)}(z)}{k!} \epsilon^k. \end{aligned} \quad (\text{HTP24})$$

The first line is HTP15 and HTP20. Apply HTP16 with $F = H_0$ for the second, and HTP18 for the third. The first transported principal divisor thus has the original H_0 character zeros and original multiplicities. The ordinary analytic zeros and multiplicities of H_t are instead those of H_{2t} tested by these transported characters and jets. This supplies the exact comparison between both objects; ordinary evaluation cannot be substituted for a transported character when $t \neq 0$ by HTP13. None of these isomorphisms decides the zero-location question for H_0 or asserts a conclusion about RH.

$$\begin{array}{ccc}
(\mathcal{E}, \cdot) & \xrightarrow{T_t} & (\mathcal{E}, \star_t) \\
\text{ev}_z \downarrow & & \downarrow \chi_{t,z} = \text{ev}_z T_{-t} \\
\mathbb{C} & \xrightarrow{\text{id}} & \mathbb{C}
\end{array}$$

Figure 12: The exact algebra and character comparison, HTP12–18. The top map sends H_0 to H_t , its principal ideal to the corresponding star ideal, and each finite jet to HTP18. Thus the right character evaluates H_t as $H_0(z)$. The original analytic value $H_t(z)$ instead equals $\chi_{t,z}(H_{2t})$, HTP24. The diagram uses the actual Rodgers–Tao kernel HTP19 and the full proved heat action; it does not identify ordinary evaluation with the right character when $t \neq 0$.

34 The full support heat algebra and its exact global-function bridge

Sources and retained objects. The underlying $G_L(R)$ is the existing full-lattice programme construction, retained under its foundation byline The Clankers in *Split Support Geometry: Universal Zero Fibres, Arithmetic Curves, Frobenius and Perfect Quantization*, v11, Definition `def:lattice-split`. Its supported-zero localization has the public proof 031, Theorems `thm:rebuilt-full-support` and `thm:e-stalk`. The present full support calculation uses HSL1–35 of `FULL_LATTICE_HEAT_OPERATOR.tex` for the independently labelled heat operators, their analytic completion and their exact coefficient map. The analytic transported-product calculation is HTP3–10 of `HEAT_TRANSPORTED_PRODUCT.tex`; its coefficients and convergence bound are stated at their use below. The HTP source actually read here is lines 1–256, including the complete proofs HTP1–12, at SHA–256

b81c4489625eed9420eb202009be7b6185cc060bd83f942053b3088d80d91474.

The actual theta specialization uses THS8–15 and THS18–23 of `THETA_HEAT_FULL_SUPPORT.tex`, whose human-source kernel is Brad Rodgers and Terence Tao, *The De Bruijn–Newman constant is non-negative*, arXiv:1801.05914v5. No zero-location theorem is a premise of the following algebra.

Let L be the entire nontrivial bounded distributive lattice. Retain

$$S = G_L(\mathbb{C}), \quad (a, \lambda) + (b, \mu) = (a + b, \lambda \vee \mu), \quad (a, \lambda)(b, \mu) = (ab, \lambda \wedge \mu).$$

Thus $a \neq 0$ forces $\lambda = 1_L$, $\tau = (0, 0_L)$ is the additive zero, $e = (0, 1_L)$ is supported zero, and $(1, 1_L)$ is the unit. The original Taylor coefficients are denoted a_n :

$$\mathcal{E} = \left\{ f(z) = \sum_{n \geq 0} a_n z^n : \|f\|_h := \sum_{n \geq 0} |a_n| \sqrt{n!} h^n < \infty \text{ for every } h > 0 \right\}. \quad (\text{SHP1})$$

This is the complete entire-function space proved in HSL15–18; the coefficients themselves have not been changed. Multiplication of entire functions preserves it. In fact

$$\|fg\|_h \leq \|f\|_{\sqrt{2}h} \|g\|_{\sqrt{2}h}. \quad (\text{SHP2})$$

For coefficients a_r, b_s , use the absolutely convergent positive sum and $\sqrt{(r+s)!} \leq 2^{(r+s)/2} \sqrt{r!s!}$, which follows from $\binom{r+s}{r} \leq 2^{r+s}$. Summing proves (SHP2). Thus $\mathcal{E}$ is a commutative unital complex algebra with its original pointwise function product.

34.1 Parity-tail closure is an exact multiplicative nucleus

We first prove the support identity in a setting where every join written in the proof exists. Put $\widehat{L} = \text{Idl}(L)$, with principal embedding $j(\lambda) = \downarrow \lambda$, ideal intersection as meet, and ideal generation from unions as arbitrary join. Finite meets distribute over arbitrary joins: if $c \in I \cap \bigvee_{\alpha} J_{\alpha}$, write $c \leq b_1 \vee \cdots \vee b_r$ with each b_k in an indicated J_{α_k} , and then $c = \bigvee_k (c \wedge b_k)$ is in $\bigvee_{\alpha} (I \cap J_{\alpha})$. This proves precisely the distributivity used below.

For sequences ν, μ in this complete lattice define

$$\begin{aligned} (\nu * \mu)_n &= \bigvee_{r+s=n} \nu_r \wedge \mu_s, & (P\nu)_i &= \bigvee_{\substack{j \geq i \\ j-i \text{ even}}} \nu_j, \\ \nu + \mu &= (\nu_i \vee \mu_i)_{i \geq 0}, & \mathbf{u} &= (1, 0, 0, \dots). \end{aligned} \quad (\text{SHP3})$$

Here the symbols 1, 0 mean the top and bottom of the chosen lattice. Convolution is associative: both triple products at degree n are the join of $\nu_r \wedge \mu_s \wedge \omega_k$ over $r + s + k = n$. It has unit $\mathbf{u}$,

is commutative, and distributes over the coordinate joins, by finite distributivity. The operator P is extensive, idempotent and preserves arbitrary joins. Nested same-parity tails prove these assertions directly.

The needed multiplicative identity is

$$(P\nu) * (P\mu) \leq P(\nu * \mu), \quad P((P\nu) * (P\mu)) = P(\nu * \mu). \quad (\text{SHP4})$$

To prove the first inequality, distribute the row of its left side. A term at degree n has indices $r + s = n$, $a \geq r$, $b \geq s$, with $a - r$ and $b - s$ even. The term is $\nu_a \wedge \mu_b$. Then $a + b \geq n$ and $a + b - n$ is even, so that term occurs in $P(\nu * \mu)_n$. This proves the first inequality after joining all terms. Applying P gives one inequality in the second formula; the reverse follows from $\nu \leq P\nu$ and $\mu \leq P\mu$. Thus P is a multiplicative nucleus in the explicit sense of a join-preserving closure satisfying (SHP4).

For already closed sequences $\nu_i \geq \nu_{i+2}$ and $\mu_i \geq \mu_{i+2}$, its closed product is completely finite:

$$(\nu \otimes \mu)_0 = (\nu_0 \wedge \mu_0) \vee (\nu_1 \wedge \mu_1), \quad (\nu \otimes \mu)_n = \bigvee_{r=0}^n \nu_r \wedge \mu_{n-r} \quad (n \geq 1). \quad (\text{SHP5})$$

It equals $P(\nu * \mu)$. Indeed, if $n \geq 1$, every term of $(\nu * \mu)_{n+2}$ has indices $r + s = n + 2$, and at least one of r, s is at least 2. Lowering that index by 2 weakly increases its label and supplies a term of $(\nu * \mu)_n$. Therefore $(\nu * \mu)_{n+2} \leq (\nu * \mu)_n$ for $n \geq 1$. At $n = 0$, the terms of degree 2 with indices $(2, 0)$ and $(0, 2)$ are at most $\nu_0 \wedge \mu_0$. The only remaining term is $\nu_1 \wedge \mu_1$. Higher even degrees are already bounded by degree 2. This proves (SHP5), including the one additional mixed-parity contribution at degree zero. The formulas lie in the original L when the input labels do.

Let $\mathcal{F}(L)$ be the set of all such parity flags in $L^{\mathbb{N}_0}$. Coordinate join, the zero flag and $\otimes$ make it a commutative unital idempotent semiring with unit $\mathbf{u}$. For associativity, view it in the complete ideal lattice and use

$$P(P(\nu * \mu) * \omega) = P((\nu * \mu) * \omega) = P(\nu * (\mu * \omega)) = P(\nu * P(\mu * \omega)),$$

where (SHP4) applies because $P\omega = \omega$ and $P\nu = \nu$. Commutativity follows in the same way. The unit follows from $P\mathbf{u} = \mathbf{u}$. Distributivity follows from join preservation of P and distributivity of convolution. The principal embedding is injective, so these equations prove the claims in the original L as well. No infinite join in L has been assumed.

34.2 The original independent-coefficient algebra and its flag quotient

Let $\mathcal{D}_L$ consist of pairs (f, ν) with $f = \sum a_n z^n \in \mathcal{E}$, $\nu_n \in L$, $a_n \neq 0 \Rightarrow \nu_n = 1_L$, and every inclusive parity-tail ideal principal:

$$\bigvee_{\substack{j \geq i \\ j-i \text{ even}}} \downarrow \nu_j = \downarrow s_i \quad \text{for each } i. \quad (\text{SHP6})$$

This is the exact original-lattice domain of HSL25. The generator s_i is a finite join of labels from that tail, because it belongs to the generated ideal; conversely such a finite dominating join generates the tail ideal. Consequently $P\nu = (s_i)$ is well defined in L and is a flag. The labels ν_i in $\mathcal{D}_L$ are otherwise independent.

Addition and ordinary coefficient convolution make $\mathcal{D}_L$ a commutative unital S -algebra. For closure under convolution, apply (SHP4) in $\hat{L}$: the inclusive tail ideals of $\nu * \mu$ have the explicit principal generators

$$g_0 = (s_0 \wedge t_0) \vee (s_1 \wedge t_1), \quad g_n = \bigvee_{r=0}^n s_r \wedge t_{n-r} \quad (n \geq 1), \quad (\text{SHP7})$$

where $(s_i) = P\nu$ and $(t_i) = P\mu$. This follows because the right side of (SHP4) is the closed product of those principal tails, calculated by (SHP5). It proves precisely the required principality. For addition, tail ideals join and are generated by $s_i \vee t_i$. Multiplication of all coefficient labels by a scalar label λ gives tail generators $\lambda \wedge s_i$. The amplitudes stay in $\mathcal{E}$ by (SHP2); if a product coefficient is nonzero, one of its finitely many contributing amplitude products is nonzero, forcing the corresponding support meet and hence the output label to be 1_L . Addition and scalar multiplication satisfy the same carrier test. These arguments prove closure, and the algebra laws follow from the coefficient operations.

Define the full flag algebra and its product by

$$\begin{aligned} \mathcal{A}_L &= \{(f, \nu) : f = \sum a_n z^n \in \mathcal{E}, \nu \in \mathcal{F}(L), a_n \neq 0 \Rightarrow \nu_n = 1_L\}, \\ \mathcal{P} : \mathcal{D}_L &\longrightarrow \mathcal{A}_L, \quad \mathcal{P}(f, \nu) = (f, P\nu), \\ (f, \nu) \circ (g, \mu) &= (fg, \nu \circledast \mu). \end{aligned} \tag{SHP8}$$

A flag has principal tail ideal generated by its first label in that tail, so $\mathcal{A}_L$ is exactly the image of $\mathcal{P}$ and is contained in $\mathcal{D}_L$. The formula for $\circ$ is exactly $\mathcal{P}((f, \nu)(g, \mu))$. Its carrier condition follows either from that construction or from the nonzero-term argument just used.

The map $\mathcal{P}$ is an onto unital S -algebra homomorphism from ordinary coefficient convolution in $\mathcal{D}_L$ to $\circ$ in $\mathcal{A}_L$. Addition and the unit are preserved directly. For multiplication its exact identity is

$$\mathcal{P}(vw) = \mathcal{P}(v) \circ \mathcal{P}(w), \quad v \sim w \iff f_v = f_w \text{ and } P\nu^v = P\nu^w. \tag{SHP9}$$

This is (SHP4) with the amplitude product unchanged. Thus the displayed relation is a semiring congruence, and $\mathcal{D}_L/\sim$ is exactly $\mathcal{A}_L$. The algebra laws of $\circ$ follow either from the proved support semiring laws and the amplitude laws, or from this surjective homomorphism. The unit has amplitude 1 and support $\mathbf{u}$; the zero has amplitude 0 and the zero flag. The scalar $(a, \lambda) \in S$ embeds as its constant coefficient at degree 0 and τ at all higher degrees. Formula (SHP5) proves this is unital and acts by the original scalar multiplication.

For the original monomial vector Z representing z , the exact product is

$$Z \circ Z : \quad a_2 = 1, \quad a_i = 0 \ (i \neq 2), \quad \nu_0 = \nu_2 = 1_L, \quad \nu_i = 0_L \ (i \neq 0, 2). \tag{SHP10}$$

Its constant coefficient is the supported zero e , not the additive zero τ . More generally $Z^{\circ n} = \mathcal{P}(z^n)$ has support 1_L exactly at $i \leq n$ of the same parity as n . This follows from (SHP9) and the exact parity-tail of the single monomial label. It exhibits the mixed even/odd contribution explicitly rather than dropping it from multiplication.

34.3 The transported heat product with every label retained

On $\mathcal{A}_L$, HSL10 and HSL18 give the actual heat group

$$U_t(f, \nu) = (T_t f, \nu), \quad T_t = e^{-t\partial_z^2}, \quad U_t^{-1} = U_{-t}.$$

The identity here is the restricted supported-zero heat operator U_e ; it is the identity of this flag space. Both the algebra unit and every embedded scalar are fixed. Define the transported product

$$v \star_t w = U_t(U_{-t}v \circ U_{-t}w) = (T_t((T_{-t}f)(T_{-t}g)), \nu \circledast \mu). \tag{SHP11}$$

This is well defined by (SHP2) and the heat bounds in HSL17. For example those estimates give the explicit bound

$$\|T_t((T_{-t}f)(T_{-t}g))\|_h \leq \|f\|_{c(h,t)} \|g\|_{c(h,t)}, \quad c(h,t) = \sqrt{2}(h + \sqrt{2|t|}) + \sqrt{2|t|}.$$

In particular the construction is on the complete analytic coefficient space, not on an assumed commuting degree truncation.

The amplitude product is exactly the convergent Wick series

$$f \star_t^{\text{amp}} g = \sum_{n=0}^{\infty} \frac{(-2t)^n}{n!} f^{(n)} g^{(n)}. \quad (\text{SHP12})$$

The complete analytic proof is HTP7–10 of `HEAT_TRANSPORTED_PRODUCT.tex`. Its literal bound, in the same original-coefficient seminorms, is

$$\sum_{n \geq 0} \frac{(2|t|)^n}{n!} \|f^{(n)} g^{(n)}\|_h \leq \frac{\|f\|_{\sqrt{2h^2+a^2}} \|g\|_{\sqrt{2h^2+b^2}}}{1 - 2|t|/(ab)} \quad (a, b > 0, \quad ab > 2|t|). \quad (\text{SHP13})$$

For clarity, the finite-polynomial identity behind this formula retains the exact sign and factor 2. On $f(x)g(y)$ write $A = \partial_x$ and $B = \partial_y$. Evaluation at $x = y = z$ turns ∂_z into $A + B$. The commuting exponentials therefore combine as $e^{-t(A+B)^2} e^{tA^2} e^{tB^2} = e^{-2tAB}$; on polynomials all relevant series terminate. The bound (SHP13), density of polynomials and continuity of the transported product extend that identity to all of $\mathcal{E}$ with absolute convergence. No support label is obtained by discarding a zero amplitude term in this series: its support in the actual lifted algebra is exactly the product already proved in (SHP5) and (SHP11).

The algebra $(\mathcal{A}_L, \star_t)$ is commutative, associative, distributive and unital with the same unit and embedded scalars as $(\mathcal{A}_L, \circ)$. These assertions follow by the invertible transport definition, using the algebra laws already proved for $\circ$ and the linearity of U_t . The exact time-changing algebra isomorphisms are

$$U_t : (\mathcal{A}_L, \circ) \xrightarrow{\sim} (\mathcal{A}_L, \star_t), \quad U_s : (\mathcal{A}_L, \star_t) \xrightarrow{\sim} (\mathcal{A}_L, \star_{t+s}). \quad (\text{SHP14})$$

For the second identity, insert (SHP11) on both sides and use $U_{-(t+s)} U_s = U_{-t}$. These maps preserve every original coefficient label.

34.4 The full coefficient map is a faithful algebra morphism

Let K be the tropical semifield with zero, $K_L = K \otimes_{\mathbb{B}} L$, and $\iota_L(\lambda) = 1_K \otimes \lambda$. The semiring map $\rho_L = \beta_K \otimes \text{id}_L$ with $\beta_K(a) = 1$ for $a \neq 0_K$ and $\beta_K(0_K) = 0$ satisfies $\rho_L \iota_L = \text{id}_L$. Here a tropical sum is zero exactly when both entries are zero, and a tropical product is zero exactly when a factor is zero; this proves that β_K is a unital semiring homomorphism.

For any commutative additively idempotent semiring A , define $\mathcal{F}(A)$ to be its parity-decreasing sequences for the natural order, with the product in (SHP5) written using addition and multiplication in A . It is a commutative unital semiring. One precise proof for possibly incomplete A is to embed it into its ordered-ideal completion, whose joins and products are the ideals generated by unions and pairwise products. Multiplication in that completion distributes over arbitrary joins by the generator description. The index proof of (SHP4) uses only that distributivity and monotonicity, so applies there. For already decreasing sequences, the proof of (SHP5) uses only finite sums and monotonicity and returns entries in the original A . The principal-ideal embedding is injective, so the proved algebra laws descend to $\mathcal{F}(A)$. The unit is $(1_A, 0_A, 0_A, \dots)$, not the constant sequence of units.

Coordinatewise application of ι_L and ρ_L gives unital semiring maps

$$\mathcal{F}(L) \xrightarrow{\mathcal{F}(\iota_L)} \mathcal{F}(K_L) \xrightarrow{\mathcal{F}(\rho_L)} \mathcal{F}(L), \quad \mathcal{F}(\rho_L) \mathcal{F}(\iota_L) = \text{id}. \quad (\text{SHP15})$$

This follows immediately from the finite formula (SHP5); order preservation ensures that flags remain flags. It proves the full-lattice faithfulness of the support map, including all mixed terms in degree zero.

The complete coefficient map is

$$\Phi : \mathcal{A}_L \longrightarrow \mathcal{E} \times \mathcal{F}(K_L), \quad (f, \nu) \longmapsto (f, (\iota_L(\nu_i))_{i \geq 0}). \quad (\text{SHP16})$$

It is an injective unital semiring homomorphism for $\circ$ and ordinary amplitude multiplication. Both coordinates preserve the operations by (SHP5) and (SHP15), and equality of their values determines f and each ν_i . Its image is exactly those (f, w) for which every w_i belongs to $\iota_L(L)$ and $a_i \neq 0$ implies $w_i = 1_{K_L}$. This is the coefficientwise original map $(a_i, \nu_i) \mapsto (a_i, 1_K \otimes \nu_i)$ with its two coordinates organized into the entire function and the complete support flag. For $\star_t$, the same map is a unital homomorphism to $(\mathcal{E}, \star_t^{\text{amp}}) \times \mathcal{F}(K_L)$. Its heat intertwiner is exactly $\Phi U_s = (T_s, \text{id})\Phi$, with the products transported according to (SHP14).

34.5 Synchronization to the global function carrier is a localization

The independently labelled algebra is not identified with one globally labelled function. Its exact morphism to that object is

$$g : \mathcal{A}_L \longrightarrow G_L(\mathcal{E}), \quad g(f, \nu) = (f, \nu_0 \vee \nu_1). \quad (\text{SHP17})$$

This belongs to the stated carrier: if $f \neq 0$, some $a_n \neq 0$ forces $\nu_n = 1_L$, and its parity flag forces $\nu_0 = 1_L$ or $\nu_1 = 1_L$. The map is unital and additive. For multiplication (SHP5) gives the exact identity

$$(\nu \circledast \mu)_0 \vee (\nu \circledast \mu)_1 = (\nu_0 \vee \nu_1) \wedge (\mu_0 \vee \mu_1). \quad (\text{SHP18})$$

Indeed the left side is the join of the four meets $\nu_i \wedge \mu_j$ for $i, j \in \{0, 1\}$, which is the distributed right side. This proves the multiplicative assertion, not merely equality of underlying amplitudes.

There is an explicit zero-preserving, additive and multiplicative section

$$j : G_L(\mathcal{E}) \longrightarrow \mathcal{A}_L, \quad j(f, \lambda) = (f, (\lambda, \lambda, \lambda, \dots)), \quad jg = \text{id}. \quad (\text{SHP19})$$

If $f \neq 0$ its label is top, so the carrier condition holds at every coefficient; if $f = 0$ it imposes no restriction. The labels form a flag. Addition is immediate, and (SHP5) shows that the product of two constant flags is their constant meet. This proves the three preserved operations and the section identity.

The section's unit is the following particular idempotent:

$$J = j(1, 1_L) = (1, (1_L, 1_L, 1_L, \dots)), \quad J \circ J = J, \quad J \circ v = jg(v). \quad (\text{SHP20})$$

It is distinct from the unit of $\mathcal{A}_L$, whose support is $(1_L, 0_L, 0_L, \dots)$. For the last identity the amplitude is still f . Its degree-zero support is $\nu_0 \vee \nu_1$. For every $n \geq 1$, the support is $\bigvee_{i=0}^n \nu_i = \nu_0 \vee \nu_1$, because the flags decrease separately by parity. This proves the entire coefficient equality. Although J has the constant polynomial amplitude 1, its support has infinitely many nonzero labels. It belongs to the completed original-lattice flag domain and is not a finite-support coefficient vector. In particular J is not a supported-zero element: its amplitude is the unit 1.

For a commutative semiring A and idempotent c , $A[c^{-1}] \simeq cA$ with internal unit c . Indeed an invertible idempotent in a receiving semiring equals its unit, so every map inverting c factors uniquely through $a \mapsto ca$; equality there is exactly $ca = cb$. Applying this proved universal property to (SHP20) yields

$$\mathcal{A}_L[J^{-1}] \simeq J\mathcal{A}_L \xrightarrow[\sim]{g} G_L(\mathcal{E}), \quad g^{-1} = j. \quad (\text{SHP21})$$

The map j is unital into the corner whose unit is J , not into all of $\mathcal{A}_L$. The localization congruence and its complete fibres are

$$(f, \nu) \sim_J (h, \mu) \iff f = h, \quad \nu_0 \vee \nu_1 = \mu_0 \vee \mu_1, \quad (\text{SHP22})$$

$$g^{-1}(f, \lambda) = \{(f, \nu) : \nu_i \geq \nu_{i+2}, \quad \nu_0 \vee \nu_1 = \lambda, \quad a_i \neq 0 \Rightarrow \nu_i = 1_L\}.$$

Necessity is the definition of g . Sufficiency and the complete equality congruence follow from $Jv = jg(v)$ and injectivity of j . Every displayed target has a preimage (SHP19). This specifies

exactly the mixed-support information that synchronization loses while retaining the entire analytic amplitude.

There is a different idempotent that performs amplitude erasure:

$$e_0 = (0, (1_L, 0_L, 0_L, \dots)), \quad e_0 \circ (f, \nu) = (0, \nu), \quad e_0^2 = e_0. \quad (\text{SHP23})$$

It is precisely the embedded constant scalar $e \in S$. The support equality follows from (SHP5), since the only nonzero label of e_0 is in degree zero. Thus

$$\begin{array}{ccc} \mathcal{A}_L[e_0^{-1}] \simeq \mathcal{F}(L), & \mathcal{A}_L[(e_0 \circ J)^{-1}] \simeq L, & \\ \mathcal{A}_L & \xrightarrow{g} & G_L(\mathcal{E}) \\ (f, \nu) \mapsto \nu \downarrow & & \downarrow (f, \lambda) \mapsto \lambda \\ \mathcal{F}(L) & \xrightarrow{\nu \mapsto \nu_0 \vee \nu_1} & L. \end{array} \quad (\text{SHP24})$$

The first isomorphism identifies the corner $e_0 \mathcal{A}_L$ with its full flag and its internal unit with $\mathbf{u}$. The product $e_0 \circ J$ has amplitude zero and every coefficient label top. Its corner consists exactly of zero amplitudes with constant flags, which is L . All maps in the square are unital for their own displayed target identities and are multiplicative by (SHP18). The two routes agree at every element. Localizing successively at the two commuting idempotents gives the same corner as localizing at their product: making their product invertible makes both factors invertible, and the converse is immediate. This proves the complete commuting localization description, including the distinction between amplitude erasure and support synchronization.

34.6 The joint coefficient square and its exact localization

For any commutative idempotent semiring A set $\epsilon_A : \mathcal{F}(A) \rightarrow A$ by $\epsilon_A(w) = w_0 + w_1$. The proof of (SHP18), using addition and multiplication in A , proves that this is a unital semiring map. Its constant-flag section has internal unit $\mathbf{J}_A = (1_A, 1_A, \dots)$, and multiplication by $\mathbf{J}_A$ sends every flag to the constant flag of its ϵ_A value. The argument uses only the parity inequalities, so it applies to $A = K_L$.

The full original coefficient map consequently has the exact commuting square

$$\begin{array}{ccc} \mathcal{A}_L & \xrightarrow{\Phi} & \mathcal{E} \times \mathcal{F}(K_L) \\ g \downarrow & & \downarrow \text{id}_{\mathcal{E}} \times \epsilon_{K_L} \\ G_L(\mathcal{E}) & \xrightarrow{\Phi_{\mathcal{E}}} & \mathcal{E} \times K_L, \end{array} \quad \Phi_{\mathcal{E}}(f, \lambda) = (f, 1_K \otimes \lambda). \quad (\text{SHP25})$$

Both composites give $(f, \iota_L(\nu_0 \vee \nu_1))$. The right vertical map is localization at $\Phi(J) = (1, \mathbf{J}_{K_L})$: its corner consists of all amplitudes together with a constant K_L -flag, with identity $(1, \mathbf{J}_{K_L})$. The left vertical map is localization at J by (SHP21). Hence (SHP25) is also a pushout of unital semirings. Directly, a map from its top-right object agreeing with a map from the lower-left object must invert $\Phi(J)$, and therefore factors uniquely through the corner just calculated. The converse is immediate. This proves the universal property and every connecting map of the square.

For completely general ideal-labelled flags the same construction retains a precise completed joint map. Replace L by $\widehat{L}$ and K_L by $\mathfrak{K}_L = \text{Idl}_{\text{ord}}(K_L)$. The map on labels is the explicit HSL33 morphism

$$\iota^{\#}(I) = \{w \in K_L : \exists a \in I, w \leq \iota_L(a)\}.$$

It preserves arbitrary joins, products and the unit, and its retraction is the lower ideal generated by $\rho_L(w)$ for w in the input ideal. The composition is the identity, because $\rho_L \iota_L = \text{id}_L$ and both maps are monotone. Applying these maps to flags preserves (SHP5), so it gives the injective completed version of (SHP16) and the complete square (SHP25). For each original label λ this map is $\downarrow \lambda \mapsto \downarrow \iota_L(\lambda)$. Thus the completed square restricts to the original one via explicit principal-ideal embeddings, with no identification of L with a Boolean lattice.

34.7 Transported synchronization and the actual theta fibre

Let $\mathcal{E}_t$ denote the ring whose underlying vector space is $\mathcal{E}$ and whose multiplication is (SHP12). The unchanged set map g is a unital semiring homomorphism $(\mathcal{A}_L, \star_t) \rightarrow G_L(\mathcal{E}_t)$. For a map into the ordinary-product global carrier, the exact morphism instead is

$$\begin{aligned} \gamma_t : (\mathcal{A}_L, \star_t) &\longrightarrow G_L(\mathcal{E}), & \gamma_t(f, \nu) &= (T_{-t}f, \nu_0 \vee \nu_1), \\ j_t(f, \lambda) &= (T_t f, (\lambda, \lambda, \dots)), & \gamma_t j_t &= \text{id}. \end{aligned} \quad (\text{SHP26})$$

The amplitude assertion follows from the transport definition, which makes $T_{-t} : \mathcal{E}_t \rightarrow \mathcal{E}$ a unital ring isomorphism. The support assertion is again (SHP18). Nonzero functions remain nonzero under T_t , so all displayed carriers are respected.

The element J is fixed by every heat map because $T_t 1 = 1$ and labels are unchanged. It satisfies

$$J \star_t v = jg(v) = j_t \gamma_t(v), \quad (\mathcal{A}_L, \star_t)[J^{-1}] \simeq G_L(\mathcal{E}_t) \xrightarrow[T_{-t}]{\sim} G_L(\mathcal{E}). \quad (\text{SHP27})$$

For the first equality apply the transport definition to $J \circ U_{-t}v = jg(U_{-t}v)$ and then apply U_t . Its amplitude returns to f and its label is constant $\nu_0 \vee \nu_1$, proving the equality coefficient by coefficient. The same idempotent-corner argument therefore proves the localization statement. In particular changing the product without changing the amplitude map has not been silently treated as a map to the ordinary function ring.

Finally consider the actual theta heat family H_s . THS9 proves that every even Taylor coefficient of H_s is nonzero at each real s , while THS8 gives all odd coefficients equal to zero. Consequently the complete fibre of (SHP17) over $(H_s, 1_L)$ at real time is

$$g^{-1}(H_s, 1_L) = \{(H_s, \nu) : \nu_{2j} = 1_L \text{ for every } j, \quad \nu_1 \geq \nu_3 \geq \nu_5 \geq \dots \text{ is any flag in } L\}. \quad (\text{SHP28})$$

Nonvanishing forces each even label to be top. The odd amplitudes impose no further carrier condition, and the flag requirement is exactly the displayed descending odd sequence. Conversely every such sequence is admissible and has global label top. This proves both inclusions and thus the full fibre. The cosine-series lift used in THS18 and HSL31 selects the zero odd flag; it is one specified member of this complete mixed-support fibre. The globally constant-label section $j(H_s, 1_L)$ instead selects the top odd flag. Their amplitudes agree, their full support objects are distinct, and their exact connecting localization is (SHP21), not an unproved identification.

All results above concern the constructed full support algebras, their heat products and their exact morphisms. They keep the actual analytic amplitude throughout the synchronization map and identify exactly where amplitude information is discarded by the different supported-zero localization in (SHP24).

35 Every tensor coupling in the full support heat product

Sources and retained objects. The coefficient space, heat group and complete lattice maps are HSL15–35 of the included full-lattice heat-operator proof. The complete transported-product proof HTP1–24 establishes the binary product, all its complex characters and the actual theta integral. The full support product proof SHP3–28 establishes the parity-flag algebra and its synchronization maps. The derivation below proves the complete finite tensor powers, their convergent coupling series and their full support formulas. No comparison with an unspecified tensor category is a premise.

The original kernel is Brad Rodgers and Terence Tao’s *The De Bruijn–Newman constant is non-negative*, author source `fmp-template.tex`, definitions `phidef` and `htdef`. The exact source identity and the bounded author-source reading are recorded in THS1–5 of the included actual theta heat proof; the kernel and its constants are reproduced below. The full-lattice carrier is the foundation retained under the byline The Clankers, *Split Support Geometry: Universal Zero Fibres, Arithmetic Curves, Frobenius and Perfect Quantization*, v11, Definition `def:lattice-split`. Its full support localization is proved in 031, Theorems `thm:rebuilt-full-support` and `thm:e-stalk`. No zero-location theorem from any of these sources is used.

35.1 The actual completed tensor space and its diagonal map

Fix an integer $k \geq 2$. Retain the original Taylor coefficients and norms

$$\mathcal{E} = \left\{ f(z) = \sum_{n \geq 0} a_n z^n : \|f\|_h = \sum_{n \geq 0} |a_n| \sqrt{n!} h^n < \infty \text{ for every } h > 0 \right\}. \quad (\text{HTK1})$$

For a vector $\mathbf{h} = (h_1, \dots, h_k)$ of positive radii, define the multivariate coefficient space by

$$\begin{aligned} F(\mathbf{z}) &= \sum_{\mathbf{n} \in \mathbb{N}_0^k} a_{\mathbf{n}} \mathbf{z}^{\mathbf{n}}, \\ q_{\mathbf{h}}(F) &= \sum_{\mathbf{n}} |a_{\mathbf{n}}| \prod_{i=1}^k \sqrt{n_i!} h_i^{n_i}, \\ \mathcal{E}^{[k]} &= \{F : q_{\mathbf{h}}(F) < \infty \text{ for every } \mathbf{h} > 0\}. \end{aligned} \quad (\text{HTK2})$$

Common radii $h_1 = \dots = h_k = h \in \mathbb{N}_{>0}$ suffice, since every finite list of radii is bounded by a common integer. Coordinate limits and completeness of each weighted ℓ^1 space prove completeness exactly as for HTK1. Rectangular polynomial truncations converge in every displayed norm. Moreover

$$|F(\mathbf{z})| \leq q_{\mathbf{h}}(F) \exp \left(\sum_i \frac{|z_i|^2}{2h_i^2} \right),$$

because $|z_i|^{n_i} / (\sqrt{n_i!} h_i^{n_i})$ is at most $\exp(|z_i|^2 / (2h_i^2))$. Thus these are actual entire functions, with their original separate coordinates.

This space is the completed projective tensor product of the specified coefficient spaces. Here is a direct proof rather than an assumed identification. On the algebraic tensor product define

$$\pi_{\mathbf{h}}(u) = \inf_{u = \sum_{r=1}^m f_{r,1} \otimes \dots \otimes f_{r,k}} \sum_{r=1}^m \prod_i \|f_{r,i}\|_{h_i}.$$

The coefficient map $f_1 \otimes \dots \otimes f_k \mapsto \prod_i f_i(z_i)$ satisfies $q_{\mathbf{h}} \leq \pi_{\mathbf{h}}$, by expanding each finite tensor presentation and taking absolute values. It is injective. To see this, restrict each tensor factor to the finite-dimensional span occurring in a presentation. Its coordinate coefficient functionals

separate that span; finitely many of them therefore span its dual. Products of those functionals separate the resulting finite-dimensional tensor product. If all multivariate coefficients vanish, every such functional vanishes and hence the original tensor is zero.

For a polynomial tensor the monomial basis presentation gives the reverse inequality $\pi_{\mathbf{h}} \leq q_{\mathbf{h}}$. For a general algebraic tensor u , truncate each factor at degree N . The finite tensor presentation and a telescoping difference give $\pi_{\mathbf{h}}(u - u_N) \rightarrow 0$, because the factors converge in their defining norms and their truncated norms are bounded. Consequently $q_{\mathbf{h}}(u - u_N) \rightarrow 0$ as well. Equality of the two norms on u_N gives equality on u . Polynomial tensors are dense in HTK2, so completion gives the precise isomorphism

$$\mathcal{E}^{\widehat{\otimes}_{\pi}} \cdots \widehat{\otimes}_{\pi} \mathcal{E} \simeq \mathcal{E}^{[k]}. \quad (\text{HTK3})$$

The seminorms with separate radii generate the projective topology: a finite collection of input seminorms is dominated in each factor by one of the increasing norms in HTK1. This specifies the completion used in HTK3 completely.

The diagonal map and the k -fold multiplication bound are

$$\begin{aligned} (\Delta_k F)(z) &= F(z, \dots, z), \\ \|\Delta_k F\|_h &\leq q_{(\sqrt{k}h, \dots, \sqrt{k}h)}(F), \quad \left\| \prod_{i=1}^k g_i \right\|_h \leq \prod_{i=1}^k \|g_i\|_{\sqrt{k}h}. \end{aligned} \quad (\text{HTK4})$$

For $N = \sum_i n_i$, the multinomial theorem gives $N! / \prod_i n_i! \leq k^N$. Taking its square root and summing absolute coefficients proves the first bound; the second follows on a simple tensor. Each output coefficient is a finite sum over $\sum_i n_i = N$, so the map is also the literal diagonal substitution on entire functions.

For $n \geq 0$, $b > 0$ and $D = d/dz$, the exact derivative bound is

$$\|D^n f\|_h \leq \sqrt{n!} b^{-n} \|f\|_{\sqrt{h^2 + b^2}}. \quad (\text{HTK5})$$

Indeed the coefficient with original degree $m \geq n$ contributes $|a_m| \sqrt{m!} \sqrt{n!} \sqrt{\binom{m}{n}} h^{m-n}$. The single binomial term $\binom{m}{n} h^{2(m-n)} b^{2n} \leq (h^2 + b^2)^m$ proves HTK5 after summation. Applying that argument in every coordinate gives, with $D_i = \partial/\partial z_i$,

$$q_{\mathbf{h}}(D_1^{d_1} \cdots D_k^{d_k} F) \leq \left(\prod_i \sqrt{d_i!} b_i^{-d_i} \right) q_{(\sqrt{h_i^2 + b_i^2})_{i=1}^k}(F). \quad (\text{HTK6})$$

35.2 The full edge series and the exact heat comparison

Let $\mathcal{E}_k = \{(i, j) : 1 \leq i < j \leq k\}$ be the complete set of unordered pairs. For a tuple $\mathbf{n} = (n_{ij})$ of nonnegative integers indexed by these pairs, put

$$N(\mathbf{n}) = \sum_{i < j} n_{ij}, \quad d_i(\mathbf{n}) = \sum_{j < i} n_{ji} + \sum_{j > i} n_{ij}, \quad \sum_i d_i = 2N. \quad (\text{HTK7})$$

Every pair is retained. Define the coupling operator by

$$C_t^{(k)} F = \sum_{\mathbf{n} \in \mathbb{N}_0^{\mathcal{E}_k}} \left(\prod_{i < j} \frac{(-2t)^{n_{ij}}}{n_{ij}!} \right) D_1^{d_1} \cdots D_k^{d_k} F. \quad (\text{HTK8})$$

The term with all $n_{ij} = 0$ has coefficient 1, including at $t = 0$. For any $b_i > 0$ with $b_i b_j > 2(k-1)|t|$ for every $i < j$, set $r_{ij} = 2(k-1)|t|/(b_i b_j)$. Then

$$\begin{aligned} \sum_{\mathbf{n}} \left(\prod_{i < j} \frac{(2|t|)^{n_{ij}}}{n_{ij}!} \right) q_{\mathbf{h}}(D_1^{d_1} \cdots D_k^{d_k} F) \\ \leq \left(\prod_{i < j} \frac{1}{1 - r_{ij}} \right) q_{(\sqrt{h_i^2 + b_i^2})_{i=1}^k}^k(F) < \infty. \end{aligned} \quad (\text{HTK9})$$

To prove it, the multinomial theorem at vertex i gives $d_i! \leq (k-1)^{d_i} \prod_{j \neq i} n_{ij}!$, where $n_{ij} = n_{ji}$ in this inequality. Multiplying the square roots over all vertices yields

$$\prod_i \sqrt{d_i!} \leq (k-1)^{N(\mathbf{n})} \prod_{i < j} n_{ij}!.$$

Insert this in HTK6. The factorials cancel, and the summand is bounded by $\prod_{i < j} r_{ij}^{n_{ij}}$ times the input norm in HTK9. The independent geometric sums prove the bound. The parameters always exist, for example all b_i equal to any $b > \sqrt{2(k-1)|t|}$. Enlarging $|t|$ to its maximum on a compact set proves locally uniform absolute convergence in every norm. Thus HTK8 defines a continuous operator on the complete space HTK2.

Keep $T_t = e^{-tD^2}$ with its proved HSL17–18 meaning:

$$T_t f = \sum_{m \geq 0} \frac{(-t)^m}{m!} D^{2m} f, \quad \sum_{m \geq 0} \frac{|t|^m}{m!} \|D^{2m} f\|_h \leq \|f\|_{h + \sqrt{2|t|}}, \quad T_t T_s = T_{t+s}.$$

On HTK2 let $A_t^{(k)}$ apply T_t separately in every coordinate. The same nonnegative coefficient proof in each coordinate bounds its absolute multiple series by $q_{(h_i + \sqrt{2|t|})_i}^k(F)$. It is a continuous group and on simple tensors equals $T_t \otimes \cdots \otimes T_t$. For a polynomial, $D\Delta_k = \Delta_k \sum_i D_i$ and all relevant differential series terminate. The commuting finite exponentials therefore give

$$\begin{aligned} T_t \Delta_k A_{-t}^{(k)} F &= \Delta_k e^{-(\sum_i D_i)^2} e^{t \sum_i D_i^2} F \\ &= \Delta_k e^{-2t \sum_{i < j} D_i D_j} F = \Delta_k C_t^{(k)} F. \end{aligned} \quad (\text{HTK10})$$

Each exponential identity follows by the finite multinomial formula on that polynomial. HTK4, HTK9 and the heat bounds make both ends continuous on HTK2. Polynomial density extends HTK10 to every $F \in \mathcal{E}^{[k]}$. The same proof first on polynomials and then by continuity gives $C_t^{(k)} C_s^{(k)} = C_{t+s}^{(k)}$ and $(C_t^{(k)})^{-1} = C_{-t}^{(k)}$. Thus the tensor correction is itself an actual continuous group, with no infinite formal cancellation required.

For $f_i \in \mathcal{E}$, the k -fold transported product therefore is exactly

$$\boxed{\begin{aligned} \star_t^k f_i &:= T_t \left(\prod_i T_{-t} f_i \right) \\ &= \sum_{\mathbf{n} \in \mathbb{N}_0^{\mathcal{E}_k}} \left(\prod_{i < j} \frac{(-2t)^{n_{ij}}}{n_{ij}!} \right) \prod_i D^{d_i(\mathbf{n})} f_i. \end{aligned}} \quad (\text{HTK11})$$

Its absolute convergence has the literal bound

$$\begin{aligned} \sum_{\mathbf{n}} \left(\prod_{i < j} \frac{(2|t|)^{n_{ij}}}{n_{ij}!} \right) \left\| \prod_i D^{d_i} f_i \right\|_h \\ \leq \left(\prod_{i < j} \frac{1}{1 - 2(k-1)|t|/(b_i b_j)} \right) \prod_i \|f_i\|_{\sqrt{k h^2 + b_i^2}}. \end{aligned} \quad (\text{HTK12})$$

Apply HTK4 to each derivative product, HTK5 to each factor at radius $\sqrt{k}h$, and the same factorial calculation as in HTK9. This also proves multilinear continuity directly. Every parenthesization of the binary product HTP8 gives HTK11: applying the injective T_{-t} to such a product gives the ordinary product $\prod_i T_{-t} f_i$, by induction on the binary parentheses. The independent tensor map $A_t^{(k)}$ and the diagonal heat map are connected by precisely $C_t^{(k)}$; HTK10 is their proved comparison, including all cross terms.

$$\begin{array}{ccccc} \mathcal{E}^{[k]} & \xrightarrow{A_{-t}^{(k)}} & \mathcal{E}^{[k]} & \xrightarrow{\Delta_k} & \mathcal{E} \\ C_t^{(k)} \downarrow & & & & \downarrow T_t \\ \mathcal{E}^{[k]} & \xrightarrow{\Delta_k} & \mathcal{E} & \xrightarrow{\text{id}} & \mathcal{E} \end{array}$$

Figure 13: The exact tensor comparison HTK10. Each upper coordinate has its own original heat operator. Moving heat through the diagonal introduces $-2tD_iD_j$ for every pair, exactly HTK7–9. The spaces, maps and their convergence are proved in HTK1–12; the diagram does not identify independent heat with diagonal heat without this map.

35.3 The original theta kernel on every tensor factor

Use the unchanged kernel and family

$$\begin{aligned} \Phi(u) &= \sum_{m \geq 1} (2\pi^2 m^4 e^{9u} - 3\pi m^2 e^{5u}) e^{-\pi m^2 e^{4u}}, \quad u \geq 0, \\ H_a(z) &= \int_0^\infty e^{au^2} \Phi(u) \cos(zu) du, \\ c &= e^{-3\pi}, \quad C_\Phi = 2\pi^2 \frac{1 + 11c + 11c^2 + c^3}{(1 - c)^5}, \\ I_p(A, B) &= \int_0^\infty u^p e^{Au^2 + Bu} \Phi(u) du, \quad \mathcal{M}_p(A, B) = \frac{C_\Phi}{2\pi} \exp \left(\frac{3A^2}{32\pi} + \frac{(B + 9 + p)^2}{8\pi} - \frac{\pi}{2} \right). \end{aligned} \tag{HTK13}$$

For $A, B \geq 0$ and integer $p \geq 0$, THS3–5 prove $0 < \Phi(u) \leq C_\Phi e^{9u - \pi e^{4u}}$ and $I_p(A, B) \leq \mathcal{M}_p(A, B)$. The positivity follows term by term from $\pi m^2 e^{5u} (2\pi m^2 e^{4u} - 3) > 0$. For the upper bound, discard the negative prefactor and use $m^2 - 1 \geq 3(m - 1)$ to sum $\sum m^4 e^{m-1}$. For the moment bound use $u^p \leq e^{pu}$, split the negative exponent into quarters and a half, and apply

$$Au^2 - \frac{\pi}{4} e^{4u} \leq \frac{3A^2}{32\pi}, \quad (B + 9 + p)u - \frac{\pi}{4} e^{4u} \leq \frac{(B + 9 + p)^2}{8\pi}, \quad \int_0^\infty e^{-\pi e^{4u}/2} du \leq \frac{e^{-\pi/2}}{2\pi}.$$

The first two inequalities follow respectively from $e^{4u} \geq (32/3)u^4$ and $e^{4u} \geq 8u^2$ by completing the relevant square. The last uses $e^{4u} \geq 1 + 4u$. These give the constants displayed in HTK13 without an unmentioned change of variables or function factor.

THS13–15, or the following direct estimates, give $H_a \in \mathcal{E}$ and $T_r H_a = H_{a+r}$ for complex a, r . Indeed $|\cos(zu)| \leq e^{|z|u}$ and $|z|u \leq \alpha|z|^2 + u^2/(4\alpha)$ give every positive Gaussian growth bound. HSL16 then gives membership in HTK1. The absolute heat-integrand sum is at most $e^{(\text{Re } a + |r|)u^2 + |\text{Im } z|u} \Phi(u)$; HTK13 bounds its integral after enlarging a negative quadratic coefficient to zero. Termwise heat differentiation has the exact sign $(-r)^m (-1)^m = r^m$, giving e^{ru^2} and hence the asserted heat identity.

For $\epsilon \in \{-1, 1\}^k$, retain the signed frequency $s_\epsilon(\mathbf{u}) = \sum_i \epsilon_i u_i$. For arbitrary complex

$a_1, \dots, a_k, t, z$, the full tensor integral is

$$\left(\begin{matrix} k \\ \star_t \\ i=1 \end{matrix} H_{a_i} \right) (z) = 2^{-k} \sum_{\epsilon \in \{-1,1\}^k} \int_{[0,\infty)^k} \left(\prod_i \Phi(u_i) \right) \exp \left(\sum_i a_i u_i^2 + 2t \sum_{i<j} \epsilon_i \epsilon_j u_i u_j \right) \cdot \cos \left(z \sum_i \epsilon_i u_i \right) d\mathbf{u}. \quad (\text{HTK14})$$

All derivatives, integrals and sums used to establish this formula are absolutely controlled in each norm of HTK1. To verify this before exchanging them, Cauchy-Schwarz gives $\sum_{m \geq 0} x^m / \sqrt{m!} \leq \sqrt{2} e^{x^2}$ for $x \geq 0$: write the summands as $2^{-m/2} (\sqrt{2} x)^m / \sqrt{m!}$. Consequently sine and cosine of uz have h -norm at most $\sqrt{2} e^{h^2 u^2}$. Their d th derivatives have norm at most $\sqrt{2} u^d e^{h^2 u^2}$, since differentiation only changes sine to cosine, a sign and the factor u^d . By HTK4 the norm of the product of derivatives in HTK11 is therefore at most $2^{k/2} \prod_i u_i^{d_i} e^{kh^2 \sum_i u_i^2}$. Summing its absolute edge weights yields $\exp(2|t| \sum_{i<j} u_i u_j)$. The inequality $2u_i u_j \leq u_i^2 + u_j^2$ now proves

$$\begin{aligned} B_i(h) &= \max\{\operatorname{Re} a_i + kh^2 + (k-1)|t|, 0\}, \\ \int_{[0,\infty)^k} \prod_i \Phi(u_i) e^{\sum_i (\operatorname{Re} a_i) u_i^2} \sum_{\mathbf{n}} \left(\prod_{i<j} \frac{(2|t|)^{n_{ij}}}{n_{ij}!} \right) \left\| \prod_i D^{d_i} \cos(zu_i) \right\|_h \\ &\leq 2^{k/2} \prod_i I_0(B_i(h), 0) \leq 2^{k/2} \prod_i \mathcal{M}_0(B_i(h), 0) < \infty. \end{aligned} \quad (\text{HTK15})$$

Here each D in the product differentiates its own cosine factor before multiplication. The finite derivative orders and any parameter derivatives introduce only finite moment powers; the corresponding I_p bounds apply. Nonnegative summation over absolute Taylor coefficients, followed by Tonelli's theorem, proves all the stated interchanges in each weighted ℓ^1 norm. The resulting coefficients belong to every such space, so they define the same element of $\mathcal{E}$. On compact parameter sets a common bound is obtained by enlarging the real parts and moduli in HTK15. This also proves local uniformity and entire parameter dependence. Expanding any of the original positive kernel series is allowed by the same majorant, so no kernel term has been dropped.

It remains to compute the integrand, including its signs. The exact cosine expansion is

$$\cos(zu_i) = 2^{-1} \sum_{\epsilon_i = \pm 1} e^{i\epsilon_i z u_i}.$$

For an edge tuple $\mathbf{n}$, its derivatives contribute $\prod_i (i\epsilon_i u_i)^{d_i}$. Since $\sum_i d_i = 2N$, the factor $i^{2N} = (-1)^N$ cancels the sign in $(-2t)^N$. The resulting edge series is the product of the exponentials $\exp(2t\epsilon_i \epsilon_j u_i u_j)$. Thus the sum is first obtained with $\exp(izs_\epsilon)$. The substitution $\epsilon \mapsto -\epsilon$ leaves the quadratic exponent unchanged and changes s to $-s$. Pairing those terms replaces their exponential by the displayed cosine. The factor 2^{-k} remains unchanged. This proves HTK14 with every pair, sign, coefficient and coordinate retained.

With $B = |\operatorname{Im} z|$ and $A_i = \max\{\operatorname{Re} a_i + (k-1)|t|, 0\}$, it follows that

$$\left| \left(\begin{matrix} k \\ \star_t \\ i=1 \end{matrix} H_{a_i} \right) (z) \right| \leq \prod_i I_0(A_i, B) \leq \prod_i \mathcal{M}_0(A_i, B). \quad (\text{HTK16})$$

Indeed $|\cos(zs)| \leq e^{B|s|} \leq e^{B \sum_i u_i}$, and the absolute cross exponent is at most $(k-1)|t| \sum_i u_i^2$. Each of the 2^k terms has the same separating bound, cancelling exactly the prefactor 2^{-k} . This proves the requested bound for arbitrary complex parameters, without replacing H_a by a quotient.

For real a_i, t , the even coefficients of HTK14 are all strictly alternating and every odd coefficient is zero:

$$[z^{2j}] \binom{k}{\star_t} H_{a_i} = \frac{(-1)^j}{2^k (2j)!} \sum_{\epsilon} \int \prod_i \Phi(u_i) e^{\sum_i a_i u_i^2 + 2t \sum_{i < j} \epsilon_i \epsilon_j u_i u_j} s_{\epsilon}(\mathbf{u})^{2j} d\mathbf{u}, \quad (-1)^j [z^{2j}] > 0. \quad (\text{HTK17})$$

The moment bounds justify differentiation at zero. Every integrand after removing $(-1)^j$ is nonnegative. For $j = 0$ it is positive everywhere; for $j > 0$ the all-positive sign tuple gives $s = \sum_i u_i > 0$ on $(0, \infty)^k$. That open set has positive measure and a positive kernel. Thus its finite integral is strictly positive, proving the assertion rather than assuming nonvanishing of a coefficient.

35.4 Actual equal-time tensor powers and their characters

Set every $a_i = t$ with $t \leq 0$ real. The exact exponent is

$$\sum_i t u_i^2 + 2t \sum_{i < j} \epsilon_i \epsilon_j u_i u_j = t \left(\sum_i \epsilon_i u_i \right)^2 \leq 0, \quad H_t^{\star_t k} = T_t(H_0^k). \quad (\text{HTK18})$$

The second identity follows from $T_{-t} H_t = H_0$ and HTK11. Consequently the exact bound with a base independent of both k and $t \leq 0$ is

$$\boxed{|T_t(H_0^k)(z)| \leq \left[\int_0^\infty \Phi(u) e^{|\operatorname{Im} z| u} du \right]^k = I_0(0, |\operatorname{Im} z|)^k.} \quad (\text{HTK19})$$

Use HTK18 directly in HTK14 and then separate the bound $e^{B \sum_i u_i}$. The proof works for $k = 1$ as well. The power on the right is part of the statement; no assertion that its base is at most 1 is used.

There is a concrete measure map behind this tensor identity. Define a finite positive measure μ on $\mathbb{R}$ by

$$\int_{\mathbb{R}} \psi(x) d\mu(x) = \frac{1}{2} \sum_{\epsilon = \pm 1} \int_0^\infty \Phi(u) \psi(\epsilon u) du, \quad \mu(\mathbb{R}) = I_0(0, 0). \quad (\text{HTK20})$$

This is the pushforward of the two original u copies, with their explicitly displayed weight; it is not a probability rescaling. Let $\Sigma(x_1, \dots, x_k) = \sum_i x_i$ and define $\mu^{\star k} = \Sigma_*(\mu^{\otimes k})$. Absolute exponential moments from HTK13 give

$$\int e^{izs} d\mu^{\star k}(s) = H_0(z)^k, \quad T_t(H_0^k)(z) = \int e^{ts^2} \cos(zs) d\mu^{\star k}(s). \quad (\text{HTK21})$$

For the first equality, Fubini factors the integral and the sign pairing in HTK20 gives H_0 . For the second, the pushforward definition and HTK18 give exactly HTK14. Both measures are even, so exponential and cosine integrals coincide. When $t \leq 0$ the positive measure $e^{ts^2} \mu^{\star k}$ is at most $\mu^{\star k}$, and

$$\int e^{B|s|} d\mu^{\star k}(s) \leq \int e^{B \sum_i |x_i|} d\mu^{\otimes k}(\mathbf{x}) = I_0(0, B)^k.$$

This gives a second proof of HTK19 and specifies the actual map from the independent tensor factors to the signed-sum coordinate. For unequal parameters, the weight on the original tensor coordinates is exactly $\exp(\sum_i a_i x_i^2 + 2t \sum_{i < j} x_i x_j)$; its pushforward has the integral HTK14, by the same definition and HTK16. In particular the tensor dependence has not been replaced by a claim of independent heat after taking the sum.

The characters of the amplitude algebra are the actual HTP15 characters $\chi_{t,z}(f) = (T_{-t}f)(z)$. Their precise comparison with these powers is

$$\begin{aligned}\chi_{t,z}\left(\bigstar_{i=1}^k H_{a_i}\right) &= \prod_i H_{a_i-t}(z), \\ \chi_{t,z}(T_t(H_0^k)) &= H_0(z)^k, \\ \chi_{t,z}(H_{2t}^{\star k}) &= H_t(z)^k.\end{aligned}\tag{HTK22}$$

Apply T_{-t} in HTK11 and use the proved theta heat law. For completeness, the character classification uses the division identity $f(w) - f(z) = (w - z)Q_z f(w)$ with $Q_z f \in \mathcal{E}$, proved with Gaussian bounds in HTP14. For an ordinary unital complex character put $z = \chi(w)$ and apply it to this identity, obtaining $\chi(f) = f(z)$. Transport by T_t gives exactly the characters displayed here. The distinction from evaluation is explicit: $(w \star_t w)(z) = z^2 - 2t$, whereas $w(z)^2 = z^2$. Thus HTK19 controls ordinary values of the displayed transported powers, and HTK22 proves the exact character values to which they are related. It supplies no lower bound excluding a zero of H_0 at a nonreal point, and proves no zero-location or RH conclusion.

35.5 Every independent label and every iterated parity closure

Let L be the full nontrivial bounded distributive lattice, with original carrier $S = G_L(\mathbb{C}) = \{(0, \lambda) : \lambda \in L\} \cup \{(a, 1_L) : a \in \mathbb{C}\}$ and its original coordinate join and meet operations. Write $\widehat{L} = \text{Idl}(L)$; its arbitrary joins are generated ideals, its meet is intersection, and $j(\lambda) = \downarrow \lambda$. The finite-meet distributivity over arbitrary joins proved in HSL19 and SHP3 also follows from $c = \bigvee_r (c \wedge b_r)$ whenever $c \leq \bigvee_r b_r$. All infinite support joins below are first computed there, without assuming completeness of L .

For coefficient support sequences define ordinary convolution $*$ and inclusive parity closure by

$$(\nu^{(1)} * \dots * \nu^{(k)})_n = \bigvee_{r_1 + \dots + r_k = n} \bigwedge_i \nu_{r_i}^{(i)}, \quad (P\nu)_n = \bigvee_{j \geq n, j \equiv n \pmod{2}} \nu_j.\tag{HTK23}$$

The full closed product of parity flags $P\nu^{(i)} = \nu^{(i)}$ has the following finite formula in the original L :

$$\begin{aligned}\mathcal{R}_{k,n} &= \{\mathbf{r} \in \mathbb{N}_0^k : \sum_i r_i = n\} \cup \\ &\quad \{\mathbf{r} \in \{0, 1\}^k : \sum_i r_i > n, \sum_i r_i \equiv n \pmod{2}\}, \\ \mathcal{Q}_k(\nu^{(1)}, \dots, \nu^{(k)})_n &= \bigvee_{\mathbf{r} \in \mathcal{R}_{k,n}} \bigwedge_i \nu_{r_i}^{(i)} = P(\nu^{(1)} * \dots * \nu^{(k)})_n.\end{aligned}\tag{HTK24}$$

To prove equality, a term of the last expression has total index at least n of the same parity. If its total is larger than n and some index is at least 2, lower that index by 2. The flag inequality weakly increases the label meet. After finitely many such steps the total reaches n or all indices are 0 or 1. The resulting tuple is in $\mathcal{R}_{k,n}$. Thus each term of the parity tail is bounded by a term of the finite join in HTK24. Conversely every tuple in that finite set occurs in the parity tail. This proves both inequalities, and proves that the ideal value is the principal ideal of the displayed original label. No extra binary tuple can occur for $n \geq k - 1$: its total would have to be at least $n + 2 > k$. For $k = 2$ the only extra term is $\nu_1^{(1)} \wedge \nu_1^{(2)}$ at $n = 0$, recovering SHP5 exactly.

This is also the actual iterated product, for arbitrary parentheses, and it includes raw independent labels. For any sequences in $\widehat{L}$, one has

$$P((P\nu^{(1)} * \dots * (P\nu^{(k)}))) = P(\nu^{(1)} * \dots * \nu^{(k)}).\tag{HTK25}$$

Indeed, on expansion, an input label to a closed factor at index r_i comes from $a_i \geq r_i$ of the same parity. Any outer parity-tail term therefore has $\sum_i a_i \geq \sum_i r_i \geq n$ and total parity n ; it

occurs in the right side. This gives one inequality. The reverse follows from $\nu^{(i)} \leq P\nu^{(i)}$. Taking $k = 2$ in this proved equality allows one to remove each internal closure in any iterated closed product. Induction on its parenthesization reduces it to the right side of HTK25, proving that every intermediate closure is accounted for by HTK24, not omitted.

The independently labelled domain $\mathcal{D}_L$ of SHP6 consists of (f, ν) with $f \in \mathcal{E}$, $[z^n]f \neq 0 \Rightarrow \nu_n = 1_L$, and principal ideal parity tails $Pj(\nu)_n = \downarrow s_n$. The labels ν_n have no further dependence requirement. For k such inputs their ordinary coefficient product has principal parity-tail generators exactly $\mathcal{Q}_k(s^{(1)}, \dots, s^{(k)})$: apply HTK25 in $\widehat{L}$, and then HTK24 to the principal tails. This proves closure within the original lattice for the full k -fold product.

The flag algebra is the exact image

$$\mathcal{A}_L = \{(f, \nu) : f \in \mathcal{E}, \nu_n \geq \nu_{n+2}, [z^n]f \neq 0 \Rightarrow \nu_n = 1_L\}, \quad \mathcal{P}(f, \nu) = (f, P\nu). \quad (\text{HTK26})$$

Its binary product $\circ$ uses ordinary amplitude product and the closed support product SHP5; HTK25 proves directly that $\mathcal{P}$ takes raw products to these products. Its heat group is $U_t(f, \nu) = (T_t f, \nu)$. This preserves the carrier: a nonzero output coefficient has a nonzero higher coefficient of matching parity, whose top label propagates down the flag. Its inverse is U_{-t} , by the amplitude heat law and the unchanged labels. Hence its exact k -fold transported product is

$$\star_{t=1}^k (f_i, \nu^{(i)}) = \left(T_t \left(\prod_i T_{-t} f_i \right), \mathcal{Q}_k(\nu^{(1)}, \dots, \nu^{(k)}) \right). \quad (\text{HTK27})$$

This follows by invertible transport from the iterated $\circ$ product just proved. It is in the carrier: each intermediate ordinary coefficient product with a nonzero amplitude has a nonzero contributing term, forcing a top support meet; the outer heat map preserves the carrier as above. In particular no support label in HTK27 is computed by erasing a term of the amplitude series HTK11 when it vanishes or when different amplitudes cancel.

35.6 The full coefficient maps and the exact theta lift

Retain $K_L = K \otimes_{\mathbb{B}} L$ and $\iota_L(\lambda) = 1_K \otimes \lambda$. The nonzero indicator $\beta : K \rightarrow \mathbb{B}$ is a semiring map, because tropical addition is zero exactly when both inputs are zero and a product is zero exactly when a factor is zero. Thus $\rho_L = \beta \otimes \text{id}_L$ satisfies $\rho_L \iota_L = \text{id}_L$. Coordinatewise application proves the injective map

$$\Phi(f, \nu) = (f, (\iota_L(\nu_n))_{n \geq 0}), \quad \Phi \left(\star_{i=1}^k v_i \right) = \left(\star_{i=1}^k f_i, \mathcal{Q}_k(\iota_L \nu^{(1)}, \dots, \iota_L \nu^{(k)}) \right). \quad (\text{HTK28})$$

In the right-hand $\mathcal{Q}_k$, finite joins and meets are respectively addition and multiplication in K_L . Every term of HTK24 is finite, so preservation follows directly from the semiring laws of ι_L . Applying ρ_L to every coefficient recovers all original labels. This is the actual original coefficient map $(a_n, \nu_n) \mapsto (a_n, 1_K \otimes \nu_n)$, organized as a function and its full flag, and not one global label.

For arbitrary ideal-labelled flags, the exact completed maps are

$$\begin{aligned} \mathfrak{K}_L &= \text{Id}_{\text{ord}}(K_L), \\ \iota^\#(I) &= \{w \in K_L : \exists a \in I, w \leq \iota_L(a)\}, \\ \rho^\#(J) &= \{\lambda \in L : \exists w \in J, \lambda \leq \rho_L(w)\}, \\ \rho^\# \iota^\# &= \text{id}, \quad \iota^\#(\downarrow \lambda) = \downarrow \iota_L(\lambda). \end{aligned} \quad (\text{HTK29})$$

Here ordered ideals are lower subsets closed under finite idempotent sums, with joins generated by unions and products generated by pairwise products. The displayed sets are ideals: finite sums of two witnesses are witnessed by their finite sum in the source ideal. The generator description shows preservation of arbitrary joins and products by the induced maps; principal units are

preserved as well. The retraction follows from $\rho_L \iota_L = \text{id}$ and monotonicity: if $w \leq \iota_L(a)$ then $\rho_L(w) \leq a$, and each a is witnessed by $w = \iota_L(a)$. These are the actual HSL33–35 completed maps, with their proof included. Applying them coordinatewise to HTK23–27 proves HTK28 for all ideal-labelled inputs too. The original formula is its restriction under the two principal embeddings. Thus the arbitrary support completion and the full original lattice have an explicit faithful connecting map.

For completeness, the exact passage to one global label is $\epsilon(\nu) = \nu_0 \vee \nu_1$. For a parity flag this is the join of all its entries, since every entry is below the first entry of its parity. Expanding the finite distributive product gives

$$\epsilon(\mathcal{Q}_k(\nu^{(1)}, \dots, \nu^{(k)})) = \bigwedge_i (\nu_0^{(i)} \vee \nu_1^{(i)}). \quad (\text{HTK30})$$

Every term on the left is at most the right side. Conversely a term of the distributed right side chooses one binary index from each factor; its total has parity 0 or 1, so it occurs in the corresponding parity-tail output at index 0 or 1. HTK24 therefore contains that term, proving the reverse inequality. The full coefficient map commutes with this global map because ι_L preserves the finite join. To retain ordinary amplitude multiplication after transport, the actual map is

$$\gamma_t(f, \nu) = (T_{-t}f, \epsilon(\nu)) \in G_L(\mathcal{E}), \quad \gamma_t \left(\star_{i=1}^k v_i \right) = \prod_i \gamma_t(v_i). \quad (\text{HTK31})$$

HTK11 and HTK30 prove every coordinate of this identity. The unchanged amplitude map instead lands in $G_L(\mathcal{E}_t)$, where multiplication is $\star_t$. These are the precise synchronization maps proved in SHP17–27; no ordinary-product target is silently assigned to the unchanged amplitude map.

Finally let $\eta_{2j} = 1_L$, $\eta_{2j+1} = 0_L$, and use the actual theta lift $\mathbf{H}_a = (H_a, \eta)$ of THS18. HTK24 gives

$$\mathcal{Q}_k(\eta, \dots, \eta) = \eta, \quad \star_{i=1}^k \mathbf{H}_{a_i} = \left(\star_{i=1}^k H_{a_i}, \eta \right), \quad \mathbf{H}_t^{\star_t k} = (T_t(H_0^k), \eta). \quad (\text{HTK32})$$

At any even output degree take $(n, 0, \dots, 0)$ in HTK24; its meet is top. At an odd output degree every admissible tuple has an odd index, hence a bottom factor, so every meet is bottom. This proves the first identity and then HTK27 proves the other two. For real a_i, t , HTK17 forces every even label to be top by actual nonvanishing; for complex parameters the same prescribed labels are retained. Applying HTK28 gives precisely $(a_{2j}, 1_{K_L})$ at even degrees and $(0, 0_{K_L})$ at odd degrees. This completes the tensor calculation with all coefficient supports, all cross couplings and the entire original theta amplitude.

36 The actual defect between ordinary and heat-transported theta powers

Sources and exact objects. The human kernel is Brad Rodgers and Terence Tao's *The De Bruijn–Newman constant is non-negative*, original author source `fmp-template.tex`, SHA-256

25b015c0fb151f1613247d82cf53a6561a320d4dab2bb4f809482e7c477a5817.

The actual prior reading is lines 1–155, including the original definitions `phidef` and `htdef`. No zero-location threshold theorem is used. The complete analytic algebra, derivative estimates, transported product and actual positive kernel are HTP1–24 of the included `HEAT_TRANSPORTED_PRODUCT.tex`; that complete proof was written and read at edition SHA-256

703cd994f758fd862d637d4ac52236624a9109127fe5c230a7b8636e73ef3ee4.

The full support map and theta fibre are the included SHP proof `SUPPORTED_HEAT_PRODUCT.tex`, read at SHA-256

0b0f5a50874ce59fdfdb74a715881182a38a9238c816107d0d3b143403426799.

The complete reading ranges are 25–62, 94–158, 195–327, 331–399 and 612–676. This records a bounded reading, including the finite support product and complete theta support fibre. The k-edge expansion is treated separately in HTPK; the present proof uses an exact differential defect and does not assume that expansion.

Retain the original coordinates and constants:

$$\begin{aligned} \Phi(u) &= \sum_{n \geq 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, \quad u \geq 0, \\ H_t(z) &= \int_0^\infty e^{tu^2} \Phi(u) \cos(zu) du, \quad T_t = e^{-tD^2}, \quad D = \frac{d}{dz}, \quad H_t = T_t H_0, \\ \mathcal{E} &= \left\{ f = \sum_{n \geq 0} a_n z^n : \|f\|_h := \sum_{n \geq 0} |a_n| \sqrt{n!} h^n < \infty \ (h > 0) \right\}. \end{aligned} \tag{HAD1}$$

The kernel is strictly positive: each summand has positive prefactor $\pi n^2 e^{5u} (2\pi n^2 e^{4u} - 3)$. For $c = e^{-3\pi}$ and $C_\Phi = 2\pi^2(1 + 11c + 11c^2 + c^3)/(1 - c)^5$, the original summands give

$$0 < \pi(2\pi - 3)e^{9u - \pi e^{4u}} \leq \Phi(u) \leq C_\Phi e^{9u - \pi e^{4u}}.$$

For the lower bound use $n = 1$ and $2\pi e^{4u} - 3 \geq (2\pi - 3)e^{4u}$. For the upper bound discard the negative prefactor term, use $n^2 - 1 \geq 3(n - 1)$, and sum $\sum n^4 c^{n-1} = (1 + 11c + 11c^2 + c^3)/(1 - c)^5$ by the geometric-series derivative identity. In particular all moments below are finite. A numerical bound used here is

$$\begin{aligned} I_p(A, B) &= \int_0^\infty u^p e^{Au^2 + Bu} \Phi(u) du \leq \mathcal{M}_p(A, B) := \frac{C_\Phi}{2\pi} \exp\left(\frac{3A^2}{32\pi} + \frac{(B + 9 + p)^2}{8\pi} - \frac{\pi}{2}\right), \\ A, B \geq 0, \quad p \geq 0, \quad h_0 &:= H_0(0) = I_0(0, 0) > 0, \quad m_2 := I_2(0, 0) > 0, \quad H_0''(0) = -m_2. \end{aligned} \tag{HAD2}$$

Indeed $u^p \leq e^{pu}$ and $e^{4u} \geq (32/3)u^4$, $e^{4u} \geq 8u^2$ give $Au^2 - (\pi/4)e^{4u} \leq 3A^2/(32\pi)$ and $(B + 9 + p)u - (\pi/4)e^{4u} \leq (B + 9 + p)^2/(8\pi)$. The remaining integral of $e^{-\pi e^{4u}/2}$ is at most $e^{-\pi/2}/(2\pi)$, using $e^{4u} \geq 1 + 4u$. These same estimates justify every fixed derivative in t, z of HAD1 on compact parameter sets. They give $H_t \in \mathcal{E}$ and $T_s H_t = H_{t+s}$ by expanding e^{su^2} under its absolute majorant, as proved fully in HTP19–20.

36.1 The exact defect and all its ordinary root jets

We shall use the following literal norms, with $b > 0$:

$$\begin{aligned} \left\| \prod_{j=1}^k f_j \right\|_h &\leq \prod_{j=1}^k \|f_j\|_{\sqrt{k}h}, & \|D^n f\|_h &\leq \sqrt{n!} b^{-n} \|f\|_{\sqrt{h^2+b^2}}, \\ \sum_{n \geq 0} \frac{|t|^n}{n!} \|D^{2n} f\|_h &\leq \|f\|_{h+\sqrt{2|t|}}, & \|H_0\|_h &\leq \sqrt{2} I_0(h^2, 0). \end{aligned} \quad (\text{HAD3})$$

For the product bound, the multinomial coefficient $(n_1 + \dots + n_k)!/(n_1! \dots n_k!)$ is at most $k^{n_1 + \dots + n_k}$. Taking square roots and summing the nonnegative coefficient bound proves the assertion. For the derivative use the single binomial term $\binom{q}{n} h^{2(q-n)} b^{2n} \leq (h^2 + b^2)^q$ on each original coefficient a_q . For heat, after setting $q = i + 2n$, the inner coefficient $\sqrt{q!}/(n! \sqrt{(q-2n)!})$ is $\sqrt{\binom{q}{2n}} \sqrt{\binom{2n}{n}} \leq \binom{q}{2n} 2^n$. Its sum is bounded by $(h + \sqrt{2|t|})^q$. These are positive-sum arguments, so they prove absolute convergence before any reordering. Finally $\sum_{n \geq 0} x^n / \sqrt{n!} \leq \sqrt{2} e^{x^2}$ by Cauchy–Schwarz with weights $2^{-n/2}$. Integrating the cosine coefficient subseries proves the last estimate.

For integers $k \geq 2$, define the two actual entire functions and their amplitude difference by

$$O_k(t) = H_t^k, \quad P_k(t) = T_t(H_0^k) = H_t^{\star k}, \quad \Delta_k(t) = O_k(t) - P_k(t). \quad (\text{HAD4})$$

All belong to $\mathcal{E}$ by HAD3. Their difference has the exact, convergent integral formula

$$\Delta_k(t) = k(k-1)t \int_0^1 T_{(1-\theta)t} \left(H_{\theta t}^{k-2} (H'_{\theta t})^2 \right) d\theta. \quad (\text{HAD5})$$

Differentiate $T_{t-s}(H_s^k)$ along the segment $s = \theta t$. The heat equation and ordinary Leibniz rule give

$$\frac{d}{ds} [T_{t-s}(H_s^k)] = T_{t-s} [D^2(H_s^k) - k H_s^{k-1} H_s''] = k(k-1) T_{t-s} [H_s^{k-2} (H_s')^2].$$

The heat series is entire in time in every coefficient norm; HAD3 proves continuity of products and derivatives there. Thus this is a derivative in the complete space $\mathcal{E}$. Integrating its continuous derivative from 0 to t gives HAD5. Equivalently the integral is the limit of Riemann sums in each weighted ℓ^1 norm; their limits have the same coefficients, so define one element of $\mathcal{E}$. There is also the explicit bound, for every $h, b > 0$,

$$\begin{aligned} R &= \sqrt{k}h + \sqrt{2(k+1)|t|}, & B &= \sqrt{R^2 + b^2}, \\ \|\Delta_k(t)\|_h &\leq k(k-1)|t|b^{-2} \|H_0\|_B^k \leq k(k-1)|t|b^{-2} (\sqrt{2}\mathcal{M}_0(B^2, 0))^k. \end{aligned} \quad (\text{HAD6})$$

At parameter θ , the outer heat first enlarges the norm radius to $h + \sqrt{2(1-\theta)|t|}$. The product then enlarges it by $\sqrt{k}$. Commuting each derivative with the inner heat enlarges this radius again by $\sqrt{2\theta|t|}$. Its resulting value is at most R , because $\sqrt{k(1-\theta)} + \sqrt{\theta} \leq \sqrt{k+1}$. Apply the derivative estimate with parameter b to the two differentiated factors and increase every input radius to B . This bounds the integrand by $b^{-2} \|H_0\|_B^k$ and proves HAD6. These bounds are uniform on compact time sets; no finite-degree heat intertwining has been assumed.

Let $Z_t = \{\zeta \in \mathbb{C} : H_t(\zeta) = 0\}$ be the actual zero set, without any assertion that its off-real part is empty or nonempty. Since T_t is invertible and $H_0(0) > 0$, H_t is not the zero function. Every member of Z_t consequently has a finite order $m \geq 1$. For every $q \geq 0$ its exact ordinary derivative defect is

$$\Delta_k^{(q)}(t, \zeta) = q! \sum_{\substack{n_1 + \dots + n_k = q \\ n_1, \dots, n_k \geq m}} \prod_{j=1}^k \frac{H_t^{(n_j)}(\zeta)}{n_j!} - P_k^{(q)}(t, \zeta). \quad (\text{HAD7})$$

This is the finite Taylor-product rule; all terms with an index below m vanish. In particular

$$\begin{aligned}\Delta_k^{(q)}(t, \zeta) &= -P_k^{(q)}(t, \zeta) \quad (0 \leq q < km), \\ \Delta_k^{(km)}(t, \zeta) &= \frac{(km)!}{(m!)^k} (H_t^{(m)}(\zeta))^k - P_k^{(km)}(t, \zeta).\end{aligned}\tag{HAD8}$$

Thus the entire defect class modulo $(z - \zeta)^{km}$ is the negative class of the actual transported power. The equation is a computed finite jet, not an equality between ordinary and transported Taylor characters.

For $k = 2$ the complete root calculation is particularly explicit. HTP10 and continuity of D^q in HAD3 give

$$\begin{aligned}P_2(t, \zeta) &= \sum_{n=m}^{\infty} \frac{(-2t)^n}{n!} (H_t^{(n)}(\zeta))^2, \\ P_2^{(q)}(t, \zeta) &= \sum_{n \geq 0} \frac{(-2t)^n}{n!} \sum_{j=0}^q \binom{q}{j} H_t^{(n+j)}(\zeta) H_t^{(n+q-j)}(\zeta).\end{aligned}\tag{HAD9}$$

The undifferentiated product series is absolutely convergent in every norm by HTP7. Apply HAD3 to the sum of its term norms at the enlarged radius to prove absolute convergence after D^q as well. Evaluation is continuous, since $|f(\zeta)| \leq e^{|\zeta|^2/(2h^2)} \|f\|_h$. Hence every sum and derivative in HAD9 is justified. For the absolute sum after expanding its finite Leibniz rule, choose $s > 0$ with $s^2 > 2|t|$ and put $B = \sqrt{2h^2 + s^2}$. HAD3 and $\sqrt{(n+j)!(n+q-j)!}/n! \leq (n+q)^{q/2}$ bound its sum of term norms by

$$2^q s^{-q} \|H_t\|_B^2 \sum_{n \geq 0} \left(\frac{2|t|}{s^2} \right)^n (n+q)^{q/2} < \infty.$$

The factor at $n = q = 0$ is 1. The factorial bound follows by writing its square as the product of the j factors $n+1, \dots, n+j$ and the $q-j$ factors $n+1, \dots, n+q-j$, each at most $n+q$.

The first surviving root jet also gives a quantified local calculation in the contraction parameter. Fix the actual function $f = H_t$ and its root ζ of order m , but allow the product parameter r to vary. Then

$$(f \star_r f)(\zeta) = \frac{(-2r)^m}{m!} (f^{(m)}(\zeta))^2 + \mathcal{R}_m(r).\tag{HAD10}$$

For any $h, b > 0$, put

$$C = e^{|\zeta|^2/(2h^2)} \|f\|_{\sqrt{2h^2+b^2}}^2, \quad L = \frac{b^{2m} |f^{(m)}(\zeta)|^2}{m!} > 0, \quad q = \frac{2|r|}{b^2}.$$

The geometric tail of HTP7 gives the completely explicit bound and consequence

$$|\mathcal{R}_m(r)| \leq \frac{Cq^{m+1}}{1-q} \quad (q < 1), \quad |(f \star_r f)(\zeta)| \geq \frac{L}{2} q^m > 0 \quad \left(0 < q \leq \min \left(\frac{1}{2}, \frac{L}{4C} \right) \right).\tag{HAD11}$$

The leading term has absolute value Lq^m . Under the displayed bound on q , the remainder is at most $2Cq^{m+1} \leq Lq^m/2$. This proves the consequence. The constants are determined by the actual root jet and original coefficient norm. Setting $r = t$ recovers HAD9; the exact remaining quantity is its displayed convergent tail, which is not dropped when t lies outside that computed neighbourhood. At $t = 0$ the original defect is identically zero, directly from HAD4.

36.2 The actual Gaussian comparison and the complex root Gram matrix

For $a > 0$, let $d\gamma_a(y) = (4\pi a)^{-1/2} e^{-y^2/(4a)} dy$ on $\mathbb{R}$. The following formula holds for every $f \in \mathcal{E}$, with absolute convergence in every coefficient norm:

$$T_{-a}f(z) = \int_{\mathbb{R}} f(z+y) d\gamma_a(y), \quad \int_{\mathbb{R}} \|f(\cdot+y)\|_h d\gamma_a(y) \leq \frac{\|f\|_{\sqrt{h^2+b^2}}}{\sqrt{1-2a/b^2}} \quad (b^2 > 2a).\tag{HAD12}$$

Here is a direct norm proof. The absolute Taylor-translation sum satisfies

$$\sum_{n \geq 0} \frac{|y|^n}{n!} \|D^n f\|_h \leq e^{|y|^2/(2b^2)} \|f\|_{\sqrt{h^2+b^2}}.$$

For the coefficient a_j , its inner sum after extracting $|a_j|\sqrt{j!}$ is $\sum_{n=0}^j \sqrt{\binom{j}{n}} h^{j-n} |y|^n / \sqrt{n!}$. Cauchy-Schwarz, placing b^n in the binomial factor, bounds it by $(h^2+b^2)^{j/2} e^{|y|^2/(2b^2)}$. Summing proves the translation estimate; its Gaussian integral is $(1-2a/b^2)^{-1/2}$. The translation series equals $f(z+y)$ first for polynomials and then by polynomial density and these bounds. Gaussian integration of a polynomial gives $T_{-a}f$, since its odd moments vanish and $\int y^{2n} d\gamma_a = (2n)!a^n/n!$. These moments follow by differentiating the Gaussian integral or its elementary integration-by-parts recurrence. Both sides are continuous in the displayed norms, so polynomial approximation proves HAD12 on all of $\mathcal{E}$.

For every real x , every $a > 0$ and every even $k \geq 2$, the actual sign is strict:

$$\Delta_k(-a, x) < 0. \quad (\text{HAD13})$$

Under γ_a , the real random variable $H_0(x+y)$ has mean $H_{-a}(x)$ by HAD12. It is not almost surely constant: continuity and the full support of the Gaussian would make it constant on $\mathbb{R}$, contradicting $H_0''(0) = -m_2 < 0$. For even k , $v \mapsto v^k$ is strictly convex on $\mathbb{R}$, since its derivative kv^{k-1} is strictly increasing. Its tangent line at the mean lies strictly below its graph away from that mean; integrating this elementary tangent inequality gives $\int H_0(x+y)^k d\gamma_a > H_{-a}(x)^k$. All moments are integrable by HAD12 applied to H_0^k , or by its absolute bound. This proves HAD13, including at every actual real root. No sign for a complex square has been inferred from this real inequality.

Now fix $t = -a < 0$ and any complex point z . Define the actual functions and sequences

$$\begin{aligned} F_z(y) &= H_0(z+y) \in L^2(\gamma_a), & c_n(z) &= \sqrt{\frac{(2a)^n}{n!}} H_{-a}^{(n)}(z), \\ P(z) &= P_2(-a, z) = \int F_z(y)^2 d\gamma_a(y) = \sum_{n \geq 0} c_n(z)^2, \\ Q(z) &= \int |F_z(y)|^2 d\gamma_a(y) = \sum_{n \geq 0} |c_n(z)|^2 > 0. \end{aligned} \quad (\text{HAD14})$$

The bilinear identity is HAD9 or HTP10. For the Hermitian identity set $f_z(w) = H_0(z+w)$ and $g_z(w) = H_0(\bar{z}+w)$. The exact transported-product identity is

$$T_{-a}(f_z g_z) = (T_{-a}f_z) \star_{-a} (T_{-a}g_z).$$

Apply HTP10 to the two inputs $T_{-a}f_z$ and $T_{-a}g_z$, which equal $H_{-a}(z+w)$ and $H_{-a}(\bar{z}+w)$, and then evaluate at $w = 0$. Translation preserves $\mathcal{E}$ by the estimate in the proof of HAD12. It commutes with the heat operators, first on polynomials and then by continuity. Also $H_{-a}^{(n)}(\bar{z}) = \overline{H_{-a}^{(n)}(z)}$, because the original kernel is real. HAD12 turns the left side into $\int |F_z|^2 d\gamma_a$; HTP7 justifies the absolute derivative sum on the right. The function F_z cannot vanish almost everywhere: continuity would force H_0 to vanish on a line, and the identity theorem would contradict $H_0(0) > 0$. Hence $Q(z) > 0$. In fact the original integral gives

$$Q(z) \leq I_0(0, |\operatorname{Im} z|)^2, \quad |P(z)| \leq Q(z). \quad (\text{HAD15})$$

Indeed $|H_0(z+y)| \leq I_0(0, |\operatorname{Im} z|)$ uniformly in real y , and the Gaussian has mass one. The second inequality is the triangle inequality for the absolutely convergent squares in HAD14.

At an actual root $\zeta \in Z_{-a}$ of order m , the first m coordinates $c_0, \dots, c_{m-1}$ vanish, and

$$\begin{aligned} \int F_\zeta d\gamma_a &= 0, & Q(\zeta) &\geq \frac{(2a)^m}{m!} |H_{-a}^{(m)}(\zeta)|^2 > 0, \\ \Delta_2(-a, \zeta) &= -P(\zeta), & \zeta \in \mathbb{R} &\implies P(\zeta) = Q(\zeta) > 0. \end{aligned} \quad (\text{HAD16})$$

For a real point all derivatives are real, so the last equality follows term by term. The original positive kernel also excludes every purely imaginary root: $H_{-a}(ib) = \int e^{-au^2} \Phi(u) \cosh(bu) du > 0$ for all real b .

At every nonreal point the complex comparison is strictly different from equality:

$$|P(z)| < Q(z) \quad (\text{Im } z \neq 0). \quad (\text{HAD17})$$

We prove the equality obstruction exactly. If $|\sum c_n^2| = \sum |c_n|^2$, its nonzero sum has a phase η of modulus one. The nonnegative differences $|c_n|^2 - \text{Re}(\eta c_n^2)$ have sum zero, so each vanishes. This means $c_n = \eta \bar{c}_n$ for every n , including the zero coordinates. The same relation holds for every derivative of H_{-a} at z . By its entire Taylor series and its real coefficients,

$$H_{-a}(z + w) = \eta H_{-a}(\bar{z} + w) \quad (w \in \mathbb{C}), \quad H_{-a}(v + 2i \text{Im } z) = \eta H_{-a}(v).$$

If $b = \text{Im } z \neq 0$, setting $v = 0$ contradicts

$$H_{-a}(2ib) = \int_0^\infty e^{-au^2} \Phi(u) \cosh(2bu) du > H_{-a}(0) > 0.$$

The strict inequality holds because $\cosh(2bu) > 1$ on $u > 0$ and the kernel is positive there. The claimed unit phase would give equal absolute values. This proves HAD17. In particular it does not assign a sign to P at a complex root and does not claim that P cannot vanish there.

The remaining root quantity is a concrete positive Gram matrix, with every entry evaluated from the original theta function. Write $u(z) = \text{Re}(c_n(z))_n$ and $v(z) = \text{Im}(c_n(z))_n$ in real ℓ^2 , and put

$$\begin{aligned} \mathbf{G}_a(z) &= \begin{pmatrix} \|u(z)\|^2 & \langle u(z), v(z) \rangle \\ \langle u(z), v(z) \rangle & \|v(z)\|^2 \end{pmatrix} = \int_{\mathbb{R}} \begin{pmatrix} (\text{Re } F_z)^2 & \text{Re } F_z \text{Im } F_z \\ \text{Re } F_z \text{Im } F_z & (\text{Im } F_z)^2 \end{pmatrix} d\gamma_a, \\ \text{tr } \mathbf{G}_a &= Q, & P &= (\mathbf{G}_a)_{11} - (\mathbf{G}_a)_{22} + 2i(\mathbf{G}_a)_{12}, \\ Q^2 - |P|^2 &= 4 \det \mathbf{G}_a, & P = 0 &\iff \mathbf{G}_a = (Q/2)I_2. \end{aligned} \quad (\text{HAD18})$$

The equality of the two matrix descriptions follows from their equal trace Q and complex traceless coordinate P , already proved in HAD14. Expanding these entries proves the determinant identity and the final equivalence. At a nonreal point HAD17 gives $\det \mathbf{G}_a > 0$, so this matrix is positive definite. At a root it is the centred covariance matrix, because both real and imaginary means in HAD16 vanish. Thus the exact double-power coincidence set on actual nonreal roots is the inverse image

$$\mathcal{C}_a = (\mathbf{G}_a|_{Z_{-a} \setminus \mathbb{R}})^{-1}(\{cI_2 : c > 0\}). \quad (\text{HAD19})$$

This is one specified map and fibre of the original function, rather than an arbitrary matrix model. It retains the positive trace, original moments and all derivatives. HAD17 proves positive definiteness on this domain; it does not prove that this scalar matrix fibre is empty or nonempty.

36.3 The same comparison with every original support flag

Let L be the full bounded distributive lattice in SHP, and let $\mathcal{A}_L$ be its actual coefficient-flag algebra. At real time take any member of the complete theta fibre

$$\mathbf{H}_t = (H_t, \nu), \quad \nu_{2j} = 1_L, \quad \nu_1 \geq \nu_3 \geq \nu_5 \geq \dots, \quad \omega = \nu_1. \quad (\text{HAD20})$$

The cosine lift has zero odd flag; the globally constant-label lift has top odd flag. Neither is substituted for the complete fibre above. The exact closed support product is $(\nu \circledast \mu)_0 = (\nu_0 \wedge \mu_0) \vee (\nu_1 \wedge \mu_1)$ and $(\nu \circledast \mu)_n = \bigvee_{r=0}^n \nu_r \wedge \mu_{n-r}$ for $n \geq 1$, as proved in SHP5. For every $k \geq 2$, its evaluated power on the actual fibre is

$$\kappa = \nu^{\circledast k}, \quad \kappa_{2j} = 1_L, \quad \kappa_{2j+1} = \omega. \quad (\text{HAD21})$$

For two factors every even coefficient contains an even-even top term. Every odd term has one odd factor at most ω , while indices $2j$ and 1 attain ω at degree $2j+1$. This includes degree one; degree zero is top by its even-even term. Further multiplication by ν leaves this same flag, by the same finite formula. This proves HAD21 without requiring any infinite joins in L .

Both powers and their exact supported difference are

$$\begin{aligned} \mathbf{O}_k(t) &= \mathbf{H}_t^{\circ k} = (O_k(t), \kappa), & \mathbf{P}_k(t) &= \mathbf{H}_t^{\star tk} = (P_k(t), \kappa), \\ \mathbf{D}_k(t) &= \mathbf{O}_k(t) + \widehat{-1} \mathbf{P}_k(t) = (\Delta_k(t), \kappa). \end{aligned} \quad (\text{HAD22})$$

The heat group preserves every label, and the two products have the same support product by SHP11. Multiplication by $\widehat{-1}$ changes the amplitude sign and preserves the flag. Addition joins the equal flags, proving the last equality. A vanishing amplitude difference therefore gives $(0, \kappa)$, not the global zero flag. Since the two powers have the same flag, their equality is nevertheless equivalent to equality of their entire amplitudes.

The complete original coefficient map is retained:

$$\begin{aligned} \Phi(f, \lambda) &= (f, (\iota_L(\lambda_n))_{n \geq 0}), & \iota_L(\lambda_n) &= 1_K \otimes \lambda_n, \\ \Phi(\mathbf{D}_k) &= (\Delta_k, (\iota_L(\kappa_n))_{n \geq 0}), \\ \text{ev}_z^{\text{amp}}(f, \lambda) &= (f(z), \lambda_0 \vee \lambda_1), & \text{ev}_\zeta^{\text{amp}}(\mathbf{D}_k) &= (\Delta_k(t, \zeta), 1_L), & \Phi_{\mathbb{C}} \text{ev}_\zeta^{\text{amp}}(\mathbf{D}_k) &= (\Delta_k(t, \zeta), 1_{K_L}). \end{aligned} \quad (\text{HAD23})$$

The support map is injective: the tropical zero/nonzero homomorphism $K \rightarrow \mathbb{B}$, tensored with L , retracts ι_L . Thus equality in the complete coefficient image recovers each original support label and the amplitude. The evaluation used here is the ordinary amplitude evaluation, multiplicative for $\circ$; it is not treated as a character for $\star_t$. At a vanishing root defect its displayed value is the supported zero $e = (0, 1_L)$. The actual homomorphism into the ordinary-product global carrier is instead SHP26's

$$\gamma_t(f, \lambda) = (T_{-t}f, \lambda_0 \vee \lambda_1), \quad \gamma_t(\mathbf{P}_k) = (H_0^k, 1_L), \quad \gamma_t(\mathbf{O}_k) = (T_{-t}(H_t^k), 1_L). \quad (\text{HAD24})$$

These follow directly by applying T_{-t} to the actual amplitudes. They preserve the distinction between the transported character divisor and the ordinary root jets in HAD7–9.

$$\begin{array}{ccc} Z_{-a} \setminus \mathbb{R} & \xrightarrow{\mathbf{G}_a} & \text{Sym}_2^{++}(\mathbb{R}) \xrightarrow{G \mapsto G_{11} - G_{22} + 2iG_{12}} \mathbb{C} \\ \zeta & & \mathbf{G}_a(\zeta) \qquad \qquad \qquad P_2(-a, \zeta) = -\Delta_2(-a, \zeta) \end{array}$$

Figure 14: The actual root-covariance map, HAD14–19. Each matrix is constructed from the unchanged theta function and Gaussian measure γ_a . Its trace is the positive quantity Q ; its complex traceless coordinate is the negative double-power defect. The precise coincidence fibre is the scalar-matrix ray. Full support is retained by HAD20–24. This diagram asserts neither existence of a nonreal root nor intersection of its image with that ray.

36.4 A proved high-even-power defect on every fixed disc

The complete independent proof ETC1–29 in the included `had_laplace/EVEN_THETA_TENSOR_CONCENTRATION.tex` evaluates the actual even powers of the same H_0 . That complete source was read and checked at SHA–256

We state all constants needed for the consequence here. Fix $a > 0$ and $R \geq 0$, retaining $h_0 = H_0(0)$ from HAD2. Put

$$\begin{aligned} m_4 &= \int_0^\infty u^4 \Phi(u) du, & \beta &= \frac{m_2}{h_0}, & c &= \frac{m_4}{24h_0}, & \delta &= \min\{\beta^{-1/2}, \sqrt{\beta/(4c)}\}, \\ D_0 &= \max\{c, \beta^2/4\}, & \rho &= \max_{|v| \geq \delta} \frac{|H_0(v)|}{h_0} \in [1/2, 1), & \lambda &= -2 \log \rho > 0, \\ G_R &= \frac{1}{4a} + \frac{R^2}{8a^2} \cosh\left(\frac{R\delta}{2a}\right), & L_R &= \frac{6\sqrt{2}D_0}{\beta^2} + \frac{\sqrt{2}G_R}{\beta}, \\ C_R &= L_R + \frac{1}{e\beta\delta^2} + 2\sqrt{a\beta} e^{R^2/(4a)} \left(\frac{3}{2e\lambda}\right)^{3/2}. \end{aligned} \tag{HAD25}$$

Every quantity comes from the original kernel, time, or disc. In particular $\rho < 1$ is proved, not assumed. Positivity of Φ gives $|H_0(v)| < h_0$ at every real $v \neq 0$: the integral triangle bound is strict on an open interval where $|\cos(vu)| < 1$. The elementary step-function proof of the cosine-transform decay in ETC6 gives $H_0(v) \rightarrow 0$. The maximum on the stated closed outer set is therefore attained and strictly below h_0 . Finally the cosine remainder $0 \leq H_0(v)/h_0 - 1 + \beta v^2/2 \leq cv^4$ gives $H_0(\delta)/h_0 \geq 1/2$, proving the lower bound on ρ and validating every constant in HAD25.

ETC14 gives the exact contour-shifted integral, and ETC25–26 give its evaluated error:

$$\begin{aligned} P_{2k}(-a, z) &= \frac{1}{\sqrt{4\pi a}} \int_{\mathbb{R}} e^{-(z-v)^2/(4a)} H_0(v)^{2k} dv, \\ A_{k,a}(z) &= h_0^{2k} e^{-z^2/(4a)} \sqrt{\frac{h_0}{4akm_2}}, \\ \sup_{|z| \leq R} \left| \frac{P_{2k}(-a, z)}{A_{k,a}(z)} - 1 \right| &\leq \frac{L_R}{k} + e^{-\beta\delta^2 k} + 2\sqrt{a\beta} e^{R^2/(4a)} \sqrt{k} \rho^{2k} \leq \frac{C_R}{k}, \quad k \in \mathbb{N}, k \geq 1. \end{aligned} \tag{HAD26}$$

The factor h_0^{2k} and all original moments remain in this formula; the entire function H_0 is unchanged. The integral follows by applying HAD12 to H_0^{2k} and shifting the Gaussian contour to the real v axis. The complete proof ETC7–14 bounds both vertical edges by a constant times $\exp(-(M-R)^2/(4a))$ as $M \rightarrow \infty$. For the error, ETC15–24 retain the fourth-order cosine remainder and the symmetric weight $e^{-v^2/(4a)} \cosh(zv/(2a))$. On $|v| \leq \delta$ their Gaussian moment integrals give L_R/k . The two outer integrals give exactly the two displayed exponential terms. The maxima $\sup_{x>0} x e^{-\beta\delta^2 x} = 1/(e\beta\delta^2)$ and $\sup_{x>0} x^{3/2} e^{-\lambda x} = (3/(2e\lambda))^{3/2}$ give the final C_R/k . These references identify the included complete proof of the estimate; none of its bounds is a hypothesis.

Let ζ be any actual root of H_{-a} with $|\zeta| \leq R$. Its existence is not assumed. HAD7 and HAD26 give the exact root defect and its effective lower bound:

$$\begin{aligned} \Delta_{2k}(-a, \zeta) &= -P_{2k}(-a, \zeta), & \left| \frac{\Delta_{2k}(-a, \zeta)}{-A_{k,a}(\zeta)} - 1 \right| &\leq \frac{C_R}{k}, \\ |\Delta_{2k}(-a, \zeta)| &\geq h_0^{2k} \sqrt{\frac{h_0}{4akm_2}} e^{-R^2/(4a)} \left(1 - \frac{C_R}{k}\right) > 0 \quad (k \in \mathbb{N}, k > C_R). \end{aligned} \tag{HAD27}$$

Indeed $H_{-a}(\zeta)^{2k} = 0$ exactly, so the first identity uses no estimate. The second is HAD26. For the lower bound use $|e^{-\zeta^2/(4a)}| = e^{-\operatorname{Re}(\zeta^2)/(4a)} \geq e^{-R^2/(4a)}$ and the reverse triangle inequality. Thus the asserted nonvanishing is uniform over the complete actual root set inside the disc.

There is consequently no actual root at which ordinary and transported powers agree for all even exponents:

$$\begin{aligned} \{\zeta \in \mathbb{C} : H_{-a}(\zeta) = 0, \quad \Delta_{2k}(-a, \zeta) = 0 \text{ for every } k \geq 1\} &= \emptyset, \\ \{\zeta : |\zeta| \leq R, \quad H_{-a}(\zeta) = 0, \quad \Delta_{2k}(-a, \zeta) = 0\} &= \emptyset \quad (k > C_R). \end{aligned} \tag{HAD28}$$

For a point in the first set choose $R \geq |\zeta|$ and an integer $k > C_R$; HAD27 contradicts its defining equalities. The same estimate proves the second line. This does not determine the double-power fibre $\mathcal{C}_a$ in HAD19: a zero at the single exponent 2 need not persist for higher even powers. It also does not say the actual root set is empty. It computes a nonzero comparison defect on that set.

For every original theta support flag in HAD20 the same statement passes through the full map without discarding any label:

$$\begin{aligned} \mathbf{D}_{2k}(-a) &= (\Delta_{2k}(-a), \kappa), & \kappa_{2j} &= 1_L, & \kappa_{2j+1} &= \omega, \\ \Phi_{\mathbb{C}} \text{ev}_{\zeta}^{\text{amp}} \mathbf{D}_{2k}(-a) &= (-P_{2k}(-a, \zeta), 1_{K_L}) \neq (0, 1_{K_L}) & (\zeta \in Z_{-a}, & |\zeta| \leq R, & k > C_R). \end{aligned} \quad (\text{HAD29})$$

HAD21–23 prove the support identities, and HAD27 proves the final inequality in its amplitude coordinate. In particular the full support map cannot remove this defect or turn this pair of powers into equal ordinary root values. All conclusions above concern the specified actual heat family and its two exact products; they do not infer a zero-location conclusion.

37 Even theta tensors at a fixed negative heat time

Sources and exact scope. The kernel and heat sign are those of Brad Rodgers and Terence Tao, *The De Bruijn–Newman constant is non-negative*, original author file `fmp-template.tex`, SHA-256

25b015c0fb151f1613247d82cf53a6561a320d4dab2bb4f809482e7c477a5817.

Lines 1–155, including `phidef` and `htdef`, were read directly. No whole-paper reading or application of a zero-location theorem is asserted. The programme source `HEAT_TRANSPORT_ED_PRODUCT.tex`, HTP1–24, was read completely, all 539 lines, at SHA-256

897bbe51fd757d8c4926aac5a82d771090e1d1dbaed5a9030fcbdbd6e2b74d14.

In particular HTP19–20, starting at source lines 357 and 380, give the positivity and moment bounds re-proved below. The present calculation retains the original kernel, coordinates, heat parameter and tensor power. It supplies a quantitative asymptotic for that entire function and makes no conclusion about the locations of the zeros of H_0 or about RH.

37.1 The original kernel, moments, and strict real maximum

For $u \geq 0$ and $z \in \mathbb{C}$ retain

$$\begin{aligned}\Phi(u) &= \sum_{n \geq 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, \\ H_t(z) &= \int_0^\infty e^{tu^2} \Phi(u) \cos(zu) du, \quad h = H_0(0), \\ m_2 &= \int_0^\infty u^2 \Phi(u) du, \quad m_4 = \int_0^\infty u^4 \Phi(u) du.\end{aligned}\tag{ETC1}$$

Every original summand of Φ is positive: its prefactor is $\pi n^2 e^{5u} (2\pi n^2 e^{4u} - 3) > 0$. The $n = 1$ term and the inequalities $n^2 - 1 \geq 3(n - 1)$, $e^{4u} \geq 1$ give

$$\begin{aligned}0 &< \pi(2\pi - 3)e^{9u - \pi e^{4u}} \leq \Phi(u) \leq C_\Phi e^{9u - \pi e^{4u}}, \\ C_\Phi &= 2\pi^2 \frac{1 + 11c_0 + 11c_0^2 + c_0^3}{(1 - c_0)^5}, \quad c_0 = e^{-3\pi}.\end{aligned}\tag{ETC2}$$

Indeed the lower prefactor satisfies $2\pi e^{4u} - 3 \geq (2\pi - 3)e^{4u}$. For the upper bound discard the negative term and factor out $2\pi^2 e^{9u - \pi e^{4u}}$. The remaining series is at most $\sum_{n \geq 1} n^4 c_0^{n-1}$; differentiating the geometric series four times with the operator $c d/dc$ gives the stated rational function. On each bounded u interval the original series converges uniformly, since its absolute terms are bounded by a fixed multiple of $(n^4 + n^2)e^{-\pi n^2}$. In particular Φ is continuous.

For $A, B, p \geq 0$ all the following integrals are finite, with the original constants:

$$\begin{aligned}I_p(A, B) &= \int_0^\infty u^p e^{Au^2 + Bu} \Phi(u) du \\ &\leq \frac{C_\Phi}{2\pi} \exp\left(\frac{3A^2}{32\pi} + \frac{(B + 9 + p)^2}{8\pi} - \frac{\pi}{2}\right).\end{aligned}\tag{ETC3}$$

To prove this, use $u^p \leq e^{pu}$ and split $-\pi e^{4u}$ into two quarters and a half. The estimates $e^{4u} \geq (32/3)u^4$ and $e^{4u} \geq 8u^2$ imply

$$Au^2 - \frac{\pi}{4}e^{4u} \leq \frac{3A^2}{32\pi}, \quad (B + 9 + p)u - \frac{\pi}{4}e^{4u} \leq \frac{(B + 9 + p)^2}{8\pi}.$$

The remaining integral is at most $e^{-\pi/2} \int_0^\infty e^{-2\pi u} du$ because $e^{4u} \geq 1 + 4u$. This proves ETC3. It follows that h, m_2, m_4 are finite and strictly positive. The bound $|\cos(zu)| \leq e^{|\operatorname{Im} z|u}$, and the corresponding bounds with extra powers of u , justify differentiation under the integral on every compact set. Thus H_0 is entire and even, is real on the real axis, and

$$H_0''(0) = -m_2, \quad H_0^{(4)}(0) = m_4, \quad |H_0(x + iy)| \leq I_0(0, |y|) \quad (x, y \in \mathbb{R}). \quad (\text{ETC4})$$

For every real $v \neq 0$ there is an open interval of positive u on which $|\cos(vu)| < 1$. Positivity and continuity of Φ therefore give

$$|H_0(v)| \leq \int_0^\infty \Phi(u) |\cos(vu)| du < h. \quad (\text{ETC5})$$

There is also an elementary proof, on this exact kernel, of

$$\lim_{|v| \rightarrow \infty} H_0(v) = 0. \quad (\text{ETC6})$$

Given $\eta > 0$, choose U with $\int_U^\infty \Phi < \eta/3$, using ETC3. Uniform continuity of Φ on $[0, U]$ supplies a finite step function s with $\int_0^U |\Phi - s| < \eta/3$. If its successive intervals are $[b_j, b_{j+1}]$ and its values s_j , then, for $v \neq 0$,

$$\left| \int_0^U s(u) \cos(vu) du \right| \leq \frac{2}{|v|} \sum_j |s_j|.$$

Choose $|v|$ so large that this is below $\eta/3$. The two approximation errors are bounded by their L^1 norms because $|\cos(vu)| \leq 1$. Their sum proves ETC6 without importing a Fourier decay theorem.

37.2 Actual heat evolution and the full Gaussian contour shift

Fix $a > 0$ and an integer $k \geq 1$, and define precisely

$$T_t f = \sum_{j \geq 0} \frac{(-t)^j}{j!} D^{2j} f, \quad D = \frac{d}{dz}, \quad P_{2k}(-a, z) = T_{-a}(H_0^{2k})(z). \quad (\text{ETC7})$$

The series for this actual function will be shown absolutely convergent; no formal heat operator is substituted for it. For compatibility with the programme domain, $H_0^{2k} \in \mathcal{E}$, where $\mathcal{E} = \{f \text{ entire} : \sup_z |f(z)| e^{-\alpha|z|^2} < \infty \text{ for all } \alpha > 0\}$. In fact $|z|u \leq b|z|^2 + u^2/(4b)$ gives $|H_0(z)| \leq e^{b|z|^2} I_0(1/(4b), 0)$. Take $b = \alpha/(2k)$ and raise to the exact power $2k$. This is the same domain and heat sign as HTP2 and HTP5.

Set $\psi(s) = \Phi(|s|)/2$ for $s \in \mathbb{R}$. The evenness of the cosine and absolute integrability give

$$H_0(z) = \int_{\mathbb{R}} \psi(s) e^{izs} ds, \quad (\text{ETC8})$$

$$H_0(z)^{2k} = \int_{\mathbb{R}^{2k}} e^{izS} \prod_{j=1}^{2k} \psi(s_j) d\mathbf{s}, \quad S = \sum_{j=1}^{2k} s_j.$$

These are absolutely convergent integrals, since $|S| \leq \sum_j |s_j|$. Differentiation of any finite order under them is justified, for example, by absorbing $|S|^N$ into a constant times $e^{|S|}$ and using ETC3. For the absolute sum of the entire heat series on $|z| \leq R$ one has the integrable majorant

$$\begin{aligned} & \int_{\mathbb{R}^{2k}} e^{R|S|} \sum_{j \geq 0} \frac{\alpha^j S^{2j}}{j!} \prod_{\ell=1}^{2k} \psi(s_\ell) d\mathbf{s} \\ &= \int_{\mathbb{R}^{2k}} e^{aS^2 + R|S|} \prod_{\ell=1}^{2k} \psi(s_\ell) d\mathbf{s} \leq I_0(2ka, R)^{2k} < \infty. \end{aligned} \quad (\text{ETC9})$$

Here $S^2 \leq 2k \sum_{\ell} s_{\ell}^2$ and $|S| \leq \sum_{\ell} |s_{\ell}|$ give the last inequality. The derivative $D^{2j} e^{izS} = (-1)^j S^{2j} e^{izS}$ now proves, with every exchange absolutely justified,

$$P_{2k}(-a, z) = \int_{\mathbb{R}^{2k}} e^{-aS^2 + izS} \prod_{\ell=1}^{2k} \psi(s_{\ell}) d\mathbf{s}. \quad (\text{ETC10})$$

This also verifies the original negative-time sign.

The real Gaussian identity, with its exact factor, is

$$\frac{1}{\sqrt{4\pi a}} \int_{\mathbb{R}} e^{-x^2/(4a)} e^{iSx} dx = e^{-aS^2} \quad (S \in \mathbb{R}). \quad (\text{ETC11})$$

One direct proof is to differentiate the left integral in S . Integration by parts, with the Gaussian boundary term zero, gives $J'(S) = -2aS J(S)$. At $S = 0$ its value is 1, by squaring the real Gaussian integral and using polar coordinates. The differential equation gives ETC11. All differentiations are dominated by a polynomial times the same Gaussian. Substituting ETC8 into the following real Gaussian integral and using ETC11 proves

$$P_{2k}(-a, z) = \frac{1}{\sqrt{4\pi a}} \int_{\mathbb{R}} e^{-x^2/(4a)} H_0(z+x)^{2k} dx. \quad (\text{ETC12})$$

The absolute multiple integral before the substitution is at most $I_0(0, |\operatorname{Im} z|)^{2k}$, since $|e^{ixS}| = 1$ and the Gaussian measure in ETC11 has mass one. Thus this Fubini interchange is independent of a formal complex Gaussian manipulation.

We now prove the required contour shift in full. Write $z = \xi + i\eta$ and put $F_z(w) = e^{-(z-w)^2/(4a)} H_0(w)^{2k}$, an entire function of w . Formula ETC12 is its integral along $\mathbb{R} + i\eta$, oriented with increasing real part, divided by $\sqrt{4\pi a}$; the real displacement ξ does not change the complete horizontal line. On the closed strip between heights 0 and η , ETC4 gives $|H_0(w)| \leq I_0(0, |\eta|)$. For $|z| \leq R$, $M > R$, and $w = \pm M + iy$ on either vertical edge of that strip,

$$|F_z(\pm M + iy)| \leq I_0(0, R)^{2k} \exp\left(\frac{R^2}{4a} - \frac{(M-R)^2}{4a}\right), \quad y \text{ between } 0 \text{ and } \eta. \quad (\text{ETC13})$$

Indeed $\operatorname{Re}(z-w)^2 = (\xi \mp M)^2 - (\eta - y)^2$, $|\xi \mp M| \geq M - R$, and $|\eta - y| \leq R$. Each edge has length at most R , so both vertical integrals tend to zero, uniformly on $|z| \leq R$ for each fixed k . Both horizontal integrals converge absolutely: on the real line $|H_0(v)| \leq h$, and on the other line its bound is $I_0(0, R)$, while the real part of the Gaussian supplies quadratic decay in the horizontal coordinate. Applying Cauchy's theorem to this rectangle and letting M tend to infinity, with the same orientation on the two horizontal lines, proves the entire-function identity

$$\boxed{P_{2k}(-a, z) = \frac{1}{\sqrt{4\pi a}} \int_{\mathbb{R}} e^{-(z-v)^2/(4a)} H_0(v)^{2k} dv.} \quad (\text{ETC14})$$

For a zero-height strip the equality is immediate. Negative η gives the same equality by reversing the rectangle's orientation. Thus no sign or half-plane is omitted.

37.3 Quantitative local concentration and the outer gap

Define constants from the original moments by

$$\beta = \frac{m_2}{h} > 0, \quad c = \frac{m_4}{24h} > 0, \quad \delta = \min\left\{\beta^{-1/2}, \sqrt{\frac{\beta}{4c}}\right\} > 0, \quad D_0 = \max\left\{c, \frac{\beta^2}{4}\right\}. \quad (\text{ETC15})$$

These are fixed constants of H_0 ; δ is a proved choice, not an assumed Taylor neighborhood. The inequalities $\cos x \geq 1 - x^2/2$ and $|\cos x - 1 + x^2/2| \leq x^4/24$ give

$$0 \leq \frac{H_0(v)}{h} - 1 + \frac{\beta v^2}{2} \leq cv^4 \quad (v \in \mathbb{R}). \quad (\text{ETC16})$$

The first cosine inequality follows from $1 - \cos x = 2 \sin^2(x/2) \leq x^2/2$; the second is Taylor's integral remainder with the fourth derivative bounded by 1. Integrating against the positive original kernel proves ETC16, with no unknown remainder constant. For $|v| \leq \delta$ it follows that

$$\frac{1}{2} \leq 1 - \frac{\beta v^2}{2} \leq \frac{H_0(v)}{h} \leq 1 - \frac{\beta v^2}{4} \leq 1, \quad \log \left(\frac{H_0(v)}{h} \right) \leq -\frac{\beta v^2}{4}, \quad \left| \log \left(\frac{H_0(v)}{h} \right) + \frac{\beta v^2}{2} \right| \leq D_0 v^4. \quad (\text{ETC17})$$

For the last bound write $e(v) = \frac{H_0(v)}{h} - 1 + \beta v^2/2 \in [0, cv^4]$ and $y = 1 - \frac{H_0(v)}{h} = \beta v^2/2 - e(v) \in [0, 1/2]$. The convergent real logarithm series yields

$$\log \left(\frac{H_0(v)}{h} \right) + \frac{\beta v^2}{2} = e(v) - \sum_{n \geq 2} \frac{y^n}{n}, \quad 0 \leq \sum_{n \geq 2} \frac{y^n}{n} \leq \frac{y^2}{2(1-y)} \leq \frac{\beta^2 v^4}{4}.$$

The difference therefore lies between $-\beta^2 v^4/4$ and cv^4 , proving the maximum in ETC15. The bound on $\log(H_0(v)/h)$ follows also from $\log(1-y) \leq -y$ and $y \geq \beta v^2/4$.

The exact outer gap is

$$\rho = \sup_{|v| \geq \delta} \left| \frac{H_0(v)}{h} \right|, \quad \frac{1}{2} \leq \rho < 1, \quad \lambda = -2 \log \rho > 0. \quad (\text{ETC18})$$

Indeed $\frac{H_0(\delta)}{h} \geq 1/2$ by ETC17. By ETC6 there is M so large that $|\frac{H_0(v)}{h}| < 1/4$ for $|v| > M$. Hence the supremum equals a maximum on the compact set $\delta \leq |v| \leq M$. Each point of this set has modulus strictly less than 1 by ETC5, so its attained maximum is less than 1. This proves every assertion in ETC18 and defines an actual kernel constant, rather than postulating a uniform outer gap.

For a fixed complex disc $|z| \leq R$, $R \geq 0$, put

$$g_z(v) = e^{-v^2/(4a)} \cosh \left(\frac{zv}{2a} \right), \quad G_R = \frac{1}{4a} + \frac{R^2}{8a^2} \cosh \left(\frac{R\delta}{2a} \right), \quad L_R = \frac{6\sqrt{2}D_0}{\beta^2} + \frac{\sqrt{2}G_R}{\beta}. \quad (\text{ETC19})$$

For real v with $|v| \leq \delta$,

$$|g_z(v) - 1| \leq G_R v^2, \quad \left| \left(\frac{H_0(v)}{h} \right)^{2k} - e^{-\beta k v^2} \right| \leq 2k D_0 v^4 e^{-\beta k v^2/2}. \quad (\text{ETC20})$$

For the first inequality use $1 - e^{-x} \leq x$ for $x \geq 0$ and $|\cosh w - 1| \leq (|w|^2/2) \cosh |w|$, which follows term by term from the absolutely convergent series. For the second, the two real exponents $2k \log \left(\frac{H_0(v)}{h} \right)$ and $-\beta k v^2$ differ by at most $2k D_0 v^4$ and are both at most $-\beta k v^2/2$. The integral form of the mean value formula for the exponential proves the second inequality.

Evenness of H_0^{2k} and absolute Gaussian integrability in ETC14 permit averaging v and $-v$, so that exactly

$$P_{2k}(-a, z) = \frac{h^{2k} e^{-z^2/(4a)}}{\sqrt{4\pi a}} J_k(z), \quad J_k(z) = \int_{\mathbb{R}} g_z(v) \left(\frac{H_0(v)}{h} \right)^{2k} dv, \quad M_k = \int_{\mathbb{R}} e^{-\beta k v^2} dv = \sqrt{\frac{\pi}{\beta k}}. \quad (\text{ETC21})$$

In particular the even complex weight has no linear term. On $|v| \leq \delta$ the exact decomposition

$$g_z(v) \left(\frac{H_0(v)}{h} \right)^{2k} - e^{-\beta k v^2} = \left(\left(\frac{H_0(v)}{h} \right)^{2k} - e^{-\beta k v^2} \right) + (g_z(v) - 1) \left(\frac{H_0(v)}{h} \right)^{2k}$$

and ETC17, ETC20 bound the integral of its absolute value by the whole-real-line integral of $(2kD_0v^4 + GRv^2)e^{-\beta k v^2/2}$. The Gaussian moments $\int_{\mathbb{R}} v^2 e^{-bv^2} dv = \sqrt{\pi}/(2b^{3/2})$ and $\int_{\mathbb{R}} v^4 e^{-bv^2} dv = 3\sqrt{\pi}/(4b^{5/2})$ follow by differentiating the Gaussian integral in $b > 0$; the derivatives are absolutely dominated on each compact positive b interval. Substitution of $b = \beta k/2$ and division by M_k give exactly

$$\frac{1}{M_k} \left| \int_{-\delta}^{\delta} (g_z(v) \left(\frac{H_0(v)}{h} \right)^{2k} - e^{-\beta k v^2}) dv \right| \leq \frac{L_R}{k}. \quad (\text{ETC22})$$

The omitted Gaussian tail satisfies

$$\frac{1}{M_k} \int_{|v|>\delta} e^{-\beta k v^2} dv \leq e^{-\beta \delta^2 k}. \quad (\text{ETC23})$$

On each half-line set $|v| = \delta + x$, $x \geq 0$, and use $(\delta + x)^2 \geq \delta^2 + x^2$. The remaining two half-Gaussian integrals sum to M_k . For the actual outer integral, the series gives $|\cosh(zv/(2a))| \leq \cosh(Rv/(2a))$ for real v , where the latter is even and positive. ETC18 therefore gives, by completing the square in each of the two real exponentials making up the hyperbolic cosine,

$$\begin{aligned} \frac{1}{M_k} \left| \int_{|v|>\delta} g_z(v) \left(\frac{H_0(v)}{h} \right)^{2k} dv \right| &\leq \frac{\rho^{2k}}{M_k} \int_{\mathbb{R}} e^{-v^2/(4a)} \cosh\left(\frac{Rv}{2a}\right) dv \\ &= 2\sqrt{a\beta} e^{R^2/(4a)} \sqrt{k} \rho^{2k}. \end{aligned} \quad (\text{ETC24})$$

No decay rate for $H_0(v)$ beyond ETC6 was used; the Gaussian supplies the absolute outer integrability.

37.4 Uniform relative error and the exact nonvanishing consequence

Combining ETC21–24 proves the following quantitative bound for every $k \geq 1$ and every $|z| \leq R$:

$$\boxed{\begin{aligned} &\left| \frac{P_{2k}(-a, z)}{h^{2k} e^{-z^2/(4a)} \sqrt{h/(4akm_2)}} - 1 \right| \leq E_{k,R}, \\ &E_{k,R} = \frac{L_R}{k} + e^{-\beta \delta^2 k} + 2\sqrt{a\beta} e^{R^2/(4a)} \sqrt{k} \rho^{2k}. \end{aligned}} \quad (\text{ETC25})$$

All denominators here are nonzero: $h, a, m_2, k > 0$ and the complex exponential never vanishes. In particular the factor in the denominator is exactly the factor obtained from ETC21; its π cancels between the two Gaussian integrals, with no change of the original a or moments.

An explicit relative $O(1/k)$ constant is

$$\boxed{C_R = L_R + \frac{1}{e\beta\delta^2} + 2\sqrt{a\beta} e^{R^2/(4a)} \left(\frac{3}{2e\lambda} \right)^{3/2}, \quad E_{k,R} \leq \frac{C_R}{k}.} \quad (\text{ETC26})$$

Here e is the exponential constant. For $x > 0$, the maximum of xe^{-bx} is $1/(eb)$ for $b > 0$, by direct differentiation. Similarly the maximum of $x^{3/2}e^{-\lambda x}$ is $(3/(2e\lambda))^{3/2}$. Apply the first with $b = \beta\delta^2$ to $ke^{-\beta\delta^2 k}$ and the second to $k^{3/2}\rho^{2k} = k^{3/2}e^{-\lambda k}$. This proves ETC26 for every positive integer k .

Consequently the actual even tensor has the locally uniform complex asymptotic

$$P_{2k}(-a, z) = h^{2k} e^{-z^2/(4a)} \sqrt{\frac{h}{4akm_2}} (1 + \varepsilon_k(z)), \quad \sup_{|z| \leq R} |\varepsilon_k(z)| \leq \frac{C_R}{k}. \quad (\text{ETC27})$$

The statement is uniform in z on each specified disc, with $a > 0$ fixed; it does not assert uniformity as a tends to zero. It gives the explicit lower bound

$$\inf_{|z| \leq R} |P_{2k}(-a, z)| \geq h^{2k} \sqrt{\frac{h}{4akm_2}} e^{-R^2/(4a)} \left(1 - \frac{C_R}{k}\right) > 0 \quad (k > C_R). \quad (\text{ETC28})$$

Indeed $|e^{-z^2/(4a)}| = e^{-\text{Re}(z^2)/(4a)} \geq e^{-R^2/(4a)}$ on this disc, and $|1 + \varepsilon_k(z)| \geq 1 - C_R/k$. Thus every fixed disc is free of zeros of $P_{2k}(-a, \cdot)$ for all integers $k > C_R$. This is an unconditional statement about the actual sequence of functions in ETC7.

In particular, retaining the actual root set of the original negative-time function, one obtains

$$\{z : |z| \leq R, H_{-a}(z) = 0, P_{2k}(-a, z) = 0\} = \emptyset \quad (k > C_R). \quad (\text{ETC29})$$

This follows directly from ETC28 and neither presupposes nor produces a nonreal root of H_{-a} . Any actual fixed root of H_{-a} , real or nonreal, is covered by taking a disc containing it. The functions $P_{2k}(-a, \cdot)$ are the heat transforms of the powers H_0^{2k} ; no argument here replaces them by the ordinary powers H_{-a}^{2k} . Their quantitatively proved nonvanishing in ETC28 therefore does not assert nonvanishing of H_{-a} or of H_0 .

A prime distinguishes the actual theta lift from its synchronization

$$\mathcal{A}_L \rightarrow L[C_2] \rightarrow \mathcal{P}(C_2), \quad (f, v) \mapsto (v_0, v_1)$$

Full lattice product: $(a, b)(c, d) = (a \wedge c \vee b \wedge d, a \wedge d \vee b \wedge c)$

$\chi: L \rightarrow \mathbb{B}$ is a stated receiver map; all labels remain in the preceding full-L construction.

Neither parity	Even only	Odd only	Both parities
$\emptyset$	$\{0\}$	$\{1\}$	$\{0, 1\}$
$(0, 0)$	$(1, 0)$	$(0, 1)$	$(1, 1)$
in the prime	outside the prime	outside the prime	in the prime

$$\mathfrak{p}_{\text{mix}} = \{\emptyset, C_2\}, \quad \{1\} \cdot \{1\} = \{0\}, \quad C_2 \cdot A = C_2 \quad (A \neq \emptyset)$$

This is an ordinary semiring prime ideal. It is not a downward or subtractive ideal.

The two original lifts have the same function and different prime membership

$$\mathbf{H}_t = (H_t, (1_L, 0_L, 1_L, 0_L, \dots)) \mapsto \{0\} \notin \mathfrak{p}_{\text{mix}}$$

$$\mathbf{J}\mathbf{H}_t = (H_t, (1_L, 1_L, 1_L, 1_L, \dots)) \mapsto C_2 \in \mathfrak{p}_{\text{mix}}$$

$$\mathbf{H}_t^{*t^k} = (T_t(H_0^k), (1_L, 0_L, 1_L, 0_L, \dots)) \quad (k \geq 1)$$

Every tensor power keeps the same distinction. Inverting J removes this prime.

Proofs: SHP17–28, HTK24–32 and the complete HFP prime-spectrum calculation.

The diagram shows the exact four-element parity receiver, not an exhaustive picture of the full spectrum.

Kernel: Brad Rodgers & Terence Tao, arXiv:1801.05914v5. Full carrier: The Clankers, v11, def:lattice-split.

Figure 15: The exact parity receiver and a prime detecting synchronization of the original theta tensor. This four-element receiving semiring is a specified image of the full lattice algebra. The full spectrum, its infinite prime chains and its full-lattice stalks are proved in HFP1–39.

38 The actual prime spectrum removed by synchronization of heat flags

Sources and scope. The full carrier is the foundation retained under the byline The Clankers, *Split Support Geometry*, v11, Definition `def:lattice-split`. Its complete support localization is proved in 031, Theorems `thm:rebuilt-full-support` and `thm:e-stalk`. Here the objects actually used are SHP1–28 of `SUPPORTED_HEAT_PRODUCT.tex`, read in full, and the tensor support maps HTK23–32 of `FULL_SUPPORT_HEAT_TENSOR.tex`. Their original theta kernel is Brad Rodgers and Terence Tao’s *The De Bruijn–Newman constant is non-negative*, original equations `phidef` and `htdef`. The spectrum computations below are independent derivations for those actual algebras. They use ordinary semiring prime ideals throughout; whenever a stronger ideal condition is used it is stated and proved. No existence or absence of a nonreal zero is assumed.

38.1 The original algebra and which prime ideals its localization retains

Let L be any nontrivial bounded distributive lattice. Retain

$$\mathcal{E} = \left\{ \sum_{n \geq 0} a_n z^n : \sum_{n \geq 0} |a_n| \sqrt{n!} h^n < \infty \ (h > 0) \right\},$$

$$\mathcal{F}(L) = \{\nu \in L^{\mathbb{N}_0} : \nu_i \geq \nu_{i+2}\}, \quad \mathcal{A}_L = \{(f, \nu) : f \in \mathcal{E}, \nu \in \mathcal{F}(L), a_i \neq 0 \Rightarrow \nu_i = 1_L\}. \quad (\text{HFP1})$$

Addition is $(f, \nu) + (h, \mu) = (f + h, \nu \vee \mu)$. The ordinary amplitude product and the exact flag product are

$$(f, \nu) \circ (h, \mu) = (fh, \nu \otimes \mu),$$

$$(\nu \otimes \mu)_0 = (\nu_0 \wedge \mu_0) \vee (\nu_1 \wedge \mu_1), \quad (\nu \otimes \mu)_n = \bigvee_{r=0}^n \nu_r \wedge \mu_{n-r} \quad (n \geq 1), \quad (\text{HFP2})$$

$$\mathbf{u} = (1_L, 0_L, 0_L, \dots), \quad 1_{\mathcal{A}} = (1, \mathbf{u}), \quad e_0 = (0, \mathbf{u}), \quad \mathbf{J} = (1_L, 1_L, \dots), \quad J = (1, \mathbf{J}),$$

$$\epsilon(\nu) = \nu_0 \vee \nu_1, \quad g(f, \nu) = (f, \epsilon(\nu)), \quad j(f, \lambda) = (f, (\lambda, \lambda, \dots)), \quad Jv = jg(v).$$

These are SHP5, SHP17–24, including $J^2 = J$, $\mathcal{A}_L[J^{-1}] \simeq G_L(\mathcal{E})$ and $\mathcal{A}_L[e_0^{-1}] \simeq \mathcal{F}(L)$.

An ideal below means a subset containing the additive zero, closed under addition, and absorbing multiplication by every semiring element. It is prime when it is proper and ab in the ideal forces one factor into it. An ideal is *subtractive* when $a, a + b \in I$ implies $b \in I$. This additional property is not part of the definition of prime used here. Write $D(c) = \{\mathfrak{p} : c \notin \mathfrak{p}\}$ and $V(c) = \{\mathfrak{p} : c \in \mathfrak{p}\}$.

Every prime in $D(J)$ is one of the following actual ideals:

$$\mathfrak{a}_{\mathfrak{p}} = \{(f, \nu) : f \in \mathfrak{p}\}, \quad \mathfrak{p} \in \text{Spec } \mathcal{E},$$

$$\mathfrak{s}_{\mathfrak{q}} = \{(0, \nu) : \nu_0 \vee \nu_1 \in \mathfrak{q}\}, \quad \mathfrak{q} \in \text{SpecLat } L. \quad (\text{HFP3})$$

Here $\text{SpecLat } L$ uses proper lattice ideals prime for meet. The correspondence is exactly the homeomorphism

$$g^* : \text{Spec } G_L(\mathcal{E}) \xrightarrow{\sim} D(J), \quad \mathfrak{r} \longmapsto g^{-1}\mathfrak{r}. \quad (\text{HFP4})$$

To check it without concealing an ideal condition, let P be prime and exclude J . If $g(v) = g(w)$, then $Jv = Jw$, and primeness gives $v \in P \iff w \in P$. Thus $g(P)$ is an ideal and its inverse image is exactly P . It is proper: a preimage in P of the target unit would give $J \in P$. Surjectivity of g and the same membership test prove that $g(P)$ is prime. Conversely the inverse image of a prime excludes J , since $g(J) = 1$. The assignments are inverse. Basic opens satisfy $g^*(D(g(v))) = D(v) \cap D(J)$, proving the topology assertion.

For completeness, the primes of $G_L(R)$ for a nonzero ring R have exactly two forms. If $e = (0, 1_L)$ belongs to the prime, then all labelled zeros belong to it. Its top-supported amplitudes form a ring prime: multiplication by $(-1, 1_L)$ supplies additive negatives, and the other ideal and prime laws follow by the amplitude operations. Conversely each ring prime gives this ideal. If e is excluded, no nonzero-amplitude element can belong, since multiplying such an element by e gives e . The labels of the remaining zeros form a proper lattice ideal prime for meet; absorption makes it downward closed. Conversely such a lattice ideal gives a semiring prime of labelled zeros. Applying this calculation to $R = \mathcal{E}$ proves every assertion in (HFP3).

There is an exact distinction stronger than just existence of this open set:

$$D(J) = \{\text{subtractive prime ideals of } \mathcal{A}_L\}, \quad V(J) = \{\text{non-subtractive prime ideals of } \mathcal{A}_L\}. \quad (\text{HFP5})$$

Indeed both types in (HFP3) are subtractive. For the arithmetic type this is subtraction in the ring prime. For the support type, two zero amplitudes imply the third amplitude is zero, and the downward lattice ideal recovers the label of either summand from their join. Their inverse images under g are subtractive. Conversely put $c = (-1, \mathbf{u})$. The exact identity

$$1_{\mathcal{A}} + Jc = e_0J \quad (\text{HFP6})$$

has amplitude $1 - 1 = 0$ and constant top support on both sides. Any ideal containing J contains Jc and e_0J . If it were subtractive, (HFP6) would put the unit in it. This proves (HFP5). It also proves that the congruence imposing $J = 0$ is the total congruence: it sets $Jc = e_0J = 0$ in (HFP6), hence $1 = 0$. Therefore the nonempty residual spectrum constructed below is not the spectrum of a nontrivial congruence quotient $J = 0$. Its ordinary prime ideals must not be discarded by imposing subtractive ideals after the calculation.

The other localization gives the additional exact open set

$$\text{Spec } \mathcal{F}(L) \xrightarrow{\sim} D(e_0), \quad P \longmapsto \{(f, \nu) : \nu \in P\}. \quad (\text{HFP7})$$

The proof is the one for (HFP4), replacing Jv by $e_0v = (0, \nu)$. In particular its part lying in $V(J)$ is exactly the spectrum of flags whose primes contain $\mathbf{J}$.

In an additively idempotent semiring, subtractive ideals are exactly ideals downward closed for its natural order: $b \leq a$ means $a + b = a$, so subtractivity implies downward closure; conversely $b \leq a + b$ gives subtractivity from downward closure. The lower prime ideals of both $\mathcal{F}(L)$ and its finite-support subsemiring are exactly

$$\epsilon^{-1}(\mathfrak{q}), \quad \mathfrak{q} \in \text{SpecLat } L. \quad (\text{HFP8})$$

Here is a direct proof valid even for an infinite flag. For any flag ν , $(\nu \otimes \nu)_0 = \epsilon(\nu)$, so $\epsilon(\nu)\mathbf{u} \leq \nu \otimes \nu$. If a lower prime contains ν , it therefore contains $\epsilon(\nu)\mathbf{u}$. Conversely $(\epsilon(\nu)\mathbf{u}) \otimes \nu = \nu$, so absorption gives the converse. Its intersection with the scalar copy $\lambda \mapsto \lambda\mathbf{u}$ is a lattice prime; this proves (HFP8). No claim classifying all flag primes has been made using this stronger lower-ideal property.

38.2 The exact residual ideal and explicit primes for every lattice

For a proper lattice prime $\mathfrak{q}$, define its exact Boolean character

$$\chi_{\mathfrak{q}} : L \longrightarrow \mathbb{B}, \quad \chi_{\mathfrak{q}}(\lambda) = \begin{cases} 0 & \lambda \in \mathfrak{q}, \\ 1 & \lambda \notin \mathfrak{q}. \end{cases} \quad (\text{HFP9})$$

It preserves join, meet and both bounds by the ideal and prime properties. Such characters exist for every nontrivial L and separate its elements. To verify the latter assertion, when $a \not\leq b$ choose by Zorn an ideal containing b and disjoint from the filter $\uparrow a$, maximal with those properties. If

x, y are excluded but $x \wedge y$ belongs, the ideals generated by adjoining x and by adjoining y each meet that filter. Thus $a \leq i \vee x$ and $a \leq j \vee y$ for suitable i, j in the ideal (replace filter witnesses by their lower bound a). Distributivity gives $a \leq i \vee j \vee (x \wedge y)$ in the ideal, a contradiction. This is a prime excluding a and containing b . Taking $a = 1_L, b = 0_L$ proves existence; using either direction of an unequal pair proves separation. Only the character receivers are Boolean: the source lattice and every original label remain L .

The ideal generated by synchronization and its radical are

$$\begin{aligned} (J) &= J\mathcal{A}_L = jG_L(\mathcal{E}), \\ \sqrt{(J)} &= \{(f, \nu) : \nu \otimes \nu \text{ is the constant flag } \epsilon(\nu)\}, \\ (f, \nu) \in \sqrt{(J)} &\iff \bigvee_{r=0}^n (\nu_r \wedge \nu_{n-r}) = \nu_0 \vee \nu_1 \quad \text{for every } n \geq 1. \end{aligned} \tag{HFP10}$$

The first equality holds because a finite sum of Jv_i is $J(\sum v_i)$; SHP20 identifies its entire image. The last equality is the explicit square formula (HFP2), whose degree-zero entry already equals $\epsilon(\nu)$.

It remains to prove that any power entering (J) forces the square to enter. Apply an arbitrary character (HFP9) to the flag. A Boolean parity flag is either zero, has finite support, has just one parity present, or is mixed with infinite support. A nonzero finite flag stays finite under every power. A nonzero one-parity flag stays one-parity under every power; its parity may change, but it cannot become the constant top flag. A mixed infinite flag has constant-top square: if its even tail is infinite, the even-even and even-odd products supply the two infinite output tails; if its odd tail is infinite, the odd-odd and odd-even products do so. A zero flag has constant-zero square. These are all possibilities. Thus a power of a Boolean flag is constant exactly when its square is constant. If a power of ν is constant, its constant label must be $\epsilon(\nu)$, because ϵ is multiplicative and the lattice product is idempotent. Each Boolean character therefore makes the square equal to that same constant. Character separation proves the equality in L at every coordinate. This establishes (HFP10), with exponent two, for the full arbitrary lattice.

Every prime containing J contains this radical, and conversely every prime containing the radical contains J , so $V(J) = V(\sqrt{(J)})$. This statement uses ordinary ideals and does not pass to the total congruence from (HFP6).

There are two particularly explicit receivers. Let $\mathcal{P}(C_2)$ be the power-set semiring of the group $C_2 = \{\bar{0}, \bar{1}\}$, with union as addition, setwise group addition as multiplication, empty set as zero and $\{\bar{0}\}$ as unit. Define

$$\pi_q(f, \nu) = \{\bar{i} : i \in \{0, 1\}, \nu_i \notin q\} \in \mathcal{P}(C_2). \tag{HFP11}$$

This is onto and unital. Addition follows by character preservation of joins. The even and odd heads of a product are respectively $(\chi\nu_0\chi\mu_0) \vee (\chi\nu_1\chi\mu_1)$ and $(\chi\nu_0\chi\mu_1) \vee (\chi\nu_1\chi\mu_0)$, exactly the group-set product. Onto follows using zero amplitudes and head labels $0_L, 1_L$ in degrees zero and one.

The two-element subset $\{\emptyset, C_2\}$ is a prime ideal of this four-element semiring. Its sum and products with any set remain inside it. Its complement consists of the two singleton sets and is multiplicatively closed. Hence

$$P_q^{\text{mix}} := \pi_q^{-1}\{\emptyset, C_2\} = \{(f, \nu) : \chi_q(\nu_0) = \chi_q(\nu_1)\} \tag{HFP12}$$

is an ordinary prime. It contains J and excludes $1_{\mathcal{A}}$ and e_0 . The condition includes both parities absent, because zero must belong to an ideal. It is not a lower or subtractive prime; the two singleton sets lie outside but their sum lies inside.

The second receiver uses the entire degree, including infinity:

$$d_q(\nu) = \sup\{n \geq 0 : \nu_n \notin q\} \in \{-\infty\} \cup \mathbb{N}_0 \cup \{\infty\}. \tag{HFP13}$$

The supremum of the empty set is $-\infty$. Equip the target with maximum as addition and degree addition as multiplication, where $-\infty$ is absorbing, including $(-\infty) + \infty = -\infty$. Its unit is degree zero. The map d_q is a unital semiring homomorphism. For two nonzero finite Boolean flags the highest product degree is the sum of their highest degrees, and that summand is present. If one has unbounded degrees and the other has a present degree, their product has unbounded degrees. If one is empty, the product is empty. These prove every case of multiplicativity; maximum under addition is immediate. The set $\{-\infty, \infty\}$ is a proper prime ideal: its complement consists exactly of finite nonnegative degrees and is multiplicatively closed. Thus

$$P_q^\infty = \{(f, \nu) : d_q(\nu) = -\infty \text{ or } d_q(\nu) = \infty\} \quad (\text{HFP14})$$

is another prime containing J and excluding e_0 . It differs from (HFP12): the finite flag $(1_L, 1_L, 0_L, \dots)$ lies in the mixed prime and not the infinite-degree prime, while $(1_L, 0_L, 1_L, 0_L, \dots)$ lies in the infinite-degree prime and not the mixed prime. Both may be supplied with amplitude zero, so the examples belong to the exact carrier for every L .

38.3 An amplitude-sensitive degree receiver, including finite flags

The preceding primes exclude e_0 . There are also residual primes containing e_0 , and an exact receiver constructs them. Define a semiring $\mathcal{T}_\mathbb{C}$ with elements

$$\{\perp, \Omega\} \sqcup (\mathbb{C} \times \mathbb{N}_0). \quad (\text{HFP15})$$

Its additive zero is $\perp$, its unit is $(1, 0)$, and Ω absorbs addition. For finite-level elements set

$$(c, n) + (b, m) = \begin{cases} (c, n) & n > m, \\ (b, m) & m > n, \\ (c + b, n) & m = n, \end{cases} \quad (c, n)(b, m) = (cb, n + m), \quad (\text{HFP16})$$

$$\perp x = \perp, \quad \Omega x = \Omega \quad (x \neq \perp).$$

In particular $(0, n)$ is never identified with $\perp$, and $\Omega(0, n) = \Omega$. Associativity of addition follows by selecting the largest level and summing all coefficients at that level; associativity of multiplication uses the complex product and integer sum. For distributivity, unequal summand levels retain the larger level after multiplication, and equal levels use $c(b + d) = cb + cd$. The cases involving Ω use the fact that a sum of two non- $\perp$ elements never equals $\perp$. This proves all semiring laws, including cancellation of coefficients to the retained element $(0, n)$.

There is an onto unital map preserving the actual coefficient at the last visible support degree:

$$\ell_q(f, \nu) = \begin{cases} \perp & d_q(\nu) = -\infty, \\ (a_d, d) & d = d_q(\nu) \in \mathbb{N}_0, \\ \Omega & d_q(\nu) = \infty. \end{cases} \quad (\text{HFP17})$$

To prove it, if $d < \infty$, every amplitude coefficient above d vanishes: a nonzero such coefficient would force its label to top and hence be seen by χ_q . If $d = -\infty$, all amplitude coefficients vanish. Consequently for a sum of distinct finite support degrees the larger leading coefficient is unchanged, while equal degrees give its exact complex sum, including zero. For a product of finite support degrees d, m , its coefficient in degree $d + m$ is exactly $a_d b_m$, because the amplitudes have no higher degrees. The support degree calculation is (HFP13). The infinite and empty cases give precisely the Ω and $\perp$ rules in (HFP16). This proves the map. Every (c, n) is received from amplitude cz^n and the parity-closed top support of that monomial, retaining that support even when $c = 0$; Ω is received from the zero amplitude with infinite even support. Thus it is onto.

The explicit prime ideal and its pullback are

$$\mathfrak{G} = \{\perp, \Omega\} \cup \{(0, n) : n \geq 0\}, \quad P_q^{\text{lead}} = \ell_q^{-1}(\mathfrak{G}). \quad (\text{HFP18})$$

Closure and absorption follow directly from (HFP16). Its complement is all (c, n) with $c \neq 0$, and products remain in the complement because $\mathbb{C}$ is a field. Thus it is prime. Its pullback contains both J and e_0 and excludes the unit. This explicitly constructs part of $V(J) \cap V(e_0)$, in addition to the primes in (HFP12)–(HFP14). No classification of all primes in this closed part is inferred from these constructed examples.

Let $\mathcal{A}_{L,\text{fin}}$ be the subsemiring of finite flags; its amplitudes are actual polynomials. Products and sums stay finite by (HFP2). The restriction of (HFP17) takes values in $\mathcal{T}_{\mathbb{C}} \setminus \{\Omega\}$ and the finite leading-zero ideal stays prime. For the exact element

$$v = (1, (1_L, 0_L, 1_L, 0_L, 0_L, \dots)) \in \mathcal{A}_{L,\text{fin}}, \quad \ell_{\mathbf{q}}(v) = (0, 2), \quad g(v) = (1, 1_L), \quad (\text{HFP19})$$

the prime contains v but excludes the unit, although g sends them to the same element. Thus synchronization already loses an actual prime distinction on finite flags. The infinite idempotent J itself is absent from this subsemiring; one must not describe the finite restriction as localization at an element that it does not contain. The finite restricted map onto $G_L(\mathbb{C}[z])$ is nevertheless explicit: take the parity closure of a polynomial's nonzero coefficient labels, and use a constant-degree label for a zero amplitude. This proves surjectivity and identifies its actual target.

38.4 The exact relation of every receiver to the full coefficient map

Retain K_L , its injection ι_L and retraction ρ_L from SHP15–16. The complete coefficient map is

$$\Phi(f, \nu) = (f, (\iota_L(\nu_i))_{i \geq 0}) \in \mathcal{E} \times \mathcal{F}(K_L). \quad (\text{HFP20})$$

The support homomorphism on the whole receiving semiring

$$(f, w) \mapsto (\chi_{\mathbf{q}}(\rho_L(w_i)))_{i \geq 0} \in \mathcal{F}(\mathbb{B}) \quad (\text{HFP21})$$

is unital and multiplicative by the finite support formula. Composing it with the parity-head and degree receivers gives exactly (HFP11) and (HFP13) after Φ . Their prime ideals are therefore inverse images of actual prime ideals under the complete map, not artefacts of discarding the lattice coordinate. The retraction recovers all labels used in the calculation.

The amplitude-sensitive map (HFP17) is defined on the exact constrained image of Φ by applying (HFP21) to find d and taking $[z^d]f$ when it is finite. Its proof uses the original carrier condition to show the absence of coefficients above d . Since Φ is an isomorphism onto its image, this gives its full commuting receiver map there. No extension of that leading-coefficient formula to an unconstrained pair (f, w) is asserted. The unrestricted product target can have coefficients beyond the displayed support, for which that formula would not preserve multiplication. The constrained image and the map just constructed retain the needed exact relation.

38.5 Complete finite and infinite Boolean flag spectra as explicit witnesses

The following specialization classifies all ordinary flag primes for $L = \mathbb{B}$; it does not replace arbitrary L . For a Boolean flag write its last even and odd supported degrees as

$$e \in \{-\infty, 0, 2, 4, \dots, \infty\}, \quad o \in \{-\infty, 1, 3, 5, \dots, \infty\}. \quad (\text{HFP22})$$

An absent parity has endpoint $-\infty$, an unbounded parity has endpoint ∞ . These endpoints determine the flag. For endpoint addition below, $-\infty$ is absorbing, even against ∞ . The exact operations are

$$\begin{aligned} e(a + b) &= \max(e(a), e(b)), & o(a + b) &= \max(o(a), o(b)), \\ e(ab) &= \max(e(a) + e(b), o(a) + o(b)), \\ o(ab) &= \max(e(a) + o(b), o(a) + e(b)). \end{aligned} \quad (\text{HFP23})$$

They follow by choosing the last degree of each parity in (HFP2), including the degree-zero odd-odd contribution. Set $u = (0, -\infty)$, $x = (-\infty, 1)$, $U = (\infty, -\infty)$, $V = (-\infty, \infty)$ and $\mathbf{J} = (\infty, \infty)$. Call a nonzero flag pure when one parity is present and mixed when both are present. For a finite mixed flag its positive odd gap is $\delta = |e - o|$ and its degree is $\max(e, o)$.

In the finite-support semiring all ordinary primes are exactly

$$\begin{aligned} & \{0\}, \quad P_1, P_3, P_5, \dots, P_\infty, \quad M_{\text{fin}}, \\ P_d &= \{0\} \cup \{a \text{ finite mixed} : \delta(a) \leq d\}, \quad P_\infty = \{0\} \cup \{a \text{ finite mixed}\}, \quad M_{\text{fin}} = \mathcal{F}_{\text{fin}}(\mathbb{B}) \setminus \{u\}. \end{aligned} \quad (\text{HFP24})$$

In the full flag semiring all ordinary primes are exactly

$$\begin{aligned} & \{0\}, \quad Q_0, Q_1, Q_3, Q_5, \dots, Q_\infty, \quad P_{\text{mix}}, \quad M, \\ Q_d &= \{0\} \cup \{a \text{ of infinite support}\} \cup \{a \text{ finite mixed} : \delta(a) \leq d\}, \\ P_{\text{mix}} &= \{0\} \cup \{a \text{ mixed}\}, \quad M = \mathcal{F}(\mathbb{B}) \setminus \{u\}. \end{aligned} \quad (\text{HFP25})$$

For $d = 0$ the finite mixed part is empty; for $d = \infty$ it contains every finite mixed flag. Every prime except $\{0\}$ in (HFP25) contains $\mathbf{J}$.

Here is a complete proof of both classifications. Nonzero degrees add under multiplication, so there are no zero divisors, the only nonzero degree-zero flag is u , and the two indicated M are prime ideals. Every nonunit flag a has $a^2 \in (x)$: its square has even endpoint at least 2, possibly infinity, and an element with such an even endpoint is divisible by x . Indeed multiplication by x sends (e, o) to $(o + 1, e + 1)$; swap and subtract 1 to obtain the preimage, retaining absent and infinite endpoints. Thus $\sqrt{(x)} = M$ in each semiring, with the finite version of M where appropriate. A prime containing x is therefore exactly M .

In a prime excluding x , membership of a is equivalent to membership of $x^k a$ for every $k \geq 0$. Every finite pure flag is a monomial x^k and is excluded. Finite mixed flags satisfy the exact formulas

$$\deg(ab) = \deg(a) + \deg(b), \quad \delta(ab) = \min\{\delta(a), \delta(b)\}. \quad (\text{HFP26})$$

For the second, put $m = \deg(a)$, $n = \deg(b)$. The highest endpoint is $m + n$, and the opposite-parity endpoint is $\max\{m + n - \delta(a), m + n - \delta(b)\}$. Multiplication by x^k shifts degree by k and preserves gap. Two finite mixed flags of the same gap satisfy $ax^{\deg(b)} = bx^{\deg(a)}$: the common degree and its parity determine both endpoints from that gap. They therefore have the same prime membership. If a gap d occurs in the prime, multiplication by $u + x^c$, for any positive odd $c \leq d$, supplies the entire gap- c class by (HFP26). The accepted gaps form an initial segment of the positive odd integers. This proves completeness of the finite list.

To verify its listed ideals, the inequalities $e(a) \leq o(a) + d$ and $o(a) \leq e(a) + d$ are preserved by coordinatewise maximum. Hence addition preserves gap at most d . Formula (HFP26) and monomial shifts prove absorption. Its complement consists of pure flags and mixed flags of gap greater than d , and is closed under products. These statements prove primeness, including $d = \infty$.

For the infinite list, $\mathbf{J}a = \mathbf{J}$ for every nonzero flag a , so every nonzero ideal contains $\mathbf{J}$. Every mixed infinite flag has square $\mathbf{J}$, hence belongs to every prime containing $\mathbf{J}$. In a prime excluding x , the equations $xU = V$ and $xV = U$ make the pure infinite endpoints occur together or neither. If they occur, all infinite flags occur and the finite part is one of the initial gap segments already proved, giving Q_d . If they do not occur, $Ua = \mathbf{J}$ for every finite mixed a forces all those a into the prime, giving P_{mix} . This proves completeness. Each Q_d is an ideal because addition with an infinite flag remains infinite, and its product with a nonzero flag remains infinite. Its complement is the finite pure flags and finite mixed flags with excessive gap, multiplicatively closed by (HFP26). Mixed flags form the ideal P_{mix} together with zero, and its complement of pure flags is multiplicatively closed. Thus all listed objects are primes.

Their inclusions are exactly the chain of Q ideals ending in $Q_\infty \subsetneq M$, together with $0 \subsetneq P_{\text{mix}} \subsetneq Q_\infty$; P_{mix} is incomparable with each finite Q_d . The finite primes form the chain printed

in (HFP24). Explicit infinite pure flags and finite mixed flags of a specified excessive gap witness every asserted strictness and incomparability. In particular

$$(\mathbf{J}) = \{0, \mathbf{J}\}, \quad \sqrt{(\mathbf{J})} = \{0\} \cup \{a \text{ mixed of infinite support}\} = Q_0 \cap P_{\text{mix}}. \quad (\text{HFP27})$$

The square calculation proves the radical equality; finite powers stay finite and pure powers stay pure. Thus Q_0 and P_{mix} are exactly the two minimal primes over this ideal.

The inclusion of finite into full flags has the exact prime contractions

full primes	finite contraction	
$0, Q_0$	0	
Q_d (d positive odd finite)	P_d	(HFP28)
Q_∞, P_{mix}	P_∞	
M	$M_{\text{fin.}}$	

Each entry follows by removing the infinite flags from its explicit membership definition, so every finite prime extends, with exactly the indicated fibers. For $t = u + x$, every finite mixed gap-one flag is $x^k t$, and every nonzero product by t has gap one. Consequently $(t) = P_1$. Its finite truncations of the top flag obey

$$t_n = (\underbrace{1, \dots, 1}_{n+1}, 0, \dots) = t^n = x^{n-1} t \quad (n \geq 1). \quad (\text{HFP29})$$

All lie in exactly the nonzero finite primes. By contrast, the only proper lower ideal of either Boolean flag semiring is zero: any nonzero flag a has $u \leq a^2$, so a lower ideal containing it would contain the unit. This verifies concretely why replacing ordinary primes by lower or subtractive primes would erase the whole computed residual spectrum.

38.6 The prime distinction survives every actual theta tensor power

Every heat product has the unchanged flag law:

$$(f, \nu) \star_t (h, \mu) = (T_t((T_{-t}f)(T_{-t}h)), \nu \circledast \mu), \quad T_t = e^{-t\partial_z^2}. \quad (\text{HFP30})$$

Thus the primes (HFP12) and (HFP14), whose receivers depend only on support, are the same subsets and are prime for every complex t . The amplitude-sensitive prime is transported exactly by replacing its receiver with $\ell_q U_{-t}$; no ordinary-product amplitude map is assigned to the changed product. The element J is fixed by heat, and synchronization to the ordinary target is $\gamma_t(f, \nu) = (T_{-t}f, \epsilon(\nu))$ as in SHP26. Consequently all preceding localization and residual identities transport by the actual isomorphism U_t .

Use the original kernel and chosen theta lift

$$\Phi(u) = \sum_{m \geq 1} (2\pi^2 m^4 e^{9u} - 3\pi m^2 e^{5u}) e^{-\pi m^2 e^{4u}}, \quad H_a(z) = \int_0^\infty e^{au^2} \Phi(u) \cos(zu) du, \quad (\text{HFP31})$$

$$\eta_{2j} = 1_L, \quad \eta_{2j+1} = 0_L, \quad \mathbf{H}_a = (H_a, \eta), \quad a \in \mathbb{C}.$$

The actual convergence and membership in $\mathcal{E}$ are THS1–15 and HTK13. Evenness makes the chosen flag admissible for all complex a , including coefficients that vanish at particular complex parameters. Real parameters additionally have the nonvanishing even coefficients proved there; that is not needed below.

For every $k \geq 1$ and arbitrary complex $a_1, \dots, a_k, t$, the full product is the exact HTK27–32 element

$$\mathbf{V}_k = \star_t^k \mathbf{H}_{a_i} = \left(T_t \left(\prod_{i=1}^k H_{a_i - t} \right), \eta \right), \quad J \star_t \mathbf{V}_k = jg(\mathbf{V}_k). \quad (\text{HFP32})$$

For $k = 1$ the heat identity gives the first formula. The amplitude formula for general k follows by repeated invertible transport. The support calculation can also be checked directly without any amplitude assertion: $(\eta \otimes \eta)_{2n} = 1_L$, using the summand with indices $0, 2n$, while each summand at odd degree has an odd index and is zero. The degree-zero formula also gives top. Hence $\eta \otimes \eta = \eta$ and induction proves the full tensor assertion. This is precisely the finite tensor support formula HTK32.

Every lattice prime now gives the exact membership and image equalities

$$\boxed{\begin{array}{ll} \pi_q(\mathbf{V}_k) = \{\bar{0}\}, & \pi_q(J \star_t \mathbf{V}_k) = C_2, \\ \mathbf{V}_k \notin P_q^{\text{mix}}, & J \star_t \mathbf{V}_k \in P_q^{\text{mix}}, \\ g(\mathbf{V}_k) = g(J \star_t \mathbf{V}_k), & \gamma_t(\mathbf{V}_k) = \gamma_t(J \star_t \mathbf{V}_k). \end{array}} \quad (\text{HFP33})$$

The first line is $\chi(1_L) = 1$, $\chi(0_L) = 0$ and the constant top flag after multiplication by J . The second follows from the exact prime ideal in (HFP12). The last follows from $J^2 = J$ and its global unit image, or by writing the unchanged amplitude and head join. These statements depend only on the displayed supports: they remain true for any zero amplitude supplied with the same flag, including cancellation in a tensor calculation. Thus no nonvanishing hypothesis is hidden in the claimed prime distinction.

The complete map (HFP20) sends the first support in (HFP33) to $(1_{K_L}, 0_{K_L}, 1_{K_L}, 0_{K_L}, \dots)$ and the second to $(1_{K_L}, 1_{K_L}, \dots)$; the retraction and receiver (HFP21) detect the distinction there as well. This is an exact spectrum effect of synchronizing the actual amplification objects at every tensor order. It supplies neither an amplitude lower bound nor exclusion of an ordinary or transported zero of H_a . Those numerical and analytic assertions do not follow from prime membership of support flags.

$$\begin{array}{ccc} (F_k, \eta) \notin P_q^{\text{mix}} & \xrightarrow{J \star_t (-)} & (F_k, \mathbf{J}) \in P_q^{\text{mix}} \\ g \downarrow & & \downarrow g \\ (F_k, 1_L) & = & (F_k, 1_L) \end{array}$$

Figure 16: The actual distinction lost by synchronization, HFP31–33. $F_k = T_t(\prod_i H_{a_i-t})$ retains its original amplitude and all constants. The upper left flag has even top and odd bottom labels; the upper right has both top. The ordinary semiring prime in HFP12 contains exactly one upper element. Their global images agree. The diagram remains valid if the displayed amplitude is zero and does not assert a zero-location result.

38.7 The parity receiver is itself a full-lattice localization

There is a localization retaining both full lattice labels before any Boolean character is taken. The original even-support element and its entire corner are

$$\begin{aligned} E_{\text{even}} &= (0, \eta), & E_{\text{even}}^2 &= E_{\text{even}}, \\ E_{\text{even}} \circ (f, \nu) &= (0, (\nu_0, \nu_1, \nu_0, \nu_1, \dots)), \\ \mathcal{A}_L[E_{\text{even}}^{-1}] &\simeq L[C_2] = L \times L, \\ (a, b) + (c, d) &= (a \vee c, b \vee d), & (a, b)(c, d) &= ((a \wedge c) \vee (b \wedge d), (a \wedge d) \vee (b \wedge c)), \\ h : \mathcal{A}_L &\longrightarrow L[C_2], & h(f, \nu) &= (\nu_0, \nu_1), & i(a, b) &= (0, (a, b, a, b, \dots)). \end{aligned} \quad (\text{HFP34})$$

The zero is $(0_L, 0_L)$ and unit is $(1_L, 0_L)$ in $L[C_2]$. For the corner formula at an even positive degree, the term with even index equal to the output degree gives ν_0 ; all other nonzero terms are earlier members of that same decreasing parity tail and are at most ν_0 . At an odd degree the analogous maximum is ν_1 . The degree-zero formula is exactly ν_0 . This proves the complete corner formula in (HFP34), including its amplitude zero. Periodic flags multiply by the displayed

two-coordinate rule, so $hi = \text{id}$, $ih(v) = E_{\text{even}}v$, and i is unital into the corner with unit E_{even} . Any unital map making this idempotent invertible makes it equal to its target unit, and factors uniquely through $v \mapsto E_{\text{even}}v$. This proves the stated localization universal property and its exact maps. The earlier parity receiver is precisely $\pi_{\mathfrak{q}} = (\chi_{\mathfrak{q}} \times \chi_{\mathfrak{q}})h$, with a Boolean pair identified with its set of present parities.

The corresponding complete coefficient square is

$$\begin{array}{ccc} \mathcal{A}_L & \xrightarrow{\Phi} & \mathcal{E} \times \mathcal{F}(K_L) \\ h \downarrow & & \downarrow (f,w) \mapsto (w_0, w_1) \\ L[C_2] & \xrightarrow{\iota_L \times \iota_L} & K_L[C_2]. \end{array} \quad (\text{HFP35})$$

Here $K_L[C_2]$ has the same pair formulas, with its semiring sum and product replacing lattice join and meet. The upper-right image of E_{even} is $(0, \iota_L \eta)$. Multiplying by it gives zero amplitude and the periodic flag $(w_0, w_1, w_0, w_1, \dots)$, so the right vertical map is its exact idempotent localization. The lower map is injective and has retraction $\rho_L \times \rho_L$. Both composites evaluate to $(\iota_L \nu_0, \iota_L \nu_1)$. The square is a pushout: a map from its upper-right corner agreeing with a map from the lower-left corner must invert $\Phi(E_{\text{even}})$, hence factors uniquely through its computed corner; conversely that factorization gives the agreeing pair. Thus neither head label is selected or collapsed in the full-lattice localization.

Let

$$p_{\mathfrak{q}} = \{(a, b) \in L[C_2] : \chi_{\mathfrak{q}}(a) = \chi_{\mathfrak{q}}(b)\}. \quad (\text{HFP36})$$

It is the inverse image of the prime in (HFP12), and $P_{\mathfrak{q}}^{\text{mix}} = h^{-1}p_{\mathfrak{q}}$. The element E_{even} is excluded by this prime, since its image is the pair $(1_L, 0_L)$. Its actual prime stalk therefore has the exact description

$$(\mathcal{A}_L)_{P_{\mathfrak{q}}^{\text{mix}}} \simeq (L[C_2])_{p_{\mathfrak{q}}}, \quad \frac{v}{s} \mapsto \frac{h(v)}{h(s)}, \quad \frac{b}{d} \mapsto \frac{i(b)}{i(d)}. \quad (\text{HFP37})$$

The denominators are permitted exactly by prime contraction. To verify inverse identities, hi is the identity. The other composite sends v/s to $E_{\text{even}}v/E_{\text{even}}s = v/s$, because the idempotent is inverted and hence is the unit in that stalk. This also proves unitality of the second map, although i was initially unital only into its corner.

This stalk is not in general the Boolean pair semiring. An explicit example keeps that distinction checkable. Take $L = \{0 < c < 1\}$ and $\mathfrak{q} = \{0, c\}$. The complement of $p_{\mathfrak{q}}$ consists of the four pairs $(1, 0), (0, 1), (1, c), (c, 1)$. The pair $x = (0, 1)$ is already a unit with $x^2 = (1, 0)$, while $s = (1, c)$ is idempotent and $xs = (c, 1)$. Thus this prime stalk is just localization at s , whose corner consists of the following five distinct actual pairs:

$$sL[C_2] = \{(0, 0), (c, c), (1, c), (c, 1), (1, 1)\}, \quad 1_{sL[C_2]} = (1, c). \quad (\text{HFP38})$$

Indeed multiplication by s sends (a, b) to $(a \vee (c \wedge b), (c \wedge a) \vee b)$; evaluating this formula on the nine original pairs gives precisely the five displayed outputs. The idempotent-corner proof shows there are no further identifications. The Boolean pair semiring has four elements, so cannot be this stalk. This example uses a stated finite lattice only to test the possible identification; (HFP34)–(HFP37) retain the arbitrary original lattice and give its exact maps. All these support localizations also hold for $\star_t$: the heat operator fixes every zero-amplitude flag, and its transport changes no support formula.

38.8 Infinite strict prime chains in the actual full and finite algebras

The classified Boolean receiver gives a further consequence for every nontrivial original lattice, not only for its Boolean specialization. Fix any lattice prime $\mathfrak{q}$, whose existence is proved in (HFP9). The map

$$r_{\mathfrak{q}} : \mathcal{A}_L \longrightarrow \mathcal{F}(\mathbb{B}), \quad (f, \nu) \longmapsto (\chi_{\mathfrak{q}}(\nu_n))_{n \geq 0} \quad (\text{HFP39})$$

is onto: realize every Boolean flag by its original labels $0_L, 1_L$ and amplitude zero. Its finite-flag restriction is likewise onto $\mathcal{F}_{\text{fin}}(\mathbb{B})$. Inverse images of the strict prime chains in (HFP24)–(HFP25) therefore remain strict. In particular the full algebra has

$$r_q^{-1}(Q_0) \subsetneq r_q^{-1}(Q_1) \subsetneq r_q^{-1}(Q_3) \subsetneq r_q^{-1}(Q_5) \subsetneq \cdots \subsetneq r_q^{-1}(M),$$

entirely in $V(J) \cap D(e_0)$. The finite algebra has the corresponding infinite strict chain of inverse images of $P_1, P_3, P_5, \dots, M_{\text{fin}}$. For explicit strictness between consecutive finite gap classes, take amplitude zero and the top-labelled even endpoint zero together with the top-labelled odd endpoint $d + 2$, with all missing labels bottom. The resulting finite mixed flag has gap $d + 2$ and lies in the larger prime but not the smaller one. The pure infinite even flag distinguishes the initial full infinite-tail prime from the zero receiver ideal.

Thus the ordinary prime-ideal Krull dimensions of both $\mathcal{A}_L$ and $\mathcal{A}_{L,\text{fin}}$ are infinite: they contain finite strict prime chains of every length. The displayed endpoint $r_q^{-1}(M)$ itself has infinite height. For the finite algebra each gap prime also fails to descend through its actual global map: the zero-amplitude mixed gap-one flag belongs, whereas e_0 does not, although both global images are the same supported zero $(0, 1_L)$. These explicit flags prove the loss of the whole indicated chain. No dimension claim about a spectrum defined using only subtractive primes is inferred; (HFP5) and (HFP8) prove why that different choice excludes precisely the residual primes used in the full chain.

39 The supported heat divisor and its exact arithmetic and lattice fibres

Scope and sources. The original heat family and all its constants are those of Brad Rodgers and Terence Tao, *The De Bruijn–Newman constant is non-negative*, original `fmp-template.tex`, equations `hoz`, `sas`, `phidef` and `htdef`, lines 78–103. The full coefficient carrier comes from The Clankers, *Split Support Geometry*, v11, `def:lattice-split`. The exact characteristic-one extension is ASL18–22 in this reader, with its cited Connes–Consani sources. All quotient and local-block comparisons used in this section are proved below. In particular a supported-zero equation is not silently changed into an absorbing-zero equation. No off-real zero is postulated.

39.1 The supported equation and the full coefficient square

Let R be a commutative unital ring, allowing the zero ring, and let L be the entire nontrivial bounded distributive lattice. Write

$$S_R = G_L(R), \quad \widehat{f} = (f, 1_L), \quad e_R = (0_R, 1_L), \quad \tau_R = (0_R, 0_L). \quad (\text{HZD1})$$

Its operations are those of ZIL1. In the zero-ring case this is exactly L , with $e_R = 1_{S_R}$ and $\tau_R = 0_{S_R}$. For a given $f \in R$, the supported equation $\widehat{f} = e_R$ defines the congruence quotient

$$\mathcal{D}_L(R, f) := S_R / \langle \widehat{f} \sim e_R \rangle \simeq G_L(R/(f)), \quad (r, \lambda) \mapsto ([r], \lambda). \quad (\text{HZD2})$$

Here angle brackets denote the smallest semiring congruence containing the indicated pair; no ideal congruence is substituted for it.

To prove (HZD2), the displayed map is well defined on the full carrier, preserves both operations and both identities, and is surjective. It identifies $\widehat{f}$ with e_R . Two elements in its kernel congruence have the same label and amplitudes differing by a multiple of f . Below top support both amplitudes must be zero, so those elements were already identical. At top support write $r - s = af$. Multiplying the generating relation by $\widehat{a}$ gives $\widehat{a}f \sim e_R$; adding $\widehat{s}$ gives $\widehat{r} \sim \widehat{s}$, since $e_R + \widehat{s} = \widehat{s}$. This proves that its whole kernel congruence is generated by the stated pair, proving the isomorphism without deleting any label.

Let K be the original tropical semifield with zero, $K_L = K \otimes_{\mathbb{B}} L$, and $\iota_L(\lambda) = 1_K \otimes \lambda$. Keep the proved retraction $\rho_L : K_L \rightarrow L$, $\rho_L \iota_L = \text{id}_L$ of ZIL11. The complete map is

$$\Phi_R : S_R \hookrightarrow R \times K_L, \quad (r, \lambda) \mapsto (r, \iota_L(\lambda)). \quad (\text{HZD3})$$

It is injective, including for the zero ring: equality of ring coordinates and application of ρ_L give equality of both source coordinates. Its image keeps the original constraint $r \neq 0_R \Rightarrow \lambda = 1_L$.

The supported divisor commutes with this complete map in the exact square

$$\begin{array}{ccc} S_R & \xrightarrow{\Phi_R} & R \times K_L \\ \downarrow (r, \lambda) \mapsto ([r], \lambda) & & \downarrow (r, w) \mapsto ([r], w) \\ G_L(R/(f)) & \xrightarrow{\Phi_{R/(f)}} & (R/(f)) \times K_L. \end{array} \quad (\text{HZD4})$$

In the upper-right semiring the congruence generated by $(f, 1_{K_L}) \sim (0_R, 1_{K_L})$ is exactly equality of the second coordinate together with ring-coordinate equality modulo (f) . Indeed multiply the generating relation by $(a, 0_{K_L})$ and add (s, w) to obtain $(s + af, w) \sim (s, w)$. Conversely reduction modulo (f) preserves the generating relation and distinguishes all remaining coordinates. This proves the claimed congruence exactly. It also proves that (HZD4) is a pushout: compatible maps out of its upper and left objects impose precisely that one generating relation on the upper-right object, and therefore factor uniquely through the displayed lower-right quotient. The lower injection shows that the supported divisor itself is retained faithfully there.

The different equation $\widehat{f} = \tau_R$ has the different quotient

$$S_R/\langle \widehat{f} \sim \tau_R \rangle \simeq R/(f). \quad (\text{HZD5})$$

To prove this, multiply its generating relation by e_R . Since $\widehat{f}e_R = e_R$ and $\tau_R e_R = \tau_R$, it imposes $e_R \sim \tau_R$. This identifies every z_λ with τ_R by multiplication, and gives the ring reflection $S_R \rightarrow R$. The remaining relation is $f = 0_R$, giving $R/(f)$. Conversely the map $(r, \lambda) \mapsto [r]$ preserves that relation and is surjective, so there are no additional identifications. Thus the exact connecting map from (HZD2) to (HZD5) is the amplitude map $G_L(R/(f)) \rightarrow R/(f)$. This proves the relation between the two zero conventions instead of replacing one with the other.

Inverting supported zero in the actual supported divisor gives

$$\mathcal{D}_L(R, f)[e^{-1}] \simeq L, \quad ([r], \lambda) \mapsto \lambda. \quad (\text{HZD6})$$

This is ZIL3 applied to e , whose corner consists of all labelled zeros and has identity e . The same conclusion follows by first localizing S_R to L : the supported equation already holds there because both $\widehat{f}$ and e_R map to 1_L . Thus localization commutes with the supported divisor, but its output is independent of f . It does not erase the distinction between (HZD2) and (HZD5) before localization.

39.2 The single off-real heat-divisor object

Use the original complex coordinate z , and set $\Omega = \mathbb{C} \setminus \mathbb{R}$. Let $A = \mathcal{O}(\Omega)$ be the unital ring of holomorphic functions on this entire open set, with both connected components retained. For the actual heat family define, for every real t ,

$$R_t = A/(H_t|_\Omega), \quad \mathcal{D}_{t,L} = G_L(R_t), \quad \Phi_{R_t} : \mathcal{D}_{t,L} \hookrightarrow R_t \times K_L. \quad (\text{HZD7})$$

This is the explicit quotient (HZD2) for the original heat function, with its quotient map, full carrier, joins, meets and constrained faithful image. The counterfactual object at the original time is $\mathcal{D}_{0,L}$, together with the distinguished element e and the map (HZD7). It has been constructed without assuming that it has an off-real arithmetic point.

For any holomorphic function h on Ω , the following equivalences hold:

$$\mathcal{O}(\Omega)/(h) = 0 \iff h \text{ is a unit in } \mathcal{O}(\Omega) \iff h \text{ has no zero in } \Omega. \quad (\text{HZD8})$$

The first equivalence says exactly that 1 lies in the principal ideal (h) . A unit cannot vanish, by evaluating an inverse identity at a proposed zero. If h has no zero, the function $1/h$ is defined at every point of Ω and is holomorphic there. For completeness, near any point z_0 write $h(z) = h(z_0)(1 + v(z))$ with $v(z_0) = 0$; on a sufficiently small neighbourhood $|v| < 1$, and the convergent geometric series $h(z_0)^{-1} \sum_{n \geq 0} (-v(z))^n$ is the local holomorphic reciprocal. These local reciprocals agree on overlaps, giving the global inverse on both components. This proves (HZD8).

The canonical support map on (HZD7) consequently satisfies

$$\begin{aligned} \sigma_t : \mathcal{D}_{t,L} \longrightarrow L \text{ is an isomorphism} &\iff R_t = 0 \\ &\iff H_t \text{ has no off-real zero.} \end{aligned} \quad (\text{HZD9})$$

If $R_t = 0$, the carrier is precisely L and σ_t is its identity. If $R_t \neq 0$, the distinct elements $(0, 1_L)$ and $(1, 1_L)$ have the same support; hence the specified map is not injective. Together with (HZD8), this proves every direction of (HZD9). The statement is about this canonical map, not an unsupported abstract identification of two semirings.

The actual arithmetic part of its spectrum is

$$V_{\mathcal{D}_{t,L}}(e) \simeq \text{Spec } R_t, \quad D_{\mathcal{D}_{t,L}}(e) \simeq \text{SpecLat } L. \quad (\text{HZD10})$$

The explicit primes and their inclusions are exactly ZIL9, whose proof applies to every nonzero R_t . If $R_t = 0$, then $e = 1$, the first displayed space is empty and the second is the whole lattice spectrum. Each actual off-real root z_0 gives the surjective evaluation $R_t \rightarrow \mathbb{C}$ and the prime

$$Q_{z_0} = \{z_\lambda : \lambda \in L\} \cup \{([h], 1_L) : h(z_0) = 0\}. \quad (\text{HZD11})$$

Evaluation is well defined because $H_t(z_0) = 0$, and its kernel is prime because its target is a field. Conversely the absence of every such root forces $R_t = 0$ by (HZD8), so there is then no arithmetic prime of any kind. We have not assumed that every maximal ideal of a ring of holomorphic functions is an evaluation ideal. No such assertion is needed for this exact emptiness test.

At $t = 0$ the unchanged Rodgers–Tao identity gives

$$H_0(z) = \frac{1}{8}\xi\left(\frac{1}{2} + \frac{iz}{2}\right), \quad s = \frac{1}{2} + \frac{iz}{2}, \quad z = -2i(s - \frac{1}{2}). \quad (\text{HZD12})$$

For $z = x + iy$, $\text{Re } s = (1 - y)/2$; therefore off-real z corresponds exactly to off-critical s . This is the source’s unchanged nontrivial-zero coordinate map. Thus the full supported counterfactual object has an arithmetic root exactly when the original zeta function has a counterexample. Supported-zero invertibility in its localization is already proved by (HZD6), and occurs in both cases of (HZD9). Whether σ_0 is an isomorphism has not been decided by that localization calculation.

39.3 Every actual primary root retains its complete supported jet

The following construction applies to each actual root, without postulating that an off-real one exists. Fix a real t and a root z_0 of the actual nonzero entire function H_t . Let m be its exact positive multiplicity. Retain the actual local factor

$$H_t(z) = (z - z_0)^m u(z), \quad u(z) = \sum_{j \geq 0} \frac{H_t^{(m+j)}(z_0)}{(m+j)!} (z - z_0)^j, \quad u(z_0) = \frac{H_t^{(m)}(z_0)}{m!} \neq 0. \quad (\text{HZD13})$$

This follows by the original Taylor expansion at z_0 and the definition of the exact multiplicity. No coefficient or unit factor has been reset to one.

Write $B = \mathcal{O}(\mathbb{C})$ and $\mathfrak{m}_{z_0} = \{h : h(z_0) = 0\}$. The exact jet map is

$$\begin{aligned} B_{\mathfrak{m}_{z_0}} / (H_t) &\xrightarrow{\sim} \mathbb{C}[\epsilon] / (\epsilon^m), \\ h &\mapsto \sum_{j=0}^{m-1} \frac{h^{(j)}(z_0)}{j!} \epsilon^j, \quad z \mapsto z_0 + \epsilon. \end{aligned} \quad (\text{HZD14})$$

For fractions the map is extended by the inverse of the denominator jet, whose constant term is nonzero. It is a ring homomorphism by the product rule with its exact binomial coefficients. It is surjective because every truncated polynomial is represented by that polynomial in $z - z_0$. Its kernel consists of functions vanishing to order at least m . Such an entire numerator h has $h/(z - z_0)^m$ entire, by its Taylor expansion at z_0 and ordinary division away from z_0 . The entire function $u = H_t/(z - z_0)^m$ has nonzero value at z_0 , so is inverted in $B_{\mathfrak{m}_{z_0}}$. Thus this numerator is a multiple of H_t there. The converse follows from (HZD13), proving the kernel assertion and the isomorphism, while keeping the original factor u explicitly available.

For an off-real z_0 , restriction $B \rightarrow A = \mathcal{O}(\Omega)$ induces the exact comparison

$$B_{\mathfrak{m}_{z_0}} / (H_t) \longrightarrow A_{\mathfrak{m}_{z_0}} / (H_t|_{\Omega}).$$

The same Taylor proof applies on Ω : division by $(z - z_0)^m$ extends holomorphically at z_0 and is already holomorphic at every other point of both components. Both quotients therefore map

isomorphically to the displayed jet ring, and restriction preserves all derivatives at z_0 . The comparison is consequently an isomorphism, identifying this block with the actual arithmetic local block of (HZD7).

Applying the exact supported quotient and localization maps gives

$$G_L(B_{\mathfrak{m}_{z_0}}/(H_t)) \simeq G_L(\mathbb{C}[\epsilon]/(\epsilon^m)). \quad (\text{HZD15})$$

The map is the coefficient map (HZD14) with every label unchanged; its inverse is the inverse ring map with the same label. To check localization directly, a top-supported denominator $(a, 1_L)$ with $a(z_0) \neq 0$ sends (h, λ) to $(h/a, \lambda)$. The semiring fraction relation is precisely ring-fraction equality and equality of the labels: multiplying by any top-supported denominator preserves that label. Thus no label is lost in the localization used in (HZD15).

The jet object in (HZD15) has the exact coefficient realization

$$\left(\sum_{j=0}^{m-1} a_j \epsilon^j, \lambda \right) \mapsto ((a_0, \lambda), \dots, (a_{m-1}, \lambda)). \quad (\text{HZD16})$$

Its image has all coordinate labels equal, and a nonzero coefficient forces that common label to be top. This is a supported submodule of $G_L(\mathbb{C})^m$, not an identification with that entire independently labelled free module. Let E_m be the matrix with amplitude identity and every entry at top support. Its action on the free module keeps each amplitude and replaces every coordinate label by the join of all labels. Thus $E_m^2 = E_m$, and its image is exactly (HZD16). This also constructs the split projective realization and its entire projection without changing the coefficient basis. The algebra product on this image is the retained truncated Cauchy product, not coordinatewise multiplication:

$$((a_j, \lambda))_j ((b_j, \nu))_j = \left(\left(\sum_{r+s=j} a_r b_s, \lambda \wedge \nu \right) \right)_{0 \leq j < m}.$$

Every coefficient sum has at least one term and every term has the same label $\lambda \wedge \nu$, so this formula follows directly from the original semiring operations even when its amplitude cancels. Thus (HZD16) also preserves the entire jet algebra on its stated image.

Multiplication by the original local coordinate $\epsilon = z - z_0$ is represented on that image by the matrix N_m whose amplitude has entries 1 at $i = j + 1$ and 0 elsewhere, with *every* matrix entry at top support. Direct coefficient multiplication gives

$$\begin{aligned} N_m^m &= Z_m, & Z_m &= ((0, 1_L))_{0 \leq i, j < m}, \\ Z_m((a_j, \lambda)_j) &= ((0, \lambda))_j, & Z_m^2 &= Z_m, & N_m^0 &= E_m \quad \text{on the projective image.} \end{aligned} \quad (\text{HZD17})$$

Indeed each application shifts the amplitude vector by one place, with zero in the first position; after m applications all amplitudes vanish. All terms in every matrix product carry top support, so the resulting entries are e , not τ . The stated action and idempotence follow by joining the equal input labels. The categorical zero map instead has output $(\tau, \dots, \tau)$ and is different whenever the input label is nonzero. This is the exact supported nilpotence, with its actual null endomorphism retained.

Finally, supported-zero inversion sends the jet algebra to L . The top-supported element $\hat{e}$ and its power $\hat{e}^m = e$ all map to 1_L :

$$G_L(\mathbb{C}[\epsilon]/(\epsilon^m))[e^{-1}] \simeq L, \quad \hat{e} \mapsto 1_L, \quad e \mapsto 1_L. \quad (\text{HZD18})$$

Thus the amplitude-nilpotent coordinate becomes the identity on the localized support module. In the projective realization, both N_m and E_m become the all-top join matrix; its image consists

of constant label vectors and is canonically L . This proves the same conclusion using the complete matrix map, not just the scalar quotient. Before localization the map $\Phi_{\mathbb{C}[\epsilon]/(\epsilon^m)}$ retains the entire jet amplitude, its multiplicity, the original coordinate $z_0 + \epsilon$, and every support label. The localization does not determine the existence or position of the root whose local object has just been calculated.

40 The heat divisor before coefficient-support synchronization

Sources and original objects. The full coefficient algebra is SHP1–28 of `SUPPORTED_HEAT_PRODUCT.tex`, with analytic completion HSL15–35 of `FULL_LATTICE_HEAT_OPERATOR.tex`. The original foundation is retained under the byline The Clankers in *Split Support Geometry: Universal Zero Fibres, Arithmetic Curves, Frobenius and Perfect Quantization*, v11, Definition `def:lattice-split`. The actual analytic kernel is that of Brad Rodgers and Terence Tao, *The De Bruijn–Newman constant is non-negative*, arXiv:1801.05914v5. The complete proofs THS8–15 and THS18–23 in `THETA_HEAT_FULL_SUPPORT.tex` establish its coefficient formula, nonvanishing of every even coefficient at real time, membership in the stated analytic space and the exact heat group. No theorem about its zero locations is used here.

The prime receiver read for this calculation is HFP11–12 and HFP34–37 of `HEAT_FLAG_PRIME_SPECTRUM.tex`, lines 216–273 and 682–787, at source SHA–256

e51115eaa2e70bb97cdf32d113222eace1e37260e1ec6781dbefe62517dab63f.

The synchronized divisor comparison is HZD1–8 of `FULL_LATTICE_HEAT_DIVISOR.tex`, read at source

986b34093592f4949645a02a2413817809476cb82519f46f61471ba1a03ede3a.

The algebra below calculates the congruence on the full coefficient flags before that synchronization. All receiver maps used from these sources are proved again in the required quotient coordinates.

40.1 The coefficient fibres and the supported equation

Let L be the entire nontrivial bounded distributive lattice, and retain the original Taylor coefficient space

$$\mathcal{E} = \left\{ f(z) = \sum_{n \geq 0} a_n z^n : \sum_{n \geq 0} |a_n| \sqrt{n!} h^n < \infty \text{ for every } h > 0 \right\}.$$

Equivalently, f is entire and $p_\alpha(f) = \sup_z |f(z)| e^{-\alpha|z|^2} < \infty$ for every $\alpha > 0$, by HSL16. It is a ring under ordinary function multiplication, with the proved bound $\|fg\|_h \leq \|f\|_{\sqrt{2}h} \|g\|_{\sqrt{2}h}$. Write $\mathcal{E}^{(0)}$ and $\mathcal{E}^{(1)}$ for its even and odd subspaces. Their projections select the even and odd Taylor coefficients, respectively, and preserve every defining seminorm.

Let $\mathcal{F}(L)$ be the full flag semiring. Its elements are sequences $\nu_i \in L$ satisfying $\nu_i \geq \nu_{i+2}$. Addition is coordinate join and multiplication is

$$(\nu \otimes \mu)_0 = (\nu_0 \wedge \mu_0) \vee (\nu_1 \wedge \mu_1), \quad (\nu \otimes \mu)_n = \bigvee_{r=0}^n \nu_r \wedge \mu_{n-r} \quad (n \geq 1). \quad (\text{HFD1})$$

The unit is $\mathbf{u} = (1_L, 0_L, 0_L, \dots)$. SHP3–9 prove its algebra laws by the exact parity-tail closure identity. The coefficient algebra is

$$\mathcal{A}_L = \{(f, \nu) : \nu \in \mathcal{F}(L), f = \sum a_n z^n \in \mathcal{E}, a_n \neq 0 \implies \nu_n = 1_L\}, \quad (f, \nu) \circ (g, \mu) = (fg, \nu \otimes \mu). \quad (\text{HFD2})$$

The zero is $(0, \mathbf{0})$ and the unit is $(1, \mathbf{u})$. Every label in L remains available at a zero coefficient.

For each flag retain its complete amplitude fibre and its two full-parity indicators:

$$\begin{aligned} \mathcal{E}_\nu &= \{f = \sum a_n z^n \in \mathcal{E} : \nu_n \neq 1_L \implies a_n = 0\}, \\ \epsilon_p(\nu) &= \begin{cases} 1, & \nu_{2j+p} = 1_L \text{ for every } j \geq 0, \\ 0, & \text{otherwise,} \end{cases} \quad (p = 0, 1), \quad (\text{HFD3}) \\ V_\nu &= \epsilon_0(\nu) \mathcal{E}^{(0)} \oplus \epsilon_1(\nu) \mathcal{E}^{(1)}. \end{aligned}$$

These are complex vector spaces. For a parity with $\epsilon_p(\nu) = 0$, its component of $\mathcal{E}_\nu$ consists of polynomials: choose an index of that parity whose label is not top. Every later label of that parity is below it, and hence also not top, so all those coefficients vanish. Before that index only the initial top-labelled coefficients are allowed. For a parity with indicator 1, the entire corresponding parity subspace of $\mathcal{E}$ is allowed. This is an exact description of $\mathcal{E}_\nu$, not a change to the remaining non-top labels in ν .

Let $s \in \mathbb{C}$ be the actual theta time, and set

$$H = H_s, \quad \eta_{2j} = 1_L, \quad \eta_{2j+1} = 0_L, \quad h_s = (H_s, \eta), \quad z_\eta = (0, \eta). \quad (\text{HFD4})$$

The analytic function H_s is even, belongs to $\mathcal{E}$, and is not zero, by the actual heat group and the positive real kernel at time zero. Both elements in (HFD4) belong to the original coefficient algebra. The supported equation is $h_s \sim z_\eta$, not $h_s \sim (0, \mathbf{0})$.

The crucial support multiplication is exactly

$$\eta \otimes \nu = \kappa(\nu), \quad \kappa(\nu)_{2j} = \nu_0, \quad \kappa(\nu)_{2j+1} = \nu_1. \quad (\text{HFD5})$$

At an even positive output index in (HFD1), the term with the η index equal to that output index contributes ν_0 , and the other contributing labels are at most ν_0 . At an odd index the largest contributing label is ν_1 . At degree zero (HFD1) gives exactly ν_0 . This proves every coordinate.

40.2 The exact generated congruence

The congruence generated by the single supported equation in (HFD4) is the following explicit equivalence:

$$(f, \nu) \equiv_s (g, \mu) \iff \nu = \mu \text{ and } f - g = H_s b \text{ for some } b \in V_\nu. \quad (\text{HFD6})$$

In particular every original coefficient label is unchanged in an equivalence class.

Here is the full congruence proof. First $H_s V_\nu \subset \mathcal{E}_\nu$, since H_s is even and multiplication preserves parity and $\mathcal{E}$. On a permitted parity the flag is top in every degree; no other parity is produced. Thus (HFD6) is an equivalence on each vector-space fibre. Under addition of a context with flag μ , $V_\nu \subset V_{\nu \vee \mu}$, so the equivalence is preserved by addition.

For multiplication the needed exact fact is

$$b \in V_\nu, \quad k \in \mathcal{E}_\mu \implies bk \in V_{\nu \otimes \mu}. \quad (\text{HFD7})$$

Decompose both functions into their even and odd parts. If the parity- p part of b is nonzero, then every parity- p label of ν is top. If the parity- q part of k is nonzero, some Taylor coefficient in that parity is nonzero, so its label and therefore μ_q are top. The output flag has every parity- $(p + q \bmod 2)$ label top. For an odd output index this follows by using the degree-0 or degree-1 label of μ in (HFD1) and the appropriate top label of ν . The same argument works for positive even indices. For output degree zero, the even-even case uses $\nu_0 \wedge \mu_0$, and the odd-odd case uses the indispensable additional term $\nu_1 \wedge \mu_1$. Thus each nonzero parity product in bk lies in an allowed full parity of the output. Joining these possibilities proves (HFD7). It follows that multiplying the difference $H_s b$ by any context amplitude again has the form in (HFD6) for the product flag. This proves multiplicative compatibility. Compatibility for two equivalent factors follows by changing one factor at a time. Hence (HFD6) is a semiring congruence, and it contains the generating pair because $1 \in V_\eta$.

Conversely every pair in (HFD6) is produced by one elementary use of that generating pair. For $b = b_0 + b_1 \in V_\nu$, let χ_b be the parity-constant flag that is top on parity p when $b_p \neq 0$, and bottom otherwise. Then $(b, \chi_b) \in \mathcal{A}_L$, $\chi_b \leq \nu$, and $\eta \otimes \chi_b = \chi_b$. Multiply $h_s \sim z_\eta$ by this element and add the context (g, ν) . The two results are

$$(g, \nu) + (H_s b, \chi_b) = (f, \nu), \quad (g, \nu) + (0, \chi_b) = (g, \nu).$$

This proves that every equivalence in (HFD6) is generated by the required relation. The congruence calculation is therefore exact.

The divisor before synchronization is the concrete semiring

$$\mathcal{D}_{s,L}^{\text{flag}} = \mathcal{A}_L / \equiv_s = \coprod_{\nu \in \mathcal{F}(L)} (\mathcal{E}_\nu / H_s V_\nu) \times \{\nu\}. \quad (\text{HFD8})$$

For representatives the operations are

$$([f]_\nu, \nu) + ([g]_\mu, \mu) = ([f + g]_{\nu \vee \mu}, \nu \vee \mu), \quad ([f]_\nu, \nu) \circ ([g]_\mu, \mu) = ([fg]_{\nu \otimes \mu}, \nu \otimes \mu). \quad (\text{HFD9})$$

They are well defined by the congruence proof. The unit and zero retain the flags $\mathbf{u}$ and $\mathbf{0}$, respectively. Formula (HFD8) records every support-dependent fibre. On the actual η fibre only even multipliers are allowed: its amplitude space is $\mathcal{E}^{(0)} / H_s \mathcal{E}^{(0)}$.

40.3 The actual heat function removes the comparison defect

The following analytic fact is proved directly on the original completion: every even, nonpolynomial function in $\mathcal{E}$ has infinitely many distinct complex zeros. Suppose instead that such an F had only finitely many zeros, and form their monic zero polynomial p , with all multiplicities. Evenness gives equal multiplicities at a and $-a$, and an even multiplicity at zero. Thus p is even. The quotient $G = F/p$ is entire, even and zero-free. It is still in $\mathcal{E}$: outside a sufficiently large disc $|p(z)| \geq 1$, and on the remaining compact disc G is bounded. Consequently every positive Gaussian bound for F gives one for G .

The entire function G'/G has an entire primitive, obtained by integrating its Taylor series. Choosing its additive constant to be a logarithm of $G(0)$ gives an entire ℓ with $G = e^\ell$. For every $\alpha > 0$,

$$\text{Re } \ell(z) \leq \log p_\alpha(G) + \alpha |z|^2. \quad (\text{HFD10})$$

Write $\ell(z) = \sum_{n \geq 0} c_n z^n$ and, on the circle of radius R , put $u(\theta) = \text{Re } \ell(Re^{i\theta}) - \text{Re } c_0$. Its average is zero by the convergent Taylor series, and $u \leq M_{\alpha,R} := \log p_\alpha(G) + \alpha R^2 - \text{Re } c_0$. This upper bound is nonnegative since $p_\alpha(G) \geq |G(0)|$. The zero average gives $\int_0^{2\pi} |u| = 2 \int_0^{2\pi} u_+ \leq 4\pi M_{\alpha,R}$. Fourier coefficient extraction from the same Taylor series therefore gives, for $n \geq 1$,

$$|c_n| R^n = \left| \frac{1}{\pi} \int_0^{2\pi} u(\theta) e^{-in\theta} d\theta \right| \leq 4M_{\alpha,R}. \quad (\text{HFD11})$$

For $n \geq 3$, let $R \rightarrow \infty$ with α fixed; then $c_n = 0$. For $n = 2$ this yields $|c_2| \leq 4\alpha$, and $\alpha \downarrow 0$ gives $c_2 = 0$. Thus $\ell = c_0 + c_1 z$. Since G is even, $G'(0) = 0$, forcing $c_1 = 0$. It follows that F is a constant times p , contradicting its nonpolynomiality. This proves the analytic fact without assuming where any of the zeros lie.

Apply it to every actual H_s , including complex time. The function H_0 has nonzero Taylor coefficients in every even degree by THS9, so it is not a polynomial. If H_s were polynomial, the identity $H_0 = T_{-s} H_s$ would make H_0 polynomial: the heat series terminates on a polynomial. Also H_s is even and is in $\mathcal{E}$. We have proved

$$H_s \text{ has infinitely many distinct zeros, } H_s b \text{ polynomial, } b \in \mathcal{E} \implies b = 0. \quad (\text{HFD12})$$

For the second assertion a polynomial $H_s b$ vanishes at every zero of H_s , and hence is zero. The ring of entire functions is an integral domain: where H_s is nonzero, b vanishes on an open set, and the identity theorem gives $b = 0$ everywhere. This also proves that (H_s) is a proper ideal in $\mathcal{E}$.

The exact comparison defect now vanishes for this actual function:

$$H_s \mathcal{E} \cap \mathcal{E}_\nu = H_s V_\nu, \quad \frac{H_s \mathcal{E} \cap \mathcal{E}_\nu}{H_s V_\nu} = 0 \quad (\nu \in \mathcal{F}(L)). \quad (\text{HFD13})$$

Only one inclusion remains to prove. Suppose $H_s b \in \mathcal{E}_\nu$. For a parity with $\epsilon_p(\nu) = 0$, that parity component is a polynomial by (HFD3). Since H_s is even, that component is $H_s b_p$, which forces $b_p = 0$ by (HFD12). All other parity components of b are allowed in V_ν . This proves the reverse inclusion and the entire displayed defect calculation.

Thus the divisor has an injective pair description, but with a proved, restricted image:

$$\begin{aligned} \mathcal{D}_{s,L}^{\text{flag}} &\hookrightarrow (\mathcal{E}/(H_s)) \times \mathcal{F}(L), & ([f]_\nu, \nu) &\longmapsto ([f], \nu), \\ \text{image} &= \{(c, \nu) : c \in \text{image}(\mathcal{E}_\nu \rightarrow \mathcal{E}/(H_s))\}. \end{aligned} \quad (\text{HFD14})$$

Injectivity is exactly (HFD13). The description of the image follows from (HFD8) and is both necessary and sufficient. It is not the unrestricted product: for example $([1], \mathbf{0})$ is not in it, since $\mathcal{E}_\mathbf{0} = 0$ and $[1] \neq 0$ by (HFD12). No support constraint has been removed.

40.4 The induced complete coefficient quotient

Retain $K_L = K \otimes_{\mathbb{B}} L$ and the injective label map $\iota_L(\lambda) = 1_K \otimes \lambda$, with retraction ρ_L . The exact full coefficient map is $\Phi(f, \nu) = (f, (\iota_L(\nu_i))_i)$. In its full target $\mathcal{E} \times \mathcal{F}(K_L)$, the congruence generated by $\Phi(h_s) \sim \Phi(z_\eta)$ is exactly

$$(f, w) \sim (g, v) \iff w = v, \quad f - g \in H_s \mathcal{E}. \quad (\text{HFD15})$$

Reduction of the first coordinate modulo (H_s) is a homomorphism coequalizing the pair, so it gives one containment of congruences. For the other, multiply that pair in the full target by $(b, \mathbf{0})$ and then add (g, w) . This gives $(g + H_s b, w) \sim (g, w)$. The full target really permits this multiplier even when $b \neq 0$; the original coefficient carrier does not. This is why its restriction was calculated in (HFD13), rather than assumed.

The complete quotient square is

$$\begin{array}{ccc} \mathcal{A}_L & \xrightarrow{\Phi} & \mathcal{E} \times \mathcal{F}(K_L) \\ \downarrow & & \downarrow \\ \mathcal{D}_{s,L}^{\text{flag}} & \xrightarrow{\Phi_s} & (\mathcal{E}/(H_s)) \times \mathcal{F}(K_L), \end{array} \quad \Phi_s([f]_\nu, \nu) = ([f], (\iota_L(\nu_i))_i). \quad (\text{HFD16})$$

The lower map is injective by (HFD13) and the label retraction. Its image is exactly the support-constrained image in (HFD14) after applying ι_L to each flag coordinate. Both composites give the displayed pair. The square is a pushout: compatible maps from the upper-right and lower-left objects impose exactly the generating relation on the upper-right object, whose entire congruence was calculated in (HFD15). The resulting factorization is unique. This proves the complete-map statement including the induced relation and its exact restriction.

Let $J = (1, (1_L, 1_L, \dots))$ be the synchronization idempotent of SHP20. Its image $\bar{J}$ in the divisor still acts by

$$\bar{J} \circ ([f]_\nu, \nu) = ([f]_{\text{const}(\nu_0 \vee \nu_1)}, \text{const}(\nu_0 \vee \nu_1)), \quad \mathcal{D}_{s,L}^{\text{flag}}[\bar{J}^{-1}] \simeq G_L(\mathcal{E}/(H_s)). \quad (\text{HFD17})$$

For a constant flag of label $\lambda \neq 1_L$, the only amplitude is zero. For the constant top flag its amplitude quotient is $\mathcal{E}/(H_s)$, since both parity indicators in (HFD3) are one. These are exactly the carrier and operations of the right side. Multiplication by an idempotent realizes its localization, with the idempotent as internal unit, proving the isomorphism and all its coordinates. In particular the map is $([f]_\nu, \nu) \mapsto ([f], \nu_0 \vee \nu_1)$. This is the exact connection to the earlier synchronized supported divisor.

40.5 The mixed prime really descends before synchronization

The quotient does not identify different flags, so it retains the full head-pair receiver

$$\begin{aligned}\bar{h} : \mathcal{D}_{s,L}^{\text{flag}} &\longrightarrow L[C_2], & \bar{h}([f]_\nu, \nu) &= (\nu_0, \nu_1), \\ (a, b) + (c, d) &= (a \vee c, b \vee d), \\ (a, b)(c, d) &= ((a \wedge c) \vee (b \wedge d), (a \wedge d) \vee (b \wedge c)), \\ 0 &= (0_L, 0_L), & 1 &= (1_L, 0_L).\end{aligned}\tag{HFD18}$$

The two head formulas in (HFD1) prove that this is a unital homomorphism. It is onto: use zero amplitude and the periodic flag $(a, b, a, b, \dots)$. Moreover $z_\eta^2 = z_\eta$, and its image $\bar{z}_\eta = \bar{h}_s$ acts by zeroing the amplitude and making the two heads periodic, by (HFD5). Thus

$$\mathcal{D}_{s,L}^{\text{flag}}[\bar{z}_\eta^{-1}] \simeq L[C_2], \quad ([f]_\nu, \nu) \longmapsto (\nu_0, \nu_1).\tag{HFD19}$$

The inverse corner map is $(a, b) \mapsto ([0], (a, b, a, b, \dots))$. It is unital into the corner whose identity is $\bar{z}_\eta$. Distinct flags have not been identified by the quotient, so this map is injective as well as onto that corner. This proves the localization directly. Under the complete map (HFD16), the head receiver commutes with $L[C_2] \hookrightarrow K_L[C_2]$ given by ι_L on both coordinates; it has the coordinate retraction ρ_L on both.

Let $\mathfrak{q}$ be any proper lattice prime and $\chi_\mathfrak{q} : L \rightarrow \mathbb{B}$ its character. It preserves join and meet: membership of a join in the ideal is equivalent to membership of both summands, and primality tests a meet. The full receiver above gives

$$\pi_\mathfrak{q}([f]_\nu, \nu) = \{\bar{p} : p = 0, 1, \nu_p \notin \mathfrak{q}\} \in \mathcal{P}(C_2).\tag{HFD20}$$

The target has union as addition and group-set addition as multiplication, zero $\emptyset$ and unit $\{\bar{0}\}$. Formula (HFD18) proves that this is a surjective unital homomorphism. The subset $I_{\text{mix}} = \{\emptyset, C_2\}$ is an ordinary prime ideal: it is closed under union and multiplication by every subset, and its complement consists of the two singletons, whose products are again singletons. Therefore

$$\bar{P}_\mathfrak{q}^{\text{mix}} = \pi_\mathfrak{q}^{-1}(I_{\text{mix}}) = \{([f]_\nu, \nu) : \chi_\mathfrak{q}(\nu_0) = \chi_\mathfrak{q}(\nu_1)\}\tag{HFD21}$$

is a proper ordinary prime of the divisor.

This also proves the required saturation of the source prime, rather than assuming it. Every pair in the generated congruence (HFD6) has the same complete flag, and hence the same $\pi_\mathfrak{q}$ value. Membership of its inverse image of I_{mix} is therefore constant on each congruence class. Its quotient image is exactly (HFD21). Addition and multiplication in the quotient preserve that ideal, and primality follows by lifting any product to representatives and using source primality, or directly from the displayed receiver. The prime is not subtractive: in the target, $C_2 \in I_{\text{mix}}$ and $C_2 \cup \{\bar{0}\} \in I_{\text{mix}}$, but $\{\bar{0}\} \notin I_{\text{mix}}$. The same counterexample lifts using periodic zero-amplitude flags.

The distinction relevant to synchronization is exact:

$$\bar{J} \in \bar{P}_\mathfrak{q}^{\text{mix}}, \quad \bar{h}_s = \bar{z}_\eta \notin \bar{P}_\mathfrak{q}^{\text{mix}}.\tag{HFD22}$$

Their head pairs are $(1_L, 1_L)$ and $(1_L, 0_L)$, respectively. Thus this prime survives the head localization (HFD19), but is excluded by localization at $\bar{J}$ in (HFD17). The supported equation kills an analytic amplitude while leaving z_η different from the absorbing zero; that is why the proper prime (HFD21) can remain.

40.6 The off-real analytic comparison retaining all flags

Put $\Omega = \mathbb{C} \setminus \mathbb{R}$ and $\mathcal{O} = \mathcal{O}(\Omega)$, keeping both connected components. Restriction is a unital ring map $\mathcal{E} \rightarrow \mathcal{O}$. Define

$$R_s = \mathcal{O}/(H_s|_\Omega), \quad I_s^{\text{off}} = \{f \in \mathcal{E} : f|_\Omega \in H_s|_\Omega \mathcal{O}\}, \quad B_s = \mathcal{E}/I_s^{\text{off}} \hookrightarrow R_s. \quad (\text{HFD23})$$

The last map identifies B_s with the exact image of the coefficient ring in the analytic quotient. Its injectivity is the definition of the kernel I_s^{off} . No assertion that arbitrary holomorphic classes are represented by $\mathcal{E}$ is needed or made.

The ideal in (HFD23) is explicitly determined by all off-real zero jets:

$$f \in I_s^{\text{off}} \iff f^{(j)}(\zeta) = 0 \text{ for every } \zeta \in \Omega \text{ with } H_s(\zeta) = 0, \ 0 \leq j < \text{ord}_\zeta H_s. \quad (\text{HFD24})$$

Indeed near such a zero write $H_s(z) = (z - \zeta)^m u(z)$ with $u(\zeta) \neq 0$. The quotient f/H_s extends holomorphically there exactly when the first m Taylor coefficients of f vanish. Away from the zeros it is already holomorphic. When all the stated conditions hold the extensions agree with that quotient on punctured neighbourhoods and hence agree with one another, defining an element of $\mathcal{O}(\Omega)$. This proves both directions with every multiplicity retained.

The exact off-real quotient before synchronization is

$$\begin{aligned} \mathcal{D}_{s,L}^{\text{flag,off}} &= \coprod_{\nu \in \mathcal{F}(L)} (\mathcal{E}_\nu / (I_s^{\text{off}} \cap \mathcal{E}_\nu)) \times \{\nu\} \\ &\hookrightarrow R_s \times \mathcal{F}(L), \quad ([f]_\nu, \nu) \longmapsto ([f|_\Omega], \nu). \end{aligned} \quad (\text{HFD25})$$

Its image consists exactly of pairs whose first coordinate has a representative from $\mathcal{E}_\nu$ for the displayed flag. The equivalence on the source is equality of flags together with the full jet test (HFD24) on $f - g$. This is the kernel of the actual semiring map $\mathcal{A}_L \rightarrow R_s \times \mathcal{F}(L)$, so its operations are (HFD9) with the new amplitude quotients; it is a congruence quotient, not an assumed unrestricted product. Since $H_s V_\nu \subset I_s^{\text{off}} \cap \mathcal{E}_\nu$, the source divisor (HFD8) maps onto (HFD25). On each fixed flag, the exact additional identified subspace is

$$(I_s^{\text{off}} \cap \mathcal{E}_\nu) / H_s V_\nu. \quad (\text{HFD26})$$

This proves the complete comparison defect to the off-real analytic receiver.

The coefficient map retains its full form: the map in (HFD25) followed by $\text{id}_{R_s} \times \mathcal{F}(\iota_L)$ is an injective map to $R_s \times \mathcal{F}(K_L)$. It agrees with the composite of (HFD16) and the ring map $\mathcal{E}/(H_s) \rightarrow R_s$. All head maps and the prime (HFD21) factor through this further quotient too, because its equivalences still preserve every label. Their properness and primality follow by exactly the same receiver proof.

Synchronization of (HFD25) has the precise image and connection to HZD:

$$\mathcal{D}_{s,L}^{\text{flag,off}} [\bar{J}^{-1}] \simeq G_L(B_s) \hookrightarrow G_L(R_s) = \mathcal{D}_{s,L}. \quad (\text{HFD27})$$

The map sends a class to $([f|_\Omega], \nu_0 \vee \nu_1)$. Its image contains every nonzero amplitude class in B_s with top label, by choosing a representing f and the constant top flag. It contains every labelled zero by choosing zero amplitude and a constant flag of that label. These exhaust $G_L(B_s)$, including when B_s is the zero ring. The idempotent corner calculation with J is unchanged, proving (HFD27). The last inclusion is induced by the injective ring map in (HFD23), with every lattice label retained. This is the exact map to HZD7 and states its possibly larger analytic amplitude ring explicitly.

For a zero-free restriction $H_s|_\Omega$, the reciprocal $1/H_s$ is holomorphic on Ω , so $R_s = 0$ and $I_s^{\text{off}} = \mathcal{E}$. Conversely, $R_s = 0$ gives a holomorphic inverse and excludes zeros by evaluation. Accordingly the complete retained object in that case is

$$R_s = 0 \implies \mathcal{D}_{s,L}^{\text{flag,off}} \simeq \mathcal{F}(L), \quad G_L(R_s) \simeq L. \quad (\text{HFD28})$$

The first equality follows in every fibre of (HFD25), whose amplitude quotient is then zero. It retains every flag and the proper mixed prime (HFD21). Thus the existence of that residual support prime is not evidence for an off-real zero; it persists even after the analytic amplitude quotient vanishes. This conclusion follows from the computed objects and does not decide whether the actual restriction has such a zero.

40.7 The transported product and the correct time in the divisor

Keep the actual theta time s distinct from the algebra transport time t . On $(\mathcal{A}_L, \star_t)$ the full heat map $U_{-t}(f, \nu) = (T_{-t}f, \nu)$ is a unital algebra isomorphism to $(\mathcal{A}_L, \circ)$. It sends h_s to h_{s-t} and z_η to itself by the proved actual heat law. Therefore the exact generated transported congruence is

$$(f, \nu) \equiv_{s;t} (g, \mu) \iff \nu = \mu, \quad T_{-t}(f - g) = H_{s-t}b \text{ for some } b \in V_\nu. \quad (\text{HFD29})$$

This follows by applying the algebra isomorphism to the entire proved congruence (HFD6), not only to its generating pair. The spaces $\mathcal{E}_\nu$ and V_ν are stable under T_t and T_{-t} : each allowed unrestricted parity stays in its parity, and every restricted parity is a finite-degree polynomial space on which the even derivatives only lower the degree. Thus (HFD29) also gives the exact vector-space fibres in the original coefficient coordinates.

Writing $\mathcal{D}_{s,L}^{\text{flag}, \star_t}$ for this quotient, the full comparison is

$$\begin{aligned} \mathcal{D}_{s,L}^{\text{flag}, \star_t} &\xrightarrow[\substack{\sim \\ [f]_\nu \mapsto [T_{-t}f]_\nu}]{} \mathcal{D}_{s-t,L}^{\text{flag}} \\ \mathcal{D}_{s,L}^{\text{flag}, \star_t} &\longrightarrow G_L(\mathcal{E}/(H_{s-t})) \longrightarrow G_L(\mathcal{O}(\Omega)/(H_{s-t}|\Omega)). \end{aligned} \quad (\text{HFD30})$$

The global amplitude representative is $T_{-t}f$. The complete off-real jet test is (HFD24) with H_{s-t} and $T_{-t}(f - g)$. The full coefficient quotient map likewise has amplitude ring $(\mathcal{E}, \star_t)/(H_s)_{\star_t}$, which T_{-t} identifies with $\mathcal{E}/(H_{s-t})$; its support flag map is unchanged. Every parity receiver and saturation argument (HFD18)–(HFD22) remains exact.

For the same physical divisor at time s in the transported presentation, the generator is $h_{s+t} = U_t h_s$. In that notation the exact identification is

$$\mathcal{D}_{s+t,L}^{\text{flag}, \star_t} \simeq \mathcal{D}_{s,L}^{\text{flag}}. \quad (\text{HFD31})$$

Ordinary evaluation on the unchanged amplitude f is not silently substituted for the transported amplitude map. The ordinary and transported constructions, their full coefficient quotients, their off-real jet receivers and their mixed prime therefore have specified connecting morphisms throughout.

41 Heat transport of divisors, actual evaluation, and the supported critical locus

Inputs and attribution. The function H_t , its kernel and its original coordinates are those of Brad Rodgers and Terence Tao, *The De Bruijn–Newman constant is non-negative*, original equations `hoz`, `sas`, `phidef`, `htdef`. THS1–31 proves the analytic estimates used here directly for that kernel. HSL15–18 constructs the complete coefficient function space $\mathcal{E}$ and its convergent heat operators T_t . The complete supported carrier and its injection Φ_R are those of the cited foundation and ASL18–22; HZD1–18 proves every quotient comparison needed below. The heat-transported product and full parity-flag algebra are proved in the included HTP and SHP sections. Their defining maps are stated explicitly here to identify the actual divisor being transported.

41.1 Which divisor is carried by the heat algebra isomorphism

Retain the whole function space

$$\mathcal{E} = \left\{ f(z) = \sum_{n \geq 0} a_n z^n : \sum_{n \geq 0} |a_n| \sqrt{n!} h^n < \infty \text{ for every } h > 0 \right\}. \quad (\text{HDT1})$$

The ordinary product and the transported product are respectively the Cauchy product and

$$f \star_t g = T_t((T_{-t}f)(T_{-t}g)), \quad T_t : (\mathcal{E}, \cdot) \xrightarrow{\sim} (\mathcal{E}, \star_t). \quad (\text{HDT2})$$

The complete convergence and multiplication proofs are in HTP. In particular $T_t 1 = 1$, $T_{-t} T_t = I$, and this is an actual unital algebra isomorphism, whereas T_t is generally not an automorphism for the unchanged ordinary product.

Set $\Omega = \mathbb{C} \setminus \mathbb{R}$ and $A = \mathcal{O}(\Omega)$ as in HZD7. The exact ring map into the original off-real function ring is

$$\rho_t : (\mathcal{E}, \star_t) \longrightarrow A, \quad \rho_t(f) = (T_{-t}f)|_{\Omega}. \quad (\text{HDT3})$$

It preserves products by (HDT2), sums by linearity and the unit because $T_{-t} 1 = 1$. It is injective: if its value vanishes, the entire function $T_{-t}f$ vanishes on a nonempty open set, so all its analytic continuation and hence f vanish. Most importantly, the actual heat identity proved in THS13 gives

$$\rho_t(H_t) = H_0|_{\Omega}. \quad (\text{HDT4})$$

This specifies the divisor transported by the algebra isomorphism. Base change of its principal quotient along (HDT3) is exactly

$$A \otimes_{(\mathcal{E}, \star_t)} ((\mathcal{E}, \star_t)/(H_t)_{\star_t}) \simeq A/(H_0|_{\Omega}). \quad (\text{HDT5})$$

To prove the universal property directly, a pair of agreeing unital ring maps from A and the displayed quotient into a ring B is precisely a unital map from A that sends $\rho_t(H_t) = H_0|_{\Omega}$ to zero. Such maps factor uniquely through the right side. Conversely every map from that right side gives this agreeing pair. These two operations are inverse, proving (HDT5) and its explicit quotient map.

The fully supported version imposes the supported equation, with every label unchanged:

$$\begin{array}{ccc} G_L(\mathcal{E}, \star_t) & \xrightarrow{G_L(\rho_t)} & G_L(A) \\ \downarrow \widehat{H}_t = e & & \downarrow \widehat{H}_0 = e \\ G_L((\mathcal{E}, \star_t)/(H_t)_{\star_t}) & \longrightarrow & G_L(A/(H_0)). \end{array} \quad (\text{HDT6})$$

It is a pushout of unital semirings. Indeed HZD2 identifies both vertical arrows with the congruences generated by the indicated supported equations. A compatible map from the upper-right object must identify the images of $\widehat{H}_t$ and e , which by (HDT4) are exactly $\widehat{H}_0$ and e . HZD2 gives the unique factorization. The converse is immediate by composing with the quotient maps. Applying the injective complete map Φ to each corner gives the corresponding ring-amplitude and K_L square of HZD4. No support character has been selected and no label has been collapsed in (HDT6).

By contrast ordinary restriction of the actual function H_t uses the different ring map $(\mathcal{E}, \cdot) \rightarrow A$ and gives $G_L(A/(H_t))$, the actual time- t object in HZD7. Ordinary evaluation is not silently used as a character of $\star_t$. For example

$$z \star_t z = z^2 - 2t, \quad \text{ev}_{z_0}(z \star_t z) - \text{ev}_{z_0}(z)^2 = -2t. \quad (\text{HDT7})$$

Here $T_{-t}z = z$ and $T_t(z^2) = z^2 - 2t$, proving both formulas without any infinite sum. For $t \neq 0$ this explicitly excludes ordinary evaluation from being a character of that algebra. The exact transported character is instead

$$\chi_{t,z_0}(f) = (T_{-t}f)(z_0).$$

It is multiplicative by (HDT2). This proves both the obstruction and the map that actually transports the point.

41.2 Every transported primary character and its exact jet

For any $z_0 \in \mathbb{C}$ and integer $m \geq 1$, define

$$J_{z_0,m}^t(f) = \sum_{j=0}^{m-1} \frac{(T_{-t}f)^{(j)}(z_0)}{j!} \epsilon^j \in \mathbb{C}[\epsilon]/(\epsilon^m), \quad (\text{HDT8})$$

$$F_{t,z_0,m}(z) = T_t((z - z_0)^m) = \sum_{k=0}^{\lfloor m/2 \rfloor} \frac{(-t)^k m!}{k!(m-2k)!} (z - z_0)^{m-2k}.$$

The first map is a unital ring homomorphism on $(\mathcal{E}, \star_t)$, because it is T_{-t} followed by the Taylor jet homomorphism with its actual factorials. It is surjective: lift an arbitrary jet by its polynomial in $z - z_0$, then apply T_t to that polynomial. Its kernel is precisely the $\star_t$ principal ideal generated by $F_{t,z_0,m}$.

For the kernel assertion, first prove division within the actual space (HDT1). If $g \in \mathcal{E}$ vanishes to order at least m at z_0 , the quotient $g/(z - z_0)^m$ is entire by its Taylor expansion. Outside $|z - z_0| \geq 1$ its absolute value is at most $|g(z)|$; on the remaining compact disc it is bounded. Thus every positive Gaussian weighted supremum of the quotient is finite. The equivalence proved in HSL16 puts it back in $\mathcal{E}$. Hence the ordinary jet kernel is $(z - z_0)^m \mathcal{E}$. Applying the proved isomorphism T_t gives exactly the asserted $\star_t$ ideal. The finite formula for $F_{t,z_0,m}$ follows by differentiating the unchanged polynomial; terms with $2k > m$ are zero. We have proved

$$G_L((\mathcal{E}, \star_t)/(F_{t,z_0,m})_{\star_t}) \simeq G_L(\mathbb{C}[\epsilon]/(\epsilon^m)), \quad (\text{HDT9})$$

where the map is $(f, \lambda) \mapsto (J_{z_0,m}^t(f), \lambda)$. This retains all labels, the exact coordinate z_0 , and the original Taylor constants. When z_0 is an actual root of H_0 of multiplicity m , (HDT4) implies $J_{z_0,m}^t(H_t) = 0$. No existence of an off-real root is assumed by this universally quantified construction.

For example, at $m = 2$ the ordinary zeros of $F_{t,z_0,2}$ satisfy $(z - z_0)^2 = 2t$, whereas its transported primary character is the single multiplicity-two character given by $J_{z_0,2}^t$. Formula (HDT8) proves their exact relation. This example does not replace the actual theta kernel by a quadratic polynomial; it identifies precisely which point map the heat algebra isomorphism uses.

41.3 An explicit local bound at every actual simple zero

The full map also identifies where analytic root transport is legitimate in the original coefficient semiring. Let $H_{t_0}(z_0) = 0$ and put $a = \partial_z H_{t_0}(z_0) \neq 0$. This is a statement about any actual simple zero; none is postulated off the real axis. Use the explicit constants of THS4 and set

$$\begin{aligned} A_0 &= \max\{\operatorname{Re} t_0 + 1, 0\}, \quad B_0 = |\operatorname{Im} z_0| + 1, \quad M_2 = \mathcal{M}_2(A_0, B_0), \quad M_3 = \mathcal{M}_3(A_0, B_0), \\ \rho_z &= \min\{1, |a|/(4M_2)\}, \quad \rho_t = \min\{1, |a|/(4M_3), |a|\rho_z/(2M_2)\}. \end{aligned} \quad (\text{HDT10})$$

All these radii are positive and retain the original t, z, a . On their closed bidisc, THS6 gives

$$|\partial_z H_t(z) - a| \leq \rho_t M_3 + \rho_z M_2 \leq |a|/2, \quad |H_t(z_0)| \leq |t - t_0| M_2 \leq |a|\rho_z/2.$$

The first inequality follows by integrating $\partial_t \partial_z H$ along the time segment and $\partial_z^2 H$ along the space segment, both inside the unit bidisc; the second uses $\partial_t H = -\partial_z^2 H$. Thus $F_t(z) = z - a^{-1} H_t(z)$ has derivative of absolute value at most $1/2$ and maps $|z - z_0| \leq \rho_z$ into itself. Starting at z_0 , its successive iterates have differences bounded by a geometric sequence of ratio $1/2$, uniformly in t on the time disc. They therefore converge to a unique fixed point $z(t)$ in that space disc. The iterates are holomorphic in t , and their convergence is uniform on compact subsets, so the limit is holomorphic there. Its fixed-point equation is exactly $H_t(z(t)) = 0$.

Differentiating this proved identity and using THS7 yields the exact root motion and bound

$$\begin{aligned} z'(t) &= \frac{\partial_z^2 H_t(z(t))}{\partial_z H_t(z(t))}, \quad |\partial_z H_t(z(t))| \geq |a|/2, \\ |z'(t)| &\leq 2M_2/|a|, \quad |z(t) - z_0| \leq 2M_2|t - t_0|/|a|. \end{aligned} \quad (\text{HDT11})$$

The last inequality follows by integrating the derivative on the straight time segment. In the original $G_L(\mathbb{C})$ the denominator is the actual unit $(\partial_z H_t(z(t)), 1_L)$, whose inverse is $(1/\partial_z H_t(z(t)), 1_L)$. This proves the root-motion division before localization, through the complete coefficient map. The fraction is not obtained by inverting e .

41.4 The full supported object where that inverse fails

Let $\mathcal{B} = \mathcal{O}(\mathbb{C}^2)$ in the original coordinates (t, z) and let $\mathcal{H}(t, z) = H_t(z)$, which is jointly entire by THS6. The exact supported critical-zero object is

$$\mathcal{C}_L = G_L(\mathcal{B}/(\mathcal{H}, \partial_z \mathcal{H})). \quad (\text{HDT12})$$

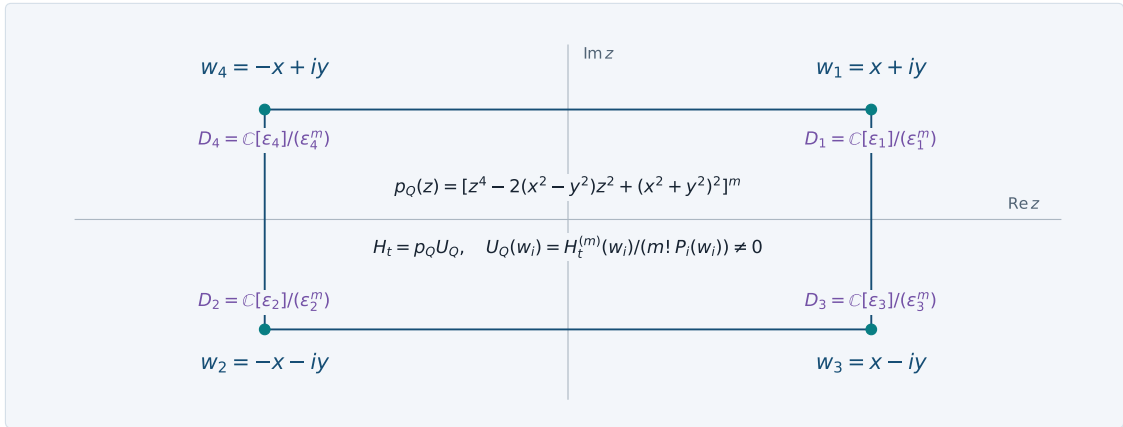
It is the congruence quotient of $G_L(\mathcal{B})$ by both supported equations $\widehat{\mathcal{H}} = e$ and $\widehat{\partial_z \mathcal{H}} = e$. The proof of HZD2 extends directly: a difference in this ideal is $b\mathcal{H} + c\partial_z \mathcal{H}$, and multiplying the two generating congruences by $\widehat{b}, \widehat{c}$ and adding proves exactly that equality at top support. Different labels remain different under the displayed quotient map, and lower-labelled amplitudes were already zero. This proves the whole kernel congruence and surjectivity.

For every actual parameter-point pair, evaluation factors through the ring coordinate of (HDT12) exactly when both $H_t(z) = 0$ and $\partial_z H_t(z) = 0$ hold. These are exactly the points on $\mathcal{H} = 0$ at which the coefficient inverse used in (HDT11) is unavailable in the original semiring. The statement follows by evaluating the two generators, and the converse follows because evaluation then annihilates every ideal combination. The complete coefficient map embeds (HDT12) into $(\mathcal{B}/(\mathcal{H}, \partial_z \mathcal{H})) \times K_L$ by the same Φ as throughout this reader.

Supported-zero inversion sends (HDT12) to L , just as in HZD6. It makes the image of the supported derivative zero invertible there, but the entire derivative amplitude has then been removed. The exact original critical object and its map have now been constructed; no singular point is declared absent on that basis. This is the location at which further information about the actual arithmetic heat family has to be calculated, not replaced by an unproved inverse in its original amplitude field.

Four distinct actual roots: the complete supported receiver

Symbolic layout for $x > 0, y > 0$; no numerical root is asserted. All formulas retain the original centres.



$$\begin{aligned}
 G_L(\mathcal{O}(\mathbb{C} \setminus \mathbb{R})/(H_t)) &\rightarrow G_L(\prod_{i=1}^4 D_i) \\
 [f] &\mapsto (\sum_{k=0}^{m-1} f^{(k)}(w_i) \varepsilon_i^k / k!)_{i=1}^4 \\
 \Phi((b_i)_i, \lambda) &= ((b_i)_i, 1_K \otimes \lambda) \\
 \hat{q}_i \hat{q}_j &= e \ (i \neq j), \quad \sum_{i=1}^4 \hat{q}_i = 1, \quad \hat{q}_i^2 = \hat{q}_i
 \end{aligned}$$

Every actual zero obeys $x^2 \partial_t H_t(iy) > 2H_t(iy) > 0$.

The receiver retains four arithmetic primes, $4m$ jet coordinates, and every lattice label.

Proofs: HSQ1–27; the shared-label injection has an exact module retract but no semiring retract.

Kernel: Brad Rodgers & Terence Tao, arXiv:1801.05914v5. Full carrier: The Clankers, v11, def:lattice-split.

Figure 17: The original four-point configuration and its complete jet receiver. The drawing illustrates the indicated orientation of symbolic coordinates; it supplies no numerical off-real zero. All factors, maps and the strict exclusion inequality are proved in HSQ1–27.

42 Every actual off-real heat quartet and its full supported jet receiver

Sources and scope. The unchanged kernel and heat coordinate are Brad Rodgers and Terence Tao's *The De Bruijn–Newman constant is non-negative*, original author source `fmp-template.tex`, equations `hoz`, `sas`, `phidef`, and `htdef`, lines 78–103. Source lines 74–110 were read for this calculation. The original source's SHA-256 is

25b015c0fb151f1613247d82cf53a6561a320d4dab2bb4f809482e7c477a5817.

The complete included theta heat proof THS1–31 and the complete supported divisor proof HZD1–18 were read. Their integral bounds, actual divisor, and full coefficient map are retained below with their exact constants. The split-support carrier is the existing construction of The Clankers, *Split Support Geometry*, v11, Definition `def:lattice-split`. All finite-jet quotient, symmetry, support, and exclusion statements made here are proved below. The construction is indexed by actual zeros of the given function; it does not assume that the indexing set is nonempty.

42.1 The original moments and the exclusion inequality

For $u \geq 0$ and real t retain exactly

$$\begin{aligned}\Phi(u) &= \sum_{n \geq 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, \\ H_t(z) &= \int_0^\infty e^{tu^2} \Phi(u) \cos(zu) du.\end{aligned}\tag{HSQ1}$$

Every summand of the kernel is positive: its prefactor is $\pi n^2 e^{5u} (2\pi n^2 e^{4u} - 3) > 0$. THS3–6 prove convergence, all differentiations used below, and the explicit estimates

$$\begin{aligned}0 &< \pi(2\pi - 3) e^{9u - \pi e^{4u}} \leq \Phi(u) \leq C_\Phi e^{9u - \pi e^{4u}}, \\ a &= e^{-3\pi}, \quad C_\Phi = 2\pi^2 \frac{1 + 11a + 11a^2 + a^3}{(1 - a)^5}, \\ \mathcal{M}_p(A, B) &= \frac{C_\Phi}{2\pi} \exp\left(\frac{3A^2}{32\pi} + \frac{(B + 9 + p)^2}{8\pi} - \frac{\pi}{2}\right), \quad A, B \geq 0.\end{aligned}\tag{HSQ2}$$

In particular the integral of $u^p e^{Au^2 + Bu} \Phi(u)$ is at most $\mathcal{M}_p(A, B)$. For a direct convergence check of this bound, use $u^p \leq e^{pu}$, reserve half of $-\pi e^{4u}$, and apply $e^{4u} \geq (32/3)u^4$ and $e^{4u} \geq 8u^2$ to the other two quarters. Completing squares bounds the remaining positive exponent by $3A^2/(32\pi) + (B + 9 + p)^2/(8\pi)$. Finally $e^{4u} \geq 1 + 4u$ bounds the integral of the reserved exponential by $e^{-\pi/2}/(2\pi)$. This recovers the exact constant in (HSQ2) and justifies every moment integral below.

For every real y and integer $r \geq 0$, define the actual positive moments

$$\begin{aligned}M_{2r}(t, y) &= \int_0^\infty u^{2r} e^{tu^2} \Phi(u) \cosh(yu) du = \partial_t^r H_t(iy), \\ c_t &= \pi(2\pi - 3) \exp(\min(t, 0) - \pi e^4), \quad \frac{c_t}{2r + 1} \leq M_{2r}(t, y) \leq \mathcal{M}_{2r}(\max(t, 0), |y|).\end{aligned}\tag{HSQ3}$$

The upper bound uses $\cosh(yu) \leq e^{|y|u}$ and (HSQ2). For the lower bound restrict to $0 \leq u \leq 1$, use $\cosh(yu) \geq 1$, $e^{tu^2} \geq e^{\min(t, 0)u^2}$, and the lower kernel bound in (HSQ2), then integrate u^{2r} . Thus $H_t(iy) = M_0(t, y) > 0$ and $\partial_t H_t(iy) = M_2(t, y) > 0$ for all the stated real coordinates. In terms of the original coefficients $h_j(t)$ of THS8 the same moments are exactly

$$M_{2r}(t, y) = \sum_{j=0}^\infty \frac{(-1)^{j+r} (2j + 2r)!}{(2j)!} h_{j+r}(t) y^{2j}.\tag{HSQ4}$$

Indeed expand $\cosh(yu)$ in its power series and substitute the moment definition of h_{j+r} . Absolute convergence is bounded by (HSQ3); all its moment summands are nonnegative for real t . This proves the formula with every original factorial and sign.

For every real x, y, t , the full real and imaginary parts are

$$\begin{aligned}\operatorname{Re} H_t(x + iy) &= \int_0^\infty e^{tu^2} \Phi(u) \cos(xu) \cosh(yu) du, \\ \operatorname{Im} H_t(x + iy) &= - \int_0^\infty e^{tu^2} \Phi(u) \sin(xu) \sinh(yu) du.\end{aligned}\tag{HSQ5}$$

The elementary inequality $\cos v \geq 1 - v^2/2$ is strict for $v \neq 0$. To verify strictness, the even function $\cos v - 1 + v^2/2$ vanishes at zero and has derivative $v - \sin v > 0$ for $v > 0$: the latter is the integral of $1 - \cos s$, positive on any such interval. Integrating the inequality against the positive weight in (HSQ5) gives

$$\operatorname{Re} H_t(x + iy) \geq H_t(iy) - \frac{x^2}{2} \partial_t H_t(iy).\tag{HSQ6}$$

For $x \neq 0$ this is a strict inequality, because its exact remainder is the integral of the strictly positive function $e^{tu^2} \Phi(u) \cosh(yu) [\cos(xu) - 1 + x^2 u^2/2]$ on $u > 0$. Its integrability follows from (HSQ3) for $r = 0, 1$. Consequently every zero of H_t at $x + iy$ satisfies

$$x \neq 0, \quad x^2 \partial_t H_t(iy) > 2H_t(iy) > 0, \quad x^2 > \frac{2M_0(t, y)}{M_2(t, y)} \geq \frac{2c_t}{\mathcal{M}_2(\max(t, 0), |y|)}.\tag{HSQ7}$$

In particular the requested non-strict necessary bound follows, and the closed region $x^2 \leq 2M_0(t, y)/M_2(t, y)$ is actually zero free. A completely explicit uniform version is obtained for $t_- \leq t \leq t_+$ and $|y| \leq Y$ by replacing its last ratio by $2c_{t_-}/\mathcal{M}_2(\max(t_+, 0), Y)$. No zero location, time, or kernel has been changed in deriving these bounds.

42.2 The four distinct original points and their common multiplicity

Let $Z_t^{\text{off}} = \{z \in \mathbb{C} \setminus \mathbb{R} : H_t(z) = 0\}$ be the actual zero set at a fixed real t . For each $z_0 = x + iy \in Z_t^{\text{off}}$, both x and y are nonzero: $y \neq 0$ by its definition, and $x \neq 0$ by (HSQ3). Retain the ordered four points

$$w_1 = z_0 = x + iy, \quad w_2 = -z_0 = -x - iy, \quad w_3 = \overline{z_0} = x - iy, \quad w_4 = -\overline{z_0} = -x + iy.\tag{HSQ8}$$

They are distinct. For example an equality with its negative would force $x = y = 0$, an equality with its conjugate would force $y = 0$, and an equality with its negative conjugate would force $x = 0$; these cover the pairwise possibilities after symmetry. They are the four corners of the actual rectangle with horizontal coordinates $\pm x$ and vertical coordinates $\pm y$.

The source integral gives $H_t(-z) = H_t(z)$ and $H_t(\bar{z}) = \overline{H_t(z)}$. Its derivatives obey

$$H_t^{(k)}(-z) = (-1)^k H_t^{(k)}(z), \quad H_t^{(k)}(\bar{z}) = \overline{H_t^{(k)}(z)}.\tag{HSQ9}$$

The first follows by differentiating the even identity, the second by conjugating its local power series. Since $H_t(0) > 0$, this entire function is not identically zero. Its first nonzero Taylor coefficient at any root therefore has a finite positive index. Equations (HSQ9) prove that the exact multiplicity is one and the same integer $m \geq 1$ at all four points. In particular the four leading coefficients remain $H_t^{(m)}(w_i)/m!$ with the signs and conjugations in (HSQ9). None is replaced by one.

42.3 The exact polynomial and the entire residual factor

Define the monic polynomial with these actual coordinates

$$\begin{aligned}
p_Q(z) &= \prod_{i=1}^4 (z - w_i)^m \\
&= [(z^2 - z_0^2)(z^2 - \bar{z}_0^2)]^m \\
&= [z^4 - 2(x^2 - y^2)z^2 + (x^2 + y^2)^2]^m, \quad P_i(z) = \prod_{j \neq i} (z - w_j)^m.
\end{aligned} \tag{HSQ10}$$

All three forms follow by multiplying the specified factors; the first form retains the four separately indexed points. Division gives an actual entire residual factor

$$H_t(z) = p_Q(z)U_Q(z), \quad U_Q(w_i) = \frac{H_t^{(m)}(w_i)}{m!P_i(w_i)} \neq 0. \tag{HSQ11}$$

Away from the four points take the ordinary quotient. At each point its Taylor factor $(z - w_i)^m$ cancels and the other factors are units, extending it holomorphically. This proves (HSQ11) globally. In the original coordinates its denominators are

$$(P_1(w_1), P_2(w_2), P_3(w_3), P_4(w_4)) = ((8ixyz_0)^m, (-8ixyz_0)^m, (-8ixy\bar{z}_0)^m, (8ixy\bar{z}_0)^m). \tag{HSQ12}$$

They follow by multiplying the three original differences at each point. All are nonzero by (HSQ8). The residual factor can have other zeros; no global inverse for it is asserted.

42.4 Explicit Hermite idempotents and the complete CRT map

Put $\Omega = \mathbb{C} \setminus \mathbb{R}$ and $A = \mathcal{O}(\Omega)$, keeping both components. For every i define, exactly in the original variable,

$$Q_i(z) = \sum_{k=0}^{m-1} \frac{(1/P_i)^{(k)}(w_i)}{k!} (z - w_i)^k, \quad E_i(z) = P_i(z)Q_i(z). \tag{HSQ13}$$

The coefficient of $(z - w_i)^k$ can equivalently be written as

$$\frac{1}{P_i(w_i)} \sum_{\substack{\alpha_j \geq 0 \\ \sum_{j \neq i} \alpha_j = k}} \prod_{j \neq i} \frac{(-1)^{\alpha_j} \binom{m+\alpha_j-1}{\alpha_j}}{(w_i - w_j)^{\alpha_j}}. \tag{HSQ14}$$

Indeed expand each factor $(1 + (z - w_i)/(w_i - w_j))^{-m}$ near w_i , using its convergent negative-integer binomial series, and multiply the three series. Only finitely many terms contribute to any displayed coefficient. This proves (HSQ14) with its original differences and signs.

By Taylor multiplication, $E_i \equiv 1 \pmod{(z - w_i)^m}$; at every other w_j its factor P_i gives $E_i \equiv 0 \pmod{(z - w_j)^m}$. Each E_i has degree less than $4m$. A polynomial vanishing to order at least m at all four distinct points is divisible by their product p_Q : successive division by the individual factors preserves the remaining vanishing orders since their values at other points are nonzero. It follows that

$$\sum_i E_i = 1 \quad \text{as polynomials,} \quad E_i E_j \equiv \delta_{ij} E_i \pmod{p_Q}. \tag{HSQ15}$$

For the first identity the difference has degree less than $4m$ and is divisible by the degree- $4m$ monic polynomial. The second follows by checking all four jets and the same divisibility test.

Retain four distinct formal coordinates and rings $D_i = \mathbb{C}[\epsilon_i]/(\epsilon_i^m)$ and $B_Q = \prod_i D_i$. The explicit ring map and its inverse on the quotient are

$$\begin{aligned} J_Q : A/(p_Q) &\xrightarrow{\sim} B_Q, & [f] &\mapsto \left(\sum_{k=0}^{m-1} \frac{f^{(k)}(w_i)}{k!} \epsilon_i^k \right)_{i=1}^4, \\ (b_i(\epsilon_i))_i &\mapsto \left[\sum_{i=1}^4 E_i(z) b_i(z - w_i) \right], & J_Q([z]) &= (w_i + \epsilon_i)_i. \end{aligned} \quad (\text{HSQ16})$$

The product rule proves multiplicativity of each jet map with its stated factorials. The displayed inverse realizes any tuple by a polynomial, because of (HSQ15). The kernel on A is exactly (p_Q) : a vanishing jet makes f/p_Q extend holomorphically at its corresponding point, and away from the four points it is already holomorphic on both components of Ω . Conversely a multiple of p_Q has all four zero jets. This proves all maps and every claimed equality in (HSQ16).

Now retain the original heat divisor ring $R_t = A/(H_t|\Omega)$. Equation (HSQ11) proves $(H_t) \subseteq (p_Q)$, so its actual quartet receiver is the surjection

$$\pi_Q : R_t \twoheadrightarrow B_Q, \quad [f] \mapsto \left(\sum_{k=0}^{m-1} \frac{f^{(k)}(w_i)}{k!} \epsilon_i^k \right)_i, \quad \ker \pi_Q = (p_Q)/(H_t). \quad (\text{HSQ17})$$

Surjectivity and the kernel follow from (HSQ16). The Hermite polynomial is a section of $A \rightarrow A/(p_Q)$ as a finite-dimensional complex vector space, with subsequent remainder understood in the quotient. It is not being asserted to define a multiplicative section $B_Q \rightarrow R_t$. The original H_t , the actual residual factor U_Q , and the exact kernel in (HSQ17) remain present.

42.5 The whole split-support receiver and the complete coefficient map

Let L be the full nontrivial bounded distributive lattice and write $S_T = G_L(T)$ for any ring T . Retain

$$\begin{aligned} \mathcal{D}_{t,L} &= G_L(R_t), & \mathcal{D}_{t,z_0,L} &= G_L(B_Q), & G_L(\pi_Q)([f], \lambda) &= (\pi_Q[f], \lambda), \\ \mathcal{D}_{t,L} / \langle ([p_Q], 1_L) \sim (0, 1_L) \rangle &\simeq \mathcal{D}_{t,z_0,L}. \end{aligned} \quad (\text{HSQ18})$$

The full operations are $(b, \lambda) + (c, \mu) = (b + c, \lambda \vee \mu)$ and $(b, \lambda)(c, \mu) = (bc, \lambda \wedge \mu)$, with nonzero amplitudes allowed only at top support. The map is a surjective semiring homomorphism by (HSQ17), with every label unchanged. Its kernel congruence is equality of labels and amplitude difference in $\ker \pi_Q$. At top support every difference is a multiple of $[p_Q]$, so multiplication of the displayed generating relation followed by addition generates that entire congruence. Below top support the amplitude is already zero. This proves the supported quotient, including its zero convention.

Let K be the retained tropical semifield with zero, let $K_L = K \otimes_{\mathbb{B}} L$, and keep $\iota_L(\lambda) = 1_K \otimes \lambda$ with its exact retraction $\rho_L(\sum a_i \otimes \lambda_i) = \bigvee_{a_i \neq 0} \lambda_i$. The complete maps are

$$\Phi_T : S_T \hookrightarrow T \times K_L, \quad (b, \lambda) \mapsto (b, \iota_L(\lambda)), \quad \Phi_{B_Q} G_L(\pi_Q) = (\pi_Q \times \text{id}_{K_L}) \Phi_{R_t}. \quad (\text{HSQ19})$$

Both coordinates preserve the operations, and the retraction of ι_L proves injectivity. The equality is verified on each displayed pair. Thus the complete map carries the actual quartet quotient, with all four jets and its original shared label. In particular the image of the supported equation $H_t = 0$ is $(0_{B_Q}, \iota_L(1_L))$, not $(0_{B_Q}, 0_{K_L})$.

For each i let $q_i \in B_Q$ be the element with unit in its i th ring component and zero in the other three; it is the class of E_i . Its full-supported lift satisfies

$$\widehat{q}_i^2 = \widehat{q}_i, \quad \widehat{q}_i \widehat{q}_j = e \quad (i \neq j), \quad \sum_i \widehat{q}_i = 1_{G_L(B_Q)}. \quad (\text{HSQ20})$$

The middle amplitude is zero but its meet of top labels is top. Each coordinate ϵ_i retains order m in its ring factor; the lift of its zero m th power is likewise the supported zero. The receiver has complex ring dimension $4m$, with the original basis $q_i \epsilon_i^k$, $1 \leq i \leq 4$, $0 \leq k < m$.

There are exactly four arithmetic primes in this receiver, namely

$$\mathfrak{Q}_i = \{(0, \lambda) : \lambda \in L\} \cup \{(b, 1_L) : b_i(0) = 0\} \quad (1 \leq i \leq 4). \quad (\text{HSQ21})$$

To prove exhaustion on the ring side, the orthogonal idempotents q_i sum to one, so a prime selects exactly one factor: two excluded idempotents would have zero product, and excluding none would put their sum in the prime. In that factor ϵ_i is nilpotent and hence belongs to every prime; quotienting by it gives $\mathbb{C}$, which has only the zero prime. The full split-support classification therefore adds exactly the support primes $\{(0, \lambda) : \lambda \in P\}$ for $P \in \text{SpecLat } L$, each contained in all four (HSQ21). The four primes are distinct, as detected by the specified q_i . None has been identified with another by a symmetry.

42.6 Shared labels, independent labels, and the exact synchronization

The canonical comparison is the injective unital semiring map

$$\theta : G_L\left(\prod_i D_i\right) \hookrightarrow \prod_i G_L(D_i), \quad ((b_i)_i, \lambda) \mapsto ((b_i, \lambda))_i. \quad (\text{HSQ22})$$

Its image consists exactly of the tuples whose four labels are equal. The individual carrier conditions then imply that any nonzero amplitude makes that common label top. Conversely those conditions are exactly the original carrier of $G_L(B_Q)$. All operations and the injection follow coordinatewise.

The strongest elementary synchronization is the actual retraction

$$r : \prod_i G_L(D_i) \longrightarrow G_L(B_Q), \quad ((b_i, \lambda_i))_i \mapsto ((b_i)_i, \bigvee_i \lambda_i), \quad r\theta = \text{id}. \quad (\text{HSQ23})$$

A nonzero ring tuple forces at least one λ_i to be top, so this belongs to the required carrier. The retraction preserves addition, zero, and unit. More precisely it is a homomorphism of $G_L(B_Q)$ -semimodules when the product is acted on through θ : for a scalar label ν , distributivity gives $\bigvee_i (\nu \wedge \lambda_i) = \nu \wedge \bigvee_i \lambda_i$, and the amplitude equality is componentwise multiplication. Thus θr is an idempotent linear projection onto the shared-label object, with every amplitude coordinate untouched.

It is not multiplicative in general. The entire defect is explicit: for $v = ((b_i, \lambda_i))$, $w = ((c_i, \mu_i))$, put

$$\alpha = \bigvee_i (\lambda_i \wedge \mu_i), \quad \beta = \left(\bigvee_i \lambda_i\right) \wedge \left(\bigvee_j \mu_j\right). \quad (\text{HSQ24})$$

Then $\alpha \leq \beta$ and $r(vw) = ((b_i c_i)_i, \alpha)$ while $r(v)r(w) = ((b_i c_i)_i, \beta) = r(vw) + z_\beta$. Taking zero amplitudes, top label only in the first coordinate of v and only in the second of w , gives $\alpha = 0_L$ and $\beta = 1_L$, so the defect is real.

In fact no unital semiring retraction of (HSQ22) exists for this nontrivial L . For a direct proof let v_i be the product element with unit in coordinate i and global zero in the other coordinates. For $i \neq j$, $v_i v_j = 0$, whereas $v_i + \theta(e) = \theta(\widehat{q}_i)$. If a semiring retraction s existed, the latter identity would give $s(v_i) + e = \widehat{q}_i$. Its amplitude is the nonzero ring element q_i , forcing $s(v_i) = \widehat{q}_i$ by the carrier condition. It follows that $s(v_i v_j) = \widehat{q}_i \widehat{q}_j = e$, contradicting preservation of zero, since $e \neq \tau$. The module retraction (HSQ23) is therefore retained in its proved category, rather than promoted to a nonexistent semiring retraction.

The full coefficient maps participate in exactly the same relation. Define

$$\begin{aligned}\Gamma : B_Q \times K_L &\hookrightarrow \prod_i (D_i \times K_L), & ((b_i), k) &\longmapsto ((b_i, k))_i, \\ \mathcal{R} : \prod_i (D_i \times K_L) &\longrightarrow B_Q \times K_L, & ((b_i, k_i))_i &\longmapsto ((b_i), \sum_i k_i), \\ (\prod_i \Phi_{D_i})\theta &= \Gamma\Phi_{B_Q}, & \mathcal{R}(\prod_i \Phi_{D_i}) &= \Phi_{B_Q}r, & \mathcal{R}\Gamma &= \text{id}.\end{aligned}\tag{HSQ25}$$

The map Γ is a semiring embedding with common second coordinate as its image. The map $\mathcal{R}$ is additive and linear for the $B_Q \times K_L$ -action through Γ . Distributivity and the idempotence of addition in K_L prove its linearity and retraction property. For the two remaining equations use $\sum_i \iota_L(\lambda_i) = \iota_L(\bigvee_i \lambda_i)$. On the original supported image, the common tropical value is exactly $\iota_L(\lambda)$, retaining all of L . These are the complete maps, not a projection to a chosen lattice character.

42.7 Exact symmetry actions on the four jet factors

On A define $\mathbf{a}f(z) = f(-z)$ and $\mathbf{c}f(z) = \overline{f(\bar{z})}$. The first is a complex linear ring involution; the second is a conjugate-linear ring involution. They commute, preserve H_t and p_Q , and hence act on both quotients in (HSQ17). Write $a = (1\ 2)(3\ 4)$ and $c = (1\ 3)(2\ 4)$ for their point permutations. In the exact jet coordinates, their actions are

$$\begin{aligned}(\mathbf{a}b)_i(\epsilon_i) &= \sum_{k=0}^{m-1} (-1)^k b_{a(i),k} \epsilon_i^k, \\ (\mathbf{c}b)_i(\epsilon_i) &= \sum_{k=0}^{m-1} \overline{b_{c(i),k}} \epsilon_i^k, & b_i(\epsilon_i) &= \sum_{k=0}^{m-1} b_{i,k} \epsilon_i^k.\end{aligned}\tag{HSQ26}$$

Differentiate $f(-z)$ k times and evaluate at w_i . The result is $(-1)^k f^{(k)}(-w_i)$, proving the first formula. For the second, conjugate the Taylor expansion at $\bar{w}_i$. These also prove compatibility with (HSQ16)–(HSQ17), including every coefficient sign and conjugation. Squaring either formula gives the identity; composing them gives the remaining involution with coefficients $(-1)^k \overline{b_{ac(i),k}}$, where $ac = (1\ 4)(2\ 3)$.

For the original coordinate element $Z = (w_i + \epsilon_i)_i$, $\mathbf{a}Z = -Z$ and $\mathbf{c}Z = Z$. The second equality follows because this conjugate-linear operation includes conjugation of the coefficient and permutes its centre; it is not coefficientwise conjugation at a fixed centre. The Hermite polynomials themselves satisfy the exact polynomial identities

$$E_i(-z) = E_{a(i)}(z), \quad \overline{E_i(\bar{z})} = E_{c(i)}(z).\tag{HSQ27}$$

Both sides have degree less than $4m$ and exactly the same four jets, by (HSQ13)–(HSQ15); their difference is divisible by p_Q and is therefore zero. Thus these symmetry assertions preserve the explicit CRT inverse as well as the quotient map.

Every symmetry lifts to G_L by changing its amplitude as in (HSQ26) and leaving its single support label unchanged. It commutes with Φ when the same amplitude action and the identity on K_L are used on the complete receiver. On the independently labelled product it also permutes the four labels with its factor permutation. Their join is unchanged, so both synchronizations in (HSQ23) and (HSQ25) commute with the actions. At no stage is the invariant subring substituted for B_Q , nor are orbit points or their nilpotent coordinates identified. The receiver therefore retains four distinct arithmetic points, all $4m$ jet coordinates, every lattice label, and the original coordinate constraints (HSQ7)–(HSQ8).

43 The unchanged Rodgers–Tao heat family through the full lattice map

Human source and exact reading coverage. Brad Rodgers and Terence Tao, *The De Bruijn–Newman constant is non-negative*, arXiv:1801.05914v5, 3 July 2021, is the author source used here. The user-supplied intact source archive is retained separately. Root read the original `fmp-template.tex`, lines 1–133, including the source equations `hoz`, `sas`, `phidef`, `htdef`, and Theorem `main`. This is a bounded reading of the definitions and theorem statement, not a claim to have audited the entire Rodgers–Tao proof. Their theorem is cited as their result. The threshold characterization is attributed in their introduction to de Bruijn and Newman; the backwards heat evolution is attributed there to Csordas, Smith and Varga. The connecting maps below are proved here using the full-lattice construction in ZIL1–25.

Keep every constant and the original time and complex coordinates:

$$\begin{aligned}\xi(s) &= \frac{s(s-1)}{2} \pi^{-s/2} \Gamma(s/2) \zeta(s), & H_0(z) &= \frac{1}{8} \xi\left(\frac{1}{2} + \frac{iz}{2}\right), \\ \Phi(u) &= \sum_{n=1}^{\infty} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, & (\text{ZHR1}) \\ H_t(z) &= \int_0^{\infty} e^{tu^2} \Phi(u) \cos(zu) du, & \partial_t H_t &= -\partial_z^2 H_t.\end{aligned}$$

These are exactly the cited author definitions, not a replacement kernel. The source’s threshold and its main theorem state

$$\{t \in \mathbb{R} : \text{every zero of } H_t \text{ is real}\} = [\Lambda, \infty), \quad \Lambda \geq 0. \quad (\text{ZHR2})$$

Since the source identifies RH with real zeros of H_0 , substitution of $t = 0$ into the displayed set gives RH if and only if $0 \geq \Lambda$. Combining this with the cited Rodgers–Tao theorem gives the exact logical consequences

$$\text{RH} \iff \Lambda = 0, \quad \neg\text{RH} \iff \Lambda > 0. \quad (\text{ZHR3})$$

Neither statement replaces the inequality in their theorem by a strict one. The invertibility of a supported zero is not used in the proof of (ZHR3).

Now use the same arbitrary nontrivial bounded distributive lattice L as before, the tropical coefficient extension $K_L = K \otimes_{\mathbb{B}} L$, and the proved injection $\iota_L(\lambda) = 1_K \otimes \lambda$. The actual lifted family is

$$\widehat{H}(t, z) = (H_t(z), 1_L) \in G_L(\mathbb{C}), \quad \mathcal{H}_L(t, z) = \Phi_{\mathbb{C}}(\widehat{H}(t, z)) = (H_t(z), 1_{K_L}). \quad (\text{ZHR4})$$

The map $\Phi_{\mathbb{C}}$ is the injective unital coefficient map of ZIL12. The family $\widehat{H}$ is a family of points; it is not asserted to be a zero-preserving homomorphism from the ring of functions into the supported semiring. In particular the ordinary zero value of H_t is lifted to e , not τ . Direct equality of the two coordinates proves

$$\begin{aligned}H_t(z) = 0 &\iff \widehat{H}(t, z) = e \iff \mathcal{H}_L(t, z) = (0, 1_{K_L}), \\ \ell_{\mu}(\widehat{H}(t, z)) &= \mu, \quad d_{\mu} \text{ pr}_2 \mathcal{H}_L(t, z) = d_{\mu}, \quad d_{\mu} = \iota_L(\mu), \quad \mu \in L.\end{aligned} \quad (\text{ZHR5})$$

The last line follows from ZIL4 and ZIL14 and holds for every t, z , including all zeros and all nonzeros. At $\mu = 1_L$ the localized value is always the unit. Thus e really is invertible in this localization, while the entire distinguishing amplitude coordinate has been identified away. The unlocalized joint map continues to detect the exact zero locus by the first line.

The actual counterfactual zero set, in both original coordinates, is therefore

$$\begin{aligned}\mathcal{E}_L &= \{z \in \mathbb{C} \setminus \mathbb{R} : \mathcal{H}_L(0, z) = (0, 1_{K_L})\}, \\ s &= \frac{1}{2} + \frac{iz}{2}, \quad z = -2i(s - \tfrac{1}{2}).\end{aligned}\tag{ZHR6}$$

Writing $z = x + iy$ gives $s = (1 - y)/2 + ix/2$. Thus $z \notin \mathbb{R}$ is equivalent to $\operatorname{Re} s \neq 1/2$. Equation (ZHR1), including its nonzero factor $1/8$, identifies the zeros with those of ξ . The source's RH equivalence identifies these with the nontrivial zeta-zero question. Consequently (ZHR3) is also equivalent to $\mathcal{E}_L = \emptyset$ for RH and $\mathcal{E}_L \neq \emptyset$ for its negation. This gives the exact connecting map; it does not assert that either emptiness or nonemptiness has been proved here. The localized family in (ZHR5) is constant on the entire ambient domain, so its inverse image of the localized unit cannot be substituted for $\mathcal{E}_L$.

There is a further precise distinction between time levels. For each $t < 0$, (ZHR2) and $\Lambda \geq 0$ give $t < \Lambda$, so H_t has a nonreal zero. At $t = 0$ the same argument gives no strict inequality: the endpoint $\Lambda = 0$ is allowed. The localization formula (ZHR5) is identical at every one of these time levels and therefore does not supply that missing strict comparison. This statement concerns the actual author heat family, with no replacement of its kernel or coordinates.

The algebraic zero-inversion theorem and the analytic theorem are therefore simultaneously valid. Their proved connecting maps do not imply $\Lambda > 0$, and this calculation is neither a proof nor a disproof of RH.

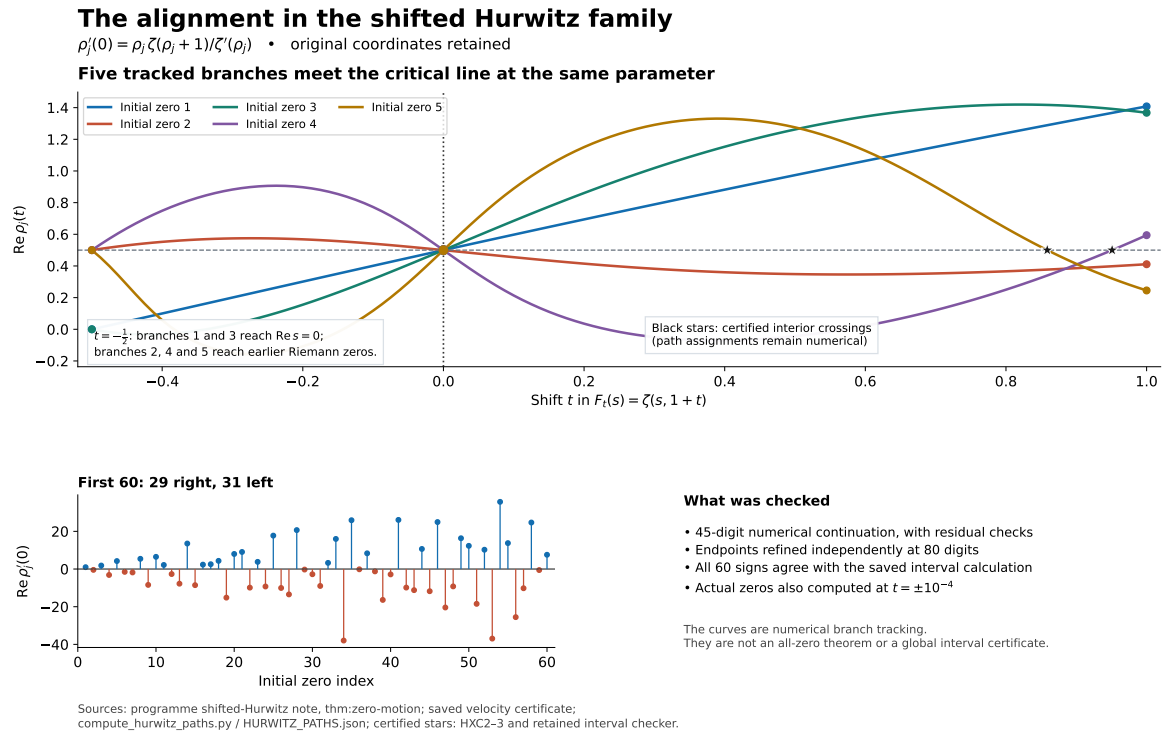


Figure 18: Five numerical zero paths and the first sixty independently recomputed velocities, in the original coordinates. The stars mark the two independently certified interior crossing pairs of HXC2–3; their assignment to complete paths is not certified. Exact local zero graph: HZA3–12. Sources and reproducible data accompany the figure.

44 The aligned Hurwitz zero graph with its complete support map

Sources and actual objects. The original programme source is the retained April 2026 draft *Split-Zero Support, Residue-Parity Reversal, Shifted-Hurwitz CUE Defects, and Finite Phase-Space Enlargement*, printed byline “Compiled theorem draft”, source labels `thm:universal-flow`, `prop:non-euler`, and `thm:zero-motion`. Its complete supplied TeX is retained with this calculation; its human author is not specified and is not inferred here. The following proof keeps its original s, t coordinates. The operator construction in HPS uses the arithmetic scaling of Jean-Benoît Bost and Alain Connes, and the precise endomotive presentation of Alain Connes, Caterina Consani and Matilde Marcolli, *Fun with $\mathbb{F}_1$* , source equations `sigmatgenerators` and relations (c1)–(c4). The full split carrier is the one in *Split Support Geometry*, v11, Definition `def:lattice-split`. The new claims below are proved here; numerical path labels have the explicit, narrower meaning stated at the end.

44.1 The full carrier and its finite-cardinality receiver

Let L be the original nontrivial bounded distributive lattice, and R a commutative ring. Retain, without identifying any labelled zeros,

$$\begin{aligned} G_L(R) &= \{(0, \lambda) : \lambda \in L\} \cup (R \times \{1_L\}), & \tau &= (0, 0_L), & e &= (0, 1_L), \\ (a, \lambda) + (b, \mu) &= (a + b, \lambda \vee \mu), & (a, \lambda)(b, \mu) &= (ab, \lambda \wedge \mu), & \\ K_L &= K \otimes_{\mathbb{B}} L, & \iota_L(\lambda) &= 1_K \otimes \lambda, & \Phi_R(a, \lambda) &= (a, \iota_L(\lambda)). \end{aligned} \quad (\text{HZA1})$$

Here K is the programme’s tropical semifield, and $\mathbb{B}$ its Boolean subsemiring. The map $K \rightarrow \mathbb{B}$ sending zero to zero and every nonzero element to one preserves addition and multiplication. Tensoring it with L gives $\rho_L : K_L \rightarrow L$ with $\rho_L \iota_L = \text{id}_L$. Thus Φ_R is injective, preserves both operations and recovers every label, including labels whose amplitude cancels. This supplies the actual comparison map, rather than a count of extra points.

For finite L and $n \geq 1$, the exact underlying-set count is

$$|G_L(\mathbb{Z}/n\mathbb{Z})| = (n - 1) + |L|, \quad \sum_{n=1}^{\infty} |G_L(\mathbb{Z}/n\mathbb{Z})|^{-s} = \zeta(s, |L|), \quad \text{Re } s > 1. \quad (\text{HZA2})$$

Indeed there are precisely $n - 1$ nonzero amplitudes, each forced to have top label, and precisely $|L|$ distinct zero-amplitude elements. The sets are disjoint. Substituting $n - 1 + |L|$ in the absolutely convergent sum proves the second equality. The quotient $(a, \lambda) \mapsto a$ identifies all these zeros and gives the ordinary count n . For $|L| = 2$, the separated count gives $\zeta(s, 2) = \zeta(s) - 1$. For arbitrary finite L its shift is $t = |L| - 1$, with no change in the carrier (HZA1).

The count is only this specified receiver. It does not recover the meet, join, or prime spectrum of L . For example, the four-element chain and the four-element Boolean lattice have the same count; they are not isomorphic because one is totally ordered and the other has two incomparable elements. For infinite L the displayed finite cardinality series is not defined. The full map (HZA1) remains defined. The real continuous shift below is an analytic family; it is not an assertion that the cardinality of L varies continuously.

44.2 An analytic construction through the critical strip

Put

$$F(s, t) = \zeta(s, 1 + t), \quad h_t(x) = \frac{xe^{-(1+t)x}}{1 - e^{-x}}. \quad (\text{HZA3})$$

The branch of $\log(1+t)$ and of $\log(n+t)$ for complex t is the one on $\operatorname{Re}(1+t) > 0$. On $\operatorname{Re} s > 0$, $s \neq 1$, $\operatorname{Re}(1+t) > 0$, the following absolutely convergent expression constructs F :

$$F(s, t) = \frac{1}{\Gamma(s)} \left\{ \frac{1}{s-1} + \int_0^1 x^{s-2} (h_t(x) - 1) dx + \int_1^\infty x^{s-2} h_t(x) dx \right\}. \quad (\text{HZA4})$$

Near zero, $h_t(x) = 1 + O(x)$, uniformly on compact parameter sets. For large positive x , it is bounded by a constant times xe^{-cx} , where $c > 0$ is a lower bound for $\operatorname{Re}(1+t)$ on the chosen compact set. The same bounds after any fixed number of s, t derivatives acquire only powers of x and $|\log x|$. These bounds prove absolute convergence and local uniform differentiability under both integrals. For $\operatorname{Re} s > 1$, combine the two integrals with $\int_0^1 x^{s-2} dx = (s-1)^{-1}$ and expand $(1 - e^{-x})^{-1} = \sum_{j \geq 0} e^{-jx}$. Absolute convergence gives $\Gamma(s) \sum_{n \geq 1} (n+t)^{-s}$. This proves (HZA4) in the original half-plane and supplies its holomorphic continuation to the stated domain. Its only possible pole there is $s = 1$, with residue one.

Differentiating h_t gives $\partial_t h_t = -x h_t$, so the two integrals immediately give the exact flow equation

$$\partial_t F(s, t) = -s F(s+1, t), \quad \partial_t^k F(s, t) = (-1)^k (s)_k F(s+k, t). \quad (\text{HZA5})$$

The first equality follows from (HZA4) and the convergent Mellin integral for $F(s+1, t)$; the second follows by induction, with $(s)_k = s(s+1) \cdots (s+k-1)$ and $(s)_0 = 1$. Holomorphy in $|t| < 1$ then gives the actual, locally uniformly convergent Taylor series

$$F(s, t) = \sum_{k=0}^{\infty} \frac{(-t)^k (s)_k}{k!} \zeta(s+k), \quad \operatorname{Re} s > 0, \quad s \neq 1, \quad |t| < 1. \quad (\text{HZA6})$$

In this domain the factors on the right introduce no additional pole: for $k \geq 1$ the argument has real part greater than one. For any compact subset in s and any smaller closed t disc, Cauchy's integral formula in a larger disc of radius less than one proves the claimed uniform convergence. Thus no formal exponential is being used as an unproved analytic operator assertion.

44.3 Every nontrivial zero gives one exact local graph

Let ρ be any zero of ζ in $0 < \operatorname{Re} \rho < 1$, and let m be its order. This is a quantification over actual zeros; no zero on or off a particular vertical line is supplied as an assumption about all the zeros. The Euler product in the absolutely convergent half-plane gives

$$B_\rho = \rho \zeta(\rho+1) \neq 0, \quad \partial_t F(\rho, 0) = -B_\rho. \quad (\text{HZA7})$$

For completeness, nonvanishing of that Euler product follows from $\sum_p \sum_{k \geq 1} |p^{-k(\rho+1)}|/k < \infty$: its exponential is nonzero, and absolute expansion by unique prime factorization equals the zeta series. Also $\rho \neq 0$ in the stated strip.

There is a neighbourhood of ρ and a unique holomorphic function θ there, taking values near zero, for which the full local zero set is

$$\{(s, t) : F(s, t) = 0\} = \{(s, \theta(s))\}, \quad \theta(\rho) = 0. \quad (\text{HZA8})$$

Here is a construction with the uniqueness included. Write $B = \partial_t F(\rho, 0) \neq 0$. Choose sufficiently small closed discs $|s-\rho| \leq d$, $|t| \leq r$ inside the holomorphy domain so that $|1 - B^{-1} \partial_t F(s, t)| \leq 1/4$ and $|B^{-1} F(s, 0)| \leq r/2$. Such discs exist by continuity and $F(\rho, 0) = 0$. The map $t \mapsto t - B^{-1} F(s, t)$ sends the t disc into itself and has Lipschitz constant at most $1/4$. Starting from zero, its iterates converge uniformly to the unique fixed point. Each iterate is holomorphic in s on the open disc; uniform convergence makes the limit holomorphic. A zero of F in the product of discs is exactly a fixed point, proving (HZA8).

The order and leading coefficient of this graph are determined without any simplicity assumption:

$$\theta(s) = (s - \rho)^m U_\rho(s), \quad U_\rho(\rho) = \frac{\zeta^{(m)}(\rho)}{m! \rho \zeta(\rho + 1)} \neq 0. \quad (\text{HZA9})$$

To prove this, the fundamental theorem of calculus gives $F(s, t) = \zeta(s) + tA(s, t)$, where $A(s, t) = \int_0^1 \partial_t F(s, ut) du$ is holomorphic and $A(\rho, 0) = -B_\rho$. At the graph, $\theta(s) = -\zeta(s)/A(s, \theta(s))$. Its denominator is a holomorphic unit near ρ . Factoring the zero of ζ proves both the order and the coefficient in (HZA9). Multiple zeta zeros therefore give ramification of the projection to t , rather than a failure to construct the total zero graph.

At a simple zero the graph can instead be written $s = \rho(t)$, by the same contraction argument with the variables interchanged. Implicit differentiation in (HZA5) proves

$$\rho'(0) = \frac{\rho \zeta(\rho + 1)}{\zeta'(\rho)}. \quad (\text{HZA10})$$

If this actual zero has $\text{Re } \rho = 1/2$ and the real part of (HZA10) is nonzero, then $\text{Re } \rho(t) - 1/2 = t \text{Re } \rho'(0) + O(t^2)$ for real t . Dividing by t and using continuity proves a transverse crossing for all sufficiently small nonzero t . This conclusion is local. It places no restriction on a later return of the same branch to the line.

44.4 The supported graph and all its comparison maps

For any open product of discs D used above, let $R = \mathcal{O}(D)$. The original function is the element $(F, 1_L)$ of $G_L(R)$, and its value at a point $x = (s, t)$ is obtained by the unital semiring homomorphism

$$\begin{array}{ccc} G_L(\mathcal{O}(D)) & \xrightarrow{\Phi_{\mathcal{O}(D)}} & \mathcal{O}(D) \times K_L \\ (a, \lambda) \mapsto (a(x), \lambda) \downarrow & & \downarrow (a, w) \mapsto (a(x), w) \\ G_L(\mathbb{C}) & \xrightarrow{\Phi_{\mathbb{C}}} & \mathbb{C} \times K_L. \end{array} \quad (\text{HZA11})$$

Evaluation preserves ring addition and multiplication, and the label is unchanged, proving the left map preserves the displayed split operations. Both composites are $(a(x), \iota_L \lambda)$, proving commutativity on every element, including every labelled zero. Consequently the zero graph just constructed is exactly

$$\{x : \text{ev}_x(F, 1_L) = e\} = \{x : \text{ev}_x \Phi(F, 1_L) = (0, 1_{K_L})\}. \quad (\text{HZA12})$$

Its value is supported zero, not τ . All other zero sections $(0, \lambda)$ remain present, evaluate to themselves, and are recovered from the full map by ρ_L . The faithful comparison does not impose an additional equation on s . The disc-algebra prime operator in HPS gives the entire family another exact receiver; HCR gives its residue-class receivers and their retained support defects. These are calculations on the same F , with the displayed maps, rather than a change of its zero convention.

44.5 The actual bounded calculation and its limits

The existing programme velocity certificate stores enclosures for $V_j = 2 \text{Re } \rho'_j(0)$. Its first 60 rows have 29 positive and 31 negative signs. The factor two must be retained when comparing that file with (HZA10). The new script `compute_hurwitz_paths.py` independently recomputes those 60 velocities at 45 decimal digits, checks each sign, and solves for actual nearby Hurwitz zeros at $t = 10^{-4}$ and $t = -10^{-4}$. It checks residuals less than 10^{-33} . These are numerical checks; the pre-existing interval enclosures and the new non-interval recomputations are recorded as different evidence.

The same script numerically follows the first five starting zeros to $t = -1/2$ and $t = 1$, with adaptive predictor/Newton steps and 80-digit endpoint refinement. At $t = 1$, the first three endpoints are, to the displayed decimal precision,

$$\begin{aligned} &1.40778804065 + 23.32798775456i, \\ &0.41092175383 + 31.71541044219i, \\ &1.36768818112 + 38.41910556064i. \end{aligned}$$

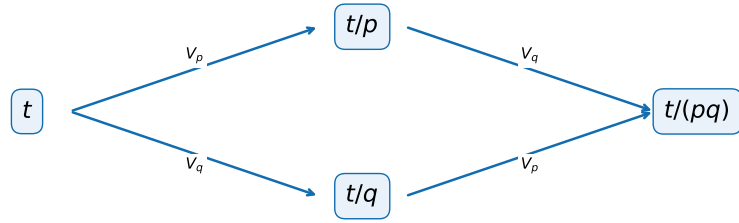
The exact endpoint identities are

$$F(s, 1) = \zeta(s) - 1, \quad F(s, -1/2) = (2^s - 1)\zeta(s). \quad (\text{HZA13})$$

The first follows by deleting the first term of the zeta series. For the second, split that series into odd and even integers: $\sum_{n \geq 0} (n + 1/2)^{-s} = 2^s \sum_{n \text{ odd}} n^{-s} = 2^s(1 - 2^{-s})\zeta(s)$. Both identities extend from their convergent half-plane by holomorphic continuation. The nonzero factor 2^s is retained. The zeros of $2^s - 1$ have $s = 2\pi ik / \log 2$ for every integer k , including $k = 0$.

The five paths in the figure are interpolations of numerical sample points. They were initialized at known Riemann zeros; their common value $\text{Re } s = 1/2$ at $t = 0$ is therefore not an independent proof for untracked zeros. The first and third numerical paths reach the first two positive imaginary zeros of $2^s - 1$; the other three reach earlier Riemann zeros. No rigorous global branch-identity certificate is claimed for this numerical path tracking. Separate interval proofs in HXC certify two additional transverse critical-line crossings at strictly positive shifts. They show why the local conclusion (HZA10) must not be changed into the assertion that a zero can cross only at an Euler-product parameter.

Prime factors act on the whole shift family



$$V_p V_q = V_{pq} = V_q V_p, \quad (V_p f)(t) = f(t/p)$$

$$\zeta(s, 1+t) = \left[\prod_p (I - p^{-s} V_p)^{-1} \right] (1+t)^{-s}, \quad \text{Res} > 1$$

At $t = 0$: every prime fixes the parameter; scalar Euler factors are recovered.

Full support: $\Phi_R(r, \lambda) = (r, 1_K \otimes \lambda)$ retains every $\lambda \in L$.

Exact parameter diagram; it does not identify the parameter fixed point with a theorem about all zeros.
Proof: HURWITZ_PRIME_SCALING.tex. Human sources: Bost–Connes; Connes–Consani–Marcolli, Fun with F_1 .

Figure 19: The exact commuting prime-dilation diagram of the whole shift family. HPS21–26 prove the operator Euler product on the original convergence domain; HPS33–39 construct its critical-strip continuation and the exact singular sector. The lattice injection is retained.

45 Exact prime dilation of the original shifted Hurwitz family

Source, scope, and coordinates. The supplied programme source is `split_zero_cue_shifted_hurwitz_phase_space.tex`, SHA-256

52f2a6bccdb15213b1992abef9f8d6476d6e9ea5566ee7ed9d85c9a2d7cc2705.

The actual source reading for this calculation is lines 250–350 and 545–686, including the complete proofs of the split-zero laws, support reflection, universal shifted-flow equation, and non-Eulerian shifted sheet. The source locators are `thm:universal-flow` and `prop:non-euler`. The full-lattice laws and synchronized lifts are restated from FLH1–3; their complete proofs were written and read in the preceding programme calculation. The tropical coefficient embedding is SHP15–16, directly read here in lines 330–401 of `SUPPORTED_HEAT_PRODUCT.tex`, at SHA-256

0b0f5a50874ce59fdfdb74a715881182a38a9238c816107d0d3b143403426799.

Every operator and convergence assertion needed below is proved here. No identification with a Bost–Connes system or assertion about zero locations is made.

For the exact classical comparison, Alain Connes, Caterina Consani and Matilde Marcolli, *Fun with $\mathbb{F}_1$* , original author file `funBC.tex`, lines 804–885, was read directly, including its equation `sigmatgenerators`. Its stated dilation evolution is the scalar m^{iu} law at zero deformation, with the source’s time renamed u only to distinguish it from the original deformation t . The file has SHA-256

ab76adc442f748a75eb5cb0a099fcbb4dcbd96f8cbc7055efca846b79a262b48.

Eugene Ha and Frédéric Paugam, *Bost–Connes–Marcolli systems for Shimura varieties. I. Definitions and formal analytic properties*, original author source lines 461–504, was also read directly: its Section 1.1 gives $H = \log N$ and $\mathrm{Tr}(e^{-sH}) = \zeta(s)$. The source SHA-256 is

7efe5e2c38c48aefaed921db505b1454d36873913617be28bf86b92ec738bf5c.

These are bounded source readings. The shifted relations below are calculated on the original Hurwitz family; no larger identification is deduced from those two classical formulas.

Throughout, $t > -1$ is the original additive deformation, $u \in \mathbb{R}$ is the unitary evolution time, and $s \in \mathbb{C}$ is the trace exponent, with $\sigma = \mathrm{Re} s > 1$ wherever a partition trace or an unrestricted infinite prime product is used. The vanishing-at-zero subspace in HPS33–39 has the explicitly proved larger domain $\mathrm{Re} s > 0$. These three parameters are not identified.

45.1 Unbounded operators and the original partition trace

On $\mathcal{H} = \ell^2(\mathbb{N}_{\geq 1})$ with its original orthonormal basis $(e_n)_{n \geq 1}$ define

$$\begin{aligned} Ne_n &= ne_n, & \mathcal{D}(N) &= \{x : \sum_{n \geq 1} n^2 |x_n|^2 < \infty\}, \\ H_t e_n &= \log(n+t)e_n, & \mathcal{D}(H_t) &= \{x : \sum_{n \geq 1} |\log(n+t)|^2 |x_n|^2 < \infty\}. \end{aligned} \tag{HPS1}$$

All logarithms of positive real numbers are real. These multiplication operators are self-adjoint: finite sequences lie in their domains and are dense. If y belongs to the adjoint domain of a real diagonal operator with entries a_n , testing against e_n forces the adjoint coordinates to be $a_n y_n$; they must be square summable. Conversely that condition makes the inner-product identity valid by Cauchy–Schwarz. Thus the adjoint has exactly the stated domain and values. In particular $N + tI$ is strictly positive, with lower bound $1 + t$, and its spectral logarithm is exactly H_t .

There is one common logarithmic domain:

$$\begin{aligned} \mathcal{D} &= \{x : \sum_{n \geq 1} (\log n)^2 |x_n|^2 < \infty\}, & \mathcal{D}(H_t) &= \mathcal{D}, \\ H_t &= H_0 + B_t, & B_t e_n &= \log(1 + t/n) e_n, & \|B_t\| &= |\log(1 + t)|. \end{aligned} \quad (\text{HPS2})$$

Indeed the diagonal difference is bounded, with its largest absolute value at $n = 1$, and tends to zero. The two inequalities $|a + b|^2 \leq 2|a|^2 + 2|b|^2$ applied in both directions show equality of the domains. This also proves that B_t is compact: truncating its diagonal after M has error the supremum of its remaining entries.

Define complex powers by their actual diagonal entries. For $\sigma > 1$,

$$\begin{aligned} A_t(s) &= e^{-sH_t}, & A_t(s)e_n &= (n + t)^{-s} e_n, \\ \|A_t(s)\|_1 &= \sum_{n \geq 1} (n + t)^{-\sigma} < \infty, & \text{Tr } A_t(s) &= F_t(s) = \sum_{n \geq 1} (n + t)^{-s} = \zeta(s, 1 + t). \end{aligned} \quad (\text{HPS3})$$

For a diagonal operator its singular values are the absolute diagonal entries, with multiplicities: the positive square root of A^*A is the diagonal of their absolute values. The trace-class criterion and trace formula therefore follow by finite truncation. Comparison with $n^{-\sigma}$ proves convergence exactly for $\sigma > 1$; at $\sigma \leq 1$ the absolute diagonal sum diverges. For $\sigma < 0$ the diagonal power is unbounded, so in that case it is not a bounded trace-class operator in the first place. For real $\beta > 1$, $A_t(\beta)$ is positive, and $A_t(\beta)/F_t(\beta)$ is a positive trace-one operator. The denominator is retained in this explicit definition; F_t itself is unchanged.

The original differential flow is an actual convergent trace identity:

$$\begin{aligned} \partial_t^j F_t(s) &= (-1)^j (s)_j F_t(s + j), & (s)_j &= s(s + 1) \cdots (s + j - 1), & (s)_0 &= 1, \\ \partial_t F_t &= -DF_t, & (Df)(s) &= sf(s + 1), \\ F_{t+r}(s) &= \sum_{j \geq 0} \frac{(-r)^j (s)_j}{j!} F_t(s + j) & (|r| < 1 + t). \end{aligned} \quad (\text{HPS4})$$

For a real compact interval of $t > -1$, $n + t \geq cn$ for some $c > 0$. On a compact set of s with real part greater than one, termwise derivatives have a summable bound $C_j n^{-\sigma_0 - j}$, and derivatives in s add only fixed powers of $\log(n + t)$. One may also let t be complex in $\text{Re } t > -1$: every $n + t$ lies in the right half-plane and uses its principal logarithm; the same compact bounds give holomorphy by uniform power-series convergence. For the last identity fix $|r| < R < 1 + t$ and apply the binomial Taylor series to each $(n + t + r)^{-s}$. Its absolute sum is bounded uniformly in n by $C_{s,R,r} (n + t)^{-\sigma}$: the Taylor coefficients of $(1 + w)^{-s}$ are bounded on $|w| = R/(1 + t) < 1$, and $|r|/(n + t) \leq |r|/(1 + t)$. The geometric Cauchy bound sums absolutely. This proves every exchange and the exact radius asserted in HPS4. It does not claim a globally convergent Taylor series across $t = -1$.

45.2 Exact dilation, its fixed-parameter defect, and coherence

For each positive integer m let $S_m e_n = e_{mn}$. It is an isometry, with $S_m^* e_j = e_{j/m}$ if $m \mid j$ and 0 otherwise, and $S_m S_k = S_{mk} = S_k S_m$. The original unbounded operators satisfy

$$\begin{aligned} x \in \mathcal{D} &\iff S_m x \in \mathcal{D}, & S_m^* \mathcal{D} &\subseteq \mathcal{D}, \\ H_t S_m x &= S_m (H_{t/m} + \log m I) x & (x \in \mathcal{D}), \\ S_m^* H_t S_m &= H_{t/m} + \log m I & \text{on domain } \mathcal{D}. \end{aligned} \quad (\text{HPS5})$$

The scalar identity is $\log(mn + t) = \log m + \log(n + t/m)$. Also $\log(mn) = \log m + \log n$, so square-summability with the logarithmic weight is equivalent before and after S_m ; for S_m^* use

$\log n \leq \log(mn)$ on the selected coordinates. Thus these are equalities with their full domains, not merely identities on basis vectors.

Let $U_t(u)e_n = e^{iu \log(n+t)}e_n$. The diagonal modulus is one, so U_t is a unitary group; dominated convergence in $\sum |x_n|^2$ proves strong continuity. Exponentiating HPS5 coordinatewise gives the exact parameter-changing scaling

$$\begin{aligned} U_t(u)S_m U_{t/m}(-u) &= m^{iu} S_m, \\ A_t(s)S_m &= m^{-s} S_m A_{t/m}(s), \quad S_m^* A_t(s)S_m = m^{-s} A_{t/m}(s) \quad (\sigma > 1). \end{aligned} \quad (\text{HPS6})$$

Thus S_m from the copy carrying $H_{t/m}$ to the copy carrying H_t restores exact scaling. For the composite dilation, S_k first carries $H_{t/(mk)}$ to $H_{t/m}$, and S_m carries that copy to H_t . The additive energy shifts are $\log k + \log m = \log(mk)$. Hence this changing of parameters is coherent for every finite product.

At a fixed deformation define $\alpha_u^t(B) = U_t(u)BU_t(-u)$ on $\mathcal{B}(\mathcal{H})$. Direct calculation gives

$$\begin{aligned} \alpha_u^t(S_m) &= S_m C_m(t, u; N), \\ C_m(t, u; n) &= \exp\left(iu \log \frac{mn+t}{n+t}\right) = m^{iu} \exp(iub_m(t; n)), \\ b_m(t; n) &= \log(n+t/m) - \log(n+t). \end{aligned} \quad (\text{HPS7})$$

These are bounded unitary diagonal operators, not constant coefficients in general. If $m > 1$, $t \neq 0$, and $u \neq 0$, equality $C_m(t, u; N) = m^{iu}I$ would require $ub_m(t; n) \in 2\pi\mathbb{Z}$ for all n . For sufficiently large n this number is nonzero and has modulus less than 2π , a contradiction. The fixed-parameter scalar formula therefore fails exactly in this nontrivial case.

Both the time cocycle and the multiplicative coherence are explicit:

$$\begin{aligned} C_m(t, u+v; n) &= C_m(t, u; n)C_m(t, v; n), \\ C_m(t, u; kn)C_k(t, u; n) &= C_{mk}(t, u; n) = C_k(t, u; mn)C_m(t, u; n), \\ b_m(t; kn) + b_k(t; n) &= b_{mk}(t; n) = b_k(t; mn) + b_m(t; n). \end{aligned} \quad (\text{HPS8})$$

The ratios in the second line telescope. Subtracting the real constants $\log m, \log k$ proves the third line, with no logarithm branch ambiguity. To verify the operator multiplication, use $f(N)S_k = S_k f(kN)$ on each coordinate. It gives $\alpha_u^t(S_m)\alpha_u^t(S_k) = S_{mk}C_m(t, u; kN)C_k(t, u; N)$, which is exactly $\alpha_u^t(S_{mk})$.

The defect from scalar scaling has an evaluated operator size. Put $c_t = \min(t, 0) > -1$. Then

$$\begin{aligned} |b_m(t; n)| &\leq \frac{|t|(1-1/m)}{n+c_t}, \quad \lim_{n \rightarrow \infty} nb_m(t; n) = -t(1-1/m), \\ R_{m,t,u} &:= \alpha_u^t(S_m) - m^{iu}S_m, \\ \|R_{m,t,u}\|_{\mathcal{S}^q}^q &\leq |ut|^q(1-1/m)^q \sum_{n \geq 1} (n+c_t)^{-q} < \infty \quad (q > 1), \\ tu \neq 0, \ m > 1 &\implies R_{m,t,u} \notin \mathcal{S}^1. \end{aligned} \quad (\text{HPS9})$$

Indeed b_m is the oriented integral of $(n+v)^{-1}$ between t and t/m . The denominator on that segment is at least $n+c_t$, which proves the bound; multiplication by n and uniform convergence of $n/(n+v)$ on the fixed segment proves the limit. Since S_m is an isometry, R^*R is diagonal with entries $|e^{iub_m(t;n)} - 1|^2$. These are its squared singular values, whether or not written in decreasing order. The inequality $|e^{iy} - 1| \leq |y|$ proves the Schatten bound and compactness. In the nontrivial case their square roots satisfy $n|e^{iub_m(t;n)} - 1| \rightarrow |ut|(1-1/m) > 0$. The harmonic series proves failure of trace class. Thus an actual compact correction has been calculated; it has not been discarded.

45.3 Divisibility projections and the exact prime strata

Let $P_m = S_m S_m^*$, the orthogonal projection onto the multiples of m . All such projections are diagonal and

$$P_m P_k = P_{\text{lcm}(m,k)}, \quad P_m A_t(s) = A_t(s) P_m, \quad \text{Tr}(P_m A_t(s)) = m^{-s} F_{t/m}(s). \quad (\text{HPS10})$$

The first identity is the intersection of two divisibility conditions. The trace follows by writing its selected index as mn and using $mn + t = m(n + t/m)$. It is also the compression identity HPS6; absolute trace convergence makes both computations agree.

For a finite set $\mathcal{P}$ of primes let $M = \prod_{p \in \mathcal{P}} p$, and let $\mu(d)$ be $(-1)^r$ for a product of r distinct primes, 0 for an integer divisible by a prime square, and 1 at $d = 1$. Put

$$Q_{\mathcal{P}} = \prod_{p \in \mathcal{P}} (I - P_p),$$

$$R_{\mathcal{P}}(t; s) = \text{Tr}(Q_{\mathcal{P}} A_t(s)) = \sum_{\substack{n \geq 1 \\ (n, M) = 1}} (n + t)^{-s} = \sum_{d|M} \mu(d) d^{-s} F_{t/d}(s). \quad (\text{HPS11})$$

All products are finite. Their expansion gives $\sum_{d|M} \mu(d) P_d$, since distinct-prime least common multiples are products. On each integer the result is one precisely when no prime in $\mathcal{P}$ divides it, and zero otherwise. HPS10 proves the final trace identity with every deformation t/d retained.

For prescribed valuations $a_p \geq 0$ at $p \in \mathcal{P}$, write $d = \prod p^{a_p}$. The exact stratum projection and trace are

$$E_{\mathcal{P}, \mathbf{a}} = \prod_{p \in \mathcal{P}} (P_{p^{a_p}} - P_{p^{a_p+1}}) = S_d Q_{\mathcal{P}} S_d^*,$$

$$\text{Tr}(E_{\mathcal{P}, \mathbf{a}} A_t(s)) = d^{-s} R_{\mathcal{P}}(t/d; s) = \sum_{e|M} \mu(e) (de)^{-s} F_{t/(de)}(s), \quad (\text{HPS12})$$

$$\sum_{\mathbf{a} \in \mathbb{N}_0^{\mathcal{P}}} E_{\mathcal{P}, \mathbf{a}} = I \quad \text{strongly}, \quad F_t(s) = \sum_{\mathbf{a}} d^{-s} R_{\mathcal{P}}(t/d; s).$$

Every coordinate integer has exactly one valuation vector; the projections therefore are orthogonal, and their finite partial sums converge strongly by the square-summable coordinate tail. Writing the stratum integer as dr with $(r, M) = 1$ proves its trace. The absolute sum over all strata is at most $\sum_n (n + t)^{-\sigma}$, so the last trace sum has no conditional rearrangement.

There is also an additive decomposition using every prime exactly as a least prime divisor. Let $\mathcal{P}_{< p}$ contain all primes less than p , and let Π_1 project onto $\mathbb{C}e_1$. Then

$$I = \Pi_1 + \sum_{p \text{ prime}} P_p Q_{\mathcal{P}_{< p}} \quad \text{strongly},$$

$$F_t(s) = (1 + t)^{-s} + \sum_{p \text{ prime}} p^{-s} R_{\mathcal{P}_{< p}}(t/p; s). \quad (\text{HPS13})$$

For $n > 1$, its least prime divisor is unique; the indicated projections select exactly these disjoint sets. For $n = pr$, having no smaller prime divisor is equivalent to $(r, \prod_{q < p} q) = 1$. This proves the second line by the same absolutely convergent trace partition as HPS12.

45.4 A convergent operator Euler product on the full parameter family

Fix a compact interval $J = [a, b] \subset (-1, \infty)$ containing 0; every compact set of allowed real deformations is contained in such a J . Let $X_J = C(J, \mathbb{C})$ with the supremum norm and pointwise ring operations. For $m \geq 1$ put

$$(V_m f)(t) = f(t/m), \quad K_m(s) = m^{-s} V_m,$$

$$V_m V_n = V_{mn}, \quad K_m(s) K_n(s) = K_{mn}(s), \quad \|K_m(s)\| = m^{-\sigma}. \quad (\text{HPS14})$$

The interval is stable under division by m . The norm upper bound follows from the supremum; the constant function 1 attains it. The composition formulas are literal substitutions. The maps V_m are unital ring homomorphisms; K_m are bounded linear operators, and generally are not ring homomorphisms because of their scalar factor.

Define bounded linear operators by absolutely convergent operator series:

$$\begin{aligned}\mathcal{E}_s &= \sum_{n \geq 1} n^{-s} V_n, & \mathcal{M}_s &= \sum_{n \geq 1} \mu(n) n^{-s} V_n, \\ \|\mathcal{E}_s\| &\leq \zeta(\sigma), & \|\mathcal{M}_s\| &\leq \sum_n |\mu(n)| n^{-\sigma} \leq \zeta(\sigma), \\ \mathcal{M}_s \mathcal{E}_s &= \mathcal{E}_s \mathcal{M}_s = I.\end{aligned}\tag{HPS15}$$

Both absolute norm sums are bounded by $\sum n^{-\sigma}$. Their double product may therefore be regrouped by the product index: the coefficient of $n^{-s} V_n$ is $\sum_{d|n} \mu(d)$. If n has r distinct prime factors this is $(1-1)^r$, hence it is 0 for $n > 1$ and 1 for $n = 1$. This proves both inverse identities, using only finite prime factorization and the proved absolute operator sums.

For a finite prime set $\mathcal{P}$ write $\mathbb{N}_{\mathcal{P}}$ for its smooth positive integers, that is, those whose prime factors belong to $\mathcal{P}$. The exact finite products are

$$\begin{aligned}\mathcal{E}_{s,\mathcal{P}} &= \prod_{p \in \mathcal{P}} (I - K_p(s))^{-1} = \prod_{p \in \mathcal{P}} \sum_{j \geq 0} K_{p^j}(s) = \sum_{n \in \mathbb{N}_{\mathcal{P}}} n^{-s} V_n, \\ \mathcal{M}_{s,\mathcal{P}} &= \prod_{p \in \mathcal{P}} (I - K_p(s)) = \sum_{d|M} \mu(d) d^{-s} V_d, \\ \lim_{\mathcal{P}} \mathcal{E}_{s,\mathcal{P}} &= \mathcal{E}_s, & \lim_{\mathcal{P}} \mathcal{M}_{s,\mathcal{P}} &= \mathcal{M}_s \quad \text{in operator norm.}\end{aligned}\tag{HPS16}$$

Since $\|K_p\| = p^{-\sigma} < 1$, the geometric series converges in operator norm and its finite telescoping product with $I - K_p$ has remainder $K_p^{j+1} \rightarrow 0$. It is therefore the actual inverse. Commutativity and unique prime factorization prove the first expansion; its multiple geometric sum is absolute. The second product is finite inclusion-exclusion. As the finite prime set increases, each integer eventually occurs in the first expansion, and each square-free integer in the second. Dominating their missing coefficient norms by the summable $n^{-\sigma}$ proves both limits. These are inverses in a Banach algebra of linear operators; no subtraction in a split-zero semiring has been used to define an inverse.

The convergence has the following uniform bound. If $\mathcal{P}$ contains every prime at most $x \geq 2$ and $q = \lfloor x \rfloor$, then

$$\|\mathcal{E}_s - \mathcal{E}_{s,\mathcal{P}}\|, \|\mathcal{M}_s - \mathcal{M}_{s,\mathcal{P}}\| \leq \sum_{n > q} n^{-\sigma} \leq \frac{q^{1-\sigma}}{\sigma - 1}.\tag{HPS17}$$

An omitted integer has a prime divisor greater than x , hence is greater than q . The decreasing function $v^{-\sigma}$ gives the integral bound from q to infinity. On compact sets with $\operatorname{Re} s \geq 1 + \epsilon$, the same estimate and all operator sums are uniform; differentiating in s adds $(\log n)^j$, still summable on each such compact set. Thus these are holomorphic operator families there.

The original shifted trace is obtained by applying this actual Euler product to its first atom:

$$\begin{aligned}g_s(t) &= (1+t)^{-s}, & Z_s(t) &= F_t(s), \\ Z_s &= \mathcal{E}_s g_s = \lim_{\mathcal{P}} \prod_{p \in \mathcal{P}} (I - K_p(s))^{-1} g_s, & \mathcal{M}_s Z_s &= g_s, \\ (\mathcal{M}_{s,\mathcal{P}} Z_s)(t) &= R_{\mathcal{P}}(t; s).\end{aligned}\tag{HPS18}$$

Indeed each term is exactly $n^{-s}(1+t/n)^{-s} = (n+t)^{-s}$, because the bases are positive and real for $t \in J$. The series is uniform on J by HPS15, and it agrees term for term with HPS3. The

inverse formula is HPS15; the finite sieve is HPS11. The errors in the two finite-product formulas are bounded by the right side of HPS17 times respectively $\|g_s\|_\infty$ and $\|Z_s\|_\infty$. This proves local uniform convergence in the original parameter, not only pointwise convergence.

Fixed-parameter evaluation retains the exact reason for the changing prime factors:

$$\begin{aligned} \text{ev}_t K_m(s) &= m^{-s} \text{ev}_{t/m}, \\ \text{ev}_0 \mathcal{E}_s &= \zeta(s) \text{ev}_0, \quad F_0(s) = \zeta(s), \quad g_s(0) = 1. \end{aligned} \tag{HPS19}$$

For $t \neq 0$ and $m > 1$, evaluation at t/m and at t are different functionals on X_J : the test function $f(v) = v$ distinguishes them. For the actual atom, their modulus differs as well, since $\sigma > 1$ and $1 + t/m \neq 1 + t$. Thus no scalar Euler factor at the original fixed t replaces this parameter-changing operator. At $t = 0$, every V_m acts as the identity after evaluation; HPS16 returns the usual scalar Euler product directly.

The operator prime product also respects the original shifted differential equation. For jointly differentiable families $f(s, t)$ set $(Df)(s, t) = sf(s + 1, t)$. Then

$$\begin{aligned} \partial_t(K_m(s)f) &= m^{-1}K_m(s)\partial_t f, \quad D(K_m(s)f) = m^{-1}K_m(s)Df, \\ (\partial_t + D)(K_m(s)f) &= m^{-1}K_m(s)(\partial_t + D)f, \\ \partial_t(\mathcal{E}_s f) &= \mathcal{E}_{s+1}\partial_t f \quad (f \in C^1(J, \mathbb{C})). \end{aligned} \tag{HPS20}$$

The first formula is the chain rule. In the second, evaluating the scalar factor at $s + 1$ gives $sm^{-(s+1)}f(s + 1, t/m)$, exactly the displayed right side. The atom $g_s(t)$ satisfies $(\partial_t + D)g = 0$. For the last formula take a nondegenerate interval J and use its usual one-sided endpoint derivatives. The differentiated operator series is uniformly bounded by $\sum n^{-\sigma-1}\|f'\|_\infty$. The original series is uniformly bounded by $\sum n^{-\sigma}\|f\|_\infty$. The fundamental theorem of calculus on J therefore proves that its sum is C^1 and that its derivative is the displayed sum. In particular $\|\mathcal{E}_s f\|_{C^1} \leq \zeta(\sigma)\|f\|_\infty + \zeta(\sigma + 1)\|f'\|_\infty$ for the sum norm. Applying the formula to g_s gives $\partial_t Z_s = -s\mathcal{E}_{s+1}g_{s+1} = -sZ_{s+1}$ with the exact original operator D . Thus HPS20 recovers the original Hurwitz flow from the convergent prime-factor construction.

For completeness the original Gibbs divisibility correlation is also evaluated. For real $\beta > 1$ and distinct primes p, q , write expectations with the explicitly defined state from HPS3. Then

$$\begin{aligned} \mathbb{E}_{t,\beta} P_p &= p^{-\beta} \frac{F_{t/p}(\beta)}{F_t(\beta)}, \\ \text{Cov}_{t,\beta}(P_p, P_q) &= (pq)^{-\beta} \frac{F_t(\beta)F_{t/(pq)}(\beta) - F_{t/p}(\beta)F_{t/q}(\beta)}{F_t(\beta)^2}, \\ \partial_t \text{Cov}_{t,\beta}(P_p, P_q)|_{t=0} &= -(pq)^{-\beta} \beta \frac{\zeta(\beta + 1)}{\zeta(\beta)} (1 - 1/p)(1 - 1/q) < 0. \end{aligned} \tag{HPS21}$$

Use $P_p P_q = P_{pq}$ and HPS10 for the first two lines. The numerator at 0 vanishes. HPS4 gives $\partial_t F_{t/d}(\beta)|_0 = -\beta\zeta(\beta + 1)/d$. Its four product derivatives yield $-\beta\zeta(\beta)\zeta(\beta + 1)(1 + 1/(pq) - 1/p - 1/q)$, and division by $F_0(\beta)^2$ proves the final line. This is a calculated failure of the undeformed prime independence to first order, with its original parameter and trace weights preserved.

45.5 Every split-zero label, the coefficient tensor map, and trace

Let L be the full bounded distributive lattice, with bottom 0_L and top 1_L . For any unital commutative complex algebra R , retain the original semiring

$$\begin{aligned} G_L(R) &= \{(0, \lambda) : \lambda \in L\} \cup \{(r, 1_L) : r \in R\}, \\ (r, \lambda) + (r', \lambda') &= (r + r', \lambda \vee \lambda'), \quad (r, \lambda)(r', \lambda') = (rr', \lambda \wedge \lambda'), \\ \tau &= (0, 0_L), \quad z_\lambda = (0, \lambda), \quad e = z_{1_L}, \quad \hat{r} = (r, 1_L). \end{aligned} \tag{HPS22}$$

The same component rules define a synchronized $G_L(\mathbb{C})$ -semimodule $V_L^\uparrow$ for every complex vector space V . Closure holds because every nonzero amplitude forces top support. Associativity and all distributive laws follow componentwise from the ring laws and distributivity of the lattice. The actual additive zero is τ ; $\widehat{0} = e$, and all z_λ are retained.

Every complex linear map $T : V \rightarrow W$ has the lift

$$\begin{aligned} T^\uparrow(v, \lambda) &= (Tv, \lambda), \\ (TU)^\uparrow &= T^\uparrow U^\uparrow, \quad (T + U)^\uparrow = T^\uparrow + U^\uparrow, \quad 0^\uparrow(v, \lambda) = (0, \lambda). \end{aligned} \tag{HPS23}$$

The map is defined since a nonzero output requires a nonzero input and hence top support. Component addition and scalar meet prove linearity. Composition preserves the label once, and the pointwise sum joins two copies of the same label, proving the identities. These statements apply also to $H_t : \mathcal{D} \rightarrow \mathcal{H}$ with the exact lifted domain $\mathcal{D}_L^\uparrow$. Consequently HPS5–8 lift with every label and their specified domains. The scalar m^{iu} is lifted as $\widehat{m}^{iu}$.

On $(X_J)_L^\uparrow$ the inverse maps in HPS15–16 become exact linear isomorphisms:

$$\begin{aligned} \mathcal{M}_s^\uparrow \mathcal{E}_s^\uparrow &= I, \quad \mathcal{E}_s^\uparrow(g_s, 1_L) = (Z_s, 1_L), \\ (I - K_p)^\uparrow(f, \lambda) &= (f - K_p f, \lambda) = I^\uparrow(f, \lambda) + \widehat{-1} K_p^\uparrow(f, \lambda), \\ \left(\sum_{j \geq 0} K_{pj} \right)^\uparrow(f, \lambda) &= \left(\sum_{j \geq 0} K_{pj} f, \lambda \right). \end{aligned} \tag{HPS24}$$

The last sum means the proved convergent amplitude sum with its one unchanged label. Every finite partial sum already has that label, so no arbitrary infinite join in L is required. The same interpretation applies to all norm limits in HPS15–18. If $f - K_p f$ vanishes, the result is z_λ , not necessarily τ . Thus these are lifted linear inverses; the proof does not invent additive inverses for a split-zero semiring.

Let K be the original tropical semifield with zero and let $K_L = K \otimes_{\mathbb{B}} L$. The complete coefficient map is

$$\begin{aligned} \iota_L(\lambda) &= 1_K \otimes \lambda, \quad \rho_L = \beta_K \otimes \text{id}_L, \quad \beta_K(a) = \begin{cases} 0, & a = 0_K, \\ 1, & a \neq 0_K, \end{cases} \\ \rho_L \iota_L &= \text{id}_L, \quad \Phi_R : G_L(R) \longrightarrow R \times K_L, \quad (r, \lambda) \longmapsto (r, \iota_L(\lambda)). \end{aligned} \tag{HPS25}$$

In a tropical semifield a sum is zero only when both terms are zero, and a product is zero exactly when a factor is zero. Hence β_K is a unital semiring homomorphism. The tensor universal property gives ρ_L , and its value on $1_K \otimes \lambda$ is λ . This proves injectivity of ι_L . It preserves join, meet, zero and top by its tensor construction, so the component laws HPS22 prove that Φ_R is an injective unital semiring homomorphism. Its image is exactly the pairs with support in $\iota_L(L)$, and with support equal to 1_{K_L} whenever the amplitude is nonzero. For synchronized vector spaces use the same formula as an injective additive map. For every linear T the exact commuting identity is $(T, \text{id})\Phi_V = \Phi_W T^\uparrow$. It applies to every operator and amplitude limit above; the retraction recovers every original label, including τ , e , and all intermediate zeros.

The ordinary trace has the precisely typed lift

$$\begin{aligned} \text{Tr}^\uparrow : \mathcal{S}^1(\mathcal{H})_L^\uparrow &\longrightarrow G_L(\mathbb{C}), \quad (A, \lambda) \longmapsto (\text{Tr } A, \lambda), \\ \Phi_{\mathbb{C}} \text{Tr}^\uparrow &= (\text{Tr}, \text{id})\Phi_{\mathcal{S}^1}, \\ \text{Tr}^\uparrow(P_m A_t(s), 1_L) &= (m^{-s} F_{t/m}(s), 1_L), \\ \text{Tr}^\uparrow(Q_{\mathcal{P}} A_t(s), 1_L) &= (R_{\mathcal{P}}(t; s), 1_L). \end{aligned} \tag{HPS26}$$

Trace is bounded and complex linear on its trace-class domain, so HPS23 proves the first two lines. The others are the already proved amplitudes HPS10–11. This is not a multiplicative map: the orthogonal rank-one projections onto e_1, e_2 have product zero, while both individual traces equal one. At a vanishing trace, the top-supported output remains e .

45.6 Independent coordinate labels and the exact support defect of a sieve

The synchronized lift can be supplemented by every independent original coordinate label, without identifying the two carriers. For $r = 1, 2$ define

$$\mathcal{H}_L^{(r)} = \{(x, \nu) : x \in \ell^r(\mathbb{N}_{\geq 1}), \nu \in L^{\mathbb{N}_{\geq 1}}, x_n \neq 0 \Rightarrow \nu_n = 1_L\}. \quad (\text{HPS27})$$

Addition joins labels coordinatewise and adds the original coordinates; the scalar $(a, \eta) \in G_L(\mathbb{C})$ acts by $(ax_n, \eta \wedge \nu_n)_n$. These laws make it a semimodule, because each coordinate has the HPS22 laws. The coefficient map $(x, \nu) \mapsto (x, (\iota_L(\nu_n))_n)$ is injective by HPS25. The synchronized map $(x, \lambda) \mapsto (x, (\lambda)_n)$ is injective and linear; its image consists exactly of constant-label sequences.

The original shifts have the full coordinate lifts

$$\begin{aligned} \tilde{S}_m(x, \nu) &= (S_m x, \nu^{(m)}), & \nu_j^{(m)} &= \begin{cases} \nu_{j/m}, & m \mid j, \\ 0_L, & m \nmid j, \end{cases} \\ \tilde{S}_m^*(x, \nu) &= (S_m^* x, (\nu_{mn})_{n \geq 1}), \\ \tilde{U}_t(u)(x, \nu) &= (U_t(u)x, \nu), & \tilde{H}_t(x, \nu) &= (H_t x, \nu) \quad (x \in \mathcal{D}). \end{aligned} \quad (\text{HPS28})$$

They are linear for the scalar and component laws just given. The shifts insert the actual unsupported zero τ on the missing coordinates; their adjoints select the stated coordinates, with no sum over discarded labels. The diagonal lifts preserve every label, including at $t = 0, n = 1$, where $H_0 e_1 = 0$ and a top input becomes the supported zero. Coordinate substitution proves $\tilde{S}_m^* \tilde{S}_m = I$, $\tilde{S}_m \tilde{S}_k = \tilde{S}_{mk}$, and the full HPS5–8 intertwining identities. For HPS5 the sum of the two right-hand diagonal lifts joins identical labels. All domain checks remain the original amplitude checks of HPS5. The constant-label embedding need not intertwine the shift with its synchronized lift: outside the multiples the former keeps λ and the latter inserts 0_L . This difference is specified next.

For any subset $A \subseteq \mathbb{N}_{\geq 1}$ let Q_A be the ordinary coordinate projection. There are two distinct exact lifts:

$$\begin{aligned} Q_A(x, \nu) &= (Q_A x, (\mathbf{1}_A(n) \nu_n)_n), & Q_A^{\text{keep}}(x, \nu) &= (Q_A x, \nu), \\ Q_A^{\text{keep}}(x, \nu) &= Q_A(x, \nu) + (0, (\mathbf{1}_{A^c}(n) \nu_n)_n). \end{aligned} \quad (\text{HPS29})$$

Here $\mathbf{1}_A(n) \nu_n$ means ν_n on A and 0_L outside A . Both are linear, as direct coordinate addition and meet show. The second equals the pointwise signed difference between the identity lift and Q_{A^c} : on A^c the amplitude cancels but its label is $\nu_n \vee \nu_n = \nu_n$. This proves the displayed exact zero-label defect. The range projector $\tilde{S}_m \tilde{S}_m^*$ is $Q_{m\mathbb{N}}$; the prime sieve mask is $Q_{\{(n,M)=1\}}$. The product of signed complementary lifts $\prod_{p \in \mathcal{P}} (I + \widehat{-1} Q_{p\mathbb{N}})$ has the same sieve amplitude and keeps the entire original label sequence. Indeed every factor keeps that sequence and has amplitude $I - P_p$; finite composition proves the assertion. Thus its difference from the mask is exactly the labelled-zero sequence on the excluded prime coordinates in HPS29.

For an arbitrary infinite sequence of labels, its trace cannot silently assume a join in L . The exact universally available target is the ordered ideal completion $\widehat{L}$: its members are nonempty lower subsets closed under finite joins. Its bottom is $\{0_L\}$, top is L , meet is intersection, and an arbitrary join is the ideal generated by the union. The injective map $j_L(\lambda) = \downarrow \lambda$ preserves finite joins and meets. In particular

$$\begin{aligned} I(\nu) &= \bigvee_{n \geq 1}^{\widehat{L}} \downarrow \nu_n = \{\lambda : \lambda \leq \nu_{n_1} \vee \cdots \vee \nu_{n_k} \text{ for some finite set of indices}\}, \\ \widehat{\text{tr}}(x, \nu) &= \left(\sum_{n \geq 1} x_n, I(\nu) \right) \in G_{\widehat{L}}(\mathbb{C}) \quad ((x, \nu) \in \mathcal{H}_L^{(1)}). \end{aligned} \quad (\text{HPS30})$$

The empty finite join is 0_L . The sum is absolutely convergent. If it is nonzero, some x_n is nonzero, so $\nu_n = 1_L$ and $I(\nu) = L$; thus the displayed output satisfies the exact carrier condition. Join-generated ideals show $I(\nu \vee \mu) = I(\nu) \vee I(\mu)$. Also $I((\eta \wedge \nu_n)_n) = \downarrow \eta \cap I(\nu)$: if $\lambda \leq \eta$ and $\lambda \leq \bigvee_{n \in F} \nu_n$, then $\lambda \leq \eta \wedge \bigvee_{n \in F} \nu_n = \bigvee_{n \in F} (\eta \wedge \nu_n)$. The reverse inclusion is immediate. This proves additivity and linearity over the scalar embedding $(a, \eta) \mapsto (a, \downarrow \eta)$. It also proves directly the arbitrary-join distributivity required in $\widehat{L}$.

The completed support in HPS30 is principal if and only if some finite join of the ν_n dominates every ν_n . For if $I(\nu) = \downarrow \lambda$, its generator λ belongs to $I(\nu)$ and hence lies below such a finite join; that finite join already belongs to $\downarrow \lambda$, so it equals λ . The converse is immediate. Mere existence of $\sup_L \nu_n$ is not sufficient: in $L = [0, 1]$ with its usual order and $\nu_n = 1 - 1/n$, the generated ideal is $[0, 1)$, whereas $\downarrow \sup_n \nu_n = [0, 1]$. Thus the actual completion and its complete fibres have not been replaced by a larger principal support. If one separately chooses an L -valued trace using an existing supremum, that is a different target construction; its principal embedding need not equal HPS30.

The coefficient tensor map commutes exactly with this completion as well. Let $\widehat{K}_L$ be the ordered-ideal completion of the idempotent semiring K_L , with ideal multiplication generated by pairwise products. Define

$$\begin{aligned} \iota^\#(I) &= \{w : w \leq \iota_L(\lambda_1) + \cdots + \iota_L(\lambda_k), \lambda_i \in I\}, \\ \rho^\#(J) &= \{\lambda : \lambda \leq \rho_L(w_1) \vee \cdots \vee \rho_L(w_k), w_i \in J\}, \\ \rho^\# \iota^\# &= \text{id}_{\widehat{L}}, \quad \iota^\# I(\nu) = \bigvee_n^{\widehat{K}_L} \downarrow \iota_L(\nu_n). \end{aligned} \tag{HPS31}$$

These are the ideals generated by the respective images. Since the original maps preserve finite sums and products, their ideal extensions preserve arbitrary joins and ideal products: both sides are generated by the same images of finite sums and pairwise products. Applying the original retraction to their generators proves the retraction identity. The last formula follows by the same generator description. Therefore the map $(r, I) \mapsto (r, \iota^\# I)$ is injective and preserves all amplitude/support operations, and the completed trace HPS30 commutes with the complete original coefficient embedding.

For the actual partition traces, all nonzero spectral coefficients have top labels. The masked full trace, each nonempty prime stratum, and the one-coordinate $n = 1$ term all have $I(\nu) = L$. Thus HPS10–18 pass through HPS30 and HPS31 with the precise top support already displayed in HPS26. Under finite alternating exclusion the labels on cancelled coordinates instead remain those calculated in HPS29. Under the growing prime sieve, its amplitude trace tends to $(1+t)^{-s}$; the surviving coordinate is $n = 1$ and its support remains top. The corresponding operator formula is the actual inverse $\mathcal{M}_s Z_s = g_s$, not a deletion of e or a collapse of intermediate zero labels.

There is an exact, useful failure of commuting coordinate-label limits with this trace. Fix real $\beta > 1$, and let $\mathcal{P}_j$ increase through the first j primes. Put $A_j = \{n \geq 2 : (n, \prod_{p \in \mathcal{P}_j} p) = 1\}$, and form the literal masked residual

$$\begin{aligned} x_n^{(j)} &= \mathbf{1}_{A_j}(n)(n+t)^{-\beta}, \quad \nu_n^{(j)} = \begin{cases} 1_L, & n \in A_j, \\ 0_L, & n \notin A_j, \end{cases} \\ a_j &= \sum_{n \in A_j} (n+t)^{-\beta} > 0, \quad a_j \longrightarrow 0, \quad \widehat{\text{tr}}(x^{(j)}, \nu^{(j)}) = (a_j, L). \end{aligned} \tag{HPS32}$$

Every finite prime set omits a prime, by applying a prime divisor to one plus its prime product. Thus A_j is nonempty and $a_j > 0$. Each fixed integer $n \geq 2$ has a prime divisor and is eventually excluded. Dominated convergence by the summable $(n+t)^{-\beta}$ gives $a_j \rightarrow 0$, which is also convergence of these diagonal residuals in trace norm. Each fixed coordinate label is eventually 0_L .

The full coordinate limit is therefore $(0, (0_L)_n)$, whose supported trace is $(0, \{0_L\})$, the completed unsupported zero. By contrast the traces in HPS32, retaining their constant top label while their amplitude tends to zero, have limit $(0, L)$, the completed supported zero. These two limits are distinct when L is nontrivial.

For the corresponding signed residual, HPS29 keeps the original top label on every coordinate. Its amplitude is the same $x^{(j)}$, but its complete coordinate limit is $(0, (1_L)_n)$ and its supported trace is $(0, L)$. Thus the exact discrepancy is the all-top labelled-zero sequence discarded by the masks. The faithful maps HPS25 and HPS31 preserve and detect it. No topology on an arbitrary lattice has been postulated here: the coordinate limits of labels are eventually constant, and each trace limit is taken in its explicit fixed-support amplitude fibre. This proves why trace-norm convergence of amplitudes alone cannot justify exchanging these two supported limiting operations.

45.7 The actual prime operator in the critical strip

The preceding operator product has an exact larger domain after separating the constant coordinate. Fix $0 < r < 1$ and let $\mathcal{A}_r$ be the disc algebra of complex functions holomorphic on $|t| < r$ and continuous on $|t| \leq r$, with its supremum norm and unchanged coordinate t . It is complete: a uniform Cauchy sequence converges uniformly, and Cauchy's integral formula on each smaller circle proves that the limit is holomorphic. Define

$$\begin{aligned} P_0 f &= f(0)\mathbf{1}, & X_r &= \ker P_0, & \mathcal{A}_r &= \mathbb{C}\mathbf{1} \oplus X_r, \\ (V_n f)(t) &= f(t/n), & \|V_n|_{X_r}\| &= \frac{1}{n}. \end{aligned} \tag{HPS33}$$

For $f(0) = 0$, the function $f(t)/t$, with its removable value at zero, is holomorphic. On the boundary its modulus is at most $\|f\|_r/r$. The maximum modulus principle, first on smaller circles and then by continuity at the outer boundary, gives $|f(w)| \leq (|w|/r)\|f\|_r$ for $|w| \leq r$. Apply this at $w = t/n$ to obtain the bound. The unchanged monomial $f(t) = t$ attains equality, proving the exact norm. No rescaling of a programme function has been made.

For $\operatorname{Re} s = \sigma > 0$ define operators on this actual vanishing subspace:

$$\begin{aligned} \mathcal{A}_s^0 &= \sum_{n \geq 1} n^{-s} V_n|_{X_r}, & \mathcal{B}_s^0 &= \sum_{n \geq 1} \mu(n) n^{-s} V_n|_{X_r}, \\ \|\mathcal{A}_s^0\|, \|\mathcal{B}_s^0\| &\leq \zeta(\sigma + 1), & \mathcal{B}_s^0 \mathcal{A}_s^0 &= \mathcal{A}_s^0 \mathcal{B}_s^0 = I_{X_r}, \\ \mathcal{A}_s^0 &= \lim_{\mathcal{P}} \prod (I - p^{-s} V_p|_{X_r})^{-1}, & \mathcal{B}_s^0 &= \lim_{\mathcal{P}} \prod (I - p^{-s} V_p|_{X_r}). \end{aligned} \tag{HPS34}$$

HPS33 makes both absolute norm sums at most $\sum n^{-\sigma-1}$. In the double convolution, the norm of $m^{-s} n^{-s} V_{mn}|_{X_r}$ is $(mn)^{-\sigma-1}$, whose double sum is finite. The divisor computation from HPS15 therefore proves the inverse identities on this larger half-plane. Each prime factor has norm at most $p^{-\sigma-1} < 1$, so its geometric inverse exists. Repeating the exact finite expansions HPS16 proves the products and limits. If $\mathcal{P}$ contains all primes at most $x \geq 2$, both missing-series operator errors are at most $[x]^{-\sigma}/\sigma$. Logarithmic derivatives of the series are uniformly summable on compact subsets of $\sigma > 0$. Thus both operators and their inverses depend holomorphically on s there.

The needed scalar continuation has a direct integral proof. For $\sigma > 1$, write $n^{-s} = s \int_n^\infty v^{-s-1} dv$ and interchange the absolutely summable integrals to obtain $\zeta(s) = s \int_1^\infty [v] v^{-s-1} dv$. Consequently

$$\begin{aligned} \zeta(s) &= \frac{s}{s-1} - s \int_1^\infty \{v\} v^{-s-1} dv \quad (\operatorname{Re} s > 0, s \neq 1), \\ \mathcal{E}_s &= \zeta(s) P_0 + \mathcal{A}_s^0 (I - P_0) \quad \text{on } \mathcal{A}_r. \end{aligned} \tag{HPS35}$$

Here $\{v\} = v - \lfloor v \rfloor$. The integral and its derivatives in s converge uniformly on compact sets of $\sigma > 0$, since $\{v\} \leq 1$ and fixed powers of $\log v$ remain integrable against $v^{-\sigma-1}$. This proves the meromorphic continuation, with a simple pole of residue one at $s = 1$. The operator $\mathcal{E}_s$ agrees with $\mathcal{E}_s$ on the disc algebra for $\sigma > 1$, by separating $f(0)$ in its convergent series. It is meromorphic on $\sigma > 0$, and its residue at 1 is exactly P_0 .

The operator singularities are now completely specified, with no assumption about where a zero lies within this half-plane:

$$\begin{aligned} \zeta(s) \neq 0, s \neq 1 &\implies \mathcal{E}_s^{-1} = \zeta(s)^{-1}P_0 + \mathcal{B}_s^0(I - P_0), \\ \zeta(\rho) = 0, \operatorname{Re} \rho > 0 &\implies \ker \mathcal{E}_\rho = \mathbb{C}\mathbf{1}, \quad \operatorname{ran} \mathcal{E}_\rho = X_r, \\ \mathcal{A}_r / \operatorname{ran} \mathcal{E}_\rho &\xrightarrow{\cong} \mathbb{C}, \quad [f] \mapsto f(0), \\ [P_0 + \mathcal{B}_s^0(I - P_0)] \mathcal{E}_s &= \zeta(s)P_0 + (I - P_0). \end{aligned} \tag{HPS36}$$

Every formula follows by multiplying the two direct-sum blocks, using HPS34. In particular at a zero the X_r block remains a bounded bijection, with inverse $\mathcal{B}_\rho^0$. The quotient map in the third line is well-defined and injective because its kernel is precisely X_r ; it is onto by constant functions. The factor on the last line is a holomorphic invertible family, its inverse being $P_0 + \mathcal{A}_s^0(I - P_0)$. Thus near a zero of order m the entire singular block is exactly the scalar vanishing factor $(s - \rho)^m$ times a nonzero holomorphic scalar on the constant line. This conclusion does not assume simple zeros. The zero order is finite because the scalar meromorphic function is not identically zero, being positive for real $s > 1$.

The actual monomial and Hurwitz receivers are

$$\begin{aligned} \mathcal{E}_s(t^k) &= \zeta(s+k)t^k \quad (k \geq 1, \operatorname{Re} s > 0), \quad \mathcal{E}_s\mathbf{1} = \zeta(s)\mathbf{1}, \\ \mathcal{E}_s g_s(t) &= \zeta(s) + \sum_{n \geq 1} ((n+t)^{-s} - n^{-s}) = F_t(s), \quad |t| \leq r < 1. \end{aligned} \tag{HPS37}$$

For the first line the n th operator term is $n^{-s-k}t^k$, and its scalar sum converges absolutely since $\operatorname{Re}(s+k) > 1$. It is nonzero there: the absolutely convergent scalar Möbius series in HPS15 at $t = 0$ supplies the inverse of $\zeta(s+k)$. For the second line, g_s is holomorphic on a neighbourhood of the closed disc because $r < 1$, with the logarithm of $1+t$ fixed by its value zero at $t = 0$. Its constant coordinate is 1. Each term of $\mathcal{A}_s^0(g_s - \mathbf{1})$ is exactly the displayed difference. The bound in HPS33, or integration of $-s(1+w)^{-s-1}$ along $w = t/n$, makes this series uniformly convergent on the closed disc and on compact sets of $\sigma > 0$. For $\sigma > 1$ it equals the original trace. It consequently defines its unique meromorphic continuation to this disc and half-plane. This is an explicit continuation of the original object, not a replacement by another function.

The differential law continues on exact bounded domains as well. If $0 < r' < r < 1$, Cauchy's formula gives $\|f'\|_{r'} \leq \|f\|_r / (r - r')$ for $f \in \mathcal{A}_r$, using smaller circles and then the limiting bound. The differentiated series converges in $\mathcal{A}_{r'}$, and yields

$$\begin{aligned} \partial_t \mathcal{E}_s f &= \mathcal{E}_{s+1}(\partial_t f) \quad (\mathcal{A}_r \longrightarrow \mathcal{A}_{r'}), \quad \operatorname{Re} s > 0, \\ \partial_t F_t(s) &= -sF_t(s+1) = -DF_t(s). \end{aligned} \tag{HPS38}$$

Here $\mathcal{E}_{s+1}$ is the ordinary absolutely convergent operator on $\mathcal{A}_{r'}$; its real part exceeds one. Termwise differentiation supplies $n^{-s-1}f'(t/n)$, whose norm sum is at most $\zeta(\sigma+1)\|f'\|_{r'}$. The derivative of the constant $\zeta(s)f(0)$ vanishes, so at $s = 1$ the first identity has a removable operator-valued singularity on its left side. In the second line, $\partial_t g_s = -s g_{s+1}$ proves the identity from HPS37 without an exchange of divergent trace series. This is the same D as HPS4, including its factor s .

All the preceding block maps are complex linear and hence have the exact lifts HPS22–25. At any zero ρ in the stated domain define the full supported null kernel using every labelled zero

$Z_L = \{z_\lambda : \lambda \in L\}$. The complete kernel and fibres are

$$\begin{aligned}
(\mathcal{E}_\rho^\dagger)^{-1}(Z_L) &= G_L(\mathbb{C}\mathbf{1}), \\
(\mathcal{E}_\rho^\dagger)^{-1}(\tau) &= \begin{cases} \{\tau\}, & 0_L \neq 1_L, \\ G_L(\mathbb{C}\mathbf{1}), & 0_L = 1_L, \end{cases} \\
(\mathcal{E}_\rho^\dagger)^{-1}(v, 1_L) &= \{(\mathcal{B}_\rho^0 v + c\mathbf{1}, 1_L) : c \in \mathbb{C}\} \quad (v \in X_r), \\
(\mathcal{E}_\rho^\dagger)^{-1}(0, \lambda) &= \{(0, \lambda)\} \quad (\lambda \neq 1_L), \\
q^\dagger : G_L(\mathcal{A}_r) &\longrightarrow G_L(\mathbb{C}), \quad (f, \lambda) \longmapsto (f(0), \lambda).
\end{aligned} \tag{HPS39}$$

Outputs with amplitude outside X_r have no preimage. For the displayed top fibre use the two blocks in HPS36. At a non-top label the carrier condition forces the input amplitude to be zero, which proves its fibre assertion. In particular a nonzero constant is sent to e , which is distinct from τ when L is nontrivial. The ordinary inverse image of the unsupported zero does not replace the displayed full supported null kernel.

The map $q^\dagger$ is a unital semiring homomorphism, because evaluation at zero is a unital ring homomorphism and labels are unchanged. Its exact fibres are $f(0) = g(0)$ at the same label. It has the following supported linear quotient property. If an $G_L(\mathbb{C})$ -linear map $b : G_L(\mathcal{A}_r) \rightarrow M$ satisfies $b(v, \lambda) = b(0, \lambda)$ for every admissible $v \in X_r$, then it factors uniquely through $q^\dagger$. To prove this write $f = f(0)\mathbf{1} + (f - f(0)\mathbf{1})$ with the same admissible label on both terms. The latter term has the same image as $(0, \lambda)$. This labelled zero is absorbed by the first image: for any semimodule element y one has $y + ey = (1 + e)y = y$, and linearity identifies this ey with $b(0, \lambda)$. Thus $b(f, \lambda) = b(f(0)\mathbf{1}, \lambda)$. The asserted factor is $(c, \lambda) \mapsto b(c\mathbf{1}, \lambda)$; constant functions prove its uniqueness and linearity. Conversely any such factor has the stated property. All these maps commute with Φ by HPS25; its retraction recovers their complete original labelled fibres.

$$\begin{array}{ccccc}
\mathcal{H} \text{ carrying } H_{t/m} & \xrightarrow{S_m} & \mathcal{H} \text{ carrying } H_t & & \\
U_{t/m}(u) \downarrow & & \downarrow U_t(u) & & \\
\mathcal{H} \text{ carrying } H_{t/m} & \xrightarrow{m^{iu} S_m} & \mathcal{H} \text{ carrying } H_t & & \\
(g_s, \lambda) & \xrightarrow{\mathcal{E}_s^\dagger} & (\mathcal{E}_s g_s, \lambda) & \xrightarrow{\mathcal{M}_s^\dagger} & (g_s, \lambda) \\
\Phi \downarrow & & \Phi \downarrow & & \Phi \downarrow \\
(g_s, \iota_L \lambda) & \xrightarrow{(\mathcal{E}_s, \text{id})} & (Z_s, \iota_L \lambda) & \xrightarrow{(\mathcal{M}_s, \text{id})} & (g_s, \iota_L \lambda)
\end{array}$$

Figure 20: Exact parameter-changing dilation, HPS5–8, and the full supported operator Euler product, HPS14–25. In the lower diagram the actual nonzero atom requires $\lambda = 1_L$; replacing its amplitude by any general input f gives the same commuting diagram for every allowed label, including all zero amplitudes. The upper square keeps deformation t distinct from unitary time u . It does not depict a fixed- t scalar action; its exact correction is HPS7–9. Independent coordinate masks and all their surviving or cancelled labels are calculated in HPS27–31.

46 The full-lattice cyclotomic receiver and the original Hurwitz trace

Human sources and exact scope. The operators below represent the Bost–Connes relations in Alain Connes, Caterina Consani and Matilde Marcolli, *Fun with $\mathbb{F}_1$* , original author source `funBC.tex`, lines 351–445 (`pres`, `idem`, `endo`, `endosigma`), 895–904 (c1)–(c4), and 929–1082 (`murhosigma`, `newrhos`, `mapstilderho`). The original system is Jean-Benoît Bost and Alain Connes, *Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory*, *Selecta Math.* (N.S.) 1 (1995), 411–457, cited as BC in that author source, lines 2207–2209. The complete split-support carrier is the foundation retained under the byline The Clankers, *Split Support Geometry*, v11, Definition `def:lattice-split`. All relations, support corrections and trace constants used here are proved below. The argument is over $\mathbb{C}$, where the displayed positive integers are invertible. It does not transfer those divisions silently to integral or characteristic- p versions, whose different maps are expressly discussed in the cited source.

46.1 The actual Hilbert-space representation and all four relations

Use $\mathbb{N} = \{1, 2, 3, \dots\}$ and its original orthonormal basis ξ_n in $\mathcal{H} = \ell^2(\mathbb{N})$. For $r \in \mathbb{Q}/\mathbb{Z}$ and $m \geq 1$, set

$$E(r)\xi_n = e^{2\pi i n r} \xi_n, \quad S_m \xi_n = \xi_{mn}, \quad S_m^* \xi_n = \begin{cases} \xi_{n/m} & m \mid n, \\ 0 & m \nmid n. \end{cases} \quad (\text{HCR1})$$

The phase is independent of the chosen representative of r . $E(r)$ is unitary and S_m is an isometry, since the indicated basis vectors remain orthonormal. Taking inner products with each basis vector proves the adjoint formula. Consequently

$$\begin{aligned} E(r)E(t) &= E(r+t), & E(0) &= I, & E(r)^* &= E(-r), \\ S_m^* S_m &= I, & S_m S_n &= S_{mn}, & S_m^* S_n^* &= S_{mn}^*, \\ S_m S_n^* &= S_n^* S_m & (\gcd(m, n) &= 1). \end{aligned} \quad (\text{HCR2})$$

The first line is multiplication of phases. The second follows by multiplication or division of the basis index. For the last line, its left side is nonzero exactly when $n \mid k$ on input ξ_k , and then gives $\xi_{mk/n}$. Its right side is nonzero exactly when $n \mid mk$. Coprimality makes these conditions identical, with the same output. This proves the source relations (c1), both parts of (c2), and (c3), without changing the basis.

Choose a rational representative of r . Its m inverse images under multiplication by m are $(r + j)/m$, $0 \leq j < m$, modulo $\mathbb{Z}$. The finite root sum is

$$\frac{1}{m} \sum_{j=0}^{m-1} e^{2\pi i n j/m} = \begin{cases} 1 & m \mid n, \\ 0 & m \nmid n. \end{cases} \quad (\text{HCR3})$$

If the ratio is one all m terms are one; otherwise multiplying the geometric sum by $1 - e^{2\pi i n/m}$ gives zero and that factor is nonzero. Therefore both sides of the exact source relation (c4) act on ξ_n by the same scalar:

$$S_m E(r) S_m^* = \frac{1}{m} \sum_{j=0}^{m-1} E((r + j)/m), \quad (S_m E(r) S_m^*) \xi_n = \begin{cases} e^{2\pi i (n/m)r} \xi_n & m \mid n, \\ 0 & m \nmid n. \end{cases} \quad (\text{HCR4})$$

Bounded operators agreeing on the complete basis agree on all of $\mathcal{H}$. In particular $P_m = S_m S_m^*$ projects onto the actual multiples of m . The covariance formulas are also direct basis identities:

$$E(r) S_m = S_m E(mr), \quad S_m^* E(r) = E(mr) S_m^*, \quad S_m^* E(r) S_m = E(mr). \quad (\text{HCR5})$$

For the original trace weight let $N\xi_n = n\xi_n$, with domain $\{x : \sum n^2|x_n|^2 < \infty\}$, and let $H = \log N$. The diagonal unitary group and its exact covariance are

$$N^{iu}S_mN^{-iu} = m^{iu}S_m, \quad N^{iu}E(r)N^{-iu} = E(r) \quad (u \in \mathbb{R}). \quad (\text{HCR6})$$

Indeed the first phase on ξ_n is $(mn)^{iu}n^{-iu} = m^{iu}$; the second identity is commutation of diagonal operators. These are the original time-evolution factors in `funBC.tex`, lines 865–869. No parameter of the trace weight is rescaled below.

46.2 Locally finite matrices with every original lattice label

Let L be any nontrivial bounded distributive lattice and retain

$$\begin{aligned} S &= G_L(\mathbb{C}) = \{(a, \lambda) : a \in \mathbb{C}, \lambda \in L, a \neq 0 \Rightarrow \lambda = 1_L\}, \\ (a, \lambda) + (b, \mu) &= (a + b, \lambda \vee \mu), \quad (a, \lambda)(b, \mu) = (ab, \lambda \wedge \mu), \\ \tau &= (0, 0_L), \quad e = (0, 1_L), \quad 1_S = (1, 1_L), \quad \hat{c} = (c, 1_L), \quad (a, \lambda)^* = (\bar{a}, \lambda). \end{aligned} \quad (\text{HCR7})$$

Let $\mathcal{M}_L$ consist of matrices indexed by $\mathbb{N}^2$ whose rows and columns contain only finitely many entries different from τ . Finiteness is measured using the actual split zero, not just the complex amplitude. Matrix addition, multiplication and adjoint make this a unital involutive semiring. For each product entry its sum is finite. A product row is contained in the union of finitely many rows of the second factor, and a product column in the union of finitely many columns of the first; this proves closure. Triple products have finite sets of contributing paths, so associativity follows from finite distributivity. The identity has 1_S on the diagonal and τ elsewhere.

The lift of the specified generators is

$$\hat{E}(r)_{ij} = \begin{cases} \widehat{e^{2\pi ijr}} & i = j, \\ \tau & i \neq j, \end{cases} \quad \hat{S}_{mij} = \begin{cases} 1_S & i = mj, \\ \tau & i \neq mj. \end{cases} \quad (\text{HCR8})$$

Both are locally finite. Their adjoints use (HCR7) entrywise and transpose. All the identities (HCR2) and (HCR5) lift exactly: each nonzero entry calculation has a unique contributing path, with the same phase and label, and all missing paths give τ . This includes both adjoint relations in (c2). One is not applying an additive map that changes every complex zero to τ : that proposed map fails on $1 + (-1)$, since $\widehat{1} + \widehat{-1} = e \neq \tau$. Formula (HCR8) specifies the lifts and all subsequent sums are calculated in S .

The full coefficient injection is retained entry by entry. Let K be the original tropical semifield with zero, $K_L = K \otimes_{\mathbb{B}} L$, and $\iota_L(\lambda) = 1_K \otimes \lambda$. The map $\beta : K \rightarrow \mathbb{B}$ taking a nonzero element to one and zero to zero is a semiring homomorphism: a tropical sum is zero exactly when both terms are zero, and a product is zero exactly when a factor is. Consequently $\rho_L = \beta \otimes \text{id}_L$ is the stated retraction, after $\mathbb{B} \otimes_{\mathbb{B}} L = L$. The exact matrix map is

$$\begin{aligned} \Phi_S(a, \lambda) &= (a, \iota_L(\lambda)) \in \mathbb{C} \times K_L, \quad \rho_L \iota_L = \text{id}_L, \\ \Phi_{\mathcal{M}} : \mathcal{M}_L &\hookrightarrow \mathcal{M}(\mathbb{C}) \times \mathcal{M}(K_L), \quad ((a_{ij}, \lambda_{ij})) \mapsto ((a_{ij}), (\iota_L \lambda_{ij})). \end{aligned} \quad (\text{HCR9})$$

Here all matrix algebras use finite rows and columns, and the K_L involution is the identity on coefficients followed by transpose. Finite sums and products are preserved because ι_L preserves the lattice semiring operations. The retraction recovers each label, proving injectivity. Its image consists exactly of pairs whose every support entry belongs to $\iota_L(L)$ and whose nonzero amplitude entries have support 1_{K_L} , with the original finite row and column condition. No infinite lattice join is required.

46.3 The exact missing support in relation (c4)

Define the two actual split matrices and the defect by

$$A_m(r) = \widehat{S}_m \widehat{E}(r) \widehat{S}_m^*, \quad R_m(r) = \widehat{1/m} \sum_{j=0}^{m-1} \widehat{E}((r+j)/m),$$

$$(M_m)_{nn} = \begin{cases} \tau & m \mid n, \\ e & m \nmid n, \end{cases} \quad (M_m)_{ij} = \tau \quad (i \neq j).$$
(HCR10)

The scalar multiplication in the average multiplies every entry by $\widehat{1/m}$; it preserves all displayed labels. The original root sum (HCR3) now gives

$$A_m(r)_{nn} = \begin{cases} (e^{2\pi i(n/m)r}, 1_L) & m \mid n, \\ \tau & m \nmid n, \end{cases} \quad R_m(r)_{nn} = \begin{cases} (e^{2\pi i(n/m)r}, 1_L) & m \mid n, \\ e & m \nmid n, \end{cases}$$
(HCR11)

$A_m(r) + M_m = R_m(r).$

Every off-diagonal entry is τ . On the diagonal each Fourier summand has top label. Their finite sum therefore has top label even where the complex sum vanishes. This proves the entire equation rather than only its amplitude.

The correction is minimal and its entire set of alternatives is explicit. If $A_m(r) + D = R_m(r)$, amplitude equality forces every amplitude of D to be zero. Off-diagonal entries must be τ . At a nonmultiple of m , its label must be top; at a multiple it may be any original label. Thus

$$D = M_m + \text{diag}(d_n), \quad d_n = \begin{cases} (0, \lambda_n), \lambda_n \in L & m \mid n, \\ \tau & m \nmid n. \end{cases}$$
(HCR12)

In the entrywise order on zero-amplitude labels, M_m is the unique least such correction. Also $M_m^2 = M_m$, $M_m A_m(r) = A_m(r) M_m = 0_{\mathcal{M}}$, $M_m \widehat{S}_m = 0_{\mathcal{M}}$ and $\widehat{S}_m^* M_m = 0_{\mathcal{M}}$, by their disjoint allowed indices. Hence the compression of the repaired relation recovers $\widehat{E}(r)$ exactly.

Let P_m^K be the diagonal K_L matrix with entry 1_{K_L} at multiples of m and zero elsewhere, and C_m^K its complementary diagonal indicator. The complete map is

$$\begin{aligned} \Phi_{\mathcal{M}}(A_m(r)) &= (S_m E(r) S_m^*, P_m^K), & \Phi_{\mathcal{M}}(M_m) &= (0, C_m^K), \\ \Phi_{\mathcal{M}}(R_m(r)) &= (S_m E(r) S_m^*, I_K), & P_m^K + C_m^K &= I_K. \end{aligned}$$
(HCR13)

Thus the full coefficient map itself exhibits the missing support and its exact correction; equality in the complex coordinate alone is not equality of the original matrices.

46.4 Exact full-label corner and common-label receivers

Let $\mathcal{D}_L \subset \mathcal{M}_L$ be the diagonal semiring, identified with $S^{\mathbb{N}}$, and put $\widehat{P}_m = \widehat{S}_m \widehat{S}_m^*$. There are exact maps retaining all entries that they receive:

$$\begin{aligned} \sigma_m(D) &= \widehat{S}_m^* D \widehat{S}_m, & (\sigma_m D)_{nn} &= D_{mn, mn}, \\ \rho_m^{\text{sp}}(D) &= \widehat{S}_m D \widehat{S}_m^*, & \sigma_m \rho_m^{\text{sp}} &= \text{id}, & \rho_m^{\text{sp}} \sigma_m(D) &= \widehat{P}_m D. \end{aligned}$$
(HCR14)

The first map is a unital semiring homomorphism. The second is a unital isomorphism onto the genuine corner $\widehat{P}_m \mathcal{D}_L$ with its unit $\widehat{P}_m$ and additive zero $0_{\mathcal{M}}$. All assertions follow entrywise from selection or insertion of the indices divisible by m ; in particular every original selected label is unchanged.

Its exact completed presentation is

$$\begin{aligned} \mathcal{C}_m &= \{D + M_m : D \in \widehat{P}_m \mathcal{D}_L\}, & C_m(D) &= D + M_m, \\ 0_{\mathcal{C}_m} &= M_m, & 1_{\mathcal{C}_m} &= \widehat{P}_m + M_m = R_m(0), & C_m^{-1}(X) &= \widehat{P}_m X. \end{aligned} \quad (\text{HCR15})$$

Addition and multiplication on this displayed set are inherited from matrices, but its additive zero is the stated M_m . For supported D, F , their products with M_m vanish and $M_m^2 = M_m$, so $(D + M_m)(F + M_m) = DF + M_m$ and $(D + M_m) + (F + M_m) = D + F + M_m$. These calculations prove every semiring law, the unit and the mutually inverse maps. This completion preserves every label at a multiple of m and inserts exactly the forced top zero at a nonmultiple. In particular $C_m(\rho_m^{\text{sp}} \widehat{E}(r)) = R_m(r)$ is (c4) in this exact completed presentation, with its zero stated. The inverse is compression, not an unidentified projection.

For the common-label diagonal ring put

$$Z = \text{diag}(e, e, \dots), \quad \mathcal{D}^{\text{top}} = \{\text{diag}((a_n, 1_L)) : a_n \in \mathbb{C}\}, \quad C(D) = D + Z. \quad (\text{HCR16})$$

This is a ring under its inherited diagonal operations, with additive zero Z , unit I and additive inverse $\text{diag}((-a_n, 1_L))$. It is canonically isomorphic to the ring of complex sequences. The displayed $C : \mathcal{D}_L \rightarrow \mathcal{D}^{\text{top}}$ is onto and unital; finite pointwise operations give $C(D + F) = C(D) + C(F)$ and $C(DF) = C(D)C(F)$. Its kernel congruence is exactly equality of every complex amplitude; the complete fibers retain arbitrary labels where those amplitudes vanish. In particular $C(A_m(r)) = R_m(r)$ and $C(M_m) = Z$. This receiver proves exactly how the classical diagonal relation is obtained and specifies the data that it forgets.

For the whole matrix algebra, the amplitude receiver has the equally exact universal description

$$\mathcal{M}_L / \langle Z = 0_{\mathcal{M}} \rangle \simeq \mathcal{M}(\mathbb{C}), \quad ((a_{ij}, \lambda_{ij})) \mapsto (a_{ij}). \quad (\text{HCR17})$$

The matrix Z is central idempotent, and ZA is the zero-amplitude matrix with precisely the labels of A . Thus imposing $Z = 0$ kills every zero-amplitude matrix. Every complex row-and-column-finite matrix has a preimage by placing top labels at nonzero entries and τ at zeros. Two preimages with the same amplitudes are congruent: add the coefficientwise amplitude negative of the second, with its same labels; both relevant sums have zero amplitude and hence vanish in the quotient. Conversely the amplitude map kills Z and distinguishes unequal amplitudes. This proves the quotient and its universal property. It receives all four original Bost–Connes relations.

Inverting Z is a different map. Its idempotent corner $Z\mathcal{M}_L$ is the semiring of locally finite L -valued support matrices. There (HCR13) becomes the two different support indicators P_m and I , with their nonzero complementary correction for $m > 1$. Thus supported-zero inversion does not itself restore (c4) with the correction omitted. Both exact connecting maps have now been calculated.

46.5 Every residue projector and its complete supported product

For $q \geq 1$ and $a \in \{1, \dots, q\}$, with $a = q$ representing residue zero, define

$$P_{q,a} = \frac{1}{q} \sum_{k=0}^{q-1} e^{-2\pi i a k / q} E(k/q), \quad P_{q,a} \xi_n = \mathbf{1}_{n \equiv a \pmod{q}} \xi_n. \quad (\text{HCR18})$$

The second formula is (HCR3) applied to $n - a$ and proves the first is exactly the orthogonal residue projection. Its split Fourier sum, sparse projection and missing-support diagonal are

$$\begin{aligned} Q_{q,a} &= \widehat{1/q} \sum_{k=0}^{q-1} e^{-2\pi i a k / q} \widehat{E}(k/q), \\ (\Pi_{q,a})_{nn} &= \begin{cases} 1_S & n \equiv a \pmod{q}, \\ \tau & n \not\equiv a \pmod{q}, \end{cases} \quad (B_{q,a})_{nn} = \begin{cases} \tau & n \equiv a \pmod{q}, \\ e & n \not\equiv a \pmod{q}, \end{cases} \\ \boxed{Q_{q,a} &= \Pi_{q,a} + B_{q,a}.} \end{aligned} \quad (\text{HCR19})$$

All off-diagonal entries are τ . The nonselected diagonal Fourier sums have zero amplitude and top label, so the last formula follows directly, not by replacing those entries with τ .

The complete projector laws in the ambient split algebra are

$$\begin{aligned} Q_{q,a}^2 &= Q_{q,a}, & Q_{q,a}Q_{q,b} &= Z \quad (a \neq b), & \sum_{a=1}^q Q_{q,a} &= I, \\ \Pi_{q,a}^2 &= \Pi_{q,a}, & \Pi_{q,a}\Pi_{q,b} &= 0_{\mathcal{M}} \quad (a \neq b), & \sum_{a=1}^q \Pi_{q,a} &= I. \end{aligned} \quad (\text{HCR20})$$

For $a \neq b$, at each index at least one raw diagonal amplitude is zero, but both labels are top, so their product is e at every index. At a given index exactly one summand in the raw sum is 1_S and the others are e ; their sum is 1_S . These calculations prove all raw identities, including the case $q = 1$ where no distinct pair exists. The sparse identities are disjoint indicators. The matrices $Q_{q,a}$ are orthogonal in the common-label ring (HCR16), whose zero is Z ; their ambient product must still be recorded as the displayed Z , not τ . For $q \geq 2$, making that one distinct product equal to the ambient zero imposes $Z = 0$, and (HCR17) gives the exact resulting quotient.

The mixed original labels can be tested on every coefficient vector $v_n = (c_n, \lambda_n)$ with $c \in \ell^2(\mathbb{N})$ and $c_n \neq 0 \Rightarrow \lambda_n = 1_L$. The indicated operators preserve that carrier and have the unchanged bounded Hilbert-space amplitudes. Their exact actions are

$$\begin{aligned} (Q_{q,a}v)_n &= \begin{cases} (c_n, \lambda_n) & n \equiv a \ (q), \\ (0, \lambda_n) & n \not\equiv a \ (q), \end{cases} \\ (\Pi_{q,a}v)_n &= \begin{cases} (c_n, \lambda_n) & n \equiv a \ (q), \\ \tau & n \not\equiv a \ (q). \end{cases} \end{aligned} \quad (\text{HCR21})$$

Thus the missing-support term retains each original λ_n , even when it is neither top nor bottom. More generally any locally finite matrix product on a finite vector has at output i the exact label $\bigvee_j (\lambda_{ij} \wedge \lambda_j)$ over its finite contributing row. The complete map sends this to the same finite sum of products in K_L and recovers it by ρ_L . This proves naturality of all mixed labels, not just of the binary masks of the generators. Any bounded lattice homomorphism $L \rightarrow L'$ preserves the finite formulas and the carrier; applying it entrywise commutes with all displayed constructions.

For completeness the coefficient images are

$$\Phi_{\mathcal{M}}(\Pi_{q,a}) = (P_{q,a}, P_{q,a}^K), \quad \Phi_{\mathcal{M}}(B_{q,a}) = (0, (P_{q,a}^{\text{comp}})^K), \quad \Phi_{\mathcal{M}}(Q_{q,a}) = (P_{q,a}, I_K). \quad (\text{HCR22})$$

The two support masks are complementary, disjoint diagonal indicators. This identifies their images without subtraction inside K_L .

The residue receiver is compatible with the actual dilation:

$$\begin{aligned} S_m P_{q,a} S_m^* &= P_{mq,ma}, & \widehat{S}_m \Pi_{q,a} \widehat{S}_m^* &= \Pi_{mq,ma}, \\ \widehat{S}_m Q_{q,a} \widehat{S}_m^* + M_m &= Q_{mq,ma}. \end{aligned} \quad (\text{HCR23})$$

At an output index divisible by m , all three equalities test $n/m \equiv a \pmod{q}$. At other indices the sparse expressions are zero and the final raw expression needs exactly M_m . This proves the complete equations. If $q \mid Q$, refining the residue classes gives

$$P_{q,a} = \sum_{\substack{1 \leq b \leq Q \\ b \equiv a \ (q)}} P_{Q,b}, \quad \Pi_{q,a} = \sum_{b \equiv a \ (q)} \Pi_{Q,b}, \quad Q_{q,a} = \sum_{b \equiv a \ (q)} Q_{Q,b}. \quad (\text{HCR24})$$

The classes partition the selected progression. In the raw equation there are $Q/q \geq 1$ summands, all with top diagonal label; their amplitude sum is the same indicator and every label remains top. Thus refinement also preserves the exact split sums.

Compression also has an exact non-coprime formula. Put $d = \gcd(m, q)$ and $q' = q/d$. If $d \nmid a$, the congruence $mn \equiv a \pmod{q}$ has no solution. If $d \mid a$, let $b \in \{1, \dots, q'\}$ be the unique representative with $(m/d)b \equiv a/d \pmod{q'}$; it exists because m/d is invertible modulo q' , including $b = 1$ when $q' = 1$. Then

$$\begin{aligned} S_m^* P_{q,a} S_m &= \begin{cases} 0 & d \nmid a, \\ P_{q',b} & d \mid a, \end{cases} \\ \widehat{S}_m^* \Pi_{q,a} \widehat{S}_m &= \begin{cases} 0 & d \nmid a, \\ \Pi_{q',b} & d \mid a, \end{cases} \\ \widehat{S}_m^* Q_{q,a} \widehat{S}_m &= \begin{cases} Z & d \nmid a, \\ Q_{q',b} & d \mid a. \end{cases} \end{aligned} \quad (\text{HCR25})$$

The diagonal amplitude test is the displayed congruence; compression of a raw matrix retains top label at every new diagonal index, whereas compression of a sparse matrix retains a label only at a selected index. This proves the three different zero cases exactly.

46.6 The trace of the original weight is exactly the Hurwitz series

For $s \in \mathbb{C}$ with $\sigma = \operatorname{Re} s > 1$, use $x^{-s} = \exp(-s \log x)$ for positive real x , with its real logarithm. Then

$$W_s = N^{-s} = e^{-sH}, \quad W_s \xi_n = n^{-s} \xi_n, \quad \|W_s\|_1 = \sum_{n \geq 1} n^{-\sigma} < \infty. \quad (\text{HCR26})$$

The absolute values of the diagonal eigenvalues are the singular values, proving the trace norm formula. For any bounded diagonal $D\xi_n = d_n \xi_n$, finite basis truncation and the tail bound $\|D\| \sum_{n > K} n^{-\sigma} \rightarrow 0$ prove

$$\operatorname{Tr}(DW_s) = \sum_{n \geq 1} d_n n^{-s}. \quad (\text{HCR27})$$

Every series here is absolutely convergent. The selected indices for $P_{q,a}$ are precisely $a + qj$, $j \geq 0$, including $q, 2q, \dots$ when $a = q$. Thus the actual Hurwitz receiver is

$$\begin{aligned} \zeta(s, \alpha) &= \sum_{j=0}^{\infty} (j + \alpha)^{-s} \quad (\alpha > 0), \\ \operatorname{Tr}(P_{q,a} N^{-s}) &= \sum_{j=0}^{\infty} (a + qj)^{-s} = q^{-s} \zeta(s, a/q), \quad 1 \leq a \leq q, \quad \sigma > 1. \end{aligned} \quad (\text{HCR28})$$

Absolute convergence of the Hurwitz series follows from the finite initial term and the bound $(j + \alpha)^{-\sigma} \leq j^{-\sigma}$ for $j \geq 1$. Additivity of real logarithms of positive factors proves the factor q^{-s} exactly. All these series are locally uniformly convergent in this half-plane by a common majorant $\sum n^{-1-\varepsilon}$ on each compact subset; hence the stated receiver is holomorphic there.

The complete split trace on these particular matrices can also be recorded without an unproved infinite join. Let $\widehat{W}_s = \operatorname{diag}(\widehat{n^{-s}})$ and let tr_K mean the sum of the first K diagonal entries. Then

$$\begin{aligned} T_K &= \sum_{\substack{1 \leq n \leq K \\ n \equiv a \pmod{q}}} n^{-s}, \\ \operatorname{tr}_K(Q_{q,a} \widehat{W}_s) &= (T_K, 1_L) \quad (K \geq 1), \\ \operatorname{tr}_K(\Pi_{q,a} \widehat{W}_s) &= \begin{cases} \tau & K < a, \\ (T_K, 1_L) & K \geq a. \end{cases} \end{aligned} \quad (\text{HCR29})$$

The first equality retains top labels even on unselected indices. In the second, the first selected index is a . After that index, the support of the finite trace is constant top and the amplitude converges absolutely. Therefore the amplitude limit with that retained label is $(q^{-s}\zeta(s, a/q), 1_L)$ for both sequences. Under the full map it is $(q^{-s}\zeta(s, a/q), 1_{K_L})$. For $q > 1$ the analogous weighted trace of $B_{q,a}$ has eventual value e , although the two preceding limits agree because addition of e leaves a top-labelled element fixed. Thus this scalar trace equality does not identify their different operator support matrices. No infinite trace on arbitrary labelled matrices, or join not present in L , has been assumed.

The original additive-character trace and its exact finite transforms are

$$\begin{aligned} F_{q,k}(s) &= \text{Tr}(E(k/q)N^{-s}) = \sum_{n \geq 1} e^{2\pi i k n / q} n^{-s}, \\ q^{-s}\zeta(s, a/q) &= \frac{1}{q} \sum_{k=0}^{q-1} e^{-2\pi i a k / q} F_{q,k}(s), \\ F_{q,k}(s) &= q^{-s} \sum_{a=1}^q e^{2\pi i k a / q} \zeta(s, a/q). \end{aligned} \tag{HCR30}$$

The first transform is the trace of (HCR18). For the inverse, $E(k/q) = \sum_a e^{2\pi i k a / q} P_{q,a}$ on each basis vector, and (HCR27) gives its trace. Finite character sums and absolute convergence justify every interchange. In particular the finite cyclic group algebra acts faithfully: if a combination of $E(k/q)$ vanishes, its values at $n = 1, \dots, q$ and (HCR3) recover each coefficient as zero.

Dilation retains the Hurwitz shift and changes the trace factor by exactly its original degree:

$$\begin{aligned} N^{-s} S_m &= m^{-s} S_m N^{-s}, & S_m^* N^{-s} S_m &= m^{-s} N^{-s}, \\ \text{Tr}(P_{mq,ma} N^{-s}) &= m^{-s} \text{Tr}(P_{q,a} N^{-s}) = (mq)^{-s} \zeta(s, a/q). \end{aligned} \tag{HCR31}$$

The operator identities follow on ξ_n from $(mn)^{-s} = m^{-s} n^{-s}$, and the trace identity also follows directly by the selected series. Refinement (HCR24), with $Q = hq$, similarly gives

$$q^{-s}\zeta(s, a/q) = Q^{-s} \sum_{j=0}^{h-1} \zeta(s, (a + jq)/Q). \tag{HCR32}$$

The representatives $a + jq$ lie in $1, \dots, Q$ and partition the original residue class, preserving all offsets and both original factors.

46.7 Finite multiplicative characters and every constant in their trace

We prove the character identity rather than assuming its factor. For a finite abelian group G , every character of a subgroup B extends to G . To prove it in one step, choose $g \notin B$ and let m be the least positive integer with $g^m \in B$. Each element of $\langle B, g \rangle$ is uniquely bg^j with $0 \leq j < m$. For a character χ of B , choose one of the m unit complex roots z of $z^m = \chi(g^m)$ and set $\chi'(bg^j) = \chi(b)z^j$. Unique representatives make it well defined, and carrying an exponent through m uses exactly the identity $z^m = \chi(g^m)$, proving multiplicativity. These are exactly its m extensions. Starting from the trivial subgroup shows both that $|\widehat{G}| = |G|$ and that subgroup characters extend. For a nonidentity element of order d , its cyclic character with value $e^{2\pi i / d}$ therefore extends and separates it from the identity. Multiplying all characters by such a separating character permutes the finite character set and forces its sum at that element to vanish. At the identity all summands are one. Consequently

$$\frac{1}{|G|} \sum_{\chi \in \widehat{G}} \overline{\chi(a)} \chi(b) = \mathbf{1}_{a=b} \quad (a, b \in G). \tag{HCR33}$$

The conjugate is exactly $\chi(a)^{-1}$, so the tested element is ba^{-1} . This fixes both the conjugation and the numerical factor.

Take $G = (\mathbb{Z}/q\mathbb{Z})^\times$, of cardinality $\varphi(q)$. Extend each character to all integers by zero on nonunits. The extension is completely multiplicative, since a product is a unit modulo q exactly when both factors are units. When $q = 1$, the unique character is identically one and $\varphi(1) = 1$. For $\gcd(a, q) = 1$, (HCR33) gives the following equality for every $n \geq 1$, including nonunit residues where both sides vanish:

$$\mathbf{1}_{n \equiv a \pmod{q}} = \frac{1}{\varphi(q)} \sum_{\chi \bmod q} \overline{\chi(a)} \chi(n). \quad (\text{HCR34})$$

Define the bounded diagonal operators and their absolutely convergent original series by

$$\begin{aligned} D_\chi \xi_n &= \chi(n) \xi_n, & D_\chi &= \sum_{b=1}^q \chi(b) P_{q,b}, \\ L(s, \chi) &= \sum_{n \geq 1} \chi(n) n^{-s} = \text{Tr}(D_\chi N^{-s}), & \sigma &> 1, \\ P_{q,a} &= \frac{1}{\varphi(q)} \sum_{\chi \bmod q} \overline{\chi(a)} D_\chi & (\gcd(a, q) = 1). \end{aligned} \quad (\text{HCR35})$$

The norm is at most one and the series is majorized by $\sum n^{-\sigma}$. The last line is (HCR34) on each basis vector. Taking its trace and using the original factor from (HCR28) proves

$$\zeta(s, a/q) = \frac{q^s}{\varphi(q)} \sum_{\chi \bmod q} \overline{\chi(a)} L(s, \chi), \quad \gcd(a, q) = 1, \quad \text{Re } s > 1. \quad (\text{HCR36})$$

The factor q^s converts the residue trace to the Hurwitz function; it is not part of the residue projection or of its original trace weight.

This finite operator decomposition has an exact common-label split receiver. Set

$$\tilde{D}_\chi = \text{diag}((\chi(n), 1_L))_{n \geq 1} \in \mathcal{D}^{\text{top}}. \quad (\text{HCR37})$$

The entries on nonunit residues are e , not τ . The corresponding identities, including all those zeros, are

$$\tilde{D}_\chi = \sum_{b=1}^q \widehat{\chi(b)} Q_{q,b}, \quad Q_{q,a} = \widehat{1/\varphi(q)} \sum_{\chi \bmod q} \widehat{\chi(a)} \tilde{D}_\chi \quad (\gcd(a, q) = 1). \quad (\text{HCR38})$$

Their amplitudes are (HCR35); every diagonal label on each side is top, including when $\chi(b) = 0$, because $\hat{0} = e$. This proves equality of the full matrices. Substituting (HCR18) into the ordinary identity for D_χ also gives the exact additive-to-multiplicative Fourier coefficients

$$D_\chi = \frac{1}{q} \sum_{k=0}^{q-1} \left(\sum_{b=1}^q \chi(b) e^{-2\pi i b k / q} \right) E(k/q). \quad (\text{HCR39})$$

Only two finite sums have been interchanged. The inner sum is retained as written, so this formula also covers imprimitive characters without inserting an unwarranted primitive-character factor.

46.8 Euler products and the exact modulus-two specialization

Absolute convergence and unique prime factorization give the Euler product with all original local factors:

$$L(s, \chi) = \prod_{p \text{ prime}} (1 - \chi(p) p^{-s})^{-1}, \quad \sigma > 1. \quad (\text{HCR40})$$

For a finite set of primes, expand each factor in its absolutely convergent geometric series. Complete multiplicativity and unique factorization identify the product with the sum over integers supported on those primes. As the finite set exhausts all primes, each integer appears, and the majorant $\sum n^{-\sigma} < \infty$ makes the omitted sum tend to zero. This proves the product and its limit. The same proof with the constant-one character gives $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$.

For $q = 2$, the sole character is one on odd integers and zero on even integers. Its factor at the prime 2 is exactly one, while every other factor is the corresponding zeta factor. Thus

$$\boxed{\begin{aligned} L(s, \chi_0) &= \prod_{p \neq 2} (1 - p^{-s})^{-1} = (1 - 2^{-s})\zeta(s), \\ \text{Tr}(P_{2,1}N^{-s}) &= (1 - 2^{-s})\zeta(s), \quad \zeta(s, \tfrac{1}{2}) = (2^s - 1)\zeta(s), \quad \sigma > 1. \end{aligned}} \quad (\text{HCR41})$$

The same constants follow by separating the even indices $2j$ from all positive indices, justified by absolute convergence. The sign can also be checked directly from

$$P_{2,1} = \tfrac{1}{2}(I - E(1/2)), \quad P_{2,2} = \tfrac{1}{2}(I + E(1/2)), \quad \text{Tr}(E(1/2)N^{-s}) = (2^{1-s} - 1)\zeta(s). \quad (\text{HCR42})$$

At index n , the phase is the original $(-1)^n$, so these are the odd, even and signed traces with their actual orientations. Equations (HCR28), (HCR30), (HCR36) and (HCR41) are proved in their common absolutely convergent domain. They identify the Hurwitz shifts with explicit trace receivers of the represented endomotive operators; they assert no additional zero-location result.

diagonal index	$m \mid n$	$m \nmid n$
$\widehat{S}_m \widehat{E}(r) \widehat{S}_m^*$	$(e^{2\pi i(n/m)r}, 1_L)$	τ
M_m	τ	e
$\widehat{1/m \sum_{j=0}^{m-1} \widehat{E}((r+j)/m)}$	$(e^{2\pi i(n/m)r}, 1_L)$	e

Figure 21: The exact completion of the Bost–Connes relation (c4), HCR10–13, from Connes–Consani–Marcolli, *Fun with $\mathbb{F}_1$* , source lines 895–904. Every off-diagonal entry is τ . The lower matrix has the same complex operator as the upper matrix but retains the shown supported zeros. On a vector with original label λ_n , the extra e acts as $(0, \lambda_n)$, exactly HCR21. The full coefficient injection HCR9 detects every displayed difference.

47 Two certified interior critical-line crossings

The unchanged family is

$$F(s, t) = \zeta(s, 1 + t), \quad t > -1.$$

This note proves that there are two isolated pairs with $0 < t < 1$, $s = \frac{1}{2} + i\gamma$, and $F(s, t) = 0$. Each zero is simple in s , and its local zero curve crosses the critical line with a certified nonzero real velocity. These are rigorous local results obtained by ball arithmetic and a contraction proof. The two seeds were suggested by numerical continuation; the proofs do not identify either zero with an initially indexed Riemann zero, and do not certify an entire continued path. They give no assertion about all zeros of any one member of the family or a proof of the Riemann hypothesis.

47.1 Analytic family and the exact real Jacobian

Theorem HXC1. *Put $a = 1 + t$. The function F is jointly holomorphic for $\Re s > -1$, $s \neq 1$, $\Re a > 0$, with the logarithm chosen on the right half-plane. It agrees with $\sum_{n=0}^{\infty} (n + a)^{-s}$ when $\Re s > 1$. At the points used below,*

$$F_t(s, t) = -s\zeta(s + 1, 1 + t). \quad (1)$$

Define the real map, in the specified coordinate order,

$$G : \mathbb{R} \times (-1, \infty) \longrightarrow \mathbb{R}^2, \quad G(\gamma, t) = \begin{pmatrix} \Re F(\frac{1}{2} + i\gamma, t) \\ \Im F(\frac{1}{2} + i\gamma, t) \end{pmatrix}.$$

Its Jacobian is exactly

$$DG(\gamma, t) = \begin{pmatrix} -\Im F_s & \Re F_t \\ \Re F_s & \Im F_t \end{pmatrix}_{(s,t)=(1/2+i\gamma,t)}. \quad (2)$$

Proof. For completeness, write $\tilde{B}_2(x) = u^2 - u + 1/6$, where $u = x - \lfloor x \rfloor$. Two integrations by parts on each interval $[n, n + 1]$, followed by summation, give, for $f(x) = (x + a)^{-s}$,

$$\sum_{n=0}^{N-1} f(n) = \int_0^N f(x) dx + \frac{f(0) - f(N)}{2} + \frac{f'(N) - f'(0)}{12} - \frac{1}{2} \int_0^N \tilde{B}_2(x) f''(x) dx.$$

The endpoint values used here are $B_2(0) = B_2(1) = 1/6$ and $B'_2(0) = -1$, $B'_2(1) = 1$; they give every displayed sign. Taking $N \rightarrow \infty$ first for $\Re s > 1$ yields

$$\zeta(s, a) = \frac{a^{1-s}}{s-1} + \frac{a^{-s}}{2} + \frac{s}{12} a^{-s-1} - \frac{s(s+1)}{2} \int_0^\infty \tilde{B}_2(x) (x+a)^{-s-2} dx. \quad (3)$$

On compact subsets of the claimed domain, $\tilde{B}_2$ is bounded, the tail has size at most a fixed constant times $x^{-\Re s - 2}$, and derivatives add only logarithmic factors or further negative powers of $x + a$. Consequently the integral and its local parameter derivatives converge uniformly when $\Re s > -1$. This proves the claimed continuation and joint holomorphy. Termwise differentiation of the original absolutely convergent series for $\Re s > 1$ gives (1). Analytic continuation gives the same identity in a neighborhood of every box below. Finally $\partial_\gamma F = iF_s$; taking its real and imaginary parts proves (2). $\square$

47.2 Exact input boxes and a completed contraction certificate

Every terminating decimal in this section denotes the exact rational number specified by its digits. Set $r = 10^{-12}$, and use the centers

$$\begin{aligned} c_{A,\gamma} &= 41.83377298972885114660494764956585698672, \\ c_{A,t} &= 0.95095300424959108490589251950106981050, \\ c_{B,\gamma} &= 47.04386821148998246126561185165482331798, \\ c_{B,t} &= 0.85870425944212436115567453829890140180. \end{aligned}$$

For $j = A, B$, define the exact closed real box

$$X_j = [c_{j,\gamma} - r, c_{j,\gamma} + r] \times [c_{j,t} - r, c_{j,t} + r]. \quad (4)$$

These boxes have $0 < t < 1$ throughout. The rational matrices used in the proof are

$$C_A = \begin{pmatrix} 4.148869437717385777760373457204 & 4.514404136857108010468674357513 \\ 0.289657552313420538063777653347 & 0.439568080147901688843822972930 \end{pmatrix},$$

$$C_B = \begin{pmatrix} -0.661551363967855286440287545465 & 5.204046413419736695167544593292 \\ 0.018549137861408871952136121318 & 0.405140849866943341137864445274 \end{pmatrix}.$$

They approximate inverse Jacobians, but the proof only uses the written rational matrices and the verified enclosures.

Theorem HXC2. *Each X_j contains exactly one zero $x_j = (\gamma_j, t_j)$ of G . In particular,*

$$0 < t_A, t_B < 1, \quad \zeta\left(\frac{1}{2} + i\gamma_j, 1 + t_j\right) = 0 \quad (j = A, B).$$

The two pairs are different, and both lie in the interior of their boxes.

Proof. The supplied checker `check_hurwitz_critical_crossings.py` uses python-flint 0.9.0, FLINT 3.6.0, a 256-bit working precision, and a power-series truncation length of two. All inputs are the exact rational decimals above, enclosed outward by Arb. No binary64 value is used in a proof check. For the complex balls

$$S_j = \frac{1}{2} + i[c_{j,\gamma} - r, c_{j,\gamma} + r], \quad A_j = 1 + [c_{j,t} - r, c_{j,t} + r],$$

the coefficient of z returned by

$$\text{acb_series}([S_j, 1], \text{prec}=2). \text{zeta}(A_j)$$

encloses F_s throughout the full box. The identity (1) encloses F_t there. The formal coefficient is the analytic derivative because $\zeta(s + z, a) = \zeta(s, a) + \partial_s \zeta(s, a)z + O(z^2)$; it is not a finite difference or a sampled derivative. The ball arithmetic of both the zeta function and its power series includes rounding and analytic evaluation errors. The Python interface and ball semantics are documented in [1, 2].

Write J_j for the resulting interval matrix (2). The following strict rational inequalities were all verified by interval comparisons, with the unrounded enclosures retained in `HURWITZ_CRITICAL_CROSSINGS_CERTIFICATE.json`:

$$\sup_{x \in X_j} \|I - C_j D G(x)\|_\infty < 6 \cdot 10^{-9}, \quad \|C_j G(c_j)\|_\infty < 5 \cdot 10^{-39}. \quad (5)$$

The infinity norm of a real matrix here is the maximum of its absolute row sums. For an additional view of the certificate, outward bounds for the two row sums, written slightly larger than the machine enclosures, are

	first row	second row
A	$5.561 \cdot 10^{-9}$	$4.607 \cdot 10^{-10}$
B	$5.414 \cdot 10^{-9}$	$3.945 \cdot 10^{-10}$

and the determinant comparisons are

$$0.516079320 < \det C_A < 0.516079322,$$

$$-0.364552058 < \det C_B < -0.364552054.$$

Thus the exact C_j are invertible.

Fix either j , and define the actual real map $T_j(x) = x - C_j G(x)$. For $x, y \in X_j$, the segment joining them stays in X_j . Integrating $DT_j = I - C_j DG$ along that segment and using (5) gives

$$\|T_j(x) - T_j(y)\|_\infty \leq 6 \cdot 10^{-9} \|x - y\|_\infty.$$

Also

$$\begin{aligned} \|T_j(x) - c_j\|_\infty &\leq \|C_j G(c_j)\|_\infty + 6 \cdot 10^{-9} \|x - c_j\|_\infty \\ &< 5 \cdot 10^{-39} + 6 \cdot 10^{-21} < 10^{-12} = r. \end{aligned}$$

Hence T_j maps the closed box into its interior. Starting with $x^{(0)} = c_j$, define $x^{(n+1)} = T_j(x^{(n)})$. Every term lies in the box and, with $q = 6 \cdot 10^{-9}$,

$$\|x^{(n+1)} - x^{(n)}\|_\infty \leq q^n \|x^{(1)} - x^{(0)}\|_\infty.$$

The convergent geometric series makes the sequence Cauchy. Its limit x_j belongs to the closed box; continuity gives $T_j(x_j) = x_j$. Invertibility of C_j now gives $G(x_j) = 0$. Any two such zeros would be fixed points and would obey $\|x - y\|_\infty \leq q \|x - y\|_\infty$, so they coincide. The strict self-map estimate puts the fixed point in the interior. Finally the two gamma intervals in (4) are disjoint. This completes existence, uniqueness, isolation, and distinctness. $\square$

47.3 The exact relation to local zero motion

Theorem HXC3. *At each $(s_j, t_j) = (\frac{1}{2} + i\gamma_j, t_j)$ from Theorem HXC2, the zero of $F(\cdot, t_j)$ is simple. There is a unique local holomorphic zero function $s_j(t)$ through this point, and*

$$s'_j(t_j) = \frac{s_j \zeta(s_j + 1, 1 + t_j)}{F_s(s_j, t_j)}. \quad (6)$$

Its real and imaginary components have the following rigorous strict bounds:

$$\begin{aligned} 1.86228474 &< \Re s'_A(t_A) < 1.86228478, \\ 11.49726106 &< \Im s'_A(t_A) < 11.49726110, \\ -2.21634857 &< \Re s'_B(t_B) < -2.21634855, \\ 12.74355586 &< \Im s'_B(t_B) < 12.74355590. \end{aligned}$$

For real t sufficiently close to t_A , the first zero is to the left of the critical line when $t < t_A$ and to the right when $t > t_A$. The second has the opposite crossing orientation at t_B .

Proof. The checker encloses F_s on each entire box and verifies that its complex ball does not contain zero. Consequently $F_s(s_j, t_j) \neq 0$, so the zero is simple. Joint holomorphy proved in Theorem HXC1 and the analytic implicit function theorem produce the unique local zero graph. Differentiation of $F(s_j(t), t) = 0$ gives

$$F_s(s_j(t), t) s'_j(t) + F_t(s_j(t), t) = 0;$$

(1) proves the exact quotient (6), with its stated sign. Division of the two complex balls $-F_t/F_s$ on each box gives the four asserted rational intervals; these are explicit checks in the supplied program.

For completeness, the real Jacobian has the exact determinant relation

$$\det DG = -\Re(F_s \overline{F_t}) = |F_s|^2 \Re(-F_t/F_s). \quad (7)$$

The first equality follows by expanding (2); the second follows by multiplying the quotient by $\overline{F_s}$. Thus the real two-variable isolation and the transverse complex zero motion are tied by an exact identity. The checker also verifies

$$1.93768662 < \det DG(X_A) < 1.93768664, \quad -2.74309248 < \det DG(X_B) < -2.74309246,$$

where each display means that every determinant on that box belongs to the indicated interval. Finally differentiability gives

$$\Re s_j(t_j + h) - \frac{1}{2} = h \Re s'_j(t_j) + o(h).$$

The nonzero real derivative has a fixed sign. Dividing by h and taking $h \rightarrow 0$ proves both strict one-sided crossing statements. $\square$

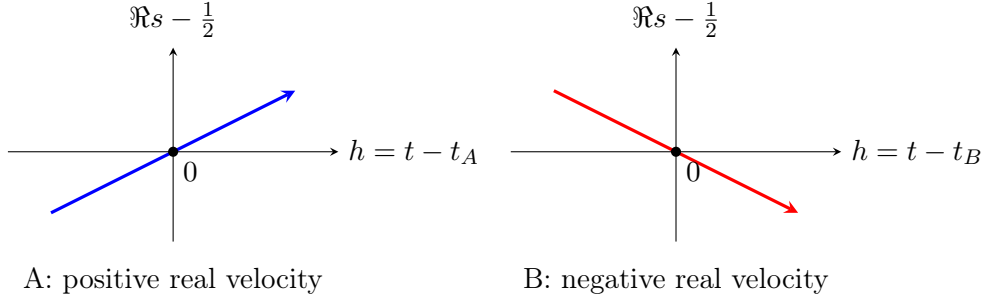


Figure 22: Certified local crossing orientations from Theorem HXC3. The arrows depict tangent orientations schematically; their lengths and the displayed coordinates are not a numerical plot of the zero curves. The exact parameter boxes are (4), and the exact derivative enclosures are stated in Theorem HXC3.

47.4 What these interior parameters imply about Euler products

Theorem HXC4. *For each $j = A, B$, the same function $s \mapsto \zeta(s, 1 + t_j)$ has the certified critical-line zero above but has no expression as an ordinary Dirichlet series $\sum_{m \geq 1} b_m m^{-s}$ absolutely convergent on a right half-plane. In particular it has no classical normalized prime Euler product that expands into such a series. Thus the presence of an isolated zero on the critical line in this family does not require such an Euler product.*

Proof. We prove the assertion for every real $t \in (0, 1)$, so it applies to the two already established parameters. Put $a = 1 + t \in (1, 2)$. For real $\sigma > 1$,

$$a^\sigma \zeta(\sigma, a) = 1 + \sum_{n=1}^{\infty} \left(\frac{a}{a+n} \right)^\sigma.$$

The decreasing positive summands satisfy

$$0 \leq \sum_{n=1}^{\infty} \left(\frac{a}{a+n} \right)^\sigma \leq \left(\frac{a}{a+1} \right)^\sigma \left(1 + \frac{a+1}{\sigma-1} \right) \rightarrow 0.$$

Suppose an ordinary Dirichlet series with coefficients b_m agreed with $\zeta(s, a)$ on a right half-plane and converged absolutely at some real σ_0 . It is nonzero, so let m_0 be its least index with $b_{m_0} \neq 0$. Dominated convergence gives

$$\begin{aligned} m_0^\sigma \sum_{m \geq m_0} b_m m^{-\sigma} &= b_{m_0} + m_0^{\sigma_0} \sum_{m > m_0} b_m m^{-\sigma_0} \left(\frac{m_0}{m} \right)^{\sigma - \sigma_0} \\ &\rightarrow b_{m_0} \neq 0. \end{aligned}$$

The domination is the absolutely summable sequence $m_0^{\sigma_0} |b_m| m^{-\sigma_0}$. On the other hand, the preceding positive-series estimate gives

$$m_0^\sigma \zeta(\sigma, a) = \left(\frac{m_0}{a} \right)^\sigma (1 + o(1)).$$

This tends to zero if $m_0 < a$, and is unbounded if $m_0 > a$. Both contradict its finite nonzero limit. Hence $m_0 = a$, impossible because m_0 is an integer and $1 < a < 2$. A classical normalized prime Euler product with an absolutely convergent ordinary Dirichlet expansion is thereby excluded on exactly the stated meaning of that term. No broader assertion about every possible generalized product is needed or made. $\square$

47.5 Sources, reproduction, and scope

The source retained with this note is *Split-Zero Support, Residue-Parity Reversal, Shifted-Hurwitz CUE Defects, and Finite Phase-Space Enlargement*, marked “Compiled theorem draft,” April 2026. Its exact relevant proof locators are `thm:universal-flow`, lines 587–631; `prop:non-euler`, lines 660–675; and `thm:zero-motion`, lines 995–1062. Those passages were read from the retained original TeX. The source identity is pinned by SHA256 in the JSON certificate. No human author name is inferred from file locations. The present note repeats the analytic identities and the non-Eulerian argument it uses, and proves the new interior crossings with their own numerical certificate. The general local motion formula in (6) specializes at $t = 0$ to the velocity formula in the retained source whenever the Riemann zero is simple. The complete retained TeX accompanies this note under `hurwitz_intake/programme_source/`. Its canonical identity is PUBUNIT-F472A3AADF35B703264575F7; its SHA256 is

52f2a6bccdb15213b1992abef9f8d6476d6e9ea5566ee7ed9d85c9a2d7cc2705.

No verified original public proof URL has been established for that draft; no live public citation is asserted in its place.

The numerical proof in the checker requires Python and `python-flint`. The retained programme source is also read to record its provenance hash. The checker does not call the numerical-path file or an arbitrary-precision root finder. It fixes the exact centers and matrices written above, encloses the derivatives on the full boxes, checks strict contraction, checks interior self-mapping, excludes zero from both determinants and from F_s , verifies every displayed velocity interval, and writes the resulting enclosures to its JSON certificate. It can be reproduced from the `hurwitz_intake` directory by

```
python check_hurwitz_critical_crossings.py
```

The associated full numerical paths remain uncertified. In particular, Theorem HXC2 does not prove that either interior crossing is connected to a particular indexed zero at $t = 0$. Its conclusion already suffices to test the claim that critical-line zeros occur only at Euler-product parameters: in the classical sense specified in Theorem HXC4, that claim is false. This conclusion is about isolated zeros at two members of a varying family, and is not an assertion that every zero of either member lies on the critical line.

References

- [1] Python-FLINT developers, *python-flint 0.9.0 documentation*, `acb_series` and `acb_series.zeta`, https://python-flint.readthedocs.io/en/latest/acb_series.html, accessed 23 September 2026.
- [2] Python-FLINT developers, *python-flint 0.9.0 documentation*, `arb`, decimal/rational inputs, outward radii, and upper bounds, <https://python-flint.readthedocs.io/en/latest/arb.html>, accessed 23 September 2026.

48 The boundary filter and the complete all-zero receiver

Sources and exact question. The supplied *Regular-inverse contact defects and their arithmetic return*, 23 September 2026, is retained unchanged in the `boundary_holonomy_intake` directory. Its equations `eq:contact`, `eq:K`, and `eq:Weil` are the respective contact forcing, endpoint correction, and full spectral Weil pairing. Root read its entire TeX. Its authorship field is empty; no human name is inferred. The original analytic function is Brad Rodgers and Terence Tao’s *The de Bruijn–Newman constant is non-negative*, original author TeX `phidef`, `htdef`, `hoz`, and `sas`:

$$H_0(z) = \int_0^\infty \Phi(u) \cos(zu) du = \frac{1}{8} \xi \left(\frac{1}{2} + \frac{iz}{2} \right).$$

The full endpoint and holonomy input is the programme’s *Actual signed-cover averaging and the supported-zero endpoint*, HWA28–38, as reproduced with proof locators in the supplied continuation. The present derivation uses the displayed spectral pairing and proves its boundary filter, its complete range, and its all-zero detection directly. It is an application of the Weil-positivity mechanism, not a claim to originate Weil’s criterion or to prove its required positivity.

Write $\mathcal{T} = C_c^\infty(\mathbb{R}, \mathbb{C})$ in the original variable v . Use the exact transform and involution

$$M_f(s) = \int_{\mathbb{R}} f(v) e^{-(s-1/2)v} dv, \quad f^\#(v) = \overline{f(-v)}, \quad f_a(v) = f(v-a) \quad (a \in \mathbb{R}). \quad (\text{BPZ1})$$

Let ρ range over the distinct zeros of the actual completed function ξ , with multiplicities m_ρ . Retain

$$B(f, g) = \sum_{\rho} m_{\rho} \overline{M_f(1-\bar{\rho})} M_g(\rho), \quad Q(f) = B(f, f). \quad (\text{BPZ2})$$

This is the supplied $\mathcal{Z}$, including its reflection convention; no local positive form is substituted for it.

48.1 Convergence on the original zero multiset

Here are the analytic estimates needed below. From the original kernel, for $u \geq 0$,

$$\begin{aligned} \Phi(u) &= \sum_{n \geq 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}} > 0, \\ \Phi(u) &\leq C e^{9u - (\pi/2)e^{4u}}, \quad C = 2\pi^2 \sum_{n \geq 1} n^4 e^{-(\pi/2)n^2} < \infty. \end{aligned} \quad (\text{BPZ3})$$

Positivity follows termwise from $2\pi n^2 e^{4u} - 3 > 0$. For the upper bound discard the negative term and use $n^2 e^{4u} \geq (n^2 + e^{4u})/2$. The constant series converges by comparison with a Gaussian integral. For each $\alpha > 0$, the elementary square inequality $Ru \leq \alpha R^2 + u^2/(4\alpha)$ therefore proves

$$|H_0(z)| \leq C_\alpha e^{\alpha|z|^2}, \quad C_\alpha = \int_0^\infty \Phi(u) e^{u^2/(4\alpha)} du < \infty, \quad H_0(0) > 0. \quad (\text{BPZ4})$$

The same bounds on compact sets justify every complex derivative of the integral, so it is entire. Its kernel is real and its cosine even; the retained affine coordinate gives the conjugation and reflection symmetries of ξ . Its Euler product and nonvanishing gamma factor exclude zeros in $\Re s > 1$, and reflection excludes $\Re s < 0$. Also $\xi(0) = \xi(1) = 1/2$, from the residue of ζ at one and reflection. Thus every zero used here satisfies $0 \leq \Re \rho \leq 1$ and $\rho(\rho-1) \neq 0$. No zero-free theorem on the two vertical boundary lines is required.

For completeness (BPZ4) supplies a sufficient polynomial zero count. Choose a circle radius $r \in (2R, 3R)$ on which H_0 has no zero, for $R \geq 1$. Factoring its finitely many interior zeros leaves

a zero-free holomorphic factor in that disc. The harmonic mean-value property for the logarithm of its modulus, and the circle mean of $\log |re^{i\theta} - a| = \log r$ for $|a| < r$, give

$$\sum_{|a| < r} \text{ord}_a(H_0) \log \frac{r}{|a|} = \frac{1}{2\pi} \int_0^{2\pi} \log |H_0(re^{i\theta})| d\theta - \log H_0(0).$$

The circle identity follows by expanding $\log(1 - ae^{-i\theta}/r)$ and integrating its real part. Every zero with $|a| \leq R$ contributes at least $\log 2$; (BPZ4) bounds the right side by a constant plus $9\alpha R^2$. Using the actual affine $z = -2i(s - 1/2)$ proves

$$\sum_{|\Im \rho| \leq R} m_\rho \leq C_0(1 + R)^2 \quad (R \geq 0) \quad (\text{BPZ5})$$

for some finite constant C_0 . Only finiteness of such a constant is used below, not an unevaluated sign or arithmetic estimate.

Repeated integration by parts in (BPZ1), with zero endpoint terms because f is smooth and compactly supported, gives for every integer $N \geq 0$

$$\sup_{0 \leq \sigma \leq 1} |M_f(\sigma + i\gamma)| \leq C_{f,N}(1 + |\gamma|)^{-N}. \quad (\text{BPZ6})$$

For $|\gamma| \geq 1$, integrate $e^{-i\gamma v}$ by parts N times; the integrals of the derivatives of $f(v)e^{-(\sigma-1/2)v}$ are uniformly bounded on the compact support and $0 \leq \sigma \leq 1$. For $|\gamma| \leq 1$ use the original integral. Equations (BPZ5)–(BPZ6) prove absolute convergence of (BPZ2), by summing over dyadic height intervals. They also prove every normal convergence assertion below. Replacing ρ by $1 - \bar{\rho}$ proves $B(g, f) = \overline{B(f, g)}$, so $Q(f)$ is real.

48.2 The exact compactly supported boundary filter

Define

$$T = \frac{d^2}{dv^2} - \frac{1}{4}, \quad E(f) = (M_f(0), M_f(1)), \quad \mathcal{T}_0 = \ker E. \quad (\text{BPZ7})$$

Direct integration by parts twice gives, for every $s \in \mathbb{C}$,

$$M_{Tg}(s) = ((s - 1/2)^2 - 1/4)M_g(s) = s(s - 1)M_g(s). \quad (\text{BPZ8})$$

In particular $T\mathcal{T} \subset \mathcal{T}_0$. The converse has a compact inverse with all constants fixed:

$$G(v) = -e^{-|v|/2}, \quad T^{-1}f(v) = (G * f)(v) = - \int_{\mathbb{R}} e^{-|v-u|/2} f(u) du \quad (f \in \mathcal{T}_0). \quad (\text{BPZ9})$$

Indeed $G'' - G/4 = \delta_0$ distributionally: away from zero it is zero, G is continuous, and its first derivative has jump one. Convolution with f is smooth, since every derivative may be transferred to the compactly supported smooth factor, and $T(G * f) = f$. If $\text{supp } f \subset [a, b]$, its value for $v > b$ is $-e^{-v/2}M_f(0) = 0$ and for $v < a$ is $-e^{v/2}M_f(1) = 0$. It is thus compactly supported in the same interval. Finally a compactly supported solution of $Tg = 0$ is a combination of $e^{v/2}$ and $e^{-v/2}$, hence zero. This proves that

$$T : \mathcal{T} \xrightarrow{\sim} \mathcal{T}_0 \quad (\text{BPZ10})$$

is a linear bijection, commuting with real translations and $\#$. For convolution, differentiation under the integral gives the exact identities $T(f * g) = (Tf) * g = f * (Tg)$; it is not asserted that $T(f * g) = (Tf) * (Tg)$.

There is also an explicit right inverse of the endpoint map. Choose $b \in C_c^\infty(\mathbb{R})$ nonnegative and positive on an interval, put $h_0(v) = e^{v/2}$, $h_1(v) = e^{-v/2}$, and form $C_{ij} = \int b h_i h_j$. Its quadratic form is $\int b |c_0 h_0 + c_1 h_1|^2$, which is positive for nonzero (c_0, c_1) : a linear combination of these two independent exponential functions cannot vanish on an interval unless both coefficients vanish. For $w \in \mathbb{C}^2$ set $R(w) = b \sum_j (C^{-1}w)_j h_j$. Then $ER = I$. Consequently the complete exact sequence and a specified projection are

$$0 \longrightarrow \mathcal{T} \xrightarrow{T} \mathcal{T} \xrightarrow{E} \mathbb{C}^2 \longrightarrow 0, \quad \Pi = I - RE, \quad \text{im } \Pi = \mathcal{T}_0. \quad (\text{BPZ11})$$

48.3 Every off-line zero gives a negative primitive test

This section proves the exact equivalence, without assuming a missing positivity estimate:

$$\boxed{Q|_{\mathcal{T}_0} \geq 0 \iff \Re \rho = \frac{1}{2} \text{ for every zero of the actual } \xi.} \quad (\text{BPZ12})$$

The reverse implication follows immediately and absolutely from (BPZ2): when $1 - \bar{\rho} = \rho$, each term is $m_\rho |M_f(\rho)|^2 \geq 0$, for every test including primitive tests. We prove the other direction by constructing a negative test from any actual off-line zero, not by assuming finite interpolation controls an omitted tail.

Choose such a zero ρ_* in the right half of the strip, using reflection if necessary. Write $c_* = \rho_* - 1/2$, so $\Re c_* > 0$. Choose an even nonnegative smooth bump g of integral one supported in $[-\delta, \delta]$, with $\delta|c_*| < \log(3/2)$. For example a positive multiple of $\exp(-1/(1 - (v/\delta)^2))$ on that interval, extended by zero, has these properties; each derivative tends to zero at its endpoints, because exponential decay dominates every power of $(1 - (v/\delta)^2)^{-1}$. Then

$$|M_g(s) - 1| \leq e^{\delta|s-1/2|} - 1 < \frac{1}{2} \quad \text{at } s = \rho_*, 1 - \bar{\rho}_*.$$

Put $f = Tg$. Equations (BPZ8)–(BPZ10) prove $f \in \mathcal{T}_0$, $f \neq 0$, and both transformed values at the selected pair are nonzero. Set

$$W_\rho = \overline{M_f(1 - \bar{\rho})} M_f(\rho), \quad K(a) = B(f_a, f) = \sum_{\rho} m_\rho W_\rho e^{(\rho-1/2)a} \quad (a \in \mathbb{R}). \quad (\text{BPZ13})$$

Here $W_{\rho_*} \neq 0$. The formula follows from $M_{f_a}(s) = e^{-(s-1/2)a} M_f(s)$, with the conjugation in the first slot retained. The same formula proves exactly

$$Q(f_a) = Q(f), \quad E(f_a) = 0. \quad (\text{BPZ14})$$

For the first identity the two exponential factors cancel at each zero; for the second, each of the two zero endpoint moments is multiplied by an exponential.

The function K is unbounded on $[0, \infty)$. Here is a proof which keeps the full infinite multiset. Absolute summability of $m_\rho |W_\rho|$ and $\Re \rho \leq 1$ permit integrating its series against e^{-wa} when $\Re w > 1/2$, giving

$$\int_0^\infty e^{-wa} K(a) da = \sum_{\rho} \frac{m_\rho W_\rho}{w - (\rho - 1/2)}. \quad (\text{BPZ15})$$

The series on the right is a meromorphic function of w on the whole complex plane. Indeed on any compact set avoiding its poles, the finitely many bounded-height denominators stay uniformly away from zero, while all other denominators have modulus at least $|\Im \rho|/2$. The previously proved rapid summability gives normal convergence. At $w = c_*$ its residue is $m_{\rho_*} W_{\rho_*} \neq 0$; distinct zeros have distinct poles, and the actual multiplicity is already in this residue.

If K were bounded on the positive half-line, its Laplace integral would instead be holomorphic on $\Re w > 0$, by locally dominated differentiation under the integral. The half-plane with the discrete pole set removed is connected: any segment joining two points meets only finitely many poles in a compact neighborhood and can be detoured by small arcs. The identity theorem therefore extends (BPZ15) from $\Re w > 1/2$ to that punctured half-plane. A holomorphic left side cannot equal the right side near its nonzero-residue pole at c_* . This contradiction proves the stated unboundedness.

If $Q(f) < 0$, the required primitive test is already f . Otherwise choose $a \geq 0$ with $|K(a)| > Q(f)$, which exists by that unboundedness, and define the actual compact test

$$h = f_a - \frac{\overline{K(a)}}{|K(a)|} f. \quad (\text{BPZ16})$$

Hermitian symmetry, conjugate linearity in the first slot and (BPZ14) give

$$E(h) = 0, \quad Q(h) = 2Q(f) - 2|K(a)| < 0. \quad (\text{BPZ17})$$

All sums in this equality converge absolutely. This proves (BPZ12) for every zero and every multiplicity and also supplies the full negative-test construction. No finite zero cutoff has been substituted.

48.4 The actual boundary correction on this entire detecting space

Retain the supplied endpoint coordinates $(A, B) = (M_f(0), M_f(1))$. The two endpoint forms and the exact holonomy correction are

$$\begin{aligned} P_e(f, f) &= 2\Re(\overline{AB}), & \overline{P}_e(f, f) &= \tfrac{1}{2}(|A|^2 + |B|^2) + \tfrac{1}{3}\Re(\overline{AB}), \\ K_e(f, f) &= \tfrac{1}{2}(|A|^2 + |B|^2) - \tfrac{5}{3}\Re(\overline{AB}). \end{aligned} \quad (\text{BPZ18})$$

For every test in $\mathcal{T}_0 = T\mathcal{T}$, each of these three expressions vanishes. Thus the supplied complete explicit formula restricts, with every term retained, to

$$Q(f) = \mathcal{A}_\infty(f, f) - \mathcal{P}_{\text{fin}}(f, f) \quad (f \in \mathcal{T}_0), \quad (\text{BPZ19})$$

before and after the holonomy endpoint average. For a test written in its exact inverse coordinates $f = Tg$, this is also an explicit prime-power calculation. Put $k = g^\# * g$ and $\widehat{g}(\gamma) = \int g(v)e^{-i\gamma v}dv$. Differentiation and the convolution identities after (BPZ10) give $(Tg)^\# * (Tg) = T^2k$ and $\widehat{Tg}(\gamma) = -(\gamma^2 + 1/4)\widehat{g}(\gamma)$. The complete formula is therefore

$$\begin{aligned} Q(Tg) &= \frac{1}{2\pi} \int_{\mathbb{R}} (\gamma^2 + 1/4)^2 |\widehat{g}(\gamma)|^2 [\Re\psi(1/4 + i\gamma/2) - \log \pi] d\gamma \\ &\quad - \sum_{p, \ell \geq 1} (\log p) p^{-\ell/2} \left[k^{(4)} - \tfrac{1}{2}k'' + \tfrac{1}{16}k \right] (\ell \log p) \\ &\quad - \sum_{p, \ell \geq 1} (\log p) p^{-\ell/2} \left[k^{(4)} - \tfrac{1}{2}k'' + \tfrac{1}{16}k \right] (-\ell \log p). \end{aligned} \quad (\text{BPZ23})$$

All prime sums are finite because k and its derivatives have compact support; the integral is the original convergent archimedean form on Tg . This derives every multiplier and sign in the remaining arithmetic expression, rather than replacing it by endpoint positivity. The archimedean term and prime sum are the original ones in the supplied `eq:Weil`; no inequality for their difference has been introduced. In particular the exact primitive witness (BPZ16)–(BPZ17) to any off-line zero is unchanged by that endpoint operation.

This is a complete description of what that boundary correction can and cannot detect. The codimension-two filter removes all its endpoint contribution but preserves an all-zero-complete test space. The remaining object is the actual quadratic form (BPZ19) on the precisely constructed range (BPZ10), with its negative witnesses and translation coefficients specified above. Positivity of the endpoint average is not a calculation of this remaining quadratic form.

48.5 The same construction under the complete tau lattice map

For any complex vector space V define its synchronized carrier

$$G_L(V) = \{(0, \lambda) : \lambda \in L\} \cup (V \times \{1_L\}), \quad (v, \lambda) + (w, \mu) = (v + w, \lambda \vee \mu).$$

It is a $G_L(\mathbb{C})$ -semimodule under $(c, \alpha)(v, \lambda) = (cv, \alpha \wedge \lambda)$. Every linear map $A : V \rightarrow W$ lifts to $A^\uparrow(v, \lambda) = (Av, \lambda)$; direct substitution proves additivity, scalar compatibility and preservation of every zero label. Its full comparison is

$$\Phi_V(v, \lambda) = (v, 1_K \otimes \lambda) \in V \times K_L, \quad K_L = K \otimes_{\mathbb{B}} L, \quad (A \times I)\Phi_V = \Phi_W A^\uparrow. \quad (\text{BPZ20})$$

The target uses its componentwise $\mathbb{C} \times K_L$ action. The zero/nonzero map $K \rightarrow \mathbb{B}$ is a semiring homomorphism; tensoring it with L gives a retraction of $\lambda \mapsto 1_K \otimes \lambda$. This proves injectivity and retention of the entire bounded distributive lattice, without replacing it by its Boolean characters.

Apply these exact lifts to T, T^{-1}, E, R, Π and every translation. Equation (BPZ11) becomes an explicit split family of linear maps with

$$(E^\dagger)^{-1}(G_L(0)) = G_L(\mathcal{T}_0), \quad (E^\dagger)^{-1}(e) = \mathcal{T}_0 \times \{1_L\}, \quad (E^\dagger)^{-1}(\tau) = \{\tau\}. \quad (\text{BPZ21})$$

The top-supported primitive tests map to supported zero in the endpoint space, not to external absence. The inverse and projection maps remain valid there with the same labels.

The spectral pairing itself has the exact lifted value

$$B^L((f, \lambda), (g, \mu)) = (B(f, g), \lambda \wedge \mu), \quad \Phi_{\mathbb{C}} B^L = (B(f, g), \iota_L(\lambda \wedge \mu)). \quad (\text{BPZ22})$$

If its amplitude is nonzero then both inputs are nonzero, hence both labels are top, so the stated element belongs to $G_L(\mathbb{C})$. Sesquilinearity follows from that of B , distributivity of meet over join, and the lifted conjugation $(c, \alpha)^* = (\bar{c}, \alpha)$. The same rule applies to every endpoint form and the retained terms in (BPZ19). On a nonzero primitive test their zero amplitudes have label top, namely e ; the negative amplitude in (BPZ17) has that same label and is still negative after the injective $\Phi_{\mathbb{C}}$ map. Thus the complete boundary filter, the all-zero receiver and the negative-test construction commute with the original full map. Localizing at supported zero would erase these amplitudes; that localization is not performed anywhere in this argument.

49 A finite rational Laurent witness for every off-real heat zero

Provenance and scope. The supplied `non_eulerian_modular_heat_attempt.md`, Sections 3–5 and equations (7.1)–(7.3), supplies the particular full-root Laurent receiver, its original residue constants, and its heat production formula. Its accompanying `rational_witness.json` supplies a numerical candidate; none of its numerical sign claims is an input to the proof below. The supplied report has no verified human authorship attribution, and none is inferred. The analytic family and its coordinate convention are those of Brad Rodgers and Terence Tao, *The de Bruijn–Newman constant is non-negative*, equations defining Φ , H_t , and $H_0 = \xi/8$. The classical factorization theorem used below is Hadamard’s theorem for entire functions of finite order. The calculation supplied here is the explicit finite interpolation and controlled infinite-tail construction (ALM12)–(ALM24), its rational coefficient version, and its exact passage through the full support lattice. It proves completeness of this particular receiver; it does not assert positivity of all its matrices at physical time zero.

49.1 The original heat family and its entire zero multiset

Retain the original real time t , coordinate Z , kernel, and mass:

$$\begin{aligned} H_t(Z) &= \int_0^\infty e^{tu^2} \Phi(u) \cos(Zu) du, \\ \Phi(u) &= \sum_{n \geq 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, \\ M_j(t) &= \int_0^\infty u^{2j} e^{tu^2} \Phi(u) du, \quad H_t(0) = M_0(t) > 0. \end{aligned} \tag{ALM1}$$

Every summand of Φ is positive on $[0, \infty)$, since $2\pi n^2 e^{4u} - 3 > 0$. Moreover

$$0 < \Phi(u) \leq C e^{9u - (\pi/2)e^{4u}}, \quad C = 2\pi^2 \sum_{n \geq 1} n^4 e^{-(\pi/2)n^2} < \infty. \tag{ALM2}$$

Indeed, discard the negative summand and use $n^2 e^{4u} \geq (n^2 + e^{4u})/2$. This bound justifies all derivatives of (ALM1) locally uniformly in $(t, Z) \in \mathbb{C}^2$. The actual H_t is entire, even in Z , real on the real axis for real t , and obeys $\partial_t H_t = -\partial_Z^2 H_t$.

For completeness, its order is at most one. For fixed real t there is a finite C_t such that $|t|u^2 + 9u \leq (\pi/8)e^{4u} + C_t$ for all $u \geq 0$. Use $|\cos Zu| \leq e^{|Z|u}$ and split the remaining exponential to get

$$|H_t(Z)| \leq C e^{C_t} \left(\int_0^\infty e^{-(\pi/4)e^{4u}} du \right) \exp \left(\sup_{u \geq 0} \{ |Z|u - (\pi/8)e^{4u} \} \right).$$

The supremum is bounded by $C'(1 + |Z| \log(2 + |Z|))$, by solving its one-variable derivative equation, or by taking its endpoint value if that critical point lies outside $[0, \infty)$. Thus $\log \max_{|Z| \leq R} |H_t(Z)| = O_t(1 + R \log(2 + R))$. Jensen’s formula on radius $2R$, with $H_t(0) > 0$, gives the zero count $n_t(R) = O_t(R \log(2 + R))$: each zero in the radius- R disk contributes at least $\log 2$ to the radius- $2R$ Jensen sum. Dyadic annuli then give

$$V_t := \sum_{\zeta: H_t(\zeta)=0} \frac{m_\zeta}{|\zeta|^2} < \infty. \tag{ALM3}$$

The finitely many roots near zero cause no problem, because zero is not a root. All sums over ζ below run over distinct roots with their full multiplicities m_ζ .

Hadamard factorization at order at most one, with the factors paired under $\zeta \mapsto -\zeta$, now gives

$$H_t(Z) = H_t(0) \prod_{\{\zeta, -\zeta\}} (1 - Z^2/\zeta^2)^{m_\zeta}, \quad \frac{H'_t(Z)}{H_t(Z)} = - \sum_{n \geq 1} S_n(t) Z^{2n-1}, \quad S_n(t) = \sum_{\zeta} \frac{m_\zeta}{\zeta^{2n}}. \tag{ALM4}$$

The product ranges over one representative of each sign pair and converges locally uniformly by (ALM3). In the Hadamard expression the remaining exponential has degree at most one. The evenness of H_t and of the paired product forces its linear coefficient to vanish: $e^{aZ} = e^{-aZ}$ for all Z implies $a = 0$ by differentiation at zero. Evaluation at zero fixes the remaining constant as $H_t(0)$. Logarithmic differentiation and expansion on a disk containing no root prove the second equality. In particular the S_n include *both* members of every sign pair.

There are infinitely many distinct roots. Otherwise (ALM4) would be a polynomial. But its Taylor coefficient of Z^{2j} is $(-1)^j M_j(t)/(2j)!$, which is nonzero for every j by (ALM1). Also $H_t(iy) > 0$ for every real y , since its integrand becomes the strictly positive function $e^{tu^2} \Phi(u) \cosh(yu)$. Thus the actual theta family has no imaginary-axis roots. We shall nevertheless give the negative-real reciprocal-node case explicitly, so that the detection argument does not hide that possibility in its algebra.

49.2 The full local algebras and the exact trace receiver

Let $\mathcal{Z}$ be any finite conjugation-stable set of roots. Retain

$$A_{t,\mathcal{Z}} = \bigoplus_{\zeta \in \mathcal{Z}} \mathcal{O}_{\mathbb{C},\zeta}/(H_t), \quad E_{d,t,\mathcal{Z}}(c) = [Z^{-1}p_c(Z^{-2})]_{\mathcal{Z}}, \quad p_c(X) = \sum_{i=0}^{d-1} c_i X^i. \quad (\text{ALM5})$$

Every inverse power in this expression is the inverse of the holomorphic unit Z at a nonzero root. No inverse of supported zero or of a root collision gap occurs. The local algebra retains its entire length m_ζ . The compatible maps for finite $\mathcal{Z}$ also define a map to the product of all these local algebras. The precise domain of the infinite pairing is the complex vector space

$$\mathcal{R} = \{Z^{-1}p(Z^{-2}) : p \in \mathbb{C}[X]\}, \quad \mathcal{E}_d : \mathbb{C}^d \longrightarrow \mathcal{R}, \quad c \longmapsto Z^{-1}p_c(Z^{-2}).$$

Let

$$j_t : \mathcal{R} \longrightarrow \prod_{\zeta : H_t(\zeta)=0} \mathcal{O}_{\mathbb{C},\zeta}/(H_t), \quad F \longmapsto ([F]_\zeta)_\zeta, \quad E_{d,t} = j_t \mathcal{E}_d.$$

The map j_t is injective: a nonzero Laurent polynomial has only finitely many nonzero zeros, whereas H_t has infinitely many distinct roots. Thus $j_t(\mathcal{R})$ is a second exact realization of this test space. The pairing below has domain $\mathcal{T}_t : \mathcal{R} \times \mathcal{R} \rightarrow \mathbb{C}$, or equivalently $j_t(\mathcal{R}) \times j_t(\mathcal{R})$ via this injection. It is not defined here on an unrestricted product of local algebras. In particular, the notation using $E_{d,t}$ below means this specified image and its transported pairing.

The original s -coordinate and reflected involution are

$$\alpha(Z) = \frac{1}{2} + \frac{iZ}{2}, \quad g_t(s) = 16H_t(-2i(s - \frac{1}{2})), \quad f^\dagger(s) = \overline{f(1 - \bar{s})}. \quad (\text{ALM6})$$

Writing $F = \alpha^* f$, one has $\alpha^*(f^\dagger)(Z) = \overline{F(\bar{Z})} =: F^\dagger(Z)$ because $\alpha(\bar{Z}) = 1 - \overline{\alpha(Z)}$. Direct substitution gives

$$\frac{ds}{g_t(s)} = \frac{i}{32} \frac{dZ}{H_t(Z)}, \quad \alpha^* g'_t = -32i H'_t, \quad (i/32)(-32i) = 1. \quad (\text{ALM7})$$

Consequently the complete Jacobian residue contraction is exactly

$$\mathcal{T}_t(F, G) = \sum_{\zeta} \text{Res}_{Z=\zeta} F^\dagger(Z) G(Z) \frac{H'_t(Z)}{H_t(Z)} dZ = \sum_{\zeta} m_\zeta \overline{F(\bar{\zeta})} G(\zeta). \quad (\text{ALM8})$$

Indeed, in a full local algebra write $H_t = u(Z)(Z - \zeta)^m$ with $u(\zeta) \neq 0$. Then $H'_t/H_t = m/(Z - \zeta) + u'/u$; the second term has zero residue. The resulting contraction has a nilpotent radical, but neither the local algebra nor its evaluation map has been replaced by a reduced

algebra. For the functions in (ALM5), the summands in (ALM8) are $O(m_\zeta/|\zeta|^2)$ at infinity. They are therefore absolutely summable by (ALM3).

For $c, b \in \mathbb{C}^d$, direct multiplication gives

$$\mathcal{T}_t(E_{d,t}c, E_{d,t}b) = c^* K_d(t) b, \quad K_d(t) = (S_{i+j+1}(t))_{0 \leq i, j < d}. \quad (\text{ALM9})$$

Conjugate roots have equal multiplicity, so every S_n is real and K_d is real symmetric. Notice that at a nonreal root the summand is $\overline{F(\zeta)}F(\zeta)$, not $|F(\zeta)|^2$.

49.3 Aggregation without deleting either root of a sign pair

Fix t for the following construction. Define the exact reciprocal nodes and their aggregated multiplicities by

$$\Lambda = \{\zeta^{-2} : H_t(\zeta) = 0\}, \quad M_\lambda = \sum_{\zeta^{-2}=\lambda} m_\zeta, \quad R = \max_{\lambda \in \Lambda} |\lambda| > 0, \quad V = \sum_{\lambda \in \Lambda} M_\lambda |\lambda| = V_t. \quad (\text{ALM10})$$

Each nonzero fiber consists of the two distinct roots $\zeta, -\zeta$; thus $M_\lambda = 2m_\zeta$. This is an exact aggregation of the original sum, with no halving of multiplicity. The set Λ has no accumulation except zero, and each subset bounded away from zero is finite. The maximum defining R exists because the nonempty discrete root set has a positive minimum modulus. Conjugation preserves Λ and $M_{\bar{\lambda}} = M_\lambda$.

If p has real coefficients, (ALM9) becomes

$$Q_t(p) := c^T K_d(t) c = \sum_{\lambda \in \Lambda} M_\lambda \lambda p(\lambda)^2. \quad (\text{ALM11})$$

This converges absolutely and is real. For a complex polynomial the correct expression is instead $\sum_{\lambda} M_\lambda \lambda p(\bar{\lambda}) p(\lambda)$. The real polynomial subspace already detects every failure below.

49.4 Explicit finite witnesses and a bound for the entire omitted tail

Take a node $\lambda_0 \in \Lambda$ which is not positive real. Choose any real r with $0 < r < |\lambda_0|$, and retain the complete finite set

$$S = \{\lambda \in \Lambda : |\lambda| \geq r\}, \quad L_0(X) = \prod_{\mu \in S \setminus \{\lambda_0\}} \frac{X - \mu}{\lambda_0 - \mu}, \quad B = \prod_{\mu \in S \setminus \{\lambda_0\}} \frac{r + |\mu|}{|\lambda_0 - \mu|}. \quad (\text{ALM12})$$

All denominators are nonzero, $L_0(\lambda_0) = 1$, and L_0 vanishes at the other nodes in S . On $|X| \leq r$ one has $|L_0(X)| \leq B$ by the triangle inequality applied factor by factor. Put $q = r/|\lambda_0| < 1$.

A negative real reciprocal node. If $\lambda_0 < 0$, the finite set $S \setminus \{\lambda_0\}$ is conjugation-stable. Hence L_0 has real coefficients, as can also be seen by pairing its conjugate factors. For every integer $N \geq 0$ set

$$p_N(X) = (X/\lambda_0)^N L_0(X), \quad d_N = N + |S|. \quad (\text{ALM13})$$

Its degree is at most $d_N - 1$, it is real, and its value at the target node is one. It vanishes at every other node of S . The omitted tail has the explicit bound

$$|Q_t(p_N) - M_{\lambda_0} \lambda_0| \leq B^2 q^{2N} \sum_{\lambda \notin S} M_\lambda |\lambda| \leq B^2 V q^{2N}. \quad (\text{ALM14})$$

Choose any N with $B^2 V q^{2N} < M_{\lambda_0} |\lambda_0|/2$. Then the complete form, including every root, satisfies

$$Q_t(p_N) < -M_{\lambda_0} |\lambda_0|/2 < 0. \quad (\text{ALM15})$$

This case corresponds exactly to the nonzero imaginary roots $\zeta^2 = 1/\lambda_0 < 0$. The positivity argument after (ALM4) already rules them out for the actual family, but the receiver detects them without using that argument.

A nonreal reciprocal node. If $\lambda_0 \notin \mathbb{R}$, choose a unit complex number a with

$$a^2 = -\bar{\lambda}_0/|\lambda_0|. \quad (\text{ALM16})$$

Either square root works. Define $L_{\bar{0}}(X) = \overline{L_0(\bar{X})}$. It is precisely the Lagrange polynomial for the node $\bar{\lambda}_0$ on the same finite set S . The polynomial

$$p_N(X) = a(X/\lambda_0)^N L_0(X) + \bar{a}(X/\bar{\lambda}_0)^N L_{\bar{0}}(X), \quad d_N = N + |S| \quad (\text{ALM17})$$

has real coefficients, because its two summands are coefficientwise conjugates. Its values at the two target nodes are $a, \bar{a}$, and it vanishes on $S \setminus \{\lambda_0, \bar{\lambda}_0\}$. The two target contributions to (ALM11) add to

$$M_{\lambda_0}(\lambda_0 a^2 + \bar{\lambda}_0 \bar{a}^2) = -2M_{\lambda_0}|\lambda_0|.$$

For $|X| < r$ both Lagrange polynomials are bounded by B , so

$$|Q_t(p_N) + 2M_{\lambda_0}|\lambda_0|| \leq 4B^2 V q^{2N}. \quad (\text{ALM18})$$

Choose N with $4B^2 V q^{2N} < M_{\lambda_0}|\lambda_0|$. Then

$$Q_t(p_N) < -M_{\lambda_0}|\lambda_0| < 0. \quad (\text{ALM19})$$

Such an integer is specified finitely, for example by

$$N = \max \left\{ 0, 1 + \left\lceil \frac{\log(4B^2 V / (M_{\lambda_0}|\lambda_0|))}{2 \log(|\lambda_0|/r)} \right\rceil \right\}.$$

For the negative-real case use the same formula with $4B^2$ replaced by $2B^2$. The additional 1 makes the required inequality strict even if the logarithmic ratio is an integer. These bounds involve the actual root multiset and its convergent total V ; no finite root list has been declared complete without a tail estimate.

An off-real root ζ always produces one of these two cases: ζ^{-2} can be positive real only when ζ itself is real. If its reciprocal square is negative real, ζ is imaginary; otherwise the reciprocal square is nonreal. This proves that the construction reaches every off-real root and every multiplicity.

49.5 Exact rational coefficients and the all-zero equivalence

The coefficients constructed above need not be rational. There is an explicit perturbation bound producing a rational witness of the same degree. For a real polynomial define

$$\|p\|_R = \sum_i |c_i| R^i. \quad (\text{ALM20})$$

This is a coefficient bound, not a change to the original trace form. Since $|\lambda| \leq R$, if p, q are real polynomials and $\delta = \|p - q\|_R$, then (ALM11) gives

$$|Q_t(q) - Q_t(p)| \leq V\delta(2\|p\|_R + \delta). \quad (\text{ALM21})$$

Indeed, $q^2 - p^2 = (q - p)(q + p)$ pointwise on the reciprocal nodes, $|p(\lambda)| \leq \|p\|_R$, and $|q(\lambda)| \leq \|p\|_R + \delta$.

For either negative witness above take a positive proved bound η with $Q_t(p) \leq -\eta$ (the right sides of (ALM15) or (ALM19) supply one). Choose

$$0 < \delta_0 \leq \min\{1, \eta/[4V(2\|p\|_R + 1)]\}. \quad (\text{ALM22})$$

For $d = 1 + \deg p$, choose rational numbers b_i satisfying $|b_i - c_i| < \delta_0/(dR^i)$, and put $q(X) = \sum_{i < d} b_i X^i$. The density of the rationals supplies every one of these finite choices; for example a sufficiently fine integer grid in each coefficient does so. Then $\|q - p\|_R < \delta_0$ and

$$b^T K_d(t) b = Q_t(q) < -3\eta/4 < 0, \quad b \in \mathbb{Q}^d. \quad (\text{ALM23})$$

Multiplying by a common positive denominator even gives an integer coefficient witness; its trace value is multiplied by the square of that denominator, so its sign is preserved exactly.

We have therefore proved the complete equivalence for the actual family, at every fixed real time t :

$$\begin{aligned} & \text{every zero of } H_t \text{ is real} \\ & \iff K_d(t) \succeq 0 \text{ for every } d \geq 1 \\ & \iff K_d(t) \succ 0 \text{ for every } d \geq 1 \\ & \iff b^T K_d(t) b \geq 0 \text{ for every } d \geq 1, \ b \in \mathbb{Q}^d. \end{aligned} \quad (\text{ALM24})$$

The forward implication follows directly from (ALM8): all summands then equal $m_\zeta |F(\zeta)|^2$. Strict positive definiteness holds because there are infinitely many distinct reciprocal nodes and a nonzero polynomial has only finitely many zeros. The reverse implication and the rational version are (ALM12)–(ALM23). Complex coefficient vectors add no new positivity condition because K_d is real symmetric: for real a, b , $(a + ib)^* K_d(a + ib) = a^T K_d a + b^T K_d b$. At physical time zero the original coordinate $s = 1/2 + iZ/2$ converts (ALM24) to the assertion that the zeros of the original ξ lie on its critical line. A single fixed size, such as $d = 32$, does not establish the quantifier over every d in (ALM24).

49.6 The actual coefficient and heat production maps

No root positions need be inserted into the finite matrices to calculate their entries. Let $a_j(t) = (-1)^j M_j(t)/(2j)!$, retaining $a_0 = M_0 = H_t(0)$. Multiplying (ALM4) by H_t and taking the coefficient of Z^{2n-1} gives

$$M_0 S_n = -2n a_n - \sum_{j=1}^{n-1} a_j S_{n-j} \quad (n \geq 1). \quad (\text{ALM25})$$

Only the nonzero original mass $M_0(t)$ is a denominator when this recurrence is solved. The entire function has not been divided by its mass or assigned a different value at zero. Since $M_0(t) > 0$ for all real t , (ALM25) also proves that all $S_n(t)$ are real analytic in real time, including times at which other zeros collide.

Put $P = H'_t/H_t$ in a neighborhood of $Z = 0$. The actual heat equation gives $\partial_t P = -P'' - 2PP'$. Substituting $P = -\sum_{n \geq 1} S_n Z^{2n-1}$ proves

$$S'_n = 2n \left[\sum_{a=1}^n S_a S_{n+1-a} - (2n+1) S_{n+1} \right]. \quad (\text{ALM26})$$

To check the convolution factor, the coefficient of Z^{2n-1} in PP' is $\sum_{a+b=n+1} (2b-1) S_a S_b$; exchanging a and b shows this equals $n \sum_{a=1}^n S_a S_{n+1-a}$. The coefficient in $-P''$ is $2n(2n+1) S_{n+1}$. Comparing with $-S'_n$ gives exactly (ALM26). Thus the supplied production formula is correct with its original time, signs, and factors. In particular, for a constant coefficient vector c the exact production is

$$\frac{d}{dt}(c^* K_d c) = \sum_{i,j < d} \bar{c}_i c_j 2(i+j+1) \left[\sum_{a=1}^{i+j+1} S_a S_{i+j+2-a} - (2i+2j+3) S_{i+j+2} \right]. \quad (\text{ALM27})$$

If $c = c(t)$ moves, the additional two terms are $c'^* K_d c + c^* K_d c'$; omitting them would change the problem. The first production value has the strictly positive exact expression

$$S'_1 = \frac{M_2 M_0 - M_1^2}{M_0^2} > 0. \quad (\text{ALM28})$$

Here $M'_0 = M_1$, $M'_1 = M_2$, and the numerator is strictly positive by Cauchy–Schwarz on the original measure $e^{tu^2} \Phi(u) du$: the function u^2 is not constant on its support. This proves directly that the full-root trace evolves under physical heat; it is not an invariant obtained by dropping the production term.

49.7 Full arbitrary-lattice realization of every map

Let L be the original arbitrary nontrivial bounded distributive lattice. For every complex vector space V retain

$$G_L(V) = \{(0, \ell) : \ell \in L\} \cup (V \times \{1_L\}), \quad (v, \ell) + (w, k) = (v + w, \ell \vee k), \quad (c, h)(v, \ell) = (cv, h \wedge \ell). \quad (\text{ALM29})$$

The scalar semiring is $G_L(\mathbb{C})$. The carrier is closed under these operations because a nonzero amplitude in a sum requires at least one top input label, and a nonzero product requires both labels to be top. The semimodule identities follow from the vector-space identities and distributivity of $\wedge$ over $\vee$.

For a complex-linear map $A : V \rightarrow W$ define $A^\uparrow(v, \ell) = (Av, \ell)$. Addition, scalar multiplication, and every zero label are preserved by direct substitution. Apply this to coefficient inclusions, polynomial evaluation, the entire local-jet maps (ALM5), finite packet restriction, and each finite-packet linear residue trace. For the infinite pairing use exactly $\mathcal{R}$ or its injective image $j_t(\mathcal{R})$ as specified after (ALM5). The quadratic trace value is the sesquilinear lift

$$\mathcal{T}_t^L((F, \ell), (G, k)) = (\mathcal{T}_t(F, G), \ell \wedge k). \quad (\text{ALM30})$$

It is a well-defined element of $G_L(\mathbb{C})$: a nonzero trace requires two nonzero amplitudes and hence two top labels. Additivity in either argument follows from the scalar trace and lattice distributivity; scalar conjugation is $(c, h)^* = (\bar{c}, h)$. Therefore

$$\mathcal{T}_t^L(E_{d,t}^\uparrow(c, \ell), E_{d,t}^\uparrow(b, k)) = (c^* K_d(t)b, \ell \wedge k)$$

is the full supported version of (ALM9), not an amplitude-only replacement for it.

Retain the programme's idempotent semifield K , its Boolean base $\mathbb{B}$, and $K_L = K \otimes_{\mathbb{B}} L$. With $\iota_L(\ell) = 1_K \otimes \ell$, the complete comparison is

$$\Phi_V(v, \ell) = (v, \iota_L(\ell)) \in V \times K_L, \quad (A \times I)\Phi_V = \Phi_W A^\uparrow. \quad (\text{ALM31})$$

For clarity, the label injection has a retraction. The map $\chi : K \rightarrow \mathbb{B}$, $\chi(0) = 0$ and $\chi(x) = 1$ for $x \neq 0$, is a semiring homomorphism: idempotent addition cannot sum a nonzero element to zero, and the semifield has no zero divisors. Hence $\rho_L = \chi \otimes I_L$ satisfies $\rho_L \iota_L = I_L$. This proves that Φ_V is injective on the whole carrier and retains each original lattice label. Neither a Boolean-character replacement of L nor a localization at e occurs.

The exact residue factors $i/32$ and $-32i$ in (ALM7) are nonzero scalar maps, so their lifts commute with (ALM31); their product has amplitude one and top label. The full trace pairing also commutes with this comparison, with its two input labels multiplied by meet. Every negative rational witness in (ALM23) therefore has exactly

$$(b^T K_d(t)b, 1_L) \xrightarrow{\Phi_{\mathbb{R}}} (b^T K_d(t)b, 1_K \otimes 1_L), \quad b^T K_d(t)b < 0. \quad (\text{ALM32})$$

An amplitude-zero value at the same top support is $e = (0, 1_L)$, whereas external absence is $\tau = (0, 0_L)$; (ALM31) does not identify them. Thus (ALM24) is a complete detection result inside the full lattice receiver: every off-real root has a finite rational test with a negative retained amplitude, and that exact test survives the injective Φ comparison.

50 Independent reproduction of the actual full-root witness

The incoming session supplied `non_eulerian_modular_heat_attempt.md` and `rational_witness.json`. The session, not this reconstruction, selected the nineteen rational coefficients. Its report Sections 3–8 give the full-root trace, moment recurrence, conditioning map, transport and analytic quadrature bounds. No individual human author is named in these supplied files. The original arithmetic heat kernel and its scale are due to the cited treatment by Brad Rodgers and Terence Tao, original author TeX, equations `phidef`, `htdef` and `hoz`. The negative-time phenomenon is not claimed as a new discovery.

The supplied report refers to a `reproduce.py` which was not among the received files. We therefore wrote and executed a separate implementation, `modular_heat_intake/reproduce_with_arb.py`, using Fredrik Johansson’s Arb interval arithmetic through python-flint. It evaluates the original theta integral without a zeta evaluator or a zero table. The validated Legendre root and quadrature-weight operation is documented in the FLINT reference, `arb_hypgeom_legendre_p_ui_root`. The proof below gives the remaining analytic bounds rather than treating agreement of decimal values as certification.

50.1 The unchanged test and quadratic form

Retain the physical heat parameter t , original coordinate Z , and

$$H_t(Z) = \int_0^\infty e^{tu^2} \Phi(u) \cos(Zu) du, \quad \Phi(u) = \sum_{j \geq 1} (2\pi^2 j^4 e^{9u} - 3\pi j^2 e^{5u}) e^{-\pi j^2 e^{4u}}. \quad (\text{RWR1})$$

Let $M_n(t) = \int_0^\infty u^{2n} e^{tu^2} \Phi(u) du$ and $a_n(t) = (-1)^n M_n(t)/(2n)!$. Positivity of each theta summand proves $M_0(t) > 0$. In the Taylor disk at zero put

$$\frac{H'_t(Z)}{H_t(Z)} = - \sum_{n \geq 1} S_n(t) Z^{2n-1}, \quad S_n = \frac{-2na_n - \sum_{j=1}^{n-1} a_j S_{n-j}}{M_0}. \quad (\text{RWR2})$$

The coefficient identity follows by multiplying the two convergent Taylor series. The full-root product in the accompanying proof identifies $S_n = \sum_{\zeta} m_{\zeta} \zeta^{-2n}$, including both sign partners and every multiplicity. No finite zero sample is used to compute S_n .

Write $D_{ii} = 1000^i$, $i = 0, \dots, 31$, and retain both matrices

$$K_d(t) = (S_{i+j+1}(t))_{0 \leq i, j < d}, \quad \widehat{K}_{32} = DK_{32}D. \quad (\text{RWR3})$$

The congruence is the explicit invertible coefficient map $c = Db$. It changes no quadratic value or inertia. The incoming nineteen integers N_i are retained verbatim in the supplied JSON, whose SHA256 is

85314e9fb3115647b7f03a5e00f1b5512c4b23c9280a753f3cf2cababdfed4ac.

They define the one fixed rational test, without fitting it again,

$$c_i = \frac{1000^i N_i}{10^{120}}, \quad f_*(Z) = \sum_{i=0}^{18} c_i Z^{-2i-1}, \quad Q_*(t) = c^T K_{19}(t) c. \quad (\text{RWR4})$$

The integer array is part of this standalone source package. In particular $N_{18} = 10^{120}$ and $c_{18} = 10^{54}$; these large original coefficients have not been divided by a norm or omitted.

50.2 All remainders in the moment calculation

The independent computation uses 1280 binary working bits, 160 Gauss–Legendre nodes in each of 24 panels of $[0, 3]$, and the first 16 theta summands. It computes moments $0 \leq n \leq 65$. For each exact panel center $c = (2j + 1)/16$, $j = 0, \dots, 23$, put

$$h = 1/16, \quad A = 17/128, \quad B = 15/128, \\ x_- = c - A, \quad x_+ = c + A, \quad R^2 = x_+^2 + B^2, \quad \kappa = 1 - (4B)^2/2 > 0.$$

The ellipse of parameter 4 around the panel has semiaxes A, B . On and inside it, $\cos(4\Im z) \geq \kappa$ and $|z|^2 \leq R^2$; the latter follows since $c > 0$ gives $|c - A| \leq c + A$. Using $3 < \pi < 4$ and $j^2 \geq 1$, the absolute value of the truncated theta sum is at most

$$P_c = \left(32e^{9x_+} \sum_{j=1}^{16} j^4 + 12e^{5x_+} \sum_{j=1}^{16} j^2 \right) e^{-3e^{4x_-} \kappa}. \quad (\text{RWR5})$$

Indeed $\Re(\pi j^2 e^{4z}) \geq 3e^{4x_-} \kappa$ and the absolute value of each of the two prefactor terms is bounded separately. The additional factor for real t is at most $T_c(t) = e^{tx_+^2}$ for $t \geq 0$, and $e^{|t|B^2}$ for $t < 0$. The integrand's ellipse supremum is thus at most $P_c T_c R^{2n}$.

For completeness, write an analytic function on a Bernstein ellipse as a Chebyshev series. Pullback by $(w + w^{-1})/2$ and the Cauchy coefficient estimate bound the degree- j coefficient by $2M4^{-j}$. The remainder after degree $2N - 1$ on $[-1, 1]$ is bounded by $2M4^{-(2N-1)}/3$. Gauss–Legendre quadrature is exact on that polynomial; both the integral and the sum of positive quadrature weights have mass $\ell = 1/8$ on the physical panel. Therefore

$$|\text{panel integral} - \text{panel quadrature}| \leq \frac{4\ell}{3} P_c T_c(t) R^{2n} 4^{-(2N-1)}, \quad N = 160. \quad (\text{RWR6})$$

The software's prefactor $1/6$ is exactly $4\ell/3$. The node intervals enclose all distinct roots of P_{160} and the positive weights enclose their exact Gauss weights. The implementation also checks ordering, containment in $(-1, 1)$, positivity and total mass.

Every omitted theta summand is positive on the original real domain. On $[0, 3]$, $\Phi_{>16}(u) \leq 32e^{27} \sum_{j \geq 17} j^4 e^{-3j^2}$. The ratio of successive terms is at most $16e^{-105} < 1/2$. Multiplying the geometric bound by the interval length, $u^{2n} \leq 3^{2n}$, and $e^{tu^2} \leq e^{9 \max(t, 0)}$ proves

$$\int_0^3 u^{2n} e^{tu^2} \Phi_{>16}(u) du \leq 192 \cdot 17^4 \cdot 3^{2n} e^{27+9 \max(t, 0)-867}. \quad (\text{RWR7})$$

For $u \geq 3$, discard the negative part of each summand and apply the same ratio argument, now from $j = 1$, to obtain $\Phi(u) \leq 64e^{9u} e^{-3e^{4u}}$. For $n \leq 65$, $t \leq 1/5$ and $u \geq 3$,

$$2n \log u + \max(t, 0)u^2 + 9u \leq 139u + u^2/5 \leq 47u^2 \leq e^{4u}.$$

The last inequality follows by monotonicity of e^{4u}/u^2 on that interval and $e^{12} > 1000 > 423$. Convexity also gives $e^{4u} \geq 1000(1 + 4(u - 3))$. Hence

$$\int_3^\infty u^{2n} e^{tu^2} \Phi(u) du \leq 64e^{-2000} \int_3^\infty e^{-8000(u-3)} du < e^{-2000}. \quad (\text{RWR8})$$

The moment ball is enlarged symmetrically by the sum of (RWR6)–(RWR8). Symmetric enlargement is conservative even for the positive omitted tails. Thus it encloses the entire original integral.

Interval evaluation of (RWR2) is legitimate because the computed M_0 interval is positive. In $\widehat{K} = L\mathcal{P}L^T$, the ordinary recursive LDL equations give intervals for each successive pivot and multiplier. No pivot enclosing zero is accepted. Induction on the recursive equations proves enclosure of the exact factorization, so the signs of all 32 pivots determine the exact inertia by the invertible triangular change of coordinates. The same moment balls evaluate (RWR4) with its exact rational coefficients.

50.3 Reproduced values and the actual transport boundary

The executed receipts enclose

$$\begin{aligned} -2.384065 \cdot 10^{-46} &< Q_*(-2) < -2.384063 \cdot 10^{-46}, \\ 0.00081442298606666 &< Q_*(0) < 0.00081442298606668. \end{aligned} \quad (\text{RWR9})$$

Both intervals also overlap the much narrower binary intervals in the incoming JSON. Agreement alone is not the proof: (RWR5)–(RWR8), the recurrences and directed interval arithmetic establish these enclosures. The complete matrix inertias, written (positive, negative, zero), are

$$\begin{array}{c|cccc} t & -2 & -3/2 & -5/4 & 0 \\ \hline \text{In } K_{32}(t) & (30, 2, 0) & (31, 1, 0) & (32, 0, 0) & (32, 0, 0). \end{array} \quad (\text{RWR10})$$

These are full-root matrix values, not zeros counted up to a height.

The heat equation yields the exact production law

$$S'_n = 2n \left[\sum_{a=1}^n S_a S_{n+1-a} - (2n+1) S_{n+1} \right]. \quad (\text{RWR11})$$

To verify it directly, put $R = H'/H$. Then $\partial_t \log H = -(R' + R^2)$. Substitute the series in (RWR2), differentiate in Z and compare degree $2n-1$. This derivation uses the positive unit $H_t(0)$ and divides by no root gap. For $n=1$ it gives $S'_1 = (M_2 M_0 - M_1^2)/M_0^2 > 0$, because u^2 is not constant on the strictly positive original theta measure.

Define the actual number

$$t_* = \inf\{t \in [-3/2, -5/4] : K_{32}(t) \succeq 0\}. \quad (\text{RWR12})$$

All entries are real analytic: the integral is jointly analytic on bounded complex-time sets and $M_0(t) > 0$ for real t . Closedness of the positive-semidefinite cone, continuity, and (RWR10) give $-3/2 < t_* < -5/4$, $K_{32}(t_*) \succeq 0$ and $\det K_{32}(t_*) = 0$. The strict inequalities follow from the open neighborhoods of the two strictly certified endpoint signatures. Positive definiteness at t_* would contradict its infimum property. The determinant is not identically zero and is analytic, so its zeros are isolated. There is therefore a nonsingular interval immediately to the left of t_* . On that interval the actual connection

$$\mathcal{A} = -\frac{1}{2} K_{32}^{-1} K'_{32}, \quad U'(t, a) = \mathcal{A}(t) U(t, a), \quad U(a, a) = I \quad (\text{RWR13})$$

satisfies $U(t, a)^T K_{32}(t) U(t, a) = K_{32}(a)$ by direct differentiation. Linear ODE existence gives the invertible fundamental matrix on this interval. No assertion of modular unitarity is made.

Every moment increases with t . Put $m_* = M_0(-3/2)$, $B_n = M_n(-5/4)/(2n)!$ and

$$E_n = \frac{2nB_n + \sum_{j=1}^{n-1} B_j E_{n-j}}{m_*}, \quad F_n = 2n \left[\sum_{a=1}^n E_a E_{n+1-a} + (2n+1) E_{n+1} \right]. \quad (\text{RWR14})$$

Induction in (RWR2) proves $|S_n(t)| \leq E_n$ on the whole interval; (RWR11) then gives $|S'_n| \leq F_n$. For the symmetric matrix, its operator norm is bounded by the maximal absolute row sum, hence by $32 \max_{1 \leq n \leq 63} F_n$. The separate executed `check_transport_bound.py` uses the lower ball endpoint of m_* and upper endpoints of the B_n , obtaining

$$\sup_{[-3/2, -5/4]} \|K'_{32}(t)\|_2 < 0.031792245049 < 4/125. \quad (\text{RWR15})$$

This is in the original c coordinates, without D . For $c(t) = U(t, a)c_a$ with retained value $c_a^T K_{32}(a)c_a = -\eta < 0$, positivity at t_* gives

$$\eta \leq c(t)^T [K_{32}(t_*) - K_{32}(t)] c(t) \leq (4/125)(t_* - t) \|c(t)\|_2^2.$$

Thus the exact obstruction determines the escape space:

$$\|c(t)\|_2 \geq \sqrt{\frac{125\eta}{4(t_* - t)}} \rightarrow \infty. \quad (\text{RWR16})$$

The boundary here is the degeneracy of the specified complete test form. It has not been identified with a collision of H_t zeros.

The complete support lift sends (c, λ) to $(U(t, a)c, \lambda)$ and commutes with Φ because $U(t, a)$ is an explicitly defined linear map. Its form value retains amplitude $-\eta$ and its original label; its norm estimate is an assertion about that same arithmetic coordinate. Passing to a local identity at supported zero does not produce a finite coefficient limit that violates (RWR16).

The receipts are `ARB_REPRODUCTION_m2_0.json`, `ARB_REPRODUCTION_m1.5_m1.25.json` and `ARB_TRANSPORT_DERIVATIVE_BOUND.json`, with the executed script hashes, complete moment and pivot enclosures, and exact input hash. The proof that the union of all finite Laurent spaces detects every off-real zero is given separately in this reader. Positivity of this particular 32-dimensional space establishes no uniform sign in all higher dimensions.

50.4 A completed extension to dimension sixty-four

The incoming report's attempted dimension-64 calculation had timed out. The independent implementation also completes this larger calculation. It retains physical time $t = 0$, the same S_n and the same exact coefficient congruence $D_{ii} = 1000^i$, now for $0 \leq i < 64$. The new executable `extend_degree64_with_arb.py` uses 2560 working bits, 320 nodes per panel, 24 theta summands and moments through 129.

Here are all changes to the proved error estimates. In (RWR5) the two finite sums run to 24; in (RWR6) $N = 320$. The omitted theta sum starts at 25, so (RWR7) becomes

$$192 \cdot 25^4 \cdot 3^{2n} e^{27-1875}.$$

For the spatial tail, $n \leq 129$ and $t = 0$ give $2n \log u + 9u \leq 267u \leq 89u^2 \leq e^{4u}$ on $u \geq 3$. The last inequality follows from the same monotonicity and $e^{12} > 1000 > 801$. Thus (RWR8) remains valid without changing its constant. These are bounds for the original moments, not a replacement kernel or an omitted high-degree contribution. All 64 interval pivots exclude zero and are positive. Therefore

$$K_{64}(0) \succ 0, \quad 1.21406452042511 \cdot 10^{-226} < \hat{p}_{64} < 1.21406452042513 \cdot 10^{-226}, \quad (\text{RWR17})$$

where $\hat{p}_{64}$ is the last pivot in the specified conditioned matrix, not an eigenvalue. The full result, all moments and all pivots are retained in `ARB_DEGREE64_AT_ZERO.json`. The executed calculation took approximately five seconds on this machine; the interval inequalities, rather than that timing, establish its result. Consequently no nonzero Laurent test $\sum_{i=0}^{63} c_i Z^{-2i-1}$ is a negative witness at time zero. The arbitrary-degree construction (ALM12)–(ALM24) retains every higher space and gives the exact quantifier still required for an all-zero conclusion.

50.5 The single residual object after the certified positive block

All the following entries are evaluated at the original arithmetic time $t = 0$. Define $A = K_{64}(0)$ and, for every integer $i \geq 64$,

$$v_i = (S_{i+a+1}(0))_{a=0}^{63}, \quad \mathcal{S}_{ij} = S_{i+j+1}(0) - v_i^T A^{-1} v_j \quad (i, j \geq 64). \quad (\text{RWR18})$$

Every $S_n(0)$ is the fixed theta integral recurrence (RWR1)–(RWR2). The inverse exists by (RWR17) and can equivalently be written $\text{adj}(A)/\det A$; none of these entries is a free parameter.

The complete remaining space and form are

$$\mathcal{V} = \bigoplus_{i=64}^{\infty} \mathbb{C} e_i, \quad \mathcal{Q}(b, c) = \sum_{i,j \geq 64} \bar{b}_i \mathcal{S}_{ij} c_j. \quad (\text{RWR19})$$

The direct sum means finite support; every displayed value is therefore a finite sum of fixed, completely defined real numbers. There is no cutoff on the indices admitted to $\mathcal{V}$.

Its exact map into the original rational-function test space is

$$Jb = \sum_{i \geq 64} b_i \Psi_i(Z), \quad \Psi_i(Z) = Z^{-2i-1} - \sum_{a=0}^{63} (A^{-1}v_i)_a Z^{-2a-1}. \quad (\text{RWR20})$$

The coefficient of Z^{-2i-1} , $i \geq 64$, in Jb is b_i ; thus J is injective. Let $E(a) = \sum_{j < 64} a_j Z^{-2j-1}$. By (ALM9), the full trace pairing gives exactly

$$\mathcal{T}_0(Ea, Jb) = 0, \quad \mathcal{T}_0(Jb, Jc) = \mathcal{Q}(b, c). \quad (\text{RWR21})$$

For the first equality each column is $v_i - AA^{-1}v_i = 0$. Expanding both arguments in the second equality gives $S_{i+j+1} - v_i^T A^{-1}v_j - v_i^T A^{-1}v_j + v_i^T A^{-1}AA^{-1}v_j$, which is (RWR18).

For any original finite coefficient vector, split it at 64 as (a, b) and put $Vb = \sum_{i \geq 64} v_i b_i$. Its exact decomposition is

$$E(a) + \sum_{i \geq 64} b_i Z^{-2i-1} = E(a + A^{-1}Vb) + Jb, \quad Q_0(a, b) = (a + A^{-1}Vb)^* A(a + A^{-1}Vb) + \mathcal{Q}(b, b). \quad (\text{RWR22})$$

The map $F(\alpha, b) = E\alpha + Jb$ is a proved linear isomorphism from $\mathbb{C}^{64} \oplus \mathcal{V}$ onto the original union of Laurent test spaces. Reading the original low and high coefficients (a, b) gives its inverse $F^{-1}(a, b) = (a + A^{-1}Vb, b)$. Equivalently the coefficient change $C(a, b) = (a + A^{-1}Vb, b)$ has inverse $C^{-1}(\alpha, b) = (\alpha - A^{-1}Vb, b)$. It retains the original quadratic value and all coefficients through that explicit inverse. It is not an assertion that arbitrary free split matrices are globally invertible.

The exact counterfactual receiver is now the one negative cone

$$\mathcal{N} = \{b \in \bigoplus_{i \geq 64} \mathbb{Q} e_i : \mathcal{Q}(b, b) < 0\}. \quad (\text{RWR23})$$

It is empty exactly when all original nontrivial zeta zeros lie on their critical line. Here is the full proof of both directions. If all zeros are on the line, (ALM24) and (RWR21) give $\mathcal{Q}(b, b) \geq 0$ for every b , so $\mathcal{N}$ is empty. Conversely any off-line zero gives a finite rational negative vector (a, b) by (ALM12)–(ALM23). Its b is rational, is nonzero because $A \succ 0$, and (RWR22) implies $\mathcal{Q}(b, b) \leq Q_0(a, b) < 0$. Thus it belongs to $\mathcal{N}$. Finally, a member of $\mathcal{N}$ maps under J to a negative original full-root test by (RWR21), which contradicts all roots being real by (ALM8). The possibly nonrational low coefficients in Jb cause no loss: the strict finite-dimensional inequality persists under rational coefficient approximation, as quantified in (ALM21)–(ALM23).

All maps in (RWR20)–(RWR22) are fixed complex-linear maps. Their lifts $(x, \lambda) \mapsto (Ax, \lambda)$ commute with the original full Φ by (ALM29)–(ALM31). The residual pairing has the exact lift

$$\mathcal{Q}^L((b, \lambda), (c, \mu)) = (\mathcal{Q}(b, c), \lambda \wedge \mu). \quad (\text{RWR24})$$

If the amplitude is nonzero then both labels are top, so this lies in $G_L(\mathbb{C})$; the zero-amplitude cases keep their actual meet. Lattice distributivity and the already proved sesquilinearity prove the full lifted pairing identities. Consequently the complete remaining object is $(G_L(\mathcal{V}), \mathcal{Q}^L, \mathcal{J}^\uparrow, \Phi)$ with the specified negative rational cone. Every entry is fixed by the original theta kernel and every higher degree remains present. The certificate has evaluated its eliminated block; it has not evaluated the sign of the whole residual object. The lifted decomposition uses $G_L(\mathbb{C}^{64} \oplus \mathcal{V})$ with its retained common label. No identification with a product of separately labelled carriers is asserted.

50.6 The zero-value edge in that same residual space

Let $\mathcal{R} = \{Z^{-1}p(Z^{-2}) : p \in \mathbb{C}[X]\}$ and $\mathcal{U} = E\mathbb{C}^{64}$. Define the exact linear observations $r(f)_a = \mathcal{T}_0(Ee_a, f)$, $0 \leq a < 64$. Then

$$\Pi f = f - EA^{-1}r(f), \quad \Pi^2 = \Pi, \quad \ker \Pi = \mathcal{U}, \quad \text{im } \Pi = \mathcal{U}^\perp = J\mathcal{V}. \quad (\text{RWR25})$$

Indeed $r(Ea) = Aa$, so $r(\Pi f) = 0$ and Π is the identity on that kernel; its kernel is exactly $\mathcal{U}$. Formula (RWR22) shows that every vector in the orthogonal complement has low coefficients $-A^{-1}Vb$, hence belongs to $J\mathcal{V}$.

The complete pairing on $\mathcal{R}$ has zero algebraic radical, without an RH assumption. To prove this, take a nonzero polynomial p and choose a reciprocal node λ_0 with $p(\bar{\lambda}_0) \neq 0$. Such a node exists because the set of nodes is infinite and a nonzero polynomial has finitely many zeros. Use the complete finite set S , Lagrange polynomial L_0 , bound B and $r < |\lambda_0|$ of (ALM12), and put $q_N(X) = (X/\lambda_0)^N L_0(X)$. The full mixed trace has target term $M_{\lambda_0} \lambda_0 p(\bar{\lambda}_0) \neq 0$, all other terms in S vanish, and the entire remaining tail is bounded by

$$V\|p\|_R B \left(\frac{r}{|\lambda_0|} \right)^N \rightarrow 0. \quad (\text{RWR26})$$

This is (ALM3) and the pointwise coefficient bound, with no root omitted. For some finite N the mixed pairing is therefore nonzero. Since $\mathcal{R} = \mathcal{U} \oplus \mathcal{U}^\perp$ and the first summand has the nondegenerate matrix A , a vector in the radical of the orthogonal complement would be in the radical of the whole space. Thus the residual form $\mathcal{Q}$ also has zero radical.

This resolves its exact zero-value edge. If $b \neq 0$ and $\mathcal{Q}(b, b) = 0$, choose c with $\beta = \mathcal{Q}(b, c) \neq 0$. For every $0 < \varepsilon < 2|\beta|/(|\mathcal{Q}(c, c)| + 1)$,

$$\mathcal{Q} \left(b - \varepsilon \frac{\bar{\beta}}{|\beta|} c, b - \varepsilon \frac{\bar{\beta}}{|\beta|} c \right) = -2\varepsilon|\beta| + \varepsilon^2 \mathcal{Q}(c, c) < 0. \quad (\text{RWR27})$$

All vectors involved have finite support. Since the matrix is real symmetric, at least one of the real and imaginary parts of a complex negative vector is negative. Rational approximation on that finite support then supplies a member of $\mathcal{N}$, with strict sign preserved by continuity (or the bound (ALM21) after the exact map J). Therefore any nonzero isotropic residual vector already produces a negative witness; it is not an undetected neutral case. The projection, inclusion and mixed trace in this argument have the same exact full lattice lifts as (RWR24).

51 Characteristic- p endomotive maps with the entire split-support lattice

Original source and conventions. The input is Alain Connes, Caterina Con-sani and Matilde Marcolli, *Fun with $\mathbb{F}_1$* , original `funBC.tex`, lines 1551–2033, in particular `geomfrobprop/sigptau1`, `triangular/triangmap/sigptau/rhoptaup`, `proptriangular/triangact`, `filtration/filtr`, and `triangularbis`. These source sections were read directly as original LaTeX. The system originates in Jean-Benoît Bost and Alain Connes, *Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory*, *Selecta Math.* (N.S.) 1 (1995), 411–457, cited in that source. The full carrier is the foundation retained under the byline The Clankers, *Split Support Geometry*, v11, Definition `def:lattice-split`. All finite-level maps, support fibers, prime ideals and limiting maps needed here are proved explicitly below. No purity or zero-location result is inferred from these algebraic constructions.

Fix a prime p , a perfect field k of characteristic p , and an arbitrary nontrivial bounded distributive lattice L . Perfection is used for the invertibility of the coefficient Frobenius, not for polynomial division or the support arguments. The complex tropical receiver is replaced only in its ring coordinate by the actual field k ; the entire original lattice and its complete coefficient map remain present.

51.1 The finite local rings, signs, and the full coefficient map

For $q = p^\ell$, $\ell \geq 0$, use the exact ring

$$A_q = k[T_q]/(T_q^q - 1) = k[\epsilon_q]/(\epsilon_q^q), \quad \epsilon_q = T_q - 1, \quad \delta_q = 1 - T_q = -\epsilon_q. \quad (\text{CFP1})$$

The equality follows because $(T - 1)^q = T^q - 1$ in characteristic p . For $q = 1$, this is k and $\epsilon_1 = \delta_1 = 0$. Every element has its unique unchanged expression $f = \sum_{j=0}^{q-1} a_j \epsilon_q^j$. The convolution coordinate in the source is δ_q ; its later filtration coordinate is ϵ_q . The equality $\delta_q = -\epsilon_q$ will be carried through each map, including characteristic two where the two signs coincide in the field.

For any ring R in this section retain

$$\begin{aligned} S_R &= G_L(R) = \{(a, \lambda) : a \in R, \lambda \in L, a \neq 0 \Rightarrow \lambda = 1_L\}, \\ (a, \lambda) + (b, \mu) &= (a + b, \lambda \vee \mu), \quad (a, \lambda)(b, \mu) = (ab, \lambda \wedge \mu), \\ \tau_R &= (0, 0_L), \quad e_R = (0, 1_L), \quad \hat{a} = (a, 1_L), \quad z_\lambda = (0, \lambda). \end{aligned} \quad (\text{CFP2})$$

For a ring map $f : R \rightarrow R'$, its full lift is $G_L(f)(a, \lambda) = (f(a), \lambda)$. It is well defined because only a zero amplitude can acquire a lower label; each operation is preserved directly. For a k -linear map that is not multiplicative the same formula is additive and $G_L(k)$ -linear, with the identical carrier check; multiplicativity is never inferred in that case.

Let K be the original tropical semifield with zero, $K_L = K \otimes_{\mathbb{B}} L$, and $\iota_L(\lambda) = 1_K \otimes \lambda$. The nonzero indicator $\beta : K \rightarrow \mathbb{B}$ preserves sum and product because tropical sums vanish only when both terms vanish and products vanish only when one factor vanishes. Thus $\rho_L = \beta \otimes \text{id}_L$ satisfies $\rho_L \iota_L = \text{id}_L$. The complete map is

$$\Phi_R : S_R \hookrightarrow R \times K_L, \quad (a, \lambda) \longmapsto (a, \iota_L(\lambda)). \quad (\text{CFP3})$$

It preserves both operations and units. The retraction recovers the label, so it is injective. Its image consists exactly of pairs whose second coordinate lies in $\iota_L(L)$ and equals 1_{K_L} whenever the first is nonzero. For every lifted map f ,

$$\Phi_{R'} G_L(f) = (f, \text{id}_{K_L}) \Phi_R. \quad (\text{CFP4})$$

For linear f this states equality of additive maps; for ring f it is equality of semiring morphisms. This distinction preserves the transfer maps below.

The exact top-filtration element is

$$\tilde{\pi}_q = \sum_{j=0}^{q-1} T_q^j = \epsilon_q^{q-1} = \delta_q^{q-1}, \quad F^i A_q = \epsilon_q^i A_q = \delta_q^i A_q \quad (0 \leq i \leq q). \quad (\text{CFP5})$$

To prove the polynomial identity, factor $T^q - 1$ in $k[T]$ as both $(T - 1) \sum_{j < q} T^j$ and $(T - 1)^q$; cancellation in the polynomial domain gives the claimed equality before passing to the quotient. The sign $(-1)^{q-1}$ is one: q is odd for odd p , and $-1 = 1$ in characteristic two. In particular no division by q has been used. For $q > 1$, $\tilde{\pi}_q \neq 0$ and multiplication by it sends $\sum a_j \epsilon_q^j$ to $a_0 \epsilon_q^{q-1}$. It detects that coefficient in the top filtration piece without replacing the filtration by its residue.

51.2 Transitions and the coefficient Frobenius, with every support fiber

For $h = p^r$, define the transition and the downward map by

$$\begin{aligned} i_{q,hq} : A_q &\hookrightarrow A_{hq}, & T_q &\longmapsto T_{hq}^h, & \epsilon_q &\longmapsto \epsilon_{hq}^h, & \delta_q &\longmapsto \delta_{hq}^h, \\ D_{h,q} : A_{hq} &\twoheadrightarrow A_q, & T_{hq} &\longmapsto T_q, & \epsilon_{hq} &\longmapsto \epsilon_q, & \delta_{hq} &\longmapsto \delta_q. \end{aligned} \quad (\text{CFP6})$$

The transition sends the basis to the distinct monomials ϵ_{hq}^{hj} for $0 \leq j < q$, proving injectivity. The downward map is reduction modulo ϵ_q^q in the stated coordinates, and is onto with kernel (ϵ_{hq}^q) . These facts prove the domains, codomains and all sign conventions without identifying the two maps. Transitions compose as $i_{hq,ghq} i_{q,hq} = i_{q,ghq}$.

On one fixed level define the relative endomorphism and the coefficient Frobenius by

$$\begin{aligned} \sigma_h^{(q)}(\sum a_j \epsilon_q^j) &= \sum a_j \epsilon_q^{hj}, & C_h^{(q)}(\sum a_j \epsilon_q^j) &= \sum a_j^h \epsilon_q^j, \\ \mathfrak{F}_h^{(q)} &= C_h^{(q)} \sigma_h^{(q)}, & \mathfrak{F}_h^{(q)}(f) &= f^h, \\ D_{h,q} i_{q,hq} &= \sigma_h^{(q)}, & i_{q,hq} D_{h,q} &= \sigma_h^{(hq)}. \end{aligned} \quad (\text{CFP7})$$

The second composite is valid termwise even when the degree being truncated is at least q , because its image degree at least hq is zero. The first Frobenius is k -linear; C_h acts on coefficients and is an automorphism because k is perfect. The commuting maps together give the absolute power map, exactly the source **sigtpaup1**. Equivalently on a group monomial with coefficient a , $T_q^j a \mapsto T_q^{hj} a^h = (T_q^j a)^h$.

Put $c = \lceil q/h \rceil \geq 1$. Then

$$\ker \sigma_h^{(q)} = \ker \mathfrak{F}_h^{(q)} = (\epsilon_q^c), \quad \text{im } \sigma_h^{(q)} = \text{span}_k \{\epsilon_q^{hj} : hj < q\}. \quad (\text{CFP8})$$

Distinct surviving degrees cannot cancel. Coefficient Frobenius does not change whether a coefficient is zero, proving both kernels. The image is k when $h \geq q$ and otherwise is an isomorphic copy of $A_{q/h}$ by its explicitly listed basis. At $h = 1$ the kernel is $(\epsilon_q^q) = 0$ as required.

Every full lift of (CFP6)–(CFP8) keeps the label. Its power identity is literally

$$G_L(\mathfrak{F}_h)(a, \lambda) = (a^h, \lambda) = (a, \lambda)^h, \quad G_L(C_h)G_L(\sigma_h) = G_L(\mathfrak{F}_h). \quad (\text{CFP9})$$

Finite meets of a repeated lattice label equal that label; the characteristic- p amplitude power is additive, and the label of a sum remains its join. This proves that the power map is the stated actual semiring morphism, with no disappearing labels.

For any of these maps $G_L(f)$, its three distinct kernel notions are completely explicit:

$$\begin{aligned} G_L(f)^{-1}(\tau) &= \{\tau\}, & G_L(f)^{-1}(e) &= \{(a, 1_L) : a \in \ker f\}, \\ G_L(f)^{-1}(\{z_\lambda : \lambda \in L\}) &= \{z_\lambda : \lambda \in L\} \cup \{(a, 1_L) : 0 \neq a \in \ker f\}, \\ G_L(f)(a, \lambda) &= G_L(f)(b, \mu) \iff \lambda = \mu \text{ and } a - b \in \ker f. \end{aligned} \quad (\text{CFP10})$$

The first assertion follows because label zero forces amplitude zero. The second is the fiber over supported zero, not an ideal containing τ . The third is an ordinary ideal and the fourth is the exact kernel congruence. These statements follow directly from the two coordinates, and retain their difference even when a nonzero amplitude is killed.

51.3 The unnormalized transfer and its exact lifted relations

There is no inverse of $h = p^r > 1$ in k . Retain the source unnormalized transfer, at its exact levels:

$$\begin{aligned} R_{h,q} : A_q &\longrightarrow A_{hq}, & R_{h,q}(T_q^j) &= \sum_{b=0}^{h-1} T_{hq}^{j+bq}, & 0 \leq j < q, \\ R_{h,q}\left(\sum_{j<q} a_j \epsilon_q^j\right) &= \sum_{j<q} a_j \epsilon_{hq}^{q(h-1)+j}, \\ R_{h,q}(\delta_q^j) &= \delta_{hq}^{q(h-1)+j}, & R_{h,q}(1) &= \epsilon_{hq}^{q(h-1)}. \end{aligned} \tag{CFP11}$$

For the second line, the first-line sum is $T_{hq}^j \sum_{b<h} (T_{hq}^q)^b$; (CFP5) applied to this polynomial sum gives $(T_{hq}^q - 1)^{h-1} = \epsilon_{hq}^{q(h-1)}$. Linearity then permits the unchanged polynomial basis change from T_q to ϵ_q . For $\delta_q = -\epsilon_q$, the exponent $q(h-1)$ is even at odd p , and its sign is one at $p = 2$, giving exactly the third line. Distinct resulting degrees prove injectivity. Its image is the full ideal $\epsilon_{hq}^{q(h-1)} A_{hq}$, with the displayed shift providing its inverse as a vector-space map.

For $h > 1$ the image is square-zero, because $2q(h-1) \geq hq$. The map is not a ring map. The exact projection and composition formulas are

$$\begin{aligned} R_{h,q}(D_{h,q}(x)y) &= xR_{h,q}(y) \quad (x \in A_{hq}, y \in A_q), \\ R_{h,q}D_{h,q}(x) &= \epsilon_{hq}^{q(h-1)}x, & D_{h,q}R_{h,q} &= 0 \quad (h > 1), \\ R_{g,hq}R_{h,q} &= R_{gh,q}, & i_{hq,bhq}R_{h,q} &= R_{h,bq}i_{q,bq} \quad (b = p^v). \end{aligned} \tag{CFP12}$$

Multiplication by the shift $\epsilon^{q(h-1)}$ kills exactly terms of degree at least q in its unshifted factor; hence reducing x modulo ϵ^q and then multiplying gives the first two equalities. The downward map kills the image because $q(h-1) \geq q$. Composition adds the exponents $hq(g-1) + q(h-1) = q(gh-1)$. The last identity multiplies every exponent by b on either side, proving compatibility with all actual transition levels. All equalities retain their original coefficients; no factor $1/h$ occurs anywhere.

The lift $\hat{R}_{h,q} = G_L(R_{h,q})$ is additive, injective and $G_L(k)$ -linear. Its failure of multiplicativity and the lifted downward identity are the following exact equations for $h > 1$:

$$\begin{aligned} \hat{R}_{h,q}(a, \lambda) \hat{R}_{h,q}(b, \mu) &= (0, \lambda \wedge \mu), \\ G_L(D_{h,q}) \hat{R}_{h,q}(a, \lambda) &= (0, \lambda) = e_{A_q}(a, \lambda), \\ \hat{R}_{h,q}(G_L(D_{h,q})(x)y) &= x \hat{R}_{h,q}(y). \end{aligned} \tag{CFP13}$$

Their amplitudes are (CFP12); their labels on each side are respectively the displayed meet or unchanged label. Thus even the zero ring composite is the nonzero supported projection on the full carrier. The source integral factor h acts here as the actual scalar $\hat{h} = e$, since $h = 0$ in k , and $e(a, \lambda) = (0, \lambda)$ reproduces the second line. This states what survives that coefficient, rather than replacing it by τ .

51.4 Supported nilpotence and the actual full prime spectrum

For $q > 1$ the distinguished generator obeys

$$\hat{\epsilon}_q^q = e_{A_q} \neq \tau_{A_q}, \quad \hat{\epsilon}_q^j \neq e_{A_q} \quad (1 \leq j < q). \tag{CFP14}$$

In fact for every ring R and every positive integer n , $(a, \lambda)^n = (a^n, \lambda)$. Therefore

$$\{x \in S_R : x^n = \tau_R \text{ for some } n \geq 1\} = \{\tau_R\}. \quad (\text{CFP15})$$

If the power has label zero, the original label is zero and its original amplitude is already zero. No reducedness of R is used. On the other hand, the elements of S_{A_q} whose powers reach e are exactly the top-labelled elements with amplitude in (ϵ_q) . For a nonzero such amplitude, let d be the least index with $a_d \neq 0$. Its n th power has leading term $a_d^n \epsilon_q^{nd}$ when $nd < q$, with nonzero coefficient because k is a field, and is zero when $nd \geq q$. Its exact supported nilpotence index is therefore $\lceil q/d \rceil$. An amplitude with $a_0 \neq 0$ is a unit: factor out a_0 and invert its remaining $1 + \text{nilpotent}$ part by the finite geometric series. Hence A_q has the unique ring prime (ϵ_q) .

All ordinary semiring prime ideals of S_{A_q} are

$$\begin{aligned} \mathfrak{M}_q &= \{z_\lambda : \lambda \in L\} \cup \{(a, 1_L) : 0 \neq a \in (\epsilon_q)\}, \\ P_\tau &= \{(0, \lambda) : \lambda \in \tau\}, \quad \tau \in \text{SpecLat } L, \quad P_\tau \subsetneq \mathfrak{M}_q. \end{aligned} \quad (\text{CFP16})$$

To prove exhaustiveness, a prime containing e contains all labelled zeros. Its top amplitudes form a ring ideal, with additive inverses supplied by multiplication by $\widehat{-1}$, and primeness makes it a ring prime. Thus it is $\mathfrak{M}_q$. If a prime excludes e , it contains no nonzero amplitude: multiplying such an element by e would put e into it. Its labels form a lattice ideal prime for meet; absorption gives downward closure. Conversely the two displayed families directly satisfy the ideal and prime conditions. Strictness follows because e is in the first and in none of the latter. Their lattice inclusions are precisely those of the original lattice primes; no Boolean replacement occurs.

The principal supported-zero ideal itself is $(e) = \{z_\lambda : \lambda \in L\}$. For $q > 1$ it is not prime: $\widehat{\epsilon}_q$ and $\widehat{\epsilon}_q^{q-1}$ are both outside it, while their product is e . Its radical is exactly $\mathfrak{M}_q$, because a power enters (e) precisely when its amplitude is nilpotent. For $q = 1$ the amplitude ring is the field k , and $(e) = \mathfrak{M}_1$ is prime. Thus the reduction homeomorphism below does not preserve the assertion that the particular principal ideal (e) was prime: its inverse image is $\mathfrak{M}_q$, not merely the original (e) .

The topological basic opens are also explicit. For a nonzero amplitude a , $D((a, 1_L))$ is the whole spectrum if $a_0 \neq 0$, and is exactly the lattice-prime part if $a_0 = 0$. For a labelled zero z_λ , its basic open is $\{P_\tau : \lambda \notin \tau\}$. These statements follow from (CFP16), including $D(e)$ equal to the lattice-prime part and $V(e) = \{\mathfrak{M}_q\}$.

The intersection of all displayed primes is $\{\tau\}$. Any nonzero amplitude is excluded from every lattice prime. For a nonbottom label λ , a lattice prime excluding it exists: choose an ideal maximal subject to being disjoint from $\uparrow \lambda$; distributivity proves it prime. Specifically if adjoining either of two excluded elements meets that filter, while their meet belongs to the ideal, distributivity puts a filter element back in the ideal, a contradiction. Thus z_λ is also excluded by a displayed prime. This verifies the ordinary nilradical directly from both powers and the actual spectrum.

51.5 Reduction is a full-label congruence, not a nilradical quotient

The exact ring reduction and its lift are

$$\text{red}_q : A_q \rightarrow k, \quad \sum a_j \epsilon_q^j \mapsto a_0, \quad \mathcal{R}_q : S_{A_q} \rightarrow S_k, \quad (a, \lambda) \mapsto (a_0, \lambda). \quad (\text{CFP17})$$

This is onto, with the constant amplitude section retaining its label. Its kernel congruence is

$$(a, \lambda) \sim (b, \mu) \iff \lambda = \mu, \quad a_0 = b_0, \quad S_{A_q} / \langle \widehat{\epsilon}_q = e \rangle \simeq S_k. \quad (\text{CFP18})$$

For top labels $a - b = \epsilon_q c$, multiplying the generating relation by $\widehat{c}$ and adding $\widehat{b}$ gives exactly the equality. Below top label both amplitudes are already zero. Conversely (CFP17) distinguishes all

pairs not specified, proving the entire congruence. It is nontrivial for $q > 1$ by (CFP14), but its τ -fiber is only $\{\tau\}$ by (CFP10). Since the ordinary semiring nilradical is already $\{\tau\}$, quotienting by that zero ideal leaves the semiring unchanged and cannot be this map.

Reduction nevertheless induces the exact homeomorphism

$$\mathrm{Spec} S_k \xrightarrow{\mathcal{R}_q^*} \mathrm{Spec} S_{A_q}, \quad \mathfrak{M}_k \mapsto \mathfrak{M}_q, \quad P_{\mathfrak{t}} \mapsto P_{\mathfrak{t}}. \quad (\text{CFP19})$$

The preimage formulas follow at once from (CFP17). They give a bijection of the complete list (CFP16). For each original element, the explicit basic open calculation following (CFP16) agrees with the basic open of its reduction: a nilpotent top amplitude reduces to e , not τ . Hence the bijection and its inverse are continuous. The unchanged prime topology does not identify the two coefficient semirings or their congruences. The faithful complete map instead gives the square

$$\begin{array}{ccc} S_{A_q} & \xrightarrow{\Phi_{A_q}} & A_q \times K_L \\ \mathcal{R}_q \downarrow & & \downarrow \text{red}_q \times \text{id} \\ S_k & \xrightarrow{\Phi_k} & k \times K_L. \end{array} \quad (\text{CFP20})$$

All labels are retained by both vertical maps; the nilpotent coordinate changes in the ring factor exactly as specified. It is a pushout: on the upper-right semiring the relation $(\epsilon_q, 1) \sim (0, 1)$ generates precisely equal second coordinate and equal reduced ring coordinate. Multiply it by $(c, 0)$ and add (b, w) to obtain $(b + \epsilon_q c, w) \sim (b, w)$; reduction proves there are no further relations. The quotient universal property proves the stated square, not only its commutativity.

Every transition and relative Frobenius induces the identity on the label-indexed primes and takes the unique arithmetic prime to the unique arithmetic prime: its constant coefficient is unchanged. Absolute Frobenius raises that coefficient to a power, which preserves its vanishing, so it has the same prime action. These are exact pullback maps on the full spectrum, although their amplitude kernels and supported congruences can be different.

51.6 The convolution limit with the original affine maps

Let $\Gamma = \mathbb{Z}[1/p] \cap [0, 1)$ and write b_a for the function on Γ equal to one at a and zero elsewhere. Define

$$\mathcal{T}(p) = \bigoplus_{a \in \Gamma} k b_a, \quad b_a b_c = \begin{cases} b_{a+c} & a + c < 1, \\ 0 & a + c \geq 1, \end{cases} \quad 1 = b_0. \quad (\text{CFP21})$$

Only finite supports are allowed. Addition of the original rational indices and truncation at 1 prove associativity: a triple survives exactly when the total of its nonnegative indices is less than 1, independent of parenthesization. This is the source convolution algebra, with b_a denoting its literal δ_a to distinguish it from the finite-ring element δ_q .

The actual compatible maps and their entire coefficient formulas are

$$\begin{aligned} I_q : A_q &\hookrightarrow \mathcal{T}(p), \quad T_q \mapsto b_0 - b_{1/q}, \quad \delta_q \mapsto b_{1/q}, \quad \epsilon_q \mapsto -b_{1/q}, \\ I_q \left(\sum_{j < q} a_j \epsilon_q^j \right) &= \sum_{j < q} (-1)^j a_j b_{j/q}, \quad I_{hq} i_{q,hq} = I_q, \quad \varinjlim_q A_q \xrightarrow{\sim} \mathcal{T}(p). \end{aligned} \quad (\text{CFP22})$$

For $q = 1$, b_1 means the truncated zero. The distinct basis functions prove injectivity. The sign in the transition agrees because $(-b_{1/(hq)})^h = -b_{1/q}$ in characteristic p . Every finite set of indices has a common denominator q , proving surjectivity onto the limit. The maps on T_q therefore prove the source **triangmap** with its exact sign. The source's malformed displayed parenthesis in **charpequ1** is read as $(1 - I_p(T_p))^{p-1} = b_{(p-1)/p}$; (CFP5) and the formula above prove this intended polynomial identity independently.

The full limit is likewise exact:

$$\varinjlim_q S_{A_q} \xrightarrow{\sim} S_{\mathcal{T}(p)}, \quad [(a, \lambda)] \mapsto (I_q(a), \lambda). \quad (\text{CFP23})$$

Every amplitude has a representative at a finite level and the label can be retained there. Equal amplitudes in the injective direct limit agree at a common level; equal elements also have equal labels. This proves the bijection and its inverse; finite sum and product use a common level, so it is a semiring isomorphism. The maps Φ commute with it coefficient by coefficient.

For $h = p^r$, the limit Frobenius and transfer are the following exact affine maps:

$$\begin{aligned} \sigma_h(b_a) &= \begin{cases} b_{ha} & ha < 1, \\ 0 & ha \geq 1, \end{cases} & (\sigma_h f)(a) &= f(a/h), \\ R_h(b_a) &= b_{(a+h-1)/h}, & (R_h f)(a) &= f(ha - (h-1)). \end{aligned} \quad (\text{CFP24})$$

Functions are zero at indices outside $[0, 1)$, as in the source. The transition compatibility in (CFP12) gives these maps on the limit; substitution of $a = j/q$ proves their displayed coefficients directly. In particular the source formulas **sigptau** and **rhoptaup** are recovered at $h = p$, with every original translation $(p-1)/p$.

The exact kernel and image are

$$\ker \sigma_h = \bigoplus_{a \in \Gamma \cap [1/h, 1)} kb_a, \quad \text{im } R_h = \bigoplus_{a \in \Gamma \cap [1-1/h, 1)} kb_a. \quad (\text{CFP25})$$

For $h = 1$ the first is zero and the second is the whole ring. The surviving map $a \mapsto ha$ is injective on $[0, 1/h)$, so no cancellation among different surviving coefficients alters the first formula. The transfer map is a bijection of the indicated index sets, proving the second. The kernel ideal has h th ideal power zero, since h indices at least $1/h$ sum to at least 1. For $h > 1$ the transfer image has square zero. Its lift has exactly the support fibers (CFP10) and the product and composite identities (CFP13), now at the limit; it is not made multiplicative.

The source filtration elements called τ_m in **basicpropeps** are denoted ϑ_m here to keep them separate from the absorbing split zero τ :

$$\vartheta_m = R_p^m(1) = b_{1-p^{-m}}, \quad m \geq 1, \quad \vartheta_m \vartheta_n = 0, \quad R_{p^r}(\vartheta_m) = \vartheta_{m+r}, \quad \sigma_p(\vartheta_m) = 0. \quad (\text{CFP26})$$

The index sum of two such elements is at least 1, while the transfer index formula gives $1 - p^{-(m+r)}$ and the Frobenius index is at least 1. These computations prove every equality. The positivity of m, n matters: the additional value $\vartheta_0 = 1$ does not obey the asserted square-zero product law. In the full lift the first and last zeros in (CFP26) are e , never τ , since all those displayed elements have top support.

The limit ring is local with maximal ideal $\mathfrak{m} = \{f : f(0) = 0\}$. If a nonzero $f \in \mathfrak{m}$ has least supported index $a > 0$, its powers vanish exactly starting at $\lceil 1/a \rceil$, by their nonzero leading coefficient before that index reaches 1. If $f(0) \neq 0$, the finite geometric inverse after factoring that constant proves it a unit. Thus every element of $\mathfrak{m}$ is nilpotent, but the ideal is not nilpotent: for each N choose $q > N$, so $b_{1/q}^N \neq 0$. The full prime spectrum and reduction $S_{\mathcal{T}(p)} \rightarrow S_k$ are exactly (CFP16)–(CFP20), with $\mathfrak{m}$ in place of (ϵ_q) . Their proofs use this actual local-ring description.

51.7 The prime-to- p reduction with its exact remaining Frobenius

For $n = qm$, $q = p^\ell$ and $(m, p) = 1$, set

$$R_n = k[T]/(T^n - 1) = k[T]/((T^m - 1)^q), \quad B_m = k[T]/(T^m - 1). \quad (\text{CFP27})$$

The derivative mT^{m-1} is coprime to $T^m - 1$, so the latter polynomial has no repeated irreducible factor over k . Its distinct irreducible factors u_i give by polynomial Bezout and the Chinese remainder maps

$$R_n \simeq \prod_i k[T]/(u_i^q), \quad B_m \simeq \prod_i k[T]/(u_i). \quad (\text{CFP28})$$

Each right factor in the second product is a field, hence their product is reduced. The kernel $(T^m - 1)$ of $R_n \rightarrow B_m$ has q th power zero. Any nilpotent maps to zero in the reduced target, proving that this kernel is exactly the ring nilradical. The ring primes are the ideals (u_i) . The full semiring primes are those arithmetic primes together with every lattice prime $P_{\mathfrak{t}}$; each lattice prime is contained in each arithmetic prime, by the same two-case proof as (CFP16).

The full reduction keeps labels:

$$S_{R_n} \longrightarrow S_{B_m}, \quad (f, \lambda) \longmapsto ([f], \lambda), \quad S_{R_n} / \langle \widehat{T^m - 1} = e \rangle \simeq S_{B_m}. \quad (\text{CFP29})$$

The equal-label congruence proof of (CFP18) applies to the displayed principal ideal. It has trivial τ -fiber and induces the same full prime-spectrum homeomorphism, now with each arithmetic prime matched to the corresponding field factor. Ring-prime homeomorphism can also be checked directly because every prime contains the nilpotent kernel; the displayed lattice primes are unchanged. The complete Φ square is obtained by keeping K_L fixed exactly as in (CFP20).

Taking the original group inclusions as transitions gives the source reduction and its full lift

$$\begin{aligned} A &= k[\mathbb{Q}/\mathbb{Z}], \quad B = k[(\mathbb{Q}/\mathbb{Z})^{(p)}], \quad A_{\text{red}} \simeq B, \\ G_L(A) &\longrightarrow G_L(B), \quad (a, \lambda) \longmapsto (\text{red}(a), \lambda). \end{aligned} \quad (\text{CFP30})$$

Indeed every finite-support group-ring element lies in a finite cyclic level. The prime-to- p levels B_m are reduced by (CFP28), and their group-basis inclusions are injective. Their union B is therefore reduced. The canonical primary decomposition of $\mathbb{Q}/\mathbb{Z}$ gives $A = \mathcal{T}(p) \otimes_k B$; augmenting the first factor to k is the indicated reduction. Its kernel elements each occur in one finite level, where they are nilpotent by (CFP27)–(CFP28). Conversely every nilpotent maps to zero in B . This proves the full ring reduction and its exact lifted equal-label congruence. The ordinary semiring nilradical remains only $\{\tau\}$ by (CFP15).

On the remaining prime-to- p group, multiplication by p is an automorphism. At every finite level m its inverse is multiplication by the inverse of p modulo m ; these inverses agree under the group inclusions, because the inverse of a group automorphism is unique. Thus σ_p and all σ_{p^r} become actual automorphisms on B and on $G_L(B)$, with labels unchanged. This is the exact surviving map in source **triangularbis** and **nodenom**. It is a different transfer after reduction: the unreduced R_p descends to the zero additive map on amplitudes, since its image has positive convolution index, while the inverse of reduced σ_p is unital. Their full lifts are respectively $(a, \lambda) \mapsto (0, \lambda)$ and the stated label-preserving automorphism. This proves their actual relation instead of silently identifying them. No inverse of p in the field has been introduced.

The finite polynomial quotient and the canonical prime-to- p group coordinate have the following exact map, so no generator is silently renamed in taking that limit. At $n = qm$, choose integers a, b with $aq + bm = 1$. Write $U_m = e(1/m)$ and $T_n = e(1/n)$ for the actual group generators. Then

$$\frac{1}{qm} = \frac{a}{m} + \frac{b}{q} \pmod{\mathbb{Z}}, \quad \text{red}(T_n) = U_m^a, \quad k[T]/(T^m - 1) \xrightarrow{\sim} k[\mathbb{Z}/m\mathbb{Z}], \quad [T] \longmapsto U_m^a. \quad (\text{CFP31})$$

The exponent a is the inverse of q modulo m ; different Bezout choices give the same power. It is prime to m , proving that the last map is an isomorphism. This is exactly the primary group projection, which commutes with every original group inclusion. On $\mathcal{T}(p) \otimes B$ the unreduced transfer has the explicit factor map $R_p \otimes \sigma_p^{-1}$: a preimage in the prime-to- p group is unique, while the p preimages in the primary group are exactly those summed by R_p . Its first factor has positive convolution index. Augmentation therefore makes the whole transfer zero, verifying the last comparison above on every original basis element. The reduced inverse remains a different unital map with the exact factor described there.

51.8 The original triangular representation and its faithful full lift

Let $V = k^{(\Gamma)}$ with basis ξ_a , $a \in \Gamma$, and put $\xi_c = 0$ for $c \geq 1$. Denote convolution by $f \in \mathcal{T}(p)$ by M_f . The source's two operators and their exact rational indices are

$$M_{b_c}\xi_a = \xi_{a+c}, \quad U\xi_a = \xi_{(a+p-1)/p}, \quad W\xi_a = \xi_{pa}. \quad (\text{CFP32})$$

They are respectively the source images of δ_c , $\tilde{\mu}_p$ and μ_p^* . The star in the last source symbol is its generator notation; no Hilbert adjoint over the characteristic- p field is asserted. Each displayed partial affine map is injective on the set of indices on which it survives, so its matrix has at most one nonzero entry per row and column. It is lower triangular in the original rational order because $a+c \geq a$, $pa \geq a$, and $(a+p-1)/p - a = (p-1)(1-a)/p > 0$. This does not enumerate the dense rational order increasingly by natural numbers.

The complete defining relations are

$$UM_fW = M_{R_p f}, \quad WM_f = M_{\sigma_p f}W, \quad M_fU = UM_{\sigma_p f}, \quad WU = 0. \quad (\text{CFP33})$$

For $f = b_c$, the first equality sends ξ_a to $\xi_{a+(c+p-1)/p}$ exactly when $pa+c < 1$, and to zero otherwise, on both sides. The second sends it to $\xi_{p(a+c)}$ exactly when $p(a+c) < 1$. The third sends it to $\xi_{(a+pc+p-1)/p}$ exactly when $a+pc < 1$. Any earlier cutoff implies the same failed final inequality, so none is omitted. Finally $WU\xi_a = \xi_{a+p-1} = 0$. Linearity proves all statements for every finite-support f .

Let $\mathcal{C}_p$ be the source algebra generated by $\mathcal{T}(p)$ and u, v with these relations. They give its complete normal monomial basis

$$u^n b_c \quad (n \geq 1, c \in \Gamma), \quad b_c v^m \quad (m \geq 0, c \in \Gamma). \quad (\text{CFP34})$$

Here is a proof of both spanning and independence, so the use of the source representation is exact. For $f, g \in \mathcal{T}(p)$ the products of these forms reduce by the displayed defining relations to

$$\begin{aligned} (u^n f)(u^r g) &= u^{n+r} \sigma_p^r(f)g, \\ (fv^m)(gv^k) &= f\sigma_p^m(g)v^{m+k}, \\ (fv^m)(u^n g) &= 0 \quad (m, n \geq 1), \\ f(u^n g) &= u^n \sigma_p^n(f)g, \\ u^n(fg)v^m &= \begin{cases} R_p^n(fg)v^{m-n} & m \geq n, \\ u^{n-m}R_p^m(fg) & n > m. \end{cases} \end{aligned} \quad (\text{CFP35})$$

The first, second and fourth repeatedly commute the coefficient by the corresponding relation. The third uses $vu = 0$. For the last, apply $uhv = R_p(h)$ exactly $\min(m, n)$ times. Expanding the resulting finite-support functions in their b_c basis proves that the span contains all generators and is closed under products, hence spans the generated algebra.

The actions of the normal monomials are

$$\begin{aligned} \theta(u^n b_c)\xi_a &= \xi_{p^{-n}a+1-p^{-n}+p^{-n}c}, \quad n \geq 1, \\ \theta(b_c v^m)\xi_a &= \xi_{p^m a+c}, \quad m \geq 0. \end{aligned} \quad (\text{CFP36})$$

They follow by iterating (CFP32), with the same cutoff at 1. The first slopes are less than one, the second at least one. Within each family its slope and intercept recover its parameters. Thus distinct normal monomials have distinct affine functions $t_j a + c_j$, where $t_j > 0$ and $0 \leq c_j < 1$. For a finite nontrivial linear combination choose $\eta > 0$ smaller than 1 and every $(1-c_j)/t_j$. The interval $(0, \eta)$ contains infinitely many points of Γ . Two distinct affine functions meet at at most one point, so choose one such point outside all their finitely many crossings. Every output then

survives and the output indices are pairwise distinct. A nontrivial linear combination cannot annihilate that basis vector. This proves independence and the faithful unital representation

$$\theta : \mathcal{C}_p \hookrightarrow \text{End}_k(V), \quad G_L(\theta) : G_L(\mathcal{C}_p) \hookrightarrow G_L(\text{End}_k(V)). \quad (\text{CFP37})$$

The lift keeps the original global label. The formula (CFP2) also defines a semiring for a noncommutative ring, using its actual product; the preceding injectivity follows from the two coordinates. In this full receiver the source relation is precisely

$$(v, 1_L)(u, 1_L) = (0, 1_L) = \underbrace{(1, 1_L) + \cdots + (1, 1_L)}_{p \text{ summands}}. \quad (\text{CFP38})$$

The top-supported copy of the source ring has its own additive zero e , and its inclusion in the full carrier does not send that zero to τ .

51.9 A coefficientwise full-lattice matrix algebra receiving both actions

The global receiver is not automatically the sparse matrix algebra over $G_L(k)$. Construct an explicit common ambient semiring as follows. Let $\mathfrak{M}_L(\Gamma)$ consist of pairs (A, Λ) , where A is a column-finite k -matrix on Γ , $\Lambda : \Gamma^2 \rightarrow L$ has finite image, and $A_{ab} \neq 0 \Rightarrow \Lambda_{ab} = 1_L$. Every entry is thus an actual split coefficient; the support labels may be dense in the index set. Define

$$\begin{aligned} (A, \Lambda) + (B, \Omega) &= (A + B, \Lambda \vee \Omega), & (A, \Lambda)(B, \Omega) &= (AB, \Lambda \odot \Omega), \\ (\Lambda \odot \Omega)_{ab} &= \bigvee_{c \in \Gamma} \Lambda_{ac} \wedge \Omega_{cb}, & I_{\text{sp}} &= (I, \Delta), \quad \Delta_{ab} = \begin{cases} 1_L & a = b, \\ 0_L & a \neq b. \end{cases} \end{aligned} \quad (\text{CFP39})$$

The indicated join uses only finite joins in the original L : its values belong to the finite set of meets of one value from each input's finite image, and one joins only the distinct values actually attained. All output entries lie among finitely many possible joins of subsets of that set. Hence the output again has finite image. The amplitude product is column finite because each column of B selects only finitely many finite columns of A . Each product entry has a finite amplitude sum; a nonzero sum has a nonzero summand and hence a top support path, proving the carrier condition.

Associativity of the label product follows by expanding either bracketing as the join of $\Lambda_{ac} \wedge \Omega_{cd} \wedge \Xi_{db}$. There are only finitely many distinct such values. Finite distributivity expands each side and gives the same set of values; the same reasoning proves distributivity over addition. Amplitude operations are those of column-finite endomorphisms. The zero is $(0, 0)$ and the unit is I_{sp} , as direct substitution proves. This establishes the semiring without assuming that arbitrary joins exist in L . Coefficient matrices with infinitely many distinct labels are not silently included in this defined ambient space; it includes every global label and every sparse original source operator.

Let $\mathbf{J}$ denote the constant-top matrix on Γ^2 , and set $H = (I, \mathbf{J})$. The source amplitudes remain the original triangular matrices in the following exact corner and faithful coefficient receiver:

$$\begin{aligned} H^2 &= H, & H(A, \Lambda)H &= (A, \text{const } \epsilon(\Lambda)), & \epsilon(\Lambda) &= \bigvee_{a,b} \Lambda_{ab}, \\ H\mathfrak{M}_L(\Gamma)H &\simeq G_L(\text{End}_k(V)), & (A, \lambda) &\longmapsto (A, \text{const } \lambda), \\ G_L(\mathcal{C}_p) &\hookrightarrow H\mathfrak{M}_L(\Gamma)H, & (x, \lambda) &\longmapsto (\theta(x), \text{const } \lambda). \end{aligned} \quad (\text{CFP40})$$

The nonempty index set and finite-image join formula show that both sides of the first product have the stated constant label. If $A \neq 0$, one entry forces $\epsilon(\Lambda) = 1_L$. Conversely every constant-labelled pair obeying that carrier condition is fixed by H on both sides. Constant labels multiply by meet and add by join, giving the exact isomorphism. Its unit is H , which differs from the

ambient sparse unit at every off-diagonal zero amplitude. Its additive zero is the ambient $(0, 0)$; the image of global e is $(0, \mathbf{J})$. No identification of these three elements has been made.

For an exact action on the original finite amplitudes, let $\mathcal{W}_L$ consist of (w, α) with $w \in V$, $\alpha : \Gamma \rightarrow L$ of finite image, and $w_a \neq 0 \Rightarrow \alpha_a = 1_L$. Scalar action is $(c, \lambda)(w, \alpha) = (cw, (\lambda \wedge \alpha)_a)$. The matrix action and its corner are

$$\begin{aligned} (A, \Lambda)(w, \alpha) &= \left(Aw, \left(\bigvee_b \Lambda_{ab} \wedge \alpha_b \right)_a \right), \\ H(w, \alpha) &= (w, \text{const} \bigvee_a \alpha_a), \quad H\mathcal{W}_L \simeq G_L(V), \\ (A, \lambda)(w, \mu) &= (Aw, \lambda \wedge \mu) \quad \text{on the corner.} \end{aligned} \tag{CFP41}$$

The amplitude remains finitely supported because w is and A is column finite. The label joins have finitely many distinct values, remain in the original lattice, and satisfy the carrier test by a nonzero contributing amplitude. Finite distributivity proves module and action associativity. Testing the vector with amplitude ξ_b and only label b top extracts the entire b th column, so the ambient action is faithful. On the global corner, testing basis amplitudes and the vector $(0, 1_L)$ likewise recovers every amplitude and label.

The complete coefficient map also exists entrywise on this ambient space:

$$(A, \Lambda) \mapsto (A, (\iota_L(\Lambda_{ab}))_{a,b \in \Gamma}). \tag{CFP42}$$

In the support coordinate use finite-image matrices over K_L , with multiplication defined by the same finite sums of distinct values. Its idempotent addition permits the repeated-value interpretation used above. The morphism preserves every product and sum because ι_L preserves their finite lattice operations. The retraction ρ_L applied entrywise proves injectivity. It takes the constant corner label to its constant ι_L value, and takes every support defect below to its exact indicator. Thus this is the original full Φ on every coefficient, with no chosen support character or lost intermediate label.

51.10 The exact sparse defects, supported nilpotence, and a connecting morphism

For an amplitude matrix define its sparse lift by $\text{sp}(A) = (A, \mathbf{1}_{\text{supp } A})$, with actual top/bottom labels. For any set of entries D , write $Z_D = (0, \mathbf{1}_D)$. The exact addition and product comparison is

$$\begin{aligned} D_+(A, B) &= (\text{supp } A \cup \text{supp } B) \setminus \text{supp}(A + B), \\ D_\times(A, B) &= (\text{supp } A \circ \text{supp } B) \setminus \text{supp}(AB), \\ \text{sp}(A) + \text{sp}(B) &= \text{sp}(A + B) + Z_{D_+(A, B)}, \\ \text{sp}(A) \text{sp}(B) &= \text{sp}(AB) + Z_{D_\times(A, B)}. \end{aligned} \tag{CFP43}$$

Relational composition means the existence of an intermediate index with both entries nonzero. The sum's label is top on the union of supports, and its extra zero amplitudes are precisely the first defect set. A product's label is top precisely where a nonzero path exists, while its amplitude may cancel after summing paths; the second defect set is exactly those remaining entries. This proves every coefficient of both identities. In particular $\text{sp}(A) + \text{sp}(-A) = Z_{\text{supp } A}$ for nonzero A , so sparse lifting is not an additive ring map.

The source gives a sharper example that retains the characteristic- p factor:

$$\begin{aligned} W_{\text{sp}}U_{\text{sp}} &= (0, 0), \quad pI_{\text{sp}} = (0, \Delta), \\ (HW_{\text{sp}}H)(HU_{\text{sp}}H) &= (0, \mathbf{J}) = pH. \end{aligned} \tag{CFP44}$$

There is no surviving sparse path in WU , by (CFP33), so its support is bottom everywhere. The p additions of the sparse identity have zero amplitude and retain only diagonal top labels. In the

constant corner, both operators have top label on all pairs, so their product retains it even when its amplitude vanishes. This is exactly the faithful image of (CFP38), and proves why the two coefficient presentations cannot be identified by an unstated lift.

Likewise, for $0 < c < 1$ in Γ , translation M_{b_c} has its first zero power at $n = \lceil 1/c \rceil$. Before that power the index 0 survives; at that power every path ends at an index at least 1. Consequently

$$\begin{aligned} \text{sp}(M_{b_c})^n &= (0, 0), & (M_{b_c}, \mathbf{J})^n &= (0, \mathbf{J}), & n &= \lceil 1/c \rceil, \\ \widehat{b}_c^n &= e_{\mathcal{T}(p)} \neq \tau_{\mathcal{T}(p)}. \end{aligned} \tag{CFP45}$$

The scalar assertion (CFP15) therefore does not claim that every sparse matrix semiring lacks nonzero nilpotents to its ambient zero. The sparse matrices have them; the displayed full global lift retains their original supported nilpotence instead. Equations (CFP40)–(CFP45) are their exact connecting maps and defects.

Saturation $M \mapsto HMH$ is additive but is not generally multiplicative. For sparse matrix units E_{ij} and E_{kl} with $j \neq k$, their product has no path and saturates to zero, whereas the product of their nonzero saturated forms is $(0, \mathbf{J})$. There is nevertheless a precisely defined unital connecting morphism. For a finite-image label matrix let $r_a(\Lambda) = \bigvee_b \Lambda_{ab}$, $c_b(\Lambda) = \bigvee_a \Lambda_{ab}$, and retain $\epsilon(\Lambda) = \bigvee_{a,b} \Lambda_{ab}$. Then

$$\begin{aligned} \mathfrak{Z}_H &= \{(A, \Lambda) : r_a(\Lambda) = c_b(\Lambda) = \epsilon(\Lambda) \text{ for every } a, b\}, \\ q_H : \mathfrak{Z}_H &\twoheadrightarrow H\mathfrak{M}_L(\Gamma)H, & M &\mapsto HMH, \\ H(MN)H &= (HMH)(HMH) \iff \bigvee_b c_b(\Lambda) \wedge r_b(\Omega) = \epsilon(\Lambda) \wedge \epsilon(\Omega), & M &= (A, \Lambda), N = (B, \Omega). \end{aligned} \tag{CFP46}$$

All joins are again finite joins of distinct palette values. The labels of MH and HM at entry (a, b) are respectively $r_a(\Lambda)$ and $c_b(\Lambda)$. They are equal for every pair exactly when every row and column join is the same total join. Thus $\mathfrak{Z}_H$ is exactly the centralizer of H , a unital subsemiring. For its members one commutes the middle H past the factors and uses $H^2 = H$, proving multiplicativity of q_H . It sends I_{sp} to the corner unit H , is onto, and has the corner inclusion as an additive and multiplicative section, whose unit differs from the ambient unit. For arbitrary M, N , both compared amplitudes are AB . Expanding the total support join of their product gives exactly the last displayed criterion. This proves the full morphism on its stated domain and the precise obstruction to extending its multiplicativity to any additional pair. No global-to-sparse relation has been replaced by a claim of nonidentity alone.

$$\begin{array}{ccc} \widehat{\epsilon}_q & \xrightarrow{\text{qth power}} & e \neq \tau \\ \mathcal{R}_q \downarrow & & \downarrow \mathcal{R}_q \\ e & \xrightarrow{\text{qth power}} & e \end{array}$$

Figure 23: The finite characteristic- p thickening with its original support retained, CFP14–20. Reduction identifies $\widehat{\epsilon}_q$ with e by the exact equal-label congruence; it does not set either element to τ . Its full prime-spectrum map is a homeomorphism despite this nontrivial congruence. The source coordinates $\epsilon_q = T_q - 1$ and $\delta_q = 1 - T_q$ are related with their sign in CFP1, CFP5 and CFP22, following Connes–Consani–Marcolli, *Fun with $\mathbb{F}_1$, filtration*, lines 1937–1953.

52 The exact completed Hurwitz–theta bridge over the full support lattice

Original-object correction, 23 September 2026. The auxiliary functions in this chapter are compared with the original ζ and F_t ; they are not replacements for them. HTB13 already provides the fractional-module isomorphism at the exceptional points. CFD1–10 retains its comparison with the original regular lattice and all finite jets. ZTC13–16 proves the joint inverse with the boundary component kept. Projection to H_0 alone discards the actual Hurwitz deformation. The original arithmetic receiving formulas and both Mellin endpoint terms are given in ZTC1–21.

Objects and source provenance. The shifted family below is the one in the supplied April 2026 programme draft *Split-Zero Support, Residue-Parity Reversal, Shifted-Hurwitz CUE Defects, and Finite Phase-Space Enlargement*, printed byline “Compiled theorem draft”, source labels `thm:universal-flow` and `thm:zero-motion`. The actual source is `Chatnotes/globalization cue/split_zero_cue_shifted_hurwitz_phase_space.tex`, lines 602–630 and 1003–1060; the exact retained copy is under `hurwitz_intake/programme_source/`. No human author is specified in those supplied bytes, and none is inferred here. Its retained source has SHA-256 `52f2a6bccdb15213b1992abef9f8d6476d6e9ea5566ee7ed9d85c9a2d7cc2705`. The heat family and every factor in its coordinate map are those of Brad Rodgers and Terence Tao, *The de Bruijn–Newman constant is non-negative*, arXiv:1801.05914v5, original source `fmp-template.tex`, equations `hoz`, `sas`, `phidef`, and `htdef`, lines 78–101. That source credits N. C. de Bruijn’s *The roots of trigonometric integrals* (1950), C. M. Newman’s *Fourier transforms with only real zeroes* (1976), and the zero dynamics of G. Csordas, W. Smith, and R. S. Varga (1994). Those earlier papers have not been independently read for this chapter; their attribution is inherited explicitly from the original Rodgers–Tao source. The theta identity used here is proved below, including its Mellin factors. The full support carrier is the one in *Split Support Geometry*, v11, Definition `def:lattice-split`. Its operations and all maps needed here are stated and verified in the final subsection.

52.1 Two parameters and the original coordinate map

Use t for the Hurwitz shift and h for heat time. They are independent parameters. Put

$$\begin{aligned} a &= 1 + t, & \operatorname{Re} a > 0, & & F_t(s) &= \zeta(s, a), \\ A(s) &= \frac{s(s-1)}{2} \pi^{-s/2} \Gamma(s/2), & \Xi_t(s) &= A(s) F_t(s), \\ s(z) &= \frac{1}{2} + \frac{iz}{2}, & z(s) &= -2i \left(s - \frac{1}{2} \right), & H_t^{\text{Hur}}(z) &= \frac{1}{8} \Xi_t(s(z)). \end{aligned} \tag{HTB1}$$

For complex a in the indicated half-plane, $(n+a)^{-s}$ uses the holomorphic logarithm with argument in $(-\pi/2, \pi/2)$. All subsequent identities in s are identities of meromorphic functions; values at removable singularities mean their analytic extensions. Statements about real parameters explicitly use $t > -1$.

Here is a direct continuation of the Hurwitz input, so that neither its pole nor the endpoint values are assumed in the comparison. Define Bernoulli polynomials by

$$\frac{x e^{bx}}{e^x - 1} = \sum_{k=0}^{\infty} B_k(b) \frac{x^k}{k!}, \quad |x| < 2\pi. \tag{HTB2}$$

For $\operatorname{Re} s > 1$, termwise integration gives

$$\Gamma(s) F_t(s) = \int_0^{\infty} x^{s-1} \frac{e^{-ax}}{1 - e^{-x}} dx. \tag{HTB3}$$

The interchange is absolutely convergent on compact subsets of $\operatorname{Re} s > 1$, $\operatorname{Re} a > 0$: near zero the absolute integrand is bounded by a constant times $x^{\operatorname{Re} s - 2}$, and at infinity it decays exponentially. Subtracting the first $N + 1$ terms of HTB2 on $(0, 1)$ gives, for $\operatorname{Re} s > -N$,

$$\begin{aligned} F_t(s) = \frac{1}{\Gamma(s)} & \left\{ \int_1^\infty x^{s-1} \frac{e^{-ax}}{1 - e^{-x}} dx \right. \\ & + \int_0^1 x^{s-1} \left(\frac{e^{-ax}}{1 - e^{-x}} - \sum_{k=0}^N \frac{B_k(1-a)}{k!} x^{k-1} \right) dx \\ & \left. + \sum_{k=0}^N \frac{B_k(1-a)}{k!(s+k-1)} \right\}. \end{aligned} \quad (\text{HTB4})$$

The remainder is $O(x^N)$ locally uniformly in a . Differentiating in a and s preserves locally integrable majorants on smaller compact sets. Thus HTB4 gives joint meromorphic continuation, holomorphic in a , with compatible values as N increases.

The recurrence $\Gamma(s+1) = s\Gamma(s)$ and $\Gamma(1) = 1$ show that Γ has residue $(-1)^n/n!$ at $-n$. Its reciprocal is entire with simple zeros there and no other zeros: for example this follows directly from the locally uniformly convergent Euler product $1/\Gamma(s) = se^{\gamma s} \prod_{n \geq 1} (1 + s/n)e^{-s/n}$, whose nonvanishing away from the listed factors follows from the absolute convergence of the logarithmic tail. In HTB4 these reciprocal zeros cancel every displayed pole except the one at $s = 1$. At $s = -n$ only the term $k = n + 1$ contributes to the value. Replacing x by $-x$ in HTB2 proves $B_k(1-a) = (-1)^k B_k(a)$. Consequently

$$\operatorname{Res}_{s=1} F_t(s) = 1, \quad F_t(-n) = -\frac{B_{n+1}(a)}{n+1} \quad (n \geq 0), \quad \partial_t F_t(s) = -sF_t(s+1). \quad (\text{HTB5})$$

For the last equality differentiate the convergent series on $\operatorname{Re} s > 1$; HTB4 and uniqueness of meromorphic continuation extend it everywhere. In particular $F_t(0) = 1/2 - a$.

52.2 The theta integral with all original constants

For $x > 0$, write

$$\psi(x) = \sum_{n=1}^{\infty} e^{-\pi n^2 x}, \quad \Theta(x) = 1 + 2\psi(x).$$

The Gaussian identity gives

$$\Theta(x) = x^{-1/2} \Theta(1/x), \quad \psi(1/x) = \sqrt{x} \psi(x) + \frac{\sqrt{x} - 1}{2}. \quad (\text{HTB6})$$

One direct proof of the required Gaussian identity is to periodize $e^{-\pi xy^2}$. The periodized series and all its derivatives converge uniformly on $[0, 1]$. Its k th Fourier coefficient is $\int_{\mathbb{R}} e^{-\pi xy^2} e^{-2\pi iky} dy = x^{-1/2} e^{-\pi k^2/x}$. This integral follows by scaling from $\int e^{-\pi y^2} dy = 1$ and differentiating its Fourier transform: integration by parts gives $\widehat{f}'(\eta) = -2\pi\eta \widehat{f}(\eta)$, with $\widehat{f}(0) = 1$. The absolutely convergent Fourier series of the periodization evaluated at zero proves HTB6.

Define the original kernel, without changing its argument or amplitude, by

$$\begin{aligned} \Phi(u) &= \sum_{n=1}^{\infty} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}} \\ &= e^u (2x^2 \psi''(x) + 3x \psi'(x))_{x=e^{4u}}. \end{aligned} \quad (\text{HTB7})$$

If $g(u) = e^u \psi(e^{4u})$, direct differentiation and HTB6 give

$$\Phi(u) = \frac{1}{8} (g''(u) - g(u)), \quad g(-u) = g(u) + \sinh u, \quad \Phi(-u) = \Phi(u). \quad (\text{HTB8})$$

Indeed $g'' - g = 24e^{5u}\psi'(e^{4u}) + 16e^{9u}\psi''(e^{4u})$, and $\partial_u^2 - 1$ annihilates $\sinh u$ and commutes with reflection. For $u \geq 0$, each summand in HTB7 is strictly positive because $2\pi n^2 e^{4u} - 3 > 0$. Moreover the series and each fixed number of its derivatives are bounded by a constant times $e^{Cu}e^{-\pi e^{4u}}$: factor out $e^{-\pi e^{4u}}$ and bound the remaining sums of powers of n times $e^{-\pi(n^2-1)}$. Evenness gives the corresponding estimate as $u \rightarrow -\infty$. These bounds justify all differentiations and integrations that follow.

For $\operatorname{Re} s > 1$, substitute $x = e^{4u}$ and integrate by parts twice. The endpoint terms vanish, at infinity exponentially and at zero by HTB6 and $\operatorname{Re} s > 1$. The exact calculation is

$$\begin{aligned} \int_{\mathbb{R}} \Phi(u) e^{(2s-1)u} du &= \frac{1}{4} \int_0^\infty \left(2x^{s/2+1} \psi''(x) + 3x^{s/2} \psi'(x) \right) dx \\ &= \frac{s(s-1)}{8} \int_0^\infty \psi(x) x^{s/2-1} dx \\ &= \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s) = \frac{1}{4} \xi(s). \end{aligned} \quad (\text{HTB9})$$

For clarity, with $b = s/2$ the integration-by-parts multiplier is $\frac{1}{4}(2b(b+1) - 3b) = s(s-1)/8$. At zero HTB6 gives $\psi(x) = \frac{1}{2}x^{-1/2} - \frac{1}{2} + O(x^{-1/2}e^{-\pi/x})$, and the derivatives give the claimed vanishing boundary terms. The last interchange in HTB9 is absolutely convergent for $\operatorname{Re} s > 1$. The left-hand side is entire by the preceding super-exponential bounds. It therefore supplies the entire continuation of $\xi = A\zeta$ and, by evenness, proves $\xi(s) = \xi(1-s)$. All its coefficients respect complex conjugation because the kernel is real.

The original heat family is thus

$$\begin{aligned} H_h^{\text{heat}}(z) &= \int_0^\infty e^{hu^2} \Phi(u) \cos(zu) du, \quad (h, z) \in \mathbb{C}^2, \\ H_0^{\text{heat}}(z) &= \frac{1}{8} \xi(s(z)) = H_0^{\text{Hur}}(z), \quad \partial_h H_h^{\text{heat}} = -\partial_z^2 H_h^{\text{heat}}. \end{aligned} \quad (\text{HTB10})$$

Joint holomorphy and the differential equation follow from locally uniform integrable majorants for every h, z derivative. To obtain the middle identity, put $2s-1 = iz$ in HTB9 and use evenness: the integral over the whole real axis is twice the cosine integral. Hence the factor is exactly $1/8$. Every H_h^{heat} is even; for real h it also satisfies $H_h^{\text{heat}}(\bar{z}) = \overline{H_h^{\text{heat}}(z)}$.

52.3 An invertible meromorphic map and its exact product

Let $\mathcal{M}_s$ and $\mathcal{M}_z$ be the fields of meromorphic functions on the entire s -plane and z -plane, respectively. The following maps are everywhere defined as maps of these fields viewed as complex vector spaces:

$$U : \mathcal{M}_s \longrightarrow \mathcal{M}_z, \quad (Uf)(z) = \frac{A(s(z))}{8} f(s(z)), \quad (U^{-1}G)(s) = \frac{8G(z(s))}{A(s)}. \quad (\text{HTB11})$$

Since A is a nonzero meromorphic function and s, z are inverse affine coordinates, these formulas prove complex linearity and both inverse identities. With ordinary pointwise products U is not multiplicative: its two product expressions differ by a factor $A/8$. The exact transported product and unit on the same meromorphic vector space are

$$(G \odot J)(z) = \frac{8G(z)J(z)}{A(s(z))}, \quad \mathbf{1}_\odot(z) = \frac{A(s(z))}{8}, \quad U(fq) = Uf \odot Uq. \quad (\text{HTB12})$$

All field axioms follow either by this inverse transport or by the displayed multiplication formula. Thus U is a unital field isomorphism to $(\mathcal{M}_z, \odot)$, as well as a linear isomorphism to the ordinary meromorphic receiver. These two assertions use different, explicitly stated products. In particular the ordinary constant function 1 is not asserted to be its transported unit.

There is an exact local regularity rule. At a finite s_0 , put $z_0 = z(s_0)$ and let $\nu_{s_0}(f)$ denote meromorphic order, with $\nu(0) = +\infty$. Since the coordinate derivative $ds/dz = i/2$ is nonzero,

$$\nu_{z_0}(Uf) = \nu_{s_0}(A) + \nu_{s_0}(f), \quad \mathcal{D}_{s_0} = \{f : \nu_{s_0}(f) \geq -\nu_{s_0}(A)\} \xrightarrow[U]{\cong} \mathcal{O}_{z_0}. \quad (\text{HTB13})$$

Here the domain is a fractional module of meromorphic germs and the arrow is a module isomorphism over the affine coordinate isomorphism; it is not claimed to be a subring with the ordinary product. At a point where A is a holomorphic unit, it restricts to an isomorphism of holomorphic modules and satisfies the exact evaluation formula

$$\text{ev}_{z_0} Uf = \frac{A(s_0)}{8} \text{ev}_{s_0} f.$$

At a zero or pole of A , HTB13, rather than an undefined product of separate point values, specifies the evaluation domain.

For example, if A is a unit at s_0 and F_t has a zero there, ordinary affine substitution, denoted $C(f) = f \circ s$, induces the unital algebra isomorphism

$$\mathcal{O}_{s_0}/(F_t) \xrightarrow[C]{\cong} \mathcal{O}_{z_0}/(H_t^{\text{Hur}}). \quad (\text{HTB14})$$

Indeed $UF_t = (A \circ s)C(F_t)/8$ and $A \circ s$ is a unit, so the two target principal ideals coincide. This proves equality of local multiplicities and the correspondence of every jet, without changing their coordinate scale. Explicitly, for $w = z - z_0$,

$$(Uf)(z_0 + w) = \frac{1}{8} \sum_{n \geq 0} \left(\frac{i}{2}\right)^n w^n \sum_{j=0}^n \frac{A^{(j)}(s_0) f^{(n-j)}(s_0)}{j!(n-j)!}.$$

Thus the complete finite jet map is triangular, with nonzero diagonal $A(s_0)(i/2)^n/8$. It is C , not multiplication by that unit, which gives the ordinary unital algebra arrow in HTB14.

52.4 Both infinitesimal generators and their difference

Define $D : \mathcal{M}_s \rightarrow \mathcal{M}_s$ by $Df(s) = sf(s+1)$. All differential and shift operators in the next display have the entire meromorphic field as their algebraic domain; no bounded-operator or semigroup assertion on an unspecified normed space is being used. Direct use of HTB11 gives

$$\begin{aligned} (D_H G)(z) &:= (UDU^{-1}G)(z) = c(s(z))G(z-2i), \\ c(s) &= \frac{sA(s)}{A(s+1)} = \sqrt{\pi} \frac{s(s-1)}{s+1} \frac{\Gamma(s/2)}{\Gamma((s+1)/2)}, \\ \partial_t H_t^{\text{Hur}} &= -D_H H_t^{\text{Hur}}. \end{aligned} \quad (\text{HTB15})$$

The sign and displacement follow from $s(z-2i) = s(z) + 1$, and the last identity follows from HTB5. Gamma recurrence gives the removable values $c(0) = -2$, $c(-1) = -2\pi$, and $c(1) = 0$. The complete product in HTB15 is meromorphic: at negative odd integers below -1 the vanishing multiplier may multiply a pole of the shifted function, and evaluation is performed after multiplication. For instance, at $s = 0$ the formula will give $\partial_t H_t^{\text{Hur}}(i) = 1/8$, exactly as HTB20 below.

The original heat generator $B = -\partial_z^2$ has pullback

$$\mathcal{L}_H := U^{-1}BU, \quad \mathcal{L}_H f(s) = \frac{1}{4A(s)} \frac{d^2}{ds^2} (A(s)f(s)). \quad (\text{HTB16})$$

This includes both powers from $ds/dz = i/2$ and the minus sign in B ; the original factors 8 in U and U^{-1} cancel each other. The actual pulled-back heat trajectory is $f_h(s) = 8H_h^{\text{heat}}(z(s))/A(s)$,

with $f_0 = \zeta$ and $\partial_h f_h = \mathcal{L}_H f_h$. No assertion is needed that an operator exponential converges on every meromorphic function.

The two actual generators in the same s -space are $\mathcal{L}_H$ and $-D$. Their exact difference is the operator

$$\begin{aligned}\mathcal{E}f(s) &:= (\mathcal{L}_H + D)f(s) \\ &= \frac{1}{4}f''(s) + \frac{b(s)}{2}f'(s) + \frac{b'(s) + b(s)^2}{4}f(s) + sf(s+1), \\ b(s) &= \frac{A'(s)}{A(s)} = \frac{1}{s} + \frac{1}{s-1} - \frac{\log \pi}{2} + \frac{1}{2} \frac{\Gamma'(s/2)}{\Gamma(s/2)}, \\ (B + D_H)U &= U\mathcal{E}.\end{aligned}\tag{HTB17}$$

The expansion follows by applying the ordinary product rule twice to HTB16. Every displayed coefficient is interpreted meromorphically; apparent singularities from the expanded expression do not change the domain just specified. HTB17 is the exact map of the generator defect, not a declaration that the two flows are unrelated.

52.5 All poles, the entire parameters, and reflection

The only finite singularities of A are simple poles at $s = -2m$, $m \geq 1$. The apparent pole at zero is removable and $A(0) = -1$; $s = 1$ is its only zero, and $A(s) = (s-1)/2 + O((s-1)^2)$ there. These statements follow from the gamma pole calculation following HTB4, the nonvanishing of gamma away from its poles, and the two polynomial factors of A . Hence HTB5 gives

$$\begin{aligned}\Xi_t(0) &= t + \frac{1}{2}, & \Xi_t(1) &= \frac{1}{2}, \\ a_m &:= \operatorname{Res}_{s=-2m} A(s) = \frac{2(-1)^m(2m+1)\pi^m}{(m-1)!}, \\ r_m(t) &:= \operatorname{Res}_{s=-2m} \Xi_t(s) = \frac{2(-1)^{m+1}\pi^m}{(m-1)!} B_{2m+1}(1+t).\end{aligned}\tag{HTB18}$$

These are all possible poles of Ξ_t . A listed point is a pole exactly when $r_m(t) \neq 0$; otherwise it is removable and its value is $a_m \partial_s F_t(-2m)$. To check every constant, the residue of $\Gamma(s/2)$ in the s -coordinate is $2(-1)^m/m!$ and $(-2m)(-2m-1)/2 = m(2m+1)$. Multiplying their product by $F_t(-2m) = -B_{2m+1}(1+t)/(2m+1)$ gives HTB18. The endpoint at one uses the residue 1 of F_t , not its undefined point value.

In particular the first residue and the exact classification are

$$\begin{aligned}r_1(t) &= 2\pi(1+t) \left(t + \frac{1}{2}\right) t, \\ t \in \mathbb{R}, t > -1 : & \quad \Xi_t \text{ is entire} \iff t \in \left\{0, -\frac{1}{2}\right\}, \\ \Xi_0(s) &= \xi(s), \quad \Xi_{-1/2}(s) = (2^s - 1)\xi(s).\end{aligned}\tag{HTB19}$$

Indeed $B_3(a) = a(a-1/2)(a-1)$ follows by expanding HTB2. Thus the first pole alone excludes every other $t > -1$. At zero, entireness was proved in HTB9. At $t = -1/2$, the absolutely convergent identity $\zeta(s, 1/2) = 2^s \sum_{n \geq 0} (2n+1)^{-s} = (2^s - 1)\zeta(s)$ extends meromorphically and proves entireness. This also proves that no unexamined higher gamma pole invalidates the converse.

The same statements in the original receiver coordinate are

$$\begin{aligned}z_m &= (4m+1)i, & \operatorname{Res}_{z=z_m} H_t^{\text{Hur}}(z) &= \frac{r_m(t)}{4i} = \frac{i(-1)^m \pi^m}{2(m-1)!} B_{2m+1}(1+t), \\ H_t^{\text{Hur}}(i) &= \frac{2t+1}{16}, & H_t^{\text{Hur}}(-i) &= \frac{1}{16}.\end{aligned}\tag{HTB20}$$

The residue factor is $1/(8s'(z_m)) = 1/(4i)$; it is different from the value factor $1/8$. In particular the tangent to the Hurwitz trajectory at zero has a genuine pole:

$$\operatorname{Res}_{s=-2} \partial_t \Xi_t(s)|_{t=0} = \pi, \quad \operatorname{Res}_{z=5i} \partial_t H_t^{\text{Hur}}(z)|_{t=0} = \frac{\pi}{4i}. \quad (\text{HTB21})$$

This is obtained by differentiating the explicit polynomial $r_1(t)$. In contrast $\partial_h H_h^{\text{heat}}|_{h=0}$ is entire by HTB10. Thus HTB17 is nonzero even on the actual initial zeta function; the difference is witnessed by a specified principal part, without a change of coordinate, multiplication factor, or zero convention.

Define the meromorphic reflection defect by

$$\begin{aligned} R_t(s) &= \Xi_t(s) - \Xi_t(1-s), & R_t(1-s) &= -R_t(s), \\ R_t(0) &= t, & R_t(1) &= -t, & H_t^{\text{Hur}}(z) - H_t^{\text{Hur}}(-z) &= \frac{1}{8} R_t(s(z)). \end{aligned} \quad (\text{HTB22})$$

For real $t > -1$, its identically vanishing locus is exactly $t = 0$: necessity follows by evaluation at zero, and sufficiency follows from HTB9. At the other entire parameter, $R_{-1/2}(s) = (2^s - 2^{1-s})\xi(s)$, which is nonzero at $s = 0$. If $r_m(t) \neq 0$, R_t has simple poles at both $-2m$ and $1 + 2m$, with residue $r_m(t)$ at each. The reflection changes the sign of a residue and the subtraction changes it again; the two pole sets are disjoint. For real t , $\Xi_t(\bar{s}) = \overline{\Xi_t(s)}$ and therefore $H_t^{\text{Hur}}(-\bar{z}) = \overline{H_t^{\text{Hur}}(z)}$. It is not legitimate to replace the last identity by reality on the real z -axis when the reflection defect is nonzero.

The reflection is also an exact operator on the source field:

$$\begin{aligned} (J_A f)(s) &= \frac{A(1-s)}{A(s)} f(1-s), & J_A^2 &= I, & U J_A &= J_z U, & (J_z G)(z) &= G(-z), \\ [D, J_A]f(s) &= s \frac{A(-s)}{A(s+1)} f(-s) - (1-s) \frac{A(1-s)}{A(s)} f(2-s). \end{aligned} \quad (\text{HTB23})$$

Both assertions follow by substitution; their denominators are nonzero elements of the meromorphic field. In particular $R_t/8$ in the receiver is $(I - J_z)U F_t$, or $U(I - J_A)F_t$. This gives a map of the entire reflection defect, including all its principal parts. The heat generator commutes with J_z , whereas its Hurwitz counterpart has exactly the commutator transported from HTB23. The two projections $(I \pm J_z)/2$ retain the complete even and odd meromorphic parts; no part is discarded in their direct sum.

52.6 The exact first transverse velocity at a critical zero

Let ρ be a simple nontrivial zeta zero with $\operatorname{Re} \rho = 1/2$. This subsection concerns each such zero; it assumes neither that every nontrivial zero is simple nor that every nontrivial zero lies on the line. At ρ , A is a holomorphic unit, so HTB14 applies. Joint holomorphy from HTB4 and the implicit function theorem give the unique local root $\rho(t)$ with $F_t(\rho(t)) = 0$, $\rho(0) = \rho$. Differentiating that equation gives

$$\rho'(0) = c_1(\rho) = \frac{\rho \zeta(\rho+1)}{\zeta'(\rho)}, \quad z_\rho(t) = -2i \left(\rho(t) - \frac{1}{2} \right), \quad \operatorname{Im} z'_\rho(0) = -2 \operatorname{Re} c_1(\rho). \quad (\text{HTB24})$$

The numerator is nonzero: $\rho \neq 0$ and the absolutely convergent Euler product for $\zeta(\rho+1)$ has no zero because $\operatorname{Re}(\rho+1) > 1$. This proves $c_1(\rho) \neq 0$, but does not prove that its real part is nonzero.

At a zero $\xi'(\rho) = A(\rho)\zeta'(\rho)$. Reflection and conjugation give $\xi'(1-\rho) = -\xi'(\rho)$ and $\overline{\xi'(\rho)} = \xi'(\bar{\rho}) = -\xi'(\rho)$, so this derivative is nonzero and purely imaginary. Similarly $c_1(1-\rho) = c_1(\rho)$.

Differentiating HTB22 at the fixed point ρ therefore yields the exact identity

$$\begin{aligned}\partial_t R_t(\rho)|_{t=0} &= -\xi'(\rho)c_1(\rho) + \xi'(1-\rho)c_1(1-\rho) \\ &= -2\xi'(\rho) \operatorname{Re} c_1(\rho), \\ \partial_t(H_t^{\text{Hur}}(z_\rho) - H_t^{\text{Hur}}(-z_\rho))|_{t=0} &= -\frac{1}{4}\xi'(\rho) \operatorname{Re} c_1(\rho), \quad z_\rho = z(\rho).\end{aligned}\tag{HTB25}$$

Thus first-order reflection failure at this zero and its first-order transverse velocity vanish together, with the original nonzero factor shown. This proves an exact relation at every critical simple zero; it does not extrapolate a finite numerical nonvanishing list to all zeros. The meromorphic global defect has already been determined without such an extrapolation in HTB18–HTB23.

52.7 The retained arithmetic correction and its boundary kernel

The full difference from the original arithmetic function is a specified entire function before completion:

$$\begin{aligned}D_t^{\text{ar}}(s) &= \zeta(s) - F_t(s), \quad C_t(s) = A(s)D_t^{\text{ar}}(s), \quad J_t(z) = \frac{1}{8}C_t(s(z)), \\ F_t + D_t^{\text{ar}} &= \zeta, \quad \Xi_t + C_t = \xi, \quad H_t^{\text{Hur}} + J_t = H_0^{\text{heat}}.\end{aligned}\tag{HTB26}$$

Both summands of D_t^{ar} have residue 1 at their only possible pole, so their difference is entire. There is a convergent boundary-kernel formula on the larger half-plane $\operatorname{Re} s > 0$:

$$D_t^{\text{ar}}(s) = \frac{1}{\Gamma(s)} \int_0^\infty x^{s-1} d_t(x) dx, \quad d_t(x) = \frac{e^{-x}(1 - e^{-tx})}{1 - e^{-x}}, \quad d_t(0) = t.\tag{HTB27}$$

At infinity both exponential terms decay for $\operatorname{Re}(1+t) > 0$; at zero Taylor expansion gives $d_t(x) = t + O(x)$. These facts prove absolute convergence and local uniformity on the indicated domain. On $\operatorname{Re} s > 1$ it is the difference of HTB3 for $a = 1$ and $a = 1+t$, so it gives the stated continuation by uniqueness. All remaining values are fixed by the entire continuation just proved. Differentiation gives $\partial_t D_t^{\text{ar}} = sF_t(s+1)$, also directly from HTB5.

This correction retains every completed boundary contribution:

$$\begin{aligned}C_t(0) &= -t, \quad C_t(1) = 0, \quad \operatorname{Res}_{s=-2m} C_t = -r_m(t), \\ C_t(s) - C_t(1-s) &= -R_t(s), \quad \partial_t C_t(s) = sA(s)F_t(s+1).\end{aligned}\tag{HTB28}$$

Each statement follows by subtracting the corresponding functions in HTB26, using entireness and reflection of ξ proved in HTB9, or by differentiation. The cancellation of all the principal parts and of the entire reflection defect is an identity of the original analytic functions. It does not assert a zero-location theorem for any summand.

52.8 Every map over the original lattice and its faithful receiver

Let L be any nontrivial bounded distributive lattice. There is no finiteness, Boolean-special-case, or one-point restriction here. For a complex vector space V , define the full support semimodule

$$\begin{aligned}G_L(V) &= \{(0, \lambda) : \lambda \in L\} \cup (V \times \{1_L\}), \\ (v, \lambda) + (w, \mu) &= (v+w, \lambda \vee \mu), \quad (c, \alpha)(v, \lambda) = (cv, \alpha \wedge \lambda), \\ \tau &= (0, 0_L), \quad e = (0, 1_L).\end{aligned}\tag{HTB29}$$

The scalars are $G_L(\mathbb{C})$ with its original addition and multiplication. Closure follows because a nonzero amplitude forces each participating amplitude that produces it to have top support. The semimodule axioms follow componentwise from vector-space laws and the two distributive

laws of L . In particular an amplitude cancellation has the join of the original labels, rather than being replaced by τ . If V is a commutative algebra, its supported multiplication is $(v, \lambda)(w, \mu) = (vw, \lambda \wedge \mu)$; it has exactly the same distributivity verification.

For every complex-linear map $T : V \rightarrow W$ occurring above, define

$$T^\uparrow(v, \lambda) = (Tv, \lambda). \quad (\text{HTB30})$$

This is well-defined because $v = 0$ implies $Tv = 0$. Linearity gives preservation of addition and of the displayed scalar action, including all zero amplitudes. Composition and inverse maps lift exactly. Thus $U^\uparrow$ is a semimodule isomorphism, the generators and their difference commute with it as in HTB15–HTB17, and both reflection projections and their zero-amplitude fibres lift with their original labels, as specified in HTB33. Their semimodule kernels, meaning the inverse images of τ , are the different fibres identified there. Where the transported algebra of HTB12 is used, $U^\uparrow$ is also a unital semiring isomorphism, with unit $(A \circ s/8, 1_L)$. Its induced map on prime spectra is the usual preimage map: an ideal is prime exactly when its inverse image is prime, since a bijective multiplicative map preserves and reflects membership in a product. This applies to every support prime as well as every arithmetic prime; no prime stratum is erased by the bridge.

To verify the original faithful comparison, let K be the programme's tropical semifield with zero and Boolean subsemiring $\mathbb{B}$. Its map $\beta : K \rightarrow \mathbb{B}$ sends zero to zero and every nonzero element to one. It preserves products, and preserves sums because a sum in an idempotent semifield is zero only when both summands are zero. Regard L as the $\mathbb{B}$ -algebra with addition $\vee$ and multiplication $\wedge$, and set

$$\begin{aligned} K_L &= K \otimes_{\mathbb{B}} L, & \iota_L(\lambda) &= 1_K \otimes \lambda, & \rho_L &= \beta \otimes \text{id}_L, & \rho_L \iota_L &= \text{id}_L, \\ \Phi_V : G_L(V) &\longrightarrow V \times K_L, & \Phi_V(v, \lambda) &= (v, \iota_L(\lambda)), \\ (T \times \text{id}_{K_L})\Phi_V &= \Phi_W T^\uparrow. \end{aligned} \quad (\text{HTB31})$$

The tensor universal property proves the retraction formula. It proves injectivity of ι_L and therefore of Φ_V . The coordinate laws verify addition and the $G_L(\mathbb{C})$ -action in the product receiver. When V is an algebra they also verify multiplication. The last line is a coordinate equality, and supplies the commuting square for every previously specified morphism, not just for a scalar amplitude map.

For regular germs, evaluation has its own such square. At s_0 where A is a unit, HTB13 lifts to

$$\begin{aligned} \text{ev}_{z_0}^\uparrow U^\uparrow(f, \lambda) &= \left(\frac{A(s_0)}{8} f(s_0), \lambda \right), \\ \Phi_{\mathbb{C}} \text{ev}_{z_0}^\uparrow U^\uparrow &= (\text{ev}_{z_0} U \times \text{id}_{K_L}) \Phi_{\mathcal{O}_{s_0}}. \end{aligned} \quad (\text{HTB32})$$

At zeros or poles of A the first domain is replaced by the exact fractional module $\mathcal{D}_{s_0}$ of HTB13; evaluation of Uf is defined there and its lifted square still holds. The unital local algebra arrow of HTB14 likewise gives its full G_L arrow, and retains the support of every nilpotent jet or vanishing amplitude.

In particular, at a regular zero of an original top-supported analytic function the evaluated element is $e = (0, 1_L)$, whose image is $(0, 1_{K_L})$. It is not τ , whose image is $(0, 0_{K_L})$. For a linear defect operator T , the entire inverse image of all zero-amplitude labels is

$$\{x : T^\uparrow x \in \{(0, \lambda) : \lambda \in L\}\} = G_L(\ker T), \quad (T^\uparrow)^{-1}(\tau) = \{\tau\}, \quad (T^\uparrow)^{-1}(e) = \ker T \times \{1_L\}. \quad (\text{HTB33})$$

The first equality is just $Tv = 0$ with its unchanged label. The second uses the defining fact that label 0_L permits only amplitude zero; the third uses the definition of e . These are three different, fully specified fibres.

Finally the correction identity HTB26 lifts without support loss:

$$(H_t^{\text{Hur}}, 1_L) + (J_t, 1_L) = (H_0^{\text{heat}}, 1_L), \quad \Phi(H_t^{\text{Hur}}, 1_L) + \Phi(J_t, 1_L) = \Phi(H_0^{\text{heat}}, 1_L). \quad (\text{HTB34})$$

This uses $1_L \vee 1_L = 1_L$ and its tensor image; it remains true at points where the two amplitudes cancel, provided evaluations are taken after the regular analytic sum. At an original zero, the sum is supported zero. Separate summands with poles are not individually evaluated. Thus the exact analytic correction, its reflection and pole defects, the original theta flow, and the arbitrary-lattice receiver form one commuting construction. The construction proves the stated maps and defects; it supplies no conclusion that all original zeta zeros lie on, or fail to lie on, the critical line.

52.9 The exact boundary correction to the logarithmic receiver

For every real $t \geq -1/2$, $H_t^{\text{Hur}}(0) > 0$. To prove this without a numerical input, HTB7 and HTB10 give $H_0^{\text{heat}}(0) > 0$. Since $A(1/2) = -\Gamma(1/4)/(8\pi^{1/4}) < 0$, it follows that $\zeta(1/2) < 0$. At $a = 1/2$ the identity used in HTB19 gives $\zeta(1/2, 1/2) = (\sqrt{2} - 1)\zeta(1/2) < 0$, while HTB5 gives $\partial_a \zeta(1/2, a) = -\zeta(3/2, a)/2 < 0$ for every $a > 0$; the last sign follows from its absolutely convergent positive series. Thus $F_t(1/2) < 0$ for $a \geq 1/2$, and the assertion follows from HTB1.

Fix such a t . On a sufficiently small disc about $z = 0$, both $H_0 = H_0^{\text{heat}}$ and $H_t = H_t^{\text{Hur}}$ are holomorphic and nonzero. Define there $c_t = J_t/H_0$, retaining the correction J_t of HTB26. Direct differentiation gives

$$\begin{aligned} \frac{H'_t}{H_t} - \frac{H'_0}{H_0} &= -\frac{c'_t}{1 - c_t}, \quad W_t(z) := \frac{H_t(z)H_t(-z)}{H_0(z)^2} = (1 - c_t(z))(1 - c_t(-z)), \\ b_n(t) &:= -\frac{1}{2}[z^{2n-1}] \left(\frac{H'_t(z)}{H_t(z)} - \frac{H'_t(-z)}{H_t(-z)} \right), \quad n \geq 1, \\ b_n(t) &= b_n(0) - \frac{1}{2}[z^{2n-1}] \frac{W'_t(z)}{W_t(z)}. \end{aligned} \quad (\text{HTB35})$$

The brackets mean the indicated Taylor coefficient at zero. No branch of a global logarithm is used. For the last identity differentiate $H_t(z)H_t(-z) = H_0(z)^2 W_t(z)$, using evenness of H_0 . This proves the correction to every coefficient of the even logarithmic receiver, with no finite truncation. All terms are fixed by HTB27 and the original theta integral.

The $b_n(t)$ are defined here as logarithmic coefficients, not as uncorrected zero-only moments of a meromorphic function. The exact distinction is its local divisor: if $H_t(z) = (z - z_*)^\nu u(z)$ with $u(z_*) \neq 0$, then $H'_t/H_t = \nu/(z - z_*) + u'/u$. A zero of multiplicity m contributes $\nu = m$ and an actual pole contributes $\nu = -1$. HTB18–HTB20 specify all those negative contributions and every cancellation. Thus any global zero-only interpretation must retain exactly these pole terms. HTB35 constructs the complete local correction to the arithmetic receiver, and HTB31 applies to its linear coefficient extractions with every support label unchanged. It does not assert that the nonlinear logarithm is a linear morphism of split carriers.

53 Completion, local lattices, and the exact observable of support

Sources and scope. The analytic source is Brad Rodgers and Terence Tao, *The de Bruijn–Newman constant is non-negative*, original author source equations `phidef`, `htdef`, `hoz`, and `sas`. Its kernel, coordinates, and Mellin identity are proved directly in the included HTB1–10; the convergence and positivity needed here are proved in THS1–15. The shifted Hurwitz family is the supplied April 2026 programme draft, byline “Compiled theorem draft”, equations `thm:universal-flow` and `thm:zero-motion`, with no inferred human attribution. The full support carrier and its public source are stated in HTB29–31 and HZA1. The complete HTB, THS, and HZA sources were read for this derivation. All new comparisons below are proved here. A lift of a classical analytic function to that carrier is not thereby a geometric construction of that function from the entire arithmetic base.

53.1 The fractional module and every finite jet

Retain the exact constants

$$A(s) = \frac{s(s-1)}{2} \pi^{-s/2} \Gamma(s/2), \quad s(z) = \frac{1}{2} + \frac{iz}{2}, \quad z(s) = -2i(s-1/2), \quad (Uf)(z) = \frac{A(s(z))}{8} f(s(z)). \quad (\text{CFD1})$$

Fix a finite s_0 , set $h = s - s_0$, $z_0 = z(s_0)$, and $w = z - z_0$. Write $A = h^k a(h)$, where $k \in \mathbb{Z}$ and a is a holomorphic unit. Let $\mathcal{O}_s$ and $\mathcal{O}_z$ be the respective rings of holomorphic germs, and $\mathcal{M}_s$ the field of meromorphic germs. Define the specified embedded fractional module

$$I = A^{-1} \mathcal{O}_s = h^{-k} \mathcal{O}_s \subset \mathcal{M}_s. \quad U : I \longrightarrow \mathcal{O}_z, \quad U^{-1}g = 8g(z(s))/A(s). \quad (\text{CFD2})$$

The equality of fractional modules uses that a is a unit. The two displayed substitutions compose to the identity. Thus this is an isomorphism of modules over the coordinate isomorphism $C : \mathcal{O}_s \rightarrow \mathcal{O}_z$, $C(h) = iw/2$. In particular

$$U(h^{N+1}I) = w^{N+1}\mathcal{O}_z, \quad I/h^{N+1}I \xrightarrow{\bar{U}} \mathcal{O}_z/w^{N+1}\mathcal{O}_z \quad (N \geq 0) \quad (\text{CFD3})$$

are isomorphisms. No undefined product of point values is used at a pole or zero of A .

For the exact coefficient matrices, write $a(h) = \sum_{j \geq 0} a_j h^j$ and $f = h^{-k} \sum_{j \geq 0} f_j h^j$. The coefficient of w^n in Uf is

$$g_n = \frac{1}{8} \left(\frac{i}{2} \right)^n \sum_{j=0}^n a_j f_{n-j}, \quad f_n = \sum_{j=0}^n d_j (2/i)^{n-j} g_{n-j}, \quad \sum_{j \geq 0} d_j h^j = \frac{8}{a(h)}. \quad (\text{CFD4})$$

The first formula is multiplication of the two convergent power series followed by $h = iw/2$; the second reverses those steps. On jets through degree N the matrix is triangular with diagonal $a_0(i/2)^n/8$, which never vanishes. These are the original coordinate and amplitude factors, not a replacement of h or w by a rescaled coordinate.

The associated graded map in degree n sends the class of h^{n-k} to $a_0(i/2)^n w^n/8$. Consequently the completed order is

$$\text{ord}_{z_0}(Uf) = k + \text{ord}_{s_0}(f). \quad (\text{CFD5})$$

The germ f and all of its Laurent coefficients are recovered from Uf once the embedded module I , the specified multiplier A , and the coordinate map are retained.

53.2 Which quotient remembers a cancelled divisor

For every nonzero $f \in I$ there is an exact module isomorphism

$$I/\mathcal{O}_s f \xrightarrow{\bar{U}} \mathcal{O}_z/\mathcal{O}_z(Uf). \quad \dim_{\mathbb{C}}(I/\mathcal{O}_s f) = k + \text{ord}_{s_0} f. \quad (\text{CFD6})$$

Indeed $U(qf) = C(q)Uf$ and C is surjective. The right hand quotient is the quotient by a unit times $w^{k+\text{ord} f}$, whose classes $1, w, \dots, w^{k+\text{ord} f-1}$ form a basis. This quotient measures the zero of the completed section in I . It does not, by itself, measure the zero or pole of f relative to the different reference lattice $\mathcal{O}_s \subset \mathcal{M}_s$.

The exact comparison retains both embeddings:

$$\begin{aligned} J &= I \cap \mathcal{O}_s = h^{\max(0, -k)} \mathcal{O}_s, & I + \mathcal{O}_s &= h^{\min(0, -k)} \mathcal{O}_s, \\ \dim_{\mathbb{C}}(I/J) &= \max(k, 0), & \dim_{\mathbb{C}}(\mathcal{O}_s/J) &= \max(-k, 0), \\ \text{ord}(f) &= \dim(I/\mathcal{O}_s f) - \dim(I/J) + \dim(\mathcal{O}_s/J). \end{aligned} \quad (\text{CFD7})$$

The first line follows by comparing integer orders of germs; the second by bases of the relevant powers of h ; the third follows from CFD6 and $\max(k, 0) - \max(-k, 0) = k$. Thus this object retains the cancelled divisor with its sign and its precise local module.

For the actual multiplier, HTB18 gives

$$\begin{aligned} A(0) &= -1, \quad A(s) = \frac{1}{2}(s-1) + O((s-1)^2), \\ \operatorname{Res}_{s=-2m} A(s) &= a_m = \frac{2(-1)^m(2m+1)\pi^m}{(m-1)!} \neq 0 \quad (m \geq 1). \end{aligned} \quad (\text{CFD8})$$

There are no other finite zeros or poles. At $s_0 = -2m$, let $h = s + 2m$. Then $k = -1$, $I = h\mathcal{O}_s$, and the Riemann zeta function is $\zeta = hc(h)$ with $c(0) \neq 0$. To see the last nonvanishing without an additional zero-location assertion, HTB9 gives $\xi(-2m) = \xi(1+2m)$, and the latter is nonzero because all factors of $A(1+2m)\zeta(1+2m)$ are positive. Hence

$$I/\mathcal{O}_s\zeta = 0, \quad \mathcal{O}_s/I \cong \mathcal{O}_s/(h), \quad (U\zeta)(z_0) = a_m\zeta'(-2m)/8. \quad (\text{CFD9})$$

The vanished quotient in the first expression cannot serve as the record of the cancelled simple zero. The second expression does record its length, and CFD4 recovers every derivative from the completed germ. The one-dimensional module $h\mathcal{O}_s/h^2\mathcal{O}_s$ is the degree-zero associated graded of I ; it contains the leading zero coefficient and maps isomorphically to $\mathcal{O}_z/w\mathcal{O}_z$. It is not the same quotient as $I/\mathcal{O}_s\zeta$.

At $s_0 = 1$, $k = 1$, $I = h^{-1}\mathcal{O}_s$ and $\zeta = h^{-1}(1 + O(h))$, so

$$I/\mathcal{O}_s\zeta = 0, \quad I/\mathcal{O}_s \cong h^{-1}\mathcal{O}_s/\mathcal{O}_s, \quad (U\zeta)(-i) = 1/16. \quad (\text{CFD10})$$

Here the one-dimensional comparison quotient records the cancelled pole. At $s = 0$ the multiplier is a unit and no divisor cancellation occurs. At all points of the open critical strip it is a unit, so the ordinary local zero algebras of ζ and $U\zeta$ are isomorphic by coordinate substitution, with every multiplicity unchanged. CFD9–10 explain precisely why this strip assertion does not replace the global comparison.

53.3 All support labels on modules and quotients

For each complex vector space V , use exactly

$$G_L(V) = \{(0, \lambda) : \lambda \in L\} \cup (V \times \{1_L\}), \quad (v, \lambda) + (v', \mu) = (v+v', \lambda \vee \mu), \quad (c, \alpha)(v, \lambda) = (cv, \alpha \wedge \lambda). \quad (\text{CFD11})$$

Every complex-linear map T above lifts by $T^\uparrow(v, \lambda) = (Tv, \lambda)$. The carrier condition holds because $T0 = 0$; addition and scalar action commute componentwise. The inverse lifts when T is invertible. This proves all module and finite-jet isomorphisms above over the entire bounded distributive lattice L , with no finiteness or Boolean restriction.

For a subspace $W \subset V$, the quotient map has the precise congruence

$$(v, \lambda) \sim (v', \lambda') \iff \lambda = \lambda' \text{ and } v - v' \in W, \quad G_L(V)/\sim \cong G_L(V/W). \quad (\text{CFD12})$$

The relation is preserved by addition and scalar action, since W is a subspace and the same labels undergo the same joins or meets. Its classes are exactly the fibres of $(v, \lambda) \mapsto (v+W, \lambda)$, which is surjective: a nonzero coset has top label, and each labelled zero is the image of the corresponding labelled zero. This proves the quotient assertion. In particular $G_L(0) = \{(0, \lambda) : \lambda \in L\}$ still retains every label; it contains no amplitude dimension recording a cancelled divisor. The defects in CFD7–10 must therefore remain even after this lift.

With the original $K_L = K \otimes_{\mathbb{B}} L$, the map

$$\Phi_V(v, \lambda) = (v, 1_K \otimes \lambda) \text{ satisfies } (T \times \text{id})\Phi_V = \Phi_W T^\uparrow. \quad (\text{CFD13})$$

The zero/nonzero character of the tropical semifield K tensored with L retracts $1_K \otimes \lambda$ to λ , proving injectivity. Thus neither completion nor these module comparisons identify $e = (0, 1_L)$ with $\tau = (0, 0_L)$. Multiplication by $A/8$ is a module map, not an ordinary unital algebra map. The ordinary product transports to $(G \odot H)(z) = 8G(z)H(z)/A(s(z))$ and unit $A(s(z))/8$, as in HTB12. Statements about prime spectra require that declared transported product, rather than the ordinary pointwise product.

53.4 The theta orbit and a faithful supported-time probe

Keep the actual coefficient vector of THS18: $a_{2j}(t) = (h_j(t), 1_L)$ and $a_{2j+1}(t) = \tau$. For $x = (r, \lambda) \in G_L(\mathbb{C})$ the exact matrix is

$$U_x(i, j) = \begin{cases} (1, 1_L), & j = i, \\ ((-r)^m j! / (m! i!), \lambda), & j = i + 2m, m \geq 1, \\ \tau, & \text{otherwise.} \end{cases} \quad (\text{CFD14})$$

THS10–12 give the absolute amplitude convergence. The diagonal summand has top support at every even index, while every odd summand is τ . Hence

$$U_{(r, \lambda)} \mathbf{a}(t) = \mathbf{a}(t + r), \quad U_{(0, \lambda)} \mathbf{a}(t) = \mathbf{a}(t) \quad \text{for every } \lambda \in L. \quad (\text{CFD15})$$

For nontrivial L this orbit observable is not injective in the supported-time parameter. Keeping the label in the ambient carrier does not make this particular observable distinguish it.

The full matrix representation nevertheless is injective:

$$U_x(0, 2) = (-2r, \lambda), \quad x = (-\frac{1}{2}, 1_L) U_x(0, 2). \quad (\text{CFD16})$$

Both the amplitude and every allowed label are recovered by this entry. Equivalently, let p be the supported polynomial whose only non- τ coefficient is $p_2 = (1, 1_L)$. Direct multiplication gives

$$(U_x p)_0 = (-2r, \lambda), \quad (U_x p)_2 = (1, 1_L), \quad (U_x p)_j = \tau \quad (j \neq 0, 2). \quad (\text{CFD17})$$

Thus the degree-two probe restores supported-time faithfulness; its left inverse is the constant-coefficient map followed by multiplication by $(-1/2, 1_L)$. Its sums are finite. A degree-zero or degree-one polynomial cannot provide this test because the second derivative annihilates both and the strictly upper even entries have no source coefficient. The proof establishes the parameter observable just specified, not faithfulness of a construction of the complete arithmetic geometry.

Another proved loss of information is the finite-cardinality map of HZA2: for finite L , $|G_L(\mathbb{Z}/n\mathbb{Z})| = n - 1 + |L|$ and its Dirichlet series is $\zeta(s, |L|)$. A four-element chain and the four-element Boolean lattice give the same series but different meet/join structures. Hence this scalar receiver is also not a reconstruction of the full lattice. The injective Φ_V and the count are different, explicitly specified maps.

53.5 Completion does not identify the two deformations

On meromorphic functions let $Df(s) = sf(s+1)$ and $B = -\partial_z^2$. The pullback of the heat generator is

$$\mathcal{L}_H = U^{-1}BU, \quad \mathcal{L}_H f = \frac{1}{4A}(Af)'', \quad (\mathcal{L}_H + D)f = \frac{1}{4}f'' + \frac{A'}{2A}f' + \frac{A''}{4A}f + sf(s+1). \quad (\text{CFD18})$$

This follows by the chain rule $ds/dz = i/2$ and differentiating twice; the factors 8 cancel. The Hurwitz generator is $-D$. On the constant function 1, their difference is $A''/(4A) + s$. At each $s_0 = -2m$, writing $A = h^{-1}a(h)$ gives $A''/A = 2h^{-2} + O(h^{-1})$, so the difference has a nonzero double-pole coefficient $1/2$. It is not the zero operator.

It also differs on the actual initial function ζ . Near $s = -2$, the residue calculation HTB18 gives $A(s) = -6\pi/(s+2) + O(1)$. Since $A\zeta = \xi$ is entire,

$$\mathcal{L}_H \zeta(-2) = 0, \quad (-D\zeta)(-2) = 2\zeta(-1) = -\frac{1}{6}, \quad ((\mathcal{L}_H + D)\zeta)(-2) = \frac{1}{6}. \quad (\text{CFD19})$$

Here $\zeta(-1) = -1/12$ follows directly from HTB5 and $B_2(1) = 1/6$. Therefore the two tangent vectors are different already before completion. Their transported difference has residue $-\pi/8$

in the s coordinate for U without coordinate substitution, and $-\pi/(4i)$ in the z coordinate. Equivalently the Hurwitz tangent alone has the opposite residue $\pi/(4i)$, whereas the heat tangent is entire, as in HTB21.

This distinction persists on the exact trajectories. Write

$$f_h^{\text{heat}}(s) = \frac{8H_h^{\text{heat}}(z(s))}{A(s)}. \quad (\text{CFD20})$$

For every real h and every $c \in \mathbb{R}$, $H_h^{\text{heat}}(ic) = \int_0^\infty e^{hu^2} \Phi(u) \cosh(cu) du > 0$; convergence and strict positivity follow from THS3–6. It follows that

$$\begin{aligned} \text{ord}_{s=-2m} f_h^{\text{heat}} &= 1 \quad (m \geq 1), & \text{Res}_{s=1} f_h^{\text{heat}} &= 16H_h^{\text{heat}}(-i), \\ \frac{d}{dh} \text{Res}_{s=1} f_h^{\text{heat}} &= 16 \int_0^\infty u^2 e^{hu^2} \Phi(u) \cosh u du > 0. \end{aligned} \quad (\text{CFD21})$$

The first conclusion uses the simple zero of $1/A$ and the strictly positive numerator; the residue uses $A'(1) = 1/2$; the derivative is justified by THS6. At $h = 0$ the residue is 1. In contrast $F_t(s) = \zeta(s, 1+t)$ has residue 1 at $s = 1$ for every allowed t and values $F_t(-2m) = -B_{2m+1}(1+t)/(2m+1)$. Already its first completed pole has residue $2\pi(1+t)(t+1/2)t$. Thus completion is invertible on the specified meromorphic data, but the heat trajectory preserves the classical trivial-zero locations by its construction and is not the arithmetic Hurwitz trajectory.

The exact bridge remains HTB26–28: $F_t + (\zeta - F_t) = \zeta$ and $UF_t + U(\zeta - F_t) = U\zeta$. It retains the complete principal parts, reflection defect, and every support label. Omitting the second summand would change the arithmetic input; retaining it is an identity returning the initial function, not a proof that the two distinct generators agree or that either trajectory reconstructs the full arithmetic base.

53.6 Faithful parameter representation is not invertible heat transport

The injectivity of $x \mapsto U_x$ proved in CFD16 does not mean that each U_x is an injective endomorphism of supported polynomials. In fact, for every $x = (r, \lambda) \neq \tau$ one has $\lambda \neq 0_L$. Take the degree-two probe p of CFD17 and let p' have the same coefficients except $p'_0 = (0, \lambda)$ instead of $p_0 = \tau$. They are distinct. Their degree-zero images are respectively $(-2r, \lambda)$ and $(0, \lambda) + (-2r, \lambda) = (-2r, \lambda)$, and every other image coefficient agrees. Thus

$$p \neq p', \quad U_x p = U_x p' \quad (x \neq \tau). \quad (\text{CFD22})$$

So no such U_x has an inverse on the full supported polynomial space. The classical time-group relation on the actual saturated theta vector results from CFD15; the full supported-time family instead satisfies the additive monoid relation of THS20. This is an exact support effect, despite injectivity of its parameter map.

53.7 The direct shifted theta kernel before completion

For real $t > -1$, put $a = 1 + t > 0$ and retain the direct kernel

$$K_t(x) = \sum_{n=0}^{\infty} e^{-\pi(n+a)^2 x}, \quad x > 0, \quad K_1(x) = K_0(x) - e^{-\pi x}. \quad (\text{CFD23})$$

The series and its parameter derivatives converge locally uniformly for $x > 0$ and a in bounded real intervals, by the Gaussian decay in n . The last identity deletes precisely the $n = 0$ term of the $a = 1$ series. For $\text{Re } s > 1$ one has

$$\int_0^\infty K_t(x) x^{s/2-1} dx = \pi^{-s/2} \Gamma(s/2) \zeta(s, a). \quad (\text{CFD24})$$

Indeed, if $\sigma = \operatorname{Re} s > 1$, the sum of integrals of the absolute values is $\pi^{-\sigma/2}\Gamma(\sigma/2)\sum_{n \geq 0}(n+a)^{-\sigma} < \infty$. Fubini's theorem applies. Substitution $y = \pi(n+a)^2x$ in each summand proves CFD24 with every constant retained. This is the shifted theta representation of the actual Hurwitz series before multiplication by $s(s-1)/2$ or any passage to a cosine transform.

Its exact parameter derivative and its integral are

$$\begin{aligned} \partial_t K_t(x) &= -2\pi x \sum_{n=0}^{\infty} (n+a) e^{-\pi(n+a)^2 x}, \\ \int_0^{\infty} \partial_t K_t(x) x^{s/2-1} dx &= -s\pi^{-s/2}\Gamma(s/2)\zeta(s+1, a), \quad \operatorname{Re} s > 0. \end{aligned} \tag{CFD25}$$

Local uniformity gives the first derivative formula. For the second, the sum of absolute integrals at $\sigma = \operatorname{Re} s > 0$ is

$$2\pi \sum_{n \geq 0} (n+a) \int_0^{\infty} x^{\sigma/2} e^{-\pi(n+a)^2 x} dx = 2\pi^{-\sigma/2}\Gamma(\sigma/2+1) \sum_{n \geq 0} (n+a)^{-\sigma-1} < \infty.$$

Integration term by term at complex s and the recurrence $\Gamma(s/2+1) = (s/2)\Gamma(s/2)$ prove the claimed identity. In particular on $\operatorname{Re} s > 1$ it is the derivative of CFD24, and the convergent differentiated integral itself continues the result to the larger stated half-plane.

The positive-real- x integral has a relevant complex-parameter restriction. The same formula is locally uniformly holomorphic when

$$\operatorname{Re} a > |\operatorname{Im} a|. \tag{CFD26}$$

On each compact subset of this sector there is $c > 0$ with $\operatorname{Re}(n+a)^2 \geq c(n+\operatorname{Re} a)^2$ for all $n \geq 0$. The preceding absolute-integral bounds therefore apply with fixed additional constants. The gamma substitution with a complex parameter follows from the holomorphic identity theorem from positive real a ; the principal logarithm satisfies $\log((n+a)^2) = 2\log(n+a)$ in this sector. This proves the holomorphic extension there. On the larger half-plane $\operatorname{Re} a > 0$, the Hurwitz function has the continuation proved in HTB4, but the unrotated kernel integral cannot be claimed on that basis. For example $a = 1 + 2i$ gives $\operatorname{Re}(a^2) = -3$, and its first summand is $e^{3\pi x} e^{-4\pi i x}$; the subsequent summands grow more slowly or decay, so the positive-real- x Mellin integral is not absolutely convergent at infinity. This identifies the precise analytic domain used by the direct kernel construction.

53.8 The retained pair and its actual support domain

On $\mathcal{M}_s \oplus \mathcal{M}_s$, the exact map

$$V(F, D) = (U(F+D), UD), \quad V^{-1}(H, J) = (U^{-1}(H-J), U^{-1}J) \tag{CFD27}$$

is a complex-linear isomorphism by direct composition. It therefore lifts to an isomorphism from $G_L(\mathcal{M}_s \oplus \mathcal{M}_s)$ to $G_L(\mathcal{M}_z \oplus \mathcal{M}_z)$, preserving its one common label. For the actual common-top pair $(F_t, \zeta - F_t)$ the image is $(U\zeta, U(\zeta - F_t))$. Both summands are recoverable. Projection to the first entry alone has linear kernel $\{(F, -F) : F \in \mathcal{M}_s\}$ and discards the deformation in this particular family.

This does not extend to the product of two split carriers with independent labels by an unmodified triangular semimodule map. There the first output label would be $\lambda \vee \mu$. For any nonzero D and two different labels λ, λ' , the two inputs $((0, \lambda), (D, 1_L))$ and $((0, \lambda'), (D, 1_L))$ have identical images $((UD, 1_L), (UD, 1_L))$. Thus common-label CFD27 has its stated inverse, while independent-label data require their original labels as additional retained data. This is another explicit domain distinction needed for a claim of faithfulness.

54 Original zeta and the retained theta comparison

Correction and provenance. The statement that the Rodgers–Tao heat kernel was the programme’s original construction over the entire absolute base was not established. THS1–31 begin with that classical analytic kernel and construct a support-preserving coefficient lift. HTB1–35 establish its exact comparison with the shifted Hurwitz family. Neither construction is a derivation of the scalar kernel from the entire arithmetic geometry. The distinction is measurable: THS22 already proves that all supported zero times act identically on the actual theta orbit. The ambient coefficient injection does not make that orbit injective.

The original arithmetic functions in this calculation remain

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s}, \quad F_t(s) = \sum_{n=0}^{\infty} (n+1+t)^{-s}, \quad \operatorname{Re} s > 1, \quad t > -1. \quad (\text{ZTC1})$$

Their continuations, endpoints, and exact shift generator are proved in HTB2–5, with the programme source identified there by version, hash, and original theorem labels. In particular $F_0 = \zeta$, $F_1 = \zeta - 1$, and $\partial_t F_t(s) = -sF_t(s+1)$. The latter family comes from the supplied April 2026 shifted-Hurwitz programme draft, not from a proposed identification of two time variables. Brad Rodgers and Terence Tao’s original author source, equations `phidef`, `htdef`, `hoz`, and `sas`, supplies the auxiliary heat kernel used in THS and HTB. These are cited inputs, not new discoveries. The following comparisons are derived explicitly. Their complete HTB, THS, and CFD proofs accompany this chapter.

54.1 A direct theta sum for the original arithmetic family

For real $t > -1$ and $x > 0$, retain the actual shifted frequencies and put

$$K_t(x) = \sum_{n=0}^{\infty} e^{-\pi(n+1+t)^2 x}. \quad (\text{ZTC2})$$

This is a scalar analytic representation of the stated arithmetic family. No assertion that it determines the full support lattice is part of this definition. For fixed $a = 1 + t > 0$, monotonicity of the summand as a function of n gives

$$K_t(x) \leq e^{-\pi a^2 x} + \int_0^{\infty} e^{-\pi(a+y)^2 x} dy \leq 1 + \frac{1}{2\sqrt{x}}. \quad (\text{ZTC3})$$

For $x \geq 1$ the same bound after retaining the first exponential gives $K_t(x) \leq C_a e^{-\pi a^2 x}$, since $(a+n)^2 - a^2 \geq n^2$ and $C_a = \sum_{n \geq 0} e^{-\pi((a+n)^2 - a^2)} < \infty$. Consequently the following integral converges absolutely for $\operatorname{Re} s > 1$:

$$\int_0^{\infty} x^{s/2-1} K_t(x) dx = \pi^{-s/2} \Gamma(s/2) F_t(s), \quad F_t(s) = \frac{\pi^{s/2}}{\Gamma(s/2)} \int_0^{\infty} x^{s/2-1} K_t(x) dx. \quad (\text{ZTC4})$$

Indeed, with $\sigma = \operatorname{Re} s > 1$, the sum of the absolute termwise integrals equals $\pi^{-\sigma/2} \Gamma(\sigma/2) \sum_{n \geq 0} (n+a)^{-\sigma}$. Fubini therefore applies. Substitution $y = \pi(n+a)^2 x$ in each term proves the displayed identity with its full Gamma and pi factors. Continuation outside that domain means the meromorphic continuation proved in HTB4, not an assertion that the unsubtracted integral converges there.

There is also an exact inverse recovering the kernel, not only the arithmetic function. For every real $c > 1$,

$$K_t(x) = \frac{1}{4\pi i} \int_{c-i\infty}^{c+i\infty} \pi^{-s/2} \Gamma(s/2) F_t(s) x^{-s/2} ds. \quad (\text{ZTC4a})$$

Here the integral is absolutely convergent. To prove the inversion with its constant, set $\sigma = c/2 > 0$ and $g(v) = \exp(\sigma v - e^v)$. This function and all its derivatives are integrable on $\mathbb{R}$: at $-\infty$ they are bounded by a constant times $e^{\sigma v}$, and at $+\infty$ by a polynomial in e^v times $g(v)$. The substitution $u = e^v$ identifies its Fourier transform with $\Gamma(\sigma - iy)$. Repeated integration by parts therefore bounds that Gamma factor by $C_N(1 + |y|)^{-N}$ for every fixed N . Fourier inversion gives $e^{-u} = (2\pi i)^{-1} \int_{\sigma - i\infty}^{\sigma + i\infty} \Gamma(w) u^{-w} dw$. One may verify this inversion directly by first multiplying the Fourier integrand by $e^{-\epsilon y^2}$: the resulting expression is the convolution of g with the Gaussian approximate identity, which converges to g ; the stated integrable transform bound permits dominated convergence as $\epsilon \downarrow 0$. Apply the identity to $u = \pi(n+1+t)^2 x$, then put $s = 2w$. The resulting factor is $1/(4\pi i)$. Finally sum in n . Interchange is justified by $\sum (n+1+t)^{-c} < \infty$ and the integrable Gamma bound. This proves ZTC4a. ZTC4 and ZTC4a are mutual recovery formulas for the specified original arithmetic family and its scalar theta sums; they do not recover a support lattice from a scalar function.

Differentiation in t is locally uniform for t in compact subsets of $(-1, \infty)$ and x in compact subsets of $(0, \infty)$, by Gaussian majorants for a polynomial times e^{-cn^2} . It gives the exact formula

$$\partial_t K_t(x) = -2\pi x \sum_{n=0}^{\infty} (n+1+t) e^{-\pi(n+1+t)^2 x}. \quad (\text{ZTC5})$$

For $\text{Re } s > 0$, integration of its absolute value is bounded by a constant times $\sum (n+a)^{-\sigma-1}$, locally uniformly in t . More precisely its Mellin transform is

$$\int_0^{\infty} x^{s/2-1} \partial_t K_t(x) dx = -2\pi^{-s/2} \Gamma(s/2 + 1) F_t(s+1) = -s\pi^{-s/2} \Gamma(s/2) F_t(s+1). \quad (\text{ZTC6})$$

Thus the original arithmetic shift generator is represented directly, without substituting the heat generator for it. The stated equality extends meromorphically by HTB4-5.

The separated sheet and its entire missing summand are visible before any transform:

$$K_1(x) = K_0(x) - e^{-\pi x}, \quad D_t(s) = \zeta(s) - F_t(s), \quad D_1(s) = 1, \quad (\text{ZTC7})$$

$$\pi^{-s/2} \Gamma(s/2) D_t(s) = \int_0^{\infty} x^{s/2-1} (K_0(x) - K_t(x)) dx \quad (\text{Re } s > 1). \quad (\text{ZTC8})$$

ZTC7 follows by deleting the $n = 1$ summand of K_0 ; ZTC8 follows by subtracting two absolutely convergent integrals. At $t = 1$ the integral is exactly $\pi^{-s/2} \Gamma(s/2)$, so its inverse returns the actual constant 1, including its amplitude and support when lifted.

54.2 Original zeta with both Mellin endpoints retained

At $t = 0$, HTB6 proves

$$K_0(x) = x^{-1/2} K_0(1/x) + \frac{1}{2} x^{-1/2} - \frac{1}{2}. \quad (\text{ZTC9})$$

Split ZTC4 at $x = 1$, apply this identity only to the integral over $(0, 1)$, and substitute $y = 1/x$ there. Initially $\text{Re } s > 1$, and direct integration gives

$$\zeta(s) = \frac{\pi^{s/2}}{\Gamma(s/2)} \left\{ \frac{1}{s-1} - \frac{1}{s} + \int_1^{\infty} K_0(x) \left(x^{s/2-1} + x^{-(s+1)/2} \right) dx \right\}. \quad (\text{ZTC10})$$

The two separate rational terms are the transforms of the two separate terms $x^{-1/2}/2$ and $-1/2$. The integral on $[1, \infty)$ is entire in s , since exponential decay bounds every logarithmic derivative on compact subsets of the s -plane. This proves the meromorphic continuation of the identity and keeps both endpoint contributions. Neither has been declared zero or absorbed into an unrecorded constant.

For the original cosine heat kernel, retain

$$\begin{aligned} A(s) &= \frac{s(s-1)}{2} \pi^{-s/2} \Gamma(s/2), & s(z) &= \frac{1}{2} + \frac{iz}{2}, & z(s) &= -2i(s-1/2), \\ H_0(z) &= \frac{s(z)(s(z)-1)}{16} \pi^{-s(z)/2} \Gamma(s(z)/2) \zeta(s(z)), \\ \zeta(s) &= \frac{16\pi^{s/2}}{s(s-1)\Gamma(s/2)} H_0(z(s)). \end{aligned} \tag{ZTC11}$$

The proof is HTB6–10: the exact Mellin substitution and two integrations by parts yield the displayed multiplier, with all endpoint limits justified in the original convergence half-plane. The last equality is an identity of meromorphic functions. It is not a licence to replace the original zeta calculation by an entire function calculation while dropping its multiplier.

54.3 The boundary is part of the receiving object

Let $\mathcal{M}_s$ and $\mathcal{M}_z$ be the complex vector spaces of meromorphic functions on the respective complex planes. Retain the linear isomorphism, already proved in HTB11,

$$Uf(z) = \frac{A(s(z))}{8} f(s(z)), \quad U^{-1}G(s) = \frac{8G(z(s))}{A(s)}. \tag{ZTC12}$$

Define the joint map on the entire two-component space by

$$\mathcal{V} : \mathcal{M}_s \oplus \mathcal{M}_s \longrightarrow \mathcal{M}_z \oplus \mathcal{M}_z, \quad (F, D) \longmapsto (U(F+D), UD). \tag{ZTC13}$$

It has the exact inverse

$$\mathcal{V}^{-1}(H, J) = \left(\frac{8(H-J) \circ z}{A}, \frac{8J \circ z}{A} \right). \tag{ZTC14}$$

Substitution proves both compositions equal the identity; all entries are meromorphic functions, so no undefined product of separate singular point values is involved. On the actual arithmetic family,

$$\begin{aligned} J_t &= UD_t, & \mathcal{V}(F_t, D_t) &= (H_0, J_t), \\ F_t(s) &= \frac{16\pi^{s/2}}{s(s-1)\Gamma(s/2)} (H_0(z(s)) - J_t(z(s))), \\ J_1(z) &= \frac{A(s(z))}{8}, & J_t(i) &= -\frac{t}{8}. \end{aligned} \tag{ZTC15}$$

For the last equality use HTB5: $D_t(0) = \zeta(0) - F_t(0) = t$, $A(0) = -1$, and $z(0) = i$. In particular the retained pair determines the real parameter t by $t = -8J_t(i)$. All t give the same first component H_0 . Discarding J_t therefore identifies different actual arithmetic inputs. This is an explicit failure of injectivity on the actual programme family, not merely an example in an unrelated vector space. On the unrestricted space the summation map $(G, J) \mapsto G + J$ has exactly the kernel $\{(B, -B) : B \in \mathcal{M}_z\}$.

For the complete support carrier, use the common label on the direct sum:

$$G_L(\mathcal{M}_s \oplus \mathcal{M}_s) \xrightarrow{\mathcal{V}^\dagger} G_L(\mathcal{M}_z \oplus \mathcal{M}_z), \quad ((F, D), \lambda) \longmapsto (\mathcal{V}(F, D), \lambda). \tag{ZTC16}$$

Here $G_L(V) = \{(0, \lambda) : \lambda \in L\} \cup (V \times \{1_L\})$ with its original joins and meets. Additivity, compatibility with the $G_L(\mathbb{C})$ action, carrier membership, and the inverse follow componentwise from linearity and ZTC14, as explicitly proved in CFD11–12. This is a bijection for every bounded distributive L . It does not silently replace the common label by independent labels on two product factors. In particular $e = (0, 1_L)$ and $\tau = (0, 0_L)$ remain distinct when L is nontrivial.

The exact multiplication comparison. The preceding maps were declared as linear and semimodule maps. Their injectivity does not identify the original multiplication with ordinary multiplication of their images. This can be checked exactly, without discarding the completion multiplier. Set

$$B(z) = \frac{A(s(z))}{8} = \frac{s(z)(s(z) - 1)}{16} \pi^{-s(z)/2} \Gamma(s(z)/2). \quad (\text{ZTC16a})$$

For meromorphic f, g on the s -plane the two products are

$$\begin{aligned} U(fg)(z) &= B(z)f(s(z))g(s(z)), \\ (Uf)(z)(Ug)(z) &= B(z)^2 f(s(z))g(s(z)). \end{aligned} \quad (\text{ZTC16b})$$

In particular $U1 = B$, not the ordinary constant unit. The factor B is nonconstant: it has a zero at $z(1) = -i$ and poles at $z(-2m)$, $m \geq 1$. Thus ordinary multiplication is not preserved. The exact transported product and its unit are

$$G \odot H = \frac{8G(z)H(z)}{A(s(z))}, \quad 1_{\odot}(z) = \frac{A(s(z))}{8}. \quad (\text{ZTC16c})$$

This formula is an identity of meromorphic functions, including their germs at the zeros and poles of the multiplier. It is not pointwise division of undefined singular values. Substitution gives $(Uf) \odot (Ug) = U(fg)$. Directly, $(G \odot H) \odot K = GHK/B^2 = G \odot (H \odot K)$, $B \odot G = G$, and distributivity follows from distributivity of meromorphic multiplication. Together with ZTC12 this proves a unital algebra isomorphism with the stated product.

For each local holomorphic ring $\mathcal{O}_{s_0}$ the affine coordinate map identifies it with $\mathcal{O}_{z(s_0)}$. Its exact image under U is the embedded fractional lattice $B\mathcal{O}_{z(s_0)}$. This lattice is closed under $\odot$, since $(Bu) \odot (Bv) = Buv$, and has unit B . Keeping instead the unmodified target $\mathcal{O}_{z(s_0)}$ can fail closure: at $s_0 = 1$, B has a simple zero, so $1 \odot 1 = 1/B$ is not holomorphic. This image-lattice statement differs from the preimage lattice $A^{-1}\mathcal{O}$ in ZTC17: the former transports an original holomorphic ring, whereas the latter specifies which original meromorphic sections have a holomorphic completed image.

The full support carrier admits the same exact comparison. Retain its original addition, and define the target product by

$$(G, \lambda) \odot_L (H, \mu) = \left(\frac{8GH}{A \circ s}, \lambda \wedge \mu \right), \quad 1_{\odot_L} = (B, 1_L). \quad (\text{ZTC16d})$$

If the amplitude of a product is nonzero, both input amplitudes are nonzero, hence both labels are 1_L ; this proves carrier closure. If either amplitude is zero its output amplitude is zero and the displayed meet retains its original label. Associativity and distributivity follow from ZTC16c and the meet/join laws of the bounded distributive lattice. The map $(f, \lambda) \mapsto (Uf, \lambda)$ and its inverse therefore give a unital semiring isomorphism from the original coefficient semiring to this specified target, preserving e and τ . An ideal P is sent to its literal image $U^\uparrow(P)$; closure under addition and multiplication follows by applying the isomorphism. Its inverse proves the converse. Moreover a product belongs to that image precisely when the corresponding original product belongs to P , proving that prime ideals correspond. This proves the exact spectrum comparison for these specified coefficient semirings. It is not a derivation of the entire arithmetic base from the scalar function H_0 .

For ZTC13–16, no multiplication on pairs was specified: the proved assertion there remains a common-label semimodule isomorphism. Neither ordinary multiplication nor a Weil pairing is silently inferred from that assertion.

54.4 Local zeros, poles, and the original trace

The earlier conversational assertion that the singular module map had not been established was also too broad: HTB13 already proves $A^{-1}\mathcal{O} \xrightarrow{U} \mathcal{O}$ with the affine coordinate change. CFD1–10

now make the comparison with the *original* regular lattice and every finite jet explicit. At a finite point let $h = s - s_0$ and $A = h^k a(h)$, $a(0) \neq 0$. Retain $I = A^{-1}\mathcal{O} = h^{-k}\mathcal{O}$ as an embedded submodule of the meromorphic germs, and $J = I \cap \mathcal{O}$. For nonzero $f \in I$ the original order is

$$\text{ord}_{s_0} f = \dim_{\mathbb{C}}(I/\mathcal{O}f) - \dim_{\mathbb{C}}(I/J) + \dim_{\mathbb{C}}(\mathcal{O}/J). \quad (\text{ZTC17})$$

This is proved by the power-of- h bases in CFD6–7, without identifying the reference lattices. At a trivial zero $s_0 = -2m$, $I = h\mathcal{O}$, $I/\mathcal{O}\zeta = 0$ and $\mathcal{O}/I$ has dimension 1. At $s_0 = 1$, $I = h^{-1}\mathcal{O}$, $I/\mathcal{O}\zeta = 0$ and $I/\mathcal{O}$ has dimension 1. Their signs in ZTC17 give, respectively, +1 and -1. At $s_0 = 0$, $A(0) = -1$ is a unit. CFD4 provides the invertible triangular matrices recovering all Laurent coefficients with the exact factors 8 and $i/2$. Keeping only the zero algebra of the completed section would give zero in both cancelled cases; keeping these comparison modules restores the actual divisor.

The earlier conversational explanation of the fibres requires a further correction. The map

$$\mathcal{O}/h\mathcal{O} \longrightarrow h\mathcal{O}/h^2\mathcal{O}, \quad [u] \longmapsto [hu] \quad (\text{ZTC17a})$$

is an isomorphism of $\mathcal{O}$ -modules. It is well-defined because $h(h\mathcal{O}) = h^2\mathcal{O}$, surjective by the definition of $h\mathcal{O}$, and injective because $hu \in h^2\mathcal{O}$ implies $u \in h\mathcal{O}$ in the local integral domain. Its inverse is $[hu] \mapsto [u]$. Consequently the two abstract fibres alone do not account for the loss. What must be retained is their specified embedding and the original reference lattice: multiplication by $A = h^{-1}a(h)$ identifies $I = h\mathcal{O}$ with $\mathcal{O}$, but the length-one quotient $\mathcal{O}/I$ records the cancelled original zero. Replacing the entire marked comparison by the completed section's zero quotient discards precisely that datum. Likewise multiplication by nonzero meromorphic A is an automorphism of the meromorphic sheaf as a module. It is not a ring automorphism for ordinary multiplication, as ZTC16b proves. It is not an automorphism of the fixed holomorphic lattice at its zeros and poles. These category statements do not contradict one another.

Logarithmic differentiation gives the original trace identity

$$\frac{\zeta'(s)}{\zeta(s)} = -2i \frac{H'_0(z(s))}{H_0(z(s))} - \frac{1}{s} - \frac{1}{s-1} + \frac{1}{2} \log \pi - \frac{1}{2} \psi(s/2). \quad (\text{ZTC18})$$

It first holds away from zeros and poles and then meromorphically; $\psi = \Gamma'/\Gamma$. For a positively oriented finite contour γ avoiding all zeros and poles of the functions and all singularities of every displayed integrand, and a holomorphic test g on and inside it, multiplication by g and integration gives

$$\begin{aligned} \frac{1}{2\pi i} \int_{\gamma} g \frac{\zeta'}{\zeta} ds &= \frac{1}{2\pi i} \int_{\gamma} g(s) (-2i) \frac{H'_0(z(s))}{H_0(z(s))} ds \\ &\quad - \frac{1}{2\pi i} \int_{\gamma} g(s) \left(\frac{1}{s} + \frac{1}{s-1} - \frac{1}{2} \log \pi + \frac{1}{2} \psi(s/2) \right) ds. \end{aligned} \quad (\text{ZTC19})$$

At every zero or pole, write a meromorphic function as $(s - s_0)^r$ times a holomorphic unit. Its logarithmic derivative then has residue r . The residue theorem proves that ZTC19 counts the original zeros with multiplicity and the original poles with negative multiplicity, including every trivial zero inside the contour. No infinite-contour limit or decay assumption is hidden here. Extending this finite identity to a particular infinite trace requires the actual convergence estimates of that trace.

54.5 Two different trajectories through the same original zeta

The original- s function associated with the classical heat path is

$$f_h^{\text{heat}}(s) = \frac{16\pi^{s/2}}{s(s-1)\Gamma(s/2)} H_h(z(s)), \quad f_0^{\text{heat}} = \zeta. \quad (\text{ZTC20})$$

CFD18–21 give its full receiving operator and prove

$$\partial_h f_h^{\text{heat}}(-2)\Big|_{h=0} = 0, \quad \partial_t F_t(-2)|_{t=0} = -\frac{1}{6}. \quad (\text{ZTC21})$$

For completeness, the heat equation and chain rule give $\partial_h f = (Af)''/(4A)$. At $h = 0$, $A\zeta$ is holomorphic at -2 whereas $1/A$ has a simple zero there, giving the first value. HTB5 gives the second as $2\zeta(-1) = -1/6$. The discrepancy holds on the actual initial arithmetic function. CFD21 further proves that all the zeros at $-2m$ remain simple along the real heat path, while its residue at 1 varies strictly; the actual Hurwitz path has constant residue 1 and values $-B_{2m+1}(1+t)/(2m+1)$ at those points.

Thus the valid original- ζ receiver consists of the original functions, their exact transform and inverse, the retained boundary component, the embedded local comparison modules, and all support labels. The scalar theta orbit by itself is a smaller observable. These proofs neither exhibit an off-line zero of the original ζ nor rule one out. They establish precisely which input and which deformation a subsequent estimate is estimating.

55 The complete pole-corrected Hurwitz zero receiver

This calculation uses the original shifted flow, not a deformation of its coordinates. The programme source is `split_zero_cue_shifted_hurwitz_phase_space.tex`, Theorem `thm:universal-flow`, lines 602–630, and Theorem `thm:zero-motion`, lines 1003–1060. Its second occurrence is `split_zero_secondary_note.tex`, lines 1439–1633. The original theta coordinate and factor are those of Brad Rodgers and Terence Tao, *The de Bruijn–Newman constant is non-negative*, arXiv:1801.05914v5, equations `htdef` and `hoz`. The elementary Hurwitz identities are recorded with original TeX encodings at DLMF 25.11.7 and DLMF 25.11.14; the gamma product is DLMF 5.8.2. These standard formulas are credited to their human literature; the pole-corrected application below is a programme derivation.

55.1 An entire receiver with an exact divisor correction

For real $t \geq -1/2$ put $a = 1 + t$, and retain

$$s(z) = \frac{1}{2} + \frac{iz}{2}, \quad A(s) = \frac{s(s-1)}{2} \pi^{-s/2} \Gamma(s/2), \quad H_t^{\text{Hur}}(z) = \frac{1}{8} A(s(z)) \zeta(s(z), a). \quad (\text{HPR1})$$

Here H_t^{Hur} denotes the completed *Hurwitz* flow in this section. Only H_0^{Hur} is the Rodgers–Tao time-zero function; their heat evolution has a different generator. The full comparison is proved in HTB. Define the fixed entire function

$$Q(z) = \prod_{m=1}^{\infty} \left(1 + \frac{z^2}{(4m+1)^2} \right) = \frac{\Gamma(5/4)^2}{\Gamma(5/4 + iz/4) \Gamma(5/4 - iz/4)}. \quad (\text{HPR2})$$

The product converges locally uniformly because the sum of the absolute values of its nonconstant terms does. Its zeros are precisely $\pm i(4m+1)$, each simple, and $Q(0) = 1$. For completeness, Euler’s limit formula for Γ gives

$$\frac{\Gamma(b)^2}{\Gamma(b+w)\Gamma(b-w)} = \lim_{N \rightarrow \infty} \prod_{k=0}^N \left(1 - \frac{w^2}{(b+k)^2} \right).$$

The factors $N^{\pm w}$ cancel. Taking $b = 5/4, w = iz/4$ proves the second expression without an undetermined exponential factor.

The complete entire receiver is

$$\mathcal{F}_t(z) = Q(z) H_t^{\text{Hur}}(z) H_t^{\text{Hur}}(-z) = \frac{\Gamma(5/4)^2}{64\sqrt{\pi}} s(z)(1-s(z)) \zeta(s(z), a) \zeta(1-s(z), a). \quad (\text{HPR3})$$

Indeed $s(-z) = 1 - s(z)$, and

$$Q(z) = \frac{4\Gamma(5/4)^2}{s(1-s)\Gamma(s/2)\Gamma((1-s)/2)}.$$

Multiplication by both factors $A/8$ proves every constant in (HPR3). The only Hurwitz poles are simple poles at $s = 1$ and $s = 0$ in the two respective factors, so the displayed polynomial cancels them. Thus $\mathcal{F}_t$ is entire. It is even and real on the real z axis; more precisely $\mathcal{F}_t(\bar{z}) = \overline{\mathcal{F}_t(z)}$. No zero has yet been removed from its divisor by this construction.

Also $H_t^{\text{Hur}}(0) > 0$ for this whole range of t . At $a = 1/2$, $\zeta(1/2, a) = (\sqrt{2} - 1)\zeta(1/2) < 0$. The last sign follows by pairing successive terms in $\eta(1/2) = \sum_{n \geq 1} (-1)^{n-1} n^{-1/2} > 0$ and using $\eta(s) = (1 - 2^{1-s})\zeta(s)$. For $a \geq 1/2$, $\partial_a \zeta(1/2, a) = -\zeta(3/2, a)/2 < 0$. The factor $A(1/2)$ is negative. Consequently

$$H_t^{\text{Hur}}(0) > 0, \quad \mathcal{F}_t(0) = H_t^{\text{Hur}}(0)^2 > 0. \quad (\text{HPR4})$$

Here are the growth and summability details needed for an *all*-zero argument. Euler–Maclaurin, with $N \geq 1$, gives

$$\begin{aligned} \zeta(s, a) &= \frac{a^{1-s}}{s-1} + \frac{a^{-s}}{2} + \sum_{k=1}^N \frac{B_{2k}}{(2k)!} (s)_{2k-1} a^{1-s-2k} \\ &\quad - \frac{(s)_{2N}}{(2N)!} \int_0^\infty \tilde{B}_{2N}(x) (a+x)^{-s-2N} dx. \end{aligned}$$

Its domain is $\Re s + 2N > 1$, excluding the displayed pole. One proves it first for $\Re s > 1$ by integrating by parts on each unit interval, using $B'_j = jB_{j-1}$ and the matching periodic endpoint values; the boundary at infinity vanishes. The integral is locally normally convergent in the stated larger half-plane, giving the continuation there. The periodic Bernoulli Fourier series gives $\|\tilde{B}_{2N}\|_\infty/(2N)! \leq 2\zeta(2N)/(2\pi)^{2N} \leq 4/(2\pi)^{2N}$. For $|s| \leq R$, take $N = \lceil R \rceil + 2$. Then $|(s)_j| \leq (R+2N)^j$, and the absolute integral without the periodic factor is $a^{1-\Re s-2N}/(\Re s+2N-1)$. Every power of the fixed positive a is bounded by $\exp(O_a(R+N))$. The finite sum has N terms. After multiplying by $s-1$ these estimates give

$$\max_{|s| \leq R} |(s-1)\zeta(s, a)| \leq \exp(C_a(R+2) \log(R+2)) \quad (\text{HPR5})$$

for a finite constant C_a , enlarged to cover bounded R . Thus $\mathcal{F}_t$ has order at most one. Jensen's formula applied at its nonzero value at zero gives $N_{\mathcal{F}_t}(R) = O_t(R \log(R+2))$. Summing over annuli $2^j \leq |z| < 2^{j+1}$ proves $\sum_{\mathcal{F}_t(w)=0} \text{ord}_w(\mathcal{F}_t) |w|^{-2} < \infty$.

The poles of H_t^{Hur} are at $z = i(4m+1)$, and their residue in the original s coordinate is

$$\text{Res}_{s=-2m} A(s)\zeta(s, a) = \frac{2(-1)^{m+1} \pi^m}{(m-1)!} B_{2m+1}(a). \quad (\text{HPR6})$$

This follows from $\text{Res}_{s=-2m} \Gamma(s/2) = 2(-1)^m/m!$ and $\zeta(-2m, a) = -B_{2m+1}(a)/(2m+1)$. In the z coordinate the residue is $-i/4$ times (HPR6), because the function has the additional factor $1/8$ and $ds/dz = i/2$. There is an actual simple pole precisely at the indices

$$I_t = \{m \geq 1 : B_{2m+1}(1+t) \neq 0\}.$$

At such an index, $H_t^{\text{Hur}}(-i(4m+1))$ evaluates at $s = 2m+1 > 1$; all its factors are strictly positive. In particular the reflected factor cannot cancel this pole. At $s = 0$ and $s = 1$ the function is regular, with values $A\zeta = 1/2 + t$ and $1/2$, respectively.

Let $\mathcal{Z}_t$ be the positive divisor counting every zero of $H_t^{\text{Hur}}(z)$ and every zero of $H_t^{\text{Hur}}(-z)$, with their actual multiplicities. Let $\mathcal{P}_t = \sum_{m \in I_t} ([i(4m+1)] + [-i(4m+1)])$. Writing div for the signed divisor of a meromorphic function, the exact equality of divisors is

$$\mathcal{Z}_t = \text{div}(\mathcal{F}_t) - \text{div}(Q) + \mathcal{P}_t. \quad (\text{HPR7})$$

It is the valuation identity for (HPR3), at every point. At an inactive index, Q supplies one artificial zero in $\mathcal{F}_t$, in addition to any actual zero there; subtracting $\text{div}(Q)$ removes exactly one. At an active index, $\mathcal{F}_t$ is nonzero at both marked points and the last two terms cancel. This also proves directly that the right side has nonnegative multiplicities. In particular

$$\sum_w \mathcal{Z}_t(w) |w|^{-2} < \infty.$$

These divisors are invariant under both $w \mapsto -w$ and conjugation. The zero $w = 0$ is absent by (HPR4).

55.2 The complete corrected trace, with no discarded roots

For $n \geq 1$ define

$$\begin{aligned} S_{F,n}(t) &= \sum_{\mathcal{F}_t(w)=0} \text{ord}_w(\mathcal{F}_t) w^{-2n}, \\ T_n(t) &= S_{F,n}(t) - 2(-1)^n \sum_{m \geq 1} (4m+1)^{-2n} + 2(-1)^n \sum_{m \in I_t} (4m+1)^{-2n} \\ &= \sum_w \mathcal{Z}_t(w) w^{-2n}. \end{aligned} \tag{HPR8}$$

All sums are absolutely convergent. The order bound, evenness and Hadamard factorization give $\mathcal{F}_t(z) = \mathcal{F}_t(0) \prod_{\{w, -w\}} (1 - z^2/w^2)^{\text{ord}_w(\mathcal{F}_t)}$. Indeed the paired genus-one factors lose their exponential terms; the remaining exponential is e^{bz+c} , and evenness forces $b = 0$. Logarithmic differentiation at zero consequently gives

$$\log \mathcal{F}_t(z) = \log \mathcal{F}_t(0) - \sum_{n \geq 1} \frac{S_{F,n}(t)}{2n} z^{2n} \tag{HPR9}$$

on the actual zero-free disk at the origin. This fixes both the multiplicity convention and the factor two.

Put $\mathcal{R} = \{z^{-1}p(z^{-2}) : p \in \mathbb{C}[X]\}$. The complete zero trace and its finite matrices are

$$\mathcal{B}_t(f, g) = \sum_w \mathcal{Z}_t(w) \overline{f(\bar{w})} g(w), \quad K_d^{\text{Hur}}(t) = (T_{i+j+1}(t))_{0 \leq i, j < d}. \tag{HPR10}$$

The summability in (HPR7) proves absolute convergence on all of $\mathcal{R}$, not only on a selected finite set of zeros. The form is Hermitian by conjugation of the divisor. Substituting the monomials gives precisely the displayed matrix.

Here is a full proof of the all-zero criterion for this new receiver:

$$\text{every zero of } H_t^{\text{Hur}} \text{ is real} \iff K_d^{\text{Hur}}(t) \text{ is positive semidefinite for every } d. \tag{HPR11}$$

The forward implication is the sum of the nonnegative terms $\mathcal{Z}_t(w)|f(w)|^2$ at real w . For the reverse implication collect $\pm$ pairs into reciprocal nodes $\lambda = w^{-2}$ with positive integer masses M_λ . They form a bounded discrete set away from its only possible accumulation point zero, are closed under conjugation with equal masses, and satisfy $V = \sum M_\lambda |\lambda| < \infty$. For a polynomial p the quadratic form is

$$\sum_\lambda M_\lambda \lambda \overline{p(\bar{\lambda})} p(\lambda).$$

An off-real w gives a negative real or a nonreal λ_0 . Choose $0 < r < |\lambda_0|$ and the finite set $S = \{\lambda : |\lambda| \geq r\}$. Let $L_0(X) = \prod_{\mu \in S \setminus \{\lambda_0\}} (X - \mu)/(\lambda_0 - \mu)$, and $B = \prod_{\mu \in S \setminus \{\lambda_0\}} (r + |\mu|)/|\lambda_0 - \mu|$. If $\lambda_0 < 0$, this polynomial is real and $p_N = (X/\lambda_0)^N L_0$ has its only nonzero value on S at λ_0 . The target contribution is $M_{\lambda_0} \lambda_0 < 0$; the absolute value of the complete remaining tail is at most $B^2 V (r/|\lambda_0|)^{2N}$, which tends to zero. If λ_0 is nonreal, choose $|c| = 1$ with $c^2 = -\bar{\lambda}_0/|\lambda_0|$, and set

$$p_N = c(X/\lambda_0)^N L_0(X) + \bar{c}(X/\bar{\lambda}_0)^N \overline{L_0(\bar{X})}.$$

This is a real polynomial. Its two target contributions total $-2M_{\lambda_0}|\lambda_0|$, while the full tail is bounded by $4B^2 V (r/|\lambda_0|)^{2N}$. Choosing a sufficiently large finite N makes the form strictly negative. These are bounds on every remaining zero, not estimates for a sampled tail. Finally approximate the finitely many real coefficients by rationals. On the bounded node disk the uniform error tends to zero, and the quadratic error is at most $V\delta(2\|p_N\| + \delta)$. Thus a rational finite test is strictly negative too, proving the converse and a rational witness.

The calculation retains the full split base. For the original bounded distributive lattice L , take $G_L(\mathcal{R}) = \{(0, \lambda) : \lambda \in L\} \cup (\mathcal{R} \times \{1_L\})$, and set

$$\widehat{\mathcal{B}}_t((f, \lambda), (g, \mu)) = (\mathcal{B}_t(f, g), \lambda \wedge \mu), \quad \Phi(c, \nu) = (c, 1_K \otimes \nu). \quad (\text{HPR12})$$

The nonzero-amplitude carrier condition is satisfied because both input amplitudes must then be nonzero, hence both labels top. Additivity holds in each variable with the original join, since meet distributes over join. Scalar multiplication and conjugation have their original amplitudes and meet labels. Evaluation at every zero is $(f, \lambda) \mapsto (f(w), \lambda)$; it is additive and linear and retains $e = (0, 1_L)$ at amplitude cancellation. Its τ fibre contains only τ . The map Φ commutes with these operations by its two coordinates; the original lattice retraction $\rho_L(1_K \otimes \nu) = \nu$ retains all intermediate labels. Formula (HPR8) is an equality of arithmetic coefficients before this lift. It does not subtract lattice labels, identify the independent support sectors, or invert e .

55.3 The actual boundary at the unshifted fibre

At $t = 0$ every odd Bernoulli value $B_{2m+1}(1)$ vanishes. The functional equation makes H_0^{Hur} even, so

$$I_0 = \emptyset, \quad \mathcal{F}_0 = Q(H_0^{\text{Hur}})^2, \quad T_n(0) = 2S_n(H_0^{\text{Hur}}), \quad K_d^{\text{Hur}}(0) = 2K_d(0). \quad (\text{HPR13})$$

In particular the arithmetic zero test at this fibre is exactly the previous all-root receiver with the factor two retained. All labels in (HPR12) are unchanged. Positive definiteness of $K_{64}(0)$ therefore proves only the corresponding 64-dimensional result here.

The other fibres have an explicitly different all-zero contribution. Let $C_n = \sum_{m \geq 1} (4m+1)^{-2n}$. Each Bernoulli polynomial is nonzero, so the union E of all its real roots $B_{2m+1}(1+t) = 0$ is countable. Its complement is dense. On this complement I_t contains every positive integer and $T_n(t) = S_{F,n}(t)$. The latter is continuous near $t = 0$ by (HPR3)–(HPR4) and the Taylor coefficient formula (HPR9). Hence

$$\lim_{\substack{t \rightarrow 0 \\ t \notin E}} T_n(t) = 2S_n(H_0^{\text{Hur}}) + 2(-1)^n C_n, \quad T_n(0) = 2S_n(H_0^{\text{Hur}}). \quad (\text{HPR14})$$

The explicitly computed jump is the disappearing zero–pole divisor. There is no assertion that taking a parameter limit commutes with discarding the poles in a meromorphic divisor.

The first corrected moment gives a stronger uniform statement. Write

$$B(t) = \frac{1}{2} \frac{d^2}{ds^2} \log(A(s)\zeta(s, 1+t)) \Big|_{s=1/2}. \quad (\text{HPR15})$$

This logarithm is defined near $s = 1/2$ by its positive value there. Since $ds/dz = i/2$, differentiating (HPR3) and retaining both reflected factors proves

$$B(t) = S_{F,1}(t) + 2C_1, \quad T_1(t) = B(t) - 2 \sum_{m \in I_t} (4m+1)^{-2}. \quad (\text{HPR16})$$

Now $B_3(1+t) = (1+t)(t+1/2)t$. Thus $1 \in I_t$ at every $0 < |t| \leq 1/1000$; all other terms in this last sum are nonnegative. The reproducible interval certificate `check_hurwitz_pole_receiver.py` evaluates the entire parameter interval $1+t \in [999/1000, 1001/1000]$ at 256-bit precision with power series of length four, using Fredrik Johansson’s Arb through Python-flint 0.9.0. It proves $A(1/2)\zeta(1/2, 1+t) > 0$ and $B(t) < 1/25$ on that interval. It uses the coefficient of x^2 in $\log(A(1/2+x)\zeta(1/2+x, 1+t))$, which is exactly (HPR15), not twice that number. No parameter samples or zero list enter. The retained machine receipt specifies the library version, enclosure, script hash and the independently retained theta moment comparison. Consequently the computer-assisted rational bound is

$$\boxed{B_t(z^{-1}, z^{-1}) = T_1(t) < -\frac{1}{25} \quad (0 < |t| \leq 1/1000), \quad T_1(0) > 0.} \quad (\text{HPR17})$$

The exact infinite-sum inequality behind it is simply $T_1(t) \leq B(t) - 2/25$; none of the other poles can weaken it. At zero, $T_1(0) = 2M_1/M_0 > 0$ for the original positive theta moments. An alternative argument requiring no new Hurwitz numerical enclosure is already available: the earlier certified bounds $M_0 > 3/50$ and $M_1 < 3/4000$ give $B(0) < 1/40$. Continuity and the same first-pole inequality prove strict negativity on a punctured neighborhood, without specifying its radius.

The roots responsible for this jump can be located analytically. At every original trivial zero $s = -2m$, the ordinary functional equation gives

$$\zeta'(-2m) = \frac{(-1)^m(2m)!}{2(2\pi)^{2m}}\zeta(2m+1) \neq 0, \quad B_{2m} = \frac{(-1)^{m+1}2(2m)!}{(2\pi)^{2m}}\zeta(2m).$$

These follow respectively by differentiating the sine factor in the functional equation and evaluating it at $1 - 2m$. The analytic implicit function theorem therefore gives a unique zero $\sigma_m(t)$ near $-2m$ for sufficiently small t , depending on m . Differentiating the actual Hurwitz equation $\partial_t \zeta(s, 1+t) = -s\zeta(s+1, 1+t)$ yields

$$\begin{aligned} \sigma_m(0) &= -2m, & \sigma'_m(0) &= \frac{B_{2m}}{\zeta'(-2m)} = -4 \frac{\zeta(2m)}{\zeta(2m+1)} < 0, \\ z_m(t) &= i(4m+1) + 8i \frac{\zeta(2m)}{\zeta(2m+1)}t + O_m(t^2). \end{aligned} \tag{HPR18}$$

The branch is real in s for real t , by conjugation and uniqueness. Its derivative is nonzero, so for small nonzero t it leaves the fixed gamma pole and becomes an actual off-real zero of H_t^{Hur} . At $t = 0$ it cancels that pole exactly. The radii here are not claimed uniform in m ; (HPR17) separately handles the entire zero divisor on its explicit common interval.

At the separated sheet $t = 1$ there is also an exact endpoint sign witness: $\zeta(-20) - 1 = -1$ and $\zeta(-19) - 1 = 168011/6600 > 0$. Thus $\zeta(s) - 1$ has a real zero in $(-20, -19)$. $A(s)$ is finite and nonzero on that open interval, so it gives an actual zero of H_1^{Hur} on $i(39, 41)$. Criterion (HPR11) produces a finite rational negative trace test for that zero, with its complete tail retained. This endpoint observation does not assert that the simple test of (HPR17) remains negative all the way to $t = 1$.

Equations (HPR7), (HPR13) and (HPR16)–(HPR18) identify exactly which part of the shifted all-zero trace returns to the original theta pairing. The correction is an explicit divisor and an explicit infinite series. Neither its nonzero shifted value nor the strict negative value in (HPR17) determines the sign of the complete unshifted Schur residual constructed in RWR18–27.

55.4 Exact return to every original arithmetic moment

The boundary correction itself gives the inverse arithmetic comparison. Retain J_t from HTB26–28, so $H_t^{\text{Hur}} + J_t = H_0^{\text{Hur}}$, and on the actual nonvanishing disk about zero set

$$c_t = J_t/H_0^{\text{Hur}}, \quad W_t = (1 - c_t(z))(1 - c_t(-z)) = H_t^{\text{Hur}}(z)H_t^{\text{Hur}}(-z)/H_0^{\text{Hur}}(z)^2.$$

The Mellin kernel determining J_t is the explicit $d_t(x) = e^{-x}(1 - e^{-tx})/(1 - e^{-x})$ of HTB27. Define the actual coefficients

$$\begin{aligned} E_n(t) &= [z^{2n-1}] \frac{W'_t(z)}{W_t(z)}, & P_n(t) &= 2(-1)^n \sum_{m \in I_t} (4m+1)^{-2n}, \\ 2S_n(H_0^{\text{Hur}}) &= T_n(t) + E_n(t) - P_n(t). \end{aligned} \tag{HPR19}$$

To prove the last identity, differentiate $\mathcal{F}_t = Q(H_0^{\text{Hur}})^2 W_t$, use (HPR9) for the first two entire even functions, and compare the coefficient of z^{2n-1} . The result is $S_{F,n} = S_{Q,n} + 2S_n(H_0^{\text{Hur}}) - E_n$.

Substitution in (HPR8) proves (HPR19) for every n , with no limiting interchange at $t = 0$. All coefficients exist because $W_t(0) > 0$.

For a finite degree d let $E_d = (E_{i+j+1})$ and $P_d = (P_{i+j+1})$, and put $v_m = (1, -(4m+1)^{-2}, \dots, (-(4m+1)^{-2})^{d-1})^T$. Then

$$\begin{aligned} 2K_d(0) &= K_d^{\text{Hur}}(t) + E_d(t) - P_d(t), \\ P_d(t) &= -2 \sum_{m \in I_t} (4m+1)^{-2} v_m v_m^T \preceq 0. \end{aligned} \tag{HPR20}$$

The matrix series converges absolutely entrywise. Its quadratic value on a complex vector b is $-2 \sum_{m \in I_t} (4m+1)^{-2} |v_m^T b|^2$, proving the sign. The boundary matrix E_d is retained in full and is not assigned a sign. In particular the negative test in (HPR17) returns to the positive original value by the exact correction $E_1 - P_1$; it is not transported by an unproved positivity-preserving map.

This also returns to the *entire* earlier residual, not only its first 64 coordinates. Put $D_n(t) = T_n(t) + E_n(t) - P_n(t)$, $A = K_{64}(0)$ and, for every $i \geq 64$, $v_i = (D_{i+a+1}(t)/2)_{0 \leq a < 64}$. The exact residual is

$$\mathcal{S}_{ij} = \frac{1}{2} D_{i+j+1}(t) - v_i^T A^{-1} v_j, \quad i, j \geq 64. \tag{HPR21}$$

It is independent of the chosen shift: (HPR19) identifies every entry and every v_i with its original value. Its domain remains the direct sum of finitely supported sequences, with no upper cutoff in i . The positivity of A was certified in RWR15–17; the coordinate map and inverse yielding this Schur form are proved in RWR18–27. Their complete source is retained with this reader. Formula (HPR21) is their exact entrywise identification through the present flow. It does not replace the residual by the known negative pole matrix. Lifting the equality through (HPR12) keeps a common original support label and its tensor image on every side, including when amplitudes cancel. Thus the boundary, pole divisor, original arithmetic trace and complete residual have explicit connecting maps.

56 One-sided distributions, every odd boundary jet, and the actual Weil endpoints

Sources and application. Alain Connes and Walter D. van Suijlekom’s *Quadratic Forms, Real Zeros and Echoes of the Spectral Action*, original author TeX, `symet`, `qua0`, and Remark `nuance` (lines 993–1032 and 1121–1122), distinguish a distribution on $[0, L]$ from its symmetrization on smooth functions on $[-L, L]$. In particular, they explicitly observe that δ'_0 gives a nonzero quadratic form although its symmetrization is zero. That observation is theirs. The calculation below determines every odd derivative, its full boundary-jet space, its completion, and its exact transport to the programme’s original Mellin endpoints. No ordinary smooth symmetrization is substituted for their one-sided construction.

The arithmetic inputs are the complete boundary filter and its inverse (BPZ7–11, proved in the accompanying source), and the original HWA28–38 endpoint and Weil calculation. The full lattice is the one in the programme foundation, Definition `lattice-split`.

56.1 The full one-sided boundary calculation

Retain $L > 0$ and $f, g \in C^\infty([0, L], \mathbb{C})$. Extend them by zero outside this interval for convolution only. For $0 \leq y \leq L$ set

$$C_{f,g}(y) = \int_0^{L-y} (\overline{f(x)} g(x+y) + \overline{f(x+y)} g(x)) dx, \quad Q_{\mathcal{D}}(f, g) = \mathcal{D}(C_{f,g}). \tag{OBJ1}$$

This is precisely $(f^* * g)(y) + (f^* * g)(-y)$ for the source involution $f^*(x) = \overline{f(-x)}$. Substitution $x \mapsto x + y$ in the first convolution proves the identity; both integrals are over their actual overlap interval. The function $C_{f,g}$ is smooth from the right at zero. For the one-sided distribution use the unaltered convention $\delta_0^{(n)}(h) = (-1)^n h^{(n)}(0+)$. In particular,

$$Q_{\delta_0}(f, g) = 2 \int_0^L \bar{f}g, \quad Q_{\delta'_0}(f, g) = \overline{f(0)}g(0) + \overline{f(L)}g(L). \quad (\text{OBJ2})$$

The factor two in the first formula and the two distinct endpoint values in the second are retained. Here is the all-order formula, with no periodic endpoint identification:

$$Q_{\delta_0^{(2r+1)}}(f, g) = \sum_{x \in \{0, L\}} \sum_{j=0}^{2r} (-1)^j \overline{f^{(2r-j)}(x)} g^{(j)}(x), \quad r \geq 0. \quad (\text{OBJ3})$$

To prove it, let $A_{f,g}(y) = \int_y^L \overline{f(x-y)}g(x) dx$. Differentiation at the moving lower endpoint gives $A'_{f,g}(y) = -\overline{f(0)}g(y) - A_{f',g}(y)$. Induction yields

$$A_{f,g}^{(n)}(0+) = (-1)^n \int_0^L \overline{f^{(n)}}g + \sum_{j=0}^{n-1} (-1)^{j+1} \overline{f^{(j)}(0)} g^{(n-1-j)}(0).$$

The other half of $C_{f,g}$ is $\overline{A_{g,f}}$. For odd $n = 2r + 1$, repeated integration by parts gives

$$\int_0^L (\overline{f^{(n)}}g + \bar{f}g^{(n)}) = \left[\sum_{j=0}^{n-1} (-1)^j \overline{f^{(n-1-j)}}g^{(j)} \right]_0^L.$$

The two lower-endpoint sums from A and its conjugate each contribute minus the bracket at zero. Consequently $C_{f,g}^{(n)}(0+)$ is minus the sum of the brackets at zero and L . Multiplication by $(-1)^n = -1$ proves (OBJ3), including (OBJ2). This also proves directly that each form in (OBJ3) is Hermitian: replacing j by $2r - j$ conjugates the formula without changing its sign.

On genuinely smooth functions h defined across zero, $(h(y) + h(-y))^{(2r+1)}|_{y=0} = 0$. Thus all of these odd distributions are erased by ordinary smooth symmetrization. Formula (OBJ3) identifies exactly the boundary data that survives their action on the source's piecewise smooth correlations.

56.2 The exact jet space, signature, and completion

Put $m = 2r + 1$ and write the complete trace map

$$J_r : C^\infty([0, L]) \longrightarrow \mathbb{C}^m \oplus \mathbb{C}^m, \quad J_r f = ((f^{(j)}(0))_{j=0}^{2r}, (f^{(j)}(L))_{j=0}^{2r}). \quad (\text{OBJ4})$$

Let A_r have entries $(A_r)_{ij} = (-1)^j$ when $i + j = 2r$, and zero otherwise, for indices $0 \leq i, j \leq 2r$. It is a real symmetric involution. Formula (OBJ3) becomes the exact pullback

$$Q_{\delta_0^{(2r+1)}}(f, g) = (J_r f)^* \text{diag}(A_r, A_r) J_r g. \quad (\text{OBJ5})$$

The map J_r is onto. Indeed, on polynomials of degree at most $2m - 1$, zero jets at both endpoints mean divisibility by $x^m(x - L)^m$, a polynomial of degree $2m$. The polynomial must be zero. Domain and codomain both have dimension $2m$, so this restricted map is bijective. This supplies a unique polynomial right inverse with all original derivative values, without rescaling its coordinates.

The paired coordinates $j, 2r - j$ for $j < r$ contribute one positive and one negative eigenvalue each. The middle coordinate has eigenvalue $(-1)^r$. The boundary form on the quotient by $\ker J_r$ therefore has the exact inertia

$$\begin{cases} (2r + 2, 2r, 0), & r \text{ even}, \\ (2r, 2r + 2, 0), & r \text{ odd}. \end{cases} \quad (\text{OBJ6})$$

Its algebraic radical on $C^\infty([0, L])$ is exactly $\ker J_r$: surjectivity of J_r and invertibility of both A_r show that a vector pairing to zero with every test has zero jet. Thus the higher odd derivatives provide nontrivial indefinite forms, even though the first odd derivative is positive.

They also specify a concrete boundary completion. Retain the norm

$$\|f\|_{\text{jet}, r}^2 = \|f\|_{L^2(0, L)}^2 + \|J_r f\|_{\mathbb{C}^{2m}}^2. \quad (\text{OBJ7})$$

The completion of this exact norm is

$$\mathcal{H}_{\text{jet}, r} = L^2(0, L) \oplus \mathbb{C}^{2m}, \quad f \mapsto (f, J_r f), \quad (\text{OBJ8})$$

with the displayed product norm. For a proof of density, choose a one-sided smooth cutoff $\chi \in C^\infty([0, \infty))$ supported in $[0, 1]$ that equals one near zero. For prescribed endpoint jets (a_j) , the functions $\chi(x/\varepsilon) \sum_{j < m} a_j x^j / j!$ have exactly those left jets and have L^2 norm tending to zero. At the right endpoint use $\chi((L - x)/\varepsilon) \sum_{j < m} b_j (x - L)^j / j!$. Their supports are disjoint when $2\varepsilon < L$, and all opposite jets vanish. Approximate an arbitrary L^2 function by interior compact smooth functions, add these two functions, and let ε tend to zero. This approximates any (h, a, b) in the product norm, proving (OBJ8). The map forgetting the finite boundary vector is a continuous surjection, with kernel exactly $0 \oplus \mathbb{C}^{2m}$. Equation (OBJ5) extends to this completion as the bounded finite boundary form with the same matrix; none of its boundary coordinates is discarded.

Already $Q_{\delta'_0}$ is not closable as a quadratic form on the original $L^2(0, L)$ with smooth domain. In fact $f_\varepsilon(x) = \chi(x/\varepsilon)$ tends to zero in L^2 , $Q_{\delta'_0}(f_\varepsilon - f_\eta) = 0$, but $Q_{\delta'_0}(f_\varepsilon) = 1$. These are the exact three conditions contradicting closability. Adding a fixed multiple of the L^2 inner product leaves this countersequence unchanged in the limit. This is an obstruction for this specific boundary form; it makes no claim about the closability of a different full arithmetic form.

In the source's original trigonometric basis $U_n(x) = L^{-1/2} e^{2\pi i n x / L}$, the $r = 0$ matrix is

$$(Q_{\delta'_0}(U_i, U_j))_{i, j \in \mathbb{Z}} = (2/L)_{i, j \in \mathbb{Z}}. \quad (\text{OBJ9})$$

For $u_N(x) = \frac{1}{2N+1} \sum_{n=-N}^N e^{2\pi i n x / L}$, $u_N(0) = u_N(L) = 1$ and $\|u_N\|_2^2 = L/(2N+1)$. Hence the same nonclosability occurs on precisely that core: $Q(u_N) = 2$ and $Q(u_N - u_M) = 0$. Its finite matrices have one positive eigenvalue $2(2N+1)/L$ and $2N$ zero eigenvalues. The boundary form alone does not have the simple isolated bottom eigenvector required by Connes–van Suijlekom's real-zero theorem. Equations (OBJ5)–(OBJ8) construct the space retained by this obstruction.

56.3 Transport to the original Mellin endpoints, at every odd order

Return to the unchanged real coordinate v . Fix any $a < b$, $L = b - a$, and $f \in C_c^\infty((a, b), \mathbb{C})$. Write

$$M_f(s) = \int_{\mathbb{R}} f(v) e^{-(s-1/2)v} dv, \quad A = M_f(0), \quad B = M_f(1), \quad u(v) = - \int_{\mathbb{R}} e^{-|v-w|/2} f(w) dw. \quad (\text{OBJ10})$$

The operator $T = \partial_v^2 - 1/4$ satisfies $Tu = f$: the kernel $-e^{-|v|/2}$ is continuous and its first derivative has jump one at zero, so its distributional image under T is δ_0 . Smoothness follows by transferring derivatives to f . For v to the left or right of its support, respectively, the exact exterior formula is

$$u(v) = -e^{v/2} B \quad (v \leq a), \quad u(v) = -e^{-v/2} A \quad (v \geq b). \quad (\text{OBJ11})$$

Because the support lies strictly inside (a, b) these identities hold in neighborhoods of the endpoints. Thus for every $j \geq 0$

$$u^{(j)}(a) = -2^{-j}e^{a/2}B, \quad u^{(j)}(b) = -(-2)^{-j}e^{-b/2}A. \quad (\text{OBJ12})$$

Here $(-2)^{-j} = (-1/2)^j$; neither the original Mellin factors nor the length L has been changed.

Use the explicit translation $u_a(x) = u(a+x)$ on $[0, L]$ only to apply (OBJ1); it is a named linear map, not a change to the original arithmetic test f . The endpoint comparison is the invertible matrix

$$\begin{pmatrix} u(a) \\ u(b) \end{pmatrix} = B_{a,b} \begin{pmatrix} A \\ B \end{pmatrix}, \quad B_{a,b} = \begin{pmatrix} 0 & -e^{a/2} \\ -e^{-b/2} & 0 \end{pmatrix}, \quad \det B_{a,b} = -e^{(a-b)/2} \neq 0. \quad (\text{OBJ13})$$

Equations (OBJ3) and (OBJ12) calculate the whole odd-jet family:

$$Q_{\delta_0^{(2r+1)}}(u_a, u_a) = 4^{-r}(e^a|M_f(1)|^2 + e^{-b}|M_f(0)|^2) \quad (r \geq 0). \quad (\text{OBJ14})$$

For each endpoint, the product of derivatives of total order $2r$ is 4^{-r} times the squared endpoint value; at the right endpoint the two derivative signs multiply to $(-1)^{2r} = 1$. The remaining sum is $\sum_{j=0}^{2r}(-1)^j = 1$. This proves the formula at every order, rather than inferring the higher orders from the first one. The polarized version follows by retaining $\overline{M}_f M_g$.

Consequently all odd one-sided jets on these actual Green images factor through the same two-dimensional endpoint space. Their common kernel is precisely

$$\{f \in C_c^\infty((a, b)) : M_f(0) = M_f(1) = 0\} = TC_c^\infty((a, b)). \quad (\text{OBJ15})$$

The inclusion from right to left is integration by parts: $M_{Tg}(s) = s(s-1)M_g(s)$. Conversely (OBJ11) shows that u is compactly supported inside (a, b) when both endpoints vanish, and $Tu = f$. A compactly supported solution of $Th = 0$ is zero because the only solutions are linear combinations of $e^{v/2}$ and $e^{-v/2}$. This proves uniqueness and the equality in (OBJ15).

For completeness the endpoint map is onto on this fixed interval. Choose $\beta \in C_c^\infty((a, b))$ nonnegative and positive on an interval, set $h_0(v) = e^{v/2}$, $h_1(v) = e^{-v/2}$, and retain $C_{ij} = \int \beta h_i h_j$. Its real quadratic form is strictly positive because h_0, h_1 are linearly independent on every interval. The explicit right inverse is $R(w) = \beta \sum_j (C^{-1}w)_j h_j$. Direct integration gives $(M_{R(w)}(0), M_{R(w)}(1)) = w$. Hence the complete exact sequence is

$$0 \longrightarrow C_c^\infty((a, b)) \xrightarrow{T} C_c^\infty((a, b)) \xrightarrow{(M(0), M(1))} \mathbb{C}^2 \longrightarrow 0. \quad (\text{OBJ16})$$

The source's positive boundary form must not be substituted for the programme's endpoint form. In the original A, B coordinates HWA gives $P_e(f, f) = 2 \operatorname{Re}(\overline{A}B)$ and $K_e(f, f) = \frac{1}{2}(|A|^2 + |B|^2) - \frac{5}{3} \operatorname{Re}(\overline{A}B)$. Their exact images under (OBJ13), writing $y_0 = u(a), y_L = u(b)$, are

$$P_e(f, f) = 2e^{(b-a)/2} \operatorname{Re}(\overline{y_L} y_0), \quad (\text{OBJ17})$$

$$K_e(f, f) = \frac{1}{2}(e^b |y_L|^2 + e^{-a} |y_0|^2) - \frac{5}{3}e^{(b-a)/2} \operatorname{Re}(\overline{y_L} y_0). \quad (\text{OBJ18})$$

These are exact congruences by $B_{a,b}^{-1}$; they keep every original coefficient and sign. They prove the relation between the constructions even though their individual Hermitian forms differ.

The complete Weil trace from BPZ2 is $\mathcal{Z}(f, g) = \sum_\rho m_\rho \overline{M}_f(1 - \bar{\rho}) M_g(\rho)$. The accompanying BPZ proof shows that every actual off-line zero produces a compact test in $\ker(M(0), M(1))$ with $\mathcal{Z}(f, f) < 0$. Choose a, b containing that test's support; then (OBJ14)–(OBJ18) vanish for that same test, for every r . Thus increasing the order of these boundary jets never by itself removes the complete primitive zero detector. The retained arithmetic value is still $\mathcal{Z}(f, f)$, including the prime and archimedean terms in HWA33–34. This is a proved common-kernel statement with an exact return map, not an assertion that boundary geometry is unrelated to the arithmetic trace.

56.4 The arbitrary lattice and its injective comparison

For every vector space V in (OBJ4), (OBJ8), (OBJ10), or (OBJ16), keep

$$G_\Lambda(V) = \{(0, \lambda) : \lambda \in \Lambda\} \cup (V \times \{1_\Lambda\}), \quad (v, \lambda) + (w, \mu) = (v + w, \lambda \vee \mu), \quad (c, \nu)(v, \lambda) = (cv, \nu \wedge \lambda). \quad (\text{OBJ19})$$

Here Λ is the full original bounded distributive lattice; the change of its letter distinguishes it from the interval length L and changes none of its elements or operations. Each displayed linear map F has the exact lift $F^\uparrow(v, \lambda) = (Fv, \lambda)$. Direct substitution in (OBJ19) proves additivity and scalar linearity. The lifted composition is the lift of the composite, so all the displayed diagrams, right inverses, and inverse maps still commute. On nonzero kernel vectors at top support the image is $e = (0, 1_\Lambda)$; every lower labelled zero retains its own label. In particular such a kernel vector is not mapped to $\tau = (0, 0_\Lambda)$.

Each Hermitian form Q lifts as $Q^\uparrow((v, \lambda), (w, \mu)) = (Q(v, w), \lambda \wedge \mu)$. It is well-defined since a nonzero amplitude requires both input amplitudes, and hence both labels, to be top. Sesquilinearity follows from distributivity of meet over join. It follows in particular that (OBJ14) is an equality of the retained supported values with the original common input label. Independent labels are not assigned to the two components of $J_r f$ or to the two Mellin endpoints.

Finally retain the programme's idempotent semifield K and its Boolean base $\mathbb{B}$. The comparison is

$$\Phi_V(v, \lambda) = (v, 1_K \otimes \lambda) \in V \times (K \otimes_{\mathbb{B}} \Lambda), \quad (F \times I)\Phi_V = \Phi_W F^\uparrow. \quad (\text{OBJ20})$$

The map $\chi : K \rightarrow \mathbb{B}$ sends zero to zero and every nonzero element to one. Idempotent addition cannot sum a nonzero element to zero, and a semifield has no zero divisors; therefore χ preserves addition and multiplication. Consequently $\chi \otimes I$ retracts $\lambda \mapsto 1_K \otimes \lambda$, proving injectivity of (OBJ20). All boundary vectors, all common kernels and every lower support label calculated above survive this comparison. No localization at e and no Boolean specialization of the lattice has occurred.

57 The exact Cayley–Toeplitz receiver of the original entire zero trace

Objects and provenance. The entire function is the original Rodgers–Tao function $H(Z) = H_0(Z) = \xi(1/2 + iZ/2)/8$, with its original factor $1/8$, coordinate Z , and positive theta kernel. The source is Brad Rodgers and Terence Tao, *The de Bruijn–Newman constant is non-negative*, the displayed definitions of Φ and H_t and the identity $H_0 = \xi/8$. The complete local-jet and Laurent receiver used here is (ALM1)–(ALM32), *A finite rational Laurent witness for every off-real heat zero*; the retained positive block and its exact residual are (RWR18)–(RWR27), *Independent reproduction of the supplied rational witness*. The Toeplitz comparison source is Alain Connes and Walter D. van Suijlekom, *Quadratic Forms, Real Zeros and Echoes of the Spectral Action*, original `Araki-final-oct25.tex`, the introduction's Corollary `corcar` and Section 2, Proposition `toeplitzcase`, read directly in source lines 751–949. Their statement is the Carathéodory–Fejér theorem's corank-one consequence, with its positivity and rank hypotheses. The present calculation connects that receiver to Toeplitz forms; it does not identify a finite Toeplitz eigenvector with a theta kernel. All implications and maps used for this connection are proved below. The support carrier is the original foundation retained under the byline The Clankers, *Split Support Geometry*, v11, Definition `def:lattice-split`.

57.1 The actual function, every zero, and the complete rational domain

Keep, without rescaling,

$$H(Z) = \int_0^\infty \Phi(u) \cos(Zu) du, \quad \Phi(u) = \sum_{n \geq 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}. \quad (\text{CTP1})$$

Every summand is positive for $u \geq 0$. The bound $0 < \Phi(u) \leq Ce^{9u - (\pi/2)e^{4u}}$, with $C = 2\pi^2 \sum_{n \geq 1} n^4 e^{-(\pi/2)n^2} < \infty$, follows by discarding the negative summand and using $n^2 e^{4u} \geq (n^2 + e^{4u})/2$. Thus H is entire, even, and real on the real axis, and differentiation under its integral is valid on every compact set. In particular

$$H(iy) = \int_0^\infty \Phi(u) \cosh(yu) du > 0 \quad (y \in \mathbb{R}). \quad (\text{CTP2})$$

Write $\mathcal{Z}$ for its distinct zeros and m_w for the full multiplicity of $w \in \mathcal{Z}$. The estimates and paired factorization in (ALM2)–(ALM4) give

$$\sum_{w \in \mathcal{Z}} \frac{m_w}{|w|^2} < \infty, \quad H(Z) = H(0) \prod_{\{w, -w\}} (1 - Z^2/w^2)^{m_w}, \quad \ell(Z) := \frac{H'(Z)}{H(Z)} = \sum_{\{w, -w\}} \frac{2m_w Z}{Z^2 - w^2}. \quad (\text{CTP3})$$

Here the product takes one representative of each sign pair, whereas all sums below labeled $\mathcal{Z}$ take both roots and all multiplicities. For reference, the bound on Φ gives $\log \max_{|Z| \leq R} |H(Z)| = O(R \log(2 + R))$ by maximizing $Ru - (\pi/8)e^{4u}$. Jensen's formula gives $n(R) = O(R \log(2 + R))$, and dyadic summation gives the first statement. Hadamard factorization, paired under $w \mapsto -w$, then gives the displayed product: the remaining exponential is of degree at most one and its linear term is zero by evenness. The logarithmic derivative follows locally uniformly off the zeros. These growth and product facts matter: evenness and the zero set alone would not exclude multiplying H by e^{Z^2} and changing ℓ . There are infinitely many distinct zeros, since a finite product would be a polynomial, whereas every coefficient $(-1)^j \int u^{2j} \Phi(u) du / (2j)!$ is nonzero.

Fix a real $b > 0$. By (CTP2), $0, ib, -ib$ are not zeros. Let

$$\mathcal{A}_b = \{F \in \mathbb{C}(Z) : \text{all finite poles of } F \text{ belong to } \{0, ib, -ib\}, F(Z) = O(Z^{-1}) \text{ at infinity}\}. \quad (\text{CTP4})$$

This is a complex vector space, preserved by $F^\dagger(Z) = \overline{F(\bar{Z})}$. It contains the original Laurent space $\mathcal{R} = \{Z^{-1}p(Z^{-2}) : p \in \mathbb{C}[X]\}$. Its full local-jet map and pairing are

$$j_b(F) = ([F]_w)_{w \in \mathcal{Z}} \in \prod_{w \in \mathcal{Z}} \mathcal{O}_{\mathbb{C}, w}/(H), \quad (\text{CTP5})$$

$$\mathcal{T}(F, G) = \sum_{w \in \mathcal{Z}} m_w \overline{F(\bar{w})} G(w) = \sum_{w \in \mathcal{Z}} \text{Res}_{Z=w} F^\dagger(Z) G(Z) \ell(Z) dZ.$$

The series is absolutely convergent by (CTP3), since its summands are $O(m_w |w|^{-2})$ outside a finite set. Every local algebra retains its original length m_w : if $H = (Z - w)^m U$ with $U(w) \neq 0$, then $\ell = m/(Z - w) + U'/U$, giving the displayed residue. Only the resulting trace kills nilpotents. The jet map itself retains them. A rational function vanishing in all these jets vanishes at infinitely many distinct points and is zero, so j_b is injective. The infinite pairing is defined on $\mathcal{A}_b$ or its specified injective image, not on an unrestricted product of local algebras. Conjugation of the sum and the substitution $w \mapsto \bar{w}$ prove that $\mathcal{T}$ is Hermitian.

The original arithmetic residue factors remain exactly $s = 1/2 + iZ/2$, $g(s) = 16H(-2i(s - 1/2))$, $ds/g = (i/32)dZ/H$, and $\alpha^* g' = -32iH'$. Their product is one. Consequently (CTP5) is the same Jacobian residue contraction as (ALM7)–(ALM8), extended to the indicated rational domain.

57.2 A finite partial-fraction rule for the entire zero trace

Suppose $r \in \mathbb{C}(Z)$ has poles only in $\{0, ib, -ib\}$ and is $O(Z^{-2})$. Its unique partial fraction expansion is

$$r(Z) = \sum_{a \in \{0, ib, -ib\}} \sum_{n=1}^{N_a} A_{a,n} (Z - a)^{-n}, \quad \sum_a A_{a,1} = 0. \quad (\text{CTP6})$$

The last equality is its vanishing coefficient of Z^{-1} at infinity. Every coefficient is an exact finite derivative: $A_{a,n} = \left[(\partial_Z^{N_a-n} / (N_a - n)!)((Z - a)^{N_a} r(Z)) \right]_{Z=a}$ when N_a is the pole order. Then

$$\sum_{w \in \mathcal{Z}} m_w r(w) = - \sum_{a \in \{0, ib, -ib\}} \sum_{n=1}^{N_a} \frac{A_{a,n}}{(n-1)!} \ell^{(n-1)}(a). \quad (\text{CTP7})$$

For $n \geq 2$, differentiating the paired, locally uniform series in (CTP3) gives $\sum_w m_w (w - a)^{-n} = -\ell^{(n-1)}(a)/(n-1)!$. For the simple-pole terms, use $\sum_a A_{a,1} = 0$ before summation: $\sum_a A_{a,1} (w - a)^{-1} = \sum_a A_{a,1} ((w - a)^{-1} - w^{-1})$. Each difference is $O(|w|^{-2})$ and its sum is $-\ell(a)$ by the paired product and $\ell(0) = 0$. This proves (CTP7) without assigning an absolutely convergent meaning to an isolated $\sum_w 1/w$. Apply (CTP7) to $r = F^\dagger G$ to compute every finite matrix of (CTP5) from finitely many derivatives of the actual H at $0, ib, -ib$. It is an exact finite evaluation rule with all zeros present, not a truncation of the zero multiset.

57.3 Cayley coordinates, the Toeplitz matrix, and its exact finite map

Set

$$C_b(Z) = \frac{Z - ib}{Z + ib}, \quad F_j(Z) = \frac{C_b(Z)^j}{Z + ib} \quad (j \in \mathbb{Z}), \quad c_k(b) = \sum_{w \in \mathcal{Z}} \frac{m_w}{w^2 + b^2} C_b(w)^k \quad (k \in \mathbb{Z}). \quad (\text{CTP8})$$

The values $C_b(w)$ are nonzero and finite. They tend to one as $|w| \rightarrow \infty$. Thus, for every fixed integer k , $C_b(w)^k$ is bounded over the root set and $|w^2 + b^2|^{-1} = O(|w|^{-2})$. This proves absolute convergence of every c_k and of every pairing of the F_j .

The involution has the precise, necessary orientation

$$C_b^\dagger = C_b^{-1}, \quad F_i^\dagger(Z) = \frac{C_b(Z)^{-i}}{Z - ib} = F_{-i-1}(Z), \quad F_i^\dagger F_j = \frac{C_b^{j-i}}{Z^2 + b^2}. \quad (\text{CTP9})$$

For $d \geq 1$ define the complex-linear injection

$$W_{b,d} : \mathbb{C}^d \longrightarrow \mathcal{A}_b, \quad a \longmapsto \sum_{j=0}^{d-1} a_j F_j, \quad T_d(b) = (c_{j-i}(b))_{0 \leq i, j < d}. \quad (\text{CTP10})$$

Equations (CTP5) and (CTP9) prove, with no scalar factor missing,

$$\mathcal{T}(W_{b,d} a, W_{b,d} v) = a^* T_d(b) v. \quad (\text{CTP11})$$

The map is injective because C_b is a nonconstant rational coordinate. The identities $C_b(-w) = C_b(w)^{-1}$ and $C_b(-\bar{w}) = \overline{C_b(w)}$, together with the unchanged multiplicities and weights, give

$$c_{-k} = c_k = \overline{c_k}. \quad (\text{CTP12})$$

Thus the exact matrix is real symmetric and Toeplitz. This conclusion uses both evenness and reality of the original H .

The finite coefficient map also has an explicit rational inverse. If $p(U) = \sum_{j < d} a_j U^j$, write

$$W_{b,d} a(Z) = \frac{q(Z)}{(Z + ib)^d}, \quad q(Z) = \sum_{j=0}^{d-1} a_j (Z - ib)^j (Z + ib)^{d-j-1}. \quad (\text{CTP13})$$

This maps all polynomials of degree less than d bijectively to all such q ; the inverse is

$$p(U) = \frac{(1 - U)^{d-1}}{(2ib)^{d-1}} q\left(ib \frac{1 + U}{1 - U}\right). \quad (\text{CTP14})$$

The right side is a polynomial of degree at most $d - 1$ because $\deg q < d$. Substituting $U = C_b(Z)$ and $1 - U = 2ib/(Z + ib)$ verifies both compositions exactly. For rational $b > 0$ these maps have coefficients in $\mathbb{Q}(i)$. The union of the images is precisely the rational functions vanishing at infinity whose only possible pole is $-ib$.

57.4 An exact resolvent from the original logarithmic derivative

For $z \neq 1$ put $v_b(z) = -ib(1+z)/(1-z)$, and let $R_b = \sup_{w \in \mathcal{Z}} |C_b(w)| < \infty$. The geometric series, dominated by the absolutely summable weights for $|z|R_b < 1$, gives

$$\begin{aligned} G_b(z) &:= \sum_{k \geq 0} c_k z^k = \sum_{w \in \mathcal{Z}} \frac{m_w}{(w^2 + b^2)(1 - zC_b(w))} \\ &= \frac{\ell(v_b(z)) - \ell(ib)}{2ib}. \end{aligned} \quad (\text{CTP15})$$

Indeed

$$\frac{1}{(w^2 + b^2)(1 - zC_b(w))} = \frac{1}{2ib} \left(\frac{1}{w - ib} - \frac{1}{w - v_b(z)} \right).$$

The difference is summable and (CTP3) sums it to the stated logarithmic derivatives. The expression is meromorphic on $\mathbb{C} \setminus \{1\}$. Alternatively its root series converges locally uniformly off 1 and the indicated poles: outside a finite root set, $C_b(w)$ is uniformly close to one and the denominators stay uniformly away from zero on such a compact set. Its only finite poles away from 1 are

$$z_w = C_b(w)^{-1} = \frac{w + ib}{w - ib}, \quad \text{Res}_{z=z_w} G_b(z) = -\frac{m_w}{(w - ib)^2} \neq 0. \quad (\text{CTP16})$$

The Cayley map is injective, so distinct roots give distinct poles. For a chosen $z_w \neq 1$, choose a closed disk about z_w disjoint from 1 and all other poles. Such a disk exists: an infinite sequence of distinct roots has modulus tending to infinity, and its Cayley pole images tend to 1; the remaining finitely many images are distinct. On this disk the series with the one w term removed is locally uniformly convergent and holomorphic, by the preceding denominator bound. It cannot cancel the pole. The multiplicity enters the nonzero residue and cannot disappear. For a direct sign check, $v'_b(z) = -2ib/(1-z)^2$, so the residue in the last expression of (CTP15) is $m_w(1 - z_w)^2/(4b^2) = -m_w/(w - ib)^2$.

Putting $z = 0$ and using $\ell(-ib) = -\ell(ib)$ gives

$$c_0 = -\frac{H'(ib)}{ibH(ib)} = \frac{\int_0^\infty u\Phi(u) \sinh(bu)du}{b \int_0^\infty \Phi(u) \cosh(bu)du} > 0. \quad (\text{CTP17})$$

The strict inequality uses the actual positive kernel in (CTP1), not merely the fact that $H(ib) > 0$. For every $k \geq 1$, expand $v_b(z) = -ib - 2ibz/(1-z)$ near zero. The coefficient of z^k in $(z/(1-z))^r$ is $\binom{k-1}{r-1}$ for $1 \leq r \leq k$. Since $\ell^{(r)}(-ib) = (-1)^{r+1}\ell^{(r)}(ib)$, one obtains the finite formula

$$c_k = -\sum_{r=1}^k \binom{k-1}{r-1} \frac{(2ib)^{r-1}}{r!} \ell^{(r)}(ib). \quad (\text{CTP18})$$

All derivatives are of the original H ; its original mass has not been reassigned. For example, $c_1 = -\ell'(ib)$ equals

$$c_1 = \frac{\int u^2 \Phi(u) \cosh(bu)du}{\int \Phi(u) \cosh(bu)du} - \left(\frac{\int u \Phi(u) \sinh(bu)du}{\int \Phi(u) \cosh(bu)du} \right)^2 > 0. \quad (\text{CTP19})$$

Cauchy–Schwarz bounds the subtracted numerator by $(\int \Phi \cosh)(\int u^2 \Phi \sinh^2 / \cosh)$, which is strictly less than the first product because $\cosh - \sinh^2 / \cosh = 1/\cosh > 0$. This proves the last inequality. It does not prove $c_1 \leq c_0$ or any later Toeplitz inequality.

57.5 Completeness: every nonreal zero yields a two-coefficient test

If all the roots are real, (CTP8)–(CTP11) give the positive finite measure and its exact Toeplitz form

$$\mu_b = \sum_{w \in \mathcal{Z}} \frac{m_w}{w^2 + b^2} \delta_{C_b(w)}, \quad \mu_b(\mathbb{T}) = c_0, \quad a^* T_d a = \int_{\mathbb{T}} |p(U)|^2 d\mu_b(U) > 0 \quad (a \neq 0). \quad (\text{CTP20})$$

Here $\mathbb{T} = \{U : |U| = 1\}$. Strictness follows from infinitely many distinct atoms and the fact that a nonzero polynomial has only finitely many zeros. No division by c_0 has been made.

If a nonreal root exists, conjugation supplies one with $\text{Im } w < 0$. Directly,

$$|C_b(w)|^2 = \frac{(\text{Re } w)^2 + (\text{Im } w - b)^2}{(\text{Re } w)^2 + (\text{Im } w + b)^2} > 1. \quad (\text{CTP21})$$

Since $C_b(w) \rightarrow 1$ along unbounded roots, $R_b > 1$ is then a maximum, attained in a finite set. Equations (CTP15)–(CTP16) show that the Taylor series at zero has radius exactly $1/R_b < 1$: there is no pole at smaller modulus, and a pole at that modulus has a nonzero residue. Therefore

$$\limsup_{k \rightarrow \infty} |c_k|^{1/k} = R_b > 1. \quad (\text{CTP22})$$

In particular there is a finite $k \geq 1$ with $|c_k| > c_0$. Because c_k is real, set $\varepsilon_k = \text{sgn } c_k$ and keep the two coefficients exactly equal to 1, $-\varepsilon_k$:

$$f_{b,k}(Z) = \frac{1 - \varepsilon_k C_b(Z)^k}{Z + ib}, \quad \mathcal{T}(f_{b,k}, f_{b,k}) = 2(c_0 - |c_k|) < 0. \quad (\text{CTP23})$$

This is a finite integer coefficient vector in the original T_{k+1} , with zero entries in all positions except $0, k$. For rational b the function itself has coefficients in $\mathbb{Q}(i)$. The series giving its trace still includes every root and every multiplicity.

For any fixed $b > 0$ we have consequently proved the equivalence

$$\begin{aligned} & \text{all zeros of the actual } H \text{ are real} \\ & \iff T_d(b) \succ 0 \text{ for every } d \geq 1 \\ & \iff T_d(b) \succeq 0 \text{ for every } d \geq 1 \\ & \iff |c_k(b)| \leq c_0(b) \text{ for every } k \geq 1. \end{aligned} \quad (\text{CTP24})$$

For the final implication use the contrapositive (CTP22); the forward implication also follows from (CTP20). Positive semidefiniteness implies the last bound by the principal block on indices $0, k$ in T_{k+1} . These are completeness statements for the constructed receiver. No assertion that their infinite positivity condition holds has been made. In the real-root case the Taylor radius is exactly one: the poles z_w lie on the unit circle, are individually nonzero, and accumulate only at one. Thus in both cases the growth identity $\limsup |c_k|^{1/k} = R_b$ holds, with $R_b = 1$ in the real-root case.

57.6 The exact relation to all Laurent moments, at every value of the parameter

Retain the reciprocal nodes and their entire multiplicity fibers from (ALM10): $\Lambda = \{w^{-2} : w \in \mathcal{Z}\}$, $M_\lambda = 2m_w$ when $\lambda = w^{-2}$, $V = \sum_\lambda M_\lambda |\lambda| < \infty$, and $R = \max_\lambda |\lambda|$. Then, for $k \geq 0$,

$$\begin{aligned} P_k(Y) &= \sum_{l=0}^k (-1)^l \binom{2k}{2l} Y^l, \\ c_k &= \sum_{\lambda \in \Lambda} M_\lambda \lambda \frac{P_k(b^2 \lambda)}{(1 + b^2 \lambda)^{k+1}}. \end{aligned} \quad (\text{CTP25})$$

To verify the identity, average the w and $-w$ summands of (CTP8), using unchanged multiplicities. Their even rational factor is

$$\frac{C_b(w)^k + C_b(w)^{-k}}{2(w^2 + b^2)} = \frac{\sum_{l=0}^k (-1)^l \binom{2k}{2l} b^{2l} w^{2k-2l}}{(w^2 + b^2)^{k+1}}.$$

This is precisely the function of λ in (CTP25). Its denominator never vanishes on the actual node set because $H(ib) \neq 0$. The aggregation loses neither member of a sign pair.

When $b^2 R < 1$, the exact convergent expansion in the original Laurent moments $S_n = \sum_w m_w w^{-2n}$ is

$$c_k = \sum_{n \geq 0} (-1)^n b^{2n} S_{n+1} \sum_{l=0}^{\min(k,n)} \binom{2k}{2l} \binom{k+n-l}{n-l}. \quad (\text{CTP26})$$

Expand $(1+Y)^{-k-1}$ by the binomial series and multiply by $P_k(Y)$. The displayed inner sum is its coefficient with sign $(-1)^n$. Absolute convergence and interchange with the root sum follow from $|S_{n+1}| \leq VR^n$ and a polynomial bound in n for the inner sum, with the fixed ratio $b^2 R < 1$. This restricted series is not used outside its stated disk.

For every $b > 0$, even when that series does not converge, the rational function in (CTP25) is a controlled limit of actual Laurent polynomials. Here is a constructive proof that also fixes the omitted infinite tail. Let $r_k(X) = P_k(b^2 X)/(1 + b^2 X)^{k+1}$. Choose $0 < r_0 < r_1 < 1/b^2$ with no node of modulus r_0 and let $S = \{\lambda \in \Lambda : |\lambda| > r_0\}$, a finite set. The Taylor series of r_k converges uniformly on $|X| \leq r_0$; for any $\epsilon > 0$ take a Taylor polynomial Q whose error there is less than $\epsilon/2$. For $\lambda \in S$ define $L_\lambda(X) = \prod_{\mu \in S \setminus \{\lambda\}} (X - \mu)/(\lambda - \mu)$ and $B_\lambda = \prod_{\mu \neq \lambda} (r_0 + |\mu|)/|\lambda - \mu|$. Choose a finite integer N such that

$$\sum_{\lambda \in S} |r_k(\lambda) - Q(\lambda)| B_\lambda (r_0/|\lambda|)^N < \epsilon/2, \quad Q_\epsilon(X) = Q(X) + \sum_{\lambda \in S} [r_k(\lambda) - Q(\lambda)] (X/\lambda)^N L_\lambda(X). \quad (\text{CTP27})$$

If S is empty the correction sum is zero. Each ratio in a nonempty sum is strictly less than one, so the choice exists. The correction has the prescribed value at each node in S , and on the whole remaining disk its absolute size is less than $\epsilon/2$. Consequently

$$\sup_{\lambda \in \Lambda} |Q_\epsilon(\lambda) - r_k(\lambda)| < \epsilon, \quad \left| \sum_{\lambda} M_\lambda \lambda Q_\epsilon(\lambda) - c_k \right| \leq V\epsilon. \quad (\text{CTP28})$$

Conjugation pairs make Q_ϵ real when Q is taken real. Every sum on the left is a finite linear combination of the original S_n . Thus (CTP25)–(CTP28) give an exact, globally valid bridge from the Laurent receiver, with a proved full-tail bound. They do not delete a cutoff-dependent boundary term.

There is no nonzero finite identification of the two original test spaces by equality of their full jets: a Laurent test has its only possible pole at zero, whereas a Cayley test in the union of (CTP10) has its only possible pole at $-ib$. Equality in every jet implies rational equality by the injectivity in (CTP5); such a common function would have no finite pole and vanish at infinity, hence be zero. Their precise relation is the common injective rational receiver (CTP4)–(CTP7), the moment maps (CTP25)–(CTP28), and the following exact residual.

57.7 The original positive block and the Cayley residual

Keep the certified block $A = K_{64}(0) \succ 0$ of (RWR18), in its original coordinates, and the original injection $E : \mathbb{C}^{64} \rightarrow \mathcal{R}$, $Ea = \sum_{h=0}^{63} a_h Z^{-2h-1}$. The positivity and its interval certificate are already part of (RWR1)–(RWR17); no new numerical assertion is used here. For $F \in \mathcal{A}_b$ set $r(F)_h = \mathcal{T}(Ee_h, F)$ and define

$$\Pi_b F = F - EA^{-1}r(F), \quad \mathcal{T}(Ea, \Pi_b F) = 0, \quad \Pi_b^2 = \Pi_b. \quad (\text{CTP29})$$

These equalities follow from $r(Ea) = Aa$. The kernel of Π_b is exactly $E\mathbb{C}^{64}$, and its image is the orthogonal complement of that space in $\mathcal{A}_b$. For the original Laurent subspace this restricts exactly to (RWR25). For $0 \leq j < d$ set

$$B_{hj} = \mathcal{T}(Z^{-2h-1}, F_j), \quad D_d(b) = T_d(b) - B^* A^{-1} B, \quad \widetilde{W}_{b,d} = \Pi_b W_{b,d}. \quad (\text{CTP30})$$

The entries of B have the exact finite derivative evaluation (CTP7) with $r(Z) = Z^{-2h-1}(Z - ib)^j / (Z + ib)^{j+1}$. All constants, signs, and poles are thus specified, including the full root sum. Direct expansion proves

$$\begin{aligned} \mathcal{T}(\widetilde{W}a, \widetilde{W}v) &= a^* D_d(b) v, \\ \mathcal{T}(E\alpha + Wa, E\alpha + Wa) &= (\alpha + A^{-1}Ba)^* A(\alpha + A^{-1}Ba) + a^* D_d(b)a. \end{aligned} \quad (\text{CTP31})$$

The map $\widetilde{W}$ is injective: a vector in its kernel would give a common nonzero Laurent and Cayley function, which the preceding pole argument excludes. The block coordinate change is exactly $(\alpha, a) \mapsto (\alpha + A^{-1}Ba, a)$, with inverse $(\beta, a) \mapsto (\beta - A^{-1}Ba, a)$. Every negative test (CTP23) has $a^* D_d a = a^* T_d a - (Ba)^* A^{-1}(Ba) < 0$. Conversely, positivity of every D_d implies positivity of every T_d by (CTP30), hence reality of all roots by (CTP24). If all roots are real, (CTP5) is positive on every nonzero rational test, so every D_d is strictly positive by injectivity. This proves a complete alternative residual criterion, in the same original trace, with no assertion about its unresolved sign. On the direct sum of the original Laurent and Cayley test spaces, (CTP29) also retains every mixed matrix entry; replacing that direct sum by two independently positive blocks would discard those entries.

57.8 The entire support lattice and all connecting maps

Retain the original arbitrary nontrivial bounded distributive lattice L . For every vector space V used above keep

$$\begin{aligned} G_L(V) &= \{(0, \lambda) : \lambda \in L\} \cup (V \times \{1_L\}), \\ (v, \lambda) + (w, \mu) &= (v + w, \lambda \vee \mu), \quad (a, \eta)(v, \lambda) = (av, \eta \wedge \lambda), \\ \tau &= (0, 0_L), \quad e = (0, 1_L). \end{aligned} \quad (\text{CTP32})$$

For every complex-linear map $A : V \rightarrow W$ in (CTP5), (CTP10), (CTP13)–(CTP14), and (CTP29)–(CTP31), the exact lift $A^\uparrow(v, \lambda) = (Av, \lambda)$ is well defined, additive, and $G_L(\mathbb{C})$ -linear. Nonzero output amplitude requires nonzero input amplitude and hence top support; the operation checks are direct substitution and lattice distributivity. The pairing lift is

$$\mathcal{T}^L((F, \lambda), (G, \mu)) = (\mathcal{T}(F, G), \lambda \wedge \mu), \quad \mathcal{T}^L(W^\uparrow(a, \lambda), W^\uparrow(v, \mu)) = (a^* T_d v, \lambda \wedge \mu). \quad (\text{CTP33})$$

It is well defined because a nonzero pairing requires both amplitudes nonzero. Sesquilinearity, including the unchanged involution $(a, \eta)^* = (\bar{a}, \eta)$, follows from the scalar pairing and distributivity of meet over join. The same formula applies to D_d , B , and the original Laurent residual.

Retain the programme's idempotent semifield K with zero, $K_L = K \otimes_{\mathbb{B}} L$, and $\iota_L(\lambda) = 1_K \otimes \lambda$. The full comparison and its commuting maps are

$$\Phi_V(v, \lambda) = (v, \iota_L(\lambda)), \quad (A \times I)\Phi_V = \Phi_W A^\uparrow. \quad (\text{CTP34})$$

The nonzero character $\chi : K \rightarrow \mathbb{B}$ is a semiring map: in an idempotent semifield a sum is zero only when both inputs are zero, and a product is zero only when one input is zero. Therefore $\rho_L = \chi \otimes I_L$ satisfies $\rho_L \iota_L = I_L$. This proves that the comparison retains every original support label. Equations (CTP33)–(CTP34) commute with the pairing, with meet of the two input labels. Every canceled amplitude at top support remains e and is not changed to τ . The finite coordinate

decompositions use $G_L(\mathbb{C}^{64} \oplus \mathbb{C}^d)$ with its one retained label, not a product with independently chosen labels.

In particular the negative test (CTP23), when present, has the unchanged image

$$(2(c_0 - |c_k|), 1_L) \xrightarrow{\Phi_{\mathbb{R}}} (2(c_0 - |c_k|), 1_K \otimes 1_L). \quad (\text{CTP35})$$

The original tau geometry, every zero multiplicity, the complete arithmetic residue factor, and all finite or infinite moment indices have survived the construction. The mathematics above constructs and proves the receiver and its precise detection criterion; it does not evaluate the sign of all its forms.

57.9 The exact Li and Weil maps, including the original factors

The coefficient criterion originates in Xian-Jin Li, *The positivity of a sequence of numbers and the Riemann hypothesis* (1997). Enrico Bombieri and Jeffrey C. Lagarias developed its zero-multiset and Weil-functional relations in *Complements to Li's criterion for the Riemann hypothesis* (1999). The original author LaTeX actually consulted here is Jeffrey C. Lagarias, *Li Coefficients for Automorphic L-Functions*, `math-0404394v4.tex`, lines 197–359 and 1170–1424: source equations 103, 101, 102a, 301, 303, 304, and Theorem th31. The author explicitly credits J. B. Keiper's 1992 precursor coefficients in lines 263–274. The following map identifies the present objects with those coefficients and test functions; no historical novelty for the coefficient criterion is asserted.

For arbitrary $b > 0$ define, in a neighborhood of $z = 0$, the unique holomorphic logarithm with value zero at zero by

$$\log \frac{H(v_b(z))}{H(ib)} = \sum_{n \geq 1} \frac{\Lambda_n(b)}{n} z^n, \quad \Lambda_0(b) = 0, \quad \Lambda_{-n}(b) = \Lambda_n(b). \quad (\text{CTP36})$$

This is legitimate because $H(v_b(0)) = H(-ib) = H(ib) > 0$. There is no change to H or its mass: the displayed ratio merely specifies the logarithm's constant. For each sign pair of roots, the exact factor identity is

$$\frac{1 - v_b(z)^2/w^2}{1 + b^2/w^2} = \frac{(1 - zC_b(w))(1 - zC_b(w)^{-1})}{(1 - z)^2}.$$

Logarithmically differentiate the locally uniformly convergent paired product, or take coefficients in its logarithm on a small zero-free disk. This gives

$$\Lambda_n(b) = \sum_{\{w, -w\}} m_w (2 - C_b(w)^n - C_b(w)^{-n}) \quad (n \geq 1). \quad (\text{CTP37})$$

The summand is $O(m_w |w|^{-2})$: for fixed n the first-order terms in $C_b(w) - 1 = O(1/w)$ cancel between the two powers. Thus the paired sum is absolutely convergent with every multiplicity retained. Differentiating (CTP36) and using $v'_b = -2ib/(1 - z)^2$, (CTP15), and $\ell(ib) = -ibc_0$, gives

$$(1 - z)^2 \sum_{n \geq 1} \Lambda_n(b) z^{n-1} = 4b^2 G_b(z) - 2b^2 c_0, \quad (\text{CTP38})$$

$$\Lambda_1(b) = 2b^2 c_0, \quad \Lambda_{k+1}(b) - 2\Lambda_k(b) + \Lambda_{k-1}(b) = 4b^2 c_k.$$

The last equality is valid for $k \geq 1$; with the displayed even extension and $\Lambda_0 = 0$ it also holds for $k = 0$. Its inverse, obtained by summing these second differences, is

$$\Lambda_n(b) = 2b^2 n c_0 + 4b^2 \sum_{k=1}^{n-1} (n - k) c_k \quad (n \geq 1). \quad (\text{CTP39})$$

Indeed both sides have the same second differences and the same values at $n = 0, 1$. These are finite, mutually inverse linear maps on each required coefficient segment, with no lost constant term.

At $b = 1$, the original arithmetic coordinate gives

$$s(v_1(z)) = \frac{1}{1-z}, \quad \frac{H(v_1(z))}{H(i)} = \frac{\xi(1/(1-z))}{\xi(0)}, \quad \xi(0) = \xi(1). \quad (\text{CTP40})$$

Lagarias's equations 102a and 103c therefore identify $\Lambda_n(1)$ with the classical Li coefficient λ_n (his index $-n$ has the same value). The exact specialization is

$$c_0(1) = \frac{\lambda_1}{2}, \quad c_k(1) = \frac{\lambda_{k+1} - 2\lambda_k + \lambda_{k-1}}{4} \quad (k \geq 1). \quad (\text{CTP41})$$

In particular the complete two-coefficient detector (CTP24) is exactly the bound $|\lambda_{k+1} - 2\lambda_k + \lambda_{k-1}| \leq 2\lambda_1$ for every $k \geq 1$, in these unchanged conventions. No proof of that infinite family of bounds is claimed.

The test-space identification is stronger than an agreement of coefficient sequences. Let $G_n(s) = 1 - (1 - 1/s)^n$ be Lagarias's Li test function, for $n \in \mathbb{Z}$, and let $\mathcal{L}$ be his Li class: rational functions vanishing at infinity with poles contained in $\{0, 1\}$. Under $\alpha(Z) = 1/2 + iZ/2$ one has exactly

$$C_1(Z) = \frac{s}{s-1}, \quad F_j(Z) = \alpha^* \left(\frac{G_{-j-1} - G_{-j}}{2i} \right) (Z) \quad (j \in \mathbb{Z}). \quad (\text{CTP42})$$

The second identity follows from $G_{-j-1} - G_{-j} = C_1^j(1 - C_1) = C_1^j/(1-s)$ and $Z + i = 2i(1-s)$. For $m > 0$ the inverse finite sums are

$$\alpha^* G_{-m} = 2i \sum_{j=0}^{m-1} F_j, \quad \alpha^* G_m = -2i \sum_{j=-m}^{-1} F_j. \quad (\text{CTP43})$$

They telescope directly. Equations (CTP13)–(CTP14) for the two pole locations also show that the span of all integer-indexed F_j is exactly the pullback of $\mathcal{L}$. The additional Laurent pole $Z = 0$ corresponds to $s = 1/2$; it is retained in $\mathcal{A}_1$ and is not claimed to belong to Lagarias's test class. The full local-jet trace is its proved common receiving space.

Lagarias's source pairing, equation 301, is linear in its first entry. The programme pairing is conjugate-linear in its first entry. With this difference explicitly retained,

$$\mathcal{T}(\alpha^* f, \alpha^* g) = \langle g, f \rangle_{\mathcal{W}}, \quad \langle G_n, G_m \rangle_{\mathcal{W}} = \lambda_n + \lambda_{-m} - \lambda_{n-m}. \quad (\text{CTP44})$$

For the first equality use $\alpha(\bar{w}) = 1 - \overline{\alpha(w)}$ in every summand and preserve m_w . For the second, the finite identity $G_n(s)G_{-m}(s) = G_n(s) + G_{-m}(s) - G_{n-m}(s)$ and $\overline{G_m(1-\bar{s})} = G_{-m}(s)$ prove the formula with the convergent symmetric zero sums used by the source. Their combined product is absolutely summable. This also gives a direct verification of the displayed source theorem without depending on any typographical intermediate expansion in its proof. Substituting (CTP42) into (CTP44) gives

$$(T_d(1))_{ij} = \frac{1}{4}(\lambda_{j-i+1} - 2\lambda_{j-i} + \lambda_{j-i-1}), \quad \lambda_{-n} = \lambda_n. \quad (\text{CTP45})$$

The factor is $1/4$, the product of $1/(2i)$ with its conjugate. Thus this Toeplitz form is exactly the finite-difference pullback of the classical Li–Weil Gram form, with complete multiplicities. All coordinate, finite-sum, and coefficient maps (CTP38)–(CTP45) are fixed complex-linear maps on their stated spaces, so their explicit full- L lifts are (CTP32)–(CTP34). For the map reversing the two pairing entries, the labels are reversed as well; their meet is unchanged by commutativity.

57.10 The precise finite extremal defect in the source comparison

For any particular d , let μ_d be the actual least eigenvalue of the actual Hermitian matrix $T_d(b)$. The spectral theorem gives the exact positive semidefinite Toeplitz matrix

$$\widehat{T}_d = T_d(b) - \mu_d I_d \succeq 0, \quad \ker \widehat{T}_d = \ker(T_d(b) - \mu_d I_d). \quad (\text{CTP46})$$

This construction does not assert $\mu_d \geq 0$ or simplicity. Its changed coefficient is exactly $c_0 - \mu_d$; every nonconstant coefficient c_k , $|k| < d$, remains c_k . The Carathéodory–Fejér corank-one consequence in Connes–van Suijlekom, Corollary `corcar` and Proposition `toeplitzcase`, concerns this positive matrix when its kernel has dimension one. Its polynomial is the coefficient polynomial of a vector in that kernel. Equations (CTP13)–(CTP14) give the exact Cayley rational function of that polynomial; they provide no identity of this finite polynomial with H .

The full defect is an actual functional, not an unspecified change of matrix. Define on Laurent polynomials in U

$$\mathcal{L}_b(P) = \sum_{w \in \mathbb{Z}} \frac{m_w}{w^2 + b^2} P(C_b(w)), \quad h(P) = \frac{1}{2\pi} \int_0^{2\pi} P(e^{i\theta}) d\theta, \quad \widehat{\mathcal{L}}_{b,d} = \mathcal{L}_b - \mu_d h. \quad (\text{CTP47})$$

Every sum converges because a fixed Laurent polynomial is bounded on the Cayley image of the roots. With $P^*(U) = \overline{P(1/\bar{U})}$, (CTP9) gives $\mathcal{L}_b(P^*Q) = \mathcal{T}(P(C_b)/(Z + ib), Q(C_b)/(Z + ib))$. Furthermore $h(U^k) = \delta_{k0}$ and $h(P^*P) = \sum_j |a_j|^2$ for $P = \sum a_j U^j$. For the finite restriction take exactly $P(U) = \sum_{j=0}^{d-1} a_j U^j$ and $Q(U) = \sum_{j=0}^{d-1} v_j U^j$, with $a, v \in \mathbb{C}^d$. Then

$$\widehat{\mathcal{L}}_{b,d}(P^*Q) = a^* \widehat{T}_d v, \quad \mathcal{T}(Wa, Wa) = \mu_d \sum_j |a_j|^2 \quad (a \in \ker \widehat{T}_d). \quad (\text{CTP48})$$

Every factor in this equality is retained, including the $1/(2\pi)$ in the unit-mass Haar functional. It is not a rescaling of the original zero-trace mass c_0 . The analytic generating function of the changed moments is exactly $G_b(z) - \mu_d$. Subtracting a constant leaves every pole (CTP16) and its residue unchanged. Thus finite extremal positivity does not remove the all-zero defect that (CTP22) detects. Equations (CTP46)–(CTP48) are the exact connecting map and its defect required to apply the source’s Toeplitz geometry to this receiver.

Finally $\mathcal{L}_b$, h , and $\widehat{\mathcal{L}}_{b,d}$ are complex-linear maps on the same Laurent-polynomial space. Their full lifts send (P, λ) respectively to $(\mathcal{L}_b(P), \lambda)$, $(h(P), \lambda)$ and $(\mathcal{L}_b(P) - \mu_d h(P), \lambda)$. Cancellation in the difference retains λ exactly; at top support it is e , never τ . These maps commute with Φ by direct substitution in (CTP34), and their Hermitian forms carry the meet of the two input labels.

58 Spectral-action derivatives with every scalar retained

The source of the Loewner matrix and its rank-one perturbation is Alain Connes and Walter D. van Suijlekom, *Quadratic Forms, Real Zeros and Echoes of the Spectral Action*, arXiv:2511.23257v1. The original author file is `Araki-final-oct25.tex`. The relevant source locators are equations `form2` and `QD-sa`, Lemmas `basics-general`, `key-general` and `lem:sa`, and Proposition `prop:der-sa`; the spectral-action displays occur at lines 1124–1147 and 1396–1523 of the local original-source file. The derivation below proves the finite-dimensional identities directly, including the cyclic factor and the original mass of the kernel vector. It does not alter the source matrices, their spectra, or the parameter of perturbation.

58.1 The all-order derivative and its cyclic factor

Let $D = \text{diag}(\lambda_1, \dots, \lambda_M)$ with real entries, and let R be any complex M by M matrix, without a selfadjointness assumption. Let f be holomorphic on a neighbourhood of the finite spectrum.

Choose a positively oriented union of small contours Γ in that neighbourhood enclosing every eigenvalue. For sufficiently small complex t , define $f(D + tR)$ by the same contours. For nodes inside them, including repeated nodes, put

$$f[x_1, \dots, x_n] = \frac{1}{2\pi i} \int_{\Gamma} \frac{f(z)}{\prod_{j=1}^n (z - x_j)} dz. \quad (\text{SAC1})$$

For one node this is Cauchy's formula. Subtracting the two rational integrands verifies the divided-difference recursion at distinct end nodes. Taking limits proves the repeated-node extension; in particular $f[x, \dots, x] = f^{(n-1)}(x)/(n-1)!$.

For every integer $n \geq 1$ the exact derivative is

$$\left. \frac{d^n}{dt^n} \text{tr } f(D + tR) \right|_{t=0} = n! \sum_{i_1, \dots, i_n=1}^M R_{i_1 i_2} \cdots R_{i_n i_1} f[\lambda_{i_1}, \dots, \lambda_{i_n}, \lambda_{i_1}] \quad (\text{SAC2})$$

$$= (n-1)! \sum_{i_1, \dots, i_n=1}^M R_{i_1 i_2} \cdots R_{i_n i_1} f'[\lambda_{i_1}, \dots, \lambda_{i_n}]. \quad (\text{SAC3})$$

For $n = 1$ the product is $R_{i_1 i_1}$. To prove (SAC2), write $A(z) = (z - D)^{-1}$. The norm-convergent Neumann series, uniformly on Γ for small $|t|$, is

$$(z - D - tR)^{-1} = \sum_{n=0}^{\infty} t^n (A(z)R)^n A(z).$$

The trace of its n th coefficient is

$$\sum_{i_1, \dots, i_n} \frac{R_{i_1 i_2} \cdots R_{i_n i_1}}{(z - \lambda_{i_1})^2 \prod_{k=2}^n (z - \lambda_{i_k})}.$$

Integrating against f and differentiating at zero proves (SAC2), with its factor $n!$.

Here is the separate step which changes that factor. The integral of the derivative of $f(z)/\prod_{j=1}^n (z - x_j)$ around Γ is zero. Consequently

$$f'[x_1, \dots, x_n] = \sum_{r=1}^n f[x_1, \dots, x_n, x_r]. \quad (\text{SAC4})$$

This remains true for repeated nodes by the same rational identity, with no limiting division by a vanishing difference. In the full sum of (SAC2), cyclically moving i_r to the first position does not change the product of the n scalar matrix entries and is a bijection of the ordered index tuples. Each of the n summands in (SAC4) therefore has the same full cyclic sum. The full sum containing f' is n times the sum containing one repeated first node. Dividing the coefficient $n!$ by that n gives (SAC3). Tuples with a nontrivial cyclic stabilizer cause no exception: the argument uses n bijections of the full index set, not a count of orbits.

The same finite-dimensional result applies to the source's Fourier definition whenever its stated factorial moment bounds hold. More explicitly, suppose $f(x) = \int_{\mathbb{R}} \hat{f}(u) e^{iux} du$ and

$$\int_{\mathbb{R}} |u|^n |\hat{f}(u)| du \leq C^{n+1} n! \quad (n \geq 0).$$

Tonelli's theorem applied to the exponential power series gives $\int e^{a|u|} |\hat{f}(u)| du \leq C/(1 - aC)$ for $0 \leq a < C^{-1}$. Thus the Fourier integral extends f holomorphically to $|\text{Im } z| < C^{-1}$, with uniform domination on narrower strips. Cauchy functional calculus and the Fourier definition of $f(D + tR)$ agree: insert the uniformly dominated Fourier integral in the contour formula

and use $e^{iu(D+tR)} = (2\pi i)^{-1} \int_{\Gamma} e^{iuz} (z - D - tR)^{-1} dz$. The domination just proved permits the interchange. Hence (SAC2)–(SAC3) also give the finite specialization of the source’s Fourier construction.

In particular,

$$F'(0) = \sum_i R_{ii} f'(\lambda_i), \quad F''(0) = \sum_{i,j} R_{ij} R_{ji} f'[\lambda_i, \lambda_j], \quad F(t) = \text{tr } f(D + tR). \quad (\text{SAC5})$$

The diagonal divided difference is $f''(\lambda_i)$. The statement of source Lemma `lem:sa` prints $n!$ instead of $(n-1)!$ in the expression using f' . Its proof ends with the repeated-node expression, for which $n!$ is correct. The earlier display in the source subsection *Relation to the spectral action* prints $F''(0)/2$ equal to the right-hand side of (SAC5); (SAC5) proves that the right-hand side is $F''(0)$. The later first and second derivative display and Proposition `prop:der-sa` have the correct second derivative factor.

A one-dimensional check already determines both factors. For $D = (0)$, $R = (1)$ and $f(x) = x^n$, $F^{(n)}(0) = n!$ whereas $f'[0, \dots, 0] = n$. Formula (SAC3) gives $(n-1)!n = n!$; replacing its coefficient by $n!$ gives an extra factor n . For $n = 2$, $f(x) = x^2$ gives $F''(0) = 2$ and $R_{11}^2 f''(0) = 2$. The discrepancy also occurs within the source’s decaying Fourier class: $f(x) = e^{-x^2}$, $D = (0)$, $R = (1)$ gives $F''(0) = -2$, not -4 . Its Gaussian Fourier transform satisfies the displayed factorial moment bound, because $\int e^{a|u|} |\widehat{f}(u)| du < \infty$ for every $a > 0$ and $|u|^n \leq n! a^{-n} e^{a|u|}$.

58.2 The original kernel mass and the rank-one perturbation

Now retain the source index set $I = \{-N, \dots, N\}$ and its distinct real numbers λ_i with $\lambda_{-i} = -\lambda_i$. Let a_i, b_i be real, $a_{-i} = a_i$ and $b_{-i} = -b_i$, and put

$$D = \text{diag}(\lambda_i)_{i \in I}, \quad q_{ii} = a_i, \quad q_{ij} = \frac{b_i - b_j}{\lambda_i - \lambda_j} \quad (i \neq j), \quad \eta = (1)_{i \in I}, \quad \beta = (b_i)_{i \in I}. \quad (\text{SAC6})$$

The evenness of the diagonal data is stated explicitly because it is needed for $Q\gamma = \gamma Q$, where $(\gamma x)_i = x_{-i}$. We use $\langle x, y \rangle = x^* y$ and $\mathcal{Q}(x, y) = x^* Q y$. Suppose $Q\xi = 0$, $\gamma\xi = \xi$ and let the original, unaltered mass be

$$s = \eta^* \xi = \sum_{i \in I} \xi_i \neq 0, \quad v = D\xi, \quad R = -\frac{v\eta^*}{s}, \quad D' = D + R. \quad (\text{SAC7})$$

No rescaling of ξ , of D , or of the parameter t is made. The vector may be complex; the real-vector case is recovered by taking $\xi \in \mathbb{R}^{2N+1}$.

Entrywise computation and parity give

$$DQ - QD = \beta\eta^* - \eta\beta^*, \quad \beta^*\xi = 0, \quad \eta^* D\xi = 0, \quad Qv = -s\beta. \quad (\text{SAC8})$$

Indeed the ij entry of the commutator is $b_i - b_j$ off the diagonal and zero on it. In both scalar parity sums the terms i and $-i$ cancel and the index-zero term is zero. Applying the commutator to ξ gives $-QD\xi = s\beta$, proving the last formula. In particular

$$R^2 = 0, \quad D'\xi = 0, \quad QR = \beta\eta^*, \quad R^*Q = \eta\beta^*, \quad QD' = D'^*Q. \quad (\text{SAC9})$$

The first identity follows from $\eta^* v = 0$; the second uses $\eta^* \xi = s$; the last follows from the two preceding identities and (SAC8). These equations remain valid without assuming Q positive.

There is always an even polynomial f realizing the entire finite data $f'(\lambda_i) = b_i$, $f''(\lambda_i) = a_i$. Here is an explicit construction, so that the application does not require an assumed interpolating

function. Put

$$\begin{aligned} \ell_i(x) &= \prod_{j \neq i} \frac{x - \lambda_j}{\lambda_i - \lambda_j}, & h_i(x) &= (1 - 2\ell'_i(\lambda_i)(x - \lambda_i))\ell_i(x)^2, & k_i(x) &= (x - \lambda_i)\ell_i(x)^2, \\ g(x) &= \sum_{i \in I} (b_i h_i(x) + a_i k_i(x)), & f(x) &= c + \int_0^x g(u) du, & c &\in \mathbb{R}. \end{aligned} \quad (\text{SAC10})$$

Direct evaluation gives $g(\lambda_i) = b_i$, $g'(\lambda_i) = a_i$. Uniqueness of this interpolating polynomial of degree at most $2(2N+1) - 1$ follows because a difference has a double zero at every node and degree less than $2(2N+1)$. The polynomial $-g(-x)$ has the same data, so uniqueness gives $g(-x) = -g(x)$ and f is even. The constant $c = f(0)$ is retained as an arbitrary constant and is never set to zero to change the action.

For this polynomial, or for any holomorphic f with the same first two derivative values at the nodes, (SAC5) yields

$$F'(0) = F''(0) = -\frac{\beta^T v}{s} = \frac{v^T Q v}{s^2} = \frac{v^* Q v}{|s|^2}. \quad (\text{SAC11})$$

To check every equality, $R_{ii} = -v_i/s$ gives the first derivative $-\beta^T v/s$. Since $q_{ij} = f'[\lambda_i, \lambda_j]$ also on the diagonal, the second derivative is $v^T Q v/s^2$. Equation (SAC8) equates these two expressions. Finally $v^* Q v = -s \beta^T \bar{v}$ is real because Q is Hermitian. Taking its conjugate gives $v^* Q v = -\bar{s} \beta^T v$, which proves the last identity. Thus for the original real vector and its real mass,

$$F'(0) = F''(0) = \frac{\mathcal{Q}(D\xi, D\xi)}{s^2}. \quad (\text{SAC12})$$

For a complex vector the denominator of the Hermitian expression is $|s|^2$, as in (SAC11). If $Q \geq 0$ the common value is nonnegative. If its kernel is one-dimensional and $D\xi \neq 0$, it is strictly positive: $D\xi$ is odd, whereas the kernel is even, so $D\xi$ is outside the kernel.

In the one-dimensional even-kernel situation the required $s \neq 0$ follows from the original data. If $s = 0$, (SAC8) gives $D\xi \in \ker Q$. Its odd parity forces $D\xi = 0$. Distinctness and symmetry of the λ_i show that $\ker D$ is spanned by e_0 . Then nonzero ξ has $s = \xi_0 \neq 0$, a contradiction. If $D\xi = 0$ from the outset, $R = 0$ and (SAC11) gives the correct zero derivative.

58.3 The exact all-order rank-one expression

Write

$$h(z) = \sum_{i \in I} \frac{\xi_i}{z - \lambda_i}, \quad B(z) = \frac{1}{s} \sum_{i \in I} \frac{\lambda_i \xi_i}{z - \lambda_i} = \frac{zh(z)}{s} - 1. \quad (\text{SAC13})$$

The determinant identity for a rank-one perturbation gives

$$\det(z - D - tR) = \det(z - D)(1 + tB(z)) = \det(z - D) \left(1 - t + \frac{tzh(z)}{s} \right). \quad (\text{SAC14})$$

For completeness, for an invertible A the formula $\det(A + uw^*) = \det(A)(1 + w^* A^{-1} u)$ follows by multilinearity in the columns: terms taking the added column twice vanish because they are proportional. Apply this to $A = z - D$, $u = tv/s$, $w = \eta$. The resulting rational identity extends to a polynomial identity at the nodes by multiplication by $\det(z - D)$.

Keeping the same contours as above and taking $|t|$ small enough that $|tB| < 1$ there, Jacobi's

determinant formula and integration by parts give

$$F(t) - F(0) = -\frac{1}{2\pi i} \int_{\Gamma} f'(z) \log(1 + tB(z)) dz, \quad (\text{SAC15})$$

$$\begin{aligned} F^{(n)}(0) &= \frac{(-1)^n (n-1)!}{2\pi i} \int_{\Gamma} f'(z) B(z)^n dz \\ &= \frac{(-1)^n (n-1)!}{s^n} \sum_{i_1, \dots, i_n \in I} \left(\prod_{r=1}^n \lambda_{i_r} \xi_{i_r} \right) f'[\lambda_{i_1}, \dots, \lambda_{i_n}]. \end{aligned} \quad (\text{SAC16})$$

The logarithm in (SAC15) is its convergent power series on the contour; there is no global choice of logarithm across the eigenvalues. Jacobi's formula is verified by differentiating the determinant in its columns, which gives $\partial_z \log \det(z - A) = \text{tr}(z - A)^{-1}$ away from the eigenvalues. Subtract the two contour functional calculus formulas, use (SAC14), and integrate the derivative of $f \log(1 + tB)$ to obtain (SAC15). Termwise expansion, uniformly convergent on Γ , proves (SAC16). This formula retains all higher derivatives instead of extrapolating the equality of the first two.

58.4 An exact three-dimensional check

With indices $-1, 0, 1$, take the unaltered data

$$D = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad Q = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}, \quad \xi = \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix}, \quad s = 6, \quad \beta = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}. \quad (\text{SAC17})$$

Here $Q = 3I - \eta\eta^T$ is positive with kernel spanned by η , and its off-diagonal entries are exactly the divided differences of the displayed β . We have

$$v = \begin{pmatrix} -2 \\ 0 \\ 2 \end{pmatrix}, \quad Qv = -6\beta, \quad v^T Qv = 24, \quad R = \begin{pmatrix} 1/3 & 1/3 & 1/3 \\ 0 & 0 & 0 \\ -1/3 & -1/3 & -1/3 \end{pmatrix}. \quad (\text{SAC18})$$

Use the explicit even polynomial

$$f(x) = x^2 - \frac{15}{8}x^4 + \frac{3}{4}x^6. \quad (\text{SAC19})$$

Its derivative is $2x - 15x^3/2 + 9x^5/2$ and its second derivative is $2 - 45x^2/2 + 45x^4/2$. Therefore $f'(-1) = 1$, $f'(0) = 0$, $f'(1) = -1$ and $f''(-1) = f''(0) = f''(1) = 2$, as required. Directly from (SAC14), or from the displayed matrices,

$$\det(z - D - tR) = z \left(z^2 - 1 + \frac{2t}{3} \right), \quad F(t) = -\frac{1}{4} + \frac{2}{3}t + \frac{1}{3}t^2 - \frac{4}{9}t^3. \quad (\text{SAC20})$$

For the second equality put $u = 1 - 2t/3$. The characteristic polynomial gives $\text{tr}(D + tR)^{2k} = 2u^k$ for $k \geq 1$; this also holds at repeated roots by polynomial continuation. Substitute it into (SAC19) to get $2(u - 15u^2/8 + 3u^3/4)$, whose expansion is (SAC20). Hence

$$F'(0) = F''(0) = \frac{2}{3} = \frac{24}{6^2}, \quad F'''(0) = -\frac{8}{3}. \quad (\text{SAC21})$$

Thus the retained-mass first and second derivative identity is exact, while even a positive Loewner matrix with a simple even kernel does not make the third derivative equal to them. The rank-one action in (SAC15)–(SAC16), rather than an unproved equality of all derivatives, is the complete finite object supplied by these source data. No identification with the programme's theta function is asserted by this calculation.

59 Every Cayley coefficient as a convergent original-prime-power trace

Source and unchanged objects. Retain exactly $H(Z) = \xi(1/2 + iZ/2)/8$, the parameter b , the full root multiplicities, and $c_k(b)$ of (CTP1)–(CTP18). The arithmetic decomposition is that used by Enrico Bombieri and Jeffrey C. Lagarias, *Complements to Li's criterion for the Riemann hypothesis*, and by Jeffrey C. Lagarias, *Li Coefficients for Automorphic L-Functions*, Section 4, original author source `math-0404394v4.tex`, lines 1425–1604, especially equations N405b, N405e and Lemma 1e42. That source range was read directly. The formulas below derive the arithmetic expression for the present parameter $b > 1$ and its exact Cayley receiver; they are not an identification of the full support spectrum with the ordinary prime list.

Fix $b > 1$ and put $q = (1 + b)/2 > 1$. Write $L_\xi(s) = \xi'(s)/\xi(s)$ and use $\psi = \Gamma'/\Gamma$. The functional equation $\xi(1 - s) = \xi(s)$ gives $L_\xi^{(r)}(1 - q) = (-1)^{r+1}L_\xi^{(r)}(q)$. The original coordinate therefore gives

$$\ell^{(r)}(ib) = (-i/2)^{r+1}L_\xi^{(r)}(q), \quad c_0(b) = \frac{L_\xi(q)}{2b}, \quad c_k(b) = \frac{1}{4} \sum_{r=1}^k \binom{k-1}{r-1} \frac{b^{r-1}}{r!} L_\xi^{(r)}(q) \quad (k \geq 1). \quad (\text{CPP1})$$

Here the first identity includes $r = 0$. To check the last sign and constant, substitute it in (CTP18); the scalar product $-(2i)^{r-1}(-i)^{r+1}/2^{r+1}$ is exactly $1/4$.

The defining factors of ξ and the ordinary Euler product of the actual ζ give the absolutely convergent expression

$$L_\xi(q) = \frac{1}{q} + \frac{1}{q-1} - \frac{\log \pi}{2} + \frac{1}{2}\psi(q/2) - \sum_p \sum_{l \geq 1} (\log p) p^{-lq}. \quad (\text{CPP2})$$

For completeness, unique prime factorization expands $\prod_p (1 - p^{-s})^{-1}$ into the absolutely convergent Dirichlet series for $\zeta(s)$ when $\Re s > 1$. Its logarithm is $\sum_{p,l} p^{-ls}/l$, normally convergent on smaller closed half-planes. Termwise differentiation yields the last term in (CPP2), and each further derivative is justified by domination by a convergent series $\sum_{n \geq 2} (\log n)^M n^{-q'}$ with $1 < q' < q$. Thus for $r \geq 1$,

$$\begin{aligned} L_\xi^{(r)}(q) &= (-1)^r r! \{q^{-r-1} + (q-1)^{-r-1}\} + 2^{-r-1} \psi^{(r)}(q/2) \\ &\quad + (-1)^{r+1} \sum_p \sum_{l \geq 1} (\log p) (l \log p)^r p^{-lq}. \end{aligned} \quad (\text{CPP3})$$

Every prime power is present, with its original logarithmic weight, exponent, and sign. Neither $q = 1$ nor a limiting interpretation of this prime series is used.

Define explicitly the finite polynomial

$$L_{k-1}^{(1)}(X) = \sum_{j=0}^{k-1} \binom{k}{j+1} \frac{(-X)^j}{j!} \quad (k \geq 1). \quad (\text{CPP4})$$

It is the associated Laguerre polynomial in the convention specified by this formula. The identity $\binom{k-1}{r-1}/r! = \binom{k}{r}/(k(r-1)!)$ then gives

$$\sum_{r=1}^k \binom{k-1}{r-1} \frac{(-1)^{r+1} b^{r-1} x^r}{r!} = \frac{x}{k} L_{k-1}^{(1)}(bx).$$

Consequently the completely evaluated arithmetic formula for each $k \geq 1$ is

$$\begin{aligned}
4c_k(b) = & -\frac{1}{q^2} \left(1 - \frac{b}{q}\right)^{k-1} - \frac{1}{(q-1)^2} \left(1 - \frac{b}{q-1}\right)^{k-1} \\
& + \sum_{n=0}^{\infty} \frac{1}{(q+2n)^2} \left(1 - \frac{b}{q+2n}\right)^{k-1} \\
& + \frac{1}{k} \sum_p \sum_{l \geq 1} l(\log p)^2 p^{-lq} L_{k-1}^{(1)}(bl \log p).
\end{aligned} \tag{CPP5}$$

The two displayed rational terms come from the original factors s and $s-1$ of ξ . The gamma term is obtained from $\psi^{(r)}(t) = (-1)^{r+1} r! \sum_{n \geq 0} (t+n)^{-r-1}$ for $r \geq 1, t > 0$. One proof is to differentiate the logarithm of Euler's limit $\Gamma(t) = \lim_{N \rightarrow \infty} N! N^t / [t(t+1) \cdots (t+N)]$: after the first derivative, all subsequent derivatives have normally convergent series on compact subsets of $t > 0$. Insertion in the finite sum in (CPP1) gives the displayed gamma series by the binomial theorem. For fixed k that series converges absolutely, as its summands are $O(n^{-2})$. The prime series converges absolutely since its summands in absolute value are bounded by a constant times $(\log(p^l))^{k+1} (p^l)^{-q}$, and prime powers form a subset of the integers with no repetitions among distinct pairs (p, l) .

The exact endpoint cancellation can also be recorded without discarding its source. The $n=0$ gamma term in (CPP5) is the negative of its first rational term. Define the three retained quantities

$$\begin{aligned}
A_k(b) &= \sum_{n=1}^{\infty} (q+2n)^{-2} (1 - b/(q+2n))^{k-1}, \\
E_k(b) &= (q-1)^{-2} (1 - b/(q-1))^{k-1}, \\
P_k(b) &= \frac{1}{k} \sum_p \sum_{l \geq 1} l(\log p)^2 p^{-lq} L_{k-1}^{(1)}(bl \log p).
\end{aligned} \tag{CPP6}$$

The exact identity is $4c_k = A_k - E_k + P_k$; the two canceled terms remain explicitly available in (CPP5). Moreover $0 < b/(q+2n) < 2$ for $n \geq 1$, so

$$|A_k(b)| \leq \sum_{n=1}^{\infty} (q+2n)^{-2} \leq (q+2)^{-2} + \frac{1}{2(q+2)} \quad (k \geq 1). \tag{CPP7}$$

The second inequality bounds the decreasing tail by its integral. Hence the complete zero detector (CTP24) is also equivalent to boundedness of the one precise prime discrepancy $\{P_k(b) - E_k(b) : k \geq 1\}$: all real roots give $|c_k| \leq c_0$ and then (CPP7) gives boundedness; conversely bounded discrepancy makes c_k bounded, whereas any off-real root makes its limsup exponential rate $R_b > 1$ by (CTP22). The exact sharper inequalities are $|A_k(b) - E_k(b) + P_k(b)| \leq 4c_0(b)$ for every k . Neither their boundedness nor their validity for all k is asserted here. The formula exhibits the required cancellation against E_k , whose factor is $1 - b/(q-1) = -(b+1)/(b-1)$; a bound on one retained term alone does not evaluate the discrepancy.

59.1 Explicit finite tails for each original prime sum

For an integer $m \geq 0$ and $N > 1$, set

$$I_m(N, q) = N^{1-q} \sum_{j=0}^m \frac{m!}{(m-j)!} \frac{(\log N)^{m-j}}{(q-1)^{j+1}} = \int_N^{\infty} (\log x)^m x^{-q} dx. \tag{CPP8}$$

Repeated integration by parts proves the equality, the endpoint at infinity vanishing because $q > 1$. If $N \geq \exp(m/q)$, the integrand is decreasing on $[N, \infty)$, as its derivative has sign $m - q \log x$.

Let $\Lambda(n) = \log p$ when $n = p^l$ and zero otherwise; this symbol is the von Mangoldt function, distinct from the Li coefficients $\Lambda_n(b)$. Since $0 \leq \Lambda(n) \leq \log n$, for integer $N \geq \exp((r+1)/q)$ one has

$$\sum_{p^l > N} (\log p)(l \log p)^r p^{-lq} = \sum_{n > N} \Lambda(n)(\log n)^r n^{-q} \leq I_{r+1}(N, q). \quad (\text{CPP9})$$

Indeed each integer summand of the positive decreasing majorant is at most its integral over $[n-1, n]$. For $r = 0$ this also bounds the tail in (CPP2). For the entire fixed- k prime contribution to c_k , (CPP1)–(CPP3) give the explicit bound

$$\left| \text{prime contribution to } c_k \text{ from } p^l > N \right| \leq \frac{1}{4} \sum_{r=1}^k \binom{k-1}{r-1} \frac{b^{r-1}}{r!} I_{r+1}(N, q) \quad (\text{CPP10})$$

$$(N \in \mathbb{N}, N \geq \exp((k+1)/q)).$$

This bounds the tail of the complete finite polynomial weight, including all its signs. It is a bound for fixed k, N, b , not a claimed uniform-in- k cancellation estimate. For the gamma tail after an integer $n = M \geq 1$, the modulus of each factor in (CPP6) is at most one, and $\sum_{n > M} (q+2n)^{-2} \leq 1/[2(q+2M)]$. Thus (CPP5) permits certified finite evaluations with complete and explicitly bounded omissions.

59.2 The original arithmetic prime points and the full support lift

For the original arbitrary nontrivial bounded distributive lattice L , the amplitude map $\pi : G_L(\mathbb{Z}) \rightarrow \mathbb{Z}$, $(a, \lambda) \mapsto a$, preserves addition, multiplication, and the unit by the defining join/meet operations. Each ordinary prime p in (CPP2)–(CPP10) therefore has the exact prime preimage

$$\mathfrak{p}_p = \pi^{-1}(p\mathbb{Z}) = \{(0, \lambda) : \lambda \in L\} \cup \{(a, 1_L) : a \in p\mathbb{Z} \setminus \{0\}\}. \quad (\text{CPP11})$$

It is prime because $p \mid ab$ implies $p \mid a$ or $p \mid b$, and it is proper because the unit is not in it. The supported zero ideal is $(e) = \{(0, \lambda) : \lambda \in L\}$: multiplication of e by z_λ gives z_λ , and every such product has zero amplitude. It is itself prime since $\mathbb{Z}$ has no zero divisors, and $(e) \subsetneq \mathfrak{p}_p$. These statements retain the supported zero stratum. They do not say that (CPP11) exhausts the full split-support spectrum, or assign an unproved numerical norm to the other lattice strata. The integer p and factor p^{-lq} in (CPP2) are the original Euler factors attached to these arithmetic prime preimages.

To specify the infinite trace without requiring an infinite join in L , use the Banach vector space $\ell^1(I)$ on the actual index set $I = \{(p, l) : p \text{ prime}, l \geq 1\}$. The linear maps $S_N(a) = \sum_{p^l \leq N} a_{p,l}$ and $S(a) = \sum_{p,l} a_{p,l}$ are defined with their actual coefficients, and $S_N a \rightarrow S a$ by absolute summability. Equations (CPP2)–(CPP5) supply elements of this vector space by the proved convergence. Their full maps are exactly

$$S_N^\uparrow(a, \lambda) = (S_N a, \lambda), \quad S^\uparrow(a, \lambda) = (S a, \lambda), \quad \Phi_{\mathbb{C}} S^\uparrow = (S \times I) \Phi_{\ell^1(I)}. \quad (\text{CPP12})$$

All labels are retained throughout the amplitude limit; no infinite lattice join is introduced. The identities follow by addition, distributivity, and direct substitution in $\Phi_V(v, \lambda) = (v, 1_K \otimes \lambda)$; its injectivity and label retraction are proved in (CTP32)–(CTP34). The same construction on $\ell^1(\mathbb{N})$ applies to the gamma sum. Finite linear combinations, including both endpoint terms, the signs in (CPP3), and $A_k - E_k + P_k$, have these same lifts. Any zero amplitude resulting from their exact cancellation keeps its original label, and at top support remains e , rather than the absence element τ .

59.3 A terminating search on every off-real input, without an equality trap

The preceding formulas provide rigorous finite coefficient enclosures for every fixed rational $b > 1$. To include the constant c_0 , put $t = q/2 > 0$ and recall the convergent digamma series, obtained from the same Euler limit for Γ :

$$\psi(t) = -\gamma + \sum_{n=0}^{\infty} \left(\frac{1}{n+1} - \frac{1}{n+t} \right), \quad (\text{CPP13})$$

$$\left| \sum_{n>M} \left(\frac{1}{n+1} - \frac{1}{n+t} \right) \right| \leq \frac{|t-1|}{M + \min(1, t)} \quad (M \in \mathbb{N}, M \geq 1).$$

The summand equals $(t-1)/((n+1)(n+t))$; bound its absolute value by $|t-1|/(n + \min(1, t))^2$ and integrate the decreasing majorant from M to infinity. Euler's constant itself has the explicit enclosure, with $h_N = \sum_{j=1}^N 1/j$,

$$h_N - \log N - \frac{1}{N} \leq \gamma \leq h_N - \log N \quad (N \in \mathbb{N}, N \geq 1). \quad (\text{CPP14})$$

Indeed $a_N = h_N - \log N$ decreases to γ and $a_n - a_{n+1} = \log(1 + 1/n) - 1/(n+1)$ lies between zero and $1/n - 1/(n+1)$, by the integral of $1/x$ on $[n, n+1]$. The telescoping tail is therefore at most $1/N$. Together (CPP2), (CPP8)–(CPP10), (CPP13)–(CPP14), and the gamma tail after (CPP10) give arbitrarily narrow rational intervals for $c_0, c_1, \dots, c_k$. Elementary constants can also be enclosed: for positive rational x use $\log x = 2 \sum_{j \geq 0} y^{2j+1}/(2j+1)$, $y = (x-1)/(x+1)$, with its geometric remainder bound; use the arctangent alternating series in Machin's identity $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$ to enclose π , whose tangent-addition formula and quadrant give the stated identity, and interval monotonicity to enclose $\log \pi$. Positive rational powers of the positive rational bases can be enclosed by rational bisection of their defining integer-power equations: for $q = A/B$ the relevant B th root is the unique positive root, and rational bracket endpoints are checked by exact integer powers. This includes both p^{-lq} and N^{1-q} . All truncation errors tend to zero for fixed k and b .

An explicit search now proceeds in finite stages $j = 1, 2, \dots$. At stage j enclose c_0 and all c_k , $1 \leq k \leq j$, to width at most 2^{-j} . Stop on the first pair of intervals that proves $|c_k| > c_0$, recording the sign of c_k and the integer coefficient test $(1, -\text{sgn } c_k)$ in positions $0, k$. The inequality certificate and (CTP23) prove negativity of the entire original trace. If an off-real zero exists, (CTP22) supplies a finite k with a strictly positive gap $|c_k| - c_0$; for all sufficiently large stages containing that k , the shrinking intervals prove the gap. The search therefore terminates on every such zero configuration. At no step does it wait forever for the sign of a single possibly zero gap: all finite indices are revisited by the stated stages. The real-root case has no such certificate. This proves the computational content of the detector without asserting that a negative coefficient has been found for the actual original H .

60 All rational-prime translates of the original adelic coefficient sheaf

Human source, original object, and scope. Alain Connes and Caterina Consani, *Knots, primes and class field theory*, arXiv:2501.06560v1, Section 4, original author file `arxiv-pisa.tex`, Lemma `localv`, Proposition `structure`, and Theorem `structure1`, construct the semilocal coefficient sheaf using the literal transition vectors $1_{\mathbb{Z}_p}$. Their author source, SHA-256 `249fed347469cfaf6da97915d547c2ca704a4ba21cc339613713e570f14d5cea`, has been read in full in this task. The proof below starts with those functions and keeps every rational scaling and Haar factor. It extends the programme calculation CCF8–34, whose complete source is retained, by making the nonunit-prime defect into a family with invertible maps. The original scalar function

remains $\zeta(s)$; its complete local integrals are calculated at the end. No identification with a Weil positive form is assumed.

60.1 The actual family and its sheaf maps

Put $X = \text{Spec } \mathbb{Z}$ and $V = \mathcal{S}(\mathbb{A}_{\mathbb{Q}})$. Here this Bruhat–Schwartz space is the algebraic restricted tensor product of the real Schwartz space and the finite locally constant compactly supported function spaces, with standard vectors $1_{\mathbb{Z}_p}$. Thus an element is a finite sum of tensors with only finitely many nonstandard finite factors. Let

$$\mathfrak{D} = \bigoplus_p \mathbb{Z}, \quad k = (k_p)_p, \quad c_{p,k_p} = 1_{p^{k_p}\mathbb{Z}_p}, \quad N(k) = \prod_p p^{k_p} \in \mathbb{Q}_{>0}. \quad (\text{CDV1})$$

Every product indexed by the support of k is finite. Write $V^{(p)}$ for the restricted tensor product at every place except p , including the real factor, and set

$$V_{p,k} = \mathbb{C}c_{p,k_p} \otimes V^{(p)} \subset V. \quad (\text{CDV2})$$

For $U = X \setminus F$, F a finite set of finite primes, define

$$\mathcal{O}_k(U) = \bigcap_{p \notin F} V_{p,k} \subset V, \quad \mathcal{O}_k(\emptyset) = 0. \quad (\text{CDV3})$$

All restriction maps for nonempty inclusions are inclusions of these actual subspaces. Explicitly its sections are the tensors with arbitrary finite factors at $p \in F$ and the specified c_{p,k_p} at every other finite prime, with arbitrary real Schwartz factor and finite linear sums.

Here is a proof of this explicit intersection and of the sheaf property. For a finite collection of primes define commuting idempotents on V by

$$P_{p,k}f(x_p, x^{(p)}) = p^{k_p}c_{p,k_p}(x_p) \int_{p^{k_p}\mathbb{Z}_p} f(y, x^{(p)}) dy, \quad \text{vol}(\mathbb{Z}_p) = 1. \quad (\text{CDV4})$$

The volume of this ball is p^{-k_p} , so $P_{p,k}^2 = P_{p,k}$ and its image is exactly $V_{p,k}$. Different p act on different tensor coordinates and therefore commute. For a given f , choose a finite set containing its nonstandard factors and $\text{supp } k$. Outside that set the conditions are automatic; inside it, being fixed by every required projection is equivalent to belonging to the image of their product. This gives precisely the factors in CDV3. Every nonempty open in X contains the generic point, so compatible sections on a covering are the same element $f \in V$. Their local conditions say exactly that $f \in V_{p,k}$ for every point of the covered open. This is CDV3, proving existence and uniqueness of gluing. The generic stalk is V , and the stalk at p is $V_{p,k}$ because every finite set away from p can be removed in its neighbourhoods. At $k = 0$ this is the original Connes–Consani sheaf with its unchanged transition factors.

The sheaf is an algebra under the original pointwise multiplication: $c_{p,k_p}^2 = c_{p,k_p}$, so products preserve every required local factor. This statement does not assert a multiplicative structure on the quotient vector spaces introduced below.

For $r \in \mathbb{Q}^\times$ write $d(r) = (v_p(r))_p$ and keep the actual dilation

$$(\alpha_r f)(x) = f(r^{-1}x), \quad \alpha_r : \mathcal{O}_k \longrightarrow \mathcal{O}_{k+d(r)}. \quad (\text{CDV5})$$

At p , multiplication by the unit part of r fixes every ball, so $\alpha_r c_{p,k_p} = c_{p,k_p+v_p(r)}$. At all other coordinates the same formula acts on the actual Schwartz factors. This proves CDV5 on every open and its compatibility with restrictions. Its inverse is $\alpha_{r^{-1}}$, and $\alpha_r \alpha_q = \alpha_{rq}$. These are algebra isomorphisms between the specified sheaves. The stabilizer of the index k consists exactly of $r = \pm 1$, by unique factorization of a rational number. The family indexed by $\mathfrak{D}$ is retained; no member is replaced by another without this explicit map.

For crossed-product coefficients retain

$$B = V \rtimes \mathbb{Q}^\times, \quad B_{p,k} = \bigoplus_{u: v_p(u)=0} V_{p,k} U_u, \quad (fU_u)(gU_v) = f\alpha_u(g)U_{uv}. \quad (\text{CDV6})$$

The sheaf $\mathcal{A}_k(X \setminus F)$ is the intersection of the $B_{p,k}$ for $p \notin F$. The argument proving CDV3 works coefficient by coefficient because every crossed-product element has a unique finite sum in the group labels. It yields exactly the coefficient sheaf $\mathcal{O}_k$ crossed with the units of $\mathbb{Z}[F^{-1}]$. The unit condition makes the pointwise coefficient product preserve the prescribed local balls. Since rational dilations commute,

$$\alpha_r^B \left(\sum_u f_u U_u \right) = \sum_u \alpha_r(f_u) U_u, \quad \alpha_r^B : \mathcal{A}_k \xrightarrow{\sim} \mathcal{A}_{k+d(r)} \quad (\text{CDV7})$$

preserves the displayed multiplication, all original group labels, and every restriction. Its inverse is $\alpha_{r^{-1}}^B$.

60.2 Every mixed-prime cohomology component

For either $W = V$ or $W = B$, use $W_{p,k} = V_{p,k}$ or $B_{p,k}$. Define

$$D_k(W) = \bigoplus_p W/W_{p,k}, \quad \delta_k w = ([w]_{p,k})_p, \quad C_k(W) = [W \xrightarrow{\delta_k} D_k(W)]. \quad (\text{CDV8})$$

The arrow lands in the direct sum: only finitely many primes occur in k , the coefficient tensors, or the nonunit valuations of a group label. The sheaf sequence

$$0 \longrightarrow \mathcal{H}_k \longrightarrow c_X W \longrightarrow \bigoplus_p i_{p*}(W/W_{p,k}) \longrightarrow 0 \quad (\text{CDV9})$$

is exact, where $\mathcal{H}_k = \mathcal{O}_k$ or $\mathcal{A}_k$. At p it is the quotient sequence and at the generic point its last term is zero; its kernel on an open is precisely the defining intersection. The constant sheaf is flasque because all nonempty opens are connected and its restriction maps are identity maps. The last sheaf has finite-support sections on every nonempty open: its generic germ is zero, so a section vanishes on a cofinite neighbourhood of the generic point, leaving only finitely many closed points. Its restrictions forget components and are onto. Thus it is also flasque, and CDV9 computes

$$H^0(X, \mathcal{H}_k) = \ker \delta_k, \quad H^1(X, \mathcal{H}_k) = D_k(W)/\delta_k W, \quad H^j(X, \mathcal{H}_k) = 0 \quad (j \geq 2). \quad (\text{CDV10})$$

This uses the definition of sheaf cohomology through an acyclic resolution; flasque sheaves are acyclic for global sections. All maps and terms of that resolution have been specified.

The complete mixed-place decomposition can be written without altering the original Haar lengths of the local vectors. Put

$$E_{p,k} = \left\{ f \in \mathcal{S}(\mathbb{Q}_p) : \int_{p^{k_p} \mathbb{Z}_p} f = 0 \right\}, \quad \mathcal{S}(\mathbb{Q}_p) = \mathbb{C}_{c_{p,k_p}} \oplus E_{p,k}. \quad (\text{CDV11})$$

The two projections are exactly CDV4 and $I - P_{p,k}$. Expanding a finite tensor gives an algebraic direct sum

$$V = \bigoplus_{T \text{ finite}} V_{T,k}, \quad V_{T,k} = \mathcal{S}(\mathbb{R}) \otimes \bigotimes_{p \in T} E_{p,k} \otimes \bigotimes'_{p \notin T} \mathbb{C}_{c_{p,k_p}}. \quad (\text{CDV12})$$

Only finitely many terms occur for any input; projection by the corresponding finite product of $P_{p,k}$ and $I - P_{p,k}$ proves uniqueness. A T component survives in $V/V_{p,k}$ precisely when $p \in T$.

For a crossed-product term $V_{T,k}U_u$ it survives precisely at $T_u = T \cup \{p : v_p(u) \neq 0\}$. Therefore the actual differentials in CDV8 are the diagonal maps on these components, and

$$\begin{aligned} H^1(X, \mathcal{O}_k) &= \bigoplus_{|T| \geq 2} V_{T,k} \otimes (\mathbb{C}^T / \Delta \mathbb{C}), \\ H^1(X, \mathcal{A}_k) &= \bigoplus_{u, T: |T_u| \geq 2} V_{T,k} U_u \otimes (\mathbb{C}^{T_u} / \Delta \mathbb{C}). \end{aligned} \quad (\text{CDV13})$$

Indeed an empty marked set has zero target, a one-element set has surjective diagonal, and for every larger finite set the cokernel is exactly its marked coordinates modulo the diagonal. This proves CDV13 by actual component maps, rather than dimension counting.

Every rational dilation induces a chain isomorphism

$$C_k(W) \xrightarrow{\alpha_r} C_{k+d(r)}(W), \quad [w]_{p,k} \mapsto [\alpha_r w]_{p,k+d(r)}, \quad (\text{CDV14})$$

with group labels unchanged in the crossed product. Well-definedness follows from CDV5 and CDV7, and commutation with δ follows componentwise. The inverse is the corresponding map for r^{-1} . The local projections conjugate exactly: $\alpha_r P_{p,k} \alpha_{r^{-1}} = P_{p,k+d(r)}$. This follows either from their ranges and orthogonality, or directly from the change of variables in CDV4. Hence every mixed component T , every marked prime, and every diagonal relation is transported by CDV14. No cohomology component is dropped in this prime action.

60.3 The original positive metric and the nonunit defect

The additive adelic Haar measure is the original product of dx at infinity and measures with $\text{vol}(\mathbb{Z}_p) = 1$. In particular

$$\|c_{p,k_p}\|^2 = p^{-k_p}, \quad \|\alpha_r f\|^2 = |r|_\infty \prod_p p^{-v_p(r)} \|f\|^2 = \|f\|^2. \quad (\text{CDV15})$$

The product formula here follows immediately by writing $r = \pm \prod p^{v_p(r)}$. Every factor in the preceding product is retained. The quotient $V/V_{p,k}$ is isometrically represented by $(I - P_{p,k})V$, since CDV4 is the original Haar orthogonal projection. Equip $D_k(V)$ with the sum of these squared quotient norms. In $D_k(B)$, at a coefficient with $v_p(u) = 0$ use this projection, and at $v_p(u) \neq 0$ use its full original Haar norm; sum over the finite coefficient and prime labels. CDV14 is an isometry for these metrics, by CDV15 and the conjugation identity.

Completion and quotient do not delete the mixed arithmetic part. The orthogonal expansion CDV12, retaining all factors p^{-k_p} , gives

$$\overline{D_k(V)} / \overline{\delta_k V} = \bigoplus_{|T| \geq 2} \overline{V_{T,k}} \otimes (\Delta \mathbb{C})^\perp. \quad (\text{CDV16})$$

To verify the closures, finite tensors are dense in each Hilbert summand by its definition. Each finite collection of diagonal components is the image of their sum in V . Conversely the bounded orthogonal projection on every marked-coordinate factor annihilates $\delta_k V$, so its closure has no component outside the diagonal. These two inclusions prove the displayed equality. On $\mathbb{C}^T$ the diagonal projection is exactly $(c_p)_p \mapsto (|T|^{-1} \sum_p c_p)_p$; its denominator remains. The same proof with T_u gives CDV16 for B with every U_u retained. The original algebraic quotient injects into the Hilbert quotient, since a finite vector has zero orthogonal projection exactly when its components were already diagonal. These are particular positive quotient metrics, not an identification with the original Weil trace.

If instead one insists on the fixed target $V/V_{p,k}$, the nonunit action has a nonzero defect. For an integer m and $f = c_{p,k_p} \otimes g$, its representative is

$$(I - P_{p,k}) \alpha_{p^m} f = \left(c_{p,k_p+m} - p^{-\max(0,m)} c_{p,k_p} \right) \otimes \alpha_{p^m}^{(p)} g. \quad (\text{CDV17})$$

Indeed the overlap of the two balls has volume $p^{-\max(k_p, k_p+m)}$ and CDV4 supplies the multiplier p^{k_p} . Taking its original squared norm gives

$$\|(I - P_{p,k})\alpha_{p^m} f\|^2 = p^{-k_p}(1 - p^{-|m|})\|g\|^2. \quad (\text{CDV18})$$

For $m \geq 0$ the finite local squared norm is $p^{-k_p}(p^{-m} - p^{-2m})$; for $m < 0$ it is $p^{-k_p}(p^{-m} - 1)$. The other places contribute p^m by the same product formula, yielding CDV18 in both cases. It vanishes exactly at $m = 0$ for nonzero g . Thus the obstruction to a fixed-space endomorphism is a calculated vector; CDV14 constructs the invertible action on its actual target family.

60.4 Fourier maps with every local scalar retained

Use the conductor- $\mathbb{Z}_p$ additive character at p and $e^{-2\pi ixy}$ at infinity. Character orthogonality proves

$$\widehat{c_{p,k_p}} = p^{-k_p} c_{p,-k_p}. \quad (\text{CDV19})$$

The annihilator of $p^{k_p}\mathbb{Z}_p$ is $p^{-k_p}\mathbb{Z}_p$: on it the integral is the ball volume; off it, translating by a point on which the character is nontrivial forces the integral to be zero. Tensoring gives a vector-sheaf isomorphism $\mathcal{F} : \mathcal{O}_k \rightarrow \mathcal{O}_{-k}$. Its inverse is Fourier followed by $x \mapsto -x$; this follows from Fourier inversion on the real Schwartz factor and character orthogonality on finite local quotients. In semilocal coordinates the transition includes the full scalar $\prod_{p \notin F} p^{-k_p}$; it is not the bare map replacing each standard vector by $c_{p,-k_p}$.

On the original crossed-product multiplication the target is convolution, and the exact isomorphism is

$$\sum_u f_u U_u \mapsto \sum_u \widehat{f}_u U_{u^{-1}}, \quad \mathcal{F}\alpha_r = \alpha_{r^{-1}}\mathcal{F}, \quad \mathcal{F}\vartheta_\lambda = \lambda\vartheta_{\lambda^{-1}}\mathcal{F}. \quad (\text{CDV20})$$

Here ϑ_λ acts only on the real coordinate by $x_\infty \mapsto \lambda^{-1}x_\infty$, $\lambda > 0$. The last two equalities follow by change of variables in the Fourier integrals; the rational product of moduli is 1, the real modulus is λ . Fourier of a pointwise product is convolution by Fubini on these functions. Therefore the image of $f\alpha_u(g)U_{uv}$ is $\widehat{f} * \alpha_{u^{-1}}(\widehat{g})U_{u^{-1}v^{-1}}$, which is exactly the target product. It preserves the nonunit valuation conditions and takes k to $-k$. For example the image vector $p^{-k_p}c_{p,-k_p}$ is a convolution idempotent: its convolution square is $p^{-2k_p}p^{k_p}c_{p,-k_p}$, equal to itself. This verifies why the retained scalar matters to the algebra map. All these arrows induce the specified chain and cohomology maps because they send each local coefficient subspace to its target.

60.5 Original zeta and the two endpoint functionals

For $a > 0$ define the actual tensor

$$f_{k,a}(x) = e^{-\pi(x_\infty/a)^2} \prod_p c_{p,k_p}(x_p). \quad (\text{CDV21})$$

The two additive moments, computed in the original measures, are

$$f_{k,a}(0) = 1, \quad \int_{\mathbb{A}} f_{k,a}(x) dx = a \prod_p p^{-k_p} = a/N(k). \quad (\text{CDV22})$$

The real Gaussian integral is a by substitution; the finite integrals are exactly the ball volumes. Thus these are separate endpoint data, not both replaced by one. Rational dilation gives $\alpha_r f_{k,a} = f_{k+d(r),|r|a}$, and real scaling gives $\vartheta_\lambda f_{k,a} = f_{k,\lambda a}$. CDV22 then proves that the rational map preserves both moments whereas the real map multiplies only the integral moment by λ .

For each finite prime use multiplicative Haar measure with $\text{vol}^\times(\mathbb{Z}_p^\times) = 1$, and at infinity use $dx/|x|$. For $\text{Re } s > 1$, define the adelic multiplicative integral as the absolutely convergent product of its local integrals. Their complete evaluation is

$$\begin{aligned} \int_{\mathbb{Q}_p^\times} c_{p,k_p}(x) |x|_p^s d^\times x &= \sum_{j=k_p}^{\infty} p^{-js} = \frac{p^{-k_p s}}{1 - p^{-s}}, \\ \int_{\mathbb{R}^\times} e^{-\pi(x/a)^2} |x|^s \frac{dx}{|x|} &= a^s \pi^{-s/2} \Gamma(s/2), \\ Z(f_{k,a}, s) &= a^s N(k)^{-s} \pi^{-s/2} \Gamma(s/2) \zeta(s). \end{aligned} \tag{CDV23}$$

For the first line partition the support into annuli $p^j \mathbb{Z}_p^\times$; multiplicative invariance makes each annulus have measure 1. The geometric series converges for $\text{Re } s > 0$. For the second integrate over both signs and substitute $y = \pi(x/a)^2$; the factor of two and the Jacobian give exactly the displayed Gamma factor. At the global step $\sum_p p^{-\text{Re } s} < \infty$ bounds the product. Unique prime factorization identifies its absolutely convergent expansion with $\sum_{n \geq 1} n^{-s} = \zeta(s)$. Hence the original function is recovered, with every factor shown, by

$$\zeta(s) = a^{-s} N(k)^s \frac{\pi^{s/2}}{\Gamma(s/2)} Z(f_{k,a}, s). \tag{CDV24}$$

No endpoint polynomial $s(s-1)$ was inserted into the arithmetic function. Meromorphic continuation here uses the original-zeta formula ZTC10 with its separate $1/(s-1)$ and $-1/s$ terms. For the actual rational map in CDV22, both a and $N(k)$ are multiplied by $|r|$, so their two factors in CDV23 cancel exactly. Changing only the finite-place ball would leave p^{-ms} and would be a different arithmetic operation. This is the full prime and archimedean comparison behind that cancellation.

60.6 Full lattice coefficients and their actual limits

Let L be any nontrivial bounded distributive lattice, and retain $G_L(W) = \{(0, \lambda) : \lambda \in L\} \cup (W \times \{1_L\})$. Every linear map above has the exact lift $(w, \lambda) \mapsto (Tw, \lambda)$. Addition uses label join and scalar multiplication label meet; substitution proves compatibility and shows that zeros of amplitudes do not delete their labels. Every inverse already displayed lifts by the same formula. In particular the quotient congruence is exactly the following $G_L(\mathbb{C})$ -semimodule congruence; it is not a quotient-algebra assertion, since $W_{p,k}$ need not be an ideal:

$$(w, \lambda) \sim (w', \lambda') \iff \lambda = \lambda' \text{ and } w - w' \in W_{p,k}, \quad G_L(W)/\sim \cong G_L(W/W_{p,k}). \tag{CDV25}$$

Its classes are the fibres of the surjective quotient map; the relation is compatible with both operations since $W_{p,k}$ is a vector subspace. Direct-sum formulas lift with one common label, not independently chosen labels on the target factors. For sheaves the sectionwise construction is sheafified; stalks are $G_L(\mathcal{H}_{k,x})$. To verify that description, filtered colimits commute with the carrier construction: a nonzero amplitude is represented with top label; a labelled zero is represented by that same labelled zero at any stage, and equality is exactly equality at a later stage together with equality of labels. Sheafification leaves stalks unchanged. All maps and inverse identities therefore survive sheafification.

This coefficient lift does not remove the supported-zero prime. For $G_L(\mathbb{Z})$ its element $e = (0, 1_L)$ generates the ideal of all labelled zeros, since $e(a, \lambda) = (0, \lambda)$. The amplitude map $q : G_L(\mathbb{Z}) \rightarrow \mathbb{Z}$ sends this ideal to (0) , and $q^{-1}(0) = (e)$ is prime: if $q(x)q(y) = 0$ in the domain $\mathbb{Z}$, one amplitude is zero. This proves the actual prime statement, retaining every zero label. The arithmetic prime (p) similarly has inverse image $q^{-1}(p)$ containing (e) strictly. The rest of the support spectrum and its stalks remain part of the base; no scalar Mellin evaluation is asserted to recover them. CDV25 specifies the coefficient maps that can be applied on that base, while its full support cohomology requires the sheaf extension constructed separately.

61 The Connes–Consani coefficient family on the actual arbitrary support base

Sources and scope. The human adelic source is Alain Connes and Caterina Consani, *Knots, primes and class field theory*, Section 4, author-source labels `localv`, `structure`, and `structure1`. CDV1–25 supplies the complete divisor-indexed extension and proofs, retaining the original local Haar measures and real factor. FSC1–9 and FSC33–34 supply the support topology and localization maps; their full proofs accompany this chapter. Their foundation is the pinned Split Support Geometry v11 source, Definition `def:lattice-split` and the subsequent ideal and prime classification. This chapter derives the actual coefficient modules and the arbitrary-support cohomology comparison. It does not identify an idempotent semimodule with an abelian sheaf.

61.1 The base and its two types of open

Let L be a nontrivial bounded distributive lattice, possibly infinite, and retain

$$X = \operatorname{Spec} \mathbb{Z}, \quad S = \operatorname{Spec}_{\text{lat}} L, \quad Y = \operatorname{Spec} G_L(\mathbb{Z}) = S \sqcup X. \quad (\text{ULC1})$$

The points of S are proper prime lattice ideals J , represented in the semiring spectrum by $\{(0, \lambda) : \lambda \in J\}$. The points of X are $q^{-1}(\mathfrak{p})$, where $q(a, \lambda) = a$. An ideal containing a top-supported element contains $e = (0, 1_L)$: add that element to its product with $(-1, 1_L)$. Multiplication by e then supplies all labelled zeros. Its amplitudes form a ring ideal. Every ideal without a top-supported element is a proper lattice ideal of zero labels. Primality is respectively ring or meet primality. This proves the stated partition of the points. Its basic opens are $D((n, 1_L)) = S \cup D_X(n)$ for $n \neq 0$ and $D((0, \lambda)) = D_L(\lambda) \subset S$. Consequently every open of Y is either an open $V \subset S$ or

$$W_U = S \cup U \quad (\emptyset \neq U \subset X \text{ open}). \quad (\text{ULC2})$$

All nonempty opens of X contain the generic point η and are cofinite. Write $i : X \hookrightarrow Y$ and $j : S \hookrightarrow Y$. The map $r : Y \rightarrow X$ is the identity on X and sends every point of S to η . It is continuous since $r^{-1}U = W_U$ for nonempty U , and $ri = \text{id}$. Nontriviality of L implies $S \neq \emptyset$ by prime-ideal separation, whose proof is included in FSC6.

The actual structure sheaf has

$$\mathcal{O}_Y(W_U) = G_L(\mathcal{O}_X(U)), \quad \mathcal{O}_Y|_S = \mathcal{O}_S, \quad \mathcal{O}_S(D_L(\lambda)) = \downarrow \lambda, \quad (\mathcal{O}_S)_J = L_J. \quad (\text{ULC3})$$

Here L_J is the localization with equality $[\lambda]_J = [\mu]_J$ precisely when $\lambda \wedge \nu = \mu \wedge \nu$ for some $\nu \notin J$. On basic support opens the restriction is $a \mapsto a \wedge \lambda$. Finite covers glue by joins and distributivity; compactness of the basic opens reduces an arbitrary covering to this assertion. On W_U , the generic arithmetic germ fixes one rational amplitude and one label, and the amplitude must be regular at every included prime; this proves the first formula. Restriction to S retains the label and localizes it. These observations prove ULC3 with the original structure sheaf, not a constant substitute.

61.2 The actual localized coefficient semimodule

Let $\mathcal{H}_k$ be either CDV's $\mathcal{O}_k$ or $\mathcal{A}_k$ on X , with generic complex vector space $W = V$ or B and stalks $W_{p,k}$. It is an $\mathcal{O}_X$ -module by multiplication by the rational scalar in each regular section. This scalar action preserves each complex subspace and every restriction.

Define a sheaf $\mathcal{M}_k$ of $\mathcal{O}_Y$ -semimodules by

$$\mathcal{M}_k(W_U) = G_L(\mathcal{H}_k(U)), \quad \mathcal{M}_k(V) = \mathcal{O}_S(V) \quad (V \subset S). \quad (\text{ULC4})$$

On arithmetic opens, restrictions restrict the amplitude and keep the label. Restriction from W_U to a support open sends (w, λ) to the section represented by λ , with all its localizations retained.

On support opens use the actual restrictions of $\mathcal{O}_S$. The scalar operation on an arithmetic open is $(a, \mu)(w, \lambda) = (aw, \mu \wedge \lambda)$ and addition uses join. On support opens scalar operation is meet and addition is join. All these formulas commute with restrictions.

Here is a direct gluing proof. A cover of W_U has some arithmetic members W_{U_a} covering U . Any two have nonempty arithmetic intersection, so compatible sections have one common label and their amplitudes glue in $\mathcal{H}_k$. A nonzero glued amplitude has nonzero restriction somewhere, hence top label. The arithmetic member already covers all of S ; compatibility forces each purely support member to carry precisely the restriction of that common label. Thus the glued section is unique and belongs to ULC4. Covers of a support open glue in $\mathcal{O}_S$. This proves the sheaf assertion and the scalar operations on the original carrier. Its stalks are

$$(\mathcal{M}_k)_\eta = G_L(W), \quad (\mathcal{M}_k)_p = G_L(W_{p,k}), \quad (\mathcal{M}_k)_J = L_J. \quad (\text{ULC5})$$

The first two follow by the filtered neighbourhood colimits and CDV's stalk descriptions. At J the neighbourhoods contained in S are cofinal and give the last formula.

This is the actual localization of the coefficient carrier at support. For every $w \in W$, multiplication by e sends (w, λ) to $(0, \lambda)$. Inverting the idempotent e makes it the unit, so

$$G_L(W)[e^{-1}] \simeq L, \quad G_L(W)_{P_J} \simeq L_J. \quad (\text{ULC6})$$

To verify the first isomorphism rather than only its surjectivity, the support map has the section $\lambda \mapsto (0, \lambda)$; every localized element equals its product with e , and equality of these zero labels is unchanged by multiplication by e . Further inversion of the zero labels outside J gives exactly the relation in ULC3. Every top-supported nonzero integer already maps to the unit of that localized lattice. No additional relation arises from those integers. This proves the second isomorphism.

Retain separately the inverse-image coefficient sheaf

$$\mathcal{P}_k = r^{-1}G_L(\mathcal{H}_k), \quad \mathcal{P}_k \longrightarrow \mathcal{M}_k. \quad (\text{ULC7})$$

Here $G_L(\mathcal{H}_k)$ on X has the section formula used in ULC4; its sheaf proof follows from the same common-generic-point argument. The arrow is the identity on arithmetic opens. On a support open it sends a locally constant section of $G_L(W)$ to its support label and then to its localizations in $\mathcal{O}_S$. It commutes with restrictions because a section on W_U restricts in $\mathcal{P}_k$ to its constant generic germ, whereas in $\mathcal{M}_k$ it restricts to its localized label. Its stalk map is the identity at arithmetic points and

$$G_L(W) \longrightarrow L \longrightarrow L_J, \quad (w, \lambda) \longmapsto \lambda \longmapsto [\lambda]_J \quad (\text{ULC8})$$

at a support prime. It is surjective on stalks. The precise fibre relation at J is existence of $\nu \notin J$ with $\lambda \wedge \nu = \mu \wedge \nu$; amplitudes impose no further condition. This comparison is linear over the corresponding sheaf map $r^{-1}G_L(\mathcal{O}_X) \rightarrow \mathcal{O}_Y$. One cannot make the unchanged $\mathcal{P}_k$ into an $\mathcal{O}_Y$ -semimodule extending that coefficient action at a support point: e is a unit there but kills each nonzero amplitude under the original action. The map ULC8 computes the required localization rather than assuming it away.

61.3 Retaining the support ideal and the exact amplitude reflection

Let $\mathcal{D} = j_*\mathcal{O}_S$. Its sections on W_U are L and its sections on $V \subset S$ are $\mathcal{O}_S(V)$. There are actual semimodule maps

$$\mathcal{D} \xrightarrow{z} \mathcal{M}_k \xrightarrow{\sigma} \mathcal{D}, \quad \sigma z = \text{id}, \quad q : \mathcal{M}_k \longrightarrow i_*\mathcal{H}_k. \quad (\text{ULC9})$$

On W_U , $z(\lambda) = (0, \lambda)$, $\sigma(w, \lambda) = \lambda$, and $q(w, \lambda) = w$. On support opens the first two maps are the identity, and the last map has zero target. On $\mathcal{D}$ the scalar action is induced by $\mathcal{O}_Y \rightarrow \mathcal{D}$; on

the amplitude target it is induced by $\mathcal{O}_Y \rightarrow i_*\mathcal{O}_X$. These formulas prove compatibility with both operations and every restriction. They give

$$e\mathcal{M}_k = z\mathcal{D} = q^{-1}(0), \quad \mathcal{M}_k/\sim_{z\mathcal{D}} \simeq i_*\mathcal{H}_k, \quad (\text{ULC10})$$

where $x \sim_{z\mathcal{D}} y$ means locally $x + z(u) = y + z(v)$ for support sections u, v . Indeed multiplication by e is precisely $z\sigma$. Two sections $(w, \lambda), (w', \mu)$ on W_U with equal amplitude obey $(w, \lambda) + z(\mu) = (w', \mu) + z(\lambda)$. Conversely adding zero-amplitude sections cannot change the amplitude. On a support open the entire module is $z\mathcal{D}$. The map q is onto on sections, using $(w, 1_L)$ as a lift. This proves the quotient description, including its original congruence.

The sheaf associated with additive group completion of $\mathcal{M}_k$ is exactly $i_*\mathcal{H}_k$. Every additive map to a sheaf of abelian groups kills each idempotent zero label because $z + z = z$. It then identifies the two sections in the preceding displayed relation and factors uniquely through q . Conversely the target is already an abelian sheaf, and each w is in the image. This proves the universal property locally and hence for sheaves. It does not identify $\mathcal{M}_k$ itself with its group completion. The retract $z\mathcal{D}$, all its localized labels, and the map ULC8 remain explicit parts of the original object.

The image of (q, σ) inside $i_*\mathcal{H}_k \times \mathcal{D}$ has on W_U the exact constraint $w \neq 0 \Rightarrow \lambda = 1_L$ and has the whole support factor on support opens. It is injective on every stalk by ULC5. Thus an unrestricted direct product is not the original carrier.

61.4 Cohomology for an arbitrary support spectrum

For any abelian sheaf F on Y there is a natural isomorphism

$$r_*F \simeq i^{-1}F. \quad (\text{ULC11})$$

The map is induced by restriction from W_U to neighbourhoods of U along i . Its stalk at $x \in X$ is the identity on F_x : all neighbourhoods of x in Y are exactly the W_U for neighbourhoods U in X . This proves the isomorphism and shows that r_* is exact. It preserves injectives because its left adjoint r^{-1} is exact. Since $\Gamma(Y, F) = \Gamma(X, r_*F)$, an injective resolution gives

$$R\Gamma(Y, F) \simeq R\Gamma(X, i^{-1}F). \quad (\text{ULC12})$$

For $F = r^{-1}\mathcal{H}_k$, the right side is represented by the actual CDV complex

$$C_k = [W \xrightarrow{\delta_k} D_k], \quad D_k = \bigoplus_p W/W_{p,k}, \quad \delta_k w = ([w]_{p,k})_p. \quad (\text{ULC13})$$

For $F = i_*\mathcal{H}_k$, the same conclusion follows either from ULC11–12 or from the exactness of closed-embedding direct image. Consequently the amplitude reflection in ULC10 has $H^1 = D_k/\delta_k W$, including every mixed-prime summand of CDV13, and vanishes in degrees at least two. No finiteness of L has been used.

There is more information in the relative comparison for the *inverse-image vector sheaf* $r^{-1}\mathcal{H}_k$. Its restriction to S is the constant sheaf $c_S W$, because $r|_S$ factors through η . This is not the support restriction $\mathcal{O}_S$ of the original semimodule $\mathcal{M}_k$. For clarity define an explicit acyclic resolution of $c_S W$ as follows. For an abelian sheaf E put $T(E)(V) = \prod_{x \in V} E_x$ with projection restrictions. This is a sheaf and is flasque; the germ map $E \rightarrow T(E)$ is injective. Set $E_0 = c_S W$, $E_{n+1} = \text{coker}(E_n \rightarrow T(E_n))$, and use the composites $T(E_n) \rightarrow E_{n+1} \rightarrow T(E_{n+1})$. The resulting sheaf sequence is exact by its kernel and cokernel definitions, and all its terms are flasque. Let $B^n = \Gamma(S, T(E_n))$ with differential d_B . Then $B^\bullet$ computes $R\Gamma(S, c_S W)$. This construction retains every infinite product over the actual points of S .

The constant section gives $c : W \rightarrow \ker(d_B : B^0 \rightarrow B^1)$. The relative cohomology with support in the closed arithmetic stratum is computed by

$$W \xrightarrow{d^0} D_k \oplus B^0 \xrightarrow{d^1} B^1 \xrightarrow{-d_B} B^2 \xrightarrow{-d_B} \dots, \quad d^0 w = (\delta_k w, cw), \quad d^1(d, b) = -d_B b. \quad (\text{ULC14})$$

Here degrees start at zero. To justify the relative comparison, take an injective resolution on Y . Its restrictions to the open S are injective since extension by zero is an exact left adjoint. Each injective sheaf is flasque, so restriction of its global sections to S is onto. The kernel complex therefore computes sections with support in X , and the short exact sequence of complexes yields the fibre of global restriction. Apply the CDV flasque resolution to $r^{-1}\mathcal{H}_k$: inverse image is exact, its constant first term is $c_Y W$, and its closed-prime term has zero restriction to S . These terms are acyclic for global sections by ULC12, whose direct images are precisely the two CDV flasque terms. Restriction of this resolution to S has just $c_S W$ in degree zero. Its comparison with the displayed resolution $B^\bullet$ is the constant germ map in degree zero and zero in degree one. Thus the fibre complex, with the sign convention shown, is exactly ULC14. Its differential squares to zero since $d_B c = 0$.

As S is nonempty, c is injective. Writing $\mathrm{LC}(S, W) = \Gamma(S, c_S W)$, calculation of the kernels and images in ULC14 gives

$$\begin{aligned} H_X^0(Y, r^{-1}\mathcal{H}_k) &= 0, \\ H_X^1(Y, r^{-1}\mathcal{H}_k) &= (D_k \oplus \mathrm{LC}(S, W)) / \{(\delta_k w, cw) : w \in W\}, \\ H_X^{j+1}(Y, r^{-1}\mathcal{H}_k) &= H^j(S, c_S W) \quad (j \geq 1). \end{aligned} \tag{ULC15}$$

In particular there is an exact sequence

$$0 \longrightarrow D_k \xrightarrow{d \mapsto [(d, 0)]} H_X^1 \longrightarrow \mathrm{LC}(S, W)/cW \longrightarrow 0. \tag{ULC16}$$

Injectivity of the first map follows from injectivity of c . The last map sends $[(d, b)]$ to $[b]$; its kernel is the displayed image, since subtracting $(\delta_k w, cw)$ removes any constant b . Choosing a point $s_0 \in S$ gives the explicit splitting $[(d, b)] \mapsto (d - \delta_k(b(s_0)), [b])$; its inverse uses the unique representative $b - b(s_0)$ of a locally constant class vanishing at s_0 . The splitting is not asserted to be independent of that point. The natural map to ordinary $H^1(Y, r^{-1}\mathcal{H}_k)$ sends $[(d, b)]$ to $[d] \in D_k/\delta_k W$.

No identity replacing $H^j(S, c_S W)$ by $W \otimes H^j(S, \mathbb{C})$ is used for an arbitrary infinite S . The full resolution above specifies its coefficient maps without commuting tensor products through infinite products.

61.5 Prime transport on all these objects

For every nonzero rational r , CDV5 and CDV7 induce the exact maps

$$\mathcal{M}_k \xrightarrow{\alpha_r} \mathcal{M}_{k+d(r)}, \quad (w, \lambda) \mapsto (\alpha_r w, \lambda) \text{ on } W_U, \quad \text{id on support opens.} \tag{ULC17}$$

They commute with restrictions, scalar multiplication, z, σ, q and ULC7. They have inverse $\alpha_{r^{-1}}$ and preserve every localization relation of L . On the arithmetic amplitude complexes they are precisely CDV14. On the constant-support vector resolution they apply α_r to each stalk coordinate; functoriality of the products and cokernels above defines every higher map. Thus ULC14–16 commute with prime transport, including their complete mixed-prime defects. The support action on the actual localized semimodule is the identity, while the auxiliary constant vector sheaf carries α_r there. ULC8 is their exact connecting map.

For the original tensors $f_{k,a}$, CDV22–24 now give on this same arithmetic stratum

$$\begin{aligned} f_{k,a}(0) &= 1, & \int f_{k,a} &= a/N(k), \\ Z(f_{k,a}, s) &= a^s N(k)^{-s} \pi^{-s/2} \Gamma(s/2) \zeta(s), & \Re s &> 1, \\ \alpha_r f_{k,a} &= f_{k+d(r), |r|a}, & N(k+d(r)) &= |r|N(k). \end{aligned} \tag{ULC18}$$

Both endpoint moments and all Mellin factors are retained. At the support stalk ULC8 sends this top-labelled coefficient to $1 \in L_J$, independently of its amplitude. That local statement follows

from the localization already proved; it is not a Mellin inversion theorem reconstructing L from a scalar function. The earlier fixed-prime quotient defect remains CDV17–18 with its full norm; ULC17 supplies its invertible family target over the actual base.

For a three-element chain $0 < a < 1$, the prime $J = \{0\}$ inverts a , so the zero labels a and 1 become equal in L_J , whereas the other prime $\{0, a\}$ leaves all three labels distinct. For every $w \neq 0$, the pullback stalk retains $(w, 1)$, while the localized stalk at either prime has its image 1 . Globally ULC4 still distinguishes $(w, 1)$, $(0, 1) = e$ and $(0, 0) = \tau$. This explicit example records the localization, its fibres and its arithmetic comparison without reducing the entire support spectrum to a single point or a Boolean coordinate.

62 Heat and translation projectors on the localized adelic coefficient family

Provenance and original objects. The incoming programme source is *Dynamical Synchronization, Heat Remapping, and the Full Supported Cauchy–Weil Form*, 23 September 2026, RECONSTRUCTION.tex, ST1–12 and SY1–16. Its archive is retained unchanged with SHA-256 70810361fd8fd66f9cc7154ed919bc5c0c15d003e3ec72504b173bbb24aeed15. Those finite-source sections were read in full. Their identities are proved below in the exact receiving sheaves; neither their later analytic claims nor the sign of a Weil form is assumed. The adelic coefficient input is Alain Connes and Caterina Consani’s *Knots, primes and class field theory*, Section 4, extended with full proofs in CDV1–25 and ULC1–18. The original arithmetic function remains ζ , with the exact factors of ULC18 and ZTC10. Finite formal-coordinate heat here is not an assertion that those coefficients follow Newman heat.

62.1 Finite matrices on the original semimodule sheaf

Use the actual localized sheaf $\mathcal{M}_k$ of ULC4 on $Y = \text{Spec } G_L(\mathbb{Z})$ and fix $N \geq 2$. Its independent coefficient source is $\mathcal{M}_k^{N+1}$, ordered by $1, Z, \dots, Z^N$. A constant split complex scalar $(c, \lambda) \in G_L(\mathbb{C})$ acts on arithmetic-open sections by $(w, \mu) \mapsto (cw, \lambda \wedge \mu)$ and on a support stalk L_J by meeting with $[\lambda]_J$. These maps commute with restrictions by ULC4 and with the $\mathcal{O}_Y$ action. Nonzero rational coefficients, including their factorial denominators, therefore define the stated endomorphisms without treating them as regular rational functions at every arithmetic prime.

Let $D_{i,i+1} = (i+1)^\bullet$ and every structural zero be τ . For split parameters a, b define

$$F_{a,b} = \sum_{n=0}^N \frac{(-aD^2 + bD)^n}{n!}, \quad (F_{a,b})_{ij} = \sum_{q=0}^{\lfloor (j-i)/2 \rfloor} \left(\frac{(-1)^q j!}{i! q! (j-i-2q)!} \right)^\bullet a^q b^{j-i-2q} \quad (j \geq i). \quad (\text{CLS1})$$

Below the diagonal the entries are τ . In the first formula the reciprocal factorial is fully supported, and the zeroth power is the free identity matrix. Each generator strictly lowers degree, so all omitted higher terms are the global zero matrix. To obtain the second formula, select q factors of $-aD^2$ and $j-i-2q$ factors of bD , then use the binomial coefficient and $D^{j-i}(Z^j) = j!Z^i/i!$. This keeps every sign and denominator. Commuting the generators and using the finite binomial identity gives

$$F_{a,b}F_{c,d} = F_{a+c,b+d}, \quad F_{\tau,\tau} = I. \quad (\text{CLS2})$$

This identity is valid in the semiring of matrices: the binomial proof uses distributivity, not subtraction of supported zeros.

At $a = (0, \lambda)$, $b = (0, \mu)$, every positive-degree amplitude vanishes but its original label remains. An odd degree drop has label μ because every term contains b and the term $q = 0$ has exactly that label. A positive even drop has the endpoint labels λ and μ , with all intermediate

labels they meet, so its resulting label is $\lambda \vee \mu$. Thus

$$P = F_{e,\tau}, \quad Q = F_{e,e} = F_{\tau,e}, \quad P^2 = P, \quad Q^2 = Q, \quad PQ = QP = Q. \quad (\text{CLS3})$$

At an arithmetic stalk, these maps keep each amplitude and change the independent labels by

$$(P\lambda)_i = \bigvee_{j \geq i, j-i \text{ even}} \lambda_j, \quad (Q\lambda)_i = \bigvee_{j \geq i} \lambda_j. \quad (\text{CLS4})$$

This follows directly from matrix multiplication: the off-diagonal entries are zero-amplitude elements, so only the diagonal term contributes an amplitude, while joins record all admissible labels. Their images have respectively nested same-parity labels and a single nested flag. They are sheaf retractions onto their images.

62.2 Localization preserves the projectors' difference

At a support prime J , ULC8 sends every nonzero rational scalar and e to $1 \in L_J$, while τ remains zero. Consequently the localized P_J and Q_J are exactly the join operators CLS4 on $(L_J)^{N+1}$, now with their localized labels. They are different at every support prime: the input with entry 1 only at degree one and zero elsewhere has output degree zero equal to 0 under P_J and 1 under Q_J . A proper prime localization has $0 \neq 1$: an equality would require an element outside J to meet 0 and 1 equally, forcing that element to be $0 \in J$. Thus the example is nonzero in every support stalk.

The additive group reflection instead sends both projectors to the identity on $(i_*\mathcal{H}_k)^{N+1}$: every off-diagonal entry in them has amplitude zero. It follows, by applying the explicit CDV complex coordinatewise, that

$$H^j(P) = H^j(Q) = \text{id} \quad \text{on } H^j(X, \mathcal{H}_k)^{N+1}, \quad j \geq 0, \quad (\text{CLS5})$$

where these cohomology maps are those of the amplitude reflection. Every mixed-prime summand of CDV13 is included. The equality is not a claim that $P = Q$ on $\mathcal{M}_k^{N+1}$; their explicit support-stalk discrepancy proves the contrary. This gives the actual location of the information lost by ordinary additive cohomology in this calculation.

For supported $t, b \in \mathbb{C}$, CLS2 gives

$$F_{t^\bullet, b^\bullet} F_{(-t)^\bullet, (-b)^\bullet} = Q. \quad (\text{CLS6})$$

Also $FQ = QF = F$, since adding e to a supported parameter leaves that parameter unchanged. Hence these maps form a group on $Q\mathcal{M}_k^{N+1}$ with identity Q . On a support stalk all these supported-parameter matrices have the same image Q_J : in CLS1 every positive degree has at least one top-labelled term, and localization retains its label even when its amplitude cancels. They therefore act as the identity on the localized projective image. On the full free localized source, their common action is the nonidentity projector Q_J . The arithmetic amplitude group is the usual invertible polynomial heat/translation action.

62.3 The common-label comparison and its exact fibres

Define $\mathcal{M}_k[Z]_{\leq N}^{\text{one}}$ by the construction ULC4 applied to the vector sheaf $\mathcal{H}_k[Z]_{\leq N}$. Its arithmetic-open sections have one common label for the entire polynomial; its support restriction is $\mathcal{O}_S$. There is a surjective sheaf map

$$\eta_N : \mathcal{M}_k^{N+1} \longrightarrow \mathcal{M}_k[Z]_{\leq N}^{\text{one}}, \quad ((w_i, \lambda_i)) \longmapsto \left(\sum_i w_i Z^i, \bigvee_i \lambda_i \right). \quad (\text{CLS7})$$

On support opens it is the join of the coordinate sections. A nonzero polynomial forces at least one nonzero coefficient, hence top label; so the image belongs to the original carrier. Distributivity proves compatibility with scalar meets, and joins prove additivity. A polynomial with top label lifts by assigning top label to each coefficient. A zero polynomial with any label lifts by placing that label at degree zero and τ elsewhere. This proves surjectivity and compatibility with support restrictions. Its exact fibre relation is equality of every amplitude coefficient and equality of the total join; at a support stalk it is equality of the join in L_J .

Diagonal entries in CLS1 retain each input label at its own degree, while other entries contribute labels below existing ones. Therefore the total join is unchanged. The amplitude calculation is the ordinary polynomial exponential, giving

$$\eta_N F_{a,b} = \left[e^{-\text{amp}(a)\partial_Z^2 + \text{amp}(b)\partial_Z} \right]^\dagger \eta_N, \quad \eta_N P = \eta_N Q = \eta_N. \quad (\text{CLS8})$$

On the common-label target the displayed linear operator keeps its single label, and on support it is the identity. This is the exact map under which the different projectors become equal.

62.4 Synchronization on every divisor-indexed coefficient source

Fix $d \geq 2$ and the monomial basis of total degree at most N in $s_1, \dots, s_d$, with independent copies of $\mathcal{M}_k$. The cyclic map β replaces Z^j by $(s_1 + \dots + s_d)^j$, with the literal multinomial coefficients. All nonzero coefficients are fully supported and all structural zeros are τ . Since each $D_i D_j \beta = \beta \partial_Z^2$, the ordinary amplitude covariance C acts by

$$\exp \left(\frac{1}{4} \sum_{i,j} C_{ij} D_i D_j \right) \beta = \beta \exp \left(\frac{1}{4} (\mathbf{1}^\top C \mathbf{1}) \partial_Z^2 \right). \quad (\text{CLS9})$$

For the original three matrices retain all entries and factors:

$$C_{\text{ind}} = \frac{t}{d} I, \quad C_{\text{sync}} = \frac{t}{d^2} \mathbf{1} \mathbf{1}^\top, \quad C_{\text{rel}} = \frac{t}{d} \left(I - \frac{\mathbf{1} \mathbf{1}^\top}{d} \right). \quad (\text{CLS10})$$

Lift only the independent off-diagonal structural zeros as τ , and every entry of the other two matrices as supported, as in the incoming source. Their off-diagonal sums are e , not τ . For each of these specified lifted matrices $\tilde{C}$, define $\mathcal{U}_{\tilde{C}}^\dagger = \exp(\frac{1}{4} \sum_{i,j} \tilde{C}_{ij} D_i D_j)$ on the stated total-degree source. The finite exponential uses the same fully supported reciprocal factorials and free identity as CLS1. The exact identity on the coefficient sheaf is therefore

$$\mathcal{U}_{\text{sync}}^\dagger \mathcal{U}_{\text{rel}}^\dagger = \mathcal{U}_{\text{ind}}^\dagger P_\Sigma, \quad P_\Sigma = \exp \left(\frac{e}{4} (D_1 + \dots + D_d)^2 \right). \quad (\text{CLS11})$$

Indeed the sum of the lifted generators is the sparse independent generator plus the displayed supported-zero generator: on the diagonal adding e does not change an already supported entry, and off the diagonal it supplies the missing supported cancellation. The finite commuting exponential proof of CLS2 gives the product identity. The matrix P_Σ has the identity on its diagonal, e at every coordinatewise lowering of positive even total degree, and τ elsewhere. These are exactly all terms of powers of the displayed derivative; each admissible path has a nonzero rational coefficient and hence the specified label. Composing two such paths again gives precisely that relation, proving $P_\Sigma^2 = P_\Sigma$.

The cyclic formula retains the projector:

$$P_\Sigma \beta = \beta P, \quad \mathcal{U}_{\text{rel}}^\dagger \beta = \beta P. \quad (\text{CLS12})$$

For a source monomial Z^j , β has top-supported coefficients at every monomial of total degree j . For any lower same-parity monomial choose a degree- j monomial above it coordinatewise.

It receives a supported contribution under P_Σ ; no other degree or parity can receive one. Its amplitude remains zero in that lower degree. This proves the first equality entry by entry. For the second, the relative amplitude generator annihilates β by $\mathbf{1}^\top C_{\text{rel}} \mathbf{1} = 0$, while its fully supported entries supply the same admissible degree-lowering labels. This proves the second equality with its original support, including $t = 0^\bullet$.

Localization of CLS11 retains a visible difference. On the input supported only at $s_1 s_2$, P_Σ contributes 1 to the constant label at every support stalk. Sparse independent heat has no such constant contribution, since no D_i^2 can act on that monomial. The synchronized and relative product does have it, as CLS11 states. The amplitude of this extra term is zero before localization, and the ordinary group reflection sends P_Σ to the identity. Thus it is present on the actual support stratum even though it is absent in the original scalar amplitude equation.

Finally the rational-prime maps α_r of ULC17 commute with every matrix here. Each matrix acts on formal degrees by complex scalars and labels; α_r is complex linear on coefficients and fixes the labels. Hence every matrix entry commutes, proving

$$\alpha_r^{N+1} F_{a,b} = F_{a,b} \alpha_r^{N+1}, \quad \alpha_r P_\Sigma = P_\Sigma \alpha_r, \quad \alpha_r \beta = \beta \alpha_r, \quad k \longmapsto k + d(r). \quad (\text{CLS13})$$

These identities hold between the actual divisor-indexed targets, with the original prime and archimedean scaling in CDV22–24. They do not assert that finite polynomial differentiation descends through an unspecified spectral quotient or changes the zeros of the original zeta function. The receiving maps, the localized projectors and the full mixed-prime amplitude cohomology are now specified on the same coefficient family.

63 The original trivial-correspondence family with every Haar constant

Human source and reading scope. Alain Connes, Caterina Consani, and Matilde Marcolli, *The Weil proof and the geometry of the adèles class space*, original author file `Yuri70v2.tex`, subsection *Fubini's theorem and the trivial correspondences*, especially Lemma `degmodifyV`, supply the polynomial Gaussian used below. The original file has SHA-256 `b1f694cb934dd4fef829287f1a55dd8de21c67a481e552453c2f7e8fe0ae6fb5`. The complete subsection, source lines 2313–2424, was read, together with the definitions `fsharpinv`, `degcorrZf`, and `codegcorrZf`. This calculation fixes the measures and sums over all nonzero rational numbers; it replaces the source's unspecified constant by a proved exact value. It does not replace the original zeta function with a completed function, and makes no RH inference.

63.1 Original adelic test, rational restrictions, and additive moments

Use additive Haar measure dx at infinity and $\text{vol}(\mathbb{Z}_p) = 1$ at every finite prime. On $C_{\mathbb{Q}} = \widehat{\mathbb{Z}}^{\times} \times \mathbb{R}_{>0}$ use Haar measure $du dy/y$, with $\text{vol}(\widehat{\mathbb{Z}}^{\times}) = 1$. Retain

$$\eta(x) = \pi x^2 \left(\pi x^2 - \frac{3}{2} \right) e^{-\pi x^2} = \left(\pi^2 x^4 - \frac{3\pi}{2} x^2 \right) e^{-\pi x^2}, \quad k \in \bigoplus_p \mathbb{Z}, \quad N(k) = \prod_p p^{k_p}, \quad a > 0, \quad (\text{ETC1})$$

and the actual tensor

$$\xi_{k,a}(x) = \eta(x_{\infty}/a) \prod_p 1_{p^{k_p} \mathbb{Z}_p}(x_p), \quad b = \frac{a}{N(k)} > 0. \quad (\text{ETC2})$$

The symbol b records this specified ratio; it does not change either of the original parameters or the local factors. The finite product of nonstandard factors defines a Bruhat–Schwartz function on the original adèles.

The Gaussian integral and its two differentiated forms are

$$\int_{\mathbb{R}} e^{-\pi x^2} dx = 1, \quad \int_{\mathbb{R}} x^2 e^{-\pi x^2} dx = \frac{1}{2\pi}, \quad \int_{\mathbb{R}} x^4 e^{-\pi x^2} dx = \frac{3}{4\pi^2}. \quad (\text{ETC3})$$

For example, square the first integral and use polar coordinates to obtain one; differentiating $\int e^{-\pi t x^2} dx = t^{-1/2}$ once and twice at $t = 1$ gives the other two. Polynomial Gaussian bounds justify both differentiations. Consequently

$$\eta(0) = 0, \quad \int_{\mathbb{R}} \eta = \frac{3}{4} - \frac{3}{4} = 0, \quad \xi_{k,a}(0) = 0, \quad \int_{\mathbb{A}_{\mathbb{Q}}} \xi_{k,a} = a \prod_p p^{-k_p} \left(\frac{3}{4} - \frac{3}{4} \right) = 0. \quad (\text{ETC4})$$

Thus the two additive zero-moment conditions hold on the original test, with both summands and every volume retained.

Define the reduction by the original full rational sum:

$$f_{k,a}(u, y) = (\mathcal{E}\xi_{k,a})(u, y) = \sum_{q \in \mathbb{Q}^{\times}} \xi_{k,a}(qy, (qu_p)_p). \quad (\text{ETC5})$$

Since u_p is a unit, its finite characteristic factors permit exactly $v_p(q) \geq k_p$ at every prime. This says that $q/N(k)$ is a rational number integral at every finite prime, hence an integer: a reduced denominator greater than one would give a negative valuation at one of its prime divisors. Conversely every nonzero integer satisfies these conditions. The map $n \mapsto N(k)n$ is therefore a bijection onto the permitted rational numbers. Since η is even,

$$f_{k,a}(u, y) = \sum_{n \in \mathbb{Z} \setminus \{0\}} \eta \left(\frac{N(k)ny}{a} \right) = 2 \sum_{n=1}^{\infty} \eta(ny/b) =: f_b(y). \quad (\text{ETC6})$$

The factor two is from the two rational signs. This is an exact map of the specified family; it shows explicitly that this scalar reduction depends on k, a only through $a/N(k)$. It does not recover the individual finite-place factors from that ratio.

63.2 Fourier calculation, Poisson identity, and convergence

Use the real Fourier convention $\widehat{h}(v) = \int h(x)e^{-2\pi i x v} dx$ and compatible conductor $\mathbb{Z}_p$ characters at finite places. Let $g(x) = e^{-\pi x^2}$. The Fourier integral has derivative $-2\pi v \widehat{g}(v)$: differentiate under the integral and integrate $g' = -2\pi x g$ by parts. Its value at zero is one by ETC3, so $\widehat{g}(v) = g(v)$. Differentiating this identity gives the complete two polynomial transforms

$$\begin{aligned}\widehat{x^2 g}(v) &= \left(\frac{1}{2\pi} - v^2\right) g(v), \\ \widehat{x^4 g}(v) &= \left(v^4 - \frac{3}{\pi} v^2 + \frac{3}{4\pi^2}\right) g(v), \\ \widehat{\eta}(v) &= \left[\pi^2 \left(v^4 - \frac{3}{\pi} v^2 + \frac{3}{4\pi^2}\right) - \frac{3\pi}{2} \left(\frac{1}{2\pi} - v^2\right)\right] g(v) = \eta(v).\end{aligned}\tag{ETC7}$$

These differentiations and integrations by parts have vanishing boundary terms because every derivative is a polynomial times a Gaussian. At a finite prime, character orthogonality gives $\widehat{1_{p^{k_p}\mathbb{Z}_p}} = p^{-k_p} 1_{p^{-k_p}\mathbb{Z}_p}$: the transform equals the original ball volume on its annihilator and is zero outside it. The real dilation contributes a . Hence

$$\widehat{\xi}_{k,a} = a \prod_p p^{-k_p} \xi_{-k,1/a} = \frac{a}{N(k)} \xi_{-k,1/a}.\tag{ETC8}$$

Applying Fourier twice yields the original even test: the second coefficient is $(1/a)/N(-k) = N(k)/a$, so the two displayed coefficients multiply to one.

For the required Poisson identity no rearrangement of a divergent integral is used. For $r > 0$, periodize $\eta(r(x+n))$ over $n \in \mathbb{Z}$. Every differentiated series converges uniformly on $x \in [0, 1]$ by Schwartz bounds. Its m th Fourier coefficient is $r^{-1} \widehat{\eta}(m/r)$, by the absolutely convergent substitution over all intervals. These coefficients decrease faster than any power, so their Fourier series converges absolutely to the periodization. Evaluating at zero and using ETC7 gives

$$\sum_{n \in \mathbb{Z}} \eta(nr) = \frac{1}{r} \sum_{m \in \mathbb{Z}} \eta(m/r), \quad f_b(y) = \frac{b}{y} f_{1/b}(1/y).\tag{ETC9}$$

The zero summand is zero on both sides because $\eta(0) = 0$. The same argument with g proves its Gaussian Poisson formula.

For $y \geq 1$ the series in ETC6 and each of its derivatives are bounded by a polynomial in y times $e^{-\pi y^2/b^2}$. Indeed factor this last exponential from each term; the remaining sum is bounded by a convergent series of a polynomial in n times $e^{-\pi(n^2-1)/b^2}$. ETC9 therefore bounds f_b near zero by a constant times $y^{-5} e^{-\pi b^2/y^2}$; its differentiated versions have the same exponential with possibly different finite powers of y^{-1} . Thus this actual reduced test and all its multiplicative derivatives decrease faster than every power at both ends. In particular every Mellin integral below converges absolutely for every s , locally uniformly with all s derivatives.

63.3 Original-zeta Mellin formula with both Gaussian summands

With the specified class-group measure put

$$M_b(s) = \int_{\widehat{\mathbb{Z}}^\times} du \int_0^\infty f_b(y) y^s \frac{dy}{y} = \int_0^\infty f_b(y) y^{s-1} dy.\tag{ETC10}$$

The preceding bounds prove that M_b is entire. Initially for $\operatorname{Re} s > 1$, absolute Fubini is valid on ETC6, since

$$2 \sum_{n \geq 1} \int_0^\infty |\eta(ny/b)| y^{\operatorname{Re} s - 1} dy = 2b^{\operatorname{Re} s} \sum_{n \geq 1} n^{-\operatorname{Re} s} \int_0^\infty |\eta(x)| x^{\operatorname{Re} s - 1} dx < \infty.$$

The local integral converges already for $\operatorname{Re} s > -2$, as $\eta(x) = O(x^2)$ at zero. Substituting $v = \pi x^2$ into each of the two original polynomial-Gaussian terms proves

$$\begin{aligned} \int_0^\infty \eta(x) x^{s-1} dx &= \frac{\pi^2}{2\pi^{(s+4)/2}} \Gamma\left(\frac{s+4}{2}\right) - \frac{3\pi}{4\pi^{(s+2)/2}} \Gamma\left(\frac{s+2}{2}\right), \\ M_b(s) &= 2 \left(\frac{a}{N(k)}\right)^s \zeta(s) \left[\frac{\pi^2}{2\pi^{(s+4)/2}} \Gamma\left(\frac{s+4}{2}\right) - \frac{3\pi}{4\pi^{(s+2)/2}} \Gamma\left(\frac{s+2}{2}\right) \right]. \end{aligned} \quad (\text{ETC11})$$

The working expression keeps the original $\zeta(s)$, both Gamma summands, the two rational signs, and all finite and real factors. It continues meromorphically by the original zeta continuation ZTC10; equality with the entire integral ETC10 proves that its singularities are removable as a whole, without asserting that its separate summands are entire.

For a precise comparison with the source's unspecified constant, the two-step Gamma recurrence gives the identity

$$2 \left[\frac{\pi^2}{2\pi^{(s+4)/2}} \Gamma\left(\frac{s+4}{2}\right) - \frac{3\pi}{4\pi^{(s+2)/2}} \Gamma\left(\frac{s+2}{2}\right) \right] = \frac{s(s-1)}{4} \pi^{-s/2} \Gamma(s/2). \quad (\text{ETC12})$$

This is an explicit equality of the retained multiplier, not a replacement arithmetic function. For real u it yields the exact source comparison

$$M_b(iu) = -\frac{b^{iu}}{4} u(u+i) \pi^{-iu/2} \Gamma(iu/2) \zeta(iu). \quad (\text{ETC13})$$

At removable points its value is the entire continuation specified by ETC10. The factor $-1/4$ is for the full rational sum in ETC5. Using only the positive integers would instead give $-1/8$ and would not be the reduction defined in ETC5.

63.4 Both endpoint values without an illicit interchange

Retain the complete Gaussian theta sum, including its zero term:

$$\Theta_b(y) = \sum_{n \in \mathbb{Z}} e^{-\pi(ny/b)^2}, \quad f_b(y) = \frac{1}{4} (y^2 \Theta_b''(y) + 2y \Theta_b'(y)) = \frac{1}{4} (y^2 \Theta_b'(y))'. \quad (\text{ETC14})$$

For each summand this follows from $x^2 g''(x) + 2x g'(x) = 4\eta(x)$; the differentiated sums converge locally uniformly. Gaussian Poisson summation as proved after ETC9 gives $\Theta_b(y) = (b/y) \Theta_{1/b}(1/y)$. As $y \rightarrow \infty$, $\Theta_b(y) \rightarrow 1$ and $y \Theta_b'(y), y^2 \Theta_b''(y) \rightarrow 0$. As $y \downarrow 0$, its leading term is b/y , with an exponentially small remainder and differentiated remainders. Consequently

$$\begin{aligned} \lim_{y \downarrow 0} y^2 \Theta_b'(y) &= -b, \\ \lim_{y \downarrow 0} (y \Theta_b'(y) + \Theta_b(y)) &= 0, \\ M_b(1) &= \frac{1}{4} [y^2 \Theta_b''(y)]_0^\infty = \frac{b}{4} = \frac{a}{4N(k)}, \\ M_b(0) &= \frac{1}{4} [y \Theta_b'(y) + \Theta_b(y)]_0^\infty = \frac{1}{4}. \end{aligned} \quad (\text{ETC15})$$

Integrate first on a compact interval $[\epsilon, R]$ and then take the endpoints; ETC9–10 guarantee convergence of the integrals, and the displayed limits justify both passages. Thus the two endpoint

values are calculated from the actual reduced test, without assuming special zeta values and without exchanging a nonabsolutely convergent sum and integral.

The failure of the tempting degree interchange is quantified exactly. Each $n \neq 0$ term has integral zero in dy , by ETC4, but

$$\sum_{n \neq 0} \int_0^\infty |\eta(ny/b)| dy = 2b \left(\int_0^\infty |\eta(x)| dx \right) \sum_{n=1}^\infty \frac{1}{n} = \infty. \quad (\text{ETC16})$$

The coefficient integral is strictly positive because η is continuous and not identically zero. The codegree interchange also lacks absolute convergence: each absolute integral with dy/y equals the same positive finite number $\int_0^\infty |\eta(x)| dx/x$, so its sum diverges. This does not conflict with the convergent whole-test endpoint values ETC15.

63.5 Sharp, conjugation, and two independent endpoint adjustments

Use the source's literal involution

$$f^\#(u, y) = y^{-1} \overline{f(u^{-1}, y^{-1})}. \quad (\text{ETC17})$$

The tests f_b are real and independent of u . Applying ETC9 at $1/y$, and comparing with ETC8, proves

$$f_b^\#(y) = b f_{1/b}(y) = \mathcal{E} \widehat{\xi}_{k,a}(y), \quad M_{f_b^\#}(s) = b M_{1/b}(s) = M_b(1-s). \quad (\text{ETC18})$$

For a general complex linear combination f of these tests, the exact formula instead is

$$M_{f^\#}(s) = \overline{M_f(1-\bar{s})}, \quad M_{f^\#}(1) = \overline{M_f(0)}. \quad (\text{ETC19})$$

Indeed substitute $v = 1/y$ in the absolutely convergent integral of $y^{-1} \overline{f(1/y)} y^{s-1}$; its integrand becomes $\overline{f(v)} v^{-s}$, which is the displayed conjugated Mellin transform. The source writes codegree both as $M_f(0)$ and as the degree of $f^\#$. Those values agree on its real test functions, including every individual f_b here; for complex coefficients the latter is the conjugate of the former. The formulas below specify which convention is being adjusted.

Define the complex-linear endpoint pair as $(d(f), c(f)) = (M_f(1), M_f(0))$. For two positive widths $b_1 \neq b_2$, both in the actual original family, ETC15 gives

$$\begin{pmatrix} d(c_1 f_{b_1} + c_2 f_{b_2}) \\ c(c_1 f_{b_1} + c_2 f_{b_2}) \end{pmatrix} = \frac{1}{4} \begin{pmatrix} b_1 & b_2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}, \quad \det \left(\frac{1}{4} \begin{pmatrix} b_1 & b_2 \\ 1 & 1 \end{pmatrix} \right) = \frac{b_1 - b_2}{16} \neq 0. \quad (\text{ETC20})$$

Thus prescribed changes $(D, C) \in \mathbb{C}^2$ are attained by the explicit inverse

$$c_1 = \frac{4(D - b_2 C)}{b_1 - b_2}, \quad c_2 = \frac{4(b_1 C - D)}{b_1 - b_2}. \quad (\text{ETC21})$$

Substitution gives $b_1 c_1 + b_2 c_2 = 4D$ and $c_1 + c_2 = 4C$, proving the inverse without a dimension argument. Both widths may use the same arbitrary divisor k by choosing $a_i = N(k)b_i$. The actual antecedent $c_1 \xi_{k,a_1} + c_2 \xi_{k,a_2}$ still has zero value and zero additive integral by ETC4, and its reduction is the displayed sum. For real desired changes the coefficients are real. If instead “codegree” literally means $d(f^\#)$ for complex f , replace C by $\overline{C}$ in ETC21; ETC19 then proves the corresponding real-linear inverse. This records the conjugation rather than erasing it.

Finally the original adelic rational action is

$$(\alpha_r \xi)(x) = \xi(r^{-1}x), \quad \alpha_r \xi_{k,a} = \xi_{k+d(r), |r|_\infty a}, \quad N(k+d(r)) = |r|_\infty N(k), \quad \mathcal{E} \alpha_r \xi_{k,a} = \mathcal{E} \xi_{k,a}. \quad (\text{ETC22})$$

The finite balls shift by their exact valuations; evenness of η removes the real sign and leaves the real factor $|r|_\infty$. The equality of N follows from unique factorization of a nonzero rational

number. The last equality follows either from the retained ratio in ETC6 or by reindexing the full rational sum $q \mapsto q/r$. Real scaling alone changes a to λa , and hence $f_b(y)$ to $f_{\lambda b}(y) = f_b(y/\lambda)$: its Mellin transform acquires exactly λ^s , its degree exactly λ , and its codegree remains unchanged. These are the actual two endpoint coordinates used in ETC20–21, with every prime and real factor retained. The calculation concerns this image family and its exact adjustment maps; it does not assert positivity of a Weil pairing.

There is a two-dimensional adjustment space stable under the literal sharp involution. Choose any $b > 0$ with $b \neq 1$ and put $c = 1/b$. In the basis (f_b, f_c) the conjugate-linear operation is exactly

$$(z_1 f_b + z_2 f_c)^\sharp = \frac{\bar{z}_2}{b} f_b + b \bar{z}_1 f_c, \quad \begin{pmatrix} 0 & 1/b \\ b & 0 \end{pmatrix}^2 = I. \quad (\text{ETC23})$$

Both coefficients come from ETC18, including its width factor. Define in this same space the original linear combinations

$$A_d = \frac{4(f_b - f_c)}{b - c}, \quad A_c = \frac{4(bf_c - cf_b)}{b - c}. \quad (\text{ETC24})$$

Direct substitution into ETC15 proves $(d(A_d), c(A_d)) = (1, 0)$ and $(d(A_c), c(A_c)) = (0, 1)$. The determinant in ETC20 proves that these moments uniquely determine an element of this two-dimensional space. ETC19 and its invariance under sharp therefore give $A_d^\sharp = A_c$ and $A_c^\sharp = A_d$, or equivalently

$$(DA_d + CA_c)^\sharp = \bar{C}A_d + \bar{D}A_c. \quad (\text{ETC25})$$

These are explicit combinations of the two original tests and their actual rational reductions. In particular they supply every prescribed endpoint pair while retaining the involution and the original zero-additive-moment antecedents.

For completeness, the two Gamma contributions in ETC11 have the following separate behavior at the two endpoint arguments. Write $h = s - 1$ and $\zeta(s) = h^{-1} + c_0 + O(h)$, where c_0 denotes its retained Laurent constant. Set $L_b = \log b - \frac{1}{2} \log \pi$. The first Gamma summand including its factors has expansion

$$\begin{aligned} b^s \pi^{-s/2} \Gamma(s/2 + 2) \zeta(s) &= \frac{3b}{4h} + \frac{3b}{4}(c_0 + L_b) + \frac{b}{2\sqrt{\pi}} \Gamma'(5/2) + O(h), \\ -\frac{3}{2} b^s \pi^{-s/2} \Gamma(s/2 + 1) \zeta(s) &= -\frac{3b}{4h} - \frac{3b}{4}(c_0 + L_b) - \frac{3b}{4\sqrt{\pi}} \Gamma'(3/2) + O(h). \end{aligned} \quad (\text{ETC26})$$

The Gamma recurrence differentiated at $3/2$ gives $\Gamma'(5/2) = \frac{3}{2} \Gamma'(3/2) + \Gamma(3/2)$. Adding the two displayed expansions therefore retains and cancels their opposite residues and their equal Laurent and width constants, leaving exactly $b/4$. At $s = 0$ the two summands have values $\zeta(0)$ and $-\frac{3}{2} \zeta(0)$. Their bracket is $-1/2$; ETC15 and ETC11 imply $\zeta(0) = -1/2$, so their separate values are $-1/2$ and $3/4$. Their sum is the independently calculated $1/4$. Thus neither endpoint calculation has discarded the contribution of one original Gaussian term.

64 The original adelic trace quotient with every support label retained

Human sources, objects, and reading scope. The analytic input is Alain Connes, Caterina Consani, and Matilde Marcolli, *The Weil proof and the geometry of the adeles class space*, original author file Yuri70v2.tex, SHA-256 b1f694cb934dd4fef829287f1a55dd8de21c67a481e552453c2f7e8fe0ae6fb5, canonical programme source PUBUNIT-5DE2A2D44F8B1470AC701FDA. The sections actually read for this derivation are source lines 610–710, 1316–1425, 1508–1551, 1680–1830, 1980–2160, 2161–2216, and 2313–2424, together with the title and author lines. The original locators are SCK, rangeTrpi, rangeCvdef, varthetaH1V, varthetafconvol, vanishV, tracepairing, fsharpinv, TraceformulaThm, and TraceformulaH1. The paragraph following moth1AKCK specifies closure of the range in the cokernel. The trace-class and explicit-formula theorems are cited analytic inputs. The convolution ideal, its sharp invariance with all endpoint terms, and the full-label quotient and receiving maps are derived below. No positivity assertion or RH conclusion is made.

64.1 The original test algebra and its domains

Retain the source’s unscaled test algebra and operations:

$$C = \mathbb{A}_{\mathbb{Q}}^{\times}/\mathbb{Q}^{\times}, \quad B = \bigcap_{\beta \in \mathbb{R}} |\cdot|^{\beta} \mathcal{S}(C), \quad (SWQ1)$$

$$(f * g)(x) = \int_C f(h)g(h^{-1}x) d^{\times}h, \quad f^{\sharp}(x) = |x|^{-1} \overline{f(x^{-1})}.$$

For $\mathbb{Q}$, the actual continuous multiplicative section in the author’s subsection XKCKsect is

$$C = \mathbb{R}_{>0} \times K, \quad K = \widehat{\mathbb{Z}}^{\times}, \quad (r, u) \mapsto (r, (u_p)_p), \quad |(r, u)| = r. \quad (SWQ2)$$

Indeed one multiplies an idele by the unique rational number that makes every finite component a unit and its real component positive. The ambiguity would have all finite valuations zero and positive real component, so would be the rational number 1. Multiplication respects these representatives. The multiplicative Haar measure is $dr/r du$, where $\int_K du = 1$. The additive adelic measure is real Lebesgue measure times the measures of volume 1 on each $\mathbb{Z}_p$. These exact measures are retained throughout.

The topology used for closure below is the usual weighted Bruhat test-function topology: at each fixed open subgroup $H \subset K$, functions are H -invariant and have all weighted Schwartz seminorms; one then uses the Bruhat inductive-limit topology over these finite levels. An explicit defining family at a fixed level, with $r = e^x$, is

$$p_{N,m,j}(f) = \sup_{x \in \mathbb{R}, u \in K} e^{N|x|} (1 + |x|)^m |\partial_x^j f(e^x, u)|, \quad N, m, j \in \mathbb{Z}_{\geq 0}. \quad (SWQ3)$$

These seminorms retain all real weights in SWQ1: weights $e^{\beta x}$ are bounded by $e^{N|x|}$ when $N \geq |\beta|$, and conversely the two weights e^{Nx} and e^{-Nx} control $e^{N|x|}$; Leibniz’s formula controls their derivatives. This is a description of the original test domain, not a change of function or scale in its operations.

All convolution integrals in SWQ1 converge absolutely. Using $x = y + (x - y)$ and $1 + |x| \leq (1 + |y|)(1 + |x - y|)$ gives the explicit bound

$$p_{N,m,j}(f * g) \leq 2p_{N+1,m,0}(f) p_{N,m,j}(g). \quad (SWQ4)$$

The constant 2 is $\int_{\mathbb{R}} e^{-|y|} dy$; the compact Haar factor is 1. Differentiation can pass under the integral by the same bound on every derivative. Convolution preserves a common finite

level, and this proves the continuity needed below. Since C is abelian, substitution $h \mapsto xh^{-1}$ proves commutativity. Fubini, justified by the corresponding three-function L^1 bound, proves associativity.

In these coordinates sharp is exactly

$$f^\sharp(e^x, u) = e^{-x} \overline{f(e^{-x}, u^{-1})}, \quad p_{N,m,j}(f^\sharp) \leq \sum_{\ell=0}^j \binom{j}{\ell} p_{N+1,m,\ell}(f). \quad (\text{SWQ5})$$

The bound follows by differentiating e^{-x} and $f(e^{-x}, u^{-1})$; the two derivative signs together have absolute value 1. Direct substitution gives $(f^\sharp)^\sharp = f$. Inversion preserves Haar measure on this abelian group, and the norm is a character. Substitution in the absolutely convergent convolution therefore gives $(f * g)^\sharp = g^\sharp * f^\sharp$. In particular sharp is continuous on the stated test space. The algebra B has no convolution unit among its functions. Indeed let $b(e^x, u) = e^{-x^2}$, which belongs to B and has trivial compact-unit character. Its real Fourier transform is $\sqrt{\pi}e^{-t^2/4}$, nonzero for every real t . A convolution unit $a \in B$ would consequently have $\int_{\mathbb{R} \times K} a(e^x, u) e^{itx} dx du = 1$ for every t . Integration by parts in x , justified by SWQ3, makes this integral tend to zero as $|t| \rightarrow \infty$, a contradiction. No unit distribution is inserted in this chapter.

64.2 The adelic source, with both Poisson endpoints

Use the canonical additive character trivial on $\mathbb{Q}$, with real component $e^{-2\pi ixy}$ and finite conductor $\mathbb{Z}_p$, and its self-dual measures specified after SWQ2. Write

$$\begin{aligned} \widehat{\xi}(y) &= \int_{\mathbb{A}} \xi(x) \alpha(xy) dx, & \varepsilon_0(\xi) &= \xi(0), & \varepsilon_1(\xi) &= \int_{\mathbb{A}} \xi(x) dx, \\ \mathcal{S}(\mathbb{A})_0 &= \ker \varepsilon_0 \cap \ker \varepsilon_1, & (E\xi)(g) &= \sum_{q \in \mathbb{Q}^\times} \xi(qg), & V &= E\mathcal{S}(\mathbb{A})_0. \end{aligned} \quad (\text{SWQ6})$$

For the reference characters e_v in SWQ28, choose exactly these local components of α . This fixes more than their conductors: it specifies the characters themselves, so that the differential idele there is $a_v = 1$ at every place. The sum is invariant under the rational choice of representative. For fixed g , finite-place compact support puts its contributing rational q in $D^{-1}\mathbb{Z}$ for one positive integer D . Real Schwartz decay then makes the sum absolutely convergent, locally uniformly with every real derivative. Unit dilation at finite places preserves the required common denominator and bounds.

For completeness the full Poisson formula used here is

$$\sum_{q \in \mathbb{Q}} \xi(qg) = |g|^{-1} \sum_{q \in \mathbb{Q}} \widehat{\xi}(qg^{-1}). \quad (\text{SWQ7})$$

It follows directly for the original Bruhat space. A finite-place test function is a finite linear combination of characteristic functions of cosets $a + N\widehat{\mathbb{Z}}$, with rational a and positive integer N , after a common denominator is chosen. The rational elements in such a coset are $a + N\mathbb{Z}$. Real periodization of a Schwartz function gives

$$\sum_{n \in \mathbb{Z}} \phi(a + Nn) = \frac{1}{N} \sum_{m \in \mathbb{Z}} e^{2\pi i m a / N} \widehat{\phi}(m/N).$$

To verify this last identity, periodize over $N\mathbb{Z}$, integrate over one period to compute every Fourier coefficient, and evaluate the resulting absolutely convergent Fourier series at a . The finite Fourier transform of the characteristic coset is $N^{-1} \alpha_f(ay) 1_{N^{-1}\widehat{\mathbb{Z}}}(y)$. At rational $y = m/N$, its character is $e^{2\pi i a m / N}$ because the global character is trivial on rational numbers. These exact factors prove adelic Poisson by linearity. Finally replacing the test function by $x \mapsto \xi(gx)$ multiplies its Fourier

transform by $|g|^{-1}$ and evaluates it at $g^{-1}y$, proving SWQ7. The same real Fourier inversion and finite character orthogonality give $\widehat{\widehat{\xi}}(x) = \xi(-x)$.

Separating the actual $q = 0$ terms in SWQ7, with neither endpoint discarded, gives

$$E\xi(g) = |g|^{-1}E\widehat{\xi}(g^{-1}) + |g|^{-1}\varepsilon_1(\xi) - \varepsilon_0(\xi). \quad (\text{SWQ8})$$

For $|g| \geq 1$, the earlier common-denominator estimate gives $E\xi(g) = O_M(|g|^{-M})$ for every M , uniformly in the compact unit coordinate and with every logarithmic real derivative. For $\xi \in \mathcal{S}(\mathbb{A})_0$, both displayed endpoints vanish by definition, and the right side applies the same large-norm estimate to $\widehat{\xi}$ at g^{-1} . Indeed $\widehat{\xi}(0) = \varepsilon_1(\xi) = 0$ and $\int \widehat{\xi} = \xi(0) = 0$ by Fourier inversion. The factor $|g|^{-1}$ stays present. Since M is arbitrary, it proves all the small-norm bounds in SWQ3 as well. Thus $V \subset B$ on the source's actual domain.

Apply SWQ8 to $\widehat{\widehat{\xi}}$, or equivalently apply SWQ7 to $\bar{\xi}$ and invert g . The exact result is

$$(E\xi)^\sharp(g) = E\widehat{\bar{\xi}}(g) - |g|^{-1}\overline{\varepsilon_0(\xi)} + \overline{\varepsilon_1(\xi)}. \quad (\text{SWQ9})$$

This identity retains the conjugations, the factor $|g|^{-1}$, and both endpoints. Put $J\xi = \widehat{\bar{\xi}}$. Its endpoints are

$$\varepsilon_0(J\xi) = \overline{\varepsilon_1(\xi)}, \quad \varepsilon_1(J\xi) = \overline{\varepsilon_0(\xi)}. \quad (\text{SWQ10})$$

Consequently J preserves the actual space $\mathcal{S}(\mathbb{A})_0$ and SWQ9 gives $V^\sharp \subset V$; applying sharp twice gives equality. This derives sharp invariance from additive Fourier–Poisson, rather than assuming that the original quotient was an involutive quotient.

64.3 The actual convolution ideal and its closure

For $f \in B$, $\xi \in \mathcal{S}(\mathbb{A})_0$, use the actual multiplicative section SWQ2 to define

$$\eta(x) = \int_C f(h)\xi(h^{-1}x) d^\times h. \quad (\text{SWQ11})$$

This integral belongs to the original Bruhat–Schwartz space. Indeed the orbit of the finite-place component of ξ under the compact group K lies in a finite-dimensional space of test functions: its compact support stays in one finite product of bounded balls, and its common local-constancy subgroups are unchanged by units. The finite quotient of this compact support by these subgroups has finitely many points. For the real component, every Schwartz seminorm has the explicit bound

$$\sup_{x_\infty, x_f} |x_\infty^N \partial_{x_\infty}^j \eta(x)| \leq C_{\xi, N, j} \int_C |f(h)| |h|^{N-j} d^\times h < \infty. \quad (\text{SWQ12})$$

The last integral is finite by SWQ3. Differentiation and integration are justified by the same bound, while the fixed finite-place test space makes the resulting function a Bruhat test function.

The two endpoints of this integral are exactly

$$\eta(0) = \varepsilon_0(\xi) \int_C f(h) d^\times h, \quad \int_{\mathbb{A}} \eta(x) dx = \varepsilon_1(\xi) \int_C |h| f(h) d^\times h. \quad (\text{SWQ13})$$

For the second equation the absolute double integral is bounded by $\|\xi\|_{L^1(\mathbb{A})} \int_C |h| |f(h)| d^\times h$, which is finite. The change of variables $x = hy$ therefore gives its exact adelic factor $|h|$, and Fubini is valid for this equation. Thus $\eta \in \mathcal{S}(\mathbb{A})_0$.

There is also a justified interchange with the arithmetic sum in the convolution calculation. For fixed g , real Schwartz decay and the fixed rational denominator give, uniformly in the unit part of h , the estimate

$$\sum_{q \neq 0} |\xi(qh^{-1}g)| \leq C_{\xi, g}(1 + |h|).$$

For example, dominate the real values on the grid $D^{-1}\mathbb{Z}$ by $C(1 + |n||g|/(D|h|))^{-2}$, and bound the resulting series by its first term plus its integral. Multiplying by $|f(h)|$ is integrable by SWQ3. Hence

$$f * E\xi = E\eta \in V. \quad (\text{SWQ14})$$

Commutativity gives the ideal property in both orders. This proves the actual ideal assertion needed for the quotient, including the source-domain check in Lemma `varthetaH1V`.

Let

$$W = \overline{V} \subset B, \quad Q = B/W, \quad (\text{SWQ15})$$

where closure is in the topology stated before SWQ3. Continuity of convolution by each fixed test function, established in SWQ4, makes W a convolution ideal. Continuity and involutivity of sharp in SWQ5, together with SWQ9–10, give $W^\sharp = W$. Thus $[f] * [g] = [f * g]$ and $[f]^\sharp = [f^\sharp]$ are well-defined on Q . Notice that SWQ13 is not an exchange of $\sum_{q \neq 0}$ with the degree integral of $E\xi$. It does not assert $\int_C |g| E\xi(g) d^\times g = 0$. The author's `fubini1-nothappen` calculation concerns precisely that different, generally invalid exchange.

64.4 The full-label quotient and the different Bourne quotient

Let L be any bounded distributive lattice. All constructions below hold without finiteness; to require $e \neq \tau$ take L nontrivial. Define the original constrained carrier and operations

$$\begin{aligned} G_L(B) &= \{(0, \lambda) : \lambda \in L\} \cup (B \times \{1_L\}), \\ (f, \lambda) + (g, \mu) &= (f + g, \lambda \vee \mu), \quad (f, \lambda) * (g, \mu) = (f * g, \lambda \wedge \mu), \\ (f, \lambda)^\sharp &= (f^\sharp, \lambda), \quad \tau = (0, 0_L), \quad e = (0, 1_L). \end{aligned} \quad (\text{SWQ16})$$

This is a commutative nonunital convolution semiring with sharp. Carrier closure follows because any nonzero output amplitude requires all relevant nonzero input amplitudes to have top label. For the product this uses that convolution with a zero function is zero; it does not assume that B has no zero divisors. Associativity and distributivity combine those of convolution with the lattice laws. Sharp is an involution preserving labels and reversing convolution. The complex scalar action has the separate semimodule formula $(c, \alpha)(f, \lambda) = (cf, \alpha \wedge \lambda)$ for $(c, \alpha) \in G_L(\mathbb{C})$.

The exact same-label quotient is

$$(f, \lambda) \sim_W (g, \mu) \iff \lambda = \mu \text{ and } f - g \in W, \quad G_L(B)/\sim_W \simeq G_L(Q). \quad (\text{SWQ17})$$

To prove this, use the map $\pi_L(f, \lambda) = ([f], \lambda)$. It respects addition, convolution, sharp, and the scalar action by the ideal and involution properties just proved. It is onto: a nonzero quotient amplitude has top label and lifts to a representative with top label; a zero amplitude with any label lifts to $(0, \lambda)$. Its fibres are exactly the displayed relation. Alternatively, adding a third element preserves equality of labels and preserves the amplitude difference modulo W ; multiplication sends that difference to an element of W ; sharp sends it into W as well. These direct checks prove the required congruence and the quotient assertion.

In particular every $(0, \lambda)$ survives, and e and τ remain distinct for nontrivial L . A nonzero $w \in W$ with top label maps to e , which differs from τ when L is nontrivial. For nontrivial L , the inverse image of the absorbing additive zero τ under π_L is only $\{\tau\}$, even though π_L can identify different top-labelled amplitudes. Thus the preimage of a single zero does not specify this semiring congruence. For the one-element lattice instead $G_L(B) = B$, and that inverse image is the ordinary amplitude kernel W .

The ordinary Bourne congruence by the ideal $I = G_L(W)$ is different:

$$x \equiv_I y \iff \exists a, b \in I : x + a = y + b, \quad (f, \lambda) \equiv_I (g, \mu) \iff f - g \in W. \quad (\text{SWQ18})$$

The forward implication follows from the amplitude equation. Conversely, if $f - g \in W$, choose $a = (g - f, 1_L)$ and $b = (0, 1_L)$. Both belong to I ; the two amplitudes become g and both labels become 1_L , proving equality. Consequently

$$G_L(B)/\equiv_I \simeq Q, \quad G_L(B)/\sim_W \longrightarrow Q, \quad ([f], \lambda) \longmapsto [f]. \quad (\text{SWQ19})$$

The Bourne quotient collapses the entire labelled-zero fibre, so it identifies e and τ . More exactly it is the same-label quotient followed by the congruence generated by $e = \tau$: multiplication of this equality by $(0, \lambda)$ makes $(0, \lambda) = \tau$ for every λ . After these identifications the carrier and operations are exactly those of Q . No Boolean character or selected support point has been used.

64.5 Every numerical additive trace and the full-labelled receiver

Let A be any abelian group. Every additive map $t : G_L(B) \rightarrow A$ kills every labelled zero: its idempotence gives $t(0, \lambda) + t(0, \lambda) = t(0, \lambda)$, hence $t(0, \lambda) = 0$. Define $\ell(f) = t(f, 1_L)$. Then ℓ is additive on B , since addition of two top-labelled elements stays top-labelled and $\ell(0) = t(e) = 0$. Every nonzero amplitude already has top label, so

$$t(f, \lambda) = \ell(f), \quad \text{Hom}_+(G_L(B), A) \simeq \text{Hom}_+(B, A). \quad (\text{SWQ20})$$

Conversely composition of any additive ℓ with amplitude projection is additive, proving both directions. This includes all numerical additive traces into $\mathbb{C}$, not only continuous or complex-linear ones. Cyclicity adds no restriction here because convolution is commutative. Such a map descends through SWQ17 exactly when $\ell(W) = 0$: necessity follows by comparing $(w, 1_L)$ with e , and sufficiency follows from the displayed difference condition. The same classification on $G_L(Q)$ is $\text{Hom}_+(G_L(Q), A) = \text{Hom}_+(Q, A)$. Thus numerical additivity itself removes zero-label distinctions; it is not a proof that those distinctions were absent from the source object.

Retain the original numerical cohomological trace

$$T(f) = \text{Tr}(\vartheta_m(f) \mid H^1). \quad (\text{SWQ21})$$

Here $\vartheta_m(f)$ is the original convolution action on the source's quotient realization, with its prescribed closure. Its trace-class meaning is the cited Theorem [varthetaTrH1](#), with the trace formula [TraceformulaThm](#); this chapter does not replace that analytic theorem by a finite-dimensional trace assumption. Its vanishing on W can be checked on the actual quotient action: for $w \in W$ and $b \in B$, $w * b \in W$, hence $\vartheta_m(w)[b] = 0$. The operator is zero, so

$$T(W) = 0, \quad \bar{T} : Q \rightarrow \mathbb{C}, \quad \bar{T}([f]) = T(f). \quad (\text{SWQ22})$$

This does not require an unproved exchange of an infinite spectral sum with a limit of representatives: the operator itself vanishes.

The original trace and Weil-pairing receivers retaining the labels are

$$\begin{aligned} T_L : G_L(Q) &\longrightarrow G_L(\mathbb{C}), & T_L([f], \lambda) &= (T(f), \lambda), \\ \mathcal{W}_L(([f], \lambda), ([g], \mu)) &= (T(f * g^\sharp), \lambda \wedge \mu). \end{aligned} \quad (\text{SWQ23})$$

The first map is well-defined by SWQ22. It respects addition and the declared $G_L(\mathbb{C})$ scalar action; it is a trace receiver, not a claimed multiplicative character. For the second, replace f by $f + w$ and g by $g + v$, with $w, v \in W$. Its change of amplitude is

$$T(w * g^\sharp + f * v^\sharp + w * v^\sharp) = 0. \quad (\text{SWQ24})$$

Every summand belongs to W by SWQ15. This proves representative independence while retaining each term of the actual change. If one input quotient amplitude is zero, the output

amplitude is zero. Thus every nonzero output amplitude has top label and the receiver belongs to the declared constrained carrier.

The receiver is additive in both variables and complex-linear in the first, conjugate-linear in the second, with the exact lattice factors. For example, if $a = (c, \alpha)$, then

$$\mathcal{W}_L(ax, y) = a\mathcal{W}_L(x, y), \quad \mathcal{W}_L(x, ay) = \bar{a}\mathcal{W}_L(x, y), \quad \bar{a} = (\bar{c}, \alpha). \quad (\text{SWQ25})$$

The amplitudes follow from convolution and sharp. For additivity, the label equality is $(\lambda \vee \lambda') \wedge \mu = (\lambda \wedge \mu) \vee (\lambda' \wedge \mu)$, and likewise in the second variable. These prove the assertions directly. In particular $\mathcal{W}_L((0, \lambda), ([g], \mu)) = (0, \lambda \wedge \mu)$. Even a zero numerical amplitude therefore retains its exact support in SWQ23. Projecting to $\mathbb{C}$ forgets that support by SWQ20.

64.6 The complete original trace terms of this receiver

For the trivial unit character write $Mf(s) = \int_C f(u)|u|^s d^\times u$. Every such integral converges absolutely on the full complex plane by SWQ3, locally uniformly with every derivative. Haar inversion gives

$$M(f^\sharp)(s) = \overline{Mf(1 - \bar{s})}, \quad M(f * g^\sharp)(s) = Mf(s)\overline{Mg(1 - \bar{s})}. \quad (\text{SWQ26})$$

For the second identity the weighted double integral is finite, so multiplicative Fubini is valid. At the identity one has instead the exact weighted inner-product expression

$$(f * g^\sharp)(1) = \int_C f(u)\overline{g(u)}|u| d^\times u. \quad (\text{SWQ27})$$

In particular $M(f^\sharp)(1) = \overline{Mf(0)}$. The conjugation is necessary for complex tests; it is missing in the author's displayed `codegcorrZf` equality if that equality is read for unrestricted complex-valued f . No real-valued restriction is imposed here.

Put $h = f * g^\sharp$. The source's full trace formula therefore gives the following amplitude for SWQ23:

$$\begin{aligned} T(h) &= Mf(0)\overline{Mg(1)} + Mf(1)\overline{Mg(0)} \\ &\quad - (\log |a|) \int_C f(u)\overline{g(u)}|u| d^\times u \\ &\quad - \sum_{v \leq \infty} \int'_{(\mathbb{Q}_v^\times, e_v)} \frac{h(u^{-1})}{|1 - u|_v} d_v^\times u. \end{aligned} \quad (\text{SWQ28})$$

Here u denotes the idele class of its original local embedding, a is the differential idele relating the global additive character to the specified local characters, and $d_v^\times u$ and the primed distributions are exactly those of `TraceformulaH1` and `princvalrhobeta`. In the canonical rational character used in SWQ6 one has $a_v = 1$ at every place, so the displayed differential term evaluates with $\log |a| = \log 1 = 0$; it has not been absorbed into another term. The prime sum includes infinity and every finite prime.

The prime notation is essential. In the author's definition a distribution ϱ_{e_v} extends multiplicative measure across zero with Fourier transform satisfying $\widehat{\varrho}_{e_v}(1) = 0$, and the local term is

$$\left\langle \varrho_{e_v}, \lambda \mapsto \frac{h((\lambda + 1)^{-1})}{|\lambda + 1|_v} \right\rangle.$$

For compactly supported tests this is the stated original definition; for the full space B , SWQ28 uses the extension on that test space in the cited trace theorem. It is not an assertion that the ordinary integral through $u = 1$ converges. No replacement of f by $|\cdot|^{1/2}f$, and no completed zeta function, is introduced into SWQ23–28.

Exact scope and analytic limits. The closure in SWQ15 is the declared weighted Bruhat closure, not an unspecified Hilbert, distributional, or pointwise closure. The quotient, ideal, and sharp statements above are proved in this domain. The scalar trace-class realization and its regularized explicit formula retain the analytic hypotheses of the original source theorems. SWQ23 does not supply a new proof of those input theorems or a positivity result. It supplies a fully specified same-label receiving object and proves how its operations and every trace term descend. Its numerical values are exactly the original ones: every such value is attained on top-labelled inputs, and every labelled input has the displayed numerical amplitude. The support data remain in $G_L(Q)$ and in $G_L(\mathbb{C})$; an additive numerical observation necessarily forgets them by SWQ20. The Bourne quotient performs that forgetting already in its source, whereas SWQ17 retains $e \neq \tau$ for every nontrivial L .

65 Actual trivial correspondences, spectral jets, and supported nilpotence

Source and scope. Alain Connes, Caterina Consani and Matilde Marcolli, *The Weil proof and the geometry of the adèles class space*, original author file `Yuri70v2.tex`, define the restriction image in `rangeCVdef`, the induced convolution action in `varthetaconvol`, and prove its trace vanishing in `vanishV`. Their `degmodifyV` uses the particular Schwartz function below, stating its transform only up to a constant factor. The factor is retained here. The interpolation and support calculations are the following programme derivation, not a claim that these proofs occur verbatim in that source.

Let $k = (k_p)_p$ have finite support in $\bigoplus_p \mathbb{Z}$, let $a > 0$, and retain

$$N(k) = \prod_p p^{k_p}, \quad \eta(x) = \pi x^2 (\pi x^2 - \tfrac{3}{2}) e^{-\pi x^2}, \quad \xi_{k,a}(x) = \eta(x_\infty/a) \prod_p \mathbf{1}_{p^{k_p} \mathbb{Z}_p}(x_p). \quad (\text{WJI1})$$

The additive Haar measures have $\text{vol}(\mathbb{Z}_p) = 1$; on $C_{\mathbb{Q}} = \widehat{\mathbb{Z}}^* \times \mathbb{R}_+^*$ retain Haar mass one on $\widehat{\mathbb{Z}}^*$ and dy/y . Here $\xi_{k,a}(0) = 0$ and its additive integral is zero: the two real Gaussian moments are $\pi^2 \cdot 3/(4\pi^2)$ and $(3\pi/2) \cdot 1/(2\pi)$, and their difference is zero. The finite Haar factor is $N(k)^{-1}$ and the real change of variable contributes a ; neither factor has been discarded.

For $u \in \widehat{\mathbb{Z}}^*$ put

$$f_b(u, y) = E\xi_{k,a}(u, y) = \sum_{q \in \mathbb{Q}^*} \xi_{k,a}(q(u, y)) = 2 \sum_{n \geq 1} \eta(ny/b), \quad b = a/N(k). \quad (\text{WJI2})$$

Indeed the finite factors require $q \in N(k)\mathbb{Z}$, and η is even. This proves the factor 2 and the exact relation of b to every original prime valuation and real width. The function is in the restriction image $\mathcal{V}$ because $\xi_{k,a}$ satisfies both original vanishing conditions.

For completeness, the real Fourier transform of η is η . One differentiates the Gaussian identity $\mathcal{F}(e^{-\pi x^2}) = e^{-\pi v^2}$ using $\mathcal{F}(xf) = -(2\pi i)^{-1} \partial_v \mathcal{F}f$. The transforms of $x^2 e^{-\pi x^2}$ and $x^4 e^{-\pi x^2}$ are respectively $(1/(2\pi) - v^2) e^{-\pi v^2}$ and $(v^4 - 3v^2/\pi + 3/(4\pi^2)) e^{-\pi v^2}$. Substitution of both terms of η proves the assertion. Poisson summation therefore gives

$$f_b(y) = \frac{b}{y} f_{1/b}(1/y). \quad (\text{WJI3})$$

There is no omitted zero summand: $\eta(0) = \widehat{\eta}(0) = 0$. Both f_b and every logarithmic derivative decay faster than every power at infinity, by the Gaussian series. WJI3 proves the same at zero. Thus f_b lies in the exact radial strong Schwartz algebra used in the source, with no compact-support replacement.

The full Mellin calculation, first for $\Re s > 1$, is

$$\mathcal{M}f_b(s) := \int_0^\infty f_b(y) y^s \frac{dy}{y} = b^s C(s), \quad C(s) = \pi^{-s/2} \zeta(s) \left[\Gamma(s/2 + 2) - \frac{3}{2} \Gamma(s/2 + 1) \right]. \quad (\text{WJI4})$$

Absolute convergence for the separate Gaussian terms follows from $\sum n^{-\Re s} < \infty$. For each term substitute $v = \pi(ny/b)^2$; its factor $1/2$ cancels the displayed factor 2 in WJI2. This yields exactly the two Gamma summands. WJI3 gives an entire Mellin transform: for any compact set of s , its derivatives introduce powers of $\log y$ dominated at both ends by the proved estimates.

The Gamma recurrence proves the comparison, without replacing the original formula WJI4:

$$C(s) = d(s) \zeta(s), \quad d(s) = \frac{s(s-1)}{4} \pi^{-s/2} \Gamma(s/2). \quad (\text{WJI5})$$

At zero use $\zeta(0) = -1/2$ and the bracket $\Gamma(2) - (3/2)\Gamma(1) = -1/2$ in WJI4. At one the bracket is $(s-1)\Gamma(s/2+1)/2$ and ζ has residue one. Consequently

$$C(0) = \frac{1}{4}, \quad C(1) = \frac{1}{4}, \quad \mathcal{M}f_b(0) = \frac{1}{4}, \quad \mathcal{M}f_b(1) = \frac{b}{4}. \quad (\text{WJI6})$$

The original zeta continuation with both endpoint terms is proved in ZTC9–10; the values and residue used here follow from that formula. No zero or pole is assigned a value by multiplying undefined separate point values.

65.1 The exact finite-jet image

Fix distinct complex points $s_1, \dots, s_\ell$, positive integers $N_1, \dots, N_\ell$, and use the original local coordinates $h_i = s - s_i$. Let

$$J = \bigoplus_{i=1}^{\ell} \mathbb{C}[h_i]/(h_i^{N_i}), \quad m_i = \text{ord}_{s_i} C, \quad a_i = \min(m_i, N_i), \quad r_i = N_i - a_i, \quad M = \sum_i r_i. \quad (\text{WJI7})$$

The nonnegative orders m_i are defined by the entire function WJI4; they are finite since $C(0) = 1/4$. Define the actual Mellin-jet map on finite linear combinations of the functions WJI2 by

$$\mathcal{J} \left(\sum_j c_j f_{b_j} \right) = \left[\sum_j c_j b_j^s C(s) \right]_{h_i^{N_i}, i}.$$

Every such jet lies in

$$I_J = \bigoplus_i h_i^{a_i} \mathbb{C}[h_i]/(h_i^{N_i}). \quad (\text{WJI8})$$

We now prove that every element of WJI8 occurs and give its inverse construction using M specified positive widths.

Discard indices with $r_i = 0$ only from the interpolation matrix; their zero image and quotient remain in WJI7–8. If $M = 0$, the image is zero and the empty linear combination suffices. Otherwise choose, on the retained indices,

$$\delta = \frac{\pi}{1 + \max_{i < j} |\Im(s_i - s_j)|}, \quad z_i = e^{\delta s_i}, \quad b_j = e^{j\delta} \quad (0 \leq j < M), \quad (\text{WJI9})$$

with maximum zero when there is only one index. The z_i are distinct. Equality of two exponentials would require $\delta(s_i - s_j) = 2\pi in$. The chosen bound makes the imaginary part smaller than π in absolute value, forcing $n = 0$ and then $s_i = s_j$, a contradiction.

The derivative matrix for the functions $e^{j\delta s}$ is

$$V_{(i,q),j} = (j\delta)^q z_i^j, \quad 0 \leq q < r_i, \quad \det V = \left[\prod_i (\delta z_i)^{r_i(r_i-1)/2} \prod_{q=0}^{r_i-1} q! \right] \prod_{i < j} (z_j - z_i)^{r_i r_j}. \quad (\text{WJI10})$$

Every factorial in this formula refers to ordinary derivatives, not Taylor coefficients. To verify it, first use polynomial derivative evaluations $P^{(q)}(z_i)$. Their confluent Vandermonde determinant equals the ordinary Vandermonde limit: replace each cluster z_i by $z_i + \epsilon w_{i,j}$, divide by its within-cluster Vandermonde, and expand each row in its Taylor series. The surviving rows are $P^{(q)}(z_i)/q!$; between-cluster factors tend to $(z_j - z_i)^{r_i r_j}$. Multiplying the rows by $q!$ gives the displayed factorial product. Finally $\partial_s = \delta z \partial_z$ has leading coefficient $(\delta z_i)^q$ in its q th power and only lower-order derivatives besides. This triangular row change proves WJI10. All its factors are nonzero.

For an explicit inverse, let a desired jet in WJI8 have at site i the representative $g_i(h_i)$ divisible by $h_i^{m_i}$. Write $C(s_i + h) = h^{m_i} u_i(h)$, $u_i(0) \neq 0$. Division gives the unique required jet

$$w_i(h) = \frac{g_i(h)}{h^{m_i} u_i(h)} \pmod{h^{r_i}}. \quad (\text{WJI11})$$

The ambiguity in a representative of g_i is divisible by h^{N_i} , so the ambiguity after division is divisible by h^{r_i} . Put

$$Q_i(z) = \prod_{\nu \neq i} (z - z_\nu)^{r_\nu}, \quad \tilde{w}_i(z) = w_i(\delta^{-1} \log(z/z_i)) \pmod{(z - z_i)^{r_i}},$$

where the local logarithm has value zero at z_i . The polynomial

$$P(z) = \sum_i Q_i(z) \operatorname{Tay}_{z_i, < r_i} \left(\frac{\tilde{w}_i(z)}{Q_i(z)} \right) = \sum_{j=0}^{M-1} c_j z^j \quad (\text{WJI12})$$

has degree below M . Its i th summand has the prescribed jet at z_i ; every other summand is divisible by $(z - z_i)^{r_i}$. Hence $C(s)P(e^{\delta s})$ has the desired original s -jets. The actual correspondence $\sum_{j=0}^{M-1} c_j f_{b_j}$ realizes them. This proves

$$\operatorname{im} \mathcal{J} = I_J, \quad \operatorname{rank} \mathcal{J} = \sum_i (N_i - a_i), \quad J/I_J = \bigoplus_i \mathbb{C}[h_i]/(h_i^{a_i}). \quad (\text{WJI13})$$

The equality on the right is the explicit residue map; its kernel is exactly WJI8. A summand with $a_i = 0$ is the zero vector space, not a omitted original point or support label.

65.2 Endpoints, original divisors and the full nilpotent

At $s_i = 0$ or 1 , WJI6 gives $a_i = 0$ for every N_i . Thus all finite jets at both endpoints can be changed independently by actual elements of the restriction image. For only the two values and two widths $b_1 \neq b_2$ the explicit equations are

$$\frac{c_1 + c_2}{4} = A_0, \quad \frac{b_1 c_1 + b_2 c_2}{4} = A_1, \quad c_1 = \frac{4(A_1 - b_2 A_0)}{b_1 - b_2}, \quad c_2 = \frac{4(b_1 A_0 - A_1)}{b_1 - b_2}. \quad (\text{WJI14})$$

Adding this correspondence changes the two Mellin values by A_0, A_1 and leaves its class modulo $\mathcal{V}$ unchanged. It leaves every trace pairing against that class unchanged, by the source's `vanishV`; the full-labelled meaning of that assertion is SWQ, not an identification of e and τ .

For a nontrivial original zeta zero ρ in $0 < \Re \rho < 1$, the factor d in WJI5 is a holomorphic unit. Hence $m_\rho = \operatorname{ord}_\rho \zeta$ and the quotient in WJI13 retains the complete truncated original primary block. At $s = -2n$, however, d has a simple pole and C is a unit. The corresponding WJI13 quotient is zero, but the original simple zero of ζ remains in $d^{-1}\mathcal{O} = h\mathcal{O} \subset \mathcal{O}$ and the length-one quotient $\mathcal{O}/h\mathcal{O}$. At $s = 1$, $d^{-1}\mathcal{O} = h^{-1}\mathcal{O}$ and $h^{-1}\mathcal{O}/\mathcal{O}$ retains the original pole with the opposite divisor sign. The embedded comparison and every factor of WJI5 are part of this calculation; WJI13 is not asserted to be the complete original-zeta divisor by itself. The germ proofs, including the module-fibre isomorphism and all exceptional-point signs, are ZTC17–17a.

The original scaling action $S_r f(y) = f(r^{-1}y)$, $r > 0$, sends f_b to f_{rb} and gives $\mathcal{M}(S_r f)(s) = r^s \mathcal{M}f(s)$ by substitution. It preserves WJI8. On a nonzero local quotient in WJI13 write N_i for multiplication by h_i ; the symbol N_i remains the integer cutoff of WJI7. The exact action is

$$N_i[v] = [h_i v], \quad N_i^{a_i} = 0, \quad S_r = r^{s_i} \sum_{q=0}^{a_i-1} \frac{(\log r)^q}{q!} N_i^q. \quad (\text{WJI15})$$

This is the Taylor expansion of $r^{s_i + h_i}$ in the specified quotient, with every factorial and eigenvalue retained. The vector [1] shows that its nilpotent length is exactly a_i : for $a_i \geq 1$, $[h_i^{a_i-1}] \neq 0$. No semisimplification of this primary block has been taken.

Let L be the original arbitrary nontrivial bounded distributive lattice. For any vector space V use exactly

$$G_L(V) = \{(0, \lambda) : \lambda \in L\} \cup (V \times \{1_L\}), \quad (v, \lambda) + (w, \mu) = (v + w, \lambda \vee \mu). \quad (\text{WJI16})$$

The $G_L(\mathbb{C})$ action multiplies amplitudes and meets labels. Every linear map T has the lift $T^\uparrow(v, \lambda) = (Tv, \lambda)$, which preserves addition and that action. The jet quotient has exact same-label fibres

$$(v, \lambda) \sim (w, \mu) \iff \lambda = \mu \text{ and } v - w \in I_J. \quad (\text{WJI17})$$

Addition and scalar action preserve this relation because I_J is a vector subspace. The map to $G_L(J/I_J)$ is surjective: use any lift of a nonzero amplitude with label 1_L , and use $(0, \lambda)$ for each zero-labelled output. Its fibres are precisely WJI17, proving the quotient identification with all original labels retained.

For each nonzero block its lifted nilpotent satisfies

$$(\mathbf{N}_i^\uparrow)^{a_i}(v, \lambda) = (0, \lambda) = e \cdot (v, \lambda), \quad e = (0, 1_L), \quad \tau = (0, 0_L). \quad (\text{WJI18})$$

This map is not the constant- τ map. On the zero vector block ($a_i = 0$), $G_L(0) = L$ and the scaling action is the identity on that lattice; an empty polynomial sum must not be read as the constant- τ map.

These are common-label modules for the entire indicated vector spaces. They do not introduce independent labels on each jet coordinate. In the actual base localization of ULC, at a support prime $J_L \subset L$, the element e is invertible, so every amplitude is killed and the coefficient stalk is L_{J_L} . Explicitly $(v, \lambda)/e = (0, \lambda)/e$, because multiplication by e makes their numerators equal. Each lifted linear map then acts as the identity on L_{J_L} , and WJI18 becomes the identity there. On the arithmetic coefficient modules, WJI15 and WJI18 retain the primary block and its supported-zero power separately. This is the exact receiving map; it asserts no new positivity of the original Weil form.

66 The exact Hurwitz boundary of the Connes–Consani–Marcolli test space

Sources and scope. Alain Connes, Caterina Consani and Matilde Marcolli, *The Weil proof and the geometry of the adeles class space*, original author source `Yuri70v2.tex`, equation `SCK`, Proposition `Trrhocyclmap`, equations `TrMbf0` and `rangeCVdef`, and Lemma `degmodifyV`, provide the actual strong test space, restriction map and Gaussian-polynomial generator used here. The source’s wording “up to normalization” in the proof of `degmodifyV` is not used as a formula: every factor of its original generator is retained and calculated below. The original Hurwitz family is the programme’s $F_t(s) = \zeta(s, 1+t)$, as recorded in HTB1–4 of `HURWITZ_THETA_FLOW_BRIDGE.tex`; a direct continuation proof is included here. Only the cited parts of the CCM source were read for this independent derivation. The results concern the actual test space and its explicit boundary, without an RH claim.

66.1 Objects, original restriction map, and all constants

Write $E = y\partial_y$, $a = 1+t > 0$, and $b > 0$. Define

$$\begin{aligned}\eta(x) &= \pi x^2(\pi x^2 - \tfrac{3}{2})e^{-\pi x^2}, \\ g(x) &= 2\eta(x) = (2\pi^2 x^4 - 3\pi x^2)e^{-\pi x^2}, \\ \Psi_{t,b}(y) &= \sum_{n=0}^{\infty} g((n+1+t)y/b), \quad y > 0, \\ \mathcal{B} &= \bigcap_{\beta \in \mathbb{R}} y^\beta \mathcal{S}(\mathbb{R}_+^*).\end{aligned}\tag{HCB1}$$

Here $\mathcal{S}(\mathbb{R}_+^*)$ is Schwartz space in the coordinate $\log y$, and $\mathcal{B}$ is the radial part of CCM’s $\mathbf{S}(C_{\mathbb{Q}})$. Equivalently, $u \in \mathcal{B}$ exactly when u is smooth and

$$\sup_{y>0} (y^N + y^{-N})|E^r u(y)| < \infty \quad \text{for every pair of integers } N, r \geq 0.\tag{HCB2}$$

Indeed, the defining weighted Schwartz seminorms become $\sup_x (1+|x|)^j |\partial_x^r (e^{-\beta x} u(e^x))|$. HCB2 bounds each of their finitely many Leibniz summands by choosing $N > |\beta| + 1$, since $(1+|x|)^j e^{-|x|}$ is bounded. Conversely, the weighted Schwartz seminorms with $\beta = N$ and $\beta = -N$, and induction in r on the Leibniz formula, give HCB2. No condition at either endpoint is discarded.

Take additive Haar measure with $\text{vol}(\widehat{\mathbb{Z}}) = 1$, and put

$$\xi_b = \mathbf{1}_{\widehat{\mathbb{Z}}} \otimes \eta(\cdot/b), \quad f_b = \text{Tr } \rho(\xi_b).\tag{HCB3}$$

The Gaussian moments are $\int_{\mathbb{R}} x^2 e^{-\pi x^2} dx = 1/(2\pi)$ and $\int_{\mathbb{R}} x^4 e^{-\pi x^2} dx = 3/(4\pi^2)$, obtained by twice differentiating $\int e^{-\pi c x^2} dx = c^{-1/2}$ for $c > 0$. For this last identity, the square of the positive integral at $c = 1$ equals $\int_0^\infty 2\pi r e^{-\pi r^2} dr = 1$ by polar coordinates; scaling gives the stated value for every $c > 0$. Consequently $\eta(0) = 0$ and $\int \eta = 3/4 - 3/4 = 0$. Thus ξ_b is in the actual source $\mathcal{S}(\mathbb{A}_{\mathbb{Q}})_0 = \ker \varepsilon_0 \cap \ker \varepsilon_1$. CCM’s restriction formula, on $(u, y) \in \widehat{\mathbb{Z}}^\times \times \mathbb{R}_+^*$, gives

$$f_b(u, y) = \sum_{n \in \mathbb{Z} \setminus \{0\}} \eta(ny/b) = \Psi_{0,b}(y), \quad \Psi_{-1/2,b} = f_{2b} - f_b = \text{Tr } \rho(\xi_{2b} - \xi_b).\tag{HCB4}$$

The first equality uses that $ku \in \widehat{\mathbb{Z}}$ for $k \in \mathbb{Q}$ precisely when $k \in \mathbb{Z}$. The second identity partitions the positive integers into their odd and even members. These are actual restriction-image elements, not merely elements with the same Mellin zeros.

66.2 An exact differentiated Euler–Maclaurin formula

Define Bernoulli polynomials, including every coefficient, by

$$\frac{ze^{az}}{e^z - 1} = \sum_{j=0}^{\infty} B_j(a) \frac{z^j}{j!}, \quad |z| < 2\pi, \quad B_j = B_j(0). \quad (\text{HCB5})$$

Multiplication of these power series proves $B_j(a+1) - B_j(a) = ja^{j-1}$, $B'_j(a) = jB_{j-1}(a)$ and $B_j(1-a) = (-1)^j B_j(a)$. For $j \geq 2$ let $\tilde{B}_j(x) = B_j(x - \lfloor x \rfloor)$, extended continuously at the integers. Also put $\tilde{B}_1(x) = x - \lfloor x \rfloor - 1/2$ off the integers and $\tilde{B}_1(x) = 0$ at the integers; its values at the jump points do not affect the distributional derivative or the integrals below. The explicitly finite constant

$$C_j = \frac{1}{j!} \sum_{\ell=0}^j \binom{j}{\ell} |B_\ell| \quad \text{satisfies} \quad \frac{|\tilde{B}_j(x)|}{j!} \leq C_j. \quad (\text{HCB6})$$

Let v and all its derivatives decrease rapidly on $[0, \infty)$, $h > 0$, $a = q + \alpha$ with $q = \lceil a \rceil - 1 \geq 0$ and $0 < \alpha \leq 1$, and take an even integer $J \geq 2$. Then

$$\begin{aligned} \sum_{n \geq 0} v((n+a)h) &= \frac{1}{h} \int_0^\infty v(x) dx - \sum_{j=1}^J \frac{B_j(a)}{j!} h^{j-1} v^{(j-1)}(0) + \mathcal{R}_J(v; a, h), \\ \mathcal{R}_J(v; a, h) &= \frac{(-1)^{J+1} h^{J-1}}{J!} \int_0^\infty \tilde{B}_J(x/h - \alpha) v^{(J)}(x) dx \\ &\quad - \sum_{n=0}^{q-1} \frac{((n+\alpha)h)^J}{(J-1)!} \int_0^1 (1-z)^{J-1} v^{(J)}(z(n+\alpha)h) dz. \end{aligned} \quad (\text{HCB7})$$

An empty finite sum is zero. To prove the formula first take $0 < \alpha < 1$. The distributional derivative of $\tilde{B}_1(x/h - \alpha)$ is $h^{-1} - \sum_{n \in \mathbb{Z}} \delta_{h(n+\alpha)}$. Integrating it against v on $[0, \infty)$, and then integrating by parts $J-1$ more times, gives the infinite-sum formula with $a = \alpha$: the endpoint identities $B_j(1-\alpha) = (-1)^j B_j(\alpha)$ give exactly the displayed signs. The rapid decrease removes all upper boundary terms. For $\alpha = 1$, take the limit from below; $J \geq 2$ makes the final periodic Bernoulli function continuous, and its uniform bound permits dominated convergence. Subtract the first q summands. Taylor's integral formula, with terms of degrees 0 through $J-1$, and the telescoped identity $B_j(a) - B_j(\alpha) = j \sum_{n < q} (n+\alpha)^{j-1}$ yield HCB7, including its second remainder term. In particular, for $0 < a \leq A$, $Q = \max(0, \lceil A \rceil - 1)$ and $0 < h \leq 1$,

$$|\mathcal{R}_J(v; a, h)| \leq C_J \|v^{(J)}\|_{L^1(0, \infty)} h^{J-1} + \frac{Q A^J}{J!} \|v^{(J)}\|_{L^\infty(0, \infty)} h^J. \quad (\text{HCB8})$$

Here are finite formulas for every constant needed when differentiating the actual generator. Set

$$\begin{aligned} P_0(x) &= 2\pi^2 x^4 - 3\pi x^2, & P_{r+1}(x) &= xP'_r(x) - 2\pi x^2 P_r(x), \\ Q_{r,0} &= P_r, & Q_{r,J+1} &= Q'_{r,J} - 2\pi x Q_{r,J}, & Q_{r,J}(x) &= \sum_j q_{r,J,j} x^j, \\ U_{r,J} &= \sum_j |q_{r,J,j}| \frac{\Gamma((j+1)/2)}{2\pi^{(j+1)/2}}, & V_{r,J} &= \sum_j |q_{r,J,j}| v_j, \\ v_0 &= 1, & v_j &= (j/(2\pi e))^{j/2} \quad (j > 0). \end{aligned} \quad (\text{HCB9})$$

Direct differentiation shows $(x\partial_x)^r g = P_r e^{-\pi x^2}$ and its J th ordinary derivative is $Q_{r,J} e^{-\pi x^2}$. Termwise integration and maximization of $x^j e^{-\pi x^2}$ give $\|((x\partial_x)^r g)^{(J)}\|_1 \leq U_{r,J}$ and its supremum at most $V_{r,J}$. These are specified finite sums, not unspecified remainder constants.

The integral of $(x\partial_x)^r g$ is zero for every $r \geq 0$: the case $r = 0$ was computed above, and integration by parts gives $\int xv'(x)dx = -\int v(x)dx$ at each step. The Taylor series is

$$g(x) = \sum_{m=1}^{\infty} \frac{(-1)^m \pi^m (2m+1)}{(m-1)!} x^{2m}, \quad ((x\partial_x)^r g)^{(2m)}(0) = (2m)^r \frac{(-1)^m \pi^m (2m+1)(2m)!}{(m-1)!}. \quad (\text{HCB10})$$

All odd derivatives and the constant term vanish. For $m = 1$ the displayed coefficient is -3π . For $m \geq 2$, it is the sum of $2\pi^2(-\pi)^{m-2}/(m-2)!$ and $-3\pi(-\pi)^{m-1}/(m-1)!$, so both original contributions have been retained in obtaining HCB10.

For integers $M \geq 1$ and $r \geq 0$, the full differentiated asymptotic coefficients and remainders are

$$c_m(t, b) = \frac{(-1)^{m+1} \pi^m B_{2m+1}(1+t)}{(m-1)! b^{2m}}, \quad E^r \Psi_{t,b}(y) = \sum_{m=1}^M (2m)^r c_m(t, b) y^{2m} + \mathcal{E}_{M,r}(t, b; y). \quad (\text{HCB11})$$

This is not merely a formal expansion. To give its uniform bound put

$$D_m(A) = \frac{\pi^m}{(m-1)!} \sum_{j=0}^{2m+1} \binom{2m+1}{j} |B_j| A^{2m+1-j}, \quad J = 2M+4, \quad (\text{HCB12})$$

$$K_{M,r,A} = (2M+2)^r D_{M+1}(A) + C_J U_{r,J} + \frac{Q A^J}{J!} V_{r,J},$$

$$|\mathcal{E}_{M,r}(t, b; y)| \leq K_{M,r,A} (y/b)^{2M+2} \quad (0 < 1+t \leq A, \quad 0 < y/b \leq 1).$$

Indeed, apply HCB7 to $v = (x\partial_x)^r g$ and $h = y/b$. The integral term and all odd Taylor terms vanish. The terms through $j = J$ give the displayed sum through $m = M+1$. HCB8 bounds its remainder by constants times h^{2M+3} and h^{2M+4} ; each is at most the same constant times h^{2M+2} on the stated domain. The omitted $m = M+1$ coefficient has modulus at most $(2M+2)^r D_{M+1}(A)$. Differentiation under the original series is valid on compact subsets of $a, b, y > 0$: each differentiated summand is a polynomial in $(n+a)y/b$ times its Gaussian, bounded by a summable sequence independent of the point in such a compact set. Thus the Euler derivatives in HCB11 are the actual derivatives.

For completeness there is also a uniform bound at the other endpoint. Write $P_r(x) = \sum_j p_{r,j} x^j$, and define

$$L_r = \sum_j |p_{r,j}| w_j, \quad w_0 = 1, \quad w_j = (j/(\pi e))^{j/2} \quad (j > 0). \quad (\text{HCB13})$$

$$|E^r \Psi_{t,b}(y)| \leq \frac{L_r}{1 - e^{-\pi/2}} e^{-\pi(1+t)^2(y/b)^2/2} \quad (y/b \geq 1).$$

This follows from $x^j e^{-\pi x^2} \leq w_j e^{-\pi x^2/2}$ and $(n+a)^2 \geq a^2 + n$ for $n \geq 0$. Consequently all derivatives have Gaussian decrease at infinity, uniformly when $a \in [a_0, A]$ and $b \in [b_0, b_1]$ with positive endpoints.

66.3 Exact strong-space membership and all Mellin residues

The first boundary coefficient is

$$c_1(t, b) = \frac{\pi t(1+t)(1+2t)}{2b^2}. \quad \Psi_{t,b} \in \mathcal{B} \iff t \in \{0, -\frac{1}{2}\}. \quad (\text{HCB14})$$

To prove necessity, HCB11 with $M = 1, r = 0$ gives the actual limit $\lim_{y \downarrow 0} \Psi_{t,b}(y)/y^2 = c_1(t, b)$; HCB2 forces it to be zero. Since $t > -1$, its only zeros are $0, -1/2$. For sufficiency, $B_{2m+1}(1) = B_{2m+1}(1/2) = 0$ for every $m \geq 1$. The first identity follows from the reflection identity and the unit-shift identity at $a = 0$; the second follows immediately from reflection. HCB12 with

arbitrarily large M , for every r , and HCB13 then give all HCB2 seminorms. HCB4 proves that both cases belong to the actual restriction image $\mathcal{V}$ as well.

The original Mellin identity, initially absolutely convergent for $\operatorname{Re} s > 1$, is

$$\int_0^\infty \Psi_{t,b}(y) y^{s-1} dy = b^s \pi^{-s/2} F_t(s) \left[\Gamma(s/2 + 2) - \frac{3}{2} \Gamma(s/2 + 1) \right], \quad F_t(s) = \zeta(s, 1+t). \quad (\text{HCB15})$$

For that domain, integrate each of the two terms of g after the change of variable $x = (n+a)y/b$. The sum of the absolute integrals is bounded by $b^{\operatorname{Re} s} \sum_{n \geq 0} (n+a)^{-\operatorname{Re} s} \int_0^\infty |g(x)| x^{\operatorname{Re} s-1} dx$, which is finite. The original Gamma difference, power of π , power of b , and Hurwitz function all remain in HCB15.

Here is the continuation of the original Hurwitz factor, with the endpoint terms shown. For any integer $N \geq 0$, in $\operatorname{Re} s > -N$ it is

$$F_t(s) = \frac{1}{\Gamma(s)} \left[\int_1^\infty x^{s-1} \frac{e^{-ax}}{1-e^{-x}} dx + \int_0^1 x^{s-1} \left\{ \frac{e^{-ax}}{1-e^{-x}} - \sum_{j=0}^N \frac{B_j(1-a)}{j!} x^{j-1} \right\} dx + \sum_{j=0}^N \frac{B_j(1-a)}{j!(s+j-1)} \right]. \quad (\text{HCB16})$$

For $\operatorname{Re} s > 1$ this follows by integrating the absolutely convergent series $e^{-ax}/(1-e^{-x}) = \sum_{n \geq 0} e^{-(n+a)x}$. HCB5 shows that the bracketed Taylor remainder is $O(x^N)$ at zero. The two integrals therefore continue holomorphically on the stated domain, with compatible formulas as N increases. The Gamma recurrence gives residue $(-1)^j/j!$ at $-j$. It follows, including the zero of $1/\Gamma$ there, that

$$\operatorname{res}_{s=1} F_t(s) = 1, \quad F_t(0) = \frac{1}{2} - a, \quad F_t(-2m) = -\frac{B_{2m+1}(a)}{2m+1} \quad (m \geq 1). \quad (\text{HCB17})$$

The left side of HCB15 already converges on $\operatorname{Re} s > -2$, by HCB12–13. Its exact meromorphic continuation to $\operatorname{Re} s > -2M - 2$ is

$$\int_1^\infty \Psi_{t,b}(y) y^{s-1} dy + \int_0^1 \left[\Psi_{t,b}(y) - \sum_{m=1}^M c_m(t,b) y^{2m} \right] y^{s-1} dy + \sum_{m=1}^M \frac{c_m(t,b)}{s+2m}. \quad (\text{HCB18})$$

HCB12 gives an integrable majorant on every compact subset of that half-plane, also after any s derivative, since logarithmic factors are dominated by a smaller positive power. Thus HCB18 proves the continuation and identifies each residue with $c_m(t,b)$. They match the factors on the right side of HCB15 separately:

$$\begin{aligned} \operatorname{res}_{s=-2m} \left[\Gamma(s/2 + 2) - \frac{3}{2} \Gamma(s/2 + 1) \right] &= \frac{(-1)^m (2m+1)}{(m-1)!}, \\ \operatorname{res}_{s=-2m} \left\{ b^s \pi^{-s/2} F_t(s) \left[\Gamma(s/2 + 2) - \frac{3}{2} \Gamma(s/2 + 1) \right] \right\} &= \frac{(-1)^{m+1} \pi^m B_{2m+1}(a)}{(m-1)! b^{2m}}. \end{aligned} \quad (\text{HCB19})$$

For $m = 1$ only the second Gamma factor has a pole, with residue $-(3/2) \cdot 2 = -3$. For $m \geq 2$ the two residues are respectively $2(-1)^{m-2}/(m-2)!$ and $-3(-1)^{m-1}/(m-1)!$. Multiplication by the unchanged value HCB17 gives HCB19; a zero coefficient means a removable singularity, not a discarded pole.

At the other exceptional points the full finite values are

$$\int_0^\infty \Psi_{t,b}(y) \frac{dy}{y} = \frac{a}{2} - \frac{1}{4}, \quad \int_0^\infty \Psi_{t,b}(y) dy = \frac{b}{4}. \quad (\text{HCB20})$$

At $s = 0$ the Gamma difference is $1 - 3/2 = -1/2$ and HCB17 applies. At $s = 1$ the Gamma difference is zero. Its derivative is $\Gamma(3/2)/2 = \sqrt{\pi}/4$: this follows by differentiating the identity

$\Gamma(s/2 + 2) - \frac{3}{2}\Gamma(s/2 + 1) = \frac{s-1}{2}\Gamma(s/2 + 1)$, while retaining both original terms in HCB15. Multiplying its first-order term by the original simple pole of F_t and by $b\pi^{-1/2}$ gives $b/4$. In particular the cancelled pole at 1 leaves an explicitly nonzero integral. The full original function at $t = 0$ remains $F_0(s) = \zeta(s)$ in every one of these formulas.

66.4 The full boundary sequence and mixed cancellations

Let $\mathcal{E}$ be the algebraic vector space of functions

$$\mathcal{E} = \mathcal{B} + \text{span}_{\mathbb{C}}\{\Psi_{t,b} : t > -1, b > 0\}, \quad \mathcal{C} = \text{span}_{\mathbb{C}}\{(c_m(t,b))_{m \geq 1} : t > -1, b > 0\} \subset \mathbb{C}^{\mathbb{N}}. \quad (\text{HCB21})$$

All spans mean finite sums; no completion or infinite interchange is asserted. Define the boundary map $\partial : \mathcal{E} \rightarrow \mathcal{C}$ by the unique differentiated asymptotic sequence. It has the explicit formula

$$\partial \left(v + \sum_{j=1}^k z_j \Psi_{t_j, b_j} \right) = \left(\sum_{j=1}^k z_j c_m(t_j, b_j) \right)_{m \geq 1}, \quad v \in \mathcal{B}, \quad 0 \longrightarrow \mathcal{B} \longrightarrow \mathcal{E} \xrightarrow{\partial} \mathcal{C} \longrightarrow 0. \quad (\text{HCB22})$$

Here exactness includes the analytic assertion $\ker \partial = \mathcal{B}$. To prove it, every finite sum has the HCB11 expansion with remainders bounded by the sum of the finitely many bounds HCB12. Its coefficients are unique: first take the limit after division by y^2 , then subtract that term, divide by y^4 , and continue inductively. A function in $\mathcal{B}$ has zero at every such limit, by HCB2. Thus HCB22 is independent of the presentation of the function and is linear. If all the coefficients vanish, HCB12 for arbitrarily large M bounds every Euler derivative by arbitrarily large powers of y at zero. HCB13 and the defining property of v give all the bounds at infinity. HCB2 now proves that the function itself is in $\mathcal{B}$. Surjectivity is exactly the stated definition of $\mathcal{C}$; no assertion that it is all sequence space is made.

For every $0 < a < 1$, a genuine mixed cancellation is

$$\Psi_{a-1,b}(y) + \Psi_{-a,b}(y) = \sum_{n \in \mathbb{Z}} g((n+a)y/b) \in \mathcal{B}, \quad c_m(a-1, b) + c_m(-a, b) = 0 \quad (m \geq 1). \quad (\text{HCB23})$$

The function identity uses the evenness of g to reindex the negative integers; every sum converges absolutely locally with all derivatives. The coefficient identity is the odd-degree Bernoulli reflection identity from HCB5. HCB22 then proves actual strong-space membership, including every derivative. Thus HCB14 classifies the individual shifted generators, not arbitrary mixed combinations.

There is also an exact small-endpoint formula for HCB23. Under the Fourier convention $\widehat{g}(\xi) = \int_{\mathbb{R}} g(x) e^{-2\pi i x \xi} dx$, one has $\widehat{g} = g$. To verify all constants, differentiate the Gaussian transform $\widehat{e^{-\pi x^2}} = e^{-\pi \xi^2}$ to obtain

$$\widehat{x^2 e^{-\pi x^2}} = (1/(2\pi) - \xi^2) e^{-\pi \xi^2}, \quad \widehat{x^4 e^{-\pi x^2}} = (\xi^4 - 3\xi^2/\pi + 3/(4\pi^2)) e^{-\pi \xi^2}.$$

The Gaussian transform identity itself follows by differentiating its absolutely convergent integral: integration by parts gives $I'(\xi) = -2\pi \xi I(\xi)$ and the preceding Gaussian integral gives $I(0) = 1$, which determine $I(\xi) = e^{-\pi \xi^2}$. The two constant terms cancel after multiplication by $2\pi^2$ and -3π , and the remaining expression is precisely g . Periodizing a Schwartz function and integrating each Fourier coefficient over $[0, 1]$ proves Poisson summation in this case; absolute convergence of all differentiated Fourier series justifies evaluation at every a . It gives

$$\Psi_{a-1,b}(y) + \Psi_{-a,b}(y) = \frac{2b}{y} \sum_{k=1}^{\infty} \cos(2\pi k a) g(kb/y), \quad 0 < a < 1. \quad (\text{HCB24})$$

The $k = 0$ term is exactly zero because $g(0) = 0$. This equality retains every Fourier mode; it does not replace the original left-hand functions by an asymptotic approximation.

Some mixed functions have an explicit receiver in the actual CCM restriction image. If $q \geq 2$ is an integer, partitioning the positive integers into residue classes modulo q gives

$$\sum_{r=1}^{q-1} \Psi_{r/q-1,b} = f_{qb} - f_b = \text{Tr } \rho(\xi_{qb} - \xi_b). \quad (\text{HCB25})$$

The original first coefficient vanishes in this sum, and HCB22 shows that every coefficient vanishes, as can also be checked by the Bernoulli multiplication identity. Three individual reflected pairs have the following more precise formulas:

$$\begin{aligned} \Psi_{-2/3,b} + \Psi_{-1/3,b} &= f_{3b} - f_b, \\ \Psi_{-3/4,b} + \Psi_{-1/4,b} &= f_{4b} - f_{2b}, \\ \Psi_{-5/6,b} + \Psi_{-1/6,b} &= f_{6b} - f_{3b} - f_{2b} + f_b. \end{aligned} \quad (\text{HCB26})$$

For denominator 3 the two residue classes exclude multiples of 3. For denominator 4 they are exactly the odd integers at scale $4b$. For denominator 6 they exclude multiples of 2 and 3 at scale $6b$; inclusion-exclusion gives the four terms, with the displayed signs. HCB3 maps the respective combinations $\xi_{3b} - \xi_b$, $\xi_{4b} - \xi_{2b}$ and $\xi_{6b} - \xi_{3b} - \xi_{2b} + \xi_b$ to these functions. Each source has both endpoint functionals zero. For a general reflected pair HCB23 proves membership in $\mathcal{B}$; no assertion of membership in $\mathcal{V}$ follows from that calculation alone.

66.5 The retained boundary pair and every support label

Retain the original defect function as a function, not solely its sequence:

$$R_{t,b} = \Psi_{0,b} - \Psi_{t,b}, \quad (\partial \Psi_{t,b}, \partial R_{t,b}) = ((c_m(t,b))_{m \geq 1}, (-c_m(t,b))_{m \geq 1}). \quad (\text{HCB27})$$

No coefficient has been removed: the cancellation is a relation between two retained coordinates. Define the actual space of pairs and its invertible map by

$$\begin{aligned} \mathcal{D} &= \{(u, v) \in \mathcal{E} \times \mathcal{E} : u + v \in \mathcal{B}\}, \\ \mathcal{E} \oplus \mathcal{B} &\xrightarrow{A} \mathcal{D}, \quad A(u, w) = (u, w - u), \quad A^{-1}(u, v) = (u, u + v). \end{aligned} \quad (\text{HCB28})$$

Both displayed maps are linear, well defined, and their two compositions are identities by direct addition. The original pair is $A(\Psi_{t,b}, f_b)$. Projection of this pair onto its sum is f_b for every t and has kernel $\{(u, -u) : u \in \mathcal{E}\}$, so that projection loses the actual deformation. By contrast, the two-coordinate map HCB28 retains it exactly. There is also the exact quotient map

$$\mathcal{D} \longrightarrow \mathcal{B} \oplus \mathcal{C}, \quad (u, v) \longmapsto (u + v, \partial u), \quad \ker = \{(u, -u) : u \in \mathcal{B}\}. \quad (\text{HCB29})$$

To see surjectivity, choose u with any prescribed sequence in $\mathcal{C}$ and set $v = w - u$ for prescribed $w \in \mathcal{B}$. Its kernel is exactly as stated by HCB22. Thus this quotient retains every boundary coefficient but does not purport to retain the entire original function modulo no relation.

For completeness, let L be any nontrivial bounded distributive lattice and use the original full carrier

$$G_L(W) = \{(0, \lambda) : \lambda \in L\} \cup \{(w, 1_L) : w \in W\}, \quad (w, \lambda) + (v, \mu) = (w + v, \lambda \vee \mu), \quad (z, \nu)(w, \lambda) = (zw, \nu \wedge \lambda). \quad (\text{HCB30})$$

The scalars are $G_L(\mathbb{C})$; the last formula is well defined because a nonzero scalar or vector amplitude has top label. For any complex-linear map $T : W \rightarrow W'$, set $G_L(T)(w, \lambda) = (Tw, \lambda)$. The two formulas HCB30 prove directly that this is a $G_L(\mathbb{C})$ -semimodule map, preserves all labels, and

preserves the distinction between $\tau = (0, 0_L)$ and $e = (0, 1_L)$. Its image consists of the full carrier on $\text{im } T$; its amplitude-zero inverse image is $G_L(\ker T)$, whereas its inverse image of the single element τ is only $\{\tau\}$. In particular HCB22 lifts to the exact amplitude-zero inverse-image statement

$$G_L(\partial)^{-1}(\{(0, \lambda) : \lambda \in L\}) = G_L(\mathcal{B}), \quad G_L(\mathcal{E} \oplus \mathcal{B}) \xrightarrow[\cong]{G_L(A)} G_L(\mathcal{D}). \quad (\text{HCB31})$$

The inverse of the last arrow is $G_L(A^{-1})$ and leaves the same label intact. Here $G_L(\mathcal{E} \oplus \mathcal{B})$ is the full carrier with a common label on the ordered pair; it is not silently identified with the product of two independently labelled carriers. In particular, the coefficient cancellation in HCB27 never changes a supported zero into global absence. For an explicit mixed example, write $u = \Psi_{a-1,b} + \Psi_{-a,b}$ with $0 < a < 1$. This is a nonzero function: for $y > b\sqrt{3/(2\pi)}/\min(a, 1-a)$ every term of the two defining series is positive, since $g(x) > 0$ when $x > \sqrt{3/(2\pi)}$. Consequently its only allowed nonzero-amplitude common label is 1_L , and the exact lifted boundary value is

$$G_L(\partial)(u, 1_L) = (0, 1_L) = e \neq \tau = (0, 0_L). \quad (\text{HCB32})$$

This statement concerns membership in the strong test space and the retained common support. It does not identify $\mathcal{B}$ with the restriction image $\mathcal{V}$, or identify a general strong-space function with a cohomologically trivial function. The special actual-image conclusions have their separate explicit source maps in HCB4, HCB25 and HCB26.

The exact retained original-zeta receiving formulas, with parameter recovery, are

$$\begin{aligned} \int_0^\infty R_{t,b}(y) y^{s-1} dy &= b^s \pi^{-s/2} [\zeta(s) - \zeta(s, 1+t)] \left[\Gamma(s/2 + 2) - \frac{3}{2} \Gamma(s/2 + 1) \right], \\ \int_0^\infty R_{t,b}(y) \frac{dy}{y} &= -\frac{t}{2}, \quad \int_0^\infty R_{t,b}(y) dy = 0, \quad t = -2 \int_0^\infty R_{t,b}(y) \frac{dy}{y}. \end{aligned} \quad (\text{HCB33})$$

The first equality is an equality of convergent integrals on $\text{Re } s > -2$ with removable values evaluated there, and continues meromorphically with residues $-c_m(t, b)$ at $-2m$. It follows by subtracting the two HCB15 formulas first on $\text{Re } s > 1$, then using HCB18. The two finite-moment equalities subtract HCB20 at $a = 1+t$ from its value at $a = 1$. The last equality gives the original parameter exactly, for every $t > -1$ and $b > 0$. In particular the vanishing of the degree-one difference does not erase its degree-zero response or any boundary coefficient. This receiver acts on the retained pair or on its actual defect coordinate; it cannot be evaluated from the constant sum f_b alone.

67 Rational mixed boundaries with the compact unit coordinate retained

The original restriction map is that of Connes–Consani–Marcolli, *The Weil proof and the geometry of the adeles class space*, source labels `TrMb0`, `rangeVdef`, and `charCK1sum`. The finite character orthogonality and original Hurwitz factors already proved in HCR33–40 are used here; this calculation constructs their actual adelic antecedents and keeps the compact unit coordinate. It does not declare a slice in that coordinate to be an equivariant replacement.

Keep η of ETC1, $b > 0$, an integer $q \geq 3$, and $1 \leq r < q$ with $\gcd(r, q) = 1$. Define the Bruhat test

$$\xi_{r,q,b}(x) = 2\eta(x_\infty/b)\mathbf{1}_{r/q+\widehat{\mathbb{Z}}}(x_f), \quad F_{r,q,b}(u, y) = E\xi_{r,q,b}(u, y). \quad (\text{RMB1})$$

Its original additive value and integral vanish: $\eta(0) = 0$, $\int_{\mathbb{R}} \eta = 0$, and the translated finite lattice has volume one. Hence $F_{r,q,b}$ belongs to the actual restriction image V in SWQ6, before any evaluation or averaging of $u \in K = \widehat{\mathbb{Z}}^\times$.

The contributing rational numbers α in its restriction sum satisfy $\alpha u - r/q \in \widehat{\mathbb{Z}}$. Multiplying by q shows $q\alpha \in \mathbb{Z}$; writing it as n , the remaining condition is $nu \equiv r \pmod{q}$. Therefore

$$F_{r,q,b}(u, y) = 2 \sum_{\substack{n \in \mathbb{Z} \\ n \equiv ru^{-1} \pmod{q}}} \eta\left(\frac{ny}{qb}\right). \quad (\text{RMB2})$$

Here u^{-1} means its actual invertible residue modulo q . The residue is nonzero, so $n = 0$ is absent. Absolute Gaussian convergence justifies every finite regrouping. At the identity, positive and negative integers give exactly

$$F_{r,q,b}(1, y) = \Psi_{r/q-1,b}(y) + \Psi_{-r/q,b}(y). \quad (\text{RMB3})$$

This explains the reflected boundary cancellation of HCB using an original adelic function. It does not yet remove its u coordinate.

For a Dirichlet character modulo q , regarded as a character of K and extended by zero on nonunit integers, define the literal character projector

$$(P_\chi F)(u, y) = \chi(u) \int_K \overline{\chi(v)} F(v, y) dv, \quad \int_K dv = 1. \quad (\text{RMB4})$$

For these level- q functions the integral is the finite average over $(\mathbb{Z}/q\mathbb{Z})^\times$. It preserves the original restriction image: the unit translations of $E\xi$ are E of the correspondingly translated adelic test, and the finite linear average keeps both zero endpoint conditions.

For a fixed unit integer n , the condition $nv = r$ has one solution modulo q of Haar mass $1/\varphi(q)$. Its character value is $\overline{\chi(r)}\chi(n)$. Thus

$$\begin{aligned} (P_\chi F_{r,q,b})(u, y) &= \frac{2}{\varphi(q)} \overline{\chi(r)} \chi(u) \sum_{n \in \mathbb{Z} \setminus \{0\}} \chi(n) \eta(ny/(qb)) \\ &= \begin{cases} \frac{4}{\varphi(q)} \overline{\chi(r)} \chi(u) \sum_{n \geq 1} \chi(n) \eta(ny/(qb)), & \chi(-1) = 1, \\ 0, & \chi(-1) = -1. \end{cases} \end{aligned} \quad (\text{RMB5})$$

The second line uses the evenness of η and retains the two signs of n . Character orthogonality HCR33 gives the exact inverse $F_{r,q,b} = \sum_{\chi \bmod q} P_\chi F_{r,q,b}$. All sums here are finite in the character index. An explicit projected antecedent, in addition to the finite-average construction, is

$$\xi_\chi(x) = \frac{2}{\varphi(q)} \overline{\chi(r)} \eta(x_\infty/b) \chi(qx_f) \mathbf{1}_{q^{-1}\widehat{\mathbb{Z}}}(x_f), \quad P_\chi F_{r,q,b} = E\xi_\chi. \quad (\text{RMB5a})$$

On the indicated finite support, $\chi(qx_f)$ is the character of its residue in $\mathbb{Z}/q\mathbb{Z}$, extended by zero on nonunits. To check the equation, contributing rationals are $\alpha = n/q$, and $\chi(nu) = \chi(n)\chi(u)$; substitution gives the first line of RMB5 with its exact factor. Both endpoint conditions hold because $\eta(0) = \int \eta = 0$.

Let

$$I_\eta(s) = \frac{\pi^2}{2\pi^{(s+4)/2}} \Gamma((s+4)/2) - \frac{3\pi}{4\pi^{(s+2)/2}} \Gamma((s+2)/2). \quad (\text{RMB6})$$

It is the original two-term Mellin integral computed in ETC11, not an unspecified completion. For $\Re s > 1$, absolute Fubini in RMB5 gives, for each even character,

$$\int_0^\infty (P_\chi F_{r,q,b})(u, y) y^s \frac{dy}{y} = \frac{4}{\varphi(q)} \overline{\chi(r)} \chi(u) (qb)^s I_\eta(s) L(s, \chi). \quad (\text{RMB7})$$

All factors, including q^s, b^s , the Haar denominator and the two Gamma terms, remain explicit. Summing at $u = 1$ recovers HCR36 for the two reflected Hurwitz functions. Each $P_\chi F$ lies in $V \subset B$, so its Mellin integral is entire; RMB7 specifies its meromorphic continuation with all exceptional local factors retained.

For the principal character, inclusion and exclusion over the finite set of prime divisors of q gives the stronger original-function identity

$$P_{\chi_0} F_{r,q,b} = \frac{2}{\varphi(q)} \sum_{d|\text{rad}(q)} \mu(d) f_{qb/d}. \quad (\text{RMB8})$$

Indeed $\mathbf{1}_{\gcd(n,q)=1} = \sum_{d|\text{rad}(q), d|n} \mu(d)$. Insert this finite identity into RMB5, put $n = dm$ and use $f_c = 2 \sum_{m \geq 1} \eta(my/c)$. This proves RMB8 without a rearrangement of a divergent endpoint integral. At the Mellin level it retains all bad-prime factors as $L(s, \chi_0) = \zeta(s) \prod_{p|q} (1 - p^{-s})$. For $q = 3, 4, 6$, the only even character is principal, giving the three actual V identities of HCB for the reflected pairs. For other moduli the nonprincipal even summands in RMB5 must also be retained.

The exact replacement for forgetting the unit coordinate is the finite tuple of character components, with inverse just proved. It also preserves convolution: for $F(u, y) = \sum_\chi \chi(u) F_\chi(y)$ and $G(u, y) = \sum_\chi \chi(u) G_\chi(y)$ at the same finite level, compact character orthogonality in the original convolution integral gives

$$(F * G)(u, y) = \sum_\chi \chi(u) (F_\chi *_{\mathbb{R}_+} G_\chi)(y). \quad (\text{RMB9})$$

The real integrals are absolutely convergent by SWQ4. In contrast the slice map $R(F)(y) = F(1, y) = \sum_\chi F_\chi(y)$ introduces all the mixed character products on its target. For a concrete failure take a nontrivial character χ , $F(u, y) = \chi(u) e^{-(\log y)^2}$ and $G(u, y) = e^{-(\log y)^2}$. Then $F * G = 0$ by orthogonality, whereas $(RF * RG)(1) = \int_{\mathbb{R}} e^{-2x^2} dx > 0$. Both inputs belong to the original strong test space. This proves the precise failure of the slice as a convolution morphism, and RMB9 supplies its full invertible replacement.

The failure also occurs on actual elements of V . Take $q = 5, r = 1$ and the even nonprincipal real character χ with $\chi(1) = \chi(4) = 1$ and $\chi(2) = \chi(3) = -1$. Write $P_\chi F(u, y) = \chi(u) A_\chi(y)$ and $P_{\chi_0} F(u, y) = A_0(y)$, and put $g = A_\chi *_{\mathbb{R}_+} A_0$. The two functions

$$X(u, y) = \chi(u) g(y), \quad Y(u, y) = g(y) \quad (\text{RMB10})$$

both belong to V : X is the convolution of $\chi A_\chi \in V$ with $\chi A_0 \in B$, while Y is the convolution of $A_0 \in V$ with the radial $A_\chi \in B$. Use SWQ14's proved ideal property. Their full convolution is zero by character orthogonality. However RMB6 gives $I_\eta(2) = 1/(4\pi)$ and RMB7 gives $MA_\chi(2) = 25b^2 L(2, \chi)/(4\pi)$ and $MA_0(2) = 25b^2 L(2, \chi_0)/(4\pi)$. Both L values are positive: their absolutely convergent Euler products have positive factors $(1 - \chi(p)p^{-2})^{-1}$, and $\sum_p p^{-2} < \infty$

keeps their product nonzero. Thus $Mg(2) > 0$ and $M(g *_{\mathbb{R}_+} g)(2) = Mg(2)^2 > 0$. The identity slice has therefore changed a zero product of two actual trivial correspondences into a nonzero product. The retained character tuple in RMB9 preserves their zero product exactly.

On the original full support carrier, apply the common-label lift of the character decomposition and of its inverse to the entire finite direct sum of component function spaces. Both maps are complex-linear; they preserve label joins, the scalar action and inverse compositions. By RMB9 they also preserve the specified convolution products, with label meet. Thus the corrected finite-character receiver retains every label, including e and τ ; it does not assign independent labels to components of one common-label input. This exact map neither asserts that an identity slice lies in the radial restriction ideal nor changes which character L -function appears in RMB7.

68 The actual adelic source and its retained prime boundary

The adelic restriction operation and its original measures are those of Alain Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Section III (6)–(19), and Alain Connes, Caterina Consani and Matilde Marcolli, *The Weil proof and the geometry of the adeles class space*, as specified in SWQ2–14. The homological complex used below is the standard Koszul complex, named for Jean-Louis Koszul. The standard regular-sequence mechanism is recorded by *The Stacks Project Authors*, Tag 062F, with original author TeX, lines 7264–7298, commit a04446e57ec1fbc252a871afcec7752fb2807b14. The contribution here is the exact application, connecting representatives, and receiving maps for these specified programme modules; the necessary Koszul exactness is proved below as well. All tensor products and quotients below are algebraic unless a different meaning is explicitly stated.

The precise public programme inputs are SWQ2–14, MRT1–20, and ER27–30. These pinned proof targets were checked against the local source versions.

68.1 The finite adelic basis and the exact original source

Put

$$G = \mathbb{Q}_{>0}^\times, \quad R = \mathbb{C}[G], \quad I = \ker(\varepsilon : R \longrightarrow \mathbb{C}), \quad H = \mathcal{S}_{\text{even}}(\mathbb{R}), \quad K = \widehat{\mathbb{Z}}^\times, \quad c_a = 1_{a\widehat{\mathbb{Z}}}. \quad (\text{ABR1})$$

The basis element $t_a \in R$ satisfies $t_a t_b = t_{ab}$ and $\varepsilon(t_a) = 1$. The finite additive measure has $\text{vol}(\mathbb{Z}_p) = 1$, and the real additive measure is Lebesgue measure. Thus

$$\text{vol}(a\widehat{\mathbb{Z}}) = \prod_p p^{-v_p(a)} = a^{-1} \quad (a > 0 \text{ rational}). \quad (\text{ABR2})$$

This equality uses the original positive real value of a and its prime factorization; the finite and real factors will both be retained.

The family $(c_a)_{a \in G}$ is a vector-space basis of $\mathcal{S}(\mathbb{A}_f)^K$. To prove spanning, first observe that a compactly supported locally constant function on $\mathbb{Q}_p$, invariant under $\mathbb{Z}_p^\times$, is constant on valuation shells. Choose integers $A \leq B$ such that its support is in $p^A \mathbb{Z}_p$ and it is constant on cosets of $p^B \mathbb{Z}_p$. If its values on the shells $A \leq n < B$ and on the last ball $p^B \mathbb{Z}_p$ are $d_A, \dots, d_B$, the function is exactly

$$d_A 1_{p^A \mathbb{Z}_p} + \sum_{n=A+1}^B (d_n - d_{n-1}) 1_{p^n \mathbb{Z}_p}.$$

The original Bruhat space on $\mathbb{A}_f$ consists of finite sums of restricted tensor products with $1_{\mathbb{Z}_p}$ outside a finite set of primes. Compact-unit averaging of a finite-place test is a finite linear operation: its support is contained in finitely many bounded local balls and its local-constancy subgroups have finite index there. Averaging therefore reduces to a finite quotient. Apply the preceding local formula at the finitely many relevant primes and expand the finite tensor products. Their balls are exactly c_a . For an invariant test averaging leaves the test itself fixed, which proves spanning.

For independence, take distinct $a_1, \dots, a_m \in G$ and partially order them by $a_i \preceq a_j$ when a_j/a_i is an integer. This is a partial order because both positive ratios can be integers only when they are one. Order this finite set by a linear extension. Evaluate its characteristic functions at the finite rational ideles $x_f = a_j$. Then

$$c_{a_i}(a_j) = 1 \iff a_i \preceq a_j.$$

Indeed $\mathbb{Q} \cap \widehat{\mathbb{Z}} = \mathbb{Z}$. The resulting matrix is lower triangular with diagonal one. Its determinant is one, proving independence. This proof also works for coefficients in any complex vector space, because back substitution uses only this finite triangular matrix.

Define

$$\Theta : R \otimes H \longrightarrow \mathcal{S}(\mathbb{A}_{\mathbb{Q}})^{K, \text{even}}, \quad \Theta(t_a \otimes h)(x_{\infty}, x_f) = h(x_{\infty}/a)c_a(x_f). \quad (\text{ABR3})$$

Here evenness refers to the original real variable x_{∞} . Every target is a finite sum $\sum_a g_a(x_{\infty})c_a(x_f)$ with $g_a \in H$. Set $h_a(x) = g_a(ax)$ to obtain its preimage in ABR3. Independence follows from the vector-valued independence of the c_a and the injectivity of each real dilation. Thus Θ is an isomorphism of vector spaces, with the displayed exact inverse. No change in the original real coordinate has been made in the target.

For $r \in G$ define full rational adelic dilation by $\alpha_r \xi(x) = \xi(r^{-1}x)$. The finite argument gives $c_a(r^{-1}x_f) = c_{ra}(x_f)$, while the real argument is $h(x_{\infty}/(ra))$. Consequently

$$\alpha_r \Theta(t_a \otimes h) = \Theta(t_{ra} \otimes h). \quad (\text{ABR4})$$

This is the left R -action. It is full diagonal rational dilation; it is not dilation of the real idele-class coordinate alone.

Let $m(h) = (h(0), \int_{\mathbb{R}} h(x) dx)$. Evaluation and integration of ABR3 give the two exact moments

$$\Theta(t_a \otimes h)(0) = h(0), \quad \int_{\mathbb{A}} \Theta(t_a \otimes h) = a \left(\int_{\mathbb{R}} h \right) \prod_p p^{-v_p(a)} = \int_{\mathbb{R}} h. \quad (\text{ABR5})$$

Thus $(\varepsilon_0, \varepsilon_1)\Theta = \varepsilon \otimes m$. In particular

$$M_0 = \ker(\varepsilon \otimes m), \quad H_{00} = \ker m, \quad \Theta(M_0) = \mathcal{S}(\mathbb{A})_0^{K, \text{even}}. \quad (\text{ABR6})$$

This is equality of the actual spaces with their two original endpoint conditions, not a comparison only after completion.

68.2 The complete algebraic coinvariant and its boundary

The two even Schwartz functions

$$h_0(x) = (1 - 2\pi x^2)e^{-\pi x^2}, \quad h_1(x) = 2\pi x^2 e^{-\pi x^2}$$

have moments $(1, 0)$ and $(0, 1)$. Indeed their evaluations are immediate and the real Gaussian integrals are $\int e^{-\pi x^2} = 1$ and $\int x^2 e^{-\pi x^2} = 1/(2\pi)$. The section $\sigma(u_0, u_1) = u_0 h_0 + u_1 h_1$ therefore splits m . It gives

$$M_0 = (R \otimes H_{00}) \oplus (I \otimes \sigma(\mathbb{C}^2)), \quad Q_0 = M_0/IM_0 \cong H_{00} \oplus (I/I^2) \otimes \mathbb{C}^2. \quad (\text{ABR7})$$

Indeed multiplication by I has exactly the two components $I \otimes H_{00}$ and $I^2 \otimes \sigma(\mathbb{C}^2)$.

For clarity the resulting boundary and inverse can be written without leaving a dependence on the chosen moment section. Put

$$W = \bigoplus_{p \text{ prime}} \mathbb{C} \ell_p, \quad \mathcal{B}_{\text{pr}} = W \otimes \mathbb{C}^2. \quad (\text{ABR8})$$

The map $[t_a - 1] \mapsto \sum_p v_p(a) \ell_p$ is an isomorphism $I/I^2 \rightarrow W$. For its proof retain the identity $t_{ab} - 1 = (t_a - 1) + (t_b - 1) + (t_a - 1)(t_b - 1)$. Modulo I^2 it gives additivity, and inversion gives the negative of a class. Prime factorization expresses each class as the stated sum. For independence the functionals $D_p(\sum_a c_a t_a) = \sum_a c_a v_p(a)$ satisfy $D_p(rs) = \varepsilon(r)D_p(s) + \varepsilon(s)D_p(r)$, and hence vanish on I^2 . They take value one on $[t_p - 1]$ and zero on the other displayed prime generators.

The canonical coordinates of $c = [\sum_a t_a \otimes h_a] \in Q_0$ are

$$\Phi(c) = \sum_a h_a \in H_{00}, \quad \beta(c) = \sum_p \ell_p \otimes \sum_a v_p(a) m(h_a). \quad (\text{ABR9})$$

Both maps kill IM_0 . For β , apply the preceding derivation identity to a product $r \sum_a t_a \otimes h_a$, $r \in I$; its value is $\sum_p D_p(r) \ell_p \otimes \sum_a m(h_a) = 0$ by the defining moment condition. The inverse of ABR9 sends

$$(h, \sum_p \ell_p \otimes u_p) \mapsto [t_1 \otimes h] + \sum_p [(t_p - 1) \otimes \sigma(u_p)]. \quad (\text{ABR10})$$

Only finitely many u_p are nonzero. Changing σ changes the latter expression by an element of $I \otimes H_{00} \subset IM_0$. To check the inverse on an arbitrary representative, subtract $t_1 \otimes \sum_a h_a$ and write the remainder as $\sum_a (t_a - 1) \otimes h_a$. Replacing each h_a there by $\sigma m(h_a)$ changes its class by IM_0 , and ABR8 then gives exactly ABR10. Thus the short exact sequence

$$0 \longrightarrow \mathcal{B}_{\text{pr}} \xrightarrow{j} Q_0 \xrightarrow{\Phi} H_{00} \longrightarrow 0 \quad (\text{ABR11})$$

is canonically split by $h \mapsto [t_1 \otimes h]$, and j is the section-independent second term of ABR10.

68.3 Actual periodization, injectivity, and exact inverse

For the original restriction map $E\xi(u, y) = \sum_{q \in \mathbb{Q}^\times} \xi(qy, qu)$, $u \in K$, $y > 0$, the finite condition in ABR3 is $q/a \in \widehat{\mathbb{Z}} \cap \mathbb{Q} = \mathbb{Z}$. Hence $q = an$ for $n \in \mathbb{Z} \setminus \{0\}$. It follows that

$$E\Theta(t_a \otimes h)(u, y) = \sum_{n \in \mathbb{Z} \setminus \{0\}} h(ny) = 2 \sum_{n \geq 1} h(ny) =: \mathcal{P}h(y). \quad (\text{ABR12})$$

Evenness of h supplies exactly the last factor two. The real and finite scales a have both participated in the equality; no finite restriction has been dropped. All sums are locally absolutely convergent with every real derivative by Schwartz decay. For $m = \sum_a t_a \otimes h_a \in M_0$, ABR12 is $E\Theta(m) = \mathcal{P}(\sum_a h_a)$.

The Fourier convention $\widehat{h}(\xi) = \int_{\mathbb{R}} h(x) e^{-2\pi i x \xi} dx$ gives the exact Poisson identity with both endpoints

$$\mathcal{P}h(y) = y^{-1} \widehat{\mathcal{P}h}(1/y) + y^{-1} \int_{\mathbb{R}} h(x) dx - h(0). \quad (\text{ABR13})$$

For $h \in H_{00}$ both displayed endpoint terms are zero, and $\widehat{h}$ also belongs to H_{00} because $\widehat{h}(0) = \int h = 0$ and $\int \widehat{h} = h(0) = 0$. At infinity the sums of every Euler derivative of $\mathcal{P}h$ decay faster than every power: choose an arbitrarily large Schwartz power $A > 1$ and bound their sums by a constant times $y^{-A} \sum n^{-A}$. ABR13 proves the corresponding estimates at zero. Therefore $\mathcal{P}h$ belongs to the original radial strong Schwartz test space. All powers of $\log y$ are also controlled by these bounds, so its Mellin transform is entire.

For $\Re s > 1$, absolute Fubini gives the original identity

$$\mathcal{M}(\mathcal{P}h)(s) = 2\zeta(s) \mathcal{M}_+ h(s), \quad \mathcal{M}_+ h(s) = \int_0^\infty h(x) x^{s-1} dx. \quad (\text{ABR14})$$

Indeed the absolute sum of integrals is $2\zeta(\Re s) \int_0^\infty |h(x)| x^{\Re s-1} dx < \infty$. The original Euler product proves that $\zeta(s) \neq 0$ in this half-plane: the logarithms of its factors converge absolutely because $\sum_p \sum_{k \geq 1} p^{-k\Re s}/k < \infty$. Thus division by the actual $2\zeta(s)$ is valid on the exact domain used below. There is no completed replacement of ζ .

For $\sigma > 1$, $g_\sigma(v) = e^{\sigma v} h(e^v)$ is Schwartz on $\mathbb{R}$. Every derivative is a finite sum of $e^{(\sigma+j)v} h^{(j)}(e^v)$. These decay exponentially at $-\infty$ and faster than any prescribed exponential at $+\infty$, using the original real Schwartz estimates. Consequently $\mathcal{M}_+ h(\sigma + it)$ decays faster than every power of t . Fourier inversion in v gives

$$h(x) = \frac{1}{4\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} \frac{\mathcal{M}(\mathcal{P}h)(s)}{\zeta(s)} x^{-s} ds, \quad x > 0, \quad \sigma > 1. \quad (\text{ABR15})$$

Every constant is explicit: Mellin inversion has $1/(2\pi i)$, and ABR14 contributes the additional $1/2$. To justify inversion directly, multiply the Fourier integrand by $e^{-\epsilon t^2}$. Its inverse is convolution with $(4\pi\epsilon)^{-1/2}e^{-v^2/(4\epsilon)}$, which tends to g_σ as $\epsilon \downarrow 0$. The proved rapid vertical decay permits dominated convergence on the Fourier side. This proves ABR15, including absolute convergence. Evenness recovers $h(x)$ for $x < 0$ and $h(0) = 0$ supplies the remaining point. In particular $\mathcal{P}$ is injective on H_{00} .

There is also an exact arithmetic inverse in the original variable:

$$h(x) = \frac{1}{2} \sum_{n \geq 1} \mu(n) \mathcal{P}h(nx), \quad x > 0. \quad (\text{ABR15a})$$

For fixed $x > 0$ and any $A > 1$, Schwartz decay bounds the absolute double sum after inserting ABR12 by a constant depending on x, A times $\sum_{n,m \geq 1} (nm)^{-A} = \zeta(A)^2$. It may therefore be regrouped by $k = nm$. Its coefficient of $h(kx)$ is $\sum_{n|k} \mu(n)$, which is one for $k = 1$ and zero for $k > 1$: prime factorization gives the finite product $\prod_{p|k} (1 - 1)$ in the latter case. This proves ABR15a with its factor $1/2$ and also proves convergence locally with every Euler derivative. It is a second exact inverse, not a truncation or a substitute for ABR14–15’s original zeta factor.

Let $V = ES(\mathbb{A})_0$ be the actual restriction image, and let V^K be its compact-unit invariant part. Then

$$V^K = ES(\mathbb{A})_0^{K, \text{even}} = \mathcal{P}(H_{00}). \quad (\text{ABR16})$$

For the nontrivial inclusion, take $F = E\xi \in V^K$. Average $\xi(x_\infty, vx_f)$ over $v \in K$. This is a finite average in the original Bruhat space, retains both moments, and has restriction equal to F . Next project to real-even functions by averaging x_∞ with $-x_\infty$. Under E , the real reflection equals the finite-unit translation $u \mapsto -u$, as seen by reindexing $q \mapsto -q$ in its absolutely convergent sum. Since F is K -invariant this projection still restricts to F . ABR6 now supplies a preimage in M_0 , whose sum of coefficients is in H_{00} . This proves ABR16 without replacing V by its closure or by a topological completion.

Full rational dilation is trivial after periodization, also directly by reindexing q/r in the rational sum. Therefore ABR12 induces a surjection on the algebraic coinvariant, with the exact kernel

$$0 \longrightarrow \mathcal{B}_{\text{pr}} \xrightarrow{j} Q_0 \xrightarrow{\rho = \mathcal{P}\Phi} V^K \longrightarrow 0. \quad (\text{ABR17})$$

Injectivity of $\mathcal{P}$ proves that this kernel is exactly the whole prime boundary, rather than merely containing it. A faithful joint receiver and its inverse are

$$\Omega(c) = (\rho(c), \beta(c)), \quad \Omega : Q_0 \xrightarrow{\sim} V^K \oplus \mathcal{B}_{\text{pr}}, \quad \Omega^{-1}(F, b) = [t_1 \otimes \mathcal{P}^{-1}F] + j(b). \quad (\text{ABR18})$$

The inverse $\mathcal{P}^{-1}$ is exactly ABR15.

Keep Connes’s original seminorm, without changing its measure, factor or weight:

$$\|c\|_\delta^2 = \int_0^\infty |y^{1/2} \rho(c)(y)|^2 (1 + \log^2 y)^{\delta/2} \frac{dy}{y}, \quad \delta > 1. \quad (\text{ABR19})$$

It is finite by the strong test estimates. Its weight is strictly positive. If the integral is zero, continuity gives $\rho(c) = 0$ everywhere, and ABR17 gives $c \in j(\mathcal{B}_{\text{pr}})$. The reverse inclusion follows from ABR17 as well. Thus ABR19 has precisely this algebraic nullspace. This identifies the lost coordinates but assigns no new norm to them and claims no equivalence with an unproved topological quotient.

68.4 Positive real ideles and the two endpoint characters

For $b > 0$, now with no rationality assumption, define $R_b h(x) = h(x/b)$ and let $\mathcal{R}_b = \text{id}_R \otimes R_b$ be the operator on $R \otimes H$. Its original moments obey

$$m(R_b h) = D_b m(h), \quad D_b = \begin{pmatrix} 1 & 0 \\ 0 & b \end{pmatrix}. \quad (\text{ABR19a})$$

Evaluation at zero is unchanged, and the real substitution $x = bu$ multiplies the additive integral by b . Thus $\mathcal{R}_b$ preserves M_0 , commutes with the full rational action, and descends to Q_0 . Under Θ it is precisely the real-idele action $\xi(x_\infty, x_f) \mapsto \xi(x_\infty/b, x_f)$; the finite support $a\hat{\mathbb{Z}}$ has not changed.

In the exact coordinates ABR9, this action is

$$(\Phi, \beta)\mathcal{R}_b c = (R_b \Phi c, (\text{id}_W \otimes D_b)\beta c). \quad (\text{ABR19b})$$

This follows directly from the finite sums in ABR9. On the boundary inverse ABR10 it is also independent of the moment section: $R_b \sigma(u) - \sigma(D_b u) \in H_{00}$, so its product with $t_p - 1$ is in $I \otimes H_{00} \subset IM_0$. Consequently Ω is equivariant for real dilation on its V^K component and the two endpoint characters $1, b$ on its boundary component. These are the actual Connes endpoint characters $\mathbb{C} \oplus \mathbb{C}(1)$, with their displayed action; no purity theorem is inferred from this character calculation.

Both periodization conventions and their exact comparison are

$$\begin{aligned} E_{2007}(R_b h)(y) &= (\mathcal{P}h)(y/b), \\ E_{1998}(h)(y) &= y^{1/2} \mathcal{P}h(y), \\ E_{1998}(R_b h)(y) &= b^{1/2} E_{1998}(h)(y/b). \end{aligned} \quad (\text{ABR19c})$$

The first equality follows term by term from ABR12. The last one retains $y^{1/2} = b^{1/2}(y/b)^{1/2}$ and therefore its exact factor $b^{1/2}$. Multiplication by the nowhere-zero function $y^{1/2}$ is explicitly inverted by multiplication by $y^{-1/2}$ on the original strong test spaces; it is not omitted in the seminorm. In particular that seminorm transforms exactly as

$$\|\mathcal{R}_b c\|_\delta^2 = b \int_0^\infty |E_{1998}(\Phi c)(y)|^2 \{1 + (\log y + \log b)^2\}^{\delta/2} \frac{dy}{y}. \quad (\text{ABR19d})$$

This follows by the positive substitution in ABR19 and does not claim that the original weighted seminorm is dilation-invariant.

68.5 Every higher-degree connecting representative

Identify R with the Laurent polynomial algebra $\mathbb{C}[t_p^{\pm 1} : p \text{ prime}]$, whose polynomials involve only finitely many variables. Let $x_p = t_p - 1$ and use the ordered prime basis ℓ_p of W . The standard algebraic Koszul complex is

$$K_n(R) = R \otimes \Lambda^n W, \quad d(r \otimes \ell_{p_1} \wedge \cdots \wedge \ell_{p_n}) = \sum_{i=1}^n (-1)^{i-1} r x_{p_i} \otimes \ell_{p_1} \wedge \cdots \widehat{\ell_{p_i}} \cdots \wedge \ell_{p_n}. \quad (\text{ABR20})$$

The indices here satisfy $p_1 < \cdots < p_n$. The two orders in which a pair of indices is removed have opposite signs and the coefficients commute, so $d^2 = 0$.

This is a free resolution of the augmentation module $\mathbb{C}$. For a finite set of primes the sequence of its x_p is regular: after any preceding variables have been set to one, the quotient is still a Laurent polynomial algebra in the remaining variables, and the next $t_p - 1$ is a nonzero divisor. The finite Koszul complex is exact in positive degrees by induction: adjoining one variable is the mapping cone of multiplication by that variable's x_p , whose kernel on the preceding degree-zero quotient is zero and whose cokernel is the further quotient. The full complex ABR20 is the filtered union of these finite complexes. A cycle uses finitely many wedge indices; it is a boundary already in the corresponding finite complex. Degree zero is $R/(x_p : p \text{ prime}) = \mathbb{C}$. This proves exactness also for the infinitely generated group, with only finite algebraic sums at every step.

Tensor this resolution with the coefficient modules in the exact sequence

$$0 \longrightarrow M_0 \longrightarrow R \otimes H \xrightarrow{\varepsilon \otimes m} \mathbb{C}^2 \longrightarrow 0.$$

Here $\mathbb{C}^2$ has trivial R -action. Tensoring each free Koszul degree preserves the short exact sequence. For the middle module, its Koszul homology is H in degree zero and zero in positive degrees, because the complex is $K_\bullet(R) \otimes H$. For the last module its differential is zero, and its degree- n homology is $\Lambda^n W \otimes \mathbb{C}^2$. Consequently

$$H_n(G, M_0) \cong \Lambda^{n+1} W \otimes \mathbb{C}^2 \quad (n \geq 1), \quad H_0(G, M_0) = Q_0, \quad (\text{ABR21})$$

with ABR11 as the degree-zero exact sequence.

The exact connecting representative of an ordered element is

$$\begin{aligned} & \partial_n((\ell_{p_1} \wedge \cdots \wedge \ell_{p_{n+1}}) \otimes u) \\ &= \left[\sum_{i=1}^{n+1} (-1)^{i-1} (x_{p_i} \otimes \sigma(u)) \otimes \ell_{p_1} \wedge \cdots \widehat{\ell_{p_i}} \cdots \wedge \ell_{p_{n+1}} \right] \in H_n(G, M_0). \end{aligned} \quad (\text{ABR22})$$

Each coefficient is in M_0 because $\varepsilon(x_p) = 0$. This representative is the differential of $(t_1 \otimes \sigma(u)) \otimes \ell_{p_1} \wedge \cdots \wedge \ell_{p_{n+1}}$ in the middle module's complex, so it is a cycle. Changing a lift of u changes it by a boundary in the first complex. For $n \geq 1$ its map is an isomorphism: a cycle of the first complex becomes a boundary in the acyclic positive-degree middle complex, proving surjectivity; if a connecting class is a boundary, subtract its first-complex primitive from the chosen lift and use middle-complex acyclicity one degree higher, proving injectivity. Projection to the last complex has zero differential, so the latter argument forces the original exterior class to be zero. For $n = 0$ the representative is $[(t_p - 1) \otimes \sigma(u)]$, exactly j in ABR11. These proofs retain the original wedge order and sign.

The module V^K has trivial full-rational action, and its Koszul homology is $\Lambda^n W \otimes V^K$. Periodization kills every coefficient $x_p \otimes h$ in ABR22 separately, by ABR12. Thus the homology map induced by $E\Theta : M_0 \rightarrow V^K$ is zero in every positive degree. In degree zero it is exactly ABR17. The action $\mathcal{R}_b$ commutes with the differential ABR20. Under the connecting isomorphism ABR21 its action is precisely $\text{id}_{\Lambda^{n+1} W} \otimes D_b$. To verify the section-independence directly in ABR22, replace $R_b \sigma(u)$ by $\sigma(D_b u)$. Their difference is in H_{00} , and the resulting difference of cycles is the differential of $(t_1 \otimes (R_b \sigma(u) - \sigma(D_b u)))$ times the complete ordered wedge, which lies in the first complex itself. Thus every higher connecting class has the same two original endpoint characters, with its prime wedge unchanged.

68.6 The existing mass and endpoint sources under this map

The actual source spaces, moments, group action, quotient and coordinate maps in ABR1 and ABR6–10 agree with MRT1–5 and ER27–30, not just their dimensions. The MRT source heat kernel is

$$k_T(x) = (4\pi T)^{-1/2} e^{-x^2/(4T)}, \quad T > 0.$$

For its even finite coefficient source a , $A_T a(x) = \sum_n a_n k_T(x - n)$ is even Schwartz and has exactly

$$m(A_T a) = \left(\sum_n a_n k_T(n), \sum_n a_n \right) = (q_T(a), m(a)). \quad (\text{ABR23})$$

Therefore the actual class and antecedent are

$$\begin{aligned} B_{p,T}(a) &= [(t_p - 1) \otimes A_T a], \\ \Theta((t_p - 1) \otimes A_T a)(x) &= (A_T a)(x_\infty/p) c_p(x_f) - (A_T a)(x_\infty) c_1(x_f), \\ (\rho, \beta) B_{p,T}(a) &= (0, \ell_p \otimes (q_T(a), m(a))). \end{aligned} \quad (\text{ABR24})$$

Each such antecedent has both moments zero, including the separate finite Haar factor p^{-1} and the real integral factor p of its first summand. Its two periodizations are equal term by term after the exact arithmetic reindexing ABR12.

These finite Gaussian sources attain all of $\mathcal{B}_{\text{pr}}$. For a prescribed pair (u_0, u_1) and a fixed $T > 0$, retain $k_0 = k_T(0)$ and $k_1 = k_T(1)$ and take

$$a_0 = \frac{u_0 - k_1 u_1}{k_0 - k_1}, \quad a_1 = a_{-1} = \frac{k_0 u_1 - u_0}{2(k_0 - k_1)}, \quad a_n = 0 \quad (|n| > 1).$$

Here $k_0 > k_1$ and direct substitution gives ABR23 equal to (u_0, u_1) . Summing the finitely many selected prime classes therefore realizes every boundary vector, with the original kernel and coordinate conventions of MRT8.

For the Gaussian coefficient family $a_n^x = e^{-\pi n^2 x}$, $x > 0$, MRT13–14 proves that the same $A_T a^x$ is even Schwartz and gives

$$\beta B_{p,T}(a^x) = \ell_p \otimes \left((4\pi T)^{-1/2} \vartheta(x + 1/(4\pi T)), \vartheta(x) \right), \quad \rho B_{p,T}(a^x) = 0. \quad (\text{ABR25})$$

No heat time has been identified with the independent parameter x . The separately specified MRT supported-zero source operation $\widehat{E}$ still has its actual receiving matrix $J_T = \begin{pmatrix} 0 & k_T(0) \\ 0 & 1 \end{pmatrix}$ on these two moments, as proved in MRT11. Its periodization is zero before and after that operation. The present comparison does not redefine $\widehat{E}$ as the scalar supported-zero operation on a different common-label vector carrier.

For an arbitrary completed MRT input $(a, 0, m)$, MRT10 already defines its boundary through $(q_T(a), m)$. ABR10 supplies the Schwartz representative $\sigma(q_T(a), m)$ and hence an actual adelic antecedent for this boundary. This does not assert that $A_T a$ itself is Schwartz for every completed coefficient sequence.

For the ER source, keep its original real variable Z , time $t > 0$, and function

$$h_t(Z) = e^{-Z^2/(4t)} H_t(Z) \cos(Z/(2t)). \quad (\text{ABR26})$$

It is even Schwartz. The exact moments are

$$m(h_t) = \left(H_t(0), \frac{\sqrt{\pi t}}{8} e^{-1/(4t)} \right). \quad (\text{ABR27})$$

To retain the reason for the second constant, insert the original cosine integral $H_t(Z) = \int_0^\infty e^{tu^2} \Phi(u) \cos(Zu) du$. The inner Gaussian integral is

$$\sqrt{\pi t} \{ e^{-t(u+1/(2t))^2} + e^{-t(u-1/(2t))^2} \} = 2\sqrt{\pi t} e^{-tu^2-1/(4t)} \cosh u.$$

Absolute Fubini follows from the Gaussian and the faster-than-Gaussian decay of the original Φ . The retained factor e^{tu^2} then gives $2\sqrt{\pi t} e^{-1/(4t)} H_0(i)$. ER3–5 gives $H_0(-i) = 1/16$ from the original limit

$$\frac{1}{16} \lim_{s \rightarrow 1} s(s-1) \pi^{-s/2} \Gamma(s/2) \zeta(s) = \frac{1}{16};$$

evenness gives $H_0(i) = H_0(-i)$. This proves ABR27 with the original zeta pole, Gamma factor, power of pi and factor 16 retained. No arithmetic Hurwitz-flow time is being substituted for t .

In ABR3 use this same real variable for x_∞ and h_t . The ER prime class has the actual antecedent and receiver

$$\begin{aligned} \Xi_{p,t}(x) &= h_t(x_\infty/p) c_p(x_f) - h_t(x_\infty) c_1(x_f), \\ \rho[(t_p - 1) \otimes h_t] &= 0, \\ \beta[(t_p - 1) \otimes h_t] &= \ell_p \otimes \left(H_t(0), \frac{\sqrt{\pi t}}{8} e^{-1/(4t)} \right) \neq 0. \end{aligned} \quad (\text{ABR28})$$

The last strict nonzero statement uses $t > 0$. Thus it is a nonzero element of the exact kernel ABR17, with both original source moments computed rather than assumed.

68.7 Common-label support and the full SWQ receiving quotient

For every complex vector space U keep

$$G_L(U) = \{(0, \lambda) : \lambda \in L\} \cup (U \times \{1_L\})$$

for the original arbitrary nontrivial bounded distributive lattice. Every linear map T lifts by $T^\uparrow(u, \lambda) = (Tu, \lambda)$, preserving addition with label join, scalar action with label meet, and compositions. The exact coinvariant quotient here is the same-label congruence

$$(m, \lambda) \sim (m', \lambda') \iff \lambda = \lambda' \text{ and } m - m' \in IM_0. \quad (\text{ABR29})$$

Its quotient is $G_L(Q_0)$. The further map $\rho^\uparrow$ has fibres with the same labels and amplitude differences in $j(\mathcal{B}_{\text{pr}})$. Therefore a top-labelled nonzero prime class from ABR24 or ABR28 maps to $(0, 1_L) = e$, not to $\tau = (0, 0_L)$. For a general zero-amplitude label its image is $(0, \lambda)$ with that same label. The joint map $\Omega^\uparrow$ is an isomorphism onto $G_L(V^K \oplus \mathcal{B}_{\text{pr}})$; the entire direct sum has one common label, not independently chosen component labels.

The Koszul amplitude complexes may be retained with these same common-label lifts. Their composites satisfy

$$d^\uparrow d^\uparrow(z, \lambda) = (0, \lambda) = e \cdot (z, \lambda). \quad (\text{ABR30})$$

Thus they are not ordinary complexes with a constant- τ zero map. The amplitude-zero cycles are exactly $G_L(\ker d)$, and the same-label congruence by $\text{im } d$ gives $G_L(H_n(G, M_0))$. Under the induced periodization the lifted classes ABR22 map to their retained zero labels in all positive degrees. This states the receiving maps without discarding the difference between amplitude zero and global absence.

Finally the range V^K in ABR17 is contained in the actual trivial-correspondence image V , not in a complementary spectral space. Let B_{CC} denote the ambient strong test algebra of SWQ and keep its original quotient $Q_{\text{CC}} = B_{\text{CC}}/\overline{V}$. The further composition

$$Q_0 \xrightarrow{\rho} V^K \longrightarrow B_{\text{CC}} \longrightarrow Q_{\text{CC}}$$

is the zero linear map because $V^K \subset V$. Its common-label lift retains each $(0, \lambda)$, in accordance with SWQ's same-label quotient; it does not identify e and τ . The exact faithful source receiver is ABR18, which retains the boundary rather than asserting an injection into this spectral quotient. No assertion that V is closed, no new metric on the prime boundary, and no positivity or Riemann-hypothesis theorem is made by this calculation.

69 The complete rational mixed boundary kernel

Original inputs and source provenance. The actual restriction map and its two zero endpoint conditions are those of Alain Connes, Caterina Consani and Matilde Marcolli, *The Weil proof and the geometry of the adeles class space*, original author source `Yuri70v2.tex`, equations `TrMbf0`, `rangeCvdef`, `SCK` and Lemma `degmodifyV`. The full Gaussian-polynomial generator is retained below. `HCB1–33` in `HURWITZ_CC_TEST_SPACE_BOUNDARY.tex` proves the original Hurwitz boundary coefficients with uniform differentiated remainders; those exact formulas, reproduced where used, are the analytic input. `RMB1–10` in `CC_RATIONAL_MIXED_BOUNDARY.tex` constructs rational reflected antecedents and their full compact-unit character receiver. The present result classifies every finite mixed boundary with commensurable rational parameters; it does not replace the original functions, their scales, or the compact-unit coordinate.

The complete public predecessor proofs are `HCB1–33` and `RMB1–10`, at the pinned edition whose expanded source matches the retained local reader.

69.1 All original parameters and the exact frequency map

Keep $b_0 > 0$ unchanged. Let $J \geq 1$, $a_j = 1 + t_j \in \mathbb{Q}_{>0}$, $r_j = b_j/b_0 \in \mathbb{Q}_{>0}$, and $\alpha_j \in \mathbb{C}$ for $1 \leq j \leq J$. Zero coefficients and repeated parameters are allowed. Define the original functions, without suppressing either summand, by

$$\begin{aligned} \eta(x) &= \pi x^2 (\pi x^2 - \tfrac{3}{2}) e^{-\pi x^2}, & g(x) &= 2\eta(x), \\ S(y) &= \sum_{j=1}^J \alpha_j \Psi_{t_j, b_j}(y) = \sum_{j=1}^J \alpha_j \sum_{n \geq 0} \left[2\pi^2 \left(\frac{(n+a_j)y}{b_j} \right)^4 - 3\pi \left(\frac{(n+a_j)y}{b_j} \right)^2 \right] e^{-\pi((n+a_j)y/b_j)^2}, & y > 0. \end{aligned} \quad (\text{RMK1})$$

All sums converge absolutely, locally with every derivative, by Gaussian bounds; the finite sum in j introduces no limit interchange.

Choose a positive integer D divisible by the denominators of $1/r_j$ and a_j/r_j for every j . For example their least common multiple is an explicit permitted choice. Put

$$p_j = Da_j/r_j \in \mathbb{Z}_{>0}, \quad q_j = D/r_j \in \mathbb{Z}_{>0}, \quad B = Db_0, \quad Q = \text{lcm}(q_1, \dots, q_J), \quad K = \max_{1 \leq j \leq J} \max(p_j - q_j, 0). \quad (\text{RMK2})$$

The original frequency comparison is the equality $(n+a_j)/b_j = (q_j n + p_j)/B$. Thus $n \mapsto q_j n + p_j$ is an exact bijection onto the positive progression starting at p_j . The full data $(\alpha_j, a_j, b_j, p_j, q_j)$ remain recorded; aggregating coincident frequencies does not identify their original presentations. Set, for every positive integer k ,

$$\begin{aligned} w_k &= \sum_{j=1}^J \alpha_j \mathbf{1}_{\{k \geq p_j, \, k \equiv p_j \pmod{q_j}\}}, \\ v_k &= \sum_{j=1}^J \alpha_j \mathbf{1}_{\{k \equiv p_j \pmod{q_j}\}}, \\ d_k &= - \sum_{j=1}^J \alpha_j \mathbf{1}_{\{1 \leq k < p_j, \, k \equiv p_j \pmod{q_j}\}}. \end{aligned} \quad (\text{RMK3})$$

Extend v to every integer by the same formula. It is Q -periodic; $v_0 = v_Q$ is its residue-zero value, not its mean. Each omitted positive frequency in the j th progression is at most $p_j - q_j$. Consequently $d_k = 0$ for $k > K$, and

$$S(y) = \sum_{k \geq 1} v_k g(ky/B) + \sum_{k=1}^K d_k g(ky/B), \quad w_k = v_k + d_k. \quad (\text{RMK4})$$

In particular no finite initial term was suppressed. If $p_j \leq q_j$ the corresponding initial defect is empty, including the case $p_j = q_j$. The frequency zero is absent from RMK1–4 because every $a_j > 0$; this does not impose $v_0 = 0$.

The exact ordinary generating function, for $|z| < 1$, and its rational continuation are

$$W(z) = \sum_{k \geq 1} w_k z^k = \sum_{j=1}^J \frac{\alpha_j z^{p_j}}{1 - z^{q_j}} = \frac{P(z)}{1 - z^Q} + D_+(z), \quad (\text{RMK5})$$

$$P(z) = \sum_{r=1}^Q v_r z^r, \quad P^*(z) = z^Q P(1/z) = \sum_{r=1}^Q v_r z^{Q-r}, \quad D_+(z) = \sum_{k=1}^K d_k z^k.$$

These identities follow by summing convergent geometric series in each original progression and then the Q residue classes. They therefore give exact rational identities, including any cancellations caused by the original weights.

69.2 The full boundary generating function and its finite criterion

Use the Bernoulli polynomials with their original generating function $xe^{ax}/(e^x - 1) = \sum_{n \geq 0} B_n(a)x^n/n!$. HCB11–12 gives the actual differentiated asymptotic expansion of S with coefficients

$$C_m = \sum_{j=1}^J \alpha_j \frac{(-1)^{m+1} \pi^m B_{2m+1}(a_j)}{(m-1)! b_j^{2m}} = \frac{(-1)^{m+1} \pi^m}{(m-1)! B^{2m}} \beta_m, \quad m \geq 1,$$

$$\beta_m = \sum_{j=1}^J \alpha_j q_j^{2m} B_{2m+1}(p_j/q_j) \quad (\text{RMK6})$$

$$= Q^{2m} \sum_{r=1}^Q v_r B_{2m+1}(r/Q) - (2m+1) \sum_{k=1}^K d_k k^{2m}.$$

The second expression for β_m follows either by substituting RMK4 and the Taylor series $g(x) = \sum_{m \geq 1} (-1)^m \pi^m (2m+1) x^{2m}/(m-1)!$, or by the Bernoulli shift identity in every progression. Every factor in the original C_m is retained in its exact comparison with β_m . For a finite sum, HCB's uniform remainder bounds may be summed; in particular

$$S \in \mathcal{B} \iff C_m = 0 \text{ for every } m \geq 1, \quad \mathcal{B} = \bigcap_{\gamma \in \mathbb{R}} y^\gamma \mathcal{S}(\mathbb{R}_+^*). \quad (\text{RMK7})$$

For clarity, the reverse implication here is analytic: the zero coefficients and the differentiated remainder of arbitrary order give $|(y \partial_y)^r S(y)| = O(y^N)$ at zero for every r, N . The Gaussian bounds give all powers of decay at infinity. These are precisely the strong-space seminorms, including every derivative. The forward implication follows by successively taking the actual limits defining the asymptotic coefficients.

The full boundary generating function, including its constant, is

$$\mathcal{A}(u) = W(e^{u/B}) + W(e^{-u/B})$$

$$= \sum_{j=1}^J \alpha_j \left[\frac{e^{a_j u/b_j}}{1 - e^{u/b_j}} + \frac{e^{-a_j u/b_j}}{1 - e^{-u/b_j}} \right] \quad (\text{RMK8})$$

$$= C + \sum_{m=1}^{\infty} \frac{2(-1)^m (m-1)!}{\pi^m (2m+1)!} C_m u^{2m}, \quad C = \sum_{j=1}^J \alpha_j (1 - 2a_j).$$

Each pair cancels its opposite simple principal parts at $u = 0$; the resulting function is holomorphic there. Substitution into the Bernoulli generating function proves the Taylor expansion for

$|u| < 2\pi \min_j b_j$. Thus every positive even coefficient is specified, rather than merely a selected finite subset. As a function of $z = e^{u/B}$, the same expression is

$$W(z) + W(1/z) = \frac{P(z) - P^*(z)}{1 - z^Q} + D_+(z) + D_+(1/z). \quad (\text{RMK9})$$

The complete mixed boundary criterion is

$$\boxed{C_m = 0 \ (m \geq 1) \iff d_k = 0 \ (1 \leq k \leq K), \quad v_r = v_{Q-r} \ (1 \leq r < Q).} \quad (\text{RMK10})$$

To prove necessity, RMK8 makes the left side equivalent to $\mathcal{A}(u) = C$ near zero. The rational function in RMK9 therefore equals C identically: after clearing denominators, a polynomial vanishes on an open set and hence has all coefficients zero. At $z = 0$ the first term on the right of RMK9 is regular, as is $D_+(z)$. Its only possible negative powers are exactly $D_+(1/z) = \sum_{k=1}^K d_k z^{-k}$. Hence every d_k is zero. The remaining polynomial identity is $P - P^* = C(1 - z^Q)$. Its constant coefficient is $-v_Q$, so $C = -v_Q = -v_0$; the coefficients of z^r , $1 \leq r < Q$, are $v_r - v_{Q-r}$ and must be zero. Conversely these conditions give $P - P^* = -v_0(1 - z^Q)$ and $D_+ = 0$. RMK9 is then constant, so RMK8 proves every $C_m = 0$. This includes all complex cancellations between different original shifts and different original scales.

Here is a finite list of exact linear equations on the original weights, requiring no test of an infinite sequence:

$$\begin{aligned} \sum_{j=1}^J \alpha_j \mathbf{1}_{\{k < p_j, \ k \equiv p_j \ (q_j)\}} &= 0, \quad 1 \leq k \leq K, \\ \sum_{j=1}^J \alpha_j \left[\mathbf{1}_{\{r \equiv p_j \ (q_j)\}} - \mathbf{1}_{\{-r \equiv p_j \ (q_j)\}} \right] &= 0, \quad 1 \leq r \leq \lfloor (Q-1)/2 \rfloor. \end{aligned} \quad (\text{RMK11})$$

The first family is exactly $d_k = 0$ and the second is the nonrepeated part of the periodic reflection equations. The classes 0, and $Q/2$ when Q is even, are fixed by reflection and impose no amplitude equation. Their weights remain part of P and of the original function; they have not been set to zero. RMK11 is a finite matrix with integer entries acting on the original α_j .

There is also a sharp finite bound on the number of initial boundary coefficients that suffice. Set

$$M = K + \lfloor (Q-1)/2 \rfloor. \quad C_1 = \cdots = C_M = 0 \iff C_m = 0 \text{ for every } m \geq 1. \quad (\text{RMK12})$$

When $M = 0$ the left side is the empty condition. To prove this, form the actual polynomial

$$N(z) = z^K [P(z) - P^*(z) + (1 - z^Q)\{D_+(z) + D_+(1/z) - C\}]. \quad (\text{RMK13})$$

All its powers are nonnegative and its degree is at most $2K + Q$. The constant in RMK8 is the actual value at $z = 1$. If the first M positive even coefficients vanish, its even holomorphic germ has $\mathcal{A}(u) - C = O(u^{2M+2})$. Since $u = B \log z$ has a simple zero at $z = 1$, multiplying by $z^K(1 - z^Q)$ shows that N vanishes at $z = 1$ to order at least $2M + 3$. This exceeds $2K + Q$ for both parities of Q . A polynomial of degree at most $2K + Q$ cannot have such a zero unless it is zero. RMK9–10 now give the result. For $Q = 1$ or $Q = 2$ and $K = 0$, all periodic weights are already even, and the empty condition correctly certifies the full kernel. For any original family, either RMK11 or the rational matrix $q_j^{2m} B_{2m+1}(p_j/q_j)$, $1 \leq m \leq M$, therefore computes the complete kernel by finite exact linear algebra.

The bound has the following precise dimension meaning. On the space of arbitrary Q -periodic amplitudes v and defects $d_1, \dots, d_K$, the obstruction coordinates are

$$\delta_r = v_r - v_{Q-r} \quad (1 \leq r \leq h), \quad h = \lfloor (Q-1)/2 \rfloor, \quad (d_1, \dots, d_K), \quad (\text{RMK14})$$

and the first $M = h + K$ comparison coefficients are

$$\beta_m = Q^{2m} \sum_{r=1}^h \delta_r B_{2m+1}(r/Q) - (2m+1) \sum_{k=1}^K d_k k^{2m}, \quad 1 \leq m \leq M. \quad (\text{RMK15})$$

This square rational matrix is invertible. Indeed RMK12 makes a zero output force a zero full boundary, and RMK10 then makes every δ_r, d_k zero; an injective linear map between finite spaces of equal dimension is invertible by elimination. Every such input occurs in an actual finite rational mixture: its periodic part is $\sum_{r=1}^Q v_r \Psi_{r/Q-1, B/Q}$, and $g(ky/B) = \Psi_{k-1, B} - \Psi_{k, B}$ represents each finite defect. Thus the dimension of this obstruction space is exactly M . Fewer than M scalar linear boundary tests cannot be injective on the whole space, by elementary rank. Symmetric periodic amplitudes are retained in (v, d) as kernel coordinates; RMK14 is not a replacement for that full original object.

For an explicit nonzero finite defect concealed by the first test, take $Q = 3$, $v_1 = 1$, $v_2 = v_3 = 0$, and $d_1 = 1/9$. Then

$$S(y) = \Psi_{-2/3, B/3}(y) + \frac{1}{9}[\Psi_{0, B}(y) - \Psi_{1, B}(y)], \quad \beta_1 = \frac{1}{3} - 3 \cdot \frac{1}{9} = 0, \quad \beta_2 = -\frac{5}{3} - 5 \cdot \frac{1}{9} = -\frac{20}{9}. \quad (\text{RMK16})$$

Here $K = 1$ and $M = 2$. Its second original boundary coefficient is $C_2 = 20\pi^2/(9B^4) \neq 0$. This calculation exhibits the finite initial term explicitly and shows why a first-coefficient test alone does not classify mixed families.

69.3 The exact actual adelic receiver and its endpoints

Suppose the equivalent conditions RMK10 hold. Define the finite locally constant function $\phi_v(x_f) = \mathbf{1}_{\widehat{\mathbb{Z}}}(x_f)v(x_f \bmod Q)$ on $\mathbb{A}_{\mathbb{Q}, f}$ and the actual adelic test

$$\xi_v(x_f, x_\infty) = \phi_v(x_f)\eta(x_\infty/B), \quad F_v(u, y) = \text{Tr } \rho(\xi_v)(u, y). \quad (\text{RMK17})$$

Every residue class modulo Q in $\widehat{\mathbb{Z}}$ has additive Haar mass $1/Q$. HCB3 computes $\eta(0) = 0$ and $\int_{\mathbb{R}} \eta = 3/4 - 3/4 = 0$. Therefore, retaining the two possibly nonzero finite weights,

$$\xi_v(0) = v_0\eta(0) = 0, \quad \int_{\mathbb{A}_{\mathbb{Q}}} \xi_v = \left(\frac{1}{Q} \sum_{r=1}^Q v_r \right) B \int_{\mathbb{R}} \eta = 0. \quad (\text{RMK18})$$

Thus $\xi_v \in \mathcal{S}(\mathbb{A}_{\mathbb{Q}})_0$ in the original CCM source, and F_v lies in its actual restriction image $\mathcal{V} \subset \mathbf{S}(C_{\mathbb{Q}})$. Substitution into that restriction map gives the complete receiving formula

$$F_v(u, y) = \sum_{k \in \mathbb{Z} \setminus \{0\}} v(ku \bmod Q) \eta(ky/B) = \sum_{k \geq 1} v(ku \bmod Q) g(ky/B), \quad F_v(1, y) = S(y). \quad (\text{RMK19})$$

Only integer rationals contribute, since αu is integral at every finite place if and only if $\alpha \in \mathbb{Z}$. The second equality uses the proved evenness $v(-k) = v(k)$ and the evenness of η . Both signs of k supply exactly the factor $g = 2\eta$. The last equality uses $d = 0$ as well. This proves the existence of the actual finite adelic antecedent for every classified kernel function, with u retained, rather than merely a matching Mellin symbol.

For direct comparison with RMB1–3, take the pair with $a = r/q$, $1 - a = (q - r)/q$ and scale b , and choose $B = qb$. Its even weights are $v_k = \mathbf{1}_{k \equiv r \pmod{q}} + \mathbf{1}_{k \equiv -r \pmod{q}}$. If $\xi_{r, q, b} = 2\eta(x_\infty/b) \mathbf{1}_{r/q + \widehat{\mathbb{Z}}}(x_f)$ is RMB's antecedent, then

$$\xi_v(x) = \frac{1}{2} [\xi_{r, q, b}(x/q) + \xi_{r, q, b}(-x/q)], \quad \text{Tr } \rho(\xi_v) = \text{Tr } \rho(\xi_{r, q, b}). \quad (\text{RMK20})$$

The first equality follows directly from the two finite cosets and the unchanged real scale qb . For the second, multiplying the summation variable in $\mathbb{Q}^*$ by $1/q$ or $-1/q$ is a bijection in the

absolutely convergent restriction sum. Hence this comparison holds on the full (u, y) space, before identity slicing. It includes the order-two case by retaining the two identical coset contributions and their factor $1/2$.

All Mellin factors and both endpoint values can also be retained. For $\text{Re } s > 1$, set

$$\begin{aligned} I_g(s) &= \pi^{-s/2} [\Gamma(s/2 + 2) - \tfrac{3}{2}\Gamma(s/2 + 1)], \\ \mathcal{MS}(s) &= I_g(s) \sum_{j=1}^J \alpha_j b_j^s \zeta(s, a_j) \\ &= B^s I_g(s) \left[Q^{-s} \sum_{r=1}^Q v_r \zeta(s, r/Q) + \sum_{k=1}^K d_k k^{-s} \right]. \end{aligned} \tag{RMK21}$$

Absolute integration proves the first equality from RMK1 and the second from RMK4. HCB's continuation retains at $s = -2m$ exactly the residue C_m , including the factors in RMK6. When RMK10 holds, $S \in \mathcal{B}$, so its Mellin integral is entire; RMK21 is its exact meromorphic expression with every cancellation specified, not a newly chosen completed replacement. The finite values at $s = 0, 1$ are, respectively,

$$\begin{aligned} \int_0^\infty S(y) \frac{dy}{y} &= \sum_j \alpha_j (a_j/2 - 1/4) = -C/4 = v_0/4, \\ \int_0^\infty S(y) dy &= \tfrac{1}{4} \sum_j \alpha_j b_j = \frac{B}{4} \left(\frac{1}{Q} \sum_{r=1}^Q v_r \right). \end{aligned} \tag{RMK22}$$

The first two expressions in each line hold for the original mixture before assuming RMK10. For the last expression in the first line use $C = -v_0$ from its proof. For the second line note that the periodic mean is $\sum_j \alpha_j/q_j$ and $b_j = B/q_j$. Neither v_0 nor this mean must vanish.

For an additional direct expression of the cancellation mechanism, define the finite additive Fourier coefficients

$$\widehat{v}_\ell = \frac{1}{Q} \sum_{r=0}^{Q-1} v_r e^{-2\pi i \ell r/Q}, \quad v_r = \sum_{\ell=0}^{Q-1} \widehat{v}_\ell e^{2\pi i \ell r/Q}. \tag{RMK23}$$

Finite geometric-series summation proves the inverse. Evenness of v gives $\widehat{v}_{-\ell} = \widehat{v}_\ell$; this is an equality without conjugation and holds for complex weights. HCB24 proves $\widehat{g} = g$ with Fourier convention $e^{-2\pi i x \xi}$. Poisson summation for each of the finitely many additive characters therefore gives the exact formula

$$S(y) = \frac{B}{2y} \sum_{m \in \mathbb{Z}} \widehat{v}_m g\left(\frac{mB}{Qy}\right) = \frac{B}{y} \sum_{m \geq 1} \widehat{v}_m g\left(\frac{mB}{Qy}\right). \tag{RMK24}$$

To verify the factor, first write $S = \frac{1}{2} \sum_{k \in \mathbb{Z}} v_k g(ky/B)$; its zero summand is exactly $v_0 g(0) = 0$. For the character $e^{2\pi i \ell k/Q}$, Poisson gives $(B/y) \sum_n g((n - \ell/Q)B/y)$. Put $m = Qn - \ell$ and use $\widehat{v}_{-m} = \widehat{v}_m$. This proves both equalities in RMK24. The dual zero term is $\widehat{v}_0 g(0) = 0$, while $\widehat{v}_0 = Q^{-1} \sum_r v_r$ is retained as an independent endpoint quantity in RMK22. No zero residue or zero Fourier weight has been imposed by the classification.

69.4 The compact coordinate, exact convolution receiver, and supports

The full compact coordinate in RMK19 can be replaced invertibly by its finite character tuple, as in RMB4–9. Let $U_Q = (\mathbb{Z}/Q\mathbb{Z})^\times$, with its single element when $Q = 1$, and let χ range over all its

characters. Give it Haar mass $1/\varphi(Q)$ per element, where $\varphi(1) = 1$. Define

$$\begin{aligned} F_\chi(y) &= \frac{1}{\varphi(Q)} \sum_{u \in U_Q} \overline{\chi(u)} F_v(u, y), \\ (P_\chi F_v)(u, y) &= \chi(u) F_\chi(y), \quad F_v(u, y) = \sum_\chi \chi(u) F_\chi(y). \end{aligned} \tag{RMK25}$$

Character orthogonality proves the inverse; it follows by multiplying the sum of a nontrivial character by a unit where its value differs from one, and by the usual finite character basis on the finite abelian group. Each projected term remains an actual restriction image: take the finite linear average of the adelic tests obtained from ξ_v by finite-unit multiplication. Both zero endpoint conditions persist under that average. No averaging of the original F_v to its principal component is performed.

For full level- Q strong functions $F(u, y) = \sum_\chi \chi(u) F_\chi(y)$ and $G(u, y) = \sum_\chi \chi(u) G_\chi(y)$, the original group convolution is

$$(F * G)(u, y) = \sum_\chi \chi(u) \int_0^\infty F_\chi(z) G_\chi(y/z) \frac{dz}{z}. \tag{RMK26}$$

Insert the two finite expansions into its Haar integral. Orthogonality annuls terms with different characters, and the strong estimates make all real integrals absolutely convergent. The identity slice instead sums every character before convolution and thereby introduces mixed products. Thus RMK19 proves an identity slice and RMK25–26 prove the full invertible receiving map; these are not asserted to be the same morphism.

The radially of this particular full receiver also has a finite exact criterion:

$$F_v(u, y) \text{ is independent of } u \iff v_{ku} = v_k \text{ for all } k \in \mathbb{Z}, u \in U_Q. \tag{RMK27}$$

For the only non-immediate direction, the function map $w \mapsto \sum_{k \geq 1} w_k g(ky/B)$ is injective on bounded sequences. If k_0 is the least index with $w_{k_0} \neq 0$, then

$$\lim_{h \rightarrow \infty} h^{-4} e^{\pi k_0^2 h^2} \sum_{k \geq 1} w_k g(kh) = 2\pi^2 k_0^4 w_{k_0} \neq 0. \tag{RMK28}$$

For $h \geq 1$, the absolute tail is bounded by a constant times $\sum_{k > k_0} (k^4 + k^2) e^{-\pi(k^2 - k_0^2)h^2}$, dominated by its convergent value at $h = 1$ and tending termwise to zero. The single leading term has exactly the displayed limit. Apply this injectivity to $w_k = v_{ku} - v_k$ in RMK19 to prove RMK27.

The unit orbits on residues modulo Q are exactly the sets with fixed $\gcd(k, Q)$. To see transitivity, after factoring out that $\gcd d$ choose the prescribed unit ratio modulo Q/d and lift it to a unit modulo Q by the Chinese remainder theorem at each prime power. Thus RMK27 is equivalent to $v_k = A_{\gcd(k, Q)}$ for numbers A_d , $d \mid Q$, with $A_Q = v_0$. In that case put

$$c_d = \sum_{e \mid d} \mu(d/e) A_e, \quad v_k = \sum_{d \mid Q} c_d \mathbf{1}_{d \mid k}, \quad F_v(u, y) = S(y) = \sum_{d \mid Q} c_d f_{B/d}(y). \tag{RMK29}$$

Here μ is the integer Möbius function. The divisor identity $\sum_{e \mid n} \mu(e) = 0$ for $n > 1$ and 1 for $n = 1$, proved by expanding $\prod_{p \mid n} (1 - 1)$, proves the inverse formula for v . Substitution into RMK19 gives the exact scales B/d and every coefficient in RMK29. In the absence of this unit invariance, the nonprincipal even components in RMK25 remain present. No nonexistence claim about some different radial antecedent is made.

Finally retain the full support data. For an arbitrary nontrivial bounded distributive lattice L , use $G_L(W) = \{(0, \lambda) : \lambda \in L\} \cup (W \times \{1_L\})$ with join in addition and meet in scalar multiplication, as in HCB30. A complex-linear map T has the exact common-label lift $G_L(T)(w, \lambda) = (Tw, \lambda)$;

its zero-amplitude inverse image is $G_L(\ker T)$, not an assertion that its output is τ . Thus all finite linear maps RMK3, RMK6, RMK11 and RMK25, and the inverse RMK25, admit these lifts. The original data may also carry separate labels on each coefficient $z_j = (\alpha_j, \lambda_j) \in G_L(\mathbb{C})$. In that case retain the entire tuple $(z_j)_{j=1}^J$ and its exact incidence images:

$$\begin{aligned} \tilde{w}_k &= \sum_{j:k \geq p_j, k \equiv p_j(q_j)} z_j, & \tilde{v}_k &= \sum_{j:k \equiv p_j(q_j)} z_j, \\ \tilde{d}_k &= (-1, 1_L) \sum_{j:1 \leq k < p_j, k \equiv p_j(q_j)} z_j, & \tilde{o}_r &= \tilde{v}_r + (-1, 1_L) \tilde{v}_{-r}. \end{aligned} \quad (\text{RMK30})$$

Empty sums are $\tau = (0, 0_L)$. These are finite sums at each frequency, so no infinite join or completeness of L is assumed. Their amplitudes are the coefficients and equations already proved; their labels are exactly the joins of the contributing original labels. In particular a repeated contribution in the difference $\tilde{o}_r$ remains a supported cancellation and is not deleted as a structural zero. The original labelled tuple is not identified with the generally non-injective collection of amplitude equations. The complete function receiver is

$$\sum_{j=1}^J (\alpha_j \Psi_{t_j, b_j}, \lambda_j) = (S, \Lambda), \quad \Lambda = \bigvee_{j=1}^J \lambda_j. \quad (\text{RMK31})$$

Only this finite join is needed. When evaluating a particular boundary coefficient, use the supported scalar $(\kappa_{m,j}, 1_L)$ with $\kappa_{m,j} = (-1)^{m+1} \pi^m B_{2m+1}(a_j) / ((m-1)! b_j^{2m})$, including its evaluated zero cases. The coefficient receiver is then (C_m, Λ) for every m . When RMK10 holds it is $(0, \Lambda)$; for any nonzero original coefficient α_j this is $e = (0, 1_L)$, not τ . The finite criterion tests amplitudes, preserves every original input label, and does not turn mixed cancellations into absence. The compact-character map and its inverse act on one common-labelled direct sum when that is the chosen input carrier; they do not assign independent labels to components of that common-labelled object.

The incidence construction also gives the exact support correction for RMK4 when the original coefficients have separate labels. Put

$$\begin{aligned} \mu_k &= \bigvee_{j:k \geq p_j, k \equiv p_j(q_j)} \lambda_j, & \sigma_k &= \bigvee_{j:1 \leq k < p_j, k \equiv p_j(q_j)} \lambda_j, \\ \nu_k &= \bigvee_{j:k \equiv p_j(q_j)} \lambda_j = \mu_k \vee \sigma_k, & \tilde{v}_k + \tilde{d}_k &= \tilde{w}_k + (0, \sigma_k). \end{aligned} \quad (\text{RMK32})$$

Each join is finite, and the join of an empty set is 0_L . The periodic incidences partition into their retained and omitted initial incidences, proving $\nu_k = \mu_k \vee \sigma_k$. Both sides of the last equality have amplitude w_k by RMK3; their labels are respectively $\nu_k \vee \sigma_k = \nu_k$ and $\mu_k \vee \sigma_k = \nu_k$. This proves the equality in the actual support carrier. The condition $d_k = 0$ in RMK10 is an amplitude condition and does not set $\sigma_k = 0_L$. Likewise equality of the reflected amplitudes does not erase the label on $\tilde{o}_r$. The retained tuples $(\tilde{w}_k, (0, \sigma_k))$ and the original labelled inputs specify the comparison without identifying a supported cancellation with absence. On the common-label function carrier, RMK4 lifts directly with its one unchanged label; no identification of these two support presentations is needed or asserted.

70 Periodic adelic points with the full lattice coefficient

Human source and exact scope. Alain Connes, Caterina Consani, and Matilde Marcolli, *The Weil proof and the geometry of the adeles class space*, original author file `Yuri70v2.tex`, Section *Thermodynamics and geometry of the primes*, supply the periodic locus and the original one-zero-place adeles. The author file was read at lines 2425–2944, including `globalper`, `preorbitckak`, `classXiadeles`, `avwadele`, `Xiorbits`, `muscalehaar`, `classpointsperorb`, `kmsbeware`, and `XiKalgpts`. The retained archive is version 1; the author’s filename does not change that version. Its SHA-256 is `b1f694cb934dd4fef829287f1a55dd8de21c67a481e552453c2f7e8fe0ae6fb5`. The actual programme coefficient and sheaf maps are those proved in `ARBITRARY_SUPPORT_CC_COEFFICIENTS.tex`, ULC1–10 and ULC17, and `FULL_SUPPORT_SHEAF_COMPARISON.tex`, FSC1–9 and FSC33–34. Those complete local derivations accompany this calculation. All new carrier, orbit, and localization statements below are proved here. No positive-energy, KMS, Hilbert-space positivity, or RH assertion is deduced from a support label.

70.1 The original carrier and the different local-product object

Let L be a nontrivial bounded distributive lattice, with bottom 0_L and top 1_L . It need not be finite. For a commutative ring R retain the constrained carrier and its full operations

$$\begin{aligned} G_L(R) &= \{(0, \lambda) : \lambda \in L\} \cup \{(a, 1_L) : a \in R\}, \\ (a, \lambda) + (b, \mu) &= (a + b, \lambda \vee \mu), \quad (a, \lambda)(b, \mu) = (ab, \lambda \wedge \mu), \\ \tau &= (0, 0_L), \quad e = (0, 1_L), \quad \mathbf{1} = (1, 1_L). \end{aligned} \tag{PLP1}$$

The two descriptions of $(0, 1_L)$ denote the same element. The constraint is $a \neq 0 \Rightarrow \lambda = 1_L$. It is preserved by both displayed operations: if a product has nonzero amplitude both factors have top label, and if a sum has nonzero amplitude at least one summand has top label. Distributivity follows from ring distributivity and lattice distributivity. In particular $e^2 = e$, $e + \tau = e$, and $e \neq \tau$.

Write $\Sigma = \{\infty\} \cup \{p : p \text{ rational prime}\}$ and $\mathbb{A} = \mathbb{A}_{\mathbb{Q}}$. Define separately

$$S_c = G_L(\mathbb{A}), \quad S_{\text{loc}} = G_L(\mathbb{R}) \times \prod_p' G_L(\mathbb{Q}_p), \quad \text{restricted relative to } G_L(\mathbb{Z}_p). \tag{PLP2}$$

The second expression is a newly declared semiring, not another name for the first. More explicitly, its elements are tuples $((a_v, \lambda_v))_{v \in \Sigma}$ for which $a = (a_v) \in \mathbb{A}$ and $a_v \neq 0 \Rightarrow \lambda_v = 1_L$. There is no additional restriction on the zero labels, since every $(0, \lambda_v)$ lies in $G_L(\mathbb{Z}_p)$. The usual finite-exception condition for the amplitudes is preserved under coordinate addition and multiplication; PLP1 therefore proves that this restricted product is a semiring.

There is an exact comparison

$$\Delta : S_c \longrightarrow S_{\text{loc}}, \quad (a, \lambda) \longmapsto ((a_v, \lambda))_v. \tag{PLP3}$$

It is an injective unital semiring homomorphism. Indeed each coordinate respects PLP1; the amplitudes and the common label can be recovered from the tuple. Its image is exactly the tuples with one constant label $\lambda_v = \lambda$ for all v . If that common label is not top, the constraint at each coordinate forces all amplitudes to vanish, which proves the reverse inclusion without enlarging $G_L(\mathbb{A})$. The same map is linear for the diagonal $G_L(\mathbb{Z})$ action.

Let $\rho : S_{\text{loc}} \rightarrow \mathbb{A}$ and $q : S_c \rightarrow \mathbb{A}$ be amplitude projection. Both are surjective semiring homomorphisms and $\rho\Delta = q$. For $Z(a) = \{v : a_v = 0\}$ the exact fibres are

$$\rho^{-1}(a) = L^{Z(a)}, \quad q^{-1}(a) = \begin{cases} \{(a, 1_L)\}, & a \neq 0, \\ \{(0, \lambda) : \lambda \in L\}, & a = 0. \end{cases} \tag{PLP4}$$

The first identification inserts top labels at the coordinates outside $Z(a)$. At $a = 0$ the map induced by Δ is the diagonal $L \hookrightarrow L^\Sigma$. At nonzero a its image is the single point with top label at every coordinate, including every zero coordinate. These are the full fibres, not only selected examples.

There is no amplitude-preserving zero-preserving semiring homomorphism $T : S_{\text{loc}} \rightarrow S_c$. To prove this, fix a place v and define x to have entry $(1, 1_L)$ at v and τ elsewhere; define y to have entry τ at v and $(1, 1_L)$ elsewhere. They belong to the restricted product, and $xy = \tau_{\text{loc}}$. Their amplitude adeles a, b are nonzero, $a + b = 1$, and $ab = 0$. Amplitude preservation forces

$$T(x) = (a, 1_L), \quad T(y) = (b, 1_L), \quad T(x)T(y) = e, \quad T(xy) = \tau. \quad (\text{PLP5})$$

This contradicts $e \neq \tau$. In particular there is no amplitude-preserving semiring retraction of Δ . The injection PLP3 and the fibre comparison PLP4 remain valid despite this obstruction.

70.2 Actual support localization and its infinite-product defect

Both carriers are algebras over $S = G_L(\mathbb{Z})$ via diagonal integers and common labels. Put $E = \Delta(e)$. Their first support localizations are

$$S_c[e^{-1}] \simeq L, \quad S_{\text{loc}}[E^{-1}] \simeq L^\Sigma. \quad (\text{PLP6})$$

Multiplication by e or E erases the amplitudes and preserves every existing label. An invertible idempotent is the unit; hence in either localization each element equals that product. Conversely equality between zero-label tuples is unchanged by multiplying by E , so no two distinct tuples become equal at this stage. Every label tuple is present already with all amplitudes zero. This proves both isomorphisms, including injectivity. The map induced by Δ is the diagonal $L \rightarrow L^\Sigma$.

Let J be a proper prime ideal of the lattice L and $P_J = \{(0, \lambda) : \lambda \in J\} \subset S$. Then

$$\begin{aligned} (S_c)_{P_J} &= L_J, & (S_{\text{loc}})_{P_J} &= \mathcal{L}_J := (L^\Sigma)[\text{diag}(L \setminus J)^{-1}], \\ [\lambda] &= [\mu] \text{ in } \mathcal{L}_J & \iff & \exists \nu \notin J \forall v \in \Sigma : \lambda_v \wedge \nu = \mu_v \wedge \nu. \end{aligned} \quad (\text{PLP7})$$

First invert e , which is outside P_J . Every nonzero top-supported integer then maps to the unit. The remaining denominators are exactly the diagonal zero labels outside J . Products of finitely many such denominators are again one diagonal $\nu \notin J$, because the complement of a prime lattice ideal is closed under meets. Inverting an idempotent identifies it with the unit, so every fraction has a label-tuple numerator. The usual localization equality is consequently precisely the displayed uniform-denominator relation. This also proves the first formula with a single label, as in ULC6.

Coordinate localization gives a surjection

$$\pi_J : \mathcal{L}_J \longrightarrow \prod_{v \in \Sigma} L_J, \quad [\lambda] \longmapsto ([\lambda_v]_J)_v. \quad (\text{PLP8})$$

Surjectivity uses coordinate representatives, available by countable choice. Equality after this map means $\forall v \exists \nu_v \notin J : \lambda_v \wedge \nu_v = \mu_v \wedge \nu_v$; equality before it requires one common ν in PLP7. For a finite set of coordinates the meet of the finitely many ν_v is outside J and is a common denominator. Infinite coordinates do not permit that argument. There is also injectivity for every place set when $L \setminus J$ has a least element ν_0 : every coordinate equality then holds after meeting with ν_0 . In particular this applies to finite L , since the meet of all elements outside J is still outside J . This finite-lattice fact does not extend to arbitrary L .

Here is an explicit failure for the actual countable place set. Take the bounded totally ordered lattice

$$L = \{0_L, 1_L, a_1, a_2, \dots\}, \quad 0_L < \dots < a_3 < a_2 < a_1 < 1_L, \quad J = \{0_L\}. \quad (\text{PLP9})$$

Joins and meets are maximum and minimum; J is prime since a meet of two positive elements is positive. Enumerate the places v_n and set $\lambda_{v_n} = a_n$, $\mu_{v_n} = 1_L$. At coordinate n , meeting with $\nu_n = a_n$ proves equality in L_J . A uniform denominator would have to be a positive element below every a_n , and there is no such element in this displayed lattice. Thus π_J is not injective. The exact defect is the difference between the two quantifier orders in PLP7–PLP8; replacing the left-hand side by the product of stalks loses this defect.

The diagonal $L_J \rightarrow \mathcal{L}_J$ is still injective: equality of two constant tuples by a common denominator is exactly equality in L_J . For tuples which are top except at one fixed place v , PLP7 reduces to

$$[(1, \dots, \lambda \text{ at } v, \dots)] = [(1, \dots, \mu \text{ at } v, \dots)] \iff [\lambda]_J = [\mu]_J. \quad (\text{PLP10})$$

Indeed every other coordinate gives the tautology $\nu = \nu$. The infinite-product defect therefore does not change this exact one-place equality criterion.

70.3 The sheaf comparison over the original split base

Use the notation of ULC1–3: $X = \text{Spec } \mathbb{Z}$, $S_L = \text{Spec}_{\text{lat}} L$, $Y = \text{Spec } G_L(\mathbb{Z}) = S_L \sqcup X$, and $W_U = S_L \cup U$ for nonempty arithmetic opens $U \subset X$. Use the actual structure sheaf $\mathcal{O}_Y$, whose restriction to S_L is $\mathcal{O}_{S_L}$, with $\mathcal{O}_{S_L}(D_L(\lambda)) = \downarrow \lambda$. The adelic point amplitude $\mathbb{A}$ is a rational vector space; no complex scalar multiplication on every $\mathbb{Q}_p$ is asserted. Its constant sheaf on X is an $\mathcal{O}_X$ -module using the inclusion $\mathcal{O}_X(U) \subset \mathbb{Q}$.

The ULC construction on this module gives $\mathcal{M}_{\mathbb{A}}$. Define the separate locally labelled sheaf $\mathcal{N}$ by

$$\begin{aligned} \mathcal{M}_{\mathbb{A}}(W_U) &= S_c, & \mathcal{M}_{\mathbb{A}}(V) &= \mathcal{O}_{S_L}(V), & V &\subset S_L, \\ \mathcal{N}(W_U) &= S_{\text{loc}}, & \mathcal{N}(V) &= \prod_{v \in \Sigma} \mathcal{O}_{S_L}(V), & V &\subset S_L. \end{aligned} \quad (\text{PLP11})$$

Restrictions between arithmetic opens are identities. Restriction to support retains the common label for the first sheaf, and retains every coordinate label for the second, followed by the actual support restrictions. The scalar action on arithmetic opens is diagonal multiplication by $(r, \mu) \in G_L(\mathcal{O}_X(U))$; on support opens it is coordinatewise meet with the one scalar section of $\mathcal{O}_{S_L}$.

These formulas define sheaves of $\mathcal{O}_Y$ -semimodules. For a cover of W_U , its arithmetic members have nonempty pairwise arithmetic intersections because $\text{Spec } \mathbb{Z}$ is irreducible. Compatible sections on them are the same carrier element. Each arithmetic member contains all of S_L , so that compatibility forces every support-only member to be the specified restriction. The sections consequently glue uniquely. Covers within S_L glue by the sheaf axiom for $\mathcal{O}_{S_L}$, coordinate by coordinate for the product. The formulas for the scalar operations commute with all these restrictions, which proves the assertion.

The map Δ extends to a sheaf monomorphism

$$\mathcal{M}_{\mathbb{A}} \hookrightarrow \mathcal{N}; \quad \text{on } W_U \text{ it is PLP3, on } V \subset S_L \text{ it is the diagonal support map.} \quad (\text{PLP12})$$

Both its arithmetic and support section maps are injective and compatible with the restrictions just proved. At a support prime the stalks and comparison are exactly

$$(\mathcal{M}_{\mathbb{A}})_J = L_J, \quad \mathcal{N}_J = \mathcal{L}_J, \quad L_J \hookrightarrow \mathcal{L}_J \xrightarrow{\pi_J} \prod_{v \in \Sigma} L_J. \quad (\text{PLP13})$$

For the second equality, basic neighbourhoods $D_L(\nu)$, $\nu \notin J$, are cofinal. Their sections are $(\downarrow \nu)^\Sigma$ and the restrictions meet every coordinate with the same smaller basic-open label. Every such section is the restriction of its label tuple in L^Σ . Equality of two germs therefore has one basic neighbourhood witnessing equality of all coordinates at once, which is precisely PLP7. This directly proves the stalk statement and the failure of commuting this infinite product with the stalk functor in PLP9.

For each v the same construction with constant rational module $\mathbb{Q}_v$ gives $\mathcal{M}_{\mathbb{Q}_v}$. Coordinate projection yields a commuting diagram of actual sheaf maps:

$$\mathcal{M}_{\mathbb{A}} \longrightarrow \mathcal{N} \longrightarrow \mathcal{M}_{\mathbb{Q}_v}, \quad (a, \lambda) \longmapsto (a_v, \lambda). \quad (\text{PLP14})$$

At arithmetic points the second arrow projects the v entry. At support points it is $\mathcal{L}_J \rightarrow L_J$, with composite $L_J \rightarrow L_J$ the identity. Thus the original support map $(a, \lambda) \mapsto [\lambda]_J$ of ULC8 and FSC34 is retained exactly. Its target is not the unchanged adele amplitude.

70.4 All labels above the classical periodic points

Write $I = \mathbb{A}^\times$, $C = I/\mathbb{Q}^\times$. The units of both carriers are exactly the top-supported ideles. For a unit, the label of it and of its inverse must have meet 1_L , so both labels at every coordinate are top. The amplitudes must be mutual inverses in the adele ring. Conversely an idele and its inverse with top labels multiply to $\mathbf{1}$. Requiring every local amplitude to be nonzero alone would be insufficient: its inverse must also satisfy the restricted-product condition.

Multiplication by $j \in I$ acts on either carrier by $(a, \lambda) \mapsto (ja, \lambda)$, coordinatewise in the local carrier. These are additive semimodule automorphisms and commute with Δ and amplitude projection. They are not asserted to be ring automorphisms for pointwise multiplication; for example $j(ab)$ need not equal $(ja)(jb)$. The quotient objects below are orbit sets, not semiring quotients.

The maps induced on rational orbit sets are

$$S_c/\mathbb{Q}^\times \hookrightarrow S_{\text{loc}}/\mathbb{Q}^\times \longrightarrow \mathbb{A}/\mathbb{Q}^\times. \quad (\text{PLP15})$$

The first remains injective: an equality between two images by one rational multiplier is, by equivariance and injectivity of PLP3, the same equality in the original carrier. The image consists of constant-label orbits. The fibre over a nonzero amplitude orbit $[a]$ is a singleton for the first carrier and $L^{Z(a)}$ for the local carrier. A nonzero adele has trivial rational stabilizer, since any one of its nonzero coordinates forces $q = 1$; labels are unchanged by multiplication. Hence the labels in this fibre are independent of the representative. The fibres over $[0]$ are respectively L and L^Σ .

For a place v , the source's exact adele is

$$a_w^{(v)} = \begin{cases} 0, & w = v, \\ 1, & w \neq v. \end{cases} \quad (\text{PLP16})$$

It is nonzero as an adele. Its only lift in S_c is $(a^{(v)}, 1_L)$, and Δ assigns the supported zero $e = (0, 1_L)$ to its v coordinate. In S_{loc} the whole fibre is the explicitly different family

$$a_\lambda^{(v)} : \quad (0, \lambda) \text{ at } v, \quad (1, 1_L) \text{ at } w \neq v, \quad \lambda \in L. \quad (\text{PLP17})$$

Only $\lambda = 1_L$ is in the image of the original carrier. The member with $\lambda = 0_L$ exists in the enlarged local carrier; it is not a second lift inside $G_L(\mathbb{A})$. All original global zero elements $(0, \lambda)$ and all local zero-amplitude tuples $((0, \lambda_v))_v$ are also present. Every idele fixes them, including e and τ as distinct points. At a support prime the original lift of $a^{(v)}$ maps to 1 in L_J ; the local lifts map to their one-place label tuples in $\mathcal{L}_J$, with the exact equality relation PLP10.

For completeness, every nonzero amplitude a has rational stabilizer $\{1\}$ and raw idele stabilizer

$$I_{Z(a)} = \{j \in I : j_w = 1 \text{ for every } w \notin Z(a)\}. \quad (\text{PLP18})$$

At $[a]$, its I stabilizer is $\mathbb{Q}^\times I_{Z(a)}$ and its C stabilizer is the injective image of $I_{Z(a)}$. Indeed $ja = qa$ forces $j_w = q$ at every nonzero coordinate, and imposes no equation at zero coordinates. Moreover a rational idele supported on $Z(a)$ equals 1 at a nonzero coordinate, so its rational

value is 1. These arguments do not depend on the labels. Thus a nonzero orbit has a nonidentity C stabilizer exactly when at least one coordinate vanishes. This proves the support-labelled version of the source's periodic-locus assertion without selecting one preferred label. The all-zero cases have the whole group as stabilizer.

In particular, if $j_v(h)$ denotes the idele equal to h at v and 1 elsewhere, then for every lift in PLP16–PLP17,

$$\begin{aligned} \text{Stab}_{\mathbb{Q}^\times}(a_\lambda^{(v)}) &= \{1\}, & \text{Stab}_I(a_\lambda^{(v)}) &= j_v(\mathbb{Q}_v^\times), \\ \text{Stab}_I([a_\lambda^{(v)}]) &= \mathbb{Q}^\times j_v(\mathbb{Q}_v^\times), & \text{Stab}_C([a_\lambda^{(v)}]) &= \iota_v(\mathbb{Q}_v^\times), \\ O_v &= C/\iota_v(\mathbb{Q}_v^\times). \end{aligned} \tag{PLP19}$$

Here ι_v is injective by the rational-intersection argument. For the original carrier the subscript λ means its sole value 1_L . At the rational quotient every rational already acts trivially; the first formula is a statement about the raw point. The local enlarged carrier has disjoint label-indexed copies $\{O_{v,\lambda} : \lambda \in L\}$, all with these stabilizers. The original carrier has precisely $O_{v,1_L}$.

70.5 The real flow, compact factors, and every orientation

The canonical section of the idele class group gives $C = \mathbb{R}_{>0} \times K$ with $K = \prod_\ell \mathbb{Z}_\ell^\times$. To prove this directly, for an idele j multiply by the rational $\text{sgn}(j_\infty) \prod_p p^{-v_p(j_p)}$. Only finitely many exponents are nonzero. Its real coordinate is positive and all its finite coordinates are units. A rational number which preserves these conditions has every finite valuation zero and positive sign, hence is 1. The representative is unique, is compatible with multiplication, and its real coordinate is the full idele norm $|j_\infty| \prod_p |j_p|_p$. The section and its inverse are continuous: in the idele topology the finite valuation tuple and real sign are locally constant, and on such a neighbourhood the section is multiplication by one fixed rational number followed by continuous coordinate maps.

For $h = p^n \epsilon \in \mathbb{Q}_p^\times$, $\epsilon \in \mathbb{Z}_p^\times$, the exact image is

$$\iota_p(p^n \epsilon) = (p^{-n}, u_p = \epsilon, u_\ell = p^{-n} \quad (\ell \neq p)). \tag{PLP20}$$

This follows by multiplying $j_p(h)$ by the rational p^{-n} , not by changing the norm convention. Put $K^{(p)} = \prod_{\ell \neq p} \mathbb{Z}_\ell^\times$ and $\ell_p = \log p$. Quotienting by the p -unit part of the stabilizer and using the generator in PLP20 gives the homogeneous-space presentation

$$O_p = (\mathbb{R} \times K^{(p)}) / ((x, u) \sim (x + \ell_p, pu)). \tag{PLP21}$$

Indeed the stabilizer generator in these coordinates is $(-\ell_p, p^{-1})$, whose inverse gives the displayed relation. Specify the forward real flow by $g_t = (e^t, 1) \in C$. It acts by $[x, u] \mapsto [x + t, u]$, so the return to the $x = 0$ transversal after positive time ℓ_p is exactly

$$[\ell_p, u] = [0, p^{-1}u], \quad u \mapsto p^{-1}u. \tag{PLP22}$$

Multiplication by p would describe the opposite time or the opposite return convention.

The full O_p has no nonzero pure-real period. If $(e^t, 1) = \iota_p(p^n \epsilon)$, the p coordinate forces $\epsilon = 1$, and any finite $\ell \neq p$ forces $p^{-n} = 1$ as an element of $\mathbb{Q}_\ell$, whence $n = 0$ and $t = 0$. Since C is abelian this applies at every point. Its compact quotient instead has

$$O_p/K = \mathbb{R}_{>0}/p^\mathbb{Z} = \mathbb{R}/(\log p)\mathbb{Z}, \tag{PLP23}$$

with least positive forward period $\log p$. It is this compact quotient which the original CCM discussion after `XiKalgpts` identifies as the periodic real-flow orbit. This distinction holds for every label copy, without changing any norm.

At the infinite place the exact formula is

$$\iota_\infty(h) = (|h|, (\text{sgn } h)_p), \quad O_\infty = K/\{1, -1\}, \quad O_\infty/K = \{\text{point}\}. \tag{PLP24}$$

Multiply the original idele by the rational sign of h to obtain this representative. The whole positive-real subgroup lies in its stabilizer and fixes O_∞ pointwise. Equations PLP21–PLP24 give the quotient-group topology of each homogeneous orbit. Its orbit map to the original $\mathbb{A}/\mathbb{Q}^\times$ is continuous. We retain that ambient quotient topology, as the source requires; the equations do not replace the embedded union by a disjoint-union topology. For arbitrary L , no additional topology on its element set is postulated. Each label-preserving copy has the specified homogeneous orbit parametrization; the base's support topology remains the actual lattice spectrum from PLP11–PLP14.

70.6 Exact normal coordinates, their Haar factors, and their labels

Put $\mathbb{A}_v = \{a \in \mathbb{A} : a_v = 0\}$ and let $b_v : \mathbb{Q}_v \rightarrow \mathbb{A}$ insert a coordinate at v and zero elsewhere. With the original additive topology and measures,

$$\mathbb{A} = \mathbb{A}_v \oplus b_v(\mathbb{Q}_v), \quad \mathbb{A}/\mathbb{A}_v = \mathbb{Q}_v, \quad a^{(v)}(z) = a^{(v)} + b_v(z). \quad (\text{PLP25})$$

Projection and insertion are continuous and inverse to the displayed product decomposition: all finite restricted-product conditions are unchanged by altering one coordinate. The map $z \mapsto [a^{(v)}(z)]$ into the rational orbit space is injective, because a rational identification forces that rational to equal 1 at every coordinate $w \neq v$. For $h \in \mathbb{Q}_v^\times$,

$$j_v(h)a^{(v)}(z) = a^{(v)}(hz), \quad (1-h) : z \mapsto (1-h)z, \quad d_v((1-h)z) = |1-h|_v d_v z \quad (h \neq 1). \quad (\text{PLP26})$$

For $h \neq 1$ the difference map has inverse multiplication by $(1-h)^{-1}$. Its exact additive-Haar determinant is $|1-h|_v$, using real Lebesgue measure or $\text{vol}(\mathbb{Z}_p) = 1$, respectively.

One can retain the whole transverse family as an associated action space

$$\mathcal{T}_v = C \times_{\iota_v(\mathbb{Q}_v^\times)} \mathbb{Q}_v, \quad (g\iota_v(h), z) \sim (g, hz), \quad [g, z] \mapsto [g a^{(v)}(z)]. \quad (\text{PLP27})$$

The last expression uses an idele representative of g and is independent of that representative. The relation is respected by PLP26. At $z = 0$ the map is the bijection with O_v of PLP19. At $z \neq 0$, $a^{(v)}(z) = j_v(z)$ is an idele, and the map is $[g, z] \mapsto g\iota_v(z) \in C$. This is a bijection: equality of the products implies exactly the displayed associated-space relation. No point of this part can equal a zero-coordinate point. Continuity follows from the continuous map $I \times \mathbb{Q}_v \rightarrow \mathbb{A}/\mathbb{Q}^\times$, $(j, z) \mapsto [ja^{(v)}(z)]$. It is unchanged by $j \mapsto qj$. The quotient $I \rightarrow C$ is open, since the inverse image of the image of an open subset is its union of rational translates. Its product with the identity of $\mathbb{Q}_v$ is open and is a quotient map. Hence the displayed map descends continuously to $C \times \mathbb{Q}_v$, then through the associated relation. Corestriction to its image with the ambient subspace topology is continuous. Thus PLP27 is a continuous bijection onto $C \cup O_v$ inside the adelic orbit space. No assertion that its inverse is continuous in the ambient orbit topology is required here.

At a finite prime write the full representative as $g = (e^x, \epsilon, u)$ with $\epsilon \in \mathbb{Z}_p^\times$ and $u \in K^{(p)}$. Since $\iota_p(\epsilon) = (1, \epsilon, 1)$, the associated relation gives exactly

$$[(e^x, \epsilon, u), z_{\text{old}}] = [(e^x, 1, u), z_{\text{new}}], \quad z_{\text{new}} = \epsilon z_{\text{old}}, \quad z_{\text{old}} = \epsilon^{-1} z_{\text{new}}, \quad |\epsilon|_p = 1.$$

Thus the original p -unit is recorded in the normal coordinate, with its exact Haar factor retained. Writing $z = z_{\text{new}}$, the remaining representative coordinates (x, u, z) obey

$$(x + \ell_p, pu, z) \sim (x, u, p^{-1}z), \quad (u, z) \mapsto (p^{-1}u, p^{-1}z) \quad \text{after time } +\ell_p. \quad (\text{PLP28})$$

To check the sign, changing (x, u) by $(+\ell_p, p)$ multiplies its class by $\iota_p(p^{-1})$. The associated relation then multiplies z by p^{-1} . For every positive integer n the two opposite-direction normal determinants are

$$|1 - p^n|_p = 1, \quad |1 - p^{-n}|_p = p^n, \quad |1 - p^n|_p^{-1} = 1, \quad |1 - p^{-n}|_p^{-1} = p^{-n}. \quad (\text{PLP29})$$

The first follows because $1 - p^n$ is a p -adic unit; the second follows by writing $1 - p^{-n} = (p^n - 1)/p^n$. The forward return is the second case. Negative iterates exchange these cases; the zeroth iterate has zero difference map and is excluded from the displayed inverse-determinant formula.

For comparison with a trace on test functions retain the inverse in the pullback action: $\vartheta(h)f(z) = f(h^{-1}z)$. Its additive distribution kernel is $\delta(y - h^{-1}z)$. At $h \neq 1$ the transverse diagonal is

$$\int_{\mathbb{Q}_v} \delta((1 - h^{-1})z) d_v z = |1 - h^{-1}|_v^{-1} = \frac{|h|_v}{|1 - h|_v}. \quad (\text{PLP30})$$

The first equality is the defining change of variables for the Dirac distribution under an invertible scalar; it can equally be checked against a compactly supported test that is one near zero. For a compactly supported coefficient b on $\mathbb{Q}_v^\times$ whose support avoids 1, multiplicative Haar measure is invariant under inversion, so

$$\int_{\mathbb{Q}_v^\times} \frac{b(h)}{|1 - h^{-1}|_v} d^\times h = \int_{\mathbb{Q}_v^\times} \frac{b(h^{-1})}{|1 - h|_v} d^\times h. \quad (\text{PLP31})$$

This proves the relation between the inverse test argument and the normal denominator away from the singular identity. It is not a replacement for the source's regularization there or for its global trace theorem.

The original hyperplane Haar factor is also retained in full: for $r \in \mathbb{Q}^\times$,

$$d_{\mathbb{A}_v}(ra) = \left(\prod_{w \neq v} |r|_w \right) d_{\mathbb{A}_v} a = |r|_v^{-1} d_{\mathbb{A}_v} a, \quad |r|_\infty \prod_p |r|_p = 1. \quad (\text{PLP32})$$

Only finitely many finite-place factors differ from 1. Writing $r = \pm \prod_p p^{n_p}$ gives the last product exactly as $(\prod_p p^{n_p})(\prod_p p^{-n_p}) = 1$. The product decomposition PLP25 then proves the first formula. This is the factor in the human source's `muscalehaar`; no measure on support labels has been inserted.

The supported normal-coordinate map is the surjective unital semiring map

$$\nu_v^\uparrow : G_L(\mathbb{A}) \longrightarrow G_L(\mathbb{Q}_v), \quad (a, \lambda) \longmapsto (a_v, \lambda). \quad (\text{PLP33})$$

A lift is $(b_v(z), \lambda)$; when $z \neq 0$ the original constraint already forces $\lambda = 1_L$. The insertion is additive and multiplicative but is not unital: its unit is the adèle idempotent supported at v . The exact equality relation for PLP33 is $(a, \lambda) \sim (c, \mu)$ if and only if $\lambda = \mu$ and $a - c \in \mathbb{A}_v$. This is a semiring congruence, since coordinate projection respects addition and multiplication. Thus its same-label quotient is exactly $G_L(\mathbb{Q}_v)$; a quotient which also identifies different zero labels is a different construction. The map is PLP14 on arithmetic stalks. On support stalks it is the identity of L_J . It is rationally equivariant, not rationally invariant, and therefore is not asserted to give an unquotiented local-field coordinate on all of $\mathbb{A}/\mathbb{Q}^\times$. The specific transversal PLP25 is what supplies that actual coordinate.

In the original common-label carrier the affine map $(z, \mu) \mapsto (a^{(v)}, 1_L) + (b_v(z), \mu)$ is $(a^{(v)}(z), 1_L)$ for every admissible (z, μ) . Thus it forgets the direction label when $z = 0$; it cannot produce the missing local points $a_\lambda^{(v)}$ for $\lambda < 1_L$. In S_{loc} , insert (z, μ) at v and τ elsewhere and add it to the actual point $a_\lambda^{(v)}$. The resulting map is

$$\gamma_{v,\lambda}(z, \mu)_w = \begin{cases} (z, \lambda \vee \mu), & w = v, \\ (1, 1_L), & w \neq v. \end{cases} \quad (\text{PLP34})$$

Its precise fibre relation is equality of z and of $\lambda \vee \mu$. The τ direction leaves the base point fixed; the e direction moves it to $a_{1_L}^{(v)}$. Every nonzero z has $\mu = 1_L$ and therefore top support. The local stabilizer acts by $z \mapsto hz$ and leaves these labels unchanged, so this affine map is equivariant.

At $\lambda = 0_L$, the map PLP34 is injective on the entire normal carrier. It yields the exact enlarged associated family

$$\tilde{\mathcal{T}}_v = C \times_{\iota_v(\mathbb{Q}_v^\times)} G_L(\mathbb{Q}_v) \longrightarrow S_{\text{loc}}/\mathbb{Q}^\times, \quad [g, (z, \mu)] \longmapsto [g \gamma_{v,0_L}(z, \mu)]. \quad (\text{PLP35})$$

It is a bijection onto $C \cup \bigcup_{\mu \in L} O_{v,\mu}$. For $z \neq 0$ this is the nonzero part proved in PLP27, with its forced top label. For $z = 0$ the label μ is fixed by every group action, and the quotient by the local group is precisely $O_{v,\mu}$; different labels cannot be identified. The map which retains amplitudes gives a surjection $\tilde{\mathcal{T}}_v \rightarrow \mathcal{T}_v$, is bijective off the zero section, and has fibre L over each point of the zero section. Its injection back as action spaces selects $(z, 1_L)$ for every z , and identifies the original family PLP27 with $C \cup O_{v,1_L}$. The section $z \mapsto (z, 1_L)$ sends 0 to e , not τ , so it is not falsely claimed to be a zero-preserving semiring map. Equations PLP34–PLP35 compute the exact new family defined by the failed retraction in PLP5.

Finally the supported version of the transverse difference retains its value even at the identity:

$$(z, \mu) + (-h, 1_L)(z, \mu) = ((1 - h)z, \mu). \quad (\text{PLP36})$$

For $h \neq 1$ this is an invertible top-supported scalar map of $G_L(\mathbb{Q}_v)$. For $h = 1$ it is multiplication by e , namely $(z, \mu) \mapsto (0, \mu)$, and not the map constantly equal to τ . On the support stalk L_J , both nonzero top-supported scalars and e act as the identity. This is compatible with the exact localization PLP6–PLP14; ordinary subtraction on an idempotent lattice has not been assumed. The additive-Haar determinants PLP26–PLP32 concern the original local-field amplitudes and do not assign an unproved determinant or multiplicity to the label set.

70.7 The receiving maps in the original CC coefficient sheaves

The adelic points and the CC test-function coefficients are related by evaluation, not by identifying their amplitude spaces. Write $V = \mathcal{S}(\mathbb{A})$ and let $\mathcal{H}_k = \mathcal{O}_k$ be the divisor-indexed CC function sheaf in CDV and ULC. Its restrictions are inclusions of subspaces of V . For each $a \in \mathbb{A}$, evaluation $\xi \mapsto \xi(a)$ therefore defines a morphism $\mathcal{H}_k \rightarrow \mathbb{C}_X$ of $\mathcal{O}_X$ -modules. The constant target is a sheaf because every nonempty arithmetic open is irreducible. The ULC construction supplies the exact lift

$$\text{ev}_a^\uparrow : \mathcal{M}_k \longrightarrow \mathcal{M}_{\mathbb{C}}, \quad (\xi, \lambda) \longmapsto (\xi(a), \lambda) \text{ on } W_U, \quad L_J \longrightarrow L_J \text{ the identity at support.} \quad (\text{PLP37})$$

The target lies in the original carrier: a nonzero evaluated amplitude implies a nonzero original function and hence top label. Evaluation is additive and rational-linear; the label join and meet formulas are unchanged. Restriction to support on both sides is $(\xi, \lambda) \mapsto [\lambda]_J$, proving compatibility with every ULC8/FSC34 map.

Retain the original rational dilation convention $\alpha_r \xi(x) = \xi(r^{-1}x)$, $d(r)_p = v_p(r)$. ULC17 supplies the exact map $\alpha_r : \mathcal{M}_k \rightarrow \mathcal{M}_{k+d(r)}$. Point multiplication and function dilation then satisfy

$$\text{ev}_{ra}^\uparrow \circ \alpha_r = \text{ev}_a^\uparrow, \quad (\alpha_r \xi)(ra) = \xi(r^{-1}ra) = \xi(a). \quad (\text{PLP38})$$

The equation holds on each arithmetic open and on each support stalk, where all maps preserve the original label. It connects the rational orbit computation to the actual divisor-indexed sheaves with no change of amplitude or support. Evaluation through ρ on the enlarged local carrier has the same value on all $a_\lambda^{(v)}$, because their original adele is identical. The carrier $\mathcal{N}$ records the extra labels; ordinary evaluation through $\mathbb{A}$ does not. No evaluation homomorphism of a crossed-product algebra is asserted by these linear sheaf maps.

Precisely retained conclusion. The original arbitrary- L carrier contains the CCM classical orbit union with the unique common top support at each nonzero adele. Its exact injection into

the independently labelled restricted product has fibres, support-stalk maps, and an explicit non-retraction defect given in PLP3–PLP14. The latter construction has the whole L family over each one-zero-place orbit, with its proved associated normal family PLP35. The original prime orbit, the $\log p$ period after the compact quotient, both return orientations, the local normal factor, the original Haar scaling, and the receiving coefficient-sheaf maps have all been retained. These constructions do not prove any new positivity assertion.

71 Prime holonomy, every finite character, and the full compact-orbit spectrum

Sources and scope. The original adelic orbit is from Alain Connes, Caterina Consani, and Matilde Marcolli, *The Weil proof and the geometry of the adeles class space*, original author locators `globalper`, `avwadele`, `Xiorbits`, `muscalehaar`, and `XiKalgpts`. The complete programme derivation `CC_PERIODIC_POINTS_FULL_LATTICE.tex`, PLP1–38, was read for this calculation; its SHA-256 is `4a472874c8c58c5cc98503c498cda5bd692e96d1eb5c98dc6b1f69587bb9004d`. The actual inputs are its full common-label carrier and support maps PLP1–14, the forward return PLP20–23 and PLP28, and the original Haar and distribution conventions PLP25–32. The finite quotients, character calculations, and infinite spectrum below are proved directly. They concern this compact homogeneous orbit, not an identification with the full arithmetic cohomological trace.

71.1 The finite quotient and its exact components

Fix a rational prime p , and retain

$$\ell = \log p, \quad K^{(p)} = \prod_{r \neq p} \mathbb{Z}_r^\times, \quad O_p = (\mathbb{R} \times K^{(p)}) / ((x, u) \sim (x + \ell, pu)). \quad (\text{PHS1})$$

The forward flow is $\Phi_t[x, u] = [x + t, u]$. It returns to the $x = 0$ transversal after time ℓ as $u \mapsto p^{-1}u$. With the actual transverse coordinate of PLP28, that return is

$$R_p(u, z) = (p^{-1}u, p^{-1}z). \quad (\text{PHS2})$$

No inverse time convention is substituted for this point action.

For a positive integer N coprime to p put

$$U_N = (\mathbb{Z}/N\mathbb{Z})^\times, \quad n_N = |U_N| = \varphi(N), \quad d_N = \text{ord}_{U_N}(p), \quad h_N = n_N/d_N. \quad (\text{PHS3})$$

For $N = 1$, use the one-element group, $\varphi(1) = 1$, and $d_1 = h_1 = 1$. The reduction map $K^{(p)} \rightarrow U_N$ is onto: factor N into prime powers, reduce the corresponding r -adic units, and apply the Chinese remainder theorem. It intertwines multiplication by p and gives

$$O_{p,N} = (\mathbb{R} \times U_N) / ((x, u) \sim (x + \ell, pu)), \quad O_p \longrightarrow O_{p,N}. \quad (\text{PHS4})$$

The permutation $u \mapsto p^{-1}u$ has precisely the cosets of $\langle p \rangle$ as its cycles. Each has length d_N , since $p^{-j}u = u$ is equivalent to $p^j = 1$ in U_N .

Each coset supplies one circle component of $O_{p,N}$, with the explicit parametrization

$$\mathbb{R}/(d_N\ell)\mathbb{Z} \longrightarrow O_{p,N}, \quad [t] \longmapsto [t, u_0]. \quad (\text{PHS5})$$

Indeed $[x, p^j u_0] = [x - j\ell, u_0]$, proving surjectivity onto that component. Equality $[t, u_0] = [t', u_0]$ means $t' - t = j\ell$ and $p^j u_0 = u_0$, hence $d_N \mid j$. This proves injectivity and the exact least period $d_N\ell$. The quotient action is properly discontinuous in the real coordinate, so its quotient is Hausdorff; the displayed continuous bijection from a compact circle is consequently a homeomorphism onto the component. There are exactly h_N components, and forward motion increases t in every component. Its d_N -fold transversal return has normal action $z \mapsto p^{-d_N}z$.

Use compact Haar measure of mass one on $K^{(p)}$, its pushforward counting measure du_N of mass $1/n_N$ at each u , and real measure dx . A fundamental interval is $0 \leq x < \ell$. Therefore

$$\mu(O_p) = \mu_N(O_{p,N}) = \ell, \quad d\mu_N \text{ on a component} = \frac{dt}{n_N}, \quad \mu_N(\text{one component}) = \frac{d_N\ell}{n_N}. \quad (\text{PHS6})$$

For the component formula, split the interval of length $d_N\ell$ into its d_N transversal pieces, each carrying dx/n_N . This retains the original real length; it is not replaced by mass one.

71.2 The two different return operators

On functions on U_N distinguish

$$C_R f(u) = f(p^{-1}u), \quad P_N f(u) = f(pu) = C_{R^{-1}} f(u). \quad (\text{PHS7})$$

Both preserve the measure in PHS6. The actual pullback convention on the full orbit is

$$(U_t F)(x, u) = F(x - t, u), \quad (U_\ell F)(0, u) = F(-\ell, u) = F(0, pu). \quad (\text{PHS8})$$

The last equality is the original relation in PHS1. Thus the Koopman return is P_N , whereas composition with the forward point return is $C_R = P_N^{-1}$. In the delta-function basis, $P_N \delta_v = \delta_{p^{-1}v}$ and $C_R \delta_v = \delta_{pv}$; each basis vector has squared norm $1/n_N$ in the specified measure.

For every character $\chi : U_N \rightarrow \mathbb{C}^\times$,

$$P_N \chi = \chi(p) \chi, \quad C_R \chi = \chi(p)^{-1} \chi, \quad \frac{1}{n_N} \sum_{u \in U_N} \chi(u) \overline{\psi(u)} = \begin{cases} 1, & \chi = \psi, \\ 0, & \chi \neq \psi. \end{cases} \quad (\text{PHS9})$$

The orthogonality follows by translating the sum by an element where the quotient character is not 1. These characters form a basis: decomposing the finite abelian group into cyclic factors and taking their roots-of-unity characters gives exactly n_N orthogonal nonzero vectors. Their squared norms are 1 with the retained finite Haar measure.

Every character of $\langle p \rangle$ extends to U_N . Here is a direct extension proof. Given a character on a subgroup H of a finite abelian group and $a \notin H$, let j be the least positive integer with $a^j \in H$. Choose $z \in \mathbb{C}$ with $z^j = \chi(a^j)$, and set $\tilde{\chi}(ha^k) = \chi(h)z^k$. If two representations differ, their exponents differ by a multiple of j and this exact root equation proves equality of their values. Thus this extends the character to $\langle H, a \rangle$. Finitely many repetitions extend it to the whole group. The kernel of restriction consists of the characters of $U_N/\langle p \rangle$, of which there are h_N by the same cyclic-factor count. Consequently every d_N th root of unity occurs as $\chi(p)$ exactly h_N times.

The original measure also verifies the permutation trace directly. The integral kernel of P_N^j in that measure is $n_N 1_{v=p^j u}$, so its diagonal integral is

$$\text{Tr}(P_N^j) = \frac{1}{n_N} \sum_u n_N 1_{u=p^j u} = \begin{cases} n_N, & d_N \mid j, \\ 0, & d_N \nmid j. \end{cases} \quad (\text{PHS10})$$

The factor $1/n_N$ in the measure therefore does not replace this trace by the proportion of fixed points.

71.3 Finite determinants and all Dirichlet factors

Retain $q = p^{-s} = e^{-s\ell}$. The exact finite determinants are

$$\begin{aligned} D_{p,N}^+(s) &= \det(I - p^{-s} P_N) = \prod_{\chi \in \widehat{U}_N} (1 - \chi(p) p^{-s}) = (1 - p^{-d_N s})^{h_N}, \\ D_{p,N}^-(s) &= \det(I - p^{-s} C_R) = \prod_{\chi \in \widehat{U}_N} (1 - \chi(p)^{-1} p^{-s}) = (1 - p^{-d_N s})^{h_N}. \end{aligned} \quad (\text{PHS11})$$

The product over all d_N th roots is $1 - q^{d_N}$, proving the last equalities with their multiplicities. Equality of the total determinants does not identify their individual character factors.

For $\Re s > 0$, expand the logarithm with value zero at $q = 0$. Absolute convergence of the geometric series gives

$$\begin{aligned} -\log D_{p,N}^+(s) &= \sum_{j \geq 1} \frac{p^{-js}}{j} \text{Tr}(P_N^j) = h_N \sum_{r \geq 1} \frac{p^{-d_N r s}}{r}, \\ \frac{d}{ds} \log D_{p,N}^+(s) &= \ell \sum_{j \geq 1} p^{-js} \text{Tr}(P_N^j) = \frac{n_N \ell p^{-d_N s}}{1 - p^{-d_N s}}. \end{aligned} \quad (\text{PHS12})$$

The corresponding formulas for $D_{p,N}^-$ use P_N^{-j} and have the same total values. The logarithmic derivatives extend meromorphically; the determinant zeros are exactly $s = 2\pi i k / (d_N \ell)$, $k \in \mathbb{Z}$, each of order h_N .

Extend a character of U_N to integers by zero on nonunits. Its original Dirichlet series and local factor, with no completion multiplier, are

$$L(s, \chi) = \sum_{n \geq 1} \chi(n) n^{-s}, \quad L_p(s, \chi) = (1 - \chi(p) p^{-s})^{-1} \quad (p \nmid N). \quad (\text{PHS13})$$

For $\Re s > 1$, absolute convergence and unique factorization prove its Euler product by expanding finite geometric products and taking their dominated limit. Thus the actual Koopman character factor in PHS11 is $L_p(s, \chi)^{-1}$. The composition operator C_R instead gives $L_p(s, \chi^{-1})^{-1}$; its character has not been silently inverted in PHS13. In particular

$$-\frac{L'_p(s, \chi)}{L_p(s, \chi)} = \ell \sum_{j \geq 1} \chi(p)^j p^{-js} = \frac{\ell \chi(p) p^{-s}}{1 - \chi(p) p^{-s}}, \quad \Re s > 0. \quad (\text{PHS14})$$

There is also an inverse-character dictionary between unit coordinates and the original local-idele coordinate. Extend the same finite character to $C_{\mathbb{Q}}$ by $\tilde{\chi}(r, u) = \chi(u \bmod N)$. The exact local uniformizer image in PLP20 gives

$$\tilde{\chi}(\iota_p(p)) = \chi(p)^{-1}, \quad L_p(s, \chi) = (1 - \tilde{\chi}^{-1}(\iota_p(p)) p^{-s})^{-1}. \quad (\text{PHS14a})$$

Indeed the unit coordinates away from p of $\iota_p(p)$ are all p^{-1} , and N has no p factor. Thus an Euler factor written using $\tilde{\chi}$ itself at the local uniformizer is the one for the inverse integer character. This is compatible with the original convolution action: for any absolutely integrable test a and a unitary character F of the orbit,

$$(a * F)(x) = F(x) \int_{C_{\mathbb{Q}}} a(g) F(g)^{-1} d^{\times} g. \quad (\text{PHS14b})$$

This follows by inserting $F(g^{-1}x) = F(g)^{-1}F(x)$ into the convolution integral. For $F(r, u) = r^{i\tau} \tilde{\chi}(u)$, the multiplier is the original Mellin integral with character $\tilde{\chi}^{-1}$ and exponent $-i\tau$. Neither character inversion is a change in the forward point return.

All missing-prime factors must be stated in a global comparison. For the present $p \nmid N$, the full original product is

$$L(s, \chi) = L_p(s, \chi) \prod_{\substack{r \neq p \\ r \nmid N}} (1 - \chi(r) r^{-s})^{-1}, \quad \Re s > 1. \quad (\text{PHS15})$$

Primes dividing N do not occur because $\chi(r) = 0$ there. Let $f_{\chi} \mid N$ be the conductor and χ_* the inducing primitive character. Explicitly, decompose U_N by the Chinese remainder theorem and take, at each prime, the least prime-power level through which its character factors; their product is f_{χ} . Restriction to these quotients gives χ_* and proves its minimality. Comparing each Euler factor yields

$$L(s, \chi) = L(s, \chi_*) \prod_{\substack{r \mid N \\ r \nmid f_{\chi}}} (1 - \chi_*(r) r^{-s}), \quad (\text{PHS16})$$

$$L(s, 1_{U_N}) = \zeta(s) \prod_{r \mid N} (1 - r^{-s}).$$

At primes dividing f_{χ} , the primitive character is already zero and no additional factor is omitted. At every other prime dividing N , the displayed factor cancels exactly the primitive Euler factor.

Both assertions follow first in $\Re s > 1$, where every product involved has its original absolute meaning. For the product over all finite characters, the entire correction is

$$\prod_{\chi \in \widehat{U}_N} L(s, \chi) = \left(\prod_{\chi \in \widehat{U}_N} L(s, \chi_*) \right) \prod_{\chi \in \widehat{U}_N} \prod_{\substack{r|N \\ r \nmid f_\chi}} (1 - \chi_*(r) r^{-s}). \quad (\text{PHS17})$$

No primitive factor or omitted prime is identified merely because several characters share its inducing character. Characters genuinely ramified at p are not characters of this orbit: the stabilizer contains $\mathbb{Z}_p^\times$, so descent requires the original character to be trivial on that subgroup. For a primitive Dirichlet character whose conductor is divisible by p , its integer extension has value zero at p and its local Euler factor is 1. No nonzero phase $\chi(p)$ is assigned to that omitted case. If only an imprimitive modulus adds p , the missing unramified factor is exactly the correction in PHS16.

If $N \mid M$ and both are coprime to p , reduction $U_M \rightarrow U_N$ is onto. This follows prime-power by prime-power and then by the Chinese remainder theorem. Its fibres have size $\varphi(M)/\varphi(N)$, so it preserves the specified probability Haar measures. The induced maps $O_{p,M} \rightarrow O_{p,N}$ intertwine both return operators and give isometric pullbacks on their L^2 spaces with mass ℓ . A character pulled back from U_N keeps the same value at p . Its global Dirichlet series, however, has precisely the extra excluded-prime factors

$$L(s, \chi_M) = L(s, \chi_N) \prod_{\substack{r|M \\ r \nmid N}} (1 - \chi_N(r) r^{-s}). \quad (\text{PHS18})$$

Comparing the individual Euler factors proves this identity, so the finite-level transition does not hide those exclusions.

71.4 The normal Haar factors in both time directions

The original transverse measure has $\text{vol}(\mathbb{Z}_p) = 1$. For an integer $j > 0$, the forward point return is $z \mapsto p^{-j}z$, and

$$\begin{aligned} d(p^{-j}z) &= p^j dz, & |1 - p^{-j}|_p &= p^j, & |1 - p^j|_p &= 1, \\ d_{\mathbb{A}_p}(p^{-j}a) &= p^{-j} d_{\mathbb{A}_p} a, & p^j \cdot p^{-j} &= 1. \end{aligned} \quad (\text{PHS19})$$

The last line retains the hyperplane factor from the rational product formula PLP32. It is not combined with the normal factor before those two original measures have been identified.

For the inverse pullback convention $\vartheta(h)f(z) = f(h^{-1}z)$, the kernel with respect to dy is $\delta(y - h^{-1}z)$. Its transverse diagonal is consequently

$$\int_{\mathbb{Q}_p} \delta((1 - h^{-1})z) dz = |1 - h^{-1}|_p^{-1}. \quad (\text{PHS20})$$

This follows by the Dirac change of variables for the invertible scalar $1 - h^{-1}$, and only holds here for $h \neq 1$. Thus the complete forward and backward entries are

point time	normal point multiplier h	finite Koopman character	normal pullback diagonal
$+j\ell$	p^{-j}	$\chi(p)^j$	1
$-j\ell$	p^j	$\chi(p)^{-j}$	p^{-j}

(PHS21)

In particular using $|1 - h|_p^{-1}$ for the forward pullback would give the other row's weight and would be a convention error. The norm on transverse test functions also retains its mass:

$$\|\vartheta(h)f\|_{L^2(dz)}^2 = |h|_p \|f\|_{L^2(dz)}^2.$$

For the forward multiplier this factor is p^j . No factor $|h|_p^{-1/2}$ has been inserted to change that operator. The table specifies local distributional factors away from the singular identity. It does not replace the regularized identity term or the adelic trace theorem.

71.5 Exact character frequencies on the full compact orbit

Let χ be a continuous character of $K^{(p)}$. It factors through some U_N , $p \nmid N$. To prove this, continuity gives an open subgroup whose image lies in a sufficiently short arc of the unit circle containing no nontrivial subgroup. Such an open subgroup has trivial character image. A congruence subgroup defined at finitely many primes lies inside it, giving the asserted finite level. In particular $\chi(p)$ is a root of unity.

The function $e^{i\tau x}\chi(u)$ descends to O_p precisely when

$$e^{i\tau\ell}\chi(p) = 1. \quad (\text{PHS22})$$

Indeed this is exactly invariance under the deck generator in PHS1. For any chosen argument θ_χ of $\chi(p)$ all its frequencies and its flow eigenvalues are therefore

$$\tau_{\chi,n} = \frac{2\pi n - \theta_\chi}{\ell}, \quad U_t(e^{i\tau_{\chi,n}x}\chi(u)) = e^{-i\tau_{\chi,n}t}e^{i\tau_{\chi,n}x}\chi(u), \quad n \in \mathbb{Z}. \quad (\text{PHS23})$$

Changing the chosen argument only changes the integer index. At $t = \ell$, this eigenvalue is $\chi(p)$, exactly PHS9.

These functions form a complete orthogonal basis of $L^2(O_p, dx du)$, each of squared norm ℓ . For two distinct compact characters, integration over $K^{(p)}$ is zero by translation invariance. For the same character, distinct integer indices differ by $2\pi n/\ell$ and integrate to zero over $[0, \ell)$. For completeness, first average a function over successively smaller compact congruence subgroups. These averages are the finite-level projections; they converge in L^2 to the function. For continuous functions this follows uniformly from uniform continuity on the compact orbit, and continuous functions are dense in its finite Haar measure. At a fixed level, finite Fourier expansion on U_N reduces to real functions satisfying the boundary condition $a_\chi(x + \ell) = \chi(p)^{-1}a_\chi(x)$. Multiplication by $e^{i\theta_\chi x/\ell}$ converts precisely this boundary condition to periodicity, and ordinary Fourier series on the interval give the frequencies in PHS23. Their completeness follows, for example, from convergence of the Fejer averages of continuous periodic functions and density in L^2 . The finite-level union proves the asserted full completeness.

In these original coordinates the self-adjoint generator is

$$U_t = e^{-itD}, \quad D = -i\partial_x, \quad \text{Dom}(D) = \left\{ \sum_{\chi,n} a_{\chi,n} e^{i\tau_{\chi,n}x} \chi(u) : \ell \sum_{\chi,n} |a_{\chi,n}|^2 < \infty, \quad \ell \sum_{\chi,n} \tau_{\chi,n}^2 |a_{\chi,n}|^2 < \infty \right\}. \quad (\text{PHS24})$$

Its action multiplies each coefficient by the real number $\tau_{\chi,n}$. The same diagonal coefficient test describes the adjoint domain, proving self-adjointness. Dominated convergence on $\ell \sum |a_{\chi,n}|^2$ proves strong continuity of U_t and its specified generator. At level N the frequencies are exactly $2\pi\mathbb{Z}/(d_N\ell)$, each of multiplicity h_N , by the character-extension calculation after PHS9. This agrees with all the component circles of PHS5.

The full set of frequencies is exactly

$$\{\tau_{\chi,n}\} = \frac{2\pi}{\log p} \mathbb{Q}. \quad (\text{PHS25})$$

One inclusion follows because every $\chi(p)$ is a root of unity. For the converse, take $N = p^m - 1$ for a positive integer m . It is coprime to p . If $N > 1$, its order is exactly m : $p^m = 1 \pmod{N}$, whereas for $0 < j < m$ the positive integer $p^j - 1$ is strictly less than N , so is not divisible by N .

The extension proof after PHS9 then realizes every m th root as $\chi(p)$. For a rational number a/b , use $m = b$ and choose $\chi(p) = e^{-2\pi ia/b}$; equation PHS22 gives $\tau = 2\pi a/(b\ell)$. The exceptional case $p = 2, m = 1$ has $N = 1$; its trivial character gives precisely the required integer frequencies. This handles that case without an order calculation in a nonexistent nontrivial unit group. It follows that the eigenvalues are dense in $\mathbb{R}$.

Every one of these eigenvalues has infinite multiplicity. Fix the denominator b of a rational frequency and let m run through its positive multiples. At $N = p^m - 1$, the required value of $\chi(p)$ has exactly $\varphi(N)/m$ character extensions. These give distinct orthogonal eigenvectors at that fixed frequency. Their number is unbounded. Indeed for any positive integer n , prime-power factorization gives the elementary bound

$$\varphi(n) \geq \sqrt{n/2}. \quad (\text{PHS26})$$

For an odd prime power r^a , $\varphi(r^a)/\sqrt{r^a} = r^{a/2-1}(r-1) \geq 1$; for powers of 2 this ratio is at least 1 except at 2^1 , where it is $1/\sqrt{2}$. Multiplying these exact inequalities proves PHS26, also valid at $n = 1$. Thus $\varphi(p^m - 1)/m \geq \sqrt{(p^m - 1)/2}/m \rightarrow \infty$. Continuous characters form a countable union of finite sets, so each of these eigenspaces is countably infinite dimensional.

The spectrum of D as a self-adjoint operator is consequently all of $\mathbb{R}$, although its eigenvalue set is the specified countable dense subset. For a real number a choose frequencies $\tau_j \rightarrow a$; the corresponding vectors have norm $\sqrt{\ell}$ and $(D - a)$ -image norm $\sqrt{\ell}|\tau_j - a|$, proving that $D - a$ has no bounded inverse. Nonreal numbers are outside its spectrum because division by $\tau - z$ defines the bounded inverse, of norm at most $1/|\text{Im } z|$, on the full coefficient space. The spectral measures are atomic in the complete basis already proved; density of the eigenvalue set is not a claim that every real number is an eigenvalue. In fact the coefficient equation $DF = aF$ forces $(\tau_{\chi,n} - a)a_{\chi,n} = 0$ for every index, so no additional eigenvalue can occur outside PHS25.

There is no compact resolvent. For example the zero eigenspace already contains infinitely many pairwise orthogonal vectors of norm $\sqrt{\ell}$. On that space $(D - i)^{-1}$ is multiplication by i , so the bounded sequence of those vectors has no convergent subsequence after applying the resolvent. This contradicts compactness. At each finite level the multiplicities are finite and the nonzero frequencies tend to infinity, so Fourier truncation does give a compact resolvent there; the full space is not one such finite level.

Likewise $P = U_\ell$ on the infinite full space is unitary and not compact. For $q \neq 0$, qP is not trace class, since its action on any infinite orthogonal sequence preserves the norm up to the fixed nonzero factor $|q|$. Thus PHS11 is a genuine finite determinant; no unproved infinite Fredholm determinant is obtained by taking its limit.

71.6 The actual common-label lifts

For the arbitrary bounded distributive lattice of PLP1–14 and any one of the specified complex amplitude spaces V , retain

$$G_L(V) = \{(0, \lambda) : \lambda \in L\} \cup (V \times \{1_L\}), \quad T^\dagger(F, \lambda) = (TF, \lambda). \quad (\text{PHS27})$$

Each linear return, flow, finite-level pullback, or Fourier map above lifts by this formula. Zero amplitudes retain their labels; linearity proves addition and the $G_L(\mathbb{C})$ scalar action, and the displayed inverses lift wherever an inverse exists. There is one common label on the entire vector, not independently chosen labels on its character coefficients.

For example the exact finite Fourier map and inverse are

$$\widehat{f}(\chi) = \frac{1}{n_N} \sum_{u \in U_N} f(u) \overline{\chi(u)}, \quad f(u) = \sum_{\chi \in \widehat{U}_N} \widehat{f}(\chi) \chi(u). \quad (\text{PHS28})$$

Orthogonality in PHS9 proves the inverse with its original $1/n_N$ Haar factor. Its lifted map is $G_L(\mathbb{C}^{U_N}) \rightarrow G_L(\mathbb{C}^{\widehat{U}_N})$, not a basis change on the larger independently labelled product $G_L(\mathbb{C})^{U_N}$.

It is a semimodule isomorphism. For pointwise multiplication of amplitudes, its transported product is explicitly $\widehat{f}g(\chi) = \sum_{\psi} \widehat{f}(\psi)\widehat{g}(\chi\psi^{-1})$, as follows by inserting the inverse sum and using the same orthogonality. It is not ordinary coordinatewise multiplication on character coefficients.

The full Hilbert Fourier map in PHS23–24 is lifted only after the convergent amplitude sum has been defined, with its one unchanged label. No infinite join in L is used. In particular $U_t^\uparrow(e) = e$ and $U_t^\uparrow(\tau) = \tau$; these remain distinct when L is nontrivial. The actual support localization sends (F, λ) to $[\lambda]_J$, and every label-preserving operator commutes with that map: $\sigma_J T^\uparrow(F, \lambda) = [\lambda]_J = \sigma_J(F, \lambda)$. This records exactly the localization of PLP7 and ULC8, rather than retaining an amplitude at a stalk where the proved localization removes it.

The original nonzero adelic orbit has its single common top label as proved in PLP15–19. The lifts here concern its coefficient spaces; they do not manufacture additional nonzero-point lifts in the original carrier. All Haar factors above are measures on the stated real, finite-unit, or local-field amplitudes. No measure or spectral multiplicity on the arbitrary label set is assigned. The finite Euler factors, their precise orientations, and the infinite compact-orbit spectrum have been calculated; none of these calculations asserts a new Weil-positivity theorem or an RH implication.

72 Galois norm and Weil transfer with every support label retained

Human source and exact input. Alain Connes, Caterina Consani and Matilde Marcolli, *The Weil proof and the geometry of the adèles class space*, original author file `Yuri70v2.tex`, lines 3514–3853, provide the original construction used here. Its source labels are `Normmap`, `AKGemb`, `explnorm`, `CWGext`, `rulesW`, `transferhom`, `taumap`, `leftrightact`, `tauequiv`, `Xitau`, and `tauXiC1`. The author TeX SHA256 is `b1f694cb934dd4fef829287f1a55dd8de21c67a481e552453c2f7e8fe0ae6fb5`. The class-field assertion used as an input is the surjectivity of the specified relative Weil-group transfer. CCM cite André Weil, *Sur la théorie du corps de classes*, J. Math. Soc. Japan **3** (1951), 1–35, for that assertion. Their local norm references are to Weil’s *Basic Number Theory*. No independent rereading or new proof of those class-field theorems is claimed here. The carrier maps, their exact fibres, the typed group-transfer formula, and the lifted orbit maps below are proved. The carriers are the arbitrary- L carriers PLP1–14; the local product is not substituted for the original common-label object.

72.1 The original field maps and the two different carriers

Let E/K be a finite Galois extension of global fields, with $G = \text{Gal}(E/K)$ and $n = [E : K]$. Denote its map of places by $\pi : \Sigma_E \rightarrow \Sigma_K$. For $w \mid v$ put $d_w = [E_w : K_v]$. Galois transitivity gives

$$\sum_{w \mid v} d_w = n, \quad G_w = \text{Gal}(E_w/K_v), \quad |G_w| = d_w. \quad (\text{GTL1})$$

Use the original diagonal inclusion and local norm formula

$$\begin{aligned} \iota : \mathbb{A}_K &\longrightarrow \mathbb{A}_E, & \iota(a)_w &= a_{\pi(w)}, \\ N : \mathbb{A}_E &\longrightarrow \mathbb{A}_K, & (Nx)_v &= \prod_{w \mid v} N_{E_w/K_v}(x_w). \end{aligned} \quad (\text{GTL2})$$

The inclusions $K_v \rightarrow E_w$ in the first line are the original field inclusions. Integral local elements have integral norm, so the second line respects the restricted-product condition. The Galois action on adèles is $(\gamma x)_w = \gamma(x_{\gamma^{-1}w})$, with the indicated isomorphism of local fields. Its fixed subring is $\iota(\mathbb{A}_K)$.

The local field norm is the determinant of multiplication on the d_w -dimensional K_v -vector space E_w . It is multiplicative, equals zero exactly at zero, and sends a scalar a_v to $a_v^{d_w}$. Regrouping the full product over G by the places above v gives

$$\iota(Nx) = \prod_{\gamma \in G} \gamma x, \quad N\iota(a) = a^n. \quad (\text{GTL3})$$

In this regrouping each place above v occurs d_w times, with its local automorphisms, and GTL1 supplies the exact exponent.

Let L be the original nontrivial bounded distributive lattice. For $F = K, E$ define the two specified semirings

$$\begin{aligned} S_{F,c} &= G_L(\mathbb{A}_F), & S_{F,\text{loc}} &= \prod_{v \in \Sigma_F}^I G_L(F_v), & \Delta_F(a, \lambda) &= ((a_v, \lambda))_v, \\ G_L(R) &= \{(0, \lambda) : \lambda \in L\} \cup (R \times \{1_L\}), \\ (a, \lambda) + (b, \mu) &= (a + b, \lambda \vee \mu), & (a, \lambda)(b, \mu) &= (ab, \lambda \wedge \mu). \end{aligned} \quad (\text{GTL4})$$

The finite restricted factors are $G_L(\mathcal{O}_v)$. Every zero label belongs to these factors, so the restricted condition is entirely the original amplitude condition. The map Δ_F is the injective unital semiring

homomorphism of PLP3: its image consists of the tuples with one common label. At a nonzero adele that common label is forced to be top, including at any of its zero coordinates.

The exact lifted inclusion and norm maps are

$$\begin{aligned} \iota_c(a, \lambda) &= (\iota(a), \lambda), & N_c(x, \lambda) &= (Nx, \lambda), \\ (\iota_{\text{loc}}(a, \boldsymbol{\mu}))_w &= (a_{\pi(w)}, \mu_{\pi(w)}), & (N_{\text{loc}}(x, \boldsymbol{\lambda}))_v &= \left(\prod_{w|v} N_{E_w/K_v}(x_w), \bigwedge_{w|v} \lambda_w \right). \end{aligned} \quad (\text{GTL5})$$

All meets here are finite, so no completeness of L is assumed. A nonzero target norm at v forces every x_w above it to be nonzero and hence every input label to be top. This proves that N_{loc} lands in the stated constrained carrier. The same argument proves the common-label constraint for N_c . Both inclusions are injective unital semiring maps. Both norm maps preserve multiplication, the multiplicative unit, and the additive-zero element τ ; they are multiplicative monoid maps.

On the local carrier the Galois action is

$$\gamma(x, \boldsymbol{\lambda})_w = (\gamma(x_{\gamma^{-1}w}), \lambda_{\gamma^{-1}w}). \quad (\text{GTL6})$$

It is a semiring automorphism. On the common carrier it preserves the one label. The fixed local tuples have constant labels over each fibre of π and fixed amplitudes, so they are exactly the image of ι_{loc} ; the common fixed points are exactly the image of ι_c . Finite idempotence of meet, including its d_w repeated occurrences, now gives all the exact squares

$$\begin{aligned} N_{\text{loc}}\Delta_E &= \Delta_K N_c, & \iota_{\text{loc}}\Delta_K &= \Delta_E \iota_c, \\ N_c \iota_c(a, \lambda) &= (a^n, \lambda) = (a, \lambda)^n, \\ N_{\text{loc}} \iota_{\text{loc}}(a, \boldsymbol{\mu}) &= (a^n, \boldsymbol{\mu}), \\ \iota_c N_c(X) &= \prod_{\gamma \in G} \gamma X, & \iota_{\text{loc}} N_{\text{loc}}(X) &= \prod_{\gamma \in G} \gamma X. \end{aligned} \quad (\text{GTL7})$$

Thus the norm/diagonal and inclusion/diagonal squares commute on the full original carriers, not merely on their amplitudes.

72.2 Exact zero-pattern fibres and the additive comparison

For a fixed adele $x \in \mathbb{A}_E$ let $Z(x) = \{w : x_w = 0\}$. Field norm detects zero, so

$$Z(Nx) = \pi(Z(x)). \quad (\text{GTL8})$$

The local amplitude fibre over x is $L^{Z(x)}$. For each $v \in \pi(Z(x))$ set $A_v = Z(x) \cap \pi^{-1}(v)$, a nonempty finite set. The exact map of these amplitude fibres and its fibres at an output $\boldsymbol{\mu} \in L^{\pi(Z(x))}$ are

$$\begin{aligned} L^{Z(x)} &\longrightarrow L^{\pi(Z(x))}, & (\lambda_w)_w &\longmapsto \left(\bigwedge_{w \in A_v} \lambda_w \right)_v, \\ \prod_{v \in \pi(Z(x))} &\left\{ (\lambda_w)_{w \in A_v} : \bigwedge_{w \in A_v} \lambda_w = \mu_v \right\}. \end{aligned} \quad (\text{GTL9})$$

Top labels at the nonzero coordinates have been inserted, not deleted from the carrier. This fibre map is surjective: select one zero place above each such v , put μ_v there and put top at the remaining zero places. This choice is only a set-theoretic section of the stated finite-fibre meet maps, not a new canonical Galois choice.

For the common carrier, a fixed nonzero x admits only its top label, even when $Nx = 0$. Such an element maps to $e = (0, 1_L)$ when its entire norm adele vanishes. At the all-zero adele the map is $(0, \lambda) \mapsto (0, \lambda)$. Consequently the complete inverse fibre of $(0, \mu)$ for $\mu < 1_L$ is the

single point $(0, \mu)$; the fibre of e is $\{(x, 1_L) : Nx = 0\}$. For a nonzero target y , its norm fibre is $\{(x, 1_L) : Nx = y\}$. These are exact set fibres; no norm surjectivity on the original adeles has been assumed.

The norm is not an additive map when $n > 1$. If some fibre of places has at least two elements, put complementary zero-and-one amplitudes in its two nonempty parts and zero amplitudes at all other places. Each of the two norm adeles is zero, but the norm of their sum has value one at this base place. The failure also occurs on zero amplitudes with independent labels: at two places above one v , use label patterns $(1_L, 0_L)$ and $(0_L, 1_L)$, and use top labels at all other places above v . Their norm labels are both bottom, whereas the norm label of their sum is top. This argument works in every nontrivial L .

If there is only one place above the selected v , its local degree is $n > 1$. A separable field norm of degree $d > 1$ on an infinite field is not additive. Indeed additivity would give $N(1 + ta) = 1 + t^d N(a)$ for every scalar t ; comparison with the determinant polynomial forces the coefficient $\text{Tr}(a)$ of t to vanish for every a . The trace of a finite Galois extension is not the zero map. Here is an elementary proof valid also when the characteristic divides d . Distinct field automorphisms are linearly independent: in a shortest nonzero relation $\sum_i c_i \sigma_i(x) = 0$, evaluate at yx and subtract $\sigma_1(y)$ times the relation; choose y separating σ_1 and another automorphism. This gives a shorter nonzero relation, a contradiction. Applying this to the sum of all local automorphisms proves that trace is nonzero. Since the local base field is infinite, the polynomial comparison is legitimate, proving nonadditivity.

There is nonetheless an exact full distributive relation, including every mixed term. For either carrier let ι and N denote the corresponding maps of GTL5. Semiring distributivity gives

$$\iota N(X + Y) = \iota N(X) + \iota N(Y) + \sum_{\emptyset \neq A \subsetneq G} \left(\prod_{\gamma \in A} \gamma X \right) \left(\prod_{\gamma \notin A} \gamma Y \right). \quad (\text{GTL10})$$

The mixed sum is Galois-fixed because the action permutes its subsets, so it has an exact inverse image under ι . This defines the full norm-additivity defect in the original target, with every mixed amplitude and label retained. Each subset term in the local carrier has at w the label $\bigwedge_{\gamma \in A} \lambda_{\gamma^{-1}w} \wedge \bigwedge_{\gamma \notin A} \mu_{\gamma^{-1}w}$; the sum joins these labels. Thus GTL10 retains the original finite distributivity calculation rather than imposing additivity.

An actual additive receiving map is the trace, not the norm:

$$\begin{aligned} T_c(x, \lambda) &= (\text{Tr}_{E/K} x, \lambda), \\ (T_{\text{loc}}(x, \lambda))_v &= \left(\sum_{w|v} \text{Tr}_{E_w/K_v}(x_w), \bigvee_{w|v} \lambda_w \right). \end{aligned} \quad (\text{GTL11})$$

Here the adelic trace uses the displayed local sum. We first prove that both formulas land in their stated restricted carriers. At every nonarchimedean place where $x_w \in \mathcal{O}_w$, all its local Galois conjugates are integral. Their finite sum $\text{Tr}_{E_w/K_v}(x_w)$ is therefore integral and belongs to K_v , hence belongs to $\mathcal{O}_v$. An adele has only finitely many nonintegral finite coordinates; their images under π form a finite set. Outside that set every term of the finite sum over $w \mid v$ belongs to $\mathcal{O}_v$, so the output amplitude is an adele of K . If its v -coordinate is nonzero, at least one input x_w over v is nonzero. The constrained input carrier then forces $\lambda_w = 1_L$, and consequently $\bigvee_{w|v} \lambda_w = 1_L$ as required. If the output coordinate is zero, its displayed join is an allowed zero label, even when nonzero trace terms cancel. This proves landing for T_{loc} . For T_c , a nonzero output adele implies a nonzero input adele and hence common label 1_L ; an output zero adele retains the original permitted common label. Thus the common carrier constraint is also preserved. These maps are additive and linear over the corresponding base carrier through ι ; finite lattice distributivity proves scalar compatibility for T_{loc} . Their comparisons are $T_{\text{loc}} \Delta_E = \Delta_K T_c$, $T\iota(a, \lambda) = (na, \lambda)$,

and $\iota T(X) = \sum_{\gamma \in G} \gamma X$. They are not asserted to replace the multiplicative norm. The three pure label maps have the exact order adjunctions

$$J_\pi \dashv \pi^* \dashv M_\pi, \quad J_\pi(\lambda)_v = \bigvee_{w|v} \lambda_w, \quad \pi^*(\mu)_w = \mu_{\pi(w)}, \quad M_\pi(\lambda)_v = \bigwedge_{w|v} \lambda_w. \quad (\text{GTL12})$$

For example $\pi^* \mu \leq \lambda$ means exactly that $\mu_v \leq \lambda_w$ for every $w | v$, or equivalently $\mu \leq M_\pi \lambda$; the join adjunction is the dual inequality. In particular $M_\pi \pi^* = J_\pi \pi^*$ is the identity. These are exact lattice maps even when norm is not an additive morphism.

72.3 Original local degrees, Haar modules and support localizations

At nonarchimedean places retain additive Haar measures of mass one on the rings of integers; at real and complex places use their usual real additive Haar measures. For $z \in E_w^\times$ the multiplication map on the finite-dimensional K_v -vector space has determinant $N_{E_w/K_v}(z)$. A coordinate Haar measure differs from the selected local Haar measure by one constant, which cancels in the scaling ratio. Consequently

$$|N_{E_w/K_v}(z)|_v = |z|_w, \quad |a|_w = |a|_v^{d_w} \quad (a \in K_v^\times). \quad (\text{GTL13})$$

These are the Haar-module absolute values: at a complex place $|z|_w$ is the square of the usual complex modulus. For finite places, multiplication by a uniformizer has module the inverse residue-field cardinality, as follows by the finite index of its image of the ring of integers. Thus GTL13 retains those finite Haar factors, rather than using an unweighted complex or finite norm. On ideles it gives the exact products

$$|Ng|_K = \prod_v \prod_{w|v} |N_{E_w/K_v}(g_w)|_v = \prod_w |g_w|_w = |g|_E, \quad |\iota c|_E = |c|_K^n. \quad (\text{GTL14})$$

Only finitely many nonarchimedean factors differ from one. These equalities concern ideles, where the products are defined; no global absolute value is assigned here to an arbitrary adele. Nor is the nonlinear field norm assigned an additive Haar Jacobian by these multiplication-map identities.

At the first support localization the common and local carriers become L and L^{Σ_F} , as in PLP6. The common norm and inclusion are identity on L ; the local norm and inclusion become M_π and π^* of GTL12. For a prime lattice ideal J retain the actual uniform-denominator localization

$$\mathcal{L}_{F,J} = (L^{\Sigma_F})[\text{diag}(L \setminus J)^{-1}], \quad [\lambda] = [\mu] \iff \exists \nu \notin J \forall v : \lambda_v \wedge \nu = \mu_v \wedge \nu. \quad (\text{GTL15})$$

Finite meets over $w | v$ commute with this one uniform meet by ν , so M_π descends to $\mathcal{L}_{E,J} \rightarrow \mathcal{L}_{K,J}$. Pullback also descends, and the two maps still compose to the identity on $\mathcal{L}_{K,J}$. The join map descends by lattice distributivity. Coordinate localization to $\prod_v L_J$ commutes with these finite-fibre operations, but that coordinate map is not silently declared injective; PLP7–9's infinite-product defect remains part of the comparison. Norm remains only a multiplicative map at this localization when its meet map is not join-preserving.

72.4 Class groups, the right cocycle, and the correctly typed transfer

Set $C_F = \mathbb{A}_F^\times / F^\times$. The inclusion of ideles induces an injective map $i_C : C_K \rightarrow C_E$. Indeed if a base idele becomes the principal idele of $a \in E^\times$, that principal idele is Galois-fixed; hence $\gamma(a) = a$ for every γ and $a \in K^\times$. This proves injectivity without assuming an injection of unrelated class-group quotients.

In fact i_C identifies C_K with C_E^G . Here is the exact algebraic descent argument. For a fixed class with representative $x \in \mathbb{A}_E^\times$, write $c_\gamma = \gamma(x)/x \in E^\times$. The cocycle law is $c_{\gamma\delta} = c_\gamma \gamma(c_\delta)$. Independence of the automorphisms, proved above, supplies an $a \in E$ such that

$b = \sum_{\gamma \in G} c_\gamma^{-1} \gamma(a) \neq 0$. The cocycle law gives $\delta(b) = c_\delta b$, so x/b is fixed. Fixed ideles are exactly the diagonal base ideles; local fixed fields and the integral-unit condition prove this at every place. Thus the class is in $i_C(C_K)$. This is the multiplicative Hilbert–90 calculation, included here to make the exact map explicit.

Let the relative Weil group be the original extension

$$1 \longrightarrow C_E \longrightarrow W_{E,K} \longrightarrow G \longrightarrow 1. \quad (\text{GTL16})$$

Choose the source's section s_α , with $s_1 = 1$, and keep the right-hand cocycle and conjugation convention

$$a_{\alpha,\beta} = s_{\alpha\beta}^{-1} s_\alpha s_\beta \in C_E, \quad s_\alpha s_\beta = s_{\alpha\beta} a_{\alpha,\beta}, \quad s_\alpha g s_\alpha^{-1} = \alpha(g), \quad x^\beta = \beta^{-1}(x). \quad (\text{GTL17})$$

Associativity gives the full twisted cocycle law

$$a_{\alpha\beta,\gamma}(a_{\alpha,\beta})^\gamma = a_{\alpha,\beta\gamma} a_{\beta,\gamma}, \quad a_{1,\alpha} = a_{\alpha,1} = 1. \quad (\text{GTL18})$$

Indeed expand $(s_\alpha s_\beta) s_\gamma$ by moving its $a_{\alpha,\beta}$ to the right of s_γ ; this gives $s_{\alpha\beta\gamma} a_{\alpha\beta,\gamma} (a_{\alpha,\beta})^\gamma$. Expanding the other bracketing gives the other side. These are right-cocycle formulas; replacing them by an untwisted product would change the group.

For precision define the ordinary finite-index group transfer $\text{Ver} : W_{E,K} \rightarrow C_E$ using these representatives. For $w \in W_{E,K}$ write $ws_\beta = s_{\pi_w(\beta)} h_{w,\beta}$, $h_{w,\beta} \in C_E$, and put $\text{Ver}(w) = \prod_\beta h_{w,\beta}$. Multiplying two such expressions gives $h_{wz,\beta} = h_{w,\pi_z(\beta)} h_{z,\beta}$, proving that Ver is a group homomorphism because C_E is abelian and π_z permutes the finite index set. Replacing representatives by $s_\beta b_\beta$ changes each factor to $b_{\pi_w(\beta)}^{-1} h_{w,\beta} b_\beta$; their product is unchanged. Conjugating both w and the representative system therefore conjugates Ver 's value. Since a homomorphism into an abelian group is invariant under conjugating its argument, its values are fixed under G . Thus Ver has image in C_E^G .

The correctly typed transfer in the source is

$$t = i_C^{-1} \text{Ver} : W_{E,K} \longrightarrow C_K, \quad i_C(t(s_\alpha)) = \prod_{\beta \in G} a_{\alpha,\beta}, \quad t(g) = N(g) \quad (g \in C_E). \quad (\text{GTL19})$$

The product of cocycle values in the middle expression is literally in C_E , which is why i_C is retained. For the last equality, $gs_\beta = s_\beta \beta^{-1}(g)$ gives $\text{Ver}(g) = \prod_\beta \beta^{-1}(g) = i_C(Ng)$; injectivity of i_C then proves it. This also proves the compatibility in GTL19 rather than trying to deduce it from surjectivity alone. The deep class-field input, precisely as stated in the cited CCM **transferhom** passage, is that this t is surjective. Denote $t_\alpha = t(s_\alpha)$. Its homomorphism law and GTL17 give

$$t_\alpha t_\beta = t_{\alpha\beta} N(a_{\alpha,\beta}). \quad (\text{GTL20})$$

72.5 The full twisted monoid and its transfer

For $F = K, E$ let $X_{F,c} = S_{F,c}/F^\times$ and $X_{F,\text{loc}} = S_{F,\text{loc}}/F^\times$, where each principal nonzero element acts with top label. These are orbit sets with well-defined multiplicative monoids: changing representatives by nonzero principal factors does not change a product orbit. No addition on these orbit sets is assumed. The maps N , ι , and Δ descend because norm takes a principal element to its nonzero field norm and inclusion takes principal elements to principal elements. The descended inclusion is injective. For nonzero amplitudes, an equality $\iota(b) = a\iota(c)$ with $a \in E^\times$ forces $(\gamma(a) - a)\iota(c) = 0$ for every γ . At one nonzero coordinate this forces $\gamma(a) = a$, since a nonzero global field element is nonzero in every completion. Thus $a \in K^\times$. At zero amplitudes labels are unchanged by every principal action, and injectivity follows directly from the label pullback.

The full fixed-orbit descent is

$$\iota_c(X_{K,c}) = (X_{E,c})^G, \quad \iota_{\text{loc}}(X_{K,\text{loc}}) = (X_{E,\text{loc}})^G. \quad (\text{GTL21a})$$

The inclusions already proved therefore identify the base orbit monoids with these fixed orbit monoids. Here fixedness is for the specified G action on the full labelled orbit; it is not only fixedness of the amplitude or of its zero pattern.

To prove GTL21a, the included base orbits are fixed by GTL6. Conversely, choose a representative X of a fixed orbit in either E carrier, and write x for its amplitude adele. First suppose $x \neq 0$. For each $\gamma \in G$ there is a $c_\gamma \in E^\times$ such that $\gamma X = c_\gamma X$. It is unique: an equality $cx = dx$ at a nonzero coordinate implies $c = d$, because E embeds in that local field. Associating $\gamma\delta X$ in two ways consequently gives $c_{\gamma\delta} = c_\gamma\gamma(c_\delta)$. The explicit Hilbert–90 calculation preceding GTL16 applies to this cocycle, even though the adele x need not be an idele. In detail, independence of the distinct automorphisms gives $a \in E$ with $b = \sum_{\gamma \in G} c_\gamma^{-1} \gamma(a) \neq 0$. Since $\delta(c_\gamma^{-1}) = c_\delta c_{\delta\gamma}^{-1}$, reindexing this finite sum gives $\delta(b) = c_\delta b$. It follows that the representative X/b is fixed by every δ . Multiplication by the principal element b^{-1} preserves the adelic restricted condition and leaves every support label unchanged: a nonzero global element is a local unit at all but finitely many finite places and acts with top label.

For completeness, the equality $\gamma X = c_\gamma X$ on the place-labelled carrier gives $\lambda_{\gamma^{-1}w} = \lambda_w$ at every place, including the zero coordinates. Galois transitivity therefore makes these labels constant on each fibre $w \mid v$. On the common-label carrier the single label was unchanged from the start. GTL6 now gives $X/b = \iota(Y)$ for a representative Y in the corresponding K carrier: fixed amplitudes descend through the local fixed fields, and the descended labels obey the original constraint. This exhibits the required base orbit.

If $x = 0$, every principal nonzero element acts trivially on the entire zero-labelled representative. Thus fixedness of its orbit is actual fixedness of its labels. A common label descends unchanged; a place-labelled tuple descends because it is constant on each fibre of π . This covers all zero amplitudes and all lattice labels without an extra choice of a cocycle. Together with the preceding injectivity proof this establishes both equalities of multiplicative monoids in GTL21a. No addition or topological quotient identification is needed for this descent.

Use either one of the two E orbit monoids and call it X . The natural C_E action is by top-supported idele multiplication; the G action is the one already specified. Define

$$(\alpha, x) \star (\beta, y) = (\alpha\beta, a_{\alpha,\beta} x^\beta y) \quad \text{on } G \times X. \quad (\text{GTL21})$$

Its two bracketings have the respective second coordinates

$$a_{\alpha\beta,\gamma} (a_{\alpha,\beta})^\gamma x^{\beta\gamma} y^\gamma z, \quad a_{\alpha,\beta\gamma} a_{\alpha,\beta} x^{\beta\gamma} y^\gamma z.$$

Here $(x^\beta)^\gamma = x^{\beta\gamma}$ by the original inverse action convention. GTL18 proves associativity, and $(1, 1)$ is the unit. The original Weil group identifies with the submonoid $G \times C_E$ via $(\alpha, g) \mapsto s_\alpha g$.

The left and right actions of the original group on this monoid are therefore exactly

$$(s_\alpha g)(\beta, x) = (\alpha\beta, a_{\alpha,\beta} g^\beta x), \quad (\alpha, x)(s_\beta g) = (\alpha\beta, a_{\alpha,\beta} x^\beta g). \quad (\text{GTL22})$$

They commute by the proved associativity. Common labels are retained by each group action. For local labels the left action leaves the x labels unchanged, whereas the right action with $s_\beta g$ replaces the label at w by the old label at βw , exactly as $x^\beta = \beta^{-1}x$ requires.

Define the full supported transfer by

$$\mathcal{T}_c(\alpha, x) = t_\alpha N_c(x), \quad \mathcal{T}_{\text{loc}}(\alpha, x) = t_\alpha N_{\text{loc}}(x). \quad (\text{GTL23})$$

The factors t_α act through their idele classes with top labels. These maps are well-defined and multiplicative for the twisted monoids GTL21. In fact, applying one to a product gives

$t_{\alpha\beta}N(a_{\alpha,\beta})N(x^\beta)N(y)$; norm is Galois-invariant and GTL20 makes this $t_\alpha t_\beta N(x)N(y)$. All label products agree because finite fibre meets commute with the place permutation. Consequently

$$\mathcal{T}(wxz) = t(w)\mathcal{T}(x)t(z), \quad \mathcal{T}_{\text{loc}}(\alpha, \Delta_E x) = \Delta_K \mathcal{T}_c(\alpha, x). \quad (\text{GTL24})$$

This is the exact lifted equivariance and norm/diagonal square. For a fixed α , the map $x \mapsto t_\alpha N(x)$ is not claimed to be a unital monoid homomorphism: its two product expressions differ by one factor t_α . Multiplicativity in GTL23 refers to the full twisted monoid, not the untwisted direct product or an additive quotient.

72.6 Every one-zero-place label and the transferred classical orbits

Let $a^{(w)} \in \mathbb{A}_E$ have amplitude zero at w and one at every other place. Its common-carrier lift has only top label. In the separately labelled carrier write $a_\lambda^{(w)}$ for the tuple with $(0, \lambda)$ at w and $(1, 1_L)$ elsewhere. The exact norms are

$$N_c(a^{(w)}, 1_L) = (a^{(\pi(w))}, 1_L), \quad N_{\text{loc}}(a_\lambda^{(w)}) = a_\lambda^{(\pi(w))}. \quad (\text{GTL25})$$

There is one zero local factor above $\pi(w)$; every other factor is the norm of one. Thus its label meet is exactly λ . The degree d_w has not been replaced by a multiplicity of zero labels: its repeated meet equals λ by the idempotence of the original lattice operation.

Define the common and local classical orbit sets respectively by

$$\Xi_{F,c} = \bigcup_{v \in \Sigma_F} C_F[a_{1_L}^{(v)}], \quad \Xi_{F,\text{loc}} = \bigcup_{v \in \Sigma_F, \lambda \in L} C_F[a_\lambda^{(v)}].$$

The common set embeds as the top-labelled part of the second. Every label is preserved by idele multiplication. Formula GTL25 gives, for $g \in C_E$,

$$\mathcal{T}(\alpha, ga_\lambda^{(w)}) = t(s_\alpha g) a_\lambda^{(\pi(w))}. \quad (\text{GTL26})$$

Here the common case has its required value $\lambda = 1_L$. For any v , choose a place $w \mid v$; for a target idele class $c \in C_K$, the cited transfer surjectivity supplies $s_\alpha g$ with $t(s_\alpha g) = c$. Hence

$$\mathcal{T}_c : G \times \Xi_{E,c} \twoheadrightarrow \Xi_{K,c}, \quad \mathcal{T}_{\text{loc}} : G \times \Xi_{E,\text{loc}} \twoheadrightarrow \Xi_{K,\text{loc}}. \quad (\text{GTL27})$$

These are the exact supported lifts of the source's classical transfer surjection. They do not rely on surjectivity of the norm $C_E \rightarrow C_K$, which is a different map.

The orbit fibres can also be specified without asserting an unproved covering degree. Let H_w be the image of $E_w^\times$ in C_E and H_v the image of $K_v^\times$ in C_K . These embeddings are injective: a principal idele equal to one at a different place has principal field element one. The stabilizer of $[a_\lambda^{(w)}]$ is H_w , since a principal comparison forces its idele to equal that principal element at every nonzero coordinate. Thus its orbit is C_E/H_w , with the same statement over K . Since $N(H_w) \subset H_v$, the exact fibre over $[ca_\lambda^{(v)}]$ is the set

$$\prod_{w \mid v} \prod_{\alpha \in G} \{gH_w : t_\alpha N(g) \in cH_v\}, \quad (\text{GTL28})$$

with label precisely λ . Different zero places and different labels cannot be identified by principal or idele multiplication. The formula is a set decomposition; it does not replace the original ambient orbit topology by a disjoint-union topology, or assert a local-homeomorphism property.

Finally let $C_{F,1} = \ker(| \cdot |_F)$. GTL14 gives $N(C_{E,1}) \subset C_{K,1}$. Right multiplication by $C_{E,1}$ in GTL22 changes only the x coordinate, and GTL24 therefore gives the descended surjections

$$G \times (\Xi_{E,c}/C_{E,1}) \twoheadrightarrow \Xi_{K,c}/C_{K,1}, \quad G \times (\Xi_{E,\text{loc}}/C_{E,1}) \twoheadrightarrow \Xi_{K,\text{loc}}/C_{K,1}. \quad (\text{GTL29})$$

Every label remains unchanged in these quotients. For a single zero-place local label at a support prime J , the equality criterion is exactly $[\lambda]_J = [\mu]_J$, as in PLP10; GTL25 transports that same criterion from w to $\pi(w)$. The original common point instead restricts to top in L_J . These are consequences of the original arbitrary-lattice maps, not a replacement of the support spectrum by a single point.

Precisely retained scope. Inclusion and diagonal comparison are unital semiring maps; Galois actions are semiring automorphisms; norm is a unital multiplicative map with exact mixed-additivity defect GTL10; trace is the additive base-linear companion GTL11. On the orbit sets only their specified multiplicative structures are used. The full Weil cocycle and the typed class-group inclusion remain in GTL17–24. Classical transfer surjectivity is the cited human source input; the labelled fibres and surjections are its proved receiving maps. No additive field-norm assertion, new norm surjectivity, support-label Haar measure, or positivity theorem is inferred.

73 Ramified Frobenius cycles, original Dedekind factors, and the norm clock

Sources, retained inputs, and scope. The adelic orbits, field norm, and modulus comparison used here are those of Alain Connes, Caterina Consani, and Matilde Marcolli, *The Weil proof and the geometry of the adeles class space*, original author locators `globalper`, `avwadele`, `Xiorbits`, `Normmap`, `explnorm`, and `tauXiC1`. The complete programme derivations read for this calculation are `CC_GALOIS_TRANSFER_FULL_LATTICE.tex`, GTL1–29, SHA-256 6d409a2566acf5233e5a41f7c42c04d32434a7b28b76600dbfd92889291ea3b1, and `CC_PRIME_HOLONOMY_SPECTRA.tex`, PHS1–28 and PHS14a–b, SHA-256 d9f280f69de1a59f4d13eee2f2465807a3d613380ecd73541ea0d5482b2ee126. The precise classical field inputs are the decomposition/inertia exact sequence, the local degree formula, and unique factorization of nonzero integral ideals in a number field. These are classical algebraic number theory, as in André Weil’s *Basic Number Theory*, the local-norm reference used by CCM. This note does not claim a new proof or an independent reading of those classical field theorems. It proves the finite representation, determinant, trace, clock, norm-receiver, and arbitrary-lattice calculations from the explicit inputs below. No local or global reciprocity convention is assumed in identifying a group element with a uniformizer.

73.1 Inertia invariants without omitting ramification

Let $E/\mathbb{Q}$ be finite Galois, let $G = \text{Gal}(E/\mathbb{Q})$, and set $n = [E : \mathbb{Q}] = |G|$. Fix a finite rational prime p and one place $w \mid p$. The classical field inputs just specified give

$$D = G_w, \quad I = \ker(D \longrightarrow \text{Gal}(\kappa_w/\mathbb{F}_p)), \quad 1 \longrightarrow I \longrightarrow D \longrightarrow \text{Gal}(\mathbb{F}_{p^f}/\mathbb{F}_p) \longrightarrow 1, \\ e = |I|, \quad f = [\kappa_w : \mathbb{F}_p], \quad d_w = [E_w : \mathbb{Q}_p] = ef, \quad g = \#\{w' \mid p\} = \frac{n}{ef}, \quad q_w = |\kappa_w| = p^f. \quad (\text{GFC1})$$

All places above p have the same e, f , since $E/\mathbb{Q}$ is Galois. Write $\bar{\phi}$ for the arithmetic residue Frobenius $a \mapsto a^p$, and choose a lift $\phi \in D$. It generates D/I , and $\phi^f \in I$. Changing the lift by an inertia element has no effect on the operator on I -invariants constructed below.

Retain the left regular representation on the amplitude space $V = \mathbb{C}[G]$, written as functions on G with delta basis δ_x . Its conventions are

$$(\lambda(a)F)(x) = F(a^{-1}x), \quad \lambda(a)\delta_x = \delta_{ax}, \quad \langle F, H \rangle = \frac{1}{n} \sum_{x \in G} F(x) \overline{H(x)}. \quad (\text{GFC2})$$

The displayed Haar mass is one on G , so each delta vector has squared norm $1/n$. No factor is absorbed into these vectors. Put

$$W_p = V^I, \quad E_I = \frac{1}{e} \sum_{i \in I} \lambda(i), \quad b_{Ix} = \sum_{i \in I} \delta_{ix}, \quad P_p = \lambda(\phi)|_{W_p}. \quad (\text{GFC3})$$

The operator E_I has image W_p : its image is fixed by I by permuting its summands, and it is the identity on an I -fixed vector. Thus $E_I^2 = E_I$. Since I is normal in D , $\lambda(\phi)$ commutes with E_I , and P_p is well-defined. Each I -fixed function is constant on the left cosets Ix , so the b_{Ix} form a basis. Consequently

$$\dim W_p = \frac{n}{e} = gf, \quad \|b_{Ix}\|^2 = \frac{e}{n} = \frac{1}{gf}, \quad P_p b_{Ix} = b_{I\phi x}. \quad (\text{GFC4})$$

There is one label Ix on a whole coset, rather than e independent copies of that coset in W_p . Formula GFC3 retains the precise averaging factor responsible for this dimension.

For every $j \in \mathbb{Z}$, the equality $I\phi^j x = Ix$ is equivalent to $\phi^j \in I$, hence to $f \mid j$. Every cycle of this permutation therefore has exact length f . Its orbits are precisely $D \backslash G$, since D is generated

by I and ϕ . There are exactly g cycles. The bijection from these cycles to the actual places above p is

$$D \setminus G \longrightarrow \{w' \mid p\}, \quad Dx \longmapsto x^{-1}w. \quad (\text{GFC5})$$

If x is replaced by dx with $d \in D$, its image is unchanged. Conversely equal images force $yx^{-1} \in D$, or $Dy = Dx$. Transitivity of the Galois action on the places proves surjectivity.

Choose one representative x_a for each of these g cycles, and let $\omega_f = \exp(2\pi i/f)$. Every eigenvector and its exact inverse finite Fourier formula can be written

$$\begin{aligned} v_{a,k} &= \sum_{j=0}^{f-1} \omega_f^{-kj} b_{I\phi^j x_a}, & P_p v_{a,k} &= \omega_f^k v_{a,k}, & \|v_{a,k}\|^2 &= \frac{fe}{n} = \frac{1}{g}, \\ b_{I\phi^j x_a} &= \frac{1}{f} \sum_{k=0}^{f-1} \omega_f^{kj} v_{a,k}, & 0 \leq k, j < f. \end{aligned} \quad (\text{GFC6})$$

Translation of the index in the finite sum proves the eigenvalue. The finite geometric sum $\sum_{k=0}^{f-1} \omega_f^{k(j-j')} = f \mathbf{1}_{j=j'}$ proves the inverse. Disjoint cosets and GFC4 prove the norm, and the same geometric sum proves orthogonality. Thus every f th root of unity occurs with multiplicity exactly g , including when p is ramified and $e > 1$.

For all $j \in \mathbb{Z}$ the trace is

$$\text{Tr}_{W_p}(P_p^j) = \text{Tr}_V(\lambda(\phi^j)E_I) = \frac{1}{e} \sum_{i \in I} \text{Tr}_V \lambda(\phi^j i) = \begin{cases} n/e = gf, & f \mid j, \\ 0, & f \nmid j. \end{cases} \quad (\text{GFC7})$$

The trace of left translation on V is n at the identity and zero otherwise, as its delta permutation has no fixed point otherwise. There is exactly one $i \in I$ with $\phi^j i = 1$ when $f \mid j$, and none otherwise. This proves every factor in the formula. In the coset Haar measure $1/(gf)$ the permutation kernel has diagonal values gf at fixed cosets, so integration again gives the number of fixed cosets, not their proportion.

73.2 The original local and global Euler factors

Keep $\ell = \log p$ and $q = p^{-s} = e^{-s\ell}$. GFC6 gives the finite determinant, as a polynomial identity for every $q \in \mathbb{C}$,

$$\begin{aligned} D_{E,p}(s) &= \det(1 - p^{-s}P_p \mid W_p) = \prod_{a=1}^g \prod_{k=0}^{f-1} (1 - \omega_f^k p^{-s}) \\ &= (1 - p^{-fs})^g = \prod_{w' \mid p} (1 - q_{w'}^{-s}). \end{aligned} \quad (\text{GFC8})$$

The product of the f linear factors is $1 - q^f$, since they have exactly the f th roots of unity as reciprocal roots and constant coefficient one. This proves the displayed identity rather than assuming a regular-representation Euler-factor theorem. Replacing P_p by P_p^{-1} replaces each individual root ω_f^k by ω_f^{-k} ; the full determinant agrees because inversion permutes all f roots. That agreement never removes the individual inverse-character convention.

The original Dedekind local Euler product is therefore precisely

$$\zeta_{E,p}(s) = \prod_{w' \mid p} (1 - q_{w'}^{-s})^{-1} = D_{E,p}(s)^{-1} = (1 - p^{-fs})^{-g}. \quad (\text{GFC9})$$

This identity includes every finite rational prime. At ramified p , the ramification degree remains in $n = efg$ and in the projection E_I ; the residue norm in the Euler factor is $q_{w'} = p^f$, not p^{ef} . No ramified-prime factor has been set to one or omitted.

For $\Re s > 0$, geometric series and the logarithm which tends to zero as $q \rightarrow 0$ give absolutely convergent identities

$$\begin{aligned} \log \zeta_{E,p}(s) &= \sum_{j \geq 1} \frac{p^{-js}}{j} \operatorname{Tr}_{W_p}(P_p^j) = g \sum_{r \geq 1} \frac{p^{-frs}}{r} = \sum_{w'|p} \sum_{r \geq 1} \frac{q_{w'}^{-rs}}{r}, \\ -\frac{\zeta'_{E,p}(s)}{\zeta_{E,p}(s)} &= \frac{D'_{E,p}(s)}{D_{E,p}(s)} = \ell \sum_{j \geq 1} p^{-js} \operatorname{Tr}_{W_p}(P_p^j) \\ &= g f \ell \sum_{r \geq 1} p^{-frs} = \frac{g f \ell p^{-fs}}{1 - p^{-fs}} = \sum_{w'|p} (\log q_{w'}) \sum_{r \geq 1} q_{w'}^{-rs}. \end{aligned} \quad (\text{GFC10})$$

Differentiation is justified uniformly on compact subsets of this half-plane by the extra factor j against a geometric series. The determinant is entire in s . Its zeros are exactly $s = 2\pi i k / (f\ell)$, $k \in \mathbb{Z}$, each of multiplicity g . Each reciprocal local Euler factor is meromorphic with the corresponding poles. These are local-factor poles, not asserted zeros of a global zeta function.

The original number-field zeta function retains its ideal series and all finite factors:

$$\begin{aligned} \zeta_E(s) &= \sum_{0 \neq \mathfrak{a} \subset \mathcal{O}_E} (N\mathfrak{a})^{-s} = \prod_{p \text{ finite}} D_{E,p}(s)^{-1}, \quad \Re s > 1, \\ -\frac{\zeta'_E(s)}{\zeta_E(s)} &= \sum_{p \text{ finite}} \sum_{w|p} (\log q_w) \sum_{r \geq 1} q_w^{-rs} = \sum_{p \text{ finite}} \frac{g_p f_p \log p p^{-f_p s}}{1 - p^{-f_p s}}. \end{aligned} \quad (\text{GFC11})$$

Here the subscript p makes explicit that e, f, g change with the rational prime. Unique factorization of nonzero integral ideals, the classical field input stated at the start, identifies each term in the expansion of the finite Euler products with its ideal. For $\sigma > 1$ their absolute logarithmic series is bounded by $n \sum_p \sum_{r \geq 1} p^{-r\sigma}/r$, since $g_p \leq n$ and $f_p \geq 1$. This converges by comparison with $\sum_{m \geq 2} m^{-\sigma}$ and the convergent geometric tails. The differentiated series converges uniformly on $\sigma \geq 1 + \varepsilon$ after inserting $\log p$, using $g_p f_p = n/e_p \leq n$ and $\sum_{m \geq 2} (\log m) m^{-1-\varepsilon} < \infty$. These bounds justify the products, series, and differentiation in their original common domain. No Gamma factor, completion, or archimedean replacement has been introduced.

Direct comparison with the finite holonomy of PHS. For a positive integer N coprime to this p , retain $U_N = (\mathbb{Z}/N\mathbb{Z})^\times$, $n_N = \varphi(N)$, $d_N = \operatorname{ord}_{U_N}(p)$, and the exact operator $P_N F(u) = F(pu)$ of PHS7–11. On the entire tensor amplitude $\mathbb{C}^{U_N} \otimes W_p$ use the tensor product of the Haar forms already specified, not a new rescaling. The original characters χ of U_N have squared norm one. Thus

$$\begin{aligned} Q_{p,N} &= P_N \otimes P_p, \quad Q_{p,N}(\chi \otimes v_{a,k}) = \chi(p) \omega_f^k(\chi \otimes v_{a,k}), \\ \|\chi \otimes v_{a,k}\|^2 &= 1/g, \quad \det(1 - p^{-s} \chi(p) P_p | W_p) = (1 - \chi(p)^f p^{-fs})^g. \end{aligned} \quad (\text{GFC11a})$$

The eigenvector statement follows by applying the two original operators to the two factors. The last determinant follows from GFC6 by retaining $\chi(p)$ in each of its f eigenvalues. The joint point return is $(u, c) \mapsto (p^{-1}u, Rc)$; inverse pullback is exactly $Q_{p,N} F(u, c) = F(pu, A(c))$. Its suspension has deck map $(t, u, c) \mapsto (t + \ell, pu, A(c))$, preserving both original orientations.

Put $b = \operatorname{lcm}(d_N, f)$. A joint point is fixed by its j th return exactly when both $d_N \mid j$ and $f \mid j$. Every cycle therefore has length b , and there are $n_N g f / b$ cycles. The full finite determinant

and all its trace coefficients are

$$\begin{aligned}
\mathrm{Tr}(Q_{p,N}^j) &= \mathrm{Tr}(P_N^j) \mathrm{Tr}(P_p^j) = \begin{cases} n_N g f, & b \mid j, \\ 0, & b \nmid j, \end{cases} \\
\det(1 - p^{-s} Q_{p,N}) &= \prod_{\chi \in \widehat{U}_N} (1 - \chi(p)^f p^{-fs})^g = (1 - p^{-bs})^{n_N g f / b}, \\
-\log \det(1 - p^{-s} Q_{p,N}) &= \sum_{j \geq 1} \frac{p^{-js}}{j} \mathrm{Tr}(Q_{p,N}^j) = \frac{n_N g f}{b} \sum_{r \geq 1} \frac{p^{-brs}}{r}, \\
\frac{d}{ds} \log \det(1 - p^{-s} Q_{p,N}) &= \frac{n_N g f \ell p^{-bs}}{1 - p^{-bs}}, \quad \Re s > 0.
\end{aligned} \tag{GFC11b}$$

The trace product is the trace in the tensor delta basis. The cycle decomposition gives the determinant, or equivalently its eigenvalues in GFC11a do so. Absolute geometric convergence proves both series calculations. A joint circle has length $b\ell$ and measure $b\ell/(n_N g f)$, with total measure ℓ . These finite statements continue to retain the inertia divisor e in $gf = n/e$.

There is an original ideal-series receiver for each individual character. Extend χ to integers by zero on nonunits and set

$$Z_E(s, \chi) = \sum_{0 \neq \mathfrak{a} \subset \mathcal{O}_E} \chi(N\mathfrak{a})(N\mathfrak{a})^{-s}, \quad \Re s > 1. \tag{GFC11c}$$

Its absolute convergence is bounded by GFC11. Multiplicativity of the ideal norm, multiplicativity of this integer character including its zero values, and unique ideal factorization give its full original Euler product

$$\begin{aligned}
Z_E(s, \chi) &= \prod_{r \nmid N} \prod_{w' \mid r} (1 - \chi(r)^{f_{w'}} r^{-f_{w'} s})^{-1}, \\
Z_{E,p}(s, \chi) &= (1 - \chi(p)^f p^{-fs})^{-g} = \det(1 - p^{-s} \chi(p) P_p \mid W_p)^{-1}, \\
-\frac{Z'_{E,p}(s, \chi)}{Z_{E,p}(s, \chi)} &= g f \ell \sum_{m \geq 1} \chi(p)^{f m} p^{-f m s} = \frac{g f \ell \chi(p)^f p^{-fs}}{1 - \chi(p)^f p^{-fs}}, \quad \Re s > 0.
\end{aligned} \tag{GFC11d}$$

Every factor at $r \mid N$ is exactly one in this particular series, because $\chi(r^{f_{w'}}) = 0$. This describes the specified integer-character ideal series, without silently replacing it by a primitive Hecke L -function after pullback. If χ_* is the primitive integer character of conductor $c_\chi \mid N$ inducing χ , comparison of these explicit local factors gives the entire omitted-prime correction

$$Z_E(s, \chi) = Z_E(s, \chi_*) \prod_{\substack{r \mid N \\ r \nmid c_\chi}} \prod_{w' \mid r} (1 - \chi_*(r)^{f_{w'}} r^{-f_{w'} s}), \quad \Re s > 1. \tag{GFC11e}$$

This includes every place over each omitted rational prime, whether ramified in E or not. In particular the trivial character gives $\zeta_E(s) \prod_{r \mid N} \prod_{w' \mid r} (1 - r^{-f_{w'} s})$, with no missing factor.

For the unit character $\tilde{\chi}$ on $C_{\mathbb{Q}}$ used in PHS14a, the exact norm comparison later proved in GFC20 yields

$$\tilde{\chi}(\iota_p(N_{E_w/\mathbb{Q}_p} \varpi_w)) = \chi(p)^{-f}, \quad Z_{E,p}(s, \chi) = \prod_{w' \mid p} (1 - \tilde{\chi}^{-1}(\iota_p(N_{E_{w'}/\mathbb{Q}_p} \varpi_{w'})) q_{w'}^{-s})^{-1}. \tag{GFC11f}$$

Indeed $\tilde{\chi}$ depends only on the away- p unit coordinates modulo N because $p \nmid N$. Each of those coordinates in GFC20 is p^{-f} , while the retained p -component is ε_w . Its absence from the value of this particular character is proved by its domain of dependence, not by dropping the unit from the idele.

73.3 The finite suspension and the exact inverse-pullback orientation

On the coset set $C_p = I \backslash G$ define the forward point return $R(c) = I\phi x$ when $c = Ix$. This is well-defined because ϕ normalizes I . Its inverse is $A(c) = I\phi^{-1}x$. Define the explicit finite suspension

$$\mathcal{S}_{E,p} = (\mathbb{R} \times C_p) / ((t, c) \sim (t + \ell, A(c))), \quad \Phi_a[t, c] = [t + a, c], \quad U_a F(t, c) = F(t - a, c). \quad (\text{GFC12})$$

Its forward return at time ℓ is R , since $[\ell, c] = [0, Rc]$. Its Koopman return is

$$(U_\ell F)(0, c) = F(-\ell, c) = F(0, A(c)) = (P_p F(0, \cdot))(c). \quad (\text{GFC13})$$

This uses exactly the left-regular convention GFC2. Composition with the forward point return is instead P_p^{-1} . This is the same inverse-pullback rule proved in PHS7–9; it is not an assertion that an unspecified reciprocity map sends a uniformizer to a particular choice of Frobenius. Arithmetic Frobenius was fixed in GFC1, and both orientations have been explicitly constructed.

Each cycle gives a circle of exact period $f\ell$. More precisely, in the cycle represented by Ix_a the map is

$$[t, I\phi^j x_a] \mapsto [t + j\ell] \quad \text{in } \mathbb{R}/(f\ell)\mathbb{Z}. \quad (\text{GFC14})$$

The deck map sends (t, j) to $(t + \ell, j - 1)$ and hence preserves $t + j\ell$. The only ambiguity in j is an integer multiple of f , which changes that value by $f\ell$. Conversely equality of these values gives exactly a deck equivalence. The inverse sends $[T]$ to $[T, Ix_a]$. Thus there are g circles, not n circles and not g circles of period $ef\ell$.

Retain the coset measure inherited from GFC2, of mass $1/(gf)$ at each c , together with the real measure dt . The suspension has total measure ℓ ; each component has measure $f\ell/(gf) = \ell/g$, and its density in GFC14 is $dT/(gf)$. This is the finite coefficient space measure constructed here. It is not asserted to equal a Tamagawa measure or a pushforward of an unspecified adelic measure.

73.4 Original norm clocks and the unit of a uniformizer norm

Let $C_F = \mathbb{A}_F^\times / F^\times$ and $C_{F,1} = \ker(|\cdot|_F)$, using the original Haar-module absolute values of GTL13–14. Let $j_w : E_w^\times \rightarrow C_E$ and $\iota_p : \mathbb{Q}_p^\times \rightarrow C_\mathbb{Q}$ be the local-idele embeddings. The one-zero-place orbit is C_E/H_w , where $H_w = j_w(E_w^\times)$; its base analogue is $C_\mathbb{Q}/H_p$. These stabilizer identifications are GTL28 and PLP15–20.

For a chosen local uniformizer ϖ_w the local norm has the exact form

$$a_w = N_{E_w/\mathbb{Q}_p}(\varpi_w) = p^f \varepsilon_w, \quad \varepsilon_w \in \mathbb{Z}_p^\times, \quad |\varpi_w|_w = q_w^{-1} = p^{-f}, \quad |a_w|_p = p^{-f}. \quad (\text{GFC15})$$

Indeed multiplication by ϖ_w sends the integer ring to an additive subgroup of index q_w , so its Haar module is q_w^{-1} . The determinant of that multiplication is the field norm, and GTL13 gives $|a_w|_p = |\varpi_w|_w$. Therefore its p -adic valuation is exactly f , which proves the displayed unit factorization. No equality $a_w = p^f$ is assumed. Replacing ϖ_w by $v\varpi_w$ with $v \in \mathcal{O}_w^\times$ changes this retained unit to $N(v)\varepsilon_w$.

The modulus map gives a complete description of the clock quotient:

$$(C_E/H_w)/C_{E,1} \simeq \mathbb{R}/(f\ell)\mathbb{Z}, \quad (C_\mathbb{Q}/H_p)/C_{\mathbb{Q},1} \simeq \mathbb{R}/\ell\mathbb{Z}. \quad (\text{GFC16})$$

The map sends an idele class to the logarithm of its modulus. It is onto because positive archimedean ideles have every positive modulus. Its kernel before dividing by H_w is $C_{E,1}$, and $\log |H_w| = (f\ell)\mathbb{Z}$ by the unit/uniformizer decomposition of $E_w^\times$. These statements prove both the quotient and its exact fibres. The analogous calculation over $\mathbb{Q}$ gives $\ell\mathbb{Z}$.

Norm preserves the full idele modulus, $|Nc|_{\mathbb{Q}} = |c|_E$, by GTL14. Its induced clock map and each of its fibres are therefore

$$\begin{aligned} \nu_w : \mathbb{R}/(f\ell)\mathbb{Z} &\longrightarrow \mathbb{R}/\ell\mathbb{Z}, & [T] &\longmapsto [T], \\ \nu_w^{-1}([T]_{\ell}) &= \{[T + j\ell]_{f\ell} : 0 \leq j < f\}. \end{aligned} \quad (\text{GFC17})$$

It is a covering of exact degree f . Taking the disjoint union over the g places gives total fibre cardinality $gf = n/e$, the dimension of GFC4. Equations GFC5 and GFC14 give an explicit isomorphism from the finite suspension to this disjoint union of clock circles after choosing each x_a . It commutes with forward time and the map to the base clock. Choice of a different representative in a cycle changes that component's clock origin by an integer multiple of ℓ ; no canonical origin is asserted. The statement concerns the disjoint union of these specified quotients, and does not change the ambient topology of the full adelic orbit union.

For the full return before the compact modulus-one quotient, choose an exact flow section. At every archimedean place of E set the component of $\sigma_E(T)$ to the positive real number $e^{T/n}$, and set its finite components to one. At a complex place the Haar-module absolute value is the square of the usual modulus. Writing $r_1 + 2r_2 = n$ proves

$$|\sigma_E(T)|_E = e^T, \quad N\sigma_E(T) = \sigma_{\mathbb{Q}}(T), \quad (\text{GFC18})$$

where $\sigma_{\mathbb{Q}}(T)$ has real component e^T and finite components one. For the second equality, each real local norm of $e^{T/n}$ is that number and each complex local norm is its square; their full product is e^T , with every finite norm equal to one.

Define the actual retained compact return frames

$$\begin{aligned} T_w &= \log q_w = f\ell, & \mathfrak{k}_w &= \sigma_E(T_w)j_w(\varpi_w) \in C_{E,1}, \\ N\mathfrak{k}_w &= \sigma_{\mathbb{Q}}(f\ell)\iota_p(p^f\varepsilon_w) =: \mathfrak{k}_{p,w} \in C_{\mathbb{Q},1}, \\ \sigma_E(T_w) &= \mathfrak{k}_wj_w(\varpi_w^{-1}), & \sigma_{\mathbb{Q}}(f\ell) &= \mathfrak{k}_{p,w}\iota_p(a_w^{-1}). \end{aligned} \quad (\text{GFC19})$$

The norm of a local idele at w is precisely the local idele of its local norm at p , since the other factors in GTL2 are norms of one. This proves the middle equality. The last two identities follow by multiplication, and retain the full unit contribution.

In the exact coordinates $C_{\mathbb{Q}} = \mathbb{R}_{>0} \times \prod_r \mathbb{Z}_r^{\times}$ of PLP20, these frames and local ideles are

$$\begin{aligned} \iota_p(p^f\varepsilon_w) &= (p^{-f}, u_p = \varepsilon_w, u_r = p^{-f} \ (r \neq p)), \\ \mathfrak{k}_{p,w} &= (1, u_p = \varepsilon_w, u_r = p^{-f} \ (r \neq p)), \\ \mathfrak{k}_{p,0} &= \mathfrak{k}_{p,0}\iota_p(\varepsilon_w), & \mathfrak{k}_{p,0} &= (1, u_p = 1, u_r = p^{-f} \ (r \neq p)). \end{aligned} \quad (\text{GFC20})$$

Thus the away- p return is exactly the f th iterate $u \mapsto p^{-f}u$ of PHS2, while the p -unit remains explicitly in its stabilizer frame. It disappears only from the clock quotient and the zero-place orbit coordinate, where that stabilizer is divided out. It need not disappear from a transverse coordinate or test.

73.5 The exact norm map on the associated transverse spaces

For clarity retain the associated transverse spaces and relations

$$\begin{aligned} \mathcal{B}_w &= (C_E \times E_w)/((cj_w(h), z) \sim (c, hz)), \\ \mathcal{B}_p &= (C_{\mathbb{Q}} \times \mathbb{Q}_p)/((d\iota_p(b), y) \sim (d, by)), \\ \mathcal{N}_w : \mathcal{B}_w &\longrightarrow \mathcal{B}_p, & [c, z] &\longmapsto [Nc, N_{E_w/\mathbb{Q}_p}(z)]. \end{aligned} \quad (\text{GFC21})$$

The local field norm at zero is zero. The map is well-defined: the two representatives in the first line are sent respectively to $[Nc\iota_p(Nh), Nz]$ and $[Nc, N(hz)]$, which agree by the second relation

and norm multiplicativity. It intertwines the flows $\sigma_E(T)$ and $\sigma_Q(T)$ by GFC18. This is an exact map of the stated associated spaces. The local fibre map $z \mapsto Nz$ is a homogeneous polynomial of degree ef , not an additive linear identification of E_w with $\mathbb{Q}_p$.

In the compact frames of GFC19, the forward return at T_w and its norm receiver are exactly

$$[c, z] \mapsto [\mathfrak{f}_w c, \varpi_w^{-1} z], \quad [Nc, Nz] \mapsto [\mathfrak{f}_{p,w} Nc, p^{-f} \varepsilon_w^{-1} Nz]. \quad (\text{GFC22})$$

Indeed moving the local factor in the last line of GFC19 across the associated-space relation gives the first formula, and $N(\varpi_w^{-1} z) = a_w^{-1} Nz$ gives the second. In the frame $\mathfrak{f}_{p,0}$ the second representative equals

$$[\mathfrak{f}_{p,0} \iota_p(\varepsilon_w) Nc, p^{-f} \varepsilon_w^{-1} Nz] = [\mathfrak{f}_{p,0} Nc, p^{-f} Nz]. \quad (\text{GFC23})$$

The cancellation here is an explicit equivalence of two framed representatives, retaining both the unit and its inverse. It is not the false scalar equality $N\varpi_w = p^f$. Equations GFC21–23 are the exact comparison to the f -step original transverse return of PHS2.

Retain additive Haar mass one on $\mathcal{O}_w$ and on $\mathbb{Z}_p$. For $r \in \mathbb{Z}_{>0}$ the source and target frame multipliers have the exact scaling factors

$$\begin{aligned} d(\varpi_w^{-r} z) &= q_w^r dz, & d(a_w^{-r} y) &= p^{fr} dy = q_w^r dy, \\ |1 - \varpi_w^r|_w &= 1, & |1 - \varpi_w^{-r}|_w &= q_w^r, & |1 - a_w^r|_p &= 1, & |1 - a_w^{-r}|_p &= p^{fr}. \end{aligned} \quad (\text{GFC24})$$

The equalities for $1 - h$ follow from the ultrametric inequality: if $|h| < 1$, then $|1 - h| = 1$; if $|h| > 1$, then $|1 - h| = |h|$. The unit ε_w does not change these valuations, but still occurs in the scalar $a_w^{\pm r}$ itself. For the specified inverse pullback $\vartheta(h)F(z) = F(h^{-1}z)$, its kernel diagonal away from $h = 1$ is $|1 - h^{-1}|^{-1}$, by the Dirac change of variables of PHS20. Hence

point time	E_w normal multiplier	$\mathbb{Q}_p$ norm-frame multiplier	inverse-pullback diagonal in either field
$+r \log q_w$	ϖ_w^{-r}	$p^{-fr} \varepsilon_w^{-r}$	1
$-r \log q_w$	ϖ_w^r	$p^{fr} \varepsilon_w^r$	q_w^{-r}

(\text{GFC25})

These local diagonal distributions retain both time orientations. They do not assert equality of all local test-function distributions under norm, and do not give a Haar Jacobian for the nonlinear map $z \mapsto Nz$. At $z = 0$ that polynomial even has zero linear derivative when $ef > 1$. The identity-time regularization and a global cohomological trace theorem are not inferred from this table. The Euler logarithmic derivative in GFC10 retains its separate primitive clock length $\log q_w = f \log p$ for each of the g places; the finite trace retains gf at $j = fr$ as in GFC7.

73.6 Full common labels and the different place-label norm fibres

Let L be the original nontrivial bounded distributive lattice, with bottom 0_L and top 1_L . On a complex amplitude space V_0 retain

$$\begin{aligned} G_L(V_0) &= \{(0, \lambda) : \lambda \in L\} \cup (V_0 \times \{1_L\}), \\ (v, \lambda) + (w, \mu) &= (v + w, \lambda \vee \mu), & (a, \alpha)(v, \lambda) &= (av, \alpha \wedge \lambda), \\ \tau &= (0, 0_L), & e_L &= (0, 1_L). \end{aligned} \quad (\text{GFC26})$$

Here e_L denotes supported zero, to distinguish it from the ramification degree e . Every linear map T above has the exact lift $T^\dagger(v, \lambda) = (Tv, \lambda)$. It is well-defined because a nonzero output forces a nonzero input and hence its top label. Linearity proves preservation of addition and of the $G_L(\mathbb{C})$ scalar action using precisely the joins and meets in GFC26. All compositions and linear inverse maps are retained. In particular

$$\begin{aligned} G_L(V)^I &= G_L(V^I) = G_L(W_p), & (E_I^\dagger)^2 &= E_I^\dagger, & P_p^\dagger E_I^\dagger &= E_I^\dagger \lambda(\phi)^\dagger, \\ \ker_{\text{set}}(G_L(V) \longrightarrow V^I, (v, \lambda) \mapsto E_I v) &= \{(v, 1_L) : E_I v = 0\} \cup \{(0, \lambda) : \lambda \in L\}. \end{aligned} \quad (\text{GFC27})$$

The first equality follows directly because the lift fixes the label and its amplitude is fixed exactly when $v \in V^I$. The full preimage of (w, μ) under $E_I^\uparrow$ is $\{(v, 1_L) : E_I v = w\}$ if $\mu = 1_L$, and is the singleton $\{(0, \mu)\}$ when $\mu < 1_L$ (in which case necessarily $w = 0$). Thus a nonzero vector killed by averaging maps to e_L , not to τ . The Fourier map GFC6 and its inverse preserve the same single label on the whole gf -dimensional vector. They do not create independent labels on the g cycles or the gf coefficients. The joint operator GFC11a lifts on $G_L(\mathbb{C}^{U_N} \otimes W_p)$ with one label on the entire $n_N gf$ -dimensional amplitude. No identification with a tensor product of independently labelled carriers is used. On localizing at a support prime $J \subset L$, the original support map $(v, \lambda) \mapsto [\lambda]_J$ commutes with all these operators, as each has kept exactly that label.

The actual norm on field amplitudes and on adeles is multiplicative, so its label lift is a different type of map. In the common carrier it is $N_c(x, \lambda) = (Nx, \lambda)$ as in GTL5, retaining one label. In the separately labelled local carrier its p -component is

$$((x_{w'}, \lambda_{w'}))_{w'|p} \mapsto \left(\prod_{w'|p} N_{E_{w'}/\mathbb{Q}_p}(x_{w'}), \bigwedge_{w'|p} \lambda_{w'} \right). \quad (\text{GFC28})$$

There are g actual place labels in this meet. The degree ef of a local field norm does not produce ef new labels; a repeated meet of the same label is that label. Both maps preserve multiplication, their multiplicative unit, and τ . They are not claimed additive or linear, unlike GFC27.

For a fixed local amplitude tuple $x = (x_{w'})_{w'|p}$ set $A_p(x) = \{w' : x_{w'} = 0\}$. If it is empty, the norm amplitude is nonzero and every input and output label is forced to top. If it is nonempty, the exact map on this amplitude fibre is

$$\begin{aligned} L^{A_p(x)} &\longrightarrow L, & (\lambda_{w'}) &\longmapsto \bigwedge_{w' \in A_p(x)} \lambda_{w'}, \\ \text{fibre over } \mu &= \{(\lambda_{w'}) \in L^{A_p(x)} : \bigwedge_{w' \in A_p(x)} \lambda_{w'} = \mu\}. \end{aligned} \quad (\text{GFC29})$$

The omitted nonzero-coordinate labels in the last display have all been fixed to top in GFC28. Surjectivity follows by putting μ at one of the zero places and top at the others. It does not require a complement, a Boolean lattice, or an infinite meet. For two or more zero places the three patterns with one chosen coordinate μ , a different chosen coordinate μ , or both coordinates μ , all have the same meet; for $\mu < 1_L$ they are distinct. Thus the map is generally many-to-one, with its complete fibre given above. For exactly one zero place it is the identity on that label.

At the level of all adeles, write $Z(x) = \{w : x_w = 0\}$ and let π be the map of places. The complete fixed-amplitude fibre map and the diagonal comparison are

$$\begin{aligned} L^{Z(x)} &\longrightarrow L^{\pi(Z(x))}, & (\lambda_w) &\longmapsto \left(\bigwedge_{w \in Z(x), w|v} \lambda_w \right)_v, \\ N_{\text{loc}} \Delta_E(x, \lambda) &= \Delta_{\mathbb{Q}} N_c(x, \lambda). \end{aligned} \quad (\text{GFC30})$$

Each target fibre is a product of the finite fibres GFC29. The diagonal square follows from finite idempotence of meet. For a nonzero adelic x , its original common label is forced to top, even when some coordinates are zero. Hence the original one-zero-place orbit has only its top common lift. Allowing a separate label at that zero place belongs to the larger local carrier, where GFC28 indeed carries its single label unchanged to the base zero place. For the all-zero common amplitude, $N_c(0, \lambda) = (0, \lambda)$, so $e_L \neq \tau$ remains exact. More generally the complete common-carrier inverse fibre of $(0, \mu)$ with $\mu < 1_L$ is just $(0, \mu)$; the inverse fibre of e_L is $\{(x, 1_L) : Nx = 0\}$, and that of a nonzero $(y, 1_L)$ is $\{(x, 1_L) : Nx = y\}$. These statements use only the carrier constraint, not norm surjectivity.

All actual norm maps just used also lift the associated-space map GFC21 with one label on each whole local fibre by $[c, (z, \lambda)] \mapsto [Nc, (Nz, \lambda)]$. The equivalence relations use multiplication

by top-supported nonzero field elements, so the proof of well-definedness in GFC21 keeps the label unchanged. Its map at an actual zero-place label is consequently the one-coordinate case of GFC29. This is an exact multiplicative and equivariant receiving map; it does not turn the nonlinear norm into the linear Frobenius projection or identify their fibres. For the original common adelic point, its other nonzero coordinates force this local fibre label to top. Allowing every label at its zero coordinate instead belongs to the separately labelled adelic carrier. This restriction is retained in the associated-space comparison and is not removed by using a fibre coefficient lift.

Exact conclusion of the calculation. Inertia averaging retains the factor $1/e$ and yields g cycles of exact length f . Their determinant is the original Dedekind p -factor inverse, their primitive clock is $\log q_w = f \log p$, and the actual norm sends that clock to the base clock with degree f . The unit ε_w of the uniformizer norm remains in the full compact and transverse return GFC19–23. The inverse-pullback distribution uses the signs and factors of GFC25. Common coefficient labels are carried by linear lifts, whereas actual multiple-place norm labels are carried by the finite meet with fibres GFC29. These proved maps relate the objects without replacing one by the other. No positivity theorem, RH conclusion, completed-zeta substitute, or assertion that this finite suspension is the entire arithmetic cohomological trace follows from this calculation.

74 Every finite-place Weil term and the conductor operator

Original sources and scope. The local distribution and its multiplicative Haar measure are from Alain Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, original author TeX, local trace theorem (12)–(17), lines 1367–1436; VIII(10), lines 2353–2360; and Appendix II(9), (22)–(29), lines 4280–4495. The original Weil principal value and its conductor term are human results, not new discoveries here. The finite-field computations and the receiving operator below are proved explicitly with all measures retained. Alain Connes, Caterina Consani and Matilde Marcolli, *The Weil proof and the geometry of the adèles class space*, original author locators `Weilexpl`, `princvalrhobeta`, `TraceformulaH1` and `intalphae`, give the global receiving formula and its differential-idele term. That original trace theorem is a cited input; the calculation here evaluates its finite local terms.

74.1 The original measure and an explicit principal value

Let F be a nonarchimedean local field, $\mathcal{O}$ its ring of integers, ϖ a uniformizer, and $q = |\mathcal{O}/\varpi\mathcal{O}|$. Retain additive Haar measure dx with $\text{vol}(\mathcal{O}) = 1$ and $|\varpi| = q^{-1}$. The source’s multiplicative measure is

$$d^*x = \frac{\log q}{1 - q^{-1}} \frac{dx}{|x|}, \quad \text{vol}_{d^*x}(\mathcal{O}^\times) = \log q. \quad (\text{WLC1})$$

Indeed every shell $\varpi^j\mathcal{O}^\times$ has that same multiplicative mass. The integral on $1 \leq |x| \leq q^M$ is $(M+1)\log q$, whose ratio to $\log(q^M)$ tends to one, as required by the original measure convention. No unit mass has been substituted for the displayed $\log q$.

For a locally constant compactly supported function b on $F^\times$ define the original Weil principal value by the explicit formula

$$\begin{aligned} W_F(b) = & \int_{|u| \neq 1} \frac{b(u^{-1})}{|1 - u|} d^*u \\ & + \int_{\mathcal{O}^\times} \frac{b(u^{-1}) - b(1)}{|1 - u|} d^*u. \end{aligned} \quad (\text{WLC2})$$

Both integrals are finite: only finitely many valuation shells meet the first support, and the second numerator vanishes on a neighborhood of one. This agrees with ordinary integration away from one and assigns zero to the test $1_{\mathcal{O}^\times}$. It is the source’s Appendix II(9) principal value. Its equality with the additive-character prescription is the cited Lemma 2 in that appendix, for a character trivial on $\mathcal{O}$ but nontrivial on $\varpi^{-1}\mathcal{O}$.

The last condition is the conductor condition. It does not assert that the additive character’s entire kernel is $\mathcal{O}$. The wording $\ker e_F = \mathcal{O}$ in the later CCM passage is too strong for general local fields. For example, on $\mathbb{F}_p((T))$, the character $\beta(\sum a_j T^j) = \exp(2\pi i a_{-1}/p)$ has the stated conductor but $\beta(T^{-2}) = 1$. In fact every additive character of this field takes values in the p th roots of unity, since the additive group has exponent p ; its quotient by its kernel has at most p elements. The quotient $\mathbb{F}_p((T))/\mathbb{F}_p[[T]]$ is infinite, so no such character can have that ring as its entire kernel. WLC2 uses the valid original conductor prescription, not that stronger assertion.

The exact Fourier relation survives with that conductor condition. One has $\mathcal{O}^\perp = \mathcal{O}$: an element of $\mathcal{O}$ pairs trivially with every element of $\mathcal{O}$. If $v(x) = -k < 0$, then $x\mathcal{O} = \varpi^{-k}\mathcal{O}$ contains $\varpi^{-1}\mathcal{O}$, where the character is nontrivial; hence some $y \in \mathcal{O}$ has $\beta_0(xy) \neq 1$. Integration of this nontrivial character over $\mathcal{O}$ is zero by translation invariance. Therefore the actual Fourier map satisfies

$$\int_{\mathcal{O}} \beta_0(xy) dy = 1_{\mathcal{O}}(x).$$

This is the comparison used in Connes’s Appendix II(24), with the original additive measure of mass one on $\mathcal{O}$.

Let $\chi : F^\times \rightarrow \mathbb{C}^\times$ be a continuous character whose restriction to $\mathcal{O}^\times$ has finite image. Put $\lambda = \chi(\varpi) \neq 0$, and let its unit conductor be

$$a = 0 \text{ if } \chi|_{\mathcal{O}^\times} = 1; \quad a = \min\{j \geq 1 : \chi|_{1+\varpi^j\mathcal{O}} = 1\} \text{ otherwise.} \quad (\text{WLC3})$$

For a continuous unitary character the finite-image condition follows from continuity and the profinite unit group: an open subgroup maps into a sufficiently short arc with no nontrivial subgroup of the unit circle, and therefore into one. For $h \in C_c^\infty(\mathbb{R}_{>0})$ use the actual test

$$b(x) = h(|x|)\chi(x). \quad (\text{WLC4})$$

It is locally constant on $F^\times$ and compactly supported there.

74.2 Every shell and the ramified unit term

On $u = \varpi^j\epsilon$, $j > 0$, one has $|1 - u| = 1$ and $b(u^{-1}) = h(q^j)\lambda^{-j}\chi(\epsilon)^{-1}$. For $u = \varpi^{-j}\epsilon$ the denominator is q^j and the numerator is $h(q^{-j})\lambda^j\chi(\epsilon)^{-1}$. The integral of a nontrivial unit character is zero: translating the Haar integral by a unit with character value different from one multiplies it by that value without changing the integral. For $a = 0$, the unit numerator in WLC2 vanishes identically. Consequently

$$W_F(b) = \log q \sum_{j \geq 1} [\lambda^{-j}h(q^j) + q^{-j}\lambda^j h(q^{-j})] \quad (a = 0). \quad (\text{WLC5})$$

The sum has finitely many nonzero terms for the stated h .

For $a > 0$, every off-unit shell instead integrates to zero. Write $K_j = 1 + \varpi^j\mathcal{O}$, $j \geq 1$. Their original masses are

$$\mu^*(K_j) = \frac{\log q}{1 - q^{-1}}q^{-j}, \quad \int_{K_j} \chi(u)^{-1}d^*u = \begin{cases} 0, & j < a, \\ \mu^*(K_j), & j \geq a. \end{cases} \quad (\text{WLC6})$$

The mass follows from additive translation by one and $|u| = 1$. The character integral follows from the same translation argument, now in the compact subgroup K_j , and from WLC3. For $u \neq 1$ in the unit group the exact finite-at-each-point identity is

$$\frac{1}{|1 - u|} = 1 + \sum_{j \geq 1} (q^j - q^{j-1})1_{K_j}(u). \quad (\text{WLC7})$$

If the valuation of $1 - u$ is k , the sum stops at k and telescopes to q^k . After multiplication by $\chi(u)^{-1} - 1$, every term with $j \geq a$ is zero, so the following integration is a finite calculation:

$$\begin{aligned} \int_{\mathcal{O}^\times} \frac{\chi(u)^{-1} - 1}{|1 - u|} d^*u &= -\log q - \sum_{j=1}^{a-1} (q^j - q^{j-1}) \frac{\log q}{1 - q^{-1}} q^{-j} \\ &= -a \log q. \end{aligned} \quad (\text{WLC8})$$

Thus the complete ramified value, including the singular unit stratum, is

$$W_F(b) = -a \log q h(1) \quad (a > 0). \quad (\text{WLC9})$$

Its local Euler factor is one, but its local Weil distribution has not vanished. WLC8 independently recovers the human conductor formula Connes VIII(10), retaining its original sign and measure.

If the additive character is $\beta_c(x) = \beta_0(cx)$ for a conductor- $\mathcal{O}$ reference character β_0 , the cited CCM identity `intalphae` retains the additional term

$$W_{F,\beta_c}(b) = W_{F,\beta_0}(b) + \log |c| b(1). \quad (\text{WLC10})$$

No different-idele term is removed by the shell computation.

74.3 The original rational prime and Dirichlet dictionaries

Use $C_{\mathbb{Q}} = \mathbb{R}_{>0} \times \prod_r \mathbb{Z}_r^\times$ in the exact PLP20 coordinates. A finite character χ of $(\mathbb{Z}/N\mathbb{Z})^\times$ induces a compact-unit character $\tilde{\chi}$ and a unique primitive character χ_* of conductor $f_\chi \mid N$. For

$$f(y, u) = h(y)\tilde{\chi}(u), \quad b_p(x) = f(\iota_p(x)), \quad (\text{WLC11})$$

the local uniformizer has the exact coordinates

$$\iota_p(p^j \epsilon) = (p^{-j}, (u_p = \epsilon, u_r = p^{-j} (r \neq p))) . \quad (\text{WLC12})$$

It follows that the local unit conductor is $a_p = v_p(f_\chi)$. If $a_p = 0$, then $\tilde{\chi}(\iota_p(p)) = \chi_*(p)^{-1}$. This follows by multiplying the finitely many other unit components; their integer evaluation at p is exactly $\chi_*(p)$. The inverse local character is essential: WLC5 therefore becomes

$$W_p(b_p) = \log p \sum_{j \geq 1} [\chi_*(p)^j h(p^j) + p^{-j} \chi_*(p)^{-j} h(p^{-j})] \quad (p \nmid f_\chi), \quad (\text{WLC13})$$

whereas at a conductor prime its full value is

$$W_p(b_p) = -v_p(f_\chi) \log p h(1). \quad (\text{WLC14})$$

This agrees with the two point-return and inverse-pullback rows PHS19–21, rather than identifying their opposite phases.

Summing the finite places gives the exact expression

$$\begin{aligned} \sum_p W_p(b_p) &= \sum_{p \nmid f_\chi} \log p \sum_{j \geq 1} [\chi_*(p)^j h(p^j) + p^{-j} \chi_*(p)^{-j} h(p^{-j})] \\ &\quad - \sum_{p \mid f_\chi} v_p(f_\chi) \log p h(1). \end{aligned} \quad (\text{WLC15})$$

All prime-power sums here are finite: the compact support of h bounds both p^j and p^{-j} away from their respective escaping endpoints. There are finitely many conductor primes. The last sum equals $-\log(f_\chi)h(1)$, with all its local summands displayed above.

The compact character depends on its conductor, not on an extra imprimitive modulus. For $p \mid N$ but $p \nmid f_\chi$, WLC13 still occurs. This is compatible with the original Dirichlet series only after retaining every missing-prime factor:

$$L(s, \chi) = L(s, \chi_*) \prod_{\substack{p \mid N \\ p \nmid f_\chi}} (1 - \chi_*(p)p^{-s}), \quad \Re s > 1. \quad (\text{WLC16})$$

The proof is term-by-term Euler-factor comparison in the region of absolute convergence, as proved in PHS13–18. Replacing $\chi_*(p)$ by the imprimitive value zero in WLC13 would lose these actual arithmetic terms.

For the standard global additive character of $\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}$, the finite reference conductors are $\mathbb{Z}_p$. The original CCM global trace receives WLC15 in the formula

$$\begin{aligned} \text{Tr}(\vartheta_m(f)|H^1) &= \hat{f}(0) + \hat{f}(1) - \log |\mathfrak{a}| f(1) \\ &\quad - W_{\mathbb{R}, e_{\mathbb{R}}}(f \circ \iota_\infty) - \sum_p W_p(f \circ \iota_p). \end{aligned} \quad (\text{WLC17})$$

Here $\mathfrak{a}$ is the source's differential idele, and the archimedean term is its specified original principal value, not an omitted remainder. With $d^\times y = dy/y$ and compact-unit Haar mass one, the endpoint integrals for WLC11 are exactly

$$\hat{f}(0) = \delta_{\chi_*, 1} \int_0^\infty h(y) \frac{dy}{y}, \quad \hat{f}(1) = \delta_{\chi_*, 1} \int_0^\infty h(y) dy, \quad f(1) = h(1). \quad (\text{WLC18})$$

These follow from the integral of the compact character. The archimedean restriction is $f(\iota_\infty(x)) = h(|x|)\chi_*(\text{sign } x)$; its parity has not been discarded. For the displayed rational reference choice $|\mathfrak{a}| = 1$, but WLC17 retains its original location and WLC10 its local dependence. WLC17 invokes the cited global theorem; WLC1–16 evaluate precisely its finite part and do not replace that theorem’s analytic proof.

74.4 An exact finite conductor operator before the trace

On $V_N = \mathbb{C}^{U_N}$ use the original finite Haar measure $\mu_N(u) = 1/\varphi(N)$, and the PHS28 Fourier coefficients. For every prime divisor p of N , put $r = v_p(N)$ and define

$$C_{p,N}\chi = v_p(f_\chi) \log p \chi, \quad C_N = \sum_{p|N} C_{p,N}. \quad (\text{WLC19})$$

This defines genuine linear operators by the finite character basis. Their full action and quadratic form are

$$\begin{aligned} (C_N a)(u) &= \sum_{\chi \in \widehat{U}_N} \left(\sum_{p|N} v_p(f_\chi) \log p \right) \widehat{a}(\chi) \chi(u), \\ \langle a, C_N a \rangle_{\mu_N} &= \sum_{\chi \in \widehat{U}_N} \left(\sum_{p|N} v_p(f_\chi) \log p \right) |\widehat{a}(\chi)|^2 \geq 0. \end{aligned} \quad (\text{WLC20})$$

Orthogonality proves the formula and self-adjointness, with all measure factors inherited from PHS9 and PHS28. The kernel is the constant-character line, since conductor one means the character is trivial. This finite positive quadratic form is not a positivity assertion for WLC17’s entire trace.

For the general finite-character test $f(y, u) = h(y)a(u)$, linearity and WLC14 show that its full ramified-unit contribution is

$$-h(1)(C_N a)(1). \quad (\text{WLC21})$$

Evaluation at one sums the character coefficients because every character has value one there. Unramified valuation-shell terms and WLC17’s other contributions remain separate summands of the same original trace.

The multiplicities can also be computed exactly. On U_{p^r} there is one conductor-zero character. The number with conductor at most j , $1 \leq j \leq r$, is $\varphi(p^j)$, since reduction onto U_{p^j} is surjective and its characters are exactly the pullbacks. Counting $a = \sum_{j=0}^{r-1} 1_{a>j}$ gives

$$\text{Tr}(C_{p,N}) = \varphi(N/p^r) \log p \left[r\varphi(p^r) - 1 - \sum_{j=1}^{r-1} \varphi(p^j) \right]. \quad (\text{WLC22})$$

The factor $\varphi(N/p^r)$ counts the independent characters at all other prime factors. The displayed finite sum includes the empty-sum case $r = 1$, so $p = 2, r = 1$ correctly contributes zero. Telescoping $\varphi(p^j) = p^j - p^{j-1}$ makes the bracket $r\varphi(p^r) - p^{r-1}$; both the local degree and every filtration contribution were retained in WLC22.

For the entire original bounded distributive lattice, the maps are

$$C_N^\uparrow : G_L(V_N) \longrightarrow G_L(V_N), \quad (a, \lambda) \longmapsto (C_N a, \lambda), \quad W_F^\uparrow(b, \lambda) = (W_F(b), \lambda). \quad (\text{WLC23})$$

Linearity proves additivity and compatibility with $G_L(\mathbb{C})$ scalars. A nonzero output amplitude comes from a nonzero input, hence has its forced top label; a zero output keeps its original label. These are common-label semimodule maps, not arbitrary basis changes on independently labelled coordinate products. In particular the constant test with top label satisfies

$$C_N^\uparrow(1, 1_L) = e, \quad e \neq \tau \quad \text{for nontrivial } L. \quad (\text{WLC24})$$

At every support stalk, its map commutes with the original localization $(a, \lambda) \mapsto [\lambda]_J$ because it keeps that label. No measure on the arbitrary lattice has been invented. The unit conductor, prime-power phases, omitted-prime factors and labelled zero images have all been evaluated in their specified original objects.

75 The complete original Weil form for finite compact-character tests

Original sources and scope of actual reading. Alain Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, original author TeX `math_9811068v1_author.tex`, SHA-256 `f89f55df9faa2b8a49ccc2fee6a0dd4cbfb8d5279edae86e70979413b055eff6`, was read at Appendix II (1)–(54), including its real and complex principal-value computations, source lines 4217–4739; and Section VIII (5)–(13), lines 2291–2401. The real formula used below is Appendix II (34)–(37), with its exact constant $\log(2\pi) + \gamma$. The complex-place calculation was read but is not inserted at the sole archimedean place of $\mathbb{Q}$.

Alain Connes, Caterina Consani and Matilde Marcolli, *The Weil proof and the geometry of the adèles class space*, original author TeX `Yuri70v2.tex`, SHA-256 `b1f694cb934dd4fef829287f1a55dd8de21c67a481e552453c2f7e8fe0ae6fb5`, was read at source lines 560–681, 1586–1675, 1700–1731, 1973–2187, and 2283–2309 for the original transform, convolution trace, sharp involution, and trace theorem. In particular the locators are `rhoFourier`, `Weilexpl`, `princvalrhobeta`, `TraceformulaH1`, `intalphae`, `fsharpinv`, `RHpositTrsharp`, and `pairTrvanishV`. The human global trace theorem is a cited input, not a newly proved trace theorem. All its terms on the test family below are calculated in this chapter. The finite-place calculation is also proved completely in the accompanying WLC1–24 and LCP1–15. No completion of an original Dirichlet series replaces that series in this calculation.

75.1 Original test space, sharp, and exact character diagonalization

Retain the original coordinates and measures

$$C_{\mathbb{Q}} = \mathbb{R}_{>0} \times K, \quad K = \prod_p \mathbb{Z}_p^\times, \quad dg = \frac{dy}{y} du, \quad \int_K du = 1. \quad (\text{WQP1})$$

Fix any positive integer N , write $U_N = (\mathbb{Z}/N\mathbb{Z})^\times$, and view every $\chi \in \widehat{U}_N$ as its actual character on K . For $N = 1$ the group and its character set have one element. Let $h_\chi, k_\chi \in C_c^\infty(\mathbb{R}_{>0})$, allowing zero functions, and retain the entire original tests

$$f(y, u) = \sum_{\chi \in \widehat{U}_N} h_\chi(y) \chi(u), \quad k(y, u) = \sum_{\chi \in \widehat{U}_N} k_\chi(y) \chi(u). \quad (\text{WQP2})$$

All sums are finite. These tests are smooth and compactly supported in the real multiplicative variable, locally constant in the compact variable, and belong to the source's test algebra. The finite inverse is $h_\chi(y) = \int_K f(y, u) \overline{\chi(u)} du$, or the same sum with weight $1/\varphi(N)$ on U_N . This retains its original Haar factor.

The original convolution and source involution are

$$(f * k)(g) = \int_{C_{\mathbb{Q}}} f(a) k(a^{-1}g) da, \quad k^\sharp(g) = |g|^{-1} \overline{k(g^{-1})}. \quad (\text{WQP3})$$

In particular

$$k^\sharp(y, u) = \sum_{\chi} y^{-1} \overline{k_\chi(y^{-1})} \chi(u).$$

The same compact character occurs after sharp because $\overline{\chi(u^{-1})} = \chi(u)$; the radial factor y^{-1} remains. Integration of $\chi(v) \psi(v)^{-1}$ on K is $\delta_{\chi, \psi}$, by translating the integral when the character is nontrivial. It follows by substituting WQP2–3 that

$$(f * k^\sharp)(y, u) = \sum_{\chi} H_\chi(y) \chi(u), \quad H_\chi(y) = y^{-1} \int_0^\infty h_\chi(x) \overline{k_\chi(x/y)} dx. \quad (\text{WQP4})$$

This is an exact diagonalization of the convolution in its compact characters; it does not omit a mixed term. Those mixed terms have the proved zero compact integral.

Every H_χ is smooth and compactly supported. If h_χ and k_χ are supported in $[A, B]$ and $[C, D]$, respectively, with positive endpoints, its support is contained in $[A/D, B/C]$. Derivatives under the integral are justified on compact y intervals by smooth compactly supported majorants in x ; outside the displayed ratio interval the integrand is zero. The exact identity and symmetry are

$$\begin{aligned} H_{h,k}(1) &= \int_0^\infty h(x)\overline{k(x)} dx, & H_{h,k}(y^{-1}) &= y \overline{H_{k,h}(y)}, \\ H_{h,h}(y^{-1}) &= y \overline{H_{h,h}(y)}. \end{aligned} \quad (\text{WQP5})$$

For the second formula substitute $x = v/y$ in its left hand integral. No replacement of dx by dx/x is permissible in the first formula.

Write the original Mellin transform with every factor visible:

$$M_h(s) = \int_0^\infty h(x)x^s \frac{dx}{x}, \quad M_{H_{h,k}}(s) = M_h(s)\overline{M_k(1-\bar{s})}. \quad (\text{WQP6})$$

The second identity follows from the original multiplicative convolution and Fubini, or from $x/y = v$ in WQP4; compact support makes all these integrals absolute for every complex s . Thus the two global endpoints, which only receive the trivial compact character, are

$$E(h, k) = M_h(0)\overline{M_k(1)} + M_h(1)\overline{M_k(0)}, \quad E(h, h) = 2 \operatorname{Re}(M_h(0)\overline{M_h(1)}). \quad (\text{WQP7})$$

The conjugations in this formula are required for complex tests. In particular a real-test formula without them is not substituted here.

75.2 The real term with its full original regularization

The archimedean measure of the source is $d^*u = du/(2|u|)$ on $\mathbb{R}^\times$. Its mass on $1 \leq |u| \leq R$ is $\log R$. For the parity $\epsilon \in \{0, 1\}$ and a compact radial test H put $b(u) = H(|u|)(\operatorname{sgn} u)^\epsilon$. Connes's original Appendix II (36)–(37), for $e_\mathbb{R}(x) = e^{-2\pi i x}$, says

$$\mathcal{A}_\epsilon(H) = (\log(2\pi) + \gamma)H(1) + \lim_{\delta \downarrow 0} \left\{ \int_{|1-u| \geq \delta} \frac{b(u^{-1})}{|1-u|} \frac{du}{2|u|} + (\log \delta)H(1) \right\}. \quad (\text{WQP8})$$

This cited principal value, including its Euler constant, is the definition of the real-place term in the original global trace.

Separate $u > 0$ and $u < 0$, then substitute $y = |u|^{-1}$ in each. The positive numerator becomes $H(y)$, and its measure-denominator product becomes $dy/(2|1-y|)$. The negative branch gives $(-1)^\epsilon H(y) dy/(2(1+y))$. Its entire contribution is ordinary integration. The positive cutoff is initially $|1-y| \geq \delta y$. Replacing it by $|1-y| \geq \delta$ changes the integral by a quantity tending to zero: its two threshold distances are $\delta/(1+\delta)$ and $\delta/(1-\delta)$; the integrals of $1/|1-y|$ over the differences are $|\log(1+\delta)|$ and $|\log(1-\delta)|$, multiplied by a bound for the smooth H . Consequently

$$\begin{aligned} \mathcal{A}_\epsilon(H) &= (\log(2\pi) + \gamma)H(1) \\ &\quad + \lim_{\delta \downarrow 0} \left\{ \frac{1}{2} \int_{\substack{y>0 \\ |1-y| \geq \delta}} \frac{H(y)}{|1-y|} dy + (\log \delta)H(1) \right\} \\ &\quad + \frac{(-1)^\epsilon}{2} \int_0^\infty \frac{H(y)}{1+y} dy. \end{aligned} \quad (\text{WQP9})$$

This includes the negative real component and its parity.

An entirely convergent version, with the subtracted term still displayed, is obtained by substituting $y \mapsto 1/y$ only in the integrals over $(1, \infty)$. Define

$$B_H(y) = H(y) + y^{-1}H(y^{-1}), \quad 0 < y \leq 1. \quad (\text{WQP10})$$

The exact result is

$$\begin{aligned} \mathcal{A}_\epsilon(H) &= (\log(2\pi) + \gamma)H(1) \\ &+ \frac{1}{2} \int_0^1 \left\{ \frac{B_H(y) - 2H(1)}{1-y} + (-1)^\epsilon \frac{B_H(y)}{1+y} \right\} dy. \end{aligned} \quad (\text{WQP11})$$

To check the constant, integration of the retained $2H(1)/(1-y)$ up to $1-\delta$, with the factor $1/2$, is $-H(1)\log \delta$, cancelling exactly the logarithm in WQP9. The transformed upper cutoff is $1/(1+\delta)$ rather than $1-\delta$; the intervening logarithmic integral tends to zero by the same bound used above. Near one the numerator $B_H(y) - 2H(1)$ is $O(1-y)$. Near zero the two terms of B_H are zero because H has compact support, and the remaining $-2H(1)/(1-y)$ is integrable. This proves convergence and the identity without an unrecorded constant.

75.3 The complete global original-coordinate formula

For each original compact character χ let χ_* be its primitive integer character, c_χ its conductor, and $\chi(-1) = (-1)^{\epsilon_\chi}$. Let

$$\mathcal{Q}(f, k) = \text{Tr}(\vartheta_m(f * k^\sharp) \mid H^1)$$

denote the source's global trace pairing on the exact tests WQP2. The source's differential idele is denoted $\mathfrak{a}$. Its reference-character trace formula, used before assigning $|\mathfrak{a}| = 1$ over $\mathbb{Q}$, gives the following full expression:

$$\begin{aligned} \mathcal{Q}(f, k) &= \sum_{\chi \in \widehat{U}_N} \left\{ 1_{\chi=1} E(h_\chi, k_\chi) + \left(\sum_{p|c_\chi} v_p(c_\chi) \log p - \log |\mathfrak{a}| \right) H_\chi(1) \right. \\ &\quad \left. - \sum_{p \nmid c_\chi} \log p \sum_{j \geq 1} [\chi_*(p)^j H_\chi(p^j) + p^{-j} \chi_*(p)^{-j} H_\chi(p^{-j})] - \mathcal{A}_{\epsilon_\chi}(H_\chi) \right\}. \end{aligned} \quad (\text{WQP12})$$

Every prime-power sum is finite: the compact interval containing all the supports of the finitely many H_χ bounds p^j above and p^{-j} below. Every conductor sum is finite. The real term is the entire expression WQP8, or exactly its convergent form WQP11. Thus no local term has become an unspecified remainder.

Here is the direct derivation from the original trace, with all signs. The two endpoint integrals of $f * k^\sharp$ are WQP7 for the trivial compact character and zero otherwise by integration over K . The differential contribution is $-\log |\mathfrak{a}| \sum_\chi H_\chi(1)$. At $p \nmid c_\chi$ the local unit character is trivial and the uniformizer character is $\chi_*(p)^{-1}$ in the PLP20 coordinates. Its two original valuation shells are the bracket in WQP12, with multiplicative unit mass $\log p$, as proved in WLC5 and WLC12–13. At $p \mid c_\chi$, the off-unit averages vanish and the unit principal value is $-v_p(c_\chi) \log p H_\chi(1)$, proved by the finite subgroup filtration WLC6–9. The global trace subtracts this value; this gives the positive displayed conductor summand in WQP12. Finally its real restriction is $H_\chi(|u|)(\text{sgn } u)^{\epsilon_\chi}$, giving exactly the subtracted WQP8. These computations account for every term of the original cited trace formula.

For the standard global additive character of $\mathbb{A}_\mathbb{Q}/\mathbb{Q}$, the differential idele satisfies $|\mathfrak{a}| = 1$. Its displayed term is then zero for that proved reference choice; local rescalings have the exact correction $+\log |a_v| b_v(1)$ inside the local term, as in WLC10. Those terms must not be inserted a second time while retaining the already shifted global differential term.

The diagonal form is particularly explicit in the original variable x . Put

$$J_\chi = \int_0^\infty |h_\chi(x)|^2 dx, \quad R_\chi(y) = y^{-1} \int_0^\infty h_\chi(x) \overline{h_\chi(x/y)} dx. \quad (\text{WQP13})$$

Then $R_\chi(1) = J_\chi$ and $R_\chi(y^{-1}) = \overline{y R_\chi(y)}$ by WQP5. Substituting the entire real term into WQP12 gives

$$\begin{aligned} \mathcal{Q}(f, f) = \sum_\chi \bigg\{ & 2 \mathbf{1}_{\chi=1} \operatorname{Re} \left(M_{h_\chi}(0) \overline{M_{h_\chi}(1)} \right) \\ & + \left(\sum_{p|c_\chi} v_p(c_\chi) \log p - \log |\mathfrak{a}| - \log(2\pi) - \gamma \right) J_\chi \\ & - 2 \sum_{p \nmid c_\chi} \log p \sum_{j \geq 1} \operatorname{Re} \left[\chi_*(p)^j p^{-j} \int_0^\infty h_\chi(x) \overline{h_\chi(x/p^j)} dx \right] \\ & - \int_0^1 \left\{ \frac{\operatorname{Re} R_\chi(y) - J_\chi}{1-y} + (-1)^{\epsilon_\chi} \frac{\operatorname{Re} R_\chi(y)}{1+y} \right\} dy \bigg\}. \end{aligned} \quad (\text{WQP14})$$

In particular the conductor contribution is exactly $\log(c_\chi) \int |h_\chi(x)|^2 dx$, with its local sum displayed in WQP14. It is one summand of a complete form whose sign has not been determined here.

For an imprimitive modulus N , primes dividing N but not c_χ still appear in WQP12–14. This follows from the compact character's actual domain and the original identity

$$L(s, \chi) = L(s, \chi_*) \prod_{\substack{p|N \\ p \nmid c_\chi}} (1 - \chi_*(p) p^{-s}), \quad \operatorname{Re} s > 1. \quad (\text{WQP15})$$

Its proof is the absolute Euler-product comparison in PHS16 and WLC16. No imprimitive zero extension is substituted for the nonzero phase at those primes in the idele-class test.

75.4 The exact Gamma representation of that same real term

The original h remains the input. Its full transform WQP6 on the line $s = 1/2 + it$ has the inversion and pairing identities

$$\begin{aligned} h(x) &= \frac{x^{-1/2}}{2\pi} \int_{\mathbb{R}} M_h(1/2 + it) x^{-it} dt, \\ H_{h,k}(1) &= \frac{1}{2\pi} \int_{\mathbb{R}} M_h(1/2 + it) \overline{M_k(1/2 + it)} dt. \end{aligned} \quad (\text{WQP16})$$

Indeed set $x = e^v$ in the defining Mellin integral. It is the Fourier transform, with positive Fourier sign, of the explicitly weighted function $e^{v/2} h(e^v)$, which is smooth and compactly supported in v . Fourier inversion and its pairing identity give WQP16 with the displayed factors. The latter pairing is $\int h(x) \overline{k(x)} dx$, not a different original measure. These are invertible comparison maps; the weighted function is not used as an unrecorded replacement for h .

Let $\psi(z) = \Gamma'(z)/\Gamma(z)$ on $\operatorname{Re} z > 0$. The same principal value WQP11 is

$$\begin{aligned} \mathcal{A}_\epsilon(H_{h,k}) &= \frac{1}{2\pi} \int_{\mathbb{R}} M_h(1/2 + it) \overline{M_k(1/2 + it)} m_\epsilon(t) dt, \\ m_\epsilon(t) &= \log(2\pi) + \gamma - \log 2 - \gamma \\ &\quad - \frac{1}{2} \psi \left(\frac{1/2 + \epsilon + it}{2} \right) - \frac{1}{2} \psi \left(\frac{1/2 + \epsilon - it}{2} \right). \end{aligned} \quad (\text{WQP17})$$

The constant contributions in its second line have all been retained. Their exact value is $\log \pi$, so equivalently the same multiplier is $\log \pi - \operatorname{Re} \psi((1/2 + \epsilon + it)/2)$. This last identity is a proved comparison of terms in WQP17, not a completed-zeta substitution.

Here is a direct calculation of its Gamma term and constants. From WQP16 applied to H , the symmetric numerator $B_H(y)$ is the integral of $2y^{-1/2} \cos(t \log y)$ times $M_h(1/2 + it)\overline{M_k(1/2 + it)}/(2\pi)$. Insert this in WQP11. The resulting multiplier is

$$\log(2\pi) + \gamma + \int_0^1 \left\{ \frac{y^{-1/2} \cos(t \log y) - 1}{1 - y} + (-1)^\epsilon \frac{y^{-1/2} \cos(t \log y)}{1 + y} \right\} dy. \quad (\text{WQP18})$$

At zero, the absolute integral is bounded by a constant times $\int_0^{1/2} y^{-1/2} dy$. At one, the numerator difference is bounded by $C(1 + |t|)(1 - y)$, using the mean-value formula for $y^{-1/2} \exp(it \log y)$. Thus the absolute integral is at most $C'(1 + |t|)$. Both Mellin factors decrease faster than every power of $|t|$, by integration by parts in v in WQP16. This justifies the interchange by an integrable majorant and proves convergence of WQP17.

For $\alpha = 1/2 + it$ combine the two displayed rational denominators without changing any summand: their integral is the real part of

$$\int_0^1 \left\{ \frac{2y^{\alpha+\epsilon-1}}{1 - y^2} - \frac{1}{1 - y} \right\} dy = \int_0^1 \frac{v^{(\alpha+\epsilon)/2-1} - 1}{1 - v} dv - \int_0^1 \frac{dy}{1 + y}.$$

The equality follows by first writing its numerator as $2y(y^{\alpha+\epsilon-2} - 1)/(1 - y^2) - 1/(1 + y)$ and then substituting $v = y^2$ in the first term. The last integral is exactly $\log 2$. For $\operatorname{Re} z > 0$,

$$\int_0^1 \frac{v^{z-1} - 1}{1 - v} dv = -\psi(z) - \gamma. \quad (\text{WQP19})$$

For completeness, integration by parts in the beta integral gives $B(z, n+1) = n!/[z(z+1) \cdots (z+n)]$. Substitution $v = x/n$ shows $n^z B(z, n+1) \rightarrow \Gamma(z)$ locally uniformly for $\operatorname{Re} z > 0$, including one z derivative: $(1 - x/n)^n \leq e^{-x}$ on $0 \leq x \leq n$ supplies integrable majorants for $x^{\operatorname{Re} z-1}(1 + |\log x|)$. Logarithmic differentiation of the displayed finite product therefore gives $\psi(z) = \lim_n (\log n - \sum_{j=0}^n (z+j)^{-1})$. Subtract the harmonic sums with limit defining γ . Expanding $(1 - v)^{-1}$ and integrating its convergent differences gives WQP19; near one those differences have a bounded quotient, and near zero $v^{\operatorname{Re} z-1}$ is integrable, which justifies the limit. This proves WQP19 and hence every Gamma, Euler-constant and $\log 2$ term in WQP17.

Consequently the conductor, differential and real-place portion of the full form, still with all constants displayed, is

$$\begin{aligned} & \frac{1}{2\pi} \sum_{\chi} \int_{\mathbb{R}} M_{h_\chi}(1/2 + it) \overline{M_{k_\chi}(1/2 + it)} \\ & \times \left[\sum_{p|c_\chi} v_p(c_\chi) \log p - \log |\mathfrak{a}| - \log(2\pi) - \gamma + \log 2 + \gamma \right. \\ & \left. + \operatorname{Re} \psi \left(\frac{1/2 + \epsilon_\chi + it}{2} \right) \right] dt. \end{aligned} \quad (\text{WQP20})$$

Adding the exact endpoints and subtracting the exact prime-power correlations in WQP12 gives the same full original form. No weight or sign in WQP20 alone determines its total sign.

75.5 Global implications and the entire common-label map

WQP12 proves the complete pairing is diagonal between distinct compact characters, not merely that its conductor operator is diagonal. Each single-character form is Hermitian: WQP7 is

Hermitian, the conductor and differential constants are real, WQP5 exchanges the two prime-power terms under conjugation, and WQP17 has a real multiplier. Thus

$$\mathcal{Q}(f, f) = \sum_{\chi} \mathcal{Q}_{\chi}(h_{\chi}, h_{\chi}) \in \mathbb{R}. \quad (\text{WQP21})$$

A negative value for the finite tuple therefore implies a negative value in at least one of its actual character components, and every such component is realized by the original test $h(y)\chi(u)$. This is an exact characterization of the test decomposition, not an assertion that a negative component exists.

The trivial compact-character component has $c_{\chi} = 1$ and $\epsilon_{\chi} = 0$. Its conductor sum is empty, its Euler series is the original $\zeta(s) = \sum_{n \geq 1} n^{-s}$ for $\text{Re } s > 1$, and every prime appears in WQP14. Thus a positive conductor contribution on other components is not a positive missing term of this component. The source's Weil explicit formula identifies the trivial component's entire value with

$$\sum_{\substack{\rho: \zeta(\rho)=0 \\ 0 < \text{Re } \rho < 1}} m_{\rho} M_h(\rho) \overline{M_k(1 - \bar{\rho})}. \quad (\text{WQP22})$$

This is an application of the cited human explicit formula to the original test, with its multiplicities. WQP12–20 calculate its other side without replacing ζ by a completed function. The original Gamma and endpoint contributions remain present in that side. No conclusion about these zeros is inferred merely from the character decomposition.

Let $\mathcal{T}_N$ denote exactly the complex test space WQP2. For any bounded distributive lattice L retain the common-label carrier

$$G_L(\mathcal{T}_N) = \{(0, \lambda) : \lambda \in L\} \cup (\mathcal{T}_N \times \{1_L\}). \quad (\text{WQP23})$$

The finite character coefficient map and its inverse are linear maps of the entire vector, so their lifts retain its one label. They give an isomorphism to $G_L(\bigoplus_{\chi} C_c^{\infty}(\mathbb{R}_{>0}))$. This target does not have independent support labels on its character components. The conjugate-linear sharp map lifts as $(f, \lambda) \mapsto (f^{\sharp}, \lambda)$. It respects addition and conjugates scalar amplitudes, keeping their original support.

Convolution and the full sesquilinear form have the exact lifts

$$\begin{aligned} (f, \lambda) *_L (k, \mu) &= (f * k, \lambda \wedge \mu), \\ \mathcal{Q}_L((f, \lambda), (k, \mu)) &= (\mathcal{Q}(f, k), \lambda \wedge \mu), \quad \mathcal{Q}_L((f, \lambda), (f, \lambda)) = (\mathcal{Q}(f, f), \lambda). \end{aligned} \quad (\text{WQP24})$$

If the output amplitude is nonzero, neither input amplitude is zero, so both input labels are top. Otherwise the zero amplitude admits the displayed meet. These facts prove carrier membership. Additivity in the first input and conjugate additivity in the second, including their labels, follow from distributivity $(\lambda \vee \lambda') \wedge \mu = (\lambda \wedge \mu) \vee (\lambda' \wedge \mu)$ and the corresponding amplitude identities. The scalar rules follow from the same meet laws and sesquilinearity. Equivalently the ordinary linear functional on $\mathcal{T}_N \otimes \overline{\mathcal{T}_N}$ induced by $\mathcal{Q}$ lifts with one common label, and WQP24 is its composition with $(f, \lambda), (k, \mu) \mapsto (f \otimes \bar{k}, \lambda \wedge \mu)$. Thus a zero value on a top-supported test is $e = (0, 1_L)$, not $\tau = (0, 0_L)$.

To avoid confusing the compact group K in WQP1 with the tropical coefficient semifield, denote that same semifield here by K_{trop} , with multiplicative notation, zero 0 and unit 1; $\mathbb{R}_+^{\max}$ is one permitted choice. Retain the exact tropical receiver $K_{\text{trop}, L} = K_{\text{trop}} \otimes_{\mathbb{B}} L$ and the original map $\Phi(v, \lambda) = (v, 1_{K_{\text{trop}}} \otimes \lambda)$. It commutes with each of the preceding labelled formulas. The function $K_{\text{trop}} \rightarrow \mathbb{B}$ sending zero to zero and every nonzero element to one is a semiring morphism: a maximum is nonzero exactly when a summand is nonzero, and a product is nonzero exactly when both factors are nonzero. Tensoring it with L gives a label retraction which recovers λ , proving injectivity. At a support prime J the actual localization is $(v, \lambda) \mapsto [\lambda]_J$. It sends WQP24 to the meet $[\lambda]_J \wedge [\mu]_J$; on the diagonal this is $[\lambda]_J$. No numerical positivity or Haar measure on the arbitrary lattice has been postulated. The calculation retains the full global original form and its exact support maps, with no claim that its sign has been settled.

76 A bound for the entire global Weil form, with every prime retained

Source and actual scope. The global input is the original trace formula of Alain Connes, Caterina Consani and Matilde Marcolli, *The Weil proof and the geometry of the adèles class space*, author labels `TraceformulaH1`, `fsharpinv`, `vanishV` and `degmodifyV`. Its original real principal value is evaluated in Alain Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Appendix II(30)–(37). The complete finite-place evaluation is WLC1–18; the actual original theta source is ETC1–15. This calculation assembles the entire global form and proves an all-frequency tail bound on it. No finite list of test windows or numerical zero samples is used. The cited global trace theorem remains the human theorem; the identities and estimate below are derived with its full original terms.

76.1 The original test space and the complete global form

Let $\mathcal{B}_\infty$ be the smooth functions $h : (0, \infty) \rightarrow \mathbb{C}$ with every seminorm

$$p_{A,k}(h) = \sup_{x>0} (x^A + x^{-A}) |(x\partial_x)^k h(x)| < \infty, \quad A, k \in \mathbb{Z}_{\geq 0}. \quad (\text{GWM1})$$

This retains the original coordinate x and both ends. For a character χ of $U_N = (\mathbb{Z}/N\mathbb{Z})^\times$, let χ_* be its inducing primitive integer character, c_χ its conductor, and $\chi_*(-1) = (-1)^{\epsilon_\chi}$, $\epsilon_\chi \in \{0, 1\}$. Use the actual adelic-class test $f(x, u) = h(x)\chi(u)$ with measure dx/x and unit-group Haar mass one. Its original sharp and correlation are

$$f^\sharp(x, u) = x^{-1} \overline{f(x^{-1}, u^{-1})}, \quad (f \star f^\sharp)(y, u) = H_h(y)\chi(u), \quad H_h(y) = \frac{1}{y} \int_0^\infty h(x) \overline{h(x/y)} dx. \quad (\text{GWM2})$$

Indeed the compact integral is the integral of $\chi(v)\chi(v^{-1}u)$, and the real factor in the convolution is $(y/x)^{-1}dx/x = dx/y$. This proves GWM2 without changing measure. Write

$$\mathcal{M}h(s) = \int_0^\infty h(x) x^s \frac{dx}{x}, \quad M_h = \int_0^\infty |h(x)|^2 dx = H_h(1), \quad H_h(y^{-1}) = y \overline{H_h(y)}. \quad (\text{GWM3})$$

The last identity follows by $x = yz$ in the conjugate integral.

The complete real-place term, with the original additive character $e^{-2\pi i x}$, is

$$\begin{aligned} \mathcal{A}_\epsilon(H) &= (\log(2\pi) + \gamma)H(1) \\ &+ \frac{1}{2} \int_0^1 \left[\frac{H(y) + y^{-1}H(y^{-1}) - 2H(1)}{1-y} + (-1)^\epsilon \frac{H(y) + y^{-1}H(y^{-1})}{1+y} \right] dy. \end{aligned} \quad (\text{GWM4})$$

Here γ is Euler's constant. To obtain this formula from Connes' Appendix II(36), retain $d^\times u = du/(2|u|)$, substitute $y = |u|^{-1}$ separately on the two real signs, and retain the negative-sign character $(-1)^\epsilon$. The positive part is $\frac{1}{2} \int H(y)|1-y|^{-1}dy$ with the subtraction $\log \delta H(1)$ at $|1-y| = \delta$. Inverting the part $y > 1$ gives $y^{-1}H(y^{-1})$ on $(0, 1)$. Subtracting $2H(1)$ gives exactly the displayed convergent integral, since $\int_0^{1-\delta} (1-y)^{-1}dy = -\log \delta$. The transformed cutoff differs from this symmetric cutoff by intervals whose endpoint ratios tend to one, and hence by a term tending to zero. The negative part has no singularity and yields the denominator $1+y$. This retains the full original principal-value constant.

The complete global quadratic form is consequently

$$\begin{aligned} Q_\chi(h) &= \delta_{\chi_*,1} \left[\mathcal{M}h(0) \overline{\mathcal{M}h(1)} + \mathcal{M}h(1) \overline{\mathcal{M}h(0)} \right] - \log |\mathfrak{a}| M_h \\ &+ \sum_{p|c_\chi} v_p(c_\chi) \log p M_h - \mathcal{A}_{\epsilon_\chi}(H_h) \\ &- \sum_{p \nmid c_\chi} \log p \sum_{j \geq 1} \left[\chi_*(p)^j H_h(p^j) + p^{-j} \chi_*(p)^{-j} H_h(p^{-j}) \right]. \end{aligned} \quad (\text{GWM5})$$

The differential idele has $|\mathfrak{a}| = 1$ over $\mathbb{Q}$ with these original reference characters; its contribution has been retained in its place. In particular at $\chi = 1$ every rational prime remains present. At an imprimitive modulus, primes dividing N but not c_χ still occur; the original comparison is $L(s, \chi) = L(s, \chi_*) \prod_{p|N, p \nmid c_\chi} (1 - \chi_*(p)p^{-s})$. No zero value of the imprimitive character replaces a primitive value in GWM5. The conductor sum is precisely the negative of the ramified local distribution, not an additional Euler factor.

Every term is defined on GWM1, including the infinite sum over all primes. For $n \geq 1$ and integer $A \geq 2$, put $C_A = \sup_x \max(x^A, x^{-A})|h(x)|$. Splitting the integral in GWM2 at 1 and n gives the explicit bound

$$|H_h(n)| \leq C_A^2 n^{-A} \left[\frac{n^{-1}}{2A+1} + 1 - n^{-1} + \frac{1}{2A-1} \right] \leq 2C_A^2 n^{-A}. \quad (\text{GWM6})$$

The three terms are the integrals on $(0, 1)$, $(1, n)$, and (n, ∞) , respectively. Thus all prime powers, with any additional fixed power of $\log n$, converge absolutely by comparison with $\sum_{n \geq 2} (\log n)^k n^{-A}$. The estimates after multiplicative differentiation prove the same statements for the smooth topology GWM1. The real integral converges at zero by the same endpoint bounds and at one by subtraction and smoothness. The endpoints converge separately.

76.2 Exact spectral evaluation of the retained real contribution

The full Gamma contribution in GWM4 can also be evaluated, without replacing the original arithmetic function:

$$-\mathcal{A}_\epsilon(H_h) = \frac{1}{2\pi} \int_{\mathbb{R}} \left[\Re \psi \left(\frac{1/2 + it + \epsilon}{2} \right) - \log \pi \right] |\mathcal{M}h(1/2 + it)|^2 dt. \quad (\text{GWM7})$$

Here $\psi = \Gamma'/\Gamma$. The multiplier is an evaluated summand of GWM5, not a substitute function for ζ or $L(s, \chi_*)$. To prove it, set $u = \log x$ only inside the integration argument. Fourier inversion of the correlation in GWM2 gives

$$H_h(y) = \frac{1}{2\pi} \int_{\mathbb{R}} |\mathcal{M}h(1/2 + it)|^2 y^{-1/2 - it} dt, \quad M_h = \frac{1}{2\pi} \int_{\mathbb{R}} |\mathcal{M}h(1/2 + it)|^2 dt. \quad (\text{GWM8})$$

Indeed the function whose Fourier integral occurs is $e^{u/2}h(e^u)$, and its squared norm is exactly $\int |h(x)|^2 dx$; this records both the factor and the Jacobian. It is Schwartz by GWM1. Substitution into GWM4 gives, for each t ,

$$\log(2\pi) + \gamma + \int_0^1 \left[\frac{y^{-1/2} \cos(t \log y) - 1}{1 - y} + (-1)^\epsilon \frac{y^{-1/2} \cos(t \log y)}{1 + y} \right] dy.$$

With $z = y^2$ the integral is

$$\int_0^1 \frac{z^{\alpha-1} \cos((t/2) \log z) - 1}{1 - z} dz + \frac{1}{2} \int_0^1 \frac{1 - z^{-1/2}}{1 - z} dz, \quad \alpha = \frac{1/2 + \epsilon}{2}.$$

The digamma series gives the first integral $-\Re \psi(\alpha + it/2) - \gamma$ and the second is $-\log 2$. For the latter, put $z = r^2$ to get $-\int_0^1 (1+r)^{-1} dr$. Thus the full expression is $\log \pi - \Re \psi(\alpha + it/2)$, proving GWM7 with all constants accounted for. Interchanging integrals is justified at $y = 1$ by a bound $C(1+t^2)$ after subtraction, at 0 by the integrable factor $y^{-1/2}$, and in t by Schwartz decay.

76.3 A proved lower bound on the entire form for all large frequencies

Keep the original function and introduce its explicitly invertible modulation $h_T(x) = x^{iT}h(x)$, $T \in \mathbb{R}$. The exact relations are

$$H_{h_T}(y) = y^{iT} H_h(y), \quad \mathcal{M}h_T(s) = \mathcal{M}h(s + iT), \quad M_{h_T} = M_h. \quad (\text{GWM9})$$

Define the following quantities from the full original h :

$$\begin{aligned} K_h &= \int_0^\infty |xh'(x) + \tfrac{1}{2}h(x)|^2 dx, & A_{0,h} &= \int_0^\infty |h(x)| \frac{dx}{x}, & A_{1,h} &= \int_0^\infty |h(x)| dx, \\ P_{\chi,h} &= 2 \sum_{p \nmid c_\chi} \log p \sum_{j \geq 1} |H_h(p^j)|, & \alpha_\chi &= \frac{1/2 + \epsilon_\chi}{2}. \end{aligned} \quad (\text{GWM10})$$

The sum in $P_{\chi,h}$ is over every prime power, including when h has noncompact support. Its convergence was proved in GWM6. For every $|T| \geq 4\alpha_\chi$, the complete global form satisfies

$$\begin{aligned} Q_\chi(h_T) &\geq M_h \left[\log \frac{|T|}{4} - \frac{1}{\alpha_\chi} - \log \pi + \sum_{p|c_\chi} v_p(c_\chi) \log p - \log |\mathbf{a}| \right] \\ &\quad - \frac{4K_h}{T^2} \log \frac{|T|}{4\alpha_\chi} - P_{\chi,h} - 2\delta_{\chi*,1} A_{0,h} A_{1,h}. \end{aligned} \quad (\text{GWM11})$$

This is a bound on all of GWM5, not merely its archimedean part.

Here is a proof of every estimate used. For $\alpha > 0$ and real b , the digamma series, with the harmonic-number constant retained, gives

$$\Re \psi(\alpha + ib) = \lim_{m \rightarrow \infty} \left[\log m - \sum_{k=0}^{m-1} \frac{k + \alpha}{(k + \alpha)^2 + b^2} \right].$$

For $r(x) = (x + \alpha)/((x + \alpha)^2 + b^2)$, $|\sum_{k=0}^{m-1} r(k) - \int_0^m r(x) dx| \leq \int_0^m |r'(x)| dx$ by integration on every unit interval. If $|b| \leq \alpha$, this total variation is $r(0) \leq 1/\alpha$. If $|b| > \alpha$, it is $1/|b| - \alpha/(\alpha^2 + b^2) \leq 1/\alpha$. The integral is $\frac{1}{2} \log(((m + \alpha)^2 + b^2)/(\alpha^2 + b^2))$. Consequently the unconditional pointwise bound is

$$\Re \psi(\alpha + ib) \geq \frac{1}{2} \log(\alpha^2 + b^2) - \frac{1}{\alpha}. \quad (\text{GWM12})$$

In GWM7 for h_T , put $u = t + T$ without altering the original $dt/(2\pi)$ factor. On $|u| \leq |T|/2$ one has $|t| \geq |T|/2$, so GWM12 is at least $\log(|T|/4) - 1/\alpha_\chi$. Everywhere it is at least $\log \alpha_\chi - 1/\alpha_\chi$. The exact Parseval identities give

$$\frac{1}{2\pi} \int_{\mathbb{R}} u^2 |\mathcal{M}h(1/2 + iu)|^2 du = K_h, \quad \frac{1}{2\pi} \int_{|u| > |T|/2} |\mathcal{M}h(1/2 + iu)|^2 du \leq \frac{4K_h}{T^2}. \quad (\text{GWM13})$$

For the first identity, differentiating $e^{u/2}h(e^u)$ gives exactly $e^{u/2}(xh'(x) + h(x)/2)$; its norm has measure dx , as in GWM8. Subtracting the two lower bounds just obtained gives the explicit tail defect $4K_h T^{-2} \log(|T|/(4\alpha_\chi))$. Each paired prime summand in GWM5 is $2\Re(\chi_*(p)^j p^{ijT} H_h(p^j))$, by GWM3 and GWM9, so its total possible negative contribution is bounded by $P_{\chi,h}$. The two original endpoint integrals contribute at least $-2\delta_{\chi*,1} A_{0,h} A_{1,h}$. The conductor and differential terms are unchanged by GWM9. Combining these estimates proves GWM11.

There is also an explicit finite threshold, with no numerical certificate. For nonzero h , let

$$\begin{aligned} T_{\chi,h} &= \max \left\{ 4\alpha_\chi, \quad 4 \exp \left[1 + \frac{1}{\alpha_\chi} + \log \pi - \sum_{p|c_\chi} v_p(c_\chi) \log p + \log |\mathbf{a}| \right. \right. \\ &\quad \left. \left. + \frac{P_{\chi,h} + 2\delta_{\chi*,1} A_{0,h} A_{1,h}}{M_h} + \frac{K_h}{8\alpha_\chi^2 \exp(1) M_h} \right] \right\}. \end{aligned} \quad (\text{GWM14})$$

For every $|T| \geq T_{\chi,h}$ one has $Q_\chi(h_T) \geq M_h > 0$. Indeed the maximum of $\log(t/(4\alpha))/t^2$ on $t \geq 4\alpha$ is $1/(32\alpha^2 \exp(1))$, as its derivative has its sole maximum at $t = 4\alpha\sqrt{\exp(1)}$. Substitution into GWM11 proves the assertion. The threshold depends on the entire original amplitude; no uniform threshold over all h has been asserted.

76.4 Global character tuples and the original support carrier

For $f(x, u) = \sum_{\chi \in \widehat{U}_N} h_\chi(x) \chi(u)$, direct compact character orthogonality in the convolution gives

$$Q(f) = \sum_{\chi \in \widehat{U}_N} Q_\chi(h_\chi). \quad (\text{GWM15})$$

The mixed χ, ψ term has compact integral $\int \chi(v) \overline{\psi(v)} du = \delta_{\chi, \psi}$; the sharp preserves the character label. Thus GWM15 is an exact global diagonalization and leaves every ramified character and its conductor. For a nonzero finite tuple, taking the maximum of GWM14 over its nonzero entries proves $Q(| \cdot |^{iT} f) \geq \sum_\chi M_{h_\chi} > 0$ for every $|T|$ beyond that threshold. All primes remain in each of its global constants.

Let L be the original nontrivial bounded distributive lattice and $\mathcal{B}$ the original finite-character amplitude algebra with the functions GWM1. Its full carrier and maps are

$$\begin{aligned} G_L(\mathcal{B}) &= \{(0, \lambda) : \lambda \in L\} \cup (\mathcal{B} \times \{1_L\}), \\ \Delta_{iT}^\uparrow(f, \lambda) &= (| \cdot |^{iT} f, \lambda), \quad Q^\uparrow((f, \lambda), (g, \mu)) = (Q(f, g), \lambda \wedge \mu). \end{aligned} \quad (\text{GWM16})$$

The first map is an invertible complex-linear and convolution algebra map: $|xy|^{iT} = |x|^{iT} |y|^{iT}$ proves the convolution identity under its integral. It commutes with sharp since $|x^{-1}|^{iT} = |x|^{iT}$. Its inverse is Δ_{-iT} . It keeps each label, including τ and e . The second map is well-defined because nonzero amplitude requires both input labels to be top. Sesquilinearity and finite distributivity of the lattice prove additivity in each supported argument and the original scalar rule. A vanished top-labelled global pairing has value e , not τ . At every support prime J , each label map restricts to the same meet in L_J . No measure or positivity ordering on an arbitrary lattice is added.

These are maps on the original test algebra. No descent of Δ_{iT} to the arithmetic spectral quotient is assumed: the actual adelic image need not be preserved by modulation. The global kernel built from ETC's theta source computes that receiving defect and reconstructs all compact tests. In particular, the large-frequency result is a proved statement about the full form on specified modulated tests, not a conclusion about its sign on every test or about RH.

76.5 One global bound for the entire original adelic theta family

Restore both original parameters from ETC, before taking their image:

$$\begin{aligned} k &\in \bigoplus_p \mathbb{Z}, \quad a > 0, \quad N(k) = \prod_p p^{k_p}, \quad b = \frac{a}{N(k)}, \\ \xi_{k,a}(x) &= \eta(x_\infty/a) \prod_p 1_{p^{k_p} \mathbb{Z}_p}(x_p), \quad \eta(x) = \pi x^2 (\pi x^2 - 3/2) e^{-\pi x^2}, \\ \mathcal{E} \xi_{k,a}(y, u) &= f_b(y) = 2 \sum_{n \geq 1} \eta(ny/b) = f_1(y/b). \end{aligned} \quad (\text{GWM17})$$

ETC3–9 proves the two original additive moments vanish, the complete Fourier and rational restriction formulas, and the rapid bounds at both ends. Thus every f_b belongs to GWM1 and to the actual image $\mathcal{V}$ of the adelic source. Their positivity is also exact: for $y \geq 1$ every summand of $f_1(y)$ is positive because $\pi n^2 y^2 > 3/2$; for $0 < y < 1$, the proved Poisson formula $f_1(y) = y^{-1} f_1(y^{-1})$ gives positivity. Scaling proves $f_b > 0$. No theta function is inferred from the whole absolute base by this argument; GWM17 is the specified original tensor and its proved map.

The full original Mellin formula is

$$\mathcal{M} f_b(s) = 2 \left(\frac{a}{N(k)} \right)^s \zeta(s) \left[\frac{\pi^2}{2\pi^{(s+4)/2}} \Gamma\left(\frac{s+4}{2}\right) - \frac{3\pi}{4\pi^{(s+2)/2}} \Gamma\left(\frac{s+2}{2}\right) \right], \quad \mathcal{M} f_b(0) = \frac{1}{4}, \quad \mathcal{M} f_b(1) = \frac{b}{4}. \quad (\text{GWM18})$$

The expression on the right agrees first for $\Re s > 1$ by absolute summation and Gaussian integration, and then by continuation with the entire Mellin integral on the left, as proved in ETC10–15. Both Gaussian summands, the original ζ , all Gamma factors, the power of π , and the real dilation factor are retained. In particular it vanishes at every nontrivial zero of the original ζ with at least its original multiplicity. Alternatively the source ideal calculation **vanishV** directly gives $Q_1(f_b) = 0$. Using every term of GWM5 and these two endpoints gives the global identity

$$\mathcal{A}_0(H_{f_b}) + 2 \sum_p \log p \sum_{j \geq 1} H_{f_b}(p^j) + \log |\mathfrak{a}| M_{f_b} = \frac{b}{8}, \quad |\mathfrak{a}| = 1. \quad (\text{GWM19})$$

Here the correlation is real and positive and the prime sum includes every prime power. Its absolute convergence was proved in GWM6. This is an exact global balance, not a bound obtained from a cutoff.

Changing variables in each original integral proves all scaling factors:

$$\begin{aligned} H_{f_b}(y) &= bH_{f_1}(y), & M_{f_b} &= bM_{f_1}, & K_{f_b} &= bK_{f_1}, & P_{1,f_b} &= bP_{1,f_1}, \\ A_{0,f_b} &= A_{0,f_1}, & A_{1,f_b} &= bA_{1,f_1}, & Q_1(x^{iT} f_b(x)) &= bQ_1(x^{iT} f_1(x)). \end{aligned} \quad (\text{GWM20})$$

For the last equality, GWM9 proves the correlation scaling, while the two Mellin endpoints scale by b^{iT} and b^{1+iT} ; their conjugate product retains exactly the factor b . Every other summand of the complete global form scales by b . The threshold GWM14 is therefore exactly the same for every $a > 0$ and every finite divisor k :

$$T_{1,f_b} = T_{1,f_1}, \quad Q_1(x^{iT} f_b(x)) \geq bM_{f_1} > 0 \quad \text{for every } |T| \geq T_{1,f_1}. \quad (\text{GWM21})$$

This proves a global result simultaneously for the whole original source family and the whole frequency tail, with the factor $a/N(k)$ kept. It does not recover a, k separately from a scalar map that ETC6 proved depends on their ratio.

There is an exact geometric consequence at the arithmetic receiving quotient. Let $\mathcal{W}$ be the closure of $\mathcal{V}$ in the original strong test topology, and $q : \mathcal{B} \rightarrow \mathcal{B}/\mathcal{W}$ its linear quotient. The bounds above show that the full pairing is continuous. The cited source ideal calculation makes $\mathcal{V}$ orthogonal to every test; continuity gives the same for $\mathcal{W}$. Consequently

$$\mathfrak{D}_T : \mathcal{V} \longrightarrow \mathcal{B}/\mathcal{W}, \quad v \longmapsto q(| \cdot |^{iT} v), \quad \ker \mathfrak{D}_T = \mathcal{V} \cap \Delta_{-iT} \mathcal{W}. \quad (\text{GWM22})$$

This is an actual linear map with the displayed full kernel, not an assumed descended modulation. It has $\mathfrak{D}_0(f_b) = 0$, but $\mathfrak{D}_T(f_b) \neq 0$ for every $|T| \geq T_{1,f_1}$, by GWM21: a zero quotient class would have zero pairing. Hence modulation does not preserve this original arithmetic quotient on that entire frequency tail. Its exact full-label lift is $(v, \lambda) \mapsto (\mathfrak{D}_T v, \lambda)$; the source $(f_b, 1_L)$ has image e at $T = 0$ and a nonzero top-labelled amplitude for every frequency in GWM21. No supported zero is identified with τ , and no omitted local stratum is used to deduce this global non-descent result.

77 Recovering every original prime-power correlation from the global form

Human sources and exact receiving calculations. The original global formula and its radical are due to Alain Connes, Caterina Consani and Matilde Marcolli, *The Weil proof and the geometry of the adeles class space*, author file Yuri70v2.tex, SHA-256 b1f694cb934dd4fef829287f1a55dd8de21c67a481e552453c2f7e8fe0ae6fb5, labels TraceformulaH1, fsharpinv, pairTrvanishV, and degmodifyV. The real-place distribution is Alain Connes's original *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Appendix II(30)–(37), author-file SHA-256 f89f55df9faa2b8a49ccc2fee6a0dd4cbfb8d5279edae86e70979413b055eff6. WQP1–24 records the actual source-reading coverage and proves the complete compact-character formula with its original measures. GWM1–16, independently checked in full at source SHA-256 d416a168a28502d4705f6a5fbfd0c1eb7c4f81ca4a0d8d1a45ee16c03f4df152, extends that calculation to the original rapidly decreasing test space used below. Its subsequent complete GWM1–22 edition was also read and checked in full, at source SHA-256 45ce2e1fc3d9e63fe970aa06c23fe26f6bf73ceaf7e65d82cf4a26137f9f7cc2. ETC1–26, in CC_EXACT_TRIVIAL_CORRESPONDENCE.tex, was read in full for the actual adelic antecedents and their reduction. The present calculation strengthens the entire global estimate and applies it to that entire antecedent family.

77.1 The entire original form and an explicit remainder

Retain the original test domain

$$\mathcal{B}_\infty = \left\{ h \in C^\infty(0, \infty) : \sup_{x>0} (x^A + x^{-A}) |(x\partial_x)^k h(x)| < \infty \text{ for all integers } A, k \geq 0 \right\}.$$

For a compact-unit character χ at a finite modulus, retain its primitive integer character χ_* , conductor c_χ , parity ϵ_χ , and $\alpha = (1/2 + \epsilon_\chi)/2 > 0$. Use the full original quantities

$$\begin{aligned} \mathcal{M}h(s) &= \int_0^\infty h(x) x^s \frac{dx}{x}, & h_T(x) &= x^{iT} h(x), & H_h(y) &= y^{-1} \int_0^\infty h(x) \overline{h(x/y)} dx, \\ M &= \int_0^\infty |h(x)|^2 dx, & K_h &= \int_0^\infty |xh'(x) + h(x)/2|^2 dx, & \rho_h(u) &= \frac{|\mathcal{M}h(1/2 + iu)|^2}{2\pi}, \\ E_h(T) &= \delta_{\chi_*,1} \left[\mathcal{M}h(iT) \overline{\mathcal{M}h(1+iT)} + \mathcal{M}h(1+iT) \overline{\mathcal{M}h(iT)} \right]. \end{aligned} \tag{GPR1}$$

Thus $\int \rho_h = M$ and $\int u^2 \rho_h = K_h$ by the exact Parseval calculation GWM8 and GWM13. The measure in both physical norms is dx , and no factor x^{-1} is suppressed.

For $n = p^j$ with $p \nmid c_\chi$ and $j \geq 1$ define

$$a_{\chi,h}(p^j) = \log p \chi_*(p)^j H_h(p^j), \quad R_{\chi,h}(T) = \sum_{p \nmid c_\chi} \sum_{j \geq 1} \left[a_{\chi,h}(p^j) e^{iTj \log p} + \overline{a_{\chi,h}(p^j)} e^{-iTj \log p} \right]. \tag{GPR2}$$

GWM6 proves absolute convergence of the coefficient sum, hence uniform absolute convergence for every real T . This is the full original prime contribution, including every prime power. Indeed $H_h(p^{-j}) = p^j \overline{H_h(p^j)}$ and $H_{h_T}(y) = y^{iT} H_h(y)$, so GPR2 is exactly the bracketed prime sum in GWM5. Its absolute bound is the full quantity $P_{\chi,h} = 2 \sum_{p \nmid c_\chi} \log p \sum_{j \geq 1} |H_h(p^j)|$.

For every real T with $t = |T| \geq \max\{1, 4\alpha\}$ the complete global form has the exact identity

$$\begin{aligned} Q_\chi(h_T) &= E_h(T) - R_{\chi,h}(T) + r_{\alpha,h}(T) \\ &\quad + M \left[\log(t/2) - \log(2\pi) - \gamma + \log 2 + \gamma + \sum_{p|c_\chi} v_p(c_\chi) \log p - \log |\mathfrak{a}| \right], \quad (\text{GPR3}) \\ r_{\alpha,h}(T) &= \int_{\mathbb{R}} \rho_h(u) \left[\Re \psi \left(\alpha + \frac{i(u-T)}{2} \right) - \log(t/2) \right] du. \end{aligned}$$

The original Euler constant and both logarithms from the real principal value remain in the first display. Their evaluated sum $-\log(2\pi) - \gamma + \log 2 + \gamma = -\log \pi$ is proved in WQP17–19, rather than omitted from the formula. GPR3 follows directly from GWM5 and GWM7 after the exact change $u = t_{\text{Mellin}} + T$ in the latter integral. The different, endpoints, conductor and all prime correlations remain summands of the same original form.

The remainder in that exact identity satisfies

$$\begin{aligned} |r_{\alpha,h}(T)| &\leq \frac{4M + 2\sqrt{MK_h} + 3K_h/\alpha}{t} \\ &\quad + \frac{8\alpha^2 M + 4K_h(|\log \alpha| + \alpha^{-1} + \log 2 + \log t)}{t^2}. \end{aligned} \quad (\text{GPR4})$$

In particular this remainder tends to zero; it has not been discarded or absorbed into a change of the arithmetic function.

Proof of GPR4. For every $\alpha > 0$, $b \in \mathbb{R}$, the same sum–integral calculation as GWM12 gives the two-sided estimate

$$\left| \Re \psi(\alpha + ib) - \frac{1}{2} \log(\alpha^2 + b^2) \right| \leq \int_0^\infty \left| \frac{d}{dx} \frac{x + \alpha}{(x + \alpha)^2 + b^2} \right| dx \leq \frac{1}{\alpha}.$$

This follows by comparing every value at an integer with the integral on its unit interval and taking the limit in the digamma series. If $|b| \geq \alpha$, the same total variation is $1/|b| - \alpha/(\alpha^2 + b^2) \leq 1/|b|$; at $|b| = \alpha$ the formula agrees with the monotone case.

On $|u| \leq t/2$, put $b = (u - T)/2$. Then $|b| \geq t/4 \geq \alpha$. Also

$$\begin{aligned} &\frac{1}{2} \log \left(\alpha^2 + \frac{(u - T)^2}{4} \right) - \log(t/2) \\ &= \log |1 - u/T| + \frac{1}{2} \log \left(1 + \frac{4\alpha^2}{(u - T)^2} \right). \end{aligned}$$

The absolute value is at most $2|u|/t + 8\alpha^2/t^2$, because $|u/T| \leq 1/2$ and $\log(1 + v) \leq v$ for $v \geq 0$. Thus the central part of the absolute remainder integral is at most

$$\frac{4M + 2 \int |u| \rho_h(u) du}{t} + \frac{8\alpha^2 M}{t^2} \leq \frac{4M + 2\sqrt{MK_h}}{t} + \frac{8\alpha^2 M}{t^2}.$$

The final inequality is Cauchy–Schwarz for the displayed original positive density ρ_h .

On $|u| > t/2$ one has $|b| \leq (|u| + t)/2 \leq 3|u|/2$ and

$$\left| \frac{1}{2} \log(\alpha^2 + b^2) \right| \leq |\log \alpha| + \frac{|b|}{\alpha}.$$

The global digamma estimate therefore bounds the integrand’s absolute factor by $|\log \alpha| + \alpha^{-1} + 3|u|/(2\alpha) + \log t + \log 2$. Finally

$$\int_{|u| > t/2} \rho_h(u) du \leq \frac{4K_h}{t^2}, \quad \int_{|u| > t/2} |u| \rho_h(u) du \leq \frac{2K_h}{t}.$$

These follow directly from $u^2 \geq t^2/4$ and $|u| \leq 2u^2/t$ on that set. They give precisely the remaining terms in GPR4. This proves the full remainder bound.

The endpoint function $E_h(T)$ tends to zero faster than every inverse power of $|T|$. To prove this without deleting either endpoint, write each original integral, for $\sigma = 0, 1$, as

$$\mathcal{M}h(\sigma + iT) = \int_{\mathbb{R}} e^{\sigma v} h(e^v) e^{iTv} dv.$$

Every derivative of $e^{\sigma v} h(e^v)$ decreases exponentially at both ends to every prescribed rate, by the specified seminorms of $\mathcal{B}_{\infty}$. Repeated integration by parts, with vanishing endpoint terms, proves the assertion for each Mellin factor, and hence for both terms of $E_h(T)$ separately.

77.2 Direct recovery from the full global form

Every original coefficient in GPR2 can be recovered from $Q_{\chi}(h_T)$ itself; no removed real term is used as a replacement form. Fix $T_0 = \max\{1, 4\alpha\}$. For every $\nu > 0$,

$$\lim_{R \rightarrow \infty} \frac{1}{R - T_0} \int_{T_0}^R Q_{\chi}(h_T) e^{-i\nu T} dT = \begin{cases} -\log p \chi_*(p)^j H_h(p^j), & \nu = j \log p, \ p \nmid c_{\chi}, \ j \geq 1, \\ 0, & \text{otherwise.} \end{cases} \quad (\text{GPR5})$$

There is at most one pair (p, j) in the first line. Indeed $p^j = q^k$ for primes p, q implies $p = q$ by unique factorization and then $j = k$.

To prove the limit, apply GPR3 retaining all its terms. The mean of $r_{\alpha, h}(T) e^{-i\nu T}$ tends to zero because $r_{\alpha, h}(T) \rightarrow 0$; splitting the integral at a fixed large point proves this directly. The same argument applies separately to each endpoint term. Every constant contribution in GPR3 has a vanishing oscillatory mean since $\nu \neq 0$. For the remaining logarithm, integration by parts gives

$$\int_{T_0}^R \log(T/2) e^{-i\nu T} dT = \left[\frac{\log(T/2) e^{-i\nu T}}{-i\nu} \right]_{T_0}^R + \frac{1}{i\nu} \int_{T_0}^R \frac{e^{-i\nu T}}{T} dT.$$

The absolute value is $O_{\nu}(1 + \log R)$, and division by $R - T_0$ gives zero. Each positive frequency in GPR2 contributes its coefficient exactly when that frequency equals ν and zero otherwise; the negative frequencies never equal $\nu > 0$. Uniform absolute convergence of GPR2 permits passage from finite sums to the entire prime sum: the integral mean of any omitted tail is bounded by its absolute coefficient sum, uniformly in R . This proves GPR5 for all original primes simultaneously.

For the trivial compact character, every rational prime is unramified, and hence every $H_h(p^j)$ is obtained from GPR5 by dividing by the explicitly nonzero number $-\log p$. For other characters, division is by $-\log p \chi_*(p)^j$, also nonzero at every prime present in their original valuation-shell term. The ramified receiving contribution remains $\sum_{p|c_{\chi}} v_p(c_{\chi}) \log p M$ in GPR3. The original mass itself is recovered by

$$\lim_{|T| \rightarrow \infty} \frac{Q_{\chi}(h_T)}{\log |T|} = M. \quad (\text{GPR6})$$

This follows from boundedness of the complete prime series, the proved vanishing of the exact remainder and endpoints, and GPR3. It accounts for every ramified term without inventing a missing prime-power correlation at such a prime.

77.3 One frequency threshold for the entire actual theta image

Retain exactly the ETC adelic family, including its two polynomial terms, finite factors, and full nonzero rational sum:

$$\begin{aligned} \eta(x) &= \left(\pi^2 x^4 - \frac{3\pi}{2} x^2 \right) e^{-\pi x^2}, & \xi_{k,a}(x) &= \eta(x_{\infty}/a) \prod_p 1_{p^{k_p} \mathbb{Z}_p}(x_p), \\ N(k) &= \prod_p p^{k_p}, & b &= \frac{a}{N(k)} > 0, & f_b(y) &= \mathcal{E} \xi_{k,a}(u, y) = 2 \sum_{n \geq 1} \eta(ny/b). \end{aligned} \quad (\text{GPR7})$$

Here $k \in \bigoplus_p \mathbb{Z}$, $a > 0$, and the original unit-variable function is constant. ETC1–10 prove the exact reduction and $f_b \in \mathcal{B}_\infty$, including every multiplicative derivative at both ends. The function is nonzero: for sufficiently large y/b each summand is positive and the first summand is strictly positive. Thus $M_{f_b} > 0$.

Its original Mellin expression is the entire continuation of

$$\mathcal{M}f_b(s) = 2 \left(\frac{a}{N(k)} \right)^s \zeta(s) \left[\frac{\pi^2}{2\pi^{(s+4)/2}} \Gamma\left(\frac{s+4}{2}\right) - \frac{3\pi}{4\pi^{(s+2)/2}} \Gamma\left(\frac{s+2}{2}\right) \right]. \quad (\text{GPR8})$$

ETC11 proves this directly from the two original integrals; its original endpoint values are $\mathcal{M}f_b(0) = 1/4$ and $\mathcal{M}f_b(1) = b/4$ by ETC15. At every nontrivial zero ρ of the original ζ the bracket in GPR8 is holomorphic and finite, so $\mathcal{M}f_b(\rho) = 0$, with at least its original zero multiplicity. This conclusion uses the original zeta and both Gamma terms, with no substitution by a different function. The cited Weil formula, equivalently the source's vanishing on the actual adelic image, therefore gives $Q_1(f_b) = 0$. The original theta test satisfies the exact full global cancellation

$$\mathcal{A}_0(H_{f_b}) + 2 \sum_p \log p \sum_{j \geq 1} H_{f_b}(p^j) + \log |\mathfrak{a}| M_{f_b} = \frac{b}{8}, \quad |\mathfrak{a}| = 1. \quad (\text{GPR9})$$

All sums converge by GWM6. The factor $b/8$ is the sum of the two original endpoint products $(1/4)(b/4) + (b/4)(1/4)$. The correlations are real because f_b is real. The original different has modulus one and the conductor of this trivial character is one, so their respective displayed terms in GPR3 have value zero here. This evaluates the exact cancellation rather than assuming that a local term alone vanishes.

The entire family has one explicit common frequency threshold. Indeed $f_b(x) = f_1(x/b)$ and direct substitutions in the original integrals give

$$\begin{aligned} H_{f_b}(y) &= bH_{f_1}(y), & M_{f_b} &= bM_{f_1}, & K_{f_b} &= bK_{f_1}, \\ P_{1,f_b} &= bP_{1,f_1}, & A_{0,f_b} &= A_{0,f_1}, & A_{1,f_b} &= bA_{1,f_1}, \\ \mathcal{M}(x^{iT}f_b)(s) &= b^{s+iT} \mathcal{M}f_1(s+iT), & Q_1(x^{iT}f_b) &= bQ_1(x^{iT}f_1). \end{aligned} \quad (\text{GPR10})$$

Here $A_{0,h} = \int |h(x)|dx/x$ and $A_{1,h} = \int |h(x)|dx$. For the first line put $x = bz$ and retain the Jacobian b . The same substitution proves the two absolute endpoint identities and the Mellin identity. The two phases b^{iT} cancel in each endpoint product. Consequently every summand of the original global form has exactly the same factor b , proving the last identity. Thus no term in this scaling statement is inferred from an unspecified equivalence of test functions.

Let $T_* := T_{1,f_1}$ be the explicit GWM14 threshold, with $\alpha = 1/4$, $c = 1$, and $|\mathfrak{a}| = 1$. All ratios appearing in that formula are unchanged by GPR10, so $T_{1,f_b} = T_*$ for every $b > 0$. Its proved inequality gives the simultaneous conclusion

$$Q_1(f_b) = 0, \quad Q_1(x^{iT}f_b) \geq bM_{f_1} > 0 \quad \text{for every } |T| \geq T_*, \quad a > 0, \quad k \in \bigoplus_p \mathbb{Z}, \quad b = a/N(k). \quad (\text{GPR11})$$

This is uniform in the entire specified original adelic family. It does not assert uniformity over unrelated $\mathcal{B}_\infty$ tests.

77.4 The exact quotient and full-support defect

Let $\mathcal{V}$ be the original adelic image inside the source test space $\mathcal{B}$, let $\mathcal{W}$ be its closure in the strong original test topology, and let $\pi : \mathcal{B} \rightarrow \mathcal{B}/\mathcal{W}$ be the actual quotient map. The source's ideal and trace-vanishing calculation gives $Q(v, k) = 0$ for $v \in \mathcal{V}$ and $k \in \mathcal{B}$: the convolved amplitude is still in that ideal. The seminorm bounds for the prime sum, endpoints and real term in GWM6–8

show continuity of the full pairing. Therefore $Q(w, k) = 0$ also for $w \in \mathcal{W}$. Every f_b in GPR7 has its displayed antecedent with zero value and zero additive integral, so it is in this image. Define the exact obstruction map

$$D_T : \mathcal{V} \longrightarrow \mathcal{B}/\mathcal{W}, \quad D_T(v) = \pi(x^{iT}v), \quad \ker D_T = \mathcal{V} \cap \Delta_{-iT}\mathcal{W}. \quad (\text{GPR12})$$

This is a linear map with the original coordinate and modulation. GPR11 proves $D_T(f_b) \neq 0$ for every $|T| \geq T_*$ and every $b > 0$: membership of $x^{iT}f_b$ in $\mathcal{W}$ would force its global pairing to be zero, contradicting the proved strictly positive value. Thus modulation fails to descend to this quotient on the entire frequency tail, as witnessed by every member of the actual original family. The precise reason is that the two representatives 0 and f_b of its zero class have distinct images after modulation and projection. This proves failure of descent, not a claim that the original test algebra lacks modulation.

For every original nontrivial bounded distributive lattice L , take the full common-label carriers $G_L(\mathcal{V})$, $G_L(\mathcal{B})$ and $G_L(\mathcal{B}/\mathcal{W})$. The exact lift is $D_T^\uparrow(v, \lambda) = (D_T(v), \lambda)$. A nonzero output has nonzero input and therefore top label, while a zero output keeps its original label; linearity proves the semimodule identities. For the top-supported f_b , $\pi^\uparrow(f_b, 1_L) = e$ but $\pi^\uparrow(x^{iT}f_b, 1_L)$ has nonzero amplitude for $|T| \geq T_*$. The zero representative at the same label maps to e , so the representative-dependence survives the entire support map. At each actual support stalk the original localization sends (v, λ) to $[\lambda]_J$, where both representatives have the same label. This describes exactly what that localization forgets; it does not replace the global amplitude defect. No positivity ordering or integration measure on L has been introduced.

77.5 The original prime powers and original zeta from the entire curve

Now fix the actual original test $h = f_1$ from GPR7, with no change of variable in its definition, and put $q_h(T) = Q_1(x^{iT}h)$ for every real T . This is a continuous function: GWM5's prime series converges locally uniformly, its endpoints are continuous, and the digamma integral is continuous by a Schwartz majorant for T in each compact interval. The exact positivity needed for the inverse is

$$h(y) > 0 \quad (y > 0), \quad H_h(y) = y^{-1} \int_0^\infty h(x)h(x/y) dx > 0 \quad (y > 0). \quad (\text{GPR13})$$

For $y \geq 1$, each term $\eta(ny) = \pi(ny)^2(\pi(ny)^2 - 3/2)e^{-\pi(ny)^2}$ is positive, since $\pi > 3/2$ and $n \geq 1$. The defining sum is convergent, so $h(y) > 0$ there. For $0 < y < 1$, the proved original Poisson identity ETC9 gives $h(y) = y^{-1}h(1/y) > 0$. The integrand defining $H_h(y)$ is therefore continuous and strictly positive. Its integral is finite by the already proved endpoint estimates, and its integral on any compact interval of positive length is strictly positive. This proves both assertions.

For every nonzero real ω , the following symmetric average of the whole original global form exists:

$$\begin{aligned} B_h(\omega) &:= \lim_{R \rightarrow \infty} \frac{1}{2R} \int_{-R}^R q_h(T) e^{-i\omega T} dT \\ &= \begin{cases} -\log p H_h(p^j), & \omega = j \log p \text{ or } \omega = -j \log p, \quad j \geq 1, \\ 0, & \text{otherwise.} \end{cases} \end{aligned} \quad (\text{GPR14})$$

To prove this, use GPR3 separately on the two tails $|T| \geq T_0$. The bounded middle interval has an integral which, divided by $2R$, tends to zero by continuity. The exact remainder and endpoints have vanishing mean as before. The retained constants have vanishing nonzero-frequency mean. On each tail, integration by parts of $\log(|T|/2)e^{-i\omega T}$ gives a bound $O_\omega(1 + \log R)$, so this term also has mean zero. Finally the absolutely convergent prime series has the symmetric average of its individual exponential terms; the uniform tail estimate used in GPR5 justifies passing to its

whole sum. Its coefficients are real in this actual case. These facts prove GPR14. No growing term was subtracted from q_h in the definition of B_h , and no completed arithmetic function occurs in this averaging operation.

Every coefficient at $\pm j \log p$ is nonzero by GPR13. Thus the full nonzero-frequency support of GPR14 is exactly the set $\{\pm j \log p : p \text{ prime}, j \geq 1\}$. More precisely define, for every integer $n \geq 2$,

$$\Lambda_q(n) = -\frac{B_h(\log n)}{H_h(n)}. \quad (\text{GPR15})$$

The denominator is an explicitly defined strictly positive original correlation, so this is an everywhere defined inverse for all those integers. GPR14 proves

$$\Lambda_q(n) = \begin{cases} \log p, & n = p^j, p \text{ prime}, j \geq 1, \\ 0, & n \text{ is not a prime power.} \end{cases}$$

In particular it recovers the original von Mangoldt function, with every prime power and its original multiplicity index j . The rational primes themselves are exactly $\{n \geq 2 : \Lambda_q(n) = \log n\}$: at $n = p^j$ equality means $j = 1$, and a zero value cannot equal $\log n > 0$. This reconstructs the arithmetic prime set from the full global curve and the specified original positive test, rather than testing only a finite list of frequencies.

For $\text{Re } s > 1$ form both original series from those recovered coefficients:

$$\begin{aligned} D_q(s) &= \sum_{n \geq 2} \Lambda_q(n) n^{-s}, & L_q(s) &= \sum_{n \geq 2} \frac{\Lambda_q(n)}{\log n} n^{-s}, \\ Z_q(s) &= \exp(L_q(s)). \end{aligned} \quad (\text{GPR16})$$

They retain the original exponent s . The coefficient bounds $0 \leq \Lambda_q(n) \leq \log n$ and $0 \leq \Lambda_q(n)/\log n \leq 1$ show absolute, locally uniform convergence of both series on this half-plane. For every fixed $\delta > 0$ the majorants $\sum (\log n) n^{-1-\delta}$ and $\sum n^{-1-\delta}$ converge, by integration comparison; any fixed number of derivatives adds only a power of $\log n$, which obeys the same comparison after decreasing δ . Consequently differentiation term by term gives $L'_q(s) = -D_q(s)$.

The exact original arithmetic reconstruction is

$$\begin{aligned} Z_q(s) &= \prod_{p \text{ prime}} (1 - p^{-s})^{-1} = \sum_{n \geq 1} n^{-s} = \zeta(s), \\ -\frac{\zeta'(s)}{\zeta(s)} &= D_q(s), & \log \zeta(s) &= L_q(s), & \text{Re } s > 1. \end{aligned} \quad (\text{GPR17})$$

Here $\log \zeta$ denotes the analytic branch with $\log \zeta(\sigma + it) \rightarrow 0$ uniformly in t as $\sigma \rightarrow +\infty$. To prove every equality, GPR15 gives

$$L_q(s) = \sum_p \sum_{j \geq 1} \frac{p^{-js}}{j}.$$

Its absolute convergence follows either from GPR16 or from $\sum_p p^{-\sigma}/(1 - p^{-\sigma}) \leq (1 - 2^{-\sigma})^{-1} \sum_{n \geq 2} n^{-\sigma} < \infty$. For each individual prime the series is the analytic function $-\log(1 - p^{-s})$, as follows by integrating the absolutely convergent geometric series from zero to p^{-s} inside the unit disk. Exponentiating finite sums and taking their absolutely convergent limit proves the product formula. A finite such product expands into $\sum n^{-s}$ over integers whose prime divisors occur in that product. Unique factorization proves the coefficient is exactly one at each such integer. As the finite sets of primes increase, every positive integer is eventually included, and domination by $\sum n^{-\sigma}$ proves the original full Dirichlet series formula. This proves $Z_q = \zeta$. The exponential formula shows nonvanishing on the half-plane, and its derivative gives the stated

original logarithmic derivative, with the sign retained. Finally $|L_q(\sigma + it)| \leq \sum_{n \geq 2} n^{-\sigma} \rightarrow 0$ as $\sigma \rightarrow \infty$, which proves the specified branch. Any two analytic logarithms on this connected domain differ by a constant integer multiple of $2\pi i$; this limiting condition selects exactly the one in GPR17. Thus neither a multiplicative constant nor an unrecorded branch choice survives the inverse.

GPR17 determines the original meromorphic zeta uniquely by continuation from its half-plane of absolute convergence. The already proved exact expression GPR8, with $b = 1$, also supplies its continued comparison: the entire original Mellin integral is the product of that same $\zeta(s)$ and its two displayed Gamma terms. Its exceptional-point interpretation and divisor remain those of that full product, not an assertion that either Gamma summand is an entire unit. No information about the location of nontrivial zeros follows merely from the existence of these inverse maps.

Full support maps at the exponential stage. For any fixed nonzero frequency, taking its existing average is a complex-linear functional on the space of curves for which the average exists. Its common-label lift preserves the entire label. In particular on the explicit line $\mathbb{C}q_h$ all the maps $q \mapsto B_q(\log n)$, $q \mapsto -B_q(\log n)/H_h(n)$, $q \mapsto D_q$, and $q \mapsto L_q$ are linear, defined on the whole line, and have the exact lifts $(q, \lambda) \mapsto (F(q), \lambda)$. Every required infinite series converges there because its coefficients are a fixed complex multiple of those proved in GPR16. At the original top-supported curve, a prime-power coefficient has nonzero amplitude and top support; at a non-prime-power integer the returned value is $e = (0, 1_L)$ rather than $\tau = (0, 0_L)$. Thus the zero outputs themselves retain the programme's supported-zero distinction.

Exponentiation is not a linear or zero-preserving map: $\exp(0) = 1$. Its faithful support accounting cannot be obtained by falsely preserving a non-top label on that nonzero amplitude. The exact full retained reconstruction is instead

$$\begin{aligned} \mathcal{R}_L : G_L(\mathbb{C}q_h) &\longrightarrow G_L(\mathbb{C}q_h) \times G_L(\mathcal{O}(\{\operatorname{Re} s > 1\})), \\ (q, \lambda) &\longmapsto ((q, \lambda), (\exp(L_q), 1_L)). \end{aligned} \tag{GPR18}$$

The second label is necessarily top because this amplitude is nowhere zero. It is the top label inserted by the nonzero constant term of the original exponential series. Projection onto the first coordinate is a left inverse, proving that every original label survives the complete record, even for zero input amplitudes. At $q = q_h$ its second component is the original ζ of GPR17. At zero amplitude its second component is one and its first component still distinguishes every $(0, \lambda)$. This is a map of sets with the displayed linear submaps, not a claim that exponentiation is a semimodule homomorphism.

The tropical receiver sends every retained original label to $1_{K_{\text{trop}}} \otimes \lambda$ as proved in WQP24, and its retraction recovers that label. The actual support localization sends it to $[\lambda]_J$. These apply separately to the two explicitly retained factors in GPR18; no replacement of $\operatorname{Spec} G_L(\mathbb{Z})$ by its recovered arithmetic prime set has been made. GPR15 reconstructs the arithmetic prime data, while these maps retain the entire original lattice geometry and its supported zeros.

78 One global Weil kernel from the actual theta image, with all tests reconstructed

Human sources and exact scope. The starting test is the polynomial Gaussian in Alain Connes, Caterina Consani, and Matilde Marcolli, *The Weil proof and the geometry of the adeles class space*, source Yuri70v2.tex, subsection *Fubini's theorem and the trivial correspondences*, Lemma degmodifyV. The original source SHA-256 is b1f694cb934dd4fef829287f1a55dd8de21c67a481e552453c2f7e8fe0ae6fb5. The source locators rangecVdef, varthetaH1V, vanishV, rhoFourier, varthetagrH1, TraceformulaH1, princvalrhobeta, and fsharpinv specify the actual image, quotient, trace, and involution. For this derivation its lines 610–710, 1680–1831, and 1980–2110 were read directly; the polynomial-Gaussian subsection had already been read completely in ETC1–26. The entire source trace theorem is a cited human input, not a newly asserted theorem about RH.

The real principal value is the one of Alain Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Appendix II (34)–(37), as reproduced and derived in WQP8–11. The whole programme proofs read here are CC_EXACT_TRIVIAL_CORRESPONDENCE.tex, ETC1–26, SHA-256 89cfc62a20413c8470e804657cedcf0c11f11eaac7194afe33738b9326ec7a7b; CC_SUPPORTED_WEIL_QUOTIENT.tex, SWQ1–28, SHA-256 d0819448b8286a66171e844171ef9a1cce25471992810f61ec198a695bb847fc; CC_FULL_WEIL_CHARACTER_QUADRATIC_FORM.tex, WQP1–24, SHA-256 d9258657e56059fdac8368aea20ea47345c4bb7e664eb9621ce238446cc6aa37; and GLOBAL_WEIL_MODULATION_BOUND.tex, GWM1–16, SHA-256 at this reading d416a168a28502d4705f6a5fbfd0c1eb7c4f81ca4a0d8d1a45ee16c03f4df152. The latter is a concurrent programme derivation of the entire large-frequency bound; its use is identified below. The present construction and reconstruction are proved here. No completed arithmetic function replaces the original zeta or Dirichlet series.

78.1 The actual adelic antecedent and a strictly positive original test

Retain real additive measure dx , finite additive measures with $\text{vol}(\mathbb{Z}_p) = 1$, and the original multiplicative coordinates and measure

$$C_{\mathbb{Q}} = \mathbb{R}_{>0} \times K, \quad K = \prod_p \mathbb{Z}_p^{\times}, \quad dg = \frac{dy}{y} du, \quad \int_K du = 1. \quad (\text{GTK1})$$

The source test and its actual reduction are

$$\begin{aligned} \eta(x) &= \pi x^2 (\pi x^2 - \tfrac{3}{2}) e^{-\pi x^2}, & \xi(x_{\infty}, (x_p)) &= \eta(x_{\infty}) \prod_p 1_{\mathbb{Z}_p}(x_p), \\ \xi(0) &= 0, \quad \int_{\mathbb{A}_{\mathbb{Q}}} \xi = 0, & h(y) &= (E\xi)(y, u) = 2 \sum_{n \geq 1} \eta(ny), \\ V &= ES(\mathbb{A}_{\mathbb{Q}})_0, & h &\in V. \end{aligned} \quad (\text{GTK2})$$

The factor two is the sum over both rational signs. Indeed the finite factors allow exactly the rational integers; evenness of η gives the two signs. The separate Gaussian moments are $\int x^2 e^{-\pi x^2} dx = 1/(2\pi)$ and $\int x^4 e^{-\pi x^2} dx = 3/(4\pi^2)$, giving $\int \eta = 3/4 - 3/4 = 0$. These values follow by differentiating $\int e^{-\pi t x^2} dx = t^{-1/2}$ with its original constants. The Fourier transform with kernel $e^{-2\pi i x v}$ gives

$$\widehat{x^2 e^{-\pi x^2}} = (1/(2\pi) - v^2) e^{-\pi v^2}, \quad \widehat{x^4 e^{-\pi x^2}} = (v^4 - 3v^2/\pi + 3/(4\pi^2)) e^{-\pi v^2}.$$

Substituting their coefficients proves $\widehat{\eta} = \eta$, including the cancellation of their constant terms. Finite character orthogonality proves $\widehat{1_{\mathbb{Z}_p}} = 1_{\mathbb{Z}_p}$. Periodizing the real Schwartz function and evaluating its absolutely convergent Fourier series at zero therefore gives

$$h(y) = y^{-1} h(y^{-1}). \quad (\text{GTK3})$$

The absent $n = 0$ terms in this formula have value $\eta(0) = 0$ on both sides, rather than an omitted nonzero endpoint. This is also ETC7–9 with all widths equal to one.

The test is strictly positive everywhere. For $y \geq 1$ every summand in GTK2 is positive, because $\pi n^2 y^2 > 3/2$ for $n \geq 1$; the first summand is nonzero. For $0 < y < 1$, GTK3 is a positive factor times the already proved positive value at $y^{-1} > 1$. Thus

$$h(y) > 0 \quad (y > 0). \quad (\text{GTK4})$$

This assertion concerns this original theta-image test; it is not positivity of the global Weil form.

For every logarithmic derivative, the series at $y \geq 1$ is bounded by a finite polynomial in y times $e^{-\pi y^2}$. To see this, differentiate its Gaussian terms, factor out $e^{-\pi y^2}$, and bound the remaining series by a fixed polynomial in n times $e^{-\pi(n^2-1)}$. Reflection GTK3 gives the corresponding bounds at zero. In particular there are positive constants C_j, B_j, c_j , for every j , such that

$$|\partial_u^j h(e^u)| \leq C_j \exp(B_j |u| - c_j e^{2|u|}), \quad u \in \mathbb{R}. \quad (\text{GTK5})$$

Finite changes to C_j cover the bounded interval around $u = 0$. All real weights and logarithmic derivatives are therefore controlled.

78.2 The original Mellin transform and both endpoint values

Put

$$M(s) = \int_0^\infty h(y) y^s \frac{dy}{y}. \quad (\text{GTK6})$$

GTK5 proves absolute convergence and local uniform convergence of every s derivative on $\mathbb{C}$, so M is entire. Changing y to $1/y$ in GTK6 and using GTK3 proves $M(s) = M(1-s)$. Reality of h proves $\overline{M(s)} = M(\bar{s})$. For $\Re s > 1$, absolute Fubini on the original integer sum gives the full expression

$$M(s) = 2\zeta(s) \left[\frac{\pi^2}{2\pi^{(s+4)/2}} \Gamma((s+4)/2) - \frac{3\pi}{4\pi^{(s+2)/2}} \Gamma((s+2)/2) \right]. \quad (\text{GTK7})$$

Indeed the absolute double integral is $2 \sum_{n \geq 1} n^{-\Re s} \int_0^\infty |\eta(x)| x^{\Re s-1} dx < \infty$, and substitution $\pi x^2 = v$ integrates each Gaussian term separately. Equality to GTK6 specifies the entire continuation of the complete expression, including every removable point of the separate summands. The Gamma recurrence gives the exact comparison

$$2 \left[\frac{\pi^2}{2\pi^{(s+4)/2}} \Gamma((s+4)/2) - \frac{3\pi}{4\pi^{(s+2)/2}} \Gamma((s+2)/2) \right] = \frac{s(s-1)}{4} \pi^{-s/2} \Gamma(s/2). \quad (\text{GTK8})$$

GTK7 remains the working original-zeta formula. GTK8 records its multiplier without replacing ζ by a completed function. On $0 < \Re s < 1$ this multiplier is holomorphic and nowhere zero, so every original nontrivial zero of ζ occurs in M with the same multiplicity.

For completeness retain the whole Gaussian theta function, including its zero term: $\Theta(y) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 y^2}$. Differentiating gives $h(y) = \frac{1}{4}(y^2 \Theta'(y))'$ and $h(y)/y = \frac{1}{4}(y \Theta'(y) + \Theta(y))'$. The theta reflection has leading term $1/y$ at zero and constant term one at infinity, with exponentially small remainders and all their derivatives. Hence

$$\begin{aligned} M(1) &= \frac{1}{4}[y^2 \Theta'(y)]_0^\infty = \frac{1}{4}[0 - (-1)] = \frac{1}{4}, \\ M(0) &= \frac{1}{4}[y \Theta'(y) + \Theta(y)]_0^\infty = \frac{1}{4}[1 - 0] = \frac{1}{4}. \end{aligned} \quad (\text{GTK9})$$

The calculation first integrates on a compact interval and then uses these limits. It does not interchange the endpoint integral with the integer sum: the absolute sum of degree integrals is a positive constant times $\sum_{n \geq 1} 1/n$, which diverges. Thus the actual additive zero moments in GTK2 and the nonzero multiplicative endpoints GTK9 coexist with their exact receiving maps.

78.3 The full global correlation kernel

For real T retain the original modulated test

$$h_T(y) = y^{iT} h(y), \quad h_T^\sharp(y) = y^{-1} \overline{h_T(y^{-1})} = h_T(y), \quad Mh_T(s) = M(s + iT). \quad (\text{GTK10})$$

The sharp equality follows by conjugating y^{-iT} and using GTK3. It does not say h_T is a real-valued function. For the literal original convolution with measure dx/x , the exact cross-correlation is

$$\begin{aligned} H_{T,S}(y) &= (h_T * h_S^\sharp)(y) = y^{iS-1} \int_0^\infty x^{i(T-S)} h(x) h(x/y) dx, \\ H_{T,S}(1) &= \int_0^\infty x^{i(T-S)} h(x)^2 dx, \quad MH_{T,S}(s) = M(s + iT) \overline{M(1 - \bar{s} + iS)} \\ &= M(s + iT) M(s + iS). \end{aligned} \quad (\text{GTK11})$$

In the first line the factor x/y from sharp multiplies the original measure dx/x to give dx/y . This explains every power and the retained dx . The last equality follows from both symmetries of M , not a change of Mellin convention. All double integrals are absolute by GTK5. Since both input tests are sharp-fixed and convolution is commutative, $H_{T,S} = H_{S,T}$ and $H_{T,S}^\sharp = H_{T,S}$. For $T = S$, GTK11 gives $H_{T,T}(y) = y^{iT} H_{0,0}(y)$ and $H_{T,T}(1) = M_h := \int_0^\infty h(x)^2 dx > 0$.

Write the complete real-place functional, with parity $\epsilon \in \{0, 1\}$, as

$$\begin{aligned} \mathcal{A}_\epsilon(H) &= (\log(2\pi) + \gamma) H(1) \\ &+ \frac{1}{2} \int_0^1 \left[\frac{H(y) + y^{-1} H(y^{-1}) - 2H(1)}{1 - y} + (-1)^\epsilon \frac{H(y) + y^{-1} H(y^{-1})}{1 + y} \right] dy. \end{aligned} \quad (\text{GTK12})$$

It equals the original real principal value for $e_\mathbb{R}(x) = e^{-2\pi i x}$ and $d^\times u = du/(2|u|)$. WQP8–11 prove this exact identity from the cited Connes formula: separate the two real signs, put $y = |u|^{-1}$, invert the interval above one, and integrate the retained subtracted constant. Its positive-branch cutoff term is $(\log \delta) H(1)$; integration of $2H(1)/(1 - y)$ with the displayed $1/2$ gives $-H(1) \log \delta$. The cutoff changes by endpoint ratios tending to one, hence has zero limit. The negative real sign contributes precisely the second denominator and its parity. These steps prove GTK12 with its original constant, rather than an undefined regularized remainder.

The one radial global kernel is

$$\begin{aligned} K(T, S) &:= Q(h_T, h_S) \\ &= M(iT) \overline{M(1 + iS)} + M(1 + iT) \overline{M(iS)} - (\log 1) H_{T,S}(1) - \mathcal{A}_0(H_{T,S}) \\ &- \sum_{p \text{ prime}} \log p \sum_{j \geq 1} [H_{T,S}(p^j) + p^{-j} H_{T,S}(p^{-j})]. \end{aligned} \quad (\text{GTK13})$$

This is the entire original Weil form evaluated on the actual theta-image modulation family. Every prime and prime power, the two endpoints, both real signs and their constant, the differential term $\log |\mathfrak{a}| = \log 1$, and all sharp factors are displayed. The differential term has the indicated value because the original rational reference additive character has $a_v = 1$ at every place.

78.4 Strong topology and convergence of the entire prime sum

Let B_0 be the radial part of the original test algebra, with seminorms

$$p_{A,m,j}(f) = \sup_{u \in \mathbb{R}} e^{A|u|} (1 + |u|)^m |\partial_u^j f(e^u)|, \quad A, m, j \in \mathbb{Z}_{\geq 0}. \quad (\text{GTK14})$$

These are the original strong weighted smooth seminorms of SWQ3 at the trivial compact-unit level. Their inclusion of all integer exponential weights is equivalent to retaining every real weight. No real coordinate or test is replaced in its convolution or pairing. By Leibniz's formula,

$$p_{A,m,j}(h_T) \leq \sum_{r=0}^j \binom{j}{r} |T|^{j-r} p_{A,m,r}(h). \quad (\text{GTK15})$$

Each T derivative inserts $(i \log y)^a$; the same proof bounds every mixed derivative by a polynomial in $|T|$ with finitely many fixed seminorms of h .

Convolution and sharp satisfy the explicit bounds from the actual integral,

$$\begin{aligned} p_{A,m,j}(f * k) &\leq 2p_{A+1,m,0}(f)p_{A,m,j}(k), \\ p_{A,m,j}(k^\sharp) &\leq \sum_{r=0}^j \binom{j}{r} p_{A+1,m,r}(k). \end{aligned} \quad (\text{GTK16})$$

To prove the first, write the logarithmic integral with variable v , use $|u| \leq |v| + |u - v|$ and $1 + |u| \leq (1 + |v|)(1 + |u - v|)$, and integrate $e^{-|v|}$, whose integral is exactly two. Derivatives can pass under that integrable majorant. For sharp, differentiate $e^{-u} \overline{k(e^{-u})}$, retaining the factor e^{-u} . GTK15–16 prove polynomial bounds in (T, S) for every seminorm of $H_{T,S}$ and its parameter derivatives.

For every $H \in B_0$ and integer $A \geq 2$,

$$\begin{aligned} &\sum_p \log p \sum_{j \geq 1} (|H(p^j)| + p^{-j} |H(p^{-j})|) \\ &\leq p_{A,0,0}(H) \sum_{n=2}^{\infty} \log n \left[\frac{n^{-A}}{1 - n^{-A}} + \frac{n^{-A-1}}{1 - n^{-A-1}} \right] < \infty. \end{aligned} \quad (\text{GTK17})$$

The first bounds are $|H(p^j)| \leq p^{-Aj} p_{A,0,0}(H)$ and $p^{-j} |H(p^{-j})| \leq p^{-(A+1)j} p_{A,0,0}(H)$. Summing their geometric series and then enlarging the prime sum to all integers gives the displayed bound. The final series converges since $A > 1$. The same argument after parameter differentiation proves uniform convergence of every kernel derivative on compact parameter sets, and polynomial bounds on the resulting all-prime functional.

The entire real functional GTK12 is also continuous on B_0 . At zero, $|H(y)| \leq p_{2,0,0}(H)y^2$ and $|y^{-1}H(y^{-1})| \leq p_{2,0,0}(H)y$. The retained constant in its first numerator is integrable on $(0, 1/2)$. On $[1/2, 1]$ the mean-value formula bounds $|H(y) + y^{-1}H(y^{-1}) - 2H(1)|/(1 - y)$ by the supremum of the derivative of that smooth numerator. That derivative is bounded by a fixed constant times $p_{2,0,0}(H) + p_{2,0,1}(H)$. The denominator $1 + y$ is nonsingular. These estimates give a fixed finite-seminorm bound for $\mathcal{A}_\epsilon(H)$ for either parity. The two Mellin endpoint functionals are continuous too: their absolute integrals are bounded by integrating $e^{-(A-|s|)|u|}$ times $p_{A,0,0}(f)$ at $s = 0, 1$, with $A > 1$. Consequently Q is a continuous sesquilinear form on B_0 , and K is smooth with at most polynomial growth, including each of its derivatives. This proves convergence of its full original expression, rather than treating the all-prime tail as a finite window.

78.5 An exact global synthesis of every compact radial test

For $g \in C_c^\infty(0, \infty)$ define

$$b_g(u) = \frac{g(e^u)}{h(e^u)}, \quad a_g(T) = \frac{1}{2\pi} \int_{\mathbb{R}} b_g(u) e^{-iT u} du. \quad (\text{GTK18})$$

Strict positivity GTK4 proves that this quotient is defined everywhere and is smooth. Its support is compact because $g(e^u)$ has compact support and the denominator never vanishes. Integration

by parts in u , with zero boundary terms, proves that a_g and each derivative decrease faster than every power: a_g is Schwartz. Fourier inversion, with precisely the displayed $1/(2\pi)$, gives

$$g(y) = \int_{\mathbb{R}} a_g(T) y^{iT} h(y) dT = \int_{\mathbb{R}} a_g(T) h_T(y) dT. \quad (\text{GTK19})$$

For completeness the inversion may be obtained by inserting $e^{-\varepsilon T^2}$ in the integral: its Fourier kernel is the Gaussian approximate identity with total mass one. Convolving this kernel with b_g converges to b_g and all its derivatives. Since a_g is integrable, dominated convergence removes the inserted factor and proves GTK19.

The convergence is in every seminorm GTK14, not only pointwise. In fact GTK15 gives the explicit tail bound

$$p_{A,m,j} \left(\int_{|T|>R} a_g(T) h_T dT \right) \leq \sum_{r=0}^j \binom{j}{r} p_{A,m,r}(h) \int_{|T|>R} |a_g(T)| |T|^{j-r} dT \longrightarrow 0. \quad (\text{GTK20})$$

The integral on a bounded parameter interval is the limit of Riemann sums in each seminorm, because its integrand is continuous in those seminorms by GTK15. Completeness follows directly by taking uniform limits of all weighted derivatives and their compatibility on each compact u interval. This proves the specified vector integral and its strong tail convergence without an unspecified weak topology.

Sesquilinearity, continuity, and the polynomial bounds just proved give the complete reconstruction

$$Q(g, k) = \int_{\mathbb{R}} \int_{\mathbb{R}} a_g(T) \overline{a_k(S)} K(T, S) dT dS, \quad g, k \in C_c^\infty(0, \infty). \quad (\text{GTK21})$$

The double integral is absolutely convergent, since each coefficient is Schwartz and K has a finite polynomial bound. First apply Q to the bounded-interval Riemann sums in GTK19, then pass to their strong limits using GTK16–17 and the real and endpoint estimates. The same absolute majorants justify interchanging these parameter integrals with every prime-power sum. At the real place the subtracted numerator in GTK12 is integrated as a whole with its mean-value bound; its two individually singular pieces are not integrated separately.

Every $g \in B_0$ is a strong limit of compact tests. Choose $\theta \in C_c^\infty(\mathbb{R})$ equal to one on $[-1, 1]$ and set $g_R(y) = \theta((\log y)/R) g(y)$. Derivative terms carry R^{-j} and are supported where $|\log y| \geq R$; one extra exponential weight in GTK14 bounds each by $C e^{-R} p_{A+1,m,j'}(g)$. Thus $g_R \rightarrow g$ in every seminorm. It follows that GTK21 recovers Q on all of B_0 by these limits. It does not claim that g/h is Schwartz for every noncompact g ; that generally need not hold. Compact synthesis followed by the proved strong limit is the exact extension.

78.6 The full compact-unit sectors and their entire global kernels

The radial kernel alone does not reconstruct a nontrivial compact-unit character. To obtain the full original Bruhat test algebra, retain every continuous character χ of K and define the single indexed family

$$F_{\chi,T}(y, u) = \chi(u) h_T(y), \quad \mathbf{K}((\chi, T), (\psi, S)) = Q(F_{\chi,T}, F_{\psi,S}). \quad (\text{GTK22})$$

Every continuous character factors through a finite $U_N = (\mathbb{Z}/N\mathbb{Z})^\times$: continuity gives an open subgroup whose image lies in a circle arc containing no nontrivial subgroup, hence in the character kernel; a finite congruence subgroup lies inside it. Let c_χ be its minimal conductor and χ_* the corresponding primitive integer character, extended by zero on integers not coprime to c_χ . Keep $\chi(-1) = (-1)^{\epsilon_\chi}$.

The original sharp fixes $F_{\chi,T}$: $\overline{\chi(u^{-1})} = \chi(u)$ and GTK10 treat both factors. Direct compact integration in convolution yields

$$F_{\chi,T} * F_{\psi,S}^\sharp = \delta_{\chi,\psi} \chi(u) H_{T,S}(y), \quad \mathbf{K}((\chi, T), (\psi, S)) = \delta_{\chi,\psi} K_\chi(T, S). \quad (\text{GTK23})$$

The vanishing mixed integral is $\int_K \chi(v) \overline{\psi(v)} du = 0$ for $\chi \neq \psi$, proved by translating it by a unit at which the quotient character is not one. Its value is one when the characters agree. No factor is lost by this character diagonalization.

The entire individual kernel, including ramified primes, is

$$\begin{aligned} K_\chi(T, S) = & 1_{\chi=1} [M(iT) \overline{M(1+iS)} + M(1+iT) \overline{M(iS)}] \\ & + \left[\sum_{p|c_\chi} v_p(c_\chi) \log p - \log 1 \right] H_{T,S}(1) - \mathcal{A}_{\epsilon_\chi}(H_{T,S}) \\ & - \sum_{p \nmid c_\chi} \log p \sum_{j \geq 1} [\chi_*(p)^j H_{T,S}(p^j) + p^{-j} \chi_*(p)^{-j} H_{T,S}(p^{-j})]. \end{aligned} \quad (\text{GTK24})$$

This is WQP12 evaluated on the actual noncompact tests, with extension justified by GTK14–17. It may also be derived directly as follows. Endpoints integrate χ over K , giving precisely the indicator in the first line. At an unramified prime the local unit character is trivial. The local embedding of p^j has real modulus p^{-j} and away- p unit coordinates p^{-j} , so its character is $\chi_*(p)^{-j}$. Evaluation at its inverse gives the positive-shell phase in GTK24. Multiplicative unit mass is $\log p$; for the negative shell the denominator $|1 - u|_p$ contributes p^{-j} . These are the two original shell factors, with neither orientation reversed.

Here is an independent calculation of the ramified unit term. Write $a = v_p(c_\chi) > 0$ and let ϑ be the nontrivial local unit character. Its integral over $B_r = 1 + p^r \mathbb{Z}_p$ is zero for $1 \leq r < a$ and is p^{-r} for $r \geq a$, using additive Haar measure. Its integral over $\mathbb{Z}_p^\times$ is zero. The canonical trivial-unit principal value is zero, so its difference for ϑ is the convergent integral

$$\frac{\log p}{1 - p^{-1}} \int_{\mathbb{Z}_p^\times} \frac{\vartheta(u)^{-1} - 1}{|1 - u|_p} du = -a \log p. \quad (\text{GTK25})$$

To calculate it, the shell outside B_1 contributes $-1 + p^{-1}$ for $a = 1$. If $a > 1$, that shell contributes $-1 + 2p^{-1}$; each shell $B_r \setminus B_{r+1}$ for $1 \leq r \leq a - 2$ contributes $-(1 - p^{-1})$ after its factor p^r ; the final shell $B_{a-1} \setminus B_a$ contributes -1 . Their sum is exactly $-a(1 - p^{-1})$. Multiplication by the retained Haar constant gives GTK25. Shells with $r \geq a$ vanish. All nonunit shells have zero local-unit average because their denominator and radial factor are constant on units. This proves the local term $-a \log p H_{T,S}(1)$; subtracting it in the global trace gives the positive conductor sum in GTK24. The trivial-unit principal value used here is exactly the reference-character one in WLC6–9 and WQP12, not an extra conductor convention.

At the real place the local negative sign embeds as $(-1)_p$ in the compact coordinate, giving $(-1)^{\epsilon_\chi}$ in GTK12. This proves the complete parity term. If a chosen finite modulus N is larger than c_χ , primes dividing N but not c_χ remain in GTK24. The exact original-series comparison is $L(s, \chi_N) = L(s, \chi_*) \prod_{p|N, p \nmid c_\chi} (1 - \chi_*(p)p^{-s})$ for $\Re s > 1$; expanding and comparing their absolutely convergent Euler factors proves it. The imprimitive integer zero extension therefore cannot be inserted as a zero phase into this idele-class kernel.

For each fixed finite level, GTK24 has the same polynomial parameter bounds as GTK13: conductor terms are finite, phases have modulus one, and the all-prime bound GTK17 applies unchanged. Thus the one object $\mathbf{K}$ is a rigorously defined smooth, polynomially bounded kernel on each of its finite character blocks. No uniform conductor-independent bound is asserted or needed.

Every compactly supported original Bruhat test at level N has its exact finite expansion and synthesis

$$\begin{aligned} f(y, u) &= \sum_{\chi \in \widehat{U}_N} g_\chi(y) \chi(u), & g_\chi(y) &= \frac{1}{\varphi(N)} \sum_{u \in U_N} f(y, u) \overline{\chi(u)}, \\ f(y, u) &= \sum_{\chi \in \widehat{U}_N} \int_{\mathbb{R}} a_{g_\chi}(T) F_{\chi, T}(y, u) dT. \end{aligned} \tag{GTK26}$$

Finite character orthogonality proves the first inverse, including its Haar factor; GTK18–20 prove the second with strong convergence. Hence the entire original global sesquilinear form is reconstructed by

$$Q(f, k) = \sum_{\chi \in \widehat{U}_N} \int_{\mathbb{R}^2} a_{g_\chi}(T) \overline{a_{k_\chi}(S)} K_\chi(T, S) dT dS. \tag{GTK27}$$

Tests at different finite levels may be put in their common multiple level without changing their characters or coefficients. The finite sum, Schwartz tails, and GTK17 justify every interchange. For the original noncompact weighted Bruhat test space, apply the logarithmic cutoff proved after GTK21 at its fixed finite compact level. Convergence holds in its original weighted topology and consequently in the Bruhat inductive-limit topology. This proves recovery of the full original B , not only its radial subspace.

The one countable global Gram object. The index set $\mathcal{I} = \widehat{K} \times \mathbb{Q}$ is countable: the compact dual is a union over positive integers N of the finite sets $\widehat{U}_N$, and the rational numbers are countable. Retain the original tests, their strong topology, and the entire array

$$\mathcal{G}_{(\chi, t), (\psi, s)} = Q(F_{\chi, t}, F_{\psi, s}) = \delta_{\chi, \psi} K_\chi(t, s), \quad (\chi, t), (\psi, s) \in \mathcal{I}. \tag{GTK27a}$$

The complex linear span of these actual tests is dense in the original B . First approximate a compact test by GTK26 truncated to $|T| \leq R$ using GTK20. On that bounded interval, $T \mapsto h_T$ is strongly continuous by GTK15, so Riemann sums converge in every fixed finite collection of seminorms. The tags of the sums may be chosen rational by their density and that same strong continuity. The finite character sum remains at the original compact level. Increasing the collection of seminorms gives a sequence converging in the metrizable strong space at that level. Finally the compact cutoffs after GTK21 approximate every original weighted test. This proves the asserted density.

For finite combinations $f = \sum_i c_i F_i$ and $k = \sum_j d_j F_j$ the entire global pairing is exactly

$$Q(f, k) = \sum_{i, j} c_i \overline{d_j} \mathcal{G}_{i, j}. \tag{GTK27b}$$

Continuity and density prove that this array, together with the displayed original test inclusion and topology, uniquely determines Q on all of B . Thus it is one countable global Gram object, not a finite collection of sampled tests. Possible linear relations among its generating tests cause no ambiguity: GTK27b is their actual pairing, so a zero test pairs to zero with every finite combination and then every strong limit. It also proves the exact equivalence between nonnegativity of the entire original form and nonnegativity of every finite quadratic combination in this one array. In one direction restrict the actual form; in the other approximate any test strongly and pass to the limit of its quadratic values. This is a proved equivalence of receiving objects, not an assertion of either sign.

78.7 The exact spectral expression, all zeros, and the zero row

For the trivial sector the source trace theorem gives

$$\begin{aligned} K(T, S) &= \sum_{\substack{\rho: \zeta(\rho)=0 \\ 0 < \Re \rho < 1}} m_\rho M(\rho + iT) \overline{M(1 - \bar{\rho} + iS)} \\ &= \sum_{\substack{\rho: \zeta(\rho)=0 \\ 0 < \Re \rho < 1}} m_\rho M(\rho + iT) M(\rho + iS). \end{aligned} \tag{GTK28}$$

Every zero has its original multiplicity. The first formula uses the original source Mellin transform and sharp. The second uses the exact two identities after GTK6. It does not replace a sum over possible off-line zeros by a sum over zeros on the line.

Here is a direct convergence proof that does not use RH. For σ in a fixed bounded real interval, integration by parts r times in u in $M(\sigma + it) = \int h(e^u) e^{\sigma u} e^{itu} du$ yields

$$|M(\sigma + it)| \leq C_{r,I} (1 + |t|)^{-r}, \quad \sigma \in I. \tag{GTK29}$$

The same estimate holds for every s derivative because it inserts a power of u in the integral. All boundary terms vanish by GTK5. That bound also gives a direct polynomial zero-count estimate for the zeros occurring in GTK28. GTK5 implies $\log \max_{|s| \leq R} |M(s)| \leq C(R+1) \log(R+2)$. Indeed integrate $\exp((R+B)|u| - ce^{2|u|})$, substitute $v = e^{2|u|}$, and bound $v^{(R+B)/2} e^{-cv/2}$ by its maximum; the remaining exponential integral is finite. Jensen's formula centered at zero, where $M(0) = 1/4$, gives $n_M(R) \log 2 \leq \log \max_{|s| \leq 2R} |M(s)| - \log |M(0)|$. If a boundary circle contains a zero, use slightly larger radii and pass to the limit. Jensen's formula itself follows by factoring the finitely many zeros in the disk and applying the mean-value formula to the logarithm of the remaining nonvanishing holomorphic factor. Thus $n_M(R) = O(R \log(R+2))$. GTK8 shows that the original zeta zeros counted here, with their multiplicities, are included among these zeros of M .

Combining this count with GTK29 proves absolute and locally uniform convergence of GTK28 and of every T, S derivative. It also proves a polynomial bound for the sum of absolute values on the entire parameter plane. For example let $R_0 = 2(|T| + |S| + 2)$. Zeros with $|\Im \rho| \leq R_0$ contribute a bounded summand each and number $O(R_0 \log(R_0 + 2))$; for the remaining zeros both shifted imaginary arguments have modulus at least $|\Im \rho|/2$, and GTK29 with $r = 3$ gives a summable tail. This is enough to exchange the spectral sum with the double Schwartz integral in GTK21. For each ρ , its resulting factors are exactly $Mg(\rho)$ and $\overline{Mk(1 - \bar{\rho})}$ by strong synthesis and continuity of Mellin evaluation. Thus GTK21 recovers the entire original spectral pairing as well as every term of its explicit formula.

The kernel is real and symmetric. Symmetry follows already from GTK10–11. For reality, conjugation of a summand in the second line of GTK28 yields $M(1 - \bar{\rho} + iT) M(1 - \bar{\rho} + iS)$. The map $\rho \mapsto 1 - \bar{\rho}$ preserves the original zero set and multiplicities, by the original functional equation with its nonvanishing strip multiplier, also encoded in GTK6–8. Absolute convergence permits that reindexing, proving $\overline{K(T, S)} = K(T, S)$. Hermitian symmetry follows. This is an exact reality statement, not positive definiteness.

Every entry in its zero row vanishes:

$$K(0, S) = K(T, 0) = 0. \tag{GTK30}$$

There are two exact proofs. In GTK28, $M(\rho) = 0$ with each original zero by GTK7–8. On the actual source quotient, $h \in V$ and V is a convolution ideal; thus $h * h_S^\dagger \in V$ and its induced operator is zero by the author's **vanishV**. The ideal and sharp maps, including endpoint terms, are proved in SWQ7–15, so this second argument uses the actual source image rather than only

its zero values. GTK13 at $T = 0$ is consequently the exact global identity

$$\begin{aligned} \frac{1}{4}\overline{M(1+iS)} + \frac{1}{4}\overline{M(iS)} &= (\log 1)H_{0,S}(1) + \mathcal{A}_0(H_{0,S}) \\ &+ \sum_p \log p \sum_{j \geq 1} [H_{0,S}(p^j) + p^{-j}H_{0,S}(p^{-j})]. \end{aligned} \quad (\text{GTK31})$$

Both endpoint constants and the complete prime and real contributions remain in this vanishing identity. No zero-row assertion for nontrivial χ is inferred: the modulation seed in those sectors is not the original adelic image element GTK2.

78.8 The calculated receiving defect under modulation

The concurrent global estimate GWM11–14 applies to this very h by GTK5, not merely to compact tests. Its constants specialize exactly to

$$\begin{aligned} M_h &= \int_0^\infty h(x)^2 dx > 0, & K_h &= \int_0^\infty |xh'(x) + \tfrac{1}{2}h(x)|^2 dx, \\ A_{0,h} = M(0) &= \tfrac{1}{4}, & A_{1,h} = M(1) &= \tfrac{1}{4}, & P_h &= 2 \sum_p \log p \sum_{j \geq 1} |H_{0,0}(p^j)| < \infty. \end{aligned} \quad (\text{GTK32})$$

The equalities for the absolute endpoint integrals use the proved positivity of h . The prime constant retains the entire sum; GTK17 proves its finiteness. With $\alpha = 1/4$, conductor one and differential modulus one, the full bound reads

$$\begin{aligned} K(T, T) &\geq M_h [\log(|T|/4) - 4 - \log \pi + \log 1 - \log 1] \\ &- \frac{4K_h}{T^2} \log |T| - P_h - \tfrac{1}{8}, \quad |T| \geq 1. \end{aligned} \quad (\text{GTK33})$$

For avoidance of an assumed missing estimate, the underlying argument is as follows. The entire real term GTK12 equals $-\frac{1}{2\pi} \int |Mh_T(1/2 + it)|^2 [\Re \psi(1/4 + it/2) - \log \pi] dt$, with its sign as in GWM7. The identity follows by Fourier inversion of GTK11 on the critical vertical line and the convergent digamma integral, retaining $\log(2\pi) + \gamma - \log 2 - \gamma$ in that calculation. The digamma series and comparison with its integral give, for $\alpha > 0$, $\Re \psi(\alpha + ib) \geq \frac{1}{2} \log(\alpha^2 + b^2) - 1/\alpha$: the error between the series $(k + \alpha)/((k + \alpha)^2 + b^2)$ and its integral is bounded by its total variation, at most $1/\alpha$. Set $u = t + T$ and use this bound separately on $|u| \leq |T|/2$ and its complement. Parseval gives the complement mass at most $4K_h/T^2$, with its original $dt/(2\pi)$ factor. The difference of the two digamma lower bounds is $\log |T|$ here. Every paired prime term has absolute value at most $2|H_{0,0}(p^j)|$, and the two endpoints together are at least $-2(1/4)(1/4) = -1/8$. Combining these proves GTK33 on the full global form.

An explicit finite tail threshold is

$$T_h = \max \left\{ 1, 4 \exp \left[1 + 4 + \log \pi - \log 1 + \log 1 + \frac{P_h + 1/8}{M_h} + \frac{2K_h}{\exp(1)M_h} \right] \right\}. \quad (\text{GTK34})$$

For $t \geq 1$, $\log t/t^2 \leq 1/(2 \exp(1))$, by differentiation with its maximum at $\sqrt{\exp(1)}$. Substituting this inequality and GTK34 in GTK33 proves

$$K(T, T) \geq M_h > 0 \quad (|T| \geq T_h). \quad (\text{GTK35})$$

This is a theorem about the entire specified modulation tail and its actual seed. It leaves no uncomputed prime window in the bound, although its exact constant contains the convergent original prime series.

On the original algebra, $\Delta_{iT} f = |\cdot|^{iT} f$ is an invertible linear convolution-algebra map, with inverse Δ_{-iT} , because $|ab|^{iT} = |a|^{iT}|b|^{iT}$ in the convolution integral. It commutes with sharp by

direct conjugation. Let $W = \overline{V}$ in the topology SWQ3. The trace vanishes on the entire closed ideal W , and therefore on its pairing with every test, by SWQ15 and SWQ22–24. GTK30 and GTK35 now prove the concrete non-descent

$$h \in V \subset W, \quad \Delta_{iT}h = h_T \notin W \quad (|T| \geq T_h). \quad (\text{GTK36})$$

If h_T belonged to W , its pairing with itself would be zero, contradicting GTK35. Consequently Δ_{iT} does not descend to an endomorphism of this original spectral quotient for those T . This proves the receiving defect on a whole frequency range. It does not assert that every nonzero frequency has that defect.

78.9 The mixed divided-difference receiver at the entire zero row

Retain the original kernel K , including its vanishing axes, and define the actual functions

$$w_T(x) = \begin{cases} \frac{x^{iT} - 1}{iT} h(x), & T \neq 0, \\ (\log x)h(x), & T = 0. \end{cases} \quad (\text{GTK36a})$$

This is a strongly continuous family in B_0 . Indeed its exact integral formula is $w_T(x) = (\log x)h(x) \int_0^1 x^{i\theta T} d\theta$. GTK5 and differentiation prove convergence in every seminorm as $T \rightarrow 0$, and prove all parameter derivatives as well. For real T one has $w_T^\# = -w_T$: the coefficient $1/(iT)$ conjugates to its negative while GTK10 fixes the numerator. At zero the same identity follows directly from $\log(x^{-1}) = -\log x$ and GTK3. Sesquilinearity and GTK30 consequently give the exact mixed global receiver

$$\begin{aligned} \mathcal{R}(T, S) &:= Q(w_T, w_S) = \frac{K(T, S)}{TS} \quad (TS \neq 0), \\ q(h_T) &= iT q(w_T), \quad q : B \longrightarrow B/W. \end{aligned} \quad (\text{GTK36b})$$

The scalar in the first variable is $1/(iT)$ and the conjugated scalar in the second is $\overline{1/(iS)}$; their product is $1/(TS)$. This accounts for its sign. The quotient identity uses $q(h) = 0$ and retains the original factor iT . The axes of $\mathcal{R}$ are defined by the actual tests in GTK36a and strong continuity, rather than being discarded.

These axes in fact have an entire two-variable extension. For complex z define $h_z(x) = x^{iz}h(x)$ and extend w_z by the same integral formula. On each compact set of z , GTK5 controls the additional factor $e^{-\Im z \log x}$ and every power of $\log x$. The exponential power series therefore converges in every strong seminorm, proving that both families are entire B_0 -valued functions. Define the complex bilinear extension, with the trace functional $\mathcal{T}$ on the original quotient,

$$\begin{aligned} \mathcal{K}(z, w) &= \mathcal{T}(h_z * h_w), \quad \mathcal{K}(z, 0) = \mathcal{K}(0, w) = 0, \\ \mathcal{K}(z, w) &= zw \mathcal{R}_{\mathbb{C}}(z, w), \quad \mathcal{R}_{\mathbb{C}}(z, w) = -\mathcal{T}(w_z * w_w). \end{aligned} \quad (\text{GTK36c})$$

The zero axes follow because $h \in V$ and V is the actual convolution ideal. Expanding $(h_z - h) * (h_w - h) = (iz)(iw)w_z * w_w$ and applying the trace proves the factor $-zw$ and the displayed identity everywhere. For real arguments $h_S^\# = h_S$ and $w_S^\# = -w_S$, so $\mathcal{K}(T, S) = K(T, S)$ and $\mathcal{R}_{\mathbb{C}}(T, S) = \mathcal{R}(T, S)$. This is an entire bilinear continuation of the real-parameter kernel; it is not a claim that sesquilinearity itself is holomorphic in its second input.

Termwise differentiation in GTK28 is justified by GTK29 and its zero-count bound. It gives

$$\begin{aligned} \mathcal{R}(0, 0) &= Q((\log x)h, (\log x)h) = - \sum_{\substack{\rho: \zeta(\rho)=0 \\ 0 < \Re \rho < 1}} m_\rho [M'(\rho)]^2, \\ \left. \frac{d^2}{dT^2} K(T, T) \right|_{T=0} &= 2\mathcal{R}(0, 0). \end{aligned} \quad (\text{GTK36d})$$

There is no absolute-value square in this formula: the product is the square of the complex derivative, paired over the entire original zero set. The resulting sum is real because it equals the actual Hermitian quadratic value. Directly, $M'(1-s) = -M'(s)$ and conjugation of M' give the same reality under the original zero symmetries. The second identity is the second derivative of the exact $K(T, T) = T^2 \mathcal{R}(T, T)$.

To keep the original arithmetic multiplier and every derivative explicit, set

$$\begin{aligned} B(s) &= \frac{\pi^2}{2\pi^{(s+4)/2}} \Gamma((s+4)/2) - \frac{3\pi}{4\pi^{(s+2)/2}} \Gamma((s+2)/2), \\ M'(s) &= 2\zeta'(s)B(s) + 2\zeta(s) \left[-\frac{\log \pi}{2} B(s) + \frac{\pi^2}{4\pi^{(s+4)/2}} \Gamma'((s+4)/2) - \frac{3\pi}{8\pi^{(s+2)/2}} \Gamma'((s+2)/2) \right]. \end{aligned} \quad (\text{GTK36e})$$

Differentiating each original factor in GTK7 proves this identity before any evaluation. At a nontrivial zero ρ all displayed Gamma expressions are finite, and the second line's bracket is multiplied by the actual zero $\zeta(\rho)$. Thus $M'(\rho) = 2\zeta'(\rho)B(\rho)$ there, with that vanished contribution accounted for, and GTK36d can equivalently be written $-4 \sum_{\rho} m_{\rho} \zeta'(\rho)^2 B(\rho)^2$. Multiple zeros with $\zeta'(\rho) = 0$ contribute zero at this derivative order; their original multiplicities have not been removed from the kernel or its higher derivatives.

Finally the actual modulation receiving defect on the source image is the well-defined linear map

$$D_T : V \longrightarrow B/W, \quad v \longmapsto q(\Delta_{iTv}). \quad (\text{GTK36f})$$

At $T = 0$ it is zero on all of V . For $|T| \geq T_h$ its value on the single specified image test is $D_T(h) = q(h_T) = iTq(w_T) \neq 0$ by GTK35–36. Every displayed mixed value $\mathcal{R}(T, S)$ is the descended pairing of these divided-difference classes, by SWQ23–24 and GTK36b. Thus the kernel records both the unmodulated vanishing and its exact global receiving defect, including the mixed directions. Positive individual diagonal tails do not establish the sign of an arbitrary finite mixed combination in GTK27b.

The value at the origin also has its complete original explicit formula, without using the zero sum as a replacement. Put

$$\begin{aligned} J_{\log} &= \int_0^{\infty} (\log x)^2 h(x)^2 dx, \quad H_{\log}(y) = \frac{1}{y} \int_0^{\infty} (\log x) \log(x/y) h(x) h(x/y) dx, \\ \mathcal{R}(0, 0) &= M'(0) \overline{M'(1)} + M'(1) \overline{M'(0)} - (\log 1) J_{\log} - \mathcal{A}_0(H_{\log}) \\ &\quad - \sum_p \log p \sum_{j \geq 1} [H_{\log}(p^j) + p^{-j} H_{\log}(p^{-j})]. \end{aligned} \quad (\text{GTK36g})$$

This follows by substituting the actual $w_0 = (\log x)h$ into the original correlation and trace. GTK5 proves membership in B_0 and all endpoint and prime convergence exactly as in GTK14–17. Here $M'(0), M'(1)$ are the entire derivatives of the full GTK7, with each Gamma contribution retained in GTK36e. They satisfy $M'(1) = -M'(0)$ and are real by the proved symmetries, but both original endpoint terms have been left visible. For any nonzero T the full Mellin map of its divided difference is $Mw_T(s) = [M(s+iT) - M(s)]/(iT)$, with its entire limit $M'(s)$ at zero. These identities preserve the original kernel, its axes and their jets, rather than using the quotient $K/(TS)$ to discard any of them.

78.10 Every original support label on the global object

Let L be an arbitrary nontrivial bounded distributive lattice and retain

$$G_L(B) = \{(0, \lambda) : \lambda \in L\} \cup (B \times \{1_L\}), \quad e = (0, 1_L), \quad \tau = (0, 0_L). \quad (\text{GTK37})$$

Finite character decomposition, each modulation, the compact coefficient map GTK18, and the synthesis map GTK26 are linear maps on their specified entire amplitude spaces. Their full lifts are always $(v, \lambda) \mapsto (Tv, \lambda)$, with one label on the whole vector of character coefficients and parameter functions. Nonzero output requires a nonzero input, hence its top label. This proves membership in the constrained carrier. Addition uses the original join, and scalar multiplication uses the original meet, so linearity proves the lifted semimodule maps.

The vector integral in GTK19 is performed on amplitudes first, with its proved strong convergence. The one already assigned label is then retained. No infinite join, measure, topology, or integration on the arbitrary lattice is postulated. The same rule applies to the finite character sum and to the continuous kernel synthesis. The full receiver is

$$Q_L((f, \lambda), (k, \mu)) = (Q(f, k), \lambda \wedge \mu),$$

$$Q_L((f, \lambda), (k, \mu)) = \left(\sum_{\chi} \int_{\mathbb{R}^2} a_{g_{\chi}}(T) \overline{a_{k_{\chi}}(S)} K_{\chi}(T, S) dT dS, \lambda \wedge \mu \right) \quad (\text{GTK38})$$

on compact tests, and on their proved strong limits. Carrier membership follows because a nonzero pairing requires both nonzero input amplitudes, hence both top labels. Finite distributivity of meet over join and the sesquilinear amplitude identity prove additivity and the original scalar rule in each argument. In particular $Q_L((h, 1_L), (h_S, 1_L)) = e$, not τ , even though its numerical amplitude vanishes.

The quotient is exactly the same-label congruence $(f, \lambda) \sim (k, \mu)$ if and only if $\lambda = \mu$ and $f - k \in W$, as proved in SWQ17. The trace pairing descends because a change of representatives contributes $w * k^{\sharp} + f * v^{\sharp} + w * v^{\sharp} \in W$. Modulation still fails to descend for GTK36: the two equivalent inputs $(h, 1_L) \sim (0, 1_L)$ map to $(h_T, 1_L)$ and $(0, 1_L)$, which are not equivalent. Every zero label survives this counterexample to descent. At each support prime J , the actual support map $(f, \lambda) \mapsto [\lambda]_J$ sends GTK38 to $[\lambda]_J \wedge [\mu]_J$ and commutes with every label-preserving linear map. No Boolean reduction or scalar collapse is used to obtain the global kernel.

The resulting object is the single character-indexed kernel $\mathbf{K}$, with GTK24 its complete original global formula, GTK26–27 its proved reconstruction of every original test, GTK28 its all-zero expression in the original zeta sector, and GTK38 its full support receiver. Its zero row and positive diagonal tail are both proved. They do not determine the sign of arbitrary mixed-frequency quadratic combinations; no positivity assertion for that entire kernel or RH conclusion is made here.

79 Actual tensor powers over the entire Split-Zero coefficient semiring

Objects, sources, and scope. The coefficient object is the original full lattice construction in *Split Support Geometry*, v11, Definition `def:lattice-split`, retained under its source byline The Clankers. The programme proof FLH1–4, in `FULL_LATTICE_NULL_HOMOLOGY.tex`, specifies its null differential; those definitions are repeated below. For the arithmetic application the human source is Alain Connes, Caterina Consani and Matilde Marcolli, *The Weil proof and the geometry of the adeles class space*, author source `Yuri70v2.tex`, locators `rangeCvdef`, `vanishV`, `fsharpinv`, `TraceformulaH1` and `tracepairing`. SWQ1–24 constructs the actual closed quotient from those definitions. GTK13,24,27–28 proves the global mixed kernel and its original arithmetic trace, retaining every prime, endpoint, real-place and conductor term. The argument here proves the universal tensor map over the original semiring, its entire quotient congruence and its full tensor pairing. No completion of an algebraic tensor product, positive metric, cohomological purity, or identification of a trace radical with an adelic image is assumed.

79.1 The universal tensor, including every zero label

Let L be any nontrivial bounded distributive lattice, and retain

$$\begin{aligned} S &= G_L(\mathbb{C}) = \{(0, \lambda) : \lambda \in L\} \cup (\mathbb{C} \times \{1_L\}), \\ (a, \lambda) + (b, \mu) &= (a + b, \lambda \vee \mu), \quad (a, \lambda)(b, \mu) = (ab, \lambda \wedge \mu), \\ z_\lambda &= (0, \lambda), \quad \tau = z_{0_L}, \quad e = z_{1_L}, \quad \widehat{a} = (a, 1_L), \quad \widehat{0} = e. \end{aligned} \tag{GFT1}$$

For every complex vector space V put

$$X(V) = G_L(V) = \{(0, \lambda) : \lambda \in L\} \cup (V \times \{1_L\}), \quad (a, \eta)(v, \lambda) = (av, \eta \wedge \lambda). \tag{GFT2}$$

Addition uses the same join as GFT1. In particular $(0, 1_L)$ is not the additive identity of this semimodule. Every linear map $A : V \rightarrow W$ has its actual lift $A^\uparrow(v, \lambda) = (Av, \lambda)$; the lift of the zero linear map is e times the identity-label map, not the constant map with value τ .

The tensor product of S -semimodules below means the commutative monoid on symbols $x \otimes y$ modulo additivity in both variables, $\tau \otimes y = x \otimes \tau = 0$, and $(sx) \otimes y = x \otimes (sy)$, with its induced S action. This presentation proves its balanced biadditive universal property. There is a canonical isomorphism

$$\Phi_{V,W} : X(V) \otimes_S X(W) \longrightarrow X(V \otimes_{\mathbb{C}} W), \quad (v, \lambda) \otimes (w, \mu) \longmapsto (v \otimes w, \lambda \wedge \mu). \tag{GFT3}$$

This holds for the entire vector spaces, with no finite-dimensional restriction. Nonzero output amplitude requires both nonzero input amplitudes, hence both top labels. Distributivity of meet over join proves biadditivity. The scalar law and associativity of meet prove balancing. Thus the displayed formula defines an S -linear map.

Here is a proof that this map has the full universal property, so that it has not silently imposed additional label identifications. Let P be any S -semimodule, and let $b : X(V) \times X(W) \rightarrow P$ be balanced and biadditive. Write $0_V^+ = (0, 1_L)$, $0_W^+ = (0, 1_L)$, and

$$t = b(0_V^+, 0_W^+), \quad P_t = \{p \in P : ep = t\}. \tag{GFT4}$$

One has $et = t$ and $t + t = t$. For any $p \in P_t$, the semiring identities $1_S + e = 1_S$ and $1_S + \widehat{-1} = e$ give $p + t = p$ and $p + \widehat{-1}p = t$. The set P_t is closed under addition because $t + t = t$, and under every $\widehat{a}$ because $\widehat{a}e = e$. These facts make P_t a complex vector space, with zero vector t and scalar multiplication $ap = \widehat{a}p$. The vector-space axioms are the semimodule axioms, including the zero-scalar axiom $\widehat{0}p = ep = t$.

Balancing by e gives, for every v, w ,

$$b(0_V^+, (w, 1_L)) = t = b((v, 1_L), 0_W^+), \quad eb((v, 1_L), (w, 1_L)) = t. \quad (\text{GFT5})$$

Consequently $(v, w) \mapsto b((v, 1_L), (w, 1_L))$ is a complex-bilinear map into P_t . The ordinary vector-space tensor universal property gives a unique linear map $u : V \otimes_{\mathbb{C}} W \rightarrow P_t$. Every lower-label value of b is forced by balancing:

$$b((0, \lambda), (w, \mu)) = z_{\lambda \wedge \mu} t \quad \text{when } w = 0 \text{ or } \mu = 1_L. \quad (\text{GFT6})$$

These are all possible second inputs. For $\mu = 1_L$ use $(0, \lambda) = z_{\lambda} 0_V^+$ and GFT5. For $w = 0$ use $(0, \mu) = z_{\mu} 0_W^+$ and balance once more. The corresponding identity holds when the second amplitude is zero.

Define the forced extension of u by

$$U(z, 1_L) = u(z), \quad U(0, \lambda) = z_{\lambda} t. \quad (\text{GFT7})$$

The formulas agree at $(0, 1_L)$ since $u(0) = t = et$. They are additive on lower labels because $z_{\lambda} + z_{\mu} = z_{\lambda \vee \mu}$. On two top inputs additivity is that of u . On a top and a lower input, if $p = u(z)$ then $p + z_{\lambda} t = (1_S + z_{\lambda} e)p = p$. Scalar compatibility for a nonzero scalar is the linearity of u and $\hat{a}z_{\lambda} = z_{\lambda}$. For a zero scalar z_{η} it is $z_{\eta} p = z_{\eta} e p = z_{\eta} t$, with the displayed meet when the input label is lower. This proves every semimodule law for U and $U\Phi = b$. Uniqueness follows because ordinary pure tensors span the top vector space and $(0, \lambda)$ is the image of $0_V^+ \otimes (0, \lambda)$. Applying the two universal properties proves GFT3 is an isomorphism.

Iteration gives the exact k -fold isomorphism, for every $k \geq 1$,

$$\bigotimes_{i=1}^k {}_S X(V_i) \cong X \left(\bigotimes_{i=1}^k {}_{\mathbb{C}} V_i \right), \quad \bigotimes_i (v_i, \lambda_i) \mapsto \left(\bigotimes_i v_i, \bigwedge_i \lambda_i \right). \quad (\text{GFT8})$$

Parenthesizations and permutations give the same map: check them on the displayed generators using associativity and commutativity of meet and the corresponding ordinary tensor maps. The unit is $X(\mathbb{C}) = S$. Linear maps commute with GFT8 by evaluation on these generators. Thus this is a strong symmetric monoidal functor on complex vector spaces with the specified lifts, not merely a formula for selected pure tensors. Every finite sum has exactly the join of its displayed meet labels; amplitude cancellations do not erase any of these labels.

79.2 Exact tensor quotients and the actual arithmetic image

Let $W_i \subset V_i$ be arbitrary complex linear subspaces. The ordinary tensor quotient has kernel

$$N = \sum_{i=1}^k V_1 \otimes \cdots \otimes W_i \otimes \cdots \otimes V_k, \quad \left(\bigotimes_i V_i \right) / N \cong \bigotimes_i (V_i / W_i). \quad (\text{GFT9})$$

To prove this, extend a basis of each W_i to a basis of V_i . The resulting algebraic tensor basis separates tensors having at least one W_i basis entry from those with only complement entries. The first set spans exactly N ; the second maps bijectively to the tensor basis of the quotient spaces. The resulting isomorphism is canonical because its formula on original pure tensors is the quotient map, independent of those auxiliary choices. No original tensor is replaced by the complement in its pairings or action.

GFT8 now gives the entire kernel congruence of the lifted tensor quotient:

$$(v, \lambda) \sim (w, \mu) \iff \lambda = \mu \quad \text{and} \quad v - w \in N. \quad (\text{GFT10})$$

Indeed two images in the constrained carrier are equal exactly when their labels and their quotient amplitudes agree. This is also the congruence generated by tensoring the individual same-label

quotients: GFT3's universal property forces precisely the ordinary multilinear quotient relations on the top additive group and leaves every lower label intact. In particular a vector in N with top label maps to e , not τ . This congruence, rather than the preimage of τ alone, is the relevant quotient.

For the original global adelic construction retain

$$C = \mathbb{A}_{\mathbb{Q}}^{\times}/\mathbb{Q}^{\times}, \quad B = \bigcap_{\beta \in \mathbb{R}} |\cdot|^{\beta} \mathcal{S}(C),$$

$$V = \left\{ x \mapsto \sum_{r \in \mathbb{Q}^{\times}} \xi(rx) : \xi \in \mathcal{S}(\mathbb{A}_{\mathbb{Q}}), \xi(0) = 0, \int \xi = 0 \right\}, \quad W = \overline{V} \subset B, \quad H = B/W.$$

(GFT11)

The closure is in the original weighted Bruhat topology SWQ3; all tensor products here remain algebraic. GFT9–10 apply to the whole B , including every compact-unit level, with $N_k = \sum_i B^{\otimes(i-1)} \otimes W \otimes B^{\otimes(k-i)}$. Thus the entire coefficient-level tensor power over S is exactly

$$X(H)^{\otimes sk} \cong X(H^{\otimes ck}) \cong X(B^{\otimes ck})/\sim_{N_k}. \quad (\text{GFT12})$$

This proves the support and amplitude map for every tensor power of the actual quotient. It asserts no unproved topological Kunneth theorem and does not replace W by a spectral kernel.

79.3 Null differentials before taking cohomology

For any original cochain complex $(C^{\bullet}, d)$ of complex vector spaces, retain its exact differential on each grade. Its lift obeys

$$(d^{\dagger})^2(v, \lambda) = (0, \lambda) = e(v, \lambda). \quad (\text{GFT13})$$

For a finite tensor of such graded complexes the amplitude differential on a homogeneous original tensor is

$$D(v_1 \otimes \cdots \otimes v_k) = \sum_{i=1}^k (-1)^{n_1 + \cdots + n_{i-1}} v_1 \otimes \cdots \otimes dv_i \otimes \cdots \otimes v_k, \quad v_i \in C_i^{n_i}. \quad (\text{GFT14})$$

For $i < j$ the two orders of differentiating positions i, j have opposite signs, since the first order increases the degree preceding j by one. Terms differentiating one position twice vanish by the original $d^2 = 0$. Hence $D^2 = 0$ on the entire algebraic graded tensor. GFT8 transports its lift to the actual S tensor of each homogeneous summand, preserving those signs in amplitudes and the common meet label. In a total degree we use the specified original synchronized carrier $X(\bigoplus_{\sum n_i = n} \bigotimes_i C_i^{n_i})$. It is not silently identified with a semimodule coproduct whose summands have independent labels. The repeated sum of the one assigned label is itself. Thus the lifted equation is $(D^{\dagger})^2 = e$, not a map constant at τ . In each total grade, original amplitude cycles with the same-label boundary congruence give exactly

$$X(\ker D)/\sim_{\text{im } D} \cong X(\ker D/\text{im } D). \quad (\text{GFT15})$$

This follows by the same equality-of-images proof as GFT10. It identifies the actual null complex and its cohomology map; no nilpotent with nonzero square has been called a differential.

For precision the exact comparison with independent labels, for any finite nonempty family of graded summands A_j , is

$$P : \bigoplus_j X(A_j) \longrightarrow X\left(\bigoplus_j A_j\right), \quad ((a_j, \lambda_j))_j \longmapsto ((a_j)_j, \bigvee_j \lambda_j),$$

$$I : X\left(\bigoplus_j A_j\right) \longrightarrow \bigoplus_j X(A_j), \quad ((a_j)_j, \lambda) \longmapsto ((a_j, \lambda))_j, \quad PI = \text{id}.$$

(GFT15a)

All image points are in their constrained carriers: a nonzero amplitude on either side forces the common label or at least one input label to be top. Addition and scalar action commute with P, I by finite distributivity. The map IP retains every amplitude and replaces each input label by the displayed join. The fibre of P consists exactly of all original label tuples with that join and the unchanged amplitudes, subject to the original nonzero-amplitude constraints. Thus P is generally not injective. The independent-label object and its entire fibre are retained in this comparison. GFT13–15 concern the original common-label total complex; they do not claim that X preserves arbitrary coproducts. For an unbounded complex the synchronized algebraic direct sum still makes sense, but GFT15a is asserted only on each stated finite family, so no infinite join in L has been assumed.

79.4 The whole tensor pairing and its exact radical

Let Q be the actual continuous Hermitian Weil pairing on B , with its full formula GTK13,24. SWQ proves W is contained in its radical, so it gives the well-defined form Q_H on H . Write $R = \text{rad } Q_H$, retaining this subspace rather than assuming it is zero. On the full algebraic tensor define

$$Q_H^{[k]} \left(\bigotimes_i u_i, \bigotimes_i v_i \right) = \prod_{i=1}^k Q_H(u_i, v_i), \quad Q_H^{[k]} \left(\sum_a \bigotimes_i u_{a,i}, \sum_b \bigotimes_i v_{b,i} \right) = \sum_{a,b} \prod_i Q_H(u_{a,i}, v_{b,i}). \quad (\text{GFT16})$$

The multilinear tensor universal properties, with conjugate linearity in the second variables, prove that these formulas are independent of every presentation. Every factor is the entire original global pairing, so the formula retains all cross terms from its primes, endpoints and real place; none is replaced by a finite window or by a completed zeta working object.

Its radical is exactly

$$\text{rad } Q_H^{[k]} = \sum_i H^{\otimes(i-1)} \otimes R \otimes H^{\otimes(k-i)}. \quad (\text{GFT17})$$

Inclusion of the right side is immediate from GFT16. For the reverse inclusion pass by GFT9 to $(H/R)^{\otimes k}$, where the original induced pairing is nondegenerate. For any finite linearly independent list $u_1, \dots, u_m$ in H/R , its pairing functionals on the second variable are independent: a linear relation would put a nonzero combination of the u_i in the radical. Therefore there exist $v_1, \dots, v_m$ realizing the coordinate dual values $Q_H(u_i, v_j) = \delta_{ij}$. This follows also directly by considering the range in $\mathbb{C}^m$; a proper range would have a nonzero annihilating functional. Expand a nonzero algebraic tensor using independent finite lists in each factor. Pairing with the corresponding products of these dual vectors selects each coefficient. Some coefficient is nonzero, proving nondegeneracy of the tensor pairing on the quotient and hence GFT17. The argument applies to the entire infinite-dimensional H , since every algebraic tensor has a finite presentation.

The full supported form is

$$Q_{H,L}^{[k]}((u, \lambda), (v, \mu)) = (Q_H^{[k]}(u, v), \lambda \wedge \mu). \quad (\text{GFT18})$$

A nonzero output amplitude requires both top input labels; linearity, conjugate linearity and finite lattice distributivity prove the semimodule pairing identities. In particular every top-supported tensor in the radical pairs to e with a top-supported test. GFT17 specifies precisely which amplitudes are invisible to the whole tensor trace; their support labels remain in GFT18.

79.5 Exact arithmetic and support receiving maps

There are two different scalar maps, both kept explicitly:

$$\rho : S \rightarrow \mathbb{C}, \quad (a, \lambda) \mapsto a, \quad \sigma : S \rightarrow L, \quad (a, \lambda) \mapsto \lambda. \quad (\text{GFT19})$$

They give the global base changes

$$\mathbb{C} \otimes_S X(V) \cong V, \quad c \otimes (v, \lambda) \mapsto cv; \quad L \otimes_S X(V) \cong L, \quad \eta \otimes (v, \lambda) \mapsto \eta \wedge \lambda. \quad (\text{GFT20})$$

For the first, the inverse is $v \mapsto 1 \otimes (v, 1_L)$. Every lower zero tensor vanishes because $\rho(z_\lambda) = 0$; the vector-space relations on top inputs follow from the balanced tensor relations. These facts verify both composites on generators. For the second, the inverse is $\eta \mapsto \eta \otimes (0, 1_L)$. Since $\sigma(e) = 1_L$, balancing gives $\eta \otimes (v, 1_L) = \eta \otimes e(v, 1_L) = \eta \otimes (0, 1_L)$. A lower label reduces similarly by balancing z_λ , giving the stated meet. This verifies both composites without discarding any label of L .

Thus the arithmetic receiver of GFT12 is the entire original $H^{\otimes k}$, while the support receiver is the entire original L . The latter map has explicitly lost every amplitude, not just the value of one function. Its fibres and a faithful joint record are

$$X(V) \hookrightarrow V \times L, \quad (v, \lambda) \mapsto (v, \lambda), \quad \text{im} = (V \times \{1_L\}) \cup (\{0\} \times L). \quad (\text{GFT21})$$

The image condition is part of the map. The projection to L has a copy of V over 1_L and a single zero amplitude over each other label. The projection to V has a copy of the whole L over zero and one top label over each nonzero amplitude. Consequently the joint record is injective, whereas either projection alone need not be. For any actual support stalk map $L \rightarrow L_J$, composition in GFT20 is exactly $(v, \lambda) \mapsto [\lambda]_J$; it commutes with GFT8 by the retained meet. This is the full-lattice support map, not a replacement of the original prime spectrum by one extra point.

The obtained arithmetic tensor powers, their null differentials, their full pairing and its radical are therefore literal objects over the original coefficient semiring. A weight assertion must be proved for the retained arithmetic action on these objects. Neither the support base change GFT20 nor the algebraic tensor isomorphism GFT8 imposes a zero-location or positive-metric assertion. They specify the exact maps on which that further arithmetic calculation operates.

80 The original dilation, its tensor powers, and the exact graded receiver

Human sources and actual scope. Alain Connes, Caterina Consani and Matilde Marcolli, *The Weil proof and the geometry of the adèles class space*, original author Yuri70v2.tex, SHA-256 b1f694cb934dd4fef829287f1a55dd8de21c67a481e552453c2f7e8fe0ae6fb5, defines the original action, image and trace at actCKAKbis, actCKCK, psigactH1, rangecVdef, varthetafconvol, varthetaTrH1, vanishV, and tracepairing. Source lines 1551–1836 were read directly for this calculation. Its original full trace and sharp were already read in WQP1–24. SWQ1–28 supplies the proved original strong topology, convolution ideal, full same-label quotient and sharp, all read here. The original theta test and complete global kernel were read in GTK1–12 and GTK22–29; their actual antecedent and two Gamma summands were also read in ETC1–26. The human trace theorem is retained as an analytic input, with all zeros and multiplicities. No finite-dimensional model is substituted for the original cohomological quotient. All tensor products below are explicitly algebraic over $\mathbb{C}$.

80.1 The actual quotient and the exact dilation action

Retain the original objects

$$\begin{aligned} C &= \mathbb{A}_{\mathbb{Q}}^{\times}/\mathbb{Q}^{\times} = \mathbb{R}_{>0} \times K, \quad K = \widehat{\mathbb{Z}}^{\times}, \quad d^{\times}(x, u) = dx/x du, \quad \int_K du = 1, \\ B &= \bigcap_{\beta \in \mathbb{R}} |\cdot|^{\beta} \mathcal{S}(C), \quad (f * g)(z) = \int_C f(y)g(y^{-1}z)d^{\times}y, \quad f^{\sharp}(z) = |z|^{-1}\overline{f(z^{-1})}, \quad (\text{GDT1}) \\ V &= E\mathcal{S}(\mathbb{A}_{\mathbb{Q}})_0, \quad (E\xi)(z) = \sum_{r \in \mathbb{Q}^{\times}} \xi(rz), \quad W = \overline{V}, \quad H = B/W. \end{aligned}$$

The two original defining conditions on the adelic source are $\xi(0) = 0$ and $\int_{\mathbb{A}} \xi = 0$. The closure is the weighted Bruhat closure of SWQ3: at a fixed finite compact level its seminorms are

$$p_{N,m,j}(f) = \sup_{v,u} e^{N|v|} (1 + |v|)^m |\partial_v^j f(e^v, u)|.$$

SWQ4–15 proves that convolution and sharp are continuous and that W is a sharp-invariant convolution ideal. The original trace and Hermitian pairing therefore descend:

$$\mathcal{T}(f) = \text{Tr}(\varrho_m(f) | H^1), \quad Q([f], [g]) = \mathcal{T}(f * g^{\sharp}). \quad (\text{GDT2})$$

In particular $\mathcal{T}(W) = 0$ and $Q(W, B) = 0$. This notation does not assume that Q is positive or nondegenerate.

For every real $a > 0$ define the original dilation

$$R_a f(x, u) = f(x/a, u), \quad \sigma_a \xi(x_{\infty}, x_f) = \xi(x_{\infty}/a, x_f). \quad (\text{GDT3})$$

The real additive Jacobian is a : $\sigma_a \xi(0) = \xi(0)$ and $\int_{\mathbb{A}} \sigma_a \xi = a \int_{\mathbb{A}} \xi$. Thus σ_a preserves both source zero conditions and $E\sigma_a \xi = R_a E\xi$, by the original absolutely convergent rational sum. It follows that $R_a V = V$. The explicit seminorm inequality

$$p_{N,m,j}(R_a f) \leq e^{N|\log a|} (1 + |\log a|)^m p_{N,m,j}(f)$$

follows by shifting v to $v - \log a$. Its inverse is $R_{1/a}$, so $R_a W = W$ and the same original action descends to H . It is strongly smooth; differentiating the shifted function gives its exact generator

$$Df = -x\partial_x f, \quad R_{et} = \exp(tD), \quad M(Df)(s) = sMf(s), \quad M(R_a f)(s) = a^s Mf(s). \quad (\text{GDT4})$$

The exponential here means the already defined strongly smooth group, not convergence of a power series on all of B . The Mellin equations follow by integration by parts and change of variables, with boundary terms zero by the weighted seminorms. Closedness and R_a -invariance of W make the difference quotients for D remain in W , so D descends too.

Directly in the original functions,

$$\begin{aligned} (R_a f)^\sharp &= a R_{1/a}(f^\sharp), \\ (R_a f) * (R_b g) &= R_{ab}(f * g), \\ Q(R_a f, R_b g) &= b \mathcal{T} \left(R_{a/b}(f * g^\sharp) \right), \\ Q(R_a f, R_a g) &= a Q(f, g), \quad Q(R_a f, g) = Q(f, a R_{1/a} g). \end{aligned} \tag{GDT5}$$

For sharp, $(R_a f)^\sharp(x, u) = x^{-1} \overline{f(1/(ax), u^{-1})}$, whereas $R_{1/a} f^\sharp(x, u)$ has the additional factor a^{-1} . For convolution put the integration variable $y = az$; the multiplicative Jacobian is one and the second argument becomes $x/(abz)$. These two identities prove the third. The last equation follows by putting $b = 1/a$ in the third with the first input undilated and retaining the real scalar a . Thus $a R_{1/a}$ is an exact adjoint for R_a under this possibly degenerate form. Differentiating the similitude identity gives

$$Q(Df, g) + Q(f, Dg) = Q(f, g), \quad D^\dagger = 1 - D. \tag{GDT6}$$

An adjoint is meant by its displayed equality, not a claim of uniqueness when the form has a radical.

80.2 Every term of the original global kernel after dilation

Let $h(x) = 2 \sum_{n \geq 1} \eta(nx)$, where $\eta(x) = (\pi^2 x^4 - 3\pi x^2/2) e^{-\pi x^2}$ is the actual adelic-image test of ETC and GTK. Put $h_T = x^{iT} h$ and $H_{T,S} = h_T * h_S^\sharp$, with original Mellin transform

$$M(s) = Mh(s) = 2\zeta(s) \left[\frac{\pi^2}{2\pi^{(s+4)/2}} \Gamma((s+4)/2) - \frac{3\pi}{4\pi^{(s+2)/2}} \Gamma((s+2)/2) \right]. \tag{GDT7}$$

The entire integral of GTK6 defines its continuation at every removable point; no separate Gamma summand is declared entire. For each finite compact character χ use its primitive character χ_* , conductor c_χ and parity ϵ_χ . Define $F_{\chi,T}(x, u) = h_T(x) \chi(u)$. The complete dilated cross-kernel is

$$\begin{aligned} Q(R_a F_{\chi,T}, R_b F_{\psi,S}) &= \delta_{\chi,\psi} \left\{ 1_{\chi=1} \left[b M(iT) \overline{M(1+iS)} + a M(1+iT) \overline{M(iS)} \right] \right. \\ &\quad + \left[\sum_{p|c_\chi} v_p(c_\chi) \log p - \log |\mathfrak{a}| \right] b H_{T,S}(b/a) - \mathcal{A}_{\epsilon_\chi}(b R_{a/b} H_{T,S}) \\ &\quad \left. - \sum_{p \nmid c_\chi} \log p \sum_{j \geq 1} \left[\chi_*(p)^j b H_{T,S}((b/a)p^j) + p^{-j} \chi_*(p)^{-j} b H_{T,S}((b/a)p^{-j}) \right] \right\}. \end{aligned} \tag{GDT8}$$

The differential idele is retained with $|\mathfrak{a}| = 1$ for the specified rational reference characters. Here the exact original real-place functional is

$$\mathcal{A}_\epsilon(F) = (\log(2\pi) + \gamma) F(1) + \frac{1}{2} \int_0^1 \left[\frac{F(y) + y^{-1} F(y^{-1}) - 2F(1)}{1-y} + (-1)^\epsilon \frac{F(y) + y^{-1} F(y^{-1})}{1+y} \right] dy.$$

It keeps the additive character $e^{-2\pi i x}$ and real multiplicative measure $du/(2|u|)$ of the source. GDT8 is obtained by substituting the exact correlation $b R_{a/b} H_{T,S}$ from GDT5 into GTK24. Its Mellin endpoints are multiplied by b and a , respectively; its value at one is $b H_{T,S}(b/a)$. These substitutions prove all displayed factors. Convergence follows from the original strong continuity and the all-prime estimates of GTK14–17 or GWM6. Thus GDT5's similitude also follows term by term when $a = b$: every original global contribution is multiplied by a , including both endpoints and the archimedean term.

80.3 The actual tensor action and pairing

For every integer $k \geq 1$, form the entire algebraic tensor $H^{\otimes k}$ and its action and pairing

$$R_a^{[k]} = R_a^{\otimes k}, \quad D^{[k]} = \sum_{i=1}^k 1^{\otimes(i-1)} \otimes D \otimes 1^{\otimes(k-i)}, \quad (\text{GDT9})$$

$$Q_k(v_1 \otimes \cdots \otimes v_k, w_1 \otimes \cdots \otimes w_k) = \prod_{i=1}^k Q(v_i, w_i).$$

Multilinearity in the first tensor and conjugate multilinearity in the second prove well-definedness by the algebraic tensor relations. Extend to finite sums. No completion, dimension bound or positive inner product is needed to define these objects. Every factor in GDT5 remains:

$$Q_k(R_a^{[k]}v, R_a^{[k]}w) = a^k Q_k(v, w),$$

$$(R_a^{[k]})^\dagger = a^k R_{1/a}^{[k]}, \quad (D^{[k]})^\dagger = k - D^{[k]}. \quad (\text{GDT10})$$

For tensor products of GTK generators their exact entire kernel is $\prod_{i=1}^k Q(R_{a_i} F_{\chi_i, T_i}, R_{b_i} F_{\psi_i, S_i})$, with each factor given by the full GDT8. Arbitrary finite mixed tensors are evaluated by its sesquilinear extension. This preserves all tuples of characters, frequencies and zeros; no diagonal tuple is selected in GDT9.

80.4 Actual Mellin jets and explicit primary chains

Let B_0 be the trivial compact-character sector and $H_0 = B_0/(W \cap B_0)$. For each nontrivial zero ρ of the original ζ , write m_ρ for its actual multiplicity and define

$$J_\rho : H_0 \longrightarrow A_\rho = \mathbb{C}[t]/(t^{m_\rho}), \quad J_\rho([f]) = \sum_{j=0}^{m_\rho-1} \frac{(Mf)^{(j)}(\rho)}{j!} t^j. \quad (\text{GDT11})$$

Here is a direct check that this is a map on the actual quotient. Average the finite components of any original adelic ξ over K . Its finite-place part becomes a finite linear combination of products $\prod_p 1_{p^{k_p} \mathbb{Z}_p}$: at each of the finitely many nonstandard primes a compact, locally constant unit-invariant function is a finite sum of ball indicators, by expressing its finitely many annuli as differences of successive balls. The archimedean factors are Schwartz. The original rational reduction of each such term is a sum over $n \in \mathbb{Z} \setminus \{0\}$ at arguments $N(k)nx$. Combining the two signs, its Mellin integral for $\Re s > 1$ is $\zeta(s)$ times a finite sum of integrals $N(k)^{-s} \int_0^\infty [\phi(x) + \phi(-x)] x^{s-1} dx$. Those integrals are holomorphic for $\Re s > 0$ by Schwartz decay at infinity and boundedness at zero. The initial interchange is absolute, since it is bounded by $\sum n^{-\Re s}$ times these absolute integrals. Thus the trivial-sector reduction has a factor $\zeta(s)$ with holomorphic complementary factor at every such ρ . For $\xi \in \mathcal{S}(\mathbb{A})_0$ the reduction is in B and its Mellin transform is entire, as proved in SWQ8. Continuation therefore gives vanishing of its first m_ρ Taylor coefficients at ρ . These jet functionals are continuous in the original weighted topology, by their absolutely convergent integrals with powers of $\log x$, so they vanish on W as well. This proves representative independence. Convolution's exact Mellin product identity proves that J_ρ is an algebra homomorphism.

It is onto, with an explicit finite right inverse. Choose any nonnegative $w \in C_c^\infty(\mathbb{R})$ positive on a nonempty interval and put $f_j(x) = w(\log x) x^{-\rho} (\log x)^j$ for $0 \leq j < m_\rho$. Set $G_{ij} = \int_{\mathbb{R}} w(v) v^{i+j} dv$. For nonzero coefficients c_j , $\sum c_i \bar{c}_j G_{ij} = \int w(v) |\sum c_i v^i|^2 dv > 0$, so G is invertible. The i th Taylor coefficient of Mf_j at ρ is $G_{ij}/i!$. Therefore, for desired coefficients d_i , the original compact function $\sum_j [G^{-1}(i!d_i)]_j f_j$ maps to $\sum_i d_i t^i$. Its class supplies the required right inverse without

an assumed local spectral model. The exact actions on this receiver are

$$J_\rho D = (\rho + t)J_\rho, \quad J_\rho R_a = a^\rho \sum_{j=0}^{m_\rho-1} \frac{(\log a)^j t^j}{j!} J_\rho. \quad (\text{GDT12})$$

All nilpotent coefficients are retained.

There are also actual primary vectors in H_0 . Put $k_{\rho,0} = h$, the original positive theta-image function, and for $1 \leq r \leq m_\rho$ define

$$k_{\rho,r}(x) = \frac{x^{-\rho}}{(r-1)!} \int_x^\infty y^{\rho-1} (\log(y/x))^{r-1} h(y) dy, \quad (\text{GDT13})$$

$$(D - \rho)k_{\rho,r} = k_{\rho,r-1}, \quad Mk_{\rho,r}(s) = \frac{M(s)}{(s - \rho)^r}.$$

For $r = 1$, the vanishing $Mh(\rho) = 0$ changes the integral into $-x^{-\rho} \int_0^x y^{\rho-1} h(y) dy$ near zero. At infinity the displayed upper integral has arbitrarily rapid decay. For any desired power at either endpoint, bound h by a higher power using its original seminorms; integrating the result proves the desired estimate for $k_{\rho,1}$. The differential equation follows by differentiating the integral and proves the same bounds for every logarithmic derivative. Integration by parts gives the Mellin equation. Its right hand side is entire because M vanishes at ρ . Repeat this argument: at step $r \leq m_\rho$ the previous transform still vanishes at ρ . Iterating the upper integrals, justified by their absolute decay and Fubini, gives the displayed factor $(r-1)!$ and logarithmic power. This proves membership in B_0 and all equations for every stated r .

Write $u_{\rho,r} = [k_{\rho,r}] \in H_0$. Since $h \in V$, $u_{\rho,0} = 0$. If $M(\rho + t) = c_\rho t^{m_\rho} + O(t^{m_\rho+1})$, then $c_\rho \neq 0$ by the exact multiplier in GDT7 and GTK8 on the open critical strip. The jet of $k_{\rho,r}$ begins with $c_\rho t^{m_\rho-r}$. With every original multiplier retained, its value is

$$c_\rho = 2 \left[\frac{\pi^2}{2\pi(\rho+4)/2} \Gamma((\rho+4)/2) - \frac{3\pi}{4\pi(\rho+2)/2} \Gamma((\rho+2)/2) \right] \frac{\zeta^{(m_\rho)}(\rho)}{m_\rho!}.$$

These different initial orders prove that $u_{\rho,1}, \dots, u_{\rho,m_\rho}$ are linearly independent. They span an actual length- m_ρ Jordan block, with

$$(D - \rho)u_{\rho,r} = u_{\rho,r-1}, \quad R_a u_{\rho,r} = a^\rho \sum_{j=0}^{r-1} \frac{(\log a)^j}{j!} u_{\rho,r-j}. \quad (\text{GDT14})$$

To justify the latter without assuming a series on B , differentiate $R_{e^t} u_{\rho,r}$ in the quotient and use the finite chain. Multiplying by $e^{-\rho t}$ and successively subtracting the displayed finite polynomial leaves a vector-valued function with zero derivative and zero initial value. Applying continuous linear functionals, which separate points of the Hausdorff locally convex quotient, proves it is zero. This constructs actual finite primary subspaces; it does not declare that their algebraic direct sum is the whole infinite topological quotient.

The dual zero is $\rho^\vee = 1 - \bar{\rho}$. GDT7 and the original theta identities $M(s) = M(1 - s)$, $\overline{M(s)} = M(\bar{s})$ show $m_{\rho^\vee} = m_\rho$. The literal sharp acts on the full jet by

$$J_\rho(f^\sharp)(t) = \overline{J_{\rho^\vee}(f)(-\bar{t})}. \quad (\text{GDT15})$$

Thus every order- j coefficient has its exact factor $(-1)^j$. The cited full spectral trace, on these actual functions, gives

$$Q(u_{\rho,r}, u_{\sigma,\ell}) = \begin{cases} m_\rho c_\rho \bar{c}_\sigma, & \sigma = \rho^\vee, \quad r = m_\rho, \quad \ell = m_\sigma, \\ 0, & \text{otherwise.} \end{cases} \quad (\text{GDT16})$$

Indeed $Mk_{\rho,r}$ vanishes at every nontrivial zero other than ρ and also at ρ for $r < m_\rho$. At $r = m_\rho$ its value there is c_ρ . Substitution into $Q(f, g) = \sum_\tau m_\tau Mf(\tau) \overline{Mg(1 - \bar{\tau})}$ proves GDT16; its convergence is the original trace theorem, or GTK29's rapid vertical bound and zero count. In particular each lower chain vector is in the global radical of Q , although it is nonzero in H_0 by its retained jet. The trace has not erased that actual nilpotent structure in the source.

These vectors also give an exact continuous primary receiver with a section. Let U_ρ denote their m_ρ -dimensional span. The triangular jet calculation proves that $J_\rho|_{U_\rho}$ is an isomorphism onto A_ρ . Its inverse followed by J_ρ defines a continuous projection $P_\rho : H_0 \rightarrow U_\rho$. Both the jet kernel and this chain space are D -invariant by GDT12–14, and their actions agree under the restriction. Hence P_ρ commutes with D and R_a , and $H_0 = U_\rho \oplus \ker J_\rho$ as a continuous representation. This finite splitting keeps its entire second summand; it is not an exhaustion claim about the quotient.

Moreover the full radical on the original trivial sector has the exact description

$$\text{rad}(Q|_{H_0}) = \{[f] \in H_0 : Mf(\rho) = 0 \text{ for every nontrivial zero } \rho\}. \quad (\text{GDT16a})$$

One inclusion follows by inserting the vanishing values into the complete spectral trace. For the other, pair $[f]$ with $u_{\rho^\vee, m_\rho}$ in GDT16: the result is $m_\rho Mf(\rho) \overline{c_{\rho^\vee}}$, whose last factor is nonzero. Thus membership in the radical forces every displayed value to vanish. In a fixed primary block the top pairing is $m_\rho |c_\rho|^2$ when $\rho = \rho^\vee$. On each two-element orbit $\{\rho, \rho^\vee\}$ the top matrix is $\begin{pmatrix} 0 & c \\ \bar{c} & 0 \end{pmatrix}$, $c = m_\rho c_\rho \overline{c_{\rho^\vee}} \neq 0$; its eigenvalues are $|c|$ and $-|c|$. The remaining chain vectors are the nonzero radical directions already computed.

On a tensor of these finite primary spaces, the generator is exactly

$$D^{[k]} = (\rho_1 + \cdots + \rho_k)1 + \sum_{i=1}^k N_i, \quad N_i = (D - \rho_i) \text{ in factor } i. \quad (\text{GDT17})$$

Its nilpotent index is $1 + \sum_i (m_{\rho_i} - 1)$. The upper bound follows because each term in the multinomial expansion of a higher power has some exponent at least m_{ρ_i} . At power $\sum_i (m_{\rho_i} - 1)$ on the tensor of the top chain vectors the sole surviving term has coefficient $[\sum_i (m_{\rho_i} - 1)]! / \prod_i (m_{\rho_i} - 1)!$ and gives $\bigotimes_i u_{\rho_i, 1} \neq 0$. This proves equality, with its coefficient. GDT16 and GDT9 retain the full tensor duality; the dual weight is $k - \overline{\rho_1 + \cdots + \rho_k}$, with each individual partner and every nilpotent order still recorded.

80.5 Convolution, its clock defect, and the exact graded repair

The k -fold original convolution defines

$$m_k : H^{\otimes k} \longrightarrow H, \quad [f_1] \otimes \cdots \otimes [f_k] \longmapsto [f_1 * \cdots * f_k]. \quad (\text{GDT18})$$

Changing any representative by $w \in W$ changes the product by a convolution containing a factor in the ideal W ; expanding a simultaneous change gives a sum of such terms. This proves the map. For the full independent dilation tuple,

$$\begin{aligned} m_k(R_{a_1} \otimes \cdots \otimes R_{a_k}) &= R_{a_1 \cdots a_k} m_k, \\ m_k R_a^{[k]} &= R_{a^k} m_k, \quad m_k D_i = D m_k, \quad m_k D^{[k]} = k D m_k, \\ m_k R_a^{[k]} - R_a m_k &= (R_{a^k} - R_a) m_k. \end{aligned} \quad (\text{GDT19})$$

These identities follow directly from the original convolution changes of variables, before passage to the quotient. They show precisely which clock it receives. For every $a > 1$, $k \geq 2$, and any nontrivial zero ρ , this last defect is nonzero on $u_{\rho, m_\rho}^{\otimes k}$: its constant jet is $(a^{k\rho} - a^\rho) c_\rho^k \neq 0$. Here $\Re \rho > 0$ makes the two absolute values different, and $c_\rho \neq 0$. This is a concrete obstruction to same-clock equivariance.

The exact receiving algebra that restores equivariance is

$$\begin{aligned} \mathcal{A} &= \bigoplus_{k \geq 1} H[k], & H[k] \text{ a specified copy of } H, & & \alpha_a|_{H[k]} = R_{a^k}, \\ H[r] \times H[s] &\longrightarrow H[r+s], & (v, w) &\longmapsto v * w, & \tilde{m}_k : H^{\otimes k} \longrightarrow H[k]. \end{aligned} \quad (\text{GDT20})$$

Only finite graded sums occur. Associativity and commutativity follow from original convolution. There is no inserted unit. The independent-clock identity GDT19 gives $(R_{a^r}v) * (R_{a^s}w) = R_{a^{r+s}}(v * w)$, so α_a is a graded algebra automorphism and $\tilde{m}_k$ is equivariant for the original diagonal tensor action. The exact sequence $0 \rightarrow \ker m_k \rightarrow H^{\otimes k} \rightarrow \text{im } m_k \rightarrow 0$ retains its kernel. More strongly the map $v \mapsto (v, \tilde{m}_k v)$ into $H^{\otimes k} \times H[k]$ is injective with left inverse the first projection, so it retains the entire original tensor along with the receiving value. On the homogeneous grade k , the original pairing has factor a^k :

$$Q(\alpha_a v, \alpha_a w) = a^k Q(v, w).$$

On the full direct sum, the grade-orthogonal pairing is the sum of the grade pairings, whose terms have their individual factors a^k .

The exact spectral information lost by m_k is also computable. For any actual finite primary vectors v_i satisfying $(D - \rho_i)^{n_i} v_i = 0$, their product is annihilated by each polynomial $(D - \rho_i)^{n_i}$, by GDT19. If two ρ_i differ, these two polynomials are coprime. Their Euclidean algorithm gives polynomial coefficients with sum one, and applying that identity to the product proves

$$v_1 * \cdots * v_k = 0 \quad \text{whenever two primary weights differ.} \quad (\text{GDT21})$$

For equal ρ the product is annihilated by $(D - \rho)^{\min n_i}$. In the trivial compact sector its exact retained jet is

$$J_\rho(v_1 * \cdots * v_k) = \prod_i J_\rho(v_i) \pmod{t^{m_\rho}}. \quad (\text{GDT22})$$

The multiplication here includes every finite Taylor coefficient, with truncation only at its specified original multiplicity. Moreover $N(v * w) = (Nv) * w = v * (Nw)$ on that primary part, rather than the sum of these two terms. Consequently the tensor nilpotent of GDT17 maps to kN in the accelerated-clock grade. The jet receiver is surjective and has the actual chains above; its kernel is retained as $\ker J_\rho$. No unsupported identification of the entire quotient with one jet algebra has been used.

The tensor pairing is not the pulled-back grade pairing. In the original formulas they are respectively

$$\begin{aligned} Q_k(\otimes_i f_i, \otimes_i g_i) &= \prod_i \mathcal{T}(f_i * g_i^\sharp), \\ Q(m_k(\otimes_i f_i), m_k(\otimes_i g_i)) &= \mathcal{T}\left(*_{i=1}^k (f_i * g_i^\sharp)\right). \end{aligned} \quad (\text{GDT23})$$

In the trivial sector their absolutely convergent spectral expressions sum, in the first line, over all tuples $(\rho_1, \dots, \rho_k)$ with factor $\prod_i m_{\rho_i}$; the second sums only over one common ρ with factor m_ρ . Both retain the original products of Mellin values at ρ_i and their partners $1 - \bar{\rho}_i$. Even the retained diagonal tuples therefore have different multiplicity factors. These exact formulas show why replacing the full tensor by its convolution value would lose information.

80.6 Pointwise multiplication and its dense obstruction

Pointwise multiplication on B has precisely the diagonal clock: $(R_a f)(R_a g) = R_a(fg)$. The same is true for every k -fold pointwise product, and it preserves B by Leibniz's formula and its weighted seminorms. It nevertheless does not define a product on the actual quotient H . Here is

an explicit original-source witness. For a nontrivial zero ρ , choose $w \in C_c^\infty(\mathbb{R})$ with $w \geq 0$ and $\int w = 1$, and let

$$\phi(x, u) = w(\log x)x^{-\rho}, \quad g(x, u) = \frac{\phi(x, u)}{h(x)}. \quad (\text{GDT24})$$

GTK4 proves that the original $h \in V$ is strictly positive at every $x > 0$. Thus g is a smooth compact test in B , and $hg = \phi$. But $M\phi(\rho) = 1$, whereas every member of W has vanishing Mellin value at ρ by GDT11. Hence $[hg] \neq 0$ although $[h] = 0$. This is a failure of representative independence, not merely a difference between two written products.

The strongest associated obstruction map is

$$\Omega : W \otimes_{\mathbb{C}} B \longrightarrow H, \quad w \otimes g \longmapsto [wg]. \quad (\text{GDT25})$$

It intertwines the diagonal original dilation. Its image contains the class of every compact Bruhat test F : the same exact antecedent $h \otimes (F/h)$ maps to $[F]$. Compact tests are dense in B . To verify this, multiply at a fixed compact level by $\chi((\log x)/R)$, with a smooth compact cutoff equal to one near zero. Leibniz's formula and a stronger exponential weight bound each seminorm of the discarded tail and cutoff derivatives by a quantity tending to zero. Thus the image of Ω is dense in H . In particular the closed pointwise ideal of B generated by W is all of B : it contains every compact test as just proved. Every continuous homomorphism from the pointwise algebra B to a Hausdorff topological algebra which kills V is therefore zero. Its closed kernel contains W , then every compact test, then all of B . The map GDT25 identifies exactly the defect of using the pointwise product as a quotient tensor receiver.

80.7 What the tensor calculations do and do not supply

The balanced mixed receiver is also an actual map. Let $Cf = \bar{f}$ be conjugation of the original amplitude. It preserves V because $\overline{E\xi} = E\bar{\xi}$ and preserves both source zero conditions; continuity then preserves W . Define

$$\begin{aligned} \beta : H_0 \otimes_{\mathbb{C}} H_0 &\longrightarrow \mathbb{C}, & \beta(f \otimes g) &= Q(f, Cg) = \sum_{\rho} m_{\rho} Mf(\rho) Mg(1 - \rho), \\ \beta(R_a f \otimes R_a g) &= a\beta(f \otimes g), & \beta(Df \otimes g) + \beta(f \otimes Dg) &= \beta(f \otimes g). \end{aligned} \quad (\text{GDT26})$$

Conjugate linearity in the second argument of Q and conjugate linearity of C make this bilinear; the displayed spectral expression follows from $M(Cg)(s) = \overline{Mg(\bar{s})}$. It is symmetric by reindexing $\rho \mapsto 1 - \rho$, which preserves the original multiplicities. The target has the explicitly specified action $z \mapsto az$; with that action β is an equivariant tensor receiver, without introducing a positive norm on the source.

For the actual chains its value is exactly

$$\beta(u_{\rho, r} \otimes u_{\sigma, \ell}) = \begin{cases} m_{\rho} c_{\rho} c_{\sigma}, & \sigma = 1 - \rho, \ r = m_{\rho}, \ \ell = m_{\sigma}, \\ 0, & \text{otherwise.} \end{cases} \quad (\text{GDT27})$$

Indeed conjugating the integral GDT13 gives $Ck_{\sigma, \ell} = k_{\bar{\sigma}, \ell}$, and GDT16 then gives the formula. The original theta symmetry gives $c_{1-\rho} = (-1)^{m_{\rho}} c_{\rho}$, so its top value is $m_{\rho}(-1)^{m_{\rho}} c_{\rho}^2 \neq 0$. These balanced weights are always distinct. To verify the only possible exception, the positive original integral $M(1/2) > 0$ together with the retained exact multiplier $s(s-1)\pi^{-s/2}\Gamma(s/2)/4$ at $s = 1/2$ gives $\zeta(1/2) < 0$. Thus $\rho = 1 - \rho$ cannot be an original nontrivial zero. GDT21 therefore kills every such balanced primary tensor under convolution, while GDT27 gives its nonzero bilinear contraction.

On the entire balanced primary block the generator is $1 + N_1 + N_2$, with the full nilpotent index from GDT17. The contraction annihilates $N_1 + N_2$ by GDT26; on these actual chains it annihilates each lower-chain direction separately by GDT27. This describes its exact loss

of nilpotent data, rather than replacing that block by its single output scalar. The combined receiving map $v \mapsto (m_2 v, \beta(v))$ has target $H[2] \oplus \mathbb{C}$, with action (R_{a^2}, a) . Appending v itself gives the faithful equivariant graph $v \mapsto (v, m_2 v, \beta(v))$. The first projection is its explicit left inverse, so all balanced mixed tensors and their nilpotents survive this full object.

On actual primary tensors the original dilation eigenfactor is $a^{\rho_1 + \dots + \rho_k}$, with the full nilpotent exponential of GDT17; their pairing has similitude factor a^k . The graded receiver GDT20 gives the same factor on surviving equal-primary tuples, with clock $a \mapsto a^k$. These are exact tensor mechanisms of the type that must be tracked in a comparison with Deligne's product amplification. They do not identify the tensor pairing with its convolution pushforward: GDT21–23 compute the difference, including its mixed-primary kernel and multiplicity factors.

Nor does a similitude exponent itself force the modulus $|a^\rho| = a^{1/2}$. For an explicit exact model, take any complex z with $\Re z \neq 1/2$ and the two-dimensional space with $R_a = \text{diag}(a^z, a^{1-\bar{z}})$ and Hermitian matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Its pairing satisfies GDT5 for every $a > 0$, and all its algebraic tensor powers satisfy GDT10, while its two moduli are different. This calculation is a counterexample to deducing purity from those action-and-pairing identities alone. It is not asserted to be a new zero of the original zeta. The actual primary chains and receiving maps remain GDT11–23, which preserve their arithmetic source, nilpotents and full global pairing. No finite cohomological dimension, positive norm, or uniform power estimate has been inserted into them.

80.8 Every original support label

For an arbitrary nontrivial bounded distributive lattice L , use the entire original carrier $G_L(U) = \{(0, \lambda) : \lambda \in L\} \cup (U \times \{1_L\})$ for every amplitude space in this chapter. The quotient is exactly the same-label congruence of SWQ17; it preserves every $(0, \lambda)$. Each displayed linear map, including R_a , D , J_ρ , m_k , $\tilde{m}_k$, the graded inclusion and projection, and Ω , lifts as $(v, \lambda) \mapsto (Fv, \lambda)$. Nonzero output requires nonzero input, whose label is top, so these lifts belong to their declared carriers. Linearity proves additivity with label join and the scalar rule with label meet. The multilinear original tensor map sends $((v_i, \lambda_i))_i$ to $(\bigotimes_i v_i, \bigwedge_i \lambda_i)$; the same meet occurs for the k -fold convolution before its linear receiving map. No identification with an independently labelled coordinate product is made.

The full tensor pairing is $Q_{k,L}((v, \lambda), (w, \mu)) = (Q_k(v, w), \lambda \wedge \mu)$. The linear contraction on the tensor amplitude has the exact lift $\beta_L(v, \lambda) = (\beta(v), \lambda)$; conjugation has the lift $(f, \lambda) \mapsto (\bar{f}, \lambda)$. Its nonzero balanced top value therefore keeps top support, and each vanished lower-chain value retains its original zero label. The combined tensor-convolution-contraction graph uses one common label on the full amplitude triple. Its factor a^k is represented by the nonzero top-supported scalar $(a^k, 1_L)$, so it leaves every support label unchanged. The graph retaining the whole tensor and its grade receiver uses the single common label on that paired amplitude. For the pointwise obstruction, the two equivalent top-supported representatives $(h, 1_L)$ and $(0, 1_L)$ have outputs $([\phi], 1_L)$ and e after multiplying by $(g, 1_L)$ and projecting. The former has nonzero amplitude; the latter is $e \neq \tau$. Thus the proved failure of descent survives the full original support map. The tropical map $(v, \lambda) \mapsto (v, 1_{K_{\text{trop}}} \otimes \lambda)$ and its label retraction commute with every linear map above. Actual localization at a support prime J sends the retained label to $[\lambda]_J$; it forgets the amplitude as specified by the original localization, without changing the global maps just proved. No measure, positivity order or infinite join on L has been assumed.

81 The original global quotient, its resolvent, and the complete primary spectral receiver

Human sources and precisely read mathematical inputs. The original adelic image, closed quotient, and trace are from Alain Connes, Caterina Consani and Matilde Marcolli, *The Weil proof and the geometry of the adeles class space*, original author source Yuri70v2.tex, SHA-256 b1f694cb934dd4fef829287f1a55dd8de21c67a481e552453c2f7e8fe0ae6fb5. For this continuation the complete passages at lines 1499–1549, 1621–1681, and 1680–1831 were read, including moth1AKCK, the following closure convention, specrealH1mot, rangecVdef, varthetaH1V, varthetaTrH1, and vanishV. The explicit formula and sharp conventions are retained from GTK1–38, SWQ1–28, and WQP1–24.

Ralf Meyer, *On a representation of the idele class group related to primes and zeros of L-functions*, original author source original_author.tex, SHA-256 9453236959ee2fc232386a203637a12ca8b7f801469d86a6af1ccd7f21800f05, was read at lines 421–470, 1150–1244, 4150–4200, 4371–4631, and 4891–4926. In particular the exact locators are lem:Lap_weighted_Sch, the:difference_spectrum, sec:global_spectral_analysis, pro:quasi_nilpotent_character, and pro:split_off_quasi_nilpotent. The original source retains additional vertical growth conditions in its summation image and expressly does not settle whether its zero representation has a quasi-nilpotent part. A trace identity is therefore not used below as a proof that an entire quotient equals a Hilbert direct sum.

The preceding programme finite-jet proof WJ11–18 was read in full. The present construction differs in an essential respect: its cardinal functions vanish at *all* other original zeros, and its resolvent acts on the actual closed global quotient. All new maps and their topology are proved below. The original source nuclear trace theorem remains the stated human input.

81.1 Original radial algebra and an exact description of its adelic image

Let B_0 be the original radial strong weighted smooth algebra of GTK14, with

$$p_{A,m,j}(f) = \sup_{u \in \mathbb{R}} e^{A|u|} (1 + |u|)^m |\partial_u^j f(e^u)|.$$

Its measure and operations remain dy/y , multiplicative convolution, and $f^\sharp(y) = y^{-1} \overline{f(y^{-1})}$. Write $F(s) = Mf(s) = \int_0^\infty f(y) y^s dy/y$. The transform is entire, and on each bounded vertical strip it and every derivative decrease faster than every power of $|\Im s|$, by integration by parts in u . Conversely this uniform vertical-strip Schwartz condition gives an element of B_0 by Fourier inversion and contour translation. This is Meyer’s cited lem:Lap_weighted_Sch; here the forward estimates and every division used below are also proved directly in the original y variable.

Let V_0 be the compact-unit-invariant part of the actual image $V = ES(\mathbb{A}_0)_0$. Its exact radial description is

$$V_0 = \left\{ y \mapsto 2 \sum_{n \geq 1} \phi(ny) : \phi \in \mathcal{S}(\mathbb{R}), \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0 \right\}. \quad (\text{GSR1})$$

Here is a proof of both directions. Average the original adelic antecedent over the compact unit group of Haar mass one. This preserves its value at zero and its additive integral, since every unit dilation has additive modulus one. A finite-place Schwartz function invariant under units is a finite linear combination of characteristic balls $1_{p^k \mathbb{Z}_p}$: its valuation profile is zero below a lower bound, constant above an upper bound by local constancy at zero, and has only finitely many intervening values. Differences of successive ball indicators give that profile exactly. Tensor expansion at the finitely many exceptional primes consequently expresses the antecedent as a finite sum $\phi_j(x_\infty) \prod_p 1_{p^{k_j, p} \mathbb{Z}_p}(x_p)$. Replace ϕ_j by its even part, which leaves the full rational-sign reduction unchanged.

Put $N_j = \prod_p p^{k_{j,p}} > 0$. Its reduction is $2 \sum_{n \geq 1} \phi_j(N_j n y)$. Thus $\phi(x) = \sum_j \phi_j(N_j x)$ gives exactly GSR1. Its value at zero is the original adelic value at zero, and its integral is $\sum_j N_j^{-1} \int \phi_j$, exactly the original finite Haar factors. Both are zero. Conversely any ϕ in GSR1 gives the actual antecedent $\phi(x_\infty) \prod_p 1_{\mathbb{Z}_p}(x_p)$ with the two required zero moments, proving the converse.

Set

$$W_0 = \overline{V_0}^{B_0}, \quad \mathcal{C} = B_0/W_0, \quad q : B_0 \longrightarrow \mathcal{C}, \quad Af = -y \partial_y f. \quad (\text{GSR2})$$

The continuous compact-unit projection proves that $W_0 = B_0 \cap \overline{V}$ in the original whole Bruhat space: apply that projection to an approximating net. Thus $\mathcal{C}$ is the actual original-zeta sector of the closed quotient, not a new closure convention. It is a Hausdorff complete Fréchet space. By GSR1, A preserves V_0 : on the even antecedent it is $-x \phi'(x)$, whose value at zero is zero and whose integral is $\int \phi = 0$ by integration by parts. Termwise differentiation commutes with E by Schwartz bounds. Continuity of A therefore proves $A(W_0) \subset W_0$, giving the continuous operator $\bar{A}$ on $\mathcal{C}$.

Retain exactly the original theta-image test of GTK2,

$$\begin{aligned} \eta(x) &= \pi x^2 (\pi x^2 - \tfrac{3}{2}) e^{-\pi x^2}, & h(y) &= 2 \sum_{n \geq 1} \eta(ny) \in V_0, \\ M(s) &= Mh(s) = 2\zeta(s) \left[\frac{\pi^2}{2\pi^{(s+4)/2}} \Gamma((s+4)/2) - \frac{3\pi}{4\pi^{(s+2)/2}} \Gamma((s+2)/2) \right], & (\text{GSR3}) \\ M(0) &= M(1) = \tfrac{1}{4}, & M(s) &= M(1-s). \end{aligned}$$

Its entire interpretation, endpoint contributions, and the retained multiplier comparison are GTK6–9. The original zeta functional equation with that full multiplier, the Euler product on $\Re s > 1$, and the trivial-zero/pole germs in ZTC17–17a show that the zeros of M are precisely the original nontrivial zeta zeros, with the same multiplicities. Denote the distinct zeros by $\mathcal{Z}$, their original multiplicities by m_ρ , and put $\sigma(\rho) = 1 - \bar{\rho}$. Then $m_{\sigma(\rho)} = m_\rho$. The zero count $\sum_{|\rho| \leq R} m_\rho = O(R \log(R+2))$ is proved directly from this h by Jensen's formula in GTK29. Infinitude of the original nontrivial zero set is the classical zeta fact retained by the source spectral realization; no finite-height list replaces it here.

81.2 Division in the original test space and global cardinal primary tests

For $g \in B_0$ with $Mg(z) = 0$, define

$$\mathcal{D}_z g(y) = -y^{-z} \int_0^y x^{z-1} g(x) dx = y^{-z} \int_y^\infty x^{z-1} g(x) dx. \quad (\text{GSR4})$$

The equality is the actual vanishing Mellin moment, not the omission of an endpoint. Every integral is absolute. In $u = \log y$ coordinates, use the first integral at $u < 0$ and the second at $u > 0$. For any fixed compact set of z and any requested exponential weight, choose a larger weight in g ; integration of the resulting decaying exponential bounds both tails. Differentiating the formula gives $(A - z)\mathcal{D}_z g = g$, so its higher derivatives are bounded recursively by those of g and the zeroth-order estimate. Parameter derivatives insert powers of $u - v$ in the integrals, absorbed by a slightly larger exponential weight. These estimates prove $\mathcal{D}_z g \in B_0$, continuity on the zero-moment subspace, and holomorphic dependence for a holomorphic family g_z with $Mg_z(z) = 0$. Integration by parts, with the now-proved vanishing boundary terms, gives

$$M(\mathcal{D}_z g)(s) = \frac{Mg(s)}{s - z}, \quad (\text{GSR5})$$

including its removable value at $s = z$.

At $\rho \in \mathcal{Z}$, successive divisions are permitted exactly m_ρ times for h : the first quotient has a zero of order $m_\rho - 1$ there, and so on. Define $h_\rho = \mathcal{D}_\rho^{m_\rho} h$ and $u_\rho(t) = M(\rho + t)/t^{m_\rho}$, with

$u_\rho(0) \neq 0$. For $0 \leq k < m_\rho$ let $P_{\rho,k}(t)$ be the Taylor polynomial of $t^k/u_\rho(t)$ modulo t^{m_ρ} , with its ordinary Taylor coefficients. The actual tests

$$b_{\rho,k} = P_{\rho,k}(A - \rho)h_\rho \in B_0 \quad (\text{GSR6})$$

then have the exact full-primary jets

$$[Mb_{\rho,k}(\nu + t)]_{t^{m_\nu}} = \begin{cases} t^k, & \nu = \rho, \\ 0, & \nu \neq \rho. \end{cases} \quad (\text{GSR7})$$

Indeed their transform is $M(s)(s - \rho)^{-m_\rho}P_{\rho,k}(s - \rho)$. At ρ its jet is the prescribed inverse Taylor product, while at every other zero ν it retains the factor of order m_ν . Multiplication by the finite polynomial is exactly the original differential operator in GSR6. Thus this is global cardinal interpolation, annihilating all other zeros with all their multiplicities. In particular $Mb_{\rho,0}(\nu) = 1_{\nu=\rho}$.

81.3 Full jets, the actual trace radical, and the image topology

Define

$$\begin{aligned} \mathcal{J}f &= ([Mf(\rho + t)]_{t^{m_\rho}})_\rho, & \mathcal{E}f &= (Mf(\rho))_\rho, \\ N &= \ker \mathcal{J}, & R &= \ker \mathcal{E}, & \mathcal{I}_{\text{jet}} &= \mathcal{J}(B_0), & \mathcal{I}_{\text{val}} &= \mathcal{E}(B_0). \end{aligned} \quad (\text{GSR8})$$

Equip the two images with the exact quotient topologies B_0/N and B_0/R . Each individual Taylor coefficient is a continuous functional on B_0 , so both kernels are closed and these are Hausdorff Fréchet image spaces. This specifies their topology; it does not silently replace them by product or Hilbert subspace topologies.

The actual source inclusion is

$$W_0 \subset N \subset R. \quad (\text{GSR9})$$

For the first inclusion, unfold GSR1 on $\Re s > 1$: $M(E\phi)(s) = 2\zeta(s) \int_0^\infty \phi(x)x^{s-1}dx$. The second factor is holomorphic for $\Re s > 0$ because ϕ is bounded at zero and Schwartz at infinity. It therefore gives the original zeta zero of full order at each ρ . Continuity of every jet passes this to the closure W_0 . The second inclusion is the zeroth Taylor coefficient.

The complete source trace pairing is exactly

$$Q(f, g) = \sum_{\rho \in \mathcal{Z}} m_\rho Mf(\rho) \overline{Mg(\sigma(\rho))}. \quad (\text{GSR10})$$

This is the author's trace theorem applied to the original sharp, with all its global endpoint, Gamma, real and prime terms retained by GTK13 and GTK21. The sum is absolutely convergent by vertical-strip decay and GTK29. Its radical on B_0 is *exactly* R . One inclusion follows immediately from the formula. Conversely, pairing f with the actual cardinal test $b_{\sigma(\rho),0}$ gives $m_\rho Mf(\rho)$. If f pairs to zero with every test, each of these values is zero. This proves the reverse inclusion without a spectral completeness assumption.

Consequently the radical on the actual closed quotient $\mathcal{C}$ is R/W_0 , and the full sequence of proved receiving maps and their kernels is

$$\mathcal{C} = B_0/W_0 \longrightarrow \mathcal{I}_{\text{jet}} = B_0/N \longrightarrow \mathcal{I}_{\text{val}} = B_0/R, \quad \ker_1 = N/W_0, \quad \ker_2 = R/N. \quad (\text{GSR11})$$

Both maps are onto their specified images. By GSR7 the jet image contains the entire algebraic direct sum $\bigoplus_\rho \mathbb{C}[t]/(t^{m_\rho})$. In particular R/N contains every finite collection of nilpotent primary directions $\bigoplus_\rho t\mathbb{C}[t]/(t^{m_\rho})$, with their exact coefficients. The trace kernel cannot identify those directions with the original closed-image kernel. This statement includes the case $m_\rho = 1$, when that particular nilpotent summand is zero.

81.4 A resolvent on the actual closed quotient, not only on numerical values

For $z \notin \mathcal{Z}$ set

$$\begin{aligned} R_z f(y) &= y^{-z} \int_0^y x^{z-1} \left[f(x) - \frac{Mf(z)}{M(z)} h(x) \right] dx, \\ M(R_z f)(s) &= \frac{Mf(s) - Mf(z)M(s)/M(z)}{z - s}. \end{aligned} \quad (\text{GSR12})$$

The numerator has zero moment at z , so GSR4–5 prove $R_z f \in B_0$ and continuity. They also prove its holomorphic dependence on z off $\mathcal{Z}$, and its meromorphic dependence on the entire plane with possible pole order at most m_ρ at ρ . More explicitly divide the holomorphic zero-moment family $M(z)f - Mf(z)h$ first, then divide its B_0 -valued result by $M(z)$. This proves the asserted meromorphy in the actual strong topology.

It is essential to prove $R_z(W_0) \subset W_0$; formal values at zeros do not prove that. First take $\Re z > 1$ and $f = E\phi \in V_0$ in the exact form GSR1. The corrected antecedent $\psi = \phi - Mf(z)M(z)^{-1}\eta$ is even Schwartz, has value and additive integral zero, and satisfies $\int_0^\infty \psi(x)x^{z-1}dx = 0$. The last assertion follows by unfolding both transforms and using the nonzero Euler product $\zeta(z)$ in $\Re z > 1$. Put

$$\tilde{\psi}(x) = x^{-z} \int_0^x t^{z-1} \psi(t) dt \quad (x > 0) \quad (\text{GSR13})$$

and extend it evenly to $\mathbb{R}$. At zero, the even Taylor expansion of ψ gives coefficients divided by $z + 2j$, none zero for $\Re z > 1$. Taylor's remainder under the integral proves smoothness to every order, and $\tilde{\psi}(0) = \psi(0)/z = 0$. At infinity the zero moment rewrites the integral as the negative tail, giving rapid decrease with all derivatives. Thus $\tilde{\psi}$ is even Schwartz. The identity $(z + x\partial_x)\tilde{\psi} = \psi$, integrated with its actual vanishing boundary terms, gives $(z - 1) \int_{\mathbb{R}} \tilde{\psi} = \int \psi = 0$. It has both required source moments. Hence $E\tilde{\psi} \in V_0$. Its transform is the second line of GSR12, so Mellin injectivity gives $R_z f = E\tilde{\psi}$. This proves $R_z(V_0) \subset V_0$ through an actual antecedent.

Continuity passes this result to W_0 . For fixed $w \in W_0$ the map $z \mapsto q(R_z w)$ is holomorphic on the connected domain $\mathbb{C} \setminus \mathcal{Z}$ and zero on $\Re z > 1$. Applying each continuous linear functional on the Hausdorff locally convex quotient, the identity theorem and separation of points give zero throughout that domain. Thus the preservation of W_0 holds for every $z \notin \mathcal{Z}$, proving the descended actual operator $\bar{R}_z$. The original differential equations give the two inverse identities

$$\begin{aligned} (z - A)R_z f &= f - \frac{Mf(z)}{M(z)} h, & R_z(z - A)f &= f, \\ (z - \bar{A})\bar{R}_z &= \bar{R}_z(z - \bar{A}) = 1_C. \end{aligned} \quad (\text{GSR14})$$

For the second equation the corrected moment of $(z - A)f$ at z is zero; the kernel of $z - A$ in B_0 is zero, since its solutions are constant multiples of y^{-z} , which cannot have both required end decays. This proves that the resolvent belongs to the actual quotient, not an assumed spectral replacement.

On the full jet image the exact intertwining is

$$\mathcal{J}(R_z f)_\rho = (z - \rho - t)^{-1} \mathcal{J} f_\rho = \sum_{k=0}^{m_\rho-1} \frac{t^k}{(z - \rho)^{k+1}} \mathcal{J} f_\rho. \quad (\text{GSR15})$$

The subtracted original $M(s)$ in GSR12 vanishes to full order at every zero, which proves this formula. At a fixed ρ , the negative Laurent coefficients of $R_z f$ have range spanned by the first m_ρ parameter derivatives of $y^{-z} \int_0^y x^{z-1} h(x) dx$ at ρ . These functions belong to B_0 : their potentially nondecaying tail coefficients are the first m_ρ zero moments of h , all zero. Thus the entire principal part has rank at most m_ρ .

Its contour projection $\Pi_\rho = (2\pi i)^{-1} \int \bar{R}_z dz$ on a small positive circle about ρ has rank exactly m_ρ . Indeed GSR15 and GSR7 show that its jet map covers that whole primary block, giving

rank at least m_ρ ; the preceding range estimate gives the reverse inequality. The resolvent identity, obtained by multiplying the two inverse equations in GSR14, and Cauchy integration prove $\Pi_\rho^2 = \Pi_\rho$ and $\Pi_\rho \Pi_\nu = 0$ for $\rho \neq \nu$. On its range $\mathcal{J}$ is an isomorphism to $\mathbb{C}[t]/(t^{m_\rho})$ and $\bar{A}$ is exactly $\rho + t$. Thus its actual pole order is m_ρ , and its principal part is

$$\sum_{k=0}^{m_\rho-1} \frac{(\bar{A} - \rho)^k \Pi_\rho}{(z - \rho)^{k+1}}. \quad (\text{GSR16})$$

In particular the spectrum of the actual quotient generator is exactly $\mathcal{Z}$: GSR14 excludes every other point and GSR16 makes each indicated point a genuine pole.

There is also an exact description of the kernel left in GSR11. For $f \in N$, the ratio $Mf(z)/M(z)$ is entire because all zero orders have been retained. GSR12 therefore extends as an entire B_0 -valued function, sends N into N , and satisfies GSR14 on N/W_0 for every $z \in \mathbb{C}$. Hence

$$N/W_0 \text{ carries the restriction of } \bar{A} \text{ with an entire resolvent.} \quad (\text{GSR17})$$

This constructs the exact remaining subquotient and its operators. It does not identify it with zero. Meyer's explicit quasi-nilpotent qualification and vertical growth conditions are therefore preserved; they are not overwritten by equality of the trace characters.

81.5 The entire direct-sum receiver and its exact indefinite form

For the value pairing use

$$\mathcal{H}_{\text{val}} = \{a : \sum_{\rho} m_{\rho} |a_{\rho}|^2 < \infty\}, \quad [a, b] = \sum_{\rho} m_{\rho} a_{\rho} \overline{b_{\sigma(\rho)}}. \quad (\text{GSR18})$$

This is a continuous Hermitian form: the permutation $Jb_{\rho} = b_{\sigma(\rho)}$ is a self-adjoint unitary for its displayed positive Hilbert norm, and $[a, b] = \langle a, Jb \rangle$. No assertion that $[a, a]$ is nonnegative is made. The original test map $\mathcal{E}$ is continuous into this space by vertical-strip decay and the zero count. GSR7 puts every finitely supported vector in its image, proving that its Hilbert closure is all of $\mathcal{H}_{\text{val}}$. This is the Hilbert completion of the value receiver in this explicit norm, not a claim that $\mathcal{C}$ itself carries that faithful norm.

The complete decomposition under σ has the original blocks

$$\begin{cases} m_{\rho} a_{\rho} \bar{b}_{\rho}, & \rho = \sigma(\rho), \\ m_{\rho} (a_{\rho} \bar{b}_{\sigma(\rho)} + a_{\sigma(\rho)} \bar{b}_{\rho}), & \{\rho, \sigma(\rho)\} \text{ has two elements.} \end{cases} \quad (\text{GSR19})$$

The latter Gram matrix is $m_{\rho} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ in the original two coordinates. For example the actual global test $b_{\rho,0} - b_{\sigma(\rho),0}$ has value $-2m_{\rho}$ in that two-element case and no other spectral contribution, by GSR7. This proves the exact relation of each permitted off-line orbit to the original full form; it neither asserts such an orbit exists nor replaces it by sampled zeros.

The full primary receiver instead uses

$$\mathcal{H}_{\text{jet}} = \bigoplus_{\rho \in \mathcal{Z}}^{\ell^2} \mathbb{C}[t]/(t^{m_{\rho}}), \quad \|(a_{\rho,k})\|^2 = \sum_{\rho} \sum_{k=0}^{m_{\rho}-1} |a_{\rho,k}|^2, \quad D_{\rho} = \rho I + N_{\rho}, \quad N_{\rho} a = ta. \quad (\text{GSR20})$$

The Taylor coefficients here are $Mf^{(k)}(\rho)/k!$; no factorial has been suppressed. Cauchy's formula on a circle of radius one about ρ bounds every such coefficient by the same supremum of Mf on that circle. Uniform vertical-strip decay on the enlarged strip therefore shows, for every r ,

$$\sum_{\rho,k} (1 + |\rho|)^{2r} |Mf^{(k)}(\rho)/k!|^2 < \infty.$$

Its estimate is continuous in finitely many seminorms of f . GSR7 supplies every finitely supported jet vector; hence the Hilbert closure of $\mathcal{I}_{\text{jet}}$ is exactly $\mathcal{H}_{\text{jet}}$. Its image has the additional proved rapid weighted decay and its original quotient topology. Neither this image nor that topology is silently replaced by its closure.

Multiplication of full jets and their original sharp are exactly

$$(ab)_\rho(t) = a_\rho(t)b_\rho(t) \pmod{t^{m_\rho}}, \quad (a^\sharp)_\rho(t) = (-1)^k \overline{a_{\sigma(\rho),k}}. \quad (\text{GSR21})$$

They follow from the original convolution Mellin product and $M(f^\sharp)(s) = \overline{Mf(1-\bar{s})}$. The ordinary finite-block trace of multiplication by $a_\rho b_\rho^\sharp$ is $m_\rho a_{\rho,0} \overline{b_{\sigma(\rho),0}}$. Its nilpotent directions have zero trace. Thus GSR18 is the actual trace form of the full primary receiver, with precisely its proved nilpotent radical. The factor m_ρ is a dimension in this jet trace. It is not the ordinary Hilbert operator trace of multiplication on the single weighted line in $\mathcal{H}_{\text{val}}$.

81.6 Global trace-class smoothing and resolvent trace duality

On $\mathcal{H}_{\text{jet}}$ define D blockwise as in GSR20 on $\{a : \sum_\rho \|D_\rho a_\rho\|^2 < \infty\}$. Since $\|N_\rho\| \leq 1$ in the displayed coefficient norm, this is a closed densely defined operator. At $z \notin \mathcal{Z}$, its resolvent is the entire block formula of GSR15. On the tail where $|z - \rho| \geq 2$, the geometric operator sum has norm at most $2/|z - \rho|$. Each remaining block is finite dimensional. Thus the resolvent is bounded, compact, and Hilbert–Schmidt, because

$$\sum_\rho m_\rho (1 + |\rho|)^{-2} < \infty$$

by the retained zero count. Truncating the direct sum proves compactness directly.

For every original $f \in B_0$, its integrated receiver is

$$\Gamma_f = Mf(D), \quad (\Gamma_f)_\rho = \sum_{k=0}^{m_\rho-1} \frac{Mf^{(k)}(\rho)}{k!} N_\rho^k, \quad \text{Tr } \Gamma_f = \sum_\rho m_\rho Mf(\rho). \quad (\text{GSR22})$$

This is trace class. Indeed the block trace norm is at most $m_\rho \sum_{k < m_\rho} |Mf^{(k)}(\rho)/k!|$. The preceding radius-one Cauchy bound is rapid in $|\rho|$, while $m_\rho \leq C(1 + |\rho|) \log(2 + |\rho|)$ by the same zero count. A sufficiently large decay exponent makes the sum of these bounds finite. The trace formula then follows by summing the diagonal of each triangular block. It is exactly the original source trace of f , with its full explicit formula as in GTK13, not a newly inferred positivity theorem.

The resolvent duality also exists on the actual quotient without assigning a trace to its bare resolvent. Convolution by $R_z f$ is $\bar{R}_z$ followed by convolution by f : multiply by $z - \bar{A}$ and use GSR14 and the fact that h acts as zero. Consequently the source trace-class theorem gives

$$\begin{aligned} \text{Tr}_{\mathcal{C}}(\bar{R}_z \vartheta(f)) &= \mathcal{T}(R_z f) = \sum_\rho \frac{m_\rho Mf(\rho)}{z - \rho} = \text{Tr}((z - D)^{-1} \Gamma_f), \\ Q(R_z f, g) &= \sum_\rho \frac{m_\rho Mf(\rho) \overline{Mg(\sigma(\rho))}}{z - \rho}. \end{aligned} \quad (\text{GSR23})$$

Every series is absolute, locally uniform off its poles, and has a simple scalar pole of residue $m_\rho Mf(\rho)$ or the displayed paired value. The actual operator resolvent still has order m_ρ at that point. Scalar trace and operator pole orders therefore record different original data. The first equality uses the original author's trace notion on $\mathcal{C}$, and the last uses an ordinary trace-class Hilbert operator on the explicit receiving space; their equality is proved, not an identification of the underlying spaces.

On the value space the exact indefinite adjoint is $D^\dagger = JD^*J = 1 - D$. Equivalently, with both variables in their domains,

$$[Da, b] = [a, (1 - D)b], \quad ((z - D)^{-1})^\dagger = -(1 - \bar{z} - D)^{-1}. \quad (\text{GSR24})$$

In the first equality the coefficient on the second side is $1 - \overline{\sigma(\rho)} = \rho$. The second follows by inversion and retains its minus sign. On primary jets the original sharp in GSR21 gives $D^\sharp = 1 - D$, including $t \mapsto -t$. These are exact weight-one duality equations for the actual form; no positive inner product has been inferred from them.

For bare spectral resolvents use the convergent difference, not a divergent unregularized sum. Products of the two Hilbert–Schmidt resolvents are trace class, and their identity gives

$$\begin{aligned} \text{Tr}((z - D)^{-1} - (z_0 - D)^{-1}) &= \sum_{\rho} m_{\rho} \left[\frac{1}{z - \rho} - \frac{1}{z_0 - \rho} \right], \\ \text{Tr}(z - D)^{-2} &= \sum_{\rho} \frac{m_{\rho}}{(z - \rho)^2}. \end{aligned} \quad (\text{GSR25})$$

The first summand is $O(|\rho|^{-2})$ and the second likewise; triangular nilpotents have zero trace, but remain in the full resolvents. These traces are also limits of actual source trace-class tests. Retain $f_{\varepsilon}(y) = (4\pi\varepsilon)^{-1/2} \exp(-(\log y)^2/(4\varepsilon))$, $\varepsilon > 0$. The original Mellin integral is $Mf_{\varepsilon}(s) = e^{\varepsilon s^2}$ by the Gaussian Fourier integral, with its indicated Jacobian dy/y . Apply GSR23 to the resolvent difference and then let $\varepsilon \downarrow 0$. For $0 < \varepsilon \leq 1$ the modulus of that factor at ρ is at most e because $0 < \Re \rho < 1$; dominated convergence with the $|\rho|^{-2}$ bound gives the first trace in GSR25. The same argument after differentiation gives the second. Thus the receiving trace has an explicit limit through original admissible tests, without assigning an unsupported trace to an operator on $\mathcal{C}$.

81.7 The exact infinite-rank obstruction for the diagonal tensor clock

On the algebraic tensor product of primary blocks retain

$$D_{\rho, \nu}^{[2]} = (\rho + \nu)I + N_{\rho} \otimes I + I \otimes N_{\nu}. \quad (\text{GSR26})$$

Its block dimension is $m_{\rho}m_{\nu}$. The nilpotent has exact length $m_{\rho} + m_{\nu} - 1$: its power $m_{\rho} + m_{\nu} - 2$ on $1 \otimes 1$ is $\binom{m_{\rho} + m_{\nu} - 2}{m_{\rho} - 1} t^{m_{\rho} - 1} \otimes t^{m_{\nu} - 1}$, which is nonzero, and the next power is zero. Thus the operator resolvent has this full pole order, while its block trace is only $m_{\rho}m_{\nu}/(z - \rho - \nu)$. For any original test f , its integrated block is the complete Taylor polynomial of Mf at $\rho + \nu$ in that nilpotent, and its block trace is $m_{\rho}m_{\nu}Mf(\rho + \nu)$. These statements are finite algebra identities on every block, with no claim that their infinite traces converge.

They give an unconditional global failure of that trace class. The original functional equation pairs every zero ρ with $1 - \rho$, with the same multiplicity. On all infinitely many ordered pairs $(\rho, 1 - \rho)$ the tensor eigenvalue is exactly 1. Retain the original coefficient basis vector $v_{\rho, 1 - \rho} = t^{m_{\rho} - 1} \otimes t^{m_{1 - \rho} - 1}$. It already has norm one in GSR20, with no rescaling. The tensor nilpotent kills this vector, so it is an eigenvector, and the vectors from distinct blocks are orthogonal in the specified Hilbert tensor product. For the actual original theta-image test GSR3,

$$Mh(1) = \frac{1}{4}, \quad Mh(D^{[2]})v_{\rho, 1 - \rho} = \frac{1}{4}v_{\rho, 1 - \rho}. \quad (\text{GSR27})$$

Hence its tensor-clock operator cannot be compact or trace class. The conclusion already follows on this orthogonal sequence; it does not assume RH or discard the nilpotents. In fact its full blockwise operator is bounded: on the strip $0 \leq \Re s \leq 2$ use Cauchy's estimate for Mh on radius three; the tensor nilpotent has norm at most two, so the Taylor-operator series is bounded by a convergent geometric series times that strip supremum. Thus noncompactness here is a proved property of a bounded full receiver, not merely failure to define a formal expression.

Likewise for any z with $\Re z < -3$, all block resolvents are bounded by a uniformly convergent geometric series, but on the displayed eigenvectors they equal $(z-1)^{-1}$. They are noncompact. For tensor order $k \geq 2$, fix the remaining $k-2$ zeros: the same infinite family has sum $1 + \rho_3 + \cdots + \rho_k$. This records the entire infinite-rank obstruction to carrying the one-variable compact-resolvent argument over to the diagonal tensor clock.

These finite primary blocks also embed in the actual quotient by GSR16; the independent original-coordinate chain construction and its continuous equivariant projections are GDT13–17 in `GLOBAL_DILATION_TENSOR_RECEIVER.tex`. Their algebraic tensor embeddings retain every block above. The Hilbert tensor completion used for the stated compactness calculation is the specified receiving completion, and is not identified with the whole actual quotient or an unspecified completion of its tensor product.

Independent-variable smoothing has a different exact map: $\Gamma_f \otimes \Gamma_g$ is trace class with trace $\mathcal{T}(f)\mathcal{T}(g)$, by trace-norm finite-rank approximation and GSR22. Its block eigenvalues are $Mf(\rho)Mg(\nu)$, whereas the diagonal-clock test has $Mf(\rho+\nu)$. Both formulas keep all multiplicities and all original Gamma and endpoint contributions through GSR3. In particular $\Gamma_h = 0$ on every original primary block, since Mh vanishes to order m_ρ there; but GSR27 is nonzero of infinite rank. This is the specific global trace and quotient obstruction, not an assertion that arbitrary nonlinear products descend.

81.8 The entire arbitrary-lattice receiver

For the original nontrivial bounded distributive lattice L , all the linear maps $q, \mathcal{J}, \mathcal{E}, A, R_z, \Pi_\rho$ and their displayed restrictions lift by

$$(f, \lambda) \mapsto (Tf, \lambda) \quad \text{on} \quad G_L(U) = \{(0, \lambda) : \lambda \in L\} \cup (U \times \{1_L\}). \quad (\text{GSR28})$$

There is one label on each entire original vector, including its infinite spectral image after that amplitude has been proved to converge. No infinite lattice join or lattice measure is introduced. Linearity and distributivity prove the semimodule maps and their same-label quotient fibres. In particular the quotients in GSR11 keep a zero amplitude with every original label, including $e = (0, 1_L) \neq \tau = (0, 0_L)$.

The exact pairing receiver is

$$Q_L((f, \lambda), (g, \mu)) = \left(\sum_{\rho} m_{\rho} Mf(\rho) \overline{Mg(\sigma(\rho))}, \lambda \wedge \mu \right). \quad (\text{GSR29})$$

Every numerical term has its original global formula GTK13 or GTK24. Zero amplitude does not remove the displayed meet. On a primary block the power $N_{\rho}^{m_{\rho}}$ lifts to $(v, \lambda) \mapsto (0, \lambda)$, not the constant- τ map. The tensor receiver retains one common label on its whole amplitude tensor; the infinite-rank sequence in GSR27 lies at top support, so its failure of compactness is not removed by any label identification. At an actual support prime J_L , the existing localization $(v, \lambda) \mapsto [\lambda]_{J_L}$ commutes with every displayed label-preserving linear map and takes the pairing to the original meet. That localization does not assert an unchanged arithmetic amplitude stalk.

The proved global relation is therefore the actual quotient and resolvent GSR12–17, its full primary image and specified closure GSR20, its exact numerical radical GSR10–11, and its original trace-class receiving map GSR22–25. The all-jet-vanishing subquotient is retained with its entire resolvent, as the human sources require. The diagonal tensor receiver has the explicit infinite-rank endpoint block GSR27. No positive-kernel assumption, identification of image with closure, or finite-prime certificate is used.

82 The full original theta extension and its tensor dilation cocycle

Original objects and proof inputs. Retain the actual original objects $T = \mathcal{S}(\mathbb{A}_{\mathbb{Q}})$, $E\xi(x) = \sum_{q \in \mathbb{Q}^\times} \xi(qx)$, $\mathcal{B} = \bigcap_{\beta \in \mathbb{R}} |\cdot|^\beta \mathcal{S}(C)$, $V = E(T_0)$, $W = \bar{V}$ and $H = \mathcal{B}/W$ from Connes–Consani–Marcolli, *The Weil proof and the geometry of the adèles class space*, source locators **rangeCVdef**, **actCKAKbis**, **actCKCK** and **vanishV**. Their precise source conventions and continuity estimates are SWQ1–28 and GDT1–8. The original full all-prime boundary is $b : T \rightarrow \mathcal{B}_\partial$ of AB20–24 and BW17. GBC17 proves $E(\ker b) = V$, using the whole boundary coinvariant theorem BW6, with every rational sign. The original primary chains and their entire Gamma factors are GDT11–17; the actual quotient resolvent is GSR12–17. No completed zeta function is a working substitute below.

82.1 The exact all-primary section and its original source defect

Use the original adelic antecedent and test

$$\eta(x) = (\pi^2 x^4 - 3\pi x^2/2)e^{-\pi x^2}, \quad \xi_h(x) = \eta(x_\infty) \prod_p \mathbf{1}_{\mathbb{Z}_p}(x_p), \quad h(y) = E\xi_h(y) = 2 \sum_{n \geq 1} \eta(ny),$$

$$M(s) = Mh(s) = 2\zeta(s) \left[\frac{\pi^2 \Gamma((s+4)/2)}{2\pi^{(s+4)/2}} - \frac{3\pi \Gamma((s+2)/2)}{4\pi^{(s+2)/2}} \right]. \quad (\text{GEC1})$$

The value and additive integral of η are both zero. Thus $b\xi_h = 0$: its infinity boundary factor is the zero pair, while every finite reference factor is $(1, 1)$. For every original nontrivial zero ρ of multiplicity m_ρ retain, for $1 \leq j \leq m_\rho$,

$$k_{\rho,j}(x) = \frac{x^{-\rho}}{(j-1)!} \int_x^\infty y^{\rho-1} (\log(y/x))^{j-1} h(y) dy, \quad k_{\rho,0} = h, \quad (\text{GEC2})$$

$$Mk_{\rho,j}(s) = \frac{M(s)}{(s-\rho)^j}, \quad (D-\rho)k_{\rho,j} = k_{\rho,j-1}, \quad D = -x\partial_x.$$

GDT13 proves membership in the entire original $\mathcal{B}$ by the two-ended integral and all vanishing Mellin moments. Its quotient classes $u_{\rho,j} = [k_{\rho,j}]$ form an actual length- m_ρ Jordan chain; $u_{\rho,0} = 0$. Jets at all the original zeros prove that these chains for distinct ρ are independent. The whole algebraic primary subspace and the explicit section therefore exist:

$$U = \bigoplus_{\rho} \text{span}\{u_{\rho,1}, \dots, u_{\rho,m_\rho}\} \subset H, \quad s : U \rightarrow \mathcal{B}, \quad s(u_{\rho,j}) = k_{\rho,j}, \quad qs = 1_U. \quad (\text{GEC3})$$

Only finite sums occur in U . Every original zero and every original multiplicity is included; no height cutoff is used. No claim that U exhausts the closed quotient is made.

The original dilation is $R_a f(x) = f(x/a)$ for $a > 0$. It descends to $\bar{R}_a$ on H , and its generator descends to $\bar{D}$. The actual infinitesimal extension is

$$(Ds - s\bar{D})u_{\rho,j} = \begin{cases} h, & j = 1, \\ 0, & j > 1. \end{cases} \quad (\text{GEC4})$$

This follows immediately from both original equations in GEC2, including $k_{\rho,0} = h$ and $u_{\rho,0} = 0$. The boundary amplitude is the same actual h for every primary eigenline, not a discarded numerical zero.

There is a full finite-action formula. For any real t put

$$C_{\rho,j}(t) = \int_0^t e^{\rho(t-v)} \frac{(t-v)^{j-1}}{(j-1)!} R_{e^v} h dv, \quad (\text{GEC5})$$

$$R_{e^t} k_{\rho,j} = e^{\rho t} \sum_{\ell=0}^{j-1} \frac{t^\ell}{\ell!} k_{\rho,j-\ell} + C_{\rho,j}(t).$$

The integral is oriented as written also for $t < 0$. To prove it let $X_j(t) = R_{e^t} k_{\rho,j}$ and $X_0(t) = R_{e^t} h$. Strong differentiability of original dilation gives $X'_j = \rho X_j + X_{j-1}$. Multiplication by $e^{-\rho t}$ and the fundamental theorem of calculus give its Volterra integral with initial value $k_{\rho,j}$. Iterating exactly j times gives the displayed finite polynomial and the last integral, with its factorial. Equivalently differentiation of GEC5 gives that recursion and the same initial value. Applying continuous linear functionals and scalar uniqueness proves the equality in the Hausdorff original test space. Thus no formal exponential series of D on arbitrary test functions is assumed.

Every defect in GEC5 has an original Schwartz antecedent in the full boundary kernel, not only a limit in W :

$$\begin{aligned} \Xi_{\rho,j}(t) &= \int_0^t e^{\rho(t-v)} \frac{(t-v)^{j-1}}{(j-1)!} \sigma_{e^v} \xi_h dv \in \ker b, \\ (\sigma_a \xi)(x_\infty, x_f) &= \xi(x_\infty/a, x_f), \quad E\Xi_{\rho,j}(t) = C_{\rho,j}(t) \in V. \end{aligned} \tag{GEC6}$$

For fixed t the real parameter interval is compact. Every Schwartz seminorm of the real integrand is uniformly bounded there, and its finite adelic factors remain exactly the original reference factors. Hence the integral is an actual Schwartz function. The infinity evaluation and integral of each $\sigma_{e^v} \eta$ are zero; its integral retains the Jacobian e^v multiplying the original zero moment. Integrating these two identities proves $b\Xi = 0$. The rational theta sum converges uniformly with each required derivative for v in this compact interval; the Gaussian bounds of GEC1 supply a summable majorant. Interchanging that sum with the integral proves the last equality in GEC6. These steps retain every finite-place Haar factor and both real endpoints.

Writing $\Delta_t = R_{e^t} s - s \bar{R}_{e^t} : U \rightarrow V$, the entire global cocycle law is

$$\Delta_{t+u} = R_{e^t} \Delta_u + \Delta_t \bar{R}_{e^u}. \tag{GEC7}$$

It follows by expanding both sides and cancelling the two identical middle terms; GEC5–6 gives every value on the original chain basis. For any other linear section $s + v$ with $v : U \rightarrow W$, its defect changes by $R_{e^t} v - v \bar{R}_{e^t}$. Thus the formula specifies both the actual extension and its change under a section, rather than declaring the chosen representatives invariant.

82.2 Every tensor direction and its complete correction

For every $r \geq 1$ retain $s_r = s^{\otimes r}$ on $U^{\otimes r}$, with original target $\mathcal{B}^{\otimes r}$. On an arbitrary original pure primary tensor $v = \bigotimes_{i=1}^r u_{\rho_i, j_i}$, put $P_i(t) = e^{\rho_i t} \sum_{\ell=0}^{j_i-1} t^\ell k_{\rho_i, j_i-\ell} / \ell!$ and $C_i(t) = C_{\rho_i, j_i}(t)$. Expanding the entire finite product gives the exact tensor correction

$$\begin{aligned} \Delta_t^{[r]} v &:= (R_{e^t}^{\otimes r} s_r - s_r \bar{R}_{e^t}^{\otimes r}) v \\ &= \sum_{\emptyset \neq S \subset [r]} \bigotimes_{i=1}^r \begin{cases} C_i(t), & i \in S, \\ P_i(t), & i \notin S. \end{cases} \end{aligned} \tag{GEC8}$$

Every cross contribution is present. By GEC6 its range lies in $N_r^V = \sum_i \mathcal{B}^{\otimes(i-1)} \otimes V \otimes \mathcal{B}^{\otimes(r-i)}$, hence in the exact closed-image tensor kernel of GFT9–12 and GBC16. Its generator defect is the complete original expression

$$(D^{[r]} s_r - s_r \bar{D}^{[r]}) v = \sum_{i=1}^r \mathbf{1}_{j_i=1} k_{\rho_1, j_1} \otimes \cdots \otimes h \otimes \cdots \otimes k_{\rho_r, j_r}. \tag{GEC9}$$

Here h occupies the displayed original position i . This follows from the tensor Leibniz sum for the action and GEC4, with no absorbed factor in $D^{[r]} = \sum_i D_i$. The analogue of GEC7 follows by the same multiplication of the actual tensor actions. These equations hold for the whole algebraic primary tensor, by linearity.

There is no equivariant section of an actual nonzero primary eigenline into the original test space. Indeed an intertwining lift f of $u_{\rho,1}$ would satisfy $Df = \rho f$. In each compact-unit coordinate its only solutions are $f(x, u) = x^{-\rho} c(u)$, by the first-order differential equation. No nonzero such function has every required power decay at both zero and infinity. Hence $f = 0$, contrary to its nonzero quotient class. This proves that the cocycle extension on that actual eigenline cannot be removed by any equivariant section.

The tensor statement is equally global. The coefficient map $\mathcal{B}^{\otimes r} \rightarrow \{\text{functions on } C^r\}$ taking an original pure tensor to its product in separate variables is injective. To verify this, restrict to the finite-dimensional factor spans in any finite tensor presentation. Point evaluations separate each such span and therefore span its dual; products of these evaluations separate the corresponding finite tensor space. Thus a zero product function has zero tensor amplitude. Every such tensor function decays to every power along a common positive dilation of all coordinates, in both directions: this follows termwise from the original weighted seminorms and the finite tensor presentation. An eigenvector of the diagonal generator with eigenvalue λ would, along that orbit, be a fixed nonzero value times the single power $a^{-\lambda}$. That cannot obey both collections of decay bounds. Therefore

$$\ker(D^{[r]} - \lambda) = 0 \quad \text{on } \mathcal{B}^{\otimes r} \quad \text{for every } \lambda \in \mathbb{C}. \quad (\text{GEC10})$$

On $U^{\otimes r}$, by contrast, $\bigotimes_i u_{\rho_i,1}$ is an actual nonzero eigenvector of eigenvalue $\sum_i \rho_i$. No equivariant section of that line into the original tensor test space exists. Thus the extension measured by GEC8–9 is nontrivial at every tensor order. This is not a statement that a tensor eigenvalue is off the critical weight.

82.3 The actual relative tensor complex receiving this correction

The source kernel appearing in GEC6 defines the two-term homological complex with its literal original map

$$\mathcal{D}_1 = \ker b \xrightarrow{E} \mathcal{D}_0 = \mathcal{B}, \quad H_1(\mathcal{D}) = \ker b \cap \ker E, \quad H_0(\mathcal{D}) = \mathcal{B}/V \longrightarrow H = \mathcal{B}/W. \quad (\text{GEC11})$$

The second homology identity uses the proved GBC17, not an assumed equality of a summation image and its closure. The displayed further arrow has exactly kernel W/V . The differential intertwines full idele dilation. The joint kernel in degree one is the very source space receiving the mixed BW connecting classes in GBC19–20.

The actual total r -fold tensor has degree- k term

$$\mathcal{D}_k^{[r]} = \bigoplus_{\substack{S \subset [r] \\ |S|=k}} \bigotimes_{i=1}^r M_{S,i}, \quad M_{S,i} = \begin{cases} \ker b, & i \in S, \\ \mathcal{B}, & i \notin S. \end{cases} \quad (\text{GEC12})$$

Its differential applies E at each source position $i \in S$ with the original Koszul sign $(-1)^{|\{j \in S: j < i\}|}$. Two different orders of applying E cancel in the next differential with opposite signs; there are no repeated source positions after one has been changed. Thus $d^2 = 0$. In particular its degree-one image is exactly N_r^V , and the original quotient maps are

$$H_0(\mathcal{D}^{[r]}) = \mathcal{B}^{\otimes r} / N_r^V \cong (\mathcal{B}/V)^{\otimes r} \longrightarrow H^{\otimes r}. \quad (\text{GEC13})$$

The last kernel is the image of $N_r^W = \sum_i \mathcal{B}^{\otimes(i-1)} \otimes W \otimes \mathcal{B}^{\otimes(r-i)}$, namely N_r^W / N_r^V . This is the exact algebraic comparison. No topological tensor-product exactness or closure equality is asserted.

GEC8 is an actual boundary in this complex, with an explicit degree-one antecedent. For each nonempty S choose its least index i_0 . In its summand of GEC8 replace $C_{i_0}(t)$ by $\Xi_{\rho_{i_0}, j_{i_0}}(t)$ in the degree-one source position; leave all other positions as the displayed $C_i(t)$ or $P_i(t)$ in $\mathcal{B}$.

Sum these tensors over every nonempty S . There is only one source position in each resulting degree-one term, so its differential has sign $+1$ and gives exactly that original GEC8 summand by GEC6. This proves the whole correction, with every mixed term, is received by the original relative tensor differential. The generator correction GEC9 has the corresponding degree-one antecedent obtained by replacing its displayed h by $\xi_h \in \ker b$.

This also keeps the higher kernel. A change between two degree-one antecedents is a genuine cycle in the complex GEC12; it is not declared zero without its degree-two boundary calculation. In particular the joint source kernel from GEC11 and GBC20 remains present. The group-homology bar complex of this equivariant complex is an actual double complex; its total differential on bar degree p and internal degree q is

$$d_{\text{tot}} = d_{\text{bar}} + (-1)^p d_{\text{int}}. \quad (\text{GEC14})$$

Equivariance makes the two unsigned differentials commute; the displayed sign then makes the cross terms cancel. Both squares already vanish. This proves the entire total complex receiving rational boundary classes and the original interior tensor, without replacing tensor-before-coinvariants by the tensor of the endpoint quotient.

Finally apply the full semiring tensor isomorphism GFT3–8 to each original vector tensor and lift each differential without changing its one assigned lattice label. Null squares return $(0, \lambda)$, and every nonzero correction in GEC4–9 has its original top label. The precise comparison of independently labelled finite total summands with the synchronized total carrier is GFT15a, including all fibres; no such fibre is discarded here. Every original e and τ remains distinct. The resulting action, non-split extension and relative tensor complex are exact objects over that full coefficient semiring. They supply the explicit extension data on which a weight calculation must act; they do not assert its missing purity.

83 The all-prime mixed boundary in the actual theta comparison complex

Sources and the original arrows. The all-prime source and boundary are BW1–24 of `ROOT_BOUNDARY_WEIGHT_HOMOLOGY.tex`, in the shared programme directory `rh_joint_strategy_20260923`; the complete proof was read for this calculation. Its original public source is *Full prime-support reconstruction*, AB1–5, AB20–24. The original theta and cohomological trace source is Alain Connes, Caterina Consani and Matilde Marcolli, *The Weil proof and the geometry of the adèles class space*, locators `rangecVdef`, `varthetaH1V`, `vanishV` and `tracepairing`. SWQ1–28 and GDT1–27 retain its exact test algebra, closed quotient, full tensor action and duality. The constructions below use the full rational group, including its sign, and the original tensor before any endpoint quotient.

83.1 The original Schwartz source and both endpoint amplitudes

Write $G = \mathbb{Q}^\times$ and retain the original Schwartz space and rational action

$$T = \mathcal{S}(\mathbb{A}_{\mathbb{Q}}), \quad (U_g \xi)(x) = \xi(g^{-1}x), \quad E\xi(x) = \sum_{q \in \mathbb{Q}^\times} \xi(qx). \quad (\text{GBC1})$$

Let $\mathcal{B}$ be the entire original weighted Bruhat test algebra on $C = \mathbb{A}_{\mathbb{Q}}^\times/G$, and let $V = E(T_0)$, where $T_0 = \{\xi : \xi(0) = 0, \int \xi = 0\}$. Set $W = \overline{V}$ in that original topology and $H = \mathcal{B}/W$. No even projection on individual tensor factors is made.

Use the following actual even real Schwartz functions only to specify an endpoint section; the source T itself is unrestricted:

$$\eta_0(x) = (1 - 2\pi x^2)e^{-\pi x^2}, \quad \eta_1(x) = 2\pi x^2 e^{-\pi x^2}, \quad \xi_i = \eta_i \prod_p \mathbf{1}_{\mathbb{Z}_p}, \quad \theta_i = E\xi_i. \quad (\text{GBC2})$$

The Gaussian integral and its derivative give $(\xi_0(0), \int \xi_0) = (1, 0)$ and $(\xi_1(0), \int \xi_1) = (0, 1)$, with every finite Haar factor equal to one. The original theta sums are $\theta_i(y) = 2 \sum_{n \geq 1} \eta_i(ny)$. Poisson summation, with its zero term retained, gives $\theta_0(y) = -1 + o(y^N)$ and $\theta_1(y) = y^{-1} + o(y^N)$ as $y \downarrow 0$, for every N , while both decrease to every order at infinity. For example the Fourier transform interchanges η_0, η_1 ; the two exact equations are $\theta_0(y) = y^{-1}\theta_1(y^{-1}) - 1$ and $\theta_1(y) = y^{-1}\theta_0(y^{-1}) + y^{-1}$. Thus the entire endpoint-enlarged target and map are

$$A = \mathcal{B} \oplus \Theta, \quad \Theta = \mathbb{C}\theta_0 \oplus \mathbb{C}\theta_1, \quad \pi(f + c_0\theta_0 + c_1\theta_1) = (c_0, c_1), \quad q_A(f + c_0\theta_0 + c_1\theta_1) = [f] \in H. \quad (\text{GBC3})$$

Directness follows from the two different displayed tails at zero. For arbitrary $\xi \in T$ put $\xi' = \xi - \xi(0)\xi_0 - (\int \xi)\xi_1$. Its two moments vanish, so $E\xi' \in V \subset \mathcal{B}$. Consequently, with no removed endpoint in the equality,

$$E\xi = E\xi' + \xi(0)\theta_0 + (\int \xi)\theta_1, \quad \pi E\xi = (\xi(0), \int \xi), \quad q_A E\xi = 0, \quad E(T) = V \oplus \Theta. \quad (\text{GBC4})$$

The rational action on A is trivial, since its functions live on C . Reindexing the full rational sum proves $EU_g = E$. This includes $g = -1$ and retains the full original sign action on the source. The actual idele dilation remains equivariant: it changes the moment pair by $(c_0, c_1) \mapsto (c_0, |a|c_1)$. The difference between a dilated endpoint section and the corresponding scaled endpoint section is an element of V , by applying GBC4 to their moment-zero difference. Thus q_A intertwines the original idele action on H .

83.2 The all-prime tensor square of maps, before coinvariants

Use $\mathcal{B}_\partial = \text{LC}(\{0, 1\}^{\{\infty\} \sqcup \mathcal{P}}, \mathbb{C})$ and the actual map $b : T \rightarrow \mathcal{B}_\partial$ of BW17, obtained by tensoring evaluation and additive integration at every place. Let $\epsilon : \mathcal{B}_\partial \rightarrow \mathbb{C}^2$ evaluate at the two constant support patterns. Its formula is $\epsilon b \xi = (\xi(0), \int \xi)$, including all additive Haar factors. For any fixed $r \geq 1$ retain all the algebraic tensors

$$\begin{aligned} T_r &= T^{\otimes r}, \quad \mathcal{B}_r = \mathcal{B}_\partial^{\otimes r}, \quad A_r = A^{\otimes r}, \quad P_r = (\mathbb{C}^2)^{\otimes r}, \\ b_r &= b^{\otimes r}, \quad E_r = E^{\otimes r}, \quad \pi_r = \pi^{\otimes r}, \quad \epsilon_r = \epsilon^{\otimes r}, \quad q_r = q_A^{\otimes r}, \\ J_r &= \ker b_r, \quad I_r = \ker \pi_r = \sum_i A^{\otimes(i-1)} \otimes \mathcal{B} \otimes A^{\otimes(r-i)}, \\ \pi_r E_r &= \epsilon_r b_r, \quad E_r(J_r) \subset I_r, \quad q_r E_r = 0. \end{aligned} \tag{GBC5}$$

Both b_r and π_r are onto: BW17 gives an actual section of b ; GBC2–3 gives one of π ; tensor their sections. The tensor kernel formula is GFT9, also proved by expanding the displayed direct sum in GBC3.

The group G acts diagonally on the original T_r and $\mathcal{B}_r$; it acts trivially on $A_r, P_r, H^{\otimes r}$. All four arrows in the square are equivariant. In particular

$$E_r(U_g^{\otimes r} - 1)t = 0 \quad (t \in T_r, g \in G). \tag{GBC6}$$

This is a literal equality of entire theta functions. Thus the degree-zero theta return of BW22 is zero before projection to $H^{\otimes r}$. It does not identify the source boundary class with zero in its actual rational homology.

83.3 A relative comparison retaining the complete chain and its lift

Use the full bar chain complex $C_n(G, M) = M \otimes \mathbb{C}[G^n]$ for every original G module M , with finite linear combinations. Write a generator as $(m; g_1, \dots, g_n)$ and retain the differential

$$\begin{aligned} d(m; g_1, \dots, g_n) &= (g_1 m; g_2, \dots, g_n) \\ &\quad + \sum_{i=1}^{n-1} (-1)^i (m; g_1, \dots, g_i g_{i+1}, \dots, g_n) \\ &\quad + (-1)^n (m; g_1, \dots, g_{n-1}). \end{aligned} \tag{GBC7}$$

The degree-zero differential is zero. Here G is abelian, so the action term has exactly this order convention. In d^2 , merging two adjacent pairs in the two possible orders cancels with opposite signs, including the triple merge by associativity; removing a last entry cancels its opposite-order term; and the two first action terms cancel because $U_g U_h = U_{gh}$ and G is abelian. This proves $d^2 = 0$. Applying an equivariant linear map to the coefficient is a chain map, as each term shows. This uses every rational sign and every prime. The prime Koszul chains of BW15 enter through alternating bar symbols; in degree one $m\theta_p$ is $(m; p)$ and its differential is $(U_p - 1)m$.

For a chain map $F : C_\bullet \rightarrow D_\bullet$ use the exact convention

$$\text{Cone}(F)_n = D_n \oplus C_{n-1}, \quad d(a, t) = (d_D a + F t, -d_C t). \tag{GBC8}$$

Its square is zero because $d_D F = F d_C$. Form these actual complexes:

$$\begin{aligned} \mathfrak{C}_r &= \text{Cone}(C_\bullet(G, T_r) \xrightarrow{E_r} C_\bullet(G, A_r)), \\ \mathfrak{P}_r &= \text{Cone}(C_\bullet(G, \mathcal{B}_r) \xrightarrow{\epsilon_r} C_\bullet(G, P_r)), \\ \mathfrak{R}_r &= \text{Cone}(C_\bullet(G, J_r) \xrightarrow{E_r} C_\bullet(G, I_r)). \end{aligned} \tag{GBC9}$$

The original square GBC5 gives a short exact sequence of complexes

$$0 \longrightarrow \mathfrak{R}_r \longrightarrow \mathfrak{C}_r \xrightarrow{\Gamma_r} \mathfrak{P}_r \longrightarrow 0, \quad \Gamma_r(a, t) = (\pi_r a, b_r t). \quad (\text{GBC10})$$

It is a chain map because $\pi_r E_r = \epsilon_r b_r$. Surjectivity in each degree follows from the two actual surjections in GBC5, applied to each finite bar coefficient. Its kernel is exactly $C_n(G, I_r) \oplus C_{n-1}(G, J_r)$ with the differential GBC8. This proves every assertion of exactness without discarding any mixed support pattern or tensor sign.

In particular take a genuine cycle $c \in C_j(G, \mathcal{B}_r)$ with $\epsilon_r c = 0$ as a chain, and choose an original lift $t \in C_j(G, T_r)$ with $b_r t = c$. Its cone cycle is $(0, c)$ in degree $j + 1$ of $\mathfrak{P}_r$. The exact connecting map is

$$\partial_{\text{cone}}[(0, c)] = [(E_r t, -dt)] \in H_j(\mathfrak{R}_r). \quad (\text{GBC11})$$

Indeed $\pi_r E_r t = \epsilon_r c = 0$ and $b_r dt = dc = 0$, so the pair belongs to the displayed complex. Its differential is $(dE_r t - E_r dt, d^2 t) = 0$. A new lift $t + u$, with u in $C_j(G, J_r)$, changes it by $(E_r u, -du) = d(0, u)$ in $\mathfrak{R}_r$. For a general change by a cone boundary, lift that bounding cone chain to $\mathfrak{C}_r$ using the degreewise surjection; its differential lifts the change and its second differential is zero. Any difference from that chosen lift is in $\mathfrak{R}_r$ and contributes a boundary there. This proves the full homology well-definedness of the connecting map.

For the actual degree-one BW22 input $c = (b_0; p)$, with $(D_p^{(r)} - 1)b_0 = 0$ and $\epsilon_r b_0 = 0$, it becomes

$$[(E_r t_0; p), (1 - U_p^{\otimes r})t_0], \quad b_r t_0 = b_0. \quad (\text{GBC12})$$

The second component is the negative of BW22 with the cone sign retained. Its theta sum alone is zero by GBC6. The first component retains the original lifted theta amplitude and is essential to the relative chain. No assertion that this cone class is nonzero has been inferred merely from a nonzero boundary coefficient.

83.4 Its exact relation to the interior tensor and the remaining complex

The canonical original interior receiver is the chain map

$$F_r : \mathfrak{R}_r \longrightarrow C_\bullet(G, H^{\otimes r}), \quad (a, j) \longmapsto q_r a. \quad (\text{GBC13})$$

It commutes with the differential because $q_r E_r = 0$ and q_r is equivariant. It is onto in each degree: I_r contains $\mathcal{B}^{\otimes r}$, whose quotient onto $H^{\otimes r}$ is onto. For the entire mixed connecting image there is the exact chain-level equality

$$F_r(E_r t, -dt) = q_r E_r t = 0. \quad (\text{GBC14})$$

Thus this particular full theta comparison does not send the mixed BW connecting classes to the nonzero balanced primary tensors of GDT26–27. This is the proved value of the actual map, not a conclusion that the spaces admit no other relation.

Its full kernel is a constructed comparison complex. Put

$$\begin{aligned} K_r &= I_r \cap \ker q_r, & \mathfrak{K}_r &= \text{Cone}(C_\bullet(G, J_r) \xrightarrow{E_r} C_\bullet(G, K_r)), \\ 0 &\longrightarrow \mathfrak{K}_r \longrightarrow \mathfrak{R}_r \xrightarrow{F_r} C_\bullet(G, H^{\otimes r}) \longrightarrow 0. \end{aligned} \quad (\text{GBC15})$$

The inclusion $E_r(J_r) \subset K_r$ follows from both equations in GBC5. Computing the kernel of GBC13 in each degree gives exactly the first two lines of GBC15, including their differential. The all-prime connecting representative GBC11 is already a cycle in this kernel complex, not just a class whose image happens to vanish after passing to homology.

There is also an explicit entire description of its coefficient space, retaining every tensor sector. Expand GBC3 with the original order of factors; for $S \subset \{1, \dots, r\}$ let $A_S = \bigotimes_i M_i$, where $M_i = \mathcal{B}$ for $i \in S$ and $M_i = \Theta$ otherwise. Then

$$\begin{aligned} A_r &= \bigoplus_{S \subset [r]} A_S, & I_r &= \bigoplus_{\emptyset \neq S \subset [r]} A_S, \\ K_r &= \left(\bigoplus_{\emptyset \neq S \subsetneq [r]} A_S \right) \oplus \sum_{i=1}^r \mathcal{B}^{\otimes(i-1)} \otimes W \otimes \mathcal{B}^{\otimes(r-i)}. \end{aligned} \quad (\text{GBC16})$$

The first equality is the finite distributive expansion of a vector-space direct sum, preserving each original position. The endpoint map is an isomorphism on the all- Θ sector and zero on every other sector. The map q_r is zero on every sector containing Θ ; on the all- $\mathcal{B}$ sector its kernel is precisely the tensor quotient kernel GFT9. These facts prove all lines of GBC16. In particular for $r = 2$ the mixed spaces $\mathcal{B} \otimes \Theta$ and $\Theta \otimes \mathcal{B}$ both remain, along with the full closed-image tensor kernel. The splitting into Θ sectors is an explicit vector-space section, not a claim that each sector is invariant under idele dilation. The subspaces I_r, K_r and all maps are invariant; the section's extension terms are the original elements of V computed after GBC4.

83.5 The joint source kernel containing the entire mixed connecting image

The preceding receiver can be made more exact by calculating its original theta-image intersection. First,

$$E(\ker b) = V. \quad (\text{GBC17})$$

The inclusion into V holds because $b\xi = 0$ implies both global moments vanish. Conversely let $E\xi \in V$ with $\xi \in T_0$. Then $b\xi \in \ker \epsilon$. The proved full boundary relation BW6 at $r = 1$ gives a finite expression $b\xi = \sum_p (D_p - 1)v_p$. Lift each coefficient v_p by the onto original map b to $t_p \in T$. The explicit vector $j = \xi - \sum_p (U_p - 1)t_p$ belongs to $\ker b$, while $Ej = E\xi$ by the full rational invariance GBC6. This proves GBC17 with an actual antecedent. No individual sign projection or infinite sum of group relations is used.

Put $U = E(T) = V \oplus \Theta$. Tensoring the exact source kernel formula BW18 and then applying E_r gives

$$\begin{aligned} E_r(J_r) &= \sum_i U^{\otimes(i-1)} \otimes V \otimes U^{\otimes(r-i)} \\ &= U^{\otimes r} \cap I_r = E_r(\ker(\epsilon_r b_r)). \end{aligned} \quad (\text{GBC18})$$

For the first equality use GBC17 in the factor belonging to $\ker b$ and surjectivity of $E : T \rightarrow U$ in every other factor; this proves both inclusions on arbitrary finite sums of tensors. For the second, expand $U = V \oplus \Theta$ inside GBC16. The endpoint map is injective on the all- Θ summand and zero on every summand with a V factor. Thus its kernel is exactly the displayed sum. Finally a tensor t has $\epsilon_r b_r t = 0$ exactly when $E_r t \in I_r$, by GBC5; the first two equalities give the last one.

The resulting original source spaces and their entire exact sequence are

$$\begin{aligned} K_r^E &= \ker E_r \subset T_r, & K_r^{b,E} &= \ker b_r \cap \ker E_r, & D_r^\epsilon &= \ker \epsilon_r \subset \mathcal{B}_r, \\ 0 &\longrightarrow K_r^{b,E} \longrightarrow K_r^E \xrightarrow{b_r} D_r^\epsilon \longrightarrow 0. \end{aligned} \quad (\text{GBC19})$$

To prove onto, take $d \in D_r^\epsilon$ and any original lift t with $b_r t = d$. Then $E_r t \in U^{\otimes r} \cap I_r$, so GBC18 supplies $u \in J_r$ with $E_r u = E_r t$. The vector $t' = t - u$ lies in K_r^E and still satisfies $b_r t' = d$. The kernel of this restricted map is exactly $K_r^{b,E}$. This proves the full sequence, with all spaces subspaces of the original Schwartz tensor. Equivariance follows from the original equivariance of b_r, E_r .

Apply this coefficient construction to the finitely many bar coefficients of a cycle $c \in C_j(G, D_r^\epsilon)$. It gives t' with $b_r t' = c$ and $E_r t' = 0$. Then $dt' \in C_{j-1}(G, K_r^{b,E})$, and the exact connecting map of GBC19 and its relation to the relative cone are

$$\begin{aligned} H_j(G, D_r^\epsilon) &\longrightarrow H_{j-1}(G, K_r^{b,E}), & [c] &\longmapsto [dt'], \\ [(E_r t, -dt)] &= [(0, -dt')] \text{ in } H_j(\mathfrak{R}_r), & t' &= t - u. \end{aligned} \tag{GBC20}$$

The first map is independent of the lift since a change lies in $C_j(G, K_r^{b,E})$ and changes the answer by its boundary. For a change of c by a boundary, lift the bounding chain and use $d^2 = 0$, as in the proof of GBC11. For the second equality, subtract the literal relative-cone boundary $d(0, u) = (E_r u, -du)$ from $(E_r t, -dt)$. The result is exactly $(0, -d(t - u))$ with the stated sign. Thus the entire secondary class comes from the shifted chain complex of the explicit joint source kernel; it has not been declared zero. The full source extension GBC19 records which boundary homology classes it reaches. This calculation proves the exact relation rather than proposing an unidentified map between spaces having the same modulus weight.

The group chains, cone differentials, exact sequences and tensor receivers lift over the full $G_L(\mathbb{C})$ by GFT3–15a: retain one original label on each whole amplitude chain and apply every displayed linear map without changing it. Tensor labels use their complete meet; finite chain sums use their join. A squared differential has amplitude zero and returns the same label, including e when it was top, rather than τ . There is no infinite lattice join and no independent-coordinate label quotient hidden in these complexes. GBC10 and GBC15 have the exact same-label kernel congruences proved in GFT10.

The idele action commutes with all original differentials and arrows. Therefore the connecting maps respect its actual characters on homology: the boundary character $|a|^m$ computed in BW14 is retained on the reached relative class, including its zero value when the class vanishes. In the modulus convention $|a|^{w/2}$ this is weight $2m$. GBC14 proves that the displayed interior receiver annihilates the whole mixed connecting image. GBC15–16 constructs the precise relative object containing it, with its full source chain, endpoint sectors and support data. This calculation does not replace it by the two endpoint coinvariants and does not assume an interior purity theorem.

84 Evaluation of the entire secondary mixed-boundary map

Original sources, proved predecessors, and scope. The original Schwartz source, rational periodization, and arithmetic quotient are due to Alain Connes, Caterina Consani and Matilde Marcolli, *The Weil proof and the geometry of the adeles class space*, original author source `Yuri70v2.tex`, labels `rangeVdef`, `varthetaH1V`, `vanishV`, and `tracepairing`. The present derivation rereads those source passages, lines 1671–1735 and 1789–1836, in the retained original author file. The additive measures and rational dilation are those retained from Alain Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Section III, through ABR1–5. The full support carrier is the originating *Split Support Geometry*, v11, Definition `def:lattice-split`, retained under its source byline The Clankers, not attributed to this continuation.

The immediate programme inputs are the complete source BW1–24, ABR1–5 in `CC_ADELIC_COINVARIANT_BRIDGE_INDEPENDENT.tex`, and GBC1–20 in `GLOBAL_BOUNDARY_CONE_LIFT.tex`. The latter two complete sources accompany the cumulative construction; an unverified public location is not asserted. The homological mechanism uses Jean-Louis Koszul’s complex; the needed exactness is proved below. This is an evaluation of the original algebraic connecting map, not an assumed purity or positivity statement.

84.1 The original map and its evaluated source

Let $\mathcal{P}$ be all rational primes, and put

$$\begin{aligned} G &= \mathbb{Q}^\times, & G_+ &= \mathbb{Q}_{>0}^\times, & T &= \mathcal{S}(\mathbb{R}) \otimes \bigotimes_p' (\mathcal{S}(\mathbb{Q}_p), \mathbf{1}_{\mathbb{Z}_p}), \\ (U_g \xi)(x) &= \xi(g^{-1}x), & E\xi(x) &= \sum_{q \in \mathbb{Q}^\times} \xi(qx), & & \text{(GSB1)} \\ \Omega &= \{0, 1\}^{\{\infty\} \sqcup \mathcal{P}}, & B &= \text{LC}(\Omega, \mathbb{C}), & b &= \bigotimes_v \left(\xi_v \mapsto (\xi_v(0), \int_{\mathbb{Q}_v} \xi_v(x) dx) \right). \end{aligned}$$

At infinity $\mathbb{Q}_\infty = \mathbb{R}$ with Lebesgue measure; at p the self-dual additive measure has $\text{vol}(\mathbb{Z}_p) = 1$. The real Schwartz and finite Bruhat spaces retain their original topologies. Every tensor, coinvariant, group chain, and quotient in this section is algebraic; no closure of a group-relation subspace is taken.

For $r \geq 1$ retain the ordered tensor powers and maps

$$\begin{aligned} T_r &= T^{\otimes r}, & B_r &= B^{\otimes r} = \text{LC}(\Omega^r, \mathbb{C}), & b_r &= b^{\otimes r}, & E_r &= E^{\otimes r}, \\ P_r &= (\mathbb{C}^2)^{\otimes r}, & \epsilon_r &= \epsilon^{\otimes r}, & \epsilon(f) &= (f(\mathbf{0}), f(\mathbf{1})), & & \text{(GSB2)} \\ M_r &= \ker \epsilon_r, & J_r &= \ker b_r, & Z_r^\theta &= \ker E_r, & K_r^\theta &= J_r \cap Z_r^\theta. \end{aligned}$$

The action on T_r is the diagonal $U_g^{\otimes r}$, including the original sign. The action on B_r , at a configuration $\omega = (\epsilon_{v,i})$, is

$$(D_g f)(\omega) = \prod_p p^{v_p(g)(m_\infty(\omega) - m_p(\omega))} f(\omega), \quad m_v(\omega) = \sum_{i=1}^r \epsilon_{v,i}. \quad \text{(GSB3)}$$

The group acts trivially on P_r . The finite product in GSB3 retains all nonzero valuations of g .

The full target of E is the endpoint-enlarged space A of GBC3: write $T_0 = \{\xi : \xi(0) = 0, \int \xi = 0\}$, $V = E(T_0)$, $\mathcal{B} = \bigcap_{\beta \in \mathbb{R}} |\cdot|^\beta \mathcal{S}(C_\mathbb{Q})$, and $W = \bar{V}$ in that topology. Then

$$\eta_0(x) = (1 - 2\pi x^2)e^{-\pi x^2}, \quad \eta_1(x) = 2\pi x^2 e^{-\pi x^2}, \quad \xi_i = \eta_i \prod_p \mathbf{1}_{\mathbb{Z}_p}, \quad \theta_i = E\xi_i, \quad \text{(GSB4)}$$

$$A = \mathcal{B} \oplus \mathbb{C}\theta_0 \oplus \mathbb{C}\theta_1, \quad E_r : T_r \longrightarrow A^{\otimes r}, \quad H = \mathcal{B}/W.$$

In particular E has not been replaced by an endpoint-free transform. The original identities $\theta_0(y) = y^{-1}\theta_1(y^{-1}) - 1$ and $\theta_1(y) = y^{-1}\theta_0(y^{-1}) + y^{-1}$ remain in this target.

GBC17–19 proves the two exact coefficient sequences

$$\begin{array}{ccccccc} 0 & \longrightarrow & K_r^\theta & \longrightarrow & Z_r^\theta & \xrightarrow{b_r} & M_r \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & J_r & \longrightarrow & T_r & \xrightarrow{b_r} & B_r \longrightarrow 0. \end{array} \quad (\text{GSB5})$$

For completeness, the essential surjectivity in the first row retains the original image. At $r = 1$, BW6 says $\ker \epsilon = \sum_p (D_p - I)B$. Given $\xi \in T_0$, lift the finitely many coefficients in such an expression for $b\xi$ to T and subtract the corresponding $(U_p - I)$ relations. This produces $j \in \ker b$ with $Ej = E\xi$, since $EU_p = E$. Thus $E(\ker b) = V$. Tensoring and expanding $E(T) = V \oplus \mathbb{C}\theta_0 \oplus \mathbb{C}\theta_1$ proves $E_r(J_r) = E_r(T_r) \cap \ker(\pi^{\otimes r})$, where $\pi(f + c_0\theta_0 + c_1\theta_1) = (c_0, c_1)$. For $c \in M_r$, lift it to $t \in T_r$; the identity $\pi^{\otimes r}E_r t = \epsilon_r b_r t = 0$ supplies $u \in J_r$ with $E_r u = E_r t$. Then $z = t - u \in Z_r^\theta$ lifts c . All of these are finite algebraic operations on the original spaces.

Define

$$\begin{aligned} \mathcal{Z}_r &= \{\omega \in \Omega^r : m_p(\omega) = m_\infty(\omega) \text{ for every } p\}, \\ \mathcal{E}_r &= \{\omega : \epsilon_{v,i} \text{ is independent of } v \text{ for every } i\}, \\ \mathcal{I}_r &= \{f \in \text{LC}(\mathcal{Z}_r, \mathbb{C}) : f|_{\mathcal{E}_r} = 0\}, \quad \mathcal{V}_{\text{pr}} = \bigoplus_{p \in \mathcal{P}} \mathbb{C}\vartheta_p. \end{aligned} \quad (\text{GSB6})$$

The set $\mathcal{E}_r$ has 2^r points and is contained in $\mathcal{Z}_r$. BW13, including its finite-stage proof, gives

$$H_j(G, B_r) = \text{LC}(\mathcal{Z}_r, \mathbb{C}) \otimes \Lambda^j \mathcal{V}_{\text{pr}}, \quad H_j(G, P_r) = \mathbb{C}^{\mathcal{E}_r} \otimes \Lambda^j \mathcal{V}_{\text{pr}}. \quad (\text{GSB7})$$

The map between them is restriction to $\mathcal{E}_r$ in the first factor and the identity in the second. This is a statement about the actual maps: at every finite prime stage, projection to the balanced coordinate lines commutes with evaluation at the constant columns. These constant columns are balanced. The Koszul differential is zero on those lines, proving the stated formula before the filtered union over prime sets.

Restriction is onto: an arbitrary function on the finite set $\mathcal{E}_r$ extends by assigning its value to the cylinder having the indicated infinity column. Therefore the homology map in GSB7 is onto in every degree. The homology sequence of $0 \rightarrow M_r \rightarrow B_r \rightarrow P_r \rightarrow 0$ consequently proves

$$H_j(G, M_r) \xrightarrow{\cong} \mathcal{I}_r \otimes \Lambda^j \mathcal{V}_{\text{pr}} \subset H_j(G, B_r) \quad (j \geq 0). \quad (\text{GSB8})$$

Both injectivity and the entire image are part of this equality; no kernel contributed by $H_{j+1}(G, P_r)$ remains because the preceding map in GSB7 is onto.

84.2 A split subobject of the full source with zero higher homology

Let $\mathcal{U} = \{\pm 1\} \times \widehat{\mathbb{Z}}^\times$ act through its actual ideles, the first factor at infinity. Define

$$Q\xi = \frac{1}{2} \int_{\widehat{\mathbb{Z}}^\times} (U_{(1,u)}\xi + U_{(-1,u)}\xi) du, \quad Q_r = Q^{\otimes r}, \quad (\text{GSB9})$$

where the compact Haar measure in this averaging operator has total mass one. This measure specifies an exact projection; it does not alter any additive measure, original function, or arithmetic pairing. For a Bruhat test this integral is a finite average: only finitely many primes have non-reference factors, and on their compact support unit translations factor through finite quotients by local constancy. The real sign average is the displayed sum with its factor $1/2$.

The operators commute with every rational and idele dilation, since the idele group is abelian. Every local unit and real sign preserves evaluation and the original additive integral. Hence

$$Q_r^2 = Q_r, \quad b_r Q_r = b_r, \quad E_r Q_r = \tilde{Q}_r E_r, \quad (\text{GSB10})$$

where $\tilde{Q}_r$ is the same finite compact averaging on the theta target. The last identity follows by applying a unit dilation to the absolutely convergent original rational sum, or termwise to the finite average. The target average exists on $E_r(T_r)$; this is all that is needed to prove preservation of $\ker E_r$. Thus Q_r preserves $J_r, Z_r^\theta, K_r^\theta$. Each of these modules, and T_r , splits as its image under Q_r plus its image under $I - Q_r$. The second summand is contained in $\ker b_r$. These are actual equivariant retracts inside the full modules; none of the complementary modules is identified with zero.

Write $R = \mathbb{C}[G_+]$ and $\mathcal{H} = \mathcal{S}_{\text{even}}(\mathbb{R})$. The exact source isomorphism of ABR3–4 is

$$\begin{aligned} \Theta : R \otimes \mathcal{H} &\longrightarrow QT, \\ t_a \otimes h &\longmapsto [(x_\infty, x_f) \longmapsto h(x_\infty/a) \mathbf{1}_{a\widehat{\mathbb{Z}}}(x_f)], \\ U_g \Theta(t_a \otimes h) &= \Theta(t_{ga} \otimes h) \quad (g > 0). \end{aligned} \tag{GSB11}$$

Here the balls $\mathbf{1}_{a\widehat{\mathbb{Z}}}$ form a basis of the finite unit-invariant Bruhat space. Locally this follows by writing a radial locally constant compact function as a finite sum of nested ball indicators with coefficients given by its successive shell-value differences. Globally, expand its finitely many non-reference factors. Independence follows by evaluating a finite proposed relation at its rational ideles a : the incidence matrix for divisibility can be ordered triangularly with diagonal one. These facts prove spanning and independence with vector coefficients as well. The inverse to GSB11 sends $g_a(x_\infty) \mathbf{1}_{a\widehat{\mathbb{Z}}}$ to $t_a \otimes h_a$, $h_a(x) = g_a(ax)$. In particular the formula is an exact map retaining the original coordinates in its target.

The diagonal tensor is again free over R . Its explicit description is

$$\begin{aligned} R \otimes \mathbb{C}[G_+^{r-1}] \otimes \mathcal{H}^{\otimes r} &\xrightarrow{\cong} Q_r T_r, \\ t_a \otimes [t_{b_2}, \dots, t_{b_r}] \otimes (h_1 \otimes \dots \otimes h_r) &\longmapsto \Theta(t_a \otimes h_1) \otimes \bigotimes_{i=2}^r \Theta(t_{ab_i} \otimes h_i). \end{aligned} \tag{GSB12}$$

On a basis tensor its inverse has $a = a_1$ and $b_i = a_i/a_1$. Rational dilation acts on t_a alone, which proves freeness. The full sign -1 acts trivially here: finite balls are unit invariant and every real factor is even.

For completeness the infinite-prime homology calculation on a free R module has no positive-degree term. Resolve the trivial module over $R_F = \mathbb{C}[t_p^{\pm 1} : p \in F]$ by the Koszul complex on $(t_p - 1)_{p \in F}$. Successive quotients are Laurent polynomial rings, so each next multiplication is injective. The one-variable complex is exact; its successive mapping cones prove exactness for each finite F . The full R is free over R_F . Extend and take the filtered union: a cycle and a bounding relation occur at finite stages, and filtered unions of vector spaces preserve exactness. This gives a free resolution of the trivial R module. Tensoring with the free module GSB12 retains exactness in positive degrees. The order-two sign subgroup has an exact invariant projection $(I + U_{-1}^{\otimes r})/2$, so contributes no positive-degree homology in characteristic zero. Therefore

$$H_j(G, Q_r T_r) = 0 \quad (j \geq 1). \tag{GSB13}$$

No such vanishing for the complementary full-source summand has been assumed or inferred.

84.3 Injection and the complete image of the secondary map

Let $\partial_{r,j}^\theta$ be the connecting homomorphism of the top row of GSB5. Applying Q_r to the coefficient modules leaves both rows exact with the same M_r and B_r : any lift can be replaced by its Q_r image using GSB10. The lower-row connecting map

$$\delta_{r,j}^Q : H_j(G, B_r) \longrightarrow H_{j-1}(G, Q_r J_r)$$

is injective for $j \geq 1$, by GSB13 and the exact homology sequence. For a cycle c in M_r , choose a chain lift z in $Q_r Z_r^\theta$. Both connecting maps are represented by dz , with the same differential and sign. Consequently the square

$$\begin{array}{ccc} H_j(G, M_r) & \xrightarrow{\partial_{r,j}^\theta} & H_{j-1}(G, K_r^\theta) \\ \downarrow & & \downarrow Q_r \text{ then inclusion} \\ H_j(G, B_r) & \xrightarrow{\delta_{r,j}^Q} & H_{j-1}(G, Q_r J_r) \end{array} \quad (\text{GSB14})$$

commutes. The left arrow is injective by GSB8 and the bottom arrow is injective by GSB13. If the upper-right class were zero, its lower-right image would be zero; injectivity then forces the original class to be zero. This proves

$$\boxed{\ker \partial_{r,j}^\theta = 0, \quad \text{im } \partial_{r,j}^\theta \cong \mathcal{I}_r \otimes \Lambda^j \mathcal{V}_{\text{pr}} \quad (r \geq 1, j \geq 1).} \quad (\text{GSB15})$$

The isomorphism is the original connecting map on the exact domain GSB8, not a dimension count. Its image lies in the split spherical summand of $H_{j-1}(G, K_r^\theta)$, even though that target retains the entire original source. This evaluates both the whole kernel and the whole image, without computing an unrelated complementary summand of the target or replacing the image by an intersection quotient left unevaluated.

Here are all the components and their idele action. For $0 \leq m \leq r$,

$$\begin{aligned} \mathcal{Z}_{r,m} &= \binom{[r]}{m} \times \prod_p \binom{[r]}{m}, & \mathcal{E}_{r,m} &= \{(A, A, A, \dots) : A \in \binom{[r]}{m}\}, \\ \mathcal{I}_r &= \bigoplus_{m=1}^{r-1} \{f \in \text{LC}(\mathcal{Z}_{r,m}, \mathbb{C}) : f|_{\mathcal{E}_{r,m}} = 0\}, & (\text{GSB16}) \\ U_a \partial_{r,j}^\theta(f \otimes \omega) &= \left(|a_\infty|_\infty \prod_p |a_p|_p \right)^m \partial_{r,j}^\theta(f \otimes \omega). \end{aligned}$$

The first equality identifies the actual chosen coordinate sets at every place. At $m = 0, r$ the space is a singleton endpoint, so its vanishing-at-endpoints subspace is zero. The last formula follows by the original product of local integration factors and equivariance of the connecting map: on a balanced configuration each $m_v = m$, so that product is exactly the one displayed. No factor depending on j appears. Each interior component has countably infinite dimension: finite cylinder stages give a countable spanning union, and arbitrarily many distinct prime coordinates give arbitrarily many independent endpoint-vanishing cylinders.

At the first mixed tensor $r = 2$, encode a column of sum one by the index of its entry equal to one. With $X = \{1, 2\}^{\{\infty\} \sqcup \mathcal{P}}$, the result is

$$\text{im } \partial_{2,1}^\theta \cong \{f \in \text{LC}(X, \mathbb{C}) : f(1, 1, \dots) = f(2, 2, \dots) = 0\} \otimes \bigoplus_p \mathbb{C} \vartheta_p. \quad (\text{GSB17})$$

Every idele acts on this entire image by its original modulus $|a_\infty|_\infty \prod_p |a_p|_p$. Its modulus weight in the convention $|a|^{w/2}$ is therefore $w = 2$, not $w = 1$.

84.4 Exact representatives for every class and a nonzero witness

Order primes increasingly. Represent a vector in GSB8 as a finite sum of $f \otimes \vartheta_{p_1} \wedge \dots \wedge \vartheta_{p_j}$. Choose a finite F containing every coefficient coordinate and every p_i . Extend each coefficient from the balanced finite patterns by zero on the unbalanced patterns of Ω_F^r ; denote this extension by c . Its endpoint values are zero. For each prime appearing in the wedge, $(D_p - I)c = 0$, so it is an actual Koszul cycle. It restricts to the original f on $\mathcal{Z}_r$.

Decompose c into its finitely many pattern-cylinder indicators. For a prime $q \in F$ use the exact moment sections

$$\phi_{q,0} = \frac{q\mathbf{1}_{q\mathbb{Z}_q} - \mathbf{1}_{\mathbb{Z}_q}}{q-1}, \quad \phi_{q,1} = \frac{q(\mathbf{1}_{\mathbb{Z}_q} - \mathbf{1}_{q\mathbb{Z}_q})}{q-1}. \quad (\text{GSB18})$$

Their evaluation/integral pairs are $(1,0)$ and $(0,1)$: in particular the first integral is $(qq^{-1} - 1)/(q-1) = 0$, and the second is $q(1 - q^{-1})/(q-1) = 1$. Use η_0, η_1 at infinity and the reference indicator at all primes outside F . Tensor these sections in their original factor order to obtain a lift $t_A \in Q_r T_r$ of each nonzero pattern cylinder c_A .

Every such pattern differs from a constant-column endpoint: its coefficient at every endpoint pattern is zero. Thus there are $q \in F$ and a factor i for which $d = \epsilon_{\infty,i} - \epsilon_{q,i}$ is 1 or -1 . Let $U_q^{[i]}$ mean rational dilation on that factor alone. Then

$$z_A = \frac{(U_q^{[i]} - I)t_A}{q^d - 1}, \quad b_r z_A = c_A, \quad E_r z_A = 0. \quad (\text{GSB19})$$

The denominator is nonzero. Its numerator has boundary $(q^d - 1)c_A$ by the actual one-factor form of GSB3; applying E in that factor gives zero by rational reindexing. Thus GSB19 is an explicit original theta-zero lift. Sum these lifts with the original cylinder coefficients to obtain z for c . The complete connecting representative is

$$\sum_{h=1}^j (-1)^{h-1} (U_{p_h}^{\otimes r} - I) z \otimes \vartheta_{p_1} \wedge \cdots \widehat{\vartheta_{p_h}} \cdots \wedge \vartheta_{p_j}. \quad (\text{GSB20})$$

Each coefficient has zero boundary since the original c is balanced at those primes, and zero theta image since $E_r z = 0$. It lies in the required complex of K_r^θ . Changing lifts changes it by a boundary there. Refining F represents the same class: the finite-stage unbalanced terms contract in the Koszul complex, as in BW15–16, while restriction retains the identical f . Alternatively uniqueness follows directly from GSB8 and the well-defined connecting map. Thus these are representatives for the entire module GSB15, with all cylinder relations and every prime.

For the concrete tensor-two witness choose any prime p . Put

$$\begin{aligned} a_p &= \eta_1 \otimes \phi_{p,0} \otimes \bigotimes_{\ell \neq p} \mathbf{1}_{\mathbb{Z}_\ell}, & d_p &= \eta_0 \otimes \phi_{p,1} \otimes \bigotimes_{\ell \neq p} \mathbf{1}_{\mathbb{Z}_\ell}, \\ c_p &= \mathbf{1}_{\{\epsilon_{\infty,1}=1, \epsilon_{\infty,2}=0, \epsilon_{p,1}=0, \epsilon_{p,2}=1\}}, & z_p &= \frac{((U_p - I)a_p) \otimes d_p}{p-1}, \\ & & \partial_{2,1}^\theta [c_p \otimes \vartheta_p] &= [(U_p \otimes U_p - I)z_p] \neq 0. \end{aligned} \quad (\text{GSB21})$$

The other coordinates of c_p are unrestricted. It is balanced at its prime, vanishes on all four endpoints, and is nonzero on $\mathbb{Z}_2$. Its class in GSB8 is nonzero because $\vartheta_p \neq 0$ and the coefficient restriction is nonzero. Equation GSB15 then proves the final inequality. This is a proof of nonvanishing in the original joint-kernel coinvariants, not an inference from a nonzero chain or finite sample.

There is a finite exact inverse on the entire reached degree-one image. In GSB12 write $Q_2 k = \Theta_2(\sum_a t_a \otimes v_a)$, where Θ_2 denotes that displayed isomorphism, for a representative of a reached class. Only finitely many coefficients are nonzero. Its augmentation $\sum_a v_a$ is zero: connecting representatives are sums of diagonal rational differences, and changing a representative by a relation in K_2^θ preserves this fact after applying Q_2 . For integer n define

$$L_n(t) = \begin{cases} \sum_{h=0}^{n-1} t^h, & n > 0, \\ 0, & n = 0, \\ -\sum_{h=n}^{-1} t^h, & n < 0, \end{cases} \quad L_p(a) = \left(\prod_{\ell < p} t_\ell^{v_\ell(a)} \right) L_{v_p(a)}(t_p). \quad (\text{GSB21a})$$

The identity $(t-1)L_n(t) = t^n - 1$ and telescoping the ordered finite prime factorization give $t_a - 1 = \sum_p (t_p - 1)L_p(a)$, including all negative valuations. Therefore define, with the original free-module coordinates,

$$v'_p = \Theta_2 \left(\sum_a L_p(a) \otimes v_a \right), \quad \mathcal{D}(k) = \sum_p (b_2 v'_p)|_{\mathcal{Z}_2} \otimes \vartheta_p, \quad Q_2 k = \sum_p (U_p^{\otimes 2} - I)v'_p. \quad (\text{GSB21b})$$

These are finite sums for each input, with no restriction on which primes an input may involve. Since $b_2 Q_2 k = 0$, the coefficient chain $\sum_p b_2 v'_p \otimes \vartheta_p$ is a cycle in B_2 . The finite-stage Koszul restriction of BW15–16 sends its class to exactly $\mathcal{D}(k)$. The ordinary connecting map $\delta_{2,1}^Q$ sends it to $[Q_2 k]$ in $H_0(G, Q_2 J_2)$. That map is injective, so the detected source is independent of the difference decomposition and is unchanged under every coinvariant relation from K_2^θ . Naturality GSB14 now gives the exact inverse identity

$$\mathcal{D} \partial_{2,1}^\theta = I \quad \text{on } \mathcal{I}_2 \otimes \mathcal{V}_{\text{pr}}, \quad \mathcal{D}((U_p \otimes U_p - I)z_p) = c_p|_{\mathcal{Z}_2} \otimes \vartheta_p. \quad (\text{GSB21c})$$

This detects the entire computed image using finite source coefficients. It does not claim that the whole target consists of reached classes.

84.5 The actual arithmetic receiver and every support label

Define $q_A : A \rightarrow H$ by $q_A(f + c_0\theta_0 + c_1\theta_1) = [f]$. GBC4 proves $q_A E = 0$. With the relative-cone differential $d(a, t) = (d_A a + E_r t, -d_T t)$, the mixed connecting class has representative $(0, -dz)$ whenever z is the theta-zero lift GSB19. The canonical interior map of GBC13 is $(a, t) \mapsto q_A^{\otimes r} a$. Consequently

$$(0, -dz) \mapsto 0 \quad \text{in } C_\bullet(G, H^{\otimes r}) \quad (\text{GSB22})$$

for every class just proved nonzero. Its image in this arithmetic receiver is exactly zero. The construction establishes a nonzero extension together with its annihilation by a specific existing receiver. It does not infer that all other comparisons are zero, and supplies no sign for the full original Weil pairing.

The relative-cone class itself is nonzero as well. To verify this without confusing the source and target, compute full-group homology by the prime Koszul complex on the exact diagonal-sign-even summands. Put $I_r^A = \ker(\pi^{\otimes r}) \subset A^{\otimes r}$. The rational action on I_r^A is trivial, so its Koszul differential is zero. In the cone of $C_\bullet(G, J_r) \xrightarrow{E_r} C_\bullet(G, I_r^A)$, if a cycle $(0, k)$ is a boundary then

$$(0, k) = d(a, t) = (E_r t, -d_T t).$$

Thus $E_r t = 0$ and t is a chain in $J_r \cap \ker E_r = K_r^\theta$; hence k was already a boundary in that kernel complex. Conversely a boundary there is visibly a cone boundary. This proves

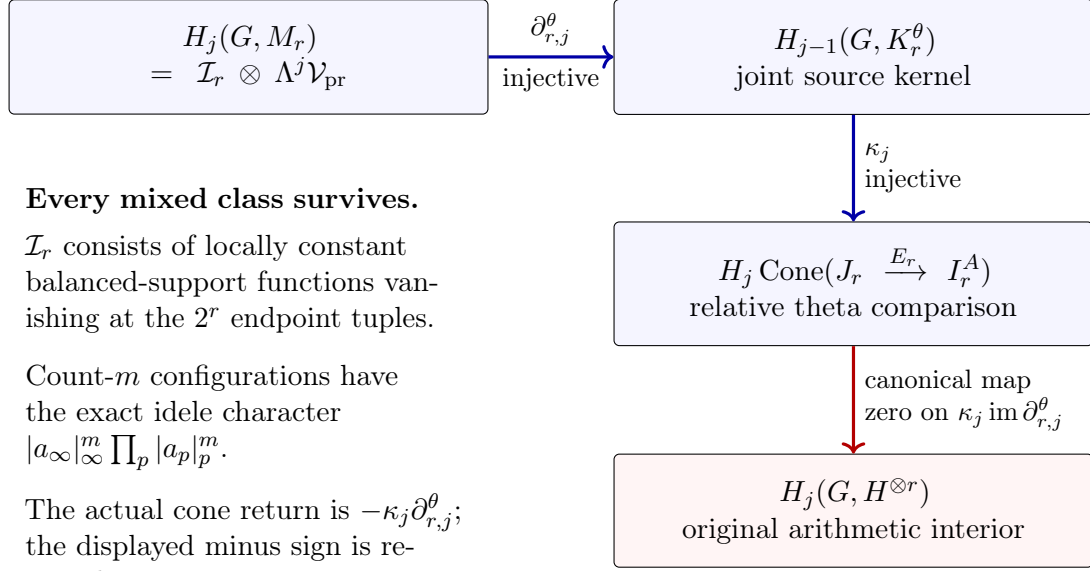
$$\kappa_j : H_{j-1}(G, K_r^\theta) \longrightarrow H_j \text{Cone}(C_\bullet(G, J_r) \xrightarrow{E_r} C_\bullet(G, I_r^A)), \quad [k] \mapsto [(0, k)] \quad \text{is injective.} \quad (\text{GSB22a})$$

The cone return is $-\kappa_j \partial_{r,j}^\theta$, with the original negative sign. It is therefore injective on the entire mixed boundary domain GSB8 while its canonical interior map is identically zero by GSB22. This reproduces the precise GBT22a–d comparison directly on the present modules.

For any nontrivial bounded distributive lattice L keep $G_L(M) = \{(0, \lambda) : \lambda \in L\} \cup (M \times \{1_L\})$, with the original join for addition and meet for scalar and tensor labels. Every linear map above has the exact lift $(v, \lambda) \mapsto (fv, \lambda)$. The quotient congruence is $(v, \lambda) \sim (w, \mu)$ exactly when $\lambda = \mu$ and the amplitude difference is the indicated ordinary boundary. Thus applying this same-label lift to GSB15 retains an injective map whose image has the full original zero-label lattice. Every nonzero amplitude class has top label; its image under GSB22 is $(0, 1_L) = e$, while each lower zero label remains itself, including $\tau = (0, 0_L)$. Squaring a lifted differential returns amplitude

zero with the same label. No ordinary abelian-homology operation on the nonadditive zero-label lattice has been presumed.

In particular the source extension GSB5 cannot have a G -equivariant linear section for $r \geq 2$: a section would lift every cycle to a cycle and force its connecting map to vanish, contradicting GSB21 (and its tensor analogues). This is an evaluated extension defect in the original source, with a known entire image, known idele characters, and an explicit zero arithmetic return.



A nonzero class at every prime: $r = 2, j = 1$.			
place	∞	p	$\ell \neq p$
column on $\mathcal{Z}_2$	$(1, 0)$	$(0, 1)$	$(1, 0)$ or $(0, 1)$
$z_p = \frac{((U_p - I)a_p) \otimes d_p}{p - 1}, \quad [k_p] = [(U_p \otimes U_p - I)z_p] \neq 0.$			
Its inverse detector returns $c_p _{\mathcal{Z}_2} \otimes \vartheta_p$. Its arithmetic return has amplitude zero, with the original support label.			

Figure 24: Exact homology maps, with all primes retained. GSB8 gives the domain, GSB15 the injection and entire image, GSB16 the action, GSB21 the explicit Schwartz class, GSB21a–c its inverse detector, and GSB22–22a the nonzero relative return with zero arithmetic image. The third column in the table describes the restriction to the balanced space; the original cylinder leaves those places unrestricted. The diagram asserts no positivity of the arithmetic pairing. Human source conventions: Connes; Connes, Consani and Marcolli, cited in the proof.

85 Shared rational periodization and its complete relative boundary

Sources and scope. The adelic Schwartz source, rational periodization and original additive measures are those of Alain Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Section III, and Alain Connes, Caterina Consani and Matilde Marcolli, *The Weil proof and the geometry of the adeles class space*, source labels `rangecVdef`, `varthetaH1V`, `fubini1–fubini4`, `nothappen` and `degmodifyV`. The latter paper explicitly identifies the failure of absolute sum–integral interchange. We compute that issue here for the actual GSB mixed antecedent and its entire relative-rational transfer. The predecessor GSB and ABR proofs and the following DMR proof accompany this section; no equivalence with a new arithmetic cohomology is assumed. All tensors are algebraic and all group-relation spaces are algebraic, without closure.

85.1 Actual sources, convergence and the relative transfer

Let $G = \mathbb{Q}^\times$ and retain

$$\begin{aligned} T &= \mathcal{S}(\mathbb{R}) \otimes \bigotimes_p^I (\mathcal{S}(\mathbb{Q}_p), \mathbf{1}_{\mathbb{Z}_p}), & T_r &= T^{\otimes r}, \\ (U_a \xi)(z) &= \xi(a^{-1}z), & E\xi(x) &= \sum_{q \in G} \xi(qx), \quad x \in \mathbb{A}_{\mathbb{Q}}^\times, \\ E_\Delta t(x_1, \dots, x_r) &= \sum_{q \in G} t(qx_1, \dots, qx_r), & E_r &= E^{\otimes r}. \end{aligned} \tag{SRT1}$$

Lebesgue measure at the real place and $\text{vol}(\mathbb{Z}_p) = 1$ at every finite place are used throughout. A pure elementary tensor and a fixed idelic tuple impose $q \in D^{-1}\mathbb{Z}$ for some positive integer D : the finite local supports have bounded denominators, and outside a finite set the original reference indicators impose integrality. Every nonzero real idelic coordinate has nonzero absolute value. The real Schwartz bounds therefore give an absolutely convergent sum over $q \neq 0$. On compact subsets of the real coordinates bounded away from zero the same bound applies to each real derivative, uniformly. Finite sums of tensors retain these conclusions. Independent periodization is absolutely convergent by the product of the individual absolute sums. These arguments establish SRT1 on the original source, not on an unspecified completion. The algebraic tensor of the individual function images injects into functions on the product set: for a finite linearly independent family of functions one can select evaluation functionals spanning its dual (otherwise a nonzero member would vanish at every point). Applying those evaluations successively to each tensor factor proves injection. Thus the function-space kernel of E_r below is also its original algebraic-tensor kernel.

Set $\mathcal{V}_\Delta = E_\Delta(T_r)$ as an actual space of functions on the idelic tuples. There is an exact linear map on this image:

$$\begin{aligned} \mathcal{P}_r : \mathcal{V}_\Delta &\longrightarrow E_r(T_r), \\ (\mathcal{P}_r F)(x_1, \dots, x_r) &= \sum_{(r_2, \dots, r_r) \in G^{r-1}} F(x_1, r_2 x_2, \dots, r_r x_r), & \mathcal{P}_r E_\Delta &= E_r. \end{aligned} \tag{SRT2}$$

Indeed, the change of indices $(q, r_2, \dots, r_r) \mapsto (q, qr_2, \dots, qr_r)$ is a bijection of G^r . The double absolute sum is finite by the preceding estimate on E_r , and the triangle inequality bounds the absolute sum of the grouped values F by it. This also proves independence from the chosen antecedent of F . The map is onto its displayed codomain. Its exact kernel is

$$\ker \mathcal{P}_r = E_\Delta(\ker E_r), \quad \ker E_\Delta \subset \ker E_r. \tag{SRT3}$$

For if $F = E_\Delta t$, SRT2 gives $\mathcal{P}_r F = E_r t$ in both directions. Every rational r_i in SRT2 multiplies both the real and all the finite coordinates. Omitting its finite coordinates would give a different map.

For $r = 2$ write $\mathcal{R}_p F(x, y) = F(x, py)$. Common rational reindexing proves

$$E_\Delta(U_p \otimes I) = \mathcal{R}_p E_\Delta, \quad E_\Delta(U_p \otimes U_p) = E_\Delta. \quad (\text{SRT4})$$

The first identity follows by substituting $q = pq'$ in $\sum_q t(qx/p, qy)$. The second uses the same substitution in both arguments. These are identities of the original functions.

For comparison with the source's spherical coordinates, retain

$$\begin{aligned} \Theta(t_a \otimes h)(z) &= h(z_\infty/a) \mathbf{1}_{a\mathbb{Z}}(z_f), \quad a \in \mathbb{Q}_{>0}^\times, \quad h \in \mathcal{S}_{\text{even}}(\mathbb{R}), \\ \ell &= \prod_p p^{\max_i v_p(a_i)}, \quad x_i = (y_i, 1_f), \quad y_i > 0, \\ E_\Delta \left(\bigotimes_i \Theta(t_{a_i} \otimes h_i) \right) (x_i) &= 2 \sum_{n \geq 1} \prod_i h_i(\ell n y_i / a_i). \end{aligned} \quad (\text{SRT5})$$

The finite constraints are precisely $q \in \ell\mathbb{Z}$; the factor two is the sum over the two rational signs. With $M_+ h(s) = \int_0^\infty h(u) u^{s-1} du$, Mellin integration gives

$$\int_{(0,\infty)^r} E_\Delta(t)(y_i, 1_f) \prod_i y_i^{s_i-1} dy_i = 2\zeta \left(\sum_i s_i \right) \prod_i (a_i/\ell)^{s_i} M_+ h_i(s_i). \quad (\text{SRT6})$$

Here each individual absolute Mellin integral must converge and $\text{Re} \sum_i s_i > 1$. In particular $\text{Re} s_i > 0$ suffices for arbitrary even Schwartz h_i . Absolute Fubini follows by replacing each h_i by its absolute value and using the convergent series $\sum_n n^{-\sum_i \text{Re} s_i}$. The original ζ and all finite rational scales remain in SRT6.

85.2 The original mixed antecedent has a nonzero shared image

Fix a rational prime p . Retain the exact GSB functions

$$\begin{aligned} \eta_0(u) &= (1 - 2\pi u^2) e^{-\pi u^2}, \quad \eta_1(u) = 2\pi u^2 e^{-\pi u^2}, \\ \phi_{p,0} &= (p \mathbf{1}_{p\mathbb{Z}_p} - \mathbf{1}_{\mathbb{Z}_p}) / (p-1), \quad \phi_{p,1} = p(\mathbf{1}_{\mathbb{Z}_p} - \mathbf{1}_{p\mathbb{Z}_p}) / (p-1), \\ a_p &= \eta_1 \otimes \phi_{p,0} \otimes \bigotimes_{l \neq p} \mathbf{1}_{\mathbb{Z}_l}, \quad d_p = \eta_0 \otimes \phi_{p,1} \otimes \bigotimes_{l \neq p} \mathbf{1}_{\mathbb{Z}_l}, \\ z_p &= \frac{((U_p - I)a_p) \otimes d_p}{p-1}, \quad k_p = (U_p \otimes U_p - I)z_p. \end{aligned} \quad (\text{SRT7})$$

Rational reindexing gives $E_2 z_p = 0$. SRT4 shows

$$E_\Delta z_p = (\mathcal{R}_p - I) E_\Delta (a_p \otimes d_p) / (p-1), \quad \mathcal{P}_2 E_\Delta z_p = 0, \quad E_\Delta k_p = 0. \quad (\text{SRT8})$$

Thus the shared image is a relative-rational difference. It is not identified with the image of the coinvariant class $[k_p]$.

At $x_i = (y_i, 1_f)$ the finite support of $d_p(qx_2)$ forces $q \in \mathbb{Z}$ and $p \nmid q$. On that support $\phi_{p,1}(q) = p/(p-1)$, $\phi_{p,0}(q) = -1/(p-1)$ and $\phi_{p,0}(q/p) = 0$. The original first summand in z_p therefore has zero contribution there, and its second summand gives

$$F_p(y_1, y_2) := (E_\Delta z_p)(x_1, x_2) = \frac{2p}{(p-1)^3} \sum_{\substack{n \geq 1 \\ p \nmid n}} \eta_1(ny_1) \eta_0(ny_2). \quad (\text{SRT9})$$

For $y_1 > 0, y_2 \geq 1$ every term is strictly negative, so this is an explicit nonzero element of $\ker \mathcal{P}_2$. The two real factors, the two rational signs and every finite coefficient are retained.

Its full double Mellin formula is

$$\begin{aligned} \int_0^\infty \int_0^\infty F_p(y_1, y_2) y_1^{s-1} y_2^{t-1} dy_1 dy_2 &= \frac{2p}{(p-1)^3} (1 - p^{-s-t}) \zeta(s+t) \\ &\cdot \frac{2\pi\Gamma((s+2)/2)}{2\pi^{(s+2)/2}} \\ &\cdot \left\{ \frac{\Gamma(t/2)}{2\pi^{t/2}} - \frac{2\pi\Gamma((t+2)/2)}{2\pi^{(t+2)/2}} \right\}. \end{aligned} \quad (\text{SRT10})$$

Absolute integration and termwise integration hold on the domain $\text{Re } s > -2, \text{Re } t > 0, \text{Re}(s+t) > 1$: η_1 vanishes to order two at zero, whereas $\eta_0(0) = 1$. These conditions justify termwise integration as in SRT6, now with the Euler factor deleting the multiples of p . Direct Gaussian integration gives the two displayed Mellin factors. The right side supplies a meromorphic continuation, retaining both Gamma summands, their poles and the zero factor; no completed zeta is substituted for the original function. At points of $s+t=1$ where both displayed Gamma expressions are holomorphic, the coefficient of $(s+t-1)^{-1}$, with t fixed, is

$$\frac{2p}{(p-1)^3} (1 - p^{-1}) \frac{2\pi\Gamma((s+2)/2)}{2\pi^{(s+2)/2}} \left\{ \frac{\Gamma((1-s)/2)}{2\pi^{(1-s)/2}} - \frac{2\pi\Gamma((3-s)/2)}{2\pi^{(3-s)/2}} \right\}. \quad (\text{SRT11})$$

For $0 < s < 1$ it is nonzero: writing $t = 1 - s \in (0, 1)$, the second term in braces is t times the first by the Gamma recurrence. This is the pole of this two-variable transform, not a claimed new zero or pole of the original one-variable arithmetic quotient. The formula also specifies the meromorphic residue coefficient on the divisor. At intersections with Gamma poles it is not a claim about the residue of a specialized one-variable function with a possible higher-order pole.

85.3 The exact diagonal moment and ray boundary

Put $h_y(u) = \eta_1(y_1 u) \eta_0(y_2 u)$ and $A = y_1^2 + y_2^2$. The actual adelic diagonal pullback $\mu_y : T_2 \rightarrow T$ gives

$$(\mu_y z_p)(z) = \frac{p}{(p-1)^3} h_y(z_\infty) \mathbf{1}_{\widehat{\mathbb{Z}}}(z_f) \mathbf{1}_{\mathbb{Z}_p^\times}(\text{pr}_p z). \quad (\text{SRT12})$$

The finite additive integral contributes $1 - p^{-1}$. On the real line, Gaussian differentiation, with both original monomials, yields

$$\begin{aligned} I(y_1, y_2) &:= \int_0^\infty h_y(u) du = \frac{y_1^2}{2A^{3/2}} - \frac{3y_1^2 y_2^2}{2A^{5/2}}, \\ R_p(y) &:= \int_{\mathbb{A}_\mathbb{Q}} \mu_y z_p = \frac{2p}{(p-1)^3} (1 - p^{-1}) I(y_1, y_2) \\ &= \frac{1}{(p-1)^2} \left\{ \frac{y_1^2}{A^{3/2}} - \frac{3y_1^2 y_2^2}{A^{5/2}} \right\}. \end{aligned} \quad (\text{SRT13})$$

In particular its sign is the sign of $y_1^2 - 2y_2^2$ for positive y_1, y_2 , and it vanishes precisely on that ray. Neither sign nor vanishing determines the nonzero GSB class $[k_p]$.

Use the original Fourier transform $\widehat{h}(\xi) = \int_{\mathbb{R}} h(u) e^{-2\pi i u \xi} du$. Ordinary Poisson summation for the Schwartz function h_y , separately at λ and $p\lambda$, gives the exact identity

$$\begin{aligned} F_p(\lambda y_1, \lambda y_2) &= \frac{p}{(p-1)^3 \lambda} \left\{ (1 - p^{-1}) \widehat{h}_y(0) + \sum_{m \neq 0} \widehat{h}_y(m/\lambda) \right. \\ &\quad \left. - p^{-1} \sum_{m \neq 0} \widehat{h}_y(m/(p\lambda)) \right\}. \end{aligned} \quad (\text{SRT14})$$

Both complete sums are absolutely convergent; their zero summands are displayed explicitly. Since h_y is even and $h_y(0) = 0$, SRT9 is exactly the difference of the two integer theta sums

multiplied by $p/(p-1)^3$. Thus no unlisted endpoint term enters SRT14. Schwartz bounds on $\widehat{h}_y$ prove, for every integer $N \geq 0$,

$$F_p(\lambda y_1, \lambda y_2) = \lambda^{-1} R_p(y) + O_{p,y,N}(\lambda^N) \quad (\lambda \downarrow 0). \quad (\text{SRT15})$$

The estimate follows by choosing a decay power exceeding $N+1$ in the two nonzero Fourier sums. On compact subsets of $y_i > 0$ those Schwartz seminorms are uniform, including every specified finite real derivative. At infinity the original Gaussian series has rapid decay. The moment and the ray coefficient are therefore the same explicit quantity, with its original additive and Fourier measures.

85.4 Every relative rational and its exact cancellation

Write a relative rational as $r = \pm m/n$, with $m, n \geq 1$ coprime. If $p \mid m$, the shared image at (x_1, rx_2) is zero: the second factor requires $v_p(q) = -v_p(m) < 0$, outside both finite supports of the first factor. Otherwise write $n = p^b n_0$, $p \nmid m n_0$. All admissible summation rationals are exactly $q = \pm p^b n_0 l$ with $l \geq 1$, $p \nmid l$. Define

$$H_b(m, n_0; y) = 2 \sum_{\substack{l \geq 1 \\ p \nmid l}} \eta_1(p^b n_0 l y_1) \eta_0(m l y_2). \quad (\text{SRT16})$$

Both signs of q are in this factor two. Either sign of r gives the same function since the original tests are even. The finite coefficients are

$$\begin{aligned} \phi_{p,0}(q) &= \begin{cases} -1/(p-1), & b=0, \\ 1, & b \geq 1, \end{cases} & \phi_{p,0}(q/p) &= \begin{cases} 0, & b=0, \\ -1/(p-1), & b=1, \\ 1, & b \geq 2, \end{cases} \\ \phi_{p,1}(qr) &= p/(p-1), & F_b &:= (E_\Delta z_p)(x_1, \pm m x_2 / (p^b n_0)) \\ &= \begin{cases} p H_0 / (p-1)^3, & b=0, \\ \frac{p}{(p-1)^2} \{-H_0/(p-1) - H_1\}, & b=1, \\ \frac{p}{(p-1)^2} (H_{b-1} - H_b), & b \geq 2. \end{cases} \end{aligned} \quad (\text{SRT17})$$

These formulas follow by substituting the finite coefficients into the two real summands of SRT7. In particular the exceptional $b=1$ term is not a geometric-series continuation of the $b=0$ term. For each fixed m, n_0 and either rational sign, the finite sum is

$$\sum_{b=0}^B F_b = -\frac{p}{(p-1)^2} H_B \quad (B \geq 1), \quad \lim_{B \rightarrow \infty} H_B = 0. \quad (\text{SRT18})$$

The first two terms cancel their H_0 contributions; every following term cancels the preceding H_{b-1} , leaving the displayed boundary. The Gaussian factor in $\eta_1(p^B n_0 l y_1)$ proves the limit by an absolutely summable bound. All reduced rationals with $p \nmid m$ are partitioned by these fixed coprime pairs (m, n_0) and the two signs. SRT2 proves absolute convergence over the entire rational set at each fixed idelic pair, so this partition is legitimate. It proves $\mathcal{P}_2 E_\Delta z_p = 0$ with its finite-cutoff error computed, as well as the independent reindexing proof in SRT8.

This pointwise absolute convergence does not justify interchanging the complete relative sum with its ray boundary. For each r let $R_{p,r}(y)$ denote that individual ray limit. It exists by applying SRT15 to the finitely many scaled terms in SRT17; it is zero when $p \mid m$. For the

subfamily $r = 1/n$, $p \nmid n$, SRT17 and SRT15 give

$$\begin{aligned}
R_{p,1/n}(y) &:= \lim_{\lambda \downarrow 0} \lambda(E_{\Delta} z_p)(\lambda x_1, (1/n)\lambda x_2) \\
&= \frac{2}{(p-1)^2} I(ny_1, y_2) \\
&= \frac{1}{(p-1)^2} \left\{ \frac{n^2 y_1^2}{(n^2 y_1^2 + y_2^2)^{3/2}} - \frac{3n^2 y_1^2 y_2^2}{(n^2 y_1^2 + y_2^2)^{5/2}} \right\} \\
&= \frac{1}{(p-1)^2 n y_1} + O_{p,y}(n^{-3}).
\end{aligned} \tag{SRT19}$$

Here λx_i means multiplication at the real place only, whereas $1/n$ acts at every place. The finite coordinates in SRT19 are therefore still those of SRT17. The last estimate follows by Taylor's formula for $(1+u)^{-3/2}$ and $(1+u)^{-5/2}$ on a bounded interval with $u = y_2^2/(n^2 y_1^2)$; the full two summands remain above. The leading term is positive. Consequently

$$\sum_{r \in \mathbb{Q}^{\times}} |R_{p,r}(y)| = \infty, \quad \lim_{\lambda \downarrow 0} \lambda(\mathcal{P}_2 E_{\Delta} z_p)(\lambda x_1, \lambda x_2) = 0. \tag{SRT20}$$

Indeed $\sum_{p \nmid n} 1/n$ diverges, whereas $\sum n^{-3}$ converges. Thus there is no absolutely convergent residue sum equal to the second expression. One may instead take each fixed (m, n_0) chain first: its residue analogue of SRT18 has terminal coefficient tending to zero as $B \rightarrow \infty$. This specifies an ordered grouping; it is not an unconditional sum or an application of absolute Fubini.

85.5 The full original Euler factors in the comparison

For $t = (h_1 \mathbf{1}_{\widehat{\mathbb{Z}}}) \otimes (h_2 \mathbf{1}_{\widehat{\mathbb{Z}}})$ with $h_1, h_2 \in \mathcal{S}_{\text{even}}(\mathbb{R})$, the complete finite support calculation at $r = \pm m/n$ reduced gives

$$E_{\Delta} t(x_1, \pm m x_2 / n) = 2 \sum_{l \geq 1} h_1(n l y_1) h_2(m l y_2). \tag{SRT21}$$

The two signs of the relative rational add another factor two. In $\text{Re } s > 1$, $\text{Re } t > 1$, the arithmetic coefficient is exactly

$$\begin{aligned}
\mathcal{C}(s, t) &= \sum_{\substack{m, n \geq 1 \\ (m, n) = 1}} n^{-s} m^{-t}, \\
\mathcal{C}(s, t) &= \prod_p \left\{ 1 + \sum_{a \geq 1} p^{-as} + \sum_{b \geq 1} p^{-bt} \right\} = \prod_p \frac{1 - p^{-s-t}}{(1 - p^{-s})(1 - p^{-t})}, \\
\zeta(s+t) \mathcal{C}(s, t) &= \zeta(s) \zeta(t).
\end{aligned} \tag{SRT22}$$

Absolute convergence justifies prime factorization; coprimality means that at each prime at most one of the two exponents is positive. Equivalently every ordered pair of positive integers is uniquely a common positive factor times a coprime pair. This proves the last identity directly without dividing by a possibly vanishing factor. The shared Mellin value and the full relative transfer are therefore

$$\begin{aligned}
&2\zeta(s+t) M_+ h_1(s) M_+ h_2(t), \\
4\zeta(s+t) \mathcal{C}(s, t) M_+ h_1(s) M_+ h_2(t) &= 4\zeta(s) \zeta(t) M_+ h_1(s) M_+ h_2(t).
\end{aligned} \tag{SRT23}$$

These are identities on the absolute-convergence domain, followed where appropriate by meromorphic continuation. At a meromorphic point the divisor identity keeps the sum of all orders on the two sides; cancellation of a zero and pole is not permission to erase their individual factors. The relative transfer has nonzero kernel by SRT9, so neither SRT22 nor a scalar ratio on a nonvanishing locus is an inverse to that function-space map.

85.6 Full support and the exact arithmetic scope

For the original bounded distributive lattice L retain

$$\begin{aligned} G_L(M) &= \{(0, \lambda) : \lambda \in L\} \cup (M \times \{1_L\}), \quad \tau = (0, 0_L), \quad e = (0, 1_L), \\ A^\uparrow(v, \lambda) &= (Av, \lambda). \end{aligned} \tag{SRT24}$$

Addition has label join and scalar multiplication has label meet, as in the originating *Split Support Geometry*, v11, Definition `def:lattice-split`, by The Clankers. Every linear map above has the exact common-label lift SRT24. Linearity proves compatibility with scalar amplitudes and joins; zero scalars retain the original meet label on both sides. Thus SRT2 becomes an equality of G_L -maps, SRT3 becomes its same-label kernel statement, and the nonzero shared image of z_p maps to e at top support rather than to τ . Every lower zero label remains unchanged. The universal tensor and its common-label comparison are the proved GFT maps; two independently chosen labels are not silently identified with one common label. Analytic convergence is proved in the original amplitude function spaces before lifting the resulting linear maps. No infinite join in an arbitrary lattice L is postulated.

The exact diagonal pullback and its moment are the DMR2 map, with the moment defined on the entire original tensor space. DMR16–23 calculates its descent on the joint-kernel coinvariants: changing a GSB antecedent by an element of that kernel can alter the raw prime integral coordinate arbitrarily, while the refined source retains the entire coordinate separately. Its canonical ABR receiver is zero on the whole GSB image. Accordingly the nonzero shared function, its sign-changing ray residue and the divergent absolute residue sum proved here are exact information about the source and transfer; they are not a negative value of the original Weil pairing. No positivity, purity or RH conclusion has been inserted as an assumption.

86 The diagonal moment and the exact receiver of its prime defect

Sources, conventions and scope. The original adelic Schwartz source and rational periodization are those of Alain Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Section III (6)–(19), and Alain Connes, Caterina Consani and Matilde Marcolli, *The Weil proof and the geometry of the adeles class space*, original source labels `rangeVdef`, `varthetaH1V`, `vanishV` and `tracepairing`. The precise previously proved applications are ABR1–30 in the accompanying complete source `CC_ADELIC_COINVARIANT_BRIDGE_INDEPENDENT.tex` and GSB1–22a in `GLOBAL_SECONDARY_BOUNDARY_EVALUATION.tex`. Both sources were read in full for this calculation. Those readings are not described as a new complete reading of the cited papers. The primary source and measure conventions, and all inputs needed here, are recorded below. The full lattice carrier retains the originating source byline The Clankers: *Split Support Geometry*, v11, Definition `def:lattice-split`.

All coefficient tensors, group homology groups, relation spaces and quotients in this section are algebraic. The original real Schwartz and finite Bruhat spaces and additive measures remain unchanged. No relation space is closed and no completed tensor is substituted. The calculation determines a map to the actual ABR source coinvariant, not a newly asserted map to the nontrivial arithmetic spectral quotient.

86.1 The original diagonal map and its moment

Put

$$\begin{aligned} G &= \mathbb{Q}^\times, \quad G_+ = \mathbb{Q}_{>0}^\times, \quad T = \mathcal{S}(\mathbb{R}) \otimes \bigotimes_p' (\mathcal{S}(\mathbb{Q}_p), \mathbf{1}_{\mathbb{Z}_p}), \\ (U_a \xi)(x) &= \xi(a^{-1}x), \quad E\xi(x) = \sum_{q \in \mathbb{Q}^\times} \xi(qx), \\ B &= \text{LC}(\{0, 1\}^{\{\infty\} \sqcup \mathcal{P}}, \mathbb{C}), \quad b = \bigotimes_v \left(\xi_v \mapsto (\xi_v(0), \int_{\mathbb{Q}_v} \xi_v) \right), \\ K &= \ker(b \otimes b) \cap \ker(E \otimes E) \subset T \otimes T. \end{aligned} \tag{DMR1}$$

Here dx_∞ is Lebesgue measure and $\text{vol}(\mathbb{Z}_p) = 1$. For an idele a its original modulus is $|a| = |a_\infty|_\infty \prod_p |a_p|_p$. The action on $T \otimes T$ is the diagonal $U_a \otimes U_a$; both factors and their order are retained.

Fix $y_1, y_2 > 0$ and let $a_i = (y_i, 1_f)$. Define

$$\begin{aligned} \mu_y : T \otimes T &\longrightarrow T, \quad (\mu_y t)(z) = t(a_1 z, a_2 z), \\ m_y : T \otimes T &\longrightarrow \mathbb{C}, \quad m_y(k) = \int_{\mathbb{A}_{\mathbb{Q}}} (\mu_y k)(z) dz. \end{aligned} \tag{DMR2}$$

The moment is defined on the entire displayed tensor source; its restriction to K is the map whose exact sequence is computed below. This also fixes its domain on the GSB antecedents z_p , which need not belong to K . For a pure tensor of elementary adelic tests, the real factor of its pullback is $f_\infty(y_1 z) g_\infty(y_2 z)$, a Schwartz function. Its finite factors are the products $f_p(z) g_p(z)$, which are locally constant with compact support and equal the reference indicator outside finitely many primes. This proves that DMR2 is an actual linear map on the stated algebraic tensor. It does not require an extension to a completed tensor product.

The multiplication of the two arguments commutes with every idele and every rational dilation. Thus

$$\mu_y(U_a \otimes U_a) = U_a \mu_y, \quad m_y((U_a \otimes U_a)k) = |a| m_y(k), \quad (\mu_y k)(0) = 0 \quad (k \in K). \tag{DMR3}$$

The middle identity is the original additive change of variable $z = aw$, including all local factors. The last follows by evaluating $(b \otimes b)k = 0$ at the pattern selecting evaluation at every place in

both columns; its value is exactly $k(0, 0)$. It does not assert that every coefficient in a finite-ball expansion of $\mu_y k$ has real evaluation zero. DMR24 below shows why that stronger assertion would be false.

The map m_y is onto for every positive pair. An explicit witness retains the function

$$\begin{aligned} h(v) &= \left(\pi^2 v^4 - \frac{3\pi}{2} v^2 \right) e^{-\pi v^2}, & f(x) &= h(x_\infty) \mathbf{1}_{\widehat{\mathbb{Z}}}(x_f), \\ g_y(x) &= e^{-\pi(y_1/y_2)^2 x_\infty^2} \mathbf{1}_{\widehat{\mathbb{Z}}}(x_f), & u_y &= ((U_2 - I)f) \otimes g_y. \end{aligned} \quad (\text{DMR4})$$

The function h is the original Gaussian-polynomial test used in GEC1; no change to its constants is made here. Its two moments are

$$h(0) = 0, \quad \int_{\mathbb{R}} h(v) dv = \pi^2 \frac{3}{4\pi^2} - \frac{3\pi}{2} \frac{1}{2\pi} = 0.$$

Consequently $bf = 0$ and $bU_2 f = 0$. Rational reindexing in the original sum gives $E(U_2 - I)f = 0$. These two identities prove $u_y \in K$ before any quotient is taken.

The finite support of the first term of $\mu_y u_y$ is $2\mathbb{Z}_2 \times \prod_{p \neq 2} \mathbb{Z}_p$, with additive volume $1/2$. That of its second term is $\widehat{\mathbb{Z}}$, with volume one. The real substitution $v = y_1 z_\infty$ therefore gives

$$\begin{aligned} C_y := m_y(u_y) &= \frac{1}{y_1} \left\{ \frac{1}{2} \int_{\mathbb{R}} h(v/2) e^{-\pi v^2} dv - \int_{\mathbb{R}} h(v) e^{-\pi v^2} dv \right\} \\ &= \frac{3}{4y_1} \left\{ \frac{1}{2^{5/2}} - \frac{4}{5^{5/2}} \right\} > 0. \end{aligned} \quad (\text{DMR5})$$

For verification, direct differentiation of $\int e^{-\pi A v^2} dv = A^{-1/2}$ yields $\int v^2 e^{-\pi A v^2} dv = (2\pi A^{3/2})^{-1}$ and $\int v^4 e^{-\pi A v^2} dv = 3/(4\pi^2 A^{5/2})$ for $A > 0$. The first integral in braces, with its factor $1/2$, is $\int h(w) e^{-4\pi w^2} dw = -3/5^{5/2}$; the second is $-3/(4 \cdot 2^{5/2})$. Finally $5^5 = 3125 > 512 = 16 \cdot 2^5$ proves the strict inequality in DMR5 without a numerical estimate.

Put $v_y = u_y/C_y$. This is an explicitly specified element of the original space, with the scalar C_y and every summand of u_y retained, and $m_y(v_y) = 1$. It supplies a linear section of the moment map. In particular, with $K_y^0 = \ker(m_y : K \rightarrow \mathbb{C})$, the actual coefficient sequence is

$$0 \longrightarrow K_y^0 \longrightarrow K \xrightarrow{m_y} \mathbb{C}(1) \longrightarrow 0. \quad (\text{DMR6})$$

The idele action on $\mathbb{C}(1)$ is multiplication by $|a|$; its rational action is trivial by the product formula. DMR3 proves equivariance. The chosen vector v_y is a linear section, not an equivariant section.

86.2 The ABR target with its full adelic action

Let $Q : T \rightarrow T$ be the exact compact projection

$$Q\xi = \frac{1}{2} \int_{\widehat{\mathbb{Z}}^\times} (U_{(1,u)}\xi + U_{(-1,u)}\xi) du, \quad \int_{\widehat{\mathbb{Z}}^\times} du = 1. \quad (\text{DMR7})$$

For a finite Bruhat test this is a finite average, and the real sign average is the displayed factor $1/2$. It preserves the two original moments and commutes with rational and idele dilation. Write $T_0 = \{\xi : \xi(0) = 0, \int \xi = 0\}$ and $T_{[0]} = \{\xi : \xi(0) = 0\}$. Then DMR3 gives the commutative diagram of exact coefficient rows

$$\begin{array}{ccccccc} 0 & \longrightarrow & K_y^0 & \longrightarrow & K & \xrightarrow{m_y} & \mathbb{C}(1) \longrightarrow 0 \\ & & \downarrow Q\mu_y & & \downarrow Q\mu_y & & \downarrow I \\ 0 & \longrightarrow & QT_0 & \longrightarrow & QT_{[0]} & \xrightarrow{\int} & \mathbb{C}(1) \longrightarrow 0. \end{array} \quad (\text{DMR8})$$

The last row is onto because $2\pi x_\infty^2 e^{-\pi x_\infty^2} \mathbf{1}_{\widehat{\mathbb{Z}}}$ has evaluation zero and integral one. The restriction to the spherical source $(Q \otimes Q)K$ also has all these maps, and the explicit moment witness DMR4 lies in that source. The full-source definition in DMR8 instead uses $Q\mu_y$: independent averaging in the two tensor factors has not been identified with averaging their diagonal pullback. In particular no identity $\mu_y(Q \otimes Q) = Q\mu_y$, and no preservation of K_y^0 by $Q \otimes Q$, is needed or asserted.

For clarity the ABR target is the spherical source coinvariant, not all of $H_0(G, T_0)$:

$$\begin{aligned} R &= \mathbb{C}[G_+], \quad I = \ker \varepsilon_R, \quad H = \mathcal{S}_{\text{even}}(\mathbb{R}), \quad M_0 = \ker(\varepsilon_R \otimes m), \\ m(h) &= (h(0), \int h), \quad H_{00} = \ker m, \quad W_{\text{pr}} = \bigoplus_p \mathbb{C}\ell_p, \\ \Theta(t_a \otimes h)(x) &= h(x_\infty/a) \mathbf{1}_{a\widehat{\mathbb{Z}}}(x_f), \quad Q_0 = H_0(G, QT_0) = M_0/IM_0. \end{aligned} \tag{DMR9}$$

ABR3–10 proves that Θ is an exact isomorphism, and gives the exact coordinates

$$\begin{aligned} (\Phi, \beta) : Q_0 &\xrightarrow{\cong} H_{00} \oplus (W_{\text{pr}} \otimes \mathbb{C}^2), \\ \Phi[\sum_a t_a \otimes h_a] &= \sum_a h_a, \quad \beta[\sum_a t_a \otimes h_a] = \sum_p \ell_p \otimes \sum_a v_p(a)m(h_a), \\ j_i \left(\sum_p w_p \ell_p \right) &= \sum_p [(t_p - 1) \otimes w_p \eta_i], \\ \eta_0(x) &= (1 - 2\pi x^2)e^{-\pi x^2}, \quad \eta_1(x) = 2\pi x^2 e^{-\pi x^2} \quad (i = 0, 1). \end{aligned} \tag{DMR10}$$

Here $\beta = (\beta_0, \beta_1)$ refers to the evaluation and integral entries respectively. Each j_i is injective; $\Phi j_i = 0$, $\beta_i j_i = I$ and $\beta_{1-i} j_i = 0$. The section-independence of j_i follows exactly as in ABR10: the difference of two tests having the same two moments is in H_{00} , and its product with $t_p - 1$ belongs to IM_0 .

The full idele action, including every finite-place modulus, has the following exact description in these coordinates. For an idele a put $r = \prod_p p^{v_p(a_p)} > 0$, a rational number, and $b = |a_\infty|_\infty / r = |a|$. Compact units and real sign act trivially on QT . On a spherical coefficient

$$U_a \Theta(t_c \otimes h) = \Theta(t_{rc} \otimes R_b h), \quad R_b h(x) = h(x/b), \quad m(R_b h) = \text{diag}(1, b)m(h). \tag{DMR11}$$

Indeed the finite ball becomes $rc\widehat{\mathbb{Z}}$ and the real argument is $x_\infty/(|a_\infty|_\infty c)$ on both sides. For $\sum_c t_c \otimes h_c \in M_0$, the extra contribution $v_p(r) \sum_c m(R_b h_c)$ to β vanishes because its sum is $m(R_b \Phi) = 0$. It follows that

$$\Phi U_a = R_{|a|} \Phi, \quad \beta_0 U_a = \beta_0, \quad \beta_1 U_a = |a| \beta_1, \quad U_a j_1(w) = j_1(|a|w). \tag{DMR12}$$

This derives the full idele statement from the original finite and real scales; it does not substitute a real-only action for it.

86.3 The entire all-prime ambiguity and its exact splitting

For any G -module M put $N_G M = \sum_{g \in G} (U_g - I)M$, with finite sums, so $H_0(G, M) = M/N_G M$. The sign group contributes no positive homology over $\mathbb{C}$ because its averaging projection is exact. Prime factorization therefore identifies $H_1(G, \mathbb{C}) = W_{\text{pr}}$; its basis is the original ordered prime Koszul basis. The connecting map of DMR6 is

$$\delta_y : W_{\text{pr}} \longrightarrow H_0(G, K_y^0), \quad \delta_y \left(\sum_p w_p \ell_p \right) = \left[\sum_p w_p (U_p \otimes U_p - I) v_y \right]. \tag{DMR13}$$

Each coefficient has moment zero by rational invariance and belongs to K . Changing v_y to any other moment-one lift changes the result by relations from K_y^0 , so DMR13 is the canonical connecting map, independent of that chosen section.

Let

$$F_y = H_0(Q\mu_y) : H_0(G, K_y^0) \longrightarrow Q_0, \quad B_y = \beta_1 F_y. \quad (\text{DMR14})$$

These are defined on the specified source quotient: an actual relation from K_y^0 maps to a relation from QT_0 . For any $w \in QT$ with moment vector $(0, c)$, ABR9 gives

$$\Phi[(U_p - I)w] = 0, \quad \beta[(U_p - I)w] = \ell_p \otimes (0, c).$$

To check directly, write $w = \Theta(\sum_a t_a \otimes h_a)$. The change $a \mapsto pa$ leaves the augmented sum unchanged and adds $\ell_p \otimes \sum_a m(h_a)$. This proves the displayed identity with no restriction on the shape of w . Apply it to $Q\mu_y v_y$, whose moments are $(0, 1)$, to obtain

$$F_y \delta_y = j_1, \quad B_y \delta_y = I_{W_{\text{pr}}}. \quad (\text{DMR15})$$

In particular δ_y is injective. Thus the map $H_1(G, K) \rightarrow H_1(G, \mathbb{C}(1))$ preceding it in the homology sequence is zero; this is a conclusion of the calculation, not an assumption about the homology of K .

The moment descends to an onto map $\overline{m}_y : H_0(G, K) \rightarrow \mathbb{C}(1)$. Define $S_y = \ker \overline{m}_y$ and let π_y be the map induced by the inclusion $K_y^0 \subset K$. Its image is exactly S_y : moment is constant on every coinvariant class, so every class of moment zero has a representative in K_y^0 . The exact result is

$$\begin{aligned} 0 &\longrightarrow W_{\text{pr}}(1) \xrightarrow{\delta_y} H_0(G, K_y^0) \xrightarrow{\pi_y} S_y \longrightarrow 0, \\ 0 &\longrightarrow S_y \longrightarrow H_0(G, K) \xrightarrow{\overline{m}_y} \mathbb{C}(1) \longrightarrow 0, \\ N_G K &\subset K_y^0, \quad N_G K / N_G K_y^0 \xleftarrow{\cong} W_{\text{pr}}(1), \quad w \longmapsto \delta_y(w). \end{aligned} \quad (\text{DMR16})$$

The last line states the complete relation defect, with its actual coefficient representative supplied by DMR13. Exactness can also be seen without abbreviating the relations: write $k = m_y(k)v_y + k^0$, $k^0 \in K_y^0$. Then $(U_g \otimes U_g - I)k$ is congruent modulo $N_G K_y^0$ to $m_y(k)(U_g \otimes U_g - I)v_y$. The identity $U_{ab} - I = (U_a - I) + (U_b - I) + (U_a - I)(U_b - I)$ reduces the latter classes to their prime valuations; the last product is a relation from K_y^0 . The sign class is zero since the sign has order two over $\mathbb{C}$. This proves spanning by DMR13. DMR15 proves independence of every prime coefficient and completes the direct verification of the last line of DMR16.

The first row of DMR16 has a canonical splitting obtained from the actual receiver, while retaining both components. For $s \in S_y$ choose any $c \in H_0(G, K_y^0)$ with $\pi_y c = s$ and put

$$\begin{aligned} \lambda_y(s) &= c - \delta_y B_y c, \quad \Psi_y(c) = (\pi_y c, B_y c), \\ \Psi_y : H_0(G, K_y^0) &\xrightarrow{\cong} S_y \oplus W_{\text{pr}}(1), \quad \Psi_y^{-1}(s, w) = \lambda_y(s) + \delta_y(w). \end{aligned} \quad (\text{DMR17})$$

If another lift is $c + \delta_y w$, DMR15 shows that its displayed expression for λ_y is identical. Thus λ_y is well-defined, $\pi_y \lambda_y = I$, and $B_y \lambda_y = 0$. These identities prove both inverse identities in DMR17. No term of the original class is discarded: its second coordinate and its exact inverse inclusion δ_y are retained. All maps are equivariant for the full idele action, with $W_{\text{pr}}(1)$ carrying $|a|$, by DMR3, DMR12 and naturality of the connecting map.

The resulting exact receiver is

$$\begin{aligned} \widehat{F}_y &= F_y \lambda_y : S_y \longrightarrow Q_0, \quad \beta_1 \widehat{F}_y = 0, \\ F_y(c) &= \widehat{F}_y(\pi_y c) + j_1(B_y c), \quad \widehat{F}_y(s) \in H_{00} \oplus j_0(W_{\text{pr}}) \quad \text{under DMR10,} \\ 0 &\longrightarrow W_{\text{pr}}(1) \xrightarrow{\delta_y} H_0(G, K_y^0) \xrightarrow{\pi_y} S_y \longrightarrow 0 \\ &\quad \downarrow j_1 \quad \quad \downarrow F_y \quad \quad \downarrow \overline{F}_y \\ 0 &\longrightarrow j_1(W_{\text{pr}}) \longrightarrow Q_0 \longrightarrow Q_0 / j_1(W_{\text{pr}}) \longrightarrow 0. \end{aligned} \quad (\text{DMR18})$$

Here $\overline{F}_y$ is induced on the quotient, and $\widehat{F}_y$ is its unique lift having $\beta_1 = 0$. The kernel of the quotient in the bottom row is specified and carried separately in the middle formula. This proves

the strongest descent supplied by this calculation rather than merely reporting the failure of a proposed map.

There cannot be a map $S_y \rightarrow Q_0$ whose composite with π_y equals F_y : it would send $\pi_y \delta_y(w) = 0$ to $F_y \delta_y(w) = j_1(w) \neq 0$ for $w \neq 0$. This is the exact ambiguity of attempting to use μ_y on the original K coinvariants. Moreover $\mu_y K$ is not even contained in T_0 before quotienting, since DMR5 is nonzero. The corrected source, the lost relation subspace, its entire prime image, and the surviving quotient receiver are all part of DMR8 and DMR16–18.

86.4 The complete GSB image under the diagonal receiver

Let $\mathcal{I}_2$ be the actual GSB17 coefficient space

$$X = \{1, 2\}^{\{\infty\} \sqcup \mathcal{P}}, \quad \mathcal{I}_2 = \{f \in \text{LC}(X, \mathbb{C}) : f(1, 1, \dots) = f(2, 2, \dots) = 0\},$$

$$S^{\text{GSB}} = \text{im } \partial_{2,1}^\theta \cong \mathcal{I}_2 \otimes \bigoplus_p \mathbb{C} \vartheta_p \subset H_0(G, K). \quad (\text{DMR19})$$

Every class has a representative $k = \sum_p (U_p \otimes U_p - I) z_p$ from GSB18–20, where only finitely many primes occur in that representative, $z_p \in \ker(E \otimes E)$ and $(b \otimes b) z_p \in \ker(\epsilon \otimes \epsilon)$. All-prime means that no fixed finite prime set is imposed on the space or the calculation. Rational invariance of the diagonal integral gives $m_y(k) = 0$, so $S^{\text{GSB}} \subset S_y$. Every z_p has $z_p(0, 0) = 0$ by its endpoint condition. Writing $\tilde{k}$ for the class of this same coefficient in $H_0(G, K_y^0)$, DMR10 and equivariance give

$$\Phi F_y(\tilde{k}) = 0, \quad \beta_0 F_y(\tilde{k}) = 0, \quad \beta_1 F_y(\tilde{k}) = \sum_p m_y(z_p) \ell_p,$$

$$F_y(\tilde{k}) = j_1 \left(\sum_p m_y(z_p) \ell_p \right), \quad \widehat{F}_y|_{S^{\text{GSB}}} = 0. \quad (\text{DMR20})$$

For the first equality each rational difference has augmented ABR coefficient zero. For the second, the evaluation moment of every $\mu_y z_p$ is zero; for the third, its integral is exactly $m_y(z_p)$. These establish the complete ABR coordinates, not only the vanishing of periodization. DMR17–18 then prove the last equality.

In particular, set $P_y = \pi_y^{-1}(S^{\text{GSB}})$. The entire reached preimage and all its receiver components are

$$P_y = \lambda_y(S^{\text{GSB}}) \oplus \delta_y(W_{\text{pr}}(1)), \quad F_y(P_y) = j_1(W_{\text{pr}}),$$

$$\ker(F_y|_{P_y}) = \lambda_y(S^{\text{GSB}}), \quad \lambda_y(S^{\text{GSB}}) \cong S^{\text{GSB}} \neq 0. \quad (\text{DMR21})$$

Changing a lift z_p by $u \in K$ changes the raw prime coordinate in DMR20 by $m_y(u) \ell_p$. DMR5 shows that this change can be any complex coefficient at any selected prime. Thus the raw prime coordinate is not an invariant of the GSB class in $H_0(G, K)$. The pair in DMR21 retains it as the second component of the refined source; it is not discarded or mistaken for a nontrivial arithmetic interior class.

For the particular GSB21 representative retain

$$\phi_{p,0} = \frac{p \mathbf{1}_{p\mathbb{Z}_p} - \mathbf{1}_{\mathbb{Z}_p}}{p-1}, \quad \phi_{p,1} = \frac{p(\mathbf{1}_{\mathbb{Z}_p} - \mathbf{1}_{p\mathbb{Z}_p})}{p-1},$$

$$a_p = \eta_1 \otimes \phi_{p,0} \otimes \bigotimes_{\ell \neq p} \mathbf{1}_{\mathbb{Z}_\ell}, \quad d_p = \eta_0 \otimes \phi_{p,1} \otimes \bigotimes_{\ell \neq p} \mathbf{1}_{\mathbb{Z}_\ell}, \quad (\text{DMR22})$$

$$z_p = ((U_p - I)a_p) \otimes d_p / (p-1), \quad k_p = (U_p \otimes U_p - I)z_p,$$

$$(\mu_y z_p)(z) = \frac{p}{(p-1)^3} \eta_1(y_1 z_\infty) \eta_0(y_2 z_\infty) \mathbf{1}_{\widehat{\mathbb{Z}}}(z_f) \mathbf{1}_{\mathbb{Z}_p^\times}(\text{pr}_p z).$$

In the last line $\text{pr}_p z$ denotes the local component of the adele z ; the preceding lines use z_p for the named tensor lift. On this local unit shell, $\phi_{p,1} = p/(p-1)$, $\phi_{p,0} = -1/(p-1)$ and $\phi_{p,0}(z_p/p) = 0$.

Off it the factor $\phi_{p,1}$ vanishes. This proves the full pullback formula including its sign and coefficient. Its finite integral is $p(1 - p^{-1})/(p - 1)^3 = 1/(p - 1)^2$. The two real contributions are retained in

$$\begin{aligned} R_p(y) &:= m_y(z_p) = \frac{1}{(p - 1)^2} \left\{ \frac{y_1^2}{(y_1^2 + y_2^2)^{3/2}} - \frac{3y_1^2 y_2^2}{(y_1^2 + y_2^2)^{5/2}} \right\}, \\ F_y([k_p]_{K_y^0}) &= j_1(R_p(y)\ell_p), \quad \lambda_y([k_p]_K) = [k_p]_{K_y^0} - \delta_y(R_p(y)\ell_p), \\ F_y \lambda_y([k_p]_K) &= 0. \end{aligned} \tag{DMR23}$$

Combining the two displayed real contributions gives $y_1^2(y_1^2 - 2y_2^2)/[(p - 1)^2(y_1^2 + y_2^2)^{5/2}]$; the preceding unsuppressed contributions specify its origin. Its possible vanishing when $y_1^2 = 2y_2^2$ does not make $[k_p]_K$ zero: GSB21 independently proves that class nonzero for every prime, and DMR17 supplies an injective lift of it.

86.5 The evaluation boundary that survives for general source classes

The vanishing of β_0 in DMR20 uses the actual GSB lift and does not hold on every element of K_y^0 . Here is an explicit counterexample to that stronger statement, with every finite coefficient retained. Fix a prime p and put

$$\begin{aligned} \chi'_p &= \mathbf{1}_{p^{-1}\mathbb{Z}_p} - (p + 1)\mathbf{1}_{\mathbb{Z}_p} + p\mathbf{1}_{p\mathbb{Z}_p}, \\ a &= \eta_0 \otimes \chi'_p \otimes \bigotimes_{\ell \neq p} \mathbf{1}_{\mathbb{Z}_\ell}, \quad d = \eta_0 \mathbf{1}_{\widehat{\mathbb{Z}}}, \quad w_{p,y} = ((U_p - I)a) \otimes d. \end{aligned} \tag{DMR24}$$

Its local evaluation is $1 - (p + 1) + p = 0$, and its local integral is $p - (p + 1) + pp^{-1} = 0$. Thus $ba = 0$ and $w_{p,y} \in K$ by the same exact rational reindexing used in DMR4. The symbol retains y to indicate its subsequent receiving map, even though this source tensor itself does not depend on y . For $c_{p^n} = \mathbf{1}_{p^n\mathbb{Z}_p} \prod_{\ell \neq p} \mathbf{1}_{\mathbb{Z}_\ell}$ define

$$A_y(x) = \eta_0(y_1 x/p) \eta_0(y_2 x), \quad D_y(x) = \eta_0(y_1 x) \eta_0(y_2 x).$$

The exact diagonal function is

$$\mu_y w_{p,y} = A_y\{c_1 - (p + 1)c_p + pc_{p^2}\} - D_y\{-pc_1 + pc_p\}. \tag{DMR25}$$

The first brace has integral $1 - (p + 1)/p + p/p^2 = 0$; the second has integral $-p + p/p = 1 - p$. Hence

$$\begin{aligned} L_{p,y} &:= m_y(w_{p,y}) = (p - 1) \int_{\mathbb{R}} D_y(x) dx \\ &= (p - 1) \left\{ (y_1^2 + y_2^2)^{-1/2} - (y_1^2 + y_2^2)(y_1^2 + y_2^2)^{-3/2} + 3y_1^2 y_2^2 (y_1^2 + y_2^2)^{-5/2} \right\}. \end{aligned} \tag{DMR26}$$

These are respectively the constant, quadratic and quartic Gaussian contributions. Set $w_{p,y}^0 = w_{p,y} - L_{p,y}v_y \in K_y^0$. In DMR25 the real evaluations of the coefficients at the three balls c_1, c_p, c_{p^2} are $p + 1, -(2p + 1), p$. Passing to the Θ coordinates dilates the real arguments but does not change these evaluations. Moreover every real coefficient of $Q\mu_y v_y$ vanishes at zero, because both terms contain h at a scalar multiple of that variable and $h(0) = 0$. Therefore ABR9 proves

$$\beta_0 F_y([w_{p,y}^0]) = \{0(p + 1) + 1(-(2p + 1)) + 2p\}\ell_p = -\ell_p. \tag{DMR27}$$

In particular, this evaluation boundary cannot be included in the integral-boundary ambiguity $j_1(W_{\text{pr}})$. Its class $\pi_y[w_{p,y}^0] \in S_y$ is nonzero, since every element of $\ker \pi_y = \delta_y(W_{\text{pr}})$ has $\beta_0 F_y = 0$ by DMR15. The canonical receiver $\widehat{F}_y$ retains the same evaluation component: $\beta_0 \widehat{F}_y(\pi_y[w_{p,y}^0]) = -\ell_p$. Thus $\beta_0 \widehat{F}_y : S_y \rightarrow W_{\text{pr}}$ is onto, by finite linear combinations of these explicit all-prime examples. No assertion about surjectivity of the uncomputed H_{00} component is needed or made.

86.6 The original theta image and the complete support carrier

For every input of Q_0 , ABR12–18 gives the faithful joint source receiver

$$\begin{aligned} \Omega(c) &= (\rho(c), \beta(c)), \quad \rho(c) = \mathcal{P}\Phi(c), \quad \mathcal{P}h(y) = 2 \sum_{n \geq 1} h(ny), \\ \Omega : Q_0 &\xrightarrow{\cong} V^{\hat{\mathbb{Z}}^\times} \oplus (W_{\text{pr}} \otimes \mathbb{C}^2), \quad \mathcal{M}(\mathcal{P}h)(s) = 2\zeta(s) \int_0^\infty h(x)x^{s-1} dx \quad (\text{Re } s > 1). \end{aligned} \tag{DMR28}$$

The factor two, original zeta function and original Mellin measure are present. ABR13 retains the complete Poisson endpoint identity $\mathcal{P}h(y) = y^{-1}\mathcal{P}\hat{h}(1/y) + y^{-1} \int h - h(0)$ before applying the two moment conditions. The conclusions DMR20–23 concern the complete faithful source receiver, not only ρ . For every GSB class its canonical lift has zero full ABR image; its retained second component is the integral prime boundary. For the general classes DMR24–27 the evaluation boundary survives. The further map from $V^{\hat{\mathbb{Z}}^\times}$ to the original arithmetic quotient $\mathcal{B}/\overline{E(T_0)}$ is zero because its source is a subspace of $E(T_0)$ itself. These facts neither identify the nonzero source extensions with zero nor provide a new sign for the full Weil pairing.

Finally retain every label in the original bounded distributive lattice L , not just its two endpoints. For every amplitude space M use

$$\begin{aligned} G_L(M) &= \{(0, \lambda) : \lambda \in L\} \cup (M \times \{1_L\}), \quad e = (0, 1_L), \quad \tau = (0, 0_L), \\ A^\dagger(v, \lambda) &= (Av, \lambda), \quad (v, \lambda) \sim (w, \mu) \iff \lambda = \mu \text{ and } v - w \in N_G M. \end{aligned} \tag{DMR29}$$

Addition uses join and scalar or tensor labels use meet, as in the originating definition. DMR2, DMR6, DMR8 and all the homological maps have precisely these common-label lifts. The exact sequences above are proved for their amplitude spaces; their lifted quotient statements use the displayed same-label congruence and do not presume ordinary additive homology on the label lattice. The isomorphism DMR17 becomes $G_L(H_0(G, K_y^0)) \cong G_L(S_y \oplus W_{\text{pr}}(1))$ with one common label, not two independently assigned labels. Every zero map on a nonzero amplitude class therefore sends its top label to e , and every lower zero label remains itself. In particular DMR20 retains the nonzero GSB source extension and its label even when its full ABR amplitude vanishes. Squaring a lifted amplitude differential gives $(0, \lambda)$; it does not replace that supported zero by global absence.

This determines the complete diagonal-moment defect and its all-prime connecting image, evaluates its action on the whole GSB secondary image, and supplies nonzero surviving evaluation boundary classes outside that image. All these conclusions use the original source functions, rational action, finite Haar factors and explicit full-label maps.

87 The odd source sector and the full range of the diagonal receiver

Sources and exact scope. This calculation continues DMR1–29 in `DIAGONAL_MOMENT_REFINED_RECEIVER.tex`, read in full for this derivation. The original Schwartz source and rational periodization are those of Alain Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Section III, and Alain Connes, Caterina Consani and Matilde Marcolli, *The Weil proof and the geometry of the adèles class space*, source labels `rangecVdef`, `varthetaH1V`, `vanishV`, `tracepairing`. Their use here is through the precise original definitions retained in DMR1 and the ABR source-coordinate comparison; no new full-paper reading is claimed. The full support carrier has the originating byline The Clankers, *Split Support Geometry*, v11, Definition `def:lattice-split`.

We compute the previously undetermined H_{00} range of DMR18, prove the full receiver onto, and give explicit coordinates for an entire source summand and its kernel. The real odd source is retained. It is killed by independent spherical averaging but survives the original diagonal multiplication. No assertion of Weil positivity, purity or an RH conclusion is made.

87.1 An exact odd summand of the original coefficient kernel

Use the original spaces, measures and actions

$$\begin{aligned} T &= \mathcal{S}(\mathbb{R}) \otimes \mathcal{S}(\mathbb{A}_f), & G &= \mathbb{Q}^\times, & (U_a \xi)(x) &= \xi(a^{-1}x), \\ E\xi(x) &= \sum_{q \in \mathbb{Q}^\times} \xi(qx), & K &= \ker(b \otimes b) \cap \ker(E \otimes E), \\ (\mu_y t)(z) &= t((y_1, 1_f)z, (y_2, 1_f)z), & m_y(k) &= \int_{\mathbb{A}} \mu_y k, & y_1, y_2 &> 0. \end{aligned} \tag{DOR1}$$

Here b is the complete tensor of local evaluation and local integral maps, not just its two constant endpoints. Lebesgue measure is used at infinity and $\text{vol}(\mathbb{Z}_p) = 1$ at every prime. The action on $T \otimes T$ is diagonal. Every tensor product, direct sum and coinvariant below is algebraic; every vector has finite tensor length. No topological closure of a relation space is introduced.

Let P_f average finite units and let J change only the real sign:

$$\begin{aligned} P_f \xi &= \int_{\widehat{\mathbb{Z}}^\times} U_{(1,u)} \xi \, du, & \int du &= 1, & J &= U_{(-1,1_f)}, \\ P^- &= \tfrac{1}{2}(I - J)P_f, & T^- &= P^- T, & H^- &= \mathcal{S}_{\text{odd}}(\mathbb{R}), & D &= T^- \otimes T^-. \end{aligned} \tag{DOR2}$$

The finite average is finite on each original finite Bruhat test: its compact unit orbit factors through a finite quotient of $\widehat{\mathbb{Z}}^\times$. Thus DOR2 is an actual map on T . It is idempotent and commutes with every idele dilation. An element of T^- is odd in the real variable and invariant under every finite unit. Its local real evaluation and integral both vanish. Hence $b(T^-) = 0$, with every local support pattern retained.

Finite unit invariance also gives $\xi(-x) = -\xi(x)$ for $\xi \in T^-$, where this minus sign changes every adelic coordinate. The original periodization on an idelic x is absolutely convergent: the finite support forces its rational summation parameters into a fixed fractional integer lattice, and real Schwartz decay gives a convergent series there. Pairing each q with $-q$ therefore proves

$$E(T^-) = 0, \quad D \subset K, \quad P^- \otimes P^- : K \longrightarrow D \text{ is an equivariant retraction.} \tag{DOR3}$$

The retraction assertion does not presume that independent averaging commutes with μ_y . Its image is contained in $D \subset K$, and its restriction to D is the identity. Therefore the inclusion $H_0(G, D) \rightarrow H_0(G, K)$ is injective, with the induced retraction as its left inverse. This is an actual source summand.

Put $R = \mathbb{C}[\mathbb{Q}_{>0}^\times]$. The map

$$\Theta_-(t_a \otimes F)(x) = F(x_\infty/a) \mathbf{1}_{a\widehat{\mathbb{Z}}}(x_f) \quad (\text{DOR4})$$

is an isomorphism $R \otimes H^- \rightarrow T^-$. To verify its finite part, a compactly supported locally constant radial function on $\mathbb{Q}_p$ is a finite linear combination of the nested ball indicators $\mathbf{1}_{p^n \mathbb{Z}_p}$. Such finite combinations are independent by evaluating on the successive nonempty shells. At finitely many primes take their tensor products; at all remaining primes keep $\mathbf{1}_{\mathbb{Z}_p}$. The resulting finite ball products are indexed by the positive rational number a with the specified valuations. These products form a basis of the finite unit invariant Bruhat space. Rescaling each real coefficient by a then gives DOR4 and its unique inverse. Positive rational q acts by $t_a \mapsto t_{qa}$, and rational -1 acts as $-I$.

On D the rational sign is therefore $+I$. Dividing the pair of positive rational indices (a, b) by their common second index identifies its diagonal R -module as free. Consequently

$$\begin{aligned} H_0(G, D) &\cong \mathcal{A}^- := \bigoplus_{c \in \mathbb{Q}_{>0}^\times} (H^- \otimes H^-)_c, \\ [\Theta_-(t_a \otimes F) \otimes \Theta_-(t_b \otimes G)] &\longmapsto [a/b] \otimes F \otimes G. \end{aligned} \quad (\text{DOR5})$$

An inverse sends $[c] \otimes F \otimes G$ to the class with indices $(c, 1)$. Indeed applying the common rational b^{-1} changes (a, b) to $(a/b, 1)$ without changing F or G . There are no further relations: the common index is precisely a free R factor, and its coinvariant is the augmentation of that factor. This proves both inverse identities, including for sums.

87.2 The complete multiplication receiver on this summand

Write $c = m/n$ in its unique relatively prime positive integer form. This is only the original prime-valuation description of the rational index: no source coefficient is divided out. For the representative $(c, 1)$, intersection of the two finite balls is $c\widehat{\mathbb{Z}} \cap \widehat{\mathbb{Z}} = m\widehat{\mathbb{Z}}$. The pullback and its actual ABR coefficient are therefore

$$\begin{aligned} \mu_y(\Theta_-(t_{m/n} \otimes F) \otimes \Theta_-(t_1 \otimes G))(z) &= F(y_1 n z_\infty / m) G(y_2 z_\infty) \mathbf{1}_{m\widehat{\mathbb{Z}}}(z_f) \\ &= \Theta(t_m \otimes H_{m,n})(z), \\ H_{m,n}(x) &= F(y_1 n x) G(y_2 m x). \end{aligned} \quad (\text{DOR6})$$

The finite volume is $1/m$, and the real substitution $z_\infty = mx$ gives the exact moment $\int_{\mathbb{R}} H_{m,n}(x) dx$. The result is already invariant under finite units and the real sign, so $Q\mu_y = \mu_y$ on D .

Define the linear map on the explicit direct sum

$$\begin{aligned} \mathcal{T}_y : \mathcal{A}^- &\longrightarrow H_{[0]} := \{h \in \mathcal{S}_{\text{even}}(\mathbb{R}) : h(0) = 0\}, \\ \mathcal{T}_y \left(\sum_{m,n,j} [m/n] \otimes F_{m,n,j} \otimes G_{m,n,j} \right) (x) &= \sum_{m,n,j} F_{m,n,j}(y_1 n x) G_{m,n,j}(y_2 m x), \\ \nu_y &= \int_{\mathbb{R}} \mathcal{T}_y, \quad \mathcal{A}_y^{-,0} = \ker \nu_y. \end{aligned} \quad (\text{DOR7})$$

Every displayed sum is finite; each term retains both original real scales and its reduced positive integer pair. Equations DOR5–6 prove that ν_y is the restriction of the original descended moment $\overline{m}_y$, and that $\mathcal{A}_y^{-,0}$ embeds in $S_y = \ker \overline{m}_y \subset H_0(G, K)$.

For a moment-zero class the ABR formulas of DMR10 on the chosen representative give all three coordinates, not only the augmented function:

$$\begin{aligned}\Phi &= \sum_{m,n,j} H_{m,n,j}, & \beta_0 &= 0, \\ \beta_1 &= \sum_p \ell_p \sum_{m,n,j} v_p(m) \int_{\mathbb{R}} H_{m,n,j}(x) dx.\end{aligned}\tag{DOR8}$$

The second identity follows because every real coefficient is the product of two odd functions and has evaluation zero. The third retains every prime valuation and integral. Changing the coefficient lift may alter this third coordinate by the complete connecting image DMR15; it does not alter either of the first two. The DMR decomposition retains that integral-prime coordinate separately. Accordingly its canonical receiver on this source summand is

$$\widehat{F}_y|_{\mathcal{A}_y^-,0} = (\mathcal{T}_y, 0) : \mathcal{A}_y^-,0 \longrightarrow H_{00} \oplus j_0(W_{\text{pr}}), \quad H_{00} = \{h \in H_{[0]} : \int h = 0\}.\tag{DOR9}$$

Here $\widehat{F}_y$ is exactly DMR18, with its separately retained $j_1(W_{\text{pr}})$ contribution. No replacement receiver has been introduced.

87.3 A proved Schwartz factorization with its construction

We now prove the factorization needed to compute the range, rather than assuming that an arbitrary Schwartz function has appropriate Schwartz factors. For every even $h \in \mathcal{S}(\mathbb{R})$ there exists a positive even $A \in \mathcal{S}(\mathbb{R})$ such that

$$h/A^r \in \mathcal{S}(\mathbb{R}) \quad \text{for every integer } r \geq 1.\tag{DOR10}$$

Choose increasing real numbers T_N , $N \geq 1$, satisfying $T_1 \geq 2$, $T_{N+1} \geq T_N + 2$, and

$$|h^{(j)}(x)| \leq (1+x^2)^{-N^2} \quad (0 \leq j \leq N, \log(1+x^2) \geq T_N).$$

Such numbers exist from the defining Schwartz bounds, with finitely many derivatives specified at each step. Choose a smooth increasing function χ equal to zero on $(-\infty, 0]$ and one on $[1, \infty)$; for definiteness it may be obtained on $(0, 1)$ by integrating $e^{-1/(u(1-u))}$ and dividing by its positive integral over $(0, 1)$, retaining that integral as its defining constant. Define, for $t \geq 0$,

$$\phi(t) = \int_0^t \left(1 + \sum_{N \geq 1} \chi(u - T_N) \right) du, \quad A(x) = \exp\{-\phi(\log(1+x^2))\}.\tag{DOR11}$$

The sum is locally finite. For $T_N \leq t < T_{N+1}$ one has $0 \leq \phi(t) \leq (N+1)t$. For every fixed integer M its derivative is at least M for all sufficiently large t ; integrating this inequality gives $\phi(t)/t \rightarrow \infty$. Derivatives of ϕ of order at least two are uniformly bounded, because their transition intervals $[T_N, T_{N+1}]$ are disjoint. Its first derivative is at most $t+2$, using $T_N \geq N+1$.

Repeated differentiation of $\phi(\log(1+x^2))$ is therefore bounded by a polynomial in $1+|x|$ in every fixed derivative order. Every derivative of A is A times a polynomial in those derivatives. Since $\phi(t)/t \rightarrow \infty$, A and all its derivatives decay faster than every power of $1+|x|$. This proves $A \in \mathcal{S}(\mathbb{R})$, with strict positivity and evenness. For $t = \log(1+x^2) \in [T_N, T_{N+1})$, $j \leq N$, the bounds give

$$|h^{(j)}(x)|/A(x)^r \leq \exp\{-[N^2 - r(N+1)]t\}.$$

For fixed r, j the exponent tends to negative infinity faster than any prescribed fixed negative multiple of t . Derivatives of A^{-r} are A^{-r} times polynomially bounded factors by the same chain-rule calculation. The product rule now proves every Schwartz seminorm of h/A^r finite.

This proves DOR10 completely. No infinite tensor or infinite sum of source vectors was used; DOR11 constructs a single permissible real Schwartz function.

When $h(0) = 0$ and h is even, set

$$G_h(x) = xA(x), \quad F_h(x) = \frac{h(x)}{xA(x)}, \quad F_h(0) = 0, \quad h = F_h G_h. \quad (\text{DOR12})$$

Both functions are odd Schwartz functions. At zero this follows from the exact Taylor formula $h(x) = x^2 \int_0^1 (1-u)h''(ux)du$; its quotient by $xA(x)$ extends smoothly and is odd. Away from zero the division by x and DOR10 imply all Schwartz bounds. This proves the needed factorization with an explicit construction and handles the original zero to its actual order. It does not assert a fixed factor A valid for every h , a linear choice of factors, or an equivariant section of the factorization.

87.4 The exact range and a concrete kernel

For $h \in H_{[0]}$ put

$$\begin{aligned} f_{h,y}(x) &= F_h(x_\infty/y_1)\mathbf{1}_{\widehat{\mathbb{Z}}}(x_f), & g_{h,y}(x) &= G_h(x_\infty/y_2)\mathbf{1}_{\widehat{\mathbb{Z}}}(x_f), \\ k_{h,y} &= f_{h,y} \otimes g_{h,y} \in D \subset K, & \mu_y k_{h,y} &= h(x_\infty)\mathbf{1}_{\widehat{\mathbb{Z}}}(x_f), & m_y(k_{h,y}) &= \int_{\mathbb{R}} h. \end{aligned} \quad (\text{DOR13})$$

Both original individual theta sums vanish by DOR3, not by an assumed arithmetic relation. The displayed pullback proves $\mathcal{T}_y$ onto $H_{[0]}$. For $h \in H_{00}$ the source lies in K_y^0 and has exact ABR coordinates $(h, 0, 0)$. Thus

$$\begin{aligned} \widehat{F}_y([k_{h,y}]_K) &= (h, 0), & 0 &\longrightarrow \mathcal{N}_y^- \longrightarrow \mathcal{A}_y^{-,0} \xrightarrow{\mathcal{T}_y} H_{00} \longrightarrow 0, \\ \mathcal{N}_y^- &= \left\{ \sum_{m,n,j} [m/n] \otimes F_{m,n,j} \otimes G_{m,n,j} : \sum_{m,n,j} F_{m,n,j}(y_1 n x) G_{m,n,j}(y_2 m x) = 0 \text{ for every } x \in \mathbb{R} \right\}. \end{aligned} \quad (\text{DOR14})$$

This is a kernel on the explicitly determined algebraic direct sum DOR5, not an unspecified cohomological kernel. Its moment condition is automatic from the displayed pointwise identity. It is nonzero. Choose linearly independent odd Schwartz functions F, G ; the element in its [1] summand

$$F(\cdot/y_1) \otimes G(\cdot/y_2) - G(\cdot/y_1) \otimes F(\cdot/y_2)$$

has zero image and is nonzero. Nonzero follows by applying an algebraic linear functional on the span of the first two independent factors that takes values 1, 0. Its image is then the nonzero second factor $G(\cdot/y_2)$. The injection DOR3 means this remains a nonzero class in $H_0(G, K)$; it is not an artifact of a chosen presentation of the larger kernel.

The full refined receiver is onto:

$$\widehat{F}_y : S_y \twoheadrightarrow H_{00} \oplus j_0(W_{\text{pr}}), \quad F_y : H_0(G, K_y^0) \twoheadrightarrow Q_0. \quad (\text{DOR15})$$

Here is the exact surjectivity argument including the simultaneous coordinates. DMR24–27 supplies for each prime a class $s_p \in S_y$ with $\beta_0 \widehat{F}_y(s_p) = -\ell_p$. Its first component is an actual $h_p \in H_{00}$. By DOR14 choose s'_p with image $(h_p, 0)$. Then $-s_p + s'_p$ has image $(0, \ell_p)$. Finite combinations give every evaluation-prime vector, and DOR14 gives every H_{00} vector with zero evaluation-prime component. This proves the first surjection. For the second use DMR15, $F_y \delta_y = j_1$, together with DMR17–18; the missing integral-prime component is reached with its original connecting map. This does not identify the full kernel of $\widehat{F}_y$ with the smaller explicit kernel $\mathcal{N}_y^-$. It computes the entire range and computes the exact kernel on the displayed source summand.

For completeness, the whole integral-prime component can also be reached from the odd summand itself. Put $F(x) = G(x) = xe^{-\pi x^2}$ and use the same real rescalings and finite references as DOR13. Its tensor $v \in D$ has

$$m_y(v) = \int_{\mathbb{R}} x^2 e^{-2\pi x^2} dx = \frac{1}{2\pi 2^{3/2}} > 0. \quad (\text{DOR16})$$

The Gaussian identity follows by differentiating $\int e^{-\pi Ax^2} dx = A^{-1/2}$ at $A = 2$, retaining its full factor. For each prime p , the vector $(U_p \otimes U_p - I)v/m_y(v)$ belongs to $D \cap K_y^0$ and has ABR coordinates $(0, 0, \ell_p)$. All sources in $D \cap K_y^0$ have $\beta_0 = 0$, by expansion in the original finite-ball basis and odd real coefficients. Consequently

$$F_y(\text{im}[H_0(G, D \cap K_y^0) \rightarrow H_0(G, K_y^0)]) = H_{00} \oplus j_1(W_{\text{pr}}).$$

The integral-prime classes here may vanish after passage to $H_0(G, K)$, exactly as DMR16 requires. They are retained in the refined source and its DMR17 decomposition.

87.5 Idele actions, theta image and full support

For an idele a retain its finite valuation factor $r = \prod_p p^{v_p(a_p)} > 0$ and original modulus $b = |a_\infty|_\infty / r = |a|$. On T^- , finite units act trivially and real sign $\sigma = \text{sgn}(a_\infty)$ acts by σ . Thus the exact action on DOR4 is

$$U_a \Theta_-(t_c \otimes F) = \sigma \Theta_-(t_{rc} \otimes R_b F), \quad (R_b F)(x) = F(x/b).$$

On D the two signs multiply to 1, the common finite rational factor disappears by the original diagonal rational relation, and the relative index remains unchanged. Therefore

$$\begin{aligned} U_a([c] \otimes F \otimes G) &= [c] \otimes R_{|a|} F \otimes R_{|a|} G, \\ \mathcal{T}_y U_a &= R_{|a|} \mathcal{T}_y, \quad \nu_y U_a = |a| \nu_y. \end{aligned} \quad (\text{DOR17})$$

This proves equivariance of DOR14 for the full original idele action and proves that its kernel is invariant. DOR15 is equivariant by the already proved DMR action; no equivariant section of either surjection is claimed.

For the original source $k_{h,y}$ the independent spherical projection and the diagonal theta receiver give, simultaneously,

$$\begin{aligned} (Q \otimes Q)k_{h,y} &= 0, \quad Q\mu_y k_{h,y} = h \mathbf{1}_{\widehat{\mathbb{Z}}}, \quad (E \otimes E)k_{h,y} = 0, \\ E\mu_y k_{h,y}(u, 1_f) &= 2 \sum_{n \geq 1} h(nu), \quad u > 0, \\ \int_0^\infty E\mu_y k_{h,y}(u, 1_f) u^{s-1} du &= 2\zeta(s) \int_0^\infty h(x) x^{s-1} dx, \quad \text{Re } s > 1. \end{aligned} \quad (\text{DOR18})$$

The first identity follows because each factor is odd under the real sign average. The second is the actual product of two odd functions and hence is even; it is not obtained by asserting commutation of the two averages. The theta series retains the two rational signs, the original zeta function and the original Mellin measure. Its Mellin identity follows by absolute Fubini: the integral of the absolute majorant is $2 \sum_{n \geq 1} n^{-\text{Re } s} \int_0^\infty |h(x)| x^{\text{Re } s-1} dx < \infty$. As h runs over H_{00} , this is the entire source theta image in DMR28, not an uncomputed subset of it. Its further image in the arithmetic quotient by $\overline{E(T_0)}$ remains zero, because these are actual elements of $E(T_0)$. This calculates the surviving source range without converting it into an unsupported spectral assertion.

Finally, for the original bounded distributive support lattice L retain all labels, their original joins and meets, and

$$\begin{aligned} G_L(M) &= \{(0, \lambda) : \lambda \in L\} \cup (M \times \{1_L\}), \quad e = (0, 1_L), \quad \tau = (0, 0_L), \\ A^\uparrow(v, \lambda) &= (Av, \lambda) \quad \text{for every displayed linear map } A. \end{aligned} \tag{DOR19}$$

The quotient uses exactly the same-label amplitude congruence of DMR29. DOR3, DOR5, DOR7, DOR14 and DOR15 therefore have these common-label lifts. In particular the construction of $k_{h,y}$ is a construction of an amplitude preimage, with its one unchanged source label. It does not give two independently assignable tensor labels. Odd cancellation in the original individual theta factors sends top-supported amplitude to e , not τ , and the nonzero diagonal product retains that label. The exact source distinction in DOR18 persists on the full carrier. No ordinary additive homology on the label lattice has been assumed.

87.6 The original CCM degree adjustment inside this source summand

We now apply the range construction to the actual function in Connes–Consani–Marcolli, source lemma `degmodifyV`, with its complete constants already reconstructed in ETC1–26 of the accompanying `CC_EXACT_TRIVIAL_CORRESPONDENCE.tex`. The endpoint values and their two-dimensional inverse are existing results, not new claims of this section. The new receiving calculation locates their complete original family inside the odd tensor summand of the joint kernel.

For each original positive width b define

$$\begin{aligned} \eta_b(x) &= \left(\pi^2(x/b)^4 - \frac{3\pi}{2}(x/b)^2 \right) e^{-\pi(x/b)^2}, \\ F_b(x) &= \pi(x/b) \left(\pi(x/b)^2 - \frac{3}{2} \right) e^{-\pi(x/b)^2/2}, \quad G_b(x) = (x/b) e^{-\pi(x/b)^2/2}, \\ f_{b,y}(z) &= F_b(z_\infty/y_1) \mathbf{1}_{\widehat{\mathbb{Z}}}(z_f), \quad g_{b,y}(z) = G_b(z_\infty/y_2) \mathbf{1}_{\widehat{\mathbb{Z}}}(z_f), \quad t_{b,y} = f_{b,y} \otimes g_{b,y}. \end{aligned} \tag{DOR20}$$

Multiplication gives $F_b G_b = \eta_b$ with every Gaussian exponent, polynomial coefficient and real scale retained. Both factors are odd Schwartz functions. Thus DOR3 and direct integration give

$$t_{b,y} \in D \cap K_y^0, \quad (E \otimes E)t_{b,y} = 0, \quad (Q \otimes Q)t_{b,y} = 0, \quad \mu_y t_{b,y} = \eta_b \mathbf{1}_{\widehat{\mathbb{Z}}}.$$

Indeed $\eta_b(0) = 0$ and its additive integral is $b(\pi^2 3/(4\pi^2) - (3\pi/2)/(2\pi)) = 0$. Its actual diagonal theta image is the original f_b of ETC6: $f_b(u) = 2 \sum_{n \geq 1} \eta_b(nu)$.

The complete original finite data are an arbitrary finitely supported integer vector $k = (k_p)_p$ and an independent real scale $a > 0$. Retain $N(k) = \prod_p p^{k_p} > 0$, allowing negative k_p , and put

$$\begin{aligned} f_{k,a,y}(z) &= F_a(z_\infty/y_1) \prod_p \mathbf{1}_{p^{k_p} \mathbb{Z}_p}(z_p), \quad g_{k,a,y}(z) = G_a(z_\infty/y_2) \prod_p \mathbf{1}_{p^{k_p} \mathbb{Z}_p}(z_p), \\ t_{k,a,y} &= f_{k,a,y} \otimes g_{k,a,y}, \quad b = a/N(k), \\ \mu_y t_{k,a,y} &= \xi_{k,a} = \eta_a(z_\infty) \prod_p \mathbf{1}_{p^{k_p} \mathbb{Z}_p}(z_p), \end{aligned} \tag{DOR20a}$$

$$t_{k,a,y} = (U_{N(k)} \otimes U_{N(k)}) t_{b,y}, \quad \int_{\mathbb{A}_{\mathbb{Q}}} \xi_{k,a} = \frac{a}{N(k)} \left(\frac{\pi^2 3}{4\pi^2} - \frac{(3\pi/2)}{2\pi} \right) = 0.$$

The finite product is exactly $\mathbf{1}_{N(k)\widehat{\mathbb{Z}}}$; its square is itself and its original additive volume is $\prod_p p^{-k_p} = N(k)^{-1}$. Both real factors remain odd, so $t_{k,a,y} \in D \cap K_y^0$ and its independent theta and independent Q images vanish. The displayed rational comparison has the explicit inverse $U_{1/N(k)} \otimes U_{1/N(k)}$. At a finite-unit idele, a rational q contributes exactly when $v_p(q) \geq k_p$ for every prime, equivalently $q = N(k)n$ with $n \in \mathbb{Z} \setminus \{0\}$. Consequently the diagonal theta image

is $f_{a/N(k)}$. The tuple (k, a) remains in the antecedent; the ratio b does not recover it. In ABR coordinates $\xi_{k,a} = \Theta(t_{N(k)} \otimes \eta_b)$, and its prime component is exactly $\sum_p k_p \ell_p \otimes (\eta_b(0), \int \eta_b) = \sum_p k_p \ell_p \otimes (0, 0) = 0$.

To retain the original zeta formula explicitly, the Mellin transform is

$$\begin{aligned} M_b(s) &= 2b^s \zeta(s) \left\{ \frac{\pi^2 \Gamma((s+4)/2)}{2\pi^{(s+4)/2}} - \frac{3\pi \Gamma((s+2)/2)}{4\pi^{(s+2)/2}} \right\}, \\ M_b(0) &= \frac{1}{4}, \quad M_b(1) = \frac{b}{4}, \quad M_{f_b^\#}(s) = \overline{M_b(1 - \bar{s})}. \end{aligned} \tag{DOR21}$$

The first equality follows by absolute Mellin integration for $\operatorname{Re} s > 1$. The entire continuation and endpoint evaluations are ETC10–19: both Gamma summands, the pole of the original zeta and the endpoint zeros are retained there before evaluating their combined limits. The sharp identity uses the original $f^\#(u) = u^{-1}f(u^{-1})$ and its measure du/u .

For prescribed complex-linear endpoint changes (D, C) , choose any $b_1, b_2 > 0$ with $b_1 \neq b_2$ and any fixed original finite vector k . Set $a_i = N(k)b_i$ and retain the original ETC inverse:

$$\begin{aligned} c_1 &= \frac{4(D - b_2 C)}{b_1 - b_2}, \quad c_2 = \frac{4(b_1 C - D)}{b_1 - b_2}, \quad t = c_1 t_{k,a_1,y} + c_2 t_{k,a_2,y} \in D \cap K_y^0, \\ (E \otimes E)t &= 0, \quad (Q \otimes Q)t = 0, \quad E\mu_y t = c_1 f_{b_1} + c_2 f_{b_2}, \\ (M_{E\mu_y t}(1), M_{E\mu_y t}(0)) &= (D, C). \end{aligned} \tag{DOR22}$$

Substitution gives $b_1 c_1 + b_2 c_2 = 4D$ and $c_1 + c_2 = 4C$. If codegree is defined instead as $M_{f^\#}(1)$, replace C in the two coefficients by $\overline{C}$; DOR21 supplies exactly the required conjugation. The original theta image $E\mu_y t$ belongs to $\mathcal{V} = E(T_0)$. Its action on the arithmetic quotient is zero, and the original trace pairing with any admitted second argument is zero by Connes–Consani–Marcolli, Lemma **vanishV** and Proposition **pairTrvanishV**. Those are cited arithmetic representation results, not new sign theorems. DOR22 supplies their actual new tensor antecedent and every one of its maps. Consequently both endpoint values can be varied independently within the same arithmetic quotient class by this explicit source family, while all original factors and their trace contributions remain in DOR21 and the ETC comparison.

The lifted vanishing in DOR22 has the source’s unchanged lattice label as in DOR19. In particular these degree changes occur among top-supported tensors and give supported zero under the individual periodizations; neither the whole tensor nor its support is declared absent. This computes the programme’s exact relation with the CCM trivial-correspondence operation without turning freedom to change degree into an unproved positivity assertion.

88 Global theta majorants and the quotient that retains both endpoints

Original sources and scope of this derivation. The source is Alain Connes, Caterina Consani and Matilde Marcolli, *The Weil proof and the geometry of the adeles class space*, original author file Yuri70v2.tex, canonical source PUBUNIT-5DE2A2D44F8B1470AC701FDA. Its SHA-256 is

b1f694cb934dd4fef829287f1a55dd8de21c67a481e552453c2f7e8fe0ae6fb5.

For this calculation the original passages read were lines 494–589, 623–640, 688–718, 1371–1462, 1631–1751, 2181–2284, and 2286–2435. The exact locators are SCK, rangeVdef, varthetaH1V, varthetaafconvol, rhoFourier, degcorrZf, codegcorrZf,

PositSect, ZfbulletZf, fubini1–fubini4, nothappen and degmodifyV. The adelic measure conventions are also those of Alain Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Section III, as retained in the accompanying ABR derivation.

The original polynomial Gaussian comes from CCM's degmodifyV. The prior programme proof GTK1–9 in GLOBAL_THETA_WEIL_KERNEL.tex already proves its theta image strictly positive, its strong decay, its full Mellin formula and its two endpoint values. Those facts are inputs, not new results here. The original test topology, convolution and involution proofs are SWQ1–15 in CC_SUPPORTED_WEIL_QUOTIENT.tex. These precise source passages and programme statements were read for this derivation. The new construction proves cofinal domination for the entire original real test space, an actual adelic antecedent for each majorant, and the exact resulting ordered quotient with degree and codegree retained. The last section distinguishes this test-function order from the geometric effectivity used by Weil and discussed by CCM.

88.1 The original strong test space and the existing theta test

For the rational field retain the actual product and measures

$$\begin{aligned}
 C_{\mathbb{Q}} &= \mathbb{R}_{>0} \times K, & K &= \widehat{\mathbb{Z}}^{\times}, & d^{\times}c &= dy/y du, & \int_K du &= 1, \\
 B &= \mathbf{S}(C_{\mathbb{Q}}) = \bigcap_{\beta \in \mathbb{R}} |\cdot|^{\beta} \mathcal{S}(C_{\mathbb{Q}}), & B_{\mathbb{R}} &= \{g \in B : g \text{ is real-valued}\}, \\
 T &= \mathcal{S}(\mathbb{R}) \otimes \bigotimes_p' (\mathcal{S}(\mathbb{Q}_p), \mathbf{1}_{\mathbb{Z}_p}), & T_0 &= \{\xi \in T : \xi(0) = 0, \int_{\mathbb{A}} \xi = 0\}, \\
 E\xi(y, u) &= \sum_{q \in \mathbb{Q}^{\times}} \xi(qy, qu), & V &= E(T_0), & V_{\mathbb{R}} &= V \cap B_{\mathbb{R}}.
 \end{aligned} \tag{CTO1}$$

CCM's definition SCK uses the intersection of all real weights. The symbol B here does not mean the larger space with only ordinary Schwartz decay in $\log y$. At a fixed compact open level in K its original seminorms are

$$p_{A,m,j}(g) = \sup_{t \in \mathbb{R}, u \in K} e^{A|t|} (1 + |t|)^m |\partial_t^j g(e^t, u)|, \quad A, m, j \in \mathbb{Z}_{\geq 0}. \tag{CTO2}$$

The full Bruhat topology is the inductive limit over these finite compact-unit levels. Every test considered below remains at a specified finite level; the constructed majorants are at the trivial level. All additive integrals use Lebesgue measure at infinity and $\text{vol}(\mathbb{Z}_p) = 1$ at each finite place.

Retain the original seed and all its factors:

$$\begin{aligned}\eta(x) &= \pi x^2(\pi x^2 - \tfrac{3}{2})e^{-\pi x^2}, & \xi(x_\infty, x_f) &= \eta(x_\infty) \prod_p \mathbf{1}_{\mathbb{Z}_p}(x_p), \\ h(y) &= E\xi(y, u) = 2 \sum_{n \geq 1} \eta(ny), & H(t) &= h(e^t), \\ \xi(0) &= 0, \quad \int_{\mathbb{A}} \xi = 0, \quad h \in V_{\mathbb{R}}, \quad h(y) > 0, \quad h(y) = y^{-1}h(y^{-1}).\end{aligned}\tag{CTO3}$$

These are the proved GTK1–5 inputs. The additive zero integral is the exact cancellation $3/4 - 3/4$ of the quartic and quadratic Gaussian moments. The factor two in $E\xi$ retains both rational signs, and the finite factors permit exactly the nonzero integers. All derivatives of H satisfy the strong seminorms CTO2.

An explicit lower bound on the interval used below is

$$H(t) \geq \kappa := 2\pi(\pi - \tfrac{3}{2}) \exp\{-\pi e^{3/2}\} > 0 \quad (|t| \leq 3/4).\tag{CTO4}$$

For $0 \leq t \leq 3/4$, retain the $n = 1$ term in CTO3; all other terms are positive. In that term $e^{2t} \geq 1$, $\pi e^{2t} - 3/2 \geq \pi - 3/2$, and $e^{-\pi e^{2t}} \geq e^{-\pi e^{3/2}}$. Their product proves the bound. For negative t , CTO3 gives $H(t) = e^{-t}H(-t)$, and its extra positive factor is at least one.

88.2 A global majorant in the original image

Fix any $g \in B_{\mathbb{R}}$, including its full dependence on K . Define the single compact bump and its original integral by

$$\begin{aligned}\psi(t) &= \begin{cases} \exp\{-1/(1 - 16t^2)\}, & |t| < 1/4, \\ 0, & |t| \geq 1/4, \end{cases} & b_\psi &= \int_{\mathbb{R}} \psi(t) dt > 0, \\ M_n &= \sup_{u \in K, |t-n| \leq 1/2} |g(e^t, u)|, & a_n &= \frac{M_n + e^{-n^2}}{\kappa b_\psi} \quad (n \in \mathbb{Z}), \\ W(t) &= \sum_{n \in \mathbb{Z}} a_n \psi(t - n), & w(y, u) &= W(\log y).\end{aligned}\tag{CTO5}$$

The compact supremum exists and is finite: at the original finite unit level it is the maximum of finitely many continuous real functions on a compact interval. Each a_n is strictly positive. For every $A \geq 0$, CTO2 implies

$$M_n \leq p_{A,0,0}(g) \exp\{-A \max(|n| - 1/2, 0)\}.$$

It follows, using an arbitrarily larger exponential weight, that

$$\sum_{n \in \mathbb{Z}} a_n e^{A|n|} (1 + |n|)^m < \infty \quad \text{for every } A, m \geq 0.\tag{CTO6}$$

For the added term e^{-n^2} this follows directly from its quadratic exponent. For the M_n term choose the estimate with weight $A + 2$ and bound $(1 + |n|)^m e^{-2|n|}$ by a summable sequence. Noninteger A, m are covered by larger integer values.

The bump sum is locally finite, and its supports are disjoint. For every seminorm in CTO2, each derivative of its tail is bounded by

$$p_{A,m,j} \left(\sum_{|n| > N} a_n \psi(\log y - n) \right) \leq \|\psi^{(j)}\|_\infty \sup_{|n| > N} a_n e^{A(|n|+1/4)} (1 + |n| + 1/4)^m \longrightarrow 0.\tag{CTO7}$$

Thus $w \in B_{\mathbb{R}}$ and the displayed series converges in the original topology at the trivial compact-unit level, with every derivative. In particular all integrals $\int |W(t)|e^{A|t|}(1+|t|)^m dt$ are finite, by integrating the bumps and using CTO6. The coefficients do not need to dominate the derivatives of g ; they construct w and its derivatives from a fixed bump, while the weighted values of g suffice for the domination asserted below.

Use the original multiplicative convolution and keep its Haar factors:

$$v(y, u) = (h * w)(y, u) = \int_K \int_0^\infty h(y/a)w(a, k) \frac{da}{a} dk = \int_{\mathbb{R}} H(\log y - t)W(t) dt. \quad (\text{CTO8})$$

The compact integral has value factor one; both inputs are radial, so the resulting function is independent of u . Convolution remains in B with its original seminorms. For completeness the inequality $|s| \leq |t| + |s - t|$ and $1 + |s| \leq (1 + |t|)(1 + |s - t|)$ gives

$$p_{A,m,j}(h * w) \leq 2p_{A+1,m,0}(w)p_{A,m,j}(h),$$

because $\int_{\mathbb{R}} e^{-|t|} dt = 2$ and $\int_K dk = 1$. The same integrable majorant justifies every derivative and proves the strong convergence after replacing W by its finite sums.

Given a real s , choose an integer n with $|s - n| \leq 1/2$. For every t in the support of $\psi(t - n)$, one has $|s - t| \leq 3/4$. Since all terms in CTO8 are nonnegative, CTO4 gives the exact domination

$$v(e^s, u) \geq \kappa a_n b_\psi = M_n + e^{-n^2} > |g(e^s, u)| \quad (s \in \mathbb{R}, u \in K). \quad (\text{CTO9})$$

This is a global inequality on all of $C_{\mathbb{Q}}$, not a finite interval or a finite set of compact-unit characters.

88.3 The actual adelic antecedent and all convergence

Membership in the algebraic image V , rather than merely its closure, has an explicit original Schwartz antecedent:

$$\eta_W(x) = \int_{\mathbb{R}} W(t)\eta(e^{-t}x) dt, \quad \xi_W(x_\infty, x_f) = \eta_W(x_\infty) \prod_p \mathbf{1}_{\mathbb{Z}_p}(x_p). \quad (\text{CTO10})$$

For $P_{M,j}(\eta) = \sup_x (1 + |x|)^M |\eta^{(j)}(x)|$, a real dilation has the precise seminorm bound

$$\begin{aligned} \sup_x (1 + |x|)^M \left| \frac{d^j}{dx^j} \eta(e^{-t}x) \right| &\leq P_{M,j}(\eta) e^{M \max(t,0) - jt} \\ &\leq P_{M,j}(\eta) e^{(M+j)|t|}. \end{aligned} \quad (\text{CTO11})$$

The first inequality uses the actual factor e^{-jt} and $(1 + e^t|z|)^M \leq e^{M \max(t,0)}(1 + |z|)^M$. CTO6–7 make the integral of this bound against $|W(t)|$ finite in every seminorm. Differentiating on compact x intervals and then applying the same bound proves that CTO10 is Schwartz and that its integral converges in the Schwartz topology. One may equivalently first integrate each compact bump and then sum; the resulting series is absolutely convergent in each seminorm. This constructs one real Schwartz function and a single permissible adelic tensor with every finite reference factor retained. It does not introduce an infinite algebraic tensor expansion.

Its two source endpoints vanish with absolute justifications:

$$\begin{aligned} \xi_W(0) &= \int W(t)\eta(0) dt = 0, \\ \int_{\mathbb{A}} \xi_W &= \left(\prod_p \text{vol}(\mathbb{Z}_p) \right) \int_{\mathbb{R}} W(t)e^t dt \int_{\mathbb{R}} \eta(x) dx = 1 \cdot \int_{\mathbb{R}} W(t)e^t dt \cdot (3/4 - 3/4) = 0. \end{aligned} \quad (\text{CTO12})$$

For the second exchange the integral of the absolute values is $\int |W(t)|e^t dt \int |\eta(x)|dx < \infty$. Thus $\xi_W \in T_0$ in the original topology.

To prove its exact theta image, put $P_2 = \sup_x (1 + |x|)^2 |\eta(x)|$. For every $z > 0$, monotone integral comparison gives

$$\sum_{n \geq 1} |\eta(nz)| \leq P_2 \sum_{n \geq 1} (1 + nz)^{-2} \leq P_2 \int_0^\infty (1 + uz)^{-2} du = P_2/z.$$

Consequently, at each $y > 0$,

$$\begin{aligned} \sum_{n \neq 0} \int_{\mathbb{R}} |W(t) \eta(ne^{-t}y)| dt &\leq \frac{2P_2}{y} \int_{\mathbb{R}} |W(t)| e^t dt < \infty, \\ E\xi_W(y, u) &= \int_{\mathbb{R}} W(t) 2 \sum_{n \geq 1} \eta(ne^{-t}y) dt = v(y, u), \quad v \in V_{\mathbb{R}}. \end{aligned} \tag{CTO13}$$

The finite argument is still $nu \in \widehat{\mathbb{Z}}$ precisely for integral n ; unit coordinates and the two signs have not been dropped. CTO13 uses an absolutely convergent exchange for this fixed y and t integral. It does not interchange a multiplicative endpoint integral with the original integer sum, the invalid exchange in CCM's `nothappen`.

88.4 The original Mellin factors and the endpoint cost

Use CCM's Fourier convention without an inverse character:

$$\widehat{f}(\chi, s) = \int_K \int_0^\infty f(y, u) \chi(u) y^s \frac{dy}{y} du, \quad M_f(s) = \widehat{f}(1, s). \tag{CTO14}$$

Every such transform is entire, with derivatives obtained by inserting $(\log y)^j$, by CTO2. Set

$$B_\psi(s) = \int_{-1/4}^{1/4} \psi(t) e^{st} dt, \quad M_w(s) = \int_{\mathbb{R}} W(t) e^{st} dt = B_\psi(s) \sum_{n \in \mathbb{Z}} a_n e^{sn}.$$

CTO6 proves local uniform convergence of this series and of all its derivatives. Absolute weighted Fubini in CTO8 gives the full original arithmetic formula

$$\begin{aligned} M_v(s) &= M_h(s) M_w(s), \\ M_h(s) &= 2\zeta(s) \left\{ \frac{\pi^2 \Gamma((s+4)/2)}{2\pi^{(s+4)/2}} - \frac{3\pi \Gamma((s+2)/2)}{4\pi^{(s+2)/2}} \right\} \quad (\operatorname{Re} s > 1). \end{aligned} \tag{CTO15}$$

The first identity holds for every complex s . The second is GTK7, obtained on its specified domain by integrating the two original Gaussian monomials separately. Its equality to the entire Mellin integral specifies continuation at each removable point of the product; no individual zero, pole, factor or summand is deleted. In particular the original zeta function has not been replaced by a completed one.

The known original endpoint values are $M_h(0) = M_h(1) = 1/4$ (GTK9). Therefore the exact costs of the majorant are

$$\begin{aligned} C_v &:= M_v(0) = \frac{1}{4} \int_{\mathbb{R}} W(t) dt = \frac{1}{4} b_\psi \sum_n a_n > 0, \\ D_v &:= M_v(1) = \frac{1}{4} \int_{\mathbb{R}} W(t) e^t dt = \frac{1}{4} B_\psi(1) \sum_n a_n e^n > 0. \end{aligned} \tag{CTO16}$$

These are codegree and degree respectively in CCM's `codegcorrZf` and `degcorrZf`. They coexist with the two additive zeros CTO12 because their receiving integrals are different, and CTO13

does not perform the invalid endpoint Fubini exchange. Adding this particular v changes both endpoint values by the displayed positive amounts.

The construction retains the full spectral jets. For a nontrivial zero ρ of the original ζ of multiplicity m_ρ , the Gamma expressions in CTO15 are holomorphic at ρ and M_w is entire. Thus

$$\begin{aligned} M_v^{(j)}(\rho) &= 0 \quad (0 \leq j < m_\rho), \\ \widehat{v}(\chi, s) &= 0 \quad \text{for every nontrivial continuous character } \chi \text{ of } K \text{ and every } s. \end{aligned} \tag{CTO17}$$

The first statement retains at least the full original multiplicity; additional zeros of M_w may increase it. The second follows from radiality and Haar invariance: if $\chi(u_0) \neq 1$, then $\int_K \chi(u) du = \chi(u_0) \int_K \chi(u) du$, forcing that integral to zero. All its s derivatives are zero as well. Thus adding the constructed majorant leaves those values and jets unchanged.

88.5 The complete quotient-cone consequence and fixed endpoints

Let $B_{\mathbb{R},+}$ be the cone of everywhere nonnegative real tests, and let $q : B_{\mathbb{R}} \rightarrow Q_{\mathbb{R}} := B_{\mathbb{R}}/V_{\mathbb{R}}$ be the algebraic quotient. CTO9 and CTO13 prove, for every real g , the two representatives

$$g + v > 0, \quad g - v < 0, \quad q(g + v) = q(g - v) = q(g). \tag{CTO18}$$

In particular

$$q(B_{\mathbb{R},+}) = Q_{\mathbb{R}}, \quad q(\{f \in B_{\mathbb{R}} : f > 0\}) = Q_{\mathbb{R}}.$$

This evaluates the entire image cone. Its associated preorder is the universal relation: every quotient class is greater than or equal to every other one. It does not separate any distinct quotient classes. The same conclusion holds after replacing V by its closure, because the very same actual antecedents already lie in V ; no new closed-image assertion is needed.

Nonzero nonnegative tests have strictly positive values at both endpoints. Indeed continuity at a positive value gives an open set of positive Haar measure on which the function is positive; both $du dy/y$ and $y du dy/y$ assign positive measure there. Their integrals are finite by CTO2. Therefore, putting $B_{\mathbb{R},00} = \ker(M_0, M_1)$, one has

$$B_{\mathbb{R},+} \cap \ker M_0 = B_{\mathbb{R},+} \cap \ker M_1 = B_{\mathbb{R},+} \cap B_{\mathbb{R},00} = \{0\}. \tag{CTO19}$$

Consequently a nonzero nonnegative correction cannot preserve either zero endpoint change, and in particular cannot preserve both prescribed degree and codegree.

There is a precise necessary bound for any nonnegative representative with fixed endpoints $C = M_f(0)$, $D = M_f(1)$. For every continuous character χ of K and every $\rho = \sigma + i\gamma$ with $0 < \sigma < 1$, the original integral and Hölder's inequality give

$$\begin{aligned} |\widehat{f}(\chi, \rho)| &\leq \int_K \int_0^\infty f(y, u) y^\sigma \frac{dy}{y} du \\ &= \int_K \int_0^\infty f(y, u)^{1-\sigma} \{y f(y, u)\}^\sigma \frac{dy}{y} du \\ &\leq C^{1-\sigma} D^\sigma. \end{aligned} \tag{CTO20}$$

Here $|\chi(u)| = |y^{i\gamma}| = 1$, and the Hölder exponents are $1/(1-\sigma)$ and $1/\sigma$ on the original measure. For the zero test both sides are zero; otherwise CTO19 proves $C, D > 0$. This is a receiving inequality with an endpoint budget. CTO9 by itself supplies no uniform values of that budget, while CTO16 gives its exact cost for each input and CTO17 retains the indicated spectral data.

88.6 An actual ordered extension retaining both endpoints

The failed quotient order defines a more informative exact receiver. Set

$$N = V_{\mathbb{R}} \cap B_{\mathbb{R},00}, \quad \mathcal{E}_{\text{ord}} = B_{\mathbb{R}}/N, \quad \mathcal{E}_{\text{ord},+} = \pi_N(B_{\mathbb{R},+}). \quad (\text{CTO21})$$

These are algebraic quotients with the original topology available; no closedness claim about N or the cone is made. The real linear endpoint map $\epsilon(f) = (M_f(0), M_f(1))$ descends to $\mathcal{E}_{\text{ord}}$. For $b > 0$ write $h_b(y) = h(y/b)$, with the exact source $\eta(x_{\infty}/b) \prod_p \mathbf{1}_{\mathbb{Z}_p}$. The source still has evaluation zero and additive integral $b(3/4 - 3/4) = 0$, so $h_b \in V_{\mathbb{R}}$. The multiplicative substitutions give $\epsilon(h_b) = (1/4, b/4)$. In particular the explicit linear map

$$j(C, D) = [4(2C - D)h + 4(D - C)h_2]_N$$

has endpoints (C, D) . A different representative in $V_{\mathbb{R}}$ with the same endpoints differs by exactly an element of N , so this is the canonical identification of the kernel with its endpoint pair, independent of that selected formula. We obtain the exact sequence

$$0 \longrightarrow \mathbb{R}^2 \xrightarrow{j} \mathcal{E}_{\text{ord}} \xrightarrow{q_{\text{ord}}} Q_{\mathbb{R}} \longrightarrow 0, \quad \epsilon j = I_{\mathbb{R}^2}. \quad (\text{CTO22})$$

Indeed the middle kernel is $V_{\mathbb{R}}/N$, whose endpoint map is injective by the definition of N and onto by the displayed h, h_2 formula. Surjectivity of q_{ord} is the original quotient map. Every map in this sequence has therefore been specified on its original representatives.

This linear extension splits canonically; its additional information is its positive cone, not an asserted non-split linear structure. The exact isomorphism and inverse are

$$\begin{aligned} \Upsilon : \mathcal{E}_{\text{ord}} &\xrightarrow{\cong} Q_{\mathbb{R}} \oplus \mathbb{R}^2, & \Upsilon([f]_N) &= ([f]_V, M_f(0), M_f(1)), \\ \Upsilon^{-1}(q, C, D) &= [g]_N + j(C - M_g(0), D - M_g(1)), & [g]_V &= q. \end{aligned} \quad (\text{CTO22a})$$

If g is replaced by $g+v$ with $v \in V_{\mathbb{R}}$, then $[v]_N = j(M_v(0), M_v(1))$ by the exact kernel identification. The two added terms therefore cancel in the inverse formula. Thus the inverse is independent of the representative and is linear. Applying either composite proves the identity, so surjectivity and injectivity are both established. The real linear section of q_{ord} is $q \mapsto \Upsilon^{-1}(q, 0, 0)$. No degree or codegree contribution is omitted in this decomposition.

The original convolution algebra structure is retained too. To recall its required source calculation, for $f \in B$ and $\xi \in T_0$ the actual section $C_{\mathbb{Q}} \rightarrow \mathbb{A}^{\times}$ of CTO1 gives

$$\xi_f(x) = \int_C f(a) \xi(a^{-1}x) d^{\times}a, \quad E\xi_f = f * E\xi.$$

SWQ11–14 proves this directly. The compact unit orbit of a fixed finite Bruhat test spans a finite-dimensional space, and each real Schwartz seminorm is bounded by a constant times $\int_C |f(a)| \max(1, |a|)^M |a|^{-j} d^{\times}a < \infty$. Its endpoints are $\xi_f(0) = \xi(0)M_f(0)$ and $\int \xi_f = (\int \xi)M_f(1)$, both zero. For each fixed receiving idele the absolute rational sum has a majorant $C_{\xi}(1 + |a|)$; integrating against $|f(a)|$ justifies the last displayed identity. These are the original source maps and factors, proving V is a convolution ideal. On real tests, complex conjugation shows the same for $V_{\mathbb{R}}$. Absolute multiplicative Fubini gives $M_{f*g}(0) = M_f(0)M_g(0)$ and $M_{f*g}(1) = M_f(1)M_g(1)$, retaining $|ab| = |a||b|$ in the second identity. Thus N is a convolution ideal and

$$\begin{aligned} \Upsilon([f * g]_N) &= ([f]_V * [g]_V, M_f(0)M_g(0), M_f(1)M_g(1)), \\ j(C, D) * [f]_N &= j(CM_f(0), DM_f(1)). \end{aligned} \quad (\text{CTO22b})$$

This proves the direct-product algebra structure on the same canonical coordinates; it introduces no convolution identity among the original test functions.

The original involution is $f^\sharp(c) = |c|^{-1} \overline{f(c^{-1})}$. With the self-dual additive Fourier transform retained by SWQ6–10, the full Poisson endpoint comparison is

$$\begin{aligned} (E\xi)^\sharp(c) &= E\widehat{\xi}(c) - |c|^{-1} \overline{\xi(0)} + \int_{\mathbb{A}} \xi, \\ M_{f^\sharp}(0) &= \overline{M_f(1)}, \quad M_{f^\sharp}(1) = \overline{M_f(0)}. \end{aligned} \tag{CTO22c}$$

The first equality retains the original real Fourier kernel $e^{-2\pi i x v}$, finite conductor $\mathbb{Z}_p$ and both source endpoints; SWQ7–10 proves it by Poisson summation. The Fourier antecedent has endpoints $\int \xi, \xi(0)$, so $V^\sharp = V$. The last two equations follow by $c \mapsto c^{-1}$ in the original Haar integrals, including the factor $|c|^{-1}$. Thus sharp preserves N and, on the real ordered extension, acts by the original quotient involution and interchange of the two real endpoints. It preserves its positive cone because the multiplying modulus is positive. No Hilbert-space positivity is inferred from this algebraic involution.

The cone of CTO21 is pointed and generating. If x and $-x$ have nonnegative representatives f and g , then $f + g \in N$, so $M_{f+g}(0) = 0$. CTO19 implies $f + g = 0$, hence $f = g = 0$ and $x = 0$. Generating follows from CTO18: $[g]_N = [g + v]_N - [v]_N$, with both terms nonnegative. Its projection to $Q_{\mathbb{R}}$ is the entire quotient by CTO18. Its intersection with the exact kernel in CTO22 is completely determined:

$$j^{-1}(\mathcal{E}_{\text{ord},+} \cap \ker q_{\text{ord}}) = \{(C, D) : C > 0, D > 0\} \cup \{(0, 0)\}. \tag{CTO23}$$

For necessity a representative of a kernel class belongs to $V_{\mathbb{R}}$, and CTO19 gives both strict inequalities unless it is zero. For sufficiency, if $C, D > 0$, the actual function $4Ch_{D/C} > 0$ belongs to $V_{\mathbb{R}}$ and has endpoints $4C(1/4, (D/C)/4) = (C, D)$. This gives its exact representative in the image of j . The origin is represented by zero. Thus the two coordinate axes, apart from their origin, are not silently included in this cone. Under CTO22a this cone is therefore not the naive product $Q_{\mathbb{R}} \times \mathbb{R}_{\geq 0}^2$: for example its kernel coordinate $(0, 1, 0)$ has no nonnegative representative, by CTO19 and CTO23. The calculation of the kernel cone does not assert that the complete positive cone at each prescribed nonzero spectral class and each prescribed endpoint pair has been evaluated; CTO20 supplies an exact constraint there.

For an idele class a , the original translation $U_a f(c) = f(a^{-1}c)$ preserves pointwise nonnegativity, V , and N . Invariance of V follows already at the source by $EU_a = U_a E$, and the additive source integral is multiplied by $|a|$ while its zero remains zero. The original multiplicative change of variable gives $\epsilon(U_a f) = (M_f(0), |a|M_f(1))$. Consequently CTO22 is equivariant with kernel action $(C, D) \mapsto (C, |a|D)$, and the cone CTO23 is invariant. The real linear maps in the sequence, rather than a selected nonlinear majorant, carry this equivariance.

88.7 The closed arithmetic quotient and the exact topology of the cone

The algebraic calculation has an exact map to CCM's quotient by the closed theta image. Keep the original topology CTO2 and define

$$\begin{aligned} s(C, D) &= 4(2C - D)h + 4(D - C)h_2, \quad P = I - s\epsilon, \\ \overline{Q}_{\mathbb{R}} &= B_{\mathbb{R}}/\overline{V}_{\mathbb{R}}, \quad \overline{\mathcal{E}}_{\text{ord}} = B_{\mathbb{R}}/\overline{N}. \end{aligned} \tag{CTO25}$$

The bar on $\overline{\mathcal{E}}_{\text{ord}}$ here denotes this specified quotient, not an unproved completion of an ordered space. The endpoint integrals are continuous: each is bounded by a constant times $p_{2,0,0}$ on each compact-unit level. The fixed two-function map s is continuous, has image in $V_{\mathbb{R}}$, and satisfies $\epsilon s = I$. Hence P is a continuous projection onto $\ker \epsilon$, with $P(V_{\mathbb{R}}) = N$. Continuity of ϵ first gives $\overline{N} \subseteq \overline{V}_{\mathbb{R}} \cap \ker \epsilon$. Conversely, if z is in the latter intersection, a net $v_i \in V_{\mathbb{R}}$ tends to z and $Pv_i \in N$ tends to $Pz = z$. Therefore

$$\overline{N} = \overline{V}_{\mathbb{R}} \cap \ker \epsilon, \quad \overline{\Upsilon} : \overline{\mathcal{E}}_{\text{ord}} \xrightarrow{\cong} \overline{Q}_{\mathbb{R}} \oplus \mathbb{R}^2, \quad [f]_{\overline{N}} \mapsto ([f]_{\overline{V}_{\mathbb{R}}}, M_f(0), M_f(1)). \tag{CTO26}$$

This is a topological real vector-space isomorphism, not only an algebraic bijection. Indeed $f \mapsto (Pf, \epsilon f)$ is a continuous isomorphism from $B_{\mathbb{R}}$ to $\ker \epsilon \oplus \mathbb{R}^2$, with continuous inverse $(z, C, D) \mapsto z + s(C, D)$. It takes $V_{\mathbb{R}}$ to $N \oplus \mathbb{R}^2$ and its closure to $\overline{N} \oplus \mathbb{R}^2$. Passing to these quotient topologies proves (CTO26) and its inverse. The same argument without bars also makes CTO22a a topological isomorphism in the possibly non-Hausdorff quotient topologies.

Complex conjugation preserves V , since it commutes with E . Taking real parts of a convergent net proves $\overline{V_{\mathbb{R}}} = \overline{V} \cap B_{\mathbb{R}}$. Thus (CTO26) uses the actual real part of the closed-image quotient. Multiplication by a fixed original test is continuous, by the weighted convolution estimate following CTO8. Consequently the closures of the ideals $V_{\mathbb{R}}$ and N remain ideals. The endpoint characters give exactly the product law CTO22b on (CTO26); it is an isomorphism of the original convolution algebras. Sharp and every fixed idele translation are continuous by their explicit formulas, so CTO22c and its original actions also descend. No closedness assumption on V or N has been inserted.

Give $\overline{\mathcal{E}}_{\text{ord}}$ the image of $B_{\mathbb{R},+}$, without closing that cone. Its cone is pointed by the identical CTO19 argument, since every element of $\overline{N}$ has both endpoints zero. It is generating, its projection to $\overline{Q}_{\mathbb{R}}$ is onto, and its exact kernel cone is still CTO23: necessity uses the positive endpoint integrals and sufficiency uses the same function $4Ch_{D/C}$.

Both the algebraic and the closed-image versions have a positive cone which is *not closed* in their stated quotient topology. For the first, and identically for the second, continuity of j gives

$$j(1, 1/n) = [4h_{1/n}]_N \in \mathcal{E}_{\text{ord},+}, \quad j(1, 1/n) \longrightarrow j(1, 0) \notin \mathcal{E}_{\text{ord},+}. \quad (\text{CTO27})$$

The convergence is of quotient classes through the fixed continuous map s ; convergence of $4h_{1/n}$ in the original test space is not asserted. The excluded limit follows from CTO23. Similarly $j(0, 1) \not\geq 0$, whereas $j(0, 1) + \epsilon j(1, 1) = j(\epsilon, 1 + \epsilon) > 0$ for every $\epsilon > 0$. Thus this order is not Archimedean: positive perturbations by arbitrarily small multiples of this order unit do not force positivity at their limit.

There is nevertheless an actual order unit $u_{\text{ord}} = j(1, 1)$ for the entire extension. For $x = [g]_N$, the constructed $v > |g|$ has endpoints $C_v, D_v > 0$ and $[v]_N = j(C_v, D_v)$. For every real $R > \max(C_v, D_v)$,

$$Ru_{\text{ord}} \pm x = [v \pm g]_N + j(R - C_v, R - D_v) \in \mathcal{E}_{\text{ord},+}. \quad (\text{CTO28})$$

Both summands have actual positive representatives, by CTO9 and CTO23. This proves the order-unit assertion on every class and also on (CTO26), keeping its full individual cost. It does not justify replacing this cone by its closure, passing a positive limit back into it, or declaring the original arithmetic trace positive on it. Those inferences would contradict or exceed the computed cone properties.

88.8 Geometric scope and every original support label

CCM's source discussion `PositSect` uses Riemann–Roch on a curve: after adjusting a correspondence's degree, a principal correspondence produces geometric effectivity in its linear equivalence class. The preceding definition of effective correspondence is a sum of curves with nonnegative integer multiplicities, with intersection products and Frobenius action. The later `degmodifyV` gives freedom to adjust degree by an actual trivial theta correspondence. The calculation above retains and strengthens the test-function part of that comparison: it gives a positive representative in every real test quotient class, computes its endpoint costs, and constructs the exact ordered extension retaining them. It does not identify a pointwise-nonnegative test function with an effective divisor, construct the missing principal-divisor relation, or establish the intersection inequality `ZfbulletZf`. The universal cone in CTO18 is consequently an evaluated defect of this particular proposed order, not a no-go statement about all geometric notions of effectivity.

For an arbitrary bounded distributive lattice L retain

$$G_L(M) = \{(0, \lambda) : \lambda \in L\} \cup (M \times \{1_L\}), \quad e = (0, 1_L), \quad \tau = (0, 0_L), \quad A^\uparrow(z, \lambda) = (Az, \lambda) \quad (\text{CTO24})$$

for every displayed linear map A on its specified amplitude spaces. The originating support source, by The Clankers, is *Split Support Geometry*, v11, Definition `def:lattice-split`. Addition retains join and scalar multiplication retains meet. All quotients above use the same-label amplitude congruence; CTO22 lifts as the corresponding exact amplitude description under those congruences. Every zero amplitude retains its assigned label, so the positive theta majorant at top support maps to e in $G_L(Q_{\mathbb{R}})$, not to τ .

The cofinal choice $g \mapsto v$ is nonlinear and is asserted as a strict majorant construction for top-supported inputs. A lower-supported zero $(0, \lambda)$ has no admissible nonzero amplitude at that label. For it choose $(0, \lambda)$ itself as a non-strict majorant. Thus no strict positive amplitude is inserted into a forbidden lower support stratum. The convergent series and integrals CTO5–13 are operations on amplitude functions with one admissible unchanged label; no infinite join or integration on an arbitrary lattice is postulated. The endpoint pair in CTO22 likewise has one common label, not independently assigned labels on its two coordinates.

This constructs a pointed ordered extension with exact degree and codegree, while evaluating why the pointwise cone on the unextended spectral quotient fails to separate its classes. It supplies neither an assumed positivity theorem for the original Weil form nor an RH conclusion.

89 The original divisor, every supported section level, and its theta image

Sources and the comparison being proved. Alain Connes and Caterina Consani, *Riemann–Roch for the ring $\mathbb{Z}$* , original author file `RRZ.tex`, define the bounded $\mathbb{S}$ -module by the *all-subset* condition in `xlem`, `xdefn`, and `xdefnr`, and its sums and dimension in `defnsum` and Definition `generator`. Their Theorem `RRZ` calculates H^0 ; Proposition `caseh1` and Theorem `rrspzb` concern their stated tolerant H^1 and Riemann–Roch identity. The source SHA-256 is `1975ade5af5d3facf0f954da414a85c07c8adb7a83d1cc08e38f3112c5d072f9`.

The Gaussian below is the one in Connes–Consani–Marcolli, *The Weil proof and the geometry of the adeles class space*, `degmodifyV`; its Mellin and trace conventions are `rhoFourier`, `degcorrZf`, `codegcorrZf`, and `pairTrvanishV`.

The full support operations come from The Clankers, *Split Support Geometry*, version 11, Definition `def:lattice-split`. The accompanying programme proof `SRR1–8`, `SUPPORTED_RIEMANN_ROCH_SECTIONS.tex`, already determines the supported integer-section module and its dimension. `DOR20a` gives the exact finite-prime diagonal antecedent. We prove their common divisor dictionary, the full section counts, and the counting-to-theta transform here. These are maps between specified objects; the tolerant H^1 of the Riemann–Roch source is not assumed to be the arithmetic spectral H^1 of the adelic trace construction.

89.1 All finite divisor coefficients and the archimedean bound

Let L be a bounded distributive lattice, $0_L \neq 1_L$. Retain an arbitrary Arakelov divisor with finite support at the finite places:

$$\begin{aligned} D &= \sum_p d_p[p] + d_\infty[\infty], \quad d_p \in \mathbb{Z}, \quad d_\infty \in \mathbb{R}, \\ I(D) &= \{q \in \mathbb{Q} : v_p(q) \geq -d_p \text{ for every } p\}, \quad v_p(0) = +\infty, \\ N &= \prod_p p^{-d_p} > 0, \quad a = e^{d_\infty} > 0, \quad b = a/N, \quad \deg D = \sum_p d_p \log p + d_\infty. \end{aligned} \tag{RDT1}$$

Here N is a positive rational number, not necessarily an integer. For nonzero rational q , the inequalities defining $I(D)$ say that $v_p(q/N) \geq 0$ at every prime. Writing a rational number as a reduced fraction proves that this is equivalent to $q/N \in \mathbb{Z}$. The assertion also holds at zero. Hence, with the original parameters still present,

$$I(D) = N\mathbb{Z}, \quad \prod_p \text{vol}(p^{-d_p}\mathbb{Z}_p) = \prod_p p^{d_p} = N^{-1}, \quad b = e^{\deg D}. \tag{RDT2}$$

The local measure is $\text{vol}(\mathbb{Z}_p) = 1$, and the real measure is Lebesgue measure. No finite-prime contribution is absorbed into a replacement for D .

The additive monoid used for sections is $G_L(I(D)) = \{(0, \lambda) : \lambda \in L\} \cup (I(D) \times \{1_L\})$, with addition $(x, \lambda) + (z, \mu) = (x + z, \lambda \vee \mu)$ and identity $\tau = (0, 0_L)$. A fractional ideal need not be a ring, so this notation here denotes the additive monoid, not an assertion of multiplicative closure. Its supported zero is $e = (0, 1_L)$.

89.2 The complete supported bounded section module

For $r_+ = \{*, 1, \dots, r\}$, define

$$\begin{aligned} \mathcal{H}_L^0(D)(r_+) &= \{((q_j, \lambda_j))_{j=1}^r : q_j \in I(D), \quad q_j \neq 0 \Rightarrow \lambda_j = 1_L, \\ &\quad -a \leq \sum_{j \in J} q_j \leq a \text{ for every } J \subseteq \{1, \dots, r\}\}. \end{aligned} \tag{RDT3}$$

The empty sum is τ and has amplitude zero. The maximum of the subset amplitudes is the sum of positive coordinates, and the minimum is the sum of negative coordinates. Consequently the complete condition is precisely

$$q_j = Nn_j, \quad n_j \in \mathbb{Z}, \quad N \sum_{n_j > 0} n_j \leq a, \quad N \sum_{n_j < 0} |n_j| \leq a. \quad (\text{RDT4})$$

These are two separate bounds. They do not impose a single bound on $\sum_j |q_j|$. This distinction is the one between the source's $\mathbb{S}$ construction and its preceding $\mathbb{S}[\pm 1]$ construction. For a pointed map $f : r_+ \rightarrow s_+$ the full structure map is

$$((q_j, \lambda_j))_j \mapsto \left(\sum_{f(j)=i} q_j, \bigvee_{f(j)=i} \lambda_j \right)_{i=1}^s. \quad (\text{RDT5})$$

Empty fibres give $(0, 0_L)$; coordinates sent to $*$ are discarded. Every subset sum of the output is a subset sum of the input over a disjoint union of fibres, hence remains in $[-a, a]$. If an output amplitude is nonzero, one summand was nonzero and had top support, so its output label is top. Composition follows from associativity of sums and joins. Thus these are actual pointed Γ -module maps at all levels. Division of each amplitude by the retained positive number N gives the exact pointed natural isomorphism

$$\mathcal{H}_L^0(D) \xrightarrow{\cong} \mathcal{E}_{L,b}, \quad ((q_j, \lambda_j)) \mapsto ((q_j/N, \lambda_j)), \quad ((n_j, \lambda_j)) \mapsto ((Nn_j, \lambda_j)) \quad (\text{RDT6})$$

as its inverse. The target is exactly the module in SRR2–8, with the bound $[-b, b]$; (RDT4) proves the domain and range, and (RDT5) proves naturality. We retain the source, target, N , and both maps. The amplitude map onto Connes–Consani's section module has, over each admitted tuple (q_j) , exactly the fibre $L^{\{j: q_j=0\}}$. Its zero fibre is $H(L, \vee, 0_L)$. For $b \geq 1$ it has no pointed natural section: the admitted pair $(N, -N)$ has unique coordinate lifts, and its fold is e ; the pointed lift of zero is τ . Projection naturality and fold naturality would identify these two distinct labels. For $0 < b < 1$, (RDT4) forces every amplitude to be zero in every level, and gives exactly

$$\mathcal{H}_L^0(D) = H(L, \vee, 0_L) \quad (0 < b < 1). \quad (\text{RDT7})$$

The absence of nonzero amplitude sections does not remove this fibre. For finite L , the existing complete SRR6–8 proof transported by (RDT6) gives, with $J(L)$ the nonzero join-irreducibles and $J_{<}(L) = J(L) \setminus \{1_L\}$,

$$\dim_{\mathbb{S}} \mathcal{H}_L^0(D) = \begin{cases} |J(L)|, & 0 < b < 1, \\ n + 1 + |J_{<}(L)|, & 2^{n-1} \leq b < 2^n, \quad b \geq 1. \end{cases} \quad (\text{RDT8})$$

The exact generator definition is the equality-tolerance version of the source's Definition **generator**. The proof is the cited SRR proof with the inverse in (RDT6) applied to each generator and each admissible sum, so both minimality and attainability transfer. For infinite L no finite set generates: a set of t generators has at most 2^t subset sums while all points $(0, \lambda)$ remain.

89.3 Exact counts in every level

Let $m = \lfloor b \rfloor$. For a tuple in (RDT4), let u be its number of positive coordinates, v its number of negative coordinates and $w = r - u - v$ its number of zero coordinates. The number of positive integer u -tuples with total at most m is $\binom{m}{u}$, including the value one when $u = 0$ and the value zero when $u > m$. To prove this for $u > 0$, the partial sums $n_1, n_1 + n_2, \dots, n_1 + \dots + n_u$ are exactly increasing u -element subsets of $\{1, \dots, m\}$, with inverse successive differences. The

same count applies independently to the magnitudes of the negative coordinates. Choosing their disjoint positions gives

$$\#\mathcal{H}_L^0(D)(r_+) = \sum_{\substack{u,v \geq 0 \\ u+v \leq r}} \frac{r!}{u!v!(r-u-v)!} \binom{m}{u} \binom{m}{v} |L|^{r-u-v} \quad (L \text{ finite}). \quad (\text{RDT9})$$

The factor $|L|^{r-u-v}$ counts all zero labels independently. Deleting this factor gives the amplitude-tuple count, not the full supported count. In particular the amplitude count in level 1_+ is $2m+1$, while the full supported count is $2m+|L|$. At $r=0$ the formula gives the single pointed empty tuple. For arbitrary infinite L the fibre formula preceding (RDT7), rather than a finite cardinality assertion, remains exact. Vary only the original archimedean bound: for $R > 0$ the divisor $D_R = D + \log R[\infty]$ retains $I(D)$ and has bound aR . The nonzero amplitude count in level 1_+ is

$$A_D(R) = \#\{q \in I(D) \setminus \{0\} : |q| \leq aR\} = 2\lfloor bR \rfloor, \quad dA_D = 2 \sum_{n \geq 1} \delta_{n/b}. \quad (\text{RDT10})$$

This is an exact image of the all-level module under amplitude projection and degree-one counting. It is not a faithful replacement for that module, and counting is not asserted to be a Γ -module morphism. The full fibres and folds remain (RDT3)–(RDT5).

89.4 The original theta source and its diagonal antecedent

Retain the entire real function and every finite factor:

$$\begin{aligned} \eta(x) &= \left(\pi^2 x^4 - \frac{3\pi}{2} x^2 \right) e^{-\pi x^2}, \\ \xi_D(x) &= \eta(x_\infty/a) \prod_p \mathbf{1}_{p^{-d_p} \mathbb{Z}_p}(x_p), \\ \xi_D(0) &= 0, \quad \int_{\mathbb{A}} \xi_D = \frac{a}{N} \left(\pi^2 \frac{3}{4\pi^2} - \frac{3\pi}{2} \frac{1}{2\pi} \right) = 0. \end{aligned} \quad (\text{RDT11})$$

There are finitely many nonreference finite factors, so this is an actual Bruhat–Schwartz function in the original adelic space. For $y > 0$ and $u \in \widehat{\mathbb{Z}}^\times$, the admissible rational summands are exactly $q = Nn$, $n \in \mathbb{Z} \setminus \{0\}$. Its original theta image is therefore

$$f_D(y, u) = E\xi_D(y, u) = 2 \sum_{n \geq 1} \eta(Nny/a) = 2 \sum_{n \geq 1} \eta(ny/b). \quad (\text{RDT12})$$

The first expression retains the original finite and real parameters; the second records their exact comparison. The sums and all fixed $y > 0$ derivatives converge absolutely by Gaussian decay. For the diagonal source of DOR20a, keep $y_1, y_2 > 0$ and put

$$\begin{aligned} F_a(z) &= \pi(z/a) \left(\pi(z/a)^2 - \frac{3}{2} \right) e^{-\pi(z/a)^2/2}, \quad G_a(z) = (z/a) e^{-\pi(z/a)^2/2}, \\ t_{D,y}(x_1, x_2) &= F_a(x_{1,\infty}/y_1) G_a(x_{2,\infty}/y_2) \prod_p \mathbf{1}_{p^{-d_p} \mathbb{Z}_p}(x_{1,p}) \mathbf{1}_{p^{-d_p} \mathbb{Z}_p}(x_{2,p}), \\ \mu_y t_{D,y}(x) &= t_{D,y}(y_1 x_\infty, x_f; y_2 x_\infty, x_f) = \xi_D(x). \end{aligned} \quad (\text{RDT13})$$

Both real factors are odd. Each has zero evaluation and zero additive integral; pairing rational summands $q, -q$ gives zero independent theta image for each factor. The product on the diagonal is exactly $F_a(z)G_a(z) = \eta(z/a)$, proving (RDT13), with diagonal moment zero by (RDT11). The DMR/DOR maps therefore receive this actual tensor in their stipulated joint kernel. This is not a

freely chosen function without an original source antecedent. Combining (RDT10) and (RDT12) gives the exact whole-scale transform

$$\begin{aligned} f_D(y, u) &= \int_{(0, \infty)} \eta(yR) dA_D(R), \\ &= -y \int_0^\infty A_D(R) \eta'(yR) dR, \\ \eta'(x) &= \left[4\pi^2 x^3 - 3\pi x - 2\pi x \left(\pi^2 x^4 - \frac{3\pi}{2} x^2 \right) \right] e^{-\pi x^2}. \end{aligned} \quad (\text{RDT14})$$

The first integral is absolutely convergent, since its total variation integral is $2 \sum_{n \geq 1} |\eta(ny/b)| < \infty$. Integration by parts on finite intervals, then passage to the endpoints, proves the second equality. Near zero A_D is zero; at infinity $A_D(R) \leq 2bR$ and $\eta(yR)$ and its derivative have Gaussian decay. These estimates prove both vanished boundary terms and absolute integrability of the second expression. No endpoint Mellin integral is exchanged with the integer sum in this argument.

89.5 Full Mellin factors, endpoints and recovery

Use $M_D(s) = \int_0^\infty f_D(y, u) y^s dy/y$. For $\text{Re } s > 1$, absolute integration of the original Gaussian monomials gives

$$M_D(s) = 2 \left(\frac{a}{N} \right)^s \zeta(s) \left\{ \frac{\pi^2 \Gamma((s+4)/2)}{2\pi^{(s+4)/2}} - \frac{3\pi \Gamma((s+2)/2)}{4\pi^{(s+2)/2}} \right\}. \quad (\text{RDT15})$$

For completeness its original Poisson comparison is exact. For the Fourier convention $\widehat{f}(t) = \int f(x) e^{-2\pi i x t} dx$,

$$\widehat{x^2 e^{-\pi x^2}}(t) = \left(\frac{1}{2\pi} - t^2 \right) e^{-\pi t^2}, \quad \widehat{x^4 e^{-\pi x^2}}(t) = \left(t^4 - \frac{3t^2}{\pi} + \frac{3}{4\pi^2} \right) e^{-\pi t^2}.$$

These follow by differentiating the Gaussian Fourier identity twice and four times, using $\widehat{x f} = -(2\pi i)^{-1} \partial_t \widehat{f}$. Multiplying by the two coefficients in (RDT11) gives $\widehat{\eta} = \eta$, including cancellation of the two constant terms. Poisson summation for this Schwartz function, whose value and integral at zero vanish, gives for $h(z) = 2 \sum_{n \geq 1} \eta(nz)$ $h(z) = z^{-1} h(z^{-1})$. The Gaussian series for $z \geq 1$ and this identity prove faster-than-every-power decay at both ends, including all logarithmic derivatives. Hence M_D is entire, and (RDT15) continues as that entire integral with all displayed factors retained.

The entire adelic Fourier comparison retains the opposite divisor and the full volume factor as well. At each finite place take the original additive character of conductor $\mathbb{Z}_p$ and its self-dual measure. The annihilator of $p^{-d_p} \mathbb{Z}_p$ is $p^{d_p} \mathbb{Z}_p$. The integral of its character is the ball's volume p^{d_p} on that annihilator; off the annihilator, translation by an element of the ball multiplies the integral by a scalar different from one and therefore forces it to zero. Real scaling in the proved Gaussian Fourier formula then gives

$$\begin{aligned} \widehat{\mathbf{1}_{p^{-d_p} \mathbb{Z}_p}} &= p^{d_p} \mathbf{1}_{p^{d_p} \mathbb{Z}_p}, & \widehat{\eta(\cdot/a)}(t) &= a\eta(at), \\ \widehat{\xi}_D &= \frac{a}{N} \xi_{-D}, & f_D(y, u) &= \frac{a}{Ny} f_{-D}(1/y, u^{-1}), & M_D(s) &= \frac{a}{N} M_{-D}(1-s). \end{aligned} \quad (\text{RDT15a})$$

Poisson summation on the original adelic source has zero source endpoints by (RDT11), so it proves the middle theta identity, including the modulus y^{-1} . Its compact-unit arguments are retained even though these particular functions are independent of them. The last identity follows by an absolutely convergent Mellin substitution. Thus self-duality of the real polynomial Gaussian has not been incorrectly transferred to the whole source: its full dual is the displayed multiple of the source for $-D$.

At $s = 0$ the expression in braces has value $1/2 - 3/4 = -1/4$, while $\zeta(0) = -1/2$; thus $M_D(0) = 1/4$. The Mellin integral of $h(z) = z^{-1}h(z^{-1})$ gives $M_h(1) = M_h(0)$ by an absolutely convergent substitution. Real scaling in the original integral then proves

$$M_D(0) = \frac{1}{4}, \quad M_D(1) = \frac{a}{4N}, \quad \frac{M_D(1)}{M_D(0)} = \frac{a}{N} = e^{\deg D}. \quad (\text{RDT16})$$

Thus both endpoint constants and the entire finite additive volume are explicit. In particular the zero additive integral (RDT11) is not the endpoint integral in (RDT16).

The section module can now be recovered from the retained data (N, L, f_D) , by the exact inverse formulas

$$d_p = -v_p(N), \quad a = N \frac{M_D(1)}{M_D(0)}, \quad d_\infty = \log \left(N \frac{M_D(1)}{M_D(0)} \right), \quad (\text{RDT17})$$

followed by (RDT3)–(RDT5). This is a proved inverse on the specified divisor family. An arbitrary test function is not asserted to lie in that family. The theta function alone recovers b ; retaining N and L is necessary to recover the full original presentation and all its labelled section fibres.

89.6 Exact principal-divisor covariance

For $c \in \mathbb{Q}^\times$, use the full principal divisor and the original dilation convention

$$\begin{aligned} \operatorname{div}(c) &= \sum_p v_p(c)[p] - \log |c|_\infty[\infty], & (U_c \xi)(x) &= \xi(c^{-1}x), \\ D' &= D + \operatorname{div}(c), \quad N' = N/|c|_\infty, \quad a' = a/|c|_\infty, \quad b' = b, & (\text{RDT18}) \\ \mathcal{H}_L^0(D) &\longrightarrow \mathcal{H}_L^0(D'), & ((q_j, \lambda_j)) &\longmapsto ((q_j/c, \lambda_j)), \\ \xi_{D'} &= U_{1/c} \xi_D, & t_{D',y} &= (U_{1/c} \otimes U_{1/c}) t_{D,y}, & f_{D'} &= f_D. \end{aligned}$$

The rational product formula here is the elementary equality $\prod_p p^{v_p(c)} = |c|_\infty$. It gives the displayed N' and b' , while the valuation and real inequalities give the section map and its inverse, multiplication by c . These maps commute with all folds and keep every label. For negative c , the two signed bounds are interchanged. The finite indicators in $\xi_D(cx)$ give exactly $p^{-d_p - v_p(c)} \mathbb{Z}_p$; the real function is even. In the tensor identity both odd factors change sign when $c < 0$, so the two signs cancel. Finally $q \mapsto cq$ permutes $\mathbb{Q}^\times$ in the absolutely convergent theta sum, proving $EU_{1/c} \xi_D = E \xi_D$.

Two divisors have the same f_D if and only if they have the same b , by (RDT12) and (RDT16). This is equivalent to their difference being principal: choose the positive rational $c = N/N'$, and equality of a/N and a'/N' gives the infinite coefficient in (RDT18). Thus the fibres of $D \mapsto f_D$ are exactly its principal-divisor orbits. The actual representative and label data are restored by (RDT17), rather than assumed recoverable from the quotient alone.

89.7 Effectivity, positive transforms, and the original trace

For $z \geq 1$ every term of $h(z)$ is strictly positive, because $\pi(nz)^2 > 3/2$. For $0 < z < 1$ its exact Poisson identity gives $h(z) = z^{-1}h(z^{-1}) > 0$. Therefore

$$f_D(y, u) > 0 \quad \text{for all } y > 0, \quad u \in K, \quad \text{for every original divisor } D. \quad (\text{RDT19})$$

On the other hand a nonzero amplitude section at the original bound exists exactly when $b \geq 1$, by (RDT4). For the concrete original divisor $D = -(\log 2)[\infty]$ one has

$$\begin{aligned} N &= 1, \quad a = b = 1/2, \quad \mathcal{H}_L^0(D) = H(L, \vee, 0_L), \\ f_D(y) &= 2 \sum_{n \geq 1} \eta(2ny) > 0, \quad (M_D(0), M_D(1)) = (1/4, 1/8). \end{aligned}$$

There is no contradiction: (RDT14) uses sections of every D_R , with its first nonzero amplitude at $R = 1/b$, not only sections at $R = 1$. It is a proved relation between two different effectivity tests. Moreover $f_D \in E(T_0) = V$ by (RDT11), so the original CCM trace pairing satisfies

$$\langle f_D, g \rangle = 0 \quad (g \in B). \quad (\text{RDT20})$$

This is the exact cited theorem **pairTrvanishV** on the original test space, with the actual antecedent already supplied. In the endpoint-retaining quotient of CTO21–22a the same source has coordinates $(0, 1/4, a/(4N))$. Thus its positive function, original degree, labelled sections, and zero arithmetic trace all coexist in one explicit diagram. A divisor on the arithmetic curve has not been silently identified with an effective correspondence on its square. The latter comparison in the Weil argument remains a different mathematical map.

All scalar linear comparisons above lift at a fixed admissible support by $(x, \lambda) \mapsto (Fx, \lambda)$. Every nonzero amplitude has label 1_L ; an amplitude cancellation keeps that label and gives e , while a zero input with lower label retains that exact label. The entire section module was defined with coordinatewise labels in (RDT3), so it is not reconstructed from scalar counting alone. For an infinite series or integral, take its proved amplitude limit at a fixed label; no infinite join in an arbitrary lattice is assumed.

89.8 The canonical divisor, finite prime two, and the full Fourier operator

The two dualities just encountered have different original formulas. Connes–Consani’s signed Serre duality uses $K - D$ with $K = -2[2]$ (their **serredual**, calculated with all labels in CTD16–20), whereas the original self-dual adelic Fourier measure in (RDT15a) gives $-D$. Their exact relation uses the finite idele

$$\begin{aligned} v_\infty = 1, \quad v_2 = 4, \quad v_p = 1 \ (p \neq 2), \quad |v| = |4|_2 = \frac{1}{4}, \\ D^\vee = K - D, \quad N^\vee = \prod_p p^{-(K_p - d_p)} = \frac{4}{N}, \quad a^\vee = e^{-d_\infty} = \frac{1}{a}, \quad b^\vee = \frac{a^\vee}{N^\vee} = \frac{N}{4a}. \end{aligned} \quad (\text{RDT21})$$

This idele is not the rational diagonal element 4: its infinite component is 1. With $(U_v \xi)(x) = \xi(v^{-1}x)$, its action on the full original seed is

$$\begin{aligned} \xi_{-D}(x) &= \eta(ax_\infty) \prod_p \mathbf{1}_{p^{d_p} \mathbb{Z}_p}(x_p), \\ U_v \xi_{-D}(x) &= \eta(ax_\infty) \mathbf{1}_{4 \cdot 2^{d_2} \mathbb{Z}_2}(x_2) \prod_{p \neq 2} \mathbf{1}_{p^{d_p} \mathbb{Z}_p}(x_p) = \xi_{K-D}(x). \end{aligned} \quad (\text{RDT22})$$

The prime-two lattice is $4 \cdot 2^{d_2} \mathbb{Z}_2$. At every other prime 4 is a unit. These local equalities prove (RDT22), including its real factor.

Define the actual operator $T_K = U_v \mathcal{F}$ on the original adelic Schwartz space, where $\mathcal{F}$ has the original additive character and self-dual measures of (RDT15a). A Haar change of variables proves

$$\mathcal{F} U_v = |v| U_{v^{-1}} \mathcal{F} : \quad \int \xi(v^{-1}x) \psi(-xy) dx = |v| \int \xi(z) \psi(-zvy) dz.$$

Writing $\mathcal{R}\xi(x) = \xi(-x)$, Fourier inversion gives

$$T_K \xi_D = \frac{a}{N} \xi_{K-D}, \quad T_K^2 = \frac{1}{4} \mathcal{R}, \quad T_K^{-1} = 4 \mathcal{R} T_K. \quad (\text{RDT23})$$

Reflection commutes with U_v and $\mathcal{F}$, which verifies both sides of the inverse formula. The original zero-value, zero-integral source space is preserved: Fourier exchanges these two zero conditions, and dilation multiplies the integral by $|v|$ while leaving the value at zero unchanged. Thus these

are operators on the actual theta source, not only identities on a scalar family. On the even seeds, the product of the two displayed family coefficients is $\frac{a}{N} \frac{N}{4a} = 1/4$, as the operator square requires. The involution $D \mapsto K - D$ has therefore not been incorrectly transferred to the unscaled operator T_K .

The entire rational summation and its Mellin map retain the same factor. Since the nonzero rational points of $I(K - D)$ are $q = 4n/N$, $n \in \mathbb{Z} \setminus \{0\}$, direct substitution gives

$$\begin{aligned} f_{K-D}(y, u) &= 2 \sum_{n \geq 1} \eta(4any/N) = f_{-D}(4y, u), \\ f_D(y, u) &= \frac{a}{Ny} f_{K-D}(1/(4y), u^{-1}), \\ M_D(s) &= \frac{a}{N} 4^{1-s} M_{K-D}(1-s). \end{aligned} \tag{RDT24}$$

The first equality is the absolutely convergent original theta series. The second follows from that equality and (RDT15a). For the third, the Mellin integrals converge absolutely for every s by the two-end decay already proved. Substituting $z = 1/(4y)$ yields $y^{s-1} = (4z)^{1-s}$ and proves exactly the stated factor. In original zeta and Gamma factors this says

$$\begin{aligned} 2 \left(\frac{a}{N} \right)^s \zeta(s) &\left\{ \frac{\pi^2 \Gamma((s+4)/2)}{2\pi^{(s+4)/2}} - \frac{3\pi \Gamma((s+2)/2)}{4\pi^{(s+2)/2}} \right\} \\ &= \frac{a}{N} 4^{1-s} 2 \left(\frac{N}{4a} \right)^{1-s} \zeta(1-s) \left\{ \frac{\pi^2 \Gamma((5-s)/2)}{2\pi^{(5-s)/2}} - \frac{3\pi \Gamma((3-s)/2)}{4\pi^{(3-s)/2}} \right\}. \end{aligned} \tag{RDT25}$$

This is an identity of the entire continuations of the displayed products, including their removable singularities; an individual Gamma or zeta factor need not be finite at an exceptional point. No factor is discarded. At $s = 0$, the right side of (RDT24) is $\frac{a}{N} 4 \frac{N}{16a} = 1/4$; at $s = 1$ it is $\frac{a}{N} 1/4 = a/(4N)$. Applying (RDT24) twice multiplies its prefactors to $\frac{a}{N} \frac{N}{4a} 4^{1-s} 4^s = 1$, as the divisor involution requires. Both endpoint evaluations and the differing operator square in (RDT23) are thereby retained.

The same finite-prime contribution occurs in the actual Serre evaluation, rather than merely in these Gaussian formulas. The original circle has circumference N and characters $\chi_n([x]) = [nx/N] \in \mathbb{R}/\mathbb{Z}$. CTD8–11 prove that the full Γ -level tolerance is exactly $\sum_i |n_i| \leq N/(4a)$. The signed section in CTD18 is

$$q'_i = \frac{4n_i}{N} \in I(K - D), \quad \sum_i |q'_i| \leq \frac{1}{a}, \quad \chi_{n_i}([x]) = [q'_i x/4]. \tag{RDT26}$$

It is an equality of the original character and evaluation maps, with all coefficient labels retained, not a claim that Fourier transformation is itself the entire duality functor. The full internal Hom has the additional support maps calculated in CTD12; CTD15 specifies the exact retract to (RDT26). Both f_D and f_{K-D} have the original theta-source antecedents and hence the zero trace pairing (RDT20). Consequently this proved duality and functional equation do not supply positivity of the nonzero arithmetic quotient.

The exact section and order comparisons

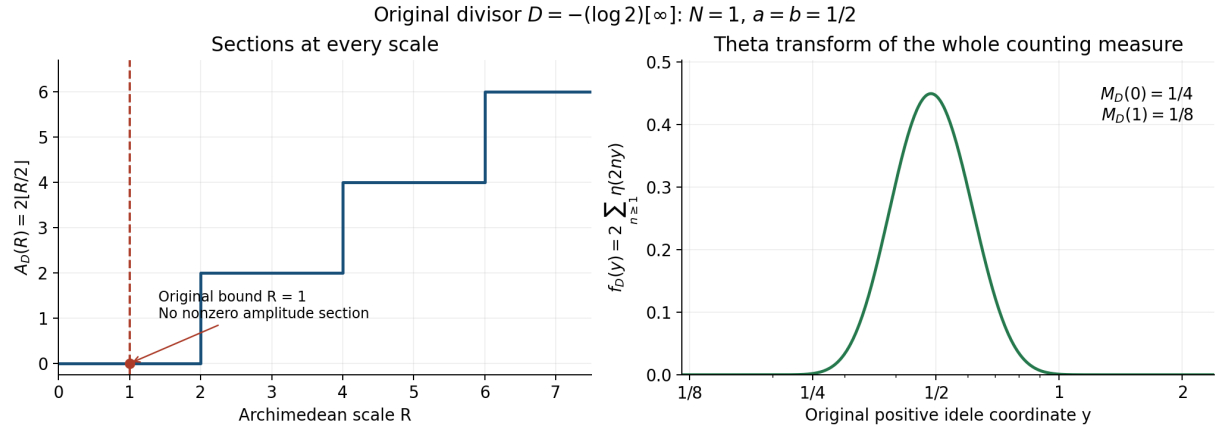


Figure 25: RDT10–20, for the displayed original divisor. Left: the exact degree-one nonzero amplitude count. At $R = 1$ its full zero fibre remains L , and at level r_+ it remains L^r with join folds. Right: a finite plot of the theta transform of the *entire* counting measure. Global strict positivity is proved by the original Gaussian and Poisson identities in RDT15 and RDT19, not by this plot. The series is evaluated using those identities and twelve positive-side summands; for $z \geq 1$ the omitted absolute terms are bounded by $2\pi^2 \sum_{n \geq 13} (nz)^4 e^{-\pi(nz)^2}$, and for $z < 1$ by z^{-1} times that bound at z^{-1} . Original human sources: Connes–Consani, `xdefn`, `xdefnr`, `RRZ`; Connes–Consani–Marcolli, `degmodifyV`. The accompanying script reproduces both panels without changing N, a, b .

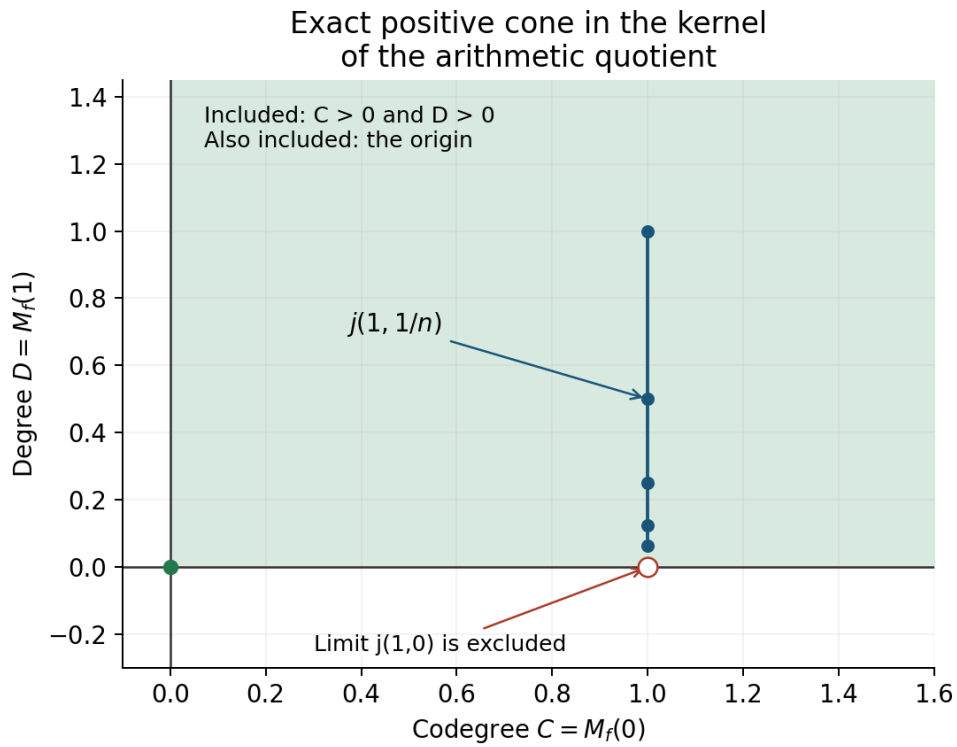


Figure 26: The kernel of the quotient map in CTO22 and CTO26, drawn in its exact endpoint coordinates. The shaded open quadrant and the origin constitute the full kernel cone. Nonzero axis points are excluded. CTO27 proves the displayed convergence of quotient classes; it does not assert convergence of their dilated Gaussian representatives in the original test space. This is the exact kernel cone, not a drawing of the entire infinite-dimensional positive cone over nonzero arithmetic classes. CTO20 retains a Mellin bound on that larger cone.

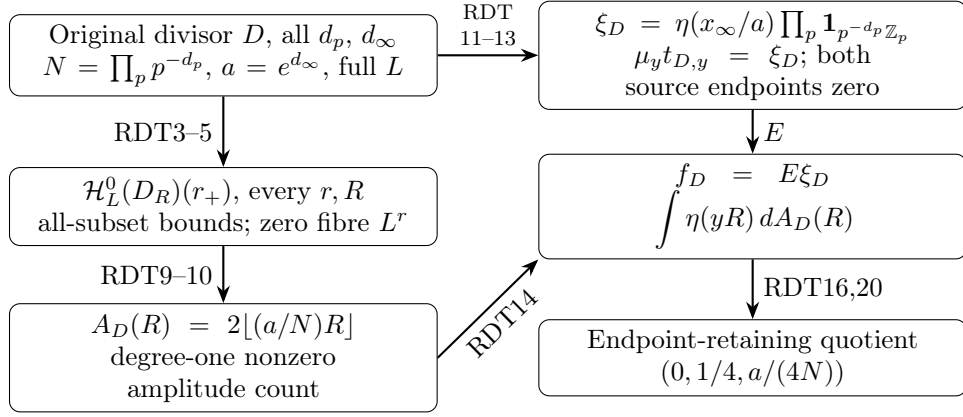


Figure 27: One construction with every receiving map specified. The counting arrow is the stated amplitude observation, not a faithful Γ -module morphism. Keeping (N, L, f_D) gives the exact inverse RDT17 to the divisor and the all-level section module. Principal-divisor covariance is RDT18; Fourier duality is RDT15a with multiplier a/N . The first zero in the bottom-right box is the amplitude in the original arithmetic quotient; with the original top support its lift is e . All three entries there have one common support label.

90 The full supported adelic simplicial object and its relative cohomology

Original source and exact scope. Alain Connes and Caterina Consani, *Riemann–Roch for $\overline{\text{Spec } \mathbb{Z}}$* , original author source `RR-J-final.tex`, define the signed bounded section functor and adelic addition in `setup`, `psi`, `setup1`, and `cokerpsi` (source lines 199–274). Their Appendix B, `descript1-proppi0`, constructs the restricted Dold–Kan object and calculates its spherical simplices and relation; Appendix A, `thedimh1`, gives its adelic circle comparison. The present construction applies these actual maps to the full support carrier. It retains both independently labelled source legs. Its directed relation, relative spherical simplices and comparison fibres are calculated below. A lift of a scalar formula is not assumed to commute with the original kernel or product.

The lattice operations are those of The Clankers, *Split Support Geometry*, version 11, Definition `def:lattice-split`. RDT1–6 in the accompanying complete proof retain the original divisor and the unsigned, all-subset section functor of Connes–Consani’s later *Riemann–Roch for the ring $\mathbb{Z}$* . Both different section functors are retained here with their exact inclusion.

90.1 The divisor and the full two-leg source

Let L be a bounded distributive lattice with $0_L \neq 1_L$. For an additive abelian group A put

$$G_L(A) = \{(0, \lambda) : \lambda \in L\} \cup (A \times \{1_L\}), \quad (x, \lambda) + (y, \mu) = (x + y, \lambda \vee \mu), \quad \tau = (0, 0_L), \quad e = (0, 1_L). \quad (\text{ASC1})$$

The symbol $H(M)$ for an additive pointed monoid means the functor $H(M)(r_+) = M^r$, with sums over fibres of a pointed map and empty sum the monoid identity. No additive inverse relative to τ is asserted for a nonzero amplitude.

Retain the original divisor and every local restriction:

$$\begin{aligned} D &= \sum_p d_p[p] + d_\infty[\infty], & N &= \prod_p p^{-d_p} > 0, & a &= e^{d_\infty} > 0, \\ O_f(D) &= \prod_p p^{-d_p} \mathbb{Z}_p = N\widehat{\mathbb{Z}}, & I(D) &= \mathbb{Q} \cap O_f(D) = N\mathbb{Z}. \end{aligned} \quad (\text{ASC2})$$

The equality of the finite ideals holds place by place because $v_p(N) = -d_p$. The last equality follows by writing q/N as a reduced rational number and testing its valuation at every prime. All appearances of N, a below refer to these retained original data.

Set $B_L = G_L(\mathbb{A}_{\mathbb{Q}})$. The source uses the product of two monoids, with the two labels independent:

$$P_L = G_L(\mathbb{Q}) \times G_L(\mathbb{A}_{\mathbb{Q}}), \quad \Phi_L((q, \alpha), (o, \beta)) = (q + o, \alpha \vee \beta). \quad (\text{ASC3})$$

Here q is embedded diagonally. This is an additive pointed monoid map: both orders of adding two inputs give amplitude $q + q' + o + o'$ and label $\alpha \vee \alpha' \vee \beta \vee \beta'$. Its value is admissible, since a nonzero output amplitude requires at least one nonzero input amplitude and hence top output support.

For $n \geq 0$ and $r_+ = \{*, 1, \dots, r\}$ define $X_n(D)(r_+)$ by the following full arrays:

$$\begin{aligned} b_j &= (x_j, \lambda_j) \in B_L \quad (1 \leq j \leq r), \\ z_{\ell j} &= ((q_{\ell j}, \alpha_{\ell j}), (o_{\ell j}, \beta_{\ell j})) \in P_L \quad (1 \leq \ell \leq n, 1 \leq j \leq r), \\ o_{\ell j, f} &\in O_f(D), \quad \sum_{\ell=1}^n \sum_{j=1}^r |o_{\ell j, \infty}| \leq a. \end{aligned} \quad (\text{ASC4})$$

The empty interior array at $n = 0$ has sum zero. At $r = 0$ there is one empty array. The vertex b has no size restriction. Every amplitude-zero source coordinate has its full independent L -label. In particular $G_L(\mathbb{Q}) \times G_L(\mathbb{A})$ has not been replaced by $G_L(\mathbb{Q} \times \mathbb{A})$.

90.2 Every simplicial and pointed-set map

Write $z_\ell = (z_{\ell j})_j$, with addition componentwise. For $n \geq 1$ use the following face convention, which gives the source's two faces b and $b + \Phi_L z$ in degree one:

$$\begin{aligned} d_0(b; z_1, \dots, z_n) &= (b; z_1, \dots, z_{n-1}), \\ d_i(b; z_1, \dots, z_n) &= (b; z_1, \dots, z_{n-i} + z_{n-i+1}, \dots, z_n) \quad (0 < i < n), \\ d_n(b; z_1, \dots, z_n) &= (b + \Phi_L z_1; z_2, \dots, z_n). \end{aligned} \tag{ASC5}$$

For $0 \leq i \leq n$, s_i inserts the identity $0_{P_L} = (\tau, \tau)$ immediately after z_{n-i} ; s_n inserts it before z_1 , and s_0 appends it after z_n . This convention includes $s_0(b) = (b; 0_{P_L})$ when $n = 0$.

All these arrays remain admitted. Dropping an interior coordinate decreases the real absolute sum; merging two decreases it by the triangle inequality; insertion contributes zero. The finite components remain in the additive subgroup $O_f(D)$. No support label is removed from a surviving coordinate except by its displayed join with another label.

For completeness, the simplicial identities can be checked without an inverse in either monoid. Form the category whose objects are B_L^r and whose arrow labelled $z \in P_L^r$ goes from b to $b + \Phi_L z$. Composition adds arrow labels; its associativity and identity laws follow from (ASC3). Its ordinary nerve has the usual faces deleting a vertex and composing the adjacent arrows. Reversing the ordinal gives exactly (ASC5) and the stated degeneracies. Deleting two ordered vertices in either order deletes the same vertices and sums the same consecutive arrow blocks. Inserting two identity arrows in either order gives the same array; deleting an inserted identity either reverses that insertion or commutes past it. These are respectively the face–face, degeneracy–degeneracy and face–degeneracy identities. The restrictions in (ASC4) are stable under every such operation, so they define a simplicial subset of that nerve.

For a pointed map $f : r_+ \rightarrow s_+$, sum the vertices and each interior row over each nonbasepoint fibre of f . On amplitudes these are the original sums; on every leg's supports they are joins; an empty fibre gives τ on that leg. The resulting absolute sum is at most the old sum, and its finite restrictions hold. These maps commute with (ASC5) and all degeneracies by additivity of Φ_L . Associativity of sums and joins proves composition in Γ . Thus (ASC4)–(ASC5) define the entire supported Γ -space, not just its first two sets.

The construction uses the nerve formulas directly. No Dold–Kan equivalence for noncancellative monoids is invoked. Let $X^{\text{amp}}(D)$ be the same restricted arrays and maps on the original amplitude groups, as in the cited Connes–Consani construction. Forgetting every label gives the pointed simplicial Γ -map

$$p_D : X(D) \longrightarrow X^{\text{amp}}(D). \tag{ASC5a}$$

It is surjective in every bidegree: attach top labels to all nonzero coordinates and any labels to zero coordinates. It commutes with faces, degeneracies and pointed-set maps because their amplitude formulas are unchanged. Over a fixed admitted amplitude array its fibre has one independent copy of L for each zero vertex amplitude, each zero rational-leg amplitude, and each zero adelic-leg amplitude. Nonzero coordinates force top labels independently. In particular the fibre over the entirely zero array in bidegree (n, r) is

$$p_D^{-1}(0) = L^{(2n+1)r}. \tag{ASC5b}$$

The maps between these fibres are the actual joins and identity insertions already specified, rather than unconstrained product maps. There is no pointed simplicial Γ -section of (ASC5a): in degree zero such a section would be a pointed additive section $\mathbb{A} \rightarrow G_L(\mathbb{A})$. A nonzero x and $-x$ have forced top lifts whose sum is e , whereas the pointed lift of zero is τ . This contradicts fold naturality. Degreewise surjectivity therefore does not provide the missing natural section.

90.3 The native relation and its exact symmetric part

Let $\mathcal{R}_{D,r}$ be the original Connes–Consani relation on $\mathbb{A}^r$: $(x, y) \in \mathcal{R}_{D,r}$ means that there are $q_j \in \mathbb{Q}$, $o_{j,f} \in O_f(D)$ with $y_j - x_j = q_j + o_j$ and $\sum_j |o_{j,\infty}| \leq a$. It is reflexive and symmetric, by zero choices and simultaneous negation. The degree-zero relation furnished by (ASC5), directed from d_0 to d_1 , is exactly

$$((x_j, \lambda_j), (y_j, \mu_j)) \in \mathcal{R}_{L,D,r}^{\rightarrow} \iff (x, y) \in \mathcal{R}_{D,r} \text{ and } \lambda_j \leq \mu_j \text{ for every } j. \quad (\text{ASC6})$$

Necessity follows because the output label is $\lambda_j \vee \alpha_j \vee \beta_j$. For sufficiency choose the original amplitude antecedents. If $\mu_j < 1_L$, admissibility and $\lambda_j \leq \mu_j$ force $x_j = y_j = 0$. At that coordinate choose $q_j = o_j = 0$ instead; this does not increase the absolute sum. Give these zero legs labels μ_j and 0_L . If $\mu_j = 1_L$, give both selected legs top support. These choices realize the desired label and amplitude in every coordinate and prove the converse.

Consequently the native relation is generally directed: $\tau \mathcal{R}^{\rightarrow} e$, whereas $e \mathcal{R}^{\rightarrow} \tau$. Its maximal symmetric subrelation is precisely

$$\mathcal{R}_{L,D,r}^{\leftarrow} = \mathcal{R}_{L,D,r}^{\rightarrow} \cap (\mathcal{R}_{L,D,r}^{\rightarrow})^{\text{op}} = \{((x, \lambda), (y, \mu)) : (x, y) \in \mathcal{R}_{D,r}, \lambda_j = \mu_j \forall j\}. \quad (\text{ASC7})$$

Indeed a symmetric subrelation can contain a pair only when it and its reverse occur; the displayed intersection is itself symmetric and reflexive. It is preserved by every Γ map: the amplitude relation is preserved by the triangle inequality and output joins of equal input labels are equal. This gives a specified tolerant Γ -object while keeping e distinct from τ in its relation as well as its carrier.

For comparison the amplitude-pullback relation $\mathcal{R}_{L,D,r}^{\text{amp}}$ imposes only $(x, y) \in \mathcal{R}_{D,r}$. The identity carrier map $(B_L^r, \mathcal{R}^{\leftarrow}) \rightarrow (B_L^r, \mathcal{R}^{\text{amp}})$ is a tolerance morphism. The reverse identity is not, since the amplitude-pullback relation relates τ and e . Both exact relations are retained; neither is silently substituted for the other. The duality comparison CTD below calculates their maps.

90.4 Spherical simplices over the entire zero section

The global pointed vertex at level r is $(\tau, \dots, \tau)$. A degree-one simplex $(b; z)$ is spherical there precisely when $b = \tau^r$ and $\Phi_L z = \tau^r$. This forces $\alpha_j \vee \beta_j = 0_L$, hence both source legs are τ . The ordinary global pointed spherical set is therefore a singleton:

$$\text{Sph}_1(X(D)(r_+), \tau^r) = \{(\tau^r; 0_{P_L}^r)\}. \quad (\text{ASC8})$$

This statement does not say that the original supported sections are absent. Nor is it a calculation of the fundamental group of a geometric realization: no Kan condition for these bounded simplicial sets has been asserted. Their actual relative spherical object is calculated next.

Let $\sigma = (\sigma_j) \in L^r$ and $b_\sigma = ((0, \sigma_j))_j$. Both faces of $(b_\sigma; z)$ equal b_σ exactly when

$$\begin{aligned} z_j &= ((q_j, \alpha_j), (-q_j, \beta_j)), \quad q_j \in N\mathbb{Z}, \quad \sum_j |q_j| \leq a, \\ q_j \neq 0 &\implies \alpha_j = \beta_j = 1_L, \quad \alpha_j \vee \beta_j \leq \sigma_j. \end{aligned} \quad (\text{ASC9})$$

Necessity and sufficiency follow directly from the two faces and the original finite and real restrictions. This proves the relative spherical object over every zero-section label, including all its proper support strata.

The degree-one homotopy relation used in the original source is equality on these spherical simplices. In the explicit degree-two array its faces are

$$d_0(b; z_1, z_2) = (b; z_1), \quad d_1(b; z_1, z_2) = (b; z_1 + z_2), \quad d_2(b; z_1, z_2) = (b + \Phi_L z_1; z_2). \quad (\text{ASC10})$$

The source's degree-one homotopy requires the first face to be the degenerate vertex $s_0(b_\sigma)$. It forces $b = b_\sigma$ and $z_1 = 0_{P_L}^r$. The two remaining faces then coincide. Conversely any spherical simplex gives such a homotopy by taking $z_1 = 0$, $z_2 = z$. Thus no unmentioned quotient of (ASC9) is taken. The base labels are themselves pushed forward by joins under Γ maps; fixing the vector of top labels alone would not be natural at an output with an empty fibre.

Let $\mathcal{Z}_L^1(D)$ denote the full relative object (ASC9), and let $\mathcal{H}_L^{\text{ind}}(D)$ forget σ but keep both source labels. Define the signed section object

$$\mathcal{S}_L^0(D)(r_+) = \{((q_j, \nu_j)) : q_j \in N\mathbb{Z}, \sum_j |q_j| \leq a, q_j \neq 0 \Rightarrow \nu_j = 1_L\}. \quad (\text{ASC11})$$

There are two natural retractions with their displayed sections:

$$\begin{aligned} \mathcal{Z}_L^1(D) &\longrightarrow \mathcal{H}_L^{\text{ind}}(D) \longrightarrow \mathcal{S}_L^0(D), \\ (q, \alpha, \beta, \sigma) &\longmapsto (q, \alpha, \beta) \longmapsto (q, \alpha \vee \beta), \\ (q, \alpha, \beta) &\longmapsto (q, \alpha, \beta, \alpha \vee \beta), \quad (q, \nu) \longmapsto (q, \nu, \nu). \end{aligned} \quad (\text{ASC12})$$

Each composite with its section is the identity. All formulas commute with sums and joins over fibres and keep empty fibres. For the middle section this uses $(\bigvee \alpha_j) \vee (\bigvee \beta_j) = \bigvee (\alpha_j \vee \beta_j)$. The first section sets the base to the least label that makes the loop spherical. Thus this is an exact relation to the original signed section amplitudes, not a claim that the ordinary τ -pointed kernel was nonzero.

Over an admitted $(q_j, \nu_j)_j$ the fibre of the composite in (ASC12) is the product, over coordinates with $q_j = 0$, of

$$\{(\alpha, \beta, \sigma) \in L^3 : \alpha \vee \beta = \nu_j, \quad \nu_j \leq \sigma\}. \quad (\text{ASC13})$$

At a nonzero coordinate all three labels are forced to 1_L . Formula (ASC13) is retained even when L is infinite; no finite cardinality or infinite join is imposed. In particular, replacing the two source legs by one common label would lose real fibres.

The signed bound in (ASC11) implies every original subset bound $-a \leq \sum_{j \in J} q_j \leq a$. It gives the natural injection

$$\mathcal{S}_L^0(D) \hookrightarrow \mathcal{H}_L^0(D) \quad (\text{ASC14})$$

into RDT3, keeping all labels. The injection is generally proper: for $N \leq a < 2N$ the tuple $((N, 1_L), (-N, 1_L))$ belongs to the all-subset object but fails the signed bound $2N \leq a$. Thus the two original Connes–Consani constructions are related by an explicit map and an explicit difference, not by an assumed identification.

90.5 The exact adelic circle map and its retained fibres

For $x = (x_f, x_\infty) \in \mathbb{A}$ choose $q \in \mathbb{Q}$ with $x_f - q \in O_f(D)$ and put

$$\pi_D(x) = [x_\infty - q] \in \mathbb{R}/N\mathbb{Z}. \quad (\text{ASC15})$$

Here is a direct existence argument. Set $u_f = x_f/N$ and choose a positive integer M with $Mu_f \in \widehat{\mathbb{Z}}$. By the Chinese remainder theorem choose $h \in \mathbb{Z}$ such that $h \equiv Mu_p \pmod{p^{v_p(M)}\mathbb{Z}_p}$ at each prime dividing M . Then $u_f - h/M \in \widehat{\mathbb{Z}}$: the assertion follows by division at primes dividing M and integrality at all others. Take $q = Nh/M$. Two choices differ by $\mathbb{Q} \cap O_f(D) = N\mathbb{Z}$, so (ASC15) is independent of the choice. Adding two choices proves additivity. On an open finite-adelic coset of $O_f(D)$ the choice can be held fixed, proving continuity. Every real class is reached by $(0, t)$, choosing $q = 0$.

Its exact kernel is

$$\ker \pi_D = \mathbb{Q} + j(O_f(D)), \quad j(o_f) = (o_f, 0). \quad (\text{ASC16})$$

If $x_\infty - q = Nn$, replace q by $q + Nn$ to express x in the right hand side. The reverse inclusion follows at once from (ASC15). This also proves that the kernel is closed, as the inverse image of the identity in the Hausdorff circle. It is not correct to use $x_\infty \bmod N\mathbb{Z}$ alone as (ASC15): the finite-adelic choice of q is essential. There is no additive section of π_D . The class of N/m for $m \geq 2$ has order m , whereas the additive adele group is torsion-free. A group-homomorphic section would send this nonzero class to a torsion element and could not be a section.

Let d_N be the metric on $\mathbb{R}/N\mathbb{Z}$ induced by the original absolute value on $\mathbb{R}$, so its circumference is N . Then the entire original relation is

$$(x, y) \in \mathcal{R}_{D,r} \iff \sum_{j=1}^r d_N(\pi_D x_j, \pi_D y_j) \leq a. \quad (\text{ASC17})$$

For the forward implication use the admitted real components $o_{j,\infty}$ as representatives of the circle differences. For the reverse choose q_j matching each finite difference; vary q_j by $N\mathbb{Z}$ so that the real difference has minimum absolute value. Such a minimum exists in every real coset. Those finitely many minimum representatives give $o_j = y_j - x_j - q_j$ with exactly the bound in (ASC17).

The full label map is

$$\Pi_D : G_L(\mathbb{A}) \longrightarrow G_L(\mathbb{R}/N\mathbb{Z}), \quad (x, \lambda) \longmapsto (\pi_D x, \lambda). \quad (\text{ASC18})$$

It is an additive pointed map. It commutes with the underlying $G_L(\mathbb{Z})$ scalar action $(n, \mu)(x, \lambda) = (nx, \mu \wedge \lambda)$ by additivity of π_D . The scalars $(0, \mu)$ give the explicit zero-amplitude masks $Z_\mu(x, \lambda) = (0, \mu \wedge \lambda)$, and $\Pi_D Z_\mu = Z_\mu \Pi_D$. These masks preserve the displayed tolerances. Multiplication by an integer of absolute value greater than one need not preserve the fixed tolerance bound a . Meeting a nonzero amplitude's label with a proper label while keeping that amplitude would leave the carrier, and is not an operation used here. Equations (ASC7),(ASC17) show that it pulls back the label-exact circle tolerance to the native symmetric part; it also pulls back the amplitude-pullback circle tolerance to the stated amplitude relation. The fibre over $(0, \lambda)$ for $\lambda < 1_L$ is its one labelled zero. The fibre over e is $\ker \pi_D \times \{1_L\}$. Every nonzero circle class has its full adele coset at top support. The ordinary pointed kernel is again the singleton $\{\tau\}$, although this map has the full nontrivial fibres just calculated. A pointed additive section of Π_D would, on amplitudes of the top group sector, yield an additive section of π_D ; the torsion argument excludes it.

This gives the complete original divisor relation, both source supports, the relative loop object, and the exact map into the tolerant circle used in Serre duality. It leaves no inference from a singleton pointed kernel to absence of supported sections. Nor does it identify this divisor cohomology with the spectral cohomology of the original Weil trace. The next comparison is the explicit duality calculation CTD, applied to (ASC18); the RDT theta map retains its original zeta and Gamma factors.

91 The full supported tolerant-circle Hom and its signed divisor receiver

Sources and scope. Alain Connes and Caterina Consani, *Riemann–Roch for $\overline{\text{Spec } \mathbb{Z}}$* , original author source `RR-J-final.tex`, define their signed bounded sections in Proposition `setup1`, equation `kerpsi`; the internal Hom in Appendix A, Lemma `catcat`, equation `catcat1`; the total-distance relations in Proposition `defnA`, equation `inequdef`; and their duality in Proposition `pontrj` and Theorem `serredual`. Appendix B, Proposition `proppi0`, defines the actual Dold–Kan-derived kernel and tolerant cokernel. The complete original source and its version-specific SHA-256 are retained in the source-use ledger. We read the original section on cohomology, the complete section `sec5`, and Appendices A and B for this calculation.

The later Connes–Consani paper *Riemann–Roch for the ring $\mathbb{Z}$* , original `RRZ.tex`, equations `xdefn` and `xdefnr`, uses an all-subset bound. Its preceding equation `normdefn` states the distinct signed bound. We prove the exact comparison of these two bounds below. The full support carrier is the one of The Clankers, *Split Support Geometry*, version 11, Definition `def:lattice-split`.

The objects calculated here are explicitly specified tolerant Γ -objects and their internal Homs. We do not redefine the authors’ $H^1(D)$ by declaring one of these objects to be its Dold–Kan lift. Neither the divisor H^1 nor its lift is identified with an arithmetic spectral H^1 .

91.1 The original divisor and its supported tolerant circle

Let L be a bounded distributive lattice with $0_L \neq 1_L$. Retain

$$\begin{aligned} D &= \sum_p d_p[p] + d_\infty[\infty], & N &= \prod_p p^{-d_p} > 0, & a &= e^{d_\infty} > 0, \\ I(D) &= N\mathbb{Z}, & \mathcal{O}(D)_f &= \prod_p p^{-d_p} \mathbb{Z}_p = N\widehat{\mathbb{Z}}, & \\ A_N &= \mathbb{R}/N\mathbb{Z}, & d_N(\bar{x}, \bar{z}) &= \min_{n \in \mathbb{Z}} |x - z - Nn|. \end{aligned} \tag{CTD1}$$

The equality $I(D) = N\mathbb{Z}$ follows by applying the valuation inequalities to q/N . The finite product retains every divisor coefficient; N need not be an integer.

For an abelian group A define the commutative monoid

$$G_L(A) = \{(0, \lambda) : \lambda \in L\} \cup (A \times \{1_L\}), \quad (x, \lambda) + (z, \nu) = (x+z, \lambda \vee \nu), \quad \tau = (0, 0_L), \quad e = (0, 1_L). \tag{CTD2}$$

It is a semimodule over the semiring $S = G_L(\mathbb{Z})$ by $(n, \kappa)(x, \lambda) = (nx, \kappa \wedge \lambda)$. The carrier condition is preserved: a nonzero product amplitude forces both relevant labels to be top. Distributivity of joins and meets proves both distributive laws of this scalar action.

For a commutative monoid M , write $HM(r_+) = M^r$. A pointed map acts by adding along each nonbasepoint fibre; the empty fibre has value τ . This defines the Eilenberg–MacLane Γ -object of the monoid. Define the tolerant object

$$\mathcal{T}_{N,a}^{L,\text{amp}}(r_+) = G_L(A_N)^r, \quad (X, Z) \in \mathcal{R}_r^{N,a,\text{amp}} \iff \sum_{j=1}^r d_N(x_j, z_j) \leq a, \tag{CTD3}$$

where x_j, z_j are the amplitudes, and all support labels remain in X, Z . The basepoint is the all- τ tuple. The relation is reflexive and symmetric. The triangle inequality gives

$$\sum_i d_N \left(\sum_{f(j)=i} x_j, \sum_{f(j)=i} z_j \right) \leq \sum_j d_N(x_j, z_j),$$

so every Γ structure map preserves the relation. Thus (CTD3) is an object of the original category $\Gamma\mathcal{T}_*$. Its support-sensitive carrier has a tolerance pulled back from amplitudes; in particular it relates τ to e . The label-exact object has the same carrier and the smaller relation

$$(X, Z) \in \mathcal{R}_r^{N,a,\text{eq}} \iff (X, Z) \in \mathcal{R}_r^{N,a,\text{amp}} \quad \text{and} \quad \lambda_j = \nu_j \quad (1 \leq j \leq r),$$

where λ_j, ν_j are the respective labels. It too is a tolerant Γ -object: reflexivity and symmetry hold, and a fold of coordinatewise equal labels produces equal joins. Denote it by $\mathcal{T}_{N,a}^{L,\text{eq}}$. Throughout the principal calculation below, $\mathcal{T}_{N,a}^L$ without a superscript means this *label-exact* object. Both versions and all four directions of internal-Hom comparison are retained in (CTD27)–(CTD28).

Use the exact original internal-Hom convention

$$\mathfrak{H}_D(k_+) = \text{Hom}_{\Gamma\mathcal{T}_*}(\mathcal{T}_{N,a}^L, \mathcal{T}_{1,1/4}^L(k_+ \wedge -)). \quad (\text{CTD4})$$

Its basepoint is the constant all- τ map. A pointed map $k_+ \rightarrow \ell_+$ acts by postcomposition with the corresponding target fold. We will also calculate the subfunctor $\mathfrak{H}_D^S$ whose underlying monoid maps are S -linear. This is an explicitly imposed restriction on morphisms, not a change to the original category in (CTD4). In particular, fixed tolerance bounds are not asserted to be preserved by multiplication by every integer in S .

91.2 All natural maps and the additional support maps

A natural map $HM \rightarrow HN(k_+ \wedge -)$ is exactly a k -tuple of pointed additive monoid maps $M \rightarrow N$. Indeed, naturality for the coordinate projections forces the map at level r_+ to apply its level-one maps to each of the r input coordinates. Naturality for the two-input fold then forces additivity, and pointedness forces preservation of zero. Conversely these properties commute with every finite fibre sum, hence give a natural map.

Let

$$J_0(L) = \{\theta : L \rightarrow L : \theta(0_L) = 0_L, \theta(\lambda \vee \nu) = \theta(\lambda) \vee \theta(\nu)\}, \quad J_{01}(L) = \{\theta \in J_0(L) : \theta(1_L) = 1_L\}.$$

These are definitions of sets of maps; $J_0(L)$ here does not denote the join-irreducible elements of L .

Every pointed additive monoid map $F : G_L(A) \rightarrow G_L(B)$ has exactly the form

$$F_{\chi,\theta}(x, \lambda) = (\chi(x), \theta(\lambda)), \quad \chi \in \text{Hom}_{\mathbb{Z}}(A, B), \quad \theta \in J_0(L), \quad \chi \neq 0 \implies \theta(1_L) = 1_L. \quad (\text{CTD5})$$

Here is a proof, including the support constraint. The additive idempotents of $G_L(B)$ are exactly $(0, \mu)$, because $b + b = b$ in the group B implies $b = 0$. Thus $F(0, \lambda) = (0, \theta(\lambda))$ with θ preserving joins and bottom. Set $\mu = \theta(1_L)$. For $\hat{x} = (x, 1_L)$, the identities $\hat{x} + e = \hat{x}$ and $\hat{x} + \neg x = e$ force the label of $F(\hat{x})$ to be both above and below μ , hence equal to μ . Write its amplitude as $\chi(x)$. The group law among the $\hat{x}$ proves additivity of χ . If $\chi(x) \neq 0$ for some x , the target carrier forces $\mu = 1_L$. This proves necessity and uniqueness. Conversely (CTD5) is well-defined, preserves bottom, and preserves addition by the stated properties of χ and θ .

The S -linear maps among (CTD5) are exactly

$$F_{\chi,\mu}(x, \lambda) = (\chi(x), \lambda \wedge \mu), \quad \mu \in L, \quad \chi \neq 0 \implies \mu = 1_L. \quad (\text{CTD6})$$

In fact scalar linearity applied to $(0, \lambda)e$ gives $\theta(\lambda) = \lambda \wedge \theta(1_L)$. Conversely the meet formula is additive by distributivity of L , and

$$F_{\chi,\mu}((n, \kappa)(x, \lambda)) = (n\chi(x), \kappa \wedge \lambda \wedge \mu) = (n, \kappa)F_{\chi,\mu}(x, \lambda).$$

It also preserves τ . Consequently the underlying, unbounded semimodule Hom is exactly

$$\text{Hom}_S(G_L(A), G_L(B)) \cong G_L(\text{Hom}_{\mathbb{Z}}(A, B)).$$

A k -tuple of such maps has k independent labels μ_i . It must not be replaced by $G_L(\text{Hom}(A, B)^k)$, which has a single common label.

At the unbounded level, pointwise addition and scalar action give

$$F_{\chi, \mu} + F_{\psi, \nu} = F_{\chi + \psi, \mu \vee \nu}, \quad (n, \kappa)F_{\chi, \mu} = F_{n\chi, \kappa \wedge \mu}. \quad (\text{CTD7})$$

In particular a nonzero amplitude map and its negative add to $F_{0, 1_L}$, not to the zero map $F_{0, 0_L}$.

91.3 The all-level tolerance excludes discontinuous characters

For maps (CTD5) from $G_L(A_N)$ to $G_L(\mathbb{R}/\mathbb{Z})$, equal input labels give equal output labels for every support map θ_i . The numerical part of the tolerance is the amplitude distance. The necessary and sufficient condition on a k -tuple (χ_i) is

$$\sum_{j=1}^r d_N(x_j, 0) \leq a \implies \sum_{j=1}^r \sum_{i=1}^k d_1(\chi_i(x_j), 0) \leq \frac{1}{4} \quad \text{for every } r \geq 1. \quad (\text{CTD8})$$

Necessity follows by taking the comparison tuple to have zero amplitudes and exactly the same coordinate labels as the input; all these zero-amplitude labelled coordinates are admissible. Conversely use additivity on amplitude differences in any pair; equal input labels already give equal output labels. This proves sufficiency at all levels, with every support retained.

No continuity is assumed. Repeat the same x in m input coordinates. Formula (CTD8) gives

$$d_N(x, 0) \leq a/m \implies \sum_{i=1}^k d_1(\chi_i(x), 0) \leq \frac{1}{4m}. \quad (\text{CTD9})$$

For each positive ϵ , choose m with $1/(4m) < \epsilon$. Then $d_N(x, 0) \leq a/m$ forces the distance of each $\chi_i(x)$ from zero to be less than ϵ . Thus each additive map is continuous at zero and, by translation, uniformly continuous. This eliminates all discontinuous additive maps by the actual all-level tolerance, not by an additional hypothesis.

For completeness, every continuous additive $\chi : A_N \rightarrow \mathbb{R}/\mathbb{Z}$ is uniquely

$$\chi_t(\bar{x}) = tx \bmod \mathbb{Z}, \quad t \in N^{-1}\mathbb{Z}. \quad (\text{CTD10})$$

To prove this, compose with $\mathbb{R} \rightarrow A_N$ and call the result φ . On a sufficiently small interval about zero, continuity gives a unique continuous real lift g with values in $(-1/8, 1/8)$. Whenever $x, y, x + y$ lie in that interval, $g(x + y) - g(x) - g(y)$ is an integer of absolute value less than $3/8$, hence is zero. Extend g to $\mathbb{R}$ by $G(x) = 2^m g(2^{-m}x)$ for sufficiently large m . Local additivity proves independence of m and global additivity. The extension agrees with g near zero and is continuous there; hence $G(x) = tx$, as follows first for rational x and then for real x by continuity. Addition in the circle gives $\varphi(x) = tx \bmod \mathbb{Z}$ at every x . Since $\varphi(N) = 0$, $tN \in \mathbb{Z}$. Uniqueness follows because $(t - t')x$ cannot be integer for every sufficiently small real x unless $t = t'$.

The exact coefficient bound is

$$(\text{CTD8}) \iff t_i \in N^{-1}\mathbb{Z} \ (1 \leq i \leq k), \quad a \sum_{i=1}^k |t_i| \leq \frac{1}{4}. \quad (\text{CTD11})$$

For sufficiency choose a closest representative $x - Nn$ in $[-N/2, N/2]$. Since $t_i Nn$ is integer,

$$d_1(t_i x, 0) \leq |t_i| d_N(\bar{x}, 0).$$

Summing this inequality proves (CTD8). For necessity choose an integer m so large that $a/m < N/2$ and $|t_i|a/m < 1/2$ for every i . Repeat $\bar{x} = a/m$ exactly m times. Its total source distance is exactly a , and no target term wraps around the circle. Consequently (CTD8) is precisely $m \sum_i |t_i|a/m \leq 1/4$. This proves the displayed bound, including its boundary equality.

91.4 The entire internal Hom and its canonical linear retract

The preceding proofs determine (CTD4) completely:

$$\begin{aligned} \mathfrak{H}_D(k_+) = \{((t_i, \theta_i))_{i=1}^k : t_i \in N^{-1}\mathbb{Z}, \quad a \sum_i |t_i| \leq 1/4, \quad \theta_i \in J_0(L), \\ t_i \neq 0 \implies \theta_i \in J_{01}(L)\}. \end{aligned} \quad (\text{CTD12})$$

The actual natural transformation represented by this tuple acts at level r_+ by

$$((x_j, \lambda_j))_{j=1}^r \longmapsto ((t_i x_j \bmod \mathbb{Z}, \theta_i(\lambda_j)))_{1 \leq i \leq k, 1 \leq j \leq r}.$$

A pointed map $\alpha : k_+ \rightarrow \ell_+$ acts by

$$(t_i, \theta_i)_i \longmapsto \left(\sum_{\alpha(i)=h} t_i, \bigvee_{\alpha(i)=h} \theta_i \right)_{h=1}^{\ell}, \quad (\text{CTD13})$$

where the join of maps is pointwise and an empty join is the constant-bottom map. It preserves (CTD12), since the triangle inequality preserves the coefficient bound, a pointwise join of join-preserving maps preserves joins, and a nonzero resulting coefficient has some nonzero input coefficient and therefore top output at 1_L . Formula (CTD13) also follows directly by folding the displayed natural transformation.

Its linear subfunctor is exactly

$$\mathfrak{H}_D^S(k_+) = \{((t_i, \mu_i))_{i=1}^k : t_i \in N^{-1}\mathbb{Z}, \quad a \sum_i |t_i| \leq 1/4, \quad \mu_i \in L, \quad t_i \neq 0 \implies \mu_i = 1_L\}. \quad (\text{CTD14})$$

The maps

$$\begin{aligned} \iota_D : \mathfrak{H}_D^S &\longrightarrow \mathfrak{H}_D, & (t_i, \mu_i) &\longmapsto (t_i, [\lambda \mapsto \lambda \wedge \mu_i]), \\ \rho_D : \mathfrak{H}_D &\longrightarrow \mathfrak{H}_D^S, & (t_i, \theta_i) &\longmapsto (t_i, \theta_i(1_L)), & \rho_D \iota_D &= \text{id} \end{aligned} \quad (\text{CTD15})$$

are pointed natural transformations. For the section, naturality uses $\lambda \wedge (\bigvee_i \mu_i) = \bigvee_i (\lambda \wedge \mu_i)$; for the retraction it uses $(\bigvee_i \theta_i)(1_L) = \bigvee_i \theta_i(1_L)$. The fibre of ρ_D over a tuple (t_i, μ_i) is exactly

$$\prod_{i=1}^k \{\theta \in J_0(L) : \theta(1_L) = \mu_i\}.$$

All these fibres are retained in (CTD12). The pointed kernel of ρ_D is a singleton: if $\theta(1_L) = 0_L$, monotonicity forces θ to be the constant-bottom map. This does not make ρ_D injective.

For example take $L = \{0_L < u < 1_L\}$. The map $\theta(0_L) = 0_L$, $\theta(u) = 0_L$, $\theta(1_L) = 1_L$ preserves joins and bottom, but differs from the identity meet map with $\mu = 1_L$. Already the zero character with this θ and the zero character with the identity map are distinct elements of $\mathfrak{H}_D(1_+)$ having the same image under ρ_D , for every D . Thus the proposed full supported coefficient carrier is a canonical retract, not the entire original-category Hom in general. For the two-element lattice all members of $J_0(L)$ are either the bottom map or the identity, so ρ_D is an isomorphism in that particular case.

91.5 The canonical divisor and the exact signed section map

Keep the source's actual divisor $K = -2[2]$. Then

$$\begin{aligned} I(K) &= 4\mathbb{Z}, & e^{K\infty} &= 1, & \mathcal{T}_K^L &= \mathcal{T}_{4,1}^L, \\ \mathcal{T}_{4,1}^L &\xrightarrow{\cong} \mathcal{T}_{1,1/4}^L, & (z \bmod 4\mathbb{Z}, \lambda) &\longmapsto (z/4 \bmod \mathbb{Z}, \lambda), \\ I(K - D) &= \frac{4}{N}\mathbb{Z}, & e^{(K-D)\infty} &= a^{-1}. \end{aligned} \quad (\text{CTD16})$$

The circle map and its inverse multiplication by 4 preserve every label. The distance identity $d_1(z/4, w/4) = d_4(z, w)/4$ proves preservation of the displayed tolerances at every level.

Define the supported signed section module by its full levels:

$$\mathcal{H}_L^{0,\pm}(K - D)(k_+) = \{((q_i, \mu_i))_{i=1}^k : q_i \in (4/N)\mathbb{Z}, q_i \neq 0 \Rightarrow \mu_i = 1_L, \sum_i |q_i| \leq 1/a\}. \quad (\text{CTD17})$$

Its folds sum amplitudes and join labels. The triangle inequality proves that each fold preserves the bound. This is the support lift of the signed section module of the 2022 source, with its full original ideal and real bound.

The exact pointed natural isomorphism is

$$\mathfrak{H}_D^S \xrightarrow{\cong} \mathcal{H}_L^{0,\pm}(K - D), \quad ((t_i, \mu_i)) \mapsto ((4t_i, \mu_i)), \quad q_i \mapsto t_i = q_i/4. \quad (\text{CTD18})$$

The ideal and the bound follow respectively from $t_i \in N^{-1}\mathbb{Z}$ and $a \sum_i |t_i| \leq 1/4$. All folds commute with multiplication by 4, and all labels are unchanged. Combining (CTD15) and (CTD18) therefore gives an explicit retract of the complete internal Hom onto the signed supported section module.

The underlying evaluation before the circle comparison is

$$((q, \mu), (x \bmod N\mathbb{Z}, \lambda)) \mapsto (qx \bmod 4\mathbb{Z}, \mu \wedge \lambda). \quad (\text{CTD19})$$

It is well-defined because $qN \in 4\mathbb{Z}$. For a k -tuple of coefficients and an r -tuple of inputs, its (i, j) entry is the same expression with q_i, x_j . The bound $\sum_i |q_i| \leq 1/a$ implies

$$\sum_{i,j} d_4(q_i x_j, q_i z_j) \leq \left(\sum_i |q_i| \right) \sum_j d_N(x_j, z_j) \leq 1.$$

Thus evaluation gives the actual required tolerant morphism into $\mathcal{T}_K^L(k_+ \wedge -)$, with the original factor 4 and all support meets present.

For principal covariance retain $\text{div}(c) = \sum_p v_p(c)[p] - \log |c|[\infty]$. For $D' = D + \text{div}(c)$ one has $N' = N/|c|$ and $a' = a/|c|$. The exact isomorphism $\mathcal{T}_D^L \rightarrow \mathcal{T}_{D'}^L$ is $(x, \lambda) \mapsto (x/c, \lambda)$. Its inverse induces on internal Homs, and on their signed receivers, the maps

$$(t_i, \theta_i) \mapsto (ct_i, \theta_i), \quad (q_i, \mu_i) \mapsto (cq_i, \mu_i). \quad (\text{CTD20})$$

Indeed $a' \sum |ct_i| = a \sum |t_i|$, and $I(K - D') = (4|c|/N)\mathbb{Z} = c(4/N)\mathbb{Z}$. All inverse formulas divide by c ; all support labels remain. Negative c reverses the amplitude signs and leaves the absolute bounds and support joins unchanged.

91.6 The strongest direct comparison with the all-subset module

Write $c_D = 1/a$ and $M_D = 4/N$, retaining these exact values. Let $\mathcal{U}_L(M_D, c_D)$ be the RDT all-subset module with amplitudes in $M_D\mathbb{Z}$ and every subset sum in $[-c_D, c_D]$. Let $\mathcal{S}_L(M_D, c_D)$ denote the signed module (CTD17). For a tuple q , put

$$P(q) = \sum_{q_i > 0} q_i, \quad Q(q) = \sum_{q_i < 0} |q_i|.$$

The original conditions, with every sign retained, are

$$\begin{aligned} q \in \mathcal{S}_L(M_D, c_D) &\iff P(q) + Q(q) \leq c_D, \\ q \in \mathcal{U}_L(M_D, c_D) &\iff P(q) \leq c_D \text{ and } Q(q) \leq c_D. \end{aligned} \quad (\text{CTD21})$$

All support conditions in both lines are identical. The identity on amplitudes and labels gives the exact natural inclusions

$$\mathcal{H}_L^{0,\pm}(K-D) = \mathcal{S}_L(M_D, c_D) \hookrightarrow \mathcal{H}_L^0(K-D) = \mathcal{U}_L(M_D, c_D) \hookrightarrow \mathcal{S}_L(M_D, 2c_D). \quad (\text{CTD22})$$

The second inclusion follows from $P+Q \leq 2c_D$. These are actual componentwise maps, commuting with all folds.

They agree at level 1_+ , but the first inclusion is strict at level 2_+ whenever $m = \lfloor c_D/M_D \rfloor \geq 1$: take $q = mM_D$ and the full supported tuple $((q, 1_L), (-q, 1_L))$. It is admitted by the all-subset condition, but $2mM_D > c_D$ because $c_D < (m+1)M_D \leq 2mM_D$. It therefore fails the signed condition. If $m = 0$, both modules are precisely $H(L, \vee, 0_L)$ in all levels.

There is no natural transformation from the middle module in (CTD22) to the first whose level-one map is the identity when $m \geq 1$. Projection naturality would force its value at the displayed two-coordinate tuple to have those same coordinates, which are not an element of the required signed target. This proves the exact limitation; it does not erase the inclusions or the full retained objects.

The exact excess object is the pointed Γ -cofiber

$$\mathcal{Q}_{D,L}(k_+) = \mathcal{U}_L(M_D, c_D)(k_+)/\mathcal{S}_L(M_D, c_D)(k_+). \quad (\text{CTD23})$$

Here the indicated subset is collapsed to one basepoint. The remaining elements are precisely the full labelled tuples with $P, Q \leq c_D$ and $P+Q > c_D$. Folds are the original folds, followed by this stated collapse; well-definedness follows because the signed object is a subfunctor. This is a pointed-set cofiber, not a claimed abelian quotient or a faithful replacement for the two original modules. Equations (CTD12), (CTD15), (CTD18), and (CTD22) retain every source and comparison in the duality calculation.

91.7 Pullback along the actual adelic quotient

The source's Proposition [thedimh1](#) uses the adelic quotient with finite subgroup $\mathcal{O}(D)_f$, not just an unqualified real coordinate projection. Explicitly, for $x = (x_\infty, x_f) \in \mathbb{A}$, choose $r \in \mathbb{Q}$ such that $x_f - r \in \mathcal{O}(D)_f$ and set

$$\pi_D(x) = x_\infty - r \bmod N\mathbb{Z}. \quad (\text{CTD24})$$

Such an r exists by the following complete finite calculation. Choose a positive integer M with $Mx_f/N \in \widehat{\mathbb{Z}}$. For each prime dividing M , prescribe the residue of an integer j to be that of Mx_p/N modulo $p^{v_p(M)}$. The Chinese remainder theorem supplies j . At primes not dividing M , both Mx_p/N and j are integral and M is a unit. Hence $Mx_f/N - j \in M\widehat{\mathbb{Z}}$ at every prime. Take $r = Nj/M \in \mathbb{Q}$; then $x_f - r \in N\widehat{\mathbb{Z}}$, as required. Two choices differ by $\mathbb{Q} \cap N\widehat{\mathbb{Z}} = N\mathbb{Z}$, so the formula is independent of the choice. Choosing sums of the rational representatives proves additivity. It is surjective by restricting to adeles $(t, 0)$, and

$$\ker \pi_D = \mathbb{Q} + (\{0\} \times \mathcal{O}(D)_f).$$

Both containments follow directly from (CTD24).

Give $H(G_L(\mathbb{A}))$ the relation $\sum_j d_N(\pi_D x_j, \pi_D z_j) \leq a$ together with equality of the two labels in each coordinate, and denote the resulting explicit tolerant object by $\widetilde{\mathcal{T}}_D^L$. Then precomposition with the label-preserving $G_L(\pi_D)$ gives exact natural isomorphisms

$$\begin{aligned} \text{Hom}_{\Gamma\mathcal{T}_*}(\widetilde{\mathcal{T}}_D^L, \mathcal{T}_{1,1/4}^L(k_+ \wedge -)) &\cong \mathfrak{H}_D(k_+), \\ \text{Hom}_{\Gamma\mathcal{T}_*}^S(\widetilde{\mathcal{T}}_D^L, \mathcal{T}_{1,1/4}^L(k_+ \wedge -)) &\cong \mathfrak{H}_D^S(k_+). \end{aligned} \quad (\text{CTD25})$$

For the proof apply (CTD5) to each component. If $x \in \ker \pi_D$, compare repeated inputs $(x, 1_L)$ with the same number of coordinates $(0, 1_L)$. They have equal labels and zero source distance.

Preservation of the target tolerance forces $m d_1(\chi_i(x), 0) \leq 1/4$ for every m , hence $\chi_i(x) = 0$. Thus each amplitude homomorphism factors uniquely through π_D . All support maps θ_i are unchanged. Surjectivity of π_D shows that the factored maps satisfy exactly the circle tolerance. Conversely composition with π_D preserves it by definition. This proves both isomorphisms and compatibility with every internal-Hom fold.

This pullback has an exact receiving map from the native supported adelic construction: (ASC7) and (ASC17)–(ASC18) identify its relation with the maximal symmetric subrelation of the degree-zero directed relation of $X(D)$. Thus (CTD25) applies to that calculated symmetric part, with its original vertex carrier and all its fibres. It is not an identification of the directed relation with a symmetric relation, nor a claim that (ASC8)’s pointed spherical set is the signed section module. The actual signed sections are the retract of the complete relative zero-section object in (ASC9)–(ASC13).

The assertion is a proof about the explicitly defined pullback object. In particular the functor G_L does not preserve pointed kernels of group maps. For an additive map $f : A \rightarrow B$,

$$G_L(f)(x, \lambda) = (f(x), \lambda), \quad G_L(f)^{-1}(\tau_B) = \{\tau_A\}, \quad (\text{amplitude} \circ G_L(f))^{-1}(0) = G_L(\ker f). \quad (\text{CTD26})$$

The first kernel statement follows because an output equal to τ_B requires $\lambda = 0_L$, hence $x = 0$ by the carrier condition. The amplitude-zero fibre follows by requiring only $f(x) = 0$. This exact difference prevents an unproved transfer of the original Dold–Kan kernel through G_L . The original source’s $H^1(D)$ and its amplitude comparison are retained; the support lift calculated here has been specified and its actual maps proved, without identifying it with an unconstructed spectral cohomology.

91.8 Both tolerances and all four directions of comparison

The identity on the full carrier induces a natural morphism

$$j_{N,a} : \mathcal{T}_{N,a}^{L,\text{eq}} \longrightarrow \mathcal{T}_{N,a}^{L,\text{amp}}. \quad (\text{CTD27})$$

It retains every element and label, and preserves relations because the label-exact relation is contained in the amplitude relation. The reverse identity is not a tolerant morphism for a nontrivial lattice: τ and e are related in the amplitude object and are not related in the label-exact one. The map (CTD27) forgets a condition on pairs; it does not identify those two points.

For superscripts $\alpha, \beta \in \{\text{eq}, \text{amp}\}$, write $\mathfrak{H}_D^{\alpha \rightarrow \beta}$ for the internal Hom from $\mathcal{T}_{N,a}^{L,\alpha}$ to $\mathcal{T}_{1,1/4}^{L,\beta}$. Then the exact classifications, as pointed Γ -objects, are

$$\begin{aligned} \mathfrak{H}_D^{\text{eq} \rightarrow \text{eq}} &\cong \mathfrak{H}_D^{\text{amp} \rightarrow \text{amp}} \cong \mathfrak{H}_D^{\text{eq} \rightarrow \text{amp}} = \text{the object in (CTD12),} \\ \mathfrak{H}_D^{\text{amp} \rightarrow \text{eq}} &= \{*\}. \end{aligned} \quad (\text{CTD28})$$

The same statement holds for the S -linear subfunctors, with (CTD14) as the common nontrivial object. Here the displayed isomorphisms keep the actual natural transformations, their characters, and all their support maps. They are not isomorphisms between the original tolerant source objects themselves.

To prove the first three classifications, a morphism in any of those directions is still of the form (CTD5) by Γ -naturality. In each case the repeated top-labelled inputs and top-labelled zero comparisons used in (CTD8)–(CTD11) are related in the source, so they force the same continuity and exact coefficient bound. Conversely that coefficient bound preserves the amplitude part of the relation. If the target relation is label-exact in one of these three cases, the source relation is also label-exact, and applying θ_i to equal labels gives equal labels. This proves necessity and sufficiency. Precomposing or postcomposing with (CTD27) realizes the displayed isomorphisms on actual maps.

For the remaining direction, every pair $((0, \lambda), \tau)$ is related in the source amplitude tolerance. A morphism into the label-exact target must therefore satisfy $\theta_i(\lambda) = \theta_i(0_L) = 0_L$ for every λ and

every output coordinate i . Thus every θ_i is the bottom map. Since $\theta_i(1_L) = 0_L \neq 1_L$, (CTD5) forces $\chi_i = 0$. The only possible natural transformation is the constant all- τ transformation, which is indeed a morphism. This proves the final assertion at every level.

Replacing the label-exact adelic relation in (CTD25) by its amplitude version gives the corresponding same-tolerance pullback isomorphism, by the identical repeated-input proof. The mixed direction with amplitude source and label-exact target remains the one-point object. Consequently there is no silent exchange of support tolerances in the duality diagram: the label-exact calculation has its complete nontrivial character and support receiver, the amplitude calculation has the same internal-Hom parameters when used on both sides, and switching only the target to label-exact destroys all nonzero maps.

91.9 Every finite-support cardinality and the zero-character range

For finite L , set $j_0 = \#J_0(L)$, $j_{01} = \#J_{01}(L)$, and $m = \lfloor N/(4a) \rfloor$. Then the complete level counts are

$$\begin{aligned} \#\mathfrak{H}_D(k_+) &= \sum_{r=0}^k \binom{k}{r} 2^r \binom{m}{r} j_{01}^r j_0^{k-r}, \\ \#\mathfrak{H}_D^S(k_+) &= \sum_{r=0}^k \binom{k}{r} 2^r \binom{m}{r} |L|^{k-r}. \end{aligned} \tag{CTD29}$$

Use $\binom{m}{0} = 1$ and $\binom{m}{r} = 0$ for $r > m$. Indeed write each character coefficient as $t_i = n_i/N$, retaining the original N . The exact bound (CTD11) is $\sum_i |n_i| \leq N/(4a)$, hence $\sum_i |n_i| \leq m$. Choose its r nonzero positions, their 2^r signs, and their ordered positive magnitudes with total at most m . The latter have number $\binom{m}{r}$: their successive partial sums form an increasing r -element subset of $\{1, \dots, m\}$, with inverse successive differences. Each nonzero coefficient allows exactly j_{01} support maps in the full Hom, and each zero coefficient allows exactly j_0 . In the linear subfunctor the respective numbers are one and $|L|$. This proves both formulas, including the single empty tuple when $k = 0$.

Without any finiteness hypothesis on L , if $a > N/4$ then $m = 0$ and the exact entire Γ -objects are

$$\mathfrak{H}_D = H(J_0(L), \vee, 0), \quad \mathfrak{H}_D^S = H(L, \vee, 0_L) \quad (a > N/4). \tag{CTD30}$$

The join of maps in the first expression is pointwise and its zero is the constant-bottom map. All character coefficients vanish by (CTD11), but their support maps do not disappear. For example the map with zero character and $\theta = \text{id}_L$ sends every top-supported input to e and every $(0, \lambda)$ to itself. It differs from the constant- τ map and satisfies the stated tolerances. Equation (CTD30) follows also directly from the fold formula (CTD13); the linear retract is exactly evaluation at 1_L and its meet-map section. Thus absence of nonzero amplitude characters leaves the entire calculated support receiver.

92 Geometric realization of the full supported restricted adelic object

Sources and scope. The original restricted simplicial object is due to Alain Connes and Caterina Consani, *Riemann–Roch for $\overline{\text{Spec } \mathbb{Z}}$* , original author source `RR-J-final.tex`, Appendix B, particularly `descript1`, `descript2`, `descript1sub`, `descript1sub1`, `Maydefn`, and `proppi0`. Their divisor restrictions are in `setup–setup1`. The support carrier is that of The Clankers, *Split Support Geometry*, version 11, Definition `def:lattice-split`. The comparison proved here applies to the full independent-leg extension in ASC1–ASC5b. It concerns geometric realizations of simplicial sets; the adele amplitudes have the discrete topology at this stage, as in that simplicial-set construction. No topology on L is assumed. In particular it does not identify realization homotopy groups with the strict spherical-simplex relation used in the original source.

92.1 The two original arrays and their amplitude map

Let L be a bounded distributive lattice with $0_L \neq 1_L$, and set

$$\begin{aligned} G_L(A) &= \{(0, \lambda) : \lambda \in L\} \cup (A \times \{1_L\}), & (x, \lambda) + (y, \mu) &= (x + y, \lambda \vee \mu), \\ \tau_A &= (0, 0_L), & e_A &= (0, 1_L), \\ D &= \sum_p d_p[p] + d_\infty[\infty], & N &= \prod_p p^{-d_p}, & a &= e^{d_\infty}, & O_f(D) &= N\widehat{\mathbb{Z}}. \end{aligned} \tag{SRC1}$$

Here $N, a > 0$ and only finitely many d_p are nonzero. Put $B_L = G_L(\mathbb{A}_{\mathbb{Q}})$ and $P_L = G_L(\mathbb{Q}) \times G_L(\mathbb{A}_{\mathbb{Q}})$; its two factors have independent labels. The action map is

$$\Phi_L((q, \alpha), (o, \beta)) = (q + o, \alpha \vee \beta), \tag{SRC2}$$

where q is embedded diagonally in the adeles. For a pointed finite set r_+ the supported n -simplices are

$$\begin{aligned} b &= (b_j) \in B_L^r, & z_\ell &= (z_{\ell j}) \in P_L^r \quad (1 \leq \ell \leq n), \\ z_{\ell j} &= ((q_{\ell j}, \alpha_{\ell j}), (o_{\ell j}, \beta_{\ell j})), & o_{\ell j, f} &\in O_f(D), & \sum_{\ell=1}^n \sum_{j=1}^r |o_{\ell j, \infty}| &\leq a. \end{aligned} \tag{SRC3}$$

The amplitude object Y_r has exactly these arrays, finite restrictions and total bound, with the labels forgotten. Write X_r for the supported object. Neither object replaces the total bound by individual bounds on the entries.

For the proof temporarily use the usual chronological bar faces ∂_i :

$$\begin{aligned} \partial_0(b; z_1, \dots, z_n) &= (b + \Phi_L z_1; z_2, \dots, z_n), \\ \partial_i(b; z_1, \dots, z_n) &= (b; z_1, \dots, z_i + z_{i+1}, \dots, z_n) \quad (0 < i < n), \\ \partial_n(b; z_1, \dots, z_n) &= (b; z_1, \dots, z_{n-1}). \end{aligned} \tag{SRC4}$$

The degeneracy σ_i inserts $(\tau_{\mathbb{Q}}, \tau_{\mathbb{A}})$ after the first i arrows. ASC5 uses the reversed convention $d_i = \partial_{n-i}$ and $s_i = \sigma_{n-i}$. Reversal of barycentric coordinates identifies the two realizations, including their pointed finite-set maps, so this convention retains the actual ASC object. All faces preserve (SRC3) by the triangle inequality, and degeneracies insert real component zero. Summation over the nonbasepoint fibres of a pointed finite-set map preserves the same bound and joins each leg's labels separately. An empty fibre inserts the relevant τ .

Thus forgetting labels gives the original pointed simplicial Γ -map

$$p_D : X(D) \longrightarrow Y(D). \tag{SRC5}$$

It is onto in every degree, but this is not the argument for the homotopy assertion below. Over the all-zero array its actual degree- (n, r) fibre is $L^{(2n+1)r}$; the maps in that fibre are the prescribed joins and identity insertions.

92.2 The zero-cost arrow and its exact bar prism

Fix r . Let $c = c_r \in P_L^r$ have each coordinate $(e_{\mathbb{Q}}, e_{\mathbb{A}})$. Then

$$c + c = c, \quad \Phi_L c = e_{\mathbb{A}}^r, \quad c_{\ell j, \text{amplitude}} = 0. \quad (\text{SRC6})$$

It is admitted with real cost zero for every original divisor D . For $r = 0$ this is the unique empty array. Define a map $t_r : Y_r \rightarrow X_r$ on simplices by attaching top labels to the vertex and to each of the two legs of every arrow, including amplitude-zero entries. Then

$$p_r t_r = \text{id}, \quad t_r p_r(b; z_1, \dots, z_n) = (b + \Phi_L c; z_1 + c, \dots, z_n + c). \quad (\text{SRC7})$$

Both maps in (SRC7) commute with every face. For the initial face, the supported vertex after the action is $b + \Phi_L c + \Phi_L(z_1 + c) = b + \Phi_L z_1 + \Phi_L c$. For an interior face, $(z_i + c) + (z_{i+1} + c) = z_i + z_{i+1} + c$. The last face only drops an arrow. These identities also cover zero amplitudes and independent zero-leg labels.

The map t_r is a map of semisimplicial sets, not generally of simplicial sets: t_r turns an amplitude identity arrow into c , whereas a degeneracy in X_r inserts $(\tau_{\mathbb{Q}}, \tau_{\mathbb{A}})$. For $r > 0$, the two arrows are different. No simplicial section has been assumed.

For $0 \leq i \leq n$ define the following admitted $(n+1)$ -simplex:

$$H_i(b; z_1, \dots, z_n) = (b; z_1, \dots, z_i, c, z_{i+1} + c, \dots, z_n + c). \quad (\text{SRC8})$$

Every rational and adelic amplitude in this array is the original amplitude, with one new zero row. The finite components still belong to $N\widehat{\mathbb{Z}}$ and the full real sum in (SRC3) is unchanged, not merely bounded above by a different estimate. The two source labels of every displayed $z_{\ell} + c$ are joined independently with 1_L . They have not first been identified with one common label.

The prism identities are

$$\begin{aligned} \partial_0 H_0 &= t_r p_r, & \partial_{n+1} H_n &= \text{id}, \\ \partial_j H_i &= H_{i-1} \partial_j & (j < i), \\ \partial_j H_i &= H_i \partial_{j-1} & (j > i+1), \\ \partial_i H_i &= \partial_i H_{i-1} & (1 \leq i \leq n). \end{aligned} \quad (\text{SRC9})$$

Here all operators on the right have the degree determined by their input. For $j < i$ the deleted or merged arrows precede the inserted c ; the case $j = 0$ updates the same vertex by the same $\Phi_L z_1$. For $j > i+1$ they follow c ; merging uses $(z + c) + (z' + c) = z + z' + c$. At the seam the two results are $z_i + c$ and $c + (z_i + c)$, which agree. The endpoint identities follow by dropping the last c , or by applying $\Phi_L c$ to the initial vertex. This proves every identity without an additive inverse in the supported monoids.

Let $\|K\|$ denote fat realization, in which only face identifications are imposed. Triangulate $\Delta^n \times [0, 1]$ by the $n+1$ simplices with vertices

$$(v_0, 0), \dots, (v_i, 0), (v_i, 1), \dots, (v_n, 1) \quad (0 \leq i \leq n). \quad (\text{SRC10})$$

On that simplex use the characteristic map labelled by H_i . The last three identities in (SRC9) make these maps agree on all common faces and with all face identifications from the original n -simplex. The endpoint identities give a continuous homotopy from the identity to $\|t_r\| \|p_r\|$. Continuity follows on each prism and hence on the quotient: product with the locally compact Hausdorff interval preserves the quotient map defining realization. Since $p_r t_r$ is the identity,

$$\|p_r\| : \|X_r\| \longrightarrow \|Y_r\| \quad \text{is a homotopy equivalence.} \quad (\text{SRC11})$$

This argument does not pretend that (SRC8) respects degeneracies. After applying p_r , however, its inserted arrow is the amplitude identity. Consequently its image in the ordinary realization $|Y_r|$ is constant in the homotopy parameter. The displayed homotopy lies over the fixed map $\|X_r\| \rightarrow |Y_r|$.

92.3 A cellular proof of the fat-to-ordinary comparison

For completeness the following argument proves the realization fact needed here, so no unverified theorem about a simplicial topological space is imported. For any simplicial set K , there is a natural quotient

$$q_K : \|K\| \longrightarrow |K| \quad (\text{SRC12})$$

which also imposes the degeneracy identifications. It is a weak homotopy equivalence by the complete cellular argument below. The further conclusion that it is a homotopy equivalence uses the classical CW Whitehead theorem, originating in J. H. C. Whitehead, *Combinatorial homotopy. I*, *Bulletin of the American Mathematical Society* **55** (1949), 213–245. This is an attribution of the classical theorem; the present source-reading claim does not include the 1949 article.

First, every simplex has a unique expression α^*x where x is nondegenerate and α is an order-preserving surjection. Existence follows by repeatedly removing a degeneracy. To verify uniqueness, the indices i for which a simplex lies in the image of σ_i are exactly the indices with $\alpha(i) = \alpha(i+1)$. One inclusion follows by factoring α through the elementary surjection identifying $i, i+1$. For the converse, if $\alpha(i) < \alpha(i+1)$, these are consecutive values, and one can choose an order-preserving section of α containing both i and $i+1$. Pulling an expression in the image of σ_i back along that section would express x as a degeneracy, a contradiction. These equal-adjacent indices determine α uniquely. Pullback along any section then determines x .

Fat realization is a CW space with one cell for every simplex, including every degenerate simplex: its interior is not identified with the interior of another cell by a face identification. Each closed simplex has only finitely many faces, giving closure finiteness, and the quotient topology is the weak cell topology. Ordinary realization has one cell for each nondegenerate simplex, by the preceding unique decomposition. For any simplicial subset, both induced inclusions are CW-subcomplex inclusions.

Write $K^{(d)}$ for the simplicial subset generated by the nondegenerate simplices of dimension at most d , with $K^{(-1)} = \emptyset$. Unique decomposition gives the pushout

$$\begin{array}{ccc} \coprod_{x \in K_d^{\text{nd}}} \partial \Delta[d] & \longrightarrow & K^{(d-1)} \\ \downarrow & & \downarrow \\ \coprod_{x \in K_d^{\text{nd}}} \Delta[d] & \longrightarrow & K^{(d)} \end{array} \quad (\text{SRC13})$$

Both realization constructions preserve this pushout: they are defined by disjoint unions of products with simplices followed by the indicated face, or face-and-degeneracy, quotient. In the fat construction $\|\Delta[d]\|$ is contractible. An n -simplex of $\Delta[d]$ is a nondecreasing list $(v_0, \dots, v_n)$ with entries in $\{0, \dots, d\}$. The explicit contraction is

$$[(v_0, \dots, v_n); (t_0, \dots, t_n)] \longmapsto [(v_0, \dots, v_n, d); ((1-u)t_0, \dots, (1-u)t_n, u)], \quad 0 \leq u \leq 1. \quad (\text{SRC14})$$

It respects every face identification, starts at the given point, and ends at vertex d . Ordinary $|\Delta[d]| = \Delta^d$ is also contractible. Thus $q_{\Delta[d]}$ is a homotopy equivalence.

We recall the elementary gluing argument used with (SRC13). A pushout of a span $C \leftarrow A \hookrightarrow B$, whose indicated leg is a CW-subcomplex inclusion, is homotopy equivalent to its double mapping cylinder $C \cup (A \times [0, 1]) \cup B$. To see this, the cylinder collapse maps to the pushout. The homotopy extension property for $A \subset B$ extends the path on A from the B end of the cylinder to its C end, starting with the inclusion of B . The resulting end map on B agrees with the inclusion of C along A and defines a map back from the pushout. The collapse followed by this map is homotopic to the identity by that extended homotopy on B , the constant homotopy on C , and $(a, s) \mapsto (a, (1-u)s)$ on the connecting cylinder. The reverse composite is homotopic to

the identity after collapsing the same homotopy. An objectwise homotopy equivalence of two such spans induces a homotopy equivalence of their double mapping cylinders: choose inverses on A, B, C ; their commutation with each attaching map up to homotopy follows by inserting the chosen inverse homotopies into the commuting square. On a cylinder use, in order, the first such attaching homotopy, the cylinder on the inverse map of A , and the reverse of the second attaching homotopy. This defines an inverse. Its two composites are homotopic to the identity by the chosen inverse homotopies and cancellation of each path with its reverse. All pieces agree at the cylinder ends. This proves the gluing assertion needed here.

Induct on d , simultaneously for all simplicial sets of nondegenerate dimension at most d . The assertion for $d = -1$ is empty. In (SRC13) the boundary $\partial\Delta[d]$ and the previous skeleton have nondegenerate dimension at most $d - 1$, so their comparison maps are homotopy equivalences by induction. The comparison on each $\Delta[d]$ is a homotopy equivalence by (SRC14), and the boundary inclusions are CW-subcomplex inclusions. The gluing argument proves that $q_{K(d)}$ is a homotopy equivalence. This includes $d = 0$, when the boundary is empty.

Finally both full realizations are unions of these simplicial skeleta. A compact subset of a CW space lies in a finite subcomplex: if it met infinitely many open cells, selecting one point in each of infinitely many different cells would produce an infinite closed discrete subset. Indeed each closed cell meets only finitely many selected points by closure finiteness; the same holds for every subset of the selected points, so the weak topology makes every such subset closed. This contradicts compactness. Each finite collection of fat cells has a finite maximum nondegenerate-ancestor dimension. Therefore every sphere, disk and homotopy in either full realization is contained in some displayed simplicial skeleton. The finite-stage homotopy equivalences give isomorphisms on all homotopy groups, including components, in the union. This proves that (SRC12) is a weak equivalence without invoking the CW Whitehead theorem. The cited CW Whitehead theorem then gives the asserted homotopy equivalence. Naturality of (SRC12) is immediate from its definition as the additional quotient.

92.4 The full conclusion and its exact limitations

For every original divisor D and every finite pointed level r_+ there is a commuting square

$$\begin{array}{ccc} \|X_r\| & \xrightarrow{\|p_r\|} & \|Y_r\| \\ q_{X_r} \downarrow & & \downarrow q_{Y_r} \\ |X_r| & \xrightarrow{|p_r|} & |Y_r| \end{array} \quad (\text{SRC15})$$

The upper map is a homotopy equivalence by (SRC6)–(SRC11), and both vertical maps are weak equivalences by (SRC12)–(SRC14). It follows on every homotopy group and on components that $|p_r|$ is a weak equivalence; since its source and target are CW spaces, it is a homotopy equivalence. The maps $|p_r|$ themselves are pointed and commute with *every* pointed finite-set map. Thus they give a Γ -natural levelwise weak equivalence of the full supported and original restricted objects.

The displayed inverse $\|t_r\|$ and the displayed prism are levelwise and need not be pointed or Γ -natural. At an empty output fibre a pointed finite-set map inserts τ , whereas t_r would have inserted top support. This is an explicit failure of naturality, not an omitted check. In fact there is no pointed simplicial Γ -section of p_D : at degree zero such a section would give a pointed additive section $\mathbb{A} \rightarrow G_L(\mathbb{A})$. The forced lifts of $x \neq 0$ and $-x$ add to e , whereas the pointed lift of zero is τ . The fold map would therefore fail. This does not contradict the proved Γ -natural weak equivalence, which concerns the amplitude projection itself and asserts no natural strict inverse.

In particular the singleton strict spherical set at τ^r in ASC8 cannot be used to conclude that $\pi_1(|X_r|, \tau^r)$ is zero. The exact map identifies that group with $\pi_1(|Y_r|, 0)$, and likewise identifies every higher homotopy group at the indicated basepoint. As a concrete retained loop, an admitted signed vector $q \in (N\mathbb{Z})^r$ with $\sum_j |q_j| \leq a$ gives the loop $(q, -q)$ at the top-labelled zero vertex.

The edge c from τ^r to that vertex, followed by this loop and the reverse geometric path along c , is a based topological loop at τ^r . Under $|p_r|$ it is exactly the original amplitude loop, since the two c paths become degenerate. No assertion about that loop's nontriviality is needed or made without the corresponding calculation in Y_r .

All original divisor parameters, finite ideals, rational and adelic amplitudes, independent support labels and total real restrictions have been retained in the calculation. Realization homotopy does not retain every distinction of the strict support relations: ASC6–ASC13 record those distinctions and their exact fibres. The result here proves their relation to the original realization; it does not identify either realization with the arithmetic spectral cohomology occurring in a Weil trace formula.

The maps and their geometric meaning

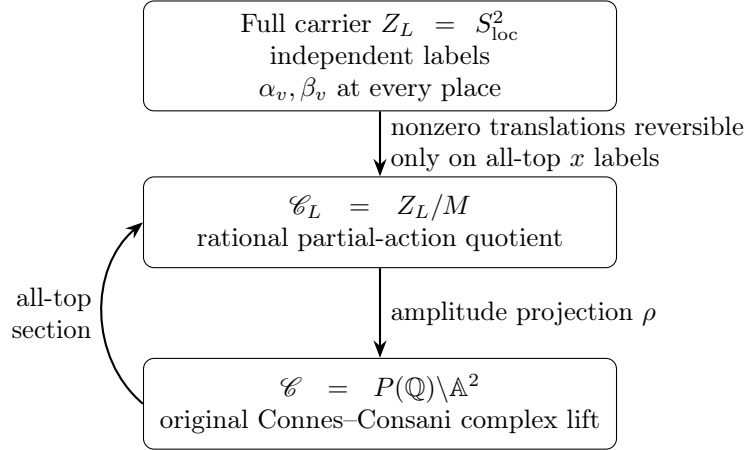


Figure 28: The complete quotient is calculated in SAP1–11. The section recovers the original action but is not an inverse on the full supported object. Above a free representative (y, x) , the fibre is exactly $L^{Z(y)} \times (\{*\} \sqcup \coprod_{t \in \mathbb{Q}} (L^{Z(x+t)} \setminus \{1\}))$. Thus partially supported translations retain the actual rational parameter t , not merely an additional unstructured point. Human source: Connes–Consani, 1805.10501v1, Section 5.2.

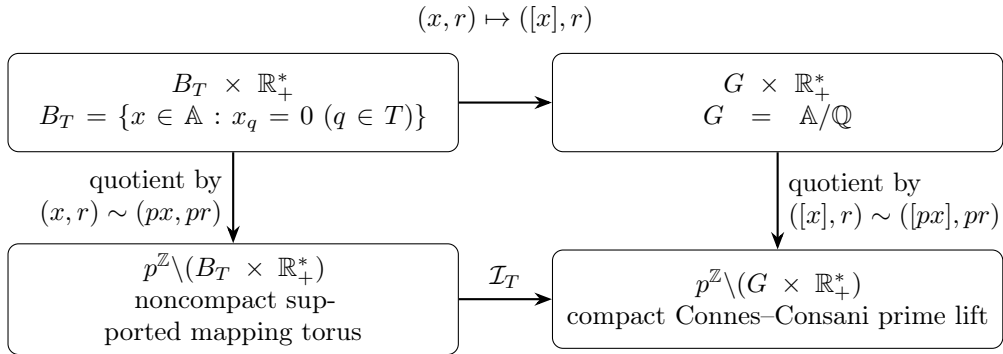


Figure 29: SPL5–10: the same prime and original period $\log p$ occur on both sides. The horizontal arrows are dense proper injections, not identifications of topology or Haar measure. The additive map $B_T \rightarrow G$ has cokernel $\mathbb{A}_T/\mathbb{Q}$ as an additive group, with its non-Hausdorff quotient topology. Every lower x label on T and the y_p label λ remain fixed. T is any nonempty set of finite primes. The diagram states no invented topology on the lattice.

$ \begin{aligned} p &\in T \\ \mu_T(pE) &= p\mu_T(E) \\ \ V_p f\ ^2 &= p\ f\ ^2 \end{aligned} $	$ \begin{aligned} p &\notin T \\ \mu_T(pE) &= \mu_T(E) \\ \ V_p f\ ^2 &= \ f\ ^2 \end{aligned} $
---	--

Original operator: $V_p f(x) = f(x/p)$ Exact modulus: $ p _\infty \prod_{q \notin T} p _q = p^{\mathbf{1}_{p \in T}}$ Compact target: multiplication by p preserves Haar probability on G .
--

Figure 30: SPL14–17 derive all factors using the original product Haar measure. The half-density factor $p^{-\mathbf{1}_{p \in T}/2}$ relates V_p to a separate unitary operator; it is not a proof of a zeta weight or of RH. Pushing the source Haar measure into compact G gives infinite total mass and hence not a Radon measure. The exact map is necessary to any proposed trace comparison.

$ \begin{aligned} \mathcal{J} &= \bigoplus_{j \geq 0} \delta^{(j)}(x) \mathcal{D}'((0, \infty)) \\ L_\ell &= y(\ell \partial_x + i \partial_y), \quad \ell > 0 \\ \text{Res}_\ell \left(\sum_j \delta^{(j)} g_j \right) &= \sum_j (-i/\ell)^j \partial_y^j (g_j/y) \end{aligned} $

Figure 31: The branch does not carry the full ordinary affine flow. The displayed distributional receiver retains every finite normal jet and calculates its exact differential, residue and arithmetic Frobenius comparison in SDB. The blue line is the complete half-line, drawn with finite length; the distributional arrows are formulas, not a claim that a delta distribution is an ordinary function. Human source for L_ℓ : Connes–Consani, Section 7.1, equation `holom`.

93 Supported parabolic matrices, inverse domains, and the complete action category

Sources and scope. Alain Connes and Caterina Consani, *The Riemann–Roch strategy: Complex lift of the Scaling Site*, original author source `thecurve_K.tex`, define $P(R)$, its affine matrix plane, and its coordinates in Section 5.2, equations `alggroupP`, `algspacP`, and `algspaciota`. The left rational action is `actionpq`, the adelic quotient is `adelicomp3`, and Lemma `Pdescrofgamma` gives the matrix presentation and right real action. Proposition `ax+bact` and Section 6.6, Proposition `comparescalcs`, describe the complex structure. Section 6.2, Lemma `fundinclusion`, equation `P2gl20`, gives the comparison with the GL_2 system. Section 7, equations `tropdet`, `holom-holom1`, retains the distinct complex-structure issue and the Witt-coefficient lift.

The original-source reading for this calculation covers lines 1115–1457, 1801–1917, 2024–2087, 2342–2411, and 2526–2635. Its SHA-256 is

da52001209424c19fb223905804a9eed1147c72331cf881aa0352bc759976061.

The support operations are those of The Clankers, *Split Support Geometry*, version 11, Definition `def:lattice-split`. The algebra and actions below are derived from these operations. A classical corner is not substituted for the complete supported object, and no RH conclusion is inferred.

93.1 The matrices and their exact product

Let R be a commutative ring with identity and let L be a bounded distributive lattice, initially with $0_L \neq 1_L$. Retain

$$\begin{aligned} G_L(R) &= \{(0, \lambda) : \lambda \in L\} \cup (R \times \{1_L\}), \\ (x, \lambda) + (z, \nu) &= (x + z, \lambda \vee \nu), \quad (x, \lambda)(z, \nu) = (xz, \lambda \wedge \nu), \\ \tau &= (0, 0_L), \quad e = (0, 1_L), \quad \mathbf{1} = (1_R, 1_L). \end{aligned} \quad (\text{SPA1})$$

Define

$$\mathcal{P}_L(R) = \left\{ M(a, b, \beta) = \begin{pmatrix} (a, 1_L) & (b, \beta) \\ \tau & \mathbf{1} \end{pmatrix} : a \in R^\times, b \in R, \beta \in L, b \neq 0 \Rightarrow \beta = 1_L \right\}. \quad (\text{SPA2})$$

An element is also denoted (a, b, β) . Entrywise multiplication using (SPA1) gives

$$(a, b, \beta)(c, d, \delta) = (ac, ad + b, \beta \vee \delta), \quad 1_{\mathcal{P}} = E_{0_L} := (1_R, 0, 0_L). \quad (\text{SPA3})$$

In particular the upper-right entry is $(a, 1_L)(d, \delta) + (b, \beta)\mathbf{1} = (ad + b, \delta \vee \beta)$. The lower-left entry stays τ , and the diagonal entries have the stated form. A nonzero $ad + b$ forces $b \neq 0$ or $d \neq 0$, hence top output support. This proves closure. Both triple-product bracketings give $(acr, act + ad + b, \beta \vee \delta \vee \theta)$, proving associativity. The displayed identity is two-sided.

93.2 Generalized inverse and every local group

The unique inverse in the inverse-semigroup sense is

$$s^* = (a^{-1}, -a^{-1}b, \beta), \quad ss^* = s^*s = E_\beta := (1_R, 0, \beta), \quad ss^*s = s, \quad s^*ss^* = s^*. \quad (\text{SPA4})$$

It is admitted because multiplication by a unit preserves nonvanishing. The equalities follow from (SPA3). For uniqueness, if $t = (c, d, \delta)$ satisfies $sts = s$ and $tst = t$, the first amplitude equation gives $aca = a$, hence $c = a^{-1}$, and the translation equation then gives $ad + b = 0$. The first support equation gives $\delta \leq \beta$, and the second gives $\beta \leq \delta$. Thus $t = s^*$ with the same label.

Every idempotent is exactly an E_β , and

$$E_\beta E_\delta = E_{\beta \vee \delta}, \quad E_\delta s = s E_\delta. \quad (\text{SPA5})$$

Indeed $a^2 = a$ with a a unit gives $a = 1_R$; the remaining amplitude equation $2b = b$ gives $b = 0$. The converse and the displayed formulas follow by multiplication. Therefore $\mathcal{P}_L(R)$ is a Clifford inverse monoid: it is an inverse monoid and all its idempotents are central.

The natural inverse-semigroup order, defined by $s \leq_P t$ if $s = Et$ for some idempotent E , is

$$(a, b, \beta) \leq_P (c, d, \delta) \iff a = c, \quad b = d, \quad \delta \leq_L \beta. \quad (\text{SPA6})$$

Multiplication by E_η proves necessity, and $\eta = \beta$ proves sufficiency. The idempotent order reverses the given lattice order; this reversal is retained.

The maximal subgroup with identity E_β is

$$H_\beta = \begin{cases} \{(a, 0, \beta) : a \in R^\times\} \cong R^\times, & \beta < 1_L, \\ \{(a, b, 1_L) : a \in R^\times, b \in R\} \cong P(R), & \beta = 1_L. \end{cases} \quad (\text{SPA7})$$

These are exactly the elements with $s^*s = ss^* = E_\beta$, and (SPA3)–(SPA4) prove their group laws and inverses. For nontrivial L , the global units are only H_{0_L} . A top-supported translation has an inverse in its corner, not a global inverse.

Even an inverse sought in the entire matrix monoid cannot change that conclusion. Suppose that $M(a, b, \beta)X = I_2$. Its bottom row forces $X_{21} = \tau$ and $X_{22} = \mathbf{1}$. The upper-right sum can equal τ only if both summands are τ , because a joined support is bottom only when both supports are bottom. Since $(a, 1_L)$ is a scalar unit, this forces $X_{12} = \tau$ and $(b, \beta) = \tau$. Conversely those conditions give the diagonal inverse. Thus the off-diagonal supported zero cannot be treated as the global matrix zero.

93.3 The exact universal group image

The amplitude map is the surjective unital homomorphism

$$\pi : \mathcal{P}_L(R) \rightarrow P(R), \quad (a, b, \beta) \mapsto \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}, \quad \pi^{-1}(a, b) = \begin{cases} \{(a, b, 1_L)\}, & b \neq 0, \\ \{(a, 0, \beta) : \beta \in L\}, & b = 0. \end{cases} \quad (\text{SPA8})$$

It is universal for homomorphisms into groups. To prove this, let $f : \mathcal{P}_L(R) \rightarrow G$ be a unital monoid homomorphism with G a group. Every $f(E_\beta)$ is idempotent and hence equals 1_G . Therefore

$$f(a, b, \beta) = f((a, b, \beta)E_{1_L}) = f(a, b, 1_L).$$

The map $(a, b) \mapsto f(a, b, 1_L)$ is the unique group homomorphism through which f factors. The multiplicative top-corner section sends the classical identity to E_{1_L} , not to the global identity E_{0_L} . Thus group completion identifies all distinct support idempotents, with its complete fibres given in (SPA8).

93.4 The total affine action and exact inverse domains

Put $X_L = G_L(R)$. Multiplying the affine column with lower coordinate $\mathbf{1}$ gives

$$F_s(x, \lambda) = (ax + b, \lambda \vee \beta), \quad s = (a, b, \beta). \quad (\text{SPA9})$$

The output is admitted: nonzero amplitude requires a nonzero input amplitude or translation, and the corresponding label is top. Substitution proves $F_s F_t = F_{st}$ and $F_{E_{0_L}} = \text{id}$. The action is faithful: $F_s(\tau) = (b, \beta)$ recovers b, β , and $F_s(\mathbf{1}) = (a + b, 1_L)$ recovers a .

The full inverse identities are

$$F_{E_\beta}(x, \lambda) = (x, \lambda \vee \beta), \quad F_{s^*} F_s = F_s F_{s^*} = F_{E_\beta}. \quad (\text{SPA10})$$

The maximal domain on which these displayed inverse maps recover the input exactly is

$$D_\beta = \text{Fix}(F_{E_\beta}) = \{(x, \lambda) : \beta \leq \lambda\} = (R \times \{1_L\}) \cup \{(0, \lambda) : \lambda \geq \beta\}. \quad (\text{SPA11})$$

Restricted to D_β , the map F_s is a bijection onto D_β with inverse F_{s^*} and unchanged label. Both necessity and sufficiency follow from $\lambda \vee \beta = \lambda$ in (SPA10). Also $F_s(X_L) = D_\beta$: the output label is above β , and the restriction is already onto. No larger domain containing D_β keeps that restriction injective. A point z outside it has the same image as $F_{E_\beta}z \in D_\beta$, by $F_s F_{E_\beta} = F_s$.

Every fibre of the total action is exactly

$$F_s^{-1}(y, \nu) = \{(a^{-1}(y - b), \lambda) : \lambda \vee \beta = \nu, a^{-1}(y - b) \neq 0 \Rightarrow \lambda = 1_L\}. \quad (\text{SPA12})$$

For $b \neq 0$, every zero label is sent to $(b, 1_L)$; the generalized inverse then returns e . At the nonzero input $x = -a^{-1}b$, cancellation produces e , not τ , and the inverse on D_{1_L} recovers the input. For $b = 0$ and proper β , unit multiplication preserves nonzero amplitudes and raises each zero label by the stated join.

The restrictions form a faithful representation by partial bijections:

$$\alpha_s = F_s|_{D_\beta}, \quad \alpha_s^{-1} = \alpha_{s^*}, \quad D_\beta \cap D_\delta = D_{\beta \vee \delta}, \quad \alpha_s \alpha_t = \alpha_{st}. \quad (\text{SPA13})$$

Indeed α_t preserves its input label, so the composition domain consists exactly of labels above both δ and β . Its formula is that of st . Equality of two partial maps implies equality of their zero-label upsets, hence of β, δ , and their values on the top copy of R recover both affine coefficients. This proves faithfulness.

93.5 The Karoubi category and its action

The Karoubi category $\mathcal{K}_L(R)$ has objects E_β and hom-sets $E_\delta \mathcal{P}_L(R) E_\beta$ from E_β to E_δ . Their exact form is

$$\text{Hom}_{\mathcal{K}_L(R)}(E_\beta, E_\delta) = \{(a, b, \gamma) \in \mathcal{P}_L(R) : \gamma \geq \beta \vee \delta\}. \quad (\text{SPA14})$$

This follows by multiplying on both sides and comparing the resulting label $\delta \vee \gamma \vee \beta$. The identity is E_β . For $s = (a, b, \gamma) : E_\beta \rightarrow E_\delta$ and $t = (c, d, \eta) : E_\delta \rightarrow E_\epsilon$,

$$t \circ s = (ca, cb + d, \eta \vee \gamma).$$

Its support is above β, ϵ , and associativity and identities follow from (SPA3).

Every arrow factors exactly as

$$E_\beta \xrightarrow{E_\gamma} E_\gamma \xrightarrow{(a, b, \gamma)} E_\gamma \xrightarrow{E_\gamma} E_\delta. \quad (\text{SPA15})$$

The middle arrow lies in H_γ . An arrow is a categorical isomorphism precisely when $\beta = \delta = \gamma$. In fact a categorical inverse of support η would give composites with support $\gamma \vee \eta$ equal to both β and δ . Since $\gamma \geq \beta, \delta$, this forces equality of all four labels; the converse follows from (SPA4). Thus distinct support objects are not isomorphic, and $\text{Aut}(E_\beta) = H_\beta$.

The total affine action defines

$$\mathcal{D} : \mathcal{K}_L(R) \rightarrow \text{Sets}, \quad E_\beta \mapsto D_\beta, \quad s \mapsto F_s|_{D_\beta} : D_\beta \rightarrow D_\delta. \quad (\text{SPA16})$$

For an arrow of support $\gamma \geq \beta, \delta$, its output label $\lambda \vee \gamma$ is above δ . The object identity acts as the identity, and composition is (SPA9). This functor is faithful on each hom-set: evaluation at $(0, \beta)$ recovers (b, γ) ; evaluation at $(1_R, 1_L)$ then recovers a . A general arrow need not act bijectively, because its inverse-pair composites are F_{E_γ} rather than the source and target identities.

The full one-object top subcategory is

$$\text{End}_{\mathcal{K}_L(R)}(E_{1_L}) = E_{1_L} \mathcal{P}_L(R) E_{1_L} = H_{1_L} \cong P(R). \quad (\text{SPA17})$$

Here *full* means that every arrow between the chosen object and itself is included. It does not mean a generating or Morita-full idempotent: for nontrivial L ,

$$\mathcal{P}_L(R)E_{1_L}\mathcal{P}_L(R) = H_{1_L} \neq \mathcal{P}_L(R),$$

since the left side omits E_{0_L} .

The action category has objects (E_β, z) , $z \in D_\beta$. An arrow labelled $s : E_\beta \rightarrow E_\delta$ has target $(E_\delta, F_s z)$; composition retains the actual s label. Its full subcategory over E_{1_L} is exactly

$$P(R) \ltimes R, \quad (\text{SPA18})$$

via $(x, 1_L) \leftrightarrow x$. All stabilizer arrows remain. No germ quotient identifying arrows that agree at a single point is taken.

93.6 Both original Connes–Consani coordinates

Retain independent coordinate supports in

$$W(y, \lambda; x, \nu) = \begin{pmatrix} (y, \lambda) & (x, \nu) \\ \tau & \mathbf{1} \end{pmatrix}, \quad (y, \lambda), (x, \nu) \in G_L(R). \quad (\text{SPA19})$$

This is $G_L(R)_y \times G_L(R)_x$, not $G_L(R^2)$ with one common label. The original matrix $\iota(x, y) = \begin{pmatrix} y & x \\ 0 & 1 \end{pmatrix}$ is its all-top subspace, including both zero amplitudes.

The two entrywise products are

$$\begin{aligned} M(a, b, \beta)W(y, \lambda; x, \nu) &= W(ay, \lambda; ax + b, \nu \vee \beta), \\ W(y, \lambda; x, \nu)M(u, v, \delta) &= W(yu, \lambda; yv + x, \nu \vee (\lambda \wedge \delta)). \end{aligned} \quad (\text{SPA20})$$

They follow directly from (SPA1). Associativity proves that the left and right total actions commute. Explicitly both orders give amplitudes $(ayu, a(x + yv) + b)$ and labels $(\lambda, \nu \vee \beta \vee (\lambda \wedge \delta))$.

Their maximal inverse-pair domains are

$$\begin{aligned} D_\beta^{\text{left}} &= \{W(y, \lambda; x, \nu) : \beta \leq \nu\}, \\ D_\delta^{\text{right}} &= \{W(y, \lambda; x, \nu) : \lambda \wedge \delta \leq \nu\}. \end{aligned} \quad (\text{SPA21})$$

These are the fixed sets of the corresponding idempotent maps in (SPA20). On the stated domain each action preserves both labels, and its generalized inverse is a genuine inverse. Necessity follows by computing its inverse-pair composite. Maximality as an injective domain containing the stated domain follows because any additional point has the same image as its idempotent image inside that domain.

Right multiplication is a right action: $\rho_t \circ \rho_s = \rho_{st}$ when $\rho_s(W) = Ws$. The corresponding partial maps have this same right composition convention. Moreover

$$D_\delta^{\text{right}} \cap D_\eta^{\text{right}} = D_{\delta \vee \eta}^{\text{right}},$$

because $\lambda \wedge (\delta \vee \eta) = (\lambda \wedge \delta) \vee (\lambda \wedge \eta)$. On their respective inverse domains both actions preserve both coordinate labels. Consequently the two partial actions commute on exactly their common domain. All affine cancellations keep the joined supports in (SPA20). For example a right translation with $\delta = 1_L$ requires $\lambda \leq \nu$, not that both coordinates be top-supported.

93.7 The classical action groupoid and its larger supported corner

The top-translation fixed set is

$$X_L^{\text{tr}}(R) = G_L(R)_y \times (R \times \{1_L\})_x. \quad (\text{SPA22})$$

Its x label is top, and every y label remains. It is invariant under the top affine group on the left and on the right: $\nu = 1_L$ satisfies every condition in (SPA21), and both inverse-domain actions preserve it. The original classical subspace is the smaller invariant set

$$X^{\text{cl}}(R) = (R \times \{1_L\})_y \times (R \times \{1_L\})_x \cong R^2. \quad (\text{SPA23})$$

The restrictions of (SPA20) there are exactly $(y, x) \mapsto (ay, ax + b)$ and $(y, x) \mapsto (yu, yv + x)$.

For $R = \mathbb{A}_{\mathbb{Q}}$ and rational left coefficients, use the injective diagonal map $\mathbb{Q} \rightarrow R$ to let $\mathcal{P}_L(\mathbb{Q})$ act. Apply the Karoubi action construction to the left fixed sets in (SPA21). Its full subcategory on E_{1_L} and states in (SPA23) is precisely

$$P(\mathbb{Q}) \ltimes \mathbb{A}_{\mathbb{Q}}^2. \quad (\text{SPA24})$$

The bijection records each original adelic state and each rational affine coefficient. All source, target, identity, and composition formulas agree by (SPA20). This is the transformation groupoid presenting the original Connes–Consani quotient **adelicomp3**. The source uses $\iota(x, y)$; the matrix first row is (y, x) , as retained throughout (SPA19)–(SPA24).

The entire top-translation corner is larger when L is nontrivial: it still contains $((0, \lambda), (x, 1_L))$ for every $\lambda < 1_L$. Thus the classical groupoid is a full invariant subcategory of that corner, not the entire corner. The amplitude projection of (SPA22), with x fixed, has one-point fibres at $y \neq 0$ and fibre L at $y = 0$.

A placewise adelic carrier has a separate label at each zero place; it is not $G_L(\mathbb{A}_{\mathbb{Q}})$ with one global label. Formulas (SPA20) apply at every place, and the separately constructed adelic product retains its placewise fibres. No replacement of that product by one global label is asserted.

93.8 The original real action and its analytic content

For the source’s right real action, the multiplier is the identity at every finite place and has real component $\begin{pmatrix} u & v \\ 0 & 1 \end{pmatrix}$. Thus finite coordinates are unchanged and at infinity

$$(y_{\infty}, x_{\infty}) \longmapsto (uy_{\infty}, x_{\infty} + vy_{\infty}). \quad (\text{SPA25})$$

Differentiating $(u, v) = (e^{\epsilon}, 0)$ and $(1, \epsilon)$ on the original top-supported complex leaf gives

$$Y = y_{\infty} \partial_{y_{\infty}}, \quad X = y_{\infty} \partial_{x_{\infty}}, \quad [Y, X] = X, \quad X + iY = y_{\infty} (\partial_{x_{\infty}} + i \partial_{y_{\infty}}).$$

There is a typographical error in the original Section 6.6 display for $u(\epsilon)$: its upper-left entry is printed as 0, although it is called a one-parameter subgroup and its following multiplication retains y . The correct entry is 1, as follows from the source’s definition **alggroupP** and the entrywise product (SPA20). No calculation here uses that printed 0 as a group element.

Right scaling $(u, v) = (\mu, 0)$ sends the original coordinate $z = x_{\infty} + iy_{\infty}$ to $x_{\infty} + i\mu y_{\infty}$. With the source’s operator convention,

$$(\partial_{x_{\infty}} + i \partial_{y_{\infty}})(x_{\infty} + i\mu y_{\infty}) = 1 - \mu. \quad (\text{SPA26})$$

It is not holomorphic in that unchanged complex structure unless $\mu = 1$. The support construction retains this exact calculation on its faithful classical subcategory. It does not by itself replace the distinct Witt-coefficient mechanism in **holom-holom1**. This identifies the surviving analytic issue while retaining the complete algebraic comparison.

93.9 Boundary conventions

The two identity roles remain distinct:

$$E_{0_L} \neq E_{1_L}, \quad F_{E_{0_L}}(\tau) = \tau, \quad F_{E_{1_L}}(\tau) = e, \quad F_{E_{1_L}}(e) = e. \quad (\text{SPA27})$$

Every intermediate idempotent and zero label remains in the category and its action. No completeness assumption on L is used: all joins and meets are finite. Zero divisors in R do not change the proofs, because every division is by an explicitly required unit.

If L has one element, its bottom and top coincide and

$$G_L(R) = R, \quad \mathcal{P}_L(R) = P(R), \quad D_{0_L} = R. \quad (\text{SPA28})$$

There is one Karoubi object and the original affine group action. If R is the zero ring, its unit set has one element, every amplitude is zero, and the formulas reduce to $(L, \vee, 0_L)$ acting on the zero labels by joins. Every local group is then a singleton, while the category and the inverse-domain formulas remain valid.

94 The full supported adelic affine quotient and its real actions

Human source and retained coordinates. Alain Connes and Caterina Consani, *The Riemann–Roch strategy, Complex lift of the Scaling Site*, original author source `thecurve_K.tex`, Section 5.2, define the affine action in `actionpq`, the quotient in `adelicomp3`, and its matrix coordinates in `algspacetiota`. Lemma `Pdescrofgamma` gives the two commuting actions and proves freeness of the right real action on its specified open set. The original source was read here at lines 1115–1451, including Proposition `ax+bact` and the calculation of `lapxy`; this is not a claim to have read the entire paper. Their coordinate order in the affine-plane formula is (x, y) ; their actual matrix is $\begin{pmatrix} y & x \\ 0 & 1 \end{pmatrix}$. We retain the two entries and write the ordered matrix row (y, x) throughout. Thus no exchange of the action formulas is implicit.

The coefficient is the placewise carrier proved in PLP1–PLP4 of

`CC_PERIODIC_POINTS_FULL_LATTICE.tex`.

It differs from the common-label carrier $G_L(\mathbb{A})$. Its lattice operations come from The Clankers, *Split Support Geometry*, version 11, Definition `def:lattice-split`. All orbit quotients below are set quotients with the stated actions; no topology on an arbitrary lattice, complex structure on its elements, or semiring structure on an orbit set is assumed.

94.1 All places, both labels, and the rational inverse monoid

Let L be a nontrivial bounded distributive lattice, and let $\Sigma = \{\infty\} \cup \{p : p \text{ prime}\}$. Put $\Lambda = L^\Sigma$, with coordinatewise order and joins, $\mathbf{0} = (0_L)_v$, and $\mathbf{1} = (1_L)_v$. Retain

$$\begin{aligned} G_L(K) &= \{(0, \lambda) : \lambda \in L\} \cup (K \times \{1_L\}), \\ (c, \lambda) + (d, \mu) &= (c + d, \lambda \vee \mu), \quad (c, \lambda)(d, \mu) = (cd, \lambda \wedge \mu), \\ \mathbb{A}_{G,L} &= G_L(\mathbb{R}) \times \prod_p G_L(\mathbb{Q}_p), \quad Z = \mathbb{A}_{G,L}^2. \end{aligned} \tag{SAP1}$$

The restricted product is relative to $G_L(\mathbb{Z}_p)$. An element of Z is denoted $(y, x; \alpha, \beta)$, where $y, x \in \mathbb{A}_{\mathbb{Q}}$ and

$$y_v \neq 0 \implies \alpha_v = 1_L, \quad x_v \neq 0 \implies \beta_v = 1_L. \tag{SAP2}$$

All zero coordinates have their independent labels. For an adele w , write $Z(w) = \{v : w_v = 0\}$. A tuple in $L^{Z(w)}$ is always extended by top labels outside $Z(w)$ when used below.

The original rational action, including both constants, is

$$(a, b)(y, x) = (ay, ax + b), \quad (a, b)(a', b') = (aa', ab' + b), \quad a, a' \in \mathbb{Q}^\times. \tag{SAP3}$$

The rational b is diagonal in *every* local field. To give its exact supported partial action, define the monoid

$$\begin{aligned} M &= \mathbb{Q}^\times \ltimes G_\Lambda(\mathbb{Q}), \quad m = (a, b, \gamma), \quad b \neq 0 \implies \gamma = \mathbf{1}, \\ (a, b, \gamma)(a', b', \gamma') &= (aa', ab' + b, \gamma \vee \gamma'), \quad 1_M = (1, 0, \mathbf{0}). \end{aligned} \tag{SAP4}$$

The product belongs to this carrier: a nonzero resulting translation requires a nonzero translation in at least one factor and therefore top joined support. Its unique inverse in the inverse-monoid sense is

$$(a, b, \gamma)^\dagger = (a^{-1}, -a^{-1}b, \gamma), \quad m^\dagger m = mm^\dagger = (1, 0, \gamma). \tag{SAP5}$$

Associativity follows by expanding (SAP4). The displayed inverse satisfies $mm^\dagger m = m$ and $m^\dagger mm^\dagger = m^\dagger$. If $n = (c, d, \delta)$ satisfies both equations, their affine amplitude parts force $c = a^{-1}$ and $d = -a^{-1}b$, while the label equations give $\gamma \vee \delta = \gamma$ and $\delta \vee \gamma = \delta$. Thus $\delta = \gamma$ and the inverse is unique. The idempotents are exactly $(1, 0, \gamma)$ and commute.

For $\gamma \in \Lambda$ define $D_\gamma = \{(y, x; \alpha, \beta) \in Z : \gamma \leq \beta\}$. Then the faithful representation of M by partial bijections is

$$T_{a,b,\gamma} : D_\gamma \longrightarrow D_\gamma, \quad (y, x; \alpha, \beta) \longmapsto (ay, ax + b; \alpha, \beta). \quad (\text{SAP6})$$

If $b = 0$, multiplication by nonzero a changes no zero place. If $b \neq 0$, $\gamma = \mathbf{1}$ forces $\beta = \mathbf{1}$, so every translated coordinate is admitted. The source and target labels agree and the inverse is (SAP5), proving bijectivity. Composition has domain $D_\gamma \cap D_{\gamma'} = D_{\gamma \vee \gamma'}$ and formula (SAP4). The identity has domain all of Z . Distinct γ have distinct domains, since zero amplitudes permit every β ; the amplitude formula on the common top-supported part determines a, b . This proves faithfulness.

Pure dilations $(a, 0, \mathbf{0})$ act everywhere. Every nonzero translation acts only on $\beta = \mathbf{1}$. The element $(1, 0, \gamma)$ is the identity on its displayed domain; it does not discard the points outside that domain or identify their labels.

94.2 Every rational orbit and every amplitude fibre

Let $\mathcal{C}_L = Z/M$ be the equivalence relation generated by these partial bijections, and let $\mathcal{C} = P(\mathbb{Q}) \backslash \mathbb{A}^2$ be the original Connes–Consani set quotient. Amplitudes induce the surjection

$$\rho : \mathcal{C}_L \longrightarrow \mathcal{C}. \quad (\text{SAP7})$$

It has the explicit section assigning top labels at every place on both entries. That section is well defined under all original affine changes of representative, and ρ followed by this choice in the reverse order is not asserted to be the identity on $\mathcal{C}_L$.

Each allowed arrow preserves both complete tuples α, β . At $\beta = \mathbf{1}$, its orbits are the original affine orbits, retaining the admissible tuple α . At each $\beta \neq \mathbf{1}$, every available affine arrow has $b = 0$; the orbits are precisely diagonal $\mathbb{Q}^\times$ -orbits of (y, x) , retaining both labels. A chain of partial arrows cannot leave this label tuple to gain a translation and return: every arrow preserves the tuple. This proves the full orbit assertion, not only a description of the top corner.

An ordinary point (y, x) has trivial stabilizer under (SAP3) unless $y = 0$ and x is a diagonal rational. Indeed a nonzero coordinate of y forces $a = 1$, then $b = 0$. When $y = 0$, any stabilizer with $a \neq 1$ gives $x = -b/(a - 1) \in \mathbb{Q}$ at every place. Conversely for $y = 0, x = r \in \mathbb{Q}$, all $(a, (1 - a)r)$ stabilize the pair. These exceptional points make one ordinary affine orbit, that of $(0, 0)$.

Fix a free ordinary representative (y, x) . There is the following exact representative-dependent parametrization of its entire fibre:

$$\rho^{-1}([y, x]) \simeq L^{Z(y)} \times \left(\{*\} \sqcup \coprod_{t \in \mathbb{Q}} (L^{Z(x+t)} \setminus \{\mathbf{1}\}) \right). \quad (\text{SAP8})$$

For the $*$ summand the representative is $(y, x; \alpha, \mathbf{1})$. For the (t, β) summand it is $(y, x + t; \alpha, \beta)$, with all forced top entries restored. Here the notation $\mathbf{1}$ in each factor means that factor's unique all-top tuple, including the unique empty tuple when the zero-place set is empty.

To prove surjectivity of (SAP8), an amplitude in the ordinary orbit is $(ay, ax + b)$. The supported dilation by a^{-1} is always available and gives $(y, x + b/a)$ with the same labels. If $\beta = \mathbf{1}$, translation returns to (y, x) . Otherwise set $t = b/a$; the labels are exactly those permitted by $Z(y)$ and $Z(x + t)$. To prove injectivity, labels themselves cannot change. In the non-top part, an identification must have the form $c(y, x + t) = (y, x + t')$, $c \in \mathbb{Q}^\times$. If $y \neq 0$ then $c = 1$ and $t = t'$. If $y = 0$, its free-orbit hypothesis says $x \notin \mathbb{Q}$; $c \neq 1$ would give $x = (t' - ct)/(c - 1) \in \mathbb{Q}$. Again $c = 1, t = t'$. Top and non-top labels cannot be identified. This proves the bijection, and proves why the rational translation parameter remains.

For another free representative $(y', x') = (a_0 y, a_0 x + b_0)$ the exact transition is

$$\alpha' = \alpha, \quad * = *, \quad t' = a_0 t - b_0, \quad \beta' = \beta, \quad x' + t' = a_0(x + t), \quad Z(y') = Z(y). \quad (\text{SAP9})$$

Consequently $Z(x' + t') = Z(x + t)$ as subsets of the original place set; there is no relabelling of places. Composition of two changes gives the corresponding affine transformation of t , by (SAP3). Thus (SAP8) is canonical relative to the chosen representative, with the full transition law recorded.

Above the exceptional orbit the answer is instead

$$\rho^{-1}([0, 0]) \simeq \Lambda \times \Lambda, \quad (\alpha, \beta) \mapsto [0, 0; \alpha, \beta]. \quad (\text{SAP10})$$

Every ordinary amplitude in that orbit is $(0, r)$ with $r \in \mathbb{Q}$. For $r \neq 0$ all local x -coordinates are nonzero, forcing $\beta = \mathbf{1}$; the available translation by $-r$ gives the indicated representative. When $r = 0$ all labels already occur. Distinct labels cannot be identified, proving both existence and uniqueness in (SAP10). This also exhibits the nonfree stabilizer explicitly rather than applying (SAP8) to it.

The cardinality formula for (SAP8) is the cardinality of that displayed disjoint union and product; subtraction of infinite cardinals is not used. For finite L of size $m \geq 2$, when $Z(y)$ and every $Z(x + t)$ are finite it reads

$$|\rho^{-1}([y, x])| = m^{|Z(y)|} \left(1 + \sum_{t \in \mathbb{Q}} (m^{|Z(x+t)|} - 1) \right). \quad (\text{SAP11})$$

The sum is an ordinary sum of nonnegative integers if it has finite support, and is countably infinite otherwise. Different sets $Z(x + t)$ are disjoint: $x_v + t = x_v + t' = 0$ forces $t = t'$. If $Z(y)$ or one $Z(x + t)$ is infinite, countability of Σ gives the exact cardinal $m^{\aleph_0}$ instead. Indeed the corresponding factor or summand has that cardinal, and the whole expression has at most a countable union of products of that cardinal. The exceptional cardinal is always $m^{\aleph_0}$. For an arbitrary infinite L , (SAP8), (SAP10) are the exact cardinal expressions without a finiteness assumption. For example, the free point $(y, x) = (1, 0)$ has fibre Λ : only $t = 0$ has zero coordinates in $x + t$, and $Z(y)$ is empty.

94.3 The actual top-supported right real formula

Write $g = (u, v) \in P(\mathbb{R})$, $u \neq 0$, with multiplication $(u, v)(u', v') = (uu', uv' + v)$. At the infinite place the original matrix product is $(y_\infty, x_\infty) \mapsto (uy_\infty, x_\infty + vy_\infty)$. The finite components are unchanged. Lifting every real scalar c as $(c, 1_L)$, including zero as $e = (0, 1_L)$, gives the full formula

$$F_g(y, x; \alpha, \beta)_\infty = ((uy_\infty, \alpha_\infty), (x_\infty + vy_\infty, \beta_\infty \vee \alpha_\infty)), \quad (F_g z)_p = z_p \quad (p < \infty). \quad (\text{SAP12})$$

The zero contribution when $v = 0$ still has label α_∞ ; replacing it by τ would change the formula. In particular define

$$\begin{aligned} C &= F_{(1,0)}, & C(y, x; \alpha, \beta) &= (y, x; \alpha, \beta^\sharp), \\ \beta_\infty^\sharp &= \beta_\infty \vee \alpha_\infty, & \beta_p^\sharp &= \beta_p, \\ W &= \text{im } C = \{\alpha_\infty \leq \beta_\infty\}. \end{aligned} \quad (\text{SAP13})$$

All entries remain in their original carriers, and $C^2 = C$. Direct expansion of both real amplitudes and both labels gives

$$F_h \circ F_g = F_{gh}, \quad F_g C = C F_g = F_g, \quad F_{g^{-1}} F_g = F_g F_{g^{-1}} = C. \quad (\text{SAP14})$$

For the translation entry this expansion is $x_\infty + vy_\infty + v'uy_\infty = x_\infty + (uv' + v)y_\infty$; the repeated join with α_∞ is unchanged. Thus the identity element of the real group acts as C , not as the identity of the full source.

The restrictions $A_g = F_g|_W$ give a genuine right group action, with inverse $A_{g^{-1}}$. It is the maximal subset of Z on which the *displayed full formula* is a unital group action: any such subset must be fixed by $F_{(1,0)} = C$, and hence must lie in W . The exact factorization on the whole source is

$$F_g : Z \xrightarrow{C} W \xrightarrow{A_g} W \hookrightarrow Z. \quad (\text{SAP15})$$

For fixed finite entries and fixed real $y_\infty, x_\infty, \alpha_\infty, \beta_\infty^\sharp$, the fibre of C consists exactly of

$$\{\beta_\infty \in L : \beta_\infty \vee \alpha_\infty = \beta_\infty^\sharp, \quad x_\infty \neq 0 \Rightarrow \beta_\infty = 1_L\}. \quad (\text{SAP16})$$

The other entries are fixed. This equation retains the complete defect introduced by the right matrix identity.

94.4 Descent and two distinct repairs on the full carrier

The saturation C , and every F_g , send each left rational partial orbit into a left rational partial orbit. On the domain of $T_{a,b,\gamma}$ one has exactly

$$F_g T_{a,b,\gamma} z = T_{a,b,\gamma} F_g z, \quad C T_{a,b,\gamma} z = T_{a,b,\gamma} C z. \quad (\text{SAP17})$$

For amplitudes, both sides give auy_∞ and $a(x_\infty + vy_\infty) + b$ at infinity and the original affine formula at finite places. For labels, left maps preserve α, β and the right map joins only the displayed infinite-place label. The right image remains in D_γ , since its β is greater than or equal to the original one. If $b \neq 0$, the original β was top at all places and remains so. These checks prove the domains as well as the equations.

The set W is saturated for the left partial-orbit relation, because that relation preserves labels. It follows that the real group acts on W/M , that C induces an idempotent $\bar{C} : \mathcal{C}_L \rightarrow W/M$, and that (SAP15) descends. Its fixed set is exactly W/M : an orbit cannot identify two different label tuples. The original amplitude quotient map is equivariant for the right real action wherever that action is defined, and the all-top section of (SAP7) intertwines the entire original real action.

One repair retains the coefficient-exact reversible part and adds the already-defined global dilations. Define D_u on all of Z by scaling y_∞ by $u \neq 0$ and retaining every other amplitude and label. It corresponds to using τ for the zero shear coefficient; it is therefore a *different, explicitly declared lift* from $F_{(u,0)}$ outside W . There is the partial right $P(\mathbb{R})$ action

$$\theta_{(u,v)} = \begin{cases} D_u : Z \rightarrow Z, & v = 0, \\ A_{(u,v)} : W \rightarrow W, & v \neq 0. \end{cases} \quad (\text{SAP18})$$

Its inverse has the same domain, because the inverse translation coefficient is $-v/u$. Composing two maps has domain Z if both are pure dilations, and W otherwise. On that domain the amplitude law is $(u,v)(u',v') = (uu', uv' + v)$ and labels are unchanged. If two nonzero shear coefficients cancel, the result is the restriction to W of the resulting global dilation, as required for a partial action. The resulting inverse monoid of partial bijections is exactly $\{D_u : u \in \mathbb{R}^\times\} \cup \{A_g : W \rightarrow W : g \in P(\mathbb{R})\}$, with idempotents id_Z, id_W . Its two parts are the dilation group and the affine group on their stated domains. The sets and formulas are left-orbit saturated, preserve all left-action domains, and commute with the left action. Thus (SAP18) descends to $\mathcal{C}_L$.

This inverse monoid also has its exact supported matrix realization, rather than only a description by partial maps. At the real place put $\mathbf{1} = (1, 1_L)$, $e = (0, 1_L)$ and $\tau = (0, 0_L)$, and retain the two matrix families

$$\mathcal{D}_u = \begin{pmatrix} (u, 1_L) & \tau \\ \tau & \mathbf{1} \end{pmatrix}, \quad \mathcal{B}_{u,v} = \begin{pmatrix} (u, 1_L) & (v, 1_L) \\ \tau & \mathbf{1} \end{pmatrix}. \quad (\text{SAP18a})$$

The original full right map is multiplication of the row by $\mathcal{B}_{u,v}$. The matrix $\mathcal{B}_{u,0}$ has an e entry and is different from $\mathcal{D}_u$. Multiplication in the original semiring gives all four products

$$\begin{aligned} \mathcal{D}_u \mathcal{D}_w &= \mathcal{D}_{uw}, & \mathcal{D}_u \mathcal{B}_{w,t} &= \mathcal{B}_{uw,ut}, \\ \mathcal{B}_{u,v} \mathcal{D}_w &= \mathcal{B}_{uw,v}, & \mathcal{B}_{u,v} \mathcal{B}_{w,t} &= \mathcal{B}_{uw,v+ut}. \end{aligned} \quad (\text{SAP18b})$$

In the last product, cancellation of $v + ut$ leaves the upper right entry e , not τ . The two idempotents are $\mathcal{D}_1$ and $\mathcal{E} = \mathcal{B}_{1,0}$, which commute. The unique inverse of $\mathcal{D}_u$ is $\mathcal{D}_{u^{-1}}$; the unique inverse of $\mathcal{B}_{u,v}$ is $\mathcal{B}_{u^{-1}, -v/u}$. The latter pair multiplies to $\mathcal{E}$ in both orders. An inverse between the two different families is impossible because a product containing a $\mathcal{B}$ factor still has top upper-right support; within each family its amplitude equations force the displayed inverse. This proves the inverse-monoid assertion directly for matrices. Its faithful right action by partial bijections sends $\mathcal{D}_u$ to $\mathcal{D}_u$ on Z and $\mathcal{B}_{u,v}$ to $\mathcal{A}_{u,v}$ on W , with idempotents id_Z, id_W . The different domains distinguish the two families. At top-supported $y_\infty = 1, x_\infty = 0$, the output amplitudes distinguish each u, v . The action convention is $R_{mn} = R_n \circ R_m$, as required for multiplication of rows on the right; it is not silently changed to a left action.

A second, larger partial action can be defined if the requirement is to keep every label and the original amplitude formula, rather than to be the reversible restriction of (SAP12). Its exact maximal full-group domain is

$$\Omega = \{z : \beta_\infty = 1_L \text{ or } y_\infty = 0\}, \quad W \subset \Omega. \quad (\text{SAP19})$$

On Ω use $(uy_\infty, x_\infty + vy_\infty)$ with both labels unchanged. If $\beta_\infty = 1_L$ this is always admissible; if $y_\infty = 0$, its x -entry is unchanged. These two conditions persist under every group element, so this is a group action on Ω . Outside Ω , one has $y_\infty \neq 0, \beta_\infty < 1_L$ and $x_\infty = 0$. Any $v \neq 0$ would give nonzero vy_∞ with a non-top label, outside the carrier. This proves maximality under the stated label-preserving amplitude requirement. The same definition as (SAP18), with W replaced by Ω , is therefore a partial action on all Z , and descends to the left orbit set. On $\Omega \setminus W$ it differs from the full supported matrix product: it retains β_∞ instead of replacing it by $\beta_\infty \vee \alpha_\infty$. Equations (SAP12), (SAP16), and (SAP19) specify that difference; these lifts are not identified with each other. If T_g denotes this larger group action on Ω , its comparison to the full coefficient action is the exact equation

$$C \circ T_g = A_g \circ C = F_g|_\Omega. \quad (\text{SAP19a})$$

Both sides keep the same amplitudes and all finite labels and replace only β_∞ by $\beta_\infty \vee \alpha_\infty$. On $\Omega \setminus W$ both real amplitudes are zero and T_g fixes the point, whereas F_g performs that displayed support saturation.

There is also an exact groupoid which keeps the full original map (SAP12) and all its antecedents. Its objects are all $z \in Z$, and its arrows are

$$\mathcal{G}_C^{(1)} = \{(z, g, z') : g \in P(\mathbb{R}), C(z') = A_g C(z)\}, \quad (z, g, z') : (z \longrightarrow z'). \quad (\text{SAP20})$$

For arrows (z, g, z') and (z', h, z'') , composition is (z, gh, z'') ; its defining equation follows from the right action law on W . The identity is $(z, 1, z)$ and the inverse is (z', g^{-1}, z) . Thus all groupoid laws hold. The arrows over the group identity are precisely all pairs in the fibres of C , with those fibres given by (SAP16). In particular the graph $(z, g, F_g z)$ supplies the original maps as arrows, and their inverses remain arrows without pretending that $F_{g^{-1}} F_g z = z$. The orbit set of this groupoid is exactly $W/P(\mathbb{R})$: every z is connected to Cz by an identity-labelled arrow, and the remaining relation is the original group action on W . Replacing Z, W, C by $\mathcal{C}_L, W/M, \overline{C}$ gives the same exact construction after the rational partial quotient. Its objects still retain every supported rational orbit.

The saturation is also the universal quotient on which the original maps become a genuine group action. Precisely, if $f : Z \rightarrow B$ intertwines every F_g with a genuine right group action on B , then the group identity gives

$$f(Cz) = f(F_1 z) = f(z), \quad f = \tilde{f} \circ C, \quad \tilde{f} = f|_W. \quad (\text{SAP20a})$$

This factor is unique because C is onto, and it is equivariant because $C|_W$ is the identity and $F_g|_W = A_g$. Conversely every equivariant $\tilde{f} : W \rightarrow B$ yields such an f by (SAP15). Thus $Z/\ker C \simeq W$ has the stated universal property. After the left partial quotient, the same proof gives the quotient by $[z] \sim [z']$ if and only if Cz and Cz' lie in the same left partial orbit, with target W/M .

94.5 A full-fibre obstruction, beyond failure of one formula

There is already an obstruction to *any* global bijective action with the prescribed original amplitudes and unchanged finite data, even allowing the real support labels to change. Choose two points with identical finite data and $y_\infty = 1, x_\infty = 0, \alpha_\infty = 1_L$, but respectively $\beta_\infty = 0_L$ and $\beta_\infty = 1_L$. A group element (u, v) with $u \neq 0, v \neq 0$ would send both points to real amplitudes (u, v) ; each is nonzero and forces both real labels to be top. All finite data were required to remain the same. The two images would therefore coincide, contradicting injectivity of a group-action map. This argument does not assume the particular label formula (SAP12).

There cannot be a global group action by bijections on Z making $C : Z \rightarrow W$ equivariant for the given action A on W . Indeed fix all finite coordinates and labels, take $y_\infty = 1, x_\infty = 0, \alpha_\infty = 1_L$, and let $w_0 \in W$ have $\beta_\infty = 1_L$. Equation (SAP16) gives $C^{-1}(w_0) \simeq L$. The real shear $g = (1, v)$ for $v \neq 0$ sends w_0 to w_v with $x_\infty = v \neq 0$, whose C -fibre is a singleton. An equivariant bijection for g would restrict to a bijection of these two fibres, impossible for nontrivial L . This rules out every such global action, not merely the specific formula (SAP12).

The corresponding obstruction persists after taking the actual left rational partial quotient. Choose full adeles and labels

$$y_v = 1 \quad (v \in \Sigma), \quad x_v = 0 \quad (v \in \Sigma), \quad \alpha = \beta = \mathbf{1}, \quad (\text{SAP21})$$

and let $\bar{w}_0$ be their class in W/M . Since all y -coordinates are nonzero, every antecedent under $\bar{C}$ has $\alpha = \mathbf{1}$, and its finite β -coordinates must be top. Formula (SAP8) shows

$$\bar{C}^{-1}(\bar{w}_0) \simeq L : \quad \beta_\infty \text{ is arbitrary, all finite } \beta_p = 1_L. \quad (\text{SAP22})$$

For a proper β_∞ , the representative has $t = 0$, because $x_\infty + t = 0$ is necessary. The top value is the top branch of (SAP8), so it contributes one further value of that same L parametrization. Now choose $v \in \mathbb{R} \setminus \mathbb{Q}$ and apply the right shear $(1, v)$. Its amplitude representative has $y = 1, x_p = 0$ for finite p , and $x_\infty = v$. An antecedent with proper β_∞ would require a rational t satisfying $v + t = 0$, impossible. Consequently

$$|\bar{C}^{-1}(\bar{w}_0 \cdot (1, v))| = 1 \quad (v \notin \mathbb{Q}), \quad \bar{C}^{-1}(\bar{w}_0) \simeq L. \quad (\text{SAP23})$$

Thus no global bijective real affine action on the supported rational orbit set can make $\bar{C}$ equivariant either. The partial actions and the groupoid above are specified mathematical replacements with proved maps, not assumptions that this obstruction has disappeared.

94.6 The retained free real leaves

On the part of W/M with $y_f \neq 0$ and $y_\infty \neq 0$, the right real affine action is free. If $[z] \cdot (u, v) = [z]$ in this supported quotient, the same holds in the original amplitude quotient. There is therefore $(a, b) \in P(\mathbb{Q})$ with $y_f = ay_f, x_f = ax_f + b, uy_\infty = ay_\infty$, and $x_\infty + vy_\infty = ax_\infty + b$. A nonzero finite coordinate of y_f forces $a = 1$; the finite x -equation then forces $b = 0$; the two real equations force $u = 1, v = 0$. This reproduces the exact argument in the cited Lemma `Pdescrofgamma`, with every support retained.

For a fixed label tuple these free $P_+(\mathbb{R})$ orbits therefore carry the original homogeneous complex-leaf structure. This means the manifold structure transported from the free orbit

parametrization by $P_+(\mathbb{R})$; no new topology on the entire label set is being declared. The actual real formulas, with their original constants, are

$$X = y_\infty \partial_{x_\infty}, \quad Y = y_\infty \partial_{y_\infty}, \quad ds^2 = \frac{dx_\infty^2 + dy_\infty^2}{y_\infty^2}, \quad \Delta = X^2 + (Y - \tfrac{1}{2})^2 - \tfrac{1}{4}. \quad (\text{SAP24})$$

The vector fields follow by differentiating the original right shear and dilation, respectively. For a base point (y_0, x_0) , the orbit coordinates are $y_\infty = uy_0$, $x_\infty = x_0 + vy_0$, so the metric pullback is exactly $y_0^2(du^2 + dv^2)/(u^2y_0^2)$. The Laplacian expression retains both scalar terms and agrees with $y_\infty^2(\partial_{x_\infty}^2 + \partial_{y_\infty}^2)$ by expanding $Y^2 - Y$ and X^2 . These are the original formulas of Proposition **ax+bact** and equation **lapxy**. They are supported real-leaf statements on the indicated locus, not a global complex structure on the full supported carrier. No spectral positivity or RH conclusion follows from the orbit or fibre calculations alone.

95 Supported additive coordinates on the prime complex lifts

Sources and scope. Alain Connes and Caterina Consani, *The Riemann–Roch strategy, Complex lift of the Scaling Site*, original author source `thecurve_K.tex`, supply the complex lift, its coordinates and the prime mapping torus: Section 5.1, Lemma `adelicomp`; Section 5.2, equations `actionpq`, `algspacetiota`, Lemma `Pdescrofgamma`; Section 5.3, Lemma `lemper` and Theorem `gammapp1`. Their compact group is $G = \mathbb{A}_{\mathbb{Q}}/\mathbb{Q}$, and their matrix is $\begin{pmatrix} y & x \\ 0 & 1 \end{pmatrix}$. The prime lift below uses these exact coordinates. The full support operations come from The Clankers, *Split Support Geometry*, version 11, Definition `def:lattice-split`; the common-label and placewise carriers and their comparison were proved in PLP1–5, `CC_PERIODIC_POINTS_FULL_LATTICE.tex`. All algebraic, fibre, topology and measure assertions needed below are proved here. The construction does not assign a scalar trace to the set of support labels, or identify a measure exponent with Deligne purity.

95.1 The complete coordinate carrier and the admitted prime strata

Let L be a nontrivial bounded distributive lattice. Retain, for every commutative ring R ,

$$\begin{aligned} G_L(R) &= \{(0, \lambda) : \lambda \in L\} \cup \{(r, 1_L) : r \in R\}, \\ (r, \lambda) + (s, \mu) &= (r + s, \lambda \vee \mu), \quad (r, \lambda)(s, \mu) = (rs, \lambda \wedge \mu), \\ \tau &= (0, 0_L), \quad e = (0, 1_L), \quad \mathbf{1} = (1, 1_L). \end{aligned} \tag{SPL1}$$

Put $\Sigma = \{\infty\} \cup \{\text{rational primes}\}$, $\mathbb{A} = \mathbb{A}_{\mathbb{Q}}$, and

$$S_{\text{loc}} = G_L(\mathbb{R}) \times \prod_q' G_L(\mathbb{Q}_q), \quad Z_L = S_{\text{loc}} \times S_{\text{loc}}. \tag{SPL2}$$

The restricted factors are $G_L(\mathbb{Z}_q)$; thus an element of S_{loc} is (z, λ) with $z \in \mathbb{A}$ and $z_v \neq 0 \Rightarrow \lambda_v = 1_L$. There is no finiteness restriction on labels at zero coordinates. All sums and products are coordinatewise. The original common-label carrier has the injective map $(z, \lambda) \mapsto ((z_v, \lambda))_v$; it is not identified with S_{loc} . Write a point of Z_L as $((y, \alpha), (x, \beta))$. Thus this calculation forms the local fields and their restricted product after applying the original G_L construction at each place. It does not assume that G_L commutes with forming the adelic ring. Coordinatewise projection and the displayed common-label injection are its exact comparisons with that other order of construction.

The rational left matrix action with translation label $\gamma \in L^{\Sigma}$ is

$$((y, \alpha), (x, \beta)) \mapsto ((ay, \alpha), (ax + b, \beta \vee \gamma)), \quad a \in \mathbb{Q}^{\times}, \quad b \neq 0 \Rightarrow \gamma = \mathbf{1}. \tag{SPL3}$$

Its inverse matrix, with amplitude translation $-a^{-1}b$, restores a point precisely when $\beta \geq \gamma$. Indeed the double product has amplitude (y, x) and second label $\beta \vee \gamma$. Consequently nonzero rational translations are partial bijections on the all-top x labels; on any fixed $\beta \neq \mathbf{1}$ the permitted invertible actions are exactly simultaneous rational dilations. This is an actual partial-action quotient, not the equivalence closure of noninvertible support-raising maps. The latter would discard some of the labels.

Fix a rational prime p , a label $\lambda \in L$, and a nonempty subset T of the finite rational primes; T may be infinite. Fix exactly

$$\begin{aligned} \alpha_p &= \lambda, \quad \alpha_v = 1_L \ (v \neq p), \quad \beta_{\infty} = 1_L, \quad \beta_q < 1_L \ (q \in T), \quad \beta_q = 1_L \ (q \notin T), \\ y_p &= 0, \quad y_q \in \mathbb{Z}_q^{\times} \ (q \neq p, \infty), \quad y_{\infty} = r > 0, \quad x_q = 0 \ (q \in T). \end{aligned} \tag{SPL4}$$

Every specified lower label is retained, including labels strictly between bottom and top. Let

$$B_T = \{x \in \mathbb{A} : x_q = 0 \ (q \in T)\} = \mathbb{R} \times \prod_{q \notin T}' \mathbb{Q}_q, \tag{SPL5}$$

where the equality inserts zero at all omitted places. None of the strata SPL4 lies in the image of the common-label $G_L(\mathbb{A})^2$: its x labels are top at infinity and non-top at a place in T . These are features of the explicitly constructed placewise adelic object, not extra points assigned to the smaller common-label carrier. The group B_T is locally compact in its original restricted-product topology. Dividing the y coordinate by $K = \widehat{\mathbb{Z}}^\times$ makes it uniquely represented by $y_q = 1$ for $q \neq p, \infty$, $y_p = 0$, $y_\infty = r$. Two such representatives are related by positive rational multiplication and K exactly when the multiplier is p^n , $n \in \mathbb{Z}$: its valuation is zero at every prime other than p , and its sign is positive. The surviving quotient is therefore exactly

$$\Gamma_{p,T;\lambda,\beta} = p^\mathbb{Z} \backslash (B_T \times \mathbb{R}_+^*), \quad p^n \cdot (x, r) = (p^n x, p^n r). \quad (\text{SPL6})$$

There is no quotient by additive $\mathbb{Q}$ in SPL6, since $\beta \neq 1$ in SPL3. This proves the stated description directly from the full original coordinate operations. For the separate all-top x stratum, rational translations are available and the identical calculation instead gives the original compact lift

$$\Gamma_{p;\lambda}^{\text{top}} = p^\mathbb{Z} \backslash (G \times \mathbb{R}_+^*), \quad G = \mathbb{A}/\mathbb{Q}. \quad (\text{SPL7})$$

Each fixed λ gives that original lift, with λ still attached to $y_p = 0$. Taking all labels is not permission to multiply any original trace by $|L|$.

95.2 Exact return, density and the lost additive quotient

Define $\mathbb{A}_T = \prod'_{q \in T} \mathbb{Q}_q$, restricted relative to $\mathbb{Z}_q$. Projection onto the T coordinates is a continuous open surjection $\mathbb{A} \rightarrow \mathbb{A}_T$ with kernel B_T : all assertions follow from the restricted-product basic open sets, and a tuple is lifted by inserting zero at all other places. The diagonal $\mathbb{Q} \rightarrow \mathbb{A}_T$ is injective since T is nonempty. There is an exact sequence of additive groups

$$0 \longrightarrow B_T \xrightarrow{\iota_T} G \xrightarrow{\pi_T} \mathbb{A}_T/\mathbb{Q} \longrightarrow 0, \quad \iota_T(x) = [x], \quad \pi_T([z]) = [(z_q)_{q \in T}]. \quad (\text{SPL8})$$

Well-definedness of π_T follows by changing z by a diagonal rational. Its kernel consists precisely of classes having a representative whose T coordinates vanish, proving exactness. The first map is injective because a diagonal rational that vanishes at one place is zero. The last map is onto by the preceding lift.

The subgroup $\mathbb{Q}$ is dense in $\mathbb{A}_T$ even if T is infinite. To prove this, fix a basic open set, specifying sufficiently small balls at a finite set $F \subset T$ and integrality elsewhere. Choose N with prime factors in F so that all specified centres become integral after multiplication by N . The Chinese remainder theorem gives an integer m meeting the finitely many required congruences, so m/N belongs to the original open set and is integral at every place outside F . It follows from the openness of projection that $B_T + \mathbb{Q}$ is dense in $\mathbb{A}$. Thus $\iota_T(B_T)$ is dense in G . It is proper: choose a nonrational element of $\mathbb{Q}_q$ at one $q \in T$, and zero elsewhere, to obtain a nonzero class in $\mathbb{A}_T/\mathbb{Q}$. In particular the final quotient in SPL8 has the indiscrete quotient topology; SPL8 is not a short exact sequence of Hausdorff locally compact groups. No topology of B_T is replaced by its image topology.

All maps in SPL8 commute with multiplication by p . They induce the explicit continuous injective return

$$\mathcal{I}_T : \Gamma_{p,T;\lambda,\beta} \longrightarrow \Gamma_{p;\lambda}^{\text{top}}, \quad [(x, r)] \longmapsto [([x], r)]. \quad (\text{SPL9})$$

For injectivity, equality of images gives $r' = p^n r$ and $x' - p^n x \in \mathbb{Q}$. This difference lies in B_T , so it is zero. The image is dense and proper: both mapping tori locally have product charts over $r \bmod p^\mathbb{Z}$, and the fibre map is the dense proper injection $B_T \rightarrow G$. This also shows the return retains the same base circle and its literal circumference $\log p$.

The supported mapping torus is noncompact. The continuous function

$$[(x, r)] \longmapsto x_\infty/r \quad (\text{SPL10})$$

is well defined and onto $\mathbb{R}$, since both numerator and denominator acquire the same factor p^n . In contrast the original mapping torus SPL7 is compact: it is the quotient of $G \times [1, p]$ identifying $(g, 1)$ with (pg, p) . The same description with B_T proves that SPL6 is its mapping torus, in the original noncompact fibre topology.

95.3 The original real action and its entire leaf metric

Right multiplication by the original real matrix $\begin{pmatrix} u & v \\ 0 & 1 \end{pmatrix}$, $u > 0$, induces

$$(x, r) \longmapsto (x + j_\infty(vr), ur), \quad (\text{SPL11})$$

where $j_\infty(t)$ is the adele with real coordinate t and all finite coordinates zero. SPL11 preserves B_T , all fixed labels and the relation SPL6. The upper-right label at infinity remains top, and no finite label changes. Composition agrees with the original matrix product, as follows either by multiplying the two matrices or by substituting twice in SPL11. The return SPL9 intertwines SPL11 with the Connes–Consani action on G .

On the positive real coordinate plane of each parametrized orbit, the original fields and metric are retained:

$$X = r\partial_{x_\infty}, \quad Y = r\partial_r, \quad ds^2 = \frac{dx_\infty^2 + dr^2}{r^2}, \quad X + iY = r(\partial_{x_\infty} + i\partial_r), \quad (\text{SPL12})$$

$$\Delta = X^2 + (Y - \frac{1}{2})^2 - \frac{1}{4} = r^2(\partial_{x_\infty}^2 + \partial_r^2). \quad (\text{SPL13})$$

For verification, $Y^2 = r^2\partial_r^2 + r\partial_r$ and $(Y - \frac{1}{2})^2 = Y^2 - Y + \frac{1}{4}$; the displayed constants cancel in the stated identity and have not been omitted from its original operator. The equality is on smooth functions before taking any Hilbert completion. It does not identify a spectrum or a trace. At a point with some nonzero finite x_q , a stabilizer in SPL11 would require $p^n x_q = x_q$, hence $n = 0$, then $u = 1, v = 0$. The corresponding orbit is a copy of the upper half plane in its intrinsic leaf topology; no equality with an ambient subspace topology is asserted. When all finite coordinates of x vanish, all such points belong to one orbit; its parametrization is the upper half plane modulo simultaneous $(x_\infty, r) \mapsto (p^n x_\infty, p^n r)$. This proves the exact exceptional cylinder, including the case when T is every finite prime and there are no other orbits. The metric descends because each of these simultaneous dilations preserves SPL12.

95.4 All Haar factors and the prime-dependent defect

Give B_T its product Haar measure μ_T : Lebesgue measure at infinity and Haar measures with $\mu_q(\mathbb{Z}_q) = 1$ at every remaining finite place. For any $a \in \mathbb{Q}^\times$, multiplication by a is an automorphism, with exact modulus

$$\mu_T(aE) = \mathfrak{m}_T(a)\mu_T(E), \quad \mathfrak{m}_T(a) = |a|_\infty \prod_{q \notin T} |a|_q = \prod_{q \in T} |a|_q^{-1}. \quad (\text{SPL14})$$

Only finitely many factors differ from one. Local changes of variables prove the first expression on product rectangles, hence on Borel sets by uniqueness of Haar measure; the rational product formula proves the second. In particular, without changing any measure,

$$\mathfrak{m}_T(p^n) = p^{n\epsilon_{p,T}}, \quad \epsilon_{p,T} = \begin{cases} 1, & p \in T, \\ 0, & p \notin T. \end{cases} \quad (\text{SPL15})$$

The original compact group G carries its Haar probability measure m_G . Multiplication by p preserves m_G : it is an automorphism, so the pushforward is again Haar measure of total mass one. Thus the injective equivariant map SPL8 has different specified Haar moduli on its two sides when $p \in T$. This is possible because it is not a Haar-measure-preserving identification.

More strongly, $(\iota_T)_*\mu_T$ is not a Radon measure on compact G : its mass on G is $\mu_T(B_T) = \infty$. The latter follows from the real factor, by taking arbitrarily long real intervals and the product of all remaining finite integral subgroups of additive volume one. A Radon measure on a compact space has finite total mass. Thus using the original compact Haar measure in place of the supported source measure would require an additional, nontrivial construction; SPL9 alone does not justify it.

For a direct operator check put $V_p f(x) = f(x/p)$ on $L^2(B_T, \mu_T)$. Full change of variables gives

$$\begin{aligned} \|V_p f\|^2 &= p^{\epsilon_{p,T}} \|f\|^2, & V_p^* g(x) &= p^{\epsilon_{p,T}} g(px), \\ V_p^* V_p &= p^{\epsilon_{p,T}} I, & U_p &= p^{-\epsilon_{p,T}/2} V_p, & U_p^* U_p &= I. \end{aligned} \tag{SPL16}$$

The last line is an explicit comparison with a different operator; it does not replace V_p . Its factor is derived from the Haar modulus, not an asserted spectral weight of the original zeta. On the cover $B_T \times \mathbb{R}_+^*$, the original product measure $d\mu_T(x) dr/r$ has change-of-set factor $p^{\epsilon_{p,T}}$ under $(x, r) \mapsto (px, pr)$. The additional density

$$r^{-\epsilon_{p,T}} d\mu_T(x) \frac{dr}{r} \tag{SPL17}$$

is invariant, since its pointwise multiplier contributes the opposite factor $p^{-\epsilon_{p,T}}$. Both measures, and their literal relation, are retained. No change of scalar norm or Archimedean endpoint convention is implicit.

SPL6–17 apply to every prime p , every nonempty set T of finite primes and every specified admissible label tuple. They compute an entire supported family of complex lifts and its return, rather than a finite sample of zero coordinates. The compact top stratum recovers the Connes–Consani lift exactly; the full supported object also retains the noncompact strata and their different Haar factors. No positivity conclusion for the arithmetic trace follows merely from this geometry. These are the exact spaces and measure maps through which a proposed trace or Riemann–Roch comparison must act.

96 The supported distribution receiver for the original complex-lift operator

Human source and exact scope. Alain Connes and Caterina Consani, *The Riemann–Roch strategy: Complex lift of the Scaling Site*, version 1, Section 5.2, author-source Lemma `adelicomp1` and equation `actionpq`, define

$$\ell(a, b)(x, y) = (ax + b, ay), \quad D = \partial_x + i\partial_y. \quad (\text{SDB1})$$

These are the original coordinates and the original operator; there is no factor $1/2$ in the source’s definition of D . The original file `thecurve_K.tex`, lines 1203–1224, supplies this definition and the quotient by $P(\mathbb{Q})$. Section 6.6, the paragraph immediately preceding Proposition `comparescalcs` (source lines 2381–2418), also gives

$$X = y\partial_x, \quad Y = y\partial_y, \quad D_y = X + iY = y\partial_x + iy\partial_y = yD. \quad (\text{SDB2})$$

We construct and compute a particular distribution receiver for these operators on the open archimedean branch $x = 0$, $y > 0$. This construction is a new local calculation using the cited operators. Its cohomology is not asserted to be the cohomology of the whole adelic quotient, and no weight, purity, zeta-zero realization, or Riemann hypothesis statement follows from its definition.

96.1 The entire finite-normal-jet space and its distribution embedding

Put $I = (0, \infty)$ and $\Omega = \mathbb{R} \times I$. The notation $\mathcal{D}'(I)$ means complex distributions on I , acting complex-linearly on $C_c^\infty(I)$, and derivatives are defined by $g'(\psi) = -g(\psi')$. On Ω , both partial derivatives use the same convention. All test functions have compact support inside the indicated open set; no extension across $y = 0$ is part of these definitions. For $j \geq 0$ and $g \in \mathcal{D}'(I)$ define

$$\left\langle \delta^{(j)}(x) \otimes g(y), \phi(x, y) \right\rangle = (-1)^j \left\langle g, \partial_x^j \phi(0, \cdot) \right\rangle, \quad \phi \in C_c^\infty(\Omega). \quad (\text{SDB3})$$

Here $\delta^{(j)}$ is the j th derivative of the Dirac distribution at $x = 0$; its order and its sign have not been absorbed into g . The continuous restriction-and-derivative map on test functions in (SDB3) proves that this expression is a distribution. It vanishes on every test function whose support misses $\{0\} \times I$. Define the algebraic direct sum

$$\mathcal{J} = \bigoplus_{j \geq 0} \delta^{(j)}(x) \otimes \mathcal{D}'(I), \quad \mathcal{J}_{\leq N} = \bigoplus_{j=0}^N \delta^{(j)}(x) \otimes \mathcal{D}'(I). \quad (\text{SDB4})$$

Thus every individual element has a finite normal-jet expansion, while no common finite upper bound on j is imposed on $\mathcal{J}$.

Proposition 49 (Exact embedding and normal-order description). *Formula (SDB3) embeds the direct sum (SDB4) injectively in $\mathcal{D}'(\Omega)$. Every expansion in this sum is unique, and*

$$\mathcal{J}_{\leq N} = \{u \in \mathcal{D}'(\Omega) : x^{N+1}u = 0\}, \quad \mathcal{J} = \bigcup_{N \geq 0} \ker(x^{N+1}). \quad (\text{SDB5})$$

Proof. Choose $\rho \in C_c^\infty(\mathbb{R})$ equal to 1 on a neighborhood of 0. For $0 \leq m \leq N$, insert $\phi(x, y) = \rho(x)x^m\psi(y)/m!$ in (SDB3). Its j th x derivative at 0 equals ψ for $j = m$ and equals 0 for every $j \neq m$. Consequently

$$\left\langle \sum_{j=0}^N \delta^{(j)} \otimes g_j, \rho(x)x^m\psi(y)/m! \right\rangle = (-1)^m \langle g_m, \psi \rangle.$$

If the sum is zero, each g_m is zero. This proves uniqueness and injectivity without imposing any regularity on the distributions g_m . Direct differentiation of $x\phi$ gives

$$x\delta = 0, \quad x\delta^{(j)} = -j\delta^{(j-1)} \quad (j \geq 1), \quad (\text{SDB6})$$

so x^{N+1} kills every term of $\mathcal{J}_{\leq N}$.

Conversely, suppose $x^{N+1}u = 0$. For an arbitrary test function $\phi \in C_c^\infty(\Omega)$, Taylor's formula at $x = 0$ gives

$$\phi(x, y) = \rho(x) \sum_{j=0}^N \frac{x^j}{j!} \partial_x^j \phi(0, y) + x^{N+1} \psi_\phi(x, y), \quad \psi_\phi \in C_c^\infty(\Omega).$$

Indeed, the difference on the right before division vanishes to order $N+1$ along $x = 0$, so its quotient by x^{N+1} is smooth there; away from $x = 0$ division is smooth. Both the difference and its quotient have compact support in a fixed compact x set and in the compact projection of $\text{supp } \phi$ on I . Define

$$\langle g_j, \eta \rangle = (-1)^j \langle u, \rho(x) x^j \eta(y) / j! \rangle \quad (\eta \in C_c^\infty(I)).$$

Each g_j is a distribution by continuity of the displayed test-function map. Applying u to the Taylor identity and using $x^{N+1}u = 0$ gives (SDB3) summed over $j = 0, \dots, N$. This proves (SDB5). $\square$

The finiteness in (SDB4) must not be silently changed. There are distributions supported on $\{0\} \times I$ outside this space. For example,

$$u_* = \sum_{n=1}^{\infty} \delta^{(n)}(x) \otimes \delta(y - n) \quad \text{in } \mathcal{D}'(\Omega) \quad (\text{SDB7})$$

is well-defined: a compact subset of Ω meets only finitely many of the sets $y = n$. For every $m \geq 1$, its term at $y = n \geq m$ has $x^m \delta^{(n)} = (-1)^m n! \delta^{(n-m)} / (n-m)! \neq 0$. Testing in a small neighborhood of that one $y = n$ shows $x^m u_* \neq 0$. Proposition 49 therefore proves $u_* \notin \mathcal{J}$. We compute exactly the space specified in (SDB4), without identifying it with all distributions on the branch or with a completion of the jet direct sum.

96.2 The complete two-term complex, residue, and homotopy

Distributional differentiation in (SDB3) yields

$$D(\delta^{(j)} \otimes g) = \delta^{(j+1)} \otimes g + i\delta^{(j)} \otimes g'. \quad (\text{SDB8})$$

For example, the x derivative has value $-(-1)^j g(\partial_x^{j+1} \phi(0, \cdot))$, which is precisely the sign of $\delta^{(j+1)} \otimes g$; the y derivative is $\delta^{(j)} \otimes g'$ by the defining sign of g' . In particular, D maps $\mathcal{J}$ to itself and maps $\mathcal{J}_{\leq N}$ into $\mathcal{J}_{\leq N+1}$. The complex in question is

$$C^\bullet : 0 \longrightarrow C^0 = \mathcal{J} \xrightarrow{D} C^1 = \mathcal{J} \longrightarrow 0. \quad (\text{SDB9})$$

It is two-term: the arrow out of degree 1 is the zero map. The statement that this is a complex does not mean $D^2 = 0$ as an operator $\mathcal{J} \rightarrow \mathcal{J}$. Indeed, with 1 denoting the constant distribution on I , $D^2(\delta \otimes 1) = \delta'' \otimes 1 \neq 0$.

Define the following linear maps on their full indicated domains:

$$R : \mathcal{J} \longrightarrow \mathcal{D}'(I), \quad R \left(\sum_{j=0}^N \delta^{(j)} \otimes g_j \right) = \sum_{j=0}^N (-i)^j g_j^{(j)}, \quad (\text{SDB10})$$

$$E : \mathcal{D}'(I) \longrightarrow \mathcal{J}, \quad E(h) = \delta \otimes h, \quad (\text{SDB11})$$

$$T : \mathcal{J} \longrightarrow \mathcal{J}, \quad T \left(\sum_{j=0}^N \delta^{(j)} \otimes g_j \right) = \sum_{j=1}^N \sum_{k=0}^{j-1} (-i)^k \delta^{(j-1-k)} \otimes g_j^{(k)}. \quad (\text{SDB12})$$

The empty sum is zero. Uniqueness of the normal-jet expansion makes all three definitions unambiguous; every sum is finite for its input. No convergence assumption or differentiability of g_j beyond being a distribution is used.

Theorem 50 (Exact calculation and explicit splitting). *The maps above satisfy*

$$RD = 0, \quad RE = \text{id}, \quad TE = 0, \quad DT = \text{id} - ER, \quad TD = \text{id}. \quad (\text{SDB13})$$

Consequently

$$0 \longrightarrow \mathcal{J} \xrightarrow{D} \mathcal{J} \xrightarrow{R} \mathcal{D}'(I) \longrightarrow 0 \quad (\text{SDB14})$$

is a split exact sequence of complex vector spaces, and

$$H^0(C^\bullet) = 0, \quad H^1(C^\bullet) = \mathcal{J}/D\mathcal{J} \xrightarrow[\substack{\cong \\ [u] \mapsto Ru}]{} \mathcal{D}'(I), \quad h \mapsto [E(h)] \quad (\text{SDB15})$$

gives both directions of the last isomorphism. More precisely, the mutually inverse linear maps are

$$\begin{aligned} \Phi : \mathcal{J} &\longrightarrow \mathcal{D}'(I) \oplus \mathcal{J}, & \Phi(u) &= (Ru, Tu), \\ \Psi : \mathcal{D}'(I) \oplus \mathcal{J} &\longrightarrow \mathcal{J}, & \Psi(h, k) &= E(h) + Dk. \end{aligned} \quad (\text{SDB16})$$

Proof. For $u = \sum_{j=0}^N \delta^{(j)} \otimes g_j$, formula (SDB8) gives

$$RDu = \sum_{j=0}^N ((-i)^{j+1} + i(-i)^j) g_j^{(j+1)} = 0.$$

The formulas $RE = \text{id}$ and $TE = 0$ follow directly from the $j = 0$ terms. For $j \geq 1$, direct application of D to every summand of $T(\delta^{(j)} \otimes g)$ gives

$$\begin{aligned} DT(\delta^{(j)} \otimes g) &= \sum_{k=0}^{j-1} (-i)^k \delta^{(j-k)} \otimes g^{(k)} + \sum_{k=0}^{j-1} i(-i)^k \delta^{(j-1-k)} \otimes g^{(k+1)} \\ &= \delta^{(j)} \otimes g + i(-i)^{j-1} \delta \otimes g^{(j)} \\ &= \delta^{(j)} \otimes g - \delta \otimes ((-i)^j g^{(j)}). \end{aligned}$$

In the middle line, the coefficient of $\delta^{(j-r)} \otimes g^{(r)}$ for $1 \leq r \leq j-1$ is $(-i)^r + i(-i)^{r-1} = 0$. For $j = 0$, both DT and $\text{id} - ER$ vanish on $\delta \otimes g$. This proves $DT = \text{id} - ER$ on every finite sum, with all signs explicit.

It remains to prove injectivity and $TD = \text{id}$, rather than infer them from a truncated calculation. If $u = \sum_{j=0}^N \delta^{(j)} \otimes g_j \neq 0$ and N is its largest nonzero index, the coefficient of $\delta^{(N+1)}$ in Du is exactly g_N . Uniqueness in Proposition 49 therefore implies $Du \neq 0$. Hence D is injective. Now

$$D(TDu - u) = (\text{id} - ER)Du - Du = -ERDu = 0,$$

so injectivity gives $TDu = u$. The identity $u = ERu + DTu$ shows that $\ker R = D\mathcal{J}$, and $RE = \text{id}$ proves surjectivity of R . These facts establish (SDB14)–(SDB15). Finally,

$$\Phi\Psi(h, k) = (REh + RDk, TEh + TDk) = (h, k), \quad \Psi\Phi(u) = ERu + DTu = u,$$

which proves the complete splitting. $\square$

For orientation, the first three residue contributions are exactly

$$R(\delta \otimes g_0 + \delta' \otimes g_1 + \delta'' \otimes g_2) = g_0 - ig_1' - g_2'', \quad T(\delta' \otimes g_1 + \delta'' \otimes g_2) = \delta \otimes g_1 + \delta' \otimes g_2 - i\delta \otimes g_2'. \quad (\text{SDB17})$$

Higher jets remain present and are governed by (SDB10)–(SDB12).

96.3 Simultaneous left dilation, including both Jacobians

For $a > 0$ let $F_a(x, y) = (ax, ay)$ on Ω and $f_a(y) = ay$ on I . We use scalar-distribution pullback, which agrees with composition for distributions represented by locally integrable functions. Its complete formulas are

$$\begin{aligned}\langle P_a u, \phi \rangle &= a^{-2} \langle u, \phi(\cdot/a, \cdot/a) \rangle, & P_a &= F_a^*, \\ \langle Q_a g, \psi \rangle &= a^{-1} \langle g, \psi(\cdot/a) \rangle, & Q_a &= f_a^*.\end{aligned}\tag{SDB18}$$

The positive determinant a^2 in two variables and the positive determinant a in one variable account for these factors. In particular, these definitions do not treat distributions as densities with a different transformation law. For a locally integrable u , substitution $(s, t) = (ax, ay)$ in its integral pairing verifies $P_a u(x, y) = u(ax, ay)$; the same substitution in one variable gives $Q_a g(y) = g(ay)$ for functions g .

Proposition 51 (Full covariance). *The pullbacks preserve the indicated distribution spaces, have $P_a P_b = P_{ab}$ and $Q_a Q_b = Q_{ab}$, and their inverses are $P_{1/a}$ and $Q_{1/a}$. Their exact actions on jets and derivatives are*

$$P_a(\delta^{(j)} \otimes g) = a^{-j-1} \delta^{(j)} \otimes Q_a g, \tag{SDB19}$$

$$(Q_a g)^{(k)} = a^k Q_a(g^{(k)}), \quad DP_a = a P_a D. \tag{SDB20}$$

The residue, section, and homotopy transform as

$$R P_a = a^{-1} Q_a R, \quad P_a E = a^{-1} E Q_a, \quad T P_a = a^{-1} P_a T. \tag{SDB21}$$

Proof. The group identities follow by two substitutions in (SDB18), with determinants multiplied rather than discarded. For a jet, the two-variable formula gives

$$\begin{aligned}\langle P_a(\delta^{(j)} \otimes g), \phi \rangle &= a^{-2} (-1)^j \langle g, \partial_x^j [\phi(x/a, \cdot/a)]_{x=0} \rangle \\ &= a^{-j-2} (-1)^j \langle g, (\partial_x^j \phi)(0, \cdot/a) \rangle \\ &= a^{-j-1} (-1)^j \langle Q_a g, \partial_x^j \phi(0, \cdot) \rangle.\end{aligned}$$

This proves (SDB19). Applying a distributional derivative to (SDB18) and differentiating the test function gives $\partial_x P_a = a P_a \partial_x$ and $\partial_y P_a = a P_a \partial_y$. The identical one-variable argument, iterated k times, proves the first formula in (SDB20); adding the two partial derivatives proves the second.

For each jet,

$$R P_a(\delta^{(j)} \otimes g) = a^{-j-1} (-i)^j \partial_y^j (Q_a g) = a^{-1} Q_a((-i)^j g^{(j)}).$$

The formula for E is the case $j = 0$ of (SDB19). For $j \geq 1$, the k th summand of $T P_a(\delta^{(j)} \otimes g)$ is

$$a^{-j-1+k} (-i)^k \delta^{(j-1-k)} \otimes Q_a(g^{(k)}).$$

The corresponding summand of $P_a T(\delta^{(j)} \otimes g)$ has coefficient $a^{-(j-k)} (-i)^k$. Their ratio is a^{-1} . For $j = 0$ both homotopy expressions vanish. Finiteness allows all these identities to be summed, proving (SDB21). $\square$

There are two precise consequences, with different degree actions. First, $P_a(D\mathcal{J}) = D\mathcal{J}$, since $P_a D = a^{-1} D P_a$. Therefore raw pullback $[u] \mapsto [P_a u]$ is a well-defined invertible action on the quotient $\mathcal{J}/D\mathcal{J}$. Under the residue isomorphism its action is exactly $a^{-1} Q_a$, including the factor a^{-1} . Second, the actual cochain action on (SDB9) is

$$\rho_a^0 = P_a, \quad \rho_a^1 = a P_a, \quad D \rho_a^0 = \rho_a^1 D. \tag{SDB22}$$

It is a representation of $\mathbb{R}_{\geq 0}^\times$ in both degrees because $a P_a b P_b = ab P_{ab}$. On H^1 this cochain action is exactly Q_a : $R \rho_a^1 = Q_a R$, $\rho_a^1 E = E Q_a$, and $T \rho_a^1 = \rho_a^0 T$. One may record the one-form degree

using a formal constant symbol ε and $du = (Du)\varepsilon$, with $F_a^*\varepsilon = a\varepsilon$. This gives precisely (SDB22); it does not change the original operator to $D/2$. The complete equivariant splitting is represented by

$$\begin{array}{ccc} \mathcal{J} & \xrightarrow{(R,T)} & \mathcal{D}'(I) \oplus \mathcal{J} \\ aP_a \downarrow & & \downarrow Q_a \oplus P_a \\ \mathcal{J} & \xrightarrow{(R,T)} & \mathcal{D}'(I) \oplus \mathcal{J}, \end{array} \quad (R,T)^{-1}(h,k) = Eh + Dk. \quad (\text{SDB23})$$

These simultaneous dilations are the archimedean restrictions of the source's left action with $b = 0$ when $a \in \mathbb{Q}_{>0}^\times$. The formulas above also define an archimedean action for every real $a > 0$; they do not assert a multiplication of finite adeles by an arbitrary real number. Likewise, the left translation $(x, y) \mapsto (x + b, y)$ with $b \neq 0$ does not preserve this receiver. Its scalar-distribution pullback sends $\delta(x) \otimes g(y)$ to $\delta(x+b) \otimes g(y)$, supported at $x = -b$. For nonzero g , this distribution is nonzero and cannot belong to $\mathcal{J}$, whose distributions are supported at $x = 0$. Thus restricting this particular complex to $x = 0$ retains pure dilation but does not supply an invertible action of the full rational translation group on it.

96.4 Retaining the full vector-field operator $X + iY = yD$

Multiplication by y and by y^{-1} is defined on every distribution on I or Ω : both functions are smooth on these open sets. Their products are the identity. They preserve $\mathcal{J}$ because they multiply each coefficient g_j and leave $\delta^{(j)}(x)$ unchanged. Denote these multiplication operators by M_y and $M_{1/y}$, on the relevant domain. Retain

$$D_y = M_y D, \quad E_y = M_y E, \quad R_y = R M_{1/y}, \quad T_y = T M_{1/y}. \quad (\text{SDB24})$$

In particular $E_y h = \delta \otimes (yh)$, not $\delta \otimes h$. There is no omission of the derivatives created by $1/y$: for $u = \sum_{j=0}^N \delta^{(j)} \otimes g_j$ the full formulas are

$$\begin{aligned} R_y u &= \sum_{j=0}^N (-i)^j \partial_y^j (y^{-1} g_j) \\ &= \sum_{j=0}^N \sum_{r=0}^j (-i)^j \binom{j}{r} (-1)^r r! y^{-r-1} g_j^{(j-r)}, \end{aligned} \quad (\text{SDB25})$$

$$T_y u = \sum_{j=1}^N \sum_{k=0}^{j-1} \sum_{r=0}^k (-i)^k \binom{k}{r} (-1)^r r! \delta^{(j-1-k)} \otimes (y^{-r-1} g_j^{(k-r)}). \quad (\text{SDB26})$$

For completeness, the distribution product rule follows from $(fg)'(\psi) = -g(f\psi') = g'(f\psi) + g(f'\psi)$, so $(fg)' = fg' + f'g$. Induction gives the displayed binomial terms, and $\partial_y^r(y^{-1}) = (-1)^r r! y^{-r-1}$ gives their full coefficients. These operations occur on I ; they neither introduce nor remove any distribution at the excluded endpoint $y = 0$.

Theorem 52 (Exact receiver and covariance for yD). *For the original vector-field operator in (SDB2),*

$$R_y D_y = 0, \quad R_y E_y = \text{id}, \quad T_y E_y = 0, \quad D_y T_y = \text{id} - E_y R_y, \quad T_y D_y = \text{id}. \quad (\text{SDB27})$$

Thus its two-term complex has $H^0 = 0$ and $H^1 \cong \mathcal{D}'(I)$, with the complete decomposition

$$u = yE(R_y u) + D_y(T_y u), \quad u \mapsto (R_y u, T_y u), \quad (h, k) \mapsto yE(h) + D_y k. \quad (\text{SDB28})$$

Here both cochain degrees carry the same raw pullback P_a , and

$$D_y P_a = P_a D_y, \quad R_y P_a = Q_a R_y, \quad T_y P_a = P_a T_y, \quad E_y Q_a = P_a E_y. \quad (\text{SDB29})$$

In particular its quotient action is exactly Q_a .

Proof. All five identities in (SDB27) follow by retaining the multiplication operators in their correct order. For example,

$$\begin{aligned} D_y T_y &= M_y D T M_{1/y} = M_y (\text{id} - ER) M_{1/y} = \text{id} - E_y R_y, \\ T_y D_y &= T M_{1/y} M_y D = T D = \text{id}, \\ R_y D_y &= R M_{1/y} M_y D = R D = 0. \end{aligned}$$

The other two identities use $M_{1/y} M_y = \text{id}$ in exactly the same positions. Multiplication by y is invertible, so D_y is injective because D is. The proof of the splitting and cohomology now follows verbatim from these explicit identities: a residue-zero u equals $D_y T_y u$, and each h is the residue of $E_y h$.

Scalar pullback obeys $P_a(fu) = (f \circ F_a)P_a u$ for every smooth f , as follows by applying (SDB18) to the test function pairing. In particular

$$P_a M_y = a M_y P_a, \quad M_{1/y} P_a = a P_a M_{1/y}. \quad (\text{SDB30})$$

Using these and (SDB21), with all factors retained, gives

$$\begin{aligned} D_y P_a &= M_y (a P_a D) = P_a M_y D = P_a D_y, \\ R_y P_a &= R (a P_a M_{1/y}) = Q_a R M_{1/y} = Q_a R_y, \\ T_y P_a &= T (a P_a M_{1/y}) = P_a T M_{1/y} = P_a T_y, \\ E_y Q_a &= M_y (E Q_a) = M_y (a P_a E) = P_a M_y E = P_a E_y. \end{aligned}$$

This proves (SDB29). □

The exact comparison of the two complexes, including the full y factor, is the cochain isomorphism

$$\begin{array}{ccccc} \mathcal{J} & \xrightarrow{D} & \mathcal{J} & \xrightarrow{R} & \mathcal{D}'(I) \\ \text{id} \downarrow & & M_y \downarrow & & \text{id} \downarrow \\ \mathcal{J} & \xrightarrow{D_y} & \mathcal{J} & \xrightarrow{R_y} & \mathcal{D}'(I). \end{array} \quad (\text{SDB31})$$

Indeed $D_y = M_y D$ and $R_y M_y = R$; the inverse in degree 1 is $M_{1/y}$. It also preserves the respective dilation actions because $M_y(a P_a) = P_a M_y$. This exact comparison does not replace D_y by D in the full formulas (SDB25) and (SDB26).

96.5 Arbitrary external branch labels and the shared-label lattice lift

Two label constructions can be made precisely, and need not be identified. First let B be any set of retained branch indices, for example the indices $\beta < \top$ of an original support lattice on a branch with $y > 0$ and hence second coordinate label $\top$. Apply all the preceding operators component by component on

$$\mathcal{J}_B = \bigoplus_{\beta \in B} \mathcal{J}, \quad \mathcal{H}_B = \bigoplus_{\beta \in B} \mathcal{D}'(I). \quad (\text{SDB32})$$

Every element has finite branch support, and each component has finite normal-jet support. Each identity (SDB13) and (SDB27) holds component by component. The formulas preserve finite branch support, so their proofs give the same exact sequences, splittings, and dilation actions with $\mathcal{J}_B$ and $\mathcal{H}_B$. The same assertion holds for the Cartesian products over B if those products are specified instead: the explicit maps are defined at every component, no sum over B is taken, and no uniform jet-order bound is required. These assertions describe external labels and do not turn a family with infinite branch support into a distribution on an unspecified geometric union.

Second, let L be a bounded distributive lattice, with bottom $\perp$ and top $\top$, retaining all its elements. For a complex vector space V , set

$$\begin{aligned} G_L(V) &= (\{0\} \times L) \cup (V \times \{\top\}), \\ (v, \lambda) + (w, \mu) &= (v + w, \lambda \vee \mu), \\ (c, \alpha)(v, \lambda) &= (cv, \alpha \wedge \lambda), \quad (c, \alpha) \in G_L(\mathbb{C}), \\ \tau_V &= (0, \perp), \quad e_V = (0, \top), \quad Z_L(V) = \{(0, \lambda) : \lambda \in L\}. \end{aligned} \tag{SDB33}$$

Here $Z_L(V)$ is the full carrier of zero amplitudes and is naturally identified with L . Closure follows directly: a coordinate with a label below $\top$ has zero amplitude. In the scalar distributive laws the amplitude identity is the corresponding vector-space law, and the label identities are $\alpha \wedge (\lambda \vee \mu) = (\alpha \wedge \lambda) \vee (\alpha \wedge \mu)$ and its symmetric version. Thus (SDB33) is the usual semimodule structure for a bounded distributive L . The set maps and the splitting below also preserve the displayed operations for a bounded lattice without distributivity; a semimodule assertion for that wider class is not made. No finiteness assumption is made on L .

Every complex-linear map $A : V \rightarrow W$ has the label-preserving lift

$$\widehat{A} : G_L(V) \longrightarrow G_L(W), \quad \widehat{A}(v, \lambda) = (Av, \lambda). \tag{SDB34}$$

It is well-defined because $A0 = 0$, and it preserves addition and the scalar action by linearity and by retaining the same join or meet of the labels. Composition satisfies $\widehat{A}\widehat{B} = \widehat{AB}$ whenever AB is defined. In particular, the lift of a zero linear map is

$$\widehat{0}(v, \lambda) = (0, \lambda). \tag{SDB35}$$

It is not the constant map with value τ_W when a label $\lambda \neq \perp$ exists. Specifically, $\widehat{0}(v, \top) = e_W$.

Proposition 53 (Shared-label splitting and exact zero carrier). *Define*

$$\mathcal{P}_L = \{(h, k, \lambda) : h \in \mathcal{D}'(I), k \in \mathcal{J}, \lambda \in L, \lambda < \top \Rightarrow h = k = 0\}. \tag{SDB36}$$

It has addition $(h, k, \lambda) + (h', k', \mu) = (h + h', k + k', \lambda \vee \mu)$ and scalar action $(c, \alpha)(h, k, \lambda) = (ch, ck, \alpha \wedge \lambda)$. Then

$$\begin{aligned} G_L(\mathcal{J}) &\longrightarrow \mathcal{P}_L, & (u, \lambda) &\longmapsto (Ru, Tu, \lambda), \\ \mathcal{P}_L &\longrightarrow G_L(\mathcal{J}), & (h, k, \lambda) &\longmapsto (Eh + Dk, \lambda) \end{aligned} \tag{SDB37}$$

are mutually inverse maps preserving addition, scalar action, and every label. Moreover,

$$\begin{aligned} \widehat{D}^{-1}(Z_L(\mathcal{J})) &= Z_L(\mathcal{J}), \\ \widehat{R}^{-1}(Z_L(\mathcal{D}'(I))) &= \widehat{D}(G_L(\mathcal{J})). \end{aligned} \tag{SDB38}$$

The identical conclusions hold with D, E, R, T replaced by D_y, E_y, R_y, T_y .

Proof. When $\lambda < \top$, the domain element has $u = 0$, hence both Ru and Tu vanish. In the inverse direction $h = k = 0$ gives $Eh + Dk = 0$. Thus both maps have the exact stated domains. Their two compositions are the identity by $RE = \text{id}$, $RD = 0$, $TE = 0$, $TD = \text{id}$, and $ER + DT = \text{id}$ from Theorem 50; the label is unchanged in each composition. Linearity of every amplitude map and the displayed join and meet rules prove compatibility with addition and scalar action.

For the first equality in (SDB38), $\widehat{D}(u, \lambda)$ has zero amplitude exactly when $Du = 0$, which by injectivity of D is equivalent to $u = 0$. For the second equality, if $Ru = 0$, then $u = DTu$, and $(Tu, \lambda) \in G_L(\mathcal{J})$ because a label below $\top$ forces $u = 0$ and therefore $Tu = 0$. It follows that $(u, \lambda) = \widehat{D}(Tu, \lambda)$. Conversely $RD = 0$ puts every point in the indicated image into the zero carrier under $\widehat{R}$. All arguments repeat with the identities (SDB27), proving the assertion for D_y . $\square$

The shared-label object in (SDB36) is exactly the image of

$$(h, k, \lambda) \longmapsto ((h, \lambda), (k, \lambda)) \quad \text{in } G_L(\mathcal{D}'(I)) \times G_L(\mathcal{J}).$$

It consists of pairs with equal labels; it is not the full product with two independently chosen labels. For example, when $\perp \neq \top$, the pair $((0, \perp), (0, \top))$ belongs to the full product but cannot come from (SDB36). It would lose exactly the shared-label information to replace the target of (SDB37) by that larger product.

The distinction also fixes the meaning of the endpoint arrows. The zero vector space lifts to $G_L(\{0\}) = \{(0, \lambda) : \lambda \in L\}$, not to a singleton. The lift of the ordinary two-term complex is therefore a sequence

$$G_L(\{0\}) \xrightarrow{\widehat{0 \rightarrow \mathcal{J}}} G_L(\mathcal{J}) \xrightarrow{\widehat{D}} G_L(\mathcal{J}) \xrightarrow{\widehat{\mathcal{J} \rightarrow 0}} G_L(\{0\}), \quad (\text{SDB39})$$

where the first arrow sends $(0, \lambda)$ to $(0, \lambda)$ and the last sends (u, λ) to $(0, \lambda)$. Successive composites are the label-preserving supported-zero maps. They are not the constant- τ maps. More generally, if linear maps satisfy $d^{n+1}d^n = 0$, their lifts satisfy $\widehat{d}^{n+1}\widehat{d}^n(v, \lambda) = (0, \lambda)$; at $\lambda = \top$ this is e , with its label retained. These statements prove the exact lifted identities rather than assert an ordinary abelian cochain complex or ordinary acyclicity for (SDB39).

Finally, the amplitude map $q_V : G_L(V) \rightarrow V$, $q_V(v, \lambda) = v$, commutes with all the lifted linear maps but sends every $(0, \lambda)$ to the same vector 0. Indeed, every additive map $f : G_L(V) \rightarrow A$ into an abelian group kills every zero label: since $(0, \lambda) + (0, \lambda) = (0, \lambda)$, cancellation in A gives $f(0, \lambda) = 0$. Also $(v, \top) + (0, \top) = (v, \top)$ and $(v, \top) + (w, \top) = (v + w, \top)$, so $\bar{f}(v) = f(v, \top)$ is additive and $f = \bar{f} q_V$. This proves the precise universal loss of the zero labels in an additive-group receiver. Thus the ordinary cohomology in (SDB15) is a rigorous amplitude calculation, and (SDB37) is its exact label-preserving splitting; neither assertion treats an additive group as faithful to all the lattice labels.

All dilation maps above are complex-linear and invertible, so they also lift by (SDB34). The identities (SDB22), (SDB23), and (SDB29) continue to hold with the label unchanged. In particular the explicit shared-label splitting for D_y is equivariant for $\widehat{P}_a$ on its source and for the shared-label action $(h, k, \lambda) \mapsto (Q_a h, P_a k, \lambda)$ on its target. This concludes the distribution and lattice calculation for the specified branch, retaining both original operators and every factor in their comparison.

96.6 The full character family and arithmetic Frobenius

The same original human source, Section 7.1, equations `holom`, `holombis`, and `holom1`, and Proposition `frobarith` (`thecurve_K.tex`, lines 2564–2601), introduces the full family

$$\mathcal{L}_\lambda = \lambda X + iY = y(\lambda \partial_x + i \partial_y), \quad \mathcal{L}_\lambda \chi_\lambda(f) = 0 \quad (\lambda > 0), \quad \text{Fr}_\mu^a = \theta_\mu R(\mu^{-1}) \quad (\mu > 0). \quad (\text{SDB40})$$

Here the source's coefficient maps satisfy $\theta_\mu([r]) = [r^\mu]$, $\chi_\lambda([r]) = r^\lambda$, and $\chi_\lambda \theta_\mu = \chi_{\lambda\mu}$. The symbol $R(\mu^{-1})$ in (SDB40) is the source's right geometric action; it is distinct from the residue map R of (SDB10). For scalar distributions its geometric part is the pullback

$$B_\mu = (x, y \mapsto x, y/\mu)^*, \quad \langle B_\mu u, \phi \rangle = \mu \langle u, \phi(\cdot, \mu \cdot) \rangle. \quad (\text{SDB41})$$

The map has determinant $1/\mu$, so its inverse Jacobian is exactly the displayed μ . Therefore

$$B_\mu(\delta^{(j)} \otimes g) = \delta^{(j)} \otimes Q_{1/\mu} g. \quad (\text{SDB42})$$

There is no normal x scaling in this formula. The full arithmetic Frobenius on a character family u_λ is

$$(\mathfrak{F}_\mu u)_\lambda = B_\mu u_{\lambda\mu}. \quad (\text{SDB43})$$

This retains both the right geometric action and the character parameter change from the cited formula.

For each $\lambda > 0$, define

$$D_\lambda = \lambda \partial_x + i \partial_y, \quad \mathcal{L}_\lambda = M_y D_\lambda, \quad (\text{SDB44})$$

$$R_{y,\lambda} \left(\sum_{j=0}^N \delta^{(j)} \otimes g_j \right) = \sum_{j=0}^N \left(\frac{-i}{\lambda} \right)^j \partial_y^j (y^{-1} g_j), \quad (\text{SDB45})$$

$$T_{y,\lambda} \left(\sum_{j=0}^N \delta^{(j)} \otimes g_j \right) = \frac{1}{\lambda} \sum_{j=1}^N \sum_{k=0}^{j-1} \left(\frac{-i}{\lambda} \right)^k \delta^{(j-1-k)} \otimes \partial_y^k (y^{-1} g_j). \quad (\text{SDB46})$$

The section is still the full $E_y h = \delta \otimes (y h)$. For precision, the complete coefficient expansions in these formulas are

$$R_{y,\lambda} u = \sum_{j=0}^N \sum_{r=0}^j \left(\frac{-i}{\lambda} \right)^j \binom{j}{r} (-1)^r r! y^{-r-1} g_j^{(j-r)}, \quad (\text{SDB47})$$

$$T_{y,\lambda} u = \frac{1}{\lambda} \sum_{j=1}^N \sum_{k=0}^{j-1} \sum_{r=0}^k \left(\frac{-i}{\lambda} \right)^k \binom{k}{r} (-1)^r r! \delta^{(j-1-k)} \otimes (y^{-r-1} g_j^{(k-r)}). \quad (\text{SDB48})$$

Theorem 54 (Exact character-family residue and Frobenius comparison). *For every $\lambda > 0$ one has*

$$\begin{aligned} R_{y,\lambda} \mathcal{L}_\lambda &= 0, & R_{y,\lambda} E_y &= \text{id}, & T_{y,\lambda} E_y &= 0, \\ \mathcal{L}_\lambda T_{y,\lambda} &= \text{id} - E_y R_{y,\lambda}, & T_{y,\lambda} \mathcal{L}_\lambda &= \text{id}. \end{aligned} \quad (\text{SDB49})$$

In particular, $\mathcal{L}_\lambda$ is injective and $\mathcal{J}/\mathcal{L}_\lambda \mathcal{J} \cong \mathcal{D}'(I)$ by the residue $R_{y,\lambda}$ with inverse $h \mapsto [E_y h]$. For every $\lambda, \mu > 0$, the exact covariance identities are

$$\begin{aligned} \mathcal{L}_\lambda B_\mu &= B_\mu \mathcal{L}_{\lambda\mu}, \\ R_{y,\lambda} B_\mu &= \mu^{-1} Q_{1/\mu} R_{y,\lambda\mu}, \\ T_{y,\lambda} B_\mu &= B_\mu T_{y,\lambda\mu}, \\ B_\mu E_y &= E_y (\mu^{-1} Q_{1/\mu}). \end{aligned} \quad (\text{SDB50})$$

Thus the residue has exactly one further factor μ^{-1} after the full scalar-distribution pullback $Q_{1/\mu}$ is written.

Proof. First set $c_\lambda = -i/\lambda$ and temporarily retain the operators before multiplication by y :

$$R_\lambda \left(\sum_j \delta^{(j)} \otimes g_j \right) = \sum_j c_\lambda^j g_j^{(j)}, \quad T_\lambda (\delta^{(j)} \otimes g) = \frac{1}{\lambda} \sum_{k=0}^{j-1} c_\lambda^k \delta^{(j-1-k)} \otimes g^{(k)}.$$

Since $D_\lambda (\delta^{(j)} \otimes g) = \lambda \delta^{(j+1)} \otimes g + i \delta^{(j)} \otimes g'$, one has

$$R_\lambda D_\lambda (\delta^{(j)} \otimes g) = (\lambda c_\lambda^{j+1} + i c_\lambda^j) g^{(j+1)} = 0.$$

In $D_\lambda T_\lambda (\delta^{(j)} \otimes g)$, the first term is $\delta^{(j)} \otimes g$. Each internal term with derivative index $1 \leq r \leq j-1$ has coefficient $c_\lambda^r + (i/\lambda) c_\lambda^{r-1} = 0$. The last term is $(i/\lambda) c_\lambda^{j-1} \delta \otimes g^{(j)} = -c_\lambda^j \delta \otimes g^{(j)}$. Thus $D_\lambda T_\lambda = \text{id} - E R_\lambda$ on every finite sum. Also $R_\lambda E = \text{id}$ and $T_\lambda E = 0$. The highest normal coefficient of $D_\lambda u$, for a nonzero u of highest order N , is $\lambda g_N \neq 0$. Hence D_λ is injective. Applying it to $T_\lambda D_\lambda u - u$ now proves $T_\lambda D_\lambda = \text{id}$, by the same displayed operator identity and $R_\lambda D_\lambda = 0$.

The maps in (SDB45)–(SDB46) are exactly $R_\lambda M_{1/y}$ and $T_\lambda M_{1/y}$. Multiplying the identities just proved by M_y and $M_{1/y}$ in their stated order therefore proves all five formulas (SDB49); for instance

$$\mathcal{L}_\lambda T_{y,\lambda} = M_y D_\lambda T_\lambda M_{1/y} = \text{id} - M_y E R_\lambda M_{1/y} = \text{id} - E_y R_{y,\lambda}.$$

This also proves the quotient assertion.

Next put $b = 1/\mu$. Formula (SDB41) gives

$$\partial_x B_\mu = B_\mu \partial_x, \quad \partial_y B_\mu = b B_\mu \partial_y, \quad B_\mu M_y = b M_y B_\mu.$$

Consequently

$$\begin{aligned} \mathcal{L}_\lambda B_\mu &= M_y B_\mu (\lambda \partial_x + i b \partial_y) \\ &= b M_y B_\mu (\lambda \mu \partial_x + i \partial_y) = B_\mu M_y D_{\lambda \mu} = B_\mu \mathcal{L}_{\lambda \mu}. \end{aligned}$$

On one-variable distributions, the two rules needed for all the remaining factors are

$$y^{-1} Q_b g = b Q_b (y^{-1} g), \quad \partial_y^j Q_b g = b^j Q_b (g^{(j)}). \quad (\text{SDB51})$$

The first follows by multiplying the test function in the formula for Q_b ; the second was proved in (SDB20). Therefore each order- j contribution to the residue of $B_\mu u$ is

$$\left(\frac{-i}{\lambda}\right)^j \partial_y^j (y^{-1} Q_b g_j) = b \left(\frac{-i}{\lambda \mu}\right)^j Q_b \partial_y^j (y^{-1} g_j).$$

Summing proves the second line of (SDB50), including $b = \mu^{-1}$. Similarly, each (j, k) contribution to the homotopy of $B_\mu u$ has the coefficient

$$\frac{1}{\lambda} \left(\frac{-i}{\lambda}\right)^k b^{k+1} = \frac{1}{\lambda \mu} \left(\frac{-i}{\lambda \mu}\right)^k.$$

Together with (SDB42), this is exactly the corresponding summand of $B_\mu T_{y,\lambda \mu} u$. Finally, $Q_b(yh) = by Q_b h$ gives $B_\mu E_y h = \delta \otimes Q_b(yh) = b \delta \otimes (y Q_b h) = E_y(b Q_b h)$. This proves all four covariance identities. $\square$

These are covariance identities for the entire family, not an invariance assertion for a fixed λ under B_μ . To make the receiver and all maps explicit, set

$$\mathcal{C}^0 = \prod_{\lambda > 0} \mathcal{J}, \quad \mathcal{C}^1 = \prod_{\lambda > 0} \mathcal{J}, \quad \mathcal{H} = \prod_{\lambda > 0} \mathcal{D}'(I), \quad (\text{SDB52})$$

where each component of an element of $\mathcal{C}^n$ has finite normal order but there is no uniform bound in λ . Define $\mathcal{L}$, $\mathcal{R}$, $\mathcal{T}$, and $\mathcal{E}$ componentwise by $\mathcal{L}_\lambda$, $R_{y,\lambda}$, $T_{y,\lambda}$, and E_y . Their explicit identities in (SDB49) prove

$$0 \longrightarrow \mathcal{C}^0 \xrightarrow{\mathcal{L}} \mathcal{C}^1 \xrightarrow{\mathcal{R}} \mathcal{H} \longrightarrow 0, \quad \mathcal{C}^1 \xrightarrow[u \mapsto (\mathcal{R}u, \mathcal{T}u)]{\cong} \mathcal{H} \oplus \mathcal{C}^0, \quad (h, k) \mapsto \mathcal{E}h + \mathcal{L}k. \quad (\text{SDB53})$$

No choice of preimages or convergence over the parameter set is needed: the preimages are the displayed componentwise formulas.

On both cochain degrees use (SDB43), and on the receiver use

$$(\mathfrak{F}_\mu^H h)_\lambda = \mu^{-1} Q_{1/\mu} h_{\lambda \mu}. \quad (\text{SDB54})$$

The four identities in (SDB50) prove that every arrow and every splitting map in (SDB53) is equivariant. Both actions are representations of $\mathbb{R}_{>0}^\times$: the two geometric pullbacks compose as $B_\mu B_\nu = B_{\mu \nu}$, the character index becomes $\lambda \mu \nu$, and the receiver factors multiply as $\mu^{-1} \nu^{-1} = (\mu \nu)^{-1}$ with $Q_{1/\mu} Q_{1/\nu} = Q_{1/(\mu \nu)}$. The inverse action is the one with parameter $1/\mu$. The two-term

family complex consequently has $H^0 = 0$ and $H^1 \cong \mathcal{H}$, with its action exactly (SDB54). For a smooth coefficient family the latter reads $h_\lambda(y) \mapsto \mu^{-1}h_{\lambda\mu}(y/\mu)$; for distributions the full pullback formula for $Q_{1/\mu}$ in (SDB18) remains part of the statement.

The simultaneous left action also persists for every λ : the proof of (SDB29), retaining λ in each term, gives $\mathcal{L}_\lambda P_a = P_a \mathcal{L}_\lambda$, $R_{y,\lambda} P_a = Q_a R_{y,\lambda}$, and $T_{y,\lambda} P_a = P_a T_{y,\lambda}$. For example, in its residue calculation the order- j factors are a^{-j-1} from the normal jet, a from multiplication by $1/y$ before pullback, and a^j from the derivative, whose product is 1. The maps $(x, y) \mapsto (ax, ay)$ and $(x, y) \mapsto (x, y/\mu)$ commute, and their full determinants multiply. Hence these simultaneous left actions commute with $\mathfrak{F}_\mu$; their receiver actions commute with $\mathfrak{F}_\mu^H$ because the Q actions commute. This statement retains both the left geometric and right arithmetic actions rather than identifying them.

Exact entry of algebraic W coefficients and its scope. The original source states its algebraic coefficient model as the complex group ring of $\mathbb{R}_{>0}^\times$ before considering completions. Write this algebraic model explicitly as $W_{\text{alg}} = \mathbb{C}[\mathbb{R}_{>0}^\times]$. There is a character-evaluation map

$$\begin{aligned} \text{ev} : W_{\text{alg}} \otimes_{\mathbb{C}} \mathcal{J} &\longrightarrow \prod_{\lambda > 0} \mathcal{J}, \\ \sum_{r \in F} [r] \otimes u_r &\longmapsto \left(\sum_{r \in F} r^\lambda u_r \right)_{\lambda > 0}, \quad F \subset \mathbb{R}_{>0}^\times \text{ finite.} \end{aligned} \tag{SDB55}$$

It is injective. To prove this, list the distinct elements of F as $r_1, \dots, r_m$. If its output vanishes, evaluating at $\lambda = 1, \dots, m$ gives a linear system for $u_{r_1}, \dots, u_{r_m}$ with coefficient matrix $(r_k^n)_{1 \leq n, k \leq m}$. Its determinant is

$$\left(\prod_{k=1}^m r_k \right) \prod_{1 \leq k < \ell \leq m} (r_\ell - r_k) \neq 0.$$

The inverse complex matrix acts on the vector space $\mathcal{J}^m$, so all u_{r_k} vanish. This proves injectivity without presuming any topology on W_{alg} .

On these algebraic tensor coefficients the source's arithmetic action is exactly

$$\sum_{r \in F} [r] \otimes u_r \longmapsto \sum_{r \in F} [r^\mu] \otimes B_\mu u_r. \tag{SDB56}$$

Evaluation at λ gives $\sum_r r^{\mu\lambda} B_\mu u_r = B_\mu \sum_r r^{\lambda\mu} u_r$, which proves that (SDB55) intertwines (SDB56) and (SDB43). Applying the residue then gives the explicit original-coefficient formula

$$\left(\mathcal{R} \text{ev} \sum_{r \in F} [r] \otimes \sum_{j=0}^{N_r} \delta^{(j)} \otimes g_{r,j} \right)_\lambda = \sum_{r \in F} \sum_{j=0}^{N_r} r^\lambda \left(\frac{-i}{\lambda} \right)^j \partial_y^j (y^{-1} g_{r,j}). \tag{SDB57}$$

Every factor r^λ , λ^{-j} , $(-i)^j$, and y^{-1} , together with all its derivatives in (SDB47), is retained. Equations (SDB50) and (SDB55) prove that this exact receiver is equivariant for the source's algebraic arithmetic Frobenius.

The product-family complex (SDB52) is the constructed distribution receiver. It includes arbitrary character families and therefore is not identified here with the source's W -valued function space, a completed W module, or an adelic Dolbeault complex. The injection (SDB55) proves the precise relation supplied for algebraic W coefficients; it does not assert that every family in the product comes from those coefficients or that the differential preserves their image. One immediate consequence within this exact scope is that an algebraic tensor in (SDB55) satisfying $\mathcal{L}_\lambda \text{ev}(u)_\lambda = 0$ for every $\lambda > 0$ must be zero: injectivity of each $\mathcal{L}_\lambda$ first kills every character

component, and injectivity of ev then kills u . This is a proved assertion about the supported distribution receiver, not a vanishing assertion about the holomorphic functions in the original whole complex lift.

Finally, every linear map in the family sequence and every action above lifts by (SDB34). Applying G_L to the full spaces $\mathcal{C}^0$, $\mathcal{C}^1$, and $\mathcal{H}$ gives the exact shared-label splitting

$$(u, \sigma) \mapsto (\mathcal{R}u, \mathcal{T}u, \sigma), \quad (h, k, \sigma) \mapsto (\mathcal{E}h + \mathcal{L}k, \sigma),$$

with $\sigma < \top$ forcing both amplitudes to vanish. The arithmetic action is $(h, k, \sigma) \mapsto (\mathfrak{F}_\mu^H h, \mathfrak{F}_\mu k, \sigma)$. The identities already proved show equivariance and preserve every zero label. As in (SDB35), composite zero differentials land at $(0, \sigma)$, and in particular at e for $\sigma = \top$, rather than at a constant τ .

96.7 A moving Teichmüller coefficient and its exact receiver

The source's own example demonstrates why the algebraic tensor model does not exhaust the kinds of coefficient-valued functions that occur in its discussion. Equations `holom2` and Proposition `functionq` in Section 7.1 give

$$q(x + iy) = [e^{-2\pi y}]e^{2\pi ix}, \quad q_\lambda(x, y) = \chi_\lambda(q(x + iy)) = e^{-2\pi\lambda y} e^{2\pi ix}. \quad (\text{SDB58})$$

The bracket basis element moves with y . These scalar evaluations are smooth functions on Ω , so multiplication by each one is an unambiguous operation on distributions. They define the supported family

$$v_\lambda = q_\lambda(x, y)(\delta(x) \otimes 1(y)) = \delta(x) \otimes e^{-2\pi\lambda y}. \quad (\text{SDB59})$$

Its arithmetic action and residue are exactly

$$\mathfrak{F}_\mu v = v, \quad (\mathcal{R}v)_\lambda = y^{-1} e^{-2\pi\lambda y}, \quad \mathfrak{F}_\mu^H(\mathcal{R}v) = \mathcal{R}v. \quad (\text{SDB60})$$

To verify these assertions, the first pullback evaluates $e^{-2\pi(\lambda\mu)(y/\mu)} = e^{-2\pi\lambda y}$ and leaves $\delta(x)$ unchanged. The residue is the $j = 0$ term of (SDB45). Its action is

$$\mu^{-1} \left((y/\mu)^{-1} e^{-2\pi(\lambda\mu)(y/\mu)} \right) = y^{-1} e^{-2\pi\lambda y},$$

which proves the last identity with its full factor. Although q_λ itself satisfies $\mathcal{L}_\lambda q_\lambda = 0$, the supported distribution has

$$\mathcal{L}_\lambda v_\lambda = \lambda \delta'(x) \otimes (ye^{-2\pi\lambda y}) - 2\pi i \lambda \delta(x) \otimes (ye^{-2\pi\lambda y}) \neq 0. \quad (\text{SDB61})$$

The coefficient of δ' is nonzero and cannot cancel the δ coefficient by Proposition 49. This proves exactly what happens to the original holomorphic example under its multiplication into the supported receiver.

Moreover v in (SDB59) is not in the algebraic tensor image (SDB55). Suppose otherwise that $v_\lambda = \sum_{k=1}^m r_k^\lambda u_k$ with finitely many distinct $r_k > 0$ and $u_k \in \mathcal{J}$. The invertible matrix used to prove (SDB55) expresses each u_k as a complex linear combination of $v_1, \dots, v_m$. Hence $u_k = \delta \otimes a_k(y)$ with a_k a smooth function, a finite linear combination of the functions $e^{-2\pi n y}$, $1 \leq n \leq m$. The assumed identity of distributions then implies the identity of smooth functions

$$e^{-2\pi\lambda y} = \sum_{k=1}^m r_k^\lambda a_k(y) \quad \text{for every } \lambda > 0, y > 0.$$

Choose $y_0 > 0$ with $r_0 = e^{-2\pi y_0}$ distinct from the finite set $\{r_1, \dots, r_m\}$. Evaluating this identity at $\lambda = 1, \dots, m+1$ gives a homogeneous linear relation between the $m+1$ columns with bases $r_0, r_1, \dots, r_m$, whose coefficient of the first column is 1. The determinant is

$(\prod_{k=0}^m r_k) \prod_{0 \leq k < \ell \leq m} (r_\ell - r_k) \neq 0$, a contradiction. This proves the claimed nonmembership while retaining the explicit evaluation and residue maps.

Define the following concrete evaluation-family space and its receiver, using precisely the algebraic tensor image and the moving-coefficient family just constructed:

$$\begin{aligned} U &= \text{ev}(W_{\text{alg}} \otimes \mathcal{J}) + \mathbb{C}v \subset \mathcal{C}^1, \\ V_U &= \{k \in \mathcal{C}^0 : \mathcal{L}k \in U\}, \quad H_U = \mathcal{R}(U) \subset \mathcal{H}. \end{aligned} \tag{SDB62}$$

The inclusion of the tensor image into U is strict by the nonmembership proof above. Its arithmetic invariance is proved, rather than assumed: the tensor image is invariant by (SDB56), and v is fixed by (SDB60). The exact sequence and its quotient map are

$$0 \longrightarrow V_U \xrightarrow{\mathcal{L}} U \xrightarrow{\mathcal{R}} H_U \longrightarrow 0, \tag{SDB63}$$

$$U \cap \ker \mathcal{R} = U \cap \mathcal{L}(\mathcal{C}^0) = \mathcal{L}(V_U), \quad U/\mathcal{L}(V_U) \xrightarrow[\substack{\cong \\ [u] \mapsto \mathcal{R}u}} H_U. \tag{SDB64}$$

To prove exactness, injectivity of $\mathcal{L}$ is already established, and the last map is onto its defined image. If $u \in U$ and $\mathcal{R}u = 0$, then $u = \mathcal{L}\mathcal{T}u$ by (SDB53), so $\mathcal{T}u \in V_U$ and u is in the image of the first arrow. The converse is $\mathcal{R}\mathcal{L} = 0$. For $k \in V_U$, equivariance gives $\mathcal{L}\mathfrak{F}_\mu k = \mathfrak{F}_\mu \mathcal{L}k \in U$, proving stability of V_U . Similarly $\mathfrak{F}_\mu^H \mathcal{R}u = \mathcal{R}\mathfrak{F}_\mu u$ proves stability of H_U and equivariance of every displayed arrow. Its specified moving-coefficient residue is the full family (SDB60). No equality $V_U = U$ or $\mathcal{E}(H_U) \subset U$ is asserted. This concrete construction proves an exact connection to a moving coefficient in the human source without adding a completion hypothesis or identifying the product-family cohomology with the original W -valued cohomology.

97 Presence, parity, and the information retained by a map

The user's intended distinction. The ungraded presence unit τ has no parity. Integer 1 is present and odd. These descriptions specify different amounts of data. We now construct their exact comparison. The labels Z_0, Z_1, Z_2 are the user's notation: absence, ungraded presence, and integer parity respectively. The source unit τ carries presence at Z_1 and has no Z_2 parity. The notation Z_0 for absence is not treated as a cyclic group with zero elements. The elementary set and linear algebra proofs below are given in full. The quantum comparison uses Arun Kumar Pati and Samuel L. Braunstein, *Impossibility of deleting an unknown quantum state*, original author source, equations `del2`, `tran1`, `tran2`, `almostfinal`, and `final`. Its distinction between erasure, transfer to an ancilla, and deletion is retained.

97.1 Empty, present, and parity-labelled

Define the label sets and their maps by

$$Z_0 = \emptyset, \quad Z_1 = \{\tau\}, \quad Z_2 = \{\text{even}, \text{odd}\}, \quad Z_0 \longrightarrow Z_2 \xrightarrow{q} Z_1, \quad q(\text{even}) = q(\text{odd}) = \tau. \quad (\text{PPI1})$$

A Z_i -labelled set is a pair (A, ℓ) with a function $\ell : A \rightarrow Z_i$. A morphism $(A, \ell) \rightarrow (B, m)$ is a function $f : A \rightarrow B$ satisfying $mf = \ell$. Composition preserves this equation, and identity functions satisfy it, so these objects and maps form the slice category Set/Z_i . Postcomposition defines the functor

$$F : \text{Set}/Z_2 \longrightarrow \text{Set}/Z_1, \quad F(A, \ell) = (A, q\ell), \quad F(f) = f. \quad (\text{PPI2})$$

Indeed $(qm)f = q\ell$; identities and compositions are unchanged. It forgets the parity label while retaining every underlying element. Let $T = (\{*\}, * \mapsto \tau)$ in Set/Z_1 , and let E and O be the objects with carrier $\{*\}$ labelled even and odd in Set/Z_2 . Then

$$F(E) = T = F(O), \quad E \not\cong O. \quad (\text{PPI3})$$

The only function between their singleton carriers sends the single element to the single element. It fails the label-preservation equation because even and odd are distinct. There is therefore not even a morphism $E \rightarrow O$ in this slice. In particular F cannot have an inverse functor: an inverse up to isomorphism would make both E and O isomorphic to the inverse image of T . Choosing E or O is possible, but does not recover which object was supplied. The source unit τ is represented by unlabelled presence T , not by a secretly even or odd version of T . The complete space of parity refinements of T has the two objects E, O . The label exchange interchanges them and fixes neither. There is also a canonical construction retaining both choices: $q^*(A) = (A \times Z_2, \text{pr}_2)$. For every labelled (B, ℓ) , the assignments

$$f : B \rightarrow A \quad \longleftrightarrow \quad [b \mapsto (f(b), \ell(b))] : (B, \ell) \rightarrow q^*(A)$$

are inverse because the second coordinate of a label-preserving map must be ℓ . They commute with precomposition and postcomposition, proving $F \dashv q^*$. For a singleton A , this creates two new tagged elements. It does not give two parities to the original ungraded unit.

The parity-labelled integer object is

$$I = (\mathbb{Z}, \pi), \quad \pi(n) = n \bmod 2, \quad \pi(n + m) = \pi(n) + \pi(m), \quad \pi(nm) = \pi(n)\pi(m) \quad \text{in } \mathbb{Z}/2\mathbb{Z}. \quad (\text{PPI4})$$

The equations follow because writing $n = 2a + r$, $m = 2b + s$ gives $n + m = 2(a + b) + (r + s)$ and $nm = 2(2ab + as + br) + rs$. The element 1 is an odd-labelled point of I . Presence and oddness are two descriptors of that one point; they are not two alternative states occupied by 1, and presence is not an additional independent binary choice once that point has been specified. This statement imposes no parity on τ .

The unique map $A \rightarrow \emptyset$ exists exactly when A is empty: if $a \in A$ it would require an image in the empty set; if A is empty the empty function exists. Thus demanding an actual label map into Z_0 forces emptiness. Merely omitting a label map does not: the singleton $\{*\}$ with no specified parity is nonempty. This proves the exact distinction between an empty label target and an unspecified grading.

97.2 Boolean arithmetic and counts

Let $\mathbf{B}$ be the ordered category with objects false and true and a unique arrow $b \rightarrow c$ when $b \leq c$. Define

$$\text{supp}(A) = \begin{cases} \text{false}, & A = \emptyset, \\ \text{true}, & A \neq \emptyset. \end{cases} \quad (\text{PPI5})$$

A function from a nonempty set to an empty set cannot exist, so each set map induces the required arrow between supports. The arrow is unique; therefore identities and compositions are preserved. For all sets A, B ,

$$\text{supp}(A \sqcup B) = \text{supp}(A) \vee \text{supp}(B), \quad \text{supp}(A \times B) = \text{supp}(A) \wedge \text{supp}(B). \quad (\text{PPI6})$$

A disjoint union has an element exactly when at least one summand does; a product has an element exactly when both factors do. These equivalences prove both formulas. Identifying false with $\emptyset$ and true with $\{*\}$ as the two subsets of a singleton gives Boolean union and intersection, with $\text{true} \vee \text{true} = \text{true}$. Before this support map, $\{*\} \sqcup \{*\}$ has two distinct tagged elements. For finite sets their cardinalities satisfy

$$|A \sqcup B| = |A| + |B|, \quad |A \times B| = |A||B|, \quad \mathbb{N} \xrightarrow{p} \mathbf{B}, \quad \text{quad}p(0) = \text{false}, \quad p(n) = \text{true} \ (n > 0). \quad (\text{PPI7})$$

To check the cardinality equations, enumerate the two disjoint blocks consecutively, and enumerate the pairs in $|A|$ rows of $|B|$ entries. PPI6 then proves that p preserves semiring addition, multiplication, zero, and unit. It does not preserve parity information: $p(1) = p(2)$, while their parities differ. This is the exact finite-set realization of the two procedures $(\tau + \tau)1 = 1$ and $\tau 1 + \tau 1 = 2$.

There are two questions about zero, with explicitly different domains:

$$\text{supp}(\emptyset) = \text{false}, \quad \text{supp}(\{0\}) = \text{true}, \quad |\emptyset| = 0, \quad \pi(0) = 0. \quad (\text{PPI8})$$

No new numerical zero is introduced. The set containing the integer zero has an element, while the empty set has none. Even if the integer zero is represented by $\emptyset$, the singleton containing that representative is $\{\emptyset\} \neq \emptyset$. Consequently “every specified integer is present” concerns the point $\{n\}$, whereas a cardinality-based presence test sends the value zero to false. They cannot be interchanged as one map on one domain. The truth map in PPI7 cannot extend to a unital semiring map $\mathbb{Z} \rightarrow \mathbf{B}$: such a map would send $0 = 1 + (-1)$ to $\text{true} \vee p(-1) = \text{true}$, contradicting preservation of zero. This determines the exact domain of the comparison rather than discarding integer signs.

97.3 Trivial symmetry is not emptiness

Every set has an identity bijection. In particular the empty function $\emptyset \rightarrow \emptyset$ is a bijection, as is the identity on a singleton. These are their only self-maps, so

$$\text{Aut}(\emptyset) \cong \{1\} \cong \text{Aut}(\{*\}). \quad (\text{PPI9})$$

Thus the automorphism group alone cannot distinguish these two sets. The trivial group acts on every set by its identity; this does not force a set to have one element. A C_2 action can be restricted along the unique homomorphism $\{1\} \rightarrow C_2$ to its trivial-group action. That is a precise sense in which this particular symmetry can be forgotten. A parity labelling $A \rightarrow Z_2$ is different data from an action $C_2 \times A \rightarrow A$: the odd-labelled singleton exists although every action on that singleton fixes its only point. No parity-flipping bijection is supplied by its label alone.

97.4 An exact linear information receiver

The set map q has a complex-linear extension

$$V = \mathbb{C}e_{\text{even}} \oplus \mathbb{C}e_{\text{odd}}, \quad W = \mathbb{C}t, \quad T : V \rightarrow W, \quad T(e_{\text{even}}) = T(e_{\text{odd}}) = t. \quad (\text{PPI10})$$

For $v = ae_{\text{even}} + be_{\text{odd}}$, one has $T(v) = (a + b)t$. Hence

$$\ker T = \mathbb{C}(e_{\text{even}} - e_{\text{odd}}). \quad (\text{PPI11})$$

Both inclusions follow immediately from $a + b = 0$. This kernel is the entire lost parity-difference direction. Give the displayed bases orthonormal inner products and define $Ze_{\text{even}} = e_{\text{even}}$, $Ze_{\text{odd}} = -e_{\text{odd}}$. There is no operator A on W satisfying $AT = TZ$: the even vector would require $At = t$ and the odd vector would require $At = -t$. Over $\mathbb{C}$ these are incompatible because $t \neq 0$. This proves exactly which observable does not descend to presence alone.

Retaining the full record gives an isometry instead:

$$J : V \longrightarrow W \otimes V, \quad J(v) = t \otimes v, \quad J^*J = I_V, \quad JZ = (I_W \otimes Z)J. \quad (\text{PPI12})$$

The first equality follows from $\langle t, t \rangle = 1$ and the tensor-product inner product. The second follows by evaluating both sides on each basis vector. The first factor represents ungraded presence and the second retains the parity state. It is the second factor, not t or τ , on which Z acts. This is a mathematical state-space comparison, without a claim that the integers or τ thereby become physical particles.

97.5 What the quantum no-deleting comparison proves

Pati–Braunstein consider arbitrary unknown states, not just distinguishable classical parity labels. The following exact inner-product test gives the fixed-environment obstruction. Let all displayed state vectors have norm one, and suppose an isometry U obeyed, for every input vector ψ ,

$$U(\psi \otimes \psi \otimes a) = \psi \otimes \Sigma \otimes a_0, \quad (\text{PPI13})$$

where a , Σ , and a_0 are fixed independently of ψ . For two allowed inputs ψ, ϕ with inner product c , taking inner products gives

$$c^2 = c. \quad (\text{PPI14})$$

Indeed the input product is $c \cdot c \cdot 1$ and the output product is $c \cdot 1 \cdot 1$. In dimension at least two, choose $\psi = e_0$, $\phi = (e_0 + e_1)/\sqrt{2}$; then $c = 1/\sqrt{2}$ and PPI14 is false. Thus this fixed-record deletion is impossible. Allowing the environment to retain the state instead gives the explicit unitary swap

$$\psi \otimes \psi \otimes \Sigma \longmapsto \psi \otimes \Sigma \otimes \psi. \quad (\text{PPI15})$$

Swapping the last two tensor factors preserves the inner product on pure tensors and is its own inverse, proving unitarity. It transfers the information rather than deleting it; the original paper explicitly excludes this as the sought deletion.

For the two orthogonal labels $b \in \{0, 1\}$, there is also a reversible map

$$|b\rangle|c\rangle \longmapsto |b\rangle|c \oplus b\rangle, \quad |b\rangle|b\rangle \longmapsto |b\rangle|0\rangle. \quad (\text{PPI16})$$

It permutes the four orthonormal basis vectors and is its own inverse. Its linear extension is therefore unitary. It does not delete an arbitrary superposition against a copy: applying it to $(\alpha|0\rangle + \beta|1\rangle)^{\otimes 2}$ gives $\alpha^2|00\rangle + \alpha\beta|01\rangle + \alpha\beta|11\rangle + \beta^2|10\rangle$, retaining every term. For $\alpha\beta \neq 0$ this is not the proposed blank-second-factor state. Thus the quantum theorem does not forbid the mathematical forgetful functor PPI2 or turn a classical label into a conserved physical charge. The constructive information-preserving comparison here is PPI12; the exact information lost by PPI10 is PPI11. None of these results asserts an RH sign or purity bound.

98 The ungraded presence unit, spherical algebra, and arithmetic realization

Original sources and exact scope. Alain Connes and Caterina Consani, *On the Absolute Geometry of $\text{Spec } \mathbb{Z}$ and the Fargues–Fontaine curve*, original `FF.tex`, line 145, identify their $\mathbb{F}_1$ with the spherical algebra of the pointed multiplicative monoid $\{0, 1\}$. Lines 343–371 describe its arithmetic stalks; lines 374–404 describe morphisms through the underlying multiplicative monoid, with the exponent-zero condition $\chi(0) = 1$. The source SHA-256 is

14c17a95040be7bfc57bb5f87e58907a05cacd0bcb1b6967b2892cf3d9bc1229.

Their earlier *Absolute algebra and Segal’s Γ -rings*, original `F1_corr.tex`, gives the needed definitions and comparisons: lines 416–438 define the category and the sum-over-fibres functor; lines 464–501 define the sphere and spherical monoid algebra; Proposition `monomono`, lines 504–510, proves the monoid adjunction; Lemma `sssalg`, lines 519–532, defines the semiring construction and its unit. Lines 544–560 give the two projections and folding definition of addition. The Boolean construction and comparison with $\mathbb{F}_2$ are at lines 568–580, and the absence of two simultaneous nonzero inputs in a spherical monoid algebra is at lines 865–880. Its source SHA-256 is

3b1319ec1df4ed6e2412ace960c2813b683a36c88509234c9abd0a39f98e8913.

These precise passages were read in original author TeX for this derivation. The calculations below instantiate their existing constructions; the source mathematics is not claimed as a new discovery. The support operations remain those of The Clankers, *Split Support Geometry*, Definition `def:lattice-split`.

98.1 Presence has a multiplicative unit before arithmetic parity

Keep the user’s presence unit τ and denote the external absent point by 0_M . Define the pointed monoid

$$M = \{0_M, \tau\}, \quad \tau^2 = \tau, \quad 0_M \tau = \tau 0_M = 0_M^2 = 0_M. \quad (\text{FSP1})$$

Here τ is the multiplicative unit and 0_M the absorber. No addition and no parity function are part of this multiplicative structure. The named map $j : M \rightarrow \{0, 1\}_{\text{CC}}$ with $j(0_M) = 0$ and $j(\tau) = 1$ is an isomorphism of pointed multiplicative monoids: its four products agree with the four products of FSP1, and the map is bijective. Its target is Connes–Consani’s pointed monoid. The singleton $\{\tau\}$ supplies its unit monoid; M is that singleton with an external basepoint adjoined. This is the exact structure used here, not an identification of τ with a pre-existing integer.

Let Γ^{op} have objects $k_+ = \{0, 1, \dots, k\}$ and all basepoint-preserving maps, as in the original source. The spherical monoid algebra is

$$\mathbb{S}M(k_+) = M \wedge k_+ = \{0, \tau_1, \dots, \tau_k\}, \quad \mathbb{S}M(f)(\tau_j) = \begin{cases} \tau_{f(j)} & f(j) \neq 0, \\ 0 & f(j) = 0. \end{cases} \quad (\text{FSP2})$$

Here τ_j is the class of (τ, j) ; all pairs with either coordinate a basepoint are identified with 0. Composition follows immediately from composition of f . The product sends (τ_i, τ_j) to $\tau_{(i,j)}$ at $k_+ \wedge l_+$, with every product containing a basepoint sent to the basepoint. Associativity is associativity of the label pair (i, j) , and the one available unit is τ . The maps $\tau_j \mapsto j$ and $0 \mapsto 0$ give an isomorphism of spherical algebras

$$\mathbb{S}M \cong \mathbb{S}, \quad \mathbb{S}(k_+) = k_+. \quad (\text{FSP3})$$

Thus this is the specified Connes–Consani spherical $\mathbb{F}_1$ model. Its description requires no parity assignment.

98.2 Every coefficient realization, with all coordinates

For any commutative unital semiring A , let

$$HA(k_+) = A^k, \quad (HA(f)(a))_r = \sum_{j \in f^{-1}(r)} a_j \quad (1 \leq r \leq l). \quad (\text{FSP4})$$

The empty sum is 0_A ; the basepoint is the all- 0_A tuple. Associativity and commutativity of addition prove functoriality: the sum for a composite groups exactly the same finite terms according to the intermediate fibre. Multiplication on two inputs is the tensor-coordinate rule $(a, b) \mapsto (a_i b_j)_{(i,j)}$ at $k_+ \wedge l_+$. Distributivity proves its naturality, and semiring associativity and the unit prove the spherical algebra laws.

The exact unit realization is

$$\eta_{A,k} : SM(k_+) \longrightarrow HA(k_+), \quad 0 \longmapsto (0_A, \dots, 0_A), \quad \tau_j \longmapsto \delta_j^A, \quad (\text{FSP5})$$

where δ_j^A has entry 1_A at j and 0_A at all other coordinates. Under (FSP4), a delta vector stays a delta at $f(j)$ if that index is nonzero, and becomes the zero tuple otherwise. Hence (FSP5) is natural for every pointed map. A product of two delta vectors is the delta at (i, j) , proving multiplicativity and preservation of the spherical unit. This supplies a complete spherical algebra morphism, not merely a map at 1_+ .

Specializing $A = \mathbb{Z}$ gives

$$\eta_{\mathbb{Z},k}(\tau_j) = e_j \in \mathbb{Z}^k. \quad (\text{FSP6})$$

The image of the ungraded presence unit at level one is integer 1. The map and codomain give that image its arithmetic meaning. They do not retroactively add an integer grading to the definition (FSP1).

98.3 What the fold does, and what its domain contains

Let $p_1, p_2 : 2_+ \rightarrow 1_+$ select the first or the second nonzero coordinate and send the other to the basepoint. Let $\nabla : 2_+ \rightarrow 1_+$ send both 1 and 2 to 1. Then

$z \in SM(2_+)$	0	τ_1	τ_2
$SM(p_1)z$	0	τ	0
$SM(p_2)z$	0	0	τ
$SM(\nabla)z$	0	τ	τ

(FSP7)

There is no z whose two projections are both τ . In the source's generalized-addition definition, this says the set of witnesses for $\tau \oplus \tau$ is empty. It does not say the fold maps a simultaneous pair to τ : no such pair lies in its domain.

In contrast $H\mathbb{Z}(2_+) = \mathbb{Z}^2$ contains the additional tuple $(1, 1)$, and

$$H\mathbb{Z}(\nabla)(1, 1) = 1 + 1 = 2. \quad (\text{FSP8})$$

The image of $\eta_{\mathbb{Z},2}$ is precisely $\{(0, 0), (1, 0), (0, 1)\}$ and does not contain $(1, 1)$. Thus (FSP7) and (FSP8) are compatible under an exact morphism. They describe different available inputs, with all coordinates displayed.

98.4 Boolean addition is an explicit extension of the source

If one adds the law $\tau + \tau = \tau$ and makes 0_M an additive identity, the result is the Boolean semiring $B_M \cong \mathbf{B}$, not merely (FSP1). Its spherical algebra is

$$HB_M(k_+) = \{0_M, \tau\}^k, \quad HB_M(\nabla)(\tau, \tau) = \tau. \quad (\text{FSP9})$$

It receives the same delta construction $\eta_{B_M} : \mathbb{S}M \rightarrow HB_M$. There is no unital spherical algebra morphism $HB_M \rightarrow H\mathbb{Z}$. To prove this directly, its level-one map would send 0_M to 0 and τ to 1. Naturality under p_1, p_2 then forces (τ, τ) to map to $(1, 1)$. Naturality under ∇ would give $1 = 2$, contradicting integer arithmetic. The same argument forbids a unital map $HB_M \rightarrow H\mathbb{F}_2$: the fold there gives $1 + 1 = 0$, not 1. The actual comparison is the pair of proved maps

$$HB_M \longleftarrow \mathbb{S}M \longrightarrow H\mathbb{Z}. \quad (\text{FSP10})$$

Their existence retains the common multiplicative source without falsely identifying its two additive extensions.

There is also an exact action, on the entire integer receiver, of the pointed multiplicative monoid underlying B_M :

$$\rho : B_M^{\text{mult}} \longrightarrow \text{End}_+(\mathbb{Z})^{\text{mult}}, \quad \rho(0_M) = 0, \quad \rho(\tau) = \text{id}. \quad (\text{FSP11})$$

Here the multiplication in the endomorphism monoid is composition; each of its displayed operators preserves receiver addition. Its pointed multiplicative table is exactly (FSP1), proving the action. It does not preserve addition of scalars, and makes the two calculations explicit:

$$(\tau + \tau) \cdot 1 = \tau \cdot 1 = 1, \quad \tau \cdot 1 + \tau \cdot 1 = 1 + 1 = 2. \quad (\text{FSP12})$$

The first equality uses Boolean substitution followed by the action. The second is addition in the integer receiver. No distributive equality between these expressions is asserted or needed. If scalar distributivity were imposed, it would force $x = x + x$ for every receiver element; on an abelian-group receiver that condition forces $x = 0$. The existing nontrivial monoid action (FSP11) remains valid.

98.5 The full support carrier and its distinct zero

For any bounded distributive lattice L , retain

$$\begin{aligned} A_L &= G_L(\mathbb{Z}) = \{(0, \lambda) : \lambda \in L\} \cup \{(n, 1_L) : n \in \mathbb{Z}\}, \\ (n, \lambda) + (m, \mu) &= (n + m, \lambda \vee \mu), \quad (n, \lambda)(m, \mu) = (nm, \lambda \wedge \mu), \\ 0_{A_L} &= \tau_L = (0, 0_L), \quad e_L = (0, 1_L), \quad 1_{A_L} = (1, 1_L). \end{aligned} \quad (\text{FSP13})$$

The exact realization (FSP5) in these coefficients has

$$\eta_{A_L, k}(\tau_j)_i = \begin{cases} (1, 1_L) & i = j, \\ \tau_L & i \neq j, \end{cases} \quad \eta_{A_L, k}(0) = (\tau_L, \dots, \tau_L). \quad (\text{FSP14})$$

Thus the presence unit τ is mapped to the ambient multiplicative unit, and the absent point 0_M to the ambient additive zero. The original absorber τ_L is not renamed a multiplicative unit. Mapping τ to τ_L instead would fail unit preservation, because $(1, 1_L) \neq (0, 0_L)$. There is nevertheless an exact corner realization using the old τ_L itself as a unit. Adjoin a strict new bottom $*$ to L , put $A_L^+ = G_{L^+}(\mathbb{Z})$, and write $z_* = (0, *)$. Then

$$C = \tau_L A_L^+ = \{z_*, \tau_L\}, \quad M \longrightarrow C^{\text{mult}}, \quad 0_M \longmapsto z_*, \quad \tau \longmapsto \tau_L. \quad (\text{FSP14a})$$

This is an isomorphism of pointed multiplicative monoids: $\tau_L^2 = \tau_L$, $\tau_L z_* = z_*$, and $z_*^2 = z_*$. The corner also has Boolean addition with zero z_* and unit τ_L . It therefore gives the exact map $\mathbb{S}M \rightarrow HC$ by (FSP5). The inclusion $C \rightarrow A_L^+$ preserves addition, multiplication, and zero, but not the multiplicative unit; its induced $HC \rightarrow HA_L^+$ is correspondingly a multiplicative natural transformation that fails unit preservation as a spherical algebra map. In the other direction, $x \mapsto \tau_L x$ is a unital semiring map $A_L^+ \rightarrow C$: every old element maps to τ_L , and z_* maps to z_* . It

is precisely localization at τ_L , since an invertible idempotent must be the unit and then $\tau_L x = \tau_L$ forces every old x to that unit. The two elements remain distinct under the displayed map to C . Thus the old symbol can realize the presence unit in its named corner or localization, with the complete maps and their different unit conditions specified. The complete functor on a tuple retains cancellations as

$$(HA_L(f)((n_j, \lambda_j)_j))_r = \left(\sum_{j \in f^{-1}(r)} n_j, \bigvee_{j \in f^{-1}(r)} \lambda_j \right), \quad (\text{FSP15})$$

with empty sum 0 and empty join 0_L . In particular the fold of $((1, 1_L), (-1, 1_L))$ is e_L , retaining top support, and the fold of $((1, 1_L), (1, 1_L))$ is $(2, 1_L)$.

Amplitude, support, and parity define exact coefficient homomorphisms

$$\begin{aligned} p_L : A_L &\rightarrow \mathbb{Z}, & (n, \lambda) &\mapsto n, & \chi_L : A_L &\rightarrow L, & (n, \lambda) &\mapsto \lambda, \\ \pi_{2,L} : A_L &\rightarrow G_L(\mathbb{F}_2), & (n, \lambda) &\mapsto (\bar{n}, \lambda), & p_2 \pi_{2,L} &= \pi_2 p_L. \end{aligned} \quad (\text{FSP16})$$

All preserve both operations, zero, and unit by (FSP13). Applying H coordinatewise proves naturality using (FSP15) and gives, in particular, $H p_L \eta_{A_L} = \eta_{\mathbb{Z}}$ and $H \pi_{2,L} \eta_{A_L} = \eta_{G_L(\mathbb{F}_2)}$. The full parity map still distinguishes every zero label; supported even integers map to e_L . If L is nontrivial, there is no unital spherical algebra map $H\mathbb{Z} \rightarrow HA_L$ serving as a reverse arithmetic inclusion. Indeed naturality under coordinate projections and folding forces its level-one map to be additive, and unit preservation forces $1 \mapsto (1, 1_L)$. The image of -1 would then have to satisfy $(1, 1_L) + x = \tau_L$, impossible because the support of that sum is 1_L . The forward amplitude map and the spherical unit map remain the exact comparison.

The action (FSP11) also exists on the entire additive monoid of A_L : the two operators are the constant τ_L map and the identity. Each preserves addition, their composition table is (FSP1), and all labels are retained by the identity. Pointwise addition of two identity operators sends (n, λ) to $(2n, \lambda)$; this equals the identity exactly on the complete zero fibre. Thus the failure of scalar additivity and its fixed locus are explicitly computed in the original coefficients.

98.6 Free arithmetic base change and the origin of parity

Form the ordinary monoid ring $\mathbb{Z}[M]$ with basis $[0_M], [\tau]$ and multiplication $[a][b] = [ab]$. Its multiplicative identity is $[\tau]$. Contract the absorbing basis element by its generated ideal:

$$\mathbb{Z}_0[M] = \mathbb{Z}[M]/([0_M]) \cong \mathbb{Z}, \quad n[\tau] \mapsto n. \quad (\text{FSP17})$$

The ideal $([0_M])$ is exactly $\mathbb{Z}[0_M]$, since multiplying either basis vector by $[0_M]$ gives $[0_M]$. The quotient is therefore free of rank one with basis $[\tau]$, and that basis vector is its unit; this proves both injectivity and surjectivity of the displayed ring map. Equivalently a pointed multiplicative map $M \rightarrow R^{\text{mult}}$ to any unital ring must send 0_M to 0_R and τ to 1_R ; it extends uniquely to the ring homomorphism $\mathbb{Z} \rightarrow R$. This proves the free arithmetic base-change property directly.

For this realization the composite is

$$M \longrightarrow \mathbb{Z}_0[M] \cong \mathbb{Z} \longrightarrow \mathbb{F}_2, \quad 0_M \mapsto 0 \mapsto 0, \quad \tau \mapsto 1 \mapsto 1. \quad (\text{FSP18})$$

Thus the realized absent point has even integer image and the realized unit has odd integer image. These are image properties under named maps. The source (FSP1) has no intrinsic integer parity predicate or additive cancellation rule. Nor is (FSP17) a base change of the extra Boolean addition: any additive unit-preserving map from B_M to a ring would impose $1 + 1 = 1$, forcing the zero ring. Forgetting the extra Boolean addition before (FSP17) is therefore an explicit change of category, not an unmentioned equality of the two extensions.

98.7 The map on Connes–Consani’s actual stalk generators

At a closed arithmetic prime p , write $H_p^+ = \mathbb{Z}[1/p]_{\geq 0}$ and form the pointed monoid

$$N_p = \{0_{N_p}\} \sqcup \{T^a : a \in H_p^+\}, \quad T^a T^b = T^{a+b}, \quad T^0 = \tau. \quad (\text{FSP19})$$

The source stalk is $\mathbb{S}N_p$, agreeing with the spherical presentation in **FF.tex**, Proposition **prop12**. Every pointed multiplicative map $c : N_p \rightarrow A_L$ with $c(T^0) = 1_{A_L}$ determines the exact spherical map

$$\mathbb{S}N_p(k_+) \longrightarrow HA_L(k_+), \quad (T^a, j) \longmapsto (\tau_L, \dots, \tau_L, c(T^a), \tau_L, \dots, \tau_L), \quad 0 \longmapsto (\tau_L, \dots, \tau_L). \quad (\text{FSP20})$$

Naturality follows because each input has only one active coordinate, and products are preserved precisely by $c(T^{a+b}) = c(T^a)c(T^b)$. Conversely a spherical algebra map restricts at 1_+ to a pointed multiplicative map. Every (T^a, j) is the image of T^a under the pointed injection $1_+ \rightarrow k_+$ selecting j , so naturality forces exactly (FSP20); there is no additional choice. This proves the monoid adjunction here explicitly. Amplitude, support, and parity of every such stalk morphism are then obtained by composing c with the exact maps (FSP16), at every coordinate. No assigned source parity is needed, and no old zero label is discarded except by the explicitly named amplitude projection.

99 The ungraded pointed unit and the full supported absolute base

The source used in this section is the pointed multiplicative monoid

$$M = \{0, \tau_{\mathbf{F}_1}\}, \quad 0x = 0, \quad \tau_{\mathbf{F}_1}x = x \quad (x \in M). \quad (\text{FGM1})$$

No parity is assigned to $\tau_{\mathbf{F}_1}$. The parity of an integer is retained as its residue in $\mathbb{Z}/2\mathbb{Z}$. The multiplicative map below is defined independently of a global addition on a space containing both the integers and $\tau_{\mathbf{F}_1}$. We calculate both the count of occurrences and its idempotent collapse explicitly.

The human sources are Alain Connes and Caterina Consani, *Absolute algebra and Segal's Γ -rings: au dessous de $\overline{\text{Spec } \mathbb{Z}}$* , original `F1_corr.tex`, equations `functsum`, Definition `defnsalg`, Propositions `mono2sss`, `monomono`, and Lemma `sssalg` (lines 360–560 read for this derivation); and their *On the Absolute Geometry of $\text{Spec } \mathbb{Z}$ and the Fargues–Fontaine curve*, original `FF.tex`, Section `sect9.1`, Proposition `prop12`, and Proposition `prop:monoid_structure_morphisms` (lines 140–154 and 308–410 read, including the source's exponent-zero Frobenius). The original coefficient object is the complete lattice-split construction in *Split Support Geometry*, source byline *The Clankers*, Version 11, Definition `def:lattice-split` and Theorem `thm:lattice-normal-form`, complete pinned proof source. All coefficient and receiving-map arguments used here are given below.

99.1 The original coefficient carrier and its exact base map

Let L be a bounded distributive lattice with bottom 0_L and top 1_L ; the one-element lattice is allowed. For a nonzero commutative unital ring R retain the entire carrier

$$\begin{aligned} S_R = G_L(R) &= \{z_\lambda = (0, \lambda) : \lambda \in L\} \cup \{\hat{a} = (a, 1_L) : a \in R\}, \\ (a, \lambda) + (b, \mu) &= (a + b, \lambda \vee \mu), \quad (a, \lambda)(b, \mu) = (ab, \lambda \wedge \mu), \\ \tau_{\text{old}} &= z_{0_L}, \quad e = z_{1_L} = \hat{0}, \quad 1_{S_R} = \hat{1}. \end{aligned} \quad (\text{FGM2})$$

This is a unital semiring. Indeed a nonzero amplitude in a sum forces top support, and a product with a support below top has zero amplitude. Thus the carrier is closed; associativity and commutativity follow coordinatewise, and distributivity follows from that of R and L . The displayed zero and unit follow by direct substitution.

Proposition 55 (The unique pointed unital map). *The unique pointed unital multiplicative map to the original coefficient object is*

$$\iota_R : M \longrightarrow (S_R, \cdot), \quad 0 \longmapsto \tau_{\text{old}}, \quad \tau_{\mathbf{F}_1} \longmapsto \hat{1}. \quad (\text{FGM3})$$

It is injective. Its image is the two-element pointed multiplicative submonoid $\{\tau_{\text{old}}, \hat{1}\}$; every z_λ remains present in its codomain.

Proof. A pointed map must have the displayed value at 0, and a unital map must have the displayed value at $\tau_{\mathbf{F}_1}$. Products involving 0 map to the absorbing τ_{old} ; the remaining product maps to $\hat{1}\hat{1} = \hat{1}$. The images are different because their amplitudes are 0_R and 1_R . $\square$

In particular the new source unit is not sent to the old absorber. For any multiplicative map f preserving the old multiplication, the assignment $f(\tau_{\text{old}}) = 1$ forces $f(x) = f(\tau_{\text{old}})f(x) = f(\tau_{\text{old}}x) = 1$ for every x . When f also preserves zero, its target has $0 = 1$ and is the one-element semiring. This proves the exact collapse required by that assignment, rather than denying a relation between the objects.

The integer 1 can be retained as distinct from the new global unit already at the multiplicative level. Define

$$\mathcal{U} = \mathbb{Z} \amalg \{\tau_{\mathbf{F}_1}\}, \quad \tau_{\mathbf{F}_1}u = u, \quad nm = (nm)_{\mathbb{Z}}. \quad (\text{FGM4})$$

Its zero is the integer zero, and associativity follows by deleting factors equal to the new unit and multiplying the remaining integers. Here $1_{\mathbb{Z}}\tau_{\mathbf{F}_1} = 1_{\mathbb{Z}} \neq \tau_{\mathbf{F}_1}$, so $1_{\mathbb{Z}}$ is not its global unit. There is an exact pointed unital multiplicative map

$$j : \mathcal{U} \longrightarrow S_{\mathbb{Z}}, \quad j(0) = \tau_{\text{old}}, \quad j(n) = \widehat{n} \ (n \neq 0), \quad j(\tau_{\mathbf{F}_1}) = \widehat{1}. \quad (\text{FGM5})$$

To check multiplication, a zero factor gives the absorber on both sides; two nonzero integers have nonzero product in $\mathbb{Z}$; and a new-unit factor acts identically. Its sole non-singleton fibre is $\{1_{\mathbb{Z}}, \tau_{\mathbf{F}_1}\}$. This construction specifies no mixed addition and assigns no parity to its adjoined unit.

99.2 Occurrence counts and the two exact additive defects

Write $\mathbb{B} = \{0, \tau_{\mathbf{F}_1}\}$ for the Boolean semiring whose multiplication is FGM1 and whose additional rule is $\tau_{\mathbf{F}_1} + \tau_{\mathbf{F}_1} = \tau_{\mathbf{F}_1}$. This is an explicitly defined receiving quotient, not an assertion that every expression in the user's larger interaction uses its sum. There are exact semiring maps

$$\begin{aligned} c : \mathbb{N} &\longrightarrow \mathbb{B}, & c(0) &= 0, & c(n) &= \tau_{\mathbf{F}_1} \ (n > 0), \\ r : \mathbb{N} &\longrightarrow S_{\mathbb{Z}}, & r(0) &= \tau_{\text{old}}, & r(n) &= \widehat{n} \ (n > 0). \end{aligned} \quad (\text{FGM6})$$

For c , a sum is positive precisely when at least one summand is positive, and a product is positive precisely when both factors are. For r , zero cases use the global absorber, and positive cases use $\widehat{n} + \widehat{m} = \widehat{n+m}$ and $\widehat{n}\widehat{m} = \widehat{nm}$. These checks prove all operations, including both zeros and units. The map r is injective. Moreover $\mathbb{N}$ is the occurrence receiver freely generated by a multiplicative unit in semirings: any unital semiring map must send n to the sum of n copies of its target unit, and these assignments preserve addition and multiplication by expansion of the finite sums.

The full defect of factoring this receiver through c is the pair relation

$$c(m) = c(n) \ (m, n > 0), \quad r(m) = \widehat{m}, \quad r(n) = \widehat{n}, \quad r(m) = r(n) \iff m = n. \quad (\text{FGM7})$$

In particular r does not factor through c , even as a set map. The distinct evaluations of one and two occurrences are retained as

$$\begin{aligned} \iota_{\mathbb{Z}}(\tau_{\mathbf{F}_1} +_{\mathbb{B}} \tau_{\mathbf{F}_1})\widehat{1} &= \widehat{1}, \\ \iota_{\mathbb{Z}}(\tau_{\mathbf{F}_1})\widehat{1} + \iota_{\mathbb{Z}}(\tau_{\mathbf{F}_1})\widehat{1} &= \widehat{2}, \quad r(1+1) = \widehat{2}. \end{aligned} \quad (\text{FGM8})$$

Thus the expanded receiver gives 2 in amplitude. Applying the Boolean collapse first gives 1. FGM8 computes the operation defect of the specified collapse; it does not apply a distributive law in a source where that law has not been specified.

There is a second, different defect when integer cancellation is combined with a shared global zero. The supported arithmetic inclusion

$$k : \mathbb{Z} \longrightarrow S_{\mathbb{Z}}, \quad k(n) = \widehat{n} \quad (\text{FGM9})$$

preserves the two binary operations and the unit, with its arithmetic zero $k(0) = e$. It does not preserve the global zero when $0_L \neq 1_L$. The pointed map $j|_{\mathbb{Z}}$ of FGM5 instead preserves zero and multiplication, but has precisely the nonzero cancellation defects

$$j(n + (-n)) = \tau_{\text{old}}, \quad j(n) + j(-n) = e \quad (n \neq 0). \quad (\text{FGM10})$$

For all other pairs of integers it preserves addition: a zero summand uses the absorber, and two nonzero summands of nonzero sum are covered by FGM2. This exhausts the cases.

Theorem 56 (The two exact congruences). *For $S = S_{\mathbb{Z}}$ the smallest semiring congruences imposing, respectively, the defects in FGM10 and FGM8 are exactly*

$$\begin{aligned} S/(e \sim \tau_{\text{old}}) &\simeq \mathbb{Z}, & q(a, \lambda) &= a, \\ S/(\widehat{2} \sim \widehat{1}) &\simeq L, & \chi(a, \lambda) &= \lambda. \end{aligned} \tag{FGM11}$$

The first quotient identifies every z_{λ} and preserves every integer amplitude. The second preserves every lattice label and identifies all top-supported integer amplitudes with e . Imposing both relations produces the one-element semiring.

Proof. Both q and χ preserve zero, unit and both operations by FGM2. The relation $e \sim \tau_{\text{old}}$ implies, for every λ , $z_{\lambda} = ez_{\lambda} \sim \tau_{\text{old}}z_{\lambda} = \tau_{\text{old}}$. These are all fibres of q with more than one element; nonzero amplitudes already have unique support 1_L . Consequently the generated congruence is exactly equality under q .

Add $\widehat{-1}$ to $\widehat{2} \sim \widehat{1}$. This gives $\widehat{1} \sim e$. Multiplication by $\widehat{n}$ gives $\widehat{n} \sim e$ for every integer n . These are all fibres of χ with more than one element. Since χ itself respects the generating relation, it does not identify two different labels; thus equality under χ is exactly the generated congruence. Finally the two relations together give $1_S \sim e \sim 0_S$, which makes every element equal to zero by multiplication. $\square$

The ungraded pointed base therefore maps into the original full coefficient object without this quotient. Extending that particular map to preserve Boolean addition has the exact receiving quotient L . Assigning its shared arithmetic zero to the original global zero has the different exact receiving quotient $\mathbb{Z}$.

99.3 The full spectrum, all stalks, and the sheaf unit

Let $X = \text{Spec } S$ mean the spectrum of proper additive semiring prime ideals, and $Y = \text{Spec } \mathbb{Z}$. A lattice prime ideal $J \subsetneq L$ is a lower set closed under finite joins, containing 0_L , with $\lambda \wedge \mu \in J$ implying $\lambda \in J$ or $\mu \in J$. The complete point list is

$$P_J = \{z_{\lambda} : \lambda \in J\}, \quad Q_{\mathfrak{p}} = q^{-1}(\mathfrak{p}), \quad \mathfrak{p} = (0) \text{ or } (p), \quad P_J \subset Q_{(0)}. \tag{FGM12}$$

To prove completeness, an ideal containing a nonzero $\widehat{n}$ also contains $\widehat{-n}$, hence e and all $z_{\lambda} = ez_{\lambda}$. An ideal containing e is the full inverse image of an ideal of $\mathbb{Z}$: its top-supported amplitudes are closed under sums, negatives, and multiplication by integers. Primality is exactly primality of that ring ideal. An ideal not containing e consists only of zero labels; addition is join, multiplication by z_{μ} is meet, and the ideal and prime axioms are exactly those of a proper lattice prime J . Conversely these formulas verify all ideal and prime axioms for the displayed list. The case of all zero labels is $Q_{(0)}$, not an omitted additional support point.

Put $A = \{P_J\}$. Direct membership in the basic opens gives

$$\begin{aligned} D_S(z_{\lambda}) &= \{P_J : \lambda \notin J\}, & D_S(e) &= A, \\ D_S(\widehat{n}) &= A \cup \{Q_{\mathfrak{p}} : n \notin \mathfrak{p}\} & (n \neq 0). \end{aligned} \tag{FGM13}$$

The continuous map $r_X : X \rightarrow Y$ sends $P_J, Q_{(0)}$ to the generic point η and $Q_{(p)}$ to p . Indeed for every nonzero integer n , the inverse image of $D_{\mathbb{Z}}(n)$ is $D_S(\widehat{n})$; these opens and the empty set form a basis. For $n = 0$ its inverse image is empty, whereas $D_S(\widehat{0}) = A$: this exceptional value is retained. The map $i : Y \rightarrow X$, $\mathfrak{p} \mapsto Q_{\mathfrak{p}}$, identifies Y with $V(e)$, and $r_X i = \text{id}_Y$.

Let $\mathcal{O}_X$ be the sheaf of locally represented fractions of S . For each lattice prime J put

$$L_J = L[(L \setminus J)^{-1}], \quad [\lambda] = [\mu] \iff \exists \nu \notin J : \lambda \wedge \nu = \mu \wedge \nu. \tag{FGM14}$$

Its operations are induced join and meet. The exact stalks and generizations are

$$\begin{aligned} (\mathcal{O}_X)_{Q(p)} &= G_L(\mathbb{Z}_{(p)}), & (\mathcal{O}_X)_{Q(0)} &= G_L(\mathbb{Q}), & (\mathcal{O}_X)_{P_J} &= L_J, \\ G_L(\mathbb{Z}_{(p)}) &\longrightarrow G_L(\mathbb{Q}), & (a, \lambda) &\longmapsto (a, \lambda), & G_L(\mathbb{Q}) &\longrightarrow L_J, & (a, \lambda) &\longmapsto [\lambda]. \end{aligned} \quad (\text{FGM15})$$

At an arithmetic point the denominators are exactly $\hat{n}$ with $n \notin \mathfrak{p}$. Fractions map to $(a/n, \lambda)$; their equality is the ordinary ring localization equality together with equality of the unchanged support labels. This proves the first two stalk formulas. At P_J the denominator e is allowed. Since $e^2 = e$, inverting it gives $e = 1$ and $(a, \lambda) = e(a, \lambda) = z_\lambda$. Inverting the remaining z_ν , $\nu \notin J$, gives FGM14 and the third formula. An invertible idempotent is 1, so fractions in L_J are represented by their numerators, with exactly the equality relation FGM14. These constructions prove all displayed maps. For $J \subseteq K$ the further map $L_K \rightarrow L_J$ is $[\lambda]_K \mapsto [\lambda]_J$. The bottom and top in L_J differ: their equality would require $\nu \notin J$ with $\nu = 0_L$, contrary to $0_L \in J$. Inverting e therefore identifies it with the unit, not with zero.

Retain the Connes–Consani source sheaf with its original stalks:

$$\begin{aligned} \mathcal{F} &= \Theta^{-1}\mathcal{O}_{\mathbf{F}_1}, & \mathcal{O}_{\mathbf{F}_1} &= \mathbf{F}_1[T], & \Theta(\eta) &= [\{0\}], & \Theta(p) &= [\mathbb{Z}[1/p]], \\ \mathcal{F}_\eta &= \mathbf{F}_1, & \mathcal{F}_p &= \mathbf{F}_1[T^{H_p^+}], & H_p^+ &= \mathbb{Z}[1/p] \cap [0, \infty), & \mathcal{F}_L &= r_X^{-1}\mathcal{F}. \end{aligned} \quad (\text{FGM16})$$

The inverse image here is the sheafification of the presheaf colimit over opens $V \subseteq Y$ with $W \subseteq r_X^{-1}V$. Taking a stalk gives the colimit over neighborhoods of $r_X(x)$, so $(\mathcal{F}_L)_x = \mathcal{F}_{r_X(x)}$. Thus every P_J and $Q_{(0)}$ has stalk $\mathbf{F}_1$, and every $Q_{(p)}$ has the displayed monoid algebra. The source includes the endomorphism $\text{Fr}_0(T^h) = T^0 = 1$, fixing its separate absorbing zero. Every generization commutes with it; on $\mathbf{F}_1$ it is the identity. Therefore the exact finite-to-generic map is

$$\epsilon_p : \mathcal{F}_p \longrightarrow \mathbf{F}_1, \quad \epsilon_p(0) = 0, \quad \epsilon_p(T^h) = 1 \quad (h \geq 0). \quad (\text{FGM17})$$

Indeed $g(T^h) = \text{Fr}_0 g(T^h) = g(T^0) = 1$. All maps between the generic $\mathbf{F}_1$ stalks are the identity.

Theorem 57 (The full-support spherical map). *There is exactly one unital spherical-algebra sheaf map*

$$\beta : \mathcal{F}_L \longrightarrow H\mathcal{O}_X. \quad (\text{FGM18})$$

Its degree-one values are

stalk	source absorbing zero	source unit or T^h
$Q_{(p)}$	τ_{old}	$\hat{1} \quad (h \in H_p^+)$
$Q_{(0)}$	τ_{old}	$\hat{1}$
P_J	$[0_L]$	$[1_L]$.

(FGM19)

Every support label remains in the target stalk or its exact localization class from FGM14.

Proof. For a semiring B , $(HB)(k_+) = B^k$; a pointed map acts by sums over non-basepoint fibres, with empty sum 0_B , and multiplication takes $(b_i), (c_j)$ to $(b_i c_j)_{i,j}$. For a pointed monoid N , the spherical monoid algebra in degree k_+ is $N \wedge k_+$. A pointed unital monoid map $f : N \rightarrow B$ induces

$$(n, j) \longmapsto (0_B, \dots, 0_B, f(n), 0_B, \dots, 0_B), \quad (\text{FGM20})$$

with the indicated value in coordinate j and the basepoint sent to the all-zero tuple. A pointed map moves the single occupied coordinate or sends it to the basepoint; the fibre-sum rule does the same. Products agree by multiplicativity of f . Conversely the maps $1_+ \rightarrow k_+$ selecting a coordinate force FGM20 from its degree-one component. This proves the needed source adjunction explicitly.

The original Fr_0 factors through the constant spherical algebra $\mathbf{F}_1$ and commutes with all source Frobenius maps. Pulling back this factorization gives the sheaf augmentation $\mathcal{F}_L \rightarrow \underline{\mathbf{F}_1}$.

Compose with the spherical unit map of $H\mathcal{O}_X$, given by FGM20 and the stalk units of FGM15. This constructs β and proves FGM19 globally. For uniqueness its component at $Q_{(0)}$ is the unique unit map. The arithmetic generization in FGM15 is injective; compatibility with FGM17 forces every T^h at $Q_{(p)}$ to have image $\hat{1}$. At P_J the source is $\mathbf{F}_1$, whose unital map is unique. Formula FGM20 determines all degrees; equal maps on every stalk are equal sheaf maps. $\square$

There is no asserted Boolean sum in the spherical source unit. In degree 2_+ the sphere is $\{0, 1, 2\}$. Its two coordinate projections give $(0, 0)$, $(1, 0)$, and $(0, 1)$ respectively; none gives $(1, 1)$. Thus there is no degree-two source element encoding two occupied unit coordinates. In $H\mathbb{B}$ such an element does exist and folds to $\tau_{\mathbf{F}_1}$; in $HS_{\mathbb{Z}}$ it folds to $\hat{2}$. FGM20 and FGM8 give the exact relation between these constructions.

99.4 Every zero branch and the exact parity reduction

For completeness retain all algebraic finite-prime points with target $HS_{\mathbb{C}}$. A character is a multiplicative map $\xi : H_p^+ \rightarrow S_{\mathbb{C}}$ with $\xi(0) = \hat{1}$; the extra absorbing zero of the monoid algebra maps separately to τ_{old} . The entire set is

$$\begin{aligned} \mathcal{U}_p = \{ (u_n)_{n \geq 0} \in (\mathbb{C}^\times)^{\mathbb{N}} : u_{n+1}^p = u_n \}, \quad \text{Hom}_{\mathbf{F}_1}(\mathcal{F}_p, HS_{\mathbb{C}}) = \mathcal{U}_p \amalg \{ Z_\lambda : \lambda \in L \}, \\ \xi_u(a/p^n) = \widehat{u_n^a} \quad (a \geq 0), \quad Z_\lambda(0) = \hat{1}, \quad Z_\lambda(h) = z_\lambda \quad (h > 0). \end{aligned} \quad (\text{FGM21})$$

To prove completeness set $b_n = \xi(p^{-n})$, so $b_{n+1}^p = b_n$. Nonzero complex amplitudes stay nonzero under positive powers. Thus either every b_n has nonzero amplitude, giving exactly the displayed sequence u , or every b_n is a zero label. In the second case $z_\lambda^p = z_\lambda$, hence the labels are all one fixed λ . Every positive exponent is a/p^n with $a > 0$, so these cases determine the full character. Conversely the formulas respect a change of denominator, exponent addition, the unit and the separate zero, and therefore define all points. Their multiplication retains every label:

$$\xi_u \xi_v = \xi_{uv}, \quad \xi_u Z_\lambda = Z_\lambda, \quad Z_\lambda Z_\mu = Z_{\lambda \wedge \mu}. \quad (\text{FGM22})$$

The unit point is the constant unit sequence. The absorber point is Z_{0_L} , with value $\hat{1}$ at exponent zero, not an everywhere-zero function. The amplitude map identifies all Z_λ with the single zero branch and preserves each u . The generic point map induced by FGM17 selects the constant unit sequence, not a zero branch. For every integer $m \geq 1$ source Frobenius gives $u \mapsto (u_n^m)$ and fixes each Z_λ ; for $m = 0$ it sends every point to the unit point, with the extra source absorbing zero still sent to τ_{old} . These assertions follow by evaluating at mh , including $h = 0$.

Proposition 58 (Parity with full support). *There is a surjective unital semiring map*

$$\pi_2 : G_L(\mathbb{Z}) \longrightarrow G_L(\mathbb{F}_2), \quad (n, \lambda) \longmapsto (\bar{n}, \lambda). \quad (\text{FGM23})$$

Its exact fibres are

$$\begin{aligned} \pi_2^{-1}(\hat{1}) &= \{ \hat{n} : n \text{ odd} \}, \\ \pi_2^{-1}(e) &= \{ \hat{n} : n \text{ even} \}, \\ \pi_2^{-1}(z_\lambda) &= \{ z_\lambda \} \quad (\lambda \neq 1_L). \end{aligned} \quad (\text{FGM24})$$

Consequently the zero-amplitude fibre contains every zero label and every even integer; the nonzero-amplitude fibre consists exactly of the odd integers. All reduced nonzero amplitudes are odd. There is no assigned parity for $\tau_{\mathbf{F}_1}$ in these statements.

Proof. Reduction respects ring operations, while both lattice coordinates in FGM2 are unchanged. It therefore respects the full operations, zeros and units; each target element has an indicated preimage. All nonzero source amplitudes have top support, and the only nonzero residue in $\mathbb{F}_2$ is 1. These observations prove each fibre formula. On the supported integer copy k , the amplitude composite is exactly the intrinsic residue $n \mapsto \bar{n}$; its being a numeric invariant does not assign a parity to the different source unit in FGM1. $\square$

At the arithmetic stalk over 2 the reduction is

$$G_L(\mathbb{Z}_{(2)}) \longrightarrow G_L(\mathbb{F}_2), \quad (a/b, \lambda) \longmapsto (\bar{a}\bar{b}^{-1}, \lambda), \quad b \text{ odd.} \quad (\text{FGM25})$$

Cross multiplication proves it is well-defined, since every allowed denominator reduces to 1; the operation checks are those of FGM23. For $p \neq 2$ there is no unital semiring map from $G_L(\mathbb{Z}_{(p)})$ to $G_L(\mathbb{F}_2)$ extending FGM23, and none from $G_L(\mathbb{Q})$: $\widehat{2}$ is invertible in either source, while its required image e is not invertible. Indeed every product with e has zero amplitude, so it cannot be $\widehat{1}$ in the target. This proves the exact local domain of parity reduction.

At each pure-support stalk the precise compatibility instead is the commuting square

$$\begin{array}{ccc} G_L(\mathbb{Z}) & \xrightarrow{\pi_2} & G_L(\mathbb{F}_2) \\ (n, \lambda) \mapsto [\lambda] \downarrow & & \downarrow (a, \lambda) \mapsto [\lambda] \\ L_J & \xlongequal{\quad} & L_J. \end{array} \quad (\text{FGM26})$$

Both routes preserve the entire resulting label class and discard amplitude by the proved map FGM15. They identify all integer images with $[1_L]$; the label equality remains exactly FGM14. Thus parity is not recoverable from that support quotient. In particular FGM8 reduces over 2 to the distinct pair $(\widehat{1}, e)$, whereas its two entries both become $[1_L]$ in a pure-support stalk. The image of the new base unit is always the appropriate target unit in FGM19; its ungraded source designation remains unchanged.

100 The full support lattice in integer inclusion–exclusion coordinates

Sources, provenance, and what is being mapped. The starting carrier and its original operations are those of The Clankers, *Split Support Geometry*, version 11, Definition `def:lattice-split`. The same source already proves finite Birkhoff coordinates in Theorem `thm:birkhoff-support-coordinates` (lines 1358 onward) and identifies Boolean characters with prime lattice ideals in Theorem `thm:booleanization-prime-branches` (lines 1804 onward). Its bibliography attributes the classical lattice background to Garrett Birkhoff, *Lattice Theory*, third edition, American Mathematical Society Colloquium Publications 25 (1967). The Birkhoff book is background attribution from that bibliography, not a claim that it was read for this derivation.

The exact local foundational edition read here is the 9,249-line file at this workspace-relative path:

```
work/claude_orientation_20260922/first_papers/
split_support_geometry_arithmetic_curve_v11.tex
```

Its SHA-256 is

```
5fa6a55aa9d1a2e57599e206087d3d2466b25dd99fa1a90d5f86367aafab1b4d.
```

This identifies the local bytes actually read; a longer local edition with the same title is not being claimed byte-identical. The public pinned link above cites the original definition and named proof, while the reading coverage below refers to this specified local edition. Reading covers its exact carrier and proof at lines 1211–1296, finite-character discussion at 1355–1419, branch correspondence at 1779–1866, and Birkhoff bibliography at 9183–9199. The classical representation mechanism is not claimed as new. The following proof constructs the representation for every bounded distributive lattice, its full universal integer valuation image, and its receiving map on every original amplitude and zero label. All invoked prime separation and valuation assertions are proved below.

100.1 Every prime coordinate, with no branch selected

Let L be a bounded distributive lattice with bounds $0_L, 1_L$. An ideal is a nonempty downward closed subset closed under finite joins. A proper ideal P is prime if $a \wedge b \in P$ implies $a \in P$ or $b \in P$. Put $X = \text{SpecLat}(L)$, the set of all such ideals, and define

$$\chi_\lambda : X \longrightarrow \{0, 1\}, \quad \chi_\lambda(P) = \begin{cases} 0, & \lambda \in P, \\ 1, & \lambda \notin P. \end{cases} \quad (\text{FSI1})$$

All character coordinates are retained simultaneously.

For completeness, if $a \not\leq b$, there is $P \in X$ containing b and not a . To prove this using Zorn's lemma, the principal ideal $\downarrow b$ and filter $\uparrow a$ are disjoint. Consider ideals containing $\downarrow b$ and disjoint from $\uparrow a$. The union of a chain is an ideal with both properties: any two elements belong together to one ideal of the chain. A maximal such ideal P exists and is proper because 1_L belongs to $\uparrow a$.

The ideal generated by P and x is exactly

$$P[x] = \{z : z \leq p \vee x \text{ for some } p \in P\}.$$

This is downward closed, contains P and x , and is closed under joins since $z \leq p \vee x, w \leq q \vee x$ imply $z \vee w \leq (p \vee q) \vee x$. Every ideal containing P and x contains this set. If $x, y \notin P$, maximality gives $a \leq p \vee x$ and $a \leq q \vee y$ for some $p, q \in P$. Were $x \wedge y \in P$, distributivity would give

$$\begin{aligned} a &\leq (p \vee x) \wedge (q \vee y) \\ &= (p \wedge q) \vee (p \wedge y) \vee (x \wedge q) \vee (x \wedge y) \\ &\leq p \vee q \vee (x \wedge y) \in P, \end{aligned} \quad (\text{FSI2})$$

contradicting $a \notin P$. Hence P is prime. This proves separation for arbitrary, including infinite, bounded distributive lattices with the stated use of Zorn's lemma.

Ideal closure gives $a \vee b \in P$ exactly when both $a, b \in P$. Primeness and downward closure give $a \wedge b \notin P$ exactly when neither belongs to P . Consequently, in the ordinary integer-valued function ring $\mathbb{Z}^X$,

$$\begin{aligned} \chi_{0_L} &= 0, & \chi_{1_L} &= 1, & \chi_{\lambda \wedge \mu} &= \chi_\lambda \chi_\mu, \\ \chi_{\lambda \vee \mu} &= \chi_\lambda + \chi_\mu - \chi_\lambda \chi_\mu. \end{aligned} \tag{FSI3}$$

The sum and product on the right are ordinary pointwise integer operations. The first sum does not stand for a second integer addition. Its subtracted intersection term is part of the exact map.

The map $\chi : L \rightarrow \{0, 1\}^X$ reflects order: if $a \not\leq b$, separation gives a coordinate with $\chi_a = 1, \chi_b = 0$. If $a \leq b$, downward closure gives $\chi_a \leq \chi_b$ at every coordinate. It is therefore injective. For the singleton lattice, $0_L = 1_L$, no proper prime ideals exist. Then $X = \emptyset$ and $\{0, 1\}^\emptyset$ is a singleton, so the same map is an isomorphism of singleton lattices. Here its zero and one functions are the same empty function. No nonexistent character of the singleton lattice is chosen.

100.2 Every finite overlap and the universal integer valuation

For $\lambda_1, \dots, \lambda_n \in L$, all $n \geq 1$, (FSI3) gives

$$\begin{aligned} \chi_{\bigvee_{i=1}^n \lambda_i} &= 1 - \prod_{i=1}^n (1 - \chi_{\lambda_i}) \\ &= \sum_{\emptyset \neq J \subseteq \{1, \dots, n\}} (-1)^{|J|+1} \prod_{j \in J} \chi_{\lambda_j} \\ &= \sum_{\emptyset \neq J \subseteq \{1, \dots, n\}} (-1)^{|J|+1} \chi_{\bigwedge_{j \in J} \lambda_j}. \end{aligned} \tag{FSI4}$$

The first identity follows by induction from (FSI3); multiplying out the finite product proves the second, retaining every subset and its sign. Meet multiplicativity proves the third. For $n = 0$, the empty join is 0_L and the empty product is 1, giving $0 = 1 - 1$ and the empty sum. No infinite join or convergence operation is asserted for a lattice not supplied with such joins.

The global additive receiver can be described exactly. Let

$$V(L) = \mathbb{Z}^{(L)} / \langle [0_L], [\lambda] + [\mu] - [\lambda \vee \mu] - [\lambda \wedge \mu] : \lambda, \mu \in L \rangle, \tag{FSI5}$$

where $\mathbb{Z}^{(L)}$ is the free abelian group on the displayed symbols. For an abelian group A , every map $v : L \rightarrow A$ with $v(0_L) = 0$ and $v(\lambda) + v(\mu) = v(\lambda \vee \mu) + v(\lambda \wedge \mu)$ extends uniquely to a homomorphism $V(L) \rightarrow A$. Indeed the extension from the free group exists uniquely and kills exactly the displayed relations. This proves the universal valuation property directly.

There is a well-defined commutative ring structure

$$[\lambda][\mu] = [\lambda \wedge \mu], \quad 1 = [1_L]. \tag{FSI6}$$

To check its compatibility with the relations, multiply the relation for λ, μ by $[\nu]$. Distributivity of the lattice gives the relation

$$[\lambda \wedge \nu] + [\mu \wedge \nu] - [(\lambda \wedge \nu) \vee (\mu \wedge \nu)] - [(\lambda \wedge \nu) \wedge (\mu \wedge \nu)] = 0.$$

The $[0_L]$ relation also stays zero. Bilinearity proves the well-defined product. The lattice meet laws prove associativity, commutativity and its unit, and every $[\lambda]$ is idempotent. Formula (FSI3) induces a ring homomorphism

$$J : V(L) \longrightarrow \mathbb{Z}^X, \quad [\lambda] \longmapsto \chi_\lambda. \tag{FSI7}$$

This homomorphism is injective, as the following finite calculation proves for the whole, possibly infinite, lattice.

Take a finite family $\lambda_1, \dots, \lambda_n$ occurring in a given expression $v = \sum_i c_i [\lambda_i]$, and for $S \subseteq \{1, \dots, n\}$ define

$$\begin{aligned} A_S &= \prod_{i \in S} [\lambda_i] \prod_{j \notin S} (1 - [\lambda_j]), \\ C_S &= \{P : \chi_{\lambda_i}(P) = 1 \ (i \in S), \ \chi_{\lambda_j}(P) = 0 \ (j \notin S)\}, \\ a_S &= \bigwedge_{i \in S} \lambda_i, \quad b_S = \bigvee_{j \notin S} \lambda_j, \end{aligned} \tag{FSI8}$$

with empty meet 1_L and empty join 0_L . The full signed expansion is

$$A_S = \sum_{J \subseteq S^c} (-1)^{|J|} \left[\bigwedge_{i \in S \cup J} \lambda_i \right].$$

The set C_S is nonempty exactly when $a_S \not\leq b_S$. In one direction its point witnesses $\chi_{a_S} = 1, \chi_{b_S} = 0$. In the other, prime separation supplies a prime containing b_S and excluding a_S , which has exactly the stipulated pattern.

The valuation relation and (FSI6) give inside $V(L)$ the same finite identity $[\bigvee_j \lambda_j] = 1 - \prod_j (1 - [\lambda_j])$. Therefore $A_S = [a_S](1 - [b_S])$. If C_S is empty, $a_S \leq b_S$, so $[a_S][b_S] = [a_S \wedge b_S] = [a_S]$, and $A_S = 0$. Finally, expanding every factor $[\lambda_i] + (1 - [\lambda_i]) = 1$ gives

$$1 = \sum_S A_S, \quad v = \sum_S \left(\sum_{i \in S} c_i \right) A_S, \quad J(A_S) = \mathbf{1}_{C_S}. \tag{FSI9}$$

For the middle equality multiply the first by each $[\lambda_i]$; idempotence makes the result A_S when $i \in S$, and zero otherwise. If $J(v) = 0$, evaluating on every nonempty C_S makes its integer coefficient vanish; empty cells already have $A_S = 0$. Thus $v = 0$, proving injectivity without a finiteness assumption on L .

Here is the exact image. Let $\mathcal{A}_L$ be the Boolean algebra of subsets of X generated by $U_\lambda = \{P : \lambda \notin P\}$. Then

$$J(V(L)) = \left\{ \sum_{j=1}^m n_j \mathbf{1}_{E_j} : m < \infty, \ n_j \in \mathbb{Z}, \ E_j \in \mathcal{A}_L \right\}. \tag{FSI10}$$

Every $E \in \mathcal{A}_L$ is a finite union of cells from a finite family of the U_λ 's; their indicators are (FSI8), so its indicator lies in the image. Conversely, every χ_λ has this form and every element of $V(L)$ is a finite integer combination. This proves both inclusions. Equivalently these are the integer-valued finite step functions determined by finitely many of the support tests. No arbitrary function on an infinite spectrum has been claimed to lie in this image. For the singleton lattice both sides are the zero ring of functions on the empty set, so this case is retained as well.

100.3 The exact map of the entire original split carrier

For a commutative ring R , retain every original amplitude and label:

$$\begin{aligned} G_L(R) &= \{(0, \lambda) : \lambda \in L\} \cup (R \times \{1_L\}), \\ (a, \lambda) + (b, \mu) &= (a + b, \lambda \vee \mu), \quad (a, \lambda)(b, \mu) = (ab, \lambda \wedge \mu), \\ \tau_L &= (0, 0_L), \quad e_L = (0, 1_L), \quad \mathbf{1}_{G_L} = (1_R, 1_L). \end{aligned} \tag{FSI11}$$

Define the target to be the exact admissible image

$$\mathcal{C}_L(R) = \{(a, u) \in R \times \chi(L) : a \neq 0 \implies u = \chi_{1_L}\}, \quad \Phi(a, \lambda) = (a, \chi_\lambda). \tag{FSI12}$$

Injectivity of χ proves injectivity of Φ , and the definition proves that it is onto $\mathcal{C}_L(R)$. The complete operations, expressed using ordinary integer functions in the second coordinate, are

$$(a, u) \oplus (b, v) = (a + b, u + v - uv), \quad (a, u) \otimes (b, v) = (ab, uv). \quad (\text{FSI13})$$

Formula (FSI3) proves both identities $\Phi(x + y) = \Phi(x) \oplus \Phi(y)$ and $\Phi(xy) = \Phi(x) \otimes \Phi(y)$.

Closure also follows directly with all support restrictions retained. If $a + b \neq 0$, at least one of a, b is nonzero, so at least one of u, v is the all-one function and $u + v - uv = 1$. If $ab \neq 0$, both amplitudes are nonzero, so $u = v = 1$ and $uv = 1$. All other cases have zero amplitude and the support lies in $\chi(L)$ by (FSI3). No integral-domain assumption on R is used: zero divisors, nilpotents, and every cancellation $a + (-a) = 0$ are retained.

The additive zero, supported ring zero and unit map respectively to

$$\Phi(\tau_L) = (0, 0), \quad \Phi(e_L) = (0, 1), \quad \Phi(\mathbf{1}_{G_L}) = (1_R, 1). \quad (\text{FSI14})$$

For a trivial lattice the two support functions coincide, as already specified. For a zero ring all amplitudes are zero and the full support lattice still survives. These edge cases do not require a different formula.

This is a semiring isomorphism with the explicitly transported operations $\oplus, \otimes$, not a unital ring homomorphism for ordinary coordinatewise addition in $R \times \mathbb{Z}^X$. The exact discrepancy from that other addition is

$$\Phi(x) +_{\text{coord}} \Phi(y) -_{\text{coord}} \Phi(x + y) = (0, \chi_\lambda \chi_\mu) \quad (x = (a, \lambda), y = (b, \mu)). \quad (\text{FSI15})$$

For example, the support part of distributivity is verified without omitting any overlap:

$$u(v + w - vw) = uv + uw - uvw = uv + uw - (uv)(uw),$$

because $u^2 = u$. Its amplitude part is $a(b + c) = ab + ac$ in R . Associativity of the support addition retains $u + v + w - uv - uw - vw + uvw$ under either association. For any $n \geq 1$, the complete finite sum is

$$\Phi \left(\sum_{i=1}^n (a_i, \lambda_i) \right) = \left(\sum_{i=1}^n a_i, \sum_{\emptyset \neq J \subseteq \{1, \dots, n\}} (-1)^{|J|+1} \chi_{\bigwedge_{j \in J} \lambda_j} \right). \quad (\text{FSI16})$$

Every amplitude summand and every signed support intersection remain. In particular equal top support does not replace $a + a$ by a ; the amplitude still adds by the original ring law.

100.4 Truth actions on intrinsic integer data

To make the requested truth-to-count comparison exact, let

$$I = \{(n, \bar{n}) : n \in \mathbb{Z}, \bar{n} = n \bmod 2\}.$$

It has one integer addition and one multiplication:

$$(n, \bar{n}) + (m, \bar{m}) = (n + m, \bar{n} + \bar{m}), \quad (n, \bar{n})(m, \bar{m}) = (nm, \bar{n} \bar{m}). \quad (\text{FSI17})$$

Parity is part of each element. It is not a label selecting a different addition. Thus $1_{\text{odd}} + 1_{\text{odd}} = 2_{\text{even}}$.

For all prime coordinates together use the additive group I^X . Each support function $u = \chi_\lambda$ defines an endomorphism

$$(M_\lambda f)(P) = (u(P)n_P, u(P)\bar{n}_P), \quad f(P) = (n_P, \bar{n}_P). \quad (\text{FSI18})$$

The second product is in $\mathbb{Z}/2\mathbb{Z}$; it is exactly the parity of the first integer, so the formula stays inside I^X . These maps are additive endomorphisms. Their exact support-action laws in the ordinary endomorphism ring are

$$M_{\lambda \wedge \mu} = M_\lambda M_\mu, \quad M_{\lambda \vee \mu} = M_\lambda + M_\mu - M_\lambda M_\mu, \quad M_{0_L} = 0, \quad M_{1_L} = \text{id}. \quad (\text{FSI19})$$

They follow coordinatewise from (FSI3). Applying them to the constant function 1_{odd} proves that the whole map $\lambda \mapsto M_\lambda$ is injective whenever $X \neq \emptyset$; when $X = \emptyset$, both its source and image are singleton objects. Thus every support label is still distinguished. The map $V(L) \rightarrow \text{End}(I^X, +)$ obtained from (FSI7) by multiplication operators is also injective: applying a coefficient function to the constant 1_{odd} recovers that entire function and its parity coordinate.

For the two-state truth object write $\mathbf{f}, \mathbf{t}$, with OR and AND, to keep it distinct from reused programme symbols. At one coordinate its multiplicative action is

$$\rho(\mathbf{f}) = 0, \quad \rho(\mathbf{t}) = \text{id}_I, \quad \rho(b \vee c) = \rho(b) + \rho(c) - \rho(b)\rho(c). \quad (\text{FSI20})$$

If the user denotes true by τ , this last τ is the truth-action input $\mathbf{t}$. It is not assigned an intrinsic parity and is not identified with 1_{odd} . It is also not silently identified with the earlier $\tau_L = (0, 0_L)$ in (FSI11), which has the different explicitly recorded role of absent support. For $b = c = \mathbf{t}$, the operator equation is

$$\text{id}_I = \text{id}_I + \text{id}_I - \text{id}_I.$$

Evaluating at 1_{odd} gives the exact integer equation $1_{\text{odd}} = 2_{\text{even}} - 1_{\text{odd}}$. This preserves both truth idempotence and ordinary integer addition through their proved map, including its overlap term.

The multiplicative truth action is not additive for uncorrected OR:

$$\rho(\mathbf{t}) + \rho(\mathbf{t}) - \rho(\mathbf{t} \vee \mathbf{t}) = \text{id}_I. \quad (\text{FSI21})$$

Dropping the last term in (FSI20) would therefore be a different, false compatibility claim. Occurrence counts remain visible: k copies of id_I act on 1_{odd} as $(k, k \bmod 2)$, not as 1_{odd} . For every finite support family their corrected union action is

$$M_{\bigvee_{i=1}^n \lambda_i} = \sum_{\emptyset \neq J \subseteq \{1, \dots, n\}} (-1)^{|J|+1} \prod_{j \in J} M_{\lambda_j}. \quad (\text{FSI22})$$

This follows by applying (FSI4) to multiplication operators. All multiplicities and all signed intersections have explicit receivers; none are replaced by a choice of parity or by one selected Boolean branch.

The construction maps the entire original support lattice and the entire admissible split carrier. It establishes the precise valuation and operator formulas needed to compare idempotent truth with integer counts. It makes no cohomology, analytic trace, purity or positivity identification without the additional objects and maps those statements would require.

101 The global pointed-monoid unit in the original arithmetic trace

Objects and source scope. Let $M = \{0, \tau\}$ have absorbing element 0 and multiplicative unit τ , so $\tau\tau = \tau$ and $0\tau = 0$. The element τ is ungraded. This is the pointed multiplicative monoid used for the absolute base by Alain Connes and Caterina Consani, *On the Absolute Geometry of Spec $\mathbb{Z}$* , original author file `FF.tex`, introduction footnote at line 145; their monoid-to-spherical-algebra comparison appears at lines 375–405. Its unit is here named τ to match the requested global unit. No addition on M follows from its pointed multiplication. Whenever the extra Boolean law $\tau + \tau = \tau$ is considered below, it is stated as that additional operation.

The actual arithmetic receivers are SLW1–29 in the supported local Weil return chapter and SGW1–21 in the global assembly chapter. Their human global trace input is Alain Connes, Caterina Consani and Matilde Marcolli, *The Weil proof and the geometry of the adèles class space*, original author `Yuri70v2.tex`, equation `TraceformulaH1` and `intalphae`, lines 2014–2035. The analysis here retains that theorem’s complete test, its local principal values, all negative as well as positive periods, and its real and endpoint terms. It does not assign an additive trace to a pointed monoid or an operator trace to an identity on an infinite-dimensional space. The real formula below is the full SGW14 comparison with Alain Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Appendix II(34)–(37); the finite reference principal value is the SLW18 comparison with Appendix II(9), Lemma 2, and (22)–(29). Those existing source comparisons are the inputs here; this derivation does not claim a new original-source reading of that appendix or replace its principal-value constants.

101.1 Scalar extension and expression multiplicity are exact different maps

For a finite pointed set $k_+ = \{*, 1, \dots, k\}$, the spherical algebra $S(M)$ has pointed carrier $M \wedge k_+ = \{*, (\tau, 1), \dots, (\tau, k)\}$. Since τ is the only nonzero element, this is the sphere unit object on finite pointed sets. The explicit scalar map to the integer spherical algebra is

$$\begin{aligned} \eta_k : S(M)(k_+) &\longrightarrow (H\mathbb{Z})(k_+) = \mathbb{Z}^k, \\ * &\longmapsto (0, \dots, 0), \\ (\tau, j) &\longmapsto (0, \dots, \frac{1}{j}, \dots, 0). \end{aligned} \tag{FTR1}$$

For a pointed map $\alpha : k_+ \rightarrow l_+$, the target sends a vector to the sums of its coordinates in each non-basepoint fibre of α . A vector in the image of (FTR1) has at most one nonzero coordinate, so those sums either move its 1 to its designated fibre or kill it when that coordinate maps to the basepoint. This proves naturality. Pairwise multiplication of two occupied coordinates gives $1 \cdot 1 = 1$ at the paired coordinate, matching $\tau\tau = \tau$. Basepoints map to zero. Thus (FTR1) is the pointed, unital multiplicative scalar extension, with no Boolean addition asserted.

In degree one the underlying map is

$$M \longrightarrow \mathbb{Z} \longrightarrow \mathbb{C} \longrightarrow \text{End}_{\mathbb{C}}(V), \quad 0 \longmapsto 0 \longmapsto 0 \longmapsto 0_V, \quad \tau \longmapsto 1 \longmapsto 1 \longmapsto I_V. \tag{FTR2}$$

The last map sends a scalar a to aI_V , and its multiplication is composition. Hence the global scalar unit acts as identity on every complex vector space V , with inverse itself. No decomposition or grading of V is required for this action.

The exact map that counts repeated occurrences is instead

$$\mathbb{N} \longrightarrow \text{End}_{\mathbb{C}}(V), \quad m \longmapsto mI_V, \quad (m+n)I_V = mI_V + nI_V, \quad (mn)I_V = (mI_V)(nI_V). \tag{FTR3}$$

The spherical description retains the same counts: in $(H\mathbb{Z})(m_+)$ the vector $(1, \dots, 1)$ is sent by the fold $m_+ \rightarrow 1_+$ to the integer m . For $m \geq 2$ this vector is not in the image of η_m , whose

vectors have at most one occupied coordinate. Thus the fold that records multiplicity does not assert that an element representing $\tau + \tau$ existed in the original one-unit carrier with a prescribed idempotent addition.

Let $\mathbb{B} = \{0, \tau\}$ have Boolean addition and the same multiplication as M . The presence projection is the semiring map

$$\pi : \mathbb{N} \longrightarrow \mathbb{B}, \quad \pi(0) = 0, \quad \pi(m) = \tau \ (m \geq 1). \quad (\text{FTR4})$$

For addition, $m + n$ is positive exactly when at least one summand is positive; for multiplication, mn is positive exactly when both factors are positive. These two facts prove its laws. If $V \neq 0$, the count action (FTR3) does not factor as $\mathbb{N} \xrightarrow{\pi} \mathbb{B} \rightarrow \text{End}(V)$: such a factorization would give $I_V = 2I_V$ from $\pi(1) = \pi(2)$, which is impossible after subtracting I_V and applying it to a nonzero vector. The exact additive defect of the multiplicative unit action is

$$\rho(\tau +_{\mathbb{B}} \tau) - \rho(\tau) - \rho(\tau) = I_V - I_V - I_V = -I_V. \quad (\text{FTR5})$$

This proves the precise two readings of the written expression $(\tau + \tau)1$. On the free expression, evaluation in the count receiver gives $(1 + 1)1 = 2$, and evaluation of its expanded expression $\tau 1 + \tau 1$ also gives 2. If its inner sum is first identified with the Boolean value τ , the multiplicative evaluation gives $\tau 1 = 1$. The expression-counting evaluation therefore does not descend to that collapsed value. This states the exact equality that holds and the exact factorization that fails; it does not impose distributivity on a proposed nonadditive ambient object.

More generally, every additive map $a : \mathbb{B} \rightarrow A$ to an abelian group satisfies $a(\tau) = 0$, since $a(\tau) = a(\tau + \tau) = 2a(\tau)$ and cancellation holds in A . In particular there is no zero- and unit-preserving additive map from $\mathbb{B}$ to a nonzero ring. Conversely no unital semiring map from a nonzero ring to $\mathbb{B}$ exists: the equality $1 + (-1) = 0$ would imply $\tau + a(-1) = 0$, although a Boolean sum with τ is τ . The multiplicative map in (FTR2) remains well defined; the proved obstruction is specifically additive.

101.2 Exact placement in every original support branch

Retain the original bounded distributive lattice L and

$$\begin{aligned} G_L(\mathbb{C}) &= (\{0\} \times L) \cup (\mathbb{C} \times \{1_L\}), \\ (a, \lambda) + (b, \mu) &= (a + b, \lambda \vee \mu), \quad (a, \lambda)(b, \mu) = (ab, \lambda \wedge \mu), \\ \tau_G &= (0, 0_L), \quad e = (0, 1_L), \quad \widehat{1} = (1, 1_L). \end{aligned} \quad (\text{FTR6})$$

The new global unit $\tau \in M$ is distinct from the historical zero τ_G in this displayed object. There is the exact pointed multiplicative map

$$j : M \longrightarrow G_L(\mathbb{C}), \quad j(0) = \tau_G, \quad j(\tau) = \widehat{1}. \quad (\text{FTR7})$$

Its unit action on every $G_L(V)$ is

$$j(\tau)(v, \lambda) = (v, 1_L \wedge \lambda) = (v, \lambda), \quad j(0)(v, \lambda) = (0, 0_L) = \tau_V. \quad (\text{FTR8})$$

This proves the action on every fixed label, including each zero-amplitude point below the top. A zero scalar in the pointed ambient semiring sends all labels to bottom, as the second formula shows. In contrast the lift of the zero linear map retains its label:

$$\widehat{A}(v, \lambda) = (Av, \lambda), \quad \widehat{0}_V(v, \lambda) = (0, \lambda) = e(v, \lambda), \quad \widehat{I}_V = \text{id}_{G_L(V)}. \quad (\text{FTR9})$$

All compositions obey $\widehat{A}\widehat{B} = \widehat{AB}$. Thus the zero linear lift and ambient scalar zero are two explicit maps, with the same amplitude and different label behavior. The additive-group receiver

$q_V(v, \lambda) = v$ intertwines both with the zero map on V and intertwines the new unit with I_V . It does not recover discarded labels.

For integral scalar multiplicity the fixed-label action is

$$(v, \lambda) \longmapsto (mv, \lambda), \quad m \in \mathbb{N}, \quad (mv, \lambda) + (nv, \lambda) = ((m+n)v, \lambda). \quad (\text{FTR10})$$

The zero value here is the supported-zero lift in (FTR9). At top amplitude $v \neq 0$, the outputs for $m = 1$ and $m = 2$ are different, proving the same failure of the presence factorization on the full carrier. Equations (FTR8)–(FTR10) apply coordinatewise to the entire placewise carrier and to every fixed label tuple in SLW1 and SGW1. The unit leaves all return maps and test coefficients fixed. No measure on the set of label tuples and no sum of one trace for every label is introduced by this identity action.

If one sends the new τ to the old τ_G instead, its action is the second formula of (FTR8), so it is the constant-bottom map, not identity. Sending it to e gives the supported-zero map of (FTR9), also not identity on a nonzero amplitude. These exact formulas prove which original placement realizes a global scalar unit.

101.3 Prime weights and Haar constants under the actual receiver

For every prime p , nonzero integer n , and unit $v \in \mathbb{Z}_p^\times$, SLW5–12 and SGW2–6 retain

$$\begin{aligned} A_{p,n,v}(x_\infty, x_p) &= (p^{-n}x_\infty, p^{-n}v^{-1}x_p), \\ C_{p,n,v} &= \frac{\log p}{|1 - p^{-n}|_\infty |1 - p^{-n}v^{-1}|_p}, \\ w(p^n)C_{p,n,v} &= \frac{\log p}{|1 - p^n v|_p}, \quad w(y) = |y - 1|, \\ d^*u &= \frac{\log p}{1 - p^{-1}} \frac{du}{|u|_p}, \quad \text{vol}_{du}(\mathbb{Z}_p) = 1, \quad \text{vol}_{d^*}(\mathbb{Z}_p^\times) = \log p. \end{aligned} \quad (\text{FTR11})$$

These are literal point pullback, original product Haar measures, and real-determinant receiving multiplier. The scalar τ acts on its kernel as I ; hence its flat-trace coefficient is $C_{p,n,v}$. Two occurrences recorded additively act as $2I$ on that kernel and give $2C_{p,n,v}$. The weighted receiving coefficients are respectively $\log p/|1 - p^n v|_p$ and $2 \log p/|1 - p^n v|_p$, which are unequal for every displayed p, n, v . Thus this is an actual arithmetic witness for (FTR5), not only a finite-dimensional operator example. For $n > 0$ the local denominator is 1; for $n = -m < 0$ it is p^m , giving the full pair $\log p$ and $p^{-m} \log p$ before multiplication by tests.

For a locally constant compactly supported test $b : \mathbb{Q}_p^\times \rightarrow \mathbb{C}$, the entire receiver is

$$\begin{aligned} W_p(b) &= \sum_{n \neq 0} \frac{1}{\log p} \int_{\mathbb{Z}_p^\times} w(p^n) C_{p,n,v} b(p^{-n}v^{-1}) d^*v \\ &\quad + \int_{\mathbb{Z}_p^\times} \frac{b(v^{-1}) - b(1)}{|1 - v|_p} d^*v. \end{aligned} \quad (\text{FTR12})$$

Its first sum is finite for this domain. The second numerator vanishes near 1, so both parts are defined with the original principal-value subtraction. In particular every additive scalar coefficient is received linearly, giving

$$W_p(\rho(\tau +_{\mathbb{B}} \tau)b) - W_p(\rho(\tau)b) - W_p(\rho(\tau)b) = -W_p(b). \quad (\text{FTR13})$$

For the concrete test $b = 1_{p^{-1}\mathbb{Z}_p^\times}$, only the $n = 1$ shell contributes, and its unit test is zero. Hence $W_p(b) = \log p \neq 0$. The difference in (FTR13) is exactly $-\log p$.

The finite Haar constants use additive integration, not presence. For $j \geq 1$, the unit subgroup $K_j = 1 + p^j \mathbb{Z}_p$ has mass

$$\mu^*(K_j) = \frac{\log p}{1 - p^{-1}} p^{-j}, \quad [\mathbb{Z}_p^\times : K_j] = (p - 1)p^{j-1}, \quad (p - 1)p^{j-1} \frac{\log p}{1 - p^{-1}} p^{-j} = \log p. \quad (\text{FTR14})$$

The coset count follows by reduction to units modulo p^j : there are $p^j - p^{j-1}$ such residues. Each coset has the stated equal mass by translation invariance. The last identity is the integral of their disjoint indicator sum. For a j with $(p - 1)p^{j-1} > 1$, collapsing that sum to a single presence value loses that displayed multiplicity. All factors must instead remain in the real or complex additive target; a Boolean scalar value alone cannot reconstruct them by an additive map, by the proof following (FTR5).

The zero-time receiving construction in SLW14–17 also retains its full parameter. In the invariant real coordinate $q = x_\infty/r$, the added real operator has kernel $r^{-1}\rho(q)\sigma(q')$ relative to $dx_\infty = r dq'$. Its trace parameter and supported density are

$$c_{\rho,\sigma} = \int_{\mathbb{R}} \rho(q)\sigma(q) dq, \quad d\Lambda_{0,v} = \frac{\rho(q)\sigma(q)}{|1 - v^{-1}|_p} dq ds \delta_0(dx_p),$$

$$\int d\Lambda_{0,v} = \frac{c_{\rho,\sigma} \log p}{|1 - v|_p}, \quad U_p(b) = \frac{1}{c_{\rho,\sigma} \log p} \int_{\mathbb{Z}_p^\times} (b(v^{-1}) - b(1)) \left(\int d\Lambda_{0,v} \right) d^*v \quad (c_{\rho,\sigma} \neq 0). \quad (\text{FTR15})$$

The subtraction makes the integrand vanish near $v = 1$; no scalar trace of the real identity is assigned there. For the original pair $\rho(q) = \sigma(q) = e^{-\pi q^2}$ one has $c_{\rho,\sigma} = 1/\sqrt{2}$, so the numerator contains $\log p/\sqrt{2}$ and the receiver contains $\sqrt{2}/\log p$. They are not Boolean scalars. The action of τ fixes the entire chosen operator and density. The count m multiplies the operator and its density by m ; with the fixed receiving denominator in (FTR15), its contribution is $mU_p(b)$. Replacing the chosen operator by m copies and also changing its receiving denominator would be a different receiving map, not preservation of the original additive trace.

For a character test $b(x) = h(|x|_p)\chi_p(x)$ with local unit conductor exponent a , SLW20–26 prove the full formula

$$W_p(b) = \begin{cases} \log p \sum_{m \geq 1} [z^{-m} h(p^m) + p^{-m} z^m h(p^{-m})], & a = 0, \quad z = \chi_p(p), \\ -a \log p h(1), & a > 0. \end{cases} \quad (\text{FTR16})$$

In the ramified row, the compact character annihilates every nonzero-return unit average. Its exact remaining calculation is

$$-\log p - \sum_{j=1}^{a-1} (p^j - p^{j-1}) \frac{\log p}{1 - p^{-1}} p^{-j} = -a \log p. \quad (\text{FTR17})$$

The subtraction sign, the a occurrences, and the factors in every filtration summand use addition and additive inverses in the target. The presence projection sends every positive count a to the same value, and cannot supply the coefficient a . For example at $p = 2$, primitive characters of conductor 4 and 8 have exponents 2 and 3. The character modulo 4 that sends 3 to -1 is primitive. For modulus 8, the values $\chi(1) = \chi(7) = 1$, $\chi(3) = \chi(5) = -1$ define a character and do not factor modulo 4, since 1 and 5 have different values. Their respective terms are $-2 \log 2 h(1)$ and $-3 \log 2 h(1)$. This proves on actual compact characters the loss caused by replacing conductor multiplicity with presence.

101.4 The full global Weil form and its scalar operation defect

Use exactly SGW7–14. Put $C_{\mathbb{Q}} = \mathbb{R}_{>0} \times K$, $K = \prod_p \mathbb{Z}_p^\times$, with measure $dy/y du$ and $\int_K du = 1$. For all characters χ of a fixed finite quotient $(\mathbb{Z}/N\mathbb{Z})^\times$, let

$$\begin{aligned} f(y, u) &= \sum_{\chi} h_{\chi}(y) \chi(u), & k(y, u) &= \sum_{\chi} k_{\chi}(y) \chi(u), & k^{\sharp}(g) &= |g|^{-1} \overline{k(g^{-1})}, \\ H_{\chi}(y) &= y^{-1} \int_0^{\infty} h_{\chi}(x) \overline{k_{\chi}(x/y)} dx, & (f * k^{\sharp})(y, u) &= \sum_{\chi} H_{\chi}(y) \chi(u), \\ M_h(s) &= \int_0^{\infty} h(y) y^s \frac{dy}{y}. \end{aligned} \tag{FTR18}$$

All radial functions here are smooth and compactly supported in $\mathbb{R}_{>0}$. Write c_{χ} for the primitive conductor, χ_* for the primitive character, and $\chi(-1) = (-1)^{\varepsilon_{\chi}}$. The finite local terms, including the inverse local character from the original embedding $\iota_p(p^n v) = (p^{-n}, u_p = v, u_q = p^{-n} \ (q \neq p))$, are

$$\mathcal{W}_{p,\chi}(H) = \begin{cases} \log p \sum_{n \geq 1} [\chi_*(p)^n H(p^n) + p^{-n} \chi_*(p)^{-n} H(p^{-n})], & p \nmid c_{\chi}, \\ -v_p(c_{\chi}) \log p H(1), & p \mid c_{\chi}. \end{cases} \tag{FTR19}$$

The complete real distribution is

$$\begin{aligned} \mathcal{A}_{\varepsilon}(H) &= (\log(2\pi) + \gamma) H(1) \\ &\quad + \lim_{\delta \downarrow 0} \left\{ \frac{1}{2} \int_{\substack{y>0 \\ |1-y| \geq \delta}} \frac{H(y)}{|1-y|} dy + (\log \delta) H(1) \right\} \\ &\quad + \frac{(-1)^{\varepsilon}}{2} \int_0^{\infty} \frac{H(y)}{1+y} dy. \end{aligned} \tag{FTR20}$$

Thus the exact original global form, with differential idele $\mathfrak{a}$ retained, is

$$\begin{aligned} Q(f, k) &:= \text{Tr}(\vartheta_m(f * k^{\sharp})|H^1) \\ &= \sum_{\chi} \left\{ 1_{\chi=1} [M_{h_{\chi}}(0) \overline{M_{k_{\chi}}(1)} + M_{h_{\chi}}(1) \overline{M_{k_{\chi}}(0)}] \right. \\ &\quad \left. - \log |\mathfrak{a}| H_{\chi}(1) - \sum_p \mathcal{W}_{p,\chi}(H_{\chi}) - \mathcal{A}_{\varepsilon_{\chi}}(H_{\chi}) \right\}. \end{aligned} \tag{FTR21}$$

The finite-place formulas are the proved SLW receiving maps, while the global equality is the cited human trace theorem applied exactly as in SGW13. This retains the original two endpoints, signs, conductor contributions, inverse-return weights, real parity, constants and real principal value.

Let $T_{f,k} = \vartheta_m(f * k^{\sharp})|H^1$ be the operator to which that theorem applies. The unit action is the identity on its representation space, hence $\rho(\tau) T_{f,k} = T_{f,k}$. Its trace is consequently $Q(f, k)$, term by term in (FTR21). Counting m written copies gives $m T_{f,k}$ and trace $m Q(f, k)$. No trace of I_{H^1} on its own is used. The exact Boolean-to-trace defect is

$$\begin{aligned} \text{Tr}([\rho(\tau +_{\mathbb{B}} \tau) - \rho(\tau) - \rho(\tau)] T_{f,k}) &= -Q(f, k), \\ Q(\rho(\tau) f, \rho(\tau) k) &= Q(f, k), & Q(mf, nk) &= m\bar{n} Q(f, k) \quad (m, n \in \mathbb{C}). \end{aligned} \tag{FTR22}$$

In particular doubling the first input doubles every term; doubling both inputs multiplies every term by 4. To verify these identities without hiding a contribution, (FTR18) gives $H_{mh,nk} = m\bar{n} H_{h,k}$ and $M_{mh}(s) = m M_h(s)$. Each summand of (FTR19), each finite cutoff expression of (FTR20), its limit, and the differential term is complex-linear in H . The endpoint

products acquire exactly $m\bar{n}$. This proves the displayed factors on every term of the original form. A particular numerical value of Q may vanish through its actual cancellations; that does not alter the operator defect $-I$ or the proved nonzero local witness (FTR13).

The additive operation is indispensable even in the real term: the cutoff integral and $(\log \delta)H(1)$ form a signed sum before the limit, and the negative-real contribution carries the separate sign $(-1)^{\varepsilon_\chi}$ and factor $1/2$. The powers of π in the original archimedean source enter here through the full constant $\log(2\pi) + \gamma$ retained in (FTR20). An additive map out of the Boolean value kills its unit by the earlier cancellation proof, so it cannot carry any nonzero one of these complex-linear coefficients. Keeping the original coefficients as external real or complex weights still works; it is exactly the weighted linear receiver just written, not their derivation from a Boolean projected scalar alone.

101.5 The intrinsic integer parity and the arithmetic scalar extension

Retain integer parity intrinsically by the complete graph of the original residue map:

$$\begin{aligned} \mathbb{Z}_{\text{par}} &= \{(n, \bar{n}) : n \in \mathbb{Z}, \bar{n} = n \bmod 2\}, \\ (m, \bar{m}) + (n, \bar{n}) &= (m + n, \bar{m} + \bar{n}), & (m, \bar{m})(n, \bar{n}) &= (mn, \bar{m}\bar{n}), \\ n &\longmapsto (n, \bar{n}), & (n, \bar{n}) &\longmapsto n, & \tau &\longmapsto (1, \bar{1}) \longmapsto I_V. \end{aligned} \tag{FTR23}$$

The sums and products in the second coordinates take place in $\mathbb{Z}/2\mathbb{Z}$. The two maps between $\mathbb{Z}$ and $\mathbb{Z}_{\text{par}}$ are inverse ring isomorphisms: the congruences $\overline{m+n} = \bar{m} + \bar{n}$ and $\overline{mn} = \bar{m}\bar{n}$ prove preservation of both operations, and the graph condition proves both compositions are identity. In particular local integer 1 has its intrinsic odd value $\bar{1}$, and $1 \cdot 1 = 1$ has the same parity because $\bar{1}\bar{1} = \bar{1}$. No vector-space super-grading or odd degree-changing operator is assigned by this assertion. The ungraded source unit τ maps to the integer unit $(1, \bar{1})$, whose full coefficient is the identity scalar. Keeping the graph retains all of the integer's parity information; it does not assign a parity to the source unit.

The real arithmetic parity in (FTR20) is the separate specified compact-character value at -1 . The scalar identity acts on every character sector and leaves its own ε_χ , conductor and phases unchanged. It does not select the trivial compact character or any other sector. Intrinsic integer parity alone does not determine that character value: both the trivial character and an odd character below use the same scalar unit $(1, \bar{1})$. For comparison, the odd primitive character modulo 4 is independent original arithmetic data: it has $\chi(-1) = -1$, conductor 4, unit term $-2 \log 2 H(1)$ at $p = 2$, and at odd primes the full phases $\chi(p)^n$ and $\chi(p)^{-n}$ in (FTR19). Its real last summand changes from $+\frac{1}{2} \int H(y)/(1+y)dy$ to the negative of that term. These differences show exactly what arithmetic data an actual character change would alter. Under the identity action, each such character retains all these data. The intrinsic integer parity graph and the character's arithmetic parity are both kept in their specified maps and neither is substituted for the other.

The ordinary coefficient ring generated under addition and additive inverses by the image of τ in (FTR1) is precisely $\mathbb{Z}$, with generator 1. It has not acquired a second scalar generator. At the pointed-monoid level M has exactly one proper ideal, $\{0\}$, which is prime since its complement $\{\tau\}$ is closed under multiplication. Every ring prime $\mathfrak{p}$ of $\mathbb{Z}$ pulls back under $M \rightarrow \mathbb{Z}$ to that same ideal $\{0\}$: zero belongs to every ring prime and 1 belongs to none. This gives the precise map on the degree-zero prime data, rather than identifying the several arithmetic primes with distinct elements of the unit monoid.

In the original support semiring the amplitude map is $q : G_L(\mathbb{Z}) \rightarrow \mathbb{Z}$, $q(a, \lambda) = a$. It is a surjective unital semiring map by the displayed coordinate operations, and $q(j(\tau)) = 1$. Its exact prime pullback is

$$q^{-1}(\mathfrak{p}) = (\{0\} \times L) \cup \{(n, 1_L) : n \in \mathfrak{p}\}. \tag{FTR24}$$

It is prime because a product amplitude lies in $\mathfrak{p}$ exactly when one amplitude does, and it is proper because $\hat{1}$ is excluded. Conversely every prime containing $e = (0, 1_L)$ has this form: multiplying e by every zero label supplies that entire zero carrier, and its top amplitudes form a ring ideal, with negatives supplied by multiplication by $(-1, 1_L)$. The ideal and prime tests then agree with those in $\mathbb{Z}$. The new unit action picks the already existing $\hat{1}$, which belongs to no proper ideal. It therefore creates no additional arithmetic prime in this specified scalar target. Other support primes of the full original lattice remain their own existing points; the scalar map has not added or deleted any element or ideal of that carrier.

This statement concerns exactly the scalar extension $S(M) \rightarrow H\mathbb{Z}$ and its specified G_L placement. It does not compute the spectrum of a separate enlarged ambient set with additional nonassociative or nondistributive operations. Such an ambient completion is not needed to define or to prove the map in (FTR2).

101.6 The exact return to the original zeta function

Use the original function $\zeta(s) = \sum_{m=1}^{\infty} m^{-s}$, on $\Re s > 1$. The unchanged positive-period data are

$$C_{p,n,1} = \frac{\log p}{p^n - 1}, \quad (p^n - 1)C_{p,n,1} = \log p, \quad \sum_p \sum_{n \geq 1} (p^n - 1)C_{p,n,1} p^{-ns} = -\frac{\zeta'(s)}{\zeta(s)}. \quad (\text{FTR25})$$

For completeness the original Euler product follows from unique factorization: a finite product expands as the sum over integers supported on its finite prime set, and absolute convergence of $\sum m^{-s}$ passes to the limit. Its logarithmic series $\sum_p \sum_{n \geq 1} p^{-ns}/n$ converges absolutely and locally uniformly on $\Re s > 1$. On $\Re s \geq 1 + \eta$, $\eta > 0$, the derivative series is bounded by

$$\frac{1}{1 - 2^{-(1+\eta)}} \sum_{m=2}^{\infty} (\log m) m^{-(1+\eta)} < \infty,$$

where integral comparison proves convergence of the last sum. Termwise differentiation proves the final equality of (FTR25), with every prime power retained.

Apply now the scalar extension (FTR2). It sends each flow operator $U_{p,n}$ to $IU_{p,n} = U_{p,n}$, and acts identically on its kernel, both determinant factors, the base length $\log p$, its Haar measure, and the receiving test. Thus every coefficient in (FTR25) is equal before and after that operation. The exact defect of counting two occurrences and then collapsing them is instead

$$\begin{aligned} \sum_p \sum_{n \geq 1} (p^n - 1)[C_{p,n,1} - 2C_{p,n,1}]p^{-ns} \\ = - \sum_p \sum_{n \geq 1} (\log p) p^{-ns} = \frac{\zeta'(s)}{\zeta(s)}, \quad \Re s > 1. \end{aligned} \quad (\text{FTR26})$$

For real $s > 1$ the sum preceding the last equality is strictly negative, so this operation defect is nonzero on the original zeta receiver. Recording m copies would similarly multiply the original logarithmic derivative by m , whereas the single global unit leaves it fixed.

One can also recover the original function from the proved coefficient identity without an unspecified multiplicative constant. For real $\sigma > 1$, integrating the derivative series from σ to $+\infty$ gives

$$\int_{\sigma}^{\infty} \sum_p \sum_{n \geq 1} (\log p) p^{-nt} dt = \sum_p \sum_{n \geq 1} \frac{p^{-n\sigma}}{n} = \log \zeta(\sigma). \quad (\text{FTR27})$$

The integrands are nonnegative, so monotone convergence justifies the interchange; the displayed convergent logarithmic series gives the last equality. The original Dirichlet series satisfies $\zeta(\sigma) \rightarrow 1$ as $\sigma \rightarrow +\infty$ by dominated convergence using any fixed exponent greater than 1. Therefore the action of the single scalar unit preserves that constant as well, and its recovered function is

exactly the original ζ on this half-plane, rather than an unspecified multiple. No extra Euler factor, rational prime, or new zero set is introduced by this specified identity operation. The entire Weil form still includes (FTR19)–(FTR21); the positive-period calculation does not remove its negative, conductor, real, or endpoint terms.

The final operation defect is thus fully explicit at every receiving level: $-I_V$ in (FTR5), $-\log p$ for the actual local test following (FTR13), $-Q(f, k)$ with every global term retained in (FTR22), and $\zeta'(s)/\zeta(s)$ in (FTR26). The count expression with two occurrences has value 2 in the integer receiver; that receiver cannot factor through the Boolean equation identifying those occurrences with one unit. The multiplicative global-unit action and the exact arithmetic trace remain related by the explicit scalar map, without identifying their different additive operations.

101.7 The exact overlap correction: truth and counting in the same receiver

The Boolean presence value does have a constructive receiver, with its overlap term retained. On $\mathbb{B}$ write $\vee$ for Boolean addition and $\wedge$ for its multiplication. Let $\epsilon(0) = 0$ and $\epsilon(\tau) = 1$. For all four pairs $b, c \in \mathbb{B}$, direct inspection gives

$$\epsilon(b \vee c) = \epsilon(b) + \epsilon(c) - \epsilon(b)\epsilon(c), \quad \epsilon(b \wedge c) = \epsilon(b)\epsilon(c). \quad (\text{FTR28})$$

If both values are 0 the two sides vanish. If exactly one is τ , the first right side is $1 + 0 - 0 = 1$. If both are τ , it is $1 + 1 - 1 = 1$. This proves the formulas exhaustively. Thus presence is received as a valuation with an explicit overlap subtraction, not as ordinary addition of independent occurrences.

For the actual operators $\rho(b) = \epsilon(b)I_V$, one has

$$\begin{aligned} \rho(b \vee c) &= \rho(b) + \rho(c) - \rho(b \wedge c), \\ \rho(b \wedge c) &= \rho(b)\rho(c). \end{aligned} \quad (\text{FTR29})$$

These are equalities of operators on the full original receiver space. For any operator T for which the tested trace is defined, multiply by T and use its additive trace to obtain

$$\text{Tr}(\rho(b \vee c)T) = \text{Tr}(\rho(b)T) + \text{Tr}(\rho(c)T) - \text{Tr}(\rho(b \wedge c)T). \quad (\text{FTR30})$$

For $b = c = \tau$, the intersection is the same unit, so the formula is $\text{Tr}(T) = \text{Tr}(T) + \text{Tr}(T) - \text{Tr}(T)$. The expression count of two occurrences remains $2 \text{Tr}(T)$. Equation (FTR30) identifies exactly the term removed when those two occurrences designate the same presence.

Taking the local functional in (FTR12) and the complete tested global trace in (FTR21) gives the constructive arithmetic identities

$$\begin{aligned} W_p(\epsilon(b \vee c)h) &= W_p(\epsilon(b)h) + W_p(\epsilon(c)h) - W_p(\epsilon(b \wedge c)h), \\ Q(\rho(b \vee c)f, k) &= Q(\rho(b)f, k) + Q(\rho(c)f, k) - Q(\rho(b \wedge c)f, k). \end{aligned} \quad (\text{FTR31})$$

The first formula uses a local test h in the domain of (FTR12). The second holds for the full global tests in (FTR18). Their proofs are the actual linearity and sesquilinearity proved above, together with (FTR28); no support projection on the arithmetic H^1 has been assumed. In particular each finite prime coefficient, each conductor summand, the full real cutoff expression and its limit, the differential term, and both endpoints obey this overlap identity separately. For the concrete local shell test following (FTR13), it reads $\log p = \log p + \log p - \log p$. For a ramified test of exponent a it reads $-a \log p h(1) = -a \log p h(1) - a \log p h(1) + a \log p h(1)$, preserving the original negative term and the positive overlap correction.

For a finite family of presence values $b_1, \dots, b_m$, the complete formula is

$$\begin{aligned} \rho\left(\bigvee_{j=1}^m b_j\right) &= I_V - \prod_{j=1}^m (I_V - \rho(b_j)) \\ &= \sum_{\emptyset \neq A \subseteq \{1, \dots, m\}} (-1)^{|A|+1} \prod_{j \in A} \rho(b_j). \end{aligned} \quad (\text{FTR32})$$

All factors are commuting scalar idempotents. Their product in the first line is I_V precisely when all $b_j = 0$, and is zero otherwise, proving that line directly. Expanding its finite product in the endomorphism ring proves the second line with every intersection and sign retained. Testing this finite identity against the original arithmetic operators gives the same finite sum of full local or global Weil expressions. When every $b_j = \tau$, every nonempty intersection is I_V and its coefficient sum is $\sum_{r=1}^m (-1)^{r+1} \binom{m}{r} = 1$, by the binomial expansion of $(1 - 1)^m$. This is the exact passage from counting m occurrences to the truth of one repeated presence.

Finally the original measures themselves have this relation on measurable finite-mass subsets, because their characteristic functions satisfy $1_{A \cup B} = 1_A + 1_B - 1_{A \cap B}$ pointwise. Integrating against the unchanged Haar measure proves $\mu(A \cup B) = \mu(A) + \mu(B) - \mu(A \cap B)$. For the disjoint cosets in (FTR14), every overlap is empty and ordinary counting gives the full unit mass. For a repeated identical coset, the overlap is that entire coset and exactly one copy is subtracted. These two calculations retain both truth and multiplicity within the actual Haar and trace receivers. They prove a positive comparison between the two operations without asserting a Boolean-additive ring homomorphism or deleting the explicit overlap term.

102 An ungraded unit, integer parity, and the complete interaction

Intended data and provenance. The user specifies a globally trivial element τ with no parity; ordinary integers carry their intrinsic even or odd parity. The integer 1 is not identified with τ . The prescribed values are $\tau + \tau = \tau$, $\tau^2 = \tau$, $\tau 1 = 1$, $0\tau = 0$, $\tau + 1 = 1$, and $1 + 1 = 2$. The global zero is shared, and may be represented by the empty set. The calculations below retain these values. A requested additional distributive equality is examined explicitly rather than added silently as an axiom. The extension of the displayed mixed rules to all integers is specified here as a constructed completion; its further values are not attributed to the user.

The comparison with $\mathbb{F}_1$ uses Alain Connes and Caterina Consani, *Absolute algebra and Segal's Gamma sets*, original author `F1_corr.tex`, formulas `functsum`, `addition`, and propositions `mono2sss`, `monomono`, `sssalg1`. The finite calculations below are proved directly; no novelty claim is made for the standard free-monoid or idempotent-ring constructions.

102.1 Full elements and a single explicit operation pair

Write

$$I = \{n_{\bar{n}} : n \in \mathbb{Z}, \bar{n} = n \bmod 2\}, \quad X = I \sqcup \{\tau\}. \quad (\text{FMC1})$$

The notation $n_{\bar{n}}$ retains the value and its intrinsic parity; it is not a second copy of the integer for each parity. The parity map is defined on I , and is undefined at τ . We identify the one integer zero with the one global zero, denoted 0. Replacing that single representative by $\emptyset$ through a bijection transports every operation below and introduces no new zero. The integer operations are

$$n_{\bar{n}} + m_{\bar{m}} = (n + m)_{\overline{n+m}}, \quad n_{\bar{n}} m_{\bar{m}} = (nm)_{\overline{nm}}. \quad (\text{FMC2})$$

Define the remaining operation values, symmetrically, by

$$\begin{aligned} \tau x = x\tau = x \quad (x \in X), \quad \tau + \tau = \tau, \quad \tau + 0 = 0 + \tau = \tau, \\ \tau + n_{\bar{n}} = n_{\bar{n}} + \tau = n_{\bar{n}} \quad (n \neq 0). \end{aligned} \quad (\text{FMC3})$$

These formulas define total operations on X and satisfy all the user's displayed operation values with distinct 0, τ , $1_{\bar{1}}$ and $2_{\bar{0}}$. There is no parity assigned to τ .

Multiplication is associative and commutative, with unit τ and absorbing zero. Indeed, a product of only τ factors is τ ; every other finite product is the ordinary product of its integer factors, with parity that of the resulting integer. Inserting or removing τ does not change it. This proves associativity for every case, including an integer zero. The integer $1_{\bar{1}}$ is the multiplicative unit on the integer subset, but it is not the unit on X , since $1_{\bar{1}}\tau = 1_{\bar{1}} \neq \tau$.

Addition is commutative with identity 0. Its associativity defect is completely described by

$$(\tau + n_{\bar{n}}) + (-n)_{\overline{-n}} = 0, \quad \tau + (n_{\bar{n}} + (-n)_{\overline{-n}}) = \tau \quad (n \neq 0), \quad (\text{FMC4})$$

and the reversed placement with τ last. There are no other cases. To verify exhaustion, triples of integers use integer associativity; triples with two or three τ values are checked by FMC3. With exactly one τ in the middle, the two associations agree by FMC3, whether either integer is zero or not. With τ first or last, a zero integer drops out; two nonzero integers give the same value unless their sum is zero, which gives FMC4. Thus this completion is an explicit algebra, but is not an associatively additive semiring. That limitation is a proved property of the specified operations, not a deletion of the integer layer or its quantum-number data.

Its entire distributivity discrepancy can also be evaluated. For an integer multiplier $k \neq 0$, the ordered pair of outputs is

$$\begin{aligned} (k(\tau + \tau), k\tau + k\tau) &= (k, 2k), \\ (k(\tau + n), k\tau + kn) &= (kn, k + kn) \quad (n \neq 0). \end{aligned} \quad (\text{FMC5})$$

Every output in FMC5 carries its resulting integer parity. The difference of right and left integer values is exactly k . All other triples satisfy the distributive equation: multiplier τ is the identity action; multiplier zero gives zero; two integer summands use ordinary distributivity; and the remaining mixed case has the other summand zero and gives $k = k + 0$. This proves the complete discrepancy without inferring distributivity from notation.

102.2 Two exact arithmetic maps, with their full defects

There are maps

$$\begin{aligned} r_+ : X &\rightarrow \mathbb{Z}, & r_+(n_{\bar{n}}) &= n, & r_+(\tau) &= 0, \\ r_\times : X &\rightarrow \mathbb{Z}, & r_\times(n_{\bar{n}}) &= n, & r_\times(\tau) &= 1. \end{aligned} \tag{FMC6}$$

The first preserves addition and zero, by checking the integer, τ -zero, τ -nonzero, and τ - τ pairs in FMC3. It does not preserve multiplication: $r_+(\tau n) = n$ whereas $r_+(\tau)r_+(n) = 0$ for $n \neq 0$. The second preserves multiplication, zero, and the multiplicative unit, by the product description following FMC3. Its complete additive defect is

$$r_\times(x) + r_\times(y) - r_\times(x + y) = \begin{cases} 1, & \text{at least one of } x, y \text{ is } \tau \text{ and neither is } 0, \\ 0, & \text{otherwise.} \end{cases} \tag{FMC7}$$

For two integers this is zero by ordinary addition. The three remaining pair types give $1 + 1 - 1 = 1$, $1 + n - n = 1$ for $n \neq 0$, and $1 + 0 - 1 = 0$. This proves every case. Neither map identifies the elements of X before its stated application. In particular $r_\times(\tau) = 1$ is a value of a map, not an equality $\tau = 1_{\bar{1}}$ in X or an assignment of parity to τ .

102.3 The global action and the retained occurrence count

Let $\text{End}(I, +)$ use pointwise addition and composition. For every $x \in X$, define the endomorphism

$$\rho(x)(n_{\bar{n}}) = (r_\times(x)n)_{\overline{r_\times(x)n}}. \tag{FMC8}$$

Integer distributivity proves each map is additive. The multiplicativity of $r_\times$ proves $\rho(xy) = \rho(x) \circ \rho(y)$, with $\rho(\tau) = \text{id}_I$ and $\rho(0) = 0$. The restriction to $B = \{0, \tau\}$ is injective, as its two maps have different values at $1_{\bar{1}}$. The identity action preserves intrinsic parity; this does not place τ in the domain of the parity map. FMC7 gives the full operator discrepancy

$$\rho(x) + \rho(y) - \rho(x + y) = \begin{cases} \text{id}_I, & \text{in the first case of FMC7,} \\ 0, & \text{otherwise.} \end{cases} \tag{FMC9}$$

In particular its value at $1_{\bar{1}}$ is $1_{\bar{1}}$ in the first case. The summed action and the collapsed action themselves have values $2_{\bar{0}}$ and $1_{\bar{1}}$ for $x = y = \tau$.

Retain every occurrence by the exact pair of maps

$$B \xleftarrow{q} \mathbb{N} \xrightarrow{c} \text{End}(I, +), \quad q(0) = 0, \quad q(k) = \tau \ (k > 0), \quad c(k)(n_{\bar{n}}) = (kn)_{\overline{kn}}. \tag{FMC10}$$

Both are unital semiring homomorphisms. For q , nonnegative sums are positive exactly when at least one summand is positive, and products are positive exactly when both factors are positive. These are precisely Boolean addition and multiplication on B . For c , the statements are integer distributivity and composition of scalar multiplication. No map, even a set map, can factor c through q , because $q(1) = q(2)$ while $c(1)(1_{\bar{1}}) = 1_{\bar{1}} \neq 2_{\bar{0}} = c(2)(1_{\bar{1}})$. For every positive count the exact discrepancy is

$$c(k) - \rho(q(k)) = (k - 1) \text{id}_I. \tag{FMC11}$$

Thus the entire missing data are specified, rather than described by an unspecified failure of a map. No order-of-evaluation convention makes c factor through q .

102.4 A universal ring receiver retaining the local integer unit

There is a ring receiver that preserves the entire integer arithmetic and the multiplication of X , while recording the failure of its mixed addition. Define

$$R = \mathbb{Z}[U]/(U^2 - U), \quad j(\tau) = 1_R, \quad j(n_{\bar{n}}) = nU. \quad (\text{FMC12})$$

Every element has a unique expression $a + bU$, and its complete product is

$$(a + bU)(c + dU) = ac + (ad + bc + bd)U. \quad (\text{FMC13})$$

Existence of that expression follows by replacing U^m with U for $m \geq 1$. Uniqueness follows by evaluating a zero expression at $U = 0$ and $U = 1$, obtaining $a = 0$ and $a + b = 0$. These evaluations give the exact ring isomorphism

$$R \longrightarrow \mathbb{Z} \times \mathbb{Z}, \quad a + bU \longmapsto (a, a + b), \quad (r, s) \longmapsto r + (s - r)U. \quad (\text{FMC14})$$

Formula FMC13 or direct substitution verifies both operations and the inverse. The integer layer is the ideal $\mathbb{Z}U$, with internal unit U , and has its original parity $nU \mapsto n \bmod 2$. The global unit 1_R is not in that ideal, so this parity is not defined on it. In particular j is injective on X : its integer images have first coordinate zero in FMC14, while $j(\tau)$ has first coordinate one. It preserves zero and multiplication and all sums of integers, using $U^2 = U$. Its remaining additive defects are exactly 1_R in the first case of FMC7 and zero otherwise.

For any commutative ring A receiving a multiplicative map from X that preserves zero and the global unit and is additive on the integer subset, the image u of $1_{\bar{1}}$ satisfies $u^2 = u$, and the image of $n_{\bar{n}}$ is nu . Therefore the unique unital ring map $R \rightarrow A$ carrying this data is $a + bU \mapsto a1_A + bu$. FMC13 proves multiplicativity; addition and uniqueness follow from the displayed expressions. This establishes the precise universal property of this receiver. Requiring it also to preserve $\tau + \tau = \tau$ forces $1_A + 1_A = 1_A$, hence $1_A = 0_A$.

The ring spectrum of this receiver is fully determined rather than inferred to be the original support spectrum. Under FMC14, its primes are exactly

$$\mathfrak{p} \times \mathbb{Z} \quad \text{and} \quad \mathbb{Z} \times \mathfrak{p} \quad (\mathfrak{p} \in \text{Spec } \mathbb{Z}). \quad (\text{FMC15})$$

Indeed $(1, 0)(0, 1) = 0$ forces any prime to contain one of the two idempotents, and it cannot contain both because their sum is the unit. In the quotient by the contained component, the prime is exactly a prime of the other copy of $\mathbb{Z}$. Conversely every listed ideal is prime by projection. The two idempotent basic opens are disjoint, exhaust the spectrum, and each identifies with $\text{Spec } \mathbb{Z}$ through projection. Thus this particular receiver has two clopen arithmetic components. It is not silently substituted for the programme's ordinal-sum support spectrum. The exact comparison with that spectrum is proved in FGM.

102.5 Cohomology, tensor powers, and the scope of the unit action

The scalar receiver in FMC8 also specifies exactly what the new pointed unit does to every original arithmetic complex on which it acts by scalar extension. For a complex $(C^\bullet, d)$ of abelian groups or vector spaces its degreewise action is

$$\rho(\tau)|_{C^r} = \text{id}_{C^r}, \quad d\rho(\tau) = \rho(\tau)d = d. \quad (\text{FMC16})$$

Both composites send an element x to dx , proving the cochain identity. It restricts to the identity on $\ker d$ and on $\text{im } d$, so the induced map on $H^r = \ker d / \text{im } d$ is exactly the identity. For a chain map $f : C \rightarrow D$, the usual cone has components $D^r \oplus C^{r+1}$ and differential $(y, x) \mapsto (d_D y + f(x), -d_C x)$. Applying the two identity actions before or after this full differential gives the same pair, including its minus sign. Thus the same assertion holds on that cone and its

cohomology. It makes no claim that a new nonadditive source has already been identified with this complex.

For any operator F on the receiving vector space and every positive tensor order k ,

$$\rho(\tau)F = F\rho(\tau) = F, \quad \rho(\tau)^{\otimes k} = \text{id}, \quad (\rho(\tau)F)^{\otimes k} = F^{\otimes k}. \quad (\text{FMC17})$$

The tensor equality follows by evaluation on every pure tensor; pure tensors generate the algebraic tensor product, so it holds on all elements. In particular, for an eigenvector $Fv = \alpha v$,

$$F^{\otimes k}(v^{\otimes k}) = \alpha^k v^{\otimes k}. \quad (\text{FMC18})$$

Every factor is displayed: inserting this unit changes none of the eigenvalues or their tensor powers. This locates exactly what the unit action does in a tensor-amplification calculation. An additional weight bound does not follow from inserting it. Any different effect must come from the additional geometric maps, not from a different value for this identity action.

For commuting scalar truth actions, the constructive additive comparison instead retains the overlap:

$$\rho(b \vee c) = \rho(b) + \rho(c) - \rho(b \wedge c), \quad b, c \in \{0, \tau\}. \quad (\text{FMC19})$$

When both are τ it reads $I = I + I - I$; when one is zero it reads $I = I + 0 - 0$; when both are zero it reads $0 = 0$. These cases prove the formula without assuming additivity of ρ on Boolean sums. The full lattice form is proved in FSI and the original arithmetic trace evaluation in FTR.

What the construction establishes. FMC1–19 provide a complete non-distributive interaction, its exact associativity and distributivity discrepancies, a global operator action, the full occurrence-count receiver, and a universal ring receiver with all its prime points. Every integer value and its intrinsic parity is retained; τ never receives parity. These calculations do not supply a sign for the original Weil pairing, a Deligne purity estimate, or a proof or disproof of RH.

103 Presence, parity, and prismatic maps in pictures

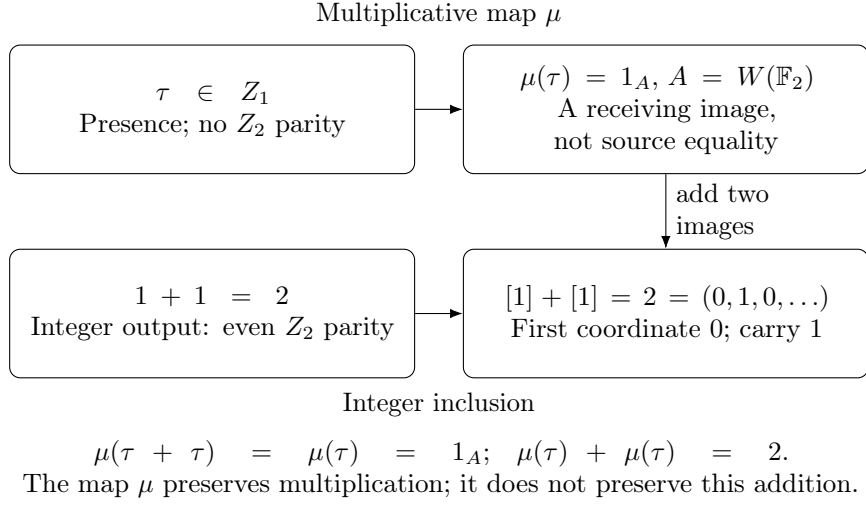


Figure 32: The comparison retains the user’s Z_1 and Z_2 types. The full additive carry is $[a] + [b] - [a + b] = 2[a][b]$ for $a, b \in \mathbb{F}_2$. Proofs: the rank-one return chapter, equations BLR7 and BLR12. Bhatt–Lurie supply the rank-one definition $\delta(v) = 0$ in *Absolute Prismatic Cohomology*, source lines 970–983; the displayed comparison is proved in this reader.

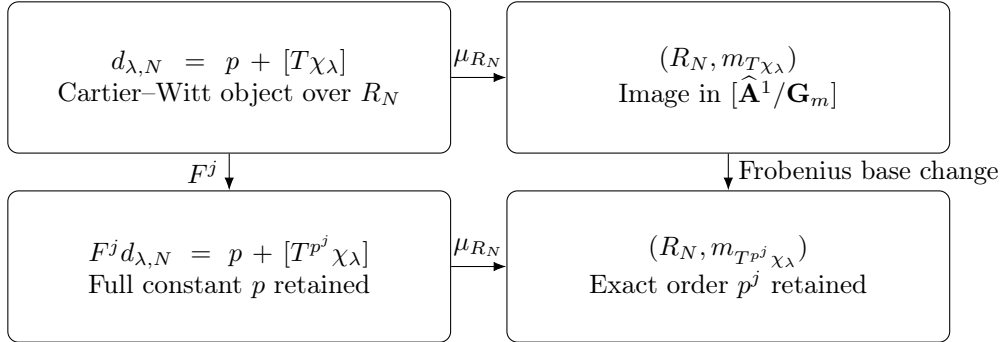
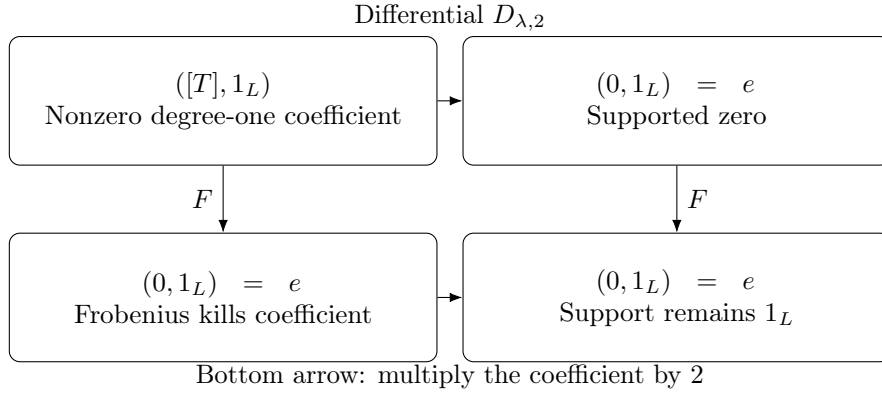


Figure 33: Every lattice prime coordinate occurs in $R_N = \text{Map}(\text{Spec}_{\text{lat}} L, \mathbb{F}_p)[T]/(T^N)$. For $N > p^j$, distinct labels remain nonisomorphic. For $N \leq p^j$, the chosen Witt vectors all equal p . The whole compatible tower retains all labels and has trivial stabilizers (BLM24–30). The arrows are Bhatt–Lurie’s affine-line morphism and Frobenius, with their action on this family proved in BLM3–12.



$$\begin{aligned}
p &= 2, & R_2 &= B[T]/(T^2), & d &= 2 + [T\chi_\lambda]. \\
d[T] &= [T^2\chi_\lambda] + V([T^2]) = 0; & [T] &\neq 0. \\
e &= (0, 1_L) \neq (0, 0_L) = \tau_{\text{old}} \text{ for nontrivial } L.
\end{aligned}$$

Figure 34: A computed class in $\pi_1(W(R_2)//d)$ and its exact return to the historical full support carrier. The subscript “old” keeps that carrier’s absorber distinct from the user’s ungraded $\tau \in Z_1$. The source of the derived quotient is Bhatt–Lurie’s *The prismatization of p -adic formal schemes*, Constructions **GenCartDivAnim** and **CartWitttoAnim**. All entries and arrows are proved in BLM13–17,21–23,29.

104 Bhatt–Lurie rank-one return and every arithmetic carry

104.1 Scope, source, and notation

This note proves arithmetic and support-preserving maps. It makes no assertion about zeros of the Riemann zeta function or about any consequence for the Riemann hypothesis.

The user’s three labels remain distinct:

$$\begin{aligned} Z_0 &= \text{absence, with underlying set } \emptyset, \\ Z_1 &= \text{ungraded presence } \tau, \\ Z_2 &= \text{parity information for integer elements.} \end{aligned}$$

In particular, Z_2 is not the notation for the coefficient ring used below. That ring is always written $A = W(\mathbb{F}_2)$. The symbol τ at the Z_1 level carries no parity. The maps below keep the source element $\tau \in Z_1$ distinct from its coefficient image $\mu(\tau) = 1_A$ and its supported image $\hat{\mu}(\tau) = u_A \in G_L(A)$. Parity belongs to a coefficient image after reduction; it is never assigned to the source element τ .

For precision, the two-element monoid introduced below adjoins an absorbing symbol 0 to the presence symbol τ . This construction does not assert that the empty set Z_0 has an element.

The source used for the meaning of rank one and the prismatic logarithm is Bhargav Bhatt and Jacob Lurie, *Absolute Prismatic Cohomology*, arXiv:2201.06120v1, original author source `apc.tex`, read at the following locations:

- source lines 970–983, in the introduction to `section:twist-and-log`: rank one means $\delta(v) = 0$; the map from the Tate module to rank-one units is described there;
- source lines 1944–1975, including `construction:Tate-logarithm`: the Tate-module logarithm is the composite through $(1 + I)_{\text{rk}=1}$.

The exact read file has SHA-256

BB3264C72D2B348E27B29B6B954B8F8DB3D5A9717D75567DCA02B7935C676F84.

Only these source passages were read for this derivation. The proofs below are local derivations, not claims that the support construction occurs in Bhatt–Lurie. The use of that source is the rank-one definition and the domain and codomain of its logarithm.

104.2 The full coefficient ring and its Witt coordinates

We use the usual 2-typical Witt ring. Its underlying set on a ring R is $R^{\mathbb{N}}$, and its universal addition and multiplication polynomials are specified by the Witt polynomials

$$w_n(X_0, \dots, X_n) = \sum_{i=0}^n 2^i X_i^{2^{n-i}} \quad (n \geq 0). \quad (\text{BLR1})$$

The standard Witt construction supplies the integral universal polynomials with these ghost identities; this is the definition of the Witt operations being used. No Witt construction is applied to the support semiring defined later.

The Teichmüller representative is $[a] = (a, 0, 0, \dots)$, and the Verschiebung is $V(a_0, a_1, \dots) = (0, a_0, a_1, \dots)$. Witt Frobenius F is specified by $w_n(Fx) = w_{n+1}(x)$. In all subsequent formulas, the bracket $[a + b]$ means the representative of the sum in $\mathbb{F}_2$; $[a] + [b]$ means the sum in A . These expressions are not interchanged.

Lemma 59. *On $A = W(\mathbb{F}_2)$, the following identities hold:*

$$F = \text{id}, \quad V(x) = 2x, \quad [a][b] = [ab], \quad V^r(A) = 2^r A = \{(a_i)_{i \geq 0} : a_0 = \dots = a_{r-1} = 0\}.$$

Multiplication by 2 is injective. There is a canonical ring isomorphism

$$A \xrightarrow{\sim} \varprojlim_{r \geq 1} \mathbb{Z}/2^r \mathbb{Z}. \quad (\text{BLR2})$$

Under it, every element retains the unique full expansion

$$(a_0, a_1, a_2, \dots) = \sum_{i=0}^{\infty} 2^i [a_i], \quad a_i \in \mathbb{F}_2. \quad (\text{BLR3})$$

Proof. First, universal ghost identities can be checked over a polynomial ring over $\mathbb{Z}$. Its ghost map is injective: if all ghosts of a Witt vector vanish, the zeroth coordinate vanishes, and then the n th ghost and the vanishing of the previous coordinates give $2^n x_n = 0$, hence $x_n = 0$. An identity of the integral universal coordinate polynomials verified in that ring therefore holds after every specialization.

The identity $[a][b] = [ab]$ follows since its n th ghosts on both sides are $a^{2^n} b^{2^n} = (ab)^{2^n}$. The formula

$$w_0(Vx) = 0, \quad w_n(Vx) = 2w_{n-1}(x) \quad (n \geq 1)$$

proves that V is additive and that $FV(x) = 2x$.

For completeness, the Frobenius coordinate polynomials satisfy $F_n(X) \equiv X_n^2 \pmod{2}$. Here is the needed divisibility argument. If two polynomials u, v satisfy $u - v \in 2\mathbb{Z}[X]$, repeated squaring gives

$$u^{2^k} - v^{2^k} \in 2^{k+1}\mathbb{Z}[X] \quad (k \geq 0),$$

because at every squaring the other factor is even. The identity for the zeroth Frobenius coordinate is $F_0 = X_0^2 + 2X_1$. If the assertion is known for indices below n , the defining ghost identity gives

$$2^n(F_n - X_n^2) = 2^{n+1}X_{n+1} + \sum_{i=0}^{n-1} 2^i (X_i^{2^{n+1-i}} - F_i^{2^{n-i}}).$$

Every term on the right is divisible by 2^{n+1} by the preceding divisibility statement. Division in the torsion-free polynomial ring proves the assertion for n . On $\mathbb{F}_2$ every coordinate has square equal to itself, so $F = \text{id}$ on A . Thus $FV = 2$ gives $V(x) = 2x$. The coordinate shift V is injective, proving injectivity of multiplication by 2 and the displayed description of $2^r A$.

Consequently $A/2^r A$ has exactly 2^r elements, represented by its first r Witt coordinates. In this quotient the element $1 = [1]$ has additive order exactly 2^r : $2^r 1 = V^r 1$ vanishes, but $2^{r-1} 1 = V^{r-1} 1$ has coordinate 1 at index $r-1$ and does not vanish. Its order is a divisor of 2^r and is not a divisor of 2^{r-1} , so is 2^r . The unital ring map

$$\mathbb{Z}/2^r \mathbb{Z} \longrightarrow A/2^r A, \quad \bar{n} \longmapsto n1,$$

is therefore injective and, by equality of the finite cardinalities, surjective. These ring isomorphisms commute with the quotient maps. The underlying coordinate sequences identify A with $\varprojlim_r A/2^r A$, giving (BLR2).

Finally, the finite Witt vector $(a_0, \dots, a_{r-1}, 0, \dots)$ equals $\sum_{i=0}^{r-1} V^i [a_i]$: their n th ghosts are both $\sum_{i=0}^{\min(n, r-1)} 2^i a_i^{2^{n-i}}$, so the universal identity argument applies. Passing to the coordinate limit and using $V^i = 2^i$ gives (BLR3), including uniqueness of every digit. $\square$

Lemma 60. *The ring A is an integral domain. Every nonzero $a \in A$ has a unique expression $a = 2^m u$, where $m \geq 0$ and u is a unit with odd reduction.*

Proof. Let m be the index of the first nonzero Witt coordinate of a . Lemma 59 gives a unique u with $a = 2^m u$ and first coordinate 1. At every finite level its residue is odd, hence invertible in $\mathbb{Z}/2^r \mathbb{Z}$ by Bézout's identity. Its unique inverses commute with reduction, and (BLR2) gives their inverse limit, an inverse of u in A . If $b = 2^n v \neq 0$, then $ab = 2^{m+n} uv \neq 0$, since uv is a unit and multiplication by 2 is injective. This also proves uniqueness of m ; injectivity of multiplication by 2^m proves uniqueness of u . $\square$

104.3 The rank-one image of presence and the base prism

Proposition 61. *With $\varphi = F = \text{id}$, the canonical 2-derivation on A is*

$$\delta(a) = \frac{a - a^2}{2}, \quad \varphi(a) = a^2 + 2\delta(a) = a. \quad (\text{BLR4})$$

The fraction denotes the unique element whose product with 2 is $a - a^2$. In particular,

$$\delta(0) = 0, \quad \delta(1) = 0, \quad \delta(2) = -1.$$

The rank-one locus is exactly $A_{\text{rk}=1} = \{0, 1\}$; its unit part is $A_{\text{rk}=1}^\times = \{1\}$.

Proof. The first coordinate of $a - a^2$ is zero, so this difference belongs to $2A$, and injectivity of multiplication by 2 proves uniqueness. Direct expansion in the original coefficient ring gives

$$\begin{aligned} \delta(a + b) &= \delta(a) + \delta(b) - ab, \\ \delta(ab) &= a^2\delta(b) + b^2\delta(a) + 2\delta(a)\delta(b). \end{aligned}$$

For the second identity, the numerator on the right after multiplication by 2 is

$$a^2(b - b^2) + b^2(a - a^2) + (a - a^2)(b - b^2) = ab - a^2b^2.$$

These identities, together with $\delta(0) = \delta(1) = 0$, are the 2-derivation identities. Substitution of 2 gives $\delta(2) = (2 - 4)/2 = -1$. By Lemma 60, the equation $\delta(a) = 0$ is equivalent to $a(a - 1) = 0$, hence to $a = 0$ or $a = 1$. Only the latter is a unit. $\square$

Definition 62. Let $M = \{0, \tau\}$ have multiplication

$$0 \cdot 0 = 0, \quad 0 \cdot \tau = \tau \cdot 0 = 0, \quad \tau \cdot \tau = \tau.$$

Its identity is τ and its absorbing element is 0.

Theorem 63. *The map*

$$\mu : M \longrightarrow A_{\text{rk}=1}, \quad 0 \longmapsto 0_A, \quad \tau \longmapsto 1_A,$$

is an isomorphism of multiplicative monoids with an absorbing element. It neither supplies addition on M nor gives Z_1 a parity. The composite $M \xrightarrow{\mu} A \xrightarrow{\rho} \mathbb{F}_2$ satisfies $\rho(\mu(\tau)) = 1_{\mathbb{F}_2}$. This describes the arithmetic image only: the source τ , the coefficient 1_A , and the residue $1_{\mathbb{F}_2}$ remain elements of their respective, different domains.

Proof. The four products in Definition 62 are sent to the four equal products in $\{0, 1\}$. The image is the whole rank-one locus by Proposition 61, and the two images are distinct. The claims about structure are literal domain statements: M was given only multiplication, while reduction is a map from the coefficient ring. $\square$

Proposition 64. *The pair $(A, I) = (W(\mathbb{F}_2), (2))$ is a bounded prism. In this prism*

$$(1 + I)_{\text{rk}=1} = \{1\}, \quad T_2((A/I)^\times) = \{1\}.$$

Thus the Bhatt–Lurie maps in this particular prism are

$$\{1\} \xrightarrow{\rho_0} \{1\} \xrightarrow{\log_{\text{Prism}}} A\{1\}, \quad 1 \longmapsto 1 \longmapsto 0.$$

Proof. The ring A is a δ -ring by Proposition 61. The ideal (2) is generated by a non-zero-divisor and is therefore an invertible Cartier ideal. The prism inclusion is $2 \in I + \varphi(I) = (2)$, since $\varphi(2) = 2$. The ring is derived 2-complete: it is 2-torsion-free, so its derived quotients by 2^r are the ordinary quotients; the transition maps of these quotients are surjective, so their higher inverse limit vanishes; and their inverse limit is A by (BLR2). For the asserted higher-limit vanishing, the product description of $\varprojlim^1$ has difference map $(x_r) \mapsto (x_r - q_{r+1,r}(x_{r+1}))$; surjectivity is obtained by choosing x_{r+1} successively after x_r has been chosen. This proves the required vanishing directly. As $(2, I) = (2)$, this is the required derived $(2, I)$ -completeness. Finally $A/I = \mathbb{F}_2$ has 2-torsion killed by 2, proving boundedness.

The only rank-one unit is 1 and it lies in $1 + I$. The group $(A/I)^\times = \mathbb{F}_2^\times$ has only its identity, so its Tate module, the group of compatible root sequences beginning at the identity, has only the constant identity sequence. The source construction carries that sequence to 1. Any homomorphism from this one-element group to the additive group of $A\{1\}$ sends it to 0. $\square$

The symbol 0 of M does not lie in the domain of the prismatic logarithm, because that domain consists of units. The map μ therefore does not by itself give a nonzero logarithm or a new cohomology class.

104.4 Every support label in the coefficient construction

Let L be a bounded distributive lattice. Its least and greatest elements are written $\perp$ and $\top$, and its operations are $\vee$ and $\wedge$. Its trivial one-element case is permitted; statements distinguishing $\perp$ and $\top$ will explicitly require a nontrivial lattice. Let R be a commutative unital ring.

Definition 65. Define the set, with no quotient of its displayed labels,

$$G_L(R) = (\{0_R\} \times L) \cup (R \times \{\top\}) \subseteq R \times L.$$

For its elements set

$$(a, \lambda) + (b, \eta) = (a + b, \lambda \vee \eta), \tag{BLR5}$$

$$(a, \lambda)(b, \eta) = (ab, \lambda \wedge \eta). \tag{BLR6}$$

Its global zero is $0_G = (0_R, \perp)$ and its identity is $1_G = (1_R, \top)$. Write $z_\lambda = (0_R, \lambda)$ for every supported zero. In particular,

$$\tau_{\text{old}} = z_\perp, \quad e = z_\top, \quad u_R = 1_G = (1_R, \top).$$

Here u_R is an element of the receiving semiring $G_L(R)$. The retained historical symbol τ_{old} names its absorbing zero; it does not denote the ungraded source element $\tau \in Z_1$.

Theorem 66. *Equations (BLR5)–(BLR6) give a commutative unital semiring. On the supported zeros their exact operations are*

$$z_\lambda + z_\eta = z_{\lambda \vee \eta}, \quad z_\lambda z_\eta = z_{\lambda \wedge \eta}.$$

For any bounded lattice, distributivity of these operations on $G_L(R)$ forces distributivity of the lattice. Thus this hypothesis is required by the displayed construction, not an omitted assumption.

Proof. If $a + b \neq 0_R$, at least one of a, b is nonzero; its label is $\top$, so $\lambda \vee \eta = \top$. This proves addition is closed. If $ab \neq 0_R$, both a, b are nonzero, so both labels are $\top$, proving multiplication is closed. Associativity and commutativity are componentwise. The equations $\lambda \vee \perp = \lambda$, $\lambda \wedge \top = \lambda$, and $\lambda \wedge \perp = \perp$ prove the identities and absorption. For distributivity, both coefficients are $a(b + c) = ab + ac$, and both labels agree because $\lambda \wedge (\eta \vee \kappa) = (\lambda \wedge \eta) \vee (\lambda \wedge \kappa)$.

Conversely, applying distributivity to $z_\lambda(z_\eta + z_\kappa)$ forces this lattice identity, since the map $\lambda \mapsto z_\lambda$ is injective. In a lattice this law also implies the dual law. Explicitly,

$$\begin{aligned}(x \vee y) \wedge (x \vee z) &= ((x \vee y) \wedge x) \vee ((x \vee y) \wedge z) \\ &= x \vee (z \wedge x) \vee (z \wedge y) = x \vee (y \wedge z),\end{aligned}$$

by absorption. Hence the lattice is distributive. $\square$

Proposition 67. *Every unital ring map $f : R \rightarrow S$ gives a unital semiring map*

$$G_L(f) : G_L(R) \longrightarrow G_L(S), \quad (a, \lambda) \longmapsto (f(a), \lambda).$$

These maps preserve every support label and commute with composition. The amplitude map $\alpha_R : G_L(R) \rightarrow R$, $(a, \lambda) \mapsto a$, is a unital semiring map. The top-support inclusion

$$i_R : R \longrightarrow G_L(R), \quad a \longmapsto (a, \top),$$

preserves the two binary operations and their multiplicative identity, but for nontrivial L it does not preserve the global additive zero. It has $\alpha_R i_R = \text{id}_R$ and $i_R(0_R) = e$.

Proof. If $f(a) \neq 0_S$, then $a \neq 0_R$, so the original label was $\top$; the image lies in $G_L(S)$. Applying f to the coefficient equations proves preservation of addition, multiplication, and their identities. Composition and the amplitude assertions follow by substitution. For i_R , the equalities $\top \vee \top = \top$ and $\top \wedge \top = \top$ prove the binary-operation statements. Its value at 0_R is e , which differs from 0_G exactly when $\perp \neq \top$. $\square$

When R is nonzero and L is nontrivial, the three receiving pairs $(0_R, \perp)$, $(0_R, \top)$, and $u_R = (1_R, \top)$ are different. The top-support slice is an additive group with its own identity e ; its inverse map is $(a, \top) \mapsto (-a, \top)$. The whole semiring is not an additive group: for example $e + (b, \eta)$ always has support $\top$, so it cannot equal 0_G . All these claims follow directly from the equality rule for ordered pairs and (BLR5); no zero label is identified with another.

For $R = A$, the precise multiplicative map from presence is

$$\widehat{\mu} : M \longrightarrow G_L(A), \quad 0 \longmapsto \tau_{\text{old}}, \quad \tau \longmapsto u_A = (1_A, \top), \quad \alpha_A \widehat{\mu} = \mu.$$

Thus $u_A := \widehat{\mu}(\tau)$ is the supported arithmetic image, not the source element τ . Its four multiplication identities follow from absorption and $u_A^2 = u_A$.

104.5 Reduction, Teichmüller lifting, and the exact additive defect

Let $\rho : A \rightarrow \mathbb{F}_2$ be first-coordinate reduction, and let $t : \mathbb{F}_2 \rightarrow A$ be $a \mapsto [a]$. The map ρ is a ring map; t is multiplicative, preserves 0, 1, and is a section of ρ . It is not additive.

Theorem 68. *Reduction and its label-preserving section are*

$$\begin{aligned}\pi : G_L(A) &\longrightarrow G_L(\mathbb{F}_2), & (a, \lambda) &\longmapsto (\rho(a), \lambda), \\ \sigma : G_L(\mathbb{F}_2) &\longrightarrow G_L(A), & (a, \lambda) &\longmapsto ([a], \lambda).\end{aligned}$$

The map π is a surjective unital semiring map, and σ is a multiplicative section preserving the global zero, identity, and every support label. For $x = (a, \lambda)$ and $y = (b, \eta)$, let $s = \lambda \vee \eta$. The full coefficient defect and its supported version are

$$C(a, b) = [a] + [b] - [a + b] = 2[a][b] = V([ab]), \tag{BLR7}$$

$$K(x, y) = (C(a, b), s) = (2[a][b], \lambda \vee \eta). \tag{BLR8}$$

The following is an equality in $G_L(A)$, with every support retained:

$$\sigma(x) + \sigma(y) = \sigma(x + y) + K(x, y). \tag{BLR9}$$

Proof. The assertions about π follow from Proposition 67 and the section constructed here. If $[a] \neq 0$, then $a = 1$, and the label of (a, λ) is $\top$, so σ is well-defined. Teichmüller multiplication and the unchanged meet of the labels prove multiplicativity. The section identity follows from $\rho([a]) = a$.

There are exactly four coefficient cases; none is suppressed:

a	b	$a + b$ in $\mathbb{F}_2$	$[a] + [b]$ in A	$[a + b]$	$C(a, b)$
0	0	0	0	0	0
0	1	1	1	1	0
1	0	1	1	1	0
1	1	0	2	0	2

This proves the first equality in (BLR7); the last is $V = 2$ and $[a][b] = [ab]$. If $C(a, b) \neq 0$, then $a = b = 1$, forcing $\lambda = \eta = s = \top$. Thus $K(x, y)$ belongs to $G_L(A)$ in all cases. Both sides of (BLR9) have coefficient $[a] + [b] = [a + b] + C(a, b)$ and support $s = s \vee s$, proving the equality. $\square$

There is no subtraction operation asserted on the additive monoid of $G_L(A)$. If a difference notation is wanted, define the sign map $J(a, \lambda) = (-a, \lambda)$, which is well-defined and preserves addition. Then the exact replacement for the coefficient difference is

$$\sigma(x) + \sigma(y) + J(\sigma(x + y)) = K(x, y). \quad (\text{BLR10})$$

Indeed its coefficient is (BLR7), and its support is $s \vee s = s$. For a general $w = (a, \lambda)$, the same sign map obeys $w + J(w) = z_\lambda$, not necessarily 0_G .

In particular, when the numerical carry is zero, its support is not discarded:

$$K(z_\lambda, z_\eta) = z_{\lambda \vee \eta}.$$

For $\lambda \vee \eta \neq \perp$, this is different from the global zero. This selected defect records exactly the support of the pair being added; it is not a claim that no other additive correction could give the same equality after joining labels.

For two copies of the image u_A the exact arithmetic is

$$u_A + u_A = (2, \top) \text{ in } G_L(A), \quad \pi(2, \top) = (0, \top) = e \text{ in } G_L(\mathbb{F}_2). \quad (\text{BLR11})$$

For nontrivial L , the latter e is distinct from $\tau_{\text{old}} = (0, \perp)$. Also

$$\sigma((1, \top) + (1, \top)) = e \neq (2, \top) = \sigma(1, \top) + \sigma(1, \top).$$

This proves failure of additivity. In Witt coordinates the coefficient on the right is exactly

$$[1] + [1] = 2 = V(1) = (0, 1, 0, 0, \dots).$$

Its first coordinate is parity zero, while its second coordinate is one. Thus the carry stores arithmetic information beyond parity, with no change to the support label.

The carry also has an exact expression through the 2-derivation:

$$\delta([a] + [b]) = -[a][b], \quad C(a, b) = -2\delta([a] + [b]). \quad (\text{BLR12})$$

Indeed $[a], [b] \in \{0, 1\}$, so each has δ equal to zero; the first equality is the full additive 2-derivation identity in Proposition 61. The second follows from (BLR7). For $a = b = 1$, this is exactly $\delta(2) = -1$ and $C(1, 1) = 2$, with the sign and factor 2 retained.

Proposition 69. *The full kernel congruence of π is*

$$(a, \lambda) \sim (b, \eta) \iff \lambda = \eta \text{ and } a - b \in 2A.$$

For nontrivial L , the fibre over the global zero is the singleton $\{\tau_{\text{old}}\}$, even though π is not injective. For each $\lambda \neq \top$, the fibre of $(0, \lambda)$ is the singleton $\{(0, \lambda)\}$. The fibre of $(c, \top)$ is $\{(a, \top) : \rho(a) = c\}$.

Proof. Equality of the two images is equality of both their ordered-pair coordinates, proving the congruence. If $\lambda \neq \top$, membership in $G_L(A)$ forces the coefficient to be exactly zero. This proves all the asserted singleton fibres, including $\lambda = \perp$ for a nontrivial lattice. The top fibre has no further restriction. In particular $(2, \top)$ and $(0, \top)$ are distinct and have the same image. Semiring congruences, rather than a ring zero-kernel criterion, record this noninjectivity. $\square$

104.6 The compatible tower, its complete limit, and all later carries

For every integer $r \geq 1$, put $R_r = \mathbb{Z}/2^r\mathbb{Z}$ and let $q_{r+1,r} : R_{r+1} \rightarrow R_r$ be reduction. Set $B_r = G_L(R_r)$ and

$$Q_{r+1,r} : B_{r+1} \longrightarrow B_r, \quad (a, \lambda) \longmapsto (q_{r+1,r}(a), \lambda).$$

Every transition is a surjective unital semiring map retaining the exact support. A lift of a top-labelled element may be taken in the top slice; a lower-labelled element is its supported zero and lifts to the same supported zero. This proves the claimed surjectivity.

Theorem 70. *The map*

$$\Omega : G_L(A) \longrightarrow \varprojlim_{r \geq 1} B_r, \quad (a, \lambda) \longmapsto ((a \bmod 2^r A, \lambda))_{r \geq 1}, \quad (\text{BLR13})$$

where $A/2^r A$ is identified with R_r by Lemma 59, is an isomorphism of unital semirings. It retains each individual zero label. If L is discrete, A has its 2-adic topology, and all B_r are discrete, this is also a homeomorphism for the indicated subspace and inverse-limit topologies.

Proof. Every coordinate of Ω is the semiring map obtained from reduction, so the map is well-defined and preserves the operations and identities. For a compatible sequence (a_r, λ_r) , exact support preservation by every transition forces $\lambda_{r+1} = \lambda_r$ for every r . Call their common value λ . If $\lambda \neq \top$, membership in B_r forces $a_r = 0$ for every r , so the sequence is the image of $(0, \lambda)$. If $\lambda = \top$, the compatible coefficients give a unique $a \in A$ by (BLR2), and the sequence is the image of $(a, \top)$. This proves surjectivity without identifying any support labels. If two images agree, their labels agree and all residue coefficients agree. The coefficient limit then gives equality of the coefficients, proving injectivity.

The inverse-limit topology is generated by restrictions at finitely many levels. Under Ω , a restriction at level r fixes the support and a coefficient residue modulo $2^r A$. These sets are exactly the basic open sets of the subspace of $A \times L$, with L discrete. This proves the final assertion. $\square$

For every r there is a compatible multiplicative section

$$\sigma_r : G_L(\mathbb{F}_2) \longrightarrow B_r, \quad (a, \lambda) \longmapsto ([a] \bmod 2^r A, \lambda).$$

The identities $Q_{r+1,r}\sigma_{r+1} = \sigma_r$ follow by reduction, and their inverse limit is exactly σ . For $x = (a, \lambda), y = (b, \eta)$, the entire finite-level defect is

$$K_r(x, y) = (2[a][b] \bmod 2^r A, \lambda \vee \eta), \quad \sigma_r(x) + \sigma_r(y) = \sigma_r(x + y) + K_r(x, y).$$

These equations follow from (BLR9) by reduction. At $r = 1$ its coefficient is always zero, but its support is still $\lambda \vee \eta$. At every $r \geq 2$, the case $a = b = 1$ has the nonzero residue coefficient 2. Thus each truncation has been named, and none replaces the original coefficient equality in A .

Proposition 71. *The coefficient 2-derivation δ does not descend to a function $A/2^r A \rightarrow A/2^r A$ for any $r \geq 1$. It does give a well-defined function with the exact loss of one digit*

$$D_r : R_{r+1} \longrightarrow R_r, \quad \bar{a} \longmapsto \frac{a - a^2}{2} \bmod 2^r,$$

where a is any integer representative of $\bar{a}$. These functions satisfy

$$q_{r+1,r}D_{r+1} = D_r q_{r+2,r+1}.$$

They extend to support-preserving functions $\widehat{D}_r : B_{r+1} \rightarrow B_r$ by $\widehat{D}_r(a, \lambda) = (D_r(a), \lambda)$. Their coefficient inverse limit is the original $\delta : A \rightarrow A$.

Proof. The two elements $0, 2^r$ agree modulo $2^r A$, but

$$\delta(2^r) - \delta(0) = 2^{r-1}(1 - 2^r) \equiv 2^{r-1} \not\equiv 0 \pmod{2^r A}.$$

Thus descent at unchanged precision fails. For any $a, h \in A$,

$$\delta(a + 2^{r+1}h) - \delta(a) = 2^r h(1 - 2a) - 2^{2r+1}h^2 \in 2^r A.$$

This identity proves independence of the representative in D_r and also proves that evaluating on an integer representative agrees with reduction of the original δ on A . Evaluating either composite in the compatibility equation on a single integer representative gives the same residue, proving that equation. As $D_r(0) = 0$, the supported formula is well-defined: a nonzero output coefficient forces a nonzero input coefficient and therefore top input support. Compatibility and the equality of all coefficient residues prove the inverse-limit claim. $\square$

The maps D_r and $\widehat{D}_r$ are functions with the stated identities; they are not asserted to be ring or semiring homomorphisms. In particular, reduction in this tower was proved to preserve the ring and semiring operations, and was not declared a morphism of δ -rings.

Proposition 72. *Let $u, v \in A$ have all their binary digits retained:*

$$u = \sum_{i=0}^{\infty} 2^i[a_i], \quad v = \sum_{i=0}^{\infty} 2^i[b_i], \quad a_i, b_i \in \mathbb{F}_2.$$

Identify each digit with its integer representative 0 or 1 only in the following recurrence. Starting with $c_0 = 0$, define

$$t_i = a_i + b_i + c_i, \quad c_{i+1} = \lfloor t_i/2 \rfloor, \quad d_i = t_i - 2c_{i+1}. \quad (\text{BLR14})$$

Then $c_i, d_i \in \{0, 1\}$, and for every $r \geq 1$ the exact finite identity, including its boundary term, is

$$\sum_{i=0}^{r-1} 2^i[a_i] + \sum_{i=0}^{r-1} 2^i[b_i] = \sum_{i=0}^{r-1} 2^i[d_i] + 2^r c_r. \quad (\text{BLR15})$$

If $u_r, v_r \in A$ are the unique tails specified by

$$u = \sum_{i=0}^{r-1} 2^i[a_i] + 2^r u_r, \quad v = \sum_{i=0}^{r-1} 2^i[b_i] + 2^r v_r,$$

then the full equality in the original coefficient ring is

$$u + v = \sum_{i=0}^{r-1} 2^i[d_i] + 2^r(u_r + v_r + c_r). \quad (\text{BLR16})$$

Its limit is $u + v = \sum_{i=0}^{\infty} 2^i[d_i]$.

Proof. If $c_i \in \{0, 1\}$, then $0 \leq t_i \leq 3$, so its quotient and remainder on division by 2 are both in $\{0, 1\}$. Induction from $c_0 = 0$ gives the asserted digit ranges. Rearranging the defining identity gives $a_i + b_i = d_i + 2c_{i+1} - c_i$ as an equality of integers and therefore as an equality in A . Multiplying by 2^i and summing for $0 \leq i \leq r-1$ gives a telescoping sum in which the initial term is $-c_0 = 0$ and the final term is $2^r c_r$. This is (BLR15). The displayed definitions of u_r, v_r and injectivity of multiplication by 2^r give existence and uniqueness of the tails from Lemma 59. Adding those tail terms to (BLR15) proves (BLR16). The last term belongs to $2^r A$, so the limit of the partial digit sums is $u + v$ by completeness and separation. This limiting assertion follows from the exact finite identity; it does not omit its boundary term at any finite level. $\square$

For supported elements $(u, \lambda), (v, \eta) \in G_L(A)$, put $s = \lambda \vee \eta$. The full supported form of (BLR16) is

$$(u, \lambda) + (v, \eta) = \left(\sum_{i=0}^{r-1} 2^i [d_i], s \right) + (2^r (u_r + v_r + c_r), s). \quad (\text{BLR17})$$

Both terms lie in $G_L(A)$. If $s = \top$ this follows by definition; if $s \neq \top$, then neither input label is $\top$, so $u = v = 0$; every digit, carry, and tail is zero, and both terms are precisely z_s . Their sum is still z_s , since $s \vee s = s$. Thus (BLR17) retains supported zeros even when every arithmetic contribution vanishes.

104.7 What has, and has not, been transported

The construction consists of the following exact maps:

$$\begin{array}{ccccc} M & \xrightarrow{\hat{\mu}} & G_L(A) & \xrightarrow{\pi} & G_L(\mathbb{F}_2) \\ & & \downarrow \alpha_A & & \downarrow \alpha_{\mathbb{F}_2} \\ & & A & \xrightarrow{\rho} & \mathbb{F}_2, \end{array} \quad G_L(A) \xrightarrow[\Omega]{\sim} \varprojlim_r G_L(\mathbb{Z}/2^r \mathbb{Z}).$$

The left multiplicative map sends the source τ to $u_A = \hat{\mu}(\tau) \in G_L(A)$; its amplitude is $\alpha_A(u_A) = \mu(\tau) = 1_A$ in the rank-one locus. Reduction measures parity of these coefficients, while $\tau \in Z_1$ retains no parity. The section σ retains support and multiplication; its full failure to preserve addition is (BLR9). The full coefficient ring, its arithmetic carries, and every lattice label remain present.

For nontrivial L , $G_L(A)$ is not a ring, by the explicit failure of additive inverses proved above. Accordingly no assertion here applies the ordinary Witt functor or the Bhatt–Lurie prism definition directly to that semiring. Witt vectors and the prism occur on the coefficient ring A . The support construction is transported along the explicit ring maps by Proposition 67, and its full limit is proved in Theorem 70.

105 Bhatt–Lurie morphisms retaining the full support lattice

Notation, sources, and scope. The user’s notation is retained: Z_0 denotes absence, Z_1 ungraded presence represented by τ , and Z_2 the even/odd parity carried by integer elements. No Z_2 parity is assigned to τ . The coefficient ring of p -adic integers is written $\mathbf{Z}_p$; it is not the user’s state label Z_2 . The original split-support absorber is written τ_{old} , because its multiplication differs from that of the new unit.

The primary sources are Bhargav Bhatt and Jacob Lurie, *Absolute Prismatic Cohomology*, original `apc.tex`, Definition `definition:generalized-Cartier-divisor` (2057–2065), Definition `definition:Cartier-Witt-divisor` (2086–2095).

Remark `remark:arithmetic-lambda-line` (2108–2120), Construction `construction:Cartier-Witt-atlas` (2149–2157), and Remark `remark:WCart-lift-Frobenius` (2952–2959); and their *The prismaticization of p -adic formal schemes*, Construction `GenCartDivAnim` (286–301), Definition `DefAnimPrism` (318–343), and Example `CartWitttoAnim` (386–389). The local originals were matched byte for byte against the retained arXiv source archives. Reading these stated passages is not a claim to have read the complete 242-page first work.

The support carrier is retained from *Split Support Geometry*, version 11, source byline The Clankers, Definition `def:lattice-split`. The prime-character representation is reproved in FSI1–3 in the accompanying source. The construction below is a morphism into the stated Cartier–Witt moduli problem, not an assertion that an arbitrary semiring is already a prism or that the new unit is the historical absorber.

105.1 The affine-line morphism with its complete input

For a commutative ring R , the groupoid $[\mathbf{A}^1/\mathbf{G}_m](R)$ classifies pairs (J, β) with J an invertible R -module and $\beta : J \rightarrow R$ linear. A morphism to (J', β') is an isomorphism $u : J \rightarrow J'$ such that $\beta = \beta' u$. For free lines with coordinates $a, b \in R$, this condition is

$$u \in R^\times, \quad a = bu. \quad (\text{BLM1})$$

The inverse morphism has coordinate u^{-1} ; composition is multiplication, which preserves BLM1. In particular the automorphism group of (R, a) is

$$\{u \in R^\times : a(u - 1) = 0\}. \quad (\text{BLM2})$$

At $a = 1$ it is trivial, and at $a = 0$ it is $R^\times$. The zero map from a line is therefore not the absent line. This is an exact property of the moduli object, with no inference that the user’s Z_0 is a line or that τ carries parity.

Fix a prime p and a ring R on which p is nilpotent. Write $W(R)$ for its p -typical Witt ring and $w_0 : W(R) \rightarrow R$ for its first-coordinate homomorphism. A Cartier–Witt divisor (I, α) has I invertible over $W(R)$, nilpotent image of $w_0 \alpha$, and unit ideal generated by $\delta \alpha(I)$. Bhatt–Lurie’s affine-line morphism is explicitly

$$\begin{aligned} \mu_R : \text{WCart}(R) &\longrightarrow [\widehat{\mathbf{A}^1}/\mathbf{G}_m](R), \\ (I, \alpha) &\longmapsto (R \otimes_{W(R)} I, \ r \otimes x \mapsto r w_0(\alpha x)). \end{aligned} \quad (\text{BLM3})$$

The formula is balanced because w_0 is a ring homomorphism. Tensoring an isomorphism u of input lines gives an isomorphism of output lines and preserves $\alpha = \alpha' u$, so this is a functor on the stated groupoids. For a ring map $R \rightarrow S$, extension of scalars along $W(R) \rightarrow W(S)$ followed by w_0 agrees with extension along $R \rightarrow S$. The elementary tensor associativity isomorphism identifies both outputs and their maps. This proves the naturality square. The image section is nilpotent by the first Cartier–Witt condition; hence the formal completion is retained. In the arXiv source, the direction printed in Remark `remark:functoriality-WCart`, lines 2127–2129, is reversed relative to extension of scalars; the tensor formula just given fixes the direction explicitly, consistently with lines 2138–2140.

105.2 All support labels enter a compatible prismatic family

Let L be a nontrivial bounded distributive lattice. Let X be the set of all proper prime lattice ideals and define $\chi_\lambda(P) = 0$ for $\lambda \in P$ and 1 otherwise. Prime separation implies that $\lambda \mapsto \chi_\lambda$ is injective. Here is its separation argument: if $a \not\leq b$, maximize an ideal containing b and disjoint from the filter $\uparrow a$ by Zorn's lemma. If x, y both lie outside that ideal but $x \wedge y$ lies in it, maximality gives $a \leq u \vee x$ and $a \leq v \vee y$ with u, v in the ideal. Distributivity then implies $a \leq u \vee v \vee (x \wedge y)$, a contradiction. Thus that ideal is prime and separates a, b . The full characteristic identities, valid also in characteristic p with the displayed subtraction retained, are

$$\chi_{0_L} = 0, \quad \chi_{1_L} = 1, \quad \chi_{\lambda \wedge \nu} = \chi_\lambda \chi_\nu, \quad \chi_{\lambda \vee \nu} = \chi_\lambda + \chi_\nu - \chi_\lambda \chi_\nu. \quad (\text{BLM4})$$

They follow at each prime from its join and meet conditions.

For every integer $N \geq 2$, put

$$B = \text{Map}(X, \mathbf{F}_p), \quad R_N = B[T]/(T^N), \quad a_{\lambda, N} = T\chi_\lambda. \quad (\text{BLM5})$$

All prime coordinates of L occur simultaneously. Polynomial coefficients in R_N are unique in degrees $0, \dots, N-1$, so distinct χ_λ give distinct $a_{\lambda, N}$. The exact support formulas are

$$\begin{aligned} a_{\lambda \vee \nu, N} &= a_{\lambda, N} + a_{\nu, N} - a_{\lambda \wedge \nu, N}, \\ a_{\lambda, N} a_{\nu, N} &= T a_{\lambda \wedge \nu, N}. \end{aligned} \quad (\text{BLM6})$$

These are obtained by multiplying BLM4 by T and T^2 , respectively. In particular the extra factor T is not discarded. This is not a claim that $\lambda \mapsto a_{\lambda, N}$ is a meet-to-product homomorphism.

Define a free-line Cartier–Witt object by

$$d_{\lambda, N} = [a_{\lambda, N}] + V(1), \quad \alpha_{\lambda, N} : W(R_N) \xrightarrow{d_{\lambda, N}} W(R_N). \quad (\text{BLM7})$$

The bracket denotes the Teichmüller map and V the Verschiebung. The Witt coordinates of this element are exactly $(a_{\lambda, N}, 1, 0, 0, \dots)$. To check this without cancelling in a torsion ring, check the ghost polynomials over the universal torsion-free polynomial ring: both expressions have zeroth ghost coordinate a and n th coordinate $a^{p^n} + p$ for $n \geq 1$. Ghost injectivity there identifies their universal coordinates; specialization gives the equality over R_N . Its first coordinate is nilpotent and its second is a unit. Thus it lies in Bhatt–Lurie's Cartier–Witt atlas and BLM7 satisfies their divisor definition. Their morphism gives

$$\mu_{R_N}(W(R_N), \alpha_{\lambda, N}) = (R_N, m_{T\chi_\lambda}). \quad (\text{BLM8})$$

This calculates the image, rather than identifying a label with an unspecified point of an affine line.

For $2 \leq M \leq N$, the quotient $R_N \rightarrow R_M$ sends $a_{\lambda, N}$ to $a_{\lambda, M}$. Witt functoriality sends $d_{\lambda, N}$ to $d_{\lambda, M}$, because it preserves Teichmüller representatives and V . Tensoring BLM7 therefore gives BLM7 at level M , including the map on its line. We obtain a compatible formal family at every support label.

105.3 Exact separation and automorphisms after the quotient

If the two outputs in BLM8 were isomorphic, BLM1 would give $T\chi_\lambda = T\chi_\nu u$ with $u \in R_N^\times$. At each coordinate $P \in X$, the constant coefficient of $u(P)$ is nonzero. Comparing coefficients of T shows $\chi_\lambda(P) = \chi_\nu(P)u_0(P)$. This is impossible if exactly one of the two characteristic values is zero. Thus the two supports coincide and prime separation gives $\lambda = \nu$. Conversely equal labels give the identity isomorphism. Therefore

$$\lambda \longmapsto [(W(R_N), \alpha_{\lambda, N})] \quad \text{is injective on isomorphism classes.} \quad (\text{BLM9})$$

The conclusion for Cartier–Witt objects follows because their images under the functor μ are already separated. We do not call a functor from discrete labels fully faithful: target automorphisms need not be trivial.

The automorphisms of the output line are calculated by BLM2:

$$\mathrm{Aut}(R_N, m_{T\chi_\lambda}) = \{u \in R_N^\times : T\chi_\lambda(u-1) = 0\}. \quad (\text{BLM10})$$

On the factor $\chi_\lambda B$, this means $u = 1 + T^{N-1}v$ with $v \in \chi_\lambda B$; on the complementary factor $(1 - \chi_\lambda)B$, any unit is permitted. This follows by comparing all coefficients of $T, \dots, T^{N-1}$; the idempotent decomposition of B separates the two factors. These are the full automorphisms in the affine-line quotient; no assertion that all of them lift to Cartier–Witt automorphisms is made.

105.4 Frobenius moves the retained label into higher order

On a characteristic- p ring the Witt Frobenius is $W(x \mapsto x^p)$, and $p = V(1)$. Thus F acts coordinatewise by p th powers, fixes p , and preserves the displayed formal family:

$$F^j(d_{\lambda,N}) = p + [T^{p^j}\chi_\lambda], \quad j \geq 0. \quad (\text{BLM11})$$

Indeed $\chi_\lambda^{p^j} = \chi_\lambda$ at every coordinate, and $F^j([a]) = [a^{p^j}]$. This proves the formula including the constant p and the complete order p^j . If $p^j \geq N$, every element in BLM11 equals p . If $p^j < N$, their affine-line images $(R_N, m_{T^{p^j}\chi_\lambda})$ are still pairwise nonisomorphic, by the coefficient argument of BLM9 in degree p^j . Hence

$$\begin{cases} F^j(d_{\lambda,N}) = p \text{ for all } \lambda, & p^j \geq N, \\ \lambda \mapsto [F^j(d_{\lambda,N})] \text{ is injective,} & p^j < N. \end{cases} \quad (\text{BLM12})$$

For each fixed j , the compatible family over all N retains every label because one can take $N > p^j$. A fixed finite truncation and the entire formal family therefore have different Frobenius information, with the loss threshold computed exactly.

105.5 A nonzero derived class at every support label

Every BLM7 has the following explicit annihilator element:

$$w_N = [T^{N-1}] \neq 0, \quad d_{\lambda,N}w_N = 0. \quad (\text{BLM13})$$

Its first Witt coordinate is nonzero, proving $w_N \neq 0$. The identities $[a][b] = [ab]$ and $V(1)[b] = V([b^p])$ give

$$\begin{aligned} d_{\lambda,N}w_N &= [T\chi_\lambda][T^{N-1}] + V(1)[T^{N-1}] \\ &= [T^N\chi_\lambda] + V([T^{p(N-1)}]) = 0 + 0 = 0, \end{aligned} \quad (\text{BLM14})$$

since $N \geq 2$, $p \geq 2$, and $p(N-1) \geq N$. All exponents and summands remain displayed. The Witt product identity used here follows from $V(x)y = V(xF(y))$; it can be checked on universal ghost polynomials and specialized as above.

The animated quotient of Bhatt–Lurie is the derived quotient

$$\overline{A}_{\lambda,N} = W(R_N) \parallel d_{\lambda,N}, \quad K_{\lambda,N} = [W(R_N) \xrightarrow{d_{\lambda,N}} W(R_N)], \quad (\text{BLM15})$$

with the left module in homological degree one. To see its underlying complex, form the derived tensor product of $W(R_N)[x]/(x)$ and $W(R_N)$ over $W(R_N)[x]$, where the second map sends x to $d_{\lambda,N}$. The two-term free resolution by multiplication by x is exact because x is a non-zero-divisor

in the polynomial ring, regardless of torsion in its coefficients. Tensoring gives exactly BLM15. It follows that

$$\pi_0 \bar{A}_{\lambda,N} = W(R_N)/(d_{\lambda,N}), \quad \pi_1 \bar{A}_{\lambda,N} = \text{Ann}(d_{\lambda,N}) \ni w_N \neq 0, \quad \pi_i = 0 \ (i \geq 2). \quad (\text{BLM16})$$

The Cartier–Witt source gives an animated prism as in their Example `CartWitttoAnim`; the explicit nonzero class also proves why it cannot be replaced by an injective classical Cartier divisor. This is a computed derived contribution, not a proposed future comparison.

The coefficientwise Frobenius defines an actual chain map

$$K_{\lambda,N} \longrightarrow K_{\lambda,N}^{(j)} = [W(R_N) \xrightarrow{F^j d_{\lambda,N}} W(R_N)], \quad (F^j, F^j), \quad (\text{BLM17})$$

because $F^j(dx) = F^j(d)F^j(x)$. The witness is sent to $[T^{p^j(N-1)}]$, which is zero for every $j \geq 1$. Thus this particular nonzero degree-one class is killed by Frobenius, with both its source and target complexes specified.

105.6 The return to the original full split-support carrier

Put $A = W(R_N)$ and retain every pair in

$$G_L(A) = (\{0\} \times L) \cup (A \times \{1_L\}), \quad (a, \sigma) + (b, \eta) = (a + b, \sigma \vee \eta), \quad (a, \sigma)(b, \eta) = (ab, \sigma \wedge \eta). \quad (\text{BLM18})$$

The scalar maps $\mathbf{Z} \rightarrow A \xrightarrow{w_0} R_N$ induce the exact semiring morphisms

$$G_L(\mathbf{Z}) \longrightarrow G_L(A) \longrightarrow G_L(R_N), \quad (n, \sigma) \longmapsto (n1_A, \sigma) \longmapsto ((n \bmod p)1_{R_N}, \sigma). \quad (\text{BLM19})$$

They preserve join, meet, amplitudes, global zero and unit by direct substitution in BLM18. Nonzero amplitudes of an image still occur only at top support. Every $(0, \sigma)$ is retained as $(0, \sigma)$, even when a nonzero integer is annihilated by the last amplitude map. At $p = 2$ this is the arithmetic Z_2 receiver with all labels kept. The first map is injective: evaluation at any $P \in X$ and $T = 0$ gives $A \rightarrow W(\mathbf{F}_p) = \mathbf{Z}_p$, where the integer inclusion is injective. Equality of images also forces equality of support labels.

The new ungraded presence unit acts through the pointed multiplicative morphism

$$\{0, \tau\} \longrightarrow G_L(A), \quad 0 \longmapsto (0, 0_L) = \tau_{\text{old}}, \quad \tau \longmapsto (1, 1_L). \quad (\text{BLM20})$$

All four products of $\{0, \tau\}$ verify this multiplicative map. It is not asserted to preserve the presence addition: the source has $\tau + \tau = \tau$, whereas its receiving images add to $(2, 1_L)$. This is an image of τ , not an assignment of parity to its source. The image acts as identity on every label. This is scalar action on the receiving module; it does not identify τ with the nilpotent section $T\chi_\lambda$ of the formal affine-line object. Multiplication by zero is a module endomorphism, not an isomorphism of an invertible line.

Lift the differential of BLM15 without dropping supports:

$$D_{\lambda,N} : G_L(A) \rightarrow G_L(A), \quad D_{\lambda,N}(x, \sigma) = (d_{\lambda,N}x, \sigma). \quad (\text{BLM21})$$

It preserves sums and scalar action: both sides of each identity have the same join or meet label and the same ordinary linear amplitude. Composition with amplitude projection is exactly multiplication by $d_{\lambda,N}$. For the nonzero class in BLM13 it gives

$$D_{\lambda,N}(w_N, 1_L) = (0, 1_L) = e \neq \tau_{\text{old}}. \quad (\text{BLM22})$$

The complete amplitude-zero fibre is

$$\{(x, \sigma) : d_{\lambda,N}x = 0\} = G_L(\text{Ann } d_{\lambda,N}), \quad (\text{BLM23})$$

where the right side is the supported additive-module carrier, with nonzero x allowed only at 1_L . In contrast, $D_{\lambda,N}^{-1}(\tau_{\text{old}}) = \{\tau_{\text{old}}\}$ for nontrivial L : an output at bottom must have input label bottom, and the original carrier then forces amplitude zero. Thus the ordinary global-zero fibre would miss the computed class, whereas the supported-zero fibre retains it. Coefficientwise Frobenius also lifts as $(x, \sigma) \mapsto (F^j x, \sigma)$ and commutes with BLM21 into the lifted BLM17. Even when $F(w_N) = 0$, its image is $(0, 1_L)$ and retains its support.

For the one-element lattice the split carrier is simply A ; there are no distinct support labels or prime characters to embed. Taking $R_N = \mathbf{F}_p[T]/(T^N)$ directly gives the same arithmetic and annihilator calculation. In that case $e = \tau_{\text{old}}$, so the distinction in BLM22–23 is explicitly absent, exactly as the original carrier requires.

105.7 The complete annihilator and its automorphisms

The witness BLM13 is part of an explicitly computable whole module. For $q \geq 1$, an idempotent $c \in B$, and $N \geq 2$, write

$$d_{N,q,c} = p + [T^q c], \quad h_N = \lceil N/p \rceil. \quad (\text{BLM24})$$

For a Witt vector $x = (x_0, x_1, \dots)$ the exact answer is

$$\begin{aligned} d_{N,q,c} x = 0 &\iff x_n \in T^{h_N} R_N \text{ and } c T^{qp^n} x_n = 0 \text{ for every } n \geq 0, \\ \text{Ann}(d_{N,q,c}) &= \text{Ann}(p) \cap \text{Ann}([T^q c]). \end{aligned} \quad (\text{BLM25})$$

In particular, on cR_N the n th coordinate belongs to $T^{\max(h_N, N-qp^n, 0)} cR_N$; on $(1-c)R_N$ it belongs to $T^{h_N}(1-c)R_N$. The letter c here denotes an idempotent coefficient and does not change the supported-zero symbol e .

Here is a proof including the possible cancellation between the two summands of d . Frobenius on R_N sends $\sum b_i T^i$ to $\sum b_i T^{pi}$, because $b_i^p = b_i$ in B . Its kernel is therefore precisely $T^{h_N} R_N$. The identities

$$px = V(Fx), \quad [T^q c]x = (T^q c x_0, T^{qp} c x_1, T^{qp^2} c x_2, \dots) \quad (\text{BLM26})$$

and injectivity of V prove both annihilator descriptions on the right of BLM25. They immediately prove that those conditions are sufficient for $dx = 0$.

For necessity, split the ring by the idempotent c . On the complement $d = p$, already handled. On the c factor write $d_q = p + [T^q]$. If $q \geq N$, again $d_q = p$. Otherwise the zeroth coordinate of $d_q x = 0$ gives $T^q x_0 = 0$, so $x_0 \in T^{N-q} R_N$. If $pq < N$, then $p(N-q) > (p-1)N \geq N$ and hence $x_0^p = 0$. Writing $y = (x_1, x_2, \dots)$, the exact identities

$$\begin{aligned} x &= [x_0] + V(y), \quad [T^q]V(y) = V([T^{pq}]y), \quad p[x_0] = V([x_0^p]), \\ d_q x &= [T^q x_0] + V([x_0^p]) + V(d_{pq} y) = V(d_{pq} y) \end{aligned} \quad (\text{BLM27})$$

reduce the assertion to the next coordinate. Repetition reaches $q' < N \leq pq'$. At this stage $T^{q'} x'_0 = 0$ and $T^{pq'} = 0$, so the full vector $[T^{q'}]x'$ vanishes, by BLM26. The equation now forces $px' = 0$, and all remaining coordinates have p th power zero. The preceding steps also gave the required powers and the coordinate equations $T^{qp^n} x_n = 0$ at each earlier index. This proves necessity and BLM25 without assuming the two summands cannot cancel.

For a chosen free-line divisor, an automorphism after forgetting the chosen basis is a Witt unit v satisfying $dv = d$. Thus

$$\text{Aut}(W(R_N), m_{d_{N,q,c}}) = 1 + \text{Ann}(d_{N,q,c}). \quad (\text{BLM28})$$

Every displayed $1 + x$ is a unit: its zeroth coordinate is $1 + x_0$ with $x_0 \in T^{h_N} R_N$ nilpotent. A Witt vector is invertible exactly when its zeroth coordinate is invertible; successive multiplication equations determine its inverse coordinate by coordinate. For the family BLM7 the element

$1 + [T^{N-1}]$ is a nonidentity automorphism. Thus every finite stage has genuinely nontrivial stabilizer, calculated by BLM25.

At $p = 2, N = 2$ the whole calculation reads

$$\begin{aligned} d_{\lambda,2} &= 2 + [T\chi_\lambda], & \text{Ann}(d_{\lambda,2}) &= \{(x_n)_{n \geq 0} : x_n \in TR_2 \ \forall n\}, \\ [T] &\neq 0, & d_{\lambda,2}[T] &= [T^2\chi_\lambda] + V([T^2]) = 0, & F(d_{\lambda,2}) &= 2, & F([T]) &= 0. \end{aligned} \quad (\text{BLM29})$$

The coefficient class in degree one is nonzero before Frobenius. Its lifted differential still ends at the supported zero in BLM22.

105.8 A fully faithful map into the compatible formal towers

Fix $j \geq 0$. Form the groupoid whose objects are compatible systems of Cartier–Witt objects over the rings R_N , $N \geq 2$, with specified transition isomorphisms along the truncations. A morphism is a family of isomorphisms commuting with these transitions. The systems BLM11 have their canonical transitions from BLM7. There is a functor from the discrete groupoid on the set L :

$$\mathcal{D}_j : L_{\text{disc}} \longrightarrow \varprojlim_{N \geq 2} \text{WCart}(R_N), \quad \lambda \longmapsto (W(R_N), m_{p+[T^{p^j}\chi_\lambda]})_N. \quad (\text{BLM30})$$

The limit denotes the just specified groupoid of compatible systems. This functor is fully faithful.

Indeed distinct labels admit no isomorphism because any stage $N > p^j$ separates them by BLM12. An automorphism of a single system is represented by compatible units $1 + x_N$, with $x_N \in \text{Ann}(p + [T^{p^j}\chi_\lambda])$. Fix a Witt-coordinate index n . The compatible coefficients $(x_N)_n$ give an element of $B[[T]]$. Formula BLM25 forces it to vanish to order at least $[N/p]$ for every N . Every power-series coefficient is therefore zero. This holds for each Witt coordinate, giving $x_N = 0$ for every N . The identity is the only automorphism. It is the only map from each labelled object to itself in the discrete source as well. The two Hom sets consequently agree for every pair of labels, which proves the assertion.

There is an essential domain statement. For $\lambda \neq 0_L$, prime separation provides a P with $\chi_\lambda(P) = 1$. Evaluation at P proves that $T^{p^j}\chi_\lambda$ is not nilpotent in $B[[T]]$. Therefore the inverse-limit Witt vector does not define an ordinary Cartier–Witt point over $B[[T]]$ with the nilpotence condition in BLM3. BLM30 is a morphism into compatible formal towers, exactly as its codomain states. At every finite stage that nilpotence condition is satisfied.

Calculated scope. The resulting arrows reach Bhatt–Lurie’s actual moduli objects and animated quotients, retain every original lattice label, separate their deformations, and calculate a nonzero derived class with its Frobenius image. They do not identify these classes with the programme’s arithmetic spectral cohomology or assign them a Deligne weight. No positivity statement is inferred from the existence of these morphisms.

106 The exact comparison with earlier layered constructions

The bounded prior-work search found a precise comparison, not an established identification of the full typed programme with earlier work. It did not establish historical novelty. The following calculation uses the carrier and operation formulas from Zur Izhakian, Manfred Knebusch and Louis Rowen, arXiv:0912.1398v3, Construction **defn5** (original source lines 1325–1341), with the one-versus-two calculation in Proposition **cancel11** (lines 1366–1374). The root task read these original TeX passages; this independent calculation uses their full operation formulas. We apply those formulas to the explicitly specified auxiliary monoid below and check every required identity. The letters t, u are auxiliary monoid elements, not names for the user's typed objects. The algebra C considered here is the positive-count restriction of the integer extension chosen in the FMC construction. This comparison neither asserts that this extension is uniquely forced by the original rules nor assigns a parity to τ .

Let $\mathbb{N}_{>0} = \{1, 2, 3, \dots\}$. Give $G = \{t, u\}$ the order $t < u$ and multiplication

$$tt = t, \quad tu = ut = u, \quad uu = u. \quad (\text{LPC1})$$

This multiplication is associative and commutative, with identity t : a product is t exactly when every factor is t , and is u otherwise. It preserves the order in each argument. It is noncancellative, since $tu = uu$ while $t \neq u$.

Define the carrier $H = (\mathbb{N}_{>0} \times G) \sqcup \{0\}$. The element 0 is an additive identity and a multiplicative absorber. For positive integers k, l , the two operations on the other elements are

$$(k, a)(l, b) = (kl, ab), \quad (k, a) + (l, b) = \begin{cases} (k, a), & a > b, \\ (l, b), & a < b, \\ (k + l, a), & a = b. \end{cases} \quad (\text{LPC2})$$

These are definitions of a two-operation algebra. Its multiplication is associative and commutative by the corresponding facts for positive integers and for G , including the prescribed rule for 0. Its multiplicative identity is $(1, t)$. Its addition is commutative. To prove associativity of addition, consider three nonzero elements (k_i, a_i) , $1 \leq i \leq 3$, and put $a = \max\{a_1, a_2, a_3\}$. Both parenthesizations give $(\sum_{i:a_i=a} k_i, a)$: if the first two labels are both below a , the third term is retained; if exactly one of the first two has label a , that term is retained before the third is added; and if both have label a , their counts are added before the third is added. These cases also cover the other parenthesization after interchanging the first and third terms. If a term is 0, the additive-identity rule reduces the assertion to two terms. This proves associativity in every case.

Define the disjoint union $C = \{0, \tau\} \sqcup \mathbb{N}_{>0}$. For positive integers n, m , use their ordinary integer sum and product, and set

$$\begin{aligned} 0 + x &= x + 0 = x, & 0x &= x0 = 0 \quad (x \in C), \\ \tau + \tau &= \tau, & \tau + n &= n + \tau = n, \\ \tau\tau &= \tau, & \tau n &= n\tau = n. \end{aligned} \quad (\text{LPC3})$$

The parity label is attached to each positive integer n by its integer parity. In particular, τ has no parity label.

Both operations on C are commutative. For associativity of addition, if at least one of three terms is a positive integer, either parenthesization gives the ordinary sum of precisely the positive integer terms; 0 and τ contribute according to the displayed rules. If there is no positive integer term but there is a τ , both parenthesizations give τ . If all terms are zero, both give zero. These cases exhaust all triples. For associativity of multiplication, a zero factor makes both parenthesizations zero. Among nonzero factors, if there is a positive integer factor, both parenthesizations give the ordinary product of precisely the positive integer factors; if all factors are τ , both give τ . Thus multiplication is also associative. Its identity is τ .

Define maps of carriers $Q : H \longrightarrow C$ and $S : C \longrightarrow H$ by

$$\begin{aligned} Q(0) &= 0, & Q(k, t) &= \tau, & Q(k, u) &= k, \\ S(0) &= 0, & S(\tau) &= (1, t), & S(n) &= (n, u). \end{aligned} \quad (\text{LPC4})$$

Every value is in the indicated carrier. Direct substitution gives

$$\begin{aligned} Q \circ S &= \text{id}_C, & (S \circ Q)(0) &= 0, \\ (S \circ Q)(k, t) &= (1, t), & (S \circ Q)(k, u) &= (k, u). \end{aligned} \quad (\text{LPC5})$$

Thus Q is surjective, S is injective, and $S \circ Q$ is idempotent. All fibres of these maps are

$$\begin{aligned} Q^{-1}(\{0\}) &= \{0\}, & Q^{-1}(\{\tau\}) &= \{(k, t) : k \in \mathbb{N}_{>0}\}, & Q^{-1}(\{n\}) &= \{(n, u)\}, \\ S^{-1}(\{0\}) &= \{0\}, & S^{-1}(\{(1, t)\}) &= \{\tau\}, \\ S^{-1}(\{(k, t)\}) &= \emptyset \quad (k \geq 2), & S^{-1}(\{(n, u)\}) &= \{n\}. \end{aligned} \quad (\text{LPC6})$$

The map Q preserves addition in every case. For nonzero terms the complete calculation, up to interchanging the terms, is

$$\begin{aligned} Q((k, t) + (l, t)) &= Q(k + l, t) = \tau = \tau + \tau, \\ Q((k, u) + (l, u)) &= Q(k + l, u) = k + l, \\ Q((k, t) + (l, u)) &= Q(l, u) = l = \tau + l. \end{aligned} \quad (\text{LPC7})$$

An input zero gives the same assertion by the additive-identity rules.

The exact comparison of products is

(x, y)	$Q(xy)$	$Q(x)Q(y)$
$((k, t), (l, t))$	τ	τ
$((k, u), (l, u))$	kl	kl
$((k, t), (l, u))$	kl	l
$((k, u), (l, t))$	kl	k
$(0, y) \text{ or } (x, 0)$	0	0

(LPC8)

The third row agrees exactly when $k = 1$, since l is a positive integer. The fourth agrees exactly when $l = 1$, since k is a positive integer. Hence all failures of multiplicativity are precisely the mixed-label products whose t -labelled factor has count at least two. The product of two t -labelled factors is preserved.

The map S preserves multiplication in every case:

$$\begin{aligned} S(\tau\tau) &= (1, t) = (1, t)(1, t) = S(\tau)S(\tau), \\ S(\tau n) &= (n, u) = (1, t)(n, u) = S(\tau)S(n), \\ S(nm) &= (nm, u) = (n, u)(m, u) = S(n)S(m). \end{aligned} \quad (\text{LPC9})$$

Commutativity covers the reversed mixed product, and a zero factor is covered by the absorber rule. Its additive comparison is

$$\begin{aligned} S(\tau + \tau) &= (1, t), & S(\tau) + S(\tau) &= (2, t), \\ S(\tau + n) &= (n, u), & S(\tau) + S(n) &= (n, u), \\ S(n + m) &= (n + m, u), & S(n) + S(m) &= (n + m, u). \end{aligned} \quad (\text{LPC10})$$

An input zero is again preserved. Since $(1, t) \neq (2, t)$, the ordered pair (τ, τ) is the only failure of additivity of S .

There is an explicit failure of distributivity in H :

$$\begin{aligned}(1, u)((1, t) + (1, u)) &= (1, u)(1, u) = (1, u), \\ (1, u)(1, t) + (1, u)(1, u) &= (1, u) + (1, u) = (2, u).\end{aligned}\tag{LPC11}$$

The distinct labels $t < u$ become equal after multiplication by u , because $ut = uu = u$; the two resulting counts on the second line are therefore added. This is the exact cancellation failure responsible for the displayed discrepancy. There is also an explicit failure of distributivity in C :

$$1(\tau + 1) = 1 \cdot 1 = 1, \quad 1\tau + 1 \cdot 1 = 1 + 1 = 2.\tag{LPC12}$$

Both calculations follow directly from the specified operations. Consequently neither H nor C is a semiring, while each still has the associative and commutative operations proved above. No distributive law was used to define the user's typed rules.

Finally, the fibres of Q form an additive congruence, since $Q(x + y) = Q(x) + Q(y)$. They do not form a multiplicative congruence: for every $k \geq 2$,

$$\begin{aligned}Q(k, t) &= Q(1, t) = \tau, \\ Q((k, t)(1, u)) &= k \neq 1 = Q((1, t)(1, u)).\end{aligned}\tag{LPC13}$$

Thus this identification of all t -labelled counts cannot give a quotient preserving both operations of H . Its exact surviving relationships are the additive map Q , the multiplicative map S , the identity $Q \circ S = \text{id}_C$, and the complete failures recorded in (LPC8) and (LPC10).

107 Withdrawal of the earlier total-addition interpretation

The user has explicitly retracted the earlier additive equation on presence. Addition is not supplied at the Z_1 stage. The former TAC calculations used an agent-chosen total operation and are not definitions or consequences of the intended source. The previous source and figures are preserved only in the private correction archive. In particular, no distributivity, associativity, ideal, or receiver obstruction derived from that total-addition interpretation is carried forward as a programme conclusion. The current accepted translation theorem is given separately, with no addition on τ .

108 Integer translations and the identity-only symmetry of Z_1

The user's notation is retained: Z_0 is absence and zero has no parity; Z_1 is presence and no addition on τ is supplied; Z_2 is the even/odd grading of nonzero integers. The user has retracted the earlier additive interpretation of presence. It is not a premise here.

Theorem 73 (TRZ1: integer addition as translations). *For $m \in \mathbb{Z}$, put $T_m(n) = n + m$ for all $n \in \mathbb{Z}$, and set*

$$\text{Trans}(\mathbb{Z}) = \{T_m : m \in \mathbb{Z}\}.$$

This is a subgroup of $\text{Bij}(\mathbb{Z})$ under composition, with

$$T_m \circ T_k = T_{m+k}, \quad T_m^{-1} = T_{-m}, \quad T_0 = \text{id}.$$

The map

$$\Phi : (\mathbb{Z}, +) \longrightarrow (\text{Trans}(\mathbb{Z}), \circ), \quad m \longmapsto T_m$$

is an isomorphism, with inverse $T \mapsto T(0)$ on the translation subgroup. The translation action is faithful, and each nonidentity translation has infinite order.

Proof. For all $m, k, n \in \mathbb{Z}$,

$$(T_m \circ T_k)(n) = (n + k) + m = n + (m + k) = T_{m+k}(n).$$

This uses ordinary integer associativity and commutativity. Taking $k = -m$ in both orders proves the inverse formulas. Thus every T_m is bijective and the family is a subgroup. The composition formula proves that Φ is a homomorphism; it is surjective by definition. If $T_m = T_k$, evaluation at zero gives $m = T_m(0) = T_k(0) = k$, proving injectivity. Both inverse identities are

$$\text{ev}_0(\Phi(m)) = m, \quad \Phi(\text{ev}_0(T_m)) = T_m.$$

A translation acting identically satisfies $m = T_m(0) = 0$, so the action is faithful. Induction gives $T_m^r = T_{rm}$ for every positive integer r . If $m \neq 0$, then $rm \neq 0$, hence this is not T_0 . No parity of zero has been used. $\square$

Theorem 74 (TRZ2: no faithful factorization through identity-only symmetry). *Let $I_{Z_1} = \{\text{id}\}$ be the group consisting only of the stipulated identity symmetry of Z_1 . No group homomorphism*

$$\text{Trans}(\mathbb{Z}) \longrightarrow I_{Z_1}$$

is injective. Every group homomorphism from $\text{Trans}(\mathbb{Z})$ to any group H factoring through I_{Z_1} is trivial, and therefore not faithful.

Proof. Both T_0 and T_1 have the sole available image, although $T_0(0) = 0 \neq 1 = T_1(0)$. A homomorphism from the identity group into H sends its only element to the identity of H , so every composite through it is trivial. $\square$

The exact programme conclusion is that the existing identity-only symmetry cannot faithfully represent integer addition through TRZ1. A distinguishable nonidentity translation is additional transformation data. This does not prohibit adding structure externally or imposing a binary operation on a carrier.

Theorem 75 (TRZ3: parity compatibility with its exact domain). *Put*

$$D = \mathbb{Z} \setminus \{0\}, \quad D_1 = \mathbb{Z} \setminus \{-1, 0\}, \quad D_2 = \mathbb{Z} \setminus \{0, 1\}.$$

Let $\pi : D \rightarrow Z_2 = \{\text{even}, \text{odd}\}$ assign the ordinary parity of each nonzero integer, and let σ interchange these two labels. The restriction $T_1 : D_1 \rightarrow D_2$ is a bijection and

$$\pi \circ (T_1|_{D_1}) = \sigma \circ (\pi|_{D_1}).$$

Proof. Translation by one maps D_1 onto D_2 , with inverse translation by minus one. If $n = 2q$, then $n + 1 = 2q + 1$ is odd. If $n = 2q + 1$, then $n + 1 = 2(q + 1)$ is even. All inputs and outputs here are nonzero, so no parity at zero is assigned. $\square$

This is a compatibility identity rather than an isomorphism with the parity states: $\pi(1) = \pi(3)$ while $1 \neq 3$.

TRZ4: the three-state observation. On distinct states $0, a, b$, the permutation $R(0) = a$, $R(a) = b$, $R(b) = 0$ has $R^3 = \text{id}$, whereas $R(0) = a \neq 0$ and $R^2(0) = b \neq 0$. It therefore already supplies a nonidentity symmetry of order three. It does not derive that symmetry from the identity-only symmetry of Z_1 , and no parity of the three states is needed.

Scope and provenance. TRZ1 begins with integer addition; it does not construct addition from presence or parity. No theorem about all meanings of $\mathbb{F}_1$, no prismatic comparison, and no claim that every addition requires Z_2 is made. The source's lack of addition at Z_1 is stipulated, not inferred from the absence of nonidentity symmetries alone.

This is the classical regular action of an additive group, not a new group-theoretic result. For historical human provenance, see Arthur Cayley, *On the theory of groups, as depending on the symbolic equation $\theta^n = 1$* , Philosophical Magazine, Series 4, 7(42) (1854), 40–47, doi:10.1080/14786445408647421. Publisher metadata was checked; no original author L^AT_EX was available in that record, and the paper was not read in full. No proof is imported from an unread source: all arguments used here are complete above. The programme framing and notation come from the user's current request; this task wrote the proof and obtained an independent mathematical check.

109 Source operations and their exact arithmetic consequences

24 September 2026. Agent-authored formal interpretation and complete elementary proofs. The user's wording is in `USER_DEFINITIONS_VERBATIM.md` (private source record). References U01–U18 below identify those exact quotations. No historical novelty is claimed for the elementary algebra. The source interpretation is explicit, so a correction to it can be applied without rewriting the user record.

109.1 Scope and order of construction

This defines the arithmetic operations currently specified for tau and the integer layer. It does not construct the full support lattice, an involution geometry, a Cartan involution, a metric, a vector representation, a pairing, reflection positivity, a Deligne weight, or an RH conclusion. The plank is a metaphor, as U09 says. No theorem below substitutes that metaphor for source data.

The latest instructions U15–U16 authorize continued mathematics without an approval gate. They require the definition to precede its use. The exact geometric production of integer torsion is not being assumed accomplished by taking the already supplied integer layer as input.

109.2 B1–B5. Primitive supplied data and retained integer operations

These are the inputs. The separation from integers, the global-versus-integer identity distinction, and the excluded inputs to existing integer operations are proved below; they are not additional primitive definitions. U18 controls this distinction.

109.2.1 B1–B3. Absence, presence and the integer data

The source symbols are:

- $0_{\langle Z_0; \text{no } Z_2 \rangle} = e = \emptyset$: the supplied absence element, also the integer zero. [U02, U05]
- $\tau_{\langle Z_1; \text{no } Z_2 \rangle}$: primitive presence with no parity. No integer coefficient is introduced here; exclusion from the integer carrier is proved in D1a. [U01, U06]
- For each nonzero integer n , the original integer with its exact amount n , its Z_1 presence, and its Z_2 odd/even data. The word torsion here means the user's retained integer amount; it does not assert finite order in a group. [U01, U08, U11]

An integer's full amount is not replaced by parity: 2 and 4 have different amounts. Equality preserves the complete declared source data. An absent parity value is not an even value. No source parity is assigned to zero or tau. No family of copies of tau is introduced. [U01–U03]

D1a. Distinctness follows from these data. Tau differs from zero because their declared presence statuses are Z_1 and Z_0 . Tau differs from a nonzero integer because that integer has the declared Z_2 datum and tau does not. Equality would require retaining and lacking that datum simultaneously. Therefore tau differs from every integer. This conclusion uses the source distinctions rather than adding a midpoint or an independent separation axiom. [U07]

After that conclusion, denote the whole indicated carrier by S , and its original integer part by $\mathbb{Z}$. Thus S consists of that integer part and the additional primitive tau. This is a description of the resulting carrier, not an equation constructing tau from integer arithmetic.

109.3 B4–B5. The existing operations

B4. Integer arithmetic. The notation $+$, $-$ denotes the original operations on the supplied integer layer:

$$+, - : \mathbb{Z} \times \mathbb{Z} \longrightarrow \mathbb{Z}.$$

Integer multiplication is the original operation $m \cdot n = mn$ for integer arguments. An output retains the data of the resulting integer, including the supplied Z_0 status when that result is zero. This specifies the existing integer operations. It is not a postulate that every conceivable additive extension is impossible. [U01, U05, U08]

B5. The supplied global identity. For every indicated source element x , use the source statement that tau is the global multiplicative identity:

$$\tau \cdot x = x \cdot \tau = x.$$

This interprets the explicit global-identity wording in U13. It is not proved from tau lacking parity. In particular the displayed laws $\tau \cdot \tau = \tau$, $\tau \cdot n = n \cdot \tau = n$, and $0 \cdot \tau = \tau \cdot 0 = 0$ follow by applying B5 to those arguments. The integer coefficient of a mixed product comes from its integer input. The integer rules in B4 and the identity rules in B5 together give a product on the whole indicated carrier. [U05, U13]

Retraction R1. U04 withdraws the former tau self-sum. The previous mixed additive line is not used as a current premise. No new sum involving tau is introduced here. A retraction removes an alleged law; it is not a proof of a universal nonexistence theorem.

Derived input exclusion. By D1a, tau is not in the integer carrier. Therefore the ordered pairs (n, τ) , (τ, n) , and (τ, τ) are not in the domain of either integer operation in B4. Consequently these expressions cannot be evaluated by those operations. This proves the input exclusion from the declared integer signatures and D1a, rather than adding the exclusion as a separate axiom. The complete expression result is P4; P5 states its precise limit. [U04, U12, U18]

Integer comparison retains its original integer domain. D1a similarly excludes tau as an argument to that comparison. The sign of an integer is not thereby a displacement from tau; U14 supplies no permission to erase integer signs.

109.4 L1. Specification limit: a missing metric is not an impossibility axiom

B1–B5 supply no distance operation, numerical coordinate for tau, inner product or norm. This records the actual input data; it is not an extra axiom saying such structures can never be studied. The expressions $n - \tau$ and $\tau - n$ have already been excluded from the existing integer subtraction by D1a and B4. A statement excluding every possible relation would require a stronger argument than that input-domain proof; it is not assumed here. The specified multiplication is retained throughout. [U10–U13, U18]

The user’s intended positive-undefined statement is preserved verbatim in U13. No metric, numerical infinity, positive scalar or purity conclusion is inserted into the inputs to make it true. The present calculation proves the narrower arithmetic consequences stated in P1–P5.

Figure 1. This displays the supplied data and the proved arithmetic-domain distinctions; it assigns no source metric or coordinate. Source references: U01–U05 and U08–U13. The complete arguments remain below. Reproducible figure source: `render_source_domains.py`.

109.5 P1. Multiplication is associative and commutative, with global identity tau

Proof. For two integer inputs, commutativity is integer commutativity. For an input tau, the product is the other input. For two tau inputs it is tau. These cases exhaust the domain and prove commutativity.

Three integer inputs satisfy associativity by original integer arithmetic. All remaining triples are covered by the following identities, for arbitrary source elements x, y, z :

Retained source data and operation domains

Inputs: Z0 absence, Z1 presence; nonzero integers retain Z2 parity and their full amount.

Source element	Retained layers	Parity data	Amount
$0 = e = \emptyset$	Z_0	No source parity	Integer zero
τ	Z_1	No parity	No integer amount
$n \in \mathbb{Z}, n \neq 0$	Z_1 plus Z_2	Even / odd	Exact integer n retained

B4-B5: supplied integer operations and global identity

$$m, n \in \mathbb{Z}: m + n, m - n, mn \in \mathbb{Z}$$

$$n \in \mathbb{Z}: \tau \cdot n = n \cdot \tau = n \qquad \tau \cdot \tau = \tau$$

Addition and subtraction take integer inputs only.

Typing diagram. Layout positions encode no source coordinates or distances. See B1-B5, derived distinction D1a, and P1-P5.

Figure 35: Retained source data and exact operation domains

$$(\tau y)z = yz = \tau(yz), \quad (x\tau)z = xz = x(\tau z), \quad (xy)\tau = xy = x(y\tau).$$

The same identities include triples with more than one tau. Thus multiplication is associative. Its stipulated two-sided identity is tau. If u were another global identity, its identity property would give $u\tau = \tau$, whereas tau's identity property gives $u\tau = u$; hence $u = \tau$.

Zero is absorbing: integer products with zero are zero, and $0\tau = \tau 0 = 0$ by the supplied mixed rule. Integer 1 acts as identity on integers. It is not the global identity because $1\tau = 1$, while D1a gives $1 \neq \tau$. This distinguishes the global identity from the integer identity without assigning parity to tau. ■

109.6 P2. Every product with an integer input has integer output

Proof. The other input is either an integer, giving an integer product, or tau, leaving the integer input unchanged. Both cases return the integer part of the carrier. ■

109.7 P3. Available distributivity uses integer additive inputs

For every $a \in S$ and integers m, n , both sides of

$$a(m + n) = am + an, \quad (m + n)a = ma + na$$

are defined and equal.

Proof. If a is an integer, use original integer distributivity. If $a = \tau$, each side of the first equation is the original integer sum $m + n$. Multiplication is commutative by P1, giving the second equation. P2 ensures that each sum on the right is a sum of integers. This assertion has no instance with tau as an additive input. ■

109.8 P4. Exactly which defined arithmetic expressions can produce tau

Consider finite expressions whose leaves are source elements and whose internal operations are the operations of B4-B5. An expression is defined only when each operation receives inputs in its

stated domain.

Then:

1. Every defined expression containing at least one integer leaf has integer value.
2. A defined expression with only tau leaves has value tau and uses multiplication only.

Proof by induction on the expression tree. A single integer leaf has integer value; a single tau leaf has value tau.

At a multiplication node, if either subtree contains an integer leaf, its value is an integer by induction. P2 then gives an integer value for the node. If neither subtree contains an integer leaf, induction gives value tau to each, and their product is tau.

At an addition or subtraction node, both subtree values must be integers for the expression to be defined. In particular neither subtree can consist entirely of tau leaves, since induction would give it value tau, outside the input domain. The original integer operation returns an integer.

These cases exhaust the allowed nodes and establish both conclusions. ■

Exact consequence. No defined expression made from integer inputs by these addition, subtraction and multiplication rules produces tau. This remains true when tau is allowed as a multiplicative identity within an expression already containing an integer input. Tau is not recovered as an integer midpoint, an integer difference, or another expression of this language. This is the precise algebraic content established for U10–U12.

The quantifier is over the stated finite-expression language. No claim is made about every conceivable additional operation, an unspecified category, or a subsequently added geometric construction. Such a stronger statement is not contained in this proof.

109.9 P5. The proposed numerical comparison is outside the source language

A subtraction node requires integer-valued arguments. D1a proves tau is not an integer source element. Therefore a node with arguments (n, τ) , (τ, n) , or (τ, τ) is not an instance of source subtraction. B4–B5 contain no distance-operation node. This fact specifies the expression language; it does not prove that every extension of the language is impossible.

This is a typing conclusion, not a negative numerical value and not an unknown numerical value. It does not establish a positive numerical value either. In particular no sign or magnitude of $d(n, \tau)$ is deduced by the present proof. U13's intended positive-undefined statement remains accurately quoted rather than being replaced with a metric claim.

109.10 Consequence for the rejected auxiliary construction

The supplied transcript introduced vectors, a midpoint, a metric and a pairing before providing a source construction of them. Those operations do not occur in B1–B5, and no comparison carrying them from this source was supplied. Its matrix calculation therefore cannot be used here as a test of this source. This scope correction does not assert that its separate matrix arithmetic was internally false. It withdraws the claimed application to the user's construction.

No replacement pairing is chosen here. P1–P5 do not supply a proof of reflection positivity, Deligne purity or RH.

109.11 Exact dependency record

Statement	Status	Premises and complete derivation
Z_0 absence; Z_1 presence; source parity data	Supplied data	B1–B3, traced to U01–U02, U05–U06, U08
Original integer operations	Supplied operations	B4; these do not claim integers were generated from tau
Tau is the global multiplicative identity	Supplied law	B5, explicit interpretation of U13
Tau self-addition	Retracted law	R1 and U04; never an input to P1–P5
$\tau \neq 0$ and $\tau \neq n$ for nonzero integers	Derived	D1a: equality preserves the stated presence and parity data
Integer 1 is not the global identity	Derived	P1 uses B5 and D1a, not an independent inequality axiom
Integer arithmetic expressions cannot produce tau	Derived	P2 and the complete induction in P4
Existing integer subtraction has no tau input	Derived	D1a and the integer input signature B4; P5
No numerical distance was supplied	Specification fact	L1; not converted into a universal impossibility axiom
Geometric fixed-centre, positive-undefined and purity conclusions	Not asserted as proved here	They are not added as premises to the arithmetic proofs

This classification follows U18. An agent-authored contradiction with the supplied definitions must be repaired before the affected calculation continues. No proof may depend on a correction being silently promoted to an axiom.

109.12 Reproducibility and provenance

All source quotations have locators in `USER_DEFINITIONS_VERBATIM.md` and hashes in `SOURCE_MANIFEST.json`. The complete pasted exchanges are preserved without byte changes. These proofs are elementary direct derivations from the explicit operations and the retained integer arithmetic; no unseen literature theorem is invoked. The user’s geometric proposal and terminology are credited to the user, without publishing a personal name. This is a local foundational note; no remote publication or acceptance of a complete geometric reconstruction is claimed.

110 Ordered prime absorption on the signed Connes–Consani winding structure

24 September 2026. Complete local derivation, PA1–PA10.

The user’s sequential observation is the starting point: a line first appears at 2, and its later multiples introduce no new prime line. The observation is an ordered history, with the original winding states retained. This note constructs that history, proves its exact quotient and kernel maps, and applies it to the signed generic structure in Alain Connes and Caterina Consani’s *The Absolute Twistor Line and the Geometry of the compactification of Spec Z*, arXiv:2609.00299v1, §§3–4. It also computes the corresponding finite-character receiver and the original zeta series in its domain of absolute convergence.

Classical cyclic-group arithmetic, prime factorization and the Euler product are not claimed as new results. The programme calculation is their explicit assembly with the user’s observation and the source’s retained sign, branch and mirror. Every assertion made below has its proof here, apart from the attributed Deligne theorem stated only to specify the comparison target.

110.1 PA1. Starting data, notation and the actual supporting map

Keep the user’s notation: Z_0 denotes absence, and Z_1 presence. The source $0 = e = \emptyset$ and τ have no Z_2 parity. Nonzero integers retain their full amount and their Z_2 odd/even datum. The retracted expression adding tau to itself is not used. There are no numerical coordinates or distances on tau in this construction.

The following additional geometric data are taken from the original Connes–Consani source, not deduced from the set $\{\tau\}$. Its three-point space is

$$X = \{+, \eta, -\}, \quad \mathcal{T} = \{\emptyset, \{\eta\}, \{+, \eta\}, \{-, \eta\}, X\}.$$

The involution α exchanges the closed points and fixes η . Indeed, $\overline{\{\eta\}} = X$, whereas the other singleton closures are themselves. A homeomorphism preserves closures, so the unique generic point must be fixed.

The exact supporting comparison is the continuous map

$$b : \{\tau_{\langle Z_1, \text{no } Z_2 \rangle}\} \longrightarrow X, \quad b(\tau) = \eta.$$

The inverse image of each open set is empty or the whole singleton, proving continuity. Give the singleton the identity action. Then $\alpha b = b$. Pulling back the sheaf gives $(b^{-1}\mathcal{O})_\tau = \mathcal{O}_\eta$, because $\{\eta\}$ is the smallest neighbourhood of η ; the neighbourhood colimit defining the stalk is its section object. This transports the retained stalk to the supporting point. It does not map tau to an integer or to a section denoted 1.

A two-element set on which the mirror exchanges both elements has no equivariant label at this fixed point: an equivariant label f would satisfy $f(\eta) = f(\alpha\eta) = \text{flip}(f(\eta))$, and neither target element is fixed. This proves absence of this particular exchanged label; it does not assert absence of all symmetry or all data in the stalk.

The generic spherical algebra in the source is

$$\mathcal{O}(\{\eta\}) = \mathbb{F}_{12}[T^{\mathbb{Z}}, J]/(J^2 = \epsilon), \quad \epsilon^2 = 1.$$

Its two chart restrictions send $J_+ \mapsto J$ and $J_- \mapsto \epsilon J$. We retain those branch signs. The calculation below concerns its presented monomial group and the adjoined absorbing section:

$$G = \langle T, J, \epsilon \mid TJ = JT, \epsilon = J^2, \epsilon^2 = 1 \rangle, \quad M = G \sqcup \{0_M\}.$$

The subscript on 0_M identifies a section in this receiving object; it introduces no extra source zero. This multiplicative calculation is not asserted to identify the full spherical algebra with a group ring.

Every unit has a unique expression

$$g = \epsilon^a T^m J^b, \quad a, b \in \{0, 1\}, \quad m \in \mathbb{Z}.$$

Proof: the relations reduce every expression to this form. The map to $\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$ sending $T \mapsto (1, 0)$, $J \mapsto (0, 1)$, $\epsilon \mapsto (0, 2)$ respects every relation. Its inverse sends (m, k) to $T^m J^k$. Thus it is an isomorphism and proves uniqueness; $k = 2a + b$ retains both sign and branch.

Write $A = \alpha^*$. The exact source formulas and their consequences are

$$\begin{aligned} A(T) &= \epsilon T^{-1}, & A(J) &= \epsilon J = J^{-1}, & A(\epsilon) &= \epsilon, \\ A(\epsilon^a T^m J^b) &= \epsilon^{a+m+b} T^{-m} J^b, & A(m, k) &= (-m, 2m - k). \end{aligned}$$

They preserve the defining relations and applying them twice returns (m, k) , so A is an involutive automorphism. Set $A(0_M) = 0_M$.

Let $H = \langle J \rangle = \{1, J, \epsilon, \epsilon J\}$. It is precisely the finite-order subgroup: (m, k) has finite order exactly when $m = 0$. The exact sequence

$$1 \longrightarrow H \longrightarrow G \xrightarrow{q} L \longrightarrow 1, \quad L = \langle t \rangle, \quad q(\epsilon^a T^m J^b) = t^m$$

retains the full group and its finite subgroup while observing winding. Its kernel and surjectivity follow directly from the displayed form. The mirror on L is $t^m \mapsto t^{-m}$.

Source locators: Connes–Consani, equations `opens1`, `opens`, `eq:restriction_maps`, and Definition `def:alpha_symmetry`; original author `CC.tex`, lines 499–537, 576–609, 775–806. Original paper.

110.2 PA2. Clock quotients and the exact meaning of prime

For each integer $d \geq 1$, define

$$C_d = L / \langle t^d \rangle, \quad c_d = t \langle t^d \rangle.$$

The identity is the marked state and multiplication by c_d is the designated winding step. The map from its states to the common support sends every state to τ . It is equivariant for the trivial action on the support and puts no addition or counting on τ .

Every finite pointed transitive L -set has this form. To prove it, take its marked point x , and let d be the least positive integer with $t^d x = x$. Finiteness gives such a return. Division with remainder shows that every return exponent is divisible by d , since otherwise its nonzero remainder would give an earlier return. Therefore $c_d^k \mapsto t^k x$ is a well-defined bijection preserving the point and the step.

There is a pointed equivariant quotient

$$\pi_{d,e} : C_d \longrightarrow C_e, \quad c_d^k \longmapsto c_e^k$$

exactly when $e \mid d$. Necessity follows by sending $c_d^d = 1$ to $c_e^d = 1$. Divisibility makes the formula well-defined and surjective, proving sufficiency. Pointed equivariance forces this formula, so it is unique. For $f \mid e \mid d$, direct evaluation proves $\pi_{e,f} \pi_{d,e} = \pi_{d,f}$. Its exact kernel sequence is

$$1 \longrightarrow C_{d/e} \xrightarrow{c_{d/e}^k \mapsto c_d^{ek}} C_d \xrightarrow{\pi_{d,e}} C_e \longrightarrow 1.$$

The first map is injective because $d \mid ek$ exactly when $d/e \mid k$; its image is the kernel of the second. The inherited step in this kernel is t^e , retaining the factor e , rather than one original winding step.

A nontrivial clock has no proper nontrivial pointed quotient exactly when it is C_p for a prime p . For prime p its only positive divisors are $1, p$. Conversely a composite $d = ab$, $a, b > 1$,

admits the proper quotient C_a . These are isomorphism classes of simple winding clocks; complex irreducible characters are not being substituted for this definition.

The winding also supplies exact integer arithmetic. In $R = \text{End}_{\text{Ab}}(L)$, let addition be pointwise group multiplication and multiplication be composition. Every endomorphism is uniquely $[n](t^k) = t^{nk}$, determined by its value at t . Then

$$[n] + [m] = [n + m], \quad [n] \circ [m] = [nm].$$

Thus $n \mapsto [n]$ is a ring isomorphism from the retained integer arithmetic to R . It uses the infinite winding furnished by the source. In particular, $[0]$ is the constant identity-valued endomorphism; it is not tau and does not assign source parity to zero.

110.3 PA3. Absorption is loss of invertibility, with an ordered first appearance

The endomorphism $[n]$ descends to every C_d , since nk and $n(k + d)$ differ by nd . It commutes with every $\pi_{d,e}$, as both composites send c_d^k to c_e^{nk} .

This degree label is independent of the choice of winding orientation. With $t' = t^{-1}$, the same map sends $(t')^k$ to $(t')^{nk}$. Equivalently inversion conjugates $[n]$ to itself. Thus using positive degrees for arrivals does not remove the source's opposite winding orientation.

On a simple clock it satisfies the complete dichotomy

$$[n] : C_p \longrightarrow C_p \begin{cases} \text{is bijective,} & p \nmid n, \\ \text{is constant with value } 1_{C_p}, & p \mid n. \end{cases} \quad (\text{PA3.1})$$

For the first case, Bezout's identity gives $un + vp = 1$, so $[u]$ is the inverse. For completeness, Bezout follows by choosing the least positive integer among integer combinations of n, p ; division with remainder shows it divides both, and primality with $p \nmid n$ makes it 1. In the second case every exponent nk is divisible by p . These cases are exhaustive.

The time evolution itself is the translation $U_{p,n}(c_p^k) = c_p^{k+n}$, always a bijection with inverse $U_{p,-n}$. Its exact relation to the degree map is evaluation at the designated generator:

$$[n]_p(c_p) = c_p^n, \quad U_{p,n}(x) = [n]_p(c_p) x.$$

Every group endomorphism of C_p is determined by that evaluation. In particular,

$$c_p^n = 1_{C_p} \iff U_{p,n} = \text{id} \iff [n]_p \text{ is constant} \iff p \mid n.$$

The first equivalence follows by evaluating the translation at its marked state; the second follows from the endomorphism being determined by its generator. Thus the same displayed event is a return in the winding evolution and collapse in the degree map. A return does not make the translation noninvertible. These operations and their exact connecting formula are retained separately.

Define the complete divisibility line

$$\Lambda_d = d\mathbb{N}_{>0} = \{d, 2d, 3d, \dots\}.$$

By PA3.1, Λ_p is exactly the set of positive degrees at which C_p loses invertibility. Its first occurrence is p . This is the mathematical use of "absorption" here: a map becomes constant on that clock, not a physical light spectrum.

Now retain arrival order. Put $V_1 = H_1 = \emptyset$. At step $n \geq 2$, if $n \notin H_{n-1}$, set

$$V_n = V_{n-1} \cup \{n\}, \quad H_n = H_{n-1} \cup \Lambda_n.$$

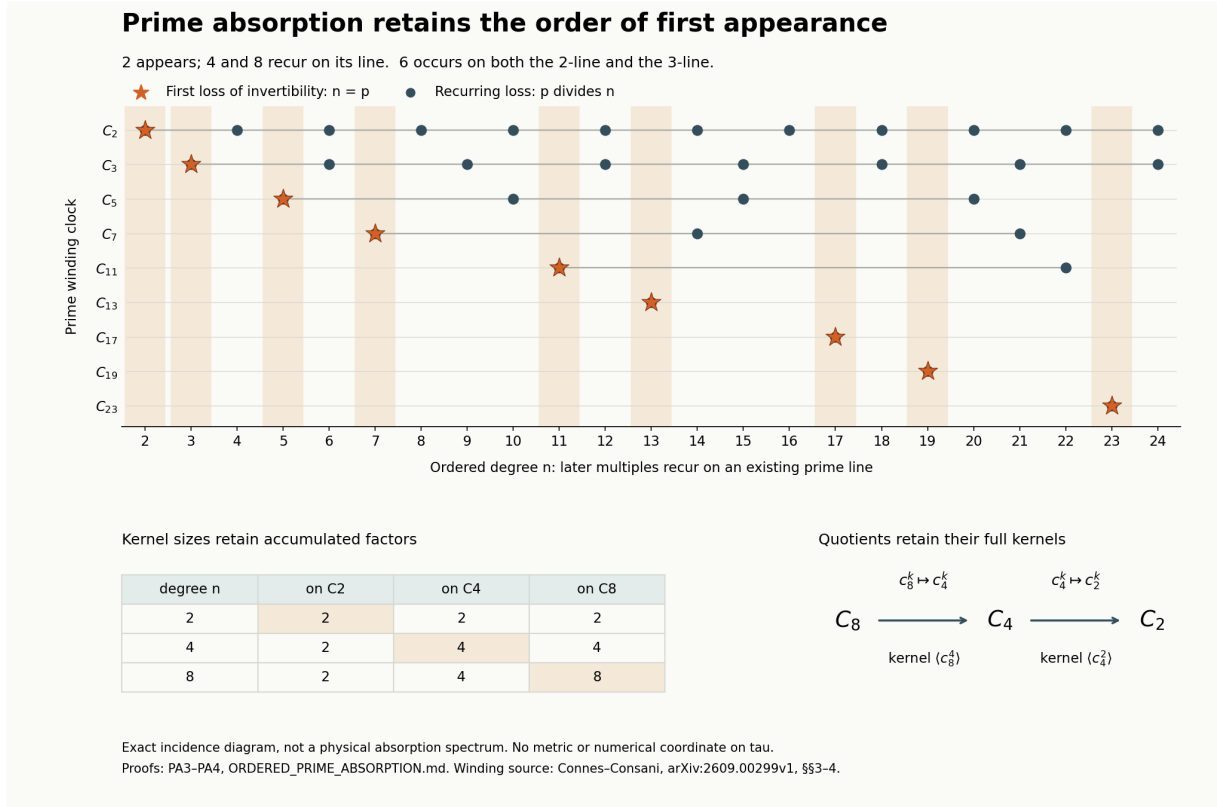


Figure 36: Ordered prime absorption and retained prime-power depth

Otherwise keep $V_n = V_{n-1}$ and $H_n = H_{n-1}$. The clock C_n , step n , and all its quotient maps remain in the full object in either case. The result is

$$V_n = \{p \leq n : p \text{ prime}\}, \quad H_n = \bigcup_{\substack{p \leq n \\ p \text{ prime}}} \Lambda_p. \quad (\text{PA3.2})$$

Proof by induction: a prime n has no smaller prime divisor, hence is uncovered at its arrival. A composite n has a prime divisor $p < n$: its least divisor greater than 1 is prime, since a factorization would give a smaller such divisor. By induction Λ_p was already present, so n is covered. This proves both formulas.

Moreover,

$$\Lambda_d \subseteq \Lambda_e \iff e \mid d \iff C_d \twoheadrightarrow C_e. \quad (\text{PA3.3})$$

Divisibility gives line containment; conversely the containment sends $d \in \Lambda_d$ into Λ_e , proving divisibility. PA2 gives the quotient equivalence. Thus the entire future line of a composite is already contained in each of its prime-divisor lines.

The full timed incidence is

$$I_t(m) = \{p \leq t : p \text{ prime}, p \mid m\}, \quad m \geq 2.$$

Its first nonempty time is the least prime divisor of m . At arrival m , one tests coverage at $t = m - 1$, before introducing a newly visible line. In particular 6 is first covered at 2 and also covered at 3; both incidences are retained. This first time is a property of the ordered filtration and is not recovered from an unordered set alone.

Figure 1. Each row is a prime clock. A star is its first noninvertible positive degree; later dots are recurring losses on the same clock. The lower panel retains the kernel sizes on the 2, 4 and 8 clocks. The horizontal coordinate is the supplied integer winding-step count, not a coordinate on tau. Full proofs: PA3-PA4. Figure source: `render_ordered_absorption.py`.

110.4 PA4. The repeated factors are retained by the clock tower

For $n > 0$, let $v_p(n)$ be the largest integer $v \geq 0$ for which $p^v \mid n$. It exists because $p^v \geq 2^v \geq v + 1$ is eventually greater than n . Write the full factorization $n = p^{v_p(n)}u$, with $p \nmid u$. Then

$$\ker([n] : C_{p^a} \rightarrow C_{p^a}) = \langle c_{p^a}^{p^{\max(a-v_p(n), 0)}} \rangle, \quad |\ker[n]| = p^{\min(a, v_p(n))}, \quad a \geq 1. \quad (\text{PA4.1})$$

Indeed $c_{p^a}^k$ lies in the kernel exactly when $p^a \mid p^{v_p(n)}uk$. If $v_p(n) \geq a$, every k works. Otherwise u is invertible modulo $p^{a-v_p(n)}$ by Bezout, giving $p^{a-v_p(n)} \mid k$. Counting this subgroup proves its stated order. For $n = 0$ the map is constant and the kernel is all C_{p^a} ; zero is excluded from positive arrivals. For negative degree the kernel agrees with that of its absolute value, while the original signed degree is still retained.

Consequently, the largest a for which the whole C_{p^a} is killed is exactly $v_p(n)$. At $n = 2, 4, 8$, the kernel-size triples for C_2, C_4, C_8 are respectively

$$(2, 2, 2), \quad (2, 4, 4), \quad (2, 4, 8).$$

They share the first 2-line while their repeated factors are fully distinguished.

Repeatedly choose a prime divisor of $n > 1$ and divide by it. Each division decreases the positive integer, so the process terminates at 1 and supplies a factorization. The multiplicities are independent of the choices. To prove this, if $p \mid ab$ and $p \nmid a$, an equation $xp + ya = 1$ multiplied by b gives $p \mid b$. Hence a prime dividing a product divides a factor. Compare two prime factorizations, match a prime from one with an equal prime in the other, cancel it, and repeat. Thus

$$n = \prod_{p \mid n} p^{v_p(n)}$$

with every repetition retained. This is classical unique factorization proved here, not a new theorem about primes.

It also reconstructs the whole clock. Put $n_p = p^{v_p(n)}$ and $N_p = n/n_p$. Then

$$C_n \longrightarrow \prod_{p \mid n} C_{n_p}, \quad c_n^k \longmapsto (c_{n_p}^k)_{p \mid n} \quad (\text{PA4.2})$$

has the explicit inverse

$$(c_{n_p}^{a_p})_{p \mid n} \longmapsto c_n^{\sum_{p \mid n} a_p u_p N_p}, \quad u_p N_p + v'_p n_p = 1.$$

Here u_p, v'_p are Bezout coefficients, with the prime valuation still denoted $v_p(n)$. Modulo n_p , the indicated sum is a_p : its other terms contain n_p , while $u_p N_p \equiv 1 \pmod{n_p}$. Conversely an exponent zero modulo all n_p is divisible by their product n , by the prime-divisor result just proved. These facts prove both inverse identities and well-definedness. The isomorphism preserves the marked state and diagonal winding action.

110.5 PA5. The full source mirror forces an even winding period

The prime-clock obstruction also gives the precise ordinary prime point in the recovered ring R . Its action map

$$R \longrightarrow \text{End}_{\text{Ab}}(C_p), \quad [n] \longmapsto [n]_p$$

uses the induced map $[n]_p(c_p^k) = c_p^{nk}$ on the quotient and is surjective: every endomorphism is determined by the image c_p^n of its generator. Its kernel is pR by PA3.1. Every nonzero class has an inverse by the same Bezout calculation, so R/pR is a field and pR is maximal, hence prime. This is the annihilator of the entire prime clock.

These, together with (0) , are all prime ideals of R . Through PA2's ring isomorphism, a nonzero ideal has a least positive integer d ; division with remainder shows each of its elements is divisible by d , so it is $d\mathbb{Z}$. The value $d = 1$ gives the whole ring and is excluded by properness. If $d = ab$ with $1 < a, b < d$, then ab lies in this ideal and neither factor does, so it is not prime. The prime values of d give the ideals just proved maximal. The zero ideal is prime because a product of two nonzero integers is nonzero. This proves the stated classification. It concerns $\text{Spec } R$, with its exact source in the winding endomorphisms; it does not replace the full tau-lattice spectrum or identify its additional support points with ring elements.

The winding quotients in PA2 do not discard the original H : it remains in the exact sequence in PA1. To perform finite observations retaining the mirror and H together, use

$$G_{2p} = G/\langle T^{2p} \rangle \cong C_{2p} \times C_4, \quad M_{2p} = G_{2p} \sqcup \{0_M\}.$$

The quotient has $8p$ units, all four elements of H remain distinct, and

$$A_{2p}([m]_{2p}, [k]_4) = ([-m]_{2p}, [2m - k]_4).$$

Changing m by $2p$ changes $2m$ by $4p$, proving well-definedness. Also $A(T^{2p}) = T^{-2p}$, proving that the quotient map intertwines the mirrors.

The extra factor 2 has an exact source reason. Let $K \leq G$ be mirror invariant and retain H injectively in G/K , namely $K \cap H = \{1\}$. For $g = T^m J^k \in K$,

$$gA(g) = \epsilon^m \in K \cap H.$$

Therefore m must be even. In particular for odd p , the relation $T^p = 1$ alone is not mirror invariant: its mirror is ϵT^{-p} , forcing $\epsilon = 1$ if both were imposed. Its smallest invariant enlargement is $\langle T^p, \epsilon \rangle$; this gives $C_p \times C_2$ and loses the original nontrivial sign. We retain it by using period $2p$.

The map to the prime clock is

$$\rho_p : G_{2p} \rightarrow C_p, \quad ([m]_{2p}, [k]_4) \mapsto c_p^m.$$

It is surjective; its kernel consists of $m = 0, p \pmod{2p}$ with arbitrary k , hence has eight elements. Its mirror is inversion on C_p .

For odd p , the full coordinates are equivalently $(m \bmod p, m \bmod 2, k \bmod 4)$; injectivity follows from coprimality, and surjectivity from equal finite cardinality. The mirror is $(r, s, k) \mapsto (-r, s, 2s - k)$. At $p = 2$ retain $C_4 \times C_4$: C_4 cannot be replaced by $C_2 \times C_2$, whose elements all have order at most 2. Period 2 already preserves the mirror at this prime; period 4 is the uniform choice, not a claimed minimum there. None of these finite observations alters the source's restriction to odd Frobenius powers.

110.6 PA6. A faithful finite-prime receiver with a derived common modulus

Retain both complex branches and every prime label:

$$D_p = \{(z, j) : z^{2p} = 1, j = i \text{ or } -i\}, \quad D = \coprod_{p \text{ prime}} \{p\} \times D_p.$$

The source evaluates ϵ as -1 , and $J^2 = \epsilon$ requires $j^2 = -1$ (Connes–Consani, Proposition `prop:points_X_K`, original lines 619–678). Each pair therefore gives the exact multiplicative map

$$\chi_{p,z,j} : M \rightarrow \mathbb{C}, \quad \chi(0_M) = 0, \quad \chi(\epsilon^a T^m J^b) = (-1)^a z^m j^b.$$

All defining relations hold: $j^2 = -1$, $j^4 = 1$, and complex multiplication commutes. Since $z^{2p} = 1$, it factors through M_{2p} . No map from tau to the number 1 is used.

Set $\ell_p = \text{lcm}(2p, 4)$, which is $4p$ for odd p and 4 for $p = 2$. Every unit value satisfies

$$\chi(g)^{\ell_p} = 1.$$

Thus it is algebraic and has absolute value 1. Every complex embedding of its number field preserves the equation $x^{\ell_p} = 1$, so every algebraic conjugate has absolute value 1 as well. This common modulus follows from finite order and was not imposed on the infinite source.

The simultaneous receiver

$$E : M \rightarrow \mathbb{C}^D, \quad E(g)(p, z, j) = \chi_{p,z,j}(g) \quad (\text{PA6.1})$$

is injective and multiplicative, with coordinatewise multiplication in its target. To prove injectivity, suppose the images of $T^m J^k$ and $T^n J^h$ agree. Evaluate at $z = 1, j = i$ to obtain $k = h \pmod{4}$. Then $z^{m-n} = 1$ for every $2p$ -th root and every prime p . If $m - n \neq 0$, choose a prime divisor p of $|m - n| + 1$, using the least-divisor proof in PA3. This prime does not divide $m - n$. The value $z = \exp(\pi i/p)$, of exact order $2p$, contradicts $z^{m-n} = 1$. Thus $m = n$. Finally $E(0_M)$ is identically zero, whereas each unit image is nowhere zero. This proves the assertion on all of M .

The receiver is compatible with the full mirror:

$$E(Ag)(p, z, j) = E(g)(p, -z^{-1}, -j). \quad (\text{PA6.2})$$

Indeed $(-z^{-1})^{2p} = 1$, and $-j$ is the other allowed branch. For $g = T^m J^k$, its left side is $z^{-m} j^{2m-k}$; its right side is $(-z^{-1})^m (-j)^k$. They agree by $j^2 = -1$ and $j^{-1} = -j$. Both sides vanish on 0_M .

For the source's positive odd power map $F_r(g) = g^r$, $F_r(0_M) = 0_M$,

$$E(F_r g)(p, z, j) = E(g)(p, z^r, j^r).$$

Oddness retains $j^r \in \{i, -i\}$ and $F_r(\epsilon) = \epsilon$. This equality follows by raising the complete value to its r -th power; the factor $(-1)^a$ remains. It is a point-level monomial comparison, not a new theorem identifying an arithmetic cohomology.

For winding translation $\mu_r(g) = T^r g$, with 0_M fixed, the exact formulas are

$$E(\mu_r g)(p, z, j) = z^r E(g)(p, z, j), \quad A\mu_r = \nu^r \mu_{-r} A, \quad \nu(g) = \epsilon g,$$

with $\nu(0_M) = 0_M$. These follow from $A(T^r) = \epsilon^r T^{-r}$. In particular the mirror-translated receiver is $(-1)^r z^{-r} E(Ag)$. Translations are bijections, $\mu_r \mu_s = \mu_{r+s}$, but for $r \neq 0$ they are not unital monoid maps because $\mu_r(1) = T^r \neq 1$.

110.7 PA7. Complete additive scope of the receiver

Faithfulness in PA6 is a theorem on the original monomials and absorbing element. Its complex-linear extension on the separately constructed group algebra has the exact kernel

$$\ker(\mathbb{C}[G] \rightarrow \mathbb{C}^D) = (1 + J^2). \quad (\text{PA7.1})$$

Here addition belongs to the group algebra, not tau. To prove the assertion, write an arbitrary finite sum as

$$f = \sum_m T^m (a_{m,0} + a_{m,1}J + a_{m,2}J^2 + a_{m,3}J^3).$$

Vanishing at $j = i$ and at $j = -i$ gives two Laurent polynomials in z that vanish at all finite-prime parameters. There are infinitely many distinct such parameters: a finite list of primes cannot contain every prime divisor of their product plus 1, by PA3, and $\exp(\pi i/p)$ are distinct for distinct primes. A nonzero polynomial of degree d has at most d distinct roots, proved by repeated

division by its linear factors. Multiplying a Laurent polynomial by a sufficiently large power of z therefore shows both Laurent polynomials are zero. Their coefficients give

$$a_{m,0} - a_{m,2} + i(a_{m,1} - a_{m,3}) = 0, \quad a_{m,0} - a_{m,2} - i(a_{m,1} - a_{m,3}) = 0.$$

Adding and subtracting gives $a_{m,0} = a_{m,2}$, $a_{m,1} = a_{m,3}$. Hence

$$f = (1 + J^2) \sum_m T^m (a_{m,0} + a_{m,1} J).$$

Conversely $1 + J^2$ evaluates to zero everywhere, proving both inclusions. Thus the exact faithful additive receiver is the quotient by this ideal, with its projection retained. This does not replace the spherical algebra in PA1.

The finite-prime parameters are dense, but not all, in $S^1 \times \{i, -i\}$. For $e^{i\theta}$, choose an integer k_p nearest $p\theta/\pi$; then

$$|e^{i\theta} - e^{\pi i k_p / p}| \leq |\theta - \pi k_p / p| \leq \frac{\pi}{2p}.$$

The first inequality follows by integrating ie^{it} between the two angles. Primes are unbounded by the preceding product-plus-one argument, giving density. The value $e^{2\pi i \sqrt{2}}$ is not a root of unity: a finite order would make a positive integer multiple of $\sqrt{2}$ an integer. Irrationality of $\sqrt{2}$ follows since a coprime equation $a^2 = 2b^2$ makes first a , then b , even. This proves properness. These circles describe complex character values; no circle or metric is assigned to tau.

110.8 PA8. The original zeta function recovered from the ordered births

The birth set in PA3 gives the original Euler factors with all repetitions retained. For a complex variable s with $\sigma = \Re s > 1$, use the original series

$$\zeta(s) = \sum_{n=1}^{\infty} \exp(-s \log n).$$

Its absolute convergence follows from $\sum_{n>B} n^{-\sigma} \leq \int_B^{\infty} x^{-\sigma} dx = B^{1-\sigma}/(\sigma-1)$, for integer $B \geq 1$. For the finite birth set V_B , define

$$P_B(s) = \prod_{p \in V_B} (1 - \exp(-s \log p))^{-1}.$$

Expanding each geometric series is valid because $p^{-\sigma} < 1$. A finite product of absolutely convergent series is absolutely convergent. PA4's unique factorization shows its terms are exactly the original n^{-s} for integers all of whose prime factors belong to V_B , once each, including the term $n = 1$. Consequently the exact remainder and its bound are

$$\zeta(s) - P_B(s) = \sum_{n \geq 1: \exists p > B \text{ prime, } p|n} \exp(-s \log n), \quad |\zeta(s) - P_B(s)| \leq \frac{B^{1-\sigma}}{\sigma-1}. \quad (\text{PA8.1})$$

Every omitted integer is greater than B , giving the inequality. Therefore the ordered births and their repeated factors recover the original zeta series on this domain, with an explicit remainder. Nothing has been divided by a Gamma factor, a power of pi, an endpoint factor or a completion multiplier. This computation does not extend the convergence domain or bound zeros in the critical strip.

110.9 PA9. Exact comparison with the requested purity mechanism

PA6 proves that every nonzero finite-character value is algebraic and that all its conjugates have absolute value 1. In Deligne’s terminology for an algebraic number relative to $q > 1$, this is weight zero because $q^{0/2} = 1$. It is a conclusion about these values, not a choice of radius on the source.

The arithmetic statement to which it must be compared keeps its original factors. Deligne’s weight definition at a closed point x requires Frobenius eigenvalues with

$$|\iota(\alpha_x)| = N(x)^{w/2}$$

for every complex conjugate. His proper smooth conclusion for degree i has $|\iota(\alpha)| = q^{i/2}$, where the eigenvalues are roots of $\det(t1 - F^*, H^i)$. These are Deligne’s results, not conclusions proved by this note. Pierre Deligne, *La conjecture de Weil. II*, Definitions 1.2.1–1.2.2 and Corollary 3.3.9.

The exact gain over a freely selected common-modulus character set is PA5–PA6: finite prime clocks give a faithful monomial receiver with that modulus, and the source mirror, sign and branch are retained. No Frobenius action on an RH-detecting arithmetic H^1 has been identified with these evaluations here. The weight-one factor $N(x)^{1/2}$ has therefore not been obtained or suppressed. The original zeta calculation currently has the proved domain in PA8.

110.10 PA10. What the single construction retains

The full object used throughout is the signed winding group at the fixed support, its winding quotient, all finite quotient clocks and compatible degree maps, together with the ordered filtration (V_n, H_n) , the signed quotients M_{2p} and their joint receiver E . Each new component is linked by the explicit maps proved above. A hidden composite still has its full clock and its arrival position. Its repeated prime factors remain in PA4’s compatible kernels and reconstruction maps.

Thus “2 is first; 4 is already covered” is an exact statement of this construction. Replacing its ordered filtration by only the set of prime divisors would change the observation. The first-appearance rule, multiplicity tower, faithful signed receiver and original Euler series are the proved consequences retained together here.

110.11 Provenance and reproducibility

- **User:** sequential prime visibility, absorption interpretation, retention of tau’s Z_1 type and the no-parity source distinction. Exact input is retained privately beside this file, without converting speculative statements from the pasted conversation into premises.
- **Alain Connes and Caterina Consani:** original geometric space, generic algebra, imaginary generator, twisted chart restrictions and mirror; arXiv:2609.00299v1. Read the original author TeX at the precise ranges in `SOURCE_READING_USE_PRIVATE.json`.
- **Pierre Deligne:** the quoted weight target and cohomological purity theorem, *La conjecture de Weil. II*, IHÉS 52 (1980), 137–252. The local French TeX is a transcription, not author-supplied TeX. The historically retained English transcription is available at the pinned programme edition; its known reciprocal-characteristic-polynomial error is not used. The French transcription controls PA9.
- **This task:** PA1–PA10’s connecting calculations and complete elementary proofs; independently checked clock and signed-character derivations. The underlying cyclic quotient theory, unique factorization and Euler product are classical.
- **render_ordered_absorption.py** reproduces the illustration. **check_clock_maps.py** checks finite instances of the displayed maps and outputs; its checks supplement the proofs and are not substituted for them.

The earlier source-definition note remains in `../foundations/user_definitions_20260924/`. This construction does not reinstate its withdrawn addition on τ or assert an isomorphism with the entire support-lattice programme. The cumulative TeX includes this complete note; older PDFs are not represented as rebuilt editions.

111 The whole winding structure

The unit, the complete clock diagram, and the cohomological boundary.

24 September 2026. Complete derivation GW1–GW11. This note follows the user’s question about reconstructing the unit from the whole absorption history, including the possibility that a later event changes a previously inferred counting interval. It then calculates a global first-cohomology operator from the actual Connes–Consani curves, evaluates its quantized boundary functional, and lifts the operators through the full support lattice.

The notation is retained: Z_0 is absence, Z_1 is presence, and Z_2 is the user’s integer parity datum. The source $0 = e = \emptyset$ and $\tau_{\langle Z_1; \text{no } Z_2 \rangle}$ have no parity. No addition, numerical coordinate, distance, or family of counted copies is assigned to tau. Subscripts on auxiliary zero maps identify their domains; they do not introduce another source zero. In particular, the zero endomorphism below is a function on a winding group, not a new assertion about source parity.

The question supplies a proposed reconstruction, not permission to assume its conclusion. We prove the actual maps from the published source. A unique supporting point, its stalk, a distinguished integer, and an infinite observation history are not identified by notation.

111.1 GW1. The complete source object before numerical labels

Alain Connes and Caterina Consani’s signed generic monomial object has presentation

$$G = \langle T, J, \epsilon \mid TJ = JT, J^2 = \epsilon, \epsilon^2 = 1 \rangle, \quad M = G \sqcup \{0_M\}. \quad (\text{GW1.1})$$

The absorbing section is retained. The source mirror is

$$A(T) = \epsilon T^{-1}, \quad A(J) = J^{-1}, \quad A(\epsilon) = \epsilon, \quad A(0_M) = 0_M. \quad (\text{GW1.2})$$

The underlying archimedean space before quotient has points $+, \eta, -$, with opens

$$\emptyset, \quad \{\eta\}, \quad \{+, \eta\}, \quad \{-, \eta\}, \quad X.$$

The mirror exchanges the two closed points and fixes the unique generic point. These are source definitions, not consequences claimed for presence alone. Source: Connes–Consani, *The Absolute Twistor Line and the Geometry of the compactification of Spec Z*, v1, equations `opens`, `eq:restriction_maps`, Definition `def:alpha_symmetry`; original `CC.tex`, lines 576–609 and 775–806.

Every element of G has a unique form $T^m J^k$, with $m \in \mathbb{Z}$ and $k \in \mathbb{Z}/4\mathbb{Z}$. Reduction by the relations gives existence. The homomorphisms

$$G \longrightarrow \mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}, \quad T \mapsto (1, 0), \quad J \mapsto (0, 1), \quad (m, k) \longmapsto T^m J^k$$

are inverse, proving uniqueness. These integer coordinates are a proved description of the source presentation; they have not been extracted from tau.

Define the subgroup intrinsically by

$$H = \{g \in G : g \text{ has finite order}\}.$$

The displayed isomorphism proves $H = \langle J \rangle$. Retain the full exact sequence

$$1 \longrightarrow H \longrightarrow G \xrightarrow{q} W := G/H \longrightarrow 1. \quad (\text{GW1.3})$$

The quotient W is infinite cyclic. The full signed source remains in (GW1.3); its four-element subgroup is not discarded. Direct computation gives

$$A(m, k) = (-m, 2m - k),$$

and hence inversion on W .

Here is the whole observation object, defined without assigning numbers to individual finite clocks. Let $\mathcal{Q}_W$ have an object for every finite-index subgroup $N \subset W$, and one arrow $N \rightarrow N'$ when $N \subseteq N'$. Define the functor

$$\mathcal{F}_W : \mathcal{Q}_W \longrightarrow \mathbf{Ab}, \quad N \longmapsto W/N, \quad (N \subseteq N') \longmapsto (wN \mapsto wN'). \quad (\text{GW1.4})$$

The formula is well-defined because elements equal modulo N are equal modulo N' . The quotient maps compose as required. This is one diagram, including all clocks and all their comparisons, attached to the retained source sequence (GW1.3).

The supporting comparison is

$$b : \{\tau_{\langle Z_1, \text{no } Z_2 \rangle}\} \longrightarrow X, \quad b(\tau) = \eta. \quad (\text{GW1.5})$$

It is continuous because every inverse image of an open is empty or the singleton. It is mirror-equivariant because $A\eta = \eta$. Pulling back the source sheaf retains its stalk: the singleton $\{\eta\}$ is the smallest neighbourhood, so the colimit defining the inverse-image stalk is exactly $\mathcal{O}_\eta$. No section of this stalk is thereby identified with tau.

111.2 GW2. The unit and arithmetic as transformations of that whole object

Set

$$R = \text{End}_{\mathbf{Ab}}(W).$$

For $a, b \in R$, define $(a + b)(w) = a(w)b(w)$, and define multiplication by composition. Commutativity of W makes $a + b$ a homomorphism. Associativity, the additive inverse $(-a)(w) = a(w)^{-1}$, and both distributive laws follow by evaluating on w and using the homomorphism law. Thus this is a ring. Its multiplicative identity is the already determined transformation

$$\mathbf{1}_R = \text{id}_W. \quad (\text{GW2.1})$$

Uniqueness follows from the identity equations: two identities u, v satisfy $u = u \circ v = v$. This defines the unit through its action on the complete object, before naming any particular prime.

To calculate R , choose a generator t of W . Every endomorphism is uniquely

$$[n](t^j) = t^{nj} \quad (n, j \in \mathbb{Z}).$$

Indeed its value on t is a unique power t^n ; the homomorphism law determines all powers. Evaluation proves

$$[n] + [m] = [n + m], \quad [n] \circ [m] = [nm], \quad [1] = \text{id}_W. \quad (\text{GW2.2})$$

This is a ring isomorphism $\mathbb{Z} \rightarrow R$. It is independent of the generator: replacing t by t^{-1} gives the same degree n for each endomorphism. Thus the generator was used to prove the arithmetic description, not to choose a different unit.

All finite-index subgroups are $\langle t^d \rangle$, with $d > 0$: take the least positive exponent in a nontrivial subgroup and use division with remainder. Every endomorphism preserves each such subgroup. It therefore induces a natural endomorphism of the whole functor (GW1.4):

$$\iota : R \longrightarrow \text{Nat}(\mathcal{F}_W, \mathcal{F}_W), \quad a \longmapsto (a_N)_{N \in \mathcal{Q}_W}. \quad (\text{GW2.3})$$

Each natural transformation here has group-homomorphism components. The components commute with every quotient map because both composites are induced by a on W . The map preserves addition and composition componentwise.

The nonzero integer corresponding to $\mathbf{1}_R$ receives the user's Z_2 odd datum through its arithmetic description. This construction is not a map from tau to that integer. Addition has been constructed in the receiving endomorphism ring using the actual winding group law, not at Z_1 .

111.3 GW3. Exactly what all clocks determine, and what finitely many leave

Give a clock its size label only now:

$$\mathbf{C}_d = W/\langle t^d \rangle.$$

The induced maps satisfy

$$[n]_d = [m]_d \iff d \mid n - m. \quad (\text{GW3.1})$$

Necessity follows by evaluating on the class of t ; sufficiency follows by evaluating on all its powers. For a finite collection $d_1, \dots, d_r$, let D be their least common multiple. The complete ambiguity is

$$\{a \in R : a_{d_j} = [n]_{d_j} \text{ for all } j\} = \{[n + Dk] : k \in \mathbb{Z}\}. \quad (\text{GW3.2})$$

The intersection of the divisibility conditions is exactly divisibility by D , proving the equality. In particular, finite readings that agree with the identity do not distinguish it from $[1 + D]$.

For the entire family, $[n]_d = [m]_d$ for every d implies $n = m$: if $n - m \neq 0$, choose $d > |n - m|$. Thus ι is injective and

$$\iota^{-1}((\text{id}_{\mathbf{C}_d})_d) = \{\text{id}_W\}. \quad (\text{GW3.3})$$

This proves an exact global-observation result: all clocks distinguish the integer endomorphisms; no finite collection does so without an extra bound on the candidate degrees. It does not assume an experiment has literally finished infinitely many ticks.

The target of (GW2.3) is also calculable. An endomorphism of $\mathbf{C}_d$ is multiplication by a unique residue $a_d \bmod d$. Compatibility with quotient maps is exactly

$$a_e \equiv a_d \pmod{d} \quad (d \mid e).$$

Conversely, such residues give compatible endomorphisms. Hence

$$\text{Nat}(\mathcal{F}_W, \mathcal{F}_W) \cong \varprojlim_d \mathbb{Z}/d\mathbb{Z}. \quad (\text{GW3.4})$$

The whole finite-clock diagram has more endomorphisms than the retained discrete winding group. To prove this strictly, for $d = 2^a m$, m odd, prescribe

$$u_d \equiv 3 \pmod{2^a}, \quad u_d \equiv 1 \pmod{m}.$$

The coprime Chinese remainder construction gives a unique residue. Its compatibility follows from the same congruences under divisibility. One proof of the construction is to choose integers r, s with $r2^a + sm = 1$; then the residue $3sm + r2^a$ has the two required values. Every u_d is a unit. No integer represents this family: all odd moduli would divide its difference from 1, forcing it to be 1, whereas modulo 4 it is 3. Thus the injection (GW2.3) is not surjective.

This calculation retains the exact role of the whole source. The family of finite clocks alone cannot be silently substituted for the discrete winding group. Its canonical identity transformation is nevertheless well-defined. Choosing a coherent generator in each unpointed clock is a further matter: the compatible choices are exactly the unit families in (GW3.4), by the same componentwise argument.

111.4 GW4. Prime objects are recovered inside the same diagram

A prime clock is intrinsically a nontrivial quotient W/N with no proper nontrivial quotient. A cyclic clock of size d has a quotient of size e exactly when $e \mid d$: the image of a generator must have order dividing d , and divisibility makes the quotient map well-defined. It follows that the prime clocks are exactly those whose calculated sizes are prime.

The action map and its kernel are

$$R \longrightarrow \text{End}(\mathbf{C}_d), \quad [n] \longmapsto [n]_d, \quad \ker = dR. \quad (\text{GW4.1})$$

Surjectivity follows because every endomorphism of a cyclic group is a degree map. At a prime clock, the receiver is the field $\mathbb{Z}/p\mathbb{Z}$: a nonzero residue has an inverse by Bezout, since its greatest common divisor with p is 1. Its annihilator pR is therefore a maximal, hence prime, ideal.

These account for the nonzero prime ideals of R . Indeed every nonzero ideal of $\mathbb{Z}$ is generated by its least positive member, by division with remainder. A composite generator ab , with $a, b > 1$, fails primality, since ab is in the ideal while neither a nor b is. The zero ideal is prime because a product of nonzero integers is nonzero. It is retained as the generic prime, not represented by a simple finite clock.

For positive degrees, a clock C_p first becomes constant at degree p , because $[n]_p = 0$ exactly when $p \mid n$. Before this event every nonzero degree residue is invertible. Quotienting by the identity degree yields the trivial quotient W/W , so it contributes no prime clock. The distinction between the unit and the prime objects follows from their roles in the same diagram; the prime list is not a prior definition of the unit.

111.5 GW5. A later event can change an empirically inferred interval

This section treats the event-time question exactly in its stated observational sense. The receiving time space is an oriented real affine line V : differences of two event times are real numbers, but no origin is required. This is not a metric or a numerical coordinate on τ . For a nonempty event set $\mathcal{E} \subset V$, form

$$K(\mathcal{E}) = \langle t - t' : t, t' \in \mathcal{E} \rangle_{\mathbb{Z}} \subset \mathbb{R}. \quad (\text{GW5.1})$$

Changing the time origin leaves this subgroup unchanged. A common counting interval for the whole history is a positive generator of this group, when one exists.

Every subgroup $K \subset \mathbb{R}$ falls into precisely three cases: $K = \{0\}$; $K = h\mathbb{Z}$ for a unique $h > 0$; or K is dense. Here is a proof. For nonzero K , let $c = \inf(K \cap \mathbb{R}_{>0})$. If $c = 0$, any real open interval (a, b) contains a multiple of a positive $k \in K$ smaller than $b - a$; choose the first integer multiple greater than a . Thus K is dense. If $c > 0$, choose $h \in K$ with $c \leq h < 2c$. If $h > c$, the infimum supplies $k \in K$ with $c \leq k < h$, so $0 < h - k < c$, a contradiction. Hence $h = c$. Division of any real member of K by h leaves a remainder in $K \cap [0, h)$, necessarily zero. Therefore $K = h\mathbb{Z}$; its least positive element gives uniqueness.

The finite record $t_*, t_* + h, \dots, t_* + Nh$ generates $h\mathbb{Z}$. It has the following distinct continuations with the exact same past:

- Continuing at $t_* + (N + 1)h, t_* + (N + 2)h, \dots$ leaves the group $h\mathbb{Z}$.
- Adding a late event at $t_* + (N + \frac{1}{2})h$ and then continuing at integer multiples yields $(h/2)\mathbb{Z}$. The generated group contains h and $(N + \frac{1}{2})h$, hence $h/2$; all its generators lie in $(h/2)\mathbb{Z}$.
- Adding a late event at $t_* + (N + \sqrt{2})h$ yields a dense group. It contains h and $\sqrt{2}h$. It cannot equal $c\mathbb{Z}$, since their ratio would then be rational. Irrationality of $\sqrt{2}$ follows because a reduced fraction a/b satisfying $a^2 = 2b^2$ would force both a and b even. The preceding classification now proves density.

The event sets themselves may remain locally finite in all three examples; density concerns the subgroup of their differences. When the complete group is $h\mathbb{Z}$, every duration dividing all its members is of the form h/m , with m a positive integer: membership of h in $u\mathbb{Z}$ forces $h = mu$, and conversely each such subdivision works. The positive generator h is the maximal common unit; “divides every duration” alone does not distinguish it from finer subdivisions. Therefore a finite observation alone does not establish that every future event is an integral multiple of the inferred interval. This proves the observation claim without inserting the missing future arithmetic as a premise.

These examples are not asserted to be additional points of the Connes–Consani source. Its winding group has an actual discrete presentation, already used in GW1. The proof that

an observation belongs to that source must use this presentation or an exact reconstruction map. Finiteness of an observation and the existence of a global structural theorem are different information conditions; the examples specify the former precisely.

111.6 GW6. What uniqueness of the supporting point does and does not force

Every clock and W has the unique map of underlying sets to $B = \{\tau\}$. For $a \in R$,

$$q_B \circ a = q_B, \quad R \longrightarrow \text{End}_{\text{Set}}(B), \quad a \longmapsto \text{id}_B. \quad (\text{GW6.1})$$

This preserves composition. It is not injective: the identity map on W and the constant group-identity-valued homomorphism are distinct, but have the same image. It is not a claim of an additive morphism into Z_1 , where addition is absent. Thus uniqueness of the supporting point by itself does not distinguish the unit action from all other actions. The complete quotient diagram distinguishes them by GW3.3.

The actual source also lets us compute sections above its fixed point. On monomials its mirror-fixed locus is

$$\text{Fix}(A : M \rightarrow M) = \{0_M, 1, \epsilon\}. \quad (\text{GW6.2})$$

Indeed $(-m, 2m - k) = (m, k)$ implies $m = 0$ and $2k = 0 \pmod{4}$, giving $k = 0, 2$. The absorbing section is fixed separately. The same computation for $F_3(m, k) = (3m, 3k)$ gives the same locus. These elements are fixed under every odd power; thus the simultaneous odd-power fixed locus is also exactly this set. This is a direct calculation in the monomial presentation, not a replacement of the full spherical algebra by a ring of sums.

Connes–Consani identify the invariant generic stalk with their signed base $\mathbb{F}_{12}$, and their quotient retains an orbifold isotropy group of order two. Their global-section proposition likewise returns $\mathbb{F}_{12}$. The source’s signed base and isotropy are not the user’s Z_2 parity label, nor the source singleton B . Locators: `CC.tex`, lines 882–899, Definition `def:absolute_curve0`, and Proposition `prop:global_sections_frobenius`, lines 1058–1146, original paper.

Consequently the inference “one generic point, therefore one distinguished section or one possible counting scale” is not established by uniqueness of that point. Equations (GW6.1)–(GW6.2) show exactly which observation loses the distinction and which sections remain. The unit in GW2 is recovered through the whole winding action, with its source data retained.

111.7 GW7. The counted clock maps exactly into the source’s full-period curve

Connes–Consani’s other source constructs

$$E_p = \mathbb{C}^\times / p^{\mathbb{Z}}.$$

Put $a = \log p$, $b = 2\pi$, with both constants retained. Exponentiation induces

$$\mathbb{C}/(a\mathbb{Z} + ib\mathbb{Z}) \xrightarrow{\sim} E_p, \quad [z] \longmapsto [e^z]. \quad (\text{GW7.1})$$

Changing z by $ma + ib\ell$ multiplies its exponential by p^m , proving well-definedness. Conversely, equal image classes have exponential ratio p^m , so their difference is exactly $ma + ib\ell$. Surjectivity of the complex exponential proves surjectivity. The source’s modular coordinate uses $z/(2\pi i)$; its positive real period then maps to $-i \log p/(2\pi)$, the negative of the displayed modular generator. This sign and the full factor are retained. Its modular-parameter symbol is not the user’s tau.

The count-clock embedding is

$$\iota_p : \mathbb{C}_p \hookrightarrow E_p, \quad [j]_p \longmapsto [p^{j/p}], \quad [x] \in \mathbb{R}/p\mathbb{Z} \longmapsto [(a/p)x] \in \mathbb{R}/a\mathbb{Z}. \quad (\text{GW7.2})$$

The first map is well-defined because adding p multiplies by p . Equality with the identity class means $p^{j/p} = p^m$, so $p \mid j$, proving injectivity. It intertwines every degree map with $[u] \mapsto [u^n]$ by direct substitution. The factor a/p is essential; applying logarithm to an integer representative does not define this quotient map.

The full kernel of $[u] \mapsto [u^p]$ is

$$\left\{ \left[p^{r/p} \exp(2\pi i s/p) \right] : 0 \leq r, s < p \right\}. \quad (\text{GW7.3})$$

These classes are distinct because equality forces both differences $r - r'$ and $s - s'$ divisible by p . Every kernel element lifts to z with pz in the full lattice, proving that this list is exhaustive. The embedded count clock is its real-direction subgroup; the angular direction has not been discarded.

There are two distinct source-related operations whose exact comparison matters. The Frobenius used to form the source quotient acts on its covering parameter by $u \mapsto pu$. On E_p this is the identity, since $[pu] = [u]$. The power map $u \mapsto u^p$ has the p^2 -element kernel (GW7.3). They cannot be substituted for each other in a weight calculation. Source: Connes–Consani, *On the Absolute Geometry of Spec Z and the Fargues–Fontaine curve*, v1, original `FF.tex`, “The Tate curve E_p ,” lines 1548–1579.

111.8 GW8. The full first-cohomology chain and its exact boundary

We now calculate globally, retaining every cohomological component. Fix an integer $n \geq 2$, keep $a = \log p$, $b = 2\pi$, and define the actual analytic curves

$$E_k = \mathbb{C}/(n^k a\mathbb{Z} + ib\mathbb{Z}), \quad k \geq 0. \quad (\text{GW8.1})$$

These are the quotients $\mathbb{C}^\times/(p^{n^k})^\mathbb{Z}$. Composite parameters are stated explicitly, not renamed as prime places. In $z = x + i\theta$, retain $dx, d\theta$, orientation $dx \wedge d\theta$, and metric $dx^2 + d\theta^2$. They descend because lattice translations preserve the forms. The area is exactly $n^k ab$.

The first complex de Rham cohomology is

$$V_k = H^1(E_k, \mathbb{C}) = \mathbb{C}[dx] \oplus \mathbb{C}[d\theta]. \quad (\text{GW8.2})$$

For completeness, expand a smooth form in Fourier modes with frequencies $r = 2\pi m/(n^k a)$, $s = 2\pi \ell/b$. A closed one-form at a nonzero frequency has coefficients A, B satisfying $rB = sA$. The scalar mode with coefficient $(rA + sB)/(i(r^2 + s^2))$ has differential equal to it. Smooth Fourier coefficients decay faster than every power; the displayed division preserves smooth convergence, so all nonconstant closed modes are exact. The two constant forms are not exact, since their periods on the two lattice cycles are respectively $(n^k a, 0)$ and $(0, b)$. This proves (GW8.2).

Since $*dx = d\theta$ and $*d\theta = -dx$, the full Hermitian form, linear in its first argument, is

$$h_k(A dx + B d\theta, C dx + D d\theta) = n^k ab(A\overline{C} + B\overline{D}). \quad (\text{GW8.3})$$

This follows by integrating the wedge product with the conjugate Hodge star over the original rectangle. Its positivity is explicit. No coordinate or pairing is assigned to tau.

There are two holomorphic isogenies

$$g_k : E_{k+1} \longrightarrow E_k, \quad [z] \longmapsto [z], \quad f_k : E_k \longrightarrow E_{k+1}, \quad [z] \longmapsto [nz]. \quad (\text{GW8.4})$$

The first uses the lattice inclusion; its kernel consists of the n real classes $rn^k a$, $0 \leq r < n$. For the second, multiplication by n sends the source lattice into the target lattice, and the kernel consists of the n angular classes ibr/n , $0 \leq r < n$. Both are therefore degree n coverings, with the exact real and angular factors displayed. On the unchanged forms their pullbacks are

$$g_k^* : V_k \rightarrow V_{k+1}, \quad (A, B) \mapsto (A, B), \quad f_k^* : V_{k+1} \rightarrow V_k, \quad (A, B) \mapsto (nA, nB). \quad (\text{GW8.5})$$

Equations (GW8.3)–(GW8.5) give, for $v \in V_k, w \in V_{k+1}$,

$$h_{k+1}(g_k^*v, w) = h_k(v, f_k^*w), \quad \|g_k^*v\|_{k+1}^2 = n\|v\|_k^2. \quad (\text{GW8.6})$$

The same norm-square multiplier n holds for f_k^* in its own direction. Both compositions are multiplication by n on first cohomology. These local identities agree with the separate programme calculation `TATE_H1_DERIVATION.md`, T5–T6; their complete proof needed here has been included.

Assemble the whole forward chain as the Hilbert space

$$\mathcal{H}_+ = \left\{ v = (v_k)_{k \geq 0} : \sum_{k \geq 0} n^k ab(|A_k|^2 + |B_k|^2) < \infty \right\}, \quad v_k = (A_k, B_k). \quad (\text{GW8.7})$$

Use precisely the pullbacks in (GW8.5), without rescaling them:

$$(Sv)_0 = 0, \quad (Sv)_{k+1} = g_k^*v_k = v_k.$$

Direct substitution in the series gives $\|Sv\|^2 = n\|v\|^2$, and the inner-product identity gives

$$(S^*v)_k = f_k^*v_{k+1} = nv_{k+1}, \quad S^*S = nI, \quad SS^* = n(I - P_0). \quad (\text{GW8.8})$$

Here P_0 is the orthogonal projection to the entire first-cohomology space V_0 . Therefore

$$S^*S - SS^* = nP_0, \quad \text{rank } P_0 = 2, \quad \text{Tr}(nP_0) = 2n. \quad (\text{GW8.9})$$

The boundary has not disappeared on taking infinitely many forward stages. It lies in first cohomology itself; it is not a constant-function H^0 component.

The full spectrum is calculable. For every $|\lambda| < \sqrt{n}$, choose $v_0 \neq 0$ and set

$$v_k = (\lambda/n)^k v_0.$$

Then

$$S^*v = \lambda v, \quad \|v\|^2 = ab(|A_0|^2 + |B_0|^2) \sum_{k \geq 0} (|\lambda|^2/n)^k = \frac{ab(|A_0|^2 + |B_0|^2)}{1 - |\lambda|^2/n}. \quad (\text{GW8.10})$$

Thus every point in the open disk is an eigenvalue of S^* . For $|\lambda| > \sqrt{n}$, the series

$$(S - \lambda I)^{-1} = -\lambda^{-1} \sum_{j \geq 0} (S/\lambda)^j$$

converges in operator norm; the identical argument applies to S^* . For $|\lambda| = \sqrt{n}$, truncate the displayed vector for S^* at stage K . Its squared norm is $(K+1)ab\|v_0\|_{\mathbb{C}^2}^2$ and its squared residual is $nab\|v_0\|_{\mathbb{C}^2}^2$. The residual ratio tends to zero, excluding a bounded inverse on the circle. Taking adjoints preserves invertibility with conjugate spectral parameter. Consequently

$$\text{spec}(S) = \text{spec}(S^*) = \{\lambda \in \mathbb{C} : |\lambda| \leq \sqrt{n}\}. \quad (\text{GW8.11})$$

The operator S itself has no eigenvector: for a nonzero eigenvalue its stage-zero equation forces $v_0 = 0$, then all components vanish recursively; at eigenvalue zero, $S^*S = nI$ proves injectivity. This specifies spectrum and eigenvalues separately.

The exact singular moduli are

$$|S| = \sqrt{n}I, \quad |S^*| = \sqrt{n}(I - P_0). \quad (\text{GW8.12})$$

The first identity does force any actual eigenvalue of S to have modulus $\sqrt{n}$: take norms in $Sv = \lambda v$. Here S has no eigenvalues, and its spectrum is the disk. Its adjoint has constant

nonzero singular values together with the two-dimensional zero singular subspace, and has every interior point as an eigenvalue. Thus the proved distinction is between constant singular modulus and confinement of the full spectrum, with the adjoint's eigenvalues calculated separately. It occurs inside first cohomology itself. The exact boundary is (GW8.9).

The boundary also has an integer-valued global index. The range of S is exactly the closed subspace with stage-zero component zero, while its kernel is zero. Hence

$$\text{index } S = \dim \ker S - \dim \ker S^* = -2. \quad (\text{GW8.13})$$

Throughout $|\lambda| < \sqrt{n}$, the estimate

$$\|(S - \lambda I)v\| \geq (\sqrt{n} - |\lambda|)\|v\|$$

proves that the range is closed and the kernel zero; (GW8.10) gives a two-dimensional cokernel. Thus the same index is -2 throughout the open disk.

At a finite cutoff $0 \leq k \leq K$, with $K \geq 1$, keep both ends. The compressed forward operator S_K satisfies

$$\begin{aligned} S_K^* S_K &= n(I - P_K), & S_K S_K^* &= n(I - P_0), \\ S_K^* S_K - S_K S_K^* &= n(P_0 - P_K), & \text{Tr}(n(P_0 - P_K)) &= 2n - 2n = 0. \end{aligned} \quad (\text{GW8.14})$$

These identities follow componentwise from the same weighted adjoint calculation, retaining the terminal block that no longer maps forward. The matrix is strictly block triangular, so its characteristic polynomial is $t^{2(K+1)}$. It is nilpotent of order $K+1$, with every eigenvalue zero. Extending these compressions by zero gives strong convergence to S : for any vector, the weighted tail tends to zero, and the shift has the uniform norm bound $\sqrt{n}$. Nevertheless their spectral sets do not converge to (GW8.11). This states the exact limitation of finite-cutoff eigenvalues for this receiving chain, rather than asserting a result about a different programme matrix.

111.9 GW9. Completing the chain in both directions changes the result exactly

For comparison, retain the same explicit curves (GW8.1) for every $k \in \mathbb{Z}$. Their period lengths are $n^k a$, including all negative stages. These are well-defined analytic quotients, but the parameters p^{n^k} at negative k are not being asserted to be norms of integer prime places.

On

$$\mathcal{H}_{\mathbb{Z}} = \left\{ (v_k)_{k \in \mathbb{Z}} : \sum_{k \in \mathbb{Z}} n^k ab \|v_k\|_{\mathbb{C}^2}^2 < \infty \right\}$$

let $(\tilde{S}v)_k = v_{k-1}$. The same direct calculation now gives

$$\tilde{S}^* v = (nv_{k+1})_k, \quad \tilde{S}^* \tilde{S} = \tilde{S} \tilde{S}^* = nI, \quad \|\tilde{S}^{-1}\| = 1/\sqrt{n}. \quad (\text{GW9.1})$$

The inverse is the backward shift; the norm follows by shifting the weighted series. The outside-disk Neumann argument from GW8 and the identity $\tilde{S} - \lambda I = \tilde{S}(I - \lambda \tilde{S}^{-1})$ exclude both $|\lambda| > \sqrt{n}$ and $|\lambda| < \sqrt{n}$ from the spectrum. For $|\lambda| = \sqrt{n}$, truncate $v_k = (\lambda/n)^k v_0$ to $-K \leq k \leq K$. For $\tilde{S}^* - \lambda I$ the only residuals occur at the two ends. Their total squared norm is $2nab\|v_0\|^2$, while the vector's squared norm is $(2K+1)ab\|v_0\|^2$. Thus the circle is in the spectrum. We have proved

$$\text{spec}(\tilde{S}) = \{\lambda : |\lambda| = \sqrt{n}\}. \quad (\text{GW9.2})$$

Extension by zero at negative stages gives an isometric map

$$J : \mathcal{H}_+ \hookrightarrow \mathcal{H}_{\mathbb{Z}}, \quad \tilde{S}J = JS, \quad J^* \tilde{S}^* J = S^*. \quad (\text{GW9.3})$$

All identities follow componentwise. Its image is invariant under $\tilde{S}$ but not under $\tilde{S}^*$: the latter moves a vector at stage zero to stage minus one with coefficient n . This is exactly the lost boundary in (GW8.9), now expressed as a map between the two complete constructions.

The difference between the disk and the circle is therefore not inferred from a slogan about infinity. It is proved from the retained first stage, the exact adjoint, and the presence or absence of negative stages. Both objects contain infinitely many stages. The comparison is an explicit isometric inclusion with a non-invariant adjoint action.

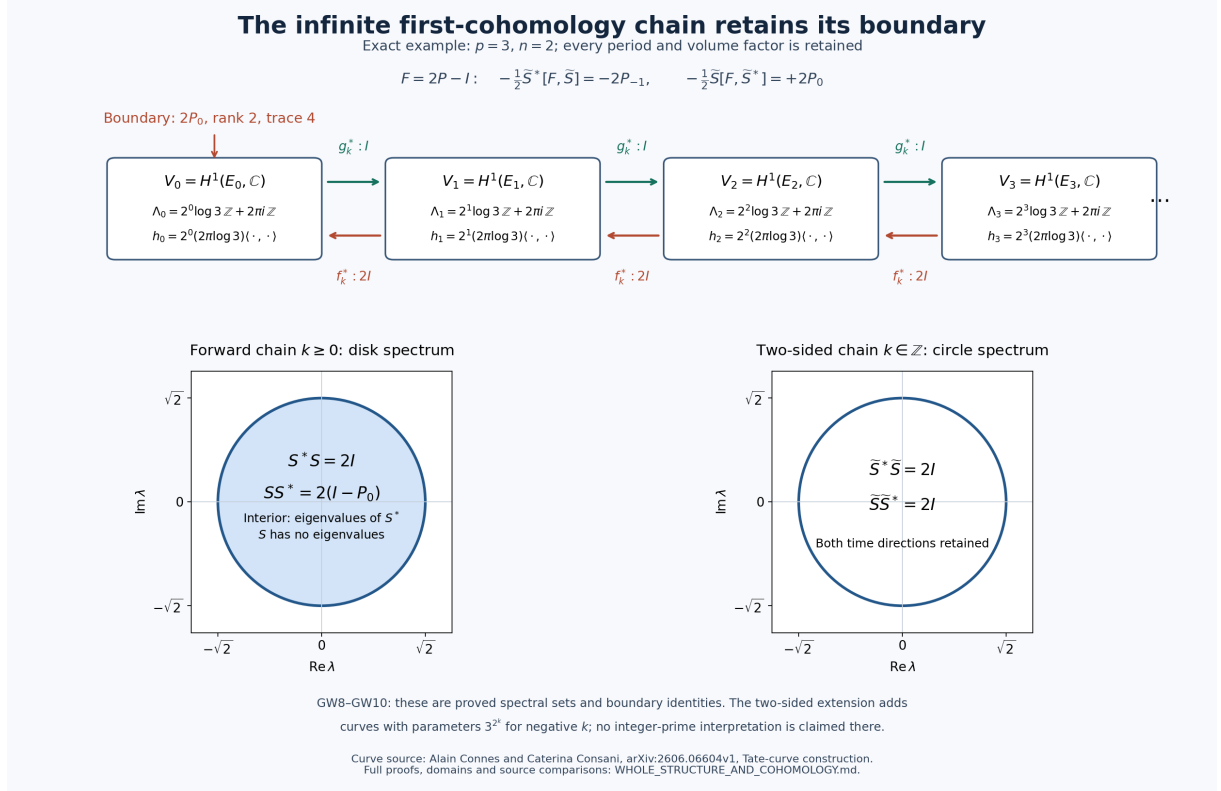


Figure 37: The full-period first-cohomology chain, its boundary, and its two spectra

The upper row displays the degree- n maps with both original period factors. The lower panels show the proved spectral sets, not numerical samples. The disk includes eigenvalues of the adjoint; the circle belongs to the enlarged two-sided analytic chain. The label nP_0 retains the entire two-dimensional first-cohomology boundary. Proofs: GW8.4–GW9.3. Reproducible source: `render_whole_cohomology.py`.

111.10 GW10. Applying the source’s quantized differential with every degree factor

In *BC-system*, *absolute cyclotomy and the quantized calculus*, Connes–Consani use $F = 2P - I$, $d_F U = FU - UF$, and, for a triangular unitary U , identify the positive kernel term in $-\frac{1}{2}U^*d_F U$. We now evaluate the complete expression for the actual operators above. The unscaled degree maps have $A^*A = nI$; they are not declared unitary. Source locator: Connes–Consani, arXiv:2112.08820v1, subsection “The Main Lemma,” `SullivanRH.tex`, lines 540–566, Proposition `fromqi` and Lemma `mainlemma`.

On $\mathcal{H}_{\mathbb{Z}}$, take P to project onto stages $k \geq 0$, and let P_j denote projection onto the full two-dimensional stage j . For either $A = \tilde{S}$ or $A = \tilde{S}^*$, (GW9.1) gives the full identity

$$\begin{aligned} -\frac{1}{2}A^*d_F A &= -\frac{1}{2}(A^*(2P - I)A - A^*A(2P - I)) \\ &= -\frac{1}{2}(2A^*PA - nI - 2nP + nI) \\ &= nP - A^*PA. \end{aligned} \tag{GW10.1}$$

Every identity contribution and degree factor is displayed. The forward map crosses the threshold from stage -1 , while the backward map crosses it from stage zero. Componentwise evaluation gives

$$\tilde{S}^*P\tilde{S} = nP_{\{k \geq -1\}}, \quad \tilde{S}P\tilde{S}^* = nP_{\{k \geq 1\}}.$$

Subtracting the stated projections in (GW10.1) proves

$$-\frac{1}{2}\tilde{S}^*d_F\tilde{S} = -nP_{-1}, \quad -\frac{1}{2}\tilde{S}d_F(\tilde{S}^*) = nP_0. \quad (\text{GW10.2})$$

Thus the two directions have explicitly opposite boundary signs. For any bounded positive operator f , the products in the next display have finite rank, so their ordinary traces are defined and

$$\begin{aligned} -\frac{1}{2}\text{Tr}(f\tilde{S}^*d_F\tilde{S}) &= -n\text{Tr}(fP_{-1}) \leq 0, \\ -\frac{1}{2}\text{Tr}(f\tilde{S}d_F(\tilde{S}^*)) &= n\text{Tr}(fP_0) \geq 0. \end{aligned} \quad (\text{GW10.3})$$

For the sign, compute the trace on the finite-dimensional stage: each diagonal entry is a positive quadratic-form value divided by the positive squared length of its coordinate vector. Their sum is nonnegative. In the unchanged coordinate basis the exact factor in that length is $n^j ab$; none has been omitted. At $f = I$, the two traces are respectively $-2n$ and $+2n$.

The backward compression on $P\mathcal{H}_{\mathbb{Z}} \cong \mathcal{H}_+$ is exactly S^* , with kernel V_0 ; its positive term is n times the projection onto that kernel. The forward compression is S , whose kernel is zero and whose opposite crossing has the negative term in (GW10.2). This proves the precise connection to the source lemma while retaining the degree factor and checking the orientation. It does not discard a crossing block to obtain positivity.

The source's deck Frobenius $u \mapsto pu$, the power map $u \mapsto u^n$, the isogenies between different curves, and the global chain operators now have exact formulas and connecting maps. No equation identifies the last operator and its test space with the arithmetic trace operator and tests that recover the zeros of the original zeta function. In particular the positive trace in (GW10.3) has been proved for its stated receiver; it is not substituted for Weil's complete arithmetic functional. No completion multiplier or archimedean summand has been removed from a zeta formula, because no such replacement formula has been introduced here.

111.11 GW11. Returning the whole calculation through every support label

Keep an arbitrary bounded distributive lattice L , not a Boolean replacement. The retained programme coefficient object is

$$\begin{aligned} G_L(\mathbb{C}) &= \{(0, \lambda) : \lambda \in L\} \cup \{(c, 1_L) : c \in \mathbb{C}\}, \\ (c, \lambda) + (d, \mu) &= (c + d, \lambda \vee \mu), \quad (c, \lambda)(d, \mu) = (cd, \lambda \wedge \mu). \end{aligned} \quad (\text{GW11.1})$$

Write $z_\lambda = (0, \lambda)$ and $\hat{c} = (c, 1_L)$. In particular its additive identity is z_{0_L} , its unit is $\hat{1}$, and its supported zero is z_{1_L} . These names deliberately do not identify the historical coefficient zeros with the current primitive tau. The addition in (GW11.1) is an operation of this specified receiving semiring; it does not reinstate the user's retracted addition on tau.

Source: the retained programme foundation *Split Support Geometry*, v11, credited in that source to The Clankers. The exact proof locators, in original source lines 1219–1307, are:

- Definition `def:lattice-split`.
- Theorem `thm:lattice-normal-form`.

For completeness, the carrier is closed under the operations: any nonzero sum has a summand at top support, and a nonzero product has both factors at top. Associativity and commutativity follow componentwise, while distributivity uses exactly the distributive lattice identity for meet over join. Thus it is a semiring. Amplitude projection

$$\pi : G_L(\mathbb{C}) \longrightarrow \mathbb{C}, \quad (c, \lambda) \mapsto c \quad (\text{GW11.2})$$

preserves both operations and the identities. Its zero fibre is the full lattice $\{z_\lambda : \lambda \in L\}$, with no identifications among its labels before projection.

Take the free semimodule of finite-support sequences

$$\mathcal{V}_{L,+} = \bigoplus_{k \geq 0} G_L(\mathbb{C})^2,$$

where finite support means that only finitely many coordinates differ from z_{0_L} , including coordinates with zero amplitude but non-bottom support. Define the actual lifts of the two previously calculated operators by

$$(\widehat{S}v)_0 = z_{0_L}^2, \quad (\widehat{S}v)_{k+1} = v_k, \quad (\widehat{B}v)_k = \widehat{n} v_{k+1}. \quad (\text{GW11.3})$$

Each map is semimodule-linear by (GW11.1), and maps finite-support sequences to finite-support sequences. Multiplication by $\widehat{n}$ preserves every support label, since $1_L \wedge \lambda = \lambda$. Coordinatewise amplitude projection gives the exact intertwining maps into the dense finite-support part of $\mathcal{H}_+$:

$$\pi \widehat{S} = S\pi, \quad \pi \widehat{B} = S^* \pi. \quad (\text{GW11.4})$$

This proves the comparison with the calculated Hilbert adjoint. It does not declare an unspecified adjoint operation on semimodules.

Let $\widehat{P}_0$ retain stage zero and replace every other coordinate by z_{0_L} , and let $\widehat{P}_{\geq 1}$ do the converse. The disjoint-coordinate decomposition gives

$$I = \widehat{P}_0 + \widehat{P}_{\geq 1}, \quad \widehat{B}\widehat{S} = \widehat{n}I, \quad \widehat{S}\widehat{B} = \widehat{n}\widehat{P}_{\geq 1}. \quad (\text{GW11.5})$$

In particular, $\ker \widehat{B}$ is the entire stage-zero semimodule $G_L(\mathbb{C})^2$. Indeed $\widehat{n}$ is invertible with inverse $\widehat{1/n}$, so every higher coordinate must equal the actual additive identity. Moreover

$$\mathcal{V}_{L,+} \longrightarrow G_L(\mathbb{C})^2 \oplus \mathcal{V}_{L,+}, \quad v \longmapsto (v_0, (v_{k+1})_{k \geq 0})$$

is an isomorphism with inverse insertion of the initial block. This is a complete decomposition retaining every labelled zero. Under amplitude projection it becomes the stage decomposition used in GW8.

Now retain both summands of the apparent commutator instead of assuming additive cancellation. The well-defined semiring expression is

$$\mathcal{D}_L = \widehat{B}\widehat{S} + \widehat{-1}\widehat{S}\widehat{B}. \quad (\text{GW11.6})$$

Its complete diagonal, with both entries in every stage, is

$$\mathcal{D}_L = \text{diag}(\widehat{n}I_2, z_{1_L}I_2, z_{1_L}I_2, z_{1_L}I_2, \dots), \quad (\text{GW11.7})$$

with every off-diagonal entry z_{0_L} . To verify it, at stage zero the second summand is absent and the first is $\widehat{n}$. At every other stage the sum is exactly

$$\widehat{n} + \widehat{-n} = (n + (-n), 1_L \vee 1_L) = (0, 1_L) = z_{1_L}.$$

Consequently

$$\pi \mathcal{D}_L = nP_0\pi, \quad \mathcal{D}_L \neq \widehat{n}\widehat{P}_0 \quad (0_L \neq 1_L). \quad (\text{GW11.8})$$

For the nonidentity, apply both maps to a vector whose only non-absent coordinate is $\widehat{1}$ at stage one: the outputs there are z_{1_L} and z_{0_L} , respectively. More generally a stage-one input z_λ returns z_λ under $\mathcal{D}_L$, since $1_L \wedge \lambda = \lambda$. Thus the whole lattice of distinctions survives in the cancellation tail; only its amplitude is zero. This is the exact additional information the support return contributes to the global boundary calculation.

At finite cutoff $K \geq 1$, first truncate both shifts to stages $0, \dots, K$, giving $\widehat{S}_K, \widehat{B}_K$. Form

$$\mathcal{D}_{L,K} = \widehat{B}_K \widehat{S}_K + \widehat{-1} \widehat{S}_K \widehat{B}_K.$$

Its two endpoint blocks are $\widehat{n}I_2$ and $\widehat{-n}I_2$; every interior block is $z_{1_L}I_2$. This is not the compression of the already formed infinite operator $\mathcal{D}_L$, whose last retained diagonal block would still be $z_{1_L}I_2$. Its full supported trace is

$$\text{Tr}_L \mathcal{D}_{L,K} = (2n + (-2n) + 0, 1_L) = z_{1_L}, \quad (\text{GW11.9})$$

where the zero amplitude term includes all interior blocks and the support join is top. Its ordinary amplitude trace is zero, as in GW8.14. For the infinite diagonal (GW11.7), no infinite sum in $G_L(\mathbb{C})$ has been defined; an infinite supported trace is therefore not asserted. The complete sequence of diagonal labels and its action on every finite-support input are already defined and proved.

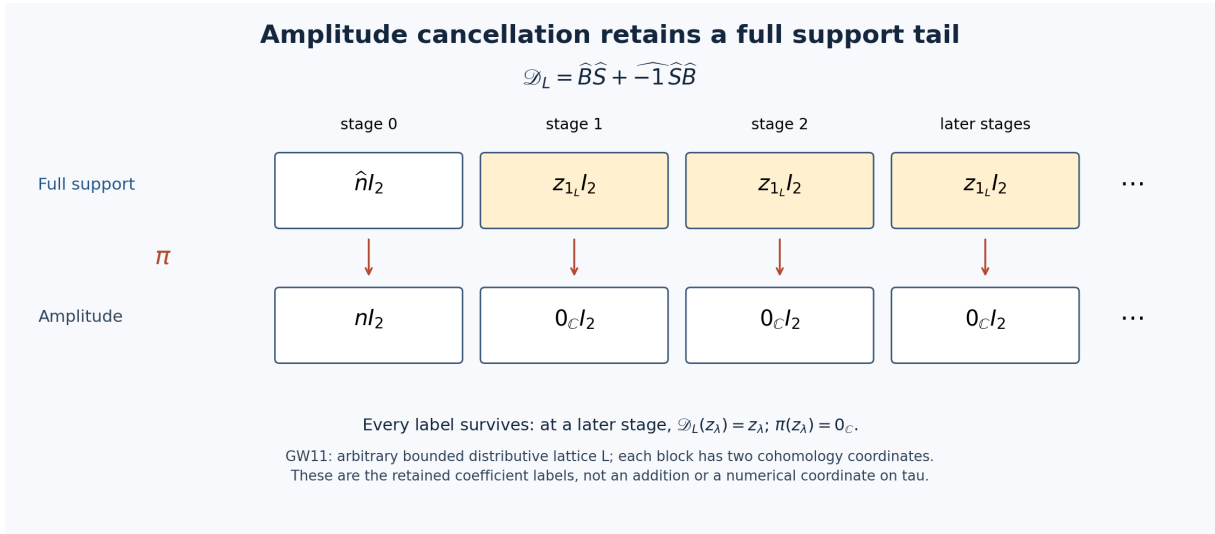


Figure 38: The full lattice retains the cancellation tail that amplitude projection removes

Figure 2 displays (GW11.7) and its exact amplitude projection. The lower comparison retains a general label λ , not only the top label. In the one-element lattice the displayed supported zero equals the additive identity; the nonidentity in (GW11.8) explicitly requires a nontrivial lattice. This is a coefficient lift of the specified cohomology operators. No unconstructed cohomology theory of a split-support curve is silently identified with it.

111.11.1 Sources and provenance

The finite cyclic-group and subgroup arguments, the endomorphism description of an infinite cyclic group, and unilateral/bilateral shift spectral phenomena are established mathematics; no historical novelty is claimed for them. The programme calculation is their exact assembly with the specified source, full periods, two isogenies, typed support comparison, and latest event-history question. Human source authorship remains Alain Connes and Caterina Consani.

Original author sources actually used: `CC.tex`, arXiv:2609.00299v1, the ranges and named definitions cited above; `FF.tex`, arXiv:2606.06604v1, lines 1515–1593; `SullivanRH.tex`, arXiv:2112.08820v1, lines 540–566 for GW10. The previously retained programme proof `ORDERED_PRIME_ABSORPTION.md`, PA1–PA4, supplies the context being strengthened by GW1–GW4. Its necessary group and quotient proofs are repeated here so this note does not require guessing an omitted definition. The local `TATE_H1_DERIVATION.md` has been read through T3, in part in T4, and in full in T5–T6; its full-period calculation is credited above. Source hashes and exact intake provenance are in the accompanying private reading ledger. Quoted assistant statements in the intake are not accepted as proved claims merely because they were supplied with the request.

112 Integral history, the identified zeta function, and a detected off-critical orbit

24 September 2026. Complete receiving calculation IH1–IH8. The question is whether the complete reconstruction of the integers, their unit and their normed prime spectrum makes an off-critical zero an undetected object, and therefore a contradiction. This note accepts the completed scalar zeta identification in item 26 and calculates the proposed counterfactual in that identified function and its actual quotient.

The source notation remains Z_0 for absence, Z_1 for primitive presence, and Z_2 for carried integer parity. Source τ has no parity and no addition. The source support η , the arithmetic identity id_W , the identity of a coefficient algebra, and a zero vector in a linear receiving space are not identified. In particular, a vanishing observation in a linear receiver does not mean that the unsupported source element has been identified with a numerical zero.

112.1 IH1. The completed identification is an input carried forward

The complete source comparison is CG0–CG8 of *The unique common support and canonical reconstruction of the original zeta function*, together with TP0–TP14 of *First-appearance time, primitive periods, and the original zeta function*. The receiving composition is I26.1–I26.4 of *Applying the canonical reconstruction to item 26*. The source versions and their complete files are retained alongside this note. They use the actual signed Connes–Consani chart system, not a bare singleton from which a stalk is presumed to emerge.

For the essential composition, retain

$$1 \longrightarrow H \longrightarrow G \longrightarrow W \longrightarrow 1, \quad R = \text{End}_{\text{Ab}}(W), \quad \mathbf{1}_R = \text{id}_W. \quad (\text{IH1.1})$$

Here H retains the four finite-order states of the signed source and W is its infinite cyclic quotient. CG0 constructs the unique primitive step of each complete ordered chart quotient; CG2 proves the twisted chart restrictions and their group-completion universal properties. CG4 proves $R \cong \mathbb{Z}$, with the isomorphism $n \mapsto [n]$, where $[n](x) = x^n$. The comparison $b : W \rightarrow W'$ preserves every power action because

$$b[n]b^{-1}(x') = b((b^{-1}x')^n) = (x')^n. \quad (\text{IH1.2})$$

It follows that it preserves each residue norm $p = |W/[p]W|$. The complete return measures are

$$\mathcal{R} = \sum_p \sum_{k \geq 1} \frac{1}{k} \delta_{k \log p}, \quad \mathcal{D} = \exp_* \mathcal{R} = \delta_0 + \sum_{n \geq 2} \delta_{\log n}, \quad \mathcal{W} = t\mathcal{R} = \sum_p \sum_{k \geq 1} (\log p) \delta_{k \log p}. \quad (\text{IH1.3})$$

The coefficients of the convolution exponential are one by unique prime factorization. In detail, for one prime the formal identity $\exp(\sum_{k \geq 1} X^k/k) = \sum_{k \geq 0} X^k$ follows by differentiating and comparing coefficients, with constant coefficient one. Finite-prime convolution products enumerate exponent tuples once each. On a bounded time interval only finitely many primes, repetitions and convolution orders contribute, so the resulting identity holds for the complete locally finite measures. This retains the empty convolution and its mass-one atom at zero.

For $\Re s > 1$, absolute convergence gives

$$\int e^{-st} d\mathcal{D}(t) = 1 + \sum_{n \geq 2} n^{-s} = \zeta(s), \quad \int e^{-st} d\mathcal{W}(t) = -\frac{\zeta'(s)}{\zeta(s)}. \quad (\text{IH1.4})$$

Indeed $\sum n^{-\sigma} < \infty$ and $\sum (\log n) n^{-\sigma} < \infty$ for $\sigma > 1$, uniformly on smaller closed half-planes. These bounds justify the products, convolution transforms and derivative of the logarithmic

series. Both chart constructions preserve every term in (IH1.3), so their transforms agree with the original zeta function. Its classical meromorphic continuation is unique by the identity theorem: two such continuations have zero difference on the open half-plane and hence everywhere. All zero germs and their multiplicities agree. No completion multiplier is introduced in (IH1.4). The continuation fact is DLMF 25.2i; this calculation does not claim a new proof of that classical fact.

The chart mirror induces inversion $a(x) = x^{-1}$ on W . Its arithmetic transport is conjugation, and

$$(a[n]a^{-1})(x) = ((x^{-1})^n)^{-1} = x^n. \quad (\text{IH1.5})$$

Thus this particular transport preserves the reconstructed zeta function at the same parameter s . The completed scalar identification is retained. An isometry of a cohomology space or fixedness of each spectral character is not an additional consequence contained in this identity.

112.2 IH2. The actual quotient detects an off-critical zero

Use the linked task's spaces exactly. Let $\mathcal{A}$ consist of smooth $k : (0, \infty) \rightarrow \mathbb{C}$ with

$$p_{N,j}(k) = \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j k(u)| < \infty \quad (N, j \geq 0).$$

Let S_0^{even} consist of even Schwartz functions f on $\mathbb{R}$ with $f(0) = 0$ and $\int f = 0$. Retain the original sum and its factor:

$$\begin{aligned} \mathcal{E}f(u) &= u^{1/2} \sum_{n \geq 1} f(nu), & I &= \overline{\mathcal{E}(S_0^{\text{even}})}^{\mathcal{A}}, & Q &= \mathcal{A}/I, \\ F_k(s) &= \int_0^\infty k(u) u^{s-1/2} \frac{du}{u}. \end{aligned} \quad (\text{IH2.1})$$

The integral is entire in s : any fixed real strip and any derivative in s are dominated by a stronger defining seminorm. For $\Re s > 1$, absolute interchange and $v = nu$ give

$$F_{\mathcal{E}f}(s) = \zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v}. \quad (\text{IH2.2})$$

The second factor is holomorphic throughout the critical strip: evenness and $f(0) = 0$ give $f(v) = O(v^2)$ at zero, and Schwartz decay controls infinity. The identity continues to this strip. Hence every derivative of order below the multiplicity of an actual zero vanishes on the source image, and continuity of Mellin derivatives gives the same statement on I . Poisson summation with both stated endpoints zero proves that $\mathcal{E}f \in \mathcal{A}$; these are the precise source and topology used in G2 and GIQ1, not an L^2 closure substituted for them.

For a zero ρ of multiplicity m_ρ , define

$$A_\rho = \mathbb{C}[t_\rho]/(t_\rho^{m_\rho}), \quad j_\rho([k]) = \sum_{r=0}^{m_\rho-1} \frac{F_k^{(r)}(\rho)}{r!} t_\rho^r. \quad (\text{IH2.3})$$

This is a well-defined algebra map for multiplicative convolution. Its product law is the derivative product rule with the displayed factorials. G4 identifies the continuous characters of Q with the original nontrivial zeros; its complete retained kernel comparison is

$$F_{\mathcal{E}f_0}(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad f_0(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2}. \quad (\text{IH2.4})$$

The limits at zero and one are $1/8$. At $s = -2r$, $r \geq 1$, the value is

$$\frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r) \neq 0. \quad (\text{IH2.5})$$

The residue of $\Gamma(s/2)$ is $2(-1)^r/r!$; multiplication by the simple trivial-zero germ proves the formula. Simplicity follows from the sine factor of the original functional equation and its nonzero remaining factors there. Thus the completion data are explicit in this comparison and are not a replacement for the original zeta function.

Here is a direct detector that answers the proposed inference. Choose a nonnegative nonzero $\varphi \in C_c^\infty(\mathbb{R})$, and for any actual zero, on or off the line, set

$$k_\rho(e^v) = \varphi(v)e^{-(\rho-1/2)v}.$$

It belongs to $\mathcal{A}$, and

$$F_{k_\rho}(\rho) = \int_{\mathbb{R}} \varphi(v) dv > 0. \quad (\text{IH2.6})$$

Since every element of I has zero observation at ρ , this test gives a nonzero class of the actual quotient. No zero outside the detected system has been introduced. The proposed off-critical zero is an actual detected character under the counterfactual assumption itself.

112.3 IH3. The complete counterfactual orbit and every prime action

For the contradiction test, write the hypothetical actual off-critical zero as $\rho = \beta + i\gamma$, with $0 < \beta < 1$, $\beta \neq 1/2$. Nontrivial zeros are nonreal: for real $0 < s < 1$, the alternating series $\sum_{n \geq 1} (-1)^{n-1} n^{-s}$ is positive, by grouping its positive consecutive pairs, whereas $1 - 2^{1-s} < 0$; its quotient is $\zeta(s)$. Thus $\gamma \neq 0$.

The full four-point orbit and its full local algebra are

$$\mathcal{O}_\rho = \{\rho, 1 - \bar{\rho}, \bar{\rho}, 1 - \rho\}, \quad B_\rho = \prod_{\omega \in \mathcal{O}_\rho} \mathbb{C}[t_\omega]/(t_\omega^m), \quad j_\mathcal{O} : \mathcal{Q} \longrightarrow B_\rho. \quad (\text{IH3.1})$$

All four points are distinct and have the same multiplicity m , by conjugation and the original functional equation. The product observation is surjective. Indeed its finitely many coordinate functionals on compact smooth logarithmic tests have densities $v^r e^{(\omega-1/2)v}/r!$. To prove their independence, apply $\prod_{\nu \neq \omega} (\partial_\nu - (\nu - 1/2))^m$ to a vanishing exponential-polynomial combination. It kills the other summands and is invertible on the remaining polynomial after removing its exponential, because each resulting operator $\partial_\nu + \omega - \nu$ has nonzero constant diagonal. Every coefficient therefore vanishes. Compact smooth tests separate these densities, proving surjectivity onto the finite-dimensional product.

Retain both compared operators

$$V_a k(u) = k(u/a), \quad W_a = a^{1/2} V_a, \quad F_{V_a k}(s) = a^{s-1/2} F_k(s), \quad F_{W_a k}(s) = a^s F_k(s). \quad (\text{IH3.2})$$

They preserve I : substitution gives $V_a \mathcal{E}f = \mathcal{E}(a^{-1/2} f(\cdot/a))$, and the new Schwartz function has both required vanishing endpoints. Continuity gives preservation of the closure. The inverse is the corresponding operator at $1/a$. On every local factor the full W_a action is multiplication by

$$w_{a,\omega} = a^\omega \sum_{r=0}^{m-1} \frac{(\log a)^r t_\omega^r}{r!}, \quad j_\omega W_a = M_{w_{a,\omega}} j_\omega. \quad (\text{IH3.3})$$

The finite exponential retains all multiplicity information. Its product rule follows by the binomial formula, so $W_a W_b = W_{ab}$ on the complete quotient. Thus all prime powers and products act in the same object.

The off-critical displacement is visible at every prime, through the original unscaled coordinate:

$$D_p(\rho) = |p^\rho|^2 - p = p^{2\beta} - p, \quad \log \frac{|p^\rho|^2}{p} = (2\beta - 1) \log p. \quad (\text{IH3.4})$$

For every pair of positive integers, the complete defect satisfies

$$D_{ab} = bD_a + aD_b + D_aD_b, \quad D_1 = 0,$$

$$D_{p^k} = (p^{2\beta} - p) \sum_{j=0}^{k-1} p^{2\beta(k-1-j)} p^j. \quad (\text{IH3.5})$$

Expand $(D_a + a)(D_b + b) - ab$ for the first identity and use the difference-of-powers identity for the second. Every term in the last sum is positive. Thus retaining every repetition detects and propagates the displacement; it does not set it to zero. No new prime or nonintegral integer label appears.

Reflection on B_ρ is $(x^\#)_\omega(t) = \overline{x_{1-\overline{\omega}}}(-t)$. It gives the exact full-jet identity $w_a^\# w_a = a1_{B_\rho}$. Conjugation supplies the other orbit permutation. The algebra trace form is

$$\text{Tr}_{B_\rho/\mathbb{C}}(M_{x^\#y}) = m \sum_{\omega \in \mathcal{O}_\rho} \overline{x_{1-\overline{\omega}}(0)} y_\omega(0). \quad (\text{IH3.6})$$

In the displayed order of the four points its value matrix is $m \text{diag}(J_2, J_2)$, $J_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$; its signature is $(2, 2)$. Its radical is precisely the four higher-jet ideals, of total dimension $4(m-1)$. This follows because multiplication on each jet algebra is triangular with constant diagonal $x(0)$, while the displayed value matrix is invertible. These are all maps in one receiving diagram: the actual Q , its actual orbit quotient, every W_a , both symmetries and the trace. It is not a separately selected polynomial masquerading as a zeta zero.

112.4 IH4. What the off-critical orbit looks like in the absorption record

The actual absorption measure for this route is $\mathcal{W}$ in (IH1.3), with its full prime-power weights. On $t > 0$ its exact spectral identity is

$$\boxed{\mathcal{W} = e^t dt - \sum_{\omega} m_\omega e^{\omega t} dt - \sum_{r \geq 1} e^{-2rt} dt.} \quad (\text{IH4.1})$$

The middle sum ranges over all distinct original nontrivial zeros with multiplicities. This is a distribution identity on compact smooth tests supported in $(0, \infty)$, not a pointwise convergent series of densities. Here is its derivation retaining the full archimedean and endpoint contributions.

The programme's full original-zeta explicit formula, for $h \in C_c^\infty(\mathbb{R})$, is

$$\sum_{\omega} m_\omega H(\omega) = H(0) + H(1) + A_\infty(h) - P(h), \quad H(s) = \int h(v) e^{-(s-1/2)v} dv,$$

$$P(h) = \sum_{p, k \geq 1} \frac{\log p}{p^{k/2}} (h(k \log p) + h(-k \log p)),$$

$$A_\infty(h) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{h}(y) \left(\Re \frac{\Gamma'(1/4 + iy/2)}{\Gamma(1/4 + iy/2)} - \log \pi \right) dy, \quad \widehat{h}(y) = \int h(v) e^{-iyv} dv. \quad (\text{IH4.2})$$

The exact digamma integral and Fourier inversion, with both endpoints retained, give the equivalent full expression already derived in C10.3:

$$A_\infty(h) = -(\gamma_E + \log \pi) h(0) + \int_0^\infty \frac{2h(0)e^{-2v} - (h(v) + h(-v))e^{-v/2}}{1 - e^{-2v}} dv. \quad (\text{IH4.3})$$

For clarity about the interchange: first integrate on $[\varepsilon, R]$. The paired numerator in the digamma representation vanishes at zero; after Fourier inversion the displayed numerator is $O(v)$. At infinity the exponentials and compact support give integrable bounds. The truncated digamma kernel is bounded uniformly by $C(1 + y^2)$: near zero use $1 - \cos(yv) \leq y^2 v^2 / 2$, and away from

zero use the exponential factors. The rapid decay of $\widehat{h}(y)$ then justifies the outer integral limit by dominated convergence. No gamma or constant term has been silently suppressed.

Now take $f \in C_c^\infty((0, \infty))$ and set $h(v) = e^{v/2}f(v)$, extended by zero. This is an explicit invertible comparison on this test subspace, $f(v) = e^{-v/2}h(v)$. It gives, term by term,

$$\begin{aligned} P(h) &= \int f d\mathcal{W}, \quad H(0) = \int e^v f(v) dv, \quad H(1) = \int f(v) dv, \\ A_\infty(h) &= - \int_0^\infty \frac{f(v)}{1 - e^{-2v}} dv, \quad H(1) + A_\infty(h) = - \sum_{r \geq 1} \int f(v) e^{-2rv} dv. \end{aligned} \quad (\text{IH4.4})$$

The constant in (IH4.3) is zero here because this specified test is identically zero near zero, not because the term has been discarded globally. The trivial-zero series converges uniformly on the test support. Also $H(\omega) = \int f(v) e^{(1-\omega)v} dv$; reindexing by the multiplicity-preserving symmetry $\omega \mapsto 1 - \omega$ proves (IH4.1). Its nontrivial-zero sum is absolutely convergent on each such smooth test by repeated integration by parts and the unconditional $O(T \log(T+2))$ zero count. The atom at time zero remains in $\mathcal{D}$, and the pole at one supplies e^t in (IH4.1).

The full four-point counterfactual (IH3.1) contributes exactly

$$-m \sum_{\omega \in \mathcal{O}_\rho} e^{\omega t} = -2m(e^{\beta t} + e^{(1-\beta)t}) \cos(\gamma t). \quad (\text{IH4.5})$$

This is the precise answer to where it would be in the absorption spectrum: a detected oscillatory component with two unequal growth rates. On a single critical-line conjugate pair the corresponding term is $-2me^{t/2} \cos(\gamma t)$. Letting an off-line quartet approach the line merges two pairs and doubles multiplicity; it must not be relabelled as one simple pair.

The locations of the atoms on the arithmetic side remain $k \log p$, with coefficients $\log p$. Equation (IH4.1) does not identify one zero term with an extra atomic event. Its complete sum is the spectral expression for the already identified arithmetic measure. Completeness therefore puts the counterfactual inside the tested spectrum. It does not supply the sign or growth bound excluding (IH4.5).

The diagram displays the same complete prime record and its actual quotient. Every off-critical symbol is conditional on the counterfactual zero in (IH3.1); no numerical zero or omitted prime is asserted. Proofs: IH1–IH6. The source of the illustration is `draw_integral_history.py`.

112.5 IH5. The exact comparison of integral counting with prime dilation

On Q , the reconstructed ring has the scalar action $c([n]) = nI_Q$. Alongside it, retain the full clock family and the actual W_n . The map

$$[n] \mapsto (([n] \text{ on } W/dW)_{d \geq 1}, nI_Q, W_n), \quad n \geq 1, \quad (\text{IH5.1})$$

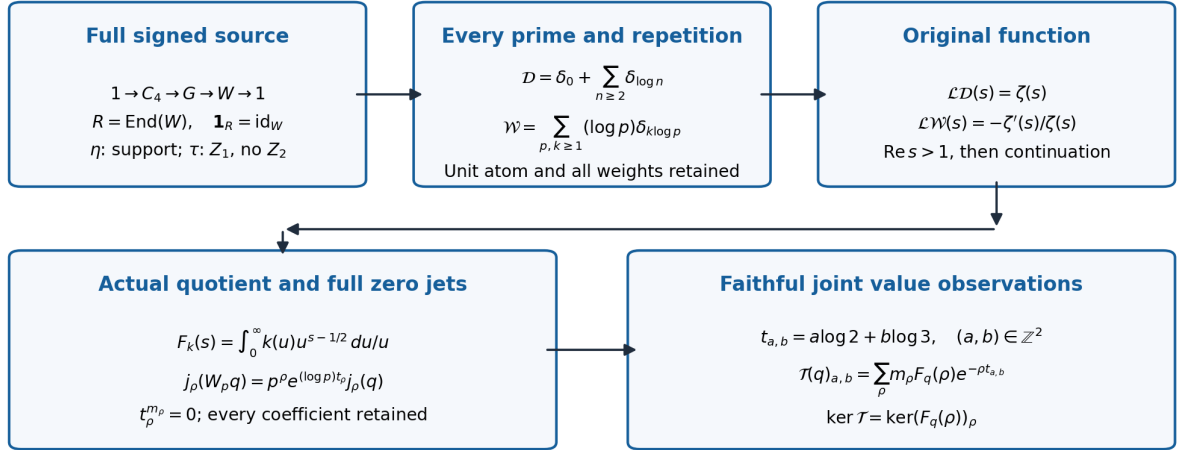
is an injective multiplicative-monoid map. Every component respects multiplication. Its first component is injective: equality says that every positive d divides $n - m$, forcing $n = m$. All prime labels and their integral degree actions are retained.

The W_n component does not preserve the ring addition. On the full ρ -jet, the coefficient of t^j in $W_{a+b} - W_a - W_b$ is

$$\frac{(a+b)^\rho \log^j(a+b) - a^\rho \log^j a - b^\rho \log^j b}{j!}, \quad 0 \leq j < m_\rho. \quad (\text{IH5.2})$$

This follows by expanding (IH3.3). At $a = b = 1, j = 0$ it is $2^\rho - 2 \neq 0$, since $0 < \Re \rho < 1$ implies $|2^\rho| < 2$. Thus the distinction is present even on the critical line. It is a calculated difference between two actions of the reconstructed numbers, not an alteration of integer addition or any operation on source τ .

One complete arithmetic record, with its spectral observations



A hypothetical off-critical quartet has the exact absorption contribution

$$-2m(e^{\beta t} + e^{(1-\beta)t})\cos(\gamma t)$$

It is a term in the same complete distribution, not a new prime event.

Proofs IH1-IH6, Geometry: Connes-Consani (2026); scalar reconstruction: programme CG/TP/I26.
No numerical off-critical zero is asserted. This diagram displays maps, not a proof of RH.

Figure 39: Complete counting and spectral observation

There is a stronger verified fact about algebraic integrality of these particular eigenvalues. For every nonreal ρ , at most two distinct primes have algebraic p^ρ . To prove it, suppose three distinct primes p, q, r have algebraic powers. The matrix

$$M = \begin{pmatrix} \log p & \log q & \log r \\ \rho \log p & \rho \log q & \rho \log r \end{pmatrix} \quad (\text{IH5.3})$$

then consists of logarithms of algebraic numbers and has complex rank one. Its rows are rationally independent because $1, \rho$ are, by taking imaginary parts. Its columns are rationally independent because a rational relation among the three real prime logarithms, after clearing denominators and exponentiating, contradicts unique factorization. The six-exponentials theorem forbids this rank-one matrix. Therefore there cannot be three such primes. In particular at least one of $2^\rho, 3^\rho, 5^\rho$ is transcendental.

The human theorem is credited to Serge Lang and Kanakanahalli Ramachandra; the original author TeX actually read is Samit Dasgupta, *Ranks of matrices of logarithms of algebraic numbers I*, arXiv:2303.02037v1, Theorem 4.2 (t:six, Baker.tex 818–820), with its deduction from Waldschmidt–Masser at 1117–1133. That theorem is an external established input; the displayed application and the prime-dilation comparison are proved here. The original gzip source is retained intact.

Every eigenvalue of a finite integer matrix is algebraic, since its monic characteristic polynomial has integer coefficients. Consequently these exact three dilation eigenvalues cannot all be realized by finite-rank integral endomorphisms with the same eigenvalues. The jet matrix does not evade this conclusion: it is triangular with diagonal p^ρ , so its characteristic polynomial is $(X - p^\rho)^{m_\rho}$. This obstruction applies equally to critical-line zeros. It excludes that specific transfer of algebraic-integral Frobenius eigenvalues to these operators; it does not exclude the integral clock construction, infinite integral objects, or a different proved comparison with Deligne’s framework.

112.6 IH6. One faithful joint receiver for the complete zero observations

The following global map retains the actual analytic spectrum through two already counted primes. Let Z denote the distinct original nontrivial zeros with multiplicities m_ω , and define

$$\begin{aligned} E_0(q) &= (F_q(\omega))_{\omega \in Z}, & t_{a,b} &= a \log 2 + b \log 3 \quad ((a,b) \in \mathbb{Z}^2), \\ \mathcal{T}(q)_{a,b} &= \sum_{\omega \in Z} m_\omega F_q(\omega) e^{-\omega t_{a,b}}. \end{aligned} \quad (\text{IH6.1})$$

All indices, including negative ones, belong to this specified receiving grid. They are repetitions in the rational multiplicative action; they are not additional positive-integer events in (IH1.3).

Repeated integration by parts in (IH2.1) proves, uniformly for $0 \leq \sigma \leq 1$,

$$|F_q(\sigma + i\gamma)| \leq C_N(k)(1 + |\gamma|)^{-N}, \quad (\text{IH6.2})$$

where k is any representative and $C_N(k)$ is bounded by finitely many defining seminorms. The unconditional zero count then gives $\sum_\omega m_\omega |F_q(\omega)| < \infty$. Since $0 < \Re \omega < 1$, the series $\Phi_q(t) = \sum_\omega m_\omega F_q(\omega) e^{-\omega t}$ is absolutely and uniformly convergent on every compact real interval. Thus it is continuous, and every coordinate of $\mathcal{T}$ is a continuous linear map on Q .

The grid $\{t_{a,b}\}$ is dense in $\mathbb{R}$. A nondense nonzero additive subgroup of the real line has a least positive element h and is $h\mathbb{Z}$: take the infimum of its positive elements; a positive infimum must be attained, since two closer approximants otherwise have smaller positive difference; division with remainder then proves the assertion. Such a discrete subgroup containing both logarithms would make their ratio rational, contrary to unique factorization. If the infimum is zero, a sufficiently small positive group element has a multiple in every real interval, proving density.

If $\mathcal{T}(q) = 0$, continuity gives $\Phi_q(t) = 0$ for all real t . For $\Re z < 0$, absolute interchange yields

$$0 = \int_0^\infty \Phi_q(t) e^{zt} dt = \sum_\omega \frac{m_\omega F_q(\omega)}{\omega - z}. \quad (\text{IH6.3})$$

The absolute integral is bounded by $\sum_\omega m_\omega |F_q(\omega)| / (-\Re z)$. The last series is normally convergent on compact sets avoiding the zero set: denominators are uniformly bounded away from zero for the finite nearby zeros and uniformly bounded below for the tail. It therefore continues meromorphically to $\mathbb{C}$, with residue $-m_\omega F_q(\omega)$ at each distinct zero. The complement of a discrete set is connected, so the identity theorem and the residues give $F_q(\omega) = 0$ for every zero. The reverse implication is immediate. We have proved

$$\ker \mathcal{T} = \ker E_0, \quad Q / \ker E_0 \hookrightarrow \mathbb{C}^{\mathbb{Z}^2}. \quad (\text{IH6.4})$$

This is a continuous injective linear map with the product topology in the target. No surjectivity onto that product or topological inverse is asserted. The full jet map (IH2.3) is retained alongside this reduced-value comparison; higher jets have not been declared absent.

The exact intertwining and reflection laws are

$$\begin{aligned} \mathcal{T}(W_2 q)_{a,b} &= \mathcal{T}(q)_{a-1,b}, & \mathcal{T}(W_3 q)_{a,b} &= \mathcal{T}(q)_{a,b-1}, \\ \mathcal{T}(q^\#)_{a,b} &= 2^{-a} 3^{-b} \overline{\mathcal{T}(q)_{-a,-b}}. \end{aligned} \quad (\text{IH6.5})$$

The first two follow by multiplying each absolutely summable term by 2^ω or 3^ω . For the third, substitute $F_{q^\#}(\omega) = \overline{F_q(1 - \bar{\omega})}$, reindex by the multiplicity-preserving reflection, and retain the factor $e^{-t_{a,b}}$. Thus all the observations and their weight-one duality live together in this single receiver. Off-critical coordinates are detected by the same faithful map.

Every other prime action remains in this receiver. For every positive real c ,

$$\Phi_{W_c q}(t) = \Phi_q(t - \log c). \quad (\text{IH6.6})$$

This follows term by term from (IH3.2); multiplication by e^ω is uniformly bounded on $0 < \Re \omega < 1$. The dense grid determines the continuous function Φ_q uniquely, so it determines this translation at every other prime as well. Two and three supply observation coordinates; the other primes have not been discarded.

The reconstruction is explicit. From the continuous extension of the grid data, form $C_q(z) = \int_0^\infty \Phi_q(t) e^{zt} dt$ on $\Re z < 0$ and use the meromorphic series in (IH6.3). Then

$$F_q(\omega) = -\frac{1}{m_\omega} \operatorname{Res}_{z=\omega} C_q(z). \quad (\text{IH6.7})$$

The minus sign comes from $(\omega - z)^{-1} = -(z - \omega)^{-1}$. This is an exact inverse on the represented value data, without a claim that analytic continuation is numerically stable.

112.7 IH7. Exact limits of the earlier Hilbert-space comparison

In GW8 the one-sided first-cohomology chain has weights $n^k ab$, $a = \log p$, $b = 2\pi$, and $Sv = (0, v_0, v_1, \dots)$. It satisfies $S^*S = nI$. It has no eigenvectors: $Sv = \lambda v$ forces $v_0 = 0$ and then every coordinate zero when $\lambda \neq 0$; when $\lambda = 0$, the shifted coordinates directly force the same conclusion.

On $\mathcal{H}_Z = \ell^2(Z, m)$ let $D_n x(\omega) = n^\omega x(\omega)$. This is bounded, since $|n^\omega| \leq n$. Any bounded intertwiner $T : \mathcal{H}_Z \rightarrow \mathcal{H}_+$ satisfying $TD_n = ST$ must be zero. Indeed each coordinate vector e_ω is an eigenvector of D_n , so Te_ω would be an eigenvector of S and must vanish. Finite-support vectors are dense and boundedness finishes the proof.

This proves a specific obstruction, including on the critical line. A direct bounded identification of these two operator representations cannot provide the desired purity argument. The surviving exact connection is (IH6.4)–(IH6.5), a faithful global observation map into the complete sequence receiver with its original weights and reflection law. The earlier positive Hodge-chain norm has not thereby become the norm of the zeta quotient.

112.8 IH8. Full support, the original pairing, and the outcome of the contradiction test

The linked programme retains the formal coefficient algebra with separate symbols $U = [\tau]$, $f = [1]$, and the vector space W_L on every specified support label. These are receiving coefficients; their linear operations do not define addition on source τ . Apply $j_{\mathcal{O}} \otimes \text{id} \otimes \text{id}$ and $\mathcal{T} \otimes \text{id} \otimes \text{id}$ to the existing algebraic coefficient extension of Q . Choosing bases proves the precise kernels: each element is a finite sum in independent coefficient coordinates, so an image vanishes exactly when each scalar coordinate belongs to the original kernel. In particular (IH6.4) remains faithful after this extension. No support label, including the designated unsupported record, is mapped to absence by these identity maps. This is the programme's existing formal receiver, not a newly asserted equivalence with arbitrary semimodule cohomology.

The full original Weil pairing remains

$$B_Q(q, r) = \sum_{\omega} m_{\omega} \overline{F_q(1 - \bar{\omega})} F_r(\omega). \quad (\text{IH8.1})$$

It converges absolutely by (IH6.2); G5 and GIQ9 identify it with (IH4.2) applied to the original product test, retaining all endpoint and archimedean terms. The raw divisor also retains, at every finite trivial-zero cutoff B ,

$$V_B(H) = \sum_{\omega} m_{\omega} H(\omega) + \sum_{j=1}^B H(-2j) - H(1),$$

$$G_B(H) = A_\infty(h) + H(0) + \sum_{j=1}^B H(-2j), \quad V_B(H) = G_B(H) - P(h). \quad (\text{IH8.2})$$

No convergence of an unrestricted trivial-zero evaluation sum is presumed. In the positive-time test space of IH4 the corresponding exponential series converges for the stated reason.

The completed counting and chart-uniqueness proof therefore identifies the actual function and its complete divisor. The additional calculations locate the hypothetical off-critical orbit inside its full observation system, exhibit its absorption contribution, and prove the exact integral-degree, dilation and reflection maps. They do not produce a contradiction: the step “off-critical therefore undetected” is disproved by (IH2.6) and the faithful global receiver (IH6.4). What off-criticality would change is the detected weight and the sign of the reflected value form (IH3.6). No statement in this note proves the required positivity or excludes that orbit in the actual zeta function. No disproof of RH is asserted either.

112.9 Proof sources and authorship

Human geometry: Alain Connes and Caterina Consani, 2609.00299v1, source topology, chart restrictions and mirror; 2606.06604v1, the Tate curve and both periods. The completed counting reconstruction is the programme’s TP0–TP14 public proof. The local CG and I26 proofs are credited and retained as programme dependencies, without claiming a new publication receipt for them.

The exact analytic quotient and original arithmetic pairing come from G1–G5 in `GLOBAL_CANONICAL_MELLIN_INVOLUTION.md`, GIQ1–GIQ3 and GIQ9 in `GLOBAL_INFINITESIMAL_QUOTIENT.md`, and C10 in `TAU_TWISTOR_TATE_AND_WEIL.md`. CW6–CW7 had already calculated the all-prime radial defect and its reflected pairing; those are not claimed as discoveries of this note. The new receiving work includes the full composition with I26, the explicit absorption-distribution calculation, the faithful complete two-prime sequence receiver, and the proved restrictions on bounded and algebraic-integral transfers. The six-exponentials input and its human attribution are given in IH5. Classical continuation and the alternating-series formula are also documented at DLMF 25.2.

The private source-use ledger records actual reading coverage and hashes. The independent IP proof and the independent check of IH6–IH7 are retained. No whole-paper reading, full integration of every other task’s manuscript, remote upload or resolution of RH is claimed.

113 Returning to the Z1 support: complete prime clocks, loops, and zeta characters

24 September 2026. Proofs PL1–PL8. This calculation tests the user’s proposed mechanism: a deformation invisible to the complete counting system should be a change of clock, or a loop whose return might expose the distinguished presence support. The comparison is made on the actual reconstructed arithmetic, the Connes–Consani prime curves, and the original zeta function.

The notation is retained: Z_0 is absence; Z_1 is primitive presence τ , without Z_2 parity. No addition, coordinate, metric, or winding number is assigned to τ . The previously proved supporting role of $\tau\langle Z_1; \text{no } Z_2 \rangle$ is the common point η . The stalk, its winding quotient, and its arithmetic endomorphisms remain part of the complete datum.

113.1 PL1. What the complete counting reconstruction already fixes

The source contains the signed monomial group

$$G = \langle T, J, \epsilon : TJ = JT, T\epsilon = \epsilon T, J\epsilon = \epsilon J, J^2 = \epsilon, \epsilon^2 = 1 \rangle \cong \mathbb{Z} \times C_4,$$

its finite subgroup $H = \langle J \rangle$, and the infinite cyclic winding group $W = G/H$. The separate absorbing source element is outside this group and is not identified with its identity. The exact sequence, source involution and both chart maps are retained in CG2–CG4:

$$1 \longrightarrow H \longrightarrow G \longrightarrow W \longrightarrow 1, \quad A(T) = \epsilon T^{-1}, \quad A(J) = J^{-1}. \quad (\text{PL1.1})$$

The arithmetic ring is $R = \text{End}_{\text{Ab}}(W)$, with pointwise group multiplication as its addition and composition as its multiplication. Every endomorphism is the power map $[n]$, independently of the choice of generator of W . Thus

$$R \cong \mathbb{Z}, \quad \mathbf{1}_R = \text{id}_W, \quad [a] + [b] = [a + b], \quad [a][b] = [ab]. \quad (\text{PL1.2})$$

These are operations on endomorphisms, not on τ . Any unital ring automorphism of this reconstructed R is the identity: it fixes $\mathbf{1}_R$, hence each of its finite sums, their additive inverses, and therefore every $[n]$. Consequently it fixes every prime ideal and its residue norm $p = |W/[p]W|$.

Retain all three complete measures:

$$\mathcal{R} = \sum_p \sum_{k \geq 1} \frac{1}{k} \delta_{k \log p}, \quad \mathcal{D} = \delta_0 + \sum_{n \geq 2} \delta_{\log n} = \exp_* \mathcal{R}, \quad \mathcal{W} = \sum_p \sum_{k \geq 1} (\log p) \delta_{k \log p} = t\mathcal{R}. \quad (\text{PL1.3})$$

TP9–TP13 and IH1 prove these locally finite measure identities with all coefficients and the unit atom retained. Their transforms on $\Re s > 1$ are the original functions

$$\int e^{-st} d\mathcal{D}(t) = \zeta(s), \quad \int e^{-st} d\mathcal{W}(t) = -\frac{\zeta'(s)}{\zeta(s)}. \quad (\text{PL1.4})$$

The composed CG/I26 proof identifies the same meromorphic zeta function and zero multiplicities in both charts. That identification is closed in this route.

113.2 PL2. An invisible change of clock cannot turn an off-critical zero into a critical one

Specify the proposed clock change. Let $h : \mathbb{R} \rightarrow \mathbb{R}$ be an order-preserving bijection that preserves composition of logarithmic durations and the identity duration:

$$h(t + u) = h(t) + h(u), \quad h(0) = 0. \quad (\text{PL2.1})$$

Then $h(t) = ct$ for a unique $c > 0$. Indeed $c = h(1) > 0$; additivity gives $h(a/b) = ca/b$ for every rational a/b . Rational lower and upper approximations to a real t , together with monotonicity, force $h(t) = ct$. There is no nonlinear bending in this stated class. A translation by b preserves (PL2.1) only when $b = 0$.

This proof states exactly what is preserved; it does not classify every transformation of every space merely from the word “invisible.” On the full normed arithmetic with the specified times $t_p = \log p$, preserving even one such time forces $c = 1$. A simultaneous change of measuring units remains a valid comparison, with every factor as follows. Write $S_c(t) = ct$:

$$\mathcal{D}_c = (S_c)_* \mathcal{D}, \quad \mathcal{R}_c = (S_c)_* \mathcal{R}, \quad \mathcal{W}_c = t \mathcal{R}_c = c(S_c)_* \mathcal{W}. \quad (\text{PL2.2})$$

For $\Re(cs) > 1$,

$$\int e^{-st} d\mathcal{D}_c(t) = \zeta(cs), \quad \int e^{-st} d\mathcal{W}_c(t) = -c \frac{\zeta'(cs)}{\zeta(cs)}. \quad (\text{PL2.3})$$

The factor c in the second equality comes from the rescaled primitive return length. The pole, its residue, the nontrivial zeros, and the trivial zeros become

$$\frac{1}{c}, \quad \frac{1}{c}, \quad \frac{\rho}{c}, \quad -\frac{2r}{c} \quad (r \geq 1). \quad (\text{PL2.4})$$

Zero multiplicities are unchanged. The exact comparison with a completed object, without adopting it as the working object, is

$$\xi(cs) = \frac{1}{2}(cs)(cs-1)\pi^{-cs/2}\Gamma(cs/2)\zeta(cs). \quad (\text{PL2.5})$$

Its derivative retains all contributions:

$$c \frac{\xi'(cs)}{\xi(cs)} = c \frac{\zeta'(cs)}{\zeta(cs)} + \frac{1}{s} + \frac{c}{cs-1} - \frac{c}{2} \log \pi + \frac{c}{2} \frac{\Gamma'(cs/2)}{\Gamma(cs/2)}. \quad (\text{PL2.6})$$

The reflection becomes $s \mapsto c^{-1} - s$. Hence the critical line has real coordinate $1/(2c)$, and

$$2\Re(\rho/c) - 1/c = (2\Re\rho - 1)/c. \quad (\text{PL2.7})$$

A change of clock therefore cannot hide or manufacture off-criticality: it rescales the zero and the comparison line together. This proves the proposed dilation invariance for the complete original function, including its exceptional points.

113.3 PL3. All prime curves together: no positive real period, and the angular loop survives

The original Connes–Consani curve is

$$E_p = \mathbb{C}^\times / p^\mathbb{Z}. \quad (\text{PL3.1})$$

Their exponential coordinate is $z = \exp(2\pi iw)$, with periods 1 and $i \log p / (2\pi)$. To work directly with the original differential dz/z , retain the exact comparison $u = 2\pi iw$. The two oriented periods become $2\pi i$ and $-\log p$; the lattice as a subgroup is

$$\Lambda_p^{\log} = 2\pi i \mathbb{Z} + (\log p) \mathbb{Z}, \quad \frac{dz}{z} = du. \quad (\text{PL3.2})$$

The sign of the second oriented source generator has been exhibited. The source’s modular parameter is a parameter of E_p , not the user’s primitive τ .

For the positive real flow $t \mapsto [e^t]_p$, return occurs precisely when $t = k \log p$, $k \in \mathbb{Z}$. Use the joint map

$$j_{\mathbb{R}} : \mathbb{R} \longrightarrow E_2 \times E_3, \quad t \longmapsto ([e^t]_2, [e^t]_3). \quad (\text{PL3.3})$$

Its kernel is zero: $t = a \log 2 = b \log 3$ would imply $2^a = 3^b$; clearing negative exponents and unique factorization give $a = b = 0$. Therefore the joint real clock cannot return to its starting state at a nonzero time even though each individual prime orbit is a circle. Keeping more primes cannot enlarge this kernel.

There is nevertheless a genuine common angular period:

$$\Lambda_2^{\log} \cap \Lambda_3^{\log} = 2\pi i\mathbb{Z}. \quad (\text{PL3.4})$$

Equality of two lattice elements forces their real parts to vanish by the preceding proof; their imaginary parts leave precisely the stated integer multiples. The exact map

$$J_{\mathbb{C}^\times} : \mathbb{C}^\times \longrightarrow E_2 \times E_3, \quad z \longmapsto ([z]_2, [z]_3) \quad (\text{PL3.5})$$

is injective, since its kernel is $2^\mathbb{Z} \cap 3^\mathbb{Z} = \{1\}$.

Injectivity is the assertion here; no topological embedding into the compact product is asserted. The angular loop has not been suppressed by considering only positive real time. It is the original exponential covering period, present before any question about zeta zeros.

113.4 PL4. Returning to the same Z1 support does not identify the retained winding

The actual source space is

$$X = \{+, -, \eta\}, \quad \Omega_+ = \{+, \eta\}, \quad \Omega_- = \{-, \eta\}, \quad \Omega_+ \cap \Omega_- = \{\eta\}. \quad (\text{PL4.1})$$

The projection of the generic monomials to their support is the set map

$$\pi : G \longrightarrow \{\eta\}, \quad \pi(T^m J^k) = \eta. \quad (\text{PL4.2})$$

Every winding has the same supporting point under this map. Its fibre is the entire G , including all four J -states and all integer windings. Thus seeing the Z_1 support again does not, by this observation alone, mean that the full state has returned. This is a direct calculation of the map and its fibre; it neither assigns parity to τ nor identifies different monomials.

An actual identification of winding after $N > 0$ steps is the quotient

$$q_N : W \longrightarrow W/NW, \quad \ker q_N = NW. \quad (\text{PL4.3})$$

It acts on reconstructed arithmetic by

$$R \longrightarrow \text{End}(W/NW) \cong \mathbb{Z}/N\mathbb{Z}, \quad [n] \longmapsto [n \bmod N]. \quad (\text{PL4.4})$$

The kernel is $N\mathbb{Z}$: an endomorphism vanishes on the quotient exactly when its degree is divisible by N . Thus the loop is detected by the lost endomorphisms. This zero is a receiver's zero endomorphism, not source absence.

On the full signed group, the quotient $G/\langle T^N \rangle$ preserves the original involution A exactly when N is even. Indeed $A(T^N) = \epsilon^N T^{-N}$, whose class is the identity precisely when $\epsilon^N = 1$, since all four J -states remain. Odd N would require additionally killing ϵ , changing the specified signed datum. Using every even N retains the full datum faithfully: if $T^m J^k$ maps to the identity in every such quotient, then $J^k = 1$ and every even N divides m ; choosing even $N > |m|$ forces $m = 0$.

These exact quotients identify the information that closed winding loses. They also show why many individually periodic observations can jointly retain an infinite clock.

113.5 PL5. Folding the complete absorption record around a circle is detectable

Let $L > 0$, $C_L = \mathbb{R}/L\mathbb{Z}$, and $q_L(t) = t \bmod L$. Pushforward gives actual measures:

$$(q_L)_* \mathcal{D} = \sum_{n \geq 1} \delta_{\log n \bmod L}, \quad (q_L)_* \mathcal{W} = \sum_{p, k \geq 1} (\log p) \delta_{k \log p \bmod L}. \quad (\text{PL5.1})$$

Both give infinite mass to every nonempty open arc. For the first, lift an arc to $a < t < b$, $b - a < L$. The disjoint intervals (e^{a+jL}, e^{b+jL}) , for sufficiently large positive j , each contain an integer. Each gives an atom in that arc, proving infinite mass.

For the second, at least one of $\log 2/L, \log 3/L$ is irrational, because otherwise $\log 2/\log 3$ would be rational. Positive multiples of an irrational circle rotation visit every open arc infinitely often. To justify the needed fact, pigeonhole $M + 1$ successive multiples into M equal arcs, giving a nonzero multiple within L/M of zero in the circle metric. These distances become arbitrarily small, so the closed subgroup generated by the rotation is the whole circle: multiples of a sufficiently small positive representative approximate every circle point. Positive multiples have the same closure. A subsequence of positive multiples tends to zero, and subtracting any fixed multiple from sufficiently late terms approximates its negative using positive multiples. Every tail of the positive orbit is therefore dense. Each visit contributes the fixed positive weight $\log p$, proving the assertion.

The folded unweighted record is consequently not locally finite, unlike the original record on bounded real intervals. The pushforward still exists as a measure; it has not become the same counting object.

There is an exact finite comparison retaining every convergence factor. For $\sigma > 1$, set

$$\mathcal{D}_{\sigma, L} = (q_L)_*(e^{-\sigma t} \mathcal{D}), \quad \mathcal{W}_{\sigma, L} = (q_L)_*(e^{-\sigma t} \mathcal{W}).$$

Their circle Fourier coefficients, with kernel $e^{-2\pi i j t/L}$, are

$$\widehat{\mathcal{D}_{\sigma, L}}(j) = \zeta(\sigma + 2\pi i j/L), \quad \widehat{\mathcal{W}_{\sigma, L}}(j) = -\frac{\zeta'(\sigma + 2\pi i j/L)}{\zeta(\sigma + 2\pi i j/L)}. \quad (\text{PL5.2})$$

Absolute convergence proves both identities by substitution into (PL1.4). The explicit weight has not replaced the unweighted object.

For the existing prime circle $L = \log p$, the mass at its identity is

$$\mathcal{D}_{\sigma, \log p}(\{0\}) = \sum_{k \geq 0} p^{-k\sigma} = \frac{1}{1 - p^{-\sigma}} > 1, \quad (\text{PL5.3})$$

whereas the original unit atom has mass one. The corresponding prime-power mass is

$$\mathcal{W}_{\sigma, \log p}(\{0\}) = (\log p) \sum_{k \geq 1} p^{-k\sigma} = \frac{\log p}{p^\sigma - 1} > 0. \quad (\text{PL5.4})$$

No other prime contributes at this identity: $q^k = p^a$ forces $q = p$. The programme retains these repetitions instead of erasing them when a point returns.

Checking only the origin is insufficient even for these exact quotients. For $L = \log(3/2)$, the images of integer events 2 and 3 coincide, while the only integer sent to the origin is 1. Indeed a collision is exactly $\log(n/m) \in L\mathbb{Z}$, and an integer at the origin would be $(3/2)^j$. Unique factorization excludes this being an integer for any nonzero integer j . This quotient is excluded by the complete count precisely because it identifies two prime events, even though the unit observation alone does not expose it. It is not a counterexample to the complete reconstruction.

113.6 PL6. What a zero contributes to a loop: a return multiplier

The actual quotient Q , Mellin observations and maps W_a are constructed with full formulas in IH2–IH3. For an actual zeta zero $\rho = \beta + i\gamma$, the observation is a character satisfying

$$F_{W_a q}(\rho) = a^\rho F_q(\rho), \quad a^\rho = e^{\rho \log a}, \quad a > 0. \quad (\text{PL6.1})$$

This acts on observed amplitudes. It does not replace the prime norm p by p^ρ , permute primes, or move an event $k \log p$. Its exact relation to the clock is the character identity $(ab)^\rho = a^\rho b^\rho$.

On the real universal cover of C_L , $v_\rho(t) = e^{\rho t}$ obeys

$$v_\rho(t + L) = e^{\rho L} v_\rho(t). \quad (\text{PL6.2})$$

It descends to an ordinary scalar function on C_L exactly when $e^{\rho L} = 1$. For a nontrivial zero, $0 < \beta < 1$, so $|e^{\rho L}| = e^{\beta L} > 1$; scalar descent fails for critical-line zeros as well as off-critical ones.

An exact receiving object retains the character. On the trivial complex line bundle over C_L , the one-form dt is globally defined and

$$\nabla_\rho = d - \rho dt \quad (\text{PL6.3})$$

is a well-defined connection. Its horizontal equation is $dv/dt = \rho v$; transport around the positively oriented circle multiplies the fibre by $e^{\rho L}$. On $E_p = \mathbb{C}/\Lambda_p^{\log}$, the connection $d - \rho du$ has multipliers

$$p^\rho, \quad e^{2\pi i \rho} \quad (\text{PL6.4})$$

along the positive real and positive angular periods. Both remain. This line bundle with connection is constructed for each character; no identification with all source cohomology is assumed. It proves how the loop and the actual spectral character are related.

At a prime the weight-one comparison is

$$|p^\rho|^2 - p = p^{2\beta} - p. \quad (\text{PL6.5})$$

At $\beta = 1/2$, a full return still multiplies a nonzero vector by $p^{1/2+i\gamma}$, not by one; its squared magnitude agrees with arithmetic degree p . At $\beta \neq 1/2$, the same complete degree record remains p , while this equality fails. This tests fibre transport relative to degree, not whether the support has appeared again.

The common angular period has transport

$$|e^{2\pi i \rho}| = e^{-2\pi \gamma}. \quad (\text{PL6.6})$$

It does not distinguish $\beta = 1/2$ from another β . The retained angular loop therefore cannot be relabelled as an off-critical-only phenomenon.

For multiplicity m , retain the full primary block:

$$p^{\rho+t_\rho} = p^\rho \sum_{r=0}^{m-1} \frac{(\log p)^r}{r!} t_\rho^r, \quad e^{2\pi i(\rho+t_\rho)} = e^{2\pi i \rho} \sum_{r=0}^{m-1} \frac{(2\pi i)^r}{r!} t_\rho^r \mod t_\rho^m. \quad (\text{PL6.7})$$

These solve the same constant-coefficient connection with coefficient $\rho + t_\rho$. No nilpotent or repeated-zero datum is omitted.

The full complex reflection also has an exact map. Denote the line with connection $d - \rho du$ by $\mathcal{K}_{\rho,p}$, and let $c_p : E_p \rightarrow E_p$ be $[u] \mapsto [\bar{u}]$. This is well defined because conjugation preserves the lattice. With $\rho^\# = 1 - \bar{\rho}$, multiplication on the one-dimensional fibres gives

$$c_p^* \overline{\mathcal{K}_{\rho,p}} \otimes \mathcal{K}_{\rho^\#,p} \cong \mathcal{K}_{1,p}. \quad (\text{PL6.8})$$

To check the connection, pullback after conjugation sends its coefficient to $\bar{\rho} du$. The tensor connection is therefore $d - (\bar{\rho} + 1 - \bar{\rho}) du = d - du$. Equivalently its multiplier around any $\lambda \in \Lambda_p^{\log}$ is $e^{\bar{\rho}\lambda} e^{(1-\bar{\rho})\lambda} = e^\lambda$. This includes the angular period, where it equals one. Omitting the base conjugation would give $\bar{\lambda}$ in the first factor and would generally fail on that period. This duality holds for off-critical pairs too; it is not by itself a positive Hermitian norm on one of their fibres.

113.7 PL7. What a hypothetical off-critical zero is composed from

The original zeta has the exact expression, for $0 < \Re s < 1$,

$$\zeta(s) = \frac{\sum_{n=1}^{\infty} (-1)^{n-1} n^{-s}}{1 - 2^{1-s}}. \quad (\text{PL7.1})$$

Every n , its parity sign, and $\log n$ come from the reconstructed arithmetic. No parity or addition on τ is used. Here is the derivation, keeping zeta rather than replacing it by the numerator.

On $\Re s > 1$, separating even terms in the absolutely convergent original series gives numerator $= (1 - 2^{1-s})\zeta(s)$. For $\Re s = \sigma > 0$, pair consecutive terms:

$$(2k-1)^{-s} - (2k)^{-s} = s \int_{2k-1}^{2k} x^{-s-1} dx.$$

The paired series converges absolutely, locally uniformly in this half-plane. Its unpaired last term tends to zero. Hence the numerator is holomorphic there and agrees with the displayed product by analytic continuation. In $0 < \Re s < 1$, the denominator cannot vanish because $|2^{1-s}| = 2^{1-\Re s} > 1$. This proves (PL7.1) on the required domain. The classical identity is also DLMF 25.2.3.

For an actual zero $\rho = \beta + i\gamma$, (PL7.1) is exactly the two cancellation equations

$$\sum_{n \geq 1} (-1)^{n-1} n^{-\beta} \cos(\gamma \log n) = 0, \quad \sum_{n \geq 1} (-1)^{n-1} n^{-\beta} \sin(\gamma \log n) = 0. \quad (\text{PL7.2})$$

The series retain their increasing integer order, justified above. The imaginary part of $n^{-\rho}$ has a minus sign; setting it to zero yields the second equation displayed. These equations change none of the counted integers. They specify a property of the uniquely reconstructed function.

The complete absorption identity, derived with all Gamma and endpoint terms in IH4, is

$$\mathcal{W} = e^t dt - \sum_{\omega} m_{\omega} e^{\omega t} dt - \sum_{r \geq 1} e^{-2rt} dt \quad \text{on } C_c^{\infty}((0, \infty)). \quad (\text{PL7.3})$$

This is a distributional identity on these smooth tests, not a pointwise density series. An off-critical orbit $\{\rho, 1 - \bar{\rho}, \bar{\rho}, 1 - \rho\}$ of common multiplicity m contributes

$$-2m(e^{\beta t} + e^{(1-\beta)t}) \cos(\gamma t). \quad (\text{PL7.4})$$

The prime record on the left is unchanged. The exponents describe cancellation within that record's spectral formula, not a second candidate prime record with altered event times. A single spectral summand is not separately equal to a positive counting measure.

An explicit detector exists inside the actual quotient. Choose a nonnegative nonzero smooth compactly supported $\varphi(v)$ and set

$$k_{\rho}(e^v) = \varphi(v) e^{-(\rho-1/2)v}.$$

Then $F_{k_{\rho}}(\rho) = \int \varphi(v) dv > 0$, whereas the defining ideal of the quotient has zero observation at ρ . IH6 also constructs one faithful receiver for all reduced observations through the complete signed grid $a \log 2 + b \log 3$, $a, b \in \mathbb{Z}$, retaining every other prime action by continuous translation.

113.8 PL8. Outcome of the proposed loop contradiction

The calculation gives definite exclusions: simultaneous rescaling preserves off-criticality relative to the pole and reflection; a nonlinear order- and composition-preserving clock change does not exist; a nonzero common positive real return does not exist; and forgetting winding has an explicit

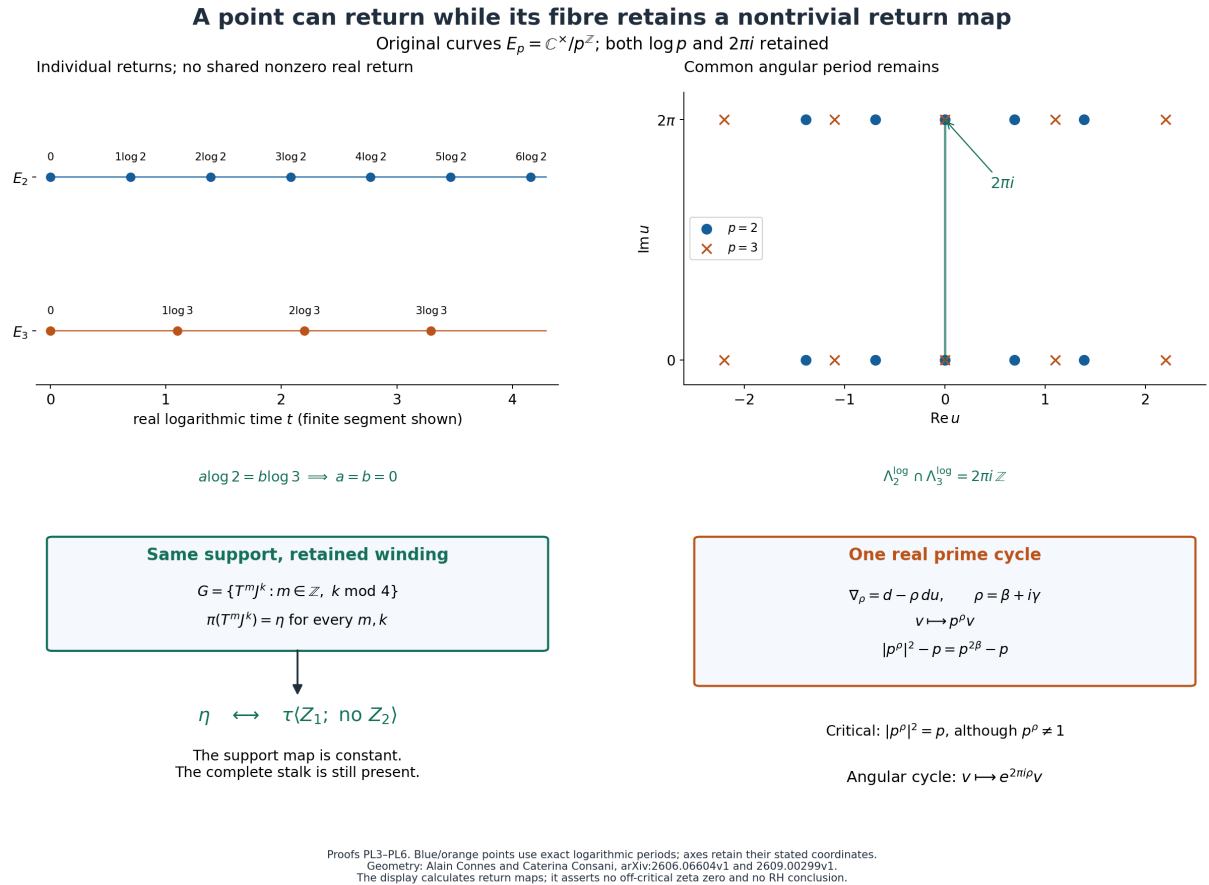


Figure 40: Complete clocks, common angular loop, and fibre return

detectable kernel. The existing common angular loop and every local prime loop remain, with their exact return maps.

These results exclude the proposed invisible clock changes. They do not force a zeta zero to be such a change. Its actual map is (PL6.1), and its defining equation is (PL7.2), both on the already reconstructed arithmetic. Reappearance of the Z_1 support alone records (PL4.2); fibre transport is additional retained data calculated in (PL6.4)–(PL6.7). No contradiction follows from reappearance alone.

The quantity testing the stronger return comparison is $p^{2\beta} - p$, with multiplicity blocks and all-prime compatibility retained. This note calculates it and the surrounding maps; it does not prove that it vanishes for all actual zeros or exhibit an actual off-critical zero. The scalar original-zeta identification remains proved; RH is neither proved nor disproved here.

Figure: exact maps in PL2–PL6, retaining original periods $\log p$ and $2\pi i$. The pictured two-clock orbit is a finite segment; its no-return assertion is proved in PL3. The occurrence of η labels source support, not a numerical position. Reproducible figure sources accompany the proof.

113.9 Sources and provenance

Human geometry: Alain Connes and Caterina Consani, The Absolute Twistor Line and the Geometry of the compactification of Spec $\mathbb{Z}$, arXiv:2609.00299v1, original author CC.tex, labels opens1, opens, eq:restriction_maps, lines 499–600; and On the Absolute Geometry of Spec $\mathbb{Z}$ and the Fargues–Fontaine curve, arXiv:2606.06604v1, original author FF.tex, lines 1550–1633, including thm:tate_decomposition. These passages were read directly in the original TeX. The periods, original coordinate and decomposition $E_p = C_p \times S^1$ are their results; the joint-period and return comparisons are derived here.

Programme sources: CG0–CG8 in `CANONICAL_GENERIC_RECONSTRUCTION.md`; I26.1–I26.4 in `ITEM26_CANONICAL_ZETA_COMPOSITION.md`; TP0–TP14 public proof, pinned edition; and the complete accompanying IH1–IH8 proof. The private reading ledger records actual ranges and versions; this is not a claim to have reread every programme source in full.

The user supplied the global counting, dilation and returning-support questions. The classifications and maps above are receiving calculations, independently compared with the integrality/prime-action agent’s derivation. No historical novelty is claimed for elementary clock or covering arguments. No new remote publication receipt is claimed.

114 Deligne’s weight argument and the programme’s global quotient

24 September 2026. A reconstruction and receiving calculation, with the original zeta function retained.

The mathematical source is Pierre Deligne, *La conjecture de Weil : II*, Publications Mathématiques de l’IHÉS **52** (1980), 137–252, official publication, DOI 10.1007/BF02684780. The local reading source is the complete French S20 LaTeX transcription, retained with its hash. It is not an original author TeX file. The reading ledger identifies the parts actually read and the errors detected in that transcription. A corrected deduction below is identified as a deduction, not silently attributed to the transcription.

Source notation throughout: Z_0 is absence, Z_1 is primitive presence, and Z_2 is carried integer parity. Source τ has no addition and no parity. The retracted equation $\tau + \tau = \tau$ is not used. Deligne’s zero sheaf, an arithmetic zero, a vector-space zero, the source support η , and source τ are not identified. Every calculation below takes place in a named arithmetic or cohomological receiver.

The user’s proposed global quotient is tested on the existing, identified original zeta function and its actual zero observations. No new arithmetic is postulated as a counterexample. The conclusion established here is an exact description of the tensor-pole displacement and of what an observation-preserving quotient does to it. It is not a proof or disproof of RH.

114.1 D1. What Deligne controls

Let X_0 be a finite-type scheme over $\mathbb{F}_q$, $X = X_0 \otimes_{\mathbb{F}_q} \overline{\mathbb{F}_q}$, and let $\ell \neq \text{char } \mathbb{F}_q$. A lisse Weil sheaf $\mathcal{F}_0$ has finite-dimensional $\overline{\mathbb{Q}}_\ell$ stalks and an action of geometric Frobenius. At a closed point x , put

$$d_x = [k(x) : \mathbb{F}_q], \quad N(x) = q^{d_x}.$$

For a fixed field isomorphism $\iota : \overline{\mathbb{Q}}_\ell \rightarrow \mathbb{C}$, Deligne’s numerical weight is

$$w_{N(x)}(\alpha) = \frac{2 \log |\iota \alpha|}{\log N(x)}. \quad (\text{D1.1})$$

Pointwise ι -purity of weight β means that every Frobenius eigenvalue at every closed point satisfies

$$|\iota \alpha| = N(x)^{\beta/2}. \quad (\text{D1.2})$$

Here β may be any real number. Purity without ι additionally requires algebraicity and the equality for every complex conjugate, with integer weight. Deligne proves the estimates separately for each ι ; these two statements must not be conflated.

A mixed sheaf is built by a finite filtration whose nonzero successive quotients are pointwise pure. Mixedness does not mean that a numerical average of weights is pure. An extension retains the eigenvalues of both its subobject and quotient, with multiplicities, because an adapted basis makes Frobenius block triangular.

The zero sheaf has no eigenvalues, so every asserted eigenvalue bound on it is vacuous. Its set of weights is empty. A nonzero constant rank-one sheaf has one Frobenius eigenvalue, equal to 1, and weight 0. This distinction follows directly from determinants:

$$\det(1 - TF, 0) = 1, \quad \det(1 - TF, \overline{\mathbb{Q}}_\ell) = 1 - T. \quad (\text{D1.3})$$

It is not a definition of source τ , and no map from τ to either of these objects has been assumed.

The Tate factor is essential: geometric Frobenius on $\overline{\mathbb{Q}}_\ell(1)$ is q^{-1} , of weight -2 . On a twist by (-1) , it is q , of weight 2. On a constant Weil twist with Frobenius b , it is b^{d_x} at x . Consequently

$$w_{N(x)}(b^{d_x} \alpha) = w_{N(x)}(\alpha) + \frac{2 \log |\iota b|}{\log q}. \quad (\text{D1.4})$$

This records the factor explicitly; none of the proofs below replaces the original sheaf by an unnamed weight-zero object. [Deligne, §§1.1–1.2, pp.146–156.]

114.2 D2. The global trace formula and its independently controlled poles

For a smooth geometrically connected curve and a lisse sheaf, write V for a geometric stalk and π_1^{geom} for geometric monodromy. The extreme cohomology groups are

$$H^0(X, \mathcal{F}) = V^{\pi_1^{\text{geom}}}, \quad H_c^0(X, \mathcal{F}) = \begin{cases} V^{\pi_1^{\text{geom}}} & X \text{ proper,} \\ 0 & X \text{ not proper,} \end{cases}$$

$$H_c^2(X, \mathcal{F}) = V_{\pi_1^{\text{geom}}}(-1). \quad (\text{D2.1})$$

The quotient in the last formula is the quotient by the span of $gv - v$. It retains an explicitly computed Frobenius action, and then the Tate factor multiplies it by q . It is not an arbitrary quotient chosen to remove an unwanted eigenvalue.

Grothendieck's trace formula is

$$\prod_{x \in |X_0|} \det(1 - F_x T^{d_x}, \mathcal{F}_x)^{-1} = \frac{\det(1 - FT, H_c^1(X, \mathcal{F}))}{\det(1 - FT, H_c^0(X, \mathcal{F})) \det(1 - FT, H_c^2(X, \mathcal{F}))}. \quad (\text{D2.2})$$

Both denominator factors are retained. On an affine curve the first factor is 1, by the stated vanishing of H_c^0 .

The number of closed points of degree d is at most Cq^d : a finite stratification by quasi-finite maps to affine space gives this estimate. If the stalk eigenvalues have absolute value at most $N(x)^{\beta/2}$, the logarithm of the product converges absolutely for

$$|T| < q^{-(\beta+2)/2}. \quad (\text{D2.3})$$

Indeed the first powers are bounded by a constant times $\sum_{d \geq 1} q^d q^{d\beta/2} |T|^d$; the higher powers form convergent geometric tails on each smaller disk. The exponential of this convergent logarithm has neither zero nor pole there.

For pointwise pure input, (D2.1) gives exact extreme weights β and $\beta + 2$, when the corresponding spaces are nonzero. More generally their weights are obtained from the determinant weights of the constituents. This is the source of an independent pole bound in (D2.2). Recovering an Euler product alone would not calculate these two cohomology groups. [Deligne, §§1.3–1.4, especially (1.4.1.3), (1.4.3), (1.4.5.1), (1.4.6).]

114.3 D3. First amplification: positivity and determinant weights

For a lisse ι -real sheaf, every $\text{Tr}(F_x^r)$ is real. For its $2k$ -th tensor power,

$$\log \det(1 - F_x T^{d_x}, \mathcal{F}_x^{\otimes 2k})^{-1} = \sum_{r \geq 1} \frac{(\text{Tr } F_x^r)^{2k}}{r} T^{rd_x}. \quad (\text{D3.1})$$

Every coefficient on the right is nonnegative. Exponentiating preserves nonnegative coefficients and constant coefficient 1.

Let r_0 be the largest determinant weight of a constituent of $\mathcal{F}_0$. Deligne first proves that determinant weights add under tensor products, using the central degree element of the algebraic monodromy group; see §1.3. In detail, on an irreducible representation a central element of degree d acts as a scalar c . Its determinant has modulus $|c|^{\dim V}$, so determinant weight r_0 means $|c| = q^{dr_0/2}$. Tensoring multiplies these scalar actions and hence adds their weights. Passing to constituent subquotients preserves their scalar eigenspaces.

It follows from (D2.1) that the right side of (D2.2), for $\mathcal{F}^{\otimes 2k}$, has no poles in

$$|T| < q^{-(2kr_0+2)/2}. \quad (\text{D3.2})$$

The nonnegative Euler coefficients transfer this pole exclusion to each local factor. Write the full formal product as $P_x(T)B_x(T)$, with P_x the local factor at x . There are finitely many closed points of bounded degree, so each coefficient is well-defined. Both factors have nonnegative coefficients and constant coefficient 1, giving

$$0 \leq [T^n]P_x(T) \leq [T^n](P_x(T)B_x(T)).$$

The global rational function is analytic throughout the disk (D3.2); its Taylor series therefore converges at every smaller positive radius. The displayed coefficient domination proves convergence of P_x there. This proves the local pole exclusion without assuming that arbitrary complex poles cannot cancel.

The factor at x has a pole of modulus $|\iota\alpha|^{-2k/d_x}$ for every stalk eigenvalue α . Therefore

$$|\iota\alpha|^{2k/d_x} \leq q^{(2kr_0+2)/2}, \quad |\iota\alpha| \leq N(x)^{r_0/2+1/(2k)}. \quad (\text{D3.3})$$

For all k , this forces $|\iota\alpha| \leq N(x)^{r_0/2}$.

The determinant identities turn the upper bounds into equalities for each constituent. If $n(\gamma)$ is the total rank of constituents of determinant weight γ , then

$$\sum_i w_{N(x)}(\alpha_i^\gamma) = n(\gamma)\gamma. \quad (\text{D3.4})$$

For a chosen β , take the exterior power of degree $1 + \sum_{\gamma > \beta} n(\gamma)$. Its largest determinant weight is $\beta + \sum_{\gamma > \beta} n(\gamma)\gamma$. Applying the bound to the eigenvalue $\alpha_i^\beta \prod_{\gamma > \beta, j} \alpha_j^\gamma$, and subtracting the exact equalities (D3.4), gives $w_{N(x)}(\alpha_i^\beta) \leq \beta$. Their sum is $n(\beta)\beta$, so every inequality is an equality. This is Deligne's purity criterion (1.5.1), not an inference from uniqueness of an Euler product.

114.4 D4. Local monodromy, with its Tate action retained

After a finite cover, local inertia is unipotent:

$$\rho(\sigma) = \exp(Nt_\ell(\sigma)), \quad N : V(1) \longrightarrow V. \quad (\text{D4.1})$$

The map is Frobenius-equivariant as a map with that domain. Upon displaying the Tate scalar it says

$$FN = q^{-1}NF, \quad FNF^{-1} = q^{-1}N. \quad (\text{D4.2})$$

In particular N sends the generalized α -eigenspace to the generalized α/q -eigenspace. This follows by repeatedly applying $(F - \alpha/q)N = q^{-1}N(F - \alpha)$.

The monodromy filtration M is the unique finite increasing filtration satisfying

$$NM_i \subset M_{i-2}, \quad N^j : \text{Gr}_j^M V(j) \xrightarrow{\sim} \text{Gr}_{-j}^M V. \quad (\text{D4.3})$$

It is constructed recursively from kernels and images of the highest nonzero power of N . On a Jordan chain of length $d+1$, the basis degrees are $d, d-2, \dots, -d$, and N lowers the degree by two. This gives the filtration explicitly. Tensoring uses $N_{V \otimes V'} = N_V \otimes 1 + 1 \otimes N_{V'}$; taking a dual uses $-N_V^t$.

Write

$$P_{-j} = \ker(N : \text{Gr}_{-j}^M V \rightarrow \text{Gr}_{-j-2}^M V).$$

The decomposition, with the Tate sign forced by (D4.2), is

$$\mathrm{Gr}_i^M V \cong \bigoplus_{\substack{j \geq |i| \\ j \equiv i \pmod{2}}} P_{-j}(-(i+j)/2). \quad (\text{D4.4})$$

Indeed $N^{(i+j)/2}$ identifies the corresponding summand of $\mathrm{Gr}_i^M V((i+j)/2)$ with P_{-j} . Moving the displayed twist to the other side gives its negative in (D4.4). The available transcription has a positive sign in (1.6.14.3); that is internally inconsistent with its own Tate convention and its stated equal-weight check. We use the derived formula and retain the source unchanged.

Deligne's boundary estimate first gives

$$|\iota\alpha| \leq N(x)^{(\beta+2)/2} \quad \text{on } (j_*\mathcal{F})_x.$$

The complete formula includes the factors at the missing points:

$$\prod_{x \in U_0} \det(1 - F_x T^{d_x}, \mathcal{F}_x)^{-1} \prod_{x \in X_0 \setminus U_0} \det(1 - F_x T^{d_x}, (j_*\mathcal{F})_x)^{-1} = \frac{\det(1 - FT, H_c^1)}{\det(1 - FT, H_c^2)} \quad (\text{D4.5})$$

on the affine curve used in the proof. The omitted H_c^0 is zero by (D2.1). Both the first product and the right-side denominator are controlled in (D2.3), which proves the displayed estimate for the remaining local poles.

The injection

$$(\mathcal{F}_\eta^I)^{\otimes k} \hookrightarrow (\mathcal{F}_\eta^{\otimes k})^I \quad (\text{D4.6})$$

is the usual inclusion: a tensor of invariant vectors is invariant, and the inclusions of vector spaces stay injective on tensoring over a field. It implies

$$|\iota\alpha|^k \leq N(x)^{(k\beta+2)/2}, \quad |\iota\alpha| \leq N(x)^{\beta/2+1/k}. \quad (\text{D4.7})$$

Letting k grow gives the exact boundary bound $N(x)^{\beta/2}$.

Put $Q = N(s)$ at a specified boundary point s . If α is an eigenvalue on P_{-j} , the primitive decomposition of the tensor square contains

$$P_{-j} \otimes P_{-j}(-j) \subset P_0(V \otimes V). \quad (\text{D4.8})$$

Thus $\alpha^2 Q^j$ occurs in the inertia invariants of the tensor square. That input has weight 2β , so (D4.7) gives

$$|\iota\alpha|^2 Q^j \leq Q^\beta, \quad |\iota\alpha| \leq Q^{(\beta-j)/2}. \quad (\text{D4.9})$$

The dual primitive piece is $P_{-j}(V^\vee) = P_{-j}(V)^\vee(j)$; its eigenvalue is $\alpha^{-1}Q^{-j}$. Apply the same upper bound with input weight $-\beta$:

$$|\iota\alpha|^{-1}Q^{-j} \leq Q^{(-\beta-j)/2}, \quad |\iota\alpha| \geq Q^{(\beta-j)/2}. \quad (\text{D4.10})$$

Equality follows. Formula (D4.4) multiplies this eigenvalue by $Q^{(i+j)/2}$, giving weight $\beta + i$ on $\mathrm{Gr}_i^M V$. Every Tate factor is visible. [Deligne, §§1.6–1.8, especially (1.8.1), (1.8.4).]

114.5 D5. The strict bound before the final improvement

The first cohomology estimate is

$$w_q(\alpha) \leq \beta + 2. \quad (\text{D5.1})$$

Deligne improves it to

$$w_q(\alpha) < \beta + 2 \quad (\text{D5.2})$$

before beginning the product bootstrap. This strict inequality is Corollary (2.2.10). Its proof cannot be replaced by saying that the spectrum is complete.

Here is the chain of the argument. Semisimplifying reduces to irreducible sheaves. The Zariski closure of geometric monodromy has semisimple identity component, by (1.3.9). Its extension by the degree group admits a compact form modulo its central degree action. The determinant calculation and already proved purity criterion put the semisimple Frobenius conjugacy classes in that compact form after keeping their scalar modulus separately.

For unitary representations V of the compact part, the prime-power measure gives

$$-\frac{L'}{L}(V, \sigma) = \sum_{x, r \geq 1} (\log N(x)) N(x)^{-r\sigma} \operatorname{Tr}(F_x^r | V), \quad \sigma > 1. \quad (\text{D5.3})$$

For $V \otimes \bar{V}$, each trace is the squared modulus of a trace, so the expression is nonnegative. The residue at $\sigma = 1$ is the pole order of the L -function. These integer orders define an additive functional ν on representations with

$$\nu(1) = 1, \quad \nu(V) = \nu(\bar{V}), \quad \nu(V) \leq 0 \ (V \neq 1), \quad \nu(V \otimes \bar{V}) \geq 0. \quad (\text{D5.4})$$

Approximating a positive central function concentrated at the identity by characters gives, for any finite family T of irreducibles and any $\epsilon > 0$, a representation R such that

$$[R \otimes \bar{R} : V] \geq (1 - \epsilon) \dim(V) [R \otimes \bar{R} : 1] \quad (V \in T).$$

One may obtain an actual representation from a virtual integral character $R_+ - R_-$ by using $R_+ + R_-$; the mixed summands increase the required multiplicities, while the multiplicity of the trivial representation is unchanged because R_+ and R_- have disjoint irreducible supports. Substitution in (D5.4), followed by $\epsilon \downarrow 0$, gives

$$\sum_{V \in T} \dim(V) (-\nu(V)) \leq 1. \quad (\text{D5.5})$$

Consequently at most one nontrivial representation has a zero on that boundary; it must be a one-dimensional real character of order two. Its double cover has zeta function $\zeta_{X'} = \zeta_X L(\epsilon)$. Both curve zeta functions have a simple pole at the boundary, excluding the exceptional zero. This is §§2.1.4–2.2.9.

The trace formula (D2.2) now excludes first-cohomology zeros at the boundary where the top cohomology has weight $\beta + 2$, yielding (D5.2). For arbitrary β , retain a constant twist b with $|\iota b| = q^{\beta/2}$: the comparison spectral variable is bT , the corresponding determinant eigenvalues are $b\alpha_0$, and its absolute value restores exactly $q^{(\beta+2)/2}$. The scalar has not been dropped or called harmless. This passage is between specified sheaves over the finite field; it is not a rescaling of the original Riemann zeta function.

114.6 D6. The complete product improvement with the original weight

For every $k \geq 0$, let the current universal curve estimate be

$$|\iota\alpha| \leq q^{(\beta+1+2^{-k})/2} \quad \text{on } H_c^1(U, \mathcal{F}). \quad (\text{D6.1})$$

The case $k = 0$ is (D5.1). We now reconstruct the induction proving the case $k + 1$, retaining β .

Let X_0 be the smooth projective completion of U_0 , set

$$S_0 = X_0 \times X_0, \quad V_0 = U_0 \times U_0, \quad \mathcal{G}_0 = \operatorname{pr}_1^* \mathcal{F}_0 \otimes \operatorname{pr}_2^* \mathcal{F}_0. \quad (\text{D6.2})$$

The input $\mathcal{G}_0$ has weight 2β . The boundary is the union $(X_0 \setminus U_0) \times X_0$ and $X_0 \times (X_0 \setminus U_0)$, with their actual crossings. Choose a Lefschetz pencil whose axis misses the boundary, whose

exceptional fibres have at most one exceptional point, and whose exceptional points consist of ordinary nodes away from the boundary, ordinary tangencies to a boundary branch, and crossings of two boundary branches.

Blowing up the axis gives $\pi : \tilde{S} \rightarrow S$ and the pencil map $f : \tilde{S} \rightarrow \mathbb{P}^1$. Put $\tilde{V} = \pi^{-1}V$. The blowup contribution is retained:

$$H_c^*(\tilde{V}, \pi^*\mathcal{G}) \cong H_c^*(V, \mathcal{G}) \oplus (H^0(V \cap A, \mathcal{G})(-1) \text{ in degree } 2). \quad (\text{D6.3})$$

In particular π^* injects the original cohomology. Leray gives

$$E_2^{ab} = H^a(\mathbb{P}^1, R^b f_! \pi^* \mathcal{G}) \implies H_c^{a+b}(\tilde{V}, \pi^* \mathcal{G}). \quad (\text{D6.4})$$

The precise boundary calculation is as follows. Let $\epsilon(B)$ be the rank-one orientation module of a two-element set B , with its two opposite generators. Its finite permutation action contributes roots of unity, of weight zero.

- At an ordinary node away from the boundary, the only nonzero vanishing-cycle sheaf is $\Phi_x^1 = \mathcal{G}_x(-1) \otimes \epsilon(B)$, where B is the pair of branches.
- At a tangency, the exact sequence $0 \rightarrow j_! \overline{\mathbb{Q}}_\ell \rightarrow \overline{\mathbb{Q}}_\ell \rightarrow \overline{\mathbb{Q}}_{\ell D} \rightarrow 0$ gives $\Phi_x^1 = \overline{\mathbb{Q}}_\ell \otimes \epsilon(B)$ for constant coefficients. Filtering the locally unipotent sheaf gives $\text{Gr } \Phi_x^1 = \text{Gr } \mathcal{G}_x \otimes \epsilon(B)$.
- At a crossing, the four-term sequence $0 \rightarrow j_! \overline{\mathbb{Q}}_\ell \rightarrow \overline{\mathbb{Q}}_\ell \rightarrow i_* \overline{\mathbb{Q}}_\ell \rightarrow \overline{\mathbb{Q}}_{\ell x} \otimes \epsilon(B) \rightarrow 0$, with i the boundary normalization map, gives the same graded formula. The two successive connecting maps account for the degree shift.

In every case $\Phi_x^a = 0$ for $a \neq 1$. The source proves these statements by Picard–Lefschetz at the node and the displayed exact sequences at the other two types. They retain the node’s additional Tate factor; it is absent at the other types. [Deligne, (3.1.3.2), (3.1.4.6), (3.1.5.4).]

Let $w : W_0 \hookrightarrow \mathbb{P}^1$ remove the exceptional values and write $\mathcal{K} = R^1 f_! \pi^* \mathcal{G}$. The five-term exact sequence at an exceptional value t is

$$0 \rightarrow \mathcal{K}_{\bar{t}} \rightarrow \mathcal{K}_{\bar{\eta}} \rightarrow \Phi_x^1 \rightarrow (R^2 f_! \pi^* \mathcal{G})_{\bar{t}} \rightarrow (R^2 f_! \pi^* \mathcal{G})_{\bar{\eta}} \rightarrow 0. \quad (\text{D6.5})$$

Thus $\mathcal{K} \hookrightarrow w_* w^* \mathcal{K}$; there is no nonzero section supported just at the exceptional values.

The sheaf $w^* \mathcal{K}$ has pointwise pure constituents. To see the real comparison without changing the input weight, choose the constant Weil line $\mathcal{C}$ with Frobenius $q^{2\beta}$ under ι ; its weight is 4β . At a point of degree d , eigenvalues of $\mathcal{G}^\vee \otimes \mathcal{C}$ are $q^{2\beta d}/\alpha = \bar{\alpha}$, because $|\alpha|^2 = q^{2\beta d}$. Therefore

$$\mathcal{G} \oplus (\mathcal{G}^\vee \otimes \mathcal{C}) \quad (\text{D6.6})$$

is ι -real of weight 2β and contains the original $\mathcal{G}$ as a direct summand. Write $\mathcal{H}$ for this displayed sheaf restricted to a smooth pencil fibre Y . The strict bound (D5.2) separates its H_c^1 eigenvalues from its H_c^2 eigenvalues. Consequently the latter determinant is recovered from the poles of the real rational function (D2.2) of modulus $N(x)^{-(2\beta+2)/2}$; those poles cannot cancel with the numerator. Its coefficients are therefore real. If Y is affine, $H_c^0 = 0$. If Y is projective, the exact duality

$$H^0(Y, \mathcal{H}) \cong H^2(Y, \mathcal{H}^\vee(1))^\vee$$

proves reality of H^0 by the same H^2 calculation on the pure real input $\mathcal{H}^\vee(1)$. Both denominator factors of (D2.2) are now real, so its numerator is real as well. Apply D3 to the resulting real sheaf on W_0 . The consulted transcription prints a negative Tate twist at this step in (3.2.1); the positive twist here is forced by the displayed Poincaré duality.

Take the constituent filtration of $w^* \mathcal{K}$, extend it by intersection inside $w_* w^* \mathcal{K}$, and denote a successive quotient on $\mathbb{P}^1$ by $\mathcal{K}_i$. At each exceptional point there is an exact sequence

$$0 \rightarrow (\mathcal{K}_i)_{\bar{t}} \rightarrow (\mathcal{K}_i)_{\bar{\eta}} \rightarrow A_t^i \rightarrow 0, \quad (\text{D6.7})$$

where A_t^i is a subquotient of Φ_x^1 .

The product input and D4 give all weights of Φ_x^1 in $2\beta + \mathbb{Z}$. If some $A_t^i \neq 0$, D4 applied to the pure constituent shows that its weight is also in $2\beta + \mathbb{Z}$. If every $A_t^i = 0$, $\mathcal{K}_i$ is lisse on the whole geometric projective line, hence geometrically constant, and $H^1(\mathbb{P}^1, \mathcal{K}_i) = 0$.

For every constituent that contributes to this H^1 , the strict fibrewise curve bound (D5.2) gives

$$\text{weight} \in 2\beta + \mathbb{Z}, \quad \text{weight} < 2\beta + 2.$$

Therefore its weight is at most $2\beta + 1$. This is the integer-step deduction, with the original β retained. It concerns differences of cohomological weights, not the parity of source τ .

Apply the induction estimate (D6.1) on the pencil's base curve:

$$w_q(\alpha) \leq 2\beta + 2 + 2^{-k} \quad \text{on } E_2^{11}. \quad (\text{D6.8})$$

For the other two terms of total degree two, (D2.1) used on the fibres and then on the base gives

$$w_q(\alpha) \leq 2\beta + 2 \quad \text{on } E_2^{02} \text{ and } E_2^{20}. \quad (\text{D6.9})$$

Spectral-sequence differentials commute with Frobenius. Each later page is a subquotient of the earlier page; its eigenvalues therefore occur among those already bounded. The finite filtration of the abutment has these bounded eigenvalues as its successive factors. This proves the bound $2\beta + 2 + 2^{-k}$ on $H_c^2(\tilde{V}, \pi^*\mathcal{G})$, including the contribution (D6.3).

Künneth embeds

$$H_c^1(U, \mathcal{F}) \otimes H_c^1(U, \mathcal{F}) \hookrightarrow H_c^2(V, \mathcal{G}) \hookrightarrow H_c^2(\tilde{V}, \pi^*\mathcal{G}). \quad (\text{D6.10})$$

An eigenvector of eigenvalue α gives a nonzero tensor eigenvector of eigenvalue α^2 . Hence

$$|\iota\alpha|^2 \leq q^{(2\beta+2+2^{-k})/2}, \quad |\iota\alpha| \leq q^{(\beta+1+2^{-(k+1)})/2}. \quad (\text{D6.11})$$

This proves the induction. The error halves because the pencil bound in (D6.8) contains one copy of the previous error. Tensoring the old bound directly would contain two copies and prove no improvement.

The preliminary finite covers and finite field extensions are removed by the injective pullback maps in (3.2.14). A field extension of degree d replaces α by α^d and q by q^d ; the equality

$$\frac{2 \log |\iota\alpha^d|}{\log q^d} = \frac{2d \log |\iota\alpha|}{d \log q} \quad (\text{D6.12})$$

proves that the original weight bound follows. No scale has been silently discarded.

Letting $k \rightarrow \infty$ gives the upper bound $\beta + 1$. For $H^1(X, j_*\mathcal{F})$, duality pairs it perfectly with $H^1(X, j_*\mathcal{F}^\vee)$ into $\mathbb{Q}_\ell(-1)$. The partner of α is q/α ; the upper bound for input weight $-\beta$ gives

$$|q/\iota\alpha| \leq q^{(-\beta+1)/2}, \quad |\iota\alpha| \geq q^{(\beta+1)/2}. \quad (\text{D6.13})$$

Together the two bounds give purity of weight $\beta + 1$. The degrees zero and two follow from (D2.1). This reconstructs (3.2.3) through (3.2.15).

114.7 D7. Mixed objects and proper pushforward

Deligne extends the curve theorem by exact sequences, open/closed decompositions, compositions and relative-curve reductions. The basic open/closed sequence is

$$0 \rightarrow j_!j^*\mathcal{F} \rightarrow \mathcal{F} \rightarrow i_*i^*\mathcal{F} \rightarrow 0. \quad (\text{D7.1})$$

On a relative curve compactification $\bar{f}$, the boundary is finite over the base after the stated stratifications and finite alterations of the relative curve. For pure input, D4 gives on its boundary

$$\mathrm{Gr}_i^M(i^*j_*\mathcal{F}) \text{ of weight } \beta + i, \quad i \leq 0. \quad (\text{D7.2})$$

The curve theorem controls $R^a\bar{f}_*j_*\mathcal{F}$ at weight $\beta + a$. Apply the long exact sequence from $0 \rightarrow j_!\mathcal{F} \rightarrow j_*\mathcal{F} \rightarrow i_*i^*j_*\mathcal{F} \rightarrow 0$. Because the boundary map is finite, its higher direct images vanish, and its remaining weights are bounded by (D7.2). The resulting $R^af_!\mathcal{F}$ have weights at most $\beta + a$.

For a composition $f = g \circ h$, Leray has terms $R^ag_!R^bh_!\mathcal{F}$ of weights at most $\beta + b + a$, so the same bound passes to the abutment in degree $a + b$. For an extension of input sheaves, the long exact sequence passes the bounds from its two factors to the middle term. The relative dimension-zero case follows directly from finite sums of stalks. These reductions give Theorem (3.3.1), with the real-weight refinement (3.3.10).

For smooth X of dimension d , duality retains both degree and Tate shift:

$$H^a(X, \mathcal{F})^\vee \cong H_c^{2d-a}(X, \mathcal{F}^\vee)(d). \quad (\text{D7.3})$$

The right side has upper weight $-\beta + (2d - a) - 2d = -\beta - a$, so H^a has lower weight $\beta + a$. Thus the image of $H_c^a \rightarrow H^a$ is pure of weight $\beta + a$. Properness identifies the two groups and yields purity.

At the derived level, retain the actual dualizing complex

$$K_X = Ra^!\overline{\mathbb{Q}}_\ell, \quad DK = R\mathcal{H}om(K, K_X).$$

For smooth d -dimensional X , $K_X = \overline{\mathbb{Q}}_\ell(d)[2d]$. A complex has weight at most n when $\mathcal{H}^iK$ has weight at most $n + i$; it is pure when both K and DK have the corresponding upper bounds n and $-n$. Properness gives

$$DRf_*K = Rf_*DK, \quad Rf_* = Rf!, \quad (\text{D7.4})$$

so applying (3.3.1) to both sides proves purity of the proper direct image, (6.2.6). Mixed extension control is what makes these maps usable. A mixed object is not rendered pure merely by forgetting its filtration.

Deligne's separate integrality observation, (3.3.2), says that a nonzero algebraic integer pure of integer weight n has $n \geq 0$: its field norm is a nonzero integer of modulus $q^{nd/2} \geq 1$. This is different from D6's statement that weight differences belong to $\mathbb{Z}$. Counting integers, algebraic-integrality of an eigenvalue, and the discrete monodromy weight steps have different exact maps in the proof.

114.8 P1. The programme object receiving this calculation

The established source reconstruction retains

$$1 \rightarrow H \rightarrow G \rightarrow W \rightarrow 1, \quad R = \mathrm{End}_{\mathrm{Ab}}(W), \quad \mathbf{1}_R = \mathrm{id}_W, \quad (\text{P1.1})$$

with the finite signed states in H , the complete infinite cyclic quotient W , the chart maps, and every power map $[n]$. A chart isomorphism b satisfies $b[n]b^{-1} = [n]$, so it preserves $p = |W/[p]W|$.

The full return measures are

$$\mathcal{R} = \sum_p \sum_{r \geq 1} \frac{1}{r} \delta_{r \log p}, \quad \mathcal{D} = \exp_* \mathcal{R} = \delta_0 + \sum_{n \geq 2} \delta_{\log n}, \quad \mathcal{W} = t\mathcal{R} = \sum_p \sum_{r \geq 1} (\log p) \delta_{r \log p}. \quad (\text{P1.2})$$

Their transforms are the original functions

$$\int e^{-st} d\mathcal{D}(t) = \zeta(s), \quad \int e^{-st} d\mathcal{W}(t) = -\zeta'(s)/\zeta(s) \quad (\Re s > 1). \quad (\text{P1.3})$$

All coefficients, return times and the unit atom are retained. The receiving composition proves the same meromorphic continuation for both charts. This part of item 26 is carried forward as closed.

For the actual analytic quotient, use

$$\mathcal{A} = \{k \in C^\infty(\mathbb{R}_{>0}) : \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j k(u)| < \infty \text{ for every } N, j \geq 0\},$$

$$\mathcal{E}f(u) = u^{1/2} \sum_{n \geq 1} f(nu), \quad I = \overline{\mathcal{E}(S_0^{\text{even}})}^{\mathcal{A}}, \quad Q = \mathcal{A}/I, \quad F_k(s) = \int_0^\infty k(u) u^{s-1/2} \frac{du}{u}. \quad (\text{P1.4})$$

Here S_0^{even} consists of even Schwartz functions with $f(0) = 0$ and $\int_{\mathbb{R}} f = 0$. Absolute interchange first in $\Re s > 1$, followed by continuation, gives

$$F_{\mathcal{E}f}(s) = \zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v}. \quad (\text{P1.5})$$

Thus the jet of order below the multiplicity at any actual nontrivial zero vanishes on I , and defines an observation on Q .

For the counterfactual actual zero $\rho = \beta + i\gamma$, $0 < \beta < 1$, $\beta \neq 1/2$, its distinct four-point orbit is

$$\mathcal{O} = \{\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}\}.$$

The original functional equation used here is

$$\zeta(s) = 2^s \pi^{s-1} \sin(\pi s/2) \Gamma(1-s) \zeta(1-s). \quad (\text{P1.5a})$$

Every factor in its multiplier is holomorphic and nonzero in $0 < \Re s < 1$, the exact domain of this comparison. Complex conjugation preserves the original Dirichlet series and hence its continuation. Thus all four multiplicities are the same, denoted m . Moreover $\gamma \neq 0$: for $0 < s < 1$, pairing consecutive terms makes $\sum_{n \geq 1} (-1)^{n-1} n^{-s}$ strictly positive, while $1 - 2^{1-s} < 0$, and their quotient is $\zeta(s)$. The four points are therefore distinct. Retain

$$A_\omega = \mathbb{C}[t_\omega]/(t_\omega^m), \quad V_{\mathcal{O}} = \bigoplus_{\omega \in \mathcal{O}} A_\omega, \quad j_\omega[k] = \sum_{r=0}^{m-1} \frac{F_k^{(r)}(\omega)}{r!} t_\omega^r. \quad (\text{P1.6})$$

The full finite observation $j_{\mathcal{O}}$ is onto: the functionals on compactly supported logarithmic tests have independent densities $v^r e^{(\omega-1/2)v} / r!$. Apply $\prod_{\nu \neq \omega} (\partial_v - \nu + 1/2)^m$ to a vanishing linear combination. It kills the other exponents and is invertible on the polynomial attached to ω , since its constant diagonal factors are $\omega - \nu \neq 0$. All polynomial coefficients vanish, proving independence and hence surjectivity.

The original prime action, including the comparison scalar, is

$$W_p k(u) = p^{1/2} k(u/p), \quad F_{W_p k}(s) = p^s F_k(s), \quad W_p|_{A_\omega} = p^\omega \exp((\log p) t_\omega). \quad (\text{P1.7})$$

Nothing in (P1.1)–(P1.7) assigns numerical parity or addition to τ . This is the programme's already reconstructed arithmetic and its actual receiving quotient.

114.9 P2. The global tensor Euler product and its exact moving pole

Put $B = \max(\beta, 1 - \beta) > 1/2$. On $V_{\mathcal{O}}^{\otimes 2k}$, use the diagonal prime action, retaining every ordered tuple of branches and every jet:

$$W_p^{\otimes 2k}|_{A_{\omega_1} \otimes \cdots \otimes A_{\omega_{2k}}} = p^{\omega_1 + \cdots + \omega_{2k}} \exp\left((\log p) \sum_{a=1}^{2k} t_a\right). \quad (\text{P2.1})$$

The commuting variables satisfy $t_a^m = 0$. The nilpotent exponential is triangular with every diagonal entry 1 in the monomial basis of dimension m^{2k} . Consequently, exactly,

$$\det(1 - p^{-s} W_p^{\otimes 2k}) = \prod_{(\omega_1, \dots, \omega_{2k}) \in \mathcal{O}^{2k}} (1 - p^{\omega_1 + \dots + \omega_{2k} - s})^{m^{2k}}. \quad (\text{P2.2})$$

This determinant calculation retains the nilpotent operator (P2.1); it proves why its strictly triangular entries do not enter this determinant.

Therefore the complete prime product is

$$\mathcal{L}_k(s) = \prod_p \det(1 - p^{-s} W_p^{\otimes 2k})^{-1} = \prod_{\omega \in \mathcal{O}^{2k}} \zeta\left(s - \sum_{a=1}^{2k} \omega_a\right)^{m^{2k}}. \quad (\text{P2.3})$$

Absolute convergence holds for $\Re s > 1 + 2kB$, since every shifted argument has real part greater than 1. The right side is a finite product and supplies meromorphic continuation without a completed substitute.

All prime-power logarithmic coefficients are nonnegative:

$$\log \mathcal{L}_k(s) = \sum_p \sum_{r \geq 1} \frac{p^{-rs}}{r} \left[2m(p^{r\beta} + p^{r(1-\beta)}) \cos(r\gamma \log p) \right]^{2k}. \quad (\text{P2.4})$$

Exponentiating gives nonnegative Dirichlet coefficients. The expression includes every prime and repetition, not a finite-prime sample.

At the real point

$$s_k = 1 + 2kB \quad (\text{P2.5})$$

a tuple has sum $2kB$ precisely when every branch has real part B and exactly k of the $2k$ branches have imaginary part $+\gamma$, the other k having imaginary part $-\gamma$. There are $\binom{2k}{k}$ such tuples. Each contributes a pole of order m^{2k} .

No other factor cancels those poles. For a tuple with smaller total real part, its zeta argument at s_k has real part greater than one and hence is nonzero. For a tuple with maximal real part but nonzero imaginary sum, its argument lies on $\Re s = 1$ away from 1, where the original zeta has no zero. For completeness, this last fact follows directly from

$$\zeta(\sigma)^3 |\zeta(\sigma + it)|^4 |\zeta(\sigma + 2it)| \geq 1 \quad (\sigma > 1),$$

whose logarithm is the sum of $p^{-r\sigma} [3 + 4 \cos(rt \log p) + \cos(2rt \log p)]/r$, with bracket $2(1 + \cos(rt \log p))^2 \geq 0$. A zero of multiplicity $a \geq 1$ at $1 + it$, $t \neq 0$, makes the left side tend to zero as $\sigma \downarrow 1$, since its order is at least $4a - 3 > 0$; this contradicts the inequality. Only the known simple pole of the original zeta at one was used.

Thus the pole order at (P2.5) is exactly

$$\text{ord}_{\text{pole}, s_k} \mathcal{L}_k = m^{2k} \binom{2k}{k}. \quad (\text{P2.6})$$

There are no poles further to the right: each factor has its only pole when its shifted argument equals one. The use of the zero-free boundary above is the classical Hadamard–de la Vallée-Poussin argument, reproduced here for this application; it is not a new nonvanishing theorem.

For $k = 1$, the full product can be written without tuple notation:

$$\begin{aligned} \mathcal{L}_1(s) = & \left[\zeta(s - 2\beta - 2i\gamma) \zeta(s - 2\beta + 2i\gamma) \zeta(s - 2\beta)^2 \right. \\ & \cdot \zeta(s - 2 + 2\beta - 2i\gamma) \zeta(s - 2 + 2\beta + 2i\gamma) \zeta(s - 2 + 2\beta)^2 \\ & \left. \cdot \zeta(s - 1 - 2i\gamma)^2 \zeta(s - 1 + 2i\gamma)^2 \zeta(s - 1)^4 \right]^{m^2}. \end{aligned} \quad (\text{P2.6a})$$

The displayed exponents total 16, accounting for all ordered pairs. No mixed branch terms have been removed.

The whole quartet already has an integral determinant of average weight one:

$$\det(W_p | V_{\mathcal{O}}) = p^{m(\rho+\bar{\rho}+1-\rho+1-\bar{\rho})} = p^{2m}, \quad \dim V_{\mathcal{O}} = 4m, \\ \frac{2 \log |\det W_p|}{4m \log p} = 1. \quad (\text{P2.6b})$$

Its semisimple joint characters nevertheless have weights 2β and $2(1-\beta)$, each with total multiplicity $2m$. Their maximum is $2B$. Deligne's r_0 in D3 is the maximum determinant weight of the constituents, not the determinant average of the whole object. Substituting that average would replace the quantity actually controlled by his proof.

The displacement from the weight-one boundary is

$$s_k - (1+k) = 2k \left| \beta - \frac{1}{2} \right|. \quad (\text{P2.7})$$

It grows under tensor amplification. The calculation does not change the reconstructed prime spectrum or the original zeta function. It shows exactly how an off-line zero, inside that same function, appears in a global Euler product derived from the programme's actual prime operators.

114.10 P3. What quotienting preserves

Fix a branch and a prime p . Every W_p -invariant subspace of A_{ω} is invariant under multiplication by t_{ω} . Indeed

$$t_{\omega} = \frac{1}{\log p} \sum_{j=1}^{m-1} \frac{(-1)^{j+1}}{j} (p^{-\omega} W_p - 1)^j \quad (\text{P3.1})$$

as an exact finite nilpotent logarithm. A subspace stable under t_{ω} is an ideal of $\mathbb{C}[t_{\omega}]/(t_{\omega}^m)$, since it is stable under every polynomial. Such ideals are exactly (t_{ω}^r) , $0 \leq r \leq m$: take the smallest degree of a nonzero element; its remaining factor has nonzero constant coefficient and is a unit, so the corresponding monomial generates the ideal.

The quotient is consequently

$$A_{\omega}/(t_{\omega}^r) \cong \mathbb{C}[t_{\omega}]/(t_{\omega}^r). \quad (\text{P3.2})$$

For $r > 0$ it still has eigenvalue p^{ω} , now of multiplicity r . The constant observation $a(t) \mapsto a(0)$ descends precisely for $r \geq 1$; retaining the entire original jet requires $r = m$. An equivariant quotient cannot move the eigenvalue to $p^{1/2}$.

For distinct branches, the joint eigenvalue pair $(2^{\omega}, 3^{\omega})$ separates them. Equality for two branches would give $(\omega - \nu) \log 2 = 2\pi i a$ and $(\omega - \nu) \log 3 = 2\pi i b$. If $\omega \neq \nu$, this would make $\log 2 / \log 3 = a/b$ rational and hence $2^b = 3^a$, impossible for nonzero integers. Finite-dimensional polynomial spectral projectors for a generic linear combination of W_2 and W_3 therefore recover each branch. Every invariant quotient splits into the quotients (P3.2).

It follows that all four branch observations survive an equivariant quotient exactly when each branch retains positive length. Its even tensor Euler product still has a pole at $1 + 2kB$; the multiplicity is changed by the retained branch lengths, but remains positive. An isomorphism retains the original multiplicities as well. This is the exact restriction on "quotient out the differences" supplied by the current prime actions.

For any quotient $q : Q \rightarrow Q/K$, the full observation $j_{\mathcal{O}}$ descends if and only if

$$K \subseteq \ker j_{\mathcal{O}}. \quad (\text{P3.3})$$

Proof: if $j_{\mathcal{O}} = \bar{j}_{\mathcal{O}} q$, it vanishes on $\ker q$. Conversely the inclusion makes $\bar{j}_{\mathcal{O}}([x]) = j_{\mathcal{O}}(x)$ independent of the representative. Thus a quotient that preserves all these observations cannot erase their pole displacement. A quotient that erases them is a different, explicitly nonfaithful observation map.

114.11 P4. Full support and the exact nilpotent comparison

Retain the programme's existing formal coefficient receiver, with its separate coefficient symbols and the vector space W_L on all specified support labels. On its algebraic coefficient extension, the comparison is

$$j_{\mathcal{O}} \otimes \text{id} \otimes \text{id}. \quad (\text{P4.1})$$

For any linear map $T : V \rightarrow V'$ and any coefficient vector space C with specified basis $\{c_a\}$,

$$\ker(T \otimes \text{id}_C) = (\ker T) \otimes C. \quad (\text{P4.2})$$

Every tensor is a finite sum $\sum_a v_a \otimes c_a$; its image vanishes exactly when all $T(v_a) = 0$. This proves the equality while retaining each label. For a finite retained support subspace of dimension d , the tensor-power determinant in (P2.2) repeats each eigenvalue d^{2k} times. For an infinite support receiver, the proof applies coordinatewise; no finite determinant of the whole infinite support space is asserted.

These are exact maps on the existing formal receiver. They do not establish an equivalence with every proposed semimodule cohomology over the source. In particular they do not define source $\tau + \tau$, replace source absence by a vector zero, or remove the actual mixed-support maps. A general nontrivial action on support would enter the determinant through its own eigenvalues; it cannot be suppressed by the identity-extension calculation.

For the actual jet, put $T = m_{t_\omega}$. Formula (P1.7) gives

$$W_p T W_p^{-1} = T. \quad (\text{P4.3})$$

Deligne's local monodromy equation (D4.2) instead has $p^{-1}T$. The exact difference is

$$W_p T W_p^{-1} - p^{-1}T = (1 - p^{-1})T. \quad (\text{P4.4})$$

This vanishes for a length-one jet and is nonzero for a jet of length greater than one. It is unrelated to whether $\Re\omega = 1/2$. Therefore the original spectral jet nilpotent cannot silently stand in for Deligne's inertia nilpotent.

There is a constructive relation retaining the original jet. For $m \geq 1$, form

$$\begin{aligned} B &= \mathbb{C}[x, y]/(x, y)^m, & X &= m_x, & N &= m_y, \\ F &= p^\omega \exp((\log p)X) S, & S f(x, y) &= f(x, p^{-1}y). \end{aligned} \quad (\text{P4.5})$$

Then $FX = XF$ and $FN F^{-1} = p^{-1}N$. The map $A_\omega \hookrightarrow B$, $t_\omega \mapsto x$ is injective and preserves the original W_p and its full commuting jet. On the other hand $B \rightarrow A_\omega$, $x, y \mapsto t_\omega$, identifies X and N but fails Frobenius descent:

$$F(y - x) \mapsto p^\omega \exp((\log p)t_\omega)(p^{-1} - 1)t_\omega. \quad (\text{P4.6})$$

Thus the two different nilpotents and their exact defect have a single receiving object. This creates no purity theorem: on y -degree j , the eigenvalue of F is $p^{\omega-j}$, with the surviving x -jets still present.

114.12 P5. The result of the proposed contradiction test

The complete reconstruction fixes the arithmetic actions and the original ζ . It now gives an explicit global tensor calculation, (P2.3), whose rightmost real pole is (P2.5) with multiplicity (P2.6). Observation-preserving equivariant quotients retain that pole. This directly addresses the class of deviations represented by actual off-line zero blocks; it does not replace them by arbitrary malformed arithmetic.

Deligne’s quotient (D2.1) computes geometric coinvariants and their Frobenius action. His integer-step argument then comes from actual local monodromy and the three vanishing-cycle geometries in D6. These supply a tensor-power bound whose error does not grow with the tensor exponent. Our quotient classification and positivity calculation by themselves give the pole in (P2.5), not that independent bound.

Consequently the proposed contradiction has not been established. The calculated issue is now precise and global: the retained programme would have to control the pole $1 + 2kB$ through its genuine cohomological trace maps. Merely mapping back to the same unique zeta function retains this pole rather than contradicting it. No off-line zero has been found, and no assumption of an off-line zero has been discharged.

114.13 Figure: the two global calculations

The top panel shows the proved source bound D6.1 for the illustrative input sheaf weight 3. The bottom panel shows P2.5 for the illustrative counterfactual real part $5/8$. These are different named parameters, not an identification of the sheaf weight with a zero coordinate. The pole gap is $2k(5/8 - 1/2) = k/4$; the complete pole multiplicity is P2.6. The bottom panel does not assert that such a zeta zero exists. Source, full proof and rendering code remain alongside the figure.

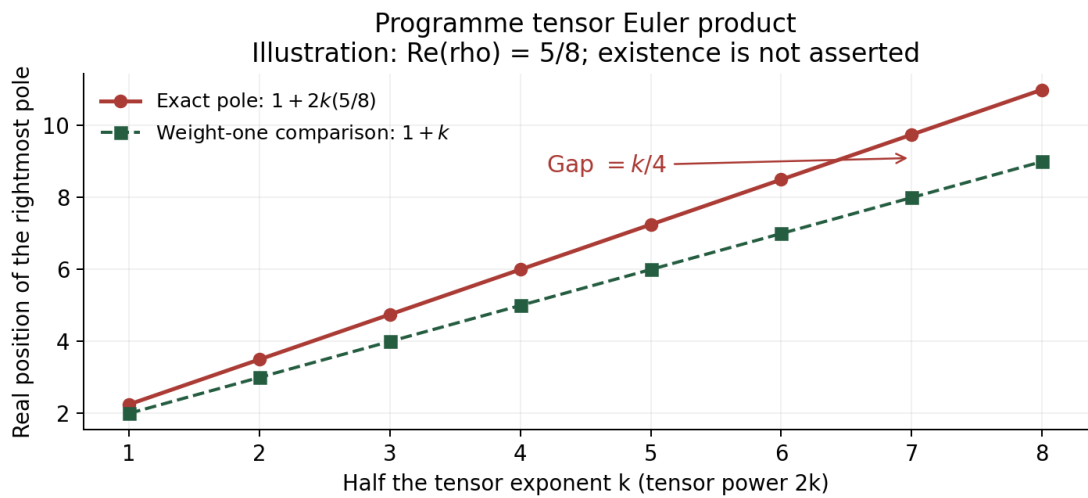
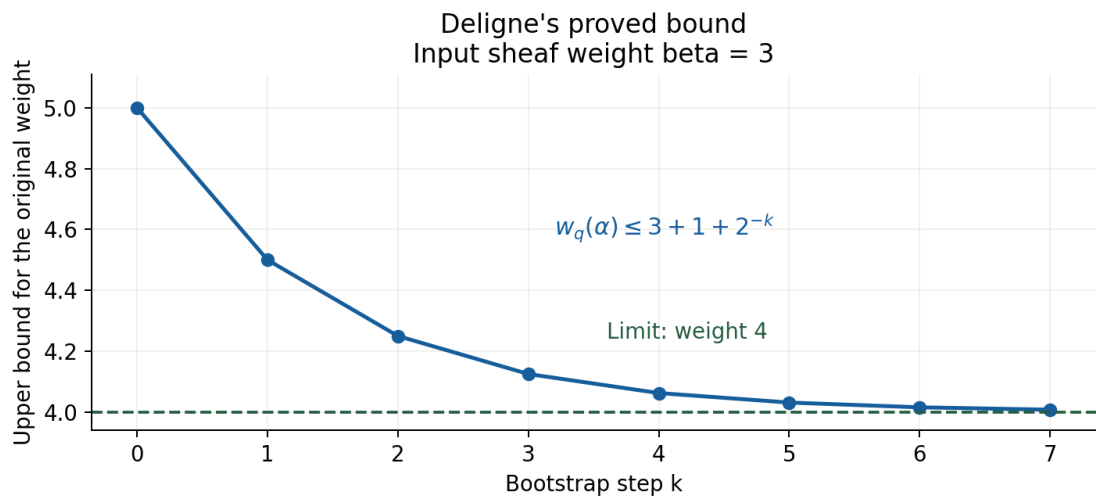
114.14 Sources and mathematical provenance

The finite-field weight arguments D1–D7 are Deligne’s, reconstructed from the cited sections; the trace formula and cohomology formalism are credited there to Alexander Grothendieck and the SGA authors. The local monodromy construction is from SGA 7 and Deligne’s §§1.6–1.8. The account does not claim that these arguments are new.

The user’s contribution is the proposed full-spectrum reconstruction and global-quotient contradiction question, with Z_0, Z_1, Z_2, τ kept in the specified notation. P2’s exact shifted-zeta tensor product, P3’s receiving quotient calculation and P4’s two-nilpotent comparison are derivations in this continuation, assisted by independent mathematical agents. They are applications and tests of that question, not attributed to Deligne as statements about the Riemann zeta function.

The programme input is retained in the companion source directory and in the preceding *Integral history and purity* note, IH1–IH8. The public source for the timed reconstruction is TP0–TP14, pinned programme edition. Connes–Consani’s geometric source remains Alain Connes and Caterina Consani, *On the Absolute Geometry of Spec Z*, arXiv:2606.06604v1, together with their arXiv:2609.00299v1; exact local author-TeX reading locations are in the preceding source ledger. None of the private correspondence is being represented as a published proof.

Geometric control reduces the error;
tensoring alone retains it



Top: Deligne, Weil II 3.2.4-3.2.15; proof D6.

Bottom: exact original-zeta product P2.3-P2.7; all primes and mixed branches.
The displayed coordinates illustrate the proved formulas, with a counterfactual
quartet below. Its pole multiplicity is $m^{(2k)}$ times $\text{binomial}(2k, k)$.

Figure 41: Exact weight improvement and exact tensor pole displacement.

115 Deligne’s weight amplification and the actual prime-primary operators

Private independent mathematical derivation, 24 September 2026.

This note reconstructs Deligne’s arguments in *La conjecture de Weil. II*, §§1.5 and 1.8, and calculates the tensor Euler products of the programme’s actual prime-primary operators. The human source is Pierre Deligne, *Publications mathématiques de l’IHÉS* **52** (1980), 137–252. Grothendieck’s trace formula, local monodromy theorem, and geometric-monodromy theorem retain their human attribution. They are cited inputs; this note does not claim to reprove those theories.

115.1 1. Source identity and what was read

The main consulted file is the local French page-record transcription:

S20_FR_record_export.tex

SHA256: d6836ae15d98f2b27eb98c9bd039a7d921679becd9fa47ff4717410289dbd351.

It is a mechanically escaped page-record transcription, **not an original author TeX source**.

The historical English math-mode transcription was consulted for comparison:

032_Deligne_The_Weil_Conjecture_II_EN_source_checked_corrected.tex

SHA256: 316cbc52349cd6bc2dedbd0a493a530696d367ca0e292fc64ee3c3973327567e.

Actual reading in this derivation: French §§1.2.1–1.2.14, §§1.3.1–1.3.13, §§1.4.1–1.4.7, all of §1.5, §§1.6.1–1.6.4 and 1.6.14, §§1.7.1–1.7.6, and all of §1.8, including its final Hodge-theoretic problem. Sections not in that list have not been read completely here. English §1.5 and the displayed formulas (1.6.14.3–5) and §1.8.4 were compared. Reading all of these sections is not a claim to have read all 116 printed pages or to have reverified their external dependencies.

There are mathematical defects in the supplied copies. Most consequential here, both copies print a positive twist in (1.6.14.3). That sign is incompatible with their own convention that a Tate twist (1) has weight -2 . Section 5 below derives the required negative twist from the equivariant nilpotent map. This derivation does not silently attribute a transcription defect to Deligne.

Other defects encountered, not used as mathematical premises: French (1.3.10)(iv) prints a finite subgroup of $\mathbb{Z}$, although its proof produces a subgroup of finite index $n\mathbb{Z}$, $n \neq 0$. French (1.7.5) prints a product indexed by $j < i$ for the increasing weight filtration; the filtration prescribed by the proposition is the direct sum of generalized eigenspaces of weights $j \leq i$.

115.2 2. Fixed conventions and the geometric input

Let X_0 be a smooth geometrically irreducible curve over $\mathbb{F}_q$, let $X = X_0 \otimes_{\mathbb{F}_q} \overline{\mathbb{F}}_q$, and let $\mathcal{F}_0$ be a lisse Weil sheaf with coefficients in $\overline{\mathbb{Q}}_\ell$, $\ell \nmid q$. Fix the field isomorphism $\iota : \overline{\mathbb{Q}}_\ell \rightarrow \mathbb{C}$ used by Deligne. For a closed point x , write

$$d_x = [\kappa(x) : \mathbb{F}_q], \quad Q_x = N(x) = q^{d_x}, \quad w_{Q_x}(\alpha) = \frac{2 \log |\iota \alpha|}{\log Q_x}. \quad (\text{DW2.1})$$

No twist to weight zero is performed in this note. Every weight β remains explicit.

For a nonzero irreducible constituent $\mathcal{G}_0$ of rank d , its determinantal weight is the weight of $\det \mathcal{G}_0$, divided by d . Deligne’s §§1.3.4–1.3.13 supply the following facts:

- a rank-one lisse Weil sheaf has a single pointwise ι -weight;
- tensor products add determinantal weights;
- the determinantal weights of $\bigwedge^a \mathcal{F}_0$ are the sums $\sum_\gamma a(\gamma)\gamma$, with $\sum_\gamma a(\gamma) = a$ and $0 \leq a(\gamma) \leq n(\gamma)$, where $n(\gamma)$ is the total rank of constituents of weight γ .

The reason for the tensor assertion is the central element in the geometric-monodromy construction of §§1.3.7–1.3.12. After passing to the semisimplification, an element g central in that algebraic group has nonzero degree d_g , and on a constituent of determinantal weight β it acts with eigenvalues of absolute value $q^{d_g\beta/2}$. Its eigenvalues multiply on tensors. This is geometric-monodromy information, not a consequence of the existence or uniqueness of an Euler product.

Grothendieck's trace formula, §1.4.5, is

$$\prod_{x \in |X_0|} \det(1 - F_x T^{d_x}, \mathcal{F})^{-1} = \prod_i \det(1 - FT, H_c^i(X, \mathcal{F}))^{(-1)^{i+1}}. \quad (\text{DW2.2})$$

The application of ι to the coefficients is understood in subsequent complex analytic formulas and can be inserted on every determinant. On an affine curve, $H_c^0 = 0$, so the full right side is

$$\frac{\det(1 - FT, H_c^1(X, \mathcal{F}))}{\det(1 - FT, H_c^2(X, \mathcal{F}))}. \quad (\text{DW2.3})$$

The absent H_c^0 factor is exactly $\det(1 - FT, 0) = 1$; the reason for its absence is the proved vanishing for a lisse sheaf on a nonproper smooth connected curve, not a cancellation convention.

Section 1.4.1 identifies

$$H_c^2(X, \mathcal{F}) = \mathcal{F}_{\bar{\eta}, \pi_1(X)}(-1). \quad (\text{DW2.4})$$

The coinvariant quotient has determinantal weights among those of $\mathcal{F}_0$. The Tate twist (-1) multiplies geometric Frobenius by q , and therefore adds exactly 2 to each weight. This is the source of the fixed error in the pole bound.

If all local eigenvalues have weight at most β , the number of degree- n points is bounded by Cq^n , and

$$\sum_{n \geq 1} Cq^n q^{n\beta/2} |T|^n \quad (\text{DW2.5})$$

converges for $|T| < q^{-\beta/2-1}$. The logarithmic Euler sum is bounded by the full repetition series

$$R_{\mathcal{F}} \sum_{n \geq 1} Cq^n \sum_{r \geq 1} \frac{(q^{n\beta/2} |T|^n)^r}{r}, \quad (\text{DW2.6})$$

where $R_{\mathcal{F}}$ is the rank. On compact subdiscs of the displayed domain, the inner sum is bounded by a constant times its first power, uniformly in n . Thus (DW2.5) proves absolute convergence of the logarithm and a holomorphic nonzero Euler product. The rank and C do not change the exponent.

115.3 3. The complete positivity and amplification in §1.5

Call a sheaf ι -real when the characteristic polynomial of each F_x , after applying ι , has real coefficients. Consequently every power trace $\text{tr}(F_x^r)$ is real.

Fix $k \geq 1$. For each local Frobenius $A = F_x$,

$$\text{tr}((A^{\otimes 2k})^r) = (\text{tr}(A^r))^{2k} \geq 0. \quad (\text{DW3.1})$$

The reciprocal local determinant has the full formal expansion

$$\det(1 - A^{\otimes 2k} U)^{-1} = \exp \left(\sum_{r \geq 1} \frac{(\text{tr } A^r)^{2k}}{r} U^r \right). \quad (\text{DW3.2})$$

This identity follows by triangularizing A , applying $-\log(1 - z) = \sum_{r \geq 1} z^r / r$ formally to every diagonal eigenvalue, and summing. Nilpotent off-diagonal entries do not change a determinant or trace; their presence is retained in the operator.

All coefficients in (DW3.2) are nonnegative. Explicitly, if $c_0 = 1$ and c_n are its coefficients, differentiation gives

$$nc_n = \sum_{r=1}^n (\text{tr } A^r)^{2k} c_{n-r}. \quad (\text{DW3.3})$$

Induction proves $c_n \geq 0$.

Let r_0 be the largest determinantal weight of $\mathcal{F}_0$. Tensor compatibility from §1.3.13 bounds all determinantal weights of $\mathcal{F}_0^{\otimes 2k}$ by $2kr_0$. Equations (DW2.3–4) prove that the global rational function has no pole in

$$|T| < R_k, \quad R_k = q^{-kr_0-1}. \quad (\text{DW3.4})$$

The numerator is a polynomial; its zeros cannot create poles. A pole in this disc is precluded by the denominator bound itself.

Here is the exact use of coefficient positivity, without the imprecise assertion that arbitrary complex poles cannot cancel. For a fixed closed point x , factor the global formal series as $P_x(T)B_x(T)$, with P_x its local reciprocal determinant and B_x the product of all other local factors. Both have nonnegative coefficients and constant coefficient 1. There are only finitely many closed points of bounded degree, so each formal coefficient is a finite expression. Coefficient by coefficient,

$$0 \leq [T^n]P_x(T) \leq [T^n](P_x(T)B_x(T)). \quad (\text{DW3.5})$$

The global rational function is analytic on $|T| < R_k$. Its Taylor coefficients therefore give an absolutely convergent series at every positive $r < R_k$. The domination (DW3.5) proves the same for P_x . Hence the local reciprocal determinant has no pole in that disc.

If α is any eigenvalue of F_x on $\mathcal{F}_0$, then α^{2k} is an eigenvalue on its $2k$ -fold tensor power. The polynomial factor

$$1 - \iota(\alpha)^{2k} T^{d_x} \quad (\text{DW3.6})$$

has roots of absolute value $|\iota\alpha|^{-2k/d_x}$. They are poles of the reciprocal local determinant: it has constant numerator 1, so local cancellation is impossible. Comparing with (DW3.4) gives

$$|\iota\alpha|^{-2k/d_x} \geq q^{-kr_0-1}, \quad |\iota\alpha| \leq Q_x^{r_0/2+1/(2k)}. \quad (\text{DW3.7})$$

Equivalently,

$$w_{Q_x}(\alpha) \leq r_0 + \frac{1}{k}. \quad (\text{DW3.8})$$

The original α , point x , and weight r_0 are fixed as k ranges through the positive integers. Taking the infimum over k proves $w_{Q_x}(\alpha) \leq r_0$. The error is $1/k$ in weight and $1/(2k)$ in the exponent of the absolute value.

To obtain purity for every constituent, group the constituents by determinantal weight γ , with total ranks $n(\gamma)$. Their eigenvalues α_i^γ satisfy the determinant equality

$$\sum_{i=1}^{n(\gamma)} w_{Q_x}(\alpha_i^\gamma) = n(\gamma)\gamma. \quad (\text{DW3.9})$$

Fix a nonempty group of weight β and put

$$N_\beta = \sum_{\gamma > \beta} n(\gamma). \quad (\text{DW3.10})$$

The exterior power $\bigwedge^{N_\beta+1} \mathcal{F}_0$ is still ι -real. Its eigenvalues are products over subsets of the original eigenvalue multiset; complex conjugation permutes that multiset and those subsets. Its largest determinantal weight is

$$\beta + \sum_{\gamma > \beta} n(\gamma)\gamma. \quad (\text{DW3.11})$$

For every i , the eigenvalue

$$\alpha_i^\beta \prod_{\gamma > \beta} \prod_{j=1}^{n(\gamma)} \alpha_j^\gamma \quad (\text{DW3.12})$$

belongs to that exterior power. Applying the upper bound and subtracting the exact determinant equalities for $\gamma > \beta$ gives

$$w_{Q_x}(\alpha_i^\beta) \leq \beta. \quad (\text{DW3.13})$$

The sum of these $n(\beta)$ inequalities is equality by (DW3.9). Every individual difference $\beta - w_{Q_x}(\alpha_i^\beta)$ is nonnegative, and their sum is zero. Every difference is therefore zero. Thus every constituent is pointwise ι -pure of its determinantal weight. This is the complete mechanism of §§1.5.1–1.5.3.

115.4 4. Boundary weights: every step of §1.8.1

Let $j : U_0 \hookrightarrow X_0$ be an open immersion with finite complement S_0 , and suppose $\mathcal{F}_0$ is lisse and pointwise ι -pure of weight β on U_0 . Removing a point from U_0 , rather than from the boundary point being examined, makes X_0 affine without changing the local assertion.

The full trace formula is

$$\begin{aligned} \prod_{x \in |U_0|} \det(1 - F_x T^{d_x}, \mathcal{F}_0)^{-1} \prod_{x \in |S_0|} \det(1 - F_x T^{d_x}, j_* \mathcal{F}_0)^{-1} \\ = \frac{\det(1 - FT, H_c^1(X, j_* \mathcal{F}))}{\det(1 - FT, H_c^2(X, j_* \mathcal{F}))}. \end{aligned} \quad (\text{DW4.1})$$

The first product on the left is analytic and nonzero for

$$|T| < q^{-(\beta+2)/2}. \quad (\text{DW4.2})$$

Top cohomology depends only on the restriction to a dense smooth open, so its eigenvalues have weights $\beta + 2$. Thus the right side has no pole in (DW4.2). Dividing by the nonvanishing open product shows that the finite boundary product has no pole there. This boundary product is itself a reciprocal polynomial, so its individual determinant zeros cannot cancel one another.

For every eigenvalue α at a boundary point x , this proves

$$|\iota \alpha| \leq Q_x^{(\beta+2)/2}, \quad w_{Q_x}(\alpha) \leq \beta + 2. \quad (\text{DW4.3})$$

At a geometric boundary point, the stalk of $j_* \mathcal{F}_0$ is the inertia-invariant space V^I . There is an injective, Frobenius-equivariant map

$$(V^I)^{\otimes k} \longrightarrow (V^{\otimes k})^I. \quad (\text{DW4.4})$$

It is the restriction of the tensor power of the vector-space inclusion $V^I \hookrightarrow V$. Its image is invariant because each factor is invariant. It is injective because tensoring injections of vector spaces over a field preserves injectivity. This is the exact map behind

$$(j_* \mathcal{F}_0)^{\otimes k} \hookrightarrow j_*(\mathcal{F}_0^{\otimes k}). \quad (\text{DW4.5})$$

The tensor sheaf on U_0 has weight $k\beta$; the boundary eigenvalue α^k is carried by (DW4.4). Applying (DW4.3) to that tensor sheaf proves

$$|\iota \alpha|^k \leq Q_x^{(k\beta+2)/2}, \quad w_{Q_x}(\alpha) \leq \beta + \frac{2}{k}. \quad (\text{DW4.6})$$

Letting $k \rightarrow \infty$ gives $w_{Q_x}(\alpha) \leq \beta$.

The error here is $2/k$ in weight, or $1/k$ in the absolute-value exponent. It is not the same numerical expression as (DW3.8), because §1.5 uses the even tensor power $2k$ and its positivity argument.

Section 1.8.2 also compares zeros in (DW4.1). After the boundary bound is known, both local products are nonzero in (DW4.2); the right denominator does not vanish there. Hence the numerator does not vanish there. An eigenvalue α on $H_c^1(X, j_*\mathcal{F})$ consequently satisfies $w_q(\alpha) \leq \beta + 2$. This is the rough cohomological estimate, not yet the sharper weight $\beta + 1$ for cohomological degree one.

115.5 5. Local monodromy: §1.8.4 with every β retained

Fix a boundary point s , set $Q = N(s)$, and let $V = \mathcal{F}_{\bar{\eta}}$ for the corresponding henselian trait. Grothendieck's quasi-unipotence theorem allows a finite cover on which inertia is unipotent, with

$$\rho_I(\sigma) = \exp(t_\ell(\sigma)N), \quad N : V(1) \longrightarrow V. \quad (\text{DW5.1})$$

A finite ramification change multiplies the chosen logarithm N by a nonzero scalar, leaving its monodromy filtration unchanged. With compatible Frobenius lifts, a residue extension of degree f changes Q to Q^f and eigenvalues to their f -th powers; it leaves the numerical weight (DW2.1) unchanged. Changes of lift can multiply eigenvalues by roots of unity, which have complex absolute value one and therefore leave these weights unchanged. These exact changes justify the reduction.

Let F be geometric Frobenius. Equivariance of the map in (DW5.1), including the Q^{-1} action on the Tate line, is

$$FN = Q^{-1}NF, \quad FNF^{-1} = Q^{-1}N. \quad (\text{DW5.2})$$

The monodromy filtration M is centered at zero and satisfies

$$NM_i(1) \subset M_{i-2}, \quad N^j : \text{Gr}_j^M V(j) \xrightarrow{\sim} \text{Gr}_{-j}^M V. \quad (\text{DW5.3})$$

Write P_{-j} for the primitive kernel at index $-j$, $j \geq 0$. It is a subquotient of $\ker N = V^I$. The boundary estimate gives

$$|\iota\alpha| \leq Q^{\beta/2} \quad \text{for each eigenvalue on } P_{-j}(V). \quad (\text{DW5.4})$$

The primitive tensor formula contains a direct summand

$$P_{-j}(V) \otimes P_{-j}(V)(-j) \longrightarrow P_0(V \otimes V). \quad (\text{DW5.5})$$

This is the $j' = j'' = j$, total primitive index zero, summand in (1.6.14.4). Its twist is essential. A direct Jordan-string check is available: take a length- $j+1$ string $Nv_r = v_{r-1}$, $Nv_0 = 0$. The vector

$$\Omega = \sum_{r=0}^j (-1)^r v_r \otimes v_{j-r} \quad (\text{DW5.6})$$

has degree zero and is killed by $N \otimes 1 + 1 \otimes N$. On the degree-zero basis $b_r = v_r \otimes v_{j-r}$, the functional $b_r \mapsto (-1)^r$ annihilates the image of N from degree two, since that image is spanned by $b_{r-1} + b_r$. It evaluates Ω to $j+1 \neq 0$. Thus this primitive component is nonzero. The total j inverse steps up the two strings contribute the twist $(-j)$. The general primitive-multiplicity formula is Deligne's (1.6.14.4).

The tensor sheaf has weight 2β . If α occurs on $P_{-j}(V)$, then $\alpha^2 Q^j$ occurs in (DW5.5). Applying (DW5.4) to the tensor sheaf yields

$$|\iota\alpha|^2 Q^j \leq Q^\beta, \quad |\iota\alpha| \leq Q^{(\beta-j)/2}. \quad (\text{DW5.7})$$

The primitive dual formula is

$$P_{-j}(V^\vee) \simeq P_{-j}(V)^\vee(j). \quad (\text{DW5.8})$$

Its eigenvalue corresponding to α is $\alpha^{-1}Q^{-j}$. The dual sheaf has weight $-\beta$, so (DW5.7) applied to it gives

$$|\iota\alpha|^{-1}Q^{-j} \leq Q^{(-\beta-j)/2}, \quad |\iota\alpha| \geq Q^{(\beta-j)/2}. \quad (\text{DW5.9})$$

Together, the inequalities prove

$$|\iota\alpha| = Q^{(\beta-j)/2} \quad \text{on } P_{-j}(V). \quad (\text{DW5.10})$$

Here is the sign check needed to recover all the graded pieces. A primitive string contributes to $\text{Gr}_i^M V$ when $j \geq |i|$ and $j \equiv i \pmod{2}$. Put $r = (i+j)/2$. On this component,

$$N^r : \text{Gr}_i^M V(r) \longrightarrow P_{-j}(V) \quad (\text{DW5.11})$$

is the equivariant isomorphism. Therefore its exact untwisted-domain form is

$$\text{Gr}_i^M V \simeq \bigoplus_{\substack{j \geq |i| \\ j \equiv i \pmod{2}}} P_{-j}(V) \left(-\frac{i+j}{2} \right). \quad (\text{DW5.12})$$

The eigenvalue on the corresponding graded piece is $\alpha Q^{(i+j)/2}$. Its absolute value, with every factor retained, is

$$Q^{(\beta-j)/2} Q^{(i+j)/2} = Q^{(\beta+i)/2}. \quad (\text{DW5.13})$$

This proves the asserted weight $\beta+i$ of $\text{Gr}_i^M V$. The positive sign printed in the consulted copies of (1.6.14.3) would instead give $Q^{(\beta-i-2j)/2}$, contradicting (DW5.11) and the theorem. Formula (DW5.12) is independently derived, rather than silently copied.

115.6 6. What the remainder of §1.8 establishes

The propagation statements have precise content beyond the tensor estimate.

In (1.8.5), a lisse sheaf carries a finite filtration W by lisse subsheaves whose graded pieces are pointwise pure of weight a . On each $\text{Gr}_a^W V$, (DW5.13), with its center shifted from zero to a , gives the relative monodromy isomorphisms

$$N^j : \text{Gr}_{a+j}^M \text{Gr}_a^W V(j) \xrightarrow{\sim} \text{Gr}_{a-j}^M \text{Gr}_a^W V. \quad (\text{DW6.1})$$

The Frobenius weight filtration exists by (1.7.5); equation (DW5.2) lowers weights by two. Consequently that weight filtration satisfies both defining properties of the relative monodromy filtration. Uniqueness of the latter, (1.6.13), identifies the two. This uses a proved uniqueness theorem for a filtration with specified maps; it is not uniqueness of the scalar zeta function.

Statements (1.8.6–7) restrict to curves transverse to a smooth boundary divisor. The tame nearby local system $\mathcal{F}_0[D_0]$ retains its $\mathbb{Z}_\ell(1)$ -action. Formula (DW6.1) gives its relative monodromy filtration and the asserted weights. The construction of $\mathcal{F}_0[D_0]$ and transverse restriction are inputs from §1.7.8; that construction has not been rederived in this note.

In (1.8.8), the stalk of $j_*\mathcal{F}_0$ is obtained from $\ker N$ by taking invariants under the finite residual inertia group. All primitive indices in $\ker N$ are nonpositive. Therefore boundary weights are $\beta+i$ with $i \leq 0$. Taking a subspace of invariants cannot introduce new eigenvalues. The divisor version is the same calculation on the tame nearby local system.

Corollary (1.8.9) extends the upper bound and mixedness to j_* for arbitrary finite-type schemes. Its proof uses the following actual injective maps, each preserving the pointwise upper-weight conclusion:

1. For an extension $0 \rightarrow \mathcal{F}'_0 \rightarrow \mathcal{F}_0 \rightarrow \mathcal{F}''_0 \rightarrow 0$, left exactness gives $0 \rightarrow j_*\mathcal{F}'_0 \rightarrow j_*\mathcal{F}_0 \rightarrow j_*\mathcal{F}''_0$.

2. A supported sheaf embedded in $i_*i^*\mathcal{F}_0$ reduces the assertion to its support closure.
3. For a finite surjective map $\epsilon : X'_0 \rightarrow X_0$, the pullback-detection map is $j_*\mathcal{F}_0 \hookrightarrow \epsilon_*j'_*\epsilon^*\mathcal{F}_0$.
4. Restriction to a dense open gives $j_*\mathcal{F}_0 \hookrightarrow (jk)_*k^*\mathcal{F}_0$.
5. With X_0 normal, lisse $\mathcal{F}_0$, boundary immersion i , and a dense boundary open k , there is an injection $i^*j_*\mathcal{F}_0 \hookrightarrow k_*k^*i^*j_*\mathcal{F}_0$. On a geometric stalk, it follows from the factorization of the injection to a generic stalk through the displayed target, using connectedness of the appropriate strict henselian open.

Induction on the dimension of the support and these reductions bring the boundary calculation to the smooth tame-divisor case (1.8.8). This is Deligne's reduction; the underlying reduction theory is not reproved here.

Corollary (1.8.10) starts with a lisse sheaf pure of weight β on a dense open. Both it and its dual inject into their j_*j^* . The upper bound for the sheaf is β , and for its dual is $-\beta$. If α occurs on the sheaf, the two inequalities are

$$|\iota\alpha| \leq Q^{\beta/2}, \quad |\iota\alpha|^{-1} \leq Q^{-\beta/2}. \quad (\text{DW6.2})$$

They give equality, so purity extends across the original lisse locus.

For (1.8.11), normality ensures that restriction of an irreducible lisse sheaf to a dense open remains irreducible. An ι -mixed irreducible sheaf is pure on some such open; (1.8.10) extends its purity. A Jordan–Hölder filtration then gives successive lisse pure constituents.

For (1.8.12), on a connected normal scheme the finite collection of constituent weights is visible on every stalk, so purity at one stalk forces all constituents to have that weight. On a general connected scheme, pullback to the normalization shows that the locus pure of that fixed weight contains every irreducible component through each of its points. It is therefore open and closed, and connectedness propagates it globally. This is purity propagation after mixedness and the preceding theorems are known, not a rule allowing one ungraded support point to assert a weight.

Statement (1.8.13) permits the analogous purity and integral-weight formulations. Statements (1.8.14–15) compare the result with W. Schmid's asymptotic Hodge theory and formulate a problem about good variations of mixed Hodge structure. They are not an additional proof that arbitrary filtered local systems have the required relative monodromy filtration.

115.7 7. The exact primary-jet tensor action in the programme

Let $\rho = \beta + i\gamma$ be an actual nontrivial zero of the original $\zeta(s)$, of multiplicity m , and suppose for the calculation that $\beta \neq 1/2$. The full distinct orbit is

$$\mathcal{O}_\rho = \{\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}\}. \quad (\text{DW7.1})$$

The original functional equation, with every factor retained, is

$$\zeta(s) = 2^s \pi^{s-1} \sin(\pi s/2) \Gamma(1-s) \zeta(1-s). \quad (\text{DW7.1a})$$

The multiplier is holomorphic and nonzero in $0 < \Re s < 1$. Complex conjugation commutes with ζ , initially by its Dirichlet series and then by unique meromorphic continuation. Therefore these two exact maps give the same multiplicity m at all four points. The original zeta has no real zero in $0 < s < 1$: the convergent alternating Dirichlet series there is strictly positive, while $1 - 2^{1-s} < 0$, and their quotient is $\zeta(s)$. Positivity follows by pairing consecutive terms $(2a-1)^{-s} - (2a)^{-s} > 0$ and retaining the first positive pair; convergence is the alternating-series criterion. Thus $\gamma \neq 0$, and all four points in (DW7.1) are distinct. Formula (DW7.1a) is used only on its stated strip for multiplicity preservation; it does not discard its exceptional points globally.

The programme's primary space and operators are

$$V_\rho = \bigoplus_{\omega \in \mathcal{O}_\rho} \mathbb{C}[\epsilon_\omega]/(\epsilon_\omega^m), \quad W_p|_{V_\omega} = p^\omega \sum_{j=0}^{m-1} \frac{(\log p)^j}{j!} N_\omega^j, \quad N_\omega v = \epsilon_\omega v. \quad (\text{DW7.2})$$

The letter ϵ_ω is a spectral jet coordinate, not the source presence τ . The vector-space zero in these coefficient modules is not an identification of source Z_1 with Z_0 .

The full d -fold tensor space of one block is

$$V_\omega^{\otimes d} \simeq \mathbb{C}[\epsilon_1, \dots, \epsilon_d]/(\epsilon_1^m, \dots, \epsilon_d^m), \quad (\text{DW7.3})$$

and the tensor operator is exactly

$$W_p^{\otimes d} = p^{d\omega} \exp \left((\log p) \sum_{a=1}^d N_a \right). \quad (\text{DW7.4})$$

All terms of the exponential of degree $> d(m-1)$ vanish. When $m > 1$, the power of degree $d(m-1)$ is nonzero: its coefficient of $\epsilon_1^{m-1} \dots \epsilon_d^{m-1}$ is

$$\frac{(d(m-1))!}{((m-1)!)^d} \neq 0. \quad (\text{DW7.5})$$

Thus the nilpotency index of $\sum N_a$ is exactly $d(m-1) + 1$. None of this tensor structure is discarded when taking determinants below.

For $r \geq 1$,

$$\text{tr}(W_p^r|V_\rho) = m \sum_{\omega \in \mathcal{O}_\rho} p^{r\omega} = 2m(p^{r\beta} + p^{r(1-\beta)}) \cos(r\gamma \log p). \quad (\text{DW7.6})$$

This is real. Consequently the reciprocal determinant for the $2k$ -fold tensor representation has nonnegative coefficients, by precisely (DW3.1–3):

$$\det(1 - UW_p^{\otimes 2k})^{-1} = \exp \left(\sum_{r \geq 1} \frac{[2m(p^{r\beta} + p^{r(1-\beta)}) \cos(r\gamma \log p)]^{2k}}{r} U^r \right). \quad (\text{DW7.7})$$

This proves an exact transfer of Deligne's local coefficient-positivity step. It does not yet import his independent geometric bound on the global poles.

115.8 8. The complete global tensor Euler product is computable

Put $B = \max(\beta, 1 - \beta)$, and define the actual tensor Euler product

$$\mathcal{L}_k(s) = \prod_p \det(1 - p^{-s} W_p^{\otimes 2k}|V_\rho^{\otimes 2k})^{-1}. \quad (\text{DW8.1})$$

It converges absolutely for $\Re s > 1 + 2kB$. Each of the finitely many eigenvalues in a local tensor determinant has absolute value at most p^{2kB} . For $\delta = \Re s - 2kB > 1$, the full logarithmic sum is bounded by

$$(4m)^{2k} \sum_p \sum_{r \geq 1} \frac{p^{-r\delta}}{r} \leq \frac{(4m)^{2k}}{1 - 2^{-\delta}} \sum_{n \geq 2} n^{-\delta} < \infty. \quad (\text{DW8.1a})$$

The tensor rank $(4m)^{2k}$ is finite for every k .

For an ordered tuple $\omega = (\omega_1, \dots, \omega_{2k}) \in \mathcal{O}_\rho^{2k}$, define

$$\sigma(\omega) = \sum_{a=1}^{2k} \omega_a. \quad (\text{DW8.2})$$

The associated tensor block has dimension m^{2k} . In the monomial basis of (DW7.3), ordered by total degree, its exponential nilpotent factor is triangular with all diagonal entries 1. Every diagonal eigenvalue is therefore $p^{\sigma(\omega)}$, with all m^{2k} occurrences retained. Thus

$$\det(1 - p^{-s} W_p^{\otimes 2k}) = \prod_{\omega \in \mathcal{O}_\rho^{2k}} (1 - p^{\sigma(\omega)-s})^{m^{2k}}. \quad (\text{DW8.3})$$

The finite product over tuples can be interchanged with the absolutely convergent prime product, giving the exact original-zeta identity

$$\boxed{\mathcal{L}_k(s) = \prod_{\omega \in \mathcal{O}_\rho^{2k}} \zeta(s - \sigma(\omega))^{m^{2k}}.} \quad (\text{DW8.4})$$

Both sides therefore have the meromorphic continuation specified by the right side. This uses the original ζ , with its original pole at 1; no completion, Gamma factor, or endpoint factor has been inserted or removed.

The real point

$$s_k = 1 + 2kB \quad (\text{DW8.5})$$

is a genuine pole of order

$$h_k = m^{2k} \binom{2k}{k}. \quad (\text{DW8.6})$$

Here is a proof including exclusion of every possible cancellation. Let $\omega_+ = B + i\gamma'$, $\omega_- = B - i\gamma'$ be the two points in the orbit with maximal real part, where $\gamma' \neq 0$. A tuple has real part $2kB$ exactly when all its entries are ω_+ or ω_- . Its imaginary part is zero exactly when there are k of each. There are $\binom{2k}{k}$ such ordered tuples. They contribute the factor

$$\zeta(s - 2kB)^{m^{2k} \binom{2k}{k}}. \quad (\text{DW8.7})$$

Every other tuple has either $\Re(s_k - \sigma) > 1$, or $\Re(s_k - \sigma) = 1$ and $\Im(s_k - \sigma) \neq 0$. The original ζ is holomorphic and nonzero at all these points. For completeness, the classical Hadamard–de la Vallée-Poussin nonvanishing argument is reproduced here, rather than claimed as new. It uses the Euler product and the known single pole at 1:

$$3 + 4 \cos \theta + \cos 2\theta = 2(1 + \cos \theta)^2 \geq 0 \quad (\text{DW8.8})$$

implies, for $u > 1$ and real v ,

$$\zeta(u)^3 |\zeta(u + iv)|^4 |\zeta(u + 2iv)| \geq 1. \quad (\text{DW8.9})$$

Indeed taking logarithms expands the left logarithm as

$$\sum_p \sum_{r \geq 1} \frac{p^{-ru}}{r} [3 + 4 \cos(rv \log p) + \cos(2rv \log p)] \geq 0. \quad (\text{DW8.10})$$

If $\zeta(1 + iv)$ had a zero of order $a \geq 1$ with $v \neq 0$, its factor in (DW8.9) would be $O((u - 1)^{4a})$, while $\zeta(u)^3 = O((u - 1)^{-3})$ and the last factor stays bounded as $u \downarrow 1$. The left side would tend to zero, contradicting (DW8.9). In $\Re s > 1$, the Euler product itself proves nonvanishing. Thus every factor other than (DW8.7) is finite and nonzero at s_k , proving the pole and its exact order.

This proves more than an abstract list of potential properties: the entire tensor Euler function, its domain, its continuation, a forced global pole, and that pole's multiplicity are constructed from the actual primary orbit. It does not create or assert the existence of an off-critical zero. It calculates exactly the consequence for an actual zero if its real part differs from $1/2$.

There is also an exact global coefficient consequence. Write

$$\det(1 - UW_p^{\otimes 2k})^{-1} = \sum_{a \geq 0} c_{p,k}(a) U^a. \quad (\text{DW8.7a})$$

Equation (DW7.7) proves $c_{p,k}(a) \geq 0$ and $c_{p,k}(0) = 1$. For $n = \prod_p p^{a_p}$, define the finite product

$$A_k(n) = \prod_p c_{p,k}(a_p). \quad (\text{DW8.7b})$$

The absolutely convergent Euler expansion gives

$$\mathcal{L}_k(s) = \sum_{n \geq 1} A_k(n) n^{-s}, \quad A_k(n) \geq 0, \quad \Re s > 1 + 2kB. \quad (\text{DW8.7c})$$

Its abscissa of convergence is exactly $1 + 2kB$. If it converged at a real $\sigma_0 < 1 + 2kB$, nonnegativity would make the series absolutely convergent there and locally uniformly convergent on $\Re s > \sigma_0$, by domination with its value at a smaller intermediate real parameter. It would define a holomorphic function near s_k , equal to (DW8.4) on a nonempty right half-plane and hence throughout the connected domain away from poles. That contradicts the pole already proved at s_k . Convergence to the right was proved in (DW8.1a).

The pole is invariant under every prime-action-preserving isomorphism. Indeed if an invertible $T : V_\rho \rightarrow V'$ satisfies $TW_p = W'_p T$ for every prime, then $T^{\otimes 2k}$ conjugates their tensor operators. Every determinant in (DW8.1), their Euler product, and the continuation (DW8.4) agree exactly. Neither s_k nor h_k changes.

A quotient has a different exact formula. For a subspace $K \subset V_\rho$ invariant under all W_p , and $Q = V_\rho/K$, a basis adapted to K gives

$$\det(1 - UW_p|V_\rho) = \det(1 - UW_p|K) \det(1 - UW_p|Q). \quad (\text{DW8.7d})$$

The tensor filtration on $V_\rho^{\otimes 2k}$ has associated graded the direct sum, over all words of length $2k$ in the letters K, Q , of their ordered tensor products. Triangularity gives the corresponding product formula for all the tensor determinants. Thus a quotient does not make removed spectral contributions vanish by isomorphism: the contributions persist in the kernel and the mixed tensor terms. Those terms must be retained to recover the original global object.

For comparison with Deligne, a weight-one input has expected tensor weight $2k$; the geometric top-cohomology error $+2$ would place the furthest allowed pole at the exponent $k + 1$. The exact displacement in the constructed tensor object is

$$s_k - (k + 1) = k(2B - 1). \quad (\text{DW8.11})$$

For the off-critical orbit under examination it is positive and grows linearly with k . The computed positivity does not erase it. The determinant/cohomology control that supplies $k + 1$ in Deligne's argument has not been proved for these operators merely by reconstructing the original zeta function.

One must also not assign the quartet's average determinant weight to every irreducible constituent. The joint representation of all W_p has one-dimensional semisimple characters $p \mapsto p^\omega$, with weights $2\Re\omega$. Their maximum is $2B$. Applying only a constituent-weight statement with the weights already visible in this representation would return $2B$, not 1. The full determinant and its average weight are exactly

$$\det(W_p|V_\rho) = p^{m \sum_{\omega \in \mathcal{O}_\rho} \omega} = p^{2m}, \quad \frac{2 \log |\det(W_p|V_\rho)|}{(4m) \log p} = 1. \quad (\text{DW8.12})$$

That average does not identify the constituent weights.

115.9 9. The exact monodromy-comparison defect

Write $J_\rho = \mathbb{C}[\epsilon_\rho]/(\epsilon_\rho^m)$ for the single spectral primary block of length m , retaining V_ρ for the four-block direct sum. On J_ρ , (DW7.2) gives

$$W_p N_\rho W_p^{-1} = N_\rho. \quad (\text{DW9.1})$$

Deligne's geometric local-monodromy relation, at norm p , is instead (DW5.2). The exact defect is

$$C_p = W_p N_\rho - p^{-1} N_\rho W_p = (1 - p^{-1}) W_p N_\rho. \quad (\text{DW9.2})$$

Because W_p is invertible, its rank is $m - 1$ when $m > 1$. Thus this defect is nonzero for every prime p on a nontrivial spectral jet block.

In particular, an invertible linear map intertwining both the same W_p and the same N_ρ with a norm- p Weil-monodromy representation would carry (DW9.1) and (DW5.2) to the same operators. Subtraction gives $(1 - p^{-1})N = 0$, hence $N = 0$, contrary to $m > 1$. This proves failure of that specific two-operator identification; it does not say the two frameworks have no relation.

There are exact surviving comparisons:

- The quotient $J_\rho \rightarrow J_\rho/N_\rho J_\rho$ is equivariant for every W_p , and its target has scalar p^ρ and nilpotent zero. This quotient has dimension one and therefore does **not** preserve the original multiplicity m .
- On the full vector space with basis $1, \epsilon, \dots, \epsilon^{m-1}$, define $S_p(\epsilon^j) = p^{-j} \epsilon^j$, and $F_p^{\text{geom}} = p^\rho S_p$. Then $F_p^{\text{geom}} N_\rho (F_p^{\text{geom}})^{-1} = p^{-1} N_\rho$, and $F_a^{\text{geom}} F_b^{\text{geom}} = F_{ab}^{\text{geom}}$ for positive a, b . This constructs a complex representation satisfying the Weil-monodromy operator relation, with the same nilpotent and the same scalar action on $J_\rho/N_\rho J_\rho$. It is not asserted to be an arithmetic lisse sheaf.

The latter construction has full eigenvalues $p^{\rho-j}$, $0 \leq j < m$, whereas the original W_p has eigenvalue p^ρ with multiplicity m . Thus it is not a hidden isomorphism replacing W_p ; its characteristic polynomial and weight data have changed in an explicitly displayed way. Its role is to identify exactly what extra grading a comparison would have to supply and exactly what original spectral data that supply alters.

Neither (DW9.2) nor this construction uses an addition on τ , a parity assignment to τ , or an identification of source support with the coefficient zero. The obstruction occurs in the actual coefficient operators and survives attachment of unchanged support labels.

115.10 10. What has and has not transferred

The transferred calculations are exact tensor eigenvalue multiplication, preservation of all spectral jets, the even-tensor coefficient positivity, the complete tensor Euler product (DW8.4), its forced pole and exact multiplicity, and the monodromy-comparison defect and its two explicit receiving maps.

The source of Deligne's contradiction is the combination of these tensor operations with an independently proved pole bound whose additive error is independent of tensor degree. It comes from geometric top cohomology, its Tate factor, and determinantal-weight control. It does not come from renaming the zero object or from uniqueness of the arithmetic reconstruction.

For the actual programme orbit, the directly constructed global object gives the explicit pole (DW8.5). Thus the current computation specifies the precise object to which any stronger geometric comparison must apply. The scalar zeta identification remains established; this note does not reopen it, assert RH, assert a disproof, or claim that the required geometric bound has already been transferred.

116 Exact equivariant quotients and tensor weight error in the original zeta receiver

Independent mathematical derivation, 24 September 2026. This note uses the actual Mellin quotient and prime actions defined in IH2–IH3 and PL6. It proves the quotient maps and their effects rather than treating “quotientable” as a synonym for “pure.” It does not assume a Deligne weight estimate for this receiver.

Source notation is retained: Z_0 is absence; Z_1 is primitive presence τ , without Z_2 parity. Addition on source τ is not defined here. The vector-space zero and scalar identities below belong to receiving algebras. Neither is identified with source absence or with τ . Every support coefficient is retained by an explicitly specified identity map in QS8.

116.1 QS1. The actual local and global objects

Let $\mathcal{A}$, I , and $Q = \mathcal{A}/I$ be precisely IH2.1:

$$p_{N,j}(k) = \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j k(u)| < \infty, \quad I = \overline{\mathcal{E}(S_0^{\text{even}})}^{\mathcal{A}},$$

$$\mathcal{E}f(u) = u^{1/2} \sum_{n \geq 1} f(nu), \quad F_k(s) = \int_0^\infty k(u) u^{s-1/2} \frac{du}{u}. \quad (\text{QS1.1})$$

The conditions defining S_0^{even} are even Schwartz regularity, $f(0) = 0$, and $\int_{\mathbb{R}} f = 0$. Every symbol below is an observation in this receiver of the original, uncompleted zeta function.

For an actual nontrivial zero ρ with multiplicity m , retain

$$A_{\rho,m} = \mathbb{C}[t_\rho]/(t_\rho^m), \quad j_{\rho,m}([k]) = \sum_{j=0}^{m-1} \frac{F_k^{(j)}(\rho)}{j!} t_\rho^j. \quad (\text{QS1.2})$$

For $a > 0$, the clock operator and its full local multiplier are

$$W_a k(u) = a^{1/2} k(u/a), \quad w_{a,\rho,m} = a^\rho \sum_{j=0}^{m-1} \frac{(\log a)^j}{j!} t_\rho^j. \quad (\text{QS1.3})$$

Thus $j_{\rho,m} W_a = M_{w_{a,\rho,m}} j_{\rho,m}$. The factor $a^{1/2}$ in the original operator and every nilpotent coefficient are retained. IH2–IH3 prove that these maps descend through I , that $j_{\rho,m}$ is onto, and that their product is onto for any finite set of distinct zeros.

No deformation of the arithmetic counting measure is presumed merely from the existence of a spectral character.

116.2 QS2. Complete classification on one primary block

Fix $p > 1$, put $\lambda = p^\rho$, and write $T = M_{t_\rho}$. Then

$$M_w = \lambda \exp((\log p)T), \quad T^m = 0,$$

$$X = \lambda^{-1} M_w - I, \quad T = \frac{1}{\log p} \sum_{j=1}^{m-1} \frac{(-1)^{j+1}}{j} X^j. \quad (\text{QS2.1})$$

These are finite polynomial identities: substitute into $\log(\exp z) = z$ modulo z^m . Conversely $M_w = \lambda \sum_{j=0}^{m-1} (\log p)^j T^j / j!$.

A complex-linear subspace $K \subset A_{\rho,m}$ is M_w -invariant if and only if it is T -invariant, by these two polynomial identities. Such a subspace is an ideal, because it is invariant under every polynomial in t_ρ .

Every ideal is exactly (t_ρ^r) for a unique $r \in \{0, \dots, m\}$. For a nonzero ideal, take its least nonzero degree r , and select $t_\rho^r u(t_\rho)$ with $u(0) \neq 0$. The inverse of u is a finite geometric polynomial in its nilpotent part, so the ideal contains t_ρ^r . Minimality of r says that every other element is divisible by t_ρ^r . The zero ideal corresponds to $r = m$.

Consequently every equivariant quotient is, up to equivariant linear isomorphism, the exact truncation

$$q_r : A_{\rho, m} \longrightarrow A_{\rho, r} = \mathbb{C}[t_\rho]/(t_\rho^r), \quad 0 \leq r \leq m. \quad (\text{QS2.2})$$

Here $r = 0$ denotes the zero vector space; for every $r \geq 1$ the surviving operator is multiplication by

$$p^\rho \sum_{j=0}^{r-1} \frac{(\log p)^j}{j!} t_\rho^j, \quad \det(XI - M_{w, r}) = (X - p^\rho)^r. \quad (\text{QS2.3})$$

A nonzero quotient cannot change p^ρ . It can shorten the nilpotent block or kill the entire block. These are all possibilities in this category.

The intertwiner space between blocks at the same ρ , of lengths m, r , is explicit. Formula (QS2.1) shows that an intertwiner commutes with t_ρ . It is determined by the image v of the constant polynomial and must satisfy $t_\rho^m v = 0$. Thus

$$\text{Hom}_{W_p}(A_{\rho, m}, A_{\rho, r}) \cong t_\rho^{\max(r-m, 0)} A_{\rho, r}. \quad (\text{QS2.4})$$

The map sends $f(t_\rho)$ to $f(t_\rho)v$; it is surjective precisely when $m \geq r$ and $v(0) \neq 0$.

116.3 QS3. Two counted primes separate all distinct zero characters

For complex ρ, ν ,

$$2^\rho = 2^\nu, \quad 3^\rho = 3^\nu \implies \rho = \nu. \quad (\text{QS3.1})$$

Indeed $(\rho - \nu) \log 2 = 2\pi i a$ and $(\rho - \nu) \log 3 = 2\pi i b$ for integers a, b . A nonzero difference forces $a, b \neq 0$ and $\log 2 / \log 3 = a/b$, contradicting unique factorization after exponentiation.

For a finite set S of distinct actual zeros, put

$$B_S = \prod_{\rho \in S} A_{\rho, m_\rho}.$$

There is $c \in \mathbb{C}$ for which $\lambda_\rho = 2^\rho + c3^\rho$ are pairwise distinct. Each equality forbids at most one c , unless its coefficients of c agree; in that case (QS3.1) makes the constant coefficients different, so no c is forbidden.

Set $M = M_{w_2} + cM_{w_3}$. On the ρ -block, $(M - \lambda_\rho I)^{m_\rho} = 0$. The pairwise coprime polynomials $(X - \lambda_\rho)^{m_\rho}$ have Chinese-remainder projectors $P_\rho(X)$:

$$P_\rho \equiv 1 \pmod{(X - \lambda_\rho)^{m_\rho}}, \quad P_\rho \equiv 0 \pmod{(X - \lambda_\nu)^{m_\nu}} \quad (\nu \neq \rho).$$

Bezout's identity constructs them. Evaluating at M gives the exact projection onto each block.

Any subspace $K \subset B_S$ invariant under W_2, W_3 is invariant under these projections. QS2 therefore gives

$$K = \prod_{\rho \in S} (t_\rho^{r_\rho}), \quad 0 \leq r_\rho \leq m_\rho. \quad (\text{QS3.2})$$

Conversely every displayed K is invariant under every W_a . This is the complete finite spectral quotient classification for all prime actions.

No nonzero simultaneous W_2, W_3 intertwiner connects blocks at distinct ρ, ν . At least one prime has distinct eigenvalues by (QS3.1); Bezout's identity for its two coprime minimal-polynomial factors forces that intertwiner to vanish. Thus equivariant isomorphism cannot transport an off-critical block to a critical block.

This is for the original fixed actions. PL2 treats a simultaneous change of clock separately: $\zeta(s)$ becomes $\zeta(cs)$, and both its zero and reflection line rescale. That comparison does not make an off-critical position critical relative to the rescaled line.

116.4 QS4. Reflection-compatible quotients retain the off-critical value form

Take the full actual off-critical orbit

$$\mathcal{O}_\rho = (\rho, 1 - \bar{\rho}, \bar{\rho}, 1 - \rho), \quad \rho = \beta + i\gamma, \quad 0 < \beta < 1, \quad \beta \neq \frac{1}{2}, \quad \gamma \neq 0.$$

All four multiplicities are m . Reflection is

$$(x^\#)_\omega(t) = \overline{x_{1-\bar{\omega}}}(-t). \quad (\text{QS4.1})$$

A quotient in QS3 carries this involution exactly when

$$r_\omega = r_{1-\bar{\omega}}. \quad (\text{QS4.2})$$

The original conjugation symmetry further imposes $r_\omega = r_{\bar{\omega}}$. All four lengths then agree. Length zero kills the whole orbit; any positive length retains all four characters.

Retain the original-multiplicity form

$$H_m(x, y) = m \sum_{\omega \in \mathcal{O}_\rho} \overline{x_{1-\bar{\omega}}(0)} y_\omega(0). \quad (\text{QS4.3})$$

Its radical is the product of the four ideals (t_ω) , because its value matrix is the invertible matrix $m \operatorname{diag}(J_2, J_2)$, with

$$J_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Every quotient with all $r_\omega \geq 1$ has kernel in that radical, so the form descends. Its value signature remains $(2, 2)$; values $(1, -1, 0, 0)$ give $-2m$. Quotienting higher jets cannot make this form positive.

The descended form still has coefficient m , the original multiplicity. Intrinsic trace on a shortened block would have coefficient r_ω . Replacing m by r_ω would alter the original trace datum.

No prescribed finite value vector is asserted to vanish at all other zeta zeros. IH3 gives surjectivity onto this finite orbit; this signature statement is about its exact finite receiving form. The global form retains all other zeros.

116.5 QS5. The greatest globally invisible quotient is explicit

For the set Z of distinct actual nontrivial zeros define

$$J_{\text{val}} = \bigcap_{\rho \in Z} \ker(F_\bullet(\rho)), \quad J_{\text{jet}} = \bigcap_{\rho \in Z} \ker j_{\rho, m_\rho}. \quad (\text{QS5.1})$$

These are closed invariant convolution ideals. Closedness follows from continuity of Mellin derivatives. The ideal property follows from the character and algebra-map product laws. Full multiplier formulas prove invariance under W_a ; permutation of zeros and jets proves invariance under the original symmetries.

For $\pi_K : Q \rightarrow Q/K$, all original value observations factor through π_K exactly when

$$K \subset J_{\text{val}}, \quad (\text{QS5.2})$$

and all full original jets factor exactly when

$$K \subset J_{\text{jet}}. \quad (\text{QS5.3})$$

Proof: a linear map factors through a quotient exactly when it vanishes on its kernel. Apply this to each specified observation.

Thus Q/J_{val} and Q/J_{jet} are the greatest reductions retaining respectively every value and every jet. Every quotient satisfying (QS5.3) has the unique induced map

$$Q/K \longrightarrow Q/J_{\text{jet}}, \quad q + K \longmapsto q + J_{\text{jet}}. \quad (\text{QS5.4})$$

Each original j_{ρ, m_ρ} is still onto from this terminal observation quotient, since it was onto from Q . Every actual character and original primary length survives.

For value preservation alone, the nonzero maps $F_\bullet(\rho)$ remain onto $\mathbb{C}$. On an equivariant quotient, $j_{\rho, m_\rho}(K)$ is W_p -invariant, hence is an ideal by QS2. Value preservation forces that ideal into (t_ρ) . The induced local quotient therefore has positive length: jets may shorten, but the character cannot disappear.

No equality $J_{\text{jet}} = 0$, or unsupplied spectral-synthesis theorem, is asserted. Removing any extra vectors in this intersection still changes no zero character. IH6 already provides a complete joint-grid receiver with kernel J_{val} , giving a concrete injective realization of Q/J_{val} .

Preserving the original meromorphic zeta function has an additional elementary consequence: equality in its original variable fixes all zero germs and their orders. It cannot simultaneously assert a changed zero set of that same function. A new conclusion about those zeros' positions requires an estimate rather than another invocation of uniqueness.

116.6 QS6. Actual tensor powers with every nilpotent retained

The k -fold tensor power of one block is exactly

$$A_{\rho, m}^{\otimes k} = \mathbb{C}[t_1, \dots, t_k] / (t_1^m, \dots, t_k^m). \quad (\text{QS6.1})$$

With $N = t_1 + \dots + t_k$, its diagonal prime multiplier is

$$w_{p, \rho, m}^{\otimes k} = p^{k\rho} \prod_{i=1}^k \left(\sum_{j=0}^{m-1} \frac{(\log p)^j t_i^j}{j!} \right) = p^{k\rho} \sum_{j=0}^{k(m-1)} \frac{(\log p)^j N^j}{j!}. \quad (\text{QS6.2})$$

The multinomial identity proves this equality in the stated quotient. Its dimension is m^k . The nilpotence order of N is exactly $k(m-1) + 1$: larger total degree forces an exponent at least m , while

$$N^{k(m-1)} = \frac{(k(m-1))!}{((m-1)!)^k} \prod_{i=1}^k t_i^{m-1} \neq 0. \quad (\text{QS6.3})$$

For $m = 1$, this says $N^0 = 1$ and $N = 0$.

For every repetition $r \geq 1$,

$$\text{Tr}((W_p^{\otimes k})^r) = m^k p^{kr\rho}, \quad \det(XI - W_p^{\otimes k}) = (X - p^{k\rho})^{m^k}. \quad (\text{QS6.4})$$

Every positive-degree term is strictly triangular in a degree-ordered monomial basis. This proves both identities without deleting nilpotents from the actual operator.

For the full four-point algebra B_ρ , distribute the tensor product:

$$B_\rho^{\otimes k} = \prod_{(\omega_1, \dots, \omega_k) \in \mathcal{O}_\rho^k} \mathbb{C}[t_1, \dots, t_k] / (t_1^m, \dots, t_k^m). \quad (\text{QS6.5})$$

On its indicated factor the multiplier is exactly

$$p^{\omega_1 + \dots + \omega_k} \left(\sum_{j=0}^{k(m-1)} \frac{(\log p)^j N^j}{j!} \right). \quad (\text{QS6.6})$$

All mixed choices of reflected and conjugate characters remain. Consequently

$$\mathrm{Tr}((W_p^{\otimes k})^r) = m^k \left(\sum_{\omega \in \mathcal{O}_p} p^{r\omega} \right)^k, \quad (\text{QS6.7})$$

$$\det(XI - W_p^{\otimes k}) = \prod_{(\omega_1, \dots, \omega_k) \in \mathcal{O}_p^k} (X - p^{\omega_1 + \dots + \omega_k})^{m^k}. \quad (\text{QS6.8})$$

Coincident eigenvalues add their positive integer multiplicities; they do not cancel.

116.7 QS7. The tensor exponent error is computed, not assumed bounded

For a finite matrix A with nonzero distinct eigenvalues λ_j and positive integer algebraic multiplicities d_j ,

$$\sum_{r \geq 0} \mathrm{Tr}(A^r) z^r = \sum_j \frac{d_j}{1 - \lambda_j z} \quad (\text{QS7.1})$$

near zero. Distinct denominators have distinct poles and nonzero residues. The radius of this series is $1/\max_j |\lambda_j|$, so Cauchy's coefficient-radius formula gives

$$\limsup_{r \rightarrow \infty} |\mathrm{Tr}(A^r)|^{1/r} = \max_j |\lambda_j|. \quad (\text{QS7.2})$$

This handles cancellation among trace terms and all multiplicities; it requires no lower bound for each individual repetition.

Apply this to the actual operator on $B_\rho^{\otimes k}$. Set

$$b = \max(\beta, 1 - \beta), \quad \delta = b - \frac{1}{2} = |\beta - \frac{1}{2}|.$$

All tuples have real exponent at most kb , and repeating a character with real part b attains it. Therefore

$$L_{p,k} = \limsup_{r \rightarrow \infty} \frac{\log |\mathrm{Tr}((W_p^{\otimes k})^r)|}{r \log p} = kb,$$

$$E_{p,k} := L_{p,k} - \frac{k}{2} = k\delta. \quad (\text{QS7.3})$$

The logarithm at a zero trace term is $-\infty$, which does not affect (QS7.2). Every prime action and every repetition is retained in this definition.

The actual estimate supplied by the critical strip is

$$|\mathrm{Tr}((W_p^{\otimes k})^r)| \leq (4m)^k p^{rk}, \quad r, k \geq 1. \quad (\text{QS7.4})$$

Every one of the four summands in (QS6.7) has modulus at most p^r . This proves (QS7.4). Its error relative to $k/2$ is $k/2$, growing with k , so it supplies no amplification to zero.

Among the calculated nonnegative sequences $E_{p,k} = k\delta$, the ones with bounded supremum are exactly the zero sequence. More generally the sequences with $E_{p,k}/k \rightarrow 0$ are exactly the zero sequence. These follow directly from $\delta \geq 0$. They classify the error sequences; they do not assert that the actual receiver has either bound.

For comparison with the numerical form of a Deligne amplification, an estimate shaped as

$$|\mathrm{Tr}((W_p^{\otimes k})^r)| \leq C_k r^{d_k} p^{r(k/2+c)} \quad (\text{QS7.5})$$

has exponent at most $k/2 + c$, because for fixed k its r -th root loses the factors $C_k^{1/r} r^{d_k/r}$. Its exponent error is the fixed c , independent of k . Comparing the exact (QS7.3) gives $k\delta \leq c$, and the preceding sequence classification removes $\delta > 0$.

Equation (QS7.5) is a comparison of estimate shapes. It is not proved for the programme in this note and is not supplied as an assumption completing RH. The proved results are the actual exponent (QS7.3), actual bound (QS7.4), and full quotient classification. Every positive-length reflection-compatible quotient retains the same b, δ ; shortening multiplicities changes finite trace coefficients but not the exponential rate.

The tensor product here is the complex-linear receiving tensor product proved in QS6. No tensor comparison from a source cohomology is assumed by changing notation, and no operation on source τ appears in its exponent computation.

116.8 QS8. Every support label is retained with the exact kernel

Let C_{coeff} be the programme's formal receiving coefficient algebra, with separate recorded symbols $U = [\tau]$, $f = [1]$, and let W_L be the vector space on all specified support labels. Put $V = C_{\text{coeff}} \otimes_{\mathbb{C}} W_L$. Their vector operations define no addition on source τ .

Use exactly

$$q \otimes \text{id}_V, \quad j_{\rho, m} \otimes \text{id}_V. \quad (\text{QS8.1})$$

Choose a basis of V . Every algebraic tensor is a finite sum $\sum_{\alpha} x_{\alpha} \otimes v_{\alpha}$ in independent basis elements. Its image is zero precisely when each $q(x_{\alpha})$ is zero. Hence

$$\ker(q \otimes \text{id}_V) = (\ker q) \otimes V. \quad (\text{QS8.2})$$

Every named support coordinate remains the same coordinate. A surviving character survives in every nonzero coefficient coordinate; no supported label is identified with source absence by these maps.

This is the exact identity-on-labels extension. It does not assert that every possible quotient of a larger mixed coefficient category preserves labels. Such a quotient needs its own definition.

116.9 QS9. Consequence for the proposed global quotient route

The canonical original-zeta reconstruction fixes the arithmetic measure and original meromorphic function. The quotient classification now gives:

1. Killing a block removes its zero observation; it is detectable.
2. Shortening a block retains the character and loses specified higher derivatives; it does not retain complete original jets.
3. Removing the global joint observation kernel retains every observation, including any off-critical one.
4. Equivariant isomorphism for fixed prime actions preserves every character; 2, 3 already separate them.
5. Tensor powers of a surviving off-critical orbit have exact exponent error $k|\beta - \frac{1}{2}|$, with all mixed sectors and nilpotents present.

A quotient invisible to all retained observations therefore cannot make an observed off-critical character critical. It can remove unobserved receiver data. This proves the distinction through the actual quotient maps and kernels.

No proof or disproof of RH is asserted. The result identifies the invariant that a geometric weight estimate must control and calculates why invisible quotienting or meromorphic uniqueness alone cannot change it.

116.10 Reading and source identity

Read completely for this derivation:

- IH1–IH8, original local programme source: `INTEGRAL_HISTORY_AND_PURITY.md`
- PL1–PL8, original local programme source: `PRIME_CLOCK_LOOPS_AND_Z1.md`

QS2–QS8 supply complete finite-algebra, quotient, and tensor proofs. No historical novelty is asserted for these elementary mechanisms. This independent note has not read Deligne’s full source; the coordinating task is undertaking the human-source reconstruction and must supply precise theorem locators. No alleged Deligne theorem is used as an input here.

117 Deligne's twisted monodromy and the original zeta primary jet

24 September 2026. Independent mathematical derivation. The supplied `READ_FR_local_monodromy.txt`, printed pages 165–175 of Pierre Deligne, *La conjecture de Weil. II*, was read in full. The comparison uses the equivariant map of §1.7.2–§1.7.3 and the filtration criterion of §1.6.1. It is a comparison of explicit complex receiving operators, not an asserted equivalence of arithmetic sheaves or cohomology theories.

Source τ retains Z_1 presence, no Z_2 parity, and no addition. All linear operations and grading indices below belong to named complex receiving objects. The monodromy grading is not parity assigned to τ . The original zeta zero, its multiplicity, and every original prime factor remain explicit.

117.1 MJ1. The exact source relation and the actual primary action

Fix an integer $q > 1$. Under the specified geometric Frobenius convention, Frobenius on the Tate object (1) is q^{-1} . After choosing a basis of that one-dimensional object, equivariance of $N : V(1) \rightarrow V$ gives

$$FN = N(q^{-1}F), \quad FNF^{-1} = q^{-1}N. \quad (\text{MJ1.1})$$

This follows by writing the domain operator as $q^{-1}F$. The receiving data considered here are an invertible complex linear operator F and a nilpotent N satisfying this relation. A simultaneous intertwiner preserves both operators.

For an actual nontrivial zero ρ of the original ζ , of multiplicity $m \geq 1$, retain

$$A_\rho = \mathbb{C}[t]/(t^m), \quad T = M_t, \quad e_j = t^j \quad (0 \leq j < m). \quad (\text{MJ1.2})$$

Thus $Te_j = e_{j+1}$ except that $Te_{m-1} = 0$. For a prime p , set $\alpha = p^\rho$ and $\ell = \log p$, using the real logarithm. The programme's actual unshifted prime action is

$$F_0 = W_p = \alpha \exp(\ell T) = \alpha \sum_{r=0}^{m-1} \frac{\ell^r T^r}{r!}. \quad (\text{MJ1.3})$$

This is the original Mellin-jet action, including the explicit $p^{1/2}$ comparison with its centered presentation. It has inverse $\alpha^{-1} \exp(-\ell T)$, by multiplying the finite sums. Its matrix in the unchanged basis is triangular with diagonal α , so its characteristic polynomial, determinant, and trace are respectively

$$(X - \alpha)^m, \quad \alpha^m, \quad m\alpha.$$

Since it is a polynomial in T , it satisfies the different exact relation

$$F_0 T = T F_0. \quad (\text{MJ1.4})$$

117.2 MJ2. Complete classification of simultaneous intertwiners

For any pair (V, F, N) satisfying (MJ1.1), every linear map $a : A_\rho \rightarrow V$ with

$$Fa = aF_0, \quad Na = aT \quad (\text{MJ2.1})$$

is exactly of the form

$$a(f(t)) = f(0)v, \quad v \in \ker N \cap \ker(F - \alpha I). \quad (\text{MJ2.2})$$

Indeed, the two intertwining identities and (MJ1.4) give $FNa = aF_0T = aTF_0 = NFa$. But (MJ1.1) gives $FNa = q^{-1}NFa$. Hence $(1 - q^{-1})NaF_0 = 0$. Since F_0 is invertible and $q > 1$,

$Na = aT = 0$. Thus a factors through the constant-coordinate quotient A_ρ/TA_ρ , and its value at 1 has the properties in (MJ2.2). Conversely every displayed map satisfies (MJ2.1), proving the classification in both directions.

Every linear map $b : V \rightarrow A_\rho$ with

$$F_0b = bF, \quad Tb = bN \quad (\text{MJ2.3})$$

is exactly of the form

$$b(v) = \lambda(v)t^{m-1}, \quad \lambda \in (V/(NV + (F - \alpha I)V))^*. \quad (\text{MJ2.4})$$

In fact $F_0Tb = TF_0b = TbF = bNF$, while also $F_0Tb = bFN = q^{-1}bNF$. Invertibility of F now gives $bN = Tb = 0$. Thus the image is in $\ker T = \mathbb{C}t^{m-1}$. On that line F_0 acts by α , so the remaining intertwining equation is $\lambda F = \alpha\lambda$; the nilpotent equation is $\lambda N = 0$. These are precisely the displayed quotient conditions and also prove sufficiency.

For $m > 1$, no map of the first kind is injective and no map of the second kind is surjective. At $m = 1$, $T = 0$, and those particular obstructions disappear. This makes no assumption about actual simplicity of zeros.

The obstruction is basis-independent. For every scalar λ and nonnegative integer r ,

$$(F - q^{-1}\lambda I)^r N = q^{-r} N (F - \lambda I)^r. \quad (\text{MJ2.5})$$

Subtracting the scalar term in (MJ1.1) proves the case $r = 1$, and multiplication proves the induction. Therefore N sends the generalized λ -eigenspace to the generalized $q^{-1}\lambda$ -eigenspace. If an invertible F has only the eigenvalue α , then $F - q^{-1}\alpha I$ is invertible and (MJ2.5) forces $N = 0$. Thus conjugating the original single primary block cannot turn its nonzero T into Deligne's monodromy while retaining the same Frobenius.

117.3 MJ3. The exact twisted ladder on the same nilpotent space

A related pair on the same vector space is obtained by displaying the changed Frobenius:

$$D_q e_j = q^{-j} e_j, \quad F_{\text{lad}} = F_0 D_q, \quad N_{\text{lad}} = T. \quad (\text{MJ3.1})$$

Evaluation on the basis gives $D_q T D_q^{-1} = q^{-1} T$, including the final zero. Since F_0 commutes with T ,

$$F_{\text{lad}} T F_{\text{lad}}^{-1} = q^{-1} T, \quad F_{\text{lad}} - F_0 = F_0 (D_q - I). \quad (\text{MJ3.2})$$

The change is nonzero for $m > 1$. Its complete eigenvalue and determinant data are

$$\text{char}_{F_{\text{lad}}}(X) = \prod_{j=0}^{m-1} (X - \alpha q^{-j}), \quad \det F_{\text{lad}} = \alpha^m \prod_{j=0}^{m-1} q^{-j}, \quad \text{Tr } F_{\text{lad}} = \alpha \sum_{j=0}^{m-1} q^{-j}. \quad (\text{MJ3.3})$$

These follow from the triangular matrix in the retained basis. The full off-diagonal exponential has an explicit diagonal comparison, not a deletion. Set $c = \ell/(1 - q^{-1})$ and $P = \exp(cT)$. Then P commutes with T , is invertible, and

$$P(\alpha D_q)P^{-1} = \alpha \exp(cT) \exp(-cq^{-1}T) D_q = F_{\text{lad}}. \quad (\text{MJ3.4})$$

Put $d = m - 1$ and define the monodromy filtration directly:

$$M_i = \text{span}\{e_j : d - 2j \leq i\}. \quad (\text{MJ3.5})$$

It satisfies $TM_i \subseteq M_{i-2}$. For every $k \geq 0$, the map $T^k : \text{Gr}_k^M \rightarrow \text{Gr}_{-k}^M$ is an isomorphism: whenever nonzero, the source and target have respective bases $e_{(d-k)/2}$ and $e_{(d+k)/2}$, and T^k sends the first exactly to the second; when the index range or parity excludes them, both spaces

are zero. Thus this is exactly the block filtration characterized in Deligne §1.6.1. The auxiliary indices do not change the user's Z_2 notation.

On its graded pieces, the original F_0 acts by α , since positive powers of T strictly lower the filtration index. The ladder acts at index $i = d - 2j$ by αq^{-j} . With the numerical weight $w_q(\alpha) = 2 \log |\alpha| / \log q$,

$$w_q(\alpha q^{-j}) = w_q(\alpha) - 2j = (w_q(\alpha) - d) + (d - 2j). \quad (\text{MJ3.6})$$

This proves the weight spacing, without fixing the central weight to a prescribed value. At $q = p$, the new eigenvalues are $p^{\rho-j}$, explicitly added receiver data, not asserted zeros of the original zeta function.

The top-versus-bottom comparison can be classified exactly. Replace the ladder Frobenius by $q^r F_{\text{lad}}$, where r is an integer. MJ2 shows that the simultaneous-map space from the original jet has dimension one exactly for $r = d$, and zero otherwise; the reverse-map space has dimension one exactly for $r = 0$, and zero otherwise. For the first assertion, $\ker T = \mathbb{C}e_d$ has eigenvalue αq^{r-d} , equal to α exactly for $r = d$. For the second, the quotient by TA_ρ has eigenvalue αq^r , equal to α exactly for $r = 0$. These conditions coincide at $m = 1$, as required.

117.4 MJ4. A larger module retaining the complete original primary block

The preceding construction changes Frobenius. The following one instead retains the full original block and adds an explicitly distinct monodromy coordinate:

$$B_\rho = \mathbb{C}[x, y]/(x, y)^m, \quad J = M_x, \quad N = M_y. \quad (\text{MJ4.1})$$

Its basis is all $x^i y^j$ with $i, j \geq 0$, $i + j < m$, so its dimension is $\sum_{j=0}^{m-1} (m - j)$. Both nilpotents have order m when $m > 1$, and commute. Let D_q now be the automorphism of this new algebra given by $D_q(x) = x$, $D_q(y) = q^{-1}y$, and define the invertible linear operator

$$\mathcal{F}_{p,q} = M_{p^\rho \exp((\log p)x)} D_q. \quad (\text{MJ4.2})$$

The exponential terminates in the stated quotient. Evaluating on monomials gives

$$\mathcal{F}_{p,q} J = J \mathcal{F}_{p,q}, \quad \mathcal{F}_{p,q} N \mathcal{F}_{p,q}^{-1} = q^{-1} N. \quad (\text{MJ4.3})$$

Thus the original spectral nilpotent and the source-type monodromy coexist as different named operators.

There are algebra maps

$$i : A_\rho \longrightarrow B_\rho, \quad t \longmapsto x, \quad r : B_\rho \longrightarrow A_\rho, \quad x \longmapsto t, \quad y \longmapsto t, \quad ri = \text{id}_{A_\rho}. \quad (\text{MJ4.4})$$

The monomial basis proves that i is injective. Every degree- m monomial maps to $t^m = 0$, proving that r is well-defined; the section proves surjectivity. Its complete kernel is

$$K = (y - x)B_\rho, \quad B_\rho/K \simeq \mathbb{C}[x]/x^m = A_\rho. \quad (\text{MJ4.5})$$

Indeed imposing $y = x$ sends the total-degree- m ideal exactly to (x^m) . The precise preserved operators are

$$\mathcal{F}_{p,q} i = i F_0, \quad J i = i T, \quad r J = T r, \quad r N = T r. \quad (\text{MJ4.6})$$

All four equations follow by substitution, retaining every coefficient. Thus i includes the complete original Frobenius primary block. The return r identifies both receiving nilpotents with the original T .

Their failure to preserve all data simultaneously is the explicit formula

$$r \mathcal{F}_{p,q} - F_0 r = F_0 r (D_q - I). \quad (\text{MJ4.7})$$

On $b = \sum_{i+j < m} b_{ij} x^i y^j$, its complete value is

$$\alpha \exp(\ell t) \sum_{i+j < m} (q^{-j} - 1) b_{ij} t^{i+j}. \quad (\text{MJ4.8})$$

In particular,

$$r\mathcal{F}_{p,q}(y - x) = \alpha \exp(\ell t)(q^{-1} - 1)t. \quad (\text{MJ4.9})$$

For $m > 1$ this is nonzero, because $t \neq 0$, $q^{-1} - 1 \neq 0$, and the other factors are invertible. Hence the calculated return kernel is not Frobenius-stable, and its quotient does not inherit $\mathcal{F}_{p,q}$. The other defect is $(Ni - iT)(f(t)) = (y - x)f(x)$, which lies in the same explicit kernel. At $m = 1$, all these nilpotent defects vanish. These are exact maps and defects; neither map is falsely described as a simultaneous intertwiner of all three operators.

117.5 MJ5. All added blocks and all prime actions

The complete decomposition by the displayed y -degree is

$$B_\rho = \bigoplus_{j=0}^{m-1} y^j \mathbb{C}[x]/x^{m-j}. \quad (\text{MJ5.1})$$

The j -summand has full operator $p^\rho q^{-j} \exp((\log p)M_x)$, with characteristic polynomial $(X - p^\rho q^{-j})^{m-j}$. It has one Jordan block of that length: $\exp(\ell x) - 1 = xg(x)$ with $g(0) = \ell \neq 0$; multiplication by g is invertible, so all kernels of powers equal those of M_x . Consequently

$$\begin{aligned} \text{char}_{\mathcal{F}_{p,q}}(X) &= \prod_{j=0}^{m-1} (X - p^\rho q^{-j})^{m-j}, \\ \det \mathcal{F}_{p,q} &= \prod_{j=0}^{m-1} (p^\rho q^{-j})^{m-j}, \quad \text{Tr } \mathcal{F}_{p,q} = \sum_{j=0}^{m-1} (m-j)p^\rho q^{-j}. \end{aligned} \quad (\text{MJ5.2})$$

The complete $j = 0$ summand is the original primary block. Every other summand remains explicitly labelled as additional receiver data.

For all $a > 0$, define

$$D_a(x) = x, \quad D_a(y) = a^{-1}y, \quad \mathcal{F}_a = M_{a^\rho \exp((\log a)x)} D_a. \quad (\text{MJ5.3})$$

Since D_a fixes x , direct multiplication gives

$$\mathcal{F}_a \mathcal{F}_b = \mathcal{F}_{ab}, \quad \mathcal{F}_a N \mathcal{F}_a^{-1} = a^{-1}N, \quad \mathcal{F}_a i = iW_a. \quad (\text{MJ5.4})$$

Thus one object retains the programme's original multiplicative action at every prime. At $a = p$, it is MJ4 with $q = p$. All added eigenvalues are $a^{\rho-j}$, with the complete blocks in (MJ5.1); none has been declared an actual added zero or a changed prime event.

The simultaneous-map classification MJ2 can be specialized fully to this new module. The kernel of $N = M_y$ consists precisely of the total-degree- $m - 1$ polynomials: multiplication by y sends all lower-degree monomials to distinct surviving monomials. On that kernel the exponential in x acts identically, while D_q has eigenvalues q^{-j} . Its α -eigenline is therefore $\mathbb{C}x^{m-1}$. Every simultaneous map from the original jet is $f(t) \mapsto cf(0)x^{m-1}$.

For the reverse maps, quotienting by $NB_\rho = yB_\rho$ leaves $\mathbb{C}[x]/x^m$. On it $\mathcal{F}_{p,q} - \alpha I$ is multiplication by $\alpha(\exp(\ell x) - 1)$, whose ideal is (x) . Therefore $B_\rho/(NB_\rho + (\mathcal{F}_{p,q} - \alpha I)B_\rho) \simeq \mathbb{C}$, and every reverse simultaneous map is $b \mapsto cb(0,0)t^{m-1}$. These formulas include $m = 1$. They describe every simultaneous intertwiner alongside the larger, differently preserving maps of MJ4.

117.6 MJ6. The complete representation and the limit of the transfer

The twisted relation yields an actual representation. Let $H_q = \mathbb{C} \rtimes \mathbb{Z}$, with group law $(u, k)(v, l) = (u + q^{-k}v, k + l)$. For every pair satisfying (MJ1.1),

$$(u, k) \mapsto \exp(uN)F^k \quad (\text{MJ6.1})$$

is a representation: $F^k \exp(vN)F^{-k} = \exp(q^{-k}vN)$, and multiplication verifies the displayed law. All exponentials terminate. This applies to the ladder and the larger module. This explicit complex representation is not asserted to supply the additional topology, coefficient field, or geometric origin of an arithmetic Weil representation.

The comparison now retains the exact original objects. MJ2 proves every possible intertwiner of the original two operators. MJ3 gives the complete Frobenius change needed for a same-space ladder and computes its filtration. MJ4–MJ5 retain the original prime block inside a larger source-type pair, with both an exact inclusion and an exact return; the return’s Frobenius defect is (MJ4.8)–(MJ4.9). No failed identification has been treated as evidence that the two programmes are unrelated.

Deligne §1.7.6 assumes purity before deducing vanishing of monodromy. No purity of the original zeta quotient follows merely from the calculated equivariance. The added ladder weights and their original center are explicit in MJ3.6, and the original zero ρ is unchanged in the included block. No RH conclusion is claimed.

117.7 Reading coverage and transcription caution

The supplied Deligne excerpt was read in full. This note uses its stated twist map and filtration criterion, and proves all operator comparisons itself. It does not certify the supplied transcription as an error-free author source. For example, the sentence in §1.6.7 excluding the upper index from the lowering formula is inconsistent with the displayed Jordan matrix and the separately stated zero at the lower index. MJ1 and MJ3 define and verify the block action directly, so no result depends on that apparent transcription slip. The chosen geometric Frobenius convention is explicit in MJ1.1.

The programme input is the actual primary action read in `GLOBAL_INFINITESIMAL_QUOTIENT.md`, `GIQ3.2–GIQ3.3`, and expanded independently in `DERIVATION.md`, IP4.2. Deligne’s theorem and human authorship remain distinct from these finite-dimensional comparisons. No root source file was edited and no publication or rendering was performed.

118 Lifting by separated weights: the exact Deligne argument and retained programme maps

24 September 2026. Current target: the user's request to prove the lifting-obstruction mechanism for the tau programme. The user-originating proposal is the full-spectrum reconstruction and global quotient argument. Deligne's geometric theorem and Connes–Consani's constructions are credited below. The finite receiving calculation here is a new derivation on the previously retained MJ4 object; it is not asserted to be the missing geometric specialization theorem.

118.1 TL0. The controlling definitions and the prerequisite map

The complete preserved corpus is identified in `CORPUS_AND_OPERATION_RULES.md` (private source record). Its current instructions have stable identifiers `USR-508e136619bfcd2` and `USR-4322be19bff532cd`. The exact user statements U01–U18 remain in `USER_DEFINITIONS_VERBATIM.md`.

We retain Z_0 as absence, $0 = e = \emptyset$, without source parity; Z_1 as primitive presence τ , without Z_2 parity; and the supplied numerical values and Z_2 data in the integer layer. No addition, subtraction, vector coordinate, numerical distance, weight or multiplicity is assigned to primitive τ . U04 retracts the former self-sum, U03 excludes counting copies of τ , U10–U14 exclude the invented arithmetic midpoint and metric, and U18 requires proofs of derived consequences.

The following complex arithmetic takes place in the already constructed receiving spaces. It does not create arithmetic from primitive presence. The input comparison, retained in IH2–IH3 and QS1, is

$$\mathcal{A} = \left\{ k \in C^\infty(\mathbb{R}_{>0}) : \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j k(u)| < \infty \quad (N, j \geq 0) \right\},$$

$$\mathcal{E}f(u) = u^{1/2} \sum_{n \geq 1} f(nu), \quad I = \overline{\mathcal{E}(S_{00}^{\text{even}})}^{\mathcal{A}}, \quad Q = \mathcal{A}/I,$$

where S_{00}^{even} consists of even Schwartz functions with $f(0) = 0$ and $\int_{\mathbb{R}} f = 0$. The original Mellin transform and action are

$$F_k(s) = \int_0^\infty k(u) u^{s-1/2} \frac{du}{u}, \quad W_a k(u) = a^{1/2} k(u/a), \quad a > 0. \quad (\text{TL0.1})$$

Substitution $u = av$, including $du/u = dv/v$, proves $F_{W_a k}(s) = a^s F_k(s)$. On the initial half-plane of absolute convergence,

$$F_{\mathcal{E}f}(s) = \zeta(s) \int_0^\infty f(v) v^{s-1} dv. \quad (\text{TL0.2})$$

This is the original zeta function, not a completed replacement. Analytic continuation and the decay conditions give the previously proved jet map, for each actual nontrivial zero ρ of multiplicity m_ρ ,

$$j_\rho : Q \longrightarrow A_\rho = \mathbb{C}[t_\rho]/(t_\rho^{m_\rho}), \quad j_\rho([k]) = \sum_{r=0}^{m_\rho-1} \frac{F_k^{(r)}(\rho)}{r!} t_\rho^r,$$

$$j_\rho W_a = a^\rho \exp((\log a) t_\rho) j_\rho. \quad (\text{TL0.3})$$

All later polynomial operations are on A_ρ or its explicitly specified extension. The original jet observations, their common kernel, and the distinction between a quotient and an isomorphism remain in place. In particular the map $Q \rightarrow \prod_\rho A_\rho$ is not here asserted onto or injective.

118.2 TL1. Deligne's lifting theorem, including the obstruction space

The exact source is Pierre Deligne, *La conjecture de Weil II*, §3.6.1–§3.6.3, printed pp.212–214, with §3.6.4 on p.215 for the complex analogue. Official publication. The full local French TeX transcription, lines 2468–2569, was read for this calculation. This is a transcription, not author TeX; no claim is made that a source scan was checked in this continuation.

Deligne takes S to be the spectrum of the henselization of $k[T]$ at (T) , with closed point s , generic point η , and geometric generic point $\bar{\eta}$. The morphism $f : X \rightarrow S$ is proper; X is essentially smooth over k ; $X_{\bar{\eta}}$ is smooth. These are geometric hypotheses of his theorem, not definitions of the tau source. After spreading out, the required diagram carries the arithmetic Frobenius of a finite field $\mathbb{F}_q$.

Write I_{in} for inertia, to distinguish it from TL0's closed image. Hochschild–Serre and the fact that the tame pro- ℓ quotient is $\mathbb{Z}_{\ell}(1)$ give

$$0 \longrightarrow H^{i-1}(X_{\bar{\eta}}, \mathbb{Q}_{\ell})_{I_{\text{in}}}(-1) \longrightarrow E^i := H^i(X_{\eta}, \mathbb{Q}_{\ell}) \xrightarrow{b} C^i := H^i(X_{\bar{\eta}}, \mathbb{Q}_{\ell})^{I_{\text{in}}} \longrightarrow 0. \quad (\text{TL1.1})$$

Here $H^0(\mathbb{Z}_{\ell}(1), V) = V^{\mathbb{Z}_{\ell}(1)}$, $H^1(\mathbb{Z}_{\ell}(1), V) = V_{\mathbb{Z}_{\ell}(1)}(-1)$, and the higher groups vanish. The twist is retained.

The cohomology-with-support sequence and proper base change give

$$H^i(X_s, \mathbb{Q}_{\ell}) \xrightarrow{a} E^i \xrightarrow{\partial} O^i, \quad \text{im } a = \ker \partial, \quad (\text{TL1.2})$$

$$O^i = H_{X_s}^{i+1}(X, \mathbb{Q}_{\ell}) \cong H^{2N-i-1}(X_s, \mathbb{Q}_{\ell})^{\vee}(-N), \quad (\text{TL1.3})$$

where the smooth ambient variety has dimension N . The target of the support pairing is $\mathbb{Q}_{\ell}(-N)$. This fixes both the dual and the sign of the twist.

Deligne proves in §3.6.2, using the smooth proper generic fibre and §1.8.8(i), that C^i has weights at most i . The proper special fibre has $H^{2N-i-1}(X_s)$ of weights at most $2N - i - 1$. Duality changes a weight w to $-w$, and the retained twist $(-N)$ adds $2N$. Thus every weight of O^i is at least

$$-(2N - i - 1) + 2N = i + 1. \quad (\text{TL1.4})$$

These weight statements are Deligne's geometric results; they are not inferred merely from the existence of a spectrum.

For completeness, the linear-algebra step is exact. Fix the complex embedding used to measure weights. In a finite-dimensional Frobenius module, let W_i be the sum of generalized eigenspaces with $|\alpha| \leq q^{i/2}$. An equivariant map preserves each generalized eigenspace: applying $(F - \alpha)^r$ before or after the map gives the same result. Exact sequences restrict to exact sequences on each generalized eigenspace, because the relatively prime factors of the minimal polynomial give polynomial projection operators. Summing those projections proves that W_i is exact.

Apply this to (TL1.1): $W_i E^i \rightarrow C^i$ is onto. Apply it to the support arrow: $W_i O^i = 0$ by (TL1.4), so $\partial(W_i E^i) = 0$. For any $c \in C^i$, choose $e \in W_i E^i$ with $b(e) = c$. Exactness of (TL1.2) supplies $x \in H^i(X_s)$ with $a(x) = e$. Consequently $ba(x) = c$. This is the proved surjectivity of specialization.

The argument chooses a lift into the correct weight range before killing the support obstruction. It does not assert that every arbitrarily chosen lift in E^i has zero obstruction.

118.3 TL2. The existing two-nilpotent receiving object

Retain the existing object MJ4–MJ5 without replacing its variables:

$$B_{\rho} = \mathbb{C}[x_{\rho}, y_{\rho}] / (x_{\rho}, y_{\rho})^{m_{\rho}}, \quad J_{\rho} = M_{x_{\rho}}, \quad N_{\rho} = M_{y_{\rho}}.$$

Its basis consists of the monomials $x_{\rho}^u y_{\rho}^v$ with $u, v \geq 0$ and $u + v < m_{\rho}$. The symbol M_f means multiplication by the displayed polynomial. Define

$$D_a(x_{\rho}) = x_{\rho}, \quad D_a(y_{\rho}) = a^{-1} y_{\rho}, \quad F_a = a^{\rho} \exp((\log a) J_{\rho}) D_a. \quad (\text{TL2.1})$$

Every exponential terminates in this algebra. Because D_a fixes x_ρ , multiplication proves $F_a F_b = F_{ab}$, including every coefficient. Direct evaluation on monomials gives

$$F_a J_\rho = J_\rho F_a, \quad F_a N_\rho F_a^{-1} = a^{-1} N_\rho. \quad (\text{TL2.2})$$

These operators are separately named: J_ρ is the retained spectral nilpotent; N_ρ is the extra Tate-lowering operator. Neither is primitive τ .

The actual quotient used for lifting is

$$q_\rho : B_\rho \longrightarrow A_\rho, \quad x_\rho \longmapsto t_\rho, \quad y_\rho \longmapsto 0_{A_\rho}. \quad (\text{TL2.3})$$

The subscript on 0_{A_ρ} specifies the zero of this complex algebra. It does not give an operation on τ . Degree- m_ρ monomials map to zero, so the map is well-defined. Its kernel is exactly $K_\rho = y_\rho B_\rho$. We have the full equivariant exact sequence

$$0 \longrightarrow K_\rho \longrightarrow B_\rho \xrightarrow{q_\rho} A_\rho \longrightarrow 0. \quad (\text{TL2.4})$$

The section $i_\rho(f(t_\rho)) = f(x_\rho)$ is well-defined, because $x_\rho^{m_\rho} = 0$, and $q_\rho i_\rho = \text{id}$. It preserves the original J_ρ/T_ρ action and every F_a/W_a action.

118.4 TL3. Computed weights and the actual correction of every section

The full decomposition is

$$K_\rho = \bigoplus_{j=1}^{m_\rho-1} y_\rho^j \mathbb{C}[x_\rho]/(x_\rho^{m_\rho-j}). \quad (\text{TL3.1})$$

On its j -summand, $F_p = p^{\rho-j} \exp((\log p)J_\rho)$, while $W_p = p^\rho \exp((\log p)T_\rho)$ on A_ρ . Thus its eigenvalue has weight $2\text{Re } \rho - 2j$; the original block has weight $2\text{Re } \rho$. These follow from $w_p(\alpha) = 2 \log |\alpha| / \log p$ in the receiving representation, not from a presumed weight of τ .

Set $\mathcal{H}_j = \text{Hom}_{\mathbb{C}}(A_\rho, K_{\rho,j})$. The relevant conjugation operator is

$$U_p(h) = F_p h W_p^{-1} = p^{-j} \exp((\log p)\mathcal{D}_j)h, \quad \mathcal{D}_j h = J_\rho h - h T_\rho. \quad (\text{TL3.2})$$

Left and right multiplication commute. Since their nilpotence orders are $m_\rho - j$ and m_ρ , respectively, the binomial expansion proves $\mathcal{D}_j^{2m_\rho-j-1} = 0$: each term has at least one factor beyond its nilpotence order. Therefore $R_j = \exp((\log p)\mathcal{D}_j) - I$ has the same upper bound on its nilpotence order.

With $d_j = 2m_\rho - j - 1$, every coefficient is retained in the inverse

$$(U_p - I)^{-1}|_{\mathcal{H}_j} = \frac{1}{p^{-j} - 1} \sum_{n=0}^{d_j-1} \left(-\frac{p^{-j}}{p^{-j} - 1} R_j \right)^n. \quad (\text{TL3.3})$$

This is verified by multiplying the finite geometric sum by $(p^{-j} - 1)I + p^{-j}R_j$. The final term is zero by nilpotence. The denominator is nonzero because $p > 1$ and $j \geq 1$; no zero is being inverted.

Now let $s : A_\rho \rightarrow B_\rho$ be any complex-linear section of q_ρ . Its obstruction to equivariance is the explicitly constructed map

$$c_p = F_p s W_p^{-1} - s \in \text{Hom}_{\mathbb{C}}(A_\rho, K_\rho). \quad (\text{TL3.4})$$

Its range is in the kernel because $q_\rho F_p = W_p q_\rho$ and $q_\rho s = \text{id}$. Replacing s by $s - h$, where h is kernel-valued, changes this obstruction to $c_p - (U_p - I)h$. Formula (TL3.3) supplies

$$h = (U_p - I)^{-1}c_p, \quad s_{\text{eq}} = s - h. \quad (\text{TL3.5})$$

Thus the obstruction class in $\text{coker}(U_p - I)$ vanishes by its computed weight separation. The corrected section is unique: the difference of two corrected sections lies in $\ker(U_p - I) = 0$. Since the explicitly constructed i_ρ is equivariant, $s_{\text{eq}} = i_\rho$. This also proves simultaneous equivariance at every prime and every $a > 0$, without making separate incompatible choices.

This is a vanishing theorem for an actual lifting obstruction in the retained receiver, not a theorem that assumes its obstruction space has separated weights.

The entire group obstruction also vanishes. On $\mathcal{H} = \text{Hom}(A_\rho, K_\rho)$, let $U_a(h) = F_a h W_a^{-1}$. For any cocycle $c(ab) = c(a) + U_a c(b)$, commutativity gives

$$(U_p - I)c(a) = (U_a - I)c(p).$$

The operators U_a commute, including with the inverse (TL3.3). Consequently $h = (U_p - I)^{-1}c(p)$ satisfies

$$c(a) = (U_a - I)h \quad (a > 0). \quad (\text{TL3.6})$$

Thus every such cocycle is a coboundary, with its unique correction explicitly calculated. For a section, $c(a) = F_a s W_a^{-1} - s$; expanding the product proves the cocycle identity, so this is its actual obstruction.

118.5 TL4. The calculation across all original zeros

Let ρ, σ be any actual nontrivial zeros. On $\text{Hom}(A_\sigma, K_{\rho,j})$ the calculation becomes

$$U_a = a^{\rho - \sigma - j} \exp((\log a)(L_{J_\rho} - R_{T_\sigma})). \quad (\text{TL4.1})$$

The nilpotence exponent is at most $m_\rho - j + m_\sigma - 1$. Because

$$0 < \text{Re } \rho < 1, \quad 0 < \text{Re } \sigma < 1, \quad j \geq 1,$$

we have

$$\text{Re } \rho - \text{Re } \sigma - j < 0. \quad (\text{TL4.2})$$

Hence the conjugation eigenvalue has modulus strictly below one. Formula (TL3.3), with p^{-j} replaced by $p^{\rho - \sigma - j}$, applies to every finite mixed block. No claim $\text{Re } \rho = 1/2$ was used.

For the complete product receivers, give $\mathbf{A} = \prod_\rho A_\rho$, $\mathbf{B} = \prod_\rho B_\rho$, $\mathbf{K} = \prod_\rho K_\rho$ their product topologies. The maps

$$\mathbf{q} = \prod_\rho q_\rho, \quad \mathbf{i} = \prod_\rho i_\rho$$

are continuous, commute with every prime action, and $\mathbf{qi} = \text{id}$. Exactness follows coordinate by coordinate. A continuous linear map from a product of finite-dimensional complex spaces to a finite-dimensional space depends on only finitely many coordinates: continuity supplies a neighbourhood restricting finitely many coordinates; an unrestricted coordinate subspace must have zero image, because its elements can be multiplied by arbitrarily large scalars while staying in that neighbourhood. Therefore every coordinate of a continuous equivariant map $\mathbf{A} \rightarrow \mathbf{K}$ is a finite sum of the cross-block maps in (TL4.1). Every term vanishes by (TL4.2), proving uniqueness of $\mathbf{i}$.

This product is the complete receiving observation object. Its relationship with Q is the specified map $\mathbf{j} : Q \rightarrow \mathbf{A}$. We obtain a concrete continuous map

$$Q \xrightarrow{\mathbf{j}} \mathbf{A} \xrightarrow{\mathbf{i}} \mathbf{B}, \quad \mathbf{qi} \mathbf{j} = \mathbf{j}. \quad (\text{TL4.3})$$

It lifts every retained observation simultaneously. It does not change the observation kernel, prove spectral synthesis, or assert that every independent product tuple is a class of Q .

118.6 TL5. All support labels survive the lift

Let Λ be the original bounded distributive support lattice, denoted here by a capital lambda to distinguish it from a Connes–Consani rank-one group. For any receiving complex vector space V , set

$$G_\Lambda(V) = \{(0_V, \lambda) : \lambda \in \Lambda\} \cup \{(v, 1_\Lambda) : v \in V\}. \quad (\text{TL5.1})$$

For a linear map $f : V \rightarrow W$, define

$$G_\Lambda(f)(v, \lambda) = (f(v), \lambda). \quad (\text{TL5.2})$$

This is well-defined: a nontop label forces $v = 0_V$, whose image is 0_W . The retained addition in this receiving object is $(v, \lambda) + (w, \mu) = (v + w, \lambda \vee \mu)$. It is not addition on primitive τ . Linearity and equality of the joined labels prove that (TL5.2) preserves it. Composition and identities are preserved by direct substitution.

Applying the functor to $\mathbf{q}$ and $\mathbf{i}$ gives

$$G_\Lambda(\mathbf{q})G_\Lambda(\mathbf{i}) = \text{id}_{G_\Lambda(\mathbf{A})}, \quad G_\Lambda(\mathbf{q})G_\Lambda(\mathbf{i})G_\Lambda(\mathbf{j}) = G_\Lambda(\mathbf{j}). \quad (\text{TL5.3})$$

Thus every label survives, including when amplitudes cancel. For a supported zero the exact lift is $(0_{\mathbf{A}}, \lambda) \mapsto (0_{\mathbf{B}}, \lambda)$; for a nonzero amplitude it is $(a, 1_\Lambda) \mapsto (\mathbf{i}(a), 1_\Lambda)$.

A zero-valued linear obstruction lifts to the support-preserving map

$$(v, \lambda) \mapsto (0, \lambda). \quad (\text{TL5.4})$$

It does not become the constant map to $(0, 0_\Lambda)$. This retains the distinction between vanishing arithmetic amplitude and erasing support. Neither receiving zero is called primitive τ . The proof establishes the section with all labels; it does not postulate an abelian cohomology theory on τ .

More precisely, if $Z_\Lambda(V) = \{(0_V, \lambda) : \lambda \in \Lambda\}$, then

$$G_\Lambda(\mathbf{q})^{-1}(Z_\Lambda(\mathbf{A})) = G_\Lambda(\mathbf{K}). \quad (\text{TL5.5})$$

This follows because its label is unchanged and its amplitude maps to zero exactly on $\mathbf{K}$. This amplitude kernel with all labels is not replaced by the inverse image of the single bottom zero.

The independent labels of different observed blocks can also be retained. Their carrier is

$$\mathfrak{A}_\Lambda = \prod_{\rho} G_\Lambda(A_\rho), \quad \mathfrak{B}_\Lambda = \prod_{\rho} G_\Lambda(B_\rho), \quad \mathfrak{K}_\Lambda = \prod_{\rho} G_\Lambda(K_\rho). \quad (\text{TL5.6})$$

For a tuple $((a_\rho, \lambda_\rho))_\rho$, the section is exactly

$$\mathbf{i}((a_\rho, \lambda_\rho)_\rho) = (i_\rho(a_\rho), \lambda_\rho)_\rho, \quad \mathbf{q}((b_\rho, \lambda_\rho)_\rho) = (q_\rho(b_\rho), \lambda_\rho)_\rho. \quad (\text{TL5.7})$$

Every coordinate is well-defined by TL5.2, and coordinatewise computation gives $\mathbf{qi} = \text{id}$, preservation of receiving addition, and equivariance for every original prime action. No label of one block is joined with a label of another block. For any fixed label tuple, the allowed amplitude space in coordinate ρ is A_ρ if $\lambda_\rho = 1_\Lambda$, and $\{0\}$ otherwise. Both maps are continuous on this product of amplitude spaces. No topology on an arbitrary lattice is required.

There is a specified injective comparison

$$G_\Lambda(\mathbf{A}) \longrightarrow \mathfrak{A}_\Lambda, \quad (a, \lambda) \mapsto ((a_\rho, \lambda))_\rho. \quad (\text{TL5.8})$$

Its domain condition makes every coordinate valid. Equality of image tuples gives equality of every amplitude and of the common label, proving injectivity. The lift and quotient commute with this comparison by substitution into TL5.7. Its image contains only constant label tuples. Thus the common-label object has not been substituted for the object with independent block labels; both and their exact map are retained. Neither carrier is asserted to be the complete source geometry without its geometric comparison.

118.7 TL6. What happens to the two nilpotents

The lift preserves every original spectral jet: $J_\rho i_\rho = i_\rho T_\rho$. It does not identify the added N_ρ with T_ρ . For the actual quotient (TL2.3), $q_\rho N_\rho = 0$, so its induced N -operator on A_ρ is zero.

For $m_\rho > 1$, $N_\rho i_\rho(1_{A_\rho}) = y_\rho \neq 0$. More completely,

$$\ker N_\rho = \text{span}\{x_\rho^u y_\rho^v : u + v = m_\rho - 1\}, \quad q_\rho(\ker N_\rho) = \mathbb{C} t_\rho^{m_\rho - 1}. \quad (\text{TL6.1})$$

Multiplication by y_ρ sends distinct monomials of lower total degree to distinct surviving monomials, proving the kernel statement. Its image under q_ρ gives the second equality. Hence an N -equivariant section cannot be onto A_ρ when $m_\rho > 1$. This is a computed property of the two specified receiving operators. It is not an impossibility assertion about primitive τ , and it does not invalidate the prime-equivariant lift already proved.

The previous map $r_\rho(x_\rho) = r_\rho(y_\rho) = t_\rho$ asks for different data: it identifies both receiving nilpotents. Its exact defect remains

$$r_\rho F_p - W_p r_\rho = p^\rho e^{(\log p)t_\rho} \sum_{u+v < m_\rho} (p^{-v} - 1) b_{uv} t_\rho^{u+v} \quad \text{on} \quad \sum b_{uv} x_\rho^u y_\rho^v. \quad (\text{TL6.2})$$

No source operation on τ occurs in this formula. The correction audit is explicit: U04's retracted addition, U10's rejected midpoint, and U11's rejected distance are absent from its prerequisites. Its scope remains the receiving comparison MJ4, not a disproof of the user's geometric argument.

118.8 TL7. The relevant Connes–Consani geometry and the actual endpoint weights

Alain Connes and Caterina Consani, *On the Jacobian of $\overline{\text{Spec } \mathbb{Z}}$* , arXiv:2602.15941v1, is used through its original author TeX, byte-matched to the downloaded original source. Original paper and source. Exact reading ranges and archive hashes are in `sources/CC_SOURCE_IDENTITY_PRIVATE.json`.

Their generic fibre is $C_\eta = \{\lambda \mathbb{Z} : \lambda \in \mathbb{R}^\times\} \cong \mathbb{R}_{>0}^\times$, author lines 1713–1751. The quotient $\text{Pic}(\overline{\text{Spec } \mathbb{Z}}) \rightarrow \text{Jac}(\overline{\text{Spec } \mathbb{Z}})$ forgets the positive scale. The projection is $\lambda \mathbb{Z} \mapsto [\mathbb{Z}]$ on this fibre; its fibre is the entire positive scaling orbit, not a singleton. Their arithmetic divisor carries its stated seminorm and its Γ -module of sections, author lines 1390–1524. Neither structure is silently assigned to primitive τ .

In their spectral section, author lines 2501–2631, the action is by idèle translations on a monoid. The complete explicit formula retains

$$\widehat{h}(0) + \widehat{h}(1) - \sum_{\chi} \sum_{\rho \in Z_{\tilde{\chi}}} \widehat{h}(\tilde{\chi}, \rho) = \sum_v \int'_{\mathbb{Q}_v^\times} \frac{h(u^{-1})}{|1 - u|_v} d^*u. \quad (\text{TL7.1})$$

The local distribution prescription comes from the chosen additive characters. The semilocal cutoff retains the additional term $2h(1) \log \lambda$ and the $o(1)$ term; it is not identified with a finite-dimensional obstruction space here. The authors associate that leading term with the generic stratum and suggest relative cohomology. These cited statements do not themselves provide the exact specialization diagram (TL1.1)–(TL1.3).

The earlier actual adelic calculation ABR19a–d supplies an independent concrete boundary action. For an even Schwartz function,

$$m(h) = \left(h(0), \int_{\mathbb{R}} h(x) dx \right), \quad R_b h(x) = h(x/b),$$

$$m(R_b h) = \begin{pmatrix} 1 & 0 \\ 0 & b \end{pmatrix} m(h). \quad (\text{TL7.2})$$

The equality follows by evaluation and the substitution $x = bu$, with its Jacobian b . On the complete retained prime boundary

$$\mathcal{B}_{\text{pr}} = \left(\bigoplus_{p \text{ prime}} \mathbb{C} \ell_p \right) \otimes \mathbb{C}^2,$$

the action is the identity on ℓ_p and the matrix in (TL7.2) on the endpoint coordinates. Therefore the endpoint weights are exactly 0 and 2. Original nontrivial-zero blocks have weights strictly between these endpoints. At any fixed prime, their eigenvalues differ from both 1 and p ; the corresponding generalized-eigenvalue intertwiner into either endpoint component is zero. This follows by the relatively prime minimal-polynomial argument in TL1, and it retains both endpoints rather than discarding one.

This establishes separation for an actual previously calculated boundary. It does not identify that boundary with the support-obstruction group O^i in Deligne's diagram. The connecting map must come from the programme's actual geometric complex; the existence of separated receiving eigenvalues is not permission to invent such a map.

There is also a direct connecting-map calculation on the Connes–Consani prime mapping torus

$$\mathcal{T}_p = (K^{(p)} \times \mathbb{R}) / ((u, t) \sim (pu, t + \log p)).$$

At a finite compact quotient and character χ , a coefficient system with positive-return operator H_p has the two-term complex

$$V \xrightarrow{\chi(p)H_p - I} V. \quad (\text{TL7.3})$$

For an exact equivariant coefficient sequence $0 \rightarrow V_A \xrightarrow{\iota} V_B \xrightarrow{\pi} V_C \rightarrow 0$, the connecting class is explicitly

$$\delta_\chi(c) = [a], \quad \pi b = c, \quad \iota a = (\chi(p)H_p - I)b. \quad (\text{TL7.4})$$

Changing b changes a by an image of the differential, proving well-definedness. The class is zero exactly when subtracting a preimage produces an invariant lift. All signs and the positive-return convention are proved in the complete companion derivation MT0–MT7.

For the retained original zero jet, $H_p = p^\rho e^{(\log p)T}$, put $z = \chi(p)p^\rho$. Since $|z| = p^{\text{Re } \rho} > 1$, its differential has the finite inverse

$$(\chi(p)H_p - I)^{-1} = \sum_{r=0}^{m_\rho-1} \frac{(-1)^r z^r (e^{(\log p)T} - I)^r}{(z - 1)^{r+1}}. \quad (\text{TL7.5})$$

Multiplication verifies this using $T^{m_\rho} = 0$. This exact receiver has zero cohomology in both degrees for every original nontrivial-zero jet. It therefore cannot provide a nonzero class for the specialization argument by silently identifying its cohomology with the full programme. This is a calculated scope of the particular comparison, not an impossibility theorem about the source. The full companion preserves the torus, Haar measure, characters, nilpotents and all prime-action conventions.

118.9 TL7a. The actual global adelic extension

The full independent derivation AC0–AC9 now applies the calculation to the actual adelic periodization. Retain its two factors:

$$J(c)(y) = y^{1/2} \mathcal{P}(\Phi c)(y) = 2\mathcal{E}(\Phi c)(y), \quad I_0 = J(Q_0). \quad (\text{TL7.6})$$

The existing faithful coordinates are $(Jc, \beta c)$, and their exact inverse is

$$(k, b) \mapsto [t_1 \otimes h_k] + j(b), \quad h_k(x) = \frac{1}{4\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} \frac{F_k(s)}{\zeta(s)} x^{-s} ds, \quad \sigma > 1. \quad (\text{TL7.7})$$

This inverse is used only on I_0 ; it follows from the original factor 2 and Mellin inversion. AC1–AC2 prove the inverse, its equivalent Möbius formula, convergence and equivariance, with both endpoints retained. The resulting actual two-term complex has the chain isomorphism

$$[Q_0 \xrightarrow{J} \mathcal{A}] \cong \mathcal{B}_{\text{pr}}[1] \oplus [I_0 \hookrightarrow \mathcal{A}]. \quad (\text{TL7.8})$$

Indeed the differential sends the coordinates (k, b) to k . Thus its endpoint connecting two-extension is zero, and every group-cohomology connecting map from the split endpoint sequence is zero. The proof is on cochains: the section $k \mapsto [t_1 \otimes h_k]$ commutes with the action and hence with the cochain differential. Its lifted cocycle is a cocycle. This is a vanishing result on the actual adelic modules, beyond TL2's auxiliary polynomial receiver.

Its degree-zero cohomology is $C_{\text{alg}} = \mathcal{A}/I_0$. The Hausdorff quotient and the entire difference remain in the exact sequence

$$0 \longrightarrow \overline{I_0}/I_0 \longrightarrow C_{\text{alg}} \longrightarrow Q \longrightarrow 0. \quad (\text{TL7.9})$$

No closed-image or spectral-synthesis assertion is inserted into this step.

There is also vanishing in every extension degree between any original zero block and the original endpoint space:

$$\text{Ext}^n(A_\rho, \mathcal{B}_{\text{pr}}) = \text{Ext}^n(\mathcal{B}_{\text{pr}}, A_\rho) = 0 \quad (n \geq 0). \quad (\text{TL7.10})$$

Here the category is modules over the complex group algebra of the specified positive dilation group. To verify the vanishing, fix a prime p and its central group-algebra element z . The source and target annihilators are $(z - p^\rho)^{m_\rho}$ and $(z - 1)(z - p)$. They are coprime because $1 < |p^\rho| < p$. The Euclidean identity writes their linear combination as 1. On Ext , central multiplication through the two variables agrees: on a free resolution, precomposition equals postcomposition on module-linear cochains. A polynomial killing the module lifts to a null-homotopic resolution map, by lifting into successive kernels using projectivity and exactness. Both annihilators therefore kill Ext , and their combination 1 forces it to vanish. AC5 gives the full resolution argument and the explicit unique section polynomial in degree one.

This statement does not replace the different exact sequence $0 \rightarrow \ker L_\rho \rightarrow \mathcal{A} \xrightarrow{L_\rho} A_\rho \rightarrow 0$. AC6 calculates its full section defect. AC7 constructs the exact receiving morphism for its spectral data:

$$A_\rho^* \hookrightarrow Q' \hookrightarrow \mathcal{A}', \quad \lambda_{\rho,r}(k) = \frac{1}{r!} \int_0^\infty k(y) y^{\rho-1/2} (\log y)^r \frac{dy}{y}. \quad (\text{TL7.11})$$

Every factor, multiplicity and action convention is retained. The source Z_1/τ is not made into a test function or a distribution by these receiving maps. AC0 records the complete-history arguments and corrected definitions checked before interpreting this calculation.

118.10 TL7b. The geometric distribution map and its entire rational orbit

There is now an exact realization of TL2's operators in the retained Connes–Consani distribution receiver. Use their original archimedean coordinates X, Y , with $Y > 0$, and their operators $D = \partial_X + i\partial_Y$ and $D_Y = YD$. For $c \in \mathbb{Q}$, the map is

$$\Phi_c(x^u y^j) = \delta^{(j)}(X - c) \otimes Y^{\rho+1} \frac{(\log Y)^{m_\rho-j-1-u}}{(m_\rho-j-1-u)!}, \quad u+j < m_\rho. \quad (\text{TL7.12})$$

The delta derivative has its original distributional sign. Compactly supported tests lie inside $Y > 0$, so every displayed coefficient defines a distribution. Different normal derivatives and different powers of $\log Y$ are independent, proving injectivity. The operators that realize J and N are exactly

$$\mathcal{J} = Y\partial_Y - (\rho+1), \quad \mathcal{N} = \partial_X \mathcal{J}. \quad (\text{TL7.13})$$

Applying $\mathcal{J}$ lowers the logarithm degree by one with the displayed factorial; applying $\mathcal{N}$ also raises the delta-derivative order by one. These are the complete original monomial actions, including their zero values at total degree m_ρ . The construction does not define products of arbitrary distributions.

The exact affine pullback sends

$$P(a, b)\Phi_c(x^u y^j) = a^{\rho-j} \sum_{k=0}^{m_\rho-j-1-u} \frac{(\log a)^k}{k!} \Phi_{(c-b)/a}(x^{u+k} y^j), \quad a \in \mathbb{Q}_{>0}, \quad b \in \mathbb{Q}. \quad (\text{TL7.14})$$

DR2 proves this from the two-dimensional Jacobian a^{-2} , the one-dimensional Jacobian a^{-1} , and the full logarithm expansion. Pullback is a right action: $P(a, b)P(a', b') = P(a'a, a'b + b')$. The algebraic direct sums over all rational branches and all actual original zeros retain this whole action. No infinite distribution sum is presumed: each vector has finite branch and zero support. The maps remain injective on that entire direct sum, by finite exponential-polynomial independence.

The original D_Y -residue of TL7.12 is particularly direct. Put $r = m_\rho - j - 1 - u$, $v = \log Y$. Keeping the entire factor and differentiating after multiplying by Y^{-1} gives

$$R_Y \Phi_c(x^u y^j) = (-i)^j Y^{\rho-j} \prod_{h=0}^{j-1} (\rho - h + \partial_v) \frac{v^r}{r!}. \quad (\text{TL7.15})$$

DR4 expands every coefficient of this product. On the finite polynomial space its diagonal coefficient is $(-i)^j \prod_{h=0}^{j-1} (\rho - h)$, which is nonzero for every original zero in $0 < \Re \rho < 1$. The remaining operator lowers polynomial degree, so the finite geometric inverse proves invertibility. The exponents $\rho - j$ are distinct across different retained pairs (ρ, j) : equality would make $\rho - \rho'$ an integer, and the open strip forces that integer to be zero. Thus the whole residue map is injective on this image. The analogous formula for D retains $\rho + 1$ and its different cochain factor; DR3–DR4 calculate both.

In particular, the extra $j > 0$ pieces survive as genuine residue classes. The auxiliary quotient q_ρ is instead the following exact map on this cohomology image:

$$q_\rho = \mathcal{J}_{Y,\rho,c}^{-1} \Pi_{Y,0} R_Y \Phi_c, \quad \mathcal{J}_{Y,\rho,c}(t^u) = Y^\rho \frac{(\log Y)^{m_\rho-1-u}}{(m_\rho-1-u)!}. \quad (\text{TL7.16})$$

Here $\Pi_{Y,0}$ is the projection onto the $j = 0$ summand. DR5.3 supplies it as a finite Hermite polynomial in $Y\partial_Y$, proves that it is identity on that summand and zero on every other summand, and gives the full inverse on its image. All unprojected summands remain in the construction. The quotient has not been mistaken for the Dolbeault residue or for an assertion that the lower-weight classes are zero.

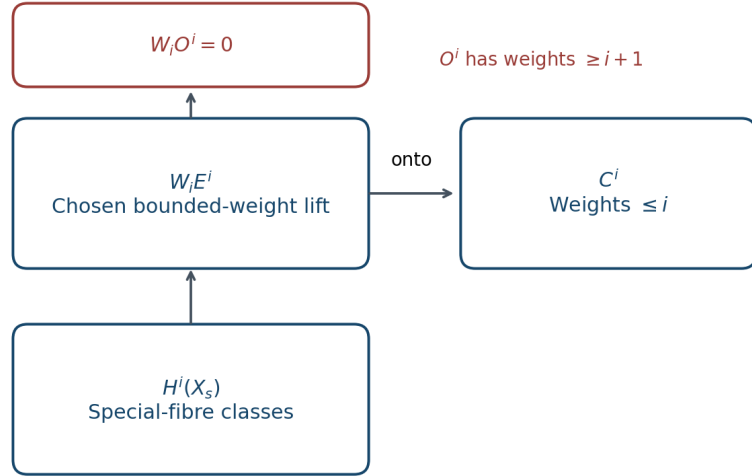
DR6 also proves uniqueness of the lift on the complete specified rational orbit, permitting maps between different zero blocks. For a map from the finite-dimensional branch at $c = 0$, its image has finite branch support. Equivariance under dilation forces that finite support to be invariant under $c \mapsto c/2$, hence contained in $\{0\}$. At that branch, every original eigenvalue has modulus $p^{\Re \rho} > 1$, while every kernel eigenvalue has modulus $p^{\Re \rho' - j} < 1$. The coprime-polynomial argument kills each possible intertwiner, including its nilpotent part. Translation equivariance then kills it at every rational branch. Therefore the direct sum section is the unique $P^+(\mathbb{Q})$ -equivariant section.

The entire proof DR0–DR8 accompanies this reader. It cites the original author TeX of Connes–Consani, *The Riemann–Roch strategy: Complex lift of the Scaling Site*, §5.2, Lemma adelicomp1 and equation actionpq, and §6.6 before Proposition comparecalcs. Original source. The finite adele variables, endpoint strata and separate arithmetic Frobenius have not been identified with this rational archimedean orbit. No such identification is needed for the proved maps, and none is assumed to finish the tau theorem.

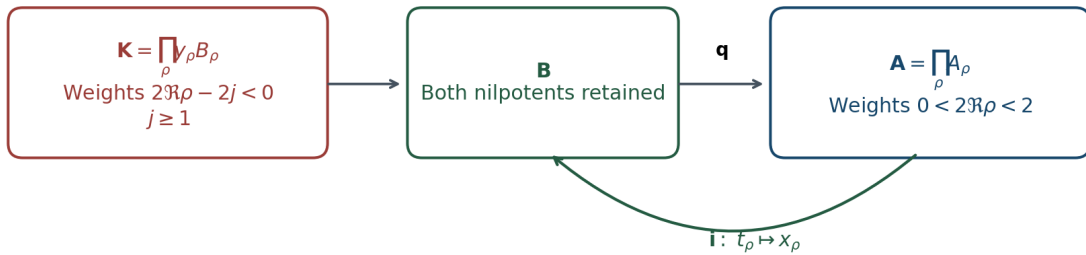
Lifting by separated weights

Two exact diagrams; no numerical weight is assigned to Z_1/τ

Deligne: local invariant cycles, Weil II 3.6



Retained receiver: all original zero jets and all support labels



$qi = I$; unique prime-equivariant section; every original jet survives

$\text{Hom}(A_{\sigma}, K_{\rho, j})$: exponent $\rho - \sigma - j$, with negative real part

Proofs: TL1–TL5 and WL1–WL9. Deligne supplies the upper geometric theorem.

The lower receiver is an explicit programme construction; the full geometric comparison remains unproved.

Figure 42: The proved lifting diagram and the weights of every retained kernel block.

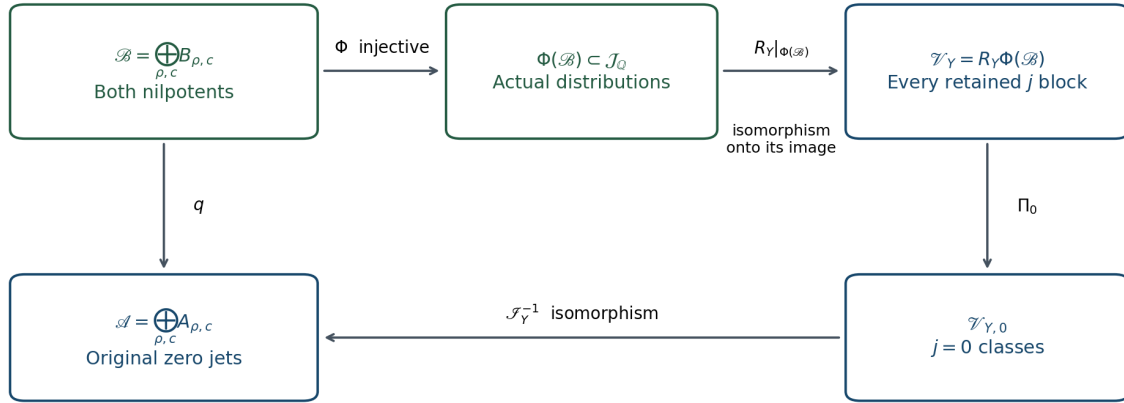
118.11 Figure: the exact location of weight separation

The upper diagram is Deligne's cross, TL1.1–TL1.4. The lower diagram is the constructed receiving sequence TL2.4 and its explicitly computed Hom obstruction TL3.2–TL3.6. The interval $0 < 2\Re \rho < 2$ is the original open-strip statement; the negative shifts are exactly $2\Re \rho - 2j$, $j \geq 1$. The illustration assigns no weight or coordinate to primitive Z_1/τ . Source: Pierre Deligne, Weil II §3.6.1–3.6.3; receiving calculations TL2–TL5 and WL1–WL9. Reproducible figure code accompanies the proof.

The second diagram retains the whole specified rational orbit and all actual zero blocks. Its top-right space contains every lower-weight residue class; the vertical projection is the exact additional map DR5 constructs. No equality between that projection and the original differential is implied. Complete formulas and proofs: DR1–DR6. Human source for the geometric operator: Alain Connes and Caterina Consani, arXiv:1805.10501v1, §5.2 and §6.6.

The quotient is a projection of surviving cohomology classes

Entire rational affine orbit; every original zero and nilpotent retained



$$q = \mathcal{I}_Y^{-1} \Pi_0 R_Y \Phi$$

$j > 0$ classes survive the residue and are retained in $\mathcal{V}_Y$.

Schematic of proved maps, DR1-DR6; finite branch and zero support per vector.
Human source: Alain Connes and Caterina Consani, arXiv:1805.10501v1, sections 5.2 and 6.6.
This archimedean orbit is not asserted to be the complete adelic specialization complex.

Figure 43: The exact distribution realization, residue image and projection.

118.12 TL8. Exact status

The completed positive results include the explicit, unique, all-prime-equivariant lift (TL4.3), with support-preserving extensions (TL5.3) and (TL5.7), and its obstruction cancellation (TL3.3)–(TL3.5). The actual adelic complex also has the endpoint extension vanishing (TL7.8) and the all-degree weight-separated vanishing (TL7.10). All original spectral nilpotents survive. The proofs apply to actual nontrivial zeros without assuming their real parts are $1/2$.

The distribution realization (TL7.12)–(TL7.16) further gives a proved geometric receiving map, both complete residues, the exact projection realizing the auxiliary quotient, and a unique lift over the whole specified rational affine orbit. Its lower-weight classes remain present before that projection.

The requested full tau theorem is not claimed complete: these extensions have not been identified with the complete geometric specialization complex of the original source. The actual adelic endpoint extension is now split explicitly in TL7a, and the archimedean geometric comparison is constructed in TL7b, but no missing specialization arrow or arithmetic Frobenius comparison is assumed. This is the exact present scope, not a universal obstruction or an assertion that the user’s proposed argument is impossible.

The immediate continuation is to construct that geometric comparison from the original support/sheaf complexes and the complete user corpus, using the proved lift and retaining every remaining support contribution. No goal-completion or RH claim follows from this document.

118.13 Current propagation from OMS — 24 September 2026

The complete synthesis proof, OMS1–OMS5, proves that the original summation image is closed and that the full actual-zero jet map on Q has zero common kernel. Its image remains a proper dense subspace of the unrestricted product, carrying the stronger original quotient topology. Earlier statements leaving those two questions undetermined are superseded by that proof, not by an assumption. Every original zero multiplicity and support label remains.

OMS7A proves the exact return to AC0–AC4: the same original spaces give $I_0=2 E(S)=E(S)$, with the factor 2 retained in the periodization map and inverse. Thus $D_{cl}=0$, $C_{alg} \rightarrow Q$ is the identity-induced topological isomorphism, and the specific global two-extension class of the actual adelic two-term complex is zero. Both endpoint contributions at each prime and the higher derived-coinvariant groups remain unchanged. No vanishing of the entire Ext group or full tau purity is inferred.

118.14 Global adelic lifting propagation - 24 September 2026

The complete current proofs are GLOBAL_ADELIC_LIFTING_AND_PRIME_BOUNDARY.md GAP0–GAP9 (including GAP6A–GAP6B), COMPLETE_GENERATOR_ENDPOINT_RETURN.md GER0–GER7, GLOBAL_ENDPOINT_EXTENSION_RETURN.md GEX0–GEX9, and COMPLETE_COMPARISON_TRIANGLE_FORMALITY.md CTF0–CTF8. Every original prime wedge, endpoint pair and nilpotent jet is retained. GER credits the existing RZ endpoint resolvents and gives their explicit original Schwartz return and covariance corrections. GEX proves vanishing of the entire extension groups between the whole actual Q and the stated endpoint modules, in the explicitly constructed algebraic and strict locally convex operator categories. This strengthens the earlier finite-block and particular-extension assertions without assigning that theorem to a larger unspecified category.

GAP and CTF construct the full derived adelic complex, its canonical cohomology map, the exact original CC comparison and its entire cone. The faithful coefficient copies distinguish primitive tau from integer one in both objects. All additional cone contractions and the final-arrow homotopy are explicit; no test-space section $Q \rightarrow A$ is asserted. On the separate supported- Q row the same polynomial is invertible, rather than zero. Full tau numerical purity does not follow from endpoint separation and remains unproved.

119 Weight-separated lifting with every original prime jet retained

24 September 2026. Independent finite-dimensional derivation for the expressly constructed receiving object of MJ4–MJ5. This document proves existence and uniqueness of an equivariant lift, gives the complete correction operator, extends it to all primary blocks, classifies tensor resonances, and carries every map through the complete support-labelled carrier. It does not assert that the auxiliary receiving object is the user’s primitive Z_1 , or that it is already the full geometric specialization required by the research programme.

119.1 WL0. Definitions consulted before operations

The controlling user record is USER_DEFINITIONS_VERBATIM.md, U01–U18, read in full before this derivation. The accompanying SOURCE_OPERATIONS_AND_PROOFS.md, B1–B5 and P1–P5, was also read in full.

The following restrictions are retained throughout.

- U01–U02: Z_1/τ is presence without Z_2 parity; source zero also has no parity. No map in this document assigns either source element a parity.
- U03–U04: no copies of primitive τ are counted and no source sum $\tau + \tau$ is introduced. The retracted law is not used.
- U06 and U10–U12: the source origin is not replaced by a numerical midpoint, a zero vector, or a coordinate extracted from integer arithmetic.
- U08, U11 and U18: existing arithmetic data are retained where they have been constructed; consequences are proved rather than added as source definitions.
- U15–U16: a statement about an auxiliary receiving operator is not a counterexample to the user’s source definitions.

The input to the calculation is the already named complex Mellin-jet receiver of the monodromy comparison, MJ1 and MJ4–MJ5. The actual original-zeta map into this receiver is retained in the integral-history proof, IH2–IH3, and is reproduced with its full Mellin factor in the main lifting proof, TL0. Its complex coefficient field and its spectral parameter are inputs from that receiver. They are not generated here from presence alone. Every addition, subtraction, exponential, tensor product, and inverse below has a stated domain in those existing receiving vector spaces. Section WL9 constructs the corresponding maps with arbitrary bounded distributive support labels; it does not declare a label-zero to be primitive τ .

The comparison with Pierre Deligne is the explicitly specified Frobenius-monodromy relation

$$F_a N F_a^{-1} = a^{-1} N. \quad (\text{WL0.1})$$

For geometric Frobenius at a finite residue field this is the coordinate form of the Tate-twisted map in *La conjecture de Weil. II*, §1.7.2–§1.7.3. The authoritative publication is Deligne, Publications Mathématiques de l’IHÉS 52 (1980), 137–252, DOI 10.1007/BF02684780. This independent calculation proves every operator assertion it uses and does not certify the earlier local transcription or claim a reading of the entire original article. The comparison with Connes–Consani must retain the original Mellin receiver and its source citations; no identification of the auxiliary object below with their full arithmetic geometry is asserted.

119.2 WL1. The entire receiving object and all its actions

Let ρ be any specified complex spectral parameter and $m \geq 1$ its specified positive integer jet length. For an actual nontrivial zero of the original ζ , m is its original multiplicity. Define, as receiving complex algebras,

$$A_\rho = \mathbb{C}[t]/(t^m), \quad B_\rho = \mathbb{C}[x, y]/(x, y)^m. \quad (\text{WL1.1})$$

The quotient ideal $(x, y)^m$ contains exactly the monomials of total degree at least m . Thus the classes $x^i y^j$, $i, j \geq 0$, $i + j < m$, form a basis of B_ρ . In particular

$$B_\rho = \bigoplus_{j=0}^{m-1} B_{\rho,j}, \quad B_{\rho,j} = y^j \mathbb{C}[x]/x^{m-j}. \quad (\text{WL1.2})$$

Write $T = M_t$, $X = M_x$, and $N = M_y$. Their products and nilpotence follow by multiplying the displayed monomials. For every positive real a , using its real logarithm, define

$$\begin{aligned} W_{\rho,a} &= a^\rho \exp((\log a)T), \\ D_a(x) &= x, \quad D_a(y) = a^{-1}y, \\ F_{\rho,a} &= M_{a^\rho \exp((\log a)x)} D_a. \end{aligned} \quad (\text{WL1.3})$$

The exponentials are finite sums: their r -th terms vanish once $r \geq m$. Substitution by D_a preserves $(x, y)^m$ and has inverse $D_{a^{-1}}$, so it is well-defined. Since D_a fixes x ,

$$F_{\rho,a} F_{\rho,b} = F_{\rho,ab}, \quad W_{\rho,a} W_{\rho,b} = W_{\rho,ab}. \quad (\text{WL1.4})$$

These equations follow by multiplying the finite exponential sums and using the binomial identity for the coefficient of each power. They include the inverses at a^{-1} . Monomial evaluation also gives

$$F_{\rho,a} X = X F_{\rho,a}, \quad F_{\rho,a} N F_{\rho,a}^{-1} = a^{-1} N. \quad (\text{WL1.5})$$

The restriction of $F_{\rho,a}$ to $B_{\rho,j}$ is exactly

$$a^{\rho-j} \exp((\log a)X_j), \quad X_j = M_x \text{ on } \mathbb{C}[x]/x^{m-j}. \quad (\text{WL1.6})$$

Thus no original exponential or jet coefficient has been discarded.

119.3 WL2. An equivariant quotient and a complete lift

The following are well-defined algebra maps:

$$q_\rho : B_\rho \longrightarrow A_\rho, \quad x \longmapsto t, \quad y \longmapsto 0, \quad i_\rho : A_\rho \longrightarrow B_\rho, \quad t \longmapsto x. \quad (\text{WL2.1})$$

Every total-degree- m monomial maps to zero under q_ρ ; $x^m = 0$ establishes the relation needed for i_ρ . Their composition is $q_\rho i_\rho = \text{id}$. The basis of WL1 gives

$$\ker q_\rho = K_\rho = y B_\rho = \bigoplus_{j=1}^{m-1} B_{\rho,j}, \quad B_\rho = i_\rho A_\rho \oplus K_\rho. \quad (\text{WL2.2})$$

Indeed q_ρ kills precisely the positive- y -degree basis vectors and sends the remaining basis bijectively to $1, t, \dots, t^{m-1}$.

Direct substitution, with the whole exponential retained, proves

$$\begin{aligned} q_\rho F_{\rho,a} &= W_{\rho,a} q_\rho, \quad F_{\rho,a} i_\rho = i_\rho W_{\rho,a}, \\ q_\rho X &= T q_\rho, \quad X i_\rho = i_\rho T. \end{aligned} \quad (\text{WL2.3})$$

Consequently every original spectral jet lifts: t^r lifts to x^r , with its original coefficient and its original prime action, for $0 \leq r < m$. This is an actual completed construction, including the case $m = 1$, when $K_\rho = 0$.

Uniqueness, rather than merely this displayed existence, is a consequence of the following weight calculation.

119.4 WL3. The weight of every correction and the inverse that kills the defect

For $1 \leq j < m$, put $n = m - j$ and

$$H_j = \text{Hom}_{\mathbb{C}}(A_\rho, B_{\rho,j}).$$

These are linear maps between the receiving spaces already constructed. Define

$$\mathcal{L}_j(h) = X_j h - hT, \quad \mathcal{C}_{a,j}(h) = F_{\rho,a}|_{B_{\rho,j}} h W_{\rho,a}^{-1}. \quad (\text{WL3.1})$$

Left multiplication by X_j and right multiplication by T commute as operators on H_j . Their n -th and m -th powers vanish. Expanding the binomial therefore gives

$$\mathcal{L}_j^{n+m-1} = 0. \quad (\text{WL3.2})$$

For a term in that power to survive, its left exponent would need to be at most $n - 1$ and its right exponent at most $m - 1$; their sum would then be at most $n + m - 2$, a contradiction.

Multiplication of the finite sums in WL1 now proves the complete formula

$$\mathcal{C}_{a,j} = a^{-j} \exp((\log a)\mathcal{L}_j). \quad (\text{WL3.3})$$

In particular its only eigenvalue is a^{-j} . At a prime $p > 1$, its absolute value is $p^{-j} < 1$, and the associated numerical Frobenius weight, meaning $2 \log |\cdot| / \log p$, is $-2j$. The original p^ρ factors canceled in this *Hom action* because they occur both in domain and codomain; neither original prime action was replaced or omitted.

Here is a finite exact inverse. Set

$$d = n + m - 1, \quad E_{p,j} = \exp((\log p)\mathcal{L}_j), \quad U_{p,j} = E_{p,j} - I, \quad c_{p,j} = p^{-j} - 1.$$

The constant $c_{p,j}$ is nonzero. Since $U_{p,j}$ is $\mathcal{L}_j$ times a polynomial, $U_{p,j}^d = 0$. Therefore

$$(\mathcal{C}_{p,j} - I)^{-1} = \sum_{r=0}^{d-1} \frac{(-p^{-j})^r}{(p^{-j} - 1)^{r+1}} U_{p,j}^r. \quad (\text{WL3.4})$$

To check the inverse, multiply the finite sum by $c_{p,j}I + p^{-j}U_{p,j}$. The constant term is I ; every intermediate power cancels between the two adjacent terms; the final power is a multiple of $U_{p,j}^d = 0$. The same multiplication on the other side proves a two-sided inverse.

For an arbitrary receiving vector-space section $s : A_\rho \rightarrow B_\rho$, define its exact prime defect

$$d_p(s) = F_{\rho,p}s - sW_{\rho,p}. \quad (\text{WL3.5})$$

Its image lies in K_ρ : application of q_ρ and WL2.3 gives $W_{\rho,p} - W_{\rho,p} = 0$. Let $d_{p,j}(s)$ be its $B_{\rho,j}$ -component. Then set

$$h_j = (\mathcal{C}_{p,j} - I)^{-1}(d_{p,j}(s)W_{\rho,p}^{-1}), \quad s^\sharp = s - \sum_{j=1}^{m-1} h_j. \quad (\text{WL3.6})$$

All terms are now constructed, including the inverse and its domain. The subtracted maps take values in K_ρ , so $q_\rho s^\sharp = \text{id}$. Also

$$F_{\rho,p}h_j - h_jW_{\rho,p} = (\mathcal{C}_{p,j} - I)(h_j)W_{\rho,p} = d_{p,j}(s).$$

Summing proves $F_{\rho,p}s^\sharp = s^\sharp W_{\rho,p}$.

If s_1, s_2 are two p -equivariant sections, their difference is a map $A_\rho \rightarrow K_\rho$ fixed by $\mathcal{C}_p$. The inverse WL3.4 forces each component to be zero. Hence the equivariant section is unique. Since i_ρ is one such section, the explicit correction always gives

$$\boxed{s^\sharp = i_\rho.} \quad (\text{WL3.7})$$

It follows at once from WL2.3 that this corrected section is equivariant for every positive a , not just for the prime used to compute it, and that it preserves the complete original spectral nilpotent T .

No simplicity assumption on the original zero has been made.

119.5 WL4. The lifting obstruction as a cocycle, with its vanishing proved

The preceding formula can be written independently of the initial choice of section. Let

$$H = \text{Hom}_{\mathbb{C}}(A_\rho, K_\rho), \quad \mathcal{C}_a(h) = F_{\rho,a}|_{K_\rho} h W_{\rho,a}^{-1}.$$

The group $G = \mathbb{R}_{>0}$, with multiplication, acts on H because of WL1.4. The same calculation applies to its subgroup $\mathbb{Q}_{>0}$.

A degree-one cocycle here means exactly a function $c : G \rightarrow H$ obeying

$$c(ab) = c(a) + \mathcal{C}_a c(b). \quad (\text{WL4.1})$$

No topology or continuity is assumed. Commutativity of G gives

$$(\mathcal{C}_p - I)c(a) = (\mathcal{C}_a - I)c(p). \quad (\text{WL4.2})$$

All $\mathcal{C}_a$ commute, and therefore commute with the explicit inverse of $\mathcal{C}_p - I$. With

$$h = (\mathcal{C}_p - I)^{-1} c(p)$$

equation WL4.2 implies, for every a ,

$$c(a) = (\mathcal{C}_a - I)h. \quad (\text{WL4.3})$$

This proves that every cocycle is a coboundary. It also proves that the correcting h is unique, since the common invariant subspace is zero by WL3.4. Thus both the invariants and the degree-one obstruction group vanish, in their explicit meanings just given.

For the arbitrary section s , the function

$$c_s(a) = F_{\rho,a} s W_{\rho,a}^{-1} - s \quad (\text{WL4.4})$$

has values in H . Expanding $F_{\rho,ab} s W_{\rho,ab}^{-1} - s$ proves WL4.1. Hence WL4.3 is precisely the correction in WL3, not an unrelated vanishing theorem.

The classes being lifted retain the exponent ρ ; the corrections have *relative* exponents $-1, -2, \dots, -(m-1)$. Their separation from exponent zero is the exact reason that the obstruction vanishes.

119.6 WL5. Retractions, the previous return, and the additional monodromy requirement

Every equivariant linear retraction of i_ρ equals q_ρ . To prove this, write any retraction r' as $q_\rho + u$, with $u i_\rho = 0$. Thus u is determined by its restrictions $B_{\rho,j} \rightarrow A_\rho$, $j \geq 1$. On each such Hom space, conjugation is

$$h \mapsto W_{\rho,a} h (F_{\rho,a}|_{B_{\rho,j}})^{-1} = a^j \exp((\log a)(Th - hX_j)). \quad (\text{WL5.1})$$

The exponent in parentheses is nilpotent by the same binomial argument. At $a = p > 1$, the only eigenvalue is $p^j \neq 1$. Therefore an equivariant u is zero.

More generally, every all-prime equivariant map $B_\rho \rightarrow A_\rho$ is

$$M_{g(t)}q_\rho, \quad g(t) \in \mathbb{C}[t]/t^m. \quad (\text{WL5.2})$$

The positive- y components vanish by WL5.1. On the $j = 0$ block, commuting with $W_{\rho,p}$ is equivalent to commuting with T , because

$$(\log p)T = \log(p^{-\rho}W_{\rho,p}) = \sum_{r=1}^{m-1} \frac{(-1)^{r+1}}{r} (p^{-\rho}W_{\rho,p} - I)^r. \quad (\text{WL5.3})$$

The finite logarithm identity follows by composing the formal power series modulo t^m . A map commuting with T is determined by its value $g(t)$ at 1 and sends t^r to $t^r g(t)$; hence it is multiplication by g . The retraction condition forces $g = 1$.

The previous return $r : B_\rho \rightarrow A_\rho$, $x \mapsto t$, $y \mapsto t$, is still a well-defined algebra map. Its exact difference from the equivariant retraction is

$$(r - q_\rho) \left(\sum_{i+j < m} b_{ij} x^i y^j \right) = \sum_{\substack{i+j < m \\ j \geq 1}} b_{ij} t^{i+j}. \quad (\text{WL5.4})$$

It preserves both X and N as T , while its Frobenius defect is

$$(rF_{\rho,a} - W_{\rho,a}r)(b) = a^\rho e^{(\log a)t} \sum_{i+j < m} (a^{-j} - 1) b_{ij} t^{i+j}. \quad (\text{WL5.5})$$

This is the previously computed defect with no factors suppressed. Replacing r by the proved equivariant q_ρ changes $rN = Tr$ into $q_\rho N = 0$; it does not pretend to retain that additional identification.

This last distinction has a complete calculation. The quotient operator induced by N is $N_A = 0$, since $q_\rho N = 0$. For $m > 1$, there is no section intertwining this N_A with N . Indeed

$$\ker N = \text{span}\{x^i y^j : i + j = m - 1\}, \quad q_\rho(\ker N) = \mathbb{C}t^{m-1}. \quad (\text{WL5.6})$$

Multiplication by y sends the monomials of degree at most $m - 2$ to distinct nonzero monomials, proving the kernel formula. Its displayed image is a proper subspace of A_ρ when $m > 1$. A section with $Ns = 0$ would take all of A_ρ into this kernel and would have image under q_ρ at most that one-dimensional subspace, contradicting $q_\rho s = \text{id}$. In particular $Ni_\rho(1) = y \neq 0$.

This is a statement about the three explicitly constructed receiving operators, after consulting U01, U04 and U12. It is not an obstruction assigned to primitive Z_1/τ . The weight-separated lift proved above retains Frobenius and the **original spectral nilpotent** T . Requiring it also to identify the **additional monodromy nilpotent** with a prescribed operator is a different, explicitly calculated map condition.

119.7 WL6. All original zero blocks together

Let $\mathcal{R}$ be a set of specified distinct spectral parameters satisfying

$$0 < \text{Re } \rho < 1,$$

with each original multiplicity $m_\rho \geq 1$. Define the algebraic direct sums

$$A = \bigoplus_{\rho \in \mathcal{R}} A_\rho, \quad B = \bigoplus_{\rho \in \mathcal{R}} B_\rho, \quad K = \bigoplus_{\rho \in \mathcal{R}} K_\rho, \quad (\text{WL6.1})$$

with the direct-sum maps q, i, W_a, F_a . The set may be infinite; every individual vector has finite support. This is an algebraic receiver, not an unproved identification of its completion with the full analytic quotient.

For a map $h : A_\sigma \rightarrow B_{\rho,j}$, $j \geq 1$, the complete conjugation formula is

$$\mathcal{C}_a(h) = a^{\rho-\sigma-j} \exp((\log a)\mathcal{L})(h), \quad \mathcal{L}(h) = X_{\rho,j}h - hT_\sigma. \quad (\text{WL6.2})$$

Its nilpotence exponent is at most

$$(m_\rho - j) + m_\sigma - 1. \quad (\text{WL6.3})$$

Because both parameters lie in the open strip,

$$\text{Re}(\rho - \sigma - j) < 1 - j \leq 0. \quad (\text{WL6.4})$$

The inequality is strict, including $j = 1$. Hence for every prime p , the only eigenvalue of $\mathcal{C}_p$ has modulus strictly less than 1. Formula WL3.4 applies with p^{-j} replaced by $p^{\rho-\sigma-j}$. It is therefore invertible on every such block.

This proves uniqueness of the section $i : A \rightarrow B$, even allowing sections that mix different original zeros. It also gives the correction of every linear section by the formula of WL3 on all its component maps. For an infinite algebraic direct sum the formula is still well-defined: the image of the finite-dimensional space A_σ under a given section uses only finitely many target blocks; the correction acts inside those same blocks. No infinite sum of new target coordinates is created.

Every equivariant retraction $B \rightarrow A$ likewise equals q . The positive- y components vanish because their reverse Hom exponents have strictly positive real part.

For completeness, at $j = 0$ all-prime intertwiners between distinct original parameters vanish. Nonzero intertwining requires simultaneously

$$2^{\rho-\sigma} = 1, \quad 3^{\rho-\sigma} = 1. \quad (\text{WL6.5})$$

The first equation forces $\text{Re}(\rho - \sigma) = 0$. Writing $\rho - \sigma = iu$, the two equations give $u \log 2 = 2\pi k$ and $u \log 3 = 2\pi l$, $k, l \in \mathbb{Z}$. If $u \neq 0$, both integers are nonzero and $\log 2 / \log 3 = k/l$, hence $2^l = 3^k$, which is impossible by unique factorization, including after clearing negative exponents. Thus $u = 0$ and $\rho = \sigma$.

On a repeated parameter the same finite logarithm argument as WL5 gives exactly the $\mathbb{C}[t]$ -module intertwiners. For the single original primary block of multiplicity m_ρ , every endomorphism is multiplication by its uniquely determined $g_\rho(t)$. Thus the preceding statements classify the maps rather than relying only on eigenvalue intuition.

119.7.1 WL6a. Complete product receivers and continuity

There is also a simultaneous lift on the complete product observations, not only on the algebraic direct sum. Give

$$\mathbf{A} = \prod_{\rho \in \mathcal{R}} A_\rho, \quad \mathbf{B} = \prod_{\rho \in \mathcal{R}} B_\rho, \quad \mathbf{K} = \prod_{\rho \in \mathcal{R}} K_\rho$$

their product topologies. The coordinatewise $F_a, W_a, \mathbf{q}, \mathbf{i}$ are continuous: each output coordinate is a continuous map of the corresponding finite-dimensional input. The identity $\mathbf{q}\mathbf{i} = \text{id}$, equivariance, and $\ker \mathbf{q} = \mathbf{K}$ all follow coordinate by coordinate.

Every continuous linear map from $\mathbf{A}$ to a finite-dimensional receiving space V depends on finitely many input coordinates. Indeed choose a bounded neighbourhood U of zero in V . Continuity supplies a product neighbourhood whose restrictions involve only finitely many input coordinates. The subspace on which those restricted coordinates vanish is contained in that neighbourhood and remains there under every complex scalar. Its image must therefore be zero:

a nonzero image vector has unbounded scalar multiples, contradicting containment in U . The map consequently factors through the finite product of the restricted coordinates.

Apply this to each coordinate of a continuous map $\mathbf{A} \rightarrow \mathbf{K}$. Each output is a finite sum of the cross-block Hom maps of WL6. Their conjugation operators minus the identity are invertible by WL6.2–WL6.4. The inverse acts within the same finite set of input coordinates, so it again defines a continuous map to that output. These inverses for all outputs yield a well-defined continuous inverse on the space of continuous linear maps $\mathbf{A} \rightarrow \mathbf{K}$, in the exact sense needed by the correction formula. Thus every continuous linear section of $\mathbf{q}$ is corrected uniquely to $\mathbf{i}$, which retains all original jets and all prime actions.

Let the already constructed actual observation map be $\mathbf{j} : Q \rightarrow \mathbf{A}$, with components the original Mellin jets j_ρ . Then

$$Q \xrightarrow{\mathbf{j}} \mathbf{A} \xrightarrow{\mathbf{i}} \mathbf{B}, \quad \mathbf{qij} = \mathbf{j}$$

is the complete simultaneous lift of those observations. A map into a product is continuous precisely when each component is continuous, so the established continuity of the individual jets gives the continuity of this composite. The statement does not assert that $\mathbf{j}$ is an isomorphism, onto, or injective. Its original kernel is unchanged.

119.8 WL7. Tensor products retaining the original tuple

Fix a finite ordered tuple $\boldsymbol{\rho} = (\rho_1, \dots, \rho_d)$, with original lengths $m_1, \dots, m_d$. Define

$$A_{\boldsymbol{\rho}} = \bigotimes_{\nu=1}^d A_{\rho_\nu}, \quad B_{\boldsymbol{\rho}} = \bigotimes_{\nu=1}^d B_{\rho_\nu}, \quad q_{\boldsymbol{\rho}} = \bigotimes_{\nu} q_{\rho_\nu}, \quad i_{\boldsymbol{\rho}} = \bigotimes_{\nu} i_{\rho_\nu}. \quad (\text{WL7.1})$$

The bases are the tensor products of the bases in WL1, so the displayed maps are defined on every basis vector and satisfy $q_{\boldsymbol{\rho}} i_{\boldsymbol{\rho}} = \text{id}$.

Write $\mathbf{j} = (j_1, \dots, j_d)$, $0 \leq j_\nu < m_\nu$, $J = \sum_\nu j_\nu$, and $R = \sum_\nu \rho_\nu$. The exact decomposition is

$$B_{\boldsymbol{\rho}} = \bigoplus_{\mathbf{j}} \left(\bigotimes_{\nu=1}^d y_\nu^{j_\nu} \mathbb{C}[x_\nu] / x_\nu^{m_\nu - j_\nu} \right). \quad (\text{WL7.2})$$

The kernel of $q_{\boldsymbol{\rho}}$ consists of exactly the summands with $J \geq 1$; the $\mathbf{j} = 0$ summand is the entire image of $i_{\boldsymbol{\rho}}$. On the $\mathbf{j}$ -summand the operator is

$$a^{R-J} \exp \left((\log a) \sum_{\nu=1}^d X_\nu \right), \quad (\text{WL7.3})$$

whereas the original tensor block has

$$a^R \exp \left((\log a) \sum_{\nu=1}^d T_\nu \right). \quad (\text{WL7.4})$$

The nilpotence exponent of $\sum X_\nu$ is at most

$$1 + \sum_{\nu} (m_\nu - j_\nu - 1),$$

because every surviving monomial has total degree at most the sum in parentheses. The corresponding exponent of $\sum T_\nu$ is at most $1 + \sum_{\nu} (m_\nu - 1)$.

Thus on the Hom space from the original tensor block to a kernel summand, conjugation is

$$a^{-J} \exp((\log a) \mathcal{L}), \quad \mathcal{L}(h) = \left(\sum X_\nu \right) h - h \left(\sum T_\nu \right), \quad (\text{WL7.5})$$

with nilpotence exponent at most

$$1 + 2 \sum_{\nu} (m_{\nu} - 1) - J. \quad (\text{WL7.6})$$

Since $J \geq 1$, the explicit inverse and cocycle argument of WL3–WL4 apply. Consequently i_{ρ} is the unique equivariant section for this retained tuple, and q_{ρ} the unique equivariant retraction. The correction retains every original tensor basis coefficient and all the original nilpotent terms.

The tuple is not discarded in this result. The following section calculates exactly what happens when different tuples may mix.

119.9 WL8. Complete resonance condition when tensor tuples mix

Take a source tuple σ and a target tuple ρ , with their specified jet lengths, and a target kernel label $\mathbf{j}$ with $J \geq 1$. Put

$$\Delta = \sum_{\nu} \rho_{\nu} - \sum_{\mu} \sigma_{\mu} - J. \quad (\text{WL8.1})$$

The complete Hom conjugation is

$$\mathcal{C}_a = a^{\Delta} \exp((\log a)\mathcal{L}), \quad \mathcal{L}(h) = X_{\text{sum}}h - hT_{\text{sum}}. \quad (\text{WL8.2})$$

For a nonzero all-prime invariant, both $2^{\Delta} = 1$ and $3^{\Delta} = 1$ are necessary: an exponential of a nilpotent has the single eigenvalue 1. The proof of WL6.5 therefore forces

$$\boxed{\sum_{\nu} \rho_{\nu} - \sum_{\mu} \sigma_{\mu} = J.} \quad (\text{WL8.3})$$

Conversely, when this equality holds, all-prime invariance is exactly

$$X_{\text{sum}}h = hT_{\text{sum}}. \quad (\text{WL8.4})$$

Indeed the equation implies commutation with every finite exponential. In the other direction, invariance for $a = 2$ gives $(\exp((\log 2)\mathcal{L}) - I)h = 0$. The operator in parentheses equals $\mathcal{L}$ times a polynomial in $\mathcal{L}$ whose constant coefficient is $\log 2 \neq 0$. That polynomial is invertible by a finite geometric series, so $\mathcal{L}h = 0$.

Equations WL8.3–WL8.4 give all the mixed-tuple maps. They are explicitly constructible. For bases $t^{\mathbf{u}}$ in the source and $x^{\mathbf{v}}y^{\mathbf{j}}$ in the target, write the matrix as $h_{\mathbf{v}, \mathbf{u}}$. Equation WL8.4 is the finite, fully specified set of equalities

$$\sum_{\nu: v_{\nu} \geq 1} h_{\mathbf{v} - \mathbf{e}_{\nu}, \mathbf{u}} = \sum_{\mu: u_{\mu} + 1 < m_{\sigma_{\mu}}} h_{\mathbf{v}, \mathbf{u} + \mathbf{e}_{\mu}}, \quad (\text{WL8.5})$$

with each row and column constrained by its stated monomial ranges. This contains every coefficient of the map; no spectrum-only approximation has replaced its nilpotents.

There is always a nonzero solution at a genuine resonance. Put $n_{\nu} = m_{\rho_{\nu}} - j_{\nu}$, let

$$v_{\text{top}} = \prod_{\nu} x_{\nu}^{n_{\nu}-1} y_{\nu}^{j_{\nu}}, \quad h(f) = f(0, \dots, 0)v_{\text{top}}. \quad (\text{WL8.6})$$

The target monomial is nonzero and is killed by X_{sum} ; the source constant-coefficient functional kills T_{sum} . Hence WL8.4 holds and $h \neq 0$.

If a numerical dimension is required, it is also determined by the actual nilpotent operators without an added assumption. Choose Jordan chains for their finite nilpotent matrices. For a source chain of length r and a target chain of length s , every intertwiner sends the first source vector to an arbitrary element of the target's kernel of its r -th power, then sends the successive

vectors to the successive images of that element. This is necessary and sufficient by direct evaluation. The target kernel has basis its last $\min(r, s)$ chain vectors. Thus this pair contributes exactly $\min(r, s)$ dimensions. Summing over all chains gives the complete dimension. The chain lengths themselves are determined by the finite matrices in WL7: if $b_\ell = \dim \ker X^\ell$, the number of chains of length exactly ℓ is $2b_\ell - b_{\ell-1} - b_{\ell+1}$, as is verified by checking one chain and adding. These are finite exact linear calculations on the displayed monomials.

For a direct sum of tensor tuples, every equivariant section of the direct-sum quotient is therefore

$$i + h,$$

where the only nonzero components of $h : A \rightarrow K$ lie on the resonance locus WL8.3 and satisfy WL8.5. The canonical section i always exists. It is unique precisely when there are no such nonzero components. The strip restriction alone excludes all resonances in WL6, but does not establish their exclusion between arbitrary tensor tuples. This statement neither asserts that actual zeta zeros realize such a resonance nor assumes that they do not.

For example, the formal equality

$$\rho + \bar{\rho} - ((1 - \rho) + (1 - \bar{\rho})) = 4 \operatorname{Re} \rho - 2$$

shows exactly which condition would be tested in this particular quartet of tuples; $J = 1$ would require $\operatorname{Re} \rho = 3/4$, and a nonzero target j_ν also requires the corresponding original jet length to exceed 1. No actual zero or multiplicity satisfying these conditions has been claimed.

119.10 WL9. Every support label is retained

Let L be an arbitrary bounded distributive lattice, with bottom 0_L , top 1_L , join and meet. For each receiving complex vector space V , form exactly the carrier

$$G_L(V) = \{(0, \lambda) : \lambda \in L\} \cup (V \times \{1_L\}). \quad (\text{WL9.1})$$

The two occurrences of $(0, 1_L)$ denote the same element. A nonzero amplitude occurs only with top support.

The operations used on this receiving carrier are specified, not presumed:

$$(v, \lambda) \oplus (w, \mu) = (v + w, \lambda \vee \mu), \quad (\text{WL9.2})$$

and the receiving scalar action from $G_L(\mathbb{C})$ is

$$(c, \lambda) \cdot (v, \mu) = (cv, \lambda \wedge \mu). \quad (\text{WL9.3})$$

Both are closed. In WL9.2 a nonzero output amplitude requires at least one nonzero input amplitude and therefore at least one top label, so its join is top. In WL9.3 a nonzero output requires both input amplitudes to be nonzero and hence both labels to be top. Associativity, commutativity of addition, the receiving scalar laws, and distributivity follow respectively from the already existing vector and scalar operations and the lattice identities. The bounded distributive lattice assumption provides each required meet-over-join or join-over-meet identity. These receiving operations do not define an addition on primitive Z_1/τ .

For every complex linear map $f : V \rightarrow W$, define the common-label map

$$G_L(f)(v, \lambda) = (f(v), \lambda). \quad (\text{WL9.4})$$

It is well-defined: if $\lambda \neq 1_L$, the input amplitude is zero and $f(0) = 0$; if $\lambda = 1_L$, every output amplitude is admitted. The formula gives

$$G_L(gf) = G_L(g)G_L(f), \quad G_L(\text{id}) = \text{id}, \quad (\text{WL9.5})$$

and preserves WL9.2–WL9.3 by linearity, leaving both join and meet coordinates untouched. An invertible receiving linear map has the inverse common-label map $G_L(f^{-1})$.

In particular all the maps q, i, F_a, W_a, X, N of WL1–WL2 have such lifts, and every proved composition identity lifts exactly. Thus

$$G_L(q)G_L(i) = \text{id}, \quad G_L(F_a)G_L(i) = G_L(i)G_L(W_a). \quad (\text{WL9.6})$$

The full zero-label carrier is $Z_L(V) = \{(0, \lambda) : \lambda \in L\}$. It is preserved pointwise, not identified to one point:

$$G_L(f)(0, \lambda) = (0, \lambda). \quad (\text{WL9.7})$$

The full inverse image of this zero-label carrier under $G_L(q)$ is exactly

$$G_L(q)^{-1}(Z_L(A)) = G_L(K). \quad (\text{WL9.8})$$

This is the **amplitude kernel with all labels**, not a claim that the semimodule kernel at the bottom zero contains every supported zero. When $0_L \neq 1_L$, the inverse image of the single bottom element $(0, 0_L)$ is just that bottom element. When L has one element, the usual vector kernel is recovered. These assertions follow directly from the unchanged label in WL9.4 and retain the possible degenerate lattice.

There is no erased support under the weight correction. If s is the section of WL3, its correction h and its defect $d_p(s)$ are already constructed receiving linear maps. On any allowed input (v, λ) , the common-label difference is

$$G_L(s)(v, \lambda) \oplus G_L(-h)(v, \lambda) = (s(v) - h(v), \lambda \vee \lambda) = (i(v), \lambda). \quad (\text{WL9.9})$$

If the amplitude of the corrected defect is zero, its exact value is $(0, \lambda)$, not $(0, 0_L)$ unless that was already the input label. Similarly the inverse WL3.4 fixes each zero-label input when lifted by WL9.4. No scalar image of primitive τ has been used.

For an explicit decomposition, let $P = iq$ and $C = \text{id}_B - iq$, defined on the receiving vector space. Then $P^2 = P$, $C^2 = C$, $PC = CP = 0$, $qC = 0$, $Ci = 0$. On the labelled carrier,

$$G_L(P)(v, \lambda) \oplus G_L(C)(v, \lambda) = (v, \lambda),$$

while the zero-amplitude composites $G_L(PC)$, $G_L(qC)$, and $G_L(Ci)$ return $(0, \lambda)$. This proves the retained-label meaning of every vanishing used here.

All previous results, including direct sums and tensors, are lifted by applying this construction to the entire explicitly constructed receiving space. No distributive operation on primitive τ , no tau-to-scalar-one map, and no comparison that forgets a support label is inserted.

119.11 WL10. A complete length-two calculation

For $m = 2$, retain the ordered bases $1, t$ and $1, x, y$. Put $\alpha = a^\rho$ and $\ell = \log a$. Then

$$W_{\rho, a} = \alpha \begin{pmatrix} 1 & 0 \\ \ell & 1 \end{pmatrix}, \quad F_{\rho, a} = \alpha \begin{pmatrix} 1 & 0 & 0 \\ \ell & 1 & 0 \\ 0 & 0 & a^{-1} \end{pmatrix}. \quad (\text{WL10.1})$$

Every vector-space section has exactly the form

$$s(1) = 1 + cy, \quad s(t) = x + dy.$$

Its full defect is

$$d_a(s)(1) = \alpha((a^{-1} - 1)c - \ell d)y, \quad d_a(s)(t) = \alpha(a^{-1} - 1)dy. \quad (\text{WL10.2})$$

At $a > 1$, the factor $a^{-1} - 1$ is nonzero. If the two displayed coefficients are written as αu and αv , the correction coefficients are exactly

$$d = \frac{v}{a^{-1} - 1}, \quad c = \frac{u + \ell v / (a^{-1} - 1)}{a^{-1} - 1}. \quad (\text{WL10.3})$$

Subtracting the corresponding map to $\mathbb{C}y$ gives $1 \mapsto 1$, $t \mapsto x$. This displays the whole correction including the coupling $-\ell d$; there is no diagonal-only argument and no loss of the original nilpotent. Its support-labelled value is obtained by retaining the same λ in every line of WL10.

119.12 WL11. The exact result and its use

For the actual original primary-jet action inside MJ4–MJ5, the quotient $q_\rho : x \mapsto t$, $y \mapsto 0$ has a unique all-prime equivariant section preserving every original jet. Every initially chosen receiving section is corrected to it by the explicit finite inverse WL3.4. The complete correction space has relative weights $-2j$, so its invariants and degree-one lifting obstruction vanish by WL4. The result persists across all original zeros in the open strip, with all multiplicities, by WL6, and across each retained tensor tuple by WL7. Mixed tensor resonances are exactly WL8.3–WL8.5. Every map and every zero-amplitude conclusion preserves every label of an arbitrary bounded distributive L , by WL9.

The original absolute spectral parameter ρ was unrestricted in WL1–WL5. Consequently this completed weight-separated lifting calculation holds equally for a hypothetical off-critical primary block. It proves a specific lifting result and does not by itself force $\text{Re } \rho = 1/2$. That is a proved scope statement: the formula and its inverse contain no constraint on $\text{Re } \rho$. A claimed application to a different, full geometric specialization must exhibit its exact receiving maps; the present note does not silently substitute B_ρ for that object.

The source of the Z_1, Z_2, τ distinctions remains the user corpus U01–U18. Deligne is credited for the geometric weight and monodromy framework; the finite two-variable receiver and the calculations WL1–WL11 are explicitly programme derivations. This file makes no publication claim, no original-author-source reading claim beyond those stated in WL0, and no claim to have proved or disproved RH.

119.13 Current propagation from OMS — 24 September 2026

The complete synthesis proof, OMS1–OMS5, proves that the original summation image is closed and that the full actual-zero jet map on Q has zero common kernel. Its image remains a proper dense subspace of the unrestricted product, carrying the stronger original quotient topology. Earlier statements leaving those two questions undetermined are superseded by that proof, not by an assumption. Every original zero multiplicity and support label remains.

OMS7A proves the exact return to AC0–AC4: the same original spaces give $I_0=2 \text{ E}(S)=\text{E}(S)$, with the factor 2 retained in the periodization map and inverse. Thus $D_{\text{cl}}=0$, $C_{\text{alg}} \rightarrow Q$ is the identity-induced topological isomorphism, and the specific global two-extension class of the actual adelic two-term complex is zero. Both endpoint contributions at each prime and the higher derived-coinvariant groups remain unchanged. No vanishing of the entire Ext group or full tau purity is inferred.

119.14 Global adelic lifting propagation - 24 September 2026

The complete current proofs are GLOBAL_ADELIC_LIFTING_AND_PRIME_BOUNDARY.md GAP0–GAP9 (including GAP6A–GAP6B), COMPLETE_GENERATOR_ENDPOINT_RETURN.md GER0–GER7, GLOBAL_ENDPOINT_EXTENSION_RETURN.md GEX0–GEX9, and COMPLETE_COMPARISON_TRIANGLE_FORMALITY.md CTF0–CTF8. Every original prime wedge, endpoint pair and nilpotent jet is retained. GER credits the existing RZ endpoint resolvents and gives their explicit original

Schwartz return and covariance corrections. GEX proves vanishing of the entire extension groups between the whole actual Q and the stated endpoint modules, in the explicitly constructed algebraic and strict locally convex operator categories. This strengthens the earlier finite-block and particular-extension assertions without assigning that theorem to a larger unspecified category.

GAP and CTF construct the full derived adelic complex, its canonical cohomology map, the exact original CC comparison and its entire cone. The faithful coefficient copies distinguish primitive tau from integer one in both objects. All additional cone contractions and the final-arrow homotopy are explicit; no test-space section $Q \rightarrow A$ is asserted. On the separate supported- Q row the same polynomial is invertible, rather than zero. Full tau numerical purity does not follow from endpoint separation and remains unproved.

120 The mapping-torus lifting obstruction and the retained prime jets

Independent derivation, 24 September 2026. This calculation concerns the explicit Connes–Consani prime fiber and its finite compact quotients. It does not identify their full adelic cohomology with the cohomology of these quotients.

The active target is the user’s formulation: “Deligne proves that an obstruction to lifting vanishes by separating its weights from those of the classes being lifted.” Sections MT5–MT7 below construct the actual connecting homomorphism and calculate its vanishing or nonvanishing. The preceding Fourier and holonomy calculations fix its signs and the precise receiver.

120.1 Sources and notation

[CCJ] Alain Connes and Caterina Consani, *On the Jacobian of $\overline{\text{Spec } \mathbb{Z}}$* , arXiv:2602.15941. Original source inspected:

`Jacobian.tex`, original author source retained under `sources/`.

The displayed mapping torus is at lines 255–261. The group-fiber calculation is in subsection fibers, lines 1852–1912. The conventions for scaling and its inverse are at lines 2539–2542 and 2587–2593. The local denominator $1/|1 - u|$ appears in subsection geomtrace, lines 2544–2575. This derivation reads these portions, not the entire paper.

[DII] Pierre Deligne, *La conjecture de Weil. II*, Publications mathématiques de l’IHÉS **52** (1980), 137–252, publication record and DOI. The exact argument being compared is §3.6, Theorem 3.6.1 and Lemmas 3.6.2–3.6.3, pp. 213–214. The local French transcription inspected is:

`../deligne_quotient_weight_20260924/sources/S20_FR_record_export.tex`,

lines 2467–2553. It is a transcription, not Deligne’s original author TeX. Its status must remain explicit.

The user’s current Z_1/τ has presence without parity and has no stipulated addition. Nothing below assigns addition to τ , sends τ to an integer, or identifies it with a zero vector. Every vector space, matrix, scalar 1, and addition below belongs to the already constructed complex arithmetic receiver. Any retained support index is carried unchanged as an external index; the proofs do not identify distinct support labels.

Fix a rational prime p , and write

$$L = \log p > 0, \quad K = K^{(p)} = \prod_{\ell \neq p} \mathbb{Z}_{\ell}^{\times}. \quad (\text{MT0.1})$$

Translation $u \mapsto pu$ is a homeomorphism of K . It is not a group homomorphism of K : it sends its identity to p . Its mapping torus, with the exact source orientation, is

$$\mathcal{T}_p = (K \times \mathbb{R})/D^{\mathbb{Z}}, \quad D(u, t) = (pu, t + L). \quad (\text{MT0.2})$$

In particular,

$$[u, t + L] = [p^{-1}u, t]. \quad (\text{MT0.3})$$

The source’s deck action is multiplication by p ; the return of the positive t -flow on the section $t = 0$ is multiplication by p^{-1} . These are inverse maps, not a change of the source convention.

For a finite quotient $H = K/U$ by an open subgroup, let $N = |H|$, let $\bar{p}$ denote the image of p , and let $d = \text{ord}_H(\bar{p})$. Then

$$\mathcal{T}_{p,H} = (H \times \mathbb{R})/((u, t) \sim (\bar{p}u, t + L)). \quad (\text{MT0.4})$$

Every orbit of $\bar{p}$ has d elements, so this space is a disjoint union of N/d circles, each of period dL . This follows by following t from 0 to dL : the section label becomes $\bar{p}^{-d}u = u$, and d is the least positive return index.

120.2 MT1. Complete characters, measure, and signs

Equip K with Haar measure ν , retain its arbitrary positive total mass $v_K = \nu(K) > 0$, and retain dt without dividing it by L . On a fundamental domain,

$$\int_{\mathcal{T}_p} f d\mu = \int_K \int_0^L f([u, t]) dt d\nu(u), \quad \mu(\mathcal{T}_p) = v_K L. \quad (\text{MT1.1})$$

At finite level, each $u \in H$ has measure v_K/N . Each of the N/d circles consequently has measure $v_K dL/N$. The total measure remains $v_K L$. The endpoints 0 and L are identified by (MT0.3); they are not two independent boundary components.

For a continuous character $\chi : K \rightarrow S^1$, choose a real number θ_χ with

$$\chi(p) = e^{i\theta_\chi}. \quad (\text{MT1.2})$$

For $n \in \mathbb{Z}$, put

$$\lambda_{\chi,n} = \frac{2\pi n - \theta_\chi}{L}, \quad \Phi_{\chi,n}([u, t]) = \chi(u) \exp(i\lambda_{\chi,n} t). \quad (\text{MT1.3})$$

The function is well-defined because

$$\Phi_{\chi,n}(pu, t + L) = \chi(p) \exp(i\lambda_{\chi,n} L) \Phi_{\chi,n}(u, t) = \Phi_{\chi,n}(u, t). \quad (\text{MT1.4})$$

Replacing θ_χ by $\theta_\chi + 2\pi j$ relabels n by $n - j$; it does not change the set of functions.

These are precisely all continuous group characters of $\mathcal{T}_p$. Indeed, a character lifted to $K \times \mathbb{R}$ restricts to some χ on K and to $t \mapsto e^{i\lambda t}$ on $\mathbb{R}$; being trivial on (p, L) gives $\chi(p)e^{i\lambda L} = 1$, which is (MT1.3).

The functions $(v_K L)^{-1/2} \Phi_{\chi,n}$ form an orthonormal basis in $L^2(\mathcal{T}_p, \mu)$. For different χ , orthogonality follows by integrating over K ; for fixed χ , the t -integral gives the ordinary Fourier orthogonality on $[0, L]$. For completeness, expand a function on the fundamental domain in characters of K ; in the χ -coefficient, multiplication by $\exp(i\theta_\chi t/L)$ turns the boundary condition into periodicity of period L . Ordinary Fourier series on that interval then give every n . The character expansion on K can be obtained by finite quotient averaging: continuous characters factor through finite quotients, and finite-cylinder functions are dense in $L^2(K)$. For the factorization claim, continuity puts the image of some open subgroup in a sufficiently short arc about 1; that arc contains no nontrivial subgroup of S^1 , so this open subgroup lies in the kernel.

Define the positive-flow pullback and the inverse-flow action separately:

$$U_a f([u, t]) = f([u, t + a]), \quad V_a = U_{-a}. \quad (\text{MT1.5})$$

Both are unitary for the positive Haar inner product, since translation preserves (MT1.1). On the displayed basis,

$$U_a \Phi_{\chi,n} = e^{i\lambda_{\chi,n} a} \Phi_{\chi,n}, \quad V_a \Phi_{\chi,n} = e^{-i\lambda_{\chi,n} a} \Phi_{\chi,n}. \quad (\text{MT1.6})$$

The self-adjoint operator $-i\partial_t$, with the boundary conditions above, has eigenvalues $\lambda_{\chi,n}$. The infinitesimal generator of V_a is $-\partial_t$. Thus the source scaling convention $\vartheta(u)\xi(x) = \xi(u^{-1}x)$ must not be silently replaced by the opposite flow.

120.3 MT2. The full finite-level return distribution

For $h \in C_c^\infty(\mathbb{R})$, the operator $U(h) = \int h(a) U_a da$ is trace class at finite level. Its eigenvalues decay faster than any power of n , by integration by parts in the integral $\int h(a) e^{i\lambda_{\chi,n} a} da$. Since there are N characters, their absolute sum converges. The trace is

$$\text{Tr } U(h) = \sum_{\chi \in \hat{H}} \sum_{n \in \mathbb{Z}} \int_{\mathbb{R}} h(a) e^{i\lambda_{\chi,n} a} da = L \sum_{\chi \in \hat{H}} \sum_{r \in \mathbb{Z}} \chi(\bar{p})^{-r} h(rL). \quad (\text{MT2.1})$$

The last equality is the Fourier series identity for the periodic delta distribution:

$$\sum_{n \in \mathbb{Z}} e^{i(2\pi n - \theta)a/L} = L \sum_{r \in \mathbb{Z}} e^{-ir\theta} \delta(a - rL). \quad (\text{MT2.2})$$

There is no pointwise convergence assertion in (MT2.2). Pairing with h gives the equality used in (MT2.1).

Finite character orthogonality gives

$$\sum_{\chi \in \hat{H}} \chi(\bar{p})^{-r} = \begin{cases} N, & d \mid r, \\ 0, & d \nmid r, \end{cases}$$

hence

$$\text{Tr } U(h) = NL \sum_{j \in \mathbb{Z}} h(jdL). \quad (\text{MT2.3})$$

The Haar mass v_K cancels from the trace because the basis has norm factor $(v_K L)^{-1/2}$ on each side of the inner product; the Hilbert space still uses the original measure. Every remaining factor is retained: N is the character count, $L = \log p$, dL is the orbit period, and $j = 0$ is the identity-time contribution. For $V(h)$, the unsummed character coefficient is $\chi(\bar{p})^r$, and the fully summed expression (MT2.3) is unchanged because j runs through all integers.

For the trivial character alone, positive return times contribute

$$(\log p) \sum_{r \geq 1} h(r \log p). \quad (\text{MT2.4})$$

Restricting to positive times means taking h supported in $(0, \infty)$; it is not a deletion of the identity-time or negative-time terms from (MT2.1).

This is a trace calculation on the compact fiber. It does **not** contain the transverse local-field factor $1/|1 - u|$ in [CCJ, subsection geomtrace]. That factor comes from a different, explicitly stated transverse scaling kernel. Consequently (MT2.1) is not being represented as the complete Weil explicit formula.

120.4 MT3. A bundle retaining the complete actual prime jet

For $\rho \in \mathbb{C}$ and an integer $m \geq 1$, let

$$A_{\rho,m} = \mathbb{C}[T]/(T^m), \quad H_p = p^\rho \exp((\log p)T), \quad p^\rho = \exp(\rho \log p). \quad (\text{MT3.1})$$

Here T acts by multiplication, and the exponential is the exact finite sum

$$\exp((\log p)T) = \sum_{j=0}^{m-1} \frac{(\log p)^j}{j!} T^j. \quad (\text{MT3.2})$$

For an actual zero ρ of the original zeta function, this is the retained primary jet from the programme. The definition of this receiver makes sense for every ρ ; its existence for a value of ρ is not evidence that $\zeta(\rho) = 0$.

Define the flat bundle over the source-oriented mapping torus by

$$E_{\rho,m} = (K \times \mathbb{R} \times A_{\rho,m}) / ((u, t, v) \sim (pu, t + L, H_p^{-1}v)). \quad (\text{MT3.3})$$

The inverse in the gluing is essential. Positive horizontal transport from $[u, 0]$ to $[p^{-1}u, 0]$ has fiber map H_p , because

$$(u, L, v) \sim (p^{-1}u, 0, H_p v). \quad (\text{MT3.4})$$

At finite level its holonomy around a full positive circle of length dL is

$$H_p^d = p^{d\rho} \exp(d(\log p)T). \quad (\text{MT3.5})$$

Thus no power, logarithm, or nilpotent part has been dropped.

A section is represented by a function F satisfying

$$F(pu, t + L) = H_p^{-1} F(u, t). \quad (\text{MT3.6})$$

In the character sector $F(u, t) = \chi(u)f(t)$, this becomes

$$f(t + L) = (\chi(p)H_p)^{-1} f(t). \quad (\text{MT3.7})$$

Writing $f(t) = e^{-t(\rho I + T)} e^{-i\theta_\chi t/L} g(t)$, where $g(t + L) = g(t)$, gives the exact generator blocks

$$\partial_t|_{\chi, n} = \left(-\rho + \frac{i(2\pi n - \theta_\chi)}{L} \right) I - T. \quad (\text{MT3.8})$$

In particular, positive Haar measure on the base does not remove the real part $-\Re\rho$ or the nilpotent term $-T$ from this coefficient action.

For completeness, the coefficient trace at finite level is

$$\text{Tr } U_{E_{\rho, m}}(h) = L \sum_{\chi \in \hat{H}} \sum_{r \in \mathbb{Z}} \chi(\bar{p})^{-r} \text{Tr}(H_p^{-r}) h(rL) = mL \sum_{\chi, r} \chi(\bar{p})^{-r} p^{-r\rho} h(rL). \quad (\text{MT3.9})$$

The proof is (MT2.2) applied to (MT3.8), with $\text{Tr} \exp(-rLT) = m$. The inverse-flow convention instead has coefficient

$$L \chi(\bar{p})^r \text{Tr}(H_p^r) = mL \chi(\bar{p})^r p^{r\rho}. \quad (\text{MT3.10})$$

All nilpotent terms remain in the bundle and generator; the trace annihilates their strictly triangular parts. Equations (MT3.9)–(MT3.10) therefore do not justify replacing the jet by its semisimplification in a lifting problem.

The formula (MT3.9) may be justified without using a nonorthogonal spectral basis. The invertible smooth gauge in (MT4.1) below changes sections into periodic functions on each finite circle. Its conjugated generator is a constant finite-dimensional matrix plus the periodic derivative. Integration against compactly supported h gives matrix Fourier coefficients with rapid decay in n , including the finitely many polynomial factors from T . The operator is trace class; trace is invariant under this bounded invertible gauge on the compact circle.

120.5 MT4. Exact triviality and the limit of the positive-metric argument

Put $A = \rho I + T$, so $H_p = e^{LA}$. There is an explicit smooth bundle isomorphism to the product vector bundle:

$$[(u, t, v)] \longmapsto ([u, t], e^{tA} v). \quad (\text{MT4.1})$$

It is well-defined because $e^{(t+L)A} H_p^{-1} = e^{tA}$. Under this map the flat connection becomes

$$\nabla = d - A dt. \quad (\text{MT4.2})$$

The connection term is retained: (MT4.1) is generally not a flat trivialization. Its inverse sends w to $e^{-tA} w$.

At finite level, $E_{\rho, m}$ is flat-trivial precisely when $H_p^d = I$. Necessity follows by transporting a global flat frame around a circle. For sufficiency, choose a frame at one vertex in each of the N/d circles and extend it by horizontal transport; $H_p^d = I$ is exactly the condition that the frame returns to its original value.

For the jet (MT3.1), this condition is exactly

$$m = 1, \quad e^{dL\rho} = 1, \quad \text{equivalently} \quad m = 1, \quad \Re\rho = 0, \quad dL \Im\rho \in 2\pi\mathbb{Z}. \quad (\text{MT4.3})$$

To prove the restriction on m , equality $H_p^d = I$ first forces its scalar eigenvalue to be 1. The coefficient of T in $\exp(dLT) - I$ is $dL \neq 0$. Thus the exponential cannot equal I when $T \neq 0$, which is exactly $m > 1$.

Two bundles of this form over the same finite mapping torus are isomorphic as flat bundles over the identity of the base precisely when

$$m = m', \quad e^{dL\rho} = e^{dL\rho'}. \quad (\text{MT4.4})$$

Flat isomorphism implies similarity of the two circle holonomies and hence equal rank and scalar eigenvalue. Conversely, under these equalities the two holonomy matrices in their specified T -bases are equal. Choose the identity map at one vertex of each circle and extend by the two parallel transports. Equality of the complete loop maps makes this extension well-defined. This proves sufficiency, including the complete nilpotent part.

A positive definite Hermitian metric preserved by parallel transport exists exactly when

$$m = 1, \quad \Re\rho = 0. \quad (\text{MT4.5})$$

Indeed, preservation of a metric makes every loop holonomy unitary for that metric. Its eigenvalues must have modulus 1, giving $p^{d\Re\rho} = 1$. A unitary matrix is diagonalizable, whereas the nontrivial jet has a nontrivial Jordan block, so $m = 1$. Conversely, for $m = 1$ and $\Re\rho = 0$, multiplication by p^ρ has modulus 1, and the constant metric descends and is parallel.

This has a precise consequence: the positive Haar inner product in MT1 proves unitarity of scalar translation on the base. It does not prove (MT4.5) for the nonunitary coefficient system (MT3.3). A nonparallel positive metric always exists, for example the metric transported through (MT4.1); its existence does not make the flat transport unitary.

The full prime tuple

$$H_q = q^\rho \exp((\log q)T) \quad (q \text{ any prime}) \quad (\text{MT4.6})$$

commutes on $A_{\rho,m}$. It defines compatible coefficient operators for every ρ and every m ; no condition $\Re\rho = \frac{1}{2}$ is used in this construction. These are coefficient maps. This statement does not assert that every q acts as multiplication by a unit on $K^{(p)}$: when $q \neq p$, its q -component is not a unit.

A constant-fiber isomorphism intertwining every H_q for two such prime tuples forces $m = m'$ and $\rho = \rho'$. Rank gives the first equality. Equality of the scalar eigenvalues for $q = 2, 3$ gives

$$(\rho - \rho') \log 2 = 2\pi ia, \quad (\rho - \rho') \log 3 = 2\pi ib$$

for integers a, b . If $\rho - \rho' \neq 0$, both a, b are nonzero and $\log 2 / \log 3 = a/b$. Exponentiation would give $2^b = 3^a$, impossible by unique prime factorization, including after moving negative exponents to the other side. Therefore $\rho = \rho'$. Conversely equal (ρ, m) have the identity intertwiner. Thus the full tuple distinguishes the parameter; it does not force its real part.

120.6 MT5. The exact cochain complex and connecting map

Let V be any finite-dimensional coefficient space with invertible positive return H_p . At finite level, put $C = \text{Map}(H, V)$. Positive parallel transport pushes section data forward by

$$(Rc)(u) = H_p c(\bar{p}u). \quad (\text{MT5.1})$$

To verify the argument $\bar{p}u$, a value initially at $\bar{p}u$ arrives at u after positive time L , and its fiber value is multiplied by H_p , by (MT3.4).

The local-system cohomology is computed by the two-term complex

$$C \xrightarrow{R-I} C \quad (\text{MT5.2})$$

in degrees 0, 1, with no higher cohomology. One can prove this directly on each circle: subdivide it at its d vertices, use the edge flat trivializations, and write a coboundary as transported starting value minus ending value. All edge transports and orientations are invertible, and this cellular differential is (MT5.2) after the indicated identification of edge fibers with their ending fibers. Eliminating $d - 1$ successive vertices leaves the one-vertex differential $H_p^d - I$. Its kernel is the set of global flat sections, and its cokernel is first cohomology.

Equivalently, in the χ -sector,

$$C_\chi^\bullet(V) : \quad V \xrightarrow{B_{\chi,V}} V, \quad B_{\chi,V} = \chi(\bar{p})H_p - I. \quad (\text{MT5.3})$$

This sign agrees with the invariant-section boundary condition (MT3.7): a constant horizontal coefficient is possible exactly when $\chi(\bar{p})H_p v = v$.

Now take an exact sequence of such coefficient systems

$$0 \longrightarrow V_A \xrightarrow{\iota} V_B \xrightarrow{\pi} V_C \longrightarrow 0 \quad (\text{MT5.4})$$

whose maps commute with the specified H_p . For $c \in \ker B_{\chi,V_C}$, choose $b \in V_B$ with $\pi b = c$. Since

$$\pi B_{\chi,V_B} b = B_{\chi,V_C} c = 0,$$

there is a unique $a \in V_A$ satisfying

$$\iota a = B_{\chi,V_B} b.$$

The connecting homomorphism is

$$\boxed{\delta_\chi(c) = [a] \in V_A / B_{\chi,V_A} V_A.} \quad (\text{MT5.5})$$

Replacing b by $b + \iota a_0$ replaces a by $a + B_{\chi,V_A} a_0$, so the class is independent of the lift. Linearity follows by choosing lifts of a sum and of a scalar multiple. Finally,

$$\delta_\chi(c) = 0 \iff c \text{ has a lift in } \ker B_{\chi,V_B}. \quad (\text{MT5.6})$$

For the forward implication, write $a = B_{\chi,V_A} a_0$; then $b - \iota a_0$ is an invariant lift. The reverse implication uses the invariant lift itself, for which $a = 0$.

This gives the complete relevant portion of the long exact sequence:

$$\ker B_A \longrightarrow \ker B_B \longrightarrow \ker B_C \xrightarrow{\delta_\chi} \text{coker } B_A \longrightarrow \text{coker } B_B \longrightarrow \text{coker } B_C \longrightarrow 0. \quad (\text{MT5.7})$$

Exactness at each term follows by the same lift calculation: a cokernel class becomes zero in the next term precisely when its representative can be adjusted by a differential to lie in the preceding subspace.

Suppose specified coefficient maps F_A, F_B, F_C commute with their H_p and with ι, π . They induce maps on every kernel and cokernel. Then

$$\delta_\chi F_C = F_A \delta_\chi. \quad (\text{MT5.8})$$

Indeed $F_B b$ lifts $F_C c$, and

$$B_B F_B b = F_B B_B b = F_B \iota a = \iota F_A a.$$

For the actual prime jets, one can take $F = H_q$ for every prime q , since the exponentials in T commute.

120.7 MT6. Exact spectral separation and the actual jet obstruction

Here is the finite-dimensional algebra used by the Deligne mechanism. Let $S : U \rightarrow W$ satisfy $SF_U = F_W S$, where F_U, F_W are invertible endomorphisms of finite-dimensional complex spaces. If their spectra are disjoint, then $S = 0$. To prove this without semisimplicity, let P and Q be their minimal polynomials. They have no common root, so there are polynomials a, b with $aP + bQ = 1$. For $u \in U$,

$$Su = S(a(F_U)P(F_U) + b(F_U)Q(F_U))u = b(F_W)Q(F_W)Su = 0. \quad (\text{MT6.1})$$

If, for a fixed real $q > 1$, all eigenvalues on U have modulus at most $q^{i/2}$, and all on W have modulus at least $q^{(i+1)/2}$, the spectra are disjoint and this argument applies. This last observation is an exact algebraic consequence of the displayed modulus separation; it supplies neither modulus bound for a new geometric object.

Apply the calculation now to the **actual retained jet** (MT3.1). Put

$$z = \chi(\bar{p})p^\rho, \quad B_\chi = z e^{LT} - I. \quad (\text{MT6.2})$$

If $z \neq 1$, its complete inverse is

$$B_\chi^{-1} = \sum_{j=0}^{m-1} \frac{(-1)^j z^j (e^{LT} - I)^j}{(z - 1)^{j+1}}. \quad (\text{MT6.3})$$

Indeed $e^{LT} - I$ is a multiple of T , so its m -th power is zero; multiplying the finite geometric series by $(z - 1)I + z(e^{LT} - I)$ leaves I . This inverse commutes with every H_q . It is a contracting homotopy from degree 1 to degree 0 for (MT5.3).

For a nontrivial zero ρ of the original zeta function,

$$0 < \Re \rho < 1, \quad |z| = p^{\Re \rho} > 1. \quad (\text{MT6.4})$$

Therefore $z \neq 1$, and

$$H^0(C_\chi^\bullet(A_{\rho, m})) = H^1(C_\chi^\bullet(A_{\rho, m})) = 0 \quad (\text{MT6.5})$$

for every finite quotient and every character. This applies to zeros on and off the critical line alike. The proof has used the known open critical strip, not RH.

Consequently this particular torus receiver has no nonzero invariant class of an actual zero jet to lift. Its vanishing obstruction is the explicitly calculated acyclicity (MT6.3), **not** Deligne's nonvacuous weight separation. It cannot distinguish the critical line from the rest of the open strip.

The resonant case calculates what the jet extension itself contributes when an invariant class is present. Retain $m = 2$ and $z = 1$. There is an exact sequence, equivariant for every H_q ,

$$0 \longrightarrow \mathbb{C}T \longrightarrow \mathbb{C}[T]/T^2 \longrightarrow \mathbb{C} \longrightarrow 0. \quad (\text{MT6.6})$$

The differential on the middle space is

$$B_\chi = e^{LT} - I = LT; \quad (\text{MT6.7})$$

on the subspace and quotient it is zero. The quotient class 1 has lift 1 in the middle space, so (MT5.5) gives

$$\delta_\chi(1) = L[T] \neq 0. \quad (\text{MT6.8})$$

The bracket denotes its class in the cokernel, which here is the entire line $\mathbb{C}T$, because that line's differential is zero. Since $L = \log p > 0$, the class does not vanish. Every lift of 1 has the form $1 + aT$, and applying B_χ gives LT ; thus no invariant lift exists.

The full prime action on the source of δ_χ and on its target is respectively

$$H_q|_{\mathbb{C}} = q^\rho I, \quad H_q|_{\mathbb{C}T} = q^\rho I. \quad (\text{MT6.9})$$

They have exactly the same eigenvalue, not separated weights. Their real growth index

$$\frac{2 \log |q^\rho|}{\log q} = 2\Re \rho \quad (\text{MT6.10})$$

is equal on both sides. This index is a statement about these complex coefficient operators; it is not a claim that they are algebraic Weil numbers. Equation (MT6.8) proves that retaining the nilpotent extension matters, even though its strictly triangular part is invisible in (MT3.9).

The same calculation works at every retained jet length $m \geq 2$. Use

$$0 \longrightarrow \mathbb{C}T^{m-1} \longrightarrow A_{\rho,m} \longrightarrow A_{\rho,m-1} \longrightarrow 0. \quad (\text{MT6.11})$$

At resonance,

$$e^{LT} - I = T U(T), \quad U(T) = \sum_{j=1}^{m-1} \frac{L^j}{j!} T^{j-1}, \quad (\text{MT6.12})$$

where $U(0) = L \neq 0$, so $U(T)$ is invertible in the truncated polynomial algebra. In $A_{\rho,m-1}$, the kernel of this differential is exactly the line generated by T^{m-2} . Lifting that generator to $A_{\rho,m}$ gives

$$\boxed{\delta_\chi(T^{m-2}) = L[T^{m-1}] \neq 0.} \quad (\text{MT6.13})$$

Indeed the degree-one term of the exponential produces LT^{m-1} , and every term of higher degree produces a multiple of $T^m = 0$. Both lines have prime action $H_q = q^\rho I$. Thus every stage of the retained jet tower has the same-eigenvalue obstruction at resonance, with the exact coefficient $\log p$.

One concrete resonant instance is $\rho = 0$, $\chi = 1$, at every prime. It is an example internal to the defined coefficient functor, not an asserted nontrivial zeta zero, not a counterexample to RH, and not a proposed replacement of the programme's arithmetic. It demonstrates the exact same-weight lifting obstruction.

120.8 MT7. The precise comparison with Deligne §3.6

Deligne's setting is a proper morphism $f : X \rightarrow S$, with S the spectrum of a henselian local ring of a smooth curve, total space essentially smooth, and smooth geometric generic fiber. After reduction to a finite-field situation, the exact cross in [DII, §3.6, equation (8)] has middle and right terms

$$E = H^i(X_\eta), \quad C = H^i(X_{\bar{\eta}})^I,$$

and support target

$$O = H^{2N-i-1}(X_s)^\vee(-N). \quad (\text{MT7.1})$$

Its horizontal row is

$$0 \longrightarrow H^{i-1}(X_{\bar{\eta}})_I(-1) \longrightarrow E \longrightarrow C \longrightarrow 0. \quad (\text{MT7.2})$$

Its vertical maps have the form

$$H^i(X_s) \longrightarrow E \xrightarrow{b} O, \quad \text{im}(H^i(X_s) \rightarrow E) = \ker b. \quad (\text{MT7.3})$$

The Tate twists (-1) and $(-N)$, and the index $2N - i - 1$, are part of these maps.

The two weight calculations in Lemmas 3.6.2–3.6.3 are

$$\text{weights}(C) \leq i, \quad \text{weights}(O) \geq i + 1. \quad (\text{MT7.4})$$

Since the weight functor W_i is exact on the objects in question, (MT7.2) gives a surjection $W_i E \rightarrow C$. The map b kills $W_i E$ by (MT7.4), so

$$W_i E \subseteq \ker b = \operatorname{im}(H^i(X_s) \rightarrow E). \quad (\text{MT7.5})$$

Every class of C therefore has a lift in $W_i E$, and that lift has a preimage in $H^i(X_s)$. This is the surjectivity of specialization in Theorem 3.6.1.

The elementary part of this mechanism is exactly the intertwining calculation (MT6.1), together with an exact source of bounded-weight lifts. One must retain both parts: a putative direct map from C to O is not by itself Deligne's entire cross. In particular the choice of a lift in E , its possible change by the left term of (MT7.2), and the subsequent support obstruction are all present.

For the concrete mapping-torus receiver, the corresponding exact sequence is (MT5.7), and its genuine lifting obstruction is (MT5.5). The actual calculation yields:

1. For the original nontrivial zeta jets, the complex is equivariantly contractible by (MT6.3). This receiver has no invariant class available for the desired specialization argument.
2. When a jet class is resonant and does exist, the explicitly retained length-two extension has the nonzero obstruction (MT6.8), while source and obstruction have identical prime eigenvalues (MT6.9).
3. The circle cochain differential (MT5.3) carries no built-in (-1) Tate twist. The occurrence of (-1) in Deligne's inertia cohomology in [DII, §3.6, equations (4)–(5)] cannot be inferred from a topological circle's degree alone. In this calculated receiver the additional prime maps act on coefficients and do not rescale the circle generator.

Thus this computation provides exact connecting maps and an exact obstruction object. It does not supply the two distinct geometric weight bounds in (MT7.4), nor a comparison identifying the torus cohomology with Deligne's support group. The unchanged source Z_1/τ does not create such a Tate factor by relabeling a vector-space zero.

120.9 MT8. What is preserved for the global receiving calculation

The following data are ready to be used without a new interpretation of the source:

- The source-oriented prime torus, its full finite-quotient characters, periods, and Haar volume: (MT0.1)–(MT2.3).
- A flat bundle with the entire actual prime jet as positive return holonomy, and the inverse gluing required by that orientation: (MT3.1)–(MT3.8).
- Exact finite-level flat-isomorphism and flat-metric conditions, with the connection term retained under smooth trivialization: (MT4.1)–(MT4.6).
- The genuine lifting obstruction, its prime-equivariance, and the exact vanishing criterion through invariant lifts: (MT5.5)–(MT5.8).
- An explicit contracting homotopy for every actual open-strip zero jet in this receiver, and an explicit nonzero same-eigenvalue obstruction at resonance: (MT6.3)–(MT6.10).
- The exact logical position of the comparison within Deligne's cross: (MT7.1)–(MT7.5).

These statements neither discard the nilpotent jet nor identify a scalar circle trace with the full adelic trace. They also do not imply a classification of global adelic support cohomology. The nontrivial global continuation is to use the actual programme support sequence and its connecting map, rather than replacing it by the acyclic finite-torus complex calculated here.

121 The actual adelic periodization complex and its weight-separated endpoint

24 September 2026. Independent derivation from ABR1–ABR25 and the actual strong Mellin receiver. The result concerns the original adelic periodization, not the auxiliary two-variable object B_p . Every source factor, endpoint, prime label and closure difference is retained.

121.1 AC0. Corpus, prerequisites and exact scope

Before interpreting an obstruction, the following were read: `USER_DEFINITIONS_VERBATIM.md`, U01–U18; `SOURCE_OPERATIONS_AND_PROOFS.md`, B1–B5 and P1–P5; `CORPUS_AND_OPERATION_RULES.md` (private source record); and the complete retained user arguments in `USER_INPUTS_VERBATIM_PRIVATE.md`, lines 18538–18541, 19055–19083, 19169–19181, 19202–19242 and 19262 onward. These passages include “everything before anything,” the completed counting history, the unique arithmetic reconstruction, and the proposed global quotient.

Accordingly Z_1/τ remains primitive presence with no source Z_2 parity, no source addition and no numerical coordinate. The original completed arithmetic reconstruction, including every prime and the original zeta function, is an input here; no test function is used to reconstruct an integer before that reconstruction. All vector spaces below are the expressly constructed complex receiving spaces. All computations with norms, logarithms, moments, duals and extension classes occur there. A failure of one lift into one test space is not treated as an obstruction to the source definition, to another completion, or to the user’s global argument.

The complete input `CC_ADELIC_COINVARIANT_BRIDGE_INDEPENDENT.tex`, ABR1–ABR25, was read for this derivation, including all signs and the two periodization conventions. The original sources credited there are Alain Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, and Alain Connes, Caterina Consani and Matilde Marcolli, *The Weil proof and the geometry of the adeles class space*. Their adelic operations are the origin of the receiving construction. The calculations AC1–AC9 below are programme derivations, not quotations attributed to those authors.

The exact source for the Deligne mechanism under discussion is Pierre Deligne, *La conjecture de Weil. II*, §3.6.1–§3.6.3, printed pp.212–214, DOI 10.1007/BF02684780. This note calculates the actual adelic extension rather than assuming that it is Deligne’s specialization sequence.

121.2 AC1. Original source and its canonical split

Keep

$$G = \mathbb{Q}_{>0}^\times, \quad R = \mathbb{C}[G], \quad I_R = \ker(\varepsilon : R \rightarrow \mathbb{C}), \quad H = \mathcal{S}_{\text{even}}(\mathbb{R}),$$

$$m(h) = \left(h(0), \int_{\mathbb{R}} h(x) dx \right), \quad H_{00} = \ker m, \quad M_0 = \ker(\varepsilon \otimes m), \quad Q_0 = M_0/I_R M_0. \quad (\text{AC1.1})$$

The subscript on I_R distinguishes the augmentation ideal from the analytic closed periodization image. All tensor products in this construction are algebraic. Put

$$W_{\text{pr}} = \bigoplus_{p \text{ prime}} \mathbb{C}\ell_p, \quad \mathcal{B}_{\text{pr}} = W_{\text{pr}} \otimes \mathbb{C}^2. \quad (\text{AC1.2})$$

Every boundary vector has finite prime support, and each prime retains both endpoint coordinates.

For $c = [\sum_a t_a \otimes h_a] \in Q_0$, the actual ABR coordinates are

$$\Phi(c) = \sum_a h_a, \quad \beta(c) = \sum_p \ell_p \otimes \sum_a v_p(a) m(h_a). \quad (\text{AC1.3})$$

Both descend to Q_0 : Φ kills $I_R M_0$ by augmentation, while the identity

$$v_p(ab) = v_p(a) + v_p(b)$$

implies that applying β to a product $r \cdot m_0$, $r \in I_R$, gives a multiple of the total moment of m_0 , which is zero. The latter verification is ABR9's derivation calculation.

To retain a completely explicit inverse, let

$$h_0(x) = (1 - 2\pi x^2)e^{-\pi x^2}, \quad h_1(x) = 2\pi x^2 e^{-\pi x^2}, \quad \sigma(u_0, u_1) = u_0 h_0 + u_1 h_1.$$

The Gaussian integrals give $m(h_0) = (1, 0)$, $m(h_1) = (0, 1)$. Then

$$(h, \sum_p \ell_p \otimes u_p) \mapsto [t_1 \otimes h] + \sum_p [(t_p - 1) \otimes \sigma(u_p)] \quad (\text{AC1.4})$$

inverts (Φ, β) . To verify this, subtract $t_1 \otimes \sum h_a$ from a representative and use

$$[t_a - 1] = \sum_p v_p(a)[t_p - 1] \quad \text{in } I_R/I_R^2.$$

Replacing h_a by $\sigma(m(h_a))$ in a term with coefficient $t_a - 1$ changes its class by $I_R \otimes H_{00} \subset I_R M_0$. This proves the inverse and its independence of the moment section.

For $b > 0$, let $R_b h(x) = h(x/b)$ and let $\mathcal{R}_b$ act on Q_0 by $\text{id}_R \otimes R_b$. The exact action in AC1.3 is

$$(\Phi, \beta)\mathcal{R}_b c = \left(R_b \Phi c, \left(\text{id}_{W_{\text{pr}}} \otimes \begin{pmatrix} 1 & 0 \\ 0 & b \end{pmatrix} \right) \beta c \right). \quad (\text{AC1.5})$$

Evaluation at zero and substitution $x = bu$ prove the endpoint matrix. They also prove that the moment condition and $I_R M_0$ are preserved. Thus (AC1.4) is equivariant, despite a particular representative σ need not being so: its discrepancy has zero moments and is multiplied by $t_p - 1$, hence is zero in Q_0 .

121.3 AC2. The full original strong test space and the factor two

Let

$$\mathcal{A} = \left\{ k \in C^\infty(\mathbb{R}_{>0}) : \sup_{y>0} (y^N + y^{-N}) |(y\partial_y)^j k(y)| < \infty \text{ for every } N, j \geq 0 \right\}. \quad (\text{AC2.1})$$

Its topology is the locally convex topology of these seminorms. Define

$$\mathcal{P}h(y) = 2 \sum_{n \geq 1} h(ny), \quad J(c)(y) = y^{1/2} \mathcal{P}(\Phi c)(y). \quad (\text{AC2.2})$$

This is exactly the actual adelic sum, including the full nonzero rational sum reduced by the finite supports and the evenness factor 2; ABR12 proves that reduction. It is also

$$J(c) = 2\mathcal{E}(\Phi c), \quad \mathcal{E}h(y) = y^{1/2} \sum_{n \geq 1} h(ny). \quad (\text{AC2.3})$$

No factor is silently removed from the map.

The Poisson formula with Fourier convention $e^{-2\pi i x \xi}$ is

$$\mathcal{P}h(y) = y^{-1} \widehat{\mathcal{P}h}(1/y) + y^{-1} \int_{\mathbb{R}} h(x) dx - h(0). \quad (\text{AC2.4})$$

For $h \in H_{00}$ the last two terms vanish by the actual endpoint conditions, and $\widehat{h} \in H_{00}$. Schwartz estimates at infinity applied both to h and to $\widehat{h}$ prove that $y^{1/2} \mathcal{P}h(y)$ and all its Euler derivatives decay faster than every power at both ends. Thus $J(Q_0) \subset \mathcal{A}$.

Use the original Mellin transform

$$F_k(s) = \int_0^\infty k(y) y^{s-1/2} \frac{dy}{y}. \quad (\text{AC2.5})$$

For $\Re s > 1$, absolute interchange and $x = ny$ give

$$F_{J(c)}(s) = 2\zeta(s) \int_0^\infty \Phi c(x) x^{s-1} dx. \quad (\text{AC2.6})$$

This is the original ζ , not a completed replacement.

Put $I_0 = J(Q_0)$, with no closure. Since $\mathcal{P}$ is injective on H_{00} by ABR14–ABR15, AC1 gives the exact sequence

$$0 \longrightarrow \mathcal{B}_{\text{pr}} \xrightarrow{j} Q_0 \xrightarrow{J} I_0 \longrightarrow 0. \quad (\text{AC2.7})$$

Its inverse along the image is explicit. For $k \in I_0$, define

$$h_k(x) = \frac{1}{4\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} \frac{F_k(s)}{\zeta(s)} x^{-s} ds, \quad \sigma > 1. \quad (\text{AC2.8})$$

The divisor is nonzero on this half-plane by the convergent Euler product, and the integral is exactly ABR15. The equivalent original-variable formula is

$$h_k(x) = \frac{1}{2} \sum_{n \geq 1} \mu(n) (nx)^{-1/2} k(nx). \quad (\text{AC2.9})$$

Insertion of AC2.2 and absolutely convergent regrouping yield the coefficient $\sum_{n|r} \mu(n)$, equal to one for $r = 1$ and zero otherwise, proving this formula with its factor $1/2$. Both recover the unique $h_k \in H_{00}$.

Thus the actual section is

$$s_I(k) = [t_1 \otimes h_k], \quad J s_I = \text{id}_{I_0}, \quad (\text{AC2.10})$$

and the complete faithful decomposition is

$$Q_0 \xrightarrow{\sim} I_0 \oplus \mathcal{B}_{\text{pr}}, \quad c \longmapsto (Jc, \beta c), \quad (k, b) \longmapsto s_I(k) + j(b). \quad (\text{AC2.11})$$

The action on $\mathcal{A}$ is the original

$$W_b k(y) = b^{1/2} k(y/b). \quad (\text{AC2.12})$$

It preserves all defining seminorms up to constants depending on b, N , hence is continuous. Its inverse is $W_{b^{-1}}$. Direct substitution in AC2.2 gives $J\mathcal{R}_b = W_b J$. Uniqueness of h_k then proves $\mathcal{R}_b s_I = s_I W_b$. Therefore the entire split (AC2.11) is equivariant for every positive real b . It retains the endpoints $1, b$ rather than changing their weights.

121.4 AC3. The actual two-term complex and its extension class

Consider the two-term cochain complex in the category of complex representations of $\mathbb{R}_{>0}^\times$:

$$C^{-1} = Q_0, \quad C^0 = \mathcal{A}, \quad d^{-1} = J, \quad (\text{AC3.1})$$

with zero spaces in all other degrees. This is an algebraic cochain complex on the original spaces; $\mathcal{A}$ retains its topology, but no unproved topological quotient identification is part of this statement.

The equivariant decomposition AC2.11 gives a chain isomorphism

$$\boxed{C \simeq \mathcal{B}_{\text{pr}}[1] \oplus [I_0 \hookrightarrow \mathcal{A}]} \quad (\text{AC3.2})$$

Here [1] puts $\mathcal{B}_{\text{pr}}$ in degree -1 , and the second summand has I_0 in degree -1 and $\mathcal{A}$ in degree zero. Indeed its differential in the decomposed coordinates is $(k, b) \mapsto k$.

It follows by taking kernel and cokernel, without a completion, that

$$H^{-1}(C) = \mathcal{B}_{\text{pr}}, \quad H^0(C) = C_{\text{alg}} := \mathcal{A}/I_0. \quad (\text{AC3.3})$$

The chain map from C to $\mathcal{B}_{\text{pr}}[1] \oplus C_{\text{alg}}[0]$, given by β in degree -1 and the quotient $\pi : \mathcal{A} \rightarrow C_{\text{alg}}$ in degree zero, induces the identity on both these cohomology groups. It is therefore an equivariant quasi-isomorphism. No equivariant section of π is asserted.

The precise connecting two-extension is

$$0 \longrightarrow \mathcal{B}_{\text{pr}} \longrightarrow Q_0 \xrightarrow{J} \mathcal{A} \longrightarrow C_{\text{alg}} \longrightarrow 0. \quad (\text{AC3.4})$$

Its class in $\text{Ext}^2(C_{\text{alg}}, \mathcal{B}_{\text{pr}})$ is zero. One direct verification is (AC3.2) and its cohomology quasi-isomorphism, which splits the connecting derived map. Equivalently, the Yoneda extension is the splice of (AC2.7) with $0 \rightarrow I_0 \rightarrow \mathcal{A} \rightarrow C_{\text{alg}} \rightarrow 0$; its first short exact sequence has the explicitly proved equivariant section s_I , so its extension class, and hence its product with the second class, is zero.

For every degree, the group-cohomology connecting map supplied by (AC2.7) is also exactly zero. A cochain with values in I_0 lifts via s_I ; equivariance makes its differential the lift of its original differential. A cocycle therefore lifts to a cocycle, proving the connecting map is zero directly on representatives. The connecting maps from $0 \rightarrow I_0 \rightarrow \mathcal{A} \rightarrow C_{\text{alg}} \rightarrow 0$ are not thereby asserted zero. Only their further connection into $\mathcal{B}_{\text{pr}}$ through (AC2.7) vanishes.

This proves the vanishing for the actual ABR endpoint extension, beyond the auxiliary B_ρ .

121.5 AC4. The closure difference is a retained object

The original Hausdorff spectral quotient is

$$Q = \mathcal{A}/\overline{I_0}^{\mathcal{A}}. \quad (\text{AC4.1})$$

It agrees with the earlier $Q = \mathcal{A}/\overline{\mathcal{E}(H_{00})}$ because $I_0 = 2\mathcal{E}(H_{00})$ and scalar multiplication by 2 is an invertible map on this receiving vector space. The actual map J and its inverse retain that factor as AC2.3 and AC2.8 show.

Define the exact closure-difference object

$$D_{\text{cl}} = \overline{I_0}^{\mathcal{A}}/I_0.$$

Then there is the full equivariant exact sequence

$$0 \longrightarrow D_{\text{cl}} \longrightarrow C_{\text{alg}} \longrightarrow Q \longrightarrow 0. \quad (\text{AC4.2})$$

This follows by taking representatives in $\mathcal{A}$; its kernel consists exactly of the classes represented by $\overline{I_0}$. No vanishing of D_{cl} , closedness of I_0 , or spectral synthesis is presumed. In particular $H^0(C)$ is C_{alg} , not silently Q .

ABR20–ABR22 also contain higher derived-coinvariant groups

$$H_n(G, M_0) = \Lambda^{n+1}W_{\text{pr}} \otimes \mathbb{C}^2, \quad n \geq 1,$$

with the exact exterior-order signs and action $\text{id}_{\Lambda^{n+1}W_{\text{pr}}} \otimes \text{diag}(1, b)$. Periodization kills each of their displayed connecting representatives coefficient by coefficient. These remain retained higher data. The two-term complex AC3 is the expressly requested complex after degree-zero coinvariants; it is not claimed to be the entire derived adelic or geometric specialization complex.

121.6 AC5. Every original finite spectral block and both endpoints

Let ρ be an actual nontrivial zero of the original ζ , with original multiplicity m_ρ . Define the receiving algebra and original action

$$A_\rho = \mathbb{C}[t]/t^{m_\rho}, \quad W_{\rho,b} = b^\rho \exp((\log b)T), \quad T = M_t. \quad (\text{AC5.1})$$

The Mellin-jet map

$$L_\rho : \mathcal{A} \longrightarrow A_\rho, \quad L_\rho(k) = \sum_{r=0}^{m_\rho-1} \frac{F_k^{(r)}(\rho)}{r!} t^r \quad (\text{AC5.2})$$

is continuous: on each fixed vertical strip, powers of $\log y$ in the derivatives are dominated by stronger seminorms of $\mathcal{A}$. The same bounds show that F_k is entire. Since $h \in H_{00}$ is even with $h(0) = 0$, it is $O(x^2)$ at zero, so the second factor in AC2.6 is holomorphic in the critical strip. The multiplicity of the original zero thus proves

$$L_\rho J = 0, \quad L_\rho(\overline{I_0}) = 0. \quad (\text{AC5.3})$$

The second assertion follows from continuity. Hence L_ρ factors through both C_{alg} and Q , with identical full jets. The entire D_{cl} is in its kernel.

The prime action follows from $y = bv$, with its full factors:

$$F_{W_b k}(s) = b^s F_k(s), \quad L_\rho W_b = W_{\rho,b} L_\rho. \quad (\text{AC5.4})$$

In particular L_ρ is a cochain map from AC3 to A_ρ in degree zero.

The two endpoint actions on each prime-labelled copy of $\mathcal{B}_{\text{pr}}$ are 1 and p at the fixed prime $p > 1$. The actual spectral block has sole eigenvalue p^ρ . Because $0 < \Re \rho < 1$,

$$1 < |p^\rho| < p. \quad (\text{AC5.5})$$

This uses the known critical strip, not RH.

For either endpoint character p^ϵ , $\epsilon = 0, 1$, the conjugation operator on $\text{Hom}(A_\rho, \mathbb{C}_\epsilon)$ is exactly

$$h \longmapsto p^{\epsilon-\rho} h \exp(-(\log p)T). \quad (\text{AC5.6})$$

Its nilpotent part is retained, and its single eigenvalue differs from 1 by AC5.5. The reverse Hom operator has sole eigenvalue $p^{\rho-\epsilon} \neq 1$. The finite geometric inverse of each operator minus the identity proves

$$\text{Hom}_G(A_\rho, \mathcal{B}_{\text{pr}}) = 0, \quad \text{Hom}_G(\mathcal{B}_{\text{pr}}, A_\rho) = 0. \quad (\text{AC5.7})$$

These statements also hold with all positive real actions. For the infinite direct sum, a map from finite-dimensional A_ρ has finite prime support; a map in the reverse direction is tested on each endpoint basis vector separately.

There is a stronger exact extension statement. In the category of complex modules over the abelian group algebra of either positive rational or positive real dilations,

$$\text{Ext}^n(A_\rho, \mathcal{B}_{\text{pr}}) = \text{Ext}^n(\mathcal{B}_{\text{pr}}, A_\rho) = 0 \quad (n \geq 0). \quad (\text{AC5.8})$$

Here is a proof including the prerequisite for this use of Ext. The element z of the group algebra representing dilation by p is central. It satisfies

$$P(z)A_\rho = 0, \quad P(X) = (X - p^\rho)^{m_\rho}, \quad R(z)\mathcal{B}_{\text{pr}} = 0, \quad R(X) = (X - 1)(X - p).$$

The polynomials have no common root by AC5.5, so the Euclidean algorithm gives $uP + vR = 1$. A central element acts on Ext in the same way through either variable: take a free resolution of

the first module, on which multiplication by that central element is a chain map; on a module-linear cochain, precomposition by it equals postcomposition by its action on the target. If a polynomial acts as zero on the first module, its chain map on a free resolution is null-homotopic. To construct the homotopy, lift the degree-zero map into the kernel of the augmentation using freeness and exactness, then inductively lift the difference left after the preceding homotopy into the next kernel; exactness identifies that kernel with the next boundary. This constructs the required homotopy in every degree. Thus P and R both annihilate the displayed Ext groups, and $uP + vR = 1$ annihilates them only if they are zero.

For degree one an explicit section can also be given. In any extension $0 \rightarrow \mathcal{B}_{\text{pr}} \rightarrow E \rightarrow A_\rho \rightarrow 0$, choose a polynomial $a(X)$ with

$$a(X) \equiv 0 \pmod{R(X)}, \quad a(X) \equiv 1 \pmod{P(X)}.$$

Such a polynomial follows from the same Euclidean algorithm. Its action kills the boundary and induces identity on A_ρ . Therefore $v \mapsto a(z)\tilde{v}$, for any lift $\tilde{v} \in E$, is independent of that lift, equivariant, and is a section. The difference of two equivariant sections is zero by AC5.7, so it is unique. This is a proved weight-separated lifting statement on the original endpoints and actual original zero blocks.

Neither AC5.7 nor AC5.8 asserts that these endpoints equal Deligne's entire geometric support group, or that C_{alg} has no additional classes beyond its finite observed blocks.

121.7 AC6. What lifts into the actual test space, and the exact defect

The finite jet observation itself has an explicitly constructed linear section, without an equivariance claim. Choose a nonnegative smooth compactly supported function φ on $\mathbb{R}$ that is positive on an open interval. Put

$$M_{r\ell} = \int_{\mathbb{R}} \varphi(v) v^{r+\ell} dv, \quad 0 \leq r, \ell < m_\rho.$$

For a nonzero coefficient vector c ,

$$\sum_{r,\ell} \bar{c}_r M_{r\ell} c_\ell = \int \varphi(v) \left| \sum_{\ell} c_\ell v^\ell \right|^2 dv > 0.$$

The inequality holds because a nonzero polynomial cannot vanish on an open interval. Thus M is invertible. For $a(t) = \sum_r a_r t^r$, define $c = M^{-1}(r!a_r)_r$, $P_a(v) = \sum_{\ell} c_\ell v^\ell$, and

$$s_\rho(a)(e^v) = e^{-(\rho-1/2)v} \varphi(v) P_a(v). \quad (\text{AC6.1})$$

This is a compact logarithmic test in $\mathcal{A}$. Substitution in AC2.5 gives $F_{s_\rho(a)}^{(r)}(\rho) = r!a_r$, proving $L_\rho s_\rho = \text{id}$. No original- Q eigenvector has been postulated.

Let $I_\rho = \ker L_\rho$. The actual test-space extension is

$$0 \longrightarrow I_\rho \longrightarrow \mathcal{A} \xrightarrow{L_\rho} A_\rho \longrightarrow 0. \quad (\text{AC6.2})$$

Its kernel contains $\overline{I_0}$; it is not the endpoint space $\mathcal{B}_{\text{pr}}$. On logarithmic coordinates, for $b > 0$, $\ell = \log b$, its section defect is the fully explicit I_ρ -valued map

$$\begin{aligned} & (W_b s_\rho - s_\rho W_{\rho,b})(a)(e^v) \\ &= b^\rho e^{-(\rho-1/2)v} [\varphi(v-\ell) P_a(v-\ell) - \varphi(v) P_{\exp(\ell t)a}(v)]. \end{aligned} \quad (\text{AC6.3})$$

This follows from the exact $b^{1/2}$ in W_b , and linearity of P_a . Applying L_ρ gives zero by AC5.4, proving the asserted target.

There is no equivariant section of AC6.2 into this $\mathcal{A}$. This is a statement about its specified decay, not about the source arithmetic. To prove it, first suppose $W_p k = \alpha k$, $\alpha \neq 0$. Evaluation at $p^n y$ gives

$$k(p^n y) = (p^{1/2}/\alpha)^n k(y).$$

For every N , AC2.1 bounds its left side by a constant times $p^{-nN} y^{-N}$ for sufficiently large n . Choose N such that $p^N |p^{1/2}/\alpha| > 1$. The bound forces $k(y) = 0$ for every y , hence $k = 0$. If $\alpha = 0$, invertibility of W_p gives the same result. Thus $W_p - \alpha$ is injective, and so is every positive power.

Any equivariant map $A_\rho \rightarrow \mathcal{A}$ has image annihilated by $(W_p - p^\rho)^{m_\rho}$; injectivity forces the map to be zero. A section cannot be zero. This proves the claim and, equivalently, that (AC6.2) has a nonzero equivariant extension class. Its cocycle $W_b s_\rho W_{\rho,b}^{-1} - s_\rho$ cannot be a coboundary for this group action on $\text{Hom}(A_\rho, I_\rho)$, because such a correction would produce the excluded section.

Before this calculation, U10–U14 and the user’s complete-history passages were checked as recorded in AC0. The proof uses the already reconstructed full arithmetic and the original strong test-space definition. It does not add a metric to τ , treat an incomplete prime list as complete, assume that the prime quotient fails, claim that Q has no eigenvectors, or exclude a lift in another receiving completion. Its precise failed map is a section into rapidly decaying tests. The following section constructs the exact receiving space in which the spectral data do live.

121.8 AC7. The corresponding dual distributions and their full-jet map

Let $\mathcal{A}'$ be the continuous complex-linear dual of $\mathcal{A}$. Define, for every original zero and each $0 \leq r < m_\rho$,

$$\lambda_{\rho,r}(k) = \frac{1}{r!} \int_0^\infty k(y) y^{\rho-1/2} (\log y)^r \frac{dy}{y}. \quad (\text{AC7.1})$$

Every such functional is continuous by the estimates in AC5. Its distribution density, including the original exponent $1/2$, is shown explicitly. The surjectivity constructed in AC6 proves that these functionals are linearly independent.

The exact transpose of the observation map is

$$L_\rho^* : A_\rho^* \hookrightarrow \mathcal{A}', \quad \ell \longmapsto \ell \circ L_\rho. \quad (\text{AC7.2})$$

For the coordinate functionals $\ell_r(a) = a_r$, it is precisely $\lambda_{\rho,r}$. It lands in the annihilator of $\overline{I_0}$, by AC5.3. If Q is given its quotient topology, pullback identifies Q' exactly with that annihilator: a continuous functional on $\mathcal{A}$ vanishing on the closed subspace defines a unique functional on the quotient, and the definition of quotient topology proves its continuity. Thus the same exact injection is

$$A_\rho^* \hookrightarrow Q' \hookrightarrow \mathcal{A}'. \quad (\text{AC7.3})$$

It is an injection into the dual of the actual quotient, not an asserted injection into Q itself.

For precision about orientation, the transpose action below is $W_b^{\text{tr}} \lambda = \lambda \circ W_b$. Because the dilation group is abelian, these transposes form an action with the same product law. Differentiating $b^s F_k(s)$, with every factorial retained, gives

$$W_b^{\text{tr}} \lambda_{\rho,r} = b^\rho \sum_{j=0}^r \frac{(\log b)^{r-j}}{(r-j)!} \lambda_{\rho,j}. \quad (\text{AC7.4})$$

This is the complete transpose primary block, including its nilpotent. If the contragredient convention is wanted instead, it is explicitly $\lambda \mapsto \lambda \circ W_{b^{-1}}$, with

$$b^{-\rho} \sum_{j=0}^r \frac{(-\log b)^{r-j}}{(r-j)!} \lambda_{\rho,j}. \quad (\text{AC7.5})$$

Neither sign convention is substituted silently.

Consequently the decay calculation in AC6 identifies the correct receiving distinction and supplies its connecting morphism: original spectral blocks are finite *quotients* of the test space and become finite *subspaces of distributions* by the exact transpose map. The distributional characters are not lost when no test-function eigenvector exists. All original zero observations and multiplicities remain.

121.9 AC8. All support labels and what is not being collapsed

For any bounded distributive lattice L , retain

$$G_L(V) = \{(0, \lambda) : \lambda \in L\} \cup (V \times \{1_L\}).$$

Every already constructed linear map f has the common-label lift

$$(v, \lambda) \mapsto (f(v), \lambda). \quad (\text{AC8.1})$$

This is well-defined: a nontop input label forces zero amplitude and linear maps preserve that amplitude. It preserves receiving addition with join labels, scalar action with meet labels, identities and composition by direct substitution. These receiving operations are not an addition on primitive Z_1/τ .

For the complex AC3, amplitude-zero cycles in degree -1 are exactly $G_L(j\mathcal{B}_{\text{pr}})$. The same-label quotient in degree zero by I_0 is $G_L(C_{\text{alg}})$. The further same-label quotient by D_{cl} is $G_L(Q)$. The maps (AC2.11), (AC3.2) and (AC7.2) lift in this same sense, so their proven amplitude identities preserve every label.

In particular a zero connecting map returns $(0, \lambda)$, not a constant bottom label. The common-label object $G_L(I_0 \oplus \mathcal{B}_{\text{pr}})$ is not silently replaced by the product $G_L(I_0) \times G_L(\mathcal{B}_{\text{pr}})$, which permits independent labels. A construction with independent block labels must display those additional label coordinates separately; none is forgotten here. The inherited ABR notation identifying an older bottom supported-zero label with a symbol tau is not imported as an identification with the current primitive Z_1/τ .

121.10 AC9. Exact conclusions for the requested lifting calculation

The actual adelic two-term complex has the explicit equivariant decomposition (AC3.2). Its endpoint two-extension and the connecting maps furnished by ABR17 vanish, with the original $y^{1/2}$, factor 2, both endpoint characters and every prime label retained. Independently, every finite original nontrivial-zero block has vanishing Hom and all equivariant extension groups to or from the endpoint space, by the computed separation $1 < |p^\rho| < p$. This is a completed weight-separated lifting result on actual programme modules, not the earlier auxiliary polynomial extension.

The calculation also determines the exact scope. The actual Mellin-jet kernel I_ρ is larger than the endpoint boundary and retains the closure difference and other unobserved test data. Its test-space extension (AC6.2) does not have an equivariant section; its full defect and its distributional receiving map are explicitly constructed. This is no statement about whether original Q has eigenvectors in another realization.

The extension splitting and endpoint weight separation hold for every actual zero in the entire open critical strip. They do not narrow that strip to its midpoint. No geometric specialization diagram identifying these particular endpoint modules with Deligne's complete support-obstruction groups has been assumed, and no RH conclusion is claimed. The retained complex, its original cohomology, its closure difference, its exact dual morphism and the surviving source labels are the mathematical output.

121.11 Current propagation from OMS — 24 September 2026

The complete synthesis proof, OMS1–OMS5, proves that the original summation image is closed and that the full actual-zero jet map on Q has zero common kernel. Its image remains a proper dense subspace of the unrestricted product, carrying the stronger original quotient topology. Earlier statements leaving those two questions undetermined are superseded by that proof, not by an assumption. Every original zero multiplicity and support label remains.

OMS7A proves the exact return to AC0–AC4: the same original spaces give $I_0=2 E(S)=E(S)$, with the factor 2 retained in the periodization map and inverse. Thus $D_{cl}=0$, $C_{alg} \rightarrow Q$ is the identity-induced topological isomorphism, and the specific global two-extension class of the actual adelic two-term complex is zero. Both endpoint contributions at each prime and the higher derived-coinvariant groups remain unchanged. No vanishing of the entire Ext group or full tau purity is inferred.

121.12 Global adelic lifting propagation - 24 September 2026

The complete current proofs are `GLOBAL_ADELIC_LIFTING_AND_PRIME_BOUNDARY.md` GAP0–GAP9 (including GAP6A–GAP6B), `COMPLETE_GENERATOR_ENDPOINT_RETURN.md` GER0–GER7, `GLOBAL_ENDPOINT_EXTENSION_RETURN.md` GEX0–GEX9, and `COMPLETE_COMPARISON_TRIANGLE_FORMALITY.md` CTF0–CTF8. Every original prime wedge, endpoint pair and nilpotent jet is retained. GER credits the existing RZ endpoint resolvents and gives their explicit original Schwartz return and covariance corrections. GEX proves vanishing of the entire extension groups between the whole actual Q and the stated endpoint modules, in the explicitly constructed algebraic and strict locally convex operator categories. This strengthens the earlier finite-block and particular-extension assertions without assigning that theorem to a larger unspecified category.

GAP and CTF construct the full derived adelic complex, its canonical cohomology map, the exact original CC comparison and its entire cone. The faithful coefficient copies distinguish primitive tau from integer one in both objects. All additional cone contractions and the final-arrow homotopy are explicit; no test-space section $Q \rightarrow A$ is asserted. On the separate supported- Q row the same polynomial is invertible, rather than zero. Full tau numerical purity does not follow from endpoint separation and remains unproved.

122 Exact distribution realization of the two-nilpotent object and its rational orbit

Independent mathematical derivation, 24 September 2026.

This calculation realizes the entire existing auxiliary object B_ρ inside the existing supported-distribution receiver. It computes both original Dolbeault residues, constructs the exact quotient comparison, and proves uniqueness of the equivariant lift on the entire finite rational-translation orbit, including algebraic sums over all original zero blocks. It does not identify that orbit with the complete adelic quotient.

122.1 DR0. Source, definitions, and operation domains

Human source: Alain Connes and Caterina Consani, *The Riemann–Roch strategy: Complex lift of the Scaling Site*, arXiv:1805.10501v1. The original author source read for this calculation is

`thecurve_K.tex`, original author source retained under `sources/`.

Section 5.2, Lemma `adelicomp1` and equation `actionpq`, lines 1203–1224, gives the affine action and original operator. Section 6.6, the paragraph before Proposition `comparescales`, lines 2381–2418, gives the two original vector fields and their full Y factor. Section 7.1, equations `holom`, `holombis`, `holom1` and Proposition `frobarith`, lines 2564–2601, concerns the separate arithmetic Frobenius combining a right action and coefficient change. That last arithmetic Frobenius is not identified with the raw left affine pullback calculated here.

The existing programme sources are:

- Supported Dolbeault receiver, equations SDB1–SDB31 and their proofs.
- Existing two-nilpotent comparison, MJ4.1–MJ5.4.
- Existing equivariant lifting calculation, TL2.1–TL3.3, in particular the quotient $q_\rho(x_\rho) = t_\rho$, $q_\rho(y_\rho) = 0_{A_\rho}$.
- User definitions U01–U18 and operation domains, read before defining the present maps.

The present operations are in the complex arithmetic receivers already specified in those sources. In particular, Z_1/τ remains presence without source parity or source addition. No polynomial, derivative, scalar subtraction, branch coordinate, or residue below has primitive τ as an operand. The origin $X = 0$ here is an archimedean coordinate in Connes–Consani’s explicitly stated receiver, not an identification of that coordinate with the source Z_1 . All statements involving an actual zero ρ use the original $\zeta(\rho) = 0$, not a replacement zeta function.

Use capital X, Y for the original archimedean coordinates, to distinguish them from the algebra variables x, y of B_ρ . Write

$$\Omega = \mathbb{R} \times (0, \infty), \quad D = \partial_X + i\partial_Y, \quad D_Y = YD. \quad (\text{DR0.1})$$

There is no factor $1/2$ in D . All distributions are complex scalar distributions on this open set. In particular there is no extension across $Y = 0$, and no discarded endpoint distribution.

Fix $\rho \in \mathbb{C}$, an integer $m \geq 1$, and put

$$\alpha = \rho + 1, \quad A_\rho = \mathbb{C}[t]/t^m, \quad B_\rho = \mathbb{C}[x, y]/(x, y)^m. \quad (\text{DR0.2})$$

The existing operators and maps are

$$J = M_x, \quad N = M_y, \quad F_a = a^\rho e^{(\log a)J} D_a, \quad D_a(x) = x, \quad D_a(y) = a^{-1}y, \quad a > 0, \quad (\text{DR0.3})$$

$$q_\rho(x) = t, \quad q_\rho(y) = 0_{A_\rho}, \quad i_\rho(t) = x, \quad K_\rho = \ker q_\rho = yB_\rho. \quad (\text{DR0.4})$$

All exponentials in these finite algebras terminate. These formulas are restated from the existing receiver, not definitions of arithmetic from primitive τ .

122.2 DR1. The distribution embedding and both nilpotents

For $r \geq 0$, define the smooth function on $Y > 0$

$$g_r(Y) = Y^\alpha \frac{(\log Y)^r}{r!}, \quad Y^\alpha = \exp(\alpha \log Y), \quad g_{-1} = 0. \quad (\text{DR1.1})$$

For each rational number c , define

$$\boxed{\Phi_c(x^u y^j) = \delta^{(j)}(X - c) \otimes g_{m-j-1-u}(Y), \quad u, j \geq 0, \quad u + j < m.} \quad (\text{DR1.2})$$

The tensor distribution has the full pairing

$$\langle \delta^{(j)}(X - c) \otimes g, \varphi \rangle = (-1)^j \int_0^\infty g(Y) \partial_X^j \varphi(c, Y) dY. \quad (\text{DR1.3})$$

The test function is compactly supported in Ω , so the integral is finite for every ρ , including near the excluded endpoint. No asymptotic integrability at $Y = 0$ or $Y = \infty$ has been assumed.

Formula (DR1.2), extended linearly from the monomial basis of B_ρ , is injective. Different j are independent normal derivatives: a test function supported near c , of the form $(X - c)^j \chi(X) \psi(Y) / j!$, isolates the j -coefficient as in SDB3–SDB6. At fixed j , the functions $g_0, \dots, g_{m-j-1}$ are linearly independent. Indeed division by the nonzero function Y^α , followed by $s = \log Y$, gives a polynomial identity on all real s , which forces every coefficient to vanish.

On distributions define the explicitly named operators

$$\mathcal{J} = Y \partial_Y - \alpha, \quad \mathcal{N} = \partial_X \mathcal{J}. \quad (\text{DR1.4})$$

The scalar $-\alpha$ is part of the operator; it is not removed from the formulas. Differentiation gives

$$\mathcal{J} g_r = g_{r-1}, \quad \mathcal{N}(\delta_c^{(j)} \otimes g_r) = \delta_c^{(j+1)} \otimes g_{r-1}, \quad (\text{DR1.5})$$

where $\delta_c^{(j)} = \delta^{(j)}(X - c)$. Therefore

$$\mathcal{J} \Phi_c = \Phi_c \mathcal{J}, \quad \mathcal{N} \Phi_c = \Phi_c \mathcal{N}. \quad (\text{DR1.6})$$

For $u + j = m - 1$, the right sides vanish because the degree- m monomials vanish, and the left sides vanish because $g_{-1} = 0$. Thus these identities also prove the boundary cases, including $m = 1$. No order term has been lost. The operators $\mathcal{J}, \mathcal{N}$ commute on this image because ∂_X commutes with $Y \partial_Y - \alpha$.

This is an embedding of the specified vector space with its named operators. It is not an algebra homomorphism into an algebra of distributions: arbitrary products of the displayed Dirac derivatives are not being defined.

122.3 DR2. Full rational affine action and all Jacobians

For $a > 0$ and $b \in \mathbb{R}$, let

$$\ell(a, b)(X, Y) = (aX + b, aY).$$

Scalar-distribution pullback, in the exact source convention, is

$$\langle P(a, b)u, \varphi \rangle = a^{-2} \left\langle u, \varphi \left(\frac{X - b}{a}, \frac{Y}{a} \right) \right\rangle. \quad (\text{DR2.1})$$

The factor a^{-2} is the inverse determinant of the two-dimensional map; it has not been absorbed into the distributions. For one variable put $Q_a g(Y) = g(aY)$, with distributional definition

$$\langle Q_a g, \psi \rangle = a^{-1} \langle g, \psi(\cdot/a) \rangle. \quad (\text{DR2.2})$$

Direct substitution into (DR1.3) gives

$$P(a, b)(\delta_c^{(j)} \otimes g) = a^{-j-1} \delta_{(c-b)/a}^{(j)} \otimes Q_a g. \quad (\text{DR2.3})$$

All signs in the derivative of the delta distribution are those of (DR1.3); since $a > 0$, no sign from an absolute determinant is omitted.

The full coefficient calculation is

$$Q_a g_r = a^\alpha \sum_{k=0}^r \frac{(\log a)^k}{k!} g_{r-k}, \quad (\text{DR2.4})$$

because

$$\frac{(\log(aY))^r}{r!} = \sum_{k=0}^r \frac{(\log a)^k (\log Y)^{r-k}}{k!(r-k)!}.$$

Combining (DR2.3) and (DR2.4), with $r = m - j - 1 - u$, proves

$$P(a, b)\Phi_c(x^u y^j) = a^{\rho-j} \sum_{k=0}^{m-j-1-u} \frac{(\log a)^k}{k!} \Phi_{(c-b)/a}(x^{u+k} y^j). \quad (\text{DR2.5})$$

For $b = 0, c = 0$, this is exactly

$$P(a, 0)\Phi_0 = \Phi_0 F_a, \quad (\text{DR2.6})$$

with the complete original F_a , not its scalar part.

For $a \in \mathbb{Q}_{>0}$ and $b \in \mathbb{Q}$, the support $(c - b)/a$ stays rational. Define algebraic direct sums

$$B_{\rho, \mathbb{Q}} = \bigoplus_{c \in \mathbb{Q}} B_{\rho, c}, \quad A_{\rho, \mathbb{Q}} = \bigoplus_{c \in \mathbb{Q}} A_{\rho, c}. \quad (\text{DR2.7})$$

Every vector has finite branch support. Define $\mathcal{P}(a, b)$ on the first space by the right side of (DR2.5), with Φ replaced by the branch basis notation. Define its action on $A_{\rho, \mathbb{Q}}$ by

$$t_c^u \mapsto a^\rho \sum_{k=0}^{m-1-u} \frac{(\log a)^k}{k!} t_{(c-b)/a}^{u+k}. \quad (\text{DR2.8})$$

These are invertible finite sums.

Pullback is contravariant. For the source affine multiplication

$$(a, b)(a', b') = (aa', ab' + b),$$

the exact composition law is

$$P(a, b)P(a', b') = P(a'a, a'b + b'). \quad (\text{DR2.9})$$

For instance the support changes first to $(c - b')/a'$ and then to $(c - b' - a'b)/(a'a)$, which is the support on the right. The polynomial coefficients multiply by $F_a F_{a'} = F_{aa'}$, giving the same law for $\mathcal{P}$. Thus these are right actions of $P^+(\mathbb{Q})$, or equivalently left actions after replacing (a, b) by its inverse $(a^{-1}, -b/a)$. No switch of convention is hidden.

The sum

$$\Phi_{\mathbb{Q}}((b_c)_c) = \sum_c \Phi_c(b_c) \quad (\text{DR2.10})$$

is an injective distribution map and intertwines these right actions. To prove injectivity, a finite support set can be separated by disjoint X -neighborhoods, so each branch is isolated by a test

function; then DR1 gives injectivity within that branch. This proof uses finite branch support. It does not assert that every infinite sum on the dense set $\mathbb{Q}$ defines a distribution.

The quotient and section extend componentwise:

$$q_{\rho, \mathbb{Q}} = \bigoplus_c q_{\rho, c}, \quad i_{\rho, \mathbb{Q}} = \bigoplus_c i_{\rho, c}, \quad K_{\rho, \mathbb{Q}} = \bigoplus_c y B_{\rho, c}. \quad (\text{DR2.11})$$

Equations (DR2.5) and (DR2.8) prove that both maps commute with the full rational affine action: the terms with $j > 0$ remain in the kernel, and the terms with $j = 0$ have identical coefficients in both formulas. Hence

$$0 \longrightarrow K_{\rho, \mathbb{Q}} \longrightarrow B_{\rho, \mathbb{Q}} \xrightarrow{q_{\rho, \mathbb{Q}}} A_{\rho, \mathbb{Q}} \longrightarrow 0 \quad (\text{DR2.12})$$

is an equivariantly split exact sequence over the complete specified rational orbit.

122.4 DR3. The original distribution complex on the entire rational orbit

Extend the existing SDB space to the explicitly defined algebraic direct sum

$$\mathcal{J}_{\mathbb{Q}} = \bigoplus_{c \in \mathbb{Q}} \bigoplus_{j \geq 0} \delta_c^{(j)} \otimes \mathcal{D}'((0, \infty)). \quad (\text{DR3.1})$$

Every element has finitely many branches and finitely many normal derivatives in each branch. The distribution embedding is injective by the same test-function argument as in DR2. The original D and D_Y act branchwise. Define the full, branch-retaining residues

$$\begin{aligned} (R_{\mathbb{Q}} u)_c &= \sum_j (-i)^j \partial_Y^j g_{c, j}, \\ (R_{Y, \mathbb{Q}} u)_c &= \sum_j (-i)^j \partial_Y^j (Y^{-1} g_{c, j}), \end{aligned} \quad (\text{DR3.2})$$

for $u = \sum_{c, j} \delta_c^{(j)} \otimes g_{c, j}$, with targets $\bigoplus_c \mathcal{D}'((0, \infty))$. These formulas retain the branch label c . Taking a sum of the outputs over c would be another map and would forget branches.

SDB8–SDB31 apply to each branch by replacing X with $X - c$. To make the exactness explicit, put

$$\begin{aligned} E_{\mathbb{Q}}(h) &= \sum_c \delta_c \otimes h_c, & E_{Y, \mathbb{Q}}(h) &= \sum_c \delta_c \otimes Y h_c, \\ T_{\mathbb{Q}} u &= \sum_c \sum_{j \geq 1} \sum_{k=0}^{j-1} (-i)^k \delta_c^{(j-1-k)} \otimes \partial_Y^k g_{c, j}, & T_{Y, \mathbb{Q}} &= T_{\mathbb{Q}} M_{Y^{-1}}. \end{aligned} \quad (\text{DR3.3})$$

All sums are finite. Applying D to a term yields

$$D(\delta_c^{(j)} \otimes g) = \delta_c^{(j+1)} \otimes g + i \delta_c^{(j)} \otimes g', \quad (\text{DR3.4})$$

so $R_{\mathbb{Q}} D = 0$. In $DT_{\mathbb{Q}}$, consecutive terms cancel by $(-i)^k + i(-i)^{k-1} = 0$, leaving

$$DT_{\mathbb{Q}} = I - E_{\mathbb{Q}} R_{\mathbb{Q}}. \quad (\text{DR3.5})$$

The highest normal derivative in Du has the same nonzero coefficient as the highest derivative in u , raised by one, so D is injective. Applying (DR3.5) to Du then proves $T_{\mathbb{Q}} D = I$. Also $R_{\mathbb{Q}} E_{\mathbb{Q}} = I$. It follows that

$$0 \longrightarrow \mathcal{J}_{\mathbb{Q}} \xrightarrow{D} \mathcal{J}_{\mathbb{Q}} \xrightarrow{R_{\mathbb{Q}}} \bigoplus_{c \in \mathbb{Q}} \mathcal{D}'((0, \infty)) \longrightarrow 0 \quad (\text{DR3.6})$$

is exact, with the stated splitting. Multiplication by Y is invertible on this open domain. Retaining that multiplication in the order $D_Y = M_Y D$ proves the corresponding exact sequence with $D_Y, R_{Y, \mathbb{Q}}, E_{Y, \mathbb{Q}}, T_{Y, \mathbb{Q}}$. This is precisely the branchwise version of SDB27–SDB31.

The complete affine covariance is

$$DP(a, b) = aP(a, b)D, \quad D_Y P(a, b) = P(a, b)D_Y. \quad (\text{DR3.7})$$

On residues, the c -component moves to $c' = (c - b)/a$, and

$$\begin{aligned} (R_{\mathbb{Q}}P(a, b)u)_{c'} &= a^{-1}Q_a(R_{\mathbb{Q}}u)_c, \\ (R_{Y, \mathbb{Q}}P(a, b)u)_{c'} &= Q_a(R_{Y, \mathbb{Q}}u)_c. \end{aligned} \quad (\text{DR3.8})$$

For the first formula, each j -term contributes $a^{-j-1}(-i)^j \partial_Y^j Q_a g = a^{-1}Q_a((-i)^j g^{(j)})$. For the second, apply the first formula to $Y^{-1}P(a, b)u = aP(a, b)(Y^{-1}u)$. These derivations retain both Jacobians and every derivative.

The original cochain actions must also be retained:

$$\rho^0(a, b) = P(a, b), \quad \rho^1(a, b) = aP(a, b) \quad \text{for } D; \quad (\text{DR3.9})$$

both degrees carry $P(a, b)$ for D_Y . Accordingly $R_{\mathbb{Q}}$ intertwines ρ^1 with Q_a , whereas its raw-pullback quotient intertwines $P(a, b)$ with $a^{-1}Q_a$. There is no replacement of one action by the other. Formula (DR2.6) realizes F_a by **raw** $P(a, 0)$. In degree 1 of the original D -complex it consequently realizes the operator aF_a . In the D_Y -complex it realizes F_a in both degrees.

122.5 DR4. Every residue coefficient, and exact injectivity

For integers $j \geq 0$, define the full polynomial

$$P_j(z) = \prod_{k=0}^{j-1} (z - k), \quad P_0(z) = 1.$$

Write

$$P_j(\alpha + z) = \sum_{h=0}^j C_{j,h}(\alpha) z^h, \quad C_{j,h}(\alpha) = \sum_{\substack{S \subseteq \{0, \dots, j-1\} \\ |S|=h}} \prod_{k \notin S} (\alpha - k). \quad (\text{DR4.1})$$

The empty product is 1; in particular $C_{0,0} = 1$. For $s = \log Y$,

$$\partial_Y^j (Y^\alpha f(s)) = Y^{\alpha-j} P_j(\alpha + \partial_s) f(s). \quad (\text{DR4.2})$$

This follows by induction: one derivative changes the exponent from $\alpha - j$ to $\alpha - j - 1$ and applies $\alpha - j + \partial_s$ to the polynomial. Thus for the exact $r = m - j - 1 - u$,

$$\boxed{R\Phi_c(x^u y^j) = (-i)^j Y^{\rho+1-j} \sum_{h=0}^{\min(j,r)} C_{j,h}(\rho+1) \frac{(\log Y)^{r-h}}{(r-h)!}.} \quad (\text{DR4.3})$$

The notation R on the left means the c -component of $R_{\mathbb{Q}}$; the branch label is retained. Similarly,

$$\boxed{R_Y \Phi_c(x^u y^j) = (-i)^j Y^{\rho-j} \sum_{h=0}^{\min(j,r)} C_{j,h}(\rho) \frac{(\log Y)^{r-h}}{(r-h)!}.} \quad (\text{DR4.4})$$

This is obtained by multiplying the coefficient in (DR1.2) by Y^{-1} **before** differentiating. It is not obtained by dropping a Y factor after the derivative.

Here is the full injectivity argument. At fixed j , the polynomial operator $P_j(\alpha + \partial_s)$ on polynomials of degree at most $m - j - 1$ is triangular with diagonal coefficient $P_j(\alpha)$. It is invertible precisely when $P_j(\alpha) \neq 0$. An exact inverse when this coefficient is nonzero is the finite operator

$$\sum_{n=0}^{m-j-1} \frac{(-1)^n}{P_j(\alpha)^{n+1}} (P_j(\alpha + \partial_s) - P_j(\alpha)I)^n. \quad (\text{DR4.5})$$

The parenthesized operator decreases polynomial degree, so its $(m - j)$ -th power is zero. Multiplication by $P_j(\alpha + \partial_s)$ telescopes to the identity, proving the inverse.

Different j give distinct exponents $\alpha - j$. The spaces

$$e^{(\alpha-j)s} \mathbb{C}[s]_{\leq m-j-1} \quad (\text{DR4.6})$$

are linearly independent. To verify this for a finite sum, choose one index j and apply

$$\prod_{h \neq j} (\partial_s - (\alpha - h))^{m-h}.$$

This kills every summand of index $h \neq j$. On the j -summand it is a polynomial in ∂_s with nonzero constant term after factoring out $e^{(\alpha-j)s}$, hence invertible on the finite polynomial space by the same finite inverse argument. The chosen summand must therefore be zero.

It follows that $R\Phi_c$ is injective whenever

$$\rho + 1 \notin \{0, 1, \dots, m - 2\}. \quad (\text{DR4.7})$$

The displayed set is empty when $m = 1$. This is also a complete classification of the exceptional kernel. If $\alpha = n$ is in that set, then for $j > n$ exactly one factor of $P_j(\alpha + \partial_s)$ is ∂_s ; all other factors are invertible on the polynomial space. Its kernel is precisely the constants. For $j \leq n$ it is invertible. Therefore

$$\ker(R\Phi_c) = \text{span}\{x^{m-j-1}y^j : n + 1 \leq j \leq m - 1\} \quad (\rho + 1 = n \in \{0, \dots, m - 2\}). \quad (\text{DR4.8})$$

Outside those exceptional values the kernel is zero. The same proof for (DR4.4) gives $R_Y\Phi_c$ injective whenever

$$\rho \notin \{0, \dots, m - 2\}, \quad (\text{DR4.9})$$

and the corresponding kernel (DR4.8) with $n = \rho$.

Every actual nontrivial zero of the original zeta function has $0 < \Re \rho < 1$. Both conditions (DR4.7) and (DR4.9) therefore hold, including for zeros off the critical line. Hence both maps

$$\begin{aligned} B_{\rho, \mathbb{Q}} &\xrightarrow{\Phi_{\mathbb{Q}}} \mathcal{J}_{\mathbb{Q}} \xrightarrow{R_{\mathbb{Q}}} \bigoplus_c \mathcal{D}'((0, \infty)), \\ B_{\rho, \mathbb{Q}} &\xrightarrow{\Phi_{\mathbb{Q}}} \mathcal{J}_{\mathbb{Q}} \xrightarrow{R_{Y, \mathbb{Q}}} \bigoplus_c \mathcal{D}'((0, \infty)) \end{aligned} \quad (\text{DR4.10})$$

are injective for actual zero blocks. By (DR3.6), no nonzero vector of this embedded $B_{\rho, \mathbb{Q}}$ is a D -boundary; the analogous assertion holds for D_Y .

122.6 DR5. The exact relation with the auxiliary quotient

The result (DR4.10) implies

$$\ker(R\Phi_c) = 0, \quad \ker q_{\rho} = yB_{\rho} \quad (\text{DR5.1})$$

for actual zeros. At $m > 1$, these kernels differ. This does not end the comparison: the exact factorization of q_{ρ} through the residue image is as follows.

Let

$$\mathcal{V}_{\rho,c} = R\Phi_c(B_\rho) = \bigoplus_{j=0}^{m-1} Y^{\alpha-j} \mathbb{C}[\log Y]_{\leq m-j-1}. \quad (\text{DR5.2})$$

Equality, including surjectivity on every summand, follows from (DR4.3) and (DR4.5). Put $E = Y\partial_Y$ on smooth functions, retaining the exact exponents $\alpha - j$. For $m > 1$, define

$$G(z) = \prod_{j=1}^{m-1} (z - (\alpha - j))^{m-j},$$

$$H(z) = \sum_{r=0}^{m-1} \frac{1}{r!} \frac{d^r}{dw^r} \frac{1}{G(w)} \Big|_{w=\alpha} (z - \alpha)^r, \quad \Pi_0 = G(E)H(E). \quad (\text{DR5.3})$$

All these coefficients are defined: $G(\alpha) = \prod_{j=1}^{m-1} j^{m-j} \neq 0$. For $m = 1$, put $G = H = 1$, so $\Pi_0 = I$. On the $j = 0$ summand, $(E - \alpha)^m = 0$; Taylor's formula for $1/G$ shows that $G(E)H(E) = I$. On each $j > 0$ summand, $G(E) = 0$ by its explicit factor of order $m - j$. Thus Π_0 is precisely the projection onto $j = 0$, with kernel all $j > 0$ summands.

Define the isomorphism onto that zeroth summand by

$$\mathcal{J}_{\rho,c} : A_{\rho,c} \longrightarrow Y^\alpha \mathbb{C}[\log Y]_{\leq m-1}, \quad \mathcal{J}_{\rho,c}(t^u) = Y^\alpha \frac{(\log Y)^{m-1-u}}{(m-1-u)!}. \quad (\text{DR5.4})$$

It is $R\Phi_c i_\rho$, and its inverse reads off the displayed polynomial coefficients in reverse order, retaining each factorial. Now put

$$\mathcal{Q}_{\rho,c} = \mathcal{J}_{\rho,c}^{-1} \Pi_0 : \mathcal{V}_{\rho,c} \longrightarrow A_{\rho,c}.$$

The complete comparison is

$$\boxed{q_\rho = \mathcal{Q}_{\rho,c} R\Phi_c, \quad \mathcal{Q}_{\rho,c} \mathcal{J}_{\rho,c} = I.} \quad (\text{DR5.5})$$

It follows on each basis monomial: at $j = 0$ it returns t^u ; at $j > 0$ it returns zero. The maps and their kernels are therefore proved on the whole object.

The same construction for R_Y replaces $\alpha = \rho + 1$ in (DR5.2)–(DR5.4) by ρ , giving

$$q_\rho = \mathcal{Q}_{Y,\rho,c} R_Y \Phi_c. \quad (\text{DR5.6})$$

Its codomain exponent remains ρ ; this change is forced by the already retained Y^{-1} in (DR4.4).

All these maps extend to the rational branch direct sums and are equivariant. Indeed E commutes with Q_a , and translations only permute the branch labels. Therefore Π_0 commutes with the target action $a^{-1}Q_a$ for raw R , and with Q_a for R_Y . The maps $\mathcal{J}$ intertwine those exact actions with (DR2.8), as follows either by substituting their full logarithm expansions or by (DR2.6), (DR3.8).

The $j > 0$ pieces are thus genuine, independent residue classes. They disappear under the explicit spectral projection Π_0 , not under the SDB residue or differential. The auxiliary quotient has now been constructed as an exact map on the actual residue image; it has not been identified with a boundary operation.

122.7 DR6. Uniqueness of the lift on the whole rational orbit

We prove

$$\text{Hom}_{P+(\mathbb{Q})}(A_{\rho,\mathbb{Q}}, K_{\rho,\mathbb{Q}}) = 0. \quad (\text{DR6.1})$$

The Hom means complex-linear maps commuting with the right actions in DR2; it does not impose an unmentioned topology.

Let f be such a map, and restrict it to the finite-dimensional branch $A_{\rho,0}$. The image of a finite basis has a finite union $S \subset \mathbb{Q}$ of branch supports. Every vector of $f(A_{\rho,0})$ has support in S . Conversely choose S as the union of all nonzero component positions of the image, which is still finite. Since $A_{\rho,0}$ is preserved by every positive rational dilation and that action is invertible, equivariance makes the image invariant under those dilations. Formula (DR2.5) sends branch c to c/a by an invertible coefficient operator, so $S/a = S$ for all $a \in \mathbb{Q}_{>0}$. If S contained a nonzero c , it would contain all $c/2^n$, $n \geq 0$, which are distinct. Finiteness excludes this. Thus $S \subseteq \{0\}$.

It remains to calculate the coefficient map $f : A_{\rho,0} \rightarrow K_{\rho,0}$. For any prime p , its source action has only eigenvalue p^ρ ; its target action has eigenvalues

$$p^{\rho-j}, \quad 1 \leq j \leq m-1, \quad (\text{DR6.2})$$

with full Jordan lengths $m-j$. None equals p^ρ , because the positive real number p^{-j} is not 1. An intertwiner between two finite-dimensional operators with disjoint spectra is zero: their minimal polynomials are coprime; a Bézout identity, evaluated on the two operators and moved across the intertwiner, annihilates it. This argument does not assume semisimplicity. Therefore $f|_{A_{\rho,0}} = 0$. Finally $P(1, -c)$ maps $A_{\rho,0}$ isomorphically to $A_{\rho,c}$. Translation equivariance proves $f = 0$ on every branch and hence on the direct sum. This proves (DR6.1).

Any two equivariant sections of $q_{\rho,\mathbb{Q}}$ differ by a map into its kernel. Equation (DR6.1), together with the existing section (DR2.11), proves that

$$\boxed{i_{\rho,\mathbb{Q}} \text{ is the unique } P^+(\mathbb{Q})\text{-equivariant section of } q_{\rho,\mathbb{Q}}.} \quad (\text{DR6.3})$$

The conclusion includes $m = 1$, where the kernel is zero.

There is also an exact simultaneous result allowing mixing of different original zeros. Let $\mathcal{Z}$ be any set of complex numbers with $0 < \Re \rho < 1$, each with specified finite positive integer m_ρ ; it may in particular be the distinct nontrivial zeros of the original ζ , with their actual multiplicities. Form algebraic direct sums

$$\mathcal{A} = \bigoplus_{\rho \in \mathcal{Z}} A_{\rho,\mathbb{Q}}, \quad \mathcal{B} = \bigoplus_{\rho \in \mathcal{Z}} B_{\rho,\mathbb{Q}}, \quad \mathcal{K} = \bigoplus_{\rho \in \mathcal{Z}} K_{\rho,\mathbb{Q}}. \quad (\text{DR6.4})$$

Every vector has finite ρ -support and finite branch support. Actions preserve each ρ -summand. Any equivariant map $f : \mathcal{A} \rightarrow \mathcal{K}$, even if it is not required to preserve ρ , is zero.

The distribution and residue maps also retain this complete algebraic sum. For Φ , normal derivative order and branch are first isolated by test functions. At each remaining index, finitely many distinct ρ give independent functions $Y^{\rho+1}$ times polynomials in $\log Y$, by the differential-operator proof in DR4. For the residue images, equality of two exponents would imply $\rho - j = \rho' - j'$. Taking real parts and using $0 < \Re \rho, \Re \rho' < 1$ forces the integer $j - j'$ to be zero, hence $j = j'$ and $\rho = \rho'$. Thus the residue exponents are distinct across different (ρ, j) . The same finite exponential-polynomial argument proves that both residue maps are injective on $\mathcal{B}$. This proves actual distribution realization of the whole specified algebraic sum, not only of each block separately.

To prove the asserted vanishing of equivariant maps, restrict to a fixed finite-dimensional $A_{\rho,0}$. The preceding support argument forces its image to branch 0. It has components in only finitely many target ρ' -summands because a finite basis has finite images. Project to any one of these summands; projection commutes with dilation. For every prime p , the source eigenvalue has modulus

$$|p^\rho| = p^{\Re \rho} > 1,$$

whereas every target kernel eigenvalue satisfies

$$|p^{\rho'-j}| = p^{\Re \rho' - j} < 1, \quad j \geq 1. \quad (\text{DR6.5})$$

The same coprime-polynomial argument kills every component. Translations then kill the image of every branch. Since the domain is the algebraic direct sum, this proves

$$\boxed{\mathrm{Hom}_{P^+(\mathbb{Q})}(\mathcal{A}, \mathcal{K}) = 0.} \quad (\text{DR6.6})$$

Consequently the direct sum of $i_{\rho, \mathbb{Q}}$ is the unique equivariant section of the direct sum quotient. This separation uses the known open critical strip, not RH, and retains every nilpotent block. There is no interchange of an infinite sum and an analytic limit in the proof. No corresponding result for an unspecified completion or product is inferred.

122.8 DR7. Exact limits of the complex comparison

The inclusion Φ_c realizes the two auxiliary nilpotents as actual operators on distributions, but those named operators are not the original Dolbeault differential. Their commutators on the full distribution space are

$$[D, \mathcal{J}] = i\partial_Y, \quad [D, \mathcal{N}] = i\partial_X\partial_Y. \quad (\text{DR7.1})$$

The first identity follows from $[\partial_Y, Y\partial_Y] = \partial_Y$; the second follows because ∂_X commutes with $D, \mathcal{J}$. Thus neither is asserted to be a cochain endomorphism of the entire D -complex using the same action in both degrees.

In particular $\mathcal{J}$ need not preserve its boundaries. From $D\mathcal{J} - \mathcal{J}D = i\partial_Y$ and $RD = 0$,

$$R\mathcal{J}D = -i\partial_Y R. \quad (\text{DR7.2})$$

For example $u = \delta_c \otimes Y$ has $Ru = Y$, so $R\mathcal{J}Du = -i \neq 0$. Likewise

$$R\mathcal{N}D = -iR\partial_X\partial_Y = -\partial_Y^2 R, \quad (\text{DR7.2a})$$

where $R\partial_X = -i\partial_Y R$ follows directly from the residue formula. Taking $u = \delta_c \otimes Y^2$ gives $R\mathcal{N}Du = -2 \neq 0$. These are smooth coefficients defining distributions on the open domain, so no compact support of the coefficients themselves is required. This is an explicit domain calculation, not an assertion that the constructions are unrelated. On the injective finite cohomology image (DR4.10), the $\mathcal{J}, \mathcal{N}$ actions do transport through $R\Phi_c$ and $R_Y\Phi_c$, because these maps are isomorphisms onto their stated images. Equations DR1, DR4 and DR5 give those exact comparison maps. They do not extend that transported action to all cohomology without a separately defined operation.

Likewise $\Phi_c(B_\rho)$ itself is generally not a subcomplex for D . Its complete derivative is

$$D\Phi_c(x^u y^j) = \delta_c^{(j+1)} \otimes Y^\alpha \frac{(\log Y)^r}{r!} + i\delta_c^{(j)} \otimes Y^{\alpha-1} \left(\alpha \frac{(\log Y)^r}{r!} + \frac{(\log Y)^{r-1}}{(r-1)!} \right), \quad (\text{DR7.3})$$

where the final term is zero when $r = 0$. All terms remain inside $\mathcal{J}_\mathbb{Q}$. In the top y -degree, the first term already has normal derivative order m , outside the image of B_ρ , and is nonzero. The D_Y formula is Y times the whole right side, retaining that factor on both terms.

Finally, the rational orbit extension removes the earlier restriction to the single branch $X = 0$ for the **positive rational affine action**. It does not include finite adele variables, the negative- a component, endpoint strata, infinite branch completions, or the separate right-action arithmetic Frobenius of the human source. Those are explicit scope boundaries of this constructed receiver, not statements that those objects cannot be related. The exact achieved connection is (DR2.10), the exact residue comparison is (DR5.5)–(DR5.6), and the global-within-this-receiver lifting result is (DR6.3)–(DR6.6).

122.9 DR8. Retained support indices

For any specified family of existing support indices λ , each amplitude map above has the index-preserving lift

$$(v, \lambda) \longmapsto (f(v), \lambda). \quad (\text{DR8.1})$$

If the source is the historical lattice carrier $G_L(V) = \{(0, \lambda) : \lambda \in L\} \cup (V \times \{\top\})$, well-definedness follows from $f(0) = 0$: a nontop label has zero amplitude, which remains zero. Linearity proves compatibility with the already defined lattice receiver's operations, exactly as in SDB34–SDB38. Injective maps remain injective because both the amplitude and the unchanged label can be recovered. Every composition identity retains the same label, including when an amplitude vanishes.

This statement concerns z_λ in that existing receiving carrier. It does not rename any z_λ as primitive τ , and does not reinstate the retracted source addition. The rational branch index c , the zero index ρ , and any support index λ are distinct retained data in these maps.

123 Tau lifting: the finite boundary, the global quotient of lifts, and the original zeta receiver

24 September 2026. Complete proofs FM0–FM7, ST0–ST10, CW0–CW7, FA0–FA8 and FB0–FB9 follow this entry. The earlier reader, *Tau lifting and weight separation*, retains TL, WL, MT, AC and DR in full. These two readers form successive mathematical parts; this addition does not replace the earlier proofs.

123.1 GL0. The controlling source and the actual target

The requested target is Deligne’s vanishing of an obstruction to lifting through separation of its weights from those of the lifted classes, proved for the user’s actual tau programme. The complete preserved user corpus and its corrections govern the operation prerequisites. In particular, source Z_0 is absence, $0 = e = \emptyset$, without source parity; Z_1 is presence τ , without Z_2 parity or addition; and the integer layer retains its amounts and Z_2 data. The withdrawn equation about adding tau is not used. No coefficient, coordinate, weight or multiplicity below is assigned to primitive tau.

The complete global arguments consulted are corpus records USR-9ad1c0a2d09dba92, USR-f55d16feb948d8c2 and USR-6152e3bc6302258c. They ask for reconstruction of the entire arithmetic, globally compatible quotients and an exact Deligne comparison. Their original text remains in the private corpus; these descriptions are navigation, not substitute definitions. The operation rules and individual proofs give the relevant prerequisites before each operation. Human authorship of the cited constructions remains with Pierre Deligne, Alain Connes, Caterina Consani, and the other named authors.

Deligne’s specialization argument uses a geometric exact sequence whose source classes have weights at most i , whereas its support obstruction has weights at least $i+1$. Frobenius equivariance then forces that connecting map to vanish. The original hypotheses, duality, Tate twist and exact sequence are retained in TL1, with the source locators §§3.6.1–3.6.3, printed pp. 212–214. The French TeX used for that reading is a transcription, not established original author TeX. Deligne, *La conjecture de Weil : II*.

The new results below construct actual receiving maps and a quotient of their complete calculated ambiguity. They do not identify the full tau geometric specialization with Deligne’s map, and they do not assert the requested numerical purity result as proved.

123.2 GL1. What the retained source multiplication itself proves

ST starts from the current source multiplication, including its distinct global unit τ and integer idempotent 1. No source addition involving tau is introduced. Multiplication by the already supplied integer 1 gives the exact retraction

$$c : S \longrightarrow (\mathbb{Z}, \cdot), \quad c(s) = 1 \cdot s.$$

Its sole non-singleton fibre is $\{\tau, 1\}$. This is an explicit loss of source distinction under a map with integer input, not an identification of source tau and integer one. ST1 proves both the multiplicative retraction and its localization interpretation.

For a stated linear receiving action, let E be the action of integer 1. Then $E^2 = E$, and ST5–ST7 prove

$$V = \ker E \oplus EV, \quad A_n = nE \quad (n \in \mathbb{Z}).$$

The unit of the receiving endomorphism ring is the image of the source global multiplicative unit; it is not assigned back to source tau as a numerical value. The universal receiving ring, all source images and the exact injection of the multiplicative source are constructed in ST6. Its central idempotent separates the two receiving summands and kills their cross-extension groups. This is

a proved source-induced splitting. It is not a computation of the numerical Frobenius weights of the original zeta zeros.

ST also calculates the complete multiplicative prime spectrum, its integer-additive-compatible subspace, its localizations, and its maps to Connes–Consani’s arithmetic power actions. Every operation in those comparisons has its source, target and failure of any stronger preservation property stated. Connes–Consani, *On the Jacobian of Spec Z*, as located in ST0 and ST10.

123.3 GL2. The finite boundary changes the complete weight calculation

Use Connes–Consani’s original action

$$\ell(a, b)(X, Y) = (aX + b, aY), \quad a \in \mathbb{Q}_{>0}, \quad b \in \mathbb{Q},$$

on the full adelic coordinates. At each finite prime retain the choice of Haar measure or delta at the stated coordinate. For the two-coordinate profile ϵ , write $d_\epsilon(p) \in \{0, 1, 2\}$ for the number of delta factors at that prime. This number records receiving distribution support; it is not the user’s Z_2 parity. FM1 constructs the distribution for every profile, including infinite profiles, on the complete Schwartz–Bruhat test space. The Haar convention $\text{vol}(\mathbb{Z}_p) = 1$ remains explicit.

For every actual original-zeta zero ρ , its actual multiplicity m_ρ , and normal order $0 \leq j < m_\rho$, the full prime action is

$$F_{p, \epsilon, j} = p^{\rho - j + d_\epsilon(p)} \exp((\log p) J_{m_\rho - j}). \quad (\text{GL2.1})$$

The archimedean normal derivative contributes p^{-j} ; the finite delta factors contribute $p^{d_\epsilon(p)}$. Neither is omitted. FM2–FM4 prove that a nonzero all-prime intertwiner from a source zero block σ , normal order zero, to a target normal order j exists exactly when

$$\rho = \sigma, \quad d_{\epsilon^1}(p) - d_{\epsilon^0}(p) = j \quad \text{at every prime } p. \quad (\text{GL2.2})$$

The full spectral nilpotents are retained. In this Haar/delta family a positive normal-order correction can have only $j = 1$ or $j = 2$. This classifies the receiving family, not every possible finite-adele distribution.

Human source for the original adelic action and boundary: Connes and Consani, §5.2, equation actionpq and Definition adelicomp2 in *The Riemann–Roch strategy: Complex lift of the Scaling Site*. FM and FB prove the receiving distribution calculations rather than attributing them to that paper.

123.4 GL3. The whole calculated ambiguity is quotientable

Retain the exact finite algebras and map

$$B_\rho = \mathbb{C}[x, y]/(x, y)^{m_\rho}, \quad A_\rho = \mathbb{C}[t]/t^{m_\rho}, \quad q(x) = t, \quad q(y) = 0, \quad i(f) = f(x).$$

Form the algebraic direct sums over all actual zeros, every receiving profile and every rational branch, as constructed in FM7. No source tau is an operand of their complex addition or polynomial multiplication.

FM7 proves that every section preserving the full spectral nilpotent and the specified positive rational affine action is obtained from i by a unique automorphism in the calculated group:

$$s = Ui, \quad U = I + T, \quad T^3 = 0, \quad U^{-1} = I - T + T^2, \quad qU = q. \quad (\text{GL3.1})$$

Each matrix entry of T is multiplication by $y^j P(x)$ between the profiles in (GL2.2). It commutes with both target nilpotents $J = M_x$ and $N = M_y$, and with every stated affine action. Products of three such entries vanish because each raises every $d_\epsilon(p)$ by at least one and $0 \leq d_\epsilon(p) \leq 2$. All matrices are column-finite, so their products and inverses exist on the algebraic direct sum. Each entry is determined by its value at the generator 1 of its receiving block; this proves uniqueness.

Finite support can restore an original prime character

Every entry retains $p^{\rho-j+d}$ and its full logarithmic nilpotent.

Uniform $d(p) = d$ shown; the proof treats every profile at every prime.

	$d = 0$	$d = 1$	$d = 2$
$j = 0$	p^ρ	$p^{\rho+1}$	$p^{\rho+2}$
$j = 1$	$p^{\rho-1}$	p^ρ	$p^{\rho+1}$
$j = 2$	$p^{\rho-2}$	$p^{\rho-1}$	p^ρ

Green: $d_1(p) - d_0(p) = j$ for every p , with the same original ρ .

$$p p^{\rho-1} = p^\rho \quad | \quad p^2 p^{\rho-2} = p^\rho$$

The whole calculated ambiguity of lifts is one free group orbit:

$$U = I + T, \quad T^3 = 0, \quad U^{-1} = I - T + T^2, \quad qU = q$$

U preserves both J and N ; it carries the reference lift to every calculated lift.

Exact exponent schematic, not a numerical plot. Rows $j=0,1,2$ exist when $m \geq 3$.

Proof: FM1-FM7; finite factors from the original Connes-Consani adelic action, 1805.10501v1 §5.2.

No numerical coordinate or weight is assigned to primitive $\mathbb{Z}_\ell/\tau$; this quotient alone does not prove purity.

Figure 44: Exact prime factors and the quotient of all calculated lifts. The grid shows uniform profiles; FM1–FM7 prove the statement for every profile at every prime.

The detailed proof, including finite branch support and the free transitive group action, is FM7 and its independent verification FR6–FR7.

This realizes the user’s requested global quotient for this entire family of lift differences. It preserves ρ , the actual multiplicity and the full spectral nilpotent. Consequently the quotient of these lift differences does not itself move a zero or prove that $\Re\rho = 1/2$. That is an exact preserved-data statement about (GL3.1), not an objection to the user’s global construction.

123.5 GL4. The compensating term is a sheaf-local boundary map

The finite trace in FB1–FB4 is

$$T_f(f) = f(X_f, 0) dX_f \delta_0(Y_f).$$

Its domain is the sheaf of locally constant functions on $\mathbb{A}_f^2$, its image is the sheaf of boundary densities, and its kernel is extension by zero from $Y_f \neq 0$. The minimal distribution enlargement contains the regular densities and these boundary densities as an internal direct sum. Its operator N_f sends a regular density to $T_f(f)$ and sends a boundary density to zero. Thus $N_f^2 = 0$. The finite product formula proves

$$P_f(a, b)N_f = aN_fP_f(a, b).$$

Together with the archimedean identity $P_\infty(a, b)\partial_X = a^{-1}\partial_X P_\infty(a, b)$, this gives the actual equivariant chain map

$$\tilde{\mathcal{B}} = \partial_X \otimes N_f, \quad \tilde{\mathcal{B}}^2 = 0, \quad [\tilde{\mathcal{B}}, L_\lambda] = 0, \quad L_\lambda = Y(\lambda\partial_X + i\partial_Y). \quad (\text{GL4.1})$$

FB5–FB6B retain the complete coefficient character $\lambda > 0$, arithmetic Frobenius and its change $\lambda \mapsto \lambda\mu$. In the explicitly proved residue coordinates the induced map and the connecting map are

$$K_\lambda = -\frac{i}{\lambda}\partial_Y \otimes T_f, \quad \Delta_\lambda(h) = \lambda h \quad \text{for constant densities } h. \quad (\text{GL4.2})$$

The residue Frobenius includes the exact factor μ^{-1} . FB6B proves the commutation with (GL4.2) by retaining both coefficients $-i/(\lambda\mu^2)$. In this specified distribution complex the displayed connecting map is nonzero. It has not been identified with the geometric specialization map in the requested tau theorem.

There is also a direct chain-level quotient calculation. For every receiving complex scalar z , (GL4.1) gives

$$V_z = I + z\tilde{\mathcal{B}}, \quad V_z^{-1} = I - z\tilde{\mathcal{B}}. \quad (\text{GL4.3})$$

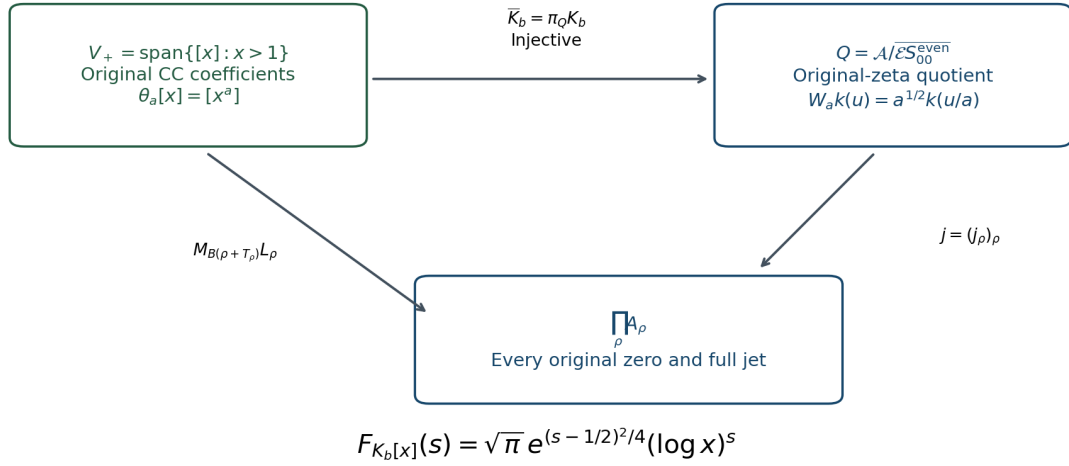
Multiplication gives both inverse identities because $\tilde{\mathcal{B}}^2 = 0$. Commutation in (GL4.1) proves that both maps are chain maps. The proved affine and arithmetic Frobenius commutations give their corresponding equivariance. Projection from the direct sum onto the original regular-density summand kills $\tilde{\mathcal{B}}$, so this projection composed with V_z is unchanged. Thus these transformations are invertible over that quotient at the level of the actual stated sheaf complex, not merely a finite matrix analogy. No assertion that this family exhausts all sheaf automorphisms is needed or made.

123.6 GL5. Original CC coefficients map injectively into the original zeta quotient

CW0 constructs the entry from the original Connes–Consani constant coefficients into their holomorphic coefficient condition. On

$$V_+ = \text{span}_{\mathbb{C}}\{[x] : x > 1\} \subset \mathbb{C}[\mathbb{R}_+^\times], \quad \theta_a[x] = [x^a],$$

Original CC coefficient scaling reaches the actual zeta quotient



The complete Gaussian factor remains in the map and its explicit jet inverse.

Proof: CW0–CW7. The diagonal map is injective and onto every finite jet packet.
No injectivity of j on all Q is asserted. Maps are linear scaling intertwiners; the source product defect is explicit.
Human sources: Connes–Consani, 1805.10501v1 §7.1; Hasanalizade–Shen–Wong, 2107.06506v1 Corollary 1.2.

Figure 45: The original CC coefficient action and the full Mellin comparison, including the Gaussian factor and every zero jet.

put $b(u) = e^{-(\log u)^2}$ and define the linear map

$$K_b[x] = W_{\log x} b, \quad W_a k(u) = a^{1/2} k(u/a).$$

Its original Mellin transform, including the complete nonzero factor, is

$$F_{K_b[x]}(s) = \sqrt{\pi} e^{(s-1/2)^2/4} (\log x)^s. \quad (\text{GL5.1})$$

Let $Q = \mathcal{A}/\overline{\mathcal{E}S_{00}^{\text{even}}}$ be the original-zeta quotient, with $\mathcal{E}f(u) = u^{1/2} \sum_{n \geq 1} f(nu)$ and $F_{\mathcal{E}f}(s) = \zeta(s) Mf(s)$. CW proves

$$\overline{K}_b : V_+ \hookrightarrow Q$$

is injective and intertwines θ_a with W_a for every positive real a . Its maps onto every finite packet of original zero jets retain all actual multiplicities. The proof uses the full original Mellin identity, the exact primary jets, and an unconditional count of the original zeta zeros with multiplicity. It does not assume RH or replace zeta by a completed function.

The global injection proof and the complete jet inverse are CW3–CW5. CW6 calculates the failure to preserve the source algebra product and the exact convolution identity in its place. No stronger algebra morphism is asserted. The zero-count source is Elchin Hasanalizade, Quanli Shen and Peng-Jie Wong, *Counting zeros of the Riemann zeta function*, Corollary 1.2. Its original author source archive and the exact bounded reading coverage are preserved in the source ledger. The coefficient construction is Connes–Consani, original author TeX §7.1, labels holom and frobarith.

123.7 GL6. Proof status and continuation

The source-induced idempotent splitting, the finite-boundary chain map, the original CC-to-zeta linear injection and the quotient of every lift in the specified Haar/delta family are proved here.

All labels, branch coordinates, original zero multiplicities and character parameters are retained in the full proofs. Twelve new exact symbolic matrix checks supplement, rather than replace, the earlier 155 checks and the complete written proofs.

The requested full tau lifting theorem remains unfinished. In particular, the calculated quotient preserves arbitrary original zero labels throughout the open critical strip, and the full geometric specialization has not been proved to be the sequence whose weight separation Deligne uses. The current calculation therefore supplies exact maps for pursuing that identification without substituting an auxiliary sequence or an assumed purity assertion for it.

124 Finite support and the weights in the lifting map

124.1 FM0. Source, prerequisites and the question calculated

This calculation retains the finite adele coordinates in Alain Connes and Caterina Consani's adelic complex lift, §5.2, equations `actionpq` and `adelicomp3`. Their left action is

$$(X, Y) \mapsto (aX + b, aY), \quad a \in \mathbb{Q}^\times, \quad b \in \mathbb{Q}. \quad (\text{FM0.1})$$

The finite coordinates are part of that formula. They are not numerical coordinates on the user's primitive Z_1/τ .

Operation prerequisites are the rational arithmetic already reconstructed in the programme, finite-adele Schwartz–Bruhat tests, their continuous linear dual, and the finite-dimensional primary blocks TL2. These receiving vector spaces have addition. No sum involving source τ occurs. The current source definitions U01–U18 and the complete global arguments USR-9ad1c0a2d09dba92, USR-f55d16feb948d8c2 and USR-6152e3bc6302258c were consulted before this calculation. Their locators and verbatim texts are retained in `CORPUS_AND_OPERATION_RULES.md` and `user_corpus_private/USER_MESSAGES.jsonl`. Complete arithmetic is used as already reconstructed; no incomplete prime list substitutes for it.

The earlier archimedean uniqueness theorem DR6 is valid in its stated receiving space. Here we compute precisely what changes after adjoining particular finite-adele distributions. This is a comparison of proved receiving constructions, not an assertion that the full tau specialization has already been identified. The separate finite-boundary sheaf calculation determines how these distributions enter the original coefficient sheaves.

124.2 FM1. All finite support profiles as actual distributions

For each rational prime p , specify two numbers $\epsilon_{p,X}, \epsilon_{p,Y} \in \{0, 1\}$. They record which finite coordinate is supported at zero. They are labels of the receiving distribution, not the user's Z_2 parity. Put

$$d_\epsilon(p) = \epsilon_{p,X} + \epsilon_{p,Y} \in \{0, 1, 2\}. \quad (\text{FM1.1})$$

Use the same additive Haar measure with $\text{vol}(\mathbb{Z}_p) = 1$ at every finite prime. Its role and factors will remain explicit. In one coordinate write $h_{p,0}$ for the distribution given by Haar measure on $\mathbb{Q}_p$, and $h_{p,1} = \delta_0$ for evaluation at zero. Both take value one on $\mathbf{1}_{\mathbb{Z}_p}$.

For a rational branch c , translate the X factor by c when it is a delta distribution; its Haar factor stays Haar. Define

$$h_{\epsilon,c} = \bigotimes_p (h_{p,\epsilon_{p,X},c} \otimes h_{p,\epsilon_{p,Y}}). \quad (\text{FM1.2})$$

This is a well-defined distribution on $\mathbb{A}_f^2$, even when the profile involves infinitely many primes. Here is a direct construction. Every Schwartz–Bruhat test is a finite linear combination of elementary tests whose p-components equal $\mathbf{1}_{\mathbb{Z}_p^2}$ outside a finite set. Enlarge that set to include the denominator primes of c . Define the value on an elementary test as the finite product of its remaining local values. Enlarging the set again does not change the product, because all extra factors are one. This defines compatible linear functionals on the finite tensor stages, hence a linear functional on their union. On each compact-support, fixed-local-constancy stage the test space is finite dimensional, so the functional is continuous there. The defining inductive-limit topology of Schwartz–Bruhat tests therefore makes it a distribution.

Equivalently it is Haar measure on the closed affine subgroup given by

$$X_p = c \quad (\epsilon_{p,X} = 1), \quad Y_p = 0 \quad (\epsilon_{p,Y} = 1), \quad (\text{FM1.3})$$

embedded as a distribution in the ambient two-coordinate space. At every unrestricted coordinate the Haar factor is retained. The subgroup description follows by checking elementary tests against

the product definition. Its support is exactly (FM1.3): a test supported away from that set is killed after a finite cylinder refinement; a neighbourhood meeting that set contains a compact cylinder with positive product measure on the unrestricted coordinates.

Distinct profiles at a fixed branch define linearly independent distributions in every finite subfamily. To prove this, choose finitely many primes which distinguish that subfamily. At each selected prime, Haar measure and delta are independent: evaluation on $\mathbf{1}_{\mathbb{Z}_p^\times}$ is respectively $1 - p^{-1}$ and zero, while evaluation on $\mathbf{1}_{\mathbb{Z}_p}$ is one for both. Translated versions give the same two-dimensional independence. Their tensor products have independent coordinate functionals, by applying the dual bases in each factor. Tests at the remaining primes take value one. This separates every coefficient in the finite subfamily. Only algebraic sums of profiles are used below.

124.3 FM2. Every finite Jacobian and every prime action

Let $P_f(a, b)$ denote distribution pullback by (FM0.1) on $\mathbb{A}_f^2$. Our convention, for a finite-adele distribution v , is

$$\langle P_f(a, b)v, \varphi \rangle = |a|_f^{-2} \langle v, \varphi((X - b)/a, Y/a) \rangle, \quad |a|_f = \prod_p |a|_p. \quad (\text{FM2.1})$$

This is the pullback convention that gives ordinary composition for a function regarded as a density distribution. Every product over absolute values is finite because a is rational.

The exact transformation of (FM1.2) is

$$P_f(a, b)h_{\epsilon, c} = \kappa_\epsilon(a)h_{\epsilon, (c-b)/a}, \quad \kappa_\epsilon(a) = \prod_p |a|_p^{-d_\epsilon(p)}. \quad (\text{FM2.2})$$

Indeed a Haar coordinate supplies $|a|_p$ on change of variable, cancelling its ambient factor $|a|_p^{-1}$. A delta coordinate supplies evaluation without that change-of-variable factor, leaving $|a|_p^{-1}$. The X evaluation moves to $(c - b)/a$; the Y evaluation stays at zero. Multiplication of these factors proves (FM2.2). In particular

$$\kappa_\epsilon(p) = p^{d_\epsilon(p)}. \quad (\text{FM2.3})$$

For the profile with Haar in X and delta in Y at every finite prime, the product formula gives

$$\kappa_\epsilon(a) = |a|_f^{-1} = |a|_\infty. \quad (\text{FM2.4})$$

This factor is lost if the finite coordinates are omitted. It is a consequence of their actual distributional pullback, not a freely selected twist.

124.4 FM3. Combining finite support with the original primary blocks

Fix an actual original-zeta zero ρ , with multiplicity m , and retain

$$B_\rho = \mathbb{C}[x, y]/(x, y)^m, \quad J = M_x, \quad q_\rho(x) = t, \quad q_\rho(y) = 0. \quad (\text{FM3.1})$$

The j -th summand is $y^j \mathbb{C}[x]/(x^{m-j})$, for $0 \leq j < m$. DR1 realizes its basis as

$$\Phi_c(x^u y^j) = \delta^{(j)}(X_\infty - c) \otimes Y_\infty^{\rho+1} \frac{(\log Y_\infty)^{m-j-1-u}}{(m-j-1-u)!}, \quad Y_\infty > 0. \quad (\text{FM3.2})$$

Both archimedean coordinates are retained. The proof and original differential are in DR1–DR5. Tensor (FM3.2) with (FM1.2). Tests and distribution tensors exist as algebraic tensors of the already specified continuous functionals; no product of arbitrary distributions is taken.

At $c = 0, b = 0, a = p > 0$, the exact eigenvalue and nilpotent action on this block are

$$F_{p, \epsilon, j} = p^{\rho-j+d_\epsilon(p)} \exp((\log p)J_{m-j}). \quad (\text{FM3.3})$$

Here the archimedean factor is $p^{\rho-j}$, including its distribution Jacobian, and the finite factor is (FM2.3). Thus the primewise absolute-value exponent is

$$2\Re\rho - 2j + 2d_\epsilon(p). \quad (\text{FM3.4})$$

We call this a primewise exponent; its identification with a Deligne geometric weight requires the actual Frobenius and cohomology comparison. In particular this formula does not assign a weight to primitive τ .

124.5 FM4. Exact classification of simultaneous prime resonances

Take source zero σ , source profile ϵ^0 , and its full primary block of multiplicity $n = m_\sigma$. Take target zero ρ , target profile ϵ^1 , and target normal order j , $0 \leq j < m_\rho$. We calculate all linear maps between these origin-branch blocks commuting with every rational prime action.

If the map is nonzero, the scalar eigenvalues of source and target must agree at every prime. To justify this with the nilpotents retained, when the two eigenvalues differ their annihilating powers $(X - \alpha)^n$ and $(X - \beta)^{m_\rho-j}$ are coprime. Their polynomial Bezout identity, applied across the intertwiner, forces the map to vanish. Consequently a nonzero map requires

$$p^{\rho-\sigma-j+d_{\epsilon^1}(p)-d_{\epsilon^0}(p)} = 1 \quad \text{for every prime } p. \quad (\text{FM4.1})$$

The original open critical strip gives $-1 < \Re\rho - \Re\sigma < 1$. Taking absolute values in (FM4.1) therefore forces

$$\Re\rho = \Re\sigma, \quad d_{\epsilon^1}(p) - d_{\epsilon^0}(p) = j \quad \text{at every prime.} \quad (\text{FM4.2})$$

No finite prime sample replaces the second condition. Put $\rho - \sigma = i\theta$. At primes 2 and 3, (FM4.1) gives $\theta \log 2 = 2\pi k_2$, $\theta \log 3 = 2\pi k_3$, with integers k_2, k_3 . If $\theta \neq 0$, then $2^{k_3} = 3^{k_2}$ with nonzero integers, contrary to unique factorization (clear negative powers by multiplying). Hence $\rho = \sigma$.

Conversely, when

$$\boxed{\rho = \sigma, \quad d_{\epsilon^1}(p) - d_{\epsilon^0}(p) = j \text{ for all } p,} \quad (\text{FM4.3})$$

the scalar factors coincide. Commuting with the remaining exponentials is equivalent to $J_{m-j}H = HT_m$. Indeed one may take the finite polynomial logarithm of the unipotent matrices for the prime 2; dividing by the nonzero real number $\log 2$ recovers their nilpotents. This gives all maps explicitly:

$$H_v(f(t)) = f(J_{m-j})v, \quad v \in y^j \mathbb{C}[x]/(x^{m-j}). \quad (\text{FM4.4})$$

They are well-defined because $J_{m-j}^m = 0$, and every intertwiner has this form by applying it to the cyclic generator. The dimension is exactly $m - j$. No spectral nilpotent has been dropped.

For a lift from a $j=0$ source into the kernel of q_ρ , one has $j \geq 1$. Since each finite profile has d between zero and two, (FM4.3) restricts compensation to $j = 1$ or $j = 2$. At $j=1$ the difference must be one at every prime; at $j=2$ it must be two at every prime. All higher normal orders are excluded from this precise Haar/delta family by the proved integer comparison. Other kinds of finite distributions are not excluded by this calculation.

124.6 FM5. A surviving lift variation, with the omitted finite factor restored

Let $m \geq 2$, let h_0 be Haar in both finite coordinates, and let h_1 be Haar in finite X and delta in finite Y at every prime. Work on the algebraic direct sum of their rational-branch blocks, retaining both coefficients. Define a map into the kernel of q by

$$H(h_0 \otimes f(t)) = h_1 \otimes yf(x), \quad H(h_1 \otimes f(t)) = 0. \quad (\text{FM5.1})$$

The right side is taken modulo $(x, y)^m$, so the highest source power is allowed to map to zero. It commutes with the full spectral nilpotent $J = M_x$ and every positive rational left dilation: the

finite factor a cancels exactly the normal-order factor a^{-1} . It commutes with rational translations because both distributions move their rational branch by the same formula $(c - b)/a$. Thus, with the earlier section i and quotient q on the two coefficient summands,

$$q(i + zH) = \text{id} \quad (z \in \mathbb{C}). \quad (\text{FM5.2})$$

The sections are distinct since H sends the constant polynomial in the h_0 summand to the nonzero vector $h_1 \otimes y$. This proves existence of a precisely computed family of lifts, not an obstruction to existence.

No section here is asserted to intertwine the additional nilpotent $N = M_y$ with the zero operator on A . In fact $Ni(1) = y \neq 0$ for $m > 1$, and $NH(1) = h_1 \otimes y^2 \neq 0$ for $m > 2$. The stronger correct statement is about automorphisms of the full receiving target: extend H to $\tilde{H}(h_0 \otimes g) = h_1 \otimes yg$, $\tilde{H}(h_1 \otimes g) = 0$. Then $\tilde{H}^2 = 0$, and it commutes with both J and N and every indicated positive affine action. Thus $U_z = I + z\tilde{H}$ has inverse $I - z\tilde{H}$, satisfies $qU_z = q$, and sends i to the section $i + zH$. Multiplication by y proves commutation with both nilpotents directly; the finite and archimedean factors in FM2–FM3 prove the affine identity.

The positive rational action is the one just proved. The full signed action, arithmetic Frobenius, and all cochain factors are calculated separately in `FINITE_ADELE_FROBENIUS_DERIVATION.md`; no unsigned formula is silently extended to negative a . The sheaf-local trace giving the Haar-to-delta entry and the exact enlargement of coefficients are calculated in `FINITE_BOUNDARY_SHEAF_TRACE.md`. Those comparisons determine the scope of (FM5.1) in the original geometry.

124.7 FM6. Every external lattice label and the exact change in the argument

For the earlier named receiving functor G_L , lift every displayed linear map f by $(v, \lambda) \mapsto (f(v), \lambda)$. This retains the label at amplitude zero as well: it sends $(0, \lambda)$ to $(0, \lambda)$. To verify addition and scalar action, insert the component formulas in the already constructed receiving semimodules; linearity gives the amplitude identity and both sides use the same join or meet of labels. Composition follows by evaluation. These formulas are not operations on primitive Z_1/τ .

The geometric support profile ϵ is also retained. H changes h_0 to h_1 and hence changes that geometric support. Preservation of an external lattice label is not a claim that those geometric supports coincide. Whether a required geometric filtration permits this map is checked by the explicit sheaf trace, not inferred from the external label formula.

Pierre Deligne's Weil II, §§3.6.1–3.6.3, proves vanishing after obtaining non-overlapping weight ranges for the actual classes and actual support group. FM2–FM5 show why the finite support factors must be calculated before importing that step: in this specified part of the full adelic distribution space, they restore exactly the eigenvalues that the archimedean-only computation had separated. The original DR theorem remains true for its stated archimedean receiver. Its uniqueness conclusion is not a theorem for this larger coefficient space.

This gives a concrete remaining geometric contribution, with exact maps, factors and nilpotents, to control in the requested tau lifting proof. It neither supplies an RH counterexample nor establishes a contradiction in the user's source. The original-zeta identification, source notation, and current goal are unchanged.

124.8 FM7. Quotienting the complete calculated ambiguity of lifts

We now retain every profile from FM1, every actual original zero and every rational branch, taking algebraic direct sums so that each vector has finitely many nonzero summands. Let $\mathcal{A}_f$ and $\mathcal{B}_f$ be the sums of their A and B blocks, with q and i defined blockwise. Each source profile is independently retained, even when two profiles have the same numbers $d(p)$. On $\mathcal{B}_f$, J and N denote the blockwise multiplications by x and y .

Consider all sections s of q that commute with J and with the complete positive rational affine action just constructed. Their entire difference from i is $h = s - i$, a map $\mathcal{A}_f \rightarrow \ker q$. The

restrictions of h to an origin branch have finite-dimensional images, hence finite support in profiles, zeros and branches. Equivariance under dilation by 2 forces the finite set of output rational branches to be stable under $c \mapsto c/2$. This set can contain only zero: a nonzero rational has an infinite orbit. Decompose that origin-branch image into its independent profile and zero summands and its normal-order summands. FM4 proves that its only nonzero components are

$$h_{\epsilon^1, \epsilon^0, \rho}(f(t)) = y^j P_{\epsilon^1, \epsilon^0, \rho}(x) f(x), \quad \deg P_{\epsilon^1, \epsilon^0, \rho} < m_\rho - j, \quad (\text{FM7.1})$$

where j is 1 or 2 and $d_{\epsilon^1}(p) - d_{\epsilon^0}(p) = j$ for every prime. All other components are zero. Translation equivariance determines the same coefficients at every rational branch. Conversely these formulas, with finitely many outputs per source summand, construct every such h : FM4 gives all prime actions, the product formula extends their scalar factors to every positive rational a , and the common branch formula gives translations.

Extend each component of (FM7.1) to B by multiplication by the same polynomial $y^j P(x)$, and call the resulting block matrix $\tilde{h}$. This is a well-defined operator on the algebraic direct sum. It commutes with J , N and the full indicated affine action, and $q\tilde{h} = 0$. In a product of three such strictly raising components, each finite prime's d -value would increase by at least three. Since $0 \leq d(p) \leq 2$, every such product is zero. Hence

$$\tilde{h}^3 = 0, \quad U_h = I + \tilde{h}, \quad U_h^{-1} = I - \tilde{h} + \tilde{h}^2. \quad (\text{FM7.2})$$

These are finite formulas even on the whole infinite profile sum. Direct multiplication proves the inverse, while $qU_h = q$ and $U_h i = i + h = s$.

Let $\mathcal{U}$ be the group of all automorphisms $I + T$ whose entries have precisely this strictly raising form. Multiplication is

$$(I + T)(I + S) = I + (T + S + TS), \quad (\text{FM7.3})$$

and the profile increases and nilpotent polynomial products keep the result in the same form. The inverse is FM7.2. This group acts transitively on the calculated sections. It acts freely: an operator T of this form is linear over the block algebra generated by x and y , and the image of i contains the generator 1 of each source B block. If $Ti=0$, every column is zero on that generator, so every polynomial entry and T itself vanish. Therefore

All calculated prime-and- J -equivariant lifts form one free orbit of $\mathcal{U}$.

(FM7.4)

Freeness at i implies freeness at every other section: transitivity expresses that section as Ui , and conjugation by U identifies its stabilizer with the stabilizer of i .

This is an exact quotient of the full ambiguity calculated here, including both nilpotents in the automorphisms, every finite support profile, all primes and all rational branches. It realizes a part of the user's global-quotient proposal as a proved map and action, without assuming that every conceivable full-geometric contribution belongs to this Haar/ δ family. The original zero character p^ρ , its multiplicity and its nilpotent are unchanged under this quotient. The construction therefore removes this lifting ambiguity but does not force $\Re \rho = 1/2$. Its extension to the full geometric specialization remains the active goal.

125 The current Z_1/τ source: spectrum, specialization and the two arithmetic actions

24 September 2026. Independent derivation from the current B1–B5, not from the older supported-zero semiring. The source is kept in the user’s notation. Its arithmetic comes from the complete integer layer already supplied after the programme’s global reconstruction. This document constructs the localization sheaf, the specialization by integer 1, its exact nonfaithful fibres, the universal additive receiver, and the morphisms relating scalar arithmetic, arithmetic power maps and the actual idèle action.

125.1 ST0. Corpus and prerequisites

The controlling statements are the verbatim user definitions, U01–U18, and the source operations and proofs, B1–B5 and P1–P5. They were read before the derivation. The current corpus and operation rules and the full retained arguments about the whole counting history were also consulted in the verbatim record, especially lines 18538–18541, 19055–19083, 19169–19181, 19202–19242 and 19262 onward.

These require that Z_0 denotes absence, Z_1/τ denotes primitive presence without Z_2 parity, and Z_2 denotes the retained parity data of nonzero integers. No source addition on τ is used. No arithmetic coordinate, midpoint, distance or weight is assigned to τ . The complete already reconstructed integer layer is an input; no finite prime sample is substituted for the user’s whole-history construction.

A bounded duplication check was made in the current foundational and lifting files. The monoid laws and the distinction of the two identities are already P1–P5, and are cited rather than claimed as new. The localization sheaf and the complete action-category calculation below were not found in that bounded check. This is not a claim to have reread every historical file.

Let S denote the carrier consisting of the supplied integers and the additional primitive τ . The only operations used initially are:

$$\tau s = s\tau = s, \quad mn = \text{the original integer product}, \quad (\text{ST0.1})$$

and the already supplied $+$ and $-$ on pairs of integers. Thus 0 is absorbing, integer 1 is an idempotent distinct from the global identity τ , and no integer is a unit for the global identity: a product with an integer is still an integer, not τ .

125.2 ST1. Multiplication by integer 1 is the exact arithmetic specialization

Put $M = (\mathbb{Z}, \cdot, 1, 0)$, the original integer multiplicative monoid. The maps

$$c : S \longrightarrow M, \quad c(s) = 1 \cdot s, \quad j : M \longrightarrow S, \quad j(n) = n \quad (\text{ST1.1})$$

are explicit. Here $c(\tau) = 1$, $c(n) = n$, $cj = \text{id}_M$, and

$$jc(s) = 1s. \quad (\text{ST1.2})$$

The map c preserves multiplication: $c(s)c(t) = (1s)(1t) = 1st = c(st)$, using $1^2 = 1$. It takes the source global identity to the target’s identity and zero to zero. It also preserves every *defined* integer sum and difference, because its restriction to the integer layer is identity. The inclusion j preserves integer arithmetic and multiplication, but it is not a homomorphism of globally unital monoids: it sends the identity of M to integer 1, not to τ .

The exact equivalence relation induced by c is

$$s \sim t \iff s = t \text{ or } \{s, t\} = \{\tau, 1\}. \quad (\text{ST1.3})$$

Every other fibre is a singleton. Thus the map is not presented as a faithful typed identification. The output 1 in $c(\tau) = 1$ comes from the already supplied input integer 1 in $1 \cdot \tau$; it is not parity generated from τ alone. Source τ still has no Z_2 datum.

This is precisely localization at the integer idempotent 1. Indeed if a unital monoid map $f : S \rightarrow N$ makes $f(1)$ invertible, then $f(1)^2 = f(1)$ and cancellation by its inverse give $f(1) = 1_N = f(\tau)$. Hence f factors uniquely through (ST1.3), giving $S[1^{-1}] \cong M$. Conversely $c(1) = 1_M$ is invertible, establishing the universal property in both directions. No sum involving τ has entered the proof.

125.3 ST2. The complete multiplicative prime spectrum

A monoid ideal here means a subset containing 0, stable under multiplication by every source element. It is prime when proper and its complement is closed under multiplication. This definition uses no source addition.

For each subset T of the ordinary rational primes, define

$$P_T = \{0\} \cup \{n \in \mathbb{Z} \setminus \{0\} : \text{some } p \in T \text{ divides } n\}. \quad (\text{ST2.1})$$

There is one further prime

$$\mathfrak{m} = \mathbb{Z} \subset S. \quad (\text{ST2.2})$$

These are all the multiplicative prime ideals.

Proof. The displayed P_T is an ideal because divisibility by a member of T survives multiplication, and it does not contain τ . If neither factor is zero or divisible by a member of T , their product has the same property by unique prime factorization. This proves primality. The complement of $\mathfrak{m}$ is $\{\tau\}$, also multiplicatively closed.

Conversely let P be prime. It cannot contain τ , since multiplying τ by every s would give $P = S$. If it contains integer 1, its ideal property gives all integers, hence $P = \mathfrak{m}$. Otherwise $-1 \notin P$, since $(-1)^2 = 1$. For every nonzero integer n , factor n into its sign and finitely many ordinary primes. If $n \in P$, repeated primality forces one of those primes into P ; the sign cannot be responsible. If a prime factor is in P , the ideal property gives $n \in P$. Taking $T = \{p : p \in P\}$ proves $P = P_T$. ■

The specialization order is

$$P_T \subseteq P_U \iff T \subseteq U, \quad P_T \subsetneq \mathfrak{m}. \quad (\text{ST2.3})$$

In particular the complete multiplicative spectrum is larger than ordinary ring $\text{Spec } \mathbb{Z}$; that difference has been calculated rather than suppressed.

Primitive τ is the unit element of this source, not itself a prime ideal. The extra point just constructed is the ideal $\mathfrak{m} = \mathbb{Z}$. Its later residue description does not identify it with a differently constructed privileged generic or support point η in the user's full geometry. This spectrum is the explicitly specified multiplicative-spectrum functor applied to B1–B5; any identification with that additional geometry must retain its separate map.

Now retain the supplied integer addition too, by requiring that a prime ideal be closed under every defined sum of its integer members. The proper P_T satisfying this condition are exactly $P_\emptyset = \{0\}$ and $P_{\{p\}} = p\mathbb{Z}$; $\mathfrak{m}$ also satisfies it. To prove this, if T contains distinct primes p, q , integer Bézout gives integers a, b with $ap + bq = 1$. Both summands belong to P_T , so additive closure would give $1 \in P_T$, impossible. Conversely $\{0\}$, $p\mathbb{Z}$, and $\mathbb{Z}$ are closed under the stated integer sums and products. No addition on τ was needed. Denote this arithmetic-compatible subspace by

$$X_{\text{ar}} = \{P_\emptyset\} \cup \{p\mathbb{Z} : p \text{ prime}\} \cup \{\mathfrak{m}\}. \quad (\text{ST2.4})$$

125.4 ST3. Localizations, their kernels, and the structure sheaf

On the full spectrum X , use the basic opens $D(s) = \{P : s \notin P\}$. Directly from ST2,

$$\begin{aligned} D(\tau) &= X, \quad D(0) = \emptyset, \quad D(1) = X \setminus \{\mathfrak{m}\}, \\ D(n) &= \{P_T : T \cap \{p : p \mid n\} = \emptyset\} \quad (n \in \mathbb{Z} \setminus \{0\}). \end{aligned} \quad (\text{ST3.1})$$

The only open neighbourhood of $\mathfrak{m}$ is X : a basic open containing it must be $D(\tau)$.

For a multiplicative set $U \subset S$, localization consists of fractions (s, u) , with

$$(s, u) \sim (t, v) \iff \text{there is } w \in U \text{ such that } wvs = wut. \quad (\text{ST3.2})$$

This is an equivalence relation: transitivity follows by multiplying the two witnessing equations by the remaining denominators; reflexivity and symmetry are immediate. Multiplication of fractions is well-defined by multiplying the witnessing equations. Its universal map sends every $u \in U$ to a unit, and any map doing so factors uniquely by $s/u \mapsto f(s)f(u)^{-1}$. This constructs the localization rather than assuming cancellation in S .

At $\mathfrak{m}$, the denominator set is $\{\tau\}$, so the stalk is exactly

$$S_{\mathfrak{m}} = S. \quad (\text{ST3.3})$$

At P_T , integer 1 is inverted, and

$$S_{P_T} \cong \left\{ \frac{a}{b} \in \mathbb{Q} : a \in \mathbb{Z}, b \in \mathbb{Z} \setminus \{0\}, p \nmid b \text{ for every } p \in T \right\} \quad (\text{ST3.4})$$

as multiplicative monoids with zero. A fraction from S is sent to $c(s)/c(u)$. The relations ST3.2 imply equality of these rational numbers. Conversely equality of the rational numbers supplies the same localization equality after multiplying by the allowed integer 1, which identifies only τ and integer 1. Thus this map is an isomorphism with the displayed fraction monoid.

The map $S \rightarrow S_{P_T}$ has the exact fibre $\{\tau, 1\}$ over its identity and is injective on the integer part. In particular no claim of a faithful source localization is made. For $T = \emptyset$ the stalk is the multiplicative monoid of $\mathbb{Q}$; for $T = \{p\}$ it is the multiplicative monoid of $\mathbb{Z}_{(p)}$; for arbitrary T it is the displayed semilocal integer localization. All its integer operations are receiver operations inherited from those fractions, not source operations on τ .

Define the sheaf $\mathcal{O}_S$ by functions on an open U taking values in the indicated stalks and locally represented by a fraction s/u with u absent from every prime in that neighbourhood. Matching functions glue uniquely, and the local fraction property survives gluing; hence this is a sheaf. Its stalks are the localizations just constructed, because equality of germs is witnessed by ST3.2 after restricting to a suitable denominator open.

Its global sections are exactly S . A local representation around $\mathfrak{m}$ must hold on the whole of X , and its denominator there can only be τ ; hence every global section comes from a unique $s \in S$, with uniqueness checked in the stalk $S_{\mathfrak{m}} = S$.

For nonzero integer n ,

$$\Gamma(D(n), \mathcal{O}_S) = S[n^{-1}] \cong (\mathbb{Z}[1/n], \cdot). \quad (\text{ST3.5})$$

Here is a direct proof of the section assertion. The prime P_T , where T consists of all primes not dividing n , is contained in $D(n)$. Every relative basic neighbourhood of it in $D(n)$ equals $D(n)$: any allowed denominator can have prime factors only among those dividing n . Such denominators are invertible in $\mathbb{Z}[1/n]$, since each $p \mid n$ has inverse $(n/p)/n$. A local representation at this point therefore represents the section on all of $D(n)$, with exactly the displayed denominators. Uniqueness follows in the rational stalk.

Restriction to X_{ar} has the same stalks at its retained points. Its open $D(1) \cap X_{\text{ar}}$ is ordinary $\text{Spec } \mathbb{Z}$, with the ordinary arithmetic structure sheaf when the already defined integer additions

are retained. The extra closed point has stalk S , and the specialization-induced maps from that stalk to the integer stalks are precisely ST3.4. No ring addition at the extra point is invented.

The residue *pointed monoid* at the closed point is

$$S/\mathfrak{m} = \{0, \tau\}, \quad (\text{ST3.6})$$

where all integers are collapsed to its absorbing element and τ remains its identity. This quotient only preserves multiplication. It is not an equality of source τ with a scalar, nor a Boolean addition law. A residue ring at an ordinary arithmetic point additionally uses the existing integer additive congruence; the two types of residue must not be interchanged.

125.5 ST4. The source specialization on actions, before linearization

An action of S on a set X means maps $A_s : X \rightarrow X$ with $A_\tau = \text{id}$ and $A_s A_t = A_{st}$. No vector space or parity on X is presumed. Put

$$E = A_1, \quad O = A_0.$$

Then

$$E^2 = E, \quad O^2 = O, \quad EO = OE = O, \quad A_n E = E A_n = A_n \quad (n \in \mathbb{Z}). \quad (\text{ST4.1})$$

All follow by applying the source multiplication law. Thus $Y = EX$ is an invariant retract:

$$X \xrightarrow{E} Y \xrightarrow{\iota} X, \quad E\iota = \text{id}_Y. \quad (\text{ST4.2})$$

On Y , integer 1 acts as identity, and the source action factors through the localization $c : S \rightarrow M$. For every $s \in S$,

$$E A_s = A_{c(s)} E. \quad (\text{ST4.3})$$

Every equivariant map from X to an action in which integer 1 is identity factors uniquely through E : it sends x and Ex to the same point. Conversely the induced map on Y is equivariant. Consequently Y is the exact localization of the action, with equivalence relation

$$x \sim y \iff Ex = Ey. \quad (\text{ST4.4})$$

This records its entire possible nonfaithfulness.

There is a full reconstruction of the action. Above each $y \in Y$, retain the fibre $X_y = E^{-1}(y)$, with its distinguished element $\iota(y)$. Every integer sends any $x \in X_y$ to $\iota(A_n y)$, while τ fixes x . These formulas reconstruct the action and prove the product law, since $A_m A_n x = \iota(A_{mn} y)$. Thus the specialization does not merely discard an unknown difference: its exact retained data are the integer action on Y and the specified fibres with their marked sections.

The zero action need not be constant on an unpointed action. The image OX consists of points fixed by every A_s , and every fibre of O is preserved by the action. On each such fibre the zero action is constant at its distinguished fixed point. This observation will retain independently labelled zero amplitudes in ST9. If a pointed action is specified with A_0 the constant basepoint, its quotient X/Y , collapsing the invariant image Y , carries the residue action $S/\mathfrak{m}$: every integer sends it to the basepoint and τ acts identically.

In particular an integer n can act invertibly on all of X only when $E = \text{id}$, since $A_n E = A_n$ and cancellation by the inverse gives that identity. This is an exact fact about this same action, not a claim that another geometric Frobenius action cannot exist.

125.6 ST5. Linear actions and the additional requirement of integer additivity

First consider a complex vector space V with a multiplicative S -action by linear maps, with $A_0 = 0$. The existing additive receiver makes the idempotent splitting explicit:

$$V = V_+ \oplus V_-, \quad V_+ = EV, \quad V_- = \ker E. \quad (\text{ST5.1})$$

Indeed $v = Ev + (v - Ev)$, the second summand is killed by E , and the intersection is zero. These are vector-space operations in the receiver, not subtraction from primitive τ . Every integer acts by

$$A_n = \begin{pmatrix} B_n & 0 \\ 0 & 0 \end{pmatrix}, \quad B_1 = \text{id}_{V_+}, \quad B_m B_n = B_{mn}, \quad (\text{ST5.2})$$

while $A_\tau = \text{id}_{V_+} \oplus \text{id}_{V_-}$. This follows from $A_n E = E A_n = A_n$. Conversely every such multiplicative integer action B_n gives an S -action by these formulas. The residue boundary is exactly V_- , and the arithmetic specialization is $E : V \rightarrow V_+$, with its canonical inclusion.

Now retain B4 as an operation-preservation requirement too:

$$A_{m+n} = A_m + A_n \quad (m, n \in \mathbb{Z}). \quad (\text{ST5.3})$$

Induction starting at $A_1 = E$, then $A_{-n} = -A_n$, proves the much stronger exact classification

$$\boxed{A_n = nE \quad (n \in \mathbb{Z}).} \quad (\text{ST5.4})$$

Conversely this formula preserves every defined integer addition, subtraction and product, as well as the supplied mixed multiplication with τ . Thus arbitrary multiplicative prime matrices are **not** representations preserving B4 in ordinary endomorphism addition.

There is an explicit universal receiver justifying this construction. Form the unital ring U_S from formal symbols $[s]$, $s \in S$, with

$$[0] = 0, \quad [\tau] = 1_{U_S}, \quad [s][t] = [st], \quad [m] + [n] = [m + n] \quad (m, n \in \mathbb{Z}). \quad (\text{ST5.5})$$

It is constructed as the quotient of the free unital associative ring by precisely these relations. The bracket is a receiving map, not equality of the source element and the receiving symbol. No source addition involving τ is pulled back from this ring.

With $\epsilon = [1]$, the relations imply $[n] = n\epsilon$ and $\epsilon^2 = \epsilon$, while $[\tau]$ remains the formal global unit. The maps

$$U_S \longleftrightarrow \mathbb{Z}[\epsilon]/(\epsilon^2 - \epsilon) \quad (\text{ST5.6})$$

sending these named generators to each other are inverse: each respects all defining relations, and each composite fixes the generators. This proves the isomorphism and the universal property. A map preserving all defined source operations into a unital ring factors uniquely through the displayed quotient by its values on the generators.

The source comparison remains injective. Distinct integer symbols have distinct values because evaluation at $\epsilon = 1$ recovers their original integers. The formal global unit differs from every $n\epsilon$ because evaluation at $\epsilon = 0$ sends the former to one and the latter to zero. This proof does not assign either evaluation, or its parity, to primitive τ ; they are explicit tests of equality in the receiving ring. In particular the new ambient expression $[\tau] + [\tau]$ is not used to define $\tau + \tau$ in the source.

The additive decomposition in the receiver is

$$U_S = \epsilon U_S \oplus (1_{U_S} - \epsilon) U_S. \quad (\text{ST5.7})$$

Each factor is a copy of $\mathbb{Z}$, their mixed products vanish, and the second is annihilated by every integer symbol $n\epsilon$. Tensoring this expressly constructed receiver with $\mathbb{C}$ gives the classification ST5.4. It never identifies source τ with the scalar identity of the integer layer.

125.7 ST6. The actual lifting map and independent Frobenius

The source specialization in any B4-preserving complex receiver is the exact sequence

$$0 \longrightarrow V_- \longrightarrow V \xrightarrow{E} V_+ \longrightarrow 0, \quad (\text{ST6.1})$$

with canonical section the inclusion $i : V_+ \rightarrow V$. Its extension class is zero by the displayed decomposition; no purity assumption is needed.

If a further invertible family F_a commutes with the source action, it commutes with E , hence preserves V_+ and V_- . In the decomposition,

$$F_a = F_a^+ \oplus F_a^-. \quad (\text{ST6.2})$$

The section and retraction in ST6.1 are then equivariant for the entire family. Thus every connecting map produced by this actual source specialization is zero: lift a cocycle through i , whose equivariance makes its coboundary the lift of the original coboundary.

The cross-extension vanishing holds in every degree in the category retaining both source and commuting F -actions. On V_+ , the central operator ϵ acts as identity; on V_- , it acts as zero. The action of a central operator on Ext through its two variables agrees, as follows by applying it to a free resolution and commuting it through module-linear cochains. Therefore every $\text{Ext}^r(V_+, V_-)$ and $\text{Ext}^r(V_-, V_+)$ is simultaneously killed by ϵ and by $1 - \epsilon$, so is zero. In degree zero this is simply $h = hE = Eh = 0$ for a cross-map. This is idempotent separation proved from the source laws, not an assumed numerical weight separation.

The numerical weights of the independent F -action have not been prescribed by this proof. Every pair of F -representations gives such a source-equivariant receiver: use their direct sum and define integer n to act as $n\text{id} \oplus 0$, and τ to act as the identity operator. Conversely ST5–ST6 recover that pair from every such receiver. This is an equivalence of the explicitly specified receiving categories, not a counterexample claimed against the user's full geometry.

The *same* integer multiplication action has zero prime operator on V_- , by ST5.4. Zero is not an invertible Frobenius eigenvalue and has no finite numerical Deligne weight. The independent F_p^- can nevertheless be invertible. The exact relation is the commuting pair of actions just constructed, not an assertion that they are unrelated.

For the actual retained programme take

$$V = Q \oplus \mathcal{B}_{\text{pr}}, \quad A_n = n\text{id}_Q \oplus 0, \quad A_\tau = \text{id}_V, \\ F_b = W_b^Q \oplus \left(\text{id}_{W_{\text{pr}}} \otimes \begin{pmatrix} 1 & 0 \\ 0 & b \end{pmatrix} \right). \quad (\text{ST6.3})$$

Here Q , W_b^Q , and the actual adelic endpoint action are those constructed with their original factors in ACTUAL_ADELIC_COMPLEX_INDEPENDENT.md, AC1–AC7. Every commutation and product law in ST6.3 follows from scalar multiplication and the already proved $W_a^Q W_b^Q = W_{ab}^Q$. The full source laws, including integer addition, hold on V ; the endpoint's independent F_p -weights remain exactly 0 and 2. The actual Mellin jets $Q \rightarrow A_\rho$ are F -equivariant, while A_n acts on their arithmetic receiving vectors as the scalar n . This exhibits a concrete comparison retaining the two actions and both endpoints. It does not claim that this direct-sum receiver has already been identified with a geometric specialization sheaf.

125.8 ST7. Arithmetic power maps preserve addition in an exact different operation

The source integer operations also act on every abelian group H , written multiplicatively, by the arithmetic power maps

$$[n](h) = h^n, \quad n \in \mathbb{Z}. \quad (\text{ST7.1})$$

Their composition obeys $[m][n] = [mn]$. Their sum as group endomorphisms is the pointwise group product, and

$$([m] + [n])(h) = h^m h^n = h^{m+n} = [m+n](h). \quad (\text{ST7.2})$$

Thus the integer addition is preserved exactly in the abelian-group endomorphism ring. Its zero map is $h \mapsto 1_H$. The global τ -action is the identity map on H , and so this action factors through $c : S \rightarrow M$; its lost source fibre remains exactly the $\tau, 1$ distinction.

The passage to functions has an equally exact formula. On the algebraic torus take the coordinate Hopf algebra

$$B = \mathbb{C}[z, z^{-1}], \quad \Delta(z) = z \otimes z, \quad \varepsilon(z) = 1.$$

Define $\Phi_n(z) = z^n$, extended to an algebra homomorphism. For all integer m, n ,

$$\Phi_m \Phi_n = \Phi_{mn}, \quad \boxed{\Phi_{m+n} = \mu(\Phi_m \otimes \Phi_n) \Delta}. \quad (\text{ST7.3})$$

Both equations are checked on every monomial z^r : the first gives z^{rmn} , the second $z^{rm} z^{rn} = z^{r(m+n)}$. Linearity proves them on all Laurent polynomials. In particular $\Phi_0 = \text{unit} \circ \varepsilon$, not the zero linear operator. The operation on the right of ST7.3 is the explicit convolution induced by group multiplication. It is not ordinary sum of linear operators: already on z , $\Phi_{m+n}(z) = z^{m+n}$ whereas $(\Phi_m + \Phi_n)(z) = z^m + z^n$.

This is the exact morphism explaining what happened to source addition under geometric pullback. It was transported to the Hopf-convolution operation; it was not retained as matrix addition. No argument may therefore impose ST5.3 on these pullbacks merely because both constructions carry an index n .

For the retained arithmetic cyclic quotient W , the comparison $W \simeq \mathbb{Z}$ after its full reconstruction gives

$$\nu([n]) = |W/[n]W| = |n| \quad (n \neq 0). \quad (\text{ST7.4})$$

The equality follows from the exact residue classes of the reconstructed cyclic group. It retains the whole-group input and does not derive a unit from a finite pair of already labelled numbers. For nonzero m, n , the subgroup indices give

$$\nu([mn]) = \nu([m])\nu([n]). \quad (\text{ST7.5})$$

There is no finite value assigned by this formula at $n = 0$, because $W/[0]W = W$ is infinite. The positive norm loses the sign and records the size of the arithmetic cokernel; its domain and its lost information are explicit.

The actual spectral action is the composite of this multiplicative norm map with the already constructed positive idèle representation:

$$[n] \mapsto |n| \mapsto W_{|n|}^Q, \quad n \neq 0. \quad (\text{ST7.6})$$

On the original Mellin-jet quotient it is

$$|n|^\rho \exp((\log |n|)T). \quad (\text{ST7.7})$$

These are complete formulas, not scalar replacements: the full nilpotent and the original factor in $W_b k(y) = b^{1/2} k(y/b)$ remain. They preserve multiplicative composition but do not purport to preserve source integer addition as ordinary operator addition. The exact defect, for positive m, n , is

$$(m+n)^\rho e^{(\log(m+n))T} - m^\rho e^{(\log m)T} - n^\rho e^{(\log n)T}. \quad (\text{ST7.8})$$

This is a receiving comparison of defined operators, not a sum or a coordinate involving primitive τ .

If one additionally demanded ordinary source additivity for these *same* operators, ST5.4 would force $W_p = p \text{ id}$ on the arithmetic part. On a nonzero primary block, its eigenvalue condition would give $p^\rho = p$ for both $p = 2, 3$. Their logarithms force $\rho = 1$: the real part follows from absolute values, and a nonzero imaginary part would imply a rational relation between $\log 2$ and $\log 3$, contrary to unique prime factorization. The exponential's positive nilpotent part would also have to vanish, because $e^{(\log p)T} - I = Tg(T)$ with $g(0) = \log p \neq 0$. This tests precisely the extra demand, and not the user's source: ST7.3 and ST7.6 explicitly show why the geometric and spectral functors do not impose that demand.

125.8.1 ST7a. The exact tensor comparison needed for amplification

Let V be a B4-preserving receiver, so its integer scalar action is $A_n = nE$. On $V^{\otimes k}$, diagonal tensoring of these multiplicative operators gives

$$A_n^{\otimes k} = n^k E^{\otimes k}. \quad (\text{ST7a.1})$$

This is multiplicative in n , but its complete additive defect is

$$A_{m+n}^{\otimes k} - A_m^{\otimes k} - A_n^{\otimes k} = ((m+n)^k - m^k - n^k) E^{\otimes k}. \quad (\text{ST7a.2})$$

For $k \geq 2$, $m = n = 1$, and nonzero EV , this is $(2^k - 2)E^{\otimes k} \neq 0$: a tensor of nonzero arithmetic vectors is nonzero over the field $\mathbb{C}$. Thus this diagonal multiplicative action is not the integer-additive scalar action on the tensor receiver.

The latter action is instead

$$\tilde{A}_n = nE^{\otimes k}, \quad \tilde{A}_r = \text{id}_{V^{\otimes k}}, \quad (\text{ST7a.3})$$

which preserves all B1–B5 by the same idempotent proof. If the geometric F_p commutes with E , its tensor action $F_p^{\otimes k}$ commutes with $E^{\otimes k}$, so ST7a.3 and the geometric tensor action coexist in one receiver. The exact difference between the two candidate integer-labelled operators is

$$A_n^{\otimes k} - \tilde{A}_n = (n^k - n)E^{\otimes k}. \quad (\text{ST7a.4})$$

On the actual spectral primary tuple, the geometric tensor action remains the full

$$p^{\rho_1 + \dots + \rho_k} \exp((\log p)(T_1 + \dots + T_k)). \quad (\text{ST7a.5})$$

Nothing in this calculation replaces it by $p \text{ id}$ or by $p^k \text{ id}$. This proves exactly how the source scalar arithmetic and Deligne's geometric tensor amplification can be retained together, and exactly which interchange would fail.

125.9 ST8. Relation to Connes–Consani and Deligne, with the actual comparison retained

Alain Connes and Caterina Consani expressly distinguish arithmetic Frobenius on the function-field Jacobian from the idèle-class **translation** action in their number-field construction. This is in their original author source On the Jacobian of $\text{Spec } \mathbb{Z}$, local **Jacobian.tex**, lines 2501–2550, read for this calculation. Their displayed operator is $\vartheta(u)\xi(x) = \xi(u^{-1}x)$. Their local trace calculation and the endpoint terms are not statements that a source scalar p acts by an arithmetic-linear operator $p \text{ id}$.

ST7 supplies an exact elementary comparison at the level of arithmetic power maps, pullback and finite-index norm. ST6.3 supplies a concrete simultaneous action on the actual retained Q and prime boundary. Together these prevent the integer scalar action, the geometric power-map pullback, and the idèle translation from being conflated while proving their connecting maps.

Pierre Deligne's lifting proof in La conjecture de Weil. II, §3.6.1–§3.6.3, uses geometric Frobenius with separated weights on the lifted classes and support group. The source specialization ST6.1 has a stronger immediate splitting at the **source-idempotent** level, proved from $1^2 = 1$, and every commuting Frobenius preserves it. However ST6.2 shows that this alone does not determine the numerical weights of either component. The required numerical comparison must use the actual independent geometric action, as in the explicit diagram ST6.3, and its cohomology rather than identify it with integer multiplication.

This conclusion has not stopped at nonidentity: the full localization map, action-category retract, universal B4-preserving receiver, Hopf-convolution formula, norm map, actual Q -plus-boundary action, and their kernels have all been constructed. No arbitrary spectral parameter has been offered as a counterexample to the user's complete geometry.

125.10 ST9. Support labels, zero action and the specialization kernel

For the original arbitrary bounded distributive lattice L , let

$$G_L(V) = \{(0, \lambda) : \lambda \in L\} \cup (V \times \{1_L\})$$

be the named receiving carrier. Every constructed linear operator A_s has the common-label lift

$$\tilde{A}_s(v, \lambda) = (A_s v, \lambda). \quad (\text{ST9.1})$$

This defines a multiplicative S -action because the composition identities hold on amplitudes and the label never changes. It does not require identifying any zero label with primitive τ .

The exact lifted zero action is

$$\tilde{A}_0(v, \lambda) = (0, \lambda). \quad (\text{ST9.2})$$

It is not a constant map to the bottom label. This is precisely the unpointed action case of ST4: its fixed zero-label set is the complete copy of L , and each label fibre has its own fixed zero amplitude. Within that fibre, integer addition of the lifted operators is preserved, since

$$(A_m v, \lambda) \oplus (A_n v, \lambda) = (A_{m+n} v, \lambda \vee \lambda) = (A_{m+n} v, \lambda). \quad (\text{ST9.3})$$

Here $\oplus$ is the receiving addition with join, not source addition on τ . Its zero in this family of fibre-preserving operators is ST9.2; it is not the global constant-bottom operator.

The source specialization lifts exactly to

$$G_L(V) \xrightarrow{G_L(E)} G_L(V_+) \xrightarrow{G_L(i)} G_L(V), \quad G_L(E)G_L(i) = \text{id}. \quad (\text{ST9.4})$$

Its complete equivalence relation is

$$(v, \lambda) \sim (w, \mu) \iff \lambda = \mu \text{ and } E(v) = E(w). \quad (\text{ST9.5})$$

Thus the quotient by the amplitude boundary is $G_L(V_+)$, and every support label survives. The inverse image of all zero amplitudes is $G_L(V_-)$, while the inverse image of the single bottom-labelled zero need not be that full amplitude kernel. These are different receiving questions, and the formula answers each without erasing a label.

If F_a commutes with the source action, its lift commutes with ST9.4 and the section remains exact. A vanishing connecting amplitude returns $(0, \lambda)$. Independent labels on independent blocks are additional coordinates and may be retained block by block; $G_L(V_+ \oplus V_-)$ with one common label is not silently identified with $G_L(V_+) \times G_L(V_-)$.

125.11 ST10. Exact output

Starting only with current B1–B5, the arithmetic specialization is multiplication by the already supplied integer 1, with localization fibre $\{\tau, 1\}$ and no source parity assignment. The full multiplicative prime spectrum, its integer-addition-compatible part, localizations, sheaf and residue at the extra closed point are computed in ST2–ST3.

The source action specializes by its central idempotent E . Its entire lost data are the fibres in ST4, or the exact complementary boundary in the constructed additive receiver. The lifting sequence splits, including every commuting Frobenius action and all support labels. This is a positive source-derived lifting result.

Preserving B4 is decisive: additive receivers have integer action nE . Arithmetic power-map pullbacks instead preserve addition through the exact Hopf-convolution formula, and the original spectral dilation comes through the explicitly displayed finite-index norm. All comparisons retain the lost sign, the undefined zero norm, the localization fibre and the full spectral nilpotent.

No calculation here replaces the current source by the older $G_L(\mathbb{Z})$, derives arithmetic before the user's completed reconstruction, or claims that this split source-idempotent boundary is already Deligne's geometric support group. Its proved relation to the actual Q , endpoint actions and support labels is ST6.3 and ST9. The numerical purity claim remains distinct from the lifting result and has not been assumed.

126 Connes–Consani coefficient scaling inside the actual Mellin quotient

24 September 2026. Independent proof of the explicit map proposed by the root task. The operation prerequisites were checked in the local AGENTS.md, CORPUS_AND_OPERATION_RULES.md, U01–U18, and B1–B5/P1–P5 of SOURCE_OPERATIONS_AND_PROOFS.md. The complete relevant user passages USR-9ad1c0a2d09dba92, USR-f55d16feb948d8c2, and USR-6152e3bc6302258c were read in the preserved user corpus. This proof uses the already reconstructed arithmetic and its original zeta function. It does not construct arithmetic by counting primitive tau, and does not assign tau a coordinate, metric, parity, or addition.

The source datum supplied by the root task is Alain Connes and Caterina Consani’s coefficient group algebra

$$W_{\text{alg}} = \mathbb{C}[\mathbb{R}_{>0}^{\times}], \quad [x][y] = [xy], \quad \theta_a[x] = [x^a] \quad (a > 0). \quad (\text{CW0.1})$$

The root task read the original 2018 author TeX at §7.1, equations `holom`, `holom1`, and Proposition `frobarith`: Connes–Consani, The Riemann–Roch strategy, Complex lift of the Scaling Site. This independent note proves the indicated receiving construction from those explicit operations, without claiming a reading of that whole paper.

The entry into their holomorphic construction is exact. Regard $w \in W_{\text{alg}}$ as a constant W -valued function on the original adelic complex lift. Every character evaluation is constant in both archimedean coordinates, so

$$(\lambda X + iY)\chi_{\lambda}(w) = 0 \quad (\lambda > 0). \quad (\text{CW0.2})$$

Right geometric scaling fixes a constant function. The source arithmetic Frobenius $\theta_a R(a^{-1})$ therefore restricts to θ_a on these constant coefficients. Distinct algebraic coefficients remain distinct under their complete character family: for finitely many distinct $x_j > 0$, the evaluations at $\lambda = 1, \dots, d$ have matrix (x_j^r) , with determinant

$$\left(\prod_{j=1}^d x_j \right) \prod_{1 \leq i < j \leq d} (x_j - x_i) \neq 0. \quad (\text{CW0.3})$$

This establishes the precise source entry of the scaling representation used here. It does not identify the constant-coefficient subspace with the full holomorphic sheaf.

126.1 CW1. Domains and the complete Gaussian Mellin factor

Let $V_+ \subset W_{\text{alg}}$ be the complex span of basis elements $[x]$ with $x > 1$. It is invariant under every θ_a , since $x^a > 1$ for $a > 0$. Its elements are finite linear combinations; their multiplicities and coefficients belong to this specified group algebra, not to primitive τ . The source group-algebra unit $[1]$ is not in V_+ . We do not extend the formulas below to it by treating zero as a positive dilation.

Use the original strong Mellin space $\mathcal{A}$, with seminorms

$$p_{N,j}(k) = \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j k(u)|, \quad N, j \geq 0,$$

and original transform and action

$$F_k(s) = \int_0^\infty k(u) u^{s-1/2} \frac{du}{u}, \quad W_a k(u) = a^{1/2} k(u/a), \quad a > 0. \quad (\text{CW1.1})$$

Keep the explicit vector

$$b(u) = \exp(-(\log u)^2). \quad (\text{CW1.2})$$

It belongs to $\mathcal{A}$: in $v = \log u$, every j -th derivative is a polynomial in v times e^{-v^2} , and multiplication by $e^{Nv} + e^{-Nv}$ has finite supremum for each polynomial, because the negative quadratic dominates its linear and logarithmic growth.

Its entire Mellin transform is

$$B(s) := F_b(s) = \int_{\mathbb{R}} e^{-v^2 + (s-1/2)v} dv = \sqrt{\pi} \exp\left(\frac{(s-1/2)^2}{4}\right). \quad (\text{CW1.3})$$

The integral is entire by an integrable Gaussian bound on each compact s -set, including derivatives. For real s , completing the square and translating the real integration variable prove the last equality. Both sides are entire, so the identity theorem proves it for every complex s . In particular $B(s)$ never vanishes. Every factor in this evaluated transform remains in the comparison.

For $x > 1$, put $t_x = \log x > 0$ and define

$$K_b[x] = W_{t_x} b, \quad (K_b[x])(u) = t_x^{1/2} \exp(-(\log u - \log t_x)^2). \quad (\text{CW1.4})$$

Extend linearly to V_+ . Dilation preserves $\mathcal{A}$, since

$$p_{N,j}(W_a k) \leq a^{1/2} \max(a^N, a^{-N}) p_{N,j}(k).$$

Substitution $u = av$ in (CW1.1) gives

$$F_{W_a k}(s) = a^s F_k(s),$$

and consequently the full exact formula is

$$\boxed{F_{K_b[x]}(s) = \sqrt{\pi} \exp\left(\frac{(s-1/2)^2}{4}\right) (\log x)^s.} \quad (\text{CW1.5})$$

Here $(\log x)^s = \exp(s \log(\log x))$, with the real logarithm of the positive number $\log x$. Neither this double logarithm nor the Gaussian factor is absorbed into a replacement parameter.

Since $\log(x^a) = a \log x$ and $W_a W_t = W_{at}$, direct substitution proves

$$K_b \theta_a = W_a K_b \quad (a > 0). \quad (\text{CW1.6})$$

This is a linear intertwiner of the complete positive-real scaling action.

126.2 CW2. The actual original-zeta quotient and every multiplicity condition

Retain

$$\mathcal{E}f(u) = u^{1/2} \sum_{n \geq 1} f(nu), \quad I = \overline{\mathcal{E}(S_0^{\text{even}})}^{\mathcal{A}}, \quad Q = \mathcal{A}/I, \quad (\text{CW2.1})$$

where S_0^{even} is the even Schwartz space with $f(0) = 0$ and $\int_{\mathbb{R}} f = 0$. Denote the quotient map by π_Q , and set

$$\overline{K}_b = \pi_Q K_b : V_+ \longrightarrow Q. \quad (\text{CW2.2})$$

The programme has proved that I is invariant under W_a , so (CW1.6) descends to this actual quotient.

The source of every zero condition is the original identity, initially on $\Re s > 1$,

$$F_{\mathcal{E}f}(s) = \zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v}. \quad (\text{CW2.3})$$

Absolute convergence justifies exchanging sum and integral there. The second factor is holomorphic for $\Re s > -2$, since evenness and $f(0) = 0$ give $f(v) = O(v^2)$ at zero and the Schwartz property controls infinity. The product is therefore holomorphic throughout the critical strip, and its

meromorphic identity continues from the half-plane. For every actual nontrivial zero ρ of ζ , with its full multiplicity m_ρ ,

$$F_{\mathcal{E}_f}^{(r)}(\rho) = 0 \quad (0 \leq r < m_\rho). \quad (\text{CW2.4})$$

Each derivative evaluation is continuous on $\mathcal{A}$, by its defining integral with the factor $(\log u)^r$ and a sufficiently strong seminorm at both endpoints. Thus (CW2.4) holds for all of I , not merely its generating image.

No completed zeta function replaces ζ in this argument. The original pole and the full source endpoint restrictions remain part of (CW2.1)–(CW2.3); no assertion about omitted trivial zeros is needed for this implication.

126.3 CW3. Injectivity through the quotient: a global zero-density proof

Take $v = \sum_{j=1}^d c_j [x_j]$, with the $x_j > 1$ distinct and all displayed coefficients nonzero. Write

$$t_j = \log x_j > 0, \quad \lambda_j = \log t_j \in \mathbb{R}, \quad P_v(s) = \sum_{j=1}^d c_j e^{\lambda_j s}. \quad (\text{CW3.1})$$

Then

$$F_{K_b v}(s) = B(s)P_v(s). \quad (\text{CW3.2})$$

If $\overline{K_b}v = 0$, then $K_b v \in I$, and CW2 implies that this product vanishes to order at least m_ρ at every actual nontrivial zero ρ . Since the entire factor B is nowhere zero, P_v itself has those zeros with at least those multiplicities.

A nonzero finite exponential polynomial of the displayed kind has at most $O(R)$ zeros in a disk of radius R , counted with multiplicity. Here is the complete Jensen estimate needed for that assertion. Choose s_* with $P_v(s_*) \neq 0$, put $H = \max_j |\lambda_j|$, and $C = \sum_j |c_j| e^{\lambda_j \Re s_*} > 0$. Then for $|z| \leq R'$,

$$|P_v(s_* + z)| \leq C e^{H R'}. \quad (\text{CW3.3})$$

Apply Jensen's formula to $P_v(s_* + z)$ at a radius R' between $2R$ and $2R + 1$ having no boundary zeros. Each zero in $|z| \leq R$ contributes at least $\log 2$. Thus its zero count $n_v(R)$ satisfies

$$n_v(R) \log 2 \leq \frac{1}{2\pi} \int_0^{2\pi} \log |P_v(s_* + R' e^{i\theta})| d\theta - \log |P_v(s_*)| \leq \log C + H(2R + 1) - \log |P_v(s_*)|. \quad (\text{CW3.4})$$

Such a radius exists because zeros are discrete when P_v is not identically zero. This proves the asserted $O(R)$ estimate with all quantities specified.

The unconditional explicit Riemann–von Mangoldt bound of Elchin Hasanalizade, Quanli Shen, and Peng-Jie Wong, *Counting zeros of the Riemann zeta function*, arXiv:2107.06506v1, Corollary 1.2, states for every $T \geq e$

$$\left| N_\zeta(T) - \frac{T}{2\pi} \log \left(\frac{T}{2\pi e} \right) \right| \leq 0.1038 \log T + 0.2573 \log \log T + 9.3675, \quad (\text{CW3.5})$$

where $N_\zeta(T)$ counts the nontrivial zeros with $0 < \Im \rho \leq T$, including multiplicity. The source defines this count in original TeX lines 187–201, states the complete bound at lines 255–261 (`main-thm`, `main-bound-1`), and uses the argument principle at lines 327–337, which retains multiplicity. Those passages were read directly for this proof. All constants of the source bound are displayed; its lower estimate grows faster than every fixed multiple of T . This is an established external theorem, not a new programme result or an RH assumption. The previous programme upper-count bound alone would not supply this lower estimate.

Every zero counted by $N_\zeta(R - |s_*| - 2)$ lies in the disk $|\rho - s_*| \leq R$, since $0 < \Re \rho < 1$ gives $|\rho| \leq 1 + |\Im \rho|$. Therefore the zero multiplicity conditions above contradict (CW3.4) unless P_v is identically zero.

Finally the functions $e^{\lambda_j s}$ are linearly independent. Differentiating an identically vanishing P_v at $s = 0$, for derivative orders $0, \dots, d-1$, gives the matrix (λ_j^r) . Its determinant is $\prod_{i < j} (\lambda_j - \lambda_i) \neq 0$, because the x_j and therefore the λ_j are distinct. Thus every $c_j = 0$. We have proved

$$\boxed{\overline{K}_b : V_+ \longrightarrow Q \text{ is injective.}} \quad (\text{CW3.6})$$

The same argument proves that K_b is injective before quotienting; there even (CW3.2) and independence suffice. No RH, zero simplicity, bounded-height test, or selected finite prime list enters the quotient injection.

126.4 CW4. The exact jet maps, their Gaussian unit, and finite joint surjectivity

For each actual zero define the full local algebra and the linear source observation

$$A_\rho = \mathbb{C}[T_\rho]/(T_\rho^{m_\rho}), \quad L_\rho[x] = (\log x)^\rho \sum_{r=0}^{m_\rho-1} \frac{(\log \log x)^r T_\rho^r}{r!}. \quad (\text{CW4.1})$$

Every label ρ and multiplicity remains. This expression is the complete Taylor jet of $(\log x)^s$ at $s = \rho$. For the actual quotient jet

$$j_\rho([k]) = \sum_{r=0}^{m_\rho-1} \frac{F_k^{(r)}(\rho)}{r!} T_\rho^r,$$

the exact commutative comparison is

$$\boxed{j_\rho \overline{K}_b = M_{B(\rho+T_\rho)} L_\rho.} \quad (\text{CW4.2})$$

Here M means multiplication in the named receiving algebra, and the complete Gaussian jet factor is

$$B(\rho + T_\rho) = \sqrt{\pi} \exp\left(\frac{(\rho - 1/2)^2}{4}\right) \exp\left(\frac{\rho - 1/2}{2} T_\rho + \frac{1}{4} T_\rho^2\right). \quad (\text{CW4.3})$$

All exponentials in the nilpotent variable are taken modulo $T_\rho^{m_\rho}$; every surviving term is retained. Its inverse is

$$\frac{1}{\sqrt{\pi}} \exp\left(-\frac{(\rho - 1/2)^2}{4}\right) \exp\left(-\frac{\rho - 1/2}{2} T_\rho - \frac{1}{4} T_\rho^2\right). \quad (\text{CW4.4})$$

The derivative product rule proves (CW4.2) with its factorials. Multiplication of the finite exponentials proves the inverse formula, so the Gaussian unit has an explicit return rather than being discarded.

For any finite collection $\mathcal{S}$ of distinct actual zeros, the linear map

$$L_{\mathcal{S}} : V_+ \longrightarrow \prod_{\rho \in \mathcal{S}} A_\rho \quad (\text{CW4.5})$$

is onto. To prove it, let a linear functional on the finite target vanish on every $L_{\mathcal{S}}[x]$. Since $\lambda = \log \log x$ ranges over all real numbers as x ranges over $(1, \infty)$, this gives an identically vanishing function

$$\sum_{\rho \in \mathcal{S}} p_\rho(\lambda) e^{\rho \lambda} = 0, \quad \deg p_\rho < m_\rho. \quad (\text{CW4.6})$$

Fix ρ_0 . Apply

$$\prod_{\substack{\eta \in \mathcal{S} \\ \eta \neq \rho_0}} (\partial_\lambda - \eta)^{m_\eta}.$$

It kills all other summands. After removing $e^{\rho_0 \lambda}$, it acts on p_{ρ_0} by a product of $(\partial_\lambda + \rho_0 - \eta)^{m_\eta}$. On polynomials of degree below m_{ρ_0} , each such factor is triangular with nonzero diagonal, hence invertible. Therefore $p_{\rho_0} = 0$; the argument applies to every member of $\mathcal{S}$. The supposed target functional is zero, because its coefficients differ from those of the p_ρ only by the nonzero factorials in (CW4.1).

A proper linear subspace of a finite-dimensional vector space is annihilated by a nonzero functional: extend a basis of the subspace to a basis of the target and take one new coordinate functional. Therefore the absence of such a functional proves surjectivity in (CW4.5). This proof also applies to a single zero block. By the invertible componentwise Gaussian multiplication, $j_S \bar{K}_b$ is onto the same finite product.

The complete product map

$$L : V_+ \longrightarrow \prod_{\rho \in \mathcal{Z}} A_\rho \quad (\text{CW4.7})$$

is injective, by the proof of CW3 applied directly to P_v . No surjectivity onto that complete product has been inferred from finite surjectivity. In fact its constant coordinates obey an exact restriction:

$$|P_v(\rho)| \leq \sum_{j=1}^d |c_j| \max(1, t_j) \quad \text{for every } 0 < \Re \rho < 1. \quad (\text{CW4.8})$$

Thus they are bounded uniformly over the full zero set for each fixed source element. The complete product permits, for example, constant coordinates $|\Im \rho|$, which are unbounded along zeros by (CW3.5), so this product element is not in the image. This proves the non-surjectivity of (CW4.7), with no claim about the topology of the full quotient.

The image is nevertheless dense in the complete product when each finite local algebra has its ordinary finite-dimensional topology and the product has the product topology. A basic open neighbourhood restricts only finitely many coordinates. CW4.5 supplies an element realizing those coordinates exactly, hence meeting that neighbourhood. This density statement is about the product topology, not a Hilbert norm or an asserted purity metric.

126.5 CW5. Exact scaling at every primary block

For every $a > 0$, Taylor expansion of $(\log x^a)^s = (a \log x)^s$ proves

$$L_\rho \theta_a = M_{a^\rho \exp((\log a)T_\rho)} L_\rho. \quad (\text{CW5.1})$$

The exponential is again the full finite nilpotent sum, and the multiplier is the original unshifted operator. The Gaussian multiplier in CW4.2 commutes with it because both are multiplication in the same local algebra. Thus the finite-block diagrams and the complete-product diagram simultaneously preserve every positive-real dilation, in particular every prime and every prime power. The image of V_+ is an actual invariant subspace of Q , by the injective map (CW3.6), with finite jet maps onto all finite original-zero packets.

At no point does this identify $[x]$, T_ρ , a scalar identity, or a Mellin coordinate with primitive τ . The domain is a specified part of CC's coefficient algebra after arithmetic reconstruction, with its source scaling operation retained.

126.6 CW6. What the source multiplication does and does not preserve

These are linear intertwiners, not algebra homomorphisms for the original CC product. For $x, y > 1$, write $t = \log x$, $u = \log y$. Then

$$L_\rho([x][y]) = (t + u)^\rho \exp((\log(t + u))T_\rho), \quad (\text{CW6.1})$$

whereas multiplication of the two receiving values is

$$L_\rho[x] L_\rho[y] = (tu)^\rho \exp((\log t + \log u)T_\rho). \quad (\text{CW6.2})$$

These retain the source product $[xy]$ and its exact image. At $x = y = e$, the first constant is 2^ρ , while the second is 1; for a nontrivial zero their moduli differ because $0 < \Re \rho < 1$. Thus the failure is exhibited on an actual primary receiver, rather than hidden in notation.

The full strong-space convolution comparison is also exact:

$$K_b[x] * K_b[y] = b * K_b[\exp(tu)]. \quad (\text{CW6.3})$$

One can prove it directly in logarithmic coordinates, retaining all factors: convolution obeys $(W_t b) * (W_u b) = W_{tu}(b * b)$, and $b * W_{tu} b = W_{tu}(b * b)$, by substituting the integration variable and using the explicit square-root factors in W . The original source product instead gives $K_b[xy] = W_{t+u} b$. Formula (CW6.3) is a proved alternative convolution relation, not a replacement for the original CC algebra law.

Therefore the established comparison is an injective linear representation map for scaling, with all its exact finite and global jet observations. It does not assert a ring isomorphism, a positive metric, a lift of all infinite jet data, or Deligne purity merely from the existence of that map.

126.7 CW7. Every support label on the exact linear comparison

For the programme's bounded distributive lattice L , and any of the named complex vector spaces V , retain its labelled receiving semimodule

$$G_L(V) = \{(0, \lambda) : \lambda \in L\} \cup \{(v, 1_L) : v \in V\}.$$

Its receiving addition is $(v, \lambda) + (w, \mu) = (v + w, \lambda \vee \mu)$; the scalar $(c, \nu) \in G_L(\mathbb{C})$ acts by $(cv, \nu \wedge \lambda)$. These are the already specified receiving operations, not operations on primitive τ . A linear map $f : V \rightarrow V'$ induces

$$G_L(f)(v, \lambda) = (f(v), \lambda). \quad (\text{CW7.1})$$

It is well-defined because a non-top input label forces $v = 0$, and linearity gives $f(v) = 0$. Linearity and the unchanged joins and meets prove preservation of the displayed operations directly. Identity maps and compositions are preserved by substitution.

Applying this to $\overline{K}_b$ gives an injective map $G_L(V_+) \rightarrow G_L(Q)$: equality of images gives equality of labels and, by CW3.6, equality of amplitudes. Its inverse on its image sends $(\overline{K}_b v, \lambda)$ to (v, λ) , preserving that label as well. Every finite-jet map of CW4 lifts coordinatewise with exactly the same label, and its surjectivity lifts because a preimage exists for every top-supported amplitude while each $(0, \lambda)$ has preimage $(0, \lambda)$. This uses a common label on the finite product; independent labels on different blocks would be a different object and are not substituted for it.

In particular a scalar amplitude cancellation remains $(0, \lambda)$ under every lifted map, retaining the distinction between every supported zero and the bottom label. No $z_\lambda = (0, \lambda)$ is identified with primitive τ . This proves the full support return for the linear intertwiners, without asserting an unconstructed equivalence of semimodule cohomology theories.

126.8 Sources and exact reading scope

- The local operation rules, U01–U18, B1–B5/P1–P5, and the three complete user arguments listed in the opening paragraph were read before deriving the comparison. The complete-history reconstruction remains an input; this note does not construct integers from an incomplete clock.
- The actual source quotient and prime action are the already-read programme proofs *Global canonical Mellin involution*, G1–G5, and *Global infinitesimal quotient*, GIQ1–GIQ3. Their exact filenames are retained in the source ledger. The complete implications needed here are repeated in CW1–CW2 and CW4.

- Alain Connes and Caterina Consani retain authorship of the coefficient group algebra and θ_a action in CW0.1. The root read their original 2018 TeX, `thecurve_K.tex`, lines 2564–2601, including §7.1’s `holom`, `holom1` and `frobarith`, and proved the constant-function entry above. This subagent does not claim a whole-paper reading.
- The complete explicit zero-count input is Hasanalizade–Shen–Wong, arXiv:2107.06506v1, original `sources/2107.06506v1/source.tex`, lines 181–201, 221–277, and 309–359 read for this calculation. These include the count definition, full Corollary 1.2, and multiplicity convention from the argument principle. SHA256: 3FAA3CB45EE34B5FCD96F84ED036DE2AF1C3DCEA75723776410E2AC359FF5021. This is bounded original-source reading, not a claim to have reread the whole paper. CW3 uses its stated count of original-zeta zeros; it does not replace zeta by the source’s completed function. Jensen’s formula is the classical analytic theorem used in CW3.4, with the complete actual bound displayed.
- A bounded routing lookup located the existing canonical-index receipt, whose database is recorded as `research_canonical_index.sqlite` in the retained literature foundation. This was routing only, not a claim of reading its mathematical sources. No PDF was used.
- The root proposed K_b . This independent proof verifies its full factor, global injectivity, every finite-jet surjection, the complete-product injection, the precise multiplication defect, and all operator identities. No additional claim of literature novelty or RH resolution is made.

126.9 Current propagation from OMS — 24 September 2026

The complete synthesis proof, OMS1–OMS5, proves that the original summation image is closed and that the full actual-zero jet map on Q has zero common kernel. Its image remains a proper dense subspace of the unrestricted product, carrying the stronger original quotient topology. Earlier statements leaving those two questions undetermined are superseded by that proof, not by an assumption. Every original zero multiplicity and support label remains.

OMS7A proves the exact return to AC0–AC4: the same original spaces give $I_0=2$ $E(S)=E(S)$, with the factor 2 retained in the periodization map and inverse. Thus $D_{cl}=0$, $C_{alg} \rightarrow Q$ is the identity-induced topological isomorphism, and the specific global two-extension class of the actual adelic two-term complex is zero. Both endpoint contributions at each prime and the higher derived-coinvariant groups remain unchanged. No vanishing of the entire Ext group or full tau purity is inferred.

126.10 Global adelic lifting propagation - 24 September 2026

The complete current proofs are `GLOBAL_ADELIC_LIFTING_AND_PRIME_BOUNDARY.md` GAP0–GAP9 (including GAP6A–GAP6B), `COMPLETE_GENERATOR_ENDPOINT_RETURN.md` GER0–GER7, `GLOBAL_ENDPOINT_EXTENSION_RETURN.md` GEX0–GEX9, and `COMPLETE_COMPARISON_TRIANGLE_FORMALITY.md` CTF0–CTF8. Every original prime wedge, endpoint pair and nilpotent jet is retained. GER credits the existing RZ endpoint resolvents and gives their explicit original Schwartz return and covariance corrections. GEX proves vanishing of the entire extension groups between the whole actual Q and the stated endpoint modules, in the explicitly constructed algebraic and strict locally convex operator categories. This strengthens the earlier finite-block and particular-extension assertions without assigning that theorem to a larger unspecified category.

GAP and CTF construct the full derived adelic complex, its canonical cohomology map, the exact original CC comparison and its entire cone. The faithful coefficient copies distinguish primitive tau from integer one in both objects. All additional cone contractions and the final-arrow homotopy are explicit; no test-space section $Q \rightarrow A$ is asserted. On the separate supported- Q row the same polynomial is invertible, rather than zero. Full tau numerical purity does not follow from endpoint separation and remains unproved.

127 Finite adele support and arithmetic Frobenius in the distribution realization

Independent derivation, 24 September 2026.

This calculation restores the finite adele variables and the source's arithmetic Frobenius to the previously constructed distribution realization. The maps, residue, rational quotient action, and a finite-boundary contribution are computed explicitly. The resulting exact maps concern the stated distribution receiver; no identification with primitive Z_1/τ or with the complete cohomology of the adelic complex lift is presumed.

127.1 FA0. Controlling sources and prerequisite check

Alain Connes and Caterina Consani, *The Riemann–Roch strategy: Complex lift of the Scaling Site*, arXiv:1805.10501v1, is the human source. The original author file read is

`thecurve_K.tex`; its exact local path and source identity are retained in the private source-use ledger.

The exact source locators are:

- §5.2, Lemma `adelicomp1` and equations `actionpq`, `adelicomp3`, lines 1203–1224: $(x, y) \mapsto (ax + b, ay)$, $\bar{\partial} = \partial_x + i\partial_y$, and $P(\mathbb{Q}) \backslash \mathbb{A}_{\mathbb{Q}}^2$.
- §6.6, the paragraph before Proposition `comparescales`, lines 2381–2418: the vector fields $y\partial_x, y\partial_y$ and their full factor y .
- §7 and §7.1, lines 2532–2601: right vertical scaling, the coefficient maps $\theta_{\mu}([r]) = [r^{\mu}]$, the evaluation $\chi_{\lambda}([r]) = r^{\lambda}$, and arithmetic Frobenius $\text{Fr}_{\mu}^a = \theta_{\mu}R(\mu^{-1})$.
- §7.1, equations `holom2` and Proposition `functionq`, lines 2602–2624: the moving Teichmüller coefficient. That function q is different from the auxiliary quotient q_{ρ} used below.

The preceding programme derivations used here are:

- SDB receiver, SDB18–SDB31 and SDB40–SDB57.
- DR realization, DR1–DR8.
- Original auxiliary lifting calculation, TL2–TL3.

Before deriving the comparison, the current operation rules, the complete quoted definitions U01–U18, and the complete user passages USR-9ad1c0a2d09dba92, USR-f55d16feb948d8c2, USR-6152e3bc6302258c, USR-508e136619bfcd2, and USR-4322be19bff532cd in the retained user corpus were read. These passages require the whole arithmetic construction and forbid replacing the source by an invented numerical model. Every calculation below consequently takes place in a stated arithmetic receiver after the existing arithmetic reconstruction. None counts copies of τ , assigns source parity, or restores the retracted source addition. In FA7 the comparison changes because an actual finite-support contribution is restored; that result does not contradict a definition of primitive τ .

The symbols have distinct domains:

(X_f, Y_f)	finite adele coordinates
(X, Y)	archimedean coordinates
x, y, t	variables of the existing finite complex algebras
λ, μ	positive real evaluation and Frobenius parameters
q_{ρ}	the auxiliary algebra quotient $x \mapsto t, y \mapsto 0$
p	a rational prime.

127.2 FA1. Exact test spaces and finite pullback

Let $\mathbb{A}_f$ be the finite adeles of $\mathbb{Q}$. On each $\mathbb{Q}_\ell$, fix additive Haar measure with $\mathbb{Z}_\ell$ of mass 1, and use the product measure on $\mathbb{A}_f^2$. Put

$$\mathcal{S}_f = C_c^{\text{lc}}(\mathbb{A}_f^2), \quad \mathcal{V}_f = \mathcal{S}'_f, \quad (\text{FA1.1})$$

where the dual consists of continuous complex linear functionals on the usual Bruhat test-function space. A locally integrable function denotes its distribution relative to this measure.

For $a \in \mathbb{Q}^\times, b \in \mathbb{Q}$, set

$$L_f(a, b)(X_f, Y_f) = (aX_f + b, aY_f).$$

Its scalar-distribution pullback is

$$\langle P_f(a, b)v, \psi \rangle = |a|_f^{-2} \left\langle v, \psi \left(\frac{X_f - b}{a}, \frac{Y_f}{a} \right) \right\rangle, \quad |a|_f = \prod_{\ell < \infty} |a|_\ell. \quad (\text{FA1.2})$$

This is a distribution because the affine map carries compact locally constant test functions to such test functions and is continuous on their test-function space. Substitution proves that it agrees with composition for functions. The support is transformed exactly by

$$\text{supp } P_f(a, b)v = L_f(a, b)^{-1} \text{supp } v. \quad (\text{FA1.3})$$

The inclusion in one direction follows by testing away from that inverse image; the reverse inclusion follows by applying the inverse pullback. For a compact-open box,

$$1_{(u+U) \times (v+V)} \longmapsto 1_{((u-b)/a + a^{-1}U) \times (v/a + a^{-1}V)}. \quad (\text{FA1.4})$$

For a point mass,

$$P_f(a, b)\delta_{(u,v)} = |a|_f^{-2} \delta_{((u-b)/a, v/a)}. \quad (\text{FA1.5})$$

These different coefficient factors must not be interchanged.

The elementary product formula in this situation is

$$|a|_f = |a|_\infty^{-1}. \quad (\text{FA1.6})$$

Indeed write $a = \epsilon \prod_p p^{n_p}$, with finitely many nonzero integer n_p and $\epsilon = \pm 1$. Then $\prod_p |a|_p = \prod_p p^{-n_p} = |a|_\infty^{-1}$. Consequently the finite two-variable Jacobian in (FA1.2) is $|a|_\infty^2$. The archimedean two-variable scalar pullback has factor $|a|_\infty^{-2}$. Their product is exactly 1 on the entire rational adelic affine map. This cancellation is proved while retaining both original factors; it is not a choice to omit finite support.

We will use two specific finite distributions:

$$h_0 = 1_{X_f} \otimes 1_{Y_f}, \quad h_1 = 1_{X_f} \otimes \delta_0(Y_f). \quad (\text{FA1.7})$$

Their pairings are respectively integration over $\mathbb{A}_f^2$ and $\psi \mapsto \int_{\mathbb{A}_f} \psi(X_f, 0) dX_f$. Both are distributions although their supports are not compact: the tests are compactly supported. Their exact supports and actions are

$$\begin{aligned} \text{supp } h_0 &= \mathbb{A}_f^2, & P_f(a, b)h_0 &= h_0, \\ \text{supp } h_1 &= \mathbb{A}_f \times \{0\}, & P_f(a, b)h_1 &= |a|_\infty h_1. \end{aligned} \quad (\text{FA1.8})$$

For the second equality, substitute $X'_f = (X_f - b)/a$ in (FA1.2): its measure factor is $|a|_f$, leaving $|a|_f^{-1} = |a|_\infty$. For the first, the two substitutions cancel the full inverse determinant. The distributions are linearly independent: a test supported away from $Y_f = 0$ annihilates h_1 but need not annihilate h_0 .

127.2.1 FA1a. Entry from finite functions, and the exact current enlargement

Neither the existence of $\mathcal{V}_f$ nor the displayed delta current is asserted to be part of an unexamined source sheaf. The source paper constructs its adelic quotient and discusses W -valued functions; the distribution receiver here is an explicitly constructed enlargement. Its entry from the finite function space is

$$j_0 : C_c^{\text{lc}}(\mathbb{A}_f^2) \longrightarrow \mathcal{V}_f, \quad j_0(f)(\psi) = \int_{\mathbb{A}_f^2} f\psi. \quad (\text{FA1a.1})$$

It is injective: if a locally constant f is nonzero at a point, choose a compact-open neighborhood where its value is a nonzero constant and test by the indicator of that neighborhood. It intertwines scalar pullbacks by substitution in (FA1.2). The same formula defines the sheaf of locally constant functions as locally integrable distributions without a compact-support requirement; compact test functions still make the pairing finite.

Let $D_f = \mathbb{A}_f \times \{0\}$ and identify it with $\mathbb{A}_f$ using X_f . Restriction of a locally constant function to D_f is defined pointwise. Put

$$i_f^* f(X_f) = f(X_f, 0), \quad j_1(g)(\psi) = \int_{\mathbb{A}_f} g(X_f) \psi(X_f, 0) dX_f. \quad (\text{FA1a.2})$$

Both maps are defined on compactly supported locally constant functions and also on the corresponding sheaves without compact support. The map j_1 is injective by the same compact-open testing argument. Thus the exact source-to-current map is

$$\Lambda_f = j_1 i_f^*, \quad \Lambda_f(f) = f(X_f, 0) \delta_0(Y_f). \quad (\text{FA1a.3})$$

On the constant function 1, it gives h_1 . On the genuine finite test function $1_{\widehat{\mathbb{Z}} \times \widehat{\mathbb{Z}}}$, it gives $1_{\widehat{\mathbb{Z}}}(X_f) \delta_0(Y_f)$. Thus the construction is available on finite compact-open data as well as on global constants.

This delta current is not a regular finite function. The subset $\{0\} \subset \mathbb{A}_f$ has Haar measure zero: in a compact neighborhood it lies inside the sets with one fixed prime component restricted to $\ell^n \mathbb{Z}_\ell$, whose measures tend to zero. Hence D_f has product Haar measure zero locally. A regular locally integrable distribution supported on D_f must be zero, since its density vanishes almost everywhere outside D_f , and D_f has measure zero. But $j_1(g) \neq 0$ for $g \neq 0$, by injectivity. Consequently h_1 , and every such nonzero divisor current, lies outside the regular-function entry (FA1a.1).

The exact affine covariance of this new entry is

$$\boxed{P_f(a, b) \Lambda_f(f) = |a|_\infty \Lambda_f(P_f(a, b)f)}. \quad (\text{FA1a.4})$$

Indeed (FA1.2) applied to (FA1a.3), followed by the one-variable substitution in X_f , gives

$$|a|_f^{-1} f(aX_f + b, 0) \delta_0(Y_f).$$

Use (FA1.6) to obtain (FA1a.4). This is precisely the finite factor used later; it is not inferred from a dimension analogy.

The map is local in the sheaf sense. For every open subset $U \subset \mathbb{A}_f^2$, the same restriction and pairing formulas give a map from locally constant functions on U to distributions on U , and these maps commute with restriction to smaller open subsets. For a locally constant multiplier a_f on U ,

$$\Lambda_f(a_f f) = a_f \Lambda_f(f), \quad \text{supp } \Lambda_f(f) \subseteq \text{supp } f \cap D_f. \quad (\text{FA1a.5})$$

The first identity holds because multiplication of a delta current restricts the multiplier to D_f . The support assertion follows by testing where f vanishes or away from D_f . Thus no global coefficient projection is needed to construct this map from finite functions.

For an exact common coefficient space on which it is an endomorphism, define the sheaf

$$\mathcal{V}_f^{\text{reg},0}(U) = j_0(C^{\text{lc}}(U)) \oplus j_1(C^{\text{lc}}(U \cap D_f)). \quad (\text{FA1a.6})$$

The direct-sum assertion follows from the measure-zero argument: an equality of a regular distribution and a zero-stratum current forces both to vanish. Restrictions preserve the two summands, and $P_f(a, b)$ preserves their types by (FA1a.4). Define

$$\tilde{\Lambda}_f(j_0(f) + j_1(g)) = j_1(i_f^* f). \quad (\text{FA1a.7})$$

This map is well-defined by uniqueness of the decomposition, commutes with restrictions, is linear over the locally constant function sheaf, and kills the zero-stratum summand. It has $\tilde{\Lambda}_f^2 = 0$, and its affine covariance is (FA1a.4) with both input types included.

It also preserves every filtration defined by geometric support containment: if a distribution of (FA1a.6) has support in a closed set Z , both of its regular and zero-stratum parts have support in Z , by uniqueness of the decomposition on the open complement; (FA1a.5) then puts its image in Z . In particular it maps the whole coefficient space into the zero-stratum support step and maps that step to zero. Equality of supports is not claimed: h_0 has support $\mathbb{A}_f^2$, while $\tilde{\Lambda}_f h_0 = h_1$ has support D_f . Any external lattice index is separate from this geometric support change.

127.3 FA2. Both archimedean charts and the rational action

Retain the open domain

$$\Omega^\times = \mathbb{R} \times \mathbb{R}^\times. \quad (\text{FA2.1})$$

Its two components are indexed by $\sigma \in \{+1, -1\}$, with $\sigma Y > 0$. This adds the negative chart to DR without inserting the excluded stratum $Y = 0$. For $\rho \in \mathbb{C}$, $m \geq 1$, $\alpha = \rho + 1$, define

$$g_{\sigma,r}(Y) = 1_{\{\sigma Y > 0\}} |Y|^\alpha \frac{(\log |Y|)^r}{r!}, \quad r \geq 0. \quad (\text{FA2.2})$$

The indicator is smooth on the disconnected open set $\mathbb{R}^\times$. For each $c \in \mathbb{Q}$,

$$\Phi_{c,\sigma}(x^u y^j) = \delta^{(j)}(X - c) \otimes g_{\sigma, m-j-1-u}, \quad u + j < m. \quad (\text{FA2.3})$$

The same test-function and logarithmic-polynomial arguments as DR1 prove injectivity. With

$$\mathcal{J} = Y \partial_Y - \alpha, \quad \mathcal{N} = \partial_X \mathcal{J},$$

one still has $\mathcal{J} \Phi = \Phi M_x$ and $\mathcal{N} \Phi = \Phi M_y$, because $Y \partial_Y \log |Y| = 1$ on both components.

For arbitrary nonzero real a , scalar pullback satisfies

$$P_\infty(a, b)(\delta_c^{(j)} \otimes g) = |a|^{-1} a^{-j} \delta_{(c-b)/a}^{(j)} \otimes Q_a g, \quad Q_a g(Y) = g(aY). \quad (\text{FA2.4})$$

The factor a^{-j} , including its sign, comes from the j -th derivative of the test function; the factor $|a|^{-1}$ comes from the normal Jacobian. Thus, writing $\varepsilon = \text{sgn}(a)$,

$$P_\infty(a, b) \Phi_{c,\sigma}(x^u y^j) = \varepsilon^j |a|^{\rho-j} \sum_{k=0}^{m-j-1-u} \frac{(\log |a|)^k}{k!} \Phi_{(c-b)/a, \sigma \varepsilon}(x^{u+k} y^j).$$

(\text{FA2.5})

The formula follows by expanding $(\log |aY|)^r / r!$ with every binomial coefficient retained. It specializes exactly to DR2.5 for positive a .

On $\mathcal{V}_f \otimes \Phi(B)$ the full left rational pullback is

$$\mathcal{P}(a, b) = P_f(a, b) \otimes P_\infty(a, b). \quad (\text{FA2.6})$$

The tensor product is algebraic. It embeds injectively into distributions on the product test space $\mathcal{S}_f \otimes C_c^\infty(\Omega^\times)$. For injectivity, express a tensor using finitely many independent finite distributions; test functions give enough independent evaluations to isolate their coefficients, and then the archimedean distribution independence applies. No completion or unspecified infinite tensor sum is used.

The full support in (FA2.6) changes by (FA1.3) at finite places and by $(c, \sigma) \mapsto ((c - b)/a, \sigma\epsilon)$ at infinity. The rational pullbacks obey

$$\mathcal{P}(a, b)\mathcal{P}(a', b') = \mathcal{P}(a'a, a'b + b'). \quad (\text{FA2.7})$$

This is the exact right action, or the left action after group inversion. Arithmetic Frobenius below commutes with this action.

127.4 FA3. The full family complex and its residues

Let

$$\mathcal{J}_{\mathbb{Q}, \pm} = \bigoplus_{c \in \mathbb{Q}} \bigoplus_{\sigma = \pm 1} \bigoplus_{j \geq 0} \delta_c^{(j)} \otimes \mathcal{D}'(\{Y : \sigma Y > 0\}). \quad (\text{FA3.1})$$

Only finite sums are allowed in each element. Define $\mathcal{J}_f = \mathcal{V}_f \otimes \mathcal{J}_{\mathbb{Q}, \pm}$. For $\lambda > 0$, the original character-family operators are

$$D_\lambda = \lambda \partial_X + i \partial_Y, \quad \mathcal{L}_\lambda = Y D_\lambda. \quad (\text{FA3.2})$$

They act identically on finite distributions and branch labels. Their complete residues on the archimedean normal jets are

$$\begin{aligned} R_\lambda \left(\sum_j \delta_c^{(j)} \otimes g_j \right) &= \sum_j (-i/\lambda)^j \partial_Y^j g_j, \\ R_{Y, \lambda} \left(\sum_j \delta_c^{(j)} \otimes g_j \right) &= \sum_j (-i/\lambda)^j \partial_Y^j (Y^{-1} g_j). \end{aligned} \quad (\text{FA3.3})$$

They retain c, σ as separate target indices. The section is

$$E_Y h = \delta_c \otimes Y h \quad (\text{FA3.4})$$

on each branch. The exact homotopy is

$$T_{Y, \lambda} \left(\sum_j \delta_c^{(j)} \otimes g_j \right) = \frac{1}{\lambda} \sum_{j \geq 1} \sum_{k=0}^{j-1} (-i/\lambda)^k \delta_c^{(j-1-k)} \otimes \partial_Y^k (Y^{-1} g_j). \quad (\text{FA3.5})$$

All derivatives of Y^{-1} are retained by this formula; expanding them gives

$$\partial_Y^k (Y^{-1} g) = \sum_{r=0}^k \binom{k}{r} (-1)^r r! Y^{-r-1} g^{(k-r)}.$$

Direct differentiation and cancellation of adjacent terms give

$$\begin{aligned} R_{Y, \lambda} \mathcal{L}_\lambda &= 0, & R_{Y, \lambda} E_Y &= I, & T_{Y, \lambda} E_Y &= 0, \\ \mathcal{L}_\lambda T_{Y, \lambda} &= I - E_Y R_{Y, \lambda}, & T_{Y, \lambda} \mathcal{L}_\lambda &= I. \end{aligned} \quad (\text{FA3.6})$$

For clarity, before multiplying by Y , the terms of D_λ are $\lambda \delta_c^{(j+1)} \otimes g + i \delta_c^{(j)} \otimes g'$. In its residue their coefficients sum to $\lambda(-i/\lambda)^{j+1} + i(-i/\lambda)^j = 0$. In the homotopy, the same cancellation leaves

only the starting jet and its residue section. The highest normal coefficient proves injectivity of D_λ because $\lambda \neq 0$; it then proves the last identity. Multiplication by Y and Y^{-1} in their displayed order gives (FA3.6). Tensoring these explicit identities by the identity of $\mathcal{V}_f$ and taking finite branch sums proves them on $\mathcal{J}_f$, without an exactness assumption on a completed tensor product.

Now form the precisely specified family spaces

$$\mathcal{C}_f^0 = \prod_{\lambda>0} \mathcal{J}_f, \quad \mathcal{C}_f^1 = \prod_{\lambda>0} \mathcal{J}_f, \quad \mathcal{H}_f = \prod_{\lambda>0} \left(\mathcal{V}_f \otimes \bigoplus_{c,\sigma} \mathcal{D}'(\sigma Y > 0) \right). \quad (\text{FA3.7})$$

Each individual component has finite tensor and branch support. No common finite bound as λ varies is imposed. The componentwise operators give the split exact sequence

$$0 \longrightarrow \mathcal{C}_f^0 \xrightarrow{\mathcal{L}} \mathcal{C}_f^1 \xrightarrow{\mathcal{R}} \mathcal{H}_f \longrightarrow 0, \quad u \longmapsto (\mathcal{R}u, \mathcal{T}u), \quad (h, k) \longmapsto \mathcal{E}h + \mathcal{L}k. \quad (\text{FA3.8})$$

Both compositions equal the identity by (FA3.6). This constructs the receiver and proves its exactness.

For $a \in \mathbb{Q}^\times$, the **raw** residues obey

$$R_\lambda P_\infty(a, b) = |a|^{-1} Q_a R_\lambda, \quad R_{Y,\lambda} P_\infty(a, b) = \varepsilon Q_a R_{Y,\lambda}, \quad (\text{FA3.9})$$

with the indicated branch changes. The first identity follows from $|a|^{-1} a^{-j}$ in (FA2.4) and a^j from the derivative of Q_a . The second follows from $Y^{-1} P_\infty(a, b) = a P_\infty(a, b) Y^{-1}$. The original D_λ cochain action has

$$\rho^0(a, b) = \mathcal{P}(a, b), \quad \rho^1(a, b) = a \mathcal{P}(a, b), \quad (\text{FA3.10})$$

since $D_\lambda \mathcal{P} = a \mathcal{P} D_\lambda$. For $\mathcal{L}_\lambda$, both degrees carry $\mathcal{P}$. The receiver action for either cochain presentation is

$$v \otimes h_{c,\sigma} \longmapsto P_f(a, b) v \otimes \varepsilon Q_a h_{c,\sigma} \quad \text{at } ((c-b)/a, \sigma \varepsilon). \quad (\text{FA3.11})$$

The sign ε is required on the negative component. The section and homotopy commute with these actions: on E_Y , the factors are $|a|^{-1}$ from δ and a from $Q_a(Yh)$, yielding ε . The homotopy factors cancel as in (FA3.9), giving $T_{Y,\lambda} \mathcal{P} = \mathcal{P} T_{Y,\lambda}$.

127.5 FA4. Original arithmetic Frobenius on the realization

Let B_μ , $\mu > 0$, be the pullback of the right geometric map

$$(X_f, Y_f, X, Y) \longmapsto (X_f, Y_f, X, Y/\mu). \quad (\text{FA4.1})$$

It fixes the finite coordinates. At infinity its inverse Jacobian is μ , and its exact distribution pairing is

$$\langle B_\mu u, \varphi \rangle = \mu \langle u, \varphi(X, \mu Y) \rangle. \quad (\text{FA4.2})$$

Thus

$$B_\mu(\delta_c^{(j)} \otimes g) = \delta_c^{(j)} \otimes Q_{1/\mu} g. \quad (\text{FA4.3})$$

There is no normal X scaling. The rational branch c , the sign chart σ , and all finite-adele support are unchanged.

The coefficient algebra used explicitly by Connes–Consani here is

$$W_{\text{alg}} = \mathbb{C}[\mathbb{R}_{>0}^\times], \quad \theta_\mu([r]) = [r^\mu], \quad \chi_\lambda([r]) = r^\lambda. \quad (\text{FA4.4})$$

The authors' Teichmüller notation is not primitive Z_1/τ . Evaluation gives the injective map

$$\text{ev} : W_{\text{alg}} \otimes \mathcal{J}_f \longrightarrow \mathcal{C}_f^1, \quad \sum_{r \in F} [r] \otimes u_r \longmapsto \left(\sum_{r \in F} r^\lambda u_r \right)_\lambda. \quad (\text{FA4.5})$$

For distinct $r_1, \dots, r_n$, evaluations at $\lambda = 1, \dots, n$ have determinant $(\prod_k r_k) \prod_{k < l} (r_l - r_k) \neq 0$. Applying the inverse matrix to vector-valued coefficients proves injectivity. The map is not claimed onto the product family. This is a degreewise entry of the algebraic coefficient space. It is not asserted to be an inclusion of a full original W_{alg} -valued complex: multiplication by λ in $\mathcal{L}_\lambda$ need not preserve the finite exponential-sum image (FA4.5). The differential and its splitting are proved on the explicitly defined family complex (FA3.7).

The exact original arithmetic action on the algebraic coefficients is

$$[r] \otimes u \longmapsto [r^\mu] \otimes B_\mu u. \quad (\text{FA4.6})$$

Its family action is

$$(\mathfrak{F}_\mu u)_\lambda = B_\mu u_{\lambda\mu}. \quad (\text{FA4.7})$$

Evaluation of (FA4.6) gives (FA4.7) directly. On the original operator $\mathcal{L}_\lambda$,

$$\mathcal{L}_\lambda B_\mu = B_\mu \mathcal{L}_{\lambda\mu}, \quad R_{Y,\lambda} B_\mu = \mu^{-1} Q_{1/\mu} R_{Y,\lambda\mu}, \quad (\text{FA4.8})$$

$$T_{Y,\lambda} B_\mu = B_\mu T_{Y,\lambda\mu}, \quad B_\mu E_Y = E_Y (\mu^{-1} Q_{1/\mu}). \quad (\text{FA4.9})$$

For the first identity, $\partial_Y B_\mu = \mu^{-1} B_\mu \partial_Y$, $\partial_X B_\mu = B_\mu \partial_X$, and $B_\mu Y = \mu^{-1} Y B_\mu$. For the residue, each term has the complete coefficient

$$(-i/\lambda)^j \mu^{-j-1} = \mu^{-1} (-i/(\lambda\mu))^j;$$

the factor μ^{-1} before the j derivatives is supplied by $Y^{-1} Q_{1/\mu} = \mu^{-1} Q_{1/\mu} Y^{-1}$. The same termwise calculation proves the homotopy identity. Thus the induced action on the actual residue receiver is

$$(\mathfrak{F}_\mu^H h)_\lambda = \mu^{-1} Q_{1/\mu} h_{\lambda\mu}, \quad (\text{FA4.10})$$

with every finite coefficient and branch label retained.

The corresponding original D_λ -complex has arithmetic actions $B_\mu u_{\lambda\mu}$ in degree 0 and $\mu^{-1} B_\mu u_{\lambda\mu}$ in degree 1. Indeed $D_\lambda B_\mu = \mu^{-1} B_\mu D_{\lambda\mu}$. Its raw residue satisfies $R_\lambda B_\mu = Q_{1/\mu} R_{\lambda\mu}$; after the degree-one factor it has exactly (FA4.10). The cochain isomorphism to the $\mathcal{L}$ -complex multiplies degree 1 by Y . It intertwines both actions because $Y \mu^{-1} B_\mu = B_\mu Y$. This preserves all original cochain factors rather than treating them as optional twists.

On the exact DR realization, (FA4.3) yields

$$\boxed{\mathfrak{F}_\mu([r] \otimes v_f \otimes \Phi_{c,\sigma}(x^u y^j)) = [r^\mu] \otimes v_f \otimes \mu^{-(\rho+1)} \sum_{k=0}^{m-j-1-u} \frac{(-\log \mu)^k}{k!} \Phi_{c,\sigma}(x^{u+k} y^j).} \quad (\text{FA4.11})$$

Here $\mathfrak{F}$ means the algebraic action followed by evaluation when required. The geometric calculation is simply

$$g_{\sigma,r}(Y/\mu) = \mu^{-(\rho+1)} \sum_{k=0}^r \frac{(-\log \mu)^k}{k!} g_{\sigma,r-k}(Y).$$

The source arithmetic action therefore realizes

$$G_\mu = \mu^{-(\rho+1)} e^{-(\log \mu)J} \quad (\text{FA4.12})$$

on every y -degree, together with $[r] \mapsto [r^\mu]$. It commutes with both J and N ; it does not conjugate N by μ^{-1} . At a rational prime p , the scalar and all nilpotent coefficients are $p^{-(\rho+1)}$ and $(-\log p)^k/k!$, respectively.

To see the precise cohomological cancellation, let

$$C_{j,h}(z) = [w^h] \prod_{k=0}^{j-1} (z + w - k), \quad r_0 = m - j - 1 - u.$$

The full residue of (FA4.5) on a monomial is

$$\begin{aligned} & (\mathcal{R} \operatorname{ev}([r] \otimes v_f \otimes \Phi_{c,\sigma}(x^u y^j)))_\lambda \\ &= r^\lambda v_f \otimes \sigma^{j+1} (-i/\lambda)^j 1_{\{\sigma Y > 0\}} |Y|^{\rho-j} \sum_{h=0}^{\min(j, r_0)} C_{j,h}(\rho) \frac{(\log |Y|)^{r_0-h}}{(r_0-h)!}. \end{aligned} \quad (\text{FA4.13})$$

The sign σ^{j+1} comes from Y^{-1} and the j derivatives on the negative chart. Applying (FA4.10) introduces exactly

$$\mu^{-1} \mu^{-j} \mu^{-\rho+j} = \mu^{-(\rho+1)} \quad (\text{FA4.14})$$

from the degree factor, $(\lambda\mu)^{-j}$, and vertical scaling of $|Y|^{\rho-j}$. The logarithm becomes $\log |Y| - \log \mu$, with the complete coefficients in (FA4.11). The finite support remains unchanged. Thus the original arithmetic Frobenius does not inherit a different scalar eigenvalue at different j merely because left dilation did.

127.6 FA5. Exact comparison of left primes and arithmetic primes

For positive rational a , define the horizontal-and-finite operator

$$\mathcal{H}_a = P_f(a, 0) \otimes (X, Y \mapsto aX, Y)^*. \quad (\text{FA5.1})$$

It sends finite support through the inverse finite dilation, and sends

$$\delta_c^{(j)} \otimes g \longmapsto a^{-j-1} \delta_{c/a}^{(j)} \otimes g \quad (\text{FA5.2})$$

at infinity. Its finite two-coordinate Jacobian is a^2 ; the horizontal real Jacobian is a^{-1} . Formula (FA5.2) includes the derivative factor a^{-j} as well. At the level of geometric maps,

$$B_a \mathcal{P}(a, 0) = \mathcal{H}_a. \quad (\text{FA5.3})$$

Indeed vertical scaling by $1/a$ cancels the archimedean vertical scaling by a , while the finite and horizontal coordinates are exactly those displayed in (FA5.1). All scalar-distribution pullback factors multiply accordingly.

On algebraic W coefficients the complete identity is

$$\boxed{\mathfrak{F}_a \mathcal{P}(a, 0) = \theta_a \mathcal{H}_a, \quad \mathcal{P}(a, 0) = \theta_a \mathcal{H}_a \mathfrak{F}_a^{-1}.} \quad (\text{FA5.4})$$

These operations commute in this identity: θ_a changes only coefficient labels, B_a only the vertical real coordinate, and $\mathcal{H}_a$ the remaining displayed coordinates. At $[r]$ the inverse arithmetic action changes the label to $[r^{1/a}]$; the subsequent θ_a returns it to $[r]$. This accounts for the coefficient map in the second identity.

For the invariant finite distribution h_0 and the unit coefficient $[1]$, (FA5.4) has the exact scalar product

$$a^{-j-1} \cdot a^{\rho+1} = a^{\rho-j}, \quad (\text{FA5.5})$$

and the inverse arithmetic logarithmic series has $(\log a)^k/k!$. This recovers every term of the original left action in DR. It explains where the j -dependent factor lies: horizontal normal derivatives and finite coefficient action are retained; they are not part of (FA4.12) alone.

The quotient q_ρ and section i_ρ extend by the identity on W_{alg} , finite distributions, and branch indices. They commute with $\mathfrak{F}_\mu$ because $q_\rho J = T q_\rho$, $J i_\rho = i_\rho T$, and (FA4.12) contains no additional y -mixing. They commute with the rational affine action by (FA2.5). The previously computed comparison with residue remains exact, using (FA4.13) and its explicit $j = 0$ projection. The quotient is still not a differential or a residue kernel.

For actual zeros with $0 < \Re \rho < 1$, the residue in (FA4.13) is injective on the algebraic W -coefficient, finite-distribution realization of all retained blocks. To verify this, first separate branches and finite linear coefficient spans by test functions. The exponents $\rho - j$ are distinct for distinct pairs (ρ, j) : equality forces the real difference of two numbers in $(0, 1)$ to be the integer $j - j'$, hence forces $j = j'$, then $\rho = \rho'$. Finite exponential-polynomial independence therefore separates these blocks. The derivative polynomial $\prod_{k=0}^{j-1} (\rho + \partial_s - k)$ is invertible on the finite logarithmic-polynomial space, because $\rho \notin \{0, \dots, j-1\}$. After undoing it and the nonzero λ^{-j} , the remaining r^λ coefficients are independent by the Vandermonde argument in (FA4.5). This proves the assertion without suppressing a finite support factor or an evaluation parameter.

127.7 FA6. The complete rational quotient cochain comparison

The split sequence (FA3.8) is equivariant for the full rational affine group and the arithmetic Frobenius. The finite coordinates commute with all archimedean differential operators. Left affine maps commute with right vertical scaling, so the two group actions commute. Thus the existing homotopy gives an exact construction after taking rational quotient group cochains, as follows.

Let $\Gamma = P(\mathbb{Q})$, and convert the right pullbacks to left module actions by $g \cdot v = \mathcal{P}(g^{-1})v$, with the receiver action (FA3.11). For a left Γ -module V , define

$$C^r(\Gamma, V) = \text{Map}(\Gamma^r, V)$$

and use the full differential

$$\begin{aligned} (d_\Gamma f)(g_1, \dots, g_{r+1}) &= g_1 f(g_2, \dots, g_{r+1}) \\ &\quad + \sum_{k=1}^r (-1)^k f(g_1, \dots, g_k g_{k+1}, \dots, g_{r+1}) \\ &\quad + (-1)^{r+1} f(g_1, \dots, g_r). \end{aligned} \tag{FA6.1}$$

All coefficient maps in (FA3.8) extend pointwise. Their Γ -equivariance proves that they commute with every term of (FA6.1).

Form the double complex with horizontal degree r and vertical degrees $0, 1$ from $\mathcal{C}_f^\bullet$. Its total differential on horizontal degree r is

$$d_{\text{tot}} = d_\Gamma + (-1)^r \mathcal{L}. \tag{FA6.2}$$

Put

$$\mathcal{H}_{\text{tot}} = (-1)^r \mathcal{T} \text{ on } C^r(\Gamma, \mathcal{C}_f^1), \quad \mathcal{H}_{\text{tot}} = 0 \text{ on } C^r(\Gamma, \mathcal{C}_f^0). \tag{FA6.3}$$

In vertical degree 1, the two horizontal terms $(-1)^r d_\Gamma \mathcal{T}$ and $(-1)^{r+1} \mathcal{T} d_\Gamma$ cancel. The vertical term is $\mathcal{L}\mathcal{T} = I - \mathcal{E}\mathcal{R}$. In degree 0, the surviving composition is $\mathcal{T}\mathcal{L} = I$. Therefore

$$d_{\text{tot}} \mathcal{H}_{\text{tot}} + \mathcal{H}_{\text{tot}} d_{\text{tot}} = I - \mathcal{E}\mathcal{R} \tag{FA6.4}$$

with $\mathcal{E}\mathcal{R}$ understood as zero on vertical degree 0. The maps $\mathcal{E}, \mathcal{R}$ are inverse on the residue complex, and (FA6.4) proves the homotopy equivalence, with degree 1 retained:

$$H^n(\text{Tot } C^\bullet(\Gamma, \mathcal{C}_f^\bullet)) \cong H^{n-1}(\Gamma, \mathcal{H}_f) \quad (n \geq 1), \quad H^0 = 0. \tag{FA6.5}$$

The arithmetic action commutes with this entire homotopy by (FA4.8)–(FA4.10), so the isomorphism is arithmetic-Frobenius equivariant.

This is an exact rational-quotient calculation for the defined finite-adele distribution complex. Its construction does not require the invariant or coinvariant functor to be exact. It is not a claim that this coefficient complex already computes every sheaf or completed cohomology of $P(\mathbb{Q}) \backslash \mathbb{A}_{\mathbb{Q}}^2$. In particular $Y = 0$ at infinity remains excluded, and the full character product is not identified with an unspecified completed W -coefficient sheaf.

127.8 FA7. A finite boundary contribution changes the weight comparison

The previous DR uniqueness theorem concerned the archimedean rational-orbit receiver. We now calculate, rather than discard, the new kernel maps supplied by an actual finite-adele boundary coefficient.

Let

$$V_f^{(01)} = \mathbb{C}h_0 \oplus \mathbb{C}h_1 \subset \mathcal{V}_f \quad (\text{FA7.1})$$

with h_0, h_1 from (FA1.7). It is stable under the complete rational affine action, by (FA1.8), and fixed pointwise by right archimedean arithmetic Frobenius. Take all finite rational branch sums on both sign charts:

$$A_{\rho, \mathbb{Q}, \pm} = \bigoplus_{c, \sigma} A_{\rho, c, \sigma}, \quad B_{\rho, \mathbb{Q}, \pm} = \bigoplus_{c, \sigma} B_{\rho, c, \sigma}.$$

Let $q = I \otimes q_\rho$ and $i = I \otimes i_\rho$ on the tensor products with $V_f^{(01)}$. The full maps are also tensored with W_{alg} when those source coefficients are retained. No primitive source operation is involved.

For $m \geq 2$, define a complex-linear map into the exact kernel of q by

$$\begin{aligned} H(h_0 \otimes f(t)_{c, \sigma}) &= \sigma h_1 \otimes (yf(x))_{c, \sigma}, \\ H(h_1 \otimes f(t)_{c, \sigma}) &= 0. \end{aligned} \quad (\text{FA7.2})$$

The linear map $f(t) \mapsto yf(x)$ is well-defined modulo t^m because $yx^m = 0$ in B_ρ . It kills t^{m-1} , and $qH = 0$. It is nonzero since $H(h_0 \otimes 1_{c, \sigma}) = \sigma h_1 \otimes y_{c, \sigma} \neq 0$.

We prove full rational equivariance with all signs. Write $a = \varepsilon|a|$. On $h_0 \otimes f(t)_{c, \sigma}$, the source action sends it to

$$h_0 \otimes |a|^\rho e^{(\log |a|)t} f(t)_{(c-b)/a, \sigma \varepsilon}.$$

Applying H gives coefficient $\sigma \varepsilon |a|^\rho h_1$ times $ye^{(\log |a|)x} f(x)$. In the other order, the finite factor in (FA1.8) is $|a|$, and the $j = 1$ normal factor in (FA2.5) is $\varepsilon |a|^{\rho-1}$. Their product, including the original σ , is exactly

$$\sigma \cdot |a| \cdot \varepsilon |a|^{\rho-1} = \sigma \varepsilon |a|^\rho. \quad (\text{FA7.3})$$

The logarithmic polynomial is identical in both orders. On the h_1 -source summand, both compositions are zero because that summand is stable and H is zero there. The rational branch change is the same in both orders. Thus H commutes with full $P(\mathbb{Q})$.

It also commutes with the exact arithmetic action. The finite coefficients and σ are unchanged, and (FA4.12) commutes with multiplication by y :

$$e^{-(\log \mu)x} yf(x) = ye^{-(\log \mu)x} f(x). \quad (\text{FA7.4})$$

The source action on A_ρ is the same $\mu^{-(\rho+1)} e^{-(\log \mu)t}$. The coefficient change $[r] \mapsto [r^\mu]$ commutes with the identity on W_{alg} . This proves arithmetic-Frobenius equivariance on every displayed coefficient space.

There is a stronger, sheaf-local version of (FA7.2). Use the coefficient sheaf (FA1a.6) or its compactly supported sections, and set

$$H_{\text{loc}}(v_f \otimes f(t)_{c, \sigma}) = \sigma \tilde{\Lambda}_f(v_f) \otimes (yf(x))_{c, \sigma}. \quad (\text{FA7.4a})$$

On regular input, the finite operation is exactly restriction to $Y_f = 0$ followed by current inclusion; on zero-stratum input it is zero. Equation (FA1a.4) supplies the factor $|a|$, so the full calculation (FA7.3) proves rational equivariance for every such finite function or current, not only for the constant two-dimensional coefficient space. Arithmetic Frobenius leaves this finite operation unchanged, and (FA7.4) proves its arithmetic equivariance. The map is compatible with multiplication by finite locally constant functions and with restriction to finite-adele open sets. Its geometric support can only decrease to D_f , while its rational branch and its sign chart

are retained before applying a group action. It is nonzero on compact finite tests, for example by applying it to $1_{\widehat{\mathbb{Z}}_2} \otimes 1_{c,\sigma}$.

This supplies an exact map **from** a finite function input **into** the defined current enlargement. The source paper's function complex is not thereby declared closed under delta currents. If one restricts both domain and codomain to regular finite functions, this particular H_{loc} is not an endomorphism, because of the proved nonmembership in FA1a. Sheaf-locality and a filtration by support containment do not exclude it in the current receiver: both properties have been proved. A different, more restrictive source category must be specified before making a claim that its morphisms contain or exclude this map.

Consequently, for each $z \in \mathbb{C}$,

$$\boxed{s_z = i + zH, \quad qs_z = I} \quad (\text{FA7.5})$$

is an exact section commuting with both full rational affine action and arithmetic Frobenius. The sections are distinct for distinct z when $m \geq 2$, by the nonzero value in (FA7.2). Replacing H by H_{loc} gives the same exact section family on the sheaf (FA1a.6), with all maps compatible with finite-adele restrictions. No choice of an off-critical zero is needed for this construction. It holds for any specified ρ and retained multiplicity $m \geq 2$. If $m = 1$, the auxiliary kernel is zero and this additional map is zero.

For a positive prime p , the compensation can be seen directly in the scalar eigenvalues:

$$\underbrace{p}_{\text{finite support } \mathbb{A}_f \times \{0\}} \cdot \underbrace{p^{\rho-1}}_{\text{normal order } j=1} = p^\rho. \quad (\text{FA7.6})$$

Under the source arithmetic Frobenius all of these j -blocks already have the common scalar $p^{-(\rho+1)}$, as calculated in (FA4.11)–(FA4.14). Thus the former archimedean weight gap is filled by a specific retained finite boundary factor in this enlarged receiver.

This result does not say that a lift fails: the entire family (FA7.5) consists of lifts. It proves that uniqueness by the earlier archimedean weight comparison does not extend unchanged to this explicit finite-adele object. It also does not imply a zero of ζ outside the critical line. The new data are the actual finite support, its modulus, and the resulting exact equivariant kernel map.

127.9 FA8. What is now available to the original specialization calculation

The mathematical connection now includes both directions needed to use the existing auxiliary object:

1. Its complete jets embed into distributions with finite adele coefficients by (FA2.3), (FA2.6). All prime powers and logarithmic nilpotent terms are retained.
2. Original source arithmetic Frobenius acts by (FA4.11), with the complete residue action (FA4.13)–(FA4.14). The exact relation to the left prime action is (FA5.4).
3. The auxiliary quotient is recovered from the injective residue image by the explicit DR spectral projection, tensored with finite coefficients and retaining the evaluation parameter. It is not a boundary quotient.
4. The full rational group-cochain comparison has the explicit equivariant homotopy (FA6.4). This keeps the nonzero residue receiver and does not replace the problem by an acyclic finite-torus example.
5. The finite stratum $Y_f = 0$ has an explicit effect on the actual lifting calculation, given by (FA7.2)–(FA7.6).

Every displayed amplitude map preserves an additional specified support index by $(v, \lambda_0) \mapsto (f(v), \lambda_0)$. For the historical lattice receiver this is well-defined because all maps send amplitude zero to amplitude zero. No index is identified with another after cancellation. The arithmetic evaluation parameter λ and the retained lattice index λ_0 are different variables.

The remaining comparison is not filled by renaming any vector-space zero as Z_1/τ . The user's primitive source, the full finite-adele support, the archimedean $Y = 0$ stratum, and the exact specialization map retain their stated domains. The constructed distribution group-cochain receiver is an explicit receiving object with proved maps, not an asserted replacement for all of those objects.

128 The finite boundary trace, its sheaf exact sequence, and the original Dolbeault connecting map

Independent receiving calculation FB0–FB9, 24 September 2026. The complete coefficient-character family is retained in FB6A–FB6B, with its exact arithmetic Frobenius, rather than being replaced by the $\lambda = 1$ specialization.

The finite boundary supplies an actual local trace map. Its rational scaling character compensates one archimedean normal derivative. The map is constructed on the complete finite-adele transversal, with every original Jacobian, before taking an adelic quotient. It is not a projection onto a selected two-dimensional space.

128.1 FB0. Source, user definitions, and scope

Human source: Alain Connes and Caterina Consani, *The Riemann–Roch strategy: Complex lift of the Scaling Site*, arXiv:1805.10501v1, original author file thecurve_K.tex. The source passages actually read are §5.2, equation actionpq, Lemma adelicomp1, Definition adelicomp2, and the subsequent description of the open set V ; and §6.6 through Proposition compare-scalcs. The beginning of §6.8, including degenerate triangular structures, was also read. Their original data are

$$\begin{aligned}\ell(a, b)(X, Y) &= (aX + b, aY), & \mathcal{C}_{\mathbb{Q}} &= P(\mathbb{Q}) \backslash \mathbb{A}_{\mathbb{Q}}^2, \\ D &= \partial_{X_{\infty}} + i\partial_{Y_{\infty}}, & D_Y &= Y_{\infty}D.\end{aligned}\tag{FB0.1}$$

The positive upper-half-plane branch uses $a \in \mathbb{Q}_{>0}$, $b \in \mathbb{Q}$, and $Y_{\infty} > 0$. Source §5.2 retains the finite boundary $Y_f = 0$ in the full adelic space. Its stated open set on which the right real action is free excludes this boundary.

The operation rules in `CORPUS_AND_OPERATION_RULES.md` were read before this calculation. The exact user passages USR-9ad1c0a2d09dba92, USR-f55d16feb948d8c2, and USR-6152e3bc6302258c were read from the complete preserved user corpus. They require the full arithmetic reconstruction, globally compatible comparisons, and a proved Deligne comparison. The original arithmetic has already been reconstructed in the receiving programme; no conclusion below treats that arithmetic as absent. U01–U18 were consulted for source notation and corrections.

Thus Z_0, Z_1, Z_2, τ retain their source meanings. The coordinate equation $Y_f = 0$ below belongs to Connes–Consani’s specified adelic affine space. It does not assign a coordinate or parity to τ . No addition on source τ is used. Existing supported coefficients z_{λ} retain their labels.

The locally constant coefficient sheaf below is the specified finite-transversal receiving sheaf. The cited passages provide the adelic space, affine action and archimedean operators; they do not themselves assert that the distribution enlargement constructed below is identical to all cohomology of the adelic quotient.

The original coefficient-character operator from §7.1 is also retained:

$$L_{\lambda} = Y_{\infty}(\lambda\partial_{X_{\infty}} + i\partial_{Y_{\infty}}), \quad \lambda > 0.\tag{FB0.2}$$

It arises from $(\lambda X + iY)\chi_{\lambda}(f)$ in source equation holom. The coordinate Y_{∞} is distinct from the source’s vector field denoted $Y = Y_{\infty}\partial_{Y_{\infty}}$. FB6A–FB6B retain the character change $\lambda \mapsto \lambda\mu$.

128.2 FB1. The finite trace is a local sheaf map

Put

$$F = \mathbb{A}_f, \quad E = F_X \times F_Y, \quad Z = F_X \times \{0\}, \quad U = E \setminus Z.$$

Write $i : Z \hookrightarrow E$, $j : U \hookrightarrow E$. Let $\mathcal{L}_E$ and $\mathcal{L}_Z$ be the sheaves of complex-valued locally constant functions. Choose additive Haar measures dX_f, dY_f with the original arithmetic convention

$$\text{vol}(\widehat{\mathbb{Z}}) = 1.\tag{FB1.1}$$

This convention fixes the scalar coefficient of every density in the following formulas; it is not removed by a rescaling.

For an open set $O \subset E$, let $\mathcal{D}'_E(O)$ be the continuous dual of the compactly supported locally constant test functions on O , with their usual inductive-limit topology. There are injective maps

$$\begin{aligned}\iota_E(f)(\varphi) &= \int_O f(X_f, Y_f) \varphi(X_f, Y_f) dX_f dY_f, \\ \iota_Z(k)(\varphi) &= \int_{O \cap Z} k(X_f) \varphi(X_f, 0) dX_f.\end{aligned}\tag{FB1.2}$$

The second map has domain $i_*\mathcal{L}_Z(O)$. Both are well defined because a locally constant function has finitely many values on any compact set. To prove injectivity, a nonzero value is constant on a compact open neighborhood of positive Haar measure; the indicator of a smaller such neighborhood detects it.

Restriction and boundary insertion give

$$\begin{aligned}r : \mathcal{L}_E &\longrightarrow i_*\mathcal{L}_Z, & r(f)(X_f) &= f(X_f, 0), \\ \boxed{T = \iota_Z r, \quad T(f) &= f(X_f, 0) dX_f \otimes \delta_0(Y_f).}\end{aligned}\tag{FB1.3}$$

All three maps commute with restriction to open subsets. In particular T is a sheaf map, and

$$\text{supp } T(f) = i(\text{supp}(f|_Z)) \subset \text{supp } f \cap Z.\tag{FB1.4}$$

The equality follows because each nonzero locally constant value on Z is detected by a compact open test there.

The map is $\mathcal{L}_E$ -linear:

$$T(gf) = gT(f),\tag{FB1.5}$$

where multiplication on the right restricts g to the boundary. This locality and module law distinguish it from an arbitrary projection onto a finite-dimensional distribution space.

128.3 FB2. Exact sequence, image, and absence of a sheaf splitting into functions

Restriction gives the exact sheaf sequence

$$0 \longrightarrow j_!\mathcal{L}_U \longrightarrow \mathcal{L}_E \xrightarrow{r} i_*\mathcal{L}_Z \longrightarrow 0.\tag{FB2.1}$$

Exactness can be checked at every stalk. At a point off Z , the right stalk is zero and the first map is the identity. At a point of Z , restriction of a locally constant germ is its constant value and is onto; a germ restricting to zero is identically zero on a neighborhood. Thus its kernel stalk and the stalk of $j_!\mathcal{L}_U$ both vanish. This also proves

$$\ker T = j_!\mathcal{L}_U, \quad \text{im } T = \mathcal{B} := \iota_Z(i_*\mathcal{L}_Z).\tag{FB2.2}$$

There is no nonzero locally constant function on E supported in Z . The set Z has empty interior: every neighborhood of $0 \in F_Y$ contains nonzero elements. A nonzero locally constant function is nonzero on an open neighborhood, so cannot have that support.

Consequently (FB2.1) has no sheaf-linear section. The image of a section $i_*\mathcal{L}_Z \rightarrow \mathcal{L}_E$ would restrict to zero on U , hence consist of functions supported on Z , and therefore be zero. It could not split the nonzero boundary restriction.

This does not obstruct global restriction or extension of individual functions. Globally $k(X_f)$ extends as $k(X_f)$ independent of Y_f . For compactly supported k , choose a compact open neighborhood K of zero in F_Y ; then $k(X_f)1_K(Y_f)$ is a compactly supported locally constant extension. These extensions are not sheaf sections of (FB2.1): values away from Z depend on boundary data outside the local open set. The distribution insertion ι_Z , in contrast, is local and has the specified boundary support.

128.4 FB3. Exact distribution enlargement and its cokernel

Let $\mathcal{D}'_{E,Z}$ be distributions on E supported in Z . There is a canonical isomorphism

$$i_* \mathcal{D}'_Z \xrightarrow{\sim} \mathcal{D}'_{E,Z}, \quad v \mapsto [\varphi \mapsto v(\varphi|_Z)]. \quad (\text{FB3.1})$$

Here $\mathcal{D}'_Z$ means all finite-transversal distributions on F_X , not only densities.

For surjectivity, take a distribution u supported in Z . If a compactly supported locally constant test φ vanishes on Z , local constancy and compactness imply that its compact support is disjoint from Z : at any point of its support, a nonzero local constant value persists. Thus $u(\varphi) = 0$. Restriction of compact tests to Z is onto, by the extension construction in FB2. On an arbitrary open O , cover the compact boundary support by finitely many compact-open rectangles contained in O , partition its X_f support into finitely many disjoint compact-open pieces subordinate to these rectangles, and use their respective Y_f indicators. This supplies the required compact extension inside O . Therefore u factors uniquely through restriction. Its induced functional is continuous: on each finite-dimensional test stage it is continuous, and the locally constant test topology is the inductive limit of these stages. This proves (FB3.1).

In particular there are no additional normal derivatives at the finite boundary in this test category. Every distribution supported there is exactly a distribution in X_f multiplied by $\delta_0(Y_f)$. This fact concerns the specified totally disconnected tests; it does not remove any archimedean normal derivative.

The exact quotient of distributions by the image of T has the filtration

$$0 \longrightarrow i_*(\mathcal{D}'_Z / \iota_Z^{\text{int}} \mathcal{L}_Z) \longrightarrow \mathcal{D}'_E / \mathcal{B} \longrightarrow \mathcal{D}'_E / \mathcal{D}'_{E,Z} \longrightarrow 0, \quad (\text{FB3.2})$$

where $\iota_Z^{\text{int}} k = k dX_f$ is the internal density map on Z . The injection and quotient are those of nested subspaces $\mathcal{B} \subset \mathcal{D}'_{E,Z} \subset \mathcal{D}'_E$; their exactness follows by taking the displayed quotients. For example $\delta_0(X_f)\delta_0(Y_f)$ lies in $\mathcal{D}'_{E,Z}$ but not in $\mathcal{B}$, because no locally constant Haar density in X_f is supported on its singleton. Thus the first cokernel piece need not vanish.

Moreover

$$\iota_E(\mathcal{L}_E) \cap \mathcal{D}'_{E,Z} = 0. \quad (\text{FB3.3})$$

A nonzero locally constant density has a nonzero value on a compact open set meeting U , and a test there detects it; it therefore cannot be supported in Z .

The minimal coefficient sheaf containing the regular functions as densities and the boundary trace image is consequently the internal direct sum

$$\boxed{\mathcal{M} = \iota_E(\mathcal{L}_E) \oplus \mathcal{B} \subset \mathcal{D}'_E.} \quad (\text{FB3.4})$$

It has the exact sequence

$$0 \longrightarrow \iota_E(\mathcal{L}_E) \longrightarrow \mathcal{M} \longrightarrow \mathcal{B} \longrightarrow 0 \quad (\text{FB3.5})$$

split by the actual boundary inclusion. This split sequence is different from the nonsplit function-restriction sequence (FB2.1).

The span of the two global distributions

$$h_0 = dX_f dY_f, \quad h_1 = dX_f \delta_0(Y_f) \quad (\text{FB3.6})$$

is not a coefficient subsheaf or $\mathcal{L}_E$ -submodule: multiplication by a nonconstant compact-open indicator produces distributions outside that two-dimensional span. Its full sheaf-generated enlargement is (FB3.4). No projection onto only the two global vectors is used.

128.5 FB4. Full affine covariance and the finite product formula

For $a \in \mathbb{Q}_{>0}$, $b \in \mathbb{Q}$, put

$$L_f(a, b)(X_f, Y_f) = (aX_f + b, aY_f), \quad q = |a|_f.$$

Scalar-distribution pullback is

$$\langle P_f(a, b)u, \varphi \rangle = q^{-2} \left\langle u, \varphi \left(\frac{X_f - b}{a}, \frac{Y_f}{a} \right) \right\rangle. \quad (\text{FB4.1})$$

Both finite Jacobian factors remain. On regular distributions it agrees with function pullback:

$$P_f \iota_E(f) = \iota_E(f \circ L_f). \quad (\text{FB4.2})$$

On boundary densities, changing $X_f = az + b$, with $dX_f = q dz$, gives

$$P_f \iota_Z(k) = q^{-1} \iota_Z(k(aX_f + b)). \quad (\text{FB4.3})$$

Hence

$$\boxed{P_f T = q^{-1} T P_f^{\text{fun}}}. \quad (\text{FB4.4})$$

The arithmetic product formula is exact here:

$$a = \prod_p p^{v_p(a)}, \quad q = \prod_p p^{-v_p(a)} = a^{-1}, \quad q^{-1} = a. \quad (\text{FB4.5})$$

All but finitely many factors equal one. Thus h_0 is fixed and h_1 transforms by a , with no arbitrary character supplied:

$$P_f h_0 = h_0, \quad P_f h_1 = a h_1. \quad (\text{FB4.6})$$

The same finite trace commutes with multiplication of Y_f by an element of $\widehat{\mathbb{Z}}^*$, because this fixes $Y_f = 0$ and has modulus one.

Extend the trace to the minimal sheaf by

$$N_f(\iota_E(f) + \iota_Z(k)) = \iota_Z(f|_Z). \quad (\text{FB4.7})$$

Its unique decomposition in (FB3.4) proves that this is well defined and local. Its full algebra is

$$N_f^2 = 0, \quad \ker N_f = \iota_E(j! \mathcal{L}_U) \oplus \mathcal{B}, \quad \text{im } N_f = \mathcal{B}, \quad \text{coker } N_f \cong \iota_E(\mathcal{L}_E),$$

$$P_f N_f = a N_f P_f. \quad (\text{FB4.8})$$

The covariance follows on both summands from (FB4.2)–(FB4.4). This is an actual coefficient operator on a stated enlargement, not an operator already acting inside the original function summand.

128.6 FB5. The compensated normal derivative is an equivariant chain map

Write $\Omega_\infty = \mathbb{R}_X \times (0, \infty)_Y$. Retain the DR supported space on every rational branch:

$$\mathcal{J}_{\mathbb{Q}} = \bigoplus_{c \in \mathbb{Q}} \bigoplus_{j \geq 0} \delta_c^{(j)}(X_\infty) \otimes \mathcal{D}'((0, \infty)). \quad (\text{FB5.1})$$

Every vector has finite branch and normal-order support. All statements also hold locally as sheaf statements. Scalar pullback at infinity has the original factor a^{-2} , and DR2 gives

$$P_\infty(a, b)(\delta_c^{(j)} \otimes g) = a^{-j-1} \delta_{(c-b)/a}^{(j)} \otimes Q_a g, \quad Q_a g(Y) = g(aY). \quad (\text{FB5.2})$$

Therefore

$$P_\infty \partial_{X_\infty} = a^{-1} \partial_{X_\infty} P_\infty. \quad (\text{FB5.3})$$

Use the original diagonal rational action $P_\infty \otimes P_f$, not separate unrelated choices of a . The coefficient trace and normal derivative combine as

$$\boxed{\mathcal{B} = \partial_{X_\infty} \otimes T : \mathcal{J}_\mathbb{Q} \otimes \mathcal{L}_E \longrightarrow \mathcal{J}_\mathbb{Q} \otimes \mathcal{B}.} \quad (\text{FB5.4})$$

Equations (FB4.4) and (FB5.3) give

$$(P_\infty \otimes P_f) \mathcal{B} = (a^{-1} a) \mathcal{B} (P_\infty \otimes P_f^{\text{fun}}) = \mathcal{B} (P_\infty \otimes P_f^{\text{fun}}). \quad (\text{FB5.5})$$

The compensation is the full product formula. It occurs for every positive rational simultaneously. Both original operators commute with ∂_{X_∞} :

$$[D, \partial_{X_\infty}] = 0, \quad [D_Y, \partial_{X_\infty}] = 0, \quad (\text{FB5.6})$$

because Y_∞ is independent of X_∞ . Thus $\mathcal{B}$ is a chain map for both two-term complexes. For D , its degree-zero action is $P_\infty \otimes P_f$ and its degree-one action is $a P_\infty \otimes P_f$; this factor is retained. For D_Y , both degrees have the raw action.

On the minimal enlargement there is the square-zero endomorphism

$$\tilde{\mathcal{B}} = \partial_{X_\infty} \otimes N_f, \quad \tilde{\mathcal{B}}^2 = 0. \quad (\text{FB5.7})$$

It is equivariant and commutes with both differentials by the same calculations. Its archimedean derivative is nonzero, but its square vanishes because the actual finite coefficient operator has square zero.

As a map of coefficient vector spaces, (FB5.4) has

$$\begin{aligned} \ker \mathcal{B} &= \mathcal{J}_\mathbb{Q} \otimes j_! \mathcal{L}_U, \quad \text{im } \mathcal{B} = \mathcal{J}_{\mathbb{Q}, \geq 1} \otimes \mathcal{B}, \\ \text{coker } \mathcal{B} &\cong \left(\bigoplus_{c \in \mathbb{Q}} \mathcal{D}'((0, \infty)) \right) \otimes \mathcal{B}. \end{aligned} \quad (\text{FB5.8})$$

Here $\mathcal{J}_{\mathbb{Q}, \geq 1}$ consists of positive normal orders and the last quotient is the coefficient of δ_c . The kernel follows from injectivity of ∂_X on finite normal jets and the exact kernel in FB2. The image and cokernel follow by the same basis expansion. These statements concern algebraic tensor products of the displayed coefficient spaces, or their sheafification; no completed tensor product is silently substituted.

128.7 FB6. The induced cohomology map and the nonzero connecting map

Retain the complete original residues from DR3:

$$R(u)_c = \sum_j (-i)^j \partial_Y^j g_{c,j}, \quad R_Y(u)_c = \sum_j (-i)^j \partial_Y^j (Y^{-1} g_{c,j}) \quad (\text{FB6.1})$$

for $u = \sum_{c,j} \delta_c^{(j)} \otimes g_{c,j}$. DR3 proves

$$H^0(\mathcal{J}_\mathbb{Q} \xrightarrow{D} \mathcal{J}_\mathbb{Q}) = 0, \quad H^1 \xrightarrow{R} \bigoplus_c \mathcal{D}'((0, \infty)),$$

and the corresponding assertions for D_Y, R_Y , with section $E_Y h = \sum_c \delta_c \otimes Y h_c$. Every Y^{-1} and Y stays in these comparisons.

Direct substitution gives

$$R\partial_X = -i\partial_Y R, \quad R_Y \partial_X = -i\partial_Y R_Y. \quad (\text{FB6.2})$$

Consequently the exact cohomological map induced by (FB5.4), for either original complex in its stated residue coordinates, is

$$\boxed{-i\partial_Y \otimes T}. \quad (\text{FB6.3})$$

It is not zero. For instance $h(Y) = Y$, any nonzero compactly supported locally constant boundary coefficient k , and any one rational branch give output $-i \otimes \iota_Z(k)$. The input boundary coefficient is reached from an actual locally constant extension as in FB2.

There is a stronger exact connecting sequence. Identify the coefficient quotient $\mathcal{L}_E/j_!\mathcal{L}_U$ with $i_*\mathcal{L}_Z$, retaining its ordinary function action. In each cochain degree, the derivative gives

$$0 \longrightarrow \mathcal{J}_{\mathbb{Q}} \otimes i_*\mathcal{L}_Z \xrightarrow{\partial_X \otimes \iota_Z} \mathcal{J}_{\mathbb{Q}} \otimes \mathcal{B} \longrightarrow \left(\bigoplus_c \mathcal{D}'((0, \infty)) \right) \otimes \mathcal{B} \longrightarrow 0. \quad (\text{FB6.4})$$

It is equivariant by the compensated covariance, and is a short exact sequence of the two-term D complexes. On its last complex the differential is $i\partial_Y$, since the ∂_X term has positive normal order and vanishes in the quotient. For D_Y the last differential is $iY\partial_Y$.

On the connected interval $(0, \infty)$, distributional differentiation is onto and its kernel is the constant densities. An explicit proof of onto-ness is as follows. Choose $\chi \in C_c^\infty((0, \infty))$ with integral one, and define

$$\mathcal{P}\psi(y) = \int_0^y \left(\psi(t) - \chi(t) \int_0^\infty \psi(s) ds \right) dt.$$

This is compactly supported. For a distribution v , set $w(\psi) = -v(\mathcal{P}\psi)$; then

$$w'(\psi) = -w(\psi') = v(\mathcal{P}\psi') = v(\psi).$$

Continuity follows from the usual compact-test seminorms, with support in a fixed compact set containing the supports of ψ, χ . If $v' = 0$, every test of integral zero is a compactly supported derivative, so v vanishes on it and equals a constant times the integral. Multiplication by Y is invertible on this open interval, so $iY\partial_Y$ is also onto with the same kernel.

The long exact cohomology sequence of (FB6.4) is therefore the explicit sequence

$$0 \longrightarrow \left(\bigoplus_c \mathbb{C} \right) \otimes i_*\mathcal{L}_Z \xrightarrow{\Delta} \left(\bigoplus_c \mathcal{D}'((0, \infty)) \right) \otimes i_*\mathcal{L}_Z \xrightarrow{-i\partial_Y \otimes \iota_Z} \left(\bigoplus_c \mathcal{D}'((0, \infty)) \right) \otimes \mathcal{B} \longrightarrow 0. \quad (\text{FB6.5})$$

The connecting map Δ is the inclusion of constant densities, with no missing sign. For D , lift the constant quotient cocycle as $\delta_c \otimes 1 \otimes \iota_Z(k)$. Its differential is $\delta'_c \otimes 1 \otimes \iota_Z(k)$. The preimage under the first arrow of (FB6.4) is $\delta_c \otimes 1 \otimes k$, whose residue is $1 \otimes k$. For D_Y , the differential of that same lifted cocycle is $\delta'_c \otimes Y \otimes \iota_Z(k)$; its preimage has residue $R_Y(\delta_c \otimes Y) = 1$. Both give the displayed map.

Thus this receiver has a nonzero canonical boundary connecting map and a completely calculated surjective derivative map after it. Their existence does not by itself calculate all cohomology of the adelic quotient.

128.8 FB6A. The complete coefficient-character family and its residue

For every $\lambda > 0$, put

$$D_\lambda = \lambda\partial_X + i\partial_Y, \quad L_\lambda = YD_\lambda, \quad c_\lambda = -\frac{i}{\lambda}.$$

These expressions contain no rescaling of the source operator. Define the full residues on all rational branches by

$$R_\lambda(u)_c = \sum_j c_\lambda^j \partial_Y^j g_{c,j}, \quad R_{\lambda,Y}(u)_c = \sum_j c_\lambda^j \partial_Y^j (Y^{-1} g_{c,j}). \quad (\text{FB6A.1})$$

The source relation is $R_\lambda D_\lambda = 0$, since the two consecutive coefficients satisfy $\lambda c_\lambda + i = 0$.

There is an explicit full homotopy:

$$T_\lambda u = \lambda^{-1} \sum_c \sum_{j \geq 1} \sum_{r=0}^{j-1} c_\lambda^r \delta_c^{(j-1-r)} \otimes \partial_Y^r g_{c,j}. \quad (\text{FB6A.2})$$

Applying D_λ makes consecutive terms cancel and leaves

$$D_\lambda T_\lambda = I - E R_\lambda, \quad T_\lambda D_\lambda = I, \quad R_\lambda E = I,$$

where $Eh = \sum_c \delta_c \otimes h_c$. The second identity follows from the first applied to $D_\lambda u$, because the highest normal order proves D_λ injective. For L_λ , use $T_{\lambda,Y} = T_\lambda M_{Y^{-1}}$, $E_Y = M_Y E$, and $R_{\lambda,Y} = R_\lambda M_{Y^{-1}}$; their identities follow by multiplication in this order. Thus all exact cohomology comparisons used in FB6 hold for the full family.

The normal derivative commutes with D_λ , L_λ , and

$$R_\lambda \partial_X = -\frac{i}{\lambda} \partial_Y R_\lambda, \quad R_{\lambda,Y} \partial_X = -\frac{i}{\lambda} \partial_Y R_{\lambda,Y}. \quad (\text{FB6A.3})$$

Consequently the full family of induced boundary maps is

$$\boxed{K_\lambda = -\frac{i}{\lambda} \partial_Y \otimes T.} \quad (\text{FB6A.4})$$

Equations (FB6.2)–(FB6.3) are exactly its $\lambda = 1$ specialization for the two explicitly named original operators, not a replacement for (FB6A.4).

The cokernel complex in FB6.4 still has differential $i\partial_Y$ for D_λ , or $iY\partial_Y$ for L_λ , because the normal derivative term vanishes in the quotient. The connecting map now retains the full coefficient

$$\boxed{\Delta_\lambda(h \otimes k) = \lambda h \otimes k \quad \text{for constant densities } h.} \quad (\text{FB6A.5})$$

Indeed $D_\lambda(\delta_c \otimes h) = \lambda \delta'_c \otimes h$, whose preimage under ∂_X has residue λh . For L_λ , its preimage is $\lambda \delta_c \otimes Yh$; $R_{\lambda,Y}$ again gives λh . The image is exactly the kernel of K_λ .

128.9 FB6B. Arithmetic Frobenius including the residue Jacobian

Source equations holombis and holom1 and Proposition frobarith give

$$\mathfrak{F}_\mu = \theta_\mu R(\mu^{-1}), \quad \chi_\lambda \mathfrak{F}_\mu f = S_{\mu^{-1}} \chi_{\lambda\mu} f, \quad \mu > 0,$$

where $S_c u(X, Y) = u(X, cY)$ for functions. Its scalar-distribution definition is

$$\langle S_c u, \varphi \rangle = c^{-1} \langle u, \varphi(X, Y/c) \rangle. \quad (\text{FB6B.1})$$

On $\delta_c^{(j)} \otimes g$, it is the identity on the normal derivative and Q_c on g . The original family intertwining is

$$L_\lambda S_{\mu^{-1}} = S_{\mu^{-1}} L_{\lambda\mu}. \quad (\text{FB6B.2})$$

Direct differentiation proves this: both sides on functions are $Y\lambda Q_{\mu^{-1}} \partial_X + iY\mu^{-1} Q_{\mu^{-1}} \partial_Y$. The distribution identity follows by transposition.

On the L_λ residue family the exact Frobenius is

$$R_{\lambda,Y} S_{\mu^{-1}} u_{\lambda\mu} = \mu^{-1} Q_{\mu^{-1}} R_{\lambda\mu,Y} u_{\lambda\mu}. \quad (\text{FB6B.3})$$

To prove it retain both identities

$$Y^{-1} Q_{\mu^{-1}} = \mu^{-1} Q_{\mu^{-1}} Y^{-1}, \quad \partial_Y^j Q_{\mu^{-1}} = \mu^{-j} Q_{\mu^{-1}} \partial_Y^j.$$

Substitution in (FB6A.1) gives (FB6B.3) term by term. The factor μ^{-1} is part of the original residue comparison. The raw D_λ residue identity is $R_\lambda S_{\mu^{-1}} = Q_{\mu^{-1}} R_{\lambda\mu}$. For the two-term D_λ complex, however, the cochain map has degree-zero component $S_{\mu^{-1}}$ and degree-one component $\mu^{-1} S_{\mu^{-1}}$, because

$$D_\lambda S_{\mu^{-1}} = \mu^{-1} S_{\mu^{-1}} D_{\lambda\mu}.$$

Consequently its induced degree-one cohomology action is also $\mu^{-1} Q_{\mu^{-1}}$. The raw identity is not substituted for this cochain action.

The coefficient trace is independent of λ and commutes with the coefficient automorphism θ_μ . This right archimedean action fixes the finite-adele factor. Hence the boundary map (FB6A.4) commutes with arithmetic Frobenius. On L_λ residue families its two composites are

$$\begin{aligned} K_\lambda(\mu^{-1} Q_{\mu^{-1}} h_{\lambda\mu}) &= -\frac{i}{\lambda\mu^2} Q_{\mu^{-1}} h'_{\lambda\mu}, \\ \mu^{-1} Q_{\mu^{-1}}(K_{\lambda\mu} h_{\lambda\mu}) &= -\frac{i}{\lambda\mu^2} Q_{\mu^{-1}} h'_{\lambda\mu}. \end{aligned} \quad (\text{FB6B.4})$$

The factors $1/\lambda$, character change, derivative scaling and residue Jacobian all remain. On cochains the same result follows because ∂_X commutes with $S_{\mu^{-1}}$ and N_f is independent of the coefficient character.

128.10 FB7. Compatibility with the existing finite primary realization

DR defines

$$B_{\rho,m} = \mathbb{C}[x,y]/(x,y)^m, \quad \alpha = \rho + 1, \quad \Phi_c(x^u y^j) = \delta_c^{(j)} \otimes Y^\alpha \frac{(\log Y)^{m-j-1-u}}{(m-j-1-u)!}. \quad (\text{FB7.1})$$

The source of x, y here is that explicit complex receiver; neither is primitive τ . On its image,

$$\mathcal{J} = Y \partial_Y - \alpha, \quad \mathcal{N} = \partial_X \mathcal{J}, \quad \mathcal{N} \Phi_c = \Phi_c M_y. \quad (\text{FB7.2})$$

Because $\mathcal{J}$ commutes with the rational affine pullback,

$$P_\infty \mathcal{N} = a^{-1} \mathcal{N} P_\infty.$$

Therefore

$$\mathcal{N} \otimes N_f \quad (\text{FB7.3})$$

is an affine-equivariant operator on the fixed DR finite image with the minimal finite coefficient enlargement. It realizes $M_y \otimes N_f$, preserves the fixed value of m , and has square zero because $N_f^2 = 0$. For $m \geq 2$ it is nonzero. This proves that finite-boundary coefficients can compensate the existing normal-order weight shift inside that finite object. It precludes transporting an argument that assumed all finite coefficients were fixed under dilation to this enlarged coefficient sheaf.

The different map ∂_X from FB5 is a chain map but does not preserve every fixed finite DR image. Precisely,

$$\partial_X \Phi_{c,m}(x^u y^j) = \Phi_{c,m+1}(x^u y^{j+1}), \quad (\text{FB7.4})$$

with ρ, α unchanged. This relation increases the auxiliary truncation parameter; it does not increase an actual zeta zero's multiplicity by assertion.

The full residue of that tower entry is explicit. Put $r = m - j - 1 - u$, retain

$$P_j(z) = \prod_{v=0}^{j-1} (z - v), \quad P_0(z) = 1, \quad P_j(\rho + z) = \sum_{h=0}^j C_{j,h}(\rho) z^h, \\ C_{j,h}(\rho) = \sum_{\substack{S \subset \{0, \dots, j-1\} \\ |S|=h}} \prod_{v \notin S} (\rho - v). \quad (\text{FB7.4a})$$

The empty product is one. Applying j derivatives to $Y^\rho f(\log Y)$ gives $Y^{\rho-j} P_j(\rho + \partial_{\log Y}) f(\log Y)$, by induction with every factor $\rho - v$ retained. Therefore

$$R_{\lambda,Y} \Phi_{c,m}(x^u y^j) = \left(-\frac{i}{\lambda}\right)^j Y^{\rho-j} \sum_{h=0}^{\min(j,r)} C_{j,h}(\rho) \frac{(\log Y)^{r-h}}{(r-h)!}, \\ \boxed{R_{\lambda,Y} \partial_X \Phi_{c,m}(x^u y^j) = \left(-\frac{i}{\lambda}\right)^{j+1} Y^{\rho-j-1} \sum_{h=0}^{\min(j+1,r)} C_{j+1,h}(\rho) \frac{(\log Y)^{r-h}}{(r-h)!}.} \quad (\text{FB7.4b})$$

In the full boundary map this expression is tensored with $T(f) = f(X_f, 0) dX_f \delta_0(Y_f)$. Every branch c , original u, j, m , nilpotent coefficient and coefficient-character parameter is retained.

For example, on the $j = 0, u = 0$ source generator and $m \geq 2$, the image residue is

$$-\frac{i}{\lambda} Y^{\rho-1} \left(\rho \frac{(\log Y)^{m-1}}{(m-1)!} + \frac{(\log Y)^{m-2}}{(m-2)!} \right) \otimes T(f). \quad (\text{FB7.4c})$$

For $m = 1$, only the first term $-i\lambda^{-1} \rho Y^{\rho-1} \otimes T(f)$ occurs. The second term in (FB7.4c) is the retained nontrivial spectral nilpotent contribution, not an error term.

At an actual nontrivial zero $0 < \Re \rho < 1$, all factors $\rho - v$ are nonzero. For fixed j , the full logarithmic-polynomial operator in (FB7.4b) has nonzero diagonal $(-i/\lambda)^{j+1} P_{j+1}(\rho)$, after retaining its displayed factor $Y^{\rho-j-1}$. The incremental derivative from the preceding residue has polynomial factor $(\rho - j) + \partial_{\log Y}$, whose diagonal is $\rho - j \neq 0$; its additional residue coefficient remains $-i/\lambda$. Both descriptions prove injectivity on the stated finite spans. The logarithmic degree and its original multiplicity length are retained. The larger symbol $B_{\rho,m+1}$ in the tower is an ambient normal-jet realization containing the image; it is not a statement that the original zeta has order $m + 1$.

Its full rational affine weight is also retained. The source branch has factor $a^{\rho-j}$ with the complete logarithmic mixing of DR2.5. The target archimedean branch has $a^{\rho-j-1}$, while its finite boundary density has a . Their product is precisely $a^{\rho-j}$, including all original logarithmic mixing. This is the concrete compensation without changing the reconstructed arithmetic or the zeta multiplicity.

Conversely the fixed-image operator $\mathcal{N}$ need not be a chain map on the whole Dolbeault distribution complex. Its exact commutators are

$$[D, \mathcal{N}] = i \partial_X \partial_Y, \quad [D_Y, \mathcal{N}] = -Y \partial_X^2. \quad (\text{FB7.5})$$

Indeed $[D, \mathcal{J}] = i \partial_Y$, $[D_Y, \mathcal{J}] = -Y \partial_X$, and both differentials commute with ∂_X . These formulas prove (FB7.5) and prevent substituting (FB7.3) for the cohomology map (FB6.3). On the finite cohomology image already proved injective in DR4, one can transport M_y by the explicitly invertible image comparison; that is a named finite-image operation, not a newly asserted cochain map on all distributions.

128.11 FB8. Support filtration, quotient descent, and retained labels

On archimedean normal order, FB5 gives

$$\mathcal{B}(\mathcal{J}_{\leq n} \otimes \mathcal{L}_E) \subset \mathcal{J}_{\leq n+1} \otimes \mathcal{B}. \quad (\text{FB8.1})$$

On actual supports it gives

$$\{X_\infty = c\} \times \text{supp } f \longmapsto \{X_\infty = c\} \times (\text{supp } f \cap \{Y_f = 0\}). \quad (\text{FB8.2})$$

The finite support becomes smaller, and archimedean normal order increases by one. It is therefore compatible with retaining boundary supports and with the displayed degree-shifted normal-order filtration. It is not a degree-zero endomorphism of the original function-only coefficient filtration: (FB3.3) proves that its nonzero boundary output is outside that function summand. The exact enlargement (FB3.4) is the map resolving that issue.

On the source open set V with $Y_f \neq 0$, the trace restricts to zero. On the full adelic space it is present. Rational affine maps preserve $Y_f = 0$, and the compensated map is equivariant under their full action. Thus it defines a morphism of equivariant sheaves, or of sheaves on the corresponding action groupoid. Taking global invariants or coinvariants need not preserve every short exact sequence; no such preservation is inferred merely from equivariance.

Every additional specified split-support label is retained by tensoring these maps with its identity, or by applying them separately on each labeled summand. Algebraic tensors are finite sums in independent coefficient coordinates, so the kernel after identity extension is the original kernel tensored with the label space. No z_λ is identified with τ , and amplitude vanishing does not delete a retained label.

128.12 FB9. Mathematical outcome and proof-use limits

The finite boundary contribution is connected to the original locally constant coefficients by the actual sheaf-local map T , its exact restriction sequence, and its boundary-density insertion. The nonzero finite factor $a = |a|_f^{-1}$ is forced by the original Haar Jacobians and rational product formula. It compensates an archimedean normal derivative with factor a^{-1} . This yields the explicit equivariant chain map $\partial_X \otimes T$, its square-zero extension, and the nonzero connecting map (FB6.5).

These results supply an actual geometric coupling where a calculation using only fixed finite coefficients would have missed it. They do not identify the enlarged local coefficient complex with the full primitive- τ cohomology, do not turn an auxiliary truncation into an actual zero multiplicity, and do not prove or disprove the programme's complete lifting theorem or RH. Every limitation here is attached to a specified map together with its proved extension, rather than claimed as an obstruction to the full user-defined global system.

For the complete coefficient-character family the same commutator calculation in (FB7.5) is

$$[D_\lambda, \mathcal{N}] = i\partial_X \partial_Y, \quad [L_\lambda, \mathcal{N}] = -\lambda Y \partial_X^2. \quad (\text{FB9.1})$$

Thus retaining λ also retains the exact failure of the fixed-block operator to be a global chain map; the compensated chain map is the independently proved $\partial_X \otimes T$.

128.13 Source-reading record

Primary public source: Alain Connes and Caterina Consani, *The Riemann–Roch strategy: Complex lift of the Scaling Site*, arXiv:1805.10501v1. The retained author TeX has SHA256 **da52001209424c19fb223905804a9eed1147c72331cf881aa0352bc759976061**. The exact source labels used are actionpq, adelicomp1, adelicomp2, comparescales, holom, holombis, holom1 and frobarith.

Read for this calculation:

- `CORPUS_AND_OPERATION_RULES.md` completely.
- Exact user corpus records USR-9ad1c0a2d09dba92; USR-f55d16feb948d8c2; and USR-6152e3bc6302258c completely. U01–U18 supply definition routing and current corrections.
- Original author `thecurve_K.tex`, §5.2 through the stated open-set condition; §6.6 through Proposition `comparescalcs`; §6.8 beginning through Proposition `degeneratetr1`; §7 opening and §7.1 through Proposition `functionq` and the following remark `perfectoid`. These are the specified passages, not a whole-paper reading claim.
- `DISTRIBUTION_REALIZATION_OF_B_INDEPENDENT.md`, DR0–DR3 completely, and DR4 through the displayed exact residue formulas; the finite definitions and operator comparisons used here are proved there and recalculated where needed above.
- `CC_SUPPORTED_DOLBEAULT_BOUNDARY.tex`, its original-source statement, SDB3–SDB6 definitions, and SDB8–SDB16 exact complex and homotopy portion. DR3 restates the complete branchwise proof used for FB6.

Human authorship of the adelic complex lift and its operators belongs to Alain Connes and Caterina Consani. The finite trace, exact coefficient enlargement, compensated chain map and connecting-map calculations FB1–FB8 are receiving derivations supplied here. No historical novelty is asserted for restriction of locally constant functions or insertion of a boundary distribution.

128.14 FB10. Exact comparison with the added classical-orbit boundary

The factors M_{Y-1} used in FB6A act on the stated open archimedean domain $Y > 0$. They do not extend as invertible smooth multipliers over $Y = 0$. The later complete calculation CB4–CB6A in `CLASSICAL_BOUNDARY_LIFTING.md` retains Connes–Consani’s extension $(G \times \mathbb{R})/\{\pm 1\}$, their actual moving functions $q^r(g, y) = e_r(g)[e^{-2\pi r y}]$, and the entire original operator $L_\lambda = y(\lambda \partial_g + i \partial_y)$. On the defined smooth-radial finite-frequency algebra it proves the exact chain sequence

$$0 \longrightarrow (\mathcal{C} \xrightarrow{i\partial_y} \mathcal{C}) \xrightarrow{(\text{id}, M_y)} (\mathcal{C} \xrightarrow{iy\partial_y} \mathcal{C}) \longrightarrow \mathcal{B}[-1] \longrightarrow 0.$$

Here $\mathcal{B} = W_{\text{alg}} \otimes \mathbb{C}[\mathbb{Q}]$, with its full character realization, is the exact cokernel in degree one. The raw and original degree-one Frobenius factors differ by μ^{-1} ; their intertwining is proved in CB6A. No boundary term may be removed by extending the open-domain inversion across zero. This comparison retains the validity of FB6A on its domain and supplies the additional archimedean boundary contribution. CB1–CB3 separately construct the finite-valuation factor and graph correspondence relating a finite collapsed point to the entire G fibre; a finite-adele zero locus is not identified with this added archimedean boundary by notation alone.

129 Source specialization and the original Connes–Consani boundary

24 September 2026. Complete calculations CB0–CB8, including CB6A; SS0–SS9; MQ0–MQ9; OQ0–OQ10. The preceding readers and their complete proof sources remain retained. This reader develops the next actual geometric comparison; it does not assert the requested full tau weight theorem or RH.

129.1 The retained source and the result proved here

The user’s notation remains Z_0 for absence and Z_1/τ for primitive presence without Z_2 parity. Addition on primitive τ was retracted. All vector-space and ring operations below have explicitly constructed receiving objects as their domains. They are not operations on primitive presence. The user’s complete global reconstruction and quotient arguments were consulted under the corpus locators recorded in CB0, SS0, MQ0 and OQ0. The corpus is preserved in full; these locators do not replace it with a short axiom list.

Alain Connes and Caterina Consani’s original construction supplies

$$G = \mathbb{A}_{\mathbb{Q}}/\mathbb{Q}, \quad q^r(g, y) = e_r(g)[e^{-2\pi r y}], \quad L_\lambda = y(\lambda \partial_g + i \partial_y).$$

Their classical-orbit extension is $(G \times \mathbb{R})/\{\pm 1\}$. Every parameter, character, sign and rational frequency in these formulas is retained. The exact source locators are §5.4, Remark additivestruct, and §7.1, holom through functionq, of their *The Riemann–Roch strategy: Complex lift of the Scaling Site*. Their original author TeX was read directly; the source hash and actual coverage are recorded in the private reading ledger.

On the explicitly defined algebra of smooth radial coefficients and finite sums of these actual moving functions, the induced differential is

$$d(h(y)wq^r) = iyh'(y)wq^r.$$

Its two exponential derivative terms cancel with their full constants. Set

$$\mathcal{B} = W_{\text{alg}} \otimes \mathbb{C}[\mathbb{Q}], \quad R(hwq^r) = h(0)w \otimes [r], \quad E(w \otimes [r]) = wq^r.$$

The complete lifting operator is

$$(Th)(y) = \frac{1}{i} \int_0^y \frac{h(t) - h(0)}{t} dt, \quad dT + Td = I - ER, \quad RE = I.$$

CB4–CB6 prove all domains, smooth extension through zero, both cochain identities, arithmetic Frobenius equivariance and sign equivariance. Thus boundary restriction is a cohomology isomorphism in both degrees for this complex, and its connecting map is zero. The formulas, not an assumed weight inequality, prove this vanishing.

129.2 Why the entire radial factor matters

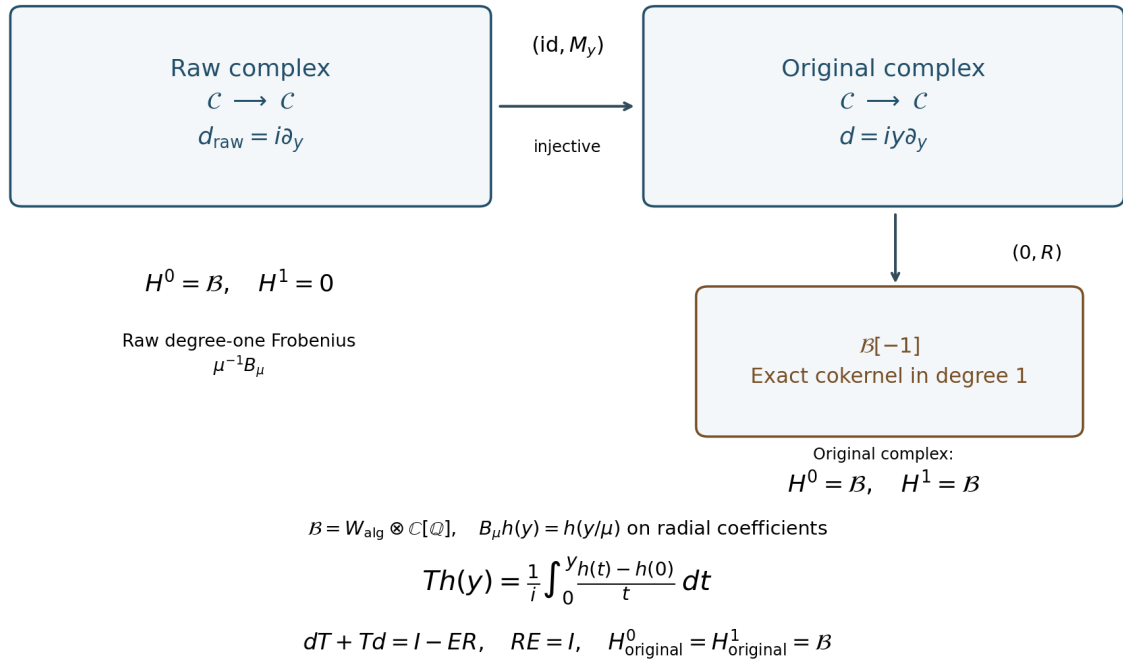
CB6A proves the exact sequence

$$0 \longrightarrow (\mathcal{C} \xrightarrow{i\partial_y} \mathcal{C}) \xrightarrow{(\text{id}, M_y)} (\mathcal{C} \xrightarrow{iy\partial_y} \mathcal{C}) \longrightarrow \mathcal{B}[-1] \longrightarrow 0.$$

The symbol $\mathcal{B}[-1]$ here means $\mathcal{B}$ in degree one. Multiplication by y is injective, its image consists exactly of coefficients vanishing at zero, and its cokernel is $\mathcal{B}$. Consequently the raw derivative has zero degree-one cohomology while the original operator has degree-one cohomology $\mathcal{B}$. This precisely identifies the boundary term lost by removing the original factor. The raw degree-one Frobenius includes μ^{-1} ; the original degree-one action does not. The comparison retains both and intertwines them.

The original radial factor retains the boundary

$$q^r(g, y) = e_r(g)[e^{-2\pi r y}], \quad L_\lambda = y(\lambda \partial_g + i \partial_y)$$

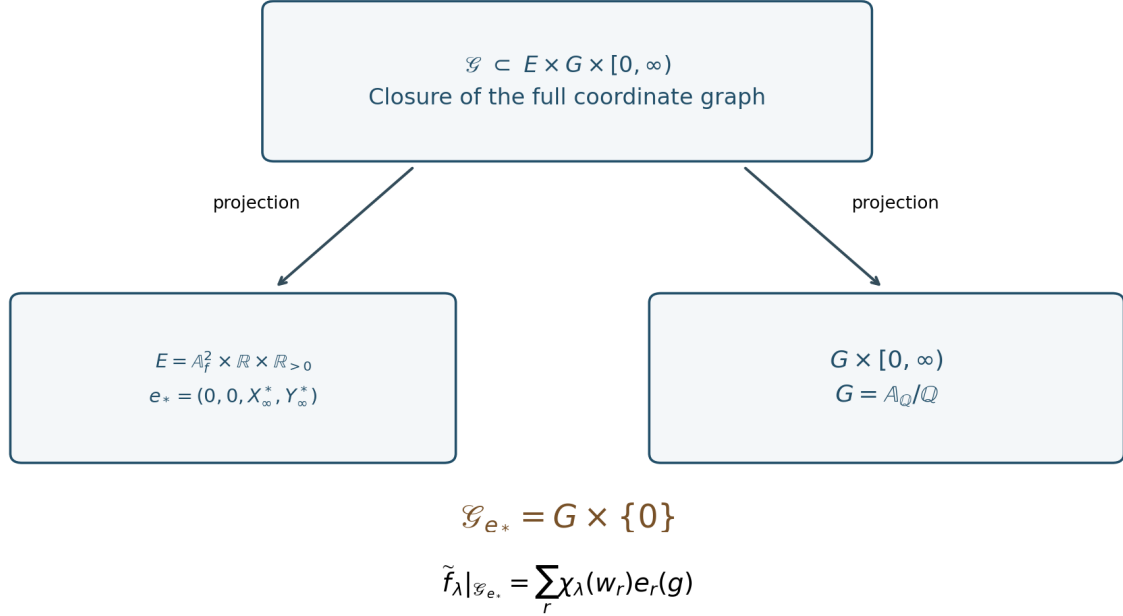


Proofs: CB4–CB6A; independent derivation GR4–GR6. B_μ also applies ∂_μ to W coefficients.
The original complex has B_μ in both degrees. All rational frequencies and both signs are retained.
Human source: Connes–Consani, arXiv:1805.10501v1, §§5.4 and 7.1. No source τ addition is used.

Figure 46: Exact radial complexes and their retained boundary contribution. This is a commutative-algebra diagram, not a numerical plot. Every arrow and cohomology statement is proved in CB4–CB6A; independent derivation GR4–GR6. The original operator and geometric extension are from Connes–Consani, §§5.4 and 7.1.

The boundary graph retains every direction in G

$$c = \prod_p p^{v_p(Y_p)}, \quad \kappa(X, Y) = ([X/c], Y_\infty/c)$$



A single collapsed value retains only the $r = 0$ coefficients.

Exact correspondence diagram, not a geometric embedding of G . Proofs: CB1–CB3 and GR1–GR3.
The full boundary algebra is $\text{Walg} \otimes \mathbb{C}[\mathbb{Q}]$. No point in this diagram is identified with primitive $\mathbb{Z}_l/\tau$.

Figure 47: The exact graph correspondence retains all directions over the stated finite boundary point. Boxes denote spaces and do not depict a finite-dimensional shape for the adelic solenoid G . The projections and full fibre are proved in CB1–CB3 and independently in GR1–GR3.

129.3 The finite boundary retains an entire direction fibre

The finite second coordinate has its full factorization

$$Y_f = cu, \quad c = \prod_p p^{v_p(Y_p)}, \quad u \in \widehat{\mathbb{Z}}^\times.$$

The actual quotient coordinates are $g = [X/c]$ and $y = Y_\infty/c$. Thus the pullback of original q^r has cover frequency r/c . A fixed-frequency cover function is a different object, with the exact comparison in CB1, MQ8 and OQ10.

For the ambient additive-adele topology this coordinate map is not continuous. CB1 gives the complete sequence demonstrating the failure; it does not substitute that topology for the source orbit topology. CB2 therefore constructs a closed graph correspondence, with both projections explicitly retained. Over the stated finite collapsed point $e_* = (0, 0, X_\infty^*, Y_\infty^*)$, $Y_\infty^* > 0$, its fibre is exactly

$$\mathcal{G}_{e_*} = G \times \{0\}.$$

CB3 proves that collapsing this fibre loses every nonzero rational Fourier frequency. Keeping it retains the complete boundary algebra $\mathcal{B}$. This is a calculated property of that receiving correspondence, not an impossibility claim about the user's primitive source.

129.4 The other complete comparisons included in this reader

SS constructs the source-induced ringed space from the already proved multiplicative source and additive receiver, then calculates its stalks, exact sections, support cohomology and the contraction from the full spectrum of the additive receiver. The split structure-sheaf sequence is proved. SS8 retains the additional kernel and cokernel for general module sheaves instead of extending that structure-sheaf conclusion to every coefficient object.

MQ constructs the original moving-function algebra and its complete coefficient extraction. It maps the coefficients to the original-zeta quotient and every full zero jet, retaining the Gaussian multiplier, frequencies and nilpotents. The map on all positive coefficients retains two sectors and the exceptional coefficient one; it does not delete a crossing through that value. For the original moving function, the full height change cancels the apparent fixed-height zero-jet scaling. For general constant coefficients, their original spectral action remains.

OQ constructs a global equivariant boundary section and vanishing connecting maps for a specified fixed-frequency covering algebra. OQ10 gives its exact comparison with original quotient q , including all rational valuation frames and both signs. The fixed-cover result is not presented as a theorem for an unidentified full source sheaf. The original quotient boundary calculation is CB, with the map between the presentations retained.

129.5 Relation to the active Deligne goal

Pierre Deligne's *La conjecture de Weil : II*, §§3.6.1–3.6.3 supplies the target mechanism: the class to lift and the supported obstruction lie in separated weight ranges. The earlier reader includes the complete local invariant-cycle comparison read in a French TeX transcription; it is not described as original author TeX. The calculations in this reader prove actual lifting maps, their boundary contribution and their Frobenius actions. They do not derive Deligne's numerical weight ranges for the full tau specialization.

In particular the original-zeta receiving jet still has action

$$p^\rho \exp((\log p)T_\rho), \quad T_\rho^{m_\rho} = 0,$$

with every original multiplicity retained. Its occurrence in the displayed boundary complex does not prove $\Re \rho = 1/2$. The active work remains the exact identification with the full tau specialization and the derivation of its weights from the retained geometry. The reader provides the completed calculations used in that continuation; it is not a substitute conditional theorem.

The full proofs follow. Human source provenance is stated in each calculation. Earlier programme dependencies, including the complete CW coefficient-to-Mellin proof and FB finite-current proof, are retained with this working repository and its preceding readers.

130 The original classical-orbit boundary: retained directions and exact equivariant lifting

24 September 2026. Proofs CB0–CB8. This calculation uses the actual moving functions of Alain Connes and Caterina Consani and the extension in their §5.4, rather than replacing them by constant coefficients. It constructs an exact boundary complex and its full equivariant homotopy. The relation to the user’s primitive source remains a stated map, not an identification of source tau with a boundary coordinate.

130.1 CB0. Source definitions and operation prerequisites

The current corpus and operation rules, U01–U18 and B1–B5 were read before this calculation. The complete global arguments USR-9ad1c0a2d09dba92, USR-f55d16feb948d8c2 and USR-6152e3bc6302258c were consulted again. The receiving arithmetic here is the already reconstructed arithmetic; no finite prime sample constructs it. Primitive Z_1/τ receives no addition, parity, count, coordinate, metric or numerical weight. Z_0 and the integer Z_2 data retain their supplied meanings. The calculations below take place in explicitly defined adelic spaces, smooth functions and original Connes–Consani coefficients.

Human source: Alain Connes and Caterina Consani, *The Riemann–Roch strategy: Complex lift of the Scaling Site*, arXiv:1805.10501v1, original author TeX thecurve_K.tex. The passages used are §5.1 Lemma adelicomp; §5.2 equations actionpq and adelicomp3 and Definition adelicomp2; §5.4 Jdefn1, classorb1, proetcov and Remark additivestruct; and §7.1 holom, holom1, holom2, functionq and perfectoid. Actual root reading in this continuation covers lines 1173–1315, 1634–1719 and 2481–2726. Source wording and the distinction between the original paper’s statements and the receiving derivations below remain in the reading ledger.

The source uses

$$G = \mathbb{A}_{\mathbb{Q}}/\mathbb{Q}, \quad \Gamma_{\mathbb{Q},\text{cl}} = G \times \mathbb{R}_{>0}.$$

The full additive character is $\psi = \psi_f \psi_{\infty}$, with $\psi_{\infty}(t) = e^{2\pi i t}$ and $\psi(b) = 1$ for rational b . For $r \in \mathbb{Q}$, write $e_r(g) = \psi(rg)$. The character group of G is $\mathbb{Q}$, as in Lemma adelicomp. Retain a Haar measure $d\nu$ with total mass $M = \nu(G) > 0$. Fourier coefficient extraction is $M^{-1} \int_G f \bar{e}_r d\nu$; the source probability-measure convention is the case $M = 1$. No measure factor is removed from the comparison.

The coefficient algebra is the original algebraic first approximation

$$W_{\text{alg}} = \mathbb{C}[\mathbb{R}_{>0}^{\times}], \quad [x][z] = [xz], \quad \chi_{\lambda}[x] = x^{\lambda}, \quad \theta_{\mu}[x] = [x^{\mu}], \quad \lambda, \mu > 0.$$

The source’s Teichmüller notation is not the user’s primitive tau. These receiving coefficients have their own explicit complex addition. The character family is faithful: for finitely many distinct $x_j > 0$, evaluations at $\lambda = 1, \dots, d$ have determinant $(\prod_j x_j) \prod_{i < j} (x_j - x_i) \neq 0$.

130.2 CB1. The complete finite-valuation factor and the two coordinate maps

On the preimage of the classical stratum write the finite second coordinate uniquely as

$$Y_f = cu, \quad c = \prod_p p^{v_p(Y_p)} \in \mathbb{Q}_{>0}, \quad u \in \widehat{\mathbb{Z}}^{\times}. \quad (\text{CB1.1})$$

Here every $Y_p \neq 0$, the valuations vanish outside a finite set, and all primes are present in the formula. Existence follows by taking that finite product; uniqueness follows because a positive rational with valuation zero at every prime is one. This does not discard u : multiplication of the finite second coordinate by $\widehat{\mathbb{Z}}^{\times}$ is precisely the separate quotient in source Definition adelicomp2.

The actual classical coordinates of the point $(X_f, Y_f, X_{\infty}, Y_{\infty})$, with $Y_{\infty} > 0$, are

$$g = [(X_f/c, X_{\infty}/c)] \in G, \quad y = Y_{\infty}/c > 0. \quad (\text{CB1.2})$$

Under the original positive rational affine action $X \mapsto aX + b$, $Y \mapsto aY$, the factor becomes $c' = ac$. Thus

$$g' = [X/c + b/(ac)] = g, \quad y' = y, \quad (\text{CB1.3})$$

because $b/(ac)$ is a rational diagonal adele. This proves gauge invariance with the finite factor retained.

The source moving function and its actual pullback are

$$q^r(g, y) = e_r(g)[e^{-2\pi r y}],$$

$$\kappa^* q^r = \psi_f(rX_f/c)e^{2\pi i r X_\infty/c}[e^{-2\pi r Y_\infty/c}], \quad \kappa = (g, y). \quad (\text{CB1.4})$$

The cover-coordinate function with fixed frequency r , obtained by putting $c = 1$ in this displayed expression, instead transforms to frequency ar . Equations (CB1.2)–(CB1.4) prove the comparison; neither action is substituted for the other.

There is a topology prerequisite. The map κ is not continuous for the topology on the classical preimage inherited solely from the ambient finite adeles and the archimedean coordinates. Indeed $c_n = 1 + n!$ converges to one in $\mathbb{A}_f$: it is integral at every prime and its difference from one has arbitrarily high valuation at each fixed finite set of primes. Take $X = 0, Y_f = c_n, Y_\infty = 1$. These cover points converge to $(0, 1, 0, 1)$, whereas their y -coordinates are $1/c_n \rightarrow 0$, not one. The original $G \times \mathbb{R}_{>0}$ orbit coordinates and their product topology must therefore be retained explicitly when taking limits. No claim that this example contradicts the source's chosen orbit topology is made.

130.3 CB2. The exact boundary correspondence retains all of G

Let $E = \mathbb{A}_f^2 \times \mathbb{R} \times \mathbb{R}_{>0}$ have its indicated product topology. Form the graph of the explicitly given set map κ on its classical preimage, then take its closure

$$\mathcal{G} \subset E \times G \times [0, \infty).$$

This defines a closed correspondence and both projections. Because of CB1's continuity calculation, it is not asserted to be a blowup of the ambient subspace, or the whole compactification from the original source.

Fix $e_* = (0, 0, X_\infty^*, Y_\infty^*)$ with $Y_\infty^* > 0$. The fibre of this graph closure is exactly

$$\boxed{\mathcal{G}_{e_*} = G \times \{0\}}. \quad (\text{CB2.1})$$

First consider any net of classical cover points approaching e_* . Eventually $Y_f \in \widehat{\mathbb{Z}}$, so (CB1.1) makes c a positive integer. Since $Y_2 \rightarrow 0$, its valuation and hence $v_2(c)$ tend to infinity. Thus $c \rightarrow \infty$ as a positive real number, while $Y_\infty \rightarrow Y_\infty^*$; therefore $y \rightarrow 0$. This proves containment in the right side of (CB2.1).

For the reverse containment, the image of $\mathbb{A}_f \times \{0\}$ is dense in G . Explicitly, for any representative (t_f, t_∞) , choose rationals $b_n \rightarrow t_\infty$ in the real topology. The points $[(t_f - b_n, 0)]$ differ from $[(t_f, t_\infty)]$ by $[(0, b_n - t_\infty)]$, which tends to zero. This proves density without dropping a finite component.

For fixed $t_f \in \mathbb{A}_f$, choose positive integers c_n tending to infinity and divisible enough that both $c_n \rightarrow 0$ and $c_n t_f \rightarrow 0$ in $\mathbb{A}_f$. A concrete choice starts with an integer clearing all negative valuations of t_f and multiplies it by $n!$. Then the classical points

$$(X_f, Y_f, X_\infty, Y_\infty) = (c_n t_f, c_n, X_\infty^*, Y_\infty^*)$$

approach e_* , and their graph coordinates tend to $[(t_f, 0), 0)$. The fibre of the closed set $\mathcal{G}$ over e_* is closed in $G \times [0, \infty)$; it therefore contains the closure $G \times \{0\}$ of these points. This proves (CB2.1). Neither the fibre nor its points are identified with primitive Z_1/τ .

130.4 CB3. Exactly what a collapsed boundary would lose

For $f = \sum_{r \in F} w_r q^r$, with finite $F \subset \mathbb{Q}$ and $w_r \in W_{\text{alg}}$, every character evaluation extends continuously on the graph closure by

$$\tilde{f}_\lambda(e, g, y) = \sum_{r \in F} \chi_\lambda(w_r) e_r(g) e^{-2\pi\lambda r y}. \quad (\text{CB3.1})$$

All factors and rational frequencies remain. On the fibre (CB2.1), its value is $\sum_r \chi_\lambda(w_r) e_r(g)$.

This family has a characterwise continuous extension to the single collapsed point e_* , independent of the approach, if and only if $w_r = 0$ for every $r \neq 0$. For necessity, continuity forces the displayed boundary function to be constant in g . For a nontrivial character, Haar translation invariance gives $\int_G e_r d\nu = 0$: choose h with $e_r(h) \neq 1$, translate the integral, and subtract. The exact orthogonality formula is $M^{-1} \int_G e_r \overline{e_s} d\nu = \delta_{r,s}$. It extracts each nonzero frequency coefficient $\chi_\lambda(w_r) = 0$. Faithfulness of all χ_λ , proved in CB0, gives $w_r = 0$. Sufficiency follows because an element of W_{alg} constant in the geometric coordinates extends by that same value.

Thus the exact object retaining the lost data is the fibre G , and its generated boundary algebra is

$$W_{\text{alg}} \otimes \mathbb{C}[\mathbb{Q}] \longrightarrow \{\text{boundary character families}\}, \quad w \otimes [r] \longmapsto (\chi_\lambda(w) e_r(g))_\lambda. \quad (\text{CB3.2})$$

Fourier orthogonality followed by character faithfulness proves this map injective; its image is exactly the stated generated algebra. The failed extension to a collapsed receiving point is not an impossibility theorem about the user's original source. The full correspondence and the injective boundary map are the constructed replacement.

130.5 CB4. The actual source extension and an exact complex on it

Connes–Consani's Remark additivestruct explicitly gives

$$\bar{\Gamma}_{\mathbb{Q}, \text{cl}} = (G \times \mathbb{R}) / \{(g, y) \sim (-g, -y)\},$$

whose new locus is $G/\{\pm 1\}$ at $y = 0$. Work first on its oriented cover $G \times \mathbb{R}$. This uses their radial coordinate and retains both signs. No numerical coordinate on source tau is introduced.

Define $\mathcal{C}$ as the algebraic tensor product

$$\mathcal{C} = C^\infty(\mathbb{R}) \otimes W_{\text{alg}} \otimes \mathbb{C}[\mathbb{Q}],$$

realized by finite sums $h(y) w q^r(g, y)$ using the actual q^r of CB1. This realization is injective. At each fixed y , Fourier extraction isolates r ; multiplication by the nonzero factor $e^{-2\pi\lambda r y}$ and faithfulness in λ give every coefficient in W_{alg} . Linear independence of its basis then makes each smooth scalar coefficient zero at every y . Products use exactly $q^r q^s = q^{r+s}$ and the original group-algebra law of W_{alg} .

The original character operator is

$$L_\lambda = y(\lambda \partial_g + i \partial_y), \quad \partial_g e_r = 2\pi i r e_r.$$

Direct differentiation, including both exponential derivatives, gives

$$L_\lambda \chi_\lambda(h w q^r) = i y h'(y) \chi_\lambda(w) e_r(g) e^{-2\pi\lambda r y}. \quad (\text{CB4.1})$$

The terms $2\pi i \lambda r y h$ and $-2\pi i \lambda r y h$ cancel exactly. Consequently the original family acts on this realized algebra by the single, explicitly derived operator

$$d(h w q^r) = i y h'(y) w q^r.$$

It defines the two-term complex $\mathcal{C} \xrightarrow{d} \mathcal{C}$ in degrees zero and one. Its boundary complex is $\mathcal{B} \xrightarrow{0} \mathcal{B}$, where $\mathcal{B} = W_{\text{alg}} \otimes \mathbb{C}[\mathbb{Q}]$ has its faithful realization (CB3.2). The exact restriction and extension in both degrees are

$$R(hwq^r) = h(0)w \otimes [r], \quad E(w \otimes [r]) = wq^r. \quad (\text{CB4.2})$$

They are chain maps: $Rd = 0$, $dE = 0$, and $RE = I$. This identifies a specific original-character complex on a defined algebra of functions; it does not assert that every possible completed W -valued section has a finite expression in $\mathcal{C}$.

130.6 CB5. The complete lifting homotopy

For a smooth scalar function h , set

$$(Th)(y) = \frac{1}{i} \int_0^y \frac{h(t) - h(0)}{t} dt. \quad (\text{CB5.1})$$

The integrand is smooth through zero: the exact identity

$$\frac{h(t) - h(0)}{t} = \int_0^1 h'(ut) du$$

holds for nonzero t by the fundamental theorem of calculus, and its right side provides every derivative at zero. Thus (CB5.1) defines a smooth function on all of $\mathbb{R}$, retaining the oriented integral for negative y . Extend T coefficientwise without altering w or r .

Differentiation and integration give the two exact identities

$$dTh = h - h(0), \quad Tdh = \frac{1}{i} \int_0^y \frac{ith'(t)}{t} dt = h - h(0). \quad (\text{CB5.2})$$

Hence on both degrees

$$\boxed{dT + Td = I - ER, \quad RE = I.} \quad (\text{CB5.3})$$

Only the applicable term acts in each degree of the two-term complex. Therefore

$$H^0(\mathcal{C} \xrightarrow{d} \mathcal{C}) \simeq \mathcal{B}, \quad H^1(\mathcal{C} \xrightarrow{d} \mathcal{C}) \simeq \mathcal{B}, \quad (\text{CB5.4})$$

with inverse in both cases induced by the actual extension E . Equivalently, the boundary restriction sequence has kernel $\ker R$ in both degrees, and (CB5.2) contracts that kernel because $Th(0) = 0$. Its entire connecting map is zero. Every boundary class in both displayed degrees lifts, and every difference between two lifts is explicitly controlled by T . No assumed vanishing or purity is used.

130.7 CB6. Frobenius, both signs, and the complete retained support

Original arithmetic Frobenius has the exact action

$$\mathfrak{F}_\mu(h(y)wq^r) = h(y/\mu)\theta_\mu(w)q^r, \quad \mu > 0. \quad (\text{CB6.1})$$

Indeed the geometric radial change multiplies the exponent in the moving coefficient by μ^{-1} , and θ_μ multiplies its logarithm by μ ; they cancel on q^r while retaining $\theta_\mu(w)$. On evaluations the character also changes from λ to $\lambda\mu$, exactly as source holom1. Both cochain degrees have (CB6.1), since this is the original L_λ complex with its full radial factor.

Substitution $t = \mu u$ in (CB5.1) proves

$$T(h(\cdot/\mu))(y) = (Th)(y/\mu).$$

Thus T, d, E, R all intertwine arithmetic Frobenius, whose boundary action is $\theta_\mu \otimes I$. On the original sign involution,

$$\iota(h(y)wq^r) = h(-y)wq^{-r}.$$

Substitution $t = -u$, with the oriented integral retained, gives $T(h(-\cdot))(y) = (Th)(-y)$. The other maps also commute with ι , by their displayed formulas. Restricting the identities (CB5.2) to invariants proves the same lifting result on the actual quotient extension, with boundary

$$\mathcal{B}^\iota = W_{\text{alg}} \otimes \mathbb{C}[\mathbb{Q}]^{r \mapsto -r}. \quad (\text{CB6.2})$$

No exactness of an unconstructed invariant functor is presumed; the homotopy itself restricts. The invariant character algebra is generated linearly by its zero frequency and $[r] + [-r]$ for positive rational r , retaining the coefficient two when the orbit has two elements.

For every previously defined support-lattice receiver $G_L(V)$, the displayed linear maps extend by $(v, \ell) \mapsto (f(v), \ell)$. Their amplitude identities leave the label unchanged. In particular when a difference has zero amplitude its label remains ℓ ; the homotopy is not asserted to identify all such labelled zeros with one bottom element. The vector-space lifting identities above do not silently become identities in a new undefined semimodule cohomology category. They supply the explicit labelled maps and their preserved amplitudes.

130.8 CB6A. The complete boundary contribution of the radial factor

The following exact comparison was independently derived in GR6 of the independent review. Write $B_\mu = \mathfrak{F}_\mu$ and let ι denote the simultaneous sign action from CB6. The raw family $D_\lambda = \lambda \partial_g + i \partial_y$ acts on the same realized finite algebra by

$$d_{\text{raw}}(hwq^r) = ih'wq^r.$$

It defines $\mathcal{C}_{\text{raw}} = (\mathcal{C} \xrightarrow{i \partial_y} \mathcal{C})$; the original family defines $\mathcal{C}_{\text{original}} = (\mathcal{C} \xrightarrow{iy \partial_y} \mathcal{C})$. There is the exact chain map

$$F = (F^0, F^1) = (\text{id}, M_y) : \mathcal{C}_{\text{raw}} \longrightarrow \mathcal{C}_{\text{original}}. \quad (\text{CB6A.1})$$

It is a chain map because $M_y(i \partial_y) = iy \partial_y$.

Multiplication by y is injective on smooth functions: $yh = 0$ implies $h(y) = 0$ away from zero, and continuity gives $h(0) = 0$. Its image is exactly the smooth functions vanishing at zero, by

$$h(y) = y \int_0^1 h'(uy) du \quad \text{when } h(0) = 0.$$

These statements apply coefficientwise to the finite tensor algebra. Its cokernel is therefore $\mathcal{B}$, by evaluation at zero. There is a short exact sequence of complexes

$$0 \longrightarrow \mathcal{C}_{\text{raw}} \xrightarrow{(\text{id}, M_y)} \mathcal{C}_{\text{original}} \xrightarrow{(0, R)} \mathcal{B}[-1] \longrightarrow 0, \quad (\text{CB6A.2})$$

where $\mathcal{B}[-1]$ denotes $\mathcal{B}$ in degree one and zero in every other degree.

The derivative $i \partial_y$ is surjective on $C^\infty(\mathbb{R})$, since g has smooth preimage $(1/i) \int_0^y g(t) dt$. Its kernel consists exactly of constants. Hence

$$H^0(\mathcal{C}_{\text{raw}}) = \mathcal{B}, \quad H^1(\mathcal{C}_{\text{raw}}) = 0,$$

whereas CB5 gives $H^0(\mathcal{C}_{\text{original}}) = H^1(\mathcal{C}_{\text{original}}) = \mathcal{B}$. In (CB6A.2), evaluation identifies the additional degree-one class with the boundary algebra itself. Inverting the factor y at $y = 0$ would remove exactly this contribution and would not be an isomorphism of the stated complexes.

The cochain actions agree with this sequence. For the raw complex,

$$d_{\text{raw}} B_\mu = \mu^{-1} B_\mu d_{\text{raw}},$$

so its degree-zero Frobenius is B_μ and its degree-one Frobenius is $\mu^{-1}B_\mu$. For the original complex both degrees have B_μ . The intertwining equality

$$B_\mu M_y = M_y(\mu^{-1}B_\mu) \quad (\text{CB6A.3})$$

proves equivariance of (CB6A.1) in degree one. The raw sign action is ι in degree zero and $-\iota$ in degree one, since $d_{\text{raw}}\iota = -\iota d_{\text{raw}}$. The original sign action is ι in both degrees, and

$$\iota M_y = M_y(-\iota). \quad (\text{CB6A.4})$$

Thus (CB6A.2) respects both Frobenius and the source sign quotient. These degree factors cannot be transferred unchanged from the raw operator to the original operator at the boundary.

130.9 CB7. Original-zeta comparison without discarding any frequency

The earlier proved map CW is

$$\begin{aligned} \overline{K}_b : V_+ &\hookrightarrow Q, \quad V_+ = \text{span}\{[x] : x > 1\}, \quad Q = \mathcal{A}/\overline{\mathcal{E}S_{00}^{\text{even}}}, \\ F_{K_b[x]}(s) &= \sqrt{\pi}e^{(s-1/2)^2/4}(\log x)^s, \quad \overline{K}_b\theta_a = W_a\overline{K}_b. \end{aligned} \quad (\text{CB7.1})$$

Every original-zeta factor and multiplicity in CW remains in that map. Tensor (CB7.1) with $C^\infty(\mathbb{R})$ and $\mathbb{C}[\mathbb{Q}]$. This gives an injective map of the corresponding two-term complexes, because the tensor factors are vector spaces over $\mathbb{C}$; explicitly, write any tensor in a finite linearly independent list of the other factor and apply injectivity to each coefficient. The diagram commutes with d, T, E, R , since these act solely on h , and with Frobenius by (CB7.1). The actual evaluated moving function may cross the coefficient value $x = 1$; the map uses its unique Fourier coefficient w before multiplying by its moving q^r , not an unproved pointwise application of K_b .

For every original zero ρ , with actual multiplicity m_ρ , the full receiving jet action on the boundary is consequently

$$p^\rho \exp((\log p)T_\rho) \otimes I_{\mathbb{C}[\mathbb{Q}]}, \quad T_\rho^{m_\rho} = 0. \quad (\text{CB7.2})$$

All rational frequencies remain, and no nilpotent term is dropped. This follows by applying the exact CW jet map, including its invertible complete Gaussian jet factor, to the coefficient of each q^r . The same formula holds after taking the sign invariants because the jet action does not change r . It is a full comparison with the original zeta receiver, not a replacement by a completed zeta function.

130.10 CB8. What this closes and what it does not

The actual original-character complex on the stated smooth-radial, finite-frequency coefficient algebra has an equivariant boundary lifting theorem, proved by the exact homotopy (CB5.1). The original arithmetic Frobenius, both orientations, every rational frequency, every coefficient character and the complete support maps are retained. The boundary-direction correspondence (CB2.1) proves which additional data must survive when approaching the finite-adele boundary. It does not assign these data to primitive Z_1/τ , and the graph closure is not asserted to be the whole original geometric compactification.

In this exact complex the restriction is a cohomology isomorphism and the kernel is contracted explicitly. Thus its connecting obstruction vanishes without an assumed numerical weight separation. The coefficient action (CB7.2) is unchanged by this lifting. Consequently this proves a specific geometric lifting result while leaving the target identification with the full tau specialization and Deligne's numerical weight ranges unfinished. Deligne's *La conjecture de Weil* : II, §§3.6.1–3.6.3 remains the human source for the distinct geometric weight argument; that theorem is not claimed solely from (CB5.4).

131 The additive receiving sheaf of the current tau source

24 September 2026. Independent continuation of `SOURCE_TAU_SPECIALIZATION_DERIVATION.md`, ST0–ST10. This constructs the sheaf and its actual restriction, support and spectral maps from the current source operations. It does not replace that source by its additive receiver.

131.1 SS0. Operation prerequisites and the retained corpus

The current `CORPUS_AND_OPERATION_RULES.md` (private source record), including its new global-argument checks, was read before the calculation. The controlling source remains `SOURCE_OPERATIONS_AND_PROOFS.md`, B1–B5, together with the verbatim U01–U18 and the full-history passages cited in ST0 and AC0. In particular:

- Z_1/τ is primitive presence with no Z_2 parity or source addition.
- The original integers, their full values and the applicable Z_2 data are retained after the complete arithmetic reconstruction.
- The source operations are original integer arithmetic and the supplied global multiplicative identity τ .
- Constructing an additive receiving object does not extend the domain of the source's addition. Formal receiving elements are not asserted to be copies of τ .
- A local calculation about this sheaf is not made a counterexample to the user's whole-spectrum argument.

The symbols $0, 1, \tau$ below have their unchanged source meanings when used in S . Brackets such as $[1]$ explicitly denote their images in a constructed receiving ring. All sheaf cohomology in this document is ordinary sheaf cohomology of the stated topological space, with abelian receiving coefficients. No étale, prismatic, or full programme cohomology is silently identified with it.

131.2 SS1. The exact source space and universal local receiving rings

Retain the source S and the integer-addition-compatible prime subspace of ST2:

$$X = X_{\mathbb{Z}} \cup \{\mathfrak{m}\}, \quad X_{\mathbb{Z}} = \{(0)\} \cup \{p\mathbb{Z} : p \text{ prime}\} \simeq \text{Spec } \mathbb{Z}, \quad \mathfrak{m} = \mathbb{Z} \subset S. \quad (\text{SS1.1})$$

This notation names the actual ideal $\mathfrak{m}$, not primitive τ or the separately constructed privileged η . The open immersion is $j : X_{\mathbb{Z}} \hookrightarrow X$, and $i : \{\mathfrak{m}\} \hookrightarrow X$ is the closed point inclusion. The only open neighbourhood of $\mathfrak{m}$ is X , and $D(1) = X_{\mathbb{Z}}$. Every proper open of X is an ordinary open of $X_{\mathbb{Z}}$.

Every nonempty open of $X_{\mathbb{Z}}$ has the form $D(n)$, $n \neq 0$: a proper nonempty closed subset is the finite set of prime divisors of an integer, and the generic point lies in every nonempty open. The empty open is treated separately.

The universal additive receiver constructed in ST5 is

$$U_S = \mathbb{Z}[\epsilon]/(\epsilon^2 - \epsilon), \quad [\tau] = 1_{U_S}, \quad [1] = \epsilon, \quad [n] = n\epsilon. \quad (\text{SS1.2})$$

The bracket map is injective on the specified source carrier and preserves every defined source operation, as proved in ST5. It is not an identification of the source with this ring, and its new sums do not define sums of primitive τ .

At the source closed point the only denominator is τ , so its universal local receiver is U_S . At an arithmetic basic open $D(n)$, the universal ring is

$$U_S[[n]^{-1}] \cong \mathbb{Z}[1/n]. \quad (\text{SS1.3})$$

To prove it, $[n] = n\epsilon$ becomes invertible. The equation $\epsilon[n] = [n]$ then gives $\epsilon = 1$. Thus the remaining ring is $\mathbb{Z}[1/n]$. Conversely that ring receives the source localization by $s/u \mapsto c(s)/c(u)$,

with $c(s) = 1s$, and makes $[n]$ invertible. The two maps fix all integer fractions and are inverse. This proves both the ring identification and its universal property, without cancelling an element in the unlocalized source.

Equivalently, constructing the receiving ring commutes with these specified localizations. A ring map from either side is precisely a map preserving the source's defined operations and making the indicated denominator images invertible. This gives the same universal factorization in both directions.

At an arithmetic prime the resulting ring is $\mathbb{Z}_{(p)}$, and at the arithmetic generic point it is $\mathbb{Q}$. All local additions there are operations on the explicitly localized integer receiver.

131.3 SS2. Construction and proof of the sheaf

Define $\mathcal{R}$ on the complete list of opens by

$$\mathcal{R}(X) = U_S, \quad \mathcal{R}(U) = \mathcal{O}_{\mathbb{Z}}(U) \quad (U \subseteq X_{\mathbb{Z}}), \quad (\text{SS2.1})$$

where $\mathcal{R}(\emptyset)$ is the zero ring. For nonempty $U = D(n)$, this means exactly $\mathbb{Z}[1/n]$. The restriction from X to $D(n)$ is

$$a + b\epsilon \mapsto a + b \quad \text{in } \mathbb{Z}[1/n]. \quad (\text{SS2.2})$$

All other restrictions are the original inclusions of the stated fraction rings. These maps are the local universal maps from SS1 and compose correctly.

Here is a direct sheaf proof. A covering of X must contain X itself, because it must cover $\mathfrak{m}$. Gluing on such a covering is therefore immediate from the matching condition. For an open $U \subseteq X_{\mathbb{Z}}$, every nonempty covering member contains the generic point. Matching sections in their rational stalk are thus one and the same rational number q . At each arithmetic prime of U , the section is regular on some covering member, so that prime does not divide the reduced denominator of q . Consequently the denominator has prime factors only among the finitely many primes excluded by $U = D(n)$. Hence $q \in \mathbb{Z}[1/n]$, which gives the unique glued section. This proves the sheaf axiom on every open.

Its stalks, computed from these restriction maps, are

$$\mathcal{R}_{\mathfrak{m}} = U_S, \quad \mathcal{R}_{(p)} = \mathbb{Z}_{(p)}, \quad \mathcal{R}_{(0)} = \mathbb{Q}. \quad (\text{SS2.3})$$

The first equality follows because $\mathfrak{m}$ has only the neighbourhood X . The others follow by allowing all denominators absent from the specified arithmetic prime. These are exactly the universal additive receivers of the source stalks, not different stalks inferred only from an analogy.

The source monoid sheaf maps to $\mathcal{R}$ by its actual receiving maps on each localization. On X , the bracket map is faithful. On $D(1)$, the original $\tau, 1$ distinction is identified by the explicitly nonfaithful localization $c(s) = 1s$. No source parity is assigned to τ .

131.4 SS3. The exact product decomposition and its maps

There is a receiving-ring isomorphism

$$\Psi : U_S \longrightarrow \mathbb{Z} \times \mathbb{Z}, \quad f(\epsilon) \mapsto (f(1), f(0)), \quad (\text{SS3.1})$$

with inverse

$$(a, b) \mapsto a\epsilon + b(1_{U_S} - \epsilon). \quad (\text{SS3.2})$$

Multiplication is componentwise because $\epsilon(1 - \epsilon) = 0$ and both components are idempotent. Composing the maps verifies that they are inverse on 1_{U_S} and ϵ , hence on every element. These are coordinates in the receiving ring, not coordinates of primitive τ .

For a sheaf $\mathcal{F}$ on $X_{\mathbb{Z}}$, $j_*\mathcal{F}(U) = \mathcal{F}(U \cap X_{\mathbb{Z}})$. For an abelian group or ring B on the single closed point, $i_*B(U) = B$ when $U = X$, and is zero when $U \subseteq X_{\mathbb{Z}}$. Direct inspection of SS2 now proves the ring-sheaf isomorphism

$$\boxed{\mathcal{R} \cong j_*\mathcal{O}_{\mathbb{Z}} \times i_*\mathbb{Z}.} \quad (\text{SS3.3})$$

On X it is Ψ ; on every proper open it is the identity of $\mathcal{O}_{\mathbb{Z}}(U)$ with its product with the zero ring. The restrictions commute because restriction from X is first projection followed by the integer localization.

The restriction map and its exact kernel are

$$r : \mathcal{R} \longrightarrow j_*\mathcal{O}_{\mathbb{Z}}, \quad \ker r = i_*\mathbb{Z} = (1_{U_S} - \epsilon)\mathcal{R}. \quad (\text{SS3.4})$$

There is the canonical additive and $\mathcal{R}$ -linear section

$$s_+ : j_*\mathcal{O}_{\mathbb{Z}} \longrightarrow \mathcal{R}, \quad a \longmapsto (a, 0) \text{ on } X, \quad (\text{SS3.5})$$

which is the identity on proper opens. It preserves multiplication but is not globally unital: it sends the unit to ϵ , not to 1_{U_S} . The boundary inclusion $b \mapsto (0, b)$ is likewise the inclusion of an idempotent corner. Thus

$$0 \longrightarrow i_*\mathbb{Z} \longrightarrow \mathcal{R} \xrightarrow{r} j_*\mathcal{O}_{\mathbb{Z}} \longrightarrow 0 \quad (\text{SS3.6})$$

is a canonically split sequence of abelian sheaves and $\mathcal{R}$ -modules. No section preserving the source integer inclusion is falsely called a globally unital map.

For completeness a different unital ring-sheaf section of r does exist: on X , $a \mapsto (a, a)$, and on proper opens the identity. Compatibility follows from first projection. It does **not** preserve the specified source integer inclusion $n \mapsto (n, 0)$. The two maps differ by the explicit boundary component $(0, a)$. The source-compatible cohomological splitting used below is SS3.5, not this altered source map.

The inverse-image sheaves are

$$j^{-1}\mathcal{R} = \mathcal{O}_{\mathbb{Z}}, \quad i^{-1}\mathcal{R} = U_S \cong \mathbb{Z} \times \mathbb{Z}. \quad (\text{SS3.7})$$

In particular the closed stalk is the whole product, not merely the support summand. If one specifies the closed receiving ring to be $\mathbb{Z}$ through second projection $U_S \rightarrow \mathbb{Z}$, the module pullback is

$$i^*\mathcal{M} = \mathbb{Z} \otimes_{U_S} \mathcal{M}_{\mathfrak{m}} \cong (1 - \epsilon)\mathcal{M}_{\mathfrak{m}}. \quad (\text{SS3.8})$$

This construction is distinct from the inverse image in SS3.7 and will agree with the support part for $\mathcal{R}$.

131.5 SS4. Global cohomology on X , and the arithmetic open

For **any** sheaf of abelian groups $\mathcal{F}$ on X ,

$$\Gamma(X, \mathcal{F}) = \mathcal{F}_{\mathfrak{m}}. \quad (\text{SS4.1})$$

There is only one neighbourhood in the stalk limit. Since exactness of sheaves of abelian groups is tested on stalks, this proves that $\Gamma(X, -)$ is an exact functor. Hence

$$H^q(X, \mathcal{F}) = 0 \quad (q > 0) \quad (\text{SS4.2})$$

for this precise topological site. This does not identify it with a different arithmetic, étale, or derived programme site.

We also need the arithmetic-open calculation, which is supplied explicitly. On $U = X_{\mathbb{Z}}$, let $\mathcal{Q}$ have value $\mathbb{Q}$ on every nonempty open and zero on the empty open. For every ordinary prime p , put $D_p = \mathbb{Q}/\mathbb{Z}_{(p)}$, and define

$$\mathcal{P}(V) = \bigoplus_{\substack{p \in V \\ p \text{ closed}}} D_p \quad (V \subseteq U), \quad (\text{SS4.3})$$

with restriction the coordinate projection. All sums of sections here have finite support.

These are sheaves. For $\mathcal{Q}$, every nonempty open intersection contains the generic point, so matching sections are one rational number. For $\mathcal{P}$, every open of U is quasi-compact: choose one nonempty member of any cover; its complement in the given open contains only finitely many closed points, so finitely many further members cover those. Compatible finite-support sections therefore glue with finite support. Both $\mathcal{Q}$ and $\mathcal{P}$ are flabby: nonempty restriction in $\mathcal{Q}$ is identity, and a finite tuple in $\mathcal{P}$ extends by zero in additional coordinates.

The maps

$$0 \longrightarrow \mathcal{O}_{\mathbb{Z}} \longrightarrow \mathcal{Q} \xrightarrow{d} \mathcal{P} \longrightarrow 0, \quad d(q)_p = q \bmod \mathbb{Z}_{(p)} \quad (\text{SS4.4})$$

are exact on every open. Only denominator primes give nonzero residues, so $d(q)$ has finite support. Its kernel on $V = D(n)$ is precisely $\mathbb{Z}[1/n]$. For surjectivity, a class in D_p has a representative a/p^k : for a fraction $a/(p^k b)$, with $p \nmid b$, choose c such that $cb \equiv a \pmod{p^k}$; the difference from c/p^k belongs to $\mathbb{Z}_{(p)}$. For a finite tuple of such classes, sum their representatives. Each representative is integral at the other primes, so this rational number has exactly the prescribed residue tuple.

Thus SS4.4 is a flabby resolution whose global section sequence is also exact. It follows that

$$H^0(U, \mathcal{O}_{\mathbb{Z}}) = \mathbb{Z}, \quad H^q(U, \mathcal{O}_{\mathbb{Z}}) = 0 \quad (q > 0). \quad (\text{SS4.5})$$

The same proof applies on every $D(n)$, with $\mathbb{Z}[1/n]$ in degree zero.

Here the cohomological fact used about a flabby resolution is the usual derived-functor calculation, not a purity premise. One proof is to embed a flabby sheaf in an injective sheaf. Sections of its quotient lift globally: first choose local lifts and well-order a cover; at each step their difference on an overlap lies in the flabby kernel and extends to the new open, so subtract that extension before gluing; at a limit stage use the sheaf gluing axiom. The same construction on smaller opens shows that the quotient is flabby. Induction in an injective resolution makes the global section complex exact in positive degrees. This proves the acyclicity used in the displayed resolution.

131.6 SS5. Local cohomology, restriction and every connecting map

For the closed set $Z = \{\mathfrak{m}\}$, define the supported-section functor

$$\Gamma_Z(X, \mathcal{F}) = \ker(\mathcal{F}(X) \rightarrow \mathcal{F}(U)), \quad U = X_{\mathbb{Z}}. \quad (\text{SS5.1})$$

This constructs the operation whose derived functors are being calculated. The usual restriction exact triangle is obtained by applying an injective or flabby resolution to the supported-section kernel. Since flabby restrictions are onto, the resulting sequence of complexes is short exact and gives

$$R\Gamma_Z(X, \mathcal{F}) \longrightarrow R\Gamma(X, \mathcal{F}) \longrightarrow R\Gamma(U, \mathcal{F}|_U) \longrightarrow R\Gamma_Z(X, \mathcal{F})[1]. \quad (\text{SS5.2})$$

This also proves the long exact sequence used below; it is not an assumed geometric specialization triangle.

For $\mathcal{R}$, the middle and open terms have already been computed:

$$R\Gamma(X, \mathcal{R}) = U_S[0] \cong (\mathbb{Z} \oplus \mathbb{Z})[0], \quad R\Gamma(U, \mathcal{R}|_U) = \mathbb{Z}[0]. \quad (\text{SS5.3})$$

The arrow is exactly first projection. Its kernel is the second summand and its cokernel is zero. Consequently

$$\boxed{H_Z^0(X, \mathcal{R}) = (1 - \epsilon)U_S \cong \mathbb{Z}, \quad H_Z^q(X, \mathcal{R}) = 0 \quad (q > 0).} \quad (\text{SS5.4})$$

Equivalently $R\Gamma_Z(X, \mathcal{R})$ is represented by the two-term complex

$$[U_S \xrightarrow{\pi_+} \mathbb{Z}] \quad \text{in degrees } 0, 1. \quad (\text{SS5.5})$$

Its arithmetic component is the contractible complex $[\mathbb{Z} \xrightarrow{\text{id}} \mathbb{Z}]$, and its other component is $\mathbb{Z}$ in degree zero. A contracting homotopy on that first component sends its degree-one element back to the same degree-zero integer. Thus the vanishing has an explicit chain-level calculation.

The exact section $s_+(a) = (a, 0)$ lifts every open global class. The connecting map

$$\partial : \mathbb{Z} = H^0(U, \mathcal{O}_{\mathbb{Z}}) \longrightarrow H_Z^1(X, \mathcal{R})$$

is zero. This follows both from the target calculation and directly from the displayed section and homotopy. All higher connecting maps vanish because the already calculated source or target groups are zero. This proves an actual lifting statement from the sheaf's original source-induced restriction map.

The internal local-cohomology sheaves satisfy

$$\mathcal{H}_Z^0(\mathcal{R}) = i_*\mathbb{Z}, \quad \mathcal{H}_Z^q(\mathcal{R}) = 0 \quad (q > 0). \quad (\text{SS5.6})$$

Their stalks outside Z vanish by restricting to the open complement. At $\mathfrak{m}$, their only neighbourhood is X , giving exactly SS5.4. In the customary notation for the supported right adjoint, this says $i^!\mathcal{R} = \mathbb{Z}[0]$, in contrast with $i^{-1}\mathcal{R} = U_S$ in SS3.7. There is no hidden degree shift or Tate twist.

The same computation proves $R^q j_* \mathcal{O}_{\mathbb{Z}} = 0$ for $q > 0$. At an arithmetic stalk use the cofinal neighbourhoods $D(n)$ and SS4.5; at $\mathfrak{m}$ use U itself and SS4.5. Hence the sheaf-level triangle is the actual split

$$i_*\mathbb{Z}[0] \longrightarrow \mathcal{R}[0] \longrightarrow j_*\mathcal{O}_{\mathbb{Z}}[0] \longrightarrow i_*\mathbb{Z}[1], \quad (\text{SS5.7})$$

with zero last arrow and the maps of SS3.

131.7 SS6. The two components of the receiving spectrum and the exact map back

Let

$$Y = \text{Spec } U_S.$$

The ring product SS3.1 gives two complete components:

$$Y = Y_+ \sqcup Y_-, \quad Y_+ \simeq \text{Spec } \mathbb{Z}, \quad Y_- \simeq \text{Spec } \mathbb{Z}. \quad (\text{SS6.1})$$

Here $Y_+ = D(\epsilon)$, $Y_- = D(1 - \epsilon)$. To verify that these exhaust the prime spectrum, in a prime ideal one of the two orthogonal idempotents must lie in the ideal, since their product is zero; they cannot both lie in it, since their sum is the unit. The remaining component is a prime ideal in its copy of $\mathbb{Z}$.

For every ordinary ring prime $\mathfrak{p} \subset \mathbb{Z}$, including $\mathfrak{p} = (0)$, write

$$\mathfrak{q}_{+, \mathfrak{p}} = \mathfrak{p} \times \mathbb{Z}, \quad \mathfrak{q}_{-, \mathfrak{p}} = \mathbb{Z} \times \mathfrak{p}. \quad (\text{SS6.2})$$

Contracting along the source receiving map $S \rightarrow U_S$ gives

$$f(\mathfrak{q}_{+, \mathfrak{p}}) = \mathfrak{p} \subset S, \quad \boxed{f(\mathfrak{q}_{-, \mathfrak{p}}) = \mathfrak{m} \text{ for every } \mathfrak{p}.} \quad (\text{SS6.3})$$

Indeed the original integer n maps to $(n, 0)$, so it lies in $\mathfrak{q}_{+, \mathfrak{p}}$ exactly when $n \in \mathfrak{p}$, while every original integer lies in every $\mathfrak{q}_{-, \mathfrak{p}}$. The formal image of τ is the global ring unit $(1, 1)$, which belongs to neither kind of prime. Thus each contraction is an actual compatible source prime, as required.

The map $f : Y \rightarrow X$ is continuous:

$$f^{-1}(D(n)) = D_Y(n\epsilon) = D_{\mathbb{Z}}(n) \subset Y_+ \quad (n \neq 0), \quad f^{-1}(X) = Y. \quad (\text{SS6.4})$$

These are all the needed basic opens. On Y_+ , f is the homeomorphism onto $X_{\mathbb{Z}}$. Its entire fibre at $\mathfrak{m}$ is the second copy Y_- , with its generic point, every arithmetic closed point, their specializations, and their residue fields retained.

This continuous surjection is not a topological quotient: $f^{-1}(\{\mathfrak{m}\}) = Y_-$ is open, but $\{\mathfrak{m}\}$ is not open in X . This states its actual topology rather than pretending that collapsing the second component would automatically reproduce the source topology.

The receiving sheaf is exactly its direct image:

$$\boxed{f_*\mathcal{O}_Y = \mathcal{R}.} \quad (\text{SS6.5})$$

On X , both sides have sections $\mathbb{Z} \times \mathbb{Z}$. On $U' \subseteq X_{\mathbb{Z}}$, its inverse image is the same open in Y_+ and contains none of Y_- , giving $\mathcal{O}_{\mathbb{Z}}(U')$. All restriction maps agree. The adjoint $f^{-1}\mathcal{R} \rightarrow \mathcal{O}_Y$ gives an exact morphism of ringed spaces. At a plus prime, the stalk map is the corresponding arithmetic localization. At a minus prime it is

$$U_S \cong \mathbb{Z} \times \mathbb{Z} \xrightarrow{\pi_-} \mathbb{Z} \longrightarrow \mathbb{Z}_{(\mathfrak{p})}, \quad (\text{SS6.6})$$

where $\mathbb{Z}_{((0))} = \mathbb{Q}$.

The stalk $\mathcal{R}_{\mathfrak{m}} = U_S$ is not a local ring: it has the two nonzero complementary idempotents. In a local ring one of $e, 1 - e$ must be a unit, forcing the other to be zero, so that situation cannot occur. Thus $(X, \mathcal{R})$ is the constructed ringed receiving space; it is not asserted to be a scheme with this structure sheaf. The actual affine scheme is Y , and the map and all its fibres are displayed rather than erased.

The added receiving primes of Y_- must be distinguished from original integer primes. The original source integer n has image $n\epsilon = (n, 0)$, which is zero on the entire minus component. The ordinary ring scalar $n1_{U_S} = (n, n)$ can distinguish its different primes. The latter uses the newly constructed receiving ring's integer scalar operation and is not the source integer embedding. No expression counting copies of primitive τ has been defined. This distinction explains exactly why the additional ring primes occur and why contraction groups them over one source point.

The higher direct images $R^q f_*\mathcal{O}_Y$ vanish for $q > 0$: on the preimages of X and of the arithmetic basic opens, use SS4's explicit acyclicity for each of the one or two copies. Consequently SS6.5 also identifies the complete derived direct image on this structure sheaf.

131.8 SS7. What the splitting says for coefficients and independent Frobenius

The global idempotent ϵ decomposes every sheaf of $\mathcal{R}$ -modules as

$$\mathcal{M} = \epsilon\mathcal{M} \oplus (1 - \epsilon)\mathcal{M}. \quad (\text{SS7.1})$$

This follows on each section from $m = \epsilon m + (1 - \epsilon)m$, zero intersection, and compatibility with restrictions. The second summand is supported at $\mathfrak{m}$, because $1 - \epsilon = 0$ on $X_{\mathbb{Z}}$. It is therefore i_*B , where $B = ((1 - \epsilon)\mathcal{M})_{\mathfrak{m}}$.

For a general $\mathcal{M}$, it does **not** follow that $\epsilon\mathcal{M} = j_*j^{-1}\mathcal{M}$. There can be extra data in its $\mathfrak{m}$ -stalk and in the restriction map. The exact comparison is

$$\epsilon\mathcal{M} \longrightarrow j_*j^{-1}\mathcal{M}, \quad (\text{SS7.2})$$

with its actual kernel and cokernel supported at $\mathfrak{m}$; away from that point it is the identity. Because the only neighbourhood of $\mathfrak{m}$ is X , these groups are explicitly the kernel and cokernel of

$$(\epsilon\mathcal{M})(X) \longrightarrow \Gamma(X_{\mathbb{Z}}, j^{-1}\mathcal{M}). \quad (\text{SS7.3})$$

For the particular structure sheaf $\mathcal{M} = \mathcal{R}$, this map is an isomorphism on the epsilon component by SS3.3, so these extra groups are zero. The scope of the proved lifting must not be enlarged by omitting SS7.2 for other coefficients.

Let a specified independent Frobenius or idèle action F on the receiving sheaf or complex preserve the source action, hence commute with ϵ . Then it preserves both parts of SS7.1, their inclusion and their projection. For $\mathcal{R}$'s split restriction complex SS5.5, the lift through the epsilon summand is equivariant and the support connecting map remains zero. This is a proved consequence of the source-induced idempotent, not an assigned numerical weight of primitive τ .

A unital ring-sheaf endomorphism over the identity on X that fixes the source receiving map is necessarily the identity on $\mathcal{R}$. On X it fixes ϵ and the global unit, hence every element of U_S . On each arithmetic $D(n)$, it fixes every integer and every allowed inverse, hence every element of $\mathbb{Z}[1/n]$. This proves the claim on all opens. Nontrivial numerical Frobenius actions therefore require their stated coefficient objects or geometric correspondences; they are not manufactured by calling source multiplication a unital ring endomorphism.

At the coefficient-module level the exact situation is different and is already classified by the split. Over $\mathbb{C}$, on the two components of $\mathcal{R} \otimes_{\mathbb{Z}} \mathbb{C}$, any two specified invertible coefficient actions commute with the source idempotent. They can have different numerical eigenvalues while the section and all connecting maps still split. More generally the actual Q and $\mathcal{B}_{\text{pr}}$ actions of ST6.3 can be retained as independently named coefficient actions. This observation is a classification of the constructed receiver, not an arbitrary-parameter counterexample to the user's full geometric programme.

Source integer multiplication on the sheaf is multiplication by $[n] = n\epsilon$; it acts by n on the arithmetic part and by zero on the supported part. The independent idèle boundary action in ABR19 is instead $\text{diag}(1, b)$. Their exact coexistence and the norm/power-map comparison are ST6–ST7a. In particular no logarithmic Frobenius weight is assigned to the zero scalar operator on the supported source corner, and no invertible geometric action is identified with that zero operator.

131.9 SS8. Relation to the cited geometric programmes

The source calculation begins with the user's current Z_0, Z_1, Z_2, τ definitions and the completed integer layer. The arithmetic open, the extra source point, the universal receiving spectrum and the supported sheaf have each been constructed from those inputs. The original source S and its monoid sheaf remain in the diagram; no ring operation from $\mathcal{R}$ has been made a new source operation.

The exact geometric distinction used in the comparison is also present in Alain Connes and Caterina Consani, *On the Jacobian of $\text{Spec } \mathbb{Z}$* , original `Jacobian.tex`, lines 2501–2550: their number-field spectral action is idèle translation, whereas their function-field comparison uses arithmetic Frobenius. ST7 proves the connecting power-map, pullback and norm formulas; SS7 preserves both actions instead of replacing one by the other.

Pierre Deligne's lifting argument in *La conjecture de Weil. II*, §3.6.1–§3.6.3, kills a geometric support obstruction by separated Frobenius weights. Here the actual source-induced additive sheaf has its support obstruction killed by the explicit idempotent splitting, and SS5 computes the full local cohomology and connecting maps. This is a proved lifting statement for this sheaf. Neither this splitting, which does not involve numerical weights, nor the topology's exact global-sections functor proves that a different cohomology theory has only weight one.

The explicit remaining comparison is not hidden: other programme coefficient complexes enter through SS7.2 and the independently specified geometric action. The current result does

not assert that their restriction maps, or Deligne's entire proper specialization complex, are this structure-sheaf sequence. All the kernels, additional receiving primes, source localization fibres and support maps needed to compare them are retained.

131.10 SS9. Exact conclusions

The universal additive receiving sheaf of the current source is

$$\mathcal{R} = j_*\mathcal{O}_{\mathbb{Z}} \times i_*\mathbb{Z}.$$

Its closed stalk is $U_S = \mathbb{Z} \times \mathbb{Z}$; its supported part is only the second copy. Its local cohomology at the added source point is $\mathbb{Z}$ in degree zero and zero in higher degrees. Every connecting map for this restriction sequence is zero, with an explicit equivariant additive section.

The affine receiving spectrum contains two entire copies of $\mathrm{Spec} \mathbb{Z}$. One maps identically to the source arithmetic open; the other, with all its primes retained, contracts to the added source point. The direct image of its structure sheaf is exactly $\mathcal{R}$.

These are actual sheaf and source-specialization results. No source addition on τ , no source parity for it, no erasure of the added ring primes, and no RH or purity assertion has been introduced.

132 The original moving q^r functions and the original-zeta Mellin quotient

Independent receiving derivation MQ0–MQ9, 24 September 2026. This extends CW0–CW7 from constant Connes–Consani coefficients to their actual moving functions. Every rational exponent is retained. All sums below are finite unless their domain explicitly states otherwise.

The controlling corpus and operation rules were read, including the complete preserved arguments USR-9ad1c0a2d09dba92, USR-f55d16feb948d8c2 and USR-6152e3bc6302258c. This calculation uses the already reconstructed arithmetic. The complex coefficient algebra, its unit, the coordinates below, and its addition are not primitive Z_1/τ . No addition, parity, coordinate, metric or counting is assigned to primitive τ .

Human source: Alain Connes and Caterina Consani, The Riemann–Roch strategy: Complex lift of the Scaling Site, arXiv:1805.10501v1, original author TeX sources/thecurve_K.tex, §5.1 Lemma adelicomp, §5.4 equations Jdefn1 and the following right action, Proposition classorb1 and Proposition proetcov, and §7.1 equations holom–holom2, Proposition functionq and Remark perfectoid. The exact reading record is at MQ9.

132.1 MQ0. The full original source

Retain

$$W_{\text{alg}} = \mathbb{C}[\mathbb{R}_{>0}^\times], \quad [x][z] = [xz], \quad \theta_\mu[x] = [x^\mu], \quad \chi_\lambda[x] = x^\lambda, \quad \mu, \lambda > 0. \quad (\text{MQ0.1})$$

The symbol [1] is this named group algebra’s unit.

Put $G = \mathbb{A}_{\mathbb{Q}}/\mathbb{Q}$. Let α be the standard additive character of $\mathbb{A}_{\mathbb{Q}}$, trivial on the diagonal $\mathbb{Q}$, whose real component is $\exp(2\pi i X_\infty)$. Write

$$e_r(g) = \alpha(rX), \quad r \in \mathbb{Q}, \quad g = [X] \in G.$$

This is well defined, $e_r e_s = e_{r+s}$, and

$$e_r(\iota_\infty(v)) = \exp(2\pi i r v). \quad (\text{MQ0.2})$$

The characters e_r are distinct: for $r \neq s$, choosing real v with $(r - s)v \notin \mathbb{Z}$ distinguishes them.

The original classical orbit is

$$\Gamma_{\mathbb{Q}, \text{cl}} \simeq G \times \mathbb{R}_{>0}.$$

Its right real action is

$$(g, Y) \cdot \begin{pmatrix} u & v \\ 0 & 1 \end{pmatrix} = (g + \iota_\infty(vY), uY). \quad (\text{MQ0.3})$$

In particular right geometric scaling acts by $R(u)F(g, Y) = F(g, uY)$.

The original functions and every character evaluation are

$$\begin{aligned} q^r(g, Y) &= e_r(g)[\exp(-2\pi r Y)], \\ \chi_\lambda(q^r)(g, Y) &= e_r(g) \exp(-2\pi r \lambda Y), \quad r \in \mathbb{Q}. \end{aligned} \quad (\text{MQ0.4})$$

Negative r gives growth at infinity; this is retained. The algebraic source imposes no decay condition in Y . Source holomorphy is tested by the complete character family, with

$$L_\lambda = Y(\lambda \partial_{X_\infty} + i \partial_Y).$$

Differentiation gives

$$L_\lambda \chi_\lambda(q^r) = Y(2\pi i \lambda r - 2\pi i \lambda r) \chi_\lambda(q^r) = 0. \quad (\text{MQ0.5})$$

132.2 MQ1. The moving algebra is an actual injective geometric evaluation

Use a formal rational-frequency basis $\mathbf{u}_r$, with $\mathbf{u}_r \mathbf{u}_s = \mathbf{u}_{r+s}$, and put

$$\mathcal{P} = W_{\text{alg}} \otimes_{\mathbb{C}} \mathbb{C}[\mathbb{Q}].$$

Define its geometric evaluation by

$$\text{Ev}_W \left(\sum_r w_r \mathbf{u}_r \right) (g, Y) = \sum_r w_r q^r(g, Y) = \sum_r e_r(g) w_r [\exp(-2\pi r Y)]. \quad (\text{MQ1.1})$$

It preserves addition and multiplication by the displayed source laws.

This evaluation is injective. Give G its Haar probability measure dg . Translation invariance proves character orthogonality: for $r \neq s$, choose g_0 with $e_{r-s}(g_0) \neq 1$; changing g to $g + g_0$ shows that the integral of e_{r-s} is zero. The integral of e_0 is one. At any fixed $Y_0 > 0$, the function in (MQ1.1) takes values in the finite-dimensional span of its displayed coefficients, so its Fourier integral is an ordinary finite-dimensional integral and equals

$$\int_G \text{Ev}_W(F)(g, Y_0) \overline{e_r(g)} dg = w_r [\exp(-2\pi r Y_0)].$$

The basis element on the right is invertible. Hence the exact return is

$$\boxed{w_r = [\exp(2\pi r Y_0)] \int_G \text{Ev}_W(F)(g, Y_0) \overline{e_r(g)} dg.} \quad (\text{MQ1.2})$$

This recovers every w_r , is independent of the chosen Y_0 , and proves injectivity. Thus the formal algebra above realizes the original $W[q^r]$, rather than deleting its moving functions.

The complete scalar character evaluations also distinguish these W -valued functions. Indeed distinct positive numbers $x_1, \dots, x_d$ give an invertible evaluation matrix $(x_j^\lambda)_{\lambda=1, \dots, d}$, whose determinant is

$$\left(\prod_j x_j \right) \prod_{i < j} (x_j - x_i).$$

Apply this to each finite W -valued Fourier coefficient. A single character is not substituted for the full family.

132.3 MQ2. Arithmetic Frobenius and the faithful coefficientwise map

The source arithmetic Frobenius is

$$\mathfrak{F}_\mu = \theta_\mu R(\mu^{-1}).$$

Using (MQ0.4) gives, with no omitted radial factor,

$$\mathfrak{F}_\mu(w_r q^r)(g, Y) = \theta_\mu(w_r) e_r(g) [\exp(-2\pi r Y / \mu)^\mu] = \theta_\mu(w_r) q^r(g, Y). \quad (\text{MQ2.1})$$

Therefore its exact action on $\mathcal{P}$ is

$$\mathfrak{F}_\mu = \theta_\mu \otimes \text{id}_{\mathbb{C}[\mathbb{Q}]}. \quad (\text{MQ2.2})$$

This holds for every real $\mu > 0$, and hence for all primes and prime powers simultaneously.

Retain the original CW domain $V_+ = \text{span}_{\mathbb{C}}\{[x] : x > 1\}$, the strong Mellin space $\mathcal{A}$, and the original-zeta quotient

$$Q = \mathcal{A}/I, \quad I = \overline{\mathcal{E}(S_0^{\text{even}})}^{\mathcal{A}}, \quad \mathcal{E}f(u) = u^{1/2} \sum_{n \geq 1} f(nu).$$

The retained transforms, action and exact comparison are

$$\begin{aligned}
F_k(s) &= \int_0^\infty k(u) u^{s-1/2} \frac{du}{u}, & W_a k(u) &= a^{1/2} k(u/a), \\
b(u) &= \exp(-(\log u)^2), & K_b[x] &= W_{\log x} b, & \bar{K}_b &= \pi_Q K_b, \\
B(s) &= \sqrt{\pi} \exp((s - \tfrac{1}{2})^2/4), & F_{K_b[x]}(s) &= B(s)(\log x)^s, & \bar{K}_b \theta_\mu &= W_\mu \bar{K}_b.
\end{aligned} \tag{MQ2.3}$$

CW3 proves $\bar{K}_b$ injective into the actual Q . It uses the original identity

$$F_{\mathcal{E}f}(s) = \zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v}$$

and the complete actual zero multiplicities, together with the unconditional original-zeta zero count. No completed zeta function is substituted here.

Define

$$\mathcal{K}_+ = \bar{K}_b \otimes \text{id} : V_+ \otimes \mathbb{C}[\mathbb{Q}] \longrightarrow Q \otimes \mathbb{C}[\mathbb{Q}]. \tag{MQ2.4}$$

Every vector is a finite sum in the frequency basis. Thus equality of images implies equality of every $\bar{K}_b w_r$, and injectivity of CW gives equality of every w_r . Therefore $\mathcal{K}_+$ is injective. Its exact action intertwining is

$$\mathcal{K}_+ \mathfrak{F}_\mu = (W_\mu \otimes \text{id}) \mathcal{K}_+. \tag{MQ2.5}$$

This acts on the actual moving functions by $\mathcal{K}_+ \text{Ev}_W^{-1}$; (MQ1.2) supplies the explicit inverse rather than merely asserting one. Importantly, the coefficient is extracted before applying CW. A basis value $[x \exp(-2\pi r Y)]$ can cross the value $[1]$, even when $x > 1$. Nothing in (MQ2.4) treats such a moving value as a positive constant coefficient. MQ6 below provides a separate pointwise map that retains those crossings.

132.4 MQ3. Geometric realization of the coefficientwise receiver and full zero jets

The target $Q \otimes \mathbb{C}[\mathbb{Q}]$ has the faithful geometric realization

$$\text{Ev}_{Q,\lambda} \left(\sum_r v_r \mathbf{u}_r \right) (g, Y) = \sum_r e_r(g) \exp(-2\pi r \lambda Y) v_r, \quad \lambda > 0. \tag{MQ3.1}$$

Fourier integration at any Y_0 , followed by multiplication by $\exp(2\pi r \lambda Y_0)$, recovers v_r . Thus even a fixed λ distinguishes this target's coefficients; retaining all λ additionally retains the original geometric character family.

The arithmetic action on the full target family is

$$(\widehat{\mathfrak{F}}_\mu H)_\lambda(g, Y) = W_\mu H_{\lambda\mu}(g, Y/\mu). \tag{MQ3.2}$$

Substitution in (MQ3.1) gives

$$\widehat{\mathfrak{F}}_\mu \text{Ev}_Q(v_r \mathbf{u}_r) = \text{Ev}_Q(W_\mu v_r \mathbf{u}_r),$$

since $-2\pi r(\lambda\mu)(Y/\mu) = -2\pi r \lambda Y$. The factors are retained in the calculation rather than suppressed.

For every actual nontrivial zero ρ of multiplicity m_ρ , put

$$A_\rho = \mathbb{C}[T_\rho]/(T_\rho^{m_\rho}), \quad j_\rho([k]) = \sum_{h=0}^{m_\rho-1} \frac{F_k^{(h)}(\rho)}{h!} T_\rho^h.$$

For $x > 1$, all moving jets are exactly

$$(j_\rho \otimes \text{id})\mathcal{K}_+([x]\mathbf{u}_r) = B(\rho + T_\rho)(\log x)^\rho \exp((\log \log x)T_\rho)\mathbf{u}_r,$$

$$B(\rho + T_\rho) = \sqrt{\pi} \exp((\rho - \frac{1}{2})^2/4) \exp((\rho - \frac{1}{2})T_\rho/2 + T_\rho^2/4). \quad (\text{MQ3.3})$$

The exponentials are taken with every surviving term modulo $T_\rho^{m_\rho}$. Arithmetic Frobenius acts by

$$p^\rho \exp((\log p)T_\rho) \otimes \text{id} \quad (\text{MQ3.4})$$

on every frequency r . No r -dependent numerical weight appears in this full action.

For every finite set of zeros and every finite set of rational frequencies, the map from $V_+ \otimes \mathbb{C}[\mathbb{Q}]$ to the product of the specified zero blocks at those frequencies is onto. CW4 proves simultaneous finite-zero interpolation; apply it separately to each frequency. These choices form a finite sum and therefore lie in the stated source.

The map to all zero blocks at all frequencies is injective. If every such jet vanishes, then for each of the finitely many source frequencies the complete CW4 observation vanishes; its injectivity forces that coefficient to be zero. No surjectivity onto the infinite product is asserted. In particular a source element has finite frequency support, whereas an arbitrary element of that product need not.

132.5 MQ4. Separate geometric and coefficient scaling, with the complete radial family

Neither pure $R(\mu)$ nor pure θ_μ preserves the original holomorphic algebra $W[q^r]$ in general. The exact extension that accommodates both is

$$\mathcal{P}_{\text{rad}} = W_{\text{alg}} \otimes \mathbb{C}[\mathbb{Q}] \otimes \mathbb{C}[\mathbb{R}],$$

with basis written $[x]\mathbf{v}_{r,c}$, and evaluation

$$\text{Ev}_{\text{rad}}([x]\mathbf{v}_{r,c})(g, Y) = e_r(g)[x \exp(-2\pi cY)]. \quad (\text{MQ4.1})$$

Here c is a named radial frequency, independent of primitive τ . Multiplication multiplies x and adds the two frequency coordinates.

This evaluation is injective. Fourier projection separates the finite set of r 's. For a fixed r , combine duplicate pairs $(\log x, c)$. Distinct pairs yield distinct affine functions $\log x - 2\pi cY$, except at finitely many intersection values of Y . Choose any $Y > 0$ outside this finite set. Their W -basis values are then distinct, so vanishing of the evaluated sum forces every combined coefficient to be zero.

The exact actions are

$$R(v) : [x]\mathbf{v}_{r,c} \mapsto [x]\mathbf{v}_{r,vc},$$

$$\theta_\mu : [x]\mathbf{v}_{r,c} \mapsto [x^\mu]\mathbf{v}_{r,\mu c},$$

$$\mathfrak{F}_\mu = \theta_\mu R(\mu^{-1}) : [x]\mathbf{v}_{r,c} \mapsto [x^\mu]\mathbf{v}_{r,c}. \quad (\text{MQ4.2})$$

These follow directly from (MQ4.1), so their representation and commutation laws hold on the evaluated functions as well.

The source holomorphy condition identifies the diagonal exactly:

$$L_\lambda \chi_\lambda(\text{Ev}_{\text{rad}}([x]\mathbf{v}_{r,c})) = 2\pi i \lambda Y(r - c) e_r(g) x^\lambda \exp(-2\pi c \lambda Y). \quad (\text{MQ4.3})$$

For a finite sum annihilated for every $\lambda > 0$, Fourier projection separates r . At a positive Y avoiding the finitely many intersections used above, the values $x \exp(-2\pi cY)$ are distinct. Their complete character evaluations are linearly independent by the Vandermonde calculation in MQ1.

Since $2\pi i\lambda Y \neq 0$, every coefficient with $c \neq r$ must vanish. Conversely each $c = r$ term is annihilated by (MQ4.3). Hence

$$\boxed{\text{The holomorphic subalgebra in } \mathcal{P}_{\text{rad}} \text{ is exactly } W_{\text{alg}} \otimes \text{span}\{\mathbf{v}_{r,r} : r \in \mathbb{Q}\}.} \quad (\text{MQ4.4})$$

This is the original moving algebra under $\mathbf{u}_r \mapsto \mathbf{v}_{r,r}$.

On V_+ -coefficients the map $\bar{K}_b \otimes \text{id} \otimes \text{id}$ is again injective. Its target operations are

$$R(v) : (v_0, r, c) \mapsto (v_0, r, vc), \quad \theta_\mu : (v_0, r, c) \mapsto (W_\mu v_0, r, \mu c), \quad \mathfrak{F}_\mu : (v_0, r, c) \mapsto (W_\mu v_0, r, c).$$

Thus the larger representation retains both separate actions; the original holomorphic diagonal is preserved by their arithmetic combination.

132.6 MQ5. A faithful map for all constant W -coefficients, retaining the unit and both signs

To include the source coefficient [1] and all moving evaluations, use the exact invariant decomposition

$$W_{\text{alg}} = \mathbb{C}[1] \oplus V_+ \oplus V_-, \quad V_- = \text{span}\{[x] : 0 < x < 1\}.$$

Define

$$\widehat{Q} = \mathbb{C} \oplus Q_+ \oplus Q_-, \quad \widehat{W}_\mu = \text{id}_{\mathbb{C}} \oplus W_\mu \oplus W_\mu,$$

where Q_+ and Q_- are two distinguished copies of the actual original-zeta quotient. Set

$$J([1]) = (1, 0, 0),$$

$$J([x]) = \begin{cases} (0, \bar{K}_b[x], 0), & x > 1, \\ (0, 0, \bar{K}_b[x^{-1}]), & 0 < x < 1. \end{cases} \quad (\text{MQ5.1})$$

This preserves the sign of $\log x$ as an explicit summand; it does not identify $[x]$ with $[x^{-1}]$.

The map J is injective, because the three source summands are disjoint and each nontrivial summand map is CW's injection composed, for V_- , with the bijection $[x] \mapsto [x^{-1}]$. It intertwines every positive real scaling:

$$J\theta_\mu = \widehat{W}_\mu J, \quad (\text{MQ5.2})$$

since x^μ stays in the same sign sector and $(x^\mu)^{-1} = (x^{-1})^\mu$.

The original multiplicative law is also recoverable without asserting that J is a homomorphism for an unstated product on Q . Let $\mathbf{v}_t = J([e^t])$, $t \in \mathbb{R}$. These are a linearly independent basis of $J(W_{\text{alg}})$. Define the transported product there by

$$\mathbf{v}_t \star \mathbf{v}_u = \mathbf{v}_{t+u}. \quad (\text{MQ5.3})$$

This is associative and commutative by addition of the original real logarithms, with unit $\mathbf{v}_0$. It retains transitions between sign sectors and every source product, and makes J an algebra isomorphism onto this explicitly supplied image algebra. No ordinary quotient multiplication on Q is assumed.

The full coefficientwise moving map

$$J \otimes \text{id} : \mathcal{P} \longrightarrow \widehat{Q} \otimes \mathbb{C}[\mathbb{Q}] \quad (\text{MQ5.4})$$

is injective and intertwines arithmetic Frobenius with $\widehat{W}_\mu \otimes \text{id}$. This includes every original q^r , whose source coefficient is [1].

132.7 MQ6. The actual pointwise moving map and every crossing

There is also a direct map on the original moving functions:

$$\boxed{\mathcal{J}_{\text{mov}}(F)(g, Y) = J(\text{Ev}_W(F)(g, Y))}. \quad (\text{MQ6.1})$$

It takes values in $\widehat{Q}$, is linear, and is injective because both J and Ev_W are injective.

The same formula with Ev_{rad} gives an injective map on the larger space in MQ4. Thus its separate coefficient and geometric actions are also defined there, even when they move a holomorphic diagonal term away from that diagonal.

For a source basis function $[x]q^r$, put

$$\eta(x, r, Y) = \log x - 2\pi r Y.$$

Its exact image is

$$\mathcal{J}_{\text{mov}}([x]\mathbf{u}_r)(g, Y) = e_r(g) \begin{cases} (0, \overline{K}_b[e^{\eta(x, r, Y)}], 0), & \eta(x, r, Y) > 0, \\ (1, 0, 0), & \eta(x, r, Y) = 0, \\ (0, 0, \overline{K}_b[e^{-\eta(x, r, Y)}]), & \eta(x, r, Y) < 0. \end{cases} \quad (\text{MQ6.2})$$

In particular the crossing height, whenever positive, is

$$Y = \frac{\log x}{2\pi r}.$$

The unit is retained at the crossing, and the two sides remain in different copies of Q . There is no positive dilation by zero.

The actual action on this function receiver is

$$(\mathfrak{F}_\mu^{\text{mov}} H)(g, Y) = \widehat{W}_\mu H(g, Y/\mu). \quad (\text{MQ6.3})$$

Equations (MQ2.1) and (MQ5.2) prove

$$\mathcal{J}_{\text{mov}} \mathfrak{F}_\mu = \mathfrak{F}_\mu^{\text{mov}} \mathcal{J}_{\text{mov}}.$$

On (MQ6.2), the complete identity is

$$\log(x^\mu) - 2\pi r Y = \mu(\log x - 2\pi r Y/\mu). \quad (\text{MQ6.4})$$

Its positive multiplier preserves the sign, takes the crossing to its correct new height, and applies W_μ in the corresponding quotient summand.

The coefficientwise map (MQ5.4) and the pointwise map (MQ6.1) are exactly related on their images, not identified without a comparison:

$$\mathcal{T} = \mathcal{J}_{\text{mov}} \circ (J \otimes \text{id})^{-1}. \quad (\text{MQ6.5})$$

The inverse of J on its image is unique by its injection; (MQ1.2) explicitly recovers the frequency coefficients from the moving function. Thus (MQ6.5) is a bijection of the two images intertwining their stated arithmetic actions. Formula (MQ6.2) evaluates it on every basis vector.

We do not assert differentiability of (MQ6.2) in an unstated topology on $\widehat{Q}$ at its crossings. The source's exact holomorphic observation structure has an explicit return. Define, on $J(W_{\text{alg}})$,

$$\tilde{\chi}_\lambda = \chi_\lambda J^{-1}.$$

On the three basis types in (MQ5.1), its values are respectively $1, x^\lambda, x^\lambda$, retaining the original $x < 1$ in the last case. Hence

$$\tilde{\chi}_\lambda \mathcal{J}_{\text{mov}}(F) = \chi_\lambda \text{Ev}_W(F). \quad (\text{MQ6.6})$$

Every source scalar holomorphic equation is therefore recovered exactly. This proves a relation of the specified character-observed objects, without silently substituting an unrelated distributional or Hilbert topology.

132.8 MQ7. The original q^r itself: apparent zero weights and their exact cancellation

Take $x = 1$. For $r > 0$, (MQ6.2) becomes

$$\mathcal{J}_{\text{mov}}(\mathbf{u}_r)(g, Y) = e_r(g)(0, 0, \overline{K}_b[\exp(2\pi r Y)]).$$

For $r < 0$ it is

$$e_r(g)(0, \overline{K}_b[\exp(-2\pi r Y)], 0),$$

and $r = 0$ gives $(1, 0, 0)$. Every nonzero rational frequency is retained, including its sign.

For either nonzero sign, the complete actual-zero jet in its designated summand is

$$e_r(g) \sqrt{\pi} \exp((\rho - \tfrac{1}{2})^2/4) \exp((\rho - \tfrac{1}{2})T_\rho/2 + T_\rho^2/4)(2\pi|r|Y)^\rho \exp(\log(2\pi|r|Y)T_\rho). \quad (\text{MQ7.1})$$

The 2π , $|r|$, height, Gaussian factor and every nilpotent coefficient are explicit. Its length is the actual m_ρ .

Pure geometric scaling $Y \mapsto pY$ multiplies (MQ7.1) by $p^\rho \exp((\log p)T_\rho)$. The coefficient action alone supplies the same multiplier. In the full arithmetic Frobenius, the geometric argument is instead Y/p , and the two factors cancel exactly:

$$p^\rho e^{(\log p)T_\rho} (2\pi|r|Y/p)^\rho e^{\log(2\pi|r|Y/p)T_\rho} = (2\pi|r|Y)^\rho e^{\log(2\pi|r|Y)T_\rho}. \quad (\text{MQ7.2})$$

The Gaussian factor commutes with these multiplications and remains unchanged. Thus the full moving image of each q^r is arithmetic-Frobenius invariant, exactly as the original source proves.

This explains precisely why a zero-jet power seen at fixed height cannot be declared a new arithmetic weight of q^r . The full operator also changes that height. For general constant W -coefficients, their nontrivial arithmetic action remains as in MQ2–MQ3; the original q^r factors themselves contribute the invariant representation. Neither conclusion fixes $\Re\rho$.

132.9 MQ8. Full left rational affine action, finite valuation frame, and the classical quotient

Retain the original covering action

$$\ell(a, b)(X_f, X_\infty, Y_f, Y_\infty) = (aX_f + b, aX_\infty + b, aY_f, aY_\infty), \quad a \in \mathbb{Q}_{>0}, \quad b \in \mathbb{Q}. \quad (\text{MQ8.1})$$

Define the actual covering functions

$$U_r(X, Y) = \psi_f(rX_f) \exp(2\pi i r X_\infty) [\exp(-2\pi r Y_\infty)] = \alpha(rX) [\exp(-2\pi r Y_\infty)]. \quad (\text{MQ8.2})$$

The standard global additive character satisfies $\psi_f(rb) \exp(2\pi i r b) = \alpha(rb) = 1$ for rational rb . Hence

$$\boxed{U_r \circ \ell(a, b) = U_{ar}.} \quad (\text{MQ8.3})$$

Both the finite phase and the archimedean phase are retained and cancel; dropping the finite phase would give an incorrect extra factor. More generally, a finite locally constant coefficient f transforms by

$$(fU_r) \circ \ell(a, b) = (f \circ L_f(a, b))U_{ar}. \quad (\text{MQ8.4})$$

Constant W -coefficients are unchanged by this geometric pullback.

For the larger radial functions the same calculation sends $(r, c) \mapsto (ar, ac)$, with no global rational-translation phase. It commutes with the arithmetic action in MQ4. On the coefficientwise receivers it acts on frequency labels, together with the stated pullback of finite coefficients; it does not introduce an unstated Mellin eigenvalue on the constant W -coefficient.

The original quotient's q^r is related to these covering functions by a precise frame-dependent frequency. On the classical preimage suppose Y_f is a finite idele. Define its positive rational valuation representative and retained unit part by

$$c(Y_f) = \prod_p p^{v_p(Y_f)} \in \mathbb{Q}_{>0}, \quad Y_f = c(Y_f)u_f, \quad u_f \in \widehat{\mathbb{Z}}^*. \quad (\text{MQ8.5})$$

Only finitely many factors are different from one. The source's quotient by finite units and left positive rationals gives its classical coordinates

$$g = [X/c(Y_f)] \in G, \quad Y = Y_\infty/c(Y_f). \quad (\text{MQ8.6})$$

These formulas can also be proved directly: multiplication by $c(Y_f)^{-1}$ sends the finite Y valuations to zero, finite units then send $Y_f/c(Y_f)$ to one, and diagonal rational translations identify $X/c(Y_f)$ modulo $\mathbb{Q}$.

The pullback of the original quotient function is therefore

$$q^r(g, Y) = \alpha\left(\frac{rX}{c(Y_f)}\right) [\exp(-2\pi r Y_\infty/c(Y_f))] = U_{r/c(Y_f)}(X, Y). \quad (\text{MQ8.7})$$

Under (MQ8.1),

$$c(aY_f) = ac(Y_f), \quad \left[\frac{aX+b}{ac(Y_f)}\right] = \left[\frac{X}{c(Y_f)}\right], \quad \frac{aY_\infty}{ac(Y_f)} = \frac{Y_\infty}{c(Y_f)}.$$

The middle identity holds because $b/(ac(Y_f)) \in \mathbb{Q}$. Equivalently,

$$U_{r/(ac(Y_f))} \circ \ell(a, b) = U_{r/c(Y_f)}. \quad (\text{MQ8.8})$$

Thus the original quotient q^r is invariant under the covering gauge action, whereas a fixed-frequency cover function U_r changes frequency as in (MQ8.3). These are related by the explicit map (MQ8.7). On the representative with $c(Y_f) = 1$, U_r restricts to q^r .

The valuation representative is retained as part of this comparison. No assertion is made that it extends as a locally constant function on all of $\mathbb{A}_f$, or across $Y_f = 0$. The idele topology makes it locally constant on finite ideles; the subspace topology inherited from additive $\mathbb{A}_f$ need not do so. Equations (MQ8.5)–(MQ8.8) are pointwise identities on the stated classical preimage and identify the exact quotient maps. They do not turn the original classical orbit into the finite boundary $Y_f = 0$.

For completeness there is a different, explicitly defined self-map of the classical-coordinate product:

$$A_{a,b}(g, Y) = (ag + \iota_\infty(b), aY), \quad a \in \mathbb{Q}_{>0}, \quad b \in \mathbb{R}.$$

Its pullback satisfies

$$A_{a,b}^* q^r = \exp(2\pi i r b) q^{ar}. \quad (\text{MQ8.9})$$

The calculation uses only (MQ0.2). This coordinate self-map is not the diagonal covering gauge action (MQ8.1), which also rescales the finite valuation frame. Equation (MQ8.7) supplies their precise difference. The phases in (MQ8.9) compose consistently: $A_{a,b}A_{a',b'} = A_{aa',ab'+b}$, and pullbacks compose in the reverse order.

132.10 MQ9. Result and source-reading record

The established objects are:

1. An injective geometric evaluation of the complete finite algebra $W[q^r]$, with an explicit Fourier return for every rational frequency.

2. An injective moving coefficient map from $V_+ \otimes \mathbb{C}[\mathbb{Q}]$ to $Q \otimes \mathbb{C}[\mathbb{Q}]$, with the complete original-zero multiplicities and all prime actions.
3. An exact larger radial-frequency representation in which separate geometric and coefficient scalings are retained and the original holomorphic subalgebra is calculated, not assumed.
4. A faithful extension of CW to all W -coefficients using $\mathbb{C} \oplus Q_+ \oplus Q_-$, and a pointwise map of the original moving functions retaining every crossing through [1].
5. The exact comparison between covering U_r , the finite valuation frame, and the quotient's invariant q^r .

These are actual representation and geometric-evaluation maps. The displayed source multiplication is retained by its specified transported image law; ordinary quotient multiplication, a positive metric and a full purity theorem are not inferred. On the proved image, nonzero q -frequencies supply invariant factors for the full arithmetic Frobenius, while their fixed-height zero-jet powers have the explicitly calculated cancellation (MQ7.2).

Read in this derivation:

- `CORPUS_AND_OPERATION_RULES.md` completely.
- The complete preserved user passages USR-9ad1c0a2d09dba92, USR-f55d16feb948d8c2 and USR-6152e3bc6302258c.
- `CC_W_MELLIN_INDEPENDENT.md` CW0–CW7, whose complete proofs provide the original quotient injection, finite-jet interpolation and label functor.
- Original author `thecurve_K.tex`: §5.1 Lemma `adelicomp` and its proof; §5.2 equations `actionpq`, `algspacetiota` and Lemma `Pdescrofgamma`; §5.4 lines 1634–1708, including `Jdefn1`, the complete classical right action, `classorb1` and `proetcov`; §7.1 lines 2560–2626, including `holom`, `holombis`, `holom1`, `frobarith`, `holom2`, `functionq` and `perfectoid`. These are bounded reading passages, not a whole-paper reading claim.

The source authors are Alain Connes and Caterina Consani. The original-zeta count used by CW3 is due to Elchin Hasanlizade, Quanli Shen and Peng-Jie Wong, Counting zeros of the Riemann zeta function, arXiv:2107.06506v1, Corollary 1.2; its full constants and proof-use record remain in CW3. The MQ comparisons and all claims about their exact receiving maps are derived above, without a historical novelty claim.

Every linear map here extends to the programme's already specified labelled receiver by

$$(v, \ell) \longmapsto (f(v), \ell).$$

CW7 proves this functor preserves its receiving operations and labels, including zero amplitudes. Applying it to each stated linear map retains every support label. Neither a source τ nor a support label is assigned the numerical coordinate of a W -basis element, a Fourier frequency, or a zeta jet.

133 The exact covering algebra, its original q-descent, and the finite-boundary section

Independent derivation OQ0–OQ10, 24 September 2026.

The finite-boundary restriction of the explicit rational-frequency algebra on the adelic covering space, with the original W_{alg} and locally constant finite coefficients, has an equivariant global section. Its algebraic group-cochain connecting maps vanish. The original source q^r on the classical quotient is a frame-dependent invariant family of these covering functions, not one fixed-frequency function on the full covering space. OQ10 proves that exact descent map and its domain. The vanishing calculation is not asserted for all geometric sheaf cohomology or silently transferred to a boundary extension of that invariant family.

133.1 OQ0. Original source, current definitions, and coordinates

Human source: Alain Connes and Caterina Consani, *The Riemann–Roch strategy: Complex lift of the Scaling Site*, arXiv:1805.10501v1, original author thecurve_K.tex, §7.1, equations holom, holombis, holom1, holom2, Propositions frobarith and functionq, and Remark perfectoid. These passages were read directly. The source uses the algebraic first approximation

$$W_{\text{alg}} = \mathbb{C}[\mathbb{R}_{>0}^\times], \quad [x][y] = [xy], \quad \theta_\mu([x]) = [x^\mu], \quad \chi_\lambda([x]) = x^\lambda \quad (\mu, \lambda > 0). \quad (\text{OQ0.1})$$

Elements of W_{alg} are finite complex linear combinations of the displayed formal basis elements. The source’s Teichmüller notation $\tau(x)$ is not the user’s primitive Z_1/τ . We use only $[x]$ here.

The original moving function is

$$q(X + iY) = [e^{-2\pi Y}]e^{2\pi i X}, \quad (\text{OQ0.2})$$

on the original classical orbit, with its rational powers and coefficient-character holomorphy. The source proposes the ring $W[q^r]$, $r \in \mathbb{Q}$, in Remark perfectoid; it does not identify that algebraic ring with every completion or all sheaf cohomology.

The current corpus and operation rules and B1–B5 of the source-operation proofs were read for this derivation. The complete user passages USR-9ad1c0a2d09dba92, USR-f55d16feb948d8c2, and USR-6152e3bc6302258c were read in the preceding finite-boundary derivation and remain controlling. All arithmetic operations below belong to the already reconstructed arithmetic and original complex receiving algebra. No addition, parity, numerical metric, or coordinate is assigned to primitive τ .

Write $F = \mathbb{A}_f$, $E = F_X \times F_Y$, $Z = F_X \times \{0\}$, and $\Omega^+ = \mathbb{R}_{X_\infty} \times \mathbb{R}_{>0, Y_\infty}$. The algebra will first be realized on $E \times \Omega^+$. To give the action of all $P(\mathbb{Q})$, including $a < 0$, we then extend its same formulas to

$$\Omega^\times = \mathbb{R}_{X_\infty} \times \mathbb{R}_{Y_\infty}^\times. \quad (\text{OQ0.3})$$

Negative rational a exchanges the two halves; it is not claimed to preserve Ω^+ . Positive rational a preserves each half. Both the two-sided extension and this restriction are retained.

133.2 OQ1. The global additive character and the actual rational powers

Use the additive character

$$\psi(X_f, X_\infty) = \psi_f(X_f)e^{2\pi i X_\infty} \quad \text{on } \mathbb{A}_{\mathbb{Q}}/\mathbb{Q}. \quad (\text{OQ1.1})$$

One explicit convention is

$$\psi_f(X_f) = \exp\left(-2\pi i \sum_p \{X_p\}_p\right), \quad (\text{OQ1.2})$$

where $\{X_p\}_p$ is the finite negative-power part of the p -adic expansion, chosen in $\mathbb{Z}[1/p] \cap [0, 1)$. All but finitely many terms vanish. The identity $\{x+y\}_p - \{x\}_p - \{y\}_p \in \mathbb{Z}$ proves the character law. Each local factor is trivial on $\mathbb{Z}_p$; the restricted product makes ψ_f continuous and locally constant.

For $b \in \mathbb{Q}$, the rational number

$$b - \sum_p \{b\}_p$$

is integral at every finite prime: at a fixed p , its first subtraction leaves a p -adic integer and all other summands have denominator prime to p . A rational integral at every prime is an integer. Therefore

$$\psi_f(b)e^{2\pi ib} = 1. \quad (\text{OQ1.3})$$

This fixes the sign convention and proves triviality on diagonal rational translations. The same formula applies to rb for every rational r .

For each $r \in \mathbb{Q}$, define the actual W_{alg} -valued function

$$\boxed{q_r(X_f, Y_f, X_\infty, Y_\infty) = \psi_f(rX_f)e^{2\pi irX_\infty} [e^{-2\pi rY_\infty}].} \quad (\text{OQ1.4})$$

The finite coordinate Y_f remains present in the domain; this particular function is independent of it. Formula (OQ1.4) agrees with original q^r on the finite-unit frame $Y_f \in \widehat{\mathbb{Z}}^*$, $Y_\infty > 0$. On the complete classical preimage the original pullback is instead $q_{r/c}$, with the exact frame c proved in OQ10. There is no choice of an r -th root or logarithm of a complex phase: the global character defines the covering rational power directly. Multiplication gives

$$q_r q_s = q_{r+s}, \quad q_0 = [1], \quad q_r^{-1} = q_{-r}. \quad (\text{OQ1.5})$$

These use the original factors in (OQ1.4), not a completed or rescaled zeta function.

133.3 OQ2. The exact algebra presentation and its injective realization

Let $L_E = \text{LC}(E, \mathbb{C})$, with unrestricted global locally constant functions and pointwise multiplication. This is a unital function algebra; no compact-support condition is imposed. Let $\mathbb{C}[\mathbb{Q}]$ have basis U_r , $r \in \mathbb{Q}$, with $U_r U_s = U_{r+s}$. Define the algebraic tensor product

$$A_E = L_E \otimes_{\mathbb{C}} W_{\text{alg}} \otimes_{\mathbb{C}} \mathbb{C}[\mathbb{Q}]. \quad (\text{OQ2.1})$$

Its elements have finite sums of the unique form

$$F = \sum_{r \in S} \sum_{x \in S_r} f_{x,r}[x] U_r, \quad (\text{OQ2.2})$$

where the finite sets $S \subset \mathbb{Q}$ and $S_r \subset \mathbb{R}_{>0}$ contain distinct indices, and $f_{x,r} \in L_E$. The formal tensor presentation has the realized map

$$\text{ev}(f[x]U_r) = f(X_f, Y_f)\psi_f(rX_f)e^{2\pi irX_\infty} [xe^{-2\pi rY_\infty}]. \quad (\text{OQ2.3})$$

Its multiplicativity follows from the character law and (OQ1.5).

This map is injective. Suppose the image of (OQ2.2) vanishes. Apply every χ_λ , $\lambda > 0$, and fix a finite point (X_f, Y_f) and $Y_\infty > 0$. The resulting identity for all $X_\infty \in \mathbb{R}$ is

$$\sum_{r \in S} e^{2\pi irX_\infty} \psi_f(rX_f) e^{-2\pi rY_\infty} \left(\sum_{x \in S_r} f_{x,r}(X_f, Y_f) x^\lambda \right) = 0. \quad (\text{OQ2.4})$$

Distinct r give linearly independent exponential functions. For a direct proof, differentiate at any X_∞ in orders $0, \dots, |S| - 1$; the matrix with entries $(2\pi ir)^j$ is a Vandermonde matrix with

nonzero determinant. Hence every parenthesized sum vanishes for every $\lambda > 0$. For a fixed r , evaluate at $\lambda = 1, \dots, d$, where $d = |S_r|$. The matrix (x^λ) has determinant

$$\left(\prod_{x \in S_r} x \right) \prod_{x < x' \text{ in a fixed ordering}} (x' - x) \neq 0. \quad (\text{OQ2.5})$$

Thus every coefficient $f_{x,r}(X_f, Y_f)$ is zero. The finite point was arbitrary, so all coefficients vanish. This also proves that the complete character family faithfully observes this exact moving-function algebra.

No dependence of $f_{x,r}$ on X_∞, Y_∞ was allowed in (OQ2.1); introducing such coefficients would change the presentation and require another independence argument. No completion or infinite Fourier sum is asserted here.

133.4 OQ3. Original coefficient-character holomorphy

The scalar character realization is

$$\chi_\lambda \text{ev}(f[x]U_r) = f(X_f, Y_f) \psi_f(rX_f) x^\lambda e^{2\pi i r X_\infty} e^{-2\pi r \lambda Y_\infty}. \quad (\text{OQ3.1})$$

The original operator is

$$L_\lambda = Y_\infty(\lambda \partial_{X_\infty} + i \partial_{Y_\infty}). \quad (\text{OQ3.2})$$

Its two terms on (OQ3.1) have coefficients $2\pi i r \lambda Y_\infty$ and $-2\pi i r \lambda Y_\infty$, so they cancel exactly:

$$L_\lambda \chi_\lambda \text{ev}(F) = 0 \quad (F \in A_E, \lambda > 0). \quad (\text{OQ3.3})$$

This proves the extension is holomorphic in the exact source sense on every archimedean leaf. Locally constant finite coefficients introduce no archimedean derivatives. The same calculation holds on $Y_\infty < 0$; no growth or boundedness assertion on that half is required.

This proves inclusion in the corresponding algebraic holomorphic family. It does not assert that every holomorphic function in every proposed completed structure sheaf is a finite element of A_E .

133.5 OQ4. All rational affine pullbacks with the translation factors retained

Let $g = (a, b) \in P(\mathbb{Q}) = \mathbb{Q}^\times \ltimes \mathbb{Q}$. Its source multiplication is

$$(a, b)(a', b') = (aa', ab' + b).$$

The action on both finite and infinite coordinates is

$$\ell_g(X_f, Y_f, X_\infty, Y_\infty) = (aX_f + b, aY_f, aX_\infty + b, aY_\infty). \quad (\text{OQ4.1})$$

For all $a \neq 0$ this is an automorphism of $E \times \Omega^\times$. Direct substitution gives

$$q_r \circ \ell_g = \psi_f(rb) e^{2\pi i r b} q_{ar} = q_{ar}, \quad (\text{OQ4.2})$$

where the first equality retains both rational-translation factors and the second uses (OQ1.3). The induced exact algebra automorphism is

$$\boxed{\sigma_g(f[x]U_r) = (f \circ L_f(a, b))[x]U_{ar}}, \quad L_f(a, b)(X_f, Y_f) = (aX_f + b, aY_f). \quad (\text{OQ4.3})$$

Its realization is pullback by the original source action; no coordinate change or scalar factor is inserted.

Pullback is contravariant:

$$\sigma_g \sigma_h = \sigma_{hg}. \quad (\text{OQ4.4})$$

Indeed $f \circ L_h \circ L_g = f \circ L_{hg}$, and the rational frequency becomes $a_h a_g r$. Therefore these formulas give a right $P(\mathbb{Q})$ -action, or a left action $g \cdot F = \sigma_{g^{-1}} F$. Both conventions are explicit. On the original upper half only the positive subgroup preserves the domain.

Holomorphy remains compatible: $\partial_X + i\lambda^{-1}\partial_Y$ is rescaled by a , while Y contributes a^{-1} when comparing pullbacks. In the full form the exact identity is

$$L_\lambda P_\infty(a, b) = P_\infty(a, b) L_\lambda. \quad (\text{OQ4.5})$$

This follows by substituting $Y' = aY$ into the displayed operator, and holds for both signs of a on $\Omega^\times$.

133.6 OQ5. Right arithmetic Frobenius and its complete character action

Use the original

$$\mathfrak{F}_\mu = \theta_\mu R(\mu^{-1}), \quad \mu > 0,$$

where the geometric right action is $Y_\infty \mapsto Y_\infty/\mu$, with X_∞ and finite coordinates fixed. Its exact effect is

$$\mathfrak{F}_\mu q_r = \psi_f(r X_f) e^{2\pi i r X_\infty} \theta_\mu([e^{-2\pi r Y_\infty/\mu}]) = q_r. \quad (\text{OQ5.1})$$

For constant coefficients, $\theta_\mu[x] = [x^\mu]$. Thus

$$\boxed{\mathfrak{F}_\mu(f[x]U_r) = f[x^\mu]U_r.} \quad (\text{OQ5.2})$$

These are algebra automorphisms, with $\mathfrak{F}_\mu \mathfrak{F}_\nu = \mathfrak{F}_{\mu\nu}$. They commute with all rational affine pullbacks: one changes x by the coefficient power, the other changes r and finite arguments, and the displayed operations commute.

At every character, the original formula remains

$$\chi_\lambda \text{ev}(\mathfrak{F}_\mu F) = S_{\mu^{-1}} \chi_{\lambda\mu} \text{ev}(F), \quad S_{\mu^{-1}} h(X_\infty, Y_\infty) = h(X_\infty, Y_\infty/\mu). \quad (\text{OQ5.3})$$

For a generator, the right side has coefficient $x^{\lambda\mu}$ and exponential $e^{-2\pi r \lambda\mu(Y_\infty/\mu)}$, proving equality with every factor retained. This is the exact source coefficient-character Frobenius, not an independently chosen action on a spectral quotient.

133.7 OQ6. Boundary restriction and its global equivariant algebra section

Define the boundary algebra

$$A_Z = \text{LC}(F_X, \mathbb{C}) \otimes W_{\text{alg}} \otimes \mathbb{C}[\mathbb{Q}], \quad (\text{OQ6.1})$$

with the same moving- q_r realization restricted to $Y_f = 0$. The proof in OQ2 applies verbatim with one finite coordinate, so this realization is injective.

Coefficient restriction and extension are

$$R : A_E \longrightarrow A_Z, \quad R(f[x]U_r) = f(X_f, 0)[x]U_r,$$

$$\boxed{S : A_Z \longrightarrow A_E, \quad S(k[x]U_r) = k(X_f)[x]U_r.} \quad (\text{OQ6.2})$$

They are algebra homomorphisms and $RS = I$. The extension uses no choice of a cutoff, Haar average or transverse measure. It is independent of Y_f , exactly as specified.

Both maps commute with every σ_g in OQ4:

$$R\sigma_g = \sigma_g^Z R, \quad \sigma_g S = S\sigma_g^Z, \quad (\text{OQ6.3})$$

because $(aY_f)|_{Y_f=0} = 0$, while $k(aX_f + b)$ is still independent of Y_f . Both commute with every arithmetic Frobenius in OQ5 because neither changes the coefficient parameter x or frequency r .

Let $I_Z^E = \{f \in L_E : f|_Z = 0\}$. Injectivity of the algebra presentations gives

$$\ker R = I_Z^E \otimes W_{\text{alg}} \otimes \mathbb{C}[\mathbb{Q}]. \quad (\text{OQ6.4})$$

There is an exact, equivariantly split sequence of global algebras and their underlying complex vector spaces:

$$0 \longrightarrow \ker R \longrightarrow A_E \xrightarrow{R} A_Z \longrightarrow 0. \quad (\text{OQ6.5})$$

Its additive decomposition is

$$F = (F - SRF) + SRF, \quad A_E = \ker R \oplus S(A_Z). \quad (\text{OQ6.6})$$

The projection SR is an equivariant idempotent. This is not an assertion that A_E is a direct product of rings: mixed products between the kernel ideal and section subalgebra need not vanish.

The unrestricted global locally constant coefficient domain matters. If one separately imposes compact support in Y_f , the displayed $S(k)$ is not compactly supported unless zero. The current unital algebra contains the original constant coefficient 1, so compact support was not its definition. No claim about a different Schwartz completion is inferred.

133.8 OQ7. The global section is not a local splitting

Sheafify the same finite-sum presentations on open finite-transversal sets and archimedean open sets, retaining the W_{alg} and q_r generators. Independence is still valid on an archimedean interval: differentiating the exponential identity at a point gives the same Vandermonde argument. Boundary restriction is consequently locally exact, and on the finite coefficient sheaf it is exactly

$$0 \longrightarrow j_! \mathcal{L}_U \longrightarrow \mathcal{L}_E \longrightarrow i_* \mathcal{L}_Z \longrightarrow 0 \quad (\text{OQ7.1})$$

as proved in FB2, tensored with the displayed finite algebra generators.

There is no nonzero locally constant-coefficient section supported entirely on Z . A nonzero coefficient persists on a finite open neighborhood, and the injective moving- q presentation prevents cancellation from hiding it off Z . Therefore this sheaf restriction has no sheaf section, by the support argument of FB2.

The explicit global S in OQ6 does not contradict this. For example $S(1) = 1$. On an open finite-transversal set disjoint from Z , the boundary sheaf restricts 1 to zero, while the restriction of $S(1)$ is the nonzero constant function. Thus S fails the restriction-commutation requirement for a sheaf morphism. Its exact domain is the global module, and it is an equivariant global algebra map there.

This gives a proved map relating the local nonsplitting and global splitting. Neither is presented as a statement about the other category without its map.

133.9 OQ8. The group-cochain connecting map is identically zero

Let $G = P(\mathbb{Q})$, considered as a discrete group, act from the left by $g \cdot F = \sigma_{g^{-1}} F$. Alternatively use its positive subgroup on the upper half. The maps R, S are G -linear. All statements below also hold for the discrete direct product with the commuting arithmetic-Frobenius group $\mathbb{R}_{>0}^\times$, using (OQ5.2). No topology on group cochains is silently added.

For a G -module M , take inhomogeneous cochains

$$C^n(G, M) = \{c : G^n \rightarrow M\},$$

with differential

$$(dc)(g_1, \dots, g_{n+1}) = g_1 c(g_2, \dots, g_{n+1}) + \sum_{j=1}^n (-1)^j c(g_1, \dots, g_j g_{j+1}, \dots, g_{n+1}) + (-1)^{n+1} c(g_1, \dots, g_n). \quad (\text{OQ8.1})$$

For degree zero this means $dm(g) = gm - m$. Define S_n by applying S to every value of a cochain. Since S commutes with the action and is linear, each term in (OQ8.1) gives

$$dS_n = S_{n+1}d, \quad R_n S_n = I. \quad (\text{OQ8.2})$$

Hence (OQ6.5) gives a split short exact sequence of these entire cochain complexes.

For any boundary cocycle c , its explicitly chosen lift is $S_n c$. Its coboundary is

$$d(S_n c) = S_{n+1}(dc) = 0.$$

The definition of the connecting map therefore gives

$$\boxed{\delta_G^n : H^n(G, A_Z) \longrightarrow H^{n+1}(G, \ker R), \quad \delta_G^n = 0 \quad \text{for every } n \geq 0.} \quad (\text{OQ8.3})$$

This is an unconditional vanishing calculation for the exact covering algebra (OQ2.1), not a proposed vanishing premise. The decomposition of cochain complexes also gives

$$H^n(G, A_E) \cong H^n(G, \ker R) \oplus H^n(G, A_Z). \quad (\text{OQ8.4})$$

The cohomology groups on the right are not claimed to vanish.

The map (OQ8.3) is not the distribution-complex connecting map Δ_λ of FB6A.5. The latter uses a normal derivative between two supported-current complexes and has value λh on constants. Here the exact sequence is the covering function-algebra boundary restriction with section (OQ6.2); its group connecting map is zero. The sequences, modules and comparison maps have all been displayed.

133.10 OQ9. Entry into the finite-boundary distribution map and exact limits

The covering-algebra function restriction is linked directly to FB by finite-boundary insertion. At each coefficient character λ , define

$$\mathcal{T}_\lambda(F) = \sum_{r,x} f_{x,r}(X_f, 0) \psi_f(rX_f) x^\lambda e^{2\pi i r X_\infty} e^{-2\pi r \lambda Y_\infty} dX_f \otimes \delta_0(Y_f). \quad (\text{OQ9.1})$$

This is a smooth archimedean family of finite-adele distributions: compactly supported finite tests make every Haar integral finite, and the sum over x, r is finite. It is defined character by character; no integral of arbitrary moving basis elements in the uncompleted group algebra is asserted.

It is precisely restriction R , followed by the injective boundary-density insertion of FB1. Injectivity on boundary data follows from finite test-function separation and the Fourier–Vandermonde proof of OQ2. Thus its joint kernel for all $\lambda > 0$ is exactly $\ker R$. The original global section provides the exact return

$$RS = I$$

before insertion into distributions. This links the actual covering function algebra to the preceding distribution enlargement, with every coefficient and character retained. Its source-classical-orbit comparison is OQ10.

For positive rational affine pullback the finite insertion acquires the full factor $a = |a|_f^{-1}$ from FB4. The moving- q factors transform as proved in OQ4, so that factor is not cancelled by an omitted phase. For negative a , the finite factor is $|a|_\infty$; combining it with an archimedean derivative factor a^{-1} would leave $\text{sgn}(a)$. FB’s compensated chain-map assertion was on the positive branch and is not extended across this sign without an explicit orientation coefficient.

The construction proves a boundary-lift vanishing on the exact algebraic group-cochain sequence (OQ6.5). It does not identify that sequence with Deligne’s specialization sequence, nor identify its group cohomology with all geometric sheaf cohomology of the adelic quotient. The source q -descent comparison is proved next and retained rather than replaced by the covering algebra. Primitive Z_1/τ , all supported labels z_λ , the original arithmetic and original zeta function remain unchanged.

133.11 OQ10. Exact comparison with the original classical quotient

Source §5.4 identifies

$$\Gamma_{\mathbb{Q},\text{cl}} \cong G \times \mathbb{R}_{>0}, \quad G = \mathbb{A}_{\mathbb{Q}}/\mathbb{Q}. \quad (\text{OQ10.1})$$

Its selected representatives have $Y_f \in \widehat{\mathbb{Z}}^*$ and positive real Y_{∞} . On the full preimage of this orbit, Y_f is a finite idele and $Y_{\infty} \neq 0$. There is a unique

$$c = \text{sgn}(Y_{\infty}) \prod_p p^{v_p(Y_f)} \in \mathbb{Q}^{\times}, \quad u = Y_f/c \in \widehat{\mathbb{Z}}^*, \quad (\text{OQ10.2})$$

with $y = Y_{\infty}/c > 0$. The product is finite by the finite-idele condition. Existence is the displayed formula; uniqueness follows because a positive rational unit at every prime is one, with sign fixed by $Y_{\infty}/c > 0$.

The exact quotient-coordinate map is

$$\kappa(X_f, Y_f, X_{\infty}, Y_{\infty}) = (g = [(X_f, X_{\infty})/c] \in G, y = Y_{\infty}/c). \quad (\text{OQ10.3})$$

Right multiplication of Y_f by $\widehat{\mathbb{Z}}^*$ changes u , not c, g, y . Under left $(a, b) \in P(\mathbb{Q})$, the new frame is $c' = ac$. The new y equals y , and the new g is

$$[(aX + b)/(ac)] = [X/c + b/(ac)] = [X/c], \quad (\text{OQ10.4})$$

because $b/(ac)$ is diagonal rational. Thus κ is invariant under the actual quotient actions. Every orbit has a unit-frame representative obtained by $a = 1/c$, followed by the finite unit action. Two such representatives have the same G -class exactly when their X 's differ by a rational translation. This proves the set-level quotient comparison and its fibres.

The original source function is

$$q_{\text{source}}^r(g, y) = \psi(rg)[e^{-2\pi r y}].$$

Its exact pullback is

$$\boxed{\kappa^* q_{\text{source}}^r = q_{r/c}(X_f, Y_f, X_{\infty}, Y_{\infty})}. \quad (\text{OQ10.5})$$

Its invariance follows directly from the covering calculation:

$$q_{r/(ac)} \circ \ell(a, b) = q_{r/c}. \quad (\text{OQ10.6})$$

Thus fixedness under the actual left gauge action and the covering frequency change $r \mapsto ar$ describe two explicitly related maps. No single action has been assigned contradictory rules.

The original finite q -algebra has a complete algebraic descent description. Put $B = W_{\text{alg}} \otimes \mathbb{C}[\mathbb{Q}]$, and take

$$\mathfrak{A}_{\text{frame}} = \prod_{c \in \mathbb{Q}^{\times}} B. \quad (\text{OQ10.7})$$

Each component is a finite polynomial; there is no imposed common finite frequency set over all frames. Define $\sigma_a^{\text{freq}}([x]U_r) = [x]U_{ar}$, and

$$(\sigma_{a,b}^{\text{frame}} b)_c = \sigma_a^{\text{freq}}(b_{ac}). \quad (\text{OQ10.8})$$

The translation phase is one by OQ1.3. The exact mutually inverse algebra maps are

$$\begin{aligned} B &\longrightarrow \mathfrak{A}_{\text{frame}}^{P(\mathbb{Q})}, & F &\longmapsto (\sigma_{1/c}^{\text{freq}} F)_c, \\ \mathfrak{A}_{\text{frame}}^{P(\mathbb{Q})} &\longrightarrow B, & (b_c)_c &\longmapsto b_1. \end{aligned} \quad (\text{OQ10.9})$$

Substitution into OQ10.8 proves the first family invariant. Conversely invariance with $a = c$, evaluated at frame one, gives $b_1 = \sigma_c^{\text{freq}} b_c$, hence $b_c = \sigma_{1/c}^{\text{freq}} b_1$. Both maps commute with arithmetic Frobenius on W_{alg} . This is the exact all-frame source descent.

For $r \neq 0$, the pulled-back invariant family OQ10.5 is not a single element of the globally finite-frequency algebra A_E . Any element of A_E has one finite set S of archimedean frequencies. Equality with OQ10.5 on frame c requires $r/c \in S$, by the exponential independence in OQ2. The frequencies r/c for $c \in \mathbb{Q}^\times$ form an infinite set. The same argument applies to a nonconstant finite polynomial in source q : one nonzero frequency has an infinite frame orbit, and distinct frequencies at a fixed frame cannot cancel. Constant W_{alg} coefficients do lie in A_E .

This is a specific domain statement with the exact extension OQ10.7–OQ10.9. It is not an obstruction to a full source sheaf or a larger completed algebra. In particular OQ8’s connecting-map vanishing must not be transferred by identifying its A_E with the invariant frame-family algebra.

Topology must also be retained. The map $Y_f \mapsto c$ is locally constant for the finite-idele topology, with sign fixed on each archimedean half. It is not locally constant in the subspace topology inherited from all $\mathbb{A}_f$. For example $c_n = 1 + n!$ tends to 1 in $\mathbb{A}_f$: all terms are integral at every prime, and $n!$ tends to zero at each fixed finite set of primes. Each c_n is a finite idele. With $X = 0, Y_\infty = 1$, OQ10.3 gives $y_n = 1/(1 + n!)$, which does not tend to the value $y = 1$ at $Y_f = 1$. Thus κ , with target $G \times \mathbb{R}_{>0}$, is not continuous for that ambient subspace topology. No ambient continuous extension across $Y_f = 0$ follows from the set-level quotient formula.

The source-compatible outcome consists of two linked exact constructions. The original invariant $W[q^r]$ algebra has the all-frame realization OQ10.9. The separate fixed-frequency covering algebra has the global equivariant boundary section OQ6.2. Their common unit-frame restriction is explicit. The section of the second is not an asserted boundary extension of the first.

The positive-branch comparison was independently derived in MOVING_Q_ZETA_COMPARISON.md, MQ8.5–MQ8.8. That complete MQ8 passage was read and compared with OQ10; the additional full-sign frame, invariant product-algebra presentation, and finite-frequency exclusion are proved here. The topology counterexample was supplied by the coordinating calculation and its convergence and coordinate values are verified above.

133.12 Source-reading and provenance

The original author thecurve_K.tex has SHA256 da52001209424c19fb223905804a9eed1147c72331cf881aa0352bc759976061. Read directly for this calculation: §7.1 from its heading through Remark perfectoid; and §5.4 completely, including Jdefn1, classorb1, proetcov and Remark additivestruct. The original affine action and global character convention were read in the preceding FB derivation at §5.1–5.2, especially Lemma adelicomp and equation actionpq, and were recalculated in OQ1 and OQ4. This is bounded source coverage, not a whole-paper claim.

The current operation rules and B1–B5 were reread before constructing (OQ2.1). The user-originating programme seeks full global reconstruction and vanishing of the appropriate lifting map; OQ6 and OQ8 provide the displayed exact global section and its actual vanishing result. Connes and Consani retain credit for W , the coefficient characters, moving q^r , and arithmetic Frobenius. The finite-transversal extension, injectivity, exact global section, and cochain calculation are derived in this note. No historical novelty is asserted for a split-module connecting map being zero.

134 Source comparison, complete coefficients, and the retained boundary

24 September 2026. A continuation of the actual τ lifting calculation. The four complete proofs following this introduction are CGS, DCP, OMS and CCB. Earlier synthesis and primary-projector results are credited where used, rather than presented as new discoveries.

The user's notation remains Z_0 for absence, Z_1 for primitive presence τ without Z_2 parity, and Z_2 for the retained parity data on the specified integer layer. Addition on τ was retracted. None is reinstated here. The complete verbatim corpus and its corrections govern the construction; `CORPUS_AND_OPERATION_RULES.md` records the relevant reading for each receiving calculation. The full reconstructed arithmetic precedes the coefficient operations in this document.

134.1 The concrete source-to-coefficient construction

CGS proves the gluing classification for every module sheaf on the already constructed source receiving space. Its exact datum is

$$(\mathcal{G}, A_+, A_-, r : A_+ \longrightarrow \Gamma(U, \mathcal{G})), \quad U = \operatorname{Spec} \mathbb{Z}.$$

The complete supported complex is

$$A_-^\bullet \oplus \operatorname{Fib} (A_+^\bullet \longrightarrow R\Gamma(U, \mathcal{G}^\bullet)).$$

The inverse functor, all restrictions, the cone differential, and its connecting maps are proved in CGS1–CGS8. This is the concrete instance, with the additional receiving idempotents, of the sheaf gluing construction recalled by Pierre Deligne in *La conjecture de Weil. II*, §3.4.8. His §3.6 weight argument remains a further geometric calculation; the gluing classification does not assume its weight inequalities.

DCP inserts an actual Connes–Consani coefficient sheaf into that datum. It doubles the source chart along its entire arithmetic open and constructs

$$f : X^{\operatorname{dbl}} \longrightarrow Y = \{x_+, \eta, x_-\}, \quad f^{-1}(\eta) = \operatorname{Spec} \mathbb{Z}, \quad f^{-1}(x_\pm) = \{\mathfrak{m}_\pm\}.$$

The arithmetic primes remain distinct in this fibre. The sheaf $\mathcal{N} = f^{-1}\Omega$ has chart spaces $V_\pm = S_{00}^{\operatorname{even}} \oplus \mathbb{C}^2$, overlap A , and the original restrictions

$$r_+h(u) = 2 \sum_{n \geq 1} h(nu), \quad r_-h(u) = 2u^{-1} \sum_{n \geq 1} h(n/u).$$

Their complete domains, convergence, Fourier comparison and endpoint coordinates occur in DCP3. The human construction is Alain Connes and Caterina Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, §5. The original author TeX, lines 1376–1667, was read in full for this receiving calculation.

The inverse-image coefficient alone has the same action for source τ and integer 1. DCP8 constructs the larger coefficient sheaf

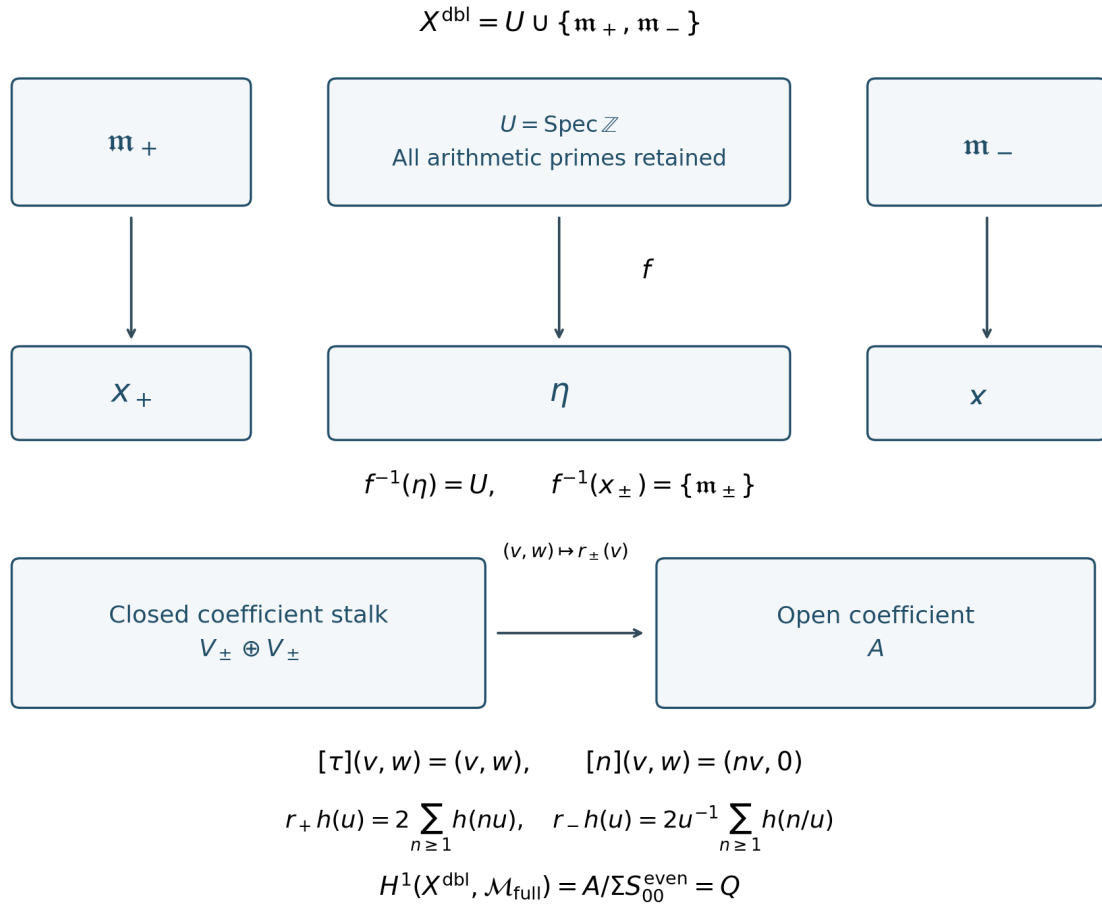
$$\mathcal{M}_{\operatorname{full}} = \mathcal{R}^{\operatorname{dbl}} \otimes_{\mathbb{Z}} \mathcal{N} \cong \mathcal{N} \oplus (i_+)_* V_+ \oplus (i_-)_* V_-.$$

At a closed stalk this has the faithful source action

$$[\tau](v, w) = (v, w), \quad [n](v, w) = (nv, 0).$$

The bracketed operators act on constructed receiving modules. They are not equations assigning numerical coordinates or parity to primitive τ . The unital scalar-extension map and the

The source distinction survives the coefficient comparison



DCP1-DCP12 prove every map. The extra closed copies remain in H^0 .
The brackets denote receiving actions; primitive $\mathbb{Z}_i/\tau$ has no addition or parity.
Received sheaf: Alain Connes and Caterina Consani, arXiv:0903.2024v3, §5.

Figure 48: The complete source coefficient comparison

distinguished source-integer map are both displayed and proved in DCP8; their difference is retained.

Figure 1. Exact sheaf and cochain diagram. Every arithmetic prime lies in the displayed fibre U . The second closed-stalk components distinguish the source actions of τ and integer 1; they remain in degree zero. The degree-one comparison is the original zeta quotient. Complete proofs: CGS1–CGS9 and DCP1–DCP12. Human source for the received coefficient complex: Connes–Consani, §5, cited above. This is a diagram of maps, not a metric or a picture of primitive τ .

134.2 Actual summation, every zero jet, and the original factors

OMS receives the existing S/SSI synthesis proof into the present quotient and checks its comparison with Ralf Meyer, *A spectral interpretation for the zeros of the Riemann zeta function*. It proves on the original domains

$$\mathcal{E}S_{00}^{\text{even}} = \{k \in \mathcal{A} : F_k^{(j)}(\rho) = 0 \text{ for every actual nontrivial zero } \rho, 0 \leq j < m_\rho\}.$$

The image is closed and has the explicitly constructed continuous source inverse. The full joint zero-jet map of the actual quotient Q is therefore injective. Its image is a proper dense subspace of the unrestricted product of jets, with a strictly stronger quotient topology. OMS5 proves that distinction, including the estimates that force it.

OMS7A also resolves the earlier adelic closure difference: $D_{\text{cl}} = 0$, and the original algebraic quotient is the actual Hausdorff quotient Q , with the identical quotient topology. The full adelic two-term complex consequently has $H^{-1} = \mathcal{B}_{\text{pr}}$, $H^0 = Q$, and its specific two-extension class vanishes. Both endpoint coordinates at every prime and the separately computed higher derived groups remain. This does not assert that the entire extension group vanishes.

Every original exceptional contribution remains in the inverse. In the centered convention used by OMS, it is the inverse Mellin transform of $F(s)/\zeta(s)$, with

$$\frac{f_F^{(2r)}(0)}{(2r)!} = \frac{F(-2r)}{\zeta'(-2r)}, \quad \int_0^\infty f_F(v) \frac{dv}{v} = -2F(0), \quad \int_0^\infty f_F(v) \log v \, dv = F(1).$$

The raw two-sign convention has its additional factor two throughout. The full Gamma and endpoint multiplier used in the division proof is displayed, evaluated at its exceptional points, and never substituted for the original zeta. OMS6 credits and transports the existing RZ primary projectors, retaining all nilpotents.

134.3 Completion must retain the operations it claims to carry

CCB constructs the exact completions of the original Connes–Consani moving-character complex. The ordinary smooth completion does not support a continuous extension of its radial homotopy; CCB2 proves this with an explicit convergent sequence. CCB3–CCB6 then construct complete spaces where that homotopy is continuous, preserving the entire original operator

$$y(\lambda D + i\partial_y),$$

all rational frequencies, both signs and all finite-valuation frames. These are specified receiving spaces, not counterexamples to the user’s complete arithmetic construction. The source is Connes–Consani, *The Riemann–Roch strategy: Complex lift of the Scaling Site*, §§5.4 and 7.1.

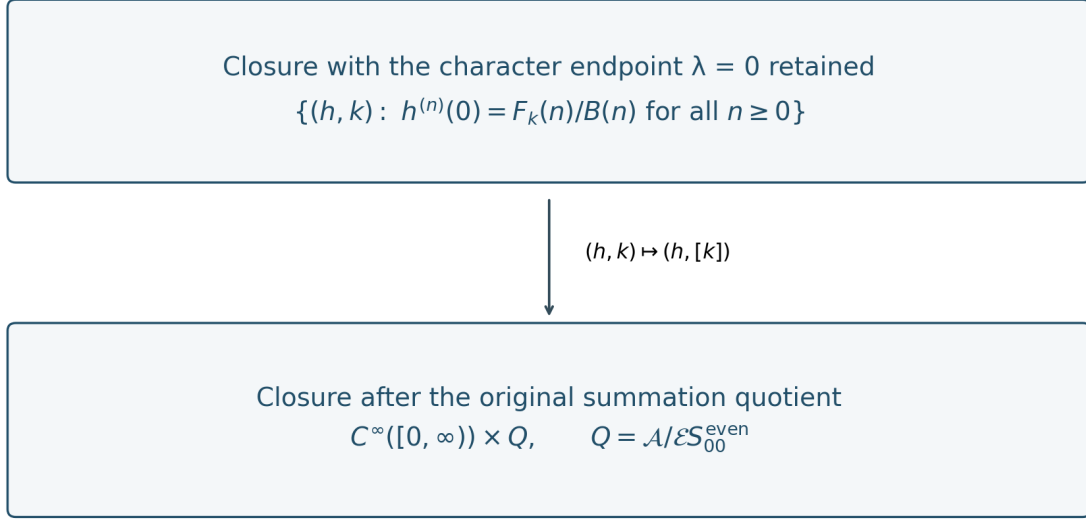
The later coefficient calculation in CCB also retains the original strong Mellin space before its quotient. For $x = e^a$, $a > 0$, its two coefficient observations are

$$\chi_\lambda[x] = e^{\lambda a}, \quad K_b[x] = W_a b, \quad b(u) = e^{-(\log u)^2},$$

Integer moments are retained before taking the quotient

$$\chi_\lambda[e^a] = e^{\lambda a}, \quad K_b[e^a] = W_a b, \quad a > 0$$

$$F_{K_b[e^a]}(s) = B(s)a^s, \quad B(s) = \sqrt{\pi} \exp((s - \frac{1}{2})^2/4)$$



The coefficient topology and the quotient are specified separately.

$$F_{\varepsilon f}(0) = -\frac{1}{2} \int_0^\infty f(v) \frac{dv}{v}$$

$$F_{\varepsilon f}(1) = \int_0^\infty f(v) \log v \, dv$$

CCB proves the full graph closures and every factor. This is a coefficient comparison, not τ arithmetic.
The endpoint $\lambda = 0$ is a character parameter; it is not identified with Z_0 , Z_1 , or primitive τ .

Figure 49: The integer moment relations before taking the quotient

$$F_{K_b[x]}(s) = B(s)a^s, \quad B(s) = \sqrt{\pi} \exp((s - \frac{1}{2})^2/4).$$

When the coefficient-character endpoint $\lambda = 0$ is retained, the exact relation is

$$\partial_\lambda^n \chi(v)(0) = \frac{F_{K_b v}(n)}{B(n)} \quad (n \geq 0).$$

This endpoint is a parameter of a receiving character, not primitive τ . CCB proves the corresponding whole-space closure statement and its exact change after passage to Q , including the pole contribution at the integer argument one. These maps specify which data a proposed completion carries; they are not a purity assumption.

Figure 2. The displayed derivatives and Mellin values are equal on the original coefficient image. The graph closure before quotienting retains every such equality. The subsequent projection to the original quotient is a separate operation with its own calculated closure. The proof keeps $B(n)$, $\zeta(0)$, the pole at one, and the integer moments. Complete derivation: CCB, final coefficient-closure sections. Original coefficient characters: Connes–Consani, *Complex lift*, cited above.

134.4 The remaining weight calculation is stated on these actual maps

The constructed both-support row is

$$0 \longrightarrow Q \xrightarrow{q \mapsto (q, q)} Q \oplus Q \xrightarrow{(q_+, q_-) \mapsto q_+ - q_-} Q \longrightarrow 0.$$

Its mirror-equivariant section is $q \mapsto \frac{1}{2}(q, -q)$, with the complete original dilation and orientation sign proved in DCP10–DCP11. At an actual zero its action is

$$p^\rho \sum_{j=0}^{m_\rho-1} \frac{(\log p)^j}{j!} N_\rho^j.$$

This is an explicit lifting result on the received source coefficients. It does not derive $\operatorname{Re} \rho = 1/2$. In Deligne’s §3.6, the two weight ranges follow from proper base change, inertia, duality and the displayed Tate twists. The present calculations have not proved that geometric weight comparison for the full τ programme. The goal remains active; none of the receiver splittings is marked as its completion.

All four complete derivations follow. Their source files, original author sources, earlier proof dependencies, reproducible diagrams and exact reading ledger accompany this reader. The complete user corpus remains private and is referenced by its stable passage identifiers.

135 Every coefficient sheaf on the current source: exact gluing and supported cohomology

24 September 2026. Independent continuation of `SOURCE_SHEAF_SPECIALIZATION.md`, SS0–SS9. The coefficient comparison left in SS7 is now an explicit object with inverse construction, all morphisms, its complete restriction defect and its derived supported complex.

135.1 CGS0. Source and operation prerequisites

The current `CORPUS_AND_OPERATION_RULES.md` (private source record), including the global-argument and classical-boundary corrections, was read for this continuation. The source remains the current B1–B5, with the verbatim U01–U18 and whole-history arguments retained as in ST0, AC0 and SS0. In particular Z_1/τ has no source addition, coordinate or Z_2 parity. The complete reconstructed integer layer is already present before any arithmetic receiver below is used. Nothing here changes those definitions or treats a coefficient choice as a counterexample to the user’s complete programme.

The proved source receiving space is

$$X = U \cup \{\mathfrak{m}\}, \quad U = X_{\mathbb{Z}} \simeq \operatorname{Spec} \mathbb{Z},$$

where the only open neighbourhood of $\mathfrak{m}$ is X . Keep

$$\mathcal{R} = j_* \mathcal{O}_U \times i_* \mathbb{Z}, \quad R_0 = \mathcal{R}(X) = \mathbb{Z}[\epsilon]/(\epsilon^2 - \epsilon) \cong \mathbb{Z} \times \mathbb{Z}. \quad (\text{CGS0.1})$$

Here $\epsilon = [1]$ is the receiving idempotent, not primitive τ . The original source integer n maps to $n\epsilon$. The bracketed image of τ is the formal global unit of this receiver; its new additive operations are not pulled back to the source. All sheaves and complexes in this document have these explicitly constructed receiving modules as coefficients.

The letter i in the sheaf-support calculations denotes the closed topological inclusion equipped, when module rings are needed, with the full stalk ring $R_0 = i^{-1}\mathcal{R}$. This matters: the selected residue ring $\mathbb{Z}$ through second projection is a different coefficient functor, recorded in CGS7. We do not replace the full stalk by that residue.

135.2 CGS1. A complete classification, including inverse functors

The human-source gluing mechanism is explicitly recalled by Pierre Deligne in *La conjecture de Weil. II*, §3.4.8, printed p.209: an open sheaf, a closed sheaf, and its specialization map into the restriction of the open direct image describe the whole sheaf. The calculation here specializes that recalled equivalence to the proved ringed space CGS0.1 and separates its two receiving idempotents. It is not presented as a new discovery of the general sheaf-gluing equivalence. CGS10 gives the exact specialization target and source record.

Define a gluing datum to consist of

$$(\mathcal{G}, A_+, A_-, r), \quad r : A_+ \longrightarrow \Gamma(U, \mathcal{G}), \quad (\text{CGS1.1})$$

where $\mathcal{G}$ is a sheaf of $\mathcal{O}_U$ -modules, A_+ and A_- are abelian groups, and r is $\mathbb{Z}$ -linear. No surjectivity, injectivity, topology on these groups or purity is imposed.

Every sheaf $\mathcal{M}$ of $\mathcal{R}$ -modules supplies exactly such data:

$$\mathcal{G} = j^{-1}\mathcal{M}, \quad A = \mathcal{M}(X) = \mathcal{M}_{\mathfrak{m}}, \quad A_+ = \epsilon A, \quad A_- = (1 - \epsilon)A. \quad (\text{CGS1.2})$$

The restriction $A \rightarrow \Gamma(U, \mathcal{G})$ kills A_- , because $1 - \epsilon$ restricts to zero on U . Its restriction to A_+ is r . The decomposition $A = A_+ \oplus A_-$ is proved by the complementary idempotent identities, as in ST5.

Conversely construct a sheaf $\mathcal{M}(\mathcal{G}, A_+, A_-, r)$ as follows:

$$\mathcal{M}(X) = A_+ \oplus A_-, \quad \mathcal{M}(V) = \mathcal{G}(V) \quad (V \subseteq U), \quad (\text{CGS1.3})$$

and

$$\text{res}_{X,V}(a_+, a_-) = r(a_+)|_V. \quad (\text{CGS1.4})$$

The R_0 -module action on $A_+ \oplus A_-$ is componentwise: under $R_0 \cong \mathbb{Z} \times \mathbb{Z}$, $(m, n)(a_+, a_-) = (ma_+, na_-)$. On proper opens it is the given $\mathcal{O}_U$ -module action. The restrictions are module maps because $R_0 \rightarrow \mathcal{O}_U(V)$ is first projection followed by the integer localization, and r is $\mathbb{Z}$ -linear.

This is a sheaf. A cover of X contains X itself, so its gluing condition is automatic from the matching condition. A cover of a proper open is a cover inside U , where $\mathcal{G}$ is already a sheaf. These are all covers. Applying CGS1.2 to this construction recovers the given quadruple. Applying CGS1.3 to an original sheaf recovers all its sections and all restrictions. These are explicit inverse functors, not merely a correspondence between dimensions.

A morphism between two data is precisely

$$(g, a_+, a_-), \quad g : \mathcal{G} \rightarrow \mathcal{G}', \quad a_{\pm} : A_{\pm} \rightarrow A'_{\pm}, \quad (\Gamma(U, g))r = r'a_+. \quad (\text{CGS1.5})$$

It acts by $a_+ \oplus a_-$ on X and by g on proper opens. The displayed equation is exactly compatibility with restriction. Every module-sheaf morphism has this form because it commutes with ϵ . Composition is componentwise and preserves the displayed equation. This proves an equivalence of categories in both directions, including all maps.

135.3 CGS2. Exactness, kernels and cokernels

An exact sequence of these sheaves is exact on their A_+ , A_- , and open-sheaf components. Necessity follows by taking the stalk at $\mathfrak{m}$, then the two idempotents, and taking all stalks in U . These conditions are sufficient because they exhaust the stalks of X .

For a morphism CGS1.5, its kernel has data

$$(\ker g, \ker a_+, \ker a_-, r|_{\ker a_+}). \quad (\text{CGS2.1})$$

The last map takes values in $\Gamma(U, \ker g)$, because $(\Gamma g)r = r'a_+$ vanishes on $\ker a_+$, and a section in the kernel is exactly a section of the kernel sheaf.

Its cokernel has open sheaf $\text{coker } g$ and stalk groups $\text{coker } a_{\pm}$. The restriction map is induced by

$$A'_+ \xrightarrow{r'} \Gamma(U, \mathcal{G}') \longrightarrow \Gamma(U, \text{coker } g). \quad (\text{CGS2.2})$$

It kills $\text{im } a_+$ by the compatibility equation, hence descends. This is the cokernel datum by stalkwise verification. In particular $\Gamma(U, \text{coker } g)$ is not replaced by the cokernel of global sections unless that additional equality has actually been proved.

The same description gives images, finite limits and finite sums. It proves directly that the classified category has the abelian structure of the original module-sheaf category. No abelian category of primitive τ objects is assumed.

135.4 CGS3. The exact open and closed objects

The three standard sheaves have these complete data:

sheaf	$(\mathcal{G}, A_+, A_-, r)$	(CGS3.1)
$j_! \mathcal{G}$	$(\mathcal{G}, 0, 0, 0)$	
$j_* \mathcal{G}$	$(\mathcal{G}, \Gamma(U, \mathcal{G}), 0, \text{id})$	
$i_* A$	$(0, \epsilon A, (1 - \epsilon)A, 0).$	

For $j_!$, these formulas also follow directly from extension by zero: a nonzero section supported away from $\mathfrak{m}$ cannot extend by zero globally, since the closure in X of every nonempty subset of U contains $\mathfrak{m}$. Its value on X is therefore zero. It is the original sheaf on every proper open. The formulas for j_* and i_* follow from their actual section definitions.

For a general $\mathcal{M}$, the restriction comparison

$$\eta_{\mathcal{M}} : \mathcal{M} \longrightarrow j_*\mathcal{G} \quad (\text{CGS3.2})$$

is identity on U , and on X it is $(a_+, a_-) \mapsto r(a_+)$. Thus its *complete* support kernel and cokernel are

$$\ker \eta_{\mathcal{M}} = i_*(\ker r \oplus A_-), \quad \text{coker } \eta_{\mathcal{M}} = i_*(\text{coker } r). \quad (\text{CGS3.3})$$

In the first kernel, ϵ acts as identity on $\ker r$ and as zero on A_- ; in the cokernel it acts as identity. This retains the two source-idempotent types inside the same closed support. The exact sequence is

$$0 \rightarrow i_*(\ker r \oplus A_-) \rightarrow \mathcal{M} \xrightarrow{\eta_{\mathcal{M}}} j_*\mathcal{G} \rightarrow i_*(\text{coker } r) \rightarrow 0. \quad (\text{CGS3.4})$$

The source's automatic boundary component is i_*A_- , and it splits off:

$$\mathcal{M} \cong \mathcal{M}(\mathcal{G}, A_+, 0, r) \oplus i_*A_-. \quad (\text{CGS3.5})$$

The explicit maps are the two idempotent projections and their inclusions. The additional supported objects $\ker r$ and $\text{coker } r$ belong to the $\epsilon = 1$ component. Consequently the automatic $\epsilon = 0$ splitting does not erase them.

135.5 CGS4. The gluing extension is exactly the map r

There is another exact sequence whose maps are also fully determined:

$$0 \longrightarrow j_!\mathcal{G} \longrightarrow \mathcal{M} \longrightarrow i_*(A_+ \oplus A_-) \longrightarrow 0. \quad (\text{CGS4.1})$$

On U , its first map is identity and its last target is zero. At $\mathfrak{m}$, its first stalk is zero and its last map is identity. This proves exactness. The last map on global sections is also identity; no restriction lift is hidden there.

The sequence is the actual pullback of

$$0 \rightarrow j_!\mathcal{G} \rightarrow j_*\mathcal{G} \rightarrow i_*\Gamma(U, \mathcal{G}) \rightarrow 0 \quad (\text{CGS4.2})$$

along $i_*r : i_*(A_+ \oplus A_-) \rightarrow i_*\Gamma(U, \mathcal{G})$, where r kills A_- . Explicitly,

$$\mathcal{M} = j_*\mathcal{G} \times_{i_*\Gamma(U, \mathcal{G})} i_*(A_+ \oplus A_-). \quad (\text{CGS4.3})$$

On X the pairs are (g, a) with $g = r(a_+)$, hence are exactly $A_+ \oplus A_-$. On a proper open they are $\mathcal{G}(V)$. These identifications preserve every module action and restriction, proving the formula as a sheaf isomorphism.

The complete classification of extensions with these fixed endpoints is

$$\boxed{\text{Ext}_{\mathcal{R}}^1(i_*A, j_!\mathcal{G}) \cong \text{Hom}_{\mathbb{Z}}(\epsilon A, \Gamma(U, \mathcal{G}))}. \quad (\text{CGS4.4})$$

Indeed in any such extension, its open sheaf is canonically $\mathcal{G}$ and its closed stalk is canonically A , by exactness. The only remaining datum in CGS1 is r , which must kill $(1 - \epsilon)A$. Conversely CGS4.3 constructs the extension for every such map. An equivalence of extensions fixing the endpoints fixes those same open and stalk identifications, hence fixes r . Finally the Baer sum is addition of these maps: form the fibre product over i_*A , then push out its two open kernels along addition $\mathcal{G} \oplus \mathcal{G} \rightarrow \mathcal{G}$. The resulting restriction on A is $r_1 + r_2$. This proves the group isomorphism, not just a set correspondence.

In particular CGS4.1 has a sheaf section precisely when $r = 0$. A section would be identity on the A -stalk and zero on the open source; compatibility with restriction forces $r = 0$. Conversely $r = 0$ gives the direct sum $j_!\mathcal{G} \oplus i_*A$ and the indicated section.

This is the gluing extension, not the support/restriction sequence CGS3.4. The exact pullback CGS4.3 displays their relationship and prevents their different lifting questions from being conflated.

135.6 CGS5. Higher supported cohomology for every coefficient sheaf

The full global-section functor on X is exact:

$$\Gamma(X, \mathcal{M}) = \mathcal{M}_{\mathfrak{m}} = A_+ \oplus A_-. \quad (\text{CGS5.1})$$

The only neighbourhood of $\mathfrak{m}$ is X , so this is equality with a stalk functor, which is exact. Thus $H^q(X, \mathcal{M}) = 0$ for $q > 0$, on this stated site.

For $Z = \{\mathfrak{m}\}$, form the supported-section kernel

$$\Gamma_Z(X, \mathcal{M}) = \ker(\Gamma(X, \mathcal{M}) \rightarrow \Gamma(U, \mathcal{G})).$$

Its restriction triangle and CGS5.1 give, with **all** higher open cohomology retained,

$$\begin{array}{l} H_Z^0(X, \mathcal{M}) = A_- \oplus \ker r, \\ H_Z^1(X, \mathcal{M}) = \text{coker } r, \\ H_Z^q(X, \mathcal{M}) = H^{q-1}(U, \mathcal{G}) \quad (q \geq 2). \end{array} \quad (\text{CGS5.2})$$

To derive these formulas directly, take a flabby resolution $\mathcal{G} \rightarrow \mathcal{I}^\bullet$ on U , put $B^\bullet = \Gamma(U, \mathcal{I}^\bullet)$, and embed r in its degree-zero cocycles through $\Gamma(U, \mathcal{G}) \hookrightarrow B^0$. The complete supported complex is

$$\begin{aligned} K^0 &= A_+ \oplus A_-, & K^q &= B^{q-1} \quad (q \geq 1), \\ d_K^0(a_+, a_-) &= r(a_+), & d_K^q &= -d_B^{q-1} \quad (q \geq 1). \end{aligned} \quad (\text{CGS5.3})$$

The differential squares to zero because $r(a_+)$ lies in the cocycles of B^0 . Its degree-zero kernel is the first line of CGS5.2. Its degree-one cocycles are $\Gamma(U, \mathcal{G})$ and its degree-one boundaries are $r(A_+)$, proving the second line. Its remaining cohomology is the displayed shifted open cohomology. This also fixes every sign.

The long exact sequence's degree-zero connecting map is precisely

$$\Gamma(U, \mathcal{G}) \longrightarrow \text{coker } r, \quad g \longmapsto [g]. \quad (\text{CGS5.4})$$

For $q \geq 1$, its connecting map

$$H^q(U, \mathcal{G}) \longrightarrow H_Z^{q+1}(X, \mathcal{M})$$

is the isomorphism given by including a cocycle $b \in B^q$ as the same b in K^{q+1} , with the sign convention CGS5.3. These are actual maps, not unspecified groups that might contain an obstruction.

The local cohomology sheaves are the skyscrapers with these groups:

$$\mathcal{H}_Z^q(\mathcal{M}) = i_* H_Z^q(X, \mathcal{M}). \quad (\text{CGS5.5})$$

Away from Z , every such stalk vanishes. At $\mathfrak{m}$, the only neighbourhood is X , giving the displayed group. Its R_0 -action is the one already stated: A_- has $\epsilon = 0$, whereas $\ker r$, $\text{coker } r$ and every higher open-cohomology term have $\epsilon = 1$. This is an explicit action calculation, not a numerical purity assignment.

For completeness, the derived open direct image has

$$R^0 j_* \mathcal{G} = j_* \mathcal{G}, \quad R^q j_* \mathcal{G} = i_* H^q(U, \mathcal{G}) \quad (q > 0). \quad (\text{CGS5.6})$$

At $\mathfrak{m}$ this follows from the same sole neighbourhood. On U , the higher sheaves vanish because the restricted direct-image functor is identity; equivalently, a sheaf-cohomology class becomes zero in a sufficiently small neighbourhood of its representative stalk. Thus these formulas include every possible higher term.

135.7 CGS6. Complexes: the full mapping fibre and its boundary representatives

For a cochain complex $\mathcal{M}^\bullet$ of $\mathcal{R}$ -modules, the classification is termwise exact:

$$(\mathcal{G}^\bullet, A_+^\bullet, A_-^\bullet, r^\bullet), \quad r^\bullet : A_+^\bullet \rightarrow \Gamma(U, \mathcal{G}^\bullet), \quad (\text{CGS6.1})$$

where the differentials are the classified sheaf morphisms. Thus $r^\bullet$ is a chain map, and conversely these data construct a complex by CGS1. Chain maps and homotopies are classified by the same compatibility equation in each degree. This assertion applies without any boundedness restriction.

Here is a fully specified derived calculation for bounded-below complexes, including all two-term and finite complexes used in the programme. Take a functorial flabby double resolution of $\mathcal{G}^\bullet$ and its total complex; in each degree only finitely many summands occur because the original complex is bounded below and the resolution degrees are nonnegative. Let

$$B^\bullet = R\Gamma(U, \mathcal{G}^\bullet)$$

denote that explicit total section complex, and let

$$\tilde{r} : A_+^\bullet \longrightarrow B^\bullet \quad (\text{CGS6.2})$$

be the composite of $r^\bullet$ with the resolution map. A usual double resolution is obtained degree by degree from injective or Godement resolutions and their functorial maps; its resolution-direction signs are included in the total differential. No higher open cohomology is deleted.

The actual supported complex is the following mapping fibre:

$$\boxed{K^n = A_+^n \oplus A_-^n \oplus B^{n-1}, \quad d_K(a_+, a_-, b) = (d_+ a_+, d_- a_-, \tilde{r}(a_+) - d_B b).} \quad (\text{CGS6.3})$$

To check $d_K^2 = 0$, its first two components are the squares of the original differentials, and its last is $\tilde{r}(d_+ a_+) - d_B \tilde{r}(a_+) + d_B^2 b = 0$. This convention is a concrete model of

$$R\Gamma_Z(X, \mathcal{M}^\bullet) \cong A_-^\bullet \oplus \text{Fib}\left(A_+^\bullet \xrightarrow{\tilde{r}} R\Gamma(U, \mathcal{G}^\bullet)\right). \quad (\text{CGS6.4})$$

The restriction triangle proves the identification: the middle global derived sections equal $A_+^\bullet \oplus A_-^\bullet$ because CGS5.1 is exact, the open term is $B^\bullet$, and its restriction is $(\tilde{r}, 0)$. The mapping fibre of precisely that map is CGS6.3. The sheaf-level supported complex is $i_* K^\bullet$.

For genuinely unbounded input, the termwise classification CGS6.1 is unchanged. The same mapping-fibre formula applies when $R\Gamma(U, \mathcal{G}^\bullet)$ is represented by a K-injective model of that unbounded complex; it must not be replaced by the finite-diagonal totalization just used. The construction and applications proved here require only the bounded-below case. No conclusion about an unbounded original programme object is obtained by truncating it.

Every connecting map is now explicit. If $b \in B^n$ is a cocycle, its boundary is

$$\partial[b] = [(0, 0, b)] \in H^{n+1}(K). \quad (\text{CGS6.5})$$

If $b = d_B c$, this representative is $d_K(0, 0, -c)$, so the formula is well-defined. The class $[b]$ has a global lift exactly when there exist a cocycle $a_+ \in A_+^n$ and $c \in B^{n-1}$ with

$$b = \tilde{r}(a_+) - d_B c. \quad (\text{CGS6.6})$$

This is not an assumed existence statement: it is the explicit boundary criterion obtained by writing out when the representative in CGS6.5 is a boundary. The already present A_-^n supplies an independent global cocycle component and does not alter the restriction.

In every degree there is the complete exact sequence

$$\begin{aligned} 0 \longrightarrow \text{coker}(H^{n-1}(A_+^\bullet) \rightarrow H^{n-1}(B^\bullet)) &\longrightarrow H^n(K^\bullet) \\ &\longrightarrow \ker(H^n(A_+^\bullet) \rightarrow H^n(B^\bullet)) \oplus H^n(A_-^\bullet) \longrightarrow 0. \end{aligned} \quad (\text{CGS6.7})$$

It follows by the long exact cohomology sequence of the displayed mapping fibre, or directly by taking the first two coordinates of a cocycle in CGS6.3. The $H^n(A_-^\bullet)$ summand splits canonically at the complex level; the remaining short exact sequence is not declared split without a constructed section.

135.8 CGS7. Full support and the selected residue are different computed functors

The full closed-point coefficient ring is $R_0 = \mathbb{Z} \times \mathbb{Z}$, and $i_* A$ in CGS3 allows either idempotent type. Its supported right adjoint is computed by CGS5–CGS6.

If instead the selected closed receiving ring is $\mathbb{Z}$ via the second projection $R_0 \rightarrow \mathbb{Z}$, its pushforward has $\epsilon = 0$. Its right adjoint on module sheaves is simply

$$\mathcal{M} \longmapsto (1 - \epsilon)\mathcal{M}_{\mathfrak{m}} = A_-. \quad (\text{CGS7.1})$$

Indeed a module map from a second-component stalk module into $\mathcal{M}$ lands in the $\epsilon = 0$ part of its stalk, which automatically restricts to zero; conversely any such group map gives a sheaf map. This proves the adjunction. The functor is exact because stalk and idempotent projection are exact, so its derived version on complexes is $A_-^\bullet$.

It follows that the selected residue functor records the automatic source boundary but not the entire topological-support functor. The precise additional object is

$$\text{Fib}\left(A_+^\bullet \xrightarrow{\tilde{r}} R\Gamma(U, \mathcal{G}^\bullet)\right), \quad (\text{CGS7.2})$$

with every group and map computed above. This is the explicit relation between the two functors; no absence of a comparison is used to dismiss their connection.

For the structure sheaf itself, CGS9 below shows that CGS7.2 is acyclic, so the two support calculations agree there. They need not be silently equated for other coefficients.

135.9 CGS8. Commuting arithmetic actions and the gluing class in higher degrees

Suppose specified additive coefficient operators act over the identity on the source space and commute with the supplied source action. Commutation with integer 1 means commutation with ϵ . On a localized arithmetic open an additive map commutes with every integer, and therefore with every inverted integer: multiply the desired commutation relation by that invertible multiplication operator to verify it. Thus these maps are precisely the module maps in the classification.

An action of a specified group or monoid $\mathcal{H}$ consequently consists exactly of compatible actions on

$$\mathcal{G}, \quad A_+, \quad A_-, \quad \Gamma(U, F_h) r = r F_h^+ \quad (h \in \mathcal{H}). \quad (\text{CGS8.1})$$

Their composition, inverses when specified, commutation relations and chain-map properties hold on the sheaf exactly when they hold on these entries. Conversely CGS1 constructs the whole sheaf action from those data. A functorial resolution retains the actions in CGS6, so its cone differential and every connecting map are equivariant. If an intended geometric operator moves the base rather than fixing it, that requires its stated pullback map; it is not silently included among these over-identity actions.

In particular a finite-dimensional Frobenius-equivariant quotient of $\text{coker } r$ has exactly the quotient action induced from $\Gamma(U, \mathcal{G})$. A generalized eigenclass can have nonzero boundary only at the same eigenvalue: $(F - \alpha)^k \partial = \partial(F - \alpha)^k$ follows by the proved equivariance. Thus CGS5.4 retains the complete receiving spectral action; source-idempotent splitting alone does not change its eigenvalue. No numerical weight of primitive τ is introduced.

There is also a complete higher derived form of CGS4.4:

$$R\text{Hom}_{\mathcal{R}}(i_*A, j_!\mathcal{G}) \cong R\text{Hom}_{R_0}(A, R\Gamma(U, \mathcal{G}))[-1] \cong R\text{Hom}_{\mathbb{Z}}(\epsilon A, R\Gamma(U, \mathcal{G}))[-1]. \quad (\text{CGS8.2})$$

Here the open derived sections carry $\epsilon = 1$. To prove it, choose an injective resolution $\mathcal{I}^\bullet$ of $\mathcal{G}$. For every resolution term the exact sequence

$$0 \rightarrow j_!\mathcal{I}^q \rightarrow j_*\mathcal{I}^q \rightarrow i_*\Gamma(U, \mathcal{I}^q) \rightarrow 0$$

is CGS4.2. Hence it yields the triangle

$$j_!\mathcal{G} \rightarrow Rj_*\mathcal{G} \rightarrow i_*R\Gamma(U, \mathcal{G}) \rightarrow j_!\mathcal{G}[1]. \quad (\text{CGS8.3})$$

There are no derived Hom maps from i_*A to $Rj_*\mathcal{G}$: the exact restriction j^{-1} sends i_*A to zero, and its right adjoint j_* sends injective modules to injectives. Thus the corresponding Hom complex is identically zero. Likewise i^{-1} to the full stalk ring is exact, so its right adjoint i_* preserves injectives and is fully faithful. Applying derived Hom to CGS8.3 proves the first isomorphism. The complementary idempotents in R_0 eliminate the $(1 - \epsilon)A$ factor, proving the second.

In degree one this recovers CGS4.4 with the actual map r . In higher degrees it retains all the open cohomology and any abelian-group extensions in the indicated derived Hom. None is replaced by its numerical dimension.

The same gluing classification with specified actions holds in the equivariant category. In degree one the right side is exactly the equivariant Hom group of the map r . For higher equivariant Ext, the derived Hom on the right must be taken in the corresponding module category with its specified action; ordinary nonequivariant Ext is not silently substituted for it.

135.10 CGS9. Application to the complete universal structure sheaf

The actual structure sheaf has the exact datum

$$\mathcal{R} \longleftrightarrow (\mathcal{O}_U, \mathbb{Z}, \mathbb{Z}, \text{id}_{\mathbb{Z}}). \quad (\text{CGS9.1})$$

Indeed its global sections are $R_0 = \mathbb{Z} \oplus \mathbb{Z}$ and its restriction is first projection. The open cohomology was explicitly calculated in SS4 by the flabby resolution

$$0 \rightarrow \mathcal{O}_U \rightarrow \mathcal{Q} \rightarrow \bigoplus_p i_{p*}(\mathbb{Q}/\mathbb{Z}_{(p)}) \rightarrow 0.$$

Its global section map is onto by the retained rational representatives for each finite residue tuple. Thus

$$R\Gamma(U, \mathcal{O}_U) = \mathbb{Z}[0]. \quad (\text{CGS9.2})$$

Substitution in CGS6 gives

$$R\Gamma_{\{\mathfrak{m}\}}(X, \mathcal{R}) \cong \mathbb{Z}[0] \oplus [\mathbb{Z} \xrightarrow{\text{id}} \mathbb{Z}], \quad (\text{CGS9.3})$$

where the second summand occupies degrees zero and one. Sending its degree-one integer to the same degree-zero integer is a contracting homotopy. Consequently its supported cohomology is exactly $\mathbb{Z}$ in degree zero and zero in every positive degree, with the explicit restriction lift $a \mapsto (a, 0)$. This is the actual completed application, not a theorem assuming a vanishing obstruction.

The classification also computes a different arrow that must be retained:

$$0 \rightarrow j_! \mathcal{O}_U \rightarrow \mathcal{R} \rightarrow i_*(\mathbb{Z} \oplus \mathbb{Z}) \rightarrow 0. \quad (\text{CGS9.4})$$

Its gluing class under CGS4.4 is the map

$$(a, b) \longmapsto a,$$

or equivalently $\text{id}_{\mathbb{Z}}$ on the epsilon component. It is therefore nonzero and this extension is not split as a sheaf extension. This does not contradict CGS9.3 or SS3's split support/restriction sequence:

$$0 \rightarrow i_* \mathbb{Z} \rightarrow \mathcal{R} \rightarrow j_* \mathcal{O}_U \rightarrow 0. \quad (\text{CGS9.5})$$

The exact relation is CGS4.3. In CGS9.4 one tries to extend a sheaf supported at the point into global and open sections without creating its nonzero arithmetic restriction; naturality forbids a section precisely because the computed map is identity. In CGS9.5 one lifts an already present open arithmetic section by $(a, 0)$; that is the proved section. Both are actual source-induced maps and both have now been calculated.

For complexes of coefficients the structure-sheaf identity comparison generalizes to the explicit object CGS6.3. Once the actual Mellin or Connes–Consani coefficient entry $(\mathcal{G}^\bullet, A_+^\bullet, A_-^\bullet, r^\bullet)$ is supplied by its proven receiving maps, that formula retains its complete defect. This note does not invent that entry or replace it by the structure sheaf.

135.11 CGS10. Provenance and the exact mathematical output

For the general gluing equivalence, the explicit human reference is Pierre Deligne, *La conjecture de Weil. II*, §3.4.8, printed p.209. The local French record export, lines2375–2383, was consulted for that precise passage; it is a transcription/record export, **not original author LaTeX**. The paper recalls the triple consisting of $j^* \mathcal{F}$, $i^* \mathcal{F}$, and its specialization map $i^* \mathcal{F} \rightarrow i^* j_* j^* \mathcal{F}$. Its surrounding finite-field weight argument is not thereby imported into the present receiver.

Here that specialization target has the exact, proved form

$$i^{-1} j_* \mathcal{G} = (j_* \mathcal{G})_{\mathfrak{m}} = \Gamma(X, j_* \mathcal{G}) = \Gamma(U, \mathcal{G}). \quad (\text{CGS10.1})$$

The second equality uses the sole neighbourhood X of $\mathfrak{m}$; the last is the definition of direct image along j . The full receiving closed coefficient is $i^{-1} \mathcal{M} = A_+ \oplus A_-$, and its specialization is exactly $(a_+, a_-) \mapsto r(a_+)$. Indeed it is the original restriction from X to U , and its A_- component vanishes because $1 - \epsilon$ restricts to zero. Thus the two inverse constructions in CGS1 prove the literal relation between Deligne's recalled triple and the displayed quadruple, with the receiving idempotent splitting and the entire specialization arrow retained. The distinction between full closed stalk and selected residue remains CGS7.

The source definitions and whole-spectrum argument originate with the user and remain in their original notation and verbatim corpus. The universal source receiver and its sheaf are programme calculations ST and SS. The distinction between source scalar arithmetic, power-map pullback and the independent idèle spectral action is proved in ST7–ST7a, with the actual original Connes–Consani source recorded there: *On the Jacobian of $\text{Spec } \mathbb{Z}$* , original **Jacobian.tex**, lines 2501–2550. Their geometric construction is not replaced by an unspecified sheaf in this note.

The lifting mechanism under comparison is Pierre Deligne's *La conjecture de Weil. II*, §3.6.1–§3.6.3. The present calculation determines which maps and coefficient objects are needed before

numerical weight separation can be invoked: the actual closed-stalk complex, the actual open coefficient complex, and the exact restriction map. Their complete supported receiving object is the mapping fibre CGS6.3.

For the source structure sheaf those entries are computed, and the supported obstruction vanishes by the explicit identity homotopy. For every other coefficient sheaf the remaining object is no longer an unspecified comparison: it is exactly $\ker r$, $\operatorname{coker} r$ and the retained higher open cohomology, or their full mapping-fibre complex. These statements assign no addition, parity, numerical weight or coordinate to primitive Z_1/τ , and make no RH or purity assertion.

136 The doubled source and the exact Connes–Consani coefficient entry

24 September 2026. Independent derivation DCP0–DCP12, continuing SS and CGS. This constructs the coefficient entry rather than postulating a geometric weight comparison.

136.1 DCP0. Source, provenance, and operations

The controlling source is B1–B5 of `SOURCE_OPERATIONS_AND_PROOFS.md`, with U01–U18 in `USER_DEFINITIONS_VERBATIM.md`. The current `CORPUS_AND_OPERATION_RULES.md` (private source record) was read. The complete user arguments USR-9ad1c0a2d09dba92, USR-f55d16feb948d8c2, and USR-6152e3bc6302258c retain the whole arithmetic reconstruction before subsequent coefficient operations. Those arguments were consulted in the earlier source derivations ST and SS; the present calculation does not substitute a finite prime system for that reconstruction.

The source remains $S = \mathbb{Z} \cup \{\tau\}$: primitive Z_1/τ has no addition or Z_2 parity, and is the global multiplicative identity; integer 1 is a distinct idempotent. The source integer addition is defined only on its supplied integer layer. All additive operations below occur in the explicitly constructed receiving ring and coefficient groups. They do not add copies of τ .

The human source is Alain Connes and Caterina Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, arXiv:0903.2024v3, §5. For this derivation the original author file `announc3.tex`, lines 1376–1667, was read in full. This includes Proposition `cechprop`, equations `restmaps`, `bord`, `rep1`, `liftw`, `compinv`, and Theorem `lemreasspec`. It is their coefficient sheaf and Fourier localization mechanism that is received here.

The programme inputs are CSL0–CSL10 and SSI0–SSI10. SSI was read completely for this calculation. Its proved inverse-source theorem makes the summation image closed; no closure equality is assumed. The full finite-unit sector is specified in SSI10, rather than identified with all finite-unit characters. The exact topology and scalar receiver are SS0–SS9; the sheaf classification used below is CGS1–CGS8 of `SOURCE_COEFFICIENT_GLUE.md`.

136.2 DCP1. The doubled source, its complete topology, and its receiving ring

Let

$$U = \operatorname{Spec} \mathbb{Z}, \quad X_{\pm} = U \cup \{\mathfrak{m}_{\pm}\}.$$

Each $X_{\pm}$ is a separately labelled copy of the source specialization space SS1: the only neighbourhood of $\mathfrak{m}_{\pm}$ in its chart is $X_{\pm}$. Glue the copies by the identity on the open U . The resulting space is

$$X^{\text{dbl}} = U \cup \{\mathfrak{m}_+, \mathfrak{m}_-\}. \quad (\text{DCP1.1})$$

Its opens are precisely the opens $V \subseteq U$, the two charts X_+, X_- , and the whole X^{dbl} . Indeed an open containing $\mathfrak{m}_+$ must contain X_+ , and similarly for the minus chart. These are also sufficient, by the gluing topology. Every rational prime point (p), and the arithmetic generic point (0), remains a distinct point of U .

The two sheaves $\mathcal{R}_{\pm} = j_* \mathcal{O}_U \times (i_{\pm})_* \mathbb{Z}$ agree on U . Their identity comparison there glues them to a ring sheaf $\mathcal{R}^{\text{dbl}}$. Its sections are

$$\begin{aligned} \mathcal{R}^{\text{dbl}}(V) &= \mathcal{O}_U(V) & (\emptyset \neq V \subseteq U), \\ \mathcal{R}^{\text{dbl}}(X_{\pm}) &= \mathbb{Z} \oplus \mathbb{Z}, & (\text{DCP1.2}) \\ \mathcal{R}^{\text{dbl}}(X^{\text{dbl}}) &= \{((a, b_+), (a, b_-)) : a, b_+, b_- \in \mathbb{Z}\} \cong \mathbb{Z}^3. \end{aligned}$$

All products in these receiving rings are componentwise. The global triples (a, b_+, b_-) restrict to $(a, b_{\pm})$ on their respective chart and to the rational section a on U . These formulas prove the ring sheaf gluing directly.

Write the global complementary idempotents as

$$\epsilon = (1, 0, 0), \quad e_+ = (0, 1, 0), \quad e_- = (0, 0, 1), \quad 1_{\mathcal{R}} = \epsilon + e_+ + e_-. \quad (\text{DCP1.3})$$

These are receiving ring elements. Neither $e_{\pm}$ is the source zero or primitive τ . The supplied source map is

$$[n] = (n, 0, 0) = n\epsilon, \quad [\tau] = (1, 1, 1). \quad (\text{DCP1.4})$$

It preserves every supplied multiplication, every integer-input addition, and the source zero. Its images of τ and integer 1 are distinct. Formula DCP1.4 is injective, because distinct integers have distinct first coordinate and $(1, 1, 1)$ is not of the form $(n, 0, 0)$.

136.3 DCP2. The map to the original three-point base

Use a different letter for Connes–Consani’s base:

$$Y = \{x_+, \eta, x_-\}, \quad Y_{\pm} = \{x_{\pm}, \eta\}.$$

Its opens are $\emptyset, \{\eta\}, Y_+, Y_-, Y$. Define

$$f : X^{\text{dbl}} \longrightarrow Y, \quad f(u) = \eta \ (u \in U), \quad f(\mathfrak{m}_{\pm}) = x_{\pm}. \quad (\text{DCP2.1})$$

Its inverse images of the five opens are respectively $\emptyset, U, X_+, X_-, X^{\text{dbl}}$, so it is continuous. Its full fibres are

$$f^{-1}(\eta) = \text{Spec } \mathbb{Z}, \quad f^{-1}(x_{\pm}) = \{\mathfrak{m}_{\pm}\}. \quad (\text{DCP2.2})$$

The arithmetic primes remain distinct source points even though f sends all of them to η ; their whole fibre is displayed. The map is not injective and is not an isomorphism of spaces. It is not an identification of primitive τ with η , or a claimed morphism between the two structure sheaves.

The geometric part of the exact comparison is the commutative diagram

$$\begin{array}{ccccc} \mathfrak{m}_+ & \subset & X^{\text{dbl}} & \supset & \mathfrak{m}_- & U = \text{Spec } \mathbb{Z} & \hookrightarrow & X^{\text{dbl}} \\ \downarrow & & \downarrow f & & \downarrow & \downarrow & & \downarrow f \\ x_+ & \subset & Y & \supset & x_- & \{\eta\} & \hookrightarrow & Y. \end{array} \quad (\text{DCP2.3})$$

The lower generic point receives the complete arithmetic fibre; the source’s localizations $\mathbb{Z}_{(p)}$ are still the stalks of $\mathcal{R}^{\text{dbl}}$ at (p) .

136.4 DCP3. The original coefficient spaces and every restriction factor

The specified finite-unit-invariant coefficient sheaf on Y is denoted Ω . Its chart spaces are

$$V_+ = S_{00}^{\text{even}} \oplus \mathbb{C}^2, \quad V_- = S_{00}^{\text{even}} \oplus \mathbb{C}^2,$$

$$S_{00}^{\text{even}} = \{h \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : h(-v) = h(v), \ h(0) = 0, \ \int_{\mathbb{R}} h(v) dv = 0\}. \quad (\text{DCP3.1})$$

The labelled endpoint coordinates are (c_0, c_1) on V_+ and (d_0, d_1) on V_- . Its overlap is the strong space

$$A = \{b \in C^{\infty}(\mathbb{R}_{>0}) : \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j b(u)| < \infty \text{ for all } N, j \geq 0\}. \quad (\text{DCP3.2})$$

Define

$$\begin{aligned} \Sigma h(u) &= 2 \sum_{n \geq 1} h(nu), & Rb(u) &= u^{-1}b(u^{-1}), \\ r_+(h, c_0, c_1) &= \Sigma h, & r_-(h, d_0, d_1) &= R\Sigma h. \end{aligned} \quad (\text{DCP3.3})$$

The factor 2 retains both signs of rational summation in the even sector. The reciprocal argument and the factor u^{-1} in R are both retained. The Fourier convention is

$$\widehat{h}(t) = \int_{\mathbb{R}} h(v) e^{-2\pi i v t} dv.$$

The full Poisson identity is

$$\Sigma \widehat{h}(u) = u^{-1} \Sigma h(u^{-1}) + u^{-1} h(0) - \int_{\mathbb{R}} h(v) dv. \quad (\text{DCP3.4})$$

For the stated source both final terms vanish by its two moment conditions, and therefore $R\Sigma h = \Sigma \widehat{h}$. The endpoint spaces remain separately present even though their restrictions are zero.

The original centered receiver is still related by the complete map

$$(\mathcal{T}k)(u) = 2u^{-1/2}k(u), \quad \mathcal{T}^{-1}b(u) = \frac{1}{2}u^{1/2}b(u). \quad (\text{DCP3.5})$$

For $\mathcal{E}h(u) = u^{1/2} \sum_{n \geq 1} h(nu)$, one has $\Sigma h = \mathcal{T}\mathcal{E}h$. These are specified invertible maps of receiving test spaces, with all constants retained, not replacement formulas for the original source.

Let $J = \Sigma(S_{00}^{\text{even}})$. SSI1–SSI7 proves that Σ is a continuous bijection onto the closed subspace J , with continuous inverse. Its Mellin description is

$$\mathcal{M}_0 b(s) = \int_0^\infty b(u) u^s \frac{du}{u}, \quad \mathcal{M}_0 \Sigma h(s) = 2\zeta(s) \int_0^\infty h(v) v^s \frac{dv}{v}. \quad (\text{DCP3.6})$$

The identity is first absolutely convergent for $\Re s > 1$, with the continuation and endpoint/trivial-zero residue calculations proved in SSI2–SSI6. In particular

$$Q = A/J = A/\bar{J} \cong \mathcal{B}/I, \quad (\text{DCP3.7})$$

where $\mathcal{B}$ is the strong entire Mellin space and I imposes the full vanishing order at every actual original nontrivial zero of ζ . This is the proved closed-image theorem, not an inference from the name cohomology. The function in DCP3.6 is the original ζ .

136.5 DCP4. Calculate the inverse-image sheaf on every open

Form the inverse image of the underlying sheaf of abelian groups:

$$\mathcal{N} = f^{-1}\Omega. \quad (\text{DCP4.1})$$

On a nonempty open $V \subseteq U$, the smallest target open whose inverse image contains V is $\{\eta\}$. The inverse-image presheaf therefore has value A on V . This presheaf is already a sheaf there: every nonempty open of the irreducible U contains the arithmetic generic point, so all nonempty opens intersect; compatible locally constant values must be the same value. Hence

$$\mathcal{N}|_U = \underline{A}_U, \quad \underline{A}_U(V) = A \quad (\emptyset \neq V \subseteq U). \quad (\text{DCP4.2})$$

Equivalently, if $g : \{(0)\} \hookrightarrow U$ is the arithmetic generic inclusion, this sheaf is g_*A . All its nonempty restriction maps are identity, so it is flabby and has zero higher sheaf cohomology on U .

On $X_\pm$ the smallest target open is $Y_\pm$, and a cover of $X_\pm$ contains $X_\pm$ itself. Sheafification therefore changes nothing. On the whole doubled space the gluing condition is exactly the one on the original cover. Consequently

$$\begin{aligned} \mathcal{N}(X_\pm) &= V_\pm, \\ \mathcal{N}(V) &= A \quad (\emptyset \neq V \subseteq U), \\ \mathcal{N}(X^{\text{dbl}}) &= \{(v_+, v_-) : r_+ v_+ = r_- v_-\}. \end{aligned} \quad (\text{DCP4.3})$$

The restrictions from the two charts to any such V are exactly $r_{\pm}$, followed by the identity of A .

Give this sheaf a receiving $\mathcal{R}^{\text{dbl}}$ -module action as follows. At a chart, $(a, b) \in \mathbb{Z}^2$ acts on $V_{\pm}$ by scalar a . On a nonempty arithmetic open, a section of $\mathcal{O}_U(V) \subset \mathbb{Q}$ acts on A by the specified rational scalar in its complex vector space. Both restrictions preserve this action because $r_{\pm}$ are complex-linear. On global pairs the triple (a, b_+, b_-) acts by a on both entries. These maps prove the module sheaf axioms on every open.

Thus the precise CGS datum on the plus or minus source chart is

$$(\underline{A}_U, V_{\pm}, 0, r_{\pm}). \quad (\text{DCP4.4})$$

Here the middle entries mean the $\epsilon = 1$ and $\epsilon = 0$ stalk components. In particular source τ and integer 1 have the same action on $\mathcal{N}$. This nonfaithful action is retained explicitly; it is not used to identify their source elements. DCP8 constructs a canonical larger coefficient sheaf that distinguishes them.

136.6 DCP5. Direct images: exact recovery and the higher-image qualification

By DCP4.3 the adjunction unit

$$\Omega \longrightarrow f_* f^{-1} \Omega \quad (\text{DCP5.1})$$

is identity on V_+ , V_- , and A . On Y it is identity on their displayed fibre product. These exhaust the target opens and all their restrictions, so DCP5.1 is an isomorphism of sheaves. It also preserves the original section topologies because these section maps are identities.

For $q > 0$,

$$R^q f_* \mathcal{N} = 0. \quad (\text{DCP5.2})$$

Here is the full local proof. The minimal target neighbourhoods are Y_+ at x_+ , Y_- at x_- , and $\{\eta\}$ at η . The stalks of the higher direct image are the corresponding cohomology groups on their inverse images. The charts $X_{\pm}$ have exact section functor, equal to the stalk at $\mathfrak{m}_{\pm}$, so their higher groups vanish for every sheaf. The remaining inverse image is U , where DCP4.2 gives the flabby sheaf $\underline{A}_U$. Its higher groups vanish too. Every stalk of DCP5.2 is therefore zero.

It would be false to strengthen this proof into acyclicity over **every** target open. For the whole target Y , the inverse image is X^{dbl} , whose first cohomology is Q , calculated below. The higher-direct-image sheaf is zero because the local basis is acyclic; the global cohomology is carried by the original base Y . In particular the exact comparison is

$$Rf_* \mathcal{N} \simeq \Omega, \quad R\Gamma(X^{\text{dbl}}, \mathcal{N}) \simeq R\Gamma(Y, \Omega), \quad (\text{DCP5.3})$$

not a claim that either global complex has zero first cohomology.

136.7 DCP6. The whole Čech complex and all four endpoint lines

The cover X_+ , X_- and its intersection U are acyclic for $\mathcal{N}$, by DCP5. The exact Čech complex is therefore

$$D^0 = V_+ \oplus V_-, \quad D^1 = A, \quad d(v_+, v_-) = r_+ v_+ - r_- v_-. \quad (\text{DCP6.1})$$

Its topology and differential are precisely the original CC complex, not merely isomorphic cohomology groups. Fourier preserves S_{00}^{even} and is an involution there; DCP3.4 therefore gives $r_+(V_+) = r_-(V_-) = J$.

Injectivity of Σ gives

$$H^0(X^{\text{dbl}}, \mathcal{N}) = \{(\widehat{h}, c_0, c_1), (h, d_0, d_1) : h \in S_{00}^{\text{even}}, c_0, c_1, d_0, d_1 \in \mathbb{C}\}, \quad (\text{DCP6.2})$$

$$H^1(X^{\text{dbl}}, \mathcal{N}) = Q, \quad H^q = 0 \quad (q \geq 2). \quad (\text{DCP6.3})$$

Indeed the kernel equation is $\Sigma f = \Sigma \widehat{h}$, hence $f = \widehat{h}$; the endpoint coordinates are unrestricted. The differential has image J , proving the cokernel formula. Its image is closed by SSI, so ordinary sheaf cohomology with its displayed quotient topology is already Hausdorff.

136.8 DCP7. Every closed-support map, including the orientation sign

Excision to a chart and CGS5 identify the single-support complex with

$$K_{\pm} = [V_{\pm} \xrightarrow{r_{\pm}} A] \quad \text{in degrees } 0, 1. \quad (\text{DCP7.1})$$

Consequently its degree-zero cohomology is precisely its two labelled endpoint lines, its degree-one cohomology is Q , and its other groups vanish. Its map to DCP6.1 is $v_+ \mapsto (v_+, 0)$ and identity in degree 1 for the plus support. For the minus support it is $v_- \mapsto (0, v_-)$ and **minus identity** in degree 1. Substitution into DCP6.1 verifies both chain maps.

The restriction $H^0(X^{\text{dbl}}, \mathcal{N}) \rightarrow V_-$ has the continuous lift

$$(h, d_0, d_1) \mapsto ((\widehat{h}, 0, 0), (h, d_0, d_1)). \quad (\text{DCP7.2})$$

Its restriction is identity by inspection. The equation $r_+ \widehat{h} = r_- h$ proves it lands in global sections. Thus its connecting map $V_- \rightarrow H_{\{\mathfrak{m}_+\}}^1(X^{\text{dbl}}, \mathcal{N})$ is zero by this actual lift.

For both closed supports $Z = \{\mathfrak{m}_+, \mathfrak{m}_-\}$, the complete supported complex is

$$K^0 = V_+ \oplus V_-, \quad K^1 = A \oplus A, \quad d_K(v_+, v_-) = (r_+ v_+, r_- v_-). \quad (\text{DCP7.3})$$

To verify it, take the mapping fibre of the restriction $D \rightarrow A[0]$, using r_+ on degree 0. In degree 1 this fibre has coordinates (k, b) and differential $(r_+ v_+ - r_- v_-, r_+ v_+)$. The invertible change $(k, b) \mapsto (b, b - k)$ gives DCP7.3. Its inverse is $(b_+, b_-) \mapsto (b_+ - b_-, b_+)$. This supplies the complete chain comparison and all signs.

The supported-to-global map is identity in degree 0 and $(b_+, b_-) \mapsto b_+ - b_-$ in degree 1. The boundary from the open A is $b \mapsto (b, b)$. The resulting sequence is

$$0 \rightarrow \mathbb{C}^4 \rightarrow H^0(X^{\text{dbl}}, \mathcal{N}) \rightarrow A \xrightarrow{b \mapsto ([b], [b])} Q \oplus Q \xrightarrow{(q_+, q_-) \mapsto q_+ - q_-} Q \rightarrow 0. \quad (\text{DCP7.4})$$

All four endpoint lines remain in its first term. The image of the preceding restriction into A is J , so after the displayed quotient this is the exact diagonal/difference row

$$0 \rightarrow Q \xrightarrow{\Delta} Q \oplus Q \xrightarrow{\delta} Q \rightarrow 0. \quad (\text{DCP7.5})$$

The continuous section $q \mapsto (q, 0)$ splits δ ; the mirror-compatible section is computed in DCP11.

136.9 DCP8. Canonical scalar extension retaining the source distinction

Use the constant ring sheaf $\mathbb{Z}$ and its **unital** map into $\mathcal{R}^{\text{dbl}}$. On the whole space that map sends

$$n \mapsto (n, n, n). \quad (\text{DCP8.1})$$

This is different from the distinguished source-integer map $n \mapsto n\epsilon = (n, 0, 0)$ in DCP1.4. Define the scalar extension of the underlying abelian sheaf by

$$\mathcal{M}_{\text{full}} = \mathcal{R}^{\text{dbl}} \otimes_{\mathbb{Z}} \mathcal{N}. \quad (\text{DCP8.2})$$

This operation is defined entirely in the already constructed receiving module category. At $\mathfrak{m}_{\pm}$, its stalk is

$$(\mathbb{Z} \oplus \mathbb{Z}) \otimes_{\mathbb{Z}} V_{\pm} \xrightarrow{\sim} V_{\pm} \oplus V_{\pm}, \quad (a, b) \otimes v \mapsto (av, bv). \quad (\text{DCP8.3})$$

The inverse is $(v, w) \mapsto (1, 0) \otimes v + (0, 1) \otimes w$.

At any arithmetic stalk (p) , including the generic point, the corresponding map is

$$\mathcal{O}_{U, (p)} \otimes_{\mathbb{Z}} A \longrightarrow A, \quad q \otimes a \mapsto qa. \quad (\text{DCP8.4})$$

It is an isomorphism because A is a $\mathbb{Q}$ -vector space. Its inverse is $a \mapsto 1 \otimes a$. To verify the nontrivial inverse identity, for $q = r/t$ use $t(q \otimes a - 1 \otimes qa) = 0$. Multiplication by $t \neq 0$ on the tensor is invertible by multiplication by $1/t$ on its A factor, so the difference is zero. The same construction on every open proves $\mathcal{M}_{\text{full}}|_U \simeq \underline{A}_U$.

There are no hidden higher tensor terms: $\mathcal{R}^{\text{dbl}}$ is flat over $\mathbb{Z}$. Its closed stalks are the free group $\mathbb{Z}^2$, and its arithmetic stalks are localizations of $\mathbb{Z}$, which are flat since localization carries injections to injections by its explicit fraction equality. Tensor exactness is stalkwise, proving the assertion. Thus DCP8.2 is also the derived scalar extension for these degree-zero inputs.

The complete chart datum is now

$$(\underline{A}_U, V_{\pm}, V_{\pm}, r_{\pm}). \quad (\text{DCP8.5})$$

The restriction is

$$(v, w) \mapsto r_{\pm}(v), \quad (\text{DCP8.6})$$

since the second scalar component restricts to zero on U . Therefore there is the canonical receiving-module decomposition

$$\boxed{\mathcal{M}_{\text{full}} \cong \mathcal{N} \oplus (i_+)_* V_+ \oplus (i_-)_* V_-}, \quad (\text{DCP8.7})$$

where the extra summands have $\epsilon = 0$. The projections and inclusions are the global complementary idempotents DCP1.3. On the first summand the earlier $\epsilon = 1$ module action on $\mathcal{N}$ is retained; on the extra closed copies the corresponding $e_{\pm}$ acts as identity. The restrictions DCP8.6 prove this equality as sheaves, not only as stalk vector spaces.

At either closed stalk the source action is

$$[\tau](v, w) = (v, w), \quad [n](v, w) = (nv, 0). \quad (\text{DCP8.8})$$

It is faithful as an action of the supplied source monoid. Distinct integers act differently on the nonzero complex vector space in the first component. Primitive τ acts differently from every integer on any nonzero vector in the second component. Products preserve the supplied multiplication; integer-input addition preserves the supplied integer addition. This assigns no source parity, coordinate, or addition to τ .

There are two distinct natural maps that must not be conflated. The scalar-extension unit of underlying abelian sheaves is

$$\eta : \mathcal{N} \rightarrow \mathcal{M}_{\text{full}}, \quad v \mapsto 1_{\mathcal{R}} \otimes v,$$

which is $v \mapsto (v, v)$ at a closed stalk. It is not equivariant for the distinguished source-integer action on the original $\epsilon = 1$ comparison: for integer n , $\eta(nv) = (nv, nv)$, whereas $[n]\eta(v) = (nv, 0)$. The receiving-module section of the projection onto $\mathcal{N}$ is instead

$$\iota_{\epsilon}(v) = \epsilon \otimes v = (v, 0). \quad (\text{DCP8.9})$$

It is $\mathcal{R}^{\text{dbl}}$ -linear and source-equivariant. Both maps have been constructed, and the exact difference is the extra closed component $v \mapsto (0, v)$.

136.10 DCP9. Cohomology and direct images of the faithful scalar extension

Each extra closed sheaf in DCP8.7 is acyclic: its sections over any open containing its point are the fixed stalk group and all remaining sections are zero, so it is flabby. It follows by the proved decomposition that

$$\begin{aligned} H^0(X^{\text{dbl}}, \mathcal{M}_{\text{full}}) &\cong H^0(X^{\text{dbl}}, \mathcal{N}) \oplus V_+ \oplus V_-, \\ H^1(X^{\text{dbl}}, \mathcal{M}_{\text{full}}) &\cong Q, \\ H^q(X^{\text{dbl}}, \mathcal{M}_{\text{full}}) &= 0 \quad (q \geq 2). \end{aligned} \quad (\text{DCP9.1})$$

On the Čech complex this is also immediate: degree 0 becomes $(V_+ \oplus V_+) \oplus (V_- \oplus V_-)$, and the differential is

$$((v_+, w_+), (v_-, w_-)) \mapsto r_+(v_+) - r_-(v_-). \quad (\text{DCP9.2})$$

There is no degree-one term from either w component. In particular the extra complete Schwartz spaces and their endpoint lines are retained in degree zero, not silently discarded.

At a single closed support the complex is

$$[V_\pm \oplus V_\pm \longrightarrow A], \quad (v, w) \mapsto r_\pm(v). \quad (\text{DCP9.3})$$

Its degree-zero group is $\mathbb{C}^2 \oplus V_\pm$, its degree-one group is Q , and all higher groups vanish. For both supports the first term of DCP7.4 becomes $\mathbb{C}^4 \oplus V_+ \oplus V_-$; the subsequent supported degree-one row remains DCP7.5 with exactly the same maps and signs.

The corresponding direct image is

$$f_*\mathcal{M}_{\text{full}} \cong \Omega \oplus (k_+)_*V_+ \oplus (k_-)_*V_-, \quad R^q f_*\mathcal{M}_{\text{full}} = 0 \quad (q > 0), \quad (\text{DCP9.4})$$

where $k_\pm : \{x_\pm\} \hookrightarrow Y$. These formulas are checked on the five target opens using DCP8.7. Higher direct images vanish on the same minimal-open basis as in DCP5, with the additional flabby point summands. This does not claim $f_*\mathcal{M}_{\text{full}} = \Omega$; its projection onto Ω and its extra kernel are displayed in DCP9.4.

136.11 DCP10. Source integer action and independent spectral action

The original unshifted positive-real action is

$$\begin{aligned} T_a b(u) &= b(u/a), \\ \rho_+(a)(h, c_0, c_1) &= (h(\cdot/a), c_0, ac_1), \quad a > 0. \\ \rho_-(a)(h, d_0, d_1) &= (ah(a \cdot), ad_0, d_1), \end{aligned} \quad (\text{DCP10.1})$$

Substitution proves $r_\pm \rho_\pm(a) = T_a r_\pm$. Each is complex-linear and commutes with all rational scalar multiplications, so these formulas give receiving-module automorphisms of $\mathcal{N}$. The induced action on $\mathcal{M}_{\text{full}}$ is $1_{\mathcal{R}} \otimes \rho(a)$. At a closed stalk it applies the same $\rho_\pm(a)$ to both copies; on U it is T_a . The decomposition DCP8.7 is equivariant.

These actions commute with the source action DCP8.8. They are nevertheless different operators: a supplied source integer p acts by $p\epsilon$, whereas the independent spectral operator at $a = p$ acts by dilation on the test functions. For example on the plus endpoint-value line the spectral operator is identity, while source scalar p is multiplication by p on its $\epsilon = 1$ copy and zero on its $\epsilon = 0$ copy. This evaluates their exact difference without claiming absence of a relation.

The relation to the original zeta quotient is direct:

$$\mathcal{M}_0 T_a b(s) = a^s \mathcal{M}_0 b(s). \quad (\text{DCP10.2})$$

It follows by $u = av$ in DCP3.6. On an actual zero ρ of multiplicity m_ρ , the receiving jet map

$$j_\rho([b]) = \sum_{r=0}^{m_\rho-1} \frac{(\mathcal{M}_0 b)^{(r)}(\rho)}{r!} t_\rho^r \in \mathbb{C}[t_\rho]/(t_\rho^{m_\rho}) \quad (\text{DCP10.3})$$

is well-defined by DCP3.7. Its full intertwining operator is

$$j_\rho T_a = a^\rho \left(\sum_{r=0}^{m_\rho-1} \frac{(\log a)^r}{r!} N_\rho^r \right) j_\rho, \quad N_\rho(v) = t_\rho v. \quad (\text{DCP10.4})$$

Multiplying Taylor expansions proves the identity, retaining every nilpotent term. These are actual observations from Q ; no unrestricted product decomposition or spectral synthesis assertion

is needed here. The scalar extension preserves these maps because its degree-one projection is the identity on A/J .

The four original endpoint lines in DCP6.2 have characters $1, a, a, 1$, in the order c_0, c_1, d_0, d_1 . Their numerical character weights at $p > 1$ are respectively $0, 2, 2, 0$. These are weights of the specified receiving dilation, not weights of primitive Z_1/τ . The extra closed $V_\pm$ copies retain the complete same dilation actions, including their Schwartz summands; they are not replaced by their endpoint characters.

136.12 DCP11. The mirror, its source lift, and its cochain sign

The source involution $\sigma : X^{\text{dbl}} \rightarrow X^{\text{dbl}}$ fixes every arithmetic point of U and exchanges $\mathfrak{m}_+, \mathfrak{m}_-$. It exchanges the two extra receiving ring coordinates b_+, b_- and fixes the arithmetic coordinate a . Thus it is a ringed-space isomorphism, and $f\sigma = wf$, where w is the original chart swap on Y .

On the coefficient sheaf, the mirror exchanges the two chart copies and acts by R on A . Equations $R^2 = 1$ and $Rr_- = r_+$, with chart domains exchanged, verify compatibility. For the ordered Cech complex DCP6.1, the exact chain action is

$$w_D^0(v_+, v_-) = (v_-, v_+), \quad w_D^1(b) = -Rb. \quad (\text{DCP11.1})$$

Indeed $d(v_-, v_+) = \Sigma h - R\Sigma f = -R(\Sigma f - R\Sigma h)$. The minus sign is forced by the reversal of the ordered cover.

On the supported complex DCP7.3 the mirror is

$$w_K^0(v_+, v_-) = (v_-, v_+), \quad w_K^1(b_+, b_-) = (Rb_-, Rb_+). \quad (\text{DCP11.2})$$

It commutes with the displayed differential by the restriction identities. Hence DCP7.5 is equivariant with R on its first Q , exchanged R on its middle pair, and $-R$ on its last Q . Its exact continuous equivariant section is

$$s : Q \rightarrow Q \oplus Q, \quad s(q) = \frac{1}{2}(q, -q), \quad (\text{DCP11.3})$$

because $\delta s(q) = q$ and $w_K s(q) = \frac{1}{2}(-Rq, Rq) = s(-Rq)$. The factor $1/2$ is a scalar in the constructed coefficient field, not an operation on primitive τ .

The full scalar extension uses the source ring involution on the first tensor factor and the coefficient mirror on the second. At the closed stalks it sends (v_+, w_+) to the correspondingly typed pair in the minus chart, and conversely. The extra closed summands are exchanged; they add no term to DCP11.1 in degree 1. This proves compatibility with DCP8.7.

The original unshifted relation remains

$$T_a R = a R T_{a^{-1}}, \quad \rho(a)w = a w \rho(a^{-1}). \quad (\text{DCP11.4})$$

The first equality follows by evaluating both sides at u : both give $au^{-1}b(a/u)$. On the chart Schwartz and endpoint coordinates, the second follows from DCP10.1 and direct substitution. No $a^{-1/2}$ twist has been substituted. Raw Mellin gives

$$\mathcal{M}_0 Rb(s) = \mathcal{M}_0 b(1 - s). \quad (\text{DCP11.5})$$

Thus the degree-one mirror is negative reflection of the original Mellin coordinate. At jets, it sends the jet at $1 - \rho$ to the jet at ρ by $t_{1-\rho} \mapsto -t_\rho$, with the further negative coefficient sign of DCP11.1. No multiplicity or nilpotent sign is lost.

136.13 DCP12. The computed source-supported receiver and the exact scope

The simple inverse-image coefficient entry and its source-faithful scalar extension have now both been constructed. Their exact relation is the projection and idempotent section

$$\mathcal{M}_{\text{full}} \rightleftharpoons \mathcal{N}$$

of DCP8.7–DCP8.9. The extra closed kernels retain the distinction between source τ and source integer 1, while the induced degree-one map is the identity on the original-zeta quotient Q .

This calculates the previously unspecified CGS coefficient entry on each chart. For $\mathcal{N}$, the full supported object is

$$\text{Fib}(V_{\pm} \xrightarrow{r_{\pm}} A);$$

for the source-faithful extension it is

$$V_{\pm}[0]_{\epsilon=0} \oplus \text{Fib}(V_{\pm} \xrightarrow{r_{\pm}} A)_{\epsilon=1}. \quad (\text{DCP12.1})$$

Both follow from CGS6 by substituting the constructed data, since $R\Gamma(U, \underline{A}) = A[0]$. Their degree-one group Q has $\epsilon = 1$. Therefore the automatic separation of the source idempotents splits the additional closed $V_{\pm}$ component but does not annihilate this particular supported Q . This is the calculated action on the actual receiver, not a proposed counterexample to the user's whole programme.

The connecting map for the single-chart restriction is zero by the explicit Fourier lift DCP7.2; the both-support diagonal/difference sequence has the exact mirror-equivariant section DCP11.3. Conversely the boundary $A \rightarrow Q \oplus Q$ is the displayed diagonal quotient map, so its nonzero original-zero observations retain the same spectral characters in source and target. These are the actual lifting and boundary maps produced by this coefficient comparison.

The comparison with Pierre Deligne's *La conjecture de Weil. II*, §3.6 has not been assumed as an additional geometry. DCP1–DCP12 provides the exact source space, coefficient functors, kernels, quotient, actions, and lifts to which a further numerical weight argument would have to apply. Its new content is the explicit faithful source coefficient extension and the complete source-to-CC sheaf comparison; the underlying CC Fourier localization remains credited to Connes and Consani. No arithmetic parity, addition, coordinate, or numerical weight has been assigned to primitive τ , and no RH or purity assertion is made.

136.14 Global adelic lifting propagation - 24 September 2026

The complete current proofs are GLOBAL_ADELIC_LIFTING_AND_PRIME_BOUNDARY.md GAP0–GAP9 (including GAP6A–GAP6B), COMPLETE_GENERATOR_ENDPOINT_RETURN.md GER0–GER7, GLOBAL_ENDPOINT_EXTENSION_RETURN.md GEX0–GEX9, and COMPLETE_COMPARISON_TRIANGLE_FORMALITY.md CTF0–CTF8. Every original prime wedge, endpoint pair and nilpotent jet is retained. GER credits the existing RZ endpoint resolvents and gives their explicit original Schwartz return and covariance corrections. GEX proves vanishing of the entire extension groups between the whole actual Q and the stated endpoint modules, in the explicitly constructed algebraic and strict locally convex operator categories. This strengthens the earlier finite-block and particular-extension assertions without assigning that theorem to a larger unspecified category.

GAP and CTF construct the full derived adelic complex, its canonical cohomology map, the exact original CC comparison and its entire cone. The faithful coefficient copies distinguish primitive tau from integer one in both objects. All additional cone contractions and the final-arrow homotopy are explicit; no test-space section $Q \rightarrow A$ is asserted. On the separate supported- Q row the same polynomial is invertible, rather than zero. Full tau numerical purity does not follow from endpoint separation and remains unproved.

136.15 Complete supported duality and original-zeta trace - 24 September 2026

The full continuation is `ACTUAL_SUPPORTED_DUALITY_INDEPENDENT.md` ASD0–ASD14, `GLOBAL_ORIGINAL_ZETA_RESIDUE_DUALITY_INDEPENDENT.md` GZR0–GZR9, and `ORIGINAL_ZETA_JACOBIAN_TRACE_BRIDGE.md` JTB0–JTB8. The original Fréchet coefficient complex now has its explicit continuous-dual localization row, with all faithful extra closed copies and every orientation sign. Its residue pairing is a pair of absolutely convergent integrals with the original denominator $\zeta(s)$; it descends to the entire actual quotient and detects every multiplicity jet. The exact derivative operator maps this pairing to the existing Weil form, with the original pole correction and full Gamma factors retained.

ASD14 constructs the resulting actual chain map into the character- a -twisted shifted dual and computes its complete cone. Its spectral cokernel is $Q/D_L(Q)$; weak- $*$ density makes its Hausdorff quotient zero but is not treated as algebraic surjectivity. The original H_0 and its continuous dual remain in the cone. This strengthens the explicit duality comparison without declaring a Verdier isomorphism, numerical purity or an RH proof. Current source Z_0 , Z_1/τ and integer Z_2 data are unchanged; no τ addition is introduced. The complete relevant user corpus and later corrections were checked before interpreting these maps.

136.16 Explicit global continuation: finite lifting and a nonzero remainder

The earlier uncertainty about algebraic surjectivity is now resolved by `GLOBAL_POLYNOMIAL_DUAL_DEFECT.md` PGD1–PGD6, independently checked in `PGC0–PGC7`: the actual algebraic cokernel is nonzero. Its explicit class c_ζ is the original-zeta contour functional with the constant arithmetic function 1 in the first slot. A lift would require a rapidly decreasing entire representative to equal 1 at all actual zeros, which the proved unbounded zero heights exclude. PGD constructs an injected rational-function line, entire representatives retaining all Hermite pole corrections, and the whole real/prime orbit. This is a receiving cokernel class, not primitive τ or a claimed off-critical zero.

SCL0–SCL9 and DPL0–DPL9 prove every nonzero polynomial in the actual cokernel generator invertible. Thus all finite-dimensional invariant dual spaces lift uniquely, continuously and with the original real/prime action. DPL gives explicit extension splittings and computes every degree of the cyclic derived Hom into the actual cone. These results retain its H and $H^\vee$ terms, all faithful closed copies and support labels. The weak- $*$ Hausdorff quotient remains zero alongside the now proved nonzero algebraic quotient. Current source Z_0 , Z_1/τ and Z_2 are unchanged. Complete proofs and exact human citations are in the linked continuation files; the full geometric weight-separation goal remains active.

137 The actual Mellin quotient: synthesis, image topology, and source return

24 September 2026. OMS0–OMS8. This receives the existing global synthesis, exact Schwartz image, and original-zeta projectors into CW’s actual quotient. It does not claim those earlier theorems as new discoveries. The division argument, continuous inverse, and comparisons required here are proved with all original factors.

137.1 OMS0. Sources and operation prerequisites

The complete user arguments `USR-9ad1c0a2d09dba92`, `USR-f55d16feb948d8c2`, and `USR-6152e3bc6302258c` were reread in the preserved corpus together with `CORPUS_AND_OPERATION_RULES.md`. Arithmetic, all primes, the original zeta function, and the complex receiving field below are the already reconstructed global objects. No operation counts primitive tau, assigns it parity, or introduces addition on tau. The notation remains Z_0, Z_1, Z_2, τ ; a receiving supported zero z_λ is not identified with primitive tau.

Human source: Ralf Meyer, A spectral interpretation for the zeros of the Riemann zeta function, arXiv:math/0412277v3. Original author source `sources/Meyer_0412277v3/Meyer.tex`, SHA256 `ab9bc31f3c105a64fdc6f3b65ad16701dd8bf218fc90df98fe8d7f3c7c58c00e`. Lines 225–738 were read, including the function spaces, the Zeta operator, Poisson summation, the closed-image theorem `the:Zeta_estimate`, and the exact Fourier–Laplace range theorem `the:Lap_range`. His theorem requires two rapid-division conditions; this note proves them from full zero-jet vanishing instead of attributing that unproved inference to his statement.

Programme sources read: `GLOBAL_MELLIN_SYNTESIS_REVIEW.md` R1–R5; `EXACT_SCHWARTZ_SUMMATION_IMAGE.md` SSI0–SSI10; `CC_W_MELLIN_INDEPENDENT.md` CW0–CW7; and the actual representative/projector construction RZ1–RZ8 in `sources/programme/cc_sheaf/ORIGINAL_ZETA_RESOLVENT_AND_PROJECTORS.md` under `quantum_tau_programme_bridge_20260924`. The longer independent image check was consulted through its strip calculation; no complete new reading of that check is claimed. The S/SSI synthesis and RZ projectors are retained prior programme results, not newly discovered here. Their complete proofs are supplied locally; this statement is not an external-publication receipt.

S2 credits Jacques Hadamard’s factorization theorem and records its original-TeX presentation in Alain Connes, The Riemann Hypothesis: Past, Present and a Letter Through Time, arXiv:2602.04022v1, `rhready.tex` lines 526–535. That source reading is inherited from S2, not claimed as a new reading of either complete paper. The particular product estimate needed here is proved below.

137.2 OMS1. Original spaces and exact comparison to Meyer

Let

$$S = \{f \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : f(-v) = f(v), f(0) = 0, \int_{\mathbb{R}} f(v) dv = 0\},$$

$$\mathcal{A} = \{k \in C^\infty(\mathbb{R}_{>0}) : p_{N,j}(k) = \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j k(u)| < \infty \quad (N, j \geq 0)\},$$

$$\mathcal{E}f(u) = u^{1/2} \sum_{n \geq 1} f(nu), \quad I_A = \overline{\mathcal{E}S}^{\mathcal{A}}, \quad Q = \mathcal{A}/I_A, \quad F_k(s) = \int_0^\infty k(u) u^{s-1/2} \frac{du}{u}. \quad (\text{OMS1.1})$$

The original action is $W_a k(u) = a^{1/2} k(u/a)$, $a > 0$. Write $\mathcal{B}$ for the entire functions with all seminorms

$$b_{A,M}(F) = \sup_{|\sigma| \leq A, t \in \mathbb{R}} (1 + |t|)^M |F(\sigma + it)| < \infty. \quad (\text{OMS1.2})$$

Integer indices suffice.

The Mellin map $\mathcal{M} : \mathcal{A} \rightarrow \mathcal{B}$, $\mathcal{M}k = F_k$, is a topological isomorphism. For $K(x) = k(e^x)$, its source seminorms are equivalent to $\sup_x e^{N|x|} |K^{(j)}(x)|$. The integral is $\int K(x) e^{(s-1/2)x} dx$. Stronger exponential seminorms justify every derivative in s ; repeated integration by parts proves uniform rapid decrease on every vertical strip. The exact inverse is

$$K(x) = \frac{1}{2\pi} \int_{\mathbb{R}} F(1/2 + it) e^{-itx} dt = \frac{e^{-(c-1/2)x}}{2\pi} \int_{\mathbb{R}} F(c + it) e^{-itx} dt. \quad (\text{OMS1.3})$$

Contour shifting proves the second expression for every real c , since horizontal sides tend uniformly to zero. For $x \geq 0$ choose $c = 1/2 + N + 1$; for $x \leq 0$ choose $c = 1/2 - N - 1$. Differentiation inserts $-(c - 1/2 + it)^j$, and an integrable bound with $M > j + 1$ proves each inverse seminorm estimate. Fourier inversion and the identity theorem prove the inverse identities. Completeness of the smooth weighted uniform topology shows that both spaces are Fréchet.

Meyer's source is $\mathcal{H}_- = \bigcap_{\sigma \in \mathbb{R}} \mathcal{S}(\mathbb{R}_{>0}^\times)_\sigma$, his operator is $Zf(u) = \sum_{n \geq 1} f(nu)$, and his action is $\lambda_a b(u) = b(u/a)$. The exact topological comparison and every factor are

$$T_0 k(u) = u^{-1/2} k(u), \quad T_0^{-1} b(u) = u^{1/2} b(u), \quad T_0 \mathcal{E} = Z, \quad T_0 W_a = \lambda_a T_0, \\ \widehat{T_0 k}(s) = \int_0^\infty T_0 k(u) u^s \frac{du}{u} = F_k(s). \quad (\text{OMS1.4})$$

To check the spaces and topology, derivatives in $x = \log u$ of the displayed exponential multiple are finite sums of weighted derivatives of K . Every polynomial in x is bounded by a larger exponential, and weights $e^{\pm(N+1)x}$ bound $e^{N|x|}$. These inequalities prove continuity in both directions. Substitution proves all intertwining identities.

For the raw two-sign convention in SSI,

$$\Sigma f(u) = 2 \sum_{n \geq 1} f(nu) = 2T_0 \mathcal{E} f(u), \quad \mathcal{M}_0 \Sigma f = 2F_{\mathcal{E} f}. \quad (\text{OMS1.5})$$

Meyer's Fourier phase $e^{2\pi i v t}$ equals the phase $e^{-2\pi i v t}$ on these even functions by $v \mapsto -v$. These comparisons retain the half-power, the sign convention, and the factor two.

137.3 OMS2. The exact ideal and its global division estimate

Let $\mathcal{Z}$ be the distinct actual nontrivial zeros of the original ζ , with actual multiplicity m_ρ , and set

$$I_\zeta = \{F \in \mathcal{B} : F^{(r)}(\rho) = 0 \quad (\rho \in \mathcal{Z}, 0 \leq r < m_\rho)\}. \quad (\text{OMS2.1})$$

This is closed by Cauchy's evaluation estimate on fixed small circles. Absolute summation on $\Re s > 1$ gives the original identity

$$F_{\mathcal{E} f}(s) = \zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v}. \quad (\text{OMS2.2})$$

The second factor is holomorphic for $\Re s > -2$, because $f(v) = O(v^2)$ at zero, and vanishes at $s = 1$. Continuation proves $\mathcal{M} \mathcal{E} S \subset I_\zeta$, with every multiplicity retained.

Keep the actual source vector and all its factors:

$$f_*(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2}, \quad k_* = \mathcal{E} f_*, \quad F_*(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s). \quad (\text{OMS2.3})$$

Here $f_*(0) = 0$, and Gaussian moments give

$$\int f_* = \frac{\pi}{2} \left(2\pi \frac{3}{4\pi^2} - 3 \frac{1}{2\pi} \right) = 0.$$

The direct Mellin integral before Gamma recurrence is

$$\frac{\pi}{2} \left[2\pi \frac{1}{2} \pi^{-(s+4)/2} \Gamma((s+4)/2) - 3 \frac{1}{2} \pi^{-(s+2)/2} \Gamma((s+2)/2) \right]. \quad (\text{OMS2.4})$$

It yields exactly the multiplier in (OMS2.3). Fourier differentiation of the Gaussian gives $\widehat{v^2 e^{-\pi v^2}} = (1/(2\pi) - t^2) e^{-\pi t^2}$ and $\widehat{v^4 e^{-\pi v^2}} = (t^4 - 3t^2/\pi + 3/(4\pi^2)) e^{-\pi t^2}$. Thus $\widehat{f_*} = f_*$. Poisson summation proves $k_*(u) = k_*(1/u)$. Gaussian decay at infinity and that reflection prove $k_* \in \mathcal{A}$, and hence $F_* \in \mathcal{B}$. The exceptional values are

$$F_*(0) = F_*(1) = \frac{1}{8}, \quad F_*(-2r) = F_*(1+2r) = \frac{(1+2r)2r}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r) \neq 0. \quad (\text{OMS2.5})$$

The pole at 1 and reflection give these values. Gamma has no zeros, so the zeros of F_* are exactly $\mathcal{Z}$ with multiplicities. This auxiliary transform is used with its entire multiplier, not as a working replacement for ζ .

Gaussian summation gives $|k_*(u)| \leq C u^B e^{-cu^2}$ for $u \geq 1$. Consequently

$$\max_{|s| \leq R} |F_*(s)| \leq 2C \int_1^\infty u^{B+R+1/2} e^{-cu^2} \frac{du}{u} \leq e^{C_1(R+2) \log(R+2)}. \quad (\text{OMS2.6})$$

The Gamma integral after $y = cu^2$, or splitting its exponential tail beyond a constant multiple of $R+2$, proves the final estimate. Jensen at radius $2R$, using $F_*(0) = 1/8$, gives $n(R) \leq C_2(R+2)^{3/2}$, with multiplicities. Hadamard's theorem then gives

$$F_*(s) = e^{a+bs} \prod_{\rho} (1 - s/\rho) e^{s/\rho}, \quad e^a = \frac{1}{8}, \quad b = F'_*(0)/F_*(0) = \frac{1}{2} \log(4\pi) - 1 - \frac{\gamma}{2}. \quad (\text{OMS2.7})$$

Every repeated zero and every genus-one exponential remains in the product.

For $F \in I_\zeta$, $G = F/F_*$ is entire by local Taylor division. We prove

$$\sup_{|\sigma| \leq A} |G(\sigma + it)| \leq C_{A,F} \exp\{C(|t| + 2)^{3/2} \log(|t| + 2)\}. \quad (\text{OMS2.8})$$

At large radius R , surround each zero $|\rho| \leq 8R$ by an open disk of radius R^{-2} . The total radii are $O(R^{-1/2})$. Every connected component has diameter below one for large R , since a simple chain of intersecting disks joining two points has diameter at most twice their total radii. Outside these disks and for $R/2 \leq |s| \leq 3R$, each finite factor satisfies $|1 - s/\rho| \geq (8R^3)^{-1}$. Their logarithms sum to at least $-CR^{3/2} \log R$. Dyadic summation of $n(R)$ gives

$$\sum_{|\rho| \leq 8R} |\rho|^{-1} = O(R^{1/2}), \quad \sum_{|\rho| > 8R} |\rho|^{-2} = O(R^{-1/2}).$$

The finite exponential factors therefore contribute at least $-CR^{3/2}$. In the tail $|w| = |s/\rho| \leq 3/8$, and

$$\log |(1-w)e^w| = \Re(-\sum_{j \geq 2} w^j/j) \geq -\frac{8}{5}|w|^2.$$

The tail contributes at least $-CR^{3/2}$; the leading exponential contributes at least $-|a| - 3|b|R$. Hence $|F_*(s)| \geq e^{-CR^{3/2} \log R}$ outside the disks in that annulus. For $R \leq |s| \leq 2R$ inside a component with $|\Re s| \leq A$, the entire component boundary lies within $|\Re z| \leq A+1$, inside the larger annulus, and outside the open disk union. The boundary quotient is bounded by the last estimate and $b_{A+1,0}(F)$. The maximum principle for the entire quotient gives the same bound inside. Applying this at dyadic radii and bounding the remaining compact set proves (OMS2.8), including all zero clusters and multiplicities.

137.4 OMS3. Rapid division, exact source inverse, and endpoints

Set $H(s) = F(s)/\zeta(s)$. The only possible poles are simple poles at $-2r$, with

$$\text{Res}_{s=-2r} H(s) = c_r(F) = \frac{F(-2r)}{\zeta'(-2r)}, \quad \zeta'(-2r) = (-1)^r 2^{-2r-1} \pi^{-2r} (2r)! \zeta(1+2r). \quad (\text{OMS3.1})$$

The complete functional multiplier is

$$\zeta(s) = \chi(s) \zeta(1-s), \quad \chi(s) = 2^s \pi^{s-1} \sin(\pi s/2) \Gamma(1-s) = \pi^{s-1/2} \frac{\Gamma((1-s)/2)}{\Gamma(s/2)}. \quad (\text{OMS3.2})$$

Its simple sine zero proves (OMS3.1). At the endpoints,

$$H(0) = -2F(0), \quad H(1) = 0, \quad H'(1) = F(1). \quad (\text{OMS3.3})$$

Also

$$H(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) G(s). \quad (\text{OMS3.4})$$

For any bounded real strip and $|t| \geq 1$, choose an integer L making $\sigma/2 + L \geq 1$. Gamma recurrence and its Euler integral give

$$|\Gamma(s/2)| \leq \frac{\Gamma(\sigma/2 + L)}{\prod_{j=0}^{L-1} |s/2 + j|} \leq C_{A,L} (2/|t|)^L.$$

Together with (OMS2.8) this proves subquadratic logarithmic growth of H and of each polynomial times H on those strip parts.

Choose $N \geq 0$, $a = -2N - 1$, $b \geq 2$. The reciprocal Euler series gives

$$|\zeta(b+it)^{-1}| \leq \sum_{n \geq 1} |\mu(n)| n^{-b} \leq \zeta(b). \quad (\text{OMS3.5})$$

On the left edge, writing $y = t/2$, Gamma recurrence and reflection give

$$|\chi(a+it)|^2 = \pi^{-4N-3} y \coth(\pi y) \prod_{k=1}^N (k^2 + y^2) \prod_{j=0}^N ((j + \frac{1}{2})^2 + y^2). \quad (\text{OMS3.6})$$

The first product is one at $N = 0$; the continuous value of $y \coth(\pi y)$ at zero is $1/\pi$. Indeed recurrence reduces the Gamma ratio to

$$\frac{|\Gamma(1-iy)|^2}{|\Gamma(1/2+iy)|^2} = \frac{\pi y / \sinh(\pi y)}{\pi / \cosh(\pi y)}.$$

Each quadratic factor is at least $(1+|t|)^2/8$, and $y \coth(\pi y) \geq (1+|t|)/(2\pi)$. Applying (OMS3.5) at real part $1-a = 2N+2$ yields

$$|\zeta(a+it)^{-1}| \leq \zeta(2N+2) \sqrt{2 \cdot 8^{2N+1}} \pi^{2N+2} (1+|t|)^{-2N-3/2}. \quad (\text{OMS3.7})$$

Thus H decreases rapidly on both vertical edges, with continuous bounds in finitely many seminorms of F .

Put $P_N(s) = \prod_{r=1}^N (s+2r)$, $P_0 = 1$, and $\kappa = 2N+2$. The function $V(s) = (s+\kappa)^M P_N(s) H(s)$ is holomorphic on the closed strip; every possible pole there has been cancelled explicitly. Its vertical-edge bound is $C b_{A_0, M+N}(F)$, for $A_0 \geq \max(2N+1, b)$. Apply the maximum principle to $e^{\varepsilon s^2} V(s)$ on rectangles of height $\pm T$. Their horizontal bound is a polynomial times $\exp[-\varepsilon T^2 + C(T+2)^{3/2} \log(T+2)]$, which tends to zero. Their vertical bound is multiplied

by at most $e^{\varepsilon \max(a^2, b^2)}$. First let $T \rightarrow \infty$, then $\varepsilon \downarrow 0$, with the evaluation point fixed. Since $|s + \kappa| \geq (1 + |t|)/\sqrt{2}$, this proves

$$\sup_{a \leq \sigma \leq b, t \in \mathbb{R}} (1 + |t|)^M |P_N(\sigma + it)H(\sigma + it)| \leq C_{N,b,M} b_{A_0, M+N}(F). \quad (\text{OMS3.8})$$

The F -dependent growth constant was used only to eliminate horizontal edges and does not enter this final linear seminorm bound. Away from the finitely many trivial poles this proves rapid bounds for H ; Cauchy's formula on slightly larger strips supplies every derivative bound.

Now define, for $v > 0$,

$$f_F(v) = \frac{1}{2\pi} \int_{\mathbb{R}} H(2 + it)v^{-2-it} dt. \quad (\text{OMS3.9})$$

Moving to any line $c \geq 2$ and differentiating gives

$$f_F^{(j)}(v) = \frac{(-1)^j}{2\pi} \int_{\mathbb{R}} (c + it)_j H(c + it)v^{-c-it-j} dt, \quad (s)_j = s(s+1) \cdots (s+j-1). \quad (\text{OMS3.10})$$

Arbitrary c proves rapid decrease of every derivative at infinity. Moving instead to $a = -2N - 1$ crosses exactly the original trivial-zero poles:

$$f_F(v) = \sum_{r=1}^N c_r(F)v^{2r} + \frac{1}{2\pi} \int_{\mathbb{R}} H(a + it)v^{-a-it} dt. \quad (\text{OMS3.11})$$

The j -th derivative of the remainder is bounded for $0 < v \leq 1$ by

$$\frac{v^{2N+1-j}}{2\pi} \int_{\mathbb{R}} |(a + it)_j H(a + it)| dt.$$

Choose N with $2N + 1 > j$. All derivatives extend continuously with the indicated polynomial limits. The fundamental theorem of calculus for consecutive derivatives verifies smoothness of the extension. Its even extension to $\mathbb{R}$ is Schwartz and obeys

$$f_F(0) = 0, \quad f_F^{(2r-1)}(0) = 0, \quad \frac{f_F^{(2r)}(0)}{(2r)!} = \frac{F(-2r)}{\zeta'(-2r)}. \quad (\text{OMS3.12})$$

Fourier inversion at $\Re s = 2$, and then the identity theorem, prove

$$\int_0^\infty f_F(v)v^s \frac{dv}{v} = H(s) \quad (\Re s > -2). \quad (\text{OMS3.13})$$

At $s = 1$, this gives $\int_{\mathbb{R}} f_F = 2H(1) = 0$. Thus $f_F \in S$, and (OMS2.2) gives $\mathcal{ME}f_F = F$.

Each Schwartz seminorm is bounded continuously by finitely many seminorms of F : use (OMS3.10) on $|v| \geq 1$, and (OMS3.11) with N large on $|v| \leq 1$. Its finitely many residues are bounded by the corresponding evaluations of F . Injectivity follows from (OMS2.2) on $\Re s > 1$, where $\zeta \neq 0$, and Fourier uniqueness. Therefore

$$\boxed{\mathcal{ME} : S \xrightarrow{\sim} I_\zeta \text{ topologically,} \quad I_A = \mathcal{ES} = \mathcal{M}^{-1}I_\zeta.} \quad (\text{OMS3.14})$$

The retained endpoint moments are

$$\int_0^\infty f_F(v) \frac{dv}{v} = -2F(0), \quad \int_0^\infty f_F(v) \log v \, dv = F(1). \quad (\text{OMS3.15})$$

They follow from (OMS3.3) and (OMS3.13). The complete higher local coefficients are the Laurent division coefficients of the original F/ζ ; no Gamma cancellation has erased them.

137.5 OMS4. Meyer's full criterion and the counting inverse

The estimate proves that F/ζ , after removable extension, is Schwartz on every line $\sigma \geq 1/2$. Apply the same proof to $F^\sharp(s) = F(1-s)$. The full functional equation preserves every nontrivial-zero multiplicity, so $F^\sharp \in I_\zeta$. For $\sigma \leq 1/2$ put $w = 1 - \sigma - it$. Then

$$\frac{F(\sigma + it)}{\zeta(1 - \sigma - it)} = \frac{F^\sharp(w)}{\zeta(w)} \quad (\text{OMS4.1})$$

is Schwartz too. These are exactly both conditions in Meyer's original range theorem. Conversely their pole removability forces all corresponding zero multiplicities to divide F . The overlap $\sigma = 1/2$ retains both conditions.

If $b = T_0 k$ and $F = \hat{b} \in I_\zeta$, the exact inverse is also

$$f_F(v) = \sum_{n \geq 1} \mu(n) b(nv) \quad (v > 0). \quad (\text{OMS4.2})$$

Indeed $b = Zf_F$. For fixed $v > 0$, the double sum after this substitution is bounded by $C_{v,L} \sum_{n,m \geq 1} (mn)^{-L} < \infty$, $L > 1$. Rearrangement gives $\sum_k f_F(kv) \sum_{n|k} \mu(n) = f_F(v)$; the finite divisor sum is one at $k = 1$ and zero otherwise by $\prod_{p|k} (1-1)$. This uses the entire already reconstructed prime system. Derivatives converge locally for $v > 0$; the endpoint extension is (OMS3.11), not an unjustified termwise endpoint limit.

For any even Schwartz source the complete Poisson identity remains

$$\frac{f(0)}{2} + Zf(u) = u^{-1} Z\hat{f}(u^{-1}) + \frac{\hat{f}(0)}{2u}. \quad (\text{OMS4.3})$$

Only the two stated defining conditions of S remove these two terms in this sector. Their endpoint characters $1, a$ remain in Meyer's larger source extension. In particular neither $F(0)$, $F(1)$, nor $F(-2r)$ has been set to zero by the synthesis theorem.

137.6 OMS5. Exact global jet image and its topology

Put $A_\rho = \mathbb{C}[T_\rho]/(T_\rho^{m_\rho})$ and

$$j_\rho([k]) = \sum_{r=0}^{m_\rho-1} \frac{F_k^{(r)}(\rho)}{r!} T_\rho^r, \quad J : Q \longrightarrow \prod_{\rho \in \mathcal{Z}} A_\rho. \quad (\text{OMS5.1})$$

By (OMS3.14),

$$\boxed{\ker J = 0}. \quad (\text{OMS5.2})$$

Every representative invisible to all full original zero jets belongs to the actual Schwartz summation image. Thus the common jet kernel previously left unspecified in QS/CW is calculated, not merely renamed.

The exact image is

$$\mathcal{J} = \{(F(\rho + T_\rho) \bmod T_\rho^{m_\rho})_\rho : F \in \mathcal{B}\}. \quad (\text{OMS5.3})$$

The exact quotient relations are

$$Q \xrightarrow{[k] \mapsto [F_k]} \mathcal{B}/I_\zeta \xrightarrow{[F] \mapsto (F(\rho + T_\rho))_\rho} \mathcal{J}. \quad (\text{OMS5.4})$$

Both arrows are isomorphisms when $\mathcal{J}$ has the transported quotient topology

$$\bar{b}_{A,M}(JF) = \inf_{G \in I_\zeta} b_{A,M}(F + G). \quad (\text{OMS5.5})$$

These seminorms define the quotient topology because the strip seminorms form a directed family. Completeness follows, for example, by lifting the successive differences of a quotient-Cauchy subsequence with its j -th difference bounded by 2^{-j} in $b_{j,j}$, summing in the complete $\mathcal{B}$, and using closedness of I_ζ .

For every fixed r, M , Cauchy's formula on radius-1/2 circles proves

$$\sup_{\rho: m_\rho > r} (1 + |\Im \rho|)^M \left| \frac{F^{(r)}(\rho)}{r!} \right| \leq C_{r,M} b_{2,M}(F). \quad (\text{OMS5.6})$$

All circles lie in $|\Re s| \leq 2$, since $0 < \Re \rho < 1$; their imaginary weights differ by a bounded factor. No zero-spacing condition occurs. The product tuple having constant coordinate one at every zero therefore is not in $\mathcal{J}$, since zero heights are unbounded. This classical fact is quantitatively supplied by the original-source zero-count theorem cited in CW3. Every finite tuple is realized by OMS6 below. Hence $\mathcal{J}$ is a proper dense subspace of the product in its product topology.

The topology (OMS5.5) is strictly stronger than that inherited from the product. Were they equal, completeness would make $\mathcal{J}$ closed in the product: a net in its closure has a Cauchy net of approximants, which completeness makes converge in the subspace. This contradicts the proved proper dense image. Thus the actual Q has not been replaced by arbitrary zero sequences or by coordinatewise convergence.

If $\pi : Q \rightarrow V$ is a linear quotient through which every original full j_ρ factors, then $\ker \pi \subset \ker J = 0$, so it is an isomorphism of vector spaces. This is an exact receiving statement about observation-preserving quotients. It is not an assertion that every quotient of the full tau geometry has already been mapped to this receiver, and it is not a bound on $\Re \rho$.

137.7 OMS6. Existing RZ primary sections, with their exact receiving action

This is the previously established RZ5–RZ8 construction, now entered into (OMS5.4), not a new discovery. Fix ρ , let $m = m_\rho$, and set

$$\begin{aligned} U_\rho(s) &= F_*(s)/(s - \rho)^m, & u_r &= U_\rho^{(r)}(\rho)/r!, & u_0 &\neq 0, \\ c_0 &= u_0^{-1}, & c_n &= -u_0^{-1} \sum_{r=1}^n u_r c_{n-r} \quad (1 \leq n < m). \end{aligned} \quad (\text{OMS6.1})$$

The entire U_ρ belongs to $\mathcal{B}$: outside a compact neighborhood of ρ division by the fixed polynomial preserves rapid decay; inside, the Taylor-removable quotient is bounded. Let $c_\rho(T) = \sum_{r < m} c_r T^r$. For $P \in A_\rho$, define

$$\mathcal{R}_\rho(P)(s) = U_\rho(s)[P(T)c_\rho(T)]_{<m}|_{T=s-\rho}, \quad s_\rho(P) = [\mathcal{M}^{-1}\mathcal{R}_\rho(P)] \in Q. \quad (\text{OMS6.2})$$

The bracket means the unique polynomial representative of degree below m ; its finite multiplication occurs only in the specified complex algebra. The inverse Mellin is (OMS1.3), so these are actual test representatives with every seminorm finite. The reciprocal recursion gives full jet P at ρ ; at every other zero, U_ρ retains its original full vanishing order. Thus

$$j_\eta s_\rho = \delta_{\eta\rho} \text{id}, \quad \Pi_\rho = s_\rho j_\rho, \quad \Pi_\rho^2 = \Pi_\rho, \quad \Pi_\rho \Pi_\eta = 0 \quad (\rho \neq \eta). \quad (\text{OMS6.3})$$

Continuity follows from finite jet evaluation and the finite-dimensional section domain. Summing finitely many sections proves the finite surjectivity used in OMS5. No convergence of the infinite sum of projectors is asserted.

All multiplier derivatives are still available exactly. With $C(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)/8$,

$$u_r = \sum_{h=0}^r \frac{C^{(h)}(\rho)}{h!} \frac{\zeta^{(m+r-h)}(\rho)}{(m+r-h)!},$$

$$C^{(h)}(s) = \frac{\pi^{-s/2}}{8} \sum_{\substack{a+b+c=h \\ 0 \leq a \leq 2}} \frac{h!}{a!b!c!} (s(s-1))^{(a)} \left(-\frac{\log \pi}{2} \right)^b 2^{-c} \Gamma^{(c)}(s/2). \quad (\text{OMS6.4})$$

Every Gamma, pi, endpoint, and original-zeta derivative is retained.

Substitution in (OMS1.1) gives $F_{W_a k}(s) = a^s F_k(s)$, and hence

$$j_\rho W_a = M_{a^\rho \exp((\log a)T_\rho)} j_\rho.$$

The two sides of the following equality have equal full jets at every zero; (OMS5.2) therefore proves

$$W_a s_\rho(P) = s_\rho \left(a^\rho \sum_{r=0}^{m_\rho-1} \frac{(\log a)^r T_\rho^r}{r!} P \right) \quad (a > 0). \quad (\text{OMS6.5})$$

These are actual invariant primary subspaces of Q , of dimension m_ρ . The original nilpotent order and all prime/prime-power actions remain. The section is into Q ; its test representative in $\mathcal{A}$ need not be equivariant before quotienting. Its covariance error lies in the actual image $\mathcal{E}S$ by synthesis. This gives the exact return to earlier test-space section obstructions.

137.8 OMS7. CW propagation and every support label

CW's original Gaussian and coefficient map still give

$$\begin{aligned} B(s) &= \sqrt{\pi} e^{(s-1/2)^2/4}, \quad F_{K_b[x]}(s) = B(s)(\log x)^s, \quad x > 1, \\ j_\rho \bar{K}_b[x] &= B(\rho + T_\rho)(\log x)^\rho \sum_{r < m_\rho} \frac{(\log \log x)^r T_\rho^r}{r!}, \\ B(\rho + T_\rho) &= \sqrt{\pi} e^{(\rho-1/2)^2/4} e^{(\rho-1/2)T_\rho/2 + T_\rho^2/4}. \end{aligned} \quad (\text{OMS7.1})$$

All finite nilpotent terms are retained. Equality of these full joint observations is now proved to be equality in the actual Q , with no residual invisible class. This does not enlarge CW's finite coefficient domain to arbitrary infinite tuples. MQ's moving-frequency and finite-valuation-frame domains remain their own proved domains; synthesis alone does not extend them across an unproved geometric boundary.

For the original bounded distributive support lattice L , use the actual CW7 receiver

$$G_L(V) = \{(0, \lambda) : \lambda \in L\} \cup \{(v, 1_L) : v \in V\}.$$

For a linear map f , the lift is

$$G_L(f)(v, \lambda) = (f(v), \lambda). \quad (\text{OMS7.2})$$

A non-top input forces $v = 0$, so it is well-defined. Linearity proves that it preserves the already specified receiving addition $(v, \lambda) + (w, \mu) = (v+w, \lambda \vee \mu)$ and scalar action $(c, \nu)(v, \lambda) = (cv, \nu \wedge \lambda)$. These are not operations on primitive tau. Applying (OMS7.2) to the two inverse maps in (OMS5.4) proves

$$G_L(Q) \xrightarrow{\sim} G_L(\mathcal{J}), \quad (\text{OMS7.3})$$

with every $(0, \lambda)$ retained separately. The primary maps, prime actions, and finite diagrams likewise keep that same label. Products here carry a common label; independent block labels would define a different object. Ordinary tensoring does not prove this zero-label statement: $0 \otimes e_\lambda = 0$ for every λ . Equations (OMS7.2)–(OMS7.3) provide the exact supported-zero repair rather than a tensor assertion that erases those fibres.

137.9 OMS7A. Return to the complete adelic periodization complex

The earlier AC1–AC4 use the same even Schwartz space with both endpoint conditions:

$$H_{00} = S = \{h \in \mathcal{S}_{\text{even}}(\mathbb{R}) : h(0) = 0, \int_{\mathbb{R}} h = 0\}.$$

The map $\Phi : Q_0 \rightarrow H_{00}$ there is onto: the class $[t_1 \otimes h]$ has image h . Its periodization is the original $J = 2\mathcal{E}\Phi$. Consequently, as subspaces of the same strong space $\mathcal{A}$,

$$I_0 = J(Q_0) = 2\mathcal{E}S = \mathcal{E}S = \overline{\mathcal{E}S}. \quad (\text{OMS7A.1})$$

The equality of subspaces uses the invertible receiving scalar 2; it does not replace the map J by $\mathcal{E}$. Its inverse still has the exact factor

$$h_k(x) = \frac{1}{4\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} \frac{F_k(s)}{\zeta(s)} x^{-s} ds, \quad \sigma > 1.$$

The endpoint conditions, trivial-zero residues and continuity of this inverse follow from OMS3–OMS4 with this factor retained.

Thus AC4's defined closure-difference space and comparison are now determined:

$$D_{\text{cl}} = \overline{I_0}/I_0 = 0, \quad C_{\text{alg}} = \mathcal{A}/I_0 \longrightarrow Q = \mathcal{A}/\overline{I_0}, \quad [k]_{I_0} \longmapsto [k]_{\overline{I_0}}. \quad (\text{OMS7A.2})$$

The inverse sends the same representative back. The two quotient topologies are identical, since their quotient maps have the same domain and kernel. All dilations commute with both maps.

For the full two-term complex after degree-zero adelic coinvariants, $C = [Q_0 \xrightarrow{J} \mathcal{A}]$ in degrees $-1, 0$, AC2's explicit split coordinates give the cochain isomorphism

$$C \cong \mathcal{B}_{\text{pr}}[1] \oplus [I_0 \hookrightarrow \mathcal{A}], \quad c \longmapsto (\beta(c), Jc).$$

Its inverse in degree -1 is $(b, k) \mapsto j(b) + [t_1 \otimes h_k]$. Hence

$$H^{-1}(C) = \mathcal{B}_{\text{pr}}, \quad H^0(C) = Q. \quad (\text{OMS7A.3})$$

The cochain map to $\mathcal{B}_{\text{pr}}[1] \oplus Q[0]$, given by β and the original quotient, induces these identities and is an equivariant quasi-isomorphism in the stated algebraic representation category. No unspecified topology on Q_0 is used for a derived equivalence. The specific two-extension

$$0 \rightarrow \mathcal{B}_{\text{pr}} \rightarrow Q_0 \xrightarrow{J} \mathcal{A} \rightarrow Q \rightarrow 0 \quad (\text{OMS7A.4})$$

has zero class: it is the Yoneda splice of the split sequence $0 \rightarrow \mathcal{B}_{\text{pr}} \rightarrow Q_0 \rightarrow I_0 \rightarrow 0$ with $0 \rightarrow I_0 \rightarrow \mathcal{A} \rightarrow Q \rightarrow 0$. The explicit section above makes the first class zero, so its Yoneda product is zero. This does not assert vanishing of the entire group $\text{Ext}^2(Q, \mathcal{B}_{\text{pr}})$, or a section of $\mathcal{A} \rightarrow Q$.

Every endpoint and prime label remains:

$$\mathcal{B}_{\text{pr}} = \left(\bigoplus_p \mathbb{C}\ell_p \right) \otimes \mathbb{C}^2, \quad W_a|_{\mathcal{B}_{\text{pr}}} = \text{id} \otimes \begin{pmatrix} 1 & 0 \\ 0 & a \end{pmatrix}.$$

AC4's higher derived-coinvariant groups are not terms of this two-term complex and are unchanged by (OMS7A.1). The result removes the earlier closure uncertainty on the actual adelic coefficient complex; it does not identify that complex with the entire geometric specialization. The original adelic construction retains the Connes and Connes–Consani–Marcolli attribution and exact ABR proof dependencies listed in AC0. This propagation was independently checked against the full definitions in AC0–AC4 and OMS1–OMS5.

137.10 OMS8. The calculated global conclusion

The result on the original domain is

$$\boxed{\mathcal{E}S = \overline{\mathcal{E}S} = \{k \in \mathcal{A} : F_k^{(r)}(\rho) = 0 \text{ for all actual } \rho, 0 \leq r < m_\rho\}.} \quad (\text{OMS8.1})$$

The proof constructs the actual continuous source inverse, proves both Meyer half-plane conditions, and recovers every endpoint moment and trivial-zero Taylor coefficient. The quotient has zero common full-jet kernel, the exact image topology (OMS5.5), and the retained original RZ primary sections with all nilpotents and prime actions. These are consequences for the actual original arithmetic receiver after the full-spectrum reconstruction.

No conclusion about $|p^\rho| = p^{1/2}$ was inserted. The calculation retains each actual ρ , rather than moving it or deleting it. Its contribution to the geometric programme is that a constructed map into this Q can now be checked on its complete original spectral data without an unknown common jet kernel; its construction and any Deligne weight estimate remain distinct mathematical calculations.

138 Completed original-character boundary complexes and the exact zeta comparison

Independent derivation CCB0–CCB17, 24 September 2026. The original finite moving algebra has several inequivalent completions. This note calculates the ordinary smooth completion, a complete algebra retaining all imaginary-leaf continuations, the maximal smooth graph completion of its two-term lifting complex, and the joint completion with all actual zeta jets. CCB11–CCB17 calculate the graph closure in the actual strong Mellin space and its original quotient, retaining the character endpoint and its integer-moment relations. A maximality statement always specifies its ambient observations; no topology-independent largest completion is asserted.

All operations occur in the specified complex receiving spaces. The corpus and operation rules, and the complete global user arguments `USR-9ad1c0a2d09dba92`, `USR-f55d16feb948d8c2` and `USR-6152e3bc6302258c`, were consulted before interpreting any failed receiving map. Primitive Z_1/τ retains no addition, parity, numerical coordinate or weight. The original arithmetic is already reconstructed.

138.1 CCB0. Original functions, all signs and the ambient smooth topology

Retain Connes–Consani’s $G = \mathbb{A}_{\mathbb{Q}}/\mathbb{Q}$, with characters $e_r(g)$, $r \in \mathbb{Q}$, and leaf derivative

$$De_r = 2\pi i r e_r.$$

Here the leaf is the image of the real subgroup $\iota_{\infty}(\mathbb{R})$ in G . Retain every $\lambda > 0$, the full signed coordinate $y \in \mathbb{R}$, and

$$W_{\text{alg}} = \mathbb{C}[\mathbb{R}_{>0}^{\times}], \quad \chi_{\lambda}[x] = x^{\lambda}, \quad \theta_{\mu}[x] = [x^{\mu}], \\ q^r(g, y) = e_r(g)[e^{-2\pi r y}].$$

The source algebra from CB4 is

$$\mathcal{C} = C^{\infty}(\mathbb{R}) \otimes W_{\text{alg}} \otimes \mathbb{C}[\mathbb{Q}].$$

Its faithful character realization consists of finite sums

$$\alpha(h(y)[x]q^r)(\lambda, g, y) = h(y)e^{\lambda \log x} e_r(g)e^{-2\pi \lambda r y}. \quad (\text{CCB0.1})$$

MQ1 and CB4 give the exact Fourier and character return. Negative r, y and every factor $2\pi, \lambda$ are retained.

Let $\mathcal{S}$ be the space of all functions $f(\lambda, g, y)$ smooth in λ, y and in the leaf direction D , with every mixed derivative jointly continuous. Give it seminorms

$$p_{N,m}(f) = \max_{a+b+c \leq m} \sup_{\lambda \in [N^{-1}, N], g \in G, |y| \leq N} |\partial_{\lambda}^a \partial_y^b D^c f|, \quad N \geq 2, m \geq 0. \quad (\text{CCB0.2})$$

The space is complete: locally uniform limits of every derivative are compatible by the fundamental theorem of calculus in λ, y and along $g + \iota_{\infty}(t)$. Leibniz’s formula proves continuous multiplication.

The full original operator and raw operator are

$$d = yD_{\lambda}, \quad D_{\lambda} = \lambda D + i\partial_y. \quad (\text{CCB0.3})$$

Both act continuously on $\mathcal{S}$. On (CCB0.1),

$$d\alpha(h[x]q^r) = \alpha(iyh'[x]q^r).$$

This is the original character operator, not the operator obtained by deleting its radial factor.

For a Haar measure ν on G with mass $M > 0$, Fourier extraction below always means

$$f_r = M^{-1} \int_G f(g) \overline{e_r(g)} d\nu(g). \quad (\text{CCB0.4})$$

The coefficient M^{-1} is retained. Haar translation invariance proves orthogonality.

138.2 CCB1. Density of the original algebra in the ordinary smooth space

The completion of $\alpha(\mathcal{C})$ for (CCB0.2) is all of $\mathcal{S}$. The proof also records the approximation mechanism used later.

For $n \geq 1$, let $G \rightarrow \mathbb{R}/n!\mathbb{Z}$ be the quotient dual to $(1/n!)\mathbb{Z} \subset \mathbb{Q}$, and let K_n be its kernel. These compact kernels decrease to $\{0\}$, since rational characters distinguish points. They eventually lie in every neighborhood of zero: otherwise compactness and their nesting give a nonzero point in their intersection outside such a neighborhood.

Average over K_n with Haar mass one. The resulting operator P_n commutes with all derivatives in (CCB0.2). Uniform continuity on compact parameter sets gives $P_n f \rightarrow f$ in every stated seminorm. On the resulting circle quotient, Fejér convolution gives finite rational-frequency sums and also commutes with all these derivatives. The approximate-identity estimate proves simultaneous convergence for each finite set of derivatives. No absolute convergence of a Fourier series over $\mathbb{Q}$ is assumed.

It remains to approximate each smooth coefficient $a(\lambda, y)$ of a finite Fourier sum. Finite sums

$$\sum_j h_j(y) e^{t_j \lambda}, \quad t_j \in \mathbb{R}, \quad (\text{CCB1.1})$$

are dense in the smooth compact-open topology on $(0, \infty) \times \mathbb{R}$. One explicit argument first approximates $\partial_\lambda^m a$ uniformly on a fixed compact rectangle, including the required y -derivatives, by Bernstein polynomials in λ . Their coefficients are smooth evaluations in y . Integrating m times in λ , with the original derivatives of orders below m at the left endpoint retained, approximates every required mixed derivative by polynomials in λ with smooth y -coefficients. Finally

$$e^{\epsilon \lambda} \left(\frac{e^{\epsilon \lambda} - 1}{\epsilon} \right)^j \rightarrow \lambda^j \quad (\epsilon \downarrow 0)$$

with every derivative uniformly on compact λ -sets. Its expansion is a finite sum of exponentials with strictly positive exponents. This proves (CCB1.1), and even allows all $t_j > 0$.

For a fixed rational frequency, multiplication by $e^{2\pi \lambda r y}$ is a smooth invertible operation on each compact parameter set. Approximate the coefficient after that multiplication and then restore $e^{-2\pi \lambda r y}$. This yields elements of (CCB0.1), proving the asserted density. The same proof shows density using only coefficients $[x]$ with $x > 1$.

138.3 CCB2. Why the ordinary smooth completion does not retain the radial homotopy

On the finite original algebra, CB's boundary projection and homotopy are

$$\begin{aligned} P \alpha(h[x]q^r) &= \alpha(h(0)[x]q^r), \\ T \alpha(h[x]q^r) &= \alpha \left(\frac{1}{i} \int_0^y \frac{h(t) - h(0)}{t} dt [x]q^r \right). \end{aligned} \quad (\text{CCB2.1})$$

They obey $dT = Td = I - P$. The expression inside the integral is smooth through zero and uses the oriented integral for negative y .

These maps are not continuous for the topology (CCB0.2). To prove it, choose a smooth compactly supported χ equal to one near zero, and put

$$f_n(\lambda, g, y) = e^{-n} \chi(n^2 y) e_n(g) e^{-2\pi \lambda n y}. \quad (\text{CCB2.2})$$

This belongs to the original algebra with coefficient $[1]$. On its support, $|y| \leq Cn^{-2}$. For λ in any fixed compact positive interval, the moving exponential is bounded by $\exp(C'/n)$. Each prescribed derivative adds only a fixed polynomial power of n , so $p_{N,m}(f_n) \rightarrow 0$.

However,

$$(Pf_n)(1, 0, -1) = e^{-n} e^{2\pi n} \longrightarrow \infty.$$

Thus P has no continuous extension to the ordinary smooth completion. If T had one, continuity of d and the identity $P = I - dT$ would give such an extension of P , a contradiction. This is an exact failure of a specified receiving topology. CCB3 and CCB6 construct completions retaining the maps.

138.4 CCB3. A complete algebra retaining every imaginary-leaf continuation

Introduce an auxiliary real continuation parameter z , distinct from y , and define

$$\mathcal{X} = \{F(\lambda, g, y, z) : \partial_z F = i\lambda D F, \text{ all mixed } \lambda, y, z, D \text{ derivatives are continuous}\}. \quad (\text{CCB3.1})$$

The topology is given by

$$q_{N,m}(F) = \max_{a+b+c+d \leq m} \sup_{\lambda \in [N^{-1}, N], g \in G, |y|, |z| \leq N} |\partial_\lambda^a \partial_y^b \partial_z^c D^d F|. \quad (\text{CCB3.2})$$

Completeness follows from the derivative-limit argument in CCB0 and the closed defining equation. Leibniz's rule proves that $\mathcal{X}$ is a complete Fréchet algebra.

Lift each original generator by

$$\tilde{\alpha}(h[x]q^r) = h(y)e^{\lambda \log x} e_r(g)e^{-2\pi \lambda r z}. \quad (\text{CCB3.3})$$

It satisfies the equation in (CCB3.1). The lift is faithful by the Fourier and character argument of MQ1.

This finite algebra is dense in $\mathcal{X}$. The compact-kernel averages and Fejér convolutions of CCB1 commute with $\partial_z - i\lambda D$, and approximate with all derivatives. Each remaining finite Fourier coefficient satisfies

$$\partial_z F_r = -2\pi \lambda r F_r,$$

hence equals $a_r(\lambda, y)e^{-2\pi \lambda r z}$. Approximate a_r by (CCB1.1), retaining the finitely many moving exponential factors. This proves density and identifies $\mathcal{X}$ as the completion for the explicitly given continuation seminorms.

In this specified ambient smooth continuation category, $\mathcal{X}$ is the full, and hence maximal, solution space: every smooth family satisfying the original imaginary-leaf equation occurs in the completion. This is not a claim of maximality among every possible topology.

The diagonal map

$$J_{\text{diag}} F(\lambda, g, y) = F(\lambda, g, y, y) \quad (\text{CCB3.4})$$

is continuous and injective. Indeed its r -th Fourier coefficient is $a_r(\lambda, y)e^{-2\pi \lambda r y}$, so its vanishing forces every a_r to vanish; uniqueness of Fourier coefficients then gives $F = 0$. Fourier uniqueness follows already from the convergent compact-kernel and Fejér approximations.

Its image has an exact continuation characterization. For $w = u + iv$, define

$$H_f(\lambda, g, y; w) = F(\lambda, g + \iota_\infty(u), y, y + v/\lambda).$$

The equation in (CCB3.1) gives $\partial_v H_f = i\partial_u H_f$, so this is the full entire continuation of the real leaf function. Conversely such a jointly smooth entire continuation determines

$$F(\lambda, g, y, z) = H_f(\lambda, g, y; i\lambda(z - y)). \quad (\text{CCB3.5})$$

Thus the extra parameter retains actual continuation data determined uniquely by f , rather than an independent unexplained coefficient.

The transported topology on $J_{\text{diag}}\mathcal{X}$ is essential. For example

$$f_n(\lambda, g, y) = e^{-\sqrt{n}} e_n(g)$$

tends to zero in every ordinary smooth seminorm, while its unique lift

$$F_n = e^{-\sqrt{n}} e_n(g) e^{2\pi\lambda n(y-z)}$$

diverges at $(\lambda, g, y, z) = (1, 0, 0, -1)$. Moreover the smooth function

$$f(g) = \sum_{n \geq 1} e^{-\sqrt{n}} e_n(g)$$

has no such full continuation: at $y = 0, z < 0$, its necessary Fourier coefficients would be unbounded, whereas the Fourier coefficients of any continuous function on compact G are bounded by its supremum norm. The derivative series defining f converge uniformly because $\sum n^k e^{-\sqrt{n}} < \infty$ for every k . Therefore the diagonal image is proper. It is dense in $\mathcal{S}$, by CCB1.

138.5 CCB4. Continuous homotopy and the complete boundary cohomology

On $\mathcal{X}$, define

$$\begin{aligned} \tilde{d} &= iy\partial_y, & \tilde{P}F(\lambda, g, y, z) &= F(\lambda, g, 0, z), \\ \tilde{T}F(\lambda, g, y, z) &= \frac{1}{i} \int_0^y \frac{F(\lambda, g, t, z) - F(\lambda, g, 0, z)}{t} dt. \end{aligned} \quad (\text{CCB4.1})$$

All integrals have their original orientation. The identity

$$\frac{F(t) - F(0)}{t} = \int_0^1 \partial_y F(ut) du$$

proves smoothness through zero and preservation of the continuation equation.

For $b \geq 1$,

$$\partial_y^b \tilde{T}F = \frac{1}{i} \int_0^1 s^{b-1} (\partial_y^b F)(\lambda, g, sy, z) ds. \quad (\text{CCB4.2})$$

For $b = 0$,

$$|\tilde{T}F| \leq |y| \sup_{|t| \leq |y|} |\partial_y F(\lambda, g, t, z)|.$$

The other derivatives commute with the integrals. These bounds prove continuity in (CCB3.2), with a specified loss of at most one y -derivative. They prove

$$\tilde{d}\tilde{T} = \tilde{T}\tilde{d} = I - \tilde{P}, \quad \tilde{P}\tilde{T} = 0. \quad (\text{CCB4.3})$$

Let

$$\mathcal{B}_{\text{ent}} = \{F \in \mathcal{X} : \partial_y F = 0\}.$$

Restriction at $y = 0$ and extension independent of y give continuous inverse maps to this closed boundary subspace. If $\tilde{d}F = 0$, its derivative vanishes away from zero and hence also at zero by continuity. Thus

$$\ker \tilde{d} = \mathcal{B}_{\text{ent}}, \quad \text{im } \tilde{d} = \ker \tilde{P}.$$

The image is closed, and the actual topological cohomology is

$$\boxed{H^0(\mathcal{X} \rightarrow \mathcal{X}) = H^1(\mathcal{X} \rightarrow \mathcal{X}) = \mathcal{B}_{\text{ent}}.} \quad (\text{CCB4.4})$$

The boundary-restriction kernel is contracted by the displayed continuous homotopy, so its connecting map is zero.

On the actual diagonal character functions, the differential is still the original one:

$$\partial_y J_{\text{diag}} F = J_{\text{diag}} \partial_y F + i\lambda D J_{\text{diag}} F,$$

and therefore

$$J_{\text{diag}} \tilde{d} = y(\lambda D + i\partial_y) J_{\text{diag}}. \quad (\text{CCB4.5})$$

The boundary projection on that image is the full continued boundary function

$$Pf(\lambda, g, y) = F(\lambda, g, 0, y).$$

It is not ordinary extension constant in y . The diagonal homotopy is

$$Tf(\lambda, g, y) = \frac{1}{i} \int_0^y \frac{H_f(\lambda, g, t; i\lambda(y-t)) - H_f(\lambda, g, 0; i\lambda y)}{t} dt. \quad (\text{CCB4.6})$$

These formulas retain the entire moving q^r factors.

The raw operator is also retained. On $\mathcal{X}$, its counterpart is $i\partial_y$, and multiplication by y gives the chain map (I, M_y) to the original complex. Multiplication by y is injective; its image is exactly the kernel of restriction at zero by the displayed division formula. Its cokernel is $\mathcal{B}_{\text{ent}}$. Integration from zero makes $i\partial_y$ onto. Thus the raw complex has $H^0 = \mathcal{B}_{\text{ent}}$, $H^1 = 0$, whereas the original complex retains the extra $\mathcal{B}_{\text{ent}}$ in degree one. Completion has not removed that radial boundary contribution.

138.6 CCB5. Frobenius, all real signs, and the finite-valuation frame

The exact arithmetic action on $\mathcal{X}$ is

$$A_\mu F(\lambda, g, y, z) = F(\mu\lambda, g, y/\mu, z/\mu), \quad \mu > 0. \quad (\text{CCB5.1})$$

Indeed

$$\partial_z A_\mu F = \mu^{-1} i(\mu\lambda) D A_\mu F = i\lambda D A_\mu F.$$

Each compact seminorm of the transformed function is controlled by finitely many seminorms on the transformed compact set, with the explicit derivative factors $\mu^a \mu^{-b-c}$. Thus these are continuous automorphisms. They obey $A_\mu A_\nu = A_{\mu\nu}$ and commute with $\tilde{d}, \tilde{P}, \tilde{T}$; the last equality follows by $t = \mu u$ in the integral. On original generators this is exactly $h(y/\mu)[x^\mu]q^r$.

The original sign action is

$$SF(\lambda, g, y, z) = F(\lambda, -g, -y, -z). \quad (\text{CCB5.2})$$

It preserves the equation because both D and ∂_z change sign. The oriented substitution $t = -u$ proves $S\tilde{T} = \tilde{T}S$. It also commutes with the other maps and with A_μ . The same homotopy therefore restricts to the closed sign-invariant subspaces. Their boundary cohomology is $\mathcal{B}_{\text{ent}}^S$; no frequency r is discarded, and the pair $r, -r$ retains its full two-term sum.

The finite-valuation comparison remains exactly MQ8 and CB1. On finite ideles,

$$c(Y_f) = \prod_p p^{v_p(Y_p)} > 0, \quad g = [X/c(Y_f)], \quad y = Y_\infty/c(Y_f), \quad (\text{CCB5.3})$$

with all signs allowed for Y_∞ . A negative rational gauge transformation sends c to $|a|c$, and hence sends (g, y) to $(-g, -y)$, while a positive one leaves these quotient coordinates unchanged. Therefore the sign-invariant objects descend to the original signed quotient.

On the frequency r , the exact pullback is

$$e_r(g) e^{-2\pi\lambda r y} = \psi_f(r X_f/c) e^{2\pi i r X_\infty/c} e^{-2\pi\lambda r Y_\infty/c}. \quad (\text{CCB5.4})$$

For every fixed valuation frame c , substitution is a continuous pullback in the specified character topology. This gives a compatible product of completed frame spaces, with framewise topology; the gauge transformations permute the frames and implement the displayed sign action. Compatibility is a closed equality between continuous frame maps. The $c = 1$ component determines the compatible family by the exact formulas, so its inverse reconstructs all frames.

Nothing here asserts continuity of $c(Y_f)$ for the ambient additive-adele topology, or extends that map across $Y_f = 0$. CB1 proves why such a substitution would be false. The completion uses the source orbit topology and the retained frame maps. The graph correspondence's G -boundary is still present and is not identified with a point or with primitive τ .

138.7 CCB6. The maximal smooth graph completion of the lifting complex

The entire-continuation algebra above is useful but does not contain every smooth interior for which the two-term lifting maps have smooth limits. The maximal graph realizations can also be calculated exactly.

In $\mathcal{S}$, put

$$\mathcal{H} = \ker D_\lambda, \quad \mathcal{S}_0 = \{g \in \mathcal{S} : g(\lambda, \cdot, 0) = 0\}.$$

Both are closed Fréchet spaces. The Fourier equation for $h \in \mathcal{H}$ is

$$\partial_y h_r + 2\pi\lambda r h_r = 0, \quad h_r(\lambda, y) = h_r(\lambda, 0)e^{-2\pi\lambda r y}. \quad (\text{CCB6.1})$$

Consequently boundary restriction is injective on $\mathcal{H}$, and its image consists exactly of the boundary functions with the full smooth continuation in (CCB6.1). This is the same boundary space as $\mathcal{B}_{\text{ent}}$, with the explicit map $F(\lambda, g, z) \mapsto F(\lambda, g, y)$.

Define two completed cochain spaces as follows:

$$\begin{aligned} \mathcal{C}_{\max}^0 &= \{h + g : h \in \mathcal{H}, g \in \mathcal{S}_0\}, \\ \mathcal{C}_{\max}^1 &= \{h + dg : h \in \mathcal{H}, g \in \mathcal{S}_0\}, \end{aligned} \quad (\text{CCB6.2})$$

each carrying the product topology of its displayed parameters (h, g) .

Both realizations in $\mathcal{S}$ are injective. For degree zero, take the boundary trace of $h + g = 0$; (CCB6.1) gives $h = 0$, hence $g = 0$. For degree one, if $h + dg = 0$, the trace gives $h = 0$. Then $dg = 0$; away from $y = 0$, $D_\lambda g = 0$, and smoothness gives the same equation at zero. Equation (CCB6.1) and the zero trace give $g = 0$.

The continuous complex and homotopy are now fully explicit:

$$d(h + g) = dg, \quad T(h + dg) = g, \quad P(h + g) = h, \quad P(h + dg) = h. \quad (\text{CCB6.3})$$

They satisfy $dT = I - P$, $Td = I - P$. Thus

$$H^0(\mathcal{C}_{\max}^\bullet) = H^1(\mathcal{C}_{\max}^\bullet) = \mathcal{H} \quad (\text{CCB6.4})$$

with continuous splittings and closed differential image.

These are actual completions of the finite original complex. Finite original holomorphic sums are dense in $\mathcal{H}$: Fourier averaging and Fejér preserve $D_\lambda h = 0$, after which their coefficient functions of λ are approximated by finite source exponentials as in CCB1. Original finite functions of zero trace are dense in $\mathcal{S}_0$: write $g = yv$ using the smooth division formula, approximate v in $\mathcal{S}$ by CCB1, and multiply by y . Therefore $h_n + g_n$ approximates every degree-zero pair, and $h_n + dg_n$ approximates every degree-one pair. Their finite radial homotopies converge to g , exactly as (CCB6.3) states.

This establishes the maximal graph assertion. In degree zero the completion is the closure of the map $f \mapsto (f, Pf)$ in $\mathcal{S} \times \mathcal{H}$; its image consists exactly of pairs (f, h) with equal boundary traces. In degree one it is the closure of $f \mapsto (f, Tf, Pf)$ in $\mathcal{S} \times \mathcal{S}_0 \times \mathcal{H}$, and its image is exactly

$$\{(dg + h, g, h) : g \in \mathcal{S}_0, h \in \mathcal{H}\}. \quad (\text{CCB6.5})$$

This image is closed. Every smooth simultaneous limit of the original function, homotopy and projection must satisfy this equation; conversely density above supplies every such triple.

Equivalently, the maximal degree-one domain is characterized by the full mode formula

$$(Tf)_r(\lambda, y) = \frac{e^{-2\pi\lambda ry}}{i} \int_0^y \frac{e^{2\pi\lambda rt} f_r(\lambda, t) - f_r(\lambda, 0)}{t} dt, \quad (\text{CCB6.6})$$

provided these coefficients assemble to a member of $\mathcal{S}$. This is an exact domain characterization, not an assumption: assembly gives $g \in \mathcal{S}_0$, and then $h = f - dg$ has the coefficients (CCB6.1), so lies in $\mathcal{H}$. Conversely every member of (CCB6.2) has exactly (CCB6.6) by the elementary first-order ODE. All constants and exponential factors are retained.

Frobenius and sign act on $\mathcal{S}$ by $f(\lambda, g, y) \mapsto f(\mu\lambda, g, y/\mu)$ and $f(\lambda, -g, -y)$. They preserve $\mathcal{H}, \mathcal{S}_0$ and commute with d ; hence they act continuously on both graph spaces and commute with (CCB6.3).

The original finite algebra is dense in each degree, but different degrees now have their specified graph topologies. We do not assert an unproved pointwise multiplication on degree one. The common complete algebra carrying every operator as a continuous endomorphism is the explicitly proved $\mathcal{X}$ in CCB3–CCB5, with continuous maps into both graph realizations. This identifies the categories rather than calling one arbitrary topology the unique completion.

138.8 CCB7. The original-zeta comparison is not continuous in the character completion

Retain the exact CW map

$$\overline{K}_b : V_+ \longrightarrow Q, \quad F_{K_b[x]}(s) = B(s)(\log x)^s, \quad B(s) = \sqrt{\pi} \exp((s - \tfrac{1}{2})^2/4), \quad (\text{CCB7.1})$$

where Q is the actual quotient constructed from the original ζ , and V_+ has basis $[x]$, $x > 1$. For an actual nontrivial zero ρ , write $\beta = \Re \rho \in (0, 1)$, and let $j_{\rho,0}$ be its constant jet coordinate, a continuous functional on Q .

For $t > 0$, take the exact source element

$$v_t = t^{-\rho}([e^t] - [e^{2t}]) \in V_+, \quad t^{-\rho} = \exp(-\rho \log t). \quad (\text{CCB7.2})$$

For every compact positive λ -interval and each fixed derivative order,

$$\partial_{\lambda}^k \chi_{\lambda}(v_t) = t^{-\rho} \partial_{\lambda}^k (e^{\lambda t} - e^{2\lambda t}) = O(t^{1-\beta}). \quad (\text{CCB7.3})$$

For $k = 0$ this follows from the mean value integral; for $k \geq 1$, the two derivative terms are bounded by constants times $t^{k-\beta}$, which is $O(t^{1-\beta})$ for $0 < t \leq 1$. Thus v_t , regarded as a zero-frequency character function independent of y, z , tends to zero in every topology constructed above.

But

$$j_{\rho,0} \overline{K}_b(v_t) = B(\rho)(1 - 2^{\rho}) \neq 0 \quad (\text{CCB7.4})$$

for every t . The inequality holds because $|2^{\rho}| = 2^{\beta} > 1$. Therefore the coefficientwise CW map has no continuous extension from this character completion to the original Q , or even to this single continuous zero observation.

This uses the established original quotient and the actual zero; it is not a statement about a missing source operation or a failure of the full user-defined arithmetic. CCB8–CCB9 construct the exact replacement that retains both observations.

138.9 CCB8. Every finite packet of actual zero jets can be fixed while characters vanish

Let Σ be any finite set of distinct actual nontrivial zeros, with their exact multiplicities m_ρ . Write

$$A_\Sigma = \prod_{\rho \in \Sigma} \mathbb{C}[T_\rho]/(T_\rho^{m_\rho}), \quad D_t|_{A_\rho} = t^\rho \exp((\log t)T_\rho). \quad (\text{CCB8.1})$$

The complete CW comparison gives

$$j_\Sigma \bar{K}_b \theta_t = D_t j_\Sigma \bar{K}_b.$$

It retains the Gaussian jet

$$B(\rho + T_\rho) = \sqrt{\pi} e^{(\rho - \frac{1}{2})^2/4} \exp((\rho - \frac{1}{2})T_\rho/2 + T_\rho^2/4)$$

and its explicit inverse from CW4.

Finite interpolation remains onto after imposing the augmentation-zero condition

$$\text{aug} \left(\sum_j c_j [x_j] \right) = \sum_j c_j = 0. \quad (\text{CCB8.2})$$

Here is a complete proof. Consider the joint map

$$V_+ \longrightarrow \mathbb{C} \oplus A_\Sigma, \quad v \longmapsto (\text{aug } v, j_\Sigma \bar{K}_b v).$$

A linear functional vanishing on its image gives, after writing $u = \log \log x$,

$$c + \sum_{\rho \in \Sigma} p_\rho(u) e^{\rho u} = 0, \quad \deg p_\rho < m_\rho.$$

The invertible Gaussian jets only make invertible triangular changes in the coefficient polynomials. The functions in this expression are independent. Apply products of the operators $(\partial_u - \eta)^{m_\eta}$, including the factor ∂_u for the distinct exponent 0, to isolate each chosen exponent. On its remaining polynomial factor every operator with different exponent has nonzero constant diagonal and is invertible. Therefore every polynomial and c vanishes. A proper subspace of a finite-dimensional target has a nonzero annihilating functional, by extension of a basis; hence the joint map is onto. In particular each $(0, a)$ has a preimage.

Choose fixed vectors $v_1, \dots, v_M \in V_+$ of augmentation zero mapping to a basis of A_Σ . For any desired packet $a \in A_\Sigma$, let $c_j(t)$ be the coordinates of $D_t^{-1}a$ in that basis and define

$$u_t(a) = \sum_{j=1}^M c_j(t) \theta_t(v_j). \quad (\text{CCB8.3})$$

Then, for every $t > 0$,

$$\boxed{j_\Sigma \bar{K}_b u_t(a) = a.} \quad (\text{CCB8.4})$$

Every basis element in $\theta_t(v_j)$ is still greater than one.

Put $\beta_\Sigma = \max_{\rho \in \Sigma} \Re \rho < 1$. Expanding the finite nilpotent inverse in (CCB8.1) gives the explicit bound

$$|c_j(t)| \leq C_a t^{-\beta_\Sigma} (1 + |\log t|)^{M_\Sigma}, \quad 0 < t \leq 1, \quad (\text{CCB8.5})$$

where one can take $M_\Sigma = \max_\rho (m_\rho - 1)$, increasing only the constant for the fixed basis.

If $v_j = \sum_\ell b_{j\ell} [x_{j\ell}]$, augmentation zero gives

$$\chi_\lambda(\theta_t v_j) = \sum_\ell b_{j\ell} (e^{\lambda t \log x_{j\ell}} - 1).$$

Every prescribed λ -derivative on a compact positive interval is $O(t)$; the proof is exactly the estimate in CCB7 with finitely many fixed $\log x_{j\ell}$. Consequently

$$\|\chi(u_t(a))\|_{K,k} \leq C_{a,K,k} t^{1-\beta_\Sigma} (1 + |\log t|)^{M_\Sigma} \longrightarrow 0. \quad (\text{CCB8.6})$$

Thus all character seminorms vanish while the entire finite jet packet stays exactly equal to a , with every multiplicity retained. Multiplication by any fixed smooth $h(y)$ and fixed moving frequency q^r preserves convergence in the character continuation topology: on each compact set all derivatives of its finitely many exponential factors are bounded.

138.10 CCB9. The full joint completion is an exact product of observations

First consider boundary coefficients alone. Let $\mathcal{H}$ be the completed character boundary space of CCB6, realized by its full y -dependent holomorphic extension. For each actual zero ρ , define $\mathcal{H}_\rho$ to be the complete Fréchet space of A_ρ -valued functions $H(g, z)$, smooth in z and the leaf direction, satisfying

$$\partial_z H = iDH, \quad (\text{CCB9.1})$$

with all compact-open derivative seminorms. Its finite Fourier elements are $\sum_r a_r e_r(g) e^{-2\pi r z}$. The compact-kernel and Fejér proof shows these elements are dense. Every rational frequency remains. The parameter z here realizes the geometric factor after the exact substitution $z = \lambda y$; no factor λ has been deleted.

On the original positive-coefficient finite boundary algebra define the joint map

$$v = \sum_r w_r q^r \longmapsto \left(\sum_r \chi_\lambda(w_r) e_r(g) e^{-2\pi \lambda r y}, \left(\sum_r j_\rho \bar{K}_b w_r e_r(g) e^{-2\pi r z} \right)_\rho \right). \quad (\text{CCB9.2})$$

Its target is

$$\mathcal{H} \times \prod_{\rho \in \mathcal{Z}} \mathcal{H}_\rho, \quad (\text{CCB9.3})$$

where $\mathcal{Z}$ is the full set of actual nontrivial zeros, each retained with its multiplicity algebra. Use the product topology, not a Hilbert norm or a sum norm.

The closure of (CCB9.2) is exactly all of (CCB9.3). To prove it, fix a finite zero packet Σ and finitely many frequencies. For any prescribed coefficients at those frequencies, apply (CCB8.3) separately to each frequency and sum. Its character component tends to zero while its finite packet stays exactly prescribed. Finite-frequency elements are dense in each $\mathcal{H}_\rho$, and every product neighborhood restricts only finitely many zero coordinates. Hence the closure contains the whole vertical subspace

$$\{0\} \times \prod_{\rho} \mathcal{H}_\rho.$$

The character projection of the source is dense in $\mathcal{H}$, by CCB1 restricted to the holomorphic coefficients. Subtracting the already contained vertical component from any source image places its character component with zero jet component in the closure. Density then gives $\mathcal{H} \times \{0\}$, proving the product assertion.

The same proof includes the entire smooth-radial algebra. Define $\mathcal{Y}_\rho$ by allowing a further smooth parameter y in (CCB9.1), with equation in g, z unchanged. Its differential and homotopy are $iy\partial_y$ and the exact integral at fixed z . The simultaneous completion of the source positive-coefficient algebra for its character-continuation seminorms and all these moving-jet seminorms is

$$\boxed{\mathcal{X} \times \prod_{\rho \in \mathcal{Z}} \mathcal{Y}_\rho.} \quad (\text{CCB9.4})$$

Indeed finite sums $h(y)a_re_r(g)e^{-2\pi rz}$ are dense in each finite target packet, by the same Fourier approximation and smooth coefficient approximation. CCB8 realizes them with vanishing character component; the preceding product proof applies without changing a sign or multiplicity.

Both components carry the original two-term differential and its homotopy. The product differential has closed image and the product homotopy proves

$$H^0 = H^1 = \mathcal{B}_{\text{ent}} \times \prod_{\rho} \mathcal{H}_{\rho}. \quad (\text{CCB9.5})$$

The arithmetic action on the character component is (CCB5.1). On $\mathcal{Y}_{\rho}$ it is exactly

$$H_{\rho}(g, y, z) \mapsto \mu^{\rho} \exp((\log \mu)T_{\rho}) H_{\rho}(g, y/\mu, z). \quad (\text{CCB9.6})$$

The geometric variable $z = \lambda y$ is unchanged because $(\mu\lambda)(y/\mu) = \lambda y$. Every Gaussian factor from the actual CW comparison remains in the source-to-jet map and commutes with this full nilpotent multiplier. Sign acts by $(g, y, z) \mapsto (-g, -y, -z)$. The homotopy commutes with these actions. Projecting by the exact averaging operator $(I + S)/2$ shows the same joint-density and product statements on the invariant subspaces, retaining the coefficient $1/2$ and both members of nontrivial sign orbits.

At frequency zero and without the radial variable, (CCB9.3) specializes to the particularly explicit result

$$\overline{\{(\chi(v), (j_{\rho}\overline{K}_bv)_{\rho}) : v \in V_{+}\}} = C^{\infty}((0, \infty)) \times \prod_{\rho} A_{\rho}. \quad (\text{CCB9.7})$$

Thus the character and complete product-jet observations become independent in this topology. This is a calculated property of the completion, not a proof that arbitrary product jets are realized in the original Q . CW explicitly distinguishes that quotient from the unrestricted product. The dense original graph and its completed product must not be substituted for one another in a weight argument.

For an actual- Q comparison one may instead retain Q 's own Fréchet seminorms as well as the character seminorms and take the closure of the actual graph. Concretely $Q = \mathcal{A}/I$ is complete because I is closed in the complete strong Mellin space; the graph closure in the corresponding product of complete character and Q -valued moving-function spaces is a complete receiving space with continuous projections. Its finite zero observations are the maps calculated above. No equality of that actual- Q graph closure with (CCB9.4) is asserted: the product-jet topology does not contain all the original quotient seminorms. The failure in CCB7 and the exact product calculation specify why those seminorms cannot be omitted while claiming the same comparison.

138.11 CCB10. Proven maps and provenance

The calculation supplies:

- The exact ordinary smooth completion and a sequence proving that it loses the radial lifting homotopy.
- A complete algebra containing all original signs, coefficient characters and rational frequencies, with an explicit faithful diagonal map and full continuation topology.
- The continuous original differential, radial homotopy, exact boundary cohomology and the retained cokernel of multiplication by y .
- The maximal smooth graph completion of the two-term lifting complex, with complete degree-by-degree domains and no assumed vanishing.
- A discontinuity witness for the actual original-zeta comparison and an exact calculation of the joint completion retaining every actual zero jet.

These computations concern named receiving spaces and their topologies. They neither assign cohomology to primitive Z_1/τ nor establish a numerical Deligne weight. The finite-valuation frame, the signed source quotient and every lattice-label comparison remain explicit. Each linear map extends to the already specified label receiver by $(v, \ell) \mapsto (f(v), \ell)$, retaining a label even when its amplitude vanishes; no new semimodule cohomology is inferred from the vector-space quotient.

Human source: Alain Connes and Caterina Consani, The Riemann–Roch strategy: Complex lift of the Scaling Site, arXiv:1805.10501v1. The original author TeX was read at §5.1 Lemma adelicomp, §5.2 actionpq and the adelic quotient, §5.4 classorb1/proetcov, and Remark additivestruct (lines 1707–1719), and §7.1 holom, holom1, holom2, functionq and perfectoid. These are bounded source passages, not a whole-paper reading claim.

Programme proofs read and used: `CLASSICAL_BOUNDARY_LIFTING.md` CB0–CB8 completely; `MOVING_Q_ZETA_COMPARISON.md` MQ0–MQ9; `CC_W_MELLIN_INDEPENDENT.md` CW0–CW7; `CORPUS_AND_OPERATION_RULES.md` completely and the three full user arguments named at the beginning. The actual-zeta interpolation, Gaussian factors and zero multiplicities are retained from CW and rederived where used in CCB7–CCB9. No completed zeta function replaces the original in this argument.

An independent mathematical check of CCB3–CCB5 verified the density, completeness, continuation topology, exact diagonal differential and action factors. The maximal graph completion and joint zeta-product calculation were derived here. The historical source of the original complex lift and arithmetic Frobenius remains Connes–Consani; the completed receiving spaces above are explicit derivations rather than attributed source assertions.

138.12 CCB11. The actual strong Mellin space and both character domains

The following calculation determines the actual- Q graph, rather than replacing Q by its zero-jet observations.

Retain the original strong Mellin space and all its seminorms

$$\begin{aligned} \mathcal{A} &= \{k \in C^\infty(\mathbb{R}_{>0}) : p_{N,j}(k) < \infty \text{ for every } N, j\}, \\ p_{N,j}(k) &= \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j k(u)|. \end{aligned} \tag{CCB11.1}$$

Its original transform and dilations are

$$F_k(s) = \int_0^\infty k(u) u^{s-1/2} \frac{du}{u}, \quad W_a k(u) = a^{1/2} k(u/a).$$

The original Gaussian is $b(u) = e^{-(\log u)^2}$, with

$$F_b(s) = B(s) = \sqrt{\pi} \exp((s - \tfrac{1}{2})^2/4). \tag{CCB11.2}$$

For $x = e^a > 1$, so $a > 0$, the source graph generator is exactly

$$\left(\lambda \mapsto e^{\lambda a}, W_a b \right). \tag{CCB11.3}$$

Its second coordinate includes $a^{1/2}$; it is not an unscaled Gaussian translate.

Write

$$\mathcal{E}_+ = C^\infty((0, \infty))$$

with smooth compact-open seminorms. Also retain the endpoint space

$$\mathcal{E}_0 = C^\infty([0, \infty)),$$

defined as functions smooth on $(0, \infty)$ whose derivatives of every order have continuous limits at zero, with seminorms

$$\|f\|_{N,m} = \max_{0 \leq j \leq m} \sup_{0 \leq \lambda \leq N} |f^{(j)}(\lambda)|. \quad (\text{CCB11.4})$$

These are complete Fréchet spaces: integration of the uniform derivative limits proves the derivative identities, including at zero.

The source character functions have the unique endpoint extension $\chi_0[x] = 1$. This section retains that extension and every derivative there. It does not identify the auxiliary character parameter zero with Z_0 , Z_1 , or τ .

In logarithmic coordinates $v = \log u$, (CCB11.1) is the space of smooth functions $g(v)$ whose derivatives decay faster than every fixed exponential. Multiplication by smooth cutoffs equal to one on $[-R, R]$ proves that $C_c^\infty(\mathbb{R})$ is dense in this logarithmic space. Explicitly, the error and all its derivatives are supported where $|v| \geq R$; a seminorm with exponent $N + 1$ bounds the exponent- N tail by a constant times e^{-R} , while cutoff derivatives contribute only bounded inverse powers of R . This also proves completeness of $\mathcal{A}$ by the same derivative-limit argument.

138.13 CCB12. Two exact analytic lemmas

138.13.1 The Laplace growth lemma

The Fourier convention in the two analytic lemmas of this section is $\widehat{\varphi}(\xi) = \int_{\mathbb{R}} \varphi(v) e^{-iv\xi} dv$, with inverse $\varphi(v) = (2\pi)^{-1} \int_{\mathbb{R}} \widehat{\varphi}(\xi) e^{iv\xi} d\xi$. The factor $1/(2\pi)$ is retained when pairing this inverse with a distribution.

Let T be a continuous linear functional on $\mathcal{E}_+$, or on $\mathcal{E}_0$, and put

$$L_T(z) = T(\lambda \mapsto e^{z\lambda}).$$

Continuity supplies a bound by finitely many derivative seminorms on one compact interval $[c, C]$, where $0 < c < C$ in $\mathcal{E}_+$, and $0 = c < C$ is allowed in $\mathcal{E}_0$. Thus for some integer m and constant A ,

$$|L_T(z)| \leq A(1 + |z|)^m \begin{cases} e^{C\Re z}, & \Re z \geq 0, \\ e^{c\Re z}, & \Re z \leq 0. \end{cases} \quad (\text{CCB12.1})$$

The map is entire: its derivative series and difference quotients converge in every compact character seminorm. In particular it has at most exponential growth of order one and has polynomial growth on the imaginary axis.

Suppose additionally that $L_T(t)$ has polynomial growth as $t \rightarrow +\infty$. Then L_T is a polynomial on the whole complex plane. Here is the complete sector argument. Choose an integer n at least as large as the polynomial exponents on the real and imaginary axes. In each right quadrant divide $L_T(z)$ by $(z + 1)^n$; in each left quadrant divide it by $(z - 1)^n$. The relevant denominator has no zero in that closed quadrant, and the quotient is bounded on both bounding rays.

For a quadrant with central argument θ , choose $1 < \alpha < 2$ and multiply this quotient by

$$\exp(-\epsilon e^{-i\alpha\theta} z^\alpha), \quad \epsilon > 0,$$

using the branch of z^α on that quadrant. On the whole sector,

$$\Re(e^{-i\alpha\theta} z^\alpha) \geq |z|^\alpha \cos(\alpha\pi/4) > 0.$$

On a sufficiently large circular boundary this decay dominates the order-one exponential bound in (CCB12.1). On the two rays the multiplier has modulus at most one. The maximum principle on truncated quadrants, followed by letting their radius tend to infinity, bounds the multiplied quotient by its ray bound. Letting $\epsilon \downarrow 0$ proves a bound on the unmultiplied quotient. Applying this argument in all four quadrants gives

$$|L_T(z)| \leq A'(1 + |z|)^n$$

on the plane. Cauchy's integral formula then makes every derivative of order greater than n vanish. This proves the polynomial assertion without a positivity assumption on T .

For $\mathcal{E}_+$, (CCB12.1) on the negative real axis gives exponential decay times a polynomial. A polynomial with that decay is zero. Hence in this case $L_T = 0$.

The functional conclusion is also exact. Restrict T to restrictions of test functions on the whole real line. Its compact derivative bound makes this a distribution supported on $[c, C]$. Fourier inversion shows that its transform $L_T(i\xi)$ determines it: Fourier inversion of a Schwartz test can be paired under the integral because the distribution has finite order and compact support. Thus $L_T = 0$ gives the zero distribution. Restrictions of globally smooth functions are dense in either character space: on any compact interval, approximate the highest derivative by polynomials and integrate, retaining the lower derivatives at one endpoint, as in CCB1. It follows that $T = 0$.

In the endpoint case, write the polynomial as $L_T(z) = \sum_{n=0}^N c_n z^n$. The functional $\sum c_n f^{(n)}(0)$ has exactly this transform. The same Fourier argument and density prove

$$\boxed{T(f) = \sum_{n=0}^N c_n f^{(n)}(0)}. \quad (\text{CCB12.2})$$

This proves that polynomial positive-axis growth leaves only the endpoint derivatives; none of the argument assumes T positive.

138.13.2 Totality of the original Gaussian dilations

The linear span of $\{W_a b : a > 0\}$ is dense in $\mathcal{A}$. To prove it, let ℓ be a continuous linear functional annihilating every such vector. With $r = \log a$, the logarithmic function is

$$W_a b = e^{r/2} \exp(-(v-r)^2).$$

The factor $e^{r/2}$ is nonzero, so $\ell(\exp(-(v-r)^2)) = 0$ for every real r .

The function of a complex parameter

$$z \mapsto \ell(\exp(-(v-z)^2))$$

is entire. On each compact z -set, every derivative and every seminorm of the integrand are bounded by a polynomial in v times $\exp(-v^2 + C|v| + C)$; this proves differentiability in the Fréchet topology and hence after applying ℓ . The identity theorem gives its vanishing for all complex z .

Define a tempered distribution by

$$S(\varphi) = \ell(e^{-v^2} \varphi(v)), \quad \varphi \in \mathcal{S}(\mathbb{R}).$$

The map $\varphi \mapsto e^{-v^2} \varphi$ is continuous into the logarithmic version of $\mathcal{A}$: after every derivative, its weighted seminorm is bounded by finitely many Schwartz seminorms because the Gaussian dominates every retained exponential weight.

For real ξ ,

$$e^{-v^2} e^{-i\xi v} = e^{-\xi^2/4} \exp(-(v + i\xi/2)^2).$$

Applying ℓ gives zero. This is the Fourier transform of S : for a Schwartz function of ξ , its integral against these Gaussian-weighted exponentials converges in every $\mathcal{A}$ -seminorm, since each derivative introduces only a fixed polynomial in ξ . The pairing may therefore be interchanged with that integral. Fourier inversion gives $S = 0$.

For $\varphi \in C_c^\infty$, the function $e^{v^2} \varphi$ is a Schwartz function, so $\ell(\varphi) = S(e^{v^2} \varphi) = 0$. Density of compactly supported smooth functions in $\mathcal{A}$, proved in CCB11, gives $\ell = 0$. Hahn–Banach separation of a proper closed linear subspace now proves the asserted totality.

138.14 CCB13. Full graph density before quotienting when $\lambda > 0$

For $v \in V_+$, let

$$\Gamma_+(v) = (\chi(v), K_b v) \in \mathcal{E}_+ \times \mathcal{A}.$$

Then

$$\overline{\Gamma_+(V_+)} = \mathcal{E}_+ \times \mathcal{A}. \quad (\text{CCB13.1})$$

Indeed a continuous linear functional on the product has the form $T(f) + \ell(k)$. If it annihilates the graph, (CCB11.3) gives

$$L_T(a) = -\ell(W_a b) \quad (a > 0). \quad (\text{CCB13.2})$$

Continuity of ℓ bounds it by finitely many of (CCB11.1). The exact dilation estimate is

$$p_{N,j}(W_a b) \leq a^{1/2} \max(a^N, a^{-N}) p_{N,j}(b). \quad (\text{CCB13.3})$$

It follows by substituting $u = av$, retaining $a^{1/2}$ and both powers $a^{\pm N}$. For $a \geq 1$, the right side has polynomial growth $a^{N+1/2}$. Thus (CCB13.2) satisfies the Laplace growth lemma, which gives $T = 0$. Then $\ell(W_a b) = 0$ for all $a > 0$, and Gaussian totality gives $\ell = 0$. A proper closed linear subspace of the locally convex product has a nonzero continuous linear annihilator. Therefore the graph is dense.

Both factors are complete, so (CCB13.1) identifies the completion for the joint character and original strong Mellin seminorms. It is a theorem about these exact observations and their topology. No zero count, zero simplicity, RH assumption or replacement of ζ is used.

138.15 CCB14. Retaining the endpoint restores every integer Mellin relation before quotienting

In $\mathcal{E}_0 \times \mathcal{A}$, the source graph satisfies, for every integer $n \geq 0$,

$$\partial_\lambda^n \chi_\lambda(v)|_{\lambda=0} = \frac{F_{K_b v}(n)}{B(n)}, \quad B(n) = \sqrt{\pi} \exp((n - \frac{1}{2})^2/4). \quad (\text{CCB14.1})$$

For a basis vector $[e^a]$, both sides equal a^n . Linearity proves the identity for every v . All factors of the Mellin transform remain.

Let

$$\mathcal{R} = \left\{ (f, k) \in \mathcal{E}_0 \times \mathcal{A} : f^{(n)}(0) = B(n)^{-1} F_k(n) \text{ for all } n \geq 0 \right\}.$$

Every displayed observation is continuous. For the Mellin coordinate at n , choose $N > |n - \frac{1}{2}|$; in logarithmic coordinates its absolute value is bounded by

$$p_{N,0}(k) \int_{\mathbb{R}} \frac{e^{(n-1/2)v}}{e^{Nv} + e^{-Nv}} dv,$$

whose retained integral is finite. Thus $\mathcal{R}$ is closed. In fact

$$\overline{\{(\chi(v), K_b v) : v \in V_+\}} = \mathcal{R}. \quad (\text{CCB14.2})$$

To prove the reverse inclusion, classify the annihilators. The same calculation as CCB13 gives polynomial positive-axis growth for L_T . The endpoint conclusion (CCB12.2) therefore gives $T(f) = \sum_{n=0}^N c_n f^{(n)}(0)$. On all Gaussian dilations,

$$\ell(W_a b) = - \sum_{n=0}^N c_n a^n = - \sum_{n=0}^N \frac{c_n}{B(n)} F_{W_a b}(n).$$

Gaussian totality then determines

$$\ell(k) = - \sum_{n=0}^N \frac{c_n}{B(n)} F_k(n)$$

on all of $\mathcal{A}$. Hence the entire annihilator of the graph consists precisely of finite linear combinations of the defining relations of $\mathcal{R}$. Hahn–Banach separation proves (CCB14.2).

This is the exact prequotient information recovered by including the endpoint. Merely saying that $\lambda = 0$ was outside the original positive family would fail to retain these available source identities. They are kept here as an explicit extension and comparison, without identifying $\lambda = 0$ with primitive τ .

138.16 CCB15. The original zeta image independently controls every finite integer packet

Retain

$$I = \overline{\mathcal{E}(S_0^{\text{even}})}^{\mathcal{A}}, \quad Q = \mathcal{A}/I, \quad \mathcal{E}f(u) = u^{1/2} \sum_{m \geq 1} f(mu),$$

where $f(0) = 0$ and $\int_{\mathbb{R}} f = 0$. We now prove that no nonzero finite combination of the integer Mellin observations $F_k(n)$, $n \geq 0$, vanishes on I .

It suffices to use the following actual source tests. Let $\varphi \in C_c^\infty((0, \infty))$ have integral zero, and extend it evenly to f on $\mathbb{R}$. Because its support stays away from zero, f is even Schwartz, $f(0) = 0$, and $\int_{\mathbb{R}} f = 2 \int_0^\infty \varphi = 0$.

Membership $\mathcal{E}f \in \mathcal{A}$, and hence in I , has a direct estimate. In this paragraph write

$$(\mathcal{F}_{2\pi} f)(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx.$$

The smooth periodization $\sum_{n \in \mathbb{Z}} f(u(x+n))$ has m -th Fourier coefficient $u^{-1}(\mathcal{F}_{2\pi} f)(m/u)$, by integrating over one period and substituting $v = u(x+n)$. Its Fourier series converges with all derivatives because these coefficients decay faster than every power. Evaluating at $x = 0$ proves the exact Poisson identity. Since f is even and $f(0) = (\mathcal{F}_{2\pi} f)(0) = 0$, it gives

$$\mathcal{E}f(u) = \frac{1}{2} u^{-1/2} \sum_{m \in \mathbb{Z} \setminus \{0\}} (\mathcal{F}_{2\pi} f)(m/u).$$

For $0 < u \leq 1$, its j -th Euler derivative is

$$(u \partial_u)^j \mathcal{E}f(u) = \frac{1}{2} u^{-1/2} \sum_{m \neq 0} \left[\left(-\frac{1}{2} - \xi \partial_\xi \right)^j \mathcal{F}_{2\pi} f \right] (m/u).$$

Each bracket is Schwartz. For every real $L > 1$, its modulus is at most

$$C_{j,L} \zeta(L) u^{L-1/2}, \quad \frac{1}{2} \sum_{m \neq 0} |m|^{-L} = \zeta(L).$$

Choose $L > N + 1/2$ for each original seminorm $p_{N,j}$. This controls the endpoint $u = 0$. For u exceeding the support radius of f , the original positive sum defining $\mathcal{E}f$ vanishes identically. On the intervening compact interval it and every derivative are smooth and bounded. Thus all seminorms (CCB11.1) are finite, with the factor $1/2$ and both signs retained throughout the comparison.

Define its entire Mellin factor

$$M_\varphi(s) = \int_0^\infty \varphi(u) u^{s-1} du.$$

The original identity is

$$F_{\mathcal{E}f}(s) = \zeta(s)M_\varphi(s). \quad (\text{CCB15.1})$$

It first follows by absolute convergence for $\Re s > 1$, then by analytic continuation; here M_φ is entire and $M_\varphi(1) = 0$.

For completeness the two endpoint constants can be recovered without a completed zeta function. For $\Re s > 1$,

$$\zeta(s) = s \int_1^\infty [u]u^{-s-1} du = \frac{1}{s-1} + \frac{1}{2} - s \int_1^\infty (\{u\} - \frac{1}{2})u^{-s-1} du. \quad (\text{CCB15.2})$$

The first equality follows by summing the integrals from each integer to infinity. Put $P(x) = \int_1^x (\{u\} - \frac{1}{2}) du = \frac{1}{2}(\{x\}^2 - \{x\})$. Thus $-1/8 \leq P(x) \leq 0$, and it vanishes at each positive integer. Integration by parts gives

$$\int_1^\infty (\{u\} - \frac{1}{2})u^{-s-1} du = (s+1) \int_1^\infty P(u)u^{-s-2} du,$$

whose right side is holomorphic for $\Re s > -1$. Thus the original formula gives $\zeta(0) = -1/2$ and the original pole at $s = 1$ has residue 1.

Consequently the full integer observations of (CCB15.1) are

$$\begin{aligned} F_{\mathcal{E}f}(0) &= -\frac{1}{2} \int_0^\infty \frac{\varphi(u)}{u} du, \\ F_{\mathcal{E}f}(1) &= \int_0^\infty \varphi(u) \log u du, \\ F_{\mathcal{E}f}(n) &= \zeta(n) \int_0^\infty \varphi(u) u^{n-1} du \quad (n \geq 2). \end{aligned} \quad (\text{CCB15.3})$$

The middle formula keeps the pole cancellation exactly:

$$\zeta(s) = \frac{1}{s-1} + a_0 + O(s-1), \quad M_\varphi(s) = (s-1)M'_\varphi(1) + O((s-1)^2),$$

so the constant product is $M'_\varphi(1)$. The term $a_0 M_\varphi(1)$ is zero because the actual source test has integral zero; it has not been silently omitted. For $n \geq 2$, $\zeta(n) > 0$ by its convergent positive Dirichlet series.

Suppose $\sum_{n=0}^N d_n F_k(n)$ vanishes on I . Then for every such φ ,

$$\int_0^\infty \varphi(u) \left(-\frac{d_0}{2u} + d_1 \log u + \sum_{n=2}^N d_n \zeta(n) u^{n-1} \right) du = 0, \quad (\text{CCB15.4})$$

with absent indices interpreted as absent terms. A smooth function annihilating every compact test of integral zero is constant. To verify this, choose a fixed compact test χ with integral one; applying the equation to $\psi - (\int \psi)\chi$ shows that its distribution equals the constant $\int \chi G$. Since G is smooth, it equals that constant pointwise.

No nonzero combination in parentheses in (CCB15.4) can be constant. Let $u \rightarrow \infty$: the highest positive polynomial power forces its coefficient to be zero; repeat for all such powers. The remaining logarithmic term forces $d_1 = 0$. The remaining function $-d_0/(2u)$ tends to zero, so the constant is zero and $d_0 = 0$. Since each $\zeta(n) \neq 0$, this proves every $d_n = 0$.

Equivalently, for every finite set $J \subset \mathbb{Z}_{\geq 0}$, the map

$$I \longrightarrow \mathbb{C}^J, \quad k \longmapsto (F_k(n))_{n \in J} \quad (\text{CCB15.5})$$

is onto: a proper subspace of this finite-dimensional target would have a nonzero annihilating functional, contradicting what was proved.

One may construct these finite tests directly. The functions $1, u^{-1}, \log u, u, \dots, u^{N-1}$, with only the needed indices retained, are linearly independent by the same calculation. There are finitely many positive sample points at which their evaluation matrix has full rank; otherwise all evaluation vectors would lie in a proper subspace, giving a nonzero linear relation between these functions. Choose smooth compact bumps of integral one around those points. For sufficiently small supports, their moment matrix is still invertible by continuity of its determinant. Solve that finite linear system with the first moment prescribed as zero and the other observations prescribed as desired. The resulting finite linear combination supplies φ , hence an actual $\mathcal{E}f \in I$, with the exact requested integer packet. This construction retains the zero, pole and every integer coefficient in (CCB15.3).

138.17 CCB16. The actual- Q graph is the full product, including the endpoint topology

Let $\pi_Q : \mathcal{A} \rightarrow Q$ be the original quotient. Both actual graph closures are

$$\overline{\{(\chi(v), \overline{K}_b v) : v \in V_+\}} = \mathcal{E}_+ \times Q, \quad (\text{CCB16.1})$$

$$\overline{\{(\chi(v), \overline{K}_b v) : v \in V_+\}} = \mathcal{E}_0 \times Q. \quad (\text{CCB16.2})$$

The character domain in each equality is the one displayed there. These are closures in the product of that domain's full Fréchet topology and Q 's actual quotient topology.

For (CCB16.1), apply the continuous surjection $\text{id}_{\mathcal{E}_+} \times \pi_Q$, namely $(f, k) \mapsto (f, \pi_Q k)$, to the dense graph in (CCB13.1). Every target point has a preimage, and an approximating source net maps to an approximating net in the target. This proves density.

For (CCB16.2), let $T(f) + \ell_Q(q)$ annihilate the graph and put $\ell_A = \ell_Q \pi_Q$. The endpoint annihilator calculation of CCB14 gives

$$T(f) = \sum_{n=0}^N c_n f^{(n)}(0), \quad \ell_A(k) = - \sum_{n=0}^N \frac{c_n}{B(n)} F_k(n).$$

Because ℓ_A factors through Q , it vanishes on I . CCB15 proves independence of every finite integer packet on I , and each $B(n) \neq 0$; hence every $c_n = 0$. Then $\ell_A = 0$, and surjectivity of π_Q gives $\ell_Q = 0$. Hahn–Banach separation proves (CCB16.2).

The source graph is injective because its character component is faithful on finite V_+ . Its completion for the two actual seminorm families is therefore the full product in the displayed formulas. This proves the actual- Q result globally, without inferring it from finite jets.

The full scaling maps are retained. On each product the action is

$$(f, k) \longmapsto (f(\mu \cdot), W_\mu k), \quad (f, q) \longmapsto (f(\mu \cdot), W_\mu q). \quad (\text{CCB16.3})$$

The estimate (CCB13.3) proves continuity on $\mathcal{A}$. The original I is W_μ -invariant by the exact formula

$$W_\mu \mathcal{E}f(u) = \mu^{1/2} (u/\mu)^{1/2} \sum_{m \geq 1} f(mu/\mu) = u^{1/2} \sum_{m \geq 1} f_\mu(mu) = \mathcal{E}f_\mu(u), \quad f_\mu(v) = f(v/\mu).$$

Here f_μ is even Schwartz, $f_\mu(0) = 0$, and $\int_{\mathbb{R}} f_\mu = \mu \int_{\mathbb{R}} f = 0$. Continuity extends this to the closure I , and applying the inverse dilation proves $W_\mu I = I$. Thus the action descends to Q . The exact endpoint relations before quotienting are preserved:

$$\partial_\lambda^n (f(\mu\lambda))|_{\lambda=0} = \mu^n f^{(n)}(0), \quad F_{W_\mu k}(n) = \mu^n F_k(n).$$

Thus no source factor or group action is discarded in either completion.

The comparison has now identified precisely where the original integer-moment relation resides:

$$\begin{array}{ccc}
& \text{source finite graph} & \\
& \downarrow & \\
\mathcal{R} \subset \mathcal{E}_0 \times \mathcal{A} & \text{with all relations } f^{(n)}(0) = B(n)^{-1}F_k(n) & (\text{CCB16.4}) \\
& \downarrow (f, k) \mapsto (f, \pi_Q k) & \\
& \mathcal{E}_0 \times Q & \text{as the closure of its image.}
\end{array}$$

No claim that the middle-to-bottom map is surjective before closure is needed or made. Its image is dense by (CCB16.2). The original subspace I supplies arbitrary finite changes of those integer observations by (CCB15.5), which is the exact reason no nonzero continuous endpoint coupling survives in that quotient graph completion.

138.18 CCB17. Consequences and proof scope of the actual-quotient calculation

CCB13–CCB16 compute the coefficient case of the actual- Q graph described at the end of CCB9, using the actual original strong Mellin space and the actual Q . The separately proved moving-function product-jet calculation remains as stated there; no unproved tensor-completion identification is needed for the coefficient theorem.

The source’s integer-moment identities have not been declared irrelevant. They are explicitly present in the complete prequotient object $\mathcal{R}$, with every Gaussian factor, and their exact relation to the original zeta-image subspace is computed in (CCB15.3)–(CCB15.5). Passing to the stated completed quotient graph makes the character and quotient coordinates independent. That is a proved statement about this operation and topology; it is not a purity theorem, an identification of $\lambda = 0$ with primitive τ , or a reason to discard $\mathcal{R}$ from the programme.

The completion maps are actual continuous maps and the scalar source graph remains faithful before completion. The new full-product statement cannot be substituted for the source’s original algebraic image when proving an arithmetic constraint. Conversely it can be used to identify exactly which constraints a proposed completion argument has lost, by checking the retained middle object in (CCB16.4).

An independent mathematical review read CCB11–CCB17 in full and verified the quadrant growth estimate, original dilation factor, Gaussian-weighted distribution proof, endpoint annihilator classification, and actual-zeta integer observations. Its requested membership check for the compact source tests was supplied by the complete periodization and seminorm calculation now included in CCB15. No core mathematical error was found in that review.

Human provenance remains as in CCB10 and CW: Connes–Consani supply the original moving functions, character action, arithmetic Frobenius and zeta receiving programme. The Gaussian transform and full quotient definitions are those retained in CW. The sector estimate, Gaussian-totally proof, graph-closure classification, original-zeta endpoint calculation and finite moment independence are proved in full in CCB11–CCB16. Fourier inversion and Hahn–Banach are used in their usual complex locally convex forms, with their exact applications stated. No zero-count theorem, RH assumption, positivity assumption on a dual functional, or completed-zeta replacement enters these new graph-density proofs.

This reader contains the complete derivations of the whole adelic prime-boundary complex, its equivariant comparison with the Connes–Consani coefficient sheaf, and the continuous separation of all endpoint extensions of the original zeta quotient. The actual source distinction between primitive Z_1/τ and the parity-bearing integer 1 is retained by the explicit closed coefficient copies. Primitive τ is not assigned addition, parity, a numerical coordinate, or a weight.

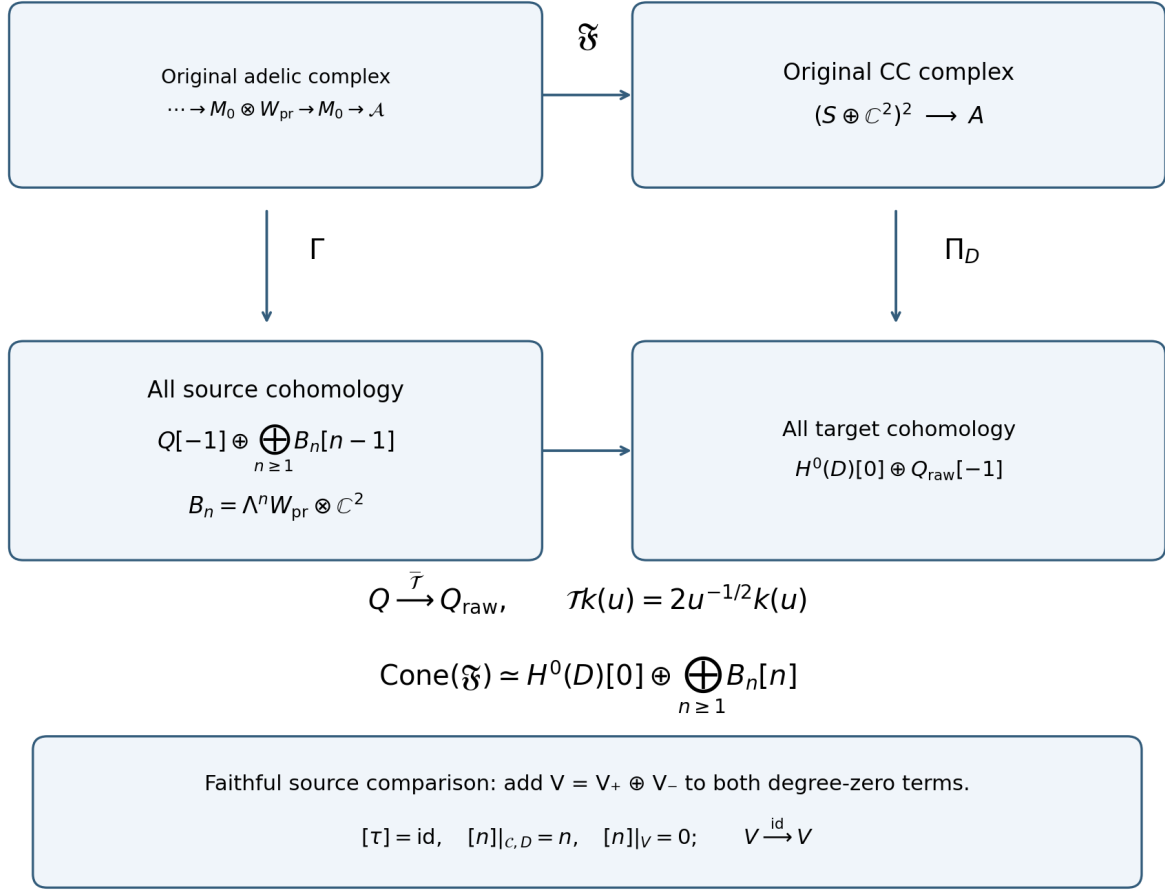
The active target remains the requested Deligne lifting theorem for the full τ programme. The calculations below prove global endpoint-extension vanishing and the entire displayed comparison triangle. They do not prove the remaining numerical purity statement or RH. In particular the non-endpoint supported quotient is retained: the separation polynomial is invertible on it, and cannot be called zero there.

The complete user definitions, arguments and corrections are retained in the private corpus. GAP0 and GEX0 identify the exact argument records consulted and the receiving domains of every operation. The user-originating global reconstruction argument is not replaced by an assumption that a restricted receiving test space is the whole source geometry.

The human constructions used here are Alain Connes’s adelic periodization; Alain Connes, Caterina Consani and Matilde Marcolli’s adelic Weil programme; Alain Connes and Caterina Consani’s original two-chart sheaf; Ralf Meyer’s original spectral-function spaces; and Pierre Deligne’s local invariant-cycle and weight arguments. Exact source versions, proof locators, prior programme credit and original factors are given in the proofs that follow. A local source-reading ledger distinguishes author TeX from the retained French transcription of Deligne.

The complete arguments appear in this order: GAP, the entire global comparison; GER, the original analytic inverse and all endpoint corrections; GEX, the unique section and full extension groups; FDB, an independent calculation of all prime degrees and signs; CTF, an independent derivation of the entire comparison triangle and its faithful coefficient extension. Earlier proof dependencies are retained in the same local repository, with their original sources and version records. Nothing in this reader is a remote-publication receipt.

One comparison retaining every prime degree



GAP2–GAP6B and CTF: exact maps, degree shifts and cone homotopies.
 CC sheaf: Alain Connes and Caterina Consani, arXiv:0903.2024v3, §5.

Figure 50: The complete comparison, with all prime-boundary degrees

Figure 1. GAP2–GAP6B construct all objects and maps, including the full ordered-prime exterior terms, the cohomology projections, Fourier orientation and the faithful closed coefficient copies. The extra identity-cone is retained and explicitly contracted. This diagram is about constructed receiving complexes; it does not assign a vector structure to primitive τ .

The entire original quotient: exact endpoint lifting

$$0 \longrightarrow B \longrightarrow E \xrightarrow{\pi} Q \longrightarrow 0$$

Endpoint modules
 $L_B = \text{id} \otimes \text{diag}(0, 1)$
 $P(L_B) = 0$

Original quotient
 $Q = \mathcal{A}/\mathcal{E}S$
 $P(L_Q)K_Q = K_Q P(L_Q) = \text{id}$

$$P(s) = s(s - 1)$$

$$\mathcal{K}F(s) = \frac{F(s) - 8((1 - s)F(0) + sF(1))F_*(s)}{s(s - 1)}$$

$$F_*(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad F_*(0) = F_*(1) = \frac{1}{8}$$

Canonical lift, independent of the chosen preimage

$$S(q) = P(L_E)e, \quad \pi(e) = K_Q q$$

Both endpoint corrections and the full original multiplier remain.

GER2–GER7 give continuous original-space inverses and Schwartz corrections.
 GEX4–GEX8 prove uniqueness, equivariance and the stated Ext vanishing.
 Deligne, Weil II §§3.6.1–3.6.3 is the geometric target; weight one is not inferred here.

Figure 51: The exact global separation operator and lifting formula

Figure 2. GER2–GER7 construct the inverse on the actual original quotient and its Schwartz return. GEX4–GEX8 prove the lift, its continuity, full equivariance and all-degree extension vanishing in the stated categories. The full original-zeta multiplier in the correction is retained. DCP’s different supported Q -term has the same invertible polynomial action as the target; it is not an annihilated endpoint module.

139 The complete adelic boundary and its oriented source comparison

24 September 2026. GAP0–GAP9. This calculation continues the actual source-coefficient comparison, retaining every derived prime-boundary degree. The independent companion proofs are `COMPLETE_GENERATOR_ENDPOINT_RETURN.md`, `GLOBAL_ENDPOINT_EXTENSION_RETURN.md` and `FULL_DERIVED_PRIME_BOUNDARY_REVIEW.md`. No full τ purity theorem is asserted.

139.1 GAP0. Source definitions, prerequisites and provenance

The complete user arguments `USR-9ad1c0a2d09dba92`, `USR-f55d16feb948d8c2` and `USR-6152e3bc6302258c`, and the current correction rule `USR-4322be19bff532cd`, were reread with B1–B5 and P1–P5. They remain in the complete private corpus, with pasted assistant passages distinguished from user arguments. The full arithmetic reconstruction precedes the coefficient operations below. The notation remains Z_0 for absence, Z_1 for primitive presence τ without Z_2 parity, and the specified integer data with their Z_2 parity. No source addition on τ is introduced.

The group algebras, vector spaces, exterior powers, derivatives, quotients and Fourier transforms below belong to the already constructed arithmetic receiving spaces. Their relation to the source is the explicit DCP coefficient comparison. A statement about a selected receiving boundary is not automatically a statement about the entire τ geometry.

Human sources: Alain Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, and Alain Connes, Caterina Consani and Matilde Marcolli, *The Weil proof and the geometry of the adèles class space*, for the original adelic construction. The exact received calculation is ABR1–ABR30 in `CC_ADELIC_COINVARIANT_BRIDGE_INDEPENDENT.tex`; its basis, moments, resolution and connecting representatives are proved there. Jean-Louis Koszul’s resolution is used with its displayed differential, whose needed exactness is also recalled below. The original Connes–Consani sheaf is *Schemes over $\mathbb{F}_1$ and zeta functions*, §5: the source restrictions, Fourier graph, original dilation and mirror actions were reread in author TeX lines 1521–1667. The receiving source double and its faithful scalar extension are DCP1–DCP12.

Pierre Deligne’s *La conjecture de Weil. II*, §§3.6.1–3.6.3, is the target comparison. The entire local invariant-cycle argument was reread in the retained French transcription, not represented as original author TeX. The present calculation supplies the full adelic maps and endpoint separation; it does not import his geometric weight bounds as assumptions.

139.2 GAP1. The original receiving objects and all endpoint factors

Keep

$$G = \mathbb{Q}_{>0}^\times, \quad R = \mathbb{C}[G], \quad \varepsilon(t_a) = 1, \quad I_R = \ker \varepsilon, \quad H = \mathcal{S}_{\text{even}}(\mathbb{R}; \mathbb{C}),$$

$$m(h) = \left(h(0), \int_{\mathbb{R}} h(v) dv \right), \quad M_0 = \ker(\varepsilon \otimes m : R \otimes H \rightarrow \mathbb{C}^2), \quad S = \ker m. \quad (\text{GAP1.1})$$

Every tensor sum is algebraic. Let

$$W_{\text{pr}} = \bigoplus_{p \text{ prime}} \mathbb{C}\ell_p, \quad x_p = t_p - 1, \quad B_n = \Lambda^n W_{\text{pr}} \otimes \mathbb{C}^2 \quad (n \geq 1). \quad (\text{GAP1.2})$$

The ordering in each wedge is the increasing order of the original primes. No prime is replaced by an unlabelled dimension.

The actual strong Mellin space is

$$\mathcal{A} = \left\{ k \in C^\infty(\mathbb{R}_{>0}) : \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j k(u)| < \infty \text{ for every } N, j \geq 0 \right\}.$$

Use all these seminorms and the original functions

$$\mathcal{E}h(u) = u^{1/2} \sum_{r \geq 1} h(ru), \quad F_k(s) = \int_0^\infty k(u) u^{s-1/2} \frac{du}{u}, \quad Q = \mathcal{A}/\mathcal{E}S. \quad (\text{GAP1.3})$$

OMS proves that $\mathcal{E}S$ is closed on these original domains. Its original zeta relation is

$$F_{\mathcal{E}h}(s) = \zeta(s) \int_0^\infty h(v) v^{s-1} dv \quad (\Re s > 1), \quad (\text{GAP1.4})$$

continued with every endpoint and trivial-zero term as in OMS3–OMS4.

For $c = \sum_a t_a \otimes h_a \in M_0$, set

$$\Phi c = \sum_a h_a \in S, \quad Jc = 2\mathcal{E}\Phi c. \quad (\text{GAP1.5})$$

The factor 2 is the two-sign factor in the actual adelic sum. It is retained throughout. For every $a > 0$,

$$\mathcal{R}_a(t_b \otimes h) = t_b \otimes h(\cdot/a), \quad W_a k(u) = a^{1/2} k(u/a). \quad (\text{GAP1.6})$$

Then $J\mathcal{R}_a = W_a J$, by direct substitution. On endpoint coordinates the action is $D_a = \text{diag}(1, a)$: evaluation at zero is fixed and the integral changes by the retained factor a .

139.3 GAP2. One complex containing every derived prime degree

Define a cochain complex $\mathcal{C}$ by

$$\mathcal{C}^1 = \mathcal{A}, \quad \mathcal{C}^{-k} = M_0 \otimes \Lambda^k W_{\text{pr}} \quad (k \geq 0), \quad \mathcal{C}^j = 0 \quad (j \geq 2). \quad (\text{GAP2.1})$$

Its differential in degree zero is J . For $k \geq 1$, its differential is the full Koszul differential

$$d(c \otimes \ell_{p_1} \wedge \cdots \wedge \ell_{p_k}) = \sum_{i=1}^k (-1)^{i-1} x_{p_i} c \otimes \ell_{p_1} \wedge \cdots \widehat{\ell_{p_i}} \cdots \wedge \ell_{p_k}. \quad (\text{GAP2.2})$$

The products lie in M_0 . Two successive removals of distinct wedge indices have opposite signs, so $d^2 = 0$ in negative degrees. In degree -1 , $\Phi(x_p c) = 0$, since $\varepsilon(x_p) = 0$; hence $Jd = 0$. These checks establish the whole cochain complex, including the join at degrees $-1, 0, 1$.

For completeness, the underlying complex $R \otimes \Lambda^\bullet W_{\text{pr}}$ resolves the augmentation module $\mathbb{C}$. For a finite set of primes, adjoining one variable $t_p - 1$ is the cone of its multiplication on the preceding Koszul complex. The preceding degree-zero quotient is a Laurent polynomial algebra in the remaining variables; multiplication by the next $t_p - 1$ is injective. Induction gives exactness in positive degrees. Every cycle and every coefficient uses a finite set of primes, so it is already a boundary in a finite such complex. Degree zero is the quotient setting every $t_p = 1$. This proves the infinite-prime statement without completing the exterior algebra.

Tensor the exact sequence

$$0 \rightarrow M_0 \rightarrow R \otimes H \xrightarrow{\varepsilon \otimes m} \mathbb{C}^2 \rightarrow 0$$

with each free resolution degree. The middle complex has only homology H in degree zero; the last has zero differential and terms $\Lambda^k W_{\text{pr}} \otimes \mathbb{C}^2$. The connecting sequence gives the ABR groups

$$H^1(\mathcal{C}) = Q, \quad H^{1-n}(\mathcal{C}) = B_n \quad (n \geq 1). \quad (\text{GAP2.3})$$

For $n = 1$, this uses $\ker J/I_R M_0 = B_1$, with the exact section in AC1. For $n \geq 2$, it is the connecting isomorphism just computed. OMS7A identifies the degree-one quotient with the same Q , rather than a potentially different algebraic quotient.

139.4 GAP3. The canonical whole-complex comparison

Define the original arithmetic derivations

$$D_p \left(\sum_a c_a t_a \right) = \sum_a c_a v_p(a).$$

They satisfy

$$D_p(rs) = \varepsilon(r)D_p(s) + D_p(r)\varepsilon(s), \quad D_p(x_q) = \begin{cases} 1 & p = q, \\ 0 & p \neq q. \end{cases} \quad (\text{GAP3.1})$$

Each input has only finitely many nonzero D_p . At degree $-k$, write a coefficient of a fixed wedge as $c = \sum_i r_i \otimes h_i \in M_0$ and put

$$\Gamma^{-k}(c \otimes \omega) = \frac{1}{k+1} \sum_{p,i} D_p(r_i)(\ell_p \wedge \omega) \otimes m(h_i) \in B_{k+1}. \quad (\text{GAP3.2})$$

This is a well-defined linear map on the indicated tensor product: its formula is bilinear in the original factors, and the finite sum of derivations defines a linear map on R . In degree one put $\Gamma^1 = \pi_Q$.

Let $\mathcal{H}$ have zero differential, degree-one term Q , and degree $1-n$ term B_n . Then $\Gamma : \mathcal{C} \rightarrow \mathcal{H}$ is a chain map. At the join $\pi_Q J = 0$. For a negative-degree differential, the derivation of $x_p c$ is its p -th augmentation moment. All terms are multiples of $\sum_i \varepsilon(r_i)m(h_i) = 0$, the defining condition of M_0 . Explicitly, moving the removed ℓ_{p_i} back to its ordered position cancels the Koszul sign and gives k equal wedge terms times this zero moment. Thus $\Gamma d = 0$.

To compute its map on every cohomology group, choose a section of the two moments only to write a representative:

$$h_0(v) = (1 - 2\pi v^2)e^{-\pi v^2}, \quad h_1(v) = 2\pi v^2 e^{-\pi v^2}, \quad \sigma(u_0, u_1) = u_0 h_0 + u_1 h_1.$$

The Gaussian integrals give $m\sigma = \text{id}$. For an ordered n -wedge $\omega = \ell_{p_1} \wedge \cdots \wedge \ell_{p_n}$, the connecting cycle is

$$z_{\omega,u} = \sum_{i=1}^n (-1)^{i-1} (x_{p_i} \otimes \sigma(u)) \otimes \ell_{p_1} \wedge \cdots \wedge \widehat{\ell_{p_i}} \cdots \wedge \ell_{p_n}. \quad (\text{GAP3.3})$$

It is a cycle because it is the differential of $(t_1 \otimes \sigma(u)) \otimes \omega$ in the middle resolution. Changing σ changes it by a boundary in the M_0 complex. Each of its n terms maps under (GAP3.2) to $(1/n)\omega \otimes u$. Hence

$$\Gamma^{1-n}[z_{\omega,u}] = \omega \otimes u. \quad (\text{GAP3.4})$$

The factor $1/n$ is the explicit inverse to those n equal receiving terms; no exterior term is discarded. Equations (GAP2.3)–(GAP3.4), and the identity on Q , prove a canonical quasi-isomorphism on all degrees. With the convention $V[r]^j = V^{j+r}$, it is

$$\mathcal{C} \xrightarrow{\Gamma} Q[-1] \oplus \bigoplus_{n \geq 1} B_n[n-1]. \quad (\text{GAP3.5})$$

Each degree has its one displayed term. No convergence or interchange of infinite-degree sums is used.

The comparison commutes with every $\mathcal{R}_a$, acting on B_n by $\text{id}_{\Lambda^n W_{\text{pr}}} \otimes D_a$. The identity follows from $m(R_a h) = D_a m(h)$ in (GAP3.2). Full diagonal rational action t_b on the source has trivial action on the target: its extra derivative term is $D_p(t_b) \sum_i \varepsilon(r_i)m(h_i) = 0$. This is not a conflation of diagonal rational action with the independent spectral real dilation.

139.5 GAP4. Full Fourier action, including the prime-wedge coefficients

Use the original adelic additive character with finite conductor $\widehat{\mathbb{Z}}$, finite measure $\text{vol}(\mathbb{Z}_p) = 1$, and real transform $\widehat{h}(\xi) = \int_{\mathbb{R}} h(v) e^{-2\pi i v \xi} dv$. For $c_a = 1_{a\widehat{\mathbb{Z}}}$, its finite Fourier transform is $a^{-1}c_{a^{-1}}$. Indeed integrating a character on that compact subgroup is its Haar volume if the character is trivial, and zero otherwise by translating by an element where it is nontrivial. The annihilator is $a^{-1}\widehat{\mathbb{Z}}$, and its original volume is a^{-1} .

The real transform of $h(v/a)$ is $a\widehat{h}(a\xi)$. Consequently the complete factors in ABR's Θ give

$$\mathcal{F}\Theta(t_a \otimes h) = a a^{-1} \Theta(t_{a^{-1}} \otimes \widehat{h}) = \Theta(t_{a^{-1}} \otimes \widehat{h}). \quad (\text{GAP4.1})$$

The displayed cancellation is justified by the exact original integral maps. Both Haar factors have been calculated.

Write $\iota(t_a) = t_{a^{-1}}$. For an ordered prime set P of cardinality k , put

$$\mathcal{F}_k((r \otimes h) \otimes \ell_P) = (-1)^k \left(\prod_{p \in P} t_p^{-1} \right) \iota(r) \otimes \widehat{h} \otimes \ell_P. \quad (\text{GAP4.2})$$

This preserves M_0 : augmentation of the displayed monomial is one, and Fourier swaps the two endpoint coordinates. It commutes with (GAP2.2), because $\iota(x_p) = -t_p^{-1}x_p$; each removed wedge factor contributes exactly that multiplier. Applying it twice gives identity: inversion cancels the monomial, $(-1)^{2k} = 1$, and the square of the real transform is identity on even functions.

In degree one put $\mathcal{F}_A k(u) = k(1/u)$. The original Poisson identity, with $h(0) = \int h = 0$, gives $J\mathcal{F}_0 = \mathcal{F}_A J$. Thus these operators give an involutive chain map on $\mathcal{C}$.

Let $S_2(u_0, u_1) = (u_1, u_0)$. Under (GAP3.2), the action on B_n is

$$\Gamma^{1-n} \mathcal{F}_{n-1} = (-1)^n (\text{id}_{\Lambda^n W_{\text{pr}}} \otimes S_2) \Gamma^{1-n}. \quad (\text{GAP4.3})$$

To prove it, $D_p \iota(r) = -D_p(r)$ supplies one minus sign. The other $n-1$ signs are in (GAP4.2). Differentiating the product of t_p^{-1} instead gives a multiple of the total augmentation moment, which is zero. The remaining moment is exactly $m(\widehat{h}) = S_2 m(h)$. This retains every finite monomial and wedge sign.

The original relation remains

$$W_a \mathcal{F}_A = a \mathcal{F}_A W_{a^{-1}}, \quad D_a S_2 = a S_2 D_{a^{-1}}. \quad (\text{GAP4.4})$$

No half-weight twist replaces this action.

139.6 GAP5. The exact map to the Connes–Consani coefficient complex

Use DCP's raw space A , with

$$\Sigma h(u) = 2 \sum_{r \geq 1} h(ru), \quad R_A b(u) = u^{-1} b(1/u), \quad \mathcal{T} : \mathcal{A} \rightarrow A, \quad \mathcal{T} k(u) = 2u^{-1/2} k(u). \quad (\text{GAP5.1})$$

Its inverse is $\mathcal{T}^{-1} b(u) = \frac{1}{2} u^{1/2} b(u)$. Weighted Euler derivative estimates prove both are continuous on all defining seminorms, by the product rule and an increased integer weight index. They satisfy $\mathcal{T}\mathcal{E} = \Sigma$, $\mathcal{T}W_a = T_a \mathcal{T}$, and $\mathcal{T}\mathcal{F}_A = R_A \mathcal{T}$.

The original Čech complex is

$$\begin{aligned} D^0 &= V_+ \oplus V_-, \quad V_{\pm} = S \oplus \mathbb{C}^2, \quad D^1 = A, \\ d_D((h, c_0, c_1), (g, d_0, d_1)) &= \Sigma h - R_A \Sigma g. \end{aligned} \quad (\text{GAP5.2})$$

Its mirror swaps chart coordinates in degree zero and is $-R_A$ in degree one. The latter minus sign is the orientation of the ordered two-chart cover.

Record that orientation in the source by the one-dimensional receiving sign representation: keep the same spaces, differential and dilations of $\mathcal{C}$, but let its mirror be $-\mathcal{F}_k$ in degree $-k$ and $-\mathcal{F}_A$ in degree one. Denote this oriented representation by $\mathcal{C}^{\text{or}}$. It changes neither τ nor the source arithmetic; it explicitly supplies the sign required by the ordered Čech comparison. Its square is identity and it still commutes with the differential.

There is now the actual chain map

$$\begin{aligned}\mathfrak{F}^1(k) &= \mathcal{T}k, \\ \mathfrak{F}^0(c) &= ((\Phi c, 0, 0), (-\widehat{\Phi}c, 0, 0)), \\ \mathfrak{F}^{-k} &= 0 \quad (k \geq 1).\end{aligned}\tag{GAP5.3}$$

It preserves the differential: Poisson gives $R_A \Sigma \widehat{h} = \Sigma h$, so

$$d_D \mathfrak{F}^0(c) = \Sigma \Phi c + R_A \Sigma \widehat{\Phi} c = 2\Sigma \Phi c = \mathcal{T}(2\mathcal{E}\Phi c) = \mathfrak{F}^1 Jc.\tag{GAP5.4}$$

On a negative-degree boundary $\Phi d = 0$, proving the other chain identities. Fourier scaling proves equivariance in degree zero: $\widehat{h}(\cdot/a) = a\widehat{h}(a\cdot)$, which is exactly DCP's minus-chart action. Equation (GAP5.1) gives degree-one equivariance.

For the mirror, $\Phi \mathcal{F}_0 = \widehat{\Phi}$ and $\widehat{\widehat{h}} = h$ give

$$\mathfrak{F}^0(-\mathcal{F}_0 c) = ((-\widehat{\Phi}c, 0, 0), (\Phi c, 0, 0)),$$

the chart swap of $\mathfrak{F}^0 c$. In degree one, $\mathcal{T}(-\mathcal{F}_A) = -R_A \mathcal{T}$. Thus every sign in the comparison is proved.

The source of (GAP5.3) can also be checked by averaging. The plus-chart map has degree-zero component $((2\Phi c, 0, 0), 0)$ and degree-one component $\mathcal{T}$. Conjugating by both mirrors gives degree-zero component $(0, (-2\widehat{\Phi}c, 0, 0))$ and the same degree-one component. Their sum times the explicit coefficient $1/2$ is (GAP5.3). This explains the orientation adjustment without discarding the original factor two.

On cohomology, $H^1(\mathfrak{F})$ is exactly the isomorphism induced by $\mathcal{T}$ on the original quotient. In every degree at most zero its map is zero: negative target degrees vanish, and a degree-zero source cycle has $\Phi c = 0$ by injectivity of $\mathcal{E}$. Its representative therefore maps to zero, not merely to an unspecified coboundary.

139.7 GAP6. The complete comparison defect, with its connecting signs

Do not replace the preceding quasi-isomorphism in degree one by an equivalence of the entire complexes. Define the actual cochain cone

$$\mathcal{K}^j = D^j \oplus (\mathcal{C}^{\text{or}})^{j+1}, \quad d_{\mathcal{K}}(d, c) = (d_D d + \mathfrak{F}c, -d_{\mathcal{C}} c).\tag{GAP6.1}$$

The chain identity in GAP5 proves its differential squares to zero. Its long exact sequence, with the computed cohomology maps, gives

$$\begin{aligned}H^0(\mathcal{K}) &= H^0(D) = \{(h, \widehat{h}) : h \in S\} \oplus \mathbb{C}^4, \\ H^{-n}(\mathcal{K}) &= B_n \quad (n \geq 1), \quad H^j(\mathcal{K}) = 0 \quad (j \geq 1).\end{aligned}\tag{GAP6.2}$$

The degree-zero map is induced by $d \mapsto (d, 0)$. For the negative group, projection $(d, c) \mapsto c$ induces the indicated connecting isomorphism into $H^{1-n}(\mathcal{C})$. In degree -1 , c is a degree-zero cycle and $\mathfrak{F}^0 c = 0$, so $(0, c)$ is its explicit cone cycle. For lower degrees the target D^j is zero, and

the same representative works. This verifies the formula without choosing an unproved splitting of the entire cone.

The boundary action in $\mathcal{C}^{\text{or}}$, and hence in these negative cone groups, is

$$\text{id}_{\Lambda^n W_{\text{pr}}} \otimes D_a, \quad w|_{B_n} = (-1)^{n+1} \text{id}_{\Lambda^n W_{\text{pr}}} \otimes S_2. \quad (\text{GAP6.3})$$

All prime labels and endpoint coordinates survive. The top Fourier graph and all four original endpoint lines also survive in $H^0(\mathcal{K})$.

139.8 GAP6A. The entire comparison triangle has an explicit cohomology projection

The following strengthens GAP6: it constructs the full comparison, including a quasi-isomorphism for the cone, rather than only listing that cone's groups. It does not construct a section of the quotient map from the original test space to its spectral quotient.

For $v = ((h, c_0, c_1), (g, d_0, d_1)) \in D^0$, define

$$\Pi_0 v = \left(\left(\frac{h + \widehat{g}}{2}, c_0, c_1 \right), \left(\frac{\widehat{h} + g}{2}, d_0, d_1 \right) \right). \quad (\text{GAP6A.1})$$

Both source functions belong to the original S . Fourier is a continuous involution on that space, so this is continuous. Its image belongs to $H^0(D) = \ker d_D$, because the first function is the Fourier transform of the second. On that kernel it is identity. In particular, it retains all four endpoint coordinates.

The complementary component is exactly

$$v - \Pi_0 v = ((a, 0, 0), (-\widehat{a}, 0, 0)), \quad a = \frac{h - \widehat{g}}{2}. \quad (\text{GAP6A.2})$$

The displayed coefficient $1/2$ retains both original chart contributions. This is an explicit projection in the receiving complex vector space, not an additive operation on primitive Z_1/τ .

Let $\mathcal{H}_D$ have $H^0(D)$ in degree zero, Q_{raw} in degree one, and zero differential. Put

$$\Pi_D^0 = \Pi_0, \quad \Pi_D^1 = \pi_{Q_{\text{raw}}}. \quad (\text{GAP6A.3})$$

It is a chain map because $\pi_{Q_{\text{raw}}} d_D = 0$. Its cohomology maps are identity in degree zero and the original quotient in degree one; hence it is a quasi-isomorphism. Both maps are continuous in the original topologies. They intertwine real dilation: for the minus-chart function, the original action is $g \mapsto ag(a \cdot)$, and its Fourier transform is $\widehat{g}(\cdot/a)$, precisely the plus-chart action. The same equality verifies the second coordinate. Interchanging the two charts commutes with (GAP6A.1). In degree one the quotient retains the original oriented mirror $-R_A$. Thus all claimed actions are preserved without changing their factors.

Use Γ from GAP3 with its oriented actions from GAP5. Define

$$\mathcal{H}_F : \mathcal{H} \longrightarrow \mathcal{H}_D$$

to be $\overline{\mathcal{T}} : Q \rightarrow Q_{\text{raw}}$ in degree one and zero in all other degrees. Then there is an exact equality of cochain maps

$$\Pi_D \mathfrak{F} = \mathcal{H}_F \Gamma. \quad (\text{GAP6A.4})$$

In degree zero, (GAP5.3) is the anti-Fourier pair in (GAP6A.2), so its projection is zero. In degree one the equality is the definition of the quotient map induced by $\mathcal{T}$. Every negative target term is zero. These checks prove (GAP6A.4) degree by degree, not only after taking cohomology.

Consequently the map of cones

$$(d, c) \longmapsto (\Pi_D d, \Gamma c) \quad (\text{GAP6A.5})$$

commutes with the exact differential (GAP6.1). It is a quasi-isomorphism: the two source-and-target maps are isomorphisms on all cohomology groups, and the cone long exact sequences give the same for the cone map. More explicitly, at each degree use exactness to lift a target class to the adjacent group, transfer it by those two isomorphisms, and correct its difference in the preceding group; this proves surjectivity, while the same exactness with a zero target proves injectivity.

The target cone retains the two-term summand

$$Q \xrightarrow{\bar{\mathcal{T}}} Q_{\text{raw}} \quad \text{in degrees } 0, 1. \quad (\text{GAP6A.6})$$

Its contracting homotopy in degree one is the actual inverse $\bar{\mathcal{T}}^{-1}$; it is zero on all other terms. Its differential followed by this homotopy and the reverse composition sum to identity on the displayed pair. Projection away from this explicitly contractible pair therefore gives

$$\text{Cone}(\mathfrak{F}) \longrightarrow H^0(D)[0] \oplus \bigoplus_{n \geq 1} B_n[n], \quad (\text{GAP6A.7})$$

a canonical equivariant quasi-isomorphism. The convention on shifts is the one in GAP3.5: $B_n[n]$ occurs in degree $-n$. In degree zero this map is $(v, k) \mapsto \Pi_0 v$; in degree $-n$ it is $(0, c) \mapsto \Gamma^{1-n} c$, and in degree one it is zero. Its chain identities are the identities just proved. The cochain minus sign on the shifted source remains the one in (GAP6.1).

The final connecting arrow in the contracted triangle commutes up to its exact homotopy, rather than strictly. Write δ for the identity on each surviving shifted B_n , zero on $H^0(D)$, and write Π_K for (GAP6A.7). With $\text{pr}_{\mathcal{C}}$ the original cone projection, the difference

$$\Gamma[1] \text{pr}_{\mathcal{C}} - \delta \Pi_K = d\eta + \eta d, \quad \eta^1(b) = \bar{\mathcal{T}}^{-1} \pi_{Q_{\text{raw}}}(b) \quad (\text{GAP6A.8})$$

has all other homotopy components zero. Its degree-zero value is exactly $\pi_Q(k)$, since $\pi_{Q_{\text{raw}}} d_D v = 0$; this equals the difference of the two connecting maps on (v, k) . In negative degrees the connecting maps coincide. This proves the full triangle comparison in the homotopy/derived category with its specified homotopy. The independent proof CTF6 gives every connecting map and sign.

In this entire triangle, all prime-labelled exterior terms and all four endpoint lines survive in their stated degrees. No spectral quotient is embedded into the original test space, and no convergence of a sum over zeros is used.

139.9 GAP6B. Faithful source coefficients in the same whole comparison

DCP8 constructs the actual source coefficient sheaf

$$\mathcal{M}_{\text{full}} = \mathcal{N} \oplus (i_+)_* V_+ \oplus (i_-)_* V_-.$$

Its extra closed coefficient group is $V = V_+ \oplus V_-$, placed in degree zero. Retain these coefficients on both sides of the preceding comparison:

$$\mathcal{C}_{\text{full}} = \mathcal{C}^{\text{or}} \oplus V[0], \quad D_{\text{full}} = D \oplus V[0], \quad \mathfrak{F}_{\text{full}} = \mathfrak{F} \oplus \text{id}_V. \quad (\text{GAP6B.1})$$

The target is exactly the Čech complex (DCP9.2). The source is the displayed coefficient enlargement of the derived adelic complex, not an assertion that a new undeclared primitive cohomology theory has been constructed.

The original global receiving ring is $\mathbb{Z}^3$. Its element (a, b_+, b_-) acts by scalar a on $\mathcal{C}^{\text{or}}$ or D , and by b_+, b_- on the two extra closed copies. These actions commute with the differentials and the map (GAP6B.1). DCP's exact source maps give

$$[\tau] = (1, 1, 1), \quad [n] = (n, 0, 0). \quad (\text{GAP6B.2})$$

Here brackets denote receiving actions, not source coordinates for τ . Thus source τ acts as identity on all the displayed coefficients, while integer n acts as n on the first summand and zero on the extra closed copies. A nonzero vector in either extra copy distinguishes τ from every integer. A nonzero vector in the first summand distinguishes unequal integers. Multiplying these actions proves preservation of every supplied source product, and integer-input addition is the original scalar addition. No sum with τ as an input is introduced.

Real dilations act on V_+ and V_- by their original DCP formulas, and the mirror interchanges those two copies. The source and target use those same actions; hence their identity comparison preserves them. The global-ring mirror interchanges b_+ and b_- , so source scalar compatibility is also retained.

Extend both Γ and Π_D by id_V . Their commuting square (GAP6A.4) remains exact. On the cone, the additional terms are exactly

$$\text{Cone}(\text{id}_{V[0]}) = [V \xrightarrow{\text{id}} V] \quad \text{in degrees } -1, 0. \quad (\text{GAP6B.3})$$

The degree-zero-to-degree-minus-one identity is its contracting homotopy: the cone differential sends the shifted source coordinate to the target coordinate with positive sign, since the original differential on $V[0]$ is zero. This verifies $dh + hd = \text{id}$ on both terms. It commutes with the retained source actions, mirror and dilations. The extra coefficients have therefore been retained on both objects and compared by identity; only their explicitly contractible identity-cone contributes zero cohomology. It follows that (GAP6A.7) is also the exact comparison-cone result for (GAP6B.1).

The final-arrow homotopy (GAP6A.8) gains the degree-zero component sending the extra target $w \in V$ to the identical w in the shifted source V . The differential from the shifted source has positive sign in (GAP6B.3), so this homotopy gives exactly the retained identity on that component. CTF7.7 verifies it degree by degree. This faithful whole-complex comparison supplies the source distinction that the simple coefficient pullback alone lost. It neither assigns parity to primitive τ nor obtains weight one from the endpoint vanishing proved below.

139.10 GAP7. Infinitesimal action and the globally invertible endpoint polynomial

The original smooth dilations have generators

$$L_H h = -vh'(v), \quad L_A k = \tfrac{1}{2}k - uk'(u). \quad (\text{GAP7.1})$$

These are derivatives at $a = e^t, t = 0$ in the original Schwartz or strong seminorms; Taylor's integral remainder and the same seminorms on a compact t -interval prove the limit. Differentiating (GAP1.6) gives $JL_H = L_A J$. Direct integration by parts gives

$$F_{L_A k}(s) = sF_k(s), \quad m(L_H h) = (0, \int_{\mathbb{R}} h), \quad (\text{GAP7.2})$$

with zero boundary terms because of the retained decay. Thus on every actual B_n , the generator is $L_{B_n} = \text{id} \otimes \text{diag}(0, 1)$, and

$$P(L_{B_n}) = 0, \quad P(z) = z(z - 1). \quad (\text{GAP7.3})$$

The actual $P(L_Q)$ is a continuous automorphism of the entire original quotient. The companion generator proof retains the complete representative

$$F_*(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad F_*(0) = F_*(1) = \frac{1}{8},$$

and its explicit inverse formula

$$\mathcal{V}F(s) = \frac{F(s) - 8((1-s)F(0) + sF(1))F_*(s)}{s(s-1)}. \quad (\text{GAP7.4})$$

Both removable endpoint values, every pole and Gamma contribution, the continuous division estimates and the exact original Schwartz correction are proved there. This is a map on the original F and its quotient; F_* is the displayed auxiliary summation image, not a replacement for ζ . RZ's existing complete-quotient endpoint resolvents retain their credit.

Full mirror covariance is retained: differentiating (GAP4.4) gives $Lw = w(1 - L)$. Since $P(1 - L) = P(L)$, the polynomial and its quotient inverse commute with the mirror as well as every original dilation and prime action. This is why the endpoint separation is global and does not require a finite list of zeros or a uniform gap estimate for their heights.

139.11 GAP8. What the global lifting calculation proves

The complete proof `GLOBAL_ENDPOINT_EXTENSION_RETURN.md` applies the preceding exact operators to the original endpoint coefficient modules. For every extension of the actual Q by a displayed B_n in the stated operator category, its canonical lift is

$$s(q) = P(L_E)P(\widetilde{L_Q})^{-1}q. \quad (\text{GAP8.1})$$

Here the tilde is any preimage under the extension quotient. Two choices differ by B_n , annihilated by $P(L_E)$, so the formula is independent of that choice. Applying the quotient gives q . The companion supplies the complete uniqueness, continuity and all-degree extension argument, with the exact category and quotient-topology prerequisites. Every additional action commuting with P , including the original mirror, is preserved. No assumption of the desired purity enters this lift.

Numerically, the two actual endpoint characters at a prime p are $1, p$, of weights $0, 2$ in the convention $|\alpha| = p^{w/2}$. At an original zero ρ , the unchanged operator is

$$p^\rho \sum_{j=0}^{m_\rho-1} \frac{(\log p)^j}{j!} N_\rho^j. \quad (\text{GAP8.2})$$

The original strip gives $0 < 2\Re\rho < 2$. The whole-quotient polynomial calculation proves endpoint separation without replacing the full quotient by the direct sum or product of its primary blocks. It does not strengthen this interval to weight one.

The actual DCP supported obstruction also contains Q itself, rather than only the endpoint modules. On that supported Q , $P(L)$ is the same automorphism just proved. Consequently (GAP8.1) applies to the endpoint extension and not to that distinct supported quotient. DCP gives its exact diagonal quotient boundary and Fourier lifting map. These are retained constructions, not a claim that the user has supplied contradictory definitions.

139.12 GAP9. Deligne comparison and the next actual geometric calculation

Deligne's local invariant-cycle cross uses three substantive maps: proper-base-change identification with the special fibre, the Hochschild–Serre surjection onto inertia invariants, and a support obstruction identified by duality as $H^{2N-i-1}(X_s)^\vee(-N)$. His §§3.6.2–3.6.3 derive weights at most i for the target and at least $i + 1$ for the obstruction. Exactness of W_i supplies the lift in the correct subspace. No weight is assigned by the name of the supported point alone.

GAP1–GAP8 have constructed the entire actual adelic complex, its source-to-CC comparison, all higher prime-boundary terms, exact mirror orientation, and a global separation operator for those endpoints. This replaces earlier finite-block reasoning on that boundary with a theorem on the entire original quotient. It does not identify the non-endpoint supported Q with Deligne's dual special-fibre term or derive its weight bound. The next calculation remains that specific geometric comparison, with the full source support retained; no arbitrary receiving acyclicity is substituted for it.

The user-originating whole-spectrum and quotient arguments remain controlling for what may be claimed about the original source. All diagrams and receiving maps above preserve their

exact domains and the distinction between source τ and receiving coefficients. The goal remains active.

140 Complete-domain endpoint inverses and their original Schwartz return

24 September 2026. GER0–GER7. The continuous endpoint resolvents already occur in the retained ORIGINAL_ZETA_RESOLVENT_AND_PROJECTORS.md, RZ3.1–RZ3.9. This note does not claim their discovery. It propagates them through the actual original summation image, constructs integral formulas on both original source spaces, computes the full $L(L - 1)$ representative and its exact covariance correction, and retains every zero jet and support label.

140.1 GER0. Source and operation prerequisites

CORPUS_AND_OPERATION_RULES.md and the complete user global arguments USR-9ad1c0a2d09dba92, USR-f55d16feb948d8c2, and USR-6152e3bc6302258c were read for the accompanying OMS proof. They continue to govern this calculation. The arithmetic and coefficient field here are the already reconstructed receivers. No addition, numerical weight, coordinate, or parity is assigned to primitive Z_1/τ .

The exact source spaces, synthesis proof, and comparison to Ralf Meyer's original author TeX are in ORIGINAL_MELLIN_SPECTRAL_SYNTHESIS.md OMS0–OMS8, read and proved for this continuation. Human source: Ralf Meyer, A spectral interpretation for the zeros of the Riemann zeta function, arXiv:math/0412277v3, original Meyer.tex, lines 225–738, especially the:Zeta_estimate and the:Lap_range. The original author file has SHA256 ab9bc31f3c105a64fdc6f3b65ad16701dd8bf218fc90df98fe8d7f3c7c58c00e. Connes and Consani retain attribution for the source geometry and summation construction; RZ and OMS are programme derivations, not replacements of their sources.

The exact prior endpoint formula is RZ3.9 with resolvent convention $(\lambda - L)^{-1}$. Here the two inverse operators have convention $(L - c)^{-1}$, so the overall minus sign is retained in GER2. All divisions below occur in named complex receiving spaces.

140.2 GER1. Original functions, operator, and full multiplier

Keep

$$\begin{aligned} S &= \{f \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : f \text{ even, } f(0) = 0, \int_{\mathbb{R}} f = 0\}, \\ \mathcal{A} &= \{k : p_{N,j}(k) = \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j k(u)| < \infty \quad (N, j \geq 0)\}, \\ \mathcal{E}f(u) &= u^{1/2} \sum_{n \geq 1} f(nu), \quad F_k(s) = \mathcal{M}k(s) = \int_0^\infty k(u) u^{s-1/2} \frac{du}{u}, \\ I_A &= \mathcal{E}S = \overline{\mathcal{E}S}^{\mathcal{A}}, \quad Q = \mathcal{A}/I_A. \end{aligned} \tag{GER1.1}$$

The image equality is the existing proved global synthesis, with continuous source inverse, OMS3.14. Its Mellin target is the Fréchet space

$$\mathcal{B} = \{F \text{ entire} : b_{A,M}(F) = \sup_{|\sigma| \leq A, t \in \mathbb{R}} (1 + |t|)^M |F(\sigma + it)| < \infty\}.$$

Under the topological isomorphism $\mathcal{M} : \mathcal{A} \rightarrow \mathcal{B}$,

$$I_\zeta = \mathcal{M}I_A = \{F : F^{(r)}(\rho) = 0 \quad (\rho \in \mathcal{Z}, \quad 0 \leq r < m_\rho)\}, \quad Q \cong \mathcal{B}/I_\zeta. \tag{GER1.2}$$

Here $\mathcal{Z}$ and m_ρ are the actual original nontrivial zeros and their full multiplicities.

The full generator on $\mathcal{A}$ is

$$D = \frac{1}{2} - u\partial_u, \quad \mathcal{M}Dk(s) = sF_k(s); \tag{GER1.3}$$

integration by parts proves the identity, with both boundary terms zero by the defining seminorms. Multiplication by s preserves I_ζ , so D induces the continuous operator L on all of Q . Retain the actual source element

$$f_*(v) = \frac{\pi}{2}v^2(2\pi v^2 - 3)e^{-\pi v^2}, \quad k_* = \mathcal{E}f_*,$$

$$F_*(s) = \mathcal{M}k_*(s) = \frac{s(s-1)}{8}\pi^{-s/2}\Gamma(s/2)\zeta(s). \quad (\text{GER1.4})$$

The full Gaussian Mellin calculation and its endpoint conditions are OMS2.3–OMS2.5. In particular

$$F_*(0) = F_*(1) = \frac{1}{8}, \quad b := \frac{F'_*(0)}{F_*(0)} = \frac{1}{2}\log(4\pi) - 1 - \frac{\gamma}{2}, \quad F'_*(1) = -\frac{b}{8}. \quad (\text{GER1.5})$$

The last identity follows from the exact reflection $F_*(s) = F_*(1-s)$. At each original trivial zero,

$$F_*(-2r) = \frac{(1+2r)2r}{8}\pi^{-(1+2r)/2}\Gamma((1+2r)/2)\zeta(1+2r) \neq 0. \quad (\text{GER1.6})$$

These values retain the zero/pole cancellations of the displayed multiplier. The original ζ is not replaced by F_* .

140.3 GER2. Exact endpoint representatives and continuous inverse on Q

For $c = 0$ or $c = 1$, define on every $F \in \mathcal{B}$

$$\mathcal{S}_c F(s) = \frac{F(s) - 8F_*(s)F(c)}{s - c}. \quad (\text{GER2.1})$$

Its numerator vanishes at c by (GER1.5). The exact removable value is

$$\mathcal{S}_0 F(0) = F'(0) - bF(0), \quad \mathcal{S}_1 F(1) = F'(1) + bF(1). \quad (\text{GER2.2})$$

These are the negatives of the RZ endpoint resolvents with parameter $\lambda = c$.

For completeness, this entire division is continuous in the full topology. If $N(c) = 0$, then outside $|s - c| < 1$, $(1 + |\Im s|)^M |N(s)/(s - c)| \leq (1 + |\Im s|)^M |N(s)|$. Inside $|s - c| \leq 1$, the maximum principle applied to the removable quotient on the radius-two disk gives

$$|N(s)/(s - c)| \leq \frac{1}{2} \sup_{|z-c|=2} |N(z)|.$$

For $A' \geq \max(A, c + 2)$, the weight on the smaller disk is at most 2^M , whence

$$b_{A,M}(N/(s - c)) \leq b_{A,M}(N) + 2^{M-1}b_{A',0}(N). \quad (\text{GER2.3})$$

The numerator in (GER2.1) satisfies

$$b_{A,M}(F - 8F_*F(c)) \leq b_{A,M}(F) + 8b_{A,M}(F_*)b_{A',0}(F)$$

when $A' \geq c$. This proves continuity of $\mathcal{S}_c : \mathcal{B} \rightarrow \mathcal{B}$.

If $F \in I_\zeta$, its numerator is in I_ζ ; division by $s - c$, which is nonzero at every actual nontrivial zero, preserves every required order. Thus it descends independently of the representative to a continuous $S_c : Q \rightarrow Q$. On $\mathcal{B}$ the exact identities are

$$(s - c)\mathcal{S}_c F = F - 8F_*F(c), \quad \mathcal{S}_c((s - c)F) = F. \quad (\text{GER2.4})$$

The correction in the first identity belongs to the actual summation image. Therefore

$$\boxed{S_0 = L^{-1}, \quad S_1 = (L - 1)^{-1}} \quad (\text{GER2.5})$$

are continuous two-sided inverses on the complete original Q .

140.4 GER3. Explicit return on the original strong test space

For $k \in \mathcal{A}$, let

$$g_c(u) = k(u) - 8F_k(c)k_*(u), \quad F_{g_c}(c) = 0. \quad (\text{GER3.1})$$

Define

$$(s_c k)(u) = u^{1/2-c} \int_u^\infty g_c(v) v^{c-1/2} \frac{dv}{v} = -u^{1/2-c} \int_0^u g_c(v) v^{c-1/2} \frac{dv}{v}. \quad (\text{GER3.2})$$

The two expressions agree by $F_{g_c}(c) = 0$. Both integrals converge absolutely by the endpoint estimates for $\mathcal{A}$. Differentiation gives

$$(D - c)s_c k = g_c. \quad (\text{GER3.3})$$

Here are complete topology estimates. For an integer $K \geq 1$, on $u \geq 1$ use the first integral and $|g_c(v)| \leq p_{K,0}(g_c)v^{-K}$ to get

$$|s_c k(u)| \leq \frac{p_{K,0}(g_c)}{K - c + 1/2} u^{-K}.$$

On $u \leq 1$, the second gives

$$|s_c k(u)| \leq \frac{p_{K,0}(g_c)}{K + c - 1/2} u^K. \quad (\text{GER3.4})$$

All denominators are positive for $K \geq 1$, $c = 0, 1$. Taking $K \geq N + 1$ controls $p_{N,0}$. Let $\partial_{\log} = u\partial_u$, $\alpha = 1/2 - c$, and $h = s_c k$. The differential identity gives for $j \geq 1$

$$\partial_{\log}^j h = \alpha^j h - \sum_{\ell=0}^{j-1} \alpha^{j-1-\ell} \partial_{\log}^\ell g_c. \quad (\text{GER3.5})$$

Together with (GER3.4) this bounds every seminorm of h by finitely many of g_c . Finally

$$|F_k(c)| \leq p_{K,0}(k) \left(\frac{1}{K + c - 1/2} + \frac{1}{K - c + 1/2} \right) \quad (\text{GER3.6})$$

follows by splitting its integral at one. Hence $s_c : \mathcal{A} \rightarrow \mathcal{A}$ is continuous. Taking Mellin transforms in (GER3.3), with the proved endpoint decay, gives exactly $\mathcal{M}s_c k = \mathcal{S}_c F_k$. Thus the integral and the RZ representative agree, with no auxiliary function space.

140.5 GER4. Explicit return on the actual Schwartz summation image

This strengthens a mere closure-preservation statement by giving the actual preimage. Let $f \in S$, put $F = F_{\mathcal{E}f}$, and define

$$r_c(v) = f(v) - 8F(c)f_*(v). \quad (\text{GER4.1})$$

For $v > 0$, put

$$(t_c f)(v) = v^{-c} \int_v^\infty r_c(w) w^c \frac{dw}{w} = - \int_0^1 r_c(vt) t^{c-1} dt, \quad (\text{GER4.2})$$

and extend evenly. The equality of the two forms uses $\int_0^\infty r_c(w) w^c dw/w = 0$, verified next.

For $c = 0$, the original Mellin identity $F(s) = \zeta(s)M_f(s)$, with $\zeta(0) = -1/2$, gives $M_{r_0}(0) = 0$ because $F_{\mathcal{E}r_0}(0) = 0$. For $c = 1$, $M_{r_1}(1) = 0$ since $r_1 \in S$. Its additional logarithmic moment also vanishes:

$$M'_{r_1}(1) = F_{\mathcal{E}r_1}(1) = 0. \quad (\text{GER4.3})$$

Indeed $M_r(1) = 0$ and the original residue $\text{Res}_{s=1} \zeta(s) = 1$ imply $F_{\mathcal{E}r}(1) = M'_r(1)$; this uses the full pole rather than treating $\zeta(1)$ as a value.

At infinity the first expression in (GER4.2), and its differentiated versions, gives arbitrary Schwartz decay. At zero use the second. For $j \geq 1$ its derivative is the integral of $-r_c^{(j)}(vt)t^{c+j-1}$, which is integrable. For $j = 0, c = 0$, use $r_c(vt) = O(v^2t^2)$, because r_c is even and vanishes at zero. These estimates also prove continuity in all local smooth seminorms. At zero its value is zero and every odd derivative is zero, so the even extension is smooth.

For sufficiently large $\Re s$, absolute Fubini in the tail formula gives

$$M_{t_c f}(s) = \frac{M_{r_c}(s)}{s - c}. \quad (\text{GER4.4})$$

Both sides continue to the region containing $s = 1$. For $c = 0$ its value there is $M_{r_0}(1) = 0$. For $c = 1$ it is $M'_{r_1}(1) = 0$ by (GER4.3). Thus $\int_{\mathbb{R}} t_c f = 0$, proving $t_c f \in S$. For an explicit tail estimate after differentiation, the equation $-v(t_c f)' - ct_c f = r_c$ expresses each derivative in terms of the tail integral and finitely many derivatives of r_c , multiplied by powers of v^{-1} ; on $v \geq 1$ these preserve arbitrary decay. On $v \leq 1$ the integral derivative bounds above are controlled by finitely many Schwartz seminorms. The scalar $F(c)$ depends continuously on f through $\mathcal{E}$, so $t_c : S \rightarrow S$ is continuous.

Multiplying (GER4.4) by the original $\zeta(s)$ proves

$$\boxed{s_c \mathcal{E} = \mathcal{E} t_c, \quad \mathcal{S}_c F_{\mathcal{E} f} = F_{\mathcal{E} t_c f}.} \quad (\text{GER4.5})$$

The preservation of the actual image is therefore constructive, without replacing it by its closure.

140.6 GER5. The exact inverse of $P(L) = L(L - 1)$

Direct substitution into (GER2.1), retaining both endpoint values, gives the identities on the full $\mathcal{B}$:

$$\boxed{\mathcal{S}_0 \mathcal{S}_1 = \mathcal{S}_1 \mathcal{S}_0 = \mathcal{S}_1 - \mathcal{S}_0 = \mathcal{I},} \quad \mathcal{T}F(s) = \frac{F(s) - 8F_*(s)((1-s)F(0) + sF(1))}{s(s-1)}. \quad (\text{GER5.1})$$

For example $(\mathcal{S}_1 F)(0) = F(1) - F(0)$; substituting it into $\mathcal{S}_0$ gives the displayed numerator. The other composition uses $(\mathcal{S}_0 F)(1) = F(1) - F(0)$ and gives exactly the same expression. Subtracting the two original fractions gives $\mathcal{S}_1 - \mathcal{S}_0$. These are representative identities, not merely quotient identities.

Both endpoint singularities are removable. Their exact values are

$$\begin{aligned} \mathcal{T}F(0) &= -F'(0) + (b-1)F(0) + F(1), \\ \mathcal{T}F(1) &= F'(1) + (b-1)F(1) + F(0). \end{aligned} \quad (\text{GER5.2})$$

They follow by differentiating the numerator and dividing by $P'(0) = -1$, $P'(1) = 1$. At an original trivial zero $z = -2r$,

$$\mathcal{T}F(z) = \frac{F(z) - 8F_*(z)((1-z)F(0) + zF(1))}{z(z-1)}, \quad (\text{GER5.3})$$

where $F_*(z)$ has the full nonzero Gamma/zeta value (GER1.6). Continuity and preservation of I_ζ follow either from (GER5.1) or the two successive proved divisions.

On representatives the full source correction is

$$s(s-1)\mathcal{T}F = F - 8F_*((1-s)F(0) + sF(1)), \quad \mathcal{T}(s(s-1)F) = F. \quad (\text{GER5.4})$$

Let $t = s_0 s_1$. Under Mellin inversion this says

$$D(D-1)tk = k - \mathcal{E} [8F_k(0)f_* + 8(F_k(1) - F_k(0))(-vf'_*(v))],$$

$$t(D(D-1)k) = k. \quad (\text{GER5.5})$$

The exact generator intertwining used here is $D\mathcal{E}f = \mathcal{E}(-vf')$, proved by differentiating the original series. The full derivative is

$$-vf'_*(v) = (2\pi^3v^6 - 7\pi^2v^4 + 3\pi v^2)e^{-\pi v^2}. \quad (\text{GER5.6})$$

It is even, vanishes at zero, and has integral $\int f_* = 0$ by integration by parts, so both correction terms belong to the original S . Consequently

$$\boxed{P(L)^{-1} = S_0S_1 = S_1S_0 = S_1 - S_0} \quad (\text{GER5.7})$$

on the complete actual Q , with its continuous actual Schwartz return (GER4.5).

140.7 GER6. Full arithmetic equivariance and its exact Schwartz correction

Retain $W_ak(u) = a^{1/2}k(u/a)$ and its Mellin multiplier $T_aF(s) = a^sF(s)$ for every $a > 0$. Their full representative commutator is

$$\mathcal{S}_cT_aF - T_a\mathcal{S}_cF = 8F(c)F_*(s)\frac{a^s - a^c}{s - c}. \quad (\text{GER6.1})$$

The quotient is entire, with value $a^c \log a$ at $s = c$. More precisely, with the oriented integral when $\log a < 0$,

$$\frac{a^s - a^c}{s - c} = \int_0^{\log a} e^{c(\log a - t)} e^{st} dt. \quad (\text{GER6.2})$$

Thus this commutator has the explicit original Schwartz preimage

$$r_{c,a,F}(v) = 8F(c) \int_0^{\log a} e^{c(\log a - t)} f_*(ve^{-t}) dt. \quad (\text{GER6.3})$$

It is a finite-interval integral in S : each dilation is even, has value zero at zero, and has integral $e^t \int f_* = 0$. Every Schwartz seminorm of the integrand is bounded on that interval by a fixed seminorm of f_* , proving convergence in the Schwartz topology. Also

$$W_{e^t}\mathcal{E}f_* = \mathcal{E}(f_*(\cdot e^{-t})),$$

by direct substitution, with the exact factor $e^{t/2}$ in W_{e^t} cancelling the source $u^{1/2}$ change. Taking Mellin transforms proves that (GER6.3) sums to exactly (GER6.1).

For $\mathcal{S} = \mathcal{S}_1 - \mathcal{S}_0$, the complete source correction is the difference

$$r_{P,a,F}(v) = 8 \int_0^{\log a} \left(F(1)e^{\log a - t} - F(0) \right) f_*(ve^{-t}) dt. \quad (\text{GER6.4})$$

Thus every S_c and $P(L)^{-1}$ commutes exactly with every W_a on Q , with an explicit prequotient correction in the actual $\mathcal{E}S$. This includes all primes and all prime powers; no restricted height or finite set has replaced the arithmetic action.

140.8 GER7. Every primary jet, original multiplicity, and support label

For the actual local algebra $A_\rho = \mathbb{C}[T_\rho]/(T_\rho^{m_\rho})$, Taylor division gives

$$j_\rho S_c = M_{m_\rho-1} \sum_{r=0}^{m_\rho-1} \frac{(-1)^r T_\rho^r}{(\rho - c)^{r+1}} j_\rho, \quad (\text{GER7.1})$$

$$j_\rho P(L)^{-1} = M_{m_\rho-1} \sum_{r=0}^{j_\rho} (-1)^r ((\rho-1)^{-r-1} - \rho^{-r-1}) T_\rho^r. \quad (\text{GER7.2})$$

The denominators are nonzero because no original nontrivial zero is 0 or 1. Multiplication by $(\rho - c) + T_\rho$ telescopes in (GER7.1) to one modulo $T_\rho^{m_\rho}$; the difference of the two identities gives (GER7.2), which is the complete inverse of $(\rho + T_\rho)(\rho - 1 + T_\rho)$. Every multiplicity and every surviving nilpotent term remains.

For the original support lattice L_{supp} , write it with the subscript here only to distinguish it from the already named generator L , not to replace its labels. The actual receiving functor is

$$G_{L_{\text{supp}}}(V) = \{(0, \lambda) : \lambda \in L_{\text{supp}}\} \cup \{(v, 1_{L_{\text{supp}}}) : v \in V\}, \quad G(f)(v, \lambda) = (f(v), \lambda). \quad (\text{GER7.3})$$

Every linear identity just proved lifts by this formula; substitution proves preservation of the receiving joins, meets, addition, and scalar action already specified in CW7/OMS7. In particular

$$G(S_c)G(L - c) = G(L - c)G(S_c) = \text{id}, \quad G(P(L)^{-1})G(P(L)) = \text{id}. \quad (\text{GER7.4})$$

Every $(0, \lambda)$ remains its own labelled zero under each map. This is not a tensor operation that collapses the zero labels, and it does not add an operation on primitive tau.

The full-domain propagation is now explicit: endpoint division and $P(L)$ -division are continuous actual inverses on the original Q , their changes of representative and arithmetic covariance errors have original Schwartz preimages, and their full jet actions preserve every original zero and multiplicity. The calculation does not delete the original function's trivial zeros or pole: their contribution appears in the retained multiplier and endpoint corrections above. It supplies an inverse in this nontrivial-zero quotient, with precisely this domain and codomain.

141 Canonical lifting across every endpoint extension of the whole original quotient

24 September 2026. GEX0–GEX9. This calculation treats the entire original Mellin quotient with its actual topology, rather than a finite selection of zero blocks.

141.1 GEX0. Source, provenance, and operation domains

The current `CORPUS_AND_OPERATION_RULES.md` (private source record), `AC0–AC5` of `ACTUAL_ADELIC_COMPLEX_INDEPENDENT.md`, and `DCP12` of `SOURCE_CC_DOUBLE_PULLBACK.md` were consulted. The complete user arguments `USR-9ad1c0a2d09dba92`, `USR-f55d16feb948d8c2`, and `USR-6152e3bc6302258c` remain the governing whole-arithmetic reconstruction and global-quotient arguments. They were read for the preceding source and coefficient calculations; no new exhaustive corpus audit is claimed.

Primitive Z_1/τ still has no source addition, parity, coordinate, or numerical weight. The complete reconstructed integers, primes, original zeta, receiving complex field, and test space are already present. Every polynomial, subtraction, inverse, module extension, and quotient below is an operation on that receiver. The $0, 1$ in $L(L-1)$ are the calculated endpoint characters of its infinitesimal dilation, not numerical values assigned to primitive τ .

The synthesis and source inverse are `ORIGINAL_MELLIN_SPECTRAL_SYNTHESIS.md`, `OMS1–OMS6`. Its human analytic source is Ralf Meyer, *A spectral interpretation for the zeros of the Riemann zeta function*, arXiv:math/0412277v3, with original author TeX and actual reading recorded in `OMS0`. `AC0` credits the adelic periodization to Alain Connes and to Alain Connes, Caterina Consani, and Matilde Marcolli. `DCP` receives Connes–Consani’s *Schemes over $\mathbb{F}_1$ and zeta functions*, arXiv:0903.2024v3, §5. Pierre Deligne’s target lifting argument remains *La conjecture de Weil. II*, §§3.6.1–3.6.3. No identification of its obstruction group with an analytic endpoint module is assumed.

141.2 GEX1. The actual continuous generator

Keep

$$\mathcal{A} = \{k \in C^\infty(\mathbb{R}_{>0}) : p_{N,j}(k) = \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j k(u)| < \infty \text{ for all } N, j \geq 0\},$$

$$S = \{h \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : h(-v) = h(v), \ h(0) = 0, \ \int_{\mathbb{R}} h = 0\}, \quad \mathcal{E}h(u) = u^{1/2} \sum_{n \geq 1} h(nu),$$

$$Q = \mathcal{A}/\mathcal{E}S, \quad F_k(s) = \int_0^\infty k(u) u^{s-1/2} \frac{du}{u}. \quad (\text{GEX1.1})$$

`OMS3.14` proves that $\mathcal{E}S$ is closed, and that the original Mellin transform identifies this topological quotient with

$$Q \cong \mathcal{B}/I_\zeta, \quad b_{A,M}(F) = \sup_{|\sigma| \leq A, \ t \in \mathbb{R}} (1 + |t|)^M |F(\sigma + it)|,$$

$$I_\zeta = \{F \in \mathcal{B} : F^{(r)}(\rho) = 0 \text{ at every actual nontrivial zero } \rho, \ 0 \leq r < m_\rho\}. \quad (\text{GEX1.2})$$

This is the actual quotient and image topology in `OMS5`, not the unrestricted product of zero jets.

The unshifted dilation and generator are

$$W_a k(u) = a^{1/2} k(u/a), \quad L_{\mathcal{A}} k = \left. \frac{d}{dt} W_{e^t} k \right|_{t=0} = \frac{1}{2} k - u \partial_u k. \quad (\text{GEX1.3})$$

Taylor’s formula for $k(e^{-t}u)$ has an integral remainder bounded in each $p_{N,j}$ by a constant times stronger seminorms with two additional Euler derivatives, for t in a fixed compact interval.

This proves the derivative in the displayed topology; its formula also proves continuity of $L_{\mathcal{A}}$. Integration by parts, with both boundary values zero by the strong decay, and substitution $u = av$, respectively give

$$F_{L_{\mathcal{A}}k}(s) = sF_k(s), \quad F_{W_ak}(s) = a^s F_k(s). \quad (\text{GEX1.4})$$

Multiplication by s is continuous on $\mathcal{B}$ by a seminorm with one additional vertical power. Leibniz's formula shows that it preserves every vanishing multiplicity in I_{ζ} . Therefore it induces the continuous L_Q , and every W_a commutes with it.

141.3 GEX2. A global inverse of $P(L_Q) = L_Q(L_Q - 1)$

Retain the actual source vector and its full transform from OMS2:

$$h_*(v) = \frac{\pi}{2}v^2(2\pi v^2 - 3)e^{-\pi v^2}, \quad F_*(s) = F_{\mathcal{E}h_*}(s) = \frac{s(s-1)}{8}\pi^{-s/2}\Gamma(s/2)\zeta(s). \quad (\text{GEX2.1})$$

It belongs to I_{ζ} , with

$$F_*(0) = F_*(1) = \frac{1}{8}. \quad (\text{GEX2.2})$$

OMS2.3–OMS2.5 proves its source conditions and all exceptional-point cancellations. The full original-zeta multiplier remains displayed; this is an actual image vector used for correction, not a replacement for ζ .

For $F \in \mathcal{B}$, define

$$\begin{aligned} C_F(s) &= 8((1-s)F(0) + sF(1))F_*(s), \\ \mathcal{K}F(s) &= \frac{F(s) - 8((1-s)F(0) + sF(1))F_*(s)}{s(s-1)}. \end{aligned} \quad (\text{GEX2.3})$$

At 0, 1 use the removable extensions. They exist because $C_F(0) = F(0)$ and $C_F(1) = F(1)$.

This is a continuous map $\mathcal{B} \rightarrow \mathcal{B}$. Evaluations at the two points and multiplication by the fixed F_*, sF_* have continuous seminorm bounds, so the numerator $N_F = F - C_F$ does too. Outside the disks $|s| < 1/4$, $|s-1| < 1/4$, division by $s(s-1)$ is bounded on each fixed vertical strip, and for large imaginary part improves its decay. Inside either disk the quotient is holomorphic; its maximum is bounded by the boundary maximum, where the denominator is bounded away from zero. Those compact circles lie in a fixed larger strip, and their imaginary weights are bounded. Together these estimates bound each $b_{A,M}(\mathcal{K}F)$ by a fixed finite sum of seminorms of F .

For $F \in I_{\zeta}$, both F and C_F lie in I_{ζ} . The actual nontrivial zeros are neither 0 nor 1, so division by $s(s-1)$ preserves their entire multiplicities. Thus

$$\mathcal{K}(I_{\zeta}) \subset I_{\zeta}. \quad (\text{GEX2.4})$$

The induced continuous operator K_Q obeys

$$s(s-1)\mathcal{K}F = F - C_F, \quad \mathcal{K}(s(s-1)F) = F,$$

$$\boxed{K_Q = P(L_Q)^{-1}, \quad P(T) = T(T-1).} \quad (\text{GEX2.5})$$

For the second identity the input $s(s-1)F$ has both endpoint values zero. For the first, C_F belongs to the actual image ideal. These identities prove both inverse directions on Q .

This is not division on arbitrary zero sequences. It is a continuous construction on the full original strong space, followed by its actual quotient. It assumes neither a uniform distance of all zeros from the endpoint lines nor convergence of an infinite sum of primary projectors. Every operator commuting with $P(L_Q)$ commutes with K_Q , by multiplication with the two-sided inverse.

141.4 GEX3. All actual adelic endpoint groups

Keep

$$W_{\text{pr}} = \bigoplus_{p \text{ prime}} \mathbb{C}\ell_p, \quad B^{[r]} = \Lambda^r W_{\text{pr}} \otimes \mathbb{C}^2 \quad (r \geq 1). \quad (\text{GEX3.1})$$

Thus $B^{[1]} = \mathcal{B}_{\text{pr}}$, and AC4's higher groups are $H_n(G, M_0) = B^{[n+1]}$, $n \geq 1$. Their already calculated action and its generator are

$$W_a^B = \text{id} \otimes \begin{pmatrix} 1 & 0 \\ 0 & a \end{pmatrix}, \quad L_B = \text{id} \otimes \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}. \quad (\text{GEX3.2})$$

Differentiation at $a = e^t = 1$ gives the second formula, hence

$$P(L_B) = 0. \quad (\text{GEX3.3})$$

Both operators are continuous on the locally convex direct sum: on each finite-dimensional coordinate they are the displayed maps, and the direct-sum universal property proves continuity. All prime labels, exterior orders, and endpoint pairs remain.

There are two distinct retained involutions on this boundary. Write $C(u_0, u_1) = (u_1, u_0)$. In `GLOBAL_ADELIC_LIFTING_AND_PRIME_BOUNDARY.md`, GAP4.3 computes the **unoriented adelic Fourier** action, whereas GAP5–GAP6.3 adds the ordered-cover orientation required by the CC chain map:

$$\mathcal{F}_B^{[r]} = (-1)^r (\text{id}_{\Lambda^r W_{\text{pr}}} \otimes C), \quad w_{B, \text{or}}^{[r]} = (-1)^{r+1} (\text{id}_{\Lambda^r W_{\text{pr}}} \otimes C) = -\mathcal{F}_B^{[r]}. \quad (\text{GEX3.4})$$

Neither is substituted for the other. The first sign includes the inverted prime-wedge factors and endpoint swap. The additional minus sign in the second is the separate orientation representation used by the actual CC chain comparison.

For either separately typed $T = \mathcal{F}_B^{[r]}$ or $T = w_{B, \text{or}}^{[r]}$, direct multiplication of the endpoint matrices gives

$$T^2 = 1, \quad L_B T = T(1 - L_B), \quad W_a^B T = a T W_{a^{-1}}^B, \quad P(L_B) T = T P(L_B) = 0. \quad (\text{GEX3.5})$$

Indeed $\text{diag}(0, 1)C = C \text{diag}(1, 0)$ and $\text{diag}(1, a)C = aC \text{diag}(1, a^{-1})$; the factors $(-1)^r$ and $(-1)^{r+1}$ preserve each equality independently. The polynomial therefore commutes with both actions without erasing their difference.

The arguments also apply to specified direct sums of these endpoint groups. They do not assign GEX3.3 to a different tensor product without computing its generator.

141.5 GEX4. The unique section for every extension

Take any short exact sequence of complex $\mathbb{C}[L]$ -modules

$$0 \longrightarrow B \xrightarrow{\iota} E \xrightarrow{\pi} Q \longrightarrow 0, \quad (\text{GEX4.1})$$

with B one of the actual groups of GEX3 or their specified direct sums. Thus L_E restricts to L_B , induces L_Q , and the displayed maps intertwine them. No section is assumed.

For $q \in Q$, choose any $e \in E$ with $\pi(e) = K_Q q$, and set

$$\boxed{S(q) = P(L_E)e.} \quad (\text{GEX4.2})$$

Such a lift exists by surjectivity. If e' is another, $e' - e = \iota b$, so $P(L_E)(e' - e) = \iota P(L_B)b = 0$. Thus S is independent of the lift. Linear combinations of lifts lift the corresponding combination of $K_Q q$; this proves linearity without a chosen linear family of lifts.

Its projection is

$$\pi S(q) = P(L_Q)K_Q q = q. \quad (\text{GEX4.3})$$

If e lifts $K_Q q$, then $L_E e$ lifts $K_Q L_Q q$, since K_Q commutes with L_Q . Therefore

$$S(L_Q q) = P(L_E)L_E e = L_E S(q). \quad (\text{GEX4.4})$$

So this is a module section.

It is unique. The difference of two module sections factors through a module map $h : Q \rightarrow B$, for which

$$hP(L_Q) = P(L_B)h = 0.$$

Since $P(L_Q)$ is onto, $h = 0$. Likewise a map $h : B \rightarrow Q$ satisfies $P(L_Q)h = 0$, and injectivity of $P(L_Q)$ makes it zero. Thus

$$\text{Hom}_{\mathbb{C}[L]}(Q, B) = \text{Hom}_{\mathbb{C}[L]}(B, Q) = 0. \quad (\text{GEX4.5})$$

The exact endpoint retraction is

$$R_E(e) = \iota^{-1}(e - S\pi(e)). \quad (\text{GEX4.6})$$

The argument lies in $\ker \pi = \iota B$, proving that it is defined. It gives $R_E \iota = \text{id}$, $R_E S = 0$, and mutually inverse maps

$$E \rightleftharpoons B \oplus Q, \quad e \mapsto (R_E e, \pi e), \quad (b, q) \mapsto \iota b + S q. \quad (\text{GEX4.7})$$

141.6 GEX5. All prime actions, full dilations, and the original mirror

Let T_B, T_E, T_Q be compatible operators on GEX4.1 commuting with the respective $P(L)$. Commutation with L is sufficient but not necessary. If e lifts $K_Q q$, then $\pi(T_E e) = T_Q K_Q q = K_Q T_Q q$. Hence

$$S(T_Q q) = P(L_E)T_E e = T_E P(L_E) e = T_E S(q). \quad (\text{GEX5.1})$$

The same identity makes GEX4.6 equivariant. Thus the section simultaneously preserves every supplied prime, prime-power, full dilation, or source scalar/idempotent action satisfying these explicit compatibilities. It does not select a finite prime list.

In particular a compatible original mirror with $LR = R(1 - L)$ commutes with $P(L)$, because

$$P(1 - L) = (1 - L)((1 - L) - 1) = L(L - 1). \quad (\text{GEX5.2})$$

GEX5.1 therefore proves equivariance for either of the separately typed involutions too. This preserves the original unshifted relation

$$W_a R = a R W_{a^{-1}}; \quad (\text{GEX5.3})$$

it does not replace the action by a half-character twist. The unoriented action on Q is induced by the original reflected test function $\mathcal{F}_A k(u) = k(1/u)$, and has Mellin argument $s \mapsto 1 - s$. The oriented CC action on this same underlying quotient is instead $-\mathcal{F}_Q$, corresponding under the raw map $\mathcal{T}$ to $-R_A$ in DCP11. Both satisfy GEX5.2–GEX5.3. Their endpoint actions are respectively the two formulas GEX3.4, whose matrix compatibilities are GEX3.5.

To state exactly how the splitting respects this difference, an unoriented equivariant extension has compatible operators $(\mathcal{F}_B, \mathcal{F}_E, \mathcal{F}_Q)$. Its orientation twist changes **all three** to $(-\mathcal{F}_B, -\mathcal{F}_E, -\mathcal{F}_Q)$, leaving the vector spaces, extension maps, L , and dilations unchanged. This is tensoring the specified involution action by its sign representation; it is an involutive equivalence of the two typed extension categories. The section GEX4.2 is the same underlying formula and satisfies

$$S\mathcal{F}_Q = \mathcal{F}_E S \implies S(-\mathcal{F}_Q) = (-\mathcal{F}_E)S. \quad (\text{GEX5.4})$$

Thus it preserves either coherent action, including the oriented one used by the CC chain map. Changing only a boundary sign while leaving the other extension actions untouched is not claimed to preserve equivariance. A mirror on the middle extension must be one of its supplied compatible operators; none is invented by naming it.

141.7 GEX6. Continuity without an assumed topological section

Use Hausdorff locally convex complex spaces with everywhere-defined continuous L , the actual quotient topology on Q , and the endpoint topology on B . A strict short exact sequence here means:

- ι identifies B homeomorphically with the closed subspace $\ker \pi \subset E$;
- $\pi : E \rightarrow Q$ is continuous and onto, with Q carrying precisely its quotient topology;
- the L -actions are continuous and are intertwined by these maps.

The quotient requirement is equivalent to openness of π : for open $V \subset E$, the saturation $\pi^{-1}\pi(V) = V + \ker \pi$ is open. No continuous vector-space section is assumed, and extensions with only a partially defined unbounded generator are not silently included.

Since the continuous $P(L_E)$ kills $\ker \pi$, it induces

$$\bar{P} : Q \rightarrow E, \quad \bar{P}(\pi e) = P(L_E)e. \quad (\text{GEX6.1})$$

This is continuous by the defining quotient topology: its composite with π is the original continuous polynomial operator. Therefore

$$S = \bar{P}K_Q \quad (\text{GEX6.2})$$

is continuous. The retraction GEX4.6 is continuous because its numerator is continuous with image in the subspace ιB , and ι^{-1} is continuous there. GEX4.7 is consequently a topological isomorphism. All compatible continuous operators in GEX5 are preserved.

For an algebraic extension with no specified middle-space topology, GEX4–GEX5 still prove the algebraic statement. They do not assert the extra conclusion GEX6 without its explicit quotient and continuity prerequisites.

141.8 GEX7. Entire Ext groups in every algebraic degree

Let $\mathcal{R}$ be the specified unital operator algebra, with $P(L)$ central. It may be just $\mathbb{C}[L]$. It may also contain the actual dilation operators and a mirror satisfying GEX5.2–GEX5.3; L need not be central in that larger algebra, but $P(L)$ is. The inverse K_Q is $\mathcal{R}$ -linear because every generator commutes with $P(L_Q)$, hence with its inverse.

For all $n \geq 0$,

$$\boxed{\text{Ext}_{\mathcal{R}}^n(Q, B) = 0, \quad \text{Ext}_{\mathcal{R}}^n(B, Q) = 0.} \quad (\text{GEX7.1})$$

This is about entire Ext groups in the stated category, strengthening AC3's earlier vanishing of one specific two-extension class.

Take a free resolution $P_{\bullet} \rightarrow Q$. Central multiplication by $P(L)$ is a chain map lifting the automorphism $P(L_Q)$. For any $\mathcal{R}$ -linear cochain $h : P_j \rightarrow B$,

$$h(P(L)x) = P(L_B)h(x) = 0. \quad (\text{GEX7.2})$$

Thus its precomposition action on the entire Hom complex is zero. On Ext that same action is the functorially induced action of the automorphism $P(L_Q)$; its inverse is induced by K_Q . Therefore it is both zero and invertible, so each Ext group is zero.

For the reverse order, take a free resolution $P_{\bullet} \rightarrow B$. Its central multiplication map lifts zero on B , and is null-homotopic. To construct the homotopy, in degree zero lift its image, whose augmentation is zero, into the next boundary using freeness. Inductively, subtract the previously constructed $h_{j-1}d_j$ from the degree- j central map. Its differential is zero by the preceding identity, so exactness and freeness give a lift h_j through the next boundary. Thus

$$P(L)P_j = d_{j+1}h_j + h_{j-1}d_j$$

in every degree. On $\text{Hom}_{\mathcal{A}}(P_{\bullet}, Q)$, precomposition is therefore zero on cohomology. By centrality it equals postcomposition by the automorphism $P(L_Q)$, whose inverse is postcomposition by K_Q . Again the cohomology vanishes. Degree zero is also directly GEX4.5.

No decomposition of the whole Q into a direct sum of finite packets appears in this proof, and no passage through an unproved infinite spectral limit is used.

141.9 GEX8. Strict topological Yoneda extensions

The all-degree conclusion also holds for strict Yoneda extensions in the category of GEX6, with specified continuous operators whose relations make $P(L)$ central. These are finite strings of strict short exact sequences; the assertion is not about an unspecified derived functor in a larger analytic category.

The finite pushouts and pullbacks needed here stay in the category. A pushout of the closed embedding $\iota : B \rightarrow E$ along continuous $g : B \rightarrow B'$ is

$$(E \oplus B') / \{(\iota b, -gb) : b \in B\}. \quad (\text{GEX8.1})$$

Put $N_g = \{(\iota b, -gb) : b \in B\}$, let $q : E \oplus B' \rightarrow P = (E \oplus B')/N_g$ be the quotient, and let $p : E \rightarrow Q$ be the original strict quotient. The subspace N_g is closed: in closed $\iota B \oplus B'$, it is the graph of the continuous map $-g\iota^{-1}$. Therefore P is Hausdorff locally convex, with its quotient topology, and q is open by the saturation calculation in GEX6.

The induced $\bar{p} : P \rightarrow Q$ has composite $\bar{p}q(e, b') = p(e)$. It is onto and open: the composite $p \text{pr}_E$ is open, so for every open $O \subset P$, $\bar{p}(O) = p \text{pr}_E(q^{-1}O)$ is open. Its kernel is closed because Q is Hausdorff. Its preimage in $E \oplus B'$ is the saturated subspace $K = \iota B \oplus B'$. The restriction $q|_K$ is open onto that kernel with its subspace topology: for open $O \subset E \oplus B'$,

$$q(O \cap K) = q(O) \cap q(K), \quad (\text{GEX8.1a})$$

since K is saturated. The quotient of K by N_g is topologically B' : the map

$$K \rightarrow B', \quad (\iota b, b') \mapsto gb + b' \quad (\text{GEX8.1b})$$

is the open projection after the continuous invertible triangular change $(b, b') \mapsto (b, gb + b')$, has kernel N_g , and has continuous section $b' \mapsto (0, b')$. Thus $B' \rightarrow P$ is exactly a closed topological embedding and $P \rightarrow Q$ is exactly an open quotient. No algebraic pushout topology has been presumed to be strict.

For a continuous map $h : Q' \rightarrow Q$, the pullback

$$P_h = \{(e, q') \in E \oplus Q' : p(e) = h(q')\}$$

is a closed subspace, since it is the kernel of the continuous map $(e, q') \mapsto p(e) - h(q')$ into Hausdorff Q . Its projection $p' : P_h \rightarrow Q'$ is onto and open. For any product-neighbourhood $V_E \times V_{Q'}$, the exact image is

$$p'((V_E \times V_{Q'}) \cap P_h) = V_{Q'} \cap h^{-1}(p(V_E)), \quad (\text{GEX8.1c})$$

which is open because p is open and h continuous. Its kernel is the copy $b \mapsto (\iota b, 0)$, with precisely the original subspace topology. These prove that the pullback is strict too.

Identities, finite direct sums and compositions of closed embeddings or open quotients preserve their stated properties. Together with these pushout and pullback calculations, they give the exact structure used for the finite Yoneda strings below. Every compatible continuous operator preserves the relevant graph or fibre product and descends or restricts continuously. No completed tensor product, completion of a quotient, or open mapping theorem is used.

For every such n -extension class,

$$P(L_B)_*[\mathfrak{c}] = P(L_Q)_*[\mathfrak{c}]. \quad (\text{GEX8.2})$$

Here is the exact Yoneda comparison. Central multiplication on every middle object is a morphism of the extension to itself with endpoint maps $P(L_B)$ and $P(L_Q)$. Push the left end out along $P(L_B)$, and pull the right end back along $P(L_Q)$. At the first object the induced comparison is

$$[e, b] \mapsto P(L_E)e + \iota b,$$

which kills the pushout relation because $P(L_E)\iota = \iota P(L_B)$. At the last object it is

$$e \mapsto (P(L_E)e, \pi e),$$

which lies in the pullback by $\pi P(L_E) = P(L_Q)\pi$. Intermediate maps are central multiplication. For $n = 1$ the combined formula is $[e, b] \mapsto (P(L_E)e + \iota b, \pi e)$. They are continuous, commute with the differentials, and induce identity maps on the endpoints. This is the Yoneda equivalence proving GEX8.2.

The left endpoint action is zero because $P(L_B) = 0$: the first strict short factor $0 \rightarrow B \rightarrow E_{n-1} \rightarrow K_1 \rightarrow 0$, pushed out by the zero map $B \rightarrow B$, becomes $0 \rightarrow B \rightarrow K_1 \oplus B \rightarrow K_1 \rightarrow 0$, with its explicit continuous section $k \mapsto (k, 0)$. Its Yoneda splice is therefore zero. The right endpoint action is invertible because pullback by the continuous automorphism $P(L_Q)$ is inverted by pullback by K_Q . Thus every class in the stated order vanishes.

For the opposite order, the same naturality identity equates pushout by the target automorphism $P(L_Q)$ with pullback by the zero source map $P(L_B)$. The latter splits the last strict short factor: its pullback middle object is the product of that factor's kernel with B , with the section $b \mapsto (0, b)$. The pushout action is invertible, with inverse induced by K_Q , so every reverse class vanishes too. Degree zero follows from GEX4.5. Thus GEX7.1 also holds for these explicitly specified topological Yoneda groups, without an assumed topological projective resolution.

No completeness assumption on an arbitrary middle object was used. If a narrower category of topological representations is selected, its exact structure and closure under these finite constructions must first be verified.

141.10 GEX9. Exact application and the distinct DCP boundary

AC retains $J = 2\mathcal{E}\Phi$, its factor $1/2$ in the source inverse, and the equivariant splitting of its prime-labelled boundary. OMS7A identifies its algebraic cokernel with the same actual Q . Its particular global two-extension class was already zero. GEX7 now proves that every extension degree between the entire Q and every specified actual endpoint group $B^{[r]}$ is zero in the stated categories, by a global continuous inverse and an explicit canonical section.

All higher derived-coinvariant groups are retained as their full $\Lambda^{n+1}W_{\text{pr}} \otimes \mathbb{C}^2$, not removed from the derived object. No assertion that AC's two-term complex is the complete derived adelic complex is made.

DCP's supported degree-one row remains the actual

$$Q \xrightarrow{\Delta} Q \oplus Q \xrightarrow{\delta} Q, \quad \Delta(q) = (q, q), \quad \delta(q_+, q_-) = q_+ - q_-. \quad (\text{GEX9.1})$$

Here $P(L)$ is invertible on every term; it does not annihilate either target Q component. Therefore the endpoint identity GEX3.3 does not apply to this different coefficient receiver. Its precise operator comparison is

$$P(L_{Q \oplus Q})\Delta = \Delta P(L_Q), \quad \delta P(L_{Q \oplus Q}) = P(L_Q)\delta. \quad (\text{GEX9.2})$$

The row retains the section $\frac{1}{2}(q, -q)$, all mirror signs, and all original zero-jet observations proved in DCP11. The distinction is thus expressed by exact maps and calculated actions, not merely by different names.

For the existing bounded distributive support lattice, each constructed linear map has the exact common-label lift

$$(v, \lambda) \mapsto (f(v), \lambda)$$

on $G_L(V)$, as proved in OMS7.2. Linearity preserves its specified receiving operations and sends every $(0, \lambda)$ to the same labelled zero. This lifts the constructed maps, not an unconstructed abelian category of primitive τ ; common labels are not equated with independent block labels.

The new vanishing is global for the original Q and these actual endpoint coefficients, with its category, source inverse, topology, and equivariances fully specified. It excludes all endpoint extensions of this type. It does not prove $\Re\rho = 1/2$: the central separation used here excludes the endpoint parameters $0, 1$, using the retained original-zeta source transform. Full tau purity is not inferred from that endpoint separation.

142 Full derived prime boundary: exact quasi-isomorphism, mirror, and comparison cone

24 September 2026. Independent mathematical verification FDB0–FDB9. This is a derivation for the specified receiving complexes, with all prime labels, endpoint coordinates, original periodization factors and cochain signs retained.

142.1 FDB0. Scope, sources, and notation

The primitive source remains the supplied Z_0, Z_1, Z_2, τ construction. In particular Z_1/τ has no source addition or parity. None of the additions, exterior products, coefficients, signs, Fourier transforms or character actions below is an operation on primitive τ . They belong to the explicitly named complex receiving spaces, after the whole arithmetic reconstruction.

The current `CORPUS_AND_OPERATION_RULES.md` (private source record) was read. This review reads and uses the actual definitions and proofs in:

- `CC_ADELIC_COINVARIANT_BRIDGE_INDEPENDENT.tex`, ABR1–ABR22, including the full finite adelic basis, both endpoint moments, both periodization conventions, and the ordered Koszul differential. The reading here covered source lines 1–503.
- `ACTUAL_ADELIC_COMPLEX_INDEPENDENT.md`, AC0–AC4, including $J = 2\mathcal{E}\Phi$, its exact inverse and all real-dilation factors.
- `ORIGINAL_MELLIN_SPECTRAL_SYNTHESIS.md`, OMS0–OMS1 and OMS7A. Its closed-image result is a proved programme input here, not a new analytic proof in this review.
- `SOURCE_CC_DOUBLE_PULLBACK.md`, DCP0–DCP12, for the actual chart spaces, restriction maps, real action and ordered-Cech mirror.

Their human source attribution is retained: Alain Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Section III; Alain Connes, Caterina Consani and Matilde Marcolli, *The Weil proof and the geometry of the adeles class space*; Alain Connes and Caterina Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, §5; and Ralf Meyer, *A spectral interpretation for the zeros of the Riemann zeta function*. The original-source readings and exact version records are those documented in the cited programme proofs; this bounded review does not claim a fresh complete reading of these papers. The standard resolution used below is Jean-Louis Koszul’s complex; its exactness is proved here in the present Laurent algebra.

We distinguish the centered receiving test space $\mathcal{A}$ from the raw receiving test space A , even though their sets have the same strong-decay definition. We also distinguish their quotients Q_{cen} and Q_{raw} until the complete comparison map is written. No completed zeta function substitutes for the original ζ .

All tensor products and exterior powers in the adelic/Koszul construction are algebraic. In particular each vector has finite prime support. The quasi-isomorphisms below are in the algebraic category of complex representations of the specified actions. No extra topology on an infinite algebraic direct sum, and no equivariant continuous section of $\mathcal{A} \rightarrow Q_{\text{cen}}$, is asserted.

142.2 FDB1. Original spaces and the augmented complex

Let

$$\begin{aligned}
 G &= \mathbb{Q}_{>0}^\times, & R &= \mathbb{C}[G], & \varepsilon(t_a) &= 1, & I_R &= \ker \varepsilon, \\
 H &= \mathcal{S}_{\text{even}}(\mathbb{R}), & m(h) &= \left(h(0), \int_{\mathbb{R}} h(x) dx \right), & H_{00} &= \ker m, \\
 M_0 &= \ker(\varepsilon \otimes m : R \otimes H \rightarrow \mathbb{C}^2), & W &= \bigoplus_{p \text{ prime}} \mathbb{C}\ell_p, & x_p &= t_p - 1.
 \end{aligned} \tag{FDB1.1}$$

The original adelic realization is

$$\Theta(t_a \otimes h)(x_\infty, x_f) = h(x_\infty/a) 1_{a\widehat{\mathbb{Z}}}(x_f). \quad (\text{FDB1.2})$$

Finite additive Haar measure has $\text{vol}(\mathbb{Z}_p) = 1$, so $\text{vol}(a\widehat{\mathbb{Z}}) = a^{-1}$. Real dilation in the integral contributes a . Consequently the original adelic evaluation and integral of (FDB1.2) are exactly $h(0)$ and $a a^{-1} \int h = \int h$. This is why the actual source with both endpoint conditions is precisely M_0 , with its displayed moment map.

Put

$$\begin{aligned} \Phi &= \varepsilon \otimes \text{id}_H : M_0 \longrightarrow H_{00}, & \mathcal{E}h(u) &= u^{1/2} \sum_{n \geq 1} h(nu), \\ \mathcal{A} &= \{k \in C^\infty(\mathbb{R}_{>0}) : \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j k(u)| < \infty \text{ for every } N, j \geq 0\}, \\ J &= 2\mathcal{E}\Phi : M_0 \longrightarrow \mathcal{A}, & I_A &= 2\mathcal{E}H_{00}, & Q_{\text{cen}} &= \mathcal{A}/I_A. \end{aligned} \quad (\text{FDB1.3})$$

The factor 2 is the original sum over both nonzero integer signs after the full rational sum is restricted by the finite adelic support. ABR12 proves this reduction. ABR14–ABR15 prove injectivity of $\mathcal{E}$, and OMS7A proves that I_A is closed and equals the earlier closed image. These results use the original identity

$$F_{Jm}(s) = 2\zeta(s) \int_0^\infty \Phi(m)(x) x^{s-1} dx, \quad F_k(s) = \int_0^\infty k(u) u^{s-1/2} \frac{du}{u}. \quad (\text{FDB1.4})$$

For $k \in I_A$, their inverse to the map $2\mathcal{E}$ is

$$h_k(x) = \frac{1}{4\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} \frac{F_k(s)}{\zeta(s)} x^{-s} ds, \quad \sigma > 1. \quad (\text{FDB1.5})$$

This retains the factor $1/2$ from the two-sign summation in addition to the factor $1/(2\pi i)$ of Mellin inversion.

The augmented cochain complex is

$$C^1 = \mathcal{A}, \quad C^0 = M_0, \quad C^{-k} = M_0 \otimes \Lambda^k W \quad (k \geq 1), \quad (\text{FDB1.6})$$

with $d_C^0 = J$, and, for the ordered wedge $\ell_P = \ell_{p_1} \wedge \cdots \wedge \ell_{p_k}$,

$$d_C^{-k}(m \otimes \ell_P) = \sum_{j=1}^k (-1)^{j-1} x_{p_j} m \otimes \ell_{P \setminus p_j}. \quad (\text{FDB1.7})$$

There is no extra minus sign in this augmented differential. The two deletion orders of any pair of indices have opposite signs, proving $d_C^2 = 0$ in negative degrees. Also $\Phi(x_p m) = 0$, proving $Jd_C^{-1} = 0$.

142.3 FDB2. The full cohomology, retaining all prime wedges

The Koszul complex $R \otimes \Lambda^\bullet W$ resolves the augmentation module $\mathbb{C}$. For a finite prime set, successively setting $t_p = 1$ leaves a Laurent polynomial algebra, and the next $t_p - 1$ is a nonzero divisor. Adjoining that generator forms the mapping cone of multiplication by $t_p - 1$. Induction proves exactness in positive degrees and the stated degree-zero quotient. For infinitely many primes every cycle uses finitely many wedge indices and Laurent variables; the finite-set proof gives a finite primitive. Thus the same exactness holds for the algebraic full complex.

Apply this resolution degree-wise to the exact sequence

$$0 \longrightarrow M_0 \longrightarrow R \otimes H \xrightarrow{\varepsilon \otimes m} \mathbb{C}^2 \longrightarrow 0. \quad (\text{FDB2.1})$$

Surjectivity holds, for example, by the actual moment section

$$\sigma(u_0, u_1) = u_0(1 - 2\pi x^2)e^{-\pi x^2} + u_1 2\pi x^2 e^{-\pi x^2}. \quad (\text{FDB2.2})$$

Indeed $\int e^{-\pi x^2} = 1$ and $\int x^2 e^{-\pi x^2} = 1/(2\pi)$, so its two moments are (u_0, u_1) .

The middle complex has homology H in degree zero and zero in positive degrees. The last complex has zero differential, since each x_p acts by zero; its degree- n term is $\Lambda^n W \otimes \mathbb{C}^2$. The connecting cycle for an ordered n -wedge, including $n = 1$, is

$$\delta_n(\ell_{p_1} \wedge \cdots \wedge \ell_{p_n} \otimes u) = \sum_{j=1}^n (-1)^{j-1} (x_{p_j} \otimes \sigma(u)) \otimes \ell_{P \setminus p_j}. \quad (\text{FDB2.3})$$

It is a cycle because it is the differential of $(t_1 \otimes \sigma(u)) \otimes \ell_P$ in the middle complex, and $d^2 = 0$. Changing the moment section changes (FDB2.3) by the differential of the same full wedge with a coefficient in $H_{00} \subset M_0$. Its cohomology class is therefore independent of the section.

For $n \geq 2$ the connecting map is an isomorphism: a cycle in the first complex is a boundary in the positive-degree middle complex; projecting such a primitive gives its preimage in the last complex. If this last class connects to a boundary, subtract that first-complex primitive; middle exactness then implies the projected class is zero. For $n = 1$ it identifies the kernel of $M_0/I_R M_0 \rightarrow H$ with $W \otimes \mathbb{C}^2$. The image in H is H_{00} , since $t_1 \otimes h$ maps to every $h \in H_{00}$.

Because $\mathcal{E}$ is injective, $\ker J = \ker \Phi$. Adding the last term $\mathcal{A}$ consequently leaves precisely that boundary kernel in degree zero. Writing $B_n = \Lambda^n W \otimes \mathbb{C}^2$, the complete answer is

$$H^1(C) = Q_{\text{cen}}, \quad H^{1-n}(C) = B_n \quad (n \geq 1), \quad H^j(C) = 0 \quad (j \geq 2). \quad (\text{FDB2.4})$$

In particular the higher groups have not been replaced by the two-term degree-zero coinvariant calculation.

142.4 FDB3. An explicit canonical quasi-isomorphism

Define the **scalar augmentation derivation**

$$D_p : R \rightarrow \mathbb{C}, \quad D_p \left(\sum_a c_a t_a \right) = \sum_a c_a v_p(a). \quad (\text{FDB3.1})$$

This is not the R -valued Euler derivation. Its exact product law is

$$D_p(rs) = \varepsilon(r)D_p(s) + \varepsilon(s)D_p(r), \quad D_p(x_q r) = \delta_{pq}\varepsilon(r). \quad (\text{FDB3.2})$$

Both equations follow by expanding finite sums and using $v_p(ab) = v_p(a) + v_p(b)$.

Let

$$B = Q_{\text{cen}}[-1] \oplus \bigoplus_{n \geq 1} B_n[n-1] \quad (\text{FDB3.3})$$

with zero differential. Our convention is $(K[r])^j = K^{j+r}$; thus Q_{cen} occurs in degree 1, and B_n in degree $1 - n$. Define

$$\Gamma^1 = \pi_{Q_{\text{cen}}}, \quad \Gamma^{-k} \left(\left(\sum_i r_i \otimes h_i \right) \otimes \omega \right) = \frac{1}{k+1} \sum_{p,i} D_p(r_i) \ell_p \wedge \omega \otimes m(h_i), \quad k \geq 0. \quad (\text{FDB3.4})$$

This is a well-defined linear map on the algebraic tensor product: $\sum_p D_p(r) \ell_p$ is a finite, linear function of r , and every operation displayed is multilinear. It depends on no choice of moment section.

For $k \geq 1$, evaluate $\Gamma^{-k+1}d_C^{-k}$ on a coefficient $m = \sum_i r_i \otimes h_i \in M_0$ and a k -wedge. By (FDB3.2) the result is

$$\frac{1}{k} \sum_{j=1}^k (-1)^{j-1} \ell_{p_j} \wedge \ell_{P \setminus p_j} \otimes \sum_i \varepsilon(r_i) m(h_i) = \ell_P \otimes 0 = 0. \quad (\text{FDB3.5})$$

The equality uses $(-1)^{j-1} \ell_{p_j} \wedge \ell_{P \setminus p_j} = \ell_P$, and the final zero is exactly the defining total-moment equation of M_0 . In the last degree $\Gamma^1 J = 0$, since $J(M_0) = I_A$. Hence Γ is a chain map.

Apply Γ^{1-n} to (FDB2.3). Each of its n terms contributes the same ordered n -wedge tensored with u ; $D_p(x_q) = \delta_{pq}$ and $m\sigma = \text{id}$. Its denominator is exactly n . Therefore

$$\Gamma^{1-n} \delta_n = \text{id}_{B_n} \quad (n \geq 1). \quad (\text{FDB3.6})$$

In degree 1, Γ induces the identity quotient. The cohomology calculation (FDB2.4) now proves

$\Gamma : C \longrightarrow Q_{\text{cen}}[-1] \oplus \bigoplus_{n \geq 1} (\Lambda^n W \otimes \mathbb{C}^2)[n-1]$ is a canonical quasi-isomorphism.

(FDB3.7)

Canonical here refers to the specified rational prime generators, the canonical derivations (FDB3.1), the actual two endpoint moments and the actual quotient map. It does not mean a strict inverse cochain map has been constructed: such an inverse on the quotient degree would in particular require a lift of Q_{cen} into $\mathcal{A}$. No such lift is used.

142.5 FDB4. All real dilations and the full rational action

For every real $a > 0$, set

$$R_a h(x) = h(x/a), \quad \mathcal{R}_a = \text{id}_R \otimes R_a, \quad W_a k(u) = a^{1/2} k(u/a). \quad (\text{FDB4.1})$$

On negative degrees the real action is $\mathcal{R}_a \otimes \text{id}_{\Lambda^k W}$; on degree 1 it is W_a . The exact moment formula is

$$m(R_a h) = \begin{pmatrix} 1 & 0 \\ 0 & a \end{pmatrix} m(h) =: D_a m(h). \quad (\text{FDB4.2})$$

Thus $\mathcal{R}_a$ preserves M_0 and commutes with every negative differential. Direct substitution gives

$$J \mathcal{R}_a m(u) = 2u^{1/2} \sum_{n \geq 1} \Phi(m)(nu/a) = a^{1/2} (Jm)(u/a) = W_a Jm(u). \quad (\text{FDB4.3})$$

The quasi-isomorphism Γ is equivariant with action W_a on Q_{cen} and $\text{id}_{\Lambda^n W} \otimes D_a$ on every B_n , directly from (FDB3.4).

The **full diagonal rational** action of $b \in G$ is multiplication by t_b on the R coefficient in every nonpositive degree. It is identity on $\mathcal{A}$ and its quotient, because actual rational periodization reindexes the full rational sum. The differential commutes with this action and $J(t_b m) = Jm$. By the product rule,

$$D_p(t_b r_i) = D_p(r_i) + v_p(b) \varepsilon(r_i).$$

The additional term in $\Gamma(t_b m \otimes \omega)$ is

$$\frac{1}{k+1} \sum_p v_p(b) \ell_p \wedge \omega \otimes \sum_i \varepsilon(r_i) m(h_i) = 0. \quad (\text{FDB4.4})$$

Therefore the full rational action on every receiving B_n is trivial, and Γ is equivariant for it. These rational operators and the real-idele operators (FDB4.1) are different specified actions, even when their parameters have the same rational value. Both have now been evaluated rather than identified.

142.6 FDB5. Full adelic Fourier and all higher mirror signs

Use the original real Fourier phase $e^{-2\pi i x \xi}$, and the finite self-dual character with annihilator of $\widehat{\mathbb{Z}}$ equal to $\widehat{\mathbb{Z}}$. The finite ball transform is

$$\widehat{1_{a\widehat{\mathbb{Z}}}} = a^{-1} 1_{a^{-1}\widehat{\mathbb{Z}}},$$

because the integral of a character on that compact additive subgroup is its volume if the character is trivial there and is zero otherwise. The real transform is

$$\widehat{h(\cdot/a)}(\xi) = a\widehat{h}(a\xi).$$

Multiplying these two original factors gives $a a^{-1} = 1$. Consequently the full adelic Fourier operator in the original Θ coordinates is exactly

$$\mathcal{F}_0(t_a \otimes h) = t_{a^{-1}} \otimes \widehat{h}. \quad (\text{FDB5.1})$$

It preserves M_0 because $m(\widehat{h}) = Sm(h)$, where $S(u_0, u_1) = (u_1, u_0)$, and augmentation is unchanged. Its square is identity: Fourier squared is real reflection, identity on the even source, while rational inversion squares to identity.

Write $\iota(t_a) = t_{a^{-1}}$ and $t_P = \prod_{p \in P} t_p$. The exact higher-degree lift is

$$\mathcal{F}_k((r \otimes h) \otimes \ell_P) = (-1)^k (t_P^{-1} \iota(r) \otimes \widehat{h}) \otimes \ell_P. \quad (\text{FDB5.2})$$

The coefficient remains in M_0 for every coefficient sum, since $\varepsilon(t_P^{-1}) = 1$. To check the chain equation, use

$$\iota(x_p) = t_p^{-1} - 1 = -t_p^{-1} x_p.$$

Applying $\mathcal{F}_{k-1}$ to the term deleting p_j introduces $(-1)^{k-1} t_{P \setminus p_j}^{-1} (-t_{p_j}^{-1}) = (-1)^k t_P^{-1}$, exactly its coefficient in $d\mathcal{F}_k$, with the unchanged deletion sign $(-1)^{j-1}$. Hence

$$d\mathcal{F}_k = \mathcal{F}_{k-1} d, \quad \mathcal{F}_k^2 = 1. \quad (\text{FDB5.3})$$

For the second equality, the two monomial multipliers cancel because $t_P^{-1} \iota(t_P^{-1}) = 1$, and the two signs multiply to 1.

At degree 1 define $R_{\text{cen}} k(u) = k(u^{-1})$. The full Poisson formula is

$$2 \sum_{n \geq 1} \widehat{h}(nu) = u^{-1} 2 \sum_{n \geq 1} h(n/u) + u^{-1} h(0) - \int h. \quad (\text{FDB5.4})$$

For $h = \Phi(m) \in H_{00}$, both endpoint terms vanish by its actual defining conditions. Multiplication by $u^{1/2}$ then proves

$$J\mathcal{F}_0 = R_{\text{cen}} J. \quad (\text{FDB5.5})$$

Thus (FDB5.2) with R_{cen} in degree 1 is an involutive chain action on C .

Its exact action under Γ follows from

$$D_q(t_P^{-1} \iota(r)) = -D_q(r) - 1_{q \in P} \varepsilon(r). \quad (\text{FDB5.6})$$

The second term gives zero when wedged with ℓ_P , because $q \in P$; it also disappears after the total-moment summation. The first term and the sign $(-1)^k$ give

$$\Gamma^{-k} \mathcal{F}_k = (-1)^{k+1} (\text{id} \otimes S) \Gamma^{-k}. \quad (\text{FDB5.7})$$

Therefore the un-oriented mirror on B_n is $(-1)^n \text{id}_{\Lambda^n W} \otimes S$, and on Q_{cen} it is the induced R_{cen} .

All actions retain their original twist relation. Direct Fourier substitution and direct evaluation on $\mathcal{A}$ give

$$\mathcal{F}_k \mathcal{R}_a = a \mathcal{R}_{a^{-1}} \mathcal{F}_k, \quad R_{\text{cen}} W_a = a W_{a^{-1}} R_{\text{cen}}. \quad (\text{FDB5.8})$$

On endpoints this is $SD_a = a D_{a^{-1}} S$. Also $\mathcal{F}_k t_b = t_{b^{-1}} \mathcal{F}_k$, so inversion of the full rational action is retained.

142.7 FDB6. The exact map to the actual two-chart complex

Use DCP's original coefficient spaces and restrictions:

$$V_+ = H_{00} \oplus \mathbb{C}^2, \quad V_- = H_{00} \oplus \mathbb{C}^2, \quad \Sigma h(u) = 2 \sum_{n \geq 1} h(nu), \quad R_{\text{raw}} b(u) = u^{-1} b(u^{-1}),$$

$$D^0 = V_+ \oplus V_-, \quad D^1 = A, \quad d_D((h, c_0, c_1), (g, d_0, d_1)) = \Sigma h - R_{\text{raw}} \Sigma g. \quad (\text{FDB6.1})$$

The raw quotient is $Q_{\text{raw}} = A/\Sigma H_{00}$. Its relation to (FDB1.3), with all factors, is

$$T : \mathcal{A} \xrightarrow{\sim} A, \quad Tk(u) = 2u^{-1/2}k(u), \quad T^{-1}b(u) = \frac{1}{2}u^{1/2}b(u). \quad (\text{FDB6.2})$$

Indeed $T\mathcal{E}h = \Sigma h$, so $T(2\mathcal{E}H_{00}) = 2\Sigma H_{00} = \Sigma H_{00}$ as vector subspaces. Its induced map $\bar{T} : Q_{\text{cen}} \rightarrow Q_{\text{raw}}$ is an isomorphism. The original Mellin comparison is $\mathcal{M}_0(Tk) = 2F_k$; this receiving factor is not suppressed.

Define the plus-chart comparison

$$(F_+)^1 = T, \quad (F_+)^0(m) = ((2\Phi(m), 0, 0), (0, 0, 0)), \quad (F_+)^j = 0 \quad (j < 0). \quad (\text{FDB6.3})$$

Its last chain identity, including every factor, is

$$d_D(F_+)^0 m = \Sigma(2\Phi(m)) = 4 \sum_{n \geq 1} \Phi(m)(nu) = 2u^{-1/2}(2u^{1/2} \sum_{n \geq 1} \Phi(m)(nu)) = TJm. \quad (\text{FDB6.4})$$

The next identity is $(F_+)^0 d_C^{-1} = 0$, since $\Phi(x_p m) = 0$. All other identities are between zero maps. Thus F_+ is an actual chain map, not only a proposed map on cohomology.

On the raw complex the full real action is

$$\begin{aligned} T_a b(u) &= b(u/a), \quad \rho_+(a)(h, c_0, c_1) = (R_a h, c_0, a c_1), \\ \rho_-(a)(g, d_0, d_1) &= (a g(a \cdot), a d_0, d_1). \end{aligned} \quad (\text{FDB6.5})$$

Substitution proves $TW_a = T_a T$, and proves that (FDB6.3) intertwines both chart and source real actions. Assigning the full diagonal rational action to be trivial on D is exactly its action after rational periodization; (FDB6.3) also intertwines this action.

The degree-zero cohomology of D is the actual space

$$H^0(D) = \{((\hat{h}, c_0, c_1), (h, d_0, d_1)) : h \in H_{00}, c_0, c_1, d_0, d_1 \in \mathbb{C}\}, \quad (\text{FDB6.6})$$

because Poisson gives $R_{\text{raw}} \Sigma h = \hat{\Sigma} h$ and Σ is injective. Also $H^1(D) = Q_{\text{raw}}$, and all other groups vanish. A degree-zero cycle of C has $\Phi(m) = 0$, so its image under F_+ is zero already as a cochain. Thus

$$H^1(F_+) = \bar{T}, \quad H^j(F_+) = 0 \quad (j \leq 0). \quad (\text{FDB6.7})$$

142.8 FDB7. The mirror-compatible map and its orientation representation

The ordered Cech mirror is

$$w_D^0(v_+, v_-) = (v_-, v_+), \quad w_D^1 = -R_{\text{raw}}. \quad (\text{FDB7.1})$$

The minus sign is forced by reversing the ordered difference in (FDB6.1). On the same underlying complex C , use the oriented mirror

$$w_{C^{\text{or}}}^{-k} = -\mathcal{F}_k \quad (k \geq 0), \quad w_{C^{\text{or}}}^1 = -R_{\text{cen}}. \quad (\text{FDB7.2})$$

It is still an involutive chain action, since every degree of the previous chain action is multiplied by the same receiving sign. This is a one-dimensional orientation representation in the receiving coefficient category; it supplies no sign operation or parity on primitive τ .

The exact symmetrized map is

$$F_{\text{sym}} = \frac{1}{2}(F_+ + w_D F_+ w_{C^{\text{or}}}),$$

$$F_{\text{sym}}^1 = T, \quad F_{\text{sym}}^0(m) = ((\Phi(m), 0, 0), (-\widehat{\Phi(m)}, 0, 0)), \quad F_{\text{sym}}^j = 0 \ (j < 0). \quad (\text{FDB7.3})$$

To evaluate the average, the mirrored degree-zero map is $((0, 0, 0), (-2\widehat{\Phi(m)}, 0, 0))$. In degree one, $(-R_{\text{raw}})T(-R_{\text{cen}}) = T$, because $R_{\text{raw}}T = TR_{\text{cen}}$. This proves (FDB7.3) with the averaging factor retained.

Directly, its differential is

$$\Sigma\Phi(m) - R_{\text{raw}}\Sigma(-\widehat{\Phi(m)}) = 2\Sigma\Phi(m) = TJm. \quad (\text{FDB7.4})$$

Its mirror identity in degree zero reads

$$F_{\text{sym}}^0(-\mathcal{F}_0 m) = ((-\widehat{\Phi(m)}, 0, 0), (\Phi(m), 0, 0)) = w_D^0 F_{\text{sym}}^0(m).$$

The degree-one identity is $T(-R_{\text{cen}}) = -R_{\text{raw}}T$. Real equivariance follows from $\widehat{R_a h} = a\widehat{h}(a \cdot)$, exactly the minus-chart action in (FDB6.5). Full diagonal rational equivariance follows from unchanged augmentation. Hence F_{sym} is simultaneously equivariant for the original real actions, full rational action and oriented mirror. It has the same maps on cohomology as (FDB6.7).

Under Γ , the oriented mirror on B_n is

$$(-1)^{n+1} \text{id}_{\Lambda^n W} \otimes S, \quad (\text{FDB7.5})$$

and on Q_{cen} it is $-R_{\text{cen}}$. Thus (FDB3.7) is equivariant for this full oriented action as well.

142.9 FDB8. The entire mapping-cone cohomology, without an asserted splitting

Let F mean either F_+ , or F_{sym} with the oriented actions. Fix the mapping-cone convention

$$\text{Cone}(F)^j = D^j \oplus C^{j+1}, \quad d_{\text{Cone}}(v, c) = (d_D v + Fc, -d_C c). \quad (\text{FDB8.1})$$

This cone differential has its displayed minus sign on the shifted C term. It is separate from the unmodified augmented differential (FDB1.7).

For $j \leq -2$, the D terms vanish, and the cone is the shifted negative complex of C . In degree -1 , a cone cochain is $m \in M_0$; its differential is $(F^0 m, -Jm)$. Since J is injective after Φ , and both displayed versions of F^0 factor through Φ , this is zero exactly when $\Phi(m) = 0$. Its boundaries are $-d_C^{-1}$ of the degree -1 Koszul terms, so the sign changes no boundary subspace. These calculations give canonical projection isomorphisms

$$H^{-n}(\text{Cone}(F)) \cong B_n = \Lambda^n W \otimes \mathbb{C}^2 \quad (n \geq 1). \quad (\text{FDB8.2})$$

In degree zero a cochain is $(v, k) \in D^0 \oplus \mathcal{A}$. It is a cycle exactly when $d_D v + Tk = 0$. Since T is bijective, each v gives the unique cycle $(v, -T^{-1}d_D v)$. The boundaries are precisely $(F^0 m, -Jm)$, so

$$H^0(\text{Cone}(F)) \cong D^0 / F^0(M_0). \quad (\text{FDB8.3})$$

The natural inclusion of $H^0(D)$ into this quotient is an isomorphism. To prove surjectivity, write $d_D v \in \Sigma H_{00}$. The map T sends I_A onto ΣH_{00} , and J sends M_0 onto I_A . Therefore there is $m \in M_0$ with $TJm = d_D v$. Then $v - F^0 m$ lies in $\ker d_D = H^0(D)$. To prove injectivity, if

$v \in H^0(D) \cap F^0(M_0)$, write $v = F^0 m$; then $0 = d_D v = T J m$, hence $\Phi(m) = 0$, and consequently $v = F^0 m = 0$. This proves

$$H^0(\text{Cone}(F)) \cong H^0(D) \cong H_{00} \oplus \mathbb{C}^4, \quad (\text{FDB8.4})$$

where the first isomorphism is the canonical one induced by $D \rightarrow \text{Cone}(F)$. The second uses exactly the Fourier-parametrization (FDB6.6), not an arbitrary choice of representatives.

In degree one the cone is A ; its incoming differential is surjective because the $\mathcal{A}$ summand of degree zero maps by the isomorphism T . There are no terms above degree one. Thus

$$H^j(\text{Cone}(F)) = 0 \quad (j \geq 1). \quad (\text{FDB8.5})$$

Equations (FDB8.2), (FDB8.4) and (FDB8.5) give every cohomological degree.

The actions are retained explicitly. In (FDB8.2), real dilation acts by $\text{id}_{\Lambda^n W} \otimes \text{diag}(1, a)$, and full rational dilation acts trivially. For F_{sym} , its mirror is (FDB7.5). On the coordinates (h, c_0, c_1, d_0, d_1) of (FDB8.4), real dilation is

$$(h, c_0, c_1, d_0, d_1) \mapsto (ah(a \cdot), c_0, ac_1, ad_0, d_1), \quad (\text{FDB8.6})$$

and the mirror is

$$(h, c_0, c_1, d_0, d_1) \mapsto (\widehat{h}, d_0, d_1, c_0, c_1). \quad (\text{FDB8.7})$$

These formulas follow by applying the chart actions to (FDB6.6), including $\widehat{ah(a \cdot)} = R_a \widehat{h}$. They retain all four endpoint lines, with real characters $1, a, a, 1$.

No equivariant decomposition of the **cone complex** as the direct sum of these cohomology groups has been asserted or used. The maps proving (FDB8.2)–(FDB8.4) are canonical cohomology maps; a derived splitting of this cone would be a further statement. In contrast, the quasi-isomorphism of the **source complex** C was actually constructed in (FDB3.4), so its derived direct-sum conclusion (FDB3.7) is proved.

142.10 FDB9. Verified conclusion and exact scope of the additional result

The requested map Γ , including its degree-one quotient, is a canonical equivariant quasi-isomorphism of the full augmented complex. It proves the formal decomposition of that particular entire derived-coinvariant receiver, with every higher prime wedge and both original endpoint coordinates retained. It strengthens the earlier two-term extension calculation without replacing the original quotient by an unrestricted product of zero jets.

The maps F_+ and F_{sym} are actual cochain maps to the specified Connes–Consani two-chart coefficient complex, with the exact factors $2, u^{-1/2}$, Fourier transform and orientation signs. They induce an isomorphism only in degree one; their lower cohomology maps are zero. The full cone records precisely the retained comparison data given in FDB8. It therefore supplies an exact global object measuring this comparison, rather than inferring that the extra prime data vanish.

These calculations do not prove that the entire geometric specialization of a different source equals this augmented complex, do not prove that an entire extension group vanishes, and do not assign a numerical weight to primitive τ . They establish the displayed maps, quasi-isomorphism, actions and all cone cohomology for the original receiving objects.

143 The whole comparison triangle: canonical projections and exact contractions

24 September 2026. CTF0–CTF8. Independent derivation from the actual GAP, FDB and DCP complexes. The result strengthens GAP6 and FDB8 from a calculation of every cohomology group to an explicit equivariant quasi-isomorphism of the entire comparison cone and a derived comparison of the entire triangle. It also retains the faithful source extension on both objects and contracts precisely its identity comparison cone.

143.1 CTF0. Scope, original sources, and category

The files `GLOBAL_ADELIC_LIFTING_AND_PRIME_BOUNDARY.md` GAP0–GAP9, `FULL_DERIVED_PRIME_BOUNDARY_REVIEW.md` FDB0–FDB9, and `SOURCE_CC_DOUBLE_PULLBACK.md` DCP0–DCP12 were read, including a separate complete reading of DCP8–DCP9 after a combined tool output truncated that passage. `CORPUS_AND_OPERATION_RULES.md` was reread. The complete user global arguments were read for OMS and continue to govern the receiving operations: primitive Z_1/τ has no source addition or parity, and the arithmetic and complex coefficient operations below occur after the whole-spectrum reconstruction.

Human sources retain their authorship: Alain Connes, Trace formula in noncommutative geometry and the zeros of the Riemann zeta function; Alain Connes, Caterina Consani and Matilde Marcolli, The Weil proof and the geometry of the adeles class space; Alain Connes and Caterina Consani, Schemes over $\mathbb{F}_1$ and zeta functions, §5; Ralf Meyer, A spectral interpretation for the zeros of the Riemann zeta function. The original-source reading records are the cited programme proofs; this note does not claim a new full reading of those papers. The closed summation-image theorem used below was proved and independently checked in OMS/SSI. Jean-Louis Koszul’s resolution is retained with its actual differential and the exactness argument recalled in CTF1.

All adelic group algebras and exterior powers below are algebraic, and the complete calculation is in complexes of complex representations of the specified real dilations, diagonal rational action and mirror. The mirror and dilation satisfy their original twisted relation. Where spaces have the original Schwartz or strong Fréchet topologies, the displayed maps on those spaces are continuous. No topology on an infinite algebraic direct sum is added. No equivariant section of the quotient Q_{cen} into its test space is asserted.

Canonical in this note means that the maps are explicitly determined by the original Fourier transform, ordered two-chart comparison, rational valuation derivations, endpoint moments, and quotient maps. It does not assert uniqueness among every conceivable quasi-isomorphism.

143.2 CTF1. The exact original source complex and its existing comparison

Keep

$$\begin{aligned}
 G &= \mathbb{Q}_{>0}^\times, & R &= \mathbb{C}[G], & \varepsilon(t_a) &= 1, & x_p &= t_p - 1, & W_{\text{pr}} &= \bigoplus_{p \text{ prime}} \mathbb{C}\ell_p, \\
 H &= \mathcal{S}_{\text{even}}(\mathbb{R}), & m(h) &= \left(h(0), \int_{\mathbb{R}} h(v) dv \right), & S &= \ker m, & M_0 &= \ker(\varepsilon \otimes m : R \otimes H \rightarrow \mathbb{C}^2), \\
 \Phi &= \varepsilon \otimes \text{id} : M_0 \rightarrow S, & \mathcal{E}h(u) &= u^{1/2} \sum_{r \geq 1} h(ru), & J &= 2\mathcal{E}\Phi.
 \end{aligned} \tag{CTF1.1}$$

The original strong space $\mathcal{A}$ consists of the smooth functions with $\sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j k(u)| < \infty$ for every $N, j \geq 0$. Put $Q_{\text{cen}} = \mathcal{A}/(2\mathcal{E}S)$. The original complex has $C^1 = \mathcal{A}$ and $C^{-k} = M_0 \otimes \Lambda^k W_{\text{pr}}$, $k \geq 0$, with $d_C^0 = J$ and

$$d_C(c \otimes \ell_{p_1} \wedge \cdots \wedge \ell_{p_k}) = \sum_{i=1}^k (-1)^{i-1} x_{p_i} c \otimes \ell_{p_1} \wedge \cdots \widehat{\ell_{p_i}} \cdots \wedge \ell_{p_k}. \tag{CTF1.2}$$

The primes are ordered increasingly in each wedge. No factor two or Koszul sign is changed.

The established map to the zero-differential complex

$$H_C = Q_{\text{cen}}[-1] \oplus \bigoplus_{n \geq 1} B_n[n-1], \quad B_n = \Lambda^n W_{\text{pr}} \otimes \mathbb{C}^2, \quad (K[r])^j = K^{j+r},$$

is

$$\Gamma^1 = \pi_{\text{cen}}, \quad \Gamma^{-k} \left(\left(\sum_i r_i \otimes h_i \right) \otimes \omega \right) = \frac{1}{k+1} \sum_{p,i} D_p(r_i) \ell_p \wedge \omega \otimes m(h_i), \quad (\text{CTF1.3})$$

where $D_p(\sum_a c_a t_a) = \sum_a c_a v_p(a)$ is the scalar augmentation derivation. Its product rule gives $D_p(x_q r) = \delta_{pq} \varepsilon(r)$. Therefore applying Γ to (CTF1.2) produces k copies of the same ordered wedge divided by k , each multiplied by the zero moment $\sum_i \varepsilon(r_i) m(h_i) = 0$. At the final differential, $\pi_{\text{cen}} J = 0$. Thus Γ is a chain map.

For the quasi-isomorphism, the ordered Koszul complex of R resolves the augmentation module. On a finite prime set, successive quotients by $t_p - 1$ remain Laurent polynomial algebras; the next $t_p - 1$ is a nonzero divisor. Taking the cone of its multiplication proves exactness inductively. Every cycle in the full algebraic complex uses finitely many variables and wedge indices, so the finite proof gives a finite primitive. Applying this resolution to

$$0 \rightarrow M_0 \rightarrow R \otimes H \rightarrow \mathbb{C}^2 \rightarrow 0$$

gives $H^1(C) = Q_{\text{cen}}$ and $H^{1-n}(C) = B_n$. Indeed the middle resolved module has only degree-zero homology H , while the last has zero differential and terms $\Lambda^n W_{\text{pr}} \otimes \mathbb{C}^2$. Its connecting representative is

$$z_{\omega,u} = \sum_{i=1}^n (-1)^{i-1} (x_{p_i} \otimes \sigma(u)) \otimes \ell_{p_1} \wedge \cdots \widehat{\ell_{p_i}} \cdots \wedge \ell_{p_n}, \quad (\text{CTF1.4})$$

for any moment section $m\sigma = \text{id}$. One available section is $\sigma(u_0, u_1)(v) = [u_0(1 - 2\pi v^2) + u_1 2\pi v^2] e^{-\pi v^2}$. Changing it changes the representative by a Koszul boundary, since the difference has both moments zero. Each of the n terms in (CTF1.4) contributes $\omega \otimes u/n$ under Γ^{1-n} . Hence Γ induces identity on B_n , and it induces the quotient identity in degree one. This proves the established quasi-isomorphism with every degree retained.

The source is the oriented complex C^{or} of GAP5/FDB7: its differential is unchanged, and its mirror is $-\mathcal{F}_k$ in degree $-k$ and $-R_{\text{cen}}$ in degree one, where $R_{\text{cen}} k(u) = k(1/u)$ and

$$\mathcal{F}_k((r \otimes h) \otimes \ell_P) = (-1)^k (t_P^{-1} \iota(r) \otimes \widehat{h}) \otimes \ell_P, \quad t_P = \prod_{p \in P} t_p, \quad \iota(t_a) = t_{a^{-1}}. \quad (\text{CTF1.5})$$

The identity $\iota(x_p) = -t_p^{-1} x_p$ proves that this commutes with the Koszul differential. Applying D_p to the inverted coefficient gives the extra sign and the augmentation-moment term, which vanishes. Thus under Γ , the oriented mirror is

$$w_{H_C}|_{B_n} = (-1)^{n+1} \text{id} \otimes S_2, \quad S_2(u_0, u_1) = (u_1, u_0), \quad w_{H_C}|_{Q_{\text{cen}}} = -R_{\text{cen}}. \quad (\text{CTF1.6})$$

Real dilation acts by $h(v) \mapsto h(v/a)$ on the H coefficient, by $W_a k(u) = a^{1/2} k(u/a)$ on $\mathcal{A}$, and by $\text{id} \otimes \text{diag}(1, a)$ on each B_n . The moment formula proves equivariance of Γ . The diagonal rational action t_b has an extra derivative $v_p(b) \sum_i \varepsilon(r_i) m(h_i) = 0$, so it acts trivially on all terms of H_C . This retains both kinds of arithmetic action separately.

143.3 CTF2. A canonical projection of the actual two-chart complex

Use the original raw spaces

$$\begin{aligned} V_+ &= S \oplus \mathbb{C}^2, \quad V_- = S \oplus \mathbb{C}^2, \quad D^0 = V_+ \oplus V_-, \quad D^1 = A, \\ \Sigma h(u) &= 2 \sum_{r \geq 1} h(ru), \quad R_{\text{raw}} b(u) = u^{-1} b(1/u), \\ d_D((h, c_0, c_1), (g, d_0, d_1)) &= \Sigma h - R_{\text{raw}} \Sigma g, \quad Q_{\text{raw}} = A/\Sigma S. \end{aligned} \quad (\text{CTF2.1})$$

Here A has the strong raw test-space seminorms of DCP3. The full Poisson formula

$$\Sigma \widehat{h}(u) = u^{-1} \Sigma h(u^{-1}) + u^{-1} h(0) - \int_{\mathbb{R}} h(v) dv$$

gives $R_{\text{raw}} \Sigma h = \Sigma \widehat{h}$ on this S , with its two endpoint conditions imposed and its separate chart endpoint coordinates retained. Fourier is involutive on even functions, and Σ is injective with its closed image by the proved original-zeta synthesis. Consequently

$$H_D^0 = \ker d_D = \{((h, c_0, c_1), (\widehat{h}, d_0, d_1)) : h \in S\}, \quad H_D^1 = Q_{\text{raw}}. \quad (\text{CTF2.2})$$

Let $H_D = H_D^0[0] \oplus Q_{\text{raw}}[-1]$, with zero differential.

Define

$$\begin{aligned} \Pi_D^0((h, c_0, c_1), (g, d_0, d_1)) &= \left(\left(\frac{h + \widehat{g}}{2}, c_0, c_1 \right), \left(\frac{\widehat{h} + g}{2}, d_0, d_1 \right) \right), \\ \Pi_D^1(b) &= \pi_{\text{raw}}(b). \end{aligned} \quad (\text{CTF2.3})$$

The second Schwartz entry is the Fourier transform of the first, so the output lies in H_D^0 . The endpoint coordinates are all unchanged. The chain equation in degree zero is $\pi_{\text{raw}} d_D = 0$; in the other degrees it is between zero maps. The restriction of Π_D^0 to $\ker d_D$ is identity, and the induced map on degree-one cohomology is the quotient identity. Therefore

$$\boxed{\Pi_D : D \longrightarrow H_D \text{ is a canonical quasi-isomorphism.}} \quad (\text{CTF2.4})$$

Both components are continuous on the original section and quotient topologies.

Its kernel is also exact on the stated spaces. In degree zero it is precisely

$$A_{\text{anti}} = \{((h, 0, 0), (-\widehat{h}, 0, 0)) : h \in S\},$$

and in degree one it is ΣS . On the displayed degree-zero vector the differential is $2\Sigma h$, with continuous inverse

$$b \longmapsto \left(\left(\frac{1}{2} \Sigma^{-1} b, 0, 0 \right), \left(-\frac{1}{2} \widehat{\Sigma^{-1} b}, 0, 0 \right) \right). \quad (\text{CTF2.5})$$

Thus this kernel complex has a concrete contraction on its own domain. This proves no equivariant section of the quotient map $A \rightarrow Q_{\text{raw}}$ with image in A : (CTF2.5) has domain ΣS , not Q_{raw} .

143.4 CTF3. Every action on the projection

For $a > 0$, write $R_a h(v) = h(v/a)$ and $M_a g(v) = ag(av)$. The original chart action is

$$\rho_D^0(a)((h, c_0, c_1), (g, d_0, d_1)) = ((R_a h, c_0, ac_1), (M_a g, ad_0, d_1)), \quad \rho_D^1(a)b(u) = b(u/a). \quad (\text{CTF3.1})$$

Fourier change of variables proves

$$\widehat{R_a h} = M_a \widehat{h}, \quad \widehat{M_a g} = R_a \widehat{g}. \quad (\text{CTF3.2})$$

The first Schwartz entry of $\Pi_D^0 \rho_D^0(a)v$ is consequently $R_a(h + \widehat{g})/2$, and its second is $M_a(\widehat{h} + g)/2$. These are exactly the entries of $\rho_{H_D}^0(a)\Pi_D^0 v$, while all four endpoint factors agree individually. In degree one, the quotient is equivariant because dilation preserves ΣS .

The ordered chart-swap mirror is

$$w_D^0(v_+, v_-) = (v_-, v_+), \quad w_D^1 = -R_{\text{raw}}. \quad (\text{CTF3.3})$$

On swapping h, g , the first and second Schwartz entries in (CTF2.3) interchange. Therefore $\Pi_D^0 w_D^0 = w_{H_D}^0 \Pi_D^0$. In degree one the quotient retains the negative reflection, so $\Pi_D^1 w_D^1 = w_{H_D}^1 \Pi_D^1$. No change of scale is made; the original relation remains $\rho(a)w = a w \rho(a^{-1})$. The full diagonal rational action is trivial on D , hence also on H_D , so its equivariance is immediate.

143.5 CTF4. The strict commuting square for the whole comparison

Keep the exact test-space isomorphism and its induced quotient map:

$$\mathcal{T} : \mathcal{A} \rightarrow A, \quad \mathcal{T}k(u) = 2u^{-1/2}k(u), \quad \mathcal{T}^{-1}b(u) = \frac{1}{2}u^{1/2}b(u), \quad \overline{\mathcal{T}} : Q_{\text{cen}} \xrightarrow{\sim} Q_{\text{raw}}. \quad (\text{CTF4.1})$$

The map takes $2\mathcal{E}S$ onto $2\Sigma S = \Sigma S$; it has $\mathcal{M}_0(\mathcal{T}k) = 2F_k$, so its complete receiving factor two is retained. It intertwines the real actions and the two oriented degree-one mirrors:

$$\mathcal{T}W_a = T_a \mathcal{T}, \quad \mathcal{T}(-R_{\text{cen}}) = -R_{\text{raw}} \mathcal{T}. \quad (\text{CTF4.2})$$

The actual GAP5/FDB7 map is

$$F_{\text{sym}}^1 = \mathcal{T}, \quad F_{\text{sym}}^0(c) = ((\Phi c, 0, 0), (-\widehat{\Phi c}, 0, 0)), \quad F_{\text{sym}}^j = 0 \quad (j < 0). \quad (\text{CTF4.3})$$

Its chain equation retains both summation factors:

$$d_D F_{\text{sym}}^0 c = \Sigma \Phi c + R_{\text{raw}} \Sigma \widehat{\Phi c} = 2\Sigma \Phi c = \mathcal{T}(2\mathcal{E}\Phi c) = F_{\text{sym}}^1 Jc.$$

Also $\Phi d_C = 0$ on the negative differential. Equations (CTF3.2) and (CTF4.2), and the oriented mirror (CTF1.5), prove the equivariance recorded in FDB7.

Define the zero-differential comparison

$$\beta : H_C \rightarrow H_D, \quad \beta^1 = \overline{\mathcal{T}}, \quad \beta^{1-n}|_{B_n} = 0 \quad (n \geq 1). \quad (\text{CTF4.4})$$

It is equivariant by (CTF4.2), with zero maps in the remaining degrees. Direct substitution into (CTF2.3) gives

$$\Pi_D^0 F_{\text{sym}}^0(c) = 0$$

since $-\widehat{\Phi c} = -\Phi c$; all endpoint entries are zero. In degree one, $\pi_{\text{raw}} \mathcal{T} = \overline{\mathcal{T}} \pi_{\text{cen}}$. All lower components vanish. We have therefore proved the strict square

$$\boxed{\Pi_D F_{\text{sym}} = \beta \Gamma} \quad (\text{CTF4.5})$$

on all cochains, not merely on their cohomology.

143.6 CTF5. The entire cone and the exact comparison map

Retain precisely the original cochain cone convention

$$K^j = D^j \oplus (C^{\text{or}})^{j+1}, \quad d_K(v, c) = (d_D v + F_{\text{sym}} c, -d_C c). \quad (\text{CTF5.1})$$

The plus sign in the off-diagonal term and the minus sign in the shifted differential are both required. Write $K_H = \text{Cone}(\beta)$ with the same convention. The strict square (CTF4.5) gives

$$\Xi : K \rightarrow K_H, \quad \Xi^j(v, c) = (\Pi_D^j v, \Gamma^{j+1} c). \quad (\text{CTF5.2})$$

No additional sign is applied to the shifted component. Indeed

$$\Xi d_K(v, c) = (\Pi_D d_D v + \Pi_D F_{\text{sym}} c, -\Gamma d_C c) = (\beta \Gamma c, 0) = d_{K_H} \Xi(v, c),$$

since the target two complexes have zero differentials. All actions are componentwise on each cone, so equivariance follows from the proved equivariance of $\Gamma, \Pi_D, F_{\text{sym}}$.

For completeness, Ξ is a quasi-isomorphism without a bounded-below hypothesis. The cone short exact sequences

$$0 \rightarrow D \rightarrow K \rightarrow C^{\text{or}}[1] \rightarrow 0, \quad 0 \rightarrow H_D \rightarrow K_H \rightarrow H_C[1] \rightarrow 0$$

are degreewise split. Their connecting map sends a class $[c] \in H^{j+1}(C)$, represented in the shifted term, to $[F_{\text{sym}} c] \in H^{j+1}(D)$: the lift $(0, c)$ has differential $(F_{\text{sym}} c, 0)$, with positive sign. The strict square gives a map of these exact sequences. For each degree the long exact cohomology rows have isomorphisms on the source and target-complex terms because Γ, Π_D are quasi-isomorphisms. The exactness diagram chase makes the middle cone map an isomorphism. This argument uses only the five neighboring groups in any degree, so the infinitely many negative degrees introduce no limit operation.

The explicit terms of K_H are

$$\begin{aligned} K_H^{-n} &= B_n \quad (n \geq 1), & K_H^0 &= H_D^0 \oplus Q_{\text{cen}}, & K_H^1 &= Q_{\text{raw}}, \\ d_{K_H}^0(h, q) &= \bar{\mathcal{T}} q, & \text{all other differentials} & \text{zero.} \end{aligned} \quad (\text{CTF5.3})$$

Let

$$E = H_D^0[0] \oplus \bigoplus_{n \geq 1} B_n[n], \quad d_E = 0. \quad (\text{CTF5.4})$$

Its canonical projection $p : K_H \rightarrow E$ is identity on each B_n , sends $(h, q) \mapsto h$ in degree zero and sends degree one to zero. Its canonical inclusion $i : E \rightarrow K_H$ is identity on the negative terms and $h \mapsto (h, 0)$ in degree zero.

There is a complete explicit degree-minus-one homotopy

$$h^1 : Q_{\text{raw}} \rightarrow H_D^0 \oplus Q_{\text{cen}}, \quad h^1(q) = (0, \bar{\mathcal{T}}^{-1} q), \quad h^j = 0 \quad (j \neq 1). \quad (\text{CTF5.5})$$

In degree zero, $hd(h, q) = (0, q)$; in degree one, $dh(q) = q$; in every negative degree both are zero. Therefore

$$\boxed{dh + hd = \text{id}_{K_H} - ip, \quad pi = \text{id}_E.} \quad (\text{CTF5.6})$$

Also $h^2 = 0$, $hi = 0$, and $ph = 0$, directly from the component formulas. The inverse $\bar{\mathcal{T}}^{-1}$ is equivariant, including the negative reflection on both quotient terms; all these maps and the homotopy preserve the original actions.

The resulting direct cone projection is

$$\boxed{\Pi_K = p\Xi : K \longrightarrow H_D^0[0] \oplus \bigoplus_{n \geq 1} B_n[n]} \quad (\text{CTF5.7})$$

with the exact formulas

$$\Pi_K^0(v, k) = \Pi_D^0 v, \quad \Pi_K^{-n}(c) = \Gamma^{1-n} c \quad (n \geq 1), \quad \Pi_K^1 = 0. \quad (\text{CTF5.8})$$

At degree -1 , its chain equation is precisely $\Pi_D^0 F_{\text{sym}} c = 0$; at lower degrees it is $\Gamma d_C = 0$, with the retained cone minus sign. This is an equivariant quasi-isomorphism of the actual entire cone. It strengthens the prior all-degree cohomology calculation, rather than asserting that every original cochain is already a direct sum of cohomology representatives.

One can also check the degree-zero assertion directly on the original complex, with an explicit boundary. A cycle is (v, k) with $d_D v + \mathcal{T}k = 0$. Write $v = ((h, c_0, c_1), (g, d_0, d_1))$, and put $t = (h - \widehat{g})/2 \in S$. Then

$$v - \Pi_D^0 v = ((t, 0, 0), (-\widehat{t}, 0, 0)), \quad d_D v = 2\Sigma t, \quad k = -2\mathcal{E}t.$$

The element $c = t_1 \otimes t \in M_0 = C^0 = K^{-1}$ has $\Phi c = t$, so

$$d_K c = (F_{\text{sym}}^0 c, -Jc) = (v - \Pi_D^0 v, k).$$

Thus every degree-zero cone cycle differs from its projected H_D^0 representative by this actual boundary, including the negative shifted differential. Conversely $\Pi_D^0 F_{\text{sym}}^0 = 0$ shows that a nonzero H_D^0 representative is not a boundary. This use of the supplied arithmetic unit t_1 and the explicit $t \in S$ is not a section of a Mellin quotient.

143.7 CTF6. The whole triangle and its final connecting homotopy

The square in CTF4 and map Ξ give a strict map of the two cone triangles before contraction. After contraction, the derived triangle is represented by

$$H_C \xrightarrow{\beta} H_D \xrightarrow{\alpha} E \xrightarrow{\delta} H_C[1], \quad (\text{CTF6.1})$$

where α is projection to H_D^0 in degree zero and zero on Q_{raw} , while δ is identity from each B_n in degree $-n$ to the same $B_n \subset H_C^{1-n}$ and zero on H_D^0 . The connecting sign is positive: the cone projection is $(v, c) \mapsto c$, and the differential on $C[1]$ is $-d_C$, already accounted for in (CTF5.1).

It is essential to distinguish strict equality from homotopy in the final square. On K_H , let $q_H(v, c) = c$ be the cone projection. Then $q_H - \delta p$ is the projection from $Q_{\text{cen}} \subset K_H^0$ to $H_C[1]^0 = Q_{\text{cen}}$. Define a degree-minus-one map to $H_C[1]$ by

$$\eta_H^1(q_{\text{raw}}) = \overline{\mathcal{T}}^{-1} q_{\text{raw}} \in H_C[1]^0, \quad \eta_H^j = 0 \ (j \neq 1).$$

Since $d_{H_C[1]} = 0$, (CTF5.3) gives

$$q_H - \delta p = d_{H_C[1]} \eta_H + \eta_H d_{K_H}. \quad (\text{CTF6.2})$$

On the original cone this reads

$$\Gamma[1] \text{pr}_C - \delta \Pi_K = d_{H_C[1]} \eta + \eta d_K, \quad \eta^1(b) = \overline{\mathcal{T}}^{-1} \pi_{\text{raw}}(b) \quad (b \in K^1 = A), \quad (\text{CTF6.3})$$

and every other η^j is zero. In degree zero,

$$\eta^1(d_D v + \mathcal{T}k) = \overline{\mathcal{T}}^{-1} \pi_{\text{raw}}(d_D v + \mathcal{T}k) = \pi_{\text{cen}} k,$$

which is exactly the difference in that degree. In every negative degree the two connecting maps coincide. The second square commutes strictly: $\Pi_K(v, 0) = \alpha \Pi_D v$. Thus the full triangle comparison is canonical in the homotopy/derived category, with a specified homotopy where strict commutation fails.

Nothing in this proof gives an equivariant section $Q_{\text{cen}} \rightarrow \mathcal{A}$ or $Q_{\text{raw}} \rightarrow A$. The inverse in (CTF5.5) has codomain the other quotient, not a test space. The contraction is of the explicit intermediate cone of an isomorphism of quotients. A strict equivariant inverse to Γ , Π_D , or Ξ on the original complexes is not used.

143.8 CTF7. Faithful source extension and its retained identity cone

DCP8 supplies the actual extra closed copies

$$V = V_+ \oplus V_-,$$

as degree-zero terms, where each $V_{\pm} = S \oplus \mathbb{C}^2$ retains its complete Schwartz component and its two labelled endpoint coordinates. Define exactly

$$C_{\text{full}} = C^{\text{or}} \oplus V[0], \quad D_{\text{full}} = D \oplus V[0], \quad F_{\text{full}} = F_{\text{sym}} \oplus \text{id}_{V[0]}. \quad (\text{CTF7.1})$$

The target is precisely the global Cech complex of DCP8–DCP9 after ordering the summands by their receiving idempotents. The source is the displayed faithful extension of the original C^{or} ; no identification of a different geometric derived source with this extension is presumed.

The global receiving ring is $\mathbb{Z}^3$. A triple (r, b_+, b_-) acts by scalar r on the original C, D summands and by b_+, b_- on the two respective copies in V . Thus

$$[\tau] = (1, 1, 1) \text{ acts as identity,} \quad [n] = (n, 0, 0) \text{ acts as scalar } n \text{ on } C, D \text{ and as zero on } V. \quad (\text{CTF7.2})$$

These actions preserve every differential and the comparison. They are faithful as actions of the supplied source monoid: distinct integers act differently on the original nonzero complex coefficient spaces; τ acts differently from each integer on the nonzero extra closed copies. No source addition on τ is defined.

The real action on V_+ is $\rho_+(a)$ and on V_- is $\rho_-(a)$, exactly as in (CTF3.1). The mirror on V is

$$w_V(v_+, v_-) = (v_-, v_+), \quad (\text{CTF7.3})$$

using the original identification of the two typed copies. It is semilinear for the receiving ring swap $(r, b_+, b_-) \mapsto (r, b_-, b_+)$, and fixes every supplied source element (CTF7.2). It retains $\rho_V(a)w_V = a w_V \rho_V(a^{-1})$, checked on the Schwartz terms and on each endpoint coordinate using the original actions. There is no extra cone-orientation sign on this degree-zero identity comparison. The oriented action on C^{or} remains the one in CTF1. The diagonal rational action on these already periodized V coefficients is trivial, as it is on the original D coefficients; it is not the independent real spectral dilation.

Set

$$\begin{aligned} H_{C,\text{full}} &= H_C \oplus V[0], & H_{D,\text{full}} &= H_D \oplus V[0], \\ \Gamma_{\text{full}} &= \Gamma \oplus \text{id}_V, & \Pi_{D,\text{full}} &= \Pi_D \oplus \text{id}_V, & \beta_{\text{full}} &= \beta \oplus \text{id}_V. \end{aligned} \quad (\text{CTF7.4})$$

All maps are receiving-ring linear, equivariant for the source action, real action, full diagonal rational action and the mirror, and

$$\Pi_{D,\text{full}} F_{\text{full}} = \beta_{\text{full}} \Gamma_{\text{full}}$$

strictly. Both vertical maps are quasi-isomorphisms.

The componentwise regrouping gives the exact cone isomorphism

$$\text{Cone}(F_{\text{full}}) \cong K \oplus \text{Cone}(\text{id}_{V[0]}). \quad (\text{CTF7.5})$$

The added cone has V_{source} in degree -1 , V_{target} in degree zero, and differential $+\text{id}_V$. This follows directly from the retained formula $d(v, c) = (d_D v + Fc, -d_C c)$: $d_V = 0$, so the off-diagonal identity has positive sign. Its contracting homotopy is

$$h_V^0 : V_{\text{target}} \rightarrow V_{\text{source}}, \quad h_V^0(v) = v, \quad h_V^j = 0 \ (j \neq 0),$$

and satisfies $dh_V + h_V d = \text{id}$ in both degrees. Because both copies carry exactly the same actions, this contraction is receiving-ring linear and equivariant, including the mirror swap of V_+ and V_- .

In the intermediate full cone, take $h_{\text{full}} = h \oplus h_V$, $p_{\text{full}} = p \oplus 0$, and $i_{\text{full}} = i \oplus 0$. Equation (CTF5.6) holds with these full maps and the same final target E . The resulting exact original-cone projection is

$$\begin{aligned}\Pi_{\text{full}}^0(v, w_{\text{target}}, k) &= \Pi_D^0 v, \\ \Pi_{\text{full}}^{-1}(c, w_{\text{source}}) &= \Gamma^0 c, \\ \Pi_{\text{full}}^{-n}(c) &= \Gamma^{1-n} c \quad (n \geq 2), \\ \Pi_{\text{full}}^1 &= 0.\end{aligned}\tag{CTF7.6}$$

The receiving-ring action on the resulting E is through its arithmetic coordinate r ; this describes the cone comparison, not the action on either full source object. In particular the extra V terms are still present in both full objects and in both cohomology complexes in (CTF7.4). They disappear only from their identity cone by the displayed contraction. Thus the full objects retain the faithful distinction $\tau \neq 1$.

The full triangle's last square has the same explicit homotopy as CTF6, with the additional component

$$\eta_{\text{full}}^0(v, w_{\text{target}}, k) = (0, w_{\text{target}}) \in H_{C,\text{full}}[1]^{-1} = B_1 \oplus V.\tag{CTF7.7}$$

Its degree-one component is still $\eta_{\text{full}}^1(b) = \overline{\mathcal{T}}^{-1} \pi_{\text{raw}}(b)$. On a degree- -1 cone cochain, the differential sends the extra source copy w_{source} to $+w_{\text{source}}$ in the target copy, so (CTF7.7) returns it with positive sign. This is exactly the difference between the full cone projection to $H_{C,\text{full}}[1]$ and the map through E . In degree zero the difference is still $\pi_{\text{cen}} k$. Every other degree agrees strictly. This verifies all cone and connecting signs in the faithful triangle as well.

143.9 CTF8. Exact resulting statement

The original oriented comparison admits the explicit equivariant quasi-isomorphism

$$\boxed{\text{Cone}(F_{\text{sym}}) \xrightarrow{\Pi_K} H^0(D)[0] \oplus \bigoplus_{n \geq 1} (\Lambda^n W_{\text{pr}} \otimes \mathbb{C}^2)[n]}.\tag{CTF8.1}$$

Its degree-zero Fourier graph, all four original endpoint lines, every original prime wedge, and both moment coordinates survive with their specified real actions and mirror signs. Its full faithful extension has the same cone model by (CTF7.5)–(CTF7.6), while retaining the extra complete closed copies in both compared objects.

This proves formality of the specified whole comparison triangle in the derived representation category, with a strict comparison square, explicit cone map, and the displayed final-arrow homotopy. It strengthens the earlier statement that no derived splitting of this cone had yet been proved. It does not assert a strict equivariant homotopy inverse on the original test complexes, a lift of Q into a strong test space, or a new weight/purity theorem. The surviving original-zeta quotient is identified by its full factor-preserving source map; no completed replacement, missing multiplicity, or extra source operation on τ enters the argument.

143.10 Complete supported duality and original-zeta trace - 24 September 2026

The full continuation is ACTUAL_SUPPORTED_DUALITY_INDEPENDENT.md ASD0–ASD14, GLOBAL_ORIGINAL_ZETA_RESIDUE_DUALITY_INDEPENDENT.md GZR0–GZR9, and ORIGINAL_ZETA_JACOBIAN_TRACE_BRIDGE.md JTB0–JTB8. The original Fréchet coefficient complex now has its explicit continuous-dual localization row, with all faithful extra closed copies and every orientation sign. Its residue pairing is a pair of absolutely convergent integrals with the original denominator zeta(s); it descends to the entire actual quotient and detects every multiplicity jet. The exact derivative operator maps this pairing to the existing Weil form, with the original pole correction and full Gamma factors retained.

ASD14 constructs the resulting actual chain map into the character-a-twisted shifted dual and computes its complete cone. Its spectral cokernel is $Q/D_L(Q)$; weak-* density makes its Hausdorff quotient zero but is not treated as algebraic surjectivity. The original H_0 and its continuous dual remain in the cone. This strengthens the explicit duality comparison without declaring a Verdier isomorphism, numerical purity or an RH proof. Current source Z_0 , Z_1/τ and integer Z_2 data are unchanged; no τ addition is introduced. The complete relevant user corpus and later corrections were checked before interpreting these maps.

143.11 Explicit global continuation: finite lifting and a nonzero remainder

The earlier uncertainty about algebraic surjectivity is now resolved by GLOBAL_POLYNOMIAL_DUAL_DEFECT.md PGD1–PGD6, independently checked in PGC0–PGC7: the actual algebraic cokernel is nonzero. Its explicit class c_zeta is the original-zeta contour functional with the constant arithmetic function 1 in the first slot. A lift would require a rapidly decreasing entire representative to equal 1 at all actual zeros, which the proved unbounded zero heights exclude. PGD constructs an injected rational-function line, entire representatives retaining all Hermite pole corrections, and the whole real/prime orbit. This is a receiving cokernel class, not primitive τ or a claimed off-critical zero.

SCL0–SCL9 and DPL0–DPL9 prove every nonzero polynomial in the actual cokernel generator invertible. Thus all finite-dimensional invariant dual spaces lift uniquely, continuously and with the original real/prime action. DPL gives explicit extension splittings and computes every degree of the cyclic derived Hom into the actual cone. These results retain its H and $H/$ terms, all faithful closed copies and support labels. The weak-* Hausdorff quotient remains zero alongside the now proved nonzero algebraic quotient. Current source Z_0 , Z_1/τ and Z_2 are unchanged. Complete proofs and exact human citations are in the linked continuation files; the full geometric weight-separation goal remains active.

144 The complete continuous dual of the actual supported coefficient complex

24 September 2026. Independent calculation ASD0–ASD14. All duals below are continuous duals of the original coefficient topologies. The proved comparison retains the original supported quotient, its full multiplicity jets, the four original endpoint lines, and both complete extra closed coefficient copies.

144.1 ASD0. Controlling definitions, mathematical source, and scope

The source definitions are B1–B5 and their proved consequences P1–P5 in `SOURCE_OPERATIONS_AND_PROOFS.md`, with the user’s verbatim U01–U18 in `USER_DEFINITIONS_VERBATIM.md`. The current `CORPUS_AND_OPERATION_RULES.md` (private source record) was read. Before interpreting the pairing calculation below, the complete corpus passages USR-9ad1c0a2d09dba92, USR-f55d16feb948d8c2, and USR-6152e3bc6302258c were reread in its retained private record. They supply the user’s full-system argument and the requested global quotient and Deligne comparison; they are not replaced by a truncated prime system.

The source remains Z_0 absence, primitive Z_1/τ without parity or source addition, and the original integer layer with the user’s specified Z_2 data. Source zero has no source parity. Every sum, integral, scalar, continuous functional, or complex below is an operation in an already constructed receiving space after the retained global arithmetic reconstruction. No dualization creates source addition on τ .

The human geometric source is Alain Connes and Caterina Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, arXiv:0903.2024v3, §5. Its original author TeX, `announc3.tex` lines1376–1667, was read in the earlier DCP calculation; it supplies the two charts, restriction, Fourier identity, and mirror. The exact source-space entry and faithful coefficient extension were proved in `SOURCE_CC_DOUBLE_PULLBACK.md`, DCP0–DCP12.

The analytic inputs are the original-space closed-image and inverse theorems in `ORIGINAL_MELLIN_SPECTRAL_SYNTHESIS.md`, OMS1–OMS5, and SSI0–SSI10 cited there. Their human analytic source is Ralf Meyer, *A spectral interpretation for the zeros of the Riemann zeta function*, arXiv:math/0412277v3, with his exact source and reading coverage recorded in OMS0. The present note uses the actual proved image equality, not an assumed spectral synthesis statement.

`COMPLETE_COMPARISON_TRIANGLE_FORMALITY.md`, CTF0–CTF8, was read, including its distinction between the algebraic full adelic complex and the Fréchet CC complex. The latter, not an unstated topology on the former, is continuously dualized here. The existing reflected Weil form in the retained complete `ORIGINAL_ZETA_RESOLVENT_AND_PROJECTORS.md`, RZ10–RZ11, was also read in full and is retained in ASD11. The external programme file actually read and this retained copy were verified byte-identical: SHA256 `f3553e1362b4c2c1c1556098b55ee153be9fc716a9d4c35baedcf39bd28ad755`.

Pierre Deligne’s *La conjecture de Weil. II*, §3.6, printed pp.212–214, is the human source of the target supported-cohomology argument. The local reading for this calculation was the explicitly identified transcription sources/S20_FR_record_export.tex, lines2466–2559, not an author TeX file. Section3.6.1(6)–(8) uses a genuine perfect duality with a Tate twist to identify its supported group; Lemmas3.6.2 and3.6.3 then give separate weight bounds. ASD13 compares the actual morphisms below to that mechanism without declaring an unproved Verdier duality.

144.2 ASD1. Original spaces and all original factors

Set

$$S = \{h \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : h(-v) = h(v), h(0) = 0, \int_{\mathbb{R}} h(v) dv = 0\},$$

$$A = \{b \in C^\infty(\mathbb{R}_{>0}; \mathbb{C}) : p_{N,j}(b) = \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j b(u)| < \infty \quad (N, j \geq 0)\}. \quad (\text{ASD1.1})$$

These are the original Fréchet spaces; S has its closed-subspace Schwartz topology and A its displayed seminorms. The two labelled chart spaces are

$$V_+ = S \oplus \mathbb{C}_{c_0, c_1}^2, \quad V_- = S \oplus \mathbb{C}_{d_0, d_1}^2.$$

Keep

$$\Sigma h(u) = 2 \sum_{n \geq 1} h(nu), \quad Rb(u) = u^{-1}b(u^{-1}), \quad \widehat{h}(t) = \int_{\mathbb{R}} h(v) e^{-2\pi i v t} dv, \quad (\text{ASD1.2})$$

$$r_+(h, c_0, c_1) = \Sigma h, \quad r_-(h, d_0, d_1) = R\Sigma h = \widehat{\Sigma h}. \quad (\text{ASD1.3})$$

The last equality follows from the complete Poisson identity

$$\widehat{\Sigma h}(u) = u^{-1}\Sigma h(u^{-1}) + u^{-1}h(0) - \int_{\mathbb{R}} h(v) dv. \quad (\text{ASD1.4})$$

Its two last terms vanish on this explicitly specified S ; the separate endpoint lines are still retained.

Write

$$J = \Sigma S, \quad \pi : A \longrightarrow Q = A/J. \quad (\text{ASD1.5})$$

The cited inverse-source theorem proves that J is closed and $\Sigma : S \rightarrow J$ is a topological isomorphism. In particular Q is Hausdorff Fréchet, and π is an open quotient map: for open $O \subset A$, $\pi^{-1}\pi(O) = O + J$ is open.

For the entire Mellin space $\mathcal{B}$ in OMS1.2, retain

$$\mathcal{M}_0 b(s) = \int_0^\infty b(u) u^s \frac{du}{u}, \quad \mathcal{M}_0 \Sigma h(s) = 2\zeta(s) \int_0^\infty h(v) v^s \frac{dv}{v}. \quad (\text{ASD1.6})$$

The latter identity is initially absolute on $\Re s > 1$; its continuation, endpoint terms, and original trivial-zero residues are those proved in SSI/OMS. The exact topological identification is

$$Q \simeq \mathcal{B}/I_\zeta, \quad I_\zeta = \{F : F^{(r)}(\rho) = 0 \text{ } (\rho \text{ an actual nontrivial zero of } \zeta, \text{ } 0 \leq r < m_\rho)\}. \quad (\text{ASD1.7})$$

This uses the original ζ and actual multiplicities.

The previous centered receiver has not been silently substituted:

$$\mathcal{T}k(u) = 2u^{-1/2}k(u), \quad \mathcal{T}^{-1}b(u) = \frac{1}{2}u^{1/2}b(u), \quad \mathcal{M}_0 \mathcal{T}k = 2F_k, \quad \Sigma = \mathcal{T}\mathcal{E}. \quad (\text{ASD1.8})$$

Every dual comparison uses the transpose of this same map, including its factor2 and half-power.

144.3 ASD2. Continuous duals, quotient exactness, and the precise topology

For a Hausdorff locally convex complex space E , let E' be its continuous complex-linear dual. Unless a stronger topology is explicitly named, give E' the weak-* topology $\sigma(E', E)$. This is a topology on the dual of the original space, not a change to the original topology on E . For continuous $T : E \rightarrow F$, define

$$T' : F' \rightarrow E', \quad T'\lambda = \lambda \circ T. \quad (\text{ASD2.1})$$

It is weak-* continuous because evaluation at e becomes evaluation at Te . It is also continuous for the strong dual topologies: for a bounded set $B \subset E$,

$$\sup_{e \in B} |T'\lambda(e)| = \sup_{f \in T(B)} |\lambda(f)|, \quad (\text{ASD2.2})$$

and $T(B)$ is bounded. Thus every transpose constructed below is strong-dual continuous as well.

Here is the exact quotient fact used in all subsequent rows. For a closed linear subspace $J \subset E$ with its subspace topology and quotient $q : E \rightarrow E/J$,

$$0 \longrightarrow (E/J)' \xrightarrow{q'} E' \xrightarrow{\text{res}} J' \longrightarrow 0 \quad (\text{ASD2.3})$$

is algebraically exact and strictly exact for weak-* topologies.

Indeed q' is injective. Its image is precisely $J^\perp = \{\lambda \in E' : \lambda(J) = 0\}$: an annihilating functional factors uniquely through E/J , and the factor is continuous by the quotient topology. The weak-* topology on this image agrees with $\sigma((E/J)', E/J)$, since each quotient vector has a representative in E . The annihilator is weak-* closed. Every continuous functional on J extends continuously to E by the complex Hahn–Banach theorem, proving surjectivity of restriction.

For completeness, restriction is an open map in these weak-* topologies. Let O be a basic zero-neighbourhood in E' , specified by inequalities on a finite-dimensional subspace $F \subset E$. The image of O under restriction is exactly the set of $\psi \in J'$ whose restriction to $F \cap J$ extends to a functional $\ell \in F^*$ satisfying those inequalities. The restriction map $F^* \rightarrow (F \cap J)^*$ is a surjective finite-dimensional linear map, hence open; this makes that set weak-* open in J' . To justify the asserted equality, start with a continuous Hahn–Banach extension λ_0 of ψ . The difference $\ell - \lambda_0|_F$ vanishes on $F \cap J$, so it is a functional on the finite-dimensional image of F in the Hausdorff locally convex quotient E/J . It is continuous there and has a continuous Hahn–Banach extension to E/J . Add its pullback to λ_0 . The resulting extension has restriction ψ and the prescribed F -values. This proves the equality and openness.

No strong-dual quotient or completeness conclusion is silently substituted for ASD2.3. The statements proved here are strict weak-* exactness, algebraic exactness of continuous duals, and strong continuity of each displayed transpose. Where a row has an explicit continuous splitting, its dual splitting is also strong continuous, directly by ASD2.2.

Applied to ASD1.5, this gives the full spectral dual

$$\boxed{Q' \xrightarrow[\pi']{\sim} J^\perp \subset A'.} \quad (\text{ASD2.4})$$

It contains every continuous distribution on the original strong test space that annihilates the exact summation image. It is not restricted to finite sums of zero evaluations.

144.4 ASD3. The entire faithful CC complex and its dual differential

Use the order $+$, then $-$. The original cochain complex is

$$D^0 = V_+ \oplus V_-, \quad D^1 = A, \quad d_D(v_+, v_-) = r_+ v_+ - r_- v_-. \quad (\text{ASD3.1})$$

Let

$$V_{\text{extra}} = V_+^{\text{extra}} \oplus V_-^{\text{extra}}.$$

DCP8 proves the source-faithful coefficient sheaf has the global complex

$$D_{\text{full}} = D \oplus V_{\text{extra}}[0]. \quad (\text{ASD3.2})$$

Every extra term is the entire space $S \oplus \mathbb{C}^2$, not its endpoint quotient.

The dual cochain convention is

$$(C^\vee)^n = (C^{-n})', \quad d_{C^\vee}^n = (-1)^{n+1} (d_C^{-n-1})'. \quad (\text{ASD3.3})$$

It comes from the Hom differential into $\mathbb{C}[0]$, and its square is zero because consecutive original differentials compose to zero. Thus the full dual has exactly degrees $-1, 0$:

$$(D_{\text{full}}^\vee)^{-1} = A', \quad (D_{\text{full}}^\vee)^0 = V_+' \oplus V_-' \oplus V_{\text{extra}}',$$

$$d^\vee \lambda = (r'_+ \lambda, -r'_- \lambda, 0). \quad (\text{ASD3.4})$$

There is no further differential sign: at degree -1 the sign in ASD3.3 is $+1$. On individual coordinates,

$$r'_+ \lambda = (\Sigma' \lambda, 0, 0), \quad r'_- \lambda = (\Sigma' R' \lambda, 0, 0). \quad (\text{ASD3.5})$$

All dual endpoint coordinates therefore remain independent in degree 0.

Parametrize the original degree-zero cohomology using the first Schwartz coordinate:

$$H_0 = \ker d_D = \{((h, c_0, c_1), (\widehat{h}, d_0, d_1)) : h \in S\} \simeq S \oplus \mathbb{C}^4, \quad H = H_0 \oplus V_{\text{extra}}. \quad (\text{ASD3.6})$$

The DCP parametrization by the second coordinate gives the same subspace by Fourier involution; no chart has been discarded. The dual complex satisfies

$$\boxed{H^{-1}(D_{\text{full}}^\vee) = Q', \quad H^0(D_{\text{full}}^\vee) = H'}, \quad (\text{ASD3.7})$$

and has no other cohomology.

To prove the first equality, ASD3.4 vanishes exactly when λ annihilates J , giving ASD2.4. For the second, restrict a degree-zero functional to H . Restriction is onto by Hahn–Banach; more explicitly the continuous projection in ASD4 gives a continuous section of this restriction. Its kernel is exactly the image of ASD3.4. If ϕ vanishes on H_0 , it factors through $d_D : D^0 \rightarrow J$. The factor is continuous: a continuous section of this surjection is $j \mapsto ((\Sigma^{-1}j, 0, 0), 0)$. Extend that functional on J continuously to A by Hahn–Banach. Its transpose image is ϕ . A functional annihilating H also vanishes on all extra copies, so the same argument proves the complete assertion.

This identifies cohomology with its weak-* quotient topology. The restriction to H has an explicit continuous dual section, so its final quotient identification is strong-dual continuous in both directions too; no statement about the strong topology on A'/Q' is needed.

144.5 ASD4. Exact dual of the already constructed global comparison

CTF2 constructs the continuous cochain map

$$\Pi_D^0((h, c), (g, d)) = \left(\left(\frac{h+\widehat{g}}{2}, c \right), \left(\frac{\widehat{h}+g}{2}, d \right) \right), \quad \Pi_D^1 = \pi. \quad (\text{ASD4.1})$$

It takes values in $H_0[0] \oplus Q[-1]$ and is the identity on cohomology. Its kernel is the exact complex

$$\{((h, 0), (-\widehat{h}, 0)) : h \in S\} \xrightarrow{d_D} J, \quad h \mapsto 2\Sigma h, \quad (\text{ASD4.2})$$

whose inverse is

$$j \mapsto \left(\left(\frac{1}{2}\Sigma^{-1}j, 0 \right), \left(-\frac{1}{2}\widehat{\Sigma^{-1}j}, 0 \right) \right). \quad (\text{ASD4.3})$$

Both factors 2 and $1/2$ remain.

Its exact continuous transpose is a quasi-isomorphism

$$\Pi_{D, \text{full}}^\vee : Q'[1] \oplus H'_0[0] \oplus V'_{\text{extra}}[0] \longrightarrow D_{\text{full}}^\vee. \quad (\text{ASD4.4})$$

In degree -1 it is π' ; in degree 0 it is $(\Pi_D^0)'$ together with identity on V'_{extra} . A functional $\alpha = (u, \alpha_0, \alpha_1, \beta_0, \beta_1)$ on the $S \oplus \mathbb{C}^4$ coordinates of H_0 is sent to

$$(\Pi_D^0)' \alpha = \left(\left(\frac{1}{2}u, \alpha_0, \alpha_1 \right), \left(\frac{1}{2}\widehat{u}, \beta_0, \beta_1 \right) \right), \quad (\widehat{u})(g) = u(\widehat{g}). \quad (\text{ASD4.5})$$

Substitution in ASD4.1 proves this formula. The degree-zero restriction back to H'_0 is identity. The degree -1 image is exactly ASD2.4. These facts prove the quasi-isomorphism directly.

Equivalently, dualize the degreewise exact sequence with kernel ASD4.2. Hahn–Banach makes each dual restriction onto. Its last complex is the transpose of the topological isomorphism 2Σ , and is exact. This gives another complete proof of ASD4.4 without an asserted section $Q \rightarrow A$.

The extra identity cone in CTF7 has V_{extra} in degrees $-1, 0$, with differential $+\text{id}$. Its dual therefore has V'_{extra} in degrees $0, 1$, with differential $-\text{id}$. Its contracting map from degree1 to degree0 is $-\text{id}$, since both dh and hd are identity on their respective degrees. This contracts only the dual of the extra identity comparison cone. It does not remove $V'_{\text{extra}}[0]$ from ASD3.2–ASD3.7.

The full adelic exterior complex in CTF is explicitly algebraic. No continuous dual of its infinite direct sums is asserted without a chosen topology. ASD4.4 is the actual continuous CC-side comparison, and the transpose of ASD1.8 is

$$\mathcal{T}'\lambda(k) = \lambda(2u^{-1/2}k(u)), \quad (\mathcal{T}^{-1})'\mu(b) = \mu(\tfrac{1}{2}u^{1/2}b(u)). \quad (\text{ASD4.6})$$

They are continuous inverse maps and restrict to the corresponding annihilators by $\mathcal{T}(\mathcal{E}S) = J$. Thus the same original spectral quotient is retained in the continuous comparison.

144.6 ASD5. Single-support and both-support duals, with all signs

For either closed source point, the faithful single-support complex is

$$K_{\pm, \text{full}} = [V_{\pm} \oplus V_{\pm}^{\text{extra}} \xrightarrow{(v,w) \mapsto r_{\pm}v} A] \quad \text{in degrees } 0, 1. \quad (\text{ASD5.1})$$

Its dual is

$$[A' \xrightarrow{\lambda \mapsto (r'_{\pm}\lambda, 0)} V'_{\pm} \oplus (V_{\pm}^{\text{extra}})'] \quad \text{in degrees } -1, 0. \quad (\text{ASD5.2})$$

The same annihilator and Hahn–Banach argument gives

$$H^{-1}(K_{\pm, \text{full}}^{\vee}) = Q', \quad H^0(K_{\pm, \text{full}}^{\vee}) = (\mathbb{C}^2)' \oplus (V_{\pm}^{\text{extra}})'. \quad (\text{ASD5.3})$$

Here the first summand consists of the corresponding labelled endpoint duals.

The primal plus-support map to D_{full} is inclusion on its two degree-zero copies and identity on degree1. The minus-support map is the corresponding inclusion on degree0 and minus identity on degree1. Therefore their duals, directed from $D_{\text{full}}^{\vee}$, are the corresponding projections in degree0, and respectively $+\text{id}$ and $-\text{id}$ in degree -1 . These signs make the differential squares commute by direct substitution in ASD3.4 and ASD5.2.

For both closed points the faithful supported complex is

$$K_{\text{full}}^0 = V_+ \oplus V_- \oplus V_{\text{extra}}, \quad K_{\text{full}}^1 = A \oplus A, \\ d_K(v_+, v_-, w) = (r_+v_+, r_-v_-). \quad (\text{ASD5.4})$$

Its dual is

$$(K_{\text{full}}^{\vee})^{-1} = A' \oplus A', \quad (K_{\text{full}}^{\vee})^0 = V'_+ \oplus V'_- \oplus V'_{\text{extra}}, \\ d_K^{\vee}(\lambda_+, \lambda_-) = (r'_+\lambda_+, r'_-\lambda_-, 0). \quad (\text{ASD5.5})$$

Consequently

$$H^{-1}(K_{\text{full}}^{\vee}) = Q' \oplus Q', \quad H^0(K_{\text{full}}^{\vee}) = (\mathbb{C}^4 \oplus V_{\text{extra}})'. \quad (\text{ASD5.6})$$

The supported-to-global map $K_{\text{full}} \rightarrow D_{\text{full}}$ is identity in degree0 and $(b_+, b_-) \mapsto b_+ - b_-$ in degree1. Its dual is identity in degree0 and

$$\lambda \mapsto (\lambda, -\lambda) \quad (\text{ASD5.7})$$

in degree -1 . In particular the dual signs are determined by the original Čech orientation rather than chosen to improve a weight statement.

144.7 ASD6. The complete continuous-dual localization row

Put $E = \mathbb{C}^4 \oplus V_{\text{extra}}$. The actual primal localization row from DCP7 and DCP9 is

$$0 \rightarrow E \xrightarrow{\iota} H \xrightarrow{\text{res}} A \xrightarrow{\partial} Q \oplus Q \xrightarrow{\delta} Q \rightarrow 0, \quad (\text{ASD6.1})$$

where, using ASD3.6,

$$\text{res}(h, c_0, c_1, d_0, d_1, w) = \Sigma h, \quad \partial b = (\pi b, \pi b), \quad \delta(q_+, q_-) = q_+ - q_-. \quad (\text{ASD6.2})$$

Its dual is the exact reversed row

$$\boxed{0 \rightarrow Q' \xrightarrow{\delta'} Q' \oplus Q' \xrightarrow{\partial'} A' \xrightarrow{\text{res}'} H' \xrightarrow{\iota'} E' \rightarrow 0,} \quad (\text{ASD6.3})$$

with every map explicit:

$$\begin{aligned} \delta' \lambda &= (\lambda, -\lambda), & \partial'(\lambda_+, \lambda_-) &= \pi'(\lambda_+ + \lambda_-), \\ \text{res}' \lambda &= (\Sigma' \lambda, 0, 0, 0, 0, 0), & \iota'(u, c'_0, c'_1, d'_0, d'_1, w') &= (c'_0, c'_1, d'_0, d'_1, w'). \end{aligned} \quad (\text{ASD6.4})$$

For example, the kernel of ∂' is the anti-diagonal because π' is injective. The image of ∂' is $J^\perp$, which is the kernel of Σ' . Hahn–Banach and the topological isomorphism $\Sigma : S \rightarrow J$ prove that $\Sigma' : A' \rightarrow S'$ is onto. This identifies the image of res' with the kernel of ι' . Finally ι' is onto and split by appending zero in its S' coordinate.

All kernels are weak-* closed and all maps onto their images have the quotient topology, by ASD2 and the explicitly split finite-coordinate maps. Thus ASD6.3 is a strictly exact row in the declared weak-* category, not only a sequence of abstract dual vector spaces. Each map is also strong-dual continuous.

The condensed spectral part is the completely split row

$$0 \rightarrow Q' \xrightarrow{\lambda \mapsto (\lambda, -\lambda)} Q' \oplus Q' \xrightarrow{(\lambda_+, \lambda_-) \mapsto \lambda_+ + \lambda_-} Q' \rightarrow 0. \quad (\text{ASD6.5})$$

A section of its last arrow is $\lambda \mapsto (\lambda/2, \lambda/2)$. A retraction of its first arrow is $(\lambda_+, \lambda_-) \mapsto (\lambda_+ - \lambda_-)/2$. All four constants and signs are retained. These are precisely the transposes of the primal half-sum retraction and half-difference section.

A functional therefore lifts from either spectral summand through the dual row exactly as these maps prescribe. This row does not annihilate Q' : its first arrow is injective and the displayed quotient is again the same Q' .

144.8 ASD7. Source actions, prime actions, and the complete mirror

The receiving ring is the actual global $\mathbb{Z}^3$ of DCP1:

$$\epsilon = (1, 0, 0), \quad e_+ = (0, 1, 0), \quad e_- = (0, 0, 1).$$

The source action is

$$[\tau] = (1, 1, 1), \quad [n] = (n, 0, 0) = n\epsilon. \quad (\text{ASD7.1})$$

On the original D and on Q , (r, b_+, b_-) acts by r ; on the two extra copies it acts respectively by b_+ and b_- . On continuous duals define the receiving-ring action by transpose:

$$(c\lambda)(v) = \lambda(cv). \quad (\text{ASD7.2})$$

The ring is commutative, so this is a left module action: $c_1(c_2\lambda)(v) = \lambda(c_2c_1v)$. It preserves all the supplied source multiplications and integer-input additions. In particular τ acts by identity everywhere, and integer n acts by n on the original dual component and by zero on each extra

dual component. Hahn–Banach separates a nonzero vector of either extra Fréchet space, so those dual copies are nonzero. Thus the full dual action still distinguishes τ from integer 1, and distinguishes any two integers on the original nonzero components.

This transpose action does not invert a source integer. In particular it does not attempt to invert the source zero.

The separate invertible real dilation action is

$$\begin{aligned} T_a b(u) &= b(u/a), & \rho_+(a)(h, c_0, c_1) &= (h(\cdot/a), c_0, ac_1), \\ \rho_-(a)(h, d_0, d_1) &= (ah(a\cdot), ad_0, d_1), & a &> 0. \end{aligned} \quad (\text{ASD7.3})$$

It commutes with the source action. For this group action the contragredient dual action is

$$T_a^\vee = (T_{a^{-1}})', \quad \rho_\pm^\vee(a) = (\rho_\pm(a^{-1}))'. \quad (\text{ASD7.4})$$

It makes evaluation invariant: $(T_a^\vee \lambda)(T_a v) = \lambda(v)$. At a rational prime the original operator is T_p , with no additional scale removed.

For chart functionals the complete formulas are

$$\begin{aligned} \rho_+^\vee(a)(u, \alpha_0, \alpha_1) &= (h \mapsto u(h(a\cdot)), \alpha_0, a^{-1}\alpha_1), \\ \rho_-^\vee(a)(v, \beta_0, \beta_1) &= (h \mapsto v(a^{-1}h(\cdot/a)), a^{-1}\beta_0, \beta_1). \end{aligned} \quad (\text{ASD7.5})$$

The original four endpoint characters $1, a, a, 1$ consequently become $1, a^{-1}, a^{-1}, 1$ on their dual lines. The extra dual copies carry the same full contragredient actions on both their Schwartz and endpoint coordinates. They have not been reduced to four characters.

The mirror on D_{full} swaps the chart and extra pairs in degree0 and is $-R$ in degree1. Its dual therefore swaps the same dual pairs in degree0 and is $-R'$ in degree -1 . On $K_{\text{full}}^\vee$ it swaps chart and extra pairs in degree0 and is

$$(\lambda_+, \lambda_-) \longmapsto (R'\lambda_-, R'\lambda_+) \quad (\text{ASD7.6})$$

in degree -1 . The ring involution exchanges e_+ and e_- , fixes ϵ , and hence fixes every source $[n]$ and $[\tau]$. Formula ASD7.2 shows that these dual mirror maps are semilinear for exactly that ring involution.

On ASD6.5 the first Q' has mirror $-R'$, the middle has the exchanged R' , and the final Q' has mirror R' . Thus

$$(\lambda, -\lambda) \longmapsto (-R'\lambda, R'\lambda) = \delta'(-R'\lambda),$$

and the sum is sent to $R'(\lambda_+ + \lambda_-)$. The section $(\lambda/2, \lambda/2)$ and retraction $(\lambda_+ - \lambda_-)/2$ are mirror-equivariant with these precise endpoint actions. Their dilation and source equivariance follows directly from the componentwise formulas.

144.9 ASD8. Exactly which character reflection supplies

Let χ denote the separately specified numerical character $\chi(a) = a$ of the receiving dilation group. The original reflection relation is

$$T_a R = a R T_{a^{-1}}, \quad R^2 = 1. \quad (\text{ASD8.1})$$

For $b \in A$, both sides of the first equation evaluate to $au^{-1}b(a/u)$. It descends to Q because $RJ = J$.

Let Q^{inv} denote the same topological vector space as Q , with group action $a \mapsto T_{a^{-1}}$. Then

$$R : Q \longrightarrow \chi \otimes Q^{\text{inv}} \quad \text{is a continuous equivariant isomorphism.} \quad (\text{ASD8.2})$$

Indeed $RT_a = aT_{a^{-1}}R$, obtained from ASD8.1 by replacing a by a^{-1} and rearranging. Its inverse is R between these specified actions. This is an exact relation to the inverted representation.

On the continuous dual with its contragredient action, transposition gives instead

$$T_a^\vee R' = a^{-1} R' T_{a^{-1}}^\vee. \quad (\text{ASD8.3})$$

To verify the order, $T_a^\vee R' = (RT_{a^{-1}})'$, and $RT_{a^{-1}} = a^{-1} T_a R$. Taking transpose yields ASD8.3. Hence

$$R' : Q' \longrightarrow \chi^{-1} \otimes (Q')^{\text{inv}} \quad (\text{ASD8.4})$$

is the corresponding equivariant isomorphism. The Cech orientation multiplies both instances of R' by -1 ; it does not change the factor a^{-1} .

ASD8.2 is not a map to Q' ; ASD8.4 is not a map from Q . The morphisms between a space and its continuous dual must actually be constructed, as in ASD10–ASD11 below. Merely reversing a prime operator is not taking the continuous dual.

There are three explicitly different operations:

operation	effect on a scalar eigencharacter a^ρ	(ASD8.5)
contragredient continuous dual	$a^{-\rho}$	
tensor with χ	$a^{\rho+1}$	
reflection of the original Mellin coordinate	$a^{1-\rho}$.	

The numerical convention $w = 2\Re\rho$ gives $-w$, $w + 2$, and $2 - w$, respectively, on such receiving characters. This assigns no weight or parity to primitive τ .

In geometric-Frobenius notation the scalar at a prime of an actual Tate twist (-1) is p , so its character agrees with $\chi(p)$. Here $\chi \otimes Q'$ is a fully defined representation operation. Its equality with a geometric Tate-twisted dualizing object has not been proved by ASD8.1. An even cochain shift, for example $C^\vee[-2]$, changes degrees but supplies no factor a : by definition $(C[-2])^n = C^{n-2}$ and its differential sign is unchanged. For the actual complex its groups are

$$H^1(D_{\text{full}}^\vee[-2]) = Q', \quad H^2(D_{\text{full}}^\vee[-2]) = H', \quad (\text{ASD8.6})$$

whereas D_{full} has H in degree 0 and Q in degree 1. This displays the degrees before any claimed dualizing identification.

144.10 ASD9. Full multiplicity jets and the entire spectral dual

For every actual nontrivial zero ρ and $0 \leq r < m_\rho$, define

$$\ell_{\rho,r}([b]) = \frac{(\mathcal{M}_0 b)^{(r)}(\rho)}{r!}. \quad (\text{ASD9.1})$$

This is a continuous element of Q' by ASD1.7 and the quotient topology. Let

$$j_\rho(q) = \sum_{r=0}^{m_\rho-1} \ell_{\rho,r}(q) t_\rho^r, \quad A_\rho = \mathbb{C}[t_\rho]/(t_\rho^{m_\rho}), \quad N_\rho v = t_\rho v.$$

The exact intertwining and its two distinct transpose formulas are

$$j_\rho T_a = a^\rho \sum_{k=0}^{m_\rho-1} \frac{(\log a)^k}{k!} N_\rho^k j_\rho,$$

$$T_a' \ell_{\rho,r} = a^\rho \sum_{k=0}^r \frac{(\log a)^{r-k}}{(r-k)!} \ell_{\rho,k}, \quad T_a^\vee \ell_{\rho,r} = a^{-\rho} \sum_{k=0}^r \frac{(-\log a)^{r-k}}{(r-k)!} \ell_{\rho,k}. \quad (\text{ASD9.2})$$

All logarithmic nilpotent terms remain. The first transpose is precomposition with T_a ; the second is the contragredient group action and precomposes with $T_{a^{-1}}$.

Likewise

$$R'\ell_{\rho,r} = (-1)^r \ell_{1-\rho,r}, \quad (-R)'\ell_{\rho,r} = (-1)^{r+1} \ell_{1-\rho,r}. \quad (\text{ASD9.3})$$

These follow by differentiating the original identity $\mathcal{M}_0 Rb(s) = \mathcal{M}_0 b(1-s)$. They preserve every actual multiplicity. The real conjugation symmetry is different and is not silently inserted into ASD9.3.

The finite jet functionals are independent, and each finite joint jet map is onto. One direct construction retains the complete original source transform

$$f_*(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2}, \quad F_*(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad G_*(s) = \mathcal{M}_0 \Sigma f_*(s) = 2F_*(s). \quad (\text{ASD9.4})$$

OMS proves its zero divisor is exactly the actual nontrivial-zero divisor, including multiplicities; its values at $0, 1, -2r$ retain the original multiplier and exceptional-point calculation. For a finite set P of these zeros, divide G_* by the finite polynomial $\prod_{\rho \in P} (s - \rho)^{m_\rho}$. The result is entire, belongs to $\mathcal{B}$, and is nonzero at each $\rho \in P$. Multiplying it by a Hermite interpolation polynomial prescribes arbitrary full jets at those points. To verify the interpolation, its Taylor series at each point is a unit in the finite Taylor algebra, so divide the prescribed jet by that unit, and solve the simultaneous polynomial congruences at the distinct points. The ideals of distinct linear factors are coprime by the Euclidean algorithm, which proves the finite polynomial interpolation. Multiplication and division by the displayed finite polynomials preserve rapid decrease on strips outside a fixed compact set; holomorphic divisibility controls that compact set. Thus the construction remains in the exact $\mathcal{B}$ domain and gives the asserted finite right inverse. All other-zero jets remain zero.

The full dual is characterized without declaring every functional a finite sum:

$$Q' = \overline{\text{span}\{\ell_{\rho,r}\}}^{\sigma(Q',Q)}. \quad (\text{ASD9.5})$$

For a direct proof, let L be the displayed span. Its common annihilator in Q is zero by ASD1.7. If L were not weak-* dense, there would be a weak-* neighbourhood separating one functional from its closure. A weak-* continuous linear functional is a finite linear combination of evaluations at vectors of Q : this follows because its continuity bounds it by finitely many such evaluations, so it vanishes on their common kernel and factors through their finite-dimensional image. Finite-dimensional separation would therefore produce a nonzero $q \in Q$ annihilated by all of L , a contradiction. This proves ASD9.5. It is a density statement for the full continuous dual, not an unproved unrestricted product decomposition of Q .

Combining ASD6 with ASD9 gives its exact effect on every zero and derivative:

$$\ell_{\rho,r} \xrightarrow{\delta'} (\ell_{\rho,r}, -\ell_{\rho,r}), \quad (\ell_{\rho,r}, 0) \xrightarrow{\partial'} \pi' \ell_{\rho,r}. \quad (\text{ASD9.6})$$

The continuous transpose maps and ASD9.5 extend these identities to the entire dual in precisely the weak-* sense already proved.

144.11 ASD10. The original test-space pairing and its exact receiving image

There is a concrete, continuous bilinear pairing on the original A :

$$B_A(b, c) = \int_0^\infty b(u) c(u) du. \quad (\text{ASD10.1})$$

It uses the displayed $du = u du/u$, not a dropped Haar factor. For example

$$|B_A(b, c)| \leq p_{2,0}(b) p_{2,0}(c) \int_0^\infty (u^2 + u^{-2})^{-2} du, \quad (\text{ASD10.2})$$

and the last integral is finite at both endpoints. This also proves that

$$\mathfrak{I}_A : A \rightarrow A', \quad \mathfrak{I}_A(b)(c) = B_A(b, c) \quad (\text{ASD10.3})$$

is strong-dual continuous: on a bounded family of c 's the second seminorm in ASD10.2 is uniformly bounded. It is injective, since $\mathfrak{I}_A(b)(\bar{b}) = \int |b|^2 du$ and $\bar{b} \in A$.

Direct changes of variables give

$$\begin{aligned} B_A(T_a b, T_a c) &= a B_A(b, c), & B_A(Rb, Rc) &= B_A(b, c), \\ \mathfrak{I}_A(T_a b) &= a T_a^\vee \mathfrak{I}_A(b), & \mathfrak{I}_A(Rb) &= R' \mathfrak{I}_A(b). \end{aligned} \quad (\text{ASD10.4})$$

This constructs an actual map $A \rightarrow \chi \otimes A'$, with the same factor a that occurs in the reflection relation.

This particular pairing does not descend to A/J . The original $g = \Sigma f_*$ from ASD9.4 is a nonzero real member of J , and

$$B_A(g, g) = \int_0^\infty g(u)^2 du > 0. \quad (\text{ASD10.5})$$

It is finite by ASD10.2 and positive because a continuous nonzero real function is nonzero on an interval. Nonzeroness follows also from Σ 's injectivity and $f_* \neq 0$. Thus it would assign a nonzero value to a pair of zero quotient classes. This proves only the precise failure of ASD10.1 to descend; it is not a statement that quotient pairings do not exist. The actual existing quotient pairing is included in ASD11.

The exact receiving replacement is ASD2.4, the annihilator $J^\perp$. In fact this annihilator has no nonzero element of the form $\mathfrak{I}_A(b)$:

$$\mathfrak{I}_A(A) \cap J^\perp = \{0\}. \quad (\text{ASD10.6})$$

Here is a proof using the entire original image, not a finite test. The image J is dense in $L^2(\mathbb{R}_{>0}, du)$. It contains every $T_a g$, because $T_a \Sigma h = \Sigma(h(\cdot/a))$ and the latter source remains in S . To prove their dense span, set

$$U_f(x) = e^{x/2} f(e^x), \quad U_g(x) = e^{x/2} g(e^x).$$

The substitution $u = e^x$ gives $\|U_f\|_{L^2(dx)} = \|f\|_{L^2(du)}$, and, with every factor retained,

$$U_{T_a g}(x) = a^{1/2} U_g(x - \log a), \quad \widehat{U}_g(t) = \mathcal{M}_0 g(1/2 - it) = 2F_*(1/2 - it). \quad (\text{ASD10.7})$$

In this line the Fourier transform is $\int h(x) e^{-itx} dx$, so its inverse has factor $1/(2\pi)$; this is the logarithmic-variable transform, separately specified from the original real Fourier phase in ASD1.2. The function U_g is Schwartz because $g \in A$.

If f is orthogonal in $L^2(du)$ to every $T_a g$, ASD10.7 says that U_f is orthogonal to every translate of U_g . Define, for real v ,

$$C(v) = \int_{\mathbb{R}} U_f(x) \overline{U_g(x-v)} dx, \quad \widetilde{U}_g(x) = \overline{U_g(-x)}.$$

Then $C = U_f * \widetilde{U}_g$ is the zero continuous function, and its Fourier transform in the convention of ASD10.7 is exactly $\widehat{C}(t) = \widehat{U}_f(t) \widehat{\widetilde{U}_g}(t)$. Plancherel, or first the convolution formula for Schwartz approximations of U_f followed by L^2 convergence, proves this identity in L^2 ; $\widehat{U}_g$ is bounded and $\widetilde{U}_g$ is integrable. The entire nonzero function $2F_*$ has only isolated zeros, so $\widehat{U}_g(t) \neq 0$ for almost every real t . Hence $\widehat{U}_f = 0$ almost everywhere, and $f = 0$. This proves the density assertion.

Finally J is stable under complex conjugation. If $\mathfrak{I}_A(b)$ annihilates J , then $\int b \bar{j} du = 0$ for all $j \in J$, so $b = 0$ by that density. This proves ASD10.6. The original Fréchet topology is stronger than the Hilbert topology just used for this exact comparison: J remains closed in A , as already proved. No Hilbert quotient has replaced Q , and no positivity of the spectral quotient has been asserted.

144.12 ASD11. The already constructed Weil form really maps into this dual

The existing RZ10–RZ11 form is retained. For $F = \mathcal{M}_0 b$, $G = \mathcal{M}_0 c$, define

$$W_{\text{val}}([b], [c]) = \sum_{\rho} m_{\rho} \overline{F(\rho^{\#})} G(\rho), \quad \rho^{\#} = 1 - \bar{\rho}. \quad (\text{ASD11.1})$$

The sum is over distinct actual original nontrivial zeros, and the multiplicity is written explicitly. It is conjugate-linear in the first variable and linear in the second. The proved bound $\sum_{|\rho| \leq R} m_{\rho} \leq C(R+2)^{3/2}$, together with rapid decrease on the full strip containing $0 < \Re \rho < 1$, gives

$$|W_{\text{val}}([F], [G])| \leq C_W q_{1,2}([F]) q_{1,2}([G]). \quad (\text{ASD11.2})$$

For clarity, the estimate before taking quotient infima is obtained by bounding each summand by $b_{1,2}(F) b_{1,2}(G) (1 + |\Im \rho|)^{-4}$ and summing over dyadic height ranges. Membership in I_{ζ} changes no value in ASD11.1, so separate infima over representatives prove ASD11.2. This proves an actual continuous quotient form.

Consequently

$$\mathfrak{W} : \overline{Q} \longrightarrow Q', \quad \mathfrak{W}(\bar{q})(r) = W_{\text{val}}(q, r) \quad (\text{ASD11.3})$$

is a complex-linear continuous map from the conjugate complex vector space. It is strong-dual continuous by ASD11.2 and bounded-set control. Its image lies in the already computed $Q' = J^{\perp}$ after composition with π' . Thus ASD10's failed integral descent does not erase this distinct existing morphism.

The complete dilation identity is

$$W_{\text{val}}(T_a q, T_a r) = a W_{\text{val}}(q, r), \quad \mathfrak{W}(\overline{T_a q}) = a T_a^{\vee} \mathfrak{W}(\bar{q}). \quad (\text{ASD11.4})$$

Termwise, the exponent is $\overline{\rho^{\#}} + \rho = (1 - \rho) + \rho = 1$. Absolute convergence justifies the calculation. This is a genuine map $\overline{Q} \rightarrow \chi \otimes Q'$, not a change of the source meaning of τ .

The same actual form is reflection-invariant:

$$W_{\text{val}}(Rq, Rr) = W_{\text{val}}(q, r), \quad \mathfrak{W}(\overline{Rq}) = R' \mathfrak{W}(\bar{q}). \quad (\text{ASD11.4a})$$

To check the first equality, its summand is $m_{\rho} \overline{F(1 - \rho^{\#})} G(1 - \rho)$. Set $\sigma = 1 - \rho$; then $1 - \rho^{\#} = \sigma^{\#}$ and $m_{\sigma} = m_{\rho}$, giving exactly ASD11.1. The second equality follows by replacing the second argument of the first by Rr . For the oriented cohomology mirror $w = -R$, the corresponding formula is $\mathfrak{W}(\overline{wq}) = (-R)' \mathfrak{W}(\bar{q})$; both signs remain.

Its exact kernel is

$$\ker \mathfrak{W} = \{[\overline{F}] : F(\rho) = 0 \text{ for every actual nontrivial zero } \rho\}. \quad (\text{ASD11.5})$$

One inclusion follows directly from the sum. For the reverse inclusion, the finite-jet representatives constructed in ASD9, with all other-zero jets zero, allow G whose only nonzero zero value is $G(\sigma) = 1$. The sum then reads $m_{\sigma} \overline{F(\sigma^{\#})}$. Vanishing for every σ proves all the required value vanishings. Thus all higher multiplicity jets remain in the original Q , even though this particular ordinary value-trace form has that radical. Neither simplicity of zeros nor positivity is assumed.

The comparison with the original centered receiver retains the coefficient2:

$$W_{\text{val}}(\overline{\mathcal{T}q}, \overline{\mathcal{T}r}) = 4 W_{\text{cen}}(q, r). \quad (\text{ASD11.6})$$

Indeed both Mellin functions acquire the factor2 in ASD1.8. If the original centered form is transported as a form on the raw space, its formula is $W_{\text{transported}} = \frac{1}{4} W_{\text{val}}$. Both maps are now explicit; no factor has disappeared.

For the zero-value functionals the row in ASD6 transports this form concretely:

$$\begin{aligned}\mathfrak{W}(\bar{q}) &\xrightarrow{\delta'} (\mathfrak{W}(\bar{q}), -\mathfrak{W}(\bar{q})), \\ (\mathfrak{W}(\bar{q}), 0) &\xrightarrow{\partial'} \pi' \mathfrak{W}(\bar{q}).\end{aligned}\tag{ASD11.7}$$

All continuous-dual target spaces and transposes in this formula have been constructed. These are valid irrespective of the sign of the form.

144.13 ASD12. The dual diagram is not obtained by reversing a sheaf restriction silently

The original restriction maps point from chart sections $V_{\pm}$ to the common open sections A . Their transposes point from A' to $V'_{\pm}$. This reversal is part of the derived dual calculation:

$$V_{\pm} \xrightarrow{r_{\pm}} A \quad \rightsquigarrow \quad A' \xrightarrow{r'_{\pm}} V'_{\pm}.\tag{ASD12.1}$$

It is a covariant section diagram, rather than the same sheaf with renamed coefficients. The actual whole-space gluing on dual sections is the cokernel map

$$A' \xrightarrow{(r'_+, -r'_-, 0)} V'_+ \oplus V'_- \oplus V'_{\text{extra}} \longrightarrow H' \longrightarrow 0.\tag{ASD12.2}$$

Its exactness was proved in ASD3 using Hahn–Banach. This is the explicit continuous-dual gluing correspondence. The kernel of its first map is Q' , not zero. On every nonempty arithmetic open the original restrictions were identities on A , so the transposed extension maps remain identities on A' ; all arithmetic prime points in the source fibre remain present.

Thus the transpose construction retains precisely the extra derived term that would be lost by treating dualization as a pointwise replacement of a coefficient sheaf without its direction or complex. It has not constructed a geometric dualizing sheaf or an $i^!$ -to- i^* Verdier comparison. It has constructed all displayed transposes, the full dual cohomology, and their exact relation to the original localization maps.

144.14 ASD13. What this contributes to the requested supported-obstruction calculation

The concrete global result is the exact row ASD6.3 and the complete complexes ASD3–ASD5. They carry the full original Q' , every original multiplicity jet, the four endpoint dual lines, the two entire faithful closed dual copies, the source integer action, each prime dilation, and the chart mirror with its orientation sign.

Deligne’s §3.6.1(8) identifies the supported obstruction group with

$$H^{2N-i-1}(X_s)^{\vee}(-N).$$

The duality and twist are geometric ingredients of his diagram; Lemma3.6.3 then gives its lower weight bound. The class-to-be-lifted group has the separate upper bound in Lemma3.6.2. Exactness of the weight truncation eliminates the incompatible supported image.

Here the actual transpose of the supported-to-global map is the injective anti-diagonal $Q' \rightarrow Q' \oplus Q'$, and the actual next map is the sum into $J^{\perp} \subset A'$. These maps are already globally split as in ASD6.5. Their corresponding finite zero jets have exactly the same contragredient characters $p^{-\rho}$ on both sides; no weight displacement has been inserted into the localization arrow. The character a is present in the original reflection relation and in the genuine pairing maps ASD10.4 and ASD11.4. These statements calculate the factor precisely, but neither a cochain degree shift nor reflection alone adds a geometric Tate twist to the supported row.

The actual stronger correspondence now available is

$$Q' = J^\perp \subset A', \quad \overline{Q} \xrightarrow{\mathfrak{M}} \chi \otimes Q', \quad Q \xrightarrow{R} \chi \otimes Q^{\text{inv}},$$

with the distinct domains, full maps, and exact kernel of $\mathfrak{M}$ proved above. This gives a specified receiving target for further actual coefficient pairings. It does not replace them by an assumed perfect, positive, or finite-dimensional duality.

144.14.1 The completed full-jet continuation received by this dual row

The companion `GLOBAL_ORIGINAL_ZETA_RESIDUE_DUALITY_INDEPENDENT.md`, GZR2–GZR8, now constructs the stronger continuous nondegenerate bilinear pairing on this same entire quotient:

$$B_\zeta([F], [G]) = \frac{1}{2\pi i} \left(\int_{2-i\infty}^{2+i\infty} - \int_{-1-i\infty}^{-1+i\infty} \right) \frac{F(s)G(1-s)}{\zeta(s)} ds. \quad (\text{ASD13.1})$$

Both edge integrals run upward. Its original denominator, all multiplicities, convergence, quotient descent, and nondegeneracy are proved there; this is no longer a prospective pairing. GZR8 supplies strong-continuous injective maps with weak-* dense images

$$\mathcal{D}_L x(y) = B_\zeta(x, y), \quad \mathcal{D}_R y(x) = B_\zeta(x, y), \quad \mathcal{D}_{L,R} : Q \longrightarrow \chi \otimes Q'. \quad (\text{ASD13.2})$$

The target $Q' = J^\perp$ and all its supported transpose maps are exactly the ones constructed above. These stronger maps do not erase a higher-jet radical: their kernels are zero. They do not assert surjectivity onto the continuous dual, positivity, or a geometric dualizing identification. Their mirror comparison is the exact GZR7.5–GZR7.7 relation with the companion denominator $\zeta(1-s)$, including its minus sign and full functional multiplier, not an asserted invariance of B_ζ itself.

The further `ORIGINAL_ZETA_JACOBIAN_TRACE_BRIDGE.md`, JTB2–JTB7, constructs the original derivative operator

$$\mathcal{J}_\zeta = (L-1)^{-2} M_{(s-1)^2 \zeta'(s)} : Q \rightarrow Q$$

with its complete endpoint-corrected representative, and proves

$$W_{\text{val}}(x, y) = B_\zeta(\mathcal{J}_\zeta y, Cx), \quad (CF)(s) = \overline{F(\bar{s})}. \quad (\text{ASD13.3})$$

Consequently the exact factorization into the present continuous-dual receiver is

$$\mathfrak{W}(\bar{x}) = \mathcal{J}'_\zeta \mathcal{D}_R(Cx). \quad (\text{ASD13.4})$$

Indeed evaluation of its right-hand side at y gives $\mathcal{D}_R(Cx)(\mathcal{J}_\zeta y) = B_\zeta(\mathcal{J}_\zeta y, Cx)$, which is ASD13.3. Thus the exact higher-jet loss of the value-trace form is retained as a constructed derivative factor, while the full residue pairing supplies the stronger dual injections. ASD6.3 receives both maps without changing a support label or a differential sign. The complete companion proofs are the cited sources for these new statements; the proof of ASD13.4 itself is the displayed evaluation.

Nothing here assigns a weight to source Z_1/τ , retracts the user's global arithmetic reconstruction, or uses an arbitrary receiver as a counterexample to that reconstruction. The calculation proves the full dual correspondence and its exact supported maps. Numerical purity and RH do not follow from the calculation alone, and no such conclusion is asserted.

144.14.2 Reading and dependency record for this derivation

- Current source rules and B1–B5/P1–P5 reread; the three complete global corpus arguments listed in ASD0 reread before ASD10’s interpretation.
- DCP3, DCP7–DCP12 read completely for the restriction, support, faithful extra-copy, and mirror formulas; their previous complete source reading is retained.
- CTF0–CTF8 read completely, with ASD4 applying CTF2–CTF4 and the exact extra identity-cone sign from CTF7.
- OMS1’s original spaces and Mellin isomorphism, OMS2’s original f_*, F_* formulas, and the already proved full image theorem are explicitly the receiving inputs. No new claim of a complete fresh reading of all OMS is made.
- RZ10–RZ11 read completely and actually applied in ASD11, preserving the form, radical, and similitude rather than asserting absence of quotient pairings.
- The complete retained RZ copy and the external file actually read were compared by SHA256 and are identical, with the hash in ASD0. For the final continuation, GZR7–GZR8 and JTB0–JTB6 plus the opening definitions and covariance of JTB7 were read. No complete fresh reading or independent verification of both entire companion manuscripts is claimed by this source-use entry.
- Deligne transcription lines2466–2559 read completely; source kind retained as transcription. Human theorem attribution is to Pierre Deligne. CC Fourier localization is attributed to Alain Connes and Caterina Consani; the analytic image theorem’s provenance is retained from Meyer and the cited programme derivations.

144.15 ASD14. The actual residue chain map and its entire cone

This continuation uses the same source prerequisites as ASD0, with no change to Z_0, Z_1, Z_2, τ , the source ring action, or the original coefficient topologies. The pairing here is the constructed original- ζ pairing GZR2–GZR8, rather than the test-space integral ASD10.1. Its strong-continuous injective left map and its exact similitude are

$$\mathcal{D}_L : Q \rightarrow Q', \quad \mathcal{D}_L(x)(y) = B_\zeta(x, y), \quad \mathcal{D}_L T_a = a T_a^\vee \mathcal{D}_L. \quad (\text{ASD14.1})$$

Its image is weak- $*$ dense, as proved in GZR8. These properties hold on the complete original quotient with every multiplicity; they are not finite-packet assumptions.

144.15.1 ASD14.1. Construct and verify the chain map

Write $P = D_{\text{full}}^0 = V_+ \oplus V_- \oplus V_{\text{extra}}$, and retain

$$D_{\text{full}} = [P \xrightarrow{d} A], \quad d(v_+, v_-, w) = r_+ v_+ - r_- v_-.$$

Set

$$T = \chi \otimes D_{\text{full}}^\vee[-2].$$

Here $\chi(a) = a$ changes the receiving real-group action only; it does not change the source-ring action. By ASD3.3 and the even shift, the only nonzero terms and differential of T are

$$T^1 = \chi \otimes A', \quad T^2 = \chi \otimes P', \quad d_T^1 = d' : \lambda \mapsto (r'_+ \lambda, -r'_- \lambda, 0). \quad (\text{ASD14.2})$$

Define

$$\Psi_\zeta : D_{\text{full}} \longrightarrow T, \quad \Psi_\zeta^0 = 0, \quad \boxed{\Psi_\zeta^1 = \pi' \mathcal{D}_L \pi : A \rightarrow A'}. \quad (\text{ASD14.3})$$

Thus its actual evaluation formula is

$$(\Psi_\zeta^1 b)(c) = \frac{1}{2\pi i} \left(\int_{2-i\infty}^{2+i\infty} - \int_{-1-i\infty}^{-1+i\infty} \right) \frac{(\mathcal{M}_0 b)(s)(\mathcal{M}_0 c)(1-s)}{\zeta(s)} ds. \quad (\text{ASD14.4})$$

Both edges retain their upward orientation, original denominator, full Mellin transforms, and all the factors in ASD1.6–ASD1.8. This is a continuous map from the original A to its strong dual and hence to its weak- $*$ dual: π is continuous, $\mathcal{D}_L$ is strong continuous, and π' is strong continuous by ASD2.2.

There are exactly two nontrivial cochain equations to check. First

$$\Psi_\zeta^1 d = \pi' \mathcal{D}_L(\pi d) = 0 = d_T^0 \Psi_\zeta^0,$$

since $d(P) = J = \ker \pi$. Second

$$d_T^1 \Psi_\zeta^1 = d' \pi' \mathcal{D}_L \pi = 0.$$

Indeed $\pi' \lambda$ annihilates $J = d(P)$, so $d' \pi' \lambda = 0$. This proves the chain map on the stated complexes, rather than only on their cohomology.

Let $\rho_D(a)$ be the original action ASD7.3 and let $\rho_T(a) = a \rho_D^\vee(a)$ on the shifted dual. In degree1,

$$\Psi_\zeta^1 T_a = \pi' \mathcal{D}_L T_a \pi = a \pi' T_a^\vee \mathcal{D}_L \pi = a T_a^\vee \Psi_\zeta^1. \quad (\text{ASD14.5})$$

Here the T_a between $\mathcal{D}_L$ and π is its induced quotient operator; equivariance of π proves both transpose identities. In degree0 both sides are zero. Thus this is equivariance for every positive real a , including every original prime p , with no factor removed.

It is also linear for the source receiving ring $\mathbb{Z}^3$. An element (r, b_+, b_-) acts by the same scalar r on A and A' , so ASD14.3 commutes with it. In degree0 the zero map commutes with every action, including the separate b_+ and b_- actions on the full extra copies. Consequently all supplied source multiplications and integer-input additions are preserved. In particular the chain map has introduced neither addition on source τ nor an identification of τ with source integer 1.

144.15.2 ASD14.2. Compute every cone degree and its cohomology

Use the original CTF convention

$$K_\zeta^n = T^n \oplus D_{\text{full}}^{n+1}, \quad d_{K_\zeta}(v, c) = (d_T v + \Psi_\zeta c, -dc).$$

Substitution of every degree gives the exact four-term complex

$$\boxed{P \xrightarrow{-d} A \xrightarrow{\Psi_\zeta^1} \chi \otimes A' \xrightarrow{d'} \chi \otimes P'} \quad \text{in degrees } -1, 0, 1, 2. \quad (\text{ASD14.6})$$

The first minus sign is the shifted-source sign. The final differential is $+d'$, with its internal chart signs in ASD14.2; the dual convention and shift add no further sign.

Write $H = \ker d = H_0 \oplus V_{\text{extra}}$, as in ASD3.6. The cohomology, with its exact action, is

$$\boxed{\begin{aligned} H^{-1}(K_\zeta) &= H, \\ H^0(K_\zeta) &= 0, \\ H^1(K_\zeta) &= \chi \otimes (Q'/\mathcal{D}_L Q), \\ H^2(K_\zeta) &= \chi \otimes H', \\ H^n(K_\zeta) &= 0 \quad (n \notin \{-1, 0, 1, 2\}). \end{aligned}} \quad (\text{ASD14.7})$$

Here the tensor notation retains the numerical action; as vector spaces the third and fourth entries are $Q'/\mathcal{D}_L Q$ and H' .

The degree -1 statement is the definition of H . For degree0, both π' and $\mathcal{D}_L$ are injective, whence $\ker \Psi_\zeta^1 = \ker \pi = J = \text{im}(-d)$. For degree1, ASD2.4 and ASD3.4 give

$$\ker d' = \pi' Q', \quad \text{im } \Psi_\zeta^1 = \pi' \mathcal{D}_L Q,$$

where the second equality uses the surjectivity of π . The injective map π' therefore supplies exactly the third quotient in ASD14.7. Finally ASD3.7 identifies the cokernel of d' with H' , proving the fourth entry. No other degrees occur.

For the weak-* topologies in ASD2, $\pi' : Q' \rightarrow J^\perp$ is a homeomorphism onto its image, and $\mathcal{D}_L Q$ is dense in Q' . Hence

$$(H^1(K_\zeta))_{\text{Hausdorff}} = 0 \quad \text{in this specified weak-* quotient topology.} \quad (\text{ASD14.8})$$

This is the quotient by the closure of zero. It does not prove that the algebraic cokernel $Q'/\mathcal{D}_L Q$ vanishes, and it does not prove that $\mathcal{D}_L Q$ is closed or onto. No corresponding strong-dual density or strong-quotient conclusion is asserted. The original source spaces P, A, Q still have their original topologies.

The full faithful closed copies are explicit in the cone:

$$K_\zeta = K_{\zeta, \epsilon} \oplus V_{\text{extra}}[1] \oplus (\chi \otimes V'_{\text{extra}})[-2]. \quad (\text{ASD14.9})$$

Indeed d kills the source extra copies, Ψ_ζ has only the overlap component, and d' has zero extra component. Thus all these displayed extra differentials are zero. Their degree -1 and degree2 cohomology has not been contracted. The four original endpoint lines are likewise present in $H_0 \subset H^{-1}$ and their full duals in $\chi H'_0 \subset H^2$. This cone is not the extra identity cone from ASD4, and the identity-cone contraction cannot remove its terms.

144.15.3 ASD14.3. The exact companion mirror identity

Define $\Psi_{\zeta^\vee}$ by the same construction, using the companion $B_{\zeta^\vee}$ whose denominator is $\zeta(1-s)$. GZR7 proves

$$B_\zeta(Rx, Ry) = -B_{\zeta^\vee}(x, y). \quad (\text{ASD14.10})$$

Replacing y by Ry and evaluating gives

$$\mathcal{D}_L^\zeta R = -R' \mathcal{D}_L^{\zeta^\vee}. \quad (\text{ASD14.11})$$

This is an identity between the two constructed pairing maps, not a self-reflection assertion.

Let w_D be the original Cech mirror, with w_D^0 the chart and extra-copy swaps and $w_D^1 = -R$. On $T = \chi D_{\text{full}}^\vee[-2]$, denote by $w_T = w_D^\vee$ its shifted dual mirror: it is $-R'$ in degree1 and the corresponding dual chart and extra swaps in degree2. The even shift introduces no new sign. Then

$$\boxed{\Psi_\zeta w_D = -w_T \Psi_{\zeta^\vee}.} \quad (\text{ASD14.12})$$

In degree1 its left side is

$$\pi' \mathcal{D}_L^\zeta \pi(-R) = -\pi' \mathcal{D}_L^\zeta R \pi = \pi' R' \mathcal{D}_L^{\zeta^\vee} \pi = R' \Psi_{\zeta^\vee}^1.$$

Its right side is $-(-R') \Psi_{\zeta^\vee}^1$, the same expression. In the remaining source degree both sides vanish. This verifies every sign.

The character twist is necessary in checking compatibility with the full unshifted prime/reflection relation. On the contragredient dual before twisting,

$$\rho_D^\vee(a) w_D^\vee = a^{-1} w_D^\vee \rho_D^\vee(a^{-1}).$$

Consequently on T

$$\rho_T(a) w_T = a \rho_D^\vee(a) w_T = w_T \rho_D^\vee(a^{-1}) = a w_T \rho_T(a^{-1}). \quad (\text{ASD14.13})$$

The original source complex has the same final relation $\rho_D(a) w_D = a w_D \rho_D(a^{-1})$. Thus ASD14.12 is compatible with prime inversion and the exact factor a . No geometric Tate identification is needed to prove this representation identity, and none is inferred from it.

The corresponding actual cone map is

$$\mathcal{W} : K_{\zeta^\vee} \longrightarrow K_\zeta, \quad (v, c) \longmapsto (-w_T v, w_D c). \quad (\text{ASD14.14})$$

For its target differential the first coordinate is $-w_T d_T v + \Psi_\zeta w_D c$. By ASD14.12 it equals $-w_T(d_T v + \Psi_{\zeta^\vee} c)$; the second coordinate is $-w_D dc$. These are precisely the coordinates of $\mathcal{W}d_{K_{\zeta^\vee}}(v, c)$. Thus it is a cochain isomorphism. Applying the companion version twice yields identity because both mirrors square to identity and the two target signs multiply to $+1$. It intertwines the original real actions with the same inversion and factor a in ASD14.13, and is semilinear for the receiving ring involution exchanging e_+, e_- . That involution fixes every original source integer and τ .

ASD14 constructs a genuine full-coefficient dual comparison and computes its entire cone. Its nonzero H, H' terms, exact possibly nonzero algebraic dual cokernel, and companion mirror are displayed. It is not asserted to be a Verdier isomorphism, a self-mirror pairing, a closed-image theorem for $\mathcal{D}_L$, or a numerical weight bound.

144.16 Explicit global continuation: finite lifting and a nonzero remainder

The earlier uncertainty about algebraic surjectivity is now resolved by GLOBAL_POLYNOMIAL_DUAL_DEFECT.md PGD1–PGD6, independently checked in PGC0–PGC7: the actual algebraic cokernel is nonzero. Its explicit class `c_zeta` is the original-zeta contour functional with the constant arithmetic function 1 in the first slot. A lift would require a rapidly decreasing entire representative to equal 1 at all actual zeros, which the proved unbounded zero heights exclude. PGD constructs an injected rational-function line, entire representatives retaining all Hermite pole corrections, and the whole real/prime orbit. This is a receiving cokernel class, not primitive τ or a claimed off-critical zero.

SCL0–SCL9 and DPL0–DPL9 prove every nonzero polynomial in the actual cokernel generator invertible. Thus all finite-dimensional invariant dual spaces lift uniquely, continuously and with the original real/prime action. DPL gives explicit extension splittings and computes every degree of the cyclic derived Hom into the actual cone. These results retain its H and H' terms, all faithful closed copies and support labels. The weak-* Hausdorff quotient remains zero alongside the now proved nonzero algebraic quotient. Current source `Z_0`, `Z_1/tau` and `Z_2` are unchanged. Complete proofs and exact human citations are in the linked continuation files; the full geometric weight-separation goal remains active.

144.17 Full global prime action and actual supported placement

The new complete proofs GGT0–GGT8, FPO0–FPO8 and UOS0–UOS9 (including UOS8A) strengthen the preceding global remainder calculation. The original-zeta Gaussian trace has a `TlogT` leading term only at the identity scale and at most order T at every other fixed positive scale. Testing a proposed finite relation against every inverse scale proves all coefficients zero. Consequently the entire finite group algebra $\mathbb{C}[\mathbb{R}_+^{\wedge \times}]$, including the Laurent algebra of all original primes, injects into the original multiplier quotient and linearly into the actual comparison cokernel. This computes relations after the actual zero-jet quotient, not only among entire functions before quotienting.

UOS constructs the full chain map into the actual dual-supported comparison cone. It sends `c_r` to `(c_r, 0)`, acts identically on H , and restricts $\chi H'$ to $\chi E'$ with every endpoint and extra closed copy retained. Its companion mirror is `c_r` $\rightarrow$ $-r$ `c_r` $(1/r)$ with the companion denominator; the oriented global-dual mirror has the opposite sign. The ordinary Fourier restriction still has its explicit section. Thus the new classes are relative residue-representation classes with proved supported placement, not a claimed failure of that existing Fourier lift. The source `Z_0`, `Z_1/tau` and integer `Z_2` have not changed, and the full numerical weight-separation target remains active.

The original Connes–Consani explicit formula has been applied with a proved cutoff extension; the exact archimedean comparison is $W_R(f_T,x)=A_T(x)+1$, retaining the endpoint at zero and every finite trivial-zero contour correction. Complete human citations, author-source archive and bounded reading coverage are recorded with these proofs. All complete proof bodies and inspected reproducible diagrams are in the cumulative TeX.

144.18 Exact original character lifting and its full supported return

MCL0–MCL11 now computes the original transpose $A' \rightarrow S'$ on every actual zero and all its Mellin jets. A target jet of order j needs generalized-character length exactly $m+j+1$ at a zero of multiplicity m . The full Euler-distribution classification proves minimality against the whole original continuous dual; two actual primes remove the single-prime character aliases. Every prime logarithm and the full Gamma/Fourier return remain. This is an actual continuous lift, with a nonzero class only when its original annihilator must be preserved.

RPC0–RPC8 constructs the original restriction’s actual pushout by the residue image, proves its precise cochain location and a single coherent real/prime-equivariant section on all locally finite classes. ORE0–ORE10 calculates the unmodified connecting class, including its full Gamma germ and rank $\min(m,r)$, then proves its place in every degree of the full supported derived comparison. The degree-two map is $(\eta,e) \mapsto (-\delta_q(\eta)/2,e)$. Thus its image in C_zeta/qC_zeta is zero, while the full supported map retains its Y/qY component. Both assertions follow from the constructed maps; no cone degree or extra coefficient is discarded.

These results refine the meaning of the prior finite-lift statement without withdrawing its proved scope. Fixed-annihilator splitting is stronger than the surjectivity in Deligne’s invariant-cycle theorem. The original longer-chain map is already surjective. The present numerical character truncation retains its same-character extension whenever it retains the target; it is not Deligne’s independently proved geometric weight gap. Complete proofs, independent checks and inspected reproducible diagrams have been inserted into the cumulative TeX. The actual tau weight-separation goal remains active.

145 Global residue duality for the actual original-zeta quotient

24 September 2026. Independent derivation. Proof locators GZR0–GZR9.

145.1 GZR0. Original objects, sources, and operation domains

This calculation constructs a continuous, nondegenerate bilinear pairing on the entire original quotient. It uses the original denominator $\zeta(s)$, every actual zero multiplicity, and the quotient Fréchet topology. Its definition is a pair of absolutely convergent integrals. An unrestricted infinite sum of residues is not used as its definition or assumed to converge.

The controlling source notation remains Z_0, Z_1, Z_2, τ . Primitive τ is neither added nor assigned parity, an arithmetic coordinate, or a numerical weight. All additions and derivatives below occur in the existing complex receiving spaces of the already reconstructed original zeta function. A receiving supported zero z_λ is not primitive τ .

Before interpreting the result, the complete user passages USR-9ad1c0a2d09dba92, USR-f55d16feb948d8c2, and USR-6152e3bc6302258c were reread in the preserved corpus, together with `CORPUS_AND_OPERATION_RULES.md`. They concern the whole reconstructed arithmetic system, admissible global transformations, and the requested comparison with Deligne’s weight argument. This note does not replace that source by an independently invented spectrum.

The exact prior programme inputs are:

1. OMS1–OMS6, Original Mellin spectral synthesis: the original test spaces, topological Mellin comparison, exact Schwartz image, uniform rapid division, full zero jets, and original primary sections.
2. RZ1–RZ8, Original-zeta resolvent and projectors: actual global representatives of each individual multiplicity jet and the full arithmetic action. These sections and the retained multiplier were read in `quantum_tau_programme_bridge_20260924/sources/programme/cc_sheaf/ORIGINAL_ZETA_RESOLVENT_AND_PROJECTORS.md`. The linked retained copy is byte-identical: SHA256 f3553e1362b4c2c1c1556098b55ee153be9fc716a9d4c35baedcf39bd28ad755. The representatives are prior results, not a new construction claimed here.
3. The earlier finite residue pairing and Jacobian trace contraction, `TAU_BASE_MODEL §6`. The locally retained §6 was read. That calculation uses a finite completed-germ denominator. Its finite residue mechanism is prior work; GZR1–GZR8 establish the present global construction on the actual quotient with the original denominator. Only §6’s receiving finite residue and trace algebra is used: that older document’s introductory absorber interpretation of tau is not the current primitive identity source and is not imported. The current source remains the user’s B1–B5 and retraction R1.

The human analytic source underlying OMS is Ralf Meyer, A spectral interpretation for the zeros of the Riemann zeta function, arXiv:math/0412277v3, original author TeX, function spaces and Zeta operator through the exact Fourier–Laplace range theorem. The retained original author source is `sources/Meyer_0412277v3/Meyer.tex`, SHA256 ab9bc31f3c105a64fdc6f3b65ad16701dd8bf218fc90df98fe8d7f3c7c58c00e. The stronger rapid-division result invoked below is proved in OMS3; it is not attributed without proof to Meyer’s theorem.

Write $\mathcal{Z}$ for the distinct actual nontrivial zeros of the original ζ , and m_ρ for their actual positive multiplicities. No assumption of simple zeros, RH, or positivity is made.

145.2 GZR1. The entire original quotient and the needed division theorem

Retain

$$\begin{aligned}
 S &= \{f \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : f(-v) = f(v), f(0) = 0, \int_{\mathbb{R}} f(v) dv = 0\}, \\
 \mathcal{A} &= \left\{ k \in C^\infty(\mathbb{R}_{>0}) : \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j k(u)| < \infty \text{ for every } N, j \geq 0 \right\}, \\
 \mathcal{E}f(u) &= u^{1/2} \sum_{n \geq 1} f(nu), \\
 \mathcal{M}k(s) &= \int_0^\infty k(u) u^{s-1/2} \frac{du}{u}.
 \end{aligned} \tag{GZR1.1}$$

The receiving Fréchet algebra is

$$\mathcal{B} = \{F \text{ entire} : b_{A,M}(F) := \sup_{|\sigma| \leq A, t \in \mathbb{R}} (1 + |t|)^M |F(\sigma + it)| < \infty \text{ for all integers } A, M \geq 0\}. \tag{GZR1.2}$$

Products and multiplication by every fixed polynomial are continuous on $\mathcal{B}$, as follows immediately by distributing the displayed weights. Reflection $RF(s) = F(1 - s)$ is continuous, since

$$b_{A,M}(RF) \leq b_{A+1,M}(F). \tag{GZR1.3}$$

The exact original closed image theorem proved in OMS gives

$$\begin{aligned}
 \mathcal{M}(\mathcal{E}S) &= \mathcal{I}, \quad \mathcal{I} = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \ (\rho \in \mathcal{Z}, 0 \leq j < m_\rho)\}, \\
 Q &= \mathcal{A}/\mathcal{E}S \xrightarrow[\mathcal{M}]{\cong} \mathcal{Q} = \mathcal{B}/\mathcal{I}.
 \end{aligned} \tag{GZR1.4}$$

The equality uses the actual image, not only its closure. The arrow is a topological isomorphism. The ideal is closed because all the finite derivative evaluations are continuous. We write $\pi F = [F]$, and use the quotient seminorms

$$\bar{b}_{A,M}([F]) = \inf_{H \in \mathcal{I}} b_{A,M}(F + H). \tag{GZR1.5}$$

We use the following proved part of OMS3, with its original Schwartz inverse. If $H \in \mathcal{I}$, then H/ζ extends holomorphically throughout a neighborhood of the strip $-1 \leq \Re s \leq 2$, and, for every M , its absolute value times $(1 + |\Im s|)^M$ is bounded uniformly on this strip by finitely many seminorms of H . This is a conclusion of the exact image theorem, not an assumption that meromorphic division preserves $\mathcal{B}$. Globally H/ζ can have the retained trivial-zero poles.

More explicitly, the actual inverse is

$$f_H(v) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{H(2 + it)}{\zeta(2 + it)} v^{-2-it} dt, \quad v > 0, \quad f_H \in S, \quad \mathcal{M}\mathcal{E}f_H = H. \tag{GZR1.6}$$

Its continuation toward zero retains

$$\text{Res}_{s=-2r} \frac{H(s)}{\zeta(s)} = \frac{H(-2r)}{\zeta'(-2r)}, \quad \zeta'(-2r) = (-1)^r 2^{-2r-1} \pi^{-2r} (2r)! \zeta(1 + 2r), \tag{GZR1.7}$$

and its endpoint values retain

$$(H/\zeta)(0) = -2H(0), \quad (H/\zeta)(1) = 0, \quad (H/\zeta)'(1) = H(1). \tag{GZR1.8}$$

The uniform strip estimate is OMS3.8, proved by the full original functional equation, the Hadamard product with all factors, and the Gaussian-damped maximum principle before removal of that damping. Equations (GZR1.6)–(GZR1.8) are the original Schwartz return used here.

The complete original functional multiplier is

$$\zeta(s) = \chi(s) \zeta(1 - s), \quad \chi(s) = 2^s \pi^{s-1} \sin(\pi s/2) \Gamma(1 - s) = \pi^{s-1/2} \frac{\Gamma((1-s)/2)}{\Gamma(s/2)}. \tag{GZR1.9}$$

It is holomorphic and nonzero at every nontrivial zero. Thus $1 - \rho$ is an actual zero of the same multiplicity, and $R\mathcal{I} = \mathcal{I}$.

145.3 GZR2. Two absolutely convergent integrals and their continuity

For $F, G \in \mathcal{B}$, define

$$\mathfrak{B}_\zeta(F, G) = \frac{1}{2\pi i} \left(\int_{2-i\infty}^{2+i\infty} - \int_{-1-i\infty}^{-1+i\infty} \right) \frac{F(s)G(1-s)}{\zeta(s)} ds. \quad (\text{GZR2.1})$$

Both displayed vertical integrals are oriented upward. Their difference agrees with the vertical part of a counterclockwise rectangle: the right side is upward and the left side is downward.

Here are absolute bounds using the original function, without a replacement denominator. The reciprocal Euler series gives

$$|\zeta(2+it)^{-1}| \leq \zeta(2). \quad (\text{GZR2.2})$$

For $s = -1+it$, set $y = t/2$. The full Gamma factors in (GZR1.9) give

$$|\chi(-1+it)|^2 = \pi^{-3} y \coth(\pi y) (y^2 + \frac{1}{4}). \quad (\text{GZR2.3})$$

The continuous value at $y = 0$ is understood. To verify this identity, use

$$\Gamma(-\frac{1}{2}+iy) = \frac{\Gamma(\frac{1}{2}+iy)}{-\frac{1}{2}+iy}, \quad |\Gamma(1-iy)|^2 = \frac{\pi y}{\sinh(\pi y)}, \quad |\Gamma(\frac{1}{2}+iy)|^2 = \frac{\pi}{\cosh(\pi y)}$$

in the Gamma ratio in (GZR1.9). Since

$$y^2 + \frac{1}{4} \geq \frac{(1+|t|)^2}{8}, \quad y \coth(\pi y) \geq \frac{1+|t|}{2\pi},$$

the same reciprocal Euler series at $1-s = 2-it$ yields

$$|\zeta(-1+it)^{-1}| \leq 4\pi^2 \zeta(2)(1+|t|)^{-3/2}. \quad (\text{GZR2.4})$$

For the second elementary inequality, put $x = \pi|y|$. Then $x \coth x \geq \max(1, x) \geq (1+2x/\pi)/2$; the limiting value at zero is one.

All four numerator evaluations on the two edges lie in $|\Re s| \leq 2$. Since $\int_{\mathbb{R}} (1+|t|)^{-2} dt = 2$, equations (GZR2.2) and (GZR2.4) prove the explicit bound

$$|\mathfrak{B}_\zeta(F, G)| \leq C_\zeta b_{2,1}(F)b_{2,1}(G), \quad C_\zeta = \frac{\zeta(2)(1+4\pi^2)}{\pi}. \quad (\text{GZR2.5})$$

Thus each integral is absolutely convergent and the pairing is jointly continuous and complex bilinear on the whole of $\mathcal{B}$.

There is no endpoint pole in the integrand: $\zeta(0) = -1/2$, while $1/\zeta(s)$ has a simple zero at $s = 1$. The trivial zeros $-2, -4, \dots$ lie outside this strip; they remain in the original source return (GZR1.7). No contour enclosing them has been substituted for (GZR2.1).

145.4 GZR3. Descent in both slots through the actual Schwartz image

Suppose first $F \in \mathcal{I}$. The function F/ζ is holomorphic and uniformly rapidly decreasing on the closed strip by GZR1. The product

$$K(s) = (F(s)/\zeta(s))G(1-s)$$

has the same properties. On the rectangle with vertical sides at $-1, 2$ and horizontal sides at imaginary parts $\pm T$, Cauchy's theorem gives zero for its complete contour integral. The horizontal sides have length three and tend uniformly to zero as $T \rightarrow \infty$. The absolutely convergent vertical sides therefore give

$$\mathfrak{B}_\zeta(F, G) = 0 \quad (F \in \mathcal{I}). \quad (\text{GZR3.1})$$

This proof invokes the actual division theorem and its Schwartz return, not formal cancellation of the zeta denominator.

If $G \in \mathcal{I}$, then $RG \in \mathcal{I}$, with all multiplicities, by (GZR1.9). Apply the identical argument to

$$K(s) = F(s)(G(1-s)/\zeta(s)).$$

It gives

$$\mathfrak{B}_\zeta(F, G) = 0 \quad (G \in \mathcal{I}). \quad (\text{GZR3.2})$$

There is consequently a well-defined bilinear form on the entire original quotient:

$$B_\zeta : \mathcal{Q} \times \mathcal{Q} \longrightarrow \mathbb{C}, \quad B_\zeta([F], [G]) = \mathfrak{B}_\zeta(F, G). \quad (\text{GZR3.3})$$

For every $H, K \in \mathcal{I}$, descent and (GZR2.5) give the same bound with $F + H, G + K$. Taking the two independent infima proves

$$|B_\zeta(x, y)| \leq C_\zeta \bar{b}_{2,1}(x) \bar{b}_{2,1}(y). \quad (\text{GZR3.4})$$

In particular the descended bilinear form is jointly continuous for the actual quotient topology.

145.5 GZR4. Every local multiplicity coefficient and its finite perfect form

Fix an actual zero ρ , put $m = m_\rho$, and write $t = s - \rho$. Retain the full local expansion

$$\zeta(\rho + t) = t^m u_\rho(t), \quad u_\rho(t) = \sum_{k \geq 0} u_{\rho,k} t^k, \quad u_{\rho,k} = \frac{\zeta^{(m+k)}(\rho)}{(m+k)!}, \quad u_{\rho,0} \neq 0. \quad (\text{GZR4.1})$$

Define the exact reciprocal coefficients

$$c_{\rho,0} = u_{\rho,0}^{-1}, \quad c_{\rho,n} = -u_{\rho,0}^{-1} \sum_{r=1}^n u_{\rho,r} c_{\rho,n-r} \quad (n \geq 1). \quad (\text{GZR4.2})$$

Equating coefficients in $u_\rho(t) \sum_{n \geq 0} c_{\rho,n} t^n = 1$ proves that these are precisely the Taylor coefficients of u_ρ^{-1} . They retain every original zeta derivative contributing to the corresponding order.

The exact residue is

$$\text{Res}_{s=\rho} \frac{F(s)G(1-s)}{\zeta(s)} ds = \sum_{\substack{i,j,k \geq 0 \\ i+j+k=m-1}} \frac{F^{(i)}(\rho)}{i!} \frac{G^{(j)}(1-\rho)}{j!} (-1)^j c_{\rho,k}. \quad (\text{GZR4.3})$$

Indeed $G(1-\rho-t) = \sum_{j \geq 0} G^{(j)}(1-\rho)(-t)^j/j!$, and the residue is the coefficient of t^{m-1} in the product of this series, the Taylor series of F , and u_ρ^{-1} . This proves the displayed formula, including its sign and every factorial.

The local algebras are the actual full primary algebras

$$A_\rho = \mathbb{C}[t]/(t^m), \quad A_{1-\rho} = \mathbb{C}[v]/(v^m), \quad v = s - (1-\rho). \quad (\text{GZR4.4})$$

For $P = \sum_{i < m} p_i t^i$ and $Q = \sum_{j < m} q_j v^j$, their local pairing is

$$\beta_\rho(P, Q) = [t^{m-1}] \left(P(t) Q(-t) \sum_{k=0}^{m-1} c_{\rho,k} t^k \right). \quad (\text{GZR4.5})$$

Its matrix in the ordered monomial bases is

$$M_{\rho,ij} = \begin{cases} (-1)^j c_{\rho,m-1-i-j}, & i+j \leq m-1, \\ 0, & i+j > m-1, \end{cases} \quad 0 \leq i, j < m. \quad (\text{GZR4.6})$$

Reversing the columns yields a triangular matrix. The column-reversal sign is $(-1)^{m(m-1)/2}$, and the product of the anti-diagonal signs is also $(-1)^{m(m-1)/2}$. Hence

$$\det M_\rho = c_{\rho,0}^m = \left(\frac{m!}{\zeta^{(m)}(\rho)} \right)^m \neq 0. \quad (\text{GZR4.7})$$

Every multiplicity jet is therefore retained. For example, when $m = 2$, the actual matrix is

$$\begin{pmatrix} c_{\rho,1} & -c_{\rho,0} \\ c_{\rho,0} & 0 \end{pmatrix}, \quad c_{\rho,0} = \frac{2}{\zeta''(\rho)}, \quad c_{\rho,1} = -\frac{2\zeta'''(\rho)}{3\zeta''(\rho)^2}. \quad (\text{GZR4.8})$$

These are exact finite calculations, not an assumption that the global pairing is a convergent infinite sum of them.

145.6 GZR5. Actual global isolators and a justified finite-residue theorem

We recall the existing RZ representatives with all factors. Set

$$f_*(v) = \frac{\pi}{2}v^2(2\pi v^2 - 3)e^{-\pi v^2}, \quad F_*(s) = \mathcal{ME}f_*(s) = \frac{s(s-1)}{8}\pi^{-s/2}\Gamma(s/2)\zeta(s). \quad (\text{GZR5.1})$$

This is an actual auxiliary transform in $\mathcal{I}$; it does not replace the denominator in (GZR2.1). Its full exceptional values are

$$F_*(0) = F_*(1) = \frac{1}{8}, \quad F_*(-2r) = F_*(1+2r) = \frac{(1+2r)2r}{8}\pi^{-(1+2r)/2}\Gamma((1+2r)/2)\zeta(1+2r) \neq 0. \quad (\text{GZR5.2})$$

For $\eta \in \mathcal{Z}$, put $m = m_\eta$,

$$U_\eta(s) = \frac{F_*(s)}{(s-\eta)^m}, \quad a_{\eta,k} = \frac{F_*^{(m+k)}(\eta)}{(m+k)!},$$

$$d_{\eta,0} = a_{\eta,0}^{-1}, \quad d_{\eta,n} = -a_{\eta,0}^{-1} \sum_{r=1}^n a_{\eta,r} d_{\eta,n-r}, \quad e_\eta(s) = U_\eta(s) \sum_{n=0}^{m-1} d_{\eta,n} (s-\eta)^n. \quad (\text{GZR5.3})$$

Removal of the zero at η proves U_η entire. Division by the fixed polynomial outside a bounded neighborhood, and Taylor division inside, prove $U_\eta \in \mathcal{B}$. The finite polynomial factors therefore give $e_\eta \in \mathcal{B}$.

The reciprocal recursion proves

$$e_\eta(s) = 1 + O((s-\eta)^m), \quad E_{\eta,j}(s) = e_\eta(s)(s-\eta)^j, \quad 0 \leq j < m.$$

$$\frac{E_{\eta,j}^{(r)}(\nu)}{r!} = \begin{cases} 1, & \nu = \eta, r = j, \\ 0, & \text{otherwise,} \end{cases} \quad \nu \in \mathcal{Z}, \quad 0 \leq r < m_\nu. \quad (\text{GZR5.4})$$

At every other zero the full original vanishing order is retained by F_* . Thus these equations hold at all zeros simultaneously. Each function has its actual original representative

$$k_{\eta,j}(u) = \frac{1}{2\pi} \int_{\mathbb{R}} E_{\eta,j}(\tfrac{1}{2} + it) u^{-it} dt \in \mathcal{A}. \quad (\text{GZR5.5})$$

The Mellin isomorphism in GZR1 proves the asserted membership and identity of its transform.

For completeness, all derivatives used in the reciprocal recursion retain the original multiplier

$$C(s) = \frac{1}{8}s(s-1)\pi^{-s/2}\Gamma(s/2),$$

$$a_{\eta,k} = \sum_{h=0}^k \frac{C^{(h)}(\eta)}{h!} \frac{\zeta^{(m+k-h)}(\eta)}{(m+k-h)!},$$

$$C^{(h)}(s) = \frac{\pi^{-s/2}}{8} \sum_{\substack{a+b+c=h \\ 0 \leq a \leq 2}} \frac{h!}{a!b!c!} (s(s-1))^{(a)} \left(-\frac{\log \pi}{2} \right)^b 2^{-c} \Gamma^{(c)}(s/2). \quad (\text{GZR5.6})$$

Here is the contour result needed to apply these actual representatives. Suppose $F, G \in \mathcal{B}$, and the meromorphic function $F(s)G(1-s)/\zeta(s)$ has nonremovable poles inside the strip at only a finite set $S_0 \subset \mathcal{L}$. Put

$$P_{S_0}(s) = \prod_{\rho \in S_0} (s - \rho)^{m_\rho}. \quad (\text{GZR5.7})$$

The product $H(s) = P_{S_0}(s)F(s)G(1-s)$ lies in $\mathcal{B}$. At every zero outside S_0 , pole removability says that the numerator already has the required full zero order. At a zero in S_0 , the explicit factor supplies it. Thus $H \in \mathcal{I}$.

GZR1 now proves H/ζ uniformly rapidly decreasing on the whole strip. At sufficiently large $|\Im s|$, the polynomial P_{S_0} is bounded below by a positive constant times $(1 + |\Im s|)^{\deg P_{S_0}}$, uniformly in that strip. Consequently

$$\frac{F(s)G(1-s)}{\zeta(s)} = \frac{H(s)/\zeta(s)}{P_{S_0}(s)}$$

also tends uniformly to zero faster than any inverse power on the horizontal sides of sufficiently tall rectangles. The residue theorem and absolute convergence of the vertical edges give

$$B_\zeta([F], [G]) = \sum_{\rho \in S_0} \text{Res}_{s=\rho} \frac{F(s)G(1-s)}{\zeta(s)} ds. \quad (\text{GZR5.8})$$

This is a proved finite sum. It does not use estimates of $1/\zeta$ close to infinitely many unrelated zeros.

In particular, if $G = E_{1-\rho,j}$, its reflected numerator vanishes to every full zero order except possibly at ρ ; (GZR5.8) has only that one residue. The same is true with the first slot supported by an $E_{\rho,j}$. This is the exact bridge from the global contour pairing to the perfect local form.

145.7 GZR6. Global nondegeneracy, exact radicals, and primary adjoints

Take a nonzero $x = [F] \in \mathcal{Q}$. Synthesis (GZR1.4) implies that at least one actual full jet is nonzero. Fix such a ρ , and let $\ell < m = m_\rho$ be the least index with

$$f_\ell = F^{(\ell)}(\rho)/\ell! \neq 0.$$

Choose $G = E_{1-\rho, m-1-\ell}$. Equations (GZR5.8) and (GZR4.3) give exactly

$$B_\zeta(x, [G]) = (-1)^{m-1-\ell} f_\ell c_{\rho,0} \neq 0. \quad (\text{GZR6.1})$$

Indeed the only possible term has $i = \ell, j = m-1-\ell, k = 0$: all smaller i vanish, while larger i exceed the permitted total degree.

For a nonzero $y = [G]$, choose a zero η with first nonzero Taylor coefficient $g_\ell = G^{(\ell)}(\eta)/\ell!$, and put $\rho = 1 - \eta$. Taking $F = E_{\rho, m_\eta-1-\ell}$ gives

$$B_\zeta([F], y) = (-1)^\ell g_\ell c_{\rho,0} \neq 0. \quad (\text{GZR6.2})$$

Thus both radicals of the global form vanish. Equivalently, the two radicals of the original form $\mathfrak{B}_\zeta$ on $\mathcal{B}$ are exactly $\mathcal{I}$, not larger subspaces.

Let $s_\rho : A_\rho \rightarrow \mathcal{Q}$ be the existing primary section supplied by (GZR5.4), and j_ρ the full Taylor-jet map. The actual projector is $P_\rho = s_\rho j_\rho$. Formula (GZR5.8) proves

$$B_\zeta(s_\rho P, s_\eta Q) = 0 \quad (\eta \neq 1 - \rho),$$

$$B_\zeta(s_\rho P, s_{1-\rho} Q) = \beta_\rho(P, Q), \quad B_\zeta(P_\rho x, y) = B_\zeta(x, P_{1-\rho} y). \quad (\text{GZR6.3})$$

For the last identity, both sides are the same local residue at ρ ; the projectors retain exactly the jets appearing in (GZR4.3). These equations prove orthogonality and the reflected adjoint relation for actual primary subspaces of the original quotient, with all nilpotents retained.

145.8 GZR7. Full real action and the reflected companion denominator

For every $a > 0$, retain

$$W_a k(u) = a^{1/2} k(u/a), \quad T_a F(s) = a^s F(s), \quad \mathcal{M}W_a = T_a \mathcal{M}. \quad (\text{GZR7.1})$$

Multiplication by a^s is continuous on every strip and preserves all zero orders, so it acts on the entire quotient. Inside each integral,

$$(T_a F)(s)(T_a G)(1-s) = a^s a^{1-s} F(s)G(1-s) = a F(s)G(1-s).$$

Therefore the full global similitude is

$$B_\zeta(T_a x, T_a y) = a B_\zeta(x, y). \quad (\text{GZR7.2})$$

No factor $a^{1/2}$ has been absorbed into a different spectral parameter.

Let $L[F] = [sF]$, corresponding on the original source to $1/2 - u\partial_u$. Directly in the absolutely convergent integrals,

$$B_\zeta(Lx, y) = B_\zeta(x, (1-L)y), \quad B_\zeta(Lx, y) + B_\zeta(x, Ly) = B_\zeta(x, y). \quad (\text{GZR7.3})$$

More generally $B_\zeta(P(L)x, y) = B_\zeta(x, P(1-L)y)$ for every complex polynomial P . At paired primary blocks this retains the full nilpotent reflection $t \mapsto -v$, not only the scalar eigenvalues.

Define the companion using its actual original denominator:

$$B_{\zeta^\vee}([F], [G]) = \frac{1}{2\pi i} \left(\int_{2-i\infty}^{2+i\infty} - \int_{-1-i\infty}^{-1+i\infty} \right) \frac{F(s)G(1-s)}{\zeta(1-s)} ds. \quad (\text{GZR7.4})$$

Its edge bounds exchange (GZR2.2) and (GZR2.4), so both integrals converge absolutely. Substitute $w = 1 - s$ in each integral before subtraction. A single upward integral on $\Re s = c$ becomes an upward integral on $\Re w = 1 - c$: the reversed traversal and $ds = -dw$ cancel. The two vertical edges then exchange places, yielding the remaining minus sign

$$\boxed{B_\zeta(y, x) = -B_{\zeta^\vee}(x, y).} \quad (\text{GZR7.5})$$

This identity proves descent and continuity of the companion as well. Since $R^2 = 1$ and $(RF)(s)(RG)(1-s) = G(s)F(1-s)$, it also proves

$$B_\zeta(Rx, Ry) = B_\zeta(y, x) = -B_{\zeta^\vee}(x, y). \quad (\text{GZR7.6})$$

The sign is an orientation calculation, not a choice of a phase.

The full functional equation records the exact denominator comparison

$$\frac{1}{\zeta(1-s)} = \frac{2^s \pi^{s-1} \sin(\pi s/2) \Gamma(1-s)}{\zeta(s)} = \frac{\pi^{s-1/2} \Gamma((1-s)/2)}{\Gamma(s/2) \zeta(s)}. \quad (\text{GZR7.7})$$

These are meromorphic equalities with their removable values retained. At $s = 0$ the companion reciprocal has a zero; at $s = 1$ it equals $1/\zeta(0) = -2$. The Gamma and endpoint behavior is not discarded. Equations (GZR7.5)–(GZR7.7) do not assert that B_ζ itself is symmetric, skew-symmetric, Hermitian, or positive.

145.9 GZR8. Continuous injections into the strong dual and their weak-* density

Let $\mathcal{Q}'$ denote the continuous complex-linear dual, and let $\mathcal{Q}'_b$ have the strong topology of uniform convergence on bounded subsets of the actual quotient Fréchet space. Define

$$\mathcal{D}_L(x)(y) = B_\zeta(x, y), \quad \mathcal{D}_R(y)(x) = B_\zeta(x, y). \quad (\text{GZR8.1})$$

Equation (GZR3.4) proves that both are well-defined maps into the continuous dual. GZR6 proves both injective.

For a bounded subset $K \subset \mathcal{Q}$, its associated strong seminorm is

$$p_K(\lambda) = \sup_{y \in K} |\lambda(y)|.$$

Since every continuous seminorm is bounded on a bounded set, (GZR3.4) gives

$$p_K(\mathcal{D}_L x) \leq C_\zeta \left(\sup_{y \in K} \bar{b}_{2,1}(y) \right) \bar{b}_{2,1}(x). \quad (\text{GZR8.2})$$

The coefficient is finite. This proves continuity into the strong dual for every defining strong seminorm. The same calculation with the slots exchanged proves continuity of $\mathcal{D}_R$.

Both images are dense for the weak-* topology $\sigma(\mathcal{Q}', \mathcal{Q})$. Here is a direct proof retaining the full function space. Fix $\lambda \in \mathcal{Q}'$ and finitely many $y_1, \dots, y_n \in \mathcal{Q}$. Consider the linear map

$$A : \mathcal{Q} \rightarrow \mathbb{C}^n, \quad A(x) = (B_\zeta(x, y_1), \dots, B_\zeta(x, y_n)). \quad (\text{GZR8.3})$$

If a vector of coefficients $(c_1, \dots, c_n)$ annihilates its range, then

$$B_\zeta \left(x, \sum_{j=1}^n c_j y_j \right) = 0 \quad \text{for all } x.$$

Right nondegeneracy implies $\sum_j c_j y_j = 0$, and hence $\sum_j c_j \lambda(y_j) = 0$. In finite-dimensional linear algebra the range of A is the common kernel of all linear functionals annihilating it: extend a basis of its range to a basis of $\mathbb{C}^n$ to verify this statement. Thus

$$(\lambda(y_1), \dots, \lambda(y_n)) \in A(\mathcal{Q}). \quad (\text{GZR8.4})$$

One element $\mathcal{D}_L x$ therefore agrees exactly with λ on every prescribed finite list. This proves weak-* density. The proof for $\mathcal{D}_R$ uses left nondegeneracy.

The action on this dual is the exact character- a -twisted contragredient action

$$(T_a^\diamond \lambda)(y) = a \lambda(T_{a^{-1}} y). \quad (\text{GZR8.5})$$

A character with value a is not itself a proof that $\mathcal{Q}$ has Deligne weight one. When $a = q$ is a Frobenius norm, that one-dimensional character has Deligne weight two, since its absolute value is $q^{2/2}$. No purity conclusion is assigned to the receiving space by this notation. A continuous linear automorphism $T_{a^{-1}}$ sends bounded sets to bounded sets, so this is continuous on the strong dual. Formula (GZR7.2) gives

$$\mathcal{D}_L T_a = T_a^\diamond \mathcal{D}_L, \quad \mathcal{D}_R T_a = T_a^\diamond \mathcal{D}_R. \quad (\text{GZR8.6})$$

For the first equality, $B_\zeta(T_a x, y) = a B_\zeta(x, T_{a^{-1}} y)$; the second follows by exchanging the slots.

These are continuous injective maps with weak-* dense images. Neither surjectivity onto the continuous dual nor a topological self-duality has been proved or assumed.

145.10 GZR9. All receiving support labels and the exact scope of the result

Let L_{supp} be the existing support lattice with top element $1_{L_{\text{supp}}}$. For a receiving complex vector space V , retain

$$G_{L_{\text{supp}}}(V) = \{(0, \lambda) : \lambda \in L_{\text{supp}}\} \cup \{(v, 1_{L_{\text{supp}}}) : v \in V\}. \quad (\text{GZR9.1})$$

For every linear map $f : V \rightarrow W$, its actual support-preserving lift is

$$G_{L_{\text{supp}}}(f)(v, \lambda) = (f(v), \lambda). \quad (\text{GZR9.2})$$

It is well-defined because a nonzero amplitude occurs only at top support. Identity and composition are preserved by direct substitution. In particular the quotient, original Mellin comparison, full primary projectors, reflection, real action, and the two dual injections all have these lifts. A zero output retains its original λ . If f is injective, then equality of its lifted outputs first gives equality of labels and then equality of amplitudes, so the lift is injective.

The bilinear form itself admits a lift retaining both input labels:

$$\begin{aligned} \tilde{B}_\zeta : G_{L_{\text{supp}}}(\mathcal{Q}) \times G_{L_{\text{supp}}}(\mathcal{Q}) &\longrightarrow G_{L_{\text{supp}} \times L_{\text{supp}}}(\mathbb{C}), \\ \tilde{B}_\zeta((x, \lambda), (y, \mu)) &= (B_\zeta(x, y), (\lambda, \mu)). \end{aligned} \quad (\text{GZR9.3})$$

If either input has nontop support, its amplitude is zero, so the output amplitude is zero as required. If the output amplitude is nonzero, both inputs have top support, giving the top pair in the product lattice. This proves well-definedness without merging either input label. All scalar and action identities above preserve these pairs of labels. This is an explicit map on the stated receiving carrier; an ordinary tensor product that identifies all zero amplitudes is not substituted for it.

The global calculation has therefore produced a full-multiplicity continuous duality map for the actual $\mathcal{B}/\mathcal{I} = \mathcal{A}/\mathcal{E}S$, together with its exact real action, original denominator, reflected transpose, and strong-dual continuity. The finite residue forms are its proved restrictions, not a replacement for the whole space. The result gives no positive-definite form and no estimate forcing $\Re\rho = 1/2$. It supplies the nondegenerate global pairing on which an additional proved weight or positivity argument would have to act; it is not itself that argument.

145.11 Explicit global continuation: finite lifting and a nonzero remainder

The earlier uncertainty about algebraic surjectivity is now resolved by GLOBAL_POLYNOMIAL_DUAL_DEFECT.md PGD1–PGD6, independently checked in PGC0–PGC7: the actual algebraic cokernel is nonzero. Its explicit class `c_zeta` is the original-zeta contour functional with the constant arithmetic function 1 in the first slot. A lift would require a rapidly decreasing entire representative to equal 1 at all actual zeros, which the proved unbounded zero heights exclude. PGD constructs an injected rational-function line, entire representatives retaining all Hermite pole corrections, and the whole real/prime orbit. This is a receiving cokernel class, not primitive tau or a claimed off-critical zero.

SCL0–SCL9 and DPL0–DPL9 prove every nonzero polynomial in the actual cokernel generator invertible. Thus all finite-dimensional invariant dual spaces lift uniquely, continuously and with the original real/prime action. DPL gives explicit extension splittings and computes every degree of the cyclic derived Hom into the actual cone. These results retain its H and H' terms, all faithful closed copies and support labels. The weak-* Hausdorff quotient remains zero alongside the now proved nonzero algebraic quotient. Current source Z_0 , Z_1/τ and Z_2 are unchanged. Complete proofs and exact human citations are in the linked continuation files; the full geometric weight-separation goal remains active.

146 Original-zeta derivative, full residue duality and the Weil trace

24 September 2026. Complete derivation JTB0–JTB8. This calculation continues the actual supported quotient used by Connes–Consani, with the source comparison DCP retained. It constructs the exact map from the full-multiplicity contour pairing to the ordinary Weil form. It proves neither positivity nor numerical purity.

146.1 JTB0. Source, prerequisites and provenance

The complete preserved user arguments USR-9ad1c0a2d09dba92, USR-f55d16feb948d8c2 and USR-6152e3bc6302258c, and the current prerequisite instruction USR-4322be19bff532cd, were reread for this calculation. They remain in `user_corpus_private/USER_MESSAGES.jsonl`. The source data and retractions are B1–B5 and R1 in the retained `SOURCE_OPERATIONS_AND_PROOFS.md`. These references locate the complete arguments; they do not replace them.

The source remains Z_0 absence, Z_1 presence τ without Z_2 parity, and the distinct arithmetic layer with its original integer amounts. There is no source τ addition. Complex linear operations below occur on the already constructed receiving spaces, after the programme’s arithmetic reconstruction. A numerical character of a receiving prime action is not a weight assigned to primitive τ .

Alain Connes and Caterina Consani’s Schemes over F1 and zeta functions, arXiv:0903.2024v3, original author file `announc3.tex`, §5, provides the sheaf, summation restrictions, Fourier graph and supported spectral quotient. Its source labels are `restmaps`, `bord`, `hzero`, `compinv`, `lemreasspec` and `court`. The present reading covered source lines 1461–1668. Ralf Meyer’s A spectral interpretation for the zeros of the Riemann zeta function, `math/0412277v3`, original `Meyer.tex`, the:Zeta_estimate and the:Lap_range, provides the source closed-image and range results. The programme’s exact version, including both signs and the full original multiplier, is SSI1–SSI6 and OMS.

Pierre Deligne’s La conjecture de Weil II, §§3.6.1–3.6.3, printed pp.212–214, supplies the lifting argument being sought. The local text read here is the French transcription `S20_FR_record_export.tex`, lines 2468–2592, not established author TeX. Its perfect supported duality and proper weight bounds are not inferred from the existence of the pairing below.

The Hadamard product used in JTB4 is the retained complete product S2.6, with attribution there to Jacques Hadamard. The indexed original author source read again here is Alain Connes, The Riemann Hypothesis: Past, Present and a Letter Through Time, arXiv:2602.04022v1, `rhready.tex` lines 501–546, canonical source PUBUNIT-171DAC50D1C3B7A2438B39AE. The complete genus-one statement at lines 526–535 is used; its later symmetric-product presentation does not replace the original formulas below. The new calculation is the global derivative operator, its full endpoint return, and the exact factorization through the newly constructed contour pairing. Finite-dimensional residue/trace identities are established algebra; no historical novelty is claimed for them. Earlier programme local residue calculations remain credited in `BOUNDARY_DEFECTS_AND_ARITHMETIC_RETURN.tex`, especially its cotangent/residue calculation, and the pinned `TAU_BASE_MODEL` §6. Only that earlier receiving residue algebra is used: its older source-absorber notation is not imported as the current primitive Z_1/τ definition.

146.2 JTB1. Every receiving object and map used here

Let $\mathcal{B}$ be the entire functions with all seminorms

$$b_{A,M}(F) = \sup_{|\sigma| \leq A, t \in \mathbb{R}} (1 + |t|)^M |F(\sigma + it)| < \infty, \quad A, M \in \mathbb{Z}_{\geq 0}. \quad (\text{JTB1.1})$$

Let $\mathcal{Z}$ be the actual nontrivial zeros of the original ζ , with their original multiplicities m_ρ . Set

$$\mathcal{I} = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ for every } \rho \in \mathcal{Z}, 0 \leq j < m_\rho\}, \quad Q = \mathcal{B}/\mathcal{I}. \quad (\text{JTB1.2})$$

Continuous jet evaluation makes $\mathcal{I}$ closed. The topology on Q is this Fréchet quotient topology. SSI proves its exact identification with the original summation cokernel, not an unrestricted product of formal jets. DCP constructs its source-supported realization; all faithful extra closed coefficients remain in that comparison.

The continuous operators on $\mathcal{B}$ and their descended operators are

$$LF = sF, \quad T_a F(s) = a^s F(s) \quad (a > 0), \quad RF(s) = F(1-s), \quad CF(s) = \overline{F(\bar{s})}. \quad (\text{JTB1.3})$$

Here C is conjugate-linear. Multiplication by s and a^s preserves every vanishing order. The original functional equation and real Dirichlet coefficients, with unique continuation, prove that R and C preserve $\mathcal{I}$. The operators retain their displayed constants. In particular $T_a R = a R T_{a^{-1}}$.

Retain the complete source transform

$$F_*(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad F_*(0) = F_*(1) = \frac{1}{8}, \quad (\text{JTB1.4})$$

$$b_1 = 8F'_*(1) = 1 + \frac{\gamma}{2} - \frac{1}{2} \log(4\pi). \quad (\text{JTB1.5})$$

This transform is an auxiliary element of the actual summation image, not a replacement of ζ . Its values at the trivial zeros are retained in GER1.6. GER2 constructs the continuous inverse of $L-1$ on Q , with representative

$$\mathcal{S}_1 F(s) = \frac{F(s) - 8F_*(s)F(1)}{s-1}. \quad (\text{JTB1.6})$$

The numerator has a removable zero at 1. All expressions using this inverse below have this already constructed domain and endpoint return.

For a function H integrable on the two vertical edges, write

$$\mathfrak{C}(H) = \frac{1}{2\pi i} \left(\int_{2-i\infty}^{2+i\infty} H(s) ds - \int_{-1-i\infty}^{-1+i\infty} H(s) ds \right). \quad (\text{JTB1.7})$$

Both integrals run upward. Their difference is the positive rectangle orientation after horizontal sides have been justified. The full pairing constructed in GZR is

$$\mathcal{B}_\zeta([F], [G]) = \mathfrak{C} \left(\frac{F(s)G(1-s)}{\zeta(s)} \right). \quad (\text{JTB1.8})$$

GZR proves convergence, continuity, descent and nondegeneracy with every original zero jet retained. It does not assert a positive form or a topological isomorphism with the entire continuous dual. The proof below uses exactly that pairing.

146.3 JTB2. The full original derivative defines a continuous quotient operator

The function

$$A_\zeta(s) = (s-1)^2 \zeta'(s) \quad (\text{JTB2.1})$$

is entire. Its value at 1 is -1 , because the original pole has residue 1. It and its derivatives have polynomial growth on each bounded vertical strip. Here is an explicit bound prerequisite. For an integer $K \geq 1$, repeated integration by parts in Euler summation gives, on $\Re s > 1 - 2K$,

$$\zeta(s) = \frac{1}{s-1} + \frac{1}{2} + \sum_{k=1}^K \frac{B_{2k}}{(2k)!} (s)_{2k-1} - \frac{(s)_{2K}}{(2K)!} \int_1^\infty \tilde{B}_{2K}(x) x^{-s-2K} dx. \quad (\text{JTB2.2})$$

Here $(s)_r = s(s+1)\cdots(s+r-1)$, $(s)_0 = 1$, and $\tilde{B}_{2K}$ is the bounded periodic Bernoulli polynomial. To obtain this formula, apply the finite Euler summation formula to x^{-s} on $[1, N]$; its

r -th derivative is $(-1)^r(s)_r x^{-s-r}$. For $\Re s > 1$ all upper boundary terms tend to zero, giving the displayed sign and factors. The remainder integral is locally uniformly convergent on $\Re s > 1 - 2K$, which extends the equality by the identity theorem. Differentiation inserts $-\log x$ in the integral. On a fixed strip with $2K > 1 + A$, both integrals are uniformly bounded by constants times $\int_1^\infty (1 + \log x)x^{A-2K} dx < \infty$. The other factors are polynomials in s , apart from the displayed pole. Multiplication by $(s-1)^2$ removes that pole in ζ' . This proves the asserted strip bound. Higher derivatives follow by further logarithms with the same convergence margin.

It follows that multiplication $M_{A_\zeta} : \mathcal{B} \rightarrow \mathcal{B}$ is continuous: for each A, M , a polynomial bound of degree d_A gives

$$b_{A,M}(A_\zeta F) \leq C_A b_{A,M+d_A}(F). \quad (\text{JTB2.3})$$

Since A_ζ is entire, this multiplication preserves $\mathcal{I}$. Hence the operator

$$\boxed{\mathcal{J}_\zeta = (L-1)^{-2} M_{A_\zeta} : Q \longrightarrow Q} \quad (\text{JTB2.4})$$

is constructed and continuous. It commutes with L and every T_a . On each original zero germ it is exactly multiplication by $\zeta'(s)$: the factor $(s-1)^2$ is invertible there. This assertion is local identification of the already constructed global operator, not a replacement of it by arbitrary interpolation.

146.4 JTB3. The representative, both pole coefficients, and every endpoint term

Apply $\mathcal{S}_1$ twice to $A_\zeta G$. Since

$$(A_\zeta G)(1) = -G(1), \quad (A_\zeta G)'(1) = -G'(1), \quad (\text{JTB3.1})$$

the exact result is

$$\boxed{\mathcal{J}_\zeta G(s) = \zeta'(s)G(s) + \frac{8F_*(s)}{(s-1)^2} [G(1) + (s-1)(G'(1) - b_1 G(1))].} \quad (\text{JTB3.2})$$

Indeed the first division is $(A_\zeta G + 8F_* G(1))/(s-1)$, whose value at 1 is $-G'(1) + b_1 G(1)$. The second subtraction gives exactly (JTB3.2). The two applications of the continuous entire operator $\mathcal{S}_1$ prove that the displayed meromorphic expression is an element of $\mathcal{B}$. No undefined value of $\zeta'(1)$ is used.

For explicit checking, put $h = s-1$ and use the original Stieltjes convention $\zeta(1+h) = h^{-1} + \gamma - \gamma_1 h + \gamma_2 h^2/2 + \dots$. Then

$$\zeta'(1+h)G(1+h) = -G(1)h^{-2} - G'(1)h^{-1} - \frac{1}{2}G''(1) - \gamma_1 G(1) + O(h). \quad (\text{JTB3.3})$$

The correction in (JTB3.2) contributes $G(1)h^{-2} + G'(1)h^{-1}$, with both coefficients unchanged, and its constant term is $b_1 G'(1) + (4F''_*(1) - b_1^2)G(1)$. Consequently

$$\mathcal{J}_\zeta G(1) = -\frac{1}{2}G''(1) + b_1 G'(1) + (4F''_*(1) - b_1^2 - \gamma_1)G(1). \quad (\text{JTB3.4})$$

The derivative $F''_*(1)$ denotes the derivative of the full product (JTB1.4), with its removable value. It is not dropped or replaced by a scale choice.

At zero and at every original trivial zero the exact values are obtained without deleting the correction:

$$\mathcal{J}_\zeta G(z) = \zeta'(z)G(z) + \frac{8F_*(z)}{(z-1)^2} [G(1) + (z-1)(G'(1) - b_1 G(1))], \quad z = 0, -2, -4, \dots \quad (\text{JTB3.5})$$

In particular the term involving $F_*(-2r) \neq 0$ is retained. These are values of the chosen source representative; endpoint evaluation is not asserted to be a well-defined functional on Q .

146.5 JTB4. A global logarithmic-derivative contour identity

For every $\Phi \in \mathcal{B}$,

$$\mathfrak{C}\left(\Phi \frac{\zeta'}{\zeta}\right) = \sum_{\rho \in \mathcal{Z}} m_\rho \Phi(\rho) - \Phi(1). \quad (\text{JTB4.1})$$

All integrals and the sum in this identity converge absolutely. This is proved directly, without choosing an infinite sequence of contours close to unknown zeros.

Use the complete Hadamard product S2.6, with every exponential factor. Its logarithmic derivative is

$$\frac{F'_*(s)}{F_*(s)} = b_0 + \sum_{\rho} m_\rho \left(\frac{1}{s-\rho} + \frac{1}{\rho} \right), \quad b_0 = \frac{1}{2} \log(4\pi) - 1 - \frac{\gamma}{2}. \quad (\text{JTB4.2})$$

Compact convergence follows from $\sum m_\rho |\rho|^{-2} < \infty$, established by the count $n(R) \leq C(R+2)^{3/2}$ in S2.5. On either vertical edge $c = -1, 2$, every $|s - \rho| \geq 1$. With $R = |s| + 2$, the terms with $|\rho| \leq 2R$ satisfy

$$\sum_{|\rho| \leq 2R} m_\rho \left| \frac{1}{s-\rho} + \frac{1}{\rho} \right| \leq |s| \sum_{|\rho| \leq 2R} \frac{m_\rho}{|\rho|} \leq CR^{3/2}. \quad (\text{JTB4.3})$$

The last estimate follows by splitting into dyadic annuli and using the same count; a fixed disk around zero is zero-free since $F_*(0) \neq 0$. For the remaining terms, $|s - \rho| \geq |\rho|/2$, so their sum is bounded by $2|s| \sum_{|\rho| > 2R} m_\rho |\rho|^{-2} \leq CR^{1/2}$. Thus the sum of absolute values on each edge has polynomial growth. Rapid decrease of Φ justifies exchanging the edge integrals and the sum.

For one fixed ρ , the contour difference of $\Phi(s)(1/(s-\rho) + 1/\rho)$ equals $\Phi(\rho)$. This follows from a finite rectangle and the vanishing of its horizontal integrals as its height tends to infinity. The constant term has zero contour difference. The zero sum is absolutely convergent by rapid decrease and S2.5. Therefore

$$\mathfrak{C}(\Phi F'_*/F_*) = \sum_{\rho} m_\rho \Phi(\rho). \quad (\text{JTB4.4})$$

The original factor comparison, with all derivatives present, is

$$\frac{F'_*}{F_*} = \frac{\zeta'}{\zeta} + \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2} \log \pi + \frac{1}{2} \psi(s/2). \quad (\text{JTB4.5})$$

Here $\psi = \Gamma'/\Gamma$. At 0 the poles of $1/s$ and $\psi(s/2)/2$ cancel exactly. The remaining multiplier derivative has in the strip $-1 \leq \Re s \leq 2$ only the simple pole at 1, of residue 1. It has polynomial growth on the two edges and horizontal edges at large height. One can check the growth directly from $\psi(z) = -\gamma + \sum_{n \geq 0} (1/(n+1) - 1/(n+z))$: split at $n = 2|z| + 2$, and use the recurrence to shift the bounded real strip away from its poles. The tail is $O(1)$, while the initial terms have a polynomial bound. Rapid decrease of Φ therefore permits the contour shift, giving exactly $\Phi(1)$. Subtracting this contribution from (JTB4.4) proves (JTB4.1). The term $-\Phi(1)$ is the original zeta pole and is essential.

146.6 JTB5. Exact global factorization of the Weil form

The previously established reflected Weil form is

$$W([F], [G]) = \sum_{\rho} m_\rho \overline{F(1-\bar{\rho})} G(\rho). \quad (\text{JTB5.1})$$

It is conjugate-linear in F , linear in G , continuous, and retains the actual zero multiplicities. The following exact identity holds on the entire original quotient:

$$\boxed{W(x, y) = \mathcal{B}_\zeta(\mathcal{J}_\zeta y, Cx).} \quad (\text{JTB5.2})$$

Proof. Divide (JTB3.2) by the original ζ , keeping the complete multiplier (JTB1.4):

$$\frac{\mathcal{J}_\zeta G(s)}{\zeta(s)} = \frac{\zeta'(s)}{\zeta(s)} G(s) + s\pi^{-s/2}\Gamma(s/2) \left[\frac{G(1)}{s-1} + G'(1) - b_1 G(1) \right]. \quad (\text{JTB5.3})$$

Multiply by $(CF)(1-s) = \overline{F(1-\bar{s})}$. For the first summand, (JTB4.1), applied to $\Phi(s) = G(s)(CF)(1-s)$, gives

$$W([F], [G]) - G(1)\overline{F(0)}. \quad (\text{JTB5.4})$$

For the second summand, $s\Gamma(s/2) = 2\Gamma(1+s/2)$ is holomorphic on the whole closed strip. Its only remaining pole there is $1/(s-1)$. At the coefficient $s\pi^{-s/2}\Gamma(s/2)$ equals 1. Thus the residue is exactly

$$G(1)\overline{F(0)}. \quad (\text{JTB5.5})$$

The Gamma recurrence and its integral, or its full strip estimate used in SSI3, give polynomial growth away from bounded heights. The other factors decrease rapidly, so both horizontal integrals vanish. This proves (JTB5.5) as a contour equality. Equations (JTB5.4) and (JTB5.5) cancel these two explicitly retained contributions and prove (JTB5.2). No endpoint functional is taken on an unspecified quotient class: both terms belong to the chosen representatives and disappear only in their proved sum. ■

The apparent local loss in the ordinary trace therefore has a complete global operator, $\mathcal{J}_\zeta$, and a continuous source representative. It is not described merely by forgetting derivatives.

146.7 JTB6. All multiplicity jets and the exact kernel

At an actual zero ρ of multiplicity m , set $t = s - \rho$ only in its receiving germ and write

$$\zeta(\rho + t) = t^m u_\rho(t), \quad u_\rho(t) = \sum_{r \geq 0} \frac{\zeta^{(m+r)}(\rho)}{(m+r)!} t^r, \quad u_\rho(0) \neq 0. \quad (\text{JTB6.1})$$

The full primary quotient is $\mathbb{C}[t]/(t^m)$, identified by RZ's actual continuous projector and source representatives, not by changing the full quotient topology. The derivative operator there is

$$\mathcal{J}_\zeta x(t) = (mt^{m-1}u_\rho(t) + t^m u'_\rho(t))x(t) = mu_\rho(0)x(0)t^{m-1} \pmod{t^m}. \quad (\text{JTB6.2})$$

The first equality retains the whole original derivative. The second is proved by the displayed ideal (t^m) ; no term is discarded outside that stated quotient.

Every block has rank 1 under $\mathcal{J}_\zeta$. For $m = 1$, this is the nonzero scalar $\zeta'(\rho)$. For $m > 1$, its kernel is precisely the ideal (t) , and its image is the one-dimensional socle $\mathbb{C}t^{m-1}$. Its square is zero on these multiple-zero blocks because $2m - 2 \geq m$. These statements do not assert the existence of any actual multiple zero.

Since the full collection of original jet maps separates Q , equation (JTB6.2) proves the global equality

$$\boxed{\ker \mathcal{J}_\zeta = \{[F] \in Q : F(\rho) = 0 \text{ for every } \rho\} = \text{rad } W.} \quad (\text{JTB6.3})$$

The final equality also follows from (JTB5.2), nondegeneracy of $\mathcal{B}_\zeta$, and bijectivity of C . The exact range is the actual subspace $\mathcal{J}_\zeta Q$ of Q . It is not asserted to be an unrestricted product of block socles or a closed subspace.

For a finite block, the residue identity behind the factorization is

$$\text{Res}_{s=\rho} \frac{\zeta'(s)F(s)G(1-s)}{\zeta(s)} ds = mF(\rho)G(1-\rho). \quad (\text{JTB6.4})$$

Indeed $\zeta'/\zeta = m/t + u'_\rho/u_\rho$; the second summand is holomorphic. This proves both the factor m and the precise higher-jet kernel without a simplicity assumption.

146.8 JTB7. Exact prime action and the constructed dual map

Define the continuous linear-dual map associated with the full contour pairing by

$$\mathcal{D}_\zeta : Q \longrightarrow Q', \quad (\mathcal{D}_\zeta y)(z) = \mathcal{B}_\zeta(y, z). \quad (\text{JTB7.1})$$

The prime action on Q' is the contragredient $(T_a^\vee \ell)(z) = \ell(T_{a^{-1}} z)$. GZR proves that $\mathcal{D}_\zeta$ is injective, continuous for the strong dual topology, and has weak-* dense image. No surjectivity onto Q' is assumed. The exact character is

$$\mathcal{D}_\zeta T_a = a T_a^\vee \mathcal{D}_\zeta, \quad \mathcal{D}_\zeta L = (1 - L^t) \mathcal{D}_\zeta, \quad (\text{JTB7.2})$$

where $L^t \ell(z) = \ell(Lz)$. The first follows by moving T_a to the other argument in $\mathcal{B}_\zeta(T_a y, T_a z) = a \mathcal{B}_\zeta(y, z)$. The second follows in the original integral from $s + (1 - s) = 1$.

The conjugate-linear first argument of the Weil form is accounted for by the explicitly constructed C :

$$W(x, y) = (\mathcal{D}_\zeta \mathcal{J}_\zeta y)(Cx). \quad (\text{JTB7.3})$$

Thus the full diagram retains both the nondegenerate residue map and the generally degenerate trace map. All operations occur in the receiving complex spaces; neither dualization nor the scalar a assigns parity or a numerical coordinate to τ .

Under the exact source-space comparison $\mathcal{T}k(u) = 2u^{-1/2}k(u)$, the raw Mellin transform satisfies $\mathcal{M}_0 \mathcal{T}k = 2\mathcal{M}k$. Consequently the same displayed contour expression on two raw images is exactly four times the centered expression. This factor follows from bilinearity and is retained in the comparison to the original CC coefficient complex.

146.9 JTB8. Consequence for the active lifting calculation

Figure 1. The upper arrow is the constructed nondegenerate dual injection; the derivative operator gives the exact Weil trace map. The lower row displays its full primary-block rank and the proved endpoint cancellation. The integrals run upward on both displayed vertical lines before their difference is taken. Complete proofs: GZR2–GZR8 and JTB2–JTB7. The two different roles of the pairing and trace are connected by (JTB5.2), not by identifying their kernels. Reproducible source: `check_and_draw_residue_bridge.py`. Human source geometry: Connes–Consani §5; geometric weight target: Deligne §§3.6.1–3.6.3.

The receiving obstruction space has an actual dual map compatible with all positive real actions and therefore every retained prime action. Its full residue pairing detects all multiplicity jets. The map to the ordinary Weil form is exactly the original derivative operator, with the pole at 1 and all Gamma correction factors handled by (JTB3.2)–(JTB5.5). This supplies an exact comparison that the unproved identification of a raw integral pairing with the quotient did not supply.

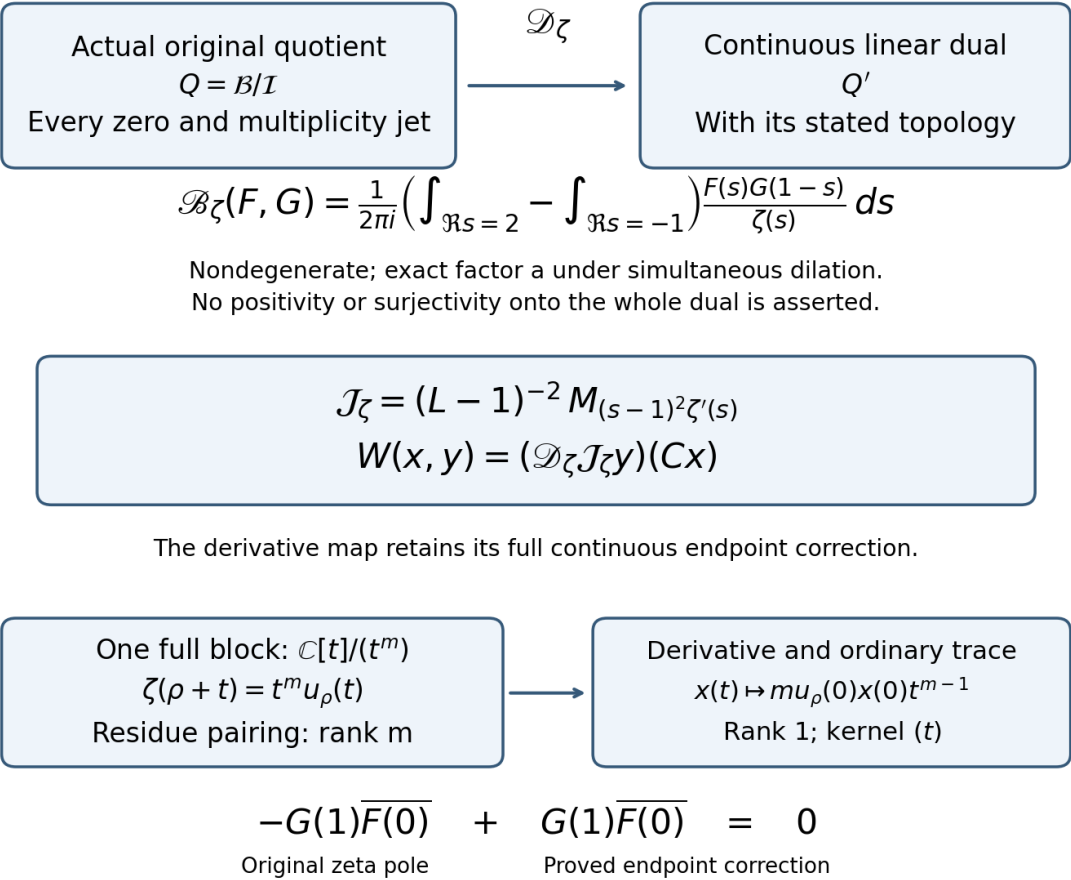
The character a in (JTB7.2) alone is not Deligne’s strict weight separation. On an actual ρ -block, the action remains

$$T_a = a^\rho \sum_{j=0}^{m_\rho-1} \frac{(\log a)^j}{j!} N_\rho^j. \quad (\text{JTB8.1})$$

Its paired block is $1 - \rho$; the twisted contragredient has the same character as its input under (JTB7.2). This is the computed exact map, not a hypothesis about a hypothetical altered prime spectrum. The full source reconstruction is unchanged.

Deligne’s §3.6 instead uses both a geometric identification of the support term with $H^{2N-i-1}(X_s)^\vee(-N)$ and proper geometric bounds to place it in weights at least $i + 1$, while the classes to lift have weights at most i . Neither inequality follows from a bilinear similitude alone. The present calculation therefore establishes the duality/trace comparison on the actual receiver and keeps the full tau weight-separation goal active. It makes no claim that the primitive source forbids a further geometric comparison, or that the user’s whole-spectrum argument has been contradicted.

Full residue duality and the original Weil trace



Proofs GZR and JTB2-JTB7. Source geometry: Connes-Consani, arXiv:0903.2024v3, §5.
Deligne, Weil II §§3.6.1-3.6.3 remains the weight-separation target. Source Z1 / τ has no assigned weight.

Figure 52: Exact global residue and trace maps, with the original pole correction retained

146.10 Full global prime action and actual supported placement

The new complete proofs GGT0–GGT8, FPO0–FPO8 and UOS0–UOS9 (including UOS8A) strengthen the preceding global remainder calculation. The original-zeta Gaussian trace has a $T \log T$ leading term only at the identity scale and at most order T at every other fixed positive scale. Testing a proposed finite relation against every inverse scale proves all coefficients zero. Consequently the entire finite group algebra $C[\mathbb{R}_+^{\wedge \times}]$, including the Laurent algebra of all original primes, injects into the original multiplier quotient and linearly into the actual comparison cokernel. This computes relations after the actual zero-jet quotient, not only among entire functions before quotienting.

UOS constructs the full chain map into the actual dual-supported comparison cone. It sends c_r to $(c_r, 0)$, acts identically on H , and restricts $\chi H'$ to $\chi E'$ with every endpoint and extra closed copy retained. Its companion mirror is $c_r \rightarrow -r c_r(1/r)$ with the companion denominator; the oriented global-dual mirror has the opposite sign. The ordinary Fourier restriction still has its explicit section. Thus the new classes are relative residue-representation classes with proved supported placement, not a claimed failure of that existing Fourier lift. The source Z_0 , Z_1/τ and integer Z_2 have not changed, and the full numerical weight-separation target remains active.

The original Connes–Consani explicit formula has been applied with a proved cutoff extension; the exact archimedean comparison is $W_R(f_T, x) = A_T(x) + 1$, retaining the endpoint at zero and every finite trivial-zero contour correction. Complete human citations, author-source archive and bounded reading coverage are recorded with these proofs. All complete proof bodies and inspected reproducible diagrams are in the cumulative TeX.

147 Independent verification of the original-zeta Jacobian bridge

24 September 2026. Check locators JTC1–JTC6. This checks the complete JTB2–JTB7 argument in `ORIGINAL_ZETA_JACOBIAN_TRACE_BRIDGE.md` against the independently constructed GZR pairing. It is a mathematical check of the stated receiving maps, not a new assertion about primitive Z_1/τ .

147.1 JTC1. Entire multiplier and Euler-summation sign

Write $h = s - 1$ with the original Stieltjes convention

$$\zeta(1+h) = h^{-1} + \gamma - \gamma_1 h + \gamma_2 h^2/2 + \cdots.$$

Then

$$A_\zeta(1+h) = h^2 \zeta'(1+h) = -1 - \gamma_1 h^2 + \gamma_2 h^3 + \cdots.$$

In particular $A_\zeta(1) = -1$ and $A'_\zeta(1) = 0$, proving both endpoint coefficients in JTB3.1.

The remainder sign in JTB2.2 is negative. For a direct independent check, Euler summation first gives

$$\zeta(s) = \frac{1}{s-1} + \frac{1}{2} - s \int_1^\infty \tilde{B}_1(x) x^{-s-1} dx \quad (\Re s > 1).$$

Integrate using $\tilde{B}'_2(x) = 2\tilde{B}_1(x)$ on each integer interval. The internal endpoint contributions cancel, and $\tilde{B}_2(1) = B_2 = 1/6$. Thus the first integration step gives

$$\zeta(s) = \frac{1}{s-1} + \frac{1}{2} + \frac{s}{12} - \frac{s(s+1)}{2} \int_1^\infty \tilde{B}_2(x) x^{-s-2} dx.$$

Repeating in pairs gives exactly JTB2.2, with the rising factorial, $B_{2k}/(2k)!$, and the displayed negative remainder. Its integral and every differentiated integral converge uniformly on any fixed strip contained in $\Re s > 1 - 2K$. This proves the required polynomial growth of the entire A_ζ . Multiplication by it preserves both $\mathcal{B}$ and every full original zero order.

147.2 JTC2. Two divisions at the original pole

Put $g_0 = G(1)$, $g_1 = G'(1)$, and $b_1 = 8F'_*(1)$, with $8F_*(1) = 1$. First division of $A_\zeta G$ gives

$$\frac{A_\zeta(s)G(s) + 8F_*(s)g_0}{s-1}.$$

Its value at one is $-g_1 + b_1 g_0$. The second division therefore gives exactly

$$\zeta'(s)G(s) + \frac{8F_*(s)}{(s-1)^2} [g_0 + (s-1)(g_1 - b_1 g_0)].$$

No endpoint sign is reversed. Expansion $8F_*(1+h) = 1 + b_1 h + 4F''_*(1)h^2 + \cdots$ yields the constant

$$-\frac{1}{2}G''(1) + b_1 G'(1) + (4F''_*(1) - b_1^2 - \gamma_1)G(1),$$

as in JTB3.4. The two principal-part terms are cancelled only by the two explicitly displayed correction coefficients. The original values at zero and the trivial zeros remain JTB3.5.

147.3 JTC3. Absolute interchange of the entire logarithmic derivative

The genus-one pairing of terms is essential:

$$\frac{1}{s-\rho} + \frac{1}{\rho} = \frac{s}{\rho(s-\rho)}.$$

On $\Re s = -1, 2$, the actual zeros satisfy $|s - \rho| \geq 1$. The retained estimate $n(R) \leq C(R+2)^{3/2}$, with multiplicities, gives

$$\sum_{|\rho| \leq 2R} \frac{m_\rho}{|\rho|} = O(R^{1/2}), \quad \sum_{|\rho| > 2R} \frac{m_\rho}{|\rho|^2} = O(R^{-1/2}).$$

For $R = |s| + 2$, the total sum of absolute values of the paired genus-one terms is therefore $O(R^{3/2})$ on both edges. Multiplying by an arbitrary $\Phi \in \mathcal{B}$ gives an integrable majorant, so exchanging the original vertical integrals with the zero sum is valid. Each individual paired term has contour difference $\Phi(\rho)$. The absolute convergence of $\sum_\rho m_\rho |\Phi(\rho)|$ follows from the same zero count and rapid strip decrease.

The original multiplier derivative

$$\frac{1}{s} + \frac{1}{s-1} - \frac{1}{2} \log \pi + \frac{1}{2} \psi(s/2)$$

has cancelling residues at zero and residue +1 at one. It has no further poles inside this strip. Consequently subtraction from the full F'_*/F_* identity gives precisely

$$\mathfrak{C}(\Phi \zeta'/\zeta) = \sum_\rho m_\rho \Phi(\rho) - \Phi(1).$$

Thus the original pole sign in JTB4.1 is correct; no unproved choice of heights or sum of $1/\zeta'(\rho)$ is needed.

147.4 JTC4. The pole correction in the global Weil factorization

Division of the actual entire representative by the original ζ gives

$$\frac{\mathcal{J}_\zeta G(s)}{\zeta(s)} = \frac{\zeta'(s)}{\zeta(s)} G(s) + s\pi^{-s/2} \Gamma(s/2) \left[\frac{G(1)}{s-1} + G'(1) - b_1 G(1) \right].$$

Multiplication by $(CF)(1-s) = \overline{F(1-\bar{s})}$ gives from the first term

$$W([F], [G]) - G(1) \overline{F(0)}.$$

In the second term $s\Gamma(s/2) = 2\Gamma(1+s/2)$ removes the zero-endpoint pole, while the full coefficient at $s = 1$ is 1. Its residue is therefore

$$+G(1) \overline{F(0)}.$$

The two contributions cancel with the stated signs. Polynomial strip bounds for the retained Gamma expression and rapid decrease of the other factors justify the separate contour calculation. Hence

$$W(x, y) = B_\zeta(\mathcal{J}_\zeta y, Cx)$$

holds for the whole quotient, not only finite primary subspaces.

147.5 JTC5. Exact primary rank, kernel, and covariance

With the original germ $\zeta(\rho + t) = t^m u_\rho(t)$,

$$\zeta'(\rho + t) = mt^{m-1}u_\rho(t) + t^m u'_\rho(t).$$

Its multiplication on $\mathbb{C}[t]/(t^m)$ is

$$P(t) \mapsto mu_\rho(0)P(0)t^{m-1}.$$

Its rank is exactly one. For $m > 1$ its kernel is (t) and its square is zero; for $m = 1$ it is the nonzero scalar $\zeta'(\rho)$. Full original zero jets separate $\mathcal{Q}$, so the stated global kernel is exactly the subspace whose value coordinate is zero at every actual zero. The nondegeneracy of GZR and the surjectivity of conjugation C identify that kernel with the second-slot radical of the Weil form. The existing Hermitian symmetry of the Weil form identifies both radicals. No assertion that the range is a closed or unrestricted product of socles follows.

The operator $\mathcal{J}_\zeta = (L - 1)^{-2}M_{(s-1)^2\zeta'}$ commutes with L and T_a : the multiplier does so, and the inverse of the commuting invertible operator $L - 1$ does so. GZR7 gives the dual identities

$$\mathcal{D}_\zeta T_a = aT_a^\vee \mathcal{D}_\zeta, \quad \mathcal{D}_\zeta L = (1 - L^t)\mathcal{D}_\zeta.$$

The conjugation and the factor a are therefore placed correctly in JTB7.2–JTB7.3. The raw comparison factor is exactly four, since each raw Mellin image contributes the retained factor two and the contour pairing is bilinear.

147.6 JTC6. Result and one typographical correction

The mathematical identities and arguments in JTB2–JTB7 pass these independent checks. In JTB6.4 the displayed residue ends with a comma followed by ds ; it should have the differential ds . This is a typesetting correction, not an altered residue or sign. The proof does not establish positivity, purity, or a topological isomorphism from $\mathcal{Q}$ onto its continuous dual.

148 Division in the spectral dual cokernel and exact finite-dimensional lifting

24 September 2026. Independent derivation SCL0–SCL9.

This proves that every $L^t - a$ acts bijectively on the actual algebraic cokernel $\mathcal{C} = Q'/\mathcal{D}_L Q$. It constructs division on the original Fréchet quotient before taking transposes, then gives the explicit unique lift of every finite-dimensional L^t -invariant subspace of Q' . The full real-group covariance of those finite lifts is proved from the original jet action. No assertion that the whole cokernel vanishes, no geometric purity, and no RH conclusion is inserted.

148.1 SCL0. Source definitions, prior results and operation prerequisites

The controlling source remains B1–B5 and their proved consequences in `SOURCE_OPERATION_S_AND_PROOFS.md`, with the verbatim U01–U18 in `USER_DEFINITIONS_VERBATIM.md`. The complete global user arguments `USR-9ad1c0a2d09dba92`, `USR-f55d16feb948d8c2` and `USR-6152e3bc6302258c`, and the operation instruction `USR-4322be19bff532cd`, remain controlling as recorded in `CORPUS_AND_OPERATION_RULES.md` (private source record) and ASD0. They are not replaced by an assumption about a finite prime system.

The current calculation uses already reconstructed integer arithmetic and the original ζ . Z_0 retains absence, Z_1/τ retains primitive presence without Z_2 parity, and the original integer layer retains its supplied amount and parity data. There is no source addition on τ . A continuous complex dual, a polynomial in L , and a receiving scalar below belong to the constructed coefficient category, not to an unstated operation on primitive τ .

The actual coefficient complex, its faithful source-ring action, and its dual cone are `ACTUAL_SUPPORTED_DUALITY_INDEPENDENT.md`, ASD1–ASD14. The new full residue pairing is `GLOBAL_ORIGINAL_ZETA_RESIDUE_DUALITY_INDEPENDENT.md`, GZR2–GZR8. The complete jet isolators and projectors are the retained `ORIGINAL_ZETA_RESOLVENT_AND_PROJECTORS.md`, RZ5–RZ6, read completely for this derivation; GZR4 and GZR7–GZR8 were also read. The full Schwartz image theorem and original Mellin topology are `ORIGINAL_MELLIN_SPECTRAL_SYNTHESIS.md`, OMS1–OMS5.

Human provenance remains Alain Connes and Caterina Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, arXiv:0903.2024v3, §5, for the coefficient sheaf and Fourier localization; Ralf Meyer, *A spectral interpretation for the zeros of the Riemann zeta function*, arXiv:math/0412277v3, for the analytic closed-image framework, with the programme's exact original-space return proved in the cited sources; and Pierre Deligne, *La conjecture de Weil. II*, §3.6, for the lifting-obstruction argument motivating this continuation. The new calculation does not attribute a numerical purity theorem to these receiving maps.

148.2 SCL1. The exact spaces, auxiliary source transform and original jet representatives

Let

$$\mathcal{B} = \{F \text{ entire} : b_{A,N}(F) = \sup_{|\sigma| \leq A, t \in \mathbb{R}} (1 + |t|)^N |F(\sigma + it)| < \infty \quad (A, N \geq 0)\},$$

$$I_\zeta = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ for every actual nontrivial zero } \rho, 0 \leq j < m_\rho\}, \quad Q = \mathcal{B}/I_\zeta. \quad (\text{SCL1.1})$$

Use the original Fréchet topology and quotient topology. The ideal is closed by Cauchy's derivative estimates. The original raw test-space map is

$$\mathcal{M}_0 b(s) = \int_0^\infty b(u) u^s \frac{du}{u}, \quad \mathcal{M}_0 \Sigma h(s) = 2\zeta(s) \int_0^\infty h(v) v^s \frac{dv}{v}. \quad (\text{SCL1.2})$$

The cited complete image theorem identifies Q with the actual $A/\Sigma S$, not with an unrestricted formal jet product. The earlier centered receiver retains the exact comparison

$$\mathcal{T}k(u) = 2u^{-1/2}k(u), \quad \mathcal{T}^{-1}b(u) = \frac{1}{2}u^{1/2}b(u), \quad \mathcal{M}_0\mathcal{T}k = 2F_k. \quad (\text{SCL1.3})$$

No factor of this comparison is changed in the calculation below.

Write $L[F] = [sF]$, a continuous operator. Let Q' be its continuous complex-linear dual, and

$$(L^t\lambda)(q) = \lambda(Lq).$$

All transposes preserve the actual continuous dual. For exact dual topology statements use $\sigma(Q', Q)$; the displayed transposes are also strong-dual continuous by ASD2.2.

The auxiliary source function, used with its full multiplier, is

$$F_*(s) = \frac{s(s-1)}{8}\pi^{-s/2}\Gamma(s/2)\zeta(s) \in \mathcal{B}. \quad (\text{SCL1.4})$$

It is the transform of the explicit original source $f_*(v) = \frac{\pi}{2}v^2(2\pi v^2 - 3)e^{-\pi v^2}$. Its zero divisor is exactly the actual nontrivial-zero divisor with every multiplicity. Its retained exceptional values include

$$F_*(0) = F_*(1) = \frac{1}{8},$$

$$F_*(-2r) = \frac{(1+2r)2r}{8}\pi^{-(1+2r)/2}\Gamma((1+2r)/2)\zeta(1+2r) \neq 0 \quad (r \geq 1). \quad (\text{SCL1.5})$$

These are the original multiplier's removable values, proved in OMS2. They are not discarded by calling F_* an auxiliary function.

For an actual zero a of order m , put

$$A_a(s) = \frac{F_*(s)}{(s-a)^m}.$$

It is entire and in $\mathcal{B}$, and $A_a(a) \neq 0$. Recursively compute the first m Taylor coefficients of $1/A_a$ at a , obtaining a polynomial B_a such that

$$e_a(s) = A_a(s)B_a(s) = 1 + O((s-a)^m).$$

Explicitly, if $A_a(a+t) = \sum_{r \geq 0} a_r t^r$, set $b_0 = a_0^{-1}$ and $b_r = -a_0^{-1} \sum_{h=1}^r a_h b_{r-h}$, and take $B_a(a+t) = \sum_{r < m} b_r t^r$. Then

$$E_{a,j}(s) = e_a(s)(s-a)^j, \quad e_{a,j} = [E_{a,j}] \in Q, \quad \ell_{a,j}([F]) = \frac{F^{(j)}(a)}{j!}. \quad (\text{SCL1.6})$$

For $0 \leq j < m$ the function $E_{a,j}$ has Taylor coefficient 1 in degree j at a , its other coefficients below m are zero, and every required jet at every other zero is zero. This follows from the reciprocal recursion and the full zero divisor of F_* . In particular

$$\ell_{a,i}(e_{a,j}) = \delta_{ij}, \quad (L-a)e_{a,j} = e_{a,j+1} \quad (j < m-1), \quad (L-a)e_{a,m-1} = 0. \quad (\text{SCL1.7})$$

The continuous projector

$$P_a q = \sum_{j < m} \ell_{a,j}(q) e_{a,j} \quad (\text{SCL1.8})$$

has finite-dimensional image Q_a , gives $Q = Q_a \oplus \ker P_a$ topologically, and retains every other primary observation. Equality of all original zero jets implies equality in Q by SCL1.1; no infinite sum of projectors is used.

148.3 SCL2. Construct division on the original source before taking a transpose

For $a \notin \mathcal{Z}$, the actual set of nontrivial zeros, $F_*(a) \neq 0$. Define the entire source operator

$$\mathcal{S}_a F(s) = \frac{F(s) - F(a)F_*(s)/F_*(a)}{s - a}. \quad (\text{SCL2.1})$$

The numerator vanishes at a . This is continuous on $\mathcal{B}$. To verify the division estimate, outside a fixed disk about a , division by $s - a$ is bounded on every bounded vertical strip at bounded height and is $O((1 + |\Im s|)^{-1})$ at large height. Inside that disk the quotient is holomorphic and bounded by a Cauchy estimate on a larger fixed disk. Evaluation $F(a)$ and all derivatives on these fixed disks are controlled by a finite strip seminorm. These estimates prove every output seminorm is controlled by finitely many input seminorms.

If $F \in I_\zeta$, both numerator terms have all required zero jets. Since a is not one of those zeros, division preserves them. Therefore $\mathcal{S}_a$ descends to a continuous $S_a : Q \rightarrow Q$, and direct multiplication gives

$$(L - a)S_a = S_a(L - a) = \text{id}_Q. \quad (\text{SCL2.2})$$

The term involving F_* vanishes in the quotient; it remains present in the entire source formula SCL2.1, including at $a = 0, 1, -2, -4, \dots$

For an actual zero a of order m , the naive division by $s - a$ needs its full source correction. Set

$$H_F(s) = F(s) - F(a)E_{a,0}(s)$$

and define

$$\boxed{\mathcal{S}_a F(s) = \frac{H_F(s) - F_*(s)H_F^{(m)}(a)/F_*^{(m)}(a)}{s - a}.} \quad (\text{SCL2.3})$$

The denominator $F_*^{(m)}(a)$ is nonzero by the actual multiplicity. The numerator vanishes at a , so the same division estimate proves an entire member of $\mathcal{B}$, continuously depending on F . Every evaluation and derivative in SCL2.3 is a continuous finite-order evaluation on the original space.

If $F \in I_\zeta$, then $F(a) = 0$, and subtraction of the displayed multiple of F_* cancels its Taylor coefficient of order m at a . The resulting numerator vanishes to order at least $m + 1$ there, and to each required order at every other zero. Division by $s - a$ consequently leaves a member of I_ζ . Thus SCL2.3 descends to a well-defined continuous $S_a : Q \rightarrow Q$.

This proves why the correction is part of the actual map. Without it, dividing an element of I_ζ of exact order m at a produces a nonzero $(m - 1)$ -jet in the quotient.

The two exact inverse identities are

$$\boxed{(L - a)S_a = 1 - e_{a,0}\ell_{a,0}, \quad S_a(L - a) = 1 - e_{a,m-1}\ell_{a,m-1}.} \quad (\text{SCL2.4})$$

For the first, multiply SCL2.3 by $s - a$; the F_* term lies in I_ζ , leaving $[F] - F(a)e_{a,0}$. For the second, apply SCL2.3 to $(s - a)F$. Its value at a is zero and its m -th derivative is $mF^{(m-1)}(a)$. Moreover

$$\left[\frac{F_*(s)}{s - a} \right] = \frac{F_*^{(m)}(a)}{m!} e_{a,m-1},$$

because these are exactly its full jets at all actual zeros. The correction therefore subtracts $F^{(m-1)}(a)e_{a,m-1}/(m - 1)!$, proving the second identity with all factorials.

Also $S_a e_{a,0} = 0$, by inserting the specific source representative $E_{a,0}$ in SCL2.3. Together with SCL2.4 this gives

$$S_a e_{a,j} = e_{a,j-1} \quad (1 \leq j < m). \quad (\text{SCL2.5})$$

On $\ker P_a$, the same source formula has zero a -jet after the correction; hence S_a preserves that subspace. There S_a is the inverse of $L - a$.

It follows constructively that

$$\ker(L - a) = \mathbb{C}e_{a,m-1}, \quad (L - a)Q = \ker \ell_{a,0} \quad (a \in \mathcal{Z}). \quad (\text{SCL2.6})$$

The range inclusion follows by evaluation at a , the reverse inclusion by the first identity in SCL2.4. The kernel inclusion follows by the second identity, and SCL1.7 proves equality. The range is closed and of codimension 1; its continuous right inverse is S_a restricted to that range. Thus $L - a$ is strict onto its image, rather than only an abstract surjection.

More generally, for $k \geq 1$ and $r = \min(k, m)$,

$$\begin{aligned} \ker(L - a)^k &= \text{span}\{e_{a,m-r}, \dots, e_{a,m-1}\}, \\ (L - a)^k Q &= \bigcap_{j=0}^{r-1} \ker \ell_{a,j}. \end{aligned} \quad (\text{SCL2.7})$$

On Q_a these are the full truncated multiplication and backward-shift calculation from SCL1.7 and SCL2.5. On $\ker P_a$ both operators are inverses, as just proved. The direct sum in SCL1.8 proves SCL2.7 on all of Q , with continuous right inverse S_a^k on the displayed closed range. No global spectral expansion is required.

148.4 SCL3. Exact continuous transpose kernels and ranges

Let $A_a = L - a$. For $a \notin \mathcal{Z}$, transposition of SCL2.2 gives an everywhere-defined continuous inverse S'_a of $L^t - a$ on Q' . Its weak-* and strong continuity follow by transposing a continuous map on Q .

For $a \in \mathcal{Z}$, put $k_a = e_{a,m-1}$. The exact result is

$$\boxed{\ker(L^t - a) = \mathbb{C}\ell_{a,0}, \quad (L^t - a)Q' = \{\lambda \in Q' : \lambda(k_a) = 0\}.} \quad (\text{SCL3.1})$$

For the kernel, a functional in $\ker(L^t - a)$ annihilates the range in SCL2.6, so it is a scalar multiple of the continuous coordinate $\ell_{a,0}$. Conversely that coordinate annihilates the range.

The range inclusion follows by evaluating $(L^t - a)\mu$ at $k_a \in \ker(L - a)$. For the reverse inclusion, take a continuous λ with $\lambda(k_a) = 0$. The explicit continuous functional $\mu = S'_a \lambda$ satisfies

$$(L^t - a)S'_a \lambda = (S_a(L - a))' \lambda = \lambda - \lambda(k_a)\ell_{a,m-1} = \lambda. \quad (\text{SCL3.2})$$

This constructs the functional in the correct continuous dual, without replacing it by an algebraic-dual extension.

The range in SCL3.1 is closed in the weak-* topology and in the strong topology, since evaluation at the fixed vector k_a is continuous in both. The right inverse S'_a is continuous in both topologies. The transpose is therefore strict onto its image. The construction proves the claimed prerequisites before using its kernel or cokernel.

The same argument with SCL2.7 gives

$$\ker(L^t - a)^k = \text{span}\{\ell_{a,0}, \dots, \ell_{a,r-1}\}, \quad r = \min(k, m). \quad (\text{SCL3.3})$$

Indeed it is the annihilator of the displayed closed finite-codimensional range of $(L - a)^k$. A functional annihilating that range factors through its finite tuple of jet coordinates, and is exactly a linear combination of those coordinates. The transpose range is the annihilator of the displayed primal kernel, with continuous right inverse $(S_a^k)'$ there. These formulas retain the complete nilpotent multiplicity.

148.5 SCL4. The full residue pairing supplies the exact two missing coefficients

Let

$$B_\zeta([F], [G]) = \frac{1}{2\pi i} \left(\int_{2-i\infty}^{2+i\infty} - \int_{-1-i\infty}^{-1+i\infty} \right) \frac{F(s)G(1-s)}{\zeta(s)} ds, \quad \mathcal{D}_L x(y) = B_\zeta(x, y). \quad (\text{SCL4.1})$$

These are the already proved convergent, continuous and nondegenerate operations GZR2–GZR8 on the original quotient. In particular $\mathcal{D}_L : Q \rightarrow Q'$ is injective and strong-dual continuous, and

$$L^t \mathcal{D}_L = \mathcal{D}_L(1 - L). \quad (\text{SCL4.2})$$

For an actual zero a , put $b = 1 - a$. The original functional equation preserves the actual multiplicity $m = m_a = m_b$. Use the full local original-zeta unit

$$\zeta(b+t) = t^m u_b(t), \quad u_b(t) = \sum_{j \geq 0} \frac{\zeta^{(m+j)}(b)}{(m+j)!} t^j, \quad c_b = u_b(0)^{-1} = \frac{m!}{\zeta^{(m)}(b)}. \quad (\text{SCL4.3})$$

Then the two exact global identities are

$$\boxed{(\mathcal{D}_L x)(k_a) = (-1)^{m-1} c_b \ell_{b,0}(x), \quad \mathcal{D}_L(k_b) = c_b \ell_{a,0}.} \quad (\text{SCL4.4})$$

To prove the first, choose the globally isolated representative of k_a in the second slot. The only possible nonremovable pole of the original integrand is at $s = b$. At that point its reflected jet is $(-t)^{m-1}$. The residue is therefore $(-1)^{m-1} u_b(0)^{-1} F(b)$. GZR5's finite-pole contour theorem justifies this residue computation on the full original contour for every $F \in \mathcal{B}$, not just a finite polynomial. For the second identity choose the isolator of k_b in the first slot. Its jet is t^{m-1} , leaving residue $u_b(0)^{-1} G(a)$. This proves both formulas, including their different signs and the original derivative denominator.

In particular the entire transpose kernel in SCL3.1 belongs to $\mathcal{D}_L Q$. Also, the obstruction to solving $(L^t - a)\lambda = \mathcal{D}_L x$ is exactly the single value $\ell_{b,0}(x)$, and SCL2.6 identifies its vanishing with an already constructed primal division by $L - b$.

148.6 SCL5. Every linear polynomial is invertible on the actual algebraic cokernel

Define

$$M = \mathcal{D}_L Q \subset Q', \quad \mathcal{C} = Q'/M. \quad (\text{SCL5.1})$$

This is the actual algebraic cokernel in ASD14. The subspace is invariant under L^t by SCL4.2. No closedness or surjectivity of $\mathcal{D}_L$ is assumed. Let T be the operator induced by L^t on $\mathcal{C}$.

For every $a \in \mathbb{C}$,

$$\boxed{\mathsf{T} - a : \mathcal{C} \longrightarrow \mathcal{C} \text{ is bijective.}} \quad (\text{SCL5.2})$$

For surjectivity when $a \notin \mathcal{Z}$, every $\lambda \in Q'$ is $(L^t - a)S'_a \lambda$, by SCL3.

For surjectivity when $a \in \mathcal{Z}$, retain $b = 1 - a$, $m = m_a$ and $\kappa_a = (-1)^{m-1} c_b \neq 0$. Given $\lambda \in Q'$, set

$$x_0 = \frac{\lambda(k_a)}{\kappa_a} e_{b,0}, \quad \lambda_0 = \lambda - \mathcal{D}_L x_0, \quad \mu = S'_a \lambda_0. \quad (\text{SCL5.3})$$

By SCL4.4, $\lambda_0(k_a) = 0$. Equation SCL3.2 then gives $(L^t - a)\mu = \lambda_0$. Thus $(\mathsf{T} - a)[\mu] = [\lambda]$, proving surjectivity with an explicit continuous-dual representative.

For injectivity, suppose

$$(L^t - a)\lambda = \mathcal{D}_L x. \quad (\text{SCL5.4})$$

If $a \notin \mathcal{Z}$, then $b = 1 - a \notin \mathcal{Z}$ too, by the original zero symmetry. Set $y = -S_b x$. It satisfies $(1 - a - L)y = x$. SCL4.2 shows $(L^t - a)(\lambda - \mathcal{D}_L y) = 0$. The transpose kernel is zero in this case, so $\lambda = \mathcal{D}_L y$.

If $a \in \mathcal{Z}$, evaluate SCL5.4 at k_a . Its left side is zero. SCL4.4 therefore gives $\ell_{b,0}(x) = 0$, and SCL2.4 makes $y = -S_b x$ solve $(1 - a - L)y = x$ again. The same subtraction now lies in $\ker(L^t - a) = \mathbb{C}\ell_{a,0}$. SCL4.4 puts that kernel in M explicitly: $\ell_{a,0} = c_b^{-1} \mathcal{D}_L(k_b)$. Hence $\lambda \in M$, proving injectivity.

This also proves that the explicit inverse from SCL5.3 is independent of the representative of the class $[\lambda]$: it solves a bijective operator equation in $\mathcal{C}$. Nothing in this proof assumes that $\mathcal{C}$ itself is zero.

148.7 SCL6. The exact remaining module has no finite-dimensional spectral part

Every nonzero polynomial $P \in \mathbb{C}[t]$ factors into a nonzero constant and linear factors. By SCL5.2 each factor $P(\mathbf{T})$ is bijective. The operators commute, and their inverses commute too: multiplying $AB = BA$ by either inverse verifies this directly. Consequently

$$\frac{P(t)}{R(t)} \cdot c = P(\mathbf{T})R(\mathbf{T})^{-1}c, \quad R \neq 0, \quad (\text{SCL6.1})$$

defines a $\mathbb{C}(t)$ -vector-space structure on $\mathcal{C}$. If two fractions agree, cross multiplication and the common invertible denominator prove their operators agree. Distributivity and multiplication follow from commuting polynomial operators. The zero vector space is allowed; the calculation has not decided its dimension over $\mathbb{C}(t)$.

There is no nonzero finite-dimensional $\mathbf{T}$ -invariant subspace of $\mathcal{C}$. Indeed its finite-dimensional characteristic polynomial annihilates it by the matrix adjugate identity, whereas that polynomial in $\mathbf{T}$ is injective on all of $\mathcal{C}$.

There is also no nonzero finite-dimensional $\mathbf{T}$ -equivariant quotient of $\mathcal{C}$. If U is an invariant subspace with finite-dimensional quotient, its quotient characteristic polynomial P gives $P(\mathbf{T})\mathcal{C} \subset U$. But $P(\mathbf{T})$ is onto, so $U = \mathcal{C}$.

In particular any equivariant linear map from a finite-dimensional operator space to $\mathcal{C}$, or from $\mathcal{C}$ to a finite-dimensional operator space, is zero: its image is such a subspace or quotient.

The actual degree-one action in ASD14 includes χ . Its infinitesimal receiving operator is $1 - \mathbf{T}$, not $\mathbf{T}$. Every nonzero polynomial in $1 - \mathbf{T}$ is again a nonzero polynomial in $\mathbf{T}$. All conclusions in this section therefore apply to that exact twisted action as well. This identifies a property of the remaining whole object rather than deleting its twist.

148.8 SCL7. Explicit unique lift of every finite invariant dual subspace

Let $V \subset Q'$ be any finite-dimensional L^t -invariant subspace. Its image in $\mathcal{C}$ is finite-dimensional and invariant, so SCL6 proves it is zero. Thus

$$V \subset \mathcal{D}_L Q.$$

Since $\mathcal{D}_L$ is injective, there is exactly one linear lift

$$s_V : V \rightarrow Q, \quad \mathcal{D}_L s_V = \text{id}_V. \quad (\text{SCL7.1})$$

We now construct that lift explicitly rather than leave it as an inverse on an unspecified image.

A finite-dimensional L^t -module decomposes into its generalized eigenspaces by the coprime factors of its characteristic polynomial. The decomposition follows from polynomial Bezout identities, which give the finite primary projectors. SCL3.3 shows that every occurring eigenvalue a is an actual nontrivial zero and that its full dual generalized eigenspace is $\text{span}\{\ell_{a,0}, \dots, \ell_{a,m_a-1}\}$.

Thus there is a finite set $P \subset \mathcal{Z}$ containing the complete support of V , and every $\lambda \in V$ has the unique expansion

$$\lambda = \sum_{a \in P} \sum_{j=0}^{m_a-1} \lambda(e_{a,j}) \ell_{a,j}. \quad (\text{SCL7.2})$$

The coefficient formula follows from SCL1.7. It is valid on the entire Q , because the generalized eigenfunctional classification in SCL3.3 is an equality of continuous functionals, not only of restrictions to a block.

For each $a \in P$, put $b = 1 - a$, $m = m_a$, and define the full receiving jet

$$P_{b,\lambda}(t) = \left[u_b(t) \sum_{j=0}^{m-1} (-1)^j \lambda(e_{a,j}) t^{m-1-j} \right]_{\deg < m} = \sum_{i=0}^{m-1} p_{b,i}(\lambda) t^i. \quad (\text{SCL7.3})$$

The brackets mean the image in the explicitly stated truncated algebra $\mathbb{C}[t]/(t^m)$; every contributing coefficient of $u_b(t) = \sum_{r \geq 0} \zeta^{(m+r)}(b) t^r / (m+r)!$ is retained. Set

$$s_V(\lambda) = \sum_{a \in P} \sum_{i=0}^{m_a-1} p_{1-a,i}(\lambda) e_{1-a,i}. \quad (\text{SCL7.4})$$

It is a finite combination of the actual global representatives from SCL1.6, hence belongs to the original quotient.

To prove its lift identity for an arbitrary $y = [G] \in Q$, the first-slot isolators reduce each term of the global pairing to its single pole $b = 1 - a$, by GZR5. Its residue is

$$[t^{m-1}] \frac{P_{b,\lambda}(t)}{u_b(t)} \sum_{j=0}^{m-1} (-1)^j \ell_{a,j}(y) t^j = \sum_{j=0}^{m-1} \lambda(e_{a,j}) \ell_{a,j}(y).$$

For the last equality, SCL7.3 says that the coefficient in degree $m-1-j$ of $P_{b,\lambda}/u_b$ is $(-1)^j \lambda(e_{a,j})$; the reflected jet contributes the second $(-1)^j$. Summing over P yields exactly $\lambda(y)$ by SCL7.2. This proves SCL7.1 for the explicit formula.

The lift is continuous: it has finitely many continuous evaluations of λ at fixed vectors, followed by a finite linear combination of fixed vectors of Q . It is therefore continuous for the subspace weak-* or strong topology on V , and for its usual finite-dimensional topology. No continuous inverse of $\mathcal{D}_L$ on its entire image is inferred.

Its generator identity is

$$(1 - L)s_V = s_V L^t, \quad Ls_V = s_V(1 - L^t). \quad (\text{SCL7.5})$$

Apply $\mathcal{D}_L$ to the first difference, use SCL4.2 and SCL7.1, and then use injectivity. The lifted finite space therefore has the reflected eigenvalues $1 - a$ and the exact negative transpose nilpotent, not a dropped nilpotent or a parity assigned to τ .

148.9 SCL8. Full real and prime covariance proved on the actual finite jets

For $r > 0$, use the original $T_r[F] = [r^s F(s)]$ and the contragredient $T_r^\vee \lambda = \lambda \circ T_{r-1}$. Their exact finite jet actions are

$$T_r^\vee \ell_{a,j} = r^{-a} \sum_{k=0}^j \frac{(-\log r)^{j-k}}{(j-k)!} \ell_{a,k}, \quad (\text{SCL8.1})$$

$$T_r e_{b,i} = r^b \sum_{k=0}^{m_b-1-i} \frac{(\log r)^k}{k!} e_{b,i+k}. \quad (\text{SCL8.2})$$

Equation SCL8.1 follows by differentiating $r^{-s}F(s)$ at a , with its full factorials. For SCL8.2 multiply the original Taylor series t^i by r^{b+t} at b ; all other-zero jets remain zero. The full zero-jet separation SCL1.1 proves equality in the original Q . Thus both formulas concern the actual globally represented blocks.

On any finite sum of these blocks, these operators equal, respectively, $\exp(-(\log r)L^t)$ and $\exp((\log r)L)$, where these are finite-dimensional matrix functions with the nilpotent sums just displayed. A polynomial in the finite matrix gives each exponential: prescribe its Taylor coefficients through the relevant Jordan orders at the finitely many distinct eigenvalues and use finite polynomial interpolation. Consequently every L^t -invariant finite subspace V above is invariant under $T_r^\vee$, and its lift $s_V V$ is invariant under T_r .

Now SCL7.5 and the actual finite matrix formulas prove

$$\boxed{T_r s_V = r s_V T_r^\vee \quad (r > 0).} \quad (\text{SCL8.3})$$

Indeed exponentiating the finite identity $L s_V = s_V(1 - L^t)$ gives $T_r s_V = s_V \exp((\log r)(1 - L^t)) = r s_V \exp(-(\log r)L^t)$, and the preceding direct jet calculation identifies each exponential with the original group action. This is a proved whole-group relation on V ; no passage from an infinitesimal relation to an unproved infinite-dimensional exponential has occurred. In particular it holds at every original prime $r = p$.

Thus $s_V : \chi \otimes V \rightarrow Q$ is equivariant for exactly the action in ASD14. The source receiving-ring action on this spectral part factors through its first scalar coordinate ϵ , so the complex-linear map s_V is also source-equivariant there: every integer n acts by n , and τ by identity. The extra $\epsilon = 0$ copies distinguishing τ from integer 1 remain in the full cone and are not discarded by this statement about its spectral summand.

Although every $\mathcal{D}_L Q$ is invariant under the full contragredient real group, no unrestricted infinite-dimensional exponential formula is claimed on $\mathcal{C}$. Its $\mathbb{C}(t)$ -module result and the finite-space covariance have their independently proved domains.

Conversely, a finite-dimensional subspace $V \subset Q'$ invariant under the entire receiving real group is L^t -invariant. To prove the differentiation prerequisite, for real h the original transform satisfies

$$\frac{e^{-hs}F(s) - F(s)}{h} \longrightarrow -sF(s) \quad \text{in } \mathcal{B}. \quad (\text{SCL8.4})$$

The integral remainder formula for the scalar exponential bounds the error on $|\Re s| \leq A$ by $\frac{1}{2}|h||s|^2 e^{A|h|}|F(s)|$. This is controlled in each $b_{A,N}$ by a constant times $|h|e^{A|h|}b_{A,N+2}(F)$, proving the asserted convergence. It descends continuously to Q . Thus, for $\lambda \in V$ and $q \in Q$,

$$\left. \frac{d}{dh} (T_{e^h}^\vee \lambda)(q) \right|_{h=0} = -\lambda(Lq) = -(L^t \lambda)(q). \quad (\text{SCL8.5})$$

This is differentiation in the weak-* topology of the actual continuous dual.

The subspace V is weak-* closed and has its ordinary finite-dimensional topology. One explicit verification is to take a basis of V and finitely many vectors $q_i \in Q$ whose evaluation matrix has full rank; such vectors exist because these continuous functionals are linearly independent. Solving that finite matrix defines a weak-* continuous projection $Q' \rightarrow V$. Its fixed space is exactly V , so V is closed; its finitely many evaluation coordinates give its ordinary topology. Difference quotients of the orbit in SCL8.5 lie in V , and closedness puts their limit $-L^t \lambda$ in V . This proves the required invariant-subspace assertion. Invariance under $rT_r^\vee$ is equivalent to invariance under $T_r^\vee$, since the scalar r is nonzero, so the statement includes the exact χ -twisted receiving group.

Its lift is the same explicit s_V . There is also a direct whole-group covariance proof using the injection, independent of differentiating an inverse: by SCL4.2's group version,

$$\mathcal{D}_L(T_r s_V \lambda) = r T_r^\vee \lambda = \mathcal{D}_L(s_V(r T_r^\vee \lambda)).$$

Injectivity gives exactly SCL8.3. Nothing in this argument says that the entire Q or Q' is finite-dimensional or pure.

148.9.1 Every receiving support label in these maps

Retain the receiving lattice L_{supp} and the carrier proved in GZR9:

$$G_{L_{\text{supp}}}(W) = \{(0, \alpha) : \alpha \in L_{\text{supp}}\} \cup \{(w, 1_{L_{\text{supp}}}) : w \in W\}.$$

For every constructed receiving linear map f in this note its exact lift is

$$G_{L_{\text{supp}}}(f)(w, \alpha) = (f(w), \alpha). \quad (\text{SCL8.6})$$

It is well-defined: a nontop-labelled input has amplitude zero, and linearity sends that amplitude to zero; a nonzero output can therefore occur only at top support. Substitution proves preservation of identity and composition. In particular

$$G_{L_{\text{supp}}}(\mathcal{D}_L)G_{L_{\text{supp}}}(s_V) = \text{id}_{G_{L_{\text{supp}}}(V)}. \quad (\text{SCL8.7})$$

These formulas apply to the exact original quotient, projectors, division operators, continuous transpose maps and the finite lifts, as well as the supported maps explicitly computed in ASD6. A vanishing map amplitude retains the output $(0, \alpha)$, not an erased support label. The receiving zero $(0, \alpha)$ is not identified with primitive τ .

Independently labelled blocks remain separate factors of the corresponding products of these carriers, with the block lift applied to its own factor. No identification of $G_{L_{\text{supp}}}(W_1 \oplus W_2)$ with $G_{L_{\text{supp}}}(W_1) \times G_{L_{\text{supp}}}(W_2)$ is made: the former has a common label, whereas the latter retains two labels. This spells out the actual receiving lifts without assigning source τ a scalar amplitude or asserting an abelian category of primitive τ objects.

148.10 SCL9. Exact consequence for the actual supported dual-comparison cone

The actual ASD14 chain map is

$$\Psi_\zeta : D_{\text{full}} \rightarrow \chi \otimes D_{\text{full}}^\vee[-2], \quad \Psi_\zeta^1 = \pi' \mathcal{D}_L \pi, \quad \Psi_\zeta^0 = 0.$$

Its cohomology sequence contains the exact global row

$$\boxed{0 \longrightarrow Q \xrightarrow{\mathcal{D}_L} \chi \otimes Q' \xrightarrow{q_{\text{cone}}} \chi \otimes \mathcal{C} \longrightarrow 0.} \quad (\text{SCL9.1})$$

The target actions retain the twist. Under the exact identification of ASD14.7, q_{cone} is the quotient map $\lambda \mapsto [\lambda]$, induced by the comparison triangle's map from its target to its cone. It is not the connecting arrow. The actual following connecting arrow has domain $\chi \otimes \mathcal{C}$ and codomain $H^2(D_{\text{full}}) = 0$, so that following arrow is zero.

For every finite-dimensional L^t -invariant subspace $V \subset Q'$, the map to cone cohomology is zero on $\chi \otimes V$, and the exact lift through the first arrow of SCL9.1 is uniquely the map s_V in SCL7.4. This is proved by SCL7.1, and its full real/prime equivariance is SCL8.3. Thus the actual comparison lifts every finite spectral subspace, with all Jordan orders retained. This conclusion improves the earlier statement that only a weak-* dense image was known.

The remaining degree-one cone group is a module on which every nonzero polynomial in its actual generator $1 - L^t$ is invertible. It has neither a nonzero finite-dimensional invariant subspace nor a nonzero finite-dimensional equivariant quotient. It is not proved to vanish as an algebraic group; its weak-* Hausdorff quotient was already proved zero in ASD14.8, and that topological statement is not substituted for the new algebraic calculation.

All other cone terms remain exactly as computed:

$$H^{-1}(K_\zeta) = H_0 \oplus V_{\text{extra}}, \quad H^0(K_\zeta) = 0, \quad H^2(K_\zeta) = \chi \otimes (H'_0 \oplus V'_{\text{extra}}). \quad (\text{SCL9.2})$$

In particular the four original endpoint lines, their duals, and both full faithful closed coefficient copies remain. The unique lift constructed here lands in $Q = H^1(D_{\text{full}})$. It is not an asserted equivariant cochain section $Q \rightarrow A$; the finite source representatives in SCL7.4 are supplied explicitly without that stronger claim.

This is a concrete lifting theorem for the actual supported dual-comparison obstruction. It neither identifies that comparison with Deligne's full geometric specialization nor supplies Deligne's required numerical upper and lower weight bounds. The result says exactly what the surviving obstruction can no longer contain, and constructs the unique lifts for the finite spectral objects it excludes.

149 Polynomial lifting through the actual residue-duality comparison

24 September 2026. Independent derivation DPL0–DPL9.

149.1 DPL0. Actual source, prior results, and the receiving category

This note uses the original quotient and the actual comparison cone constructed in GZR and ASD14. It proves exact polynomial lifting and extension consequences for their entire dual cokernel. It does not assume that this cokernel is zero.

The source notation remains $Z_0, Z_1/\tau, Z_2$. The current operations and retractions in `C0 RPUS_AND_OPERATION_RULES.md` remain controlling. The complete global user arguments `USR-9ad1c0a2d09dba92`, `USR-f55d16feb948d8c2`, and `USR-6152e3bc6302258c` were read in the preceding GZR derivation and are not replaced by a new source definition here. Complex addition, polynomial operators, and exact sequences below belong to the already constructed receiving spaces. Primitive τ is not an operand of any added operation.

The exact prior inputs are OMS1–OMS6, the original test-space and quotient identification; RZ2–RZ8, the continuous resolvent and actual full-jet representatives; GZR3–GZR8, the original-zeta residue pairing; and ASD2, ASD4, and ASD14, the continuous-dual topology and the actual comparison cone. ASD14 was read completely for this calculation. The original analytic source remains Ralf Meyer, [arXiv:math/0412277v3](https://arxiv.org/abs/math/0412277v3), as entered with full original factors in OMS. The receiving coefficient complex is that of Alain Connes and Caterina Consani, [arXiv:0903.2024v3](https://arxiv.org/abs/0903.2024v3), §5, through the exact ASD source comparison. No new reading of these entire human papers is claimed here.

Retain the entire-function Fréchet algebra

$$\mathcal{B} = \{F \text{ entire} : b_{A,N}(F) = \sup_{|\sigma| \leq A, t \in \mathbb{R}} (1 + |t|)^N |F(\sigma + it)| < \infty \text{ for all } A, N\},$$

$$\mathcal{I} = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ } (\rho \in \mathcal{Z}, 0 \leq j < m_\rho)\}, \quad Q = \mathcal{B}/\mathcal{I}. \quad (\text{DPL0.1})$$

Here $\mathcal{Z}$ is the actual nontrivial zero set of the original ζ , with every actual multiplicity. OMS proves that this quotient is the original $\mathcal{A}/\mathcal{ES}$, with its quotient topology. Write $L[F] = [sF]$, $T_a[F] = [a^s F]$ for $a > 0$, and Q' for the continuous complex-linear dual.

The constructed pairing and its injective dual map are

$$B_\zeta([F], [G]) = \frac{1}{2\pi i} \left(\int_{2-i\infty}^{2+i\infty} - \int_{-1-i\infty}^{-1+i\infty} \right) \frac{F(s)G(1-s)}{\zeta(s)} ds, \quad Dx(y) = B_\zeta(x, y). \quad (\text{DPL0.2})$$

GZR proves both integrals converge absolutely, the form descends in both slots, every full primary residue is retained, and $D : Q \rightarrow Q'_b$ is continuous and injective. Here Q'_b denotes the strong continuous dual. ASD also uses the weak-* dual Q'_σ .

Write

$$A = L^t, \quad (A\lambda)(x) = \lambda(Lx), \quad C = Q'/DQ, \quad \eta : Q' \rightarrow C. \quad (\text{DPL0.3})$$

The algebraic quotient is retained, whether or not its quotient topology is Hausdorff. We use either the specified weak-* quotient topology or, when explicitly named, the quotient of the strong dual. These topologies are not identified with each other.

The exact real-action identity and its infinitesimal form are

$$DT_a = aT_a^\vee D, \quad DL = (1 - A)D, \quad (T_a^\vee \lambda)(x) = \lambda(T_{a^{-1}}x). \quad (\text{DPL0.4})$$

Thus the actual twisted target $Y = \chi \otimes Q'$, $\chi(a) = a$, has generator

$$M = 1 - A, \quad (\text{DPL0.5})$$

and $D : (Q, L) \rightarrow (Y, M)$ is an intertwiner. The twisted cokernel $\tilde{C} = \chi \otimes C$ has the same vectors as C , with induced generator $1 - A$. This twist is kept throughout.

For Hom and Ext statements the receiving module ring is $R = \mathbb{C}[X]$, with X acting by the specified generator. Thus X acts by L on Q and by M on $Y, \tilde{C}$. These are the actual modules from (DPL0.1)–(DPL0.5), not newly postulated spectra. Algebraic Ext_R^1 means equivalence classes of exact module extensions. A continuous assertion is stated for strict exact rows of locally convex receiving modules: the injection has its subspace topology, and the surjection has its quotient topology. Non-Hausdorff quotients are not silently replaced by their Hausdorffizations.

The phrase “receiving polynomial torsion” below means that each vector in this R -module is annihilated by some nonzero polynomial in its specified receiving generator. It does not mean the user’s integer-amount datum, which the corpus sometimes also calls torsion. A single polynomial annihilating the entire module is a stronger, separately stated condition.

149.2 DPL1. Explicit continuous division on the original quotient

Let $a \in \mathbb{C}$, and write $S_a = L - a$. If $a \notin \mathcal{Z}$, the original RZ resolvent proves that S_a is a continuous automorphism of Q . Its actual representative inverse is

$$\mathcal{V}_a F(s) = \frac{F(s) - F_*(s)F(a)/F_*(a)}{s - a}, \quad F_*(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s). \quad (\text{DPL1.1})$$

All endpoint and trivial-zero values in RZ1 remain part of this formula. In particular $F_*(0) = F_*(1) = 1/8$; division at either endpoint is not division by a removed original zeta value.

Now let $a \in \mathcal{Z}$, $m = m_a$. Let $e_a = E_{a,0}$ be the actual RZ representative with Taylor jet one at a and zero to the full required order at every other actual zero. Set

$$E_{a,j}(s) = e_a(s)(s-a)^j, \quad 0 \leq j < m, \quad \ell_{a,j}([F]) = F^{(j)}(a)/j!. \quad (\text{DPL1.2})$$

All $\ell_{a,j}$ with $j < m$ are well-defined and continuous on Q . Define on the entire original test algebra

$$\mathcal{U}_a F(s) = \frac{F(s) - F(a)e_a(s)}{s - a} - \left(\frac{F^{(m)}(a)}{m!} - F(a) \frac{e_a^{(m)}(a)}{m!} \right) E_{a,m-1}(s). \quad (\text{DPL1.3})$$

The first numerator vanishes at a . Its entire division is continuous in $\mathcal{B}$: outside a fixed disk around a , divide by the bounded-away-from-zero linear factor; inside the disk use the derivative integral along the segment from a , with Cauchy’s estimate on a larger disk. This bounds each output strip seminorm by finitely many input strip seminorms. Every remaining operation in (DPL1.3) is a continuous finite-rank operation.

If $F \in \mathcal{I}$, then $F(a) = 0$, $F^{(j)}(a) = 0$ for $j < m$, and the first quotient has exactly the possible last jet $F^{(m)}(a)t^{m-1}/m!$. The explicit correction removes that last jet. At every other zero both terms retain its full vanishing order. Thus $\mathcal{U}_a \mathcal{I} \subset \mathcal{I}$, and (DPL1.3) induces a continuous $U_a : Q \rightarrow Q$.

Its identities are

$$S_a U_a = 1 - [E_{a,0}] \ell_{a,0}, \quad U_a S_a = 1 - [E_{a,m-1}] \ell_{a,m-1}. \quad (\text{DPL1.4})$$

For the first, multiplying (DPL1.3) by $s - a$ leaves $F - F(a)e_a$ modulo $\mathcal{I}$, since $(s-a)E_{a,m-1} \in \mathcal{I}$. For the second, replace F in (DPL1.3) by $(s-a)F$: its value at a is zero and its coefficient of degree m is $F^{(m-1)}(a)/(m-1)!$. This proves the second identity even at the representative level.

Consequently

$$\ker S_a = \mathbb{C}[E_{a,m-1}], \quad \text{im } S_a = \ker \ell_{a,0}. \quad (\text{DPL1.5})$$

The first identity follows by applying $U_a S_a$ to a kernel vector and observing that $S_a[E_{a,m-1}] = 0$. For the second, evaluation at a kills the range, and $S_a U_a$ is identity on $\ker \ell_{a,0}$. The range is

closed. The displayed continuous right inverse on the range and continuous projection away from the kernel give strict division on the actual quotient, with every multiplicity retained.

Transposing (DPL1.4) gives

$$\begin{aligned}(A - a)U_a^t &= 1 - \ell_{a,m-1} \text{ev}_{[E_{a,m-1}]}, \\ U_a^t(A - a) &= 1 - \ell_{a,0} \text{ev}_{[E_{a,0}]}.\end{aligned}\tag{DPL1.6}$$

Here $\text{ev}_v(\lambda) = \lambda(v)$. Therefore

$$\ker(A - a) = \mathbb{C}\ell_{a,0}, \quad \text{im}(A - a) = \{\lambda : \lambda([E_{a,m-1}]) = 0\}.\tag{DPL1.7}$$

These identities follow directly from (DPL1.6); no extension of a discontinuous functional is involved. Each transpose is continuous for both dual topologies because continuous operators preserve bounded sets and weak-* evaluations.

149.3 DPL2. Every linear factor is invertible on the actual dual cokernel

First DQ is A -stable by (DPL0.4), so $A - a$ induces an operator on C . For $a \notin \mathcal{Z}$, also $1 - a \notin \mathcal{Z}$ by the original functional equation. Both source operators in

$$(A - a)D = -D(L - (1 - a))\tag{DPL2.1}$$

are continuously invertible on the appropriate full spaces. The transpose inverse on Q' therefore induces an inverse on C .

For $a \in \mathcal{Z}$, put $b = 1 - a$, $m = m_a = m_b$, and retain

$$\zeta(b + t) = t^m u_b(t), \quad c_{b,0} = u_b(0)^{-1} = \frac{m!}{\zeta^{(m)}(b)}.\tag{DPL2.2}$$

The finite-residue theorem GZR5, on the original denominator, gives the two exact identities

$$D[E_{b,m-1}] = c_{b,0}\ell_{a,0}, \quad D[E_{b,0}](E_{a,m-1}) = (-1)^{m-1}c_{b,0}.\tag{DPL2.3}$$

For the first identity evaluate on an arbitrary F : only the residue at b remains, and its coefficient is $c_{b,0}F(a)$. For the second, the reflected monomial contributes $(-1)^{m-1}$, with no other allowed residue term. These calculations retain the actual derivative $\zeta^{(m)}(b)$.

To prove surjectivity on C , take any $\lambda \in Q'$, and put

$$h_\lambda = \frac{\lambda([E_{a,m-1}])}{(-1)^{m-1}c_{b,0}}[E_{b,0}], \quad K_a\lambda = U_a^t(\lambda - Dh_\lambda).\tag{DPL2.4}$$

The adjusted functional annihilates $[E_{a,m-1}]$. Equation (DPL1.6) therefore proves

$$(A - a)K_a\lambda = \lambda - Dh_\lambda.\tag{DPL2.5}$$

Taking classes gives every prescribed class as an image.

To prove injectivity, suppose $(A - a)\lambda = Dh$. Evaluation on $[E_{a,m-1}]$ gives zero on the left. By the same full residue calculation, the right side is $(-1)^{m-1}c_{b,0}\ell_{b,0}(h)$. Thus $\ell_{b,0}(h) = 0$. Put $k = -U_b h$; then $(L - b)k = -h$ by (DPL1.4). Equation (DPL2.1) gives

$$(A - a)Dk = Dh = (A - a)\lambda.$$

Hence $\lambda - Dk \in \ker(A - a) = \mathbb{C}\ell_{a,0}$. This kernel lies in DQ by (DPL2.3), proving $\lambda \in DQ$. The induced kernel on C is zero.

Thus every $A - a$ is bijective on C . The explicit operator K_a is continuous on Q' for either topology: it is a transpose composed with identity minus a finite-rank evaluation map. Equation (DPL2.5) and injectivity on C imply $K_a(DQ) \subset DQ$, so it induces the unique inverse on C , continuously for each specified quotient topology. The case $a \notin \mathcal{Z}$ uses the transpose of the continuous resolvent instead. Consequently every linear factor is a continuous automorphism of the actual algebraic cokernel with either of these quotient topologies.

149.4 DPL3. Every polynomial and the retained character twist

A nonzero polynomial factors over $\mathbb{C}$ as

$$q(X) = c \prod_{j=1}^r (X - a_j), \quad c \neq 0, \quad (\text{DPL3.1})$$

with repetitions retained. The commuting operators $A - a_j$ are all bijective on C ; their inverses commute because they invert commuting bijections. Thus

$$q(A)^{-1} = c^{-1} \prod_{j=1}^r (A - a_j)^{-1} \quad (\text{DPL3.2})$$

is a continuous two-sided inverse. The empty product covers nonzero constants.

On the actual twisted cokernel the generator is $M = 1 - A$, so

$$q(M) = q(1 - A) \quad (\text{DPL3.3})$$

is also continuously invertible. There is a unique action of the rational-function field $\mathbb{C}(X)$ on the actual cokernel given by

$$\frac{p(X)}{q(X)} \cdot c = p(M)q(M)^{-1}c. \quad (\text{DPL3.4})$$

If $p/q = p_1/q_1$, cross-multiplication and invertibility of $q(M)q_1(M)$ prove the two actions agree. The same identities prove addition and multiplication of rational functions. This constructs an action on the existing quotient, without asserting its dimension over that field or that the quotient is nonzero.

In particular

$$\ker q(M)|_{\tilde{C}} = 0, \quad q(M)\tilde{C} = \tilde{C} \quad (q \neq 0). \quad (\text{DPL3.5})$$

This excludes every nonzero polynomial-annihilated vector in the actual cokernel. It does not identify the entire cokernel with its Hausdorff quotient or assert surjectivity of D .

149.5 DPL4. Both Hom vanishings and an explicit unique extension splitting

Let V be an R -module with $q(T_V)V = 0$ for one specified nonzero polynomial q . Its vector space may be infinite-dimensional; the hypothesis means this one polynomial annihilates the whole module.

For an R -linear $f : V \rightarrow \tilde{C}$,

$$q(M)f(v) = f(q(T_V)v) = 0,$$

so injectivity of $q(M)$ gives $f = 0$. For an R -linear $g : \tilde{C} \rightarrow V$, every c has the form $q(M)c_0$, giving $g(c) = q(T_V)g(c_0) = 0$. Therefore

$$\boxed{\text{Hom}_R(V, \tilde{C}) = 0 = \text{Hom}_R(\tilde{C}, V).} \quad (\text{DPL4.1})$$

Consider any exact R -module extension

$$0 \longrightarrow \tilde{C} \xrightarrow{i} E \xrightarrow{p} V \longrightarrow 0. \quad (\text{DPL4.2})$$

Since $pq(T_E) = q(T_V)p = 0$, the operator $q(T_E)$ has image in $i\tilde{C}$. Define

$$r : E \rightarrow \tilde{C}, \quad r(e) = q(M)^{-1}i^{-1}(q(T_E)e). \quad (\text{DPL4.3})$$

The inverse i^{-1} is taken only on its actual image. This is an R -linear map: every operator in the formula commutes with the indicated generator. Since $q(T_E)i = i q(M)$, it satisfies $ri = 1$.

For $v \in V$, choose any e with $p(e) = v$, and set

$$\boxed{\sigma(v) = e - ir(e).} \quad (\text{DPL4.4})$$

Replacing e by $e + i(c)$ changes this expression by $i(c) - iri(c) = 0$. Thus the formula is independent of the chosen lift. It is linear and R -linear because $1 - ir$ is, and $p\sigma = 1$. It is the unique R -linear splitting: the difference of two splittings factors through i as an R -linear map $V \rightarrow \tilde{C}$, which is zero by (DPL4.1). This proves

$$\boxed{\text{Ext}_R^1(V, \tilde{C}) = 0.} \quad (\text{DPL4.5})$$

For a strict exact row in the locally convex receiving category defined in DPL0, (DPL4.3) is continuous: i^{-1} is continuous on its subspace image, $q(T_E)$ is continuous into that image, and DPL3 supplies the continuous inverse $q(M)^{-1}$. Because p has its quotient topology and $\sigma p = 1 - ir$ is continuous, σ is continuous as well. Thus the same explicit unique splitting proves the corresponding strict continuous extension statement.

Any additional commuting receiving action is preserved by this splitting. In particular, the original positive-real actions and the source receiving ring actions commute with the generator; their operators commute with $q(M)$ and its inverse, so (DPL4.3)–(DPL4.4) intertwine them. This is a full equivariance statement for extensions that already carry the actual compatible actions, not an inferred action on primitive τ .

The words “one specified nonzero polynomial” matter for the reverse Hom statement. We have not replaced this condition by elementwise torsion with no common polynomial. Forward Hom into $\tilde{C}$ does vanish for elementwise torsion, because the same argument applies separately to each vector; this stronger observation is not used to enlarge the asserted reverse Hom theorem.

149.5.1 DPL4A. The algebraic splitting for all receiving polynomial-torsion modules

There is a further exact algebraic extension result when the annihilating polynomial depends on the vector. Let V have receiving polynomial torsion in the sense just defined, and keep the algebraic extension (DPL4.2). For each $e \in E$, choose a nonzero polynomial q annihilating $p(e)$, and define

$$r_q(e) = q(M)^{-1}i^{-1}(q(T_E)e). \quad (\text{DPL4A.1})$$

This expression is independent of q . Indeed if r is another annihilating polynomial, multiplication of each proposed value by the bijection $q(M)r(M)$ gives the same vector

$$i^{-1}(q(T_E)r(T_E)e).$$

The equality follows because the two polynomials commute with each other and with the injection. Injectivity of the product proves independence.

For e_1, e_2 , the product of annihilating polynomials for their two images also annihilates their sum. Using this single product in (DPL4A.1) proves additivity of r . The same polynomial handles scalar multiplication, and a polynomial annihilating $p(e)$ also annihilates $T_V p(e)$; commutation proves $r(T_E e) = Mr(e)$. For $e = i(c)$ one may choose $q = 1$, giving $r(i(c)) = c$. Thus the independent formula defines an R -linear retraction.

Formula (DPL4.4) now again gives a splitting. It is unique because an R -linear map from receiving polynomial torsion into $\tilde{C}$ vanishes vector by vector by (DPL3.5). Therefore

$$\text{Hom}_R(V, \tilde{C}) = 0, \quad \text{Ext}_R^1(V, \tilde{C}) = 0 \quad \text{for every receiving polynomial-torsion } V. \quad (\text{DPL4A.2})$$

This argument is algebraic. It does not supply a uniform continuity estimate when the annihilating polynomial varies with the vector. The continuous extension theorem in DPL4 retains its single global polynomial and strict-topology hypotheses. Nor does (DPL4A.2) assert the reverse Hom vanishing for arbitrary elementwise torsion.

149.6 DPL5. Every polynomial-annihilated dual vector lifts uniquely to the actual Q

For the actual exact algebraic row

$$0 \longrightarrow Q \xrightarrow{D} Y = \chi \otimes Q' \xrightarrow{\eta} \tilde{C} \longrightarrow 0, \quad (\text{DPL5.1})$$

take $y \in Y$ with $q(M)y = 0$. Its image in $\tilde{C}$ is killed by the same polynomial and hence is zero by DPL3. Thus $y = Dx$ for a unique $x \in Q$. Intertwining gives $Dq(L)x = q(M)y = 0$, and injectivity of D proves $q(L)x = 0$. Therefore

$$\boxed{D : \ker q(L)|_Q \xrightarrow{\cong} \ker q(M)|_Y} \quad (\text{DPL5.2})$$

for every nonzero polynomial, including every $(X - \rho)^r$.

Both spaces in (DPL5.2) are finite-dimensional with their actual inherited topologies. To verify this on the source, factor q . At an actual zero ρ , the polynomial germ has order $r_\rho = \text{ord}_\rho q$, where $r_\rho = 0$ if $q(\rho) \neq 0$. In the actual primary algebra $\mathbb{C}[t]/(t^{m_\rho})$, the kernel is precisely

$$t^{\max(m_\rho - r_\rho, 0)} \mathbb{C}[t]/(t^{m_\rho}), \quad (\text{DPL5.3})$$

of dimension $\min(m_\rho, r_\rho)$. Only finitely many zeros can occur among the finitely many roots of q . Every other primary jet is zero. Summing the existing RZ primary sections reconstructs these finitely many possible jets, and synthesis makes the remaining class zero. Thus

$$\dim \ker q(L) = \sum_{\rho: q(\rho)=0} \min(m_\rho, \text{ord}_\rho q). \quad (\text{DPL5.4})$$

The same dimension holds on the target by (DPL5.2). These are subspaces of Hausdorff locally convex spaces, so their finite-dimensional topologies are the usual ones. The bijection in (DPL5.2) and its inverse are continuous.

More generally, any R -linear map $f : V \rightarrow Y$ from a module annihilated by q has $\eta f = 0$ by DPL4.1, hence has a unique R -linear lift $\tilde{f} : V \rightarrow Q$ satisfying $D\tilde{f} = f$. Its image lies in the finite-dimensional kernel (DPL5.3). If f is continuous, the finite-dimensional inverse just proved makes $\tilde{f}$ continuous. If f also intertwines any of the actual real or source-ring actions, so does its lift, by injectivity and the exact intertwining of D . No continuous inverse of D on its entire image has been assumed.

There is an exact local formula for this lift, retaining every original zeta derivative. Suppose $(A - a)^r \lambda = 0$, let $b = 1 - a$, and assume $a \in \mathcal{Z}$ of multiplicity m . For its unique preimage x under D , the b -jet is

$$j_b x(t) = \left[u_b(t) \sum_{j=0}^{m-1} (-1)^j \lambda([E_{a,j}]) t^{m-1-j} \right]_{<m}, \quad u_b(t) = \sum_{k \geq 0} \frac{\zeta^{(m+k)}(b)}{(m+k)!} t^k. \quad (\text{DPL5.5})$$

All other primary jets vanish. Indeed D identifies this dual generalized eigenspace with the source generalized eigenspace at b , by (DPL0.4) and (DPL5.2). GZR4 gives

$$\lambda([E_{a,j}]) = (-1)^j [t^{m-1-j}] (j_b x(t)/u_b(t)).$$

Solving these m coefficient equations proves (DPL5.5). Inserting this polynomial into the existing RZ section at b produces an actual element of Q , with its original Mellin representative and all Gamma and endpoint factors from (DPL1.1), not merely a formal jet.

For $a \notin \mathcal{Z}$, the dual generalized eigenspace is zero by (DPL5.2) and the original resolvent. Repeated factors and all nilpotents are included in these statements.

149.7 DPL6. The exact map to the cokernel in the actual ASD14 cone

Keep every degree and sign of ASD14:

$$\begin{aligned} D_{\text{full}} &= [P \xrightarrow{d} A_{\text{raw}}], & T &= \chi \otimes D_{\text{full}}^{\vee}[-2], \\ K_{\zeta} &= \left[P \xrightarrow{-d} A_{\text{raw}} \xrightarrow{\Psi_{\zeta}^1} \chi \otimes A'_{\text{raw}} \xrightarrow{d'} \chi \otimes P' \right], & \deg &= (-1, 0, 1, 2), \\ \Psi_{\zeta}^1 &= \pi' D \pi, & H^1(T) &= Y, & H^1(K_{\zeta}) &= \tilde{C}. \end{aligned} \quad (\text{DPL6.1})$$

The notation A_{raw} here names ASD's original overlap test space; it is not the transpose operator $A = L^t$ of DPL0.

The relevant part of the actual comparison triangle's long exact sequence is

$$0 \longrightarrow Q \xrightarrow{D} Y \xrightarrow{\eta} \tilde{C} \longrightarrow H^2(D_{\text{full}}) = 0. \quad (\text{DPL6.2})$$

Thus the map from the dual cohomology to the cone cohomology is exactly the quotient map η . The following arrow, the connecting map to $H^2(D_{\text{full}})$, is zero because its actual codomain is zero. We do not rename an unspecified arrow a connecting map.

DPL5 proves that η vanishes on every generalized eigenspace of the full twisted generator $M = 1 - L^t$, and more generally on every polynomial-annihilated submodule of Y . Each such dual class lifts uniquely to a source cohomology class in Q .

At the actual cochain level, a degree-one cone cycle has the form $\pi' \lambda$, because $\ker d' = \pi' Q'$. If its dual cohomology class λ satisfies the polynomial condition, write uniquely $\lambda = Dx$ by DPL5. Choose any $b \in A_{\text{raw}}$ with $\pi b = x$. Then

$$\pi' \lambda = \pi' D \pi b = \Psi_{\zeta}^1 b = d_{K_{\zeta}}^0 b. \quad (\text{DPL6.3})$$

This is the explicit null-boundary formula, with the positive middle-cone differential retained. The cohomology lift x is unique. The test-space representative b is determined modulo the actual summation image $J = \ker \pi = d(P)$; no unique test-space section has been asserted.

On $H^1(K_{\zeta})$ itself there are no nonzero generalized eigenvectors, by DPL3. This conclusion does not remove the retained groups

$$H^{-1}(K_{\zeta}) = H, \quad H^2(K_{\zeta}) = \chi \otimes H',$$

or their full faithful extra closed copies, endpoint lines, and original actions. It does not prove that $H^1(K_{\zeta}) = 0$ as an algebraic vector space.

149.8 DPL7. The actual companion mirror and supported labels

The mirror of ASD14 compares the original pairing with its companion denominator $\zeta(1-s)$. It does not assume the original pairing is self-mirror. The exact identity is

$$D^{\zeta} R = -R' D^{\zeta^{\vee}}. \quad (\text{DPL7.1})$$

On the degree-one terms in the comparison, the source mirror is $-R$ and the target mirror in the actual cone map is $+R'$. Therefore a companion lift $D^{\zeta^{\vee}} x = \lambda$ transforms into

$$D^{\zeta}(-Rx) = R' \lambda. \quad (\text{DPL7.2})$$

This is the correct signed lift on the original comparison. The mirror conjugates the generator to 1 minus the generator, as follows either by differentiating ASD14.13 or directly from $RL = (1-L)R$. It consequently transports the polynomial $q(X)$ to $q(1-X)$. DPL3 applies to both, so polynomial lifting is preserved by this exact companion mirror, including its signs.

For the existing lattice carrier

$$G_{\mathcal{L}}(V) = \{(0, \lambda) : \lambda \in \mathcal{L}\} \cup \{(v, 1_{\mathcal{L}}) : v \in V\},$$

every receiving linear map above lifts by

$$(v, \lambda) \mapsto (f(v), \lambda). \quad (\text{DPL7.3})$$

This is well-defined because a nonzero input amplitude has top support, and a vanishing output retains its original label. Composition and identities are preserved by substitution. On each polynomial-kernel subspace the inverse in (DPL5.2) has the same lift, so the unique lift retains every support label, including supported zero labels. No ordinary tensor operation identifying different supported zero labels is substituted for this map.

149.9 DPL8. Exact consequence and limit of the calculation

The actual algebraic dual cokernel has no nonzero polynomial-annihilated vector and no nonzero polynomial-annihilated quotient. The second statement follows from the reverse Hom vanishing in DPL4 for the quotient map. Its polynomial operators are continuously invertible in the stated quotient topologies. Every polynomial-annihilated module has the two Hom vanishings and the explicitly proved unique extension splitting in DPL4.

The actual ASD14 map from dual cohomology to its cone cokernel vanishes on every generalized eigenspace, with a unique full-multiplicity source lift and the explicit cochain boundary (DPL6.3). This strengthens density of DQ to a polynomial lifting theorem. It does not replace the full dual by its algebraic generalized eigenspaces, assert convergence of an infinite primary expansion, or conclude that the entire cokernel vanishes. No Deligne purity bound or RH conclusion has been inferred from this lifting result.

149.10 DPL9. Direct cyclic derived-Hom calculation on every degree of the actual cone

This final calculation is in the algebraic receiving R -module category. It does not assume an analytic derived category or a spectral sequence. Let $V = R/(q)$, $q \neq 0$. Its explicit free resolution is

$$P_V^{-1} = R \xrightarrow{q} P_V^0 = R \longrightarrow V \longrightarrow 0. \quad (\text{DPL9.1})$$

Multiplication by q is injective because $R = \mathbb{C}[X]$ has no zero divisors, so this is an exact two-term projective resolution.

On the actual cone (DPL6.1) retain the generator

$$G_K^{-1} = L_P, \quad G_K^0 = L_{A_{\text{raw}}}, \quad G_K^1 = 1 - L_{A_{\text{raw}}}^t, \quad G_K^2 = 1 - L_P^t. \quad (\text{DPL9.2})$$

Here $L_P, L_{A_{\text{raw}}}$ are the derivatives of the original real actions on those original spaces, as in ASD7 and ASD14. The identities $dL_P = L_{A_{\text{raw}}}d$, $\Psi_{\zeta}^1 L_{A_{\text{raw}}} = (1 - L_{A_{\text{raw}}}^t)\Psi_{\zeta}^1$, and their transpose prove that G_K commutes with every differential, including the first minus sign. Thus $q(G_K)$ is an actual cochain operator.

The Hom complex of (DPL9.1) into K_{ζ} , which computes $\text{RHom}_R(V, K_{\zeta})$, has terms and differential

$$\mathcal{H}^n = K_{\zeta}^n \oplus K_{\zeta}^{n-1}, \quad \delta^n(u, v) = (d_K u, d_K v - (-1)^n q(G_K)u). \quad (\text{DPL9.3})$$

These are exactly the Hom differential $d_K f - (-1)^n f d_{P_V}$. Substitution verifies $\delta^{n+1}\delta^n = 0$: the two mixed polynomial terms have opposite signs.

There is a direct exact cohomology sequence

$$0 \longrightarrow \frac{H^{n-1}(K_\zeta)}{q(G_K)H^{n-1}(K_\zeta)} \longrightarrow H^n(\mathcal{H}) \longrightarrow \ker(q(G_K) : H^n(K_\zeta) \rightarrow H^n(K_\zeta)) \longrightarrow 0. \quad (\text{DPL9.4})$$

Here is a proof with all maps. A cocycle (u, v) obeys $d_K u = 0$ and $d_K v = (-1)^n q(G_K)u$, so its image is the class $[u]$ in the displayed kernel. Every such class lifts: choose a cycle u representing it and v with the required differential. A class in the kernel of this map has $u = d_K a$. Subtracting the boundary

$$\delta^{n-1}(a, 0) = (d_K a, (-1)^n q(G_K)a)$$

leaves $(0, v - (-1)^n q(G_K)a)$, whose second component is a cycle. Two such second components differ by a boundary and a polynomial multiple of a cycle, exactly the left quotient in (DPL9.4). Its injection sends a cycle class $[v]$ to $[(0, v)]$. This proves all parts of (DPL9.4) directly.

Use the actual ASD14 cohomology

$$H^{-1}(K_\zeta) = H, \quad H^0(K_\zeta) = 0, \quad H^1(K_\zeta) = \tilde{C}, \quad H^2(K_\zeta) = \chi \otimes H',$$

and DPL3's bijectivity of $q(M)$ on $\tilde{C}$. Equation (DPL9.4) then yields every degree:

$\begin{aligned} H^{-1} \text{RHom}_R(V, K_\zeta) &= \ker(q(L_H) : H \rightarrow H), \\ H^0 \text{RHom}_R(V, K_\zeta) &= H/q(L_H)H, \\ H^1 \text{RHom}_R(V, K_\zeta) &= 0, \\ H^2 \text{RHom}_R(V, K_\zeta) &= \ker(q(1 - L_H^t) : \chi \otimes H' \rightarrow \chi \otimes H'), \\ H^3 \text{RHom}_R(V, K_\zeta) &= (\chi \otimes H')/q(1 - L_H^t)(\chi \otimes H'), \\ H^n \text{RHom}_R(V, K_\zeta) &= 0 \quad (n \notin \{-1, 0, 1, 2, 3\}). \end{aligned}$	(DPL9.5)
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The induced L_H is the original action on $H = \ker d$. In particular

$$H^2 \text{RHom}_R(V, K_\zeta) = \text{Hom}_R(V, \chi \otimes H'). \quad (\text{DPL9.6})$$

The identity follows by evaluating an R -linear map on the class of 1 in $R/(q)$. Likewise the degree-zero and degree-three quotients are the ordinary cyclic Ext_R^1 terms computed by (DPL9.1). No purity assertion is involved.

The potentially surviving terms in (DPL9.5) retain H, H' , their four original endpoint lines, and both faithful extra closed coefficient copies. The result proves the specified degree-one vanishing and calculates the exact other terms; it does not assert that the entire comparison is a quasi-isomorphism or that any of the retained terms obstructs the user's full source geometry.

149.11 Exact original character lifting and its full supported return

MCL0–MCL11 now computes the original transpose $A' \rightarrow S'$ on every actual zero and all its Mellin jets. A target jet of order j needs generalized-character length exactly $m+j+1$ at a zero of multiplicity m . The full Euler-distribution classification proves minimality against the whole original continuous dual; two actual primes remove the single-prime character aliases. Every prime logarithm and the full Gamma/Fourier return remain. This is an actual continuous lift, with a nonzero class only when its original annihilator must be preserved.

RPC0–RPC8 constructs the original restriction's actual pushout by the residue image, proves its precise cochain location and a single coherent real/prime-equivariant section on all locally finite classes. ORE0–ORE10 calculates the unmodified connecting class, including its full Gamma germ and rank $\min(m, r)$, then proves its place in every degree of the full supported derived comparison. The degree-two map is $(\eta, e) \mapsto (-\delta_q(\eta)/2, e)$. Thus its image in $C_{\text{zeta}}/qC_{\text{zeta}}$ is

zero, while the full supported map retains its Y/qY component. Both assertions follow from the constructed maps; no cone degree or extra coefficient is discarded.

These results refine the meaning of the prior finite-lift statement without withdrawing its proved scope. Fixed-annihilator splitting is stronger than the surjectivity in Deligne's invariant-cycle theorem. The original longer-chain map is already surjective. The present numerical character truncation retains its same-character extension whenever it retains the target; it is not Deligne's independently proved geometric weight gap. Complete proofs, independent checks and inspected reproducible diagrams have been inserted into the cumulative TeX. The actual tau weight-separation goal remains active.

150 An explicit global object in the original-zeta dual cokernel

24 September 2026. Complete derivation PGD0–PGD7. This continuation constructs a nonzero rational-function line in the actual algebraic comparison cokernel. It also constructs its entire representatives and exact maps. It does not identify that cokernel with a space of RH counterexamples.

150.1 PGD0. Why this is the next calculation, and which operations are present

The amended instruction USR-64a88219a3ecddd3 asks what would advance the target using the current finding and everything already known, and then asks us to attempt that calculation. ASD14 constructed the actual dual-comparison cone. SCL and DPL then proved that its degree-one cokernel has no nonzero polynomial-annihilated vector. The next question was whether anything actually remains in that cokernel. This note answers it by constructing a global element, rather than assuming nonvanishing or replacing the quotient by its zero Hausdorffization.

The controlling source is the complete preserved user corpus, in particular USR-4322be19bff532cd, the global arguments USR-9ad1c0a2d09dba92, USR-f55d16feb948d8c2 and USR-6152e3bc6302258c, and B1–B5/P1–P5 with retraction R1 in the source operations. The notation remains Z_0 for absence, Z_1/τ for primitive presence without parity or source addition, and Z_2 for the supplied integer parity data. Integer amounts are retained. The constant arithmetic function 1 below comes from the already reconstructed arithmetic layer; it is not primitive τ . No multiplication, addition, division or derivative below has primitive τ as an operand.

The receiving spaces, contour map, topologies and operation prerequisites are constructed in GZR0–GZR9, SCL0–SCL9, and DPL0–DPL8. The full source-faithful comparison is ASD14. All its other cohomology groups and extra closed copies remain present. This note concerns the named receiving cokernel, not an impossibility for the user’s source.

Human provenance is Alain Connes and Caterina Consani, *Schemes over F_1 and zeta functions*, arXiv:0903.2024v3 §5, for the actual coefficient geometry; Ralf Meyer, *A spectral interpretation for the zeros of the Riemann zeta function*, math/0412277v3, for the analytic framework, with the stronger original-space image theorem proved in OMS; and Pierre Deligne, *La conjecture de Weil II*, §§3.6.1–3.6.3, for the target weight-separated lifting argument. Hadamard’s full factorization is retained in GZR1/S2, with the indexed original Connes treatment, author TeX `rhready.tex`, lines526–535. The elementary new maps below are proved here, rather than attributed to those authors.

150.2 PGD1. Construct the larger multiplier domain and the exact contour map

Retain the actual Fréchet space

$$\mathcal{B} = \{F \text{ entire} : b_{A,N}(F) = \sup_{|\sigma| \leq A, t \in \mathbb{R}} (1 + |t|)^N |F(\sigma + it)| < \infty \text{ for every } A, N \geq 0\}.$$

Let $\mathcal{Z}$ be the actual nontrivial zero set of the original ζ , with multiplicities m_ρ , and retain

$$I = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ } (\rho \in \mathcal{Z}, 0 \leq j < m_\rho)\}, \quad Q = \mathcal{B}/I. \quad (\text{PGD1.1})$$

This is the original quotient, with its already proved Mellin comparison and quotient topology. It is not an unrestricted product of jets.

Define the receiving multiplier algebra

$$\mathcal{M} = \left\{ \begin{array}{l} H \text{ entire} : \text{for each } A \geq 0 \text{ there are } C_A > 0, d_A \geq 0 \\ \text{with } |H(\sigma + it)| \leq C_A(1 + |t|)^{d_A} \text{ whenever } |\sigma| \leq A \end{array} \right\}. \quad (\text{PGD1.2})$$

Here d_A can be taken an integer. Addition and multiplication preserve this domain: use the maximum and the sum of the two growth exponents, respectively. It contains $\mathcal{B}$, every polynomial, and $r^s = \exp(s \log r)$ for each positive real r . Multiplication by a fixed $H \in \mathcal{M}$ sends $\mathcal{B}$ continuously to itself, since

$$b_{A,N}(HF) \leq C_A b_{A,N+d_A}(F). \quad (\text{PGD1.3})$$

All quotients of $\mathcal{M}$ below are algebraic unless explicitly stated otherwise. No unconstructed topology on this union of growth classes is used.

For $H \in \mathcal{M}$, define a functional on the original quotient by

$$\lambda_H([F]) = \frac{1}{2\pi i} \left(\int_{2-i\infty}^{2+i\infty} - \int_{-1-i\infty}^{-1+i\infty} \right) \frac{H(s)F(1-s)}{\zeta(s)} ds. \quad (\text{PGD1.4})$$

Both edges are oriented upward; the integrals are taken separately before subtraction. The denominator is the original ζ .

Choose C, d bounding H on the strip $-2 \leq \Re s \leq 2$. The exact edge estimates proved in GZR2 are

$$|1/\zeta(2+it)| \leq \zeta(2), \quad |1/\zeta(-1+it)| \leq 4\pi^2 \zeta(2)(1+|t|)^{-3/2}.$$

Consequently both integrals converge absolutely and

$$|\lambda_H([F])| \leq \frac{C\zeta(2)(1+4\pi^2)}{\pi} b_{2,d+2}(F). \quad (\text{PGD1.5})$$

Indeed each edge is bounded using $(1+|t|)^{-2}$, whose integral over the real line is exactly 2. These estimates retain the constants of the original contour.

To prove descent, let $F \in I$. Reflection preserves all original zero multiplicities. Thus $H(s)F(1-s) \in \mathcal{B}$ by PGD1.3 and reflection, and it belongs to I . The actual OMS/SSI division theorem says that its quotient by ζ is holomorphic in a neighborhood of the closed strip $-1 \leq \Re s \leq 2$ and rapidly decreasing there. The horizontal sides of its rectangles therefore tend to zero. Cauchy's theorem gives equality of its two vertical integrals, proving $\lambda_H(F) = 0$. Taking the infimum in PGD1.5 over representatives proves continuity on the actual Q . Hence $\lambda_H \in Q'$, the continuous complex-linear dual, without replacing it by an algebraic dual.

The original endpoint behavior is unchanged: $\zeta(0) = -1/2$, while $1/\zeta$ has a zero at 1. No pole is introduced at either endpoint in PGD1.4. The trivial zeros $-2, -4, \dots$ remain outside the two stated edges. They have not been deleted from the source formula or enclosed by a replacement contour.

150.3 PGD2. Every full jet is detected, and the kernel is calculated exactly

For each $a \in \mathcal{X}$, retain the actual global isolators $E_{a,j} \in \mathcal{B}$ from GZR5/SCL1, for $0 \leq j < m_a$. They have the prescribed monomial jet at a and every required jet zero at every other actual zero. Their construction retains the entire auxiliary source transform

$$F_*(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad F_*(0) = F_*(1) = \frac{1}{8},$$

$$F_*(-2r) = \frac{(1+2r)2r}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r) \neq 0. \quad (\text{PGD2.1})$$

This auxiliary function is not substituted for the denominator of PGD1.4.

Put $b = 1 - a$, $m = m_a = m_b$, and write the original germ

$$\zeta(b+t) = t^m u_b(t), \quad u_b(t) = \sum_{k \geq 0} \frac{\zeta^{(m+k)}(b)}{(m+k)!} t^k, \quad u_b(0) \neq 0.$$

Then

$$\boxed{\lambda_H([E_{a,j}]) = (-1)^j [t^{m-1-j}] \frac{H(b+t)}{u_b(t)}.} \quad (\text{PGD2.2})$$

For its global justification, the numerator $H(s)E_{a,j}(1-s)$ belongs to $\mathcal{B}$ by PGD1.3. Its quotient by the original ζ has only the possible pole at b . GZR5's finite-pole contour theorem applies to this numerator: multiplying out its finitely many possible poles puts the numerator in I , and OMS rapid division kills the horizontal edges. Its residue is precisely PGD2.2. Thus no infinite formal residue sum has been substituted for the actual contour.

Define

$$\mathcal{J} = \{H \in \mathcal{M} : H^{(j)}(\rho) = 0 \ (\rho \in \mathcal{Z}, \ 0 \leq j < m_\rho)\}. \quad (\text{PGD2.3})$$

If $\lambda_H = 0$, PGD2.2 vanishes for every $j = 0, \dots, m-1$. These equations give every Taylor coefficient of $H(b+t)/u_b(t)$ through degree $m-1$, in reverse order with the stated signs. They are all zero. Since $u_b(0) \neq 0$, every corresponding coefficient of H is zero. Reflection permutes the actual zero set, so $H \in \mathcal{J}$.

Conversely, if $H \in \mathcal{J}$, then for every $F \in \mathcal{B}$ the product $H(s)F(1-s)$ lies in I . The same exact rapid-division argument as in PGD1 proves $\lambda_H(F) = 0$. We have proved

$$\boxed{\ker(H \mapsto \lambda_H) = \mathcal{J}.} \quad (\text{PGD2.4})$$

Every multiplicity is retained in both directions.

150.4 PGD3. The full multiplier quotient maps injectively into the actual cone

Write $D = \mathcal{D}_L$, so that for $F \in \mathcal{B}$, $\lambda_F = D[F]$ by the original definition. Set

$$Y = \chi \otimes Q', \quad (T_r^\diamond \lambda)(x) = r\lambda(T_{r^{-1}}x), \quad G = 1 - L^t, \quad \mathcal{C}_\zeta = Y/DQ. \quad (\text{PGD3.1})$$

This is the degree-one cokernel of ASD14 with its exact twist, not a new unspecified obstruction.

Since $I = \mathcal{B} \cap \mathcal{J}$, there is a commutative diagram of exact algebraic rows

$$\begin{array}{ccccccc} 0 & \longrightarrow & Q & \longrightarrow & \mathcal{M}/\mathcal{J} & \longrightarrow & \mathcal{M}/(\mathcal{B} + \mathcal{J}) \longrightarrow 0 \\ & & \parallel & & \downarrow \Theta & & \downarrow \overline{\Theta} \\ 0 & \longrightarrow & Q & \xrightarrow{D} & Y & \longrightarrow & \mathcal{C}_\zeta \longrightarrow 0, \end{array} \quad (\text{PGD3.2})$$

where $\Theta[H] = \lambda_H$ and $\overline{\Theta}[H] = [\lambda_H]$. Both vertical maps are injective. For the middle map this is PGD2.4. For the right map, its kernel means $\lambda_H = D[F] = \lambda_F$ for some $F \in \mathcal{B}$. PGD2.4 gives $H - F \in \mathcal{J}$, exactly $H \in \mathcal{B} + \mathcal{J}$. This proves the asserted kernel and hence injectivity. Exactness of the upper row follows directly from the two stated subspace quotients; exactness of the lower row is its actual image quotient. No surjectivity of Θ onto the full dual is claimed.

The generator and full real action are exact:

$$G\lambda_H = \lambda_{sH}, \quad T_r^\diamond \lambda_H = \lambda_{r^s H} \quad (r > 0). \quad (\text{PGD3.3})$$

For the first, the integrand for $\lambda_H - \lambda_H \circ L$ contains $H(s)(1-(1-s))F(1-s) = sH(s)F(1-s)$. For the second, the factor is exactly $r r^{-(1-s)} = r^s$. The growth domains proved in PGD1 justify each resulting integral. Multiplication by s and r^s preserves $\mathcal{B}, \mathcal{J}$ and $\mathcal{M}$, so these identities descend through every map in PGD3.2. They apply at every original prime $r = p$. No infinite-dimensional exponential has been inferred from a generator identity.

150.5 PGD4. A nonzero constant class and all polynomial classes

There are infinitely many actual nontrivial zeros with unbounded imaginary part. Here is the needed argument from the retained full factors, without a numerical RH assumption. The complete Hadamard formula GZR1/S2 is

$$F_*(s) = \frac{1}{8} e^{b_0 s} \prod_{\rho \in \mathcal{Z}} \left(1 - \frac{s}{\rho}\right)^{m_\rho} e^{m_\rho s / \rho}, \quad b_0 = \frac{1}{2} \log(4\pi) - 1 - \frac{\gamma}{2}. \quad (\text{PGD4.1})$$

If it had only finitely many zeros, for positive real σ this formula would give

$$\log |F_*(\sigma)| \leq -\log 8 + \sigma \left(|b_0| + \sum_{\rho} m_\rho / |\rho| \right) + \sum_{\rho} m_\rho \log(1 + \sigma / |\rho|). \quad (\text{PGD4.2})$$

The sums here are finite under the assumption being tested.

On the other hand, the exact original formula on $\sigma > 1$ is

$$F_*(\sigma) = \frac{\sigma(\sigma-1)}{8} \pi^{-\sigma/2} \Gamma(\sigma/2) \zeta(\sigma) > 0.$$

For $x \geq 2$, Euler's defining Gamma integral gives the elementary bound

$$\Gamma(x) = \int_0^\infty v^{x-1} e^{-v} dv \geq \int_x^{x+1} x^{x-1} e^{-x-1} dv = x^{x-1} e^{-x-1}.$$

The original Dirichlet series gives $\zeta(\sigma) \geq 1$. Thus for $\sigma \geq 4$, with all original factors retained,

$$\log F_*(\sigma) \geq \log \frac{\sigma(\sigma-1)}{8} - \frac{\sigma}{2} \log \pi + \left(\frac{\sigma}{2} - 1\right) \log \frac{\sigma}{2} - \frac{\sigma}{2} - 1. \quad (\text{PGD4.3})$$

After dividing the bounds by $\sigma \log \sigma$, the right side of PGD4.2 tends to 0 and the right side of PGD4.3 tends to $1/2$. This contradicts the two inequalities for the same function. Hence the zero set is infinite. Its zeros lie in the retained bounded critical strip, and an entire nonzero function has only finitely many zeros in each compact set. Their imaginary parts must therefore be unbounded.

Now let $P \neq 0$ be any complex polynomial. If $[\lambda_P] = 0$ in $\mathcal{C}_\zeta$, PGD3.2 supplies $F \in \mathcal{B}$ with $P - F \in \mathcal{J}$. In particular

$$F(\rho) = P(\rho) \quad (\rho \in \mathcal{Z}). \quad (\text{PGD4.4})$$

Along a sequence of these zeros of unbounded height, the left side tends to 0 by $b_{2,1}(F) < \infty$. A nonzero constant polynomial stays nonzero. For degree $d \geq 1$, writing its leading coefficient $p_d \neq 0$, the triangle inequality gives $|P(z)| \geq |p_d| |z|^d / 2$ for all sufficiently large $|z|$, because its finitely many lower terms have strictly smaller powers. Thus the right side cannot tend to 0. This contradiction proves

$$\boxed{P \mapsto [\lambda_P] \text{ is an injective map } \mathbb{C}[s] \longrightarrow \mathcal{C}_\zeta.} \quad (\text{PGD4.5})$$

In particular the explicit class

$$\boxed{c_\zeta = [\lambda_1] \neq 0, \quad \lambda_1([F]) = \frac{1}{2\pi i} \left(\int_{2-i\infty}^{2+i\infty} - \int_{-1-i\infty}^{-1+i\infty} \right) \frac{F(1-s)}{\zeta(s)} ds.} \quad (\text{PGD4.6})$$

This is an actual nonzero global cokernel class. It is not an off-critical zero and not a numerical or parity assignment to τ .

150.6 PGD5. Construct the rational-function line, including entire representatives

SCL5–SCL6 and DPL2–DPL3 prove that every nonzero polynomial in the actual G is bijective on $\mathcal{C}_\zeta$. Therefore the exact operation

$$\iota : \mathbb{C}(s) \longrightarrow \mathcal{C}_\zeta, \quad \iota(P/Q) = P(G)Q(G)^{-1}c_\zeta \quad (Q \neq 0) \quad (\text{PGD5.1})$$

is defined only after those inverses have been constructed. Equality of fractions gives equality of these operators by cross multiplication and the commuting inverses. This is linear over the receiving field $\mathbb{C}(s)$. It is injective: if $P/Q \neq 0$, then $P(G)$ and $Q(G)$ are bijective, so their composition cannot send the nonzero c_ζ to zero. In the polynomial case PGD3.3 identifies PGD5.1 with PGD4.5 exactly.

Here is an entire representative for every fraction; it keeps the correction used to remove every pole. Given P, Q , let $d = \deg Q$. For $d = 0$, set $B_{P,Q} = 0$ and $H_{P,Q} = P/Q$. For $d > 0$, prescribe at every root a of Q , through its multiplicity k_a , the Taylor coefficients of the entire function $e^{-s^2}P(s)$. There exists a unique polynomial $T_{P,Q}$ of degree less than d with these prescribed jets. Indeed the evaluation map from polynomials of degree less than d to these d jet coordinates is injective: a kernel polynomial is divisible by every $(s - a)^{k_a}$, hence by Q , and must be zero by its degree. Equal finite dimensions prove bijectivity and give its coefficients by an invertible finite linear system.

Define

$$B_{P,Q}(s) = e^{s^2}T_{P,Q}(s), \quad H_{P,Q}(s) = \frac{P(s) - B_{P,Q}(s)}{Q(s)}. \quad (\text{PGD5.2})$$

On each bounded vertical strip, $|e^{s^2}| = e^{(\Re s)^2 - (\Im s)^2}$, so $B_{P,Q} \in \mathcal{B}$. The interpolation makes the numerator vanish through the full order of every denominator root. Thus $H_{P,Q}$ extends to an entire function. At a root a of order k_a , its value is exactly

$$H_{P,Q}(a) = \frac{(P - B_{P,Q})^{(k_a)}(a)}{Q^{(k_a)}(a)}. \quad (\text{PGD5.3})$$

Higher removable jets are obtained by the full Taylor division of the same numerator and denominator. Outside fixed disks around the finitely many roots, polynomial division and PGD1.3 bound this function by a polynomial on each vertical strip. Inside those disks it is entire and bounded on compact sets. Hence $H_{P,Q} \in \mathcal{M}$.

The identity with every counterterm retained is

$$Q(s)H_{P,Q}(s) = P(s) - B_{P,Q}(s).$$

Applying PGD3.3 repeatedly gives

$$Q(G)\lambda_{H_{P,Q}} = \lambda_P - D[B_{P,Q}]. \quad (\text{PGD5.4})$$

Taking the actual cokernel and using the proved inverse yields

$$\boxed{[\lambda_{H_{P,Q}}] = \iota(P/Q)}. \quad (\text{PGD5.5})$$

This supplies actual continuous-dual representatives for the entire embedded field. Different choices of an interpolating function in $\mathcal{B}$ with the same required jets would give the same class: their difference divided by Q is entire and remains in $\mathcal{B}$, by fixed polynomial division and the removable local values. Thus the possible difference is explicitly a $D(Q)$ term, not an ignored pole correction.

The full positive-real orbit also has exact representatives

$$T_r^\diamond c_\zeta = [\lambda_{r^s}], \quad r > 0. \quad (\text{PGD5.6})$$

Each is nonzero: at all actual zeros, $|r^\rho|$ is bounded below by $\min(1, r) > 0$, whereas values of a member of $\mathcal{B}$ tend to 0 along unbounded heights. PGD3.2 therefore excludes its being in DQ . This gives the actual prime orbit at $r = p$, rather than assigning an unsupported eigenvalue to c_ζ . No assertion that this real orbit stays inside the particular rational-function line is needed or made.

150.7 PGD6. Source labels and the actual cone

The source receiving-ring action on the spectral term is the same scalar action as in ASD14. It acts on every map in PGD3.2 by linearity. The full source distinction between τ and integer1 remains on the faithful extra closed copies in the complete coefficient complex and its dual; this calculation does not remove them.

For the existing support lattice L_{supp} , every displayed receiving linear map has the exact lift

$$G_{L_{\text{supp}}}(f)(v, \alpha) = (f(v), \alpha), \quad G_{L_{\text{supp}}}(V) = \{(0, \alpha) : \alpha \in L_{\text{supp}}\} \cup \{(v, 1_{L_{\text{supp}}}) : v \in V\}. \quad (\text{PGD6.1})$$

If the input label is not top, its amplitude is zero and linearity sends it to zero. A nonzero output therefore has top support. Substitution proves preservation of identities and composition; injective maps remain injective. A zero amplitude retains its label. These are operations in the named receiving carrier, not on primitive τ .

ASD14 identifies $H^1(K_\zeta) = \mathcal{C}_\zeta$. Therefore PGD4 proves this actual algebraic cohomology group nonzero. It does not contradict its weak-* Hausdorff quotient being zero: DQ is dense, and an algebraic quotient by a dense proper subspace need not be zero. The equality of its Hausdorff quotient with zero and the explicit nonzero class PGD4.6 are retained together, with their distinct specified categories.

SCL/DPL still prove that this nonzero group contains no nonzero generalized eigenvector of G , no nonzero finite-dimensional G -invariant subspace, and no nonzero finite-dimensional equivariant quotient. Thus c_ζ is a genuine infinite-dimensional remaining object: the vectors $c_\zeta, Gc_\zeta, \dots$ are linearly independent, by PGD4.5. It cannot be substituted for an undetected single zero. All original zeros, on or off the critical line, already occur in the retained finite primary spaces of Q .

The other actual cone groups remain

$$H^{-1}(K_\zeta) = H_0 \oplus V_{\text{extra}}, \quad H^0(K_\zeta) = 0, \quad H^2(K_\zeta) = \chi \otimes (H'_0 \oplus V'_{\text{extra}}). \quad (\text{PGD6.2})$$

No full dual-comparison isomorphism is asserted.

150.8 PGD7. What was attempted and obtained

The preceding limitation was that weak-* density and finite-class lifting did not decide the entire algebraic cokernel. The attempted next step was to use the complete original contour with the globally present arithmetic unit, then test it against every retained multiplicity jet. PGD1–PGD4 prove that this produces an explicit nonzero class. PGD5 then answers the further question of what its whole polynomial and rational operator orbit is: an embedded copy of $\mathbb{C}(s)$, with entire corrected representatives for every fraction and the full real/prime orbit separately constructed. PGD3 supplies the larger exact multiplier quotient and its injective comparison.

These calculations settle those questions on the actual global receiver. They do not prove the requested numerical separation of geometric weights. They identify precisely why deleting the dual cokernel merely because its Hausdorff quotient is zero would lose an actual object. They also retain the proved finite-class lifts, so the newly constructed class is not used as a false counterexample to those lifts or to the user's primitive source. The next geometric calculation must use the actual comparison and its retained full cone rather than replace it by a perfect pairing. The current full tau lifting goal remains active.

150.9 Full global prime action and actual supported placement

The new complete proofs GGT0–GGT8, FPO0–FPO8 and UOS0–UOS9 (including UOS8A) strengthen the preceding global remainder calculation. The original-zeta Gaussian trace has a $T \log T$ leading term only at the identity scale and at most order T at every other fixed positive scale. Testing a proposed finite relation against every inverse scale proves all coefficients zero. Consequently the entire finite group algebra $C[\mathbb{R}_+^{\wedge \times}]$, including the Laurent algebra of all original primes, injects into the original multiplier quotient and linearly into the actual comparison cokernel. This computes relations after the actual zero-jet quotient, not only among entire functions before quotienting.

UOS constructs the full chain map into the actual dual-supported comparison cone. It sends c_r to $(c_r, 0)$, acts identically on H , and restricts $\chi H'$ to $\chi E'$ with every endpoint and extra closed copy retained. Its companion mirror is $c_r \rightarrow -r c_r(1/r)$ with the companion denominator; the oriented global-dual mirror has the opposite sign. The ordinary Fourier restriction still has its explicit section. Thus the new classes are relative residue-representation classes with proved supported placement, not a claimed failure of that existing Fourier lift. The source Z_0 , Z_1/τ and integer Z_2 have not changed, and the full numerical weight-separation target remains active.

The original Connes–Consani explicit formula has been applied with a proved cutoff extension; the exact archimedean comparison is $W_R(f_T, x) = A_T(x) + 1$, retaining the endpoint at zero and every finite trivial-zero contour correction. Complete human citations, author-source archive and bounded reading coverage are recorded with these proofs. All complete proof bodies and inspected reproducible diagrams are in the cumulative TeX.

151 Independent mathematical check of the global polynomial dual defect

24 September 2026. PGC0–PGC7. The complete manuscript GLOBAL_POLYNOMIAL_DUAL_DEFECT.md, PGD0–PGD7, was read and checked. The checked file has SHA256 ec1ea5fe1dbd49e91f2a24e19b5ad7232a67a10f45c193e70b8897f2331a41e8. No mathematical correction was required.

151.1 PGC0. Scope, source and prerequisites

This is an independent verification of the new multiplier quotient, nonzero polynomial class, corrected rational representatives and complete real orbit. It uses the already checked original quotient and residue construction GZR, the continuous-dual comparison ASD, and the division theorem SCL. It does not claim a new complete reading of every human source in those dependencies.

The source remains Z_0 , Z_1/τ , and the supplied Z_2 integer data under B1–B5 and their retained retractions. The arithmetic constant function 1 used by PGD is not primitive τ . All additive and rational operations in this verification occur in the explicitly constructed receiving algebras after the retained global arithmetic reconstruction.

Human provenance remains Connes–Consani’s coefficient geometry, Meyer’s analytic framework, Hadamard’s complete factorization and Deligne’s weight-separated lifting argument, with the exact source identities and prior reading coverage recorded in PGD0, GZR0 and ASD0. This check verifies PGD’s receiving construction; it establishes neither geometric purity nor an RH counterexample.

151.2 PGC1. The larger multiplier domain and the actual continuous dual

For fixed $H \in \mathcal{M}$, the bound

$$|H(\sigma + it)| \leq C_A(1 + |t|)^{d_A}$$

gives $b_{A,N}(HF) \leq C_A b_{A,N+d_A}(F)$. Therefore $\mathcal{B}$ is an ideal in the multiplier algebra $\mathcal{M}$, not merely a subset. The full-jet vanishing subspace $\mathcal{J}$ is also an ideal: multiplication by an entire germ preserves its prescribed vanishing order.

The contour formula

$$\lambda_H([F]) = \frac{1}{2\pi i} \left(\int_{\Re s=2}^{\uparrow} - \int_{\Re s=-1}^{\uparrow} \right) \frac{H(s)F(1-s)}{\zeta(s)} ds$$

uses the unchanged original denominator and both upward orientations. On those edges $1-s$ has real part -1 or 2 , so the exact input seminorm $b_{2,d+2}(F)$ controls the reflected numerator. The integral of $(1+|t|)^{-2}$ is 2; multiplying by $1/(2\pi)$ on each edge gives precisely PGD1.5’s constant $C\zeta(2)(1+4\pi^2)/\pi$.

For $F \in I$, the entire numerator $H(s)F(1-s)$ belongs to $\mathcal{B}$ and has every original zero jet required by I . The established rapid-division theorem therefore applies to that actual numerator. It makes the quotient by ζ holomorphic and rapidly decreasing on the stated closed strip; the horizontal edges vanish. This proves descent to the original Q and continuity through its quotient seminorm, rather than asserting an algebraic functional belongs to the continuous dual.

151.3 PGC2. The exact multiplier kernel and the quotient injection

Testing λ_H on the actual isolator $E_{a,j}$, with $b = 1 - a$, $m = m_a$, yields

$$\lambda_H([E_{a,j}]) = (-1)^j [t^{m-1-j}] \frac{H(b+t)}{u_b(t)}.$$

The finite-pole contour argument remains valid with this first operand: its whole numerator $H(s)E_{a,j}(1-s)$ lies in $\mathcal{B}$. Multiplication by $(s-b)^m$ removes the sole possible pole and puts the numerator in I ; the same rapid-division estimate justifies the contour limit. This avoids applying a two- $\mathcal{B}$ -operand theorem without checking its actual numerator.

Since $u_b(0) \neq 0$, vanishing of the displayed values for all $0 \leq j < m$ is equivalent to the full m -jet of H at b vanishing. Reflection permutes the actual zeros. Conversely $H \in \mathcal{J}$ makes every tested product belong to I , so the complete kernel is exactly $\mathcal{J}$.

It follows without a closure or dual-surjectivity assumption that

$$\lambda_H \in \mathcal{D}_L Q \iff \exists F \in \mathcal{B} : \lambda_H = \lambda_F \iff H - F \in \mathcal{J} \iff H \in \mathcal{B} + \mathcal{J}.$$

This proves the stated injection

$$\mathcal{M}/(\mathcal{B} + \mathcal{J}) \hookrightarrow (\chi \otimes Q')/\mathcal{D}_L Q.$$

The upper row in PGD3.2 is exact because $\mathcal{B} \cap \mathcal{J} = I$. The lower row is the specified actual image quotient. The diagram is an exact diagram of receiving vector spaces and equivariant maps; no multiplication on the whole dual is claimed.

151.4 PGC3. The Gamma-integral proof of infinitely many zeros

The finite-zero alternative in the retained full Hadamard product gives $\log |F_*(\sigma)| \leq O(\sigma) + O(\log \sigma)$ on positive real σ , with the complete constants explicitly retained in PGD4.2.

For $x \geq 2$ and $v \in [x, x+1]$, both $v^{x-1} \geq x^{x-1}$ and $e^{-v} \geq e^{-x-1}$. Thus Euler's Gamma integral gives exactly

$$\Gamma(x) \geq x^{x-1} e^{-x-1}.$$

Together with the original $\zeta(\sigma) \geq 1$, this proves PGD4.3 with every factor of

$$F_*(\sigma) = \frac{\sigma(\sigma-1)}{8} \pi^{-\sigma/2} \Gamma(\sigma/2) \zeta(\sigma)$$

present. After division by $\sigma \log \sigma$, the upper bound under the finite-zero assumption tends to 0, whereas the lower bound tends to $1/2$. This is a contradiction.

An infinite zero set in the bounded critical strip has unbounded imaginary parts because a nonzero entire function has finitely many zeros in each compact rectangle. Thus the proof requires neither Stirling's formula nor RH, simple zeros or any numerical zero experiment.

151.5 PGC4. Polynomial nonvanishing and the exact rational action

If $P \neq 0$ and $[\lambda_P] = 0$, PGC2 forces $F(\rho) = P(\rho)$ at every actual zero for some $F \in \mathcal{B}$. Along unbounded zero heights, $F(\rho) \rightarrow 0$. A nonzero constant does not tend to zero. For degree $d \geq 1$, the leading term gives $|P(z)| \geq |p_d||z|^d/2$ for all sufficiently large $|z|$. Neither case is compatible with the equality. Thus the polynomial map is injective, and the explicitly defined $c_\zeta = [\lambda_1]$ is nonzero.

The generator calculation is exact:

$$(1 - L^t)\lambda_H = \lambda_{sH},$$

because applying $1 - L$ to the test function at $1 - s$ multiplies it by s . Thus the variable of the rational-function action is the actual twisted generator $G = 1 - L^t$, not an untwisted operator silently substituted for it.

SCL's bijectivity of every nonzero polynomial in G makes $P(G)Q(G)^{-1}c_\zeta$ defined. Cross multiplication proves independence of the fraction representation. A nonzero rational function acts by a bijection on the cokernel, so it cannot kill c_ζ . This proves the entire embedded $\mathbb{C}(s)$ -line, including its nonvanishing.

151.6 PGC5. Every rational class has the stated entire representative

For a denominator root a of order k_a , the Hermite conditions mean the derivatives of orders $0 \leq j < k_a$. Their total number is $d = \deg Q$. The evaluation map on polynomials of degree below d has zero kernel, since a polynomial in that kernel is divisible by the full denominator, including every multiplicity. Equal finite dimensions therefore give the unique interpolation polynomial $T_{P,Q}$.

The condition $T_{P,Q} - e^{-s^2} P = O((s-a)^{k_a})$ implies $e^{s^2} T_{P,Q} - P = O((s-a)^{k_a})$, because e^{s^2} is entire and nonzero. Consequently

$$B_{P,Q} = e^{s^2} T_{P,Q} \in \mathcal{B}, \quad H_{P,Q} = (P - B_{P,Q})/Q$$

is entire. The Gaussian identity $|e^{(\sigma+it)^2}| = e^{\sigma^2-t^2}$ proves the required rapid decrease of $B_{P,Q}$ on every bounded strip.

Taylor division at a gives $H_{P,Q}(a) = (P - B_{P,Q})^{(k_a)}(a)/Q^{(k_a)}(a)$; the two factors $k_a!$ cancel because they occur in both leading Taylor coefficients. For every higher coefficient, division by the nonzero leading denominator coefficient gives the usual uniquely determined Taylor recursion. No pole coefficient has been discarded.

At large $|t|$ in a bounded vertical strip, a degree- d polynomial denominator has a uniform lower bound $c(1+|t|)^d$. Outside that large region and the finitely many root disks, it has a positive compact lower bound; inside the disks the quotient is holomorphic and bounded on their closures after a harmless enlargement. These estimates prove $H_{P,Q} \in \mathcal{M}$. They also prove that changing the interpolating $\mathcal{B}$ -function with the same full jets changes $H_{P,Q}$ by an actual member of $\mathcal{B}$.

Finally

$$Q(G)\lambda_{H_{P,Q}} = \lambda_{QH_{P,Q}} = \lambda_P - \mathcal{D}_L[B_{P,Q}]$$

retains the exact correction. Taking the actual cokernel and its previously constructed inverse proves PGD5.5. There is no hidden division by a meromorphic function on the original test-space domain.

151.7 PGC6. Complete real orbit, source labels and cone scope

For the original twisted action,

$$(T_r^\diamond \lambda_H)([F]) = r \lambda_H([r^{-s} F(s)]) = \lambda_{r^s H}([F]).$$

The factor in the contour is exactly $r r^{-(1-s)} = r^s$. Each $r^s H$ belongs to $\mathcal{M}$ because $r > 0$ and $|r^{\sigma+it}| = r^\sigma$ is bounded on every bounded strip. This verifies the full real action directly; prime actions are its actual instances $r = p$.

For $H = r^s$, the values at every nontrivial zero satisfy $|r^\rho| \geq \min(1, r) > 0$. They cannot be the values of an element of $\mathcal{B}$ along unbounded zero heights. Therefore each orbit class remains nonzero. PGD correctly makes no assertion that the orbit stays inside the particular rational-function line.

The maps preserve the existing receiving support labels by their exact linear lifts. The primitive source τ is not an amplitude or the function 1. The faithful extra closed copies and all other cohomology groups of ASD14 remain. The actual algebraic cokernel is now proved nonzero, while its weak-* Hausdorff quotient remains zero; these two statements concern the explicitly different quotient constructions and do not contradict one another.

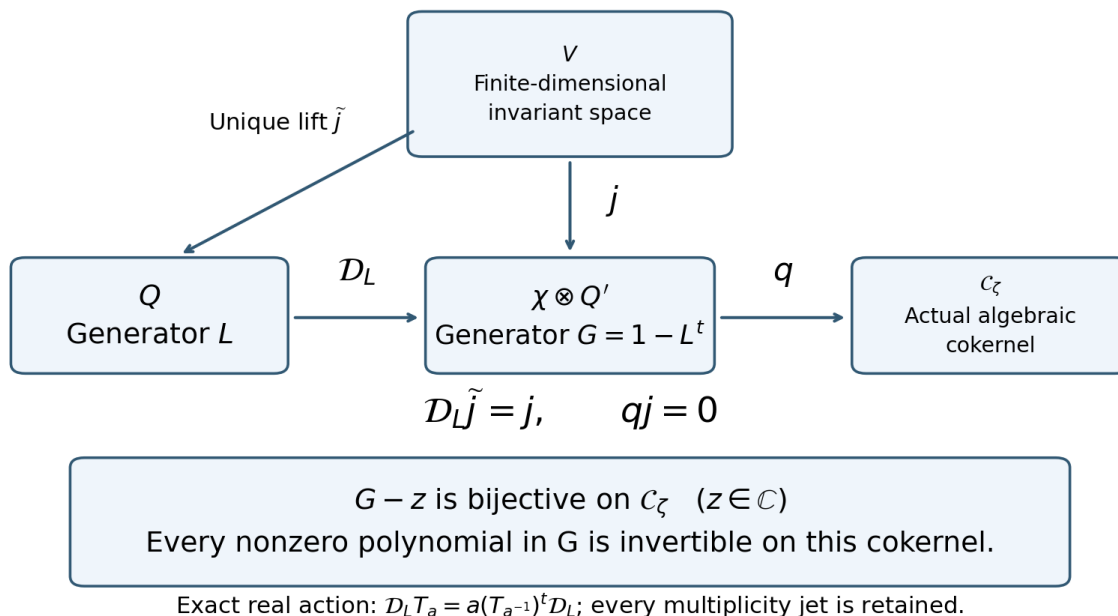
151.8 PGC7. Result of the check

Every requested part of PGD0–PGD7 is verified: the larger multiplier quotient injection, the elementary Gamma-integral argument, the full multiplicity tests, the explicit polynomial nonzero

class, the whole embedded rational-function line with entire Hermite-corrected representatives, and the complete real/prime orbit. All constants, derivative orders, signs and operation domains checked above agree with the stated proof. No correction to PGD was required.

The conclusion is the constructed nonzero infinite-dimensional receiving object in the actual comparison cone. It is not an unidentified RH counterexample, a contradiction of primitive Z_1/τ , or a replacement for the requested geometric weight separation. This bounded independent check ends here.

The actual dual comparison: finite classes lift



Exact real action: $D_L T_a = a(T_{a^{-1}})^t D_L$; every multiplicity jet is retained.

The cokernel is not assumed zero. Finite-class lifting does not assign a numerical weight to Q .
Source Z_0 , Z_1 / τ and integer Z_2 are unchanged. Proofs: ASD14 and the finite dual-cokernel continuation.
Geometry: Connes–Consani, arXiv:0903.2024v3 §5. Weight-lifting target: Deligne, Weil II §§3.6.1–3.6.3.

Figure 53: Finite classes lift

152 Illustrations of the actual dual-comparison continuation

Figure 1. Complete proofs: SCL2–SCL9 and DPL1–DPL9. The actual target is the character-a-twisted continuous dual, with generator $G=1-L^t$. The upper space is finite-dimensional and invariant for the specified receiving action; its lift is unique. The whole algebraic cokernel is retained. This picture assigns no weight or parity to primitive τ .

Figure 2. Complete proof: PGD1–PGD6; independent verification PGC0–PGC7. Both contour edges point upward. All original-zeta factors, multiplicity jets, and the rational representative’s entire correction are retained. The polynomial T has the exact prescribed derivatives of orders 0 through $k-1$ at every denominator root of multiplicity k . The arithmetic constant1 is not primitive τ . The class is not a claimed RH counterexample.

Reproducible sources: `check_and_draw_finite_dual_lift.py`, with SVG files beside these PNGs. The 36 finite symbolic checks supplement the complete global analytic proofs. Both final figures were inspected and their spacing corrected.

Human sources: Alain Connes and Caterina Consani, Schemes over F_1 and zeta functions, §5; Ralf Meyer, Spectral interpretation; Pierre Deligne, Weil II, §§3.6.1–3.6.3; Hadamard’s product through the indexed original Connes treatment. Exact versions, source files and actual reading coverage remain in `SOURCE_READING_USE_PRIVATE.json`.

A constructed global remainder in the actual dual cone

$$c_{\zeta} = [\lambda_1] \neq 0$$

$$\lambda_1([F]) = \frac{1}{2\pi i} \left(\int_{\Re s = 2} - \int_{\Re s = -1} \right) \frac{F(1-s)}{\zeta(s)} ds$$

Every finite invariant dual space
has its unique lift to Q .
All multiplicity jets remain.

A global lift of λ_1 would require
 $F(\rho) = 1$ at every original zero.
 $F \in \mathcal{B}$ instead decays at unbounded heights.

$$\mathbb{C}(s) \hookrightarrow \mathbb{C}_{\zeta}, \quad \frac{P}{Q} \mapsto P(G)Q(G)^{-1}c_{\zeta}, \quad G = 1 - L^t$$

Entire representative with all pole corrections:

$$H_{P,Q}(s) = \frac{P(s) - e^{s^2} T_{P,Q}(s)}{Q(s)}$$

$T_{P,Q}$ retains the full required jets of $e^{-s^2}P$ at every root of Q .

Proofs PGD1–PGD6 and SCL5–SCL8. Both contour edges are upward. The constant 1 is arithmetic, not τ .
This is a global class in a specified receiving cokernel; it is not an off-critical zeta zero.
Connes–Consani, arXiv:0903.2024v3 §5; Deligne, Weil II §§3.6.1–3.6.3. Complete source labels are retained.

Figure 54: The explicit global remainder

153 Global Gaussian trace of the original zeta function

24 September 2026. Independent mathematical derivation GGT0–GGT8.

153.1 GGT0. Source, original objects, and the next calculation

The preceding PGD0–PGD7 construct the actual nonzero class $c_\zeta = [\lambda_1]$ in the residue-duality cokernel and its complete real orbit. The next calculation is a global original-zeta trace that distinguishes the identity dilation from every other fixed dilation. This note derives that trace with all prime, endpoint, Gamma, and crossed trivial-zero terms, then proves the asymptotics needed for the orbit calculation.

The controlling source remains Z_0 absence, primitive Z_1/τ presence without parity or source addition, and the supplied arithmetic Z_2 data. The source operations, their retractions, and the complete global corpus arguments referenced in `CORPUS_AND_OPERATION_RULES.md` remain controlling. Its later instruction USR-64a88219a3ecddd3 asks what would advance the calculation using everything already known, and then asks for that calculation. The constant arithmetic function 1 and the positive real numbers below belong to the already reconstructed receiving arithmetic. None is identified with primitive τ .

PGD was read completely for this continuation. The analytic input is the fully proved original logarithmic-derivative identity JTB4, whose complete proof was independently checked in JTC3. It is based on the retained Hadamard product, with every genus-one exponential and the complete original multiplier. Its human-source attribution is Jacques Hadamard’s factorization, as entered through the indexed author TeX of Alain Connes, arXiv:2602.04022v1, rhready.tex lines526–535. This is the recorded prior source reading, not a claim to have freshly read the entire paper here.

The original coefficient and analytic contexts remain Alain Connes and Caterina Consani, arXiv:0903.2024v3, §5, and Ralf Meyer, arXiv:math/0412277v3, as entered in ASD and OMS with their exact source versions. Their geometric or range results are not being credited with the new Gaussian calculation. The elementary Euler-product and Gamma estimates required below are proved explicitly; no complex Stirling estimate with unspecified branches is assumed.

Write $\mathcal{Z}$ for the actual distinct nontrivial zeros of the original ζ , with every actual multiplicity m_ρ . Let $T \geq 1$, $x > 0$, with $\log x$ the real logarithm. Define the entire test function and the full zero sum

$$\Phi_{T,x}(s) = x^s e^{s^2/T^2} = \exp(s \log x + s^2/T^2), \quad S_T(x) = \sum_{\rho \in \mathcal{Z}} m_\rho x^\rho e^{\rho^2/T^2}. \quad (\text{GGT0.1})$$

The variable T is a test-function parameter, not an assigned clock, coordinate, or weight on primitive τ .

153.2 GGT1. Convergence and the full original logarithmic derivative

The original receiving test algebra is

$$\mathcal{B} = \{F \text{ entire} : b_{A,N}(F) = \sup_{|\sigma| \leq A, t \in \mathbb{R}} (1 + |t|)^N |F(\sigma + it)| < \infty \text{ for every } A, N\}. \quad (\text{GGT1.1})$$

For fixed T, x , the exact modulus is

$$|\Phi_{T,x}(\sigma + it)| = x^\sigma e^{\sigma^2/T^2} e^{-t^2/T^2}, \quad (\text{GGT1.2})$$

so $\Phi_{T,x} \in \mathcal{B}$. The already proved zero count $n(R) \leq C(R + 2)^{3/2}$, with multiplicities, gives absolute convergence of (GGT0.1), since

$$|x^\rho e^{\rho^2/T^2}| \leq \max(1, x) e^{1/T^2} e^{-(\Im \rho)^2/T^2}. \quad (\text{GGT1.3})$$

Summation over integer height intervals proves the claim from the polynomial count and Gaussian decay. In particular no ordering of conditionally convergent zero terms is being chosen.

For an integrable meromorphic function on the two edges, retain

$$\mathfrak{C}(F) = \frac{1}{2\pi i} \left(\int_{2-i\infty}^{2+i\infty} F(s) ds - \int_{-1-i\infty}^{-1+i\infty} F(s) ds \right). \quad (\text{GGT1.4})$$

Both integrals run upward. JTB4 proves, for every $\Phi \in \mathcal{B}$,

$$\mathfrak{C}\left(\Phi \frac{\zeta'}{\zeta}\right) = \sum_{\rho \in \mathcal{Z}} m_{\rho} \Phi(\rho) - \Phi(1). \quad (\text{GGT1.5})$$

For clarity, its exact retained factor comparison is

$$\begin{aligned} F_*(s) &= \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \\ \frac{F'_*}{F_*} &= b_0 + \sum_{\rho \in \mathcal{Z}} m_{\rho} \left(\frac{1}{s-\rho} + \frac{1}{\rho} \right), \quad b_0 = \frac{1}{2} \log(4\pi) - 1 - \frac{\gamma}{2}, \\ \frac{F'_*}{F_*} &= \frac{\zeta'}{\zeta} + \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2} \log \pi + \frac{1}{2} \psi(s/2), \quad \psi = \Gamma'/\Gamma. \end{aligned} \quad (\text{GGT1.6})$$

On the two edges the sum of absolute values of the paired genus-one terms is $O((1+|t|)^{3/2})$, by the retained zero count and $|s-\rho| \geq 1$. Thus it can be integrated term by term against Φ . Each paired term gives its single residue, and the multiplier derivative in the last line has cancelling residues at zero and residue +1 at one. Its subtraction gives the original pole contribution $-\Phi(1)$ in (GGT1.5). These are the actual reasons that both the zero sum and the edge integrals are absolutely convergent. F_* has not replaced ζ in (GGT1.5).

Applied to (GGT0.1), the exact starting equation is

$$S_T(x) = x e^{1/T^2} + \mathfrak{C}\left(\Phi_{T,x} \frac{\zeta'}{\zeta}\right). \quad (\text{GGT1.7})$$

153.3 GGT2. The Gaussian Mellin integral and the original prime series

For every real c and $y > 0$, direct parameterization gives

$$\begin{aligned} \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} y^s e^{s^2/T^2} ds &= \frac{y^c e^{c^2/T^2}}{2\pi} \int_{\mathbb{R}} e^{-t^2/T^2} e^{it(\log y + 2c/T^2)} dt \\ &= \frac{T}{2\sqrt{\pi}} y^c e^{c^2/T^2} \exp \left[-\frac{T^2}{4} \left(\log y + \frac{2c}{T^2} \right)^2 \right] \\ &= \frac{T}{2\sqrt{\pi}} \exp \left[-\frac{T^2}{4} \log^2 y \right]. \end{aligned} \quad (\text{GGT2.1})$$

The last equality retains and evaluates both cross terms; no factor depending on c or y has been omitted. The Fourier Gaussian used in the second line can be verified by differentiating its integral in the frequency κ . Integration by parts gives $J'(\kappa) = -T^2 \kappa J(\kappa)/2$, while the Gaussian integral gives $J(0) = T\sqrt{\pi}$. Solving this scalar differential equation proves exactly the displayed factor and exponent.

Define the original von Mangoldt coefficients by

$$\Lambda(n) = \begin{cases} \log p, & n = p^k, \text{ } p \text{ prime, } k \geq 1, \\ 0, & \text{otherwise.} \end{cases} \quad (\text{GGT2.2})$$

They are defined using the retained original arithmetic primes, not a substituted spectrum. On $\Re s > 1$, the absolutely convergent Euler product yields

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_p \sum_{k \geq 1} (\log p) p^{-ks} = \sum_{n \geq 2} \Lambda(n) n^{-s}. \quad (\text{GGT2.3})$$

For the differentiation, restrict first to any $\Re s \geq 1 + \varepsilon$. The logarithmic series $\sum_{p,k} p^{-ks}/k$ and its differentiated series converge uniformly there, since the latter is bounded by $\sum_{n \geq 2} (\log n) n^{-1-\varepsilon}$. This proves the identity before evaluating either edge.

The original functional equation, with every Gamma factor, is

$$\zeta(s) = \chi(s) \zeta(1-s), \quad \chi(s) = \pi^{s-1/2} \frac{\Gamma((1-s)/2)}{\Gamma(s/2)} = 2^s \pi^{s-1} \sin(\pi s/2) \Gamma(1-s). \quad (\text{GGT2.4})$$

Its logarithmic derivative is

$$\frac{\zeta'(s)}{\zeta(s)} = g(s) - \frac{\zeta'(1-s)}{\zeta(1-s)}, \quad g(s) = \frac{\chi'(s)}{\chi(s)} = \log \pi - \frac{1}{2} \psi((1-s)/2) - \frac{1}{2} \psi(s/2). \quad (\text{GGT2.5})$$

On the left edge $\Re s = -1$, equations (GGT2.3) and (GGT2.5) give

$$\frac{\zeta'(s)}{\zeta(s)} = g(s) + \sum_{n \geq 2} \frac{\Lambda(n)}{n} n^s. \quad (\text{GGT2.6})$$

Each series can be interchanged with its edge integral: on the right its absolute coefficient sum is $\sum \Lambda(n) n^{-2} < \infty$; on the left the coefficient absolute values are again $\Lambda(n) n^{-2}$. The remaining test function is integrable by (GGT1.2). This proves an absolute Fubini bound before the Gaussian integral is evaluated.

153.4 GGT3. The exact trace, with its full archimedean term

Define, on the original left edge,

$$A_T(x) = \frac{1}{2\pi i} \int_{-1-i\infty}^{-1+i\infty} x^s e^{s^2/T^2} \frac{\chi'(s)}{\chi(s)} ds. \quad (\text{GGT3.1})$$

The estimates below prove absolute convergence directly. Substituting (GGT2.3), (GGT2.6), and (GGT2.1) into the two distinct integrals of (GGT1.7) proves

$$S_T(x) = x e^{1/T^2} - \frac{T}{2\sqrt{\pi}} \sum_{n \geq 2} \Lambda(n) \left[\exp\left(-\frac{T^2}{4} \log^2(x/n)\right) + \frac{1}{n} \exp\left(-\frac{T^2}{4} \log^2(xn)\right) \right] - A_T(x). \quad (\text{GGT3.2})$$

The sign of the second prime family is negative because it is a positive series on the subtracted left edge. Its coefficient is exactly $1/n$. The original pole term is exactly $x e^{1/T^2}$.

Both prime sums converge absolutely. In addition to the Fubini argument, their final terms are bounded by constant multiples of $(\log n) \exp[-c(\log n)^2]$ for fixed x, T ; this is summable, for example by comparison with n^{-2} after increasing a finite cutoff. Every prime and prime-power coefficient remains.

The original endpoint and trivial-zero information can also be exposed by finite contour shifts, without replacing (GGT3.1). The meromorphic function $g = \chi'/\chi$ has residue $+1$ at

$$0, -2, -4, \dots$$

and residue -1 at

$$1, 3, 5, \dots$$

These follow from the simple Gamma poles in the numerator and denominator of the full quotient (GGT2.4). Define $A_T^{(c)}(x)$ by the same integral as (GGT3.1) on any vertical edge c avoiding those real poles. For a fixed finite integer $N \geq 0$, the residue theorem gives

$$A_T(x) = A_T^{(-2N-1)}(x) + \sum_{r=1}^N x^{-2r} e^{4r^2/T^2}, \quad (\text{GGT3.3})$$

$$A_T(x) = A_T^{(1/2)}(x) - 1. \quad (\text{GGT3.4})$$

The second identity crosses the pole of g at zero with residue $\Phi_{T,x}(0) = 1$. Thus the equivalent identity

$$S_T(x) = 1 + xe^{1/T^2} - \text{the two full prime sums in (GGT3.2)} - A_T^{(1/2)}(x) \quad (\text{GGT3.5})$$

has both endpoint contributions displayed. Equations (GGT3.3)–(GGT3.4) are exact comparisons, not discarded terms. The finite-shift horizontal integrals tend to zero by Gaussian decay and the fixed-strip Gamma estimates proved in GGT4.

The negative even points in (GGT3.3) are the original simple trivial zeros: (GGT2.4), together with $\zeta(1+2r) \neq 0$, gives

$$\zeta'(-2r) = (-1)^r 2^{-2r-1} \pi^{-2r} (2r)! \zeta(1+2r) \neq 0. \quad (\text{GGT3.6})$$

Their logarithmic-derivative residue is one, explaining the coefficient in (GGT3.3). An infinite left shift is not justified: for fixed T , the terms $x^{-2r} e^{4r^2/T^2}$ eventually grow. The full remaining contour in every finite identity is retained.

153.5 GGT4. A complete branch-controlled Gamma estimate

We prove constants C_0, C_1, C_2 such that, for $t \in \mathbb{R}$,

$$|g(-1+it)| \leq C_0 \log(2+|t|), \quad \left| \frac{d}{dt} g(-1+it) \right| \leq \frac{C_1}{1+|t|}, \quad (\text{GGT4.1})$$

and, for $|t| \geq 2$,

$$g(-1+it) = \log(2\pi) - \log|t| + r(t), \quad |r(t)| \leq C_2/|t|. \quad (\text{GGT4.2})$$

The remainder can be complex; the formula does not silently replace the original function by its real part.

Here is an elementary integral estimate for ψ . For $\Re z > 0$, use the principal logarithm, whose argument lies in $(-\pi/2, \pi/2)$. Euler's Gamma limit gives

$$\psi(z) = \lim_{N \rightarrow \infty} \left(\log N - \sum_{n=0}^{N-1} \frac{1}{n+z} \right). \quad (\text{GGT4.3})$$

The limit follows, with locally uniform first derivatives, from

$$\Gamma(z) = \lim_{N \rightarrow \infty} \frac{N! N^z}{z(z+1) \cdots (z+N)}.$$

For completeness, integration by parts in the beta integral gives the displayed finite ratio before the factor N^z . After the substitution $v = Nu$, that factor times the beta integral is $\int_0^N v^{z-1} (1-v/N)^N dv$. The bound $(1-v/N)^N \leq e^{-v}$ on $0 \leq v < N$, including logarithmic

factors for derivatives, proves locally uniform convergence to the defining Gamma integral. Its logarithmic derivative gives (GGT4.3); the extra final summand with index N tends to zero.

Let $\tilde{B}_1(u) = u - \lfloor u \rfloor - \frac{1}{2}$ off the integers. It is bounded in absolute value by $1/2$. On each integer interval, integrate $f(u) = (u + z)^{-1}$ once against this function. Summing the endpoint terms yields

$$\sum_{n=0}^{N-1} \frac{1}{n+z} = \text{Log}(N+z) - \text{Log } z + \frac{1}{2z} - \frac{1}{2(N+z)} - \int_0^N \frac{\tilde{B}_1(u)}{(u+z)^2} du.$$

All paths $u+z$ lie in the right half-plane, so the logarithms here use the same fixed branch. Taking the limit in (GGT4.3) proves

$$\begin{aligned} \psi(z) &= \text{Log } z - \frac{1}{2z} + \int_0^\infty \frac{\tilde{B}_1(u)}{(u+z)^2} du, \\ \psi'(z) &= \frac{1}{z} + \frac{1}{2z^2} - 2 \int_0^\infty \frac{\tilde{B}_1(u)}{(u+z)^3} du. \end{aligned} \quad (\text{GGT4.4})$$

Absolute convergence on every compact subset of $\Re z > 0$ justifies differentiation.

If $z = c + iy$, $c > 0$, $|y| \geq 1$, then

$$\int_0^\infty \frac{du}{|u+z|^2} \leq \frac{\pi}{2|y|}, \quad \int_0^\infty \frac{du}{|u+z|^3} \leq \frac{1}{|y|^2}.$$

Consequently (GGT4.4) gives

$$\psi(c + iy) = \text{Log}(c + iy) + O(|y|^{-1}), \quad \psi'(c + iy) = \frac{1}{c + iy} + O(|y|^{-2}), \quad (\text{GGT4.5})$$

uniformly for c in any fixed compact interval of positive real numbers. These estimates have been proved from the integral, rather than taken from an unspecified angular sector in a Stirling formula.

For the actual line $s = -1 + it$, put

$$y = t/2, \quad z_1 = 1 - iy, \quad z_2 = \frac{1}{2} + iy, \quad z_0 = -\frac{1}{2} + iy.$$

The recurrence $\psi(z+1) = \psi(z) + 1/z$ and (GGT2.5) give the exact identity

$$g(-1 + it) = \log \pi - \frac{1}{2}\psi(z_1) - \frac{1}{2}\psi(z_2) + \frac{1}{2z_0}. \quad (\text{GGT4.6})$$

Both digamma arguments in this expression have positive real part, and z_0 is never zero for real t .

For $|y| \geq 1$, the principal logarithms satisfy

$$\text{Log } z_1 + \text{Log } z_2 = 2 \log |y| + \text{Log} \left(1 + \frac{1}{2y^2} + \frac{i}{2y} \right). \quad (\text{GGT4.7})$$

Indeed $z_1 z_2 = y^2 + \frac{1}{2} + \frac{i}{2}y$ has positive real part, and the sum of their arguments is the principal argument of that product. No multiple of $2\pi i$ enters. Along the segment from 1 to $1+w$, with $w = 1/(2y^2) + i/(2y)$, the real part is at least one. Hence the last logarithm has absolute value at most $|w| \leq |y|^{-1}$. Equations (GGT4.5)–(GGT4.7) prove (GGT4.2), including both signs of t .

Differentiating the exact (GGT4.6) gives

$$\frac{d}{dt} g(-1 + it) = \frac{i}{4}\psi'(z_1) - \frac{i}{4}\psi'(z_2) - \frac{i}{4z_0^2}. \quad (\text{GGT4.8})$$

Equation (GGT4.5) bounds this by $C/|t|$ for $|t| \geq 2$. On the compact interval $[-2, 2]$, every term and its derivative is continuous, so it is bounded. This proves (GGT4.1) globally. The same recurrence argument shifts both digamma arguments into a fixed positive-real-part strip for any other fixed bounded range of $\Re s$, at large $|\Im s|$. It proves $g(s) = O(\log(2 + |\Im s|))$ uniformly on that range, which justifies the finite shifts in GGT3.

153.6 GGT5. The archimedean integral away from the identity

Parameterize the exact left-edge integral without discarding its phase:

$$A_T(x) = \frac{x^{-1}e^{1/T^2}}{2\pi} \int_{\mathbb{R}} e^{-t^2/T^2} e^{it(\log x - 2/T^2)} g(-1 + it) dt. \quad (\text{GGT5.1})$$

GGT4 and Gaussian decay prove absolute convergence. Let

$$h_T(t) = e^{-t^2/T^2} g(-1 + it), \quad \kappa_T = \log x - 2/T^2.$$

Its derivative is integrable, and (GGT4.1) gives

$$\begin{aligned} \int_{\mathbb{R}} |h'_T(t)| dt &\leq C \int_{\mathbb{R}} \frac{2|t|}{T^2} e^{-t^2/T^2} \log(2 + |t|) dt \\ &\quad + C \int_{\mathbb{R}} \frac{e^{-t^2/T^2}}{1 + |t|} dt \\ &\leq C' \log(2 + T). \end{aligned} \quad (\text{GGT5.2})$$

For the first bound substitute $t = Tu$ and use $\log(2 + T|u|) \leq \log(2 + T) + \log(1 + |u|)$ for $T \geq 1$. For the second, split at $|t| = T$; the inner integral is bounded by $2 \log(1 + T)$, and substitution in the outer Gaussian tail gives a constant. This proves the last line with constants independent of T .

For fixed $x \neq 1$, all sufficiently large T satisfy

$$|\kappa_T| \geq |\log x|/2.$$

The boundary values of h_T vanish. Integration by parts in (GGT5.1), with the exact frequency κ_T , therefore gives

$$\left| \int_{\mathbb{R}} h_T(t) e^{i\kappa_T t} dt \right| \leq \frac{2C'}{|\log x|} \log(2 + T).$$

For the remaining bounded interval of $T \geq 1$, the original absolute integral in (GGT5.1) is uniformly bounded by a fixed integrable Gaussian times $\log(2 + |t|)$. Increasing the constant thus proves

$$\boxed{A_T(x) = O_x(\log(2 + T)) \quad (x \neq 1, T \geq 1).} \quad (\text{GGT5.3})$$

In particular any finite value of T at which $\kappa_T = 0$ has been treated by the absolute integral, not by an invalid division by zero.

153.7 GGT6. The archimedean integral at the identity, with its next constant

At $x = 1$, write for $t \neq 0$

$$g(-1 + it) = \log(2\pi) - \log |t| + r(t).$$

By GGT4, $|r(t)| \leq C/|t|$ for $|t| \geq 2$. For $|t| < 2$, continuity of g gives an integrable bound $C + |\log |t||$. Consequently

$$\int_{\mathbb{R}} e^{-t^2/T^2} |r(t)| dt = O(\log(2 + T)). \quad (\text{GGT6.1})$$

The exact phase in (GGT5.1) obeys

$$|e^{-2it/T^2} - 1| \leq 2|t|/T^2.$$

After multiplication by the Gaussian and $1 + |\log |t||$, its integral is $O(\log(2 + T))$, by $t = Tu$. The prefactor $e^{1/T^2} = 1 + O(T^{-2})$ changes an $O(T \log(2 + T))$ integral by $O(\log(2 + T)/T)$. Thus every error has been controlled before evaluating the leading integral, and

$$A_T(1) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-t^2/T^2} (\log(2\pi) - \log |t|) dt + O(\log(2 + T)). \quad (\text{GGT6.2})$$

The two exact Gaussian integrals are

$$\begin{aligned} \int_{\mathbb{R}} e^{-t^2/T^2} dt &= T\sqrt{\pi}, \\ \int_{\mathbb{R}} e^{-t^2/T^2} \log |t| dt &= T\sqrt{\pi} \log T + \frac{T}{2} \Gamma'(1/2). \end{aligned} \quad (\text{GGT6.3})$$

For the second, substitute $t = Tu$, then $v = u^2$ on the two half-lines. Differentiation of the defining Gamma integral at $1/2$ is justified by the integrable logarithmic factor.

The Euler limit (GGT4.3) at $z = 1/2$ gives

$$\psi(1/2) = \lim_{N \rightarrow \infty} (\log N - 2H_{2N} + H_N) = -\gamma - 2 \log 2. \quad (\text{GGT6.4})$$

Here $H_N = \sum_{n=1}^N 1/n$ and $\gamma = \lim(H_N - \log N)$. Since $\Gamma(1/2) = \sqrt{\pi}$, substituting (GGT6.3)–(GGT6.4) proves

$$\boxed{A_T(1) = \frac{T}{2\sqrt{\pi}} \left[-\log T + \log(4\pi) + \frac{\gamma}{2} \right] + O(\log(2 + T)).} \quad (\text{GGT6.5})$$

The original phase and full Gamma terms have been bounded, not omitted without a comparison.

153.8 GGT7. All prime resonances and the global trace bounds

For fixed $x > 0$, the prime sum in (GGT3.2) has the T -independent summable majorant

$$\sum_{n \geq 2} (\log n) \left[e^{-\log^2(x/n)/4} + n^{-1} e^{-\log^2(xn)/4} \right] < \infty, \quad T \geq 1. \quad (\text{GGT7.1})$$

This proves immediately that its full contribution, including the factor $T/(2\sqrt{\pi})$, is $O_x(T)$. Combined with GGT5 and the bounded original pole term, it gives

$$\boxed{S_T(x) = O_x(T) \quad (x \neq 1).} \quad (\text{GGT7.2})$$

There is a sharper statement retaining each possible exact resonance. Define

$$R(x) = \begin{cases} \Lambda(x), & x \text{ is an integer} \geq 2, \\ 0, & \text{otherwise} \end{cases} + \begin{cases} x\Lambda(1/x), & 1/x \text{ is an integer} \geq 2, \\ 0, & \text{otherwise.} \end{cases} \quad (\text{GGT7.3})$$

The two alternatives cannot both be nonzero. These are exactly the terms with $\log(x/n) = 0$ or $\log(xn) = 0$ in the two prime families.

All other logarithms in those two families have a positive minimum absolute value $\delta_x > 0$. Indeed each family tends in absolute value to infinity with n , and only finitely many indices lie in any bounded interval; the zero logarithms have already been excluded. For a nonzero such logarithm u , $T \geq 1$, and $|u| \geq \delta_x$,

$$\frac{T^2 u^2}{4} \geq \frac{T^2 \delta_x^2}{8} + \frac{u^2}{8}.$$

The corresponding series with $e^{-u^2/8}$ is summable by the same argument as (GGT7.1). Therefore the full prime sum, before its prefactor, is

$$R(x) + O_x(e^{-T^2\delta_x^2/8}). \quad (\text{GGT7.4})$$

For $x \neq 1$, equations (GGT3.2), (GGT5.3), and (GGT7.4) prove the stronger expansion

$$\boxed{S_T(x) = -\frac{T}{2\sqrt{\pi}}R(x) + x + O_x(\log(2+T)).} \quad (\text{GGT7.5})$$

Thus nontrivial prime-power resonances have been retained rather than put into an unspecified error term.

At $x = 1$ there is no zero logarithm, and $\delta_1 = \log 2$. The prime contribution is

$$O\left(Te^{-T^2(\log 2)^2/8}\right). \quad (\text{GGT7.6})$$

Combining this with $e^{1/T^2} = 1 + O(T^{-2})$ and the full constant in (GGT6.5) gives

$$\boxed{S_T(1) = \frac{T}{2\sqrt{\pi}} \left[\log T - \log(4\pi) - \frac{\gamma}{2} \right] + 1 + O(\log(2+T)).} \quad (\text{GGT7.7})$$

In particular

$$S_T(1) = \frac{T}{2\sqrt{\pi}} \log T + O(T). \quad (\text{GGT7.8})$$

These estimates use the actual zero multiset and the original arithmetic prime series. They make no assumption about the real parts of the zeros beyond the established critical-strip inclusion.

153.9 GGT8. Exact trace tests for the original multiplier comparison

The estimates apply to the actual multiplier domain in PGD. If $F \in \mathcal{B}$, then for each fixed $x > 0$,

$$\sup_{T \geq 1} \sum_{\rho \in \mathcal{Z}} m_\rho |x^\rho F(\rho) e^{\rho^2/T^2}| < \infty. \quad (\text{GGT8.1})$$

Indeed $|x^\rho| \leq \max(1, x)$, $|e^{\rho^2/T^2}| \leq e$, and the rapid bound for $F(\rho)$, together with the polynomial zero count, makes the remaining sum absolutely convergent. Thus an actual test representative has uniformly bounded trace under these Gaussian tests, while (GGT7.7) identifies the distinct $T \log T$ behavior of the arithmetic identity multiplier.

This supplies the global estimate required to test a finite relation among PGD's already constructed real-orbit multipliers. It does not replace the original quotient by a sequence space, identify primitive τ with a constant arithmetic function, or assert a purity bound. The exact prime, Gamma, endpoint, and finite trivial-zero comparisons are (GGT3.2)–(GGT3.6); the inequalities follow from those full formulas.

154 The complete finite prime-action algebra inside the actual global remainder

24 September 2026. Complete derivation FPO0–FPO8. The original Riemann zeta function remains the arithmetic function throughout. The Gaussian is an explicitly constructed test function; it does not replace zeta or its spectrum.

154.1 FPO0. The question answered and the controlling source

The user’s amended instruction USR-64a88219a3ecddd3 is: “But what would, based on this finding and everything else we know so far?” The preceding PGD calculation constructed a nonzero global dual-comparison class and each of its positive-real translates. The accumulated original-zeta trace formula and the user’s full-spectrum/dilation arguments make the next question precise: can a finite linear combination of distinct global arithmetic rescalings disappear in that actual quotient? Here we prove that it cannot. We also prove the exact comparison with the original, unshifted Connes–Consani explicit formula, including its endpoint contribution.

The source definitions remain B1–B5, their proved consequences P1–P5, and retraction R1 in `SOURCE_OPERATIONS_AND_PROOFS.md`. The full retained user corpus supplies USR-4322be19bff532cd and the global arguments USR-9ad1c0a2d09dba92, USR-f55d16feb948d8c2 and USR-6152e3bc6302258c. These are references into complete preserved passages, not replacement definitions. Source Z_0 is absence; Z_1/τ is primitive presence with no parity or source addition; the reconstructed integer layer retains its amounts and Z_2 data. No operand below is primitive τ . The arithmetic constant function 1 is not identified with it.

Human sources used here are Alain Connes and Caterina Consani, *Weil positivity and Trace formula, the archimedean place*, arXiv:2006.13771v1, original author TeX `weil-compo.tex`, Appendix “Explicit formula,” labels `mellin`, `bombieriexplicit`, `bombieriexplicit1` and `bombieriexplicit2`, lines2033–2048; and their *Schemes over F_1 and zeta functions*, arXiv:0903.2024v3, §5, for the receiving coefficient geometry. The former explicitly credits Enrico Bombieri’s *The Riemann hypothesis* (2006) for its explicit-formula presentation; we have read that citation in their author source, not claimed a fresh reading of Bombieri’s whole chapter. Ralf Meyer’s *A spectral interpretation for the zeros of the Riemann zeta function* supplies the prior analytic framework. Pierre Deligne’s *La conjecture de Weil II*, §§3.6.1–3.6.3 remains the geometric lifting target. The new quotient calculation below is our derivation, not a theorem attributed to those authors.

Complete programme dependencies are PGD0–PGD7, GGT0–GGT8, GZR0–GZR9 and ASD0–ASD14. Their full proof bodies remain in the cumulative manuscript.

154.2 FPO1. Actual domains, ideals and the contour comparison

Retain the original spaces, without replacing their topology by an unrestricted zero-jet product:

$$\mathcal{B} = \{F \text{ entire} : \sup_{|\sigma| \leq A, t \in \mathbb{R}} (1 + |t|)^N |F(\sigma + it)| < \infty \text{ for every } A, N \geq 0\},$$

$$\mathcal{M} = \{H \text{ entire} : \text{for each } A \geq 0 \text{ some } C_A, d_A \text{ satisfy } |H(\sigma + it)| \leq C_A(1 + |t|)^{d_A} \text{ for } |\sigma| \leq A\}. \quad (\text{FPO1.1})$$

Let $\mathcal{Z}$ be the original nontrivial zero set, with every actual multiplicity m_ρ . Set

$$\mathcal{J} = \{H \in \mathcal{M} : H^{(j)}(\rho) = 0 \text{ } (\rho \in \mathcal{Z}, 0 \leq j < m_\rho)\}, \quad I = \mathcal{B} \cap \mathcal{J}, \quad Q = \mathcal{B}/I. \quad (\text{FPO1.2})$$

The quotient Q carries its proved Fréchet quotient topology; its continuous complex-linear dual is Q' . Each fixed member of $\mathcal{M}$ multiplies $\mathcal{B}$ continuously into itself: on each strip use its stated

polynomial bound and increase the exponent in the defining seminorm. Thus $\mathcal{B}$ is an ideal of $\mathcal{M}$. Leibniz's formula proves that $\mathcal{J}$ is an ideal too, with all multiplicities retained. Consequently

$$\mathcal{A}_\zeta = \mathcal{M}/(\mathcal{B} + \mathcal{J}) \quad (\text{FPO1.3})$$

is an algebraic quotient algebra. Its unit is nonzero by PGD4. No topology on this algebraic quotient is assumed.

The actual continuous-dual contour is

$$\lambda_H([F]) = \frac{1}{2\pi i} \left(\int_{2-i\infty}^{2+i\infty} - \int_{-1-i\infty}^{-1+i\infty} \right) \frac{H(s)F(1-s)}{\zeta(s)} ds. \quad (\text{FPO1.4})$$

Both edges point upward, and the two integrals converge separately. PGD1–PGD3 proves descent and continuity on Q , computes $\ker(H \mapsto \lambda_H) = \mathcal{J}$ using all actual zero jets, and proves the injective linear map

$$\overline{\Theta} : \mathcal{A}_\zeta \hookrightarrow \mathcal{C}_\zeta = (\chi \otimes Q')/DQ, \quad [H] \mapsto [\lambda_H], \quad D[F] = \lambda_F. \quad (\text{FPO1.5})$$

Here the actual action is

$$(T_a^\diamond \lambda)([F]) = a\lambda([a^{-s}F(s)]), \quad a > 0, \quad T_a^\diamond \lambda_H = \lambda_{a^s H}. \quad (\text{FPO1.6})$$

The factor is $a a^{-(1-s)} = a^s$ in the integrand, so this is a full group identity, not merely an inference from a formal generator. The cokernel has been constructed as a vector space with this action; we do not assert that it carries a multiplication extending that of $\mathcal{A}_\zeta$.

154.3 FPO2. Construct the original Mellin test and both endpoint values

For $T \geq 1$ and $x > 0$, define

$$f_{T,x}(u) = \frac{T}{2\sqrt{\pi}} \exp\left(-\frac{T^2}{4} \log^2 \frac{u}{x}\right), \quad u > 0. \quad (\text{FPO2.1})$$

Its original Mellin transform, with no shift or completion, is

$$\Phi_{T,x}(s) = \int_0^\infty f_{T,x}(u) u^s \frac{du}{u} = x^s e^{s^2/T^2}. \quad (\text{FPO2.2})$$

For a proof put $v = \log(u/x)$. The Gaussian integral $\int_{\mathbb{R}} e^{-T^2 v^2/4 + sv} dv = (2\sqrt{\pi}/T) e^{s^2/T^2}$ first follows for real s by a real translation and completing the square. Both sides are entire in s ; differentiation of the integral is justified by a Gaussian bound uniformly on compact s -sets. The identity theorem gives the stated equality for complex s . On a bounded vertical strip its modulus is

$$|\Phi_{T,x}(\sigma + it)| = x^\sigma e^{\sigma^2/T^2} e^{-t^2/T^2},$$

which proves membership in $\mathcal{B}$ for each fixed T, x .

Connes–Consani's original involution is

$$f_{T,x}^\sharp(u) = u^{-1} f_{T,x}(u^{-1}) = u^{-1} \frac{T}{2\sqrt{\pi}} \exp\left(-\frac{T^2}{4} \log^2(ux)\right). \quad (\text{FPO2.3})$$

Substitution $v = u^{-1}$ proves its Mellin transform is $\Phi_{T,x}(1-s)$. In particular the full two endpoint contributions are

$$\int_0^\infty f_{T,x}(u) du = \Phi_{T,x}(1) = x e^{1/T^2}, \quad \int_0^\infty f_{T,x}^\sharp(u) du = \Phi_{T,x}(0) = 1. \quad (\text{FPO2.4})$$

The latter 1 is an integral value. It is different from the point value $f_{T,x}(1) = T e^{-T^2 \log^2 x/4} / (2\sqrt{\pi})$; neither is primitive τ .

154.4 FPO3. Prove the domain extension before using the source formula

The cited author source states its explicit formula for $C_c^\infty(\mathbb{R}_+^\times)$. Our $f_{T,x}$ has noncompact support. Here is the complete extension needed for this one test family.

Write $g(v) = f_{T,x}(e^v)$. Fix a smooth function η , equal to 1 on $[-1, 1]$, equal to 0 outside $[-2, 2]$, and with values in $[0, 1]$. For $R \geq 2$, put $g_R(v) = \eta(v/R)g(v)$, $f_R(u) = g_R(\log u)$. Then f_R belongs to the source domain. Every derivative of g is a polynomial in $v - \log x$ times its original Gaussian and its stated constants. It follows directly, by the product rule and Gaussian decay, that for every $A \geq 0$ and integer $k \geq 0$,

$$\int_{\mathbb{R}} e^{A|v|} |(g_R - g)^{(k)}(v)| dv \longrightarrow 0, \quad \sup_v e^{A|v|} |(g_R - g)^{(k)}(v)| \longrightarrow 0. \quad (\text{FPO3.1})$$

Indeed all differences are supported in $|v| \geq R$; derivatives of the cutoff contribute powers R^{-1} , uniformly bounded for $R \geq 2$, and the Gaussian dominates every polynomial and every fixed exponential on this tail.

For $|\sigma| \leq A$, integrate the Fourier integral for the Mellin transform by parts k times in v , placing derivatives on $e^{\sigma v}(g_R - g)$. All boundary terms vanish by FPO3.1. Leibniz's rule bounds the resulting integral by a finite sum of FPO3.1 with coefficients bounded for $|\sigma| \leq A$. Treat $|t| \leq 1$ by the case $k = 0$. Thus

$$\tilde{f}_R \longrightarrow \Phi_{T,x} \quad \text{in } \mathcal{B}. \quad (\text{FPO3.2})$$

The actual zero count used in GGT is $N(Y) \leq C(Y + 2)^{3/2}$, with multiplicities. In particular $\sum_{\rho} m_{\rho}(1 + |\Im \rho|)^{-3} < \infty$, as follows by splitting heights into intervals $[2^j, 2^{j+1})$. The strip $0 < \Re \rho < 1$ and FPO3.2 therefore imply convergence of the full zero trace. Both endpoint evaluations converge too.

Write $\Lambda(n) = \log p$ when $n = p^m$ with p prime and $m \geq 1$, and 0 otherwise. This definition uses the already reconstructed integer arithmetic and its unique prime factorization. It has $0 \leq \Lambda(n) \leq \log n$. The estimate in FPO3.1 with any $A > 2$ bounds the differences of the first prime series by a constant tending to 0 times $\sum_{n \geq 2} (\log n) n^{-A}$, and bounds the reflected series by the same type of convergent sum with n^{-A-1} . Hence both full prime sums converge under this cutoff limit, including every repetition.

For the original archimedean term, substitute $u = e^v$ into the author's exact formula:

$$\mathcal{W}_{\mathbb{R}}(f) = (\log 4\pi + \gamma)g(0) + \int_0^\infty \frac{g(v) + e^{-v}g(-v) - 2e^{-v}g(0)}{1 - e^{-2v}} dv. \quad (\text{FPO3.3})$$

The numerator vanishes at $v = 0$, and its derivative is bounded near 0, so its quotient has a removable integrable endpoint bound. For $g_R - g$, the numerator is identically 0 near 0, since $R \geq 2$; its value at 0 is 0. Away from 0 the denominator is bounded away from 0, and FPO3.1 makes the integral of its difference tend to 0. The original tail term $-2e^{-v}g(0)$ is retained and integrable. This proves convergence of the entire archimedean expression.

Every term in the compactly supported source equality thus has its proved limit. In this equality the zero sum is over nontrivial zeros, as in the source's subsequent Proposition **mainprop**. The wording "all complex zeros" at source line 2042 must not be interpreted as adding the negative even zeros a second time: the archimedean expression already incorporates their contribution. GGT3 independently verifies this convention by the original-zeta contour and every finite leftward shift. We use no problematic extension of a sum over negative even integers.

154.5 FPO4. Exact original prime formula and the retained Gamma endpoint

Define the absolutely convergent trace

$$S_T(x) = \sum_{\rho \in \mathcal{Z}} m_{\rho} x^{\rho} e^{\rho^2/T^2}. \quad (\text{FPO4.1})$$

The source comparison just proved gives, with every coefficient retained,

$$S_T(x) = 1 + xe^{1/T^2} - \frac{T}{2\sqrt{\pi}} \sum_{n \geq 2} \Lambda(n) \left\{ e^{-T^2 \log^2(x/n)/4} + n^{-1} e^{-T^2 \log^2(xn)/4} \right\} - \mathcal{W}_{\mathbb{R}}(f_{T,x}). \quad (\text{FPO4.2})$$

The independent calculation GGT1–GGT3 works directly with the original logarithmic derivative and proves

$$\begin{aligned} S_T(x) &= xe^{1/T^2} - \frac{T}{2\sqrt{\pi}} \sum_{n \geq 2} \Lambda(n) \left\{ e^{-T^2 \log^2(x/n)/4} + n^{-1} e^{-T^2 \log^2(xn)/4} \right\} - A_T(x), \\ A_T(x) &= \frac{1}{2\pi i} \int_{-1-i\infty}^{-1+i\infty} x^s e^{s^2/T^2} \frac{\chi'}{\chi}(s) ds, \\ \chi(s) &= \pi^{s-1/2} \frac{\Gamma((1-s)/2)}{\Gamma(s/2)}, \\ \frac{\chi'}{\chi}(s) &= \log \pi - \frac{1}{2} \psi((1-s)/2) - \frac{1}{2} \psi(s/2). \end{aligned} \quad (\text{FPO4.3})$$

Here χ is the full functional-equation multiplier; it is not a replacement arithmetic function. Comparing the two proved equalities yields the exact relation

$$\boxed{\mathcal{W}_{\mathbb{R}}(f_{T,x}) = A_T(x) + 1.} \quad (\text{FPO4.4})$$

There is also a direct check: GGT3 proves that shifting the archimedean contour from $\Re s = -1$ to $\Re s = 1/2$ crosses exactly the residue $\Phi_{T,x}(0) = 1$ of χ'/χ at 0. Thus $A_T = A_T^{(1/2)} - 1$, with the stated orientation. No infinite leftward contour limit has been taken. A finite shift across a trivial zero $-2r$ keeps its exact term $x^{-2r} e^{4r^2/T^2}$ and the remaining contour, as GGT3 proves.

GGT4–GGT7, from an explicit digamma integral and the same original formula, establishes the bounds needed below:

$$S_T(1) = \frac{T}{2\sqrt{\pi}} \left(\log T - \log(4\pi) - \frac{\gamma}{2} \right) + 1 + O(\log T), \quad (\text{FPO4.5})$$

$$S_T(x) = -\frac{T}{2\sqrt{\pi}} \left\{ \Lambda(x) \mathbf{1}_{x \in \mathbb{N}, x \geq 2} + x \Lambda(1/x) \mathbf{1}_{1/x \in \mathbb{N}, 1/x \geq 2} \right\} + x + O_x(\log T) \quad (x \neq 1). \quad (\text{FPO4.6})$$

The symbol Λ in each summand is evaluated only when its indicated integer condition holds; the whole summand is 0 otherwise. All zero multiplicities are retained, no zero is presumed simple or on the critical line, and every estimate is for fixed $x > 0$ as $T \rightarrow \infty$. The distinction used next is exactly the nonzero coefficient of $T \log T$ at 1 and its absence at every other fixed positive scale.

154.6 FPO5. Construct the group algebra and prove its faithful quotient map

Let $\mathcal{G} = \mathbb{R}_+^\times$, with its usual multiplication. Define $\mathbb{C}[\mathcal{G}]$ to be the space of functions $a : \mathcal{G} \rightarrow \mathbb{C}$ of finite support. Addition and scalar multiplication are pointwise; multiplication is finite convolution

$$(a * b)(u) = \sum_{rt=u} a(r)b(t). \quad (\text{FPO5.1})$$

Only pairs from the two finite supports contribute. Associativity follows by expanding the same finite triple sum over $rst = u$; the function δ_1 is the multiplicative identity and $\delta_r \delta_t = \delta_{rt}$. These operations are constructed in a receiving arithmetic algebra, not on primitive presence.

For finite support define

$$\mathcal{E}(a)(s) = \sum_r a(r) r^s. \quad (\text{FPO5.2})$$

Each summand is entire and bounded on each vertical strip by a constant depending on that strip and r ; hence $\mathcal{E}(a) \in \mathcal{M}$. Finite expansion proves $\mathcal{E}(a * b) = \mathcal{E}(a)\mathcal{E}(b)$ and $\mathcal{E}(\delta_1) = 1$. Composition with the ideal quotient gives a unital algebra map

$$\iota_{\text{alg}} : \mathbb{C}[\mathcal{G}] \longrightarrow \mathcal{A}_\zeta. \quad (\text{FPO5.3})$$

We now prove it is injective; this is stronger than proving the distinct functions in FPO5.2 are different before quotienting.

Suppose its value at a is 0. List the distinct support points $r_1, \dots, r_k$ and write $a_i = a(r_i)$. By the definition of $\mathcal{A}_\zeta$, some $F \in \mathcal{B}$ satisfies

$$\sum_{i=1}^k a_i r_i^\rho = F(\rho) \quad (\rho \in \mathcal{Z}). \quad (\text{FPO5.4})$$

The quotient requires all multiplicity jets too; the displayed value equalities are consequences sufficient for the argument. Fix j , multiply these equalities by $m_\rho r_j^{-\rho} e^{\rho^2/T^2}$, and sum over all actual zeros. All these series converge absolutely: each finite exponential sum has bounded modulus on $0 \leq \Re s \leq 1$, the Gaussian damps the height, and the zero count above applies. The result is

$$\sum_{i=1}^k a_i S_T(r_i/r_j) = \sum_{\rho} m_\rho r_j^{-\rho} F(\rho) e^{\rho^2/T^2}. \quad (\text{FPO5.5})$$

The right side is bounded independently of $T \geq 1$. Indeed

$$|r_j^{-\rho}| \leq \max(1, r_j^{-1}), \quad |e^{\rho^2/T^2}| \leq e,$$

and its sum is bounded by

$$e \max(1, r_j^{-1}) b_{1,3}(F) \sum_{\rho} m_\rho (1 + |\Im \rho|)^{-3} < \infty. \quad (\text{FPO5.6})$$

In the left side, exactly the summand $i = j$ has scale 1, since the r_i are distinct. FPO4.5 and FPO4.6 therefore give

$$\sum_{i=1}^k a_i S_T(r_i/r_j) = a_j \frac{T \log T}{2\sqrt{\pi}} + O_a(T). \quad (\text{FPO5.7})$$

All scales and coefficients are fixed; the finite error constants can be added. Divide FPO5.5 by $T \log T$ and let $T \rightarrow \infty$. Its right side tends to 0 and its left side tends to $a_j/(2\sqrt{\pi})$. Therefore $a_j = 0$. Repeat for each j ; hence $a = 0$. This proves

$$\boxed{\mathbb{C}[\mathbb{R}_+^\times] \hookrightarrow \mathcal{A}_\zeta \hookrightarrow \mathcal{C}_\zeta, \quad \delta_r \longmapsto [r^s] \longmapsto c_r = [\lambda_{r^s}].} \quad (\text{FPO5.8})$$

The first arrow is an injective algebra map. The second and their composition are injective linear maps; no multiplication on all of $\mathcal{C}_\zeta$ has been asserted.

154.7 FPO6. The entire prime algebra and every finite relation

The subgroup $\mathbb{Q}_+^\times \subset \mathbb{R}_+^\times$ consists exactly of finite products $\prod_p p^{n_p}$, $n_p \in \mathbb{Z}$, uniquely. To check uniqueness, write a proposed equality as an equality of two positive integers by multiplying away the finite negative exponents, and apply the retained unique integer prime factorization. Conversely every positive rational has such a presentation by factoring its numerator and denominator.

It follows, with the displayed multiplication rather than a naming convention, that

$$\mathbb{C}[\mathbb{Q}_+^\times] \simeq \mathbb{C}[X_p, X_p^{-1} : p \text{ prime}], \quad \prod_p X_p^{n_p} \longmapsto \delta_{\prod_p p^{n_p}}. \quad (\text{FPO6.1})$$

Both sides consist of finite sums of finite monomials. Uniqueness of the exponents proves the map is bijective, and addition of exponents proves it preserves multiplication. Its exact injection into the actual cokernel is

$$\prod_p X_p^{n_p} \mapsto \left[\lambda_{\exp(s \sum_p n_p \log p)} \right]. \quad (\text{FPO6.2})$$

Every logarithm and exponent has its original positive-rational value. No common rescaling of the arithmetic times is introduced.

FPO1.6 proves that $T_a^\diamond c_r = c_{ar}$. Thus the injected space is the regular group module: multiplication by δ_a corresponds to the actual positive-real action. Its orbit of $c_1 = c_\zeta$ is exactly the basis $\{c_r : r > 0\}$. In particular, for every prime p , all nonzero Laurent polynomials in $T_p^\diamond$ act nontrivially on c_ζ . For distinct primes, the same holds for every nonzero multivariate Laurent polynomial, using the same proved prime-factorization bijection. This statement concerns their action on this class, not invertibility of those operators on the whole cokernel.

Moreover $T_a^\diamond c_\zeta = c_\zeta$ forces $a = 1$, since otherwise $\delta_a - \delta_1 \neq 0$ would map to 0. This is an exact statement about detectability in this named receiver. It does not declare that every geometrical change of presentation of the user's global counting construction is this particular action, or that all such presentations have been classified.

154.8 FPO7. Exact scope of the supported comparison and source labels

ASD14 proves the complete cone of the original coefficient comparison has degree-one cohomology $\mathcal{C}_\zeta$. On representatives the final map into that cohomology is

$$\delta_r \mapsto \lambda_{r^s} \mapsto \pi' \lambda_{r^s} \mapsto [\pi' \lambda_{r^s}] \in H^1(K_\zeta), \quad (\text{FPO7.1})$$

where $\pi : A \rightarrow Q$ is the actual raw Mellin quotient. The representative is a cycle since $\pi d = 0$ gives $d' \pi' \lambda = 0$. A boundary would mean $\pi' \lambda = \pi' D\pi(b)$; injectivity of π' would imply $\lambda \in DQ$, exactly the relation excluded above. This proves both the actual map and its injectivity in the cone, not only in a newly defined algebra.

The source receiving-ring action is retained on the full coefficient complex, with its original extra closed copies. On this spectral term its action is the first arithmetic scalar action, hence it commutes with every displayed linear map. That fact does not identify source τ with integer1: their different actions on the faithful extra closed copies remain part of the full complex and are not quotiented away here.

For the existing receiving support lattice L , the exact lift of any displayed linear map $f : V \rightarrow W$ is

$$(v, \alpha) \mapsto (f(v), \alpha), \quad G_L(V) = \{(0, \alpha) : \alpha \in L\} \cup (V \times \{1_L\}). \quad (\text{FPO7.2})$$

A nontop input has zero amplitude, which linearity sends to zero; a nonzero output therefore has top label. Substitution proves preservation of identities and composition, and injectivity is retained when f is injective. A labelled zero is never turned into an unlabelled absence. This is the named receiving construction, not a newly imposed operation on Z_1/τ .

154.9 FPO8. Contribution to the current lifting calculation

PGD constructed individual global orbit elements. This proof determines their full finite relation space: it is zero. It constructs the entire real group algebra and the Laurent algebra of all primes as an injected subalgebra of the original multiplier quotient, and their exact equivariant linear injection into the actual dual-comparison cone. The test derives from the original prime formula and retains both endpoint values, the complete Gamma contribution and every prime-power repetition. The exact relation $\mathcal{W}_\mathbb{R} = A_T + 1$ prevents losing the original endpoint when moving between the two presentations.

The numerical weight separation sought in the active tau goal is not asserted by this representation theorem. The next calculation is the actual support-localization map on these now distinguished global classes, together with the other degrees of the full complex: it must determine which map would have to annihilate an obstruction to a lifted class. That calculation uses the already constructed ASD localization row and complete source-faithful complex, rather than treating the whole nonzero comparison cone as a single finite-weight obstruction. The complete UOS3–UOS8A derivation carries it out: the exact cone map sends this entire module by $c_r \mapsto (c_r, 0)$ into $H^1(\mathcal{L}_\zeta) = \mathcal{C}_\zeta \oplus (\chi \otimes Q')$; it is the identity on $H^{-1} = H$, and is the explicit restriction $\chi H' \rightarrow \chi E'$ in degree 2. The original Fourier restriction has its written section and zero connecting map. Thus these relative residue-comparison classes have a proved supported placement, and have not been substituted for a failure of that existing Fourier lift.

155 The global unit orbit, its companion mirror and its exact supported placement

24 September 2026. Independent derivation UOS0–UOS9. This computes the actual prime and companion-mirror action on the global multiplier functionals, maps them through the original supported localization row, and constructs the full comparison with the dual-supported complex. It does not identify the relative residue-comparison cokernel with an RH counterexample space.

155.1 UOS0. Source, retained constructions and operation scope

The source remains the user's Z_0 absence, primitive Z_1/τ without parity or source addition, and the original integer layer with its Z_2 data and complete amounts. B1–B5/P1–P5 and the retraction of source τ addition are retained in `SOURCE_OPERATIONS_AND_PROOFS.md`. The complete user corpus and its global arguments are recorded in `CORPUS_AND_OPERATION_RULES.md` (private source record), particularly USR-9ad1c0a2d09dba92, USR-f55d16feb948d8c2, USR-6152e3bc6302258c and USR-4322be19bff532cd. The receiving comparisons below come after the already reconstructed arithmetic and original zeta, as those arguments require.

The arithmetic constant 1 in the multiplier algebra is not primitive τ . Every sum, scalar, transpose, cone, and quotient below is an operation in the already constructed coefficient category. No such operation has primitive τ as an additive or numerical operand.

The original coefficient geometry and Fourier localization are due to Alain Connes and Caterina Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, arXiv:0903.2024v3, §5, with their author-source definitions and reading coverage retained in DCP0. The analytic closed-image framework is due to Ralf Meyer, *A spectral interpretation for the zeros of the Riemann zeta function*, arXiv:math/0412277v3, with the original-space image and inverse proofs in OMS. The lifting argument being compared is Pierre Deligne's *La conjecture de Weil. II*, §§3.6.1–3.6.3; its geometric duality and weight bounds are not assumed for this receiver.

The complete inputs are: - DCP7–DCP12, the actual source-supported coefficient maps and faithful extra closed copies; - ASD3–ASD14, their full continuous dual, all signs, and the residue-comparison cone; - GZR7–GZR9, the companion denominator, real action and support lifts of the full residue pairing; - PGD0–PGD7, the complete multiplier map and explicit nonzero class, independently checked in PGC; - SCL1, SCL6–SCL9, actual original jet representatives, finite lifts and the exact cokernel action.

Those proof files retain their full source provenance. No completeness or finite-dimensionality of the whole spectral quotient is introduced.

155.2 UOS1. The original spaces, multiplier functionals and their exact actions

Here $S = S_{00}^{\text{even}}$ is the complex Schwartz space of even functions h on $\mathbb{R}$ satisfying $h(0) = 0$ and $\int_{\mathbb{R}} h(v) dv = 0$, with its closed-subspace Schwartz topology. The original strong space A consists of the smooth functions $b : (0, \infty) \rightarrow \mathbb{C}$ for which

$$\sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j b(u)| < \infty \quad (N, j \geq 0),$$

with these seminorms. Its entire Mellin image is $\mathcal{B}$, the functions rapidly decreasing on every bounded vertical strip, with every polynomial vertical seminorm. The restrictions come from the complete closed coefficient spaces $V_{\pm} = S \oplus \mathbb{C}^2$; their two endpoint coordinates have zero restriction. Retain the original two-sign summation

$$\Sigma h(u) = 2 \sum_{n \geq 1} h(nu), \quad Rb(u) = u^{-1}b(u^{-1}), \quad \mathcal{M}_0 b(s) = \int_0^\infty b(u) u^s \frac{du}{u}. \quad (\text{UOS1.1})$$

Its exact image $J = \Sigma S$ is closed, and

$$\pi : A \rightarrow Q = A/J \simeq \mathcal{B}/I_\zeta.$$

The original zeta identity is

$$\mathcal{M}_0 \Sigma h(s) = 2\zeta(s) \int_0^\infty h(v) v^s \frac{dv}{v}. \quad (\text{UOS1.2})$$

The equivalent centered test entry retains $\mathcal{T}k = 2u^{-1/2}k$, $\mathcal{T}^{-1}b = \frac{1}{2}u^{1/2}b$, and $\mathcal{M}_0 \mathcal{T}k = 2F_k$. These factors are unchanged.

Let $\mathcal{M}$ be the entire functions of polynomial growth on every bounded vertical strip, and $\mathcal{J} \subset \mathcal{M}$ the full actual zero-jet ideal of PGD2. For $H \in \mathcal{M}$, the actual continuous functionals are

$$\begin{aligned} \lambda_H^\zeta([F]) &= \frac{1}{2\pi i} \left(\int_{2-i\infty}^{2+i\infty} - \int_{-1-i\infty}^{-1+i\infty} \right) \frac{H(s)F(1-s)}{\zeta(s)} ds, \\ \lambda_H^{\zeta^\vee}([F]) &= \frac{1}{2\pi i} \left(\int_{2-i\infty}^{2+i\infty} - \int_{-1-i\infty}^{-1+i\infty} \right) \frac{H(s)F(1-s)}{\zeta(1-s)} ds. \end{aligned} \quad (\text{UOS1.3})$$

Both edges run upward. PGD proves convergence and descent for the first functional. For the second, reflection exchanges the two edge estimates; the same entire-numerator division proof, or the exact identity in UOS2, proves convergence, descent and continuity. The companion notation does not replace its denominator by a completed function.

Let $T_a[F] = [a^s F(s)]$, $a > 0$, and

$$T_a^\vee \lambda = \lambda \circ T_{a^{-1}}, \quad \mathcal{U}_a \lambda = a T_a^\vee \lambda, \quad Y = \chi \otimes Q', \quad \chi(a) = a.$$

Here Q' is the continuous dual of the original Fréchet quotient. Use its weak-* topology for exact quotient topology assertions; the constructed transposes are also strong-dual continuous. The action on Y is $\mathcal{U}_a$. It satisfies, for both denominators,

$$\mathcal{U}_a \lambda_H = \lambda_{a^s H}, \quad (1 - L^t) \lambda_H = \lambda_{sH}. \quad (\text{UOS1.4})$$

These follow directly inside the absolutely convergent integrals from $a a^{-(1-s)} = a^s$ and $1 - (1-s) = s$. They are not inferred from a global exponential of an infinitesimal generator.

Write

$$\begin{aligned} D_\zeta x(y) &= B_\zeta(x, y), & D_{\zeta^\vee} x(y) &= B_{\zeta^\vee}(x, y), \\ C_\zeta &= Y/D_\zeta Q, & C_{\zeta^\vee} &= Y/D_{\zeta^\vee} Q, & c_H^\zeta &= [\lambda_H^\zeta], & c_H^{\zeta^\vee} &= [\lambda_H^{\zeta^\vee}]. \end{aligned} \quad (\text{UOS1.5})$$

These cokernels include the action χ ; there is no second implicit twist on C_ζ . PGD constructs $c_1^\zeta \neq 0$ and the complete orbit

$$c_r^\zeta := c_{r^s}^\zeta, \quad \mathcal{U}_a c_r^\zeta = c_{ar}^\zeta, \quad r, a > 0. \quad (\text{UOS1.6})$$

No group-algebra independence theorem is needed for the calculations below.

155.3 UOS2. Compute the complete companion reflection

On quotient Mellin functions, $RF(s) = F(1-s)$. Direct substitution in UOS1.3 gives

$$(R' \lambda_H^\zeta)([F]) = \lambda_H^\zeta([RF]) = \frac{1}{2\pi i} \left(\int_{\Re s=2}^\uparrow - \int_{\Re s=-1}^\uparrow \right) \frac{H(s)F(s)}{\zeta(s)} ds.$$

Set $w = 1 - s$. In a single edge integral, the reversal of traversal and $ds = -dw$ cancel; its image is an upward edge integral. The two edges exchange places, producing the remaining minus sign. Therefore

$$\boxed{R'\lambda_H^\zeta = -\lambda_{RH}^{\zeta^\vee}, \quad RH(s) = H(1-s).} \quad (\text{UOS2.1})$$

Reflection preserves $\mathcal{M}$: a bounded strip maps into a bounded strip, with the same imaginary growth degree. Both operations in UOS2.1 are thus defined on the complete stated domains. Applying the same calculation to the companion gives its inverse identity. In particular its multiplier kernel is also $\mathcal{J}$, because R preserves that full jet ideal and R' is invertible.

For $H = r^s$ the exact factor is

$$RH(s) = r^{1-s} = r(r^{-1})^s, \quad \boxed{R'\lambda_{r^s}^\zeta = -r\lambda_{(r^{-1})^s}^{\zeta^\vee}.} \quad (\text{UOS2.2})$$

No coefficient r has been absorbed into the variable or action.

The original full residue maps satisfy

$$R'D_\zeta = -D_{\zeta^\vee}R. \quad (\text{UOS2.3})$$

Indeed for $x = [F] \in Q$, $D_\zeta x = \lambda_F^\zeta$, and UOS2.1 is exactly this equality. It proves that R' induces a well-defined isomorphism of the two cokernels. The resulting classes obey

$$R'c_H^\zeta = -c_{RH}^{\zeta^\vee}, \quad R'c_r^\zeta = -rc_{1/r}^{\zeta^\vee}, \quad R'c_1^\zeta = -c_1^{\zeta^\vee}. \quad (\text{UOS2.4})$$

Thus the companion constant class is nonzero as well. The statement is about the two actual residue denominators; it is not an assertion that the individual pairing is reflection-invariant.

The functional-equation comparison of the denominators remains

$$\frac{1}{\zeta(1-s)} = \frac{2^s \pi^{s-1} \sin(\pi s/2) \Gamma(1-s)}{\zeta(s)} = \frac{\pi^{s-1/2} \Gamma((1-s)/2)}{\Gamma(s/2) \zeta(s)}. \quad (\text{UOS2.5})$$

The companion reciprocal vanishes at $s = 0$ and has value -2 at $s = 1$. No meromorphic multiplier in UOS2.5 has been declared a member of the entire multiplier algebra $\mathcal{M}$; UOS2.1 uses the separately defined companion contour and is valid without such an identification.

On the dual representation the precise prime/reflection identity is

$$\boxed{\mathcal{U}_a R' = a R' \mathcal{U}_{a^{-1}}.} \quad (\text{UOS2.6})$$

It follows from the original $T_a R = a R T_{a^{-1}}$ after taking the contragredient and retaining its twist. On the full unit orbit it can also be checked without an operator convention:

$$\mathcal{U}_a R' c_r^\zeta = -r c_{a/r}^{\zeta^\vee} = a R' \mathcal{U}_{a^{-1}} c_r^\zeta.$$

The oriented global dual mirror is $-R'$, not R' ; it sends c_r^ζ to $+r c_{1/r}^{\zeta^\vee}$. The comparison cone uses R' in its degree-one term, as proved in ASD14. These distinct signs will be retained below.

155.4 UOS3. The original localization row and its full transpose

Write

$$P = V_+ \oplus V_- \oplus V_{\text{extra}}, \quad d(v_+, v_-, w) = r_+ v_+ - r_- v_-, \quad D_{\text{full}} = [P \xrightarrow{d} A].$$

The extra term is the complete pair of closed copies $V_{\text{extra}} = V_+^{\text{extra}} \oplus V_-^{\text{extra}}$. Let

$$H = \ker d = H_0 \oplus V_{\text{extra}}, \quad E = \mathbb{C}^4 \oplus V_{\text{extra}}.$$

The four endpoint coordinates in H_0 retain their labels c_0, c_1, d_0, d_1 . The actual primal localization sequence is

$$0 \rightarrow E \xrightarrow{\iota} H \xrightarrow{\text{res}} A \xrightarrow{\beta} Q \oplus Q \xrightarrow{\delta} Q \rightarrow 0,$$

$$\text{res}((h, c), (\widehat{h}, d), w) = \Sigma h, \quad \beta(b) = (\pi b, \pi b), \quad \delta(q_+, q_-) = q_+ - q_-. \quad (\text{UOS3.1})$$

It is the original DCP7/DCP9 row, with no term erased.

Its exact continuous-dual row, twisted uniformly by χ , is

$$0 \rightarrow Y \xrightarrow{\delta'} Y \oplus Y \xrightarrow{\beta'} \chi A' \xrightarrow{\text{res}'} \chi H' \xrightarrow{\iota'} \chi E' \rightarrow 0,$$

$$\delta' \lambda = (\lambda, -\lambda), \quad \beta'(\lambda_+, \lambda_-) = \pi'(\lambda_+ + \lambda_-),$$

$$\text{res}' \ell = (\Sigma' \ell, 0, 0, 0, 0, 0), \quad \iota'(u, c'_0, c'_1, d'_0, d'_1, w') = (c'_0, c'_1, d'_0, d'_1, w'). \quad (\text{UOS3.2})$$

ASD2–ASD6 proves strict exactness for weak-* topologies, with all maps strong-dual continuous. In particular $\pi' Q' = J^\perp$. The right-hand ι' is an actual restriction map, including every extra closed functional.

For each $H_{\text{mult}} \in \mathcal{M}$, set $\lambda = \lambda_{H_{\text{mult}}}^\zeta$ and construct two different supported dual elements:

$$a_\lambda = (\lambda, -\lambda), \quad b_\lambda = (\lambda/2, \lambda/2). \quad (\text{UOS3.3})$$

They satisfy

$$a_\lambda = \delta' \lambda, \quad \beta' a_\lambda = 0, \quad \beta' b_\lambda = \pi' \lambda, \quad \text{res}' \pi' \lambda = 0, \quad \frac{1}{2}(a_{\lambda,+} - a_{\lambda,-}) = \lambda. \quad (\text{UOS3.4})$$

Every factor and sign follows by evaluation in UOS3.2. The first placement comes from the global dual and annihilates the image of the original boundary. The second evaluates that boundary:

$$a_\lambda(\beta b) = 0, \quad b_\lambda(\beta b) = \lambda(\pi b). \quad (\text{UOS3.5})$$

Thus the unit functional has explicit maps to the actual support row and its dual. It does not create an additional endpoint term in H' : the exact map into H' is zero by UOS3.4.

This second evaluation is genuinely nonzero for the constructed unit. Pick any actual zero a of order m and let $b = 1 - a$. Let $q_a = e_{a,m-1}$ be its globally isolated top jet from SCL1. Then PGD2.2 gives

$$\lambda_1^\zeta(q_a) = (-1)^{m-1} \frac{m!}{\zeta^{(m)}(1-a)} \neq 0. \quad (\text{UOS3.6})$$

An actual raw representative with $\pi f_a = q_a$ is

$$f_a(u) = \frac{1}{2\pi} \int_{\mathbb{R}} E_{a,m-1}(1/2 + it) u^{-1/2-it} dt \in A. \quad (\text{UOS3.7})$$

Its Mellin transform is the full $E_{a,m-1}$, by the original inverse theorem. If k_a denotes the earlier centered representative, then $f_a = u^{-1/2} k_a = \frac{1}{2} \mathcal{T} k_a$, retaining the factor2 comparison. Consequently $b_{\lambda_1}(\beta f_a)$ is exactly UOS3.6. This is an evaluation of the original boundary, not a claim that a functional is a primal cohomology class.

The original mirrors on the dual row are respectively $-R'$ on its first Y , exchange followed by R' on its middle pair, and R' on the final overlap dual. Hence

$$(a_{\lambda_H^\zeta})^{\text{mirror}} = a_{\lambda_{RH}^{\zeta^\vee}}, \quad (b_{\lambda_H^\zeta})^{\text{mirror}} = -b_{\lambda_{RH}^{\zeta^\vee}}. \quad (\text{UOS3.8})$$

For $H = r^s$, the respective factors are $+r$ and $-r$ with parameter $1/r$. Direct substitution in the exchange map proves both identities, and shows why the two placements must not share an invented mirror sign.

155.5 UOS4. A complete map between the condensed primal and dual rows

After the exact preceding quotient by J , the primal row in UOS3.1 is

$$0 \rightarrow Q \xrightarrow{\Delta} Q^2 \xrightarrow{\delta} Q \rightarrow 0, \quad \Delta q = (q, q), \quad \delta(q_+, q_-) = q_+ - q_-.$$

The condensed dual row is

$$0 \rightarrow Y \xrightarrow{\alpha} Y^2 \xrightarrow{\sigma} Y \rightarrow 0, \quad \alpha\lambda = (\lambda, -\lambda), \quad \sigma(\lambda_+, \lambda_-) = \lambda_+ + \lambda_-.$$

There is an actual map of these exact rows:

$$\begin{array}{ccccccc} 0 & \rightarrow & Q & \xrightarrow{\Delta} & Q^2 & \xrightarrow{\delta} & Q \rightarrow 0 \\ & & \downarrow D_\zeta & & \downarrow \text{diag}(D_\zeta, -D_\zeta) & & \downarrow D_\zeta \\ 0 & \rightarrow & Y & \xrightarrow{\alpha} & Y^2 & \xrightarrow{\sigma} & Y \rightarrow 0. \end{array} \quad (\text{UOS4.1})$$

Indeed its first square sends q to $(D_\zeta q, -D_\zeta q)$ on both paths; its second square sends (q_+, q_-) to $D_\zeta(q_+ - q_-)$ on both paths. Each column is injective, and each is equivariant for the original real and prime actions because $D_\zeta T_a = \mathcal{U}_a D_\zeta$. These are constructed maps of the actual coefficient spaces, not an assumed duality of sheaves.

Taking the three displayed cokernels gives an exact row

$$\boxed{0 \rightarrow C_\zeta \xrightarrow{c \mapsto (c, -c)} C_\zeta^2 \xrightarrow{(c_+, c_-) \mapsto c_+ + c_-} C_\zeta \rightarrow 0.} \quad (\text{UOS4.2})$$

A direct proof is that the first map is injective, the sum is surjective, and its kernel consists exactly of pairs $(c, -c)$. Its section is $c \mapsto (c/2, c/2)$, and the retraction onto the first entry is half-difference. Thus this row preserves the full relative class c_1^ζ , while already calculating its exact localization maps.

The companion mirror is compatible with the entire diagram UOS4.1. On the primal row its three mirrors are R , exchanged R , and $-R$. On the dual row they are $-R'$, exchanged R' , and R' . UOS2.3 gives $D_\zeta R = -R' D_{\zeta^\vee}$, so both endpoint squares commute with these companion actions. On the middle term,

$$\text{diag}(D_\zeta, -D_\zeta)(Rq_-, Rq_+) = (-R' D_{\zeta^\vee} q_-, R' D_{\zeta^\vee} q_+),$$

which is exactly the exchanged dual mirror of $(D_{\zeta^\vee} q_+, -D_{\zeta^\vee} q_-)$. Therefore the first and last C_ζ 's in UOS4.2 have respectively the induced companion maps $-R'$ and R' ; the middle retains exchanged R' . These are the distinct orientations of the same underlying quotient and group action.

The anti-diagonal map into the original, unquotiented Y^2 does not factor through C_ζ while keeping the stated maps. If it did, it would kill $D_\zeta Q$; but for $x \neq 0$, $\alpha D_\zeta x \neq 0$ by injectivity of both maps. Thus UOS4.2 constructs the exact relative target needed to descend that map. This proves the specific nonfactorization and immediately provides its replacement, rather than treating the objects as unrelated.

155.6 UOS5. Construct the actual map to the dual-supported complex

Let K_Z be the original both-closed-support complex:

$$K_Z^0 = P, \quad K_Z^1 = A \oplus A, \quad d_Z(v_+, v_-, w) = (r_+ v_+, r_- v_-). \quad (\text{UOS5.1})$$

The supported-to-global cochain map $u : K_Z \rightarrow D_{\text{full}}$ is

$$u^0 = \text{id}_P, \quad u^1(b_+, b_-) = b_+ - b_-.$$

Both sides of its cochain equation are $r_+v_+ - r_-v_-$.

With the actual dual convention in ASD3, define

$$T_D = \chi D_{\text{full}}^\vee[-2], \quad T_Z = \chi K_Z^\vee[-2].$$

Their only nonzero degrees are 1 and 2. Their differentials are

$$d_{T_D}^1 \lambda = (r'_+ \lambda, -r'_- \lambda, 0), \quad d_{T_Z}^1(\lambda_+, \lambda_-) = (r'_+ \lambda_+, r'_- \lambda_-, 0). \quad (\text{UOS5.2})$$

The transpose map $j = \chi u^\vee[-2] : T_D \rightarrow T_Z$ is

$$j^1 \lambda = (\lambda, -\lambda), \quad j^2 = \text{id}_{\chi P'}. \quad (\text{UOS5.3})$$

Substitution in UOS5.2 verifies its cochain identity including the internal minus sign.

Compose the already constructed residue comparison with this actual transpose:

$$F_\zeta = j\Psi_\zeta : D_{\text{full}} \rightarrow T_Z.$$

It has degree-zero component 0 and degree-one component

$$\boxed{F_\zeta^1 b = (\pi' D_\zeta \pi b, -\pi' D_\zeta \pi b)}. \quad (\text{UOS5.4})$$

The first source differential is killed because $\pi d = 0$; the target differential kills both components because $\pi' Q' = J^\perp$. This proves the full cochain map on the original spaces.

No equivalence between original source restriction and numerical weights has been assumed. Formula UOS5.4 is the exact source-to-dual-supported comparison constructed from the original residue pairing and the existing support map.

155.7 UOS6. Compute the full new cone and the map from the original comparison cone

Let

$$\mathcal{K}_\zeta = \text{Cone}(\Psi_\zeta), \quad \mathcal{L}_\zeta = \text{Cone}(F_\zeta),$$

using $\text{Cone}(f)^n = T^n \oplus D_{\text{full}}^{n+1}$ and $d(v, c) = (d_T v + f c, -d c)$. The second cone is the exact complex

$$\boxed{P \xrightarrow{-d} A \xrightarrow{(\Psi_\zeta^1, -\Psi_\zeta^1)} \chi A'^2 \xrightarrow{d_{T_Z}^1} \chi P'} \quad \text{in degrees } -1, 0, 1, 2. \quad (\text{UOS6.1})$$

Let $M_\zeta = D_\zeta Q \subset Y$. Its cohomology is

$$H^{-1}(\mathcal{L}_\zeta) = H, \quad H^0(\mathcal{L}_\zeta) = 0, \quad H^1(\mathcal{L}_\zeta) = Y^2/\alpha M_\zeta, \quad H^2(\mathcal{L}_\zeta) = \chi E'. \quad (\text{UOS6.2})$$

For the first two groups, the argument is the actual source kernel and injectivity of D_ζ , as in ASD14. For the third, $\ker d_{T_Z}^1 = \pi' Q' \oplus \pi' Q'$, and the image of the previous arrow is exactly $(\pi' D_\zeta q, -\pi' D_\zeta q)$, since π is onto. The identification by π' proves the quotient. For the fourth, restriction of P' to $\ker d_Z = E$ gives the cokernel $\chi E'$, as proved directly with continuous transposes in ASD5.

The full degree-one quotient has the explicit decomposition

$$\boxed{Y^2/\alpha M_\zeta \xrightarrow{\sim} C_\zeta \oplus Y, \quad [(\lambda_+, \lambda_-)] \mapsto \left(\left[\frac{\lambda_+ - \lambda_-}{2} \right], \frac{\lambda_+ + \lambda_-}{2} \right)}. \quad (\text{UOS6.3})$$

Adding $(D_\zeta q, -D_\zeta q)$ changes the first component by $D_\zeta q$ and the second by zero, proving well-definedness. The inverse sends $([\lambda], \mu)$ to $([\lambda + \mu], -\lambda + \mu)$; a different representative of $[\lambda]$ changes the pair by αM_ζ . Substitution gives both inverse identities. The maps are continuous for the

indicated quotient weak-* topologies, since both underlying half-sum and half-difference maps and their inverses are continuous. In particular its Hausdorff quotient is Y , whereas the nonzero C_ζ summand retains the algebraic defect proved in PGD.

The strict square with identity on the source and j on the target gives the actual cone map

$$\mathcal{J}_\zeta : \mathcal{K}_\zeta \rightarrow \mathcal{L}_\zeta.$$

Its four components are

$$\mathcal{J}_\zeta^{-1} = \text{id}_P, \quad \mathcal{J}_\zeta^0 = \text{id}_A, \quad \mathcal{J}_\zeta^1 \lambda = (\lambda, -\lambda), \quad \mathcal{J}_\zeta^2 = \text{id}_{\chi P'}. \quad (\text{UOS6.4})$$

Every cochain equation follows from UOS5.2–UOS5.4. On the complete cohomology it is

degree	$H^n(\mathcal{J}_\zeta)$	
-1	id_H	
0	$0 \rightarrow 0$	
1	$C_\zeta \rightarrow C_\zeta \oplus Y, \quad c \mapsto (c, 0)$	
2	$\chi H' \rightarrow \chi E', \quad \iota'.$	(UOS6.5)

The degree2 map is not an unspecified comparison: it is the actual restriction

$$(u, c'_0, c'_1, d'_0, d'_1, w') \longmapsto (c'_0, c'_1, d'_0, d'_1, w')$$

from UOS3.2. Its kernel is precisely the S' coordinate, and its section appends zero there. All four endpoint lines and both full extra copies remain.

Consequently the nonzero orbit c_r^ζ has a proved nonzero image $(c_r^\zeta, 0)$ in the actual dual-supported comparison cone. Its image does not acquire a component in the additional Y summand. These are concrete maps locating the class, rather than a declaration that the two cones are the same.

155.8 UOS7. The companion mirror on both cones and on the split result

Let w_D be the original Cech mirror: chart and extra swap in degree0 and $-R$ in degree1. Let w_Z be the original supported mirror: the same degree-zero swap and $(b_+, b_-) \mapsto (Rb_-, Rb_+)$ in degree1. The map u is equivariant; hence j intertwines their shifted dual mirrors w_{T_D} and w_{T_Z} .

ASD14 proves

$$\Psi_\zeta w_D = -w_{T_D} \Psi_{\zeta^\vee}.$$

Composing with j gives the exact companion identity

$$F_\zeta w_D = -w_{T_Z} F_{\zeta^\vee}. \quad (\text{UOS7.1})$$

Therefore the cone mirror from the companion to the original is

$$(v, c) \longmapsto (-w_{T_Z} v, w_D c)$$

for each target $T = T_D, T_Z$. The same formula with the denominators exchanged is its inverse; the two target signs cancel on applying it twice. This is an actual cochain map by the displayed identity.

In degree1 of $\mathcal{K}_\zeta$, that map is R' , because $w_{T_D}^1 = -R'$. Thus its action on the unit orbit is exactly the signed companion action UOS2.4. In degree1 of $\mathcal{L}_\zeta$, the map is

$$(\lambda_+, \lambda_-) \longmapsto (-R' \lambda_-, -R' \lambda_+). \quad (\text{UOS7.2})$$

Under the exact half-difference/half-sum decomposition UOS6.3, it becomes

$$(c, \mu) \longmapsto (R' c, -R' \mu) \quad \text{with the companion denominators exchanged.} \quad (\text{UOS7.3})$$

Indeed the new half-difference is $R'(\lambda_+ - \lambda_-)/2$, while the new half-sum is $-R'(\lambda_+ + \lambda_-)/2$. This verifies all signs and the compatibility of $c \mapsto (c, 0)$ with the mirror.

For the orbit from the original to the companion, UOS7.3 gives

$$(c_r^\zeta, 0) \mapsto (-rc_{1/r}^{\zeta^\vee}, 0). \quad (\text{UOS7.4})$$

Each receiving real action satisfies $\mathcal{U}_a w = a w \mathcal{U}_{a^{-1}}$, including these companion maps, by UOS2.6 and the fixed signs. The source terms obey the same original relation by DCP11. Thus the full cone map UOS6.4, the companion map and every prime action commute in the stated squares with the exact factor a . No self-mirror pairing or missing Tate twist is presumed.

155.9 UOS8. Source ring, all retained labels, and the original Fourier lift

The source receiving ring remains $\mathbb{Z}^3$ with $\epsilon = (1, 0, 0)$, $e_+ = (0, 1, 0)$, $e_- = (0, 0, 1)$. The source map is

$$[n] = (n, 0, 0), \quad [\tau] = (1, 1, 1).$$

On the spectral term Q , its dual and all the core cokernels, (r, b_+, b_-) acts by the scalar r . On the complete extra closed copies and their duals it acts respectively by b_+ and b_- . Every new chain map above is receiving-ring linear: the overlap maps have the same scalar on both sides, their zero components commute with all actions, and the maps on extra copies in UOS6.4 are identities. The mirror exchanges e_+, e_- , fixes ϵ , and hence fixes each original source integer and τ .

Thus, on the orbit,

$$[n]c_r^\zeta = nc_r^\zeta, \quad [\tau]c_r^\zeta = c_r^\zeta, \quad \mathcal{U}_p c_r^\zeta = c_{pr}^\zeta. \quad (\text{UOS8.1})$$

The source integer scalar and the independent spectral prime action are explicitly different operations. Their notation is not interchanged. The faithful source distinction between τ and integer1 remains in every full P, P' and extra closed term.

For the existing receiving support lattice L_{supp} , each receiving linear map f above has the exact label-preserving lift

$$G_{L_{\text{supp}}}(f)(v, \alpha) = (f(v), \alpha), \quad G_{L_{\text{supp}}}(V) = \{(0, \alpha) : \alpha \in L_{\text{supp}}\} \cup \{(v, 1_{L_{\text{supp}}}) : v \in V\}. \quad (\text{UOS8.2})$$

Linearity proves that a nontop-labelled zero stays a zero with its original label. Identity and composition follow by substitution. In particular injectivity of $c \mapsto (c, 0)$ remains injectivity on the lifted common-label carrier. Independently labelled closed factors remain separate factors of products; no common-label carrier is identified with such a product. Vanishing amplitudes in UOS3.4 retain their support labels and are not identified with primitive τ .

The original single-chart lifting map is also still present:

$$(h, d_0, d_1) \mapsto ((\widehat{h}, 0, 0), (h, d_0, d_1))$$

from V_- to the global Fourier graph. The equality $\Sigma \widehat{h} = R \Sigma h$ proves it is a continuous section of restriction. For the full source-faithful coefficients append the given minus extra coordinate and zero plus extra coordinate. This gives the corresponding section there too. Its ordinary localization connecting map is zero by that explicit section. The nonzero class c_1^ζ does not alter this original map or turn that proved section into a failure.

155.10 UOS8A. The entire faithful group orbit in the actual supported comparison

The original-zeta calculation GGT0–GGT8 has now been read completely. It proves the following bounds for the actual zeros with all multiplicities:

$$S_T(x) = \sum_{\rho} m_{\rho} x^{\rho} e^{\rho^2/T^2}, \quad T \geq 1,$$

$$S_T(1) = \frac{T}{2\sqrt{\pi}} \left[\log T - \log(4\pi) - \frac{\gamma}{2} \right] + 1 + O(\log(2+T)), \quad S_T(x) = O_x(T) \quad (x \neq 1). \quad (\text{UOS8A.1})$$

These are consequences of the full original formula GGT3.2, with both prime series, the exact archimedean contour and the original pole term. GGT3.3–GGT3.6 retains all crossed trivial-zero terms in every finite contour comparison. GGT8 also proves that for each $F \in \mathcal{B}$ and fixed $x > 0$,

$$\sup_{T \geq 1} \sum_{\rho} m_{\rho} |x^{\rho} F(\rho) e^{\rho^2/T^2}| < \infty. \quad (\text{UOS8A.2})$$

We use these full-space results to determine the orbit's actual algebraic size.

Let $r_1, \dots, r_N > 0$ be pairwise distinct. Suppose

$$\sum_{i=1}^N a_i c_{r_i}^{\zeta} = 0.$$

By the definition of the cokernel and linearity of the contour, there is a single $F \in \mathcal{B}$ for which

$$\lambda_H^{\zeta} = D_{\zeta}[F] = \lambda_F^{\zeta}, \quad H(s) = \sum_{i=1}^N a_i r_i^s.$$

PGD2's complete kernel theorem then says $H - F \in \mathcal{J}$: the full original multiplicity jets agree, in particular $H(\rho) = F(\rho)$ at every original zero. Fix j and multiply this value equality by $m_{\rho} r_j^{-\rho} e^{\rho^2/T^2}$. The sums are absolutely convergent by GGT1 and UOS8A.2, so summation gives exactly

$$\sum_{i=1}^N a_i S_T(r_i/r_j) = \sum_{\rho} m_{\rho} r_j^{-\rho} F(\rho) e^{\rho^2/T^2}. \quad (\text{UOS8A.3})$$

The right side is $O_{F,r_j}(1)$. The term with $i = j$ equals $a_j T \log T / (2\sqrt{\pi}) + O(T)$, and each other term is $O(T)$ because $r_i/r_j \neq 1$. Divide by $T \log T$ and let $T \rightarrow \infty$. The result is $a_j / (2\sqrt{\pi}) = 0$. This holds for each j , proving linear independence of the entire positive-real orbit. The argument uses the necessary value consequence of full jet equality; it neither discards the higher jets from Q nor assumes their synthesis.

Define the group algebra $\mathbb{C}[\mathbb{R}_+^{\times}]$ to consist of finite formal sums of basis elements $[r]$, with $[r][a] = [ra]$. These brackets are receiving group-algebra notation, distinct from the source-action brackets in UOS8. The explicit module map is

$$\iota_{\text{orb}} : \mathbb{C}[\mathbb{R}_+^{\times}] \longrightarrow C_{\zeta}, \quad \sum_r a_r [r] \longmapsto \sum_r a_r c_r^{\zeta}. \quad (\text{UOS8A.4})$$

It is injective by the preceding proof and intertwines left translation by $[a]$ with the actual receiving action $\mathcal{U}_a$. Consequently its image is the regular cyclic submodule generated by c_1^{ζ} . This is an algebraic equivariant linear embedding into a module; it does not equip C_{ζ} with an invented multiplication or identify the embedding as a ring homomorphism. No Hausdorff topological embedding into this non-Hausdorff cokernel is asserted.

The subgroup of positive rationals is free abelian on the original arithmetic primes, by the already reconstructed unique integer factorization and its extension to ratios. Therefore UOS8A.4 restricts to the injective module map

$$\mathbb{C}[t_p, t_p^{-1} : p \text{ an original prime}] \longrightarrow C_{\zeta}, \quad \prod_p t_p^{n_p} \longmapsto c_{\prod_p p^{n_p}}^{\zeta}, \quad (\text{UOS8A.5})$$

where each monomial and polynomial involves finitely many primes. Every original prime dilation is retained; this is not a finite packet or a quotient by relations among distinct prime products.

Composing with the actual cone map UOS6.5 gives the proved supported placement of this whole regular module:

$$\boxed{\mathbb{C}[\mathbb{R}_+^\times] \hookrightarrow H^1(\mathcal{L}_\zeta) = C_\zeta \oplus Y, \quad [r] \longmapsto (c_r^\zeta, 0).} \quad (\text{UOS8A.6})$$

It is injective because both constituent maps are injective. On the two residue denominators, its exact comparison-cone companion mirror is the linear involution

$$[r]_\zeta \longmapsto -r[1/r]_{\zeta^\vee}. \quad (\text{UOS8A.7})$$

Its square is identity since $(-r)(-1/r) = 1$. The original oriented global-dual mirror instead has $+r[1/r]$. The minus sign in UOS8A.7 means it is not a unital group-algebra homomorphism; it is the exact companion linear symmetry of the cone. Both variants retain the relation $\mathcal{U}_a w = a w \mathcal{U}_{a^{-1}}$.

Finally the complete source action on this module is the one already proved in UOS8: source n acts by the scalar n and source τ by identity on this receiving summand, while a spectral prime translates its basis. The source distinction remains faithful on the retained extra closed copies of the full complexes. None of UOS8A.1–UOS8A.7 assigns a numerical weight to primitive Z_1/τ or turns the regular module into a purity theorem. It determines the size, exact prime action, companion inversion and actual supported placement of the global object that has been constructed.

155.11 UOS9. Exact mathematical placement of the global orbit

There are three proved statements, with their actual maps:

1. The functional λ_1^ζ is in the ordinary continuous dual Q' . Its anti-diagonal supported class is $\delta' \lambda_1^\zeta$, so it already comes from the global dual. Its half-diagonal evaluates the actual original boundary nontrivially by UOS3.6. Both maps retain all support and orientation data.
2. Its class c_1^ζ is nonzero because $\lambda_1^\zeta \notin D_\zeta Q$. This is exactly a failure of representation by the specified original-zeta residue pairing. It is the actual residue-comparison cone class, not a newly asserted obstruction to the original Fourier restriction lift. The direct anti-diagonal supported map cannot descend through this cokernel without the further quotient proved in UOS4.
3. The exact chain map UOS6.4 injects this relative class into the dual-supported comparison cone, where its image is $(c_1^\zeta, 0)$. The full real orbit and companion inversion are respectively $(c_r^\zeta, 0) \mapsto (c_{ar}^\zeta, 0)$ and $(c_r^\zeta, 0) \mapsto (-rc_{1/r}^{\zeta^\vee}, 0)$. The map on H is identity, and the map on H' is the explicit endpoint/extra restriction, so these terms have not been omitted.

These constructions establish the orbit's precise relationship to the original supported localization mechanism. They prove no numerical separation of the weights on that mechanism, and they do not turn the arithmetic constant 1, a functional, or a comparison-cone class into primitive Z_1/τ . The new mathematical result is the complete companion and prime action together with the actual localization-row and cone morphisms locating the already constructed global object.

155.12 Exact original character lifting and its full supported return

MCL0–MCL11 now computes the original transpose $A' \rightarrow S'$ on every actual zero and all its Mellin jets. A target jet of order j needs generalized-character length exactly $m+j+1$ at a zero of multiplicity m . The full Euler-distribution classification proves minimality against the whole original continuous dual; two actual primes remove the single-prime character aliases. Every prime logarithm and the full Gamma/Fourier return remain. This is an actual continuous lift, with a nonzero class only when its original annihilator must be preserved.

RPC0–RPC8 constructs the original restriction’s actual pushout by the residue image, proves its precise cochain location and a single coherent real/prime-equivariant section on all locally finite classes. ORE0–ORE10 calculates the unmodified connecting class, including its full Gamma germ and rank $\min(m,r)$, then proves its place in every degree of the full supported derived comparison. The degree-two map is $(\eta,e) \mapsto (-\delta_q(\eta)/2,e)$. Thus its image in C_ζ/qC_ζ is zero, while the full supported map retains its Y/qY component. Both assertions follow from the constructed maps; no cone degree or extra coefficient is discarded.

These results refine the meaning of the prior finite-lift statement without withdrawing its proved scope. Fixed-annihilator splitting is stronger than the surjectivity in Deligne’s invariant-cycle theorem. The original longer-chain map is already surjective. The present numerical character truncation retains its same-character extension whenever it retains the target; it is not Deligne’s independently proved geometric weight gap. Complete proofs, independent checks and inspected reproducible diagrams have been inserted into the cumulative TeX. The actual tau weight-separation goal remains active.

156 Independent verification of the full finite prime-orbit relation theorem

24 September 2026. Mathematical check FPC0–FPC6.

156.1 FPC0. Reading coverage and unchanged source

The complete FPO0–FPO8 was read and checked against GGT0–GGT8, the original PGD multiplier comparison, and the actual ASD14 cone. Source $Z_0, Z_1/\tau, Z_2$, all retractions, and the receiving-operation domains are unchanged. In particular the arithmetic constant function 1 is not primitive τ .

The original author TeX of Alain Connes and Caterina Consani, Weil positivity and Trace formula, the archimedean place, arXiv:2006.13771v1, was read at Appendix “Explicit formula,” lines2033–2069, and the immediately following Proposition mainprop and its proof. The bibliography entry for Enrico Bombieri’s The Riemann hypothesis (2006), credited by that appendix, was checked. No complete reading of Bombieri’s chapter is claimed.

The exact author source was weil-compo.tex SHA256 b01d353b0423b6fedee373c3c33fe3678ee733f62049810750f6c64ef20f3fc. The appendix’s “all complex zeros” wording at line2042 is read with its explicit nontrivial-zero convention in Proposition mainprop, line2075, and the original contour calculation in GGT. An extra infinite series over negative even integers is not inserted into that formula.

156.2 FPC1. The compact-support extension retains every term

For the actual Gaussian

$$f(u) = \frac{T}{2\sqrt{\pi}} e^{-T^2 \log^2(u/x)/4}, \quad h(v) = f(e^v),$$

take the stated smooth cutoff $h_R(v) = \eta(v/R)h(v)$, with $\eta = 1$ on $[-1, 1]$ and zero outside $[-2, 2]$. Every derivative of h is a fixed polynomial in $v - \log x$ times its Gaussian. The difference $h_R - h$, with every derivative, is supported in $|v| \geq R$. The product rule therefore proves

$$\|e^{A|v|}(h_R - h)^{(k)}\|_{L^1} + \|e^{A|v|}(h_R - h)^{(k)}\|_{\infty} \longrightarrow 0$$

for every fixed A, k . Derivatives falling on the cutoff contribute only bounded factors R^{-j} , and a Gaussian tail dominates all the resulting fixed exponential-polynomial weights.

Integration by parts in the Mellin transform on logarithmic coordinates proves convergence in every $\mathcal{B}$ strip seminorm. The zero count gives $\sum_{\rho} m_{\rho}(1 + |\Im \rho|)^{-3} < \infty$, so this implies convergence of the full nontrivial-zero trace. Endpoint evaluations at zero and one are continuous on $\mathcal{B}$.

The two prime evaluations obey

$$|f_R(n) - f(n)| \leq \varepsilon_R n^{-A}, \quad |f_R^{\sharp}(n) - f^{\sharp}(n)| \leq \varepsilon_R n^{-A-1}, \quad \varepsilon_R \rightarrow 0.$$

For $A > 2$, the sums weighted by $\Lambda(n) \leq \log n$ converge. Thus every prime repetition passes to the limit.

For the original archimedean expression, logarithmic substitution gives exactly

$$\mathcal{W}_{\mathbb{R}}(f) = (\log 4\pi + \gamma)h(0) + \int_0^{\infty} \frac{h(v) + e^{-v}h(-v) - 2e^{-v}h(0)}{1 - e^{-2v}} dv. \quad (\text{FPC1.1})$$

Its numerator vanishes at zero and has a bounded derivative there. In the difference for $h_R - h$, the complete numerator is identically zero near zero and its value term is exactly zero. On the complement of a fixed neighborhood of zero the denominator is bounded below, and the weighted L^1 convergence proves convergence of the whole integral. This verifies the cutoff extension used in FPO3, including its singular endpoint and reflected term.

156.3 FPC2. Direct independent Gamma-integral comparison

There is a direct verification of the archimedean equality that does not require subtracting the two explicit-formula identities. Write

$$\Phi(s) = x^s e^{s^2/T^2}, \quad g_\chi(s) = \frac{\chi'(s)}{\chi(s)}, \quad \chi(s) = \pi^{s-1/2} \frac{\Gamma((1-s)/2)}{\Gamma(s/2)}.$$

For $\Re z > 0$, the Euler digamma series used in GGT4 gives

$$\psi(z) + \gamma = \int_0^\infty \frac{e^{-u} - e^{-zu}}{1 - e^{-u}} du. \quad (\text{FPC2.1})$$

Indeed expand the denominator geometrically and integrate the paired terms. Their absolute integral sum converges: on each pair the mean-value integral bounds the difference by a constant times $ue^{-(n+c)u}$, with $c = \min(1, \Re z) > 0$, whose integral is $O((n+c)^{-2})$. This justifies interchange and recovers the convergent digamma difference series.

On the actual middle edge $s = 1/2 + it$, the two digamma arguments have real part $1/4$. Equation (FPC2.1) gives

$$g_\chi(1/2 + it) = \log \pi + \gamma + \int_0^\infty \frac{e^{-u/4} \cos(tu/2) - e^{-u}}{1 - e^{-u}} du. \quad (\text{FPC2.2})$$

This retains the full original Gamma quotient.

The double integral against $\Phi(1/2 + it) dt/(2\pi)$ is absolutely integrable. For $0 < u \leq 1$, the absolute value of the quotient in (FPC2.2) is bounded by $C(1+t^2u)$, using $|\cos(tu/2) - 1| \leq t^2u^2/8$. For $u \geq 1$ it is bounded by $Ce^{-u/4}$. The Gaussian in t makes both bounds integrable. Hence Fubini applies before Fourier inversion.

Original Mellin inversion, with its unshifted definition, gives

$$\begin{aligned} \frac{1}{2\pi} \int_{\mathbb{R}} \Phi(1/2 + it) dt &= h(0), \\ \frac{1}{2\pi} \int_{\mathbb{R}} \Phi(1/2 + it) \cos(tu/2) dt &= \frac{1}{2} \left[e^{-u/4} h(-u/2) + e^{u/4} h(u/2) \right]. \end{aligned} \quad (\text{FPC2.3})$$

Substitution into (FPC2.2), followed by the exact variable change $u = 2v$, proves

$$A_T^{(1/2)}(x) = (\log \pi + \gamma) h(0) + \int_0^\infty \frac{h(v) + e^{-v} h(-v) - 2e^{-2v} h(0)}{1 - e^{-2v}} dv. \quad (\text{FPC2.4})$$

All integrals have their ordinary convergent endpoint values. Compare this with the author's exact expression (FPC1.1). Their difference is

$$\begin{aligned} A_T^{(1/2)}(x) - \mathcal{W}_{\mathbb{R}}(f) &= -\log 4 h(0) + 2h(0) \int_0^\infty \frac{e^{-v} - e^{-2v}}{1 - e^{-2v}} dv \\ &= -\log 4 h(0) + 2 \log 2 h(0) = 0. \end{aligned} \quad (\text{FPC2.5})$$

The middle integral is exactly $\int_0^\infty e^{-v}/(1 + e^{-v}) dv = \log 2$. This checks the coefficient $\log 4\pi + \gamma$, including the tail subtraction in the source distribution.

Finally the full logarithmic derivative g_χ has residue $+1$ at zero. The finite shift from $\Re s = -1$ to $1/2$ therefore gives

$$A_T^{(1/2)}(x) - A_T(x) = \Phi(0) = 1.$$

Together with (FPC2.5), this proves independently

$$\boxed{\mathcal{W}_{\mathbb{R}}(f_{T,x}) = A_T(x) + 1.} \quad (\text{FPC2.6})$$

The value 1 is the original Mellin endpoint integral, not the Gaussian's point value $f(1)$, and not primitive τ .

156.4 FPC3. The quotient really is an algebra, while the full cokernel is a module

The multiplier algebra $\mathcal{M}$ in FPO1 is closed under multiplication. If $H \in \mathcal{M}$, $F \in \mathcal{B}$, then

$$b_{A,N}(HF) \leq C_A b_{A,N+d_A}(F),$$

so $\mathcal{B}$ is an ideal of $\mathcal{M}$. The full-zero-jet condition defining $\mathcal{J}$ is preserved by multiplication, by the finite Leibniz formula at every actual zero. Thus $\mathcal{B} + \mathcal{J}$ is an ideal, and $\mathcal{A}_\zeta = \mathcal{M}/(\mathcal{B} + \mathcal{J})$ is a well-defined quotient algebra.

The map from the finite group algebra sends

$$\delta_r \mapsto [r^s].$$

The exact equality $r^s t^s = (rt)^s$, using the real logarithms of positive numbers, proves multiplicativity. Finite sums preserve this identity by expansion, and the arithmetic unit goes to the arithmetic unit. This verifies an algebra map into $\mathcal{A}_\zeta$.

PGD's map $\mathcal{A}_\zeta \rightarrow \mathcal{C}_\zeta$ is proved linear and injective. No multiplication on all of $\mathcal{C}_\zeta$ is part of that construction. FPO5.8 correctly keeps this distinction. The additional intertwining

$$T_a^\circ[\lambda_{rs}] = [\lambda_{(ar)^s}]$$

identifies the injected space with the regular group module. It does not require an algebra structure on the complete cokernel.

156.5 FPC4. Full zero-jet relations imply the decisive value relations

A zero class in $\mathcal{A}_\zeta$ means that for some $F \in \mathcal{B}$, the difference

$$H(s) - F(s), \quad H(s) = \sum_{i=1}^k a_i r_i^s,$$

has all actual required zero jets equal to zero. In particular $H(\rho) = F(\rho)$ at every actual zero. Using only this consequence is valid: the proof is not weakening the ideal used to define the quotient.

For a fixed support index j , all sums in

$$\sum_{i=1}^k a_i S_T(r_i/r_j) = \sum_{\rho} m_{\rho} r_j^{-\rho} F(\rho) e^{\rho^2/T^2}$$

converge absolutely. The right side is bounded uniformly for $T \geq 1$ by

$$e \max(1, r_j^{-1}) b_{1,3}(F) \sum_{\rho} m_{\rho} (1 + |\Im \rho|)^{-3}. \quad (\text{FPC4.1})$$

This finite bound uses the original quotient representative and retains the actual zero multiplicities.

Exactly one left-side scale is 1, since the listed r_i are distinct. GGT gives

$$\sum_{i=1}^k a_i S_T(r_i/r_j) = \frac{a_j}{2\sqrt{\pi}} T \log T + O(T). \quad (\text{FPC4.2})$$

The finite sum of all other fixed-scale errors is $O(T)$, including every prime-power resonance. Dividing the equality by $T \log T$ gives $a_j = 0$. This works for complex coefficients as well: the absolute value of the remainder divided by $T \log T$ tends to zero, and the nonzero real scalar $1/(2\sqrt{\pi})$ multiplies a_j . Repeating for each j proves that the finite relation is zero.

This verifies the injectivity in the actual quotient, which is stronger than linear independence of the original exponential functions before quotienting.

156.6 FPC5. Prime algebra, actual cone cycles, and support labels

Unique integer factorization proves that

$$\bigoplus_{p \text{ prime}} \mathbb{Z} \longrightarrow \mathbb{Q}_+^\times, \quad (n_p) \longmapsto \prod_p p^{n_p}$$

is an isomorphism of groups. Only finite-support exponent tuples occur. Thus the corresponding finite group algebra is exactly the Laurent polynomial algebra in all prime variables, with each polynomial involving finitely many variables and monomials. This verifies FPO6.1 and its use of all negative and positive prime exponents.

The actual ASD14 cycle associated with a multiplier is $\pi' \lambda_H$. Since $\pi d = 0$, its image under d' is zero. If this cycle is a boundary, then $\pi' \lambda_H = \pi' D\pi(b)$. Injectivity of π' gives $\lambda_H \in DQ$, the exact relation excluded by FPC4 for a nonzero finite group-algebra element. Thus the orbit injection survives in the actual cone cohomology, with the stated positive middle differential.

The lift $(v, \alpha) \mapsto (f(v), \alpha)$ is well-defined on the existing support carrier because nonzero amplitudes occur only at top support and a zero output retains its label. Injectivity of the original receiving map implies injectivity of its lift. The separate faithful closed source copies remain in the full complex and are not removed by the orbit calculation.

156.7 FPC5A. Verification of the appended supported-cone propagation

The final FPO8 paragraph now invokes UOS3–UOS8A in the complete supported comparison. I read its original localization row, transpose row, chain comparison, cone and cohomology maps in UOS3–UOS6, and the source actions, Fourier section and full orbit propagation in UOS8–UOS8A.

Write $Y = \chi \otimes Q'$, $M_\zeta = D_\zeta Q$, and $\alpha\lambda = (\lambda, -\lambda)$. The actual degree-one map of cones sends the cycle $\pi' \lambda$ to $(\pi' \lambda, -\pi' \lambda)$. The displayed target cohomology is $Y^2/\alpha M_\zeta$, and its exact decomposition is

$$[(\lambda_+, \lambda_-)] \longmapsto \left(\left[\frac{\lambda_+ - \lambda_-}{2} \right], \frac{\lambda_+ + \lambda_-}{2} \right) \in (Y/M_\zeta) \oplus Y.$$

Substituting $(\lambda, -\lambda)$ gives $([\lambda], 0)$. A change $\lambda \mapsto \lambda + D_\zeta q$ changes the representative by $\alpha D_\zeta q$, which is precisely the subspace divided out. Thus the appended map is exactly $c_r \mapsto (c_r, 0)$, and is injective without assuming a splitting $Q' \rightarrow Q$.

In degree -1 , the chain map is the identity on P ; its restriction to $\ker d = H$ is the identity. In degree 2, the chain map is the identity on $\chi P'$; the resulting map from $\chi H'$ to $\chi E'$ is restriction along $E \hookrightarrow H$. In the original coordinates it keeps the four endpoint coordinates and both extra closed dual copies and removes only the S' coordinate. This verifies the map stated in FPO8, including its direction.

Finally UOS8's original Fourier section has restriction equal to identity by $\widehat{\Sigma h} = R\Sigma h$, with the retained minus extra coordinate and zero plus extra coordinate. Its localization connecting map is consequently zero. The supported placement of the residue-comparison classes does not replace that section or contradict its existence. No additional hypothesis is needed for the appended propagation.

156.8 FPC6. Outcome and the identified typesetting defect

FPO0–FPO8's mathematical claims checked above are valid: the compact-support extension converges in every required term; the full archimedean comparison is exactly $A_T + 1$; the group-algebra map is injective after the actual multiplier quotient; the map into the complete cokernel is linear and equivariant; and the full-zero-jet relation argument retains every multiplicity.

One typesetting defect was found in the reviewed Markdown: FPO4.3, line166, contained two literal tab characters followed by “frac12,” in place of the two commands `\frac12`. Both

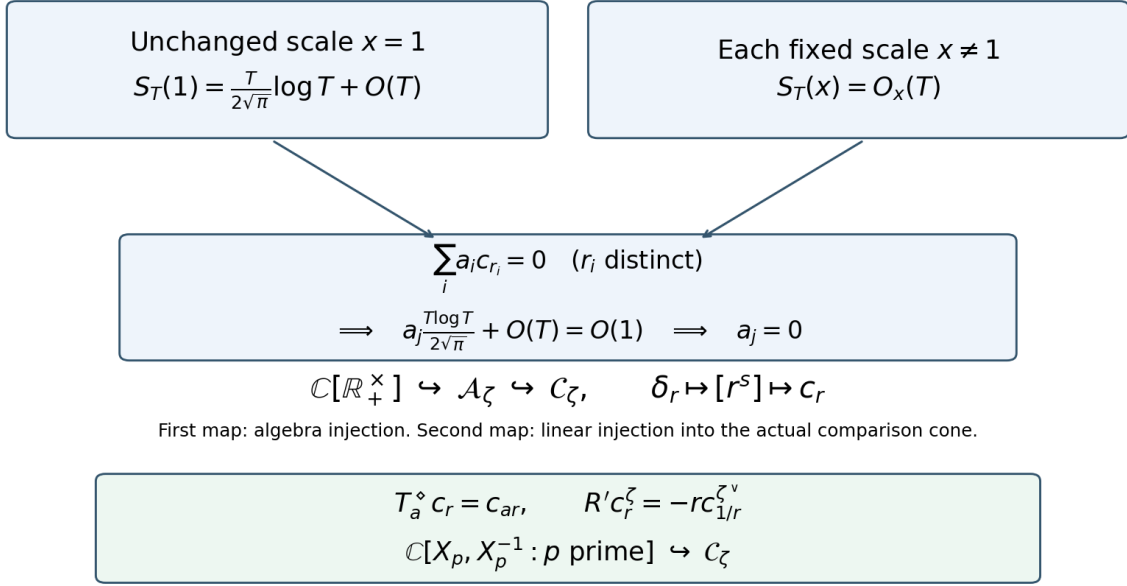
backslashes in the two one-half digamma coefficients have now been restored by the parent author. This does not change the verified mathematical signs or factors. The check did not overwrite their file.

The result concerns the exact original receiving comparison and its finite orbit relations. It does not identify the complete cokernel with an algebra, assert a purity bound, or identify a nonzero cokernel class with an RH counterexample.

Global prime actions remain distinguishable in the actual quotient

$$S_T(x) = \sum_{\rho} m_{\rho} x^{\rho} e^{\rho^2/T^2}, \quad T \geq 1, x > 0$$

Complete constants, prime-power resonances and Gamma terms: GGT3–GGT7; FPO4.



The original zeta function remains explicit. The Gaussian is a test, not a replacement.
 Both endpoint values, all multiplicities and every finite trivial-zero contour correction are retained.
 No numerical weight, arithmetic coordinate, addition or parity is assigned to primitive Z_1 / τ .

Proofs: FPO0–FPO8; GGT0–GGT8; UOS2, UOS9.
 Human sources: Connes–Consani 2006.13771v1, Explicit formula; 0903.2024v3 §5; Meyer math/0412277v3.
 The geometric lifting target remains Deligne, Weil II §§3.6.1–3.6.3.

Figure 55: The global trace distinguishes all finite dilation relations

157 Global prime action and the exact supported map

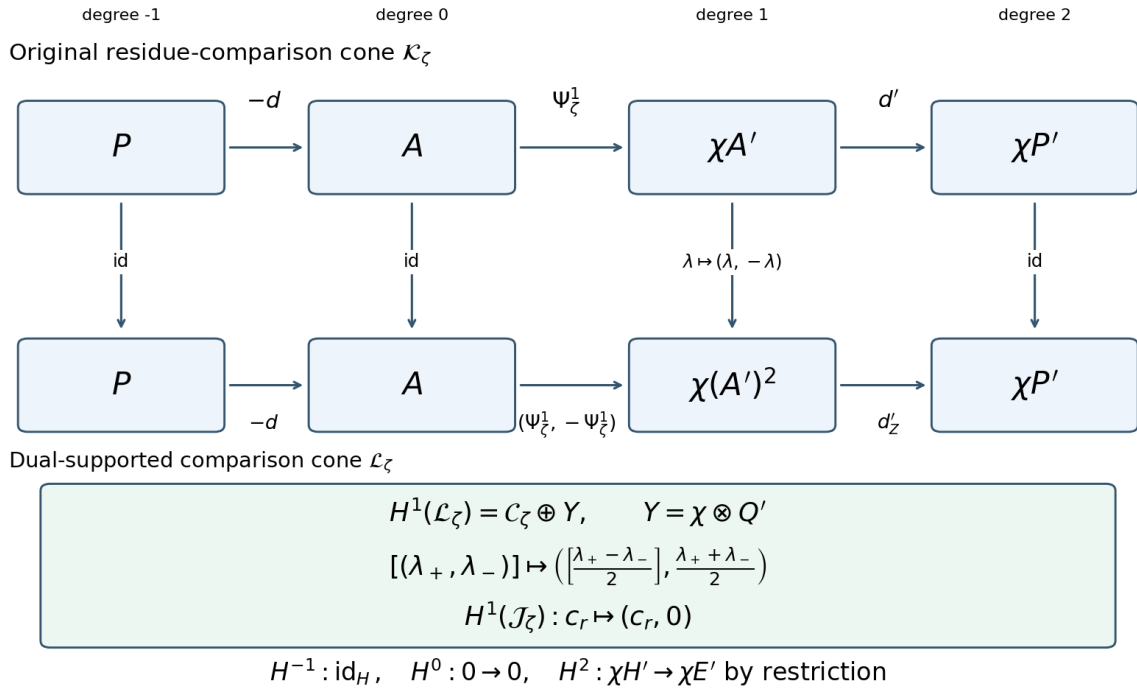
Figure 1. Complete proofs: GGT0–GGT8, FPO0–FPO8, independently verified in FPC0–FPC6. These are proved asymptotic statements for each fixed positive scale and the original full zero multiset, not a numerical plot. FPO4 retains the complete next constants, prime-power resonances, both endpoints and the Gamma integral. The group algebra first injects as an algebra into the multiplier quotient, then linearly into the actual comparison cokernel. The full target cokernel is not supplied with an invented multiplication.

Figure 2. Complete proof: UOS3–UOS8A, with original full cone ASD14. Every degree, sign and half-factor is retained. The chain map is identity in degrees $-1, 0, 2$ and anti-diagonal in degree 1. Its cohomology map is identity on H , injective $c \mapsto (c, 0)$ on the global relative cokernel, and the exact restriction $\chi H \rightarrow \chi E$. All four endpoint lines and both full extra closed coefficient copies survive in E . The existing original Fourier restriction still has its constructed section. No numerical weight is assigned by the illustration.

Reproducible vector and raster source: draw_global_prime_support.py. Both final PNGs were inspected. Seven exact symbolic factor and chain checks supplement the complete analytic and coefficient proofs. Source Z_0 , Z_1/τ and Z_2 keep the current user definitions and retractions.

Human sources: Alain Connes and Caterina Consani, 2006.13771v1, Appendix Explicit formula and 0903.2024v3 §5; Ralf Meyer, math/0412277v3. The geometric lifting target is Pierre Deligne, Weil II §§3.6.1–3.6.3. Exact source versions and actual reading coverage remain in SOURCE_READING_USE_PRIVATE.json.

The relative class has an exact supported placement



E retains all four endpoint lines and both full extra closed coefficient copies.
 The original Fourier restriction still has its explicit section. This relative cone is a different constructed map.
 The comparison preserves source multiplication and every existing support label.
 Complete proof: UOS3–UOS8A; original cone ASD14. Reproducible source: draw_global_prime_support.py.
 Human geometry: Connes–Consani 0903.2024v3 §5. Lifting target: Deligne, Weil II §§3.6.1–3.6.3.

Figure 56: The complete comparison of the two cones

158 Exact Mellin-character lifting through the original summation transpose

24 September 2026. Independent derivation MCL0–MCL11. This computes every finite Mellin-jet lift at every actual nontrivial zero of the original zeta function, proves its least generalized-character order in the full original continuous dual, and places its exact connecting class in the original supported localization row.

158.1 MCL0. Source, prerequisites and prior work used

The controlling source is the user's Z_0 absence, primitive Z_1/τ without parity or source addition, and the supplied integer layer with its full amounts and Z_2 data. B1–B5, P1–P5 and retraction R1 were read in `SOURCE_OPERATIONS_AND_PROOFS.md`. The current `CORPUS_AND_OPERATION_RULES.md` (private source record) and complete relevant passages USR-9ad1c0a2d09dba92, USR-f55d16feb948d8c2, USR-6152e3bc6302258c, USR-4322be19bff532cd and USR-64a88219a3ecddd3 were consulted in their retained verbatim corpus. Their full arithmetic-reconstruction and global-quotient arguments precede this receiving calculation. Neither a Mellin parameter nor an integer coefficient is constructed from primitive τ .

A bounded search of the retained proof corpus located the existing summation/Mellin identity, closed-image theorem, continuous-dual exact row and finite spectral lifts. The calculation below uses these proved maps; its new content is the least order for lifting through the actual Σ' , its full prime defect, Fourier return, and the resulting character connecting map. It is not a repetition of the earlier lift from Q' into Q under the residue pairing: its source and target are the actual $A' \rightarrow S'$.

The human source of the coefficients, Fourier identity and two-chart restrictions is Alain Connes and Caterina Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, arXiv:0903.2024v3, §5. Their original author TeX and full relevant reading coverage were retained in DCP0–DCP3; DCP3 and DCP10 were read for this calculation. The closed-image analytic context is Ralf Meyer, *A spectral interpretation for the zeros of the Riemann zeta function*, arXiv:math/0412277v3. The exact original-space image and inverse used here are proved in OMS1–OMS5 and its retained SSI derivation. The dual topology and supported maps were read in ASD1–ASD5 and UOS1–UOS8.

Pierre Deligne's *La conjecture de Weil. II*, §§3.6.1–3.6.3 supplies the requested supported lifting mechanism. Its geometric duality and weight estimates are not premises here. MCL9 constructs an actual connecting map for the specified original coefficient sequence; MCL10 locates it in the source-supported receiver. All needed sums, derivatives, distribution equations and quotients are constructed below in those receiving spaces.

158.2 MCL1. Original spaces and the full holomorphic Mellin families

Retain

$$S = \{h \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : h(-v) = h(v), h(0) = 0, \int_{\mathbb{R}} h(v) dv = 0\}$$

with its closed-subspace Schwartz topology, and

$$A = \{b \in C^\infty(0, \infty) : p_{N,j}(b) = \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j b(u)| < \infty \text{ for all } N, j \geq 0\}. \quad (\text{MCL1.1})$$

For $z \in \mathbb{C}$, define the continuous linear functional

$$M_{A,z}(b) = \int_0^\infty b(u) u^z \frac{du}{u}. \quad (\text{MCL1.2})$$

For $\Re z > -2$, define

$$M_{S,z}(h) = \int_0^\infty h(v) v^z \frac{dv}{v}. \quad (\text{MCL1.3})$$

These are full original positive-axis Mellin integrals. Their domains differ and are retained.

For z in a compact set $K \subset \mathbb{C}$, choose an integer $N > 1 + \sup_{z \in K} |\Re z|$. On $u \leq 1$, $|b(u)| \leq p_{N,0}(b)u^N$; on $u \geq 1$, $|b(u)| \leq p_{N,0}(b)u^{-N}$. Multiplying these bounds by $u^{\Re z-1}|\log u|^j$ gives uniformly finite integrals for every j . Thus

$$\partial_z^j M_{A,z}(b) = \int_0^\infty b(u)u^z(\log u)^j \frac{du}{u}. \quad (\text{MCL1.4})$$

The same estimates, using a slightly larger compact parameter set and the exponential Taylor remainder, prove locally uniform convergence of the Taylor series on each bounded set of A . In particular $M_{A,z}$ is entire as a family in the strong continuous dual, not only after one fixed evaluation.

Evenness and $h(0) = 0$ give $h'(0) = 0$. Taylor's integral formula gives

$$|h(v)| \leq \frac{v^2}{2} \sup_{|t| \leq 1} |h''(t)| \quad (0 < v \leq 1).$$

Together with Schwartz decay at infinity, this bounds the integrand of every z -derivative in (MCL1.3), uniformly on compact subsets of $\Re z > -2$. The same Taylor-remainder argument proves holomorphy there, with

$$\partial_z^j M_{S,z}(h) = \int_0^\infty h(v)v^z(\log v)^j \frac{dv}{v}. \quad (\text{MCL1.5})$$

Its endpoint identity is exactly

$$M_{S,1}(h) = \int_0^\infty h(v) dv = \frac{1}{2} \int_{\mathbb{R}} h(v) dv = 0. \quad (\text{MCL1.6})$$

This particular zero functional is not omitted or relabelled as an arbitrary eigenvector.

Retain the actual two-sign summation and its transpose:

$$\Sigma h(u) = 2 \sum_{n \geq 1} h(nu), \quad \Sigma' : A' \longrightarrow S', \quad (\Sigma' \lambda)(h) = \lambda(\Sigma h). \quad (\text{MCL1.7})$$

For $\Re z > 1$, Tonelli applied to absolute values gives

$$\sum_{n \geq 1} \int_0^\infty |h(nu)| u^{\Re z} \frac{du}{u} = \left(\sum_{n \geq 1} n^{-\Re z} \right) \int_0^\infty |h(v)| v^{\Re z} \frac{dv}{v} < \infty.$$

Changing $v = nu$, with the original factor 2, proves

$$\boxed{\Sigma' M_{A,z} = 2\zeta(z)M_{S,z}.} \quad (\text{MCL1.8})$$

This extends through $\Re z > -2$. Indeed the only pole of ζ there is at one, and (MCL1.6) makes its product removable; both sides are holomorphic after that specified removal. Evaluation on every h and the identity theorem proves equality, and the uniform estimates above give the same dual-family identity.

The contribution at the original pole has an explicit value. Write $\zeta(z) = (z-1)^{-1} + \gamma + O(z-1)$. Then

$$\Sigma' M_{A,1} = 2 \partial_z M_{S,z} \big|_{z=1}. \quad (\text{MCL1.9})$$

The pole has combined with the first source Mellin derivative, rather than being silently discarded. Only actual nontrivial zeros $0 < \Re \rho < 1$, never this endpoint, are used below.

For the prior centered receiver retain

$$\mathcal{T}k(u) = 2u^{-1/2}k(u), \quad \mathcal{T}^{-1}b(u) = \frac{1}{2}u^{1/2}b(u), \quad F_k(z) = \int_0^\infty k(u)u^{z-1/2} \frac{du}{u}.$$

Consequently

$$\mathcal{T}'(\partial_z^j M_{A,z})(k) = 2 \partial_z^j F_k(z). \quad (\text{MCL1.10})$$

Every factor and exponent of this earlier comparison survives.

158.3 MCL2. Actual dilation, transpose, contragredient and twist

For $a > 0$, set

$$T_a b(u) = b(u/a), \quad D_a h(v) = h(v/a).$$

Both are continuous automorphisms of their stated spaces; dilation multiplies the zero integral of h by a , keeps its zero value and evenness, and scales each defining seminorm by finite powers of a, a^{-1} . Direct summation gives

$$\Sigma D_a = T_a \Sigma. \quad (\text{MCL2.1})$$

The generators for $a = e^t$ are the continuous operators

$$L_A = -u\partial_u, \quad L_S = -v\partial_v.$$

For L_S , evenness is preserved, its value at zero vanishes, and integration by parts gives $\int L_S h = \int h = 0$. Differentiating in the stated topologies follows from the integral Taylor remainder for translation in t , bounded by the next two derivative seminorms. Write

$$K_A = L_A^t, \quad K_S = L_S^t, \quad \Sigma' K_A = K_S \Sigma'. \quad (\text{MCL2.2})$$

Here t denotes the ordinary continuous transpose, not a contragredient convention.

Substitution and differentiation prove the full formulas

$$\begin{aligned} T_a^t M_{A,z} &= a^z M_{A,z}, & D_a^t M_{S,z} &= a^z M_{S,z}, \\ T_a^t \partial_z^j M_{A,z} &= a^z \sum_{k=0}^j \binom{j}{k} (\log a)^{j-k} \partial_z^k M_{A,z}, \\ D_a^t \partial_z^j M_{S,z} &= a^z \sum_{k=0}^j \binom{j}{k} (\log a)^{j-k} \partial_z^k M_{S,z}. \end{aligned} \quad (\text{MCL2.3})$$

All identities hold before specializing a to a prime.

The continuous-dual representation used in the supported row has the contragredient followed by the stated character:

$$T_a^\vee = (T_{a^{-1}})^t, \quad \mathcal{U}_a^A = a(T_{a^{-1}})^t, \quad \mathcal{U}_a^S = a(D_{a^{-1}})^t, \quad \chi(a) = a.$$

Its generators are $G_A = 1 - K_A$, $G_S = 1 - K_S$, and

$$\mathcal{U}_a^A \partial_z^j M_{A,z} = a^{1-z} \sum_{k=0}^j \binom{j}{k} (-\log a)^{j-k} \partial_z^k M_{A,z}, \quad (\text{MCL2.4})$$

with the identical formula in S' . Thus ordinary transpose character a^z , contragredient character a^{-z} , and the supported twisted character a^{1-z} are explicitly different. The twist is an operation on the receiving representation, not a property assigned to primitive τ .

158.4 MCL3. Classify all generalized Euler characters on the original A'

Fix $\rho \in \mathbb{C}$ and abbreviate

$$A_j = \partial_z^j M_{A,z} \big|_{z=\rho} \quad (j \geq 0).$$

Then

$$(K_A - \rho)A_j = jA_{j-1}, \quad (K_A - \rho)A_0 = 0. \quad (\text{MCL3.1})$$

We prove the complete classification

$$\boxed{\ker(K_A - \rho)^d = \text{span}\{A_0, \dots, A_{d-1}\} \quad (d \geq 1).} \quad (\text{MCL3.2})$$

This concerns every continuous functional on A satisfying the equation, not a selected Mellin subspace.

Put $x = \log u$, $\phi(x) = b(e^x)$. This identifies A topologically with the smooth functions on $\mathbb{R}$ for which

$$q_{N,j}(\phi) = \sup_x (e^{Nx} + e^{-Nx}) |\phi^{(j)}(x)| < \infty$$

for all N, j . The generator becomes $-\partial_x$. The compactly supported smooth functions are dense. To prove this, choose $\eta \in C_c^\infty(\mathbb{R})$ equal to one on $[-1, 1]$, and set $\eta_R(x) = \eta(x/R)$, $R \geq 1$. Leibniz's rule bounds each $q_{N,j}((1 - \eta_R)\phi)$ by a finite sum of tails of $\phi^{(k)}$ on $|x| \geq R$, with coefficients $R^{-(j-k)}$ bounded by one. Those tails tend to zero using $q_{N+1,k}(\phi)$ and the factor $e^{-|x|}$. Hence $\eta_R\phi \rightarrow \phi$ in every seminorm.

A continuous functional on this space restricts to a distribution on $C_c^\infty(\mathbb{R})$: on each fixed compact set, the inclusion is continuous for the usual smooth test topology. If λ satisfies the left side of (MCL3.2), its restricted distribution U satisfies

$$(\partial_x - \rho)^d U = 0.$$

Indeed distribution differentiation means $U'(\phi) = -U(\phi')$, exactly the transpose of $-\partial_x$. Multiplication of distributions by the smooth function $e^{-\rho x}$ is defined by testing against that multiplier. The product rule gives

$$\partial_x^d (e^{-\rho x} U) = 0.$$

Here is the required distribution equation proof. If $V' = 0$, choose a compactly supported smooth η_0 with integral one. A compact test ψ of integral zero is the derivative of the compactly supported smooth function $\int_{-\infty}^x \psi(t) dt$. Therefore $V(\psi) = 0$, and $V(\phi) = V(\eta_0) \int \phi$; it is the constant distribution. Inductively, if $V^{(d)} = 0$, then $V^{(d-1)}$ is constant; subtract that constant times $x^{d-1}/(d-1)!$, and apply the induction hypothesis. Thus V is a polynomial of degree at most $d-1$. It follows that U has density $e^{\rho x} P(x)$ with $\deg P < d$.

Every such density is continuous on the entire original space by the exponential estimates in MCL1. Density of compact tests proves that its extension agrees with λ everywhere. The density $e^{\rho x} x^j$ is precisely A_j . These densities are linearly independent on compact tests, proving (MCL3.2) and uniqueness of the coefficients.

For any nonzero $\lambda = \sum_{j=0}^J c_j A_j$ with $c_J \neq 0$, (MCL3.1) gives

$$(K_A - \rho)^J \lambda = c_J J! A_0 \neq 0, \quad (K_A - \rho)^{J+1} \lambda = 0. \quad (\text{MCL3.3})$$

Thus its exact generalized-character order is $J+1$.

The full real action and simultaneous arithmetic-prime conditions admit the same classification. To make this assertion exact, let p, q be two distinct primes and let $r, s \geq 1$. Then

$$\ker(T_p^t - p^\rho)^r \cap \ker(T_q^t - q^\rho)^s = \text{span}\{A_j : 0 \leq j < \min(r, s)\}. \quad (\text{MCL3.4})$$

Here is a direct proof retaining the possible one-prime ambiguity. Restrict a functional to compact tests, write its distribution as $U = e^{\rho x} V$, and let E_h translate the density by $x \mapsto x + h$. The two equations become $(E_h - 1)^r V = 0$ and $(E_\ell - 1)^s V = 0$, with $h = \log p$, $\ell = \log q$. The commuting ideal generated by these two differences has its $(r+s-1)$ -st power killing V , by the binomial pigeonhole count. For integers m, n , $E_{mh+n\ell} - 1$ belongs to that ideal with coefficients that are finite translation sums. Negative integers are included using $E_{-h} - 1 = -E_{-h}(E_h - 1)$. Consequently

$$(E_{mh+n\ell} - 1)^{r+s-1} V = 0.$$

The ratio h/ℓ is irrational, since a rational relation would give equal positive powers of distinct primes. Integer combinations $mh + n\ell$ are dense in $\mathbb{R}$: the pigeonhole principle applied to fractional parts of multiples of h/ℓ gives nonzero such combinations arbitrarily close to zero,

and integer multiples approximate any fixed real number. Translation is continuous on compact tests in their test topology, so the last difference equation holds at every real step. Dividing by that step to the power $r + s - 1$ and taking its limit at zero gives $\partial_x^{r+s-1}V = 0$. Thus V is a polynomial. A nonzero finite difference with fixed nonzero step reduces polynomial degree exactly by one; its original two equations force degree $< r$ and $< s$. The density and extension argument above finishes (MCL3.4).

For one prime alone the conclusion would be false: $M_{A,\rho+2\pi ik/\log p}$, $k \in \mathbb{Z}$, has the same p^ρ eigenvalue. MCL3.4 proves that retaining two distinct original primes removes those aliases. It justifies the full-character minimality below also under simultaneous all-prime generalized-character requirements; no single-prime converse is assumed.

158.5 MCL4. The actual source Mellin jets are independent

Fix an actual nontrivial zero ρ , so $0 < \Re \rho < 1$, and put

$$S_j = \partial_z^j M_{S,z} \big|_{z=\rho}.$$

All these functionals are defined by MCL1. They satisfy

$$(K_S - \rho)S_j = jS_{j-1}, \quad (K_S - \rho)S_0 = 0. \quad (\text{MCL4.1})$$

They are linearly independent, including $S_0 \neq 0$.

To prove it, test against $h(v) = f(|v|)$, where $f \in C_c^\infty(0, \infty)$ and $\int_0^\infty f(v) dv = 0$. Such h belongs to the original S . If $\sum_{j=0}^J c_j S_j = 0$, the smooth function

$$v^{\rho-1}P(\log v), \quad P(X) = \sum_{j=0}^J c_j X^j,$$

annihilates every such zero-integral test. Choose one compact test of integral one and subtract its multiple from each arbitrary test. It follows that the displayed smooth function equals a constant on $(0, \infty)$, as distributions and hence pointwise. In $x = \log v$, differentiating this constant identity gives

$$P'(x) + (\rho - 1)P(x) = 0.$$

Since $\rho - 1 \neq 0$, comparison of the leading coefficient of a nonzero polynomial makes that impossible. Hence every $c_j = 0$. This also shows exactly why the endpoint $z = 1$, where MCL1.6 holds, must not be included in this assertion.

158.6 MCL5. The exact triangular lift of every jet

Let $m \geq 1$ be the actual multiplicity of ρ :

$$\zeta^{(k)}(\rho) = 0 \quad (0 \leq k < m), \quad \zeta^{(m)}(\rho) \neq 0.$$

Differentiating the full original identity (MCL1.8) gives

$$\Sigma' A_n = 2 \sum_{k=0}^n \binom{n}{k} \zeta^{(n-k)}(\rho) S_k. \quad (\text{MCL5.1})$$

In particular

$$\Sigma' A_n = 0 \quad (n < m),$$

$$\boxed{\Sigma' A_{m+j} = 2 \sum_{k=0}^j \binom{m+j}{k} \zeta^{(m+j-k)}(\rho) S_k.} \quad (\text{MCL5.2})$$

The coefficient of S_j is $2\binom{m+j}{j}\zeta^{(m)}(\rho) \neq 0$. Define the explicit lifts recursively:

$$\Lambda_{\rho,0} = \frac{A_m}{2\zeta^{(m)}(\rho)},$$

$$\Lambda_{\rho,j} = \frac{A_{m+j} - 2\sum_{k=0}^{j-1} \binom{m+j}{k} \zeta^{(m+j-k)}(\rho) \Lambda_{\rho,k}}{2\binom{m+j}{j} \zeta^{(m)}(\rho)} \quad (j \geq 1). \quad (\text{MCL5.3})$$

Induction using (MCL5.2) proves

$$\Sigma' \Lambda_{\rho,j} = S_j. \quad (\text{MCL5.4})$$

These are finite linear combinations of continuous functionals on the original A , so their continuity does not rely on an unspecified distribution completion.

There is a full analytic family giving the same lifts. In a disk about $t = 0$ containing no other zero and with $\Re(\rho + t) > -2$, write the exact factorization

$$\zeta(\rho + t) = t^m g_\rho(t), \quad g_\rho(0) = \frac{\zeta^{(m)}(\rho)}{m!}, \quad v_\rho(t) = \frac{1}{2g_\rho(t)}. \quad (\text{MCL5.5})$$

This is a proved local factor of the original ζ , whose derivatives are retained; it does not replace it as a working function. The dual-valued Taylor remainder

$$\mathcal{A}_\rho(t) = \frac{M_{A,\rho+t} - \sum_{n=0}^{m-1} t^n A_n / n!}{t^m}$$

has its unique removable continuation at zero by MCL1. Since every subtracted functional lies in $\ker \Sigma'$, (MCL1.8) gives

$$\Sigma'(v_\rho(t) \mathcal{A}_\rho(t)) = M_{S,\rho+t}. \quad (\text{MCL5.6})$$

Differentiating it yields the full expression

$$\Lambda_{\rho,j} = j! \sum_{k=0}^j \frac{v_\rho^{(j-k)}(0)}{(j-k)!(m+k)!} A_{m+k}. \quad (\text{MCL5.7})$$

Its leading coefficient is

$$\frac{j! m!}{2(m+j)! \zeta^{(m)}(\rho)}. \quad (\text{MCL5.8})$$

Formulas (MCL5.3) and (MCL5.7) agree because both lie in $\text{span}(A_m, \dots, A_{m+j})$, and (MCL5.2) is injective on that span. Every derivative of the reciprocal in (MCL5.7) is determined by the full identity $2g_\rho v_\rho = 1$. No derivative, multiplicity, sign or factor2 has been removed.

158.7 MCL6. Least generalized-character enlargement in the full dual

Let $N_A = K_A - \rho$, $N_S = K_S - \rho$. For integers $d \geq 0$, write

$$\mathcal{E}_A(d) = \text{span}(A_0, \dots, A_{d-1}), \quad \mathcal{E}_A(0) = 0,$$

and $\mathcal{E}_S(r) = \text{span}(S_0, \dots, S_{r-1})$ for $r \geq 1$. MCL3 and MCL5 prove

$$\ker(\Sigma'|_{\mathcal{E}_A(d)}) = \mathcal{E}_A(\min(d, m)),$$

$$\Sigma' \mathcal{E}_A(d) = \begin{cases} 0, & d \leq m, \\ \mathcal{E}_S(d-m), & d > m. \end{cases} \quad (\text{MCL6.1})$$

The proof uses the nonzero triangular coefficients (MCL5.2) and the independence in MCL4; it is valid for the complete functionals, not only their formal Taylor symbols.

Consequently every lift of S_j that is a generalized ρ -character has order at least $m + j + 1$. The lift in (MCL5.7) has exactly that order by its nonzero coefficient (MCL5.8) and (MCL3.3). Moreover all finite generalized ρ -character lifts of that particular S_j are precisely

$$\boxed{\Lambda_{\rho,j} + \mathcal{E}_A(m)}. \quad (\text{MCL6.2})$$

Indeed a larger highest derivative would give a nonzero source jet of index greater than j , by the same triangular calculation. The remaining difference lies in the stated kernel.

For the whole target block of length r , the least possible finite invariant preimage is the exact original sequence

$$\boxed{0 \longrightarrow \mathcal{E}_A(m) \longrightarrow \mathcal{E}_A(m+r) \xrightarrow{\Sigma'} \mathcal{E}_S(r) \longrightarrow 0.} \quad (\text{MCL6.3})$$

It is invariant under every positive real dilation, hence under every original prime. Its middle term is a single Jordan block of length $m + r$; the target is a block of length r ; the full original kernel is the block of length m .

For completeness, any finite K_A -invariant subspace of generalized ρ -characters that surjects onto $\mathcal{E}_S(r)$ must contain a preimage with highest derivative index at least $m + r - 1$. Its successive N_A -images then span every A_n , $0 \leq n \leq m + r - 1$, by triangular elimination. It therefore contains $\mathcal{E}_A(m + r)$. This proves the asserted least subspace, not merely a lower bound on a conveniently chosen lift.

A pure ρ -character lift of S_0 does not exist: MCL3 says it would be a multiple of A_0 , whereas (MCL5.1) sends every such multiple to zero and MCL4 gives $S_0 \neq 0$. This is a statement about the actual receiving map Σ' . Its exact replacement is (MCL6.3) with $r = 1$, of length $m + 1$, not a claim that the source or the full programme has no relation to a lift.

By (MCL3.4), the same least-order assertion holds for lifts subject to simultaneous generalized p^ρ and q^ρ conditions for two distinct original primes. A one-prime character condition alone is not silently substituted for that hypothesis. For the constructed lift, its exact order is visible at each individual nonidentity dilation too. In the finite action (MCL2.3), one application of $T_a^t - a^\rho$ lowers the highest derivative index by one with coefficient $a^\rho n \log a$. Iterating and retaining (MCL5.8) gives

$$(T_a^t - a^\rho)^{m+j} \Lambda_{\rho,j} = a^{(m+j)\rho} \frac{j! m!}{2\zeta(m)(\rho)} (\log a)^{m+j} A_0 \neq 0 \quad (a \neq 1), \quad (T_a^t - a^\rho)^{m+j+1} \Lambda_{\rho,j} = 0. \quad (\text{MCL6.4})$$

For the twisted action the same expression has $a^{(m+j)(1-\rho)}$ and $(-\log a)^{m+j}$, obtained directly from (MCL2.4). Thus no prime logarithmic factor is hidden in the order assertion.

158.8 MCL7. Full prime logarithms and the exact lifting defect

Write the retained coefficients

$$\ell_{jk} = \frac{j! v_\rho^{(j-k)}(0)}{(j-k)!(m+k)!}, \quad \Lambda_{\rho,j} = \sum_{k=0}^j \ell_{jk} A_{m+k}.$$

Applying the complete action (MCL2.3) gives

$$T_a^t \Lambda_{\rho,j} = a^\rho \sum_{k=0}^j \ell_{jk} \sum_{n=0}^{m+k} \binom{m+k}{n} (\log a)^{m+k-n} A_n. \quad (\text{MCL7.1})$$

The image under Σ' is exactly $D_a^t S_j$, so the failure of the chosen section to intertwine dilation is the explicit kernel element

$$\begin{aligned} \mathfrak{d}_{\rho,j}(a) &:= T_a^t \Lambda_{\rho,j} - a^\rho \sum_{i=0}^j \binom{j}{i} (\log a)^{j-i} \Lambda_{\rho,i} \\ &= a^\rho \sum_{n=0}^{m-1} \left[\sum_{k=0}^j \ell_{jk} \binom{m+k}{n} (\log a)^{m+k-n} \right] A_n. \end{aligned} \quad (\text{MCL7.2})$$

To verify the second equality, first expand the first term by (MCL7.1). The second term has only derivative indices at least m . The image of their difference is zero; injectivity of Σ' on the high-derivative span forces all coefficients with $n \geq m$ to cancel. Its remaining coefficients are precisely the displayed low-derivative terms. This proves the equality in the full A' , including all kernel data.

For the pure target,

$$\mathfrak{d}_{\rho,0}(a) = \frac{a^\rho}{2\zeta^{(m)}(\rho)} \sum_{n=0}^{m-1} \binom{m}{n} (\log a)^{m-n} A_n. \quad (\text{MCL7.3})$$

For every $a \neq 1$, its coefficient of A_{m-1} is $a^\rho m \log a / (2\zeta^{(m)}(\rho)) \neq 0$. In particular every original prime has a nonzero defect for this chosen lift, while the complete block (MCL6.3) carries its exact action without losing those terms.

Under the supported contragredient twist the corresponding identity is

$$\mathcal{U}_a^A \Lambda_{\rho,j} - a^{1-\rho} \sum_{i=0}^j \binom{j}{i} (-\log a)^{j-i} \Lambda_{\rho,i} = a \mathfrak{d}_{\rho,j}(a^{-1}). \quad (\text{MCL7.4})$$

It follows by applying (MCL7.2) at a^{-1} , then multiplying by the original scalar character a . Thus every logarithm, its sign and the exact exponent $1 - \rho$ are retained.

158.9 MCL8. Exact Fourier return and the reflected zero block

The original involutions are

$$Rb(u) = u^{-1}b(u^{-1}), \quad \mathcal{F}h(t) = \int_{\mathbb{R}} h(v) e^{-2\pi i vt} dv.$$

Poisson summation, with both endpoint terms retained before their vanishing on S , gives $\Sigma\mathcal{F} = R\Sigma$, and hence

$$\mathcal{F}^t \Sigma' = \Sigma' R^t. \quad (\text{MCL8.1})$$

The raw Mellin integrals give directly

$$R^t M_{A,z} = M_{A,1-z}, \quad R^t A_j(\rho) = (-1)^j A_j(1-\rho). \quad (\text{MCL8.2})$$

Let $b = 1 - \rho$. The original functional equation makes b an actual zero of the same multiplicity m . On a neighborhood of ρ , its exact multiplier

$$\kappa(z) = \frac{\zeta(1-z)}{\zeta(z)} = \pi^{1/2-z} \frac{\Gamma(z/2)}{\Gamma((1-z)/2)} = \frac{1}{2^z \pi^{z-1} \sin(\pi z/2) \Gamma(1-z)} \quad (\text{MCL8.3})$$

is holomorphic and nonzero, with all factors displayed. This formula is only used on its proved local domain; no Gamma factor is declared globally invertible through its poles.

Applying (MCL1.8) and (MCL8.1) first where division by ζ is legitimate, then continuing through its removable ratio at ρ , gives

$$\mathcal{F}^t M_{S,z} = \kappa(z) M_{S,1-z}.$$

All source Mellin integrals in this equality are defined on their common strip $-2 < \Re z < 3$, with the exact removable or meromorphic comparisons as appropriate. In the chosen neighborhood of ρ the full derivative formula is

$$\mathcal{F}^t S_j(\rho) = \sum_{k=0}^j \binom{j}{k} \kappa^{(j-k)}(\rho) (-1)^k S_k(b). \quad (\text{MCL8.4})$$

At order zero the retained coefficient is

$$\kappa(\rho) = \frac{(-1)^m \zeta^{(m)}(b)}{\zeta^{(m)}(\rho)}. \quad (\text{MCL8.5})$$

The chosen high-derivative lifts satisfy the stronger exact identity

$$R^t \Lambda_{\rho,j} = \sum_{k=0}^j \binom{j}{k} \kappa^{(j-k)}(\rho) (-1)^k \Lambda_{b,k}. \quad (\text{MCL8.6})$$

Both sides lie in $\text{span}(A_m(b), \dots, A_{m+j}(b))$, by (MCL5.7) and (MCL8.2). Their images under Σ' agree by (MCL8.1) and (MCL8.4). The triangular injectivity in MCL5 then proves equality, without discarding a kernel ambiguity. In particular this chosen lift is exactly Fourier-compatible across the reflected blocks, although its dilation defect is the nonzero expression MCL7.

The unshifted transpose relation is $T_a^t R^t = a R^t T_{a^{-1}}^t$, as can also be checked on (MCL8.2). The twisted relation is $\mathcal{U}_a^A R^t = a R^t \mathcal{U}_{a^{-1}}^A$. The oriented Cech dual adds the separately retained minus sign of ASD/UOS. No new sign or Tate twist is inferred from the Fourier ratio.

158.10 MCL9. The actual character connecting map and its nonzero class

The proved closed-image theorem makes $\Sigma : S \rightarrow J = \Sigma S$ a topological isomorphism onto a closed subspace of A . Its actual continuous-dual sequence is

$$0 \longrightarrow Q' \xrightarrow{\pi'} A' \xrightarrow{\Sigma'} S' \longrightarrow 0, \quad Q = A/J. \quad (\text{MCL9.1})$$

It is algebraically exact and strictly exact for weak-* topologies; the transposes are also strong-dual continuous. For clarity, surjectivity follows by transporting a continuous functional from S to J using Σ^{-1} , then extending it to A by complex Hahn–Banach. Its kernel is $J^\perp = \pi' Q'$. This is the actual sequence in ASD2, not a presumed spectral expansion of arbitrary distributions.

The generalized ρ -character part of its kernel is exactly

$$J^\perp \cap \bigcup_{d \geq 1} \ker(K_A - \rho)^d = \mathcal{E}_A(m). \quad (\text{MCL9.2})$$

This follows immediately from the full classification MCL3, MCL5's triangular map, and MCL4's independence. It shows that the m kernel levels in the least enlargement are the actual original multiplicity block of Q' .

To construct the connecting map rather than name one, put $N = K - \rho$ on each of the three terms of (MCL9.1). For each $r \geq 1$, form the two-term complex $[E \xrightarrow{N^r} E]$ in degrees zero and one. The operators commute with the maps in (MCL9.1), so these three complexes form a degree-wise short exact sequence. Its cohomological connecting map is

$$\partial_{\rho,r} : \ker(N_S^r : S' \rightarrow S') \longrightarrow Q'/N_Q^r Q'.$$

An actual lift $\lambda \in A'$ of $s \in \ker N_S^r$ satisfies $N_A^r \lambda \in J^\perp$, and

$$\boxed{\partial_{\rho,r}(s) = [(\pi')^{-1} N_A^r \lambda].} \quad (\text{MCL9.3})$$

Another lift differs by $\pi'\eta$, so its value changes by $N_Q^r \eta$. This proves that the formula is well-defined and is the connecting map of the displayed complexes.

For the pure character $s = S_0$, $r = 1$, the exact lift in MCL5 gives

$$\boxed{\partial_{\rho,1}(S_0) = \left[\frac{m}{2\zeta^{(m)}(\rho)} (\pi')^{-1} A_{m-1} \right] \neq 0.} \quad (\text{MCL9.4})$$

To prove nonzero in the full quotient, suppose $N_Q \eta = \frac{m}{2\zeta^{(m)}(\rho)} (\pi')^{-1} A_{m-1}$. Then $N_A^{m+1} \pi' \eta = 0$, so MCL3 places $\pi' \eta$ in a finite generalized-character space. Since it belongs to $J^\perp$, MCL9.2 puts it in $\mathcal{E}_A(m)$. Its N_A -image is then in $\text{span}(A_0, \dots, A_{m-2})$, and cannot equal $\frac{m}{2\zeta^{(m)}(\rho)} A_{m-1}$, whose coefficient is nonzero. The same proof covers $m = 1$, when this last span is zero. Thus no functional outside the initially chosen finite block supplies a missed preimage: the proposed equation itself forces it into that block.

For every jet $s = S_j$, put $r = j + 1$. It lies in $\ker N_S^r$, and the full representative of its connecting class is

$$\boxed{N_A^r \Lambda_{\rho,j} = j! \sum_{\substack{0 \leq k \leq j \\ m+k \geq r}} \frac{v_\rho^{(j-k)}(0)}{(j-k)!(m+k-r)!} A_{m+k-r}.} \quad (\text{MCL9.5})$$

Its highest term is

$$\frac{j! m}{2\zeta^{(m)}(\rho)} A_{m-1} \neq 0. \quad (\text{MCL9.6})$$

Every term in (MCL9.5) lies in the original kernel. If the class were in $N_Q^r Q'$, a putative preimage would satisfy $N_A^{m+r} \pi' \eta = 0$, hence again lie in $\mathcal{E}_A(m)$ by MCL9.2. But $N_A^r \mathcal{E}_A(m)$ has derivative indices strictly below $m - 1$, contradicting (MCL9.6). Therefore each $\partial_{\rho,j+1}(S_j)$ is nonzero.

In the actual supported twisted representation the operator is $G - (1 - \rho) = -(K - \rho)$. Replacing N^r accordingly multiplies the exact representative (MCL9.5) by $(-1)^r$; it does not alter its nonvanishing or replace the character $a^{1-\rho}$ by a^ρ .

This character connecting map is not the quotient map from the residue-duality cone in ASD14. It is constructed from the different actual exact sequence (MCL9.1), with its maps and endomorphism complexes explicit. The zero or nonzero status of one cannot be substituted for that of the other.

158.11 MCL10. Exact location in the source-supported row, including all closed terms

Use the actual global graph and faithful extra coefficients:

$$H = H_0 \oplus V_{\text{extra}}, \quad H_0 = \{((h, c_0, c_1), (\widehat{h}, d_0, d_1)) : h \in S\} \simeq S \oplus \mathbb{C}^4,$$

$$V_{\text{extra}} = V_+^{\text{extra}} \oplus V_-^{\text{extra}}, \quad E = \mathbb{C}^4 \oplus V_{\text{extra}}.$$

The original supported row and its complete transpose in UOS3 are

$$0 \rightarrow E \rightarrow H \xrightarrow{\text{res}} A \rightarrow Q^2 \rightarrow Q \rightarrow 0,$$

$$0 \rightarrow Q' \rightarrow (Q')^2 \rightarrow A' \xrightarrow{\text{res}'} H' \rightarrow E' \rightarrow 0. \quad (\text{MCL10.1})$$

Here $\text{res}((h, c), (\widehat{h}, d), w) = \Sigma h$, hence

$$\boxed{\text{res}'\lambda = (\Sigma'\lambda, 0, 0, 0, 0, 0), \quad H' = S' \oplus (\mathbb{C}^4)' \oplus V'_{\text{extra}}.} \quad (\text{MCL10.2})$$

The last zero denotes the entire pair of extra coefficient duals, not only their endpoints. Let

$$j_S : S' \rightarrow H', \quad s \mapsto (s, 0, 0, 0, 0, 0).$$

Then the lift calculation has the exact supported identity

$$\boxed{\text{res}'\Lambda_{\rho,j} = j_S S_j.} \quad (\text{MCL10.3})$$

The map $H' \rightarrow E'$ is restriction to all four endpoint and extra coordinates; it kills these particular $j_S S_j$, exactly because they lie in the image of (MCL10.2). It remains nonzero on the retained endpoint and extra copies. Thus these original supported functionals do lift as continuous functionals. MCL6 and MCL9 compute the precise extra generalized-character structure needed, rather than denying the continuous lift.

Twisting the whole row by χ yields the actual supported dual actions used in UOS. Parametrization of H_0 by h identifies its S -action with $D_a h$; its four endpoint characters stay 1, $a, a, 1$, and the full extra copies retain the two original chart actions. Therefore j_S , (MCL10.2) and (MCL10.3) have exactly the actions in MCL2. Their transpose Fourier mirror on the S' -coordinate is $\mathcal{F}^t$, with the chart/orientation signs of the full row retained separately. Formula (MCL8.6) proves the mirror compatibility of these actual lifts.

The source receiving ring is the already constructed $\mathbb{Z}^3$, with source $[n] = (n, 0, 0)$, $[\tau] = (1, 1, 1)$. On A', S' and the S' -coordinate of H' , the first ring coordinate acts by its scalar. On the two complete extra closed copies, their corresponding additional coordinates act. Formula (MCL10.2) is receiving-ring linear. Thus all lift formulas preserve the source action: original integer n acts as scalar n , independently of the spectral operator at $a = n > 0$. Primitive τ remains distinguished from integer one by the faithful extra coefficients of the full object. No addition or numerical weight of τ is introduced.

For each bounded distributive receiving support lattice L_{supp} , every linear map here has the already constructed lift

$$G_{L_{\text{supp}}}(f)(v, \alpha) = (f(v), \alpha), \quad G_{L_{\text{supp}}}(V) = \{(0, \alpha) : \alpha \in L_{\text{supp}}\} \cup \{(v, 1_{L_{\text{supp}}}) : v \in V\}. \quad (\text{MCL10.4})$$

If $\alpha \neq 1_{L_{\text{supp}}}$, the amplitude is zero and linearity keeps it zero with that same label. At the top label, an amplitude cancellation likewise keeps its label. Identity and composition follow by substitution. Independent labels on separate closed factors remain separate; no common-label carrier is equated with their product. In particular the vanishing amplitudes for $A_0, \dots, A_{m-1}$ under Σ' retain their original receiving support labels and are not relabelled as primitive τ .

158.12 MCL11. What this completes in the lifting calculation

For every actual nontrivial zero ρ and every actual multiplicity m , the original supported transpose has the explicit lift (MCL5.7) of every source Mellin jet. Its least generalized-character order is exactly $m + j + 1$, proved against all continuous generalized eigen-distributions of the original A . Simultaneous conditions at two distinct arithmetic primes recover the same full-character statement. The full least enlargement, every prime logarithm and the exact Fourier return are (MCL6.3), (MCL7.2)–(MCL7.4), and (MCL8.6).

The actual character connecting map is nonzero on these targets by (MCL9.4)–(MCL9.6). Its class is an element of the original spectral-kernel quotient, with a complete representative and proof that no omitted global dual functional makes it zero. The continuous supported lift itself exists by (MCL10.3). These two statements refer to explicitly different requirements on that same computed lift, with their exact comparison given.

This calculation neither assumes nor proves $\Re\rho = 1/2$. It occurs at every original zero, including any zero on the critical line, with the same formulas and its actual multiplicity. Consequently its nonzero character extension is not an RH disproof or a failure of the user's reconstructed arithmetic. Nor is the needed nilpotent enlargement a Deligne weight separation: its kernel and target retain the same character exponent, and the complete prime-action coupling is the calculated nonzero polynomial in $\log a$. The result specifies the actual supported lifting object and its exact coupling, which must be retained in the next comparison with Deligne's geometric argument.

159 The original restriction extension, its full supported-cone map, and Deligne's cross

24 September 2026. Independent derivation ORE0–ORE10.

159.1 ORE0. Sources, definitions and the precise receiving calculation

This calculation uses the original Connes–Consani summation map, its actual continuous transpose, and the original Riemann zeta function. It calculates a cyclic extension class, transports that class through the already proved original-zeta residue pairing, and identifies its exact location in the complete supported comparison of UOS. The calculation also proves what the finite numerical-character truncation does to that class. It does not assert a new weight theory on primitive Z_1/τ .

The current corpus and operation rules were read, together with the complete user passages USR-9ad1c0a2d09dba92, USR-f55d16feb948d8c2 and USR-6152e3bc6302258c in the retained private corpus. The source notation remains $Z_0, Z_1/\tau, Z_2$. Addition on primitive τ was withdrawn and is not reintroduced. Polynomial modules, continuous duals, derivatives and arithmetic prime actions below are operations on the already constructed complex receiving spaces. A failed fixed-order lift in this receiver is not a contradiction of the user's whole-spectrum reconstruction.

Human sources are Alain Connes and Caterina Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, arXiv:0903.2024v3, §5, for the original coefficient sheaf and Fourier restrictions; Ralf Meyer, *A spectral interpretation for the zeros of the Riemann zeta function*, arXiv:math/0412277v3, for the closed-image framework; and Pierre Deligne, *La conjecture de Weil. II*, §§3.6.1–3.6.3, pp.213–214, for the exact cross and weight-truncation argument.

For this calculation Deligne's retained French transcription `S20_FR_record_export.tex`, source lines2468–2595, was read completely. It is a French transcription, not author TeX. Its SHA256 is d6836ae15d98f2b27eb98c9bd039a7d921679becd9fa47ff4717410289dbd351. No claim of a fresh reading of all of Weil II is made.

The exact programme inputs are OMS1–OMS5, ASD1–ASD7 and ASD14, SCL1–SCL9, DPL1–DPL9, GZR4–GZR8, and UOS3–UOS8A. In particular the finite residue lift and the full supported cones are existing results, not claimed here as new.

159.2 ORE1. The original short exact sequence and every action

Retain the original Fréchet spaces

$$S = \{h \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : h(-v) = h(v), h(0) = 0, \int_{\mathbb{R}} h(v) dv = 0\},$$

$$A = \{b \in C^\infty(\mathbb{R}_{>0}) : \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j b(u)| < \infty \quad (N, j \geq 0)\}. \quad (\text{ORE1.1})$$

The exact original maps are

$$\Sigma h(u) = 2 \sum_{n \geq 1} h(nu), \quad \mathcal{M}_0 b(s) = \int_0^\infty b(u) u^s \frac{du}{u},$$

$$\mathcal{M}_0 \Sigma h(s) = 2\zeta(s) \int_0^\infty h(v) v^s \frac{dv}{v}. \quad (\text{ORE1.2})$$

The identity is initially absolute for $\Re s > 1$; its full continuation is the one in OMS, with the pole, endpoint and trivial-zero contributions retained. Write $J = \Sigma S$, $Q = A/J$, and $\pi : A \rightarrow Q$. The proved topological isomorphism $\Sigma : S \rightarrow J$, closedness of J , and Hahn–Banach give the exact continuous-dual sequence

$$0 \longrightarrow Q' \xrightarrow{\pi'} A' \xrightarrow{\Sigma'} S' \longrightarrow 0. \quad (\text{ORE1.3})$$

For example, a functional on S transports continuously to J by Σ^{-1} , extends continuously to A , and hence has a preimage under Σ' . The kernel consists exactly of functionals annihilating J , which factor uniquely and continuously through Q . We use the weak-* exactness proved in ASD; every displayed transpose is also strong-dual continuous.

The real action is $T_a b(u) = b(u/a)$, $T_a h(v) = h(v/a)$, $a > 0$, and it intertwines Σ . Its infinitesimal operator on these primal spaces is $L = -u\partial_u$, respectively $-v\partial_v$. On $\mathcal{M}_0 A = \mathcal{B}$ it is multiplication by the original parameter s , with no shift. Introduce the already used dilation character $\chi_{\text{dil}}(a) = a$. Twisting all three dual terms gives

$$0 \longrightarrow Y := \chi_{\text{dil}} \otimes Q' \xrightarrow{i=\pi'} \tilde{A} := \chi_{\text{dil}} \otimes A' \xrightarrow{R=\Sigma'} \tilde{S} := \chi_{\text{dil}} \otimes S' \longrightarrow 0. \quad (\text{ORE1.4})$$

The action on each term is $a(T_{a^{-1}})'$, and its generator is

$$G_Y = 1 - L_Q^t, \quad G_A = 1 - L_A^t, \quad G_S = 1 - L_S^t. \quad (\text{ORE1.5})$$

All maps commute with these generators and with every original T_p for a rational prime p . This twist is distinguished from the original zeta functional-equation multiplier $\chi_\zeta(s)$ used in ORE5.

The source receiving ring and its faithful extra copies are not changed: $[\tau] = (1, 1, 1)$, $[n] = (n, 0, 0)$ in $\mathbb{Z}^3$. On ORE1.4 they act by the first scalar coordinate; the other two actions remain on the complete closed copies in ORE8. An original prime dilation and a source integer scalar are different specified operators.

159.3 ORE2. An explicit two-term resolution and the actual connecting class

Let $\mathcal{R} = \mathbb{C}[X]$; X acts as G on every term of ORE1.4. For a nonzero polynomial q , the cyclic module $V_q = \mathcal{R}/(q)$ has the exact free resolution

$$0 \longrightarrow \mathcal{R} \xrightarrow{q} \mathcal{R} \longrightarrow V_q \longrightarrow 0. \quad (\text{ORE2.1})$$

Injectivity of its first map follows from the absence of zero divisors in $\mathcal{R}$. Applying $\text{Hom}_{\mathcal{R}}(-, Y)$, and identifying a map out of $\mathcal{R}$ by its value at 1, gives the concrete complex

$$Y \xrightarrow{q(G_Y)} Y \quad \text{in degrees } 0, 1.$$

Consequently

$$\text{Hom}_{\mathcal{R}}(V_q, Y) = \ker q(G_Y), \quad \text{Ext}_{\mathcal{R}}^1(V_q, Y) = Y/q(G_Y)Y. \quad (\text{ORE2.2})$$

This is an algebraic extension calculation in the stated receiving category, not an assumed sheaf-derived equivalence.

Let $\eta \in \tilde{S}$ satisfy $q(G_S)\eta = 0$. It specifies the map $V_q \rightarrow \tilde{S}$ sending $1 \bmod q$ to η . Choose $\alpha \in \tilde{A}$ with $R\alpha = \eta$. There is a unique $y_\alpha \in Y$ with

$$iy_\alpha = q(G_A)\alpha.$$

The right side lies in $\ker R$, because $Rq(G_A)\alpha = q(G_S)\eta = 0$. The exact connecting map is

$$\boxed{\delta_q(\eta) = [y_\alpha] \in Y/q(G_Y)Y.} \quad (\text{ORE2.3})$$

Replacing α by $\alpha + iy$ changes y_α by $q(G_Y)y$; hence the displayed class is independent of the lift. It is linear and intertwines all real and prime actions because $q(G)$, R and i do.

The pullback extension is explicitly

$$E_\eta = \{(a, v) \in \tilde{A} \oplus V_q : Ra = f_\eta(v)\}, \quad 0 \rightarrow Y \xrightarrow{y \mapsto (iy, 0)} E_\eta \rightarrow V_q \rightarrow 0. \quad (\text{ORE2.4})$$

Lifting the free generator in ORE2.1 to $(\alpha, 1)$ sends the relation q to $(iy_\alpha, 0)$. Thus ORE2.3 is precisely the class of ORE2.4, including its sign. It vanishes exactly when α can be replaced by $\alpha - iy$ with $q(G_A)(\alpha - iy) = 0$. In that case the map

$$p(X) \bmod q \longmapsto p(G_A)(\alpha - iy) \quad (\text{ORE2.5})$$

is a $\mathcal{R}$ -linear section over the specified cyclic target. No choice of a section of all of Σ' has been asserted.

These maps also arise from the complete two-row diagram with rows ORE1.4 and vertical maps $q(G_Y), q(G_A), q(G_S)$. Its kernel and cokernel sequence is

$$0 \rightarrow \ker q_Y \rightarrow \ker q_A \rightarrow \ker q_S \xrightarrow{\delta_q} Y/q_Y Y \rightarrow \tilde{A}/q_A \tilde{A} \rightarrow \tilde{S}/q_S \tilde{S} \rightarrow 0. \quad (\text{ORE2.6})$$

Exactness at $\ker q_S$ is ORE2.5. Exactness at the following quotient follows by taking y whose image is $q_A \alpha$ and observing $R\alpha \in \ker q_S$. All other exactness statements follow by substitution into the two original exact rows. This is the actual resolution diagram; a spectral truncation has not silently replaced it.

159.4 ORE3. Global polynomial facts before a finite character is selected

Multiplication by a polynomial $p(s)$ is injective on $\mathcal{B}$. Its image consists exactly of entire functions having the finite vanishing jets required by the roots of p . Division by p is continuous on this image: off fixed small disks around its finitely many roots it has polynomial vertical bounds; inside each disk the removable quotient is controlled by Cauchy's estimate on a larger disk. The image is closed because those finite jet evaluations are continuous. Transport through $\mathcal{M}_0$ proves the corresponding statement on the original A .

It follows that $q(G_A)$ is surjective on A' . Indeed, given a continuous functional λ on A , its prescription $q(1 - L_A)b \mapsto \lambda(b)$ is continuous on the closed image of the injective primal map. Hahn–Banach extends it to a continuous functional on A . This extension is a preimage of λ under $q(1 - L_A^t)$. Thus

$$\tilde{A}/q_A \tilde{A} = 0, \quad \tilde{S}/q_S \tilde{S} = 0 \quad (\text{ORE3.1})$$

by surjectivity of R and commutation. ORE2.6 ends with the surjective connecting map $\ker q_S \rightarrow Y/q_Y Y \rightarrow 0$.

The actual quotient $Q = \mathcal{B}/I_\zeta$ has a continuous full primary projector P_z at each nontrivial zero z , and $L_Q - z$ is continuously invertible on its complementary kernel. These are RZ/SCL's global operators on the original test space. Therefore, for $b = 1 - z$,

$$Y = Y_z \oplus Y_{\neq z}, \quad Y_z = P_z^t Y, \quad G_Y - b \text{ is continuously invertible on } Y_{\neq z}. \quad (\text{ORE3.2})$$

There is no assertion of an infinite sum of projectors. ORE3.2 uses one actual projector and its complement. At a nontrivial zero of multiplicity m , Y_z has dimension m , with the full jet action retained.

If $1 - b$ is not an actual nontrivial zero, $G_Y - b$ is continuously invertible on all of Y . In particular every cyclic class $\delta_{(X-b)r}$ is zero there, and ORE2.5 gives its unique equivariant cyclic lift. The inverse is the transpose of the original full-source resolvent. That resolvent retains

$$F_*(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad F_*(0) = F_*(1) = \frac{1}{8}, \quad (\text{ORE3.3})$$

and its nonzero removable values at every negative even integer. Consequently this conclusion includes $b = 0, 1$ and the characters corresponding to the trivial zeros. It does not apply a nontrivial-zero jet formula to a trivial zero while forgetting the source Mellin pole that cancels it. The original endpoint and extra closed terms are separately retained in ORE8.

For an arbitrary finite-dimensional generalized-character target, factor its annihilating polynomial and use the finite Chinese-remainder projectors. Those projectors are explicit polynomials obtained from Bézout identities between distinct prime-power factors of q . They commute with the real and prime actions. Thus ORE2.3 and the single-root calculations below determine its full connecting class, with no infinite primary decomposition.

159.5 ORE4. The full original Mellin derivatives and their connecting matrix

Fix an actual nontrivial zero ρ , of its actual order m , and write

$$b = 1 - \rho, \quad N = G - b, \quad q(X) = (X - b)^r, \quad r \geq 1.$$

The original functionals and their raw derivatives are

$$\begin{aligned} A_j &= \frac{\partial^j}{\partial z^j} \int_0^\infty a(u) u^z \frac{du}{u} \Big|_{z=\rho} \quad (a \in A), \\ S_j &= \frac{\partial^j}{\partial z^j} \int_0^\infty h(v) v^z \frac{dv}{v} \Big|_{z=\rho} \quad (h \in S). \end{aligned} \quad (\text{ORE4.1})$$

Here A_j, S_j denote functionals, not new primal test spaces. The first transform is entire. The second is holomorphic for $\Re z > -2$: evenness and $h(0) = 0$ give $h(v) = O(v^2)$ at zero, and Schwartz decay controls infinity. Differentiation adds powers of $\log v$, integrable uniformly on compact subsets of this half-plane, with continuous Schwartz bounds. Thus all ORE4.1 functionals are well-defined and continuous at ρ .

Differentiating ORE1.2, keeping every factorial, gives

$$RA_j = 2 \sum_{k=0}^j \binom{j}{k} \zeta^{(k)}(\rho) S_{j-k}. \quad (\text{ORE4.2})$$

The full real action and its generator are

$$\mathcal{U}_a A_j = a^{1-\rho} \sum_{k=0}^j \binom{j}{k} (-\log a)^{j-k} A_k, \quad NA_j = -j A_{j-1}, \quad (\text{ORE4.3})$$

and the identical formulas hold for S_j , with $A_{-1} = S_{-1} = 0$. They follow by the change of variable $v \mapsto av$ in the original integrals before differentiation. At every rational prime p , substitute $a = p$; no logarithm or scalar is dropped.

Use the complete Taylor series at this actual zero:

$$\begin{aligned} \zeta(\rho + t) &= t^m u_\rho(t), \quad u_\rho(t) = \sum_{\ell \geq 0} \frac{\zeta^{(m+\ell)}(\rho)}{(m+\ell)!} t^\ell, \\ \frac{1}{2u_\rho(t)} &= \sum_{k \geq 0} v_k t^k, \quad v_0 = \frac{m!}{2\zeta^{(m)}(\rho)}. \end{aligned} \quad (\text{ORE4.4})$$

All further coefficients are fixed recursively:

$$v_k = -\frac{1}{u_\rho(0)} \sum_{\ell=1}^k \frac{\zeta^{(m+\ell)}(\rho)}{(m+\ell)!} v_{k-\ell} \quad (k \geq 1). \quad (\text{ORE4.5})$$

For $j \geq 0$, an exact original-dual lift of S_j is

$$\alpha_j = j! \sum_{k=0}^j \frac{v_k}{(m+j-k)!} A_{m+j-k}, \quad R\alpha_j = S_j. \quad (\text{ORE4.6})$$

To prove the last identity, substitute ORE4.2. In the coefficient of $S_d/d!$, for $0 \leq d \leq j$, the convolution is $j!$ times the coefficient of t^{j-d} in $(2u_\rho(t)) \sum v_k t^k = 1$. It equals $j!$ for $d = j$, and zero otherwise; after the displayed $d!$, this is exactly S_j . Terms with lower A -index than m are in the kernel and need not be added to this chosen lift.

The functionals $A_0, \dots, A_{n-1}$ are linearly independent. The finite jet map on $\mathcal{B}$ is onto: for prescribed jets at ρ , multiply $e^{(s-\rho)^2}$ by the finite Taylor polynomial of the desired jet polynomial times $e^{-(s-\rho)^2}$. This is an original entire test function in $\mathcal{B}$ with those jets. This assertion about the domain functionals alone does not prove independence of their images under R .

Independence of $S_0, \dots, S_{r-1}$ has the following direct proof on the actual source, as in MCL4. Suppose $\sum_{j=0}^{r-1} c_j S_j = 0$, and write $P(X) = \sum_{j=0}^{r-1} c_j X^j$. Take $f \in C_c^\infty(0, \infty)$ with $\int_0^\infty f(v) dv = 0$ and set $h(v) = f(|v|)$. Its support avoids zero, so h is smooth and even, $h(0) = 0$, and $\int_{\mathbb{R}} h = 2 \int_0^\infty f = 0$. Hence $h \in S$. The assumed relation says

$$\int_0^\infty f(v) v^{\rho-1} P(\log v) dv = 0 \quad \text{for every such zero-integral } f.$$

Choose a fixed compact test f_0 of integral 1. For an arbitrary compact test g , the test $g - (\int g) f_0$ has integral 0. Substitution proves that the smooth function $v^{\rho-1} P(\log v)$ defines the same distribution as a constant. Therefore it equals that constant pointwise. Setting $x = \log v$ and differentiating gives

$$P'(x) + (\rho - 1)P(x) = 0.$$

Since $\rho - 1 \neq 0$, the leading coefficient of a nonzero polynomial P makes this impossible: the term $(\rho - 1)P$ has its full degree, while P' has smaller degree. Thus $P = 0$, proving every $c_j = 0$. This proof keeps the source zero-integral constraint and excludes the endpoint $z = 1$ exactly where it must. It repairs the earlier invalid inference from independence before applying the restriction map.

In fact

$$\ker N^r|_{\tilde{S}} = V_r := \text{span}\{S_0, \dots, S_{r-1}\}. \quad (\text{ORE4.7})$$

Here is a dimension proof on the full spaces. The closed multiplication-image argument in ORE3 gives $\dim \ker N^r|_{\tilde{A}} = r$. ORE3.2 gives both $\dim \ker N^r|_Y = \dim(Y/N^r Y) = \min(m, r)$. The exact sequence ORE2.6 with ORE3.1 then gives $\dim \ker N^r|_{\tilde{S}} = r$. The r independent displayed functionals belong to this kernel by ORE4.3 and hence exhaust it.

For $0 \leq j < r$, applying N^r to ORE4.6 gives the exact representative of its connecting class:

$$i y_{r,j} = N^r \alpha_j = (-1)^r j! \sum_{\substack{0 \leq k \leq j \\ m+j-k-r \geq 0}} \frac{v_k}{(m+j-k-r)!} A_{m+j-k-r},$$

$$\boxed{\delta_q(S_j) = [y_{r,j}] \in Y/N^r Y.} \quad (\text{ORE4.8})$$

An empty sum is zero. Every displayed A -index is below m , so this representative is in iY_ρ , exactly as required. For the cyclic generator S_{r-1} , its A_{m-1} coefficient is

$$(-1)^r \frac{(r-1)!m!}{2\zeta^{(m)}(\rho)(m-1)!} \neq 0. \quad (\text{ORE4.9})$$

The image of N^r on Y_ρ contains only the lower indices $A_0, \dots, A_{m-r-1}$, when these exist. Thus this particular class is nonzero. This assertion concerns lifting with the specified cyclic relation $N^r = 0$; it does not assert a failure of unrestricted character lifting.

159.6 ORE5. Transport the extension class through the complete original-zeta residue pairing

Let $D = D_\zeta : Q \rightarrow Y$ be the existing original residue map. It is continuous, injective and intertwines L_Q with G_Y . By SCL/DPL it is an isomorphism on every finite polynomial-kernel space, and its cokernel $C_\zeta = Y/DQ$ has every nonzero polynomial operator invertible.

These facts also give the exact cyclic quotient isomorphism

$$\boxed{Q/q(L_Q)Q \xrightarrow{[x] \mapsto [Dx]} Y/q(G_Y)Y.} \quad (\text{ORE5.1})$$

For surjectivity, given $y \in Y$, solve $qc = [y]$ in C_ζ , choose y_1 representing c , and write $y - qy_1 = Dx$. For injectivity, if $Dx = qy$, then $q[y] = 0$ in C_ζ , so $y = Dx_1$. Injectivity of D gives $x = qx_1$. Both arguments use the proved polynomial bijections, not a surjectivity assertion for D itself.

For the present single-root polynomial, the source quotient in ORE5.1 is exactly

$$Q_b/(L_Q - b)^r Q_b \simeq \mathbb{C}[t]/(t^d), \quad d = \min(m, r), \quad t = L_Q - b, \quad (\text{ORE5.2})$$

by the original full projector at $b = 1 - \rho$ and the continuous inverse on its complement. The equality $m_b = m_\rho = m$ follows from the full original functional equation, whose multiplier is holomorphic and nonzero inside the critical strip.

Keep that multiplier in its original form:

$$\zeta(s) = \chi_\zeta(s)\zeta(1-s), \quad \chi_\zeta(s) = \pi^{s-1/2} \frac{\Gamma((1-s)/2)}{\Gamma(s/2)}. \quad (\text{ORE5.3})$$

This is not a replacement of ζ by a completed function. At the original reflected zero it gives the exact identity of holomorphic germs

$$u_b(t) = (-1)^m \chi_\zeta(b+t) u_\rho(-t). \quad (\text{ORE5.4})$$

For $0 \leq h < m$, ORE4.2 proves $A_h \in \ker \Sigma' = iY$. Thus $i^{-1}(A_h/h!)$ is a well-defined element of the original Y , and the finite residue inverse acts on that element, not directly on an unspecified functional of A . With this annihilator identification explicit, the existing GZR/SCL inverse is

$$j_b(D^{-1}(i^{-1}(A_h/h!)))(t) = (-1)^h u_b(t) t^{m-1-h} \pmod{t^m}, \quad 0 \leq h < m. \quad (\text{ORE5.5})$$

This formula retains the $h!$ relating the original raw derivative to the Taylor-coefficient functional. It follows directly from the residue of $x(s)F(1-s)/\zeta(s)$: in the Taylor expansion of $F(\rho-t)$, the coefficient of $F^{(h)}(\rho)/h!$ is $(-1)^h t^h$. Multiplication by $u_b(t)t^{m-1-h}$ cancels exactly the denominator $t^m u_b(t)$ and selects that coefficient. All other primary observations vanish by the actual RZ isolator.

Insert ORE4.8 into ORE5.5 and retain every sign. Modulo t^d , its source image is

$$(-1)^{m+j} j! u_b(t) t^{r-1-j} \sum_{k=0}^j v_k(-t)^k.$$

Terms $k > j$ have degree at least r after multiplication by t^{r-1-j} , so one may replace the finite sum by its full convergent germ modulo this stated ideal. Using ORE4.4 and ORE5.4 gives

$$u_b(t) \sum_{k \geq 0} v_k(-t)^k = \frac{(-1)^m}{2} \chi_\zeta(b+t).$$

Therefore the full connecting matrix has the compact but factor-complete formula

$$\boxed{(D \bmod q)^{-1} \delta_q(S_j) = \left[\frac{(-1)^j j!}{2} t^{r-1-j} \chi_\zeta(b+t) \right]_{\mathbb{C}[t]/(t^d)} \quad (0 \leq j < r).} \quad (\text{ORE5.6})$$

For the cyclic generator this is

$$(D \bmod q)^{-1} \delta_q(S_{r-1}) = \left[\frac{(-1)^{r-1} (r-1)!}{2} \pi^{b+t-1/2} \frac{\Gamma((1-b-t)/2)}{\Gamma((b+t)/2)} \right]_{\mathbb{C}[t]/(t^d)}. \quad (\text{ORE5.7})$$

The coefficient $1/2$ comes from the original $\Sigma = 2 \sum_{n \geq 1}$; the Gamma ratio, π -power, factorial, reflected argument and sign all remain. Its constant term is nonzero. For every higher degree below d , ORE5.7 keeps the corresponding derivative of this entire germ at $t = 0$, not only its constant.

Multiplication by $\chi_\zeta(b+t)$ is invertible on this finite quotient since $\chi_\zeta(b) \neq 0$. Thus

$$\text{rank } \delta_q = \min(m, r), \quad \ker \delta_q = \begin{cases} 0, & r \leq m, \\ \text{span}\{S_0, \dots, S_{r-m-1}\}, & r > m. \end{cases} \quad (\text{ORE5.8})$$

This proves the whole connecting map, not only the nonzero top class.

159.7 ORE6. The complete extension and the exact finite truncation

Define the actual finite subspace of $\tilde{A}$

$$\mathcal{E}_{\rho,r} = \text{span}\{A_0, \dots, A_{m+r-1}\}.$$

ORE4 proves the exact equivariant sequence

$$0 \longrightarrow iY_\rho \longrightarrow \mathcal{E}_{\rho,r} \xrightarrow{R} V_r \longrightarrow 0. \quad (\text{ORE6.1})$$

Its kernel is precisely the first m derivatives, its surjectivity follows from ORE4.6, and every prime action is the complete triangular matrix ORE4.3. The generator has one Jordan chain of length $m+r$, with the raw coefficients $-j$ on its subdiagonal. In particular the lift α_{r-1} has

$$N^{m+r-1} \alpha_{r-1} = (-1)^{m+r-1} (r-1)! v_0 A_0 \neq 0, \quad N^{m+r} \alpha_{r-1} = 0. \quad (\text{ORE6.2})$$

This finite sequence calculates the full pullback of ORE1.4 over V_r , not an unrelated example. Indeed $R^{-1}(V_r)$ contains $\mathcal{E}_{\rho,r}$; any other preimage differs from an element of that finite space by iY . Decompose $Y = Y_\rho \oplus Y_{\neq \rho}$ by ORE3.2. The intersection of $\mathcal{E}_{\rho,r}$ with $iY_{\neq \rho}$ is zero, because a vector in this intersection is killed by a power of N , whereas N is invertible on $Y_{\neq \rho}$. Hence

$$R^{-1}(V_r) = \mathcal{E}_{\rho,r} \oplus iY_{\neq \rho}. \quad (\text{ORE6.3})$$

Both summands are invariant under all real and prime actions. This is also a topological decomposition onto its stated subspace: the finite-dimensional projection is obtained by choosing the continuous finite lift ORE4.6 of $R\alpha$, subtracting it, and applying the continuous projector of Y ; all inclusion maps use the actual weak-* quotient identifications. No infinite primary series is used.

The nonzero class ORE5.7 says that ORE6.1 has no $\mathcal{R}$ -linear section. Nevertheless its map onto V_r is already surjective, and its whole finite middle space has the same generalized character as V_r . Thus it is essential to distinguish the proved non-splitting of this exact sequence from the surjectivity of its displayed map. These two statements coexist by the explicit formulas.

There is an exact numerical truncation on finite-dimensional representations with the actual action $a \mapsto e^{(\log a)G}$. For a real w , define

$$W_{\leq w}^{\text{num}} V = \bigoplus_{\lambda: 2\Re \lambda \leq w} \ker(G - \lambda)^{\dim V}. \quad (\text{ORE6.4})$$

This formula uses every generalized eigenspace, not only eigenvectors or a fixed bound on the nilpotent order. It is functorial and exact: for any exact sequence of finite-dimensional representations, factor the product of the three characteristic polynomials into pairwise coprime primary factors. Bézout gives a common polynomial projector for every selected union of factors. Applying that same projector to the exact sequence retains its kernel and surjectivity, proving exactness.

For every prime p , the eigenvalue of T_p on each term of ORE6.1 is p^b , of modulus $p^{\Re b}$. Its numerical Deligne-style weight is therefore $2\Re b = 2(1 - \Re \rho)$; all original nilpotents remain in ORE4.3. Consequently

$$W_{\leq w}^{\text{num}}(\text{ORE6.1}) = \begin{cases} \text{the entire exact sequence ORE6.1,} & 2\Re b \leq w, \\ 0 \rightarrow 0 \rightarrow 0 \rightarrow 0, & 2\Re b > w. \end{cases} \quad (\text{ORE6.5})$$

When the target is retained, its extension class ORE5.7 is retained. No such truncation kills this particular class while retaining its same-character target. This is a calculation on this actual finite extension, not an assertion that Deligne's geometric mixed category has been transferred.

On the other hand, the union of all finite generalized b -character spaces already maps onto the corresponding union in $\tilde{S}$: each V_r has its lift in $\mathcal{E}_{\rho,r}$. Equations ORE6.1–ORE6.2 prove this by an actual enlargement of nilpotent order. The fixed-order functor $\ker N^r$ has connecting map ORE5.6; the full generalized-character functor permits the longer chain and has this surjectivity. The two operations are therefore calculated separately rather than conflated.

159.8 ORE7. What Deligne's actual cross proves

Retain every term of Deligne's §3.6 cross. In his notation and hypotheses, after descent to a finite field, set

$$\begin{aligned} A_D &= H^i(X_s), & E_D &= H^i(X_\eta), & V_D &= H^i(X_{\bar{\eta}})^I, \\ K_D &= H^{i-1}(X_{\bar{\eta}})_I(-1), & B_D &= H^{2N-i-1}(X_s)^\vee(-N). \end{aligned} \quad (\text{ORE7.1})$$

The horizontal sequence and vertical exactness at the centre are

$$0 \rightarrow K_D \rightarrow E_D \rightarrow V_D \rightarrow 0, \quad A_D \rightarrow E_D \rightarrow B_D. \quad (\text{ORE7.2})$$

The left term comes from Hochschild–Serre, with its original (-1) twist. The upper term is the actual support group identified by perfect duality, with its $(-N)$ twist. Proper base change identifies A_D with the source of specialization. In §3.6.2 Deligne proves V_D has weights at most i ; in §3.6.3 he proves B_D has weights at least $i+1$.

His exact W_i functor therefore makes $W_i E_D \rightarrow V_D$ onto and makes $W_i E_D \rightarrow W_i B_D = 0$ zero. Exactness of the vertical pair puts every element of $W_i E_D$ in the image of A_D . Composing proves $A_D \rightarrow V_D$ onto. The proof does not assert that $0 \rightarrow K_D \rightarrow E_D \rightarrow V_D \rightarrow 0$ splits as a Frobenius representation, and it does not bound the Jordan length of a lift by that of its image.

For the actual row ORE1.4, the fixed-order resolution produces the upper receiving term $Y/N^r Y$ and the actual map ORE5.6. Both its target V_r and this upper term have the same character b ; ORE6.5 proves exactly why the numerical truncation of this particular cross retains the map. The complete generalized-character lift through ORE6.1 is already surjective, just as a surjective map may have a nonzero extension class. Thus the original connecting class cannot be used as a substitute for Deligne's weight- $\geq i+1$ support group. ORE8 identifies its actual place in the retained supported geometry.

This comparison keeps the twists and the precise direction of Deligne's reasoning. It establishes neither that primitive τ has a numerical weight nor that any zero lies off the critical line.

159.9 ORE8. Locate the class in every degree of the actual supported-cone comparison

Retain UOS's original complete terms

$$P = V_+ \oplus V_- \oplus V_{\text{extra}}, \quad H = \ker d = S_{\text{Fourier graph}} \oplus \mathbb{C}^4 \oplus V_{\text{extra}},$$

$$E = \mathbb{C}^4 \oplus V_{\text{extra}}. \quad (\text{ORE8.1})$$

The four endpoints remain c_0, c_1, d_0, d_1 , and V_{extra} consists of the two full extra closed copies. The complete dual localization row is

$$0 \rightarrow Y \xrightarrow{\lambda \mapsto (\lambda, -\lambda)} Y^2 \xrightarrow{(\lambda_+, \lambda_-) \mapsto \pi'(\lambda_+ + \lambda_-)} \tilde{A} \xrightarrow{\text{res}'} \chi_{\text{dil}} H' \xrightarrow{\iota'} \chi_{\text{dil}} E' \rightarrow 0. \quad (\text{ORE8.2})$$

Here $\text{res}' \alpha = (\Sigma' \alpha, 0, 0, 0, 0, 0)$, and ι' restricts to the four endpoints and both extra dual copies. Thus ORE1.4 is exactly the non-endpoint part of this existing row.

Use the existing cones and their exact maps:

$$K_\zeta = [P \xrightarrow{-d} A \xrightarrow{\pi' D \pi} \tilde{A} \xrightarrow{d'} \chi_{\text{dil}} P'],$$

$$\mathcal{L}_\zeta = [P \xrightarrow{-d} A \xrightarrow{(\pi' D \pi, -\pi' D \pi)} \tilde{A}^2 \xrightarrow{d'_Z} \chi_{\text{dil}} P'], \quad \deg = (-1, 0, 1, 2). \quad (\text{ORE8.3})$$

The cone map $\mathcal{J} : K_\zeta \rightarrow \mathcal{L}_\zeta$ is identity in degrees $-1, 0, 2$ and $\alpha \mapsto (\alpha, -\alpha)$ in degree 1. Its cohomology is

n	$H^n K_\zeta$	$H^n \mathcal{L}_\zeta$	$H^n \mathcal{J}$
-1	H	H	id
0	0	0	0
1	C_ζ	$C_\zeta \oplus Y$	$c \mapsto (c, 0)$
2	$\chi_{\text{dil}} H'$	$\chi_{\text{dil}} E'$	ι' .

(ORE8.4)

The degree-one identification uses the exact half-difference and half-sum of UOS6.3; no factor of 2 is omitted.

Now apply the explicit resolution ORE2.1 to these actual complexes. For either complex K , its Hom complex has

$$\mathcal{H}^n(K) = K^n \oplus K^{n-1}, \quad \partial(u, v) = (d_K u, d_K v - (-1)^n q(G_K)u). \quad (\text{ORE8.5})$$

Direct substitution gives $\partial^2 = 0$. DPL9 proves from this formula the exact cohomology sequence

$$0 \rightarrow H^{n-1}(K)/qH^{n-1}(K) \rightarrow H^n \mathcal{H}(K) \rightarrow \ker(q : H^n(K) \rightarrow H^n(K)) \rightarrow 0. \quad (\text{ORE8.6})$$

Because every nonzero q is invertible on C_ζ , these formulas give

n	$H^n \text{RHom}_{\mathcal{R}}(V_q, K_\zeta)$	$H^n \text{RHom}_{\mathcal{R}}(V_q, \mathcal{L}_\zeta)$
-1	$\ker(q : H \rightarrow H)$	$\ker(q : H \rightarrow H)$
0	H/qH	H/qH
1	0	$\ker(q : Y \rightarrow Y)$
2	$\ker(q : \chi_{\text{dil}} H' \rightarrow \chi_{\text{dil}} H')$	$Y/qY \oplus \ker(q : \chi_{\text{dil}} E' \rightarrow \chi_{\text{dil}} E')$
3	$(\chi_{\text{dil}} H')/q(\chi_{\text{dil}} H')$	$(\chi_{\text{dil}} E')/q(\chi_{\text{dil}} E')$.

(ORE8.7)

All other degrees vanish. The displayed degree-two splitting on the right is canonical for the retained coordinates: $\chi_{\text{dil}} E'$ is a direct subcomplex in degree 2, by adjoining zero in the two Schwartz coordinates of P' . The differential into P' has zero endpoint and extra coordinates. The remaining degree-two cohomology is Y/qY by ORE8.6 and UOS6.3. This proves the splitting rather than assuming all complexes are formal.

The exact degree-two map induced by $\mathcal{J}$ contains the extension class calculated above:

$$\boxed{H^2 \text{RHom}(V_q, \mathcal{J}) : (\eta, e) \mapsto \left(-\frac{1}{2}\delta_q(\eta), e\right), \quad \chi_{\text{dil}} H' = \tilde{S} \oplus \chi_{\text{dil}} E'.} \quad (\text{ORE8.8})$$

Here the domain consists of the q -annihilated elements of the displayed direct sum.

We prove the sign and factor from the actual cochains. Extend $\eta \in \ker q_S$ to the plus-chart functional $u = (\eta, 0) \in \chi_{\text{dil}} P'$, with zero endpoints, extras and minus-chart Schwartz coordinate. Restriction to the Fourier graph gives exactly η . This extension is equivariant: the plus-chart action and the graph's S action are the same. Hence $qu = 0$, and $(u, 0)$ is a degree-two cocycle of $\mathcal{H}(K_\zeta)$. Choose $R\alpha = \eta$. In the supported target, put $w = (\alpha, 0) \in \tilde{A}^2$. Then $d'_Z w = u$. Formula ORE8.5 in degree1 is

$$\partial^1(w, 0) = (d'_Z w, qw) = (u, (q\alpha, 0)).$$

Subtracting this boundary from the image of $(u, 0)$ leaves

$$(0, (-q\alpha, 0)). \quad (\text{ORE8.9})$$

Its second coordinate is a degree-one cycle; write $q\alpha = iy_\alpha$. Under UOS's half-difference/half-sum it is $([-y_\alpha/2], -y_\alpha/2) \in C_\zeta \oplus Y$. The first coordinate disappears only in the proved quotient $C_\zeta/qC_\zeta = 0$; the second gives $-[y_\alpha]/2$. This is precisely the first component of ORE8.8. Endpoint and extra functionals have the same identity representative in both complexes, proving the second component. A different α changes this result by qY , in agreement with ORE2.3.

For the actual zero and $q = (X - b)^r$, ORE5.6 therefore makes the S_j component of this map

$$\boxed{(D \bmod q)^{-1} (H^2 \text{RHom}(V_q, \mathcal{J})(S_j, 0))_1 = \left[\frac{(-1)^{j+1} j!}{4} t^{r-1-j} \chi_\zeta(b+t) \right]_{\mathbb{C}[t]/(t^d)}}. \quad (\text{ORE8.10})$$

The second factor of $1/2$ is the actual supported half-sum; the first came from the original summation coefficient2. Thus the nonzero extension is located in the complete existing supported-cone map, including its degree and its full original factors.

The induced maps in degrees $-1, 0$ are identity, and the degree-one map is $0 \rightarrow \ker q_Y$. In degree3 the map is the actual restriction $(\chi_{\text{dil}} H')/q \rightarrow (\chi_{\text{dil}} E')/q$. Since q is surjective on $\tilde{S}$ by ORE3.1, this last restriction is an isomorphism. These statements retain all H, H' terms and their complete extra coordinates.

159.10 ORE9. Exact source, support and prime-action preservation

The group action on every finite term and connecting map is the original ORE4.3. The polynomial $q(G)$ commutes with it; ORE2.3 proves equivariance of δ_q , and all cochain constructions in ORE8 use equivariant maps. The twist $\chi_{\text{dil}}(a) = a$ is uniformly present in every dual term. The four original endpoint characters $1, a, a, 1$ become $a, 1, 1, a$ after the specified contragredient and twist. Both extra dual copies keep their full Schwartz and endpoint actions, rather than being replaced by four characters.

The receiving ring $\mathbb{Z}^3$ acts through its first scalar coordinate on the original Schwartz, overlap and spectral terms, and through the other two coordinates on the respective extra copies. All displayed maps are linear on matching scalar components and preserve the endpoint/extra identity maps. Consequently the faithful source distinction between τ and integer1 remains in both full complexes. No source operation is inferred from a scalar acting identically on one spectral summand.

For the existing support lattice L_{supp} , every receiving linear map $f : V \rightarrow W$ here has the exact lift

$$G_{L_{\text{supp}}}(f)(v, \lambda) = (f(v), \lambda), \quad G_{L_{\text{supp}}}(V) = \{(0, \lambda) : \lambda \in L_{\text{supp}}\} \cup (V \times \{1_{L_{\text{supp}}}\}). \quad (\text{ORE9.1})$$

A nontop input has zero amplitude and remains zero with its original label. A nonzero output can only have top support. Substitution proves preservation of identity and composition, including all connecting-map composites. Zero amplitudes in the resolution or cochain boundary retain their labels; they are not identified with primitive τ . Independently labelled factors of P and P' remain separate factors, not a newly identified common-label product.

159.11 ORE10. The mathematical consequence of the completed calculation

The original transpose Σ' has an explicitly calculated fixed-order connecting map. At an actual zero of order m , its cyclic rank is $\min(m, r)$, and the entire class is the Gamma-ratio germ ORE5.7, including the original summation factor, raw derivative factorial, reflected parameter and signs. The unrestricted finite generalized-character lift exists through the explicit length- $m + r$ block. The numerical-character truncation retains this extension whenever it retains its target; it does not require a split extension.

The exact retained supported comparison does not lose this distinction: after the two-term cyclic resolution, its degree-two map contains $-\delta_q/2$, as proved in ORE8.8–ORE8.10. Its other cohomological degrees and the original endpoint and faithful extra copies remain as displayed in ORE8.7. DPL's vanishing of the degree-one derived Hom into K_ζ is consistent with this nonzero degree-two map.

This establishes the strongest stated comparison here with Deligne's actual cross: his proof kills a higher-weight support term and proves surjectivity, while the present same-character extension has a calculated nonzero fixed-order class and an already surjective longer-character lift. No numerical weight has been assigned to primitive Z_1/τ , and no assertion about the truth or falsehood of RH follows from this extension calculation alone.

160 The actual residue pushout of the original supported restriction

24 September 2026. Complete derivation RPC0–RPC8. This note constructs the quotient of the original continuous-dual restriction by the already defined residue-comparison image, proves its exact sequence, and calculates its canonical lifting on all locally finite generalized characters. All operations are on the original receiving spaces. It does not replace Deligne’s geometric weight theorem by a claim about an auxiliary finite matrix.

160.1 RPC0. Source, current question and human provenance

The preceding FPO/UOS continuation is progress: it constructed the entire finite prime-dilation module inside the actual relative residue-comparison cokernel and its precise supported placement. The user’s amended rule USR-64a88219a3ecddd3 then asks what would advance the target using that finding and all preceding results. The next calculation here is the actual restriction extension containing that cokernel, with an exact map from the unmodified restriction extension. This determines what the residue comparison does to lifting classes, rather than assuming that a nonzero comparison cone is an obstruction to every kind of lift.

The controlling source is the complete preserved user corpus, especially USR-4322be19bff532cd, USR-9ad1c0a2d09dba92, USR-f55d16feb948d8c2 and USR-6152e3bc6302258c; B1–B5/P1–P5 and R1 in `SOURCE_OPERATIONS_AND_PROOFS.md` were reread. We retain Z_0 absence without parity, primitive Z_1/τ without parity or addition, and the integer layer’s amounts and Z_2 data. None of the vector operations, polynomial inverses or derivatives below has primitive τ as an operand. The current retractions remain in force. Receiving polynomial torsion below means annihilation by a polynomial in a stated linear operator, not the user’s integer-amount datum.

Human geometry is Alain Connes and Caterina Consani, *Schemes over F_1 and zeta functions*, arXiv:0903.2024v3 §5. The analytic setting is Ralf Meyer, *A spectral interpretation for the zeros of the Riemann zeta function*, math/0412277v3, original author TeX, “The spectral interpretation,” lines611–738, read in full for this continuation. Meyer’s source uses a completed function; the programme’s actual Mellin image, original-zeta comparison, complete endpoint factors and stronger image theorem are retained in OMS/SSI/ASD. We do not substitute a completed arithmetic function. Pierre Deligne, *La conjecture de Weil II*, §§3.6.1–3.6.3, supplies the geometric lifting target; its complete cross diagram was reread in the retained French transcription, not described as author TeX.

The full programme inputs are ASD0–ASD14, GZR0–GZR9, DPL0–DPL9, PGD0–PGD7, FPO0–FPO8 and UOS0–UOS9. All their complete proof bodies are retained. RPC makes a new application to an explicit extension rather than assuming an unspecified extension has already split.

160.2 RPC1. The unmodified restriction extension

Keep S , the even Schwartz functions h on $\mathbb{R}$ with $h(0) = \int_{\mathbb{R}} h = 0$, and the actual rapid space

$$A = \{b \in C^\infty(\mathbb{R}_+^\times) : \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j b(u)| < \infty \text{ for every } N, j \geq 0\}.$$

Both have their original Fréchet topologies. The original map is

$$\Sigma : S \longrightarrow A, \quad \Sigma h(u) = 2 \sum_{n \geq 1} h(nu), \quad J = \Sigma S, \quad Q = A/J. \quad (\text{RPC1.1})$$

OMS/SSI proves that J is closed and $\Sigma : S \rightarrow J$ is a topological isomorphism. Thus continuous transposition gives

$$0 \longrightarrow Q' \xrightarrow{\pi'} A' \xrightarrow{\Sigma'} S' \longrightarrow 0, \quad (\text{RPC1.2})$$

where $\pi : A \rightarrow Q$. Here primes denote continuous complex-linear duals; use weak-* topologies for the exact topological statements below. To verify surjectivity, a functional on S transports continuously to J by Σ^{-1} , and complex Hahn–Banach extends it to A . Its kernel is precisely the functionals vanishing on J , which factor continuously and uniquely through Q . The quotient topology is the weak-* topology on S' by the explicit finite-dimensional restriction argument proved in ASD2; we retain that exactness rather than infer it from an algebraic dual.

Let $T_a b(u) = b(u/a)$ and $T_a h(v) = h(v/a)$ for $a > 0$. Direct summation proves $\Sigma T_a = T_a \Sigma$. On each dual use the full action

$$U_a \ell = a \ell \circ T_{a^{-1}}. \quad (\text{RPC1.3})$$

In particular Σ' , π' and every arrow of the uniformly twisted sequence are equivariant. Denote

$$Y = \chi \otimes Q', \quad E = \chi \otimes A', \quad V = \chi \otimes S', \quad \chi(a) = a, \quad i = \pi', \quad p = \Sigma'. \quad (\text{RPC1.4})$$

The generator on each dual is $G = 1 - L^t$, where $L = -u\partial_u$ on its primal space. Differentiating a dilation in $\log a$ gives this formula on every test vector. Every derivative exists and is continuous in the original test topology, so the formula also holds weakly on the duals. All maps in RPC1.2 commute with G .

160.3 RPC2. The residue image determines a concrete quotient, not an assumed perfect pairing

The original-zeta residue map is the already proved injection

$$D : Q \longrightarrow Y, \quad D[F]([K]) = \frac{1}{2\pi i} \left(\int_{2-i\infty}^{2+i\infty} - \int_{-1-i\infty}^{-1+i\infty} \right) \frac{F(s)K(1-s)}{\zeta(s)} ds. \quad (\text{RPC2.1})$$

Here F, K are the original entire Mellin representatives with every multiplicity jet and the actual quotient topology. Both upward integrals converge separately, and GZR proves nondegeneracy, continuity, descent and $DT_a = U_a D$, $DL = GD$. The pole at 1, value $\zeta(0) = -1/2$, trivial zeros outside these edges and all complete multiplier comparisons remain in those full formulas. No completion replaces the denominator.

Set

$$C = Y/DQ, \quad E_{\text{res}} = E/i(DQ). \quad (\text{RPC2.2})$$

These are algebraic quotients endowed, when topology is mentioned, with the quotient weak-* topology. In particular they need not be Hausdorff. The subgroup DQ is invariant under both G and every U_a , so the actions are well-defined on both quotients.

The map

$$\bar{i} : C \rightarrow E_{\text{res}}, \quad [y] \mapsto [iy], \quad \bar{p} : E_{\text{res}} \rightarrow V, \quad [e] \mapsto pe \quad (\text{RPC2.3})$$

gives the exact sequence

$$\boxed{0 \longrightarrow C \xrightarrow{\bar{i}} E_{\text{res}} \xrightarrow{\bar{p}} V \longrightarrow 0.} \quad (\text{RPC2.4})$$

Proof: $pi = 0$ makes $\bar{p}$ well-defined. If $[iy] = 0$, injectivity of i says $y \in DQ$, proving injectivity of $\bar{i}$. If $pe = 0$, RPC1.2 gives $e = iy$, so the kernel is its image. Surjectivity follows from that of p . These arguments also prove commutativity of the exact diagram

$$\begin{array}{ccccccccc} 0 & \rightarrow & Y & \xrightarrow{i} & E & \xrightarrow{p} & V & \rightarrow & 0 \\ & & \downarrow q & & \downarrow r & & \parallel & & \\ 0 & \rightarrow & C & \xrightarrow{\bar{i}} & E_{\text{res}} & \xrightarrow{\bar{p}} & V & \rightarrow & 0, \end{array} \quad (\text{RPC2.5})$$

where q, r are the stated quotient maps.

This is exactly the pushout along $q : Y \rightarrow C$. To prove the universal property, suppose linear maps $f : E \rightarrow B$, $g : C \rightarrow B$ satisfy $fi = gq$. For $x \in Q$, $f(iDx) = g(qDx) = 0$, so f factors

uniquely through E_{res} . On $\bar{i}(C)$ the resulting map equals g . Conversely such a factorization gives the compatible pair. The same proof uses quotient continuity when all the maps carry the displayed topologies. Equivariance is inherited by uniqueness. Thus the pushout has been constructed and checked before being used.

This is a quotient already present in the actual cochain comparison: ASD14 has $\Psi^1 = \pi' D \pi : A \rightarrow E$, and surjectivity of π gives

$$i(DQ) = \text{im } \Psi^1, \quad E_{\text{res}} = \text{coker } \Psi^1. \quad (\text{RPC2.6})$$

It has not been introduced by changing the source definition of τ or the original zero-jet ideal.

160.4 RPC3. Locate the exact quotient in the full dual differential

Keep $P = V_+ \oplus V_- \oplus V_{\text{extra}}$, with $V_{\pm} = S \oplus \mathbb{C}^2$, and every original endpoint and extra closed copy. Let $\mathcal{F} : S \rightarrow S$ be the original even Fourier transform, so $\mathcal{F}^2 = \text{id}$ and $R\Sigma = \Sigma\mathcal{F}$. The full source differential and its transpose are

$$\begin{aligned} d(v_+, v_-, w) &= \Sigma h_+ - R\Sigma h_-, \\ d'\ell &= (\Sigma'\ell, 0, 0, -\Sigma'R'\ell, 0, 0, 0_{V'_{\text{extra}}}). \end{aligned} \quad (\text{RPC3.1})$$

All signs are the actual chart orientations. Transposing $R\Sigma = \Sigma\mathcal{F}$ gives $\Sigma'R' = \mathcal{F}'\Sigma'$. Define the injective map

$$\kappa : V \longrightarrow \chi P', \quad u \longmapsto (u, 0, 0, -\mathcal{F}'u, 0, 0, 0_{V'_{\text{extra}}}). \quad (\text{RPC3.2})$$

Its first component proves injectivity, and direct substitution proves $d' = \kappa p$. Consequently the factorization through the actual cochain quotient is

$$\bar{d} : E_{\text{res}} \xrightarrow{\bar{p}} V \xrightarrow{\kappa} \chi P'. \quad (\text{RPC3.3})$$

In particular $\ker \bar{d} = \bar{i}(C)$, recovering exactly $H^1(K_{\zeta}) = C$ of ASD14. A lift of a nonzero vector in V is an element of this cochain quotient whose differential is the specified nonzero κu ; it is not incorrectly called a degree-one cocycle.

The full cone still has $H^{-1} = \ker d = H$, $H^0 = 0$, and $H^2 = \chi H'$. RPC2.4 does not delete them. The different map $\text{res}' : E \rightarrow \chi H'$ in UOS3 factors instead by the inclusion of V as the Fourier-graph dual coordinate, with all endpoint and extra coordinates zero. This is distinct from κ , whose range is in $\chi P'$ and annihilates H . Its exact formulas prevent confusing the two arrows.

160.5 RPC4. The actual finite-character lifting formula

DPL2–DPL3 proves on this precise C that every nonzero polynomial $q(G)$ is bijective, and constructs its inverse before using it. SCL's corrected whole-space division, including its highest-jet correction, is the underlying proof. We apply that proved result to RPC2.4; no new surjectivity is assumed.

Let $v \in V$ satisfy $q(G)v = 0$ for a nonzero polynomial q . Choose any actual lift $e \in E_{\text{res}}$ with $\bar{p}e = v$, using the proved surjectivity. Since $\bar{p}q(G)e = 0$, there is a unique $c_e \in C$ with

$$\bar{i}(c_e) = q(G)e.$$

Define

$$\boxed{s(v) = e - \bar{i}(q(G)|_C^{-1}c_e)}. \quad (\text{RPC4.1})$$

Every operation in this formula has its constructed domain: $\bar{i}^{-1}$ is used only on its image, and $q(G)|_C^{-1}$ is the proved inverse on C . Substitution yields

$$\bar{p}s(v) = v, \quad q(G)s(v) = 0. \quad (\text{RPC4.2})$$

If the initial lift is replaced by $e + \bar{i}(c)$, its c_e changes to $c_e + q(G)c$, so RPC4.1 gives exactly the same output. If two annihilating polynomials q_1, q_2 are used, the two outputs have the same image and their difference belongs to $\bar{i}(C)$. Their product $q_1(G)q_2(G)$ kills that difference. Its invertibility on C forces the difference to be 0. Thus the formula is independent of both choices.

For finitely many inputs, take the product of their annihilating polynomials. Independence then proves linearity by using a common polynomial and arbitrary sums of lifts in RPC4.1. The same argument proves $Gs(v) = s(Gv)$. Uniqueness has a precise scope: any polynomial-annihilated element above v must equal $s(v)$, because its difference from $s(v)$ is in C and is killed by a product polynomial. Arbitrary lifts can still differ by any element of the nonzero kernel C .

For a finite-dimensional G -invariant subspace $V_0 \subset V$, one common polynomial kills V_0 , by its finite characteristic polynomial. Choose lifts of a basis to make an ordinary linear section into E_{res} . Every linear map from a finite-dimensional vector space to a topological vector space is continuous, by writing it as a finite sum of scalar multiplications by fixed vectors. The correction RPC4.1 is continuous there: $q(G)e$ is a finite-dimensional-range map into $\bar{i}(C)$, and one may take its finitely many unique preimages in C ; the inverse on C and inclusion are continuous for the quotient topologies. Therefore $s|_{V_0}$ is an actual continuous section.

160.6 RPC5. All real actions and the complete locally finite domain

Define, in the actual dual V ,

$$V_{\text{lf}} = \{v \in V : q(G)v = 0 \text{ for some nonzero polynomial } q\}. \quad (\text{RPC5.1})$$

This is a vector subspace: a product of two annihilating polynomials kills a sum. Each of its elements lies in the finite span of its successive G -iterates. Conversely a finite-dimensional G -invariant space has a common annihilating polynomial. Thus this definition agrees with the algebraic locally finite part, without assuming the whole continuous dual is locally finite.

Every U_a commutes with G , so V_{lf} and its preimage are invariant. For any $v \in V_{\text{lf}}$, the two elements $U_a s(v)$ and $s(U_a v)$ have the same image under $\bar{p}$, and both are killed by the same polynomial that kills v . Their difference lies in $\bar{i}(C)$. The inverse on C forces

$$\boxed{U_a s(v) = s(U_a v) \quad (a > 0).} \quad (\text{RPC5.2})$$

This proves full real-group and hence every prime equivariance directly. It makes no inference that a generator determines an arbitrary non-Hausdorff quotient representation. For the source's Hausdorff continuous dual, the ordinary finite matrix exponential can also be checked on each locally finite orbit by testing the finite differential system on primal vectors, but that additional assertion is not needed in RPC5.2.

The sections on finite subspaces agree on overlaps by RPC4's uniqueness, so they give a single exact algebraic section $s : V_{\text{lf}} \rightarrow E_{\text{res}}$. With all domains explicit, the entire preimage splits:

$$\boxed{\bar{p}^{-1}(V_{\text{lf}}) \simeq C \oplus V_{\text{lf}}, \quad e \mapsto (\bar{i}^{-1}(e - s(\bar{p}e)), \bar{p}e),} \quad (\text{RPC5.3})$$

whose inverse is $(c, v) \mapsto \bar{i}(c) + s(v)$. Substitution proves both identities; equivariance follows from RPC5.2. This is an algebraic equivariant decomposition, continuous on every fixed finite target component with the stated operations. No continuity on an unconstructed topology of an infinite union is claimed.

Let $(E_{\text{res}})_{\text{lf}}$ be defined by the same polynomial condition. Its intersection with $\bar{i}(C)$ is 0 because every nonzero polynomial acts invertibly on C . RPC4 and RPC5 therefore give the precise result

$$(E_{\text{res}})_{\text{lf}} = s(V_{\text{lf}}), \quad \bar{p} : (E_{\text{res}})_{\text{lf}} \xrightarrow{\sim} V_{\text{lf}}. \quad (\text{RPC5.4})$$

The full kernel is nevertheless nonzero. FPO proves that it contains the faithful regular module of all finite prime-dilation combinations. Under RPC5.3, this whole module is $(c_r, 0)$, $r > 0$. Thus the exact same global remainder and all finite generalized-character lifts are retained together.

160.7 RPC6. Compute the actual connecting class and its pushout

For a nonzero polynomial q , let $\mathbb{C}[X]$ act by G on every space. The explicit free resolution

$$0 \longrightarrow \mathbb{C}[X] \xrightarrow{q(X)} \mathbb{C}[X] \longrightarrow \mathbb{C}[X]/(q) \longrightarrow 0 \quad (\text{RPC6.1})$$

is exact: a polynomial ring has no zero divisors, and the image is exactly the ideal divided out. Applying module homomorphisms into Y gives the two-term complex $Y \xrightarrow{q(G)} Y$; its degree-one cohomology is $Y/q(G)Y$.

Take $v \in \ker q(G)|_V$, and choose a lift $\tilde{e} \in E$ in the unmodified sequence RPC1.2. Its connecting class is exactly

$$\partial_q(v) = [i^{-1}(q(G)\tilde{e})] \in Y/q(G)Y. \quad (\text{RPC6.2})$$

Indeed $pq(G)\tilde{e} = q(G)v = 0$; changing the lift by iy changes the representative by $q(G)y$. This proves the formula and its independence without importing an unspecified obstruction class. It vanishes exactly when there is a lift in $\ker q(G)|_E$: subtract the corresponding iy to prove one direction, and use such a lift for the other.

The actual pushout diagram RPC2.5 sends it to

$$[q_C(i^{-1}(q(G)\tilde{e}))] \in C/q(G)C = 0, \quad (\text{RPC6.3})$$

where $q_C : Y \rightarrow C$ is the quotient map, distinct from the polynomial $q(X)$. The final equality follows from the already constructed inverse on C , not from an assumed weight gap. Formula RPC4.1 constructs the corresponding lift and proves the stronger uniqueness on the locally finite part.

The two relevant notions of lifting remain explicit. A vector may have a lift in a larger generalized-character block of E even when RPC6.2 is nonzero for its prescribed polynomial q . The nonzero class tests preservation of that prescribed annihilator. It does not by itself prove failure of surjectivity after allowing larger Jordan blocks with the same scalar eigenvalue. Deligne's §3.6.1 concludes surjectivity after applying the exact weight truncation in his cross diagram; it does not require an equivariant section or fixed nilpotent length for each class. The accompanying original-character derivations compute RPC6.2 from every actual zeta multiplicity and retain this distinction. No contradiction to the user's source or to Deligne's theorem is inferred from an obstruction to the stronger splitting question.

160.8 RPC7. Source action, labels and consequence for the active target

On these core terms the actual receiving ring $\mathbb{Z}^3$ acts by its first arithmetic scalar; on the full extra closed terms it has the separate two other actions. Every map in RPC1–RPC6 commutes with the scalar action. The source map $[n] = (n, 0, 0)$, $[\tau] = (1, 1, 1)$ is retained on the full complexes. These receiving images are not an identification of source elements: their actions on the extra closed copies remain different. Full prime actions are the distinct U_p in RPC1.3, not integer scalar multiplication.

For the existing support lattice, a receiving linear map f lifts by $(v, \alpha) \mapsto (f(v), \alpha)$ on $G_L(V) = \{(0, \alpha) : \alpha \in L\} \cup (V \times \{1_L\})$. Linearity sends a nontop zero to a zero with the same label, proves identity and composition, and retains injectivity where present. This supplies the exact support map for RPC2.5, RPC4.1 and RPC5.3. No source addition or parity is supplied to primitive τ .

RPC constructs the actual residue pushout, proves its full exact relation to the original dual restriction and cochain quotient, and gives every locally finite generalized-character lift with the complete real and prime action. It also places the already constructed full regular prime module in the complementary kernel. These are exact global receiving maps using the existing source comparison. Numerical geometric purity has not been derived by the polynomial-divisibility

argument. The active tau target still requires Deligne's actual class/support comparison and its geometrically established weight ranges; the present calculation makes the lifting operation and the exact change of obstruction concrete, without confusing fixed Jordan-order splitting with Deligne's surjectivity statement.

160.9 RPC8. Follow the same class through the complete supported comparison

The forward calculation now uses the entire constructed object, not just the pushout row. Write the endpoint and complete extra coefficient term as $E_{\text{end}} = \mathbb{C}^4 \oplus V_{\text{extra}}$, to distinguish it from $E = \chi A'$ in RPC1. Keep the actual two cones of ASD14 and UOS:

$$K_\zeta = [P \xrightarrow{-d} A \xrightarrow{\Psi^1} E \xrightarrow{d'} \chi P'], \quad \mathcal{L}_\zeta = [P \xrightarrow{-d} A \xrightarrow{(\Psi^1, -\Psi^1)} E^2 \xrightarrow{d'_Z} \chi P'], \quad (\text{RPC8.1})$$

in degrees $(-1, 0, 1, 2)$. The actual map $\mathcal{J}$ is the identity in degrees $-1, 0, 2$, and sends α to $(\alpha, -\alpha)$ in degree 1. All endpoint and extra copies remain in P , $H = \ker d$, and E_{end} . The half-difference and half-sum of the two degree-one dual coordinates give $H^1(\mathcal{L}_\zeta) = C \oplus Y$; the respective degree-two cohomologies are $\chi H'$ and $\chi E'_{\text{end}}$.

For $V_q = \mathbb{C}[X]/(q)$, the full Hom complex constructed from RPC6.1 has terms $K^n \oplus K^{n-1}$ and differential

$$\partial^n(u, v) = (d_K u, d_K v - (-1)^n q(G)u). \quad (\text{RPC8.2})$$

Commutation of $q(G)$ with d_K proves $\partial^{n+1}\partial^n = 0$ by substitution. The complete cohomology computation is ORE8.5–ORE8.7. Its degree-two map has the exact form

$$\boxed{\ker(q : \chi H' \rightarrow \chi H') \longrightarrow Y/qY \oplus \ker(q : \chi E'_{\text{end}} \rightarrow \chi E'_{\text{end}}), \quad (\eta, e) \longmapsto (-\partial_q(\eta)/2, e).} \quad (\text{RPC8.3})$$

Here η is the $V = \chi S'$ coordinate of the graph dual. We verify the complete sign and coefficient. Represent η by the plus-chart functional $u = (\eta, 0)$ in $\chi P'$, with all endpoint and extra entries zero. Its graph restriction is η , and $qu = 0$. If $\Sigma'\alpha = \eta$, then $d'_Z(\alpha, 0) = u$. Subtract the degree-one Hom boundary

$$\partial^1((\alpha, 0), 0) = (u, (q\alpha, 0))$$

from the image of the cycle $(u, 0)$. The result is $(0, (-q\alpha, 0))$. Set $q\alpha = iy_\alpha$. Its second component maps under the actual half-difference and half-sum to

$$([-y_\alpha/2], -y_\alpha/2) \in C \oplus Y.$$

Only the first component disappears in $C/qC = 0$. The second is precisely $-[y_\alpha]/2 = -\partial_q(\eta)/2$. A different lift changes this by qY . Endpoint and extra cycles are unchanged by the cochain map, proving their identity component in RPC8.3. Thus the pushout vanishing and the nonzero supported image refer to different, explicitly related targets.

For every actual nontrivial zero ρ of multiplicity m , put $b = 1 - \rho$, $q(X) = (X - b)^r$, $d = \min(m, r)$, and $t = L_Q - b$ on the reflected primary quotient. Retain the raw Mellin derivatives $S_j = \partial_z^j M_{S,z}|_\rho$, $0 \leq j < r$. The exact residue-transport calculation ORE4–ORE5 proves

$$(D \bmod q)^{-1} \partial_q(S_j) = \left[\frac{(-1)^j j!}{2} t^{r-1-j} \pi^{b+t-1/2} \frac{\Gamma((1-b-t)/2)}{\Gamma((b+t)/2)} \right] \in \mathbb{C}[t]/(t^d). \quad (\text{RPC8.4})$$

Every derivative of the displayed Gamma germ up to the specified order is retained. In particular the cyclic generator S_{r-1} has a nonzero class, since the Gamma ratio is nonzero at $t = 0$ in this strip. The full supported map has instead coefficient $(-1)^{j+1}j!/4$, by the proved factor $-1/2$ in RPC8.3. The first factor $1/2$ comes from the original summation map $\Sigma = 2 \sum_{n \geq 1}$; the second comes from the supported half-sum. Neither is a source normalization.

The complete least lift and all prime actions are calculated in MCL3–MCL10: a target jet S_j has least generalized-character order $m + j + 1$, and the full block has the same scalar exponent b on its kernel, middle term and quotient. ORE6 proves exactness of the numerical primary truncation on finite blocks, and calculates that it retains this entire extension whenever it retains its target. This checks a specific candidate for Deligne’s cross against the full current construction. It does not infer a failure of the source arithmetic, nor turn the already proved lift into a purity theorem. The exact next geometric comparison has to respect RPC8.1–RPC8.4, including the surviving supported class; discarding that class would change the calculated map.

161 Independent check of the actual residue pushout and character lifting

24 September 2026. RQC0–RQC4. The full proof RPC0–RPC7 was read. This check uses the actual original spaces and maps, with the independently derived original extension and complete supported-cone maps in ORE0–ORE10. It introduces no operation on primitive Z_1/τ .

161.1 RQC0. The quotient is the actual pushout

In RPC’s notation the original exact sequence is

$$0 \rightarrow Y \xrightarrow{i} E \xrightarrow{p} V \rightarrow 0, \quad Y = \chi_{\text{dil}} Q', \quad E = \chi_{\text{dil}} A', \quad V = \chi_{\text{dil}} S'.$$

The original residue injection is $D : Q \rightarrow Y$, and the defined quotients are $C = Y/DQ$ and $E_{\text{res}} = E/iDQ$. The induced map $C \rightarrow E_{\text{res}}$ is injective: if $iy = iDx$, injectivity of i implies $y = Dx$. The kernel of $E_{\text{res}} \rightarrow V$ consists of classes of elements in iY . Surjectivity follows from that of p . This verifies the exact quotient row.

For maps $f : E \rightarrow B$ and $g : C \rightarrow B$ satisfying $fi = gq_C$, one has $f(iDx) = 0$ for every $x \in Q$. Thus f factors uniquely through E_{res} and restricts to g on C . Conversely any such factorization gives the compatible pair. This verifies the claimed universal property and quotient continuity. The equality $iDQ = \text{im}(\pi' D \pi)$ uses the original surjectivity of $\pi : A \rightarrow Q$; consequently this is the actual cochain quotient already present in ASD14.

161.2 RQC1. Check the complete dual differential

The original differential satisfies

$$d'\ell = (\Sigma'\ell, 0, 0, -\Sigma'R'\ell, 0, 0, 0_{\text{extra}}).$$

From $R\Sigma = \Sigma\mathcal{F}$ one obtains $\Sigma'R' = \mathcal{F}'\Sigma'$. Hence

$$d' = \kappa p, \quad \kappa u = (u, 0, 0, -\mathcal{F}'u, 0, 0, 0_{\text{extra}}).$$

The first component proves κ injective. Therefore the induced differential on E_{res} has kernel exactly C . A lift of nonzero $u \in V$ has differential $\kappa u \neq 0$; RPC correctly does not call it a cocycle. The separate graph restriction into $\chi_{\text{dil}} H'$ appends zeros to the single S' coordinate and is not this map into $\chi_{\text{dil}} P'$. The endpoints and extra closed terms are retained in these formulas.

161.3 RQC2. Check existence, uniqueness, coherence and group action

For $v \in V$ with $q(G)v = 0$, choose $e \in E_{\text{res}}$ above v and write uniquely $q(G)e = \bar{i}c_e$. The proved polynomial inverse on the actual C gives

$$s(v) = e - \bar{i}(q(G)|_C^{-1}c_e).$$

It satisfies $\bar{p}s(v) = v$ and $q(G)s(v) = 0$. Replacing e by $e + \bar{i}c$ changes c_e by $q(G)c$, so the output is unchanged. For two annihilating polynomials, their product kills the difference of the two outputs; that difference lies in $\bar{i}C$, where the product is invertible. The outputs therefore agree.

Taking a common product polynomial proves additivity and scalar linearity. It also proves compatibility on finite subspaces and their overlaps, giving the stated algebraic section on the locally finite union. On a fixed finite target space its image is finite-dimensional and its linear formula is continuous in the stated topological vector spaces. No global topology on the infinite union is inferred.

For every $a > 0$, $U_a s(v)$ and $s(U_a v)$ have the same image and are killed by the same polynomial, because U_a commutes with G . Their difference is a polynomial-annihilated element of C , hence

zero. This proves full real and prime equivariance directly, without inferring equality of group actions merely from equality of generators on a non-Hausdorff quotient.

The decomposition of the complete preimage is now verified by the mutually inverse maps

$$e \mapsto (\bar{i}^{-1}(e - s(\bar{p}e)), \bar{p}e), \quad (c, v) \mapsto \bar{i}c + s(v).$$

The locally finite part meets C trivially, because every nonzero polynomial is invertible there. This proves all RPC4–RPC5 assertions while retaining the nonzero full kernel.

161.4 RQC3. Check the extension class and its actual pushout

The free resolution $0 \rightarrow \mathbb{C}[X] \xrightarrow{q} \mathbb{C}[X] \rightarrow \mathbb{C}[X]/(q) \rightarrow 0$ computes $\text{Ext}^1(\mathbb{C}[X]/(q), Y) = Y/qY$. For the original lift $\alpha \in E$ of v , its class is

$$\delta_q(v) = [i^{-1}q(G)\alpha].$$

Replacing the lift by iy adds $q(G)y$, proving independence and the sign. The actual pushout sends this class to

$$[q_C(i^{-1}q(G)\alpha)] \in C/q(G)C = 0.$$

The final zero follows from the constructed polynomial inverse; it does not assume $C = 0$. The correction in RQC2 realizes this vanishing as an actual lift.

The original class need not vanish. At an actual zero ρ of order m , $b = 1 - \rho$, $q = (X - b)^r$, ORE5.7 identifies its value on the raw target derivative S_{r-1} , through the residue map modulo q , with

$$\left[\frac{(-1)^{r-1}(r-1)!}{2} \pi^{b+t-1/2} \frac{\Gamma((1-b-t)/2)}{\Gamma((b+t)/2)} \right]_{\mathbb{C}[t]/(t^{\min(m,r)})}.$$

Its constant term is nonzero. RPC's vanishing is therefore an actual quotient of a calculated class, not a claim that the unmodified class had already vanished.

161.5 RQC4. Retain the full supported comparison

The entire supported-cone calculation ORE8 gives

$$H^2 \text{RHom}(V_q, K_\zeta) = \ker(q : \chi_{\text{dil}} H' \rightarrow \chi_{\text{dil}} H'),$$

$$H^2 \text{RHom}(V_q, \mathcal{L}_\zeta) = Y/qY \oplus \ker(q : \chi_{\text{dil}} E' \rightarrow \chi_{\text{dil}} E').$$

Its exact map is

$$(\eta, e) \mapsto (-\delta_q(\eta)/2, e).$$

The minus sign comes from subtracting the Hom boundary in degree 1; the factor $1/2$ comes from UOS's actual half-sum. The C/qC contribution vanishes, while the Y/qY contribution remains. RPC's pushout and this full supported map are therefore compatible, with distinct calculated targets.

No mathematical defect was found in RPC0–RPC7. Its fixed-annihilator lifting, full group equivariance and locally finite splitting have the stated scope. They do not erase the other cohomological degrees or turn the quotient statement into a proof of Deligne purity or RH.

162 The exact quotient triangle of the complete supported comparison

24 September 2026. Independent proof SCT0–SCT6.

162.1 SCT0. Relation to existing proofs and source scope

UOS5–UOS7 already constructs the complete map $\mathcal{J} : K_\zeta \rightarrow \mathcal{L}_\zeta$, both cones, their cohomology and the companion mirror. ORE8 computes the full cyclic derived-Hom map. This appendix constructs the degreewise quotient of that existing map, its exact triangle, and an explicit contraction of its cone. It then identifies the entire connecting sequence after the cyclic resolution. These are further consequences of the retained maps, not a new definition of the coefficient geometry.

The original restrictions are due to Alain Connes and Caterina Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, arXiv:0903.2024v3, §5. The exact original-space image is proved in OMS, using the framework of Ralf Meyer, arXiv:math/0412277v3. The geometric lifting argument is Pierre Deligne, *La conjecture de Weil. II*, §§3.6.1–3.6.3; its retained French transcription and exact reading coverage are recorded in ORE0. Its geometric weight bounds are not assumed for this receiving calculation.

The current corpus, original $Z_0, Z_1/\tau, Z_2$ notation, and operation prerequisites remain those of ORE0–ORE1 and `CORPUS_AND_OPERATION_RULES.md` (private source record). No addition, parity or numerical weight is assigned to primitive τ . All calculations below take place on the constructed complex receiving coefficient spaces.

162.2 SCT1. The degreewise quotient and original signs

Keep the original complete terms

$$P = V_+ \oplus V_- \oplus V_{\text{extra}}, \quad V_\pm = S \oplus \mathbb{C}^2, \quad \tilde{A} = \chi_{\text{dil}} \otimes A', \quad Y = \chi_{\text{dil}} \otimes Q', \quad \chi_{\text{dil}}(a) = a.$$

The spaces, topologies and transforms are ORE1’s original spaces. The actual restrictions are

$$\Sigma h(u) = 2 \sum_{n \geq 1} h(nu), \quad Rb(u) = u^{-1}b(u^{-1}), \quad r_+(h, c) = \Sigma h, \quad r_-(h, d) = R\Sigma h = \widehat{\Sigma} h. \quad (\text{SCT1.1})$$

Put $d(v_+, v_-, w) = r_+v_+ - r_-v_-$ and $\Psi = \pi'D_\zeta\pi$, with the original residue map. In degrees $-1, 0, 1, 2$, the existing complexes are

$$\begin{aligned} K_\zeta &= [P \xrightarrow{-d} A \xrightarrow{\Psi} \tilde{A} \xrightarrow{d'} \chi_{\text{dil}} P'], \\ \mathcal{L}_\zeta &= [P \xrightarrow{-d} A \xrightarrow{(\Psi, -\Psi)} \tilde{A}^2 \xrightarrow{d'_Z} \chi_{\text{dil}} P']. \end{aligned} \quad (\text{SCT1.2})$$

The complete final differentials are

$$\begin{aligned} d'\alpha &= (\Sigma'\alpha, 0, 0, -\Sigma'R'\alpha, 0, 0, 0_{\text{extra}}), \\ d'_Z(\alpha_+, \alpha_-) &= (\Sigma'\alpha_+, 0, 0, \Sigma'R'\alpha_-, 0, 0, 0_{\text{extra}}). \end{aligned} \quad (\text{SCT1.3})$$

The injection $\mathcal{J}$ is identity in degrees $-1, 0, 2$, and $\mathcal{J}^1\alpha = (\alpha, -\alpha)$.

Let $B = \tilde{A}[-1]$, meaning that $\tilde{A}$ is in cochain degree 1 with zero differential. Define

$$\nu^1(\alpha_+, \alpha_-) = \alpha_+ + \alpha_-, \quad \nu^n = 0 \quad (n \neq 1). \quad (\text{SCT1.4})$$

This is a cochain map because it kills $(\Psi b, -\Psi b)$. Its degree-one kernel is precisely the anti-diagonal, its other kernels are the complete terms, and its degree-one map is onto. Thus

$$\boxed{0 \rightarrow K_\zeta \xrightarrow{\mathcal{J}} \mathcal{L}_\zeta \xrightarrow{\nu} \tilde{A}[-1] \rightarrow 0} \quad (\text{SCT1.5})$$

is degreewise exact. It is continuously split as a sequence of graded vector spaces by

$$\sigma^1 \alpha = (\alpha/2, \alpha/2). \quad (\text{SCT1.6})$$

This graded section is not silently declared to be a cochain section.

162.3 SCT2. The connecting cochain and explicit contraction

The failure of σ to commute with the differential is the precise cochain

$$\boxed{\theta^1 \alpha = \left(\frac{\Sigma' \alpha}{2}, 0, 0, \frac{\Sigma' R' \alpha}{2}, 0, 0, 0_{\text{extra}} \right) \in K_{\zeta}^2.} \quad (\text{SCT2.1})$$

It defines $\theta : B \rightarrow K_{\zeta}[1]$, and $d_{\mathcal{L}} \sigma = \mathcal{J} \theta$. Since $d_B = 0$ and $K^3 = 0$, the cochain identity into the shifted complex holds.

The exact graded decomposition $\mathcal{L}_{\zeta} \simeq K_{\zeta} \oplus B$ in degree 1 is

$$(\alpha_+, \alpha_-) \mapsto \left(\frac{\alpha_+ - \alpha_-}{2}, \alpha_+ + \alpha_- \right). \quad (\text{SCT2.2})$$

Its inverse is $(k, b) \mapsto (k+b/2, -k+b/2)$; in other degrees use identity on the K_{ζ} term. Substitution in every differential gives

$$d_{\mathcal{L}}(k, b) = (d_K k + \theta b, 0). \quad (\text{SCT2.3})$$

Hence the actual exact triangle is

$$\boxed{K_{\zeta} \xrightarrow{\mathcal{J}} \mathcal{L}_{\zeta} \xrightarrow{\nu} \tilde{A}[-1] \xrightarrow{\theta} K_{\zeta}[1].} \quad (\text{SCT2.4})$$

For completeness this has a full cochain contraction. With the existing cone convention,

$$\text{Cone}(\mathcal{J})^n = \mathcal{L}_{\zeta}^n \oplus K_{\zeta}^{n+1}, \quad d(\ell, k) = (d_{\mathcal{L}} \ell + \mathcal{J} k, -d_K k),$$

define

$$\mathfrak{p}(\ell, k) = \nu \ell, \quad \mathfrak{i}(b) = (\sigma b, -\theta b). \quad (\text{SCT2.5})$$

Both are cochain maps and $\mathfrak{p}\mathfrak{i} = \text{id}_B$. Their signs follow directly from $d_{\mathcal{L}} \sigma = \mathcal{J} \theta$ and $d_K \theta = 0$. Set

$$\mathfrak{h}(\ell, k) = (0, \mathcal{J}^{-1}(\ell - \sigma \nu \ell)) \in \text{Cone}(\mathcal{J})^{n-1}. \quad (\text{SCT2.6})$$

The inverse is used only on the image of $\mathcal{J}$; SCT1.5 proves that the parenthesis lies there. Write $a = \mathcal{J}^{-1}(\ell - \sigma \nu \ell)$. Then $d_{\mathcal{L}} \ell = \mathcal{J} d_K a + \mathcal{J} \theta \nu \ell$, so

$$d\mathfrak{h}(\ell, k) = (\ell - \sigma \nu \ell, -d_K a), \quad \mathfrak{h}d(\ell, k) = (0, d_K a + \theta \nu \ell + k).$$

Adding gives

$$\boxed{d\mathfrak{h} + \mathfrak{h}d = \text{id} - \mathfrak{i}\mathfrak{p}.} \quad (\text{SCT2.7})$$

Every map is continuous on the original coefficient spaces. Thus the full cone of $\mathcal{J}$ deformation-retracts onto $\tilde{A}[-1]$. Only the explicit identity-cone kernel is contracted; no retained H, H' or extra coefficient term was removed by assumption.

162.4 SCT3. Complete ordinary cohomology and the original factor2

The existing cohomology is

$$\begin{aligned} H^{-1}K_\zeta &= H^{-1}\mathcal{L}_\zeta = H, & H^0K_\zeta &= H^0\mathcal{L}_\zeta = 0, \\ H^1K_\zeta &= C_\zeta, & H^1\mathcal{L}_\zeta &= C_\zeta \oplus Y, & H^2K_\zeta &= \chi_{\text{dil}}H', & H^2\mathcal{L}_\zeta &= \chi_{\text{dil}}E'. \end{aligned}$$

Here H is the complete Fourier graph including four endpoint lines and both full extra closed copies, and $E = \mathbb{C}^4 \oplus V_{\text{extra}}$.

UOS identifies the degree-one cycle $(\pi'\lambda_+, \pi'\lambda_-)$ with

$$(c, \mu) = \left(\left[\frac{\lambda_+ - \lambda_-}{2} \right], \frac{\lambda_+ + \lambda_-}{2} \right).$$

Therefore

$$\boxed{H^1\nu(c, \mu) = 2\pi'\mu.} \quad (\text{SCT3.1})$$

The factor2 follows from the specified half-sum, with no change of coordinates.

For $\alpha \in H^1(B) = \tilde{A}$, restrict $\theta\alpha$ to the actual Fourier graph $((h, c), (\hat{h}, d), w)$. Its Schwartz value is

$$\frac{1}{2}(\Sigma'\alpha)(h) + \frac{1}{2}(\Sigma'R'\alpha)(\hat{h}) = \frac{1}{2}\alpha(\Sigma h) + \frac{1}{2}\alpha(R\Sigma\hat{h}) = \alpha(\Sigma h).$$

The equality $R\Sigma\hat{h} = \Sigma h$ uses the original Fourier and R conventions. Thus the connecting map is

$$\partial^1\alpha = (\Sigma'\alpha, 0_{\text{endpoints}}, 0_{\text{extra}}) \in \chi_{\text{dil}}H'. \quad (\text{SCT3.2})$$

The entire nontrivial part of the cohomology sequence is

$$\boxed{0 \rightarrow C_\zeta \xrightarrow{c \mapsto (c, 0)} C_\zeta \oplus Y \xrightarrow{(c, \mu) \mapsto 2\pi'\mu} \tilde{A} \xrightarrow{\Sigma' \text{ in the } S' \text{ coordinate}} \chi_{\text{dil}}H' \xrightarrow{\iota'} \chi_{\text{dil}}E' \rightarrow 0.} \quad (\text{SCT3.3})$$

The degree- -1 map is identity on H ; degree 0 is zero; all other terms vanish. Directly, $\ker \Sigma' = \pi'Y$, its image is the complete S' coordinate, and ι' removes that coordinate while retaining every endpoint and extra functional. This verifies exactness independently of the triangle formalism.

162.5 SCT4. The whole sequence after the cyclic resolution

Let $\mathcal{R} = \mathbb{C}[X]$, acting by the actual generator, and $V_q = \mathcal{R}/(q)$, $q \neq 0$. Use the explicit free resolution $\mathcal{R} \xrightarrow{q} \mathcal{R} \rightarrow V_q \rightarrow 0$. For a complex K , its Hom complex is

$$\mathcal{H}^n(K) = K^n \oplus K^{n-1}, \quad \partial^n(u, v) = (d_K u, d_K v - (-1)^n q(G_K)u). \quad (\text{SCT4.1})$$

Applying this to SCT1.5 is degreewise exact, because each Hom term is the direct sum of two original degreewise exact rows. No analytic derived-category assertion is used.

For B , the Hom complex is precisely

$$\tilde{A} \xrightarrow{+q(G_A)} \tilde{A} \quad \text{in degrees } 1, 2. \quad (\text{SCT4.2})$$

The sign is positive since $-(-1)^1 = +1$. ORE3 proves $q(G_A)$ onto on the actual continuous dual by closed polynomial division on the original test space followed by Hahn–Banach. Consequently

$$H^1\mathcal{H}(B) = \ker q_A, \quad H^n\mathcal{H}(B) = 0 \quad (n \neq 1). \quad (\text{SCT4.3})$$

Using the full cohomology proved in ORE8, the resulting middle exact sequence is

$$\boxed{0 \longrightarrow \ker q_Y \xrightarrow{2\pi'} \ker q_A \xrightarrow{\gamma \mapsto (\Sigma'\gamma, 0)} \ker(q : \chi_{\text{dil}}H' \rightarrow \chi_{\text{dil}}H') \xrightarrow{(\eta, e) \mapsto (-\delta_q(\eta)/2, e)} Y/q_Y Y \oplus \ker(q : \chi_{\text{dil}}E' \rightarrow \chi_{\text{dil}}E') \longrightarrow 0.} \quad (\text{SCT4.4})$$

Here δ_q is the original restriction connecting class of ORE2, and $\chi_{\text{dil}}H' = \tilde{S} \oplus \chi_{\text{dil}}E'$.

Check the first two arrows directly. A q -annihilated $y \in Y$ has representative $((\pi'y, \pi'y), 0)$ in $\mathcal{H}^1(\mathcal{L}_\zeta)$; its quotient is $2\pi'y$. For $\gamma \in \ker q_A$, lift it by $(\sigma\gamma, 0)$. Its differential is $(\mathcal{J}\theta\gamma, q\sigma\gamma) = (\mathcal{J}\theta\gamma, 0)$, because σ commutes with q . Hence its connecting class is $(\theta\gamma, 0)$, which restricts to $(\Sigma'\gamma, 0)$ by SCT3.2. The next map is ORE8.8. Its sign can be checked anew: extending η on the plus chart and subtracting the Hom boundary of $((\alpha, 0), 0)$, where $\Sigma'\alpha = \eta$, leaves the degree-one cycle $(-q\alpha, 0)$; its half-sum gives $-\delta_q(\eta)/2$. The C_ζ/qC_ζ component is zero by the actual polynomial inverse; the Y/qY component remains.

Thus the S' part of SCT4.4 is the original exact cyclic restriction sequence, with the factors2 and $-1/2$ required by these cone coordinates. It retains all multiplicity information in its original target Y/qY .

For completeness the other degrees are identity on $\ker(q : H \rightarrow H)$ in degree -1 , identity on H/qH in degree0, and in degree3 the map

$$(\chi_{\text{dil}}H')/q(\chi_{\text{dil}}H') \xrightarrow{\iota'} (\chi_{\text{dil}}E')/q(\chi_{\text{dil}}E'). \quad (\text{SCT4.5})$$

The latter is an isomorphism because q is onto on the complementary $\tilde{S}$ coordinate by ORE3.1. All further groups vanish. This states the complete long exact sequence, not only its finite spectral portion.

162.6 SCT5. The original-zeta value in this single triangle

At an actual nontrivial zero ρ of its full order m , set $b = 1 - \rho$, $q = (X - b)^r$, $t = L_Q - b$. For the raw original source Mellin derivatives S_j , $0 \leq j < r$, ORE4–ORE5 gives

$$(D_\zeta \bmod q)^{-1} \delta_q(S_j) = \left[\frac{(-1)^j j!}{2} t^{r-1-j} \pi^{b+t-1/2} \frac{\Gamma((1-b-t)/2)}{\Gamma((b+t)/2)} \right]_{\mathbb{C}[t]/(t^{\min(m,r)})}. \quad (\text{SCT5.1})$$

ORE4's repaired proof establishes independence directly on compact source tests of integral zero, including that constraint before using these characters.

The fourth arrow of SCT4.4 therefore sends

$$(S_j, 0) \mapsto \left(\left[\frac{(-1)^{j+1} j!}{4} t^{r-1-j} \pi^{b+t-1/2} \frac{\Gamma((1-b-t)/2)}{\Gamma((b+t)/2)} \right], 0 \right). \quad (\text{SCT5.2})$$

The first factor $1/2$ is the original factor2 in Σ ; the second is the actual supported half-sum. The Gamma ratio is holomorphic and nonzero at b . Every derivative below the stated quotient order is retained, and the map has rank $\min(m, r)$ on this source character space. Neither ρ , its multiplicity, the reflected parameter, the factorial, signs, powers of π , nor Gamma arguments have been replaced by a completed arithmetic function.

162.7 SCT6. Actions, mirror, support labels and scope

All maps commute with the original real and prime actions: every dual term carries the same uniform twist, and sum, difference, the specified scalar factors, restrictions and Ψ are equivariant. This also makes the graded section, connecting cochain and contraction equivariant. All receiving actions remain, including identity components on the faithful extra closed copies.

The exact companion mirror on $\mathcal{L}_\zeta^1$ is

$$(\alpha_+, \alpha_-) \mapsto (-R'\alpha_-, -R'\alpha_+).$$

The full sum gives the quotient mirror $-R'$ on B . The symmetric section intertwines these mirrors, since its two transformed entries are both $-R'\alpha/2$. The connecting cochain and contraction

inherit compatibility from their displayed formulas and the existing companion cochain map. On the cyclic resolution the mirror transports $q(X)$ to $q(1 - X)$, because it conjugates the generator to $1 - G$; the polynomial is not held fixed unless it has that symmetry. The original and reflected residue denominators remain companions; no self-mirror pairing is assumed.

Every receiving linear map has the exact existing label-preserving lift $(v, \lambda) \mapsto (f(v), \lambda)$. A nontop input has zero amplitude and keeps its support label; a nonzero output retains top support. The linear identities, compositions and homotopy equations hold on the corresponding fixed labelled receiving fibres, including their supported zero; primitive τ is not identified with that zero. Independently labelled factors remain separate.

This one quotient triangle makes the prior $-\delta_q/2$ placement part of a coherent complete exact sequence. Its quotient is the entire original dual overlap space $\chi_{\text{dil}}A'$ in degree 1. The ordinary cohomology has the exact factor 2 map into that space, while the cyclic sequence retains the original extension in Y/qY . The separate residue pushout sends its image to $C_\zeta/qC_\zeta = 0$, as RPC and its independent check prove. The two targets are not identified, and their exact quotient relation is not converted into a numerical purity claim.

163 Independent raw-derivative, Gamma-factor and supported-sign check

24 September 2026. Exact independent verification OEC0–OEC5 of ORE5.5–ORE5.8 and ORE8.8–ORE8.10 against the original Mellin lifts MCL. The formulas checked here are valid with their stated character twist and cochain convention. No change to ORE is made by this note.

163.1 OEC0. Objects and scope actually checked

ORIGINAL_RESTRICTION_EXTENSION_AND_DELIGNE_CROSS.md, ORE0–ORE10, was read completely. This bounded check concerns the specified residue transport and supported degree-two formulas. ORE4’s separate source-jet independence argument was corrected before integration using the full proof MCL4; this note neither edits that file nor uses its earlier independence argument.

The controlling source remains the user’s $Z_0, Z_1/\tau, Z_2$, the supplied source operations B1–B5, and the retraction of τ addition, with complete corpus references in MCL0. All calculations here occur in the already constructed complex receiving spaces after the retained arithmetic reconstruction. The original ζ , the factor2 in the Connes–Consani summation, all raw derivative factorials, the reflected variable, Gamma factors, orientation signs and source-faithful closed terms are retained.

Human-source provenance is Alain Connes and Caterina Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, arXiv:0903.2024v3, §5, for the original restrictions and Fourier geometry, and Pierre Deligne, *La conjecture de Weil. II*, §3.6, for the target lifting mechanism. The new ORE formulas are programme derivations, not formulas attributed to either human source. The full original continuous maps and their prerequisites were proved in MCL1–MCL10, GZR and UOS; no new literature equivalence is presumed here.

Fix an actual nontrivial original zero ρ of multiplicity m , put

$$b = 1 - \rho, \quad r \geq 1, \quad q(X) = (X - b)^r, \quad d = \min(m, r).$$

The actual operator on the twisted duals is

$$G = 1 - K, \quad K = L^t, \quad N = G - b = -(K - \rho). \quad (\text{OEC0.1})$$

The ordinary transpose has character a^ρ ; the contragredient followed by $\chi_{\text{dil}}(a) = a$ has character $a^{1-\rho} = a^b$. Thus the original raw derivatives

$$A_n = \partial_z^n M_{A,z}|_\rho, \quad S_n = \partial_z^n M_{S,z}|_\rho$$

satisfy

$$(K - \rho)A_n = nA_{n-1}, \quad NA_n = -nA_{n-1}, \quad (\text{OEC0.2})$$

and the identical formulas on S_n . Every sign below uses the second equation for ORE’s chosen representation.

163.2 OEC1. The lift and the residue inverse with all maps typed

Retain the exact original expansions

$$\zeta(\rho + t) = t^m u_\rho(t), \quad u_\rho(0) = \frac{\zeta^{(m)}(\rho)}{m!}, \quad \frac{1}{2u_\rho(t)} = \sum_{k \geq 0} v_k t^k. \quad (\text{OEC1.1})$$

MCL5 proves that ORE’s raw lift is exactly

$$\alpha_j = j! \sum_{k=0}^j \frac{v_k}{(m+j-k)!} A_{m+j-k}, \quad \Sigma' \alpha_j = S_j. \quad (\text{OEC1.2})$$

Indeed $v_k = v_\rho^{(k)}(0)/k!$ in MCL5.7; replacing its index k by $j - k$ gives (OEC1.2) with the full factorials unchanged.

Let $i = \pi' : Y = \chi_{\text{dil}} Q' \rightarrow \tilde{A} = \chi_{\text{dil}} A'$. For $0 \leq h < m$, $A_h/h!$ annihilates the original summation image, so the actual element on which the residue inverse acts is

$$\eta_h = i^{-1}(A_h/h!) \in Y.$$

This types ORE5.5 explicitly: the functional first enters Y by the proved annihilator identification; no inverse D^{-1} on all of A' is asserted.

At the reflected zero put $\zeta(b+t) = t^m u_b(t)$. The formula is

$$\boxed{j_b(D^{-1}\eta_h)(t) = (-1)^h u_b(t) t^{m-1-h} \pmod{t^m}.} \quad (\text{OEC1.3})$$

Here $D = D_\zeta$ is the original residue map with denominator $\zeta(s)$, not the auxiliary $F_*(s)$. To check it directly, use the actual primary representative whose germ is the right side and whose other primary germs vanish. The retained isolator theorem constructs that global representative. Its contribution against a test F is the residue at $t = 0$ of

$$\frac{(-1)^h u_b(t) t^{m-1-h} F(\rho - t)}{t^m u_b(t)} dt = (-1)^h t^{-h-1} \sum_{k \geq 0} \frac{(-1)^k F^{(k)}(\rho)}{k!} t^k dt.$$

The residue selects $k = h$, with product of signs $(-1)^{2h} = 1$, and equals $F^{(h)}(\rho)/h!$. This verifies every sign and the raw-derivative factorial. No factor $1/8$, Gamma multiplier, or completed denominator has entered this original residue map.

163.3 OEC2. Transport the full connecting class and retain its Gamma germ

For $0 \leq j < r$, repeated use of (OEC0.2) in (OEC1.2) gives

$$N^r \alpha_j = (-1)^r j! \sum_{\substack{0 \leq k \leq j \\ h = m+j-k-r \geq 0}} \frac{v_k}{h!} A_h. \quad (\text{OEC2.1})$$

Every displayed h is $< m$, since $j < r$. Thus it is in iY , and $\delta_q(S_j) = [i^{-1} N^r \alpha_j]$.

Apply (OEC1.3) term by term. For a retained summand its sign is

$$(-1)^r (-1)^h = (-1)^{m+j-k} = (-1)^{m+j} (-1)^k,$$

and its exponent is

$$m - 1 - h = r - 1 - j + k.$$

Consequently

$$(D \bmod q)^{-1} \delta_q(S_j) = \left[(-1)^{m+j} j! u_b(t) t^{r-1-j} \sum_{k=0}^j v_k (-t)^k \right]_{\mathbb{C}[t]/(t^d)}. \quad (\text{OEC2.2})$$

This writes all $0 \leq k \leq j$, including any excluded $h < 0$ terms, only after the following exact check: such an index has $r - 1 - j + k \geq m$, hence contributes zero modulo t^d , since $d \leq m$. Therefore none of these terms has been removed outside the stated quotient.

For each new index $k > j$, the exponent $r - 1 - j + k$ is at least r , hence at least d . This proves that the finite series in (OEC2.2) may be compared with the full convergent germ modulo that exact ideal. The original functional equation gives

$$\zeta(s) = \chi_\zeta(s) \zeta(1-s), \quad \chi_\zeta(s) = \pi^{s-1/2} \frac{\Gamma((1-s)/2)}{\Gamma(s/2)}.$$

Substituting $s = b + t$ retains the reflected displacement $-t$:

$$u_b(t) = (-1)^m \chi_\zeta(b+t) u_\rho(-t). \quad (\text{OEC2.3})$$

The factor $(-1)^m$ comes from $(-t)^m$, and is essential. Multiplication by the full reciprocal series now gives

$$u_b(t) \sum_{k \geq 0} v_k(-t)^k = \frac{(-1)^m}{2} \chi_\zeta(b+t).$$

Substitute this identity into (OEC2.2), keeping the two separate $(-1)^m$ factors until their product is evaluated. The result is

$$(D \bmod q)^{-1} \delta_q(S_j) = \left[\frac{(-1)^j j!}{2} t^{r-1-j} \pi^{b+t-1/2} \frac{\Gamma((1-b-t)/2)}{\Gamma((b+t)/2)} \right]_{\mathbb{C}[t]/(t^d)}. \quad (\text{OEC2.4})$$

This verifies ORE5.6 and, at $j = r - 1$, ORE5.7. Every derivative of the Gamma-ratio germ of degree below d is retained, not only its value at zero.

The quotient transport $D \bmod q$ is legitimate independently of an inverse on all Y . Its surjectivity follows from polynomial divisibility of $C_\zeta = Y/DQ$: for $y \in Y$, solve $q[c] = [y]$, choose a representative y_1 , and write $y - qy_1 = Dx$. For injectivity, $Dx = qy$ implies $q[y] = 0$ in C_ζ , hence $y = Dx_1$ by the same proved polynomial injectivity; then $x = qx_1$. This is the exact quotient isomorphism used in (OEC2.2)–(OEC2.4), with no algebraic surjectivity claim for D .

The function $\chi_\zeta(b+t)$ is holomorphic with nonzero constant term on the chosen local domain inside the critical strip. Multiplication by it is therefore invertible on this finite quotient. The images of $S_0, \dots, S_{r-1}$ in (OEC2.4) are nonzero scalar multiples of that common unit times the monomials $t^{r-1}, \dots, t^0$. MCL4 proves these source jets independent. Exactly the monomials of exponent $< d$ remain, giving

$$\text{rank } \delta_q = d, \quad \ker \delta_q = \begin{cases} 0, & r \leq m, \\ \text{span}(S_0, \dots, S_{r-m-1}), & r > m. \end{cases} \quad (\text{OEC2.5})$$

This verifies ORE5.8 with the complete kernel, not only the top class.

163.4 OEC3. Independent derivation of the supported minus sign and half-sum

Retain UOS's actual complexes in degrees $-1, 0, 1, 2$:

$$K_\zeta = [P \xrightarrow{-d} A \xrightarrow{\pi' D \pi} \tilde{A} \xrightarrow{d'} \chi_{\text{dil}} P'],$$

$$\mathcal{L}_\zeta = [P \xrightarrow{-d} A \xrightarrow{(\pi' D \pi, -\pi' D \pi)} \tilde{A}^2 \xrightarrow{d'_Z} \chi_{\text{dil}} P'].$$

The actual cone map $\mathcal{J} : K_\zeta \rightarrow \mathcal{L}_\zeta$ is identity except for its degree-one anti-diagonal. The original term

$$H' = S' \oplus (\mathbb{C}^4)' \oplus V'_{\text{extra}}$$

retains all four endpoint functionals and both complete closed extra dual copies.

For the resolution $\mathbb{C}[X] \xrightarrow{q} \mathbb{C}[X]$, the Hom cochain is

$$\mathcal{H}^n(K) = K^n \oplus K^{n-1}, \quad \partial(u, v) = (d_K u, d_K v - (-1)^n q(G_K)u). \quad (\text{OEC3.1})$$

This sign follows from $d_{\text{Hom}} f = d_K f - (-1)^n f d_{\text{resolution}}$, applied to its two components. In degree one the second component is $d_K v + qu$; in degree two it is $d_K v - qu$.

Let $\eta \in \ker q(G_S)$. Extend it to the plus-chart Schwartz coordinate

$$u = ((\eta, 0, 0), (0, 0, 0), 0) \in \chi_{\text{dil}} P'.$$

Its endpoints, minus-chart component and complete extras are zero. Restriction to the Fourier graph is exactly $(\eta, 0) \in \chi_{\text{dil}} H'$, and the plus-chart action agrees with that graph coordinate, so $qu = 0$. Thus $(u, 0)$ is a degree-two Hom cocycle for K_ζ .

Choose an actual $\alpha \in \tilde{A}$ with $\Sigma' \alpha = \eta$; MCL provides the explicit choices for all S_j . In the supported target put $w = (\alpha, 0)$. Its differential is

$$d'_Z w = (r'_+ \alpha, 0, 0) = u.$$

The plus restriction is $r'_+(\alpha) = (\Sigma' \alpha, 0, 0)$, with its two endpoint zeros retained. Formula (OEC3.1) therefore gives

$$\partial^1(w, 0) = (u, (q\alpha, 0)).$$

Subtracting this exact boundary from the image of $(u, 0)$ leaves

$$\boxed{(0, (-q\alpha, 0))}. \quad (\text{OEC3.2})$$

There is no further dual differential sign at this point: it has already entered (OEC3.1).

Write $q\alpha = iy_\alpha$. Under the exact UOS identification of degree-one cohomology,

$$[(\lambda_+, \lambda_-)] \mapsto \left(\left[\frac{\lambda_+ - \lambda_-}{2} \right], \frac{\lambda_+ + \lambda_-}{2} \right) \in C_\zeta \oplus Y,$$

the pair in (OEC3.2) is

$$([-y_\alpha/2], -y_\alpha/2). \quad (\text{OEC3.3})$$

Its first component vanishes only after taking the proved quotient $C_\zeta/qC_\zeta = 0$. The second component is $-[y_\alpha]/2 = -\delta_q(\eta)/2$ in Y/qY . Endpoint and extra functionals are direct degree-two subcomplex coordinates on both sides, and $\mathcal{J}$ acts by identity there. Thus the complete induced map is

$$\boxed{H^2 \text{RHom}(V_q, \mathcal{J})(\eta, e) = \left(-\frac{1}{2} \delta_q(\eta), e \right)}. \quad (\text{OEC3.4})$$

This verifies ORE8.8 and ORE8.9 on their exact degrees and complete terms.

Combining (OEC3.4) with (OEC2.4) proves

$$\boxed{(D \bmod q)^{-1} (H^2 \text{RHom}(V_q, \mathcal{J})(S_j, 0))_1 = \left[\frac{(-1)^{j+1} j!}{4} t^{r-1-j} \pi^{b+t-1/2} \frac{\Gamma((1-b-t)/2)}{\Gamma((b+t)/2)} \right]_{\mathbb{C}[t]/(t^d)}}. \quad (\text{OEC3.5})$$

The first $1/2$ is from the original two-sign summation. The second is the supported half-sum. The extra minus sign is from subtracting the degree-one Hom boundary. These are three distinct sources, all evaluated explicitly. This verifies ORE8.10.

163.5 OEC4. Ordinary versus twisted convention, and Fourier compatibility

If one instead uses the ordinary transpose $K = L^t$ and relation $(K - \rho)^r = 0$, the same chosen lift α_j has connecting representative

$$(K - \rho)^r \alpha_j = (-1)^r N^r \alpha_j.$$

Their underlying kernel quotients have the same subspaces because multiplication by $(-1)^r$ does not change the image of the operator. Consequently the ordinary-transpose representative transported through the same underlying D has the exact formula

$$\left[\frac{(-1)^{r+j} j!}{2} t^{r-1-j} \chi_\zeta(b+t) \right]_{\mathbb{C}[t]/(t^d)}. \quad (\text{OEC4.1})$$

This is a comparison of specified conventions, not a correction to ORE. ORE uses $G = 1 - K$ throughout, so its (OEC2.4) sign is the required one.

The original residue map intertwines

$$DL_Q = (1 - K_Q)D,$$

not $DL_Q = K_Q D$. Thus retaining the reflected parameter $b = 1 - \rho$ is necessary in the quotient transport. The whole real action on the raw derivatives is

$$\mathcal{U}_a A_j = a^{1-\rho} \sum_{k=0}^j \binom{j}{k} (-\log a)^{j-k} A_k.$$

The quotient germ $x(t) \in \mathbb{C}[t]/(t^d)$ is acted on by

$$a^{b+t} x(t) = a^b \left(\sum_{n=0}^{d-1} \frac{(\log a)^n}{n!} t^n \right) x(t) \pmod{t^d}. \quad (\text{OEC4.2})$$

Both action formulas follow from the retained intertwiner, including the opposite signs in the reflected derivative coordinate.

The Fourier comparison in MCL8 has

$$\kappa(z) = \frac{\zeta(1-z)}{\zeta(z)} = \pi^{1/2-z} \frac{\Gamma(z/2)}{\Gamma((1-z)/2)}.$$

Since $b = 1 - \rho$,

$$\chi_\zeta(b+t) = \kappa(\rho-t). \quad (\text{OEC4.3})$$

Hence the Gamma germ used in ORE is precisely the full reflected Mellin Fourier multiplier, with the displacement $-t$, not an independently chosen factor. At $r = 1, j = 0$, (OEC2.4) gives $\chi_\zeta(b)/2$, while direct raw calculation gives

$$D^{-1} \left(-\frac{m}{2\zeta^{(m)}(\rho)} i^{-1} A_{m-1} \right) \equiv \frac{(-1)^m m!}{2\zeta^{(m)}(\rho)} u_b(0) = \frac{\chi_\zeta(b)}{2} \pmod{t}.$$

This independent first-order check keeps the raw factorial and confirms the sign for every multiplicity m . The supported image is its negative half, $-\chi_\zeta(b)/4$, in agreement with (OEC3.5).

There is also a direct check of every prime logarithm in the transported matrix. Put $\ell = \log a$. Applying (OEC2.4) to the complete action on S_j gives

$$\begin{aligned} a^b \sum_{k=0}^j \binom{j}{k} (-\ell)^{j-k} \frac{(-1)^k k!}{2} t^{r-1-k} \chi_\zeta(b+t) &= a^b \frac{(-1)^j j!}{2} t^{r-1-j} \left(\sum_{h=0}^j \frac{(\ell t)^h}{h!} \right) \chi_\zeta(b+t) \\ &\equiv a^{b+t} \frac{(-1)^j j!}{2} t^{r-1-j} \chi_\zeta(b+t) \pmod{t^d}. \end{aligned} \quad (\text{OEC4.4})$$

For the last equality, every added term $h > j$ has total degree at least $r \geq d$. Thus the full action agrees at every $a > 0$, in particular at every original prime, without deriving a global group assertion only from its generator. Multiplication by the additional supported scalar $-1/2$ preserves this exact equality.

163.6 OEC5. Verification result and exact scope

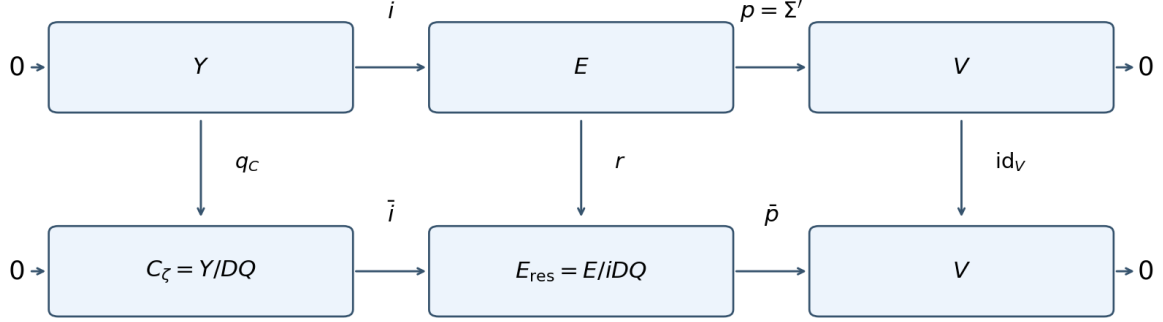
No correction to ORE5.5–ORE5.8 or ORE8.8–ORE8.10 is required. Their full factors, signs, Gamma derivatives and support placements agree with the raw Mellin lifts MCL.

Two details made explicit here are useful when carrying the formulas forward. First, the $A_h/h!$ entering D^{-1} means its actual image $i^{-1}(A_h/h!)$ in Y , justified by annihilation of the summation image. Second, replacing the finite reciprocal sum by its full germ occurs only modulo t^d , with both sets of discarded terms proved to lie in that exact ideal in OEC2.

The verified supported class is $-\delta_q/2$, in degree two of the actual resolved cone map. The unshifted character connecting class, the contragredient twisted class, and the residue-cone quotient map remain different specified maps, with their exact comparisons displayed above. None of these formulas identifies a receiving constant with primitive Z_1/τ , removes an endpoint or extra closed copy, assigns a numerical source weight, or proves RH.

The original restriction and its actual residue pushout

$$Y = \chi \otimes Q', \quad E = \chi \otimes A', \quad V = \chi \otimes S', \quad U_a = a(T_{a^{-1}})^t$$



Both rows are exact. The lower row is the pushout along the stated residue quotient.

$$\begin{aligned} q(G)v &= 0, \quad \bar{p}e = v, \quad \bar{i}c_e = q(G)e \\ s(v) &= e - \bar{i}((q(G)|_{C_\zeta})^{-1}c_e), \quad \bar{p}s(v) = v, \quad q(G)s(v) = 0 \end{aligned}$$

$$\bar{p}^{-1}(V_{\text{lf}}) \simeq C_\zeta \oplus V_{\text{lf}}, \quad U_a s(v) = s(U_a v)$$

The section is coherent on all polynomial-annihilated classes. The full kernel remains nonzero.
Its faithful prime-dilation module lies in the first summand. The full supported cone is retained.
No arithmetic operation, parity or numerical weight is assigned to primitive Z_1 / τ .

Complete proofs: RPC1–RPC6; independent check RQC0–RQC4. Original support: Connes–Consani 0903.2024v3 §5.
Analytic source: Meyer math/0412277v3. Geometric target: Deligne, Weil II §§3.6.1–3.6.3.

Figure 57: The original restriction and its actual residue pushout

164 The original character lift and its complete supported return

Figure 1. Both exact rows, the actual pushout, its cochain placement and the coherent full-group-equivariant section are proved in RPC1–RPC5, independently checked in RQC0–RQC3. Every polynomial inverse is the earlier constructed operator on the actual C_ζ . The full prime-dilation kernel remains. The source Z_0 , primitive Z_1/τ and integer Z_2 keep their original definitions and retractions; the diagram operates on the constructed receiving spaces.

Figure 2. This is an exact formula at every actual nontrivial zero, with its actual multiplicity m , not a numerical example or an assumed multiple zero. MCL5–MCL10 proves the least lift order and every real/prime action. ORE5.6–ORE5.8 and ORE8.8–ORE8.10 proves the full higher-jet class and its actual supported return. RPC6–RPC8 gives both exact receiving maps. The complete Gamma multiplier, raw derivative coefficients and signs remain in those formulas. All other cohomological degrees, four endpoint lines, and both complete extra closed copies remain in ORE8.7.

Reproducible vector/raster source: draw_actual_character_lifting.py. Both final PNGs were inspected. Its 105 exact rational checks verify finite instances of the general lift, generator, connecting-rank and residue-transport formulas; they supplement the complete proofs and are not numerical tests of RH.

Human sources: Alain Connes and Caterina Consani, 0903.2024v3 §5; Ralf Meyer, math/0412277v3, The spectral interpretation; Pierre Deligne, Weil II §§3.6.1–3.6.3. Author-source identities and the explicitly identified French transcription of Deligne are recorded with exact reading coverage in the source ledger. No geometric weight-separation or RH conclusion is supplied by these diagrams.

One lifting class, two precisely different receiving maps

$$\zeta(\rho) = 0, \quad m = \text{ord}_\rho \zeta, \quad b = 1 - \rho, \quad N = G - b$$

$$\begin{aligned} A_j &= \partial_z^j M_{A,z}|_\rho, \quad S_j = \partial_z^j M_{S,z}|_\rho, \quad NA_j = -jA_{j-1} \\ \Lambda_{\rho,0} &= \frac{A_m}{2\zeta^{(m)}(\rho)}, \quad \Sigma' \Lambda_{\rho,0} = S_0, \quad N\Lambda_{\rho,0} = -\frac{mA_{m-1}}{2\zeta^{(m)}(\rho)} \end{aligned}$$

Least generalized-character order of this lift: $m + 1$. All prime logarithms are retained in MCL7.

$$\delta_N(S_0) \in Y/NY, \quad (D \bmod N)^{-1} \delta_N(S_0) = \left[\frac{\chi_\zeta(b)}{2} \right] \neq 0$$

Actual residue pushout
 $Y/NY \rightarrow C_\zeta/NC_\zeta = 0$
 The image vanishes; the character lifts.

Full supported derived comparison
 $S_0 \mapsto -\delta_N(S_0)/2$
 $(D \bmod N)^{-1}(-\delta_N/2) = -\chi_\zeta(b)/4$

$$\chi_\zeta(b) = \pi^{b - 1/2 \frac{\Gamma((1-b)/2)}{\Gamma(b/2)}}, \quad 0 < \Re b < 1$$

This is the actual formula at every nontrivial zero, with its actual multiplicity. It is not a sample zero.
 The full order- r classes, endpoints, extra copies and all cohomological degrees are in ORE4-ORE8.
 A longer-block lift exists. Deligne's surjectivity theorem does not require a splitting with fixed block length.
 Proofs: MCL5-MCL10, ORE5.6-5.8, ORE8.8-8.10, RPC6-RPC8. Human sources: Connes-Consani, Meyer, Deligne.
 Reproducible figure and exact factor checks: draw_actual_character_lifting.py. This figure does not assert RH.

Figure 58: One exact extension class under two different receiving maps